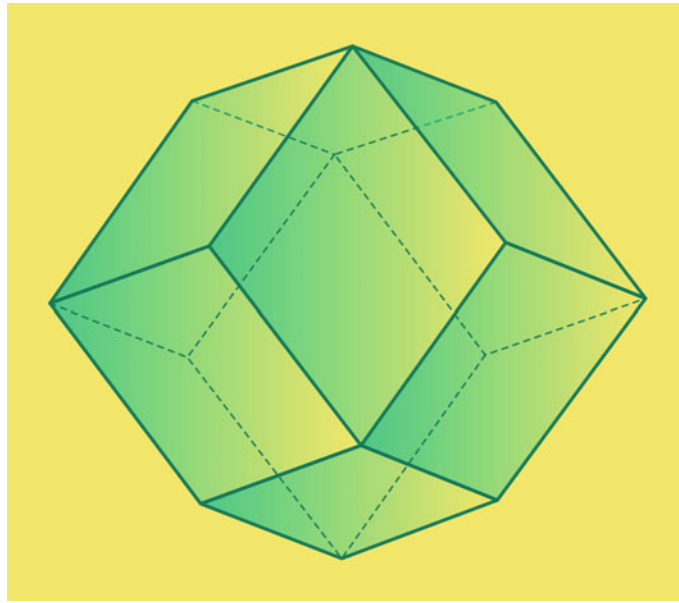
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Vitor Balestro
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Convexity from the Geometric Point of View

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This book is dedicated to our wives Mariana, Angelika, and Isabel.

Preface

The intuitive notion of *convexity* can be roughly described as follows: a set C is *convex* if the segment connecting any two points of C is contained in C. This fairly simple notion gives rise to a very extensive theory with a plethora of remarkable results known as *convex geometry* (or shorter *convexity*). As it is known today, this field has its historical origins already in ancient times, for example combined with names like Archimedes and Zenodorus. In more modern times, fundaments were laid between the second half of the nineteenth century and the first half of the twentieth century, mainly in the works of mathematicians like J. Steiner, H. Brunn, H. Minkowski, W. Blaschke, T. Bonnesen, W. Fenchel, H. Busemann, H. Hadwiger, and A. D. Alexandrov. Since then, convex geometry has been immensely developed, and nowadays it is a very active and fruitful research area, having also a lot of ramifications and relations to other areas of pure and applied mathematics (such as functional analysis, convex analysis, discrete geometry, optimization, and location science). It is natural that an area of knowledge having such depth and finding such a widespread interest stimulates the appearance of books with specific objectives and scopes, and presenting different approaches and viewpoints.

We hope that this book gives a comprehensive introduction to the "common core" of convex geometry. It is designed to be the most didactic and self-contained text that we are able to deliver, in such a way that an interested reader can read it cover-to-cover with relatively few pre-requisites (namely, standard analysis and linear algebra). We also tried to make it sufficiently complete such that the reader can acquire the needed "toolkit" to later specialize in some research subfield of convexity. Therefore, this book is mainly directed at graduate and advanced undergraduate students who want to pursue research in convexity. However, we also believe that it can be a good reference source for active researchers in convex geometry and in related areas, because we present proofs of important results in a very detailed manner and give also notes to each chapter that survey results, concepts, and methods reaching also the research front.

More than a few words are necessary regarding the existing literature in the area. As one comprehensive starting point in the twentieth century, besides the basic work of Minkowski (see [1805] and [1801]) and Blaschke's popular book [358], we should mention the monograph [446] of Bonnesen and Fenchel. On the other hand, roughly at the end of the century, we see the undoubtedly most important reference in the field, namely Schneider's outstanding book [2165]. This is a very comprehensive and deep monograph on convex geometry, providing glimpses of many ramifications of the theory. Another fundamental reference is Gruber's book [1070], which is more oriented to the discrete viewpoint of the theory (despite having also a very good overview of the basic theory). In his monograph [930], Gardner presents the field of geometric tomography which is especially rich in deep problems and results from convex geometry. Thompson gives in [2359] a comprehensive overview of Minkowski geometry, i.e., the geometry of finite-dimensional real Banach spaces which has also deep connections to the geometry of convex sets. Groemer's book [1042] covers that part of convexity which can be attacked with tools like Fourier series and spherical harmonics, and various

surveys in the proceedings [1415], edited by Klee, were fundamental for the development of the whole field. The field of convex polytopes, even more combined with discrete aspects, is excellently represented by the books [1085] of Grünbaum, [2558] of Ziegler, and [242] of Barvinok (the latter also discussing general convex bodies). Further important and nice references are Ball's booklet [199] and the very recent monograph [1263] of Hug and Weil, both also suitable as lecture notes. Moreover, the books [496] by Brazitikos et al., [133] and [134] by Artstein-Avidan, Giannopoulos, Milman, as well as [1712] by Martini, Montejano-Peimbert, and Oliveros present introductions to convex geometry as pre-requisite to neighbouring disciplines or more specialized topics (namely, asymptotic geometric analysis and the more narrow subfield treating bodies of constant width, respectively). By no means, this list of important references is complete.

Our book is (as far as it is possible for us!) completely focused on being a didactic and sufficiently complete introduction to convex geometry. In view of many thematically related books available in the literature, the crucial point justifying a book like ours can perhaps be described as follows: It contains the basic tools and notions of modern convex geometry presented in a way which is comprehensive, self-contained, and sufficiently complete such that, at the same time,

(a) it can be read even without the assistance of a specialist, and without the necessity of digging through the literature to get a clear acquaintance with concepts and proofs, and

(b) it is sufficiently broad such that the reader can have at least a glimpse of the whole breadth of the field and, in various subfields, can see the research front due to presented notes.

Nowadays, a newcomer has often to find paths through a large amount of scattered specialist literature in order to acknowledge the main ideas and techniques permeating a studied field, and this holds, of course, also for convex geometry. Due to this, it is our objective to provide a "stable bridge" which brings the reader, with a few pre-requisites, towards enough knowledge for starting to read research papers, and think also about research problems.

Each of the 12 chapters of the book (except for the last one) has the same structure: it contains the so-called *theory part*, consisting of several sections which are ordered by themes and are written in the theorem-proof-remarks style. It is important to say that, in great parts, the presented proofs are not original, but taken from the references mentioned above. Of course, the presentation and writing of them are new, and we have modified many details of the original proofs. But the main ideas behind most of the results were taken from the existing references. Of course, we give them always the due credit. Then, after this theory part, a section with a list of *exercises* is added. As is usual, some of them are simple verifications, and some of them are quite challenging. A good amount of exercises was taken from recent research papers, and, in some cases, they can even be thought of as very short introductions to notions which were not dealt with in the theory part (such as certain functional extensions of some geometric inequalities). Moreover, each chapter has a *notes section*, where the literature concerning the topics of the chapter is presented. We hope these comprehensive notes may provide glimpses of where the literature stands, and how it did come to the present stage. To give a picture that is as complete as possible, in most cases these notes are structured due to the following priority sequence of described publications: (parts of) books and proceedings, surveys, expository papers, and (selected) recent research papers. Which combination is chosen depends on the existing references. For example, in some cases the existent literature is so rich that only book chapters and surveys can be cited (bringing more would go beyond the scope of the monograph). In the same cases (and for the same reason), here and there undefined notions occur. However, it is easy to find their meanings. The reader should also look below at the description of the second chapter: there is something said about the content of the notes in this chapter.

Also, we added an *appendix containing the basic measure theory* (and geometric measure theory) that a convex geometer might find useful to know. Actually, this is not completely true because we use a few results which are not covered there (namely, the *Riesz representation theorem* for measures and some results for convergence of measures). We give detailed treatment to Hausdorff and Lebesgue measures, and we also explain why it does make sense to work with the $(n-1)$-dimensional Hausdorff measure in the boundary of a convex body. It is interesting to notice that all the theories could be developed using uniquely the Hausdorff measures, and this might even be our personal choice, since their definitions are very easy and broad, and they also give "for granted" the notion of "volume for objects of different dimensions". However, since a great part of the literature also deals with the classical Lebesgue measure, we think that it is adequate to work also with it when it is possible. It is useful to keep in mind that all concepts of area and volume appearing throughout the text can be (equivalently) defined using only Hausdorff measures.

In Chapter 1, we give a very short introduction to *Convex functions*, only containing the material that we need to study convex sets (except for the notes). To write this chapter, we were strongly inspired by Schneider's book [2165] and also by the book [1846] by Mordukhovich and Nam (we thank them for kindly sending us an ahead-of-print version of the book). For the sake of a rough description, we study early properties and regularity properties of convex functions, and also the extension of the gradient that they admit (the so-called *subgradient*). More than being an introduction to the related area of convex analysis, these concepts and results are useful to investigate support and gauge functions of convex sets. We also briefly study the *Legendre–Fenchel transform*, which gives a notion of duality for convex functions, and the *Prékopa–Leindler inequality*. Our main reference for this inequality is Artstein-Avidan et al. [133]. The reader should be aware of the fact that this inequality is not exactly an inequality for convex functions, but we use it later to obtain a simple proof of the classical *Brunn–Minkowski inequality*, and this is the reason why we present it in this chapter. The notes completing this chapter collect basic references (due to the large size of this field, only books and surveys) on convex functions and their applications, and an overview of the literature on the Prékopa–Leindler inequality is given, too.

The first steps towards the geometric approach to *Convex sets* are given in Chapter 2. We study elementary concepts such as convex and affine hulls, metric projections, support (by hyperplanes) and separation, and the basic notion of *convex body*, which is the most important concept in convex geometry. We also introduce important functions related to convex sets, such as the *support*, *gauge and radial functions*, as well as the idea of *polarity*. Moving forward, we introduce the normal cones at boundary points, which are related to the notion of support by hyperplanes, and use them to describe the *boundary structure of convex bodies*. A very important tool in studying convex geometry is endowing the collection of compact subsets of $\mathbb{R}^n$ with a metric (namely, the so-called *Hausdorff metric*), and we also introduce and explore this idea. The last section of this chapter can be seen as an introduction to symmetrization processes. We study the *Steiner symmetrization*, which is very useful in a lot of problems (for example, we use it to give a proof of Urysohn's inequality). For this chapter we followed multiple references, such as Artstein-Avidan [133], Gruber [1070], and Schneider [2165]. In the notes to this chapter, all these basics are not discussed (except Hausdorff metric and Steiner symmetrization), since they are, in many cases, folklore and very common. Thus, instead of discussing them also in notes, we inserted here (as subsection 2.9.1) notes on *special types of convex bodies*, following the idea of the handbook article [1177] by Heil and Martini. These special types of convex bodies are simplices, cubes, Platonic and Archimedean solids, regular polytopes and related polytope classes, Euclidean balls, ellipsoids, centrally symmetric convex bodies, bodies of constant width, diametrically complete sets, reduced bodies, zonotopes, and zonoids.

One of the important subfields of the area of convex geometry is the general theory of *convex polytopes* which, roughly speaking, are the extensions of polygons and (three-dimensional) polyhedra to higher dimensional vector spaces. Chapter 3 is presented to provide a brief introduction to this theory. We study the basic properties of polytopes and their description in terms of faces and normal cones. Further on, we introduce the concept of *strongly isomorphic polytopes*, which is used in the proof of the classical *Alexandrov–Fenchel inequality* for mixed volumes. It is important to say that this is a very introductory material, and that the general theory of convex polytopes is clearly much more extensive than the content of this chapter. As a criterion, we have included only the ideas which are necessary in other parts of the book (mainly in the theory of mixed volumes). The main reference for this chapter is Schneider [2165]. The notes mainly refer to the literature on the approximation of convex bodies by polytopes; related book parts, surveys, and recent research articles are presented.

In Chapter 4, we discuss *Volume and area*, which are central concepts in convex geometry (and this is why we dedicate an entire chapter for that, while one may consider the theory developed there also as a pre-requisite). The way that volumes and areas are regarded in convexity is heavily dependent on measure theory because these notions are mostly defined in terms of Lebesgue measures and Hausdorff measures (in contrast to what is usually done in differential geometry, for example, where volumes and areas are obtained by integration of differential forms). For this reason, the first section of the chapter gives an overview of these concepts, fixing ideas and concepts but without presenting formal proofs. On the other hand, for the reader who feels the necessity of a deeper understanding, we include an appendix with "everything that a convex geometer should know about (geometric) measure theory". We also discuss the so-called *principle of Cavalieri*, which relates the volume of a convex body to the (smaller dimensional) volumes of sections by affine subspaces, and derive some other formulas. Further on, we study the volume and area of polytopes and measures on the boundary of convex bodies (in particular, we prove that some measures defined in different ways on the unit sphere coincide). The notes to this chapter contain a survey on Cavalieri's principle, on volumes of cubes and (sections of) simplices, and on related general polytope theory.

Another topic which is very central for convex geometry is that of *geometric inequalities*. These are inequalities which relate geometric quantities such as volume, area, in- and circumradius, diameter, (minimal or mean) width, and so on. We dedicate Chapter 5 to the study of the classical geometric inequalities of convex geometry, and we give multiple proofs for the *classical Brunn–Minkowski inequality*, the "strongest" (because it clarifies the equality case) being based on Artstein-Avidan et al. [133]. We follow this reference also for the proofs of the *inequalities of Blaschke–Santaló and Urysohn*. Later we observe that the latter can also be obtained via the theory of mixed volumes. For the proof of the *Rogers–Shephard inequality*, our main reference is Schneider [2165]. The comprehensive notes present the literature on the Brunn–Minkowski inequality, the isoperimetric inequality, difference bodies and the Rogers–Shephard inequality, the Mahler product and the Blaschke–Santaló inequality, as well as mean width and Urysohn's inequality.

The powerful theory of *Mixed volumes* is developed in Chapters 6, 7, and 8. The first of these chapters is dedicated to the basic theory: we prove the *polynomiality of the volume of Minkowski combinations of compact convex sets* and define mixed volumes as the coefficients of these polynomials (following Gruber [1070]). After that, we also make clear that mixed volumes can be seen as functionals defined on (Cartesian products of) collections of compact convex sets, as it is the approach by Martini et al. [1712]. At this first moment, we also prove the *Minkowski inequalities for mixed volumes* and investigate the so-called *quermassintegrals*. These can be seen as specializations of mixed volumes which carry geometric information about convex bodies. They have the disadvantage of being "dimension-dependent" concepts, and to overcome this problem, McMullen made

a slight modification in their definition to introduce the *intrinsic volumes*. These are also (briefly) studied in Chapter 6. We also derive some important formulas for quermassintegrals, such as *Cauchy's surface area formula* (the proof contained here is, to the best knowledge of the authors, new). On the other hand, Chapter 7 is dedicated to introduce the *Mixed surface area measures*, which are Borel measures on the unit sphere $\mathbb{S}^{n-1}$ given by mixed volumes of convex bodies. In this chapter we also present the classical *Minkowski existence theorem*, which characterizes the Borel measures of $\mathbb{S}^{n-1}$ that are *surface area measures of convex bodies*. Last, we show how some mixed surface area measures are related to the differential geometric concept of *curvature* (in the differentiable case, of course). In Chapter 8, we give a proof (which is taken from Cordero-Erausquin et al. [659] and Hug and Weil [1263]) of the important *Alexandrov–Fenchel inequality*, one of the deepest results in the theory. We also give some consequences of this inequality, and we briefly discuss the equality case (a complete characterization is not yet settled in the literature). The notes to the first of these three chapters refer to *mixed volumes*, *related Minkowski inequalities*, *and quermassintegrals*. In the first two cases, they are restricted to book parts surveys, and for quermassintegrals also selected recent research papers are collected. In the second chapter notes about *Minkowski's existence theorem* and *surface area and curvature measures* can be found, and both parts take book chapters, surveys, and research papers into consideration. The third chapter contains notes on *Wulff shapes* and on the *Alexandrov–Fenchel inequality*; both parts are dominated by research papers.

It is an important aspect of the theory to understand what happens to some concepts, objects, and quantities related to convex bodies under *affine transformations* of the considered ambient vector space (which, for us, are compositions of translations and *invertible* linear transformations). It is natural to investigate this since, in some sense, affine transformations "preserve the geometry" (for example, they always preserve parallelism and concurrence, and also preserve volume up to a constant scalar multiplication). We dedicate Chapter 9 to this investigation. There we prove the existence and uniqueness of the *John* and the *Löwner ellipsoids*, and we also give a characterization of the case where the John ellipsoid is the Euclidean unit ball. This characterization is known as *John's theorem*. We present a proof based on Gruber's book [1070], and also a beautiful partial proof extracted from Artstein-Avidan et al. [133]. A consequence of this theorem for the so-called *Banach–Mazur distance* between convex bodies is also derived. Moving forward, we investigate the behavior of some parameters of convex bodies under affine transformations (namely, surface area, mean width, and quermassintegrals). The notes of this chapter refer to the *Löwner–John ellipsoids* (and other associated ellipsoids) as well as *positions of convex bodies*. In both cases expositions are given, but also many research papers and interesting applications are discussed.

In the literature, one can find a lot of affine constructions related to convex bodies having particular interesting features, and in Chapter 10 we introduce and study some of them. Namely, we look at *zonotopes* and *zonoids*, as well as at *projection bodies*, *centroid bodies*, and *intersection bodies* associated with a given arbitrary convex body. There is a particular importance of projection bodies through convexity, perhaps because they are related to the very natural problem of finding information about a convex body from information about its projections. This is a typical task of *geometric tomography*. Therefore we cite here Gardner's book [930], which is our main reference for this chapter. Some important inequalities involving these bodies are also derived (such as the *Petty projection inequality*, which is stronger than the classical isoperimetric inequality). The notes consist of three parts referring to zonotopes, zonoids, and bodies associated with a given convex body. For zonotopes and zonoids comprehensive surveys are given, including book parts and expositions, but also the research development after older basic surveys.

The last but one chapter of the book (Chapter 11) is a collection of (relatively) recent research topics and results which rely heavily on the "core" theory developed in the previous chapters. We start this chapter by studying the *dual mixed volumes* and the L_p *Brunn–Minkowski theory*, which can be seen as extensions of the classical Brunn–Minkowski theory. They were introduced in the 1970s and in the 1990s, respectively, by Lutwak (and then investigated by many other researchers). Next we introduce the topic of *valuations* which, in brief, can be described as functionals that have a kind of "geometric meaning" (and that are defined over suitable families of sets, e.g., convex sets or polytopes). There is a vast, deep, and exciting literature on valuations, but here we restrict ourselves to give a complete and detailed proof of the classical *theorem of Hadwiger*, which states that, under certain conditions, any valuation defined on the family of compact convex sets is a combination of intrinsic volumes (for this task, we follow the book [1263] of Hug and Weil, where this theorem is brilliantly presented). The last topic that we deal with is the *Bourgain–Milman inequality*, which is related to the classical Mahler conjecture. The proof that we follow is via *isomorphic symmetrizations*, and it is based on Artstein-Avidan et al. [133]. It is important to mention that the choice of the topics in this chapter is guided by the knowledge and interest of the authors, and therefore a lot of further nice and important recent developments are not covered (as a straightforward example we can cite the applications of Fourier analysis to convex geometry). The notes to this chapter are dedicated to three important notions: dual mixed volumes, valuations, and the Bourgain–Milman inequality. In the first two parts, survey-like representations, book parts, and recent research contributions are presented, treating also generalizations, refinements, special aspects, and applications of these two notions. Examples of respective notions are dual Brunn–Minkowski theory, dual Orlicz–Minkowski inequality, tensor valuations, and valuations in the theory of (lattice) polytopes, in stochastic geometry and in local stereology. In the third part, a relatively short representation of the important publications around the Bourgain–Milman inequality is given.

Now we come to the last chapter. While we were working on the book, one of the most impressive experiences we had was the unexpected observation how much, especially within the last decades, the field of convexity has grown, including also its increasing interlocking with neighboring disciplines. This motivated us to show somehow this strong development, e.g., by presenting famous contributors and their motivations, by showing a sequence of really fundamental publications and describing the interactions with neighboring areas. We observed the fast growth of overlaps with these disciplines and wanted to present also this. But since we understood very soon that we are not able to write a "history of convexity" (this demands wider methods of representation and more comparative care with mathematical and historical details), we decided to present all this knowledge at least in the style of concatenated building blocks and historical sketches. This representation starts with Greek antiquity and important progress from the seventeenth to the nineteenth century, and it continues (regarding the twentieth and twenty-first centuries) with information about important contributors, essential publications (research monographs, textbooks, problem books, proceedings, related handbooks, surveys, etc.), and closely related fields (discrete geometry, polytope theory, convex algebraic geometry, differential geometry, integral and stochastic geometry, convex analysis, and asymptotic convex geometry)

We finish this preface with some explanation about the title of this book. Despite being undoubtedly a book on convex geometry, we believe that, in principle, geometry is rather a *method* than a *subject*. Having this in mind, we think that studying convex geometry is one of the possible paths to understand the idea of convexity in its full scope and strength. We hope, then, that this book will be useful for anyone who wants to dive into this fantastic world, and not only for those who want to pursue research in convex geometry itself. Readers with the aim to capture the powerful

geometric side of convexity might certainly learn methods and get abilities which can be successfully applied in different fields of pure and applied mathematics.

The authors wish to thank André Martini for producing the 12 plates which are placed at the beginning of each chapter.

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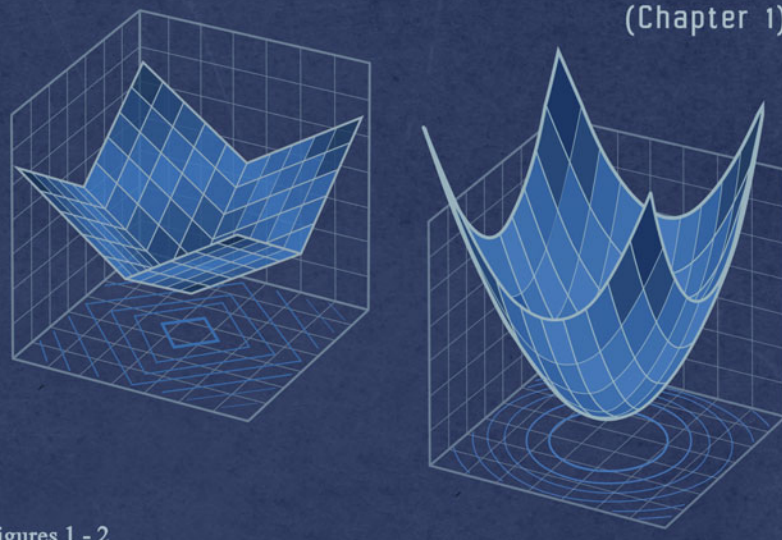
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PLATE 1

Convex functions

Two types of convex functions

(Chapter 1)



Figures 1 - 2

Multifocal ellipses as level curves of convex functions

(Fermat-Torricelli problem, see 2.9.1.6)

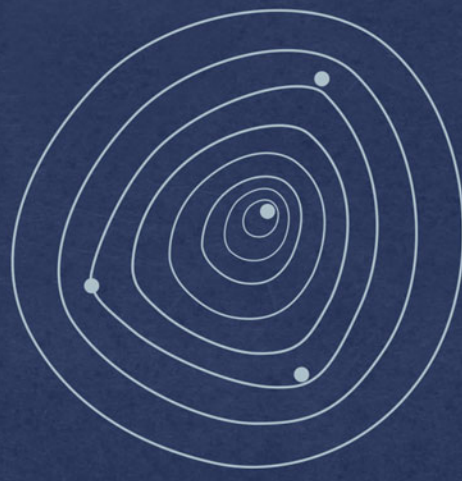


Figure 3

Convex functions ▸ (see Chapter 1 and 12.4.6)

Chapter 1
Convex functions

"Mathematics is the gate and the key to science."

— Roger Bacon

Convex functions and convex sets are somewhat interdependent. When studying convex sets, some convex functions play a very important role (mainly the *support function* of a convex set, or the *radial function* of a star-shaped set). Conversely, some geometric concepts from the theory of convex sets (such as *support* and *separation*) are very useful to understand convex functions. Plate 1 shows the graphs of two types of convex functions and (below) the level curves of a convex function used in location science (and generalizing the notion of ellipse). In this book, we are mainly interested in the geometric aspects of convexity, and hence we only introduce some basic facts on convex functions as a preliminary step. This is the reason why we start with convex functions.

As we will see in this chapter, convex functions have a lot of remarkable properties. In particular, they have "good" regularity features, and even in the nonsmooth points we can define some notion of "gradient" (the so-called *subgradient*). We also introduce some features of convex functions which are important in convex geometry, such as the *Jensen inequality* and the *Legendre transform*.

Also in this chapter we present a proof of the *Prékopa–Leindler inequality*. It is important to observe that this inequality is **not** restricted to convex functions (rather, we assume a certain "log-concavity" property of the involved functions). This inequality can be seen as a functional version of the so-called *Brunn–Minkowski inequality*, which is the most important inequality in convex geometry. We actually use the Prékopa–Leindler inequality to give a very easy proof of the Brunn–Minkowski inequality in Chapter 5, but we also give a proof of the latter that does not rely on the former. The Prékopa–Leindler inequality is also a central result to understand the geometry of the so-called *log-concave functions*. In some sense, and in some contexts, these functions "replace" convex sets leading to functional versions of geometric results. Such versions are discussed in several exercises, almost at the end of this chapter and also throughout the whole book.

The chapter ends with notes which give an overview to the literature treating the mentioned topics. We hope that, particularly regarding convex functions, this survey is also very useful for readers interested in convex analysis.

1.1 First steps

We work in the space $\mathbb{R}^n$, for some $n \in \mathbb{N}$, with its standard topology given by the *open balls* $B(x, \rho) = B_{\mathbb{R}^n}(x, \rho) := \{z \in \mathbb{R}^n : |x - z| < \rho\}$ for $x \in \mathbb{R}^n$ and $\rho > 0$. Here, $|\cdot| := \sqrt{\langle \cdot, \cdot \rangle}$ denotes the

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_1

usual *Euclidean norm* in $\mathbb{R}^n$, given by the *canonical inner product* $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, defined as

$$\langle (x_1, \ldots, x_n), (y_1, \ldots, y_n) \rangle := x_1 y_1 + \ldots + x_n y_n.$$

The (closed) *unit ball* and *unit sphere* of $\mathbb{R}^n$ are, respectively, the sets

$$B^n := \{x \in \mathbb{R}^n : |x| \leq 1\} \text{ and}$$
$$\mathbb{S}^{n-1} := \{x \in \mathbb{R}^n : |x| = 1\}.$$

The metric (and topology) considered on $\mathbb{S}^{n-1}$ will **always** be the one given from the norm of $\mathbb{R}^n$. This means that the distance between points $x, y \in \mathbb{S}^{n-1}$ is always assumed to be simply $|x - y|$. The *closed ball* with center $x \in \mathbb{R}^n$ and radius $\rho > 0$ is the closure of the open ball $B(x, \rho)$:

$$\mathrm{cl}(B(x, \rho)) = \{z \in \mathbb{R}^n : |x - z| \leq \rho\}.$$

This will be often denoted as $x + \rho B^n$. This terminology comes from the operations of *Minkowski sum* and *scalar multiplication* of sets defined as

$$A + B := \{x + y \colon x \in A \text{ and } y \in B\} \text{ and}$$
$$\lambda A := \{\lambda x : x \in A\},$$

respectively, for $A, B \subseteq \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. We will deal with these concepts in greater detail later.

For the convenience of the reader, we also recall some topological definitions that will be used throughout the text.

(i) The *interior* of a set $A \subseteq \mathbb{R}^n$, denoted by $\mathrm{int}A$, is the set of points $x \in A$ for which there is a number $\rho > 0$ such that $B(x, \rho) \subseteq A$. We say that a set is *open* if it coincides with its interior.

(ii) The *closure* of $A \subseteq \mathbb{R}^n$, denoted by $\mathrm{cl}A$, is the set of points $x \in \mathbb{R}^n$ for which there exists a sequence $(x_m)_{m \in \mathbb{N}}$ of points of A such that $x_m \to x$ as $m \to \infty$. We say that a set is *closed* if it is equal to its closure.

(iii) The *boundary* of $A \subseteq \mathbb{R}^n$, denoted by $\mathrm{bd}A$ or ∂A, is the set of points $x \in \mathbb{R}^n$ with the property that for every $\varepsilon > 0$ the open ball $B(a, \varepsilon)$ contains points of both A and $\mathbb{R}^n \setminus A$.

Remark 1.1.1. A standard topology argument gives that for any set $A \subseteq \mathbb{R}^n$ we have $\mathrm{cl}A = \mathrm{int}A \cup \partial A$. This equality will be used throughout the text.

In general, a *linear transformation* (or *linear map*) between vector spaces V and W is a map $T \colon V \to W$ such that $T(u + \lambda v) = T(u) + \lambda T(v)$ for any $u, v \in V$ and every $\lambda \in \mathbb{R}$. A *linear transformation of* $\mathbb{R}^n$ is a linear transformation $T \colon \mathbb{R}^n \to \mathbb{R}^n$. A linear map is called an *isomorphism* if it is invertible. An *affine transformation* (*map*) *of* $\mathbb{R}^n$ is a map $A \colon \mathbb{R}^n \to \mathbb{R}^n$ having the form $A(\cdot) = x_0 + T(\cdot)$ for some $x_0 \in \mathbb{R}^n$ and some isomorphism $T \colon \mathbb{R}^n \to \mathbb{R}^n$. In other words, an affine transformation is a composition of an isomorphism of $\mathbb{R}^n$ with a translation.

Before studying convex functions, we shall start by the foundational geometric idea behind convexity theory. A subset $A \subseteq \mathbb{R}^n$ is called *convex* if

$$(1 - \lambda)x + \lambda y \in A$$

for any $x, y \in A$ and $\lambda \in [0,1]$. Of course, $A \subseteq \mathbb{R}^n$ is convex if and only if $(1-\lambda)A + \lambda A = A$ for every $\lambda \in [0,1]$. In other words, a subset of $A \subseteq \mathbb{R}^n$ is *convex* when it contains the segment $\operatorname{seg}[x, y]$ for any $x, y \in A$ (see Figure 1.1.1).

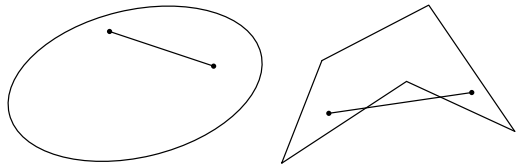


Fig. 1.1.1 A convex set and a non-convex set.

The *epigraph* of a function $f : \mathbb{R}^n \to (-\infty, +\infty]$ is the set

$$\operatorname{epi}(f) = \{(x,t) \in \mathbb{R}^{n+1} : f(x) \leq t\},$$

and f is said to be *convex* if its epigraph is a convex subset of $\mathbb{R}^{n+1}$. A function $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ is said to be *proper* if $\{f = +\infty\} := \{x \in \mathbb{R}^n : f(x) = +\infty\} \neq \mathbb{R}^n$, and *real-valued* (or *finite*) if $f(x) < +\infty$ for any $x \in \mathbb{R}^n$, that is, if the co-domain of f is (contained in) $\mathbb{R}$. Next, we present an equivalent definition for convexity of a function.

Lemma 1.1.1. *A function $f : \mathbb{R}^n \to (-\infty, +\infty]$ is convex if and only if*

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y)$$

for any $x, y \in \mathbb{R}^n$ and $\lambda \in [0,1]$. See Figure 1.1.2 for an illustration.

Proof. First, assume that f is convex. Hence, since $(x, f(x))$ and $(y, f(y))$ are obviously points of the convex set $\operatorname{epi}(f)$ for any $x, y \in \mathbb{R}^n$, we have that

$$(1-\lambda)(x, f(x)) + \lambda(y, f(y)) \in \operatorname{epi}(f)$$

for any $x, y \in \mathbb{R}^n$ and every $\lambda \in [0,1]$. The inequality follows immediately.

For the converse, let (x,t) and (y,s) be arbitrary points of $\operatorname{epi}(f)$. Then we have the inequalities $f(x) \leq t$ and $f(y) \leq s$. We have to prove that

$$(1-\lambda)(x,t) + \lambda(y,s) \in \operatorname{epi}(f)$$

for every $\lambda \in [0,1]$. Since $(1-\lambda)(x,t) + \lambda(y,s) = ((1-\lambda)x + \lambda y, (1-\lambda)t + \lambda s)$, this is the same as proving that

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)t + \lambda s.$$

But since $(1-\lambda)f(x) \leq (1-\lambda)t$ and $\lambda f(y) \leq \lambda s$, we get from the hypothesis that

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y) \leq (1-\lambda)t + \lambda s,$$

as we wished. This concludes the proof. □

Corollary 1.1.1 (Discrete Jensen's inequality). *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function. Then we have*

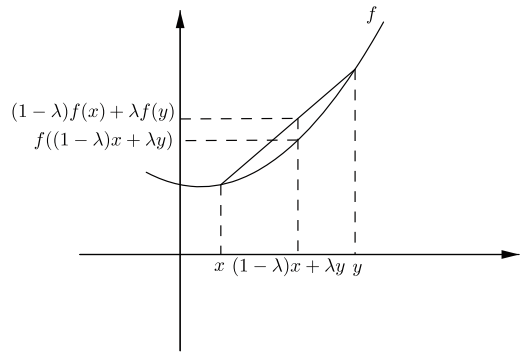


Fig. 1.1.2 How a convex function looks like.

$$f(\lambda_1 x_1 + \ldots + \lambda_m x_m) \leq \lambda_1 f(x_1) + \ldots + \lambda_m f(x_m)$$

for any $m \in \mathbb{N}$*, any points* $x_1, \ldots, x_m \in \mathbb{R}^n$*, and any numbers* $\lambda_1, \ldots, \lambda_m \in [0, +\infty)$ *such that* $\lambda_1 + \ldots + \lambda_m = 1$.

Proof. The case where $m = 1$ is trivial, and hence we assume that $m \geq 2$. We proceed by induction on m. The case $m = 2$ is precisely the previous lemma. Hence we assume that the inequality is valid for $m \in \mathbb{N}$. Let $x_1, \ldots, x_{m+1} \in \mathbb{R}^n$ be arbitrary points, and $\lambda_1, \ldots, \lambda_{m+1} \in [0, +\infty)$ be such that $\lambda_1 + \ldots + \lambda_{m+1} = 1$. Assume that $\lambda_{m+1} \neq 1$, since otherwise the desired inequality is trivial. We write

$$\sum_{j=1}^{m+1} \lambda_j x_j = \lambda_{m+1} x_{m+1} + (1 - \lambda_{m+1}) \sum_{j=1}^{m} \frac{\lambda_j}{1 - \lambda_{m+1}} x_j$$

and observe that the sum of the coefficients in the last sum equals 1:

$$\sum_{j=1}^{m} \frac{\lambda_j}{1 - \lambda_{m+1}} = \frac{\lambda_1 + \ldots + \lambda_m}{1 - \lambda_{m+1}} = \frac{1 - \lambda_{m+1}}{1 - \lambda_{m+1}} = 1.$$

Therefore, using the inequalities for 2 and for m points, respectively, we get

$$f(\lambda_1 x_1 + \ldots + \lambda_{m+1} x_{m+1}) \leq \lambda_{m+1} f(x_{m+1}) + (1 - \lambda_{m+1}) f\left(\sum_{j=1}^{m} \frac{\lambda_j}{1 - \lambda_{m+1}} x_j \right) \leq$$

$$\leq \lambda_{m+1} f(x_{m+1}) + (1 - \lambda_{m+1}) \sum_{j=1}^{m} \frac{\lambda_j}{1 - \lambda_{m+1}} f(x_j) = \lambda_1 f(x_1) + \ldots + \lambda_{m+1} f(x_{m+1}),$$

and this finishes the proof.

□

Example 1.1.1 (Affine functions). An *affine function* is a function $f : \mathbb{R}^n \to \mathbb{R}$ of the form $f(x) = \langle x, u \rangle + \alpha$, for some $u \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. Any affine function is a convex proper function. Indeed, if $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$, then

$$f((1 - \lambda)x + \lambda y) = \langle (1 - \lambda)x + \lambda y, u \rangle + \alpha = (1 - \lambda)\langle x, u \rangle + \lambda \langle y, u \rangle + (1 - \lambda)\alpha + \lambda\alpha =$$
$$= (1 - \lambda)(\langle x, u \rangle + \alpha) + \lambda(\langle y, u \rangle + \alpha) = (1 - \lambda) f(x) + \lambda f(y).$$

Proposition 1.1.1. *Assume that $f_1, f_2 : \mathbb{R}^n \to (-\infty, +\infty]$ are convex functions. Then $f_1 + f_2$ is a convex function. Moreover, if $\{f_j : j \in J\}$ is any family of convex functions, then its* superior envelope $\sup f_j : \mathbb{R}^n \to (-\infty, +\infty]$ *defined as*

$$\sup f_j(x) = \sup\{f_j(x) : j \in J\}$$

for each $x \in \mathbb{R}^n$ is also a convex function.

Proof. The fact that the sum of convex functions is convex is obvious, and the proof is left for the reader. Let $\{f_j : j \in J\}$ be a family of convex functions. If $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$, then

$$\sup f_j((1-\lambda)x + \lambda y) = \sup\{f_j((1-\lambda)x + \lambda y) : j \in J\} \leq$$
$$\leq (1-\lambda)\sup\{f_j(x) : j \in J\} + \lambda \sup\{f_j(y) : j \in J\} = (1-\lambda)\sup f_j(x) + \lambda \sup f_j(y),$$

where the inequality above comes from the fact that $f_j((1-\lambda)x + \lambda y) \leq (1-\lambda)f_j(x) + \lambda f_j(y)$ for any $j \in J$, and from elementary facts of the supremum.

□

As an easy consequence of the definition, we have that the *sublevel sets* $\{f \leq t\} := \{x \in \mathbb{R}^n : f(x) \leq t\}$ (and also sets $\{f < t\}$ defined similarly) of a convex function $f : \mathbb{R}^n \to (-\infty, +\infty]$ are convex subsets of $\mathbb{R}^n$ (the converse is not true). The *effective domain* of a function $f : \mathbb{R}^n \to (-\infty, +\infty]$ is the set

$$\mathrm{dom}(f) = \{x \in \mathbb{R}^n : f(x) < +\infty\}.$$

It is easy to see that if f is convex, then $\mathrm{dom}(f)$ is a convex set. If $x, y \in \mathrm{dom}(f)$ and $\lambda \in [0, 1]$, then we have $f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y) < +\infty$. Therefore, we have indeed that $\mathrm{seg}[x, y] \subseteq \mathrm{dom}(f)$.

The *convex hull* of a subset $A \subseteq \mathbb{R}^n$ is the set of the *convex combinations* of the points of A:

$$\mathrm{conv}(A) := \left\{ \sum_{j=1}^{m} \lambda_j x_j : x_j \in A \text{ and } \lambda_j \geq 0 \text{ for any } j = 1, \ldots, m, \text{ and } \sum_{j=1}^{m} \lambda_j = 1 \right\}. \tag{1.1.1}$$

An illustration of this concept is given in Figure 1.1.3. See also Figure 1.1.4.

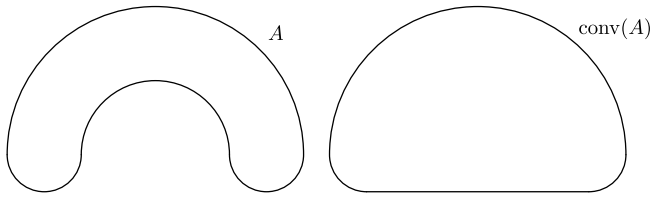


Fig. 1.1.3 A set and its convex hull.

Lemma 1.1.2. *The convex hull of any set is a convex set.*

Proof. Let $A \subseteq \mathbb{R}^n$, and assume that $x = \sum_{j=1}^{m} \beta_j x_j$ and $y = \sum_{i=1}^{k} \alpha_i y_i$ are points of $\mathrm{conv}(A)$, where all the coefficients and points are given as in (1.1.1). If $\lambda \in [0, 1]$, then

$$(1-\lambda)x+\lambda y=\sum_{j=1}^{m}(1-\lambda)\beta_j x_j+\sum_{i=1}^{k}\lambda\alpha_i y_i,$$

and we claim that this last combination is a point of $\operatorname{conv}(A)$. Indeed, we have that $(1-\lambda)\beta_j \geq 0$ for each $j=1,\dots,m$ and $\lambda\alpha_i \geq 0$ for every $i=1,\dots,k$. Further, the sum of all the coefficients is

$$\sum_{j=1}^{m}(1-\lambda)\beta_j+\sum_{i=1}^{k}\lambda\alpha_i=\sum_{j=1}^{m}\beta_j+\lambda\left(\sum_{j=1}^{m}\beta_j-\sum_{i=1}^{k}\alpha_i\right)=1+\lambda(1-1)=1.$$

Hence $(1-\lambda)x+\lambda y \in \operatorname{conv}(A)$, and the proof is done.

□

An *affine combination* of a set of points $\{x_1,\dots,x_m\} \in \mathbb{R}^n$ is a sum of the form

$$\sum_{j=1}^{m}\lambda_j x_j,$$

where each λ_j is a real number, and $\lambda_1+\ldots+\lambda_m=1$. We say that the points $x_1,\dots,x_m \in \mathbb{R}^n$ are *affinely independent* if none of them is an affine combination of the others. Observe that the cardinality of an affinely independent set in $\mathbb{R}^n$ is at most $n+1$, because if $x_0,\dots,x_m \in \mathbb{R}^n$ are $m+1$ affinely independent points, then the m vectors $x_1-x_0, x_2-x_0,\dots,x_m-x_0$ are linearly independent (see Proposition 2.1.3).

The convex hull of a finite set of points is called a *polytope* (or *convex polytope*). See Figure 1.1.4. A *k-simplex* is the convex hull of $k+1$ affinely independent points (see Figure 1.1.5). In this case, the number k is the *affine dimension* of the simplex. A simplex of maximal dimension (that is, an n-simplex in $\mathbb{R}^n$) is shortly called a *simplex*.

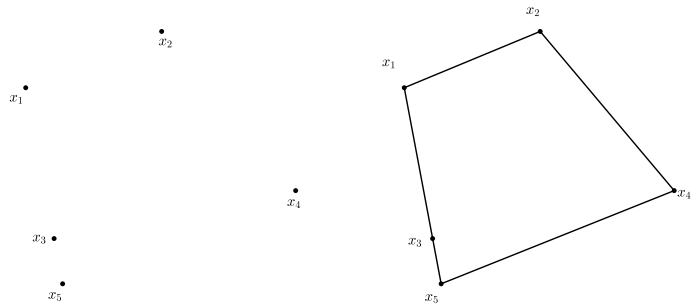


Fig. 1.1.4 The convex hull of finitely many points (that is, a *polytope*).

For the convenience of the reader, we recall that a function $f\colon A \subseteq \mathbb{R}^n \to (-\infty,\infty]$ is said to be a *Lipschitz function* if there exists a (non-negative) constant $c \in \mathbb{R}$ such that for any $x, y \in A$ we have that

$$|f(x)-f(y)| \leq c|x-y|.$$

Theorem 1.1.1. *Let $f : \mathbb{R}^n \to (-\infty,+\infty]$ be a convex function. Then f is continuous in the interior of its effective domain. Moreover, f is a Lipschitz function on any compact subset of* $\operatorname{int}(\operatorname{dom}(f))$.

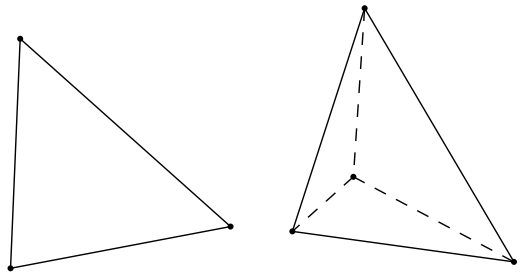


Fig. 1.1.5 A 2-simplex and a 3-simplex.

Proof. Let us start by proving that f is continuous at any point $x_0 \in \operatorname{int}(\operatorname{dom}(f))$. Choose a number $\rho > 0$ and a simplex $S = \operatorname{conv}\{x_1, \ldots, x_{n+1}\}$ such that $x_0 \in \operatorname{int}(S) \subseteq S \subseteq \operatorname{int}(\operatorname{dom}(f))$ and $\operatorname{cl}(B_{\mathbb{R}^n}(x_0, \rho)) \subseteq S$. Thus, for each $x \in S$ we can write

$$x = \sum_{j=1}^{n+1} \lambda_j x_j,$$

with $\lambda_j \geq 0$ for each $j \in \{1, \ldots, n+1\}$ and $\lambda_1 + \ldots + \lambda_{m+1} = 1$. From Jensen's inequality (Corollary 1.1.1) we get that

$$f(x) \leq \sum_{j=1}^{n+1} \lambda_j f(x_j) \leq \max\{f(x_1), \ldots, f(x_{n+1})\} =: M.$$

Now, if $y \in \operatorname{cl}(B_{\mathbb{R}^n}(x_0, \rho))$, we may write $y = x_0 + \alpha z$ for some number $\alpha \in [0, 1]$ and some vector $z \in \mathbb{R}^n$ with $|z| = \rho$. Since $y = (1 - \alpha)x_0 + \alpha(x_0 + z)$, convexity implies that

$$f(y) \leq (1 - \alpha)f(x_0) + \alpha f(x_0 + z) = f(x_0) + \alpha(f(x_0 + z) - f(x_0)) \leq f(x_0) + \alpha(M - f(x_0)),$$

where the last inequality follows from the fact that $x_0 + z \in \operatorname{cl}(B_{\mathbb{R}^n}(x_0, \rho))$ $(\subseteq S)$. Therefore, we have

$$f(y) - f(x_0) \leq \alpha(M - f(x_0)).$$

We also obtain

$$x_0 = \frac{1}{1 + \alpha} y + \frac{\alpha}{1 + \alpha}(x_0 - z),$$

and hence, again by convexity, it follows that

$$(1 + \alpha)f(x_0) \leq f(y) + \alpha f(x_0 - z) \leq f(y) + \alpha M,$$

where the last inequality comes from the fact that $x_0 - z \in \operatorname{cl}(B_{\mathbb{R}^n}(x_0, \rho))$ $(\subseteq S)$. Consequently, we have

$$f(x_0) - f(y) \leq \alpha(M - f(x_0)),$$

from which we get

$$|f(y) - f(x_0)| \leq \alpha(M - f(x_0)),$$

for any $y \in \mathrm{cl}(B_{\mathbb{R}^n}(x_0, \rho))$. Finally, we notice that $\alpha = |y - x_0|/\rho$ and rewrite the inequality above as

$$|f(y) - f(x_0)| \leq \frac{M - f(x_0)}{\rho}|y - x_0|,$$

and this clearly proves that f is continuous at $x_0 \in \mathrm{int}(\mathrm{dom}(f))$.

Now let $C \subseteq \mathrm{int}(\mathrm{dom}(f))$ be a compact subset. Since its distance to the exterior of $\mathrm{int}(\mathrm{dom}(f))$ is strictly positive (because it is the distance between a compact set and a disjoint closed set), we have that there exists a number $\rho > 0$ such that

$$C_\rho := C + \mathrm{cl}(B_{\mathbb{R}^n}(0, \rho)) \subseteq \mathrm{int}(\mathrm{dom}(f)).$$

Since f is continuous on $\mathrm{int}(\mathrm{dom}(f))$ and C_ρ is clearly compact, we get that $|f|$ attains a maximum value (M_0, say) on C_ρ. If x, y are any distinct points in C, then we have

$$z = y + \frac{\rho}{|y - x|}(y - x) \in \mathrm{cl}(B_{\mathbb{R}^n}(y, \rho)) \subseteq C_\rho.$$

Rewriting the equality as $|y - x|z = (|y - x| + \rho)y - \rho x$, we have that $y = (1 - \lambda)x + \lambda z$, where

$$\lambda = \frac{|y - x|}{|y - x| + \rho}.$$

Therefore, from the convexity of f we get that

$$f(y) \leq (1 - \lambda)f(x) + \lambda f(z) = f(x) + \lambda(f(z) - f(x)) \leq f(x) + 2M_0\lambda.$$

This implies

$$f(y) - f(x) \leq 2M_0\lambda = \frac{2M_0|y - x|}{|y - x| + \rho} \leq \frac{2M_0}{\rho}|y - x|.$$

Interchanging the roles of x and y, and repeating the argument, we have that $f(x) - f(y) \leq \beta|y - x|$ for some constant $\beta \in \mathbb{R}$ which does not depend on x and y. It follows that f is Lipschitzian on C, as we wished to prove.

□

Another important feature of convex functions is that any local minimum is a global minimum. This is formally stated and proved next.

Proposition 1.1.2. *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function. Assume that $x_0 \in \mathbb{R}^n$ is a point with the following properties:*

(i) $f(x_0) < +\infty$,

(ii) x_0 *minimizes f in some non-degenerate ball $B_{\mathbb{R}^n}(x_0, \rho)$.*

Then $f(x_0) \leq f(x)$ for every $x \in \mathbb{R}^n$. In other words, if x_0 is a local minimum, then x_0 is a global minimum.

Proof. From the hypothesis, we have that $f(x_0) \leq f(x)$ whenever $|x - x_0| \leq \rho$. Then assume that $|x - x_0| > \rho$, and let

$$y = \frac{\rho}{|x - x_0|}\, x + \left(1 - \frac{\rho}{|x - x_0|}\right) x_0.$$

We clearly have that $|y - x_0| = \rho$, from which it follows that $f(x_0) \leq f(y)$. Hence, convexity of f gives that

$$f(x_0) \leq f(y) \leq \frac{\rho}{|x - x_0|}\, f(x) + \left(1 - \frac{\rho}{|x - x_0|}\right) f(x_0),$$

implying immediately that $f(x_0) \leq f(x)$ by rearranging the terms.

□

Remark 1.1.2. A function $f\colon \mathbb{R}^n \to [-\infty, +\infty)$ is said to be *concave* if $-f$ is convex. It follows immediately that any local maximum of a concave function is a global maximum.

A function $f : \mathbb{R}^n \to (-\infty, +\infty]$ is *lower semi-continuous* or *closed* if its epigraph is a closed subset of $\mathbb{R}^{n+1}$ (see Figure 1.1.6). Below, we give several equivalent definitions for lower semi-continuity.

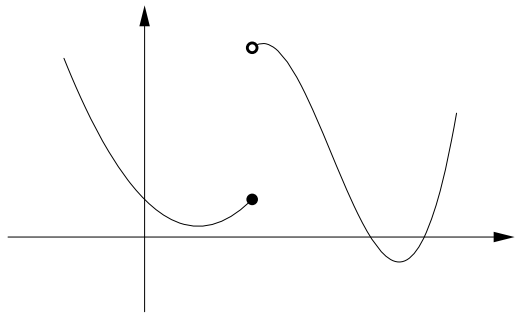


Fig. 1.1.6 A lower semi-continuous function.

Proposition 1.1.3. *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be a function. Then the following properties are equivalent:*

(a) *f is lower semi-continuous,*

(b) *for any $t \in \mathbb{R}$ the sublevel set $\{f \leq t\}$ is closed,*

(c) *for every $x \in \mathbb{R}^n$ and every $\varepsilon > 0$ there exists a neighborhood V of x such that $f(y) \geq f(x) - \varepsilon$ for any $y \in V$, and*

(d) *for any $x \in \mathbb{R}^n$ and any sequence $(x_n)_{n\in\mathbb{N}}$ in $\mathbb{R}^n$ such that $x_n \to x$ the inequality*

$$\liminf_{n\to\infty} f(x_n) \geq f(x)$$

holds.

Proof. We start with **(a)⇒(b)**. Given an arbitrary $t \in \mathbb{R}$, let $(x_n)_{n\in\mathbb{N}}$ be a sequence in the sublevel set $\{f \leq t\}$ such that $x_n \to x$ for some $x \in \mathbb{R}^n$. Since $f(x_n) \leq t$ for any $n \in \mathbb{N}$, it follows that $(x_n, t)_{n\in\mathbb{N}}$ is a sequence of points in $\mathrm{epi}(f)$, and of course we have $(x_n, t) \to (x, t)$. Since $\mathrm{epi}(f)$ is closed, we have that $(x, t) \in \mathrm{epi}(f)$, yielding $x \in \{f \leq t\}$.

Now we prove **(b)⇒(c)**. Assume that **(c)** is not true. Then there exist a point $x \in \mathbb{R}^n$ and a number $\varepsilon > 0$ with the property that every neighborhood V of x contains a point y such that $f(y) \leq f(x) - \varepsilon$. In particular, for any $n \in \mathbb{N}$ we can choose a point $y_n \in B_{\mathbb{R}^n}(x, 1/n)$ such that $f(y_n) \leq f(x) - \varepsilon$. It follows that $(y_n)_{n\in\mathbb{N}}$ is a sequence in the sublevel set $\{f \leq f(x) - \varepsilon\}$, which converges to $x \notin \{f \leq f(x) - \varepsilon\}$, and hence the sublevel set is not closed. This contradiction proves the implication.

To show **(c)⇒(d)**, notice that if $x_n \to x$ and $\liminf_{n\to\infty} f(x_n) < f(x)$, then there exists a subsequence (x_{n_k}) such that $f(x_{n_k}) \to f(x) - 2\varepsilon$ for some $\varepsilon > 0$. It follows that we have $f(x_{n_k}) < f(x) - \varepsilon$ for k sufficiently large, and this contradicts **(c)**.

It remains to prove that **(d)⇒(a)**. Let $(x_n, t_n)_{n\in\mathbb{N}}$ be a sequence in $\operatorname{epi}(f)$ such that $(x_n, t_n) \to (x, t)$. Hence we have that $x_n \to x$ and $t_n \to t$. From the hypothesis, we get that

$$f(x) \leq \liminf_{n\to\infty} f(x_n) \leq \liminf_{n\to\infty} t_n = t,$$

and therefore $(x, t) \in \operatorname{epi}(f)$. This proves that $\operatorname{epi}(f)$ is closed, and hence f is lower semicontinuous. □

Next we present some important properties of lower semi-continuous functions.

Proposition 1.1.4. **(i)** *The sum of lower semi-continuous functions is lower semi-continuous.*

(ii) *Let* $\{f_j : \mathbb{R}^n \to (-\infty, +\infty] : j \in J\}$ *be a family of lower semi-continuous functions. Then its superior envelope* $\sup_{j\in J} f_j : \mathbb{R}^n \to (-\infty, +\infty]$ *is lower semi-continuous.*

(iii) *If* $f : \mathbb{R}^n \to (-\infty, +\infty]$ *is lower semi-continuous and* $C \subseteq \mathbb{R}^n$ *is compact, then* $\inf_{x\in C} f(x)$ *is attained for some point of* C.

Proof. Let $f, g : \mathbb{R}^n \to (-\infty, +\infty]$ be lower semi-continuous functions. If $x_n \to x$ in $\mathbb{R}^n$, then

$$\liminf_{n\to\infty}(f + g)(x_n) \geq \liminf_{n\to\infty} f(x_n) + \liminf_{n\to\infty} g(x_n) \geq f(x) + g(x) = (f + g)(x),$$

and this shows **(i)**.

For **(ii)**, observe that $\sup_{j\in J} f_j(x) \leq t$ if and only if $f_j(x) \leq t$ for every $j \in J$. Therefore, we get

$$\{\sup_{j\in J} f_j \leq t\} = \bigcap_{j\in J} \{f_j \leq t\}.$$

It follows that each sublevel set $\{\sup_{j\in J} f_j \leq t\}$ is an intersection of closed sets, and hence is closed.

It remains to prove **(iii)**. Assume that $f : \mathbb{R}^n \to (-\infty, +\infty]$ is lower semi-continuous and that $C \subseteq \mathbb{R}^n$ is compact. First, let us show that $\inf_{x\in C} f(x) > -\infty$. Indeed, assume that there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in C such that $f(x_n) \to -\infty$. Passing to a subsequence if necessary, we can assume that $x_n \to x_0$ for some $x_0 \in C$. Therefore, for sufficiently large n we have that $(x_n, f(x_0) - 1) \in \operatorname{epi}(f)$. On the other hand, it is clear that $(x_n, f(x_0) - 1) \to (x_0, f(x_0) - 1) \notin \operatorname{epi}(f)$, which contradicts the fact that f is lower semi-continuous. Now assume that $(x_n)_{n\in\mathbb{N}}$ is a sequence in C such that

$$\lim_{n\to\infty} f(x_n) = \inf_{x\in C} f(x).$$

Again considering a subsequence if necessary, we may assume that $x_n \to x_0$ for a point $x_0 \in C$. Therefore, since $(x_n, f(x_n))_{n\in\mathbb{N}}$ is a sequence of points in $\operatorname{epi}(f)$ and

$$(x_n, f(x_n)) \to (x_0, \inf_{x\in C} f(x)),$$

it follows from the closedness of $\operatorname{epi}(f)$ that $f(x_0) \leq \inf_{x\in C} f(x)$. Since the inequality $f(x_0) \geq \inf_{x\in C} f(x)$ is obvious we get the desired equality.

□

1.2 Regularity

In this section we will discuss some differentiability properties of convex functions. We start with the case where the domain is one-dimensional. If $f\colon \mathbb{R} \to \mathbb{R}$ is such a function, then its *lateral derivatives* at a point $x \in \mathbb{R}$ are defined (and denoted) by

$$f'_{\pm}(x) := \lim_{t\to 0^{\pm}} \frac{f(x+t) - f(x)}{t}.$$

Recall that f is said to be *differentiable* at $x \in \mathbb{R}$ if the lateral derivatives exist and are equal.

Also, we say that f is *left continuous* if $\lim_{x\to x_0^-} f(x) = f(x_0)$ for any $x_0 \in \mathbb{R}$. Similarly, f is said to be *right continuous* if $\lim_{x\to x_0^+} f(x) = f(x_0)$ for every $x_0 \in \mathbb{R}$.

Theorem 1.2.1. *Let $f\colon \mathbb{R} \to (-\infty, +\infty]$ be a convex function. Then, in $\operatorname{int}(\operatorname{dom}(f))$, we have the following:*

(i) *the lateral derivatives f'_+ and f'_- exist and are non-decreasing.*

(ii) *the inequality $f'_-(x) \leq f'_+(x)$ holds for every $x \in \operatorname{int}(\operatorname{dom}(f))$, and equality holds except for at most countably many points.*

(iii) *the map f'_- and f'_+ are left continuous and right continuous, respectively.*

For an illustration of **(i)** and **(ii)** see Figure 1.2.1 below.

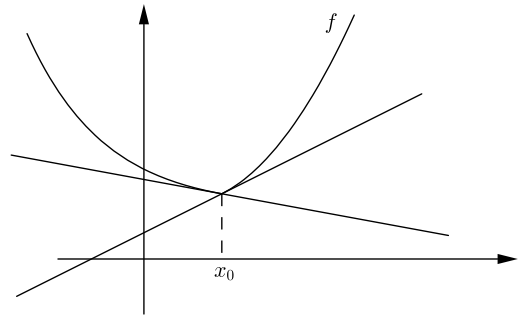


Fig. 1.2.1 A point where $f'_-(x_0) < f'_+(x_0)$.

Proof. Let $x \in \operatorname{int}(\operatorname{dom}(f))$, and let $0 < t < s$. Assume also that s is small enough such that $(x - s, x + s) \in \operatorname{int}(\operatorname{dom}(f))$. We write

$$f(x-t) = f\Big(\frac{s-t}{s}x + \frac{t}{s}(x-s)\Big) \leq \frac{s-t}{s}f(x) + \frac{t}{s}f(x-s),$$

and reorganizing the terms we get

$$\frac{f(x-s)-f(x)}{-s} \leq \frac{f(x-t)-f(x)}{-t}. \tag{1.2.1}$$

Analogously, if $0 < \lambda < \mu$ are such that $(x-\mu, x+\mu) \in \operatorname{int}(\operatorname{dom}(f))$, then

$$f(x+\lambda) = f\Big(\frac{\mu-\lambda}{\mu}x + \frac{\lambda}{\mu}(x+\mu)\Big) \leq \frac{\mu-\lambda}{\mu}f(x) + \frac{\lambda}{\mu}f(x+\mu),$$

yielding the inequality

$$\frac{f(x+\lambda)-(x)}{\lambda} \leq \frac{f(x+\mu)-f(x)}{\mu}. \tag{1.2.2}$$

Roughly speaking, inequalities (1.2.1) and (1.2.2) say that the slopes of secant lines of the graph of a convex function are monotone. These inequalities, as well as inequality (1.2.5) below, are illustrated in Figure 1.2.2.

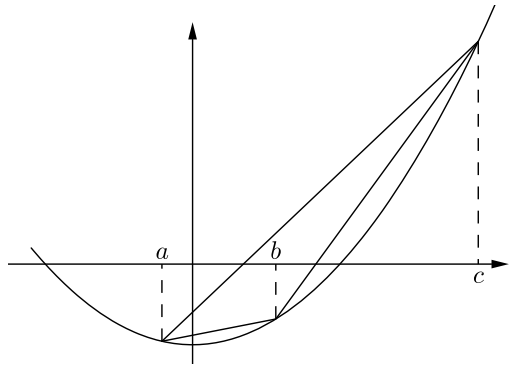


Fig. 1.2.2 We have (1.2.2), (1.2.1), and (1.2.5) for $x = a$, $x = c$, and $x = b$, respectively.

Observe that inequality (1.2.1) gives that the map

$$(0, \varepsilon) \ni t \mapsto \frac{f(x-t)-f(x)}{-t} \tag{1.2.3}$$

is non-increasing (for small $\varepsilon > 0$). Since

$$f'_-(x) = \lim_{\lambda \to 0^-} \frac{f(x+\lambda)-f(x)}{\lambda} = \lim_{t \to 0^+} \frac{f(x-t)-f(x)}{-t},$$

all we have to do for showing that the limit above exists is to verify that the function given in (1.2.3) is bounded from below. On the other hand, inequality (1.2.2) gives that the map

$$(0, \varepsilon) \ni \lambda \mapsto \frac{f(x+\lambda)-f(x)}{\lambda} \tag{1.2.4}$$

is non-decreasing. Thus, in order to prove that the limit

$$f'_+(x) = \lim_{\lambda \to 0^+} \frac{f(x+\lambda)-f(x)}{\lambda}$$

exists it suffices to show that (1.2.4) is bounded from above. We can obtain the desired bounds for (1.2.3) and (1.2.4) with a single argument. Let $\lambda, t > 0$ be arbitrary numbers such that $x-t, x+\lambda \in \operatorname{int}(\operatorname{dom}(f))$.

Then

$$\frac{\lambda}{t+\lambda}f(x)+\frac{t}{t+\lambda}f(x)=f(x)=f\Big(\frac{\lambda}{t+\lambda}(x-t)+\frac{t}{t+\lambda}(x+\lambda)\Big)\leq$$
$$\leq\frac{\lambda}{t+\lambda}f(x-t)+\frac{t}{t+\lambda}f(x+\lambda).$$

Rearranging the terms, we get the inequality

$$\frac{f(x-t)-f(x)}{-t}\leq\frac{f(x+\lambda)-f(x)}{\lambda}. \tag{1.2.5}$$

It follows that $f'_-(x)$ and $f'_+(x)$ exist. Actually, we can also use this inequality to prove that f'_- and f'_+ are non-decreasing on $\operatorname{int}(\operatorname{dom}(f))$. Indeed, it follows immediately that $f'_-(x)\leq f'_+(x)$, and if $x,z\in\operatorname{int}(\operatorname{dom}(f))$ are such that $x<z$, then we have

$$f'_-(x)\leq f'_+(x)\leq\frac{f(z)-f(x)}{z-x}\leq f'_-(z)\leq f'_+(z). \tag{1.2.6}$$

To check the inequalities in the middle, simply notice that $z=x+\lambda$ for $\lambda=z-x>0$, and $x=z-t$ for $t=\lambda$ (which is positive). Hence we get the desired from the fact that the maps (1.2.3) and (1.2.4) are non-increasing and non-decreasing, respectively.

From (1.2.6), it comes straightforwardly that if f'_- (or f'_+) is continuous at x, then $f'_-(x)=f'_+(x)$. Since f'_- (f'_+) is non-decreasing, we get that f'_- has at most countably many discontinuity points. Hence, the equality $f'_-(x)=f'_+(x)$ holds for at most countably many points of $\operatorname{int}(\operatorname{dom}(f))$. This settles **(i)** and **(ii)**.

It remains to show **(iii)**. We prove only the case for f'_+, since the other case is analogous. Let $x\in\operatorname{int}(\operatorname{dom}(f))$, and first observe that the restriction of f'_+ values which are greater than x is non-decreasing and bounded from below by $f'_+(x)$. Hence the limit $\lim_{z\to x^+}f'_+(z)$ exists. Using the continuity of f, for $x<y$ we get that

$$\frac{f(y)-f(x)}{y-x}=\lim_{z\to x^+}\frac{f(y)-f(z)}{y-z}\geq\lim_{z\to x^+}f'_+(z), \tag{1.2.7}$$

where the last inequality holds since for each $x<z<y$ we have that

$$f'_+(z)\leq\frac{f(y)-f(z)}{y-z}.$$

Observe that we are using the second inequality in (1.2.6) with other letters for the variables. Letting $y\to x^+$ in (1.2.7), we get

$$f'_+(x)=\lim_{y\to x^+}\frac{f(y)-f(x)}{y-x}\geq\lim_{z\to x^+}f'_+(z).$$

Finally, since f'_+ is non-decreasing, we have that $f'_+(x)\leq\lim_{z\to x^+}f'_+(z)$. Both inequalities yield

$$f'_+(x)=\lim_{z\to x^+}f'_+(z),$$

and hence f'_+ is right continuous at any point $x\in\operatorname{int}(\operatorname{dom}(f))$.

□

Corollary 1.2.1. *If $f\colon\mathbb{R}\to(-\infty,+\infty]$ is a convex function differentiable on* $\operatorname{int}(\operatorname{dom}(f))$*, then f is continuously differentiable (that is, of class C^1) on* $\operatorname{int}(\operatorname{dom}(f))$.

Proof. This comes immediately from the previous theorem: if f is differentiable, then $f'_- = f'_+ := f'$ on $\operatorname{int}(\operatorname{dom}(f))$. Since f'_- is left continuous and f'_+ is right continuous, it follows that f' is both left and right continuous, and hence continuous.

□

Proposition 1.2.1. *Let $J \subseteq \mathbb{R}$ be an interval, and let $f : J \to \mathbb{R}$ be a differentiable function. Then f is convex if and only if f' is non-decreasing.*

Proof. If f is convex, then f' is non-decreasing by Theorem 1.2.1 **(i)** (notice that $f' = f'_+ = f'_-$ because f is differentiable).

Now assume that f' is non-decreasing. Let $x, y \in J$ be distinct points, and let $\lambda \in (0, 1)$. By the mean value theorem, there exists a number $\mu_1 \in [x, (1-\lambda)x + \lambda y]$ such that

$$f'(\mu_1) = \frac{f((1-\lambda)x + \lambda y) - f(x)}{\lambda(y - x)},$$

and there exists a number $\mu_2 \in [(1-\lambda)x + \lambda y, y]$ for which

$$f'(\mu_2) = \frac{f(y) - f((1-\lambda)x + \lambda y)}{\lambda(y - x)}.$$

Since f' is non-decreasing, we have that $f'(\mu_2) \geq f'(\mu_1)$. This inequality gives immediately that $f((1-\lambda)x + \lambda y) \geq (1-\lambda)f(x) + \lambda f(y)$, and hence f is convex.

□

Corollary 1.2.2. *If $J \subseteq \mathbb{R}$ is an interval and $f : J \to \mathbb{R}$ is twice differentiable, then f is convex if and only if $f'' \geq 0$.*

We say that a line r *supports* a subset $A \subseteq \mathbb{R}^2$ if $r \cap A \neq \emptyset$ and A is contained in one of the two closed half-spaces determined by r (this notion is extended to higher dimensional spaces and hyperplanes in the next section). In this case, if $a \in r \cap A$, then we also say that r *supports A at a*. The next result is a useful geometric interpretation of the lateral derivatives of a convex function.

Proposition 1.2.2. *Let $f : \mathbb{R} \to (-\infty, +\infty]$ be a convex function, and let $x_0 \in \operatorname{int}(\operatorname{dom}(f))$. Then the line*

$$r(f, x_0, m) := \{(x, y) \in \mathbb{R}^2 : y - f(x_0) = m(x - x_0)\},$$

which is the line through the point $(x_0, f(x_0))$ whose inclination is m, supports $\operatorname{epi}(f)$ at $(x_0, f(x_0))$ if and only if $f'_-(x_0) \leq m \leq f'_+(x_0)$.

As a consequence, we have that for any $x_0 \in \operatorname{int}(\operatorname{dom}(f))$ the epigraph of f is supported by a non-vertical line at the point $(x_0, f(x_0))$. If $x_0 \in \operatorname{dom}(f)$, then $\operatorname{epi}(f)$ is still supported by a line at $(x_0, f(x_0))$, but we cannot guarantee that this line is non-vertical.

Remark 1.2.1. The lines drawn in Figure 1.2.1 represent the cases $m = f'_-(x_0)$ and $m = f'_+(x_0)$, and for $f'_-(x_0) < m < f'_+(x_0)$ the corresponding line lies "between" them. Hence it is geometrically clear that these lines support the epigraph of f.

Proof. First, assume that $f'_-(x_0) \leq m \leq f'_+(x_0)$. Let $x \in \operatorname{int}(\operatorname{dom}(f))$ be such that $x < x_0$. Then, by inequality (1.2.6), we have that

$$\frac{f(x_0) - f(x)}{x_0 - x} \leq f'_-(x_0) \leq m.$$

Thus, we have that $f(x) \geq f(x_0) + m(x - x_0)$. On the other hand, if $x > x_0$, then we have

$$\frac{f(x)-f(x_0)}{x-x_0} \geq f'_+(x_0) \geq m,$$

from which we also get the inequality $f(x) \geq f(x_0) + m(x - x_0)$. If $(x, y) \in \mathrm{epi}(f)$, then

$$y \geq f(x) \geq f(x_0) + m(x - x_0),$$

and hence all points of $\mathrm{epi}(f)$ lie in one of the half-spaces determined by $r(f, x_0, m)$, namely in the "upper" half-space.

For the converse, we suppose that $m < f'_-(x_0)$, because the case where $m > f'_+(x_0)$ is analogous. Of course, there exists $x < x_0$ such that

$$m < \frac{f(x_0)-f(x)}{x_0-x} \leq f'_-(x_0).$$

It follows from the first inequality above that $f(x) < f(x_0) + m(x - x_0)$. Now, if $z > x_0$, then

$$m < f'_-(x_0) \leq f'_+(x_0) \leq \frac{f(z)-f(x_0)}{z-x_0},$$

where we again used (1.2.6). Thus, $f(z) > f(x_0) + m(z - x_0)$. Consequently, the points $(x, f(x))$ and $(z, f(z))$ are points of the epigraph of f which lie in distinct half-planes determined by $r(f, x_0, m)$. Then this line does not support $\mathrm{epi}(f)$.

For any given $x_0 \in \mathrm{int}(\mathrm{dom}(f))$, we have that $f'_-(x_0) \leq f'_+(x_0)$. It follows that the epigraph of f is supported by some line in $(x_0, f(x_0))$.

It remains to deal with the case where x_0 is a non-interior point of $\mathrm{dom}(f)$. Since $\mathrm{dom}(f)$ is a convex set of $\mathbb{R}$, it clearly holds that $\mathrm{dom}(f)$ is an interval. The only possible non-interior points are its endpoints. So, we first write $\mathrm{dom}(f) = [\alpha, \beta)$, with β possibly being $+\infty$. The epigraph of f is obviously supported by the line $x = \alpha$. The case where β is the endpoint is analogous.

A remaining comment is due. In some cases, we may guarantee that f is supported by a non-vertical line in boundary points of $\mathrm{dom}(f)$. Assuming again that $\mathrm{dom}(f) = [\alpha, \beta)$, observe that any sequence $(x_m) \in (\alpha, \beta)$ converging to α is a sequence of points of $\mathrm{int}(\mathrm{dom}(f))$, and the lateral derivatives $f'_-(x_m)$ form a non-increasing sequence. As a consequence, $f'_-(x_m)$ has a limit (c, say) as $m \to +\infty$, which may possibly be $-\infty$. If $c > -\infty$ and f is lower semi-continuous at α, then for any $(x, t) \in \mathrm{epi}(f)$ we have

$$t \geq f(x_m) + f'_-(x_m)(x - x_m),$$

and hence it is clear that

$$t \geq \liminf_{m\to+\infty} f(x_m) + c(x - \alpha) \geq f(\alpha) + c(x - \alpha)$$

for any $(x, t) \in \mathrm{epi}(f)$. Thus, the line $t - f(\alpha) = c(x - \alpha)$ supports the epigraph of f at the point $(\alpha, f(\alpha))$. On the other hand, if $c = -\infty$, then one may easily check that $\mathrm{epi}(f)$ is supported by the line $x = \alpha$ at $(\alpha, f(\alpha))$.

□

As an immediate consequence, we have that the epigraph of a convex function $f\colon \mathbb{R} \to (-\infty, +\infty]$ is supported by some line at each point $(x, f(x))$ such that $x \in \mathrm{dom}(f)$). This leads to a functional version of Jensen's inequality that we state and prove now. Recall that a *probability space* is a measure space (X, Σ, μ) such that $\mu(X) = 1$. In this case, we also say that μ is a *probability measure*. To fix the notation, from now on we denote by $\mathrm{im}(f)$ the image of a function f.

Theorem 1.2.2 (Jensen's inequality)**.** *Let $D \subseteq \mathbb{R}$ be an open interval, and let $\varphi\colon D \to \mathbb{R}$ be a convex function. If X is a topological space with a Borel probability measure and $f\colon X \to D$ is an integrable function, then*

$$\varphi\left(\int_X f\,d\mu\right) \leq \int_X \varphi \circ f\,d\mu. \tag{1.2.8}$$

Equality holds if and only if φ is affine over $f(X \setminus Y)$ for some $Y \subseteq X$ such that $\mu(Y) = 0$.

Remark 1.2.2. Notice that equality holds if f is constant μ-almost everywhere. Indeed, equality condition holds trivially in this case.

Proof. Let

$$x_0 := \int_X f\,d\mu$$

be the mean of f. We claim that $x_0 \in D$. Indeed, we clearly have

$$c_0 := \inf\{f(x) : x \in X\} \leq \int_X f\,d\mu \leq \sup\{f(x) : x \in X\} =: c_1, \tag{1.2.9}$$

and $c_0, c_1 \in \mathrm{cl}(D)$. If $c_0, c_1 \in D$, then we are done. If $c_0 \notin D$ and equality holds in the left-hand inequality, then

$$\int_X f - c_0\,d\mu = 0,$$

and once $f(x) - c_0 \geq 0$ for every $x \in X$ we must have $f(x) = c_0$ for almost every $x \in X$. This is in contradiction to the fact that $c_0 \notin D$. Hence if $c_0 \notin D$ the left-hand inequality in (1.2.9) is strict. By an analogous argument, we get that if $c_1 \notin D$, then the right-hand side inequality in (1.2.9) is also strict. It follows that $x_0 \in D$, even if c_0 or c_1 are not points of D.

Because $x_0 \in D = \mathrm{int}(\mathrm{dom}(\varphi))$, we have that φ is supported by a line $x \mapsto mx + b$ at $(x_0, \varphi(x_0))$. This means that

(i) $mx_0 + b = \varphi(x_0)$ and

(ii) $\varphi(x) \geq mx + b$ for every $x \in D$.

Thus,

$$\begin{aligned}\int_X \varphi \circ f(x)\,d\mu(x) \geq \int_X mf(x) + b\,d\mu(x) = m\int_X f(x)\,d\mu(x) + b = \\ = mx_0 + b = \varphi(x_0) = \varphi\left(\int_X f\,d\mu(x)\right),\end{aligned} \tag{1.2.10}$$

and this concludes the proof of the inequality.

Now notice that equality in (1.2.8) obviously holds if φ is an affine function or if f is constant. Conversely, if equality holds, then we have from the first inequality in (1.2.10) that

$$\varphi \circ f(x) = mf(x) + b$$

for every $x \in X \setminus Y$ for some $Y \subseteq X$ such that $\mu(Y) = 0$. This clearly happens if φ, over $f(X \setminus Y)$, is the affine function $\varphi(z) = mz + b$.

□

Remark 1.2.3. We can extend Jensen's inequality (with the equality case characterization) for the case where $D \subseteq \mathbb{R}$ is any interval and $\varphi\colon D \to (-\infty, +\infty]$ is a convex function which is not necessarily real-valued. In this case, we only need to demand that

$$x_0 = \int_X f \, d\mu \in \operatorname{int}(\operatorname{dom}(\varphi)),$$

and repeat verbatim the proof above (of course, without the part where we prove that $x_0 \in \operatorname{int}(\operatorname{dom}(\varphi))$, since now we assume that as a hypothesis).

It is also worth observing that if x_0 is not a point of the effective domain of φ, then the left-hand side of (1.2.8) is $+\infty$, and hence there is no sense in bounding it from above. However, in this case it is possible to have $+\infty$ on both sides of (1.2.8) with a non-constant function f and a non-affine convex function φ (see Exercise 1.11).

We go now to the n-dimensional situation. If $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ is a convex function and $x \in \operatorname{int}(\operatorname{dom}(f))$, then for any non-zero vector $u \in \mathbb{R}^n$ the (*positive*) *one-sided directional derivative*

$$f'_+(x, u) = \lim_{\lambda \to 0^+} \frac{f(x + \lambda u) - f(x)}{\lambda}$$

exists. Indeed, for small $\varepsilon > 0$ the function $g\colon (-\varepsilon, \varepsilon) \to \mathbb{R}$ given as $g(\lambda) = f(x + \lambda u)$ is a one-variable convex function, and

$$f'_+(x, u) = \lim_{\lambda \to 0^+} \frac{f(x + \lambda u) - f(x)}{\lambda} = \lim_{\lambda \to 0^+} \frac{g(\lambda) - g(0)}{\lambda} = g'_+(0).$$

One can analogously define the *negative one-sided derivative*

$$f'_-(x, u) := \lim_{\lambda \to 0^-} \frac{f(x + \lambda u) - f(x)}{\lambda}.$$

However, it is more convenient to work only with f'_+, since f'_- can be re-obtained as

$$\begin{aligned} f'_-(x, u) = \lim_{\lambda \to 0^-} \frac{f(x + \lambda u) - f(x)}{\lambda} = \lim_{t \to 0^+} \frac{f(x - tu) - f(x)}{-t} = \\ = -\lim_{t \to 0^+} \frac{f(x + t(-u)) - f(x)}{t} = -f'_+(x, -u). \end{aligned}$$

We extend the definition of the one-sided directional derivative for the zero vector by setting $f'_+(x, 0) = 0$. Doing this, we define the *one-sided derivative map of* f *at* x to be the map $f'_+(x, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$. In what follows, recall that a function $g\colon \mathbb{R}^n \to \mathbb{R}$ is *positively homogeneous* if $g(\lambda x) = \lambda g(x)$ for every $x \in \mathbb{R}^n$ and any $\lambda \geq 0$.

Proposition 1.2.3. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function. For each $x \in \operatorname{int}(\operatorname{dom}(f))$ the one-sided derivative of f at x satisfies the properties*

i. (subadditivity) *$f'_+(x, u + v) \leq f'_+(x, u) + f'_+(x, v)$ for any $u, v \in \mathbb{R}^n$.*

ii. (positive homogeneity) *$f'_+(x, \lambda u) = \lambda f'_+(x, u)$ for any $\lambda \geq 0$ and $u \in \mathbb{R}^n$.*

Proof. For the subadditivity we simply notice that

$$f'_+(x,u+v) = \lim_{\lambda\to 0^+} \frac{f(x+\lambda(u+v))-f(x)}{\lambda} \le \lim_{\lambda\to 0^+} \frac{f\left(\frac{1}{2}(x+2\lambda u)+\frac{1}{2}(x+2\lambda v)\right)-f(x)}{\lambda} \le$$
$$\le \lim_{\lambda\to 0^+} \frac{\frac{1}{2}f(x+2\lambda u)+\frac{1}{2}f(x+2\lambda v)-\frac{1}{2}f(x)-\frac{1}{2}f(x)}{\lambda} =$$
$$= \lim_{\lambda\to 0^+} \frac{f(x+2\lambda u)-f(x)}{2\lambda} + \lim_{\lambda\to 0^+} \frac{f(x+2\lambda v)-f(x)}{2\lambda} = f'_+(x,u)+f'_+(x,v),$$

where in the first inequality we have used the convexity of f, and the last equality is justified by a simple change of variables. Now we prove that $f'_+(x,\cdot)$ is positively homogeneous, using again a change of variables. Let $u \in \mathbb{R}^n$ and $\lambda > 0$. Then

$$f'_+(x,\lambda u) = \lim_{t\to 0^+} \frac{f(x+t\lambda u)-f(x)}{t} = \lambda \lim_{s\to 0^+} \frac{f(x+su)-f(x)}{s} = \lambda f'_+(x,u),$$

where $s = \lambda t$. The case $\lambda = 0$ is immediate since, by definition, $f'_+(x,0) = 0$.

□

A *real-valued* function (that is, a function which does not assume the value $+\infty$) which is subadditive and positively homogeneous is called a *sublinear function*. That is, a sublinear function is a function $g\colon \mathbb{R}^n \to \mathbb{R}$ such that

(i) $g(x+y) \le g(x)+g(y)$ and

(ii) $g(\lambda x) = \lambda g(x)$

for every $x, y \in \mathbb{R}^n$ and any $\lambda \ge 0$ (observe that, in particular, every linear function is sublinear). It is very important to have in mind that, from this definition, every time that we refer to a function as "sublinear", we have that this function is real-valued.

It is immediate that every sublinear function is convex, and that any linear function is, in particular, sublinear. Consequently, any sublinear function is a convex function whose effective domain is $\mathbb{R}^n$. Once any convex function is continuous in the interior of its effective domain, we get that every sublinear function is continuous.

Before we continue to study the one-sided derivative of convex functions, we investigate some properties of sublinear functions which will be important later. If $g\colon \mathbb{R}^n \to \mathbb{R}$ is a sublinear function, then any vector $u \in \mathbb{R}^n$ for which the equality $g(u) = -g(-u)$ holds is called a *linearity direction* of g. The set of linearity directions is denoted by $\operatorname{lin}(g)$. What justifies this name is that if $u \in \operatorname{lin}(g)$, then g is clearly homogeneous on $\operatorname{span}\{u\}$, meaning that $g(\lambda u) = \lambda g(u)$ for every $\lambda \in \mathbb{R}$ (that is, this equality occurs even if λ is a negative number).

Proposition 1.2.4. *Let $g\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function. Then, for any $x \in \mathbb{R}^n$, the function $g'_+(x,\cdot)\colon \mathbb{R}^n \to \mathbb{R}$ is sublinear and the following properties hold:*

(a) $g'_+(x,\lambda x) = \lambda g(x)$ *for every $x \in \mathbb{R}^n$ and any $\lambda \in \mathbb{R}$,*

(b) $g'_+(x,u) \le g(u)$ *for every $x, u \in \mathbb{R}^n$,*

(c) $\operatorname{lin}(g)$ *is a vector subspace of* $\mathbb{R}^n$,

(d) $\operatorname{lin}(g'_+(x,\cdot)) \supseteq \operatorname{lin}(g) + \operatorname{span}\{x\}$ *for every $x \in \mathbb{R}^n$,*

(e) $\operatorname{lin}(g) = \mathbb{R}^n$ *if and only if g is linear.*

Proof. The fact that $g'_+(x,\cdot)$ is sublinear comes immediately from Proposition 1.2.3, since every sublinear function is, in particular, convex. For **(a)**, let $x \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. We simply notice that for any $\lambda \in \mathbb{R}$, if $t > 0$ is small enough, then $1 + t\lambda > 0$. Hence

$$g'_+(x,\lambda x) = \lim_{t\to 0^+} \frac{g(x+t\lambda x) - g(x)}{t} = \lim_{t\to 0^+} \frac{(1+t\lambda)g(x) - g(x)}{t} = \lim_{t\to 0^+} \frac{t\lambda g(x)}{t} = \lambda g(x).$$

The proof of **(b)** relies only on the following calculation, where we use both the subadditivity and the positive homogeneity of g:

$$g'_+(x,u) = \lim_{t\to 0^+} \frac{g(x+tu) - g(x)}{t} \leq \lim_{t\to 0^+} \frac{g(x) + g(tu) - g(x)}{t} = \lim_{t\to 0^+} \frac{tg(u)}{t} = g(u).$$

To prove **(c)** it is sufficient to show that $u+v \in \operatorname{lin}(g)$ whenever $u, v \in \operatorname{lin}(g)$, since we already know that g is homogeneous in any direction of $\operatorname{lin}(g)$. First observe that, if $u, v \in \operatorname{lin}(g)$, then

$$g(u+v) \leq g(u) + g(v) = -g(-u) - g(-v) \leq -g(-u-v).$$

On the other hand,

$$0 = g(u+v+(-u-v)) \leq g(u+v) + g(-u-v),$$

from where it follows that $-g(-u-v) \leq g(u+v)$. These two inequalities imply that $g(u+v) = -g(-u-v)$, and hence $u+v \in \operatorname{lin}(g)$.

Now we prove **(d)**. It is already clear from **(a)** that $\operatorname{span}\{x\} \subseteq \operatorname{lin}(g'_+(x,\cdot))$. Thus we only have to prove that any linearity direction of g is also a linearity direction of $g'_+(x,\cdot)$. For any $u \in \operatorname{lin}(g)$ and every $t \in \mathbb{R}$ we have

$$\begin{aligned} 2g(x) = g(2x) = g(x+tu+x-tu) \leq g(x+tu) + g(x-tu) \leq \\ \leq 2g(x) + t(g(u) + g(-u)) = 2g(x), \end{aligned}$$

from which it follows that $2g(x) = g(x+tu) + g(x-tu)$. As a consequence, we get that

$$\frac{g(x+tu) - g(x)}{t} + \frac{g(x-tu) - g(x)}{t} = 0,$$

whenever $t \neq 0$. Letting $t \to 0^+$ in this equality, we obtain

$$\begin{aligned} g'_+(x,u) + g'_+(x,-u) = \lim_{t\to 0^+} \frac{g(x+tu) - g(x)}{t} + \lim_{t\to 0^+} \frac{g(x+t(-u)) - g(x)}{t} = \\ = \lim_{t\to 0^+} \left[\frac{g(x+tu) - g(x)}{t} + \frac{g(x-tu) - g(x)}{t} \right] = 0, \end{aligned}$$

and hence $g'_+(x,u) = -g'_+(x,-u)$.

It remains to prove **(e)**. Notice that one of the implications is obvious: if g is linear, then all directions are linearity directions. Hence we assume that $\operatorname{lin}(g) = \mathbb{R}^n$. In this case, it is immediate by the definition that g is homogeneous, and thus we have to prove that it is additive. For that sake, we let $u, v \in \mathbb{R}^n$ and write

$$g(u) + g(v) \geq g(u+v) = -g(-u-v) \geq -g(-u) - g(-v) = g(u) + g(v).$$

This concludes the proof.

□

If $f\colon \mathbb{R}^n \to (-\infty, \infty]$ is a convex function, $x \in \operatorname{int}(\operatorname{dom}(f))$, $u \in \mathbb{R}^n$, and $f'_+(x,u) = f'_-(x,u)$, then we define

$$f'(x,u) := f'_+(x,u),$$

and we call this number the *two-sided directional derivative of f at x in the direction of u*. In this case, we also say that the *two-sided directional derivative of f at x in the direction of u exists*, or that *f has a two-sided derivative at x in the direction of u*. The reader has to be aware of the fact that some authors restrict these definitions for unit vectors, but we are not doing this.

Assume that f has a two-sided derivative at x in the direction of u. Then we have that $f'_+(x,u) = -f'_+(x,-u)$, and consequently u is a linearity direction of the one-sided derivative map $f'_+(x,\cdot)$. The linearity directions of the one-sided derivative map $f'_+(x,\cdot)$ are precisely the directions for which the two-sided directional derivative of f at x exists.

Recall that a function $f\colon \mathbb{R}^n \to \mathbb{R}$ is said to be *differentiable* (or *differentiable in the strong sense*) at a point $x \in \mathbb{R}^n$ if there exists a linear map $df_x\colon \mathbb{R}^n \to \mathbb{R}$ with the property that

$$\lim_{v \to 0} \frac{f(x+v) - f(x) - df_x(v)}{|v|} = 0,$$

where the limit is taken in $\mathbb{R}^n$, of course. In this case, the map df_x is called the *differential* of f at x. It is immediate from the definition that if f is differentiable at x, then it has two-sided directional derivative at x for any $u \in \mathbb{R}^n$, and $f'(x,u) = df_x(u)$.

A convex function is differentiable at "almost all" points of $\mathbb{R}^n$. To make this sentence formal, we need some background on basic measure theory (see Appendix A).

Theorem 1.2.3. *A convex function $f\colon \mathbb{R}^n \to \mathbb{R}$ is differentiable almost everywhere in $\mathbb{R}^n$ (with respect to the Lebesgue measure).*

Proof. Let $\rho > 0$. Since the closed ball ρB^n is compact, we get that f is Lipschitz in this ball. Hence f is Lipschitz in the open ball $B(0_{\mathbb{R}^n}, \rho)$. From Rademacher's classical theorem (see [2249]), we get that f is differentiable at almost every point of $B(0_{\mathbb{R}^n}, \rho)$. Once $\mathbb{R}^n$ can be written as a countable union of closed balls, it follows that the set of points where f is not differentiable has zero measure. □

Let $\{e_1, \ldots, e_n\}$ be the canonical basis of $\mathbb{R}^n$. For a given $i \in \{1, \ldots, n\}$, if the two-sided derivative of f in the direction of e_i exists, then

$$\partial_i f(x) := f'(x, e_i)$$

is called the (*first order*) *i-th partial derivative of f at x*. In this case, we also say that *f has the i-th partial derivative*.

Proposition 1.2.5. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function, and let $x \in \operatorname{int}(\operatorname{dom}(f))$. Then the following statements are equivalent:*

(i) *$f'_+(x,u) = f'_-(x,u)$ for every $u \in \mathbb{R}^n$ (in other words, f has all two-sided derivatives at x).*

(ii) *f has all partial derivatives at x.*

(iii) *f is differentiable at x.*

In this case, we have that $f'(x,u) = df_x(u)$ for any $u \in \mathbb{R}^n$.

Remark 1.2.4. Observe that, from Item **(e)** of Proposition 1.2.4, we have that **(i)** is also equivalent to $f'_+(x,\cdot)$ being a linear map.

Proof. First notice that **(i)⇒(ii)** is immediate from the definition (each partial derivative is a two-sided derivative). The implication **(iii)⇒(i)** is also not too difficult. From the positive homogeneity of f'_+, we can assume that u is unit. If f is differentiable at x, then putting $v = \lambda u$ yields

$$0 = \lim_{\lambda\to 0^+} \frac{f(x+\lambda u) - f(x) - df_x(\lambda u)}{\lambda},$$

implying immediately that $df_x(u) = f'_+(x,u)$. On the other hand, the equality

$$0 = \lim_{\lambda\to 0^-} \frac{f(x+\lambda u) - f(x) - df_x(\lambda u)}{-\lambda}$$

gives that

$$f'_-(x,u) = \lim_{\lambda\to 0^-} \frac{f(x+\lambda u) - f(x)}{\lambda} = \lim_{\lambda\to 0^-} \frac{-\lambda df_x(u)}{-\lambda} = df_x(u).$$

It follows that $f'_+(x,u) = df_x(u) = f'_-(x,u)$, and hence f has all two-sided derivatives at x. Observe carefully that we did not use the convexity hypothesis to prove these implications, but that we will do so to prove the remaining implication **(ii)⇒(iii)**.

If f has partial derivatives of first order at a given point $x \in \mathbb{R}^n$, then we can define the *gradient* of f at x to be the vector

$$\nabla f(x) := \sum_{j=1}^{n} \partial_j f(x)\, e_j.$$

Define the linear transformation $T\colon \mathbb{R}^n \to \mathbb{R}$ by

$$Tv := \langle \nabla f(x), v\rangle,$$

for every $v \in \mathbb{R}^n$. We claim that

$$\lim_{v\to 0} \frac{f(x+v) - f(x) - Tv}{|v|} = 0.$$

Indeed, for simplicity we write $g(v) := f(x+v) - f(x) - Tv$. Of course, g is finite in an open ball $B_{\mathbb{R}^n}(0,\rho)$, where $\rho > 0$ is such that $B_{\mathbb{R}^n}(x,\rho) \subseteq \operatorname{int}(\operatorname{dom}(f))$. Also, g is convex on this open ball, because it is the sum of the convex function $v \mapsto f(x+v) - f(x)$ with the affine function $v \mapsto Tv$. Write in coordinates $v = \sum_{j=1}^n \alpha_j e_j$, and let $J = \{j : \alpha_j \neq 0\}$. From discrete Jensen's inequality, and recalling that $g(0) = 0$, we get

$$g(v) = g\left(\sum_{j=1}^n \alpha_j e_j\right) = g\left(\frac{1}{n}\sum_{j=1}^n n\alpha_j e_j\right) \le \frac{1}{n}\sum_{j=1}^n g(n\alpha_j e_j) = \frac{1}{n}\sum_{j\in J} g(n\alpha_j e_j) =$$
$$= \sum_{j\in J} \alpha_j \frac{g(n\alpha_j e_j)}{n\alpha_j}.$$

Let $w = \sum_{j=1}^n \beta_j e_j$, where $\beta_j = 0$ if $j \notin J$, and

$$\beta_j = \frac{g(n\alpha_j e_j)}{n\alpha_j}$$

if $j \in J$. Notice that the right-hand term in the inequality above is precisely the inner product $\langle v, w\rangle$. Thus,

$$g(v) \le \langle v, w\rangle \le |v| \cdot |w| \le |v| \cdot \sum_{j=1}^{n} |\beta_j| = |v| \cdot \sum_{j\in J} \left|\frac{g(n\alpha_j e_j)}{n\alpha_j}\right|,$$

where we used the classical Cauchy–Schwarz inequality, and also the inequality $|w| \le \sum_{j=1}^{n} |\beta_j|$. It follows that

$$\frac{g(v)}{|v|} \le \sum_{j\in J} \left|\frac{g(n\alpha_j e_j)}{n\alpha_j}\right|, \tag{1.2.11}$$

whenever $v \in B_{\mathbb{R}^n}(0,\rho) \setminus \{0\}$. Using the same argument for the vector $-v$, we get immediately the inequality

$$\frac{g(-v)}{|v|} \le \sum_{j\in J} \left|\frac{g(-n\alpha_j e_j)}{n\alpha_j}\right|. \tag{1.2.12}$$

Moreover, we observe that the convexity of f gives

$$\begin{aligned} g(v) + g(-v) &= f(x+v) + f(x-v) - 2f(x) = \\ &= f(x+v) + f(x-v) - 2f\Big(\frac{1}{2}(x+v) + \frac{1}{2}(x-v)\Big) \ge \\ &\ge f(x+v) + f(x-v) - 2\Big(\frac{1}{2}f(x+v) + \frac{1}{2}f(x-v)\Big) = 0, \end{aligned}$$

yielding

$$g(v) \ge -g(-v). \tag{1.2.13}$$

Combining (1.2.11), (1.2.12), and (1.2.13) yields

$$-\sum_{j\in J} \left|\frac{g(-n\alpha_j e_j)}{n\alpha_j}\right| \le \frac{-g(-v)}{|v|} \le \frac{g(v)}{|v|} \le \sum_{j\in J} \left|\frac{g(n\alpha_j e_j)}{n\alpha_j}\right|. \tag{1.2.14}$$

On the other hand, when $v \to 0$ we have that $n\alpha_j \to 0$ for each $j \in J$. Hence, setting $t = n\alpha_j$, we calculate

$$\begin{aligned} \lim_{t\to 0} \frac{g(te_j)}{t} &= \lim_{t\to 0} \frac{f(x+te_j) - f(x) - \langle \nabla f(x), te_j\rangle}{t} = \\ &= \lim_{t\to 0} \frac{f(x+te_j) - f(x)}{t} - \partial_j f(x) = f'(x, e_j) - \partial_j f(x) = 0, \end{aligned}$$

and, of course, by a similar calculation we also have

$$\lim_{t\to 0} \frac{g(-te_j)}{t} = 0,$$

for each $j \in J$. Thus, all the terms in the sums on the left-hand and right-hand sides of (1.2.14) converge to zero as $v \to 0$. It follows that

$$\lim_{v \to 0} \frac{g(v)}{|v|} = 0,$$

and this shows that f is differentiable at x. Notice that we also proved that $df_x = T$.

□

Corollary 1.2.3. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function, and let $x \in \operatorname{int}(\operatorname{dom}(f))$. Then f is differentiable at x if and only if the one-sided derivative map $f'_+(x, \cdot) : \mathbb{R}^n \to \mathbb{R}$ is linear.*

Proof. If f is differentiable at x, then we have from Proposition 1.2.5 that $f'_+(x, \cdot) = f'(x, \cdot) = df_x$, and hence the one-sided derivative map is linear.

On the other hand, if $f'_+(x, \cdot)$ is linear, then we have that $f'_+(x, u) = -f'_+(x, -u) = f'_-(x, u)$ for any $u \in \mathbb{R}^n$, from where it follows that f has all two-sided directional derivatives at x. From Proposition 1.2.5 we get that f is differentiable.

□

As we did it for intervals in $\mathbb{R}$, the definition of convex functions can be extended when the domain $\mathbb{R}^n$ is replaced by an open convex set $D \subseteq \mathbb{R}^n$. Let $U \subseteq \mathbb{R}^n$ be an open set. A function $f\colon U \to \mathbb{R}$ is *continuously differentiable* (or of *class* C^1) if its partial derivatives exist and are continuous in U. It is a well-known fact from standard analysis that this condition implies that f is (strongly) differentiable in each point of U.

We say that a function $f\colon U \to \mathbb{R}$ is *twice continuously differentiable* (or of *class* C^2) if it is C^1 and its partial derivatives $\partial_1 f, \ldots, \partial_n f$ have continuous partial derivatives in U. These derivatives are called the *second-order partial derivatives* of f (hence it is common to say that a function is twice continuously differentiable when it has continuous second-order partial derivatives). The i-th partial derivative of the partial derivative $\partial_j f$ at a point $x \in U$ is denoted by $\partial_i \partial_j f(x)$ or $\frac{\partial f}{\partial x_i \partial x_j}(x)$. This definition extends naturally for any $k \geq 2$.

Theorem 1.2.4. *Let $U \subseteq \mathbb{R}^n$ be an open convex set, and let $f\colon U \to \mathbb{R}$ be a twice differentiable function. Then f is convex if and only if its* Hessian matrix

$$\operatorname{hess} f_x = \Big[\partial_i \partial_j f(x)\Big]_{i,j=1,\ldots,n}$$

is positive semi-definite for each point $x \in D$.

Proof. First, observe that f is convex if and only if for any $x, y \in D$ the map $g\colon (-\varepsilon, 1+\varepsilon) \to \mathbb{R}$ given by $g(\lambda) = f((1-\lambda)x + \lambda y)$ is convex (here $\varepsilon > 0$ is just any number such that $(1-\lambda)x + \lambda y \in D$ for any $-\varepsilon < \lambda < 1 + \varepsilon$). Observe that we can write $(1-\lambda)x + \lambda y = x + \lambda(y - x)$, and this allows us to interpret the derivative of g as a directional derivative of f (in the direction $y - x$).

From Corollary 1.2.2 it follows that it suffices to prove that $\operatorname{hess} f_x$ is positive semi-definite for any $x \in D$ if and only if all of these maps satisfy $g'' \geq 0$. Even better, we claim that it is enough to prove that each of these functions satisfies $g''(0) \geq 0$. In fact, if $\mu \in (0, 1)$, then, writing $z = x + \mu(y - x)$ and defining $h(t) = f((1-t)z + ty)$, we have that

$$g(\mu + \lambda) = h\left(\frac{\lambda}{1-\mu}\right).$$

It follows that $g''(\mu) = h''(0)$ and hence $g''(\mu) \geq 0$ if and only if $h''(0) \geq 0$. With this in hand, let us calculate $g''(0)$. Writing in coordinates $x = \sum_{j=1}^n \alpha_j e_j$ and $y = \sum_{j=1}^n \beta_j e_j$, we get

$$g(\lambda) = f\left(x + \sum_{j=1}^{n} \lambda(\beta_j - \alpha_j)e_j\right),$$

and thus

$$g'(\lambda) = \sum_{j=1}^{n}(\beta_j - \alpha_j)f'(x + \lambda(y - x), e_j) = \sum_{j=1}^{n}(\beta_j - \alpha_j)\partial_j f\left(x + \sum_{i=1}^{n} \lambda(\beta_i - \alpha_i)e_i\right).$$

Finally,

$$g''(\lambda) = \sum_{j=1}^{n}(\beta_j - \alpha_j)\sum_{i=1}^{n}(\beta_i - \alpha_i)\partial_i\partial_j f\left(x + \sum_{i=1}^{n} \lambda(\beta_i - \alpha_i)e_i\right).$$

Consequently, for $\lambda = 0$ we get

$$g''(0) = \sum_{i,j=1}^{n}(\beta_j - \alpha_j)(\beta_i - \alpha_i)\partial_i\partial_j f(x) = \langle y - x, \operatorname{hess} f_x(y - x)\rangle.$$

It is immediate from the equality above that if $\operatorname{hess} f_x$ is positive semi-definite for any $x \in D$, then $g''(0) \geq 0$ for all functions g constructed as at the beginning of the proof. For the converse, notice that if $u \in \mathbb{R}^n$, then $x + \varepsilon u \in D$ for small $\varepsilon > 0$. Letting $y = x + \varepsilon u$ in the equality above, we get

$$g''(0) = \langle \varepsilon u, \operatorname{hess} f_x(\varepsilon u)\rangle = \varepsilon^2\langle u, \operatorname{hess} f_x(u)\rangle,$$

and this implies that $\langle u, \operatorname{hess} f_x(u)\rangle \geq 0$ for any $u \in \mathbb{R}^n$. This proof is finished.

□

Remark 1.2.5. On the other hand, it is clear that if the Hessian matrix is negative semi-definite everywhere, then f is concave (because in this case the Hessian matrix of $-f$ is always positive semi-definite, and hence $-f$ is convex).

1.3 Subgradients and subdifferentials

Let $f\colon U \to \mathbb{R}$ be a function, where U is an open set of $\mathbb{R}^n$. If f is differentiable at x, then the gradient of f at x (see the proof of Proposition 1.2.5 for the definition) is the unique vector $\nabla f(x)$ of $\mathbb{R}^n$ for which the equality

$$df_x(v) = \langle \nabla f(x), v\rangle$$

holds for any $v \in \mathbb{R}^n$. Notice that the following holds for a convex function f due to Proposition 1.2.5: if f has all partial derivatives at a point $x \in \operatorname{int}(\operatorname{dom}(f))$, then f is differentiable at x, and hence we can define the gradient by the equality above. As a consequence, in this case the gradient does not depend on the choice of an orthonormal basis.

Proposition 1.3.1. *Let $D \subseteq \mathbb{R}^n$ be a convex open set, and assume that $f\colon D \to \mathbb{R}$ is a differentiable function. Then the following statements are equivalent:*

(a) *f is convex,*

(b) *$f(y) - f(x) \geq \langle \nabla f(x), y - x \rangle$ for any $x, y \in D$,*

(c) *$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0$ for every $x, y \in D$.*

Proof. If **(a)** holds, then for any $x, y \in D$ the equality $f((1-\lambda)x + \lambda y) = f(x + \lambda(y - x))$ implies

$$\langle \nabla f(x), y - x \rangle = \lim_{\lambda \to 0} \frac{f(x + \lambda(y - x)) - f(x)}{\lambda} \leq \lim_{\lambda \to 0} \frac{(1-\lambda)f(x) + \lambda f(y) - f(x)}{\lambda} = \\ = \lim_{\lambda \to 0} \frac{\lambda(f(y) - f(x))}{\lambda} = f(y) - f(x),$$

and that is **(b)**.

Now we prove the implication **(b)**⇒**(c)**. If **(b)** holds then, replacing x and y, we have the inequality

$$f(x) - f(y) \geq \langle \nabla f(y), x - y \rangle.$$

This, with the "original" inequality in **(b)** leads to

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle = \langle \nabla f(y), y - x \rangle - \langle \nabla f(x), y - x \rangle \geq \langle \nabla f(y), y - x \rangle + f(x) - f(y) \geq \\ \geq f(y) - f(x) + f(x) - f(y) = 0,$$

and hence we have the desired inequality. The reader should notice that we changed the signs of the inequalities that we have used.

It remains to prove that **(c)** implies **(a)**. Fix (distinct) points $x, y \in D$ and let $g \colon [0, 1] \to \mathbb{R}$ be given as

$$g(\lambda) = f((1-\lambda)x + \lambda y).$$

To simplify the notation, write $z_\lambda := (1-\lambda)x + \lambda y$. Of course, g is differentiable, and by the chain rule its derivative is given as

$$g'(\lambda) = \sum_{j=1}^{n} (y_j - x_j)\, \partial_j f(z_\lambda) = \langle \nabla f(z_\lambda), y - x \rangle.$$

Notice that if $0 \leq \lambda_0 < \lambda_1 \leq 1$, then

$$z_{\lambda_1} - z_{\lambda_0} = (1-\lambda_1)x + \lambda_1 y - (1-\lambda_0)x - \lambda_0 y = (\lambda_1 - \lambda_0)(y - x),$$

from which we get that

$$g'(\lambda_1) - g'(\lambda_0) = \langle \nabla f(z_{\lambda_1}) - \nabla f(z_{\lambda_0}), y - x \rangle = \frac{\langle \nabla f(z_{\lambda_1}) - \nabla f(z_{\lambda_0}), z_{\lambda_1} - z_{\lambda_0} \rangle}{\lambda_1 - \lambda_0} \geq 0,$$

by the hypothesis **(c)**. It follows that g' is non-decreasing in $[0, 1]$. From Proposition 1.2.1 we get that g is convex. Thus,

$$f((1-\lambda)x + \lambda y) = g(\lambda) \leq (1-\lambda)g(0) + \lambda g(1) = (1-\lambda)f(x) + \lambda f(y).$$

This proves that f is convex.

□

A convex function is not necessarily differentiable at every point of its effective domain. Nevertheless, these notions can be extended to the nondifferentiable case. Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function, and let $x \in \mathrm{dom}(f)$. Inspired by the geometric intuition for the gradient of a differentiable convex function given in Item **(b)** of Proposition 1.3.1, we define the *subdifferential* of f at x to be the set

$$\partial f(x) = \{v \in \mathbb{R}^n : f(y) \geq f(x) + \langle v, y - x\rangle \text{ for every } y \in \mathbb{R}^n\}.$$

Each vector of $\partial f(x)$ is said to be a *subgradient* of f at x. The set

$$\partial f := \{(x, v) \in \mathbb{R}^n \times \mathbb{R}^n : v \in \partial f(x)\}$$

is called the *subdifferential of* f.

The subgradient has an interesting geometric interpretation that we will discuss now. First, let us fix some notation and concepts. For any natural number d, a *hyperplane* of $\mathbb{R}^d$ (or simply a *hyperplane* when there is no possibility of confusion on the dimension of the ambient space) is a set of the form

$$\mathbf{h}(u, \alpha) := \{v \in \mathbb{R}^d : \langle u, v\rangle = \alpha\}, \tag{1.3.1}$$

where u is a non-zero vector of $\mathbb{R}^d$, α is a real number and $\langle \cdot, \cdot \rangle$ denotes the usual Euclidean inner product of $\mathbb{R}^d$. The vector u is said to be *normal* (or *orthogonal*) to the hyperplane and, if u is a unit vector, then the number α is clearly the (signed) distance from $\mathbf{h}(u, \alpha)$ to the origin. A hyperplane $\mathbf{h}(u, \alpha)$ defines two (closed) *half-spaces*:

$$\mathbf{h}^+(u, \alpha) := \{v \in \mathbb{R}^d : \langle u, v\rangle \geq \alpha\}, \text{ and}$$
$$\mathbf{h}^-(u, \alpha) := \{v \in \mathbb{R}^d : \langle u, v\rangle \leq \alpha\}.$$

It is immediate that $\mathrm{int}(\mathbf{h}^+(u, \alpha)) = \{v \in \mathbb{R}^d : \langle u, v\rangle > \alpha\}$, and that we also have a similar characterization for the interior of $\mathbf{h}^-(u, \alpha)$. It is very common to denote a hyperplane simply by $\mathbf{h}$, and the respective determined closed half-spaces by $\mathbf{h}^+$ and $\mathbf{h}^-$. This terminology will be extensively used throughout the text. Notice carefully that we always can write $\mathbb{R}^d$ as the disjoint unions $\mathbf{h}^+ \cup \mathrm{int}(\mathbf{h}^-)$ and $\mathbf{h}^- \cup \mathrm{int}(\mathbf{h}^+)$.

Let $A \subseteq \mathbb{R}^d$. We say that a hyperplane $\mathbf{h}(u, \alpha) := \mathbf{h}$ *supports* A at a point $x \in A$ if $x \in A \cap \mathbf{h}$ and A is contained in one of the half-spaces determined by $\mathbf{h}$ ($\mathbf{h}^+$ or $\mathbf{h}^-$). In this case, we say that $\mathbf{h}$ is a *supporting hyperplane* of A. If $A \subseteq \mathbb{R}^d$ is supported by a hyperplane $\mathbf{h}(u, \alpha)$ at a point $x \in A$, then we say that the vector u is *normal* to A at x. Moreover, if $A \subseteq \mathbf{h}^-(u, \alpha)$, then we say that u is an *outer normal* of A at x (see Figure 1.3.1). Otherwise we say that u is an *inner normal* of A at x. Notice very carefully that if $\mathbf{h}(u, \alpha)$ supports A at $x \in A$, then $\alpha = \langle u, x\rangle$.

Remark 1.3.1. In this chapter, we usually work with functions $f : \mathbb{R}^n \to (-\infty, +\infty]$, thus having subsets of $\mathbb{R}^{n+1}$ as epigraphs. Then the regarded supporting hyperplanes are hyperplanes of $\mathbb{R}^{n+1}$. On the other hand, in the next chapter we study the geometry of subsets of $\mathbb{R}^n$, and hence the considered supporting hyperplanes will be hyperplanes of $\mathbb{R}^n$. For this reason, we introduced generically hyperplanes of $\mathbb{R}^d$, where d is any natural number.

Proposition 1.3.2. *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function, and let $x \in \mathrm{dom}(f)$. Then $v \in \partial f(x)$ if and only if $\mathrm{epi}(f)$ is supported at $(x, f(x))$ by the hyperplane*

$$\mathbf{h} := \mathbf{h}((v, -1), \langle v, x\rangle - f(x)) \subseteq \mathbb{R}^{n+1},$$

where $(v, -1)$ is the vector of $\mathbb{R}^{n+1}$ whose n first entries form the vector $v \in \mathbb{R}^n$ and the $(n+1)$-th entry is -1. Even more, in this case we have that $(v, -1)$ is an outer normal of $\mathrm{epi}(f)$ at $(x, f(x))$.

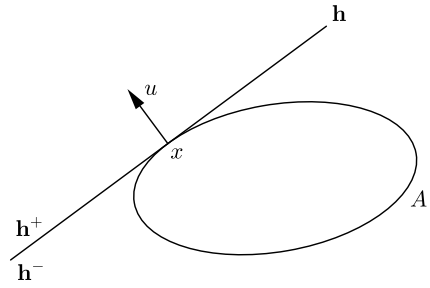


Fig. 1.3.1 The vector u is an outer normal of A at x.

Proof. Rearranging the terms in the definition, we have that $v \in \partial f(x)$ if and only if

$$\langle (y, f(y)), (v, -1) \rangle \leq \langle v, x \rangle - f(x),$$

for every $y \in \mathbb{R}^n$, where $(y, f(y))$ denotes the vector of $\mathbb{R}^{n+1}$ whose n first entries form y and the last entry is $f(y)$. Notice carefully that we are using the same notation for the standard inner product in $\mathbb{R}^n$ and in $\mathbb{R}^{n+1}$.

The inequality above is equivalent to $(y, f(y)) \in \mathbf{h}^-$, and we claim that it holds for every $y \in \mathbb{R}^n$ if and only if $\mathrm{epi}(f) \subseteq \mathbf{h}^-$. Indeed, if the inclusion holds, then particularly we have that the graph of f is contained in $\mathbf{h}^-$, which yields the inequality for any $y \in \mathbb{R}^n$. On the other hand, assume that the inequality holds for any $y \in \mathbb{R}^n$. Hence if $(y, \beta) \in \mathrm{epi}(f)$, then $\beta \geq f(y)$, and consequently

$$\langle (y, \beta), (v, -1) \rangle = \langle y, v \rangle - \beta \leq \langle y, v \rangle - f(y) \leq \langle (y, f(y)), (v, -1) \rangle \leq \langle v, x \rangle - f(x).$$

Since $(x, f(x)) \in \mathrm{epi}(f) \cap \mathbf{h}$, it follows that $v \in \partial f(x)$ if and only if $\mathbf{h}$ supports $\mathrm{epi}(f)$ at $(x, f(x))$, as we wanted to prove. Also, in this case we have that $\mathrm{epi}(f) \in \mathbf{h}^-$, and then $(v, -1)$ is an outer normal of $\mathrm{epi}(f)$ at $(x, f(x))$.

□

Corollary 1.3.1. *If $x \in \mathrm{int}(\mathrm{dom}(f))$ and $\mathbf{h} = \mathbf{h}((w, \beta), \alpha)$ supports $\mathrm{epi}(f)$ at $(x, f(x))$, then $\beta \neq 0$, and*

$$\mathbf{h}((w, \beta), \alpha) = \mathbf{h}((v, -1), \langle v, x \rangle - f(x))$$

for some $v \in \partial f(x)$. In particular, $\mathrm{epi}(f)$ is supported by a unique hyperplane at $(x, f(x))$ if and only if f has a unique subgradient at x.

Remark 1.3.2. In the equality above, it is possible that (w, β) and $(v, -1)$ have opposite orientations, but still the hyperplanes are the same (as sets).

Proof. If $\mathbf{h}$ supports $\mathrm{epi}(f)$ at $(x, f(x))$, then the assumption that $\beta = 0$ yields

$$\alpha = \langle (w, 0), (x, f(x)) \rangle = \langle w, x \rangle.$$

On the other hand, up to changing the signs of (w, β) and α, we may assume that $\mathrm{epi}(f) \subseteq \mathbf{h}^-$, and since $x \in \mathrm{int}(\mathrm{dom}(f))$, we get that $x + \varepsilon w \in \mathrm{dom}(f)$ for some small $\varepsilon > 0$. Hence $(x + \varepsilon w, f(x + \varepsilon w)) \in \mathrm{epi}(f) \subseteq \mathbf{h}^-$. But

$$\langle (w, 0), (x + \varepsilon w, f(x + \varepsilon w)) \rangle = \langle w, x \rangle + \varepsilon |w|^2 > \langle w, x \rangle = \alpha,$$

implying that $(x+\varepsilon w, f(x+\varepsilon w)) \in \operatorname{int}(\mathbf{h}^+)$, which is a contradiction. Therefore, $\beta \neq 0$ (geometrically, this means that a supporting hyperplane cannot be *vertical*). Moreover, since $(x, f(x)) \in \mathbf{h}$, we have

$$\alpha = \langle (w,\beta), (x, f(x)) \rangle = \langle w, x \rangle + \beta f(x),$$

and hence

$$-\frac{\alpha}{\beta} = \left\langle -\frac{w}{\beta}, x \right\rangle - f(x).$$

Writing $-w/\beta = v$, we observe that

$$\mathbf{h}((w,\beta),\alpha) = \mathbf{h}((v,-1), \langle v, x \rangle - f(x)).$$

It follows from Proposition 1.3.2 that $v \in \partial f(x)$.

□

The subdifferential of a convex function f at a given point $x \in \operatorname{int}(\operatorname{dom}(f))$ describes all the supporting hyperplanes to the epigraph of f at the point $(x, f(x))$. Intuitively, one can think of a point where f is differentiable as a point at which the epigraph of f is supported by a unique hyperplane. This is true, indeed, as we will show in the next theorems. It also justifies why subgradients are, in some sense, an extension of the concept of gradient.

Lemma 1.3.1. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex function, and let $x \in \operatorname{int}(\operatorname{dom}(f))$. For any $w \in \mathbb{R}^n$, we have the inequality*

$$f(x+w) - f(x) \geq f'_+(x, w).$$

Proof. If $0 < \lambda < 1$, we write $f(x+\lambda w) = f(\lambda(x+w) + (1-\lambda)x)$, and

$$f'_+(x,w) = \lim_{\lambda \to 0^+} \frac{f(x+\lambda w) - f(x)}{\lambda} = \lim_{\lambda \to 0^+} \frac{f(\lambda(x+w)+(1-\lambda)x) - f(x)}{\lambda} \leq$$
$$\leq \lim_{\lambda \to 0^+} \frac{\lambda f(x+w) + (1-\lambda) f(x) - f(x)}{\lambda} = \lim_{\lambda \to 0^+} \frac{\lambda(f(x+w) - f(x))}{\lambda} = f(x+w) - f(x).$$

This concludes the proof.

□

Theorem 1.3.1. *Let $f\colon \mathbb{R}^n \to (-\infty, \infty]$ be a convex function. For any point $x \in \operatorname{int}(\operatorname{dom}(f))$ we have that $\partial f(x) \neq \emptyset$, and that*

$$f'_+(x,u) = \max\{\langle v, u \rangle : v \in \partial f(x)\}$$

for every $u \in \mathbb{R}^n$.

Proof. First of all, observe that for every $\lambda > 0$ and $v \in \partial f(x)$ we have

$$f(x+\lambda u) - f(x) \geq \langle v, \lambda u \rangle,$$

yielding

$$\frac{f(x+\lambda u) - f(x)}{\lambda} \geq \langle v, u \rangle.$$

Letting $\lambda \to 0^+$, we get that $f'_+(x,u) \geq \langle v, u \rangle$. Consequently,

$$\sup\{\langle v, u\rangle : v \in \partial f(x)\} \leq f'_+(x, u).$$

Thus we only have to prove that $\partial f(x) \neq \emptyset$, and that there is some subgradient $v \in \partial f(x)$ such that $\langle v, u\rangle = f'_+(x, u)$. For that sake, assume first that $u \neq 0$. Let $\{e_1, \ldots, e_n\}$ be a basis of $\mathbb{R}^n$ such that $e_1 = u$, and put $g_0(\cdot) := f'_+(x, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$. Define inductively the maps

$$\begin{aligned} g_1(\cdot) &:= (g_0)'_+(e_1, \cdot), \\ g_2(\cdot) &:= (g_1)'_+(e_2, \cdot), \\ &\vdots \\ g_n(\cdot) &:= (g_{n-1})'_+(e_n, \cdot). \end{aligned}$$

Each function g_j is sublinear (because it is the one-sided derivative of a sublinear function, and any sublinear function is convex), and from Item **(b)** of Proposition 1.2.4 we get the inequalities $g_0 \geq g_1 \geq \ldots \geq g_n$. Moreover, from Items **(d)** and **(e)** of Proposition 1.2.4 we have that g_n is a linear functional. The Riesz identification given by the standard inner product guarantees that there exists $v \in \mathbb{R}^n$ such that

$$g_n(\cdot) = \langle v, \cdot\rangle.$$

We will show that $v \in \partial f(x)$. Indeed, for any $x, y \in \mathbb{R}^n$ we have

$$\langle v, y - x\rangle = g_n(y - x) \leq g_0(y - x) = f'_+(x, y - x) \leq f(y) - f(x),$$

where the last inequality comes from Lemma 1.3.1. Notice that this guarantees that $\partial f(x) \neq \emptyset$. It remains to prove that $f'_+(x, u) = \langle v, u\rangle$. To do so, we use Item **(a)** of Proposition 1.2.4, the inequality $g_0 \geq g_1 \geq g_n$, and the linearity of g_n to obtain

$$g_n(u) \leq g_0(u) = -(g_0)'_+(u, -u) = -(g_0)'_+(e_1, -u) = -g_1(-u) \leq -g_n(-u) = g_n(u),$$

from where we get that $g_n(u) = g_0(u) = f'_+(x, u)$. This concludes the proof for the case $u \neq 0$, since $\langle v, u\rangle = g_n(u)$. To finish the whole proof, observe that we already have $\partial f(x) \neq \emptyset$, and hence $\max\{\langle v, 0_{\mathbb{R}^n}\rangle : v \in \partial f(x)\} = 0 = f'_+(x, 0_{\mathbb{R}^n})$.

□

Remark 1.3.3. Notice that since $\partial f(x) \neq \emptyset$ whenever $x \in \operatorname{int}(\operatorname{dom}(f))$, we get from Proposition 1.3.2 that for every $x \in \operatorname{int}(\operatorname{dom}(f))$ the epigraph of f is supported at $(x, f(x))$ by some (non-vertical) hyperplane.

Theorem 1.3.2. *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function, and let $x \in \operatorname{int}(\operatorname{dom}(f))$. Then f is differentiable at x if and only if $\partial f(x)$ is a singleton. In this case, the unique subgradient of f at x is the gradient.*

Proof. First, let us assume that f is differentiable at x. Then, if $v \in \partial f(x)$, we have the inequality

$$f(x + \lambda u) \geq f(x) + \langle v, \lambda u\rangle,$$

for every $u \in \mathbb{R}^n$ and $\lambda > 0$. Rearranging the terms, we get immediately that

$$\langle v, u\rangle \leq \frac{f(x + \lambda u) - f(x)}{\lambda},$$

and taking the limit as $\lambda \to 0^+$, we get that $\langle v, u\rangle \leq f'_+(x,u) = f'(x,u)$. Since this holds for any $u \in \mathbb{R}^n$, we also have that

$$\langle v, -u\rangle \leq f'_+(x,-u) = -f'_-(x,u) = -f'(x,u),$$

and hence $\langle v, u\rangle \geq f'(x,u)$. It follows that $\langle v, u\rangle = f'(x,u)$ for any $u \in \mathbb{R}^n$. Consequently, if $v, w \in \partial f(x)$, then $\langle v - w, u\rangle = 0$ for every u, and thus $v = w$. It proves that if f is differentiable at x, then $\partial f(x)$ is a singleton. To check that in this case the gradient is the unique subgradient, we simply notice that if $\{v\} = \partial f(x)$, then

$$\langle v, u\rangle = f'(x,u) = df_x(u) = \langle \nabla f(x), u\rangle,$$

for any $u \in \mathbb{R}^n$.

Now we prove the converse. Assume that $\partial f(x) = \{v_0\}$. From Theorem 1.3.1, we have

$$\begin{aligned} f'_+(x,-u) = \max\{\langle v, -u\rangle : v \in \partial f(x)\} = \langle v_0, -u\rangle = -\langle v_0, u\rangle = \\ = -\max\{\langle v, u\rangle : v \in \partial f(x)\} = -f'_+(x,u), \end{aligned}$$

for any $u \in \mathbb{R}^n$. This gives that $f'_+(x,u) = f'_-(x,u)$ for every $u \in \mathbb{R}^n$, and from Proposition 1.2.5 we get that f is differentiable at x.

□

It is also useful to relate the subdifferentials of a convex function to the outer normal vectors of their sublevel sets. This is done next.

Proposition 1.3.3. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function, and assume that $f(x) = c$ is not a minimum of f. Then*

(i) *$x \in \partial\{f \leq c\}$ and*

(ii) *if $u \in \partial f(x)$, then u is an outer normal vector of $\{f \leq c\}$ at x.*

Proof. For **(i)**, all we have to do is to prove that every neighborhood of x has some point where f is strictly greater than c. To do so, since c is not a minimum of f, we may let $x_0 \in \mathbb{R}^n$ be such that $f(x_0) < c$. Given $\varepsilon > 0$, we let $\delta > 0$ be such that

$$x + \delta(x - x_0) \in B(x, \varepsilon),$$

and a simple calculation shows that we may write $x = (1-\lambda)x_0 + \lambda(x + \delta(x - x_0))$ for

$$\lambda = \frac{1}{\delta + 1}.$$

From convexity, we have that

$$f(x) \leq (1-\lambda)f(x_0) + \lambda f(x + \delta(x - x_0)),$$

and rearranging the terms we get

$$f(x + \delta(x - x_0)) \geq \frac{f(x)}{\lambda} - \frac{1-\lambda}{\lambda} f(x_0) = (\delta + 1)c - \delta f(x_0) > (\delta + 1)c - \delta c = c.$$

This shows that every neighborhood of x has a point of $\mathbb{R}^n \setminus \{f \leq c\}$. Notice that, geometrically, we "extended a little bit" a segment which connects x_0 (which is a point where $f < c$) to x.

To prove **(ii)**, first notice that if $0_{\mathbb{R}^n} \in \partial f(x)$ then we would have $f(y) \geq f(x) = c$ for every $y \in \mathbb{R}^n$, which is a contradiction since c is not a minimum. Then any subgradient of f at x is non-zero. We have to show that if $u \in \partial f(x)$, then the hyperplane $\mathbf{h}(u, \langle u, x\rangle)$ supports $\{f \leq c\}$ at x. We assume by contradiction that this is not true, that is, we let $z \in \mathbb{R}^n$ be a point such that

$$z \in \{f \leq c\} \cap \operatorname{int}\big(\mathbf{h}^+(u, \langle u, x\rangle)\big).$$

This leads to $f(z) \leq c$ and $\langle u, z\rangle > \langle u, x\rangle$. We write the latter as $\langle u, z - x\rangle > 0$ and use the assumption that $u \in \partial f(x)$ to get

$$f(z) \geq f(x) + \langle u, z - x\rangle > f(x) = c,$$

which is a contradiction.

□

Summarizing, for a convex function $f\colon \mathbb{R}^n \to (-\infty, +\infty]$, differentiability at a point $x \in \operatorname{int}(\operatorname{dom}(f))$ is equivalent to both $\operatorname{epi}(f)$ having a unique supporting hyperplane at $(x, f(x))$ and the subdifferential of f at x being a singleton. Also, if we restrict ourselves to convex functions $f\colon \mathbb{R}^n \to \mathbb{R}$, then we can establish a relation between the subdifferential at any non-minimizer point of f with the outer normal vectors of the corresponding sublevel set at this point.

From the Lipschitz property on compact subsets of the effective domain, and from Rademacher's classical theorem (see [2249]) we have that a convex function is differentiable almost everywhere in its effective domain (with respect to the usual Lebesgue measure in $\mathbb{R}^n$). Geometrically, this means that the set of points where $\operatorname{epi}(f)$ has more than one supporting hyperplane has zero (Lebesgue) measure. We finish this section by proving that, as in the one-dimensional case, any differentiable convex function must be of class C^1.

Theorem 1.3.3. *Let $U \subseteq \mathbb{R}^n$ be convex and open, and let $f\colon U \to \mathbb{R}$ be a convex function. If f is differentiable in U, then f is of class C^1 in U.*

Proof. It clearly suffices to prove that the gradient of f is continuous as a map $\nabla f\colon U \to \mathbb{R}^n$. Let (x_n) be a sequence of points in U such that $x_n \to x$ for some $x \in U$. For each $n \in \mathbb{N}$, denote $v_n = \nabla f(x_n)$. Then Proposition 1.3.1 gives that

$$f(z) - f(x_n) \geq \langle v_n, z - x_n\rangle$$

for each $z \in U$. From Theorem 1.1.1, we may fix a compact neighborhood $V \subseteq U$ of x (a closed ball, say) where f is Lipschitz (with constant c_0, say), and up to discarding finitely many points, we may assume that $x_n \in \operatorname{int} V$ for every $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, we choose a point $z_n \in V$ such that $z_n - x_n$ points in the direction of v_n. Doing this, we get

$$|v_n||z_n - x_n| = \langle v_n, z_n - x_n\rangle \leq f(z_n) - f(x_n) \leq c_0|z_n - x_n|,$$

from where we get that $(v_n)_{n\in\mathbb{N}}$ is a bounded sequence of vectors. Therefore, to prove that v_n converges to $\nabla f(x)$, it is sufficient to show that any converging subsequence of (v_n) converges to $\nabla f(x)$. Indeed, let $(v_{n_k})_{k\in\mathbb{N}}$ be a subsequence of (v_n) which converges to a vector $v \in \mathbb{R}^n$. From the continuity of f, we have that

$$f(z) - f(x) \geq \langle v, z - x\rangle$$

for any $z \in U$. It follows that v is a subgradient of f at x. Since f is differentiable at x, we must have $v = \nabla f(x)$, and the proof is complete.

□

1.4 Duality

Recall that a function $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ is called *closed* if it is lower semi-continuous (which means that its epigraph is closed in $\mathbb{R}^{n+1}$; see Section 1.1). This terminology will be extensively used in this section.

For a proper function $f\colon \mathbb{R}^n \to (-\infty, +\infty]$, not necessarily closed or convex, we define the *conjugate function* $f^*\colon \mathbb{R}^n \to (-\infty, +\infty]$ to be

$$f^*(v) := \sup\{\langle v, x\rangle - f(x) : x \in \mathbb{R}^n\} = \sup\{\langle v, x\rangle - f(x) : x \in \mathrm{dom(f)}\},$$

for each $v \in \mathbb{R}^n$. The function f^* is also called the *Legendre transform* of f. Observe that if f is convex, then f^* is also convex, because

$$\langle (1-\lambda)v + \lambda w, x\rangle - f(x) = (1-\lambda)(\langle v, x\rangle - f(x)) + \lambda(\langle w, x\rangle - f(x)),$$

for any $v, w \in \mathbb{R}^n$ and $\lambda \in [0, 1]$, and hence

$$\begin{aligned} f^*((1-\lambda)v + \lambda w) &= \sup\{(1-\lambda)v + \lambda w, x\rangle - f(x) : x \in \mathrm{dom}(f)\} = \\ &= \sup\{(1-\lambda)(\langle v, x\rangle - f(x)) + \lambda(\langle w, x\rangle - f(x)) : x \in \mathrm{dom}(f)\} \leq \\ &\leq (1-\lambda)\sup\{\langle v, x\rangle - f(x) : x \in \mathrm{dom}(f)\} + \lambda \sup\{\langle w, x\rangle - f(x) : x \in \mathrm{dom}(f)\} = \\ &= (1-\lambda)f^*(v) + \lambda f^*(w). \end{aligned}$$

At first glance, this definition may seem meaningless. However, we can give its nice geometric interpretation. The conjugate function describes the half-spaces which contain $\mathrm{epi}(f)$ in the following sense: given a *non-vertical* direction w of $\mathbb{R}^{n+1}$ (meaning that its $(n+1)$-st coordinate is not zero), how far can one move the hyperplane orthogonal to w before it reaches $\mathrm{epi}(f)$? We deal with the finite and infinite cases (for convex functions which are not necessarily closed) separately in the next two propositions. Before this, we need a technical lemma.

Lemma 1.4.1. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a proper closed convex function. Then there exists a point $x_0 \in \mathrm{dom}(f)$ such that $\mathrm{epi}(f)$ is supported by a non-vertical hyperplane at $(x_0, f(x_0))$.*

Proof. The set $\mathrm{epi}(f)$ is closed convex and non-empty, and hence from Proposition 2.2.4 ahead we have that it is the intersection of all its *supporting half-spaces* (if $\mathbf{h}$ is a supporting hyperplane of a set A, then the corresponding *supporting half-space* of A is the closed half-space determined by $\mathbf{h}$ which contains A). If $\mathrm{epi}(f)$ is supported only by vertical hyperplanes, then $(x, t_0) \in \mathrm{epi}(f)$ implies that $(x, t) \in \mathrm{epi}(f)$ for every $t \in \mathbb{R}$. On the other hand, fixing a point $x_0 \in \mathrm{dom}(f)$, we get that $(x_0, t) \notin \mathrm{epi}(f)$ if $t < f(x_0)$. This contradiction concludes the proof. □

Remark 1.4.1. It is possible to do better by proving that $\mathrm{epi}(f)$ is supported by a non-vertical hyperplane at each point $(x_0, f(x_0))$ such that x_0 is in the *relative interior* of $\mathrm{dom}(f)$. This concept is introduced in the next chapter. We will not show it here because we only need the weaker variant proved in the lemma above for the purposes of this section.

Proposition 1.4.1. *Assume that $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ is a convex function, and let $v \in \mathbb{R}^n$ be such that $f^*(v) < +\infty$. Then*

$$f^*(v) = \inf\{\alpha \in \mathbb{R} : \mathrm{epi}(f) \subseteq \mathbf{h}^-((v, -1), \alpha)\}.$$

An illustration is given in Figure 1.4.1.

Proof. For any $(x,t) \in \operatorname{epi}(f)$, we have the inequality

$$\langle (v,-1),(x,t)\rangle = \langle v,x\rangle - t \leq \langle v,x\rangle - f(x) \leq f^*(v).$$

Thus, if $\alpha \geq f^*(v)$, we get that $\operatorname{epi}(f) \subseteq \mathbf{h}^-((v,-1),\alpha)$. Consequently,

$$f^*(v) \geq \inf\{\alpha \in \mathbb{R} : \operatorname{epi}(f) \subseteq \mathbf{h}^-((v,-1),\alpha)\}.$$

If the strict inequality holds, then there exists $\alpha_0 < f^*(v)$ such that $\operatorname{epi}(f) \subseteq \mathbf{h}^-((v,-1),\alpha_0)$. This leads to

$$f^*(v) > \alpha_0 \geq \langle (v,-1),(x,f(x))\rangle = \langle v,x\rangle - f(x),$$

for every $x \in \operatorname{dom}(f)$. Taking the supremum on the right-hand side, we obtain

$$f^*(v) = \sup\{\langle v,x\rangle - f(x) : x \in \operatorname{dom}(f)\} \leq \alpha_0 < f^*(v),$$

which is a contradiction. This concludes the proof.

□

Remark 1.4.2. As a consequence, we have that $\operatorname{epi}(f) \subseteq \mathbf{h}^-((u,-1),f^*(u))$ whenever $u \in \operatorname{dom}(f^*)$. Even more, assuming also that f is closed, it holds that if u lies in the relative interior of $\operatorname{dom}(f^*)$, then the hyperplane $\mathbf{h}((u,-1),f^*(u))$ supports $\operatorname{epi}(f)$ at some (boundary) point of $\operatorname{epi}(f)$. That is, the intersection of $\operatorname{epi}(f)$ and $\mathbf{h}((u,-1),f^*(u))$ is non-empty. For a proof, see Lemma 1.6.16 in [2167].

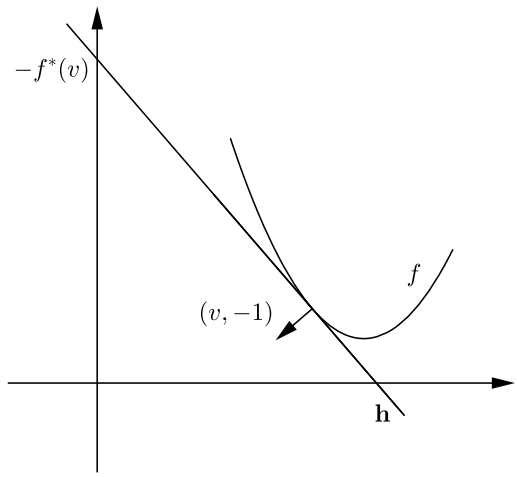


Fig. 1.4.1 Notice that $\mathbf{h}$ crosses the x_{n+1} axis at the point $(0,-f^*(v))$. In some sense, $f^*(v)$ measures how far from the origin is the supporting hyperplane of $\operatorname{epi}(f)$ whose outer normal vector is $(v,-1)$.

Proposition 1.4.2. *Let $f\colon \mathbb{R}^n \to (-\infty,+\infty]$ be a proper closed convex function. Then $f^*(v) = +\infty$ if and only if* $\operatorname{epi}(f)$ *is not contained in any half-space whose bounding hyperplane has $(v,-1)$ as normal vector.*

Proof. First, assume that $\operatorname{epi}(f) \subseteq \mathbf{h}^-((v,-1),\alpha)$ for some $\alpha \in \mathbb{R}$ (the existence of such half-space is guaranteed by Lemma 1.4.1). Then, for any $x \in \operatorname{dom}(f)$, we have

$$\langle v,x\rangle - f(x) = \langle (v,-1),(x,f(x))\rangle \leq \alpha.$$

Thus, $f^*(v) = \sup\{\langle v, x\rangle - f(x) : x \in \mathrm{dom}(f)\} \leq \alpha < +\infty$. This shows that if $f^*(v) = +\infty$, then $\mathrm{epi}(f)$ is not contained in any half-space whose bounding hyperplane is normal to $(v, -1)$.

Now assume that $f^*(v) < +\infty$. Recall that if (x, t) is any point in the epigraph of f, then in particular we have that $x \in \mathrm{dom}(f)$, and

$$\langle (v, -1), (x, t)\rangle = \langle v, x\rangle - t \leq \langle v, x\rangle - f(x) \leq \sup\{\langle v, x\rangle - f(x) : x \in \mathrm{dom}(f)\} = f^*(v).$$

This gives $\mathrm{epi}(f) \subseteq \mathbf{h}^-((v, -1), f^*(v))$.

□

Remark 1.4.3. The inclusion $\mathrm{epi}(f) \subseteq \mathbf{h}^-((v, -1), \alpha) =: \mathbf{h}^-$ is equivalent to the statement that the affine function $g\colon \mathbb{R}^n \to \mathbb{R}$ given by $g(x) = \langle v, x\rangle - \alpha$ satisfies $g \leq f$ (from now on, the notation $g \leq f$ involving functions will **always** mean that $g(x) \leq f(x)$ for any $x \in \mathbb{R}^n$). Indeed, this follows from the easy equivalence

$$\langle (v, -1), (x, y)\rangle \leq \alpha \Leftrightarrow \langle v, x\rangle - \alpha \leq y.$$

If $\mathrm{epi}(f) \subseteq \mathbf{h}^-$, then for any $x \in \mathrm{dom}(f)$ we have $\langle (v, -1), (x, f(x))\rangle \leq \alpha$ from which we get $g(x) \leq f(x)$. On the other hand, if $g \leq f$, then for any $(x, y) \in \mathrm{epi}(f)$ we have $y \geq f(x) \geq g(x) = \langle v, x\rangle - \alpha$, and the equivalence above gives that $\mathrm{epi}(f) \subseteq \mathbf{h}^-$.

This observation allows us to write a convex closed function as the supremum of a family of affine functions.

Proposition 1.4.3. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex closed function. Then for each $x \in \mathbb{R}^n$ we have*

$$f(x) = \sup\{g(x) : g \in \mathrm{Aff}(\mathbb{R}^n) \text{ with } g \leq f\},$$

where $\mathrm{Aff}(\mathbb{R}^n)$ *stands for the family of affine functions* $g\colon \mathbb{R}^n \to \mathbb{R}$.

Proof. Recall that a hyperplane in $\mathbb{R}^{n+1}$ is said to be *vertical* if the $(n+1)$-th coordinate of its normal vector is 0. The half-spaces determined by a vertical hyperplane will be also called *vertical*.

Since $\mathrm{epi}(f)$ is a closed convex set, it is the intersection of all half-spaces containing it (this will be proved later in Proposition 2.2.4). Hence we may write

$$\mathrm{epi}(f) = \bigcap \left\{\mathbf{h}^-((v, \beta), \alpha) : \mathrm{epi}(f) \subseteq \mathbf{h}^-((v, \beta), \alpha)\right\}.$$

We will prove that $\mathrm{epi}(f)$ is indeed the intersection of all **non-vertical** half-spaces containing it, which we denote by

$$U = \bigcap \left\{\mathbf{h}^-((v, -1), \alpha) : \mathrm{epi}(f) \subseteq \mathbf{h}^-((v, -1), \alpha)\right\}.$$

In other words, since $\mathrm{epi}(f) \subseteq U$, it suffices to prove that $U \setminus \mathrm{epi}(f) = \emptyset$. After that, we will show that this geometric fact yields the desired.

If $(x_0, \beta_0) \in U \setminus \mathrm{epi}(f)$, then there exists a vertical half-space containing $\mathrm{epi}(f)$ which does not contain (x_0, β_0). Let $\mathbf{h}_1^- := \mathbf{h}^-((v_1, 0), \alpha_1)$ be a vertical half-space such that $\mathrm{epi}(f) \subseteq \mathbf{h}_1^-$ and $(x_0, \beta_0) \notin \mathbf{h}_1^-$, and define the affine function $g_1\colon \mathbb{R}^n \to \mathbb{R}$ by $g_1(x) = \langle v_1, x\rangle - \alpha_1$. Observe that if $x \in \mathrm{dom}(f)$, then $(x, f(x)) \in \mathrm{epi}(f)$, and hence $(x, f(x)) \in \mathbf{h}_1^-$. Consequently,

$$\langle (v_1, 0), (x, f(x))\rangle \leq \alpha_1,$$

immediately implying that $g_1(x) \leq 0$. Now, using Lemma 1.4.1, we let $x_1 \in \mathrm{dom}(f)$ be such that there is a non-vertical hyperplane which supports $\mathrm{epi}(f)$ at $(x_1, f(x_1))$. We may write

$$\operatorname{epi}(f) \subseteq \mathbf{h}_2^- := \mathbf{h}^-((v_2, -1), \alpha_2)$$

for some $v_2 \in \mathbb{R}^n$ and $\alpha_2 \in \mathbb{R}$. Define the affine function $g_2 \colon \mathbb{R}^n \to \mathbb{R}$ by $g_2(x) = \langle v_2, x\rangle - \alpha_2$. The inclusion above gives immediately that $g_2(x) \leq f(x)$ for any $x \in \mathbb{R}^n$. Now, let $g \colon \mathbb{R}^n \to \mathbb{R}$ be the affine function given as

$$g := \lambda g_1 + g_2,$$

where $\lambda > 0$ is a number large enough such that $g(x_0) > \beta_0$. Notice that, since $(x_0, \beta_0) \notin \mathbf{h}_1^-$, we have that $\langle v_1, x_0\rangle > \alpha_1$, implying that $g_1(x_0) > 0$. Hence we can choose such a number λ.

For any $x \in \operatorname{dom}(f)$, we have that $g(x) = \lambda g_1(x) + g_2(x) \leq g_2(x) \leq f(x)$. Therefore, the inequality $g \leq f$ holds for any $x \in \mathbb{R}^n$, and hence

$$\operatorname{epi}(f) \subseteq \mathbf{h}^- := \mathbf{h}^-((\lambda v_1 + v_2, -1), \lambda\alpha_1 + \alpha_2).$$

That is, $\mathbf{h}^-$ is a non-vertical half-space containing $\operatorname{epi}(f)$. In particular, we have $U \subseteq \mathbf{h}^-$. On the other hand, since $g(x_0) > \beta_0$, we get that $(x_0, \beta_0) \notin \mathbf{h}^-$, and hence $(x_0, \beta_0) \notin U$. This contradiction gives $U \setminus \operatorname{epi}(f) = \emptyset$.

Now we will show how the characterization of the epigraph of f as an intersection of all non-vertical half-spaces containing it yields the desired. First of all, assume that $f(x_0) = +\infty$. Then for any $t \in \mathbb{R}$ we get that $(x_0, t) \notin \operatorname{epi}(f)$. This means that for any $t \in \mathbb{R}$ there exist a vector $v_t \in \mathbb{R}^n$ and a number α_t such that $\operatorname{epi}(f) \subseteq \mathbf{h}_t^- := \mathbf{h}^-((v_t, -1), \alpha_t)$ with the property that $(x_0, t) \notin \mathbf{h}^-((v_t, -1), \alpha_t)$. Consequently,

$$\alpha_t < \langle (v_t, -1), (x_0, t)\rangle = \langle v_t, x_0\rangle - t.$$

Let $g_t \colon \mathbb{R}^n \to \mathbb{R}$ be the affine function given by $g_t(x) = \langle v_t, x\rangle - \alpha_t$. The inclusion $\operatorname{epi}(f) \subseteq \mathbf{h}_t^-$ gives immediately that $g_t \leq f$, and the inequality above implies $g_t(x_0) > t$. Thus,

$$\sup\{g(x_0) : g \in \operatorname{Aff}(\mathbb{R}^n) \text{ with } g \leq f\} \geq \sup\{g_t(x_0) : t \in \mathbb{R}\} \geq \sup\{t : t \in \mathbb{R}\} = +\infty,$$

and hence the case where $f(x_0) = +\infty$ is settled. Now we assume that $f(x_0) < +\infty$. It is clear that

$$f(x_0) \geq \sup\{g(x_0) : g \in \operatorname{Aff}(\mathbb{R}^n) \text{ with } g \leq f\} =: c.$$

Let $\mathbf{h} = \mathbf{h}((v, -1), \alpha)$ be any (non-vertical) hyperplane such that $\operatorname{epi}(f) \subseteq \mathbf{h}^-$, and let $g_0 \in \operatorname{Aff}(\mathbb{R}^n)$ be given as $g_0(x) = \langle v, x\rangle - \alpha$ for each $x \in \mathbb{R}^n$. Since $\operatorname{epi}(f) \subseteq \mathbf{h}^-$, we have that $g_0 \leq f$. Therefore,

$$c = \sup\{g(x_0) : g \in \operatorname{Aff}(\mathbb{R}^n) \text{ and } g \leq f\} \geq g_0(x_0).$$

Thus,

$$\langle (x_0, c), (v, -1)\rangle = \langle x_0, v\rangle - c = g_0(x_0) + \alpha - c \leq g_0(x_0) + \alpha - g_0(x_0) = \alpha,$$

and hence $(x_0, c) \in \mathbf{h}^-((v, -1), \alpha)$. Once this holds for any non-vertical half-space containing $\operatorname{epi}(f)$, we get that (x_0, c) is contained in the intersection of all those half-spaces. Because this intersection equals $\operatorname{epi}(f)$, it follows that

$$(x_0, c) \in \operatorname{epi}(f).$$

This leads to $c \geq f(x_0)$, and this concludes the proof.

□

Next, we justify why conjugation is a dual notion (at least for the lower semi-continuous case).

Theorem 1.4.1. *Let $f : \mathbb{R}^n \to (-\infty, +\infty]$ be a convex proper function. Then f^* is a convex closed proper function. Moreover, if f is convex and closed, then $f^{**} = f$.*

Proof. We already know that f^* is a convex function. For each fixed $x \in \mathrm{dom}(f)$ we have that $v \mapsto \langle v, x\rangle - f(x)$ is an affine function, and hence it is, in particular, lower semi-continuous. It follows from the definition that f^* is the supremum of a family of lower semi-continuous functions, and therefore it is lower semi-continuous (see Proposition 1.1.4). To check that f^* is proper, we use an argument similar to that of Lemma 1.4.1 (here we are not assuming that f is closed). The set $\mathrm{cl}(\mathrm{epi}(f))$ is closed and convex, and it is not difficult to see that it is not $\mathbb{R}^n$. Indeed, assume by contradiction that $\mathrm{cl}(\mathrm{epi}(f)) = \mathbb{R}^n$, and let $x_0 \in \mathbb{R}^n$ be such that $f(x_0) = t_0 < +\infty$. Then the points of the line $\{(x, t_0) : t \in \mathbb{R}\}$ may be approximated by points of $\mathrm{epi}(f)$. Hence we may find a sequence $(x_n)_{n\in\mathbb{N}}$ in $\mathrm{dom}(f)$ such that $x_n \to t_0$ and $f(x_n) \to -\infty$, and this contradicts the fact that f is Lipschitz in the compact subset $\{x_n : n \in \mathbb{N}\} \cup \{x_0\}$ of $\mathrm{dom}(f)$.

Once $\mathrm{cl}(\mathrm{epi}(f))$ is closed, convex, and properly contained in $\mathbb{R}^n$, by the same argument as in Lemma 1.4.1, we get that $\mathrm{cl}(\mathrm{epi}(f))$ is supported by some non-vertical hyperplane. Thus, we may assume that

$$\mathrm{epi}(f) \subseteq \mathrm{cl}(\mathrm{epi}(f)) \subseteq \mathbf{h}^-((v, -1), \alpha)$$

for some vector $v \in \mathbb{R}^n$ and some number $\alpha \in \mathbb{R}$. Then, for every $x \in \mathrm{dom}(f)$, we have

$$\langle (x, f(x)), (v, -1)\rangle \leq \alpha,$$

and this gives

$$\langle v, x\rangle - f(x) \leq \alpha$$

for each $x \in \mathrm{dom}(f)$. It follows that $f^*(v) \leq \alpha$, and hence f^* is proper.

It remains to prove the "duality" claim. To do so, observe first that $(u, \alpha) \in \mathrm{epi}(f^*)$ if and only if the affine map $g(x) = \langle x, u\rangle - \alpha$ is such that $g(x) \leq f(x)$ for every $x \in \mathbb{R}^n$. Indeed, if $(u, \alpha) \in \mathrm{epi}(f^*)$, then

$$\alpha \geq f^*(u) = \sup\{\langle x, u\rangle - f(x) : x \in \mathbb{R}^n\},$$

implying that

$$\alpha \geq \langle x, u\rangle - f(x),$$

for every $x \in \mathbb{R}^n$, which yields the desired inequality. On the other hand, if $\langle x, u\rangle - \alpha \leq f(x)$ for every $x \in \mathbb{R}^n$, then it is clear that $\alpha \geq f^*(u)$, and thus $(u, \alpha) \in \mathrm{epi}(f^*)$. As a consequence, using Proposition 1.4.3 we have that

$$f(x) = \sup\{g(x) : g \in \mathrm{Aff}(\mathbb{R}^n),\ g \leq f\} = \sup\{\langle u, x\rangle - \alpha : (u, \alpha) \in \mathrm{epi}(f^*)\}, \tag{1.4.1}$$

for each $x \in \mathbb{R}^n$. Now notice that the conjugate function of a given convex function f can be equivalently defined as

$$f^*(v) = \sup\{\langle v, x\rangle - \beta : (x, \beta) \in \mathrm{epi}(f)\},$$

for each $v \in \mathbb{R}^n$. Therefore,

$$f^{**}(x) = \sup\{\langle x, u\rangle - \alpha : (u, \alpha) \in \mathrm{epi}(f^*)\}.$$

In view of equality (1.4.1) above, this yields the desired result.

□

1.5 The Prékopa–Leindler inequality

The Prekópa–Leindler is an important functional inequality which will be used later to prove the classical Brunn–Minkowski inequality. This section is devoted to state and prove this inequality. We will assume that the reader has some basic knowledge on the theory of Lebesgue measure and integration (for that, see Appendix A). We also need the following "technical" inequality.

Lemma 1.5.1. *Let $a, b > 0$ and let $\lambda \in (0,1)$. Then*

$$\lambda a + (1-\lambda)b \geq a^{\lambda} b^{1-\lambda}.$$

Equality holds for some $\lambda \in (0,1)$ if and only if $a = b$.

Proof. This comes from the convexity of the exponential map $t \mapsto e^t$. Indeed, this function is convex since its second derivative is strictly positive (see Corollary 1.2.2). Therefore,

$$e^{\lambda x + (1-\lambda)y} \leq \lambda e^x + (1-\lambda)e^y,$$

for any $x, y \in \mathbb{R}$ and any $\lambda \in (0,1)$. For $a, b > 0$, set $x = \log a$ and $y = \log b$. This yields the desired inequality. If $a \neq b$, then $\log a \neq \log b$, and hence for any $\lambda \in (0,1)$ the strict convexity of the exponential map gives

$$a^{\lambda} b^{1-\lambda} = e^{\lambda \log a + (1-\lambda)\log b} < \lambda e^{\log a} + (1-\lambda)e^{\log b} = \lambda a + (1-\lambda)b,$$

from which equality holds if and only if $a = b$.

□

In addition, we need a result which is essentially the mentioned Brunn–Minkowski inequality in one dimension, as it will be clear later. Below, $\mathcal{L}^1$ stands for the usual Lebesgue measure of $\mathbb{R}$.

Lemma 1.5.2. *Let $A, B, C \subseteq \mathbb{R}$ be non-empty (Borel) measurable sets, and let $0 < \lambda < 1$ be such that $C \supseteq (1-\lambda)A + \lambda B$. Then*

$$\mathcal{L}^1(C) \geq (1-\lambda)\mathcal{L}^1(A) + \lambda \mathcal{L}^1(B).$$

Proof. From Corollary A.1.1, we have that the measure of a Borel set in $\mathbb{R}$ is the supremum of the measures of the compact sets that it contains. Hence, we may assume that A and B are compact (since in the non-compact case we simply approximate them by respective compact subsets). Let $q = \max A$ and $p = \min B$. Geometrically, the right endpoint of the set $A - q$ and the left endpoint of the set $B - p$ are both 0, and then the same holds for the sets $(1-\lambda)(A-q)$ and $\lambda(B-p)$, respectively. Therefore, we have that

(i) $\mathcal{L}^1((1-\lambda)(A-q) \cup \lambda(B-p)) = (1-\lambda)\mathcal{L}^1(A-q) + \lambda\mathcal{L}^1(B-p)$, since the intersection of both sets has zero Lebesgue measure (it is just the point 0), and

(ii) $(1-\lambda)(A-q) + \lambda(B-p) \supseteq (1-\lambda)(A-q) \cup \lambda(B-p)$, since 0 is in both sets $(1-\lambda)(A-q)$ and $\lambda(B-p)$.

It follows that

$$\mathcal{L}^1((1-\lambda)(A-q)+\lambda(B-p)) \geq \mathcal{L}^1((1-\lambda)(A-q)\cup\lambda(B-p)) = \\ = (1-\lambda)\mathcal{L}^1(A-q)+\lambda\mathcal{L}^1(B-p).$$

On the other hand, letting $r=(1-\lambda)q+\lambda p$, it is clear that

$$(1-\lambda)(A-q)+\lambda(B-p) = (1-\lambda)A+\lambda B-r,$$

from where we get that

$$C-r \supseteq (1-\lambda)A+\lambda B-r.$$

Using the fact that the Lebesgue measure is translation-invariant, we arrive at

$$\mathcal{L}^1(C) = \mathcal{L}^1(C-r) \geq \mathcal{L}^1((1-\lambda)A+\lambda B-r) = \\ = \mathcal{L}^1((1-\lambda)(A-q)+\lambda(B-p)) \geq (1-\lambda)\mathcal{L}^1(A-q)+\lambda\mathcal{L}^1(B-p) = \\ = (1-\lambda)\mathcal{L}^1(A)+\lambda\mathcal{L}^1(B),$$

and the proof is complete.

□

In what follows, we consider that $\mathbb{R}^n$ is equipped with the usual Lebesgue measure. Hence all the integrations will be considered with respect to this measure. We refer the reader to Appendix A for more on measure theory.

Theorem 1.5.1 (Prékopa–Leindler)**.** *Let $f,g,h\colon \mathbb{R}^n \to [0,+\infty)$ be measurable functions, and assume that f and g are integrable (with respect to the Lebesgue measure of $\mathbb{R}^n$). If for a given $\lambda\in(0,1)$ the inequality*

$$h(\lambda x+(1-\lambda)y) \geq f(x)^\lambda g(y)^{1-\lambda}$$

holds for almost every $x,y\in\mathbb{R}^n$, then

$$\int_{\mathbb{R}^n} h \geq \left(\int_{\mathbb{R}^n} f\right)^\lambda \left(\int_{\mathbb{R}^n} g\right)^{1-\lambda}.$$

Proof. We prove the inequality by induction on the dimension of the domain. First, we consider the case where $n=1$. We assume that f and g are not the zero function, since otherwise there is nothing to prove. We also claim that we may assume that f and g are bounded. Indeed, if one of them is not bounded (f, say), then from the definition of the Lebesgue integral we may let (f_m) be a sequence of bounded, measurable, non-negative functions such that $f_m \leq f$ and

$$\int_{\mathbb{R}} f_m \to \int_{\mathbb{R}} f$$

as $m\to\infty$. Because $f_m \leq f$ (and all involved functions are non-negative), then the hypothesis inequality holds true if we replace f by f_m. Consequently, if we prove the result for the case where f and g are bounded, then we have the general case straightforwardly by approximations.

Now that we are assuming that f and g are bounded and not identically zero, observing that the homogeneity is the same in both inequalities of the enunciation we conclude that we may also assume that $\sup f = \sup g = 1$.

Let $t\geq 0$. If $x,y\in\mathbb{R}$ are such that $f(x)\geq t$ and $g(y)\geq t$, then $h((1-\lambda)x+\lambda y)\geq t$. Thus, we have the inclusion

$$\{z \in \mathbb{R}\colon h(z) \geq t\} \supseteq (1-\lambda)\{x \in \mathbb{R}\colon f(x) \geq t\} + \lambda\{y \in \mathbb{R}\colon g(y) \geq t\}. \tag{1.5.1}$$

From Lemma 1.5.2, this leads to

$$\mathcal{L}^1(\{h \geq t\}) \geq (1-\lambda)\mathcal{L}^1(\{f \geq t\}) + \lambda\mathcal{L}^1(\{g \geq t\}), \tag{1.5.2}$$

where the reader should notice that we adopted a reduced notation for the sets in (1.5.1). Finally, using this inequality together with Lemma A.5.2 and Lemma 1.5.1, we arrive at

$$\begin{aligned}\int_{\mathbb{R}} h = \int_0^\infty \mathcal{L}^1(\{h \geq t\})\,dt \geq \int_0^\infty (1-\lambda)\mathcal{L}^1(\{f \geq t\}) + \lambda\mathcal{L}^1(\{g \geq t\})\,dt = \\ = (1-\lambda)\int_0^\infty \mathcal{L}^1(\{f \geq t\})\,dt + \lambda\int_0^\infty \mathcal{L}^1(\{g \geq t\})\,dt = \\ = (1-\lambda)\int_{\mathbb{R}} f + \lambda\int_{\mathbb{R}} g \geq \left(\int_{\mathbb{R}} f\right)^{1-\lambda}\left(\int_{\mathbb{R}} g\right)^{\lambda},\end{aligned}$$

and this settles the case $n = 1$.

Now let $n \geq 2$ and assume that the Prékopa–Leindler inequality holds in dimension $n-1$. Let $h, f, g\colon \mathbb{R}^n \to [0, +\infty)$ be as in the enunciation of the theorem. We write $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$ to represent a vector $u \in \mathbb{R}^n$ by $(x, t) \in \mathbb{R}^{n-1} \times \mathbb{R}$. For any $t \in \mathbb{R}$, define the functions $f_t, h_t, g_t\colon \mathbb{R}^{n-1} \to [0, +\infty)$ as $h_t(x) := h(x, t)$, and f_t and g_t in the same way. From standard integration theory, we have that these functions are measurable, and that f_t and g_t are integrable (for almost all $t \in \mathbb{R}$, at least). For any $t_0, t_1 \in \mathbb{R}$ and any $x, y \in \mathbb{R}^{n-1}$ we have the inequality

$$\begin{aligned}h_{\lambda t_0+(1-\lambda)t_1}(\lambda x + (1-\lambda)y) = h(\lambda(x, t_0) + (1-\lambda)(y, t_1)) \geq f(x, t_0)^\lambda g(y, t_1)^{1-\lambda} = \\ = f_{t_0}(x)^\lambda g_{t_1}(y)^{1-\lambda},\end{aligned}$$

and the Prékopa–Leindler inequality for dimension $n-1$ gives that

$$\int_{\mathbb{R}^{n-1}} h_{\lambda t_0+(1-\lambda)t_1} \geq \left(\int_{\mathbb{R}^{n-1}} f_{t_0}\right)^\lambda \left(\int_{\mathbb{R}^{n-1}} g_{t_1}\right)^{1-\lambda}$$

for every $t_0, t_1 \in \mathbb{R}$ and any $\lambda \in (0, 1)$. Now define the function $H\colon \mathbb{R} \to [0, +\infty)$ by

$$H(t) := \int_{\mathbb{R}^{n-1}} h_t,$$

and the functions $F, G\colon \mathbb{R} \to [0, +\infty)$ analogously. Again we have from standard measure and integration theory that H is measurable, and F and G are integrable. The inequality above reads

$$H(\lambda t_0 + (1-\lambda)t_1) \geq F(t_0)^\lambda G(t_1)^{1-\lambda}.$$

Therefore, we can apply the one-dimensional Prekópa–Leindler inequality to obtain

$$\int_{\mathbb{R}} H \geq \left(\int_{\mathbb{R}} F\right)^\lambda \left(\int_{\mathbb{R}} G\right)^{1-\lambda}.$$

Finally, by Fubini's theorem we have the equality

$$\int_{\mathbb{R}} H = \int_{\mathbb{R}} \left(\int_{\mathbb{R}^{n-1}} h_t\right) dt = \int_{\mathbb{R}^n} h,$$

and analogously we have

$$\int_{\mathbb{R}} F = \int_{\mathbb{R}^n} f, \text{ and } \int_{\mathbb{R}} G = \int_{\mathbb{R}^n} g.$$

Substituting these equalities in the inequality for the integrals of H, F, and G above yields the n-dimensional Prékopa–Leindler inequality, as we wanted.

□

Later we will use the Prékopa–Leindler inequality to give a short proof of the Brunn–Minkowski inequality. But now we will present another application of this inequality. A function $f\colon \mathbb{R}^n \to [0, +\infty)$ is *log-concave* if $f = \exp(-\varphi)$ for some convex function $\varphi\colon \mathbb{R}^n \to (-\infty, +\infty]$. In what follows, we make the convention that $\exp(-\infty) = 0$. The name is justified because a function $g\colon \mathbb{R}^n \to [-\infty, +\infty)$ is said to be *concave* if $-g$ is convex.

Lemma 1.5.3. *A function $f\colon \mathbb{R}^n \to [0, +\infty)$ is log-concave if and only if*

$$f(\lambda x + (1-\lambda)y) \geq f(x)^{\lambda} f(y)^{1-\lambda},$$

for any $x, y \in \mathbb{R}^n$ and every $\lambda \in (0, 1)$.

Proof. If $f = \exp(-\varphi)$ for some convex function $\varphi\colon \mathbb{R}^n \to (-\infty, +\infty]$, then for any $x, y \in \mathbb{R}^n$ and $\lambda \in (0, 1)$ we have $-\varphi(\lambda x + (1-\lambda)y) \geq -\lambda\varphi(x) - (1-\lambda)\varphi(y)$, and hence

$$\begin{aligned} f(\lambda x + (1-\lambda)y) = \exp(-\varphi(\lambda x + (1-\lambda)y)) &\geq \exp(-\lambda\varphi(x) - (1-\lambda)\varphi(y)) = \\ &= \exp(-\varphi(x))^{\lambda} \exp(-\varphi(y))^{1-\lambda} = f(x)^{\lambda} f(y)^{1-\lambda}. \end{aligned}$$

Now assume that the inequality holds. Under the convention that $\log 0 = -\infty$, we can define the composition $\varphi = -\log \circ f\colon \mathbb{R}^n \to (-\infty, +\infty]$. All we have to prove is that φ is a convex function. For $x, y \in \mathbb{R}^n$ and $\lambda \in (0, 1)$ we have

$$\begin{aligned} \varphi(\lambda x + (1-\lambda y)) = -\log\Big(f(\lambda x + (1-\lambda)y)\Big) &\leq -\log\Big(f(x)^{\lambda} f(y)^{1-\lambda}\Big) = \\ &= -\lambda \log f(x) - (1-\lambda)\log f(y) = \lambda\varphi(x) + (1-\lambda)\varphi(y), \end{aligned}$$

and this concludes the proof.

□

Observe that the product of log-concave functions is always log-concave. The announced application of the Prékopa–Leindler inequality is that the log-concavity property is also preserved under *convolution*.

Proposition 1.5.1. *Let $f, g\colon \mathbb{R}^n \to [0, +\infty)$ be integrable log-concave functions. Then their* convolution *$f * g\colon \mathbb{R}^n \to [0, +\infty]$, defined as*

$$f * g(x) = \int_{\mathbb{R}^n} f(x - y) g(y)\, dy,$$

is also a log-concave function, provided $f * g < +\infty$ in $\mathbb{R}^n$.

Proof. Let $x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in (0, 1)$ be arbitrarily given. Define the functions $H, F_1, F_2\colon \mathbb{R}^n \to [0, +\infty)$ by

$$H(y) = f(\lambda x_1 + (1-\lambda)x_2 - y)g(y),$$
$$F_1(y) = f(x_1 - y)g(y), \text{ and}$$
$$F_2(y) = f(x_2 - y)g(y),$$

for each $y \in \mathbb{R}^n$. Notice that the log-concavity of f and g yields that for any $w, z \in \mathbb{R}^n$ we have

$$H(\lambda w + (1-\lambda)z) = f(\lambda(x_1 - w) + (1-\lambda)(x_2 - z))g(\lambda w + (1-\lambda)z) \geq$$
$$\geq f(x_1 - w)^\lambda f(x_2 - z)^{1-\lambda} g(w)^\lambda g(z)^{1-\lambda} = F_1(w)^\lambda F_2(z)^{1-\lambda}.$$

Thus, the Prékopa–Leindler inequality applied for the functions H, F_1, and F_2 leads to

$$f * g(\lambda x_1 + (1-\lambda)x_2) = \int_{\mathbb{R}^n} f(\lambda x_1 + (1-\lambda)x_2) - y)g(y)\,dy =$$
$$= \int_{\mathbb{R}^n} H(y)\,dy \geq \left(\int_{\mathbb{R}^n} F_1(w)\,dw\right)^\lambda \left(\int_{\mathbb{R}^n} F_2(z)\,dz\right)^{1-\lambda} =$$
$$= \left(\int_{\mathbb{R}^n} f(x_1 - w)g(w)\,dw\right)^\lambda \left(\int_{\mathbb{R}^n} f(x_2 - z)g(z)\,dz\right)^{1-\lambda} = (f * g(x_1))^\lambda (f * g(x_2))^{1-\lambda}.$$

Since $x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in (0,1)$ are arbitrary, it follows from Lemma 1.5.3 that $f * g$ is log-concave. □

1.6 Exercises

Exercise 1.1. *Let $D \subseteq \mathbb{R}^n$ be a non-empty convex set, and assume that $f\colon D \to (-\infty, +\infty]$ is a convex function. Prove that the extension $\bar{f}\colon \mathbb{R}^n \to (-\infty, +\infty]$ of f given by*

$$\bar{f}(x) = \begin{cases} f(x) & \text{if } x \in D \\ +\infty & \text{if } x \in \mathbb{R}^n \setminus D \end{cases}$$

is convex.

Exercise 1.2. *Give an example of a convex function $f\colon \mathbb{R} \to (-\infty, +\infty]$ which is not continuous when restricted to its effective domain.*

Remark 1.6.1. Remember that a convex function is always continuous in the *interior* of its effective domain.

Exercise 1.3. *Provide an example of a continuous convex function which is not globally Lipschitz.*

For what follows, we need a new definition. A function $f\colon \mathbb{R}^n \to [-\infty, +\infty)$ is said to be *upper semi-continuous* if the function $-f$ is lower semi-continuous.

Exercise 1.4. *Let $f\colon \mathbb{R}^n \to [-\infty, +\infty)$. Prove that the following statements are equivalent:*

(i) *f is upper semi-continuous.*

(ii) *For each $t \in \mathbb{R}$, the* upper level set

$$\{f \geq t\} := \{x \in \mathbb{R}^n : f(x) \geq t\}$$

is closed.

(iii) *For every $x \in \mathbb{R}^n$ and any $\varepsilon > 0$ there exists a neighborhood V of x such that $f(y) \leq f(x) + \varepsilon$ for any $y \in V$.*

(iv) *For each $x \in \mathbb{R}^n$ and every sequence $(x_k)_{k\in\mathbb{N}}$ converging to x it holds that*

$$\limsup_{k\to\infty} f(x_k) \leq f(x).$$

Exercise 1.5. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function. Prove that if there exists a number $c \in \mathbb{R}$ such that $f(x) \leq c$ for all $x \in \mathbb{R}^n$, then f is constant.*

Exercise 1.6. *Let $f\colon \mathbb{R}^n \to [-\infty, +\infty]$. Prove that*

$$f(x) = \inf\left\{t \in \mathbb{R} : (x,t) \in \operatorname{epi}(f)\right\}$$

for each $x \in \mathbb{R}^n$, where we use that $\inf \emptyset = +\infty$ and $\inf(\mathbb{R}) = -\infty$. As a consequence, show that if $f, g\colon \mathbb{R}^n \to [-\infty, +\infty]$ are such that $\operatorname{epi}(f) = \operatorname{epi}(g)$, then $f = g$.

Remark 1.6.2. The original definition of epigraph was given for functions onto $(-\infty, +\infty]$, but this definition clearly extends without any modification to the case of functions onto $[-\infty, +\infty]$.

Exercise 1.7. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a function. Prove that there exists a unique function $\operatorname{cl} f\colon \mathbb{R}^n \to [-\infty, +\infty]$ whose epigraph is $\operatorname{cl}(\operatorname{epi}(f))$. Verify that $\operatorname{cl} f$ is lower semi-continuous.*

Remark 1.6.3. The function $\operatorname{cl} f$ is called the *lower semi-continuous hull* of f.

Exercise 1.8. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a convex function. Show that $\operatorname{cl}(\operatorname{epi}(f))$ is a convex subset of $\mathbb{R}^{n+1}$. In particular, conclude that if $\operatorname{cl} f > -\infty$ in $\mathbb{R}^n$, then $\operatorname{cl} f$ is convex.*

Remark 1.6.4. The condition $\operatorname{cl} f > -\infty$ appears here because according to our definition only functions $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ may be convex. This also happens in some of the next exercises.

Exercise 1.9. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex function. Prove that*

$$\partial(\operatorname{epi}(f)) = \left\{(x, f(x)) : x \in \mathbb{R}^n\right\} \text{ and}$$
$$\operatorname{int}(\operatorname{epi}(f)) = \left\{(x,t) : x \in \mathbb{R}^n \text{ and } t > f(x)\right\}.$$

Remark 1.6.5. We do not need convexity here: these equalities hold for any continuous function $f\colon \mathbb{R}^n \to \mathbb{R}$. Recall that if $f\colon \mathbb{R}^n \to \mathbb{R}$ is convex, then f is continuous in $\mathbb{R}^n$ because the interior of the effective domain of f is $\mathbb{R}^n$.

Exercise 1.10. *Let $D \subseteq \mathbb{R}^n$ be a convex set, and let $f\colon D \to \mathbb{R}$ be a continuous function. Assume that for each $x \in D$ there exists an affine function $g\colon \mathbb{R}^n \to \mathbb{R}$ and an open set $U \subseteq \mathbb{R}^n$ containing x such that $f(x) = g(x)$ and $f(z) \geq g(z)$ for any $z \in U \cap D$. Prove that f is convex.*

Exercise 1.11. *Exhibit a probability space (X, Σ, μ), an interval D, a function $f\colon X \to D$ which is not constant μ-almost everywhere, and a non-affine convex function $\varphi\colon D \to (-\infty, +\infty]$ such that both sides of (1.2.8) are $+\infty$.*

Exercise 1.12. *Use the discrete Jensen inequality to prove the* inequality of arithmetic and geometric means*, which states that for any $c_1, \ldots, c_m$ such that $c_1 + \ldots + c_m = 1$, and for every $t_1, \ldots, t_m > 0$ it holds that*

$$c_1 t_1 + \ldots + c_m t_m \geq t_1^{c_1} \ldots t_m^{c_m}. \tag{1.6.1}$$

Hint: Consider the logarithm function.

Exercise 1.13. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a proper convex function, and let $x \in \operatorname{int}(\operatorname{dom}(f))$ be such that $c = f(x)$ is not a global minimum of f. Show that for any $u \in \partial f(x)$,*

(a) *$f'_+(x, u) > 0$ and*

(b) *$\sup\{f'_-(x, u+v) : v \in u^\perp\} \leq |u|^2 \leq \inf\{f'_+(x, u+v) : v \in u^\perp\}$, where $u^\perp := \mathbf{h}(u, 0)$.*

Exercise 1.14. *Let $g\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function, and assume that there exists $x \in \mathbb{R}^n$ such that $g(x) = c \neq 0$.*

(a) *Prove that $\{g > c\} \neq \emptyset$ and $\{g < c\} \neq \emptyset$.*

(b) *Show that $\{g \leq c\}$ is a convex set.*

(c) *Prove that $x \in \partial\{g \leq c\}$.*

Remark 1.6.6. In the next chapter, it will be proved that every proper convex set of $C \subseteq \mathbb{R}^n$ has non-empty boundary ∂C and that C is supported by a hyperplane at each point of ∂C. As a consequence of the previous exercise, we have that $\{g \leq c\}$ is supported by some hyperplane at x whenever g is sublinear and $g(x) = c \neq 0$. The reader should keep this in mind for the next exercise.

Exercise 1.15. *Let $g\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function, and let $x \in \mathbb{R}^n$ be a point such that $g(x) = c \neq 0$. Prove that g is differentiable at x if and only the sublevel set $\{g \leq c\}$ is supported by a unique hyperplane at x.*

Exercise 1.16. *Give an example of a convex function f which admits a non-minimizer point $x \in \operatorname{int}(\operatorname{dom}(f))$ and a vector $u \in \partial f(x)$ such that the equality $f'_-(x, u) = 0$ holds.*

Exercise 1.17 (Rockafellar's theorem). *A set $C \subseteq \mathbb{R}^n \times \mathbb{R}^n$ is said to be* cyclically monotonic *if for any $m \in \mathbb{N}$ and every pairs $(x_1, v_1), \ldots, (x_m, v_m) \in C$ we have*

$$\langle x_2 - x_1, v_1\rangle + \langle x_3 - x_2, v_2\rangle + \ldots + \langle x_m - x_{m-1}, v_{m-1}\rangle + \langle x_1 - x_m, v_m\rangle \leq 0.$$

Prove that a non-empty set $C \subseteq \mathbb{R}^n \times \mathbb{R}^n$ is cyclically monotonic if and only if there exists a proper convex function $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ such that $C \subseteq \partial f$.

Exercise 1.18. *Give an example of a real-valued, convex and continuous function $f\colon \mathbb{R}^n \to \mathbb{R}$ such that f^* is not real-valued.*

Exercise 1.19. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be any given function. Fix $x_0 \in \mathbb{R}^n$ and let $\bar{f}(x) := f(x - x_0)$ for each $x \in \mathbb{R}^n$. Show that*

$$\bar{f}^*(x) = \langle x, x_0\rangle + f^*(x)$$

for every $x \in \mathbb{R}^n$. Also, if $a \in \mathbb{R}$ and $\tilde{f}(x) := a + f(x)$, verify that

$$\tilde{f}^*(x) = -a + f^*(x)$$

for any $x \in \mathbb{R}^n$.

Exercise 1.20. *Let $A \subseteq \mathbb{R}^n$ be non-empty. The* indicator function *of A is the function $I_A^\infty\colon \mathbb{R}^n \to [0, +\infty]$ given as*

$$I_A^\infty(x) = \begin{cases} 0 \text{ if } x \in A \\ +\infty \text{ if } x \in \mathbb{R}^n \setminus A \end{cases}.$$

(a) *Prove that I_A^∞ is convex if and only if A is convex.*

(b) *Show that I_A^∞ is lower semi-continuous if and only if A is closed.*

(c) *Prove that*

$$(I_A^\infty)^*(v) = \sup\{\langle x, v\rangle : x \in A\}$$

for each $v \in \mathbb{R}^n$.

Remark 1.6.7. In other words, the conjugate of the indicator function is the *support function*, which will be intensively studied in the next chapter.

Exercise 1.21. *Let $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ be a proper convex function. Show that*

$$(\lambda f)^*(\cdot) = \lambda f^*\left(\frac{\cdot}{\lambda}\right)$$

for every $\lambda > 0$.

Most of the exercises below were taken from (or inspired in) some recent papers dealing with functional versions of geometric inequalities, such as [376], [638], and [641]. Some of them were taken from [2167]. More references in this direction are given in Remark 1.6.17 below.

Exercise 1.22. *Let $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ be functions. The* infimal convolution *of f and g is the function $f\square g\colon \mathbb{R}^n \to [-\infty, +\infty]$ defined as*

$$f\square g(x) = \inf\left\{f(z) + g(x - z) : z \in \mathbb{R}^n\right\}$$

for each $x \in \mathbb{R}^n$. We also define the Minkowski sum *(or simply* sum*) of two sets $A, B \subseteq \mathbb{R}^n$ as*

$$A + B := \{a + b : a \in A \text{ and } b \in B\}.$$

(a) *Assume that $f\square g > -\infty$ in $\mathbb{R}^n$. Prove that if f and g are convex, then $f\square g$ is convex.*

(b) *For any subsets $A, B \subseteq \mathbb{R}^n$, prove that $I_A^\infty \square I_B^\infty = I_{A+B}^\infty$.*

(c) *Give an example of functions $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ which are proper, convex and closed, but are such that $f\square g$ is proper and convex, but not closed.*

Remark 1.6.8. The Minkowski sum of sets is a very important concept in convex geometry. It will be better investigated in the next chapter.

Exercise 1.23. *Let $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ be functions such that $f\square g(x) > -\infty$ for each $x \in \mathbb{R}^n$. Prove that*

$$\operatorname{dom}(f\square g) = \operatorname{dom}(f) + \operatorname{dom}(g).$$

Exercise 1.24. *Let $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ be proper functions. Prove that*

$$\operatorname{epi}(f\square g) = \operatorname{epi}(f) + \operatorname{epi}(g).$$

Verify that if one of the functions is not proper, then both sides of the above are empty.

Remark 1.6.9. This result has a clear geometric interpretation: if $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ are proper, then $f\square g\colon \mathbb{R}^n \to [-\infty, \infty]$ is the (unique) function whose epigraph is $\operatorname{epi}(f) + \operatorname{epi}(g)$. Because of this, the infimal convolution is sometimes called *epi-sum*.

Exercise 1.25. *Let $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ be proper functions. Prove that if $f\square g(x) > -\infty$ for every $x \in \mathbb{R}^n$, then $(f\square g)^* = f^* + g^*$.*

Exercise 1.26. *The* Asplund product *of two functions $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ is the function $f \star g\colon \mathbb{R}^n \to (-\infty, +\infty]$ defined as*

$$f \star g(x) := \sup\{f(z)g(x - z) : z \in \mathbb{R}^n\}$$

for each $x \in \mathbb{R}^n$.

(a) *The* characteristic function *of a subset $A \subseteq \mathbb{R}^n$ is the function $1_A\colon \mathbb{R}^n \to \{0, 1\}$ given as*

$$1_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in \mathbb{R}^n \setminus A \end{cases}.$$

Show that $1_A \star 1_B = 1_{A+B}$ for any sets $A, B \subseteq \mathbb{R}^n$.

(b) *Assume that $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ are non-negative functions. Prove that if f is convex or g is convex, then $f \star g$ is convex.*

(c) *Let $\varphi, \psi\colon \mathbb{R}^n \to (-\infty, +\infty]$ be functions, and consider the functions $f = e^{-\varphi}$ and $g = e^{-\psi}$. Prove that*

$$f \star g = e^{-\varphi\square\psi}.$$

Conclude that if f and g are log-concave and $\varphi\square\psi > -\infty$ in $\mathbb{R}^n$, then $f \star g$ is also log-concave.

Remark 1.6.10. We also have convexity for $f \star g$ if one of the functions is convex, the other is non-negative, and $f \star g > -\infty$.

Exercise 1.27. *Remembering the convention that $e^{-\infty} = 0$, verify that $1_A = e^{-I_A^\infty}$. Conclude that 1_A is a log-concave function if and only if A is a convex set. Also, use this equality to give a solution to Item* **(a)** *of Exercise 1.26.*

Exercise 1.28. *For a non-negative function $f\colon \mathbb{R}^n \to [0, +\infty]$ and a number $\lambda > 0$, define the function $\lambda \cdot f\colon \mathbb{R}^n \to [0, +\infty]$ given as*

$$\lambda \cdot f(x) = f\left(\frac{x}{\lambda}\right)^\lambda$$

for each $x \in \mathbb{R}^n$.

(a) *For any set $A \subseteq \mathbb{R}^n$ prove that $\lambda \cdot 1_A = 1_{\lambda A}$.*

(b) *For any non-negative functions $f, g\colon \mathbb{R}^n \to [0, +\infty]$ and any number $\lambda \in (0, 1)$, show that*

$$[\lambda \cdot f] \star [(1 - \lambda) \cdot g](\lambda x + (1 - \lambda)y) \geq f(x)^\lambda g(y)^{1-\lambda}$$

holds for any $x, y \in \mathbb{R}^n$.

(c) *Conclude that if f and g are integrable and $[\lambda \cdot f] \star [(1 - \lambda) \cdot g]$ is measurable, then*

$$\int_{\mathbb{R}^n} [\lambda \cdot f] \star [(1-\lambda) \cdot g]\, dx \geq \left(\int_{\mathbb{R}^n} f\, dx \right)^{\lambda} \left(\int_{\mathbb{R}^n} g\, dx \right)^{1-\lambda}$$

for every $\lambda \in (0,1)$.

Remark 1.6.11. Notice that Item **(c)** leads to an interpretation of the Prékopa–Leindler inequality as an inequality for a type of convolution.

Exercise 1.29. *Let* $f \in \mathrm{LC}(\mathbb{R}^n)$. *Prove that for any* $\alpha, \beta > 0$ *it holds that*

$$(\alpha \cdot f) \star (\beta \cdot f) = (\alpha + \beta) \cdot f$$

Exercise 1.30. *Let* $\varphi \colon \mathbb{R}^n \to (-\infty, +\infty]$ *be a function. Prove that* φ *is lower semi-continuous if and only if* $e^{-\varphi}$ *is upper semi-continuous. Conclude that the functions of* $\mathrm{LC}(\mathbb{R}^n)$ *are upper semi-continuous.*

In what follows, we recall that, from the definition of the integral of simple functions, we have

$$\mathrm{vol}_n(A) = \int_{\mathbb{R}^n} 1_A\, dx,$$

for any Borel (or even Lebesgue) measurable set $A \subseteq \mathbb{R}^n$, where $\mathrm{vol}_n(\cdot)$ denotes the usual Lebesgue measure and dx stands for the integration with respect to that measure (these notations will **always** be adopted in this book). The reader who is not familiar with basic measure and integration theory may find an introduction to this subject in Appendix A.

Exercise 1.31. *Using Exercise 1.28, show that*

$$\mathrm{vol}_n(\lambda A + (1-\lambda)B) \geq \mathrm{vol}_n(A)^{\lambda} \mathrm{vol}_n(B)^{1-\lambda} \tag{1.6.2}$$

for any compact sets $A, B \subseteq \mathbb{R}^n$ *and every number* $\lambda \in (0,1)$.

Remark 1.6.12. Inequality (1.6.2) is the classical *Brunn–Minkowski inequality*. This is the most important inequality in convex geometry, and it will be intensively studied in Section 5.1.

Exercise 1.32. *Prove that the function*

$$\varphi(x) = \frac{|x|^2}{2}$$

is the unique proper function from $\mathbb{R}^n$ *to* $(-\infty, +\infty]$ *such that* $\varphi^* = \varphi$.

Exercise 1.33. *Let* $\varphi \colon \mathbb{R}^n \to (-\infty, +\infty]$ *be a function, and assume that* $f = e^{-\varphi}$ *does not vanish identically. We define the* support function *of* f *to be the function* $h_f \colon \mathbb{R}^n \to (-\infty, +\infty]$ *given as*

$$h_f := (-\log f)^*.$$

(a) *Show that we indeed have* $h_f > -\infty$ *everywhere.*

(b) *Prove that*

$$h_{\lambda \cdot f} = \lambda h_f$$

for any number $\lambda > 0$, *where* $\lambda \cdot f$ *stands for the scalar multiplication defined in Exercise 1.28.*

(c) *Let* $A \subseteq \mathbb{R}^n$. *Show that*

$$h_{1_A}(u) = \sup\{\langle u, x\rangle : x \in A\}$$

for any $u \in \mathbb{R}^n$.

(d) *Let* $\varphi, \psi : \mathbb{R}^n \to (-\infty, +\infty]$ *be proper functions such that* $\varphi \square \psi > -\infty$ *in* $\mathbb{R}^n$. *Letting* $f = e^{-\varphi}$ *and* $g = e^{-\psi}$, *prove that*

$$h_{f \star g} = h_f + h_g.$$

Exercise 1.34. *Let* $\varphi : \mathbb{R}^n \to (-\infty, +\infty]$ *be a lower semi-continuous convex function such that*

$$\lim_{|x| \to +\infty} \varphi(x) = +\infty,$$

meaning that for every number $c \in \mathbb{R}$ *there exists a number* $\rho > 0$ *such that* $\varphi(x) > c$ *whenever* $|x| > \rho$.

(a) *Prove that there exist constants* $a > 0$ *and* $b \in \mathbb{R}$ *such that*

$$\varphi(x) \geq a|x| + b$$

for every $x \in \mathbb{R}^n$.

(b) *Let* $f = e^{-\varphi}$. *Show that*

$$\int_{\mathbb{R}^n} f \, dx < +\infty.$$

Remark 1.6.13. A log-concave function is always measurable because for any $a \in \mathbb{R}$ the set $\{f \geq a\}$ is convex (and hence Lebesgue measurable; see Corollary A.4.3). It is relevant to know when the integral of a log-concave function f is finite (that is, f is *integrable*; see Appendix A), since when one studies log-concave functions from the geometric perspective, the total integral "plays the role" of volume. For more in this direction, we refer the reader to [637].

Now that we gave a sufficient condition under which a log-concave function has "finite volume", we investigate when this "volume" is positive. For that sake, we define the *support* of a function $f : \mathbb{R}^n \to (-\infty, +\infty]$ to be the set

$$\operatorname{spt}(f) := \big\{x \in \mathbb{R}^n : f(x) \neq 0\big\}.$$

Notice that for any functions $\varphi : \mathbb{R}^n \to (-\infty, +\infty]$ and $f = e^{-\varphi}$ we have that $\operatorname{spt}(f) = \operatorname{dom}(\varphi)$, because $f(x) \neq 0$ if and only if $\varphi(x) \neq +\infty$.

Exercise 1.35. *Let* $\varphi : \mathbb{R}^n \to (-\infty, +\infty]$ *be a convex function, and let* $f = e^{-\varphi}$. *Using the fact that any convex set with empty interior has zero Lebesgue measure, prove that*

$$\int_{\mathbb{R}^n} f \, dx = 0$$

if and only if $\operatorname{int}(\operatorname{dom}(\varphi)) = \emptyset$ *(or, equivalently, if and only if* $\operatorname{int}(\operatorname{spt}(f)) = \emptyset$*).*

Remark 1.6.14. Since $\operatorname{dom}(\varphi)$ is convex, it holds that $\operatorname{int}(\operatorname{dom}(\varphi)) = \emptyset$ if and only if $\operatorname{dom}(\varphi)$ is contained in some hyperplane of $\mathbb{R}^n$. This will be made clear in Chapter 2. It turns out that any subset of a hyperplane of $\mathbb{R}^n$ has zero Lebesgue measure. This is explained in Corollary A.3.2 and Remark A.3.1.

Now we define the following families of functions:

$$\mathrm{Cvx}(\mathbb{R}^n) := \Big\{\varphi\colon \mathbb{R}^n \to (-\infty,+\infty] : \varphi \text{ is closed, convex and proper}\Big\},$$
$$\mathrm{Cvx}_\infty(\mathbb{R}^n) := \Big\{\varphi\colon \mathbb{R}^n \to (-\infty,+\infty] : \varphi \in \mathrm{Cvx}(\mathbb{R}^n) \text{ and } \lim_{|x|\to+\infty} \varphi(x) = +\infty\Big\},$$
$$\mathrm{LC}(\mathbb{R}^n) := \Big\{f\colon \mathbb{R}^n \to [0,+\infty) : f = e^{-\varphi} \text{ for some } \varphi \in \mathrm{Cvx}(\mathbb{R}^n)\Big\} \text{ and}$$
$$\mathrm{LC}_\infty(\mathbb{R}^n) := \Big\{f\colon \mathbb{R}^n \to [0,+\infty) : f = e^{-\varphi} \text{ for some } \varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)\Big\}.$$

Notice that, in particular, Exercise 1.34 gives that any function of $\mathrm{LC}_\infty(\mathbb{R}^n)$ is integrable (measurability comes from Remark 1.6.13).

Exercise 1.36. *Let $\varphi\colon \mathbb{R}^n \to (-\infty,+\infty]$ be proper and lower semi-continuous, and assume that*

$$\lim_{|x|\to+\infty} \varphi(x) = +\infty,$$

in the same sense as in Exercise 1.34. Prove that φ attains a global minimum. Conclude that every $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$ attains a global minimum, and that every function $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ attains a global maximum.

Exercise 1.37. **(a)** *Verify that $\mathrm{Cvx}(\mathbb{R}^n)$ is closed under conjugation; that is, if $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$, then $\varphi^* \in \mathrm{Cvx}(\mathbb{R}^n)$.*

(b) *Show that $\mathrm{Cvx}_\infty(\mathbb{R}^n)$ is not closed under conjugation; that is, give an example of a function $\varphi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$ such that $\varphi^* \notin \mathrm{Cvx}_\infty(\mathbb{R}^n)$.*

(c) *Prove that $\mathrm{Cvx}_\infty(\mathbb{R}^n)$ is closed for infimal convolution, that is, show that if $\varphi, \psi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$, then $\varphi\square\psi \in \mathrm{Cvx}_\infty(\mathbb{R}^n)$. Check that this is not true for $\mathrm{Cvx}(\mathbb{R}^n)$.*

(d) *Verify that $\mathrm{LC}_\infty(\mathbb{R}^n)$ is closed for Asplund product.*

Remark 1.6.15. If we denote by $F\colon \mathrm{Cvx}(\mathbb{R}^n) \to \mathrm{Cvx}(\mathbb{R}^n)$ the map which associates each $\varphi \in \mathrm{Cvx}(\mathbb{R}^n)$ with its conjugate φ^*, then from Theorem 1.4.1 we have that F is an *involution*, meaning that F is a bijection such that $F \circ F = \mathrm{Id}_{\mathrm{Cvx}(\mathbb{R}^n)}$.

Exercise 1.38. *Verify that $h_f \in \mathrm{Cvx}(\mathbb{R}^n)$ for every $f \in \mathrm{LC}(\mathbb{R}^n)$. Let $H\colon \mathrm{LC}(\mathbb{R}^n) \to \mathrm{Cvx}(\mathbb{R}^n)$ be the mapping which associates each $f \in \mathrm{LC}(\mathbb{R}^n)$ with its support function $H(f) := h_f \in \mathrm{Cvx}(\mathbb{R}^n)$. Prove that H is a bijection.*

Exercise 1.39. *Let $f\colon \mathbb{R}^n \to [0,+\infty)$, and write $f = e^{-\varphi}$. We define the* polar function *of f to be the function*

$$f^\circ := e^{-\varphi^*} = e^{-h_f}.$$

(a) *Prove that the map which associates each function $f \in \mathrm{LC}(\mathbb{R}^n)$ with its polar function is an involution of $\mathrm{LC}(\mathbb{R}^n)$.*

(b) *Show that for any $f \in \mathrm{LC}(\mathbb{R}^n)$ we have the equality*

$$f^\circ(y) = \inf\left\{\frac{e^{-\langle x,y\rangle}}{f(x)} : x \in \mathbb{R}^n\right\}$$

for every $y \in \mathbb{R}^n$, where the quotient must be interpreted as $+\infty$ whenever $f(x) = 0$.

(c) *Prove that for* $f = e^{-|x|^2/2}$ *we have* $f^\circ = f$. *Even more, show that this function is the unique function of* $\mathrm{LC}(\mathbb{R}^n)$ *having this property.*

Exercise 1.40. *Let* $f\colon \mathbb{R}^n \to [0,+\infty]$ *be a non-negative function, and put* $\bar{f}(x) = f(-x)$ *for each* $x \in \mathbb{R}^n$. *Define the function* Δf *by*

$$\Delta f := \left(\frac{1}{2}\cdot f\right) \star \left(\frac{1}{2}\cdot \bar{f}\right).$$

(a) *Prove that*

$$\Delta f(x) = \sup\left\{\sqrt{f(z)f(-y)} : z, y \in \mathbb{R}^n,\ x = \frac{z+y}{2}\right\}$$

for each $x \in \mathbb{R}^n$. *Conclude that* Δf *is even.*

(b) *Verify that if* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, *then* $\bar{f} \in \mathrm{LC}_\infty(\mathbb{R}^n)$.

(c) *Show that if* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, *then* $\Delta f \in \mathrm{LC}_\infty(\mathbb{R}^n)$.

(d) *For any sets* $A, B \subseteq \mathbb{R}^n$, *denote by* $A - B$ *the* Minkowski difference *(or simply* difference*) of* A *and* B, *which is the Minkowski sum of* A *and* $-B$. *Show that*

$$\Delta 1_A = 1_{\frac{A-A}{2}}$$

for any set $A \subseteq \mathbb{R}^n$.

Remark 1.6.16. The subcase $A - A$ is called the *difference body*, and $\frac{1}{2}(A-A)$ the *central symmetral*, of A. We will return to these concepts in Chapters 2 and 10.

For the next exercise, observe that if $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, then Δf is integrable because it is also a function of $\mathrm{LC}_\infty(\mathbb{R}^n)$. See the comment right above Exercise 1.37.

Exercise 1.41. **(a)** *If* f *is a non-negative function, show that*

$$\mathrm{spt}(\Delta f) = \frac{\mathrm{spt}(f) - \mathrm{spt}(f)}{2}.$$

(b) *Letting* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, *prove that*

$$\int_{\mathbb{R}^n} f\,dx = 0$$

if and only if

$$\int_{\mathbb{R}^n} \Delta f\,dx = 0.$$

Hint: Using Item **(a)** *and Remark 1.6.14, prove that the interior of* $\mathrm{spt}(\Delta f)$ *is empty if and only if the interior of* $\mathrm{spt}(f)$ *is empty.*

(c) *Assuming again that* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, *prove the inequalities*

$$\int_{\mathbb{R}^n} f\,dx \le \int_{\mathbb{R}^n} \Delta f\,dx \le 2^n \int_{\mathbb{R}^n} f\,dx.$$

Remark 1.6.17. The right-hand inequality above was originally proved by Colesanti in [636], and it can be seen as a functional version of the geometric *Rogers–Shephard inequality*, which will be studied in Section 5.3. A weaker variant of this geometric inequality is given in the next exercise. Recently, a lot of nice functional inequalities related to classical convex geometry inequalities have been studied. In this direction, we refer the reader to [73], [74], [79], [77], [80], [1128], [1130], and [1129], among others.

Exercise 1.42. *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact convex set. Prove that*

$$\mathrm{vol}_n(K - K) \leq 4^n \mathrm{vol}_n(K).$$

1.7 Notes

In the following notes we present important references (monographs and survey-like papers) treating the notions which are basic for our first chapter. In this way, we present also the developments around convex functions. Also the Prékopa–Leindler inequality and extensions of it are discussed. For the sake of convenience, we give here once more short definitions of some notions which are fundamental for everything discussed in the following. First we refer to (graphs of) functions. Namely, a *convex function* is a function whose epigraph is a convex set. The *Fréchet derivative*, defined on Banach spaces, is commonly used to generalize the derivative of a real-valued function of a single real variable to the case of a vector-valued function of multiple real variables on such spaces, and to define the functional derivative used widely in the calculus of variations. The more general *Gâteaux derivative*, extending the classical directional derivative in differential calculus also to infinite-dimensional spaces, is defined for functions between locally convex topological vector spaces. Like the Fréchet derivative on a Banach space, the Gâteaux differential is often taken to formalize the functional derivative commonly used in the calculus of variations and physics. Roughly speaking, the notion of *subgradient* is an extension of the usual gradient for points where the considered function is not differentiable. The *subdifferential* at a given point is the set of subgradients at this point, and for a convex function it parametrizes the supporting hyperplanes of the epigraph. Let E be a Banach space, E^* its dual, and let (y, x) be the value of a linear functional $y \in E^*$ at an element $x \in E$. An operator A, in general nonlinear and acting from E into E^*, is called a *monotone operator* if

$$\mathrm{Re}(Ax_1 - Ax_2, x_1 - x_2) \geq 0$$

for any $x_1, x_2 \in E$.

The *Fenchel conjugate* f^* of a function f "captures" the geometric information of the distance to the origin of supporting hyperplanes of the epigraph of f, and the Fenchel conjugacy for convex and lower semi-continuous (i.e., *closed*) functions is a *dual* concept in the sense that the bi-conjugate of such a function equals itself. Here the reader is referred to the (also historically very interesting) paper [1387] of Kiselman, where the genesis of the notion of Fenchel conjugacy and, more generally, duality in convex geometry and convex analysis is discussed in detail. In convex analysis, many generalized convexity notions were investigated, and we give here a typical example. Namely, various results of the theory can be studied in the more general context of *quasi-convex* functions, which are functions for which

$$f((1-\lambda)x + \lambda y) \leq \max\{f(x), f(y)\}$$

for every $x, y \in \mathbb{R}^n$ and $\lambda \in (0, 1)$. Similarly, sometimes the log-concave functions can be replaced by *quasi-concave* functions, which are defined by the condition

$$f((1-\lambda)x + \lambda y) \geq \min\{f(x), f(y)\}$$

for each $\lambda \in (0, 1)$ and all $x, y \in \mathbb{R}^n$. For example, Milman and Rotem study in [1795] a theory of mixed volumes (so-called *mixed integrals*) for spaces of quasi-concave functions, instead of restricting themselves to the case of log-concave functions.

In convex analysis, a *pseudo-convex function* is a function that behaves like a convex function with respect to determining its local minima, but need not really be convex. For example, a differentiable function is pseudo-convex if it increases in any direction where it has a positive directional derivative.

Roughly speaking, a *matroid* is a finite set together with a generalization of a concept from linear algebra satisfying a set of properties for this concept. For example, the finite set could be the rows of a matrix, and the generalizing concept could be linear dependence and independence of any subset of

rows of the matrix. And a *submodular set function* is a set function whose value, informally, has the property that the difference in the incremental value of the function, that a single element makes when added to an input set, decreases if the size of the input set increases.

1.7.1 Properties of convex functions

At first glance, one might think that the notion of *convex function* describes a clear and relatively elementary topic, with the effect that, maybe, there are not too many recent monographs or surveys on it being close to the research front. But this is by no means true, as the following exposition will demonstrate. It turns out that there are thousands of publications treating convex functions in different ways. Thus, the only way to reflect suitably the amount of knowledge on this notion is to reduce the discussed topics and references in such a way that they refer to the contents presented in Chapter 1, and that *only monographs and surveys*, strongly related to our intentions, are taken into consideration. This is the only chance to grasp somehow everything in any way. But as the reader will see, already this selected material yields a long list of (also very recent) references.

Convex functions play an essential role in different mathematical disciplines, for example in (geometric) function theory, convex analysis, (convex) optimization, convex geometry, calculus of variations, numerical analysis, functional analysis, asymptotic geometric analysis, approximation theory, geometric inequalities, probability theory, PDE, and others. Furthermore, discrete versions can be found in special fields of discrete mathematics, e.g., in combinatorial optimization. Convex functions are also important in physics and applied sciences, like engineering and economics.

SOME HISTORY

The initiator of the notion of "convex function" was perhaps Jensen [1301]. He stated in 1906: "It seems to me that the notion of convex function is just as fundamental as positive function or increasing function. If I am not mistaken in this, the notion ought to find its place in elementary expositions of the theory of real functions". A comprehensive representation of their history, investigated properties and applications until 1946 is given in the expository paper [265] of Beckenbach, and the development of their applications till 1954 was additionally discussed in [1018] by Green. One further starting point to study convex functions systematically (and, more general, convex analysis and closely related parts of optimization) was given by lecture notes of Fenchel (cf. [855]) and by works of Moreau (see [1849]), Rockafellar (cf. [2044], [2046], and [2047]), Ioffe ([1274] and [1273]), and others (see also Laurent [1517] and the first edition of [809] by Ekeland and Teman), in the 1960s and early 1970s. Later on, convex functions became popular as studied objects or tools in many further mathematical disciplines and models. In the years after the appearance of these mentioned books and papers, at first no books dealing exclusively with convex functions appeared, and so famous monographs on inequalities (see, e.g., Hardy et al. [1163], Mitrinović [1812], and Beckenbach and Bellman [266], later also Pecarić et al. [1953]) overtook the task to give related broad outlines. Since it is invariant under many types of transformations, the convexity property of functions and domains successively created a rich structure of results and geometric methods to obtain an own interesting mathematical theory of convex functions, with elegant proof techniques (which in many cases do not depend on other fields). This theory finally got its own place in geometric function theory and, due to many inspiring impulses, has a very great network of applications (most notably, in optimization theory). Today we have a large variety of theoretically and also practically efficient methods to study convex functions, convex sets and their applications, and from our point of view, especially the interplay with convex analysis and convex geometry seems to be most interesting (at least in our opinion). Convex analysis is the topic discussed in the chapter above and within subsection 12.4.6, and convex geometry is the field discussed in the whole book.

BASICS ON CONVEX FUNCTIONS

We will start with presenting basic monographs and surveys which treat convex functions that appeared shortly after Green [1018] as well as Beckenbach and Bellman [266] (see above) and are completely dedicated to certain types or natural generalizations of convex functions, or which contain important chapters on this notion. From the beginning, and mainly due to Rockafellar's influential book [2046] (and forerunners of it, e.g., surveys like [2044] and [1274]), one cannot separate this development from that of convex analysis; i.e., several publications that we have to mention here are also well known as books about convex analysis (and vice versa). Thus, this and the next subsection are closely intertwined. The decisions to discuss such references here or in the next subsection often depended only on nuances of emphasis. Presenting such books and surveys, we try to order them, in both subsections, in two combined ways: due to suitable notions or guidelines (like "duality" or "generalized convexity notions"), and on the other hand trying to proceed (at least within these building blocks) chronologically.

In Rockafellar's already mentioned monograph the notion of *convex function* plays a fundamental role. The author lays emphasis on the facts that convex functions on $\mathbb{R}^n$ can be identified with their epigraphs (i.e., convex sets in $\mathbb{R}^{n+1}$), while the convex sets in $\mathbb{R}^n$ can be identified with certain convex functions on $\mathbb{R}^n$, called their indicators. For example, *directional derivatives, subgradients* as well as *(sub)differentials* of convex functions are further basic themes in this very comprehensive book, being fundamental for the new field of *convex analysis*. Here we should also mention the book [2304] by Stoer and Witzgall; its fourth chapter contains the classical material about convex functions in $\mathbb{R}^n$, also directional derivatives and differentiability, but not subgradients. Going back to Fenchel, *conjugate functions* of arbitrary and particular (e.g., polyhedral, or positively homogeneous) functions and some *generalized convexity concepts* (described by notions like quasi-convex, pseudo-convex, and strictly quasi-convex) applied to functions are mainly discussed here. Another early book on convex functions themselves is [2038] by Roberts and Varberg, suitable for undergraduates, graduates, and also mathematicians who want to enter the field. For example, the authors study convex functions in normed linear spaces, related classes of functions and related inequalities. Barbu and Precupanu [226] study basic tools and methods from convex analysis which are useful for problems from convex optimization in infinite-dimensional Banach spaces, with a particularly deep study of needed properties of convex functions. Here we mention also the nice booklet [2365] of van Tiel, containing a careful study of convex functions and related concepts and notions (in chapters VI and VII): differentiability, subcontinuity, Fenchel duality in Banach spaces, and so on. Also the textbook [1587] of Liu contains, nicely presented and based on Rockafellar's monograph [2046], large portions of classical material on convex functions, namely again notions and tools like conjugate functions, subgradients and differentiability, also referring to quasi-convex and pseudo-convex functions.

DIFFERENTIABILITY

A further very broad and deep monograph on convex functions is [466] by Borwein and Vanderwerff, whose Euclidean part is based on [2046] and [2038], but goes deeper. For example, a complete presentation of the topic of *differentiability*, including results of Rademacher and Alexandrov, is given. Polyhedral convex functions, functions of eigenvalues, deep extensions to Banach spaces, duality relations for semi-definite optimization and between various strict convexity notions, the corresponding smoothness concepts, deeper properties of the subdifferential operator, convex functions on lattices, Legendre functions in Banach spaces, and characterizations of finite-dimensional Banach spaces via function properties are further selected topics from this important book. Each chapter presents interesting examples, and more than 600 exercises are included, ranging (in view of difficulty steps) from early graduate to the research level. A further book, strongly related to differentiability of convex

functions is Giles [970], which is discussed (together with further references) in more detail below and also in our next subsection. Results on the differentiability and extendibility of convex functions on Banach spaces are surveyed by Borwein, Montesinos, and Vanderwerff in [465], and Giles [971] gives an interesting survey on the history of *Gâteaux and Fréchet differentiability* of real-valued functions on Banach spaces, where convex functions are particularly taken into consideration. On the other hand, results on *nondifferentiability points* of convex functions in n-dimensional space are collected by Ambrosio in [102].

Another deep and essential monograph on convex functions is the book [1971] of Phelps, based on developments of differentiability properties of such functions and Lipschitzian operators on infinite-dimensional spaces. It is strongly related to fields like *nonlinear functional analysis* and topics like *monotone operators, differentiability of vector measures, variational principles*, and *Asplund spaces*. Asplund started to investigate classes of Banach spaces, now called after him, in which the continuous convex functions are Fréchet differentiable at the points of dense subsets of their domains; analogously, for the case of Gâteaux differentiability one gets so-called *weak Asplund spaces*. Clearly, an interesting question is that for characterizations of weak Asplund spaces. For example, a classical theorem of Mazur says that every separable Banach space is a weak Asplund space. The investigations of lower semi-continuous convex functions in [1971] guarantee that subdifferentials still exist for them. Phelps describes the *variational principle* of Ekeland, Borwein, and Preiss (see our next subsection) and its application in Asplund spaces, and he discusses the duality between Asplund spaces and the spaces with the Radon–Nikodym property. The monograph of Phelps is on the level of researchers and graduates, and we underline its close relation to the already above-mentioned book [970] of Giles, which also treats convexity in Banach spaces and types of differentiable convex functions in close connection with the notion of Asplund spaces. This largely self-contained book could lead, also due to interesting research problems and exercises in it, (post-)graduates conveniently to the research level; see also above and our next subsection. Also for Fabian [828] the connection between differentiability properties of convex functions and Asplund spaces is the main topic. Among many other subjects, the author discusses the developments and efforts to get characterizations of weak Asplund spaces. Still treating properties of Banach spaces, more special themes are investigated in books from Russia, namely the classification of convex functions convenient for the study of types of Orlicz spaces in [1467], and their application for studying entire functions of several variables in [1668]. (Note that an *Orlicz space* is a type of function space which was introduced by Birnbaum and Orlicz in 1931. Itself being a Banach space, it generalizes the classical notion of Lebesgue spaces.)

CONVEX FUNCTIONS ON MANIFOLDS

With an extensive bibliography the monograph [2387] of Udriste comprehensively treats *convex function theory on Riemann manifolds*. Besides the results related to this main content and those more applied in nature, also numerical optimization methods on Riemann manifolds are described. In addition, the author surveys various notions from dynamical systems related to convexity, and an appendix refers also to *convexity theory on Finsler manifolds*. We continue with Butnariu and Iusem [554], where the notion of *totally convex functions in Banach spaces* (in the infinite-dimensional case a strengthening of strictly convex ones) is in the focus. The authors find a unified way to present results on totally convex functions in Banach spaces and their applications for computing common fixed points of measurable families of operators, and for optimization methods in infinite dimensions. They succeed in presenting unifying connections between functional analysis and finite-dimensional convex programming. Also different types of algorithms (like the augmented Lagrange algorithm) are shown to be applicable to wider problem classes than known before.

(GENERALIZED) CONVEXITY NOTIONS

The first edition of the next monograph that we mention here, namely [1894] by Niculescu and Persson, appeared already in 2006. It is a graduate textbook on convex functions, where the used notion of convexity is not restricted to this property of the epigraph, but defined in terms of averages. For example, among these *convexity notions defined by means* the so-called midpoint-convexity is then related to the arithmetic mean of the endpoints of a segment, and this is replaced also by other means yielding further convexity notions. Regarding convex functions in this framework and also in normed linear spaces, the authors investigate differentiability types and further properties, important inequalities (Hardy–Littlewood–Polya, Prékopa–Leindler, Borell–Brascamp–Lieb, etc.), separation theorems, Hahn–Banach and Krein–Milman theorems, the use of convexity to solve variational problems, and so on. At the end of each chapter exercises and valuable historical remarks are given, like also notes regarding further developments. In the second edition (cited here) three new chapters are added, e.g., dealing with notions like *convex spectral functions, matrix convexity,* and *geodesic convexity*. Also in [497] of Breckner and Trif, written mainly for students, the idea of using means to get convexity notions is used. The main focus of this book is the study of special topics treating convex functions and related functional equations. In particular, Chapter 8 contains surprisingly many results referring to singularities and equicontinuity of families of functions. Here we also mention the book [685], where other generalizations of convex functions and sets are introduced and studied, e.g., based on a convexity notion which even does not preserve connectedness. In several ways one can define such *non-connected convexity*: for instance, via axiomatic convexity, convexity in nonlinear structures, the union of convex sets, and so on. The book can also be seen as a related state-of-the-art survey; additionally it contains a brief history of convexity and links to applications of convex functions of such a generalized type. Shortly we mention also the book [632] mainly referring to Lipschitz functions. In Chapter 3 their relations to convex functions are clarified.

Maybe this is the right place to cite also the "Handbook of Generalized Convexity and Generalized Monotonicity" [1131], and to refer to the next subsection where generalized convexity notions are also discussed. As they are applied in optimization, economics, management science, and engineering, generalized convex functions share (depending on the needs) only some of the characteristic properties of convex functions. On the other hand, also *generalized monotone maps* have attracted attention. They are related to generalized convex functions like monotone maps are related to convex functions.

In [1131], by Hadjisavvas et al., 14 survey papers on the area of generalized convexity or generalized monotonicity are collected, written by a broad group of experts in these fields and offering comprehensive representations of the most significant advances in these directions. Some keywords might suffice to describe essential parts of the contents: *quasi-convex analysis, differentiability of quasi-convex functions, abstract convexity, generalized convexity in vector optimization, generalized convex duality, generalized monotonicity,* and *nonsmooth analysis*. Regarding the latter field, we also refer to our next subsection and the monograph [2134] of Schirotzek, which contains a nice representation of classical results on convex functions and convex analysis (e.g., Clarke and Michel–Pernot subdifferentials are carefully presented) as starting points for a convenient presentation of the steps of development of nonsmooth analysis. Here we also mention the survey [1959] of Penot, where generalized convexity notions are characterized via directional derivatives and subdifferentials; also a survey of characterizations of nondifferentiable generalized convex functions in the light of nonsmooth analysis is presented. In Sándor's expository papers [2121] and [2123] further concepts of generalized convex functions are discussed, based on notions like invexity and pseudo-invexity. For instance, these considerations yield a generalization of the Banach–Steinhaus theorem. The survey [1348] on the relations between convex functions and generalized convexity notions is discussed in more detail in our next subsection. Examples of further books related to convex functions are [2527] of Zalinescu and [1201] and [1202] by Hiriart-Urruty and Lemaréchal.

Now we concentrate more on *survey-like papers* which refer to further interesting aspects or applications of convex functions. Most of the topics are also discussed in the books presented above (and therefore also mentioned already), but regarding references the surveys go often beyond this.

FUNCTIONAL ANALYSIS

Topics from *functional analysis* are referred to in the following expository papers: The survey [143] of Asplund treats parts of the theory of convex functions important for the field of *monotone mappings*. V. Soltan [2269] gives a possible *axiomatic approach* to the theory of (generalized) convex functions, based on a generalization of the notion of convex combination of two points, and he surveys results on the most important properties of these generalized convex functions and of convex sets connected with them. The nice survey [459] of Borwein deals with *characterizations of types of normed spaces* by properties of convex sets and convex functions. The presented results on this topic are, in some cases, also given with proofs. Some of the characterized types of normed spaces are finite-dimensional, reflexive, and separable spaces as well as Asplund spaces. The survey [468] of Borwein and Zhu presents, among many other related topics, also characterizations of similar types of Banach spaces, also obtained via properties of convex functions. Azé and Penot [167] deal with the field of *convergence of convex sets and convex functions*, including also new results. The first part contains definitions of various known hyperspace topologies (such as the Joly topology, the Attouch–Wets topology, and the Mosco topology), and the second part treats limits of graphs and level sets of continuous linear maps with respect to the above-mentioned topologies. In the third part some results related to the continuity of the Legendre–Fenchel transform are given, again referring to the mentioned topologies, and in the final part continuity of usual operations with respect to set convergence are discussed.

In [462], Borwein and Giladi study the existence and "shape" of nearest point sets in Banach spaces. The expository paper collects known facts and possible methods to attack questions in this area. Since distance functions can sometimes be presented as so-called delta-convex functions, (sub)differentiability results applied to delta-convex functions are suggested as such possible methods. Lassonde [1512] proves some theorems from functional analysis for sublinear functionals in real vector spaces and deduces results on subdifferentials of convex functions, the separation of convex functions by affine functions, and the decomposition of the infimum of the sum of convex functions. He also discusses situations in which certain convex functions are automatically continuous on the core of their domains, applying this to further theorems from functional analysis. The survey [168] excellently summarizes results on *DC functions* (which are the difference of two continuous convex functions). They refer to compositions of the DC mapping, differentiability conditions, respective subdifferentials, interesting examples from variational calculus, a characterization of infinite-dimensional spaces by positive DC functions, and many further topics.

APPROXIMATION THEORY

Clearly, convex functions are also interesting from the viewpoint of *approximation theory*. In problems from numerical integration, often bounds for derivatives of unknown functions are not available. In general, the convexity property of an investigated function yields smaller error bounds than, for example, its monotony or positivity. In this framework, quadrature errors for certain types of functions are a well-known topic. An interesting survey on the related *quadrature theory of convex functions* is [1965] by Petras. Applications of classical quadrature rules, such as trapezoidal and midpoint rules or the Gaussian quadrature, are presented and compared with quadrature formulas which are optimal for certain types of convex functions. In the survey [1100] of Guessab, the *approximation of a convex function* on arbitrary convex polytopes by means of a one-parameter family of approximation schemes is studied. Since they are based on the use of barycentric coordinates, these schemes are called barycentric approximation schemes. The convexity assumption is exploited to establish upper and lower (pointwise) bounds on the error in the approximation of arbitrary convex functions. For the subcase of a convex twice continuously differentiable function, an optimal error estimate is obtained

(where Delaunay triangulations play an essential role). There is a comprehensive list of references, also referring to various related problems.

INEQUALITIES

Obviously, an important subfield of studying convex functions is that of *inequalities*. Besides books like [1163], [1812], and [1953] there are also important surveys in this direction. We start with the paper [765] by Dragomir et al., in which inequalities are surveyed which are functional generalizations of Jessen's classical inequality. Dvorsky and Gwinner [782] give first an overview to generalized convexity notions in calculus of variations and nonlinear elasticity, and in this context also a related generalization of Jensen's inequality is obtained. Further on, new results related to nonlinear elasticity are presented. In the two surveys [2120] and [2122], Sándor presents various concepts and results referring to inequalities for convex functions. These include extensions of Jensen's inequality (in its discrete and integral form) as well as *Hadamard's integral inequality* for convex and generalized convex functions. Tasković [2346] first gives an extensive survey on several generalizations of convex functions. Using this starting point he derives new results, among them being a generalization of the Hahn–Banach theorem, some representation and majorization theorems for general convex functions, and several inequalities for them. There are many generalizations of *subadditive and super-additive inequalities for concave and convex functions*, and in this framework also interesting inequalities for matrices came to be. The authors of [152] survey some of these matrix inequalities and present many further results. Applying classical inequalities for convex functions to suitably specified function types, Li and Yeh [1569] survey and prove some related inequalities. Rajba [2017] exposes theorems on stochastic convex ordering and their applications to functional inequalities for convex functions.

1.7.2 The Prékopa–Leindler inequality and related topics

Now we discuss results related to our Section 1.5, and we leave, due to the specific nature of the subject, the principle to cite and discuss only books and surveys. The integral inequality of Prékopa and Leindler (see Theorem 1.5.1) goes back to the papers [1995] and [1996] of Prékopa, and [1539] of Leindler. It turns out that the Brunn–Minkowski inequality on the volume of sum sets has the *Prékopa–Leindler inequality* as a functional counterpart. Brascamp and Lieb [490] extended the Prékopa–Leindler inequality to other types of convex combinations of non-negative, measurable functions, e.g., strengthening both the Prékopa–Leindler and the *Brunn–Minkowski inequality* by introducing the notion of essential addition. This was finally yielding the famous *Brascamp–Lieb inequality*, which is also discussed in several monographs on convex sets; see, e.g., Chapter 1 of Brazitikos et al. [496] and Chapter 14 of [2248] by Simon. For another early generalization we refer to Borell [454]; see also below. Brascamp and Lieb [490] prove various moment inequalities for such functions and confirm interesting related inequalities.

The first discussion of the equality case in the Euclidean setting can be found in [770]. Also Imoru [1268] derived far-reaching related extensions, and in Lecture 5 of Ball [199] it is once more nicely presented how the Brunn–Minkowski inequality, offered as an integral inequality, turns out to be a special case of the Prékopa–Leindler inequality; see also Chapter XIII of the book [1813] and the nice, extensive survey [929] of Gardner. There, in a wider form, once more the relations of the Brunn–Minkowski theorem with the inequalities of Prékopa–Leindler, and Brascamp–Lieb and its reverse are nicely presented. This *reverse of the Brascamp–Lieb inequality* was proved by Barthe [233]. He also gave a new proof of the multidimensional Brascamp–Lieb inequality and complete solutions to the equality cases. Geometrically, this yields new characteristic properties of simplices and parallelotopes. These and many further results by Barthe are presented in his habilitation which

is summarized in [234] and devoted to the more general convolution functional inequalities (to which also the inverse Brascamp–Lieb inequality belongs). Thus, also many further results, strongly related to our considerations here can be found in [234].

Bobkov and Ledoux [377] deal with probability measures and study the question of an analogue of the Brascamp–Lieb inequality involving entropy instead of variance. Their investigations lead to a *logarithmic Sobolev inequality for uniformly convex bodies*, encompassing a concentration inequality for unit spheres due to Gromov and Milman. Their paper [377], partially also a survey, interestingly links its main subject with various related topics. Several applications of the Brunn–Minkowski inequality in the Prékopa–Leindler form are developed. For example, a way is shown how the Brascamp–Lieb inequality for strictly convex potentials can be deduced from the Prékopa–Leindler theorem. Deriving the logarithmic Sobolev inequality for uniformly convex potentials, the authors get, more generally, results even for arbitrary norms.

Mainly interested in the well-known KLS conjecture and its relations to further conjectures, Alonso-Gutiérrez and Bastero [76] also present the Prékopa–Leindler inequality as a reverse of the Hölder's known inequality, deducing from it the Brunn–Minkowski inequality as the version of the Prékopa–Leindler inequality for characteristic functions. Then the classical isoperimetric inequality and the Borell inequality are derived within this framework.

THE INEQUALITY ON MANIFOLDS AND APPLICATIONS

An n-dimensional spherical version of the Prékopa–Leindler inequality is given in [655], and it is noted there that an analogous result can be obtained for the n-dimensional *hyperbolic space*. In [661] by Cordero-Erausquin et al., several types of inequalities are established on *Riemann manifolds*, like the Brunn–Minkowski inequality in its functional form (of Prékopa–Leindler) and generalizations of it, using also probability densities. Notions from *differential geometry* (like cut locus, Jacobi fields, and curvature types) play an essential role there, and deep connections between transportation science, differential geometry, and geometric inequalities are the main topic of this nice paper. We note that nowadays Borell's similar generalization given in [454] is taken more into consideration than in the past, yielding also the better designation "Borell–Brascamp–Lieb inequality"; by Cordero-Erausquin et al. this nomenclature is suggested in [661]. Continuing the investigations of Borell–Brascamp–Lieb inequalities on (complete) Riemann manifolds, Balogh and Kristály [204] provide estimates for the respective deficit there, and they use optimal mass transportation between suitable marginal probability measures for this. In these terms, the equality cases in Borell–Brascamp–Lieb inequalities (including the Brunn–Minkowski and Prékopa–Leindler subcases) are characterized, and a new proof of Dubuc's characterization of the equality case (see again Dubuc [770]) in the Euclidean setting is obtained. The survey-like paper [656] goes into the same direction, also discussing *weighted Prékopa–Leindler inequalities on Riemann manifolds and related Sobolev-type inequalities*. The starting point of the more recent developments around the *mass transportation problem* (recall the name *Monge–Kantorovich theory*, going back to Monge and reformulated by Kantorovich as a convex programming problem; see the survey [867] of Feyel) is the existence and uniqueness of an optimal transportation measure given by the graph of the gradient of a convex function. From such extensions to probability measures, also alternative proof ways for the Prékopa–Leindler and Brascamp–Lieb inequalities were derived; see again [867] and also [1734]. In [202] a stability version of the Prékopa–Leindler inequality for non-negative integrable functions with positive integrals on $\mathbb{R}$ is obtained. Further on, in [203] a *stability version of the Prékopa–Leindler inequality* (in one dimension) is used as a tool to obtain a stability version of the Blaschke–Santaló inequality, and then a stability version of the Prékopa–Leindler inequality for even functions in higher dimensions is derived. Bucur et al. [522] give two refinements of the classical Prékopa–Leindler inequality, which are obtained by adding an extra-term depending on a distance modulo translations. The first refinement is based on [202], and the second and new one-dimensional refinement involves the transport map onto the Lebesgue measure.

FUNCTIONAL VERSIONS AND RELATED TOPICS

Considering the class of quasi-concave functions defined on the Euclidean space, Bobkov et al. [376] give a *functional extension of quermassintegrals* to prove functional counterparts of several of their important features. Among them they establish concavity inequalities for quermassintegrals and for other (also hyperbolic) functionals, which generalize the Prékopa–Leindler and Brascamp–Lieb inequalities (without using induction on the dimension or mass transportation). Lehec [1527] gives a short proof of the Brascamp–Lieb theorem (see Brascamp and Lieb [490]) and of the Barthe theorem (cf. [233]) using probabilistic methods.

It is well known that the Brunn–Minkowski inequality can be obtained from the Prékopa–Leindler inequality by using the characteristic functions of bounded measurable sets in $\mathbb{R}^n$. Avkhadiev and Timergaliev [164] derive Brunn–Minkowski type inequalities for three new functionals in the Prékopa–Leindler inequality, and thus they get Brunn–Minkowski type inequalities for conformal and Euclidean characteristics of domains. In Section 2 of the excellently written lecture notes [1097] by Guédon et al., one can find a geometric version of the Brascamp–Lieb inequality together with a mass transportation proof using ideas of Barthe, whereas Section 3 refers to the Brunn–Minkowski inequality and its relations to the Prékopa–Leindler inequality and Borell's generalization of it (see once more Borells work [454]). Also based on this generalization of Borell (with a stochastic approach), Cordero-Erausquin and Maurey [660] extend the Prékopa–Leindler inequality to an abstract setting, including various known and new functional inequalities on Euclidean spaces. The two papers [641] of Colesanti et al. as well as [1191] of Hernández Cifre and Yepes Nicolás contain results on improved Prékopa–Leindler inequalities and Brunn–Minkowski inequalities under the assumption that the participating two convex bodies (resp. functions) have common projections onto linear subspaces. For the special case of hyperplane projections the Prékopa–Leindler inequality admits even a linear refinement, and it is sufficient that both compared projections (not necessarily equal as functions) have the same integral; see once more Colesanti et al. [641].

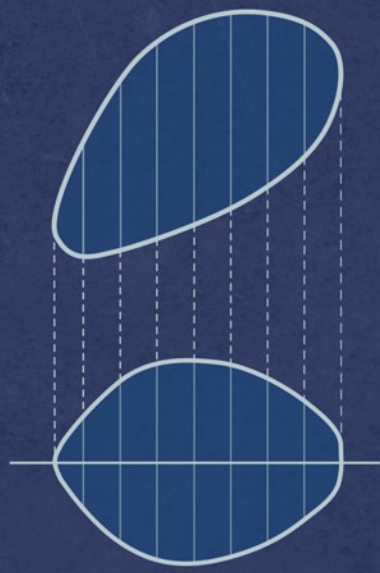
The survey [1659] of Madiman, Melbourne, and Xu refers to the recent interplay between *information theory and convexity*, particularly around entropy power inequalities. In its introductory part the Brunn–Minkowski and the Prékopa–Leindler inequality are presented, showing common approaches to get analytic, information theoretic, and probabilistic analogues of geometric results on convex sets. Another survey, namely studying the geometric theory of log-concave functions (i.e., functions that can be written in the form e^{-u} with u a convex function), is [637] due to Colesanti, bringing together also inequalities treating *functional versions of area measures and mixed volumes*, and also of *valuations* on such functions. The role of the Prékopa–Leindler inequality within this framework is taken into consideration. Finally we mention that the Prékopa–Leindler and the Borell–Brascamp–Lieb inequalities are also discussed in some of the monographs mentioned at other places of our book, like in [1894] of Niculescu and Persson, and in Chapter 1 of the book [133] of Artstein-Avidan, Giannopoulos, and Milman.

Crasta and Fragalà [684] introduce a new operation between integrable functions in n-space, where the Lebesgue integral of this geometric combination yields the geometric mean of the two combined integrals. This way they obtain a *new functional inequality of Prékopa–Leindler type*. Applying this to the combination of measurable sets, one gets a set whose volume equals the geometric mean of the two separate volumes. Based on this new operation, a new proof of the log-Brunn–Minkowski inequality for unconditional convex bodies is obtained.

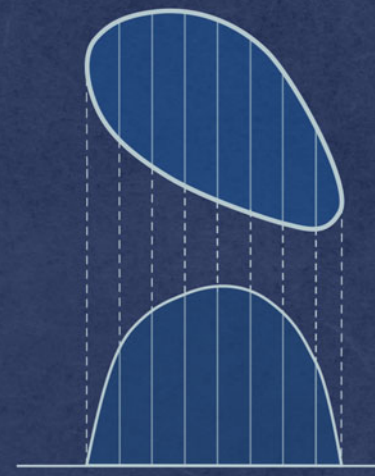
PLATE 2

Symmetrizations and similar procedures

Steiner symmetrization



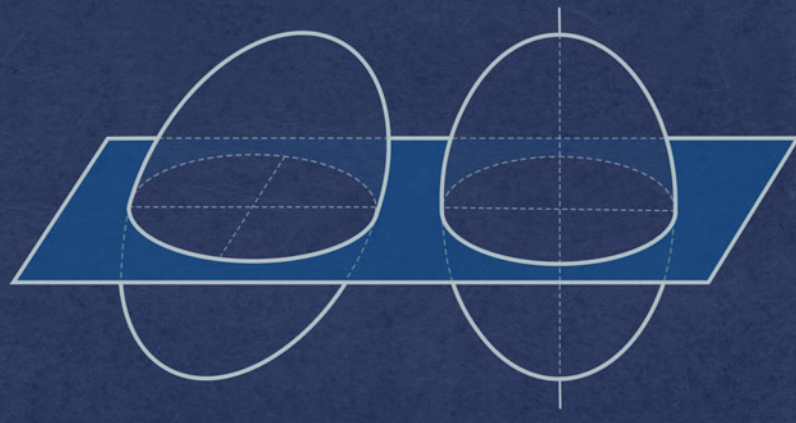
Shakedown (or X-ray)



Figures 4 - 5

Figure 6

Schwarz symmetrization



Symmetrizations and similar procedures ▸ (see, e.g., 2.7 and 2.9.3)



Chapter 2
Convex sets

"The amazing amount of underbrush that has to be cleared away,..., indicates, to this author at least, that the geometry of finite-dimensional convex sets is still quite imperfectly known."

— Juan Jorge Schäffer

Convexity is a very intuitive and geometric notion, but plays a fundamental role in many (also abstract) branches of mathematics. In vector spaces, this intuitive notion can be easily defined in a formal way: a *convex set* is a set which completely contains any segment connecting two of its points. This is the most central concept in this book. Some remarkable aspects of this idea are discussed in the *Notes* section at the end of this chapter, and a historical overview of the theory is given in Chapter 12. For now, we highlight that the simple notion of convexity gives rise to a very rich and deep theory, as it particularly becomes clear also in this chapter.

In Section 2.1, we introduce some basic notions, such as equivalent definitions of convexity and some of its topological and combinatorial aspects. Moving forward, in Section 2.2 we comprehensively discuss the notions of *support* and *separation*. The first one refers to the intuitive fact that there is always a hyperplane "settled" on a convex set in each of its boundary points, while the second one is related to the (also intuitive) idea that two disjoint convex sets can always be "separated" by a hyperplane lying "between" them.

As was mentioned in Chapter 1, it is often useful to associate certain functions with convex sets, and this approach is taken in Section 2.3. As is the case with convex functions, convex sets also have some remarkable regularity properties. In Section 2.4, we prove that the boundary of a "full-dimensional" convex set is smooth "almost everywhere" (the precise definitions and statements will be given later, of course). For that sake, we introduce and investigate the so-called *convex cones*, which are of independent interest (e.g., they will be important in Chapter 3).

In Section 2.5, we introduce the so-called *Hausdorff metric*, which is quite useful in convexity. This is a metric defined in the family of non-empty compact subsets of a vector space (or, more generally, of any metric space, but we do not use this generalization here). It happens that the subfamily of the non-empty compact sets which are also convex is closed with respect to this metric, and this qualifies the Hausdorff metric to become a powerful tool for approximations of convex sets. Even more, most of the geometric functionals defined for convex sets are continuous in the Hausdorff metric, as is briefly discussed in Section 2.6.

We finish Chapter 2 by studying a useful symmetrization technique known as *Steiner symmetrization*. This method has two properties which are very useful: first, it preserves (Lebesgue) volume, and

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_2

second, one can find for any given compact, convex set with positive volume (i.e., for a *convex body*) a sequence of directions for which the corresponding symmetrals converge to a Euclidean ball. Because of these features, Steiner symmetrizations will be used in Chapter 5 to prove some classical geometric inequalities (such as the *isoperimetric inequality*). Plate 2 shows the Steiner symmetrization of a planar convex domain and two related procedures, the last one taking place in 3-space.

Of course, from now on the entire book is dedicated to the study of convex sets. Thus, the reader must keep in mind that this chapter brings only introductory aspects of the theory which, however, are necessary steps to reach later much deeper concepts and results.

2.1 Basic notions

We start this chapter by revisiting some geometric notions which were briefly introduced in Chapter 1. Next we first provide some interpretation of the convex hull of a set $A \subseteq \mathbb{R}^n$, which was defined in (1.1.1), and we start with an equivalent definition of convexity involving combinations of more than two points.

Lemma 2.1.1. *A set $K \subseteq \mathbb{R}^n$ is convex if and only if*

$$\lambda_1 x_1 + \ldots + \lambda_m x_m \in K$$

for any points $x_1, \ldots, x_m \in K$ and any non-negative numbers $\lambda_1, \ldots, \lambda_m \geq 0$ such that $\lambda_1 + \ldots + \lambda_m = 1$, where m is any natural number. In other words, K is convex if and only if any convex combination of points of K is also a point of K.

Proof. The "only if" part trivially follows from the definition of convexity. The proof of the "if" part is obtained by induction on m. If K is convex, then any convex combination of 2 points lies in K. We assume, as induction hypothesis, that any convex combination of $m-1$ points of K is a point of K. Thus, given a convex combination

$$x := \lambda_1 x_1 + \ldots + \lambda_m x_m$$

of m points of K, we write

$$x = \lambda_1 x_1 + (1-\lambda_1)\left(\frac{\lambda_2}{1-\lambda_1} x_2 + \ldots + \frac{\lambda_m}{1-\lambda_1} x_m\right),$$

and we notice that

$$y := \frac{\lambda_2}{1-\lambda_1} x_2 + \ldots + \frac{\lambda_m}{1-\lambda_1} x_m$$

is a convex combination of $m-1$ points of K, since we clearly have that all the coefficients are non-negative, and their sum equals 1. The induction hypothesis ensures that $y \in K$, and then $x = \lambda_1 x_1 + (1-\lambda_1) y \in K$. It follows that any convex combination of m points of K belongs to K, and the proof is complete.

□

Proposition 2.1.1. *Let A be a subset of $\mathbb{R}^n$. Then the convex hull* $\mathrm{conv}(A)$ *of A is the smallest (regarding inclusion) convex set containing A, meaning that*

$$\mathrm{conv}(A) = \bigcap \{C \supseteq A : C \text{ is convex}\}.$$

Also, $K \subseteq \mathbb{R}^n$ is a convex set if and only if $K = \text{conv}(K)$.

Proof. The inclusion

$$\text{conv}(A) \subseteq \bigcap \{C \supseteq A : C \text{ is convex}\}$$

means that the convex hull of A is contained in any convex set which contains A. The proof follows from Lemma 2.1.1. Indeed, if C is a convex set and $A \subseteq C$, then any convex combination of points of A is a convex combination of points of C and, by the mentioned lemma, any convex combination of points of C is a point of C. It follows that $\text{conv}(A) \subseteq C$. The reverse inclusion

$$\text{conv}(A) \supseteq \bigcap \{C \supseteq A : C \text{ is convex}\}$$

is obvious, since $\text{conv}(A)$ is a convex set (see Lemma 1.1.2) which obviously contains A.

It remains to prove that $K \subseteq \mathbb{R}^n$ is convex if and only if $K = \text{conv}(K)$. If this equality holds, then we have immediately that K is convex, because $\text{conv}(K)$ is convex. Now assume that K is convex. The inclusion $K \subseteq \text{conv}(K)$ is immediate from the definition of the convex hull, and the reverse inclusion $\text{conv}(K) \subseteq K$ is a straightforward consequence of Lemma 2.1.1.

□

An *affine subspace* of $\mathbb{R}^n$ is a translate of a vector subspace (sometimes we use the term *associated vector subspace* to the affine subspace). That is, a set $S \subseteq \mathbb{R}^n$ is an affine subspace when there exist a point $x_0 \in \mathbb{R}^n$ and a vector space $V \subseteq \mathbb{R}^n$ such that

$$S = V + x_0 := \{v + x_0 : v \in V\};$$

see Figure 2.1.1. In this case we clearly have that $x_0 \in S$ (because $0_{\mathbb{R}^n} \in V$), and we also can write $V = S - x_0 := \{x - x_0 : x \in S\}$. Next we prove that this definition is "independent of the considered translation".

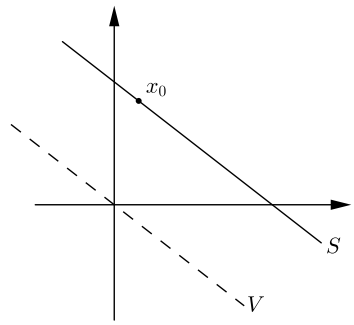


Fig. 2.1.1 The affine subspace $S = V + x_0$.

Lemma 2.1.2. *If S is an affine subspace such that $S = V + x_0$ for some subspace $V \subseteq \mathbb{R}^n$ and some $x_0 \in S$, then $V = S - x_1$ for any $x_1 \in S$.*

Proof. An arbitrary point of $S - x_1$ can be written as $z - x_1$, where $z \in S$. Since $x_1 \in S$, we have that $x_1 - x_0 \in S - x_0 = V$. On the other hand, $z - x_0 \in S - x_0 = V$. Thus, writing $z - x_1 = (z - x_0) + (x_0 - x_1)$ we have that $z - x_1 \in V$, because it is the sum of two vectors of V. It follows that $S - x_1 \subseteq V$.

Now we prove the reverse inclusion. Let $x_1 \in S$ and $v \in V$. Since $v = (v + x_1) - x_1$, all we have to prove is that $v + x_1 \in S$. For this purpose, we write $v + x_1 = [v + (x_1 - x_0)] + x_0$, and since $S = V + x_0$,

it suffices to prove that $v + (x_1 - x_0) \in V$. Recalling that V is a vector subspace, we only need to show that $x_1 - x_0 \in V$, and this comes immediately from the fact that $x_1 \in S$. Indeed, this fact leads to $x_1 - x_0 \in S - x_0 = v$.

□

Example 2.1.1. A hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha)$ is an affine subspace of dimension $n - 1$. It is easy to see that $\mathbf{h}$ is a vector subspace of $\mathbb{R}^n$ if and only if $\alpha = 0$, and that $\operatorname{span}\{u\}$ is the orthogonal complement of the vector subspace associated with $\mathbf{h}$. This justifies why we call u the *orthogonal* (or *normal*) vector of $\mathbf{h}$ (as was defined in Chapter 1). In this case, any non-zero scalar multiple of u is also said to be orthogonal (or normal) to $\mathbf{h}$.

We define the *affine hull* $\operatorname{aff} A$ of a subset A of $\mathbb{R}^n$ as the set of all *affine combinations* of the points of A:

$$\operatorname{aff} A := \left\{ \sum_{j=1}^{m} \lambda_j x_j : x_j \in A \text{ and } \lambda_j \in \mathbb{R} \text{ for } j = 1, \ldots, m, \text{ and } \sum_{j=1}^{m} \lambda_j = 1 \right\}.$$

Intuitively, the affine hull of two different points is the line passing through them, the affine hull of three non-collinear points is the two-dimensional plane which contains these points, and so on; see Figure 2.1.2.

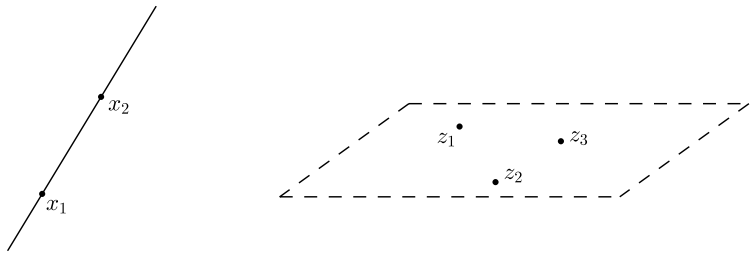


Fig. 2.1.2 Affine hulls of affinely independent points.

Lemma 2.1.3. *Let $A \subseteq \mathbb{R}^n$. Then* $\operatorname{aff}(\operatorname{aff} A) = \operatorname{aff} A$. *In other words, any affine combination of points of* $\operatorname{aff} A$ *can be written as an affine combination of points of A.*

Proof. If $z \in \operatorname{aff}(\operatorname{aff} A)$, then we may write z as an affine combination of points of $\operatorname{aff} A$:

$$z = \alpha_1 y_1 + \ldots + \alpha_s y_s.$$

For each $j = 1, \ldots, s$, the point $y_j \in \operatorname{aff} A$ is an affine combination of $m(j) \in \mathbb{N}$, say, points $x_1^j, \ldots, x_{m(j)}^j$ of A:

$$y_j = \lambda_1^j x_1^j + \ldots + \lambda_{m(j)}^j x_{m(j)}^j.$$

Thus

$$z = \alpha_1(\lambda_1^1 x_1^1 + \ldots + \lambda_{m(1)}^1 x_{m(1)}^1) + \ldots + \alpha_s(\lambda_1^s x_1^s + \ldots + \lambda_{m(s)}^s x_{m(s)}^s),$$

and this is an affine combination of points of A, because the sum of the coefficients equals 1. Indeed,

$$\alpha_1\lambda_1^1 + \ldots + \alpha_1\lambda_{m(1)}^1 + \ldots + \alpha_s\lambda_1^s + \ldots + \alpha_s\lambda_{m(s)}^s =$$
$$= \alpha_1(\lambda_1^1 + \ldots + \lambda_{m(1)}^1) + \ldots + \alpha_s(\lambda_1^s + \ldots + \lambda_{m(s)}^s) = \alpha_1 + \ldots + \alpha_s = 1.$$

Hence $z \in \operatorname{aff} A$, and this gives the inclusion $\operatorname{aff}(\operatorname{aff} A) \subseteq \operatorname{aff} A$. The reverse inclusion is obvious, since $A \subseteq \operatorname{aff} A$.

□

In the next proposition, we prove that any affine hull is an affine subspace. This allows us to define the *dimension of a set*.

Proposition 2.1.2. *Let A be a non-empty subset of $\mathbb{R}^n$. For any point $x_0 \in \operatorname{aff} A$ the set*

$$\operatorname{aff} A - x_0 := \{x - x_0 : x \in \operatorname{aff} A\}$$

is a vector subspace of $\mathbb{R}^n$.

Remark 2.1.1. By Lemma 2.1.2, the dimension of $\operatorname{aff} A - x_0$ is independent of the chosen point $x_0 \in \operatorname{aff} A$, and this number is the *dimension of the affine subspace* $\operatorname{aff} A$. We use the same notation as for the dimension of a vector subspace, and we denote the dimension of $\operatorname{aff} A$ by $\dim(\operatorname{aff} A)$. We also call this number the (*affine*) *dimension of* A. That is, the (*affine*) *dimension* of a given set is the dimension of its affine hull; see Figure 2.1.3. This is denoted by $\dim A$.

Proof. First of all, it is clear that $0_{\mathbb{R}^n} \in \operatorname{aff} A - x_0$. If $z - x_0 \in \operatorname{aff} A - x_0$ and $\lambda \in \mathbb{R}$, then we write

$$\lambda(z - x_0) = [\lambda z + (1 - \lambda)x_0] - x_0$$

and observe that $\lambda z + (1 - \lambda)x_0 \in \operatorname{aff} A$, because $z, x_0 \in \operatorname{aff} A$ and the sum of the coefficients equals 1 (hence $\lambda z + (1-\lambda)x_0 \in \operatorname{aff}(\operatorname{aff} A) = \operatorname{aff} A$, the equality given by Lemma 2.1.3). Thus, $\lambda(z - x_0) \in \operatorname{aff} A - x_0$, and we get that $\operatorname{aff} A - x_0$ is closed under scalar multiplication.

It only remains to prove that $\operatorname{aff} A - x_0$ is closed under vector addition. Let $z - x_0, y - x_0 \in \operatorname{aff} A - x_0$. We write $z - x_0 + y - x_0 = (z + y - x_0) - x_0$ and observe simply that $z + y - x_0 \in \operatorname{aff} A$, since $z, y, x_0 \in \operatorname{aff} A$ and the sum of the coefficients is $1 + 1 - 1 = 1$. The proof is finished.

□

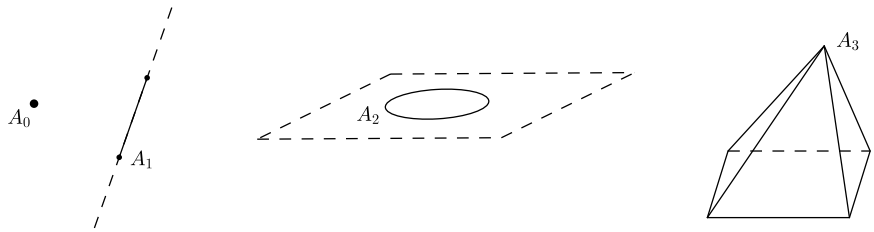


Fig. 2.1.3 Sets of dimension 0, 1, 2, and 3, respectively.

We remind the reader that in Chapter 1 we defined the notion of affinely independent points. Namely, the points $\{x_0, \ldots, x_m\} \subseteq \mathbb{R}^n$ are *affinely independent* if none of them is an affine combination of the other points; see Figure 2.1.4. For the sake of completeness, we explore this idea a little bit more.

Proposition 2.1.3. *The $m + 1$ points of the set $\{x_0, \ldots, x_m\} \subseteq \mathbb{R}^n$ are affinely independent if and only if the vectors $x_1 - x_0, \ldots, x_m - x_0$ are linearly independent. In this case, we have that $\dim(\operatorname{aff}\{x_0, \ldots, x_m\}) = m$.*

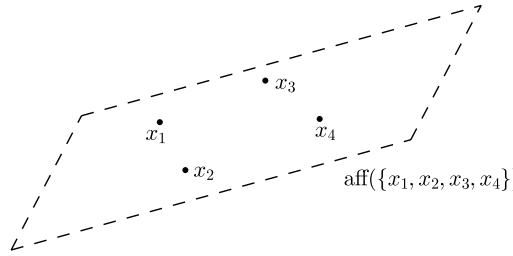


Fig. 2.1.4 The points $\{x_1, x_2, x_3\}$ are affinely independent, while $\{x_1, x_2, x_3, x_4\}$ are not.

Proof. If $x_1 - x_0, \ldots, x_m - x_0$ are not linearly independent, then there exist non-simultaneously null numbers $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$ such that

$$\lambda_1(x_1 - x_0) + \ldots + \lambda_m(x_m - x_0) = 0_{\mathbb{R}^n},$$

and we may assume that, without loss of generality, $\lambda_1 \neq 0$. Then we re-write the equality above as

$$x_1 = -\frac{\lambda_2}{\lambda_1}x_2 - \frac{\lambda_3}{\lambda_1}x_3 - \ldots - \frac{\lambda_m}{\lambda_1}x_m + \left(\frac{\lambda_2 + \ldots + \lambda_m}{\lambda_1} + 1\right)x_0,$$

and the right-hand side is an affine combination of the points $x_0, x_2, \ldots, x_m$, because the sum of the coefficients is 1. It follows from the definition that $x_0, \ldots, x_m$ are not affinely independent.

Now assume that $x_0, \ldots, x_m$ are not affinely independent. Then one of these points can be written as an affine combination of the others:

$$x_k = \lambda_0 x_0 + \ldots + \lambda_{k-1}x_{k-1} + \lambda_{k+1}x_{k+1} + \ldots + \lambda_m x_m,$$

where $\lambda_0 + \ldots + \lambda_{k-1} + \lambda_{k+1} + \ldots + \lambda_m = 1$. Observing that $\lambda_0 x_0 = [1 - (\lambda_1 + \ldots + \lambda_{k-1} + \lambda_{k+1} + \ldots + \lambda_m)]x_0$, we reorganize the equality above to obtain

$$\begin{aligned} 0_{\mathbb{R}^n} &= \lambda_0 x_0 + \ldots + \lambda_{k-1}x_{k-1} - x_k + \lambda_{k+1}x_{k+1} + \ldots + \lambda_m x_m = \\ &= \lambda_1(x_1 - x_0) + \ldots - (x_k - x_0) + \lambda_{k+1}(x_{k+1} - x_0) + \ldots + \lambda_m(x_m - x_0), \end{aligned}$$

and this is a non-trivial linear combination of $x_1 - x_0, \ldots, x_m - x_0$ (the coefficient of $x_k - x_0$ is non-zero) resulting in $0_{\mathbb{R}^n}$. It follows that these vectors are not linearly independent.

□

Corollary 2.1.1. *Let $S \subseteq \mathbb{R}^n$ be an affine subspace. Then the maximum cardinality that a set of affinely independent points of S has is* $\dim(S) + 1$.

Proof. By the definition of affine subspace, we let $S = V + x_0 := \{x + x_0 : x \in V\}$, where V is a vector subspace of $\mathbb{R}^n$ (notice carefully that, in this case, we have $x_0 \in S$). If $\{v_1, \ldots, v_m\}$ is a basis of V, then $\dim S = m$, and from the previous proposition the points $x_0, v_1 + x_0, \ldots, v_m + x_0$ are affinely independent points of S. It follows that there exists a set of $\dim S + 1$ affinely independent points in S.

It remains to prove that there is no affinely independent set in S with more than $\dim S + 1$ points. Indeed, if $\{x_0, \ldots, x_k\} \subseteq S$ is an affinely independent set with $k > m$, then $U := \{x_1 - x_0, \ldots, x_k - x_0\}$ is a set of k linearly independent vectors, all of them in the vector subspace $S - x_0$. Hence we have $\dim S \geq k > m$, a contradiction.

□

Summarizing, we have that the dimension of a subset A is the affine dimension of the affine subspace $\mathrm{aff}A$, which is the maximum number of affinely independent points that one can take in $\mathrm{aff}A$, minus 1. The next proposition states that the affine dimension of A is, in fact, the maximum number of affinely independent points that one can take **in** $\boldsymbol{A}$, minus 1. In other words, passing to the affine hull "does not create larger affinely independent sets".

Proposition 2.1.4. *Let A be a non-empty subset of $\mathbb{R}^n$. If $\dim A = m$, then there exists a set of $m+1$ affinely independent points in A. Moreover, there is no affinely independent subset in A with more than $m+1$ points.*

Proof. We start by the last claim, which is easier. If there is an affinely independent subset of A with more than $m+1$ points, then this is, in particular, an affinely independent subset of $\mathrm{aff}A$, and hence Corollary 2.1.1 gives that $\dim(\mathrm{aff}A) > m$.

If $\dim A = m$, then there exist $m+1$ affinely independent points in $\mathrm{aff}A$. We have to show that these points can be taken in A, instead. Assume, by contradiction, that we have at most $k+1$ affinely independent points in A, with $k < m$, and let $\{x_0, \ldots, x_k\} \subseteq A$ be affinely independent points. Then any point of A can be written as an affine combination of $\{x_0, \ldots, x_k\}$, since otherwise we would have a larger set of affinely independent points. We claim that

$$\mathrm{aff}A - x_0 \subseteq \mathrm{span}\{x_1 - x_0, \ldots, x_k - x_0\} =: V.$$

Indeed, if $z \in \mathrm{aff}A$, then there exist points $y_1, \ldots, y_s \in A$ and numbers $\alpha_1, \ldots, \alpha_s \in \mathbb{R}$ such that $\alpha_1 + \ldots + \alpha_s = 1$ and $z = \alpha_1 y_1 + \ldots + \alpha_s y_s$. Each y_j is a point of A, and hence can be written as an affine combination of the points $\{x_0, \ldots, x_k\}$. For each $j = 1, \ldots, s$, write y_j as the affine combination

$$y_j = \lambda_0^j x_0 + \lambda_1^j x_1 + \ldots + \lambda_k^j x_k.$$

This leads immediately to

$$z = \alpha_1(\lambda_0^1 x_0 + \ldots + \lambda_k^1 x_k) + \ldots + \alpha_s(\lambda_0^s x_0 + \ldots + \lambda_k^s x_k),$$

and this is a combination of the points $x_0, \ldots, x_k$. To prove that this is an affine combination, one only has to verify that the sum of the coefficients equals 1. But this comes from the equalities $\alpha_1 + \ldots + \alpha_s = 1$ and $\lambda_0^j + \ldots + \lambda_k^j = 1$ for each $j = 1, \ldots, s$:

$$(\alpha_1\lambda_0^1 + \alpha_1\lambda_1^1 + \ldots + \alpha_1\lambda_k^1) + \ldots + (\alpha_s\lambda_0^s + \ldots + \alpha_s\lambda_k^s) =$$
$$= \alpha_1(\lambda_0^1 + \ldots + \lambda_k^1) + \alpha_2(\lambda_0^2 + \ldots + \lambda_k^2) + \ldots + \alpha_s(\lambda_0^s + \ldots + \lambda_k^s) = \alpha_1 + \alpha_2 + \ldots + \alpha_s = 1.$$

Therefore, any given point $z \in \mathrm{aff}A$ can be written as an affine combination of $\{x_0, \ldots, x_k\}$:

$$z = \lambda_0 x_0 + \ldots + \lambda_k x_k.$$

Finally, we write

$$z - x_0 = \lambda_0 x_0 + \ldots + \lambda_k x_k - (\lambda_0 + \ldots + \lambda_k)x_0 =$$
$$= \lambda_1(x_1 - x_0) + \lambda_2(x_2 - x_0) + \ldots + \lambda_k(x_k - x_0),$$

and hence $z - x_0 \in V$. This shows that $\mathrm{aff}A - x_0 \subseteq V$. Therefore,

$$\dim A = \dim(\mathrm{aff}A) \leq \dim V = k < m,$$

which is a contradiction. Hence A contains a subset of $m+1$ affinely independent points.

□

Scholium 2.1.1. Let $A \subseteq \mathbb{R}^n$ be non-empty. If $\{x_0, \ldots, x_m\} \subseteq A$ is any maximal affinely independent set, then $\operatorname{aff} A = \operatorname{aff}\{x_0, \ldots, x_m\}$.

Proof. This is a scholium of the last proof, since we showed that any point $z \in \operatorname{aff} A$ can be written as an affine combination of the points of an arbitrarily fixed maximal affinely independent set $\{x_0, \ldots, x_k\} \subseteq A$.

□

Let $A \subseteq \mathbb{R}^n$ be such that its affine hull has dimension $\dim(\operatorname{aff} A) = k < n$ (keep always in mind that, in this case, we also write $\dim A = k$). It is easy to see that $\operatorname{int} A = \emptyset$ (here, and always, $\operatorname{int} A$ denotes the interior of A on the topology of $\mathbb{R}^n$), because any open ball of $\mathbb{R}^n$ clearly contains $n + 1$ affinely independent points. However, considering the topology induced on $\operatorname{aff} A$, one can regard the interior (and also the closure and the boundary) of A "as a k-dimensional set".

Recall that the *induced topology* given on any affine subspace $S \subseteq \mathbb{R}^n$ (more generally, on any subset of $\mathbb{R}^n$) is obtained by defining the open sets to be the intersections of the open sets of $\mathbb{R}^n$ with S. With this in mind, we define the *relative interior* and the *relative boundary* of $A \subseteq \mathbb{R}^n$ as the interior and the boundary of A in the induced topology on $\operatorname{aff} A$, respectively. These sets are denoted by $\operatorname{relint} A$ and $\operatorname{relbd} A$, respectively. In other words,

(a) $x \in \operatorname{relint} A$ if and only if $x \in \operatorname{aff}(A)$ and there exists a number $\rho > 0$ such that $B(x, \rho) \cap \operatorname{aff} A \subseteq A$, where we remind the reader that $B(x, \rho)$ denotes the open ball of $\mathbb{R}^n$ with center x and radius $\rho > 0$ (see Figure 2.1.5).

(b) $x \in \operatorname{relbd} A$ if and only if for every $\rho > 0$ the set $B(x, \rho) \cap \operatorname{aff} A$ contains points of A and of $(\operatorname{aff} A) \setminus A$.

Observe that we clearly can write

$$A \subseteq \operatorname{relint} A \cup \operatorname{relbd} A,$$

and this union is disjoint. Moreover, equality holds if A is closed. Of course, one can also think of defining the *relative closure* of A as the closure of A in the induced topology of $\operatorname{aff} A$, but this clearly coincides with the closure of A in the topology of $\mathbb{R}^n$.

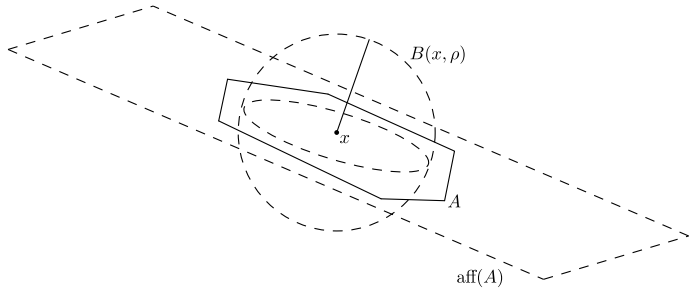


Fig. 2.1.5 A point x in the relative interior of A.

Remark 2.1.2. There is also an alternative (and equivalent) way to deal with these ideas. If $\dim(\operatorname{aff} A) = k$, then we can easily identify $\operatorname{aff} A$ with $\mathbb{R}^k$. Indeed, just choose a point $x_0 \in \operatorname{aff} A$ and an isomorphism

between the k-dimensional vector subspace $\operatorname{aff} A - x_0$ and $\mathbb{R}^k$. Such isomorphism naturally induces the topology of $\mathbb{R}^k$ on $\operatorname{aff} A - x_0$, and hence on $\operatorname{aff} A$. Of course, this topology coincides with the ambient topology (of $\mathbb{R}^n$) on $\operatorname{aff} A$.

As (geometric) examples, observe that the relative interior of a (non-degenerate) segment $\operatorname{seg}[x, y] \subseteq \mathbb{R}^n$ is the *open segment* $\operatorname{seg}(x, y) := \{(1-\lambda)x + \lambda y : \lambda \in (0, 1)\} = \operatorname{seg}[x, y] \setminus \{x, y\}$. If $n \geq 2$ and $x_1, x_2, x_3 \in \mathbb{R}^n$ are non-collinear points, then the relative interior of $\operatorname{conv}\{x_1, x_2, x_3\}$ is the triangle whose vertices are x_1, x_2, and x_3 without the edges (see Remark 2.1.3 after the next theorem).

Theorem 2.1.1. *Any non-empty convex set of $\mathbb{R}^n$ has non-empty relative interior.*

Proof. Let $C \subseteq \mathbb{R}^n$ be convex and non-empty, and assume that $\dim C = k$. Then, from Proposition 2.1.4, we can take $k+1$ affinely independent points $\{x_0, \ldots, x_k\}$ in C. Hence we have that $\operatorname{conv}\{x_0, \ldots, x_k\} \subseteq C$. We claim that any convex combination

$$x = \lambda_0 x_0 + \ldots + \lambda_k x_k \in \operatorname{conv}\{x_0, \ldots, x_k\}$$

with the property that all coefficients are strictly positive is such that $x \in \operatorname{relint} C$. The vectors $x_1 - x_0, \ldots, x_k - x_0$ are linearly independent in $\mathbb{R}^n$, and hence (if necessary) we can take $n-k$ vectors $v_1, \ldots, v_{n-k} \in \mathbb{R}^n$ such that $\{x_1 - x_0, \ldots, x_k - x_0, v_1, \ldots, v_{n-k}\}$ is a basis of $\mathbb{R}^n$. We endow $\mathbb{R}^n$ with the norm

$$||\alpha_1(x_1 - x_0) + \ldots + \alpha_k(x_k - x_0) + \alpha_{k+1}v_1 + \ldots + \alpha_n v_{n-k}||_\infty = \max\{|\alpha_j| : j = 1, \ldots, n\},$$

and we denote by $B_\infty(z, \rho)$ the open ball with center at z and radius $\rho > 0$ with respect to this norm. Since any norm in $\mathbb{R}^n$ gives the same topology as the standard Euclidean norm, it follows that the relative interior of any subset $A \subseteq \mathbb{R}^n$ is the interior of A in the topology given by the norm $||\cdot||_\infty$ on $\operatorname{aff} A$.

Let $x = \lambda_0 x_0 + \ldots + \lambda_k x_k$ be a convex combination such that $\lambda_j > 0$ for each $j = 1, \ldots, k$. If $z \in \operatorname{aff} C \cap B_\infty(x, \rho)$, the following is true.

(i) z is an affine combination of $\{x_0, \ldots, x_k\}$ (see Remark 2.1.1):

$$z = \beta_0 x_0 + \ldots + \beta_k x_k \quad \text{and}$$

(ii) recalling that $\lambda_0 + \ldots + \lambda_k = 1$ and $\beta_0 + \ldots + \beta_k = 1$, and using these equalities to write $x - x_0 = \lambda_1(x_1 - x_0) + \lambda_2(x_2 - x_0) + \ldots + \lambda_k(x_k - x_0)$, and $z - x_0 = \beta_1(x_1 - x_0) + \ldots + \beta_k(x_k - x_0)$, we have

$$||z - x||_\infty = ||(z - x_0) - (x - x_0)||_\infty = \max\{|\beta_j - \lambda_j| : j = 1, \ldots, k\} < \rho.$$

Therefore, choosing $0 < \rho \leq \min\{\lambda_j : j = 1, \ldots, k\}$ we have that $0 \leq \lambda_j - \rho < \beta_j$ for every $j = 1, \ldots, k$, and this means that $z \in \operatorname{conv}\{x_0, \ldots, x_k\}$. Indeed, z is an affine combination of $\{x_0, \ldots, x_k\}$ with non-negative coefficients, and hence a convex combination of $\{x_0, \ldots, x_k\}$. Finally, this gives the inclusion

$$B_\infty(x, \rho) \cap \operatorname{aff} C \subseteq \operatorname{conv}\{x_0, \ldots, x_k\}, \tag{2.1.1}$$

and since $\operatorname{conv}\{x_0, \ldots, x_k\} \subseteq C$, we get that $x \in \operatorname{relint} C$.

□

Remark 2.1.3. We proved more than the fact that any non-empty convex set has non-empty relative interior. What we really showed is that if $x_0, \ldots, x_k$ are affinely independent points, then any convex

combination of these points whose coefficients are all positive is a point of the relative interior of $\mathrm{conv}\{x_0, \ldots, x_k\}$. Indeed, it is clear that $\mathrm{aff} C = \mathrm{aff}(\mathrm{conv}\{x_0, \ldots, x_k\})$, and hence (2.1.1) gives

$$B_\infty(x, \rho) \cap \mathrm{aff}(\mathrm{conv}\{x_0, \ldots, x_k\}) \subseteq \mathrm{conv}\{x_0, \ldots, x_k\},$$

implying that $x \in \mathrm{relint}(\mathrm{conv}\{x_0, \ldots, x_k\})$.

Theorem 2.1.2 (Carathéodory's theorem)**.** *Let $A \subseteq \mathbb{R}^n$. Any $x \in \mathrm{conv} A$ can be written as a convex combination of affinely independent points of A.*

Proof. Let $x \in \mathrm{conv} A$. Then x can be written as a convex combination of points of A:

$$x = \lambda_0 x_0 + \ldots + \lambda_m x_m.$$

Assume that the number $m + 1$ is minimal (that is, x cannot be written as a convex combination of less than $m + 1$ points of A). This means that, in particular, each λ_j is positive. If the points $x_0, \ldots, x_m$ are affinely independent, there is nothing to prove, and hence we may assume that they are affinely dependent. This means that one of them can be written as an affine combination of the others, and we may assume that, without loss of generality,

$$x_0 = \alpha_1 x_1 + \ldots + \alpha_m x_m,$$

for $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$ such that $\alpha_1 + \ldots + \alpha_m = 1$. Of course, at least one of the α_j's is positive. Hence the number

$$\min\{\lambda_j/\alpha_j : \alpha_j > 0\}$$

is the minimum of a non-empty finite set. We may assume that, without loss of generality, this is attained for $j = 1$. Consequently,

$$\frac{\lambda_1}{\alpha_1} \leq \frac{\lambda_j}{\alpha_j} \tag{2.1.2}$$

for every $j = 1, \ldots, m$ such that $\alpha_j > 0$. Now we re-write the decomposition of x_0 above as

$$x_1 = \frac{x_0}{\alpha_1} - \frac{\alpha_2}{\alpha_1} x_2 - \ldots - \frac{\alpha_m}{\alpha_1} x_m.$$

Replacing x_1 by the right-hand side above in the decomposition of x as a convex combination of $\{x_0, \ldots, x_m\}$, and rearranging the terms, we get

$$x = \left(\lambda_0 + \frac{\lambda_1}{\alpha_1}\right) x_0 + \left(\lambda_2 - \frac{\lambda_1 \alpha_2}{\alpha_1}\right) x_2 + \ldots + \left(\lambda_m - \frac{\lambda_1 \alpha_m}{\alpha_1}\right) x_m.$$

We claim that this is a convex combination of the points $x_0, x_2, \ldots, x_m$. Indeed, if $\alpha_j > 0$, then the coefficient $\lambda_j - \frac{\lambda_1 \alpha_j}{\alpha_1}$ is positive because of (2.1.2). If $\alpha_j \leq 0$, then we also have that $\lambda_j - \frac{\lambda_1 \alpha_j}{\alpha_1}$ is positive because it is a sum of non-negative numbers, and at least one of them is positive (recall that $\lambda_j > 0$ for all j). Further, it is clear that the coefficient of x_0 in the equality above is also positive. It remains to prove that the sum of the coefficients equals 1. This is a consequence of the following calculation:

$$\left(\lambda_0 + \frac{\lambda_1}{\alpha_1}\right) + \left(\lambda_2 - \frac{\lambda_1\alpha_2}{\alpha_1}\right) + \ldots + \left(\lambda_m - \frac{\lambda_1\alpha_m}{\alpha_1}\right) =$$

$$= \lambda_0 + \lambda_2 + \ldots + \lambda_m + \frac{\lambda_1}{\alpha_1}[1 - (\alpha_2 + \ldots + \alpha_m)] =$$

$$= \lambda_0 + \lambda_2 + \ldots + \lambda_m + \frac{\lambda_1}{\alpha_1}\alpha_1 = \lambda_0 + \lambda_1 + \ldots + \lambda_m = 1.$$

This shows that the point x can be written as a convex combination of m points. This contradicts the minimality of $m + 1$, and the proof is complete.

□

Remark 2.1.4. What we have really proved is that if $x = \lambda_0 x_0 + \ldots + \lambda_m x_m$ is a minimal representation of $x \in \mathrm{conv}A$ as a convex combination of points of A, then the points $x_0, \ldots, x_m$ are affinely independent.

We proceed to investigate more relations between convexity and (relative) topology. The next lemma can be interpreted as follows: in a convex set, a segment from an interior point to a boundary point remains in the interior until it reaches the boundary.

Lemma 2.1.4. *Let $C \subseteq \mathbb{R}^n$ be a convex set. If $x \in \mathrm{relint}C$ and $y \in \mathrm{cl}C$, then*

$$\mathrm{seg}[x, y) := \{(1 - \lambda)x + \lambda y : \lambda \in [0, 1)\} \subseteq \mathrm{relint}C.$$

Proof. In view of Remark 2.1.2, we can assume that $\dim C = n$ (that is, it is maximal), and replace the relative interior by the usual interior. In fact, if $\dim C = k < n$, we simply identify $\mathrm{aff}C$ with $\mathbb{R}^k$, and consequently C is identified with a convex set of $\mathbb{R}^k$ having maximal dimension.

First assume that $y \in C$. Let $z = (1 - \lambda)x + \lambda y$ for some $\lambda \in [0, 1)$, and since $x \in \mathrm{int}C$, let $\rho > 0$ be such that $B(x, \rho) \subseteq C$. We claim that $B(z, (1 - \lambda)\rho) \subseteq C$. To prove this claim, observe that if $w \in B(z, (1 - \lambda)\rho)$, then we may write

$$w = z + (1 - \lambda)u$$

for some vector $u \in \mathbb{R}^n$ such that $|u| < \rho$. Hence $x + u \in C$, because $x + u \in B(x, \rho) \subseteq C$. Now we write

$$w = z + (1 - \lambda)u = (1 - \lambda)x + \lambda y + (1 - \lambda)u = (1 - \lambda)(x + u) + \lambda y,$$

and the latter is a convex combination of points of C. Since C is convex, it follows that $w \in C$. This argument is illustrated in Figure 2.1.6 below.

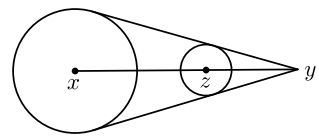


Fig. 2.1.6 The "region" between the segments from y tangent to the ball centered in x lies in C, and hence the ball centered in z also does.

Now let $y \in \mathrm{cl}C$, and still assume that $z = (1 - \lambda)x + \lambda y$ for some $\lambda \in [0, 1)$, and that $B(x, \rho) \subseteq C$. Since y is a point of the closure of C, we have that there exists a point $v \in C \cap B\left(y, \frac{(1-\lambda)\rho}{\lambda}\right)$. Similarly as before, we may write

$$v = y + \frac{(1-\lambda)}{\lambda}u$$

for some $u \in \mathbb{R}^n$ such that $|u| < \rho$. We re-write z as

$$z = (1-\lambda)x + \lambda y = (1-\lambda)x + \lambda\left(v - \frac{1-\lambda}{\lambda}u\right) = (1-\lambda)(x-u) + \lambda v,$$

and the latter is a convex combination of the points $x-u$ and v of C (observe that $x-u \in B(x,\rho)$). It follows that $z \in C$. This shows the inclusion

$$\text{seg}[x,y) \subseteq C,$$

but we want $\text{seg}[x,y) \subseteq \text{int}C$. If $z \in \text{seg}[x,y)$, with $z = (1-\lambda)x + \lambda y$ for some $0 \leq \lambda < 1$, then for $\lambda < \varepsilon < 1$ the point $y_1 = (1-\varepsilon)x + \varepsilon y$ is such that

$$z \in \text{seg}[x,y_1] \subseteq \text{seg}[x,y).$$

Hence $y_1 \in C$, and replacing y by y_1 in the first argument, we get $z \in \text{int}C$.

□

Corollary 2.1.2. *Let $C \subseteq \mathbb{R}^n$ be a non-empty convex set. Then $C \cap \text{relbd}C \subseteq \text{relbd}(\text{cl}C)$.*

Proof. Let $x \in C \cap \text{relbd}C$, and fix any point $z \in \text{relint}C$. For any given $\varepsilon > 0$, consider the point

$$y = x + \frac{\varepsilon}{2|x-z|}(x-z).$$

Then it is clear that $y \in B(x,\varepsilon) \cap \text{aff}C$, and we claim that $y \notin \text{cl}(C)$. Indeed, observe that

$$x \in \text{seg}[z,y).$$

Thus, if $y \in \text{cl}(C)$, then Lemma 2.1.4 yields that $x \in \text{relint}C$, which is a contradiction (recall that a point cannot lie simultaneously in the relative interior and in the relative boundary of a set). Therefore, for every $\varepsilon > 0$ the set $B(x,\varepsilon) \cap \text{aff}C$ has points of $\text{cl}C$ (indeed, $x \in C$) and of $\text{aff}C \setminus \text{cl}C$, and this means that $x \in \text{relbd}(\text{cl}C)$.

□

Before moving forward, we introduce some new important terminology. We define the *vector sum* or *Minkowski sum* of sets $A, B \subseteq \mathbb{R}^n$ as

$$A + B = \{a + b : a \in A \text{ and } b \in B\}.$$

This can be (geometrically) interpreted as the union of all translates of B by the points of A (and vice versa); see also Figure 2.1.7. Moreover, the *scalar multiplication* is defined for a given set $A \subseteq \mathbb{R}^n$ and for some $\lambda \in \mathbb{R}$ as

$$\lambda A = \{\lambda a : a \in A\};$$

see Figure 2.1.8.

Further, a combination of the form $\lambda_1 A_1 + \ldots + \lambda_m A_m$ where $A_1, \ldots A_m \subseteq \mathbb{R}^n$ and $\lambda_1, \ldots, \lambda_m \geq 0$ is called a *Minkowski combination*. Still in this section these concepts will be used to prove that the convex hull of an open set is open, and will be omnipresent throughout the text. To understand a little more about them, we have the following lemmas.

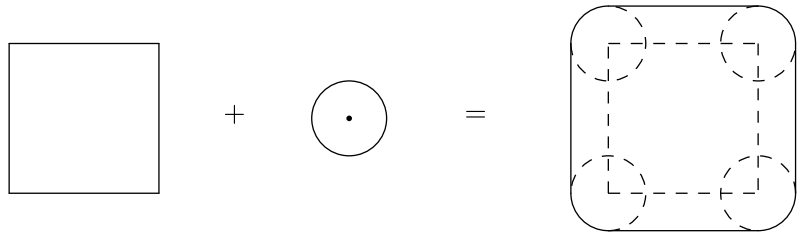


Fig. 2.1.7 The Minkowski sum of a square and a ball.

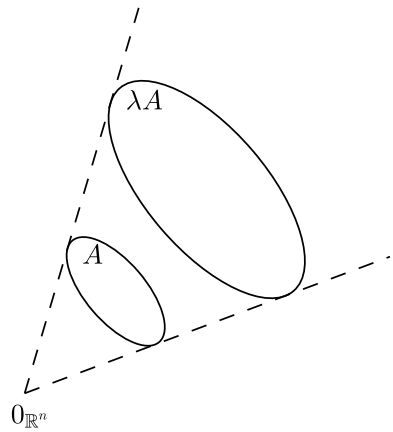


Fig. 2.1.8 Scalar multiplication of a set.

Theorem 2.1.3. ***(i)*** *If $K \subseteq \mathbb{R}^n$ is convex, then* relint$K$ *and* cl$K$ *are convex.*

(ii) *If $A \subseteq \mathbb{R}^n$ is* relatively open*, that is, if $A =$* relint$A$*, then* conv$A$ *is relatively open.*

(iii) *The convex hull of a compact set is compact.*

Remark 2.1.5. The convex hull of a closed set is **not** necessarily closed. A classical example is the convex hull of the union of a line with a point (which does not belong to the line, of course).

Proof. **(i)**: To prove that cl$K$ is convex, let $x, y \in \text{cl}K$ and $\lambda \in (0,1)$. If $(x_k)_{k\in\mathbb{N}}$ and $(y_k)_{k\in\mathbb{N}}$ are sequences of points of K such that $x_k \to x$ and $y_k \to y$, then the sequence $z_k := (1-\lambda)x_k + \lambda y_k$ clearly converges to $(1-\lambda)x + \lambda y$. Hence cl$K$ is convex.

To prove that relint$K$ is convex, we use Lemma 2.1.4. Assume that $x, y \in \text{relint}K$. Then, for some $\rho > 0$, we have $\text{cl}(B(y,\rho)) \cap \text{aff}(K) \subseteq K$. It follows that

$$y_0 := y + \frac{\rho}{|y-x|}(y-x) \in K,$$

and this implies that $\text{seg}[x, y_0) \in \text{relint}K$. Since we clearly have that $\text{seg}[x,y] \subseteq \text{seg}[x,y_0)$, we get that $\text{seg}[x,y] \subseteq \text{relint}K$, and this proves the convexity of relint$K$.

(ii): Again, in view of Remark 2.1.2, we can assume that $\dim A = n$, and hence A is open in $\mathbb{R}^n$ (instead of relatively open). Let $x \in \text{conv}A$ be written as a convex combination of points of A:

$$x = \lambda_1 x_1 + \ldots + \lambda_m x_m.$$

Since A is open, for each $j = 1, \ldots, m$ there exists a number $\rho_j > 0$ such that $B(x_j, \rho_j) \subseteq A$. Of course, we have the inclusion

$$\lambda_1 B(x_1, \rho_1) + \ldots + \lambda_m B(x_m, \rho_m) \subseteq \mathrm{conv} A,$$

since each point of the left-hand side is a convex combination of points of A. Moreover, we clearly have $x \in \lambda_1 B(x_1, \rho_1) + \ldots + \lambda_m B(x_m, \rho_m)$. Due to Lemma 2.1.7 this latter set is open, and this proves that $\mathrm{conv} A$ is open.

(iii): Assume that $C \subseteq \mathbb{R}^n$ is a compact set, and let $(x_k)_{k\in\mathbb{N}}$ be a sequence of points in $\mathrm{conv} C$. As an immediate consequence of Theorem 2.1.2 we have that each point x_k can be written as a convex combination of $n + 1$ (or less) points of C. Hence, for each $k \in \mathbb{N}$ we write such a linear combination as

$$x_k = \lambda_0^k x_0^k + \ldots + \lambda_n^k x_n^k.$$

Since $(x_0^k)_{k\in\mathbb{N}}$ is a sequence of points in C, and $(\lambda_0^k)_{k\in\mathbb{N}}$ is a sequence of numbers in $[0, 1]$, it follows that passing through a subsequence we may assume that $\lambda_0^k x_0^k \to \lambda_0 x_0$ for some $\lambda_0 \in [0, 1]$ and some $x_0 \in C$. From this subsequence, one can extract a subsequence such that $\lambda_1^k x_1^k \to \lambda_1 x_1$ for some $\lambda_1 \in [0, 1]$ and $x_1 \in C$ (notice that we are abusing a little of the notation and denote the subsequence as the original sequence). Repeating this step until we reach $\lambda_n^k x_n^k$, we get a subsequence of x_k, which we still denote by x_k, such that

$$x_k \to \lambda_0 x_0 + \ldots + \lambda_n x_n,$$

and the point $\lambda_0 x_0 + \ldots + \lambda_n x_n$ is a convex combination of points of C. Of course, this holds since we have

$$\lambda_0 + \ldots + \lambda_n = \lim_{k\to\infty} (\lambda_0^k + \ldots + \lambda_n^k) = 1,$$

due to the fact that $\lambda_0^k + \ldots + \lambda_n^k = 1$ for each $k \in \mathbb{N}$. It follows that any sequence of points in $\mathrm{conv} C$ has a converging subsequence, and this gives that $\mathrm{conv} C$ is compact.

□

Lemma 2.1.5. *For any $A, B \subseteq \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ we have that*

$$\mathrm{conv}(A + B) = \mathrm{conv} A + \mathrm{conv} B \quad \text{and}$$
$$\mathrm{conv}(\lambda A) = \lambda \mathrm{conv} A.$$

Proof. First, if $x \in \mathrm{conv}(A + B)$, then

$$x = \lambda_1(a_1 + b_1) + \ldots + \lambda_m(a_m + b_m),$$

where $\lambda_1, \ldots, \lambda_m \geq 0$, $\lambda_1 + \ldots + \lambda_m = 1$, $a_1, \ldots, a_m \in A$, and $b_1, \ldots, b_m \in B$. Simply rearranging the terms, we get

$$x = (\lambda_1 a_1 + \ldots + \lambda_m a_m) + (\lambda_1 b_1 + \ldots + \lambda_m b_m) \in \mathrm{conv} A + \mathrm{conv} B,$$

and this proves the inclusion $\mathrm{conv}(A + B) \subseteq \mathrm{conv} A + \mathrm{conv} B$. Now let $x \in \mathrm{conv} A + \mathrm{conv} B$. We write

$$x = (\lambda_1 a_1 + \ldots + \lambda_m a_m) + (\alpha_1 b_1 + \ldots + \alpha_k b_k),$$

that is, a sum of a convex combination of points of A with a convex combination of points of B. Since $\lambda_1 + \ldots + \lambda_m = 1$ and $\alpha_1 + \ldots + \alpha_k = 1$, we get

$$x = \sum_{i,j} \lambda_i \alpha_j (a_i + b_j),$$

and this is clearly a convex combination of points of $A + B$. Hence we have the remaining inclusion $\mathrm{conv}A + \mathrm{conv}B \subseteq \mathrm{conv}(A + B)$. The equality $\mathrm{conv}(\lambda A) = \lambda \mathrm{conv}A$ is trivial.

□

Remark 2.1.6. In particular, we have that the Minkowski sum of convex sets is convex. Indeed, if A and B are convex, then

$$\mathrm{conv}(A + B) = \mathrm{conv}(A) + \mathrm{conv}(B) = A + B.$$

It is immediate that convexity is also preserved under scalar multiplication.

Lemma 2.1.6. *Let $\mu, \lambda > 0$ and $A \subseteq \mathbb{R}^n$. We have the inclusion $(\mu + \lambda)A \subseteq \mu A + \lambda A$. If A is convex, then the reverse inclusion also holds. Moreover, if $\mu A + \lambda A \subseteq (\mu + \lambda)A$ for every $\mu, \lambda \in A$, then A is convex.*

Proof. The inclusion $(\mu + \lambda)A \subseteq \mu A + \lambda A$ is straightforward. Hence we assume that A is convex and let $z \in \mu A + \lambda A$. Then there exist $x, y \in A$ such that $z = \mu x + \lambda y$. We write

$$z = \mu x + \lambda y = (\mu + \lambda)\left(\frac{\mu}{\mu + \lambda} x + \frac{\lambda}{\mu + \lambda} y\right).$$

To prove that $z \in (\mu + \lambda)A$, it suffices to show that the point

$$\frac{\mu}{\mu + \lambda} x + \frac{\lambda}{\mu + \lambda} y$$

is in A. But this comes immediately from the convexity of A. Observe that the coefficients are positive (because $\mu, \lambda > 0$), and that they sum up to 1.

Now we assume that $\mu A + \lambda A \subseteq (\mu + \lambda)A$ for any $\mu, \lambda > 0$. Let $x, y \in A$ and $\beta \in (0, 1)$ be arbitrarily given. Since $(1 - \beta)A + \beta A \subseteq A$, we get that $(1 - \beta)x + \beta y \in A$, and this shows that A is convex.

□

Lemma 2.1.7. *Let $A, B \subseteq \mathbb{R}^n$ and $\lambda \neq 0$. If A is open, then $A + B$ and λA are open, too.*

Proof. Let $a + b \in A + B$, where $a \in A$ and $b \in B$. If A is open, we may take a number $\rho > 0$ such that $B(a, \rho) \subseteq A$. Hence $B(a, \rho) + b \subseteq A + B$. Finally, $B(a, \rho) + b$ is clearly an open ball, since $B(a, \rho) + b = B(a + b, \rho)$. For the other claim we simply notice that if $B(a, \rho) \subseteq A$, then $B(\lambda a, |\lambda|\rho) \subseteq \lambda A$.

□

Lemma 2.1.8. *If $A \subseteq \mathbb{R}^n$ is compact, then for any $\lambda \geq 0$ the set λA is also compact. Moreover, the Minkowski sum of compact sets is compact.*

Proof. If $(x_m)_{m \in \mathbb{N}}$ is any sequence in λA, then $(x_m/\lambda)_{m \in \mathbb{N}}$ is a sequence in A. If A is compact, then this latter sequence has a subsequence $(x_{m_k}/\lambda)_{k \in \mathbb{N}}$ converging to a point of A (z, say). Thus, $(x_{m_k})_{k \in \mathbb{N}}$ converges to λz, which is a point of λA. This shows that λA is compact.

Now let $A, B \subseteq \mathbb{R}^n$ be compact sets, and let $(z_m)_{m \in \mathbb{N}}$ be a sequence in $A + B$. For each $m \in \mathbb{N}$, write $z_m = x_m + y_m$ with $x_m \in A$ and $y_m \in B$. Once A and B are compact, we can extract a subset of indices

$\{m_k\} \subseteq \mathbb{N}$ such that both subsequences (x_{m_k}) and (y_{m_k}) converge (to $x \in A$ and $y \in B$, respectively, say). Then $x_{m_k} + y_{m_k}$ converges to $x + y \in A + B$, and this proves that $A + B$ is compact.

□

A *convex body* is a set $K \subseteq \mathbb{R}^n$ which is convex, compact, and has non-empty interior. In view of Theorem 2.1.1, saying that a convex set has non-empty interior is the same as saying that its dimension is maximal. Throughout the text, the set of convex bodies in $\mathbb{R}^n$ will be denoted by $\mathcal{K}^n$, and the set of convex bodies which contain the origin as an interior point will be denoted by $\mathcal{K}^n_0$.

We say that a set A is *centrally symmetric* if there exists a point $x_0 \in A$ such that $A - x_0 = -(A - x_0)$. In this case we also say that A is *symmetric with respect to* x_0, or *centered at* x_0. If $A = -A$, this is the subcase that A is symmetric with respect to the origin; then we also say that A is *centered at the origin*. Observe that, for any set $A \subseteq \mathbb{R}^n$, we have that $A - A := A + (-A) = \{a_1 - a_2 : a_1, a_2 \in A\}$ is centered at the origin (this set is often called the *difference body* of A; see Figure 2.1.9). For distinct sets $A, B \subseteq \mathbb{R}^n$, we still write $A - B := A + (-B)$.

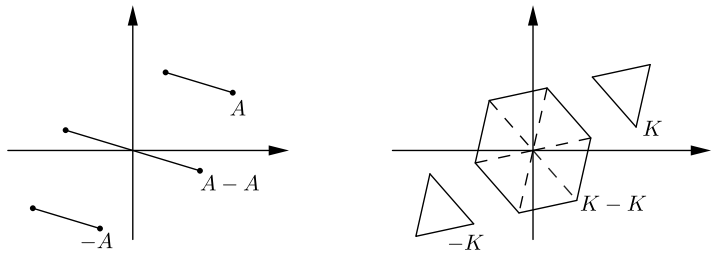


Fig. 2.1.9 Illustrations of difference bodies.

To finish this section, we present two classical theorems which give a glimpse of the combinatorial aspects of convex geometry, namely *Radon's theorem* and *Helly's theorem*.

Theorem 2.1.4 (Radon's theorem)**.** *Let $P \subseteq \mathbb{R}^n$ be a finite set of affinely dependent points. Then there exist sets $A, B \subseteq \mathbb{R}^n$ such that $A \cap B = \emptyset$, $\operatorname{conv}(A) \cap \operatorname{conv}(B) \neq \emptyset$ and $P = A \cup B$.*

Proof. Let $P = \{x_1, \ldots, x_m\}$. Once these points are affinely dependent, one can choose numbers $\lambda_1, \ldots, \lambda_m$, not all zero, such that

$$\sum_{j=1}^{m} \lambda_j x_j = 0_{\mathbb{R}^n} \quad \text{and} \quad \sum_{j=1}^{m} \lambda_j = 0.$$

Renumbering the terms if necessary, we may assume that $\lambda_j > 0$ for $j = 1, \ldots, k$ and $\lambda_j \leq 0$ for $j = k+1, \ldots, m$. Notice that since the sum of the λ_j's is zero and at least one of them is non-zero, we have that at least one of them is positive and at least one of them is negative. Define

$$A := \{x_1, \ldots, x_k\} \text{ and}$$
$$B := \{x_{k+1}, \ldots, x_m\}.$$

We have immediately that $A \cap B = \emptyset$ and $P = A \cup B$. Hence we only have to prove that $\operatorname{conv}(A) \cap \operatorname{conv}(B) \neq \emptyset$. For that sake, let

$$\lambda := \lambda_1 + \ldots + \lambda_k = -(\lambda_{k+1} + \ldots + \lambda_m) > 0,$$

and define

$$x := \sum_{j=1}^{k} \frac{\lambda_j}{\lambda} x_j.$$

Then we clearly have that $x \in \mathrm{conv}(A)$. On the other hand, since

$$\lambda_1 x_1 + \ldots + \lambda_k x_k = -\lambda_{k+1} x_{k+1} - \ldots - \lambda_m x_m,$$

we get

$$x = \sum_{j=k+1}^{m} \frac{-\lambda_j}{\lambda} x_j,$$

and this shows that $x \in \mathrm{conv}(B)$.

□

Theorem 2.1.5 (Helly's theorem)**.** *Let $m \geq n+1$ be a natural number and $\mathcal{A} = \{A_1, \ldots, A_m\}$ be a family of non-empty convex sets of $\mathbb{R}^n$. If any $n+1$ of these sets have a common point, then all of them have a common point.*

Proof. The result is obviously true for $m = n+1$. We proceed by induction on m, assuming that the assertion is true for some $m \geq n+1$. We will prove that it also holds for $m+1$. Let $\mathcal{A} = \{A_1, \ldots, A_{m+1}\}$ be a family of $m+1$ non-empty convex sets of $\mathbb{R}^n$, and assume that any $n+1$ of these sets have a common point. In particular, for each $j = 1, \ldots, m+1$ the family $\mathcal{A}_j = \{A_1, \ldots, A_{j-1}, A_{j+1}, \ldots, A_{m+1}\}$ is a family of m non-empty convex sets of $\mathbb{R}^n$ having the property that any $n+1$ of them have a common point. By the induction hypothesis, for each $j \in \mathbb{R}^n$ there exists a point

$$x_j \in A_1 \cap \ldots \cap A_{j-1} \cap A_{j+1} \cap \ldots \cap A_{m+1}.$$

The points $\{x_1, \ldots, x_{m+1}\}$ are affinely dependent because $m+1 \geq n+2$. Therefore, Radon's theorem (Theorem 2.1.4 above) implies that, renumbering the points if necessary, we may let

$$x \in \mathrm{conv}\{x_1, \ldots, x_k\} \cap \mathrm{conv}\{x_{k+1}, \ldots, x_{m+1}\}$$

for some $1 \leq k \leq m$. For each $j = 1, \ldots, m+1$, the point x_j belongs to all sets of $\mathcal{A}$ except for A_j. Therefore, $x_1, \ldots, x_k \in A_{k+1} \cap \ldots \cap A_{m+1}$. Once each of these sets is convex, we have

$$x \in \mathrm{conv}\{x_1, \ldots, x_k\} \subseteq A_{k+1} \cap \ldots \cap A_{m+1}.$$

Similarly, $x_{k+1}, \ldots, x_{m+1} \in A_1 \cap \ldots \cap A_k$. Thus,

$$x \in \mathrm{conv}\{x_{k+1}, \ldots, x_{m+1}\} \subseteq A_1 \cap \ldots \cap A_k.$$

It follows that x is a point of all sets of $\mathcal{A}$. This concludes the proof.

□

Remark 2.1.7. Helly's theorem also holds for infinite families of non-empty compact convex sets. That is, if $\mathcal{A} \subseteq CC^n$ is such that any $n+1$ sets of $\mathcal{A}$ have non-empty intersection, then all sets of $\mathcal{A}$ have non-empty intersection. Indeed, in this case the "finite Helly theorem" gives that any finite subfamily of $\mathcal{A}$ is non-empty, and hence standard topology arguments yield that the intersection of all sets of $\mathcal{A}$ is non-empty.

Other results with a "combinatorial flavor" are given in Exercises 2.3 and 2.4. For more on this direction, we refer the reader to the book [431].

2.2 Support and separation

We start this section by recalling some concepts given in Section 1.3. We say that a hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha) \subseteq \mathbb{R}^n$ *supports* a given set $A \subseteq \mathbb{R}^n$ if A is contained in one of the closed half-spaces determined by $\mathbf{h}$, and $A \cap \mathbf{h} \neq \emptyset$. In this case, if $z \in A \cap \mathbf{h}$, then we say that $\mathbf{h}$ *supports* A *at* z, and it is readily seen that z is a boundary point of A. Also, if $A \subseteq \mathbf{h}^-(u, \alpha)$, then we say that u is an *outer* (or *exterior*) normal of A at z. On the other hand, if $A \subseteq \mathbf{h}^+(u, \alpha)$, then u is an *inner normal* of A at z. Observe very carefully that any positive multiple of an outer normal (resp. inner normal) is also an outer normal (resp. inner normal).

We will also explore the notion of *separation* of sets by hyperplanes. For sets $A, B \subseteq \mathbb{R}^n$, we say that

(i) A and B are *separated* by the hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha) \subseteq \mathbb{R}^n$ if $A \subseteq \mathbf{h}^+$ and $B \subseteq \mathbf{h}^-$, or vice versa,

(ii) this separation is *proper* if A and B are not simultaneously contained in $\mathbf{h}(u, \alpha)$,

(iii) A and B are *strictly separated* by $\mathbf{h}(u, \alpha)$ if $A \subseteq \operatorname{int}(\mathbf{h}^-)$ and $B \subseteq \operatorname{int}(\mathbf{h}^+)$, or the contrary, and

(iv) A and B are *strongly separated* by $\mathbf{h}(u, \alpha)$ if there exists a number $\varepsilon > 0$ such that the hyperplanes $\mathbf{h}(u, \alpha - \varepsilon)$ and $\mathbf{h}(u, \alpha + \varepsilon)$ separate A and B.

Of course, we have that **(iv)**⇒**(iii)**⇒**(ii)**⇒**(i)**. We illustrate the concept of strong separation in Figure 2.2.1. In Figure 2.2.2, we exhibit disjoint sets which cannot be separated (notice that one of them is not convex).

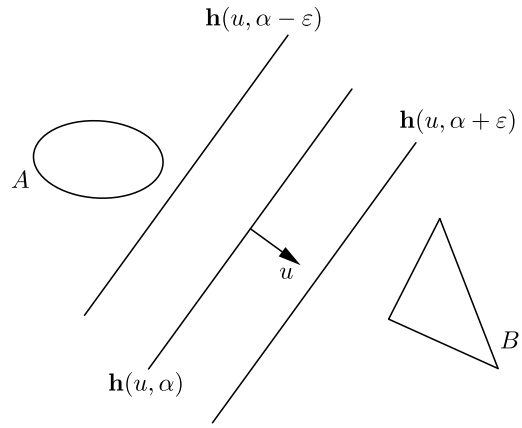


Fig. 2.2.1 The sets A and B are strongly separated by $\mathbf{h}(u, \alpha)$.

Lemma 2.2.1. *Assume that* $\mathbf{h} := \mathbf{h}(u, \alpha)$ *supports* A *at* z. *Then* u *is an outer normal if and only if* $\mathbf{h}$ *(properly) separates* $u + z$ *and* A.

Proof. Since $z \in \mathbf{h}$, we have that $\langle u, z \rangle = \alpha$. This leads to

$$\langle u, u + z \rangle = |u|^2 + \alpha > \alpha,$$

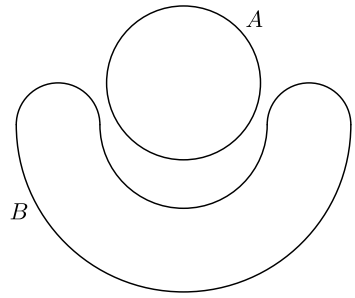


Fig. 2.2.2 The sets A and B cannot be separated by a hyperplane.

from where it follows that $u + z \in \operatorname{int}(\mathbf{h}^+)$. Hence $A \subseteq \mathbf{h}^-$ if and only if $\mathbf{h}$ separates A and $u + z$. In this case, the separation is proper (because $u + z \notin \mathbf{h}$).

□

The notions of support and separation have remarkably good properties when they are restricted to convex sets. To explore these features, we first introduce the concept of *metric projection* of a point to a set. We define the *distance* between two sets $A, B \subseteq \mathbb{R}^n$ as

$$\operatorname{dist}(A, B) := \inf\{|a - b| : a \in A,\ b \in B\},$$

where we recall that $|\cdot|$ denotes the standard Euclidean norm in $\mathbb{R}^n$. The *distance from a point to a set* is the distance between two sets such that one of them is a singleton. We write

$$\operatorname{dist}(x, A) := \operatorname{dist}(\{x\}, A),$$

for any point $x \in \mathbb{R}^n$ and every set $A \subseteq \mathbb{R}^n$. It is well-known from basic topology that if A is closed, then the distance is indeed attained, that is, there exists a point $z \in A$ such that $|z - x| = \operatorname{dist}(x, A)$. A point $z \in A$ with this property is called a *metric projection of x onto A*. Observe that we can also characterize a metric projection of x onto A as a point $z \in A$ with the property that

$$|x - z| \leq |x - y|$$

for every $y \in A$. It is clear that if $x \in A$, then the unique metric projection of x onto A is x itself. As it was mentioned before, for a point $x \in \mathbb{R}^n \setminus A$ we can guarantee the existence of a metric projection provided A is closed. Next we prove that convexity yields uniqueness.

Lemma 2.2.2. *Let $K \subseteq \mathbb{R}^n$ be a non-empty closed, and convex set. Any point $x \in \mathbb{R}^n$ has a unique metric projection onto K. Moreover, if $x \in \mathbb{R}^n \setminus K$, then the metric projection of x onto K is a point of ∂K.*

Remark 2.2.1. From now on, we will always denote the metric projection of a point $x \in \mathbb{R}^n$ onto a closed convex set $K \subseteq \mathbb{R}^n$ as $p_K(x)$.

Proof. Let $z, w \in K$ be metric projections of $x \in \mathbb{R}^n$ onto K, and let $\rho := |x - z| = |x - w| = \operatorname{dist}(x, K)$. From the convexity of K we have that $y := \frac{1}{2}(z + w)$ is a point of K,

$$\left|\frac{1}{2}(z + w) - x\right| = \left|\frac{1}{2}(z - x) + \frac{1}{2}(w - x)\right| \leq \frac{1}{2}|z - x| + \frac{1}{2}|w - x| \leq \rho,$$

and that the first inequality is strict if $z - x$ and $w - x$ are linearly independent. If $z \neq w$, then $z - x$ and $w - x$ are linearly independent. In fact, otherwise, since $|z - x| = |w - x|$, we would have that x

is the midpoint of the segment connecting z and w (from convexity this gives that x is a point of K, a contradiction). Hence $\left|\frac{1}{2}(z+w)-x\right| < \rho$. This contradicts the fact that $\operatorname{dist}(x,K)=\rho$.

Now we prove the other claim. From basic topology we have that $K = \operatorname{int}K \cup \partial K$ (because K is closed; see Remark 1.1.1). Thus, for $\operatorname{int}K = \emptyset$ there is nothing to prove. For the other case, assume that $x \notin K$, and let $z \in \operatorname{int}K$ be any interior point of K. Let

$$\lambda_0 := \sup\{\lambda \in [0,1] : (1-\lambda)z + \lambda x \in K\}.$$

Since K contains a ball centered at z we have that $0 < \lambda_0 < 1$. Also, it is easy to see that $w := (1-\lambda_0)z+\lambda_0 x \in \partial K$, and because w is a point of the relative interior of $\operatorname{seg}[x,z]$ we get $|x-w| < |x-z|$. Consequently, for each interior point of K we can find a boundary point of K with smaller distance to x, and this concludes the proof.

□

Remark 2.2.2. The converse is also true, that is, if a closed set $A \subseteq \mathbb{R}^n$ is such that each point of $\mathbb{R}^n$ has a unique metric projection onto A, then A is convex. For a proof, we refer the reader to [2167].

It follows that a closed convex set $K \subseteq \mathbb{R}^n$ defines a map $p_K \colon \mathbb{R}^n \to K$ which associates each point $x \in \mathbb{R}^n$ with its metric projection onto K.

Proposition 2.2.1. *Let $K \subseteq \mathbb{R}^n$ be a non-empty, closed, and convex set. The distance and metric projection functions of K are* contracting, *that is,*

$$|\operatorname{dist}(x,K) - \operatorname{dist}(y,K)| \le |x-y| \text{ and}$$
$$|p_K(x) - p_K(y)| \le |x-y|,$$

for any $x, y \in \mathbb{R}^n$. In particular, these functions are continuous.

Proof. We start with the distance function, which is easier. We may assume that, without loss of generality, $\operatorname{dist}(y,K) \ge \operatorname{dist}(x,K)$, and we write $y = x+z$ for some $z \in \mathbb{R}^n$. Then

$$|\operatorname{dist}(x,K) - \operatorname{dist}(y,K)| = \operatorname{dist}(y,K) - \operatorname{dist}(x,K) = |y - p_K(y)| - |x - p_K(x)| \le$$
$$\le |y - p_K(x)| - |p_K(x) - x| = |x + z - p_K(x)| - |p_K(x) - x| \le |z| = |x-y|.$$

Notice that we used the inequality

$$|y - p_K(y)| = \operatorname{dist}(y,K) = \inf\{|y-w| : w \in K\} \le |y - p_K(x)|,$$

since $p_K(x) \in K$. We also used the triangle inequality.

Now we go to the metric projection. Of course, we may assume that $p_K(y) \neq p_K(x)$, since otherwise the inequality is obvious. Let $v := p_K(x) - p_K(y)$, and define $f \colon [0,1] \to \mathbb{R}$ by

$$f(t) := |x - p_K(x) + tv|^2.$$

Observe that $p_K(x) - tv = (1-t)p_K(x) + tp_K(y)$, and hence $p_K(x) - tv \in K$ for every $t \in [0,1]$. It follows that

$$f(0) = |x - p_K(x)|^2 = \operatorname{dist}(x,K)^2 \le |x - p_K(x) + tv|^2 = f(t)$$

for any $t \in [0,1]$. In other words, $t = 0$ is a local minimum of f, and since f is clearly differentiable, we get that $f'(0) \ge 0$. This gives

$$0 \le f'(0) = \frac{d}{dt}\langle x - p_K(x) + tv, x - p_K(x) + tv\rangle\Big|_{t=0} = 2\langle x - p_K(x), v\rangle.$$

Hence we have that $\langle x - p_K(x), v\rangle \ge 0$. Similarly, we obtain that $\langle y - p_K(y), v\rangle \le 0$. Let

$$\mathbf{h}_1 := \{z \in \mathbb{R}^n : \langle z, v\rangle = \langle p_K(x), v\rangle\}, \text{ and}$$
$$\mathbf{h}_2 := \{z \in \mathbb{R}^n : \langle z, v\rangle = \langle p_K(y), v\rangle\},$$

that is, the hyperplanes orthogonal to v passing through $p_K(x)$ and $p_K(y)$, respectively. The inequality $\langle x - p_K(x), v\rangle \ge 0$ gives that $x \in \mathbf{h}_1^+$, and the inequality $\langle y - p_K(y), v\rangle \le 0$ yields $y \in \mathbf{h}_2^-$; see Figure 2.2.3. Consequently,

$$|x - y| \ge \operatorname{dist}(\mathbf{h}_1, \mathbf{h}_2) = |p_K(x) - p_K(y)|,$$

where the last equality comes immediately from the fact that the distance between $\mathbf{h}_1$ and $\mathbf{h}_2$ is attained by any perpendicular segment connecting these two hyperplanes.

□

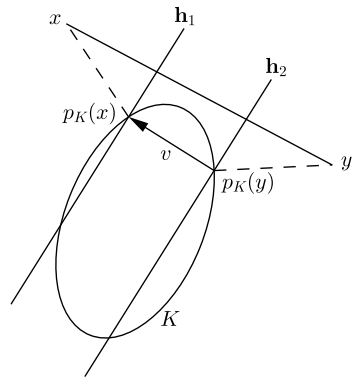


Fig. 2.2.3 Proof of Proposition 2.2.1.

Lemma 2.2.3. *The distance function* $\operatorname{dist}(\cdot, K)\colon \mathbb{R}^n \to \mathbb{R}$ *for a convex set* $K \subseteq \mathbb{R}^n$ *is convex.*

Proof. We assume that K is closed (so we can talk about metric projections). The general case follows immediately replacing K by its closure. For any $x, y \in \mathbb{R}^n$ and every $\lambda \in (0, 1)$ we have that $(1 - \lambda)p_K(x) + \lambda p_K(y) \in K$, and hence

$$\operatorname{dist}((1 - \lambda)x + \lambda y, K) \le |(1 - \lambda)x + \lambda y - [(1 - \lambda)p_K(x) + \lambda p_K(y)]| \le$$
$$\le (1 - \lambda)|x - p_K(x)| + \lambda|y - p_K(y)| = (1 - \lambda)\operatorname{dist}(x, K) + \lambda\operatorname{dist}(y, K).$$

□

Let $K \subseteq \mathbb{R}^n$ be a closed and convex set, and let $x \in \mathbb{R}^n \setminus K$. We already know that there is a unique metric projection $p_K(x) \in \partial K$ of x onto K, and of course the distance between x and K is positive. The next lemma states that the segment joining x to $p_K(x)$ "hits K orthogonally".

Lemma 2.2.4. *Let* $K \subseteq \mathbb{R}^n$ *be a non-empty closed convex set. For every* $x \in \mathbb{R}^n \setminus K$, *the vector*

$$u_K(x) := \frac{x - p_K(x)}{\operatorname{dist}(x, K)} \in \mathbb{S}^{n-1} \tag{2.2.1}$$

is an outer normal of K at $p_K(x)$. See Figure 2.2.4

Remark 2.2.3. It is usual to work with the normalization of $x - p_K(x)$ as above, but of course the vector $x - p_K(x)$ is also an outer normal of K at $p_K(x)$. Indeed, recall that any positive multiple of an outer normal is also an outer normal.

Proof. For simplicity, we write $u_K(x) = u$ and $\langle u_K(x), p_K(x)\rangle = \alpha$. We have to prove that $K \subseteq \mathbf{h}^-(u, \alpha)$. Assume that there exists a point $y \in K$ such that $y \notin \mathbf{h}^-$, that is, $y \in \operatorname{int}(\mathbf{h}^+)$. Then we have $\langle y, u\rangle \geq \alpha = \langle u, p_K(x)\rangle$. Hence $\langle y - p_K(x), u\rangle \geq 0$, and since $\operatorname{dist}(x, K)u = x - p_K(x)$, we get that

$$\langle y - p_K(x), x - p_K(x)\rangle \geq 0.$$

On the other hand, the convexity of K guarantees that $\operatorname{seg}[p_K(x), y] \subseteq K$, and since $p_K(x)$ is the metric projection of x onto K we get that $t = 0$ is a local minimum of the function $f\colon [0, 1] \to \mathbb{R}$ given as

$$f(t) := |x - [(1 - t)p_K(x) + ty]|^2.$$

Then we have $f'(0) \geq 0$, and this gives

$$0 \leq \langle x - p_K(x), -(y - p_K(x))\rangle = -\langle x - p_K(x), y - p_K(x)\rangle.$$

Therefore, we have that $\langle x - p_K(x), y - p_K(x)\rangle = 0$. But this clearly gives that $y \in \mathbf{h}(u, \alpha)$, contradicting the hypothesis that $y \in \operatorname{int}(\mathbf{h}^+)$. Hence $K \subseteq \mathbf{h}^-$.

□

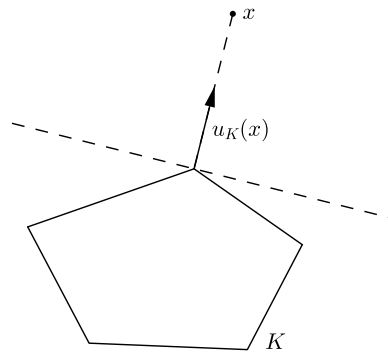


Fig. 2.2.4 The least distance is "realized" at a normal direction.

Corollary 2.2.1. *Let $K \subseteq \mathbb{R}^n$ be a closed, convex, and non-empty set. If $x \notin K$, then x and K can be strongly separated.*

Proof. The vector $u := x - p_K(x)$ is a outer normal, from which it follows that $K \subseteq \mathbf{h}^-(u, \alpha)$, where $\alpha = \langle u, p_K(x)\rangle$. From Lemma 2.2.1, we have that $x = p_K(x) + (x - p_K(x)) \in \mathbf{h}^+(u, \alpha)$. Let $2\delta = \operatorname{dist}(x, K)^2$. We claim that $\mathbf{h}(u, \alpha + \delta)$ strongly separates K and x. First observe that

$$\langle u, x\rangle = \langle u, x - p_K(x)\rangle + \langle u, p_K(x)\rangle = |x - p_K(x)|^2 + \alpha = \alpha + 2\delta.$$

Taking any number $0 < \varepsilon < \delta$, we have that $K \subseteq \mathbf{h}^-(u, \alpha) \subseteq \mathbf{h}^-(u, \alpha + \delta - \varepsilon)$ and $\langle u, x\rangle = \alpha + 2\delta > \alpha + \delta - \varepsilon$, implying $x \in \mathbf{h}^+(u, \alpha + \delta - \varepsilon)$. This shows that $\mathbf{h}(u, \alpha + \delta - \varepsilon)$ separates x and K. Similarly, we prove that $\mathbf{h}(u, \alpha + \delta + \varepsilon)$ separates x and K. It follows that $\mathbf{h}(u, \alpha + \delta)$ strongly separates x and K. □

Lemma 2.2.5. *Let K be closed, convex, and non-empty, and assume that $x \notin K$. If*

$$y \in p_K(x) + \mathbb{R}^+ u_K(x) := \{p_K(x) + t u_K(x) : t \geq 0\} = \{p_K(x) + t(x - p_K(x)) : t \geq 0\},$$

then $p_K(y) = p_K(x)$.

Proof. As in the proof of Proposition 2.2.1, let $v := p_K(x) - p_K(y)$ passing through $p_K(x)$ and $p_K(y)$. Repeating the calculations of the mentioned proof, we get

$$\begin{aligned} \langle x - p_K(x), v\rangle &\geq 0 \text{ and} \\ \langle y - p_K(y), v\rangle &\leq 0. \end{aligned}$$

Since $y \in p_K(x) + \mathbb{R}^+ u_K(x)$, it follows that $y - p_K(x)$ is a positive multiple of $x - p_K(x)$. Hence $\langle y - p_K(x), v\rangle \geq 0$. Summarizing, we have the inequalities

$$\begin{aligned} \langle y - p_K(y), p_K(x) - p_K(y)\rangle &\leq 0 \text{ and} \\ \langle y - p_K(x), p_K(x) - p_K(y)\rangle &\geq 0. \end{aligned}$$

The second one can be written as $\langle p_K(x) - y, p_K(x) - p_K(y)\rangle \leq 0$. This, together with the first inequality above, leads to

$$|p_K(x) - p_K(y)|^2 = \langle p_K(x) - p_K(y), p_K(x) - p_K(y)\rangle \leq 0,$$

and this gives $p_K(x) = p_K(y)$.

□

We go now to our main result of this section: the fact that a non-empty convex closed set K has a supporting hyperplane at each boundary point. Our approach is via metric projections, since from Lemma 2.2.4 we get that it suffices to show that any boundary point of K is the metric projection of some point $x \in \mathbb{R}^n \setminus K$ onto K.

Proposition 2.2.2. *Let $K \subseteq \mathbb{R}^n$ be a non-empty, closed, and convex set which is contained in the interior of some closed ball B. Then*

$$\partial K = \{p_K(y) : y \in \partial B\},$$

that is, every point of the boundary of K is the metric projection of some point of the boundary of B onto K.

Proof. We prove the inclusion $\partial K \subseteq \{p_K(y) : y \in \partial B\}$, since we already have the reverse inclusion. Let $x \in \partial K$ be arbitrarily taken. For each $j \in \mathbb{N}$, choose a point $x_j \in B \setminus K$ such that $|x - x_j| < 1/j$. Since the metric projection is contracting and $x = p_K(x)$, we have

$$|x - p_K(x_j)| = |p_K(x) - p_K(x_j)| \leq |x - x_j| < \frac{1}{j}. \tag{2.2.2}$$

Now, for each $j \in \mathbb{N}$, let y_j be the intersection of ∂B with the ray $p_K(x_j) + \mathbb{R}^+ u_K(x_j)$. Then $(y_j)_{j\in\mathbb{N}}$ is a sequence in the compact set ∂B, and hence we may assume that y_j converges to some point $y \in \partial B$

(passing to a subsequence, if necessary). From Lemma 2.2.5 we get that $p_K(y_j) = p_K(x_j)$ for each $j \in \mathbb{N}$, and since p_K is continuous, we get

$$p_K(y) = \lim_{j\to\infty} p_K(y_j) = \lim_{j\to\infty} p_K(x_j) = x,$$

where the equality comes from (2.2.2). This concludes the proof.

□

Theorem 2.2.1. *A non-empty, closed, and convex set $K \subseteq \mathbb{R}^n$ has a supporting hyperplane at each boundary point.*

Proof. Assume first that K is bounded, and let B be a ball such that $K \subseteq \text{int}B$. For an arbitrarily given $x \in \partial K$, Proposition 2.2.2 guarantees that there exists a point $y \in \partial B$ such that $x = p_K(y)$. Now, Lemma 2.2.1 gives that the vector

$$u_K(y) = \frac{y - p_K(y)}{\text{dist}(y, K)} = \frac{y - x}{\text{dist}(y, K)}$$

is an outer normal of K at x. This means that the hyperplane orthogonal to $u_K(y)$ supports K at x.

Now we go to the other case. If K is not bounded and $x \in \partial K$, then the intersection $L := K \cap \text{cl}(B(x, 1))$ is a closed, convex, bounded, and non-empty set. Hence L has a supporting hyperplane $\mathbf{h}$ at x (observe that x is clearly a point of the boundary of L: every neighborhood of x contains points which are not in K, and hence also not in L). We may assume that $L \subseteq \mathbf{h}^-$. We will prove that indeed we have $K \subseteq \mathbf{h}^-$. In fact, assume that there exists a point $y \in K \setminus \mathbf{h}^-$. If $|y - x| \leq 1$, then $y \in \text{cl}(B(x, 1))$, and hence $y \in L$, from where we get $y \in \mathbf{h}^-$, a contradiction. Thus, we must have $|y - x| > 1$. Let $z := \frac{y-x}{|y-x|}$. We have that

$$x + z = \left(1 - \frac{1}{|y - x|}\right) x + \frac{1}{|y - x|} y,$$

implying $x + z \in \text{seg}(x, y)$. Since K is compact and $x, y \in K$, we have that $z \in K$. Moreover, noticing that $|(x + z) - x| = |z| = 1$, we get that $x + z \in \text{cl}(B(x, 1))$, and hence $x + z \in L$. However, we claim that the open segment $\text{seg}(x, y)$ is entirely contained in $\text{int}(\mathbf{h}^+)$. In fact, setting $\mathbf{h} = \mathbf{h}(u, \alpha)$, and letting $\lambda \in (0, 1)$, we get

$$\begin{aligned}\langle (1 - \lambda)x + \lambda y, u\rangle = (1 - \lambda)\langle x, u\rangle + \lambda\langle y, u\rangle = (1 - \lambda)\alpha + \lambda\langle y, u\rangle > \\ > (1 - \lambda)\alpha + \lambda\alpha = \alpha,\end{aligned}$$

where the last inequality is strict because $y \notin \mathbf{h}^-$, and this means that $y \in \text{int}(\mathbf{h}^+)$. Since $\text{seg}(x, y) \subseteq \text{int}(\mathbf{h}^+)$, we have that $x + z \in \text{int}(\mathbf{h}^+)$. This contradicts the fact that $x + z \in L$, because $L \subseteq \mathbf{h}^-$.

□

Proposition 2.2.3. *If $C \subseteq \mathbb{R}^n$ is compact and non-empty, then any non-zero vector $u \in \mathbb{R}^n$ is an outer normal of C.*

Proof. Let $u \in \mathbb{R}^n$ be a non-zero vector. Since C is compact, there exists a point $x \in C$ such that

$$\langle x, u\rangle = \max\{\langle z, u\rangle : z \in C\}.$$

Set $\mathbf{h} := \mathbf{h}(u, \langle x, u\rangle)$. It is clear that $K \subseteq \mathbf{h}^-$ and that $x \in C \cap \mathbf{h}$. Hence u is an outer normal of C at x.

□

Remark 2.2.4. The function $h_C(u) = \max\{\langle z, u\rangle : z \in C\}$, defined for $u \in \mathbb{R}^n$, is the *support function* of C. This function will be studied in the next section. Also, observe that we do not need the convexity hypothesis for this last result. What convexity gives is that we have a supporting hyperplane at each boundary point.

Corollary 2.2.2. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For any $u \in \mathbb{S}^{n-1}$, there exist precisely two hyperplanes orthogonal to u supporting K (see Figure 2.2.5).*

Proof. By the previous proposition, the vectors u and $-u$ are outer normals of K (at points $z_1, z_2 \in \partial K$ respectively, say). Let $\mathbf{h}_1 := \mathbf{h}(u, \alpha)$ and $\mathbf{h}_2 := \mathbf{h}(-u, \beta)$ be the hyperplanes orthogonal to u (hence being parallel) which support K at z_1 and z_2, respectively, with $K \subseteq \mathbf{h}_1^- \cap \mathbf{h}_2^-$ (geometrically, this means that K is contained in the "slab" determined by $\mathbf{h}_1$ and $\mathbf{h}_2$). It is clear that $\mathbf{h}_1 \neq \mathbf{h}_2$, since otherwise we would have that $K \subseteq \mathbf{h}_1$, and this contradicts the maximality of the dimension of K (recall that K is a convex body). This proves that K is supported by at least two distinct hyperplanes orthogonal to u. Now we prove that there is no hyperplane orthogonal to u different to $\mathbf{h}_1$ and $\mathbf{h}_2$ and supporting K. Let $\mathbf{h} := \mathbf{h}(u, \gamma)$ be a hyperplane orthogonal to u which supports K, and assume that $K \subseteq \mathbf{h}^-$, that is, $\langle z, u\rangle \leq \gamma$ for any $z \in K$ (the case $K \subseteq \mathbf{h}^+$ is completely analogous). We cannot have $\gamma > \alpha$, since in this case the point $x \in \partial K$ where $\mathbf{h}(u, \gamma)$ supports K would verify $\langle x, u\rangle = \gamma > \alpha$. This would contradict the inclusion $K \subseteq \mathbf{h}_1^-$. Similarly, we get that we also cannot have $\gamma < -\beta$. It follows that $-\beta \leq \gamma \leq \alpha$. Assume that we have the strict inequalities $-\beta < \gamma < \alpha$. Then $z_1 \in \operatorname{int}(\mathbf{h}^+)$ and $z_2 \in \operatorname{int}(\mathbf{h}^-)$, and this contradicts the inclusion $K \subseteq \mathbf{h}^-$. Hence $\gamma = \alpha$ or $\gamma = -\beta$, that is, $\mathbf{h} = \mathbf{h}_1$ or $\mathbf{h} = \mathbf{h}_2$. Notice carefully that $\mathbf{h}_2 = \mathbf{h}(-u, \beta) = \mathbf{h}(u, -\beta)$.

□

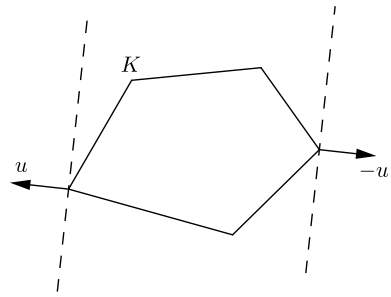


Fig. 2.2.5 Parallel supporting hyperplanes orthogonal to u.

If $A \subseteq \mathbb{R}^n$ is supported at $x \in A$ by a hyperplane $\mathbf{h}$, then the half-space determined by $\mathbf{h}$ which contains A is called a *supporting half-space* of A.

Proposition 2.2.4. *A closed, convex, and non-empty subset of $\mathbb{R}^n$ is the intersection of its supporting half-spaces.*

Proof. To fix the notation, we have to prove that if K is a closed, convex, and non-empty subset of $\mathbb{R}^n$, then

$$K = \bigcap \{\mathbf{H} : \mathbf{H} \text{ is a supporting half}-\text{space of } K\},$$

where the reader may notice that we are using capital boldface letters for half-spaces, in contrast to small boldface letters for hyperplanes. Since it is obvious that K is contained in the intersection of its supporting half-spaces, we only have to prove the reverse inclusion. Let $x \in \mathbb{R}^n$ be a point which belongs to every supporting half-space of K. If $x \notin K$, then writing $x = p_K(x) + x - p_K(x)$ and

recalling that $x - p_K(x)$ is an outer normal of K at x, we have from Lemma 2.2.1 that the hyperplane $\mathbf{h}$ normal to $x - p_K(x)$ and passing through $p_K(x)$ properly separates x and K. Hence x does not belong to the half-space determined by $\mathbf{h}$ which contains K, and this is a contradiction.

□

Corollary 2.2.3. *Let $K \subsetneq \mathbb{R}^n$ be a closed, convex, and non-empty set. Then*

$$K = \bigcap \left\{ \mathbf{H} : \mathbf{H} \text{ is a half}-\text{space such that } K \subseteq \mathbf{H} \right\}.$$

Proof. Let $\mathbf{H}_1 \subseteq \mathbb{R}^n$ be a half-space such that $K \subseteq \mathbf{H}_1$, and assume that $\mathbf{H}_1 = \mathbf{h}_1^-$, where $\mathbf{h}_1 := \mathbf{h}(u, \alpha_1)$. Once K is closed, we may let

$$\alpha_2 := \min \left\{ \beta \in \mathbb{R} : K \subseteq \mathbf{h}^-(u, \beta) \right\}.$$

The hyperplane $\mathbf{h}_2 := \mathbf{h}(u, \alpha_2)$ supports K at some boundary point $x \in \partial K$ with $K \subseteq \mathbf{h}_2^-$. Since it is clear that $\alpha_2 \leq \alpha_1$, we have that $\mathbf{H}_2 := \mathbf{h}(u, \alpha_2)^- \subseteq \mathbf{H}_1$.

Summarizing, we proved that for each half-space $\mathbf{H}_1$ containing K, there exists a supporting half-space $\mathbf{H}_2$ of K such that $\mathbf{H}_2 \subseteq \mathbf{H}_1$, that is, $\mathbf{H}_1 \cap \mathbf{H}_2 = \mathbf{H}_2$. Consequently,

$$\begin{aligned} &\bigcap \left\{ \mathbf{H} : \mathbf{H} \text{ is a half}-\text{space such that } K \subseteq \mathbf{H} \right\} = \\ &= \bigcap \left\{ \mathbf{H} : \mathbf{H} \text{ is a supporting half}-\text{space of } K \right\}, \end{aligned}$$

and this concludes the proof.

□

We go now to our main theorem about separation, which states that two convex sets with empty intersection can be strongly separated provided one of them is closed, and the other one is compact. We will obtain that as an easy consequence of a previous result taken together with the following lemma.

Lemma 2.2.6. *Let $A, B \subseteq \mathbb{R}^n$ be non-empty sets. Then A and B can be strongly separated if and only if $0_{\mathbb{R}^n}$ and $A - B$ can be strongly separated.*

Proof. Assume first that A and B can be strongly separated by a hyperplane $\mathbf{h}(u, \alpha)$. Without loss of generality, we may assume that $A \subseteq \mathbf{h}^-(u, \alpha - \varepsilon)$ and $B \subseteq \mathbf{h}^+(u, \alpha + \varepsilon)$ for some $\varepsilon > 0$. Then, for any point $a - b \in A - B$, we have

$$\langle a - b, u \rangle = \langle a, u \rangle - \langle b, u \rangle \leq \alpha - \varepsilon - (\alpha + \varepsilon) = -2\varepsilon.$$

Thus, $A - B \subseteq \mathbf{h}^-(u, -2\varepsilon)$. Since we obviously have that $0_{\mathbb{R}^n} \in \mathbf{h}(u, 0)$, we get that the hyperplane $\mathbf{h}(u, -\varepsilon)$ strongly separates $A - B$ and $0_{\mathbb{R}^n}$.

Now assume that $A - B$ and $0_{\mathbb{R}^n}$ can be strongly separated by a hyperplane $\mathbf{h}(u, \alpha)$. Without loss of generality, we may assume that $0_{\mathbb{R}^n} \in \mathbf{h}^+(u, \alpha + \varepsilon)$ and $A - B \subseteq \mathbf{h}^-(u, \alpha - \varepsilon)$ for some $\varepsilon > 0$. This inclusion means that

$$\langle a - b, u \rangle \leq \alpha - \varepsilon$$

for every $a \in A$ and any $b \in B$. On the other hand, the fact $0_{\mathbb{R}^n} \in \mathbf{h}^+(u, \alpha + \varepsilon)$ gives that $0 = \langle 0_{\mathbb{R}^n}, u \rangle \geq \alpha + \varepsilon$. Hence,

$$\langle a, u \rangle \leq \langle b, u \rangle + \alpha - \varepsilon \leq \langle b, u \rangle - 2\varepsilon,$$

for each $a \in A$ and every $b \in B$. Consequently, the numbers $c := \sup_{a\in A}\langle a, u\rangle$ and $d := \inf_{b\in B}\langle b, u\rangle$ are well-defined, and $c \leq d - 2\varepsilon$ (which gives that $\varepsilon \leq (d-c)/2$). Moreover, it is clear that $A \subseteq \mathbf{h}^-(u, c)$ and $B \subseteq \mathbf{h}^+(u, d)$. Let $\beta := (c + d)/2$. We claim that $\mathbf{h}(u, \beta)$ strongly separates A and B. Indeed, since

$$\beta - \varepsilon = \frac{c+d}{2} - \varepsilon \geq \frac{c+d}{2} + \frac{c-d}{2} = c,$$

we get that $A \subseteq \mathbf{h}^-(u, \beta - \varepsilon)$. On the other hand,

$$\beta + \varepsilon \leq \frac{c+d}{2} + \frac{d-c}{2} = d,$$

from where we get that $B \subseteq \mathbf{h}^+(u, \beta + \varepsilon)$. This concludes the proof.

□

Theorem 2.2.2. *Let $K, L \subseteq \mathbb{R}^n$ be convex and non-empty sets, and assume that K is compact and that L is closed. If $K \cap L = \emptyset$, then K and L can be strongly separated.*

Proof. It is easy to see that if K is compact and L is closed, then $K - L$ is closed. Indeed, let $(x_j - y_j)_{j\in\mathbb{N}}$ be a sequence in $K - L$ (with $x_j \in K$ and $y_j \in L$) converging to $z \in \mathbb{R}^n$. We must prove that $z \in K - L$. Since K is compact, there is a subsequence $(x_{j_k})_{k\in\mathbb{N}}$ converging to a point $x \in K$. Hence y_{j_k} converges to $y := z - x$, and since L is closed we get $y \in L$. Thus, $z = x - y \in K - L$.

Due to $K \cap L = \emptyset$ we have that $0_{\mathbb{R}^n} \notin K - L$. Since $K - L$ is closed and convex, it follows from Corollary 2.2.1 that $K - L$ and $0_{\mathbb{R}^n}$ can be strongly separated. This, together with the previous lemma, gives the desired.

□

There are a number of other theorems which give weaker separation properties by assuming weaker hypotheses on the involved convex sets. Next we investigate some of these results. The first of them states that two convex bodies which intersect only at a common boundary point have a common supporting hyperplane at this point.

Let K be a closed and convex set with non-empty interior, and let L be a closed and convex set. Assume that $K \cap L = \{x_0\}$ and that x_0 is a boundary point of K. If K is supported by a unique hyperplane $\mathbf{h}$ at x_0, with $K \subseteq \mathbf{h}^-$, then $\mathbf{h}$ also supports L at x_0, with $L \subseteq \mathbf{h}^+$.

Proposition 2.2.5. *Let K and L be convex bodies in $\mathbb{R}^n$ such that $K \cap L = \{x_0\}$, and assume that K is supported by a unique hyperplane $\mathbf{h}$ at x_0, with $K \subseteq \mathbf{h}^-$. Then $\mathbf{h}$ also supports L at x_0, with $L \subseteq \mathbf{h}^+$ (see Figure 2.2.6).*

Proof. First observe that if $\{x_0\} = K \cap L$, then it is clear that $x_0 \in \partial K \cap \partial L$. If there is a point $x \in L \cap \operatorname{int}(\mathbf{h}^-)$, then $\operatorname{seg}[x_0, x] \subseteq L$. Since $x_0 \in \mathbf{h}$, we get that $x \neq x_0$. Hence we must have $\operatorname{seg}[x_0, x] \cap K = \{x_0\}$, because any other point in this intersection would be a point in $K \cap L$ different from x_0. It follows that the line through x and x_0 intersects K only at x_0, and hence there exists a hyperplane containing this line which supports K at x_0. Since this hyperplane is not $\mathbf{h}$, this contradicts the fact that K is supported by a unique hyperplane at x_0.

□

Remark 2.2.5. Later, we will need a variant of this result. Let $K \subseteq \mathbb{R}^n$ be closed, convex with non-empty interior, and let x_0 be a boundary point of K where K is supported by a unique hyperplane $\mathbf{h}$ (with $K \subseteq \mathbf{h}^-$, say). If $L \subseteq \mathbb{R}^n$ is closed and convex, and if $K \cap L = \{x_0\}$, then $\mathbf{h}$ also supports L (with $L \subseteq \mathbf{h}^+$). The proof is essentially the same as above.

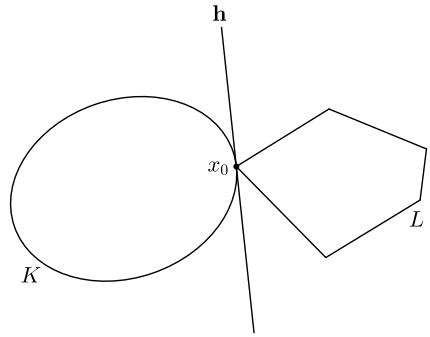


Fig. 2.2.6 The unique supporting hyperplane of K at x_0 also supports L at x_0.

In Theorem 2.2.2, we give sufficient conditions under which two convex sets can be strongly separated. Now we investigate proper separation of convex sets, which is a weaker notion. First, we state and prove a technical lemma.

Lemma 2.2.7. *Let $K \subseteq \mathbb{R}^n$ be any convex set and let $x_0 \in \mathbb{R}^n \setminus K$. Then, K and $\{x_0\}$ can be separated by a hyperplane which contains x_0.*

Proof. We divide the proof in two cases. In the first one, we assume that $x \notin \mathrm{cl}K$. In this case, $\{x_0\}$ is compact, $\mathrm{cl}K$ is closed, and both sets are convex. Thus, by Theorem 2.2.2 we have that $\{x_0\}$ and $\mathrm{cl}K$ can be strongly separated. It is immediate to check that any hyperplane which separates these two sets also separates $\{x_0\}$ and K. If $\mathbf{h} := \mathbf{h}(u, \alpha)$ is such a hyperplane, then we may assume that $\langle u, x_0\rangle \geq \alpha$ and $\langle u, z\rangle \leq \alpha$ for every $z \in K$. Thus, the hyperplane $\mathbf{h}(u, \langle u, x_0\rangle)$ contains x_0 and separates K and $\{x_0\}$ because $K \subseteq \mathbf{h}^-(u, \langle u, x_0\rangle)$.

Now we assume that $x \in \mathrm{cl}K$. Since $x \notin K$, it follows that $x \in \partial K$, and hence $\mathrm{cl}K$ is supported at x by some hyperplane $\mathbf{h}$. It is readily seen that $\mathbf{h}$ separates $\{x_0\}$ and K: indeed, $\mathrm{cl}K$ (and consequently also K) is contained in one of the closed half-spaces determined by $\mathbf{h}$, and $\{x_0\}$ is contained in $\mathbf{h}$ itself. □

Theorem 2.2.3. *Let $C, D \subseteq \mathbb{R}^n$ be non-empty convex sets. If*

$$\mathrm{relint}\, C \cap \mathrm{relint}\, D = \emptyset,$$

then C and D can be properly separated. The converse is also true.

Proof. Assume that C and D are properly separated by a hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha)$ with $C \subseteq \mathbf{h}^-$ and $D \subseteq \mathbf{h}^+$. Suppose that there exists a point $x \in \mathrm{relint}C \cap \mathrm{relint}D$. In particular, $x \in C \cap D \subseteq \mathbf{h}^- \cap \mathbf{h}^+ = \mathbf{h}$. Let z be any point of C. From the convexity of C, we have that the segment $\mathrm{seg}[x, z]$ is contained in C, and once $x \in \mathrm{relint}C$ we get that $\mathrm{seg}[x, x - \varepsilon(z - x)] \subseteq C$ for some small $\varepsilon > 0$. Thus, if $z \in \mathrm{int}(\mathbf{h}^-)$, then $\langle u, z\rangle < \alpha$, and consequently

$$\langle u, x - \varepsilon(z - x)\rangle = \alpha + \varepsilon\big(\alpha - \langle u, z\rangle\big) > \alpha.$$

This gives that $x - \varepsilon(z - x) \in \mathrm{int}(\mathbf{h}^+)$, a contradiction to the fact that $x - \varepsilon(z - x) \in C$ because $C \subseteq \mathbf{h}^-$. It follows that $C \subseteq \mathbf{h}$, and by the same argument we get that $D \subseteq \mathbf{h}$. Hence the separation is not proper.

Now we assume that $\mathrm{relint}C \cap \mathrm{relint}D = \emptyset$. Then we let $A := \mathrm{relint}C - \mathrm{relint}D$ (which is convex, because it is the Minkowski sum of convex sets), and it is readily seen that $0_{\mathbb{R}^n} \notin A$. From Lemma 2.2.7 we obtain that $\{0_{\mathbb{R}^n}\}$ and A can be separated by a hyperplane $\mathbf{h} := \mathbf{h}(u, 0)$, and without loss of generality we may assume that $A \subseteq \mathbf{h}^-$. Let

$$\lambda := \inf\{\langle z, u\rangle : z \in D\},$$

and notice that we do not know *a priori* whether $\lambda > -\infty$ or not. If this is the case, then we obviously have $D \subseteq \mathbf{h}^+(u, \lambda)$. Suppose that there exists a point $y_0 \in C$ such that $\langle y_0, u\rangle > \lambda$ (and observe that this holds for any point of C in the case where $\lambda = -\infty$), and fix any point $y_1 \in \operatorname{relint} C$. From Lemma 2.1.4, we have that

$$\operatorname{seg}[y_1, y_0) \subseteq \operatorname{relint} C.$$

Taking a point y_2 in this segment (and hence in $\operatorname{relint} C$) which is sufficiently close to y_0, from the continuity of the inner product we may assume that $\langle y_2, u\rangle > \lambda$. Now choose a point $z_0 \in D$ such that $\langle z_0, u\rangle < \langle y_2, u\rangle$ (such a point exists since otherwise $\langle y_2, u\rangle$ would be a greater lower bound for $\{\langle z, u\rangle : z \in D\}$ than λ, which is the infimum of this set). Now we repeat the previous argument: we fix any point $z_1 \in \operatorname{relint} D$ and notice that $\operatorname{seg}[z_1, z_0)$ is contained in $\operatorname{relint} D$. This guarantees that we may choose a point $z_2 \in \operatorname{relint} D$ such that $\langle z_2, u\rangle < \langle y_2, u\rangle$. It follows that $\langle y_2 - z_2, u\rangle > 0$, but this is a contradiction because $y_2 - z_2 \in A$ and $A \subseteq \mathbf{h}^-(u, 0)$. Consequently, we indeed have that $\lambda > -\infty$ and also that there is no point of C such that $\langle y_0, u\rangle > \lambda$. Hence $C \subseteq \mathbf{h}^-(u, \lambda)$.

Now we know that $\mathbf{h}(u, \lambda)$ separates C and D, because $D \subseteq \mathbf{h}^+(u, \lambda)$ and $C \subseteq \mathbf{h}^-(u, \lambda)$. We move forward to prove that this separation can be considered to be proper. Indeed, assume that $C \cup D \subseteq \mathbf{h}(u, \lambda)$. Then we may identify $\operatorname{aff}(C \cup D)$ with $\mathbb{R}^k$ for some $1 \leq k < n$ and repeat the argument to find a $(k-1)$- dimensional affine subspace $\mathbf{g}$ separating (the images of) C and D. Hence $\mathbf{g}$ is identified with a $(k-1)$-dimensional affine subspace of $\mathbb{R}^n$ which is contained in $\operatorname{aff}(C \cup D)$, and then we may choose a hyperplane $\mathbf{h}_1$ containing $\mathbf{g}$ which is not $\mathbf{h}(u, \lambda)$, and which does not contain $\operatorname{aff}(C \cup D)$. It is easy to see that $\mathbf{h}_1$ properly separates C and D (actually, any hyperplane which contains $\mathbf{g}$ and which is not $\mathbf{h}(u, \lambda)$ properly separates C and D).

□

2.3 Gauge and support functions

Let $A \subseteq \mathbb{R}^n$ be a non-empty set. The *support function* of A is the function $h_A \colon \mathbb{R}^n \to (-\infty, \infty]$ defined as

$$h_A(u) = h(A, u) := \sup\{\langle z, u\rangle : z \in A\}.$$

Throughout the text, both notations introduced above will be used. It is easy to see that h_A is positively homogeneous, that is, $h_A(\lambda u) = \lambda h_A(u)$ for any $u \in \mathbb{R}^n$ and every $\lambda \geq 0$. Notice that, for this reason, h_A is completely determined by its restriction to the unit sphere. Conversely, a function $f \colon \mathbb{S}^{n-1} \to (-\infty, +\infty]$ corresponds to a unique positively homogeneous function $\tilde{f} \colon \mathbb{R}^n \to (-\infty, +\infty]$, which is the *radial extension* of f:

$$\tilde{f}(tu) = t f(u),$$

for each $u \in \mathbb{S}^{n-1}$ and every $t \geq 0$. Hence there is no essential difference in defining the support function of a set A over $\mathbb{R}^n$ or over $\mathbb{S}^{n-1}$. We mainly regard support functions as defined over $\mathbb{R}^n$, but sometimes we consider only their restrictions to the domain $\mathbb{S}^{n-1}$. By the context we always make clear what case we are considering.

Lemma 2.3.1. *The support function $h_A \colon \mathbb{R}^n \to \mathbb{R}$ of a non-empty set $A \subseteq \mathbb{R}^n$ is convex and lower semi-continuous.*

Proof. The support function h_A is the superior envelope of the family of functions $\{f_z : z \in A\}$ defined as $f_z(u) = \langle z, u\rangle$. Each of these functions is convex and continuous, and hence h_A is convex and lower semi-continuous (see Chapter 1).

□

Remark 2.3.1. Since h_A is positively homogeneous and convex, it is easy to see that h_A is sublinear whenever it is finite-valued (recall that, for us, a sublinear function is always finite-valued). Indeed, we have

$$h_A(u+v) = 2h_A\left(\frac{u}{2}+\frac{v}{2}\right) \leq 2\left[\frac{1}{2}h_A(u)+\frac{1}{2}h_A(v)\right] = h_A(u)+h_A(v),$$

for any $u, v \in \mathbb{R}^n$.

Lemma 2.3.2. *Let $A \subseteq \mathbb{R}^n$ be a non-empty set. Then, $h_A(u) < +\infty$ for every $u \in \mathbb{R}^n$ if and only if A is bounded.*

Proof. First assume that A is bounded, and let $\rho > 0$ be such that $A \subseteq \rho B^n$. For any $u \in \mathbb{R}^n$ and $z \in A$ we have that

$$\langle z, u\rangle \leq |z||u| \leq \rho|u|,$$

from which we get that $h_A(u) \leq \rho|u| < +\infty$.

Now assume that $h_A(u) < +\infty$ for every $u \in \mathbb{R}^n$, and let $\{e_j\}_{j=1}^n$ be the canonical basis of $\mathbb{R}^n$. Then we have that $\beta := \max\{h_A(\pm e_j) : 1 \leq j \leq n\}$ is finite. If $z = \alpha_1 e_1 + \ldots + \alpha_n e_n \in A$, then

$$\pm\alpha_j = \langle z, \pm e_j\rangle \leq h_A(\pm e_j) \leq \beta,$$

for each $j = 1, \ldots, n$. Hence we get that $|\alpha_j| \leq \beta$ for every $j = 1, \ldots, n$. Thus,

$$|z| = \sqrt{\alpha_1^2 + \ldots + \alpha_n^2} \leq \beta\sqrt{n},$$

implying that $A \subseteq \beta\sqrt{n}B^n$.

□

The main geometric interpretation of the support function of a (compact) set is that it "parametrizes" the supporting hyperplanes of this set.

Proposition 2.3.1. *Let $C \subseteq \mathbb{R}^n$ be a compact, non-empty set. We have that $h_C(u) = \alpha$ if and only if C is supported by the hyperplane $\mathbf{h}(u, \alpha)$, with $C \subseteq \mathbf{h}^-(u, \alpha)$. Moreover, if $\mathbf{h}(u, \alpha)$ supports C, then it supports C at $z_0 \in \partial C$ if and only if $\alpha = \langle z_0, u\rangle$.*

Proof. Indeed, if $C \neq \emptyset$ is compact, then every non-zero vector is an outer normal of C (see Proposition 2.2.3). Hence, given a non-zero vector $u \in \mathbb{R}^n$, we have that $C \subseteq \mathbf{h}^-(u, \beta)$ for some $\beta \in \mathbb{R}$. This inclusion means that $\langle z, u\rangle \leq \beta$ for every $z \in C$, from where we get

$$h_C(u) = \sup\{\langle z, u\rangle : z \in C\} \leq \beta < \infty.$$

Also, of course $h_C(0_{\mathbb{R}^n}) = 0$. Now assume that $h_C(u) = \alpha$. Then $\langle z, u\rangle \leq \alpha$ for every $z \in C$, and this means that $K \subseteq \mathbf{h}^-(u, \alpha)$. Moreover, there exists a sequence $(z_j)_{j\in\mathbb{N}}$ of points in C such that $\langle z_j, u\rangle \to \alpha$ when $j \to \infty$. Since C is compact, we may assume that $z_j \to z$, for some $z \in C$ (passing to a subsequence if necessary). Since it is clear that $\langle z, u\rangle = \alpha$, we get that $z \in C \cap \mathbf{h}(u, \alpha)$. This, with the inclusion $C \subseteq \mathbf{h}^-(u, \alpha)$, gives that $\mathbf{h}(u, \alpha)$ supports C (at z).

For the converse, assume that C is supported by $\mathbf{h}(u,\alpha)$ at a point $z_0 \in \partial C$ with $C \subseteq \mathbf{h}^-(u,\alpha)$. This last inclusion gives immediately that $h_C(u) \leq \alpha$, and the fact that $z_0 \in C \cap \mathbf{h}(u,\alpha)$ leads to $h_C(u) \geq \langle z_0, u\rangle = \alpha$. This concludes the proof.

□

Corollary 2.3.1. *If K is compact, convex, and non-empty, then*

$$K = \bigcap_{u\in\mathbb{R}^n} \mathbf{h}^-(u, h_K(u)) = \bigcap_{u\in\mathbb{S}^{n-1}} \mathbf{h}^-(u, h_K(u)).$$

Proof. From Proposition 2.2.4 we have that K is the intersection of its supporting half-spaces which, from the last proposition, are of the form $\mathbf{h}(u, h_K(u))$, for $u \in \mathbb{R}^n$. The fact that we can consider only $u \in \mathbb{S}^{n-1}$ comes from the equality $\mathbf{h}^-(\lambda u, h_K(\lambda u)) = \mathbf{h}^-(\lambda u, \lambda h_K(u)) = \mathbf{h}^-(u, h_K(u))$, which holds for every $\lambda > 0$. Also, observe that $\mathbf{h}^-(0_{\mathbb{R}^n}, h_K(0_{\mathbb{R}^n})) = \mathbf{h}^-(0_{\mathbb{R}^n}, 0) = \mathbb{R}^n$. In other words, $u = 0_{\mathbb{R}^n}$ makes no difference in the intersection above.

□

Remark 2.3.2. Another way to look at the support function of a compact set C is the following: let $u \in \mathbb{S}^{n-1}$. By Proposition 2.2.3 we get that u is an outer normal of C at some point $z_0 \in \partial C$. From Proposition 2.3.1, the value of $h_C(u)$ is precisely the (signed) length $\langle z_0, u\rangle$ of the projection of z_0 on the direction of u or, alternatively, the (signed) distance from the hyperplane orthogonal to u and passing through z_0 to the origin. See Figure 2.3.1.

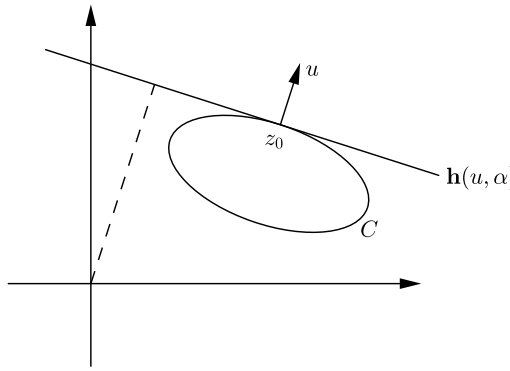


Fig. 2.3.1 $h_C(u) = \alpha = \langle z_0, u\rangle$, the (signed) distance from $\mathbf{h}(u,\alpha)$ to the origin.

This remark yields yet another feature of support functions (but only for convex bodies). In what follows, we recall that a convex body $K \subseteq \mathbb{R}^n$ is supported by precisely two hyperplanes orthogonal to each given direction (see Corollary 2.2.2).

Lemma 2.3.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For each $u \in \mathbb{S}^{n-1}$ the number*

$$w_K(u) := h_K(u) + h_K(-u)$$

equals the distance between the two supporting hyperplanes of K which are orthogonal to u. See Figure 2.3.2.

Remark 2.3.3. The number $w_K(u)$ defined above is called the *width* of K in the direction of u.

Proof. Let $u \in \mathbb{S}^{n-1}$. First notice that the sublinearity of h_K gives

$$h_K(u) + h_K(-u) \geq h_K(u-u) = h_K(0_{\mathbb{R}^n}) = 0.$$

Assume that u is an outer normal of K at $z_1 \in \partial K$ and that $-u$ is an outer normal of K at $z_2 \in \partial K$. The hyperplanes $\mathbf{h}_1$ and $\mathbf{h}_2$ orthogonal to u which support K pass through z_1 and z_2, respectively. From the last remark, we have

$$h_K(u) + h_K(-u) = \langle z_1, u\rangle + \langle z_2, -u\rangle = \langle z_1 - z_2, u\rangle,$$

and the latter is the distance between $\mathbf{h}_1$ and $\mathbf{h}_2$, because it is (the absolute value of) the projection in the direction of u of a vector connecting a point of $\mathbf{h}_2$ to a point of $\mathbf{h}_1$.

□

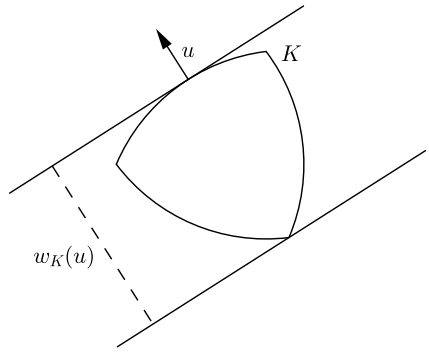


Fig. 2.3.2 The width of K in the direction of u.

We can explicitly describe the support functions of some classes of convex sets. For further applications, we will do that for polytopes and for balls centered at the origin.

Lemma 2.3.4. *Let $P = \mathrm{conv}\{x_1, \ldots, x_k\}$ be a polytope in $\mathbb{R}^n$. Then, the support function of P is given by*

$$h_P(u) = \max\{\langle x_j, u\rangle : 1 \leq j \leq k\}$$

for each $u \in \mathbb{R}^n$ (see Figure 2.3.3). In particular, the support function of a singleton $\{z\} \subseteq \mathbb{R}^n$ is

$$h_{\{z\}}(u) = \langle z, u\rangle.$$

Proof. The proof is straightforward: let $\lambda_1, \ldots, \lambda_k \geq 0$ be such that $\lambda_1 + \lambda_2 + \ldots + \lambda_k = 1$. Then for any $u \in \mathbb{R}^n$ we have

$$\langle \lambda_1 x_1 + \ldots + \lambda_k x_k, u\rangle = \lambda_1\langle x_1, u\rangle + \ldots + \lambda_k\langle x_k, u\rangle \leq$$
$$\leq (\lambda_1 + \ldots + \lambda_k)\max\{\langle x_j, u\rangle : 1 \leq j \leq k\} = \max\{\langle x_j, u\rangle : 1 \leq j \leq k\}.$$

Hence $h_P(u) \leq \max\{\langle x_j, u\rangle : 1 \leq j \leq k\}$. For the reverse inequality, simply notice that each x_j is a point of P, and hence $h_P(u) \geq \langle x_j, u\rangle$ for every $1 \leq j \leq k$.

□

Remark 2.3.4. Notice carefully that the support function of a singleton is linear. The converse is also true: if the support function of a convex set is linear, then this set is a singleton. This will be made clear in the proof of Theorem 2.3.1.

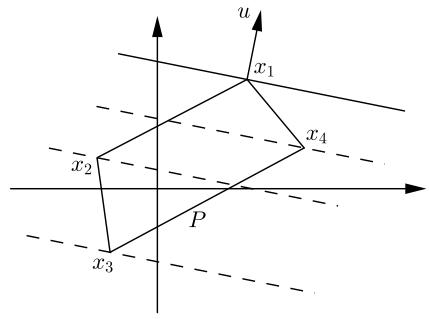


Fig. 2.3.3 The support function of a polytope can be determined by looking only at its vertices.

Corollary 2.3.2. *For any $x \in \mathbb{R}^n$, the support function of the segment* $\mathrm{seg}[-x, x]$ *is given as*

$$h_{\mathrm{seg}[-x,x]}(z) = |\langle x, z\rangle|$$

for each $z \in \mathbb{R}^n$.

Proof. Notice that $\mathrm{seg}[-x, x] = \mathrm{conv}\{-x, x\}$. Thus,

$$h_{\mathrm{seg}[-x,x]}(z) = \max\left\{\langle x, z\rangle, \langle -x, z\rangle\right\} = |\langle x, z\rangle|.$$

□

Lemma 2.3.5. *Recall that B^n denotes the closed unit ball of $\mathbb{R}^n$ with respect to the Euclidean metric. We have $h_{B^n}(u) = 1$ for every $u \in \mathbb{S}^{n-1}$.*

Proof. For each $u \in \mathbb{S}^{n-1}$, we have

$$h_{B^n}(u) = \sup\{\langle u, v\rangle : v \in B^n\} \leq \sup\{|u||v| : v \in B^n\} = 1,$$

and since $u \in B^n$ we get $h_{B^n}(u) \geq \langle u, u\rangle = |u|^2 = 1$.

□

Next we investigate some more properties of the support function. We are mainly concerned with compact convex sets, but the reader may notice that some properties hold without the compactness hypothesis. Hence this hypothesis will be dropped whenever this does not make the proof overly complicated for our purposes.

Proposition 2.3.2. *Let $K, L \subseteq \mathbb{R}^n$ be non-empty sets, and assume that $\lambda \geq 0$. Then we have the following:*

(i) $h_{K+\lambda L} = h_K + \lambda h_L$.

(ii) $h_{-K}(u) = h_K(-u)$ *for every $u \in \mathbb{R}^n$.*

(iii) *If $K \subseteq L$, then $h_K \leq h_L$.*

If we assume in addition that K and L are compact and convex, then we also have

(iv) *If $h_K \leq h_L$, then $K \subseteq L$.*

(v) *$h_K|_{\mathbb{S}^{n-1}}$ is a positive constant function if and only if K is a closed ball centered at the origin.*

Remark 2.3.5. The support function of a Minkowski sum of convex sets is illustrated in Figure 2.3.4, and property **(iii)** is illustrated in Figure 2.3.5.

Proof. For **(i)** we simply notice that, for every $u \in \mathbb{R}^n$,

$$h_{K+\lambda L}(u) = \sup\{\langle x + \lambda y, u\rangle : x \in K \text{ and } y \in L\} =$$
$$= \sup\{\langle x, u\rangle : x \in K\} + \lambda \sup\{\langle y, u\rangle : y \in L\} = h_K(u) + \lambda h_L(u).$$

To prove **(ii)**, we only have to observe that

$$h_{-K}(u) = \sup\{\langle -z, u\rangle : z \in K\} = \sup\{\langle z, -u\rangle : z \in K\} = h_K(-u),$$

for each $u \in \mathbb{R}^n$.

Property **(iii)** is immediate from the definition, and **(iv)** is a consequence of Corollary 2.3.1. Indeed, the inequality $h_K(u) \leq h_L(u)$ implies the inclusion $\mathbf{h}^-(u, h_K(u)) \subseteq \mathbf{h}^-(u, h_L(u))$. If this holds for every $u \in \mathbb{R}^n$, then we get

$$K = \bigcap_{u\in\mathbb{R}^n} \mathbf{h}^-(u, h_K(u)) \subseteq \bigcap_{u\in\mathbb{R}^n} \mathbf{h}^-(u, h_L(u)) = L.$$

Now we prove **(v)**. From Lemma 2.3.5 and **(i)**, we have that the support function of a closed ball ρB^n ($\rho > 0$) restricted to the unit sphere is $h_{\rho B^n}|_{\mathbb{S}^{n-1}} = \rho h_{B^n}|_{\mathbb{S}^{n-1}} = \rho$. Conversely, if $h_K|_{\mathbb{S}^{n-1}} = \rho > 0$, then $h_K = h_{\rho B^n}$ over $\mathbb{R}^n$ (recall that the support function is positively homogeneous). This gives that $h_K \leq h_{\rho B^n}$ and $h_K \geq h_{\rho B^n}$. From **(iv)** we get that $K \subseteq \rho B^n$ and $\rho B^n \subseteq K$, and the proof is complete. □

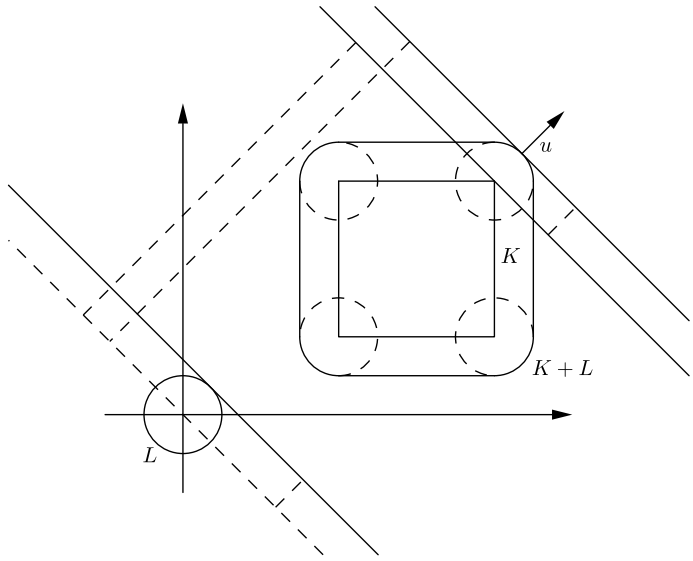


Fig. 2.3.4 The support function of the Minkowski sum $K + L$.

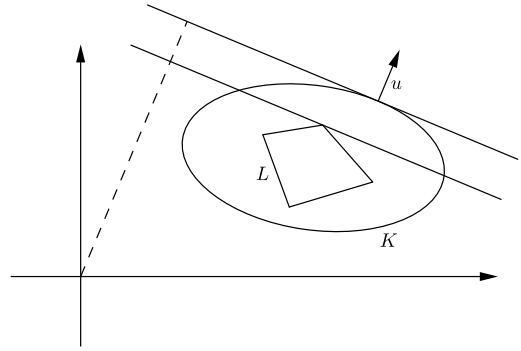


Fig. 2.3.5 The monotonicity of the support function with respect to inclusion.

Remark 2.3.6. Since h_K is positively homogeneous, saying that $h_K|_{\mathbb{S}^{n-1}} = \rho$ for some constant $\rho > 0$ is the same as saying that $h_K(u) = \rho|u|$ for every $u \in \mathbb{R}^n$.

Corollary 2.3.3. *Let $K, L \subseteq \mathbb{R}^n$ be non-empty, compact, and convex sets. Then, $h_K = h_L$ if and only if $K = L$.*

Proof. This follows immediately from the Items **(iii)** and **(iv)** of the last proposition.

□

We already know that the support function of a compact, convex, and non-empty set is sublinear (because it is finite-valued, positively homogeneous, and convex). Next we prove that any sublinear function is the support function of some of these sets. In some sense, this means that a compact, convex, and non-empty set is "represented" by its support function.

Theorem 2.3.1. *Let $h\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function. Then there exists a unique compact, convex, and non-empty set $K \subseteq \mathbb{R}^n$ such that h is the support function of K.*

Proof. Let

$$K := \bigcap_{u \in \mathbb{R}^n} \mathbf{h}^-(u, h(u)) = \{x \in \mathbb{R}^n : \langle x, u\rangle \leq h(u)\ \forall\ u \in \mathbb{R}^n\}.$$

This set is clearly convex. Moreover, K is closed, since it is the intersection of closed sets. It is also easy to see that K is bounded. Indeed, if $\{e_j\}_{j=1}^n$ is the usual canonical basis of $\mathbb{R}^n$, and if $\beta := \max\{h(\pm e_j) : j = 1, \ldots, n\}$, then

$$K \subseteq \bigcap_{j=1}^n \mathbf{h}^-(\pm e_j, h(\pm e_j)) \subseteq \bigcap_{j=1}^n \mathbf{h}^-(\pm e_j, \beta).$$

Consequently, if $\alpha_1 e_1 + \ldots + \alpha_n e_n \in K$, then $|\alpha_j| \leq \beta$ for each $j = 1, \ldots, n$. Hence

$$|\alpha_1 e_1 + \ldots + \alpha_n e_n| = \sqrt{\alpha_1^2 + \cdots + \alpha_n^2} \leq \beta\sqrt{n},$$

from which we get that $K \subseteq \beta\sqrt{n}B^n$.

As usual, we denote by h_K the support function of K. For every $u \in \mathbb{R}^n$ we have that $\langle x, u\rangle \leq h(u)$ for each $x \in K$, and hence

$$h_K(u) = \sup\{\langle x, u\rangle : x \in K\} \leq h(u).$$

We still have to prove the reverse inequality and that K is non-empty. To do so, we proceed by induction on the codimension of the set $\mathrm{lin}(h)$ of linearity directions of h. First observe that if the codimension of $\mathrm{lin}(h)$ is zero, then h is linear, and hence we may write $h(\cdot) = \langle z_0, \cdot\rangle$ for some $z_0 \in \mathbb{R}^n$. The condition $\langle x, u\rangle \leq h(u)$ becomes $\langle x - z_0, u\rangle \leq 0$. Thus,

$$K = \{x \in \mathbb{R}^n : \langle x - z_0, u\rangle \leq 0 \ \forall\, u \in \mathbb{R}^n\},$$

and this gives $K = \{z_0\}$ (indeed, if $x \neq z_0$, then $\langle x - z_0, x - z_0\rangle > 0$). Moreover, the support function of K is

$$h_K(u) = \sup\{\langle x, u\rangle : x \in K\} = \langle z_0, u\rangle = h(u),$$

for each $u \in \mathbb{R}^n$. Now assume that $\dim(\mathrm{lin}(h)) = k < n$, and that for the case where the dimension of the space of linearity directions of the considered sublinear function is greater than or equal to $k + 1$ we have that $K \neq \emptyset$ and $h_K(u) \geq h(u)$ for every $u \in \mathbb{R}^n$. (This is our induction hypothesis: notice that increasing the dimension means decreasing the codimension.) Choose a point $z \in \mathbb{R}^n \setminus \mathrm{lin}(h)$. By Proposition 1.2.3 we have that $h'_+(z, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$ is also a sublinear function, and we define

$$L := \{x \in \mathbb{R}^n : \langle x, u\rangle \leq h'_+(z, u) \ \forall\, u \in \mathbb{R}^n\}.$$

By the previous arguments, we know that L is closed, convex, bounded, and that $h_L(u) \leq h'_+(z, u)$ for every $u \in \mathbb{R}^n$. From Proposition 1.2.4 it follows that

$$\mathrm{lin}(h'_+(z, \cdot)) \supseteq \mathrm{lin}(h) + \mathrm{span}\{z\},$$

from where we get that $\dim(\mathrm{lin}(h'_+(z, \cdot))) \geq k + 1$. Hence the induction hypothesis guarantees that we have $L \neq \emptyset$ and $h_L(u) = h'_+(z, u)$ for any $u \in \mathbb{R}^n$. For a point $y \in L$, we have

$$\langle y, u\rangle \leq h'_+(z, u) \leq h(u)$$

for every $u \in \mathbb{R}^n$, where we used again Proposition 1.2.4 in the last inequality. It follows that $L \subseteq K$, and hence $K \neq \emptyset$ (also, $h_K \geq h_L$, by Proposition 2.3.2). Even more, taking $u = -z$ and fixing any point $y \in L$ we get

$$\langle y, -z\rangle \leq h'_+(z, -z) = -h(z),$$

where in the last equality we used Proposition 1.2.4 one more time. Hence

$$h_K(z) \geq h_L(z) \geq \langle y, z\rangle = -\langle y, -z\rangle \geq -(-h(z)) = h(z).$$

Since z was taken arbitrarily in $\mathbb{R}^n \setminus \mathrm{lin}(h)$, we get that $h_K \geq h$ in this set. It remains to prove this inequality for the case where $u \in \mathrm{lin}(h)$. If $u \in \mathrm{lin}(h)$, then $u \in \mathrm{lin}(h'_+(z, \cdot))$, and this gives

$$h_K(u) \geq h_L(u) = h'_+(z, u) = -h'_+(z, -u) \geq -h(-u) = h(u).$$

The proof is finished.

□

Remark 2.3.7. In Exercise 2.49, a characterization of the compact convex set which has a sublinear function h as support function is given in terms of $\mathrm{epi}(h)$. In Exercise 2.31, we investigate the conditions that h has to be the support function of a convex body.

As we have seen in the beginning of this section, sometimes it is more convenient to regard the restrictions of support functions to $\mathbb{S}^{n-1}$. In this case, from the previous theorem we get immediately that a function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$ is the support function of a compact, convex, and non-empty subset of $\mathbb{R}^n$ if and only if its radial extension is sublinear.

A function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$ which is continuous and positive, but whose radial extension is not sublinear, can still be associated with a certain convex body called the *Wulff shape* of f. This construction will be introduced and investigated in Section 8.1.

A comment is due here. From Remark 2.3.1 and Lemma 2.3.2 we get that the support function of any non-empty bounded set is sublinear. However, in the last theorem we proved that the sublinear functions are associated with compact and convex sets. Consequently, the support function of a non-convex or non-compact set must be also the support function of some compact and convex set. What happens is that two bounded sets have the same support function if and only if the respective closures of their convex hulls are the same, as we will explain next.

Proposition 2.3.3. *Let $A \subseteq \mathbb{R}^n$ be a non-empty set. Then, $h_{\mathrm{cl}(A)} = h_A$ and $h_{\mathrm{conv}(A)} = h_A$. The second equality is illustrated in Figure 2.3.6.*

Proof. It is clear that $h_A \leq h_{\mathrm{cl}(A)}$, because $A \subseteq \mathrm{cl}(A)$. If the inequality is strict for some $u \in \mathbb{R}^n$, then there exist a point $x \in \mathrm{cl}(A)$ and a number $\varepsilon > 0$ such that $\langle x, u\rangle > \langle z, u\rangle + \varepsilon$ for all $z \in A$. This is a contradiction: since $x \in \mathrm{cl}(A)$, there exists a sequence $(z_j)_{j\in\mathbb{N}}$ in A such that $z_j \to x$. Hence $\langle z_j, u\rangle \to \langle x, u\rangle$.

Now we look at the other equality. It is clear that $h_A \leq h_{\mathrm{conv}(A)}$. To prove the reverse inequality, let $u \in \mathbb{R}^n$, and $\lambda_1 x_1 + \ldots + \lambda_k x_k \in \mathrm{conv}(A)$ be a convex combination of points of A. We have

$$\langle \lambda_1 x_1 + \ldots + \lambda_k x_k, u\rangle = \lambda_1\langle x_1, u\rangle + \ldots + \lambda_k\langle x_k, u\rangle \leq (\lambda_1 + \ldots + \lambda_k)h_A(u) = h_A(u).$$

Since the inequality above holds for any convex combination of points of A, passing to the supremum we get that $h_{\mathrm{conv}(A)}(u) \leq h_A(u)$.

□

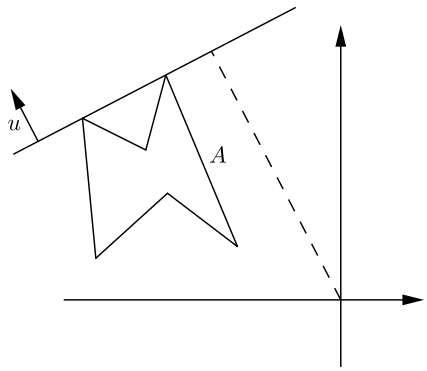


Fig. 2.3.6 The same hyperplane orthogonal to u supports A and its convex hull.

Corollary 2.3.4. *Let $A, B \subseteq \mathbb{R}^n$ be non-empty, bounded sets. Then $h_A = h_B$ if and only if* $\mathrm{cl}(\mathrm{conv}(A)) = \mathrm{cl}(\mathrm{conv}(B))$.

Proof. If $\mathrm{cl}(\mathrm{conv}(A)) = \mathrm{cl}(\mathrm{conv}(B))$, then

$$h_A = h_{\mathrm{conv}(A)} = h_{\mathrm{cl}(\mathrm{conv}(A))} = h_{\mathrm{cl}(\mathrm{conv}(B))} = h_{\mathrm{conv}(B)} = h_B.$$

If $h_A = h_B$ then we have that $h_{\mathrm{cl}(\mathrm{conv}(A))} = h_{\mathrm{cl}(\mathrm{conv}(B))}$, and the sets $\mathrm{cl}(\mathrm{conv}(A))$ and $\mathrm{cl}(\mathrm{conv}(B))$ are compact and convex. From Corollary 2.3.3 we get that $\mathrm{cl}(\mathrm{conv}(A)) = \mathrm{cl}(\mathrm{conv}(B))$.

□

We warn the reader that it is common to find in the literature the support function defined for linear functionals rather than for vectors. Letting

$$(\mathbb{R}^n)^* := \{f\colon \mathbb{R}^n \to \mathbb{R} : f \text{ is linear}\}$$

be the *dual space* of $\mathbb{R}^n$, one defines the support function of a non-empty set $A \subseteq \mathbb{R}^n$ to be $\tilde{h}_A(f) := \sup\{f(z) : z \in A\}$, for $f \in (\mathbb{R}^n)^*$. What we do is simply to identify each linear functional of $(\mathbb{R}^n)^*$ with a vector of $\mathbb{R}^n$ by the usual Riesz identification given by the standard inner product. That is, we have $h_A(u) = \tilde{h}_A(f)$ if $f(\cdot) = \langle u, \cdot\rangle$. Having this in mind, it is also natural to identify hyperplanes of $\mathbb{R}^n$ with sets of the form $\{x \in \mathbb{R}^n : f(x) = \alpha\}$ for $f \in (\mathbb{R}^n)^*$ and $\alpha \in \mathbb{R}$.

Let $A \subseteq \mathbb{R}^n$ be a non-empty set. The *polar set* of A is the set

$$A^\circ := \{x \in \mathbb{R}^n : \langle x, z\rangle \leq 1 \ \forall\, z \in A\} = \{x \in \mathbb{R}^n : \sup_{z\in A}\langle x, z\rangle \leq 1\} = \{x \in \mathbb{R}^n : h_A(x) \leq 1\};$$

see Figure 2.3.7.

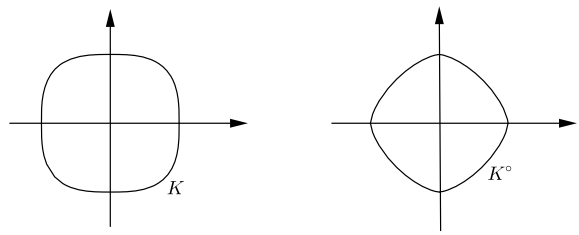


Fig. 2.3.7 The convex body $K = B_{\ell_3} := \{(x, y) : |x|^3 + |y|^3 \leq 1\}$ and its polar set.

As in the case of the support function, sometimes the polar set of A is defined as the subset A^* of $(\mathbb{R}^n)^*$ defined by $A^* := \{f \in (\mathbb{R}^n)^* : f(z) \leq 1 \text{ for all } z \in A\}$. Again, it is easy to see that A° is the image of A^* under the Riesz identification between $(\mathbb{R}^n)^*$ and $\mathbb{R}^n$ given by the standard inner product. As an example that we will use further, we describe the polar of a closed ball centered at the origin.

Example 2.3.1. Let $\rho > 0$. Then $(\rho B^n)^\circ = \frac{1}{\rho}B^n$. Indeed, if $x \in (\rho B^n)^\circ$ is non-zero (the vector $0_{\mathbb{R}^n}$ is clearly in both sets), then we have that $\rho x/|x| \in \rho B^n$, and hence

$$1 \geq \left\langle x, \frac{\rho x}{|x|}\right\rangle = \rho|x|.$$

This gives that $|x| \leq 1/\rho$. Therefore, we have the inclusion $(\rho B^n)^\circ \subseteq \frac{1}{\rho}B^n$. On the other hand, if $z \in \frac{1}{\rho}B^n$, then $|z| \leq 1/\rho$, and for every $x \in \rho B^n$ we have from the Cauchy–Schwarz inequality that

$$\langle x, z\rangle \leq |x||z| \leq \rho\frac{1}{\rho} = 1,$$

from where we get the inclusion $z \in (\rho B^n)^\circ$.

Lemma 2.3.6. *Let $A \subseteq \mathbb{R}^n$ be a non-empty set. Then its polar set A° is closed and convex. Moreover, we have that $0_{\mathbb{R}^n} \in A^\circ$.*

Proof. The fact that $0_{\mathbb{R}^n} \in A^\circ$ is straightforward, as well as the convexity. Let $(x_j)_{j\in\mathbb{N}}$ be a sequence of points in A° such that $x_j \to x$, as $j \to \infty$, for some $x \in \mathbb{R}^n$. Since for each $z \in A$ we have

$$\langle x, z\rangle = \lim_{j\to\infty} \langle x_j, z\rangle \leq 1,$$

we get that $x \in A^\circ$, and this shows that the polar set is closed.

□

Remark 2.3.8. Another way to prove this lemma is by noticing that A° is always an intersection of closed half-spaces:

$$A^\circ = \bigcap_{z\in\mathbb{A}} \mathbf{h}^-(z, 1).$$

Another immediate consequence of this viewpoint is the fact that polarity reverses inclusion, that is, if $A \subseteq B$, then $B^\circ \subseteq A^\circ$.

Recall that we denote by $\mathcal{K}_0^n$ the family of convex bodies of $\mathbb{R}^n$ which contain the origin as an interior point, and denote by $\mathcal{Q}_0^n$ the family of closed, convex sets of $\mathbb{R}^n$ containing the origin (we obviously have $\mathcal{K}_0^n \subseteq \mathcal{Q}_0^n$). Observe that Lemma 2.3.6 guarantees that $A^\circ \in \mathcal{Q}_0^n$ for every non-empty set $A \subseteq \mathbb{R}^n$. In what follows, the set-valued map $\mathrm{pol}\colon 2^{\mathbb{R}^n} \setminus \{\emptyset\} \to \mathcal{Q}_0^n$ which associates each non-empty set with its polar set (that is, $\mathrm{pol}(A) = A^\circ$) is called the *polarity map*. We also recall that an *involution* is a map whose inverse is the map itself.

Proposition 2.3.4. *The restriction of the polarity map to $\mathcal{Q}_0^n$ is an involution of $\mathcal{Q}_0^n$. In other words, $\mathrm{pol}|_{\mathcal{Q}_0^n} : \mathcal{Q}_0^n \to \mathcal{Q}_0^n$ is a bijection, and $(C^\circ)^\circ = C$ for every $C \in \mathcal{Q}_0^n$.*

Proof. Let $C \in \mathcal{Q}_0^n$. If $x \in C$, then by the definition of the polar set we have that $\langle x, z\rangle \leq 1$ for all $z \in C^\circ$, and hence $x \in (C^\circ)^\circ$. This proves the inclusion $C \subseteq (C^\circ)^\circ$. To prove the reverse inclusion, let $x \in (C^\circ)^\circ$ and assume that $x \notin C$. From Theorem 2.2.2 we get that x and C can be strongly separated by some hyperplane $\mathbf{h} = \mathbf{h}(u, \alpha)$. We may assume that $\langle x, u\rangle > \alpha$ and $\langle z, u\rangle < \alpha$ for every $z \in C$. In other words,

$$h_C(u) \leq \alpha < \langle x, u\rangle.$$

Since $0_{\mathbb{R}^n} \in C$, we have that $h_C(u) \geq 0$. If $h_C(u) > 0$, then there is a number $\beta > 0$ such that $h_C(\beta u) = 1$. Hence $\beta u \in C^\circ$, but $\langle x, \beta u\rangle = \beta\langle x, u\rangle > \beta h_C(u) = h_C(\beta u) = 1$, from which we get that $x \notin (C^\circ)^\circ$, a contradiction. If $h_C(u) = 0$, then $h_C(\beta u) = 0$ for every $\beta > 0$, and this implies that $\beta u \in C^\circ$ for every $\beta > 0$. On the other hand, from the inequality $h_C(u) < \langle x, u\rangle$ we obtain that $\langle x, u\rangle > 0$. Therefore, we have that $\langle x, \beta u\rangle > 1$ for $\beta > 0$ large enough. Again, we obtain the contradiction $x \notin (C^\circ)^\circ$. It follows that $x \in C$, and this gives the desired inclusion $(C^\circ)^\circ \subseteq C$.

The fact that $\mathrm{pol}|_{\mathcal{Q}_0^n}$ is a bijection is a consequence of what we have just proved. Indeed, for any $C \in \mathcal{Q}_0^n$ we have that $\mathrm{pol}(C^\circ) = (C^\circ)^\circ = C$, and this settles the surjectiveness. For the injectiveness, simply notice that if $C^\circ = D^\circ$, then $C = (C^\circ)^\circ = (D^\circ)^\circ = D$.

□

Lemma 2.3.7. *If $K \in \mathcal{K}_0^n$, then $K^\circ \in \mathcal{K}_0^n$. The converse is also true.*

Proof. Let $K \in \mathcal{K}_0^n$. Since any polar set is closed and convex, we only have to prove that K° is bounded and contains the origin as an interior point. To prove that K° is bounded, simply choose a number

$\rho > 0$ such that $\rho B^n \subseteq K$. Since the polarity map reverses inclusion, we get that $K^\circ \subseteq (\rho B^n)^\circ = \frac{1}{\rho} B^n$, where the last equality is given by Example 2.3.1.

The fact that K° contains the origin as an interior point comes from the boundedness of K. Indeed, there exists a number $\alpha > 0$ such that $K \subseteq \alpha B^n$, and hence $\frac{1}{\alpha} B^n = (\alpha B^n)^\circ \subseteq K^\circ$. This implies, in particular, that $(K^\circ)^\circ \in \mathcal{K}_0^n$ whenever $K^\circ \in \mathcal{K}_0^n$. Proposition 2.3.4 gives that $(K^\circ)^\circ = K$, and hence we have the converse.

□

Corollary 2.3.5. *The restriction* $\mathrm{pol}|_{\mathcal{K}_0^n}$ *of the polarity map to* $\mathcal{K}_0^n$ *is an involution of* $\mathcal{K}_0^n$.

Proof. Since $\mathcal{K}_0^n \subseteq \mathcal{Q}_0^n$, we already have that $\mathrm{pol}|_{\mathcal{K}_0^n}$ is a bijection onto its range, and that $(K^\circ)^\circ = K$ for every $K \in \mathcal{K}_0^n$. Hence we only have to verify that the range of $\mathrm{pol}|_{\mathcal{K}_0^n}$ is $\mathcal{K}_0^n$. This comes immediately from Lemma 2.3.7: if $K \in \mathcal{K}_0^n$, then $K^\circ \in \mathcal{K}_0^n$ and $\mathrm{pol}(K^\circ) = K$.

□

We define the *gauge function* (or the *Minkowski functional*) of a body $K \in \mathcal{K}_0^n$ to be the function $||\cdot||_K : \mathbb{R}^n \to \mathbb{R}$ defined as

$$||x||_K := \inf\{\lambda > 0 : x \in \lambda K\};$$

see Figure 2.3.8. Also, for any non-zero vector $x \in \mathbb{R}^n$, the ray starting at $0_{\mathbb{R}^n}$ and passing through x intersects the boundary of K at a unique point (which is not $0_{\mathbb{R}^n}$, since this is an interior point of K). Hence there exists a unique positive number $\rho_K(x)$ such that $\rho_K(x)x \in \partial K$. The function $\rho_K : \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to (0, +\infty)$ defined in this way is called the *radial function* of K. It is readily seen that the radial function can be equivalently defined as

$$\rho_K(u) = \sup\{\lambda \geq 0 : \lambda u \in K\}, \tag{2.3.1}$$

for each $u \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$.

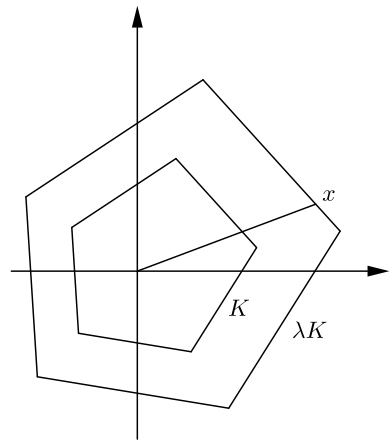


Fig. 2.3.8 The gauge function of K. Here $||x||_K = \lambda$.

Proposition 2.3.5. *Let K be a convex body containing the origin as an interior point. Then the gauge function $||\cdot||_K$ is a non-negative and finite-valued function satisfying the following conditions:*

(i) $||x||_K \geq 0$ *for all $x \in \mathbb{R}^n$, with equality if and only if $x = 0_{\mathbb{R}^n}$,*

(ii) $||\alpha x||_K = \alpha ||x||_K$ *for every* $\alpha \geq 0$,

(iii) $||x + y||_K \leq ||x||_K + ||y||_K$ *for any* $x, y \in \mathbb{R}^n$.

Proof. The fact that $|| \cdot ||_K$ is non-negative is straightfoward. Since K has the origin as an interior point, we assume from now on that $r > 0$ is such that $rB^n \subseteq K$. For any non-zero $x \in \mathbb{R}^n$ we have that $rx/|x| \in rB^n \subseteq K$, from which we get that

$$x \in \frac{|x|}{r} K,$$

and hence $||x||_K \leq |x|/r$. This, together with the obvious equality $||0_{\mathbb{R}^n}||_K = 0$, gives that the gauge function is finite-valued.

For **(i)**, the only non-trivial part is that the equality $||x||_K = 0$ only happens when $x = 0_{\mathbb{R}^n}$. To prove this, let $x \in \mathbb{R}^n$ be a non-zero vector. Then the number $\rho_K(x) > 0$ is such that $\rho_K(x)x \in \partial K$. We claim that if $\lambda < 1/\rho_K(x)$, then $x \notin \lambda K$. In fact, if $x \in \lambda K$, then $x/\lambda \in K$. From Lemma 2.1.4, we have that $\text{seg}[0_{\mathbb{R}^n}, x/\lambda) \subseteq \text{int}K$. Since $\lambda < 1/\rho_K(x)$, we have that $\rho_K(x) < 1/\lambda$, implying that $\rho_K(x)x \in \text{seg}[0_{\mathbb{R}^n}, x/\lambda)$. Hence we would have that $\rho_K(x)x$ is an interior point of K, a contradiction. We get that $||x||_K \geq 1/\rho_K(x) > 0$, since $x \notin \lambda K$ for any $\lambda < 1/\rho_K(x)$.

To prove **(ii)** simply notice that, for any $\alpha > 0$, we have that $\alpha x \in \lambda K$ if and only if $x \in \frac{\lambda}{\alpha} K$, and then

$$||\alpha x||_K = \inf\{\lambda > 0 : \alpha x \in \lambda K\} = \inf\left\{\lambda > 0 : x \in \frac{\lambda}{\alpha} K\right\} =$$
$$= \inf\{\alpha\mu > 0 : x \in \mu K\} = \alpha \inf\{\mu > 0 : x \in \mu K\} = \alpha ||x||_K.$$

It remains to prove **(iii)**, which is called the *triangle inequality*. Let $x, y \in \mathbb{R}^n$, and assume that $x \in \alpha K$ and $y \in \beta K$ for some numbers $\alpha, \beta \geq 0$. Then, $x + y \in \alpha K + \beta K$, and by Lemma 2.1.6 we get that $x + y \in (\alpha + \beta)K$. It follows that

$$||x + y||_K \leq \alpha + \beta$$

whenever $x \in \alpha K$ and $y \in \beta K$. Thus, passing to the infimum of the numbers $\alpha, \beta \geq 0$ with these respective properties, we obtain that $||x + y||_K \leq ||x||_K + ||y||_K$.

□

Remark 2.3.9. Notice that, in particular, the gauge function of a body $K \in \mathcal{K}_0^n$ is sublinear. Also, it only fails to be a norm because it is not homogeneous, but only positively homogeneous. Next we investigate under which additional hypothesis on K we have that its gauge function is homogeneous.

Corollary 2.3.6. *Let* $K \in \mathcal{K}_0^n$. *Then* K *is centered at the origin if and only if its gauge function is a norm. In this case, the unit ball of the norm is* K.

Proof. Let $x \in \mathbb{R}^n$. If K is centered at the origin, then $x \in \lambda K$ if and only if $-x \in \lambda K$. This means that the sets $\Lambda_x := \{\lambda > 0 : x \in \lambda K\}$ and $\Lambda_{-x} := \{\lambda > 0 : -x \in \lambda K\}$ are the same. In particular, they have the same infimum, and hence we get that $||x||_K = ||-x||_K$. Now let $\alpha < 0$. Then

$$||\alpha x||_K = ||(-\alpha)(-x)||_K = -\alpha ||-x||_K = |\alpha|\, ||x||_K.$$

Since we already have that $||\alpha x||_K = \alpha ||x||_K$ whenever $\alpha \geq 0$, we get that $|| \cdot ||_K$ is homogeneous. This, with the other properties proved in the last proposition, gives that $|| \cdot ||_K$ is a norm. The fact that the unit ball of this norm is K comes from the trivial equality

$$K = \{x \in \mathbb{R}^n : ||x||_K \leq 1\}.$$

Finally, assume that $||\cdot||_K$ is a norm. Hence, $||x||_K = ||-x||_K$ for every $x \in \mathbb{R}^n$, and this implies that $||x||_K \leq 1$ if and only if $||-x||_K \leq 1$. By the equality above, this means that $x \in K$ if and only if $-x \in K$, and then K is centered at the origin.

□

Next we relate the gauge, support, and radial functions of a convex body containing the origin as an interior point. In particular, we will show that the support function of a convex body $K \in \mathcal{K}_0^n$ is the same as the gauge function of its polar body K°. We also state and prove an inequality that will be useful later.

Proposition 2.3.6. *Let $K \in \mathcal{K}_0^n$. The following properties hold:*

(i) $h_K(\cdot) = ||\cdot||_{K^\circ}$,

(ii) $||x||_K = 1/\rho_K(x)$ *for every* $x \neq 0_{\mathbb{R}^n}$,

(iii) $\langle x, y\rangle \leq ||x||_K\, ||y||_{K^\circ}$ *for every* $x, y \in \mathbb{R}^n$.

Moreover, if x and y are non-zero, then equality in **(iii)** *holds if and only if y is an outer normal of K in the point $x/||x||_K$.*

Proof. We start by **(i)**, and since the equality is obvious for $x = 0_{\mathbb{R}^n}$, we assume that x is a non-zero vector. If $x \in \lambda K^\circ$ for some $\lambda > 0$, then $x/\lambda \in K^\circ$. Thus,

$$\left\langle \frac{x}{\lambda}, z \right\rangle \leq 1$$

for every $z \in K$. This is the same as saying that $\langle x, z\rangle \leq \lambda$ for all $z \in K$, and hence

$$h_K(x) = \sup\{\langle x, z\rangle : z \in K\} \leq \lambda.$$

Passing to the infimum of the values $\lambda > 0$ such that $x \in \lambda K^\circ$, we get the inequality $h_K(x) \leq ||x||_{K^\circ}$. To prove the reverse inequality, first observe that $h_K(x) > 0$. Indeed, we are assuming that $x \neq 0_{\mathbb{R}^n}$, and since K is a convex body we have that $\alpha x \in K$ for some $\alpha > 0$ (let $rB^n \subseteq K$ for some $r > 0$ and choose $\alpha = r/|x|$, for example). Then $h_K(x) \geq \langle x, \alpha x\rangle = \alpha|x|^2 > 0$. We have $\langle x, z\rangle \leq h_K(x)$ for every $z \in K$, and hence

$$\left\langle \frac{x}{h_K(x)}, z \right\rangle \leq 1$$

for any $z \in K$. Consequently, we have that $x/h_K(x) \in K^\circ$, and hence $x \in h_K(x)K^\circ$. Therefore,

$$||x||_{K^\circ} = \inf\{\lambda > 0 : x \in \lambda K^\circ\} \leq h_K(x).$$

Now we prove **(ii)**. Let $x \neq 0_{\mathbb{R}^n}$ and recall that $\rho_K(x)$ is the unique positive number such that $\rho_K(x)x \in \partial K$. This gives, in particular, that

$$x \in \frac{1}{\rho_K(x)} K,$$

implying the inequality $||x||_K \leq 1/\rho_K(x)$. If the inequality is strict, then there exists a number $\mu < 1/\rho_K(x)$ such that $x \in \mu K$. Hence $x/\mu \in K$, and from Lemma 2.1.4 we get that $\text{seg}[0_{\mathbb{R}^n}, x/\mu) \subseteq$

int$K$. Since $\mu < 1/\rho_K(x)$, we have that $\rho_K(x) < 1/\mu$, and then $\rho_K(x)x \in \text{seg}[0_{\mathbb{R}^n}, x/\mu)$. This is a contradiction, because $\rho_K(x)x$ is not an interior point of K (it is a boundary point, instead). Therefore, $||x||_K = 1/\rho_K(x)$.

Next we prove inequality **(iii)**. The equality case clearly occurs if $x = 0_{\mathbb{R}^n}$ or $y = 0_{\mathbb{R}^n}$. We assume that x and y are non-zero. Observe that

$$\frac{x}{||x||_K} \in K,$$

and

$$\frac{y}{||y||_{K^\circ}} \in K^\circ.$$

Then, by the definition of polarity, we get

$$\left\langle \frac{x}{||x||_K}, \frac{y}{||y||_{K^\circ}} \right\rangle \leq 1.$$

This gives the desired inequality.

Now assume that y is an outer normal of K at $x/||x||_K$. The hyperplane supporting K with outer normal y is $\mathbf{h}(y, h_K(y))$. Equality case in **(iii)** is equivalent to

$$||y||_{K^\circ} = h_K(y) = \left\langle \frac{x}{||x||_K}, y \right\rangle,$$

and this is the same as saying that y is an outer normal of K at $x/||x||_K$.

□

Remark 2.3.10. There is another geometric interpretation for the equality in **(i)**: one can clearly construct a body in $\mathcal{K}_0^n$ by determining the intersections of its boundary with the rays $\mathbb{R}_+u$ for $u \in \mathbb{S}^{n-1}$ (that is, how do we "walk" in a given direction before touching the boundary) for each $u \in \mathbb{S}^{n-1}$. Thus, for each $u \in \mathbb{S}^{n-1}$ the point $\mathbb{R}_+u \cap \partial K^\circ$ is the *inversion* of the intersection of the ray $\mathbb{R}_+u$ with the supporting hyperplane of K having u as outer normal. This is illustrated in Figure 2.3.9.

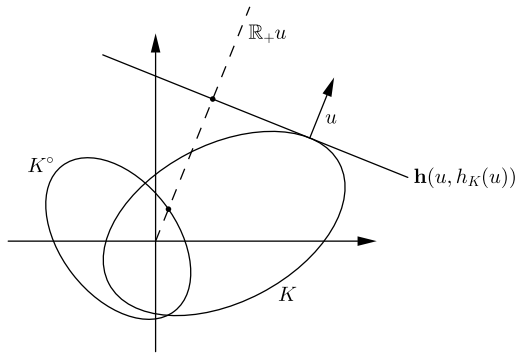


Fig. 2.3.9 $u/h_K(u) \in \mathbb{R}_+u \cap \partial K^\circ$ is the inversion of $h_K(u)u \in \mathbb{R}_+u \cap \mathbf{h}(u, h_K(u))$.

It might be also useful to understand the gauge functional of a convex body from the differential viewpoint. We discuss this in the next proposition. In what follows, $\mathcal{L}^n$ stands for the usual Lebesgue

measure of $\mathbb{R}^n$ (see Appendix A for more on measure theory). We also mention that the next proposition can be partially obtained as a consequence of Exercise 1.15.

Proposition 2.3.7. *Let $K \subseteq \mathbb{R}^n$ be a convex body containing the origin as an interior point, and denote its gauge function as*

$$\gamma_K(x) := ||x||_K := \inf\{\lambda > 0 : x \in \lambda K\},$$

for each $x \in \mathbb{R}^n$. Then γ_K is differentiable $\mathcal{L}^n$-almost everywhere. We have that γ_K is differentiable at $x \in \partial K$ if and only if it is differentiable at tx for any $t > 0$. In this case, we have the equality

$$\nabla\gamma_K(tx) = \nabla\gamma_K(x). \tag{2.3.2}$$

Moreover, if γ_K is differentiable at $x \in \partial K$, then K has a unique outer normal unit vector $u_K(x)$ at x, and

$$\nabla\gamma_K(x) = h_K(u_K(x))^{-1}u_K(x). \tag{2.3.3}$$

Consequently, $\nabla\gamma_K(x)$ is non-zero and

$$|\nabla\gamma_K(x)| = h_K(u_K(x))^{-1}, \tag{2.3.4}$$

whenever γ_K is differentiable at $x \in \partial K$.

Remark 2.3.11. The only reason why we introduce the new notation γ_K for the gauge function is because it is more convenient to write $\nabla\gamma_K(x)$ instead of $\nabla||\cdot||_K(x)$ to represent the gradient evaluated at x.

Proof. We prove that γ_K is Lipschitz. To do so, let $\rho > 0$ be any number such that $K \subseteq \rho B^n$. Hence $x \in \lambda K$ implies $x \in \lambda\rho B^n$, from which we get that $\gamma_K(\cdot) \leq \gamma_{\rho B^n}(\cdot) = \rho|\cdot|$. Thus, from the triangle inequality for the gauge function, we obtain

$$|\gamma_K(x) - \gamma_K(y)| \leq \gamma_K(x - y) \leq \rho|x - y|$$

for any $x, y \in \mathbb{R}^n$. Therefore, from Rademacher's classical theorem (see [2249]), which states that every Lipschitz function of $\mathbb{R}^n$ is differentiable $\mathcal{L}^n$-almost everywhere, we get that γ_K is differentiable (in the strong sense, that is, not only in the sense of partial derivatives) almost everywhere.

To prove the other claim, assume that γ_K is differentiable at $x \in \partial K$, let $t > 0$ be arbitrarily given, and let $T\colon \mathbb{R}^n \to \mathbb{R}$ be the linear map defined as $T(\cdot) = \langle\nabla\gamma_K(x), \cdot\rangle$. We calculate

$$\lim_{h\to 0_{\mathbb{R}^n}} \frac{\gamma_K(tx + h) - \gamma_K(tx) - T(h)}{|h|} = \lim_{h\to 0_{\mathbb{R}^n}} \frac{t\big(\gamma_K(x + h/t) - \gamma_K(x) - \langle\nabla\gamma_K(x), h/t\rangle\big)}{|h|} =$$
$$= \lim_{v\to 0_{\mathbb{R}^n}} \frac{\gamma_K(x + v) - \gamma_K(x) - \langle\nabla\gamma_K(x), v\rangle}{|v|} = 0,$$

where the last equality comes from the differentiability of γ_K at x. Hence γ_K is differentiable at tx, and the derivative of γ_K at tx is given as

$$\big(d\gamma_K\big)_{tx}(\cdot) = \langle\nabla\gamma_K(x), \cdot\rangle.$$

This leads immediately to the equality $\nabla\gamma_K(tx) = \nabla\gamma_K(x)$. The converse is completely analogous. Observe carefully that the unique property of γ_K that we used here is that it is positively homogeneous of degree 1.

The advantage of what we have just proved is that it allows us to understand the behavior of $\nabla\gamma_K(\cdot)$ in the entire $\mathbb{R}^n$ looking only for the points of ∂K, and this is the reason why we enunciate the other claims only for points of ∂K.

Assume that γ_K is differentiable at $x \in \partial K$ and let $u \in \mathbb{S}^{n-1}$ be any unit outer normal vector of K at x (we do not know *a priori* that this is unique). Thus, K is supported at x by the hyperplane $\mathbf{h} := \mathbf{h}(u, h_K(u))$ and $\mathrm{int}K \subseteq \mathbf{h}^-$. If $v \in \mathbb{R}^n$ is such that $\langle u, v\rangle = 0$, then $x + \lambda v \in \mathbf{h}$ for each $\lambda \in \mathbb{R}$. Hence $x + \lambda v \notin \mathrm{int}K$ (geometrically, we are "moving" through a supporting hyperplane, and hence we cannot "enter" the interior of K). Hence the inequality

$$\gamma_K(x + \lambda v) \geq 1$$

holds for every $\lambda > 0$. Now, since $\langle \nabla\gamma_K(x), v\rangle$ is the directional derivative of γ_K at x and observing that we obviously have $\gamma_K(x) = 1$, we get

$$\langle \nabla\gamma_K(x), v\rangle = \lim_{\lambda\to 0^+} \frac{\gamma_K(x + \lambda v) - \gamma_K(x)}{\lambda} = \lim_{\lambda\to 0^+} \frac{\gamma_K(x + \lambda v) - 1}{\lambda} \geq 0.$$

On the other hand,

$$\langle \nabla\gamma_K(x), v\rangle = \lim_{\lambda\to 0^-} \frac{\gamma_K(x + \lambda v) - \gamma_K(x)}{\lambda} = \lim_{\lambda\to 0^-} \frac{\gamma_K(x + \lambda v) - 1}{\lambda} \leq 0.$$

Therefore, we have that $\langle \nabla\gamma_K(x), v\rangle = 0$ whenever $\langle u, v\rangle = 0$. Consequently, we get that $\nabla\gamma_K(x) = \alpha u$ for some $\alpha \in \mathbb{R}$. In other words, if u is an outer normal unit vector of K at x, then $\nabla\gamma_K(x)$ is a multiple of u. This means that if K admits two distinct unit outer normal vectors at x, then $\nabla\gamma_K(x) = 0$. Moreover, we have

$$\langle \nabla\gamma_K(x), x\rangle = \lim_{\lambda\to 0} \frac{\gamma_K(x + \lambda x) - \gamma_K(x)}{\lambda} = \lim_{\lambda\to 0} \frac{1 + \lambda - 1}{\lambda} = 1,$$

yielding two consequences:

(i) u is the unique unit outer normal vector of K at x. Indeed, the equality above guarantees that $\nabla\gamma_K(x) \neq 0$.

(ii) $1 = \langle \nabla\gamma_K(x), x\rangle = \langle \alpha u, x\rangle = \alpha h_K(u)$.

Once K is a convex body containing the origin as an interior point we have $h_K(u) > 0$. Therefore, $\nabla\gamma_K(x) = \alpha u = h_K(u)^{-1}u$, and the proof is complete.

□

A boundary point x of a convex body K is said to be a *smooth point* (or *regular point*) if K has a unique unit outer normal vector at x (or, equivalently, if K is supported by a unique hyperplane at x); see Figure 2.3.10. A convex body K is said to be *smooth* (or *regular*) if all points of ∂K are smooth. As a consequence of the previous proposition, we have a very useful correspondence between the smooth points of a convex body and the points where its gauge function is differentiable.

Theorem 2.3.2. *For any convex body $K \subseteq \mathbb{R}^n$ which has the origin as an interior point we have*

$$\{\text{points of } \partial K \text{ where } \gamma_K \text{ is differentiable}\} = \{\text{smooth points of } \partial K\}.$$

Proof. The inclusion

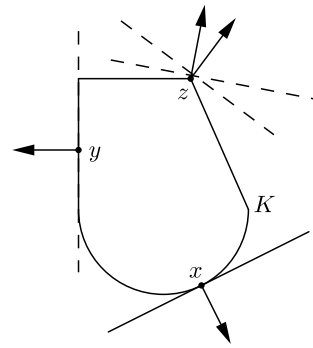


Fig. 2.3.10 The boundary points x and y are smooth, while z is not.

$$\{\text{points of } \partial K \text{ where } \gamma_K \text{ is differentiable}\} \subseteq \{\text{smooth points of } \partial K\}$$

was proved in Proposition 2.3.7. Hence we only need to prove the reverse inclusion. Let $x \in \partial K$ be a smooth point, and let $u \in \mathbb{S}^{n-1}$ be the unit outer normal of K at x. If $v \in \partial\gamma_K(x)$, then from Proposition 1.3.3 we have that v is an outer normal vector of the sublevel set $\{\gamma_K \leq 1\} = K$ at x. Therefore, we may write $v = \alpha u$ for some $\alpha > 0$. From the sublinearity of γ_K, it is easy to see that

$$(\gamma_K)'_+(x, x) = -(\gamma_K)'_+(x, -x) = \gamma_K(x).$$

These equalities, together with Theorem 1.3.1, yield

$$\begin{aligned} 1 = \gamma_K(x) = (\gamma_K)'_+(x, x) \geq \langle v, x\rangle = \alpha\langle u, x\rangle \text{ and} \\ -1 = -\gamma_K(x) = (\gamma_K)'_+(x, -x) \geq \langle v, -x\rangle = \alpha\langle u, -x\rangle. \end{aligned}$$

Because $\langle u, x\rangle = h_K(u) > 0$, these inequalities lead to

$$\frac{1}{\langle u, x\rangle} \leq \alpha \leq \frac{1}{\langle u, x\rangle},$$

and hence α is uniquely determined. It follows that $\partial\gamma_K(x)$ is a singleton, and thus Theorem 1.3.2 gives that γ_K is differentiable at x.

□

Remark 2.3.12. Clearly, we also have

$$\{\text{points of } t\partial K \text{ where } \gamma_K \text{ is differentiable}\} = \{\text{smooth points of } t\partial K\},$$

for every $t > 0$. Indeed, it is immediate that $tz \in t\partial K$ is a smooth point of tK if and only if z is a smooth point of K, and from Proposition 2.3.7 we have that γ_K is differentiable at tz if and only if γ_K is differentiable at z.

This may be interpreted as a correspondence between the geometric structure of K and the regularity of its gauge function γ_K. We can even go further: by Theorem 1.3.3 differentiable convex functions are C^1, and this leads to a stronger regularity property of gauge functions of smooth convex bodies. This is explained below.

Corollary 2.3.7. *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point. The gauge function γ_K is C^1 over $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ if and only if K is smooth.*

Proof. If γ_K is C^1 over $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, then in particular we have that γ_K is differentiable at each point of ∂K, and it follows from Theorem 2.3.2 that all points of ∂K are smooth.

Conversely, if K is smooth, then γ_K is differentiable at each point of ∂K. From Proposition 2.3.7, it follows that γ_K is differentiable at the point tx for each $t > 0$ and $x \in K$, and this gives that γ_K is differentiable for every point of $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Notice that, at this point, we still did not have proved that γ_K is continuously differentiable in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ (but only differentiable). This last step follows from Theorem 1.3.3: if $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, then we let $\varepsilon > 0$ be such that $B(x, \varepsilon) \subseteq \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. The restriction of γ_K to $B(x, \varepsilon)$ is a convex differentiable (finite) function, and hence it is C^1. In particular, the derivative of γ_K is continuous at x, and this concludes the proof.

□

Because of the duality relation $h_K = \gamma_{K^\circ}$, which holds for any convex body $K \subseteq \mathbb{R}^n$ containing the origin as an interior point, we may use these results on the regularity of gauge functions to investigate also the regularity of support functions. As we will see next, the differentiability of h_K is not related to the smoothness of K, but to another geometric property: the so-called *strict convexity*. A convex body $K \subseteq \mathbb{R}^n$ is said to be *strictly convex* if its boundary contains no line segments. A different characterization is useful.

Lemma 2.3.8. *A convex body $K \subseteq \mathbb{R}^n$ is strictly convex if and only if each $u \in \mathbb{S}^{n-1}$ is a unit outer normal of K at a unique point $z \in \partial K$.*

Proof. If K is not strictly convex, then we may let $\operatorname{seg}[x, y] \subseteq \partial K$ for $x \neq y$. Then $z := (x + y)/2$ is a boundary point of K, and hence K is supported by a hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha)$ at z (with $K \subseteq \mathbf{h}^-$, say). Hence we have

$$\alpha = \langle u, z \rangle = \left\langle u, \frac{x + y}{2} \right\rangle = \frac{1}{2}\langle u, x \rangle + \frac{1}{2}\langle u, y \rangle.$$

Because x and y are points of K and $K \subseteq \mathbf{h}^-$, we also have that $\langle u, x \rangle \leq \alpha$ and $\langle u, y \rangle \leq \alpha$. It follows that $\langle u, x \rangle = \langle u, y \rangle = \alpha$, and hence u is a unit outer normal of K at x and y.

Now assume that $u \in \mathbb{S}^{n-1}$ is a unit outer normal of K at distinct points $x, y \in \partial K$. In this case, it is clear that K is supported at x and y by the same hyperplane $\mathbf{h} := \mathbf{h}(u, \alpha)$. Thus, $x, y \in K \cap \mathbf{h}$, and by convexity it follows that $\operatorname{seg}[x, y] \subseteq K \cap \mathbf{h}$. Once $\mathbf{h}$ cannot have interior points of K, we obtain that $\operatorname{seg}[x, y] \subseteq \partial K$, and then K is not strictly convex.

□

Proposition 2.3.8. *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point. Then K is strictly convex if and only if K° is smooth.*

Remark 2.3.13. Of course, by duality we also have that K is smooth if and only if K° is strictly convex.

Proof. Assume that K is not strictly convex, and let $\operatorname{seg}[x, y] \subseteq \partial K$. As in the proof of Lemma 2.3.8, we let $\mathbf{h} := \mathbf{h}(u, \alpha)$ be a supporting hyperplane of K at $z := (x + y)/2$ (with $K \subseteq \mathbf{h}^-$, say), and we obtain that $\mathbf{h}$ also supports K at x and y. Thus,

$$\langle x, u \rangle = \langle y, u \rangle = h_K(u) = \gamma_{K^\circ}(u).$$

The point $\hat{u} := u/\gamma_{K^\circ}(u)$ is in ∂K°, and

$$\langle x, u \rangle = \left\langle x, \frac{u}{\gamma_{K^\circ}(u)} \right\rangle = 1,$$

and since $x \in K$ we have that $\langle x, z \rangle \leq 1$ for every $z \in K^\circ$ (because $K^{\circ\circ} = K$). It follows that K° is supported by the hyperplane $\mathbf{h}(x, 1)$ at $\hat{u}$, with x as outer normal. By the same argument, K° is also

supported by $\mathbf{h}(y, 1)$ at $\hat{u}$, with y as outer normal. Once $0_{\mathbb{R}^n}$ is an interior point of K and x and y are boundary points of K, we must have that x and y are linearly independent. Then, we get that $x/|x|$ and $y/|y|$ are distinct unit outer normal vectors of K° at $\hat{u}$, and hence K° is not smooth.

Now assume that K° is not smooth, and let $z \in \partial K^\circ$ be a point where K has two distinct unit outer normal vectors (u and v, say). Thus, $\gamma_K(u) = h_{K^\circ}(u) = \langle z, u\rangle$ and $\gamma_K(v) = h_{K^\circ}(v) = \langle z, v\rangle$. Let

$$\hat{u} := \frac{u}{\gamma_K(u)} = \frac{u}{\langle z, u\rangle} \quad \text{and}$$
$$\hat{v} := \frac{v}{\gamma_K(v)} = \frac{v}{\langle z, v\rangle}.$$

We have that $\hat{u}$ and $\hat{v}$ are distinct points of ∂K, and we claim that z is an outer normal of K on both. Indeed, it is clear that $\langle z, \hat{u}\rangle = 1$, and since $z \in K^\circ$ we have $\langle x, z\rangle \leq 1$ for every $x \in K$. Hence the hyperplane $\mathbf{h}(z, 1)$ supports K at $\hat{u}$, with outer normal z. By the same argument, we obtain that $\mathbf{h}(z, 1)$ also supports K at $\hat{v}$, with z being an outer normal. Repeating the argument in the proof of Lemma 2.3.8, we get that $\operatorname{seg}[\hat{u}, \hat{v}] \subseteq \partial K$, and then K is not strictly convex.

□

Remark 2.3.14. In this proof, we explored a certain duality between the outer normal vectors of K and the points of K°. This relation is made more precise in Exercise 2.42. The reader may notice that this also appears in the middle of the proof of Proposition 2.3.9 below.

This duality between smoothness and strict convexity given by the polarity explains why the regularity of support functions is related to strict convexity. This is clarified next.

Proposition 2.3.9. *The support function h_K of a convex body $K \subseteq \mathbb{R}^n$ is C^1 over $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ if and only if K is strictly convex. In this case, for each $u \in \mathbb{S}^{n-1}$ we have that $\nabla h_K(u) \in \partial K$, and this is the unique point of ∂K where u is an outer normal of K.*

Remark 2.3.15. To avoid any confusion we emphasize that, here, ∇ still stands for the usual gradient in $\mathbb{R}^n$, as before. That is, $\nabla h_K(u)$ is the gradient of h_K in $\mathbb{R}^n$ evaluated at a point u of $\mathbb{S}^{n-1}$. We are not considering the gradient of the restriction of h_K to $\mathbb{S}^{n-1}$.

Proof. Assume first that K contains the origin in its interior. If h_K is C^1 in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, then once $h_K = \gamma_{K^\circ}$ we have that K° is smooth. Thus, K is strictly convex. Conversely, if K is strictly convex, then K° is smooth, and hence $h_K = \gamma_{K^\circ}$ is C^1 over $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$.

Now assume that K is strictly convex and let $u \in \mathbb{S}^{n-1}$, and write $\hat{u} = u/\gamma_{K^\circ}(u)$. Hence $\hat{u} \in \partial K^\circ$, and from Proposition 2.3.7 we have

$$\nabla h_K(u) = \nabla \gamma_{K^\circ}(u) = \nabla \gamma_{K^\circ}(\hat{u}) = h_{K^\circ}(u_{K^\circ}(\hat{u}))^{-1} u_{K^\circ}(\hat{u}),$$

where $u_{K^\circ}(\hat{u})$ denotes the unique unit outer normal of K° at $\hat{u}$. For simplicity, write $u_{K^\circ}(\hat{u}) = v$. Then, recalling that $h_{K^\circ} = \gamma_K$, we may rewrite the equality above as

$$\nabla h_K(u) = \frac{v}{\gamma_K(v)},$$

and it follows that $\nabla h_K(u) \in \partial K$. Because $\hat{u} \in K^\circ$, we have that $\langle x, \hat{u}\rangle \leq 1$ for every $x \in K$. In other words,

$$\langle x, u\rangle \leq \gamma_{K^\circ}(u) = h_K(u)$$

for every $x \in K$. On the other hand, since v is an outer normal of K° at $\hat{u}$, we have

$$\langle v, \hat{u}\rangle = h_{K^\circ}(v) = \gamma_K(v),$$

and hence

$$\langle \nabla h_K(u), u\rangle = \left\langle \frac{v}{\gamma_K(v)}, \gamma_{K^\circ}(u)\hat{u}\right\rangle = \frac{\langle v, \hat{u}\rangle}{\gamma_K(v)}\gamma_{K^\circ}(u) = h_K(u).$$

Summarizing, we have that $K \subseteq \mathbf{h}^-(u, h_K(u))$ and $\nabla h_K(u) \in \partial K \cap \mathbf{h}(u, h_K(u))$. Thus, u is an outer normal of K at $\nabla h_K(u)$.

Now we go for the general case, that is, the case where K does not necessarily contain the origin as an interior point. In this case, we let $x_0 \in \mathbb{R}^n$ be such that $0_{\mathbb{R}^n} \in \operatorname{int}(K + x_0)$. It is clear that K is strictly convex if and only if $K + x_0$ is strictly convex, and since

$$h_{K+x_0}(\cdot) = h_K(\cdot) + h_{\{x_0\}}(\cdot) = h_K(\cdot) + \langle x_0, \cdot\rangle,$$

we have that h_K is C^1 in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ if and only if the same holds for h_{K+x_0}. This shows that K is C^1 in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ if and only if K is strictly convex.

On the other hand, if K is strictly convex and $0_{\mathbb{R}^n} \in \operatorname{int}(K + x_0)$, then for each $u \in \mathbb{S}^{n-1}$ we have that $\nabla h_{K+x_0}(u)$ is the unique point of $\partial(K + x_0)$ where u is an outer normal of $K + x_0$. As an immediate consequence, we get that the unique point of ∂K where u is an outer normal is

$$z := \nabla h_{K+x_0}(u) - x_0.$$

We may compute the gradient of h_{K+x_0} as

$$\nabla h_{K+x_0} = \nabla(h_K + \langle x_0, \cdot\rangle) = \nabla h_K + x_0,$$

since it is easy to see that the gradient of the linear function $\langle x_0, \cdot\rangle$ is x_0. Thus,

$$z = \nabla h_{K+x_0}(u) - x_0 = \nabla h_K(u) + x_0 - x_0 = \nabla h_K(u),$$

and this concludes the proof.

□

Remark 2.3.16. Since $\nabla h_K(u)$ is the point of ∂K where u is an outer normal of K, we have the equality $\langle \nabla h_K(u), u\rangle = h_K(u)$. This type of equality is characteristic for homogeneous functions; see, e.g., Lemma 8.2.2.

2.4 Convex cones and boundary structure

The *positive hull* of a subset $A \subseteq \mathbb{R}^n$ is the set of *positive combinations* of the points of A:

$$\operatorname{pos}(A) := \left\{ \sum_{j=1}^{m} \lambda_j x_j : \lambda_j \geq 0 \text{ and } x_j \in A \text{ for each } 1 \leq j \leq m \right\}.$$

A *convex cone* is a convex set $C \subseteq \mathbb{R}^n$ having the property that $\lambda x \in C$ whenever $\lambda \geq 0$ and $x \in C$. A set $C_0 \subseteq \mathbb{R}^n$ is said to be a *convex cone with apex at* $x_0 \in \mathbb{R}^n$ if $C_0 = x_0 + C$ for some convex cone C (hence what we call "convex cone" are the convex cones with apex at the origin).

An *open convex cone* is an open convex set $C \subseteq \mathbb{R}^n$ with the property that $\lambda x \in C$ whenever $x \in C$ and $\lambda > 0$ (this new definition is necessary because the origin is a point of any non-empty convex cone). On the other hand, notice carefully that a convex cone is not necessarily closed, and a *closed convex cone* is simply a convex cone which is (topologically) closed. Similarly to what we have defined

previously, a set $C_0 \in \mathbb{R}^n$ is an *open convex cone with apex at* $x_0 \in \mathbb{R}^n$ if $C_0 = x_0 + C$ for some open convex cone $C \subseteq \mathbb{R}^n$.

Lemma 2.4.1. *Let $C \subseteq \mathbb{R}^n$ be a non-empty convex set. Then C is a convex cone if and only if $C = \mathrm{pos}(C)$.*

Proof. Let C be a non-empty convex cone, and let

$$x = \lambda_1 x_1 + \ldots + \lambda_m x_m$$

be a positive combination of points of C, where we assume that at least one of the λ_j's is non-zero. Let $\lambda = (\lambda_1 + \ldots + \lambda_m)^{-1}$. Then λx is a convex combination of points of C, and hence $\lambda x \in C$. Since C is a convex cone, we get that $x = \lambda^{-1}(\lambda x) \in C$. It follows that $\mathrm{pos}(C) \subseteq C$ (observe that $0_{\mathbb{R}^n}$ is trivially in both sets). Since it is obvious that $C \subseteq \mathrm{pos}(C)$, we have the first implication.

Now let C be a non-empty convex set such that $C = \mathrm{pos}(C)$. If $x \in C$ and $\lambda \geq 0$, then $\lambda x \in \mathrm{pos}(C)$, and hence $\lambda x \in C$. This shows that C is a convex cone.

□

Corollary 2.4.1. *Let $C \subseteq \mathbb{R}^n$ be a non-empty convex set. Then C is a convex cone with apex at $x_0 \in \mathbb{R}^n$ if and only if $C - x_0 = \mathrm{pos}(C - x_0)$.*

Proof. The proof is straightforward.

□

In the next proposition we collect some properties of positive hulls of convex sets which will be useful later.

Proposition 2.4.1. *Let $C \subseteq \mathbb{R}^n$ be a non-empty convex set. Then we have*

(a) $\mathrm{pos}(C) = \{\lambda x : \lambda \geq 0 \text{ and } x \in C\}$. *In particular,* $\mathrm{pos}(C)$ *is convex;*

(b) $\mathrm{relint}(\mathrm{pos}(C)) \cup \{0_{\mathbb{R}^n}\} = \mathrm{pos}(\mathrm{relint} C)$.

Proof. In both cases **(a)** and **(b)** it is clear that $0_{\mathbb{R}^n}$ lies in all involved sets, and hence we will always work with non-zero vectors without further comments. The inclusion $\{\lambda x : \lambda \geq 0 \text{ and } x \in C\} \subseteq \mathrm{pos}(C)$ is immediate, and holds even if C is not convex. To prove the reverse inclusion, we let $z = \lambda_1 x_1 + \ldots \lambda_m x_m$ be any non-zero point of $\mathrm{pos}(C)$, with $x_1, \ldots, x_m \in C$ and $\lambda_1, \ldots, \lambda_m \geq 0$, and observe that in this case $\lambda := \lambda_1 + \ldots + \lambda_m > 0$ (because z is non-zero). Writing

$$z = \lambda \left(\frac{\lambda_1}{\lambda} x_1 + \ldots + \frac{\lambda_m}{\lambda} x_m \right),$$

we get that the term between the parentheses is a convex combination of points of C, hence a point of C. This gives the inclusion $\mathrm{pos}(C) \subseteq \{\lambda x : \lambda \geq 0 \text{ and } x \in C\}$ and the proof of the equality in **(a)** is complete. To verify that $\mathrm{pos}(C)$ is convex whenever C is convex, let $\lambda_1 x$ and $\lambda_2 y$ be (distinct) arbitrary points in $\mathrm{pos}(C)$, with $x, y \in C$ and $\lambda_1, \lambda_2 \geq 0$ (at least one of them is non-zero). For any $\lambda \in (0, 1)$, write

$$(1-\lambda)\lambda_1 x + \lambda\lambda_2 y = \big((1-\lambda)\lambda_1 + \lambda\lambda_2\big) \left[\frac{(1-\lambda)\lambda_1}{(1-\lambda)\lambda_1 + \lambda\lambda_2} x + \frac{\lambda\lambda_2}{(1-\lambda)\lambda_1 + \lambda\lambda_2} y \right],$$

and since C is convex the expression inside the brackets is a point of C. Together with the fact that $(1-\lambda)\lambda_1 + \lambda\lambda_2 > 0$ we get that $(1-\lambda)\lambda_1 x + \lambda\lambda_2 y \in \mathrm{pos}(C)$, and this settles **(a)**.

Now we prove **(b)**. Let $z = \lambda_0 x \neq 0_{\mathbb{R}^n}$ with $\lambda_0 > 0$ and $x \in \operatorname{relint} C$ (that is, z is an arbitrary point of $\operatorname{pos}(\operatorname{relint} C)$). To prove that $z \in \operatorname{relint}(\operatorname{pos}(C))$, we assume by contradiction that $z \in \operatorname{relbd}(\operatorname{pos}(C))$. Since

$$\operatorname{relint}\{z\} \cap \operatorname{relint}(\operatorname{pos}(C)) = \emptyset,$$

from Theorem 2.2.3 we have that $\{z\}$ and $\operatorname{pos}(C)$ can be properly separated by a hyperplane $\mathbf{h}$. We may assume that $\operatorname{pos}(C) \subseteq \mathbf{h}^-$, and since $z \in \operatorname{pos}(C) \cap \mathbf{h}^+$ we get that $z \in \mathbf{h}$. Hence $\mathbf{h}$ is, in particular, a supporting hyperplane of $\operatorname{pos}(C)$ at z. Once the separation is proper, there exists a point $y_0 \in \operatorname{relint}(\operatorname{pos}(C)) \cap \operatorname{int}(\mathbf{h}^-)$, because otherwise we would have $\operatorname{pos}(C) \subseteq \mathbf{h}$. It follows that

$$\operatorname{relint}(\operatorname{pos}(C)) \subseteq \operatorname{int}(\mathbf{h}^-). \tag{2.4.1}$$

Indeed, if there is a point $y_1 \in \operatorname{relint}(\operatorname{pos}(C)) \cap \mathbf{h}$, then for some small $\varepsilon > 0$ we have that $y_1 + \varepsilon(y_1 - y_0)$ is a point which lies simultaneously in $\operatorname{pos}(C)$ and $\operatorname{int}(\mathbf{h}^+)$, a contradiction.

We may let $\mathbf{h} = \mathbf{h}(u, \alpha)$. We will show that $0_{\mathbb{R}^n} \in \mathbf{h}$ or, equivalently, that $\alpha = 0$. To do so, we observe that $\langle u, z\rangle = \alpha$ (because $z \in \mathbf{h}$, of course) and $\lambda z \in \operatorname{pos}(C) \subseteq \mathbf{h}^-$ for any $\lambda \geq 0$, and hence

$$\lambda\alpha = \lambda\langle u, z\rangle = \langle \lambda z, u\rangle \leq \alpha$$

for every $\lambda > 0$. This is only possible if $\alpha = 0$. As a consequence, $0 = \langle u, z\rangle = \lambda_0\langle u, x\rangle$, from which we get that $x \in \mathbf{h}$. This is a contradiction with (2.4.1), because $x \in \operatorname{relint} C \subseteq \operatorname{relint}(\operatorname{pos}(C))$. Thus, we have that $\operatorname{pos}(\operatorname{relint} C) \subseteq \operatorname{relint}(\operatorname{pos}(C))$.

To prove the converse, let $z \in \operatorname{relint}(\operatorname{pos}(C))$ be non-zero. We first notice that it suffices to prove that $\operatorname{relint}(\mathbb{R}_+ z) \cap \operatorname{relint}(C) \neq \emptyset$, because if y is a point of this intersection, then $y = \lambda_1 z$ for some $\lambda_1 > 0$ and $y \in \operatorname{relint} C$. Thus, $z = \frac{1}{\lambda_1} y \in \operatorname{pos}(\operatorname{relint} C)$. Assume by contradiction that $\operatorname{relint}(\mathbb{R}_+ z) \cap \operatorname{relint}(C) = \emptyset$. Then, using Theorem 2.2.3 again, we get that there exists a hyperplane $\mathbf{h}$ properly separating $\mathbb{R}_+ z$ and C (with $C \subseteq \mathbf{h}^-$, say). Recalling that z is a point of the relative interior of $\operatorname{pos}(C)$, we have, in particular, that $z \in \operatorname{pos}(C)$, and hence $z = \lambda_0 x$ for some $\lambda_0 > 0$ (recall that z is non-zero) and some $x \in C$. Therefore, $x \in \mathbb{R}_+ z \cap C$, which leads to $x \in \mathbf{h}$. Writing $\mathbf{h} = \mathbf{h}(u, \alpha)$, we have thus

$$\lambda\alpha = \lambda\langle x, u\rangle = \langle \lambda x, u\rangle \geq \alpha$$

for every $\lambda > 0$, where the last inequality comes from the obvious fact that $\lambda x \in \mathbb{R}_+ z$ for all $\lambda > 0$. This inequality has two important consequences: that $\alpha = 0$ and that $\mathbb{R}_+ z \subseteq \mathbf{h}$. Once the separation is proper, we get that $C \cap \operatorname{int}(\mathbf{h}^-) \neq \emptyset$, and similarly to what we have argued before, this leads to $\operatorname{relint} C \subseteq \operatorname{int}(\mathbf{h}^-)$. In particular,

$$\operatorname{pos}(C) \cap \operatorname{int}(\mathbf{h}^-) \neq \emptyset, \tag{2.4.2}$$

because $\operatorname{relint} C \subseteq \operatorname{pos}(C)$. We also have the inclusion

$$\operatorname{pos}(C) \subseteq \mathbf{h}^-, \tag{2.4.3}$$

because $C \subseteq \mathbf{h}^-$ gives that $\langle w, u\rangle \leq 0$ for all $w \in C$, and this yields $\langle \lambda w, u\rangle \leq 0$ for every $\lambda > 0$ and $w \in C$. We claim that (2.4.2) and (2.4.3) lead to

$$\mathbf{h} \cap \operatorname{relint}(\operatorname{pos}(C)) = \emptyset. \tag{2.4.4}$$

In fact, if we assume that there exists a point $y_0 \in \mathbf{h} \cap \operatorname{relint}(\operatorname{pos}(C))$, then, choosing any point $y_1 \in \operatorname{pos}(C) \cap \operatorname{int}(\mathbf{h}^-)$, we would have $y_0 + \varepsilon(y_0 - y_1) \in \operatorname{pos}(C) \cap \operatorname{int}(\mathbf{h}^+)$ for some $\varepsilon > 0$, which

contradicts (2.4.3). Finally, (2.4.4) contradicts our assumption that $z \in \text{relint}(\text{pos}(C))$ and the fact that $\mathbb{R}_+ z \subseteq \mathbf{h}$. Hence the proof is complete.

□

Remark 2.4.1. Actually, the positive hull of *any* set is convex (even more, the positive hull of any set is a convex cone). See Exercise 2.44.

Let K be a non-empty, compact, and convex set, and let $x \in K$. The *support cone of K at x* is the set

$$\text{SC}_K(x) := \text{cl}\Big(\{\lambda(z - x) : z \in K \text{ and } \lambda > 0\}\Big) = \text{cl}\left(\bigcup_{\lambda>0} \lambda(K - x)\right);$$

see Figure 2.4.1. Here we observe that if $x \in \text{relint}(K)$, then $\text{SC}_K(x) = \text{aff}(K) - x$, and the converse is also obviously true. We also define the *normal cone* of K at a point $x \in \text{relbd}K$ to be the set

$$\text{NC}_K(x) := \{u \in \mathbb{R}^n : u \text{ is an outer normal of } K \text{ at } x\} \cup \{0_{\mathbb{R}^n}\}, \tag{2.4.5}$$

where we recall that u is an outer normal of x at K if

$$K \subseteq \mathbf{h}^-(u, \langle u, x\rangle),$$

and in this case we also have that $h_K(u) = \langle u, x\rangle$. Next we state and prove a useful equivalent definition for the normal cone.

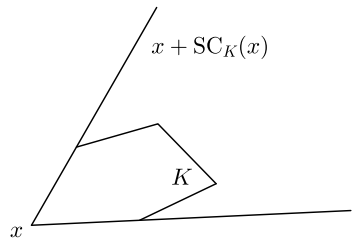


Fig. 2.4.1 The (translated) support cone of K at x.

Lemma 2.4.2. *For any non-empty compact convex set $K \subseteq \mathbb{R}^n$ and every $x \in \text{relbd}K$ we have the equality*

$$\text{NC}_K(x) = \{u \in \mathbb{R}^n : \langle u, z\rangle \le \langle u, x\rangle \text{ for every } z \in K\}.$$

Proof. If $u \in \mathbb{R}^n$ is a non-zero vector such that $\langle u, z\rangle \le \langle u, x\rangle$ for every $z \in K$, then $K \subseteq \mathbf{h}^-(u, \langle u, x\rangle)$, and hence u is an outer normal of K at x. It follows that $u \in \text{NC}_K(x)$. Conversely, if $u \in \text{NC}_K(x)$ is non-zero, then u is an outer normal of K at x. Thus, we have $K \subseteq \mathbf{h}^-(u, \langle u, x\rangle)$, and this gives that $\langle u, z\rangle \le \langle u, x\rangle$ for any $z \in K$. This concludes the proof of the first equality, since $0_{\mathbb{R}^n}$ is obviously an element of both sets.

□

Proposition 2.4.2. *Let $K \subseteq \mathbb{R}^n$ be a non-empty, compact, and convex set. For any $x \in \text{relbd}K$ the support cone and the normal cone of K at x are closed convex cones, and*

$$\text{NC}_K(x) = \text{SC}_K(x)^\circ.$$

Proof. The support cone is obviously a non-empty closed set, and it is also clear that it coincides with its positive hull. To verify that it is convex, let $\lambda_1, \lambda_2 > 0$, $z_1, z_2 \in K$ and $\lambda \in (0,1)$. Then $(1-\lambda)\lambda_1 + \lambda\lambda_2 > 0$, and hence

$$(1-\lambda)\lambda_1(z_1 - x) + \lambda\lambda_2(z_2 - x) =$$
$$= ((1-\lambda)\lambda_1 + \lambda\lambda_2)\left[\frac{(1-\lambda)\lambda_1}{(1-\lambda)\lambda_1 + \lambda\lambda_2} z_1 + \frac{\lambda\lambda_2}{(1-\lambda)\lambda_1 + \lambda\lambda_2} z_2 - x\right].$$

Since K is convex we have that

$$\frac{(1-\lambda)\lambda_1}{(1-\lambda)\lambda_1 + \lambda\lambda_2} z_1 + \frac{\lambda\lambda_2}{(1-\lambda)\lambda_1 + \lambda\lambda_2} z_2 \in K,$$

and this gives that the set

$$\{\lambda(z - x) : \lambda > 0 \text{ and } z \in K\}$$

is convex. This implies that its closure also has this property. It follows that $\mathrm{SC}_K(x)$ is a closed convex cone.

It is immediate by Lemma 2.4.2 that the normal cone $\mathrm{NC}_K(x)$ is a convex set which coincides with its positive hull (any positive multiple of an outer normal is also an outer normal), and then we only have to prove that it is closed. This comes straightforwardly from Lemma 2.4.2 together with the continuity of the inner product: if $(u_j)_{j\in\mathbb{N}}$ is a sequence in $\mathrm{NC}_K(x)$ such that $u_j \to u$ in $\mathbb{R}^n$, then for each $z \in K$ we have that

$$\langle u_j, z\rangle \leq \langle u, x\rangle$$

for any $j \in \mathbb{N}$, and (passing to the limit as $j \to \infty$) this leads to $\langle u, z\rangle \leq \langle u, x\rangle$. Since this holds for any $z \in K$, we get that $u \in \mathrm{NC}_K(x)$.

Now we prove the equality. If $u \in \mathrm{SC}_K(x)^\circ$, then $\langle u, \lambda(z-x)\rangle \leq 1$ for every $z \in K$ and any $\lambda > 0$. Writing this inequality as

$$\langle u, z - x\rangle \leq \frac{1}{\lambda},$$

and making $\lambda \to +\infty$, we obtain that $\langle u, z - x\rangle \leq 0$ for every $z \in K$. By Lemma 2.4.2 this means that $u \in \mathrm{NC}_K(x)$. Conversely, assume that $u \in \mathrm{NC}_K(x)$. Then we have that $\langle u, z - x\rangle \leq 0$ for every $z \in K$. Therefore, we have

$$\langle u, \lambda(z-x)\rangle = \lambda\langle u, z - x\rangle \leq 0 < 1$$

whenever $z \in K$ and $\lambda > 0$. By continuity of the inner product, if $(\lambda_j(z_j - x))_{j\in\mathbb{N}}$ is a sequence converging to $y \in \mathbb{R}^n$, then we also have that $\langle u, \lambda(z-x)\rangle \leq 0 < 1$. Consequently, we have that $u \in \mathrm{SC}_K(x)^\circ$, and this concludes the proof.

□

Remark 2.4.2. Let $C \subseteq \mathbb{R}^n$ be a convex cone. Then the polar of C can be equivalently defined by the equality

$$C^\circ = \{u \in \mathbb{R}^n : \langle u, x\rangle \leq 0 \text{ for all } x \in C\}.$$

Indeed, the inclusion "$\supseteq$" is obvious. On the other hand, if $u \in C^\circ$ and $x \in C$ is non-zero, then $\langle u, \lambda x\rangle \leq 1$ for every $\lambda > 0$. This leads to $\langle u, x\rangle \leq 0$. This interpretation can be visualized in Figure 2.4.2.

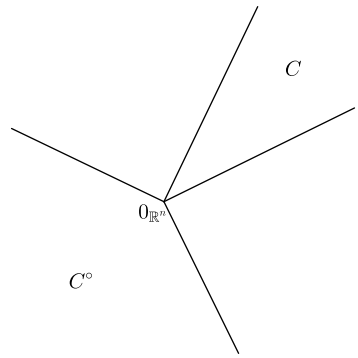


Fig. 2.4.2 The convex cone C and its polar cone C°.

We also have a useful characterization of the normal cone only in metric terms. This is presented next.

Proposition 2.4.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For each $x \in \partial K$ we have*

$$x + \mathrm{NC}_K(x) = \{z \in \mathbb{R}^n : p_K(z) = x\},$$

where p_K denotes the usual metric projection onto K.

Proof. Let $u \in \mathrm{NC}_K(x)$. Then the hyperplane $\mathbf{h} := \mathbf{h}(u, \langle u, x\rangle)$ supports K at x. We claim that $p_K(x+u) = x$. Assume that this is not the case, that is, $p_K(x+u) = y \neq x$. Since $y \in \partial K \subseteq \mathbf{h}^-(u, \langle u, x\rangle)$, the segment $\mathrm{seg}[x + u, y]$ intersects $\mathbf{h}$ at some point z (which is not x). Then

$$|x + u - y| \geq |x + u - z| > |x + u - x|,$$

where the last inequality justifies because the distance from $x + u$ to $\mathbf{h}$ is attained (only) at x, because u is normal to $\mathbf{h}$. This contradicts the fact that y is the metric projection of $x + u$ onto K. Hence $p_K(x + u) = x$, and this yields the inclusion

$$x + \mathrm{NC}_K(x) \subseteq \{z \in \mathbb{R}^n : p_K(z) = x\}.$$

The reverse inclusion comes immediately from Lemma 2.2.4. See also Figure 2.4.3.

□

Observe that, for us, the support cone has apex at the origin. The cone $x + \mathrm{SC}_K(x)$ has apex at x, and the property that $K \subseteq x + \mathrm{SC}_K(x)$; see again Figure 2.4.1. Next we relate this cone with supporting half-spaces of K at x.

Proposition 2.4.4. *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact convex set, and let $x \in \mathrm{relbd}(K)$. Then $x + \mathrm{SC}_K(x)$ is the intersection of the half-spaces supporting K whose bounding hyperplanes contain x.*

Proof. Assume that $K \subseteq \mathbf{h}^-(u, \langle u, x\rangle)$. For any $z \in K$ and every $\lambda \geq 0$, we have

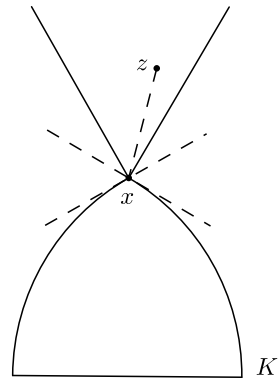


Fig. 2.4.3 $z \in x + \mathrm{NC}_K(x)$ and $p_K(z) = x$.

$$\langle u, x + \lambda(z - x)\rangle = \langle u, x\rangle + \lambda\langle u, z\rangle - \lambda\langle u, x\rangle \leq \langle u, x\rangle,$$

since $\langle u, z\rangle \leq \langle u, x\rangle$ because $z \in K$. It follows that $x + \lambda(z - x) \in \mathbf{h}^-(u, \langle u, x\rangle)$. By continuity of the inner product, we get that

$$\mathrm{cl}\Big(\{x + \lambda(z - x) : z \in K \text{ and } \lambda > 0\}\Big) \subseteq \mathbf{h}^-(u, \langle u, x\rangle).$$

Since we obviously have that the set on the left-hand side above is $x + \mathrm{SC}_K(x)$, and because we fixed an arbitrary supporting half-space whose bounding hyperplane contains x, we obtain that

$$x + \mathrm{SC}_K(x) \subseteq \bigcap \Big\{\text{supporting half}-\text{spaces of } K \text{ at } x\Big\}.$$

For the reverse inclusion, let z_0 be a point in the intersection of all supporting half-spaces of K at x, but assume that $z_0 \notin x + \mathrm{SC}_K(x)$. Since the support cone is closed and convex, it follows that z_0 and $x + \mathrm{SC}_K(x)$ can be strongly separated by some hyperplane, and in particular we may assume that there exists a number $\alpha \in \mathbb{R}$ and a non-zero vector $u \in \mathbb{R}^n$ such that

$$\langle u, z\rangle < \alpha < \langle u, z_0\rangle,$$

for every $z \in x + \mathrm{SC}_K(x)$. Equivalently, what we have is that $z_0 \in \mathrm{int}(\mathbf{h}^+(u, \alpha))$ and $x + \mathrm{SC}_K(x) \subseteq \mathrm{int}(\mathbf{h}^-(u, \alpha))$. Notice that, in particular, we have that $\langle u, x\rangle < \alpha$ because x is a point of $x + \mathrm{SC}_K(x)$, and this gives that $z_0 \in \mathrm{int}(\mathbf{h}^+(u, \langle u, x\rangle))$.

We claim that $K \subseteq \mathbf{h}^-(u, \langle u, x\rangle)$. Indeed, if there exists a point $z_1 \in K$ such that $\langle u, z_1\rangle \geq \alpha$, then

$$\langle u, z_1 - x\rangle = \langle u, z_1\rangle - \langle u, x\rangle > \alpha - \alpha = 0,$$

and this gives that

$$\langle x + \lambda(z_1 - x), u\rangle = \langle u, x\rangle + \lambda\langle u, z_1 - x\rangle > \alpha,$$

for $\lambda > 0$ large enough, since $\langle u, z_1 - x\rangle > 0$. This contradicts the fact that $x + \mathrm{SC}_K(x) \subseteq \mathrm{int}(\mathbf{h}^-(u, \alpha))$.

Consequently, $\mathbf{h}^-(u, \langle u, x\rangle)$ is a supporting half-space of K at x which does not contain z_0, and this contradicts the assumption on z_0. Hence $z_0 \in x + \mathrm{SC}_K(x)$, and this concludes the proof.

□

It is very common that we do not need **all** the supporting half-spaces at x to determine the support cone. Consider, for example, that K is a polygon in $\mathbb{R}^2$ and x is one of its vertices. Then $x + \mathrm{SC}_K(x)$

is clearly the intersection between the half-spaces determined by the two sides of ∂K containing x. In this case we also can describe the normal cone in easier terms.

Proposition 2.4.5. *Let K be a non-empty compact convex set, and assume that for a given $x \in \operatorname{relbd}(K)$ one can write*

$$x + \mathrm{SC}_K(x) = \bigcap_{j\in L} \mathbf{h}^-(u_j, \langle u_j, x\rangle),$$

where L is some set of indices. Then

$$\mathrm{NC}_K(x) = \mathrm{cl}\Big(\mathrm{pos}\{u_j : j \in L\}\Big).$$

Proof. The assumed equality yields

$$\mathrm{SC}_K(x) = \bigcap_{j\in L} \mathbf{h}^-(u_j, \langle u_j, x\rangle) - x = \bigcap_{j\in L} \mathbf{h}^-(u_j, 0),$$

where the last equality justifies because all half-spaces in the original intersection have x as a point of their respective bounding hyperplanes, and hence the respective translations (by $-x$) will contain the origin. It follows from Proposition 2.4.2 that

$$\mathrm{NC}_K(x) = \left(\bigcap_{j\in L} \mathbf{h}^-(u_j, 0)\right)^\circ.$$

Let $\{u_1, \ldots, u_m\}$ denote any finite subset of $\{u_j : j \in L\}$, and let $\lambda_1, \ldots, \lambda_m \geq 0$. If $z \in \bigcap_{j\in L} \mathbf{h}^-(u_j, 0)$, then $\langle u_k, z\rangle \leq 0$ for each $k = 1, \ldots, m$, and this leads to

$$\langle \lambda_1 u_1 + \ldots + \lambda_m u_m, z\rangle = \lambda_1\langle u_1, z\rangle + \ldots + \lambda_m\langle u_m, z\rangle \leq 0.$$

That is, if $u \in \mathrm{pos}\{u_j : j \in \lambda\}$, then $\langle u, z\rangle \leq 0$ for any $z \in \bigcap_{j\in L} \mathbf{h}^-(u_j, 0)$. By Remark 2.4.2 it follows that

$$\mathrm{pos}\{u_j : j \in L\} \subseteq \left(\bigcap_{j\in L} \mathbf{h}^-(u_j, 0)\right)^\circ,$$

and since the right-hand side is a closed set, it follows that the inclusion holds true if we pass to the closure on the left-hand side. The reverse inclusion is equivalent to

$$\Big[\mathrm{cl}\Big(\mathrm{pos}\{u_j : j \in L\}\Big)\Big]^\circ \subseteq \bigcap_{j\in L} \mathbf{h}^-(u_j, 0),$$

because polarity reverses inclusion and is an involution when restricted to the family of closed convex sets containing the origin (see Proposition 2.3.4). If

$$v \in \Big[\mathrm{cl}\Big(\mathrm{pos}\{u_j : j \in L\}\Big)\Big]^\circ,$$

then, in particular, we have that $\langle v, u_j\rangle \leq 0$ for every $j \in L$. This gives that

$$v \in \bigcap_{j\in L} \mathbf{h}^-(u_j, 0),$$

and this concludes the proof.

□

Let $K \subseteq \mathbb{R}^n$ be a convex body. Recall that a *smooth point* (or *regular point*) of ∂K is a point where K is supported by a unique hyperplane (or, equivalently, where K has a unique unit outer normal vector), and that a convex body is said to be *smooth* (or *regular*) if all of its boundary points are smooth. A boundary point which is not smooth is said to be *singular*. Also, we say that K is *strictly convex* if ∂K does not contain a line segment.

Lemma 2.4.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then $x \in \partial K$ is a smooth point if and only if $\mathrm{NC}_K(x)$ is a ray.*

Proof. If $x \in \partial K$ is a smooth point, then there exists a unique supporting hyperplane $\mathbf{h}(u, \alpha)$ of K at x, and hence from Proposition 2.4.5 we obtain

$$\mathrm{NC}_K(x) = \mathrm{cl}\Big(\mathrm{pos}\{u\}\Big) = \{\lambda u : \lambda \geq 0\}.$$

On the other hand, if $x \in \partial K$ is not a smooth point, then K has at least two distinct supporting hyperplanes $\mathbf{h}(u, \alpha)$ and $\mathbf{h}(u, \beta)$ at x, and then $\mathrm{NC}_K(x)$ contains $\mathrm{pos}\{u, v\}$. Therefore, $\mathrm{NC}_K(x)$ is not a ray.

□

Remark 2.4.3. Another way to state this result is as follows: a point $x \in \partial K$ is singular if and only if $\dim \mathrm{NC}_K(x) \geq 2$, where we remind the reader that the *dimension* of a set is the dimension of its affine hull.

The next question that appears is how "big" can the set of singular points of the boundary of a convex body be? This can be answered from different viewpoints (to define what "big" means), and we will provide an answer in terms of the Hausdorff measure. To do so, we remind the reader that a set is *σ-finite* if it can be written as a countable union of sets of finite measure.

Theorem 2.4.1. *The set of singular points of the boundary of a convex body has σ-finite $(n-2)$-dimensional Hausdorff measure.*

Proof. The proof relies on the fact that the $(n-2)$-dimensional Hausdorff measure of an $(n-2)$-simplex is finite. This is easy to verify, since we can naturally identify an $(n-2)$-simplex of $\mathbb{R}^n$ with a compact subset of $\mathbb{R}^{n-2}$, which thus has finite Lebesgue measure. Since this measure coincides with the $(n-2)$-dimensional Hausdorff measure we have the desired. For the details of this argument, we refer the reader to Corollary A.4.4.

Let K be any convex body. We let $\mathcal{S}^{n-2}(\mathbb{R}^n \setminus K)$ be the (countable) collection of all $(n-2)$-simplices in $\mathbb{R}^n \setminus K$ whose vertices are rational points. For each singular point $x \in \partial K$, we have that $\mathrm{NC}_K(x)$ has dimension greater than or equal to 2, from which we can choose a simplex $S \in \mathcal{S}^{n-2}(\mathbb{R}^n \setminus K)$ such that $(x + \mathrm{NC}_K(x)) \cap S \neq \emptyset$. From Proposition 2.4.3, we have that the metric projection of each point of this intersection onto K is x. Therefore, we have the inclusion

$$\{x \in \partial K : x \text{ is singular}\} \subseteq \bigcup \Big\{p_K(S) : S \in \mathcal{S}^{n-2}(\mathbb{R}^n \setminus K)\Big\}.$$

Observe carefully that the union on the left-hand side is countable because $\mathcal{S}^{n-2}(\mathbb{R}^n \setminus K)$ is countable. Since the metric projection is a contraction (see Proposition 2.2.1), we finally get that

$$\mathcal{H}^{n-2}(p_K(S)) \leq \mathcal{H}^{n-2}(S) < +\infty,$$

and hence we have that the set of singular points of ∂K is contained in a countable union of sets with finite $(n-2)$-dimensional Hausdorff measure, as we wanted.

□

Corollary 2.4.2. *The set of singular points of the boundary of a convex body has zero $(n-1)$-dimensional Hausdorff measure. In other words, the unit outer normal is unique for almost all points of ∂K (with respect to $\mathcal{H}^{n-1}$, of course).*

Proof. This follows immediately from Theorem 2.4.1 together with Proposition A.4.1. Indeed, the set of singular points of ∂K is contained in a countable union of sets which have zero $(n-1)$-dimensional Hausdorff measure (namely, the metric projections of the $(n-2)$-simplices of $\mathbb{R}^n \setminus K$ with rational vertices, as described in the proof of the mentioned theorem).

□

Theorem 2.4.2. *The boundary of a smooth convex body is a C^1 submanifold of $\mathbb{R}^n$.*

Proof. The boundary of a convex body can be locally parametrized as the graph of a convex function (with open $(n-1)$-dimensional convex domain). The differentiability of such a function comes from smoothness: its epigraph has a unique supporting hyperplane at each point. Since any differentiable convex function is C^1 (Theorem 1.3.3), the desired follows.

□

A boundary point x of a set A is said to be an *exposed point* if there exists a supporting hyperplane $\mathbf{h}$ of A such that $\mathbf{h} \cap A = \{x\}$. An *extreme point* of A is a point which is not in the interior of a segment of A. In other words, $x \in A$ is an extreme point of A if it cannot be written as $x = (1-\lambda)y + \lambda z$ for $y, z \in A$ and $\lambda \in (0,1)$. Next we verify that every exposed point of a convex set is an extreme point, but the converse is not necessarily true. In what follows, we denote the set of extreme points of a set A by $\mathrm{ext}A$ and the set of exposed points of A by $\mathrm{exp}A$.

Lemma 2.4.4. *Let $C \subseteq \mathbb{R}^n$ be a convex set. Then any exposed point of C is an extreme point of C.*

Remark 2.4.4. The converse is not true. To verify this, draw in $\mathbb{R}^2$ a "bone", that is, a rectangle with semi-circles attached in opposite sides, as in Figure 2.4.4. Doing this, the vertices of the rectangle (which are also the endpoints of the diameters of the semi-circles) are extreme points which are not exposed.

Proof. If $x \in K$ is not an extreme point, then there exist point $y, z \in K$ and $\lambda \in (0,1)$ such that $x = (1-\lambda)y + \lambda z$. Then any supporting hyperplane $\mathbf{h}$ of K such that $x \in \mathbf{h}$ must contain also the segment $\mathrm{seg}[y,z]$. Hence x is not an exposed point.

□

Theorem 2.4.3 (Minkowski's theorem)**.** *Let $C \subseteq \mathbb{R}^n$ be a non-empty compact convex set. Then C is the convex hull of its extreme points, that is, $C = \mathrm{conv}(\mathrm{ext}C)$. Moreover, the set $\mathrm{ext}C$ is the smallest set (with respect to inclusion) having this property.*

Proof. To prove the first claim, we proceed by induction on the (affine) dimension of C. If $\dim C = 0, 1$, the result is obvious (in the first case, C is a point, and in the second case C is a line segment). Assume that the result holds true for non-empty compact convex sets of dimension $0, 1, \ldots, k-1$, for some natural number $2 < k \leq n$, and let C be a compact convex set of dimension k. Notice that the notions of extreme point and convex hull remain unaltered under a natural identification of $\mathrm{aff}C$ with $\mathbb{R}^k$, and hence we can regard C as a convex body of $\mathbb{R}^k$.

Given a point $x \in C$, we have to prove that $x \in \mathrm{conv}(\mathrm{ext}C)$. If x is a boundary point, then there exists a supporting hyperplane $\mathbf{h}$ (in $\mathbb{R}^k$) of C such that $x \in C \cap \mathbf{h}$. The intersection $C \cap \mathbf{h}$ has the

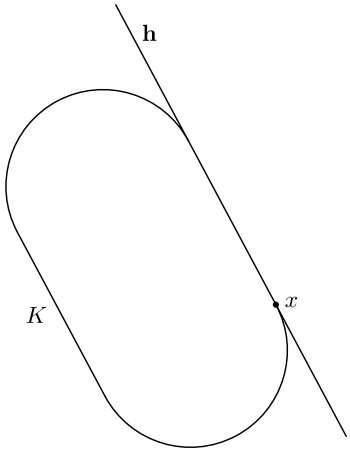


Fig. 2.4.4 The point x is extreme but not exposed.

following properties:

(i) it is a convex set, because it is the intersection of convex sets;

(ii) it is compact, since it is the intersection of a compact set with a closed set;

(iii) it has affine dimension smaller than k, because it is contained in $\mathbf{h}$, which is an affine subspace of dimension $k-1$.

Thus, from our induction hypothesis we get that $C \cap \mathbf{h} = \text{conv}(\text{ext}(C \cap \mathbf{h}))$. On the other hand, every extreme point of $C \cap \mathbf{h}$ is an extreme point of C. Indeed, if $z \in C \cap \mathbf{h}$ but $z \notin \text{ext}C$, then z can be written as $z = (1-\lambda)w + \lambda y$ for some $w, y \in C$ and $\lambda \in (0,1)$, and since $\mathbf{h}$ is a supporting hyperplane of C, we must have $w, y \in \mathbf{h}$ (otherwise we would have w and y lying in distinct open half-spaces determined by $\mathbf{h}$). Hence, z is not an extreme point of $C \cap \mathbf{h}$.

Summarizing, we have proved that any boundary point x of C is a convex combination of extreme points of $C \cap \mathbf{h}$, and that extreme points of $C \cap \mathbf{h}$ are extreme points of C. It follows that x is a convex combination of extreme points of C.

Now we consider the case where $x \in \text{int}C$. In this case, we can choose points $y, z \in \partial C$ and $\lambda \in (0,1)$ such that $x = (1-\lambda)y + \lambda z$ (simply consider the intersections of any line through x with ∂C). Since y and z are boundary points, it follows from the previous case that they can be written as convex combinations of extreme points, say, $y = \alpha_1 y_1 + \ldots + \alpha_m y_m$ and $z = \beta_1 z_1 + \ldots + \beta_s z_s$. Then

$$x = (1-\lambda)\alpha_1 y_1 + \ldots + (1-\lambda)\alpha_m y_m + \lambda\beta_1 z_1 + \ldots + \lambda\beta_s z_s,$$

and the right-hand side is a convex combination of points in ext$C$. Hence $x \in \text{conv}(\text{ext}C)$.

To prove the second claim, we first observe that if x is an extreme point of C, then $C \setminus \{x\}$ is convex. We prove this fact by contradiction: let $x \in \text{ext}C$ and assume that $C \setminus \{x\}$ is not convex. Then there exist points $y, z \in C \setminus \{x\}$ such that the segment $\text{seg}(y,z)$ contains a point w which is not in $C \setminus \{x\}$. By the convexity of C, we must have $w \in C$, and hence it follows that $w = x$. Then $x \in \text{seg}(y,z)$, and this contradicts the fact that x is an extreme point of C.

Now let A be a proper subset of ext$C$. Hence we may choose a point $x \in \text{ext}C$ such that $x \notin A$. We have $A \subseteq C \setminus \{x\}$, and this leads to $\text{conv}A \subseteq \text{conv}(C \setminus \{x\}) = C \setminus \{x\}$, where the last equality is given by the previous observation. This shows that the convex hull of any proper subset of ext$C$ is not C, and the proof is complete.

□

We finish this section with a concept that will be very important to understand the structure of polytopes later. Let $C \subseteq \mathbb{R}^n$ be a convex non-empty set. A convex subset $F \subseteq C$ is a *face* of C if every segment $\text{seg}[x, y] \subseteq C$ with $F \cap \text{seg}(x, y) \neq \emptyset$ is contained in F. A face which is not the empty set or C itself is called a *proper face* of C (it is obvious that C is a face of itself). Next we state and prove some properties of faces.

Proposition 2.4.6. *Let $C \subseteq \mathbb{R}^n$ be a non-empty closed convex set. Then we have*

(a) *If F is a face of C and G is a face of F, then G is a face of C.*

(b) *If $F \subseteq B \subseteq C$ and F is a face of C, then F is a face of B.*

(c) *If F is a proper face of C, then $F \subseteq \text{relbd}C$.*

Proof. For **(a)**, let $\text{seg}[x, y] \subseteq C$ be a segment such that $\text{seg}(x, y) \cap G \neq \emptyset$. Then, since $G \subseteq F$, we have that $\text{seg}(x, y) \cap F \neq \emptyset$. Once F is a face of C, we obtain that $\text{seg}[x, y] \subseteq F$. Now we use the fact that G is a face of F and recall that $\text{seg}(x, y) \cap G \neq \emptyset$ to obtain that $\text{seg}[x, y] \subseteq G$. This proves that G is a face of C.

Now we show **(b)**. Assume that $\text{seg}[x, y] \subseteq B$ and $\text{seg}(x, y) \cap F \neq$. Since $B \subseteq C$, and F is a face of C, we get that $\text{seg}[x, y] \subseteq F$. Hence F is a face of B.

To prove **(c)**, first observe that if F is a proper face, then there exists some point $z \in \text{relbd}C \setminus F$ (this comes from the convexity of F: if F contains $\text{relbd}C$, then it contains the convex hull of $\text{relbd}C$, which is C). If $x \in F \cap \text{relint}C$, then let

$$y = x + \varepsilon(x - z)$$

for some small $\varepsilon > 0$ such that $y \in C$. Hence the segment $\text{seg}[y, z]$ is such that $x \in F \cap \text{seg}(y, z)$, but $\text{seg}[y, z]$ is not contained in F (because $z \notin F$). This contradicts the fact that F is a face, and hence we get $F \cap \text{relint}C = \emptyset$.

□

Remark 2.4.5. Observe carefully that a point $x \in C$ is an extreme point if and only if $\{x\}$ is a face of C.

We also have a characterization of faces which will be useful later. We present it now.

Lemma 2.4.5. *Let $C \subseteq \mathbb{R}^n$ be a non-empty convex set. A convex subset $F \subseteq C$ is a face if and only if $x, y \in C$ and $(x + y)/2 \in F$ imply $x, y \in F$.*

Proof. First assume that F is a face of C. If $x, y \in C$ and $(x + y)/2 \in F$, then $F \cap \text{seg}(x, y) \neq \emptyset$, and from the definition we get that $\text{seg}[x, y] \subseteq F$. Thus, $x, y \in F$.

Conversely, assume that $F \subseteq C$ is a convex subset with the property that $x, y \in C$ and $(x + y)/2 \in F$ imply $x, y \in F$. Let $x, y \in C$ be such that $F \cap \text{seg}(x, y) \neq \emptyset$. We have to prove that $\text{seg}[x, y] \subseteq F$. Choose a point $z \in F \cap \text{seg}(x, y)$. If $z = (x + y)/2$, then $x, y \in F$, and consequently $\text{seg}[x, y] \subseteq F$. If this is not the case, then we may assume that, without loss of generality, $|z - y| < |z - x|$, that is, z is closer to y than to x. Doing this, we have that there exists a point $z_1 \in \text{seg}(x, y)$ such that $z = (z_1 + y)/2$. Then $z_1, y \in F$. If z_1 is closer to x than to z, then there is a point $w_1 \in \text{seg}(x, y)$ such that $z_1 = (x + w_1)/2$; this gives $x, w_1 \in F$, and we are done. If this is not the case, we choose a point $z_2 \in \text{seg}(x, y)$ such that $z_1 = (z_2 + z)/2$. From this we get $z_2, z \in F$. After finitely many steps, we arrive at a point $z_k \in \text{seg}(x, y)$ which is closer to x than to z (at each step we "walk" in the direction of x a

distance of $|z - y|$). Hence we may choose $w_k \in \text{seg}(x, y)$ such that $w_k = (x + z_k)/2$. This yields, in particular, that $x \in F$, and hence we have that $\text{seg}[x, y] \subseteq F$.

□

A face F of a closed convex set C is said to be *exposed* if it can be written as $F = C \cap \mathbf{h}$ for some supporting hyperplane $\mathbf{h}$ of C (in particular, any exposed face is closed, since it is the intersection of two closed sets). To convince the reader that this is not the case for all faces (and hence this new definition makes sense), we describe how to obtain a non-exposed face of a closed convex body: consider, again, the "bone" C given by a rectangle with semi-circles attached at two opposite sides (illustrated in Figure 2.4.4). Recall that the vertices of the rectangle become extreme (but not exposed) points. Hence each of these vertices is a face, but none of them can be written as the intersection of C with a supporting hyperplane. Indeed, any supporting hyperplane containing one of these vertices intersects C along a segment of the boundary (namely, one of the sides of the original rectangle without an attached semi-circle).

Whereas not every proper face of a closed convex set is exposed, we can show that every proper face is contained in some proper exposed face. In fact, let F be a proper face of a closed convex set C. Since $F \cap \text{relint}C = \emptyset$, there exists a hyperplane $\mathbf{h}$ properly separating F and C (see Theorem 2.2.3). Assuming that, without loss of generality, $\text{relint}C \subseteq \mathbf{h}^-$ and $F \subseteq \mathbf{h}^+$, we have that $C \subseteq \mathbf{h}^-$ (because $\mathbf{h}^-$ is a closed set). Thus, since $F \subseteq C$, we get that $F \subseteq \mathbf{h}^+ \cap \mathbf{h}^- = \mathbf{h}$, from which we get that F is contained in the exposed face $C \cap \mathbf{h}$. This exposed face is proper because the separation is proper, and hence $C \nsubseteq \mathbf{h}$.

Now we prove that, somehow conversely, any intersection of a convex closed set C with a supporting hyperplane is an exposed face of C.

Proposition 2.4.7. *Let $C \subseteq \mathbb{R}^n$ be a non-empty closed convex set. Then the intersection of C with a supporting hyperplane $\mathbf{h}$ is an exposed face of C.*

Proof. All we have to prove is that $C \cap \mathbf{h}$ is a face of C. Assume that, without loss of generality, $C \subseteq \mathbf{h}^-$, and let $x, y \in C$ be such that $(x + y)/2 \in C \cap \mathbf{h}$. Since $x, y \in C$, we have that $x, y \in \mathbf{h}^-$, and hence we clearly must have $x, y \in \mathbf{h}$. Therefore, $x, y \in C \cap \mathbf{h}$, and from Lemma 2.4.5 we get that $C \cap \mathbf{h}$ is a face.

□

Remark 2.4.6. In some texts, the intersection of a set C with a supporting hyperplane $\mathbf{h}$ is called a *support set* of C. By the previous proposition, here we will simply call it an exposed face.

From now on we denote by $\mathcal{F}(C)$ the collection of faces of a non-empty closed convex set $C \subseteq \mathbb{R}^n$. Next we state and prove some other properties of faces which will be important later.

Theorem 2.4.4. *For any non-empty closed convex set $C \subseteq \mathbb{R}^n$ we have the following:*

(i) *if $F_1, F_2 \in \mathcal{F}(C)$, then $F_1 \cap F_2 \in \mathcal{F}(C)$;*

(ii) *if $F_1, F_2 \in \mathcal{F}(C)$ are distinct, then $\text{relint}F_1 \cap \text{relint}F_2 = \emptyset$;*

(iii) *for every relatively open convex set $A \subseteq C$ there exists a unique face F of C such that $A \subseteq \text{relint}F$; and*

(iv) *C equals the union of the disjoint collection*

$$\{\text{relint}F : F \in \mathcal{F}(C)\}.$$

Proof. To prove **(i)**, we obviously may assume that $F_1 \cap F_2 \neq \emptyset$. Assume that $x, y \in C$ and $\mathrm{seg}(x, y) \cap (F_1 \cap F_2) \neq \emptyset$. Then $\mathrm{seg}(x, y) \cap F_1 \neq \emptyset$ and $\mathrm{seg}(x, y) \cap F_2 \neq \emptyset$. It follows from the definition that $\mathrm{seg}[x, y] \subseteq F_1 \cap F_2$, and this shows that $F_1 \cap F_2$ is a face of C.

For **(ii)**, assume that there is a point $z \in \mathrm{relint}F_1 \cap \mathrm{relint}F_2$. Without loss of generality, we may fix a point $y \in F_1 \setminus F_2$ (observe that $y \neq z$). Since $y, z \in F_1$ and $z \in \mathrm{relint}F_1$, we have that $z + \varepsilon(z - y) \in \mathrm{relint}F_1$ for some small fixed $\varepsilon > 0$. Writing $x = z + \varepsilon(z - y)$, we have $z \in \mathrm{seg}(x, y)$. Now notice that $\mathrm{seg}[x, y] \subseteq C$ and $F_2 \cap \mathrm{seg}(x, y) \neq \emptyset$ (because it contains z). Once F_2 is a face of C, it follows that $\mathrm{seg}[x, y] \subseteq F_2$. This is a contradiction because $y \notin F_2$.

Now we prove **(iii)**. Let $A \subseteq C$ be relatively open, and let F be the intersection of all faces which contain A. From **(i)**, which can be easily adapted for infinite intersections of elements of $\mathcal{F}(C)$, we have that $F \in \mathcal{F}(C)$, and obviously $A \subseteq F$. Assume that there is a point $x \in A \setminus \mathrm{relint}F$. By Theorem 2.2.3, there is a hyperplane $\mathbf{h}$ properly separating $\{x\}$ and F (with $F \subseteq \mathbf{h}^-$, say). Since $x \in F$ (because $x \in A$), it follows that $\mathbf{h}$ is a supporting hyperplane of F at x, and because the separation is proper, we have that $F \nsubseteq \mathbf{h} \cap F$. Observe that $\mathrm{aff}(A) \subseteq \mathbf{h}$, because if this is not true, then any neighborhood of $\mathrm{aff}(A)$ around x has points of $\mathrm{int}(\mathbf{h}^+)$, and since A is relatively open, we get that $A \cap \mathrm{int}(\mathbf{h}^+) \neq \emptyset$. But $A \subseteq F \subseteq \mathbf{h}^-$, which is a contradiction. Thus, in particular we have that $A \subseteq \mathbf{h}$. It follows that $A \subseteq \mathbf{h} \cap F$, and once $\mathbf{h} \cap F$ is a face of C (see Item **(a)** of Proposition 2.4.6), we get that $\mathbf{h} \cap F \supseteq F$ (because F is the intersection of all faces which contain A). This contradiction gives that $A \subseteq \mathrm{relint}F$. The fact that F is the unique face with this property comes straightforwardly from **(ii)**.

Finally, from **(ii)** we have that the collection $\{\mathrm{relint}F : F \in \mathcal{F}(C)\}$ consists of mutually disjoint sets. Also, since each point $x \in C$ is relatively open, we get from **(iii)** that C is contained in the union of the sets of this collection. This gives **(iv)**.

□

Corollary 2.4.3. *Let C be a non-empty, closed, and convex set. Any point in* $\mathrm{relbd}C$ *is in the relative interior of some proper face $F \in \mathcal{F}(C)$.*

Proof. Let $x \in C$. Then $\{x\}$ is relatively open, and due to Item **(iii)**, there exists a face F of C such that $x \in \mathrm{relint}F$. If $x \in \mathrm{relbd}C$, then $x \notin \mathrm{relint}C$. Hence $F \neq C$, and this gives that F is a proper face.

□

2.5 The Hausdorff metric

In this section, we regard the space $\mathcal{K}^n$ of convex bodies of $\mathbb{R}^n$ as a metric space. As will be clear, this is a powerful technique in convex geometry. For example, it often allows us to solve problems by approximating arbitrary convex bodies by "well-behaving" convex bodies, such as polytopes or smooth bodies. Recall that we have a notion of *distance* between sets in $\mathbb{R}^n$ given as

$$\mathrm{dist}(A, B) = \inf\{a - b : a \in A \text{ and } b \in B\},$$

for any given $A, B \subseteq \mathbb{R}^n$. However, this is not completely satisfactory, since it is easy to see that this does not define a metric. For example, $\mathrm{dist}(A, B) = 0$ whenever $A \cap B \neq \emptyset$.

The space $\mathcal{K}^n$ of convex bodies of $\mathbb{R}^n$ is a subset of the space of non-empty compact sets, and it is more convenient to define a metric in this more general family. To fix notations and ideas, we highlight below the families which we will work with from now on.

$$C^n := \Big\{\text{non-empty compact subsets of } \mathbb{R}^n\Big\},$$
$$CC^n := \Big\{\text{non-empty compact convex subsets of } \mathbb{R}^n\Big\} \text{ and}$$
$$\mathcal{K}^n := \Big\{\text{convex bodies of } \mathbb{R}^n\Big\}.$$

The third family above was already defined at the end of Section 2.1. We emphasize that these notations will be used through all the book. We define the *Hausdorff distance* (or *Hausdorff metric*) between two sets $L, C \in C^n$ by

$$d_{\mathcal{H}}(L, C) := \inf\{\lambda \geq 0 : L \subseteq C + \lambda B^n \text{ and } C \subseteq L + \lambda B^n\}, \tag{2.5.1}$$

where $B^n := \{x \in \mathbb{R}^n : |x| \leq 1\}$ is the closed unit ball of $\mathbb{R}^n$ (see Figure 2.5.1). It comes immediately from compactness that this infimum exists and is finite, and also that it can be replaced by the minimum (see Exercise 2.53). Of course, we always have $\mathcal{K}^n \subseteq CC^n \subseteq C^n$. Hence the spaces $\mathcal{K}^n$ and CC^n naturally become metric spaces with the Hausdorff metric $d_{\mathcal{H}}$ induced from C^n.

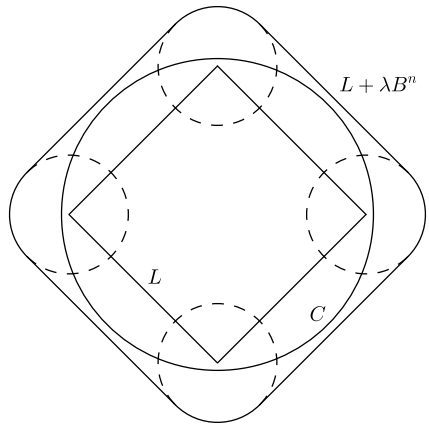


Fig. 2.5.1 We have $C \subseteq L + \lambda B^n$, where C is the big ball, L is the square, and λ is the radius of the small balls.

Remark 2.5.1. The reader must always keep in mind that

$$L + \delta B^n = \{x \in \mathbb{R}^n : \operatorname{dist}(x, L) \leq \delta\}.$$

Therefore, stating that $C \subseteq L + \delta B^n$ is the same as saying that all points of C are at distance at most δ to L.

Proposition 2.5.1. *The Hausdorff distance $d_{\mathcal{H}}$ is a metric in C^n.*

Proof. The symmetry is obvious. Also, it is clear that $d_{\mathcal{H}}(L, C) \geq 0$ for any $L, C \in C^n$, and that $d_{\mathcal{H}}(L, L) = 0$ for any $L \in C^n$. Now let $L, C \in C^n$ be compact sets such that $L \neq C$. Without loss of generality, assume that there is a point $x \in L \setminus C$. From compactness, we have that

$$\operatorname{dist}(x, C) := 2\delta > 0.$$

Therefore, L is not contained in $C + \delta B^n$. It follows that $d_{\mathcal{H}}(L, C) > \delta > 0$.

It remains to prove the triangle inequality. Let $L, C, F \in \mathcal{C}^n$. If $\alpha, \delta \geq 0$ are such that

$$L \subseteq F + \delta B^n, \text{ and } F \subseteq L + \delta B^n,$$
$$F \subseteq C + \alpha B^n, \text{ and } C \subseteq F + \alpha B^n,$$

then we get immediately that

$$L \subseteq C + (\alpha + \delta)B^n, \text{ and } C \subseteq L + (\alpha + \delta)B^n.$$

Consequently, $d_{\mathcal{H}}(L, C) \leq \alpha + \delta$ for any $\alpha, \delta \geq 0$ such that $d_{\mathcal{H}}(L, F) \leq \delta$ and $d_{\mathcal{H}}(F, C) \leq \alpha$. Passing to the minimum of α and δ with these properties, we get that $d_{\mathcal{H}}(L, C) \leq d_{\mathcal{H}}(L, F) + d_{\mathcal{H}}(F, C)$.

□

Next we give some equivalent definitions for the Hausdorff distance. Notice that, dealing with compact sets, any supremum and infimum can be replaced by maximum and minimum, respectively.

Proposition 2.5.2. **(i)** *If $L, C \in \mathcal{C}^n$, then*

$$d_{\mathcal{H}}(L, C) = \max\left\{\sup_{x \in L} \operatorname{dist}(x, C), \sup_{y \in C} \operatorname{dist}(y, L)\right\}. \tag{2.5.2}$$

(ii) *If K and T are compact, convex and non-empty, then*

$$d_{\mathcal{H}}(K, T) = \max\left\{|h_K(u) - h_T(u)| : u \in \mathbb{S}^{n-1}\right\},$$

where h_K and h_T denote the support functions of K and T, respectively.

(iii) *If $K, T \in \mathcal{K}^n$, then we also have*

$$d_{\mathcal{H}}(K, T) = d_{\mathcal{H}}(\partial K, \partial T). \tag{2.5.3}$$

Proof. We start by proving **(i)**. Observe that if

$$\max\left\{\sup_{x \in L} \operatorname{dist}(x, C), \sup_{y \in C} \operatorname{dist}(y, L)\right\} = \delta,$$

then

$$\sup_{x \in L} \operatorname{dist}(x, C) \leq \delta$$

for all $x \in L$. Consequently, $L \subseteq C + \delta B^n$. Analogously, we have that $C \subseteq L + \delta B^n$. This leads to $d_{\mathcal{H}}(L, C) \leq \delta$, that is,

$$d_{\mathcal{H}}(L, C) \leq \max\left\{\sup_{x \in L} \operatorname{dist}(x, C), \sup_{y \in C} \operatorname{dist}(y, L)\right\}.$$

Conversely, if $L \subseteq C + \alpha B^n$, then $\operatorname{dist}(x, C) \leq \alpha$ for any $x \in L$. Indeed, every point $x \in L$ belongs to some ball of radius α centered in some point of C. Therefore,

$$\sup_{x \in L} \operatorname{dist}(x, C) \leq \alpha.$$

This leads to the following: if $d_{\mathcal{H}}(L, C) \leq \alpha$, then we have that $L \subseteq C + \alpha B^n$ and $C \subseteq L + \alpha B^n$ (see again Exercise 2.53), and thus

$$\max\left\{\sup_{x\in L} \operatorname{dist}(x, C),\ \sup_{y\in C} \operatorname{dist}(y, L)\right\} \leq \alpha.$$

Passing to the minimum we get

$$\max\left\{\sup_{x\in L} \operatorname{dist}(x, C),\ \sup_{y\in C} \operatorname{dist}(y, L)\right\} \leq d_{\mathcal{H}}(L, C),$$

and this concludes the proof of **(i)**.

For **(ii)**, observe that if $\alpha \geq 0$ is such that $d_{\mathcal{H}}(K, T) \leq \alpha$, then $K \subseteq T + \alpha B^n$, and hence for any $u \in \mathbb{S}^{n-1}$ we have

$$\begin{aligned} h_K(u) = \sup\{\langle x, u\rangle : x \in K\} \leq \sup\{\langle x, u\rangle : x \in T + \alpha B^n\} = \\ = \sup\{\langle z, u\rangle : z \in T\} + \sup\{\langle \alpha v, u\rangle : v \in B^n\} = \alpha + \sup\{\langle z, u\rangle : z \in T\} = \\ = \alpha + h_T(u). \end{aligned}$$

Since we also have that $T \subseteq K + \alpha B^n$, the same inequality holds if we interchange K and T. Therefore, we get that $|h_K(u) - h_T(u)| \leq \alpha$ for every $u \in \mathbb{S}^{n-1}$, and thus

$$\max\{|h_K(u) - h_T(u)| : u \in \mathbb{S}^{n-1}\} \leq d_{\mathcal{H}}(K, T).$$

Now, if $\alpha = \max_{u\in\mathbb{S}^{n-1}} |h_K(u) - h_T(u)|$, then we have in particular that

$$h_K(u) \leq h_T(u) + \alpha,$$

for any $u \in \mathbb{S}^{n-1}$. Assume that $x \in K$ is such that $x \notin T$. Then, taking the metric projection, $z_0 := p_T(x)$ we have that $\operatorname{dist}(x, T) = |x - z_0|$, and the outer normal of T at z_0 is the vector

$$u := \frac{x - z_0}{|x - z_0|}.$$

Since the supremum in the definition of the support function in a given direction $u \in \mathbb{S}^{n-1}$ is always assumed in the boundary point where u is the outer normal we get that $h_T(u) = \langle z_0, u\rangle$. Writing $x = z_0 + |x - z_0|u$ and using the inequality $h_K(u) \leq h_T(u) + \alpha$, we get

$$h_T(u) + \alpha \geq h_K(u) \geq \langle x, u\rangle = \langle z_0, u\rangle + |x - z_0| = h_T(u) + \operatorname{dist}(x, T).$$

This implies that $\operatorname{dist}(x, T) \leq \alpha$. It follows that $x \in T + \alpha B^n$, and this shows that $K \subseteq T + \alpha B^n$ (observe that the case where $x \in T$ is obvious). It is clear that everything also remains true if we interchange K and T, and hence we also have $T \subseteq K + \alpha B^n$. Consequently, $d_{\mathcal{H}}(K, T) \leq \alpha = \max_{u\in\mathbb{S}^{n-1}} |h_K(u) - h_T(u)|$.

Now we prove **(iii)**. We will show that, for a given $\alpha \geq 0$,

$$K \subseteq T + \alpha B^n \text{ and } T \subseteq K + \alpha B^n \iff \partial K \subseteq \partial T + \alpha B^n \text{ and } \partial T \subseteq \partial K + \alpha B^n.$$

Observe that this equivalence implies immediately that $d_{\mathcal{H}}(K, T) = d_{\mathcal{H}}(\partial K, \partial T)$. Indeed, this equivalence gives that if $d_{\mathcal{H}}(K, T) \leq \lambda$, then $d_{\mathcal{H}}(\partial K, \partial T) \leq \lambda$, and vice versa.

Assume that $\alpha \geq 0$ is such that $K \subseteq T + \alpha B^n$ and $T \subseteq K + \alpha B^n$. Then, since $\partial K \subseteq K$, we get that for any $x \in \partial K$ there exists a point $y \in T$, a number $0 \leq \beta \leq \alpha$ and a vector $u \in B^n$ such that

$x = y + \beta u$. If $x \notin \text{int}(T)$, then the intersection $\text{seg}[x, y] \cap \partial T$ contains a point (z, say), and we clearly have that $x \in \partial T + \alpha B^n$ (because $x = z + \beta_0 u$ for some $0 \leq \beta_0 \leq \beta \leq \alpha$). If $x \in \text{int}(T)$, then the ray starting at x in the direction of an outer normal of K at x intersects ∂T at a point (z_0, say), and we have $\text{dist}(z_0, K) = |z_0 - x|$. Since $T \subseteq K + \alpha B^n$, we get that $|z_0 - x| \leq \alpha$, from which $x \in \partial T + \alpha B^n$. It follows that $\partial K \subseteq \partial T + \alpha B^n$, and analogously we prove that $\partial T \subseteq \partial K + \alpha B^n$.

Now suppose that we have the inclusions $\partial K \subseteq \partial T + \alpha B^n$ and $\partial T \subseteq \partial K + \alpha B^n$. Let $x \in K$. If $x \notin \text{int}(T)$, then $\text{dist}(x, T) = |x - p_T(x)|$, where $p_T(x) \in \partial T$ denotes the usual metric projection of x onto T. The ray starting at $p_T(x)$ with direction $x - p_T(x)$ intersects ∂K at a point (z, say), and we clearly have $\text{dist}(z, T) = |z - p_T(x)|$. From $\partial T \subseteq \partial K + \alpha B^n$ we get that $|z - p_T(x)| \leq \alpha$, yielding $|x - p_T(x)| \leq \alpha$. Thus $x \in T + \alpha B^n$. On the other hand, if $x \in \text{int}(T)$, then $x \in T \subseteq T + \alpha B^n$. It follows that $K \subseteq T + \alpha B^n$, and the proof of the inclusion $T \subseteq K + \alpha B^n$ is obviously the same.

□

Remark 2.5.2. Notice that the proof of **(i)** is merely a consequence of the fact that, for any $\delta \geq 0$, the inclusion $L \subseteq C + \delta B^n$ is equivalent to

$$\sup_{x \in L} \inf_{y \in C} |x - y| \leq \delta.$$

Moreover, the proof of **(ii)** is a consequence of the equivalence between the inclusion $K \subseteq T + \alpha B^n$ and the inequality

$$h_K(u) \leq h_T(u) + \alpha, \quad \forall u \in \mathbb{S}^{n-1},$$

valid for any convex bodies $K, T \subseteq \mathbb{R}^n$ and any number $\alpha \geq 0$.

Remark 2.5.3. In view of **(ii)**, we get that the assignment $K \mapsto h_K$ is an isometric embedding of $(CC^n, d_{\mathcal{H}})$ onto $C(\mathbb{S}^{n-1})$ endowed with the *maximum norm*:

$$||f||_{C(\mathbb{S}^{n-1})} := \max\{|f(u)| : u \in \mathbb{S}^{n-1}\}, \quad \forall\, f \in C(\mathbb{S}^{n-1}).$$

The *diameter* of a subset D of a metric space (M, d) is the number

$$\text{diam}(D) := \sup\{d(x, y) : x, y \in D\}. \tag{2.5.4}$$

If this number is finite, then we say that D is a *bounded set*. If $\mathcal{D} \subseteq C^n$ is a subset, then there are two possible notions of "boundedness" for $\mathcal{D}$: $\mathcal{D}$ can be bounded in the Hausdorff metric, that is, $\text{diam}_{\mathcal{H}}(\mathcal{D}) < +\infty$ (where $\text{diam}_{\mathcal{H}}$ denotes the diameter in the Hausdorff distance), or $\mathcal{D}$ can be bounded as a family of subsets of $\mathbb{R}^n$. Next we prove that these two notions coincide.

Proposition 2.5.3. *Let $\mathcal{D} \subseteq C^n$ be a non-empty subset. Then,* $\text{diam}_{\mathcal{H}}(\mathcal{D})$ *is finite if and only if there exists a number $\rho > 0$ such that $D \subseteq \rho B^n$ for each $D \in \mathcal{D}$. In other words, a subset of C^n is bounded in the Hausdorff metric if and only if it is bounded as a collection of subsets of $\mathbb{R}^n$.*

Proof. First, assume that $\text{diam}_{\mathcal{H}}(\mathcal{D}) < m < +\infty$, and let $D_0 \in \mathcal{D}$. For every $D \in \mathcal{D}$ we have that

$$d_{\mathcal{H}}(D, D_0) \leq \text{diam}_{\mathcal{H}}(\mathcal{D}) < m,$$

from which every $D \in \mathcal{D}$ is such that $D \subseteq D_0 + mB^n$. Since $D_0 + mB^n$ is clearly a compact set, there exists $\rho > 0$ such that $D_0 + mB^n \subseteq \rho B^n$. Hence $\mathcal{D}$ is bounded as a collection of subsets of $\mathbb{R}^n$.

Now assume the converse, that is, suppose that there exists $\rho > 0$ such that $D \subseteq \rho B^n$ for every $D \in \mathcal{D}$. Let $D_0, D_1 \in \mathcal{D}$. Then $D_1 \subseteq \rho B^n \subseteq D_0 + 2\rho B^n$. The second inclusion follows from the fact that if $x_0 \in D_0$ is any fixed point, and $y \in \rho B^n$, then $|y - x_0| \leq 2\rho$ (since both are points of ρB^n), and hence we have $y \in x_0 + 2\rho B^n \subseteq D_0 + 2\rho B^n$. On the other hand, by the same argument we get

the inclusion $D_0 \subseteq D_1 + 2\rho B^n$. It follows that $d_{\mathcal{H}}(D_0, D_1) \leq 2\rho$. Since $D_0, D_1 \in \mathcal{D}$ were arbitrarily taken, we have

$$\operatorname{diam}_{\mathcal{H}}(\mathcal{D}) = \sup\{d_{\mathcal{H}}(D_0, D_1) : D_0, D_1 \in \mathcal{D}\} \leq 2\rho < +\infty,$$

and this concludes the proof.

□

Now we investigate topological properties of the metric space $(C^n, d_{\mathcal{H}})$. A sequence $(C_j)_{j\in\mathbb{N}}$ of sets is *decreasing* if $C_j \supseteq C_{j+1}$ for each $j \in \mathbb{N}$.

Proposition 2.5.4. *Let $(C_j)_{j\in\mathbb{N}}$ be a decreasing sequence in C^n. We have*

$$\lim_{j\to\infty} C_j = \bigcap_{j=1}^{\infty} C_j,$$

where the limit is taken in the Hausdorff metric.

Proof. From standard topology arguments, we get that the set

$$C := \bigcap_{j=1}^{\infty} C_j$$

is compact and non-empty. What we have to prove is that for any $\varepsilon > 0$ there exists a number $j_0 \in \mathbb{N}$ such that $j > j_0$ implies $d_{\mathcal{H}}(C, C_j) < \varepsilon$. If this is false, then there exist a number $\varepsilon_0 > 0$ and a sequence $(j_k)_{k\in\mathbb{N}}$ of natural numbers such that $j_k \to +\infty$ and $d_{\mathcal{H}}(C_{j_k}, C) > \varepsilon_0$ for any $k \in \mathbb{N}$.

On the other hand, the sequence (C_j) is decreasing and $C \subseteq C_j$ for any $j \in \mathbb{N}$. Thus, $d_{\mathcal{H}}(C, C_j) = \min\{\lambda \geq 0 : C_j \subseteq C + \lambda B^n\}$ and $d_{\mathcal{H}}(C, C_m) \leq d_{\mathcal{H}}(C, C_j)$ if $m \geq j$. Consequently, we have that $d_{\mathcal{H}}(C, C_j) > \varepsilon_0$ for any $j \in \mathbb{N}$. This means that $C_m \not\subset C + \varepsilon_0 B^n$ for every $m \in \mathbb{N}$. Therefore, writing $A_m := C_m \setminus \operatorname{int}(C + \varepsilon_0 B^n)$ we get a decreasing sequence $(A_m)_{m\in\mathbb{N}}$ of compact non-empty sets. Hence the intersection

$$A := \bigcap_{m=1}^{\infty} A_m$$

is compact and non-empty. On the other hand,

$$A = \bigcap_{m=1}^{\infty} C_m \setminus \operatorname{int}(C + \varepsilon_0 B^n) = \left(\bigcap_{m=1}^{\infty} C_m\right) \setminus \operatorname{int}(C + \varepsilon_0 B^n) = C \setminus \operatorname{int}(C + \varepsilon_0 B^n) = \emptyset,$$

which is a contradiction. Notice that the last equality comes from the immediate fact that $C \subseteq \operatorname{int}(C + \varepsilon_0 B^n)$.

□

For the sake of convenience, we remind the reader that a metric space (M, d) is *complete* if every Cauchy sequence in (M, d) converges.

Theorem 2.5.1. *The metric space $(C^n, d_{\mathcal{H}})$ is complete.*

Proof. Let $(C_j)_{j\in\mathbb{N}}$ be a Cauchy sequence in $(C^n, d_{\mathcal{H}})$. We start by proving that

$$C := \bigcup_{j=1}^{\infty} C_j$$

is a bounded subset of $\mathbb{R}^n$. Let $m_0 \in \mathbb{N}$ be such that $m, j \geq m_0$ imply $d_{\mathcal{H}}(C_j, C_m) < 1$. Then, for every $j \geq m_0$, we have that $C_j \subseteq C_{m_0} + B^n$. It follows that

$$C \subseteq \left(\bigcup_{j=1}^{m_0-1} C_j \right) \cup (C_{m_0} + B^n),$$

and the right-hand side is a finite union of compact sets, therefore being bounded. Now, for each $m \in \mathbb{N}$, define

$$D_m := \mathrm{cl}\left(\bigcup_{j=m}^{\infty} C_j \right).$$

Each of these sets is bounded and closed, hence compact. Also, it is clear that the sequence $(D_m)_{m\in\mathbb{N}}$ is decreasing. Therefore, the intersection

$$D := \bigcap_{m=1}^{\infty} D_m$$

is compact and non-empty. From Proposition 2.5.4 we have that $D = \lim_{m\to\infty} D_m$, where the limit is taken in the Hausdorff metric. Consequently, for every $\varepsilon > 0$ there exists $m_1 \in \mathbb{N}$ such that $m \geq m_1$ implies $d_{\mathcal{H}}(D_m, D) < \varepsilon$. Thus,

$$C_m \subseteq \mathrm{cl}\left(\bigcup_{j=m}^{\infty} C_j \right) = D_m \subseteq D + \varepsilon B^n, \tag{2.5.5}$$

whenever $m \geq m_1$. On the other hand, one can choose a number $m_2 \geq m_1$ such that $k, j \geq m_2$ imply $d_{\mathcal{H}}(C_k, C_j) < \varepsilon$. It follows that $C_k \subseteq C_j + \varepsilon B^n$ whenever $j, k \geq m_2$, and hence

$$\bigcup_{i=k}^{\infty} C_i \subseteq C_j + \varepsilon B^n$$

if $k, j \geq m_2$. Taking the closure on both sides, and noticing that the right-hand side is a closed set, we get that $D_k \subseteq C_j + \varepsilon B^n$ for any $k, j \geq m_2$. Since $D \subseteq D_k$ for every k, we get that $D \subseteq C_j + \varepsilon B^n$ for any $j \geq m_2$. But recalling that $m_2 \geq m_1$, we get from (2.5.5) that if $j \geq m_2$, then we also have $C_j \subseteq D + \varepsilon B^n$. Then it follows that $d_{\mathcal{H}}(C_j, D) < \varepsilon$ whenever $j \geq m_2$. Finally, this gives that $\lim_{m\to\infty} C_m = D$, and this shows that $(C_j)_{j\in\mathbb{N}}$ converges in $(C^n, d_{\mathcal{H}})$.

□

A very important topological property of the Hausdorff metric is that every bounded sequence of $(C^n, d_{\mathcal{H}})$ has a convergent subsequence. We will prove this next, as a consequence of the *Ascoli–Arzelá theorem*. The reader must be aware that most authors present a more constructive proof. For the convenience of the reader, we recall a concept from standard analysis: let $X \subseteq \mathbb{R}^n$ and let $(f_j \colon X \to \mathbb{R})_{j\in\mathbb{N}}$ be a sequence of functions. We say that (f_j) *converges uniformly* to a function $f \colon X \to \mathbb{R}$ if for any $\varepsilon > 0$ there exists $j_0 \in \mathbb{N}$ such that $|f_j(x) - f(x)| < \varepsilon$ for any $j > j_0$ and $x \in X$. Notice that this is equivalent to convergence in the maximum norm defined in Remark 2.5.3.

Theorem 2.5.2 (Ascoli–Arzelá theorem). *Let $X \subseteq \mathbb{R}^n$ be a compact set, and assume that $(f_j \colon X \to \mathbb{R})_{j\in\mathbb{N}}$ be a sequence of functions which is*

(i) uniformly bounded*, meaning that there exists a number $m \in \mathbb{R}$ such that $|f_j(x)| < m$ for any $j \in \mathbb{N}$ and $x \in X$;*

(ii) equicontinuous*, meaning that for every $\varepsilon > 0$ there exists $\delta > 0$ such that $x, y \in X$ and $|x - y| < \delta$ imply $|f_j(x) - f_j(y)| < \varepsilon$ for any $j \in \mathbb{N}$.*

Then, (f_j) has a subsequence which converges uniformly to some continuous function $f \colon X \to \mathbb{R}$.

Proof. See [2075].

□

Theorem 2.5.3. *Every bounded sequence of $(C^n, d_{\mathcal{H}})$ has a convergent subsequence.*

Proof. From Proposition 2.5.3 we have that it is equivalent to regard the sequence as bounded in the Hausdorff metric, or as a bounded collection of subsets of $\mathbb{R}^n$. Hence we assume that $(C_j)_{j\in\mathbb{N}}$ is a sequence of compact sets such that there exists $m > 0$ with $C_j \subseteq mB^n$ for every $j \in \mathbb{N}$. For each $j \in \mathbb{N}$, define the function $g_j \colon mB^n \to \mathbb{R}$ as

$$g_j(x) := \operatorname{dist}(x, C_j),$$

for each $x \in mB^n$. In other words, g_j is the distance function to C_j restricted to mB^n. It is clear that $g_j(x) \leq 2m$ for any $j \in \mathbb{N}$ and $x \in mB^n$, from which the sequence $(g_j)_{j\in\mathbb{N}}$ of functions is uniformly bounded. We claim that this sequence is also equicontinuous. In fact, let $x, y \in mB^n$ be such that $|x - y| < \delta$. Fix any $j \in \mathbb{N}$ and, by compactness, let $x_0, y_0 \in C_j$ be points where the distances from x and y to C_j, respectively, are attained. That is, we have $g_j(x) = |x - x_0|$ and $g_j(y) = |y - y_0|$. Thus,

$$\begin{aligned} g_j(y) = \operatorname{dist}(y, C_j) \leq |y - x_0| \leq |y - x| + |x - x_0| < \delta + g_j(x), \text{ and} \\ g_j(x) = \operatorname{dist}(x, C_j) \leq |x - y_0| \leq |x - y| + |y - y_0| < \delta + g_j(y). \end{aligned}$$

Then we have that $x, y \in mB^n$ and $|x - y| < \delta$ imply $|g_j(x) - g_j(y)| < \delta$, and this gives that (g_j) is equicontinuous. By the Ascoli–Arzelá theorem, it follows that (g_j) has a subsequence, which we will denote by $(g_k)_{k\in\mathbb{N}}$ for the sake of simplicity, converging uniformly to a continuous function $g \colon mB^n \to \mathbb{R}$. The set $C := g^{-1}(0)$ is clearly closed (this comes from the continuity of g), and bounded (since it is contained in mB^n), hence compact. We claim that $(C_k)_{k\in\mathbb{N}}$ converges to C in the Hausdorff metric. First of all, we need to prove the inequality

$$\operatorname{dist}(x, C) \leq g(x), \tag{2.5.6}$$

for any $x \in mB^n$. This inequality is obvious if $x \in C$, since both sides are zero in this case. Thus, let $x \in mB^n \setminus C$, and for each $k \in \mathbb{N}$, let $x_k \in C_k$ be such that $g_k(x) = |x - x_k|$. Passing to a subsequence if necessary, we may assume that x_k converges to some point $x_0 \in mB^n$ (indeed, (x_k) is a sequence in the compact set mB^n). Since $g_k \to g$ uniformly, in particular we have that $g_k(x) \to g(x)$, and hence

$$g(x) = \lim_{k\to\infty} g_k(x) = \lim_{k\to\infty} |x - x_k| = |x - x_0|.$$

On the other hand, for each $k \in \mathbb{N}$ we have $g_k(x_k) = 0$. We clearly have that $g_k(x_k) \to g(x_0)$, and this gives $g(x_0) = 0$. Hence $x_0 \in C$. It follows that $\operatorname{dist}(x, C) \leq |x - x_0| = g(x)$, as we have claimed.

Let $\varepsilon > 0$ be arbitrarily given. Since $g_k \to g$ uniformly, there exists $k_0 \in \mathbb{N}$ such that $k > k_0$ implies $|g_k(x) - g(x)| < \varepsilon$ for any $x \in mB^n$. In this case, if $x \in C$, then

$$\operatorname{dist}(x, C_k) = |g_k(x) - g(x)| < \varepsilon,$$

and hence $x \in C_k + \varepsilon B^n$. It follows that if $k > k_0$, then $C \subseteq C_k + \varepsilon B^n$. Now, if $k > k_0$ and $x \in C_k$, then from (2.5.6) we get

$$\operatorname{dist}(x, C) \leq g(x) = |g_k(x) - g(x)| < \varepsilon,$$

and then we have $x \in C + \varepsilon B^n$. In other words, if $k > k_0$, then $C_k \subseteq C + \varepsilon B^n$. Finally, this gives that if $k > k_0$, then $d_{\mathcal{H}}(C, C_k) \leq \varepsilon$, and this shows that (C_k) converges to C in the Hausdorff metric.

□

In what follows, we remind the reader that a metric space (M, d) is said to be *locally compact metric space* if every point $x \in M$ admits an open neighborhood and a non-empty compact set such that $x \in U \subseteq K$.

Corollary 2.5.1. *Every closed bounded subset of $(C^n, d_{\mathcal{H}})$ is compact. In particular, $(C^n, d_{\mathcal{H}})$ is locally compact.*

Proof. This is an immediate consequence of the previous theorem, and it follows uniquely from metric space theory arguments (meaning that this holds for any metric space with the property that any bounded sequence has a convergent subsequence). If $\mathcal{D} \subseteq C^n$ is closed and bounded, then from boundedness we get that any sequence in $\mathcal{D}$ has a convergent subsequence. From closedness we get that the limit of such a subsequence belongs to $\mathcal{D}$. It follows that $\mathcal{D}$ is sequentially compact, and hence compact.

To check that $(C^n, d_{\mathcal{H}})$ is locally compact, for a given $C_0 \in C^n$ we simply consider a number $\rho > 0$ and the open ball

$$B_{\mathcal{H}}(C_0, 2\rho) := \{C \in C^n : d_{\mathcal{H}}(C_0, C) < 2\rho\}.$$

The set $\operatorname{cl}(B_{\mathcal{H}}(C_0, \rho))$ is closed and bounded, hence compact. Now we simply notice that $C_0 \in B_{\mathcal{H}}(C_0, \rho) \subseteq \operatorname{cl}(B_{\mathcal{H}}(C_0, 2\rho))$, and we have the desired.

□

We recall that the collection of all non-empty convex compact subsets of $\mathbb{R}^n$ is denoted as CC^n. Next we explore a very important topological feature that CC^n has as a subset of $(C^n, d_{\mathcal{H}})$.

Proposition 2.5.5. *The family CC^n of non-empty compact convex subsets of $\mathbb{R}^n$ is a closed subset of $(C^n, d_{\mathcal{H}})$.*

Proof. We will show that $C^n \setminus CC^n$ is open (with respect to the Hausdorff metric, of course). If $K \in C^n \setminus CC^n$, then there exist points $x, y \in K$ and a number $\lambda \in (0, 1)$ such that $z = (1-\lambda)x + \lambda y \notin K$. Even more, since K is compact, there also exists a number $\varepsilon > 0$ such that $B(z, \varepsilon) \cap K = \emptyset$. Let $L \in C^n$ be such that $d_{\mathcal{H}}(K, L) < \varepsilon/2$. It follows that $K \subseteq L + \delta B^n$ for some $0 < \delta < \varepsilon/2$, and hence there exist points $\bar{x}, \bar{y} \in L$ such that $|x - \bar{x}| < \varepsilon/2$ and $|y - \bar{y}| < \varepsilon/2$. Putting $\bar{z} = (1 - \lambda)\bar{x} + \lambda\bar{y}$, we have

$$|\bar{z} - z| = |(1 - \lambda)(\bar{x} - x) + \lambda(\bar{y} - y)| \leq (1 - \lambda)|x - \bar{x}| + \lambda|y - \bar{y}| < \varepsilon/2.$$

We claim that $\bar{z} \notin L$. In fact, if $\bar{z} \in L$, then, because $L \subseteq K + \delta B^n$ for some $0 < \delta < \varepsilon/2$, we get that there exists a point $w \in K$ such that $|w - \bar{z}| < \varepsilon/2$. Consequently, we have from the triangle inequality that $|w - z| < \varepsilon$, and this is a contradiction to the fact that $B(z, \varepsilon) \cap K = \emptyset$. Finally, the fact that $\bar{z} \notin L$ means that L is not convex. Hence,

$$B_{\mathcal{H}}(K, \varepsilon/2) := \{T \in C^n : d_{\mathcal{H}}(T, K) < \varepsilon/2\} \subseteq C^n \setminus CC^n,$$

and this shows that $C^n \setminus CC^n$ is open. Thus, CC^n is closed.

□

Notice that dropping the hypothesis of the sets having non-empty interior is necessary. Any sequence of closed balls with the same center $p \in \mathbb{R}^n$ and whose radii shrink to 0 converges to $\{p\}$ in the Hausdorff metric. In other words, the dimension of a sequence of convex bodies can "collapse" in the limit. However, an easy observation yields that the family $\mathcal{K}^n$ of convex bodies is dense in CC^n, as we will see next.

Lemma 2.5.1. *The family $\mathcal{K}^n$ is dense in CC^n with respect to the Hausdorff metric.*

Proof. Let $C \in CC^n$. For any $m \in \mathbb{N}$, define $C_m := C + \frac{1}{m}B^n$. It is immediate to see that each C_m is a convex body, and that $C_m \to C$ in the Hausdorff metric.

□

As a consequence of Proposition 2.5.5 we have a very important theorem, which states that any bounded sequence (as a family of subsets of $\mathbb{R}^n$, or in the Hausdorff metric, see Proposition 2.5.3) of non-empty compact convex sets in $\mathbb{R}^n$ admits a subsequence which converges in the Hausdorff metric to a set of CC^n. In particular, any bounded sequence of convex bodies has a subsequence which converges in the Hausdorff metric to a non-empty compact convex set. Despite the fact that this is an easy consequence of the previous results, we name it a theorem because of its importance.

Theorem 2.5.4 (Blaschke's Selection Theorem)**.** *Let $(C_j)_{j\in\mathbb{N}}$ be a bounded sequence in CC^n. Then, there exists a subsequence $(C_{j_k})_{k\in\mathbb{N}}$ and a set $C \in CC^n$ with the property that $C_{j_k} \to C$ in the Hausdorff metric.*

Proof. A bounded sequence (C_j) in CC^n is, in particular, a bounded sequence in $(C^n, d_{\mathcal{H}})$. It follows from Theorem 2.5.3 that (C_j) admits a subsequence (C_{j_k}) converging to some $C \in C^n$. From Proposition 2.5.5 we get that $C \in CC^n$.

□

2.6 Approximations and continuity

Certain classes of convex bodies (such as polytopes, for example) are dense in $\mathcal{K}^n$ with respect to the Hausdorff metric. This fact is widely used to prove properties for general convex bodies by doing the calculations in those classes which are easier to deal with. The principal aims of this section are to present this approximation, and to prove that the Lebesgue volume is a continuous functional on CC^n with respect to the Hausdorff metric. We start proving the simple fact that the usual operations in CC^n are continuous in the Hausdorff metric.

Lemma 2.6.1. *Let $C_m \to C$ and $D_m \to D$ in CC^n with respect to the Hausdorff metric. Then, for every $\lambda \in \mathbb{R}$ we have that $C_m + \lambda D_m \to C + \lambda D$ in the Hausdorff metric.*

Proof. Simply notice that for any $u \in \mathbb{S}^{n-1}$ we may write

$$\begin{aligned}\left|h_{C_m+\lambda D_m}(u) - h_{C+\lambda D}(u)\right| &\leq |h_{C_m}(u) - h_C(u)| + |\lambda|\left|h_{D_m}(u) - h_D(u)\right| \leq \\ &\leq ||h_{C_m} - h_C||_{C(\mathbb{S}^{n-1})} + |\lambda|||h_{D_m} - h_D||_{C(\mathbb{S}^{n-1})}.\end{aligned}$$

The bound on the right-hand side does not depend on u, and goes to 0 as $m \to +\infty$. From Proposition 2.5.2 we get the desired.

□

Now we provide a useful criterion for convergence of convex, compact sets in the Hausdorff metric.

Theorem 2.6.1. *We have that* $\lim_{j\to+\infty} K_j = K$ *in* $(CC^n, d_{\mathcal{H}})$ *if and only if the following hold:*

(a) *For each* $x \in K$ *there exists a sequence* $(x_j)_{j\in\mathbb{N}}$ *such that* $x_j \in K_j$ *for each* $j \in \mathbb{N}$ *and* $x_j \to x$.

(b) *Any convergent sequence* $(x_{i_j})_{j\in\mathbb{N}}$ *with* $x_{i_j} \in K_{i_j}$ *for each* $i_j \in \mathbb{N}$ *has its limit in* K.

Proof. Assume first that $K_j \to K$ in $(CC^n, d_{\mathcal{H}})$. For a point $x \in K$, let $x_j := p_{K_j}(x)$ for each $j \in \mathbb{N}$. Therefore, $x_j \in K_j$ for every $j \in \mathbb{N}$, and also

$$|x - x_j| = \inf_{z\in K_j} |x - z| = \operatorname{dist}(x, K_j) \leq \sup_{x\in K} \operatorname{dist}(x, K_j) = d_{\mathcal{H}}(K, K_j).$$

Since $d_{\mathcal{H}}(K, K_j) \to 0$, we get that **(a)** holds.

Now assume that $(x_{i_j})_{j\in\mathbb{N}}$ is a converging sequence such that $x_{i_j} \in K_{i_j}$ for each $j \in \mathbb{N}$, and let $x := \lim_{j\to\infty} x_{i_j}$. Suppose that $x \notin K$. Then there exists a number $\rho > 0$ such that $\operatorname{dist}(x, K) > 2\rho > 0$, and consequently we get $B_{\mathbb{R}^n}(x, \rho) \cap (K + \rho B^n) = \emptyset$. However, since $x_{i_j} \to x$ we have that $|x - x_{i_j}| < \rho$ for sufficiently large j, and the convergence $K_{i_j} \to K$ yields that $K_{i_j} \subseteq K + \rho B^n$ for $j \in \mathbb{N}$ large enough. Hence the intersection $B_{\mathbb{R}^n}(x, \rho) \cap (K + \rho B^n)$ is non-empty, and this contradiction shows that $x \in K$. Thus, **(b)** holds.

To prove the converse, assume that **(a)** and **(b)** hold. We proceed by contradiction. Suppose that $\varepsilon > 0$ is a number for which there exists a sequence $i_j \to +\infty$ of natural numbers such that $K \not\subset K_{i_j} + \varepsilon B^n$ for any i_j. Hence, for any $j \in \mathbb{N}$ we can choose a point $z_{i_j} \in K$ such that $z_{i_j} \notin K_{i_j} + \varepsilon B^n$, which means that $\operatorname{dist}(z_{i_j}, K_{i_j}) \geq \varepsilon$. From the compactness of K, passing to a subsequence if necessary we may assume that $z_{i_j} \to z \in K$. From **(a)** there exists a sequence $x_{i_j} \in K_{i_j}$ such that $x_{i_j} \to z$. Since both sequences (x_{i_j}) and (z_{i_j}) have the same limit, we must have $|x_{i_j} - z_{i_j}| \to 0$ as $j \to \infty$, but

$$|x_{i_j} - z_{i_j}| \geq \operatorname{dist}(z_{i_j}, K_{i_j}) \geq \varepsilon,$$

for any $j \in \mathbb{N}$. Thus we have the desired contradiction, and hence for any $\varepsilon > 0$ there exists a number $j_0 \in \mathbb{N}$ such that $K \subseteq K_j + \varepsilon B^n$ for every $j > j_0$.

Now assume that there exists a number $\varepsilon > 0$ and a sequence $i_j \to \infty$ of natural numbers such that $K_{i_j} \not\subset K + \varepsilon B^n$ for each $j \in \mathbb{N}$. Hence we may choose a sequence $z_{i_j} \in K_{i_j}$ such that $z_{i_j} \notin K + \varepsilon B^n$ for each $j \in \mathbb{N}$. This means that $\operatorname{dist}(z_{i_j}, K) \geq \varepsilon$ for every i_j. Letting $x \in K$ be any point, by hypothesis **(a)** we can take a sequence $x_m \in K_m$ such that $x_m \to x$. Thus, we have $x_{i_j} \in K + \varepsilon B^n$ for sufficiently large j. Since $z_{i_j} \in K_{i_j} \setminus (K + \varepsilon B^n)$ and $x_{i_j} \in K + \varepsilon B^n$ (for sufficiently large i_j) we get from convexity that the segment connecting x_{i_j} and z_{i_j} contains a point $y_{i_j} \in K_{i_j} \cap \partial(K + \varepsilon B^n)$. Since $\partial(K + \varepsilon B^n)$ is compact, we have that y_{i_j} has a subsequence which converges to a point y of $\partial(K + \varepsilon B^n)$. However, since $y_{i_j} \in K_{i_j}$, we have from **(b)** that $y \in K$, which is a contradiction. Consequently, for every $\varepsilon > 0$ there exists a number $j_1 \in \mathbb{N}$ such that $K_j \subseteq K + \varepsilon B^n$ whenever $j > j_1$. It follows that $K_j \to K$ in $(CC^n, d_{\mathcal{H}})$.

□

It is also useful to understand how the Hausdorff metric behaves under unions of (finitely many) sets of C^n. We investigate this next.

Lemma 2.6.2. *For any* $C, D, K, L \in C^n$, *we have that*

$$d_{\mathcal{H}}(C \cup K, D \cup L) \leq \max\left\{d_{\mathcal{H}}(C, D), d_{\mathcal{H}}(K, L)\right\}.$$

In particular, if $(C_k)_{k\in\mathbb{N}}$ *and* $(L_k)_{k\in\mathbb{N}}$ *are sequences in* C^n *converging to* $C \in C^n$ *and* $L \in C^n$, *respectively, then the sequence* $(C_k \cup L_k)$ *converges to* $C \cup L$.

Proof. Simply notice that if $\varepsilon_1, \varepsilon_2 > 0$ are such that $C \subseteq D + \varepsilon_1 B^n$ and $K \subseteq L + \varepsilon_2 B^n$, then

$$C \cup K \subseteq (D + \varepsilon_1 B^n) \cup (L + \varepsilon_2 B^n) \subseteq (D \cup L) + \max\{\varepsilon_1, \varepsilon_2\} B^n.$$

Analogously, if $D \subseteq C + \varepsilon_1 B^n$ and $L \subseteq K + \varepsilon_2 B^n$, then it follows that $D \cup L \subseteq C \cup K + \max\{\varepsilon_1, \varepsilon_2\} B^n$. Letting $\varepsilon_1 \to d_{\mathcal{H}}(C, D)$ and $\varepsilon_2 \to d_{\mathcal{H}}(K, L)$, we get the desired inequality. The other claim is an immediate consequence of this inequality.

□

Recall that a *polytope* is the convex hull of finitely many points of $\mathbb{R}^n$. A polytope is always a set of CC^n. Indeed, a polytope $P = \operatorname{conv}\{x_1, \ldots, x_m\}$ is convex by definition, and it is bounded because

$$|\lambda_1 x_1 + \ldots + \lambda_m x_m| \leq (\lambda_1 + \ldots + \lambda_m) \max\{|x_j| : j = 1, \ldots, m\} = \max\{|x_j| : j = 1, \ldots, m\}$$

for any convex combination $\lambda_1 x_1 + \ldots + \lambda_m x_m$ of $\{x_1, \ldots, x_m\}$. The argument to verify that P is closed is a standard one. Let $(\lambda_1^k x_1 + \ldots + \lambda_m^k x_m)_{k \in \mathbb{N}}$ be a sequence of points in P converging to $x \in \mathbb{R}^n$. For each $1 \leq j \leq m$ the sequence $(\lambda_j^k)_{k \in \mathbb{N}}$ of numbers is bounded, and hence has a converging subsequence. Since we have finitely many sequences, proceeding inductively we may assume that $\lambda_j^k \to \lambda_j$ as $k \to \infty$ for each $j = 1, \ldots, m$. It is easily seen that $x = \lambda_1 x_1 + \ldots + \lambda_m x_m$, and that this is a convex combination of $\{x_1, \ldots, x_m\}$.

Theorem 2.6.2. *For every $C \in CC^n$ and any $\varepsilon > 0$ there exists a polytope $P \subseteq CC^n$ such that $P \subseteq C \subseteq P + \varepsilon B^n$.*

Proof. Since C is compact, we may choose finitely many points $x_1, \ldots, x_m \in C$ such that

$$C \subseteq \bigcup_{j=1}^{m} (x_j + \varepsilon B^n).$$

Then we simply define $P := \operatorname{conv}\{x_1, \ldots, x_m\}$. The inclusion $P \subseteq C$ comes immediately from the convexity of C, and $C \subseteq P + \varepsilon B^n$ comes straightforwardly from the inclusion above. This construction is illustrated in Figure 2.6.1.

□

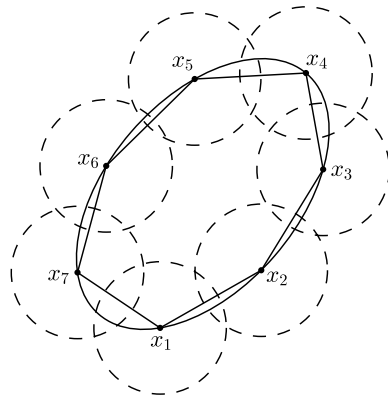


Fig. 2.6.1 The approximation of a convex body by a polytope.

Remark 2.6.1. Of course, if $P \subseteq C \subseteq P + \varepsilon B^n$, then $d_{\mathcal{H}}(C, P) \leq \varepsilon$.

Corollary 2.6.1. *The metric space $(CC^n, d_{\mathcal{H}})$ is separable.*

Proof. Let $\varepsilon > 0$ be arbitrarily given. As in Theorem 2.6.2 above, assume that $x_1, \dots, x_m \in C$ are such that

$$C \subseteq \bigcup_{j=1}^{m} x_j + \frac{\varepsilon}{2} B^n.$$

For each x_j, we may choose $z_j \in \mathbb{Q}^n$ such that

$$x_j \in z_j + \frac{\varepsilon}{2} B^n. \tag{2.6.1}$$

Notice that, in this case, we also have

$$z_j \in x_j + \frac{\varepsilon}{2} B^n \subseteq C + \frac{\varepsilon}{2} B^n \subseteq C + \varepsilon B^n. \tag{2.6.2}$$

Let $P = \operatorname{conv}\{z_1, \dots, z_m\}$. From (2.6.1), we have that

$$C \subseteq \bigcup_{j=1}^{m} z_j + \frac{\varepsilon}{2} B^n + \frac{\varepsilon}{2} B^n = z_j + \varepsilon B^n \subseteq P + \varepsilon B^n.$$

On the other hand, since $C + \varepsilon B^n$ is convex, (2.6.2) gives that

$$P = \operatorname{conv}\{z_1, \dots, z_m\} \subseteq C + \varepsilon B^n.$$

Hence, for any $\varepsilon > 0$ and for each $C \in CC^n$ there exists a polytope P which is the convex hull of points of $\mathbb{Q}^n$ and such that $d_{\mathcal{H}}(C, P) \leq \varepsilon$. Consequently, the countable set

$$\Big\{ \operatorname{conv}\{z_1, \dots, z_m\} : m \in \mathbb{N} \text{ and } z_j \in \mathbb{Q}^n \text{ for all } 1 \leq j \leq m \Big\}$$

is dense in CC^n with respect to the Hausdorff metric.

□

Recall that the polarity map is the map $\operatorname{pol}\colon 2^{\mathbb{R}^n} \setminus \{\emptyset\} \to 2^{\mathbb{R}^n}$ which associates each subset of $\mathbb{R}^n$ with its polar set. We already know that the restriction of the polarity map to the collection of convex bodies containing the origin as an interior point (denoted by $\mathcal{K}_0^n$) is an involution (Corollary 2.3.5). Next we prove that this is continuous in the Hausdorff metric. The reader may notice carefully that the space $\mathcal{K}_0^n$ endowed with the Hausdorff distance $d_{\mathcal{H}}$ is **not** a complete metric space.

Proposition 2.6.1. *The restriction of the polarity map to $\mathcal{K}_0^n$ is continuous in the Hausdorff metric.*

Proof. Let $(K_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{K}_0^n$ converging in the Hausdorff metric to a body $K \in \mathcal{K}_0^n$. We have to prove that $K_j^\circ \to K^\circ$ in the Hausdorff distance. We divide the proof into steps. Before starting, we put (to simplify the notation)

$$\gamma_K := \max\{|x| : x \in K\},$$

for each $K \in \mathcal{K}_0^n$. Observe that, in this case, we have $K \subseteq \gamma_K B^n$.

First step. We prove that for any $K \in \mathcal{K}_0^n$ and every $\varepsilon > 0$ we have

$$d_{\mathcal{H}}(K^\circ, (K + \varepsilon B^n)^\circ) \leq \varepsilon (\gamma_{K^\circ})^2.$$

Since $(K + \varepsilon B^n)^\circ \subseteq K^\circ$, all we have to prove is that

$$K^\circ \subseteq (K + \varepsilon B^n)^\circ + \varepsilon(\gamma_{K^\circ})^2 B^n,$$

and by Item **(iii)** of Proposition 2.5.2, it is enough to prove that for every point $z \in \partial K^\circ$ there exists a point $w \in \partial(K + \varepsilon B^n)^\circ$ such that $|z - w| \leq \varepsilon(\gamma_{K^\circ})^2$. To do so, notice that a given point $z \in \partial K^\circ$ can be written as

$$z = \frac{u}{||u||_{K^\circ}}$$

for some $u \in \mathbb{S}^{n-1}$. Geometrically, z is the point where the ray starting at $0_{\mathbb{R}^n}$ and going in the direction of u intersects ∂K°. We also let

$$w = \frac{u}{||u||_{(K+\varepsilon B^n)^\circ}}$$

be the point where this ray intersects the boundary of $(K + \varepsilon B^n)^\circ$. From Proposition 2.3.2 and Proposition 2.3.6 we get

$$||u||_{(K+\varepsilon B^n)^\circ} - ||u||_{K^\circ} = h_{K+\varepsilon B^n}(u) - h_K(u) = h_K(u) + \varepsilon h_{B^n}(u) - h_K(u) = \varepsilon.$$

In particular, we have that $||u||_{(K+\varepsilon B^n)^\circ} > ||u||_{K^\circ}$ (this can be obtained immediately from the inclusion $(K + \varepsilon B^n)^\circ \subseteq K^\circ$). This leads to the inequalities

$$\frac{1}{||u||_{K^\circ}} = |z| \leq \gamma_{K^\circ} \text{ and}$$
$$\frac{1}{||u||_{(K+\varepsilon B^n)^\circ}} < \frac{1}{||u||_{K^\circ}} \leq \gamma_{K^\circ}.$$

Combining these, we finally arrive at

$$|z - w| = \left| \frac{u}{||u||_{K^\circ}} - \frac{u}{||u||_{(K+\varepsilon B^n)^\circ}} \right| = \frac{||u||_{(K+\varepsilon B^n)^\circ} - ||u||_{K^\circ}}{||u||_{(K+\varepsilon B^n)^\circ} ||u||_{K^\circ}} =$$
$$= \frac{\varepsilon}{||u||_{(K+\varepsilon B^n)^\circ} ||u||_{K^\circ}} < \varepsilon(\gamma_{K^\circ})^2,$$

and this concludes the first step.

Second step. Now we prove that if $K, L \in \mathcal{K}_0^n$ are such that $d_{\mathcal{H}}(K, L) \leq \varepsilon$, then

$$d_{\mathcal{H}}(K^\circ, L^\circ) \leq \varepsilon(\gamma_{L^\circ})^2 + 2\varepsilon(\gamma_{K^\circ})^2.$$

In fact, if $d_{\mathcal{H}}(K, L) \leq \varepsilon$, then

$$K \subseteq L + \varepsilon B^n \subseteq K + 2\varepsilon B^n,$$

and this leads to

$$(K + 2\varepsilon B^n)^\circ \subseteq (L + \varepsilon B^n)^\circ \subseteq K^\circ.$$

As an immediate consequence, it follows that (see Remark 2.6.2 below)

$$d_{\mathcal{H}}((L+\varepsilon B^n)^\circ, K^\circ) \leq d_{\mathcal{H}}((K+2\varepsilon B^n)^\circ, K^\circ) \leq 2\varepsilon(\gamma_{K^\circ})^2,$$

and using the first step together with triangle inequality we finally get

$$d_{\mathcal{H}}(K^\circ, L^\circ) \leq d_{\mathcal{H}}(L^\circ, (L+\varepsilon B^n)^\circ) + d_{\mathcal{H}}((L+\varepsilon B^n)^\circ, K^\circ) \leq \varepsilon(\gamma_{L^\circ})^2 + 2\varepsilon(\gamma_{K^\circ})^2,$$

as we have claimed.

Third step. Now we have to formalize the intuition that we can "control" the size of γ_{L° in terms of K if L is "sufficiently close" to K. To do so, we fix a number $\rho > 0$ such that $\rho B^n \subseteq \mathrm{int}K$. By doing so, we get that

$$h_K(u) - h_{\rho B^n}(u) > 0$$

for every $u \in \mathbb{S}^{n-1}$. By continuity and compactness, the map $\mathbb{S}^{n-1} \ni u \mapsto h_K(u) - h_{\rho B^n}(u)$ has a positive minimum:

$$c_0 := \min\{h_K(u) - h_{\rho B^n}(u) : u \in \mathbb{S}^{n-1}\} > 0. \tag{2.6.3}$$

We claim that if $d_{\mathcal{H}}(K, L) < c_0$, then $\rho B^n \subseteq L$. Indeed, from Item **(ii)** of Proposition 2.5.2 we have that $h_L(u) \geq h_K(u) - c_0$ for all $u \in \mathbb{S}^{n-1}$. Hence, from the definition of c_0, we obtain

$$h_L(u) \geq h_K(u) - c_0 \geq h_K(u) - (h_K(u) - h_{\rho B^n}(u)) = h_{\rho B^n}(u)$$

for any $u \in \mathbb{S}^{n-1}$. It follows from Item **(iv)** of Proposition 2.3.2 that $\rho B^n \subseteq L$. Observe carefully that this inclusion leads to $L^\circ \subseteq (\rho B^n)^\circ = \frac{1}{\rho} B^n$, and hence

$$\gamma_{L^\circ} \leq \frac{1}{\rho}.$$

Fourth step. Let $(K_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{K}_0^n$ converging to a body $K \in \mathcal{K}_0^n$. Fix $\rho > 0$ such that $\rho B^n \subseteq \mathrm{int}K$, and let c_0 be as in (2.6.3). For a given $\varepsilon > 0$, let $n_0 \in \mathbb{N}$ be such that

$$d_{\mathcal{H}}(K_j, K) \leq \min\{\varepsilon, c_0\}$$

for every $j \geq n_0$. From the second step, we get

$$d_{\mathcal{H}}(K_j^\circ, K^\circ) \leq \varepsilon(\gamma_{K_j^\circ})^2 + 2\varepsilon(\gamma_{K^\circ})^2$$

whenever $j \geq n_0$. The third step gives that $\gamma_{K_j^\circ} \leq 1/\rho$ for every $j \geq n_0$, and hence we get

$$d_{\mathcal{H}}(K_j^\circ, K^\circ) \leq \varepsilon\left(\frac{1}{\rho^2} + 2(\gamma_{K^\circ})^2\right)$$

if $j \geq n_0$. Since ρ and γ_{K° only depend on K, the inequality above yields that $K_j^\circ \to K^\circ$ as $j \to \infty$, and this concludes the proof.

□

Remark 2.6.2. If $A \subseteq B \subseteq C$, then we clearly have that $d_{\mathcal{H}}(B, C) \leq d_{\mathcal{H}}(A, C)$. Indeed, if $A \subseteq C + \varepsilon B^n$ and $C \subseteq A + \varepsilon B^n$, then we have

$$B \subseteq C \subseteq A + \varepsilon B^n \subseteq C + \varepsilon B^n \text{ and}$$
$$C \subseteq A + \varepsilon B^n \subseteq B + \varepsilon B^n.$$

It is easy to see that convexity and compactness are preserved under linear transformations, and hence the image of a set of $\in CC^n$ under a linear map is also a set of CC^n. The image of a set under a linear map is called a *linear image*. When studying the *positions* of a convex body (which are the images of the given convex body under *affine transformations*, that is, compositions of translations with isomorphisms), it is important to know whether a sequence of linear images of a given body converges in the Hausdorff metric. We investigate it now.

For the convenience of the reader, we introduce some definitions and results in linear algebra. We denote the space of linear transformations $T\colon \mathbb{R}^n \to \mathbb{R}^n$ by $\mathrm{L}(n)$. This is clearly a finite-dimensional vector space, and we always consider that this is endowed with the norm

$$||T||_{\mathrm{L}(n)} := \max\left\{|Tu| : u \in B^n\right\} = \max\{|Tu| : u \in \mathbb{S}^{n-1}\} = \max\left\{\frac{|Tu|}{|u|} : u \neq 0_{\mathbb{R}^n}\right\}.$$

It is readily seen that this is indeed a norm in $\mathrm{L}(n)$, and that both equalities above in fact hold. Moreover, the last equality gives straightforwardly the inequality

$$|Tu| \leq ||T||_{\mathrm{L}(n)}|u|$$

for every $u \in \mathbb{R}^n$. For us, *convergence* in $\mathrm{L}(n)$ means convergence in the norm $||\cdot||_{\mathrm{L}(n)}$. That is, a sequence $(T_m)_{m\in\mathbb{N}}$ *converges* to a linear map $T \in \mathrm{L}(n)$ if $||T_m - T||_{\mathrm{L}(n)} \to 0$ as $m \to \infty$.

Another concept that we need to borrow from linear algebra is that of the *adjoint* of a linear map. Given $T \in \mathrm{L}(n)$, the *adjoint* of T is the unique map $T^* \in \mathrm{L}(n)$ such that

$$\langle Tx, y\rangle = \langle x, T^*y\rangle. \tag{2.6.4}$$

The existence and uniqueness of the adjoint are easy to prove: indeed, simply let T be the linear transformation whose canonical matrix representation is the transpose of the canonical matrix representation of T. For us, the importance of the adjoint comes mostly from the equality

$$\begin{aligned} h_{TA}(u) = \sup\left\{\langle x, u\rangle : x \in TA\right\} = \sup\left\{\langle Tz, u\rangle : z \in A\right\} = \\ = \sup\left\{\langle z, T^*u\rangle : z \in A\right\} = h_A(T^*u), \end{aligned} \tag{2.6.5}$$

which holds for any $A \subseteq \mathbb{R}^n$, $T \in \mathrm{L}(n)$ and $u \in \mathbb{R}^n$. Below we collect some properties of the adjoint that will be important later.

Lemma 2.6.3. *Let $T, S \in \mathrm{L}(n)$ and $\alpha \in \mathbb{R}$. Then the following statements hold.*

(i) $T^{**} := (T^*)^* = T$,

(ii) $(T + \alpha S)^* = T^* + \alpha S^*$,

(iii) $||T^*||_{\mathrm{L}(n)} = ||T||_{\mathrm{L}(n)}$ *and*

(iv) $\det(T^*) = \det(T)$. *In particular, T^* is an isomorphism if and only if T is an isomorphism.*

Proof. For **(i)**, notice that for any $x, y \in \mathbb{R}^n$ we have

$$\langle Tx, y\rangle = \langle x, T^*y\rangle = \langle (T^*)^*x, y\rangle,$$

and hence $\langle (T^{**} - T)x, y\rangle = 0$ for every $x, y \in \mathbb{R}^n$. It is clear that this is only possible if $T^{**} = T$, once any vector of the image of $T^{**} - T$ lies in the orthogonal complement of $\mathbb{R}^n$.

Now we go for **(ii)**. For any $x, y \in \mathbb{R}^n$, we calculate

$$\begin{aligned}\langle (T + \alpha S)^* x, y\rangle = \langle x, (T + \alpha S)y\rangle = \langle x, Ty\rangle + \langle \alpha x, Sy\rangle = \\ = \langle T^* x, y\rangle + \langle S^*(\alpha x), y\rangle = \langle (T^* + \alpha S^*)x, y\rangle,\end{aligned}$$

and, as before, this is only possible if $(T + \alpha S)^* = T^* + \alpha S^*$. Notice that **(ii)** means that the adjoint seen as a map $^*\colon \mathrm{L}(n) \to \mathrm{L}(n)$ is a linear map.

To prove **(iii)**, observe that for any $u \in \mathbb{S}^{n-1}$ we have

$$|T^* u|^2 = \langle T^* u, T^* u\rangle = \langle u, TT^* u\rangle \leq |u|\,|T(T^* u)| \leq ||T||_{\mathrm{L}^n}\,|T^* u|,$$

and this gives that $|T^* u| \leq ||T||_{\mathrm{L}(n)}$ for each $u \in \mathbb{S}^{n-1}$. Passing to the maximum as $u \in \mathbb{S}^{n-1}$ we arrive at $||T^*||_{\mathrm{L}(n)} \leq ||T||_{\mathrm{L}(n)}$. This also leads to $||T^{**}||_{\mathrm{L}(n)} \leq ||T^*||_{\mathrm{L}(n)}$, and using **(i)** the desired follows.

Last, to show **(iv)** we only need to remember that the canonical matrix representation of T^* is the transpose of the canonical matrix representation of T, and hence T and T^* have the same determinant. □

Proposition 2.6.2. *Let $(T_m)_{m\in\mathbb{N}}$ be a sequence in $\mathrm{L}(n)$ which converges to $T \in \mathrm{L}(n)$ in the norm defined above. Then for any $C \in CC^n$ the sequence $(T_m C)_{m\in\mathbb{N}} \subseteq CC^n$ converges in the Hausdorff metric to $TC \in CC^n$.*

Proof. First notice that by the sublinearity of h_C we have

$$h_C(u) = h_C(u - v + v) \leq h_C(u - v) + h_C(v)$$

for any $u, v \in \mathbb{R}^n$. This leads to $h_C(u) - h_C(v) \leq h_C(u - v)$.

Now fix an arbitrary number $m \in \mathbb{N}$. Let $u \in \mathbb{S}^{n-1}$ be such that $u \notin \ker(T_m^* - T^*)$, and assume that $h_{T_m C}(u) \geq h_{TC}(u)$. Then, using the inequality derived above and the homogeneity of h_C, we arrive at

$$\begin{aligned}|h_{T_m C}(u) - h_{TC}(u)| = h_{T_m C}(u) - h_{TC}(u) = h_C(T_m^* u) - h_C(T^* u) \leq \\ \leq h_C((T_m^* - T^*)u) = |(T_m^* - T^*)u|\, h_C\left(\frac{(T_m^* - T^*)u}{|(T_m^* - T^*)u|}\right) \leq \\ \leq ||T_m^* - T^*||_{\mathrm{L}(n)}|u| \sup\left\{h_C(v) : v \in \mathbb{S}^{n-1}\right\} = c||T_m^* - T^*||_{\mathrm{L}(n)} = c||T_m - T||_{\mathrm{L}(n)},\end{aligned}$$

where $c := \sup\left\{h_C(v) : v \in \mathbb{S}^{n-1}\right\}$ is finite because h_C is continuous (every sublinear function is continuous) and $\mathbb{S}^{n-1}$ is compact. We also should have in mind that the constant c does not depend on m (or even on T), but only on C. If we still have that $u \in \mathbb{S}^{n-1}$ and $u \notin \ker(T_m^* - T^*)$, but assume now that $h_{TC}(u) \geq h_{T_m C}(u)$, then we repeat the argument interchanging the roles of T and T_m to obtain

$$\begin{aligned}|h_{T_m C}(u) - h_{TC}(u)| = h_{TC}(u) - h_{T_m C}(u) = h_C(T^* u) - h_C(T_m^* u) \leq \\ \leq h_C((T^* - T_m^*)u) \leq c||T^* - T_m^*||_{\mathrm{L}(n)} = c||T_m^* - T^*||_{\mathrm{L}(n)} = c||T_m - T||_{\mathrm{L}(n)}.\end{aligned}$$

On the other hand, if $u \in \mathbb{S}^{n-1} \cap \ker(T_m^* - T^*)$, then

$$|h_{T_m C}(u) - h_C(u)| \leq h_C((T_m^* - T^*)u) = h_C(0_{\mathbb{R}^n}) = 0 \leq c||T_m^* - T^*||_{\mathrm{L}(n)} = c||T_m - T||_{\mathrm{L}(n)}.$$

Summarizing, the inequality $|h_{T_m C}(u) - h_{TC}(u)| \leq c||T_m - T||_{\mathrm{L}(n)}$ holds for every $u \in \mathbb{S}^{n-1}$. Consequently, using Item **(ii)** of Proposition 2.5.2 we get

$$d_{\mathcal{H}}(T_mC, TC) = \sup\left\{|h_{T_mC}(u) - h_{TC}(u)| : u \in \mathbb{S}^{n-1}\right\} \leq c||T_m - T||_{\mathrm{L}(n)},$$

and once $T_m \to T$ in $\mathrm{L}(n)$ it follows that $T_mC \to TC$ in the Hausdorff metric.

□

In what follows, we fix the terminology (*n-dimensional*) *volume* for the standard Lebesgue measure $\mathcal{L}^n$ on $\mathbb{R}^n$ (see Appendix A). We also introduce the notation $\mathrm{vol}_n(\cdot) := \mathcal{L}^n(\cdot)$, which is somewhat more consistent with our viewpoint. As we will see, the volume is a continuous functional on $(CC^n, d_{\mathcal{H}})$, but before this we need an auxiliary lemma.

Lemma 2.6.4. *Let $K \in \mathcal{K}^n$ and $L \in CC^n$ be such that $L \subseteq \mathrm{int}K$. There exists a number $\beta > 0$ such that every $C \in CC^n$ with $d_{\mathcal{H}}(K, C) < \beta$ satisfies $L \subseteq C$.*

Proof. The inclusion $L \subseteq \mathrm{int}K$ gives that $h_K(u) > h_L(u)$ for each $u \in \mathbb{S}^{n-1}$. Indeed, we have trivially that $h_K(u) \geq h_L(u)$, and if equality holds, then K and L would have a boundary point in common. It follows that the function

$$\mathbb{S}^{n-1} \ni u \mapsto h_K(u) - h_L(u)$$

attains a positive minimum β, say. Then, $\beta \leq h_K(u) - h_L(u)$ for every $u \in \mathbb{S}^{n-1}$. If $C \in CC^n$ is such that $d_{\mathcal{H}}(K, C) < \beta$, then $|h_K(u) - h_C(u)| < \beta$ for every $u \in \mathbb{S}^{n-1}$ (see Item **(ii)** of Proposition 2.5.2). In particular, we have that $h_K(u) - \beta < h_C(u)$. These inequalities lead to

$$h_L(u) \leq h_K(u) - \beta < h_C(u)$$

for any $u \in \mathbb{S}^{n-1}$. From Proposition 2.3.2, the inequality above gives the inclusion $L \subseteq C$.

□

To prove that the volume is continuous in the Hausdorff metric, we will need some properties of the Lebesgue measure of $\mathbb{R}^n$ (which are adequately described whenever they are needed). These properties are mainly discussed in Appendix A, but the ones which are more related to convexity are given in Chapter 4 (such as the volume of the unit ball, for example). The reader who does not have formal knowledge in measure theory may read the proof of the next theorem without knowing the proofs of the mentioned properties, which will be given later in the book.

We also need a concept from linear algebra which is, in general, very important for us. A linear transformation $U\colon \mathbb{R}^n \to \mathbb{R}^n$ is called an *orthogonal transformation* if it preserves the inner product, that is, if $\langle Ux, Uz\rangle = \langle x, z\rangle$ for any $x, z \in \mathbb{R}^n$. Orthogonal transformations and their properties are omnipresent in this book, and hence we collect some facts about them that will be used throughout the text. Of course, the reader who is familiar with linear algebra may skip this part.

Lemma 2.6.5. *Let $U \in \mathrm{L}(n)$. The following statements are equivalent:*

(i) *U is orthogonal,*

(ii) *U is an* isometry*, meaning that $|Ux| = |x|$ for every $x \in \mathbb{R}^n$, and*

(iii) $U^*U = \mathrm{Id}_{\mathbb{R}^n}$,

As a consequence, we have that every orthogonal transformation is an isomorphism whose determinant has absolute value 1.

Proof. We prove that **(ii)** and **(iii)** are equivalent to **(i)**. The implication **(i)**⇒**(ii)** is immediate. For **(ii)**⇒**(i)**, observe that for any $x, y \in \mathbb{R}^n$ we have

$$|x+y|^2 = |U(x+y)|^2 = \langle U(x+y), U(x+y)\rangle = |Ux|^2 + |Uy|^2 + 2\langle Ux, Uy\rangle = \\ = |x|^2 + |y|^2 + 2\langle Ux, Uy\rangle,$$

and since $|x+y|^2 = |x|^2 + 2\langle x, y\rangle + |y|^2$ we get that $\langle Ux, Uy\rangle = \langle x, y\rangle$. Notice that this proves that every orthogonal transformation is an isomorphism, because every isometry is clearly injective (and any injective linear transformation of $\mathbb{R}^n$ is also surjective).

Now we prove that **(i)** implies **(iii)**. For any $x, y \in \mathbb{R}^n$, we have that

$$\langle x, y\rangle = \langle Ux, Uy\rangle = \langle x, U^*Uy\rangle,$$

and hence $\langle x, (U^*U - \mathrm{Id}_{\mathbb{R}^n})y\rangle = 0$ for all $x, y \in \mathbb{R}^n$. Thus, $U^*U = \mathrm{Id}_{\mathbb{R}^n}$. On the other hand, if **(iii)** holds, then for $x, y \in \mathbb{R}^n$ we have

$$\langle x, y\rangle = \langle x, U^*Uy\rangle = \langle Ux, Uy\rangle,$$

where we used the fact that $U^{**} = U$. Hence U is orthogonal, and this proves the remaining implication **(iii)⇒(i)**.

□

Remark 2.6.3. If U is orthogonal, then we also have that $UU^* = \mathrm{Id}_{\mathbb{R}^n}$. Indeed, since we know that U is invertible, the equality $U^*U = \mathrm{Id}_{\mathbb{R}^n}$ leads to $U^* = U^{-1}$, and hence

$$\mathrm{Id}_{\mathbb{R}^n} = UU^{-1} = UU^*.$$

Thus, yet another equivalence is that $U \in \mathrm{L}(n)$ is orthogonal if and only if U is invertible and $U^{-1} = U^*$.

Any isomorphism of $\mathbb{R}^n$ which carries an orthonormal basis onto another orthonormal basis is clearly an isometry, and hence it is an orthogonal transformation (the converse is also true, of course). For that reason, sometimes we think of orthogonal transformations as "well-behaved" changes of variables. In fact, an orthogonal transformation preserves distances (because it is an isometry) and volumes (since its determinant equals 1 or -1; see Theorem A.3.3). Even more, it is easy to see that if $\mathbf{g}_1$ and $\mathbf{g}_2$ are k-dimensional vector subspaces of $\mathbb{R}^n$, then there exists an orthogonal transformation U such that $U(\mathbf{g}_1) = \mathbf{g}_2$. To verify this, simply fix an (ordered) orthonormal basis B_1 of $\mathbb{R}^n$ whose first k vectors span $\mathbf{g}_1$, an (ordered) orthonormal basis B_2 whose first k vectors span $\mathbf{g}_2$, and consider the (ordered) orthogonal transformation which carries B_1 onto B_2.

Theorem 2.6.3. *The n-dimensional volume in $\mathbb{R}^n$ restricted to CC^n is a continuous functional with respect to the Hausdorff metric.*

Proof. We start by proving the continuity for the case where $\mathrm{vol}_n(K) = 0$. If we have this equality, then K does not have maximal affine dimension (otherwise, it would contain a ball, and this would lead to positive volume; see Corollary 4.3.1 ahead), and hence K is contained in some hyperplane, which we may assume that it is the hyperplane $\{(x_1, \ldots, x_{n-1}, 0) : x_1, \ldots, x_{n-1} \in \mathbb{R}\} \simeq \mathbb{R}^{n-1}$. This is done by an orthogonal transformation composed with a translation, thus preserving volume and distances (see Lemma A.3.2 for the translation-invariance of the Lebesgue volume). From compactness, we have that K is contained in some $(n-1)$-dimensional cell:

$$K \subseteq \prod_{j=1}^{n-1} [a_j, b_j].$$

If $L \in CC^n$ is such that $d_{\mathcal{H}}(K, L) = \delta < 1$, then we have that

$$L \subseteq K + \delta B^n \subseteq \left(\prod_{j=1}^{n-1} [a_j - 1, b_j + 1] \right) \times [-\delta, \delta],$$

from which it follows that

$$\mathrm{vol}_n(L) \leq 2\delta \prod_{j=1}^{n-1} (b_j - a_j + 2),$$

since it is immediate from the definition that the Lebesgue measure is monotone under inclusion. The product above only depends on K. Then, for a given number $\varepsilon > 0$, we choose $\delta > 0$ such that

$$\delta < \min \left\{ 1, \varepsilon \left(2 \prod_{j=1}^{n-1} (b_j - a_j + 2) \right)^{-1} \right\},$$

and the argument above shows that if $L \in CC^n$ is such that $d_{\mathcal{H}}(K, L) < \delta$, then $|\mathrm{vol}_n(L) - \mathrm{vol}_n(K)| = \mathrm{vol}_n(L) < \varepsilon$.

Now we prove the case where $\mathrm{vol}_n(K) > 0$. In this case, K has maximal affine dimension, and hence it has non-empty interior. Up to translation (which preserves volume), we may assume that $0_{\mathbb{R}^n} \in \mathrm{int} K$, and hence we may fix a number $r > 0$ such that $rB^n \subseteq \mathrm{int} K$. For an arbitrarily given $\varepsilon > 0$, choose a number $\lambda > 1$ with the property that $(\lambda^n - 1)\lambda^n \mathrm{vol}_n(K) < \varepsilon$ (the existence of such number $\lambda > 1$ is guaranteed by the continuity of the function $t \mapsto (t^n - 1)t^n$). By Lemma 2.6.4, there exists a number $\beta > 0$ such that $rB^n \subseteq L$ whenever $d_{\mathcal{H}}(K, L) < \beta$. Since it is clear that any number $0 < \alpha < \beta$ also has this property, we may assume that $\beta \leq (\lambda - 1)r$. Then, if $L \in CC^n$ and $d_{\mathcal{H}}(K, L) < \beta$, we have $rB^n \subseteq L$ and

$$K \subseteq L + \beta B^n \subseteq L + (\lambda - 1) r B^n \subseteq L + (\lambda - 1)L = \lambda L,$$

and the same argument gives that $L \subseteq \lambda K$. Moreover, the dilation map $x \mapsto \lambda x$ is a linear map whose determinant is λ^n, and hence Theorem A.3.3 gives that $\mathrm{vol}_n(\lambda K) = \lambda^n \mathrm{vol}_n(K)$ and $\mathrm{vol}_n(\lambda L) = \lambda^n \mathrm{vol}_n(L)$. Therefore, we have that $\mathrm{vol}_n(K) \leq \lambda^n \mathrm{vol}_n(L)$ and $\mathrm{vol}_n(L) \leq \lambda^n \mathrm{vol}_n(K)$. These inequalities give

$$\begin{aligned} \mathrm{vol}_n(K) - \mathrm{vol}_n(L) &\leq (\lambda^n - 1)\mathrm{vol}_n(L) \leq (\lambda^n - 1)\lambda^n \mathrm{vol}_n(K) \text{ and} \\ \mathrm{vol}_n(L) - \mathrm{vol}_n(K) &\leq (\lambda^n - 1)\mathrm{vol}_n(K) \leq (\lambda^n - 1)\lambda^n \mathrm{vol}_n(K), \end{aligned}$$

where the last inequality comes from $\lambda^n > 1$. Finally, this yields

$$|\mathrm{vol}_n(K) - \mathrm{vol}_n(L)| \leq (\lambda^n - 1)\lambda^n \mathrm{vol}_n(K) < \varepsilon,$$

and this concludes the proof.

□

2.7 Steiner symmetrizations

In this section we describe the *Steiner symmetrization*, which is a useful tool to study geometric inequalities. Let $K \subseteq \mathbb{R}^n$ be a convex body, and let $u \in \mathbb{R}^n$ be a given non-zero vector. Denote by $u^\perp$ the orthogonal hyperplane of u (observe that $u^\perp = \mathbf{h}(u, 0)$; both notations will be used throughout the

text). We have that $\mathbb{R}^n$ can be written as the direct sum $\mathbb{R}^n = u^\perp \oplus \operatorname{span}\{u\}$. Thus, any point $z \in \mathbb{R}^n$ can be written as $z = v + \alpha u$ for some (unique) $v \in u^\perp$ and $\alpha \in \mathbb{R}$. Having this in mind, the *projection onto* $u^\perp$ is the map $\operatorname{proj}_{u^\perp} \colon \mathbb{R}^n \to u^\perp$ defined for each $z = v + \alpha u \in \mathbb{R}^n$ as

$$\operatorname{proj}_{u^\perp}(v + \alpha u) = v.$$

Of course, if u is unit, then the decomposition $z = v + \alpha u$ reads $z = (z - \langle z, u\rangle u) + \langle z, u\rangle u$ (that is, $v = z - \langle z, u\rangle u$ and $\alpha = \langle z, u\rangle$), and hence

$$\operatorname{proj}_{u^\perp}(z) = z - \langle z, u\rangle u.$$

Still assuming that u is a non-zero vector, the *reflection across* $u^\perp$ is the map $\operatorname{ref}_{u^\perp} \colon \mathbb{R}^n \to \mathbb{R}^n$ defined for any $z = v + \alpha u$ as

$$\operatorname{ref}_{u^\perp}(v + \alpha u) = v - \alpha u.$$

If u is unit, then, since we already know that $v = z - \langle z, u\rangle u$ and $\alpha = \langle z, u\rangle$, we simply have

$$\operatorname{ref}_{u^\perp}(z) = z - 2\langle z, u\rangle u.$$

We say that a set $A \subseteq \mathbb{R}^n$ is *symmetric with respect to* $u^\perp$ if $\operatorname{ref}_{u^\perp}(A) = A$. All of these concepts can be directly extended to the case where $u^\perp$ is replaced by a hyperplane $\mathbf{h} := \mathbf{h}(u, \beta)$ which is not necessarily a vector subspace. In this case, we still have that each point $z \in \mathbb{R}^n$ can be written as $z = x + \alpha u$ for some (unique) $x \in \mathbf{h}(u, \beta)$ and $\alpha \in \mathbb{R}$. Hence we simply let $\operatorname{proj}_{\mathbf{h}}(x + \alpha u) = x$ and $\operatorname{ref}_{\mathbf{h}}(x + \alpha u) = x - \alpha u$. We also say that a set A is *symmetric with respect to* $\mathbf{h}$ if $\operatorname{ref}_{\mathbf{h}}(A) = A$.

Now let $u \in \mathbb{S}^{n-1}$. The *Steiner symmetral* of K in the direction of u is the set $S_u(K)$ with the following properties:

(i) for each $x \in u^\perp$, $(x + \mathbb{R}u) \cap S_u(K) \neq \emptyset$ if and only if $(x + \mathbb{R}u) \cap K \neq \emptyset$,

(ii) for any $x \in \operatorname{proj}_{u^\perp}(K)$, the intersection $(x + \mathbb{R}u) \cap S_u(K)$ is a (possibly degenerate) closed segment whose midpoint is x, and

(iii) $\operatorname{length}((x + \mathbb{R}u) \cap K) = \operatorname{length}((x + \mathbb{R}u) \cap S_u(K))$, for every $x \in \operatorname{proj}_{u^\perp}(K)$.

See Figure 2.7.1 for an illustration. Since the Steiner symmetral is defined by a construction, we get immediately that it is well-defined and unique. Also, it is clear that $S_u(K)$ is symmetric with respect to $u^\perp$.

A very useful feature is that $S_u(K)$ is also convex, as we will see next.

Proposition 2.7.1. *For any convex body $K \subseteq \mathbb{R}^n$ and any $u \in \mathbb{S}^{n-1}$ the Steiner symmetral $S_u(K)$ is a convex body. Also,*

$$\operatorname{vol}_n(K) = \operatorname{vol}_n(S_u(K)).$$

Proof. Let us start proving that $\operatorname{int}(S_u(K)) \neq \emptyset$. Let $\rho > 0$ and $z_0 \in K$ be such that the open ball $B(z_0, 2\rho)$ is contained in K. Then the compact ball $\operatorname{cl}(B(z_0, \rho))$ is contained in K, and its orthogonal projection onto $u^\perp$ is the $(n-1)$-dimensional closed ball $\operatorname{proj}_{u^\perp}(\operatorname{cl}(B(z_0, \rho))) = \operatorname{cl}(B_{u^\perp}(x_0, \rho))$. Let

$$2r := \min\{\operatorname{length}((x + \mathbb{R}u) \cap K) : x \in \operatorname{cl}(B_{u^\perp}(x_0, \rho))\},$$

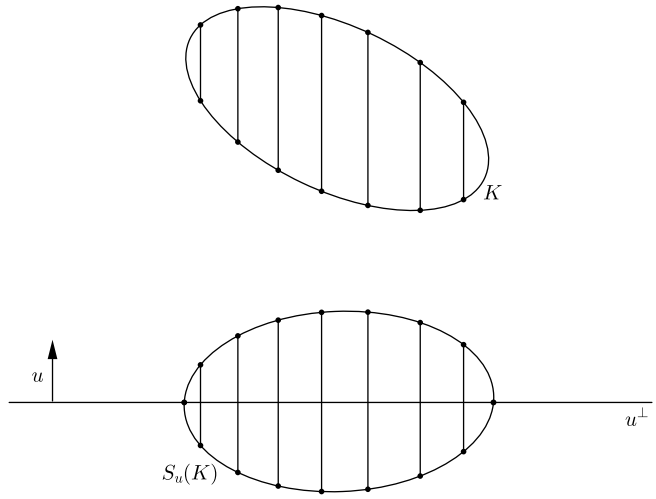


Fig. 2.7.1 The Steiner symmetral of K in the direction of u.

that is, $2r$ is the minimum length among all segments of K in the direction of u which intersect $\mathrm{cl}(B(z_0, \rho))$. The existence of this minimum follows from a simple compactness argument. Consequently, $S_u(K)$ contains the cylinder

$$\{x + \alpha u : x \in \mathrm{cl}(B_{u^\perp}(x_0, \rho)) \text{ and } -r \leq \alpha \leq r\},$$

where $B_{u^\perp}(x_0, \rho) = u^\perp \cap B(x_0, \rho)$. Hence $S_u(K)$ contains an open ball. It follows that $\mathrm{int}(S_u(K)) \neq \emptyset$.

For the compactness of $S_u(K)$, first notice that this is clearly bounded. Indeed, if

$$2M = \max\{\mathrm{length}(K \cap (x + \mathbb{R}u)) : x \in u^\perp\},$$

then we have that

$$S_u(K) \subseteq \{x + \alpha u : x \in \mathrm{proj}_{u^\perp}(K), \text{ and } -M \leq \alpha \leq M\},$$

and since $\mathrm{proj}_{u^\perp}(K)$ is obviously bounded (K is bounded) we get that $S_u(K)$ is bounded. To prove that $S_u(K)$ is closed, let $(y_n)_{n\in\mathbb{N}}$ be a sequence in $S_u(K)$ such that $y_n \to y$. Since the projections of y_n onto $u^\perp$ lie in the compact set $\mathrm{proj}_{u^\perp}(K)$, we get that $(y + \mathbb{R}u) \cap S_u(K) \neq \emptyset$. Due to the fact that this intersection is a closed segment it follows that $y \in S_u(K)$.

Now we prove the convexity of the Steiner symmetral. Let $x_1, y_1 \in S_u(K)$ and $\lambda \in (0, 1)$ be given arbitrarily, and write $z_1 = (1 - \lambda)x_1 + \lambda y_1$. Let $x, y, z \in u^\perp$ be the orthogonal projections of x_1, y_1 and z_1 onto $u^\perp$, respectively. Also, denote by ℓ_x the length of $(x + \mathbb{R}u) \cap K$ (and do the same for y and z). To show that $z_1 \in S_u(K)$, all we have to prove is that

$$\ell_z \geq (1 - \lambda)\ell_x + \lambda\ell_y.$$

Indeed, z_1 lies in the trapezoid whose parallel sides are the segments of $S_u(K)$ passing through x and y. The right-hand side of the inequality above is the length of the intersection of this trapezoid with the segment of $S_u(K)$ passing through z. Since the inequality above follows immediately from

$$(1 - \lambda)(K \cap (x + \mathbb{R}u)) + \lambda(K \cap (y + \mathbb{R}u)) \subseteq K \cap (z + \mathbb{R}u),$$

which comes from the convexity of K (and from $z = (1 - \lambda)x + \lambda y$), we get that $S_u(K)$ is a convex set.

The claim on the volumes of K and $S_u(K)$ comes easily from Cavalieri's principle (4.2.2) which will be proved later:

$$\begin{aligned}\mathrm{vol}_n(K) &= \int_{u^\perp}\left(\int_{(x+\mathbb{R}u)\cap K} dt\right) d\mathcal{H}^{n-1}(x) = \int_{u^\perp} \ell_x \, d\mathcal{H}^{n-1}(x) = \\ &= \int_{u^\perp}\left(\int_{(x+\mathbb{R}u)\cap S_u(K)} dt\right) d\mathcal{H}^{n-1}(x) = \mathrm{vol}_n(S_u(K)),\end{aligned}$$

where we still denote by ℓ_x the length of the segment $(x + \mathbb{R}u) \cap K$ for each $x \in u^\perp$.

□

Next we investigate some other properties of the Steiner symmetrization that will be useful for us later. We begin by understanding how the Steiner symmetrals behave under inclusion, addition, and scalar multiplication. We also will prove that the diameter is monotone under the symmetrization process.

Proposition 2.7.2. *Let $K, L \in \mathcal{K}^n$ and let $\lambda \in \mathbb{R}$. Then the following properties hold:*

(i) *$K \subseteq L$ implies $S_u(K) \subseteq S_u(L)$,*

(ii) *$S_u(\lambda K) = \lambda S_u(K)$,*

(iii) *$S_u(K + L) \supseteq S_u(K) + S_u(L)$, and*

(iv) $\mathrm{diam}(S_u(K)) \leq \mathrm{diam}(K)$.

Proof. For **(i)**, simply notice that if $K \subseteq L$, then for every $x \in u^\perp$ we have that $(x+\mathbb{R}u)\cap K \subseteq (x+\mathbb{R}u)\cap L$. Similarly, **(ii)** comes from the equality $(\lambda K) \cap (x + \mathbb{R}u) = \lambda(K \cap (x + \mathbb{R}u))$.

For **(iii)**, let $x_1 + \lambda_1 u \in S_u(K)$ and $x_2 + \lambda_2 u \in S_u(L)$, with $x_1 \in \mathrm{proj}_{u^\perp}(K)$ and $x_2 \in \mathrm{proj}_{u^\perp}(L)$. For each $x \in \mathrm{proj}_{u^\perp}(K)$, define

$$\begin{aligned} i_{K,u}(x) &:= \inf\{t \in \mathbb{R} : x + tu \in K\}, \text{ and} \\ s_{K,u}(x) &:= \sup\{t \in \mathbb{R} : x + tu \in K\}, \end{aligned}$$

and consider the analogous functions for the bodies L and $K+L$. The difference between these numbers equals the length of $(x + \mathbb{R}u) \cap K$. Therefore, we get

$$\begin{aligned} \frac{i_{K,u}(x_1) - s_{K,u}(x_1)}{2} \leq \lambda_1 \leq \frac{s_{K,u}(x_1) - i_{K,u}(x_1)}{2} \text{ and} \\ \frac{i_{L,u}(x_2) - s_{L,u}(x_2)}{2} \leq \lambda_2 \leq \frac{s_{L,u}(x_2) - i_{L,u}(x_2)}{2}. \end{aligned}$$

Since the projection is linear, we get that $x_1 + x_2 \in \mathrm{proj}_{u^\perp}(K + L)$. Moreover, we have

$$\begin{aligned} x_1 + x_2 + (i_{K,u}(x_1) + i_{L,u}(x_2))u \in K + L \text{ and} \\ x_1 + x_2 + (s_{K,u}(x_1) + s_{K,u}(x_2))u \in K + L, \end{aligned}$$

from where it follows that

$$\begin{aligned} i_{K+L,u}(x_1 + x_2) \leq i_{K,u}(x_1) + i_{L,u}(x_2) \text{ and} \\ s_{K+L,u}(x_1 + x_2) \geq s_{K,u}(x_1) + s_{L,u}(x_2). \end{aligned}$$

Hence, adding the inequalities for λ_1 and λ_2 we get

$$\frac{i_{K+L,u}(x_1+x_2)-s_{K+L,u}(x_1+x_2)}{2} \leq \lambda_1+\lambda_2 \leq \frac{s_{K+L,u}(x_1+x_2)-i_{K+L,u}(x_1+x_2)}{2}.$$

Consequently, we have $x_1+x_2+(\lambda_1+\lambda_2)u \in K+L$. This proves the inclusion $S_u(K)+S_u(L) \subseteq S_u(K+L)$.

The strategy to prove **(iv)** is to show that for any pair of points $x, y \in S_u(K)$ there exist points $x_0, y_0 \in K$ such that $|x_0-y_0| \geq |x-y|$. Indeed, if $x, y \in S_u(K)$, then let $x_1 = \mathrm{proj}_{u^\perp}(x)$ and $y_1 = \mathrm{proj}_{u^\perp}(y)$. Then $|x-y|$ is less than or equal to the length of the diagonal of the (isosceles) trapezoid whose bases are $(x_1+\mathbb{R}u)\cap S_u(K)$ and $(y_1+\mathbb{R}u)\cap S_u(K)$. Now observe that joining the endpoints of the segments $(x_1+\mathbb{R}u)\cap K$ and $(y_1+\mathbb{R}u)\cap K$ yields a convex quadrilateral whose (at least) one of the diagonals has length greater than or equal to the length of the diagonal of the trapezoid in $S_u(K)$. Hence, just let x_0 and y_0 be the endpoints of this diagonal, and we have the desired.

□

Remark 2.7.1. In view of **(i)** and **(ii)**, we may say that the Steiner symmetrization is *monotone for inclusion* and *homogeneous for scalar multiplication*.

Proposition 2.7.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body, and let $u \in \mathbb{S}^{n-1}$. For any orthogonal transformation $T\colon \mathbb{R}^n \to \mathbb{R}^n$ we have that*

$$S_{Tu}(TK) = T(S_u(K)).$$

Proof. Since T preserves inner products, we have $(Tu)^\perp = T(u^\perp)$. This gives $\mathrm{proj}_{(Tu)^\perp}(TK) = T(\mathrm{proj}_{u^\perp}(K))$. Assume that $x_0 \in \mathrm{proj}_{u^\perp}(K)$, and $\mathrm{seg}[x_0-ru, x_0+ru]$ is the corresponding segment in $S_u(K)$. Then, $\mathrm{seg}[Tx_0-rTu, Tx_0+rTu]$ is a segment in $T(S_u(K))$. Moreover, $Tx_0 \in \mathrm{proj}_{(Tu)^\perp}(TK)$, and since T is an isometry, $\mathrm{seg}[Tx_0-rTu, Tx_0+rTu]$ is the corresponding segment in $S_{Tu}(TK)$. This gives the desired equality.

□

Corollary 2.7.1. *Any Steiner symmetral of a convex body centered at the origin is also centered at the origin.*

Proof. If K is centered at the origin, then $K=-K$. Hence,

$$S_u(K) = S_u(-K) = S_{-u}(-K) = -S_u(K),$$

where in the last equality we used $T=-\mathrm{Id}_{\mathbb{R}^n}$ in Proposition 2.7.3. We also used the obvious fact that $S_u = S_{-u}$, because $u^\perp = (-u)^\perp$.

□

Now we go to the most important result about Steiner symmetrization, which states, roughly speaking, that for each convex body $K \in \mathcal{K}^n$ one may find a set of directions such that the respective iterated Steiner symmetrals converge in the Hausdorff metric to a Euclidean ball (with the same volume as K). We fix some terminology. If $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ are finitely many directions and K is a convex body, we say that $S_{u_1} \circ S_{u_2} \circ \ldots \circ S_{u_m}(K)$ is an *iterated Steiner symmetral* of K. We define the set

$$\mathcal{S}(K) := \{S_{u_1} \circ S_{u_2} \circ \ldots \circ S_{u_m}(K) : m \in \mathbb{N} \text{ and } u_1, \ldots, u_m \in \mathbb{S}^{n-1}\},$$

which consists of all iterated Steiner symmetrals of a given convex body $K \in \mathcal{K}^n$. We will prove an important theorem which states that we can always extract a sequence in $\mathcal{S}(K)$ which converges to a ball. Before this, we need a technical lemma.

For a given fixed $u \in \mathbb{S}^{n-1}$, we denote by $S_u : \mathcal{K}^n \to \mathcal{K}^n$ the map which associates each convex body of $\mathbb{R}^n$ with its Steiner symmetral in the direction of u. Next we prove that this map is continuous with respect to the Hausdorff metric restricted to $\mathcal{K}^n$. However, the reader must be aware of the fact that this does not remain true if we replace $\mathcal{K}^n$ by CC^n (notice that the Steiner symmetrization process can be extended verbatim for sets in CC^n). In this direction, we refer the reader to Exercise 2.57. We start the proof of the continuity of S_u over $\mathcal{K}^n$ with a useful technical lemma.

Lemma 2.7.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point, and assume that $(K_j)_{j\in\mathbb{N}}$ is a sequence of sets in CC^n. Then, we have that $K_j \to K$ in the Hausdorff metric if and only if for every $0 < \varepsilon < 1$ there exists a number $j_0 \in \mathbb{N}$ such that*

$$(1-\varepsilon)K \subseteq K_j \subseteq (1+\varepsilon)K$$

for any $j > j_0$.

Proof. Since K has the origin as an interior point, we may choose $\alpha, \beta > 0$ such that

$$\alpha B^n \subseteq \operatorname{int}(K) \subseteq K \subseteq \beta B^n.$$

Assume that $K_j \to K$ and let $\delta > 0$ be arbitrarily given. From Lemma 2.6.4, there is a number $j_0 \in \mathbb{N}$ for which $j > j_0$ implies that $\alpha B^n \subseteq K_j$. Enlarging j_0 if necessary, we may also assume that $d_{\mathcal{H}}(K, K_j) < \delta$ whenever $j > j_0$. This leads to

$$K \subseteq K_j + \delta B^n = K_j + \frac{\delta}{\alpha}\alpha B^n \subseteq \left(1 + \frac{\delta}{\alpha}\right) K_j$$

and

$$K_j \subseteq K + \delta B^n \subseteq \left(1 + \frac{\delta}{\alpha}\right) K.$$

Thus, given an arbitrary $0 < \varepsilon < 1$, letting δ be small enough such that

$$\frac{\delta}{\alpha} < \varepsilon \quad \text{and} \quad \left(1 + \frac{\delta}{\alpha}\right)^{-1} > 1 - \varepsilon,$$

we get that $(1-\varepsilon)K \subseteq K_j \subseteq (1+\varepsilon)K$.

For the converse, let $0 < \varepsilon < 1$ and let $\delta > 0$ be such that $(1-\varepsilon)^{-1} = 1+\delta$. Notice that $K_j \subseteq (1+\varepsilon)K$ implies

$$K_j \subseteq K + \varepsilon\beta B^n,$$

and $(1-\varepsilon)K \subseteq K_j$ leads to

$$K \subseteq \frac{1}{1-\varepsilon} K_j = (1+\delta)K_j = K_j + \delta K_j \subseteq$$
$$\subseteq K_j + \delta(K + \varepsilon\beta B^n) \subseteq K_j + \delta(\beta B^n + \varepsilon\beta B^n) = K_j + \delta\beta(1+\varepsilon)B^n.$$

Let $\gamma > 0$ be arbitrarily given. We clearly may fix a number $0 < \varepsilon < 1$ small enough such that

$$\max\{\varepsilon\beta, \delta\beta(1+\varepsilon)\} < \gamma.$$

Hence, letting j_0 be sufficiently large such that $(1-\varepsilon)K \subseteq K_j \subseteq (1+\varepsilon)K$ for any $j > j_0$, we get that

$$K \subseteq K_j + \gamma B^n \quad \text{and} \quad K_j \subseteq K + \gamma B^n$$

for all $j > j_0$. In other words, we have that $d_{\mathcal{H}}(K, K_j) \leq \gamma$ whenever $j > j_0$. This proves that $K_j \to K$ in the Hausdorff metric.

□

Remark 2.7.2. Observe that this lemma does not hold for the case where K is a set of CC^n which is not full-dimensional.

Proposition 2.7.4. *Let $u \in \mathbb{S}^{n-1}$ be fixed, and assume that $\mathcal{K}^n$ is endowed with the Hausdorff metric. Then, the map $S_u : \mathcal{K}^n \to \mathcal{K}^n$ is continuous.*

Proof. It suffices to prove that for every $K \in \mathcal{K}^n$ and any sequence $(K_j)_{j\in\mathbb{N}}$ in $\mathcal{K}^n$ with $K_j \to K$ it holds that $S_u(K_j) \to S_u(K)$. For that sake, since translating K just yields a translation of $S_u(K)$, we clearly may assume that K contains the origin as an interior point. In this case, letting $\varepsilon > 0$ be arbitrarily given, once $K_j \to K$, Lemma 2.7.1 gives that $(1-\varepsilon)K \subseteq K_j \subseteq (1+\varepsilon)K$ for all sufficiently large j. From Items **(i)** and **(ii)** of Proposition 2.7.2, this leads to

$$(1-\varepsilon)S_u(K) \subseteq S_u(K_j) \subseteq (1+\varepsilon)S_u(K)$$

whenever j is sufficiently large. From the other direction of Lemma 2.7.1, this gives that $S_u(K_j) \to S_u(K)$, provided we know that $S_u(K)$ has the origin as an interior point. This is indeed true: since K has the origin as an interior point, there exists $\alpha > 0$ such that $\alpha B^n \subseteq K$. Thus, $S_u(\alpha B^n) \subseteq S_u(K)$, and since $S_u(\alpha B^n) = \alpha B^n$ we get that $0_{\mathbb{R}^n}$ is an interior point of $S_u(K)$.

□

Theorem 2.7.1. *For any convex body $K \subseteq \mathbb{R}^n$ the set $\mathcal{S}(K)$ contains a sequence which converges in the Hausdorff metric to a ball centered at the origin.*

Remark 2.7.3. In this case, the ball has the same volume as K, because the Lebesgue volume is continuous in the Hausdorff metric, and it is preserved by Steiner symmetrizations.

Proof. From Proposition 2.7.1, we know that $\mathcal{S}(K)$ is, in particular, a collection of compact sets, and we will verify that this is bounded (in the sense that all sets of $\mathcal{S}(K)$ are contained in the same closed ball; see Proposition 2.5.3). To do so, simply observe that $S_u(\rho B^n) = \rho B^n$ for each $u \in \mathbb{S}^{n-1}$ and $\rho > 0$, and choose $r > 0$ such that $K \subseteq rB^n$. From Item **(i)** of Proposition 2.7.2, we get that any set of $\mathcal{S}(K)$ is contained in rB^n, and hence the collection of iterated Steiner symmetrals of a convex body is bounded. For any $L \in C^n$, define

$$r(L) := \min\{\rho > 0 : L \subseteq \rho B^n\},$$

that is, $r(L)$ is the smallest radius of a ball centered at the origin and containing L (the existence of the minimum follows from the fact that L is compact). This defines a functional $r : C^n \to \mathbb{R}$ which is clearly continuous in the Hausdorff metric. Indeed, if $d_{\mathcal{H}}(L, C) < \varepsilon$, then

$$L \subseteq C + \varepsilon B^n \subseteq r(C)B^n + \varepsilon B^n = (r(C) + \varepsilon)B^n,$$

from where we get that $r(L) \leq r(C) + \varepsilon$. From the other inclusion $C \subseteq L + \varepsilon B^n$ we obtain analogously that $r(C) \leq r(L) + \varepsilon$. It follows that $|r(C) - r(L)| < \varepsilon$, and this proves the claimed continuity. Let

$$r_0 := \inf\{r(L) : L \in \mathcal{S}(K)\},$$

and observe that $r_0 > 0$ since otherwise we would have a sequence in $\mathcal{S}(K)$ with volume converging to 0 (recall that all iterated Steiner symmetrals of K have the same volume as K). From the definition

of infimum, we can choose a sequence $(K_j)_{j\in\mathbb{N}}$ in $\mathcal{S}(K)$ such that $r(K_j)\to r_0$. Since $\mathcal{S}(K)$ is bounded we may, using the Blaschke selection theorem, assume that $K_j\to K_0$ for some $K_0\in\mathcal{K}^n$ (passing to a subsequence if necessary). Notice that we can guarantee that K_0 is a convex body because $\mathrm{vol}_n(K_0)=\lim_{j\to\infty}\mathrm{vol}_n(K_j)=\mathrm{vol}_n(K)>0$. Moreover, since the functional r is continuous in the Hausdorff metric, we have that $r(K_0)=\lim_{j\to\infty}r(K_j)=r_0$.

Let $B_0:=r_0B^n$, that is, B_0 is the closed ball centered at $0_{\mathbb{R}^n}$ with radius r_0. Assume that $K_0\neq B_0$. Since $K_0\subseteq B_0$ (because $r(K_0)=r_0$), we must have that the reverse inclusion does not hold, and hence there exists a point $z\in\partial B_0$ such that $z\notin\partial K_0$. Since K_0 is compact, we may choose a number $\rho_1>0$ such that the closed ball $B_1:=\mathrm{cl}(B(z,\rho_1))$ does not intersect K_0. In particular, $\partial B_0\cap B_1\cap K_0=\emptyset$. We claim that we also have

$$\partial B_0\cap B_1\cap S_u(K_0)=\emptyset \tag{2.7.1}$$

for any $u\in\mathbb{S}^{n-1}$. Indeed, if there exists a point $w\in\partial B_0\cap B_1\cap S_u(K_0)$, then the reflection $\mathrm{ref}_{u^\perp}(w)$ of w with respect to the hyperplane $u^\perp$ (orthogonal to u and containing the origin) is such that $\mathrm{ref}_{u^\perp}(w)\in\partial B_0\cap S_u(K_0)$, because B_0 and $S_u(K_0)$ are symmetric with respect to $u^\perp$. Hence both w and $\mathrm{ref}_{u^\perp}(w)$ are points of $S_u(K_0)$, and this is a contradiction, since with $w\notin K_0$ (once $\partial B_0\cap B_1\cap K_0=\emptyset$) and $K_0\subseteq B_0$, we get that the length of the intersection $\mathrm{seg}[w,\mathrm{ref}_{u^\perp}(w)]\cap K_0$ is less than $|w-\mathrm{ref}_{u^\perp}(w)|$. Notice very carefully that this argument holds whenever L and B are, respectively, a convex body such that $L\subseteq B_0$ and a closed ball centered at ∂B_0 having the property that $\partial B_0\cap B\cap L=\emptyset$. That is, assuming that B is a closed ball centered at ∂B_0 we have

$$L\in\mathcal{K}^n,\ L\subseteq B_0,\ \partial B_0\cap B\cap L=\emptyset\Longrightarrow\partial B_0\cap B\cap S_u(L)=\emptyset\ \ \forall\, u\in\mathbb{S}^{n-1}. \tag{2.7.2}$$

Still denoting by $\mathrm{ref}_{u^\perp}(\cdot)$ the reflection of a point or of a set through $u^\perp$ (and noticing that it is an involution as a map of $\mathbb{R}^n$), we also claim that

$$\partial B_0\cap\mathrm{ref}_{u^\perp}(B_1)\cap S_u(K_0)=\emptyset. \tag{2.7.3}$$

In fact, since ∂B_0 and $S_u(K_0)$ are symmetric with respect to $u^\perp$, we have that if there exists $y\in\partial B_0\cap\mathrm{ref}_{u^\perp}(B_1)\cap S_u(K_0)$, then $\mathrm{ref}_{u^\perp}(y)\in\partial B_0\cap B_1\cap S_u(K_0)$, and this contradicts (2.7.1). Again, we repeat the argument to obtain a more general version. If B is a closed ball centered at ∂B_0, then

$$L\in\mathcal{K}^n,\ L\subseteq B_0,\ \partial B_0\cap B\cap S_u(L)=\emptyset\Longrightarrow\partial B_0\cap\mathrm{ref}_{u^\perp}(B)\cap S_u(L)=\emptyset. \tag{2.7.4}$$

Now we take finitely many balls $B_2,\dots,B_m$ centered at points of ∂B_0 and having the same radius as B_1 such that $\partial B_0\subseteq B_1\cup B_2\cup\ldots\cup B_m$. Because B_0 is symmetric with respect to any hyperplane through the origin, for each $j=2,\dots,m$ there exists a hyperplane $\mathbf{h}_j$ containing the origin such that $\mathrm{ref}_{\mathbf{h}_j}(B_1)=B_j$. Indeed, simply define $\mathbf{h}_j$ to be the hyperplane orthogonal to the segment connecting the centers of B_1 and B_j and passing through the midpoint of this segment. Assume that each hyperplane $\mathbf{h}_j$ is orthogonal to the vector $u_j\in\mathbb{S}^{n-1}$, and define $S:=S_{u_m}\circ S_{u_{m-1}}\circ\ldots\circ S_{u_2}$. We claim that $S(K_0)\cap\partial B_0=\emptyset$. To prove this, we proceed inductively. First observe that by (2.7.1) and (2.7.3) we have

$$\partial B_0\cap B_1\cap S_{u_2}(K_0)=\emptyset\ \text{ and} \tag{2.7.5}$$

$$\partial B_0\cap B_2\cap S_{u_2}(K_0)=\emptyset, \tag{2.7.6}$$

where the second equality justifies because $B_2=\mathrm{ref}_{u_2^\perp}(B_1)$. With equality (2.7.5) above in mind, we use (2.7.2) for $L=S_{u_2}(K_0)$, $B=B_1$ and $u=u_3$ to obtain

$$\partial B_0\cap B_1\cap S_{u_3}(S_{u_2}(K_0))=\emptyset,$$

and from that we use (2.7.4), recalling that $B_3 = \mathrm{ref}_{u_3^\perp}(B_1)$, to get

$$\partial B_0 \cap B_3 \cap S_{u_3}(S_{u_2}(K_0)) = \emptyset.$$

From the second equality (2.7.6), we apply (2.7.4) with $L = S_{u_2}(K_0)$, $B = B_2$ and $u = u_3$, yielding

$$\partial B_0 \cap B_2 \cap S_{u_3}(S_{u_2}(K_0)) = \emptyset.$$

Hence we have that $\partial B_0 \cap B_j \cap S_{u_3}(S_{u_2}(K_0)) = \emptyset$ for $j = 1, 2, 3$. Now we assume as our induction hypothesis that

$$\partial B_0 \cap B_j \cap S_{u_{m-1}}(S_{u_{m-2}}(\ldots(S_{u_2}(K_0))\ldots)) = \emptyset$$

for $j = 1, 2, \ldots, m-1$. Then, applying (2.7.2) for $u = u_m$ and $L = B_j$, we get that

$$\partial B_0 \cap B_j \cap S(K_0) = \emptyset$$

for $j = 1, 2, \ldots, m-1$. Finally, since $B_m = \mathrm{ref}_{u_m^\perp}(B_1)$, we use (2.7.4) to obtain

$$\partial B_0 \cap B_m \cap S(K_0) = \emptyset.$$

Summarizing, we have $\partial B_0 \cap B_j \cap S(K_0) = \emptyset$ for every $1 \le j \le m$, and this gives that $S(K_0) \cap \partial B_0 = \emptyset$ (because the balls cover ∂B_0).

The fact that $S(K_0) \cap \partial B_0 = \emptyset$ gives $r(S(K_0)) < r_0$. In fact, observe that $S(K_0)$ is compact, and since it does not intersect the boundary of B_0 we can "shorten" the radius of B_0 before touching the boundary of $S(K_0)$. Passing to a subsequence at each step when necessary, we may assume that all the sequences

$$\begin{aligned}
(S_{u_2}(K_j))_{j\in\mathbb{N}},\\
(S_{u_3} \circ S_{u_2}(K_j))_{j\in\mathbb{N}},\\
\vdots\\
(S_{u_m} \circ S_{u_{m-1}} \circ \ldots \circ S_{u_2}(K_j))_{j\in\mathbb{N}} = (S(K_j))_{j\in\mathbb{N}}
\end{aligned}$$

converge in the Hausdorff metric. Now we recall that $K_j \to K_0$, and proceeding inductively using Proposition 2.7.4 this gives that $S(K_j) \to S(K_0)$. Hence, $r(S(K_j)) < r_0$ for sufficiently large j. This contradicts the definition of the number r_0, because $S(K_j) \in \mathcal{S}(K)$ for any $j \in \mathbb{N}$. Consequently, we get that $K_0 = B_0$, and this shows that $(K_j)_{j\in\mathbb{N}}$ converges to a ball centered at the origin.

□

2.8 Exercises

Exercise 2.1. *Let $A \subseteq \mathbb{R}^n$ be a non-empty set. Prove that* conv$A$ *is the union of all simplices whose vertices are points of A.*

Exercise 2.2. *Let $x_1, \ldots, x_m \in \mathbb{R}^n$ and let* **h** *be a supporting hyperplane of* conv$\{x_1, \ldots, x_m\}$. *Prove that*

$$\big(\mathrm{conv}\{x_1, \ldots, x_m\}\big) \cap \mathbf{h} = \mathrm{conv}\big(\{x_1, \ldots, x_m\} \cap \mathbf{h}\big).$$

Exercise 2.3 (Carathéodory's colorful theorem)**.** *Let $A_1, \dots, A_{n+1}$ be non-empty finite subsets of $\mathbb{R}^n$ such that $0_{\mathbb{R}^n}$ lies in the convex hull of each one of them. Prove that there exist points $x_1 \in A_1$, $x_2 \in A_2, \dots, x_{n+1} \in A_{n+1}$ such that $0_{\mathbb{R}^n} \in \operatorname{conv}\{x_1, \dots, x_{n+1}\}$.*

Remark 2.8.1. We interpret each set A_j as a "color". Hence the origin can be written as a combination having all "colors".

Exercise 2.4. *Let $K \subseteq \mathbb{R}^n$ be a non-empty convex set, and let $\mathcal{A} = \{A_1, \dots, A_m\}$ be any finite family of non-empty convex subsets of $\mathbb{R}^n$. Prove that if any fixed $n+1$ elements of $\mathcal{A}$ are simultaneously intersected by some translate of K, then there exists a translate of K which intersect all sets of $\mathcal{A}$.*

Exercise 2.5. (a) *Let $C, D \subseteq \mathbb{R}^n$ be closed, convex and non-empty, and assume that $C \cup D$ is convex. Prove that for each $c \in C$ and $d \in D$, there exists a number $\lambda \in [0, 1]$ such that*

$$(1 - \lambda)c + \lambda d \in C \cap D.$$

Conclude, in particular, that $C \cap D \neq \emptyset$.

(b) *Give an example of a pair of non-empty convex sets $C, D \subseteq \mathbb{R}^n$ such that C is closed, D is not closed, $C \cup D$ is convex and $C \cap D = \emptyset$.*

Exercise 2.6. *Let $K, L \subseteq \mathbb{R}^n$ be closed, convex and non-empty. Prove that if $K \cup L$ is convex, then*

$$d(K \cup L, x) = \min\{d(K, x), d(L, x)\} \text{ and}$$
$$d(K \cap L, x) = \max\{d(K, x), d(L, x)\}$$

for any fixed $x \in \mathbb{R}^n$.

Remark 2.8.2. In particular, if $K \cup L$ is convex we have that

$$d(K \cup L, x) + d(K \cap L, x) = d(K, x) + d(L, x).$$

We say that the function $CC^n \ni K \mapsto d(K, x)$ has the *valuation property*.

Exercise 2.7. (a) *Verify that the Minkowski sum of bounded sets is bounded.*

(b) *Prove that the Minkowski sum of a closed set with a compact set is closed.*

(c) *Let K be a convex body and L be a non-empty compact and convex set. Show that $K + L$ is a convex body.*

(d) *Give an example of two closed sets whose sum is not closed.*

Exercise 2.8. *Let $A, B, C \subseteq \mathbb{R}^n$ be arbitrary non-empty sets. Prove that*

$$(A \cup B) + C = (A + C) \cup (B + C) \text{ and}$$
$$(A \cap B) + C \subseteq (A + C) \cap (B + C).$$

Show that in the second line equality holds if we assume that A, B are closed convex sets such that $A \cup B$ is convex, and that C is convex.

Exercise 2.9. *Let $A, B \subseteq \mathbb{R}^n$ be non-empty sets. Verify that*

$$(A \cup B) + (A \cap B) \subseteq A + B.$$

Prove that equality holds in the case where A and B are closed and convex and $A \cup B$ is convex.

Exercise 2.10. *Let K be a convex set. Prove that a point $x \in K$ is extreme if and only if $K \setminus \{x\}$ is convex.*

Exercise 2.11. *Let $A, B \subseteq \mathbb{R}^n$ be non-empty, compact and convex. Prove that any extreme point of $A + B$ is a sum of an extreme point of A and an extreme point of B.*

Let $C \subseteq \mathbb{R}^n$ be a non-empty, closed, and convex set, and let $x \in C$. A *cap of C around x* is a set having the form $A \cap \mathbf{h}^+$, where $\mathbf{h}^+$ is a closed half-space containing x in its interior.

Exercise 2.12. *If $C \subseteq \mathbb{R}^n$ is non-empty, closed, and convex, prove that $x \in C$ is an extreme point of C if and only if every open ball of $\mathbb{R}^n$ centered at x contains a cap of C around x.*

Exercise 2.13. *Let $A \subseteq \mathbb{R}^n$ be bounded. We define the* circumradius *of A to be the number*

$$r_{\text{out}}(A) := \inf\left\{r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } A \subseteq x + rB^n\right\},$$

that is, the circumradius of A is the infimum of the radius of the closed balls containing A. If $\text{int}(A) \neq \emptyset$, *then we define the* inradius *of A as*

$$r_{\text{in}}(A) := \sup\left\{r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } x + rB^n \subseteq A\right\}.$$

In other words, the inradius of A is the supremum of the radius of the closed balls contained in A. If $\text{int}(A) = \emptyset$, *then we simply define $r_{\text{in}}(A) = 0$.*

(a) *Assume that $A \subseteq \mathbb{R}^n$ is bounded and contains at least 2 points. Prove that the circumradius of A is positive and that it is attained, that is, show that there exists a point $x \in \mathbb{R}^n$ such that $A \subseteq x + r_{\text{out}}(A)B^n$.*

(b) *Give an example of a bounded set $A \subseteq \mathbb{R}^n$ with non-empty interior whose inradius is not attained, that is, such that there is no point $x \in \mathbb{R}^n$ for which $x + r_{\text{in}}(A)B^n \subseteq A$.*

(c) *On the other hand, prove that the inradius is attained for any compact set having non-empty interior.*

Remark 2.8.3. If $A \subseteq \mathbb{R}^n$ is a bounded set containing at least two points, then any ball containing A and whose radius is the circumradius of A is called a *circumball* of A.

Exercise 2.14. (a) *Let $A \subseteq \mathbb{R}^n$ be a bounded set containing at least 2 points. Prove that there is a unique circumball of A.*

(b) *Give an example of a compact set $A \subseteq \mathbb{R}^n$ having non-empty interior which contains more than one closed ball whose radius is $r_{\text{in}}(A)$.*

Exercise 2.15. *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact convex set. Prove that the center of the circumball of K lies in K.*

For the next exercise, recall that the *width* of a set $K \in CC^n$ in a given direction $u \in \mathbb{S}^{n-1}$ is the number $w_K(u) := h_K(u) + h_K(-u)$. See Lemma 2.3.3 for the geometric meaning of this definition.

Exercise 2.16. (a) *Let $K \subseteq \mathbb{R}^n$ be compact and convex. Prove that*

$$r_{\text{out}}(K) = \inf\left\{r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } h_{K-x}(u) \leq r \text{ for all } u \in \mathbb{S}^{n-1}\right\}.$$

(b) *Assume that $K \subseteq \mathbb{R}^n$ is a convex body. Show that*

$$r_{\text{in}}(K) = \sup\left\{r > 0 : \exists\, x \in \mathbb{R}^n \text{ such that } h_{K-x}(u) \geq r \text{ for all } u \in \mathbb{S}^{n-1}\right\}.$$

(c) *Let $K \subseteq \mathbb{R}^n$ be a convex body, and recall that $w_K(u)$ stands for the width of K in the direction of $u \in \mathbb{S}^{n-1}$. Prove that*

$$2r_{\text{in}}(K) \leq w_K(u) \leq 2r_{\text{out}}(K)$$

for every $u \in \mathbb{S}^{n-1}$.

Exercise 2.17. *Let $K \in CC^n$ be arbitrarily given.*

(a) *Prove that*

$$2r_{\text{out}}(K) \geq \max\left\{w_K(u) : u \in \mathbb{S}^{n-1}\right\},$$

and that equality holds if K is centrally symmetric. Give an example of a convex body for which the equality does not hold.

(b) *Conclude that*

$$r_{\text{out}}(K - K) \leq 2r_{\text{out}}(K),$$

and that equality holds if K is symmetric.

A convex body $K \subseteq \mathbb{R}^n$ is said to be of *constant width* if its width function $w_K(u)$ is constant for $u \in \mathbb{S}^{n-1}$. In other words, we say that K is a body of constant width when it has the same width in every direction. As an alternative terminology, in this case we also say that K **has** constant width.

Exercise 2.18. (a) *Prove that a convex body $K \subseteq \mathbb{R}^n$ has constant width if and only if $K - K$ is a ball (centered at the origin, of course). In this case, verify also that the radius of $K - K$ equals the width of K.*

(b) *Conclude that any centrally symmetric convex body of constant width is a ball.*

Exercise 2.19. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin.*

(a) *Show that the circumball of K is centered at the origin and that the inradius of K is attained for some ball centered at the origin.*

(b) *Prove that*

$$r_{\text{in}}(K^\circ) = \frac{1}{r_{\text{out}}(K)},$$

and derive an expression for $r_{\text{out}}(K^\circ)$ in terms of $r_{\text{in}}(K)$.

Exercise 2.20. *Regard the inradius and the circumradius as functionals defined over the family CC^n of non-empty compact convex subsets of $\mathbb{R}^n$. Prove that these functionals are continuous in the Hausdorff metric.*

Exercise 2.21. (a) *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact and convex set which is symmetric with respect to a hyperplane $\mathbf{h}(u, \beta)$. Prove that for any $u \in \mathbb{S}^{n-1}$ the Steiner symmetral $S_u(K)$ of K with respect to u is a translate of K.*

(b) *Conclude that any Steiner symmetral of any ball is a ball having the same radius.*

(c) *Prove that the inradius and the circumradius are respectively non-decreasing and non-increasing with respect to Steiner symmetrization.*

Exercise 2.22. **(a)** *Verify that the Minkowski sum of two closed balls B_1 and B_2 is a closed ball whose radius equals the sum of the radii of B_1 and B_2.*

(b) *Let $A, B \subseteq \mathbb{R}^n$ be bounded sets. Prove that*

$$r_{\text{out}}(A + B) \leq r_{\text{out}}(A) + r_{\text{out}}(B) \text{ and}$$
$$r_{\text{in}}(A + B) \geq r_{\text{in}}(A) + r_{\text{in}}(B).$$

Provide examples where the inequalities are strict.

Exercise 2.23. *Let $K \subseteq \mathbb{R}^n$ be a non-empty convex set. Prove that*

(a) $\text{relint}(K) = \text{relint}(\text{cl}(K))$ *and*

(b) $\text{cl}(K) = \text{cl}(\text{relint}(K))$.

Exercise 2.24. **(a)** *Prove that for any affine subspace $S \subseteq \mathbb{R}^n$, $x \in S$ and $\varepsilon > 0$, we have*

$$\text{aff}(B(x, \varepsilon) \cap S) = S.$$

(b) *For any $A \subseteq \mathbb{R}^n$, show that the equality* $\text{relint}(\text{relint}A) = \text{relint}A$ *holds (in other words,* $\text{relint}A$ *is relatively open).*

(c) *Let $K \subseteq \mathbb{R}^n$ be convex. Prove that*

$$\text{relbd}(\text{relint}(K)) = \text{relbd}(K) = \text{relbd}(\text{cl}(K)).$$

Exercise 2.25. *Let $K \subseteq \mathbb{R}^n$ be a compact convex set. Show that* $K = \text{conv}(\text{relbd}K)$.

Exercise 2.26. **(a)** *Let $A, B \subseteq \mathbb{R}^n$ be non-empty. Prove that*

$$\text{aff}(A + B) = \text{aff}(A) + \text{aff}(B).$$

(b) *Verify that for any $x \in \mathbb{R}^n$ we have* $\text{aff}(x+A) = x+\text{aff}(A)$. *Conclude that* $\text{relint}(x+A) = x+\text{relint}(A)$.

Exercise 2.27. **(a)** *Let $A, B \subseteq \mathbb{R}^n$ be non-empty sets. Prove that A and B can be properly separated if and only if $A - B$ and $\{0_{\mathbb{R}^n}\}$ can be properly separated.*

(b) *Assume that $x \in \text{relint}(A)$. Show that x and A cannot be properly separated.*

(c) *If A and B are convex, show that*

$$\text{relint}(A + B) \subseteq \text{relint}(A) + \text{relint}(B).$$

Exercise 2.28. *Let $A \subseteq \mathbb{R}^n$ be a closed non-empty set. Show that for each $x \notin A$ one has*

$$\text{dist}(x, A) = \sup\left\{r > 0 : (x + rB^n) \cap A = \emptyset\right\}.$$

Verify also that the set of metric projections of x onto A is the intersection of A with the ball $x + \text{dist}(x, A)B^n$.

Exercise 2.29. *Let $A \subseteq \mathbb{R}^n$ be a closed set having non-empty interior. Assume that for each boundary point x of A there exists a hyperplane supporting A at x. Prove that A is convex. Also, show that the hypothesis that A has non-empty interior cannot be dropped.*

Exercise 2.30. *The objective of this exercise is to prove that any closed set of $\mathbb{R}^n$ for which metric projections are unique is convex. We divide it into steps. In what follows, $A \subseteq \mathbb{R}^n$ is a closed non-empty set.*

(a) *Show that if A is not convex, then there exist points $x, y \in A$ such that $A \cap \operatorname{seg}[x, y] = \{x, y\}$.*

(b) *Verify that if x and y are as given in Item* **(a)***, then there exists a number $\rho > 0$ such that*

$$\left(\frac{x+y}{2} + \rho B^n\right) \cap A = \emptyset.$$

(c) *Letting x, y and ρ be as given in Items* **(a)** *and* **(b)***, write*

$$B_0 := \frac{x+y}{2} + \rho B^n,$$

and let

$$\mathcal{B} := \left\{B : B \text{ is a closed ball such that } B \supseteq B_0 \text{ and } (\operatorname{int} B) \cap A = \emptyset\right\}.$$

Prove that $\mathcal{B}$ contains a ball B_1 with maximal radius, and that this radius is greater than ρ.

(d) *Verify that the intersection $\partial B_0 \cap \partial B_1$ is either empty or a singleton.*

Now assume that A has the property that for each point $x \in \mathbb{R}^n$ there exists a unique point $y \in A$ such that $|x - y| = \operatorname{dist}(x, A)$.

(e) *If B_1 is the closed ball defined in Item* **(c)***, show that $B_1 \cap A$ is non-empty and consists of a unique point w, say.*

(f) *Let w be the point given in Item* **(e)***. If $\partial B_0 \cap \partial B_1 \neq \emptyset$, having Item* **(d)** *in mind we let z be the unique point of this intersection. Otherwise, we let $z = (x + y)/2$. Prove that in both cases there exists a number $\varepsilon > 0$ such that $B_1 + \varepsilon(z - w) \subseteq \mathcal{B}$.*

(g) *Show that $\mathcal{B}$ contains a ball whose radius is greater than the radius of B_1. Conclude that this contradiction yields the convexity of A.*

Exercise 2.31. *Let $h\colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function.*

(a) *Prove that h is the support function of a convex body which contains the origin as an interior point if and only if $h(u) > 0$ for every $u \in \mathbb{S}^{n-1}$.*

(b) *Show that h is the support function of a convex body if and only if $h(u)+h(-u) > 0$ for each $u \in \mathbb{S}^{n-1}$.*

(c) *Let K be the non-empty convex compact set whose support function is h. Verify that K is centered at the origin if and only if h is even.*

Exercise 2.32. *Prove that if $K \in CC^n$ is centered at a point $x \in K$, then K is not centered with respect to any other point. Conclude that if K is centered at the origin, then a translate of K by a non-zero vector is not centered at the origin.*

Exercise 2.33. *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point, and let γ_K be the gauge function of K. Prove that*

$$\partial\gamma_K(0_{\mathbb{R}^n}) = K^\circ.$$

Conclude that γ_K is not differentiable at the origin.

Exercise 2.34. *Let $C \subseteq \mathbb{R}^n$ be a non-empty compact convex set.*

(a) *Verify that there exists $u_0 \in \mathbb{S}^{n-1}$ such that*

$$h_C(u_0) = \sup\left\{h_C(u) : u \in \mathbb{S}^{n-1}\right\}.$$

Check also that this supremum is positive if $C \neq \{0_{\mathbb{R}^n}\}$.

(b) *Let u_0 be as in Item* **(a)***, and assume that u_0 is an outer normal of C at $z_0 \in \partial C$. Prove that $z_0 = h_C(u_0)u_0$.*

(c) *Let z_0 be as in Item* **(b)***. Show that z_0 is an exposed point of C.*

For the next exercise, recall that f^* denotes the *conjugate function* of f (see Section 1.4 for definitions and properties).

Exercise 2.35. *Let $K \subseteq \mathbb{R}^n$ be a non-empty, closed, and convex set.*

(a) *Verify that $K = \mathrm{dom}(h_K^*)$.*

(b) *Prove that $\partial h_K(0_{\mathbb{R}^n}) = K$, where we recall the reader that $\partial h_K(0_{\mathbb{R}^n})$ denotes the subdifferential of h_K at $0_{\mathbb{R}^n}$.*

Exercise 2.36. *Let $K \subseteq \mathbb{R}^n$ be a non-empty, compact and convex set. For each non-zero $u \in \mathbb{R}^n$, let*

$$F(K,u) := \{x \in K : \langle x, u\rangle = h_K(u)\}.$$

(a) *Assume that $\mathbf{h}$ is the supporting hyperplane of K whose outer normal is u. Prove that $\mathbf{h} \cap K = F(K,u)$.*

(b) *Conclude that $F(K,u)$ is a face of K for any non-zero vector $u \in \mathbb{R}^n$.*

(c) *Prove that $\partial h_K(u) = F(K,u)$ for each non-zero vector $u \in \mathbb{R}^n$.*

Exercise 2.37. *Let $C \in CC^n$. Prove that*

$$(h_C)'_+(u,v) = h_{F(C,u)}(v)$$

for any $u, v \in \mathbb{S}^{n-1}$. In other words, the right-sided derivative of h_C at u in the direction of v equals the support function of the exposed face $F(C,u)$ evaluated at v.

Exercise 2.38. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Define*

$$K_f := \mathrm{cl}(\mathrm{dom}(f^*)),$$

and notice that K_f is non-empty, closed, and convex.

(a) *Prove that K_f is bounded if and only if f is globally Lipschitz.*

(b) *Show that*

$$f(y) - f(x) \leq h_{K_f}(y - x)$$

for any $x, y \in \mathbb{R}^n$.

Hint. For Item **(a)**, use Exercise 2.23 to show that K_f is bounded if and only if relint(dom(f^*)) is bounded. Then, use Remark 1.4.2 and Exercise 1.9 to conclude that if f is Lipschitz, then relint(dom(f^*)) is bounded.

Exercise 2.39. *Let* $f\colon \mathbb{R}^n \to \mathbb{R}$ *be a convex and globally Lipschitz function, and let* K_f *be as in Exercise 2.38 .*

(a) *Prove that*

$$h_{K_f}(u) = \lim_{t\to\infty} \frac{f(x + tu)}{t}$$

for any $x, u \in \mathbb{R}^n$.

(b) *Show that if* f *is differentiable, then*

$$\nabla f(\mathbb{R}^n) = \operatorname{int}(K_f),$$

where $\nabla f(\mathbb{R}^n)$ *denotes the image of the gradient map* $\nabla f\colon \mathbb{R}^n \to \mathbb{R}^n$.

Exercise 2.40. *Prove that if* $f, g\colon \mathbb{R}^n \to \mathbb{R}$ *are convex and globally Lipschitz, then*

$$K_{\alpha f+\beta g} = \alpha K_f + \beta K_g$$

for any $\alpha, \beta \geq 0$.

Exercise 2.41. (a) *Let* $K, L \subseteq \mathbb{R}^n$ *be convex bodies such that* $L \subseteq K$ *and assume that* $x \in \partial K \cap \partial L$. *Show that any supporting hyperplane of* K *at* x *is also a supporting hyperplane of* L *at* x.

(b) *Let* $K \subseteq \mathbb{R}^n$ *be a convex body and assume that* $rB^n \subseteq K$. *Prove that if* $x \in r\mathbb{S}^{n-1} \cap \partial K$, *then* $x/|x|$ *is the unique unit outer normal vector of* K *at* x.

(c) *Under the same hypotheses as in Item* **(b)**, *prove that*

$$\frac{x}{r|x|} \in \frac{1}{r}\mathbb{S}^{n-1} \cap \partial K^\circ.$$

Exercise 2.42. *Assume that* $K \subseteq \mathbb{R}^n$ *is a convex body having the origin as an interior point, and let* $x_0 \in \partial K$ *and* $y_0 \in \partial K^\circ$. *Prove that* y_0 *is an outer normal of* K *at* x_0 *if and only if* x_0 *is an outer normal of* K° *at* y_0.

Exercise 2.43. (a) *Let* $L \in \mathcal{K}_0^n$ *be such that* $L = L^\circ$. *Prove that* $L = B^n$.

(b) *Conclude that if* $K \in \mathcal{K}_0^n$ *is such that* $K^\circ = \alpha K$ *for some* $\alpha > 0$, *then* K *is a ball centered at the origin.*

Exercise 2.44. (a) *Let* $A, B \subseteq \mathbb{R}^n$ *be non-empty sets. Verify that* $\operatorname{pos}(A + B) \subseteq \operatorname{pos}(A) + \operatorname{pos}(B)$.

(b) *Show that* $\operatorname{pos}(A) = \operatorname{pos}(\operatorname{conv}(A))$ *for any non-empty set* $A \subseteq \mathbb{R}^n$.

(c) *Check that the positive hull of any set is convex. Conclude that the positive hull of any set is a convex cone.*

Exercise 2.45. *Let $A \subseteq \mathbb{R}^n$ be a non-empty set. Prove that the positive hull of A is the intersection of all convex cones of $\mathbb{R}^n$ which contain A.*

Exercise 2.46. (a) *Let $C \subseteq \mathbb{R}^n$ be a convex cone (not necessarily closed). Verify that its polar set C° is a closed convex cone.*

(b) *Let $C \subseteq \mathbb{R}^n$ be a non-empty, closed, and convex cone. Show that $C^{\circ\circ} = C$.*

(c) *Let A be a non-empty set. Prove the following expression for the polar set of* $\operatorname{pos}(A)$*:*

$$\operatorname{pos}(A)^\circ = \{u \in \mathbb{R}^n : \langle u, x\rangle \leq 0 \text{ for all } x \in A\}.$$

Exercise 2.47. *Let $A \subseteq \mathbb{R}^n$ be a compact non-empty set with the property that $0_{\mathbb{R}^n} \notin \operatorname{conv} A$. Prove that* $\operatorname{pos}(A)$ *is closed. Give an example of a closed set F such that* $\operatorname{pos}(F)$ *is not closed.*

Exercise 2.48. *Let $K \subseteq \mathbb{R}^n$ be a non-empty compact convex set, and let $h_K \colon \mathbb{R}^n \to \mathbb{R}$ be its support function. Define the set*

$$C_K := \{\lambda(x, -1) \in \mathbb{R}^n \times \mathbb{R} : x \in K \text{ and } \lambda \geq 0\}.$$

Verify that C_K is a closed convex cone, and prove that $\operatorname{epi}(h_K) = C_K^\circ$.

Exercise 2.49. *Let $h \colon \mathbb{R}^n \to \mathbb{R}$ be a sublinear function.*

(a) *Show that* $\operatorname{epi}(h)$ *is a convex cone.*

(b) *Letting $(x_1, \ldots, x_{n+1})$ be coordinates in $\mathbb{R}^{n+1}$, and $T \colon \mathbb{R}^{n+1} \to \mathbb{R}^n$ be the map defined as $T(x_1, \ldots, x_n, x_{n+1}) = (x_1, \ldots, x_n)$, define*

$$K := T\big(\operatorname{epi}(h)^\circ \cap \{x_{n+1} = -1\}\big).$$

Prove that K is a non-empty, compact convex set.

(c) *Show that h is the support function of K.*

Exercise 2.50. *Assume that $C_m \to C$ in CC^n with respect to the Hausdorff metric. Prove that for any linear map $T \colon \mathbb{R}^n \to \mathbb{R}^n$ we have that $TC_m \to TC$.*

Exercise 2.51. *Prove that if $(C_m)_{m\in\mathbb{N}}$ is a sequence of centrally symmetric sets in CC^n converging in the Hausdorff distance to a set $C \in CC^n$, then C is also centrally symmetric. Moreover, show that if all C_m are symmetric with respect to the same point $z_0 \in \mathbb{R}^n$, then z_0 is also the center of C.*

Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. The orthogonal complement $u^\perp$ is a hyperplane of $\mathbb{R}^n$, and it can be seen as normed vector space with the induced norm of $\mathbb{R}^n$. It is clear that the unit ball of this norm is $B^n \cap u^\perp =: B_{u^\perp}$. Hence we may introduce a Hausdorff metric in $u^\perp$ by letting

$$d_{\mathcal{H},u^\perp}(C, L) := \inf\big\{\lambda > 0 : C \subseteq L + \lambda B_{u^\perp} \text{ and } L \subseteq C + \lambda B_{u^\perp}\big\}.$$

The objective of the next exercise is, roughly speaking, to check that convergence in the Hausdorff metric of $\mathbb{R}^n$ implies convergence of the projections on $u^\perp$ in the Hausdorff metric of $u^\perp$.

Exercise 2.52. *Let $(C_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{C}^n$ converging in the Hausdorff metric of $\mathbb{R}^n$ to $C \in \mathcal{C}^n$, and let $u \in \mathbb{S}^{n-1}$. Prove that the sequence $(\mathrm{proj}_{u^\perp}(C_m))_{m\in\mathbb{N}}$ converges to $\mathrm{proj}_{u^\perp}(C)$ in the Hausdorff metric of $u^\perp$.*

Exercise 2.53. *Let $L, C \subseteq \mathbb{R}^n$ be non-empty compact sets. Prove that indeed we have*

$$0 \le d_{\mathcal{H}}(L, C) < \infty.$$

Also, writing $d = d_{\mathcal{H}}(L, C)$, show that

$$L \subseteq C + dB^n \ \text{ and } \ C \subseteq L + dB^n.$$

Remark 2.8.4. This justifies why the "infimum" can be replaced by the "minimum" in the definition of the Hausdorff metric given in (2.5.1).

Exercise 2.54. **(a)** *Let K and L be non-empty compact sets. Prove that*

$$d_{\mathcal{H}}(K, L) = \max\Big\{ \max\big\{\mathrm{dist}(x, L) : x \in K\big\},\ \max\{\mathrm{dist}(y, K) : y \in L\}\Big\}.$$

(b) *Now assume that K and L are convex bodies. Show that*

$$d_{\mathcal{H}}(K, L) = \max\Big\{ \max\big\{\mathrm{dist}(x, \partial L) : x \in \partial K\big\},\ \max\{\mathrm{dist}(y, \partial K) : y \in \partial L\}\Big\}.$$

(c) *Let K be a compact convex set, and let B be its circumball. Prove that*

$$d_{\mathcal{H}}(K, B) \le r_{\mathrm{out}}(K).$$

Give an example where equality holds.

Exercise 2.55. *Let $\mathbb{B} \subseteq \mathbb{R}^n$ be a convex body containing the origin as an interior point, and define the* relative Hausdorff metric with respect to $\mathbb{B}$ *as*

$$d_{\mathcal{H},\mathbb{B}}(L, C) := \inf\{\lambda \ge 0 : C \subseteq L + \lambda\mathbb{B} \text{ and } L \subseteq C + \lambda\mathbb{B}\},$$

for $L, C \in \mathcal{C}^n$. Prove that this distance induces the same topology on $\mathcal{C}^n$ as the usual Hausdorff metric $d_{\mathcal{H}}$.

Exercise 2.56. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that the Steiner symmetral of K in the direction $u = e_n$ can be expressed as*

$$S_{e_n}(K) = \left\{\left(x, \frac{t-s}{2}\right) : (x, t),\ (x, s) \in K\right\}.$$

Exercise 2.57. *Notice that the definition of Steiner symmetrization can be extended verbatim for sets of $\mathcal{CC}^n$. Fixing $u \in \mathbb{S}^{n-1}$ we may see the Steiner symmetrization in $u^\perp$ as a map $S_u \colon \mathcal{CC}^n \to \mathcal{CC}^n$. Show that this map does not need to be continuous in the Hausdorff metric.*

Hint: Consider the segments $\mathrm{seg}[(0, 0), (1/m, 1)]$ in $\mathbb{R}^2$, and take symmetrizations in the x-axis (which is $e_2^\perp$, of course).

Exercise 2.58. *In this exercise, $K \subseteq \mathbb{R}^n$ is an arbitrary set of $\mathcal{CC}^n$, and $u \in \mathbb{S}^{n-1}$ is any unit vector. The* Minkowski symmetral of K in the direction of u *is defined as the set*

$$M_u(K) := \frac{1}{2}\big(\mathrm{ref}_{u^\perp}(K) + K\big).$$

(a) *Verify that $M_u(K)$ is symmetric with respect to $u^\perp$.*

(b) *Show that the procedure of Minkowski symmetrization is* idempotent*, that is, we always have*

$$M_u(M_u(K)) = M_u(K).$$

(c) *Check that the Minkowski symmetrization is monotone for inclusion and homogeneous for scalar multiplication.*

(d) *Prove that the Minkowski symmetrization is translation-invariant in the following sense: for each $x \in \mathbb{R}^n$, the symmetral $M_u(x + K)$ of $x + K$ is a translate of the symmetral $M_u(K)$ of K.*

(e) *Show that*

$$h(\mathrm{ref}_{u^\perp}(K), x) = h(K, \mathrm{ref}_{u^\perp}(x))$$

for every $x \in \mathbb{R}^n$. Conclude that $h_{M_u(K)}(v) = h_K(v)$ whenever $v \in u^\perp$.

(f) *Prove that we always have $S_u(K) \subseteq M_u(K)$ if K is a convex body and $u \in \mathbb{S}^{n-1}$.*

Exercise 2.59. *Let $(K_j)_{j\in\mathbb{N}}$ be a sequence of convex bodies such that $K_j \to K$ for some $K \in CC^n$ in the Hausdorff metric. If $u \in \mathbb{S}^{n-1}$ is such that the Steiner symmetrals $S_u(K_j)$ converge to some $L \in CC^n$, prove that $L \subseteq S_u(K)$.*

2.9 Notes

In these notes we will not present the specific literature on basic properties of convex sets discussed in the Sections 2.1 till 2.4, since these are mainly folklore. Instead of this we give (in 2.9.1) an overview to *special types of convex bodies*, showing the richness of the whole field also from this side. After that we present (fitting to the Sections 2.5 and 2.7) *notes on the Hausdorff metric and Steiner symmetrization*. Aspects of approximation procedures are widespread and not only concentrated on Chapter 2 in the present book; therefore also here we wrote no corresponding notes.

2.9.1 Special types of convex bodies

Following the intention of having a handbook article titled "Special convex bodies" (see the survey [1177] of Heil and Martini), we want to collect in subsection 2.9.1 basic references (mainly, but not only book chapters, surveys, and partially also research papers) on special types of convex bodies, like simplices or other interesting polytope classes, ellipsoids, centrally symmetric bodies, and so on. Often these special classes represent extremal types of bodies (e.g., regarding affine isoperimetric inequalities), interesting unit balls of normed spaces, and bodies suitable for approximation procedures, or they show nicely non-intuitive phenomena in high dimensions. A few of these classes are also presented in "own sections and subsections" of our book (like *zonotopes and zonoids* in Chapter 10), or subcases of them even fill whole sections (like the *Löwner–John ellipsoids* in Chapter 9). In such cases we refer here to the respective book parts. And in general we go, within this subsection, not too deep into details, since then the list of citations would become too large! It is our goal to give a balanced panorama covering most of the knowledge about these classes of convex bodies.

2.9.1.1 Simplices

The reader should notice that simplices occur explicitly also in other parts of our book; see, e.g., subsection 4.7.3 on volumes (of sections) of simplices, or our next subsection 2.9.1.2 on vertex embeddings in, or triangulations of cubes. We will not repeat references given there and even study here only partial themes that are away from these topics. Our references presented here should mainly reflect tools and motivate questions suitable for studying simplices from different geometric sides, or to inspire work on them close to convexity. It is impossible to be complete; there are too many publications on the geometry of simplices. We try to give a good selection of interesting topics and results, and we give also references on the geometry of simplices themselves (which might, for some readers, "not directly belong to convexity", but, e.g., to discrete geometry). We hope that also these results might deliver ideas and tools for future research on simplices in the spirit of discrete and convex geometry.

N-dimensional simplices are those convex n-polytopes which have the smallest possible vertex (or facet) number, namely $n+1$. (Lower dimensional analogues with $k+1$ vertices are then k-simplices in their own affine hull.) Thus, in n-space they are representable as convex hulls of $n+1$ affinely independent points, and all their k-dimensional faces are k-simplices, $0 \le k \le n-1$. They are n-pyramids over each of their facets (and by this property characterized), and the two- and three-dimensional subcases are called *triangles* and *tetrahedra*, respectively. Due to this definition, they occur in many subfields and applications, for instance referring to triangulations, simplicial complexes, as simple or simplicial polytopes, in the simplex algorithm, and as extremal polytopes or even extremal convex bodies regarding many problems.

BOOKS, BOOK CHAPTERS, AND SURVEYS

In most basic books on convexity (e.g., [930], [1070], [2167], [133], and [134]) there are no chapters dedicated to simplices; the results and problems for which simplices are important (for example, as extremal bodies) are distributed over parts of individual book chapters. Nevertheless, we have the following references on books, book chapters, and surveys dedicated to simplices.

As a motivating starting point we take Section 2 of the handbook article [1177] of Heil and Martini. Except for older surveys, book parts, and a few further exceptions, we take only references into consideration which appeared after this handbook article. The main aspects studied in [1177] are *characterizations of finite-dimensional simplices* (e.g., by intersections of homothets), *measures of symmetry, geometric inequalities*, and several further viewpoints. Also characterizations of *regular simplices* are given.

Regarding older books or book chapters dedicated to (tetrahedral and) simplex geometry, we cite here in chronological order Simon [2250], Roché and Comberousse [2042], Court [666], Couderc and Ballicioni [664], Thébault [2357], Satyanarayana [2119], Baston [252], Benson [273], Grünbaum [1085], Boltyanski and Yaglom [436], Kelly and Weiss [1359], Lay [1521], and Mitrinović et al. [1814]. In some of these publications one has to work (because of the variety of properties treated) with the index of terms, in others (partial) chapters on simplices are contained. By no means, this list is complete.

The booklet [1009] is a good source for material useful for the notion of *infinite-dimensional simplices* based on results like the Choquet–Meyer existence and uniqueness theorem, or on the notion of *Choquet simplices*. In view of the latter, we cite here also the excellent survey [2272] of V. Soltan. This exposition shows various geometric results on *Choquet simplices in finite dimensions*, and it summarizes existing results and presents inspiring open problems.

Another interesting survey of V. Soltan, namely [2273], treats in Section 7 characterizations of simplices via *affine diameters* of convex bodies (i.e., segments which are the longest chords of these bodies in their own direction).

The main part of the book [1537] of Leichtweiss on affine geometry of convex bodies is its third chapter. It treats *geometric inequalities* for the equiaffine entities; in various situations the cases of equality lead to (ellipsoids and) simplices; see also Lutwak's survey [1634] on *affine isoperimetric inequalities*.

The geometry of *simplices in normed spaces* presents an interesting subfield of Banach space theory. Sections 3, 4, 5, and 7 of the basic survey [1722] of Martini et al. contain related material, mainly on the planar case, but also on results in higher dimensions. Corresponding themes are for instance *equilaterality, in- and circumscription*, and the *triangle inequality*.

Schneider's educational talks [2160] are a valuable source for getting an overview of results and open problems on simplices as extremal bodies in the spirit of central themes of convexity. The author treats such results and problems in the light of *measures of symmetry* with applications, *stability results*, and various *affine inequalities*. The Minkowski measure of symmetry and the difference body measure of symmetry are presented in detail, and Grünbaum's survey [1080] is mentioned as an important source for more related material. Applications of the difference body measure of symmetry are taken from *stochastic geometry*, and a longer part is dedicated to affine inequalities. These are the two-sided inequality for the volume of the polar projection body, the difference body inequality, Zhang's projection inequality, inequalities related to in- and circumscribed ellipsoids of extremal volume, the Blaschke–Santaló inequality, and the L^p-Busemann–Petty centroid inequality. The role of Barthe's results regarding the Brascamp–Lieb inequality and its reverse is properly appreciated, and in a final part *random polytopes* are discussed.

An n-simplex in Euclidean n-space is said to be *orthocentric* if its $n+1$ altitudes have a point in common, called the *orthocenter* of that simplex. In [799], Edmonds et al. derive various results about orthocentric simplices. In the introductory part of this paper, a survey on the rich geometry of orthocentric simplices (studied by many mathematicians) is given. Collecting various theorems on and properties of triangles, tetrahedra, and higher dimensional simplices, one observes the following phenomenon: many theorems from planar triangle geometry have natural extensions/analogues in higher dimensions if the considered simplices are orthocentric! And this is much less the case for general simplices. This interesting observation is discussed in the exposition [1144].

In addition, one should particularly emphasize the work of Fiedler on *simplex geometry with methods from matrix theory and graphs*. He has written a large number of papers on the geometry of n-simplices, and all this material is summarized and discussed, also with new viewpoints (since most of the papers are older), in his wonderful monograph [868]. This book unifies various (otherwise scattered) results and cross-connections between them. More precisely, it establishes strong relations between linear algebra, n-dimensional elementary geometry (of simplices), and graph theory. Used concepts and tools are Gram matrices, interesting solution methods for large systems of linear equations, and eigenvalue problems from linear algebra, many simplex-related concepts from Euclidean geometry (circumcenters, dihedral angles, etc.), and tools like signed graphs from graph theory. The author investigates also *special types of simplices*, like orthocentric or right ones, and in Chapter 5 he shows that his methods also deserve to be used for studying other geometric objects: finite point sets, simplicial cones, spherical cones, and also interesting types of degenerate simplices. To complete the description of Fiedler's work on n-simplices, we shortly mention one paper that appeared after [868]: In [869] he studies, based on similar methods, problems on *pairs of simplices*.

Hampton [1149] nicely surveys useful formulas from simplex geometry, mainly (but not only) in dimensions two and three, and treating *distances, areas, and cosines of angles*. Most of the presented formulas are well known, but some novel identities are also given.

Finally, we mention here the expository paper [1143] showing (by means of the median triangle theorem) a smooth way from planar and spatial to *higher dimensional (simplex) geometry*. It contains intriguing questions in this direction and shows nicely the way of analogizing theorems to get results in higher dimensions.

Now we continue with the announced collection of *research articles*, first regarding Euclidean spaces, and then with respect to normed spaces.

PROPERTIES OF SIMPLICES

Napoleon's theorem says that if one erects (to the exterior) equilateral triangles on the sides of an arbitrary (possibly even degenerate) triangle, then their three circumcenters form the vertex set of an equilateral triangle. A direct extension to n-space is clearly not possible, but if a triangle xyz has its Fermat–Torricelli point f (= point of minimal distance sum to the vertices) in its interior, then Napoleon's theorem can be formulated without these erected equilateral triangles and can also be extended to higher dimensions. Namely, the vertices x', y', z' of this obtained regular triangle are also the intersection points of the lines fx, fy, fz with the circumcircles of the triangles fyz, fxz, fxy, or the points on the three rays from x, y, z to f for which the distance equalities $|xx'| = |yy'| = |zz'| = |fx| + |fy| + |fz|$ hold, respectively. The first viewpoint is extended to n-space, $n \geq 3$, by Martini and Weissbach in [1724], the second by Hajja, Martini, and Spirova in [1145].

In contrast to the edge-lengths of n-simplices (see [724]), any *dihedral angle* is determined by the other $(n^2+n-2)/2$ dihedral angles. Maehara and Martini [1665] show that for all $n \geq 2$ there are infinitely many similarity classes of n-simplices all whose dihedral angles are rational multiples of π. A basic role in this paper play *Hill simplices* which are defined below. And recalling that a finite set of real numbers is π-independent if the numbers and π are linearly independent over the rational field, they show that for $n \geq 3$ there are uncountably many mutually non-similar n-simplices whose

dihedral angles and π are linearly independent in this sense. They form even a dense set in the family of all simplices regarding the Hausdorff distance. Clearly, if three real numbers a, b, c lie within the interval $[\lambda, 1]$ for $1/2 < \lambda < 1$, they are the edge-lengths of some triangle. This can be analogously extended to higher dimensional simplices (see again [724]), and when, e.g., λ comes close to $1/2$ in the Euclidean plane, increasingly "thin" simplices occur. If, conversely, longest and shortest edges have almost equal lengths, almost regular simplices (being "thick" in comparison with their edge-lengths) come to be. Based on these facts, Dekster [722] obtains lower bounds for the thickness of n-simplices whose edge-lengths come from a suitably given interval. Analogous results on hyperbolic and spherical spaces are given, too.

Frankl and Vojtech [900] prove the following *Ramsey-type result* for subsets X of Euclidean space. For any n-dimensional simplex with circumradius r and any $\rho > 0$ there exists $s = s(\rho, X) > 1$ such that for every $m \geq n$ and every k-colouring of the m-dimensional sphere of radius $r + \rho$, where $k \leq s^m$, there exists a monochromatic subset of the sphere which is congruent to the set X. They even prove a stronger technical Ramsey property of simplices.

For an n-dimensional convex body K of mean width $M(K)$, let $M(S_K)$ denote the *minimal mean width* of all simplices circumscribed about K. Let, further on, S denote a regular simplex circumscribed about the unit n-ball. Böröczky and Schneider [412] prove the inequality $2M(S_K) \leq M(K)M(S)$, where equality is obtained iff K is a ball. Since the used averaging technique works only for exceptional polytopes, the dual problem (to find among all simplices in a given ball those ones of maximal mean width) remains open. However, an analogous result, namely on circumscribed parallelepipeds and with balls replaced by constant width bodies, is also established.

Krantz et al. [1464] study *centroids of different k-skeletons of simplices* in Euclidean n-space assuming constant density. First the well-known fact is shown that the centroids of the full simplex and of its vertex set coincide. It is shown that the centroids of the other k-skeleta, $1 \leq k \leq n-1$, are generically different. Types of simplices are presented for which all the centroids for all values of k coincide. For example, for any $n = pq - 1$, where p and q are integers larger than 2, there exists a non-regular n-simplex for which the centroids of all k-skeletons coincide, $1 \leq k \leq n$. Further related results are obtained and also applied to verify the existence of a non-regular 4-simplex all whose centroids coincide. Regarding many further results on centroids the reader should consult the (relatively unknown) book [193].

Let S_I be the *convex hull of the incenters of the facets* of an n-simplex S. Ma, Zhao, and Yuan [1656] prove that the volume ratio of S and S_I satisfies the inequality $n^n \leq V(S)/V(S_I)$, with equality iff S is regular. Similar results are derived for the convex hull of the touching points of the insphere, and for the convex hull of the circumcenters of the facets of S. Let S be a *minimal area n-simplex* circumscribed about an n-dimensional strictly convex body K, and h be the homothety with ratio $1/(1-n)$ whose center is the centroid of a facet of S. Extending results for two and three dimensions, Miernowski et al. [1779] show that the touching points of K with S are precisely the tangency points of the spheres inscribed to the images $h(S)$ with S.

The *Gergonne and Nagel centers* of a triangle are the intersections of lines from the vertices to the points where the incircle and excircles touch the sides, respectively. Unfortunately, there is no direct higher dimensional extension since the analogous lines for n-simplices, $n \geq 3$, do not have a common point. But Hajja [1139] gives equivalent definitions such that analogues in higher dimensions lead to points satisfying these definitions and being unique for all simplices. Corresponding results are then derived.

Horváth [1234] derives tight inequalities for the *volume of the convex hull of two congruent and intersecting Euclidean n-simplices*. He starts with a new proof of a related result of Rogers and Shephard and proves then an explicit formula for this volume when the simplices are obtained from each other by reflection in a hyperplane. In particular, for regular simplices this formula implies that the volume of the convex hull is at most twice the dimension-fold volume of the starting simplex. A similar inequality is obtained for more general simplices.

Combining several metric quantities, Wang and Yang [2429] derive *Bonnesen-style inequalities* referring to n-dimensional simplices in the spirit of isoperimetric problems. Based on this, they get *characterizations of regular simplices*, and related reverse Bonnesen-style inequalities are obtained, too.

CHARACTERIZATIONS

Collecting and sharpening earlier results (e.g., due to Butler and Schneider), Martini [1706] confirms some simplex characterizations in an elementary way. Let K be an n-dimensional convex body, and xy be a chord of K having maximal length in its own direction u (thus, xy is an affine diameter of K). Then K is said to be an *oblique double cone* with respect to u if every two-dimensional half-plane with bounding line passing through x and y cuts out of K a (possibly degenerate) triangle. It is shown that the following (and further) properties are equivalent to the fact that K is an n-simplex: (1) For each direction, K is an oblique double cone. (2) For every translation vector t creating non-empty intersection $K \cap (K+t)$, this intersection is either a point or positively homothetic to K. (3) Each Steiner symmetral has exactly two extreme points outside the corresponding symmetrization space. (4) The projection body and the difference body of K are polar reciprocals with respect to the centred sphere of radius $\sqrt{n \cdot V(K)}$. The second characterization is particularly interesting for the notion of *Choquet simplices*, and V. Soltan [2271] presents characterizations of homothetic simplices by intersections of the type $K_1 \cap (K_2 + t)$, where the convex bodies K_1, K_2 belong to at most countably many homothety equivalence classes.

Here we also cite Barthe's nice result (see [232]) that each convex body in Euclidean n-space has an affine image whose *mean width* is not larger than that of a *regular simplex* of the same volume.

We say that, for K an n-dimensional convex body, a set X of exterior points *sees* the whole boundary of K if for any boundary point y there is some $x \in X$ such that K has empty intersection with the open segment xy. Martini and Soltan [1715] show that for each exterior point x_1 of K there are n further exterior points $x_2, ..., x_{n+1}$ such that the whole set $x_1, x_2, ..., x_{n+1}$ sees the whole boundary of K. It is proved that a convex body K is an n-simplex if and only if for each exterior point there is a second exterior point such that both points together see the whole boundary of K.

The classical *Bézout inequality in algebraic geometry* relates the degrees of hypersurfaces to the degree of their intersection. It has an implication for *mixed volumes* in convexity. Namely, for n-dimensional convex bodies L, M one has $V(L, M, S[n-2])V_n(S) \leq V(L, S[n-1])V(M, S[n-1])$ if S is a simplex. It is conjectured that, for all convex bodies L, M, this inequality is only true if S is indeed a simplex. This is confirmed by Saroglou, Soprunov, and Zvavitch [2116] for all polytopes instead of S, and additionally it is proved that if S is a convex body having a boundary point not lying in a boundary segment of S, then there exist convex bodies L, M for which the inequality above does not hold. The same authors study in [2117] the Bézout-type inequality $V(L_1, \ldots, L_n)V_n(K) \leq V(L_1, K[n-1])V(L_2, \ldots, L_n, K)$ for mixed volumes which characterizes n-simplices. This means: If K is so that this inequality is satisfied for all n-dimensional convex bodies $L_1, ..., L_n$, then K has to be an n-simplex. To get this result, the authors use new interesting results on *differentiability of the support functions of Wulff shapes*. They introduce also a class of convex bodies each member of which fails to satisfy the inequality above.

K. J. Böröczky and Hug [403] derive *stability versions of Ball's reverse isoperimetric inequality* and of the corresponding inequality for isotropic measures. For this purpose, they consider affine invariant distances between convex bodies, based on the Banach–Mazur and on the volume difference distance. Also the spherical Hausdorff distance is used when discussing a characterization of regular simplices and a related stability result. The paper gives an excellent overview to this subject.

Galicer et al. [919] present an asymptotic estimate on the volume ratio $V(S)/V(K)$ for K an n-dimensional convex body and S a simplex containing K. Let both have the same centroid; then $(V(S)/V(K))^{1/n} \leq cn^{1/2}$, where $c > 0$ is an absolute constant. A dual result on simplices contained in

K is also given, and a reason for the condition of equal centroids is justified. It is related to the Mahler product of S and its polar.

The paper [1425] of Kobos on a variation of the *Banach–Mazur distance* (considering positive and negative homothetic copies of the participating convex bodies) goes back to Grünbaum's famous survey [1080] on measures of symmetry; see also Chapter 3 of Toth's book [2381]. Namely, Grünbaum introduced the following distance notion for arbitrary n-dimensional convex bodies K and L: $d_G(K, L) = \inf\{|r| : K'' \subset L' \subset rK'\}$, with the infimum taken over all affine images K' and L' of K and L, respectively. Based on an analogous result on $d_G(K, L)$, Gordon et al. prove in [1006] that the Banach–Mazur distance between a convex body L and a symmetric convex body K is optimally bounded by n; namely, the distance between an n-simplex and any symmetric convex body equals n. Jiménez and Naszódi [1314] show that if one of K and L is smooth or strictly convex, the assumption that the discussed variation of the Banach–Mazur distance equals n implies that K or L is a simplex. Kobos [1425] derives a related stability result referring to $d_G(K, L)$ for the smooth case, which implies further interesting observations. In [1426] he gives a related planar result, yielding the triangles as extremal figures.

SPECIAL TYPES OF SIMPLICES

Frankl and Maehara [897] prove the following: Any finite point set F of cardinality at least 5 has the property that if all triangles formed by triples from F have the same positive area, then F is the vertex set of a *regular simplex*. They give also a result on k-distance sets built from such point sets, in terms of different triangle areas. Martini [1708] extends the characterization of regular simplices to spherical and hyperbolic space. McMullen [1747] calls an n-simplex *k-equiareal* if all its k-dimensional faces have the same k-volume (thus, due to [897] any 2-equiareal n-simplex for $n \geq 4$ is regular). Various properties of k-equiareal simplices are proved in [1747]. For example, an n-simplex is $(n-1)$-equiareal iff the incenter and the centroid coincide. Also, the author constructs $(n-2)$-equiareal n-simplices not being regular and gives examples of $(n-2)$-equiareal n-simplices not being $(n-1)$-equiareal. One central question with respect to $3 \leq k \leq n-2$ and $n \geq 5$ is whether an $(n-2)$-equiareal n-simplex is necessarily $(n-1)$-equiareal. All constructed examples have this property, but there is no proof. There exist such non-regular simplices for $n = 5$, $n \geq 7$, and $n = 11$ having no symmetry property. It is also shown that if the central question above would have a positive answer, then every k-equiareal n-simplex would be regular for $1 \leq k \leq n-3$.

An n-simplex S is said to be *replicating* if one can dissect it into at least two simplices such that all simplex tiles are mutually congruent and similar to S. A *Hill-simplex* is the convex hull of vectors $0, v_1, v_1 + v_2, \ldots, v_1 + v_2 + \ldots + v_n$, with linear independence of $v_1, v_2, \ldots, v_n$, which have equal lengths and the same angles between every two of them. Due to equi-decomposability properties, this notion (in low dimensions due to Hill) was introduced for n-space by Hadwiger [1132]. Each Hill n-simplex is replicating, implying that all such simplices tile n-space. Hertel (see [1193] and [1194]) derived several results in this direction and conjectured that a tetrahedron is replicating iff it is a Hill tetrahedron. Constructing counterexamples to this conjecture, Maehara shows in [1661] that for any $n \geq 3$ there are replicating n-simplices which are not Hill simplices. Maehara [1662] calls a simplex *reflection-symmetric* if it is invariant under some orientation reversing isometry. He gives characterizations of all Hill simplices that are reflection-symmetric. To succeed with this, he explicitly provides all isometries that preserve any given Hill simplex, finding also which of them are orientation-reversing.

Replacing the notion of equiareality by *congruence*, Martini and Wenzel [1726] prove that an n-simplex is *regular* if for some k with $1 \leq k \leq n-2$ all its k-dimensional faces are congruent. Additionally it is shown that this is not the case for $k = n-1 > 2$. Thus, in this setting the simplices with congruent facets (called *equifacetal simplices*) remain interesting. Edmonds wrote three interesting papers about this class of simplices. In [795] he shows that the isometry group of every equifacetal simplex acts *vertex-transitively* (and facet-transitively). As a corollary, each isometry of an equifacetal

simplex has only one fix-point: the centroid (i.e., the barycenter of the vertices). Conversely, he conjectures that any simplex whose isometry group acts vertex-transitively is equifacetal. There are further results, for example on combinatorial types of simplices (defined as partitions of its edge sets into equivalence classes regarding lengths). A related characterization result for even dimensions is derived. The paper contains also a nice survey on related results and interesting open problems. In [796] the *set of centers* of a simplex is shown to be a subset of the fixed point set of its group of isometries. It is a single point iff the group acts transitively on the vertices. From this it follows that an equifacetal simplex has a unique center. The *center conjecture* says: If a simplex has a unique center, then it is equifacetal. The author proves a strong version of this conjecture for the dimensions at most six. Associated with any equifacetal n-simplex is also a well-defined partition of n that counts the number of edges of each possible length incident at a given vertex. The *partition problem* asks for a characterization of the partitions coming from equifacetal simplices. In [797] this problem is resolved, and equifacetal simplices that are in this sense extremal are characterized.

Clearly, a triangle is regular/equilateral if two of its classical centers (meaning the centroid, incenter, circumcenter, or orthocenter) coincide. Edmonds et al. [798] have the aim to extend this concept to higher dimensions; they ask what consequences the *coincidence of two or even more centers* (including also the Fermat–Torricelli point, having minimal distance sum to the vertices) can have for the *facial structure* of the considered simplices. (Note that the orthocenter, in general for $n \geq 3$ not existing, is replaced by the so-called *Monge point*.) In dimension three characterizations of isosceles tetrahedra are obtained, and for $n \geq 4$ several approaches show that a more complicated machinery is necessary to get analogous results. For example, if two of the centers coincide, one can geometrically describe consequences for their facial structure, but one cannot usually infer much about other centers. In addition, the authors study the geometric importance of the cevians through a given center having the same length, and they give a modern and unifying presentation, including also the large set of known results in the plane and in 3-space. After giving a survey on *orthocentric simplices* (i.e., on at least three-dimensional simplices whose altitudes are concurrent), the authors of [799] present various stronger results for this subfamily of all simplices. For example, they show that orthocentric simplices in which either all faces are congruent, or any pair of centres (centroid, circumcentre, incentre, orthocentre) coincide, must be *regular*, i.e., have all its edges of the same length. The main tools are Gram matrices and a characterization of the barycentric coordinates of the orthocenter. Also in [1140] the author investigates the *degree of regularity* as a consequence of the coincidence of two or three classical centers, with special emphasis on edge-incentric simplices (these are simplices admitting a sphere that touches all their edges internally).

An n-simplex is called *biregular* if the set of fixed points of its motion group is a line (or, in degenerate cases, a point). Solving an old open problem at least for orthocentric simplices, Edmonds et al. [800] show the following: the *incenter* of an orthocentric simplex S of arbitrary finite dimension lies on its *Euler line* if and only if this simplex can be expressed as the join of two regular simplices, with all edges connecting the two corresponding components having the same length. These are exactly the orthocentric simplices whose group of isometries has as a fixed point set a line (or a point, for the subcase of regular simplices).

Bédaride and Rao [267] study the *billiard flow inside a regular n-simplex*. (A few related results on tetrahedra are known, but as it seems, nothing is known about periodic orbits of n-polytopes for $n > 3$.) They confirm the existence of two types of *periodic trajectories*. The first one has period $n + 1$ and hits each facet once. The second one has period $2n$ and hits one of the facets n times, while hitting any other facet once. In both cases, the exact coordinates for the points where the trajectory hits the boundary of the simplex are determined. Further on, the convex hull of the periodic orbit of length $n + 1$ is looked at, and it is shown that the orbits obtained for $n = 3$ are stable with respect to perturbations of the respective tetrahedron.

The two papers [1141] and [1142] treat the following essential types of n-simplices: *orthocentric simplices* (with concurrent altitudes), *circumscriptible simplices* (for which there exists an $(n - 1)$-

sphere that touches its edges internally), *isodynamic simplices* (for which the cevians that join the vertices to the incenters of the opposite facets are concurrent), and *isogonic simplices* (a higher dimensional generalization of isogonic tetrahedra for which the cevians that join the four vertices to the points where the insphere touches the opposite faces are concurrent). In [1141] it is shown that the intersection between any two such families yields precisely the class of n-kites (simplices with n of their vertices forming a regular $(n-1)$-simplex whose vertices are equidistant from the remaining vertex), and in [1142] several characterizations of these families are proved or reproved.

Al-Afifi et al. [34] generalize *Pompeiu's theorem* (saying that the distances from the vertices of an equilateral triangle to any point in its plane yield the side lengths of some triangle) to given regular n-simplices. The introduction of [34] contains the classical Möbius–Pompeiu theorem and an extension to regular n-simplices due to Fiedler. Giving another generalization, the authors then show that for a regular n-simplex, $n \geq 2$, and a point x in its affine hull, the distances from x to the vertices of the given simplex can be taken as the volumes of the facets of some n-simplex iff x is not a vertex of the given regular simplex. For this purpose, they prove a *higher dimensional analogue of the triangle inequality*, yielding a condition for positive numbers to be volumes of the facets of an n-simplex.

APPLICATIONS

Lutwak [1637] shows that the following *containment result* holds for two n-dimensional convex bodies K_1 and K_2: If each simplex containing K_2 contains also some translate of K_1, then K_2 has to contain a translate of K_1. Now let $V_1(S, K)$ denote the mixed volume of a simplex S and a convex body K taken once. As a corollary of the above results the author derives: If K_1, K_2 are two convex bodies satisfying $V_1(S, K_1) \leq V_1(S, K_2)$ for any simplex S, then some translate of K_1 is contained in K_2. Klain [1400] derives a somehow dual result, relating containment of K_1 and K_2 to inscribed simplices (or even polytopes). He shows equivalence of the following two statements for $1 \leq d \leq n$: (1) For each polytope P contained in K_1 and having at most $d+1$ vertices, K_2 contains a translate of P. (2) For each d-dimensional subspace D of n-space the orthogonal projection of K_2 onto D contains a translate of the respective projection of K_1 onto D. For $d = n$ this means: K_2 contains a translate of K_1 iff for any polytope P contained in K_1 and having at most $n+1$ vertices, K_2 contains a translate of P.

It is clear that many results on *simplices in normed spaces* are "automatically" close to convexity since the geometry of the *convex* unit ball plays a role there.

As already mentioned above, the survey [1722] contains a lot of basic material about the geometry *triangles and simplices in finite-dimensional normed spaces*. We want to collect here further contributions which are (with a few exceptions) not cited there or in other parts of our book.

PROPERTIES AND CHARACTERIZATIONS OF SIMPLICES IN NORMED SPACES

Averkov and Martini [163] determine the *complete system of inequalities for normalized area and side lengths of triangles in normed planes*, including the cases of equality. After discussing also possible extensions to n-dimensional simplices in normed spaces, they derive the tight lower bound zero for the normalized volume of an arbitrary n-dimensional simplex.

Averkov [156] introduces a Minkowskian area measure and a Minkowskian analogue of trilinear coordinates to study Minkowskian balls tangent to all affine hulls of facets of n-dimensional simplices. Based on this, he studies also special types of simplices, e.g., having equal Minkowskian heights, or medians of equal Minkowskian lengths.

Alonso et al. [67] classify all possible locations of circumcenters of non-degenerate triangles in arbitrary normed planes. They do this based on a complete picture of all possible shapes of the intersection of two norm circles. And in [68] they similarly classify all possible locations of minimal enclosing norm circles with respect to non-degenerate triangles in arbitrary normed planes, this time based on a new classification of triangles via their "angles". Continuing this view with circumcenters

and circumspheres of simplices in higher dimensions, Leopold and Martini [1547] derive results on *Minkowskian analogues of Euler lines and Feuerbach spheres*. Using duality in a natural way, they get also theorems on *angular bisectors* as well as *in- and exspheres of simplices* in the dual normed space. Also a special *type of regularity* is studied, obtained when the vertex centroid is one of its circumcenters. It is shown that such n-simplices are the only ones having the property that the Euler line collapses. And in [1548] the same authors continue their studies regarding similar notions for simplices in n-dimensional normed spaces: the Monge lines of an n-simplex with a circumcenter, its Monge point, Euler lines, Feuerbach spheres once more, and further notions. Thus, they obtain various results that generalize their Euclidean counterparts to the geometry of finite-dimensional normed spaces.

Inspired by [1294], the authors of [989] study (the uniqueness of) *circumcenters for gauges* (i.e., for convex unit balls with dropped symmetry demand). They show that every bounded set has a unique circumcenter iff each set of cardinality at most n has a unique circumcenter. Equivalent conditions in terms of geometric properties of the unit ball are presented, and the following problem (posed for $k \in \{0, \ldots, n-2\}$) is solved: When are the dimensions of the sets of all circumcenters of arbitrary bounded sets not larger than k?

The *Erdős–Mordell inequality* says that the sum of the distances from an interior point x of a triangle T to its vertices is larger than or equal to twice the sum of the distances from x to the sides of T. Ghandehari and Martini [954] extend this result to arbitrary normed planes and give then analogous extensions of other related inequalities, for example a version for polygons. Finally, related inequalities for simplices in normed spaces of arbitrary finite dimension are obtained.

Schneider [2164] studies *analogues of Jung's and Steinhagen's theorem in n-dimensional normed spaces* (which are due to Bohnenblust, Leichtweiss, and Eggleston) estimating the ratios of circumradius and diameter, as well as minimal width and inradius, of convex bodies. Equality holds if the studied body is a simplex and the unit ball its difference body, and if (vice versa) equality holds, then it is a simplex, but the unit ball is not unique (up to affinities). The author proves *stability versions of these inequalities* and uses also a measure of symmetry which characterizes simplices. The mentioned quantities are expressed in terms of a single function of two convex bodies, yielding the possibility to use covering arguments.

SPECIAL TYPES AND EXAMPLES IN NORMED SPACES

Chakerian and Ghandehari [595] study the *Fermat–Torricelli problem* (finding the point of minimal distance sum to m given points) for $n+1$ given points in a strictly convex and smooth normed space of dimension $n \geq 2$, using the geometry of the simplex that forms the convex hull of this set. They use an extension of Viviani's theorem (referring to convex polytopes for which the sum of the Minkowskian distances to the affine hulls of their facets is the same for all interior points) and study special types of Minkowskian simplices (with coinciding incenter and centroid, and with equal Minkowskian altitudes). A comprehensive representation of the Fermat–Torricelli problem (also in normed spaces) is Chapter II of the monograph [432], and for gauges (convex unit balls without demanding symmetry) the paper [1294] of Jahn, Martini, and Richter.

Recall that a convex body is called *reduced* if it does not properly contain a convex body of the same minimum width; see Plate 6 for reduced bodies in normed spaces with unit ball B. In [1719] it is shown that no Euclidean simplex is reduced (except for the regular triangle in the planar case), and the proof is based on the fact that if a simplex would have that property, it would be necessarily $(n-1)$-equiareal. In normed spaces and also for gauges, reduced simplices exist. Martini and Wu [1727] construct a Minkowski space and a diametrically complete simplex in it which is not reduced. Inspired also by this example, Brandenberg et al. [487] ask whether a convex body, which is diametrically complete and reduced with respect to some gauge body (asymmetric norm), is necessarily of constant width. They confirm this for simplices and for bodies having a smooth extreme point.

EQUILATERALITY IN NORMED SPACES

Brass [491] improves the lower bound for Petty's problem on the existence of *equilateral simplices in normed spaces*, asking whether such an n-space contains $n + 1$ points at pairwise distance one. He shows that for each k, there exists a certain $n(k)$ such that each normed space of dimension $n(k)$ contains k points at pairwise distance one. In the proof Dvoretzky's theorem is used, and then it is shown that for nearly Euclidean normed spaces it is always possible to extend an equilateral simplex of at most n points to an equilateral simplex of $n + 1$ points. Also Brouwer's fixed point theorem is successfully used. Independently, Dekster [723] proved that if a normed n-space is sufficiently close to a Euclidean space then it contains an equilateral set of n points. "Sufficiently close" means that the Banach–Mazur distance from the Euclidean space is at most a number like $1 + 1/n$. Again a topological fixed point theorem is needed. In the survey [2330] of Swanepoel, results on the largest size of equilateral sets in normed spaces are collected, and many interesting, related research problems are posed. The representation is also related to topics like *minimal surfaces* and *low-distortion embeddings of finite metric spaces into normed spaces*. Concrete contents are the mentioned equilaterality bounds, Petty's related theorem, Cayley–Menger theory, the theorem of Brass and Dekster (see above), embeddings, and relations to approximation theory. The following question is an example from the problem list: Does there exist an *infinite equilateral set* in any separable, infinite-dimensional normed space? The interesting exposition [2332] of the same author contains, among many other topics, also updated information on equilateral sets in normed spaces.

Brass [491] constructs an example of a normed 4-space such that for some pair of points therein no extension to further vertices of a regular simplex is possible. Motivated by this, Makeev [1682] proves that in each normed 4-space there exists a regular 4-simplex. Moreover, let $r \geq 3/4$ be a positive number. Then in any hyperplane there is a crowd of four points being equilateral, and there is a fifth point at the r-fold distance to all these four points. A convex polytope in a finite-dimensional normed space is called *subequilateral* if the norm-length of each of its edges is equal to its diameter. In [2331] it is shown that in Euclidean n-space the only subequilateral polytopes are equilateral simplices. Väisälä [2390] reproves Petty's result for normed 3-spaces alternatively. He proves that the intersection of any two spheres in every normed n-space, $n \geq 3$, is connected and gets the result (that there is an equilateral 3-simplex in any normed 3-space) as a corollary. Also in [1683], Makeev studies simplices in normed spaces, namely regarding the question whether $n + 1$ points in Euclidean n-space, $n \geq 3$, can be isometrically embedded into any n-dimensional real normed apace. For $n = 3$ he reduces the problem suitably to the case where four points lie in a 2-plane. And also for arbitrary n a certain reduction is found.

2.9.1.2 Cubes

It is clear that with an *n-dimensional cube* in Euclidean n-space we mean the Minkowski sum of n pairwise orthogonal segments of equal lengths (a Schlegel diagram presenting the 4-cube is given in Plate 3, at the beginning of the third chapter).

Here we start with a few publications in the form of books or surveys, to present after them selected research papers, ordered by certain viewpoints and then also chronologically. Again we do not discuss topics which are discussed in other parts of our book.

Curiosities of the 3-cube (like dissections, slices, colorings, Pohlke's theorem, geodesics, etc.) are nicely collected in the booklet [2020] of Ranucci and Rollins. A similar reference is Ehrenfeucht's booklet [806] collecting various results on cubes, such as on colorings, or on Rupert's problem (see below), and the expository paper [2445] of Weiss should also be mentioned here. Refreshing

discussions of topics like *geodesics, quasi-geodesics* (applying Snellius' refraction law), *billiards*, and further related topics in and on 3- and 4-cubes are given there.

In Ball's booklet [199] the extremum property of the cube in the symmetric case of the *reverse isoperimetric problem* is nicely discussed (see Lecture 6 there). For new contributions extending his well-known result on *maximal hyperplanar cube slices* we refer to the recent paper [1287] of Ivanov and Tsiutsiurupa, containing the important references.

In his booklet [2571], Zong discusses essential topics related to geometric and combinatorial properties of unit cubes. He successfully demonstrates how different mathematical branches come together when cubes are studied or used (see also Zong's brilliant survey [2570], somehow preparing [2571]). These topics are volumes of cross-sections (showing also how log-concave measures and the Brascamp–Lieb inequality can help) and of projections of cubes, simplices inscribed to the cube, minimal number of simplices in a triangulation of the cube (and the role of hyperbolic geometry in this setting), bounds for the number of faces of 0/1 polytopes, the role of group rings regarding Minkowski's conjecture and Furtwängler's conjecture, and graph-theoretical methods in view of Keller's conjecture on twins in cube tilings. For these themes, the state of the art is presented, and in some cases the methods are also discussed more precisely.

Here we also mention the lecture notes [2082] of Ryabogin and Zvavitch, in which the *Blaschke–Santaló inequality* and *Mahler's conjecture* are nicely presented. The latter states that the volume product takes, in the symmetric case, its minimum at cubes. Proofs are given for the two-dimensional case, for unconditional convex bodies, and for the local minimality of the unit cube.

SECTIONS OF CUBES

Now we switch to the interesting topic of *cube slicing*, mainly concentrating on *hyperplanar sections of cubes*.

An early contribution in this direction is Naumann's paper [1878]. For instance, Naumann asks in which way regular polytopes can occur as (central) sections or projections of other regular polytopes. For example, it is shown that the n-cube can be obtained as "centroid section" of a $(2n-1)$-dimensional regular simplex.

Of course, cube slicing has also to do with *section functions* in the spirit of *geometric tomography*. Therefore such problems are also discussed in Gardner's book [930]; see the Chapters 7 and 8 there (in particular Note 7.8), and also Section 2.3 of the survey [1710]). Further on, Koldobsky discusses this topic in Chapter 7 of his book [1444], like also Zong in [2571]. Ball [194] confirmed a conjecture of Hensley showing that the sharp upper bound for the volume of hyperplane sections of a unit cube is $\sqrt{2}$. This result has applications also regarding the *Busemann–Petty problem*! Namely, Ball [195] observed that it provides a negative answer to the Busemann–Petty problem for the dimensions $n \geq 10$. Chakerian and Logothetti [598] collect various nice results and ideas around the *geometry and combinatorics of hyperplanar cube-slicing*. For example, they study all hyperplane sections normal to the main diagonal of the given n-cube which contain vertices, and based on the number of those vertices and the shape of such slices, they get interesting relations between the sequences of slices and binomial coefficients as well as Pascal's triangle. The volumes of the slices and also of respective slabs are investigated, and interesting related probabilistic problems are posed. Ball's result on maximal cube slices and its consequences for the Busemann–Petty problem are discussed, too.

Saks [2089] surveys *combinatorial problems related with sections, partitions, and coverings of the n-cube*. A subset of the vertex set is said to be covered by an affine polynomial P if P is zero on all elements of this subset. Bounds on the maximum number of vertices which can be covered by skew hyperplanes ("skew" means that all coefficients are different from 0) and on the minimum number of skew hyperplanes needed to cover the whole vertex set are derived. The maximum number of edges that can be sliced by a single hyperplane is exactly determined, and bounds are given for the minimum number of hyperplanes needed to slice each edge. Interpreting vertices and edges as

zero- and one-dimensional cubes, also higher dimensional problems in this sense are studied. Frank and Riede [896] succeed in deducing an elementary formula for the *volume of an arbitrary hyperplane slice* of the n-cube.

Barthe and Koldobsky [237] study the *volume of intersections of unit cubes with slabs of fixed width* which are centred at the midpoint of the cube. Confirming a related conjecture of Milman, they show that if the width is at most $3/4$, then slabs parallel to facets yield the minimum. For large width, the slabs orthogonal to the main diagonal become asymptotically minimal as the dimension goes to infinity. Moody et al. [1842] investigate *extremal volumes of non-central slices of the unit n-cube*. "Non-central sections" means that the intersecting hyperplanes have to be at certain distance $t > 0$ from the center. In general, such a consideration seems to be new, and this is also the case with non-central slices of the unit n-cube by slabs. In this paper a full solution for extremal one-dimensional slices, and a partial solution for extremal hyperplane slices for the case $t > \sqrt{n-1}/2$ is given. Minimal volume slices of the n-cube by slabs of width $2t$, with the same bound for t, are shortly discussed, too. Regarding slabs that intersect a cube, also the paper [1433] of König and Koldobsky should be mentioned, and regarding demanded distances to the center also the more recent paper [1432] of König. For $n = 2$ and $n = 3$, these authors study volumes of such slabs as a function of the width in the determined normal direction. There exist, for the volume function of the slabs, very many local extrema and saddle points, and the authors present *general formulae for volumes of slices and slabs*. And in [1434] the same authors extend Ball's result on maximal cube sections to certain product measures which include Gaussian type ones. They also study in [1435] the $(n-2)$-dimensional surface area of central hyperplanar sections of the unit n-cube. The maximum is reached perpendicularly to the vector $(1, 1, 0, \ldots, 0)$, and using this result, it is shown that the answer to an analogue of the Busemann–Petty problem for surface area is negative for $n \geq 14$. The authors study also the *complex case* of their investigations.

Aliev [59] derives a sharp upper bound for the volume of hyperplane sections of the n-cube, suitably normalized with the help of the l_1- and the l_2-norm. Bartha et al. [231] study *volumes of central slices of unit n-cubes orthogonal to their main diagonals*. It is shown that they give, for $n \geq 3$, a strictly monotonically increasing function of the dimension. The authors use an integral formula going back to Pólya.

Ivanov and Tsiutsiurupa [1287] discuss *maximal volumes of k-dimensional sections of the unit n-cube*. Also based on older results of Meyer and Pajor [1770], they obtain for $n \geq 2$ a necessary and sufficient condition for a k-subspace to be a local maximizer of this function; they formulate this in a geometric way. Melbourne and Roberto [1764] derive a quantitative form of Ball's theorem on cube slicing in n-space. As a consequence of this, the authors derive a quantitative form of Bobkov's and Chistyakov's *min entropy power inequality*, allowing to characterize equality cases.

PROJECTIONS

After discussing sections, we come now to results on *projections of n-cubes*.

Using a theorem of Jacobi on determinants, McMullen [1741] shows the following: If Z_k and Z_{n-k} are orthogonal projections of the unit n-cube onto totally orthogonal subspaces of dimensions k and $n-k$ of the Euclidean n-space, then the k-volume of Z_k and the $(n-k)$-volume of Z_{n-k} are equal. An extension to so-called strongly associated zonotopes is given, too. Schnell [2184] extends this result suitably to affine cubes, and he asks for a corresponding statement on suitably related pairs of general zonotopes. Chakerian and Filliman [594] obtain several *inequalities* for the k-dimensional Lebesgue measure of the orthogonal projection of the unit n-cube into a k-dimensional subspace. It is shown that the studied problem is symmetric in k and $n-k$. For $k = 2$ and $k = n-2$ they give a sharp upper bound and derive also further inequalities for general k. Filliman [874] presents various results on zonotopes, among them also being some about *orthogonal k-dimensional projections of n-cubes* with $1 \leq k \leq n$. He gets a necessary condition for a k-zonotope to have the largest volume among all k-dimensional

orthogonal projections of the n-cube, and several further results on orthogonal shadows of n-cubes are obtained. For example, it is shown that the rhombic triacontahedron has the largest 3-volume among all orthogonal projections of a 6-cube into 3-space.

Ostrovskii [1927] investigates *minimal-volume projections of unit n-cubes* in k-dimensional linear subspaces. He proves that a k-zonotope is linearly equivalent to a minimal-volume projection of a unit n-cube if and only if it is linearly equivalent to the zonotope spanned by the multiples of rows of a totally unimodular $n \times r$ matrix of rank k. He continues in [1926] in proving that for every subspace L, the set of minimal-volume projections of unit n-cubes in L contains a parallelepiped. As the main result it is shown that for some subspaces there exist minimal-volume projections that are far away from parallelepipeds with respect to the Banach–Mazur distance.

The focus of Paouris et al. [1939] is the *volume of random projections of the unit n-cube*. More precisely, the authors succeed in deriving a *central limit theorem* for the volume of the orthogonal projection of an n-cube onto a random subspace of fixed dimension k, distributed according to the appropriate Haar probability measure as $n \to \infty$.

Naumann [1879] uses vector stars and properties of zonotopes to characterize *orthogonal projections of n-cubes*. This can be seen as generalization of *Pohlke's theorem on axonometry*. And also Bergold [289] slightly generalizes Pohlke's theorem: He shows that two arbitrary parallelepipeds in 3-space can always be turned in such a way that they appear identical, i.e., that their (parallel) projections onto a plane perpendicular to the direction of sight are similar. And one of the parallelepipeds could be a planar skew image, meaning a suitable zonogon. Maehara [1663] proves that for any $k \geq 2$ and $n \geq k$, there is an n-dimensional unit cube which is projected to a regular $2k$-gon by an orthogonal projection from n-space onto a two-dimensional subspace. The author studies these "shadows" if the dimension n is increasing.

VERTEX EMBEDDINGS: SIMPLICES IN CUBES

A famous problem, sometimes described as *vertex embedding problem regarding regular polytopes*, asks for the dimensions in which a regular n-polytope P can be embedded into a regular n-polytope Q of different type such that each vertex of P is also vertex of Q. Such pairings of different types of regular polytopes are discussed in the book [2189] of Schoute and by Adams et al. [16]. In the latter paper it is nicely shown how this circle of problems brings together methods from geometry, combinatorics, matrix theory, algebraic topology, and number theory. It turns out that the subcase "P is an n-simplex and Q an n-cube" is particularly interesting, since it is equivalent to the question for the *existence of Hadamard matrices*, for Jung constants of the form $n/(n+1)$ in l_1^n, and for a problem from coding theory (see Medyanik [1761], Dol'nikov [759], and Wallis [2414]). (A *Hadamard matrix* is a square matrix whose entries are either $+1$ or -1 and whose rows are mutually orthogonal.) Many references to this theme can be found on page 76 of Martini's survey [1709]. The more recent and widespread literature about Hadamard matrices and their applications can be found in the monographs [1228] of Horadam, [1516] of Launey and Flannery, and [2213] of Seberry. For example, Seberry and Yamada [2124] as well as Hedayat and Wallis [1175] present many constructions of Hadamard matrices or of error-correcting codes, showing also applications in statistics.

We continue with some references dedicated directly to *vertex embeddings of regular n-simplices into n-cubes* and which have some historical value or are important in a more recent sense. It is conjectured that such embeddings exist whenever $n + 1 = m$ is divisible by 4. It was pointed out already by Barrau [229] that an orthogonal matrix with m rows and entries either $+1$ or -1 provides coordinates for the vertices of an $(m-1)$-dimensional regular simplex inscribed in an $(m-1)$-dimensional cube (see also §3 of the interesting paper [672] by Coxeter). Gruner [1090] shows that such a matrix can be constructed whenever $m = k^2$, where each of the numbers $k+1$ and $k-1$ is a power of a prime. Dedo [721] uses the geometric problem to derive results on Hadamard matrices, and he pays special attention to the derivation of circulant matrices when $n = m - 1$ is a prime of the

form $4k-1$. From the viewpoint of *computational convexity*, Gritzmann and Klee study the existence of Hadamard matrices in subsection 9.6 of their exposition [1026], showing also the connection to the problem of *largest k-simplices in n-cubes*. For $k = n$, this problem is equivalent to the *Hadamard determinant problem* which also subsumes the problem on the existence of Hadamard matrices. The problem on largest simplices in cubes is also the subject of the expository, comprehensive papers [1029] of Gritzmann, Klee, and Larman as well as [1252] due to Hudelson, Klee, and Larman. For all n, there the cases $k = 2$ and $k = 3$ are solved, and various further (computational) results are obtained. Interesting conjectures are stated, and also other problems regarding simplices and cubes are posed. In the computational parts, polytopes are studied as convex hulls of finitely many points like also as intersections of finitely many half-spaces.

Based on the theory of Gauss sums, Medyanik [1760] proves that if $2n-1$ is a prime number or a product of two twin primes, then a half-circulant Hadamard matrix of order $4n$ exists, and one can inscribe to a $(4n-1)$-dimensional cube a regular simplex of the same dimension. The paper contains also computational results. Interpreting the vertex set V of the n-cube as a vector space over the field of order 2, Wenzel [2453] shows the following: the regular n-simplex can be inscribed to the n-cube such that its vertices form a subgroup of V precisely if $n+1$ is a power of 2. In addition, a related result on Hamming codes is proved.

TRIANGULATIONS OF CUBES

Our next part treats *triangulations of cubes*, i.e., minimal (regarding cardinality) dissections of the n-cube into n-simplices all whose vertices are vertices of the cube. This minimal number is sometimes also called *simplexity of the cube*. A good overview to this theme is given by Section 6.3 of the monograph [1598] of De Loera, Rambau, and Santos, and also Chapters 3 and 4 of Zong's book [2571] and his survey [2570] should be recommended here. We reflect now various interesting contributions. Mara [1694] assumes that cutting off 2^{n-1} cube vertices and to dissect the convex hull of the remaining points yields a minimal partition. He conjectures to obtain this way a partition into $(n! + s^{n-1})/2$ simplices. The assumption and the conjecture are true for $n \leq 3$, for $n = 4$ the conjecture is true (presupposing the assumption), and for $n = 5$ the author found a partition into 68 simplices. His geometric method is based on classifying simplices of a triangulation with respect to the number of interior facets that they have in the dissection. Cottle [662] and J. F. Sallee [2095] continue these investigations. Let $v(T)$ be the number of used simplices of a triangulation T, and $f(n)$ be the minimum of $v(T)$. One can easily find a T with $v(T) = n!$ implying $f(T) \leq n!$. In particular, J. F. Sallee [2095] gives an elegant construction showing that $f(T) \leq P_n$ with $P_1 = 1$ and P_i satisfying the recursion $P_{n+1} = (n+1)(P_n + 2^{n-1}) + 2^{n+1} - (n-1)$, asymptotically bounded from above by $0.477n!$. A lower bound on $f(n)$ is also given, showing a connection to Hadamard's inequality for the *determinant of a Hadamard matrix* and therefore to upper bounds for simplex volumes and lower bounds for the number of used simplices.

Haiman [1138] presents an elegant way to triangulate the kn-cube if the triangulation of the n-cube is assumed to be known. More precisely it is shown that if the n-cube is dissected into $T(n)$ simplices, then the kn-cube can be partitioned into $\varrho^{kn}(kn)!$ simplices, where $\varrho = (T(n)/n!)^{1/n} < 1$. Based on the ideas from [2095], Hughes [1264] classifies the simplices of a triangulation due to their relation to the skeleton of the triangulated n-cube, establishing also a linear programming problem whose optimal objective value is a lower bound for the simplexity of the n-cube, efficient for lower dimensions. Again the relation to the question for maximum-volume simplices in the n-cube is discussed, its equivalence to the Hadamard maximum determinant problem, having the existence-problem of Hadamard matrices as an important subcase. Hughes and Anderson [1265] continue these investigations giving also a nice exposition of the techniques and further results in this direction. For example, a triangulation of the 7-cube into 1493 simplices is presented and shown to be smallest possible, like also the known partition

of the 6-cube into 308 simplices. Dissections of the n-cube more general than triangulations are also discussed.

Smith [2265] shows that the *minimal cardinality of triangulations of the n-cube* is, surprisingly, bounded from below by the ratio of the hyperbolic volume of the ideal cube to the ideal regular simplex. Orden and Santos [1917] study triangulations of products of two polytopes, one of low and one of high dimension, with few simplices. It turns out that their results can be applied to get (asymptotically) efficient triangulations of the n-cube. Bliss and Su [370] study decompositions of the n-cube into simplices which meet only along faces, but may involve additional vertices (not only those of the participating cube). For such triangulations, they describe a continuous map from the n-cube to itself, piecewise linear on the simplices and transforming the triangulation into a simplicial cover. They use linear programming for the cases $4 \leq n \leq 12$, and the result 16 for $n = 4$ shows that Mara's triangulation (see again [1694]) is indeed minimal. Investigating geometric properties of prismoids (i.e., of polytopes all whose vertices lie in two parallel hyperplanes), Glazyrin [980] finds a non-trivial *lower bound for triangulating the n-cube*. In [489] the authors consider cube triangulations with simplices having only nonobtuse dihedral angles; the standard triangulation into $n!$ simplices is a common example. The authors show that, surprisingly, for any dimension larger than two there is only one further nonobtuse triangulation of the n-cube, which they also explicitly construct. Seacrest and Su [2212] study products of simplices, called *simplotopes* (the cube, as product of 1-simplices, is an example). Based on ideas of Bliss [370], a so-called covering approach is used to derive lower bounds for the size of simplicial covers and triangulations of simplotopes. For example, it is shown that a minimal triangulation of a product of two simplices is given by a vertex triangulation (without interior vertices). For products of more than two simplices, bounds for products of segments and triangles are obtained.

Also the paper [1725] of Martini and Wenzel is related to vertex embedding of simplices into cubes and to triangulations of cubes. The authors study different types of packings and coverings of the n-cube by rectangular simplices whose vertices are taken from the vertex set of the cube.

KELLER'S CONJECTURE

We continue with *Keller's conjecture* from 1930: If the Euclidean n-space is tiled by congruent parallel unit n-cubes, then some pair of cubes will share a complete facet. Lagarias and Shor [1488] disproved this conjecture for $n \geq 10$; see also the Mathematical Review MR1155280 of this essential paper. Various further papers on this topic appeared; for developments around Kellers conjecture we recommend to look at Zong's excellent survey [2570] and Chapter 8 of his book [2571]. We add here some references which appeared after these publications of Zong, i.e., after 2006.

Lysakowska and Przeslawski [1645] study a strengthening of Keller's conjecture: Every Keller tiling of n-space has a column of cubes. They prove it for $n \leq 6$ and consider also combinatorial extensions. Lysakowska [1644] continues to study this stronger form of the Keller conjecture. Based on a lemma working in 4-space, interesting new methods are proposed, and Keller's conjecture about columns is reproved for dimensions smaller than 5. For the open case $n = 7$ a former result of Kisielewicz (see [1388]) narrows the range of possible counterexamples to a relatively small number of cases. And Kisielewicz and Lysakowska [1390] give a further reduction of that range. They avoid the use of computer-aided calculations, and their proof can be seen as an impressive further step towards the complete solution of Keller's conjecture. Also Kisielewicz [1389] gives partial results on Keller's conjecture in the only remaining dimension 7. He presents once more an overview to the different steps of resolving the problem, finally succeeding with confirmation for all dimensions $n \leq 7$, and with the refutation for all dimensions $n \geq 8$. The very final step $n = 7$ was solved by Brakensiek et al. [481]. They succeed in the graph-theoretic way: three graphs related to Keller's conjecture for $n = 7$, namely $G_{7,3}, G_{7,4}$, and $G_{7,6}$, are investigated and yield the conclusion that Keller's conjecture is false for $n = 7$ if and only if at least one of the graphs contains a clique of size $2^7 = 128$. The authors of [481] present an automated method to solve this conjecture by encoding the existence of such a clique

as a propositional formula, and so they show that no such clique exists, completely solving Keller's conjecture! Finally we mention the paper [1229] of Horak and Kim, in which it is shown that analogues of Keller's conjecture for tiles of more complex shapes cannot be verified.

DECOMPOSING CUBES INTO CUBES

The following notion goes back to Hadwiger and is also discussed by Meier [1763]: Let $D(n)$ be the set of integers k such that the n-cube has a *decomposition into k smaller, homothetical n-cubes*. For example, $D(2)$ consists of all natural numbers except 2, 3, and 5. The reader should also consult Section C4 in the problem book [687] of Croft, Falconer, and Guy (and the whole chapter C for related problems). Meier proves in [1763] that for any dimension n, there is an $m_0(n)$ such that for any $m \geq m_0(n)$ the n-cube can be partitioned into m smaller cubes. Various interesting questions occur around this notion.

Let n be fixed and $\varepsilon > 0$. Frankl, Meir, and Pach [898] show that, for each sufficiently large $m \geq m_0(n, \varepsilon)$, the unit n-cube can be dissected into precisely m smaller cubes such that the ratio of the edge-length of the largest to the edge-length of the smallest partitioning cube is at most $1 + \varepsilon$. In addition, the authors verify that for every $m \geq m_0$ there is a decomposition with the mentioned properties by cubes with at most $n + 2$ different edge-lengths. And without the demand of roughly equal edge-lengths, it suffices to use cubes of three sizes. Frankl and Pach [899] continue by showing that if an n-cube is decomposed into m cubes whose edge-lengths belong to a certain interval, then m is a perfect nth power, and all participating cubes are of the same size. Hudelson [1250] studies the smallest positive integer $c(n)$ such that for all $m \geq c(n)$ an n-cube can be dissected into m homothetical n-cubes of not necessarily equal size. Exact values of and upper bounds on $c(n)$ are given, and also an asymptotic result is obtained. Improving one of these bounds, Hudelson, Connor, and Marmorino [1251] continue these investigations.

FURTHER RESULTS

We continue with some articles discussing nice *single results*. Sarges [2110] asks which subsets of the vertex set of an n-cube can be deleted such that the convex hull of the remaining vertices still has the same minimal-volume ellipsoid containing it. He presents all such sets in dimensions 2, 3, and 4, and for a large family of further dimensions he finds the minimal number of vertices of the cube necessary for guaranteeing this property. He proves that if for $n = 4(m + 1)$ a Hadamard matrix exists, then one can find this minimal number for the neighboring dimensions $n = 4m, 4m + 1, 4m + 2$, and $4m + 3$. Also Shnurnikov [2239] deals with *separations of vertex sets of an n-cube* with the help of the insphere S of this cube. He confirms a related conjecture for $n \leq 6$: Any tangent hyperplane of S strictly separates at most $2n-2$ vertices of the cube. And for general n he constructs examples showing the separation of precisely $2n - 2$ cube vertices. For $n \geq 3$ the hyperplanes orthogonal to the radius vectors of the vertices of the cube are shown to separate less than $2n - 2$ vertices.

A. Bezdek et al. [311] investigate *Rupert's classical problem* (see also Section B4 in the problem book [687]): to "drill" a hole into a cube such that a congruent copy of that cube can move through that hole via translation. Nieuwland's related question asks for the size of the largest cube that makes this procedure still possible (thus looking for suitable homothets). The authors call a solid *Rupert* if it has a hole through which a congruent copy of it may pass via translation. They give a useful discussion on the history of that problem. Whereas bodies of constant width clearly cannot have this translative property, the situation with convex polyhedra is interesting. The authors verify that cubes and all rectangular boxes have Rupert passages in any direction not parallel to their faces. For the cube it is assumed without proof that the solution of Nieuwland's problem is a tunnel perpendicular to the largest square contained by the cube, and that this holds analogously for all other rectangular boxes. Huber, Shultz, and Wetzel [1249] succeed in showing that also the n-dimensional cube is Rupert. Here we mention also Ehrenfeucht's booklet [806] in which questions around Nieuwland's problem

are discussed; an example of a cube of edge-length 31, which moves Rupert like through a cube of edge-length 30, is given there. Further related results are discussed in our subsection on Platonic and Archimedean polyhedra (see, e.g., the contribution [589] of Chai, Yuan, and Zamfirescu).

Here one can also mention that the shape of a suitably modified n-cube, called the *Klee–Minty cube* (see, e.g., Deza et al. [739]), was successfully used in optimization to show that in the worst case (also using the worst pivoting rule) the simplex algorithm needs exponentially many steps before delivering an optimal solution. This nice example created a lot of research, subsequently yielding polynomial-time algorithms for linear programming.

Further interesting topics around cubes that the reader might look at (with the help of MathSciNet or Zentrablatt) are *cube colorings, cubes and graphs, coverings, dissections of cubes, packings of or in cubes, the Banach–Mazur distance*, and various further ones.

2.9.1.3 Platonic and Archimedean solids

A three-dimensional convex polyhedron P is called a *regular polyhedron* if it is *vertex-transitive* (i.e., the group of isometric symmetries of P acts transitively on the vertices of P) and *its facets are regular and congruent polygons*; see Plate 12 at the beginning of Chapter 12. It turns out that, following this definition, already the summation of facet angles around vertices is sufficient to see that there are at most five regular polyhedra. Their constructions show then that "at most" in this sentence can be replaced by "exactly"; see, e.g., 12.5.4 and 12.5.5 in Berger's book [283] for constructions of all five polyhedra, and page 350 in the survey [1177] for the *regular icosahedron* bounded by 20 equilateral triangles. The other four Platonic solids are the *regular tetrahedron* (bounded by four equilateral triangles), the *cube* (bounded by six squares), the *regular octahedron* (a double pyramid bounded by eight equilateral triangles), and the *Platonic dodecahedron* (also called *pentagon dodecahedron* and bounded by twelve regular pentagons). There are various further ways to define them, and these beautiful geometric objects have very many properties and applications! Therefore numerous mathematical notions, methods, and concepts can be explained or even "trained" with the help of these polyhedra and their modifications and generalizations. Examples of such notions, methods, etc. are giving *exact* definitions of mathematical notions (see Grünbaum's article [1086]), or give a sense of the beauty of fundamental notions like *regularity* or *symmetry* (for the latter two notions, see, e.g., the classical books of Coxeter [670] and Weyl [2457], respectively).

It makes sense to give here, at the beginning, some definitions of families of polyhedra, classifying notions and properties. The *five regular polyhedra*, defined above, are also called *Platonic solids*. Neglecting the demand that the facets be congruent yields the 13 (*Archimedean polyhedra* (among them being, e.g., the cuboctahedron) plus Kepler's *infinite series of prisms and antiprisms*, together often called *semiregular convex polyhedra* (but see below for varying definitions of these and further notions). For example, replacing *vertex transitivity* (sometimes also called *isogonality*) by congruence of vertex figures (around the vertices), one gets additionally the *14th Archimedean solid*, and neglecting even the latter congruence criterion completely (but still demanding regularity and congruence of the facets) yields the class of *deltahedra*. The duals of the Archimedean solids are called *dual-Archimedean* or *Catalan solids*, and strengthening the congruence of their facets even to *facet transitivity* regarding their symmetry group, we get the so-called *isohedra*. Uniform polytopes are defined recursively: A convex polygon is uniform if it is regular. For n at least 3, a convex n-polytope is *uniform* if it is vertex-transitive and its facets are uniform. For the subfamily of *semiregular polytopes* the facets are demanded to be regular. For *regular-faced polyhedra*, all the facets are regular, and in dimension 3 the new polyhedra are also called *Johnson–Zalgaller polyhedra*. Two of the Archimedean solids are *chiral* polyhedra, meaning that their symmetry groups have two orbits on the flags such that any two adjacent flags are in distinct orbits (two flags are adjacent if they differ in precisely one element; this

can be a facet, an edge, or a vertex). These are the *snub cube* and the *snub dodecahedron*, and chirality yields, in each case, two *enantiomorphic forms*.

Neglecting convexity yields the possibility of *stellation* (i.e., by prolonging facets to new intersection points, the facets become star-shaped polygons); with the Platonic solids as starting point this yields the four *regular star-polyhedra*, also called *Kepler–Poinsot polyhedra*. In 4-space, ten regular star polytopes exist, also called the *Schläfli–Hess polytopes*. Also the following notion is interesting in polytope theory: a *regular compound* is a collection of copies of a regular polytope Q, each of which is inscribed in a fixed regular polytope P so that a subgroup of the symmetry group of P is transitive both on the vertices of P and on the copies of Q (*inscribed* means here that the vertex set of Q is a subset of the vertex set of P). Many further types of polytopes and related notions are discussed and used in the surveys [1709] and [2202].

PUBLICATIONS ON THE HISTORY OF PLATONIC AND ARCHIMEDEAN SOLIDS

The five regular convex polyhedra (see again Plate 12 at the beginning of Chapter 12) have a long history. They are called also Platonic solids since Plato mentioned them already in his *Timaeos*; but they were known already before, e.g., in megalithic time (see the historical contributions by Waterhouse [2433] and Critchlow [686], but also the critical view of Lloyd [1592] regarding related discoveries in Neolithic Scotland). One important step in Euclid's *Elements* is given by the constructions of the Platonic solids in Book XIII. Here we mention Artmann's book [131], directly related to Euclid's Elements and giving in Chapters 29 and 30 a nice overview and interesting comments about what is collected in Book XIII treating regular polyhedra and related notions (e.g., also the *deltahedra* are mentioned; see here below). Artmann discusses also applications and occurrences of these solids in nature, art, and philosophy. He nicely presents the construction of the Platonic dodecahedron and discusses also Greek architecture, emphasizing the importance of proportion in Greek culture, and continues with archaeological finds of objects having the shapes of dodecahedra and icosahedra. Also directly based on Euclid's Book XIII, Borceux [451] dedicates his Section 3.13 to the constructions and vertex coordinates of the Platonic solids. Luca Pacioli (1509) gave the most elegant construction of the icosahedron (cf. page 16 in Stillwell's book [2302]). Campedelli [564] presents a historical overview of contributions to the knowledge about regular polyhedra connected with certain cultures and famous names: the Egyptians and Etruscans, Pythagoras, Theaetetus, Euclid, Pappus, Fibonacci, Luca Pacioli, and Kepler. Some further references on constructions of regular polyhedra are [1594], [2450], [15], [494] (here §6.1.4), and [1198].

Langermann and Hogendijk present in [1493] material on regular polyhedra going back to works of Apollonius and Hypsicles, discovered in various copies of Euclid's Elements, like of Book XIV. An interesting historical recension of the astronomer al-Maghribī, who worked in the thirteenth century in Damascus, is nicely discussed by Hogendijk [1213]. He refers to Book XIV (appended by Hypsicles) and the anonymous Book XV (written in Arabic) of Euclid's Elements. Both books contain many interesting metrical properties of Platonic solids normed by being inscribed into the same sphere. It contains a list (in modern notation) of the 37 propositions into which it is divided. Langermann [1492] presents a related edition and translation of a corresponding Hebrew text treating the Platonic solids, translated from the Arabic and, as he believes, descending ultimately from a Greek original. In his historical paper [130], Artmann gives a brief outline of the history of Platonic solids somehow starting with Theaetetus (killed in a battle in 369 BC), but also underlining the occurrence of the pentagon dodecahedron around 2000 BC and through the Renaissance and other periods; also Klein's description of the groups which are the possible rotation groups of Platonic solids is presented.

We switch now to references more explicitly treating the history of the *Archimedean solids*.

It is well-known that Archimedes' lost work on the (semiregular or) Archimedean polyhedra is known from Pappus (see the booklet [1942] of Papadatos). In modern terms, an *Archimedean solid* (or a *semiregular polyhedron*) is a convex polyhedron in 3-space all whose faces are regular polygons and

all whose vertices form one orbit under isometric symmetries of the polyhedron. Thus, the congruence of faces (as demanded in case of Platonic solids) is neglected. All thirteen such polyhedra (except for Kepler's prisms and antiprisms) were enumerated by Archimedes, and in [1942] a brief account of Archimedes' life and works, a historical part with the text of Pappus' account, a free translation into modern Greek, and also a discussion of Euler's law on the face numbers of polyhedra and Descartes' role related to this are given. The main part of [1942] consists of the constructions and metrical properties of all Archimedean solids, including the relationship of the sides of the polygons forming the faces to the radius of the circumsphere. There are three excellent figures for each polyhedron. Going over representations due to Kepler and many others, we mention here the comprehensive modern discussion of Brückner; see pp. 132-140 in his impressive monograph [507].

Fields [871] analyses the *rediscovery of the Archimedean solids* in the period of Renaissance, connected with names like Piero della Francesca, Luca Pacioli, Leonardo da Vinci, Albrecht Dürer, Daniele Barbaro, and Johannes Kepler. He presents also a short history of these polyhedra in antiquity. Among the works studied in [871] one can find "De divina proportione" by Luca Pacioli, Dürer's famous "Underweysung", and Kepler's "Harmonice Mundi". Field's paper is somehow continued by Schreiber et al. [2190]: in Vienna a set of 40 printing blocks for woodcuts were found, exhibiting the nets of all regular and semiregular polyhedra; some of them show how to construct semiregular polyhedra from regular ones.

Malkevitch nicely summarizes the history of (special types of) polyhedra in [1687], presenting 25 basic milestones connected with names like Archimedes, Pappus, Dürer, Kepler, Descartes, Euler, Poinsot, Cauchy, Catalan, Brückner, Sommerville, Dehn, Steinitz, A. D. Alexandrov, Dantzig, Grünbaum, and others. A lot of literature on the Platonic solids and their history is also collected in Toepell's survey [2374]. Cromwell [689] gives a complete picture of Kepler's contributions to the theory of polyhedra, and also of his definitions and methods. After presenting his precursors Euclid, Pacioli, Dürer, and Jamnitzer, he discusses regular, rhombohedral, Archimedean, star-shaped, and further classes of polyhedra, like also semiregular prisms, antiprisms, and compounds. The paper refers also to Kepler's famous *Mysterium Cosmographicum* where he proposes that the distance relations between the six planets known at that time could be presented with the help of the five Platonic solids via an insphere–circumsphere system, all enclosed within a sphere presenting the orbit of Saturn. In his book [870], Field comprehensively treats the cosmological work of Kepler, especially also this Mysterium Cosmographicum. And also other polyhedral discoveries (Archimedean solids, regular star polyhedra,...) are discussed there. Somehow related to [870] and [689], we might mention here also the book [2222] of Senechal and Galiulin, where (as a historical review) further similar discoveries of certain polytope classes by the famous Russian crystallographer Fedorov are nicely and analogously collected.

Svobodová [2329] presents an interesting historical article on convex and non-convex regular polyhedra. The first section treats archeological stone finds in Scotland, Plato's philosophical meditations on polyhedra, constructions, and the first (but lost) *description by Archimedes of the 13 Archimedean solids* as well as the first still extant characterization of them by Pappus. In the second part, *Kepler's contributions* (prisms, antiprisms, and star polyhedra) are discussed, like also in Cromwell's paper [689]. And in the third chapter *Descartes' theorem* on the sum of vertex-deficit in convex polyhedra is given, like also the theorem stating the number of internal angles of polyhedra, knowing their sum and the number of faces. This implies Euler's polyhedral formula (100 years before Euler), which is also recalled; see also Federico's book [835] related to this. Poinsot's procedure of *stellation*, the role of Schläfli, and applications in nature and art are discussed, too. Further related articles on the history of Archimedean solids are [347] and [348] by Binder and Schmitt. For Kepler's contributions to the class of semiregular polyhedra we also refer to [1368].

Gurin and Zalgaller (see [1114] and [1113]) give a (historical) exposition on how the class of (additional) three-dimensional *regular-faced polyhedra* (also called *Johnson–Zalgaller polyhedra*) came to be (see Plate 11 at the beginning of Chapter 11).

An elementary way to derive the 13 Archimedean solids together with Kepler's two series of prisms and antiprisms is shown by Villarino [2403].

BOOKS OR BOOK CHAPTERS

The Platonic solids and their interesting properties and applications are discussed in *many classical books*. We will now present an appropriate choice from this set. And we give *short* descriptions since many of these books are mentioned already in other book parts.

We start with Klein's famous book [1422] which was reprinted in 1993. Many experts see Klein's lectures on the icosahedron, first published in 1884, as one of the most influential books in the twentieth century, mainly due to its connecting character regarding geometry (e.g., symmetry groups of the regular polyhedra), algebra (treating equalities of fifth degree), and (complex) analysis. The first two viewpoints form the main parts of the book; the second part contains the result that the solution of an equality of fifth degree (with complex coefficients) can be traced back to an icosahedral equality (determined by transformations which remain fixed under the icosahedral group). And this icosahedral equality can be solved with the help of hypergeometric series and elliptic integrals. The interest in Klein's book is still great, e.g., due to the study of Klein singularities, Hilbert–Blumenthal modular surfaces, the connection between Platonic solids and complex, simple Lie groups of certain types, and various further recent lines of research. Such developments impressively show how the *regular polyhedra can combine previously separated subfields of mathematics*. A nicely written introduction by Slodowy guarantees the reader an excellent approach to Klein's text. We cite here also the second edition of the book [2380] since Toth makes in it direct reference to Klein's book. Toth's section on Platonic solids contains a complete classification of finite Möbius groups in the style of Klein, and a quick and modern representation of Klein's main result from [1422] is now, in the second edition, in a single large section, and the author has chosen a geometric approach.

Our next reference here is Schoute's monograph [2189]. The whole sixth chapter is dedicated to the regular polyhedra, e.g., to projections and sections of them, lengths of diagonals, angles, coordinates, and symmetry properties; a short section on the semiregular polyhedra is also given. Hilbert and Cohn-Vossen study the Platonic solids and their projections in §14 and §23 of their famous monograph [1197], and Sommerville deals with regular polyhedra at the beginning of Chapter X of his book [2281].

Coxeter's famous book [670] is most essential here. The author treats regular polyhedra, Archimedean solids, regular polytopes, and various related topics. Compared with older books (of Brückner, Schoute, Sommerville, etc.), the author gives a more complete treatise, and he gives also results not yet published. Schläfli symbols and graph-theoretical representations are used, and the rotation groups of the regular polyhedra are determined. The kaleidoscope derivation of groups generated by reflections and their relation to the regular polyhedra are studied, and the Kepler–Poinsot polyhedra are obtained via natural extensions of faces to non-convex ones. Zonohedra and in particular the translative space-fillers are discussed, and in higher dimensions regular, quasi-regular, and semiregular polytopes are studied.

In his very popular general geometry book [673], Coxeter discusses the Platonic solids in Chapter 10; partial themes are projections, models, metrical information, and polarity.

The whole chapter V of the classical book [851] of L. Fejes Tóth is dedicated to *characterizations and extremal properties of Platonic solids among convex polyhedra*. These refer to spherical caps, approximation problems, volumes of polyhedra circum- and inscribed to spheres, isoperimetric problems, certain nets on spheres, total edge-lengths, and further concepts. The surveys [850], [885], and [886] somehow complete this essential book chapter.

The really nice book [696] of Cundy and Rollett presents, excellently illustrated, the Platonic solids, the Kepler–Poinsot polyhedra, the Archimedean, and the Catalan polyhedra. In addition, also

stellations of Archimedean polyhedra, certain compounds, and deltahedra are studied. Many metrical informations are given, and useful notions (like duality, or construction principles) are explained.

Section 32 of Lyusternik's book [1646] is mainly devoted to the *regular star polyhedra and the Archimedean solids*; also the *pseudorhombicuboctahedron* (the so-called 14th Archimedean solid) is explicitly presented.

In his second classical book [847], L. Fejes Tóth presents a survey of regular and semiregular polyhedra, and of parallelohedra in Part I. Also the regular polytopes of higher dimensions, especially in dimension four, are investigated there. And in Part II the regular polyhedra occur as *extremal polyhedra* regarding volume estimates, surface area, and edge curvature, as well as isoperimetric problems.

Roman [2057] shows the constructions (from nets) and a large variety of properties of the *Platonic, Archimedean, Catalan, and Kepler–Poinsot polyhedra*. Interesting applications in crystallography are given, too. Cadwell [559] dedicates the whole first chapter to regular polyhedra, including the Kepler–Poinsot polyhedra and the *regular compounds*.

In Section 19.1 of his famous book [1085], also Grünbaum discusses regular polyhedra and related concepts, particularly laying emphasis on *clear definitions*.

Holden's book [1224] describes and illustrates (with beautiful photographs of models) a large variety of three-dimensional geometric figures, most of them consisting of plane faces bounded by straight lines. The Platonic and Kepler–Poinsot polyhedra are among them, and also many variations obtained by procedures like *truncation, stellation, dualization, and compounding* are nicely presented.

In the books [2450] and [2452], Wenninger gives exact instructions for *constructing* 119 polyhedra (including also the uniform polyhedra, which are presented in the basic paper [682] of Coxeter, Longuet-Higgins, and Miller). In the second book, dual polyhedra (like isohedral solids) and stellated forms are comprehensively presented. For example, a careful description of the process of reciprocation with respect to a sphere is given. And in the foreword to [2450] Coxeter says: "But photographs cannot really show the models in their full splendour. The most complicated 'snub' solids are not only extremely difficult to make but also highly decorative: a perfect instance of the connection between truth and beauty."

We shortly mention the book [2448] of Wells, possibly more dedicated to crystallographers than to mathematicians. The author studies three-dimensional nets as (discrete) finite or infinite graphs, and three-dimensional polyhedra as polyhedral surfaces, again finite or infinite. He intends *uniform three-dimensional nets and uniform three-dimensional polyhedra* as generalizations of the Platonic polyhedra and regular tessellations of the plane. The book contains very many illustrative figures. And we mention also the book [15] of Adam and Wyss, collecting many information on and giving nice figures of the Platonic, Archimedean, Kepler–Poinsot and Catalan solids, like also of compounds; hints for creating models are given, and many metrical properties are reflected. Similarly, Pugh [2002] collects material on Platonic, Archimedean, Catalan, Johnson–Zalgaller, Kepler–Poinsot and further (also non-convex) solids; also Buckminster Fuller's geodesic polyhedra are presented.

In Berger's books [282] and [283], the Platonic solids and related topics (like *constructions and group-theoretical aspects*) are studied in Section 1.8 and also in Sections 12.5.4 and 12.5.5.

For the year 1991, the beautiful book [1349] of Kappraff was chosen by the National Association of Publishers (USA) as the best book in "Mathematics, chemistry, and astronomy—professional and reference". It is an introduction into design science which the author describes by citing A. Loeb: "Just as the grammar of music consists of harmony, counterpoint, and form which describes the structure of a composition, so spatial structures, whether crystalline, architectural, or choreographic, have their grammar which consists of such parameters as symmetry, proportion, connectivity, stability, etc.". These notions are obviously close to geometry, and the author sees the book as a collection of special topics in ancient and modern geometry. Close to our viewpoint here are the parts describing the planetary system of Kepler, special simplices called Schläfli's orthoschemes, transformations of the Platonic solids, Archimedean solids, Buckminsterfullerene, translative space-fillers, and golden

isozonohedra. This is an inspiring book not only for *geometers*, but also for *architects, designers, engineers, natural scientists*, and *computer scientists*.

Böhm and Quaisser [382] study *Platonic solids, Kepler–Poinsot polyhedra, Archimedean, and Catalan solids*, and *regular polytopes* in higher dimensions. Various properties (allowing different definitions) are discussed, like also symmetry groups and applications in crystallography.

In Cromwell's book [690], regular polyhedra and regular star-polyhedra are (also historically) treated, and it is said that three out of these four Kepler–Poinsot polyhedra were anticipated by others than Kepler and Poinsot. Archimedes had already found the semiregular polyhedra named after him; it is shown how all these figures attracted Renaissance artists, who were in great parts also familiar with the necessary mathematical knowledge. It is underlined that Kepler, while showing that planetary orbits are ellipses, still attempted to relate the sizes of these orbits to the regular solids. It is also shown that Cauchy established completeness of the classification of the four regular star-polyhedra.

Hartshorne discusses *definitions, constructions, and various properties of the Platonic solids* in Chapter 8 of his book [1167]. A representation of the semiregular polyhedra (including the pseudorhombicuboctahedron and Kepler's prisms and antiprisms) and constructions of *regular-faced polyhedra* follow (see again Plate 11 at the beginning of Chapter 11 below), like also a discussion of related symmetry groups.

The first part of Rajwade's book [2019] is dedicated to convex 3-polytopes with the following regularity properties: (A) All faces are regular polygons; (B) all faces are congruent; (C) all vertex figures are regular; (D) all solid angles of P are congruent. The author discusses all possible combinations and the relations between these regularity criteria. If, for example, (A) and (C), then all four conditions are satisfied. This leads him to a hierarchical classification of polytope classes in 3-space, yielding the Platonic solids, the regular prisms and antiprisms, the Archimedean polyhedra, the deltahedra (different from those which are Platonic), and the 92 *regular-faced polyhedra* of Johnson and Zalgaller. Also the regular star-polyhedra are briefly discussed, together with an application referring to planar tilings. It should be noted that the classification in the survey [1709], referring also to corresponding classes of n-dimensional polytopes, somehow generalizes the concept of Rajwade. There one can find many more related references. (The second part of [2019] refers to Hilbert's third problem.)

Tapp [2344] nicely studies the notion of symmetry on a broad fundament. Chapter 8 leads to the Platonic solids, which are discussed quite into detail, also using graphs and the Euler characteristic. Geometric figures having Archimedean solids as background are also presented.

In his very instructive mathematical book [153], Aumann nicely reaches his goal to explain Archimedes' mathematical thinking to a broader audience. He discusses mathematical achievements with regard to the area of a parabolic segment, the position of its center of mass, the cylinder, the cone, the sphere, Archimedean spirals, results on the volume and the center of mass of a paraboloid of revolution, and floating bodies. The last chapter is completely dedicated to the *Platonic and Archimedean solids*.

The next two references refer to *paperfolding*. Gurkewitz and Arnstein [1115] treat the relationship between origami, Archimedean solids (with the duality principle), and the buckyball (the famous carbon-60 molecule). And the second chapter of Friedman's comprehensive monograph [903] on the *history of paperfolding* (mainly in Europe, from Dürer's contribution in the sixteenth century until today) contains material about Platonic and Archimedean solids. Namely, Dürer used arrangements of edge-joined polygons on a two-dimensional surface that could be folded along their edges to get such polyhedra. Also the related contribution of Pacioli is acknowledged.

SURVEYS

A broad survey on Platonic, Archimedean, Catalan (or dual-Archimedean) polyhedra, related compounds, and stellations is given by Aschkinuse [141]. For example, it is one of the rare references

where one can see figures of all representatives of the first three mentioned types together in one publication. The survey starts also with many theorems and observations from the general theory of (convex) polyhedra.

Kajikawa [1337] surveys relations between *quasi-crystals, Penrose tilings, regular and semiregular polyhedra* in 3-space, in particular treating the isocahedron and dodecahedron. For example, he divides the icosahedron into ten pieces (called modules), and he continues by recombining such modules, thus getting several types of new polyhedra and of space-filling patterns. The close relation of such structures to crystallography is discussed, too.

The authors of [1711], [1709], and [2202] survey the knowledge on various classes of (mainly convex) n-dimensional polytopes satisfying certain degrees of regularity. For example, these are regular, semiregular and uniform polytopes, regular-faced polytopes, congruent-faced polytopes, regular star-polytopes, compounds of regular polytopes, pathways on the boundaries of regular polytopes (geodesics, Klee–Minty cube,...), regular polytopes in metrical convex geometry, and polytope classes obtained via combinations of the above criteria. In the survey [1711], more interesting research results are mentioned, and in [1709] and [2202] the hierarchical structure of these polytope classes is more discussed, in the second one emphasizing also the symmetry groups. Everywhere the three-dimensional situation is also treated, such that the presented family of classes contains also Platonic, Archimedean, Catalan, regular-faced, isohedral, and further polyhedra.

In the survey [1204] van Hoboken shows relations between classical objects in different areas of mathematics having the underlying structure of simply laced Dynkin diagrams. Thus it is shown where in mathematics Kleinian singularities occur. The author discusses Platonic solids, binary polyhedral groups, Kleinian singularities, and simply laced Lie algebras. In the final part also Slodowy's book [2261] is taken as fundament.

Schulte [2202] presents a comprehensive and clearly structured *survey on many classes of polytopes with regularity properties*. He classifies (clearly not only convex) polytopes that are regular, semiregular, uniform, complex regular, and abstract regular. Also regular star-polytopes are presented. For these and more classes he presents related basic notions, the symmetry groups, enumerations, constructions, and interesting properties.

Having in mind the *skeletal approach to polyhedral structures*, Schulte and Weiss [2203] give a nice survey on types of polyhedra in Euclidean space that have regularity properties like the classical Platonic solids, but which need no longer be convex (see also [2202]). Therefore also the star-polyhedra of Kepler and Poinsot and the infinite regular skew-polyhedra of Petrie and Coxeter are studied. The authors are inspired by Grünbaum's article [1084] (e.g., with notions like the well-defined Grünbaum–Dress polyhedra). A lot of history is discussed, like also modern concepts (mainly in view of the important book [1755] of McMullen and Schulte).

SELECTED RESEARCH PAPERS ON REGULAR POLYHEDRA

We continue with selected, more recent research papers.

The research papers to be mentioned first refer to *definitions* of regular and related types of polyhedra. In this exceptional case (treating definitions) we go a little more back. An early paper is from 1945: Jensen [1302] obtains twelve regular polyhedra in 3-space via the following consideration: Each of the regular star-polyhedra has the same vertices as a Platonic solid, its "case", and the same face-planes as a Platonic solid, its "core". Continuing with polyhedra which can be discontinuous, but still have a Platonic case and core, three additional compounds are obtained, namely Kepler's *stella octangula* and a pair of *enantiomorphous* figures, each built from five tetrahedra. In his book XIII, Euclid asserts that "no other figure, besides the said five [Platonic] figures, can be constructed which is obtained by equilateral and equiangular figures equal to one another." This definition is incomplete since for triangular faces additional polyhedra enrich the described class; and this error can be found again and again in various school books. Freudenthal and van der Waerden [902] (see also [1086]) derived all five additional convex polyhedra (following that definition besides the Platonic solids) and

called the whole class the class of *deltahedra*. The most simple non-Platonic representative is the double pyramid formed by the union of two regular, congruent tetrahedra glued together face by face. And also the pentagonal double pyramid belongs to this family. Completing this somehow, Cundy [695] presents 17 non-convex deltahedra, any of them having two types of vertices, and with faces not crossing each other.

van Kol [1440] shows criticism against the following definition: *a polyhedron is regular if it has congruent regular faces and the same face number at every vertex*. Namely, this definition allows a non-convex icosahedron. He says that adding the word *convex* would validate that definition, due to a theorem of Cauchy on congruence or enantiomorphy via dihedral angles. Grünbaum [1086] discusses an enduring error regarding the definition of Archimedean solids and uniform polyhedra, thus also in relation to the 14th Archimedean solid. First he gives a *local criterion*: All faces are regular polygons, and the cyclic arrangement of the faces around each vertex is the same. He then gives a *global criterion*, namely: All faces are regular polygons, and all vertices form one orbit under isometric symmetries of the polyhedron. Since in the time of Archimedes group theory was not known, he proposes to call the polyhedra defined with the help of the local condition *Archimedean solids*, and those satisfying the global condition *uniform or semiregular polyhedra*. The *14th Archimedean solid*, namely the non-uniform pseudorhombicuboctahedron, shows clearly that this distinction is essential. There are further informations in [1086] about enduring errors. Itoh and Ohtsuka [1281] extend the definition of convex regular polyhedra using criteria like the following one: any two vertices have isometric neighborhoods. The main result is then a classification of simple regular polyhedral BP-complexes, being two-dimensional homogeneous, locally finite cell complexes equipped with piecewise linear metrics satisfying some further conditions; the boundaries of Platonic solids are special cases. For many more examples of polytope classes obtained from the Platonic solids and their higher dimensional analogues by modifying definitions we refer to the surveys [1709] of Martini and [2202] of Schulte.

Regarding *geodesics* on the boundaries of convex polytopes we cite the three papers [663],[906] and [2382]. In the first paper, the authors characterize and describe the solutions to the isoperimetric problem for some (classical) piecewise smooth manifolds, and in particular for the three simple regular polyhedra. For this they use a classification of constant geodesic curvature curves. Based on this paper, the authors of the second article show, e.g., the following: Any geodesic on a *regular tetrahedron* is non-self-intersecting, and there are closed geodesics of arbitrarily large length. On the *unit cube*, there are exactly three non-self-intersecting, closed geodesics, and two are planar. For the *regular octahedron*, a complete description of non-self-intersecting and self-intersecting closed geodesics is given, and for the regular icosahedron a partial description of all closed geodesics and a complete description of non-self-intersecting ones is obtained. The authors do not cite the older paper [1989] of Post in which (among other nice results) the following statement is proved: A bounded closed convex polyhedron is a regular tetrahedron if and only if it contains an infinite, non-self-intersecting geodesic line. We come to the paper [2382] of Troubetzkoy: he shows that for the cube and the Platonic tetrahedra, octahedra, and icosahedra there are no geodesics from a vertex to itself.

There are several interesting geometric problems which can be obtained by modifying or generalizing the question of *holding convex polyhedra by circular rings* (i.e., to attach a hinged circular ring of suitable size such that it cannot slip out of the polyhedron or convex body in 3-space). In [1664] Maehara and Martini survey (till 2017) results coming from this and also from more general questions, such as netting arbitrary convex bodies by strings, or caging them by polyhedra. In addition they develop a unified concept and new methods to establish a systematic representation of the existing results. This way, they derive also new related results. More recent contributions in this direction are due to Fruchard, Yuan, Zamfirescu, and Zhang (see [905], [2512], and [2513]).

Now we mention results on *embeddings and deformations of convex polyhedral surfaces* in 3-space. Bleecker [362] considers such surfaces whose dihedral angles are strictly smaller than π and verifies that there is a small isometric deformation of them to a nearby polyhedral surface (necessarily non-

convex) having strictly larger volume. He shows a similar result for the boundaries of the Platonic solids and gives estimates for the gain of volume. Shtogrin [2241] applies results of Pogorelov about special isometric transformations for cylindrical surfaces that the boundaries of the Platonic solids admit piecewise smooth bendings; continuing this, he confirms in [2240] that the surfaces of the Platonic solids admit piecewise smooth isometric embeddings without flat pieces.

In view of *algebraic geometry* and related topics, relationships between Platonic solids and Lie groups via Kleinian singularities are discussed by Slodowy, in his expository paper [2262] (see also [1204]). First he gives also a historical overview, and at the end also later developments are shown. In Lamotke's book [1491] on singularity theory, fundamental results from commutative algebra and differential topology, on several complex variables and on regular polyhedra with applications to isolated singularities are presented. In the first chapter, the general concepts of convex polytopes and their regularity are introduced, and the connections between the regular polyhedra and the finite groups of rotations are clearly established; also the classification of the finite subgroups of SO(3) is presented. Battaglia and Prato [254] describe symplectic and complex toric spaces associated with the five Platonic solids. It turns out that the tetrahedron and the cube are rational and simple and that the regular octahedron is not simple. Further on, the Platonic dodecahedron is not rational, and the icosahedron is neither simple nor rational.

There are also very many results characterizing regular polyhedra as *extremal convex bodies or polyhedra*, namely in the spirit of *geometric inequalities*. Most of the results in this spirit are due to L. Fejes Tóth and the mathematicians from the school that he created. Above we mentioned already Chapter V of the book [851] and Part II of [847], like also the surveys [850], [885], and [886]. We add here Fejes Tóth's expository papers [845], [846], and [849], the latter dedicated to Coxeter presenting also results due to him. These papers refer also to regular star-polyhedra. The type of questions and results presented there refers, e.g., to maximum volume for a given surface and a given number of facets, or to inequalities taking also quantities like total edge-length, inradii, and circumradii into consideration.

Breda d'Azevedo and Jones [498] investigate *hypermaps on surfaces* which are called regular if their full automorphism group, including both orientation-preserving and orientation-reversing automorphisms, acts transitively on the set of hyperedge-vertex-face incidence triples. In particular, regular hypermaps are classified whose full automorphism group is isomorphic to the symmetry group of a Platonic solid. Also the Kepler–Poinsot polyhedra play a role in this framework.

We continue with some results in the spirit of *discrete geometry*. Schmutz Schaller [2139] presents results on polygonal tilings of the sphere and in hyperbolic geometry. Within this framework he gives two examples of classical mathematics showing that their Euclidean variant has a hyperbolic analogue with a richer structure and more beauty. The chosen examples refer to generalizations of the Platonic solids and to lattice sphere-packing problems. Let A be a well-defined family of polyhedra. A set B of polyhedra is called an *element set* for A if every polyhedron in A is the union of a finite number of polyhedra from B (identified along facets); each polyhedron of the element set B is said to be an element of A. Akiyama et al. [28] determine one element set for A being the family of regular polyhedra in 3-space, and they show that this element set is best possible. Namely, it achieves the minimum in terms of cardinality among all the element sets for A. A related indecomposability notion is introduced, too. In [30] Akiyama and Sato extend the concept of element sets to higher dimensions. In particular, they show that in 4-space the element number of the six regular polytopes there is at least 2, and that in n-space for $n > 4$ the element number of the family of all regular n-polytopes there is 3, unless $n + 1$ is a square number. Wassell and Benito [2432] emphasize that the following number has so far remained undiscovered: the edge-length ratio between a Platonic dodecahedron and its dual and vertex-to-face circumscribed icosahedron, namely one third of the golden ratio. They underline that this ratio completes an interesting triplet of ratios for vertex-to-face dual pairings when the outer regular polyhedron is the tetrahedron, octahedron, and icosahedron: $1 : 3$, $\sqrt{2} : 3$, and this new third of the golden ratio. The author of [1999] deals with the development of solvable examples for discrete

symmetries for a particle in 3-space, and in particular for a one-particle Hamiltonian with interactions centered at the vertices of the five regular polyhedra.

Düvelmeyer [772] tries to confirm a conjecture of Swanepoel: for a convex 3-polytope, centred at the origin, let $T(P)$ denote the total edge-length measured in the norm that has P as polyhedral unit ball. Is the minimum of $T(P)$ for all polyhedral unit balls reached by the Platonic dodecahedron? Perhaps confirming this conjecture, in [772] many examples are calculated, e.g., from the following classes: zonohedra, regular and Archimedean solids, and Catalan solids.

Baez [179] describes two constructions of the E_8-lattice from the icosahedron, one via a subring of the quaternions, and the other via Klein singularities. Mathematical notions coming here together are the golden ratio, quaternions, the quintic equation, the 600-cell, and the Poincaré 3-sphere.

Many further information and results (of very different types) about Platonic and Archimedean solids are collected in the two surveys [1711] (see here in particular the subsections 1.1, 1.4, 1.5 and the parts of Chapter 2 which refer to three dimensions) and [1709] (Sections 4 and 5).

SELECTED RESEARCH PAPERS ON ARCHIMEDEAN SOLIDS

For the classification of all *convex* polyhedra whose symmetry group acts transitively on the vertices we refer to [2040]. A three-dimensional polyhedron is called *uniform* if its faces are regular (and not necessarily convex) polygons and its symmetry group is transitive on its vertices. Coxeter et al. [682] obtain, following that definition, 53 representatives; among them are of course the Archimedean solids. Sopov [2282] and Skilling [2259] show that this list is complete.

We start with some *non-mathematical applications*. In the nicely illustrated article [508], Brun studies *shapes of viruses* constructed with the help of Platonic and Archimedean solids. Starting with the observation that the unit spheres around the vertices of such polyhedra having edge-length 2 touch each other at the edge-midpoints, he continues with smaller balls creating rows along the edges, and then even fills the faces with further balls. So he gets (as an extreme example) an almost spherical cluster of 1472 balls having the shape of a virus. And he also investigates a packing of balls (of different sizes) being a stereographic projection of the family of inspheres of the cells of the regular 120-cell in 4-space (see also p. 19 of the book [2448]). Duggan and Firby [773] give upper bounds on the *electrical capacitance of the regular and semiregular polyhedra* in 3-space. Lowest energy geometries for *atomic clusters* are studied in [230], and the authors propose a new structure for the 38-atom cluster; it has the shape of the Archimedean tetrakaidecahedron. Bérczi [275] studies Platonic and Archimedean polyhedra and related tessellations from the viewpoint of *technology*. Using truncation procedures, he summarizes the relations between these structures and higher dimensional analogues. Coming from *chemistry*, Balaban and Bonchev [184] consider an ordering of Platonic and Archimedean polyhedra due to *numerical indicators*. For example, certain types of angles and measured sphericity are collected and compared to get some *complexity hierarchy*. The authors state that this is possibly the first such ordering of the Archimedean solids. This paper is completed by [2411], where solid angles of the two snub Archimedean polyhedra have been calculated and integrated into the whole list given for the Platonic and Archimedean polyhedra. Similarly, Rajtmajer et al. [2018] study the complexity of the Archimedean polyhedra via the complexity index based on the augmented vertex-degree and the total number of walks, trails, and paths of their Schlegel graphs. It turns out that the rhombicuboctahedron, the icosidodecahedron, the rhombicosidodecahedron, the snub cuboctahedron, and the snub icosidodecahedron yield the most complex structures. The so-called Hasse diagram is used to illustrate this. Regarding *dipole interactions*, Batle, Shutov, and Maleev [253] study systems of particles suitably presenting the Platonic, Archimedean, and Catalan solids. They emphasize that this was not done before.

In a rarely cited series of papers Hohenberg studies metrical and projective properties as well as generalizations of the *snub cube and the snub dodecahedron* (see [1215], [1217], [1218], [1219], [1216], [1220], and [1221]). For similar extensions of (chiral) Archimedean solids we refer to Bilinski

[342], where further related papers are cited. Weissbach and Martini [2446] show that the snub cube and snub dodecahedron are characterized among all Archimedean solids by the following property: they have edge-lengths which are *not constructible* (in the sense of straightedge-and-compass constructions) from their circumradii. On the other hand, the vertices of these two Archimedean solids can be constructed via *paper folding* on the faces of a cube and dodecahedron, respectively. This is shown by Hartl and Kwickert [1165]; they present explicit folding constructions and show also the construction rules for the other Archimedean polyhedra.

We come now to selected *graph-theoretic results*. In the paper [1660] of Maehara the *sphericity of a graph G* is defined as the smallest integer n for which the nodes of G can be embedded in Euclidean n-space such that the Euclidean distance of a pair of nodes is smaller than 1 if and only if the nodes are adjacent in G. With a few exceptions, he presents the sphericity of the graphs of semiregular polyhedra in 3-space. Gordon [1004] shows how an arrangement of interlocking gears yields related *planar bipartite regular and biregular graphs*. Their parameters are determined, and their relations to the Archimedean polyhedra are discussed. Jendrol' and Schlötter [1298] consider the minimal number k such that each *coloring of the vertex set* of a convex 3-polytope, with at least k colors, yields a face all whose vertices have different colors. This number is determined for the Platonic solids, prisms, antiprisms, and ten Archimedean solids. For the other three Archimedean polyhedra estimates are given. In the dual graph of an (Archimedean) polyhedron P (i.e., a graph where each vertex presents a facet of P) two vertices are adjacent if the corresponding faces have an edge in common, and a *Hamiltonian cycle* on an Archimedean solid is defined as a Hamiltonian cycle in this dual graph. Ehrenborg [805] shows that a Hamiltonian cycle on an Archimedean solid is free of twisting, meaning that when the cycle returns to its starting facet, then this facet has the same orientation that it had at the beginning.

The authors of [891] continue and complete their work [890] examining the *design spectra* for the vertex-edge graphs of Archimedean polyhedra, ending also with first explicit examples of designs of the snub dodecahedron. Kovič [1462] looks at all centrally symmetric polyhedra among the convex and regular-faced (or Johnson–Zalgaller) 3-polytopes; these are four Platonic solids, nine Archimedean solids, thirteen Johnson–Zalgaller solids, and representatives from the two infinite families of prisms and antiprisms. He studies their projective images, with special emphasis on the structure of their flag graphs and dependencies among the vectors recording the number of faces of each type. Damasco and Frettlöh [704] study *perfect colorings of graphs*, e.g., determining the lists of all color adjacency matrices of perfect colorings with two, three, or four colors for k-regular connected graphs, $k = 2, 3, 4$. Based on this, they determine all perfect colorings of the edge graphs of the Platonic solids with two, three, or four colors; some open problems are raised.

We continue with results in the spirit of *discrete geometry*. A *dissection* of an n-polytope into simplices is called *mismatching* if not any pair of the participating simplices intersects at a common face. Besides various results (also for n-polytopes), De Loera et al. [1599] obtain for $n = 3$ precise sizes of mismatching dissections (in terms of the vertex numbers) for special types polyhedra, like also prisms, antiprisms, Platonic and Archimedean solids. The authors point out that such results are related to classical fields like Hilbert bases, algebraic geometry, and integer programming. Subbotin [2321] defines a convex polyhedron P to be *class I strongly symmetric* if the perpendicular bisecting plane of each edge is the mirror of a reflection symmetry of P, and to be *class II strongly symmetric* if the bisecting plane of each plane angle is also such a mirror regarding P. He succeeds in giving a complete classification of these two families. The first class contains the Platonic solids, 11 Archimedean polyhedra, an infinite series of prisms, and several polyhedra with congruent semiregular faces (i.e., these faces have an even number of congruent edges and angles alternating between two congruence classes). The second class contains only the five Platonic solids, the cuboctahedron, the icosidodecahedron, and two Catalan solids: the rhombic dodecahedron and the rhombic triacontahedron. With similar results, Subbotin investigates in [2322] convex polyhedra that are strongly symmetric with respect to rotation of the faces. The concept of Subbotin is strongly related to the following notion investigated

by Harazishvili [1159]: a point set in Euclidean n-space with the property that each hyperplane of symmetry for any two of its points is a hyperplane of symmetry for the whole set is called *perfectly symmetric* (see also Theorem I.1.3 in the book [1137]). In [1159] all perfectly symmetric sets are classified.

Embedded in more general results, Torquato and Jiao [2379] get an upper bound on the maximal *packing density* of nonspherical particles, and they apply this to Platonic and Archimedean solids like also ellipsoids. When the ratio of the circumradius to the inradius is not too large, the upper bounds are relatively tight and thus close to the corresponding densities of the *optimal lattice packings of the centrally symmetric Platonic and Archimedean solids*. The conjecture is posed that the densest packings of Platonic and Archimedean solids with central symmetry are given by their corresponding densest lattice packings (an analogue of Kepler's famous conjecture for these convex bodies; of course, the truncated tetrahedron has to be considered separately). Applications in crystallography are also discussed.

Demaine et al. [727] investigate *unfoldings of convex polyhedra*. They show that any convex polyhedron P may be unfolded to one planar piece, and then refolded to a different convex polyhedron. Assuming that the unfolding allows to cut only edges of P, several polyhedra thus being *edge-refold rigid* (i.e., all their unfoldings lead back to the same original) can be identified. The authors show that 11 of the 13 Archimedean polyhedra are edge-refold rigid. Also Horiyama and Shoji [1230] deal with unfoldings (by cutting edges), asking for the *number of non-isomorphic unfoldings* for arbitrary (i.e., also non-convex) polyhedra. In particular, they describe the non-isomorphic unfoldings of all regular-faced convex polyhedra (including also the Platonic and Archimedean solids, the semiregular prisms and antiprisms), Catalan solids, bipyramides, and trapezohedra. The number of non-isomorphic unfoldings are unexpectedly high.

Now we come to *group-theoretical results*. Koca et al. [1427] deal with the determination of the regular and semiregular polytopes via *Coxeter–Dynkin diagrams*, more precisely with getting the vertices of the Platonic and Archimedean solids with the help of the corresponding Coxeter diagrams, in the context of the quaternionic representations of the root systems and the *Coxeter–Weyl groups*. They show that the polyhedra possessing the tetrahedral, octahedral, and icosahedral symmetries are related to the Coxeter–Weyl groups representing the symmetries of the mentioned diagrams, and thus yield the vertex sets of all Platonic and Archimedean solids, except for the two chiral polyhedra. In [1430], a new approach, also to construct the chiral Archimedean solids and their duals, is presented. Fritz and Hauser [904] use the classical idea of *constructing algebraic surfaces with many singularities via finite group actions* to create symmetric surfaces. Taking Platonic and Archimedean polyhedra as corresponding starting points, they call the corresponding surfaces *Platonic stars*. Explicit equations determining them are given, and beautiful constructions of such stars are presented, together with many geometric properties that they have.

Aravind [121] studies the *degree of sphericity of convex bodies* in the following sense. He obtains closed form expressions for the surface area and volume of any Archimedean solid in terms of the integers describing the type and number of regular polygons ordered around every vertex. The Catalan solids are studied analogously. Nine of these polyhedra have larger sphericity than the truncated icosahedron (famous as buckyball).

The authors of [1680] derive a class of *Archimedean solids in hyperbolic space*. Using also projection methods, they are constructed from semiregular hyperbolic plane tilings in a way as also semiregular polyhedra in Euclidean space could be constructed with the help of semiregular spherical tilings. The paper contains interesting techniques for constructing semiregular hyperbolic tilings, in parts from the regular ones, and partly also using Fuchsian groups. Many figures nicely illustrate the results.

The following result is known for more than three centuries: one can create a suitable tunnel in a cube such that another cube of the same size can slide through; thus the cube has the *Prince Rupert's property*; see B4 in the problem book [687] of Croft, Falconer, and Guy. It is clear that one can ask which other types of convex bodies have this property. For instance, it is obvious that bodies of constant

width cannot have it, but for regular polytopes, or semiregular ones, this is an interesting question. More than fifty years ago, Scriba showed that the Platonic tetrahedron and octahedron have this property. Jerrard, Wetzel, and Yuan [1313] verify that the regular dodecahedron and icosahedron also have this property, and an enlarging ratio is given for all these polyhedra such that even the correspondingly larger homothet is still functioning. Chai, Yuan, and Zamfirescu [589] confirm that eight of the thirteen Archimedean solids have the Rupert property, giving also an analogous scaling constant, and they pose the conjecture (see also [1313]) that all convex polytopes have Rupert's property. The Archimedean truncated tetrahedron is added to this list by Lavau [1519] and by Hoffmann [1211].

A map on a surface is called *vertex-transitive* if its automorphism group acts transitively on the set of its vertices, and it is called *semi-equivelar* if the cyclic arrangement of faces around each vertex is the same. Vertex-transitive maps are semi-equivelar, the converse is not true. Datta and Maity [712] classify all *semi-equivelar spherical maps* showing that these are precisely the boundaries of Platonic solids, Archimedean solids, regular prisms and antiprisms, and the boundary of the pseudorhombicuboctahedron. From this they conclude that all of them, except the pseudorhombicuboctahedron, are vertex-transitive.

In the paper [987], Göring and Winter investigate mainly convex 3-polytopes whose symmetry group is *edge-transitive*, but not vertex-transitive. They show that these are certain $2k$-gons in the plane and in dimension 3 two dual-Archimedean polyhedra, namely the rhombic dodecahedron and rhombic triacontahedron. Certain proof parts involve classical arguments and techniques also used for the classification of Archimedean solids.

2.9.1.4 Regular, semiregular, and uniform polytopes

We will now discuss the higher dimensional analogues of Platonic and Archimedean polyhedra; i.e., from now on the dimension of our space is at least four. To stay in close connection with convexity, we restrict ourselves to *convex* polytopes of these types. Further classes (like non-convex ones, or abstract regular polytopes) we largely ignore, or mention them only very shortly. Two possible definitions of regular polytopes are those: A convex n-polytope is regular if it is vertex-transitive and has regular, congruent facets, or if its symmetry group acts transitively on its flags (which are sequences of faces, each contained in the next with exactly one face from every dimension), thus giving it the highest degree of symmetry.

One of the nicest discoveries of classical geometry was made by Schläfli in the middle of the nineteenth century (see his book [2135], which appeared later, and the nice article [1357] by Kellerhals). Suitably analogizing the definitions of regular polygons ($n = 2$) and regular polyhedra ($n = 3$), he found the complete family of *regular polytopes* in Euclidean n-space for $n \geq 4$. These are the analogues of the regular tetrahedron, octahedron and cube in any dimension $n \geq 4$ (called *regular simplex, regular cross-polytope, and hypercube*, respectively), and three very interesting regular 4-polytopes: Schläfli's *24-cell* (bounded by 24 regular octahedra, being centrally symmetric and nevertheless self-dual), the *120-cell* (bounded by 120 Platonic dodecahedra, having 600 vertices and seen as analogue of the pentagon dodecahedron), and the *600-cell* (bounded by 600 regular tetrahedra, having 120 vertices and presenting the analogue of the regular icosahedron).

VISUALIZATIONS, BOOKS AND BOOK CHAPTERS, AND SURVEYS

Equivalent definitions, necessary conditions, different methods of construction, and several (also analytical) approaches can be found in the books or expository articles of Schoute (§7 and §8 of [2189]), Sommerville (Chapter X of [2281]), Coxeter (§7 of [670] and §22 of [673]), L. Fejes Tóth (Section I.5 of [847]), Franz [901], Rosenfeld and Yaglom [2059], Berger (Chapter 12 of [283] and Sections VIII.11 and VIII.12 of [286]), Banchoff (Chapters 5 and 6 of [207]) as well as Böhm and Quaisser

(Section 3.1 of [382]). An important historical representation to be mentioned here is Coxeter's article [672] on research related to polytopes in the Netherlands. For more modern developments, e.g., on abstract regular polytopes, the basic books [1755] of McMullen and Schulte (on *abstract regular polytopes*) and [1751] of McMullen are trend-setting.

Interesting *figures or visualizations* of these polytopes (e.g., Schlegel diagrams, other types of projections, or nets) are given by Schoute (§7 of [2189]), Hilbert and Cohn-Vossen (§32 of [1197]), Coxeter (Plates I–VIII in [670] and §22 of [673]), Franz [901], Berger (Chapter 12 of [283]), and Banchoff (again Chapters 5 and 6 of [207]). Note that Plate 3 (at the beginning of our Chapter 3) shows Schlegel diagrams of the 4-cube and of a truncated 4-simplex. An inspiring refinement of the technique of Schlegel diagrams (of the 24-cell) due to Banchoff is presented in [206], and the booklet [1593] of Lo Jacomo is particularly interesting regarding visualization of the six four-dimensional regular polytopes. For example, in nice drawings the vertex set of Schläfli's 24-cell is arranged as a union of the vertex sets of three cross-polytopes, and for the vertex set of the 600-cell first an appropriate point on each edge of the 24-cell is chosen, and then 24 points are suitably added. Also the relations of the vertex sets of the regular 4-polytopes to finite groups of quaternions and the notion of duality in this setting are studied. In addition, Stillwell's exposition [2303] contains wonderful figures.

As a collection of excellent articles, we mention also the book [676]. It contains 26 papers of Coxeter, including a previously unpublished article on the regular 24-cell. Viewpoints and themes that are emphasized in this collection are Coxeter's *complete enumeration and description of the finite reflection groups*, the *groups of unitary reflections* and their relevance for the classification of the regular complex polytopes, *Coxeter diagrams* and their applications in geometry and algebra, Coxeter's contributions to *sphere packings and lattices*, and his research on *semiregular polyhedra and polytopes*. The collection of articles turned out surprisingly well, and the editors added a biographical sketch (six pages), a list of Coxeter's publications (214 items), a list of Coxeter's Ph.D. students, and an index.

Here also the three surveys [1709], [1711], and [2202] of Martini and Schulte should once more be mentioned. They present much material about properties of regular, semiregular and uniform polytopes, and also on related polytope classes obtained by weakening related definitions. Such polytope classes are called *regular-faced* or *congruent-faced* ones (meaning convex polytopes all whose facets are regular or congruent), *isogonal* (= vertex-transitive), *uniform (= vertex-transitive with uniform facets*, the latter recursively defined), *isohedral* (= facet-transitive) polytopes, and various further families. Stillwell [2303] presents a wonderful exposition on the last 170 years research with respect to regular polytopes in the context of algebra, geometry, topology, and also computer graphics. He starts with Hamilton and his *quaternions*, goes on with Cayley's *lifting of the rotation groups* of the Platonic solids to the quaternions. Schläfli and the discovery of the higher dimensional regular polytopes in the mid-1800s are a next point, and the connection of the three complicated regular 4-polytopes with the rotation groups of the regular 3-polytopes via quaternions is also shown. Further highlights are combined with topology: Poincaré showed that his homology sphere has a non-trivial fundamental group since it maps onto the *icosahedral group*; and thirty years later Seifert and Threlfall confirmed that the Poincaré sphere is representable as a projection of the 120-cell. As already said above, this survey is illustrated in a wonderful way.

RESEARCH PAPERS ON CONVEX REGULAR POLYTOPES

Emert and Nelson [811] show that for a regular n-polytope the *surface area is a derivative of the volume with respect to the inner radius* (with the latter meaning the minimal distance from the centroid of the polytope to its boundary). Further on, conditions are given which have to be satisfied by arbitrary convex polytopes to ensure the same relation.

Buekenhout and Parker [524] classify the *nets* (also called *unfoldings*) of the regular convex polytopes of dimension at most 4 under the isometry group of the polytope. This is equivalent to

classifying the spanning trees of the facet-adjacency graph under its automorphism group. The authors do this for all such polytypes, and in the case of the four-dimensional 600-cell, the number of (in this sense different) unfoldings is $2^{188} \cdot 3^{102} \cdot 5^{20} \cdot 7^{36} \cdot 11^{48} \cdot 23^{48} \cdot 29^{30}$, possibly one of the largest *specific* numbers in the mathematical literature.

Based on Coxeter's theory of regular polytopes (see [670]), Lucić [1609] concentrates on *all symmetries of the six four-dimensional regular polytopes*. As an example: he shows in a detailed manner that the symmetry group of the 120-cell consists of 1138 rotations, 6061 double rotations, 60 reflections, 7140 rotatory reflections, and the identity. Miyazaki [1815] presents the *stellations of the 120-cell* yielding three of the ten *Schläfli–Hess star-polytopes* (so the *regular star polytopes* in 4-space are usually called). Via stellation of the 24-cell, a four-dimensional compound being the analogue of Kepler's stella octangula is obtained. Dealing with *densest ball packings in spherical 3-space*, Coxeter [674] uses central projections of inspheres of facets of the 120-cell onto the concentric unit 3-sphere S. Reducing radii suitably, he succeeds with inserting 720 additional balls such that finally 840 balls in S create a packing of density 0.68865...

Regular polytopes can also be nicely combined with arrangements of hyperplanes. A *Grünbaum arrangement on a regular polytope P* in Euclidean n-space is a set of hyperplanes in the respective projective augmentation which contains the facet hyperplanes of P and also some of its hyperplanes of mirror symmetry, and perhaps the hyperplane at infinity. Markert and Wetzel [1695] deal with the question when such arrangements are simplicial. In view of a complete solution, they confirm that no Grünbaum arrangement on the 120-cell and on the 600-cell is simplicial. Wetzel [2456] gives a wider overview on this (meanwhile completely solved) problem. A graph-theoretical problem is solved by M. Deza and Grishukhin [744]: being a regular graph of degree 4 with 600 vertices, the skeleton of the 120-cell is not a 5-gonal graph and therefore also not an l_1-graph.

Weiss [2444] presents the regular 24-cell as an orthogonal projection of a semiregular 6-polytope, where three vertices of that 6-polytope are projected into the center of the 24-cell. Inspired by the infinite sequences of regular polytopes such that each polytope is the *vertex figure* of the next one in the sequence (regular simplices and the regular cross-polytopes in dimensions 2,3,4,...), Coxeter [679] deals with analogous sequences of uniform polytopes which are due to Gosset. Among the studied vertex figures cubes and also 24-cells occur.

Iwasaki et al. [1288] compute polynomial invariants for certain finite reflection groups and are able to determine then the solution space of functions with a mean value property related to the Platonic dodecahedron and icosahedron in 3-space, and to the 24-cell, 120-cell, and 600-cell in 4-space. Brandenberg [482] determines the *inner and outer j-radii* (i.e., the radii of the largest j-balls contained in j-sections, and the radii of the smallest j-balls containing the projections of the polytope onto j-subspaces, respectively) of those three types of regular polytopes which exist for any $n \geq 3$. Kabluchko et al. [1335] investigate point sets on the unit sphere whose convex hull has *maximal mean width*. After studying regular simplices and cross-polytopes, similar results on further regular polytopes are obtained and related formulae (conjectured by Finch) are confirmed.

The *spectrum of a polytope P* is the set of eigenvalues which are the roots of the characteristic polynomial of the adjacency matrix of the graph formed by the vertices and edges of P. Saldanha and Tomei [2091] determine the *spectra of the adjacency graphs of all regular polytopes* (including multiplicities). It turns out that the eigenvalues are algebraic integers of degree at most 3, and the structure of their eigenspaces is presented, too. The most work is necessary for the four-dimensional regular polytopes. Let Q denote a point set in Euclidean space. The set Q is said to be *Ramsey* if for every k there is a sufficiently large $m = m(Q, k)$ such that in every k-coloring of Euclidean n-space there is a monochromatic congruent copy of Q. The main problem is to characterize all Ramsey sets. For example, it is known that all regular polygons and all regular polyhedra are Ramsey. Cantwell [575] extends this to all regular polytopes.

Regarding interesting results on *truncations of the 600-cell*, needed for classifying *regular-faced n-polytopes*, we refer to the respective discussion below, and to the papers [365], [366], and [780]. The

latter paper shows: to get regular-faced 4-polytopes as truncations of the 600-cell, one has to cut off up to 24 icosahedral pyramids at non-adjacent vertices of the 600-cell, thus yielding regular-faced 4-polytopes with tetrahedral and icosahedral facets. It is shown that there are 314.248.344 isomorphism classes of such polytopes.

Effentrenger and Kühnel [801] study polyhedral $2k$-manifolds which are subcomplexes of boundary complexes of regular polytopes P. They are called k-Hamiltonian if they contain the whole k-skeleton of P. Continuing successful investigations of the cube and the simplex, the authors continue with cross-polytopes and the remaining regular 4-polytopes. By their results the existence of 1-Hamiltonian surfaces is decided for all regular polytopes, and 2-Hamiltonian 4-manifolds in n-dimensional cross-polytopes are investigated, too. Finally they decide all cases of 2-Hamiltonian 4-manifolds in regular n-polytopes for $n \leq 9$.

A *d-code in a graph* is a set of vertices such that all pairwise distances are at least d. In the paper [719], Debroni et al. deal with maximum d-code orders. Their calculations of the maximum independent set order of the 120-cell first led to the upper bound of 221. Further computations gave the maximum independent set order of the 120-cell with 220, and finally all maximum d-code orders of the icosahedron, dodecahedron, 24-cell, 600-cell, and the 120-cell are reported.

For a convex n-polytope P, we write $v(P)$ for its vertex set, and $P \subset Q$ when $v(P) \subset v(Q)$. Akiyama et al. [26] call a convex n-polytope *universal* for the family $F(n)$ of all regular n-polytopes if $P \subset Q$ for all $P \in F(n)$. In Coxeter's book [670] it is implicitly said that the 120-cell is universal for $F(4)$. In [26] a simple proof of this statement is given, based on metrical properties. Also in [2136] the 120-cell plays the main role, namely in connection with *burr puzzles*. In addition, the authors study the combinatorics of the 120-cell, demonstrating also how this 4-polytope can be decomposed into spheres and rings of dodecahedra.

Using also quaternions, McMullen [1750] presents six new examples of regular compounds in 4-space and shows, adding them to the representation of Coxeter in [670], also completeness. Five of the new compounds are inscribed in the 120-cell, and they consist of 720 copies of the 4-simplex, 120 copies of the 4-simplex, 75 copies of the 4-crosspolytope, 75 copies of the 4-cube, and 25 copies of the 24-cell, respectively. Being dual to the last but one case, the sixth compound consists of 25 copies of the 24-cell inscribed in the 600-cell.

Böröczky et al. [384] study *optimal packings of congruent spherical balls combined with simplicial regular polytopes*. These are in all dimensions the regular simplices and crosspolytopes, and additionally the icosahedron and the 600-cell. The authors nicely summarize former results with the vertices of these polytopes as centers of balls from the studied packings, and they give a stability version of the above result. It is remarkable that the used methods differ for different polytopes. With linear algebra the simplex case and with linear programming that of cross-polytopes are investigated; the icosahedron and the 600-cell are based on the simplex bound.

Denney et al. [733] study the geometry of the 600-cell, proving a statement of Schoute on the ways to subdivide its vertex set into vertex sets of five 24-cells. Based on a combinatorial method, they suitably use the actions of the non-crystallographic Coxeter group of this polytope, which is also explicitly studied with further consequences.

Using the notion of moving creases to change polyhedral faces, the problem on *continuous flattening of polyhedra* with non-rigid faces was solved for all convex polyhedra, and it was extended to continuous flattenings of the 2-skeleton of higher dimensional polytopes. In this form it was solved for all regular polytopes, except for the 24-cell, the 120-cell, and the 600-cell. Let P_0 denote the 2-skeleton of the 24-cell, and P_1 be a flat-folded state of P_0. Itoh and Nara [1280] present a continuous flattening motion for the 2-skeleton of the 24-cell which is related to the Jitterbug by Buckminster Fuller. More precisely, they construct a continuous family $\{P_t : 0 \leq t \leq 1\}$ of polyhedra P_t such that every P_t is isometric to P_0 regarding the intrinsic metric, and that a continuous folding process from P_0 onto each of the facets of the regular 24-cell exists such that 72 edges are rigid and each of the remaining 24 edges is

folded at its midpoint (the 24 faces are rigid during that motion). Similar further results are derived, and the list of references gives valuable information on related topics.

Rastanawi et al. [2021] investigate smoothness properties of *realisation spaces* of convex polytopes and show that the realisation space of the self-dual, (combinatorially) regular 24-cell is not smooth at any point corresponding to its realisation as a projectively regular polytope. Various further interesting results are given there, e.g., that there are three 4-polytopes with 8 vertices also not satisfying natural expectations.

RESEARCH PAPERS ON SEMIREGULAR POLYTOPES

Let us start here with a historical work: Polo Blanco [1982] presents Alicia Boole Stott (1860-1940) as an amateur mathematician specialized in four-dimensional geometry who studied 3-sections of regular 4-polytopes and found many of the semiregular 4-polytopes. In [1982] her results are nicely illustrated by pictures of original models and drawings.

For $n \geq 3$ we define a convex n-polytope to be *semiregular* if it is vertex-transitive and has regular facets. So it is clear that for $n \geq 3$ only in the case $n = 3$ uniform (see the definition above) and semiregular n-polytopes coincide; for $n \geq 4$ the semiregular n-polytopes present a proper subfamily of the uniform ones. Moreover, contrary to uniform polytopes the family of semiregular polytopes is getting, with increasing dimension, more and more "narrow". For $n \geq 4$ these n-polytopes were first listed by Gosset [1008] in 1900; the completeness of Gosset's list was confirmed by Makarov [1679] for $n = 4$, and by G. Blind and R. Blind in [367] for $n \geq 4$. Apart from the regular polytopes there exist three such polytopes in 4-space and exactly one for every $n \in \{5, \ldots, 8\}$. The facet families of the four-dimensional representative are (5 tetrahedra, 5 octahedra), (24 icosahedra, 120 tetrahedra), and (120 icosahedra, 600 octahedra). And for the respective n-polytopes with $5 \leq n \leq 8$, the facet families consist only of cross-polytopes and simplices.

Continuing [2091], where spectra of regular polytopes are treated, Saldanhe and Tomei derive in [2092] the spectra of all remaining semiregular polytopes (i.e., the set of eigenvalues which are the roots of the characteristic polynomial of the adjacency matrix of the polytope graph). They list all their characteristic polynomials and the numerical values of the corresponding eigenvalues and show that the algebraic degree of these eigenvalues is at most 5, achieved by two 3-polytopes.

Miyazaki [1816] studies three-dimensional orthogonal projections of regular and semiregular 4-polytopes. The projected facets are smoothly deformed from regular into non-regular shapes when the original 4-polytope is rotated. The variable polyhedral forms coming to be this way can be found in nature, art, and technology.

Möller [1818] describes all convex *uniform and Archimedean 4-polytopes* (for a complete picture regarding used methods, see also his PhD [1819]). To show completeness of the list of 41 Archimedean 4-polytopes, he considers the larger class of uniform 4-polytopes (64 known representatives), to reach the other number by certain reduction procedures (for example, carefully studying edge neighborhoods). It turns out that, besides the infinite classes of uniform 4-polytopes, namely the 4-prisms based on 3-prisms or antiprisms, and the biprisms, the class of convex uniform 4-polytopes consists of the six regular 4-polytopes, four prisms based on regular 3-polytopes, 13 prisms based on Archimedean 3-polytopes, and 41 other polytopes. The latter are completely identified and called Archimedean 4-polytopes. This is the first published proof of the result; Conway has undertaken an earlier, but similar approach (see [678]). Nicolovius [1893] investigates the same class of 4-polytopes, presenting methods for constructing such polytopes. The famous "grand antiprism" is also studied. Generating systems for related symmetry groups, starting with the regular simplex, cross-polytope, 24-cell, and 600-cell, are presented, and hierarchical structures within the investigated family of polytopes are studied, too.

Koca et al. [1428] investigate the *grand antiprism*, a famous uniform 4-polytope going back to Conway and Guy (see [649]), its 100 vertices and its symmetry group (of order 400) in terms of

quaternions with unit norm. Besides the discussion of group-theoretic properties, Coxeter diagrams and further notions and tools, the grand antiprism (with its 100 vertices, 300 tetrahedral and 20 pentagonal antiprisms as facets) and its dual polytope are constructed with quaternions as a subset of the vertices of the 600-cell, represented by the quaternionic elements of the binary icosahedral group. With analogous methods, the same authors investigate in [1429] the snub 24-cell, a semiregular 4-polytope with 96 vertices, 432 edges, 480 2-faces, and 144 facets; 120 of the latter are tetrahedra, and 24 icosahedra. This polytope and its dual are again represented in terms of quaternions, and the relations to the 600- and the 120-cell are discussed. Dechant [720] uses Clifford algebra to present group-theoretic calculations. Via some induction theorem he is able to study geometric aspects of the 600-cell and related polytopes and their symmetries; also the grand antiprism and the snub 24-cell belong to them.

Berestovskiĭ and Nikonorov [276] study *finite homogeneous metric subspaces* of Euclidean spaces. Such spaces are represented by the vertex sets of a compact, convex polytope with an isometry group that is transitive on the vertex set. In each case, all vertices lie on a sphere. The complete description of the metric properties of the vertex sets of regular and semiregular Euclidean polytopes from the viewpoint of normal homogeneity and Clifford–Wolf homogeneity has been done by the authors. Here they survey the corresponding classification along with other important properties of finite homogeneous metric spaces, in particular treating corresponding polytopes in Euclidean space. Completing the presented classification results, the authors also describe important tools that are used in this framework. A related, very recent paper is [277].

RESEARCH PAPERS ON REGULAR-FACED POLYTOPES

An interesting superset of the family of semiregular polytopes is that of *regular-faced polytopes*, i.e., of convex polytopes having regular facets. We start with the three-dimensional case. In 1966, Johnson presented in [1320] a list of *regular-faced 3-polytopes* whose completeness was verified by Zalgaller [2519]. There are, besides Platonic and Archimedean solids and the infinite classes of prisms and antiprisms, 92 representatives without vertex-transitivity. Berman [292] published a comprehensive overview, with nice photographs, a list of names (which is due to Johnson), nets, and many metrical information about all these polyhedra, and also in the book [2002] of Pugh a lot of material on regular-faced polyhedra is collected. Here one should also mention Graziotti's nice book [1017], with excellent photographs of models of the Platonic and Archimedean polyhedra, as well as of several further representatives; also nets for the construction of such solids are given.

Further interesting papers on this class of convex polyhedra are published by Gagnon [916], Valette [2392] and, regarding unfoldings, Horiyama and Shoji [1230]. An interesting historical overview on developments around regular-faced polyhedra, and polyhedra whose faces are composed of regular polygons, is written by Gurin and Zalgaller [1114]; see also [1113]. The authors extend their studies to tessellations of 3-space by regular-faced polyhedra, and they also classify the *convex polyhedra with regular vertices* (for such vertices all planar angles have to be equal, like also all dihedral angles at edges incident with the vertex). Timofeenko published the two papers [2370] and [2369] on so-called *composite regular-faced polyhedra*. Namely, if a regular-faced polyhedron can be partitioned by some plane into two regular-faced polyhedra, then it is said to be composite. Except for infinite series of prisms and antiprisms, there are 28 non-composite polyhedra in the family of Johnson and Zalgaller. In [2370] Timofeenko indicates the exact coordinates of the vertices of non-composite regular-faced polyhedra (except for trivial cases, like in the family of Platonic solids). The results in [2369] refer to a similarly defined class of polyhedra.

In higher dimensions, R. Blind [368] found all convex n-polytopes, $n \geq 4$, with regular, congruent facets. These are (besides the regular polytopes) for $n \geq 5$ only the bipyramids which are unions of two regular simplices. In 4-space, these polytopes are either regular or bounded only by tetrahedra. In [369] Blind continued in classifying those n-polytopes, $n \geq 4$, all whose facets are regular, but no

longer necessarily congruent. The interesting cases are mainly pyramids and semiregular polytopes. An interesting subfamily is given by polytopes obtained by cutting off from the 600-cell icosahedral pyramids, whose apices have to be pairwise non-adjacent vertices of this regular polytope. This cut-off procedure is only possible for at most 24 of the 120 vertices. This family was investigated in detail by Kirrmann [1386]. He gave a detailed description of the congruence classes in this family, getting for instance, if $k = 6$, 45775 congruence types, 41551 of which have trivial symmetry. However, complete results he found only for $k \leq 6$ and $21 \leq k \leq 24$. This is completed by Dutour-Sikirić and Myrvold [780]. Using delicate and hard computer investigations, they succeed to fill the gap by generating all independent sets of the 120-vertex graph of the 600-cell. The family of all regular-faced 4-polytopes having only tetrahedral facets is studied in [365].

OUTLOOK: UNIFORM POLYTOPES

As in parts already mentioned, criteria like regular-faced, congruent-faced, vertex-transitive, facet-transitive, edge-transitive, duality, etc. can be suitably weakened, strengthened, and combined to get interesting polytope classes (which in most cases certainly exist already in the literature). One of the most interesting classes of convex polytopes in this direction is that of *convex uniform polytopes*. These polytopes have vertex-transitivity and uniform facets (recursively defined, starting with regular polygons), and research on them goes back to the nineteenth century, combined with names like Gosset, Schoute, Alicia Boole Stott, Elte, Coxeter, Conway, and many others (see page 80 of the survey [1709]). Replacing "uniform facets" by "regular facets", we get semiregularity which is much more narrow (see above). A famous uniform 4-polytope is the *grand antiprism* discovered in 1965 by Conway and Guy (see [649]). It has 100 of the 120 vertices of the 600-cell, 500 edges, 20 pentagonal and 700 triangular 2-faces, and 20 pentagonal antiprisms and 300 tetrahedra as facets. In every vertex two pentagonal antiprisms with a common pentagonal face and 12 tetrahedra come together, the symmetry group is of order 400 and can be generated by two reflections and two half-turns (see Coxeter [677], pp. 588-590, and [1321], §4.3, for more details). The first book describing the complete list of the four-dimensional Archimedean polytopes is [650], but see also the articles [1818] and [1893] discussed above in the subsection on semiregular polytopes. The three basic papers [669], [677], and [678] of Coxeter are (in different ways) basic for the concept of uniform polytopes. And there is even a book manuscript on uniform polytopes due to Johnson (cf. [1321]), but as the subject itself it is, unfortunately, not finished.

Manipulating Coxeter diagrams with the help of Wythoff's construction (this is described in Coxeter's monograph [670]), Monson [1829] can provide a family of orthogonal projections of one uniform polytope onto another. After clarifying theoretical things, he presents various interesting examples concerning uniform polytopes or honeycombs in Euclidean and non-Euclidean spaces.

2.9.1.5 Euclidean balls and spheres

The knowledge around balls and spheres is essential for developing spherical convex geometry, but there are also notions and problems in classical convexity needing this knowledge directly (e.g., ball polytopes, and ball packings). Also here we start with *books or book chapters*, to continue with *surveys* and *selected research papers*. Throughout our text here, the notions "circles", "spheres", and "balls" are always meant as *Euclidean circles, spheres, and balls*; thus, here we omit usually the word "Euclidean". And clearly we do not repeat topics that are comprehensively discussed in other parts of our book (like the isoperimetric problem).

CIRCLES, SPHERES, AND BALLS IN BOOKS AND BOOK CHAPTERS

Two essential starting points for the geometry of *circles, spheres, and balls* are the monographs [653] of Coolidge and [358] of Blaschke. Although more than hundred years old, the book [653] gives still interesting and broad information on the *historical and mathematical developments around circles, spheres, and balls*, and it shows also various important and early ideas from the *theory of algebraic curves and surfaces*. Circles, spheres, and balls are studied from the viewpoints of *analytic, projective, and differential geometry*, and as special themes one can mention: the *group of inversions, Apollonius' problem, Laguerre inversions and associated groups*, and *differential geometry*, e.g., of circle systems. Blaschke's book [358] is described in our historical part, and as an update of it in our direction here we mention §16 of the monograph [446] of Bonnesen and Fenchel.

In Chapter 4 of the famous book [1197] of Hilbert and Cohn-Vossen the authors list eleven properties of the ball (not all suitable for characterization theorems), which treat themes from convexity and also differential geometry close to convexity (see §32 there). Subjects like *the isoperimetric one, constant width, and constant perimeter* are examples. Pedoe [1954] wrote a nice booklet concentrating solely on circles and emphasizing fundamental properties important for college geometry, *non-Euclidean geometries, complex numbers, Möbius transformations, the Poincaré model, polygon theory*, and many further applications. A similar book is Schwerdtfeger's [2211], containing also three-dimensional topics (like spherical geometry, stereographic projection, etc.). Bigalke wrote the comprehensive book [338] on various aspects of the *geometry of balls and spheres in 3-space*, presenting many theorems, problems and applications of *spherical geometry* (particularly broad in *astronomy*), *isometries of balls, properties of spheres in the sense of differential geometry*, a chapter on *crystallography*, and partitions of balls following several symmetry conditions.

Mainly two chapters of Berger's books [282] and [283] contain related material, namely Chapter 10 (the parts on circles and spheres, including *inversion, pencils of circles, properties of spheres*, etc.) and Chapter 18 (*spherical geometry, convex spherical polygons and Cauchy's lemma, the Möbius group*, etc.). In the analogous chapters of the problem book [288] one finds corresponding problems and solutions. And in Berger's book [286], Chapters II (also *spatial inversion, conformal group of the sphere, circle configurations*, etc.) and Chapter III (*spherical geometry, Möbius group and applications, kissing numbers*, etc.) might partially be also interesting for our theme here. Related to spheres, two further important problems should be mentioned here: *Blaschke's conjecture on "Wiedersehensflächen"* (see, e.g., p. 422 of the monograph [287] of Berger and Gostiaux), and *Ulam's floating body problem* (cf. part A6 in the problem book [687] and Section 10.6 in Schneider's monograph [2167]; a more recent contribution, with many references, is Florentin et al. [882]). And due to the frequent occurrence of balls as equality cases in geometric inequalities, we cite here also Chapter 2 of the book [531] of Burago and Zalgaller.

Popko [1987] wrote a book on *spherical geometry and spherical subdivision technique* which is suitable also for non-experts in mathematics and geometry, e.g., for designers and architects. It contains many nice figures and examples, and it shows the ways to various real-life applications. Among further topics, it treats *spherical subdivisions with symmetry properties (construction of geodesic domes), basic geometry of spheres* (e.g., subjects like polygons and angles on the sphere, excess and defects of a triangle), *symmetry groups* and relations to Platonic and Archimedian solids, trigonometric formulas for spheres, and the stereographic projection. Here we mention also the forthcoming monograph [1667] of Maehara and Martini, treating the geometry of spheres and spherical geometry (in the classical sense). Finally, we should also refer to the volume discussion for balls in subsection 4.7.2 of our book here.

SURVEY-LIKE PAPERS

Giering [967] collects *characterizations of spheres and balls in elementary geometry and convexity* (for instance that the ball is the only convex body all whose orthogonal hyperplane projections are congruent) as well as in differential geometry (e.g., showing that the sphere is the only smooth convex

surface whose mean curvature H and Gaussian curvature G satisfy a relation of the form $H = f(G)$ with $f'(G) \leq 0$). This survey is strongly related to §16 in [446] and §32 in [1197]. (As good completion the review Zbl.0236.53008 of Simon is recommended.) A small expository contribution is Section 1.2 of the handbook article [1177] of Heil and Martini. We mention also Burau's overview [533] treating the works of Blaschke and his followers on the *differential geometry of spheres*. And a survey similar to Giering's [967] is [339] by Bigalke. The characterizations of balls and spheres in 3-space as special bodies or surfaces discussed there refer to *elementary geometry, extremal properties*, and *notions from differential geometry* (e.g., main curvatures or the geodesic curvature).

The remainder of this part we fill with *research papers* (roughly in chronological order) containing natural *characterizations of circles, spheres, and balls* which are close to convexity. To go further (offering not only characterizations) would go beyond the scope of this subsection, yielding too many references.

CHARACTERIZATIONS

Improving also results of A. D. Alexandrov, Schneider [2140] proves the following: If k_1, k_2 are the principal curvatures of a closed convex surface K of class C^2 and p is its support function, then if $(pk_1 - c)(pk_2 - c) \leq 0$ (with c a constant) holds everywhere on K, this surface is a sphere, centred at the origin with $c = 1$. This implies that if there is a function $F(x, y)$, simultaneously monotone in both variables, not identically zero on a domain, and if $F(pk_1, pk_2) = 0$, then the corresponding surface is a sphere. Related to this, Schneider [2143] shows that if a three-dimensional convex body satisfies a similar condition and admits some circular orthogonal projection, then it is either a ball or a solid circular cylinder with two half-balls placed at its ends. Further on, Schneider [2144] shows the following: If K is an n-dimensional smooth convex body, $n \geq 3$, such that each Jordan domain in its boundary strictly halving the curvature of K halves also its surface area, then K is a ball. And in [2146] Schneider proves the following for an n-dimensional convex body K: If for $m = \{0, 1, 2, ..., n-2\}$ and a a real number $C_m(K, \cdot) = aC_{n-1}(K, \cdot)$, where the values $C_m(K, \cdot)$ are suitably chosen measures on Borel sets in n-space, then K is a ball; hence a convex body with one curvature measure being a constant multiple of the boundary volume measure is a ball. Kohlmann [1439] extends this result to linear combinations of curvature measures with positive coefficients. Generalizations and extensions to characterizations of metric balls in spheres and hyperbolic space are derived in [1438].

Sine [2255] proves that every three-dimensional convex body is a ball if every internally reflected light ray lies in a two-dimensional plane; for the n-dimensional case, where $n \geq 3$, see [2256].

Lickorish, and Świerczkowski [1570] answer affirmatively a question of Steinhaus: Given a three-dimensional convex body K and a level board L with circular hole H. Assume that if K is arbitrarily placed above L, one can *translate* it into a position above H such that it falls into H to fit there exactly. Is K necessarily a ball?

Kovalev [1460] gives a characterization of the n-dimensional ball that we describe here for the special case $n = 2$: Given a line l, a point $x \in l$, and a convex body K such that the *nearest point* from K to x is unique and not from the normal to l at x. It turns out that the only convex body that can rotate freely through a full turn, in each position touching l at x and K at the point nearest x, is a circular disc.

Montejano [1830] gives a nice characterization of balls using the notion of *constant width*: If an n-dimensional convex body K has the property that for some point x and some $k, 2 \leq k < n$, each intersection of K with a k-flat passing through x is a set of constant width (with respect to this k-flat), then K is a ball. For x from the interior of K, $n = 3$, and certain differentiability conditions, this characterization is due to Süss. Boskoff [471] proves the intuitively clear fact that a convex body having in every normal direction a hyperplane of symmetry is a ball. A rectangle, circumscribed about a circular disc, will always touch the disc at the midpoint of each side of it. Groemer [1041] proves that this property is characteristic for the circular discs among all planar convex domains. He shows even that if the contact point of a convex domain with the side of every circumscribed rectangle is always close to the midpoint, then the domain is close to a circular disc in the Hausdorff metric. This holds

also in higher dimensions: a convex body K is a ball iff it touches the center of each facet of every circumscribed box. And an analogous stability result holds, too.

Howard [1239] succeeded in confirming a long-standing conjecture: Any three-dimensional convex body of *constant width and constant brightness* is a ball. Interesting generalizations (to higher dimensions and relative differential geometry), also further on combined with ball characterizations, are obtained in [1240] and [2299]; see also the older interesting paper [1117].

Jerónimo-Castro, Montejano, and Morales-Amaya [1310] call a point x *point of revolution* with respect to a convex body K in 3-space if every section of K through x has an axis of symmetry passing through x. A line l is called a *false axis of revolution* if each point of it is a point of revolution for K, but l is not an axis of revolution. The authors prove that only the balls have a false axis of revolution. Let C denote a strictly convex, C^2-smooth, closed surface in 3-space, and H be the set of exterior points x of C with the following property: all tangent segments from x going to C have the same length. It is proved by Jerónimo-Castro et al. [1311] that if H is either a closed surface containing C or a plane, then C is a sphere. And it is shown that the planar situation is quite different, and the authors present also a result about sets of constant width in the hyperbolic plane.

Nitsche [1901] calls a convex domain K in the plane *isoptic* for a circle C if K subtends the same angle at all points of C. He proves that K is a circle if it is isoptic for two different concentric circles. Kurusa and Ódor [1481] extend this result to n-dimensional convex bodies K, analogously having two concentric spheres exterior to K. By the same assumptions regarding spatial angles, K is then a ball concentric to these two spheres. García-Jiménez et al. [928] show that a centrally symmetric convex body is a ball if one of its isoptic surfaces is homothetic to its boundary. This work is generalized in [1305]: a general n-dimensional convex body is a ball if one of its isoptic surfaces is a homothetical copy of its boundary. Kurusa and Ódor [1480] study problems of characterizing balls and, more generally, of determining n-dimensional convex bodies by their sections (not necessarily central) and caps. The main result is if a convex body K and a ball have *equal constant volume of caps and equal constant areas of sections* with respect to the supporting hyperplanes of a ball, then K is identical to the ball.

Jerónimo-Castro and Makai [1308] present an almost complete *characterization of pairs of closed convex domains* K and L with non-empty interiors in n-dimensional Euclidean, spherical, or hyperbolic space, namely for those pairs whose intersection $\varphi(K)\cap\psi(L)$ is centrally symmetric for all orientation-preserving congruences φ and ψ. This is motivated by High's result that for any plane convex body K the following implication holds: if the intersection of any two congruent copies of K is centrally symmetric, then K is a circle (see [1196]). The authors present more (also higher dimensional) results and treat also a somehow dual question, yielding that K and L are two congruent balls.

Bialy [326] deals with the notion of *Gutkin billiard table* in Euclidean n-space which is bounded by a smooth hypersurface H whose principal curvatures are everywhere positive such that there is some $\mu \in (0, \pi/2)$ yielding the following additional property: if for a pair of points $x, y \in S$ the angle between the vector from x to y and the tangent hyperplane of S at x equals μ, then the angle between the vector from x to y and the tangent hyperplane of S at y also equals μ. This is strongly related to questions around *floating bodies in neutral equilibrium*. Bialy shows that the only Gutkin billiard tables in Euclidean 3-space are spheres. He continues in [327] with extending this sphere characterization to dimensions $n \geq 3$. For the proof he uses the geometry of bodies of constant width and some symplectic geometry.

Myroshnychenko et al. [1868] present, within the family of star bodies, a couple of *characterizations of balls by symmetry properties of sections or projections*. For this purpose they define *complete symmetry* of an n-dimensional star body K demanding that the centroid of K is the origin and that each ellipsoid whose symmetry group contains that of K is a ball. It is shown that an at least three-dimensional star body with a continuous radial function and central sections that are all completely symmetric is a ball centred at the origin.

Jerónimo-Castro and Yee-Romero [1312] present two ball characterizations in 3-space. Let K be a three-dimensional convex body and suppose that v is an outward normal of K at its boundary point x. A plane parallel to v and passing through x is called a *normal plane of* K. The three-dimensional ball

is the only convex body with the property that each normal plane bisects the volume (resp. surface area). Second, the ball is the only three-dimensional convex body K such that for two suitable values w_1, w_2 the surface area of the intersection between any slab of width w_i and the boundary of K is a constant c_i with $i = 1, 2$. The paper [1099], also cited in our subsection on ellipsoids, contains a ball characterization treating centroids of sections. In [1306] circular discs are characterized among planar smooth and strictly convex bodies. Namely, let x, y be two boundary points of such a body K. We call the chord xy an α-chord of K if the tangents to K at x and y intersect at an angle α. The authors restrict their investigations to planar convex bodies K with α-chords having constant length and prove that they are circular discs when one of the properties (1), (2), or (3) is satisfied: (1) K is of constant width, (2) the α-isoptic of K is a circle, (3) K is equal to its image under rotation by an angle $\pi - \alpha$. They deduce that planar convex bodies of C^2 class having either $\pi/3$-chords of constant length or $\pi/2$-chords of constant length are also circular discs.

Jerónimo-Castro [1304] shows the following characterization of circular discs among plane convex bodies K: assume that for $\lambda < 1$ and each point x such that $\lambda K + x$ is exteriorly tangent to K, the common supporting cone of K and $\lambda K + x$ has a constant spatial angle α, say. If K is of constant width, then it is even a circular disc, and if K is centrally symmetric, then the same property follows. In the paper [988], also cited in our subsection on ellipsoids, the authors derive characterizations of balls (or also of ellipsoids) when sections or projections are studied for pairs of nested convex bodies, meaning that one of two considered convex bodies lies completely in the interior of the other.

The following two nice characterizations are proved by Jerónimo-Castro et al. [1307]: For $m \geq 3$, let K be a planar convex body with the property that the midpoints of all m sides of any equiangular m-gon circumscribed about K belong to K; then K is a circular disc. Also the three-dimensional analogue of the subcase $m = 3$ is proved: a three-dimensional convex body containing the centers of all facets of all its circumscribed regular tetrahedra is a ball.

2.9.1.6 Ellipsoids

Ellipsoids are surfaces that can be obtained from a Euclidean sphere via an affine transformation, or deforming this sphere using directional scalings; the same name is also used for the solid convex hulls of these surfaces. Using these convex hulls, it turns out that ellipsoids are special convex bodies which are very important in convexity. They occur as Löwner–John ellipsoids and further ellipsoids associated with convex bodies, as extremal bodies regarding many problems and (affine isoperimetric) inequalities, as unit balls of Euclidean geometry (important in functional analysis and asymptotic convexity), as convex bodies characterized via hyperplanar shadow boundaries, hyperplanar sets of midpoints of parallel chords, a false center, projections and hyperplanar sections, and via further concepts. As little addendum, we survey at the end results on multifocal ellipses and their applications. Different types of ellipsoids are shown in Plate 9 (at the beginning of Chapter 9).

GENERAL REFERENCES

In convexity, those *conics* and (higher dimensional) *quadrics* are interesting which have convexity properties, and these are mainly (but not only) *ellipsoids*; *paraboloids* are other examples. (It is clear that, as with the notion of convex surfaces in differential geometry, we do not emphasize every time whether with an ellipsoid we mean the whole convex body (as convex hull of its boundary) or only the boundary of this convex body. The context will always clarify this.) To find sufficiently many geometric properties of conics and quadrics, and therefore to have enough knowledge about possible tools for characterizing them from the affine, projective, and Euclidean point of view, we recommend the following *general references* on geometric properties of ellipsoids and also quadrics. We start with the historically basic book [652] of Coolidge, besides ancient achievements showing

also the developments in the descriptive–projective direction from Desargues to von Staudt, and in the algebraic direction from Fermat and Descartes to Euler. Further on, more modern methods and contents, like also *invariant theory*, are discussed, too. Next we mention Berger's comprehensive geometry book [283], meaning Chapters 14-17, and we refer also to the same chapter numbers in the associated problem book [288]. And we add here Chapter IV of Berger's more recent book [286]. Also Pedoe [1955] gives a broad presentation of conics and quadrics in his Chapter IX. Further on, in the general geometry book [1991] of Prasolov and Tikhomirov conics and quadrics are discussed in Chapter 4 and in the fourth addendum.

Furthermore, in Borceux's geometric trilogy, knowledge about conics and quadrics is broadly presented in the following parts: Sections 4.6-4.10 of [451], Sections 1.10-1.15 and 2.23-2.26 of [452], and Sections 3.3, 5.11, and 5.12 in [453]. And finally we mention the more modern and very comprehensive books [978] of Glaeser et al. and [1911] of Odehnal et al., treating conics and quadrics, respectively. The proofs therein are mainly synthetic (rather than analytic), and both books contain many fascinating figures, but also formal parts. For example, the second book contains the linear algebraic approach to quadrics (also projective and affine) in n-dimensional projective space, together with *correlations and polarities*, and with an interesting section treating *projective models of non-Euclidean geometries*.

For discussing *properties and characterizations of ellipsoids from the viewpoint of convexity* and differential geometry, Part VII of the appendix of Blaschke's classical book [358] and §16 of the monograph [446] of Bonnesen and Fenchel are two early expository publications which can be conveniently taken as *historical starting points*.

BOOKS AND BOOK CHAPTERS

We continue with *book chapters* (and even two books) dedicated to ellipsoids in convexity.

These two starting points (Part VII of [358] and §16 of [446]) deal, e.g., with Brunn's characterization via midpoints of parallel chords and Blaschke's characterization via hyperplanar shadow boundaries. Leichtweiss presents in Sections 2.10 and 4.15 of his book [1537] on the *affine geometry of convex bodies* several characterizations of ellipsoids and simplices. Chapter V of Barvinok's book [242] is dedicated to *ellipsoids in convexity and optimization*. For example, the author discusses the maximum volume ellipsoid and the approximation of convex bodies by polynomial hypersurfaces (i.e., also by ellipsoids), norm approximations, the ellipsoid method, the Gaussian measure on the Euclidean space, and applications to low rank approximations of matrices. Gardner [930] presents many results on ellipsoids in his Section 9.2 on *affine isoperimetric problems*; see also Note 9.4 there and Lutwak's related handbook survey [1634]. One sees in collected form that ellipsoids are extremal convex bodies regarding various affine isoperimetric inequalities, e.g., for the *Busemann–Petty centroid inequality, Petty's (affine) projection inequality, Busemann's intersection inequality*, and the *Blaschke–Santaló inequality*.

In his book [2167], Schneider treats ellipsoids in Sections 10.2 an 10.12. In Section 10.2 he collects characterizing properties of ellipsoids, e.g., by hyperplanar shadow boundaries or the hyperplanar positions of midpoints of families of parallel chords. And in Section 10.12 *ellipsoids associated with convex bodies* are discussed, such as *Löwner–John ellipsoids*, the *Binet ellipsoid*, the *Legendre ellipsoid*, and (as a "dual" counterpart to the 2-centroid body) the *LYZ-ellipsoid*. The so-called L_p *John ellipsoid*, a common generalization of John's ellipsoid and the LYZ ellipsoid, is also discussed, and the notes to Section 10.2 collect references to many extensions and modifications of these basic ellipsoids associated with convex bodies. The monograph [1713] of Martini, Montejano, and Oliveros contains with Section 2.12 a collection of characterizations of ellipsoids (like again that treating hyperplanar shadow boundaries, or the false center theorem), and the notes to this section contain also interesting references.

Also in the theory of *normed linear spaces* ellipsoids play an important role in various directions, e.g., regarding *characterizations of Euclidean spaces*, with respect to *Dvoretzky's theorem* and related

results, in view of *affine isoperimetric inequalities, John's theorem, and Löwner ellipsoids*, and also regarding *Milman ellipsoids*. These (and more) aspects are, e.g., discussed in Sections 3.3, 3.4, and 5.5 of Thompson's monograph [2361], mainly in Chapter 1 of the monograph [496] of Brazitikos et al., and in Chapters 2 and 5 of Artstein-Avidan et al. [133]. And here is the right place to cite also the two important surveys [960] of Giannopoulos and Milman and [1579] of Lindenstrauss and Milman, discussing, among many other aspects, also these topics (see in addition our related subsection of 12.4). Also *characterizations of inner product spaces* (see Amir's book [104]) and *orthogonality types in normed spaces* (cf. the surveys [69] and [70], both of Alonso, Martini, and Wu) yield many characterizations of ellipsoids.

The book [2371] of Todd is devoted to various theoretical and algorithmic aspects that *minimum-volume ellipsoids* have. Although, at first glance, the subject itself is seemingly a "relatively narrow" one, it shows its broadness by bringing together basic ideas and tools from very different fields, such as *(matrix) optimization, convex analysis, linear algebra*, and *geometry*. A variety of essential applications is also given, and an interesting modification of the minimum-volume enclosing ellipsoid problem is proposed, which suitably uses area instead of volume. In the last chapter related problems (and respective algorithms) are discussed, like *conditional minimum-volume ellipsoids, approximations by parallelotopes*, and *maximum-volume ellipsoids* inscribed in a polytope. The book is certainly of interest for researchers in *operations research, theoretical statistics, complexity theory, computational geometry*, and further related fields. Another book on ellipsoids is the forthcoming monograph [1833] of Montejano. It contains sections dealing with many *characterization theorems, complex ellipsoids, minimal and maximal ellipsoids, the Banach conjecture, ellipsoids in Minkowski, Hilbert and differential geometry*, and also *Helly-type theorems for ellipsoids*.

SURVEYS

The first exposition to be mentioned is [1074] of Gruber and Höbinger, concentrated on characterizations of ellipsoids in the spirit of convexity (e.g., via midpoints of parallel chord families, shadow-boundaries, projections, intersections with hyperplanes, and several extremum properties). The next step, Petty's survey [1969], also contains various of these characterizations, but additionally further results on extremal problems, playing a role in Minkowski geometry, Hilbert geometry, and affine differential geometry. Regarding the geometry of finite-dimensional normed spaces (Minkowski geometry) we mention here also the paper [553] of Busemann and Petty, containing 10 *Busemann–Petty problems* from which, seemingly, only one is completely solved (but see Alfonseca et al. [57]). Possible solutions of these problems are combined with characterizations of ellipsoids and therefore important for *characterizations of the Euclidean geometry* within that framework. Section 3 of the handbook article [1177] continues these survey-like representations with collecting (also more recent) characterizations of ellipsoids as well as other results on them, referring to *approximations, stereology, numerical mathematics*, and *functional analysis*.

V. Soltan (see [2276] and [2278]) nicely surveys various *characteristic properties of convex quadrics* within the family of convex hypersurfaces. The types of convex quadrics are *ellipsoids, sheets of elliptic hyperboloids of two sheets, elliptic paraboloids, nappes of elliptic cones*, and *unbounded cylinders based on convex quadrics in* $(n-1)$*-dimensional spaces*. As geometric conditions (also) used in this framework we mention for example: *symmetry of hyperplanar sections, equivalence of parallel sections up to a homothety or up to an affinity, hyperplanarity of shadow boundaries by parallel or central illumination, hyperplanarity of midpoint surfaces or of intersections of the boundary with the boundary of homothets*, and further criteria. The author gives the most recent extensions and generalisations of related classical results, and he poses interesting open problems. In most cases proofs are not included, but with the detailed and very useful list of references the author gives a complete picture of the little field. The articles are very informative, motivating, and full of geometric flavor.

CHARACTERIZATIONS OF ELLIPSOIDS

We continue now with (more recent) *research papers on ellipsoids in convex geometry*; however, because of the abundance of existing literature we limit ourselves to *characterization theorems*. And in view of surveys and book parts, where older results are completely collected and discussed, we choose here as starting point the paper [1834] of Montejano and Morales-Amaya, in which the authors generalize several classical characterisations of ellipsoids. They elegantly do this with the concept of "polarity of convex sets" and use this to show that many known results are related to the unifying concept of *projective centre of symmetry*. Based on this, it became conveniently possible to find new characterisations and to pose interesting new problems.

For K an n-dimensional convex body and H a hyperplane, K is said to be *H-homothetic* (*H-elliptic*) if all $(n-1)$-sections of K with hyperplanes parallel to H are homothetic (respectively, $(n-1)$-ellipsoids). Alonso and Martín [63] prove that if there exist non-parallel hyperplanes H_1, H_2 such that K is H_i-homothetic and H_i-elliptic for $i = 1, 2$, then K is an ellipsoid. It is also shown that this is not true if only one of the two properties holds. Moreover, a centrally symmetric convex body K being H_i-elliptic for pairwise non-parallel hyperplanes H_1, H_2, H_3 is an ellipsoid. The latter statement does not hold without assuming central symmetry. It is well known that a convex body K is an ellipsoid if and only if its intersection with each hyperplane is an ellipsoid (or empty). How can one reduce the family of intersecting hyperplanes, still to get this result? Let l be a linear variety of dimension between zero and $n-2$. We say that K is *elliptic through l* if for each hyperplane H with $l \subset H$ intersecting the interior of K, $H \cap K$ is an ellipsoid. Generalizing Busemann's ellipsoid characterization for l being a point, the following result is obtained by the same authors in [64]: If l_1, l_2 are two linear varieties of maximal dimension $n-2$ such that none is contained in the other one, l_1 meets the interior of K, and K is elliptic through l_1 as well as l_2, then K is an ellipsoid. They continue in [65] with no longer assuming that one variety meets the interior of K, but that they support K.

The *false centre theorem* is due to Aitcheson, Petty, and Rogers. It says that for $n \geq 3$ an n-dimensional convex body K with a "false center", p (i.e., p is not the centre of K, but each section of K through p is centrally symmetric) is an ellipsoid. Montejano and Morales-Amaya (see [1836] and [1835]) give, using also two-dimensional sections, a new and elegant proof of the false centre theorem. Further on, they obtain a *variation of the false center theorem*, where not all sections pass through a given point. They strongly use the characterization via hyperplanar shadow boundaries and some equichordal property of the sections.

Characterizations of ellipsoids via affine isoperimetric inequalities were mentioned already above; see once more Section 9.2 in Gardner's book [930], Lutwak's handbook survey [1634], and Chapter 3 in the monograph [1537] of Leichtweiss.

V. Soltan [2274] extends the ellipsoid characterization via hyperplanarity of chord midpoints by removing the boundedness condition, thus characterizing (the convex hulls of) quadric surfaces within the family of closed convex sets distinct from the whole space and from a half-space. Extending an older result of Bianchi and Gruber to *unbounded convex sets* (allowing correspondingly the more general notion *quadric* instead of ellipsoid), V. Soltan [2275] continues these investigations by showing that the following holds: If for an n-dimensional convex body K and a continuous function f on the unit sphere S for every direction $u \in S$ the hyperplane $\{x : x \cdot u = f(u)\}$ lies in K or intersects the boundary of K along an $(n-1)$-dimensional convex quadric, then K itself is a *convex quadric*.

Meyer and Reisner [1773] show that an n-dimensional convex body K is an ellipsoid iff for any $(n-1)$-dimensional subspace Q the centroids of the sections $(x + M) \cap K$ are collinear when x runs through all vectors which guarantee that the intersection of $(x + Q)$ with the interior of K is non-empty. Jerónimo-Castro [1303] derives characterizations of ellipsoids, extending characterizations via midpoint surfaces, shadow boundaries, and those from [1773], but mainly for the planar case. Larman, Montejano, and Morales-Amaya [1498] show the following: Consider a convex body K and an interior point x of it. Assume that in each direction one can continuously choose a section of K

being a translate of the corresponding parallel section of K through x. The authors show that if all these pairs of sections are almost everywhere different, then K is an ellipsoid. Inspired by a so-called Chakerian–Klamkin characterization of central symmetry and using the notions of *poles and polar hyperplanes*, Jeronimo-Castro and Montejano derive in [1309] new and surprising characterizations of ellipsoids and of central symmetry.

Lángi [1495] proves two characterization theorems: In a normed space with unit ball B, a convex body K and B are homothetic ellipsoids if for any x outside of K and in any 2-plane intersecting the interior of K and passing through x there are two tangent segments of the same length (in the sense of the norm) from x to K. Second, if in the two-dimensional situation B is strictly convex and smooth, then it is an ellipse iff in this norm billiard angular bisectors coincide with *angular bisectors* in the sense of the norm, called Busemann and Glogovskij bisectors. Magnanini and Marini [1670] show the following: For K an n-dimensional convex body containing the origin, a measurable set G with positive Lebesgue measure in Euclidean n-space is called *uniformly K-dense* if, for any fixed $r > 0$, the measure of $G \cap (x + rK)$ is constant when x moves on the boundary of G. Among various related results, they prove that G is uniformly K-dense iff K and G are *homothetic ellipses*. The following result in three dimensions is derived by Guerrero-Zarazua, Jerónimo-Castro, and Jimenez-Lopez in [1099]: For a convex body K and a line l meeting K consider the locus of centroids of all sections of K with 2-planes containing l. If this locus is a planar differentiable simple and closed curve not containing segments, then K is an ellipsoid. The authors add also a condition that even the ball can be characterized this way. D.-S. Kim et al. obtain in [1378] some *ellipsoid characterizations in terms of volume properties* of strictly convex bodies, and a related problem concerning elliptic paraboloids is raised, too. For various known characterizations of (balls and) ellipsoids, the investigated convex bodies are given by their *section or projection functions*, or by *support cones*. González-Garcá et al. [988] get new characterizations of (the Euclidean ball and) ellipsoids by assuming that the sections or projections are considered for a pair of nested convex bodies (one contained in the interior of the other). In addition, support cones and shadow boundaries for central illumination with apices from the boundary of the outer body are investigated.

An interesting and more recent application field of convexity is given by the *foundations of quantum mechanics*. For example, probabilistic mixtures of states are formalized as convex combinations of states, and therefore the notion of convexity makes sense and is essential in related studies. In this framework, Kimura and Nuida [1383] study fundamental properties of convex sets, affinities, and the separation of simplices and balls. In addition, some characterizations of ellipsoids are obtained.

Boman (cf. [438] proves the following result: If the *Radon transform* of a compactly supported distribution $f \neq 0$ in Euclidean n-space is supported on the set of tangent hyperplanes to the boundary of a bounded convex domain D, then this boundary encloses an ellipsoid. First it is shown that D must be equal to an ellipsoid in some neighbourhood of every point, from which it follows that it is globally equal to an ellipsoid. The symmetric subcase of this deep result (assuming that D is symmetric) is proved by Boman in the separate paper [437].

Bracho and Montejano [480] prove important subcases of the *complex Banach conjecture* (yielding Hilbert spaces as special finite-dimensional normed spaces via isometrically isomorphic subspaces), and they use for this a new characterization of *complex ellipsoids*. Similarly, also the paper [450] contributes to the Banach conjecture, in real spaces, strongly using the characterization of n-dimensional ellipsoids, $n > 4$, as the only symmetric convex bodies all whose hyperplane sections are linearly equivalent affine bodies of revolution. Continuing again with complex ellipsoids, Arocha, Bracho, and Montejano [127] derive a characterization of them which has no analogue over the real numbers: if in a complex space each complex line meets an n-dimensional convex body in a disk, then this body is a complex ellipsoid. And in [126] the same authors prove that if an n-dimensional convex body in complex space has two complex maximal inscribed ellipsoids, then they are translates of each other. They prove the analogous result on complex minimal circumscribed ellipsoids, and from this they get the analogue of Brunn's characterization for complex ellipsoids. A review of what is known about

complex ellipsoids from the convexity point of view is given by Arocha et al. in [128]. In particular, the recent state regarding the complex Banach conjecture is outlined.

MULTIFOCAL ELLIPSES AND THEIR GENERALIZATIONS

We finish the subsection on ellipsoids considering the extension of ellipses to *multifocal ellipses and ellipsoids*. These are curves in the plane or, in higher dimensions, hypersurfaces defined analogously to the ellipsoids via constant distance sums, but for an arbitrary finite number of given foci. Due to their natural definition, various alternative names for them occur: *polyellipses, polyconics, m-ellipses* (with m foci), and so on. Plate 1 (Fig. 3) shows polyellipses. It turns out that the references on this topic can only be found relatively scattered in the literature. Therefore we will cite here the most important contributions as complete as possible and study now the literature treating so-called m-ellipsoids with m foci. They consist of all points having the same distance sum to the m given foci. Due to their importance in *geometric optics*, these curves and surfaces were studied already by Descartes and v. Tschirnhaus, including an analogue of the classical *gardener's construction*. Simple constructions of tangents and normals, and related properties of them one finds already in the books [2096] of Salmon (p. 318), [1952] of Peano (p. 140), and [2130] of Scheffers (p. 123). Much later, they were presented in Section A32 of the problem book [687] of Croft, Falconer, and Guy. Compared with this, an earlier paper on *weighted m-ellipsoids*, clarifying various basic properties like strict convexity and positions of singularities, is [1870] by Sz.-Nagy ("weighted" means here that, for measuring distances, every focus gets a positive, real weight). In [1871] the same author continues with weights not necessarily positive, and he gives numerous applications of this extension. It is also clear that multifocal curves and surfaces are the *level curves and level surfaces* of the famous Fermat–Torricelli problem (yielding the minimum distance sum to the foci), one of the oldest and most important problems in *location science*.

Hartman and Valentine [1166] investigate volume, surface area, etc., of the convex hull K_x of the union of two convex sets R and S one of which can be translated by the vector x (yielding S_x, say). For the planar subcase, they give the following generalization of the construction of an ellipse: Let R be a compact, convex set, L_x the length of the boundary of K_x, and λ the length of the boundary of R. Then the set of points $C : L_x \leq a$, with $a > \lambda$, is a smooth, strictly convex set. In addition, for every boundary point x of C the tangent line to C at x creates equal angles with the generators of the set K_x. A similar result is derived for the area of K_x (instead of L_x). Zamfirescu [2521] generalizes these results establishing that *multifocal ellipsoids of convex sets* (instead of points) are differentiable boundaries of convex bodies. In the planar case, he looks for differentiability properties of analogously generalized hyperbolas. Melzak and Forsyth [1768] allow also *weights with different signs* and even replacement of foci by point sets, and they offer the connections to natural optimization problems. Established properties (which we formulate here in classical terms with foci) refer to strict convexity, smoothness (except in given foci), tangents and normals, the shape of these curves with arbitrarily small parameters (i.e., close to the object with minimum distance sum presented by the famous Fermat–Torricelli set), and possible ways of construction.

In the paper [816] interesting approximation properties of multifocal ellipses are derived. Namely, assume a fixed value for the distance sum and that the foci can move continuously over the plane. Then the limiting curve of the corresponding sequence of multifocal ellipses can have only a single straight segment, and this is only the case when the foci tend partially to infinity and partially to points of the line containing the segment. Thus, e.g., a regular triangle is not approximable by multifocal ellipses (see [815] and A35 in [687]). Also in [1049] several properties of weighted multifocal ellipsoids and their generalizations are obtained and classified. Namely, if negative weights are allowed, then convexity and compactness of multifocal ellipsoids are not guaranteed. Also *extensions to normed spaces*, focal points at infinity, and the problem of minimal distance sum (Fermat–Torricelli problem) are discussed. And generalized parabolas and hyperbolas are studied, too. In [1050] the authors

continue by establishing various topological, numerical, and even physical properties (e.g., referring to reflections and mechanical problems), investigating also the minimum problem, and taking *strictly convex norms with certain differentiability properties* and their consequences into consideration. Sekino [2218] presents properties of multifocal ellipses as level curves of the function f of distance sums, particularly studying the relation between critical points of f and foci. The paper contains many examples, figures, and suggestions for students. Also Varga and Vincse [2394] study multifocal ellipses in the Euclidean plane from the viewpoint of differential geometry. For example, their *curvature behavior* is studied in detail. In particular, a lower and an upper bound for the curvature involving explicit constants are given. These bounds depend on the number of foci, the rate of the level, and the global minimum of the function of distance sums. As an application, a new proof of the following result (see [815] and A35 in the problem book [687] of Croft, Falconer, and Guy) is given: the approximation of a regular triangle by multifocal ellipses is only possible with an absolute error which cannot be deleted; there is no way to exceed it, also not with an arbitrarily large number of foci.

From the viewpoint of *algebraic geometry*, polyellipses and their generalizations in higher dimensions are investigated in the paper [1895] of Nie, Parrilo, and Sturmfels. More precisely, the degree of the polynomial equations defining an m-ellipse is given, and these polynomial equations are presented as determinants of a symmetric matrix of linear polynomials. These results are generalized to the weighted cases and m-ellipsoids in n-space, and *geometric applications of semi-definite programming* are obtained. Starting with bifocal curves defined by constant sum, absolute difference, product, or ratio of distances to two given points (foci) and yielding, respectively, *ellipses, hyperbolas, Cassini curves*, and *Apollonius circles*, Jahn, Martini, and Richter [1294] extend this setting to *arbitrarily many foci, higher dimensions, norms*, and even *gauges* (i.e., they use symmetric, but also asymmetric distances). In addition, the foci can be replaced by convex sets. Some of the obtained results generalize facts from Euclidean geometry, and others are in sharp contrast to this. For generalized polyellipses, for example notions like *bisectors, generalized Birkhoff orthogonality*, and *confocality* play an essential role.

Regarding *computational geometry*, the two very recent papers [1334] by Junginger et al. and [1693] by Mantas et al. should be mentioned here. The authors investigate interesting subdivision methods and the ε-approximative side of the Fermat–Torricelli problem and of the construction of (generalized) polyellipses, the latter considered as level sets of the first problem.

2.9.1.7 Centrally symmetric convex bodies

In the field of convexity one can find also many characterizations, extremal properties, and applications of *centrally symmetric convex bodies*. A convex body K is said to be *centrally symmetric* if it has a center that bisects every chord of K passing through it, or if there is a translate K' of K such that $K' = -K'$. If in this situation $K' = K - x$, then x is the *center of symmetry* of K. Some of the essential keywords in this direction are geometry of *unit balls of normed spaces* (see, e.g., the books of Schäffer [2126], Thompson [2361], Artstein-Avidan, Giannopoulos, and Milman [133] and [134]), *characterizations* within the family of all convex bodies (e.g., via sections or projections; see Gardner's book [930]), measures of symmetry (cf. Toth's book [2381]), problems solved only for this subfamily (Hadwiger's covering problem in 3-space, or Mahler's conjecture), symmetrization procedures (e.g., central symmetrization and Blaschke symmetrization), degrees of symmetry (e.g., zonotopes as totally symmetric convex polytopes), and so on. This list is far from being complete since it is impossible to present "the whole picture". There are some representations of the knowledge around centrally symmetric convex bodies which are broad, but otherwise the results are very scattered in the literature. Such broad representations one can find in the following selected references.

BOOK CHAPTERS

Book chapters dedicated to this subject are for example §14 in the monograph [446] of Bonnesen and Fenchel (separated into *characterizations* and themes related to *lattice points*), Chapter 2 of Lyusternik's book [1646] (dedicated to *polyhedra*, e.g., with centrally symmetric faces, and *partition and packing problems*, like also to *convex bodies in lattices*), several *research problems* especially in Chapter A of the problem book [687], Chapter 3 in Toth's monograph [2381] (together with Grünbaum's survey [1080] presenting *measures of symmetry*), and Chapter 3 (on projections) as well as Chapter 7 (on sections) of Gardner's book [930], dealing with *tomographic problems*. In particular, the third chapter of [2381] should be emphasized here: measures of symmetry of convex bodies and the stability of the corresponding geometric inequalities are discussed. For example, the centroid Minkowski measure, Winternitz measures, the L_p-Minkowski measure, and the volume ratio of Rogers and Shephard are studied, like also Groemer's stability results on the Brunn–Minkowski inequality, Schneider's stability estimate for the Minkowski–Radon inequality, or problems related to the Banach–Mazur distance. Alexandrov-type theorems for sections and related characterizations of central symmetry are also presented in the subsections 13.1.4 and 13.1.5 of the book [1713]. Various further book parts of different authors discuss specified themes like the mentioned symmetrization procedures, and many results on centrally symmetric convex polytopes are discussed in Grünbaum's book [1085] (see the many entries in the index of terms).

SURVEYS

As surveys in this direction we cite the handbook articles of Heil and Martini [1177] (see Section 4), Gritzmann and Wills [1034] (cf. Section 2), and Chapter 4.9 (namely, [997] by Goodey and Weil) as well as [2176] of Schneider and Weil (like [997] referring to degrees of symmetry). Révész and Sarantopoulos [2028] survey results which are connected with measures of symmetry and the extension of the well-known Minkowski functional to arbitrary, not necessarily symmetric convex bodies in normed spaces (*gauges*). They collect related sources which are scattered in the literature and show nicely the connections to and applications in *approximation theory*. Discussed properties of the extended Minkowski functional include convexity, global Lipschitz boundedness, linear growth and approximation of the classical Minkowski functional of the central symmetrization of convex bodies. Many observations are new; this refers also to equivalences of various notions and properties. It is shown that various notions proved to be surprisingly relevant and effective in problems of approximation theory. In Section 4 of Soltan's survey [2273] on affine diameters (which are, in their own direction, longest chords) of convex bodies several results on centrally symmetric convex bodies are collected. In addition, the reader is referred to suitable parts of the survey [1157] on *star-shaped sets* and other parts of our book here, for instance to the Sections 5.3 (with the subsection 5.7.3), 10.2, and the subsections 10.6.1 and 12.4.7.

Due to the described situation, from now on we concentrate (mainly, but not only) on characterizations and interesting properties of centrally symmetric convex bodies, flanked by results on symmetry measures and coming from combinatorial geometry.

CHARACTERISTIC PROPERTIES

G. Fejes Tóth and Kemnitz extend in [840] results of Rogers, Larman, and Tamvakis by showing that the following holds for convex domains K in planes of constant curvature: Let f be a strictly increasing function defined for $t \geq 0$, p, p' be different interior points of K, and m be the midpoint of the segment pp'. If for any line l passing through p and meeting the boundary of K in q and r, its reflection image l' through p' meeting the boundary of K in q' and r' is such that $f(pq) + f(pr) = f(p'q') + f(p'r')$, then K has m as center of symmetry.

Fáry and Rédei [832] showed that any convex body contains a *unique centrally symmetric subset of maximal volume*. Moszyńska and Zukowski [1858] study the analogous uniqueness question for

the subsets invariant under a subgroup of the isometries. They succeed to show that the uniqueness property characterizes the subgroup generated by central symmetry. For other subgroups, examples are constructed having many invariant subsets of maximal volume. But for strictly convex bodies and for any subgroup, the uniqueness of this subset is established.

Montejano [1831] proves that a convex body K is centrally symmetric if all its *orthogonal projections* onto certain families of linear subspaces are centrally symmetric. He gives analogous results for K being of constant width or being a solid sphere. Boltyanski and Jerónimo Castro [426] extend these results giving a number of related characterizations of central symmetry. And they "dualize" also a nice result of Chakerian and Klamkin [597]: If C is a closed, bounded subset of Euclidean n-space, and for every three-point subset T of C there is a translate of $-T$ also contained in C, then C is centrally symmetric. In the mentioned dualization, the three-point condition is suitably replaced by a three-supporting-hyperplanes condition. Using the *projective setting of cross-ratios*, Jerónimo-Castro and Montejano prove further Chakerian–Klamkin type theorems in [1309].

Martini, Makai, and Ódor prove in [1677] that an n-dimensional convex body K is centred at the origin iff the $(n-1)$-dimensional volume of the intersection of K and an arbitrary hyperplane attains, among translates, its maximum when the hyperplane passes through the origin. In addition: For the convex body K, let k with $0 \leq k \leq n-2$ be an integer. Then the quermassintegral W_k of the intersection of K and an arbitrary hyperplane attains, among translates, its maximum if the hyperplane passes through the origin. (For the proof of the main result the authors used a new type of an integro-differential transform *generalizing the spherical Radon transform*; this method is deepened in [1676].) In [1677] the question was posed whether the converse holds true, and in [1675] the authors prove an infinitesimal variant of the reverse implication, for small C^2-perturbations of the unit ball if $k = 1$, and for small C^3-perturbations if $2 \leq k \leq n-2$. Inspired by the main result from [1677], Ryabogin and Yaskin [2078] replace in the formulation "if all $(n-1)$-volume functions of parallel hyperplane sections of a convex body have a critical value at the origin" the convex body by a *star body*, and the intersecting hyperplanes by *circular hypercones*, getting an analogous characterization: The star body K is origin-symmetric if all the $(n-1)$-volume functions of intersections with K have critical values when the cone degenerates to a hyperplane. The proof uses *Fourier transforms of distributions*. In [1678] a simplified approach is given.

Averkov [157] shows that a convex body K in a normed plane is centrally symmetric iff every chord of K *bisecting the Minkowskian perimeter* of K is necessarily an *affine diameter* of K. This result can be further used to get characterizations of centrally symmetric members within classes of special convex bodies (for instance, characterizing Minkowskian unit circles within the class of planar bodies of constant Minkowskian width).

Averkov et al. [162] extend a result of Schneider to normed spaces: If some *Minkowskian surface area* (e.g., that of Busemann or Holmes–Thompson) of a convex body K is halved by each *shadow boundary* of K with respect to *parallel illumination*, then the body K is centrally symmetric. This holds also for parallel illumination from "almost all directions".

As "negative results", Nazarov, Ryabogin, and Zvavitch [1882] show that if $n \geq 4$ is even, then one can find two essentially different convex bodies such that the volumes of their maximal sections, central sections, and projections coincide for all directions. These bodies constructed in [1882] all have rotational symmetry, and to get the results the authors use *layer cake integration and the Borsuk–Ulam Theorem*.

Artstein-Avidan asked the following: Does every planar convex body K have a point x such that the union of K and its reflection in x is convex? We say that if the union of K and its reflection in x is convex, then x is called a *convexity point of* K. With these notions the following is proved by Schneider [2169]: a planar convex body which is not centrally symmetric has three affinely independent convexity points.

Morales-Amaya, Jerónimo-Castro, and Verdusco Hernández [1843] prove the following: Let K be an at least three-dimensional, strictly convex body, and L be a hypersurface which is the image of an

embedding of the unit sphere, such that K is contained in the interior of L. Assume that, for any $x \in L$, there exists $y \in L$ such that the two supporting cones of K with apices x and y differ by a central symmetry. Then the point sets K and L are centrally symmetric and concentric.

MEASURES OF SYMMETRY

The notion of *measure of symmetry* is meant to denote a real-valued, non-negative function defined for convex sets, with value 0 if and only if the sets are centrally symmetric, and possibly satisfying additional criteria. For example, the value of such function should be larger for a set which is less symmetric than for a more symmetric one. For a discussion on the notion itself we recommend the introduction of Grünbaums survey [1080]. Thinking in the opposite direction, one can analogously fix the notion of *measure of asymmetry*.

As we said already above, two fundamental citations regarding measures of (a)symmetry are Grünbaum's survey [1080] and Chapters 3 and 4 of Toth's monograph [2381]. The references below are particularly not cited in the more recent bibliography of [2381].

Kaiser [1336] considers three measures of symmetry for planar convex bodies which are connected with functionals related to *chords passing through one point*, thus connected with so-called *critical points* that come closest to be centers of symmetry of these sets. The author develops an algorithm to calculate each of the three functionals for convex polygons. This can then be used to extend the results, in an approximative sense, to arbitrary convex planar bodies.

For given convex bodies, Tuzikov and Margolin [2386] introduce *measures of rotational and reflection symmetries*. These are based on corresponding symmetrization procedures (namely, a generalization of Minkowski symmetrization for rotational symmetry, and Blaschke's symmetrization for reflection symmetry). The symmetry measures are then defined via volume comparisons, and fast computations of symmetrization transforms are discussed; it turns out that Fourier transforms are very suitable.

Groemer [1045] studies two new measures of symmetry characterizing compact cones as the most asymmetric convex bodies. Here *compact cones* are meant to be the convex hull of the union of a convex compact subset of a hyperplane and an exterior point of this hyperplane. Stability estimates for both measures are investigated, giving a measure for the deviation of a convex body from a compact cone.

Let K_g be the image of a planar convex body K under reflection in a straight line g. Then the convex hull of K and K_g is the smallest (by inclusion) convex body whose axis of symmetry is g and which contains K. Lassak and Nowicka [1507] treat the ratio of the area of K to the minimum area of the mentioned convex hull and are interested in lower estimates for the family of all planar convex bodies K. They solve the subcase of triangles. Interestingly they continue their studies in [1508].

For a convex body K, Meyer et al. introduce in [1777] *affine measures of symmetry*, in each case based on a pair of affine covariant points of K, such as the *centroid, Santaló point*, or the *centers of the John or Löwner ellipsoids*. The introduced measures are essentially the appropriately normalized Euclidean distances of such point pairs, and for K sufficiently symmetric, this distance will be zero. Using probabilistic arguments, they show that for the centroid and Santaló point there are examples of convex bodies for which the above normalized distance is positive uniformly in the dimension. Weaker results are obtained for the centers of John's and Löwner's ellipsoids. Mordhorst [1844] proves Grünbaum's conjecture that the set of affine invariant points of a convex body coincides with the set of points which are invariant under all affine linear symmetries of this body, and he discusses the literature on these and related problems. Here a symmetry is understood as an affine invariant transformation leaving the considered convex body invariant. Safronenko [2086] continues these investigations giving bounds on the studied measure for the pairings of centroid and John point, as well as centroid and Löwner point. Guédon and Litvak [1096] introduce an affine invariant symmetry measure called *symmetric average*, and they obtain simplices and centrally symmetric convex bodies

as extremal cases. Similarly, Guo [1112] studies a family of affine invariant p-measures of symmetry, where the extreme case $p = +\infty$ yields the classical Minkowski measure of symmetry. Also he obtains characterizations of the simplex and the centrally symmetric case.

In terms of electrostatic p-capacity, Jin [1316] introduces the notion of *Orlicz–Minkowski capacity measure* and proves that it is indeed a measure of asymmetry in terms of volume. He verifies affine invariance, sharp bounds on this measure, and considers also the L_p-Minkowski case.

USING PROPERTIES OF CENTRALLY SYMMETRIC CONVEX BODIES

There are problems in convexity that can be sensibly posed only for the subfamily of centrally symmetric bodies. Two prominent examples are the *projection problem of Shephard* (see, e.g., Note 4.10 in Gardner's book [930]) and the *Busemann–Petty problem on sections* (cf. Notes 8.9 and 8.10 in Gardner's monograph [930] and Section 10.11 in Schneider's book [2167]). In other cases, problems on all convex bodies can successfully be reduced to the centrally symmetric subcase and (hopefully) solved there, like the *slicing problem* (see, for the recent state, [1704]). And one can also consider the centrally symmetric case on its own, besides the general case; a good example here is *Mahler's conjecture* on the sharp lower bound of the *volume product of a symmetric body and its polar*, still open for $n \geq 4$ (essential recent contributions are presented in [1276] and [894]; see also the exposition [409] where also the general case is treated). More precisely, the two papers [1276] and [894] contain elegant proofs of the centrally symmetric, three-dimensional case, yielding the sharp lower bound $32/3$ attained iff K or its polar is a parallelepiped.

Santos [2106] proves that each two-dimensional, convex, and centrally symmetric body K can be inscribed in an ellipse whose boundary is touched by K at ≥ 4 points. This yields a new characterization of ellipses as special centrally symmetric convex bodies creating a *Minkowski distance* that interestingly generalizes a property of the Euclidean distance. The discovered property can also be used in the topological study of *Voronoi diagrams*.

Makeev and Netsvetaev [1684] construct three-dimensional convex polyhedra such that some of their similar or affine images can be inscribed in (or circumscribed about) *any* centrally symmetric convex body K. They show that for centrally symmetric K either an affine-regular dodecahedron is inscribed in K, or that there are two affine-regular dodecahedra P_1 and P_2 with the property that nine pairs of opposite vertices of P_i, with $i = 1, 2$, are boundary points of K. The remaining two vertices of P_1, say, are exterior points of K, while the remaining two vertices of P_2 lie in the interior of K.

Jiménez and Naszódi [1314] study a variation of the *Banach–Mazur distance* where one of the two involved convex bodies has to be centrally symmetric. The derived inequality becomes an equality when one body is a simplex and the other one is centrally symmetric. The main result is if K is a convex body, L a strictly convex or a smooth convex body, and the distance d under consideration satisfies $d(K, L) = n$, then K is a simplex. Markessinis and Valettas [1697] derive upper bounds on the trace distance between several positions of n-dimensional centrally symmetric convex bodies. More precisely, they consider the following positions for such a body of volume 1: *isotropic position* (minimizing the second moment of the respective Euclidean norm), *minimal surface area position, minimal mean width position, John position* (maximizing the inradius), and *Löwner position* (minimizing the circumradius). The trace distance is suitably measured between two such positions, and since it is not symmetric, 20 upper bounds have to be searched which is perfectly done. Estimates are also contained. Kobos [1426] continues this work, also considering a distance notion very close to the Banach–Mazur distance (considering positive and negative homothets) and being the same if both participating bodies are centrally symmetric. For extensions, also here positions are used.

Aramyan [119] studies the *sine representation for the support function* of centrally symmetric convex bodies, getting also an *inversion formula* for the sine-transform. This way he obtains a subclass of the family of centrally symmetric convex bodies which is dense in the class of all of them.

In the context of *isoperimetric problems*, the authors of [574] deal with the bisections of a centrally symmetric planar convex body which minimize the maximum relative diameter functional. Namely, if K is a convex body in the Euclidean plane, then a curve γ from K with endpoints in the boundary of K divides K into two parts K_1 and K_2. The *maximum relative diameter of K associated with γ* is defined as the maximum of the diameters of K_1 and K_2. Necessary and sufficient conditions for minimizing bisections are given, and the behavior of a so-called standard bisection (in general not minimizing) is also analyzed (see the related paper [573]). The authors discuss also similar concepts, presenting an interesting list of references.

COMBINATORIAL GEOMETRY

Studying the well-known notion of *Helly dimension* of certain classes of convex bodies (see the famous survey [708] of Danzer, Grünbaum, and Klee), Kincses [1384] gives a complete characterization of the centrally symmetric convex bodies of Helly dimension three and four. For such results on centrally symmetric convex bodies we also refer to §30 of the book [431].

Let K be an origin-symmetric n-dimensional convex body, $K_p = K + p$ and $K_q = K + q$ be two translates of K, and F be a family of translates $K_r = K + r$ of K forming a packing. For every K_r we have points $x \in K_p$ and $y \in K_q$ such that the intersection of the segment xy and K_r is non-empty, and K_r overlaps neither K_p nor K_q. The bodies K_p, K_q are *visible from each other* if there are $x \in K_p$ and $y \in K_q$ such that xy intersects no K_r. If this is not the case, K_p and K_q are *concealed from each other*. The *concealment number* of K is the infimum of $\lambda > 0$ such that two bodies K_p and K_q with λ as lower bound on the distance of p and q can be concealed from each other by suitable families F with the properties explained above. Alonso, Martini, and Spirova [66] derive various results on concealment numbers—besides visibility in combinatorial geometry some background is also given by the geometry of normed spaces.

The following *Helly-type problem* is due to K. J. Swanepoel: a set is said to be in *centrally symmetric convex (c.s.c.) position* if it is contained in the boundary of a centrally symmetric convex body. Is there a number k such that for any set A in the plane the following implication is true: if every k-element subset of A is in c.s.c. position, then A is also in c.s.c. position? For $k \leq 5$ this is true; Garber and Roldán-Pensado [923] show that there exists a planar set of 9 points that are not in c.s.c. position although every 8 of them are. Thus it follows that if such a number k exists, it has to be at least 9. It is also shown that if A is a simple planar closed convex curve, then the answer is positive for $k = 6$.

Continuing the Part I of their contribution [1877], Naszodi and Swanepoel improve a result of Polyanskii: a family of homothets of an origin-symmetric n-dimensional convex body K is said to form a *Minkowski arrangement* if no homothet contains the center of any other homothetical member in its interior. Polyanskii [1985] shows that the largest number of homothetical copies in a pairwise intersecting Minkowski arrangement is bounded from above by 3^{n+1}. Naszodi and Swanepoel confirm that such an arrangement has at most $2 \cdot 3^n$ members. With similar methods they reprove also the following result of Polyanskii: If $K_1, \ldots, K_m$ is a sequence of homothets of an origin-symmetric convex body K such that for any $i < j$ the center of K_j lies on the boundary of K_i, then $m = O(3^n n)$. It is clarified that such results have applications for so-called *k-distance sets in normed spaces*.

Let K be an arbitrarily given smooth and strictly convex body centred at the origin. Dewar [736] proves that a *packing of homothetical copies* of K with randomly chosen radii (regarding the associated norm) will create a (2,2)-sparse planar contact graph, and that there is a comeagre set of such bodies where each (2,2)-sparse planar graph can be realized as the contact graph of a stress-free packing of homothets of them.

2.9.1.8 Bodies of constant width

It is well known that the *width* of an n-dimensional convex body K in direction u is the distance of the two parallel supporting hyperplanes of K which are orthogonal to u. A convex body K is said to be of *constant width* if, for each direction u, the value of this width function on the unit sphere is the same. For example, any ball is of constant width, but also each Reuleaux polygon. Sets of constant width have a long history, combined with names like Euler, Minkowski, Lebesgue, Blaschke, Besicovitch, and many other famous mathematicians. Plate 7 at the beginning of Chapter 7 shows examples.

EXPOSITORY REPRESENTATIONS

Expository representations collecting the comprehensive literature on this notion and its generalizations are §15 of the monograph [446] of Bonnesen and Fenchel (containing already material also on related notions like completeness), the impressive survey [596] by Chakerian and Groemer (the first publication showing the surprising broadness of this little field, also containing discussions of related notions), Section 5 of the handbook article [1177] of Heil and Martini (updating [596] and saying more about the relatively new concept of reducedness), Section 2 of the survey [1720] (concentrating on constant width sets in normed spaces), and the whole monograph [1713] of Martini, Montejano, and Oliveros, largely extending the number of aspects and viewpoints regarding sets of constant width, and presenting an almost complete list of related references. Besides these expositions, various convexity books contain chapters or distributed contributions dedicated to the field. As examples we emphasize the wonderful classic book [2487] of Yaglom and Boltyanskii (in particular, §7 and §8), or Section 3.2, the Notes 3.3, and several further parts of Gardner's monograph [930]. We mention also subsection 12.3.8 in our book here. For constant width bodies in spaces of constant curvature the reader is referred to Lassak's recent survey [1504] (Section 2.10) and to pp. 240-243 within the book [1713].

Since the monograph [1713] appeared in 2019 and, in general, sets of constant width occur as popular research topic, various interesting contributions appeared since then. Due to this fact, we decided to offer here, as an update, most of the research articles which came up after the appearance of [1713]. Here they are.

RESEARCH PAPERS

It is clear that in finite-dimensional normed spaces a set of constant width has always an interior point. In *infinite-dimensional spaces* it is not necessarily so. Maluta and Yost [1689] prove that every real Banach space with an unconditional basis can be equivalently renormed such that it contains a *set of constant width with empty interior*. Thus, constant width sets without interior points can exist in reflexive Banach spaces. The authors prove also that in the isometric case the normal structure is not only sufficient but also necessary for the situation that the class of *diametrically complete sets* coincides with the family of sets of constant radius from the boundary of the set. Interesting open questions are posed.

A family F of lines in the plane is said to be *outwardly simple* if each point exterior to some circle (including points at infinity) belongs to exactly one member of F. Hence there is one member of F in any direction. Euler introduced a formula that yields, using outwardly simple line families, a construction of the boundary of each planar constant width set. With the help of some geometric ideas, Kawohl and Sweers [1351] show that a modification of this formula describes each planar set of constant width. *Approximations of arbitrary constant width sets* by the following special cases are given: by Reuleaux polygons, by bodies of constant width whose boundaries consist of circular arcs of radii equal to the width, and by sets of constant width having analytic boundary. *Barbier's theorem* on the circumference and further classical results are reproved, like also observations of Blaschke. One

can summarize that the authors succeed to get a lot of classic results on constant width sets by new and simpler proofs, all this obtained from a new point of view.

Gao, Zhang, and Zhou [922] prove that convex curves whose support functions satisfy certain conditions are polynomial curves of degree at most 6, and they extend a method of Rabinowitz to calculate polynomial representations for convex curves. This yields constructions of several *polynomial convex curves having constant width*.

Hug and Schneider [1260] extend results on certain *motion invariant measures* of pairs of lines, which meet a planar convex body, to higher dimensions. Then pairs of hyperplanes or lines meeting a convex body are taken into consideration. The obtained results are especially simple iff the studied convex body is of constant width (in the first case), and of constant brightness (in the second one).

Montejano et al. [1837] study the *graphs of Reuleaux polyhedra*, which have to be planar, 3-connected, and strongly self-dual. They investigate the question when these conditions are sufficient and reduce the problem of constructing Reuleaux polyhedra to that of constructing metric embeddings for strongly involutive self-dual graphs. The conjecture is posed that all strongly involutive self-dual graphs admit metric embeddings into 3-space, and it is proved that, given a strongly involutive self-dual graph, there exists a (not necessarily injective) metric mapping of this graph. The authors give also algorithms delivering each such graph with up to 14 vertices, and corresponding metric embeddings are constructed. However, the statement that any such graph is realizable as a Reuleaux polyhedron remains a conjecture. In combination with older results, they also construct (numerically, but with very high precision) *many new examples of three-dimensional bodies of constant width*. Bisztriczky and Oliveros [352] continue the work in [1837] bringing up parts of the constructions there to *higher dimensions*. First, certain self-dual $(n+1)$-polytopes are constructed from self-dual n-polytopes, where involutory self-duality is preserved. Then infinite families of examples are described, and the preservation of metric embeddability is also studied. It turns out that such polytopes can create Reuleaux polytopes, whose relations to Meissner polytopes are investigated, too.

Via a generalization of *Holditch's theorem* in two-dimensional constant curvature manifolds, Monterde and Rochera [1838] get generalized versions of *Barbier's theorem for constant width curves* and of Steiner's formulae for parallel curves. The derivation of these results is possible since they obtain a general formula related to the geodesic curvatures of the studied curves.

Maluta [1688] shows that any *infinite-dimensional Banach space* contains compact, symmetric, convex sets with interior points, but having no diametral points. In a *reflexive Banach space*, a point in the boundary of a constant width set belongs to a *diametral pair* iff it is a support point of this set. Due to this, all boundary points of such a set with interior points belong to diametral pairs. If in some reflexive space the whole boundary consists of diametral pairs, then constant width sets are not characterized within the family of all compact, convex sets with interior points. Some further related results on reflexive spaces are obtained.

Henrot and Lucardesi [1186] solve the following *shape optimization problem*: among the planar sets having prescribed constant width and inradius, find that domain which minimizes area! For each choice of width and inradius, the authors give an explicit characterization of the optimal sets. The uniqueness of them within the family of *Reuleaux polygons* and the smallest value of area are given, too.

Inspired by a well-known property of the n-dimensional Euclidean ball, Jerónimo-Castro [1305] studies the following problem: For K an n-dimensional convex body and any x such that the homothet $\lambda K + x$ is outwardly tangent to K, the common support cone of K and this homothet of it has a constant solid angle. Is K then a Euclidean n-ball? He gives a positive answer in the planar situation, for the cases that K is of *constant width* or centrally symmetric.

Alfonseca et al. [55] use and prove properties and characterizations of sets of constant width and discs in the *hyperbolic plane*. In particular, they prove that the validity of any two of the following properties characterizes the disc in a hyperbolic plane: origin-symmetry; constant width; constant projection lengths on all lines through a point; constant section lengths on all lines through a point.

Horváth [1236] compares four different concepts for measuring the *breadth or width* of a convex body in *hyperbolic geometry* and studies their properties, finally introducing a new such concept. Defining then the three concepts *bodies of constant with, bodies of constant diameter*, and bodies having the *constant shadow property*, he clarifies the relation between constant diameter and the constant shadow property, showing also that both properties are stronger than constant width. The author also proves a variant of Blaschke's theorem related to minimal width.

For a planar convex body K, Januszewski and Zielonka [1296] define the parameter $\lambda_2(K)$ as the largest value of a scalar $\lambda > 0$ such that two directly congruent copies of the homothet λK can be packed into K. The double inequality $1/2 \leq \lambda_2(K) \leq \sqrt{2}/2$ is established, and the problem is posed to describe the whole family of all planar convex bodies K satisfying the equality $\lambda_2(K) = 1/2$. It is proved that bodies of constant width and regular polygons are members of this searched family.

Rochera [2043] investigates three types of curves, among them being *ϕ-isoptics* (as locus of all points from which a given curve C is seen under a constant angle $\pi - \phi$) and closed *curves of constant ϕ-width* (named so iff their ϕ-isoptic is a circle). The author gives explicit expressions for these types of curves, interesting examples, integral formulas for enclosed areas, and a Barbier-type theorem. Also a characterization of curves of constant ϕ-width in terms of properties of an angle function is given.

Let xy be a double normal of a planar convex body K. If an ellipse E, containing x and y, is contained in K, then E is said to be *surrounded* by the boundary of K. And if K is contained in the convex hull of E, then E is said to be *surrounding* K. Zamfirescu [2524] investigates surrounding and surrounded ellipses and circles for arbitrary convex bodies K, e.g., also for bodies of constant width.

In the Euclidean plane the defect in the *isoperimetric inequality* can be calculated in terms of the signed areas of some singular sets. The authors of [2109] study this problem in *normed planes* with piecewise smooth unit circles and the corresponding class of admissible curves. For such a curve, the mentioned singular sets are given as projections in the subspaces of symmetric and constant width admissible curves. Based on this concept, improved isoperimetric inequalities are derived, and their equality cases hold for symmetric curves or curves of constant width.

We want to mention here also the inspiring book [42] of Alencar et al., mainly dedicated to the *differential geometry of planar curves*. The authors present many inspiring geometric and also topological results on plane curves in a classical, appealing way. Very nice examples and over 200 figures show the beauty of this subject. Topics being close to convexity are Jordan's curve theorem, the isoperimetric inequality, convex curves, and curves of constant width.

The famous theorem that the *Reuleaux triangle* has minimal area among all planar domains of fixed constant width has been extended also to the spherical and hyperbolic case. K. J. Böröczky and Sagmeister [411] add *stability versions* for all three types of surfaces of constant curvature. The authors provide also an overview of properties of constant width bodies in spaces of constant curvature and add, for the hyperbolic case, a characterization result in terms of horospheres.

Bondarenko et al. [441] continue the work of Bezdek and Kiss on the *X-ray number of a constant width body K* (= smallest number of directions such that each point of K belongs to some line in one of these directions that passes through interior points of K) and also on the famous related *illumination number*; see the survey [1716] for these and related quantities. They construct coverings of the unit sphere with congruent spherical caps which imply, together with results of Schramm for higher dimensions, the complete confirmation of the X-ray conjecture and the illumination conjecture for bodies of constant width.

Finally we mention here the very recent survey of Lassak [1504] on *width and thickness of convex sets in spherical space*. He comprehensively discusses known results on reduced bodies, constant width bodies, Wulff shapes, and diametrically complete sets in spherical space; impressively many recent results are cited. This valuable collection nicely completes the notes on constant width in non-Euclidean geometries (different to Minkowski spaces) surveyed in the monograph [1713]; see pp. 240-243 there.

2.9.1.9 Complete sets and completions

Although there are many results about complete sets, this subsection is very short. The reason is the comfortable situation to have the very comprehensive and recent survey [1172] on complete sets; and also the recent book [1713] contains much related material.

A bounded subset of Euclidean n-space is called *complete* (or *diametrically complete*) if it cannot be enlarged without increasing its diameter. And for an n-dimensional convex body, any complete set containing it and having the same diameter as it is called a *completion* of this body. It is known that every convex body has at least one completion, and those convex bodies having a unique completion are particularly interesting (having several applications, some of them unexpected). And it turns out that in Euclidean spaces the classes of complete sets and constant width sets coincide. For normed planes the same equivalence holds, but in normed spaces of dimensions $n \geq 3$ this is in general no longer true. Norms that still satisfy this equivalence are called *perfect norms*, and to characterize them is an interesting open problem.

BOOKS, BOOK CHAPTERS, AND SURVEYS

Expositions on complete sets in Euclidean and normed spaces, containing together possibly all essential references and results, are the following ones: Bonnesen and Fenchel prove in §15 of their monograph [446] Meissner's theorem (saying that each complete set is of constant width) and Reidemeister's theorem (characterizing the bodies of constant width as bodies of constant diameter). Section 4 of the survey [596] of Chakerian and Groemer as well as subsection 5.4 of the handbook article [1177] enrich the collection of results and problems on complete sets obtained till 1993, and subsection 2.5 of the survey [1720] of Martini and Swanepoel completes this picture with results on completeness in Minkowski spaces until 2004. In the book [1713], the increased amount of results till 2019 is reflected in Chapter 7, in the Notes to Chapter 7 (pp. 159-161), in Sections 10.3 and 10.4, and in the Notes to Chapter 10 (pp. 231-235). However, the most complete and most recent survey-like representation of results on complete sets is the (already mentioned) exposition [1172] of He, Martini, and Wu. And an increased number of recent results on completeness in spherical space is summarized in Lassak's survey [1504].

2.9.1.10 Reduced bodies

Also here the shortness of the contribution is due to the existence of comprehensive expositions on reduced sets, namely the relatively recent two surveys [1505] and [1506]. In addition, also the book [1713] contains much of this material and gives an update till 2019.

Recall that a convex body is said to be *reduced* if any convex, proper subset of it has smaller minimal width. This concept goes back to Heil (1974). Each convex body contains a reduced body of the same minimal width. It is clear that the equilateral triangle is reduced, but the regular tetrahedron (like any n-simplex for $n \geq 3$) is not. In Euclidean spaces regular odd-gons and all bodies of constant width are reduced, and the family of reduced bodies forms a proper superset of that of diametrically complete bodies, the latter being identical with the family of bodies of constant width. Reduced bodies occur in various containment problems combined with minimal volumes. So the (unknown) solutions to the following natural "dualization" of the isodiametric problem have to be reduced: Which n-dimensional convex bodies ($n \geq 3$) of fixed minimal width have minimal volume? (In the plane, this problem is solved by the equilateral triangle.) Since the notion of width function of convex bodies can be suitably extended to normed spaces, also reducedness can be successfully carried over to Minkowski geometry, but there the situation is more difficult. This is due to the fact that in normed spaces completeness and constant width are no longer equivalent notions, and that there are even complete bodies not being

reduced. For example, the question occurs whether, in general normed spaces, a convex body which is reduced and (diametrically) complete is necessarily of constant width! In general this is open, but it was shown that the implication holds for a large class of convex bodies, namely for those having a smooth extreme point. Plate 6 (at the beginning of Chapter 6) shows examples of reduced convex bodies for norms with unit ball B.

BOOKS, BOOK CHAPTERS, AND SURVEYS

All the above results can be found in the following references. The first explicit mention of reducedness as a concept in a survey-like publication can be found in [1177] (end of subsection 5.4, p. 365). Regarding normed spaces, this continues in subsection 2.12 of the survey [1720]. The two surveys [1505] (presenting all results in Euclidean spaces) and [1506] (surveying reducedness in normed spaces) are completely dedicated to that notion and update everything till 2014. And in the monograph [1713], results on reducedness are completely represented in two chapters, again separated for Euclidean and for normed spaces: in Section 7.4, in the Notes to Chapter 7 (pp. 162-164), and in Section 10.5 and the Notes to Chapter 10, pp. 235-236, respectively. By this, everything is updated to the state of 2019. And regarding reduced convex bodies in spherical spaces, Lassak's recent comprehensive survey [1504] should be consulted; see Sections 2.6, 2.7, 2.8, and 2.9 there. Finally we should mention that reducedness is also discussed in our subsection 12.4.7 here, but only shortly.

RESEARCH PAPERS

After the book [1713] only some research papers on reducedness in spherical spaces appeared. But these are collected and discussed in Lassak's survey [1504].

2.9.1.11 Zonotopes and zonoids

Below we give short descriptions of important facts and references referring to zonotopes and zonoids. This shortness has its reasons in the much broader representations given in Chapter 10.

ZONOTOPES: INTRODUCTORY PART

Zonotopes are well known as convex polytopes that are *vector sums of finitely many line segments*; popular examples are affine cubes (parallelepipeds), rhombic dodecahedra and rhombic triacontrahedra (as known Catalan solids), the translative space-fillers in 3-space (often called parallelotopes; cf. Plate 10 at the beginning of Chapter 10), and the class of permutahedra. Their extreme importance in convexity is due to the fact that they have *many different applications* (metrical, combinatorial, topological ones, and also in econometrics) and *forms of representations*. The class of n-zonotopes coincides with the class of *totally symmetric n-polytopes* (i.e., with those convex n-polytopes all whose faces, of any dimension, have themselves a center of symmetry). For this property the central symmetry of all k-faces for some $2 \leq k \leq n-2$ is sufficient.

If zonotopes are centered at the origin and n-dimensional, they coincide with the class of *projection bodies* of n-dimensional convex polytopes (i.e., for any convex n-polytope P the support function value of its projection body, being such a zonotope, yields the $(n-1)$-volume of the orthogonal projection of P onto a hyperplane H in the normal direction of H). Further on, they are important in *location science* (for the minsum hyperplane problem), they can (like the more general *belt polytopes*) be represented via *arrangements of hyperplanes in projective $(n-1)$-space*, bringing them in connection with *oriented matroids*, they occur in the field of *scissors congruence* (equi-decomposability to the cube), they play

an important role regarding *approximation theory*, and there are many further aspects that could be mentioned here.

The derivations of the above-mentioned properties and results can be found in the following references, or in the broader Section 10.2 and subsection 10.6.1 in our book here.

ZONOTOPES: SURVEYS AND BOOK PARTS

The most important *surveys* discussing many aspects from the geometry and applications of zonotopes (and also of zonoids) are the following expositions: [422] by Bolker, [2176] by Schneider and Weil, and the handbook article [997] by Goodey and Weil (see, further on, also the paper [1705]), of course together with the subsection 10.6.1 of this book here.

Important *books* containing Sections or subsections treating zonotopes are due to Björner et al. [354] (Section 2.2, *oriented matroids*), Ziegler [2560] (Lecture 7, *zonotopal geometry*), Thompson [2361] (in several parts, as *projection bodies, isoperimetrices, and unit balls of normed spaces*), Barvinok [242] (posing interesting *problems on zonotopes* in Chapters VI and VII), Grünbaum ([1085], second edition, Section 18.4: update on *arrangements of hyperplanes*), Gardner [930] (Sections 4.1 and 4.2 and corresponding Notes, Chapters 8 and 9; mainly as *projection bodies* and as extremal cases in *geometric inequalities*), Schneider [2167] (Section 3.5 and its Notes, regarding *Minkowski classes*), and de Concini and Procesi [645] (Section 2.3, as *shadows of cubes*, regarding *tilings, and lattices*). The more general class of *belt polytopes* (i.e., of those convex polytopes having a segment summand parallel to each of their edges) is treated in Chapters VI and VII by Boltyanski et al. [431]; see also Fig. 28 in Plate 10. More such references (plus the more recent research papers related to zonotopes) are broadly discussed in our subsection 10.6.1; cf. also Section 10.2.

ZONOIDS: INTRODUCTORY PART

Zonoids are the *limits of zonotopes in the Hausdorff metric*, and they can be characterized as *ranges of vector measures*. Any ellipsoid is a zonoid, like also the spindle-shaped convex body of revolution obtained by rotating a suitable piece of the cosine curve; interesting types of zonoids are depicted in Plate 8. Full-dimensional zonoids centered at the origin occur as *projection bodies* of convex bodies, and also each *centroid body* is a zonoid. The polar of a zonotope cannot be a zonotope, but the Euclidean n-space contains (refuting a conjecture of Bolker) centrally symmetric, convex bodies which, together with their *polars*, are zonoids, but are not ellipsoids. There are several interesting results also about *generalized zonoids*, an interesting class of convex bodies "between" zonoids and centrally symmetric convex bodies. Another interesting extension of the notion of zonoids is that of *belt bodies*. And also for zonoids there are interesting approximation results and problems, and many applications in analysis, integral and stochastic geometry are established.

These properties and results above can be found in the following references, or in the related comprehensive representation, namely in Section 10.2 and subsection 10.6.2 in this book here.

ZONOIDS: SURVEYS AND BOOK PARTS

Also for zonoids the two most important *surveys*, from the viewpoint of convexity, are [2176] of Schneider and Weil and the handbook article [997] of Goodey and Weil. The paper [422] of Bolker is an essential forerunner. Furthermore, Gruber [1059] and Lindenstrauss and Milman [1579] discuss *approximation and symmetrization problems around zonoids*. In the handbook article [1447], Koldobsky and König deal with approximations of zonoids by zonotopes. In the handbook article [2182] of Schneider and Wieacker translative integral geometry and associated *auxiliary zonoids* are discussed, and in the deep survey [2161], Schneider discusses the developments around generalizations of the classical Crofton formula from integral geometry in Euclidean space to *finite-dimensional*

normed spaces and projective Finsler spaces, showing also connections to zonoids. More surveys are presented in our subsection 10.6.2 below.

We switch to respective *book parts*. In Thompson's monograph [2361] on Minkowski geometry zonoids occur at many places: e.g., in Sections 5.4 and 6.2 (to get *isoperimetrices*), in Section 6.4 (*integral geometry*), in Sections 2.3 and 6.5 (*inequalities*), and also in further ones. Gardner [930] discusses zonoids mainly in Chapter 4 (as *projection bodies*) and in Chapter 9 (as *centroid bodies* and in *affine isoperimetric inequalities*). In Schneider's book [2167] Section 3.5 deals comprehensively with zonoids, and they are also treated by Brazitikos et al. [496] in Chapter 4 (regarding the *hyperplane conjecture*) and Chapter 7 (as *centroid bodies*). Artstein-Avidan et al. investigate *projection bodies* in Section 2.5 of [133], and in subsection 4.6.2 of [134] regarding *Shephard's projection problem*. Again we emphasize that many more references (and also various recent research papers) are given below, in our subsection 10.6.1 (see also Section 10.2).

2.9.2 The Hausdorff metric in convexity

The notions of *Hausdorff metric* and *Hausdorff distance* are useful tools in convex geometry, in order to get, e.g., a metric space of convex bodies and in connection with Blaschke's selection theorem. It is, as distance of two sets X and Y, defined via parallel bodies obtained as Minkowski sums of balls and the considered sets. More precisely, for B being the unit ball of Euclidean n-space, the Hausdorff distance of the sets X and Y is the largest lower bound of the numbers ρ for which $X + \rho B \supset Y$ and $Y + \rho B \supset X$ holds. And if X and Y are compact, this largest lower bound is a minimum. This notion is introduced in many books and expository papers. Below we first list those books where its *definitions, properties, and applications* are presented in a broader and geometric way. As the name above suggests, Hausdorff studied first this distance type, but one should mention that Pompéiu [1986] considered earlier a non-symmetric version. Since also Blaschke studied this "deviation" already early in [358], also the name *Pompéiu–Hausdorff–Blaschke distance* is occasionally used (see, e.g., page 7 of Toth's book [2381]). Nevertheless, in the theory of convex sets Blaschke was the first who used this concept, namely in §17 and §18 of his classical book [358].

BOOK PARTS

In §17 and §18 of his book [358], Blaschke introduces the "deviation" or "neighbouring measure" as the measure given above, and in §18 he gives an easy proof of his related *selection theorem*. Also based on the Hausdorff distance, a detailed introduction to *approximation properties* of convex bodies with Blaschke's selection theorem is given in §14 of the monograph [1531] of Leichtweiss. Benson [273], in Section 18, and Guggenheimer [1102], at the end of Section 5, nicely present the concept and its way to Blaschke's selection theorem; interesting related exercises are given. On the same level, really nice representations are given by Kelly and Weiss [1359], Section 6.3, Lay [1521], Sections 14 and 15, and Webster [2437], Section 2.7. Also the general geometry book of Berger treats the Hausdorff distance in even three parts: regarding the definition and basic properties in Section 9.11, for the notion of volume in Section 9.12, and for approximation of convex bodies by polytopes in Section 12.9. Similarly, for the *convergence of convex polytopes to general convex bodies*, also Ewald studies the Hausdorff distance in Section 2 of Chapter IV of his monograph [826]. In Section 2.3 of [1042], Groemer comprehensively discusses metrics for sets of convex bodies, particularly the Hausdorff one. With detailed proofs he shows the way to certain interesting inequalities, to Blaschke's selection theorem and to basic theorems on approximation topics. Also an interesting discussion on the role of *parallel bodies* themselves is given.

In Chapter 4 of his book [1943] on metric geometry and convexity (in the tradition of Menger, Busemann, and A. D. Alexandrov), Papadopoulos introduces and studies Hausdorff and Busemann–Hausdorff distances. Relatively short representations of the role of Hausdorff metric in convexity are given in the monographs [930] (Section 0.4) and [1070] (Section 6.1), whereas Toth [2381] gives a wider discussion in Part B of Section 1.1. The most comprehensive representation of results around the Hausdorff distance and the metric space of non-empty compact (convex) sets created by it is Section 1.8 of Schneider's monograph [2167]. Many basic theorems are proved there, showing the *completeness of this space, properties of its compact elements, the closedness of the subspace of convex bodies, Blaschke's selection theorem, the role of separation properties and sequences of support functions, approximation topics, and the continuity of the volume functional*. The comprehensive notes to this section collect related results in many directions, different to what was mentioned here up to now. For example, the author surveys the connections of the following topics to the notion of Hausdorff metric: *convergence results, topological aspects of respective metric spaces, embedding theorems, further metrics* (e.g., the symmetric difference metric), *isometries, metric convexity* (due to Menger), *metric entropy, affine equivalence classes of convex bodies and of compact sets, applications of the Steiner point, and Minkowski additive selections*. The whole Section 3.1 of the book [1263] of Hug and Weil is dedicated to the Hausdorff metric and the corresponding metric space of convex bodies; interesting exercises are given, too.

Finally we mention here the book [2220] of Sendov, giving a systematic representation of the fundamental problems and results regarding the *approximation of functions and point sets based on the Hausdorff distance*, mainly of the respective Bulgarian school. The author presents the development of about 20 years in this direction (see also the expositions [2219] and [1958]). However, since the discussed Hausdorff approximation refers to bounded, closed, and connected point sets, the relation to convex geometry is only conditionally available.

SURVEYS

Gruber, in his handbook survey [1065] on spaces of convex bodies, studies in Section 4 some suitable metrics (Hausdorff, Eggleston, symmetric difference, Banach–Mazur, etc.) and gives several related *characterization theorems*. For example, he characterizes isometries with respect to the Hausdorff metric and the symmetric difference metric, and in Section 5 Blaschke's selection theorem is shortly surveyed. Also in his handbook article [1059] on *approximation of convex bodies*, the Hausdorff distance plays, besides other metrics, an important role, e.g., in view of *asymptotic estimates* (in addition, cf. [1066] and [1067]). And in Section 8 of his survey [1062] on *Baire category results* he discusses approximation of convex bodies by polytopes; again results using the Hausdorff distance are presented. Rubinov and Vladimirov [2069] give a survey on set differences referring to n-dimensional compact convex sets and metric spaces of convex compacta, with applications to *various types of functions* (e.g., quasi-differentiable functions), to *set-valued mappings*, and *nonsmooth optimization*. At first, they study ways of defining such set differences, where also differences of support functions play an important role. Then close relations between the Hausdorff metric and two further similar metrics on the family of convex compacta are discussed. These metrics are used to develop a general concept of *derivative of set-valued mappings* with values from the set of all convex compacta. Properties like *convergence, continuity, and differentiability* regarding these metrics, particularly the subcase of polyhedral-valued mappings, are investigated, always in comparison with the Hausdorff metric.

In the *combinatorial geometry of convex bodies* and related subfields there are (searched) numbers associated with convex bodies. Examples are *maximal (respectively, minimal) congruent/translative/lattice packing (respectively, covering) densities, different types of kissing numbers*, and numbers like those coming from *Hadwiger's covering conjecture and Borsuk's partition problem*. Since the methods and techniques used for them are not "unified", although the same objects are studied, Zong [2574] tries to develop a theory that gives to all these problems a common fundament

in the setting of spaces of convex bodies, of centrally symmetric convex bodies, and of affine equivalence classes of convex bodies. For this purpose, the Hausdorff metric, the Asplund metric, and the Banach–Mazur metric are basically used.

RESEARCH ARTICLES

Now we present some recent research articles strongly related to the notion of Hausdorff metric and not cited in the references above. Vassallo [2396] presents a result for gauges, i.e., he goes two ways of *approximating compact sets by balls of Minkowski spaces* whose unit ball B need not be centered at the origin. The first way is based on the Hausdorff metric.

Groemer [1044] derives several explicit *inequalities for pairs of convex bodies* combining the Hausdorff distance between them with their symmetric difference and a corresponding surface area deviation measure. Quantities occurring (besides the three mentioned metrics) in these inequalities are volume and surface area of the unit ball, the larger diameter, and the inradius of the intersection (if this is not empty). As tools, for instance, *quermassintegrals* and *radial distance* between both bodies are used.

K. Böröczky [390] studies the *approximation of n-dimensional smooth convex bodies by polytopes* for the Hausdorff metric, the Banach–Mazur distance, and the Schneider distance. He gives explicit arguments for the Hausdorff metric, and the changes for the other two metrics are then indicated. For $n = 3$ also a lower bound on the Hausdorff distance is given.

In [1436], König studies the *computational complexity of determining the Hausdorff distance of two n-dimensional polytopes* given in different presentations. It turns out that this can be computed, for most common norms, in polynomial time if both polytopes are given in vertex representation. If at least one polytope is given in half-space representation, then this problem is, in general, NP-hard. The author considers also a matching problem searching a homothetical transformation of a convex body minimizing the Hausdorff distance to another body. On these lines also John-type and Helly-type results are derived.

Ritoré and Vernadakis [2037] deal with *isoperimetric inequalities* regarding convex bodies and give first some technical results on the *connection between Hausdorff convergence and Lipschitz convergence* for a sequence of convex bodies. More precisely, they show that a sequence of convex bodies converging, in the Hausdorff metric, to a convex body also converges in Lipschitz distance. From this a uniform non-optimal isoperimetric inequality for all convex bodies follows, with bounded quotient circumradius/inradius. And if a sequence of convex sets converges in the Hausdorff distance, then the sequence of isoperimetric regions converges also for the Hausdorff distance, together with their boundaries. Further results on isoperimetric questions are obtained.

Brunel [509] studies the Hausdorff distance between a random polytope, defined as the convex hull of i.i.d. random points, and the convex hull of the support of their distribution. As examples we mention *uniform distributions on convex bodies or on the boundary of convex bodies, and projections of uniform distributions* on higher dimensional convex bodies. For uniform distributions, these random polytopes achieve their best statistical exactness under the Hausdorff metric if the boundary of the support is smooth, and its worst statistical precision when the support is a polytope. An extension to a rescaled version of the Hausdorff metric for uniform distributions is given, and estimations on functionals of the support of a distribution, like mean width and diameter, are presented.

The authors of [1468] ask whether for two planar regions A and B, with Hausdorff distance 1, a third one exists, called S, with Hausdorff distance 1/2 to and from A and B? The existence of such a set, from now on called S, is confirmed, and it is shown that, depending on convexity of A and/or B, the set S can be convex, connected, or disconnected. The authors extend such *Hausdorff middles* to more than two given sets, explaining how to approximate or compute such a middle shape. Bounds on the Hausdorff distance between the middle set and the input sets are presented, too.

For G a subgroup of affine transformations, a *G-invariant Minkowski class* is a class of convex bodies closed in the Hausdorff metric and under Minkowski addition, dilatation, and transformations in G. Molchanov and Nagel [1823] deal with the question of how rich a G-invariant Minkowski class (usually also allowing differences of support functions) generated by an essentially smaller family of convex bodies could be, namely when G is a group of diagonal matrices.

2.9.3 Steiner symmetrizations

Steiner symmetrization is a useful tool in convexity; it was introduced in 1838 by Steiner (see [2293], but also [1192]) to solve the *isoperimetric problem*. This type of symmetrization fits best into the overlap of classical convexity and *geometric tomography*, as can be seen in Section 1.2 of Gardner's book [930] for the planar case; see also Chapter 2 for extensions to higher dimensions and the Notes to Chapters 1 and 2. (There one can find also a short, but very interesting description of Steiner's life and work.) In geometric tomography it is common to call the *Steiner symmetrals* (as the symmetrized copies of original convex bodies are usually named) also $X-rays$. Here also the so-called *shake* or "shakedown procedure" ("Schüttelung" in German) should be mentioned (see Note 2.4 of [930], and the end of Section 10.3 in [2167]). For tomographic problems it yields practically "the same X-ray", but for geometric problems (like the isoperimetric one), where the shape itself is essential, its use is quite different, since it presents a type of *antisymmetrization*. Besides geometric and tomographic viewpoints, the following survey (based on former overviews and papers) takes two further mathematical areas into consideration (in which Steiner symmetrizations became particularly important): analysis and PDE, in the first case often for functions, and in the second case going back to the inspiring monograph [1984] of Polya and Szegö.

In many books and surveys dedicated to convex geometry, one can find short sections introducing and applying also Steiner symmetrization. In the following references one can find more comprehensive representations of this transform, its properties, and applications.

BOOK CHAPTERS

We start with *books* which contain whole sections or even chapters on Steiner symmetrization.

Besides Sections 1 and 2 from Gardner's monograph [930], we list the following books. Blaschke [358] dedicates the whole of §19 to this type of symmetrization, but see also §15, §22, and §25 for various applications. Also Bonnesen and Fenchel [446] treat Steiner symmetrizations in the extra Section 40. Hadwiger [1134] derives properties of the Steiner symmetrization in subsections 4.5.1 and 6.4.4 (here discussing the *continuous Steiner symmetrization as needed in integral geometry*); see also Chapter 2 of his popular booklet [1133]. In Eggleston's book [803] the fourth section of Chapter 5 is completely devoted to *basic properties of Steiner symmetrals*.

Also Benson dedicates a whole section (Section 17 of his book [273]) to Steiner symmetrization and gives interesting exercises, and Guggenheimer discusses Steiner symmetrization in connection with *Brunn's theorem, the Löwner ellipsoid, and the isoperimetric problem* in Section 14 of [1102]. Interesting problems conclude this section. In §18 of his monograph [1531], Leichtweiss gives a very detailed, comprehensive representation of the Steiner symmetrization of compact sets, and in §19 a similarly detailed discussion of the *Schwarz symmetrization* (or Schwarz surrounding) at a line in n-space follows (see also Section 9.3 in [1070]). This notion comes from the following natural generalization called *k-symmetrization* of a compact set C. Here k-dimensional parallel flats, with $1 \le k \le n-1$, which are totally orthogonal to the symmetrizing $(n-k)$-flat F, replace the intersecting lines in the classical symmetrization procedure, and F replaces the hyperplane from this procedure (cf. Section 2.1 of [930]). Then the intersection of these parallel flats with C is replaced by an $(n-k)$-ball

of the same $(n-k)$-volume whose midpoint is their intersection with F. So we obtain for $k=1$ the classical case of Steiner symmetrization, and for $k=n-1$ the *Schwarz symmetrization*. We notice that the Schwarz symmetrization is comprehensively discussed already in §21 of Blaschke's book [358], in connection with Brunn's theorem, and in Section 41 of [446] (related to the Brunn–Minkowski theorem) by Bonnesen and Fenchel. It is also treated at the end of Section 6.5 of Webster's book [2437].

Coming back to Steiner symmetrizations, we mention §9 of Chapter 2 (treating the *Brunn–Minkowski inequality in spherical and hyperbolic space*) of the monograph [531] of Burago and Zalgaller, where k-symmetrizations and further related types of symmetrizations are discussed, also in non-Euclidean spaces. In his two-volume book [282] and [283], Berger discusses Steiner symmetrization not only once. In Section 9.13 he introduces this transform and derives, e.g., the *isodiametric inequality*, in Section 12.11 it is studied as a tool for solving the *isoperimetric problem*, and in Section 12.12 interesting related exercises are collected. Also in his book [286], Berger discusses the Steiner and the Schwarz symmetrization in subsection VII.5.A; see also V.11.

In the whole Section 6.6 of Webster's book [2437] properties of Steiner symmetrizations are studied, also regarding applications, basic geometric inequalities, and some nice exercises given there. Here we mention also the book [1465] of Krantz and Parks treating *fundamental geometric tools needed in geometric analysis*. After Chapter 6 of this book, completely dedicated to convexity, Chapter 7 is particularly devoted to Steiner symmetrization, its properties, and applications. Steiner symmetrization and Schwarz symmetrization are treated nicely by Moszyńska in Sections 4.2 and 4.3 of her book [1857].

In Gruber's monograph [1070], Chapter 9.1 refers to properties of Steiner symmetrization, including also the *polar body* in this setting and the sphericity theorem of Gross as one way (besides Blaschke's) to the *isodiametric, isoperimetric, and Brunn–Minkowski inequality*. (These inequalities are derived in [1070], but the equality cases are not studied there.) In Schneider's monograph [2167] the whole Section 10.3 is dedicated to Steiner symmetrization. The author discusses *iterated symmetrizations, convergence to balls, a typical proof scheme for applications, related characterizations of ellipsoids, the shaking procedure*, and (in the notes) numerous further topics and references. Hug and Weil [1263] treat Steiner symmetrization in their Section 3.6, proving various properties and giving nice applications in the form of important *geometric inequalities*. Namely, based on Steiner symmetrization they nicely prove the isoperimetric, the Brunn–Minkowski, and the isodiametric inequality (see also again [1070]).

BOOKS FROM OTHER FIELDS

Some *books from other fields* containing chapters or sections on Steiner symmetrizations should be mentioned, too. For example, we should start with two classical books close to *mathematical physics and isoperimetry*, namely that of Polya and Szegö [1984] and that of Bandle [208], in which Steiner symmetrization and Schwarz symmetrization are discussed as important tools regarding isoperimetric problems. The book [2285] of Sperb is devoted to *partial differential equations, isoperimetric inequalities, and related geometrical problems*. This book shows also the interplay between isoperimetric inequalities and the *maximum principle*. In Chapter 11 a method for estimating the *Dirichlet integral* related to Steiner symmetrizations is discussed. The book [1350] of Kawohl is also related to *PDE*, where methods of *rearranging functions* play the main role. The variety of different methods of rearrangements gets here a unified treatment, and the Steiner symmetrization is considered as one of these important methods.

The classical book [1571] of Lieb and Loss is an introductory text on *analysis* starting with the most elementary facts on measure theory, the Fourier transform, common function spaces, Sobolev spaces and inequalities, potential theory, calculus of variations, and further topics. In the third chapter, the authors define Steiner and Schwarz symmetrization in view of this material, with a mixture of

geometry and integration theory, for *rearranging inequalities*. Baernstein's monograph [176] offers a comprehensive and deep study of symmetrizations, also mainly in view of several subfields of *analysis*. Also *non-Euclidean geometries* occur there. The author collects a broad variety of topics and tools in one unified framework; its complete Chapter 6 is dedicated to Steiner symmetrizations.

Kesavan [1371] gives an introduction to symmetrizations and isoperimetric problems, which is on the level of an advanced graduate course. In addition, one of the main topics of the book is the symmetrization of Schwarz. Its theory is developed, and interesting applications are given. The book [2313] of Stroock is a graduate text mainly on *measure theory*, in which applications in several partial fields of analysis are given. Hereby Steiner symmetrization, the isodiametric inequality, and surface measures play an essential role.

In his book [768], Dubinin studies (among other topics) symmetrization in *complex function theory*. In Chapter 4 several types of symmetrizations are discussed, among them also being Steiner symmetrization.

The classical book [1557] on *global affine differential geometry* gives, in Chapter 7, an introduction to *equiaffine convex geometry*. There also a part on Steiner symmetrization is presented.

SURVEYS ON STEINER AND SCHWARZ SYMMETRIZATION

Talenti's handbook article [2340] contains a thorough treatment of the *classical isoperimetric theorem in Euclidean space*. Giving a nice introduction into their basic properties, the author uses so-called Caccioppoli sets (a synonym is sets of [locally] finite perimeter). Also properties of Steiner symmetrals are carefully derived and then, partially, applied as tools. Therefore, the whole Section 3.8 of [2340] is devoted to Steiner symmetrization. Also Sangwine-Yager's handbook survey [2098] on *mixed volumes* should be mentioned here: its Section 5 is dedicated to symmetrization types, among them also being Steiner and Schwarz symmetrization.

In the exposition [175] Baernstein presents a unified approach to the study of different kinds of symmetrizations, like also the Steiner, Schwarz spherical cap and hyperbolic symmetrization in the study of PDEs. This is motivated by the fact that for *rearranging and symmetrizing given functions*, similar theorems hold as for various symmetrizations. This yields easier approaches to comparison results and a priori estimates for solutions of PDEs. General integral inequalities involving integrals of functions and their symmetrizations are also proved. Similarly, Kesavan [1370] brings together various results on *comparison theorems* and *isoperimetric inequalities associated with elliptic PDEs*, which can be obtained via Schwarz symmetrization.

One section of Klartag's survey [1403] treats *minimal numbers of Steiner or Minkowski symmetrizations* which transform a convex body into another one being close to the Euclidean ball, with an error bounded by some given $\varepsilon > 0$.

Fusco [913] gives a broad survey on applications of (Steiner) symmetrizations, such as the *solution of the isoperimetric problem, applications to PDEs, calculus of variations, and to integral inequalities* (e.g., of Poincaré and Sobolev).

As the headline says, the author of the (strongly historical) survey [1192] concentrates on *elementary geometric methods used to solve the isoperimetric problem*, and in this setting also proofs are given. The author emphasizes Gergonne's contribution here, who published, under the name "a subscriber" in *Ann. Math. Pures Appl. [Ann. Gergonne]* **4** (1813/14), 338-343 (see MR1555764), a proof of the isoperimetric problem based on Steiner symmetrization, as used later also by Steiner. But Steiner did never cite Gergonne. The author sums up that we will probably never know whether Steiner knew this paper of Gergonne.

Solynin presents with [2280] a research paper, but at the same time this paper is a broad and nice exposition on important aspects of Steiner symmetrization, its generalizations and applications. The material is clearly presented in 12 parts. First a one-parameter family of 1-symmetrizations (of functions and sets) is considered, and second continuous k-symmetrizations are applied and suitably used, for any $1 \leq k \leq n$. This yields a continuous path along which all basic characteristics of sets

and functions vary monotonically, finally leading to *continuous versions of certain convolution type inequalities and comparison theorems for solutions of various PDEs*.

In the survey [560] of Cagnetti classical and recent results on rigidity problems in symmetrization theory are reviewed. The studied symmetrization types are Steiner symmetrization and the Ehrhard symmetrization, and *rigidity* means here to understand whether the symmetrization procedure is strictly monotone when applied to objects which are still not symmetric (e.g., the perimeter regarding Steiner symmetrization). In particular, *measure-theoretical and geometric features* of those sets whose perimeters are invariant with respect to the used symmetrization procedures are searched. The author clearly introduces the needed tools and notions, and the background results related to the two types of symmetrizations are comprehensively presented. He shows the recent characterizations of sets whose perimeter is invariant under these symmetrizations. Interesting open problems conclude the paper.

SELECTED RESEARCH ARTICLES

We continue with selected *research papers on Steiner symmetrization* which are mainly not cited in the above book and survey references, and we start with Tsolomitis [2384]. He deals with the problem of how often a symmetrization procedure has to be applied to a given n-dimensional convex body such that the finally resulting body is in some well-defined sense suitably close to a Euclidean ball. More precisely, he asks for an *explicit upper bound on the needed number of symmetrizations*, depending only on n and the desired distance. The used symmetrization procedure is the classical k-symmetrization (see above, Section 2.1 of Gardner's book [930] and the strongly related paper [1530] of Leichtweiss, not cited in the books above). For the most interesting two subcases Tsolomitis shows that very few Steiner symmetrizations suffice to bring a symmetric convex body K close to a Euclidean ball, and that very few Schwarz symmetrizations suffice to bring K to a distance of at most $1 + \varepsilon$ from the ball, for every given ε.

Burchard [534] studies continuity, smoothing, and convergence properties of Steiner symmetrization in higher dimensions and shows that this transform is continuous in *Sobolev spaces* $W^{1,p}$, $1 \leq p < \infty$, in all dimensions. It follows that spherical symmetrization cannot be approximated in $W^{1,p}$ by sequences of Steiner symmetrizations. The paper [534] contains further interesting related results.

It is well known that Steiner symmetrization does not increase the perimeter of n-dimensional sets and that, moreover, sets preserving their perimeter (often also used for surface area in higher dimensions) under Steiner symmetrization need not have Steiner symmetry before the symmetrization procedure. (This is in contrast to Schwarz symmetrization, where only balls preserve perimeter.) In [620] Chlebík, Cianchi, and Fusco successfully study for which sets the perimeter is preserved under Steiner symmetrization. They give a characterization, for which geometric measure theory is used. Here we also mention the paper [625], where the case of equality is discussed for the analogous symmetrization of *functions* (rather than of sets).

Leichtweiss [1538] investigates analogues of Steiner symmetrals in *spherical and hyperbolic geometry*. The correspondingly defined transforms preserve volume, but in general not convexity. In a detailed manner, the author studies the planar case. He also derives a condition for the preservation of local convexity.

Bianchi et al. [330] consider the *convergence of iterates of Steiner symmetrals* applied to non-empty compact sets in Euclidean n-space. Whereas in such sequences any given compact set converges to a ball for most sequences of directions, they diverge for direction sequences whose differences are square summable. The latter can be changed if the successive symmetrizations are followed by suitable rotations. The limit of this modified iteration need not be an ellipsoid or a convex set.

Barchiesi, Cagnetti, and Fusco study in [227] Steiner k-symmetrizations. They characterize the sets whose perimeter is preserved under these symmetrizations (see also [620] discussed above), and they use regularity properties of the barycenter of the sections generated by the symmetrization procedure. The classical Steiner symmetrization (i.e., $k = 1$) is particularly studied. With their results one can

obtain a *quantitative isoperimetric estimate for convex sets*, which is possibly the first result of this type regarding Steiner symmetrization. The inequality $P(S(A)) \leq P(A)$, where $P(A)$ denotes the perimeter of A and $S(A)$ is the Steiner symmetral of A, is proved if the one-dimensional sections of A are segments and, for higher codimension, if A is a convex set.

Burchard and Chambers [535] prove that Steiner symmetrizations of every compact subset of Euclidean n-space in n linearly independent directions yield a set of finite perimeter. The proof is based on estimating the outer Minkowski perimeter. Interesting applications are derived, too.

Generalizing earlier results, Volčič [2408] shows the following: If S denotes a countable and dense set of directions in Euclidean n-space, then there exists an ordering in this direction set such that the sequence of corresponding Steiner symmetrals converges, from any compact set, in the Hausdorff metric to the ball of equal volume centered at the origin.

In the important paper [332], Bianchi et al. emphasize that they are not aware of *characteristic properties of Steiner symmetrizations*, and one aim of them is to provide some. These characterizations relate to a general notion called k-symmetrization (with respect to an k-dimensional subspace H) which is also introduced in [332]. For example, the authors show that Steiner symmetrization is the unique $(n-1)$-symmetrization of convex bodies in n-space that is monotonic, volume preserving, and either invariant on H-symmetric spherical cylinders or projection invariant. Further properties considered in this interesting paper regarding such characterizations are preservation of suitable functionals, idempotence, and several invariance properties regarding H-symmetry, projections, and translations. Eight main such properties are determined as the most useful ones, and it is checked which of them are, or are not, enjoyed by the most popular symmetrizations. The latter are classical transforms like central, Steiner, Schwarz, Minkowski, and Blaschke symmetrizations. And even more symmetrizations are checked, like further generalizations of the Steiner and the Minkowski symmetrization. Results similar to those on Steiner symmetrals are obtained for Minkowski symmetrization, where "volume preserving" is replaced by "mean width preserving". Also duality aspects are studied.

From a result of Kiselman it follows that understanding the *smoothness of Steiner symmetrals* is a hard problem, since the Steiner symmetral of a convex body of class C^∞ need not even be of class C^2. Lin [1573] shows that if the boundary of the original convex body is of C^2 class and of positive Gaussian curvature, then the same holds for its Steiner symmetral. The author poses the problem to study analogous questions for higher regularity classes.

Meyer and Reisner [1775] show that the ellipsoids are the only local maximizers of the *volume product* of a convex body and its polar reciprocal. Strengthening previous results, their technique is strongly based on shadow systems and Steiner symmetrization.

Motivated by former results on sequences of Steiner symmetrizations of convex sets converging to a Euclidean ball, Florentin and Segal [883] ask for the *rate of such convergence*. They provide a direct estimate for it in the Nikodym pseudo-metric and give also further related results.

In [1590] the authors prove that any n-dimensional *ellipsoid* can be transformed by $n-1$ Steiner symmetrizations into a Euclidean ball. The proof is constructive, identifying the directions of these successive symmetrizations. More precisely, an algorithm performing these $n-1$ iteration steps is presented.

Buttazzo and Pratelli [555] treat the interplay between *discrete and continuous Steiner symmetrizations*, where polyhedral sets play an essential role. This yields a sharp characterization of the *Blaschke–Santaló diagram* of torsion and eigenvalue.

Bianchi, Gardner, and Gronchi continue their important contributions from [332] in the more recent paper [333]. Here they study further the Steiner and the Schwarz symmetrization and numerous "relatives" (such as Minkowski-, Blaschke-, inner and outer rotational symmetrizations) of them. Their main interest refers to the *convergence of successive symmetrals* generated by a sequence of k-dimensional subspaces. They call such a sequence *universal for a family of given sets* if the successive symmetrals of any set in this family converge to a ball centered at the origin. Combining two types of results, new universal sequences for the main symmetrizations, for all valid dimensions k of the

subspaces, are discovered. Among other results, finite sets F of subspaces are found such that reflection symmetry (or rotational symmetry) with respect to each subspace in F implies full rotational symmetry. Furthermore, it is shown that for Steiner, Schwarz, and Minkowski symmetrizations a sequence of generating k-dimensional subspaces is universal for the class of compact sets if and only if it is universal for the class of compact convex sets. The authors derive many further essential results and give also a comprehensive view on related studies.

PLATE 3

Schlegel diagrams - a look into 4-space

Schlegel diagram of the 4-cube

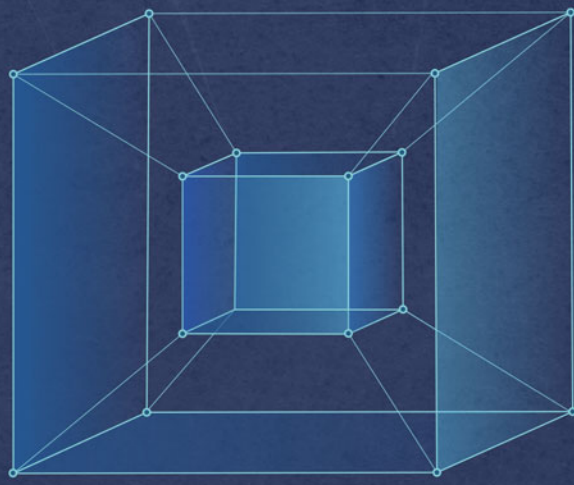
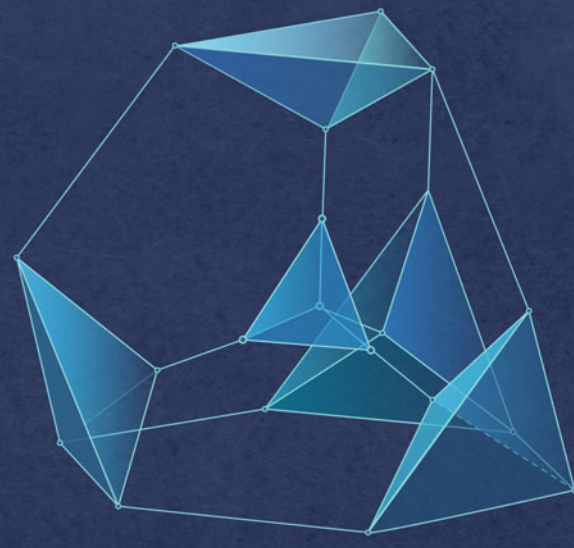


Figure 7

Figure 8

Schlegel diagram of a truncated 4-simplex



Schlegel diagrams - a look into 4-space ▸ (see, e.g., 2.9.1.4)

Chapter 3
A first look into polytopes

"Geometry existed before creation."
— Plato

Being objects of the elementary geometry, polygons in $\mathbb{R}^2$ and polyhedra in $\mathbb{R}^3$ can be intuitively described as convex figures whose boundaries are composed solely by "flat" pieces. These objects, as well as their higher dimensional analogues, can be easily characterized as the convex hulls of finitely many points. This is the definition of *convex polytopes*, which we simply call *polytopes*.

There is a very vast and rich theory on convex polytopes, and we refer the reader to the great monographs [1085] of Grünbaum and [2560] of Ziegler, which give more complete introductions to the subject. Here, we restrict ourselves to the basic theory that we need to develop the Brunn–Minkowski theory (for that sake, we follow mainly Schneider [2167]). Namely, when studying mixed volumes (and also other concepts), it is often easier to consider first polytopes, and then to obtain the general case by approximations. This is the approach taken by Artstein-Avidan et al. [133], Gruber [1070], Martini et al. [1713], and Schneider [2167], and we will follow these references.

In this spirit, we introduce basic facts about polytopes, and we briefly investigate their facial structures. We show that polytopes can be characterized as (bounded) intersections of finitely many half-spaces. As a consequence, we get the particularly useful fact that a polytope can be described by a certain set of normal directions together with their respective "weights".

Moving forward, we investigate *strongly isomorphic polytopes* which, roughly speaking, are equivalence classes of polytopes with "the same combinatorial type". This is an important step in the proof of the Alexandrov–Fenchel inequality that will be given later (in Chapter 8). Last, we introduce some results for approximating general convex bodies by polytopes. Again, the subject of approximations by polytopes is vast and interesting by itself, but here we only present the results that will be needed later in the book.

Plate 3 shows a way to look at polytopes in 4-space, the so-called *Schlegel diagram*. For example, the small cube in the first figure there presents the "back side" of the depicted 4-cube.

3.1 Basic concepts

Recall that a *polytope* (or *convex polytope*) in $\mathbb{R}^n$ is the convex hull of finitely many points. A polytope is called *proper* if it is also a convex body (that is, if it has maximal affine dimension). From now on, the collection of all polytopes of $\mathbb{R}^n$ will be denoted by $\mathcal{P}^n$. Observe that **the polytopes are precisely the compact convex sets with finitely many extreme points**. This is an easy consequence of Theorem

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_3

2.4.3: if $P = \text{conv}\{x_1, \ldots, x_m\}$, then the extreme points of P must be among $x_1, \ldots, x_m$. On the other hand, any compact convex set which has finitely many extreme points is the convex hull of finitely many points, hence a polytope. The extreme points of a polytope P (which are also its exposed points, as it will be made clear by Corollary 3.1.2 below) are called the *vertices* of P. Next, we will prove that all proper faces of a polytope are exposed.

Proposition 3.1.1. *Consider a polytope $P \subseteq \mathbb{R}^n$, and let F be an exposed face of P. If F_1 is an exposed face of F, then F_1 is an exposed face of P (see Figure 3.1.1).*

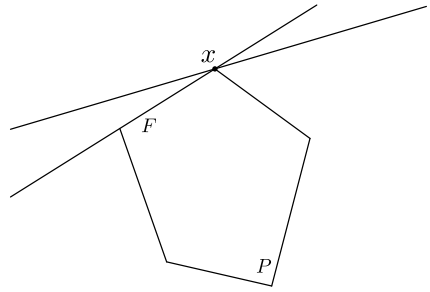


Fig. 3.1.1 The vertex x is an exposed face of the edge F, which is an exposed face of P.

Proof. In other words, we assume that $F = P \cap \mathbf{h}$ for some supporting hyperplane $\mathbf{h}$ of P and $F_1 = F \cap \mathbf{h}_1$ for some supporting hyperplane $\mathbf{h}_1$ of F, and we have to prove that there exists a supporting hyperplane $\mathbf{h}_0$ of P such that $F_1 = P \cap \mathbf{h}_0$. We assume that $\mathbf{h}_1 \neq \mathbf{h}$, since otherwise the result is redundant.

Of course, we may assume that $0_{\mathbb{R}^n} \in F_1$. Hence we write $\mathbf{h} = \mathbf{h}(u, 0) = u^\perp$ for some $u \in \mathbb{S}^{n-1}$, and we can assume that $P \subseteq \mathbf{h}^-$. We also write $\mathbf{h}_1 = \mathbf{h}(v, 0)$ (with $F \subseteq \mathbf{h}_1^-$) for some $v \in \mathbb{S}^{n-1}$, and we claim that we may suppose that $v \in u^\perp$ $(= \mathbf{h})$. In fact, since $\mathbf{h_1} \neq \mathbf{h}$, then the intersection $\mathbf{h} \cap \mathbf{h}_1$ is an $(n-2)$-dimensional vector subspace, and then we may replace $\mathbf{h}_1$ by the $(n-1)$-dimensional vector subspace which contains $\mathbf{h}_2 := \mathbf{h} \cap \mathbf{h}_1$ and is orthogonal to u (or, equivalently, replace v by its projection onto $u^\perp$). Let

$$\alpha_0 := \max\left\{-\frac{\langle x, v\rangle}{\langle x, u\rangle} : x \in (\text{ext}P) \setminus F\right\},$$

where we notice very carefully that $\langle x, u\rangle \neq 0$ if $x \in (\text{ext}P) \setminus F$, since $F = P \cap u^\perp$. It follows that $\langle x, u\rangle < 0$ for every $x \in (\text{ext}P) \setminus F$, because $\text{ext}P \subseteq P \subseteq \mathbf{h}^-$. We also observe that α_0 is well-defined, because $\text{ext}P$ is a finite set.

Choose any positive number $\alpha > \alpha_0$. We claim that the hyperplane $\mathbf{h}_0 := \mathbf{h}(\alpha u + v, 0)$ has the desired properties. First, observe that $0_{\mathbb{R}^n} \in P \cap \mathbf{h}_0$. Now notice that if $x \in (\text{ext}P) \setminus F$, then

$$\langle x, \alpha u + v\rangle = \alpha\langle x, u\rangle + \langle x, v\rangle < \alpha_0\langle x, u\rangle + \langle x, v\rangle \leq 0,$$

where in the first inequality we recall that $\langle x, u\rangle < 0$ and the last inequality comes from the definition of α_0. On the other hand, if $z \in F$, then we have

$$\langle z, \alpha u + v\rangle = \langle z, v\rangle \leq 0,$$

where the equality justifies since $F \subseteq \mathbf{h} = u^\perp$ and the inequality comes from $F \subseteq \mathbf{h}_1^-$. Since

$$\text{ext}P \subseteq ((\text{ext}P) \setminus F) \cup F,$$

we have from Minkowski's theorem (Theorem 2.4.3) that all points of P can be written as convex combinations of the points of $(\text{ext}P) \setminus F$ together with points of F. Thus, from the inequalities above we have that $\langle x, \alpha u + v\rangle \leq 0$ for every $x \in P$, and hence $\mathbf{h}_0$ is indeed a supporting hyperplane of P.

It remains to prove that $F_1 = P \cap \mathbf{h}_0$. For that, we recall that $F_1 = P \cap \mathbf{h} \cap \mathbf{h}_1$. Hence, if $x \in F_1$, then $x \in P$ and $x \in \mathbf{h} \cap \mathbf{h}_1$. The latter means that $\langle x, u\rangle = \langle x, v\rangle = 0$, from where we get that $\langle x, \alpha u + v\rangle = 0$. Therefore, $x \in \mathbf{h}_0$. This shows that $F_1 \subseteq P \cap \mathbf{h}_0$. Now we prove the reverse inclusion assuming, by contradiction, that $P \cap \mathbf{h}_0 \nsubseteq F_1 = P \cap \mathbf{h} \cap \mathbf{h}_1$. If this is the case, then we may choose a point $x \in P \cap \mathbf{h}_0$ such that $x \notin \mathbf{h} \cap \mathbf{h}_1$. It follows that $\langle x, u\rangle \neq 0$ or $\langle x, v\rangle \neq 0$, and we claim that any of these implies that **both** hold. Indeed, this comes from $x \in \mathbf{h}_0$, which means that $\alpha\langle x, u\rangle + \langle x, v\rangle = 0$ (here we also recall that α is positive). Since $P \subseteq \mathbf{h}^-$ and $\langle x, u\rangle \neq 0$, we get that $\langle x, u\rangle < 0$, and consequently we obtain $\langle x, v\rangle > 0$. Thus

$$-\frac{\langle x, v\rangle}{\langle x, u\rangle} \leq \alpha_0 < \alpha = -\frac{\langle x, v\rangle}{\langle x, u\rangle},$$

and this contradiction concludes the proof.

□

Corollary 3.1.1. *Every proper face of a polytope is an exposed face.*

Proof. We proceed by induction on the dimension of the polytope P. A zero-dimensional polytope does not have proper faces, and if the dimension of P is 1, then P is a segment and the proper faces are the endpoints, which are clearly exposed (simply consider hyperplanes which are orthogonal to the segment). Now assume that $\dim P = k > 1$, and that the result is true for polytopes whose dimension is smaller than k. Let F_1 be a proper face of P. From the discussion before Proposition 2.4.7, we have that F_1 is contained in some proper exposed face F of P. We claim that F_1 is a face of F. In fact, if $x, y \in F$ are such that $(x + y)/2 \in F_1$, then in particular $x, y \in P$, and since F_1 is a face of P we get that $x, y \in F_1$. If $F_1 = F$, there is nothing to prove. If $F_1 \neq F$, then F_1 is a proper face of F, and since $\dim F < k$ (because F is a proper face of P) we get from the induction hypothesis that F_1 is an exposed face of F. Then, from the previous proposition, we get that F_1 is an exposed face of P.

□

Remark 3.1.1. If P is a proper polytope, then it is clear that, conversely, every exposed face of P is a proper face.

Corollary 3.1.2. *Any extreme point of a polytope is an exposed point. In other words, for every vertice x of a polytope P there exists a supporting hyperplane $\mathbf{h}$ of P such that $\{x\} = P \cap \mathbf{h}$.*

Proof. We know that a point x of any given closed convex set C is extreme if and only if $\{x\}$ is a face of C. Since any proper face of a polytope is exposed, the result follows.

□

Corollary 3.1.3. *The relative boundary of a polytope is the union of its proper faces.*

Proof. This comes straightforwardly from Item **(b)** of Proposition 2.4.6 and Corollary 2.4.3.

□

Summarizing, any proper face of P is the intersection of P with some supporting hyperplane (because it is exposed). Conversely, from Proposition 2.4.7 we have that any intersection of P with some of its supporting hyperplanes is a face. Hence the proper faces of a polytope P are precisely the intersections of P with its supporting hyperplanes (which do not contain P). We can go a little further and prove that these proper faces are also polytopes.

Lemma 3.1.1. *If* $\mathbf{h}$ *is a supporting hyperplane of a polytope* $P = \text{conv}\{x_1, \ldots, x_m\}$*, then*

$$P \cap \mathbf{h} = \text{conv}\Big(\mathbf{h} \cap \{x_1, \ldots, x_m\}\Big). \tag{3.1.1}$$

Proof. The inclusion $\text{conv}(\mathbf{h} \cap \{x_1, \ldots, x_m\}) \subseteq P \cap \mathbf{h}$ is immediate. To prove the reverse inclusion, we first claim that $\mathbf{h} \cap \{x_1, \ldots, x_m\} \neq \emptyset$. In fact, once $\mathbf{h} := \mathbf{h}(u, \alpha)$ supports P, we may assume that $x_1, \ldots, x_m \in \mathbf{h}^-$. If $\mathbf{h} \cap \{x_1, \ldots, x_m\} = \emptyset$, then $\langle u, x_j \rangle < 0$ for every $1 \leq j \leq m$. Let $\beta := \max\{\langle x_j, u \rangle : 1 \leq j \leq m\}$. Then $\beta < 0$, and for any $\lambda_1, \ldots, \lambda_m \geq 0$ with $\sum_{j=1}^m \lambda_j = 1$ we have

$$\langle \lambda_1 x_1 + \ldots + \lambda_m x_m, u \rangle = \lambda_1 \langle x_1, u \rangle + \ldots + \lambda_m \langle x_m, u \rangle \leq \beta(\lambda_1 + \ldots + \lambda_m) = \beta < 0,$$

yielding that $P = \text{conv}\{x_1, \ldots, x_m\} \subseteq \text{int}(\mathbf{h}^-)$, and this contradicts the fact that $\mathbf{h}$ is a supporting hyperplane of P.

Now that we know that the intersection between $\mathbf{h}$ and $\{x_1, \ldots, x_m\}$ is non-empty, by renumbering the points if necessary we may assume that $\mathbf{h} \cap \{x_1, \ldots, x_m\} = \{x_1, \ldots, x_k\}$. Thus, $\langle x_j, u \rangle = \alpha$ for all $1 \leq j \leq k$. If $k = m$, then there is nothing to prove. So we assume that $k < m$. Let $x \in P \cap \mathbf{h}$ (in particular, $\langle x, u \rangle = \alpha$). Since x is a point of P, we may write

$$x = \alpha_1 x_1 + \ldots + \alpha_m x_m,$$

with $\alpha_j \geq 0$ for every $1 \leq j \leq m$ and $\sum_{j=1}^m \alpha_j = 1$. We have to prove that $\alpha_{k+1} = \ldots = \alpha_m = 0$. To do so, we notice that

$$\begin{aligned} \alpha = \langle x, u \rangle = \alpha_1 \langle x_1, u \rangle + \ldots + \alpha_k \langle x_k, u \rangle + \alpha_{k+1} \langle x_{k+1}, u \rangle + \ldots + \alpha_m \langle x_m, u \rangle = \\ = \alpha(\alpha_1 + \ldots + \alpha_k) + \alpha_{k+1} \langle x_{k+1}, u \rangle + \ldots + \alpha_m \langle x_m, u \rangle = \\ = \alpha + \alpha_{k+1}(\langle x_{k+1}, u \rangle - \alpha) + \ldots + \alpha_m(\langle x_m, u \rangle - \alpha). \end{aligned}$$

Therefore,

$$\alpha_{k+1}(\langle x_{k+1}, u \rangle - \alpha) + \ldots + \alpha_m(\langle x_m, u \rangle - \alpha) = 0.$$

Since each term between parentheses is negative, this equality is only possible if $\alpha_{k+1} = \ldots = \alpha_m = 0$. This concludes the proof.

□

Remark 3.1.2. It is not difficult to see that in (3.1.1) one can replace $\mathbf{h} \cap \{x_1, \ldots, x_m\}$ by $\mathbf{h} \cap \text{ext}P$. In fact, recalling that $\text{conv}\{x_1, \ldots, x_m\} = \text{conv}(\text{ext}P)$ we get

$$P \cap \mathbf{h} = \text{conv}(\text{ext}P) \cap \mathbf{h} = \text{conv}(\mathbf{h} \cap \text{ext}P),$$

where in the second inequality we are simply using (3.1.1) to a different set of points (which is, by the way, a subset of $\{x_1, \ldots, x_m\}$, as it was stated in Theorem 2.4.3).

Remark 3.1.3. Note that Lemma 3.1.1 should be compared with Exercise 2.2.

Hence any face of a polytope is indeed also a polytope. Moreover, since any face of P is the convex hull of a subset of $\text{ext}P$, it follows that a polytope has "at most" finitely many faces. Actually, a face of P is the convex hull of the points of $\text{ext}P$ which lie in F, as we prove next.

Lemma 3.1.2. *Let* F *be a face of a polytope* $P \subseteq \mathbb{R}^n$*. Then*

$$\text{ext}F = F \cap \text{ext}P.$$

As a consequence, $F = \text{conv}(F \cap \text{ext}P)$.

Proof. The inclusion $F \cap \mathrm{ext}P \subseteq \mathrm{ext}F$ is straightforward, because any point which does not lie in the relative interior of a segment of P also does not lie in the relative interior a segment of F.

For the reverse inclusion, let $z \in \mathrm{ext}F$. If $z \notin \mathrm{ext}P$, then there exist $x, y \in P$ such that $z \in \mathrm{seg}(x, y)$. Since F is a face, we get that $\mathrm{seg}[x, y] \subseteq F$. Then z belongs to the relative interior of a segment of F, contradicting the fact that $z \in \mathrm{ext}F$.

For the last claim, we simply use Minkowski's theorem (Theorem 2.4.3) and write $F = \mathrm{conv}(\mathrm{ext}F) = \mathrm{conv}(F \cap \mathrm{ext}P)$.

□

Any face of a polytope P whose affine dimension equals $n - 1$ is called a *facet* of P, and for $n - 2$ we call it *subfacet*. The one-dimensional faces are called *edges*.

We define polytopes as the convex hulls of finitely many points, but there is also another useful equivalent definition: the polytopes are the convex bodies which can be written as the intersection of finitely many closed half-spaces. We discuss this next.

Proposition 3.1.2. *A polytope $P \subseteq \mathbb{R}^n$ is the intersection of finitely many closed half-spaces.*

Proof. We clearly may assume that P is proper, because any affine subspace can be written as the intersection of finitely many closed half-spaces. Let $F_1, \ldots, F_m$ be the facets of P, and for each $j \in \{1, \ldots, m\}$ let $\mathbf{h}_j$ be the (unique) supporting hyperplane such that $F_j = \mathbf{h}_j \cap P$ and $P \subseteq \mathbf{h}_j^-$. Then we trivially have the inclusion $P \subseteq \mathbf{h}_1^- \cap \ldots \cap \mathbf{h}_m^-$, and we claim that the reverse inclusion also holds. To prove that, let $x \in \mathbb{R}^n \setminus P$, and define

$$A = \bigcup_{x_1, \ldots, x_{n-1} \in \mathrm{ext}P} \mathrm{aff}\{x, x_1, \ldots, x_{n-1}\},$$

where we recall that $\mathrm{exp}P$ is the set of vertices of P. In other words, A is the union of the affine hulls determined by x and $n - 1$ vertices of P. In particular, A is a union of finitely many affine subspaces with non-maximal dimension, and hence A does not contain an open ball. It follows that we may choose a point $y \in (\mathrm{int}P) \setminus A$. The segment $\mathrm{seg}[x, y]$ intersects the boundary of P at a point z, say, and Corollary 3.1.3 guarantees that z belongs to some proper face F of P. We will show that F is indeed a facet of P. If this is not the case, each point of F is in the convex hull of at most $n - 1$ vertices of P, which we denote by $z_1, \ldots, z_j$ (with $j \leq n - 1$), and hence z is a convex combination of these vertices. It follows that

$$y \in \mathrm{aff}\{x, z_1, \ldots, z_j\} \subseteq A,$$

and this is a contradiction. Hence F is a facet of P, and therefore $F = F_k$ for some $k = 1, \ldots, m$. Since $y \in \mathrm{int}P \subseteq \mathrm{int}\mathbf{h}_k^-$, we get that $x \notin \mathbf{h}_k^-$. This gives the inclusion $\mathbf{h}_1^- \cap \ldots \cap \mathbf{h}_m^- \subseteq P$, and the proof is complete.

□

Notice that we do not only prove that a proper polytope P is the intersection of finitely many supporting half-spaces, but also that these half-spaces can be assumed to be precisely the half-spaces determined by the supporting hyperplanes of the **facets** of P. If F is a facet of P, then $F = P \cap \mathbf{h}$ for some supporting hyperplane $\mathbf{h}$ of P (which is unique, because the intersection of two distinct hyperplanes cannot have affine dimension equal to $n - 1$). We write $\mathbf{h} = \mathbf{h}(u, \alpha)$ for some $u \in \mathbb{S}^{n-1}$ and $\alpha \in \mathbb{R}$. Up to changing the sign of u if necessary, we assume that $P \subseteq \mathbf{h}^-$ (by doing so, u is unique), and hence u is an outer normal of P at each point of F. For this reason, we call u the (*unit*) *outer normal of the facet* F. Of course, in this case we also have $\alpha = h_P(u)$. Hence the fact that P is the intersection of the half-spaces determined by its facets can be summarized as follows.

Corollary 3.1.4. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope whose facets are $F_1, \dots, F_m$. Denoting by u_j the outer normal of the facet F_j for each $j \in \{1, \dots, m\}$, we have*

$$P = \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_P(u_j)). \tag{3.1.2}$$

Proof. Simply notice that, in the proof of the previous proposition, and because of the terminology introduced above, we have $\mathbf{h}_j^- = \mathbf{h}^-(u_j, h_P(u_j))$ for each $j = 1, \dots, m$.

□

Remark 3.1.4. Because of (3.1.2), we get that a proper polytope P is uniquely determined by the set $\{u_1, \dots, u_m\} \subseteq \mathbb{S}^{n-1}$ of outer normal vectors of the facets and by the corresponding set of numbers $\{h_P(u_1), \dots, h_P(u_m)\}$. Geometrically, we may determine a polytope uniquely if we know what the "directions of its bounding hyperplanes" (given by the vectors u_j) and their correct translations (given by the numbers $h_P(u_j)$) are.

Corollary 3.1.5. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope. Then any proper face of P is contained in some facet of P.*

Proof. As usual, we let $F_1, \dots, F_m$ be the facets of P, and denote by $u_1, \dots, u_m$ their respective unit outer normals. Then we have equality (3.1.2), and $F_j = P \cap \mathbf{h}(u_j, h_P(u_j))$ for each $j = 1, \dots, m$. Observe that

$$\operatorname{int} P = \operatorname{int}\left(\bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_P(u_j))\right) \supseteq \bigcap_{j=1}^{m} \operatorname{int}\big(\mathbf{h}^-(u_j, h_P(u_j))\big),$$

and this gives that a boundary point of P cannot lie simultaneously in all strict half-spaces $\operatorname{int}\big(\mathbf{h}^-(u_j, h_P(u_j))\big)$. It follows that

$$\partial P \subseteq \left(\bigcup_{j=1}^{m} \mathbf{h}(u_j, h_P(u_j))\right).$$

Intersecting both sides with P we get

$$\partial P = \partial P \cap P = P \cap \left(\bigcup_{j=1}^{m} \mathbf{h}(u_j, h_P(u_j))\right) = \bigcup_{j=1}^{m} P \cap \mathbf{h}(u_j, h_P(u_j)) = \bigcup_{j=1}^{m} F_j,$$

and this means that the boundary of P is the union of its facets. In particular, we have that each vertex of P (a zero-dimensional face) is contained in some facet of P.

Let F be a proper face of P with dimension greater than 0, and choose $x \in \operatorname{relint} F$. Since x is a boundary point of P, we get that there exists $j \in \{1, \dots, m\}$ such that $x \in F_j$. Hence $x \in \mathbf{h}(u_j, h_P(u_j))$, and as usual we denote this hyperplane by $\mathbf{h}_j^-$. We claim that $F \subseteq \mathbf{h}_j$. We already have that $F \subseteq P \subseteq \mathbf{h}_j^-$, so we assume (by contradiction) that there exists a point $z \in F \cap \operatorname{int}(\mathbf{h}_j^-)$. Since F is convex, we have that $\operatorname{seg}[z, x] \subseteq F$. Hence any (relatively) open ball of $\operatorname{aff} F$ with center in x must contain a segment of the form $\operatorname{seg}[x, \delta(z - x)]$ for some $\delta > 0$, and thus it also contains the segment $\operatorname{seg}[x, -\delta(z - x)]$. Since $x \in \operatorname{relint} F$, there is some of these relatively open balls with center in x which is contained in F. Consequently, we get that there exists $\varepsilon > 0$ such that $x + \varepsilon(x - z) \in F$. Once $F \subseteq P$, we have $x + \varepsilon(x - z) \in P$, but on the other hand

$$\langle x + \varepsilon(x - z), u_j \rangle = \langle x, u_j \rangle + \varepsilon(\langle x, u_j \rangle - \langle z, u_j \rangle) = h_P(u_j) + \varepsilon(h_P(u_j) - \langle z, u_j \rangle) > \\ > h_P(u_j),$$

because $z \in \operatorname{int}(\mathbf{h}_j^-)$, and this means that $\langle z, u_j \rangle < h_P(u_j)$. This inequality gives that $x + \varepsilon(x - z) \in \operatorname{int}(\mathbf{h}_j^+)$, and this contradicts the inclusion $P \subseteq \mathbf{h}_j^-$. Hence $F \subseteq \mathbf{h}_j$, and this yields $F \subseteq P \cap \mathbf{h}_j = F_j$. □

Intuitively, the boundedness of a polytope means that it has to have a "sufficiently large" number of facets (for example, in $\mathbb{R}^2$ there are no polygons with less than three edges). Next we formalize this notion, also giving an application for the characterization of the polar of a proper polytope containing the origin as an interior point.

Proposition 3.1.3. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope with facets $F_1, \ldots, F_m$ whose corresponding unit outer normals are $u_1, \ldots, u_m$. Then we have*

$$\operatorname{pos}\{u_1, \ldots, u_m\} = \mathbb{R}^n. \tag{3.1.3}$$

If $0_{\mathbb{R}^n} \in \operatorname{int}(P)$, then $h_P(u_j) > 0$ for all $j = 1, \ldots, m$ and

$$P^\circ = \operatorname{conv}\left\{\frac{u_1}{h_P(u_1)}, \ldots, \frac{u_m}{h_P(u_m)}\right\}. \tag{3.1.4}$$

In particular, the polar of a proper polytope having the origin as an interior point is also a proper polytope with the origin as an interior point.

Proof. It is clear that a translation does not change the outer normals of the facets of a polytope, and hence we assume that $0_{\mathbb{R}^n} \in P$ to prove (3.1.3). Assume that $C := \operatorname{pos}\{u_1, \ldots, u_m\} \neq \mathbb{R}^n$. Then, since C is convex and closed, we have that its boundary is non-empty. Given a point $y \in \partial C$, there exists a hyperplane $\mathbf{h} = \mathbf{h}(u, \alpha)$ which supports C at y. We claim that $\alpha = 0$ (equivalently, that $0_{\mathbb{R}^n} \in \mathbf{h}$). In fact, up to changing the sign of u, we may assume that $C \subseteq \mathbf{h}^-$. Since $y \in C$, we have that $\lambda y \in C$ for every $\lambda \geq 0$, and hence $\langle \lambda y, u \rangle \leq \alpha = \langle y, u \rangle$ for any $\lambda \geq 0$. If $\alpha \neq 0$, then this inequality yields $\lambda \leq 1$ in the case where $\alpha > 0$ and $\lambda \geq 1$ if $\alpha < 0$. It follows that indeed $\alpha = 0$.

Now we observe that $\langle u_j, u \rangle \leq 0$ for each $j = 1, \ldots, m$ (because $u_j \in C \subseteq \mathbf{h}^-(u, 0)$ for each j). As a consequence

$$\langle u_j, \beta u \rangle \leq 0 = \langle u_j, 0_{\mathbb{R}^n} \rangle \leq \sup\{\langle u_j, z \rangle : z \in P\} = h_P(u_j)$$

for each $\beta \geq 0$, where we remember that we are assuming that $0_{\mathbb{R}^n} \in P$. Therefore, by (3.1.2) we get

$$\beta u \in \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_P(u_j)) = P$$

for every $\beta > 0$, and this contradicts the boundedness of P. This proves (3.1.3).

Now we prove (3.1.4). If $0_{\mathbb{R}^n} \in \operatorname{int} P$, then P contains a (closed) ball δB^n for some $\delta > 0$. Hence $h_P(u_j) \geq \langle \delta u_j, u_j \rangle = \delta > 0$ for each $j = 1, \ldots, m$. For simplicity of the notation, let

$$v_j = \frac{u_j}{h_P(u_j)},$$

for each j. If $\lambda_1 v_1 + \ldots + \lambda_m v_m$ is any convex combination, then for every $z \in P$ we have

$$\langle \lambda_1 v_1 + \ldots + \lambda_m v_m, z\rangle = \frac{\lambda_1}{h_P(u_1)}\langle u_1, z\rangle + \ldots + \frac{\lambda_m}{h_P(u_m)}\langle u_m, z\rangle \leq \lambda_1 + \ldots + \lambda_m = 1,$$

because $\langle u_j, z\rangle \leq h_P(u_j)$ for each j. It follows that any convex combination of $v_1, \ldots, v_m$ is in the polar body P°, that is,

$$\operatorname{conv}\{v_1, \ldots, v_m\} \subseteq P^\circ.$$

To prove the reverse inclusion, we show that $\mathbb{R}^n \setminus \operatorname{conv}\{v_1, \ldots, v_m\} \subseteq \mathbb{R}^n \setminus P^\circ$. Assume that $y \notin \operatorname{conv}\{v_1, \ldots, v_m\}$. Then there exists a hyperplane $\mathbf{h} = \mathbf{h}(v, \alpha)$ strongly separating y and $\operatorname{conv}\{v_1, \ldots, v_m\}$. Up to changing the sign of v, we may assume that the separation goes as

$$\langle y, v\rangle > \alpha, \quad \text{and} \quad \langle z, v\rangle < \alpha \ \ \forall\, z \in \operatorname{conv}\{v_1, \ldots, v_m\}. \tag{3.1.5}$$

Since

$$\operatorname{pos}\{v_1, \ldots, v_m\} = \operatorname{pos}\{u_1, \ldots, u_m\} = \mathbb{R}^n,$$

we claim that there exist numbers $\lambda_1, \ldots, \lambda_m \geq 0$, *not all simultaneously zero*, such that

$$0_{\mathbb{R}^n} = \sum_{j=1}^{m} \lambda_j v_j. \tag{3.1.6}$$

In fact, choose any non-zero vector $w \in \mathbb{R}^n$. Then both w and $-w$ are in $\operatorname{pos}\{v_1, \ldots, v_m\}$, from which it follows that there are numbers $\alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_m \geq 0$ such that

$$w = \sum_{j=1}^{m} \alpha_j v_j \quad \text{and} \quad -w = \sum_{j=1}^{m} \beta_j v_j.$$

Hence we simply put $\lambda_j = \alpha_j + \beta_j$. It is clear that some of the α_j's have to be non-zero, as well as some of the β's (because w is non-zero).

Once the λ_j's are not all zero (and all of them are non-negative), we may divide both sides of (3.1.6) by $\sum_{j=1}^m \lambda_j$, and we get that $0_{\mathbb{R}^n} \in \operatorname{conv}\{v_1, \ldots, v_m\}$. Thus, $0 = \langle 0_{\mathbb{R}^n}, v\rangle < \alpha$. Consequently, if we put $\tilde{v} = v/\alpha$, we get from (3.1.5) that $\langle z, \tilde{v}\rangle < 1$ for every $z \in \operatorname{conv}\{v_1, \ldots, v_m\}$. In particular, this gives that $\langle v_j, \tilde{v}\rangle < 1$ for any $j = 1, \ldots, m$, and since $h_P(u_j) > 0$, we get

$$\langle u_j, \tilde{v}\rangle = h_P(u_j)\langle v_j, \tilde{v}\rangle < h_P(u_j)$$

for each $j = 1, \ldots, m$. Thus,

$$\tilde{v} \in \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_P(u_j)) = P.$$

On the other hand, (3.1.5) also gives $\langle y, \tilde{v}\rangle > 1$, and hence $y \notin P^\circ$. This shows the remaining inclusion $P^\circ \subseteq \operatorname{conv}\{v_1, \ldots, v_m\}$.

□

We have proved previously that any polytope is the intersection of finitely many supporting half-spaces. We show now, as a consequence of the proposition above, that the converse is also true.

Corollary 3.1.6. *A bounded intersection of finitely many closed half-spaces of $\mathbb{R}^n$ is a polytope.*

Proof. Let $K = \bigcap_{j=1}^{m} \mathbf{h}_j^-$ be a bounded intersection of finitely many closed half-spaces. Without loss of generality, we assume that K has maximal affine dimension (otherwise we would restrict ourselves to $\mathrm{aff}K$). Hence K is a convex body (it is bounded, closed, convex, and has maximal affine dimension). It is clear that we may assume that $0 \in \mathrm{int}K$, because a convex set is a polytope if and only if any of its translates is a polytope. Thus writing $\mathbf{h}_j = \mathbf{h}(u_j, h_K(u_j))$ for each j, we get that all numbers $h_K(u_j)$ are positive. Therefore, using the same idea of the proof of Proposition 3.1.3, we obtain

$$K^\circ = \mathrm{conv}\left\{\frac{u_1}{h_K(u_1)}, \ldots, \frac{u_m}{h_K(u_m)}\right\},$$

which means that K° is a polytope, which is proper and contains the origin as an interior point (because K is a convex body with the origin in its interior). Consequently, from (3.1.2) it follows that K° can be written as the intersection of the supporting half-spaces of its facets:

$$K^\circ = \bigcap_{j=1}^{r} \mathbf{h}^-(v_j, h_{K^\circ}(v_j)).$$

Since $0_{\mathbb{R}^n} \in \mathrm{int}K^\circ$, we get that $h_{K^\circ}(v_j) > 0$ for each $j = 1, \ldots, r$. Finally, applying (3.1.4) again, we arrive at

$$K = K^{\circ\circ} = \mathrm{conv}\left\{\frac{v_1}{h_{K^\circ}(v_1)}, \ldots, \frac{v_r}{h_{K^\circ}(v_r)}\right\},$$

and this shows that K is a polytope.

□

Corollary 3.1.7. *Any non-empty intersection of finitely many polytopes is a polytope.*

Proof. Let $P_1, \ldots, P_m \subseteq \mathbb{R}^n$ be polytopes such that $P := P_1 \cap \ldots \cap P_m \neq \emptyset$. Each P_j is the intersection of finitely many bounding closed half-spaces, and hence P is a (bounded) intersection of finitely many half-spaces. Thus, P is a polytope.

□

We also have that the family $\mathcal{P}^n$ of polytopes of $\mathbb{R}^n$ is closed under Minkowski sums and scalar multiplications. We prove this next.

Lemma 3.1.3. *The Minkowski sum of finitely many polytopes is a polytope, and a scalar multiple of a polytope is a polytope.*

Proof. The fact that any multiple of a polytope is also a polytope is obvious, and hence we just have to prove that the Minkowski sum of finitely many polytopes is a polytope. It is clearly enough to prove that this holds for the Minkowski sum of two polytopes. Let P and Q be polytopes. Then P and Q have, each of them, at most finitely many extreme points.

We claim that any extreme point of $P+Q$ can be written as the sum of an extreme point of P with an extreme point of Q (compare with Exercise 2.11). Indeed, let x be an extreme point of $P+Q$, and write $x = a + b$ for some $a \in P$ and $b \in Q$. Suppose that $a \notin \mathrm{ext}(P)$. Then we may write $a = \lambda z_1 + (1-\lambda)z_2$ for some $\lambda \in (0,1)$ and some $z_1, z_2 \in P$. Thus, we have

$$x = \lambda z_1 + (1-\lambda)z_2 + b = \lambda(z_1 + b) + (1-\lambda)(z_2 + b),$$

and once $z_1 + b$ and $z_2 + b$ are points of $P + Q$, we get that x lies in the interior of a segment of $P + Q$, contradicting the fact that x is an extreme point of $P + Q$. This contradiction gives that $a \in \mathrm{ext}(P)$. Analogously, we prove that $b \in \mathrm{ext}(Q)$.

It is clear that $P + Q$ is a non-empty compact convex set, and from the argument above it follows that $P + Q$ has at most finitely many extreme points. Hence $P + Q$ is a polytope (recall that polytopes are precisely the non-empty compact convex sets having finitely many exposed points).

□

3.2 Faces and normal cones

Roughly speaking, this section is devoted to understanding how one can describe the structure of a polytope by looking at its (proper) faces. We already know that any proper face F of a polytope P can be written as $F = P \cap \mathbf{h}$ for some supporting hyperplane $\mathbf{h}$ of P. From Proposition 2.2.3, we have that for each non-zero vector $u \in \mathbb{R}^n$, P has a supporting hyperplane whose outer normal is u, which is clearly unique. We denote this by $\mathbf{h}(P, u)$. From uniqueness, we also can describe the face $P \cap \mathbf{h}(P, u)$ in the following way:

$$P \cap \mathbf{h}(P, u) = \big\{\text{boundary points of } P \text{ where } u \text{ is an outer normal}\big\}, \tag{3.2.1}$$

and the reader should keep this in mind from now on.

In what follows, for a given point x of a polytope P, we define the *smallest face of P containing x* to be the intersection of all faces of P that contain x. This is denoted by $\mathrm{SF}_P(x)$.

Lemma 3.2.1. *Let P be a polytope and $F \subseteq P$ be a proper face. For any $x \in P$, we have that $x \in \mathrm{relint} F$ if and only if $\mathrm{SF}_P(x) = F$. See Figure 3.2.1.*

Proof. Since $G := \mathrm{SF}_P(x)$ is an intersection of faces of P, we have that G is a face of P. We clearly have that $G \subseteq F$ (because F is among the faces which intersect up to give G). From Item **(b)** of Proposition 2.4.6, we get that G is a face of F. If $G \neq F$, then this face is proper, and hence from Item **(c)** of this same proposition, we obtain that $G \subseteq \mathrm{relbd}(F)$. It follows that $x \notin \mathrm{relint}(F)$, which is a contradiction. This shows that $G = F$.

For the converse, assume that $\mathrm{SF}_P(x) = F$. First, notice that if $x \in \mathrm{relint}(P)$, then P is the unique face containing x, and hence $\mathrm{SF}_P(x) = P$. Now let $x \in \mathrm{relbd}(P)$. By Corollary 2.4.3, x lies in the relative interior of some proper face G of P, from which we have just proved, $\mathrm{SF}_P(x) = G$, and this gives $F = G$. Hence $x \in \mathrm{relint}(F)$.

□

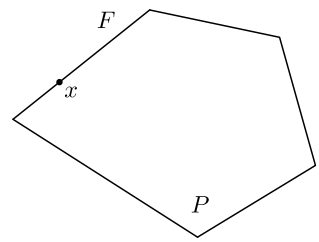


Fig. 3.2.1 The smallest face containing x is the edge F, since x is in the relative interior of F.

It is very useful to relate the concept of the smallest face of a point with that of the normal cone of a point, which was introduced in (2.4.5). We do this below.

Lemma 3.2.2. *If $x, y \in \partial P$ then $\mathrm{NC}_P(x) = \mathrm{NC}_P(y)$ if and only if $\mathrm{SF}_P(x) = \mathrm{SF}_P(y)$.*

Proof. If $\mathrm{NC}_P(x) = \mathrm{NC}_P(y)$, then any supporting hyperplane of P at x is a supporting hyperplane of P at y, and vice versa. It follows that x and y are contained in precisely the same faces of P, and this gives $\mathrm{SF}_P(x) = \mathrm{SF}_P(y)$.

Conversely, assume that $\mathrm{SF}_P(x) = \mathrm{SF}_P(y)$. Let $\mathbf{h}$ be any supporting hyperplane of P at x. Then x is a point of the face $P \cap \mathbf{h}$, and from the definition $\mathrm{SF}_P(x)$ we get that $P \cap \mathbf{h} \supseteq \mathrm{SF}_P(x)$. Then $y \in \mathrm{SF}_P(y) \subseteq P \cap \mathbf{h}$, and this gives that $\mathbf{h}$ also supports P at y. Consequently, any supporting hyperplane of P at x is also a supporting hyperplane of P at y, and this yields the inclusion $\mathrm{NC}_P(x) \subseteq \mathrm{NC}_P(y)$. The reverse inclusion follows analogously (starting with a supporting hyperplane at y).

□

We define and investigate an extension of the concept of normal cone of a point. Let $P \subseteq \mathbb{R}^n$ be a polytope, and let $F \subseteq P$ be a face. The *normal cone of the face* F is the set

$$\mathrm{NC}_P(F) = \mathrm{NC}_P(x),$$

where x is any point in $\mathrm{relint}F$. Below we prove that this definition does not depend on the chosen point $x \in \mathrm{relint}F$.

Corollary 3.2.1. *If F is a proper face of a polytope P and $x, y \in \mathrm{relint}F$, then $\mathrm{NC}_P(x) = \mathrm{NC}_P(y)$.*

Proof. We use Lemma 3.2.1 and Lemma 3.2.2. If $x, y \in \mathrm{relint}F$, then $\mathrm{SF}_P(x) = \mathrm{SF}_P(y) = F$. Consequently, $\mathrm{NC}_P(x) = \mathrm{NC}_P(y)$.

□

Remark 3.2.1. Given proper faces F, G of P we have that $\mathrm{NC}_P(F) = \mathrm{NC}_P(G)$ implies $F = G$. In fact, if $\mathrm{NC}_P(F) = \mathrm{NC}_P(G)$, then taking points $x \in \mathrm{relint}F$ and $y \in \mathrm{relint}G$ the definition gives $\mathrm{NC}_P(x) = \mathrm{NC}_P(y)$. From Lemma 3.2.1 we obtain that $\mathrm{SF}_P(x) = \mathrm{SF}_P(y)$, and that $\mathrm{SF}_P(x) = F$ and $\mathrm{SF}_P(y) = G$. Thus, $F = G$.

We know that any proper face F of a proper polytope P is contained in some facet of P (see Corollary 3.1.5). Next we will show that the normal cone of F is described by the outer normal vectors of the facets which contain F, and this will have a lot of important consequences. Before presenting that, we need a geometric lemma (which can also be used to prove that the normal cone of a face is well-defined).

Lemma 3.2.3. *Let F be a proper face of a polytope $P \subseteq \mathbb{R}^n$. If $x \in \mathrm{relint}F$ and $\mathbf{h}$ is a supporting hyperplane of P at x, then $F \subseteq \mathbf{h}$.*

Proof. Write $\mathbf{h} = \mathbf{h}(u, \alpha)$, and assume that $P \subseteq \mathbf{h}^-$. Up to a translation, we clearly may assume that $\alpha > 0$. By contradiction, assume also that $y \in F$ but $y \notin \mathbf{h}$. Then, since y is a point of P, we must have $y \in \mathrm{int}(\mathbf{h}^-)$, and this gives $\langle y, u\rangle < \alpha$. Since $x \in \mathrm{relint}F$, there exists a number $\varepsilon > 0$ such that

$$\mathrm{aff}(F) \cap B(x, \varepsilon) \subseteq F. \tag{3.2.2}$$

On the other hand, once $x, y \in F \subseteq \mathrm{aff}(F)$, we get that

$$x + \frac{\varepsilon}{2} \cdot \frac{x - y}{|x - y|} \in \mathrm{aff}(F) \cap B(x, \varepsilon),$$

and because $F \subseteq P \subseteq \mathbf{h}^-$, we obtain from (3.2.2) that

$$x + \frac{\varepsilon}{2} \cdot \frac{x - y}{|x - y|} \in \mathbf{h}^-. \tag{3.2.3}$$

But since $\alpha = \langle x, u\rangle$, we have

$$\left\langle x + \frac{\varepsilon}{2} \cdot \frac{x-y}{|x-y|}, u \right\rangle = \alpha + \frac{\varepsilon}{2|x-y|}\Big(\langle x, u\rangle - \langle y, u\rangle\Big) = \alpha + \frac{\varepsilon}{2|x-y|}\Big(\alpha - \langle y, u\rangle\Big) > 0,$$

and, since we are assuming that $\alpha > 0$, this is a contradiction to (3.2.3). Thus, we indeed have that $F \subseteq \mathbf{h}$.

□

Corollary 3.2.2. *Let $v \in \mathbb{S}^{n-1}$ and let F be a proper face of a polytope $P \subseteq \mathbb{R}^n$. Then $F \subseteq \mathbf{h}(P, v)$ if and only if $v \in \mathrm{NC}_P(F)$. This is also equivalent to* $\mathrm{aff}F \subseteq \mathbf{h}(P, v)$.

Proof. If $F \subseteq \mathbf{h}(P, v)$, then $\mathbf{h}(P, v)$ supports P at a point of $\mathrm{relint}F$. Then we have $v \in \mathrm{NC}_P(F)$ directly from the definition of normal cone of a face. Conversely, if $v \in \mathrm{NC}_P(F)$, then v is an outer normal of P at some point $z \in \mathrm{relint}F$. Hence $\mathbf{h}(P, v)$ is a supporting hyperplane of P at z, and it follows from the previous lemma that $F \subseteq \mathbf{h}(P, v)$.

The inclusion $\mathrm{aff}F \subseteq \mathbf{h}(P, v)$ trivially implies that $F \subseteq \mathbf{h}(P, v)$. On the other hand, if $F \subseteq \mathbf{h}(P, v)$, then $\mathbf{h}(P, v)$ is an affine subspace containing F, and hence it contains $\mathrm{aff}F$.

□

Theorem 3.2.1. *Let P be a proper polytope whose facets are $F_1, \ldots, F_m$, with corresponding unit outer normal vectors $u_1, \ldots, u_m$. For any proper face F of P we have*

$$\mathrm{NC}_P(F) = \mathrm{pos}\{u_i : F \subseteq F_i\}.$$

See Figure 3.2.2.

Proof. Of course, we may assume that, without loss of generality, $0_{\mathbb{R}^n} \in \mathrm{int}P$. By doing so, we have $h_P(u_j) > 0$ for each $j \in \{1, \ldots, m\}$. For the sake of simplicity, we put

$$v_j = \frac{u_j}{h_P(u_j)},$$

for each $j \in \{1, \ldots, m\}$. From (3.1.4) we have that $P^\circ = \mathrm{conv}\{v_1, \ldots, v_m\}$. For a proper face F of P, we define the *conjugate face* of F to be

$$\hat{F} := \{z \in P^\circ : \langle z, y\rangle = 1 \text{ for all } y \in F\}.$$

This is clearly closed and convex. Now we divide the proof of the theorem into two parts, by proving two claims. The first one describes the conjugate face of F in terms of the facets of P containing F.

1st claim. $\hat{F} = \mathrm{conv}\{v_i : F \subseteq F_i\}$.

We first notice that the right-hand side is non-empty, because every face of P is contained in some facet of P (in particular, proving the set equality above gives that the conjugate face of a proper face is non-empty). Recall that we denote the supporting hyperplane of the facet F_i by $\mathbf{h}_i = \mathbf{h}(u_i, h_P(u_i))$ and observe that

$$\begin{aligned} v_i \in \hat{F} \iff \langle v_i, y\rangle = 1 \ \forall\, y \in F \iff \langle u_i, y\rangle = h_P(u_i) \ \forall\, y \in F \iff \\ \iff y \in \mathbf{h}_i \ \forall\, y \in F \iff F \subseteq \mathbf{h}_i \iff F \subseteq F_i, \end{aligned} \tag{3.2.4}$$

where the last equivalence comes because we already have that $F \subseteq P$ and $F_i = P \cap \mathbf{h}_i$. It follows that $\{v_i : F \subseteq F_i\} \subseteq \hat{F}$, and from the convexity of $\hat{F}$ this yields

$$\mathrm{conv}\{v_i : F \subseteq F_i\} \subseteq \hat{F}.$$

To prove the reverse inclusion, assume that there exists a point $z \in \hat{F}$ such that $z \notin \operatorname{conv}\{v_i : F \subseteq F_i\}$. Since $z \in P^\circ$, we may use (3.1.4) to write z as a convex combination of $v_1, \ldots, v_m$:

$$z = \lambda_1 v_1 + \ldots + \lambda_m v_m.$$

The fact that $z \notin \operatorname{conv}\{v_i : F \subseteq F_i\}$ gives that there exists some $k \in \{1, \ldots, m\}$ such that $\lambda_k > 0$ and $F \not\subseteq F_k$. From (3.2.4) it follows that $v_k \notin \hat{F}$, and hence there exists $y \in F$ such that $\langle y, v_k \rangle \neq 1$. This gives $\langle y, u_k \rangle \neq h_P(u_k)$, and since $y \in P$ (and P is contained in $\mathbf{h}_k^-$) we must have $\langle y, u_k \rangle < h_P(u_k)$. Thus, $\langle y, v_k \rangle < 1$, and this yields

$$\begin{aligned} \langle z, y \rangle = \lambda_1 \langle v_1, y \rangle + \lambda_2 \langle v_2, y \rangle + \ldots + \lambda_k \langle v_k, y \rangle + \ldots + \lambda_m \langle v_m, y \rangle \leq \\ \leq \lambda_1 + \ldots + \lambda_k \langle v_k, y \rangle + \ldots + \lambda_m < \lambda_1 + \ldots + \lambda_k + \ldots + \lambda_m = 1, \end{aligned}$$

where we carefully remember that $\langle v_i, y \rangle \leq 1$ for every $i \in \{1, \ldots, m\}$ (in fact, this inequality is equivalent to $\langle y, u_i \rangle \leq h_P(u_i)$, and this holds for all points of P). Hence we have the strict inequality $\langle z, y \rangle < 1$, which contradicts $z \in \hat{F}$. Consequently, $\hat{F} \subseteq \operatorname{conv}\{v_i : F \subseteq F_i\}$.

2nd claim. $\mathrm{NC}_P(F) = \operatorname{pos}(\hat{F})$.

Let $x \in \operatorname{relint} F$. By the definition, we have $\mathrm{NC}_P(F) = \mathrm{NC}_P(x)$. Let $u \in \mathrm{NC}_P(x)$, and assume that $u \neq 0_{\mathbb{R}^n}$ (it is clear that the zero vector lies in both sets). Then there exists a hyperplane $\mathbf{h} = \mathbf{h}(u, \alpha)$ supporting P at x, with $P \subseteq \mathbf{h}^-$, say. In this case, we have $\alpha = \langle x, u \rangle = h_P(u)$. Since $x \in \operatorname{relint} F$, we get from Lemma 3.2.3 that $F \subseteq \mathbf{h}$. Then $\langle y, u \rangle = \alpha = h_P(u)$ for every $y \in F$, and letting $v = u/h_P(u)$, this gives

$$\langle y, v \rangle = 1$$

for every $y \in F$. Thus, $v \in \hat{F}$, and then $u = h_P(u)v \in \operatorname{pos}(\hat{F})$. Hence the inclusion $\mathrm{NC}_P(F) \subseteq \operatorname{pos}(\hat{F})$ follows.

Now let $u \in \operatorname{pos}(\hat{F})$. We write $u = \lambda_1 z_1 + \ldots + \lambda_j z_j$, where $\lambda_i \geq 0$ and $z_i \in \hat{F}$ for each $i = \{1, \ldots, j\}$. We have $\langle y, z_i \rangle \leq 1$ for every $y \in P$ and any $i \in \{1, \ldots, j\}$, because each z_i is a point of P°. Fix $x \in \operatorname{relint} F$ and observe that, by doing so, we have $\langle z_i, x \rangle = 1$ for every $i \in \{1, \ldots, j\}$. For every $y \in P$, we have

$$\langle u, x - y \rangle = \langle \lambda_1 z_1 + \ldots + \lambda_j z_j, x - y \rangle = \sum_{i=1}^{j} \lambda_i (\langle z_i, x \rangle - \langle z_i, y \rangle) \geq 0,$$

because $\langle z_i, x \rangle - \langle z_i, y \rangle = 1 - \langle z_i, y \rangle \geq 0$ for each $i \in \{1, \ldots, j\}$ (and because we also have $\lambda_i \geq 0$, of course). Therefore, from Lemma 2.4.2 we get $u \in \mathrm{NC}_P(x)$. This gives $\operatorname{pos}(\hat{F}) \subseteq \mathrm{NC}_P(x) = \mathrm{NC}_P(F)$, and the claim is proved.

Finally, the first and the second claim yield

$$\mathrm{NC}_P(F) = \operatorname{pos}(\hat{F}) = \operatorname{pos}(\operatorname{conv}\{v_i : F \subseteq F_i\}),$$

and the equalities

$$\operatorname{pos}(\operatorname{conv}\{v_i : F \subseteq F_i\}) = \operatorname{pos}\{v_i : F \subseteq F_i\} = \operatorname{pos}\{u_i : F \subseteq F_i\}$$

are easy to show (each u_i is a positive multiple of the respective v_i). This concludes the proof.

□

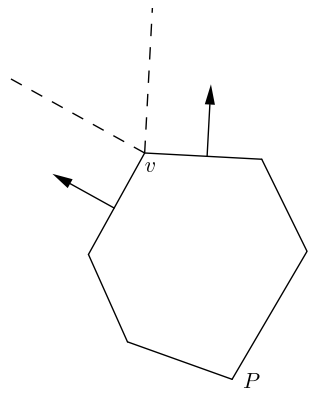


Fig. 3.2.2 The normal cone at v (translated to have apex at v, for illustrative reasons) is the positive hull of the outer normal vectors of the edges which contain v.

This theorem has several consequences that we explore next. They give us information about the boundary structure of a polytope. The first one is that any proper face of a polytope can be described in terms of the facets in which it lies.

Corollary 3.2.3. *A proper face F of a proper polytope P is the intersection of the facets of P which contain F.*

Proof. As usual, let $F_1, \dots F_m$ be the facets of P and, for each $j \in \{1, \dots, m\}$, denote by u_j the unit outer normal vector of F_j. Define the set $J = \{i : F \subseteq F_i\}$, and let

$$G = \bigcap_{i \in J} F_i.$$

That is, G is the intersection of all facets containing F. Since the intersection of faces is a face (see Item **(i)** of Theorem 2.4.4), we get that G is a face. Also, we trivially see that $F \subseteq G$. By Theorem 3.2.1 we have that

$$\mathrm{NC}_P(F) = \mathrm{pos}\{u_i : i \in J\}.$$

We claim that G is contained in exactly the same faces as F, that is, $J = \{i : G \subseteq F_i\}$. Indeed, it is clear that $G \subseteq F_i$ for every $i \in J$. If there is $s \notin J$ such that $G \subseteq F_s$, then $F \subseteq G \subseteq F_s$, and this contradicts the fact that $s \notin J$. Hence, applying Theorem 3.2.1 for G we get

$$\mathrm{NC}_P(G) = \mathrm{pos}\{u_i : i \in J\} = \mathrm{NC}_P(F),$$

and from Remark 3.2.1 we obtain that $F = G$.

□

Still another consequence of Theorem 3.2.1 is that we can relate inclusion of faces with inclusion of normal cones.

Lemma 3.2.4. *Let F and G be proper faces of a proper polytope P. Then $G \subseteq F$ if and only if* $\mathrm{NC}_P(F) \subseteq \mathrm{NC}_P(G)$.

Proof. As usual, let us denote the facets of P by $F_1, \dots, F_m$ and their respective unit outer normal vectors by $u_1, \dots, u_m$. If $G \subseteq F$, then any facet containing F also contains G. Thus,

$$\mathrm{NC}_P(F) = \mathrm{pos}\{u_i : F \subseteq F_i\} \subseteq \mathrm{pos}\{u_i : G \subseteq F_i\} = \mathrm{NC}_P(G).$$

For the converse, assume that $F \subseteq F_{i_0}$. Then $u_{i_0} \in \mathrm{NC}_P(F)$, and hence $u_{i_0} \in \mathrm{NC}_P(G)$. By the definition of normal cone of a face, u_{i_0} is an outer normal of P at each point of $\mathrm{relint}G$, from where we get

$$\mathrm{relint}G \subseteq P \cap \mathbf{h}(P, u_{i_0}) = F_{i_0}.$$

It follows immediately that $G \subseteq F_{i_0}$. Summarizing, we have proved that every facet containing F also contains G. Thus,

$$G = \bigcap_{G \subseteq F_i} F_i \subseteq \bigcap_{F \subseteq F_i} F_i = F,$$

and the proof is complete.

□

Remark 3.2.2. In view of Remark 3.2.1, we have that if the inclusion $G \subseteq F$ is proper, then the inclusion $\mathrm{NC}_P(F) \subseteq \mathrm{NC}_P(G)$ is also proper. The converse is also true.

Corollary 3.2.4. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope. For any proper face $F \subseteq P$ we have the equality*

$$\mathrm{NC}_P(F) = \bigcap_{x \in \mathrm{ext}F} \mathrm{NC}_P(x).$$

See Figure 3.2.3.

Proof. If $x \in \mathrm{ext}F$, then $x \in \mathrm{ext}P$ (from Lemma 3.1.2) and hence $\{x\}$ is a face of P contained in F (recall that any extreme point is a face). Thus, $\mathrm{NC}_P(F) \subseteq \mathrm{NC}_P(x)$. It follows that

$$\mathrm{NC}_P(F) \subseteq \bigcap_{x \in \mathrm{ext}F} \mathrm{NC}_P(x).$$

Now we prove the reverse inclusion. Let $u \in \bigcap_{\mathrm{ext}F} \mathrm{NC}_P(x)$. Then u is an outer normal vector at each point of $\mathrm{ext}F$, from which we get that $\mathrm{ext}F \subseteq \mathbf{h}(P, u)$. It follows immediately that $F \subseteq \mathbf{h}(P, u)$. Therefore, fixing z as any point in the relative interior of F we have that $z \in \mathbf{h}(P, u)$, and this gives that u is an outer normal of P at z. Thus, $u \in \mathrm{NC}_P(z) = \mathrm{NC}_P(F)$. This concludes the proof.

□

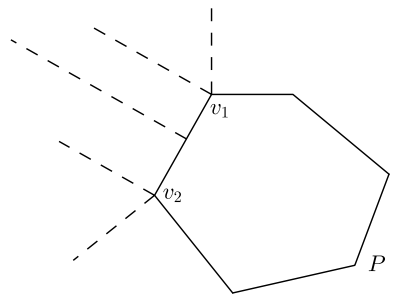


Fig. 3.2.3 The normal cone of the edge $\mathrm{seg}[v_1, v_2]$ is the intersection of the normal cones at the vertices v_1 and v_2. In the figure, all normal cones are translated (by the definition they have apex at the origin).

We continue in the direction of studying the relation between normal cones of different faces, using Theorem 3.2.1 as our main tool. Before moving forward, we need a technical lemma about the relative interior of positive hulls.

Lemma 3.2.5. *Let $u_1, \ldots, u_k \neq 0_{\mathbb{R}^n}$. If $\lambda_1, \ldots, \lambda_k > 0$, then*

$$\lambda_1 u_1 + \ldots + \lambda_k u_k \in \operatorname{relint}(\operatorname{pos}\{u_1, \ldots, u_k\}).$$

Proof. For simplicity, write $u := \lambda_1 u_1 + \ldots + \lambda_1 u_k$. Assume first that $u \neq 0_{\mathbb{R}^n}$. It is clear that $\operatorname{pos}\{u_1, \ldots, u_k\} = \operatorname{pos}(\operatorname{conv}\{u_1, \ldots, u_k\})$ and hence, in view of Item **(b)** of Proposition 2.4.1, it is enough to prove that

$$u \in \operatorname{pos}(\operatorname{relint}(\operatorname{conv}\{u_1, \ldots, u_k\})).$$

By Item **(a)** of the mentioned proposition, there exist a number $\lambda > 0$ and a point $v \in \operatorname{conv}\{u_1, \ldots, u_k\} =: P$ such that $u = \lambda v$. We write

$$v = \beta_1 u_1 + \ldots + \beta_k u_k$$

and observe that $\beta_j = \lambda_j/\lambda > 0$ for each $j \in \{1, \ldots, k\}$. Assume, without loss of generality, that $u_1, \ldots, u_m$ are the extreme points of P. Then by Minkowski's theorem (Theorem 2.4.3) we have that $P = \operatorname{conv}\{u_1, \ldots, u_m\}$. Consequently, the remaining points $u_{m+1}, \ldots, u_k$ can be written as convex combinations of $\{u_1, \ldots, u_m\}$. Thus, we can write

$$v = \alpha_1 u_1 + \ldots + \alpha_m u_m$$

with $\alpha_j > 0$ for every $j \in \{1, \ldots, m\}$. From Corollary 2.4.3, any given point v_1 in the relative boundary of $\operatorname{conv}\{u_1, \ldots, u_m\}$ must be in some proper face, which is exposed (because every face of a polytope is exposed), and hence has the form $\operatorname{conv}(\mathbf{h} \cap \{u_1, \ldots, u_m\})$ for some hyperplane $\mathbf{h}$; see (3.1.1). The hyperplane $\mathbf{h}$ cannot contain all u_j's, since otherwise the face $P \cap \mathbf{h}$ would not be a proper face. It follows that when we write v_1 as a convex combination of $\{u_1, \ldots, u_m\}$, some of the coefficients must be zero. This gives that $v \in \operatorname{relint}(\operatorname{conv}\{u_1, \ldots, u_m\})$, and the case for $u \neq 0_{\mathbb{R}^n}$ is settled.

Now assume that $u = 0_{\mathbb{R}^n}$. Putting $\lambda = \lambda_1 + \ldots + \lambda_k$, we have

$$\frac{u}{\lambda} = \frac{\lambda_1}{\lambda} u_1 + \ldots + \frac{\lambda_k}{\lambda} u_k.$$

All coefficients are positive and sum up to 1. By the same argument as before, we have that

$$0_{\mathbb{R}^n} = \frac{u}{\lambda} \in \operatorname{relint}(\operatorname{conv}\{u_1, \ldots, u_k\}).$$

There exists a number $\varepsilon > 0$ such that

$$B(0_{\mathbb{R}^n}, \varepsilon) \cap \operatorname{aff}(\operatorname{conv}\{u_1, \ldots, u_k\}) \subseteq \operatorname{conv}\{u_1, \ldots, u_k\} \big(\subseteq \operatorname{pos}\{u_1, \ldots, u_k\} \big).$$

Therefore, it is enough to prove that

$$\operatorname{aff}(\operatorname{conv}\{u_1, \ldots, u_k\}) = \operatorname{aff}(\operatorname{pos}\{u_1, \ldots, u_k\}).$$

We obviously have $\operatorname{aff}(\operatorname{conv}\{u_1, \ldots, u_k\}) \subseteq \operatorname{aff}(\operatorname{pos}\{u_1, \ldots, u_k\})$. For the reverse inclusion, in view of Lemma 2.1.3 it suffices to show that

$$\operatorname{pos}\{u_1, \ldots, u_k\} \subseteq \operatorname{aff}(\operatorname{conv}\{u_1, \ldots, u_k\}).$$

To verify this, let $v \in \text{pos}\{u_1, \ldots, u_k\}$ be non-zero (it is clear that $0_{\mathbb{R}^n}$ belongs to both sets). Then $v = \lambda v_0$ for some $\lambda > 0$ and $v_0 \in \text{conv}\{u_1, \ldots, u_k\}$, where we are recalling that $\text{pos}\{u_1, \ldots, u_k\} = \text{pos}(\text{conv}\{u_1, \ldots, u_k\})$. Thus, $v = \lambda v_0 + (1-\lambda)0_{\mathbb{R}^n}$, implying that v is a point of $\text{aff}(\text{conv}\{u_1, \ldots, u_k\})$. This concludes the proof.

□

Proposition 3.2.1. *Let F, G be proper faces of a proper polytope P. If $\text{NC}_P(F) \neq \text{NC}_P(G)$, then*

$$\text{relint}(\text{NC}_P(F)) \cap \text{relint}(\text{NC}_P(G)) = \emptyset. \tag{3.2.5}$$

In particular, if G is properly contained in F, then

$$\text{NC}_P(F) \subseteq \text{relbd}(\text{NC}_P(G)). \tag{3.2.6}$$

Even more, in this case $\text{NC}_P(F)$ is a proper face of $\text{NC}_P(G)$.

Proof. Of course, we may assume that $0_{\mathbb{R}^n} \in \text{int}P$ (because normal cones are invariant under translations of the polytope). In this case, the polar body P° is also a proper polytope containing $0_{\mathbb{R}^n}$ as an interior point. We will need two auxiliary results that we state and prove now.

Auxiliary result A. Let P be a polytope such that $0_{\mathbb{R}^n} \in \text{relint}P$. If F is a proper face of P, then $0_{\mathbb{R}^n} \notin \text{relint}(\text{pos}(F))$. To prove this, of course we may assume that $\dim(P) = n$ (since otherwise we work within $\text{aff}P$). Since F is a proper face, it is contained in some supporting hyperplane $\mathbf{h}$ of P. Once $0_{\mathbb{R}^n} \in \text{int}(P)$, we have that $\mathbf{h}$ does not contain $0_{\mathbb{R}^n}$. Let $\mathbf{g}$ be the hyperplane parallel to $\mathbf{h}$ passing through the origin. It is clear that $\text{pos}(F)$ lies entirely in one of the half-spaces determined by $\mathbf{g}$. Because it is obvious that $\text{aff}(\text{pos}(F))$ is not the hyperplane $\mathbf{g}$, any ball of $\text{aff}(\text{pos}(F))$ around $0_{\mathbb{R}^n}$ must have points of both half-spaces determined by $\mathbf{g}$ (see Figure 3.2.4 below). Hence $0_{\mathbb{R}^n}$ is not a point of $\text{relint}(\text{pos}(F))$.

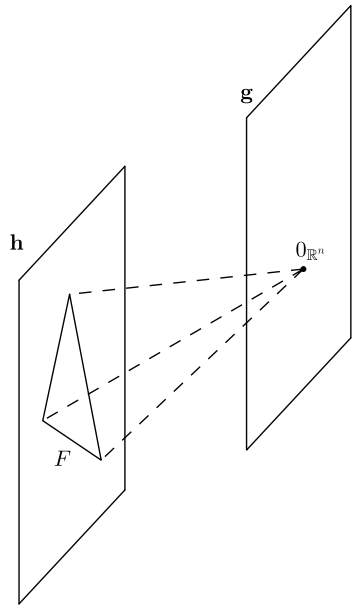


Fig. 3.2.4 Observe that the affine hull of $\text{pos}(F)$ contains the pyramid with apex at $0_{\mathbb{R}^n}$ and basis F.

Auxiliary result B. If F is a proper face of a proper polytope P, then $0_{\mathbb{R}^n} \notin \operatorname{relint}(\mathrm{NC}_P(F))$. In fact, if $0_{\mathbb{R}^n} \in \operatorname{relint}(\mathrm{NC}_P(F))$, then taking any non-zero vector $u \in \mathrm{NC}_P(F)$, we have $-u \in \mathrm{NC}_P(F)$. This means that both u and $-u$ are outer normal vectors of P at any point $x \in \operatorname{relint} F$. Thus, $P \subseteq \mathbf{h}^-(u, \langle u, x\rangle) \cap \mathbf{h}^-(-u, \langle -u, x\rangle) = \mathbf{h}(u, \langle u, x\rangle)$, and this is a contradiction to the fact that P is proper.

We also need some additional information about conjugate faces.

(i) If F is a proper face of P, then the conjugate face

$$\hat{F} = \{z \in P^\circ : \langle z, w\rangle = 1 \text{ for all } w \in F\}$$

is a proper subset of P° (we recalled the definition for the convenience of the reader). In fact, if $\hat{F} = P^\circ$, then choosing any non-zero $w \in F$ we have that $\langle z, w\rangle = 1$ for all $z \in P^\circ$. This is a contradiction to the fact that P° contains a ball centered at the origin.

(ii) For any proper face $F \subseteq P$, we have that $\hat{F}$ is a (proper) face of P°. To verify this, let $x, y \in P^\circ$ and assume that there is a point $z \in \operatorname{seg}(x, y) \cap \hat{F}$. Then there exists a number $\lambda \in (0, 1)$ such that $z = (1 - \lambda)x + \lambda y$. Fix any point $w \in F$. Since $x, y \in P^\circ$, we have that $\langle x, w\rangle \leq 1$ and $\langle y, w\rangle \leq 1$. If one of these inequalities is strict, $\langle x, w\rangle < 1$, say, then

$$1 = \langle z, w\rangle = (1 - \lambda)\langle x, w\rangle + \lambda\langle y, w\rangle < (1 - \lambda) + \lambda\langle y, w\rangle \leq 1 - \lambda + \lambda = 1,$$

which is a contradiction. It follows that $x, y \in \hat{F}$, and since we already know that $\hat{F}$ is convex, we get that this is a face of P°. The fact that $\hat{F}$ is proper comes from **(i)**.

Now assume that F and G are proper faces such that $\mathrm{NC}_P(F) \neq \mathrm{NC}_P(G)$. By the 2nd claim in the proof of Theorem 3.2.1, we have that $\operatorname{pos}(\hat{F}) \neq \operatorname{pos}(\hat{G})$, because $\mathrm{NC}_P(F) = \operatorname{pos}(\hat{F})$ and $\mathrm{NC}_P(G) = \operatorname{pos}(\hat{G})$. This leads immediately to $\hat{F} \neq \hat{G}$. It follows from Item **(ii)** of Theorem 2.4.4 that $\operatorname{relint}(\hat{F}) \cap \operatorname{relint}(\hat{G}) = \emptyset$. If there exists a point

$$z \in \operatorname{relint}(\operatorname{pos}(\hat{F})) \cap \operatorname{relint}(\operatorname{pos}(\hat{G})),$$

then z is non-zero by the auxiliary result A that we proved earlier and **(ii)**. From Item **(b)** of Proposition 2.4.1 we get

$$z \in \operatorname{pos}(\operatorname{relint}(\hat{F})) \cap \operatorname{pos}(\operatorname{relint}(\hat{G})),$$

and hence there exist points $z_1 \in \operatorname{relint}(\hat{F})$, $z_2 \in \operatorname{relint}(\hat{G})$ and numbers $\lambda_1, \lambda_2 > 0$ such that $\lambda_1 z_1 = z = \lambda_2 z_2$. It follows that the ray $\mathbb{R}_+ z$ intersects ∂P° at two distinct points z_1 and z_2 (notice that $z_1 \neq z_2$ because $\operatorname{relint}(\hat{F}) \cap \operatorname{relint}(\hat{G}) = \emptyset$), which is a contradiction. Once $\mathrm{NC}_P(F) = \operatorname{pos}(\hat{F})$ and $\mathrm{NC}_P(G) = \operatorname{pos}(\hat{G})$, we get (3.2.5).

We already know that if G is properly contained in F, then $\mathrm{NC}_P(F)$ is properly contained in $\mathrm{NC}_P(G)$. Hence these are distinct normal cones, and from what we have just proved it follows that their relative interiors are disjoint. Thus,

$$\operatorname{relint}(\mathrm{NC}_P(F)) \subseteq \operatorname{relbd}(\mathrm{NC}_P(G)),$$

and since $\operatorname{relbd}(\mathrm{NC}_P(G))$ is closed we get (3.2.6).

It remains to prove that $\mathrm{NC}_P(F)$ is a face of $\mathrm{NC}_P(G)$. By renumbering the facets of P, we may assume that $\{1, \ldots, k\} = \{1 \leq i \leq m : F \subseteq F_i\}$. That is, $F_1, \ldots, F_k$ are the (only) facets of P which

contain F. Since G is contained in F, we have that G is also contained in $F_1, \ldots, F_k$. But the inclusion is proper, and hence G must be contained in at least one more facet (otherwise Corollary 3.2.3 would lead to $G = F$). Thus, renumbering (if necessary) the remaining facets of P we may assume that $\{1, \ldots, k, k+1, \ldots, s\} = \{1 \leq i \leq m : G \subseteq F_i\}$. In other words, $F_1, \ldots, F_k, F_{k+1}, \ldots, F_s$ are the (only) facets which contain G. Then we have

$$\begin{aligned} \mathrm{NC}_P(F) &= \mathrm{pos}\{u_1, \ldots, u_k\} \text{ and} \\ \mathrm{NC}_P(G) &= \mathrm{pos}\{u_1, \ldots, u_k, u_{k+1}, \ldots, u_s\}. \end{aligned}$$

To prove that $\mathrm{NC}_P(F)$ is a face of $\mathrm{NC}_P(G)$, let $u, v \in \mathrm{NC}_P(G)$ be such that $\mathrm{seg}(u, v) \cap \mathrm{NC}_P(F) \neq \emptyset$. We write

$$\begin{aligned} u &= \alpha_1 u_1 + \ldots + \alpha_k u_k + \alpha_{k+1} u_{k+1} + \ldots + \alpha_s u_s, \text{ and} \\ v &= \beta_1 u_1 + \ldots + \beta_k u_k + \beta_{k+1} u_{k+1} + \ldots + \beta_s u_s, \end{aligned}$$

where $\alpha_1, \ldots, \alpha_s, \beta_1, \ldots, \beta_s \geq 0$. Since $\mathrm{seg}(u, v) \cap \mathrm{NC}_P(F) \neq \emptyset$, there exists a number $\lambda \in (0, 1)$ such that $(1-\lambda)u + \lambda v \in \mathrm{NC}_P(F)$. Then we may write

$$(1-\lambda)u + \lambda v = \lambda_1 u_1 + \ldots + \lambda_1 u_k,$$

where $\lambda_1, \ldots, \lambda_k \geq 0$. For each $1 \leq i \leq s$, we let $\xi_i := (1-\lambda)\alpha_i + \lambda\beta_i$ (≥ 0). Hence the equality above reads

$$\xi_1 u_1 + \ldots + \xi_k u_k + \xi_{k+1} u_{k+1} + \ldots + \xi_s u_s = \lambda_1 u_1 + \ldots + \lambda_k u_k.$$

We assume by contradiction that we have $u \notin \mathrm{NC}_P(F)$ or $v \notin \mathrm{NC}_P(F)$. This yields that some of the α_i's or some of the β_i's are non-zero, and this gives that some ξ_i is non-zero. Letting $\{i_1, \ldots, i_p\} = \{k+1 \leq i \leq s : \xi_i > 0\}$, we may write the equality above as

$$\xi_{i_1} u_{i_1} + \ldots + \xi_{i_p} u_{i_p} = (\lambda_1 - \xi_1)u_1 + \ldots + (\lambda_k - \xi_k)u_k. \tag{3.2.7}$$

The left-hand side is a point of $\mathrm{relint}(\mathrm{pos}\{u_{i_1}, \ldots, u_{i_p}\}) = \mathrm{relint}(\mathrm{NC}_P(F_{i_1} \cap \ldots \cap F_{i_p}))$ because all coefficients are strictly positive (see Lemma 3.2.5). The set $F_{i_1} \cap \ldots \cap F_{i_p}$ is a non-empty face of P, once $G = F_1 \cap \ldots \cap F_k \cap F_{k+1} \cap \ldots \cap F_s \subseteq F_{i_1} \cap \ldots \cap F_{i_p}$ (and, of course, we are using the fact that any intersection of faces is a face). By the auxiliary result B, we get that $\xi_{i_1} u_{i_1} + \ldots + \xi_{i_p} u_{i_p} \neq 0_{\mathbb{R}^n}$. Consequently, at least one of the coefficients on the right-hand side of the equality above is non-zero. We also claim that they cannot be all negative (meaning that at least one of them is positive). Indeed, let $\{j_1, \ldots, j_q\} = \{1 \leq j \leq k : \lambda_j - \xi_j < 0\}$. If there is no positive coefficient on the right-hand side of the above, then we would have

$$(\xi_{j_1} - \lambda_{j_1})u_{j_1} + \ldots + (\xi_{j_q} - \lambda_{j_q})u_{j_q} + \xi_{i_1} u_{i_1} + \ldots + \xi_{i_p} u_{i_p} = 0_{\mathrm{R}^n}.$$

The left-hand side, having all coefficients positive, is a point of

$$\begin{aligned} &\mathrm{relint}(\mathrm{pos}\{u_{j_1}, \ldots, u_{j_q}, u_{i_1}, \ldots, u_{i_p}\}) = \\ &= \mathrm{relint}(\mathrm{NC}_P(F_{j_1} \cap \ldots \cap F_{j_q} \cap F_{i_1} \cap \ldots \cap F_{i_p})), \end{aligned}$$

and we would have a contradiction with the auxiliary result B. Here, we emphasize once again that all facets in the above belong to the collection $\{F_1, \ldots, F_s\}$, and hence their intersection contains G.

Thus, we let $\{l_1, \ldots l_r\} = \{1 \leq l \leq k : \lambda_l - \xi_l > 0\}$, and we know that this set is non-empty. By doing that, we may rewrite equality (3.2.7) as

$$(\xi_{j_1} - \lambda_{j_1})u_{j_1} + \ldots + (\xi_{j_q} - \lambda_{j_q})u_{j_q} + \xi_{i_1}u_{i_1} + \ldots + \xi_{i_p}u_{i_p} = \\ = (\lambda_{l_1} - \xi_{l_1})u_{l_1} + \ldots + (\lambda_{l_r} - \xi_{l_r})u_{l_r},$$

where all coefficients on both sides are strictly positive. Consequently, the left-hand side is a point of $\operatorname{relint}(\operatorname{pos}\{u_{j_1}, \ldots, u_{j_q}, u_{i_1}, \ldots, u_{i_p}\})$ and the right-hand side lies in $\operatorname{relint}(\operatorname{pos}\{u_{l_1}, \ldots, u_{l_r}\})$. Therefore, letting $G_1 := F_{j_1} \cap \ldots \cap F_{j_q} \cap F_{i_1} \cap \ldots \cap F_{i_p}$ and $G_2 := F_{l_1} \cap \ldots \cap F_{l_r}$ we have

$$\operatorname{relint}(\mathrm{NC}_P(G_1)) \cap \operatorname{relint}(\mathrm{NC}_P(G_2)) \neq \emptyset.$$

It follows that $\mathrm{NC}_P(G_1) = \mathrm{NC}_P(G_2)$. In particular, $u_{i_1} \in \mathrm{NC}_P(G_2)$. This means that u_{i_1} is an outer normal vector of P at each point of $\operatorname{relint}(G_2)$. On the other hand, once F_{i_1} is a facet whose unit outer normal vector is u_{i_1}, we get that $F_{i_1} = P \cap \mathbf{h}(P, u_{i_1})$, and hence F_{i_1} contains all points with u_{i_1} as outer normal. This gives that $\operatorname{relint}(G_2) \subseteq F_{i_1}$, and because any facet is closed, we obtain $G_2 \subseteq F_{i_1}$. Since $1 \leq l_1, \ldots, l_r \leq k$, we have

$$F = F_1 \cap \ldots \cap F_k \subseteq F_{l_1} \cap \ldots \cap F_{l_r} = G_2 \subseteq F_{i_1},$$

but once $k + 1 \leq i_1 \leq s$ this is a contradiction (remember that $F_1, \ldots, F_k$ are the only facets containing F). Hence $u, v \in \mathrm{NC}_P(F)$, and this concludes the proof.

□

Remark 3.2.3. Of course, more generally we have that if $G \subseteq F$, then $\mathrm{NC}_P(F)$ is a face of $\mathrm{NC}_P(G)$. The case where the inclusion is proper is settled in the proposition above, and if $G = F$, then the normal cones are the same (and any closed convex set is a face of itself).

From now on, we will use the following notation:

$$F(P, u) := P \cap \mathbf{h}(P, u),$$

for each polytope $P \subseteq \mathbb{R}^n$ and every $u \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Thus, $F(P, u)$ is the intersection of P with its supporting hyperplane which has u as outer normal. Of course, $F(P, u)$ is a face of P, and by (3.1.5) it consists of all (boundary) points of P where u is an outer normal. In particular, we always have that $u \in \mathrm{NC}_P(F(P, u))$.

Lemma 3.2.6. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope, $F \subseteq P$ a face, and $u \in \mathrm{NC}_P(F)$ a non-zero vector. Then $u \in \operatorname{relint}(\mathrm{NC}_P(F))$ if and only if $F = F(P, u)$.*

Proof. First observe for any proper face F, $u \in \mathrm{NC}_P(F)$ implies $F \subseteq F(P, u)$. Indeed, saying that $u \in \mathrm{NC}_P(F)$ means that u is an outer normal of P at every point in $\operatorname{relint} F$. Thus, $\operatorname{relint} F \subseteq P \cap \mathbf{h}(P, u) = F(P, u)$. Since $P \cap \mathbf{h}(P, u)$ is closed, we obtain that $F \subseteq F(P, u)$. As a consequence, from Lemma 3.2.4 we have that $\mathrm{NC}_P(F(P, u)) \subseteq \mathrm{NC}_P(F)$.

As usual, let $F_1, \ldots, F_m$ be the facets of P, and let $u_1, \ldots, u_m$ be their respective unit outer normal vectors. Let $I_F := \{i : F \subseteq F_i\} =: \{i_1, \ldots, i_k\}$. Recalling that $\mathrm{NC}_P(F) = \operatorname{pos}\{u_i : i \in I_F\}$, we have that if $u \notin \operatorname{relint}(\mathrm{NC}_P(F))$, then in the expression $u = \lambda_1 u_{i_1} + \ldots + \lambda_k u_{i_k}$ at least one of the coefficients is zero (see Lemma 3.2.5). Therefore, taking the indices of the non-zero coefficients we obtain a **proper** subset J of I_F such that

$$u \in \operatorname{relint}(\operatorname{pos}\{u_i : i \in J\}) = \operatorname{relint}(\mathrm{NC}_P(G)),$$

where G is the face of P given by

$$G := \bigcap_{i \in J} F_i.$$

It follows that $F \subseteq G$, and we claim that the inclusion is proper. In fact, if $F = G$, then $\mathrm{NC}_P(F) = \mathrm{NC}_P(G)$, and hence $u \in \mathrm{relint}(\mathrm{NC}_P(F))$, a contradiction. Now we notice that, by the argument in the beginning of the proof, we have that $G \subseteq F(P,u)$ because $u \in \mathrm{NC}_P(G)$. Hence the inclusion $F \subseteq F(P,u)$ is proper, from where we get that $F \neq F(P,u)$.

Now assume that $F \neq F(P,u)$. Once we already know that $F \subseteq F(P,u)$, this means that the inclusion is proper. By Remark 3.2.2 we get that the inclusion $\mathrm{NC}_P(F(P,u)) \subseteq \mathrm{NC}_P(F)$ is proper. Hence the normal cones $\mathrm{NC}_P(F(P,u))$ and $\mathrm{NC}_P(F)$ are distinct. Proposition 3.2.1 gives that $\mathrm{NC}_P(F(P,u)) \subseteq \mathrm{relbd}(\mathrm{NC}_P(F))$. Since $u \in \mathrm{NC}_P(F(P,u))$, this leads to $u \notin \mathrm{relint}(\mathrm{NC}_P(F))$.

□

Now we state and prove an intuitive result regarding the *dimensions of faces and their normal cones* which will be useful later.

Proposition 3.2.2. *Let F be a proper face of a proper polytope $P \subseteq \mathbb{R}^n$ whose facets are $F_1, \ldots, F_m$ (with corresponding unit outer normal vectors $u_1, \ldots, u_m$, as usual). If $\{1 \leq i \leq m : F \subseteq F_i\} = \{i_1, \ldots, i_k\}$, then we have*

(a)

$$\bigcap_{v \in \mathrm{NC}_P(F)} \mathbf{h}(P,v) = \bigcap_{j=1}^{k} \mathbf{h}(P,u_{i_j}),$$

(b)

$$\mathrm{aff}\, F = \bigcap_{j=1}^{k} \mathbf{h}(P,u_{i_j}) \text{ and}$$

(c) $\dim(F) + \dim(\mathrm{NC}_P(F)) = n$.

Proof. For **(a)**, we first notice that the inclusion

$$\bigcap_{v \in \mathrm{NC}_P(F)} \mathbf{h}(P,v) \subseteq \bigcap_{j=1}^{k} \mathbf{h}(P,u_{i_j}),$$

because $u_{i_j} \in \mathrm{NC}_P(F)$ for each $1 \leq j \leq k$. To prove the converse inclusion, observe that without loss of generality we can assume that $0_{\mathbb{R}^n} \in \mathrm{relint} F$ (since outer normal vectors are invariant under translation). In this case, if $z \in \bigcap_{j=1}^{k} \mathbf{h}(P,u_{i_j})$, then $\langle z, u_{i_j} \rangle = 0$ for every $1 \leq j \leq k$. Once $\mathrm{NC}_P(F) = \mathrm{pos}\{u_{i_1}, \ldots, u_{i_k}\}$, then any $v \in \mathrm{NC}_P(F)$ can be written as

$$v = \lambda_1 u_{i_1} + \ldots + \lambda_k u_{i_k}$$

for non-negative coefficients $\lambda_1, \ldots, \lambda_k$. Hence it is clear that $\langle z, v \rangle = 0$, from which we get that $z \in \mathbf{h}(P,v)$. Since $v \in \mathrm{NC}_P(F)$ is arbitrary we get that $z \in \bigcap_{v \in \mathrm{NC}_P(F)} \mathbf{h}(P,v)$, and this settles **(a)**.

To prove **(b)**, we should keep in mind the fact that $\{\mathbf{h}(P,v) : v \in \mathrm{NC}_P(F)\}$ is precisely the collection of supporting hyperplanes of P which contain $\mathrm{aff}\, F$ (see Corollary 3.2.2). In view of this, it is clear that we have the inclusion

$$\mathrm{aff}\, F \subseteq \bigcap_{v \in \mathrm{NC}_P(F)} \mathbf{h}(P,v).$$

If the converse inclusion is not true, then using **(a)** we get that $\operatorname{aff} F$ must be contained in the supporting hyperplane of some facet distinct from $F_{i_1}, \dots, F_{i_k}$ (because any face of P is obtained as an intersection of facets). This contradiction gives the desired equality.

It remains to prove **(c)**. This follows from **(b)** and from standard linear algebra. It is easy to see that $\operatorname{aff}\big(\operatorname{pos}\{u_{i_1}, \dots, u_{i_k}\}\big) = \operatorname{span}\{u_{i_1}, \dots, u_{i_k}\}$. Thus, we have

$$\begin{aligned}\dim(F) + \dim(\mathrm{NC}_P(F)) = \dim(\operatorname{aff} F) + \dim(\operatorname{pos}\{u_{i_1}, \dots, u_{i_k}\}) = \\ = \dim \bigcap_{j=1}^{k} \mathbf{h}(P, u_{i_j}) + \dim(\operatorname{span}\{u_{i_1}, \dots, u_{i_k}\}) = n,\end{aligned}$$

and this concludes the proof.

□

3.3 Strongly isomorphic polytopes

Later, our approach to the Alexandrov–Fenchel inequality will be chosen via *strongly isomorphic polytopes* (as in [2167] and [133]). Thus, this section is dedicated to present all the facts that we will need about this concept. At this point, we remind the reader that $\mathbf{h}(P, u)$ denotes the supporting hyperplane of P whose outer unit normal vector is u. Thus, $P \cap \mathbf{h}(P, u)$ is a face of P. Remember also that we will use the notation

$$F(P, u) := P \cap \mathbf{h}(P, u)$$

to denote the face of P whose unit outer normal vector is u. It is straightforward to notice that for a boundary point x, we have that $u \in \mathrm{NC}_P(x)$ if and only if $u \in F(P, u)$. We say that two polytopes $P_1, P_2 \subseteq \mathbb{R}^n$ are *strongly isomorphic* if

$$\dim(F(P_1, u)) = \dim(F(P_2, u))$$

for every $u \in \mathbb{S}^{n-1}$. This is obviously an equivalence relation in $\mathcal{P}^n$, and we denote the equivalence class of a polytope P by $[P]_{\mathrm{si}}$.

Lemma 3.3.1. *If $P_1, P_2 \subseteq \mathbb{R}^n$ are strongly isomorphic polytopes, then $F(P_1, u)$ and $F(P_2, u)$ are strongly isomorphic for each $u \in \mathbb{S}^{n-1}$.*

Proof. Let $v \in \mathbb{S}^{n-1}$ be arbitrarily given, and let $G_i = F(F(P_i, u), v)$ for $i = 1, 2$. Assuming (without loss of generality) that $0_{\mathbb{R}^n} \in G_1 \cap G_2$, we may repeat the construction in the proof of Proposition 3.1.1 for G_1 and G_2 and conclude that if $\alpha > 0$ is large enough, then the hyperplane $\mathbf{h}_0 = \mathbf{h}(\alpha u + v, 0)$ is a supporting hyperplane of both P_1 and P_2, and that $G_i = P_i \cap \mathbf{h}_0$ for $i = 1, 2$. Since P_1 and P_2 are strongly isomorphic, we get that $\dim(G_1) = \dim(G_2)$, and this shows that $F(P_1, u)$ and $F(P_2, u)$ are strongly isomorphic.

□

Next we characterize strongly isomorphic polytopes by means of the normal cones of their vertices.

Proposition 3.3.1. *Let $P_1, P_2 \subseteq \mathbb{R}^n$ be proper polytopes. Then P_1 and P_2 are strongly isomorphic if and only if*

$$\big\{\mathrm{NC}_{P_1}(x) : x \in \operatorname{ext} P_1\big\} = \big\{\mathrm{NC}_{P_2}(y) : y \in \operatorname{ext} P_2\big\}. \tag{3.3.1}$$

Remark 3.3.1. The equality in (3.3.1) is an equality between the collections of the normal cones at vertices of P_1 and P_2. Geometrically, we are saying that P_1 and P_2 are strongly isomorphic when for every vertex x_1 of P_1 there exists a vertex x_2 of P_2 such that the respective normal cones are identical, and vice versa.

Proof. In view of Proposition 3.2.2, the normal cone of a vertex has total dimension. In this case, the relative interior of $\mathrm{NC}_P(x)$ is the usual interior. As an immediate consequence of Lemma 3.2.6, we have the following.

Auxiliary result: if P is a polytope, $x \in \mathrm{ext}P$ and $u \in \mathrm{NC}_P(x)$, then $u \in \mathrm{int}\big(\mathrm{NC}_P(x)\big)$ if and only if $F(P,u) = \{x\}$.

Assume first that P_1 and P_2 are strongly isomorphic. Let $x \in \mathrm{ext}P_1$, and let $u \in \mathrm{int}\big(\mathrm{NC}_{P_1}(x)\big)$. By the auxiliary result, $F(P_1,u) = \{x\}$. Hence

$$\dim\big(F(P_2,u)\big) = \dim\big(F(P_1,u)\big) = 0.$$

That is, $F(P_2,u)$ is a vertex y (say) of P_2. Even more, from the auxiliary result we have that $u \in \mathrm{int}\big(\mathrm{NC}_{P_2}(y)\big)$.

At this point, we have proved that an outer normal u of P_1 at a vertex $x \in P_1$ which lies in the interior of $\mathrm{NC}_{P_1}(x)$ is an outer normal of P_2 at some vertex $y \in P_2$ (and, moreover, u is an interior point of the normal cone of P_2 at y). It remains to consider the case where the outer normal is not an interior point of the respective normal cone. For that sake, we continue assuming that $x \in \mathrm{ext}P_1$, and we fix a vector $u \in \mathrm{int}\big(\mathrm{NC}_{P_1}(x)\big)$. Then $u \in \mathrm{int}\big(\mathrm{NC}_{P_2}(y)\big)$ for some vertex $y \in \mathrm{ext}P_2$. For any given $v \in \mathrm{NC}_{P_1}(x)$, we have that

$$v_\lambda := (1-\lambda)u + \lambda v \in \mathrm{int}\big(\mathrm{NC}_{P_1}(x)\big)$$

for every $\lambda \in [0,1)$. This comes directly from Lemma 2.1.4. Thus, by the previous argument, we obtain that for each $\lambda \in [0,1)$ the face $F(P_2,v_\lambda)$ is a vertex y_λ (say) of P_2, and

$$v_\lambda \in \mathrm{int}\big(\mathrm{NC}_{P_2}(y_\lambda)\big).$$

Since $v_0 = u \in \mathrm{int}\big(\mathrm{NC}_{P_2}(y)\big)$, it follows that

$$s := \sup\{\lambda \in [0,1) : v_\lambda \in \mathrm{NC}_{P_2}(y)\} > 0.$$

Moreover, once $\mathrm{NC}_{P_2}(y)$ is closed, we have that $v_s \in \mathrm{NC}_{P_2}(y)$. In particular, $y \in F(P_2,v_s)$. On the other hand, it is easy to see that $v_s \notin \mathrm{int}\big(\mathrm{NC}_{P_2}(y)\big)$, because otherwise we would have $v_\lambda \in \mathrm{NC}_{P_2}(y)$ for some $\lambda > s$. It follows that $v_s \in \partial\big(\mathrm{NC}_{P_2}(y)\big)$. Due to the auxiliary result, this means that $F(P_2,v_s)$ has some point other than y, and hence $F(P_2,v_s)$ is not a vertex. Since we already know that $F(P_2,v_\lambda)$ is a vertex for each $\lambda \in [0,1)$, we obtain that $s = 1$. Consequently,

$$v_\lambda \in \mathrm{NC}_{P_2}(y)$$

for every $\lambda \in [0,1)$. Using again the fact that $\mathrm{NC}_{P_2}(y)$ is closed we get that $v = v_1 \in \mathrm{NC}_{P_2}(y)$. Thus, we have the inclusion

$$\mathrm{NC}_{P_1}(x) \subseteq \mathrm{NC}_{P_2}(y).$$

For the reverse inclusion one just has to repeat the proof interchanging the roles of P_1 and P_2. This gives (3.3.1).

Now we prove the converse. Let $u \in \mathrm{S}^{n-1}$ be arbitrarily given, and for the sake of simplicity write $F_1 = F(P_1, u)$ and $F_2 = F(P_2, u)$. Choose $x \in \mathrm{ext}F_1$. From Lemma 3.1.2 we have that x is a vertex of P_1, and hence (3.3.1) yields that there exists $y \in \mathrm{ext}P_2$ such that $\mathrm{NC}_{P_1}(x) = \mathrm{NC}_{P_2}(y)$. Recalling that $F(P_1, u)$ is the set of all boundary points of P_1 where u is an outer normal, we get that $u \in \mathrm{NC}_{P_1}(x)$. This gives $u \in \mathrm{NC}_{P_2}(y)$, from where we obtain that $y \in F_2$. Once $\{x\}$ is a face contained in F_1, by Lemma 3.2.4 we get that $\mathrm{NC}_{P_1}(F_1) \subseteq \mathrm{NC}_{P_1}(x)$, and analogously we have the inclusion $\mathrm{NC}_{P_2}(F_2) \subseteq \mathrm{NC}_{P_2}(y)$. To emphasize that the normal cones of P_1 at x and of P_2 at y are the same, we write

$$C := \mathrm{NC}_{P_1}(x) = \mathrm{NC}_{P_2}(y).$$

Thus, we have that $\mathrm{NC}_{P_1}(F_1) \subseteq C$ and $\mathrm{NC}_{P_2}(F_2) \subseteq C$. From Proposition 3.2.1 (and Remark 3.2.3) we get that $\mathrm{NC}_{P_1}(F_1)$ and $\mathrm{NC}_{P_2}(F_2)$ are faces of C. On the other hand, using Lemma 3.2.6 again, we obtain

$$u \in \mathrm{relint}\big(\mathrm{NC}_{P_1}(F_1)\big) \cap \mathrm{relint}\big(\mathrm{NC}_{P_2}(F_2)\big),$$

and once the relative interiors of distinct faces of a closed convex set are always disjoint (see Theorem 2.4.4), it follows that $\mathrm{NC}_{P_1}(F_1) = \mathrm{NC}_{P_2}(F_2)$. Finally, Proposition 3.2.2 gives

$$\begin{aligned}\dim(F(P_1, u)) = \dim(F_1) = n - \dim(\mathrm{NC}_{P_1}(F_1)) = n - \dim(\mathrm{NC}_{P_2}(F_2)) = \\ = \dim(F_2) = \dim(F(P_2, u)),\end{aligned}$$

and this shows that P_1 and P_2 are strongly isomorphic.

□

This proposition is important to prove that a class of strongly isomorphic polytopes is closed under Minkowski addition and positive scalar multiplication. We will prove this next, but first we need to understand how the faces of a polytope behave under these operations. In what follows, we extend the notations which we introduced for polytopes to arbitrary compact convex sets and write $\mathbf{h}(C, u)$ for the supporting hyperplane of C having u as outer normal, and $F(C, u) = C \cap \mathbf{h}(C, u)$. It is immediate that we also have the equality

$$F(C, u) = \{x \in C : \langle x, u\rangle = h_C(u)\},$$

because $\mathbf{h}(C, u) = \mathbf{h}(u, h_C(u))$.

Lemma 3.3.2. *Let C and D be non-empty compact convex sets. For any $\lambda \geq 0$ and $u \in \mathbb{S}^{n-1}$ we have*

$$\begin{aligned}\mathbf{h}(\lambda C + D, u) = \lambda\mathbf{h}(C, u) + \mathbf{h}(D, u) \ \ \textit{and} \\ F(\lambda C + D, u) = \lambda F(C, u) + F(D, u).\end{aligned}$$

Proof. We first prove that $\mathbf{h}(u, \lambda\alpha + \beta) = \lambda\mathbf{h}(u, \alpha) + \mathbf{h}(u, \beta)$ for any $\alpha, \beta \in \mathbb{R}$. If $x \in \mathbf{h}(u, \alpha)$ and $y \in \mathbf{h}(u, \beta)$, then

$$\langle \lambda x + y, u\rangle = \lambda\langle x, u\rangle + \langle y, u\rangle = \lambda\alpha + \beta,$$

and hence $\lambda x + y \in \mathbf{h}(u, \lambda\alpha + \beta)$. This shows the inclusion $\lambda\mathbf{h}(u, \alpha) + \mathbf{h}(u, \beta) \subseteq \mathbf{h}(u, \lambda\alpha + \beta)$. For the reverse inclusion, let $z \in \mathbf{h}(u, \lambda\alpha + \beta)$ be arbitrarily given and fix any point $x \in \mathbf{h}(u, \alpha)$. Let $y = z - \lambda x$. Then

$$\langle y, u\rangle = \langle z - \lambda x, u\rangle = \langle z, u\rangle - \lambda\langle x, u\rangle = \lambda\alpha + \beta - \lambda\alpha = \beta,$$

and hence $y \in \mathbf{h}(u,\beta)$. Consequently, $z = \lambda x + y \in \lambda\mathbf{h}(u,\alpha) + \mathbf{h}(u,\beta)$. This gives $\mathbf{h}(u,\lambda\alpha+\beta) \subseteq \lambda\mathbf{h}(u,\alpha) + \mathbf{h}(u,\beta)$.

We know that $\mathbf{h}(C,u) = \mathbf{h}(u,h_C(u))$. From the properties of support functions (see Proposition 2.3.2), and from what we just have proved, we get

$$\begin{aligned}\mathbf{h}(\lambda C + D, u) &= \mathbf{h}(u, h_{\lambda C+D}(u)) = \mathbf{h}(u, \lambda h_C(u) + h_D(u)) = \\ &= \lambda\mathbf{h}(u, h_C(u)) + \mathbf{h}(u, h_D(u)) = \lambda\mathbf{h}(C,u) + \mathbf{h}(D,u).\end{aligned}$$

This settles the first equality.

The easiest way to get the second equality is to describe the directional derivative of the support function of a compact convex set in terms of the support functions of its exposed faces. We have the following:

$$(h_C)'_+(u,v) = h_{F(C,u)}(v) \tag{3.3.2}$$

for any $C \in CC^n$ and every $u, v \in \mathbb{S}^{n-1}$. To prove this, we first notice that $(h_C)'_+(u,\cdot)$ is sublinear for each fixed $u \in \mathbb{S}^{n-1}$ (because h_C is convex; see Proposition 1.2.3), and hence it is the support function of some compact convex set K_u (by Theorem 2.3.1). From Item **(b)** of Proposition 1.2.4, we obtain

$$h_{K_u}(v) = (h_C)'_+(u,v) \le h_C(v)$$

for any $v \in \mathbb{S}^{n-1}$. Hence $K_u \subseteq C$ (see Proposition 2.3.2). Thus, for any $z \in K_u$ we have $\langle z,u\rangle \le h_C(u)$. From Item **(a)** of Proposition 1.2.4, we obtain

$$(h_C)'_+(u,-u) = -(h_C)'_+(u,u) = -h_C(u).$$

On the other hand,

$$\langle z,-u\rangle \le h_{K_u}(-u) = (h_C)'_+(u,-u),$$

and hence $\langle z,-u\rangle \le -h_C(u)$. This, together with the inequality $\langle z,u\rangle \le h_C(u)$, leads to $\langle z,u\rangle = h_C(u)$. Hence $z \in F(C,u)$, and we have the inclusion $K_u \subseteq F(C,u)$.

Now let $y \in F(C,u)$. We have that $\langle y,u\rangle = h_C(u)$. Also, since $y \in C$, we have that $\langle y,v\rangle \le \sup\{\langle z,v\rangle : z \in C\} = h_C(v)$ for every $v \in \mathbb{R}^n$. For any $w \in \mathbb{S}^{n-1}$ and $\lambda > 0$, putting $v = u + \lambda w$ in this inequality yields

$$h_C(u+\lambda w) \ge \langle y, u+\lambda w\rangle = h_C(u) + \lambda\langle y,w\rangle.$$

Consequently,

$$\langle y,w\rangle \le \frac{h_C(u+\lambda w) - h_C(u)}{\lambda}.$$

Letting $\lambda \to 0^+$ we obtain that $\langle y,w\rangle \le (h_C)'_+(u,w) = h_{K_u}(w)$ for every $w \in \mathbb{S}^{n-1}$. This leads to $h_{\{y\}} \le h_{K_u}$, and from Item **(iv)** of Proposition 2.3.2 we obtain that $y \in K_u$. Hence $F(C,u) = K_u$, and this gives (3.3.2).

Finally, using the properties of support functions (and the fact that directional differentiation is clearly linear in the functions), it follows that

$$\begin{aligned}h_{F(\lambda C+D,u)}(v) &= (h_{\lambda C+D})'_+(u,v) = (\lambda h_C + h_D)'_+(u,v) = \\ &= \lambda(h_C)'_+(u,v) + (h_D)'_+(u,v) = \lambda h_{F(C,u)}(v) + h_{F(D,u)}(v) = h_{\lambda F(C,u)+F(D,u)}(v)\end{aligned}$$

for each $v \in \mathbb{S}^{n-1}$. Hence the sets $F(\lambda C + D, u)$ and $\lambda F(C, u) + F(D, u)$ have the same support function. Since they are clearly compact and convex, from Corollary 2.3.3 we get the equality $F(\lambda C + D, u) = \lambda F(C, u) + F(D, u)$.

□

Theorem 3.3.1. *Let P and Q be proper polytopes which are strongly isomorphic. For any $\lambda \geq 0$ we have that $\lambda P + Q$ is strongly isomorphic to P and Q.*

Proof. We first consider the case for positive scalar multiplication. If $\lambda > 0$, then for every $u \in \mathbb{S}^{n-1}$ we have that $F(\lambda P, u) = \lambda F(P, u)$. Since it is clear that the affine dimension of a set does not change under positive scalar multiplication, it follows that $\dim(F(\lambda P, u)) = \dim(F(P, u))$. Hence λP and P are strongly isomorphic.

Therefore, it is enough to prove that the Minkowski sum of strongly isomorphic polytopes P and Q is strongly isomorphic to P and Q. Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. For each $x \in \operatorname{ext}F(P, u)$ there exists a point $y \in \operatorname{ext}Q$ such that $\mathrm{NC}_P(x) = \mathrm{NC}_Q(y)$. Once x is a point of $F(P, u)$, we have that $u \in \mathrm{NC}_P(x)$, implying that $u \in \mathrm{NC}_Q(y)$. Therefore, $y \in F(Q, u)$, and hence $y \in (\operatorname{ext}Q) \cap F(Q, u) = \operatorname{ext}F(Q, u)$. Thus, to each point $x \in \operatorname{ext}F(P, u)$ there corresponds a point $y \in \operatorname{ext}F(Q, u)$ such that $\mathrm{NC}_P(x) = \mathrm{NC}_P(y)$, and we repeat the argument interchanging x and y to get that the converse is also true. Consequently, using Corollary 3.2.4 we obtain

$$\mathrm{NC}_P(F(P, u)) = \bigcap_{x \in \operatorname{ext}F(P,u)} \mathrm{NC}_P(x) = \bigcap_{y \in \operatorname{ext}F(Q,u)} \mathrm{NC}_Q(y) = \mathrm{NC}_Q(F(Q, u)).$$

The property of being strongly isomorphic is clearly invariant under translations, and hence we may assume that $0_{\mathbb{R}^n} \in F(P, u) \cap F(Q, u)$. Using the equality above and the Items **(a)** and **(b)** of Proposition 3.2.2 we get

$$\begin{aligned} \operatorname{aff}F(P, u) = \bigcap_{v \in \mathrm{NC}_P(F(P,u))} \mathbf{h}(P, v) = \bigcap_{v \in \mathrm{NC}_Q(F(Q,u))} \mathbf{h}(v, 0) = \\ = \bigcap_{v \in \mathrm{NC}_Q(F(Q,u))} \mathbf{h}(Q, v) = \operatorname{aff}F(Q, u). \end{aligned}$$

In particular, all the points of $F(Q, u)$ can be written as affine combinations of the points of $F(P, u)$, and this gives $\operatorname{aff}(F(P, u) + F(Q, u)) = \operatorname{aff}(F(P, u))$. Consequently, we have that $\dim(F(P, u) + F(Q, u)) = \dim(F(P, u))$. Finally, using Lemma 3.3.2 we obtain

$$\dim\big(F(P + Q, u)\big) = \dim\big(F(P, u) + F(Q, u)\big) = \dim\big(F(P, u)\big),$$

and since $u \in \mathbb{S}^{n-1}$ is arbitrary we get that $P + Q$ and P are strongly isomorphic (and these are also strongly isomorphic to Q, of course).

□

Simultaneous approximations of a finite collection of bodies by strongly isomorphic polytopes play an important role in the proof of the Alexandrov–Fenchel inequality (this will be given in Section 8.2). The existence of such approximations will be given in the next section, which is dedicated precisely to approximations by polytopes. The results below are preliminary steps to derive the existence of simultaneous approximations by strongly isomorphic polytopes. We follow closely [2167] in this regard.

We actually need a little more: we will also demand the polytopes to be simple. A proper polytope $P \subseteq \mathbb{R}^n$ is said to be *simple* if each of its vertices lies in exactly n of its facets. It is clear that if a polytope is simple, then each k-face F lies in precisely $n - k$ facets (for any $k = 0, \ldots, n - 1$).

Otherwise, to obtain a vertex v of P we would need to intersect F with at least k more facets, and hence v would lie in more than n facets. This property has the feature of being invariant under "small perturbations", as we will see next.

Proposition 3.3.2. *Let $P \subseteq \mathbb{R}^n$ be a simple proper polytope, and assume that $u_1, \ldots, u_m$ are the unit outer normal vectors of the facets $F_1, \ldots, F_m$ of P. Then there exists a number $\beta > 0$ such that the polytope*

$$P_0 := \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_P(u_j) + \alpha_j)$$

is simple and strongly isomorphic to P whenever $|\alpha_j| < \beta$ for every $1 \leq j \leq m$.

Proof. Assume that $\mathrm{ext}P = \{x_1, \ldots, x_k\}$ are the vertices of P. It is easy to see that we may choose a number $\rho > 0$ with the property that for each $j = 1, \ldots, k$ the ball $B(x_j, \rho)$ touches only the supporting hyperplanes of the facets of P which contain x_j (see, e.g., Exercise 3.3).

Notice that each ball $B(x_j, \rho)$ has a positive distance from all the hyperplanes containing facets of P which do not contain x_j. Having this in mind, we may choose $\beta > 0$ small enough such that if $|\alpha_i| < \beta$ for every $i = 1, \ldots, m$ it holds that:

(i) for each vertex x_j, the intersection

$$\bigcap_{F_i \ni x_j} \mathbf{h}(u_i, h_P(u_i) + \alpha_i)$$

is a point which lies in $B(x_j, \rho)$. We denote this point by x_j'. Notice that we are just doing "small perturbations" in the supporting hyperplanes of facets which intersect up to x_j.

(ii) for each $i = 1, \ldots, m$, the hyperplane $\mathbf{h}(u_i, h_P(u_i) + \alpha_i)$ intersects the ball $B(x_j, \rho)$ if and only if $x_j \in F_i$. In other words, the perturbation of $\mathbf{h}(u_i, h_P(u_i))$ is small enough to guarantee that the new hyperplane does not intersect balls which $\mathbf{h}(u_i, h_P(u_i))$ does not touch.

It follows that, if $|\alpha_i| < \beta$ for any $i = 1, \ldots, m$, then the polytope P_0 as in the statement is the polytope having $x_1', \ldots, x_k'$ as vertices. For each $j = 1, \ldots, k$, once x_j and x_j' are intersections of exactly the same hyperplanes up to translation, we get that P and P_0 have the same normal cones at x_j and x_j'. From Proposition 3.3.1 we get that P and P_0 are strongly isomorphic. Moreover, once each vertex of P_0 lies in the same number of facets as the corresponding vertex of P, we get that P_0 is simple.

□

Lemma 3.3.3. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope, not necessarily simple. Assume that $F_1, \ldots, F_m$ are the facets of P, with corresponding unit outer normal vectors $u_1, \ldots, u_m$. Then, for each $\beta > 0$ we may choose numbers $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$ such that*

(a) *$|\alpha_j| < \beta$ for any $j \in \{1, \ldots, m\}$,*

(b) *the intersection*

$$P_0 := \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_P(u_j) + \alpha_j)$$

is a simple proper polytope whose facets have $u_1, \dots, u_m$ as unit outer normal vectors, and

(c) *each normal cone of a vertex of P_0 is contained in the normal cone of some vertex P.*

Proof. First we just do the same construction as in Proposition 3.3.2. Then, by appropriately choosing the α's, we may "discard" some facets in order that each vertex of P_0 is contained in precisely n facets. Finally, notice that the outer normal vectors of the facets containing each vertex of P_0 are among the outer normal vectors of the facets containing the corresponding vertex in P. From Theorem 3.2.1 we get Item **(c)**.

□

3.4 Approximations by polytopes

In Theorem 2.6.2, we described how one can approximate (in the Hausdorff metric) a convex body by an inner polytope. For further applications we will need "stronger" ways of approximating convex bodies by polytopes, and we develop these ways in this section. The reader should notice that the topic of approximations by polytopes is interesting in itself, and there is a vast literature on it. Following the philosophy of this chapter, here we restrict ourselves to the results that will be needed later. We start by improving the mentioned theorem to guarantee that a convex body can be approximated by a *proper* polytope.

Proposition 3.4.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For any $\varepsilon > 0$ there exists a proper polytope $P \subseteq K$ such that $d_{\mathcal{H}}(K, P) \leq \varepsilon$.*

Proof. By Theorem 2.6.2, for a given $\varepsilon > 0$ we can choose a polytope P_0 such that $P_0 \subseteq K \subseteq P_0 + \varepsilon B^n$. Observe that this gives $d_{\mathcal{H}}(K, P_0) \leq \varepsilon$ but P_0 is not necessarily proper. Write $P_0 := \operatorname{conv}\{x_1, \dots, x_k\}$. We simply choose $n+1$ affinely independent points $\{z_1, \dots, z_{n+1}\} \subseteq K$ and put $P := \operatorname{conv}\{x_1, \dots, x_k, z_1, \dots, z_{n+1}\}$. Hence P is clearly a proper polytope. Once it is obvious that $P_0 \subseteq P \subseteq K$ we have $P \subseteq K \subseteq P + \varepsilon B^n$ and, consequently, $d_{\mathcal{H}}(K, P) \leq \varepsilon$.

□

Corollary 3.4.1. *For any convex body $K \subseteq \mathbb{R}^n$ and every $\varepsilon > 0$ there exists a proper polytope P such that $K \subseteq P$ and $d_{\mathcal{H}}(K, P) \leq \varepsilon$.*

Proof. Since any translate of a polytope is also a polytope, we may assume that K contains the origin as an interior point. Given $\varepsilon > 0$, from Proposition 2.6.1 we have that there exists $\delta > 0$ such that $d_{\mathcal{H}}(K^\circ, L) \leq \delta$ implies $d_{\mathcal{H}}(K, L^\circ) \leq \varepsilon$. Using Proposition 3.4.1, we let $Q \subseteq K^\circ$ be a proper polytope such that $d_{\mathcal{H}}(K^\circ, Q) \leq \delta$. We also may assume that Q has the origin as an interior point (for example, "add" to Q the intersection points of the coordinate axes of $\mathbb{R}^n$ with K°). Hence $P = Q^\circ$ is a proper polytope such that $d_{\mathcal{H}}(K, P) \leq \varepsilon$, and since polarity reverses inclusion we also have that $K \subseteq P$.

□

The characterization of polytopes given in (3.1.2) yields a type of approximation of convex bodies by polytopes which will be useful later. We describe it now.

Proposition 3.4.2. *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point. There exists a number $m_0 \in \mathbb{N}$ such that for any $m \geq m_0$ we may choose a proper polytope $P_m \subseteq \mathbb{R}^n$ with the property that*

$$P_m \subseteq K \subseteq (1 + 1/m)P_m.$$

See Figure 3.4.1.

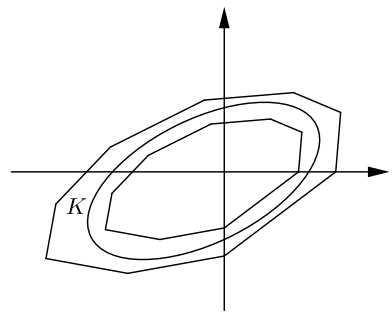


Fig. 3.4.1 The polytopes contained and containing K are homothetical.

Proof. We first need an auxiliary result. This is a particular case of Lemma 2.6.4, but nevertheless we present a different proof (which is more geometric in nature).

Auxiliary result. Let $\rho > 0$ be such that $\rho B^n \subseteq \text{int}K$. There exists a number $\delta > 0$ such that $\rho B^n \subseteq L$ whenever L is a convex body with $L \subseteq K$ and $d_{\mathcal{H}}(K, L) < \delta$.

We prove that the (positive) number

$$\delta = \frac{1}{2}\text{dist}(\partial K, \rho B^n)$$

has the desired property. In fact, let $L \subseteq K$ be a convex body such that $d_{\mathcal{H}}(K, L) < \delta$. Assume, by contradiction, that there exists a point $x \in \rho B^n \setminus L$. Then x has a unique metric projection $p_L(x) \in \partial L$. Let z be the intersection of the ray $p_L(x) + \mathbb{R}_+(x - p_L(x))$ with ∂K (see Figure 3.4.2). From Lemma 2.2.5, we have

$$\text{dist}(z, L) = |z - p_L(x)| > |z - x| \geq \text{dist}(\partial K, \rho B^n) = 2\delta.$$

On the other hand, since $d_{\mathcal{H}}(K, L) < \delta$, we get that $z \in K \subseteq L + \delta B^n$, yielding $\text{dist}(z, L) < \delta$. This gives a contradiction, and hence the proof of the auxiliary result is complete.

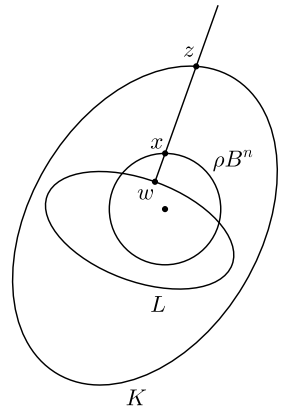


Fig. 3.4.2 In the figure, we represent $w = p_L(x)$.

Now we proceed to the proof of the proposition itself. Fix a natural number $m_0 > \rho/\delta$. For any $m \geq m_0$, let P_m be a proper polytope such that $P_m \subseteq K$ and

$$d_{\mathcal{H}}(K, P_m) < \frac{\rho}{m}.$$

The existence of such a polytope is guaranteed by Proposition 3.4.1. Notice that, in particular,

$$d_{\mathcal{H}}(K, P_m) < \frac{\rho}{m} \leq \frac{\rho}{m_0} < \rho\frac{\delta}{\rho} = \delta,$$

and hence $\rho B^n \subseteq P_m$. Letting $u_1, \ldots, u_k$ be the outer normal unit vectors of the facets of P_m, this inclusion gives

$$h_{P_m}(u_j) \geq \rho \tag{3.4.1}$$

for each $j = 1, \ldots, k$. Due to (3.1.2), the polytope P_m can be written as

$$P_m = \bigcap_{j=1}^{k} \mathbf{h}^-(u_j, h_{P_m}(u_j)),$$

and it is immediate to derive the equality

$$\left(1 + \frac{1}{m}\right)P_m = \bigcap_{j=1}^{k} \mathbf{h}^-\left(u_j, \left(1 + \frac{1}{m}\right)h_{P_m}(u_j)\right). \tag{3.4.2}$$

Now let $z \in K$ be arbitrarily given. Once $d_{\mathcal{H}}(K, P_m) < \rho/m$ we have the inclusion $K \subseteq P_m + (\rho/m)B^n$. Thus, we may write

$$z = y + \frac{\rho}{m}v$$

for some $y \in P_m$ and some $v \in B^n$. For each $j = 1, \ldots, k$, the Cauchy–Schwarz inequality gives $\langle u_j, v\rangle \leq 1$. This, together with (3.4.1), yields

$$\langle z, u_j\rangle = \langle y, u_j\rangle + \frac{\rho}{m}\langle u_j, v\rangle \leq h_{P_m}(u_j) + \frac{\rho}{m} \leq \left(1 + \frac{1}{m}\right)h_{P_m}(u_j)$$

for every $j = 1, \ldots, k$. From equality (3.4.2), it follows that

$$z \in \left(1 + \frac{1}{m}\right)P_m,$$

and hence $K \subseteq (1 + 1/m)P_m$. This concludes the proof since the inclusion $P_m \subseteq K$ was already guaranteed by the construction of P_m.

□

Remark 3.4.1. Actually, for *any* natural number m we may choose a proper polytope such that $P_m \subseteq K \subseteq (1+1/m)P_m$. In fact, if $P_{m_0} \subseteq K \subseteq (1+1/m_0)P_{m_0}$, then $0_{\mathbb{R}^n} \in (1+1/m_0)P_{m_0}$. Hence $0_{\mathbb{R}^n} \in P_{m_0}$. This leads to

$$(1 + 1/m_0)P_{m_0} \subseteq (1 + 1/m)P_{m_0}$$

whenever $m < m_0$. Thus, in this case we simply put $P_m = P_{m_0}$ to obtain $P_m \subseteq K \subseteq (1 + 1/m)P_m$.

Next we state and prove a result on simultaneous approximations of compact convex sets having convex union by polytopes which also have convex union. This is taken from [1263], and will be useful in Section 11.3.

Theorem 3.4.1. *Let $C_1, \ldots, C_m \in \mathcal{CC}^n$ be such that $C_1 \cup \ldots \cup C_m \in \mathcal{CC}^n$. For any $\varepsilon > 0$ there exist polytopes $P_1, \ldots, P_m$ such that $C_j \subseteq P_j \subseteq C_j + \varepsilon B^n$ for each $j \in \{1, \ldots, m\}$ and $P_1 \cup \ldots \cup P_m$ is a polytope.*

Proof. For simplicity, we write $C := C_1 \cup \ldots \cup C_m$. Using Proposition 3.4.1, we choose a polytope D such that $D \subseteq C \subseteq D + \varepsilon B^n$. From Corollary 3.4.1, for each $j \in \{1, \ldots, m\}$ we may fix a polytope E_j such that $C_j + \varepsilon B^n \subseteq E_j \subseteq C_j + 2\varepsilon B^n$. In particular, notice that $C_j \subseteq E_j$. Next, for each $j \in \{1, \ldots, m\}$, let $Q_j := D \cap E_j$. Then it is clear that each Q_j is a polytope (see Corollary 3.1.7). The inclusions obtained by the constructions give immediately

$$D = D \cap C = D \cap (C_1 \cup \ldots \cup C_m) \subseteq D \cap (E_1 \cup \ldots \cup E_m) = \\ = \bigcup_{j=1}^{m} (D \cap E_j) = \bigcup_{j=1}^{m} Q_j \subseteq D,$$

that is,

$$D = \bigcup_{j=1}^{m} Q_j.$$

Let $j \in \{1, \ldots, m\}$, and $x \in C_j$ be arbitrarily given. Then $x \in C$, and because $C \subseteq D + \varepsilon B^n$ it follows that $x \in D + \varepsilon B^n$. Thus, we may fix $y \in D$ such that $|x - y| \leq \varepsilon$, and this yields that $y \in C_j + \varepsilon B^n$. Since $C_j + \varepsilon B^n \subseteq E_j$, we get that $y \in E_j$, and since $y \in D$ we arrive at $y \in D \cap E_j = Q_j$. Using again the fact that $|x - y| \leq \varepsilon$, we have that $x \in Q_j + \varepsilon B^n$. Since $x \in C_j$ was arbitrarily taken, this argument leads to the inclusion

$$C_j \subseteq Q_j + \varepsilon B^n,$$

for each $j \in \{1, \ldots, m\}$. Moreover, from the inclusions $Q_j \subseteq E_j \subseteq C_j + 2\varepsilon B^n$, we may repeat the inclusion above and add an inclusion on the right-hand side to write

$$C_j \subseteq Q_j + \varepsilon B^n \subseteq C_j + 3\varepsilon B^n,$$

for every $j \in \{1, \ldots, m\}$. Now fix any polytope R such that $\varepsilon B^n \subseteq R \subseteq 2\varepsilon B^n$, and define $P_j := Q_j + R$ for each $j \in \{1, \ldots, m\}$. From the inclusions above we have

$$C_j \subseteq Q_j + \varepsilon B^n \subseteq Q_j + R = P_j \subseteq Q_j + 2\varepsilon B^n = (Q_j + \varepsilon B^n) + \varepsilon B^n \subseteq C_j + 4\varepsilon B^n,$$

for any $j \in \{1, \ldots, m\}$. Therefore, replacing ε by $\varepsilon/4$, we get that the polytopes $P_1, \ldots, P_m$ satisfy the desired approximation condition. Hence it only remains to prove that $P_1 \cup \ldots \cup P_m$ is convex. To do that, we simply notice that

$$P_1 \cup \ldots \cup P_m = (Q_1 + R) \cup \ldots \cup (Q_m + R) = (Q_1 \cup \ldots \cup Q_m) + R = D + R,$$

and because the Minkowski sum of polytopes is a polytope (see Lemma 3.1.3) we get that $P_1 \cup \ldots \cup P_m$ is a polytope. The proof is complete.

□

We finish this section by stating and proving a result on simultaneous approximations of convex bodies by simple and strongly isomorphic polytopes which will be useful to prove the classical Alexandrov–Fenchel inequality in Section 8.2.

Theorem 3.4.2. *For any convex bodies $K_1, \dots, K_m \in \mathcal{K}^n$ there exist sequences $(P_1^j)_{j\in\mathbb{N}}, \dots, (P_m^j)_{j\in\mathbb{N}}$ such that*

(i) *$P_1^j, \dots, P_m^j$ are simple and strongly isomorphic for each $j \in \mathbb{N}$, and*

(ii) *for each $i = 1, \dots, m$ we have that $P_i^j \to K_i$ in the Hausdorff metric as $j \to \infty$.*

Proof. It suffices to prove that for any $\varepsilon > 0$ there exist polytopes $P_1, \dots, P_m$ which are simple and strongly isomorphic with $d_{\mathcal{H}}(K_i, P_i) < \varepsilon$ for each $i = 1, \dots, m$. To do that, we start with approximations given as in Proposition 3.4.1: we let $Q_1, \dots, Q_m$ be proper polytopes such that $d_{\mathcal{H}}(Q_i, K_i) < \varepsilon/2$ for each $i = 1, \dots, m$. Put $P := Q_1 + \dots + Q_m$, and let P_0 be a simple polytope constructed from P as in Lemma 3.3.3. For each $i \in \{1, \dots, m\}$, we let $P_i := Q_i + \alpha P_0$, where α is small enough such that $d_{\mathcal{H}}(P_i, Q_i) < \varepsilon/2$. Notice that this gives

$$d_{\mathcal{H}}(K_i, P_i) < \varepsilon$$

for all $i \in \{1, \dots, m\}$. If x is a vertex of some P_i, then $x = z + \alpha w$ for some $z \in \operatorname{ext}(Q_i)$ and $w \in \operatorname{ext}(P_0)$. Thus, Exercise 3.5 yields

$$\mathrm{NC}_{P_i}(x) = \mathrm{NC}_{Q_i}(z) \cap \mathrm{NC}_{\alpha P_0}(\alpha w) = \mathrm{NC}_{Q_i}(z) \cap \mathrm{NC}_{P_0}(w). \tag{3.4.3}$$

From Lemma 3.3.3 we know that

$$\mathrm{NC}_{P_0}(w) \subseteq \mathrm{NC}_P(p) \tag{3.4.4}$$

for some vertex p of P. Also, since $P = Q_1 + \dots + Q_m$, we have that $p = q_1 + \dots + q_m$ for suitable $q_k \in \operatorname{ext}(Q_k)$, $k = 1, \dots m$ and, using Exercise 3.5 again, we arrive at

$$\mathrm{NC}_P(p) = \mathrm{NC}_{Q_1}(q_1) \cap \dots \cap \mathrm{NC}_{Q_m}(q_m).$$

In particular,

$$\mathrm{NC}_P(p) \subseteq \mathrm{NC}_{Q_i}(q_i). \tag{3.4.5}$$

It follows that

$$\mathrm{NC}_{P_i}(x) \subseteq \mathrm{NC}_{Q_i}(z) \cap \mathrm{NC}_{Q_i}(q_i).$$

This gives that $z = q_i$. Indeed, since x is a vertex of P_i we have that the left-hand side of the above is n-dimensional. If $q_i \neq z$, then Proposition 3.2.1 yields that the right-hand side cannot be n-dimensional, which is a contradiction.

As a consequence of the equality $q_i = z$, from (3.4.4) and (3.4.5) we get that $\mathrm{NC}_{P_0}(w) \subseteq \mathrm{NC}_P(p) \subseteq \mathrm{NC}_{Q_i}(z)$. Hence (3.4.3) gives that

$$\mathrm{NC}_{P_i}(x) = \mathrm{NC}_{P_0}(w).$$

Once x is an arbitrary vertex of P_i, we get from Proposition 3.3.1 that P_i is strongly isomorphic to P_0. Since P_0 is simple, this also means that P_i is simple. Summarizing, the polytopes $P_1, \dots, P_m$ are

strongly isomorphic and simple, and we already have that $d_{\mathcal{H}}(K_i, P_i) < \varepsilon$ for each $i \in \{1, \ldots .m\}$. The proof is complete.

□

3.5 Exercises

The main sources for the exercises in this section are [1183] and [2559]. We strongly recommend the interested reader to take a look at these references.

Exercise 3.1. *Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Verify that for any polytope $P \subseteq \mathbb{R}^n$ we have that $T(P)$ is also a polytope.*

Exercise 3.2. *Let $\pi\colon \mathbb{R}^{2n} \to \mathbb{R}^n$ be given as*

$$\pi(x, z) = x + z,$$

for any $x, z \in \mathbb{R}^n$. Verify that for any sets $A, B \subseteq \mathbb{R}^n$ it holds that

$$\pi(A \times B) = A + B.$$

Use this fact to give another proof of the fact that the Minkowski sum of polytopes is a polytope.

Exercise 3.3. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope having vertices $\{x_1, \ldots, x_k\}$. Prove that there is a number $\rho > 0$ with the property that for any $j = 1, \ldots, k$ the ball $B(x_j, \rho)$ only touches the facets of P which contain x_j.*

An intersection of finitely many closed half-spaces in $\mathbb{R}^n$ is called a *polyhedron*. In view of Corollary 3.1.6, the difference between polyhedrons and polytopes is that the first ones are not necessarily bounded. The exercise below consists in a useful characterization of polyhedrons in terms of standard linear algebra.

Exercise 3.4. *Verify that a set $P \subseteq \mathbb{R}^n$ is a polyhedron if and only if there exists a number $m \in \mathbb{N}$, an $m \times n$ matrix M, and a vector $v \in \mathbb{R}^m$ such that*

$$P = \left\{x \in \mathbb{R}^n : Mx \leq v\right\},$$

where the inequality is taken componentwise.

Exercise 3.5. *Let $K, L \in CC^n$. Prove that for any $x \in \operatorname{relbd}(K)$ and every $y \in \operatorname{relbd}(L)$ it holds the equality*

$$\mathrm{NC}_{K+L}(x + y) = \mathrm{NC}_K(x) \cap \mathrm{NC}_L(y).$$

Exercise 3.6. *Let $K \subseteq \mathbb{R}^n$ be compact and convex. Prove that*

$$\operatorname{diam}(K) = \max\{|x - z| : x, z \in \operatorname{ext} K\}.$$

Conclude that the diameter of a polytope is attained by a pair of vertices.

Exercise 3.7. *Let $P \subseteq \mathbb{R}^n$ be a polytope, and let F, G be faces of P. Prove that if $G \subseteq F$, then G is a face of F.*

A *lattice* is an algebraic structure consisting of a set X and two binary commutative and associative operations $\vee$ (*join*) and $\wedge$ (*intersection*) such that

(i) $x \vee (x \wedge y) = x$ and

(ii) $x \wedge (x \vee y) = x$

for any $x, y \in X$. The objective of the next exercise is to verify that the faces of a polytope form a lattice with suitable operations.

Exercise 3.8. *Let $P \subseteq \mathbb{R}^n$ be a polytope, and denote by $\mathcal{F}(P)$ the family of the faces of P. For any $F, G \in \mathcal{F}(P)$, define the binary operations* intersection *and* join *respectively by*

$$F \wedge G := F \cap G \quad \text{and}$$
$$F \vee G := \bigcap \big\{H \in \mathcal{F}(P) : F, G \subseteq H\big\}.$$

(a) *Check that $\big(\mathcal{F}(P), \wedge, \vee\big)$ is a lattice. This is called the* face lattice of P.

(b) *Verify that $\emptyset$ and P satisfy $\emptyset \vee F = F$ and $P \wedge F = F$ for any $F \in \mathcal{F}(P)$. That is, the sets $\emptyset$ and P are* identity elements *of the join and the intersection operations, respectively.*

Remark 3.5.1. A *bounded lattice* is a lattice which contains identity elements for the join and intersection operations. In view of **(b)**, the face lattice of a polytope is always a bounded lattice. The identity elements of the join and intersection operations are called the *bottom of the lattice* and the *top of the lattice*, respectively.

Exercise 3.9. *Let $P \subseteq \mathbb{R}^n$ be a polytope, and let $F, G \subseteq \mathbb{R}$ be such that $G \subseteq F$. Assume that* $\dim(G) = j$ *and* $\dim(F) = k$. *Prove that there are faces $F_{j+1}, \ldots, F_{k-1}$ of P such that* $\dim(F_i) = i$ *for each $i = j + 1, \ldots, k - 1$, and*

$$G \subseteq F_{j+1} \subseteq \ldots \subseteq F_{k-1} \subseteq F.$$

Recall that a *partial order* in a set X is a binary relation (which, for simplicity, we will denote by $\leq$) which is reflexive, anti-symmetric, and transitive, meaning respectively that

(i) $x \leq x$ for all $x \in X$,

(ii) for any given $x, y \in X$, $x \leq y$ and $y \leq x$ imply $x = y$, and

(iii) given $x, y, z \in X$, if $x \leq y$ and $y \leq z$, then $x \leq z$.

A set X endowed with a partial order is called a *partially ordered set* (or a *poset*).

Exercise 3.10. *Let $(X, \wedge, \vee)$ be a lattice. Define a binary operation in X writing $x \leq y$ whenever $x = x \wedge y$. Prove that this binary operation is a partial order in X. Show also that for the face lattice of a polytope, this partial order is simply the inclusion.*

Let $(X, \wedge, \vee)$ be a bounded lattice (see Remark 3.5.1 above). As is usual in the literature, we will denote the identity element of the join by 0 and the identity element of the intersection by 1 (in the case of the face lattice of a polytope P, we have that $0 = \emptyset$ and $1 = P$, or course). An element $x \in X \setminus \{0\}$ is said to be an *atom* of X if there is no element $z \in X \setminus \{0, x\}$ such that $0 \leq z$ and $z \leq x$. Moreover, an element $a \in X$ is called a *co-atom* of X if there is no $b \in X \setminus \{a, 1\}$ such that $a \leq b$ and $b \leq 1$.

A bounded lattice is said to be *atomic* if each of its elements is the join of finitely many atoms, and *co-atomic* if each of its elements is the intersection of finitely many co-atoms.

Exercise 3.11. *Let $P \subseteq \mathbb{R}^n$ be any given proper polytope.*

(a) *Show that the atoms of $\mathcal{F}(P)$ are precisely the vertices of P, and that the co-atoms of $\mathcal{F}(P)$ are exactly the facets of P.*

(b) *Prove that the face lattice of P is atomic and co-atomic.*

We say that a bounded lattice $(X, \wedge, \vee)$ is *complemented* if for each $x \in X$ there exists an element $y \in X$ such that $x \wedge y = 0$ and $x \vee y = 1$.

Exercise 3.12. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope. Show that $\mathcal{F}(P)$ is complemented.*

Recall that C^n stands for the family of non-empty compact subsets of $\mathbb{R}^n$, and let $\mathbb{Z}^n := \{(z_1, \ldots, z_n) : z_1, \ldots, z_n \in \mathbb{Z}\}$ be the *grid* of *integer points* of $\mathbb{R}^n$. For each $A \in C^n$, let

$$L(A) := \sum_{x \in A \cap \mathbb{Z}^n} 1.$$

That is, $L(A)$ is the number of integer points of $\mathbb{R}^n$ which lie in A.

Exercise 3.13. **(a)** *Regard $L\colon C^n \to \mathbb{R}$ to be defined as above. Check that L has the* valuation property*: for any $A, B \in C^n$, it holds that*

$$L(A \cup B) + L(A \cap B) = L(A) + L(B).$$

(b) *Prove that*

$$L(m[0,1]^n) = (m+1)^n$$

for any $m \in \mathbb{N}$.

Remark 3.5.2. A *lattice polytope* is a polytope whose vertices are integer points. For any lattice polytope P, the number of integer points of mP is polynomial on $m \in \mathbb{N}$. Such a polynomial is known as the *Ehrhart polynomial* of P. In Item **(b)** above, we are computing precisely the Ehrhart polynomial of the unit cube of $\mathbb{R}^n$. For more in this direction, we refer the reader to the survey [405].

Let m, n be given natural numbers, P be a proper polytope in $\mathbb{R}^n$ and Q be a proper polytope in $\mathbb{R}^m$. A map $\pi\colon P \to Q$ is said to be a *projection of polytopes* if

(i) π is the restriction to P of a map having the form $x \mapsto Ax + b$, where $A\colon \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation and $b \in \mathbb{R}^m$;

(ii) $\pi(P) = Q$.

Exercise 3.14. *Let $\pi\colon P \to Q$ be a projection of polytopes. Prove that if F is a face of Q, then $\pi^{-1}(F)$ is a face of P.*

For the next exercise, we introduce a terminology which will be also used later in this book. A set having the form

$$H := \mathbf{h}^-(u, 0) \cap \mathbb{S}^{n-1}$$

for some $u \in \mathbb{S}^{n-1}$ is called a *closed hemisphere* of $\mathbb{S}^{n-1}$.

Exercise 3.15. *Let $E = \{u_1, \dots, u_m\} \subseteq \mathbb{S}^{n-1}$ be a non-empty finite set which is not contained in a closed hemisphere of $\mathbb{S}^{n-1}$.*

(a) *Show that* $\operatorname{pos}(E) = \mathbb{R}^n$.

(b) *Prove that for any numbers $\alpha_1, \dots, \alpha_m > 0$ it holds that the polyhedron*

$$P := \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, \alpha_j)$$

is a polytope.

Hint: In Item **(a)**, verify that if $\operatorname{pos}(E) \neq \mathbb{R}^n$, then $\operatorname{pos}(E)$ has a supporting hyperplane (see Theorem 2.2.1).

Remark 3.5.3. The construction in Item **(b)** is a *Wulff shape*. This type of construction will be studied later, in Section 8.1.

We say that a proper polytope $P \subseteq \mathbb{R}^n$ is *circumscribed* around a convex body $K \subseteq \mathbb{R}^n$ if $K \subseteq P$ and K intersects each facet of P.

Exercise 3.16. *Let $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ be mutually distinct vectors such that $\operatorname{pos}\{u_1, \dots, u_m\} = \mathbb{R}^n$. Prove that there exists a polytope $P \subseteq \mathbb{R}^n$ circumscribed around B^n whose facets have precisely $u_1, \dots, u_m$ as unit outer normal vectors.*

Exercise 3.17. *Prove that any Steiner symmetral of a polytope is a polytope.*

Exercise 3.18. *Let $P \subseteq \mathbb{R}^n$ be a polytope having at least two vertices and whose circumball is $x_0 + rB^n$. Show that $x_0 \in \operatorname{conv}\big((\operatorname{ext} P) \cap \partial(x_0 + rB^n)\big)$. In particular, conclude that $x_0 \in P$.*

Remark 3.5.4. In other words, the center of the circumball of a polytope is in the convex hull of the vertices of P which are in the boundary of the circumball.

Exercise 3.19. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope whose facets are $F_1, \dots, F_m$ with corresponding supporting hyperplanes $\mathbf{h}_1, \dots, \mathbf{h}_m$. Let x an interior point of P, and assume that*

$$\min\big\{\operatorname{dist}(x, \mathbf{h}_j) : 1 \leq j \leq m\big\} = \operatorname{dist}(x, \mathbf{h}_k)$$

for some $1 \leq k \leq m$. Show that the orthogonal projection of x onto $\mathbf{h}_k$ lies in the relative interior of F_k.

Exercise 3.20. *Let $x_1, \dots, x_k \in r\mathbb{S}^{n-1}$ for some $r > 0$, and let $P = \operatorname{conv}\{x_1, \dots, x_k\}$. Prove that $r_{\text{out}}(P) = r$ if and only if $0_{\mathbb{R}^n} \in P$.*

Exercise 3.21. *Let $P \subseteq \mathbb{R}^n$ be a polytope having at most $n+1$ vertices. Prove that*

$$r_{\text{out}}(P) \leq \left(\frac{n}{2n+2}\right)^{1/2} \operatorname{diam}(P). \tag{3.5.1}$$

Remark 3.5.5. In particular, from Exercise 3.6 we have that (3.5.1) holds also if we replace P by any finite set containing at most $n+1$ points (observe that the circumradius is clearly monotone under inclusion).

Exercise 3.22. *Show that (3.5.1) actually holds for any bounded set $A \subseteq \mathbb{R}^n$.*

Hint. Use Theorem 2.1.5 and Remark 2.1.7.

Exercise 3.23. *Let X be an n-simplex in $\mathbb{R}^n$. Prove that there exists a unique ball $B \subseteq X$ touching every facet of X.*

Exercise 3.24. *Let $n \geq 1$, and let $P \subseteq \mathbb{R}^n$ be a proper polytope. Show that we may decompose P as the union of n-dimensional simplices $X_1, \ldots, X_m$ such that for any $i \neq j$ we have that the intersection $X_i \cap X_j$ has dimension smaller than n.*

Exercise 3.25. **(a)** *Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear isomorphism, and let $u \in \mathbb{S}^{n-1}$, $\alpha \in \mathbb{R}$. Check that*

$$T(\mathbf{h}(u,\alpha)) = \mathbf{h}\big((T^{-1})^* u, \alpha\big).$$

(b) *Conclude that if a set $E \subseteq \mathbb{R}^n$ is supported by the hyperplane $\mathbf{h}(u,\alpha)$ at a point x, then TE is supported by $\mathbf{h}\big((T^{-1})^* u, \alpha\big)$ at the point Tx.*

(c) *Let $P \subseteq \mathbb{R}^n$ be a proper polytope whose facets are $F_1, \ldots, F_m$, having corresponding unit outer normal vectors $u_1, \ldots, u_m$. Prove that for any linear isomorphism $T\colon \mathbb{R}^n \to \mathbb{R}^n$, the polytope $T(P)$ has facets $T(F_1), \ldots, T(F_m)$ with corresponding unit outer normal vectors $(T^*)^{-1}u_1/|(T^*)^{-1}u_1|, \ldots, (T^*)^{-1}u_m/|(T^*)^{-1}u_m|$.*

Exercise 3.26. *Let $K, L \in CC^n$. We say that L is* locally embeddable *in K if for every point $x \in \partial K$ there exists a point $y \in L$ and a neighborhood U of y such that*

$$(L \cap U) + x - y \subseteq K.$$

Let $P \subseteq \mathbb{R}^n$ be a polytope. Prove that any positive scalar multiple of P is locally embeddable in P.

The next exercise elucidates an interesting geometric property relating the zeros of a complex polynomial with the zeros of its derivative. No prior knowledge in complex analysis is needed, just simple differentiation rules.

Exercise 3.27. *Let p be a polynomial in one complex variable. Prove that the zeros of the derivative of p are contained in the convex polygon determined by the zeros of p.*

For the next exercise, use the fact that the graph of vertices and edges of any polytope is connected (for a proof, see [2560]).

Exercise 3.28. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope.*

(a) *Prove that the conjugate face of a vertex of P° is a facet of P and that the conjugate face of an edge of P° is a $(n-2)$-dimensional face of P.*

(b) *Let F, G be facets of P. Show that there exists a sequence $F_1, \ldots, F_r$ of facets of P such that $F_1 = F$, $F_r = G$ and $F_i \cap F_{i+1}$ is $(n-2)$-dimensional for any $i = 1, \ldots, r-1$.*

3.6 Notes

The basic concepts in the theory of convex polytopes are, in great parts, folklore and therefore less the subject of current research. Thus, we decided to write in this chapter only notes about the special themes of *strongly isomorphic polytopes* (which are, roughly speaking, equivalence classes of convex polytopes having the same combinatorial type) and the essential subfield of *approximation methods and results* regarding polytopes.

3.6.1 Strongly isomorphic polytopes

Two convex polytopes are called *strongly isomorphic* if for any direction vector u the dimensions of their corresponding faces (i.e., of the intersections of the polytopes with their supporting hyperplanes normal to u) are equal. Such polytopes play an important role in proofs of the Alexandrov–Fenchel inequality.

They are mentioned shortly in the second edition of Grünbaum's book [1085], namely in Sections 3.4 (page 50) and 3.6 (page 52c). There also other names (instead of "strongly isomorphic") are mentioned, namely *related, normally equivalent, analogous*, and *locally similar*. And on page 52c the reader is referred to Lecture 7 in Ziegler's book [2560] and to the Notes for Section 2.4 of Schneider's monograph [2167]. The latter is perhaps the most comprehensive source which is easily accessible: At the end of Section 2.4, Schneider mentions further alternative names, namely (in addition to the above) *strongly combinatorially isomorphic* and *strongly combinatorially equivalent*, and he proves several theorems and lemmas on strongly isomorphic polytopes (referring to the subcase of simple polytopes and to the approximation of convex bodies by strongly isomorphic polytopes). The latter goes back to Alexandrov (1937), possibly one of the oldest references on this polytope class (see Alexandrov's article on mixed volumes in [38]). And Shephard [2229], studying properties of Minkowski sums of polytopes, uses the "locally similar" notion. In Lemma 2.4.11 from [2167], that characterizes strongly isomorphic polytopes, Schneider goes back to Meyer [1778] (Corollary 2.7) and Kallay [1343].

The equivalence class of a strongly isomorphic polytope P is called its *α-type*. If the α-type of P is uniquely determined by the system of outer normal vectors of P, then P is called *monotypic*. An important paper on monotypic polytopes showing many properties of them and of related families is [1754] by McMullen, Schneider, and Shephard. One of these families is that of *primitive polytopes*: the removal of any facet of such a polytope would yield an unbounded polyhedral set. Another type, namely that of *strongly monotypic polytopes*, is also introduced. Occurring in four classes, the monotopic polytopes in 3-space are completely classified. Three of the mentioned four classes of monotypic 3-polytopes consist of strongly monotypic polytopes, and a characterization of the strongly monotypic 3-polytopes in the fourth class is given by Borowska and Grzybowski in [457], which completes the classification of strongly monotypic polytopes in Euclidean 3-space.

Sallee sets (see Sallee [2094]) are compact convex sets A such that for all sets B the Minkowski difference $A - B$ of A and B is a summand of A. Therefore the family of Sallee sets characterizes the Minkowski subtraction, and the *Sallee polytopes* are interestingly related to monotypic polytopes. Borowska and Grzybowski [458] consider sets with the *intersection property* in Euclidean 3-space: A convex body K has this property if the intersection of any family of its translates is a Minkowski summand of K. They prove that the class of all 3-polytopes that have the intersection property coincides with that of all strongly monotypic polytopes.

Fourneau [893] introduces a metric on the family of translation classes of n-dimensional convex polytopes related to decomposability. This metric yields a *characteristic property of the family of*

strongly isomorphic polytopes, and the connected components of dimension one correspond to the translation classes of indecomposable polytopes.

Using his polytope algebra and weights on polytopes, McMullen [1745] presents an *algebraic proof of the second Minkowski inequality* for strongly isomorphic, simple n-dimensional polytopes.

In [2154] Schneider gives a comprehensive update on results around the *Alexandrov–Fenchel inequality*. In addition, he proves the following: Let in this classical inequality $V(K, L, K_3, ..., K_n)^2 \geq V(K, K, K_3, ..., K_n)V(L, L, K_3, ..., K_n)$ the convex bodies $K_3, ..., K_n$ be strongly isomorphic polytopes. Then equality in this inequality is equivalent to a special condition on the support functions of K and L. And if, additionally, K and L are strongly isomorphic to the polytopes K_i, then K and L are homothets. Also Cordero-Erausquin et al. [659] deal with that inequality. They offer a *new Alexandrov–Fenchel-type inequality* based on induction and using some algebraic and convexity properties. Also here, in one important proof step, they assume that the participating convex bodies are *simple and strongly isomorphic polytopes*.

The n-dimensional *associahedron* came up in various contexts and with many applications. Ceballus et al. [585] study various constructions of it, including one previously unpublished construction by Santos. They want to enumerate the non-equivalent realizations given by the so-called Hohlweg–Lange construction and the construction of Santos. In this concept two convex polytopes are said to be "equivalent" if their normal fans are linearly isomorphic, i.e., if they are strongly isomorphic.

Generalized permutahedra can be obtained from the ordinary n-dimensional permutohedron (the convex hull of the permutations on $n + 1$ elements) by moving its vertices in such a way that the edge directions remain, while faces are allowed to merge and disappear during the procedure. Supina [2328] studies the Hopf monoid of the generalized permutohedra that are invariant under the action of the symmetric group. The vertices of these special generalized permutohedra are obtained when the symmetric group acts on a point; they are called orbit polytopes. It is shown that, up to *strong isomorphism*, orbit polytopes are in one-to-one correspondence with integer compositions. Thus, the set of integer compositions can be endowed with a Hopf algebra structure.

3.6.2 Approximation of convex bodies by polytopes

There are many contributions to this interesting subfield, such that in this part we mainly reduce to books and surveys, completed by a *selection* of recent research papers. In great parts, the latter ones are given in chronological order. We start with *parts of monographs* dedicated to such approximation results.

BOOK PARTS

The booklet [831] of Faro Rivas treats *approximation of centrally symmetric convex bodies* in finite-dimensional spaces by so-called *polynomial bodies* (representable by homogeneous polynomials of even degree). First useful tools are introduced (support functions, Blaschke's selection theorem, approximation by convex polyhedra, linear uniform approximation, certain uniqueness theorems, etc.), and then properties of polynomial bodies, symmetric star-shaped bodies, and different kinds of convergence are discussed. In the last two chapters various conditions for *outer and inner approximation of convex bodies by polytopes*, ellipsoids, algebraic bodies, and so on are given. A useful list of references on approximation of convex bodies is given.

Hadwiger presents in subsection 6.1.3 of his classical monograph [1134] two fundamental theorems about approximating a general convex body K by polytopes: the first treats the approximation of K from outside and inside, and the second the approximation of K lying "between" two homothetical copies of a polytope.

In Chapter III, §14, of his book [1537], Leichtweiss carefully lays out the basics of a little theory of approximating convex bodies based on the *Hausdorff metric*, and in particular he treats also polytope approximation.

Groemer [1042] discusses in his Section 2.3 on metrics for families of convex bodies useful tools needed in this field and a basic lemma on the *approximation of arbitrary convex bodies by special ones*. In other parts of his book, different approximation results are discussed, e.g., the approximation of support functions by spherical harmonics.

Section 6.3 of Moszyńska's monograph [1857] refers to the following type of approximation of convex bodies by polytopes: placing the boundaries of given convex bodies suitably "between" the boundaries of homothetical approximating polytopes.

Gruber discusses approximation of convex bodies in Chapter 11 of his monograph [1070]. In Section 11.1 he treats approximation results regarding John's ellipsoid, where also the Banach–Mazur compactum and Ball's reverse isoperimetric inequality occur, and in Section 11.2 he presents asymptotic results on *approximating convex bodies by circumscribed polytopes having at most m facets*, or by *inscribed polytopes with at most m vertices*. Related isoperimetric problems for polytopes and an interesting heuristic principle are discussed, too.

The book [2179] of Schneider and Weil contains a broad representation of results on *approximation of convex bodies by polytopes* and related topics, namely in its subsection 8.2.4 and the notes thereof. This book part can be taken as one essential starting point for our representation here, reflecting with numerous references the state of the art in several related directions, roughly until 2008.

In Schneider's book [2167], the *approximation of convex bodies by polytopes* is discussed in Section 1.8 (see Theorem 1.8.19 and Note 20 of that Section), and also in Section 3.4 on approximation of convex bodies in view of *Minkowski addition* (where the emphasis is on participating *smooth convex bodies*) one can find, as subcases there, some material on polytope approximation. And finally, also the last note to Section 2.4 refers to polytope approximation.

At the end of the notes to Chapter 8 of the book [1713] of Martini, Montejano, and Oliveros, many results on the *approximation of bodies of constant width* are surveyed, for instance also treating "polygon-like sets" such as Reuleaux polygons in the plane, and further piecewise spherical sets, also in higher dimensions.

We continue with survey-like or expository papers.

SURVEYS AND EXPOSITIONS

An early related presentation is the exposition of Affentranger [21], which surveys problems on *random approximation* of the following type: Let K be an n-dimensional convex body, and consider the convex hull of m independent random points $\{x_1, ..., x_m\}$ distributed identically and uniformly in the interior of K. One defines a random variable for this convex hull which can be something like *number of vertices, volume, mean width*, etc., and the author is interested in the mathematical expectation of this random variable. The survey excellently presents problems of this type, including various unsolved problems interesting from the viewpoint of convex geometry and also stochastic geometry.

Gruber's article [1053] collects results on the *approximation of convex bodies by polytopes*, mainly in the *Hausdorff metric* or the *symmetric difference metric*. The used polytopes belong to restricted families, having for instance vertex numbers bounded from above. In this setting, the author discusses dense classes of special polytopes, properties of best approximating polytopes, approximation by random polytopes, and various further aspects or concepts. Also related *Baire category results* are presented. Then, as approximating sets, special types of convex bodies are taken into consideration: simplices, parallelepipeds, balls, and ellipsoids. The paper shows that there are much less approximation results treating smooth convex bodies.

Schneider [2150] studies convex hulls of independent, identically distributed random points whose distribution depends somehow on a given (smooth) convex body K; also planar results are obtained,

e.g., on random polygons generated by intersecting random supporting half-planes of K. The second part of [2150] is a survey on results on convex hulls of random points in n-dimensional convex bodies, particularly on results on the asymptotic behaviour of expectations regarding several geometrically interesting functionals.

Continuing this and more general than in former such articles, Gruber surveys in his handbook article [1059] problems of *approximation of certain classes of convex bodies by bodies of other types*, e.g., by smooth bodies or polytopes with a prescribed vertex or facet number. Again the used metrics are mainly *Hausdorff metric* or *symmetric difference metric*, but also other ones occur, e.g., *Eggleston's metric, surface deviation*, and *Banach–Mazur distance*. Also algorithmic and asymptotic aspects are discussed, and further special classes of convex bodies (like zonoids) are discussed in this setting. For several results the main proof ideas are clarified.

Complementing this handbook article, Gruber surveys in [1066] again results on the *approximation of convex bodies by convex polytopes* with a fixed upper bound of vertices or facets, discussing the cases of inscribed and circumscribed polytopes, respectively. Again or similarly, *Hausdorff metric, surface deviation, symmetric difference metric, Banach–Mazur distance*, and *Schneider distance* are used, various upper bounds, asymptotic estimates, and algorithmic approximations are studied, and outlines of proofs are given. And in [1067], Gruber compares *best and random approximation of n-dimensional convex bodies by inscribed polytopes* with vertex number tending to infinity, and by *circumscribed polytopes* with facet number tending to infinity. Here also the *Nikodým metric, measures based on other quermassintegrals*, and the above-mentioned distances are taken into consideration, and related expectations for random polytopes are studied. Furthermore, results on best and worst approximable convex bodies, algorithms for constructing approximating polytopes, and the related approximation by *floating bodies* are presented.

From the viewpoint of *Baire categories*, Section 8 of Gruber's handbook article [1062] treats approximation of convex bodies by polytopes. A couple of related theorems is given, based on the Hausdorff metric, on the symmetric difference, and the Banach–Mazur distance. Also Zamfirescu's surveys [2522] and [2523] on Baire category results refer to approximation of convex bodies, in particular also by polytopes.

A *cap* of spherical radius α on the unit sphere is the set of points within spherical distance α from a given point on the sphere. *Cap coverings* are coverings of the sphere by caps following various demands. And a *Macbeath region* is defined as follows: Let K be an n-dimensional convex body and x a point; then $K \cap (2x - K)$ is the Macbeath region around x. Bárány (see [211]) surveys results whose proofs use cap coverings and Macbeath regions as tools. Among other topics, he discusses in this sense random and lattice polytopes, and related approximation results. Similarly, in [214], Bárány gives a nice introduction to problems around *floating bodies* and the cap-covering technique, using them as tools. With such methods, he gives proofs of various results, e.g., on random polytopes. In the final part he applies the cap-covering technique to *approximation of convex bodies by polytopes and lattice polytopes*.

In their expository paper [248], Barvinok and Veomett discuss the problem how well a convex body in n-dimensional vector space V can be approximated by a set X for which the membership question (does a given point $x \in V$ belong to X?) can be answered in an efficient way (namely in time polynomial in n). In this setting, *approximations of convex bodies by ellipsoids, by algebraic hypersurfaces, by projections of polytopes with controlled number of facets*, and by a *section of the cone of positive semi-definite quadratic forms* are discussed.

Schütt [2197] surveys results on best and random approximation of n-dimensional (smooth) convex bodies by polytopes with given vertex number, in particular comparing the dependence on n in asymptotic formulae (for this and more, see also the papers [1060], [1061], [979], and [1610] of Gruber, Glasauer, and Ludwig). An improvement of a result of Ludwig et al. [1621] regarding the unit ball is conjectured, and a brief sketch of a proof (using random polytopes) is given.

Also the surveys [2026] of Reitzner and [1255] of Hug, described more in our subsection on integral and stochastic geometry, should be mentioned here, since they reflect a lot of related material.

Also the more recent survey [1997], due to Prochno et al., goes into this direction. The selected topics discussed there refer to the *volume of the symmetric difference*, and to polytope approximation where the polytopes are obtained as convex hulls of suitable finite point sets. This leads the reader to the notions of *affine surface area* and *floating bodies*. After dealing with optimal approximations of convex bodies by polytopes, random approximation is presented, namely by convex hulls of stochastically independent random points (either from the interior or from the boundary of the convex body). Also results on *best approximation of polytopes by polytopes* (with a smaller vertex number) and *random approximation of polytopes by polytopes* are surveyed.

Werner [2455] discusses results on *floating bodies*, their variants (such as *surface bodies*), *weighted floating bodies*, and applications thereof. For example, representations of the *affine surface area* and of the L_p *affine surface areas* by limit relations for volumes of convex bodies are given. These allow extensions to convex bodies without the (before necessary) smoothness assumption. Also results on best and random approximations of convex bodies by polytopes are discussed, whose degree of approximation is measured by the *volume difference*.

Finally we cite several more recent *research papers* on approximation by polytopes. They are chronologically ordered.

RESEARCH PAPERS

Schütt [2195] investigates the approximation of convex bodies by polytopes with respect to the *symmetric difference metric*. For K an n-dimensional convex body and $t > 0$, its convex floating body K_t is obtained by cutting off from K all calottae of volume t. It is shown that one may place between K and K_t a convex polytope P_m, whose vertex number m is bounded from above. Thus, geometric information about the well-approximating polytopes is obtained. Also upper bounds on the right order for the volume difference of K and P_m are given. Using the related notion of *illumination body*, similar results for circumscribed convex polytopes with m facets are obtained.

K. J. Böröczky [389] studies the *approximation of smooth convex bodies* (being also of class C^2) by *polytopes with given vertex or facet number*. This continues a series of results due to Schneider, Gruber, Ludwig, Glasauer, and the author himself (referring to asymptotic formulae) and of Gruber, Schneider, and Glasauer (with respect to the shape of the best approximating polytopes); see the survey [1067] of Gruber. Here the author shows that in all these cases the positivity assumption regarding Gaussian curvature can be omitted. Similarly, Gruber [1069] estimates the error of asymptotic formulae for *volume approximation of sufficiently smooth convex bodies by polytopes*, assuming that the number of their facets tends to infinity. And K. J. Böröczky [390] investigates n-dimensional convex bodies ($n > 2$) with C^3 boundary whose Gauss curvature is everywhere positive. This is done with respect to the Hausdorff distances between these bodies and the approximating polytopes, which have at most m vertices (m facets). The goal is to bound the errors occurring in the existing asymptotic formulae from above. The *Banach–Mazur metric* and the *Schneider distance* are also taken into consideration.

K. J. Böröczky and Reitzner [410] consider m independent points in the boundary of a convex body K of C^2 class, and the convex polytope P being the intersection of the supporting half-spaces of K determined by these points. Regarding outer approximation of K by P, they derive *asymptotic expansions for the expectation of the volume, the surface area, and the mean width of* P. The authors compare their results also with analogous findings on the corresponding best *inner approximations by polytopes*.

For *given edge number*, K. J. Böröczky et al. [400] verify an asymptotic estimate for the *Hausdorff distance* between a three-dimensional convex body K with boundary of class C^2 and its *best approximating inscribed (respectively, circumscribed) convex polyhedron*. There are only a few results on

approximation with polytopes of given edge number, but a further paper related to this topic is [401], also referring to the three-dimensional situation and taking the *volume as approximating measure*.

K. J. Böröczky and Csikós [394] study the approximation of n-dimensional convex bodies having C^2 boundary by circumscribed polytopes of given facet number in terms of *minimal surface area*. As a step to the general results, the authors look first also at the analogous approximation of paraboloids. Further on, they present *shape properties* of the circumscribed polytopes with given facet number having minimal surface area and approximating such bodies in terms of *Hausdorff distance*. An asymptotic formula in terms of the facet number is given for the difference of the surface areas of the polytope and the approximated body.

Based on well-known classical results of Renyi and Sulanke, K. J. Böröczky and Schneider [413] derive upper and lower bounds of optimal orders for the *mean width difference* between an n-dimensional convex body and its approximating polytope being the intersection of m independent random closed half-spaces as m tends to infinity. For *simplicial polytopes* in this setting, a precise asymptotic formula is obtained.

From the *computational viewpoint*, Arya et al. [138] treat approximation of convex bodies by polytopes with suitable bounds on their vertex or facet numbers, where also the *width* of the convex body has to be bounded from below. An analogy to investigations of the concept of *Mahler volume* (treating the product of the volumes of a convex body and its polar body) is found, and furthermore improvements of the best known search times for approximate nearest neighbour searching in finite dimensions are obtained.

Since for projections the membership oracle can be constructed efficiently, Litvak, Rudelson, and Tomczak-Jaegermann [1584] deal with the question to what extent an arbitrary convex body can be well approximated by *projections of polytopes with few facets*. They give an affirmative solution to a problem of Barvinok and Veomett (see [248]) by showing that in general an n-dimensional convex body cannot be approximated by a projection of a section of a simplex of subexponential dimension. A lower estimate for the *minimal Banach–Mazur distance* between a certain convex symmetric body and a projection of a polytope with given facet number m is presented. This bound is sharp for all $m > n$ up to logarithmic terms.

Studying a *stationary Poisson hyperplane process* X of given directional distribution and intensity in Euclidean n-space with $n \geq 2$, Hug and Schneider [1258] fix a convex body K (by generalizing the zero cell of X) and consider the intersection of all closed half-spaces bounded by hyperplanes of X and containing the convex body K. Regarding *Hausdorff distance*, they study how good K can be approximated by these random polytopes when the intensity increases. It is also investigated how the approximation depends on the directional distribution, also in view of properties of K. In the strongly related paper [1261], the same authors derive various interesting theorems on random polytopes obtained by approximating convex bodies, with hyperplanes from Poisson hyperplane processes, regarding the convergence of their intensity. As a considered functional also the mean width is chosen.

Barvinok [244] deals with the well-known problem to find, for a convex body K and a given $\varepsilon > 0$, a convex polytope P with the smallest possible vertex number such that $P \subseteq K \subseteq (1+\varepsilon)P$ is guaranteed. He derives a series of related results on polytope approximation whose quality is bounded from above by various parameters. Also the centrally symmetric case is studied. Strongly related is the paper [1876] by Naszódi et al., where *ε-approximation regarding the Banach–Mazur distance* is done and a result of Barvinok from [244] is improved (also deleting the condition of central symmetry).

Lassak [1502] studies estimates related to the approximation of a convex body K by polytopes with respect to *minimal width* (also called thickness) and *diameter*. He describes how well the approximation of K by an inscribed (resp., circumscribed) polytope P with m vertices (resp., facets) is. The quality of approximation is measured by the *ratio of the minimal widths of P and K*, and the same is done for their *diameters*. Also some specific approximations are discussed. The centrally symmetric subcase is taken into extra consideration, and also a surprising singular result is obtained: the best approximation of a circle by an inscribed quadrangle is obtained by a non-regular 4-gon.

Arya, da Fonseca, and Mount [139] investigate the *combinatorial complexity of polytope approximation of convex bodies* in the following sense: Given some $\varepsilon > 0$, determine a polytope of minimal combinatorial complexity, here being the total number of faces of all dimensions of the approximating polytope, whose Hausdorff distance from a convex body K is at most $\varepsilon \cdot \operatorname{diam}(K)$. Improving earlier results significantly, they show that there exists an approximation polytope whose complexity is $\hat{O}(1/\varepsilon^{(n-1)/2})$, where $\hat{O}$ conceals a polylogarithmic factor in $1/\varepsilon$. Important tools for getting this result are Macbeath regions and economical cap covering (developed further by Bárány and Larman; see also the survey [211] of Bárány, already presented above). A further recent overview to investigations of this type is given by Arya et al. in [140].

Bonnet [447] obtains bounds, with respect to the *Hausdorff metric*, for the best approximation of convex bodies by circumscribed polytopes with a given number of facets, where elongation of the approximated body (measured by some isoperimetric ratio) plays an essential role.

Thäle et al. [2356] study the approximation of an n-dimensional convex body K with twice differentiable boundary and positive Gaussian curvature by the convex hull of $N \geq n+1$ independent random points, uniformly distributed in K. They prove *central limit theorems for all intrinsic volumes* of these polytopes as $N \to \infty$. The paper gives a nice overview on former related results, and the authors point out that they give a convenient and transparent proof of the central limit theorems for the intrinsic volumes.

For a convex body K with twice differentiable boundary having positive curvature, Grote and Werner [1052] deduce an upper bound for the approximation of K by a polytope P_f (in arbitrary position) with a given vertex number. Their results refer to the *symmetric difference metric*. The approximating polytope is derived by a random construction involving a probability measure on the boundary of K, with density f with respect to the surface area measure. The dependence on the number of vertices is shown to be optimal. By Grote et al. [1051] such investigations on approximations are continued. Namely, here the main result is an upper bound on the *surface area deviation* between a convex body K with twice differentiable boundary and a polytope with a sufficiently large given vertex number in terms of the *n-affine surface area* of K.

As an example for *numerical approximation algorithms* fitting to our subsection here we should cite also Kamenev's contribution [1345], treating approximations of convex bodies in Euclidean n-space by polytopes. For algorithmical approximation the accuracy is, in the *Hausdorff metric*, conversely proportional to the number of vertices (facets) in the degree of $2/(n-1)$. Numerical approximation algorithms can be *adaptive* (when the vertices or facets of the participating polytopes are constructed successively, depending on the information obtained during the process of approximation), and they can be *non-adaptive* (when parameters of algorithms are defined with the help of available priory information). Important tools are *Minkowskian support and distance functions*. For example, in optimal non-adaptive algorithms for the approximation of smooth convex bodies, calculations based on support functions and extremal curvature information are used. Since there are no optimal non-adaptive algorithms for the approximation of arbitrary convex bodies based on support function calculation, Kamenev studies optimal non-adaptive algorithms for this general case by using *projection functions*.

Naszódi [1875] investigates the following approximation problem: Choose m points independently and uniformly in a centered convex body K, and consider the random polytope P_m being their convex hull. The author estimates the probability that, additionally, P_m contains some smaller copy of K. He gives an explicit constant guaranteeing that this holds true with at least a certain probability.

Continuing investigations of Hoehner, Ludwig, Schütt, and Werner, Kur [1479] gives an improvement regarding the approximation of a convex body K by an n-dimensional polytope P, $n \geq 2$, with at most $N \geq 10n$ facets regarding the *volume of the symmetric difference* of K and P. The obtained upper bounds are optimal up to absolute constants. In addition, the author proves an analogous statement for the *symmetric difference of the surface area*. These results have consequences for the approximation of the n-dimensional Euclidean ball by polytopes with an exponential number of facets, with respect to the distances already mentioned here.

Using three distance notions, Besau, Hoehner, and Kur [304] get asymptotically sharp bounds for approximations of the Euclidean ball or smooth convex bodies by polytopes in high dimensions. The first one is based on *intrinsic volumes*, and the second distance notion is based on the *Wills functional*. Surprisingly, in the first case a polytope is found which is simultaneously optimal for almost all intrinsic volumes, up to absolute constants. In the third case, the authors establish asymptotic formulae for the best approximation of smooth convex bodies by polytopes with a distance notion induced by *dual volumes*, coming from Lutwak's dual Brunn–Minkowski theory.

PLATE 4

Cavalieri's principle

Cylinders with equal volumes

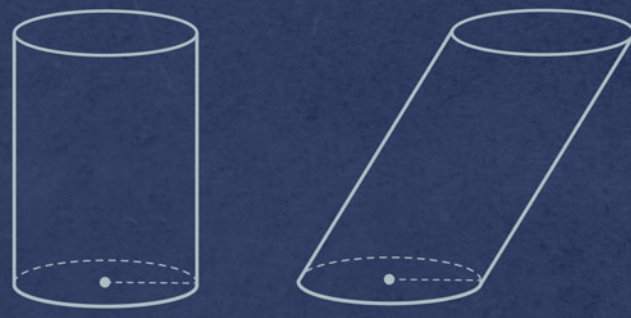
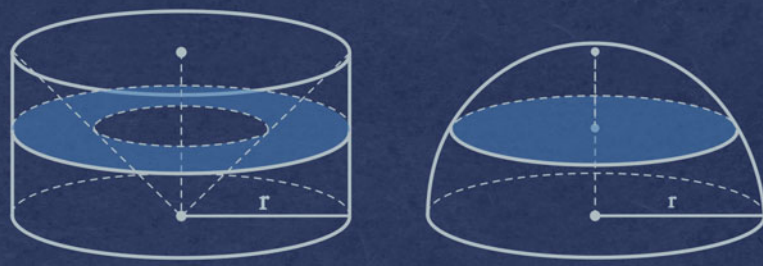


Figure 9

Figure 10

Archimedes



Cavalieri's principle ► (see, e.g., 4.2 and 4.7.1)

Chapter 4
Volume and area

"Mathematics compares the most diverse phenomena and discovers the secret analogies that unite them."

— Joseph Fourier

The notions of "length", "area", and "volume" describe and yield central topics in various mathematical fields, especially also in geometry. Whereas everyone has an idea of what they outline, in modern mathematics these notions have abstract, formal meanings, which allow us not only to have sharp definitions, but also to introduce their extensions to spaces of dimensions larger than 3. That is, in an n-dimensional space, we have precise notions of *k-dimensional volume* for each integer $1 \leq k \leq n$, and it is usual to reserve the terms "area" and "volume" for the $(n-1)$-dimensional and n-dimensional cases, respectively.

The precise concepts that we need come from classical measure theory (for us, this field contains Lebesgue integration theory). The most versatile concept in this regard is that of *Hausdorff measures*, which allows us to define the "k-dimensional volume" of any geometric object for any non-negative number k (although it might be meaningless, in some cases). This concept would be enough for our purposes in this book; however, for "flat objects" it is also possible to work with *Lebesgue measures*, and we also use these measures because they are widespread in the literature.

Of course, our interest in classical measure theory is restricted to the tools that we need to study convex geometry. For that reason, we added an appendix to this book (Appendix A) for readers who want to learn in detail (almost) everything that we use from measure theory. As an alternative, in Section 4.1 below we give an overview on how we want to look at measure theory. This is recommended for those who already have good background in measure theory, or those who even do not want to go through all the details of the proofs.

In Section 4.2, we present the classical principle of Cavalieri in the "format" and notation that we need. Roughly speaking, these results open ways of computing volumes of sets in terms of (smaller dimensional) volumes of their respective sections. Plate 4 shows this principle, in particular the famous observation of Archimedes that the two geometric figures below have, at correspondingly equal heights, also equal section areas. In Section 4.3 we obtain some expressions for volumes of convex bodies and, in particular, we compute the volume of the unit ball of $\mathbb{R}^n$ in terms of the *Gamma function*. The special case of polytopes is studied in Section 4.4.

Last, the objective of Section 4.5 is to explain how we measure areas of "pieces" of the boundary of a convex body. More specifically, we show that, for this sake, the $(n-1)$-dimensional Hausdorff measure works fine. We also pay particular attention to the case of $\mathbb{S}^{n-1}$ (which is the boundary of the unit ball B^n, of course), giving a formal proof of the fact that the $(n-1)$-dimensional Hausdorff

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_4

measure on $\mathbb{S}^{n-1}$ is the same as the *spherical Lebesgue measure*. Of course, also for this and related notions we will give precise definitions below.

4.1 Overview

This section is intended to be expository rather than formal. We want to explain what we mean when we talk about volumes and areas, and we want to do so by providing precise definitions and results, but without some of the proofs (these will come in Appendix A or later in the main text, and we refer to them when it is due). The reason is that a great part of these proofs rely on the vast world of measure theory (classical and geometric), and this is not the main scope of this book. Our objective here is that the reader can continue the reading of the book with a good intuition about volumes and areas (and knowing all the precise definitions and terminology, which is very important), but not necessarily having gone through the technical details of the proofs.

For the reader interested in a deeper knowledge on the tools which are necessary to study volumes and areas of convex bodies, we have included Appendix A. There, the reader will find a short introduction to Lebesgue measures and Hausdorff measures. We collect the definitions (strongly encouraging the reader to consider them before proceeding, on the other hand, this is not strictly necessary) and the results that we need throughout the text.

In $\mathbb{R}^n$, the *n-dimensional volume* (or *Lebesgue volume*) is the standard Lebesgue measure $\mathcal{L}^n(\cdot)$ (which coincides with the n-dimensional Hausdorff measure; see Theorem A.4.2). The n-dimensional volume will also be denoted by $\mathrm{vol}_n(\cdot)$ or by $\mathcal{H}^n(\cdot)$, which is the standard notation for the n-dimensional Hausdorff measure. From standard measure theory we have

$$0 < \mathrm{vol}_n(K) < \infty$$

for every convex body $K \subseteq \mathbb{R}^n$, since any convex body is compact and has non-empty interior (see Corollary 4.3.1). Also from standard measure theory we have that $\mathrm{vol}_n(A) = 0$ whenever A does not have maximal affine dimension (this is proved in Corollary A.3.2).

Moving forward, we want to find a way to regard measures of non-maximal dimension. This is commonly done in standard calculus courses, for example, where we often measure lengths ("one-dimensional volumes") and areas ("two-dimensional volumes") in $\mathbb{R}^3$, but we do so by using differential methods. We want to extend these notions, to deal both with higher dimensions and with objects which are "not differentiable in nature". Starting with the case of subsets of lower dimensional affine subspaces, we have two ways of defining the "k-dimensional volume" of a subset of a k-dimensional affine subspace of $\mathbb{R}^n$ ($0 < k < n$):

(i) each k-dimensional affine subspace $\mathbf{g}$ of $\mathbb{R}^n$ ($0 < k < n$) is naturally identified with $\mathbb{R}^k$ by an affine isometry $T\colon \mathbb{R}^k \to \mathbf{g}$, and hence the usual Lebesgue measure of $\mathbb{R}^k$ can be induced on $\mathbf{g}$. This idea is explicitly presented in Construction A.3.1.

(ii) If $A \subseteq \mathbb{R}^n$ is contained in a k-dimensional affine subspace of $\mathbb{R}^n$, then we simply consider the *k-dimensional volume* of A to be its k-dimensional Hausdorff measure with respect to the metric of $\mathbb{R}^n$.

Both ways clearly coincide and we explain briefly why (the formal proof is given in Corollary A.4.4): since T is an isometry, and in $\mathbb{R}^k$ the Lebesgue measure and the k-dimensional Hausdorff measure are the same, we have $\mathcal{H}^k_{\mathbb{R}^n}(A) = \mathcal{H}^k_{\mathbb{R}^k}(T(A)) = \mathcal{L}^k_{\mathbb{R}^k}(T(A))$, where $\mathcal{L}$ denotes the Lebesgue measure and the subscript means the space from where each measure is being taken. The advantage of **(ii)** is that the definition is simpler, but **(i)** does not rely on the "machinery" of Hausdorff measures.

However, we cannot restrict to lower dimensional objects which are contained in lower dimensional affine subspaces. Consider, for example, the boundary of a convex body of $\mathbb{R}^n$, which is not contained in an $(n-1)$-dimensional affine subspace, but yet is intuitively an "$(n-1)$-dimensional" object. Now we cannot use the Lebesgue measure in $\mathbb{R}^{n-1}$, but we can still define the $(n-1)$-*dimensional volume* as the $(n-1)$-dimensional Hausdorff measure (of $\mathbb{R}^n$). We prove in Theorem 4.4.1 that if $K \subseteq \mathbb{R}^n$ is a convex body, then $\mathcal{H}^{n-1}(\partial K)$ is a positive (finite) number. This is called the (total) *surface area* of K. We use the notation

$$\mathrm{vol}_{n-1}(\partial K) := \mathcal{H}^{n-1}(\partial K).$$

More generally, we can use the Hausdorff measures (of $\mathbb{R}^n$) to define k-dimensional volumes of subsets of $\mathbb{R}^n$ which are not contained in k-dimensional affine subspaces. We presented this discussion here to motivate the definitions, and also to clarify the particular cases where the Lebesgue measure can be used, but we can simplify the definitions summarizing everything as follows.

Definition 1. For $0 \leq k \leq n$, the *k-dimensional volume* of a subset $A \subseteq \mathbb{R}^n$ is its k-dimensional Hausdorff measure $\mathcal{H}^k(A)$ with respect to the canonical metric of $\mathbb{R}^n$. We will often use the notation

$$\mathrm{vol}_k(\cdot) := \mathcal{H}^k(\cdot).$$

Now we take a second look at the (total) surface area of a convex body $K \subseteq \mathbb{R}^n$. By our definition, it is the $(n-1)$-dimensional Hausdorff measure $\mathcal{H}^{n-1}(\partial K)$, but there are other reasonable ways to "capture" this idea. We describe some of them now.

The Minkowski surface area: Let $K \subseteq \mathbb{R}^n$ be a convex body. The *Minkowski surface area of K* is defined as

$$S_M(K) := \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(K + \varepsilon B^n) - \mathrm{vol}_n(K)}{\varepsilon}. \tag{4.1.1}$$

Notice very carefully that we only need the Lebesgue measure in $\mathbb{R}^n$ to define the Minkowski surface area of a convex body. This is related to the concept of *Minkowski content*, but we will not present it here in detail. We refer the reader to [834].

Approximation by polytopes: For any convex body $K \subseteq \mathbb{R}^n$ we define the number

$$S_P(K) := \sup\{\mathcal{H}^{n-1}(P) : P \subseteq K \text{ is a proper polytope}\}, \tag{4.1.2}$$

where we abuse a little of the notation and replace $\mathcal{H}^{n-1}(\partial P)$ by $\mathcal{H}^{n-1}(P)$; this will be done throughout the text sometimes without further comments. The intuition of this concept is that we can calculate the surface area of K by approximating it by inner polytopes. Notice that this is an extension of the very common idea of defining the length of a curve in terms of polygonal approximations. The fact that $S_P(K)$ is well-defined (that is, the supremum is not ∞) comes from the very intuitive fact that, under the convexity hypothesis, surface area is monotone with respect to inclusion. This is proved in Corollary A.4.7; see also (4.4.2).

The advantage of these concepts is that they provide ways to define the surface area of convex bodies which are independent of Hausdorff measures. For the case of approximation by polytopes this follows from **(i)** above (see also Remark 4.4.1).

It turns out that these three introduced surface area concepts coincide for the boundary of a convex body $K \subseteq \mathbb{R}^n$, that is,

$$S_M(K) = S_P(K) = \mathcal{H}^{n-1}(\partial K).$$

Actually, we will obtain the first equality later as a consequence of the continuity of mixed volumes (see Remark 6.2.2), and the second equality is proved in Theorem 4.4.1. We often denote the surface area of a convex body K by $S(K)$.

We introduced these notions because they appear naturally in the theory of mixed volumes, which we will study in Chapter 6, as well as in other parts of the theory (and the respective literature). Sometimes it is easier to consider instead of one simply another definition. For example, we can use the Minkowski surface area to relate easily the area of the unit sphere with the volume of the unit ball:

$$\begin{aligned} \mathrm{vol}_{n-1}(\mathbb{S}^{n-1}) &= \lim_{\varepsilon\to 0^+} \frac{\mathrm{vol}_n(B^n+\varepsilon B^n)-\mathrm{vol}_n(B^n)}{\varepsilon} = \\ &= \lim_{\varepsilon\to 0^+} \frac{[(1+\varepsilon)^n-1]\mathrm{vol}_n(B^n)}{\varepsilon} = n\mathrm{vol}_n(B^n). \end{aligned} \tag{4.1.3}$$

Here, the reader may notice that $\mathrm{vol}_{n-1}(\mathbb{S}^{n-1})$ stands for $\mathcal{H}^{n-1}(\mathbb{S}^{n-1})$, as our definition of $\mathrm{vol}_k(\cdot)$ states, but we are using the mentioned fact that for the boundary of any convex body the total $(n-1)$-dimensional Hausdorff measure coincides with the Minkowski surface area.

In what follows, we will perform some integrations over $\mathbb{S}^{n-1}$, and also over boundaries of convex bodies. To do so, we need to know with respect to what Borel measure we are integrating. In most cases, we consider the respective restrictions of the usual $(n-1)$-dimensional Hausdorff measure of $\mathbb{R}^n$. The restriction of $\mathcal{H}^{n-1}(\cdot)$ to $\mathbb{S}^{n-1}$ (as used in the integration above) is often called the *spherical Lebesgue measure*. To integrate over $\mathbb{S}^{n-1}$, sometimes we also consider the *spherical probability measure* (which is the normalized restriction of the $(n-1)$-dimensional Hausdorff measure to $\mathbb{S}^{n-1}$; this is always denoted by σ). By considering integration over $\mathbb{S}^{n-1}$, we can derive equality (4.1.3) using integration by polar coordinates (see Corollary A.6.1). From now on, we always denote by $1_A\colon \mathbb{R}^n \to \{0,1\}$ the *characteristic function* of the set A, which is defined as $1_A(x)=1$ if $x\in A$ and $1_A(x)=0$ if $x\notin A$. We have

$$\begin{aligned} \mathrm{vol}_n(B^n) &= \int_{\mathbb{R}^n} 1_{B^n}(x)\,dx = \int_0^{+\infty}\left(\int_{t\mathbb{S}^{n-1}} 1_{B^n}(y)\,d\mathcal{H}^{n-1}(y)\right)dt = \\ &= \int_0^1\left(\int_{t\mathbb{S}^{n-1}} d\mathcal{H}^{n-1}\right)dt = \int_0^1 \mathcal{H}^{n-1}(t\mathbb{S}^{n-1})\,dt = \mathcal{H}^{n-1}(\mathbb{S}^{n-1})\int_0^1 t^{n-1}\,dt = \\ &= \frac{1}{n}\mathcal{H}^{n-1}(\mathbb{S}^{n-1}), \end{aligned}$$

where the third equality comes from the fact that $1_{B^n}(y)=1$ for all $y\in t\mathbb{S}^{n-1}$ if $0\le t\le 1$, and $1_{B^n}(y)=0$ for every $y\in t\mathbb{S}^{n-1}$ whenever $t>1$. Also, the equality $\mathcal{H}^{n-1}(t\mathbb{S}^{n-1}) = t^{n-1}\mathcal{H}^{n-1}(\mathbb{S}^{n-1})$ comes from Corollary A.4.6. We emphasize that throughout this book the notation dx **always** stands for integration with respect to the Lebesgue measure of the corresponding Euclidean space $\mathbb{R}^n$ (the same for dy, dt, and so on). That is, if we are integrating over (a subset of) $\mathbb{R}^n$, then we replace $d\mathcal{L}^n(x)$ by dx. Above we also replace $d\mathcal{L}^1(t)$ by dt when we are integrating over (a subset of) $\mathbb{R}$.

At this point, we recommend the reader to take a look at Appendix A, which provides a rough introduction to Lebesgue measures and Hausdorff measures. But it is perfectly possible to go through this book without mastering this theory.

4.2 Cavalieri's principle

In this section, we collect some results known under the common name *Cavalieri's principle*. Roughly speaking, this is a technique to find the volume of a set by "slicing" it by affine subspaces of smaller dimension (see Plate 4 at the beginning of this chapter). They are, in general, obtained from Fubini's theorem (Theorem A.5.3 and, for Lebesgue measures, Corollary A.5.1). We restrict ourselves to the case of convex bodies, since this is the case which is important for us. Again, here we will use a lot of results from measure theory. These can be found in Appendix A, and alternatively the reader may only use these formulas without reading the proofs in details.

In what follows, we denote the hyperplane $\mathbf{h}(u, 0)$ by $u^\perp$ (because this hyperplane is the orthogonal space of u, of course). If $u \in \mathbb{S}^{n-1}$, then for each $t \in \mathbb{R}$ we also write

$$\mathbf{h}(u, t) = \{u^\perp + tu\} := \{x + tu : x \in u^\perp\}.$$

This justifies why these sets clearly coincide. If u is not unit, then $\{u^\perp + tu\}$ is still an affine subspace orthogonal to u, but it equals $\mathbf{h}(u, t|u|^2)$.

Moving forward, we remember that we denote the usual orthogonal projection onto $u^\perp$ by $\mathrm{proj}_{u^\perp}(\cdot)\colon \mathbb{R}^n \to u^\perp$. We also remind the reader that a linear map $T\colon \mathbb{R}^n \to \mathbb{R}^n$ is an *orthogonal transformation* if

$$\langle Tx, Ty\rangle = \langle x, y\rangle$$

for any $x, y \in \mathbb{R}^n$. The set of orthogonal transformations of $\mathbb{R}^n$ is denoted by $\mathrm{O}(n)$. It is easy to see that any orthogonal transformation is an isometric isomorphism which preserves orthogonality and volume (this follows from Theorem A.3.10 since the absolute value of the determinant is 1). Moreover, it is clear that for any $u \in \mathbb{S}^{n-1}$ there exists $T \in \mathrm{O}(n)$ such that $T(u) = e_n$. These facts and the terminology will be used below and throughout the text.

Theorem 4.2.1. *For any convex body $K \subseteq \mathbb{R}^n$ and $u \in \mathbb{S}^{n-1}$ we have*

$$\mathrm{vol}_n(K) = \int_{-h_K(-u)}^{h_K(u)} \mathrm{vol}_{n-1}\big(K \cap \{u^\perp + tu\}\big)\, dt. \tag{4.2.1}$$

See Figure 4.2.1

Proof. Let $K \subseteq \mathbb{R}^n$ be a convex body, and let $u \in \mathbb{S}^{n-1}$. We first assume that $u = e_n$, and for each $t \in \mathbb{R}$ the hyperplane $u^\perp + tu$ is naturally identified with $\mathbb{R}^{n-1}$ by the projection onto the $(n-1)$-first entries. Let $K(t) := K \cap \{u^\perp + tu\}$, which is naturally identified with a subset of $\mathbb{R}^{n-1}$ under the projection described previously, and observe that $x = (z, t) \in K$ if and only if $-h_K(-u) \le t \le h_K(u)$ and $z \in K(t)$. Thus, using the fact that the Lebesgue measure $\mathcal{L}^n$ is a product measure, we get from Fubini's theorem (Corollary A.5.1)

$$\begin{aligned}
\mathrm{vol}_n(K) = \mathcal{L}^n(K) &= \int_{\mathbb{R}^n} 1_K(x)\, d\mathcal{L}^n(x) = \\
&= \int_{\mathbb{R}^{n-1}\times\mathbb{R}} 1_{K(t)}(z)\cdot 1_{[-h_K(-e_n),h_K(e_n)]}(t)\, d\mathcal{L}^{n-1}(z)\, d\mathcal{L}^1(t) = \\
&= \int_{-h_K(-e_n)}^{h_K(e_n)} \left(\int_{\mathbb{R}^{n-1}} 1_{K(t)}(z)\, d\mathcal{L}^{n-1}(z)\right) d\mathcal{L}^1(t) = \int_{-h_K(-e_n)}^{h_K(e_n)} \mathcal{L}^{n-1}(K(t))\, d\mathcal{L}^1(t) = \\
&= \int_{-h_K(-e_n)}^{h_K(e_n)} \mathrm{vol}_{n-1}(K \cap \{e_n^\perp + te_n\})\, dt.
\end{aligned}$$

Observe that we are identifying the affine hyperplane $e_n^\perp + te_n$ with $\mathbb{R}^{n-1}$, and this is why we can regard $K(t)$ as a subset of $\mathbb{R}^{n-1}$ and measure it in the Lebesgue measure of $\mathbb{R}^{n-1}$ (in view of Construction A.3.1). Moreover, as it is clear from Theorem A.4.2 and Corollary A.4.4, the Lebesgue measures can be replaced by the respective Hausdorff measures of the same dimension. For the general case, let $T \in \mathrm{O}(n)$ be such that $T(u) = e_n$. Then it is clear that $T(u^\perp + tu) = e_n^\perp + te_n$ for each $t \in \mathbb{R}$, and since T is an isometry and the $(n-1)$-dimensional Lebesgue measures on the affine subspaces $\{u^\perp + tu\}$ and $\{e_n^\perp + te_n\}$ are the respective restrictions of the $(n-1)$-dimensional Hausdorff measure to these spaces, we get that $\mathrm{vol}_{n-1}(TK \cap \{e_n^\perp + te_n\}) = \mathrm{vol}_{n-1}(K \cap \{u^\perp + tu\})$. Therefore,

$$\mathrm{vol}_n(K) = \mathrm{vol}_n(TK) = \int_{-h_{TK}(-e_n)}^{h_{TK}(e_n)} \mathrm{vol}_{n-1}(TK \cap \{e_n^\perp + te_n\})\, dt = \\ = \int_{-h_K(u)}^{h_K(u)} \mathrm{vol}_{n-1}(K \cap \{u^\perp + tu\})\, dt,$$

where we also use the immediate equality $h_{TK}(Tv) = h_K(v)$.

□

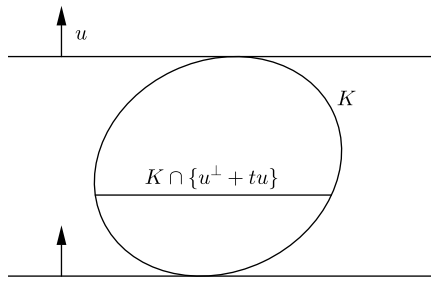


Fig. 4.2.1 As t ranges from $-h_K(-u)$ to $h_K(u)$, we have that $K \cap \{u^\perp + tu\}$ "sweeps" K.

Remark 4.2.1. Another proof of this equality is given in Theorem A.6.3 as a consequence of the co-area formula.

We also can calculate the volume of a convex body by "slicing" it with segments parallel to a given direction. For any $z \in \mathrm{proj}_{e_n^\perp}(K)$, we let

$$m_u(z) = \min\{t \in \mathbb{R} : z + tu \in K\} \text{ and}$$
$$M_u(z) = \max\{t \in \mathbb{R} : z + tu \in K\}.$$

Geometrically, these numbers parametrize the "first" and "last" points where a line in the direction of u drawn through a point of $\mathrm{proj}_{u^\perp}(K)$ touches K.

Theorem 4.2.2. *For any convex body $K \subseteq \mathbb{R}^n$ and $u \in \mathbb{S}^{n-1}$ we get*

$$\mathrm{vol}_n(K) = \int_{\mathrm{proj}_{u^\perp}(K)} M_u(y) - m_u(y)\, d\mathcal{L}^{n-1}(y). \tag{4.2.2}$$

Remark 4.2.2. We emphasize again that the symbol $\mathcal{L}^{n-1}$ on the right-hand side of the above equality stands for the $(n-1)$-dimensional Lebesgue measure on $u^\perp$ (in the sense of Construction A.3.1), and can be replaced by the $(n-1)$-dimensional Hausdorff measure; see Corollary A.4.4 again.

Proof. First, we give a proof for the case where u is in the direction of e_n. Remembering again that by Construction A.3.1 we can consider that $e_n^\perp$ is endowed with the Lebesgue measure of $\mathbb{R}^{n-1}$, we get

$$\begin{aligned}\mathrm{vol}_n(K) &= \int_{\mathbb{R}^n} 1_K \, d\mathcal{L}^n = \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} 1_K(z,t)\, d\mathcal{L}^1(t) d\mathcal{L}^{n-1}(z) = \\ &= \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} 1_{\mathrm{proj}_{e_n^\perp}(K)}(z) 1_{[m_{e_n}(z), M_{e_n}(z)]}(t)\, d\mathcal{L}^1(t) d\mathcal{L}^{n-1}(z) = \\ &= \int_{\mathrm{proj}_{e_n^\perp}(K)} \int_{m_{e_n}(z)}^{M_{e_n}(z)} d\mathcal{L}^1(t) d\mathcal{L}^{n-1}(z) = \int_{\mathrm{proj}_{e_n^\perp}(K)} M_{e_n}(z) - m_{e_n}(z)\, d\mathcal{L}^{n-1}(z).\end{aligned}$$

For the general case, we let $T \in \mathrm{O}(n)$ be such that $T(u) = e_n$. Then $T(u^\perp) = T(e_n^\perp)$, and since we clearly have that $Ty + te_n \in K$ if and only if $y + tu \in T^{-1}K$, we get

$$\mathrm{proj}_{e_n^\perp}(K) = \mathrm{proj}_{Tu^\perp}(K) = T\big(\mathrm{proj}_{u^\perp}(T^{-1}K)\big)$$

and

$$\begin{aligned} m_{e_n}(Ty) &= m_u(y) \ \text{ and} \\ M_{e_n}(Ty) &= M_u(y)\end{aligned}$$

for every $y \in \mathrm{proj}_{u^\perp}(T^{-1}K)$. Once T is an isometry and the Lebesgue $(n-1)$-dimensional measures on $u^\perp$ and $e_n^\perp$ coincide with the respective restrictions of the $(n-1)$-dimensional Hausdorff measure (which is invariant under isometries) to these spaces, we get that $\mathcal{L}^{n-1}_{e_n^\perp} = T_* \mathcal{L}^{n-1}_{u^\perp}$. Thus, using formula (A.2.4) we obtain

$$\begin{aligned}\mathrm{vol}_n(T^{-1}K) = \mathrm{vol}_n(K) &= \int_{\mathrm{proj}_{e_n^\perp}(K)} M_{e_n}(z) - m_{e_n}(z)\, d\mathcal{L}^{n-1}(z) = \\ &= \int_{T(\mathrm{proj}_{u^\perp}(T^{-1}K))} M_{e_n}(z) - m_{e_n}(z)\, d\mathcal{L}^{n-1}(z) = \\ &= \int_{\mathrm{proj}_{u^\perp}(T^{-1}K)} M_{e_n}(Ty) - m_{e_n}(Ty)\, d\mathcal{L}^{n-1}(y) = \int_{\mathrm{proj}_{u^\perp}(T^{-1}K)} M_u(y) - m_u(y)\, d\mathcal{L}^{n-1}(y).\end{aligned}$$

Since this equality holds for every convex body $K \subseteq \mathbb{R}^n$, we may replace K by TK to obtain finally

$$\mathrm{vol}_n(K) = \int_{\mathrm{proj}_{u^\perp}(K)} M_u(y) - m_u(y)\, d\mathcal{L}^{n-1}(y),$$

and this holds for every convex body $K \subseteq \mathbb{R}^n$ and any $u \in \mathbb{S}^{n-1}$.

□

We can go further and obtain a formula for the volume of a convex body in terms of slices by k-dimensional subspaces, thus generalizing the previous theorems. Since we will not work with support functions and intersections with lines, we also may drop the hypothesis of the considered set being a convex body.

Theorem 4.2.3. *Let* $\mathbf{g}$ *be a* k*-dimensional vector subspace of* $\mathbb{R}^n$*,* $k \in \{1, \ldots, n-1\}$*, and denote its orthogonal complement by* $\mathbf{g}^\perp$*. For any measurable set* $A \subseteq \mathbb{R}^n$ *with finite measure we have*

$$\mathrm{vol}_n(A) = \int_{\mathrm{proj}_{\mathbf{g}^\perp}(A)} \mathrm{vol}_k\big(A \cap \{z + \mathbf{g}\}\big)\, d\mathcal{L}^{n-k}(z),$$

where $\mathrm{vol}_k = \mathcal{L}^k$ *denotes the* k*-dimensional Lebesgue measure induced in* $\mathbf{g}$ *and* $\mathcal{L}^{n-k}$ *is the* $(n-k)$*-dimensional Lebesgue measure in* $\mathbf{g}^\perp$*; see again Construction A.3.1.*

Remark 4.2.3. Once more, we mention that the Lebesgue measures on $\mathbf{g}$ and $\mathbf{g}^\perp$ can be replaced by $\mathcal{H}^k$ and $\mathcal{H}^{n-k}$, respectively (see Corollary A.4.4).

Proof. The proof follows from Corollary A.5.1, in a similar way as we did it above. We naturally identify the affine subspaces parallel to $\mathbf{g}$ with $\mathbb{R}^k$, and the affine subspaces parallel to $\mathbf{g}^\perp$ with $\mathbb{R}^{n-k}$. Letting x, y, z be coordinates in $\mathbb{R}^n$, $\mathbb{R}^k$ and $\mathbb{R}^{n-k}$, respectively, we obtain that $x = (z, y) \in A$ if and only if $z \in \mathrm{proj}_{\mathbf{g}^\perp}(A)$ and $y \in A \cap \{z + \mathbf{g}\}$. Thus,

$$\mathrm{vol}_n(A) = \int_{\mathbb{R}^n} 1_A(x)\, d\mathcal{L}^n(x) = \int_{\mathbb{R}^{n-k}} \int_{\mathbb{R}^k} 1_A(z, y)\, d\mathcal{L}^k(y) d\mathcal{L}^{n-k}(z) =$$
$$= \int_{\mathrm{proj}_{\mathbf{g}^\perp}(A)} \int_{\mathbb{R}^k} 1_{A\cap\{z+\mathbf{g}\}}(y)\, d\mathcal{L}^k(y) d\mathcal{L}^{n-k}(z) = \int_{\mathrm{proj}_{\mathbf{g}^\perp}(A)} \mathrm{vol}_k\big(A \cap \{z + \mathbf{g}\}\big)\, d\mathcal{L}^{n-k}(z),$$

and this concludes the proof.

□

4.3 Some formulas for volumes

This section is devoted to derive some formulas relating the volume of convex bodies to other geometric objects. We start by obtaining a formula for the area of the unit sphere $\mathbb{S}^{n-1}$ and for the volume of the unit ball $B^n := \{x \in \mathbb{R}^n : |x| \leq 1\}$. Throughout the text, we will follow Schneider [2167] and use the notations

$$\kappa_n := \mathrm{vol}_n(B^n) \text{ and}$$
$$\omega_n := \mathrm{vol}_{n-1}(\mathbb{S}^{n-1}) := \mathcal{H}^{n-1}(\mathbb{S}^{n-1}),$$

for each $n \in \mathbb{N}$. The reader should know that some authors use different notation (for example, some authors use ω_n to denote the volume B^n instead of its area).

Proposition 4.3.1. *For every* $n \in \mathbb{N}$ *we have*

$$\omega_n = \frac{2\pi^{n/2}}{\Gamma\left(\frac{n}{2}\right)}$$

and

$$\kappa_n = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)},$$

where $\Gamma\colon (0, +\infty) \to \mathbb{R}$ *is the* Gamma function *given as*

$$\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t}\, dt$$

for any $x > 0$.

Proof. The strategy is to evaluate the integral

$$\int_{\mathbb{R}^n} e^{-|x|^2}\,dx$$

in two different ways. First, writing the points of $\mathbb{R}^n$ in coordinates $x = (x_1, \dots, x_n)$, Fubini's theorem yields

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-|x|^2}\,dx &= \int_{\mathbb{R}^n} e^{-x_1^2-\dots-x_n^2}\,dx_1\dots dx_n = \\ &= \left(\int_{-\infty}^{+\infty} e^{-x_1^2}\,dx_1\right)\dots\left(\int_{-\infty}^{+\infty} e^{-x_n^2}\,dx_n\right) = \pi^{n/2},\end{aligned} \tag{4.3.1}$$

where we are using the well-known Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-t^2}\,dt = \sqrt{\pi}.$$

For the sake of convenience, we prove this equality. Using Fubini's theorem for Lebesgue measures (Corollary A.5.1), polar coordinates, and a change of variables, we compute

$$\begin{aligned}\left(\int_{-\infty}^{\infty} e^{-t^2}\,dt\right)^2 &= \left(\int_{-\infty}^{+\infty} e^{-x^2}\,dx\right)\left(\int_{-\infty}^{+\infty} e^{-y^2}\,dy\right) = \int_{\mathbb{R}^2} e^{-x^2-y^2}\,dxdy = \\ &= \int_0^{+\infty}\int_0^{2\pi} e^{-r^2} r\,d\theta dr = 2\pi\int_0^{+\infty} e^{-r^2} r\,dr = \pi\int_{-\infty}^{0} e^{s}\,ds = \pi.\end{aligned}$$

Before moving forward, we need a property of the Gamma function which is related to the fact that it extends the factorial of natural numbers. For every $\alpha > 0$ we have

$$\Gamma(\alpha+1) = \alpha\Gamma(\alpha). \tag{4.3.2}$$

The proof of this equality relies on a simple integration by parts. We compute

$$\Gamma(\alpha+1) = \int_0^{+\infty} e^{-t}t^{\alpha}\,dt = -e^{-t}t^{\alpha}\Big|_0^{+\infty} + \int_0^{+\infty} \alpha t^{\alpha-1}e^{-t}\,dt = \alpha\Gamma(\alpha).$$

Finally, we use integration by polar coordinates and change of variables to get

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-|x|^2}\,dx &= \int_0^{+\infty}\int_{\mathbb{S}^{n-1}} e^{-r^2} r^{n-1}\,d\mathcal{H}^{n-1}dr = \mathcal{H}^{n-1}(\mathbb{S}^{n-1})\int_0^{+\infty} e^{-r^2}r^{n-1}\,dr = \\ &= \mathcal{H}^{n-1}(\mathbb{S}^{n-1})\int_0^{+\infty} e^{-t}t^{\frac{n-1}{2}}\frac{dt}{2t^{1/2}} = \frac{\omega_n}{2}\int_0^{+\infty} e^{-t}t^{\frac{n}{2}-1}\,dt = \frac{\omega_n}{2}\Gamma\left(\frac{n}{2}\right).\end{aligned}$$

This and (4.3.1) yield immediately the desired expression for ω_n. For κ_n, we simply have to use equalities (4.1.3) and (4.3.2).

□

For further reference, we summarize these equalities below:

$$\omega_n := \mathrm{vol}_{n-1}(\mathbb{S}^{n-1}) := \mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = n\mathrm{vol}_n(B^n) = n\kappa_n = \frac{n\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} = \frac{2\pi^{n/2}}{\Gamma\left(\frac{n}{2}\right)}. \tag{4.3.3}$$

Corollary 4.3.1. *Any non-degenerate ball in $\mathbb{R}^n$ has positive (finite) Lebesgue volume. Consequently, we have*

$$0 < \mathrm{vol}_n(C) < \infty$$

for any set $C \subseteq \mathbb{R}^n$ which is compact and has non-empty interior (in particular, this holds for every convex body).

Proof. A closed non-degenerate ball of $\mathbb{R}^n$ can be written as $x_0 + \rho B^n$ for some point $x_0 \in \mathbb{R}^n$ and some number $\rho > 0$. Theorem A.3.3 and the property that the Lebesgue measure is translation-invariant yield

$$\mathrm{vol}_n(x_0 + \rho B^n) = \mathrm{vol}_n(\rho B^n) = \rho^n \mathrm{vol}_n(B^n) > 0.$$

Here, we are observing that ρB^n is the image of B^n under the dilation map $x \mapsto \rho x$, which is clearly a linear map whose determinant is ρ^n.

The case for open balls follows straightforwardly from the fact that the volume of the open ball $B(x_0, \rho)$ is the same as the volume of its closure $x_0 + \rho B^n$. This is proved in Proposition 4.5.4 ahead.

For the remaining claim, simply consider non-degenerate balls B_1 and B_2 such that $B_1 \subseteq C \subseteq B_2$. Hence

$$0 < \mathrm{vol}_n(B_1) \leq \mathrm{vol}_n(C) \leq \mathrm{vol}_n(B_2) < \infty.$$

This concludes the proof.

□

Recall that the spherical probability measure (which we always denote by σ) is just the usual $(n-1)$-dimensional Hausdorff measure of $\mathbb{S}^{n-1}$ normalized. That is, $\sigma = \frac{1}{\omega_n}\mathcal{H}^{n-1}$. From now on, sometimes we will use this probability measure instead of $\mathcal{H}^{n-1}$ to integrate over $\mathbb{S}^{n-1}$. Notice that

$$\int_{\mathbb{S}^{n-1}} f \, d\mathcal{H}^{n-1} = \omega_n \int_{\mathbb{S}^{n-1}} f \, d\sigma, \tag{4.3.4}$$

for any integrable function $f \colon \mathbb{S}^{n-1} \to \mathbb{R}$. It follows that we can write the integration formula in polar coordinates as

$$\begin{aligned} \int_{\mathbb{R}^n} f(x)\, dx &= \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} t^{n-1} f(tu)\, d\mathcal{H}^{n-1}(u)\, dt = \\ &= n\mathrm{vol}_n(B^n) \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} t^{n-1} f(tu)\, d\sigma(u) dt. \end{aligned} \tag{4.3.5}$$

As a consequence of the integration by polar coordinates, we have a formula to calculate the volume of a convex body $K \subseteq \mathbb{R}^n$ in terms of its gauge function. We shall assume that K contains the origin as an interior point, but we also recall that the Lebesgue measure is invariant under translations.

Proposition 4.3.2. *Let $K \subseteq \mathbb{R}^n$ be a convex body containing the origin as an interior point, and recall that $||\cdot||_K \colon \mathbb{R}^n \to [0, +\infty)$ denotes the* gauge function of *K, which is given as*

$$||x||_K = \inf\{\lambda > 0 : x \in \lambda K\}.$$

Then,

$$\mathrm{vol}_n(K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} ||u||_K^{-n} \, d\mathcal{H}^{n-1}(u) = \mathrm{vol}_n(B^n) \int_{\mathbb{S}^{n-1}} ||u||_K^{-n} \, d\sigma(u). \tag{4.3.6}$$

Proof. Remember that the *radial function of K* is the function $\rho_K \colon \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to [0, +\infty)$ given as

$$\rho_K(u) = \sup\{s \geq 0 : su \in K\},$$

for each $u \in \mathbb{S}^{n-1}$. From Item **(ii)** of Proposition 2.3.6, we have $\rho_K(u) = 1/||u||_K$. It is also immediate that, in polar coordinates, we have $su \in K$ if and only if $0 \leq s \leq \rho_K(u)$. Then

$$\mathrm{vol}_n(K) = \int_{\mathbb{R}^n} 1_K \, dx = \int_{\mathbb{S}^{n-1}} \int_0^{\infty} s^{n-1} 1_K(su) \, ds \, d\mathcal{H}^{n-1}(u) =$$

$$= \int_{\mathbb{S}^{n-1}} \int_0^{\rho_K(u)} s^{n-1} \, ds \, d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-1}} \frac{\rho_K(u)^n}{n} \, d\mathcal{H}^{n-1}(u) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} ||u||_K^{-n} \, d\mathcal{H}^{n-1}(u).$$

Observe that Fubini's theorem (Theorem A.5.3) justifies the changing in the order of integration in the second equality (with respect to the original formula for integration by polar coordinates).

The other equality comes immediately when we replace $\mathcal{H}^{n-1}(\cdot)$ by $\omega_n \sigma(\cdot)$ and recall that $\omega_n = n\kappa_n = n\mathrm{vol}_n(B^n)$.

□

Remark 4.3.1. Equality (4.3.6) gives a way of calculating the volume of a convex body K (containing the origin as an interior point) in terms of its gauge function. But since $||\cdot||_K = \rho_K^{-1}$, we also have

$$\mathrm{vol}_n(K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_K(u)^n \, d\mathcal{H}^{n-1}(u) = \kappa_n \int_{\mathbb{S}^{n-1}} \rho_K(u)^n \, d\sigma(u), \tag{4.3.7}$$

which is a formula for the volume of K in terms of the radial function. This was actually derived before (4.3.6) in the previous proof. Even more, since $||\cdot||_K = h_{K^\circ}$ (see Item **(i)** of Proposition 2.3.6) we get the variant

$$\mathrm{vol}_n(K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{K^\circ}(u)^{-n} \, d\mathcal{H}^{n-1}(u) = \kappa_n \int_{\mathbb{S}^{n-1}} h_{K^\circ}(u)^{-n} \, d\sigma(u). \tag{4.3.8}$$

With new tools in hands, later we will derive more formulas for the volume of a convex body. For example, we mention Theorem 7.2.5, which relies on some theory of mixed volumes and mixed surface area measures.

4.4 Volumes and areas of polytopes

Now we specialize our study of volumes and areas to the case of proper polytopes. Throughout this section, we will make heavy use of Lebesgue and Hausdorff measures. We emphasize again that the interested reader can find a formal introduction to these topics in Appendix A. We also refer to [255] and [834] for a more complete approach. However, it is perfectly possible to proceed without going through the technicalities of the proofs in this section.

Since a proper polytope is a convex body, its *volume* is defined to be simply its Lebesgue measure. As usual, we denote it by $\mathcal{L}^n(P)$, where $\mathcal{L}^n$ stands for the Lebesgue measure on $\mathbb{R}^n$, or more often by $\mathrm{vol}_n(P)$, and since P is compact and has non-empty interior we have that $0 < \mathrm{vol}_n(P) < \infty$. Moreover, it is useful to have in mind that, by Theorem A.4.2, this is the same as the n-dimensional Hausdorff measure $\mathcal{H}^n(P)$.

The *area* (or *surface area*) of P is defined to be the $(n-1)$-dimensional Hausdorff measure $\mathcal{H}^{n-1}(\partial P)$ of its boundary. It is very common to denote the $(n-1)$-Hausdorff measure on $\mathbb{R}^n$ by $\mathrm{vol}_{n-1}(\cdot)$, and we will adopt this notation. Next we show that the area of a polytope can also be calculated in terms of the measures of its facets.

Lemma 4.4.1. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope. Then*

$$\mathrm{vol}_{n-1}(\partial P) = \sum_{F \text{ facet of } P} \mathrm{vol}_{n-1}(F). \tag{4.4.1}$$

That is, the surface area of a proper polytope equals the sum of the $(n-1)$*-dimensional Hausdorff measures of its facets.*

Proof. We first show that intersections of facets have zero measure. Indeed, let $F_1, \ldots, F_m$ be distinct facets (with $m \geq 2$, and whose unit outer normal vectors are $u_1, \ldots, u_m$, respectively). Then

$$F_1 \cap \ldots \cap F_m \subseteq \mathbf{h}(P, u_1) \cap \ldots \mathbf{h}(P, u_m),$$

and standard linear algebra gives that the latter is an affine subspace whose dimension is less than $n-1$. From Corollary A.4.5, we get that $\mathcal{H}^{n-1}(F_1 \cap \ldots \cap F_m) = 0$. Let $\mathcal{I}$ denote the (finite) collection of intersections of two or more facets of P. Once the boundary of P is the union of its facets, and because all involved sets are Borel measurable (and hence $\mathcal{H}^{n-1}$-measurable), we get

$$\mathcal{H}^{n-1}(\partial P) = \sum_{F \text{ facet of } P} \mathcal{H}^{n-1}(F) - \sum_{G \in \mathcal{I}} \mathcal{H}^{n-1}(G) = \sum_{F \text{ facet of } P} \mathcal{H}^{n-1}(F),$$

and this concludes the proof.

□

Remark 4.4.1. As was mentioned in Section 4.1, equality (4.4.1) has the consequence of allowing us to define the surface area of a polytope without using Hausdorff measures, but only Lebesgue measures. Indeed, each facet F of P is contained in some hyperplane $\mathbf{h}$, which is isometrically isomorphic to $\mathbb{R}^{n-1}$. Hence, as is explained in Construction A.3.1, the Lebesgue measure of $\mathbb{R}^{n-1}$ can be induced on $\mathbf{h}$ (denote this by $\mathcal{L}^{n-1}_{\mathbf{h}}$, say) and used to measure F. Then we define the surface area of P to be the sum of these measures of the facets. In Corollary A.4.4, we prove that $\mathcal{L}^{n-1}_{\mathbf{h}}(F) = \mathcal{H}^{n-1}(F)$, and it follows that this new definition coincides with the original one. We emphasize again that the details are given in Appendix A, but the reader can move forward only having the intuition about these notions.

From this argument, we get in particular that

$$0 < \mathcal{H}^{n-1}(\partial P) < \infty \tag{4.4.2}$$

for any proper polytope $P \subseteq \mathbb{R}^n$. In fact, each facet F of P is a compact set whose affine dimension is $n-1$, thus being identified with a compact subset of $\mathbb{R}^{n-1}$ which has non-empty interior, and hence having finite positive Lebesgue measure. This is very important for us, because this gives that the $(n-1)$-dimensional Hausdorff measure of *any* convex body is finite and positive, as we prove in Theorem 4.4.1 below. Therefore, it is indeed "reasonable" to use the measure $\mathcal{H}^{n-1}$ as a Borel measure in ∂K, as is discussed in the beginning of Section 4.5.

From now on, we will use the following terminology: a proper polytope $P \subseteq K$ is called an *inner polytope* (of K), and a proper polytope $Q \supseteq K$ is called an *outer polytope* (of K). We also recall that

$$S_P(K) := \sup\left\{\mathcal{H}^{n-1}(\partial P) : P \text{ is an inner polytope of } K\right\}$$

is the surface area of K obtained via approximations by inner polytopes defined in Section 4.1. At this point, it is interesting to give a formal proof that this is indeed well-defined (this fact was only mentioned in the referred section). The argument is as follows: if P_{out} is any outer polytope of K, then by (4.4.2) and Corollary A.4.7 the number $\mathcal{H}^{n-1}(\partial P_{\text{out}})$ is a finite positive upper bound for the areas of all inner polytopes of K. Hence the supremum in the definition of $S_P(K)$ is indeed finite.

Theorem 4.4.1. *For any convex body* $K \subseteq \mathbb{R}^n$ *we have*

(i) $0 < \mathcal{H}^{n-1}(\partial K) < \infty$ *and*

(ii) $S_P(K) = \mathcal{H}^{n-1}(\partial K)$.

Proof. For **(i)**, simply choose proper polytopes P_{in} and P_{out} such that $P_{\text{in}} \subseteq K \subseteq P_{\text{out}}$. Hence, from Corollary A.4.7, we get

$$0 < \mathcal{H}^{n-1}(\partial P_{\text{in}}) \leq \mathcal{H}^{n-1}(\partial K) \leq \mathcal{H}^{n-1}(\partial P_{\text{out}}) < \infty.$$

Now we prove **(ii)**. Since any Hausdorff measure is obviously invariant under translations, we may assume that K has the origin as an interior point. Since $\mathcal{H}^{n-1}(\partial P_{\text{in}}) \leq \mathcal{H}^{n-1}(\partial K) \leq \mathcal{H}^{n-1}(\partial P_{\text{out}})$ holds for any polytopes $P_{\text{in}} \subseteq K \subseteq P_{\text{out}}$, we get

$$\sup\left\{\mathcal{H}^{n-1}(\partial P_{\text{in}}) : P_{\text{in}} \subseteq K \text{ is a proper polytope}\right\} \leq \mathcal{H}^{n-1}(\partial K) \leq$$
$$\leq \inf\left\{\mathcal{H}^{n-1}(\partial P_{\text{out}}) : P_{\text{out}} \supseteq K \text{ is a proper polytope}\right\}.$$

Notice carefully that the left-hand side is $S_P(K)$. Our strategy is to prove that the left- and right-hand sides of the above coincide. For that, it is enough to prove that for any $\varepsilon > 0$ there exist polytopes $P_{\text{in}} \subseteq K \subseteq P_{\text{out}}$ such that

$$\mathcal{H}^{n-1}(\partial P_{\text{out}}) - \mathcal{H}^{n-1}(\partial P_{\text{in}}) < \varepsilon.$$

The existence of such polytopes will be derived as a consequence of the following approximation result: for any convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point and each sufficiently large $m \in \mathbb{N}$, there exists a proper polytope P_m such that $P_m \subseteq K \subseteq (1 + 1/m)P_m$. This is proved in Proposition 3.4.2.

For an arbitrarily given $\varepsilon > 0$, we let $m \in \mathbb{N}$ be large enough such that we can choose a proper polytope P_m with $P_m \subseteq K \subseteq (1 + 1/m)P_m$, and also

$$(1 + 1/m)^{n-1} < 1 + \frac{\varepsilon}{\mathcal{H}^{n-1}(\partial K)}.$$

Thus, from Corollary A.4.6 we obtain

$$\mathcal{H}^{n-1}\big((1 + 1/m)\partial P_m\big) - \mathcal{H}^{n-1}(\partial P_m) = \big[(1 + 1/m)^{n-1} - 1\big]\mathcal{H}^{n-1}(\partial P_m) <$$
$$< \frac{\varepsilon}{\mathcal{H}^{n-1}(\partial K)}\mathcal{H}^{n-1}(\partial P_m) < \varepsilon,$$

and hence we have the desired.

□

Scholium 4.4.1. We may define $S_P(K)$ equivalently as

$$S_P(K) := \inf\left\{\mathcal{H}^{n-1}(\partial P) : P \supseteq K \text{ is a proper polytope}\right\},$$

that is, as the infimum of the areas of the outer polytopes of K.

Next we derive a useful formula to compute the volume of a proper polytope in terms of the $(n-1)$-dimensional volumes of its facets. First, we need a useful lemma.

Lemma 4.4.2. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope, and for each facet F, let u_F denotes the unit outer normal vector. Then,*

$$\sum_{F \text{ facet of } P} u_F \mathrm{vol}_{n-1}(F) = 0_{\mathbb{R}^n}. \tag{4.4.3}$$

As a consequence, the summation

$$\sum_{F \text{ facet of } P} h_P(u_F)\mathrm{vol}_{n-1}(F)$$

is invariant under any translation of P.

Proof. Let $x \in \mathbb{R}^n$ be non-zero. From standard linear algebra, the (signed) volume of the projection of F onto $x^\perp$ is

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{x^\perp}(F)\big) = \langle x, u_F\rangle \mathrm{vol}_{n-1}(F).$$

Summing on all facets of P, we will obtain the volume of the projection of P onto $x^\perp$ twice, but with reverse signs. Thus,

$$0 = \sum_{F \text{ facet of } P} \langle x, u_F\rangle \mathrm{vol}_{n-1}(F) = \left\langle x, \sum_{F \text{ facet of } P} u_F \mathrm{vol}_{n-1}(F)\right\rangle.$$

Since this equality holds for every $x \in \mathbb{R}^n$ we get the first claim.

For the other claim, first recall that the $(n-1)$-dimensional volume is translation-invariant. For any $x \in \mathbb{R}^n$ we have

$$\sum_{F \text{ facet of } P} h_{P+x}(u_F)\mathrm{vol}_{n-1}(F+x) = \sum_{F \text{ facet of } P} (\langle x, u_F\rangle + h_P(u_F))\mathrm{vol}_{n-1}(F) =$$

$$= \sum_{F \text{ facet of } P} h_P(u_F)\mathrm{vol}_{n-1}(F) + \left\langle x, \sum_{F \text{ facet of } P} u_F \mathrm{vol}_{n-1}(F)\right\rangle = \sum_{F \text{ facet of } P} h_P(u_F)\mathrm{vol}_{n-1}(F).$$

□

Theorem 4.4.2. *The volume of a proper polytope $P \subseteq \mathbb{R}^n$ is given by*

$$\mathrm{vol}_n(P) = \frac{1}{n}\left(\sum_{F \text{ facet of } P} h_P(u_F) \cdot \mathrm{vol}_{n-1}(F)\right),$$

where $u_F \in \mathbb{S}^{n-1}$ denotes the unit normal of the facet F ("outer" with respect to P).

Proof. Let $\mathbf{h}$ be an $(n-1)$-dimensional affine subspace of $\mathbb{R}^n$, and let $x_0 \notin \mathbf{h}$. Let $E \subseteq \mathbf{h}$ be a non-empty compact set, and let

$$C := \{x_0 + \lambda(z - x_0) : z \in E \text{ and } \lambda \in [0,1]\},$$

which is the bounded (not necessarily convex) cone with basis E and apex x_0. We will prove that

$$\mathrm{vol}_n(C) = \frac{\mathrm{dist}(x_0, \mathbf{h})}{n}\mathrm{vol}_{n-1}(E).$$

Since volumes and distances are invariant under translations and under orthogonal transformations, we can assume that $x_0 = 0_{\mathbb{R}^n}$ and that $\mathbf{h}$ is parallel to $e_n^\perp := \{(x_1, \ldots, x_{n-1}, 0)\}$ (with $\mathbf{h} = e_n^\perp + de_n$ for some $d > 0$, say). Hence we can use Cavalieri's principle (see (4.2.1) and Theorem A.6.3) by slicing

C with the hyperplanes $e_n^{\perp} + te_n$, where $0 \leq t \leq d$ (each of which is identified with $\mathbb{R}^{n-1}$, of course). It is clear that the intersection of C with $e_n^{\perp} + te_n$ equals $\frac{t}{d}E$, and thus

$$\mathrm{vol}_n(C) = \int_0^d \int_{\mathbb{R}^{n-1}} 1_{\frac{t}{d}E}\, d\mathcal{L}^{n-1} dt = \int_0^d \mathrm{vol}_{n-1}\Big(\frac{t}{d}E\Big)\, dt = \\ = \frac{\mathrm{vol}_{n-1}(E)}{d^{n-1}} \int_0^d t^{n-1}\, dt = \frac{d}{n}\mathrm{vol}_{n-1}(E).$$

Since we clearly have that $\mathrm{dist}(0_{\mathbb{R}^n}, \mathbf{h}) = d$ the result follows.

Now let P be a proper polytope. The idea is to "decompose" P as a union of cones with the same apex, and each of which having a facet as basis. Since volume is translation-invariant, and with the help of Lemma 4.4.2, which shows that the right-hand side of (4.4.3) is also translation-invariant, we may assume that $0_{\mathbb{R}^n} \in \mathrm{int}P$. Let $F_1, \ldots, F_m$ be the facets of P, let $u_1, \ldots, u_m$ be their respective outer normal vectors, and write $\mathbf{h}_j := \mathbf{h}(P, u_j)$ for every $1 \leq j \leq m$. Also for each $1 \leq j \leq m$ let

$$C_j := \big\{\lambda z : z \in F_j \text{ and } \lambda \in [0,1]\big\}.$$

Because the boundary of P equals $F_1 \cup \ldots \cup F_m$, we have that $P = C_1 \cup \ldots \cup C_m$. We also claim that any non-empty intersection between distinct C_j's is contained in some vector subspace whose dimension is not total. In fact, it is easy to see that

$$C_{j_1} \cap \ldots \cap C_{j_k} = \big\{\lambda z : z \in F_{j_1} \cap \ldots \cap F_{j_k} \text{ and } \lambda \in [0,1]\big\} \subseteq \\ \subseteq \mathrm{aff}\big(\{0_{\mathbb{R}^n}\} \cup (F_{j_1} \cap \ldots \cap F_{j_k})\big),$$

and the intersection of two or more distinct facets is a face whose dimension is less than $n-1$. Then the affine hull on the right-hand side of the above has dimension smaller than n. Hence any intersection between the C_j's has zero n-dimensional Lebesgue measure (observe also that all these sets are measurable because they are compact). Thus,

$$\mathrm{vol}_n(P) = \sum_{j=1}^m \mathrm{vol}_n(C_j) = \sum_{j=1}^m \frac{\mathrm{dist}(0_{\mathbb{R}^n}, \mathbf{h}_j)}{n} \mathrm{vol}_{n-1}(F_j) = \frac{1}{n}\sum_{j=1}^m h_P(u_j)\mathrm{vol}_{n-1}(F_j),$$

where the last equality comes from the straightforward fact that $\mathrm{dist}(0_{\mathbb{R}^n}, \mathbf{h}_j) = h_P(u_j)$. Figure 4.4.1 illustrates the two-dimensional case.

□

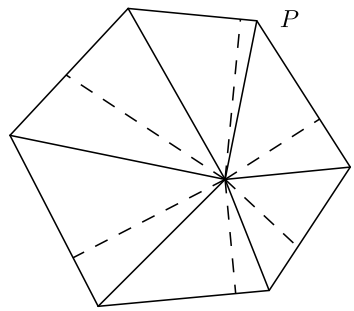


Fig. 4.4.1 The dotted lines are the respective heights of the triangles, which are given in terms of the support function in the case where the apex is the origin.

Example 4.4.1 (Pyramids and orthogonal simplices) A polytope

$$P = \operatorname{conv}\{x_1, x_2, \ldots, x_m\} \subseteq \mathbb{R}^n$$

is a *pyramid* (or *n-dimensional pyramid*) if, up to reordering the points $x_1, \ldots, x_m$, we have that $\operatorname{aff}\{x_2, \ldots, x_m\} = \mathbf{h}(u, \beta)$ for some $u \in \mathbb{S}^{n-1}$ and some $\beta \in \mathbb{R}$, and $x_1 \notin \mathbf{h}(u, \beta)$. In this case, we also say that x_1 is the *apex* of P.

In other words, a polytope is a pyramid when it is proper and all of its vertices except for one lie in the same hyperplane of $\mathbb{R}^n$. The volume of a pyramid $P := \operatorname{conv}\{x_1, \ldots, x_m\}$ with apex at x_1 is given as

$$\operatorname{vol}_n(P) = \frac{\operatorname{dist}(x_1, \mathbf{h})}{n} \operatorname{vol}_{n-1}(\operatorname{conv}\{x_2, \ldots, x_m\}),$$

where $\mathbf{h} := \mathbf{h}(u, \beta) = \operatorname{aff}\{x_2, \ldots, x_m\}$. This is a consequence of the calculation of the volume of the set C in the beginning of the proof of Theorem 4.4.2 for the case where the basis E is a polytope.

A consequence of this formula is that we can explicitly calculate the volume of the *standard orthogonal n-simplex*

$$\begin{aligned} \Delta_n := \operatorname{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\} = \\ = \left\{(t_1, \ldots, t_n) \in \mathbb{R}^n : t_j \geq 0 \text{ for } j = 1, \ldots, n \text{ and } \sum_{j=1}^{n} t_j \leq 1\right\} \end{aligned} \tag{4.4.4}$$

of $\mathbb{R}^n$. Figure 4.4.2 illustrates the standard orthogonal 3-simplex. We claim that

$$\operatorname{vol}_n(\Delta_n) = \frac{1}{n!}. \tag{4.4.5}$$

To prove this, we proceed by induction on the dimension. The formula is clearly true if $n = 2$, and we assume that this is also true for $n \in \mathbb{N}$. Hence the standard orthogonal $(n+1)$-simplex is the set $\Delta_{n+1} = \operatorname{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_{n+1}\}$, which can be seen as a pyramid with apex at e_{n+1}, because

$$\operatorname{aff}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\} = \{(t_1, \ldots, t_n, 0) : t_1, \ldots, t_n \in \mathbb{R}\} = \mathbf{h}(-e_{n+1}, 0) =: \mathbf{h}.$$

It is clear that the distance from e_{n+1} to $\mathbf{h}$ is 1. The facet $\operatorname{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\}$ is naturally identified with the standard orthogonal n-simplex of $\mathbb{R}^n$, and hence

$$\operatorname{vol}_{n+1}(\Delta_{n+1}) = \frac{1}{n+1} \operatorname{vol}_n(\operatorname{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\}) = \frac{1}{n+1} \operatorname{vol}_n(\Delta_n) = \frac{1}{(n+1)!}.$$

This proves the desired. In Exercise 4.2 the reader is invited to compute the surface area of the standard orthogonal n-simplex.

4.5 Measures on the boundary

At this point, we already discussed the *total* surface area of convex bodies. In summary, we introduced three ways of "calculating" the total surface area of a convex body $K \subseteq \mathbb{R}^n$: the Minkowski surface area $S_M(K)$, the total $(n-1)$-dimensional Hausdorff measure $\mathcal{H}^{n-1}(\partial K)$, and the area obtained via approximation by inner polytopes $S_P(K)$. These concepts coincide, and we already proved that $S_P(K) = \mathcal{H}^{n-1}(\partial K)$, but the proof of the equality $S_P(K) = S_M(K)$ will only be given in Section 6.2.

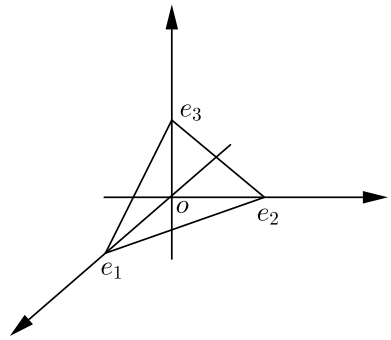


Fig. 4.4.2 The standard orthogonal 3-simplex. Notice that the origin is labeled as o.

Now a question is: what if we want to measure the area of a *subset* of the boundary of K? To do so, we need to endow ∂K with a *Borel measure*, and this section is dedicated to discuss how to do this in a "reasonable" way. More than in the other sections of this chapter, here we heavily use the language and machinery of measure theory. The technicalities and details of the proofs of the results of this section can be avoided with no serious harm to the understanding of the rest of the book.

The most natural way of defining a Borel measure on the boundary of a convex body K is to consider the restriction of $\mathcal{H}^{n-1}$ to ∂K, where $\mathcal{H}^{n-1}$ denotes the $(n-1)$-dimensional Hausdorff measure of $\mathbb{R}^n$. We clarify how this is done. First of all, we have that

$$\mathcal{H}^{n-1} \colon \mathcal{B}_{\mathbb{R}^n} \to [0, \infty]$$

is a Borel measure on $\mathbb{R}^n$, where $\mathcal{B}_{\mathbb{R}^n}$ denotes the Borel σ-algebra of $\mathbb{R}^n$. Now we regard ∂K as a metric space by endowing it with the induced metric from $\mathbb{R}^n$ (that is, the distance between $x, y \in \partial K$ is just $|x - y|$). Being then a metric space, ∂K has a Borel σ-algebra $\mathcal{B}_{\partial K}$. From Proposition A.1.2 we have

$$\mathcal{B}_{\partial K} = \{B \cap \partial K : B \in \mathcal{B}_{\mathbb{R}^n}\},$$

and since $\partial K \in \mathcal{B}_{\mathbb{R}^n}$ (because it is compact), we also have that $\mathcal{B}_{\partial K} \subseteq \mathcal{B}_{\mathbb{R}^n}$. By the mentioned proposition, the restriction

$$\mathcal{H}^{n-1}\big|_{\partial K} \colon \mathcal{B}_{\partial K} \to [0, \infty]$$

is a Borel measure on ∂K. This is how we "induce" $\mathcal{H}^{n-1}$ and see it as a Borel measure on ∂K.

Observe that, defining things this way, the measure $\mathcal{H}^{n-1}\big|_{\partial K}(A)$ of a given Borel set $A \in \mathcal{B}_{\partial K}$ is just $\mathcal{H}^{n-1}(A)$, which is simply the $(n-1)$-dimensional Hausdorff measure of A in $\mathbb{R}^n$ (A is also a Borel set of $\mathbb{R}^n$). For this reason, we warn the reader that the restriction of $\mathcal{H}^{n-1}$ to the boundary of any convex body will be denoted simply by $\mathcal{H}^{n-1}$ when there is no possibility of confusion (that is, the subindex indicating the restriction to ∂K will be omitted).

Alternatively, one could construct the $(n-1)$-dimensional Hausdorff outer measure of the metric space $(\partial K, |\cdot|)$ from the beginning, and consider it over $\mathcal{B}_{\partial K}$. This would lead to the same Borel measure, as it is proved in Proposition A.4.3, and hence we will stick to our original approach by restriction. There is no fundamental difference between both ways, and we simply choose the one that we believe to be the clearer.

Using this measure only makes sense for us because $0 < \mathcal{H}^{n-1}(\partial K) < \infty$, since otherwise the measure $\mathcal{H}^{n-1}\big|_{\partial K}$ would be clearly useless for our purposes. The fact that $\mathcal{H}^{n-1}(\partial K)$ is a positive (finite)

number for any convex body $K \subseteq \mathbb{R}^n$ is proved in Theorem 4.4.1. From a geometric perspective, it is also due to imagining that an "area measure" on ∂K should attribute a positive value for any set with non-empty interior. This is indeed true for the case of $\mathcal{H}^{n-1}\big|_{\partial K}$, but before proving that we need to investigate the specific case where $K = \mathbb{S}^{n-1}$. As was mentioned in Section 4.1, throughout the text we perform a lot of integrations over the unit sphere, and hence it is adequate to know how the measures considered in such integrations behave. We have some different ways of endowing $\mathbb{S}^{n-1}$ with a measure.

(1) One can simply consider the $(n-1)$-dimensional Hausdorff measure $\mathcal{H}^{n-1}$ of $\mathbb{R}^n$ restricted to $\mathbb{S}^{n-1}$ (as was explained above).

(2) Given a Borel set $A \subseteq \mathbb{S}^{n-1}$, we may put $\nu_{\mathbb{S}^{n-1}}(A) := n\mathcal{L}^n\big(\{tx : x \in A \text{ and } t \in [0,1]\}\big)$, where we remind the reader that $\mathcal{L}^n$ denotes the usual Lebesgue measure in $\mathbb{R}^n$. The measure constructed in this way is often called the *spherical Lebesgue measure* (or *conical measure*).

Alternatively,

(3) We may regard $\mathbb{S}^{n-1}$ as a topological group and construct a *Haar measure* (which is unique up to constant scalar multiplication).

It turns out that these three measures coincide (up to the "correct" normalization of the measure constructed in **(2)**, of course), and we will prove this now. We will focus on **(1)** and **(2)**, since we do not study in detail the construction of the Haar measure in this book. We derive some early properties of these measures.

Proposition 4.5.1. *The measures $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ and $\nu_{\mathbb{S}^{n-1}}$ have the following properties.*

(i) *Both are regular finite Borel measures.*

(ii) *They are* $\mathrm{O}(n)$*-invariant.*

Remark 4.5.1. We say that a Borel measure μ on $\mathbb{S}^{n-1}$ is $\mathrm{O}(n)$-invariant if $\mu(UA) = \mu(A)$ for any $U \in \mathrm{O}(n)$ and every $A \in \Sigma_{\mathbb{S}^{n-1}}$. This is the same as saying that the push-forward measure $U_*\mu$ equals μ itself for any $U \in \mathrm{O}(n)$.

Proof. The fact that $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ is a Borel measure was proved previously. For $\nu_{\mathbb{S}^{n-1}}$ this can be directly checked. In fact, if $A \subseteq \mathbb{S}^{n-1}$ is a Borel set, then $\{tx : x \in A \text{ and } t \in [0,1]\}$ is clearly a Borel set of $\mathbb{R}^n$. The σ-additivity comes from the σ-additivity of $\mathcal{L}^n$.

The finiteness of $\nu_{\mathbb{S}^{n-1}}$ comes immediately from the fact that $\mathcal{L}^n(B^n) < +\infty$. The fact that $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ is finite comes from Item **(i)** of Theorem 4.4.1.

Now it follows from Remark A.1.7 that both measures are regular Borel measures, and this settles **(i)**.

It remains to prove the invariance under orthogonal maps. For that sake, first remember that the Hausdorff measure $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ is constructed with respect to the Euclidean metric of $\mathbb{R}^n$ induced on $\mathbb{S}^{n-1}$, and any map $T \in \mathrm{O}(n)$ is an isometry with respect to this metric. Thus, since any Hausdorff measure is invariant under isometries, we get that $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ is $\mathrm{O}(n)$-invariant. For the measure $\nu_{\mathbb{S}^{n-1}}$, we observe that for any $T \in \mathrm{O}(n)$ and $A \subseteq \mathbb{S}^{n-1}$,

$$\{tz : z \in T(A) \text{ and } t \in [0,1]\} = T\big(\{tx : x \in A \text{ and } t \in [0,1]\}\big).$$

Then, since $|\det(T)| = 1$, we get

$$\nu_{\mathbb{S}^{n-1}}(T(A)) = n\mathcal{L}^n\big(T\big(\{tx : x \in A \text{ and } t \in [0,1]\}\big)\big) =$$
$$= |\det(T)| n\mathcal{L}^n\big(\{tx : x \in A \text{ and } t \in [0,1]\}\big) = \nu_{\mathbb{S}^{n-1}}(A),$$

for every (Borel) set $A \subseteq \mathbb{S}^{n-1}$. This concludes the proof.

□

Remark 4.5.2. In view of the $\mathrm{O}(n)$-invariance of $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$, it follows that for any integrable function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$ and for every $T \in \mathrm{O}(n)$ we have the equality

$$\int_{\mathbb{S}^{n-1}} f \, d\mathcal{H}^{n-1} = \int_{\mathbb{S}^{n-1}} f \circ T \, d\mathcal{H}^{n-1}.$$

This equality will be used throughout the text a lot of times.

A measure ν in a metric space (X, d) is said to be *uniformly distributed* if it attributes the same positive value for any open ball of a given positive radius. In other words, a measure is uniformly distributed if

$$0 < \nu(B(x,r)) = \nu(B(y,r)) < \infty$$

for all points $x, y \in X$ and every number $0 < r < \infty$, where $B(x,r)$ denotes the metric open ball of center x and radius r.

Lemma 4.5.1. *Any finite Borel measure $\nu\colon \mathbb{S}^{n-1} \to [0, +\infty)$ which is $\mathrm{O}(n)$-invariant and such that $\nu(\mathbb{S}^{n-1}) > 0$ is uniformly distributed. In particular, the measures $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ and $\nu_{\mathbb{S}^{n-1}}$ are uniformly distributed.*

Proof. For any $x \in \mathbb{S}^{n-1}$ and $r > 0$, denote by

$$B_{\mathbb{S}^{n-1}}(x,r) := \{z \in \mathbb{S}^{n-1} : |z - x| < r\} = \{z \in \mathbb{R}^n : |z| = 1 \text{ and } |z - x| < r\}$$

the open ball of center x and radius r in the considered metric of $\mathbb{S}^{n-1}$ (which, as we remember, is the Euclidean metric of $\mathbb{R}^n$ induced on $\mathbb{S}^{n-1}$; see Figure 4.5.1).

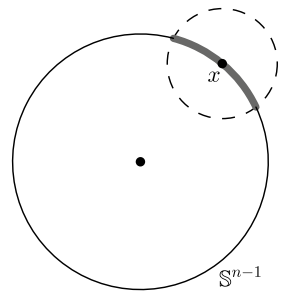


Fig. 4.5.1 The highlighted arc (without the "endpoints") is $B_{\mathbb{S}^{n-1}}(x,r)$. Of course, in this case we have that the dimension n is 2.

Let $x, y \in \mathbb{S}^{n-1}$ and $r > 0$ be arbitrarily given. Choose any $T \in \mathrm{O}(n)$ such that $T(x) = y$. Since T is a bijective isometry of $\mathbb{R}^n$, it follows straightforwardly from the characterization of the open balls of $\mathbb{S}^{n-1}$ above that

$$B_{\mathbb{S}^{n-1}}(y,r) = T\big(B_{\mathbb{S}^{n-1}}(x,r)\big).$$

Consequently, the $\mathrm{O}(n)$-invariance of ν yields

$$\nu\big(B_{\mathbb{S}^{n-1}}(y,r)\big) = \nu\big(T\big(B_{\mathbb{S}^{n-1}}(x,r)\big)\big) = \nu\big(B_{\mathbb{S}^{n-1}}(x,r)\big).$$

Since ν is finite, we trivially have that $\nu\big(B_{\mathbb{S}^{n-1}}(x,r)\big) < +\infty$ for every $x \in \mathbb{S}^{n-1}$ and $r > 0$. Therefore, it only remains to prove that ν attributes a positive value for any ball with positive radius. To do so, observe that for any $r > 0$ we can cover $\mathbb{S}^{n-1}$ with finitely many balls of radius r:

$$\mathbb{S}^{n-1} = \bigcup_{j=1}^{m} B_{\mathbb{S}^{n-1}}(x_j, r),$$

and each one of them has the same measure $h(r)$, say. Thus,

$$0 < \nu(\mathbb{S}^{n-1}) \leq \sum_{j=1}^{m} \nu\big(B_{\mathbb{S}^{n-1}}(x_j,r)\big) = mh(r),$$

yielding $h(r) > 0$. Then every open ball with radius r has positive measure.

Finally, now the fact that both $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ and $\nu_{\mathbb{S}^{n-1}}$ are uniformly distributed follows directly from Proposition 4.5.1, and the proof is complete.

□

We may use the theorem known as *Christensen's lemma*. We will state and prove it based on [624]. We need two technical tools: one is Fubini's theorem (Theorem A.5.3), and the other is *Fatou's lemma* (Theorem A.2.3).

Theorem 4.5.1 (Christensen's lemma)**.** *Let (X,d) be a separable metric space. If μ and ν are uniformly distributed regular Borel measures on (X,d), then there exists a constant c such that $\mu = c\nu$.*

Proof. Let $g, h\colon (0,+\infty) \to \mathbb{R}$ be the functions given as $g(r) = \mu(B(x,r))$ and $h(r) = \nu(B(x,r))$, respectively. Here $B(x,r)$ denotes the open ball of center x and radius r of X (we make this disclaimer because we usually reserve this notation to denote open balls in $\mathbb{R}^n$). Let $U \subseteq X$ be any non-empty open set. Then, for $r > 0$ small enough we have that $U \cap B(x,r) = B(x,r)$, from where

$$\lim_{n\to\infty} \frac{\nu(U \cap B(x,1/n))}{h(1/n)} = 1.$$

Therefore, observing that $1_{B(x,1/n)}(y) = 1_{B(y,1/n)}(x)$ for any $x, y \in X$, we have

$$\begin{aligned}
\mu(U) &= \int_U \lim_{n\to\infty} h(1/n)^{-1}\nu(U\cap B(x,1/n))\,d\mu(x) \leq \\
&\leq \liminf_{n\to\infty}\left[\int_X h(1/n)^{-1}\nu(U\cap B(x,1/n))\,d\mu(x)\right] = \\
&= \liminf_{n\to\infty}\left[h(1/n)^{-1}\int_X \nu(U\cap B(x,1/n))\,d\mu(x)\right] = \\
&= \liminf_{n\to\infty}\left[h(1/n)^{-1}\int_U\int_U 1_{B(x,1/n)}(y)\,d\nu(y)d\mu(x)\right] = \\
&= \liminf_{n\to\infty}\left[h(1/n)^{-1}\int_U\int_U 1_{B(y,1/n)}(x)\,d\mu(x)d\nu(y)\right] = \\
&= \liminf_{n\to\infty}\left[h(1/n)^{-1}\int_U \mu(B(y,1/n))\,d\nu(x)\right] = \\
&= \liminf_{n\to\infty}\left[h(1/n)^{-1}\int_U g(1/n)\,d\nu(x)\right] = \nu(U)\liminf_{n\to\infty}\frac{g(1/n)}{h(1/n)}.
\end{aligned} \tag{4.5.1}$$

Since $\mu(U) > 0$ (because U contains an open ball, which has positive measure by hypothesis), we get

$$0 < \liminf_{n\to\infty}\frac{g(1/n)}{h(1/n)} \leq \infty.$$

Interchanging μ and ν and repeating the calculation above we obtain

$$\nu(U) \leq \mu(U)\liminf_{n\to\infty}\frac{h(1/n)}{g(1/n)}. \tag{4.5.2}$$

Observe that these estimates show that if $\mu(U) = \infty$, then $\nu(U) = \infty$, and vice versa. Hence, we may assume that $\mu(U) < \infty$ and $\nu(U) < \infty$. Since we also have $\nu(U) > 0$, it follows that

$$c := \liminf_{n\to\infty}\frac{g(1/n)}{h(1/n)} < \infty.$$

Indeed, otherwise we would have $\nu(U) \leq 0$ in (4.5.2). From inequalities (4.5.1) and (4.5.2) we get $\mu(U) = c\nu(U)$ for any non-empty open set $U \subseteq X$. Because μ and ν are regular Borel measures, we finally get

$$\begin{aligned}
\mu(E) = \inf\{\mu(U) : U \text{ is open and } E \subseteq U\} &= \inf\{c\nu(U) : U \text{ is open and } E \subseteq U\} = \\
&= c\inf\{\nu(U) : U \text{ is open and } E \subseteq U\} = c\nu(E)
\end{aligned}$$

for any Borel set $E \subseteq X$. This concludes the proof.

□

It follows from Proposition 4.5.1 and Christensen's lemma that $\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$ and $\nu_{\mathbb{S}^{n-1}}$ differ by a constant scalar multiplication. It is actually not difficult to find out the constant which expresses the difference between these measures. In view of (4.1.3) we get

$$\nu_{\mathbb{S}^{n-1}}(\mathbb{S}^{n-1}) = n\mathcal{L}^n(B^n) = n\mathrm{vol}_n(B^n) = \mathrm{vol}_{n-1}(\mathbb{S}^{n-1}) = \mathcal{H}^{n-1}(\mathbb{S}^{n-1}),$$

implying that $\nu_{\mathbb{S}^{n-1}} = \mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}$. Of course, the Haar measure "correctly" normalized also agrees with these.

The *spherical probability measure* σ on $\mathbb{S}^{n-1}$ is the (unit) normalization of any of these measures. Say,

$$\sigma := \frac{1}{\mathcal{H}^{n-1}(\mathbb{S}^{n-1})}\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}.$$

Also this concept is used throughout the text.

Recall that for any convex body $K \subseteq \mathbb{R}^n$ we **always** regard ∂K as a metric space with the induced Euclidean metric of $\mathbb{R}^n$. By doing so, the ball from ∂K (as a metric space) of center $x \in \partial K$ and radius $r > 0$ is given as $B_{\partial K}(x, r) = \partial K \cap B(x, r)$. A subset $A \subseteq \partial K$ is *open* if there exists an open set $U \subseteq \mathbb{R}^n$ such that $A = U \cap \partial K$ or, equivalently, if for any $x \in A$ there exists a number $r > 0$ such that $B_{\partial K}(x, r) \subseteq A$. It is also useful to keep in mind the standard topological fact that any continuous map defined for $\mathbb{R}^n$ is still continuous when restricted to ∂K (or to any other subset of $\mathbb{R}^n$ provided it is endowed with the induced metric).

Proposition 4.5.2. *If $A \subseteq \mathbb{S}^{n-1}$ is open and non-empty, then $0 < \mathcal{H}^{n-1}(A) < +\infty$.*

Proof. We already know that $\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) < +\infty$. Hence, $\mathcal{H}^{n-1} < +\infty$, and we only have to prove the other inequality. For that sake, it is clearly enough to prove that

$$\mathcal{H}^{n-1}(B_{\mathbb{S}^{n-1}}(x, r)) > 0$$

for any $x \in \mathbb{S}^{n-1}$ and every $r > 0$, since any open subset $A \subseteq \mathbb{S}^{n-1}$ contains a ball of $\mathbb{S}^{n-1}$. But this was proved in Lemma 4.5.1, and hence we are done.

□

We can extend this result to any convex body, as it was mentioned in the beginning of the section as a justification for adopting the $(n-1)$-dimensional Hausdorff measure to regard areas of subsets of ∂K. We started with the particular case of the unit sphere because this is needed in our proof of the general case.

Proposition 4.5.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For any open and non-empty set $A \subseteq \partial K$ we have $0 < \mathcal{H}^{n-1}(A) < +\infty$.*

Proof. We start by proving that $\mathcal{H}^{n-1}(A) > 0$. Since $\mathcal{H}^{n-1}$ is invariant under translations (it is clear from the definition that any Hausdorff measure is invariant under isometries, but this is also discussed in Lemma A.4.2), we may assume that $0_{\mathbb{R}^n} \in \text{int}K$. Let $r > 0$ be such that $rB^n \subseteq \text{int}K$. We let $\text{rd}_{r\mathbb{S}^{n-1}} : \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to r\mathbb{S}^{n-1}$ be the *radial projection* onto $r\mathbb{S}^{n-1}$, which is defined as

$$\text{rd}_{r\mathbb{S}^{n-1}}(x) := \frac{rx}{|x|}$$

for each $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Now consider the restriction $g := \text{rd}_{r\mathbb{S}^{n-1}}\big|_{\partial K} : \partial K \to r\mathbb{S}^{n-1}$ (we call it g to simplify the notation). We have the following:

(i) The map g is Lipschitz. To prove this, we first notice that g equals the restriction of the usual metric projection $p_{rB^n} : \mathbb{R}^n \to r\mathbb{S}^{n-1}$ to ∂K. In fact, $g(x)$ is always a multiple of x, and hence for any $x \in \partial K$ the (non-degenerate) segment connecting x to $g(x)$ is normal to $r\mathbb{S}^{n-1}$ at $g(x)$; see Figure 4.5.2. Now we use Proposition 2.2.1 to conclude that g is Lipschitz.

(ii) The map g is a bijection with continuous inverse. The fact that g is bijective comes straightforwardly from the fact that any ray starting at the origin touches ∂K in exactly one point. To verify that g^{-1} is continuous, the easier way is to describe it explicitly as

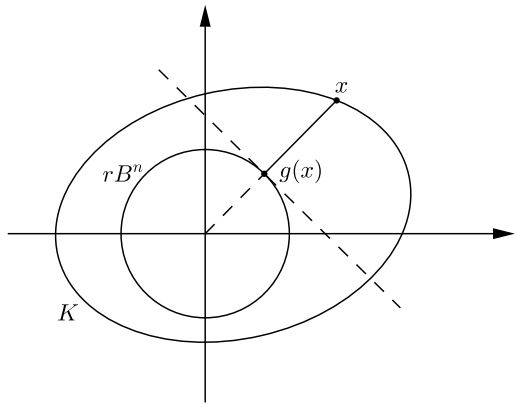


Fig. 4.5.2 $g(x)$ has the direction of x.

$$g^{-1}(y) = \frac{y}{||y||_K},$$

for each $y \in r\mathbb{S}^{n-1}$. Since the gauge function $||\cdot||_K$ is convex and finite (hence continuous), we get indeed that g^{-1} is continuous.

Now let $A \subseteq \partial K$ be any open and non-empty set. From **(ii)**, we get that $g(A)$ is an open and non-empty subset of $\mathbb{S}^{n-1}$. Thus, since g is Lipschitz, we can use Theorem A.4.3 to obtain

$$0 < \mathcal{H}^{n-1}(g(A)) \leq \mathrm{Lip}(g)^{n-1}\mathcal{H}^{n-1}(A),$$

where $\mathrm{Lip}(g)$ denotes the *Lipschitz constant* of g and the first inequality comes from Proposition 4.5.2, of course. This gives that $\mathcal{H}^{n-1}(A) > 0$, as we wanted to prove.

To prove that $\mathcal{H}^{n-1}(A) < +\infty$, we only have to consider a ball rB^n containing K and use Corollary A.4.7. We have

$$\mathcal{H}^{n-1}(A) \leq \mathcal{H}^{n-1}(\partial K) \leq \mathcal{H}^{n-1}(r\mathbb{S}^{n-1}) < +\infty.$$

□

Intuitively, the boundary of a convex body is an "$(n-1)$-dimensional object". Hence we may expect that the n-dimensional volume of a convex body equals the n-dimensional volume of its interior. Using a similar argument as we did above, we prove that this is indeed true. We also keep in mind that the Lebesgue measure coincides with the n-dimensional Hausdorff measure in $\mathbb{R}^n$ (this is proved in Theorem A.4.2).

Proposition 4.5.4. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then*

$$\mathcal{H}^n(\mathrm{int}K) = \mathcal{H}^n(K),$$

and the same holds for the Lebesgue measure.

Proof. From Proposition 4.5.3 and Proposition A.4.1, we have that $\mathcal{H}^n(\partial K) = 0$. Hence, we just write

$$\mathcal{H}^n(K) = \mathcal{H}^n((\mathrm{int}K) \cup \partial K) = \mathcal{H}^n(\mathrm{int}K) + \mathcal{H}^n(\partial K) = \mathcal{H}^n(\mathrm{int}K).$$

□

4.6 Exercises

Exercise 4.1. *Verify that the surface area of a convex body is invariant under orthogonal transformations and under translations. That is, show that if $S(\cdot)$ denotes the surface area, then $S(U(K) + x_0) = S(K)$ for any convex body $K \subseteq \mathbb{R}^n$, every $U \in \mathrm{O}(n)$ and all $x_0 \in \mathbb{R}^n$.*

Hint: Use the Minkowski surface area, which is defined in (4.1.1).

Remark 4.6.1. Geometrically, this confirms the intuitive idea that the surface area does not change under "rigid motions".

Exercise 4.2. *Recall that the* standard orthogonal n-simplex *of $\mathbb{R}^n$ is the set*

$$\Delta_n := \mathrm{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\}.$$

Prove that the surface area of Δ_n is

$$S(\Delta_n) = \frac{n + \sqrt{n}}{(n-1)!}.$$

Exercise 4.3. *The* regular n-dimensional cross-polytope *is the polytope of $\mathbb{R}^n$ given as*

$$P_\Delta^n := \mathrm{conv}\big\{ \pm e_1, \ldots, \pm e_n \big\}.$$

(a) *Show that P_Δ^n is the unit ball of the ℓ_1-*norm *of $\mathbb{R}^n$, which is defined as*

$$||(x_1, \ldots, x_n)||_1 := |x_1| + \ldots + |x_n|$$

for each $(x_1, \ldots, x_n) \in \mathbb{R}^n$.

(b) *Prove that the volume of P_Δ^n is given by*

$$\mathrm{vol}_n(P_\Delta^n) = \frac{2^n}{n!}.$$

(c) *Show that the surface area of P_Δ^n is*

$$S(P_\Delta^n) = \frac{2^n \sqrt{n}}{(n-1)!}.$$

Exercise 4.4. *Prove that the circumradius and the inradius of the regular n-dimensional cross-polytope P_Δ^n are 1 and $1/\sqrt{n}$, respectively.*

Exercise 4.5. (a) *Show that the polar body of the regular cross-polytope P_Δ^n is the* cube

$$Q^n := [-1, 1]^n = \big\{(x_1, \ldots, x_n) : -1 \le x_j \le 1 \ \ \forall 1 \le j \le n\big\}.$$

(b) *Verify that Q^n is the unit ball of the ℓ_∞-*norm *of $\mathbb{R}^n$, which is defined as*

$$||(x_1, \ldots, x_n)||_\infty := \max\big\{|x_j| : 1 \le j \le n\big\}.$$

(c) *Prove that the surface area of Q^n is*

$$S(Q^n) = n2^n.$$

For the next exercise, recall that $\Gamma\colon (0,+\infty) \to \mathbb{R}$ is the *Gamma function* defined in Proposition 4.3.1. Remember also that κ_n denotes the volume of the unit ball of $\mathbb{R}^n$. From the mentioned proposition, we have an expression for κ_n in terms of the Gamma function.

Exercise 4.6. (a) *Verify that*

$$\Gamma(1/2) = \sqrt{\pi}.$$

(b) *Show that if n is even, then*

$$\kappa_n = \frac{\pi^{n/2}}{(n/2)!},$$

and that if n is odd, then writing $n = 2k+1$, we have

$$\kappa_n = \frac{2k!(4\pi)^k}{(2k+1)!}.$$

Derive similar expressions for ω_n.

Exercise 4.7. (a) *Show that*

$$\kappa_n = \frac{2\pi}{n}\kappa_{n-2}$$

for every $n \geq 2$.

(b) *Prove that*

$$\kappa_n\kappa_{n-1} = \frac{2^n\pi^{n-1}}{n!}$$

for each $n \geq 1$.

Exercise 4.8. *Recalling that $\omega_n = n\kappa_n$, prove that $\kappa_n \to 0$ and $\omega_n \to 0$ as $n \to +\infty$.*

Exercise 4.9. *Let $u \in \mathbb{S}^{n-1}$ and let $E \subseteq u^\perp$ be a measurable set. Show that for any vector $v \notin u^\perp$, we have*

$$\mathrm{vol}_n\big(E + \mathrm{seg}[0_{\mathbb{R}^n}, v]\big) = |\langle u, v\rangle|\mathrm{vol}_{n-1}(E).$$

Remark 4.6.2. The objective of this exercise is to give a formal proof of the fact that the volume of the cylinder $E + \mathrm{seg}[0_{\mathbb{R}^n}, v]$ equals the "area of the basis multiplied by the height".

The *centroid* of a convex body $K \subseteq \mathbb{R}^n$ is the point $c(K) \in \mathbb{R}^n$ defined as

$$c(K) := \frac{1}{\mathrm{vol}_n(K)}\int_K x\,dx,$$

where dx stands for integration with respect to the usual Lebesgue measure. The integration is taken in each coordinate, in the sense that

$$\int_K x\,dx = \left(\int_K x_1\,dx, \int_K x_2\,dx, \ldots, \int_K x_n\,dx\right).$$

Exercise 4.10. **(a)** *Let $K \subseteq \mathbb{R}^n$ be a convex body whose centroid is x_0. For any $z \in \mathbb{R}^n$, prove that the centroid of $z + K$ is $z + x_0$.*

(b) *Assume that $K \subseteq \mathbb{R}^n$ is a convex body whose centroid is $x_0 \in \mathbb{R}^n$. Show that $x_0 \in \operatorname{int}(K)$.*

(c) *Show that $c(TK) = T(c(K))$ for any $K \in \mathcal{K}^n$ and every $T \in \mathrm{GL}(n)$.*

Exercise 4.11. *Let $K \subseteq \mathbb{R}^n$ be a centrally symmetric convex body with $K - x_0 = -(K - x_0)$. Prove that x_0 is the centroid of K.*

Exercise 4.12. *Let $c\colon \mathcal{K}^n \to \mathbb{R}^n$ be the map which associates each convex body $K \subseteq \mathbb{R}^n$ with its centroid $c(K)$. Prove that this map is continuous in the Hausdorff metric.*

Hint. Use the Dominated Convergence Theorem (Theorem A.2.4).

Exercise 4.13. *Let $K \in \mathcal{K}^n$. Prove that the centroid $c(K)$ of K is the unique point $p \in \mathbb{R}^n$ with the property that*

$$f(p) = \frac{1}{\operatorname{vol}_n(K)} \int_K f(x)\,dx \tag{4.6.1}$$

for every affine function $f\colon \mathbb{R}^n \to \mathbb{R}$. In other words, the mean of any affine function over K equals the value of f at the centroid $c(K)$ of K, and $c(K)$ is the unique point of $\mathbb{R}^n$ having this property.

Remark 4.6.3. Actually, assuming that (4.6.1) holds for all **linear** functions $f\colon \mathbb{R}^n \to \mathbb{R}$ is sufficient to characterize p as the centroid of K.

Exercise 4.14. *Parametrize $\mathbb{S}^1$ by $u(\theta) := (\cos\theta, \sin\theta)$, $\theta \in [0, 2\pi)$, and let $K \subseteq \mathbb{R}^2$ be a convex body having the origin as an interior point.*

(a) *Assume that K is strictly convex. Let $h_K(\theta) := h_K(u(\theta))$ be the restriction of h_K to $\mathbb{S}^1$. Prove that*

$$x(\theta) := h_K(\theta)(\cos\theta, \sin\theta) + h_K'(\theta)(-\sin\theta, \cos\theta), \quad \theta \in [0, 2\pi),$$

is a parametrization of ∂K. Check that for each $\theta \in [0, 2\pi)$ it holds that $(\cos\theta, \sin\theta)$ is the unit outer normal vector of K at the point $x(\theta)$.

(b) *Show that if h_K is C^2, then the area of K is given by*

$$\operatorname{vol}_2(K) := \frac{1}{2}\int_0^{2\pi} h_K(\theta)^2 - h_K'(\theta)^2\,d\theta,$$

where $d\theta$ stands for the usual Lebesgue measure in the interval $[0, 2\pi)$.

Hint: For Item **(a)**, write the gradient ∇h_K in polar coordinates (ρ, θ) and use Proposition 2.3.9. For Item **(b)**, use the classical Green formula for the area of a parametrized curve.

The objective of our next exercise is the so-called *Brunn concavity principle*, which was used to give the first proof of the Brunn–Minkowski inequality (see Section 5.1 and Exercise 5.17). As the main ingredient, the reader may use the fact that a modification in the proof of Theorem 2.7.1 (with the help of Exercise 2.59) leads to the following stronger result.

Theorem 4.6.1 (Steiner–Schwarz). *Let $F \subseteq \mathbb{R}^n$ be a k-dimensional vector subspace, and let $K \subseteq \mathbb{R}^n$ be a convex body. There exists a sequence of successive Steiner symmetrizations of K in directions*

$u \in F$ which converge to a body $\tilde{K}$ with the property that for any $x \in F^{\perp}$, $\tilde{K} \cap (x+F)$ is a k-dimensional ball centered at x, and

$$\mathrm{vol}_k(\tilde{K} \cap (x+F)) = \mathrm{vol}_k(K \cap (x+F)).$$

Exercise 4.15 (Brunn concavity principle)**.** *Let $F \subseteq \mathbb{R}^n$ be a k-dimensional vector subspace, and let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that the function $f\colon F^{\perp} \to \mathbb{R}$ defined as*

$$f(x) = \mathrm{vol}_k(K \cap (x+F))^{1/k}$$

is concave on its support.

Exercise 4.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin.*

(a) *Show that*

$$\max\left\{\mathrm{vol}_{n-1}(K \cap (u^{\perp} + tu)) : t \in \mathbb{R}\right\} = \mathrm{vol}_{n-1}(K \cap u^{\perp})$$

for each $u \in \mathbb{S}^{n-1}$. In other words, the section of K parallel to $u^{\perp}$ having maximal area is the one which passes through the origin.

(b) *Verify that the inequality*

$$h_K(u) \geq \frac{\mathrm{vol}_n(K)}{2\mathrm{vol}_{n-1}(K \cap u^{\perp})}$$

holds for each $u \in \mathbb{S}^{n-1}$.

(c) *Prove that*

$$h_K(u) \leq \frac{n\mathrm{vol}_n(K)}{2\mathrm{vol}_{n-1}(K \cap u^{\perp})}$$

for all $u \in \mathbb{S}^{n-1}$.

(d) *Discuss the equality cases in the inequalities given in Items* **(b)** *and* **(c)***.*

The *Gaussian measure* of $\mathbb{R}^n$ is the Borel measure γ_n whose density is

$$G(x) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{|x|^2}{2}}. \tag{4.6.2}$$

That is, for each Borel subset $A \subseteq \mathbb{R}^n$, the measure γ_n is defined by

$$\gamma_n(A) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_A e^{-\frac{|x|^2}{2}}\, dx.$$

It is clear from standard measure theory that the integral of a Borel measurable function $f\colon \mathbb{R}^n \to [-\infty, +\infty]$ with respect to γ_n is given by

$$\int_{\mathbb{R}^n} f(x)\, d\gamma_n(x) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(x) e^{-\frac{|x|^2}{2}}\, dx.$$

Exercise 4.17. (a) *Verify that the Gaussian measure is a probability measure.*

(b) *Check that γ_n is invariant under orthogonal transformations.*

It is obvious that the Gaussian measure is not translation-invariant. However, we have a bound for the measure of translations of a body centered at the origin. This is in the exercise below.

Exercise 4.18. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin. Prove that for any $y \in \mathbb{R}^n$ the inequality*

$$\gamma_n(y + K) \geq e^{-|y|^2/2}\gamma_n(K)$$

holds.

Exercise 4.19 (Markov's inequality)**.** *Let (X, μ) be a probability space, and let $f\colon X \to \mathbb{R}$ be a μ-measurable function. Prove that*

$$\mu\Big(\big\{x \in X : |f(x)| \geq \varepsilon\big\}\Big) \leq \frac{1}{\varepsilon}\int_X |f(x)|\, d\mu(x)$$

for every $\varepsilon > 0$.

Hint: Define $\phi\colon X \to \mathbb{R}$ as

$$\phi(x) = \begin{cases} \varepsilon, & \text{if } |f(x)| \geq \varepsilon \\ 0, & \text{if } |f(x)| < \varepsilon \end{cases}.$$

Notice that $\phi \leq |f|$ and integrate both with respect to μ.

Exercise 4.20. *Prove that the Gaussian measure γ_n is* log-concave *in the following sense: for any Borel subsets $A, B \subseteq \mathbb{R}^n$ and for every $\lambda \in [0, 1]$ we have that*

$$\gamma_n((1 - \lambda)A + \lambda B) \geq \gamma_n(A)^{1-\lambda}\gamma_n(B)^{\lambda}.$$

The density of the Gaussian measure is a log-concave function, and the reader may ask whether a log-concave density always yields a log-concave measure. This is true, in fact, and the reader is invited to prove that in the exercise below.

Exercise 4.21. *Let $\phi\colon \mathbb{R}^n \to [0, +\infty)$ be a log-concave function such that*

$$0 < I(\phi) := \int_{\mathbb{R}^n} \phi(x)\, dx < +\infty.$$

Let μ_ϕ be the Borel measure in $\mathbb{R}^n$ whose density with respect to the usual Lebesgue measure is $I(\phi)^{-1}\phi(\cdot)$. That is,

$$\mu_\phi(A) = \frac{1}{I(\phi)}\int_A \phi(x)\, dx$$

for each Borel set $A \subseteq \mathbb{R}^n$. Prove that μ_ϕ is a log-concave measure.

Exercise 4.22. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin.*

(a) *Prove that for every $t > 1$ the inclusion*

$$\frac{2}{t+1}(\mathbb{R}^n \setminus tK) + \frac{t-1}{t+1}K \subseteq \mathbb{R}^n \setminus K$$

holds.

(b) *Write* $\gamma_n(K) = c$. *Use Item* **(a)** *and the log-concavity of* γ_n *to show that*

$$\gamma_n(tK) \geq 1 - c\left(\frac{1-c}{c}\right)^{\frac{t+1}{2}}$$

for any $t > 1$.

Remark 4.6.4. The bound given in Item **(b)** is only useful if $c > 1/2$, because if $c \leq 1/2$, then the trivial bound

$$\gamma_n(tK) \geq \gamma_n(K) = c$$

is sharper. In fact, it is easy to see that the inequality

$$c \geq 1 - c\left(\frac{1-c}{c}\right)^{\frac{t+1}{2}}$$

is equivalent to

$$(1-c)^{\frac{t-1}{2}} \geq c^{\frac{t-1}{2}},$$

which holds for $c \in (0,1)$ if and only if $c \leq 1/2$. Observe also that to have $\gamma_n(tK) \geq \gamma_n(K)$ we just need that $K \subseteq tK$. Hence we do not need to have that K is origin-symmetric; it suffices to consider that K is any convex body containing the origin.

Exercise 4.23. *Let* $C \subseteq \mathbb{R}^n$ *be a closed set such that* $\gamma_n(C) > 0$. *Prove that*

$$\int_{\mathbb{R}^n} e^{\frac{1}{4}d(x,C)^2}\, d\gamma_n(x) \leq \frac{1}{\gamma_n(C)},$$

where we remind the reader that $d(x,C) := \inf\{d(x,z) : z \in C\}$.

Hint: Use the Prékopa–Leindler inequality (Theorem 1.5.1) for the functions $f(x) = e^{\frac{1}{4}d(x,C)^2}G(x)$, $g(x) = 1_C(x)G(x)$, $h(x) = G(x)$, and for $\lambda = 1/2$.

Exercise 4.24. *Let* $C \subseteq \mathbb{R}^n$ *be a closed set such that* $\gamma_n(C) \geq 1/2$, *and set*

$$C_\varepsilon := \{x \in \mathbb{R}^n : d(x,C) \leq \varepsilon\}.$$

Prove that for any $\varepsilon \geq 0$ *we have that*

$$\gamma_n(C_\varepsilon) \geq 1 - 2e^{-\frac{\varepsilon^2}{4}}.$$

Remark 4.6.5. This can be seen as a *concentration inequality* since, in some sense, it states that the Gaussian measure is "concentrated around" any given closed set having measure at least $1/2$.

We briefly investigate (abstract) concentration of measures below. For more in this direction, we refer the reader to [133] (which is the source of most exercises below). Let (X,d) be a metric space, and denote by $\mathcal{B}(X)$ its Borel σ-algebra. Assume that $\mu\colon \mathcal{B}(X) \to [0,1]$ is a Borel probability measure on X. In this case, we say that (X,d,μ) is a *metric probability space*. The *concentration function* of (X,d,μ) is the function $\alpha_\mu\colon [0,+\infty) \to \mathbb{R}$ defined as

$$\alpha_\mu(t) := \sup\left\{1 - \mu(A_t) : A \in \mathcal{B}(X) \text{ and } \mu(A) \geq 1/2\right\},$$

where $A_t := \{x \in X : d(x, A) \leq t\}$.

Exercise 4.25. *Show that for any metric probability space (X, d, μ) the concentration function is non-increasing. Prove also that*

$$\lim_{t \to +\infty} \alpha_\mu(t) = 0.$$

Let (X, d, μ) be a metric probability space, and let $f : X \to \mathbb{R}$ be any measurable function. The *mean* (or *expected value*) of f is the number

$$\mathbb{E}(f) := \int_X f \, d\mu. \tag{4.6.3}$$

A *median* (or *Lévy mean*) of f is a number $\operatorname{med}(f)$ such that

$$\mu\big(\{f \geq \operatorname{med}(f)\}\big) \geq \frac{1}{2} \quad \text{and} \quad \mu\big(\{f \leq \operatorname{med}(f)\}\big) \geq \frac{1}{2}.$$

It is not difficult to see that a median does not need to be unique, and its existence is also not completely trivial. In the next exercise, we provide some conditions which are sufficient to ensure the existence and uniqueness of the median.

Exercise 4.26. *Let (X, d, μ) be a metric probability space having the following properties:*

(i) *(X, d) is a connected metric space.*

(ii) *$\mu(A) > 0$ for every non-empty open set $A \subseteq X$.*

Prove that any continuous function $f : X \to \mathbb{R}$ has a unique median.

For simplicity, from now on we say that a function $f : (X, d) \to \mathbb{R}$ is 1-*Lipschitz* if $|f(x) - f(y)| \leq d(x, y)$ for any $x, y \in X$. In the exercise below, we relate these notions to the concentration function of (X, d, μ). Intuitively, we have that the value of $\alpha_\mu(t)$ "controls" the measure of the set of points of X for which f has distance greater than t to a median.

Exercise 4.27. *Let (X, d, μ) be a metric probability space, and assume that $f : X \to \mathbb{R}$ is a 1-Lipschitz function. Prove that*

$$\mu\Big(\big\{x \in X : |f(x) - \operatorname{med}(f)| > t\big\}\Big) \leq 2\alpha_\mu(t) \tag{4.6.4}$$

for any median $\operatorname{med}(f)$ of f and all $t \geq 0$.

Exercise 4.28. *Let (X, d, μ) be a metric probability space, and assume that there exist numbers $\alpha, t > 0$ such that*

$$\mu\Big(\big\{x \in X : |f(x) - \operatorname{med}(f)| > t\big\}\Big) \leq \alpha$$

for any 1-Lipschitz function $f : X \to \mathbb{R}$. Show that $\alpha_u(t) \leq \alpha$.

Hint: For a given Borel set $A \subseteq X$ such that $\mu(A) \geq 1/2$, consider the distance function $f(x) = d(x, A)$.

Inequality (4.6.4) can be seen as a bound for the probability of *deviation from the median*. Our next objective is to derive an inequality for the *deviation from the mean* for 1-Lipschitz functions with respect to the Gaussian measure. This is made in Exercise 4.31. Before this, we bring Exercises 4.29 and 4.30 for the auxiliary results.

We recall that the *infimal convolution* of functions $f, g\colon \mathbb{R}^n \to (-\infty, +\infty]$ is the function $f\square g\colon \mathbb{R}^n \to [-\infty, +\infty]$ given as

$$f\square g(x) = \inf\big\{f(z) + g(x-z) : z \in \mathbb{R}^n\big\}.$$

We refer the reader to Exercises 1.22, 1.23, 1.24, and 1.25 for more on the infimal convolution.

Exercise 4.29. *Let $\varphi\colon \mathbb{R}^n \to \mathbb{R}$ be the function given as $\varphi(x) = |x|^2/4$, and let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a measurable bounded function.*

(a) *Define the functions $\bar{f}, g, h\colon \mathbb{R}^n \to \mathbb{R}$ as*

$$\bar{f}(x) = f(x) + \frac{|x|^2}{2},\ g(x) = -f\square\varphi(x) + \frac{|y|^2}{2} \text{ and } h(x) = \frac{|x|^2}{2}.$$

Show that

$$h\left(\frac{x+y}{2}\right) \le \frac{\bar{f}(x) + g(y)}{2}$$

for any $x, y \in \mathbb{R}^n$.

(b) *Prove that*

$$\left(\int_{\mathbb{R}^n} e^{f\square\varphi}\, d\gamma_n\right)\left(\int_{\mathbb{R}^n} e^{-f}\, d\gamma_n\right) \le 1.$$

Exercise 4.30. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a bounded function which is 1-Lipschitz with respect to the Euclidean distance of $\mathbb{R}^n$, and let $t > 0$ be arbitrarily given. Also, still assume that $\varphi\colon \mathbb{R}^n \to \mathbb{R}$ is the function given as $\varphi(x) = |x|^2/4$.*

(a) *Verify that the infimum in the definition of $(tf)\square\varphi(x)$ is attained. That is, check that for any $x \in \mathbb{R}^n$ there exists $y = y(x,t) \in \mathbb{R}^n$ such that $(tf)\square\varphi(x) = tf(y) + \varphi(x-y)$.*

(b) *Using the 1-Lipschitz hypothesis, show that*

$$(tf)\square\varphi(x) \ge tf(x) - t|x-y| + \frac{|x-y|^2}{4},$$

where y is as given in Item **(a)**.

(c) *Studying the second-degree polynomial $\alpha \mapsto \alpha^2/4 - t\alpha$, conclude that*

$$(tf)\square\varphi(x) \ge tf(x) - t^2$$

for any $x \in \mathbb{R}^n$.

(d) *Prove that*

$$\int_{\mathbb{R}^n} e^{tf}\, d\gamma_n \int_{\mathbb{R}^n} e^{-tf}\, d\gamma_n \le e^{t^2}.$$

Exercise 4.31. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a bounded 1-Lipschitz function.*

(a) *Use Exercise 4.30 and Jensen's inequality to prove that*

$$\mathbb{E}\big(\exp(t(f-\mathbb{E}(f))\big) \le e^{t^2}$$

for every $t > 0$, where we recall that $\mathbb{E}$ is the expected value *as defined in (4.6.3).*

(b) *Using Item* **(a)**, *show that for every $t, s > 0$ the inequality*

$$\gamma_n\big(\{x \in \mathbb{R}^n : f(x) - \mathbb{E}(f) \ge s\}\big) \le e^{t^2 - ts}$$

holds.

(c) *Prove that*

$$\gamma_n\big(\{x \in \mathbb{R}^n : |f(x) - \mathbb{E}(f)| \ge s\}\big) \le 2e^{-s^2/4}$$

for any $s > 0$.

For the sake of completeness, we introduce now some other concepts and some terminology from probability theory (as a very brief introduction). It is usual to use the symbol $\mathbb{P}$ to represent a probability measure, and hence we denote a probability space by $(\Omega, \mathbb{P})$. In the language of probability theory, a measurable function $X\colon \Omega \to \mathbb{R}$ is also called a *random variable*. The *variance* of a random variable X is defined as

$$\mathrm{Var}[X] := \mathbb{E}\big((X - \mathbb{E}(X))^2\big).$$

The *standard deviation* of X, denoted by σ_X, is defined to be the square root of the variance. That is,

$$\sigma_X := \sqrt{\mathrm{Var}[X]}.$$

Exercise 4.32. *Let X be a random variable in a probability space $(\Omega, \mathbb{P})$.*

(a) *Check that*

$$\mathrm{Var}[X] = \mathbb{E}(X^2) - \mathbb{E}(X)^2.$$

(b) *Verify that for any constants $a, b \in \mathbb{R}$ we have*

$$\mathrm{Var}[aX + b] = a^2 \mathrm{Var}[X].$$

Finally, let X be a random variable on a probability space $(\Omega, \mathbb{P})$, and assume that there is a measurable function $f_X\colon \mathbb{R} \to \mathbb{R}$ with the property that

$$\mathbb{P}\big(a \le X \le b\big) = \int_a^b f_X(t)\, dt$$

for any $a, b \in \mathbb{R}$ (where dt stands for integration with respect to the usual Lebesgue measure of $\mathbb{R}$). In this case, we say that f_X is the *probability density function* of X.

Exercise 4.33 (Chebyshev)**. (a)** *Let X be a random variable in a probability space $(\Omega, \mathbb{P})$. Prove that for every $k > 0$ the inequality*

$$\mathbb{P}\Big(\big\{x \in \Omega : |X(x) - \mathbb{E}(X)| \geq k\big\}\Big) \leq \frac{\mathrm{Var}[X]}{k^2}$$

holds. Notice that this can be seen as an inequality for the deviation from the mean.

(b) *There is a way of restating the inequality above in terms of standard deviations. Show that for every $c > 0$ we have*

$$\mathbb{P}\Big(\big\{x \in \Omega : |X(x) - \mathbb{E}(X)| \geq c\sigma_X\big\}\Big) \leq \frac{1}{c^2}.$$

A convex body $K \subseteq \mathbb{R}^n$ is said to be *isotropic* if it has its centroid at the origin and if there exists a constant $\rho \in \mathbb{R}$ such that

$$\int_K \langle x, u\rangle^2 \, dx = \rho^2$$

for every $u \in \mathbb{S}^{n-1}$, where the integration is taken with respect to the usual Lebesgue measure of $\mathbb{R}^n$, of course. It is worth mentioning that most part of the authors demand additionally that $\mathrm{vol}_n(K) = 1$, but we do not do it here. Hence, for us, the isotropic property is invariant under rescaling.

Exercise 4.34. *Let $K \subseteq \mathbb{R}^n$ be a convex body whose volume is* 1. *The restriction of the usual Lebesgue measure endows a probability measure on K, and hence we regard (K, vol_n) as a probability space.*

(a) *For each $u \in \mathbb{S}^{n-1}$, let $X_u \colon K \to \mathbb{R}$ be the random variable defined as*

$$X_u(x) = \langle x, u\rangle$$

for each $x \in K$. Verify that X_u has probability density function $f_u \colon \mathbb{R} \to \mathbb{R}$ given as

$$f_u(t) = \mathrm{vol}_{n-1}\big(K \cap \{u^\perp + tu\}\big),$$

for $t \in \mathbb{R}$.

(b) *Show that if K is isotropic, then each random variable X_u as defined in Item* **(a)** *has variance ρ^2.*

4.7 Notes

As also discussed below in subsection 4.7.2, Chapter 13 of Matoušek's monograph [1732] gives a nice overview on *volume phenomena in high dimensions*. The "non-intuitive" topics discussed there include themes like the ball inscribed to the cube, balls in certain packings, distribution of mass, the Busemann–Petty problem, nets in spheres (defined via distances), volume approximation (e.g., of the ball by polytopes with at most m vertices), simplices in spheres, and further problems. A part of the polytope-related results reflected in [1732] is also taken into consideration in the following list of results, treating *volumes of polytopes*. Another basic reference for problems connected with volume is the book [323] by K. Bezdek and Lángi. The authors collect research problems from discrete geometry treating volume as the essential notion, calling this direction *volumetric discrete geometry*. Examples are isoperimetric-like inequalities for certain classes of polytopes, the Kneser–Poulsen conjecture stating that the volume of the union (or of the intersection) of an arbitrary arrangement of balls decreases under a contraction of the set of centers, and the study of certain translative packings. And we refer also to the nice exposition [1235] of Horváth in which various long-standing open problems connected with convex-hull-type volume functions (such as the *covariogram*) are collected.

4.7.1 Cavalieri's principle

These notes now refer to the more historical side of a useful principle named after Cavalieri, the famous mathematician and astronomer from the seventeenth century. Cavalieri's principle (roughly saying that the volume of a solid in space can be computed as the integral of the volumes of parallel slices of it; see also Plate 4 at the beginning of this chapter) is geometrically obvious, and from the viewpoint of integration theory it occurs as a *special case of Fubini's theorem*. Section 1.2 of Gardner's book [930] shows comprehensively the strong relations between *Cavalieri's principle* and basic notions from *convexity and geometric tomography*, such as *Steiner symmetrization* and *X-rays*; see also Sections 5.2, 5.3, and 5.4 on strongly related concepts like point X-rays, projective X-rays, and an application regarding projection formulas.

Its importance is honoured by a variety of applications in several books of different nature, such as Struik's historical collection [2314] (see Chapter IV, pp. 209-218), Ewald's book [823] on *geometric analysis*, Ambartzumian's monograph [98] on *geometric probability* (showing how strong its basic idea of factorization is related to Cavalieri's principle), Kieu's book [1377] with motivations from *stereology*, the interesting collection [945] of Gay (see p. 98), and Chapters 8 and 12 of Havil's nice booklet [1170].

HISTORICAL ASPECTS

Particularly interesting is the monograph [2225] of Shen, Crossley, and Lun, the English translation of the Chinese classic *The Nine Chapters on the Mathematical Art*, originally written during the Han dynasty in the last two centuries BCE and in the modern version containing many comments to later works. In its Chapter 4, Chinese investigations of volume problems are collected. Based on such investigations, the Chinese mathematician Liu Hui from the third century applied a principle that, similar to that of Cavalieri, led him to the ratio of the volumes of a ball and the intersection of two perpendicular cylinders circumscribing this ball. Additional hints are given by the translators, also regarding Archimedes and later related contributions. Liu Hui was not able to finish this work; only in the fifth century Zu Geng, also applying a Cavalieri-like principle, solved Liu's problem deriving the

volume of a ball. Additional historical papers on these contributions of Chinese mathematicians are due to Lam and Shen [1489], Horng [1231], and Straffin [2311].

Also very interesting from the historical point of view is Körle's book [1437] on the history of analysis, where Cavalieri is nicely discussed in the second part, in one of the 26 self-standing chapters there. The second edition of Pugh's textbook [2003] contains various elegant *picture proofs* (e.g., of the *monotone and dominated convergence theorems*); in particular, Chapter 5 contains a *one-line/one-picture proof of Fubini's theorem from Cavalieri's principle*! Alsina and Nelson [90] study problems and results on figures and surfaces in 3-space, emphasizing typical techniques used there. In Chapter 5 of [90], plane sections of solids and Cavalieri's principle are considered.

SELECTED RESEARCH ARTICLES OR CLOSE-TO-RESEARCH PAPERS

We come now to some selected research articles on different aspects of Cavalieri's principle.

Two early elementary contributions to the tomographic notion of *X-rays*, long before Gardner's book [930], are given by Struik [2315] and Giering [968]. For example, in [2315] it is shown that any two triangles $A_1A_2A_3$ and $B_1B_2B_3$ having the same area in the Euclidean plane can be positioned such that the three lines A_iB_i are parallel and the triangles are Cavalieri-equal (i.e., have the same Steiner symmetral in the mentioned line direction). Moreover, due to Desargues' theorem even an equiaffine reflection exists that interchanges the two triangles.

Terregino published the two nice papers [2354] and [2355] on Cavalieri's most known work *Geometria Indivisibilibus Continuorum, Nova Quadam Ratione Promota* (Bologna, 1635), where the constant-ratio variant of his principle and the three-dimensional analogue with plane sections are considered. It is shown how Cavalieri's second formulation of his principle, given at the beginning of Book VII of his *Geometria* and discussed in the second paper [2355], circumvents philosophical problems with an early formulation concerning the notion of infinity.

An early example discussing the connection between Cavalieri's principle and the theory of *reconstructive tomography* is the paper [525] of Bukhstaber and Gindikin. Interestingly, the authors discuss also non-medical *applications* of tomography, e.g., from radio-astronomy and the structure of viruses (we refer also to the Czech translation [518]). In his important historical article [2515], Yushkevich surveys the development of the notion of *limit* (and of strongly related concepts) up to Weierstrass. He poses the conjecture that Archimedes might have been aware of Cavalieri's principle, and he also underlines that the Chinese formulated it already in the fifth century AD (see above). In this survey an impressive collection of (also other) problems and foundations of concepts is presented.

Based on a result of Freiling who gives an ingenious argument which seems to establish that the continuum hypothesis is false, Simms shows that a weak consequence of his argument in [2247] is equivalent to the negation of the continuum hypothesis and finds, finally, Freiling's argumentation (though very attractive) not as utterly convincing. A suitable application of Cavalieri's principle and an extension of Fubini's theorem play a central role in the argumentation of both papers.

Using Cavalieri's principle, Apostol and Mnatsakanian [114] discuss how to find *volumes without computation*. For instance, a nice example is the quadrature of the sine curve via transformation into a three-dimensional problem. And the reviewer mentions that there exists a comprehensive collection of quadratures without computations in the papers of Roberval (Académie des Sciences) from the late 1600s.

Calling it *Cavalieri integration*, Ackermann et al. develop in [14] a new integration technique with the help of Cavalieri's principle. Using non-rectangular integration strips, such integrals differ geometrically from Riemann integrals. This opens possibilities to calculate areas of regions that cannot be so easily measured via Riemann integrals. An interesting paper discussing properties of convex polytopes which are Cavalieri-equal regarding parallel k-dimensional affine flats is [1690] by Malyshev.

4.7.2 Volumes of unit balls and related topics

In Section 4.3, the *volume of the Euclidean unit ball* is discussed. We want to present some contributions related to this subject. It is known that extensions of certain phenomena occurring there (e.g., under extensions to general convex bodies) are important in asymptotic convexity and Banach space theory. These are discussed in the corresponding chapters of our book.

SOME PHENOMENA

It is folklore that, as $n \to \infty$, the volume of the unit ball of Euclidean n-space goes down to zero and simultaneously concentrates on the thin shell near the bounding sphere. There are many different approaches to this fact, based on methods from Euclidean, discrete and convex geometry, functional analysis, probability theory, combinatorics, and so on. We add some (mostly recent) references. Li [1567] uses regularity theory of elliptic partial differential equations as well as calculus of variations to offer two further elegant approaches. Related to this, in one of the best mathematical writings of 2012 Hayes [1171] nicely describes the behaviour of an n-cube of side length 2 and an inscribed n-ball of radius 1 when the dimension increases (see also the corresponding review MR2976645 in the Mathematical Reviews). The maximality of dimension 5 regarding the volume of the ball, and of dimension 7 for its surface area, are elegantly described, and since the ball remains inscribed, already for dimension $n = 100$, say, the part of the cube being inside the ball is close to zero, although this ball further on touches every facet of the cube and no larger ball fits inside the cube. The ball essentially "disappears"! This is also nicely discussed in Section 13.1 of Matoušek's monograph [1732]. Rabiei [2008] discusses related *mass concentration phenomena in balls*, and he gives also exact formulas for the mass content of *ball caps* and upper bounds for related volume content ratios; see again also Chapter 13 in [1732]. A further related paper, recommendable for undergraduate students and giving also a recursion formula for the ball volume, is [974] of Gipple. And a further very nice paper is [2263] due to Smith and Vamanamurthy, showing different derivations for the volume $V_n(r)$ of the n-dimensional ball of radius $r > 0$, for n even and odd, and including also related convergence statements. (It should be noted that there are very many such derivations in the literature.)

RELATED INEQUALITIES

We continue with some *inequalities* bounding the volume of the Euclidean unit ball. Improving former results (also of Klain and Rota), Alzer (see [94] and [95]) uses properties of *the gamma function and its logarithmic derivative* for proving sharp upper and lower bounds on this volume, or also quotients of it using the values for different dimensions. In the second paper, also a related interesting double inequality is derived. Similarly, Chen and Lin [606] as well as Yin [2506] present sharp upper and lower bounds on ratios of suitably combined ball volumes of different dimensions. The classical notion of *ball number*, suitably combining the volume and the surface area of a ball of radius $r > 0$, is extended by Richter and Schicker [2029] to *normed spaces* having the Platonic solids (including also the non-symmetric tetrahedron) as unit balls. The introductory part gives an overview regarding further results in this direction. In the expository paper [107] on analytical inequalities involving special functions from classical analysis and geometric function theory, Anderson et al. give the gamma function, generalizations of it, and the volume of the unit ball a proper place.

PROPERTIES OF RELATED FUNCTIONS AND RELATED HYPERSURFACES

Guo and Qi [1107] generalize some *monotonicity and concavity properties* of several functions strongly related to the unit-ball volume, and properties of related sequences are also derived; see also

the related papers [1106] of Guo and Qi as well as [2006] of Qi, Wei, and Guo. And Mortici [1852], besides establishing new bounds for ratios involving the volume of the unit ball, shows also related monotonicity properties. At the end of the paper, the *surface area of the unit ball* is discussed, too. And also the following results have to do with volume and area. Dodd and Coll [758] construct hypersurfaces in Euclidean n-space which satisfy higher dimensional analogues of the *equal area zones property* of the sphere. This says that the surface area of a zone between two parallel hyperplanes depends only on the distance between the hyperplanes. These new hypersurfaces are hypersurfaces of revolution, and they are called *equizonal n-ovaloids*. Their description in [758] is very detailed, and interesting further problems are posed there. Also the volume of them is given, and the authors prove that the ratio of this volume for radius 1 to that of the unit ball approaches π as $n \to \infty$. It is also shown that equizonal n-ovaloids are of class C^∞ for even dimension, and only of class $C^{(n-1)/2}$ if n is odd. Coll et al. [642] improve these results by showing that equizonal n-ovaloids are analytic for even dimension, and for odd n everywhere analytic except at their two poles (where they are of class C^{n-1}). The used methods seem to be of further interest, for determining degrees of smoothness of rotational surfaces determined by piecewise-defined profile curves, and in analogous situations in higher dimensions.

Our next reference is elementary and treats folklore, but since it nicely combines volume, area, and Cavalieri's principle, we cite it here. Namely, Aberra and Agrawal [8] discuss surfaces of the type of *Torricelli's trumpet* (or Gabriel's horn), which come to be by rotation of the function $f(x) = x^{-p}$, $x \geq 1$, with $1/(n-1) < p \leq 1/(n-2)$, around the x-axis in Euclidean n-space. Such surfaces became famous since they create the "painter's paradox", due to the fact that they have infinite surface area but finite volume. (Another representation, showing nicely the usage of Cavalieri's principle, is given in Chapter 8 of Havil's book [1170].) Besides this, Aberra and Agrawal [8] give the well-known formulas for the volume and surface area of the n-ball, and in every case a first tentative formula is derived with the help of a *generalized Cavalieri principle*. Having done this, the authors calculate this formula applying cylindrical coordinates.

At the end of Section 4.3, the *volume formula for general convex bodies* in terms of their *gauge functions and radial functions* is given. This, together with the computation of the volume of a convex body in terms of its *support function*, is discussed at the end of Section 1.7 in Schneider's monograph [2167].

4.7.3 Volumes (of sections) of simplices

Section 4.4 clearly shows the importance of *volumes of simplices and pyramids* for the computation of the *volume of general convex polytopes*. Therefore we collect here some related results. Most references treat simplices; we could not find many referring to pyramids (except for various historical articles). Other results on volumes of simplices, not presented here, treat geometric inequalities (see, e.g., the articles of Florian and Lutwak in the handbook [1077], and the exposition [2160] on the homepage of Schneider), and there are further fields. And also Hilbert's third problem and equi-decomposability of simplices or polytopes should be mentioned here (cf., for example, Boltyanski's book [423]). For much further material on the geometry of simplices, we also refer to part 2.9.1.1 of this book here.

POLYTOPE VOLUMES VIA SIMPLEX VOLUMES

Hohenbalken [1214] describes subdivisions of an arbitrary convex polytope P into a minimal number of disjoint simplices having one vertex of P in common. This yields a natural approach to P, its volume, and its centroid. Roos [2058] studies n-simplices, via respective inequalities, as intersections of half-spaces. He gives an *explicit formula for their volumes* which is in contrast to usual recursive formulas or algorithms to get the volumes of convex polytopes, like those given in [634] and

[1509]. Namely, Cohen and Hickey [634] use, for getting the volume of an n-polytope P defined by linear systems of inequalities, the summation of volumes of simplices whose union forms P and which are determined by a fixed vertex and n vertices from a corresponding triangulation of the boundary of P. Using "pyramids over faces" instead of simplices, Lasserre [1509] gives a better algorithm for the volume of P which can be used even if P has redundant constraints, or empty interior. For further such results see subsection 4.7.4.

In view of getting the volume of a convex n-polytope P in a faster way than using simplicial subdivisions, Speevak [2283] introduces special types of "pyramids". These are convex n-polytopes with m vertices, p of which are contained in 2-flat L, whereas the remaining $m - p$ vertices lie outside of L and are affinely independent. An algorithm turns the vertex polygon in L into a triangle of equal area, and P becomes an n-simplex of equal volume. So the volume calculation for these polytopes is *drastically simplified*, and one might hope that similar ideas can be used for general polytopes, too. Samarin [2097] shows that any *bounded polyhedron* in 3-space can be isometrically deformed (meaning that the geodesic distance of boundary points remains fixed) such that the enclosed volume increases. The proof, of course using also non-convex polyhedra, is based on such deformations of pyramids (without change of their bases). Here we refer also to the related paper [60] of Allgower and Schmidt (see subsection 4.7.3).

Using linear algebra and calculus, Klebaner et al. [1412] find formulas for the volumes of $(n - k)$-faces, $0 \leq k < n$, embedded in an n-simplex S given by its $n + 1$ bounding hyperplanes. For example, for $n = 3$, the 3-volume of the tetrahedron S is expressed by formulae containing also the areas of its faces and the edge-lengths. And Volenec et al. [2409] derive inequalities that contain the volume and products of k-volumes of k-faces of an n-simplex. As applications, two inequalities are generalized which maximize the volume of one or two simplices in terms of their edge-lengths.

SPECIAL TYPES OF SIMPLICES AND POLYTOPES

Slepian [2260] solves, for a point x in Euclidean n-space and positive numbers $d_i > 0$ [$e_i > 0$] for $i = 1, ..., n + 1$, two problems: Find the *maximum [minimum] volume of an n-simplex* determined by the vertices $p_i, i = 1, ..., n + 1$, such that the distance from x to p_i is d_i [whose bounding hyperplanes have distance e_i to x]. Both problems are solved, and if all distances d_i are equal, the *regular simplex* yields the maximum.

Buchholz [516] calls a tetrahedron a *perfect pyramid* if its edge-lengths, face areas, and the volume are all integers. First studying only edge-lengths and volume, he shows that there are several types (in the sense of equality of certain edges) that give rise to infinitely many solutions. Adding the criterion of integer face areas, one might expect that only a small part of them remains. It is shown that there are still infinitely many perfect pyramids, and it is also verified that there are infinitely many dimensions in which all regular simplices with rational edge-lengths have rational volume. Inspired also by the *Diophantine problem* of the existence of triangles with rational edges and rational area, Chisholm and MacDougall [618] continue these investigations. They restrict to tetrahedra whose six edge-lengths form an arithmetic progression and confirm that no such tetrahedron is perfect. Then they search such tetrahedra with either *rational volume* or at least one face with *rational area*. The used formula for the volume in terms of the edge-lengths, which is homogeneous of degree 6, brings the volume problem down to questions like this: Do specific curves of genus 2 or 1 have rational points? The first case turns out to be locally insolvable at some prime, and in the second case these curves are elliptic and can be studied with computer software. Somehow analogously, the face-area problem comes down to the existence of rational points on curves of genus 1.

Emert and Nelson [811] show that for a *regular convex n-polytope* there is a nice relation between its volume and surface area, namely that the *surface area is a derivative of the volume with respect to the inner radius*. The latter is the minimal distance from the center of the polytope to its boundary.

In addition, conditions are discussed that have to be satisfied by arbitrary polytopes to get the same relation.

Weakly related to these special types of convex polyhedra is the so-called *Kirkman icosahedron*, having many surprising properties which are investigated by Fetter [866]. This centrally symmetric, convex, and three-dimensional polyhedron P is self-dual with 20 faces, 38 edges, and 20 vertices, has a sphere tangent to all its edges, and can be constructed with integer vertex coordinates, integer edge-lengths, and integer volumes of the pyramids obtained as convex hulls of the center of P, and all its faces. Some further properties are derived, too.

To draw the reader's attention to n-dimensional geometry, Hatada [1168] poses eight interesting problems treating n-dimensional simplices; they refer also to characterizations of regular simplices. Investigating simplices with special properties of their cevians, the paper [1143] of Hajja et al. is related; it is also written for *popularizing geometry in higher dimensions via simplices*.

Jarry-Bolduc et al. [1297] study *uniform simplices* (whose vertices have the same distances to the centroid c and satisfy a certain pairwise angle condition with respect to c). They give a construction of such simplices and show that they have the largest normalized volume of any simplex. This property makes such simplices useful for certain degenerate situations. Based on this construction of uniform simplices, the authors construct also a uniform minimal positive basis.

VOLUME FORMULAS AND SIMPLEX SLICES

Filliman [875] deals with the following two problems: For P a convex n-polytope and L a k-dimensional subspace, give a formula for the volume $V(P \cap L)$ in terms of Plücker coordinates of L. And second, find conditions for those sections for which $V(P \cap L)$ is stationary. The first problem, yielding a formula for the *volume of the polar dual* of a polytope in terms of the extreme points of the original polytope, is generally (and therefore also for simplices) solved, and the second at least for simplices. The motivations come from *isoperimetric problems*. Among other results, Webb [2434] shows (in two ways) that an $(n-1)$-slice of a regular n-simplex S, which passes through the centroid of S, has largest $(n-1)$-volume exactly if it contains $n-1$ vertices of S. There are relations to an interpolation problem.

Regarding the unsolved question of *minimal central hyperplane sections of simplices*, Filliman [875] says that his methods might help to show that the section parallel to a facet is minimal. Brzezinski [513] verifies a lower bound differing from the conjectured minimal volume by a factor of approximately 1.27. (Also k-dimensional sections of n-balls are studied in [513].) Dirksen [756] studies *k-dimensional central sections of simplices*, and also non-central ones. A general formula for the intersection of a regular n-simplex and a k-dimensional subspace is given, and Webb's formula (see [2434]) is generalized. Lassere [1511] investigates simplex slices and simplex sections (with the latter meaning their intersections with half-spaces) discussing formulae for their respective volumes. Several tools and methods play a role in this paper: univariate B-splines, Laplace transform techniques, and multivariate integrals. Dirksen [756] confirms that also for hyperplanes of fixed small distances from the centroid one containing $n-1$ vertices still gives the maximum. With respect to *minimal sections*, he derives a result that supports the corresponding conjecture, and he verifies the conjecture for up to 4 dimensions. With the help of the *Brascamp–Lieb inequality*, he also gives a bound on volumes generated by k-dimensional sections.

Continuing partially Dirksen's work [756] (regarding distances of intersecting hyperplanes from the centroid), König studies in [1432] sections of a regular n-simplex S by hyperplanes H having distance t from the centroid c of S. For example, he proves that, for t larger than the distance between c and the midpoints of the edges of S, the maximal volume of $S \cap H$ is achieved when H is orthogonal to the affine hull of c and a vertex of S. In addition it is shown that, in ranges of smaller values for t, the volume of $S \cap H$, when H is orthogonal to the affine hull of c and some vertex, is sometimes

locally maximal and sometimes locally minimal. Further related results are given in his paper [1432], containing also many interesting results on hyperplane sections of cubes and cross-polytopes.

In terms of edge-lengths, Cho [621] gives a volume formula for a 3-simplex which can be represented as a facet of a rectangular 4-simplex. *Heron's formula* (regarding facets) is implicitly used. The authors of [1169] use special types of matrices to obtain (among various further applications) a generalization of *Heron's formula* for the volume of a general simplex, and conditions for its existence. Cho [623] simplifies the Cayley–Menger approach to the simplex volume via square matrices based on edge-lengths by reducing the number of rows. And in [622] he presents a *generalized Heron formula* for the volume of an n-simplex by using multidimensional scaling.

For an n-simplex S and a hyperplane H, the set $H \cap S$ is called a *medial section* of S, if the distance between H and any vertex of S is the same constant. Extending a result of Yetter [2502] on tetrahedra, Talata [2338] presents a simple formula for the $(n-1)$-volume of any medial section of S in terms of the edge-lengths of S and obtains further related results. Yetter [2503] subdivides the set of medial sections into (k,l)-classes, meaning that k vertices of S lie in one and $l = n+1-k$ vertices in the other half-space regarding H. He proves that for any n-simplex the ratio of the sum of the squared volumes of the medial (k,l)-sections to the sum of the squared volumes of the medial (κ,λ)-sections does not depend on the geometry of the simplex, but only on n, k, and κ. He derives a formula for that ratio.

Gaifullin [917] is motivated by the classical *Cayley–Menger approach* to the volume $V(S)$ of an n-simplex S presenting $V(S)$ in terms of the edge-lengths. He searches analogues between $V(S)$ and k-volumes of k-faces of S, laying emphasis on the case $k = 2$. He succeeds in showing that for $n \geq 4$ the square of $V(S)$ satisfies a polynomial relation whose coefficients depend on the squares of the areas of the 2-faces of S. Further on, the minimal degree of such a relation is computed, and it is shown that there is a monic polynomial relation whose coefficients depend on the areas of its 2-faces iff the dimension $n \geq 6$ is even.

Akopyan [31] studies the *circumcenter of mass* of simplicial polytopes, defined as an affine combination of the circumcenters of the simplices in a triangulation of the polytope and weighted by their volumes. He gives new short proofs of properties of this center, using integral formulas. For example, he uses the integral of the power of the points of a simplex S regarding the circumsphere of S. It is related to the *volume* of S and the distance between the circumcenter and the centroid.

In [869] Fiedler proves a basic theorem on *volumes of two simplices*, one contained in another and even for different dimensions. Some interesting geometric problems are posed.

Minemyer [1802] shows a new way to compute the distance between two points in an n-simplex S in terms of barycentric coordinates, where only the edge-lengths of S are used. Among various applications of this method, there is a nice formula for computing the volume of S.

ASSOCIATED BODIES

The *illumination body* of a convex body K is defined via the difference (bounded from above by a constant σ) between the volumes of the convex hull of K and some point, and K itself. If K is a polytope, then this difference is a polynomial in σ of degree n. Werner [2454] determines the coefficients of this polynomial for K being an n-simplex.

Cheung and Xiong [616] derive a formula relating the *chord power integrals* of an n-simplex S and the *dual quermassintegrals of its difference body*. This allows to use the volume of a simplex S for expressing the *volume of its difference body and of the polar projection body* of S.

INEQUALITIES AND EXTREMUM PROBLEMS

Hug and Schneider [1256] present a stability result estimating the Banach–Mazur distance of a convex body K from a simplex if the *volume ratio* of K (between the smallest volume of circumscribed

ellipsoids and the largest volume of inscribed ellipsoids, raised to a power of $1/n$) is close to its maximum, which is reached exactly by simplices.

Of course, Mahler's conjecture on the lower bound of the *volume product* of a convex body and its polar is one of the famous open problems in convexity (the upper bound is regulated by the famous Blaschke–Santaló inequality); we refer to Section 10.7 in Schneider's book [2167] for a comprehensive representation. It is conjectured that in the general case precisely the simplices hold this lower bound sharply (some progress in the subcase of centrally symmetric convex bodies was done; see, e.g., the work [1379] of Kim and Reisner). And K. J. Böröczky et al. [409] give an excellent survey on the state of the art (in 2013) with that problem.

Horváth [1234] deduces tight inequalities holding for the volume of the convex hull of two intersecting and congruent n-simplices, reproving also a nice result of Rogers and Shephard. He gives an explicit formula for the case that one of the simplices is obtained from the other by reflection in a hyperplane. Also regular simplices are discussed in this setting, and an upper bound for arbitrary simplices is obtained, too.

Schwartz [2210] shows two non-trivial results: first he proves that for a tetrahedron $T = X_1X_2X_3X_4$ with edge-lengths $\|X_iX_j\| = d_{ij}$ for all distinct i and j from $\{1, 2, 3, 4\}$, there is a tetrahedron $T' = Y_1Y_2Y_3Y_4$ with $\|Y_iY_j\| = d_{ij} + 1$ for all distinct i and j from $\{1, 2, 3, 4\}$. Second, the *volume ratio* of T and T' satisfies $V(T')/V(T) \geq (1 + 6/s)^3$, where s denotes the total edge-length of T. The proof of the inequality here follows from the positivity of a certain quantity involving the Cayley–Menger determinant and one of its directional derivatives.

Saroglou [2114] continues the work [383] of K. Böröczky, Bárány, Makai, Jr., and Pach. Namely, the simplex was conjectured to be the extremal convex body for the following two problems: What is the minimal possible value of the volume ratio $\max_K V(K)/V(C)$ for K ranging over all symmetric convex bodies contained in C, and what is the maximal possible volume of the Blaschke body of a convex body of volume 1? K. Böröczky et al. [383] show that if the simplex solves the first question, then also the second, and Saroglou [2114] verifies that both questions yield exactly the same solution.

Tordesillas and How [2378] consider all possible polynomials of degree n with real-valued coefficients, where $t \in [a, b]$ with $a < b$. Without loss of generality, they consider $a = 0$ and $b = 1$ and study a basis of a vector space of all possible nth-degree polynomials on t. They consider a simplex which is the convex hull of the columns of the matrix whose (i, j)th element is the coefficient of the jth degree on the polynomial p_i. They look for the simplex S of a basis of the vector space of nth-degree polynomials such that S has the *smallest volume* containing a given nth-degree polynomial curve. Further on, it is verified that this basis also obtains the nth-degree polynomial curve with largest convex hull enclosed in any given n-simplex. Numerically they obtain global optimal solutions for $n = 1, 2, 3$, and feasible solutions for all n.

VOLUME APPROXIMATION WITH PARTICIPATING SIMPLICES

In the following we give a survey of results treating *volume approximation* of convex bodies by simplices or vice versa. For the participating simplices the studied versions of such approximations are "from inside", "from outside", or considered "in between" (using homothetical copies), with maximum/minimum variants.

We start with *maximum volumes* and *simplices contained in convex polytopes*. Gritzmann, Klee, and Larman [1029] deal with the famous problem of studying *largest k-simplices contained in an n-dimensional convex body K* or *in an n-polytope P*; "largest" is meant regarding k-volume V (or k-measure). The authors call such k-simplices *stable* (resp., *rigid*) if the relations $V(S) \geq V(S')$ (resp., $V(S) > V(S')$) hold for any k-simplex S' that is derived from S by a movement of a single vertex to another position in K or P. The authors deduce many results on *existence, uniqueness, and computation of largest simplices*, and of stable as well as rigid simplices. They lay special emphasis on the *computational complexity* of such problems when the participating polytopes are given by their

vertex sets or as intersections of finitely many half-spaces. The paper concludes with a large selection of open problems and an interesting list of related references. Hudelson et al. [1252] reduce, coming from this, K or P to n-cubes, and the subcase of *largest n-simplices in n-cubes* is particularly studied, due to its strong relation to the famous *existence problem of Hadamard matrices*. The authors give partial results to it and show its relationship to applications like D-optimal weighting designs for chemical balances and spring balances. They also pose interesting conjectures, present several further problems on *simplices in cubes*, and give an interesting list of references. Largest simplices in cubes are also discussed in the very nice survey [2570] of Zong. Akopyan and Glazyrin [33] use their methods developed for problems from combinatorial geometry (e.g., around Radon's theorem) to extend results of Lassak [1499] on parallelotopes in simplices, or also of the following type: To each n-simplex of maximum volume contained in $[0,1]^n$ belongs exactly one unit segment parallel to the ith axis, for $i \in \{1,...,n\}$. By Nikolov [1896] as well as Di Summa et al. [748] the problem of finding *largest volume simplices in convex hulls of finitely many points* is computationally attacked. For example, Nikolov [1896] gives, for k-simplices, a *deterministic approximation algorithm*. This approximation is obtained by rounding solutions to a generalization of the D-optimal design problem; and this is equivalent to the dual of some smallest enclosing ellipsoid problem.

Regarding volumes of *simplices contained in convex bodies* we cite the following interesting references. McKinney [1736] considers convex bodies K of volume $V(K)$, centered at x and having suitably inscribed simplices. He calls a simplex S *maximal in* K if one of its vertices is x and the other vertices are boundary points of K. He shows that $V(S) \geq (n!\pi_n)V(K)$ with equality iff K is an ellipsoid, where π_n stands for the volume of the unit ball. From this he concludes that for any centrally symmetric convex body K whose polar reciprocal K^* has volume π_n, there is an affinity such that the respective image of K is contained in a cube of edge-length 2. Tsintsifatis [2383] elegantly shows that a centrally symmetric convex body which contains a simplex S has at least the n-fold volume of S. This nice result is sharp for dimensions two and three. Lassak [1501] obtains the following nice result: For an arbitrary n-dimensional convex body K and each simplex S contained in K with maximal volume, the homothetical copy of S with ratio $n+2$ and centered at the barycenter of S will contain the body K. As one can show with the help of certain double cones, this ratio cannot be improved. Furthermore, Lassak [1500] proves that for a simplex of maximum possible volume contained in the unit ball of an n-dimensional real Banach space the following holds: the sum of the reciprocal values of the widths (in the sense of the norm) in directions of the normals of the facets equals n. For $n = 2$ it is additionally shown that all three sides of any triangle of maximal area, contained in the unit disk of a normed plane, are of lengths (again in the sense of the norm) at least $\sqrt{2}$. With regular octagons as unit disks one can show that this value cannot be increased.

Questions around maximum volume of *polytopes in simplices* are treated in Lassak's paper [1499], for the case of affine cubes (parallelotopes). For a simplex S we write $x_1,...,x_n$ for the vectors determining the edges of S starting from one of its vertices called 0. It is clear that an n-parallelotope with vertex 0 and edges along the vectors x_i is contained in S and has the $n!/n^n$-fold volume of S. Lassak proves that there is no n-parallelotope with larger volume contained in S. He gives a description of all n-parallelotopes having largest volume and being contained in S; an interesting discussion of motivations, related results, and open problems (e.g., on maximally contained k-parallelotopes) is added. In this direction, Nevskii [1890] shows the following: If S is an n-simplex containing the n-cube $[0,1]^n$, then, for some $i = 1,...,n$, S contains an interval of length n which is parallel to the ith coordinate of the axis. The author underlines that this has applications for the *linear interpolation of continuous functions* of n variables. This is continued in [1889], with computing lengths and endpoints of the longest segments in n-simplices for prescribed directions (see also the references there for further results). Regarding n-cubes contained in n-simplices we refer to Nevskii's paper [1887]: It is shown there that the minimal $\sigma > 0$, guaranteeing that a homothetical copy of an n-simplex S with ratio σ contains $[0,1]^n$, equals the sum of the n values $1/d_i(S)$, where the $d_i(S)$ stand for the longest segments in S parallel to the ith coordinate axes.

Next, we deal with *convex bodies optimally contained in simplices*. Generalizing the results from [1887], Nevskii [1888] derives, suitably using barycentric coordinates regarding an n-simplex S, the minimal ratio of homothety such that a corresponding homothetical copy of S contains a given convex body. With an elegant proof, Kanazawa shows in [1346] that for each n-dimensional convex body K there exists an n-simplex containing K whose volume is at most n^{n-1} times that of K. Galicer, Merzbacher, and Pinasco [919] give an asymptotic estimate for the ratio of the volumes of an n-dimensional convex body K and the simplices S containing K. More precisely, they confirm the existence of a simplex S containing K and sharing the barycenter with K such that $(V(S)/V(K))^{1/n} \leq cn^{1/2}$, where c is an absolute constant and the exponent $1/2$ cannot be improved. Somehow dually, they show that for a convex body K with 0 as barycenter there is a simplex $S \subset K$ with the same barycenter such that $(V(S)/V(K))^{1/n} \geq dn^{-1/2}$, where d is a constant. Mahler's volume product plays a role in the proof.

And now we cite some papers related to types of *convex bodies whose boundaries are enclosed between two convex sets*, in particular also simplices between two homothets of a convex body. Results in this direction are related to the *Banach–Mazur distance*, to *measures of symmetry*, and also to interesting *approximation problems* (see [1080], [2167], and [2381]). We recall that the *Banach–Mazur distance* $d(K, C)$ of K and C is defined as the infimum $r > 0$ such that $K^* \subset C^* \subset rK^*$, where K^* and C^* are positive homothets of K and C, respectively. And with $d_G(K, C)$ we denote a similar distance introduced by Grünbaum and considering positive as well as negative homothetical copies of K and C for the infimum. Based on John's well-known result (see Theorem 10.12.2 in [2167]) it is clear that if B is a ball, then $d(K, B) \leq n$ for each convex body K, with equality if K is a simplex. Leichtweiss [1529] and, independently, Palmon [1933] show that this equality case holds only for simplices, and Jiménez and Naszodi [1314] give a generalization of this result (see also below). Gordon, Litvak, Meyer, and Pajor confirm in [1006] (as consequences of other results) an old conjecture of Grünbaum (treating the variation d_G of d): any two n-dimensional convex bodies K and C have positions K^* and C^* such that $K^* \subset C^* \subset -nK^*$. This means that the distance between a convex body C and a symmetric convex body K is bounded by n, which is optimal; it is proved that this distance between the simplex and any centrally symmetric convex body equals n. Continuing [1006] and [1314], also Kobos [1426] investigates the distance d_G. Namely, Jiménez and Naszódi [1314] conjecture that if $d_G(K, C) = n$ for two (not necessarily centrally symmetric) convex bodies, then either K or C is a simplex, and this is confirmed there for C being either smooth or strictly convex. Kobos derives a partial affirmative answer for the planar case. For more related considerations, we refer to Section 10.12 in Schneider's monograph [2167].

FURTHER RESULTS

There are very many further results on simplices in combination with volume. But instead of starting new themes, then certainly presenting them in an incomplete way, we are content with one nice, surprising result, namely the following.

The Notes 2.4 in Gardner's book [930] and the end of Chapter 10 as well as its note 11 in Schneider's book [2167] are dedicated to the *shaking procedure* (shakedown or shake) which goes back to Blaschke (the two books mentioned here also contain some history). It works like *Steiner's symmetrization* with only one difference: the segments which are moved in their affine hull g until their midpoints reach the defining hyperplane H (at which we symmetrize), are moved in g only until their "first point" meets H. The union of all these moved segments is then the shaked set, and the volume clearly remains fixed under this procedure. An early contribution is due to Schöpf [2188], and one of the nicest results around this transform (also applicable to show the Brunn–Minkowski theorem for compacta) is the following, due to Campi, Colesanti, and Gronchi [565]: For every compact set K in n-space, there are $n + 1$ hyperplanes such that K can be transformed into an n-simplex by a sequence of iterated shakings at these $n + 1$ hyperplanes.

4.7.4 Volumes and areas of polytopes

In this subsection we start with results related to *volumes of convex polytopes*, but will not discuss lattice point problems related to volume. For basic literature on these topics, we refer to our sections on *polytopes* and on *discrete geometry* in the last chapter. And also in the former subsection on simplex volumes, results are cited that are related to our discussion here. To make a repetition as small as possible, only a few sentences refer to the latter.

TRIANGULATIONS OF POLYTOPES, ETC.

Namely, to find a start we begin with the paper [634] of Cohen and Hickey. They propose two algorithms to get the volume $V(P)$ of a convex n-polytope P. The first method is exact, and the second converges to the exact solution costing additional computer time. Continuing [634], Lasserre [1509] presents an algorithm which is more efficient. The methods in both papers are based on determinants and the matrix representation $\{x : Ax \leq b\}$. It is shown how one can derive an analytical expression for $V(P)$ as a function of b. We also mention once more Speevak's paper [2283] with a simplification of the simplicial subdivision of (special) polytopes, to get their volume faster.

Related to the partition of polytopes into simplices, Pokrovskii [1981] studies special k-multiple coverings of n-polytopes, in particular of n-cubes or n-simplices by n-simplices. He shows that for cubes such a covering has necessarily a multiplicity which is a multiple of $n!k$ and, for related coverings of simplices by simplices, a multiplicity which is a multiple of k.

Allgower and Schmidt [60] present two methods for computing the volume of any closed bounded n-polytope P having an orientable boundary triangulated into a set of simplices of dimension $n-1$. The first method yields the volume $V(P)$ as sum of volumes of suitable oriented n-simplices, and the second presents (as a generalization of the trapezoidal rule for getting the area of polygons) $V(P)$ as suitable sum of the signed volumes of n-prisms. The reader might notice the weakly related paper [2097] of Samarin (see subsection 4.7.3).

As partitioning method (like branch and bound) in global optimization, *bisection of simplices* is a common method since more than half a century. Horst [1232] slightly extends the definition of partitions showing then in which sense the bisection method is not optimal. He emphasizes that in this framework partitions of polytopes into simplices have important applications, e.g., in computational convexity (see below here, under the computational aspects), and particularly for the *computation of polytope volumes via simplex partitions* (where the polytopes are given by their vertex sets). He proposes various ways of partitions and gives also corresponding complexity results. Finally he discusses also an approach to the volume computation of polytopes given by a system of affine inequalities.

Schneider [2172] continues investigations of Esterov [820] (discussed below) providing an elementary approach to the volume of convex polytopes as interesting special case of Esterov's result. He finds a convenient way to decompose the facets of a studied polytope P into (positively or negatively oriented) special simplices, called *orthoschemes*, and finds therefore a natural approach to the volume of P as *sum (depending on orientations and signs) of volumes of orthoschemes*. Again using such simplicial decompositions, he finally uses his approach to get an analogous, more general formula for *mixed volumes*.

The second part of the handbook article [1183] by Henk et al. refers to *convex polytopes*. The authors treat the following subthemes: *volume and surface area, mixed volumes* (here also Schneider's summation formula and McMullen's volume formula for a special zonotope), *quermassintegrals, and intrinsic volumes*. A final subsection "sources and related material" collects related books and expositions with a short description of some of them, and cross-connections to other articles in this handbook are given, too. Each section starts with necessary definitions, and presents basic theorems

(without proofs) and fundamental related formulas. Clear examples and open problems can be found, and the authors demonstrate a successful, understandable style of presentation.

Abrosimov and Mednykh [9] survey results on *volume calculations for polyhedra in Euclidean, spherical and hyperbolic spaces*; see also [10]. They present a proof for the *volume formula of orthoschemes in hyperbolic and spherical 3-space* honoring also the work of Bolyai, Lobachevsky, Schläfli, and Coxeter, and they discuss results for *ideal tetrahedra* (with ideal vertices). For general tetrahedra, the recent history is presented, with relations to Hilbert's third problem and further related questions. Also the 2006 re-discovery of Sforza's forgotten work from 1907 (namely his formulas for the *volumes of hyperbolic and spherical tetrahedra*) is discussed, and the authors present a new proof (since then for the first time in print). This is related to Seidel's conjecture (1986), namely that the volume of an *ideal hyperbolic tetrahedron* can be expressed as a function of the determinant and the permanent of its Gram matrix. The further development around this problem is presented, too. In addition, *convex hyperbolic quadrilaterals inscribed in a circle, horocycle, or one branch of equidistant curves* are studied, like also several versions of the *Brahmagupta formula for the area of such quadrilaterals*. Analogously, a formula for the area of *hyperbolic trapezoids* is given.

COMPUTATIONAL ASPECTS

Bárány and Füredi [222] study the *computational complexity* of finding upper and lower bounds $V_u(K)$ and $V_l(K)$ for the volume of an n-dimensional convex body K. They prove that, for any polynomial-time algorithm to get them, one has $V_u(K)/V_l(K) \geq (cn/\log n)^n$ for some instances K. The authors use *oracles* to represent the studied convex bodies K; these oracles answer affirmatively when a point belongs to K, and provide a separating hyperplane if not. (For such oracles in general we refer to Chapter 1 of the famous book [1047] of Grötschel, Lovász, and Schrjiver.) Similar results on the width of K are obtained, too. The subsequent paper [223] refers (like also [222]) to the *volume approximation of the unit ball by polytopes*. Using also random polytopes, the authors investigate how good an n-polytope with m vertices can approximate the unit n-ball if m is a function of n (e.g., a polynomial in n), and n tends to infinity. Related are also results in the paper [577] of Carl and Pajor and an elegant construction of Kochol [1431], using a suitably scaled integer lattice.

Dyer and Frieze [786] study the *computation of the volume* $V(P)$ of convex n-polytopes given as convex hulls of their vertices or via linear inequalities (i.e., as a list of facets). They show that for both representations this problem is at least as hard as *computing the permanent of a matrix*. On the other hand, in the first case one can compute $V(P)$ in polynomial time by using a #P-complete oracle. Dyer and Frieze state their results relative to #P. Khachiyan [1372] confirms #P-hardness of the volume problem for polytopes with given list of vertices. Further on, he shows that finding the volume of the intersection of a cube and two half-spaces is #P-hard as well. Lawrence [1520] develops a novel algorithm for *computing the volume of a simple polytope from its simplex algorithm tableau*. This approach is not polynomially bounded as one would expect. Namely, it requires the enumeration of the vertices of the polytope and the computation of the sum of numbers computed at each vertex. Khachiyan summarizes the results mentioned above, as well as more related material, in the survey [1373] on *volume computation of convex n-polytopes*. He starts with volume formulae regarding the intersections of polytopes and half-spaces, shows then that the computation of volumes of specified rational polytopes is #P-hard, and discusses also the approximation of polytope volumes to a given relative accuracy. Extending the problems from polytopes to convex bodies, he reflects also the work [787] of Dyer, Frieze, and Kannan. They present the first polynomial-time randomized algorithm which can *approximate the volume of convex bodies with arbitrary precision*. Many improvements and innovations followed then; see, e.g., the important contribution by Kannan, Lovász, and Simonovits [1347]. For example, the authors of [1347] successfully introduce a new and helpful random walk called "speedy", and running-time improvements are given. The survey [2253] of Simonovits, also treating n-dimensional convex bodies regarding volume, summarizes these developments, emphasizing

the non-technical details and clearly showing motivations for certain steps and methods. Subjects and tools like (approximate) sandwiching, types of used oracles, isotropic positions, high-dimensional phenomena of convex bodies and their Löwner–John ellipsoids, random walks in convex bodies, etc. are comprehensively discussed. Here we mention also an interesting work of Betke and Henk [307]: using a so-called weak separation oracle, they develop a deterministic polynomial-time algorithm yielding an *approximation of the volume of a convex body* with an approximation factor $n!(1+\varepsilon)$, for any $\varepsilon > 0$.

The survey [1027] of Gritzmann and Klee gives a deep insight into problems around *volumes of convex bodies and convex polytopes* in the spirit of *computational convexity*, a field located in the intersection of discrete mathematics, convexity, mathematical programming, and computer science. The difference to *computational geometry* is that here the underlying structure is *convexity in real Banach spaces* of finite, but in general *not restricted dimension*. (In computational geometry the dimension is assumed to be fixed.) Regarding polytope volumes, important subjects presented in [1027] are *representations of polytopes, triangulations, lattice point enumeration, deterministic methods of volume computation, related intractability results, approximations, oracle methods, projections and sections, unions and intersections, and various applications* (e.g., treating lattice polytopes, zonotopes and mixture management, stochastic programming, permanents, unimodular matroids, partial orders, and linear extensions). Special types of convex bodies and polytopes are investigated, and a large part of the exposition deals also with mixed volumes. Here we can also mention once more Horst's contribution [1232] cited above (under triangulations), since it is also related to computational convexity. The survey [1267] deals with problems around *triangulations of convex polytopes* and in view of optimal algorithms for enumeration problems (it is therefore also close to the book [1598]). Among other topics, *Delaunay triangulations, enumeration of triangulations of polytopes with given vertex set, regular (or coherent) triangulations, and the secondary polytopes* coming from [948] are treated.

FURTHER RELATED RESULTS ON VOLUMES OF POLYTOPES

For ρ being a measure over n-space, Gluskin [982] searches, among all parallelepipeds with prescribed altitudes and centered at the origin, the one with minimal ρ-volume. He shows that for a large class of spherically symmetric measures *orthogonal parallelepipeds* (boxes) yield the solution. The author discusses interesting applications, and his paper contributes also to the *dual* of the approximation of balls by polytopes with at most m vertices. Namely, this dual setting studies the *minimal volume of the intersection of m slabs*, also deeply investigated by Bourgain, Lindenstrauss, and Milman in [476].

The next paper we mention here refers to *mixed volumes of polytopes*, but is (via Schneider's work [2172] discussed above) also connected with volumes of polytopes. Namely, for a given family of convex n-polytopes, Esterov [820] proves that there is a suitable linear function on the space of conewise polynomial functions that sends the *product of their support functions* to the mixed volume of these polytopes. Thus, the *mixed volume of the polytopes depends only on the product of their support functions*, not on the individual support functions. He gives an explicit formula for this function and shows that it yields several interesting applications in *toric and tropical geometry*. Esterov's paper is carefully written and completed by nice examples.

Finally, we mention here the recent book [2041] of Robins, treating *Fourier transforms on polytopes*. Its Chapters 4, 7, and 8 contain material referring (also) to polytope volumes. In Chapter 4, Minkowski's theorem on convex bodies and a classical result of Siegel are derived, yielding an *identity between Fourier transforms of convex bodies and their volumes*. And in Chapters 7 and 8, the *discretization of volume* (via integer points and the Ehrhart polynomial, or related to the angle polynomial) is discussed. The book contains many nice examples and exercises; solutions and hints are given in the final chapter.

SURFACE AREA OF POLYTOPES

We now come to results related to *surface areas of convex polytopes* and start with Minkowski's existence theorem and Alexandrov's uniqueness theorem for polytopes. Any convex n-polytope is, up to translation, uniquely determined by the outer normal vectors of all its facets such that in each case their length represents the area of the corresponding facet. And on the other hand, every *vector star* not contained in a half-subspace and summing up to zero determines, in the described manner, a translation class of convex n-polytopes. Thus, the *surface area of convex polytopes* is the total length of all these vectors.

A good reference for these very often cited results is the appendix part A.4 of Gardner's monograph [930], where also algorithmical results are nicely described, namely how one can reconstruct a representative of the convex n-polytopes from its "vector star". In Schneider's monograph, Minkowski's existence theorem for polytopes is proved in subsection 8.2.1.

The following references collect results which are closely related to properties of convex polytopes and their vector stars.

Let $q(1), ..., q(n)$ denote positive numbers such that $q(i_1) + ... + q(i_{n-1}) < q(i_n)$ for each permutation $i_1, i_2, ..., 1_n$ of $1, 2, ..., n$. Larman [1497] shows that then the numbers $q(i)$ present the areas of the facets of some n-dimensional convex polytope, for each $n \geq 2$. Independently, Purdy [2005] proves the same result.

Besides other results, Böröczky et al. [383] verify the following two statements: Let P be a not necessarily convex n-polytope and write $v_i, i = 1, 2, ..., m$, for the outer normal vector to the ith facet F_i of P whose length equals the area of F_i. Due to Minkowski, there is a unique (up to translation) convex polytope P^* such that the vectors v_i correspond to the facets of P^* in an analogous way. It is shown that the volume of P is bounded from above by the volume of P^*, and that equality holds exactly when P is convex. Further on, assume that P, P' are convex n-polytopes with the property that for any direction u the sum $S(u)$ of areas of the facets of P lying in hyperplanes perpendicular to u, and the corresponding sum $S'(u)$ for P', satisfy $S(u) \geq S'(u)$. Then the ratio of the volumes of P' and of P is bounded from above by some $f(n) < 2n/(n-1)$ depending only on n.

Continuing work of A. D. Alexandrov, Zalgaller [2520] presents an algorithm for constructing a convex 3-polytope P with prescribed facet normals and *facet perimeters*. Since this is not always possible, he gives necessary and sufficient conditions for the facet normals and perimeters to guarantee the existence of P. The values of the edge-lengths of P are expressed as linear combinations of the facet perimeters. Martinez-Maure [1703] goes into the same direction and succeeds in proving that a convex 3-polytope is uniquely determined by the directions and perimeters of its facets. He derives five conditions which are necessary for the existence of a convex 3-polytope with m facets whose unit normals and perimeters are prescribed. Their sufficiency is also shown. A classical theorem of Steinitz, geometric duality, and Minkowski's related theorem (replacing perimeters by areas) are used to get the result. Extensions to higher dimensions would be the next step.

Motivated by *exploratory data analysis* and based on Minkowski's existence and uniqueness theorem, Bonetti and Vitale [442] study the representation of sets of n-dimensional data using convex n-polytopes. It turns out that for increasing sample size the sequence of the corresponding polytopes converges almost surely to a limiting convex body. This means that here a strong law of large numbers holds. By its surface area measure, this body is connected with the underlying distribution of the sample. The authors prove also a central limit theorem for the *surface area measure of polytopes*.

A bounded measurable set Ω of measure 1 is called *spectral* if the Hilbert space $L_2(\Omega)$ has an orthonormal basis consisting of exponentials $e\lambda(x) = \exp(2\pi i(\lambda, x))$. Kolountzakis and Papadimitrakis [1454] prove the following: Let Ω be a not necessarily convex polytope with the property that there is a direction u such that the total area of the facets of Ω with outer normal pointing in the positive direction of u is strictly larger than the total area of the facets with outer normal pointing in the negative

direction of u. Then Ω is not spectral, which means Ω cannot tile space by translation, and a spectral polytope has to be centrally symmetric.

L. Fejes Tóth [848] studies the following problem for an arbitrary convex n-polytope P with m facets: for every facet F of P, take the ratio of the $(n-1)$-volume of F to the *maximal* $(n-1)$-volume generated by a parallel hyperplane section of P. Denoting with $S_n(P)$ the sum of these m ratios, it is shown in [848] that $\sqrt{8} \leq S_2(P) \leq 4$, and the lower bound is improved by Nikonorov and Rasskazova in [1898] to 3. Vasin and Nikonorov [2395] extend this problem suitably to n-dimensional convex bodies and give some estimates. They show that for any dimension a convex body K exists whose analogously defined number $S_n(K)$ takes the largest (respectively, smallest) possible value.

Abrosimov et al. [11] collect many historical facts and connections concerning problems of maximizing and minimizing volumes of polytopes whose facet areas are prescribed (62 references). Besides the many new results contained in [11], this paper can also be recommended as a comprehensive exposition to this subject. The main result is the following: For $m > n \geq 3, \varepsilon > 0$ and $S_m \geq S_{m-1} \geq \ldots \geq S_1 > 0$ with $S_m < S_1 + \ldots + S_{m-1}$, there exists a convex n-polytope whose m facets have areas $S_1, \ldots, S_m$ and whose volume is less than ε. The authors give three different proofs of this theorem and derive also strongly related results on spherical and hyperbolic spaces. Some necessary conditions for the existence of a convex polytope with given facet areas and partial results on sufficient conditions for the existence of tetrahedra are obtained.

Maehara and Martini [1664] study a natural transform of convex n-polytopes which, seemingly, was not considered before. It is defined as follows: The *outer normal transform* P^* *of a convex polytope* P is the convex hull of the outer normal *unit* vectors of all facets of P. It is clear that P^* is a polytope inscribed (with every vertex) to the unit sphere, and that in general P^{**} differs from P since this transform is *not the usual polarity*. The two central questions in [1664] are: Under what conditions does $P = P^{**}$ hold, and under what conditions is $P^* = P$ (a so-called *self-duality*) obtained? It is proved that if P can be inscribed to the unit sphere and is *equiradial* (i.e., the circumradii of all its facets are equal), then $P^{**} = P$. Moreover, the converse holds if P or P^* has at most $2n$ vertices. It is conjectured that $P^{**} = P$ is equivalent to the equiradiality of P. Regarding $P = P^*$, the authors get partial results, e.g.: they recursively construct, for all $m > n \geq 2$, a convex n-polytope with m vertices being equiradial and self-dual. Further related results are given, and they mainly treat the three-dimensional situation.

PLATE 5

Difference bodies

The cuboctahedron as difference body of a tetrahedron

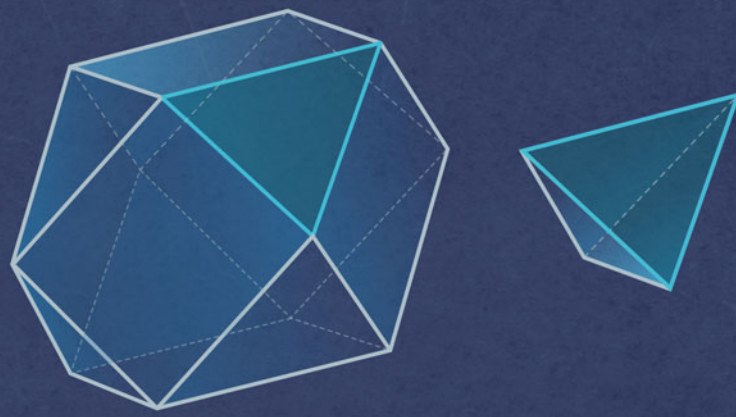
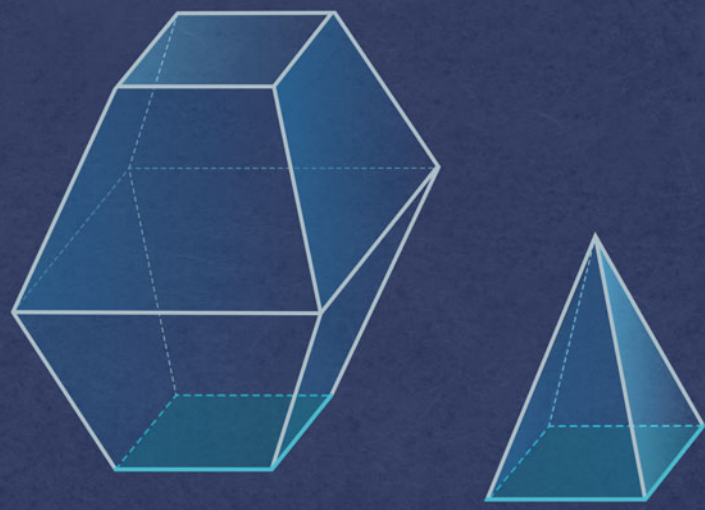


Figure 11

Figure 12

The difference body of a square pyramid



Difference bodies of convex polyhedra ► (see, e.g., 5.3 and 5.7.2)

Chapter 5
Classical inequalities

"Go down deep enough into anything,
and you will find mathematics."

— Dean Schlicter

In this chapter, we collect some classical inequalities for geometric quantities of convex bodies (such as volume and surface area, for example). We start with the classical *Brunn–Minkowski inequality*, which is the most central inequality in convex geometry. A proof that holds also for non-convex compact sets can be very easily obtained from the Prékopa–Leindler inequality. However, this proof does not characterize the equality case. So we give a second proof by induction on the dimension.

In general terms, the classical *isoperimetric inequality* bounds the surface area of a set in terms of the ball of the same volume. In case of convex bodies, we get this as a consequence of the Brunn–Minkowski inequality. For the sake of completeness, we also give proof based on Steiner symmetrizations. The latter has the disadvantage of not characterizing the equality case, but by the Brunn–Minkowski approach, we can do that.

We also introduce the *Urysohn inequality* and the *Rogers–Shephard inequality*. The first one bounds the mean width of a convex body in terms of its volume, and the second one relates the volume of a convex body with the volume of its difference body. Plate 5 shows *difference bodies* of two polyhedra; they can be seen as unions of all translates of these polyhedra containing a fixed point (e.g., the origin). Moreover, we state and prove the *Blaschke–Santaló inequality*, which gives an (optimal) upper bound for the product of the volume of a convex body centered at the origin with the volume of its polar. The lower bound of this product is the topic of the famous *Mahler conjecture* (we briefly discuss it in Section 5.4).

It is worth mentioning that some of these inequalities will be proved in different ways later (namely, after we have developed the theory of *mixed volumes*). Their presentation here is not redundant, however. We believe the approach of this chapter, which is more geometric in nature, is an adequate introduction to the topic of geometric inequalities (rather than presenting it directly as a consequence of more powerful techniques).

5.1 The Brunn–Minkowski inequality

In this section, we prove some versions of the classical Brunn–Minkowski inequality. We start proving it as an easy consequence of the Prékopa–Leindler inequality (see Theorem 1.5.1). Before this, let us fix some ideas. From the definition of the integral of simple functions (see Appendix A), we have

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_5

$$\mathrm{vol}_n(A) = \int_{\mathbb{R}^n} 1_A \, dx,$$

where $1_A \colon \mathbb{R}^n \to \mathbb{R}$ is the *characteristic function* defined as

$$1_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases}.$$

Theorem 5.1.1 (Brunn–Minkowski, first version)**.** *Let $K, L \subseteq \mathbb{R}^n$ be compact non-empty sets, and let $\lambda \in (0,1)$. Then*

$$\mathrm{vol}_n(\lambda K + (1-\lambda)L) \geq \mathrm{vol}_n(K)^{\lambda} \, \mathrm{vol}_n(L)^{1-\lambda}. \tag{5.1.1}$$

Proof. Since K and L are compact, the characteristic functions of the sets K, L and $\lambda K + (1-\lambda)L$ are integrable. Also, it is easy to see that

$$1_{\lambda K + (1-\lambda)L}(\lambda x + (1-\lambda)y) \geq 1_K(x)^{\lambda} 1_L(y)^{1-\lambda}$$

for any $x, y \in \mathbb{R}^n$. Indeed, the only case where the right-hand side is not 0 is when $x \in K$ and $y \in L$. If this happens, then $\lambda x + (1-\lambda)y \in \lambda K + (1-\lambda)L$, and hence the left-hand side is 1. Therefore, we can apply the Prékopa–Leindler inequality for the functions $1_{\lambda K + (1-\lambda)L}$, 1_K, and 1_L to obtain

$$\mathrm{vol}_n(\lambda K + (1-\lambda)L) = \int_{\mathbb{R}^n} 1_{\lambda K + (1-\lambda)L} \geq \left(\int_{\mathbb{R}^n} 1_K \right)^{\lambda} \left(\int_{\mathbb{R}^n} 1_L \right)^{1-\lambda} = \mathrm{vol}_n(K)^{\lambda} \, \mathrm{vol}_n(L)^{1-\lambda},$$

and this is the desired inequality.

□

Once $\mathcal{C}^n$ (i.e., the family of non-empty compact subsets of $\mathbb{R}^n$) is closed for convex combinations, we can extend the concavity definition for functions defined on $\mathcal{C}^n$. We say that a function $f \colon \mathcal{C}^n \to \mathbb{R}$ is *concave* if

$$f((1-t)K + tL) \geq (1-t)f(K) + t f(L)$$

for every $K, L \in \mathcal{C}^n$ and any $t \in [0,1]$.

Corollary 5.1.1 (Brunn–Minkowski, second version)**.** *The inequality*

$$\mathrm{vol}_n(K+T)^{1/n} \geq \mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(T)^{1/n}. \tag{5.1.2}$$

holds for any compact non-empty sets $K, T \subseteq \mathbb{R}^n$.

Proof. We may assume that $\mathrm{vol}_n(K)$ and $\mathrm{vol}_n(T)$ are strictly positive, since otherwise the inequality is obvious (observe that $K+T$ contains a translated copy of K and a translated copy of T). Let

$$K_1 := \mathrm{vol}_n(K)^{-1/n} \, K, \quad \text{and} \quad T_1 := \mathrm{vol}_n(T)^{-1/n} \, T,$$

and also put

$$\lambda := \frac{\mathrm{vol}_n(K)^{1/n}}{\mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(T)^{1/n}}.$$

It is clear that $\mathrm{vol}_n(K_1) = \mathrm{vol}_n(T_1) = 1$. Therefore, the first version of the Brunn–Minkowski inequality (Theorem 5.1.1) gives

$$\mathrm{vol}_n(\lambda K_1 + (1-\lambda)T_1) \geq 1.$$

On the other hand, we have the equality

$$\lambda K_1 + (1-\lambda)T_1 = \frac{K+T}{\mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(T)^{1/n}}.$$

From this, we get immediately

$$\mathrm{vol}_n(K+T) \geq \left(\mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(T)^{1/n}\right)^n,$$

and the desired inequality follows.

□

Corollary 5.1.2 (Brunn–Minkowski, third version)**.** *For any compact non-empty sets $K, T \subseteq \mathbb{R}^n$ and for every $\lambda \in (0,1)$, we have that*

$$\mathrm{vol}_n(\lambda K + (1-\lambda)T)^{1/n} \geq \lambda \mathrm{vol}_n(K)^{1/n} + (1-\lambda)\mathrm{vol}_n(T)^{1/n}. \tag{5.1.3}$$

In other words, the n-th root of the n-dimensional volume restricted to the family of compact non-empty sets of $\mathbb{R}^n$ is a concave function.

Proof. This follows immediately from the second version of the Brunn–Minkowski inequality (Corollary 5.1.1) applied to the sets λK and $(1-\lambda)T$:

$$\begin{aligned}\mathrm{vol}_n(\lambda K + (1-\lambda)T)^{1/n} &\geq \mathrm{vol}_n(\lambda K)^{1/n} + \mathrm{vol}_n((1-\lambda)T)^{1/n} = \\ &= \lambda \mathrm{vol}_n(K)^{1/n} + (1-\lambda)\mathrm{vol}_n(T)^{1/n}.\end{aligned}$$

□

Remark 5.1.1. Notice that the three versions of the Brunn–Minkowski inequality stated and proved above are equivalent, because (5.1.3) together with Lemma 1.5.1 imply (5.1.1).

All versions of the Brunn–Minkowski inequality studied so far have the advantage of not being restricted to convex bodies. However, when we consider this restriction, we obtain information about the equality case. This is what we do next, but we need to fix some concepts before.

From now on, we say that two sets $A, B \subseteq \mathbb{R}^n$ are *homothetical* if there exists a point $x_0 \in \mathbb{R}^n$ and a number $\lambda \geq 0$ such that $A = x_0 + \lambda B$. If $A = \lambda B$ for some $\lambda > 0$, then we say that A is a *dilation* of B (in this case, we clearly have that B is also a dilation of A).

Theorem 5.1.2 (Brunn–Minkowski, fourth version)**.** *Let K_0 and K_1 be convex bodies in $\mathbb{R}^n$. Then for any $\lambda \in (0,1)$, we have*

$$\mathrm{vol}_n((1-\lambda)K_0 + \lambda K_1)^{1/n} \geq (1-\lambda)\mathrm{vol}_n(K_0)^{1/n} + \lambda \mathrm{vol}_n(K_1)^{1/n}, \tag{5.1.4}$$

and equality holds for some $\lambda \in (0,1)$ if and only if K_0 and K_1 are homothetical.

Proof. Notice that we already have the inequality as a consequence of the (more general) Corollary 5.1.2. Nevertheless, we will provide another proof, which goes by induction on n. The case $n = 1$ is obvious, since K_0 and K_1 are simply compact intervals, and the n-dimensional volume is the length.

Assume that $n \geq 2$ and that the result holds for dimension $n-1$. Since the volume is invariant under translations, we can assume also that both K_0 and K_1 have their respective centroids at the origin. We first prove the result for convex bodies K_0, K_1 such that $\mathrm{vol}_n(K_0) = \mathrm{vol}_n(K_1) = 1$. For a given $\theta \in \mathbb{S}^{n-1}$, and $j = 0, 1$, we define

$$f_j(t) := \mathrm{vol}_{n-1}\Big(K_j \cap \mathbf{h}(\theta, t)\Big)$$

and

$$g_j(t) := \mathrm{vol}_n\Big(K_j \cap \mathbf{h}^-(\theta, t)\Big),$$

for $\inf_{x\in K_j}\langle x,\theta\rangle < t < \sup_{x\in K_j}\langle x,\theta\rangle$. That is, f_j is the area of the section of K_j determined by the hyperplane $\mathbf{h}(\theta, t)$, and g_j is the volume of the portion of K_j "below" this hyperplane. To simplify the notation, we recall that $h_K(\theta) = \sup_{x\in K}\langle x,\theta\rangle$ and $-h_K(-\theta) = \inf_{x\in K}\langle x,\theta\rangle$ for any convex body K. These functions are clearly continuous and positive. Cavalieri's principle given in (4.2.1) and Theorem A.6.3 yield the equality

$$g_j(t) = \int_{-h_{K_j}(-\theta)}^{t} f_j(s)\, ds,$$

implying that g_j is differentiable in the interval $(-h_{K_j}(-\theta), h_{K_j}(\theta))$, with $g_j'(t) = f_j(t)$. Consequently, g_j is strictly increasing. Let $h_j\colon (0,1) \to (-h_{K_j}(-\theta), h_{K_j}(\theta))$ be the inverse of g_j, observe that h_j is clearly a strictly increasing bijection, and notice that

$$h_j'(u) = \frac{1}{g_j'(h_j(u))} = \frac{1}{f_j(h_j(u))},$$

for each $0 < u < 1$. Also, define $h_\lambda := (1-\lambda)h_0 + \lambda h_1$ for each $\lambda \in [0,1]$. Let

$$K_\lambda := (1-\lambda)K_0 + \lambda K_1, \quad \text{and}$$
$$K_\lambda(u) := K_\lambda \cap \mathbf{h}(\theta, h_\lambda(u)) = K_\lambda \cap \{y + h_\lambda(u)\theta : y \in \theta^\perp\},$$

for every $\lambda \in [0,1]$ and $0 < u < 1$, where $\theta^\perp := \{x \in \mathbb{R}^n : \langle x,\theta\rangle = 0\}$ (indeed, of course $\mathbf{h}(\theta, t) = \theta^\perp + t\theta$). At this point, we observe that from item **(a)** of Proposition 2.3.2, we have

$$h_\lambda(0) = (1-\lambda)h_0(0) + \lambda h_1(0) = -(1-\lambda)h_{-K_0}(-\theta) - \lambda h_{-K_1}(-\theta) = -h_{-K_\lambda}(-\theta) \ \textit{and}$$
$$h_\lambda(1) = (1-\lambda)h_0(1) + \lambda h_1(1) = (1-\lambda)h_{K_0}(\theta) + \lambda h_{K_1}(\theta) = h_{K_\lambda}(\theta).$$

We also claim that for any $\lambda \in [0,1]$ and $u \in (0,1)$ the inclusion

$$(1-\lambda)K_0(u) + \lambda K_1(u) \subseteq K_\lambda(u)$$

holds. Indeed, if $y_0, y_1 \in \theta^\perp$, then

$$(1-\lambda)(y_0 + h_0(u)\theta) + \lambda(y_1 + h_1(u)\theta) = (1-\lambda)y_0 + \lambda y_1 + h_\lambda(u)\theta,$$

and $(1-\lambda)y_0 + \lambda y_1 \in \theta^\perp$. This, with the equality $K_\lambda = (1-\lambda)K_0 + \lambda K_1$, proves the claim. Hence, using Cavalieri's principle (4.2.1) to calculate the n-dimensional volume of K_λ by integrating the $(n-1)$-dimensional volumes of its "slices" parallel to $\theta^\perp$, and doing the change of variables $t = h_\lambda(u)$, we get

$$\mathrm{vol}_n(K_\lambda) = \int_{-h_{K_\lambda}(-\theta)}^{h_{K_\lambda}(\theta)} \mathrm{vol}_{n-1}(K_\lambda \cap (\theta^\perp + t\theta))\, dt =$$

$$= \int_0^1 \mathrm{vol}_{n-1}(K_\lambda \cap (\theta^\perp + h_\lambda(u)\theta))h'_\lambda(u)\, du = \int_0^1 \mathrm{vol}_{n-1}(K_\lambda(u))h'_\lambda(u)\, du \geq$$

$$\geq \int_0^1 \mathrm{vol}_{n-1}((1-\lambda)K_0(u) + \lambda K_1(u)) \left(\frac{1-\lambda}{f_0(h_0(u))} + \frac{\lambda}{f_1(h_1(u))} \right) du.$$

Now, by the induction hypothesis, we get the inequality

$$\mathrm{vol}_{n-1}((1-\lambda)K_0(u) + \lambda K_1(u))^{\frac{1}{n-1}} \geq$$

$$\geq (1-\lambda)\mathrm{vol}_{n-1}(K_0(u))^{\frac{1}{n-1}} + \lambda \mathrm{vol}_{n-1}(K_1(u))^{\frac{1}{n-1}} = (1-\lambda) f_0(h_0(u))^{\frac{1}{n-1}} + \lambda f_1(h_1(u))^{\frac{1}{n-1}},$$

where the last equality justifies because, for $j = 0, 1$, we have

$$K_j(u) = K_j \cap \{y + h_j(u)\theta : y \in \theta^\perp\} = \{x \in K_j : \langle x, \theta \rangle = h_j(u)\},$$

implying that $\mathrm{vol}_{n-1}(K_j(u)) = f_j(h_j(u))$. Plugging together the last two inequalities yield

$$\mathrm{vol}_n(K_\lambda) \geq$$

$$\geq \int_0^1 \left((1-\lambda) f_0(h_0(u))^{\frac{1}{n-1}} + \lambda f_1(h_1(u))^{\frac{1}{n-1}} \right)^{n-1} \left(\frac{1-\lambda}{f_0(h_0(u))} + \frac{\lambda}{f_1(h_1(u))} \right) du.$$

Now we use Lemma 1.5.1 twice to obtain the inequalities

$$(1-\lambda) f_0(h_0(u))^{\frac{1}{n-1}} + \lambda f_1(h_1(u))^{\frac{1}{n-1}} \geq f_0(h_0(u))^{\frac{1-\lambda}{n-1}} f_1(h_1(u))^{\frac{\lambda}{n-1}}, \text{ and} \tag{5.1.5}$$

$$\frac{1-\lambda}{f_0(h_0(u))} + \frac{\lambda}{f_1(h_1(u))} \geq \frac{1}{f_0(h_0(u))^{1-\lambda}} \frac{1}{f_1(h_1(u))^{\lambda}}. \tag{5.1.6}$$

Taking the $(n-1)$-th power of the first inequality above, and multiplying by the second one, we get

$$\left((1-\lambda) f_0(h_0(u))^{\frac{1}{n-1}} + \lambda f_1(h_1(u))^{\frac{1}{n-1}} \right)^{n-1} \left(\frac{1-\lambda}{f_0(h_0(u))} + \frac{\lambda}{f_1(h_1(u))} \right) \geq 1,$$

for every $u \in (0, 1)$. Consequently,

$$\mathrm{vol}_n((1-\lambda)K_0 + \lambda K_1) = \mathrm{vol}_n(K_\lambda) \geq \int_0^1 1\, du = 1,$$

and this settles the inequality in the case $\mathrm{vol}_n(K_0) = \mathrm{vol}_n(K_1) = 1$. Suppose that equality holds for some $\lambda \in (0, 1)$. Hence, we have equality in (5.1.5) and (5.1.6), yielding $f_0(h_0(u)) = f_1(h_1(u))$ for every $u \in (0, 1)$. Therefore,

$$h'_0(u) = \frac{1}{f_0(h_0(u))} = \frac{1}{f_1(h_1(u))} = h'_1(u),$$

for any $u \in (0, 1)$. It follows that $h_0 - h_1$ is a constant function. Since we are assuming that K_0 and K_1 have their centroid at the origin, we make the change of variables $t = h_j(u)$ again to obtain

$$0 = \left\langle \int_{K_j} x\,dx, \theta \right\rangle = \int_{K_j} \langle x, \theta\rangle\,dx = \int_{-h_{K_j}(-\theta)}^{h_{K_j}(\theta)} t f_j(t)\,dt =$$
$$= \int_0^1 h_j(u) f_j(h_j(u)) h_j'(u)\,du = \int_0^1 h_j(u)\,du.$$

Since this equality holds for both $j = 0$ and $j = 1$, we get that $h_0 = h_1$. In particular,

$$h_{K_0}(\theta) = \lim_{u\to 1} h_0(u) = \lim_{u\to 1} h_1(u) = h_{K_1}(\theta).$$

This equality holds for any $\theta \in \mathbb{S}^{n-1}$, and then K_0 and K_1 have the same support function. It follows from Proposition 2.3.2 that $K_0 = K_1$.

It remains to consider the case where the volumes of K_0 and K_1 are not necessarily 1. We skip the proof of the inequality, because this is Corollary 5.1.2. Assume that the equality

$$\mathrm{vol}_n((1-\lambda)K_0 + \lambda K_1)^{1/n} = (1-\lambda)\mathrm{vol}_n(K_0)^{1/n} + \lambda \mathrm{vol}_n(K_1)^{1/n}$$

holds for some $\lambda \in (0,1)$. Let $T_0 := (1-\lambda)K_0$ and $T_1 := \lambda K_1$. Then, the equality above reads

$$\mathrm{vol}_n(T_0 + T_1)^{1/n} = \mathrm{vol}_n(T_0)^{1/n} + \mathrm{vol}_n(T_1)^{1/n}.$$

Now let

$$C_0 := \frac{T_0}{\mathrm{vol}_n(T_0)^{1/n}},$$
$$C_1 := \frac{T_1}{\mathrm{vol}_n(T_1)^{1/n}}, \text{ and}$$
$$\lambda_0 := \frac{\mathrm{vol}_n(T_1)^{1/n}}{\mathrm{vol}_n(T_0)^{1/n} + \mathrm{vol}_n(T_1)^{1/n}}.$$

Notice that C_0 and C_1 are convex bodies of unit volume, and $\lambda_0 \in (0,1)$. From the equality of the volumes of T_0 and T_1, it follows that

$$\mathrm{vol}_n((1-\lambda_0)C_0 + \lambda_0 C_1)^{1/n} = \frac{\mathrm{vol}_n(T_0+T_1)^{1/n}}{\mathrm{vol}_n(T_0)^{1/n} + \mathrm{vol}_n(T_1)^{1/n}} = 1,$$

and hence $C_0 = C_1$. Therefore, K_0 and K_1 are homothetical.

□

Remark 5.1.2. Notice that if equality holds in (5.1.4) for **some** $\lambda \in (0,1)$, then the fact that K_0 and K_1 are homothetical implies that equality holds for **every** $\lambda \in (0,1)$.

We finish this chapter by highlighting a useful viewpoint: the Brunn–Minkowski inequality means that the function

$$f(t) = \mathrm{vol}_n((1-t)K + tL)^{1/n}$$

is concave for $t \in [0,1]$ (even when K and L are not convex, but only compact). In the case where K and L are convex bodies, then f is linear precisely when K and L are homothetical.

5.2 The isoperimetric inequality

One of the most important geometric inequalities is the so-called *isoperimetric inequality*, which states that among all convex bodies having the same volume, the (adequately normalized) Euclidean ball has the least surface area.

In Section 4.1, we introduced three equivalent types of surface area of a convex body (because of this equivalence, we refer to any of them simply as *surface area*). To study the isoperimetric inequality, it is more convenient to work with the concept of Minkowski surface area:

$$S(K) := \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(K + \varepsilon B^n) - \mathrm{vol}_n(K)}{\varepsilon}.$$

Below we collect some important facts about the surface area of convex bodies.

Lemma 5.2.1. *The surface area is continuous on $\mathcal{K}^n$ with respect to the Hausdorff metric. Moreover, we have that $S(\alpha K) = |\alpha|^{n-1} S(K)$ for any $K \in \mathcal{K}^n$ and every $\alpha \in \mathbb{R}$.*

Proof. The continuity of the surface area will actually be proved later, as a consequence of the continuity of mixed volumes and of the expression (6.1.2).

To verify the other claim, we may assume that $\alpha > 0$. In fact, the case $\alpha = 0$ is trivial and the surface area is clearly invariant under changing the sign of the body. From Theorem A.3.3, we have that $\mathrm{vol}_n(\alpha K) = \alpha^n \mathrm{vol}_n(K)$, and hence we calculate

$$S(\alpha K) = \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(\alpha K + \varepsilon B^n) - \mathrm{vol}_n(\alpha K)}{\varepsilon} =$$
$$= \alpha^{n-1} \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(K + (\varepsilon/\alpha) B^n) - \mathrm{vol}_n(K)}{\varepsilon/\alpha} = \alpha^{n-1} S(K),$$

where the last equality is justified by a simple change of variables in the limit.

□

Theorem 5.2.1 (Isoperimetric inequality)**.** *For any convex body $K \subseteq \mathbb{R}^n$, we have*

$$\left(\frac{S(K)}{S(B^n)} \right)^{\frac{1}{(n-1)}} \geq \left(\frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)} \right)^{\frac{1}{n}}.$$

Equality holds if and only if K is homothetical to B^n.

The fact that equality holds if K is homothetical to B^n is a direct consequence of the equality $S(B^n) = n\mathrm{vol}_n(B^n)$ obtained in (4.1.3).

Remark 5.2.1. We can interpret the isoperimetric inequality in two ways: first, among all convex bodies with a given prescribed volume v, the unique (up to translations) which has the least surface area is the multiple of the Euclidean ball whose volume is v. Second, among all convex bodies with a given prescribed surface area s, the unique (up to translations) which has the greatest volume is the multiple of the Euclidean ball with a surface area equal to s.

We give two proofs for the isoperimetric inequality. The first one involves Steiner symmetrizations, which were introduced in Section 2.7. The second one follows straightforwardly from the Brunn–Minkowski inequality. The advantage of the second proof is that we can characterize the equality case from it.

The key ingredient for the first proof is that the surface area of a convex body is non-increasing under Steiner symmetrizations, as it is explained in the lemma below. In what follows, recall that we denote the Steiner symmetral of a convex body $K \subseteq \mathbb{R}^n$ with respect to a direction $u \in \mathbb{S}^{n-1}$ by $S_u(K)$.

Lemma 5.2.2. *For any convex body $K \subseteq \mathbb{R}^n$, $u \in \mathbb{S}^{n-1}$ and $\varepsilon > 0$, we have*

$$S_u(K) + \varepsilon B^n \subseteq S_u(K + \varepsilon B^n).$$

As a consequence, we get

$$S(S_u(K)) \leq S(K).$$

Proof. Indeed, from Proposition 2.7.2, and since B^n is invariant under any Steiner symmetrization, we have

$$S_u(K + \varepsilon B^n) \supseteq S_u(K) + S_u(\varepsilon B^n) = S_u(K) + \varepsilon B^n.$$

This yields that $\mathrm{vol}_n(S_u(K) + \varepsilon B^n) \leq \mathrm{vol}_n(S_u(K + \varepsilon B^n))$. Because Steiner symmetrizations preserve volume, we obtain

$$\begin{aligned} S(S_u(K)) &= \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(S_u(K) + \varepsilon B^n) - \mathrm{vol}_n(S_u(K))}{\varepsilon} = \\ = \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(S_u(K) + \varepsilon B^n) - \mathrm{vol}_n(K)}{\varepsilon} &\leq \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(S_u(K + \varepsilon B^n)) - \mathrm{vol}_n(K)}{\varepsilon} = \\ &= \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(K + \varepsilon B^n) - \mathrm{vol}_n(K)}{\varepsilon} = S(K). \end{aligned}$$

This concludes the proof.

□

First proof of Theorem 5.2.1 (without equality case). In view of Theorem 2.7.1 (and Remark 2.7.3), we may consider a sequence of iterated Steiner symmetrals of K converging in the Hausdorff metric to the ball

$$\left(\frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)}\right)^{\frac{1}{n}} B^n,$$

which is the ball centered at the origin and having the same volume as K. From Lemma 5.2.2, the surface area is non-increasing along the considered sequence. Since by Lemma 5.2.1 we have that $S(\cdot)$ is continuous in the Hausdorff metric and homogeneous of degree $n-1$, we get

$$S(K) \geq S\left(\left(\frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)}\right)^{\frac{1}{n}} B^n\right) = \left(\frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)}\right)^{\frac{n-1}{n}} S(B^n),$$

and hence the desired inequality follows. Notice that the limit in the second line is just a derivative, and hence this could just be computed as so.

□

Second proof of Theorem 5.2.1 (with equality case). We can obtain the isoperimetric inequality using only the Brunn–Minkowski inequality. Indeed, from Corollary 5.1.1 we have

$$S(K) = \lim_{\varepsilon\to 0^+} \frac{\mathrm{vol}_n(K+\varepsilon B^n) - \mathrm{vol}_n(K)}{\varepsilon} \geq \lim_{\varepsilon\to 0^+} \frac{\left(\mathrm{vol}_n(K)^{1/n} + \varepsilon \mathrm{vol}_n(B^n)^{1/n}\right)^n - \mathrm{vol}_n(K)}{\varepsilon} =$$

$$= \lim_{\varepsilon\to 0^+} \frac{1}{\varepsilon}\left[\left(\sum_{j=0}^{n} \binom{n}{j} \mathrm{vol}_n(K)^{\frac{n-j}{n}} \mathrm{vol}_n(B^n)^{\frac{j}{n}} \varepsilon^j\right) - \mathrm{vol}_n(K)\right] = n\mathrm{vol}_n(K)^{\frac{n-1}{n}} \mathrm{vol}_n(B^n)^{\frac{1}{n}},$$

and this gives the isoperimetric inequality.

Now notice that if the equality holds, then we have

$$\lim_{\varepsilon\to 0^+} \frac{\mathrm{vol}_n(K+\varepsilon B^n) - \mathrm{vol}_n(K)}{\varepsilon} = n\mathrm{vol}_n(K)^{\frac{n-1}{n}} \mathrm{vol}_n(B^n)^{\frac{1}{n}}.$$

Let $g\colon [0,1] \to \mathbb{R}$ be given $g(\varepsilon) = \mathrm{vol}_n(K+\varepsilon B^n)$. The equality above means that

$$g'(0) = n\mathrm{vol}_n(K)^{\frac{n-1}{n}} \mathrm{vol}_n(B^n)^{\frac{1}{n}}.$$

From the Brunn–Minkowski inequality, it is easy to see that the function $f\colon [0,1] \to \mathbb{R}$ given by $f(\varepsilon) = g(\varepsilon)^{1/n}$ is concave. Indeed, for any $s, t, \lambda \in [0,1]$, we have

$$f((1-\lambda)s + \lambda t) = \mathrm{vol}_n(K + ((1-\lambda)s + \lambda t)B^n)^{1/n} =$$

$$= \mathrm{vol}_n((1-\lambda)(K+sB^n) + \lambda(K+tB^n))^{1/n} \geq$$

$$\geq (1-\lambda)\mathrm{vol}_n(K+sB^n)^{1/n} + \lambda \mathrm{vol}_n(K+tB^n)^{1/n} = (1-\lambda)f(s) + \lambda f(t).$$

Consequently, the graph of f lies below each of its tangent lines. In particular, if we consider the tangent line at $\varepsilon = 0$, we get

$$f(t) \leq f(0) + tf'(0)$$

for all $t \in [0,1]$. Once

$$f'(0) = \frac{1}{n} g(0)^{1/n-1} g'(0) = \mathrm{vol}_n(B^n)^{1/n},$$

for $t = 1$ this is the same as

$$\mathrm{vol}_n(K+B^n)^{1/n} \leq \mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(B^n)^{1/n}.$$

This gives equality in the Brunn–Minkowski inequality, and hence we have that K is homothetical to B^n.

□

5.3 Difference bodies and the Rogers–Shephard inequality

Recall that the *difference body* of a convex body $K \subseteq \mathbb{R}^n$ is the set

$$K - K := \{x - z : x, z \in K\},$$

see also Plate 5 at the beginning of this chapter. And note also that its homothetical copy $\frac{1}{2}(K-K)$ is called the *central symmetral* of K. The set $K-K$ is clearly a convex body centered at the origin, and

an application of the Brunn–Minkowski inequality gives immediately a lower bound for its volume in terms of the volume of K:

$$\mathrm{vol}_n(K-K) = 2^n \mathrm{vol}_n\left(\frac{1}{2}K + \frac{1}{2}(-K)\right) \geq 2^n \mathrm{vol}_n(K)^{1/2}\mathrm{vol}_n(-K)^{1/2} = 2^n \mathrm{vol}_n(K). \tag{5.3.1}$$

We claim that equality holds if and only if K is symmetric with respect to some point. Indeed, if K is symmetric with respect to $x_0 \in \mathbb{R}^n$, then by definition we have $K - x_0 = -(K - x_0)$, from which we get that $K = -K + 2x_0$. Hence K and $-K$ are homothetical, and equality holds in the Brunn–Minkowski inequality (see Theorem 5.1.2).

On the other hand, if equality holds in (5.3.1), then K and $-K$ are homothetical, meaning that there exist a point $x_0 \in \mathbb{R}^n$ and a number $\lambda > 0$ such that $K = x_0 + \lambda(-K)$. In this case, since $\mathrm{vol}_n(K) = \mathrm{vol}_n(-K)$, we must have $\lambda = 1$. Thus, the previous equality reads $K = x_0 - K$. This is equivalent to $K - x_0/2 = -(K - x_0/2)$, and hence K is symmetric with respect to $x_0/2$.

The Rogers–Shephard inequality, which we aim to prove in this section, provides an upper bound for the volume of the difference body $K - K$ which depends only on the volume of K and on the dimension. We follow the proof in [2167], and for that purpose, we need a technical lemma.

Lemma 5.3.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body which contains the origin as an interior point, and let $f\colon K \to \mathbb{R}$ be a concave and non-negative function. If $g\colon [0,+\infty) \to \mathbb{R}$ is increasing, then*

$$\int_K g \circ f(x)\,dx \geq n\mathrm{vol}_n(K)\int_0^1 g(tf(0_{\mathbb{R}^n}))(1-t)^{n-1}\,dt.$$

If g is strictly increasing, then equality holds if and only if $f|_{\partial K} = 0$ and $f|_{\mathrm{seg}[0_{\mathbb{R}^n},z]}$ is linear for every $z \in \partial K$.

Proof. Define the function $h\colon K \to \mathbb{R}$ as

$$h(x) = \Big(1 - ||x||_K\Big) f(0_{\mathbb{R}^n}).$$

We claim that $f(x) \geq h(x)$ for every $x \in K$. This comes from the hypotheses on f:

$$f(x) = f\left(\Big(1-||x||_K\Big)0_{\mathbb{R}^n} + ||x||_K \frac{x}{||x||_K}\right) \geq$$

$$\geq \Big(1-||x||_K\Big) f(0_{\mathbb{R}^n}) + ||x||_K f\left(\frac{x}{||x||_K}\right) \geq \Big(1-||x||_K\Big) f(0_{\mathbb{R}^n}) = h(x).$$

Here we used the immediate fact that $0 \leq ||x||_K \leq 1$ for every $x \in K$ (in order to use the concavity of f). Since g is increasing, this leads to $g \circ f(x) \geq g \circ h(x)$ for every $x \in K$. Now we write K in polar coordinates as $x = ru$, where $u \in \mathbb{S}^{n-1}$ and $0 \leq r \leq \rho_K(u)$. By doing so, we notice that $||x||_K = ||ru||_K = r||u||_K = r/\rho_K(u)$, and integrate

$$\int_K g \circ f(x)\,dx \geq \int_K g \circ h(x)\,dx =$$

$$= \int_{\mathbb{S}^{n-1}} \int_0^{\rho_K(u)} r^{n-1} g\left(\Big(1 - r/\rho_K(u)\Big) f(0_{\mathbb{R}^n})\right)\,dr\,d\mathcal{H}^{n-1}(u).$$

Now we use the change of variables $r = (1-t)\rho_K(u)$ to obtain

$$\int_K g \circ f(x)\,dx \geq \int_{\mathbb{S}^{n-1}} \int_0^1 (1-t)^{n-1} \rho_K(u)^{n-1} g\Big(t f(0_{\mathbb{R}^n})\Big) \rho_K(u)\,dt\,d\mathcal{H}^{n-1}(u) =$$
$$= \left(\int_{\mathbb{S}^{n-1}} \rho_K(u)^n\,d\mathcal{H}^{n-1}(u)\right)\left(\int_0^1 (1-t)^{n-1} g\Big(t f(0_{\mathbb{R}^n})\Big)\,dt\right),$$

where we used Fubini's theorem. Finally, we remember that $\rho_K(u) = ||u||_K^{-1}$ and use (4.3.7) to write

$$\int_{\mathbb{S}^{n-1}} \rho_K(u)^n\,d\mathcal{H}^{n-1}(u) = n\mathrm{vol}_n(K),$$

where the last equality justifies because $\mathrm{vol}_{n-1}(\mathbb{S}^{n-1}) = n\mathrm{vol}_n(B^n)$, and σ denotes the spherical probability measure (defined in Section 4.5). Hence the proof of the inequality is complete. If g is strictly increasing and equality holds, then $f = h$, and this gives that $f|_{\partial K} = h|_{\partial K} = 0$ (because $||x||_K = 1$ whenever $x \in \partial K$), and that f is linear over any segment $\mathrm{seg}[0, z]$ for $z \in \partial K$ (because h is clearly linear over such a segment).

□

Theorem 5.3.1 (Rogers–Shephard inequality)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then*

$$\mathrm{vol}_n(K - K) \leq \binom{2n}{n} \mathrm{vol}_n(K).$$

Proof. We start with the following useful equivalent definition of the difference body:

$$K - K = \big\{x \in \mathbb{R}^n : K \cap (K + x) \neq \emptyset\big\}.$$

In fact, if $x \in K - K$, then we may write $x = y - z$ for $y, z \in K$. By doing so, we have that $y = z + x$, and hence $y \in K \cap (K + x)$. On the other hand, if $y \in K \cap (K + x)$, then there exists $z \in K$ such that $y = z + x$, from where we get $x = y - z$ with $y, z \in K$.

We claim that for any $x, z \in K - K$ and every $\lambda \in [0, 1]$, we have the inclusion

$$(1-\lambda)\big(K \cap (K + x)\big) + \lambda\big(K \cap (K + z)\big) \subseteq K \cap \big(K + (1-\lambda)x + \lambda z\big). \tag{5.3.2}$$

To prove that, let $y \in K \cap (K + x)$ and $w \in K \cap (K + z)$. First observe that $y, w \in K$, implying that $(1-\lambda)y + \lambda w \in K$. This proves that

$$(1-\lambda)\big(K \cap (K + x)\big) + \lambda\big(K \cap (K + z)\big) \subseteq K.$$

For the remaining inclusion, write $y = x_0 + x$ for $x_0 \in K$ (because $y \in K + x$), and $w = z_0 + z$ for some $z_0 \in K$. Hence

$$(1-\lambda)y + \lambda w = (1-\lambda)(x_0 + x) + \lambda(z_0 + z) = (1-\lambda)x_0 + \lambda z_0 + (1-\lambda)x + \lambda z,$$

and since it is clear that $(1-\lambda)x_0 + \lambda z_0 \in K$, we get that $(1-\lambda)y + \lambda w \in K + (1-\lambda)x + \lambda z$. Thus, we have (5.3.2). From that, together with the Brunn–Minkowski inequality (Corollary 5.1.2), we get the inequality

$$\mathrm{vol}_n\big(K \cap \big(K + (1-\lambda)x + \lambda z\big)\big)^{1/n} \geq \mathrm{vol}_n\big((1-\lambda)\big(K \cap (K + x)\big) + \lambda\big(K \cap (K + z)\big)\big)^{1/n} \geq$$
$$\geq (1-\lambda)\mathrm{vol}_n\big(K \cap (K + x)\big)^{1/n} + \lambda\mathrm{vol}_n\big(K \cap (K + z)\big)^{1/n},$$

that is, the function $f\colon K - K \to [0, +\infty)$ given as

$$f(x) = \mathrm{vol}_n\big(K \cap (K+x)\big)^{1/n}$$

is concave. By Lemma 5.3.1 applied for f over $K-K$, and with $g(t) = t^n$, we get

$$\begin{aligned}\int_{K-K} \mathrm{vol}_n\big(K\cap(K+x)\big)\,dx &= \int_{K-K}\Big[\mathrm{vol}_n\big(K\cap(K+x)\big)^{1/n}\Big]^n\,dx \geq \\ &\geq n\mathrm{vol}_n(K-K)\int_0^1 \big[t\mathrm{vol}_n(K)^{1/n}\big]^n(1-t)^{n-1}\,dt = \\ = n\mathrm{vol}_n(K-K)\mathrm{vol}_n(K)\int_0^1 t^n(1-t)^{n-1}\,dt &= \binom{2n}{n}^{-1}\mathrm{vol}_n(K-K)\mathrm{vol}_n(K),\end{aligned}$$

where the last equality comes from a straightforward integral by parts calculation. On the other hand, since $\mathrm{vol}_n\big(K\cap(K+x)\big) = 0$ if $x \notin K-K$, we have

$$\begin{aligned}\int_{K-K}\mathrm{vol}_n\big(K\cap(K+x)\big)\,dx &= \int_{\mathbb{R}^n}\mathrm{vol}_n\big(K\cap(K+x)\big)\,dx = \\ = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} 1_{K\cap(K+x)}(z)\,dz\,dx &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} 1_K(z)1_{K+x}(z)\,dz\,dx = \\ = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} 1_K(z)1_K(z-x)\,dz\,dx &= \int_{\mathbb{R}^n}\left(\int_{\mathbb{R}^n} 1_K(z-x)\,dx\right)1_K(z)\,dz = \\ &= \mathrm{vol}_n(K)\int_{\mathbb{R}^n} 1_K(z)\,dz = \mathrm{vol}_n(K)^2.\end{aligned} \tag{5.3.3}$$

The desired inequality comes immediately from this equality together with the previous inequality, and hence we are done.

□

Remark 5.3.1. Observe that equality holds if and only if $K\cap(K+x)$ and K are homothetical for each $x \in K-K$ (because of the equality case in the Brunn–Minkowski inequality). This characterizes the simplex, but we will skip the proof.

5.4 The Mahler product and the Blaschke–Santaló inequality

Recall that we denote by $\mathcal{K}_0^n$ the collection of convex bodies of $\mathbb{R}^n$ which contain the origin as an interior point. Due to Corollary 2.3.5, we have that if $K \in \mathcal{K}_0^n$, then K° is also a convex body containing the origin as an interior point. The *Mahler product* of a convex body $K \in \mathcal{K}_0^n$ is the quantity defined as

$$s(K) := \mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ).$$

The classical Blaschke–Santaló inequality gives an (optimal) upper bound for the Mahler product of a convex body centered at the origin. In Section 11.5, we prove the Bourgain–Milman inequality, which provides a (not necessarily optimal) lower bound for the Mahler product of a convex body centered at the origin. It is also worth mentioning that a very important open problem in convex geometry is the so-called *Mahler conjecture*, which states that the Mahler product is minimized among the convex bodies centered at the origin by the cube $Q_n := [-1,1]^n$, thus providing the optimal lower bound $4^n/n!$ for $s(K)$.

We start by proving that $s(K)$ is invariant under isomorphisms (even if K is not centered at the origin). In what follows, recall that the *adjoint* of an isomorphism $T\colon \mathbb{R}^n \to \mathbb{R}^n$ is the (unique)

isomorphism $T^*\colon \mathbb{R}^n \to \mathbb{R}^n$ with the property that

$$\langle Tx, z\rangle = \langle x, T^* z\rangle$$

for every $x, z \in \mathbb{R}^n$ (see Lemma 2.6.3 for some properties of the adjoint). If $T = T^*$, then we say that T is *self-adjoint* (or *symmetric*). It is a well-known result of standard linear algebra that $\det(T) = \det(T^*)$ for any isomorphism $T\colon \mathbb{R}^n \to \mathbb{R}^n$.

Lemma 5.4.1. *Let K be a convex body containing the origin as an interior point and $T\colon \mathbb{R}^n \to \mathbb{R}^n$ an isomorphism. Then*

$$(TK)^\circ = (T^*)^{-1}K^\circ.$$

As a consequence, $s(TK) = s(K)$.

Proof. We have that $z \in (TK)^\circ$ if and only if $\langle y, z\rangle \leq 1$ for every $y \in TK$, which happens if and only if $\langle Tx, z\rangle \leq 1$ for any $x \in K$. This last condition is equivalent to having $\langle x, T^*z\rangle \leq 1$ for every $x \in K$, and this is the same as saying that $T^*z \in K^\circ$. Finally, $T^*z \in K^\circ$ if and only if $z \in (T^*)^{-1}K^\circ$.

To prove the other claim, remember that $\mathrm{vol}_n(TA) = |\det(T)|\mathrm{vol}_n(A)$ for any Lebesgue measurable set $A \subseteq \mathbb{R}^n$ and every linear map $T\colon \mathbb{R}^n \to \mathbb{R}^n$, and that $\det(T^{-1}) = \det(T)^{-1}$ for any linear isomorphism. Hence, we merely calculate

$$\begin{aligned} s(TK) &= \mathrm{vol}_n(TK)\mathrm{vol}_n((TK)^\circ) = |\det(T)|\mathrm{vol}_n(K)\mathrm{vol}_n((T^*)^{-1}K^\circ) = \\ &= |\det(T)\det((T^*)^{-1})|\mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ) = \left|\frac{\det(T)}{\det(T^*)}\right| s(K) = s(K), \end{aligned}$$

where the last inequality justifies since $\det(T) = \det(T^*)$.

□

The natural question that arises is whether there are lower and upper bounds for the Mahler product. An optimal lower bound is still unknown, and a famous conjecture attributed to Mahler asks whether this is the Mahler product of a cross-polytope.

Conjecture 1 (Mahler's conjecture). For any convex body $K \subseteq \mathbb{R}^n$ centered at the origin, we have

$$\mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ) \geq \frac{4^n}{n!}.$$

The Bourgain–Milman inequality, which we prove in Section 11.5 using techniques of asymptotic geometric analysis, provides a lower bound for the Mahler product which depends on the volume of the n-dimensional ball and on the n-th power of an independent constant.

Theorem 5.4.1 (Bourgain–Milman inequality)**.** *There exists a number $c > 0$ such that*

$$\mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ) \geq c^n \mathrm{vol}_n(B^n)^2$$

for every $n \in \mathbb{N}$ and any convex body $K \subseteq \mathbb{R}^n$ centered at the origin.

On the other hand, the optimal upper bound for the Mahler product is known, as well as the equality cases. We are investigating this now. Before doing that, we introduce a new notion: an *ellipsoid* of $\mathbb{R}^n$ is an image of the unit ball B^n by an affine transformation of $\mathbb{R}^n$.

Theorem 5.4.2 (Blaschke–Santaló inequality)**.** *For any convex body $K \subseteq \mathbb{R}^n$ centered at the origin, we have*

$$\mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ) \le \mathrm{vol}_n(B^n)^2.$$

Equality holds if and only if K is an ellipsoid.

Remark 5.4.1. Because of the invariance of the Mahler product under invertible linear transformations, an equivalent formulation is as follows: we have $\mathrm{vol}_n(K^\circ) \le \mathrm{vol}_n(B^n)$ whenever $\mathrm{vol}_n(K) = \mathrm{vol}_n(B^n)$, with equality if and only if K° is an ellipsoid.

Proof. We follow the proof in [133], which is based on the following fact: for any convex body $K \subseteq \mathbb{R}^n$ centered at the origin and every $u \in \mathbb{S}^{n-1}$ we have

$$\mathrm{vol}_n(K^\circ) \le \mathrm{vol}_n((S_u(K))^\circ), \tag{5.4.1}$$

where we remind the reader that $S_u(K)$ denotes the Steiner symmetral of K with respect to u. We prove this inequality now. Due to Proposition 2.7.3, we have that $S_{Tu}(TK) = T(S_u(K))$ for any orthogonal transformation $T\colon \mathbb{R}^n \to \mathbb{R}^n$, and because of that we may assume that $u = e_n$. By doing so, the last coordinate of a vector of $\mathbb{R}^n$ is its projection onto the direction of u, and we may write,

$$S_u(K) = \left\{\left(x, \frac{t-s}{2}\right) : (x,t),\ (x,s) \in K\right\},$$

where we write $(x,t) \in \mathbb{R}^{n-1} \times \mathbb{R} = \mathbb{R}^n$ (see Exercise 2.56). Consequently, we have the following description for the polar of $S_u(K)$:

$$(S_u(K))^\circ = \left\{(z,r) : \langle z, x\rangle + \frac{r(t-s)}{2} \le 1 \text{ whenever } (x,t),\ (x,s) \in K\right\}, \tag{5.4.2}$$

where we are denoting the inner product of $\mathbb{R}^{n-1}$ also by $\langle\cdot,\cdot\rangle$. The projection $\mathrm{proj}\colon \mathbb{R}^n \to \mathbb{R}^{n-1}$ given by $\mathrm{proj}_{n-1}(x_1,\dots,x_{n-1},x_n) = (x_1,\dots,x_{n-1})$ identifies naturally any hyperplane orthogonal to e_n with $\mathbb{R}^{n-1}$. Indeed,

$$\mathbf{h}(e_n, r) = \{(x_1,\dots,x_{n-1},r)\} \simeq \mathbb{R}^{n-1},$$

where $(x_1,\dots,x_{n-1},r) \mapsto (x_1,\dots,x_{n-1})$. Using this viewpoint, we naturally identify sections of a set $E \subseteq \mathbb{R}^n$ with subsets of $\mathbb{R}^{n-1}$:

$$E(r) := \{x \in \mathbb{R}^{n-1} : (x,r) \in E\} = \mathrm{proj}_{n-1}(E \cap \mathbf{h}(e_n, r)).$$

Hence, for any $r \in \mathbb{R}$, we have

$$E^\circ(r) = \{x \in \mathbb{R}^{n-1} : (x,r) \in E^\circ\} = \{x \in \mathbb{R}^{n-1} : \langle (x,r),(z,s)\rangle \le 1\ \forall\ (z,s) \in E\} =$$
$$= \{x \in \mathbb{R}^{n-1} : \langle x, z\rangle + rs \le 1\ \forall\ (z,s) \in E\},$$

and we claim that

$$\frac{1}{2}(K^\circ(r) + K^\circ(-r)) \subseteq (S_u(K))^\circ(r). \tag{5.4.3}$$

To prove this, let $y \in K^\circ(r)$ and $z \in K^\circ(-r)$. For $(x,t),(x,s) \in K$ arbitrarily given, we have

$$\langle (y,r),(x,t)\rangle = \langle y, x\rangle + rt \le 1 \text{ and}$$
$$\langle (z,-r),(x,s)\rangle = \langle z, x\rangle - rs \le 1.$$

Dividing these inequalities by 2 and summing up them yields

$$\left\langle \frac{y+z}{2}, x \right\rangle + \frac{r(t-s)}{2} \leq 1.$$

Since (x,t) and (x,s) were taken arbitrarily in K, we get from (5.4.2) that

$$\left(\frac{y+z}{2}, r \right) \in (S_u(K))^\circ,$$

and this leads immediately to $(y+z)/2 \in (S_u(K))^\circ(r)$. Thus we have the inclusion (5.4.3). Observe carefully that this inclusion means that, in particular, each section of $(S_u(K))^\circ$ parallel to $e_n^\perp$ has $(n-1)$-dimensional volume greater than or equal to the $(n-1)$-dimensional volume of $\frac{1}{2}(K^\circ(r) + K^\circ(-r))$. By Cavalieri's principle (4.2.1), also proved in Theorem A.6.3, we get

$$\mathrm{vol}_n((S_u(K))^\circ) = \int_{-\infty}^{\infty} \mathrm{vol}_{n-1}((S_u(K))^\circ(t))\, dt \geq \int_{-\infty}^{\infty} \mathrm{vol}_{n-1}\left(\frac{K^\circ(t) + K^\circ(-t)}{2} \right) dt. \tag{5.4.4}$$

Recalling that K is centered at the origin, we have that K° is also centered at the origin, and hence it is easy to see that $K^\circ(-r) = -K^\circ(r)$, implying that the bodies (of $\mathbb{R}^{n-1}$) $K^\circ(r)$ and $K^\circ(-r)$ have the same volume. Therefore, by the Brunn–Minkowski inequality (Theorem 5.1.1), we get

$$\mathrm{vol}_{n-1}\left(\frac{K^\circ(r) + K^\circ(-r)}{2} \right) \geq \mathrm{vol}_{n-1}(K^\circ(r))^{1/2} \mathrm{vol}_{n-1}(K^\circ(-r))^{1/2} = \mathrm{vol}_{n-1}(K^\circ(r)).$$

Hence we may estimate the last integral in (5.4.4) to obtain

$$\mathrm{vol}_n((S_u(K))^\circ) \geq \int_{-\infty}^{\infty} \mathrm{vol}_{n-1}(K^\circ(t))\, dt = \mathrm{vol}_n(K^\circ),$$

where we used Cavalieri's principle again in the last equality. This gives inequality (5.4.1).

Now observe that, in order to prove the Blaschke–Santaló inequality, we may restrict ourselves to the case where $\mathrm{vol}_n(K) = \mathrm{vol}_n(B^n)$, because the Mahler product is invariant under isomorphisms, and hence we can simply take the "correct" dilation of K. By Theorem 2.7.1, there exists a sequence $(u_j)_{j\in\mathbb{N}}$ in $\mathbb{S}^{n-1}$ with the property that the sequence of convex bodies given by the recurrence

$$\begin{aligned} K_1 &= S_{u_1}(K), \\ K_m &= S_{u_m}(K_{m-1}) \ \forall\, m \geq 2 \end{aligned}$$

is such that $K_j \to B^n$ in the Hausdorff metric as $j \to \infty$. Then we also have that $K_j^\circ \to B^n$ (see Proposition 2.6.1; since K is centered at the origin, each K_j is centered at the origin, having thus the origin as an interior point), and from (5.4.1), we get the successive inequalities

$$\begin{aligned} \mathrm{vol}_n(K^\circ) \leq \mathrm{vol}_n((S_{u_1}(K))^\circ) = \mathrm{vol}_n(K_1)^\circ \leq \mathrm{vol}_n((S_{u_2}(K_1))^\circ) = \mathrm{vol}_n(K_2^\circ) \leq \\ \leq \ldots \leq \mathrm{vol}_n(K_m^\circ) \leq \mathrm{vol}_n((S_{u_m+1}(K_m))^\circ) = \mathrm{vol}_n(K_{m+1}^\circ) \leq \ldots. \end{aligned}$$

Hence, the volumes of K° and K_j° form (in this order) a non-decreasing sequence of positive numbers which converge to $\mathrm{vol}_n(B^n)$ as $j \to \infty$ (recall that the Lebesgue volume is continuous with respect to the Hausdorff metric). It follows that $\mathrm{vol}_n(K^\circ) \leq \mathrm{vol}_n(B^n)$, and this concludes the proof.

We briefly discuss equality (for the case where $n \geq 3$), without giving a completely formal proof (this relies on a certain characterization of ellipsoids which is not in the scope of this book). If equality holds, then we have equality in the Brunn–Minkowski inequality

$$\mathrm{vol}_{n-1}\left(\frac{K^\circ(r) + K^\circ(-r)}{2} \right) \geq \mathrm{vol}_{n-1}(K^\circ(r))^{1/2} \mathrm{vol}_{n-1}(K^\circ(-r))^{1/2},$$

and hence we obtain that $K^\circ(r)$ is a translate of $K^\circ(-r) = -K^\circ(r)$, where this equality follows from the assumption that K is symmetric with respect to the origin. This gives that $K^\circ(r)$ is symmetric with respect to some of its points. Indeed, we have that $K^\circ(r) = x_0 - K^\circ(r)$ for some x_0, and hence

$$K^\circ(r) - \frac{x_0}{2} = -\left(K^\circ(r) - \frac{x_0}{2}\right).$$

Then $K^\circ(r)$ is symmetric with respect to $x_0/2$. This shows that the intersection of K° with any hyperplane orthogonal to e_n is centrally symmetric. Since we can obviously repeat the proof replacing e_n by any vector $u \in \mathbb{S}^{n-1}$, it follows that any hyperplane section of K° is centrally symmetric, and this characterizes ellipsoids in the cases $n \geq 3$ (see [1074]). From Lemma 5.4.1, we have that K° is an ellipsoid centered at the origin if and only if K is an ellipsoid centered at the origin. For more on the equality (including the case where $n = 2$), we refer the reader to [1772] and [1970].

□

There is also a version of the Blaschke–Santaló inequality for arbitrary convex bodies. The extension of the concept of the Mahler product to this context is the following:

$$s(K) := \inf\{\mathrm{vol}_n(K)\mathrm{vol}_n(K - z)^\circ : z \in \mathrm{int}K\}.$$

This is clearly an affine invariant (see Exercise 5.14), and it is known that there exists a unique point $z_0 \in \mathrm{int}K$ where this infimum is attained (a path to a proof of this fact is given in Exercise 5.16). This point is known as the *Santaló point* of K. In this general case, we have that $s(K) \leq \mathrm{vol}_n(B^n)^2$ for every convex body $K \subseteq \mathbb{R}^n$, with equality if and only if K is an ellipsoid. We refer the reader to [1772] for a proof.

5.5 Mean width of a convex body and Urysohn's inequality

Let $K \in CC^n$. Recall that the *width* of K in the direction $u \in \mathbb{S}^{n-1}$ is the number $w_K(u) = h_K(u) + h_K(-u)$, where h_K denotes the support function of K. Geometrically, this is the distance between the two supporting hyperplanes of K which are orthogonal to u (see Lemma 2.3.3). We define the *mean width* $w(K)$ of K to be the average of the width function $w_K : \mathbb{S}^{n-1} \to \mathbb{R}$:

$$w(K) := \frac{1}{\mathrm{vol}_{n-1}(\mathbb{S}^{n-1})} \int_{\mathbb{S}^{n-1}} w_K(u)\, d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-1}} w_K(u)\, d\sigma(u),$$

where we remind the reader that σ denotes the standard probability measure on $\mathbb{S}^{n-1}$, which is the normalized $(n-1)$-dimensional Hausdorff measure (see Section 4.5). Observe that we also have the equalities

$$w(K) = \int_{\mathbb{S}^{n-1}} w_K(u)\, d\sigma(u) = \int_{\mathbb{S}^{n-1}} h_K(u) + h_K(-u)\, d\sigma(u) = 2\int_{\mathbb{S}^{n-1}} h_K(u)\, d\sigma(u).$$

Even more, in the case where K contains the origin as an interior point, we have $h_K(u) = ||u||_{K^\circ}$ for every $u \in \mathbb{S}^{n-1}$, and hence

$$w(K) = 2\int_{\mathbb{S}^{n-1}} ||u||_{K^\circ}\, d\sigma(u).$$

Notice that this leads to the *mean width function* $w : CC^n \to (0, +\infty)$ which is clearly translation invariant and homogeneous of degree 1. We also have that the mean width function is continuous

in the Hausdorff metric. In fact, the continuity comes from the fact that if $d_{\mathcal{H}}(K,L) < \varepsilon$, then $|h_K(u) - h_L(u)| < \varepsilon$ for every $u \in \mathbb{S}^{n-1}$ (see Proposition 2.5.2), and hence

$$|w(K) - w(L)| \leq 2 \int_{\mathbb{S}^{n-1}} |h_K(u) - h_L(u)| \, d\sigma(u) < 2 \int_{\mathbb{S}^{n-1}} \varepsilon \, d\sigma(u) = 2\varepsilon\sigma(\mathbb{S}^{n-1}) = 2\varepsilon.$$

At this point, we introduce a terminology which will be used throughout the text. Let $\mathcal{D} \subseteq 2^{\mathbb{R}^n}$ be a family of subsets of $\mathbb{R}^n$, and $\mathcal{G}$ be a group of linear transformations of $\mathbb{R}^n$ such that $TA \in \mathcal{D}$ for any $T \in \mathcal{G}$ and every $A \in \mathcal{D}$. We say that a given functional $\varphi \colon \mathcal{D} \to \mathbb{R}$ is *$\mathcal{G}$-invariant* if $\varphi(TA) = \varphi(A)$ for every $A \in \mathcal{D}$ and any $T \in \mathcal{G}$. A functional which is invariant under the group of affine transformations (which, as we highlight again, are compositions of translations with linear **isomorphisms** of $\mathbb{R}^n$) is said to be *affine invariant*.

We claim that the mean width is $\mathrm{O}(n)$-invariant (geometrically, the mean width does not change under rotations and reflections). Indeed, since for any $U \in \mathrm{O}(n)$, we have that $U^* \in \mathrm{O}(n)$, and because σ is an $\mathrm{O}(n)$-invariant measure, we may write

$$\begin{aligned} w(UK) = 2 \int_{\mathbb{S}^{n-1}} h_{UK}(u) \, d\sigma(u) = 2 \int_{\mathbb{S}^{n-1}} h_K(U^* u) \, d\sigma(u) = \\ = 2 \int_{\mathbb{S}^{n-1}} h_K(v) \, d\sigma(v) = w(K), \end{aligned} \tag{5.5.1}$$

where the second equality comes from (2.6.5).

Urysohn's inequality, which we aim to prove in this section, answers the following question: *what is the minimum mean width of a convex body with a given prescribed volume?* The geometric inspiration comes from the intuition that, when we prescribe volume and we "enlarge" the width in a given direction, then we have to "shorten" the width in another one. It happens that one cannot make the mean width arbitrarily small by doing so. In fact, we will prove that the minimum mean width among the convex bodies of a given prescribed volume is precisely the diameter of the ball of the same volume. We will show this as a consequence of the fact that the mean width is non-increasing under Steiner symmetrizations (thus following [133] again).

Proposition 5.5.1. *For any convex body $K \subseteq \mathbb{R}^n$ and any $u \in \mathbb{S}^{n-1}$, we have $w(S_u(K)) \leq w(K)$.*

Proof. We may clearly assume that $u = e_n$, because if $T \in \mathrm{O}(n)$ is such that $Tu = e_n$, then, since the mean width is $\mathrm{O}(n)$-invariant,

$$w(TK) = w(K) \text{ and } w(S_{Tu}(Tu)) = w(T(S_u(K)) = w(S_u(K)).$$

Thus, $w(S_{Tu}(TK)) \leq w(TK)$ if and only if $w(S_u(K)) \leq w(K)$. Therefore, we let $u = e_n$ and write (as in the proof of Theorem 5.4.2)

$$S_u(K) = \left\{ \left(x, \frac{t-s}{2} \right) : (x,t), (x,s) \in K \right\}.$$

For any $v \in \mathbb{S}^{n-1}$, we have

$$\begin{aligned} h_{S_u(K)}(v) = \max \left\{ \left\langle v, \left(x, \frac{t-s}{2} \right) \right\rangle : (x,t), (x,s) \in K \right\} \leq \\ \leq \frac{1}{2} \max\{\langle (x,t), v \rangle : (x,t) \in K\} + \frac{1}{2} \max\{\langle (x,-s), v \rangle : (x,s) \in K\} = \\ = \frac{1}{2} h_K(v) + \frac{1}{2} \max\{\langle (x,s), \mathrm{ref}_{u^\perp}(v) \rangle : (x,s) \in K\} = \frac{1}{2} h_K(v) + \frac{1}{2} h_K(\mathrm{ref}_{u^\perp}(v)). \end{aligned}$$

Finally, since the reflection $\mathrm{ref}_{u^\perp}\colon \mathbb{R}^n \to \mathbb{R}^n$ is a linear isomorphism which preserves $\mathbb{S}^{n-1}$ and whose determinant is -1, it follows that

$$w(S_u(K)) = 2\int_{\mathbb{S}^{n-1}} h_{S_u(K)}(v)\,d\sigma(v) \le \int_{\mathbb{S}^{n-1}} h_K(v) + h_K(\mathrm{ref}_{u^\perp}(v))\,d\sigma(v) =$$
$$= \frac{w(K)}{2} + \int_{\mathbb{S}^{n-1}} h_K(\theta)d\sigma(\theta) = \frac{w(K)}{2} + \frac{w(K)}{2} = w(K),$$

where we made the change of variables $\theta = \mathrm{ref}_{u^\perp}(v)$. This concludes the proof.

□

Theorem 5.5.1 (Urysohn's inequality)**.** *For any convex body $K \subseteq \mathbb{R}^n$, we have the inequality*

$$w(K) \ge 2\left(\frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)}\right)^{1/n}. \tag{5.5.2}$$

Equality holds if and only if K is a ball.

Remark 5.5.1. It is obvious that equality holds for balls, but here we will not prove that it holds *only* for balls. Later, we will use the *1st Minkowski inequality* to give another proof of (5.5.2), with the advantage that the equality case will follow easily. See the comment below (6.4.2).

Proof. Since both sides of the inequality are homogeneous of degree 1, it suffices to assume that $\mathrm{vol}_n(K) = \mathrm{vol}_n(B^n)$, and, consequently, to prove that $w(K) \ge 2$. To do so, let $(K_m)_{m\in\mathbb{N}}$ be a sequence of iterated Steiner symmetrizations of $K_1 := K$ converging to B^n (see Theorem 2.7.1). From the previous proposition, we have that

$$w(K) \ge w(K_2) \ge w(K_3) \ge \ldots \ge w(K_m) \ge \ldots,$$

which means that $(w(K_m))_{m\in\mathbb{N}}$ is a non-increasing sequence. Since the mean width is continuous in the Hausdorff metric, it follows that $w(K_m) \to w(B^n) = 2$ when $m \to \infty$. Thus, $w(K) \ge 2$, as we wanted to prove.

□

We also can obtain an inequality relating the mean width of a convex body (having the origin as an interior point) to the volume of its polar body. We have

$$w(K) \ge 2\left(\frac{\mathrm{vol}_n(B^n)}{\mathrm{vol}_n(K^\circ)}\right)^{1/n}. \tag{5.5.3}$$

Indeed, this follows straightforwardly from (4.3.8) and from Jensen's inequality (Theorem 1.2.2) applied for $t \mapsto t^{-1/n}$ (which is convex for $t > 0$):

$$\frac{w(K)}{2} = \int_{\mathbb{S}^{n-1}} h_K(u)\,d\sigma(u) \ge \left(\int_{\mathbb{S}^{n-1}} h_K(u)^{-n}\,d\sigma(u)\right)^{-1/n} = \left(\frac{\mathrm{vol}_n(K^\circ)}{\kappa_n}\right)^{-1/n} =$$
$$= \left(\frac{\mathrm{vol}_n(B^n)}{\mathrm{vol}_n(K^\circ)}\right)^{1/n}.$$

Notice that equality in (5.5.3) holds if and only if we have equality in Jensen's inequality. Since the map $t \mapsto t^{-1/n}$ is not affine over any interval, we get that h_K must be constant. This characterizes the balls centered at the origin.

It is also worth observing that inequality (5.5.2) *for bodies centered at the origin* follows from (5.5.3) and the Blaschke–Santaló inequality. Indeed, the latter can be written as

$$\frac{\mathrm{vol}_n(B^n)}{\mathrm{vol}_n(K^\circ)} \geq \frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)},$$

and hence (5.5.2) follows from (5.5.3). Notice that the equality case also follows in this particular context (of bodies centered at the origin), since equality in (5.5.2) implies equality in (5.5.3).

5.6 Exercises

Exercise 5.1. *Let $A, B \subseteq \mathbb{R}^n$ be arbitrarily given non-empty compact sets. Prove that for every $\alpha, \beta > 0$ the inequality*

$$\mathrm{vol}_n(\alpha A + \beta B)^{1/n} \geq \alpha \mathrm{vol}_n(A)^{1/n} + \beta \mathrm{vol}_n(B)^{1/n}$$

holds. Show that if A and B are convex bodies, then equality holds for some $\alpha, \beta > 0$ if and only if A and B are homothetic.

Exercise 5.2. *Let K and L be convex bodies in $\mathbb{R}^n$. Prove that the equality*

$$\mathrm{vol}_n((1-\lambda)K + \lambda L) = \mathrm{vol}_n(K)^{1-\lambda}\mathrm{vol}_n(L)^{\lambda}$$

holds for some $\lambda \in (0,1)$ if and only if L is a translated copy of K.

Exercise 5.3. *Recall that $w(K)$ denotes the mean width of a set $K \in CC^n$. Verify that*

$$|w(K) - w(L)| \leq 2d_{\mathcal{H}}(K, L)$$

for any $K, L \in CC^n$. Conclude that the mean width is uniformly continuous in the Hausdorff metric.

Exercise 5.4. *Prove that the Minkowski symmetrization introduced in Exercise 2.58 preserves mean width. That is, show that for any $K \in CC^n$ and any $u \in \mathbb{S}^{n-1}$, we have*

$$w(M_u(K)) = w(K).$$

Exercise 5.5. *The ℓ-*norm *of a convex body $K \in \mathcal{K}_0^n$ is defined as*

$$\ell(K) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} ||x||_K e^{-\frac{|x|^2}{2}}\, dx.$$

Prove that for any $K \in \mathcal{K}_0^n$, we have the equality

$$2\ell(K) = \ell(B^n)w(K^\circ).$$

Exercise 5.6. *Let $K \subseteq \mathbb{R}^n$ be a convex body.*

(a) *Show that $K - K$ contains two copies of $-K$ (or K) which intersect only on their boundaries.*

(b) *Conclude from item* **(a)** *that*

$$\mathrm{vol}_n(K - K) \geq 2\mathrm{vol}_n(K),$$

which is a weaker bound than the one obtained by the Brunn–Minkowski inequality.

Exercise 5.7 (The Steiner point)**.** *We define the* Steiner point *of a given set* $K \in CC^n$ *as*

$$\mathrm{st}(K) := \frac{1}{\kappa_n} \int_{\mathbb{S}^{n-1}} h_K(u) u \, d\mathcal{H}^{n-1}(u).$$

Of course, this defines a map $\mathrm{st}\colon CC^n \to \mathbb{R}^n$*, which we call the* Steiner point map*.*

(a) *Show that the Steiner point map is* Minkowski linear*, meaning that* $\mathrm{st}(\lambda K + L) = \lambda \mathrm{st}(K) + \mathrm{st}(L)$ *for any* $\lambda \in \mathbb{R}$ *and every* $K, L \in CC^n$*.*

(b) *Prove that the Steiner point map is* translation-equivariant*, meaning that* $\mathrm{st}(x_0 + K) = x_0 + \mathrm{st}(K)$ *for any* $x_0 \in \mathbb{R}^n$ *and all* $K \in CC^n$*.*

(c) *Let* $K \in CC^n$ *be centrally symmetric with respect to* $x_0 \in \mathbb{R}^n$*. Prove that* $\mathrm{st}(K) = x_0$*.*

(d) *Verify that the Steiner point map is* $\mathrm{O}(n)$-equivariant*, meaning that* $\mathrm{st}(TK) = T(\mathrm{st}(K))$ *for any* $T \in \mathrm{O}(n)$ *and every* $K \in CC^n$*.*

(e) *Show that for any* $K, L \in CC^n$ *the estimate*

$$|\mathrm{st}(K) - \mathrm{st}(L)| \leq n d_{\mathcal{H}}(K, L)$$

holds. Conclude that the Steiner point map is uniformly continuous in the Hausdorff metric.

Remark 5.6.1. In [2142], Schneider proved that (for dimension $n \geq 2$) the Steiner point map is the unique map $f\colon CC^n \to \mathbb{R}^n$ which is Minkowski linear, translation-equivariant, $\mathrm{SO}(n)$-equivariant, and continuous in the Hausdorff metric.

Exercise 5.8. *Let* $K, L \subseteq \mathbb{R}^n$ *be convex bodies. Prove that the inequality*

$$\frac{\mathrm{vol}_n(K + L)}{\mathrm{vol}_n(K - L)} \leq \frac{1}{2^n} \binom{2n}{n}$$

holds.

Hint: Notice that $(K + L) - (K + L) = (K - L) - (K - L)$ and use the inequalities for the volume of difference bodies given in Section 5.3.

Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Define

$$M(K, L) := \sup_{x \in K + L} \mathrm{vol}_n(K \cap (x - L)).$$

Notice that the supremum is attained for some $x_0 \in \mathbb{R}^n$, since the function $x \mapsto \mathrm{vol}_n(K \cap (x - L))$ is clearly continuous and $K + L$ is compact. For $\theta \in [0, 1]$, the *θ-convolution body* of K and L is the set

$$K +_\theta L := \left\{ x \in K + L : \mathrm{vol}_n(K \cap (x - L)) \geq \theta M(K, L) \right\}.$$

The objective of the next exercises is to study the early properties of the θ-convolution bodies. For more in this direction, we refer the reader to [85]. Moreover, some interesting generalizations can be found in [78]. We start by investigating some properties of $M(K, L)$.

Exercise 5.9. *Let $K, L \in \mathcal{K}^n$, $\lambda \geq 0$ and $z \in \mathbb{R}^n$. Show that*

(a) $M(K, L) = M(L, K)$,

(b) $M(z + K, L) = M(K, L)$,

(c) $M(\lambda K, \lambda L) = \lambda^n M(K, L)$.

Exercise 5.10. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Prove that if $K = -L$ or if K and L are origin-symmetric, then*

$$M(K, L) = \mathrm{vol}_n(K \cap (-L)).$$

Exercise 5.11. *Let $K, L \in \mathcal{K}^n$, $\theta \in [0, 1]$, $x \in \mathbb{R}^n$ and $\lambda \geq 0$. Prove that*

(a) $K +_\theta L = L +_\theta K$,

(b) $(\lambda K) +_\theta (\lambda L) = \lambda(K +_\theta L)$,

(c) $(x + K) +_\theta L = x + (K +_\theta L)$.

Exercise 5.12. **(a)** *Let $K, L \in \mathcal{K}^n$ and assume that*

$$\sup_{x \in K+L} \mathrm{vol}_n(K \cap (x - L)) = \mathrm{vol}_n(K \cap (x_0 - L))$$

for some $x_0 \in K + L$. Let $\tilde{L} := -x_0 + L$. Verify that

$$\sup_{x \in K+\tilde{L}} \mathrm{vol}_n(K \cap (x - \tilde{L})) = \mathrm{vol}_n(K \cap (-\tilde{L})).$$

(b) *Having item* **(a)** *in mind, prove that for any $\theta_1, \theta_2, \lambda_1, \lambda_2 \in [0, 1]$ such that $\lambda_1 + \lambda_2 \leq 1$, we have*

$$\lambda_1(K +_{\theta_1} \tilde{L}) + \lambda_2(K +_{\theta_2} \tilde{L}) \subseteq K +_\theta \tilde{L},$$

where $\theta \in [0, 1]$ is such that $1 - \theta^{\frac{1}{n}} = \lambda_1(1 - \theta_1^{\frac{1}{n}}) + \lambda_2(1 - \theta_2^{\frac{1}{n}})$.

(c) *Conclude that for any $K, L \in \mathcal{K}^n$ and every $\theta \in [0, 1]$ the θ-convolution body $K +_\theta L$ is a convex set.*

Exercise 5.13. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies such that*

$$M(K, L) = \mathrm{vol}_n(K \cap (-L)).$$

Prove that for every $x \in K +_1 L$, the intersection $K \cap (x - L)$ is a translate of $K \cap (-L)$.

Exercise 5.14. *Verify that the quantity*

$$s(K) := \inf\{\mathrm{vol}_n(K)\mathrm{vol}_n(K - z)^\circ : z \in \mathrm{int}K\}$$

defined for any convex body $K \subseteq \mathbb{R}^n$ is indeed an affine invariant.

Exercise 5.15. *Let $a < 0 < b$, and consider the interval $K = [a, b] \subseteq \mathbb{R}$. Determine $z \in \mathbb{R}$ such that $(K - z)^\circ$ has the minimum length.*

Exercise 5.16 (The Santaló point)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body and define $f\colon \operatorname{int}(K) \to (0,+\infty)$ by*

$$f(z) = \operatorname{vol}_n((K-z)^\circ).$$

(a) *Prove that f is strictly convex.*

(b) *Show that f goes to infinity as z approaches the boundary of K in the following sense: for every $c > 0$, there exists $\varepsilon > 0$ such that if $z \in \operatorname{int}(K)$ and $\operatorname{dist}(z, \partial K) < \varepsilon$, then $f(z) > c$.*

(c) *Conclude that there is a unique point $z_0 \in \operatorname{int}(K)$ such that*

$$\operatorname{vol}_n((K-z_0)^\circ) = \inf\left\{\operatorname{vol}_n((K-z)^\circ) : z \in \operatorname{int}(K)\right\}.$$

(d) *Prove also that f is differentiable and compute $f'(z,v)$ for $z \in \operatorname{int}(K)$ and $v \in \mathbb{R}^n$.*

Hint. Use equality (4.3.8).

Exercise 5.17. *Give a proof of Corollary 5.1.1 restricted to convex bodies based on the Brunn concavity principle, which was introduced in Exercise 4.15.*

The objective of the next exercise is to prove the *Grünbaum lemma* which states, roughly speaking, that any hyperplane section through the centroid of a convex body divides it in two parts of "more or less" the same volume. As hints for this exercise, we suggest the reader to take a look at the proof of Theorem 5.1.2 and to have the Brunn concavity principle in mind.

Exercise 5.18 (Grünbaum)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body having volume 1 and whose centroid is the origin, and let $u \in \mathbb{S}^{n-1}$.*

(a) *Let $M > \max\{h_K(-u), h_K(u)\}$, and define $g\colon \mathbb{R} \to \mathbb{R}$ given as*

$$g(t) = \operatorname{vol}_n\big(K \cap \mathbf{h}^+(u,-t)\big).$$

Verify that $g(t) = 0$ for $t \le -M$ and that $g(t) = 1$ for $t \ge M$.

(b) *Show that*

$$\int_{-M}^{M} t g'(t)\,dt = 0.$$

Using integration by parts, conclude that

$$\int_{-M}^{M} g(t)\,dt = M.$$

(c) *Prove that g is a log-concave non-decreasing function. As a consequence, derive the inequality*

$$g(t) \le g(0)e^{\alpha t},$$

where $\alpha = g'(0)/g(0)$.

(d) *Choosing M large enough such that $1/\alpha < M$, check that $g(0) \ge 1/e$. Conclude that*

$$\operatorname{vol}_n\big(K \cap \mathbf{h}^+(u,0)\big) \ge \frac{1}{e}.$$

The objective of the next exercise is to provide a version of the Rogers–Shephard inequality for the case of differences of distinct bodies.

Exercise 5.19. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies, and recall that we denote $K - L := K + (-L)$.*

(a) *Prove that*

$$K - L = \big\{x \in \mathbb{R}^n : K \cap (L + x) \neq \emptyset\big\}.$$

(b) *Check that for any $x, z \in K - L$ and every $\lambda \in [0, 1]$ the inclusion*

$$(1 - \lambda)\big(K \cap (L + x)\big) + \lambda\big(K \cap (L + z)\big) \subseteq K \cap \big(L + (1 - \lambda)x + \lambda z\big)$$

holds.

(c) *Verify that the support of the function*

$$g_{K,L}(x) := \mathrm{vol}_n(K \cap (L + x))^{1/n}$$

is contained in $K - L$. Show that $g_{K,L}$ is concave in $K - L$.

Remark 5.6.2. The function $g_{K,L} \colon K - L \to \mathbb{R}$ defined above is called the *cross covariogram* of K and L.

Exercise 5.20. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies, and let $\bar{g}_{K,L} \colon \mathbb{R}^n \to \mathbb{R}$ be defined as*

$$\bar{g}_{K,L}(x) := g_{K,L}(x)^n = \mathrm{vol}_n(K \cap (L + x)),$$

for each $x \in \mathbb{R}^n$. Show that $\bar{g}_{K,L}$ is log-concave.

Hint: Write $\bar{g}_{K,L}$ as a convolution of characteristic functions, and recall that characteristic functions are log-concave and that the convolution of log-concave functions is log-concave.

A function $f \colon \mathbb{R}^n \to [0, +\infty)$ is said to be *unimodal* if for each $t \geq 0$ the level set $\{f \geq t\}$ is convex. The objective of the next exercise is to prove *Anderson's theorem*, which, roughly speaking, states that the integral of a non-negative, even and unimodal function over a centrally symmetric convex body K does not decrease when K is translated towards the origin.

Exercise 5.21 (Anderson's theorem)**. (a)** *Let $f \colon \mathbb{R}^n \to \mathbb{R}$ be a non-negative measurable function. Prove that*

$$f(x) = \int_0^{+\infty} 1_{\{f \geq t\}}(x)\, dt$$

for each $x \in \mathbb{R}^n$.

(b) *Let K and L be origin-symmetric convex bodies in $\mathbb{R}^n$. Show that*

$$\int_K 1_L(x + cy)\, dx \geq \int_K 1_L(x + y)\, dx$$

for any $y \in \mathbb{R}^n$ and every $0 \leq c \leq 1$.

(c) *Let $f \colon \mathbb{R}^n \to \mathbb{R}$ be a non-negative, even, unimodal and integrable function, and let K be an origin-symmetric convex body in $\mathbb{R}^n$. Prove that*

$$\int_K f(x+cy)\,dx \geq \int_K f(x+y)\,dx$$

for each $y \in \mathbb{R}^n$ *and all* $0 \leq c \leq 1$.

Exercise 5.22. *Prove the following extension of the Rogers–Shephard inequality:*

$$\mathrm{vol}_n(K-L) \leq \binom{2n}{n} \frac{\mathrm{vol}_n(L)\mathrm{vol}_n(K)}{\max_{x\in K-L} \mathrm{vol}_n(K\cap(L+x))}$$

for any convex bodies $K, L \subseteq \mathbb{R}^n$.

Exercise 5.23 (Simultaneous convergence of symmetrizations)**.** *Let* $K, L \subseteq \mathbb{R}^n$ *be convex bodies, and recall that for each* $u \in \mathbb{S}^{n-1}$, *we denote the Steiner symmetral of a given* $M \in \mathcal{K}^n$ *in* $u^\perp$ *by* $S_u(M)$. *Prove that there exists a sequence* $(u_j)_{j\in\mathbb{N}}$ *such that the iterations* $S_{u_m} \circ \ldots \circ S_{u_1}(K)$ *and* $S_{u_m} \circ \ldots \circ S_{u_1}(L)$ *converge to balls as* $m \to \infty$.

Exercise 5.24. *Using Exercise 5.23, prove, for convex bodies, the version of the Brunn–Minkowski inequality given in Corollary 5.1.1.*

Exercise 5.25. (a) *Let* $u, v \in \mathbb{S}^{n-1}$ *be orthogonal. Assume that* $K \subseteq \mathbb{R}^n$ *is a convex body symmetric with respect to* $v^\perp$, *that is,* $\mathrm{ref}_{v^\perp}(K) = K$. *Show that* $S_u(K)$ *is also symmetric with respect to* $v^\perp$.

(b) *Let* $K \subseteq \mathbb{R}^n$ *be any convex body. Prove that* $S_{e_n} \circ \ldots \circ S_{e_1}(K)$ *is symmetric with respect to the origin.*

(c) *Show that if* $L \subseteq \mathbb{R}^n$ *is a convex body symmetric with respect to the origin, then*

$$L \subseteq \frac{\mathrm{diam}L}{2} B^n.$$

(d) *Let* $K \subseteq \mathbb{R}^n$ *be any convex body. Prove the inequality*

$$\mathrm{vol}_n(K) \leq \frac{\kappa_n}{2^n} (\mathrm{diam}K)^n.$$

This is known as the isodiametric inequality.

Remark 5.6.3. Notice that the isodiametric inequality gives that the maximum ratio between the volume of a convex body in $\mathbb{R}^n$ and the n-th power of its diameter is attained for B^n. Actually, this is attained for any ball, since this ratio is clearly invariant by translations and dilations.

Recall that the *regular n-dimensional cross-polytope* is the polytope

$$P_\Delta^n := \mathrm{conv}\{\pm e_1, \ldots, \pm e_n\} := \{(x_1, \ldots, x_n) \in \mathbb{R}^n : |x_1| + \ldots + |x_n| \leq 1\}.$$

A *cross-polytope of* $\mathbb{R}^n$ is an affine image of the regular n-dimensional cross-polytope. That is, a cross-polytope of $\mathbb{R}^n$ is a set having the form $x_0 + T(P_\Delta^n)$ for some $x_0 \in \mathbb{R}^n$ and some $T \in \mathrm{GL}(n)$.

Exercise 5.26. (a) *Let* $K \subseteq \mathbb{R}^n$ *be a convex body. Prove that there is a unique cross-polytope* P_0 *contained in* K *maximizing the volume of the cross-polytopes contained in* K.

(b) *Assume that* $K \subseteq \mathbb{R}^n$ *is a convex body symmetric with respect to a point* $x_0 \in K$. *Show that the cross-polytope of maximal volume contained in* K *is also centered at* x_0.

(c) *Let* $K \subseteq \mathbb{R}^n$ *be a convex body centered at the origin, and assume that the regular n-dimensional cross-polytope is the cross-polytope of maximum volume contained in* K. *Prove that* $K \subseteq [-1, 1]^n$.

Hint: For item **(a)**, use the same strategy as in the proof of Theorem 9.1.2 ahead. The solution of item **(b)** is similar to the proof of Lemma 9.2.1.

Exercise 5.27 (Mahler)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin. Prove that*

$$\mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ) \geq \frac{4^n}{(n!)^2}.$$

Notice that this inequality is much weaker than that of Mahler's conjecture.

Exercise 5.28. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a continuous function which is strictly positive over $\mathbb{S}^{n-1}$. Use Jensen's inequality to prove that*

$$\int_{\mathbb{S}^{n-1}} f(u)\, d\mathcal{H}^{n-1}(u) \geq (\omega_n)^{(n+1)/n} \left(\int_{\mathbb{S}^{n-1}} f(u)^{-n}\, d\mathcal{H}^{n-1}(u) \right)^{-1/n}.$$

Show that if f is sublinear, then this inequality can be derived from inequality (5.5.3).

For the next exercise, recall that we denote by f^* the *conjugate* of a given function f. See Section 1.4 for the definition and properties.

Exercise 5.29. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a subilinear function which is strictly positive over $\mathbb{S}^{n-1}$. Prove that*

$$\int_{\mathbb{S}^{n-1}} f(u)\, d\mathcal{H}^{n-1} \geq n\kappa_n^{(n-1)/n} \left(\int_{\mathbb{R}^n} e^{-f^*(x)}\, dx \right)^{1/n}.$$

A set $K \in CC^n$ can be represented by its characteristic function 1_K, which is log-concave. This observation leads to extensions of geometric quantities defined for convex bodies to log-concave functions. In [2064], Rotem introduced the following notion of *mean width for a log-concave function* f:

$$\tilde{w}(f) := \frac{4}{n(2\pi)^{n/2}} \int_{\mathbb{R}^n} h_f(x) e^{-\frac{|x|^2}{2}}\, dx, \tag{5.6.1}$$

where h_f is the support function of f, defined in Exercise 1.33 as $h_f := (-\log f)^*$. It is worth mentioning here that we actually use a normalization different to that of [2064] (to be specific, we multiply theirs by 2). Also, in the exercises below, we restrict ourselves to the family

$$\mathrm{LC}_\infty(\mathbb{R}^n) :=$$
$$:= \left\{ f = e^{-\varphi} : \varphi \text{ is closed, convex, proper, and such that } \lim_{|x|\to+\infty} \varphi(x) = +\infty \right\}.$$

in order to avoid technicalities. In the next few exercises, we investigate this definition. Our first task is to show that there exists a constant $c_n > 0$ (which only depends on the dimension) such that

$$\tilde{w}(1_K) = c_n w(K)$$

for any $K \in CC^n$. This justifies why we may see $\tilde{w}(\cdot)$ as an extension of the "geometric" mean width to log-concave functions.

Exercise 5.30. *Let $K \in CC^n$. Prove that*

$$\tilde{w}(1_K) = \frac{2(2\pi)^{\frac{1}{2}}\kappa_n}{(n+1)\kappa_{n+1}} w(K).$$

Hint: You can use the equality

$$\int_0^{+\infty} s^k e^{-as^2}\, ds = \frac{\Gamma((k+1)/2))}{2a^{(k+1)/2}}, \tag{5.6.2}$$

which holds for any $a > 0$ and every non-negative integer k. Use also (4.3.3).

Next, we investigate some other properties of $\tilde{w}$. Among them there is a type of linearity for "reasonable" operations in $\mathrm{LC}_\infty(\mathbb{R}^n)$. We also compute the mean width of the Gaussian function

$$G(x) := e^{-\frac{|x|^2}{2}},$$

which, in some sense, "plays the role" of the unit ball. Indeed, from item **(c)** of Exercise 1.39, we have G, which is the unique self-polar function of $\mathrm{LC}_\infty(\mathbb{R}^n)$. To compute the integral of G, notice that the change of variables $y = x/\sqrt{2}$ leads to

$$\int_{\mathbb{R}^n} G\, d\mathcal{L}^n = \int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}}\, dx = \int_{\mathbb{R}^n} e^{-|y|^2} 2^{n/2}\, dy = (2\pi)^{n/2}, \tag{5.6.3}$$

where the last equality follows from (4.3.1). For the Exercises 5.31, 5.33, and 5.34 below, we use heavily the definitions introduced and investigated in the exercises of Chapter 1. In most parts, the precise references are given. However, if there is some confusion regarding notations, we advise the reader to take a look at those exercises.

Exercise 5.31. (a) *Recall that $f \star g$ denotes the* Asplund product *defined in Exercise 1.26, and that $\lambda \cdot f$ stands for the scalar multiplication introduced in Exercise 1.28. Verify that*

$$\tilde{w}(f \star (\lambda \cdot g)) = \tilde{w}(f) + \lambda \tilde{w}(g)$$

for any $f, g \in \mathrm{LC}_\infty(\mathbb{R}^n)$ and every $\lambda > 0$.

(b) *Show that $\tilde{w}(G) = 2$.*

(c) *Let $x_0 \in \mathbb{R}^n$, $U \in \mathrm{O}(n)$, and put $T(\cdot) := x_0 + U(\cdot)$. Show that for any $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$, we have*

$$h_{f\circ T}(x) = -\langle U(x), x_0\rangle + h_f(U(x))$$

for every $x \in \mathbb{R}^n$. Conclude that $\tilde{w}(f \circ T) = \tilde{w}(f)$. In other words, $\tilde{w}$ is translation invariant and $\mathrm{O}(n)$-invariant, in some sense.

Our next objective is to prove a version of the Urysohn inequality for $\tilde{w}$. This is taken from [2064]. We need some auxiliary results. The first one is known as *Shannon's inequality*, and it is a corollary of Jensen's inequality.

Exercise 5.32 (Shannon's inequality)**.** *Let $p, q\colon \mathbb{R}^n \to [0, +\infty)$ be non-negative Lebesgue integrable functions. Assume that p is strictly positive, that*

$$\int_{\mathbb{R}^n} p(x)\, dx = 1,$$

and that

$$\int_{\mathbb{R}^n} q(x)\, dx > 0.$$

Prove that

$$\log\left(\int_{\mathbb{R}^n} q(x)\,dx\right) \geq \int_{\mathbb{R}^n} \log\left(\frac{q(x)}{p(x)}\right) p(x)\,dx,$$

where we assume that $\log(0) = -\infty$. *Show that equality holds if and only if there exists* $\alpha > 0$ *such that* $q = \alpha p$ *Lebesgue-almost everywhere.*

Hint: Use Jensen's inequality (Theorem 1.2.2 and Remark 1.2.3).

We will also need an important functional version of the Blaschke–Santaló inequality. The proof is not easy enough to be inserted as an exercise. We state it below.

Theorem 5.6.1 (Theorem 1.3 in [136])**.** *Let* $f\colon \mathbb{R}^n \to [0, +\infty)$ *be any measurable function such that*

$$0 < \int_{\mathbb{R}^n} f < +\infty.$$

There exists a point $x_0 \in \mathbb{R}^n$ *such, that letting* $\bar{f} := f(x - x_0)$, *the inequality*

$$\int_{\mathbb{R}^n} \bar{f} \int_{\mathbb{R}^n} \bar{f}^\circ \leq (2\pi)^n \tag{5.6.4}$$

holds, where $\bar{f}^\circ$ *is the polar function of* $\bar{f}$, *as defined in Exercise 1.39. That is,* $\bar{f}^\circ = e^{-h_{\bar{f}}}$.

Despite not being directly necessary for the Urysohn inequality, it is also useful to understand what happens when f is of the form $f(x) = Ce^{-\frac{|x-x_0|^2}{2}}$. This is made next.

Exercise 5.33. *Let* $a \in \mathbb{R}$ *and put*

$$\bar{\varphi}(x) = a + \frac{|x - x_0|^2}{2}.$$

Prove that

$$\int_{\mathbb{R}^n} e^{-\bar{\varphi}} \int_{\mathbb{R}^n} e^{-\bar{\varphi}^*} \geq (2\pi)^n,$$

with equality if and only if $x_0 = 0_{\mathbb{R}^n}$.

Now we are in the position to state the Urysohn inequality for $\tilde{w}$.

Exercise 5.34 (Urysohn's inequality for $\tilde{w}$)**.** *Let* $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ *be such that*

$$\int_{\mathbb{R}^n} f^\circ > 0,$$

where we remind the reader that $f^\circ := e^{-h_f}$. *Prove that*

$$\tilde{w}(f) \geq 4\log\left(\frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} f\right)^{\frac{1}{n}} + 2.$$

Show that equality holds if and only if

$$f(x) = Ce^{-\frac{|x+x_0|^2}{2}}$$

for some $C > 0$ *and some* $x_0 \in \mathbb{R}^n$.

5.7 Notes

The topics discussed in this chapter are really of *central importance* in convexity. Therefore, we decided to use in the notes the same structure of headlines as in the main text, thus leaving out nothing.

5.7.1 The Brunn–Minkowski inequality

For two n-dimensional convex bodies K_1 and K_2, each having unit volume, the Minkowski sum $1/2(K_1 + K_2)$ has at least volume one; and volume one is reached iff K_1 and K_2 are translates of each other. This is the simplest version of the famous *Brunn–Minkowski theorem*. Since the literature on this theme is very extensive, we have to give it up to cite also research papers.

BOOK PARTS

Comprehensive and important references, treating this basic theorem with the famous inequality coming from it, its various extensions, applications, reverse version, etc., are presented in the following books (also this is only a selection): Blaschke's classical book [358] (third part), §11 and §12 of the monograph [446] of Bonnesen and Fenchel, Chapters 3, 4, and 8 like also Appendix B.2 of Gardner's book [930], Chapters 2, 6, and 7 of Thompson's book [2361], Chapter 2 of Burago and Zalgaller [531], Schneider's monograph [2167] (in particular, but not only, Chapters 7 and 9), Chapters 1, 3, and 8 of the monograph [133] of Artstein-Avidan et al., and the long Chapter 3 of the recent text book [1263] of Hug and Weil, treating various aspects of the Brunn–Minkowski theory.

Many more books contain smaller sections dedicated to the Brunn–Minkowski theorem, nevertheless reflecting its importance in a good and elegant way. Examples in this direction are: Chapter 5 of Eggleston's book [803], Chapter 5 of Hadwiger's book [1134], also Chapter 4 in Hadwiger's booklet [1133], Sections 21 and 22 of Benson's monograph [273], Chapter 4 of Guggenheimer's book [1102], Section 6.5 of Webster's monograph [2437], Section 3.5 of Toth [2381], and Section 2.7 of Martini et al. [1713]. Three further books to be mentioned here are Pisier [1974], Brazitikos et al. [496], and Artstein-Avidan et al. [134], but see also Chapter 6 of Krantz and Parks [1466]. In Chapter 7 of Pisier's book [1974], dealing with deep results relating the volume of centred convex bodies with the geometry of finite-dimensional Banach spaces, the *inverse Brunn–Minkowski theorem* is comprehensively studied. In the introductory Chapter 1 of Brazitikos et al. [496], the authors treat among other basic notions from convex geometry (like mixed volumes and classical positions of convex bodies) also the *Brunn–Minkowski inequality and its functional forms*.

And Krantz and Parks [1466] can be considered as a companion of Federer's book [834] studying developments around geometric measure theory and topics like covering theorems, isoperimetric inequalities, Sobolev spaces, Steiner symmetrization, the close relations between convex domains and convex functions, leading also to a nice treatment of *higher-order convexities*. Particularly, in Chapters 6 and 7, the *isodiametric, isoperimetric and Brunn–Minkowski inequalities* are dealt with. And Artstein-Avidan et al. [134] give in Chapter 9 *functional versions of fundamental geometric inequalities*, such as the inequalities of *Brunn–Minkowski and its reverse*, Urysohn, Blaschke–Santaló, Rogers–Shephard, and others. We refrain from a concrete description of the contents of all these books referring to the Brunn–Minkowski inequality, since the detailed description of topics discussed in Gardner's comprehensive survey [929], coming next below, covers really many of these aspects. However, in addition, we strongly recommend to read Section 7.1 of Schneider's book [2167] (with notes, of course), Chapter 3 of the textbook [1263] of Hug and Weil, and Chapters 1, 3, and 8 of Artstein-Avidan et al. [133].

Besides these books, an interesting sequence of *surveys and expository papers* is dedicated to this subject and should also be (partially) presented here. As already said, citing single research papers would go beyond our scope—there are too many of them.

SURVEYS AND EXPOSITORY PAPERS

Above all in this direction, we cite Gardner's beautiful paper [929]. This brilliant and extensive survey reveals the state of the art (in 2002) regarding the Brunn–Minkowski inequality, its extensions, applications, and relations to other basic analytical and geometric inequalities. The latter are, e.g., the *Prékopa–Leindler inequality* (as a reverse form of Hölder's inequality) and the *Brascamp–Lieb inequality*, as well as its reverse. Inequalities of Barthe and Milman (e.g., the *reverse Brunn–Minkowski inequality*) and their role in local Banach space theory are also discussed. It is shown how some of these inequalities, considered as analogues and versions of the Brunn–Minkowski inequality, can be extended to *non-Euclidean geometries* of different types, such as spherical, hyperbolic, Minkowski spacetime, Riemannian, and Finsler geometry. Connections to *algebraic geometry* (since the Alexandrov–Fenchel inequality can be deduced from the Hodge index theorem) are also discussed. A *discrete version of Brunn–Minkowski* opens the door to *discrete mathematics*, with relations to *combinatorics and graph theory*. Also, the list of discussed *applications* is impressive, e.g., the author discusses applications referring to *Wulff shapes* of crystals, *McCann's work on gases* along with *transport of mass* (also used by Barthe in his proof of the Brascamp–Lieb inequality and of inequalities due to himself), *log-concavity* of a convolution of log-concave functions, *probability theory and statistics*, Anderson's theorem on *multivariate unimodality*, the *entropy power inequality* of information theory, and *logarithmic Sobolev inequalities*. Sketches of proofs are presented, and analogues and variants of the Brunn–Minkowski inequality are comprehensively discussed.

Also the survey [1964] of Peri on *analytical aspects of convexity* is a good reference since it also contains (besides many other central themes in convexity) a nicely written part on mixed volumes and the *Brunn–Minkowski theorem*.

Firey observed that, for $p \geq 1$, the sum of pth powers of two support functions is again a *pth power of a support function*. Based on this result, Lutwak (see [1634] and [1635], where Firey is cited) developed respective *analogues of mixed volumes*, getting also integral representations involving certain powers of the corresponding support functions. This leads to certain *analogues of the Alexandrov–Fenchel inequalities and the Brunn–Minkowski theorem*, a result which was also (by different techniques) derived already by Firey. This yields, within this theory, also analogues of notions like *surface area measures*.

In the *Handbook of Convex Geometry*, one can find several surveys discussing also the Brunn–Minkowski inequality from different points of view. As *a consequence of the Alexandrov–Fenchel inequality* and with Blaschke's proof using the procedure of Schwarz rounding, it is presented in the survey [2098] of Sangwine-Yager on mixed volumes. Also, the two handbook surveys [1040] and [1039] of Groemer refer to the Brunn–Minkowski theorem; in Section 3 of [1040] related stability estimates are given (see also Section 3.5 of Toth's book [2381]), and in the second survey, it belongs to the *geometric applications of spherical harmonics*. For example, it is mentioned that a generalization of Wirtinger's inequality was used by Schneider [2153] to verify a stability version of the Brunn–Minkowski theorem for the mean projection measures and more general types of mixed volumes. In their survey [1579] on the local theory of normed spaces, Lindenstrauss and Milman present a deep inequality of Borell estimating the volume of the intersection of a symmetric convex body C and a homothet of a general convex body K from below in terms of the volume of K. This was derived from the classical Brunn–Minkowski inequality. And in the survey [2443] of Weil and Wieacker on stochastic geometry, Vitale's *Brunn–Minkowski theorem for random convex bodies* is mentioned.

The exposition [2155] of Schneider on *polytopes and the Brunn–Minkowski theory* shows a nice collection of results in this spirit; the author concentrates on problems from the *Brunn–Minkowski*

theory which can be investigated particularly successfully when the considered convex bodies are polytopes. The problems treated for polytopes refer to topics like *Minkowski addition, addition and decomposition involving intermediate area measures, volumes of projections, special representations of mixed volumes*, and the *Alexandrov–Fenchel inequality*. This is a welcome completion of the first edition of the author's book, normally discussing these topics for general convex bodies. In a similar spirit, McMullen shows in the final part of [1748] how an algebraic proof of the classical Brunn–Minkowski inequality for convex polytopes can be obtained from the valuation theory on convex polytopes.

In the expository paper [658], Cordero-Erausquin and Klartag discuss the relations between *spectral estimates of Hörmander-type, degenerate Monge-Ampère equations* and, as third point, *geometric inequalities related to log-concavity*, like also the Brunn–Minkowski inequality.

The most relevant functionals in the space of convex bodies (such as quermassintegrals) satisfy a Brunn–Minkowski-type inequality. Having in mind when homogeneous functionals satisfy such a property of Brunn–Minkowski type, Colesanti et al. [639] deal with the reverse question: which properties must an appropriate functional have to satisfy a Brunn–Minkowski-type inequality? They answer this question for certain types of functionals, where the area measure and a support function property play an important role. As a consequence, they get a *characterization of translation invariant, continuous valuations* which are homogeneous of degree $n - 1$ and satisfy a Brunn–Minkowski type inequality.

The survey [1659] of Madiman et al. refers to relations between *information theory and convexity*; this can be represented by *entropy power inequalities* and their reverse versions, for distributions under convexity constraints (like log-concavity). First, the authors survey several classical inequalities that are connected with the entropy power inequalities considered here, such as *the Brunn–Minkowski inequality, its functional version*, and also the *Prékopa–Leindler inequality*. Various versions of the entropy power inequalities for the classical Shannon entropy as well as Rényi entropy are considered, and they present analogues of the reverse Brunn–Minkowski inequality, including to invoke linear transformations (to put random vectors in a "good" position). In addition, the authors study relations between integral and entropic liftings based on the *Blaschke–Santaló inequality* and certain concepts of *duality*.

Berndtsson [296] surveys results on positivity of vector bundles, inspired by the theorems of Brunn–Minkowski and Prékopa–Leindler. He discusses also *applications to complex analysis, Kähler geometry, and algebraic geometry*; see also [295].

5.7.2 The isoperimetric inequality

The *isoperimetric inequality*, considered in the plane, is a geometric inequality involving the perimeter of a planar set and its area, and in n-dimensional space the perimeter is replaced by surface area, and area by volume. It completes the *isoperimetric problem*, again first posed in the plane, asking for a planar figure of smallest possible perimeter when the area is fixed (in higher dimensions analogously formulated for surface area and volume). Again we are only able to present (besides historical remarks) book parts and survey-like papers, since the number of research contributions would be too large.

HISTORICAL REMARKS

A historically famous variant of the isoperimetric problem is the so-called *problem of Dido* (the legendary founder and first queen of Carthage), thus leading us back to ancient times. The solution to the isoperimetric problem is given by the circle, and in higher dimensions by balls, and the solution of the planar case was known already in ancient Greece; but the first rigorous proof of this

statement was obtained only in the 19th century. The isoperimetric problem is one of the oldest mathematical problems from convexity, and also due to very many applications it has a very rich history! We mention a few historical articles reflecting this history with an impressive diversity. Gandz [920] discusses the isoperimetric speculations of the ancient Greeks, and Gericke [951] analyses this early history especially regarding synthetic methods and the important contribution of Zenodorus. Later contributions (showing also the role of scholastic stagnation and the ideas of Steiner) are also discussed. Another important historical reference is Porter's contribution [1988].

Danilova [707] concentrates on the period from the 16th to the 20th century. First it is shown that famous scientists like T. Bradwardine, C. Clavius and G. Galilei stayed within Zenodorus' theory, and that then G. Cramer (1752) made a breakthrough by using a symmetrization method which later was attributed to Steiner. Using this way, he verified an isoperimetric result on polygons. The contributions of further authors are recognized, among them L'Huillier (1782), to whom in [707] the priority in the application of this symmetrization procedure is given. In addition, there is an anonymous contribution to isoperimetric and isepiphanic problems in Gergonne's Annales of 1814; the author was Gergonne himself, and also he used this type of symmetrization before Steiner. Due to these contributions, the name "Steiner symmetrization" is not justified.

We continue with the work [1192] of Herrero Piñeyro, also supporting this fact and gathering (with proofs) methods from elementary geometry for solving the isoperimetric problem. He begins with Zenodorus and ends with two of Bonnesen's strong forms of the isoperimetric inequality. Further papers that have to be mentioned here are Osserman's article [1924] and, comparing the related work of various mathematicians, [372]. Indeed, in the latter paper, an impressive sequence of proof methods and related famous names is given. This list of approaches includes: Zenodorus's polygon proof, Steiner's four-hinge proof, the calculus of variations proof due to Weierstrass, Edler's finite existence proof, Carathéodory's convergence proof, the vector analysis proof of Gromov, Schmidt's projection proof, Lawlor's dissection proof, Hurwitz's Fourier series proof, the power series proof of Carleman, Demar's convexity proof, Bol's parallel curve proof, and Santaló's integral geometric proof. The theoretical steps needed to understand the used techniques are briefly introduced, and nice figures support this clear style. The last basically historical article that we mention here is [628] due to Cinti. It starts with the ancient Greeks and ends with the 20th century, emphasizing also applications (regarding extensions to anisotropic structures, relations to crystal shapes, and the theory of optimal transportation, etc.).

BOOKS AND BOOK CHAPTERS

We start now with books devoted completely, or in form of chapters, to the isoperimetric problem, its extensions and applications. Three classical starting points are the monographs [358] of Blaschke, [445] of Bonnesen, [446] of Bonnesen and Fenchel, and the earlier booklet [2009] of Rademacher and Toeplitz. The first two parts (chapters) of [358] are dedicated to carefully prepared proof details regarding the isoperimetric property, to come up finally with the solution of the isoperimetric problem in the plane and in space. The charming booklet [445] deals with *sharpenings of the isoperimetric problem in two and three dimensions*, and in [446], Section 57 gives a comprehensive report on related results. The famous booklet [2009] treats the isoperimetric problem in Section 18, where Steiner's four-hinge proof is presented.

A classical book on *isoperimetric inequalities in mathematical physics* is [1984] by Pólya and Szegö; among many other topics, the authors present chapters treating geometric subjects like nearly circular and nearly spherical domains, ellipsoids and lens, several types of symmetrization, and membranes and plates.

There are many variants of the isoperimetric problem and similar questions when, e.g., one restricts the problem to certain subfamilies of the family of all convex bodies, like special types of polygons, polyhedra and further classes of geometric figures, yielding then extremal figures having a high degree of regularity. Breathing this spirit, the classical books [851] and [847] of L. Fejes Tóth contain many

results in this direction, and also in the encyclopedic book [1814] by Mitrinović et al. many results of this type are collected and discussed. For example, in [847], such variants of the isoperimetric problem can be found in the subsections II.1.3 (planar case), II.2.1 (spherical case), and II.4.3 (isoperimetric problem for polyhedra). And in [851], the sections §6 of Chapter I, §4 of Chapter II, and §7 of Chapter V have to be emphasized.

The next classical book treating the isoperimetric problem is Hadwiger's monograph [1134], and also his booklet [1133] has to be cited here. Particularly, Chapters 4 and 5 of [1134] contain material related to the variants of the isoperimetric problem and the tools needed for their verifications. This refers to various symmetrization types of compact sets, volume and surface area, Minkowski's relative surface area, the Brunn–Minkowski inequality, the isoperimetric theorem as an inequality for relative surface areas and measures of two arbitrary compact sets, and the isoperimetric property of the sphere. In several chapters before this part of the book, the author prepares the needed tools (e.g., measure theoretic aspects, and also many geometric properties of special convex bodies and polyhedra). Also, Chapter 4 of [1133] contains related material, e.g., the representation of shapenings of isoperimetric inequalities for convex polyhedra with fixed facet number.

In the booklet [1354] of Kazarinoff, a broad discussion of *isoperimetric theorems, Dido's problem, and Steiner symmetrization* is given.

Almost together with or after Hadwiger's books, various monographs on convex sets appeared which contain representations of the isoperimetric problem and related questions, mainly (but not only) prepared for undergraduates. We want to mention some of them. Thus, this list is by no means complete: Section 5 of the book [2487] of Yaglom and Boltyanski, Chapter 6 of Eggleston [803], Part XII of Valentine [2391], and Chapter 4 of Benson [273]; later examples are Section 13 of Lay [1521] and Sections 6.5 and 6.6 of Webster [2437].

Santaló discusses the isoperimetric problem in his book [2103], more precisely in Sections 4 and 5 of Chapter 3 and in Sections 5 and 7 of Chapter 7. There the reader can find *several proofs, strengthenings* (like Bonnesen's inequality), and a discussion about the *isoperimetric deficit*.

The classical monograph [531] of Burago and Zalgaller is an especially comprehensive book regarding *variants and generalizations of isoperimetric inequalities*. The classical isoperimetric inequality and numerous generalizations are discussed in Chapter 1, where inequalities for planar figures and for surfaces are considered. These include the classical improvements of Bonnesen and isoperimetric inequalities on *two-dimensional manifolds*, also with Alexandrov-type results on closed surfaces of bounded curvature, and Osserman-type results on extremal regions in minimal surfaces. Related results on *polyhedral surfaces* are also there. In Chapter 2, the Brunn–Minkowski inequality, its generalizations and the corresponding relations to the isoperimetric problem are studied. The classical case, yielding the Euclidean ball as extremal figure, is treated, and *extensions to spherical, Lobachevsky and finite-dimensional normed spaces* are investigated, too. This chapter contains also various representations of *symmetrization procedures* needed in this framework. In the third chapter isoperimetric inequalities under *different definitions of measure* are studied, and in the fifth and sixth chapter isoperimetric inequalities involving *mean curvature* occur. Chapter 6 contains also results on isoperimetry in *spaces of negative curvature*. Due to the great extent of the field, many of the topics can only be sketched by the authors. Nevertheless, they provide clear statements with detailed backgroundgiving and, in many cases, basic ideas for proofs.

Also, in Section 6.4 of the book [1359] of Kelly and Weiss, the isoperimetric problem is discussed.

The monograph [208] of Bandle stands in the tradition of the classical monograph [1984] treating *isoperimetric problems from mathematical physics* and summarizes respective developments since 1950. First geometric inequalities in Euclidean space and on surfaces with Riemannian metrics are given, and in the second chapter, topics like *Schwarz symmetrization and applications, Green functions, and torsional rigidity* are presented. The next chapter is devoted to estimating *eigenvalues of elliptic operators* under certain boundary conditions, where also Beltrami operators and general second-order differential equations occur. The last chapter contains *linear and nonlinear boundary value problems*

of elliptic and parabolic types. With these contents this important book shows PDE as an important application field for isoperimetric problems.

Here we mention also the book [1557] of Li, Simon, Zhao and Hu on global differential geometry. Its Chapter 7 is an introduction to *equiaffine geometry*, including also the *affine isoperimetric inequality* and Steiner symmetrization.

Buldygin and Kharazishvili discuss in [526] mainly *applications of the Brunn–Minkowski inequality*, some of them far away from classical convexity. However, in the introductory part of this book, they also study *convex polyhedra and isoperimetric inequalities*.

In Groemer's book [1042], the isoperimetric inequality and extensions of it occur two times: first as *geometric applications of Fourier series* (Chapter 4: two-dimensional applications, including two approaches to the isoperimetrical inequality following Hurwitz), and then as *applications of spherical harmonics* (Chapter 5: strengthened versions of isoperimetric inequalities which can be interpreted as stability results). The results occuring in these chapters are surveyed in Groemer's handbook article [1039], see also [1040].

Thompson's book [2361] on the geometry of finite dimensional, real Banach spaces deals with the isoperimetric problem two times: In Chapter 4, the planar case is studied, and in Chapter 5, the higher-dimensional situation. In the notes to Chapter 4, it is correctly said that Busemann solved the problem in the general case. In the plane, the solution to the problem (called *isoperimetrix*) is homothetical to the polar reciprocal of the unit circle, rotated through $\pi/2$. And in higher dimensions, Thompson discusses the problem based on different notions of Minkowskian area, namely: the *Busemann definition*, the *Holmes–Thompson definition*, the *Benson definition*, and the *definition by means of the Löwner ellipsoid*. It turns out that in the first case (together with the second perhaps being most popular in this count) the isoperimetrix is the *polar of the intersection body of the unit ball*, and in the second case, the isoperimetrix is the *projection body of the polar of the unit ball* (and thus a zonoid). Various nice figures of three-dimensional examples embellish the fifth chapter of [2361]. Clearly, the two-dimensional solution as described above will be obtained by using the same two definitions (i.e., the planar versions of intersection and projection body).

The main purpose of Bogachev's important monograph [418] is the presentation of the state of the theory of *Gaussian measures*; one very important and interesting part deals with Gaussian measures on linear topological spaces. In Chapter 4, the author discusses inequalities and related estimates. Gaussian symmetrizations yield Ehrhard's inequality and *isoperimetric inequalities*, the latter being a main ingredient in estimating large deviations. This is the right place to cite also the paper [1514] of Latala, excellently surveying important inequalities related to Gaussian measures. These are mainly the isoperimetric inequality, Ehrhard's inequality, and Bobkov's inequality.

Bobkov and Zegarlinski [378] study (among other topics) *isoperimetric inequalities for certain probability measures*. First, modified forms of logarithmic Sobolev inequalities are investigated, and then such inequalities are studied in the class of *uniform distributions on convex bodies*, in Euclidean spaces. This way, the authors particularly refine a known result of Kannan, Lovasz and Simonovits on *Cheeger isoperimetric constants*.

The book [1371] of Kesavan provides an introduction to *symmetrization methods and isoperimetric problems*, emphasizing the role that symmetry plays in this framework. First basic notions are clarified, among them the classical isoperimetric inequalities in two and three dimensions, the *symmetrization of Schwarz*, and the relationship between the classical isoperimetric inequality and *Sobolev's inequality* or that of Pólya and Szegö. Then the relations of the latter to the classical isoperimetric inequality are considered, like also *optimization problems* that gave much motivation for the classical *developments of symmetrization techniques*. Also, Talenti's theorem on a priori bounds on a nonlinear elliptic problem from an appropriately symmetrized problem is presented, and various isoperimetric inequalities involving positive solutions of some nonlinear problems are shown, too. The book has a suitable level for an advanced graduate course.

Also, in Gruber's monograph [1070], one can find material on the isoperimetric inequality and related results, e.g., *isoperimetric problems for polytopes, lattice polytopes and applications to linear optimization*; see Sections 8.3 and 9.2.

Ritoré and Sinestrari present in [2036] expanded revisions of their lectures given in Bellaterra (Barcelona) in 2008, during the "Advanced Course on Geometric Flows and Hyperbolic Geometry". The second part, interesting for us here and written by Ritóre, has the headline "Geometric Flows, Isoperimetric Inequalities and Hyperbolic Geometry". Two-dimensional isoperimetric inequalities are proved, and known results on applications of the *mean curvature flow* are reviewed, to obtain analogous *isoperimetric inequalities in higher dimensions*. Related applications to *hyperbolic geometry* are observed.

The monograph [473] of Boucheron et al. refers to *concentration inequalities* for functions of independent random variables. As is well known, these inequalities have a number of applications in *quantitative analysis, high-dimensional geometry and asymptotic convexity, probability theory, mathematical statistics, random matrix theory, algorithmic theory*, and further fields. We cite here also the book [1097] of Guédon et al., which nicely shows the connections between concentration inequalities and asymptotic convexity. In Chapter 7 of [473], some aspects of the rich relationships between *isoperimetric problems and concentration inequalities* are presented. It is shown how the classical isoperimetric theorem follows from the Brunn–Minkowski theorem, and the authors study *isoperimetric inequalities on the surface of the Euclidean ball*, the *vertex isoperimetric theorem on the binary hypercube* and its relationship to *concentration inequalities*, and *Talagrand's convex distance inequality* and its applications to convex Lipschitz functions and a bin packing problem. The exercises contain interesting problems on, for example, further inequalities, convex distance and configuration functions. The authors succeed with a clear exposition of the fundaments, and in the same style they go up to recent results. Students like also experienced researchers will benefit from reading this important book.

In the first chapter of the basic monograph [133] of Artstein-Avidan et al., the Brunn–Minkowski inequality is used to prove an *isoperimetric inequality for convex bodies*, and in the second chapter, Ball's *reverse isoperimetric inequality* is proved. The third chapter treats *isoperimetric inequalities in different spaces* and also related *isomorphic versions*, e.g., the classical isoperimetric inequality on the sphere and an isomorphic version of it, the isoperimetric inequality for Gaussian measure and the isoperimetric inequality in the discrete cube.

The book [323] collects problems from *discrete geometry*; we cite it here since it contains inequalities of isoperimetric-type holding for special families of convex polytopes. E.g., the n-dimensional convex polytopes all whose m-faces touch the Euclidean sphere present such a family. This setting yields also the problems of finding the polytopes of minimum/maximum volume which are circumscribed about/inscribed in a sphere.

Also, in the very recent monograph [1263] of Hug and Weil, it is shown how the *isoperimetric inequality* can be obtained as consequence of the Brunn–Minkowski theorem (see Section 3.4), and in Section 3.6, the *Steiner symmetrization* is excellently discussed (as Example 3.2 there, the isoperimetric inequality is once more taken into consideration). It is also shown that the isoperimetric inequality and the *isodiametric inequality* both follow, via Minkowski's inequality, from the Brunn–Minkowski inequality.

We found only one example of *proceedings* of a conference devoted explicitly to isoperimetric-like inequalities. It is [1238] of Houdré et al., and besides various papers close to convexity it contains also some contributions related to quantitative isoperimetric inequalities, isoperimetric sets of radially symmetric measures, semigroup proofs related to concentration and isoperimetry, and isoperimetric bounds on convex manifolds.

ESSENTIAL SURVEYS

As a connecting link between the monographs [1984] and [208], the comprehensive survey [1951] of Payne reflects the state of the art (1967) regarding *isoperimetric inequalities in mathematical physics* and their (also numerical) *applications*. More precisely, the studied inequalities refer, e.g., to the following topics: electrostatic capacity, polarization, virtual mass, vibration of membranes with fixed and with free boundary, torsional rigidity, and vibration of incompressible elastic media. The role of uniqueness, stability and covergence criteria is emphasized, and many new methods are presented (not always with rigorous verification). Interesting conjectures are posed, and the bibliography (almost 200 references) shows impressively the progress in the field since [1984]. Bandle [209] surveys *geometrical and analytical isoperimetric inequalities*, mostly in the plane and on two-dimensional surfaces. Her survey appeared 16 years after [1951] (thus also after Bandle's book [208]) and is written with similar contents and motivations, namely to show also the state of the art (1983).

Completing the contributions in the monographs [851], [847], and [323] treating isoperimetric-like problems restricted to certain classes of polytopes, the survey [849] of L. Fejes Tóth nicely discusses the circle of ideas around *Steiner's isoperimetric conjectures for the Platonic polyhedra*. Analogous topics in three-dimensional *elliptic and hyperbolic space* are also discussed.

Petty [1968] exposes a variety of interesting geometric extremal problems connected with *affine isoperimetric inequalities*, in a setting including both Minkowski geometry and relative differential geometry. Among the applications, a lower bound on the isoperimetric deficit is obtained, and characterizations of ellipsoids and bodies of constant brightness are derived, too.

Also, in the short but interesting exposition [1880] of Naveira and Segura problems on isoperimetric inequalities are discussed. For example, the authors discuss *relations between geodesic balls and the isoperimetric problem*, obtaining an expression relating the scalar curvature to the isoperimetric deficit of geodesic balls in arbitrary Riemannian manifolds. Mixed volumes play also a role in this article.

In [2337], Talagrand collects (mathematical and physical) problems that can be formulated like isoperimetric problems, yielding then the corresponding *isoperimetric inequalities*. The author studies such an approach to probabilistic problems and gives a related overview, showing that in such a way various sharp results can be obtained.

Wills [2462] surveys *isoperimetric-type inequalities in the spirit of discrete geometry*, involving quermassintegrals of convex bodies and satisfying certain lattice-point constraints, see also Section 3 of the handbook article [1034] of Gritzmann and Wills.

We come now to some articles from the "Handbook of Convex Geometry", starting with the paper [2340] of Talenti. The author gives a nice and thorough treatment of the *isoperimetric problem in its classical form*, thus referring to the n-dimensional Euclidean space. The ancient part of the story and the names of important later contributors are given, and for the planar version the author presents an essentially analytic proof going back to Hurwitz and Lebesgue. For the general case in higher dimensions, Talenti uses the concept of *Caccioppoli sets* (i.e., sets of finite perimeter) and compactness properties of the considered sets which follow from this, and he clarifies existence. The concept of Caccioppoli sets is nicely introduced, and *Steiner symmetrization* is used to show that the ball is the unique set of smallest surface area enclosing the given volume. This paper is not a survey, but a successful representation of a famous theorem in mathematics. Nevertheless, the reader finds about 60 references. More a survey is Talenti's second contribution in this handbook, namely the article [2339]. Referring mainly to *isoperimetric theorems in mathematical physics*, this article is in the tradition of the books [1984] and [208], as well as the survey [1951] by Paine. Besides or combined with interesting historical comments, a review of symmetrization and rearrangement techniques is given, and physical quantities mainly studied are capacity, torsional rigidity and the fundamental frequency of vibrating membranes and plates.

Florian's survey [885] reflects many results and open problems around geometric inequalities for convex sets, comparing quantities like perimeter, area, volume, surface area, total mean curvature, total edge length, inradius and circumradius in two- and three-dimensional Euclidean, hyperbolic or spherical spaces. Sometimes an outlook to higher dimensions is given. Including the section with

isoperimetric results on polyhedra of special combinatorial type, various results in this survey are strongly related to the isoperimetric problems in discrete geometry.

Lutwak's survey [1633] gives a very interesting, pioneering summary of deep and important *affine isoperimetric inequalities* which are combined with many essential and central convexity notions (like intersection body, projection body, centroid body, etc.). Additionally, they create also various natural, but deep and still open problems. These inequalities create interesting cross-connections within the structure that they form. The excellent selection of Lutwak refers to *sharp* affine isoperimetric inequalities (and not to *asymptotic* ones, as we see them also, e.g., in the books [1974] of Pisier and [133] as well as [134] by Artstein-Avidan et al.). Famous examples of such inequalities discussed in [1633] are: *Petty's projection and affine projection inequalities, Busemann's intersection inequality, Petty's geominimal surface area inequality, the affine isoperimetric inequality*, and the *Blaschke–Santaló inequality*. Open problems (e.g., treating *affine quermassintegrals* or inequalities complementary to given ones) are gathered in the last section. Related developments after 1993 and up to date, and also parallel developments in the asymptotic sense, can be followed via numerous articles in the GAFA series, and also in books like [133] and [134].

In his handbook article [1040] on *stability of geometric inequalities*, Groemer discusses the isoperimetric inequality in two sections. In Section 2.1, he shows that one can interpret the well-known *Bonnesen inequality* as a stability result for the isoperimetric inequality (planar case). And Section 5.2 contains results on the *stability of higher-dimensional isoperimetric inequalities*, based on quermass-integrals.

Also, the second handbook article [1039] of Groemer, treating *Fourier series and spherical harmonics in convexity*, contains related results on the isoperimetric inequality. In Section 2.1, the auhor underlines that Hurwitz' proof of the isoperimetric inequality is possibly the earliest example for using Fourier series as tool in a proof of a geometric theorem; therefore, this proof is also given there, and later also a strengthened version is derived. After this, in Section 3.7, results of Groemer and Schneider are cited which yield stability versions of a whole sequence of inequalities depending on *mean projection measures* (including the classical isoperimetric inequality).

The survey [325] of Bezrukov deals with isoperimetric problems in discrete spaces, more precisely with different methods and results of such type within the vertex set of the n-cube. He discusses the problem to characterize the sets with smallest boundary and largest set of inner points with respect to some suitable metric (usually the Hamming one). A comprehensive survey of related results is given. Also, Minkowski's notion of surface area and the Brunn–Minkowski inequality in discrete sense play a role here, and three applications from computer science are discussed, too.

Escobar [817] exposes relations between the *classical isoperimetric inequality*, eigenvalues for the Laplacian with respect to the Neumann boundary condition, and *Sobolev inequalities on curved spaces*. These results imply interesting properties of compact Riemann manifolds.

In his handbook article [200] on convexity and functional analysis, Ball presents various interesting results on important inequalities, volumes of sections, plank problems, etc.; most important here is: in the first part, he discusses (besides the Brascamp–Lieb inequality) the famous *reverse isoperimetric inequality*.

Another article from the same handbook is Schechtman's survey [2129] on *concentration results*. The article also deals with *approximate isoperimetric inequalities* and respective applications in the asymptotic theory of finite-dimensional spaces. He introduces the *isoperimetric problem in probability metric spaces*. In Section 2, various inequalities for probability spaces and some geometric inequalities are treated, among them *Levy's isoperimetric inequality* on the Euclidean sphere.

The survey [2035] of Ritoré and Ros presents isoperimetric problems in 3-space. After presenting briefly some proofs of the classical problem, they offer a version in the spirit of "free boundary problems", and then discuss the following problem: Among surfaces enclosing and separating two given volumes, find the ones with the least possible total area. The solution is the "double bubble", and the authors discuss new ideas, conjectures and questions around this subject.

In [235], the isoperimetric problem is formulated in the general setting of a metric space, supplied with a Borel (probability) measure. Barthe surveys recent developments on the *isoperimetric problem for product probability measures*. Also, Steiner symmetrization is used, and an interesting open problem on the isoperimetric properties of half spaces in connection with isotropic convex bodies is posed.

Vempala [2400] surveys progress in a field that he calls *algorithmic convex geometry*; it includes algorithms for sampling, optimization, etc., and it uses mathematical tools like *isoperimetric and concentration inequalities*.

The paper [271] of Benguria is based on lectures that the author gave at the Arizona School of Analysis in 2010 and during the International Conference in Carthage (Tunisia, 2010) on the *isoperimetric problem* of the ancient Queen Dido. The lectures given as an intensive course for graduate students in connection with the International Conference in the Tunis Science City are published as [272]. It turns out that [271] and [272] are predominantly identical. The mathematical contents mainly refer to *isoperimetric-type inequalities in the spirit of mathematical physics*.

5.7.3 Difference bodies and the Rogers–Shephard inequality

The *difference body* DK of an n-dimensional convex body K, which is clearly centred at the origin, can be defined in different ways: It is the Minkowski sum of K and its reflection $-K$ at the origin (thus, $DK = K + (-K)$), the union of all translates of K containing the origin, the union of all translation vectors such that the corresponding translates of K have non-empty intersection with K, or the union of all difference vectors of position vectors of points from K. One can also get it in the following ways: as union of all translates of affine diameters of K containing the origin, or by its support function if it coincides with the width function of K. Further on, it is the homothetical copy (by factor 2) of the *central symmetral* of K, which therefore is the only convex body centred at the origin that has the same width function as K. For all these types of generation, see also Plate 5 above.

BOOK PARTS AND SURVEYS

Broader discussions of difference bodies and central symmetrals can be found in the following monographs: Section 33 and Section 53 of Bonnesen-Fenchel [446] (there DK is called the *vector body* or the width body of K, like also in the booklet [2404] by Vincensini, which collects the material of seven papers of Vincensini on related topics), §20 of the monograph [1531] of Leichtweiss, Section 3.13 of his second book [1537], Section 3.2, Note 3.5 and Note 8.12 of Gardner's book [930], Section 2.6 of Groemer's monograph [1042], Sections 9.5 and 30.1 of Gruber's book [1070] (also considering packings of central symmetrals), Section 10.1 and its notes of Schneider's monograph [2167], Section 3.8 of Toth's book [2381] (in view of measures of symmetry), and Chapters 10 and 13 of [1713]; see also the surveys [1710] of Martini (where the difference body is studied in view of many informations about one-dimensional inner and outer cross-section measures) and [5] of Abardia-Evéquoz.

The most comprehensive and informative presentation of results around difference bodies seems to be that one in Section 10.1 of Schneider's book [2167], possibly completed by the mentioned parts of Gardner's book [930]. Important topics and keywords which are strongly related to difference bodies and discussed in these references are: their properties important for *geometric tomography* (based on one-dimensional inner and outer cross-section measures), relations to other bodies associated with a given convex body via section or projection functions (like, e.g., *projection bodies* or *cross-section bodies*), the *covariogram, irreducible sets*, and *bodies of constant width*, also in normed spaces.

Here we mainly try to collect related references that are not given in these sources. In the *Handbook of Convex Geometry* [1077], difference bodies and central symmetrals occur mainly in the expositions of Sangwine-Yager (on symmetrizations in connection with mixed volumes), Groemer (on stability of

geometric inequalities), Lutwak (on affine isoperimetric inequalities), and Heil and Martini (on special convex bodies, mainly in connection with characterizations of simplices or central symmetry).

CHARACTERIZATIONS

First we collect *characterizations of the difference body operator*. Gardner, Hug, and Weil [934] systematically study and generalize an interesting operation called M-addition; they characterize continuous and GL(n)-covariant operations between compact convex sets in terms of this operation which is based on difference bodies. This leads to certain origin-symmetrizations, i.e., to give a mapping from given compact convex sets (or even starshaped sets) to origin-symmetric compact convex sets (or origin-symmetric starshaped sets, respectively). A nice characterization of the difference body operator is obtained by Gardner et al. in [934]: an origin-symmetrization of compact convex sets is continuous in the Hausdorff metric, GL(n)-covariant, and translation invariant iff it is of the form λDK for some non-negative λ and DK as difference body of the given set K. Abardia-Evéquoz and Saorín Gómez [7] present a general classification result for operators of type $sK + t(-K)$ on convex bodies K, with non-negative factors s and t. This implies that the difference body operator is the only continuous and GL(n)-covariant operator from the space of general convex bodies to that of origin-symmetric convex bodies satisfying a Rogers–Shephard type inequality. The paper also contains nice and illustrative examples, clearly showing the necessity of all mentioned conditions. And in [6], Abardia-Evéquoz classify all continuous, translation invariant Minkowski valuations Φ for which upper and lower bounds on the volume of $\Phi(K)$ in terms of the volume of the given convex body K are given. These results extend known characterization results on difference bodies.

THE ROGERS–SHEPHARD INEQUALITY AND RELATED INEQUALITIES

We discuss now the famous *Rogers–Shephard inequality* (and generalizations of it) treating the upper bound on the affine invariant ratio of the volumes of a convex body K and its difference body DK. It is well known that this is a *measure of symmetry* (see [1080] and [2381]), thus having for K exactly the centrally symmetric bodies (lower bound) and the simplices (upper bound) as extremal cases; the case of simplices was verified by Rogers and Shephard in [2054] and is therefore known as *Rogers–Shephard inequality*. This famous inequality was extended by the same authors in [2055] to any pair K, F of convex bodies, with equality iff $K = -F$ is a simplex. A strengthening (in terms of mixed volumes) of the Rogers–Shephard inequality is the still unsettled *Godbersen conjecture*, see [985] of Godbersen. Artstein-Avidan et al. [132] improve the best upper bound for it, and they give a new conjecture which, if true, would imply Godbersen's conjecture. Chakerian [591] gives a new proof of the Rogers–Shephard inequality clarifying its relation to the Brunn–Minkowski theorem, and he generalizes it using methods of integral geometry. K. Böröczky [388] investigates various generalizations of the Rogers–Shephard inequality and the Godbersen conjecture. And in Sections 5-6 of that paper, the mean projections and the affine surface area of the difference body are studied.

Gardner and Zhang introduce in [943] the *radial pth mean body* R_pK of an n-dimensional convex body K as follows: In direction u, for $p > -1$ the radial function of R_pK is the pth mean of the distances of points of K to the boundary of K in direction u. Extending this concept to the cases $p = -1, \infty$ by letting R_pK in the first case be the *polar projection body*, and in the second, the *difference body* DK of K, the authors obtain an interesting sequence of inclusions of these bodies associated with K with corresponding simplex characterizations, yielding also the Rogers–Shephard inequality and the *reverse Petty projection inequality* with their equality cases. Also, the importance of the bodies R_pK for the *covariogram function* of K is discussed. Interested in continuum percolation, Jonasson [1325] studies a version of the respective Boolean (or Poisson blob) continuum model where, at each point of a Poisson point process in the Euclidean plane with a certain intensity, a copy of a given compact convex

set with fixed rotation is placed. His investigations also yield a generalization of the Rogers–Shephard inequality, which is then used to obtain inequalities applicable to continuum percolation models.

Alonso-Gutiérrez et al. [82] derive certain Rogers–Shephard type inequalities for convex bodies. It is possible to recover from this functional versions of the classical Rogers–Shephard inequalities. And in [83], Alonso-Gutiérrez et al. obtain Rogers–Shephard inequalities for the *Wills functional* of a convex body K (defined as the sum of the intrinsic volumes of K). For getting them, they first prove some results related to the Wills functional of the projections of K onto hyperplanes.

The *stability of the Rogers–Shephard inequality* is the main topic of Böröczky's paper [392], see also [388]. The author derives lower and upper bounds of the Banach–Mazur distance of the given convex body to a simplex when the two sides of the inequality differ by a factor of $1-\varepsilon$. It is shown that these bounds are essentially optimal. Stability versions of further inequalities are also proved, e.g., referring to the polar projection body. Here we cite also two older papers of Groemer on stability results treating central symmetrizations. In [1038] Groemer deduces, as a consequence of other results, a stability result for the behaviour of the quermassintegrals under the procedure of central symmetrization, and in [1043], he deduces a sharp upper bound on translative L_2-distance between a plane convex body and its central symmetral.

Based on the notion of difference function for real-valued, non-negative, and log-concave functions, Colesanti [636] deals with several inequalities which can be viewed as *functional forms of the Rogers–Shephard inequality* in convex geometry. Equality conditions are clarified, and another inequality, concerning the volume of $K\cup(-K)$, where K is a convex body containing the origin, is also investigated. Using embeddings of convex bodies into log-concave functions obtained by taking the characteristic function or the exponential of the pointwise negative of the Minkowski functional, in [79] and [74] by Alonso-Gutiérrez et al., the concept for two convex bodies given by Rogers and Shephard [2055] is extended to log-concave functions. The authors establish several inequalities involving *integrals of log-concave functions* and fix equality cases. And in [73], a *reverse Rogers–Shephard inequality for log-concave functions* is proved. Also, Roysdon [2066] obtains Rogers–Shephard type inequalities for certain concave functions, as consequences of his sectional Rogers–Shephard type inequality for measures with radially decreasing densities. In [78], Alonso-Gutiérrez et al. study the class of *convolution bodies*, and it turns out that some derived volume inequalities, obtained there for the limiting convolution body and for the sum of two convex bodies, can be seen as extensions of the Rogers-Shephard inequality.

We continue with *further related inequalities*. In [1181], *inequalities connecting successive minima with inner and outer radii of symmetric convex bodies* are established; these imply also inequalities treating general convex bodies in which their difference bodies occur. Besides results on *quermassintegrals and dual quermassintegrals of difference bodies*, which also can be derived from more general and known results, the paper [2421] of Wang contains some nice related characterizations of special types of convex bodies, such as *simplices, ellipsoids and centrally symmetric convex bodies*. Rudelson [2073] estimates the volume of the section of the difference body DK by a linear subspace S via the maximal volume of sections of a given convex body K parallel to S.

GENERALIZATIONS OF DIFFERENCE BODIES AND VALUATIONS

Now we discuss results on L_p *difference bodies*, *Orlicz difference bodies*, and related *valuations*. The notion of L_p-difference body of a convex body as the Firey L_p-combination of the body and its reflection at the origin was introduced by Lutwak, and (complementary to asymmetric centroid bodies and asymmetric projection bodies, see [1126]) it was extended to the more general asymmetric L_p difference bodies by Wang and Ma [2427]. Bianchini and Colesanti [335] establish the L_p difference body inequality for two dimensions, with equality if the given convex figure is a triangle with one vertex at the origin (their methods are not extendable to higher dimensions). In [2427], several properties of asymmetric L_p difference bodies are derived, e.g., the extremal values of the volumes

of asymmetric L_p difference bodies and their polars, respectively. We now come to *Orlicz difference bodies*. And here, the important work [935] of Gardner, Hug, and Weil should be cited; namely, the development of the *Orlicz Brunn–Minkowski theory*, introduced by Lutwak, Yang, and Zhang in [1640] and [1641], was strongly influenced by [935], since in the latter paper, an appropriate Orlicz addition is introduced and the Orlicz Brunn–Minkowski inequality is established. Based on a more specific notion of *Orlicz addition* and analogously to the main result from Bianchi and Colesanti [335], Jin and Yuan [1317] and, independently, Chen, Xu, and Yang [607] established two-dimensional Orlicz versions of the difference body inequality; equality in the plane holds again for triangles with one vertex at the origin. The lower bound on the volume ratio of the difference body and its original body was also investigated in the Orlicz setting. Using the Orlicz–Brunn–Minkowski inequality, Shi, Wang, and Ma [2235] give an Orlicz analogue of this lower bound, and for the same purpose Guo and Chen [1108] successfully use *Steiner symmetrization* to obtain also a characterization of Orlicz difference bodies. Classifying the $SL(n)$-compatible *Orlicz valuations*, Li and Leng [1561] show that the Orlicz projection body operator, the Orlicz centroid body operator, and the *Orlicz difference body operator* are not Orlicz valuations. The latter property (namely, that the Orlicz difference body operator is not an Orlicz valuation) is essential for characterizing the identity operator and the reflection operator. In [406], K. J. Böröczky and Ludwig give a complete classification of the translation invariant Minkowski valuations on lattice polytopes that intertwine the special linear group over the integers. It turns out that in the equivariant case, the only such valuations are *generalized difference bodies combined with multiples of the discrete Steiner point*. Finally, we mention here the paper [1601] by Lou and Li, where the more general notion of *asymmetric Orlicz difference bodies* is introduced.

COMPLEX DIFFERENCE BODIES

There are also results on *difference bodies in complex vector spaces*, where two convex bodies are involved. Abardia [1] gives a *complete classification of continuous, translation invariant Minkowski valuations* (based on Minkowski addition) which are covariant under the complex special linear group; it turns out that these valuations are *complex difference bodies*. Also, the paper [3] of Abardia-Evéquoz and Saorín Gómez treats the notion of complex difference body $D_C K$ of a convex body K (with respect to a convex body C). The authors present a systematic study of the geometric properties and inequalities which are satisfied by $D_C K$, in particular, also studying properties that are different to their analogues established for the classical concept of difference bodies. For example, they show that the dimension of $D_C K$ depends on the position of K and how to get it precisely, yielding the *non-existence of a Rogers–Shephard type inequality*. Despite this latter property, nevertheless sharp lower bounds for quermassintegrals are obtained. Also, in general, $D_C K$ is no longer centrally symmetric. Via *spherical harmonics*, characterizations of those bodies for which the complex difference body is a ball are derived, and it is shown that $D_C K$ is a polytope iff the two involved bodies are polytopes. Further conditions for $D_C K$, for instance, to be a ball, a zonotope, or a set of constant width are derived, and among other tools also spherical harmonics are used in the proofs.

COVARIOGRAMS AND RELATED TOPICS

We come now to the notion of covariogram, also comprehensively discussed in Section 10.1 of [2167] and in Note 9.3 of Gardner's book [930]. The *covariogram* of a convex body K in Euclidean space is the function $g_K(x)$ that equals the volume of the intersection of K and $K + x$. Precisely when this intersection is non-empty, the point x belongs to the *difference body of* K. In 1986, Matheron [1730] conjectured that the covariogram determines uniquely a planar convex body, up to translation and reflection. A complete positive answer for the planar case was given only in 2009 by Averkov and Bianchi in [159]. And as the headline says, in [269], Benassi et al. extend the class of sets in which a planar convex body is determined by its covariogram. Gardner, Gronchi, and Zong [933] develop a

theory for subsets of integer lattices which is analogous to the field *geometric tomography* manifested in Gardner's book [930]. This "parallel theory" is more general than *discrete tomography* (which is normally restricted to discrete X-rays). Due to their aim, the authors study topics as, for example, Minkowski sums, reconstruction of a convex lattice set from its projections onto hyperplanes, *discrete and continuous covariograms*.

For dimensions $n > 2$, there are both positive and negative results regarding Matheron's conjecture, and Averkov and Bianchi [158] try to sharpen existing results by considering subsets of the support of the covariogram such that the restriction to these subsets identifies certain geometric properties of K. Some results, positive or negative, are proved for arbitrary n, and others are brought into the Baire category setting. Bianchi and Averkov [160] study also extended concepts of covariograms as *valuations*; namely, the function $g_K\varphi$, associating to each x the value $\varphi(K \cap (K + x))$, yields a real-valued, even, translation invariant valuation φ, monotone regarding inclusions and vanishing on singletons. For example, this valuation can be a *quermassintegral* (e.g., surface area or mean width). A positive solution to the analogous problem in this setting is provided under different assumptions, including the case where K is a polygon and φ is either strictly monotone or is the width in a given direction. It is also shown that for every $n > 2$ there are examples where K is determined by its covariogram, but at the same time, not by its width covariogram.

Older than the notion of covariogram is that of the *convex hull function* going back to the famous paper [832] of Fáry and Rédei. Instead of the volume of the intersection of K and $K + x$, one considers the volume of the convex hull of the union of K and $K + x$. In the two very recent papers [1235] and [1237], Horváth and Lángi somehow revitalize such volume problems (namely, besides looking at this convex hull function also studying some other volume problems). There are many long-standing open problems, and in [1235] Horváth concentrates on the three-dimensional case.

Agaronyan and Oganyan [22] derive a formula for calculating the probability that a random segment of given direction and length lies entirely in an n-dimensional convex body in terms of the covariogram of that body. Covariograms of finite sets in n-space are functions whose value at x is the cardinality in the intersection of the convex body K and $K + x$. Two sets are said to be *homometric* if their covariograms coincide, and *non-trivially homometric* if they do not coincide up to translations and point reflections. In [161], pairs of non-trivially homometric sets are studied, which are *lattice-convex* (i.e., intersections of lattices with convex sets). We emphasize once more that Section 10.1 and its notes from Schneider's monograph [2167] cover many further results in this direction, also about covariograms.

FURTHER RESULTS

Results on further *bodies associated with a given convex body* follow now. Makai and Martini [1674] have shown that if the *chordal symmetral* (see Section 5.1 of Gardner's book [930]) and the difference body of a convex body K are constant multiples of each other, then K is origin symmetric; and if both are origin-symmetric balls, then K is an origin-symmetric ball. An n-dimensional convex polytope P is called a (VP)-polytope if, for every $k = 0, 1, \cdots, n$, the k-dimensional volume of the orthogonal projection of P into each k-dimensional subspace L is equal to the $(n - k)$-dimensional volume of the projection of P into the orthogonal complement of L. McMullen's and Schneider's investigations on (VP)-polytopes (see [1742] and [2157]) are also related to the question *for which polytopes their projection body and their difference body are homothets*. The question for which convex bodies K the projection body and the difference body are polar to each other is still unsettled. If K is a polytope, then only simplices have this property (cf. Martini [1707]), and in general also ellipsoids have it. But only ellipsoids there?

A convexity notion important for normed spaces is that of irreducible sets, going back to Shephard [2230] and deeply investigated by Yost [2507]. A convex body K centred at the origin is called *irreducible* if it is not the difference body of some body that is not a homothet of K; otherwise it

is a *reducible set*. In terms of normed spaces, exactly irreducible sets (taken as unit balls) represent those normed spaces in which the only *constant width sets* are balls. For sets of constant width (also in normed spaces), we refer to the monograph [1713], and to the notes of Chapter 10 where these ideas are reflected. In the notes of Section 3.2 of [2167], irreducible sets and related notions are comprehensively discussed.

5.7.4 The Mahler product and the Blaschke–Santaló inequality

The *Blaschke–Santaló inequality* states that if a convex body K in Euclidean n-space has its centroid at the origin, then the product of volume of K and its polar body is bounded from above by the squared volume of the unit ball, with equality if and only if K is an ellipsoid. Together with the knowledge about lower bounds (Mahler's conjecture, etc.), this inequality has deep applications in *convex geometry, Banach space theory, differential geometry, differential equations, geometry of numbers, stochastic geometry*, and other fields, and it goes back to Blaschke [355] (for two and three dimensions) inspired by his investigation in affine differential geometry. Santaló [2100] presented the n-dimensional case with $n \geq 2$, and the equality case was completely solved by Petty [1970] when he studied the equality case in the *affine isoperimetric problem*. In this subsection, we are able to cite (besides book parts and surveys) also research papers.

It should be noted that the studies around the Blaschke–Santaló inequality and its variants are simply summarized under the thematic "headline" *volume product*. We note that after Santaló and Petty more direct proofs came to be, including the equality case. Using tools from classical convexity (like, e.g., Steiner symmetrization), for centrally symmetric convex bodies this was done by Saint-Raymond [2088] as well as Meyer and Pajor [1771], and for the general case by Meyer and Pajor [1772]. The *lower bounds for the volume product* led to the popular and still unsettled *conjecture of Mahler*, see Mahler's contributions [1671], [1672], and our discussions below. Such lower bounds are identified with the so-called *reverse Blaschke–Santaló inequality* (or *inverse Santaló inequality*). The asymptotic version, also called *Bourgain–Milman inequality*, was settled in [478] by Bourgain and Milman (see, e.g., Chapter 1 in Pisier's book [1974]), and it was reproved, with a better participating constant, by G. Kuperberg [1476].

Survey-like presentations of the recent state around the Blaschke–Santaló inequality and the Mahler problem, also for the non-symmetric case, are given in the papers [409] and [1673].

MONOGRAPHS AND BOOK PARTS

We start with citing monographs in which this inequality is proved and/or results around it are obtained and discussed. For example, Pisier's book [1974] contains already a discussion of the "inverse Blaschke–Santaló inequality" due to J. Bourgain and V. D. Milman, see also [478] and below. In Thompson's book [2361] on *Minkowski spaces*, the *inequalities of Blaschke–Santaló and Mahler–Reisner* (the latter for zonoids) occur several times; namely, in Sections 2.3 and 2.6 (where also some history is given), in Sections 6.5 and 6.6 (as tool needed for deriving bounds on the surface area of the unit ball and in connection with *isoperimetrices*), and in Section 8.2 combined with *trigonometry in normed planes*. Leichtweiss discusses the Blaschke–Santaló inequality as a consequence of the *affine isoperimetric inequality* in Section 3.12 of his book [1537], showing also the reverse direction of this connection. Presenting proofs regarding these connections, he continues with a comprehensive discussion of known results on sharp lower bounds for the volume product. Here also proofs are given, and related results are surveyed.

In Section 9.2 of Gardner's monograph [930], a proof for centered convex bodies is given, and in Section 9.3 this inequality is used as a tool to get *volume estimates from projection functions*.

Further on, in Note 9.5, the Blaschke–Santaló inequality is discussed within the framework of *affine isoperimetric inequalities* (see also Lutwak's survey [1633]) and generalizations of itself. Gruber [1070] proves or discusses in Section 9.2 various inequalities, among them the *Blaschke–Santaló inequality and Mahler's inequality*.

Schneider approaches the Blaschke–Santaló inequaliy in the classical way (initiated by Blaschke) in Section 10.5 of his book [2167], and in Section 10.7, more direct proofs are discussed, which only use classical tools from convexity (like Steiner symmetrization and the Brunn–Minkowski theorem), for centred and also for arbitrary non-symmetric convex bodies. Also, the lower bound for the volume product is comprehensively discussed, including *Mahler's conjecture, Hanner polytopes, non-sharp lower bounds*, additional observations in two dimensions, the *solution for zonoids, the solution for convex bodies with sufficiently large orthogonal symmetry group*, and results on *functional Mahler inequalities*. In the notes to Section 10.7 of [2167], further results are surveyed, among them a thorough study of the planar case (for convex bodies with m-fold symmetry) given in [409] of Böröczky et al. (this paper contains also a comprehensive survey on the volume product in general, see below).

Another important book studying the Blaschke–Santaló inequality and related results is Artstein-Avidan et al. [133]. In Chapter 1, connections between duality and volume are discussed, and a proof of the Blaschke–Santaló inequality is given. In Chapter 8, M-ellipsoids are studied as key notion in the proof of the *reverse Brunn–Minkowski enequality* and the *Bourgain–Milman inequality*, which presents an isomorphic version of the *reverse Blaschke–Santaló inequality*. This is also discussed in the earlier books [1974] of Pisier (where the first eight chapters mainly deal with three fundamental results, among them the Bourgain–Milman inequality) and in Chapter 1 of Brazitikos et al. [496]. In Section 3.6 of [1263], Hug and Weil present applications of the Steiner symmetrization, among them the *isodiametric inequality and the Blasche-Santaló inequality* (for centred convex bodies).

SURVEYS

In the *Handbook of Convexity* the Blaschke–Santaló inequality occurs (as Section 11) in the survey of Lutwak [1633], where a brief discussion of the history, of proof variants and strengthenings is given. It is also shown how Blaschke and Santaló got the result by using the affine isoperimetric inequality. In Section 12 (Open Problems) of Lutwak's paper [1633], the state of knowledge regarding the *complementary Blaschke–Santaló inequality* and also *Mahler's conjecture* are discussed. Also, Lutwak's older related exposition [1628] should be mentioned here. He derives a general affine isoperimetric inequality which contains, as subcase, Berwald's generalization of an inequality of Winternitz; and he shows that this inequality implies the Blaschke–Santaló inequality. On the other hand, using Hölder's integral inequality in combination with the Blaschke–Santaló inequality, the *general affine isoperimetric inequality* is obtained.

Back to the handbook: In Gruber's survey [1063] on the geometry of numbers, the Blaschke–Santaló inequality and Mahler's conjecture are presented in Section 3, as one of the fundamental theorems there. And in view of Banach space theory, Lindenstrauss and Milman discuss the Blaschke–Santaló inequality in subsection 4.3 of [1579], as consequence of Milman's known theorem on M-ellipsoids.

We mention also once more the research paper [409] of Böröczky et al. (see also [1673] by Makai) since it contains a comprehensive survey on the volume product (all important cases, also the non-symmetric one, are comprehensively discussed there).

In the introductory part of [334], Bianchi and Kelly give a nice overview to the history of the Blaschke–Santaló inequality and related results. Then, as a novelty, they present a *Fourier analytic proof of the Blaschke–Santaló inequality* and go on to derive further results, e.g., a related variational inequality.

The main topic of Colesanti's important exposition [637] is the geometric theory of *log-concave functions*, i.e., functions which can be written in the form e^{-u} with u a convex function. Presenting properties of the *convex cone of log-concave functions*, the author succeeds to show the analogies

between convex bodies and log-concave functions. Different duals of log-concave functions, some *functional versions of area measures and mixed volumes* for log-concave functions, and *valuations* of such functions are discussed. In particular, the Prékopa–Leindler and the Blaschke–Santaló inequality are presented within that framework.

SELECTED RESEARCH ARTICLES

We continue with a *selection of research articles* that are (with a few exceptions) not covered by the books and surveys above. For this representation, we have further subthemes.

GENERALIZATIONS, REFINEMENTS, AND VARIANTS

Meyer and Pajor [1771] establish the Blaschke–Santaló inequality for centred convex bodies K via the property that the *dual of a Steiner symmetral* of K has larger volume than the dual of K. Hug [1253] derives a representation of the *extended affine surface area* allowing to verify that it is not decreasing with respect to Steiner symmetrization. Thus, one gets an *affine isoperimetric inequality*, and the Blaschke–Santaló inequality follows easily. For both inequalities, he also gets characterizations of the equality cases.

Lutwak and Zhang [1636] try to establish a new family of analytical inequalities, one member of this family being the *Blaschke–Santaló inequality*.

After the proof of Bourgain and Milman, some other approaches to the *reverse Santaló inequality* were given, like also lower bounds for the constant occuring there. Giannopoulos et al. [961] present an alternative proof of the reverse Santaló inequality. Their approach is based on some properties of the isotropic position of a convex body and uses the fact that every convex body with *bounded isotropic constant* satisfies the reverse Santaló inequality. To get this, the authors use only typical tools from convexity and log-concave measures.

Cordero-Erausquin et al. [657] derive sharp inequalities for the volume of the polar set of random sets, yielding a *random extension of the Blaschke–Santaló inequality*. The studied random sets are random linear images of a given unconditional convex body. As tools, the authors use Shephard's *shadow systems and Busemann-type inequalities*.

Bucur and Fragalà [523] are interested in the λ_1-product functional associating with a given convex body K the λ_1-product of K and its polar; here λ_1 denotes *the first eigenvalue of the Dirichlet Laplacian*. The first main result is obtained with the help of the classical Blaschke–Santaló inequality and further inequalities, and it says that the minimum of this λ_1-product functional in the class of centrally symmetric convex bodies is attained by balls. The second main result gives the lower bound for a *related volume-product problem* based on the class of invertible diagonal transformations in the plane. Within the class of unconditional convex bodies (a really necessary restriction), here the extremal figure is the square.

Ma [1652] studies a *reverse Orlicz Blaschke–Santaló inequality* in two dimensions and gets triangles as extremal figures, like also parallelograms under the restriction to central symmetry. Also, a nice characterization of ellipsoids in the studied framework is obtained.

Alexander et al. [52] are interested in the maximum of the *volume product* among convex polytopes with a bounded vertex number. They prove a combinatorial formula characterizing a polytope of maximal volume product and show that the supremum is reached by a simplicial polytope with exactly $m \geq n+1$ vertices, say. Based on this, they also present a proof of a result of Meyer and Reisner for the planar case, namely that the regular polygon has maximal volume product among all polygons with at most m vertices. In particular, also the case of n-dimensional polytopes with $n+2$ vertices is taken care of.

As main results, Hoehner [1206] proves Blaschke–Santaló type inequalities and inverse Santaló type inequalities for the *extremal general affine surface areas*. The subcase of extremal L_p affine

surface area was introduced by Giladi et al. (2020), and this article here can be regarded as an Orlicz extension of Giladi's work.

Alonso-Gutiérrez et al. [83] give new upper and lower bounds on the *Wills functional* (i.e., on the sum of the intrinsic volumes) of a convex body, and they reprove also a known upper bound due to McMullen. Besides Jensen's inequality, in the proof they use a functional version of the *Blaschke–Santaló inequality*.

Li, Schütt and Werner [1559] extend notions from convexity to the setting of functions, first concentrating on *log-concave functions*. For example, the notion of symmetry (in the setting of convex bodies) is replaced by the notion of evenness (in the functional setting). Two well-known types of affine invariant points in convex geometry are the *centroid* and the *Santaló point*, and the corresponding affine isoperimetric inequalities are the *Busemann-Petty centroid inequality* and the *Blaschke–Santaló inequality*, respectively. For the mentioned extension, definitions of affine contravariant points and affine covariant mappings for log-concave functions are given. The authors show that the *centroid and the Santaló point of a log-concave function* are examples of affine contravariant points in this concept, and that modern notions like *floating function, John function, and Löwner function* are examples of affine covariant mappings.

Kolesnikov and Werner [1453] introduce the *Blaschke–Santaló inequality for many functions* and show its importance in the geodesic barycenter problem from optimal transportation theory. The proof of the inequality needs also the Prékopa–Leindler inequality for many of these functions, to which some refinements are added. For the equality characterization, the authors have to work out an equality characterization of the Prékopa–Leindler inequality first. A generalization of the Blaschke–Santaló inequality is obtained, and the main results are related to various further topics, e.g., the barycenter problem for measures. Also, a monotonicity property of the multimarginal Blaschke–Santaló functional is proved.

MAHLER'S CONJECTURE AND RELATED TOPICS

It is well known that the point which, taken as origin, yields the minimum product of the volumes of a convex body K and its polar, is called the *Santaló point* of K. For all convex bodies with the Santaló point at the origin, Reisner, Schütt and Werner [2025] consider the least value of the volume product. They show that a body at which the minimum is attained cannot have a nonzero generalized Gauss curvature at any single boundary point. This result is strongly related to Mahler's conjecture that among centrally symmetric bodies the minimum is attained for a cube and its polar set.

Saroglou [2113] successfully studies *shadow systems of convex bodies*, defined by Shephard as follows: Parallel projection of a convex body K onto a hyperplane H yields a shadow of K in H which is a convex set, too. Letting the light direction vary, a so-called shadow system of K, consisting of convex regions in H, is obtained. Saroglou first generalizes related results of Rogers and Shephard, and via a particular shadow system, he then gives a simple proof of Reisner's confirmation of *Mahler's conjecture for zonoids*. A further problem from convexity is also verified: assume that all Steiner symmetrals of K for an open and non-empty set of directions have a center of symmetry. Then K has to be centrally symmetric, too.

In Sections 3 and 4 of their Lecture Notes [2082], Ryabogin and Zvavitch prove the *Blaschke–Santaló inequality* and deeply discuss the *Mahler conjecture*, more precisely its proof in two dimensions and in the case of unconditional convex bodies. Also, the *local minimality of the unit cube* is discussed, and the validity of Mahler's conjecture for *zonotopes and zonoids* takes place, too.

Mahler's conjecture asks whether the cube is a minimizer for the volume product of a symmetric convex body and its polar. In the subclass of *unconditional convex bodies* (i.e., of convex bodies symmetric with respect to all coordinate hyperplanes), Mahler's conjecture was confirmed by Saint-Raymond [2088]. Kim and Zvavitch [1381] prove related stability theorems. In particular, they show that any centred convex body sufficiently close to an unconditional body satisfies the reverse Blaschke–

Santaló inequality. Improving related results of Böröczky and Makai, Ivaki [1282] derives a stability version of the Blaschke–Santaló inequality in the plane.

Böröczky et al. [409] discuss the volume-product inequalities for planar convex bodies K and their polars, due to Mahler and Reisner, Mahler and Meyer, and Eggleston, respectively. The sharp lower bounds they discuss further are: 8 for convex figures centred at the origin (equality only for parallelograms), $27/4$ if the origin is interior point (equality only for a triangle with centroid at the origin), and $2m\sin(2\pi/m)$ if K has m-fold rotational symmetry about the origin (with equality only for the regular m-gon centered at the origin). A fourth lower bound refers to the volume product of K and the polar of its central symmetral, being 6 (with equality only for triangles). For all four cases, the authors present also stability versions, regarding the convex body as well as the center of polarity.

As a result related to the *volume product*, Zhao [2538] detects the minimum for the product of *the kth width-integrals of a convex body K and its polar*, and he does the same for the product of the mixed width-integrals of K and its polar. The least upper bound for the product of the kth width-integrals of a convex body and its polar is correspondingly derived in [2539].ł

Błocki [375] surveys Nazarov's approach (using several complex variables) to the *Mahler conjecture from convex analysis*. He also shows (numerically) that Nazarov's proof of the *Bourgain–Milman inequality* cannot be improved to obtain the Mahler conjecture.

Harrell et al. [1164] prove a concavity result for the volume product due to which they can retrieve a result of Reisner et al. (see [2025]), strongly supporting Mahler's conjecture that the cube is a global minimizer of the volume product.

Akopyan et al. [32] study *billiard trajectories in convex bodies* whose length is measured by a *gauge* (i.e., an asymmetric norm). The authors prove a lower bound result for the length of the shortest closed billiard trajectory, and its relation to the *general Mahler problem* is discussed.

The paper [1652] of Ma should also be cited here, since it contains also lower-bound results with respect to the volume product.

One of the most important results around volume products was recently obtained by Iriyeh and Shibata. In [1276], they succeed in *proving the Mahler conjecture for centrally symmetric convex bodies in 3-space*! More precisely, they show that the volume product is sharply bounded from below by $32/3$ with *equality iff either the original body or its polar is a parallelepiped*.

5.7.5 Mean width and Urysohn's inequality

The *mean width of a convex body K* is an interesting quantity of K, really having diverse and interesting properties and applications. Since the *width* of K in direction u yields a translation invariant valuation, the *mean width* of K does the same. As a Minkowski additive, continuous and invariant under rigid motions map it is a constant multiple of the intrinsic length V_1 (or the quermassintegral W_{n-1}) of K, and Hadwiger proved this theorem on p. 213 of his classical book [1134]. This proof is contained in subsection 6.1.7 of [1134], thus located within Chapter 6 on integral geometry of convex bodies, and even the complete subsection is dedicated to mean width.

BOOK PARTS AND SURVEYS

This theorem on V_1, many further properties and applications of the notion of mean width, and Hadwiger's related characterization theorem on additive, continuous, and invariant under rigid motions functionals (also referring to the other quermassintegrals) are discussed in various further basic convexity books and surveys. For example, Leichtweiss reproves this theorem and Hadwiger's famous characterization theorem in §17 of his monograph [1531].

The theorem on V_1 is also presented in McMullen's handbook survey [1745] (Section 5 there) on *valuations*. Groemer treats the notion of mean width in Section 2.4 of his book [1042], yielding results as *multiple of a mean projection measure*, and in Section 5.8 regarding *applications for spherical harmonics* (see also his handbook article [1039]). Gruber discusses mean width in the framework of *intrinsic volumes* in Section 6.4 of his monograph [1070], Gardner does the same in Appendix A.5 of his book [930]. Chapter 1 of the book [496] of Brazitikos et al. presents the basics of asymptotic convexity, among them also *classical positions of convex bodies*, and, in particular, *minimal mean width and minimal surface area positions*.

Schneider discusses mean width in his book [2167] four times: as *mean value of the width function* (using the support function) in Section 1.7, as *example for special Minkowski additive maps* (see above) in Section 3.3, via *formulae involving curvature functions* (approach via quermassintegrals) in subsection 5.3.1, p. 297, and in the setting of *log-concave functions* in Section 9.5.

In Chapter 6 of their book [151], Aubrun and Szarek study stochastic processes (typically Gaussian ones) and combine the supremum of them with the mean width of a convex body. Artstein-Avidan, Giannopoulos and Milman study, e.g., mean width in the following parts of their book [133]: Section 2.2 (minimal mean width positions are studied, where position means an affine image), Chapter 6 (investigation of the product of the mean width of a symmetric convex body K and the mean width of its polar body), Chapter 7 (dimensions of Euclidean sections), and Chapter 9 (Gaussian proofs of Dvoretzky's theorem and mean width estimates). In particular, it is shown in Chapter 6 that there is a special *position* of K such that the mentioned product can be bounded by constant times $\log n$. In Chapter 7, it is shown that this product is important for studying the dimension of Euclidean sections in the sense of asymptotic convexity. This chapter starts with the following essential statement of Milman: The diameter of a random section of proportional dimension of a high-dimensional symmetric convex body is *controlled by the mean width* of this body. In subsection 9.1.5 of the second volume [134], for instance, the mean width of log-concave functions is introduced, and so the authors present a *functional Urysohn inequality*. The youngest monograph in this sequence here is [1263] of Hug and Weil; in Remark 3.21 the authors show how "mean width" is related to the first intrinsic volume.

SELECTED RESEARCH ARTICLES

We continue with *selected research articles* on the *mean width* notion that are mostly not cited in the books and surveys above. They are ordered by subthemes. It is natural that there is some overlap between the references chosen here and those in our subsection 12.4.5. As an example, we refer to the monographs cited there and, e.g., also to Affentranger's survey [21], where various results about the *mean width* of convex bodies are cited.

INEQUALITIES

Petty [1968] establishes a number of geometric inequalities, among them also one between *mean width and surface area* of convex bodies. He clarifies also the cases of equality. Similarly, Chakerian proves in [592] two inequalities on convex bodies in which *volume, surface area, mean width, general quermassintegrals and orthogonal projections* of the investigated convex bodies onto hyperplanes play a role. Both inequalities are derived via symmetrization procedures yielding bodies of revolution whose axes are parallel to the direction of projection.

For K an n-dimensional convex body and L a randomly rotated and shifted integer lattice, Ngoc and Reitzner [1891] study the *mean width* $w(K_L)$ of the random convex body K_L obtained as the convex hull of the intersection of K and L. Interesting inequalities are derived, which contain the *mean width difference*. The obtained upper bound can be generalized to all intrinsic volumes. It turns out that the asymptotic order of the mean width difference is maximized by the order obtained by polytopes, and minimized by the order for smooth convex sets.

The *mean distance* for an n-dimensional convex body K is the expected Euclidean distance of two independent and uniformly distributed random points from K. Bonnet et al. [448] obtain optimal lower and upper bounds for the ratio between this quantity and the first intrinsic volume of K (i.e., the normalized *mean width*), and extremal cases are discussed.

SPECIAL CONVEX BODIES

Interested in polyhedral universal covers, Makeev proves in [1681] the following theorem holding in Euclidean n-space: Around a convex body of mean width w, one can circumscribe an n-dimensional simplex having w as diameter of the inscribed ball.

Barthe shows in [232] that for a convex body K in Euclidean n-space the following holds: if the unit ball is the maximum volume ellipsoid contained in K, then the *mean width* of K is smaller than the mean width of any regular simplex circumscribed about this ball.

Generalizing a known result of Jung, K. Bezdek and Connelly [319] prove that the *translation cover of minimum mean width* for all sets of diameter one in Euclidean n-space is the n-dimensional ball of diameter $\sqrt{2n/(n+1)}$.

K. Bezdek [313] deals with quermassintegrals and the *strong dodecahedral conjecture*. This states that the regular dodecahedron has minimal intrinsic k-volume among the *Voronoĭ polyhedra of unit ball packings*. Here the case $k = 1$, i.e., the *mean width* of the Voronoĭ polyhedra, is investigated. The author succeeds in giving an interesting lower bound.

K. J. Böröczky and Schneider [412] prove the following nice results: For an n-dimensional convex body K, let $m(T_K)$ denote the *minimal mean width of all simplices circumscribed about* K, and $w(K)$ be the *mean width of* K. Furthermore, let T_n be a regular simplex circumscribed about the n-dimensional unit ball. The authors show that $m(T_K) \leq 1/2w(K)m(T_n)$, where equality holds if and only if K is a ball. The methods of the authors also yield an analogous result for circumscribed parallelepipeds, where the balls have to be replaced by *bodies of constant width*. The authors do not treat the long-standing dual problem: finding those simplices contained in a given ball which have maximal mean width.

Kabluchko et al. [1335] discuss the following question: how to arrange $n + 1$ points on the $(n - 1)$-dimensional sphere such that the *mean width* of their convex hull is maximized. It is conjectured that the arrangement has to be regular. They reformulate the problem in terms of intrinsic volumes and show that the conjecture is asymptotically true. Then they explain how the first intrinsic volume of the cross-polytope with $2n$ vertices is related to the first intrinsic volume of the regular simplex with a suitable vertex number. Finally, the results and developed methods are applied to regular polytopes, establishing several formulas which were conjectured by Finch.

Alonso-Gutiérrez and Prochno [88] investigate a class of suitably defined random convex sets that "interpolate" between polytopes and zonotopes. The main object is a random convex body defined as a unit ball of a suitable norm. The introduced construction gives a rich class of convex bodies. For example, there can occur a symmetric convex body which is dual to a convex hull of random vectors (i.e., dual to the classical model), but on the other hand, the unit ball in this model can be the polar of a zonotope. Together with various further investigations, the authors study the geometry of these sets in terms of *support functions and mean width*.

The topic studied by Bäsel [177] is refreshingly concrete: the author approaches the *mean width* of the *oloid* (which is the convex hull of two circles with equal radius in perpendicular planes such that the center of each circle lies on the other circle) in two ways: first via the integral of mean curvature, and then directly. He continues with the surface area and the volume of the parallel body, and he derives the *expectations of the mean width*, the surface area and the volume of the intersections of a fixed oloid and a moving ball, and of a fixed and a moving oloid.

In the paper [398], Böröczky et al. begin with three extremal properties of regular simplices: they *maximize the mean width* of convex bodies whose John ellipsoid is the Euclidean unit ball; they maximize the ℓ-norm of convex bodies whose Löwner ellipsoid is the Euclidean unit ball; and they *minimize the mean width* of convex bodies whose Löwner ellipsoid is the Euclidean unit ball. First, stronger stability versions of these statements are proved. Then, further stability strengthenings of related geometric inequalities (referring, e.g., to the mean width and to the Brascamp–Lieb inequality) are given.

Lángi [1496] studies isoperimetric-type problems for the class of *three-dimensional parallelotopes* (i.e., for those zonotopes which fill 3-space by translation). Among other results, he shows that, assuming unit volume, in this class, the regular truncated octahedron is the one with *minimal mean width*.

PROJECTIONS

It is well known that Minkowski showed the following: if all orthogonal projections of a three-dimensional convex body K have the same perimeter, then K is of *constant width*. Goodey and Groemer [992] tried to get a stability statement from this, and also to extend the dimension. They succeeded in getting the following: if two centrally symmetric bodies have projections which all have approximately the same *mean width*, then the two bodies are approximately the same, up to translation. This is related to work of Bourgain and Lindenstrauss [474] who considered, instead of mean width, the volume of orthogonal projections.

Inspired by the *Loomis–Whitney inequality*, Campi and Gronchi [571] study estimates (from above) of *intrinsic volumes of convex bodies* in terms of sums of intrinsic volumes of finitely many orthogonal projections of these bodies. Among various other results, they get a sharp estimate of the *mean width* of a convex body in terms of the mean widths of the coordinate projections. An analogous estimate for intrinsic volumes of any order is conjectured and discussed.

APPROXIMATION ASPECTS

Schneider and Wieacker [2181] consider m random points in a convex body K, independently and uniformly distributed. (The body K should satisfy a smoothness assumption.) Generalizing well-known results of Rényi and Sulanke, they derive, for large m, an asymptotic expression for the expectation of a random variable which here is the *mean width of the convex hull* of those m points. These considerations are continued by Buchta, Müller and Tichy [521]. Here the authors give explicit formulas for the expected values of the *surface area, mean width*, and *number of facets* with respect to the convex hull of the m analogously random points. This is continued by Schneider [2149]: For a convex body K in Euclidean n-space, let $w(K)$ denote its *mean width*, and $w_m(K)$ be the *mean width of the convex hull* of m random points, chosen independently and uniformly in K. Regarding expectation we set $T_m(K) = w(K) - E(w_m(K))$. In two inequalities, the author gives upper and lower bounds for $T_m(K)$ which show: for general convex bodies the considered approximation is not worse than for polytopes (right-hand side of the inequality, presenting the upper bound) and not better than for smooth convex bodies (left-hand side, lower bound). Again extending planar results of Rényi and Sulanke, Böröczky and Schneider [413] define a *random polytope* P^m to be the intersection of m independent closed halfspaces containing a convex body K, having an isotropic and (in a specified manner) uniform distribution. (Usually in the literature such a random polytope P_m is defined as the convex hull of m independent, uniformly distributed, random points from K.) They ask for the expectation $Ew(P^m)$ of the *mean width* w of the random polytope and derive upper and lower bounds of optimal orders concerning $Ew(P^m) - w(K)$. If K is a simplicial polytope, stronger results are obtained; namely, in this case, they get a precise asymptotic formula. In addition, the authors emphasize various interesting comparisons between results on P^m and P_m, using polarity with respect to the unit sphere.

Böröczky et al. [385] show the following: For $r > 1$, let $F_n(r)$ be the family of n-dimensional convex bodies which contain the n-dimensional unit ball and whose extreme points are of distance at least r from the origin. The authors study how good the unit ball can be approximated by members from $F_n(r)$ regarding volume, surface area and mean width. Asymptotic formulas for the error of approximation are proved, and good estimates on the involved constants (depending on the dimension) are obtained.

Böröczky et al. [399] obtain asymptotic results on the approximation orders of general convex bodies by random polytopes. For K an n-dimensional convex body, let K_m be the convex hull of a random sample of m independent points in K, and let $w(\cdot)$ denote the *mean width*. The authors generalize the asymptotic formula for the expectation $w(K) - w(K_m)$ and the strong law of large numbers for $w(K_m)$, earlier known when the boundary of K has positive Gaussian curvature.

There are various ways of producing circumscribed random polytopes K^n containing a given convex body K which approximate K as n tends to infinity. For Böröczky et al. [396], the circumscribed polytope is the intersection of closed halfspaces whose bounding hyperplanes are randomly chosen, and the main aim of the authors is to find asymptotic formulas for the expectation of the *mean width difference* $w(K) - w(K^n)$ and for the expectation of the *number of facets* of a circumscribed random polyhedral set. These results have been achieved by establishing general results on weighted volume approximation; no regularity or curvature assumptions on K are required.

Alonso-Gutiérrez and Prochno [87] study asymptotic distributions for the convex hull of $2m$ pairwise symmetric, uniformly distributed random points in the unit ball of l_p. Among other results, they show how the expected support function of this convex hull in any given direction behaves. It follows that the *expected mean width* behaves like $\sqrt{m}$; similar results for general convex bodies are also obtained. As a novelty of this framework, the authors use probabilistic estimates in terms of *Orlicz norms*.

Besau, Hoehner, and Kur [304] establish estimates for the asymptotically best approximation of the Euclidean unit ball by polytopes under a notion of distance induced by the *intrinsic volumes* (thus the mean width case is contained). They introduce a notion of *distance between convex bodies* that is induced by the Wills functional, and this is applied to derive asymptotically sharp bounds for approximating high-dimensional balls by polytopes. Finally, they establish asymptotic formulas for the best approximation of smooth convex bodies by polytopes with respect to a distance induced by *dual volumes*, a notion which goes back to Lutwak's dual Brunn–Minkowski theory.

POSITIONS

Markessinis, Paouris, and Saroglou [1696] compare some classical positions (the M-position, the minimal surface area position, the *minimal mean width position* etc.) of convex bodies. They provide examples of convex bodies which prove that the minimal surface area position, the *minimal mean width position*, the John position, etc., are not necessarily M-positions. They also construct examples in minimal surface area position with the worst *mean width or minimal hyperplane projection behaviour*. They give quantitative results on these examples, and also the sharpness of such results is studied.

E. Milman improves in [1785] an upper bound for the (half) *mean width* of n-dimensional origin-symmetric convex bodies K in isotropic position, in terms of the isotropic constant L_K of K and a universal constant. This improved bound implies that the isotropic position is (up to the L_K term) an almost 2-regular M-position. The bound extends to any arbitrary position, depending on a certain weighted average of the eigenvalues of the covariance matrix. Furthermore, it applies to the *mean width of L_p-centroid bodies*, extending also a related sharp upper bound of Paouris. The question whether one can remove the L_K term from the new bound is essentially equivalent to the *slicing problem*. In addition, the author uses these results to prove new estimates for *covering numbers*.

Ma [1651] studies the *minimal L_p-mean width of convex bodies*. Uniqueness questions, characterization theorems, and minimal L_p-mean width positions are typical topics treated in this paper, and the author shows to be an expert regarding developments in this direction. Continuing these investigations, he extends in [1653] the notion of *minimal L_p-mean width* of convex bodies (including the classical

minimal mean width) to the *Orlicz–Brunn–Minkowski–Firey theory* (for subcases see the references and [1651]), thus introducing the *Orlicz mean width* of an n-dimensional convex body. Related minimization problems are solved and show that the extremal bodies satisfy isotropic conditions of a suitable measure. The author also derives a stability result for L_p-centroid bodies, and new applications of the minimal Orlicz mean width position are discussed.

URYSOHN'S INEQUALITY

A famous inequality between volume and mean width of convex bodies is *Urysohn's inequality* (see [2389]), yielding the result that, if the volume is fixed, *balls* minimize *mean width*. This result, the respective inequality and generalizations thereof are cited or proved in various convexity books; see, e.g., Section 56 of Bonnesen and Fenchel [446], Note 9.6 in Gardner's book [930], Section 8.3 in Gruber's monograph [1070], Section 7.2 in Schneider's book [2167], and Chapter 1 of Artstein-Avidan et al. [133].

Using dual mixed volumes, Lutwak [1625] derives the *duals of Bieberbach's and Urysohn's inequalities* and obtains, further on, that among all convex bodies of given mean width only the ball has the property that its polar has minimal quermassintegrals, of any index. This implies a sharp lower bound for the *product of the mean widths of a convex body and its polar body*. One subcase of this result yields a new proof of Urysohn's inequality.

In [1628], Lutwak then considers a geometric inequality (which he calls general affine isoperimetric inequality) and shows a way to get the *Blaschke–Santaló inequality* from it. A special case of this is *Urysohn's inequality*.

For centrally symmetric convex bodies, Milman [1789] proves a variant of Urysohn's inequality and uses this to get an easier proof of a joint result with Pisier (in which also maximum volume ellipsoids and cotype-2-constants play an important role).

Groemer [1037] proves a stability version of Minkowski's inequality on mixed volumes. As corollaries, he gets *stability versions for the isoperimetric and for Urysohn's inequality*. Here we also mention the subsection 5.3 in Groemer's handbook survey [1040].

As isoperimetric-type result, Gao et al. [921] prove that among all (spherically) convex bodies in spherical space and with prescribed volume, the total measure of great hyperspheres hitting them becomes minimal if and only if these bodies are spherical caps. This gives a *spherical counterpart of Urysohn's inequality* in Euclidean case. *Hyperbolic Urysohn-type inequalities* are also obtained.

Klartag and Milman [1411] develop an isomorphic theory for high-dimensional log-concave measures (analogous to the corresponding theory for convex bodies). Notions like duality and Minkowski sum are described for log-concave functions, and in this context also the *reverse versions of the Brunn–Minkowski and of the Blaschke–Santaló inequality* are proved. Finally, a *functional version of the Urysohn inequality* is presented.

Hug and Schneider [1257] prove *inequalities for mixed volumes of zonoids* with isotropic generating measures. It turns out that, for zonoids, a subcase yields the reverse of extensions of Urysohn's inequality, between the mean width and other intrinsic volumes. The equality case characterizes parallelepipeds, and this is applied to a question from stochastic geometry.

In [2064], Rotem suitably defines the mean width of a log-concave function, and he uses that definition to establish the so-called *functional Urysohn inequality*, characterizing also the log-concave functions which yield the equality case.

Coming from stability versions of isoperimetric and Brunn–Minkowski inequalities, the works [2215] of Segal and [955] of Ghilli and Salani present *stability versions of Urysohn's inequality*.

In [1938], the authors derive a new quantitative reverse inequality located between the *approximate reverse Urysohn inequality* and the *sharp reverse Urysohn inequality for zonoids*, the latter due to Hug and Schneider (see [1257]).

Continuing [921], Hack and Pivovarov [1127] derive *randomized versions of Urysohn-type inequalities in spherical and hyperbolic spaces*. In addition, a randomized version of the spherical Blaschke–Santaló inequality is obtained. In order to prove these results, the authors use the two-point symmetrization procedure together with rearrangement inequalities; this allows also a methodologically new view onto the Euclidean case.

PLATE 6

Reduced convex bodies

A reduced compact cone

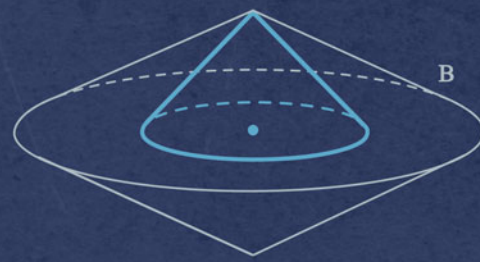
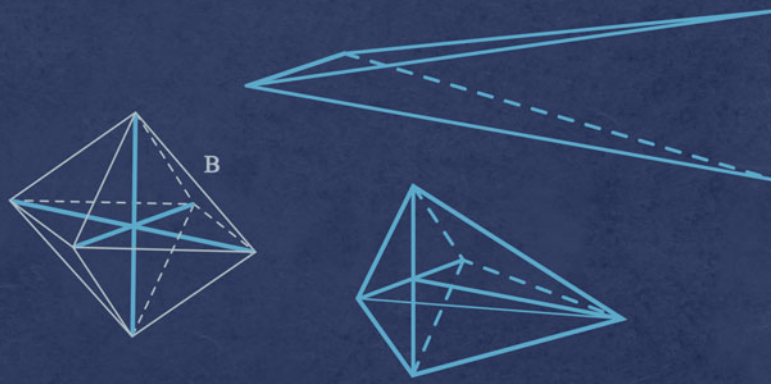


Figure 13

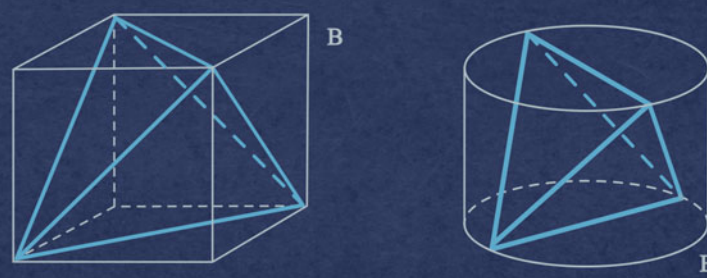
Reduced polyhedra in the Manhattan norm

Figure 14



Reduced tetrahedra

Figure 15



Reduced convex bodies in normed spaces with unit ball B ► (see, e.g., 2.9.1.10)

Chapter 6
Mixed volumes

"Mathematics is the most beautiful and most powerful creation of the human spirit."

— Stefan Banach

Mixed volumes are the basis of a deep and far-reaching theory with many geometric consequences. It arised originally from the problem of finding information about the volume

$$\mathrm{vol}_n(tK + sC),$$

where $K, C \in CC^n$ (the collection of non-empty compact convex sets of $\mathbb{R}^n$) and $s, t \geq 0$ in terms of the volumes of K and C. It turns out that the n-dimensional volume of any Minkowski combination (that is, a combination where all scalars are non-negative) of compact convex sets is polynomial on the scalars, and the coefficients of these polynomials are what we call *mixed volumes*.

This chapter is devoted to construct the "basic" theory of mixed volumes. Namely, we prove the mentioned polynomiality of the volume and investigate the (more or less) immediate consequences. Here we follow the approach of Gruber [1070], where the mixed volumes are defined in terms of the coefficients of the aforementioned polynomials. However, there are several other equivalent ways of defining mixed volumes, and we also discuss some of them.

A good illustration of the power of the mixed volumes theory appears already in Section 6.3, where the so-called *Minkowski 1st and 2nd inequalities* are stated and proved. In a certain sense, they come from the first and second order variations of mixed volumes, respectively. As we will see in this chapter, the classical isoperimetric inequality and the Urysohn inequality (for mean width) can be re-obtained from the 1st Minkowski inequality. Later, in Chapter 8, we will introduce a much deeper generalization of these inequalities, known as the *Alexandrov-Fenchel inequality*.

A certain specialization of the concept of mixed volume gives rise to important functionals defined for compact convex sets known as *quermassintegrals* (or *intrinsic volumes*, if we consider a suitable normalization). These functionals bring geometric information about convex bodies (such as mean width and surface area, for example). More generally, the quermassintegrals of a given compact convex set C are related to the mean values of the projections of C onto vector subspaces of fixed dimensions. A particularly important consequence of this approach, with a very clear geometric meaning, is the so-called *Cauchy surface area formula*, which represents the surface area of a convex body in terms of the $(n-1)$-dimensional volumes of its projections onto hyperplanes.

Plate 6 shows several *reduced convex bodies* in three-dimensional normed spaces with unit ball B. This means that each proper convex subset of these bodies has a smaller *minimal width* in the

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_6

corresponding norm. It is interesting that the "long tetrahedron" in Fig. 14 there can be (in a suitable way) arbitrarily prolonged.

6.1 Polynomiality of the volume

Recall that we consider the collection CC^n of non-empty compact convex sets of $\mathbb{R}^n$ endowed with the operations of Minkowski addition and scalar multiplication. It is not difficult to see that $(CC^n, +)$ is a commutative semigroup with cancellation law. In fact, if $C, D, E \in CC^n$ are such that $C + D = C + E$, then $h_{C+D} = h_{C+E}$, and it follows from item **(i)** of Proposition 2.3.2 and Corollary 2.3.3 that $D = E$.

The key point of the theory of mixed volumes is that the volume of a Minkowski combination of compact convex sets is a polynomial of degree n on the coefficients, as it is stated in the following theorem attributed to Minkowski. The objective of this chapter is to prove this theorem.

Theorem 6.1.1. *Let $C_1, \ldots, C_m \in CC^n$ be pairwise distinct. There are coefficients $V(C_{j_1}, \ldots, C_{j_n})$, one for each n-tuple $1 \leq j_1, \ldots, j_n \leq m$, which are symmetric in the indices and such that*

$$\mathrm{vol}_n(t_1C_1 + \ldots + t_mC_m) = \sum_{j_1,\ldots,j_n=1}^{m} V(C_{j_1}, \ldots, C_{j_n})t_{j_1} \ldots t_{j_n} \tag{6.1.1}$$

for $t_1, \ldots, t_m \geq 0$.

Remark 6.1.1. Even more than being a polynomial of degree n, the theorem gives that the function $(t_1, \ldots, t_m) \mapsto \mathrm{vol}_n(t_1C_1 + \ldots + t_mC_m)$ defined for $t_1, \ldots, t_m \geq 0$ is a *homogeneous polynomial*, meaning that all of its terms have the same degree (which is, of course, n in our case).

Remark 6.1.2. Although we are saying that the polynomial in the left-hand side of (6.1.1) has degree n, it can also be the zero polynomial. This happens, for example, if all sets C_j are contained in a same hyperplane of $\mathbb{R}^n$.

Thus, what we will prove is that $\mathrm{vol}_n(t_1C_1 + \ldots + t_mC_m)$ is a homogeneous polynomial of degree n (or the zero polynomial). Hence we define each $V(C_{j_1}, \ldots, C_{j_n})$ as the respective coefficient of the term $t_{j_1} \ldots t_{j_n}$ divided by the number of permutations of $\{j_1, \ldots, j_n\}$ (by doing that, we have symmetry). The coefficient $V(C_{j_1}, \ldots, C_{j_n})$, with $1 \leq j_1, \ldots, j_n \leq m$, is called the *mixed volume* of the (non-empty) compact convex sets $C_{j_1}, \ldots, C_{j_n}$.

Theorem 6.1.1 also works if $C_1, \ldots, C_m$ are not pairwise distinct (in fact, this will be a consequence of the continuity of the mixed volumes that we will prove later), but the requirement that they are pairwise distinct guarantees that each mixed volume corresponds to a single monomial in the polynomial. Also, it will be clear later that the number $V(C_{j_1}, \ldots, C_{j_n})$ depends only on the sets $C_{j_1}, \ldots, C_{j_n}$ (see Remark 6.2.1). It follows that the mixed volumes are well defined from the polynomial given in Theorem 6.1.1.

Before proceeding to the proof of the theorem, we will investigate some particular cases to illustrate the geometric information that the mixed volumes carry. First, observe that since

$$t^n \mathrm{vol}_n(C) = \mathrm{vol}_n(tC) = V(C, \ldots, C)t^n,$$

we get that for each $C \in CC^n$ the mixed volume $V(C, \ldots, C)$ is simply the volume of C. Now, if $K \subseteq \mathbb{R}^n$ is a convex body, then by (6.1.1) we write

$$\begin{aligned}\mathrm{vol}_n(K+\varepsilon B^n) &= V(K,\dots,K) + \binom{n}{1}V(K,\dots,K,B^n)\varepsilon + \\ &+\binom{n}{2}V(K,\dots,K,B^n,B^n)\varepsilon^2 + \binom{n}{3}V(K,\dots,K,B^n,B^n,B^n)\varepsilon^3 + \dots \\ &\dots + \binom{n}{n-1}V(K,B^n\dots,B^n)\varepsilon^{n-1} + V(B^n,\dots,B^n)\varepsilon^n,\end{aligned}$$

where we notice that the coefficient of K equals 1 and the binomials appear because the mixed volumes are symmetric in the entries. And then we can collect together the ones with the same entries but in different orders. Hence a simple calculation shows that the (total) surface area of K is given by

$$S(K) = \lim_{\varepsilon\to 0}\frac{\mathrm{vol}_n(K+\varepsilon B^n)-\mathrm{vol}_n(K)}{\varepsilon} = nV(K,\dots,K,B^n), \tag{6.1.2}$$

where the mixed volume has $n-1$ copies of K, of course. Observe that we used the definition of the Minkowski surface area, which agrees with the $(n-1)$-dimensional Hausdorff measure of ∂K (this is proved in Remark 6.2.2; see also Section 4.1).

Let us take a brief look in the two-dimensional case. In this case, the mixed volume is also called *mixed area*. If $K, L \in CC^n$, then we may write

$$\mathrm{vol}_2(tK+sL) = t^2V(K,K) + 2tsV(K,L) + s^2V(L,L) = t^2\mathrm{vol}_2(K) + 2tsV(K,L) + s^2\mathrm{vol}_2(L)$$

for $s,t \geq 0$. For $s = t = 1$, we naturally relate the mixed volume $V(K,L)$ to the volumes of K, L and $K+L$:

$$V(K,L) = \frac{\mathrm{vol}_2(K+L)-\mathrm{vol}_2(K)-\mathrm{vol}_2(L)}{2}. \tag{6.1.3}$$

This is a particular case of the *polar decomposition formula* (6.2.1) which we will state and prove later. As we will see later, mixed volumes are invariant under translations. Hence, to illustrate the equality above, we may use Figure 6.1.1 below.

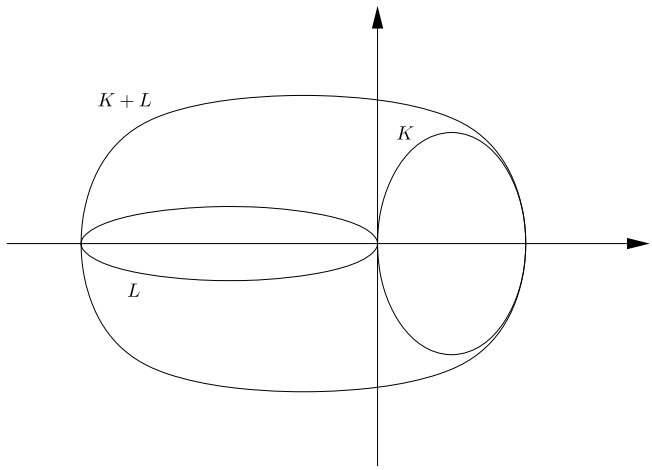


Fig. 6.1.1 The mixed area $V(K,L)$ is half the area of the region of $K+L$ outside K and L.

To prove Theorem 6.1.1, our strategy is to consider first polytopes and to use then approximations. Recall that the collection of all polytopes of $\mathbb{R}^n$ is denoted by $\mathcal{P}^n$, and that a polytope which is also a convex body (that is, a polytope which has maximal dimension) is called a *proper polytope*. In what follows, we remind the reader that the polytopes are the compact convex sets with finitely many extreme points (see the paragraph above Proposition 3.1.1).

We need some technical results about polytopes. In what follows, for $C \in CC^n$ and $u \in \mathbb{S}^{n-1}$, we denote by $\mathbf{h}(C,u)$ the hyperplane which supports C having u as unit outer normal. That is, from Proposition 2.3.1 we simply have

$$\mathbf{h}(C,u) = \mathbf{h}(u, h_C(u)) \left(:= \left\{ z \in \mathbb{R}^n : \langle z, u \rangle = h_C(u) \right\} \right),$$

where we also recalled the notation for hyperplanes introduced in (1.3.1) for the convenience of the reader (the relations between supporting hyperplanes and support functions are clarified in Sections 2.2 and 2.3). The (closed) half-spaces determined by $\mathbf{h}(C,u)$ are denoted by $\mathbf{h}^-(C,u)$ and $\mathbf{h}^+(C,u)$, with u being an outer normal of $\mathbf{h}^-(C,u)$. Consequently, we have that $C \subseteq \mathbf{h}^-(C,u)$.

Lemma 6.1.1. *For any polytopes $P_1, \ldots, P_m$ and numbers $t_1, \ldots, t_m \geq 0$, the set*

$$P := t_1 P_1 + \ldots t_m P_m$$

is a polytope. In other words, $\mathcal{P}^n$ is subset of CC^n which is closed under Minkowski addition and scalar multiplication.

Proof. We obviously may assume that all the coefficients are positive, since otherwise we simply discard the ones which are 0. The strategy is to prove that P has finitely many extreme points. Let x be an extreme point of P, and assume that $x = t_1 x_1 + \ldots + t_m x_m$, where $x_j \in P_j$ for each $j = 1, \ldots, m$. We claim that each x_j is an extreme point of P_j. Indeed, if for some $j = 1, \ldots, m$ the point x_j is not an extreme point of P_j, then there exist points $y, z \in P_j$ and a number $\lambda_0 \in (0,1)$, such that $x_j = \lambda_0 y + (1-\lambda_0) z$. Hence, x is a point of the relative interior of the segment parametrized as

$$[0,1] \ni \lambda \mapsto t_1 x_1 + \ldots + t_{j-1} x_{j-1} + t_j(\lambda y + (1-\lambda)z) + t_{j+1} x_{j+1} + \ldots + t_m x_m,$$

and this segment lies in P because $\lambda y + (1-\lambda)z \in P_j$ for every $\lambda \in [0,1]$. It follows that, in fact, x_j is an extreme point of P_j for each $j = 1, \ldots, m$. Since each polytope P_j has finitely many extreme points, we get that P has finitely many extreme points, and hence it is a polytope.

□

Next we verify that Minkowski combinations of polytopes do not create "many" new outer normals to the facets. Before doing so, we need an auxiliary lemma.

Lemma 6.1.2. *Let $C_1, \ldots, C_m \in CC^n$, $t_1, \ldots, t_m \geq 0$ and $u \in \mathbb{S}^{n-1}$. Letting $C := t_1 C_1 + \ldots + t_m C_m$, we have that*

$$C \cap \mathbf{h}(C,u) = t_1 \Big(C_1 \cap \mathbf{h}(C_1,u) \Big) + \ldots + t_m \Big(C_m \cap \mathbf{h}(C_m,u) \Big).$$

In particular, if $P_1, \ldots, P_m$ are polytopes and $t_1, \ldots, t_m \geq 0$, then any face of $P := t_1 P_1 + \ldots + t_m P_m$ is of the form $t_1 F_1 + \ldots + t_m F_m$, where each F_j is a face of P_j.

Proof. Again, we clearly can assume that $t_j > 0$ for each $j = 1, \ldots, m$. First consider

$$z = t_1 z_1 + \ldots + t_m z_m,$$

where $z_j \in C_j \cap \mathbf{h}(C_j, u)$ for each j. The fact that $z \in C$ is immediate. Since $z_j \in C_j \cap \mathbf{h}(C_j, u)$, we have that the support function $h_{C_j}(u)$ is attained at z_j, that is, $\langle z_j, u\rangle = h_{C_j}(u)$. Hence

$$\langle z, u\rangle = \langle t_1 z_1 + \ldots + t_m z_m, u\rangle = \sum_{j=1}^{m} t_j h_{C_j}(u) = h_C(u),$$

and it follows that $\mathbf{h}(C, u)$ intersects C at z (see Remark 2.3.2). Thus, $z \in C \cap \mathbf{h}(C, u)$.

Conversely, let $x \in C \cap \mathbf{h}(C, u)$. Since $x \in C$, we may write $x = t_1 x_1 + \ldots + t_m x_m$, where $x_j \in C_j$ for each $j = 1, \ldots, m$. We claim that $x_j \in \mathbf{h}(C_j, u)$ for each j. If this is not true, then for some $j = 1, \ldots, m$ we have $x_j \in \mathrm{int}(\mathbf{h}^-(C_j, u))$, because $C_j \subseteq \mathbf{h}^-(C_j, u)$. This means that $\langle x_j, u\rangle < h_{C_j}(u)$. Consequently,

$$\begin{aligned} h_C(u) = \langle x, u\rangle = \langle t_1 x_1 + \ldots + t_m x_m, u\rangle \leq t_1 h_{C_1}(u) + \ldots + t_j \langle x_j, u\rangle + \ldots + t_m h_{C_m}(u) < \\ < t_1 h_{C_1}(u) + \ldots + t_j h_{C_j}(u) + \ldots + t_m h_{C_m}(u) = h_C(u), \end{aligned}$$

and so we arrive at a contradiction. Therefore, $x_j \in \mathbf{h}(C_j, u)$ for each $j = 1, \ldots, m$ and this concludes the proof.

□

Proposition 6.1.1. *Let $P_1, \ldots, P_m \in \mathcal{P}^n$ be fixed. Then there exists a finite set $N \subseteq \mathbb{S}^{n-1}$ containing the unit outer normals of the facets of all proper polytopes of the form $t_1 P_1 + \ldots + t_m P_m$ with $t_1, \ldots, t_m \geq 0$.*

Proof. Assume that $P := t_1 P_1 + \ldots + t_m P_m$ is a proper polytope, and let $F \subseteq P$ be a facet with $F = P \cap \mathbf{h}$ for some supporting hyperplane $\mathbf{h} := \mathbf{h}(P, u)$, $u \in \mathbb{S}^{n-1}$. From Lemma 6.1.2, we have that

$$P \cap \mathbf{h}(P, u) = t_1 \Big(P_1 \cap \mathbf{h}(P_1, u)\Big) + \ldots + t_m \Big(P_m \cap \mathbf{h}(P_m, u)\Big),$$

that is, each facet of P is a combination of faces of the P_j's (recall that any face of a polytope P is given by the intersection of P with some supporting hyperplane, see Corollary 3.1.1).

Now we prove the following: if for each $j = 1, \ldots, m$ we have a face F_j of P_j and a number $t_j \geq 0$ with the property that $t_1 F_1 + \ldots + t_m F_m$ is an $(n-1)$-dimensional polytope contained in a hyperplane $\mathbf{h}_0$, then $F_1 + \ldots + F_m$ is also an $(n-1)$-dimensional polytope contained in $\mathbf{h}_0$. To show that this is true, we clearly may assume that $t_j > 0$ and that $0_{\mathbb{R}^n} \in \mathrm{relint} F_j$ for each $j = 1, \ldots, m$. In this case, we have

$$\begin{aligned} \mathbf{h}_0 = \mathrm{span}(t_1 F_1 + \ldots + t_m F_m) = \mathrm{span}(t_1 F_1) + \ldots + \mathrm{span}(t_m F_m) = \\ = \mathrm{span}(F_1) + \ldots + \mathrm{span}(F_m) = \mathrm{span}(F_1 + \ldots + F_m), \end{aligned}$$

and this gives the desired.

Finally, these arguments combined give the following: if $F = P \cap \mathbf{h}(P, u)$ is a facet of P (with outer normal $u \in \mathbb{S}^{n-1}$), then F is of the form $t_1 F_1 + \ldots + t_m F_m$ for some faces $F_j \subseteq P_j$ $(j = 1, \ldots, m)$, and u is normal to the (unique) hyperplane which contains $F_1 + \ldots + F_m$. It follows that the cardinality of the set of outer normals of facets of polytopes having the form $t_1 P_1 + \ldots + t_m P_m$ (with $t_1, \ldots, t_m \geq 0$) is not greater than the cardinality of the sums of type $F_1 + \ldots + F_m$ where each F_j is a face of P_j. The latter is finite, because each polytope P_j has at most finitely many faces. This concludes the proof.

□

Since our strategy to prove Theorem 6.1.1 is to consider first polytopes and to use then approximations, we will need a result regarding convergence of polynomials. We state and prove it next.

Proposition 6.1.2. *Let $(p_j)_{j\in\mathbb{N}}$ be a sequence of homogeneous polynomials of degree $n \in \mathbb{N}$ on m variables $(t_1, \ldots, t_m)$, and assume that (p_j) converges pointwise for a function f when $t_1, \ldots, t_m \geq 0$. Then, when restricted to $t_1, \ldots, t_m \geq 0$, f is a homogeneous polynomial whose degree is n, or f is identical to zero.*

Proof. We start with the case where $(p_j)_{j\in\mathbb{N}}$ is a sequence of (not necessarily homogeneous) polynomials of degree less than or equal to $n \in \mathbb{N}$ in one variable t. Assume that, for $t \geq 0$, the sequence (p_j) converges pointwise to a function f. For each $j \in \mathbb{N}$ and $t \geq 0$ write

$$p_j(t) = a_{0j} + a_{1j}t + a_{2j}t^2 + \ldots + a_{nj}t^n.$$

Letting $t = 0, 1, \ldots, n$ we get, for each fixed $j \in \mathbb{N}$, the following equalities:

$$\begin{aligned} a_{0j} &= p_j(0) \\ a_{0j} + a_{1j} + \ldots + a_{nj} &= p_j(1) \\ a_{0j} + 2a_{1j} + \ldots + 2^n a_{nj} &= p_j(2) \\ &\vdots \\ a_{0j} + na_{1j} + \ldots + n^n a_{nj} &= p_j(n), \end{aligned}$$

and this can be written in matrix form as

$$\begin{bmatrix} 1 & 0 & \ldots & 0 \\ 1 & 1 & \ldots & 1 \\ 1 & 2 & \ldots & 2^n \\ \vdots & \vdots & \ldots & \vdots \\ 1 & n & \ldots & n^n \end{bmatrix} \cdot \begin{bmatrix} a_{0j} \\ a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{bmatrix} = \begin{bmatrix} p_j(0) \\ p_j(1) \\ p_j(2) \\ \vdots \\ p_j(n) \end{bmatrix}. \tag{6.1.4}$$

The $n \times n$ matrix above is non-singular, because it is a Vandermonde matrix. Hence each a_{kj} can be written as a linear combination of the p_j's in the right-hand side of the equality above. Even more, since the $n \times n$ matrix does not depend on j, we get that the coefficients of this linear combination also do not. If follows that there exist $(n+1)^2$ coefficients A_{ki}, for $0 \leq k, i \leq n$, such that

$$a_{kj} = A_{k0}p_j(0) + A_{k1}p_j(1) + A_{k2}p_j(2) + \ldots + A_{kn}p_j(n) \tag{6.1.5}$$

for every $j \in \mathbb{N}$ and any $0 \leq k \leq n$. Indeed, the A_{ki}'s are nothing but the entries of the inverse Vandermonde matrix in (6.1.4). Now we make the limit as $j \to \infty$ in (6.1.5) for each $k = 1, \ldots, n$, to obtain

$$a_k := \lim_{j\to\infty} a_{kj} = A_{k0}f(0) + A_{k1}f(1) + \ldots + A_{kn}f(n).$$

Then, for each fixed $t \geq 0$, we have that $p_j(t) = a_{0j} + a_{1j}t + \ldots + a_{nj}t^n$ converges to $a_0 + a_1t + \ldots + a_nt^n$. Since we have from the hypothesis that $p_j(t)$ converges to $f(t)$, we get from the uniqueness of the limit that

$$f(t) = a_0 + a_1t + \ldots + a_nt^n,$$

and hence f is a polynomial whose degree is not greater than n. At this point, observe very carefully that if each p_j is homogeneous of degree n, then f is homogeneous of degree n or it is the zero polynomial.

The restriction of what we have just proved for the case where all polynomials of the sequence $(p_j)_{j\in\mathbb{N}}$ are homogeneous of degree n gives the proposition for the case where $m = 1$ (and the degree is arbitrary). To prove the general case, we proceed by induction on the number of variables m. Let $m > 1$, and assume that the statement is true for homogeneous polynomials on $m-1$ or less variables (of any degree).

Let $(p_j)_{j\in\mathbb{N}}$ be a sequence of homogeneous polynomials of degree n on m variables $(t_1,\dots,t_m)$, and assume that (p_j) converges pointwise to a function f when $t_1,\dots,t_m \geq 0$. Notice that each polynomial p_j can be written as

$$\begin{aligned} &p_j(t_1,\dots,t_m) = \\ &= p_{0j}(t_2,\dots,t_m) + p_{1j}(t_2,\dots,t_m)t_1 + p_{2j}(t_2,\dots,t_m)t_1^2 + \dots + p_{nj}(t_2,\dots,t_m)t_1^n, \end{aligned}$$

where each $p_{kj}(t_2,\dots,t_m)$ is either 0 or a homogeneous polynomial of degree $n-k$ in the variables $t_2,\dots,t_m$. For each fixed $(m-1)$-tuple $(t_2,\dots,t_m)$ of non-negative numbers, we define

$$p_j^{t_2,\dots,t_m}(t_1) := p_j(t_1,t_2,\dots,t_m),$$

for $t_1 \geq 0$, which is then a polynomial of one variable, and whose degree is not greater than n. We also let $f^{t_2,\dots,t_m}(t_1) := f(t_1,t_2,\dots,t_m)$, and the pointwise convergence $p_j \to f$ gives that

$$p_j^{t_2,\dots,t_m}(t_1) \to f^{t_2,\dots,t_m}(t_1)$$

as $j \to \infty$, for each $t_1 \geq 0$. It follows from the first part of the proof that $f^{t_2,\dots,t_m}$ is a polynomial of degree less than or equal to n in the variable t_1. Hence we write

$$f^{t_2,\dots,t_m}(t_1) = a_0^{t_2,\dots,t_m} + a_1^{t_2,\dots,t_m}t_1 + a_2^{t_2,\dots,t_m}t_1^2 + \dots + a_n^{t_2,\dots,t_m}t_1^n. \tag{6.1.6}$$

Consequently, we get that

$$p_{kj}(t_2,\dots,t_m) \to a_k^{t_2,\dots,t_m}$$

as $j \to \infty$, for each $k = 0,\dots,n$. From the induction hypothesis, the convergence above gives that, for each k,

$$q_k(t_2,\dots,t_m) := a_k^{t_2,\dots,t_m}$$

is a homogeneous polynomial whose degree is $n-k$ on the variables $(t_2,\dots,t_m)$, or it is the zero polynomial. Replacing these polynomials in (6.1.6) yields that f is a homogeneous polynomial of degree n or it is identical to zero, as we wanted to prove.

□

Now we are almost in the position to prove Theorem 6.1.1. The only remaining tool that we need is the formula for the volume of a proper polytope $P \subseteq \mathbb{R}^n$:

$$\mathrm{vol}_n(P) = \frac{1}{n}\left(\sum_{F \text{ facet of } P} h_P(u_F)\mathrm{vol}_{n-1}(F)\right),$$

where $u_F \in \mathbb{S}^{n-1}$ denotes the unit normal of the facet F ("outer" with respect to P). This formula is proved in Theorem 4.4.2.

Proof of Theorem 6.1.1: We first prove the theorem for polytopes, and the strategy to do so is to proceed by induction on the dimension. If $n = 1$, then the polytopes are compact intervals, and the result is obvious. So we assume that (6.1.1) holds for any polytopes and non-negative scalars in dimension $n-1$, for some $n > 1$.

Let $P_1, \ldots, P_m$ be polytopes in $\mathbb{R}^n$, and let N be as given in Proposition 6.1.1. That is, N is a finite subset of $\mathbb{S}^{n-1}$ containing the unit outer normals of all the facets of the combinations $t_1P_1 + \ldots + t_mP_m$ with $t_1, \ldots, t_m \geq 0$. Notice that for any m-tuple $t_1, \ldots, t_m \geq 0$ we may write the volume of $P := t_1P_1 + \ldots + t_mP_m$ as

$$\mathrm{vol}_n(P) = \mathrm{vol}_n(t_1P_1 + \ldots + t_mP_m) = \frac{1}{n}\sum_{u\in N} h_P(u)\mathrm{vol}_{n-1}(P \cap \mathbf{h}(P,u)),$$

because whenever $u \in N$ is not the outer normal of a facet of P, we will simply have that $\mathrm{vol}_{n-1}(P \cap \mathbf{h}(P,u)) = 0$. For each $j = 1, \ldots, m$ denote $P_j \cap \mathbf{h}(P_j,u) = F_j(u)$. Lemma 6.1.2 gives that $P \cap \mathbf{h}(P,u) = t_1F_1(u) + \ldots + t_mF_m(u)$. On the other hand, we know that $h_P = t_1h_{P_1} + \ldots + t_mh_{P_m}$. Hence the formula of the volume of P reads

$$\mathrm{vol}_n(t_1P_1 + \ldots + t_mP_m) =$$
$$= \frac{1}{n}\sum_{u\in N}\Big(t_1h_{P_1}(u) + \ldots + t_mh_{P_m}(u)\Big)\mathrm{vol}_{n-1}(t_1F_1(u) + \ldots + t_mF_m(u)),$$

and since we can identify each $F_j(u)$ with a polytope in $\mathbb{R}^{n-1}$ without changing area, it follows from the induction hypothesis that $\mathrm{vol}_{n-1}(t_1F_1(u) + \ldots + t_mF_m(u))$ is a homogeneous polynomial of degree $n-1$ with symmetric coefficients denoted by $V^{(n-1)}(F_{i_1}(u), \ldots, F_{i_{n-1}}(u))$, say (or the zero polynomial, see Remark 6.1.2). Hence from the equality above we get immediately that $\mathrm{vol}_n(t_1P_1 + \ldots + t_mP_m)$ is a homogeneous polynomial of degree n (or the zero polynomial) with symmetric coefficients $V(P_{i_1}, \ldots, P_{i_n})$ given as

$$V(\mathrm{per}(P_j, P_{i_1}, \ldots, P_{i_{n-1}})) = \frac{1}{n}\sum_{u\in N} h_{P_j}(u)V^{(n-1)}(F_{i_1}(u), \ldots, F_{i_{n-1}}(u)) \tag{6.1.7}$$

for any $1 \leq j, i_1, \ldots, i_{n-1} \leq m$, where $\mathrm{per}(P_j, P_{i_1}, \ldots, P_{i_{n-1}})$ is a permutation of $P_j, P_{i_1}, \ldots, P_{i_{n-1}}$ (the $V^{(n-1)}$'s are symmetric in the entries by the induction hypothesis). Then the proof for polytopes is complete.

Now we deal with the general case using approximations. Let $C_1, \ldots, C_m \in CC^n$ and, for each C_k, choose a sequence $(P_{kj})_{j\in\mathbb{N}}$ of polytopes converging to C_k in the Hausdorff metric as $j \to \infty$ (see Theorem 2.6.2). We already know that for each $j \in \mathbb{R}^n$ we may write

$$\mathrm{vol}_n(t_1P_{1j} + t_2P_{2j} + \ldots + t_mP_{mj}) = \sum_{i_1,\ldots,i_m=1}^{m} V(P_{i_1j}, P_{i_2j}, \ldots, P_{i_nj})t_{i_1}t_{i_2}\ldots t_{i_n},$$

for $t_1, \ldots, t_m \geq 0$, where the right-hand side is a homogeneous polynomial of degree n as in (6.1.1). For the sake of simplicity of the notation, we denote this polynomial as

$$p_j(t_1, \ldots, t_m) := \sum_{i_1,\ldots,i_m=1}^{m} V(P_{i_1j}, P_{i_2j}, \ldots, P_{i_nj})t_{i_1}t_{i_2}\ldots t_{i_n}.$$

For each fixed m-tuple $t_1, \ldots, t_m \geq 0$ the sequence of polytopes $t_1P_{1j} + \ldots + t_mP_{mj}$ converges to $t_1C_1 + \ldots + t_mC_m$ in the Hausdorff metric as $j \to \infty$. Since the n-dimensional volume is continuous in the Hausdorff metric (see Theorem 2.6.3), we get that

$$\mathrm{vol}_n(t_1P_{1j} + \ldots + t_mP_{mj}) \to \mathrm{vol}_n(t_1C_1 + \ldots + t_mC_m)$$

as $j \to \infty$. Hence $(p_j)_{j\in\mathbb{N}}$ is a sequence of homogeneous polynomials of degree n converging pointwise to the function $(t_1, \ldots, t_m) \mapsto \mathrm{vol}_n(t_1C_1 + \ldots + t_mC_m)$ when $t_1, \ldots, t_m \geq 0$. From Proposition 6.1.2 we get that $\mathrm{vol}_n(t_1C_1 + \ldots + t_mC_m)$ is a polynomial of degree n or it is identical to zero when restricted to $t_1, \ldots, t_m \geq 0$. This concludes the proof.

□

6.2 Properties of mixed volumes

This section is devoted to investigate properties of mixed volumes. We start by proving a useful formula which determines a mixed volume in terms of n-dimensional volumes.

Theorem 6.2.1 (Polar decomposition). *For any $C_1, \ldots, C_n \in CC^n$ (not necessarily pairwise distinct), we have*

$$n!V(C_1, \ldots, C_n) = \sum_{k=1}^{n}(-1)^{n+k} \sum_{1\leq j_1<\ldots<j_k\leq n} \mathrm{vol}_n\left(\sum_{m=1}^{k} C_{j_m}\right). \tag{6.2.1}$$

Proof. We first interpret the right-hand side of (6.2.1) as a function f defined for n-tuples of sets in CC^n:

$$f(C_1, \ldots, C_n) = \sum_{k=1}^{n}(-1)^{n+k} \sum_{1\leq j_1<\ldots<j_k\leq n} \mathrm{vol}_n\left(\sum_{m=1}^{k} C_{j_m}\right).$$

For each fixed n-tuple $C_1, \ldots, C_n$ we get from Theorem 6.1.1 that the map p defined as

$$p(t_1, \ldots, t_n) = f(t_1C_1, \ldots, t_nC_n) = \sum_{k=1}^{n}(-1)^{n+k} \sum_{1\leq j_1<\ldots<j_k\leq n} \mathrm{vol}_n\left(\sum_{m=1}^{k} t_{j_m}C_{j_m}\right)$$

is a homogeneous polynomial of degree n (or the zero polynomial) for $t_1, \ldots, t_n \geq 0$. In fact, each n-dimensional volume in the right-hand side of the above is a homogeneous polynomial of degree n (or the zero polynomial). Now we observe that putting $C_1 = \{0_{\mathbb{R}^n}\}$, the function f becomes a telescope sum (we multiply by $(-1)^{n+1}$ below simply to begin with a plus sign):

$$(-1)^{n+1} f(\{0_{\mathbb{R}^n}\}, C_2, \ldots, C_n) = \sum_{2 \le j_1 \le n} \mathrm{vol}_n(C_{j_1}) -$$

$$- \left[\sum_{2 \le j_1 \le n} \mathrm{vol}_n(\{0_{\mathbb{R}^n}\} + C_{j_1}) + \sum_{2 \le j_1 < j_2 \le n} \mathrm{vol}_n(C_{j_1} + C_{j_2}) \right] +$$

$$+ \left[\sum_{2 \le j_1 < j_2 \le n} \mathrm{vol}_n(\{0_{\mathbb{R}^n}\} + C_{j_1} + C_{j_2}) + \sum_{2 \le j_1 < j_2 < j_3 \le n} \mathrm{vol}_n(C_{j_1} + C_{j_2} + C_{j_3}) \right] -$$

$$\ldots$$

$$(-1)^{3n} \left[\sum_{2 \le j_1 < \ldots < j_{n-2} \le n} \mathrm{vol}_n(\{0_{\mathbb{R}^n}\} + C_{j_1} + \ldots + C_{j_{n-2}}) + \mathrm{vol}_n(C_2 + \ldots + C_n) \right] +$$

$$+ (-1)^{3n+1} \mathrm{vol}_n(C_2 + \ldots + C_n) =$$

$$= 0,$$

where the reader may notice carefully that we decomposed each k-th term of the first sum in the definition of f into two parts: one having the terms with C_1, and the other with the terms without it.

It follows immediately that $p(0, t_2, \ldots, t_n) = 0$ for every $t_2, \ldots, t_n \ge 0$ (simply replace C_j by $t_j C_j$ in the previous argument). Since p is a homogeneous polynomial of degree n, this means that the coefficients of all terms which do not have t_1 are zero. Repeating the argument for each one of the other variables, we get that the coefficient of a term of p where any of the variables is missing is 0. Therefore, p has the form

$$p(t_1, t_2, \ldots, t_n) = c t_1 t_2 \ldots t_n$$

for some number $c \ge 0$. Hence we may write

$$c t_1 t_2 \ldots t_n = \sum_{k=1}^{n} (-1)^{n+k} \sum_{1 \le j_1 < \ldots < j_k \le n} \mathrm{vol}_n \left(\sum_{m=1}^{k} t_{j_m} C_{j_m} \right), \tag{6.2.2}$$

and we claim that the coefficient of the term $t_1 \ldots t_n$ in the right-hand side is $n! V(C_1, \ldots, C_n)$. Indeed, by Theorem 6.1.1 we have that the unique n-dimensional volume above which "produces" a term of the form $t_1 \ldots t_n$ is $\mathrm{vol}_n(t_1 C_1 + \ldots + t_n C_n)$, since in every other n-dimensional volume some of the variables is missing. Further, the coefficient of the term $t_1 \ldots t_n$ in the polynomial $\mathrm{vol}_n(t_1 C_1 + \ldots + t_n C_n)$ is $n! V(C_1, \ldots, C_n)$, since we have to consider the $n!$ permutations of the entries of the mixed volume.

Consequently, $c = n! V(C_1, \ldots, C_n)$, and hence making $t_1 = t_2 = \ldots = t_n = 1$ in (6.2.2) leads to the desired equality. The proof is complete.

□

Remark 6.2.1. Recall that we defined mixed volumes as coefficients of polynomial representations of n-dimensional volumes of Minkowski combinations of compact convex sets. Hence, it was not completely clear until now whether a mixed volume depends only on its entries (and not on all sets involved in the initial combination). For example, if $C_1, C_2, C_3 \in CC^n$, then the mixed volume $V(C_1, C_1, \ldots, C_1, C_2)$ appears as the coefficient of the term $t_1^{n-1} t_2$ in both polynomials $\mathrm{vol}_n(t_1 C_1 + t_2 C_2)$ and $\mathrm{vol}_n(t_1 C_1 + t_2 C_2 + t_3 C_3)$. The previous theorem clarifies this: from equality (6.2.1) we get that indeed the mixed volumes depend only on their respective entries (of course, we could also just let

$t_3 = 0$ in the former polynomial). In other words, we can regard the mixed volume in $\mathbb{R}^n$ as a function $V\colon CC^n \times \ldots \times CC^n \to \mathbb{R}$ defined for n-tuples of sets in CC^n.

Corollary 6.2.1. *As explained before, regard the mixed volume V of $\mathbb{R}^n$ as a function defined for n-tuples of sets in CC^n. We have the following items:*

(i) *V is continuous in each argument with respect to the Hausdorff metric.*

(ii) *V is translation-invariant in each argument, that is,*

$$V(C_1 + \{x_1\}, \ldots, C_n + \{x_n\}) = V(C_1, \ldots, C_n)$$

for any $C_1, \ldots, C_n \in CC^n$ and every $x_1, \ldots, x_n \in \mathbb{R}^n$.

(iii) *For any $T \in \mathrm{GL}(n)$ and every $C_1, \ldots, C_n \in CC^n$ we have*

$$V(TC_1, \ldots, TC_n) = |\det T| V(C_1, \ldots, C_n).$$

In particular, V is $\mathrm{SL}(n)$-invariant (and also invariant under linear maps whose determinant is -1, of course).

(iv) *V is non-negative.*

Proof. The properties **(i)**, **(ii)** and **(iii)** are immediately obtained from equality (6.2.1) as follows: for **(i)** we simply recall that the n-dimensional volume is continuous in the Hausdorff metric (see Theorem 2.6.3), and **(ii)** is a consequence of the fact that the n-dimensional volume is translation-invariant. To prove **(iii)**, just notice that for every $1 \leq k \leq n$ and any $1 \leq j_1 < \ldots < j_k \leq n$, we have

$$\mathrm{vol}_n\left(\sum_{m=1}^{k} TC_{j_m}\right) = \mathrm{vol}_n\left(T\sum_{m=1}^{k} C_{j_m}\right) = |\det T| \mathrm{vol}_n\left(\sum_{m=1}^{k} C_{j_m}\right)$$

whenever $T \in \mathrm{GL}(n)$.

Now we prove **(iv)**. From **(ii)** and from the continuity of mixed volumes, it is enough to consider polytopes having the origin in their respective relative interiors. The case where $n = 1$ is obvious (in this case, the mixed volume is the usual length of segments). For the other cases, we proceed by induction using formula (6.1.7), observing that $h_P(u) \geq 0$ whenever the origin lies in the relative interior of P.

□

In what follows, we will work mainly with functionals which are *positively linear* (resp. *positively multilinear*), meaning that they are linear (resp. multilinear) for combinations with non-negative scalars (i.e., for *Minkowski combinations*). We also say that these functionals are *Minkowski linear* (resp. *Minkowski multilinear*). As we will see next, mixed volumes are positively multilinear, and the same holds for the most important maps defined for sets of CC^n with which we will work from now on (such as *mixed surface area measures*, for example). For that reason, when there is no possibility of confusion, we will refer to positive linearity or multilinearity simply as "linearity". As an illustrative example we have: when we refer to the "linearity of mixed volumes" we mean positive multilinearity, of course.

Proposition 6.2.1. *The mixed volume $V\colon CC^n \times \ldots \times CC^n \to \mathbb{R}$ is positively linear in each argument, meaning that*

$$V(tC + D, C_2, \ldots, C_n) = tV(C, C_2, \ldots, C_n) + V(D, C_2, \ldots, C_n)$$

for any $C, D, C_2, \ldots, C_n \in CC^n$ and every $t \geq 0$ (the linearity in the other arguments comes from the linearity in the first one with symmetry).

Proof. The polynomial representations of

$$\begin{aligned} &\mathrm{vol}_n(t_1(tC + D) + t_2C_2 + \ldots + t_nC_n) \text{ and} \\ &\mathrm{vol}_n((t_1t)C + t_1D + t_2C_2 + \ldots + t_nC_n) \end{aligned}$$

are identical. Of course, the coefficient of $t_1 \ldots t_n$ in the first one is $n!V(tC + D, C_2, \ldots, C_n)$. Now we look to the second one as a polynomial on the variables $t_1t, t_1, t_2, \ldots, t_n$. By doing so, the terms in which the product $t_1 \ldots t_n$ appears are the ones given respectively by the n-tuples $C, C_2, \ldots, C_n$ and $D, C_2, \ldots, C_n$:

$$n!V(C, C_2, \ldots, C_n)(t_1t)t_2 \ldots t_n + n!V(D, C_2, \ldots, C_n)t_1 \ldots t_n.$$

Consequently,

$$n!V(tC + D, C_2, \ldots, C_n) = n!tV(C, C_2, \ldots, C_n) + n!V(D, C_2, \ldots, C_n),$$

and the proof is complete. □

Our next objective is to prove the following monotonicity property for mixed volumes.

Theorem 6.2.2. *Assume that $C, D \subseteq CC^n$ are such that $C_1 \subseteq D_1$. Then*

$$V(C, C_2, \ldots, C_n) \leq V(D, C_2, \ldots, C_n)$$

for every $C_2, \ldots, C_n \subseteq CC^n$. By symmetry, the same holds for the other arguments.

To prove this theorem, we need some auxiliary results. While Schneider [2167] uses the concept of *mixed surface area measures*, which will be introduced in Section 7.1, we follow Gruber [1070]. The first result that we state and prove gives an expression for a mixed volume involving an arbitrary compact convex set and a polytope.

Lemma 6.2.1. *If $C \in CC^n$ and $P \in \mathcal{P}^n$, then*

$$V(C, P, \ldots, P) = \frac{1}{n} \sum_{F \text{ facet of } P} h_C(u_F)\mathrm{vol}_{n-1}(F), \tag{6.2.3}$$

where u_F denotes the unit outer (with respect to P) normal vector of F, as usual. The two-dimensional case is illustrated in Figure 6.2.1.

Proof. We start proving the following: let G be a face of P with $\dim G \leq n - 2$. Then there exists a constant $c > 0$ such that

$$\mathrm{vol}_n(G + \varepsilon B^n) \leq c\varepsilon^2 \tag{6.2.4}$$

for $\varepsilon > 0$ small enough, where we recall that B^n denotes the Euclidean unit ball of $\mathbb{R}^n$. To prove that we identify some $(n - 2)$-dimensional affine subspace containing G with $\mathbb{R}^{n-2}$, and hence we may assume that $G \subseteq \mathbb{R}^{n-2}$. Fix a number $\varepsilon_0 > 0$ and let $K \subseteq \mathbb{R}^{n-2}$ be any compact convex set such that $G + \varepsilon_0 B^{n-2} \subseteq K$. Thus, for every $0 < \varepsilon < \varepsilon_0$, we have

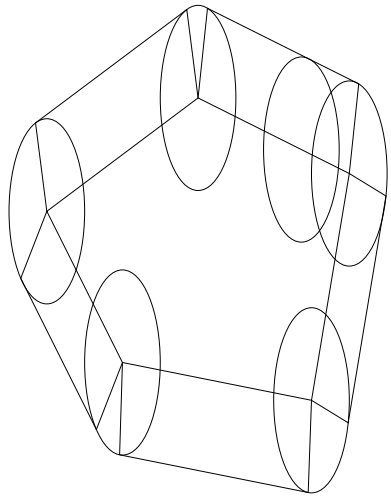


Fig. 6.2.1 We have the sum $C+P$, where P is the inner polytope and C is the ovaloid (having the origin as an interior point, say). The mixed volume $V(C,P)$ corresponds to the mean of the areas of the parallelograms above the edges of P. Notice that the height of each parallelogram is given in terms of the support function of C as in formula (6.2.3).

$$G+\varepsilon B^n \subseteq (G+\varepsilon B^{n-2})\times \varepsilon B^2 \subseteq (G+\varepsilon_0 B^{n-2})\times \varepsilon B^2 \subseteq K\times \varepsilon B^2,$$

and since the Lebesgue measure is a product measure, this inclusion gives

$$\mathrm{vol}_n(G+\varepsilon B^n) \leq \mathrm{vol}_n(K)\mathrm{vol}_n(\varepsilon B^2) = \varepsilon^2\Big(\mathrm{vol}_n(K)\mathrm{vol}_n(B^2)\Big).$$

Then we simply put $c = \big(\mathrm{vol}_n(K)\mathrm{vol}_n(B^2)\big)$, and inequality (6.2.4) holds for every $0<\varepsilon<\varepsilon_0$ (notice that c does not depend on ε, but only on the dimension n and on ε_0). Observe that, in particular, we obtain

$$\lim_{\varepsilon\to 0}\frac{\mathrm{vol}_n(G+\varepsilon B^n)}{\varepsilon} = 0. \tag{6.2.5}$$

On the other hand, Theorem 6.1.1 gives that

$$V(C,P,\ldots,P) = \frac{1}{n}\lim_{\varepsilon\to 0}\frac{\mathrm{vol}_n(P+\varepsilon C)-\mathrm{vol}_n(P)}{\varepsilon}.$$

Indeed, to obtain this equality, we simply repeat the calculations which give (6.1.2) replacing K by P and B^n by C.

We claim that we may assume that C contains the origin, and to prove this it is sufficient to show that the right-hand side of (6.2.3) is invariant under a translation of C (because we already know that the mixed volume is translation invariant). For any $x\in\mathbb{R}^n$, we have

$$\sum_{F \text{ facet of } P} h_{C+x}(u_F)\mathrm{vol}_{n-1}(F) =$$

$$= \sum_{F \text{ facet of } P} h_C(u_F)\mathrm{vol}_{n-1}(F) + \sum_{F \text{ facet of } P}\langle x,u_F\rangle\mathrm{vol}_{n-1}(F),$$

and from Lemma 4.4.2 the last sum equals 0.

Finally, assuming that C contains the origin, we divide $P + \varepsilon C$ into three parts: P itself, the orthogonal cylinders having the facets as basis, and the remaining parts (this is an intuitive argument, but can be easily formalized using metric projections). The height of the cylinder whose basis is the facet F is clearly $h_{P+\varepsilon C}(u_F) - h_P(u_F)$ and hence the volume of the part given by the cylinders can be calculated as

$$\varepsilon \cdot \sum_{F \text{ facet of } P} h_C(u_F)\mathrm{vol}_{n-1}(F),$$

and the remaining part, which we will call R, is clearly contained in the union of sets of the form $G + \varepsilon C$, where G is a face of P whose dimension is not greater than $n-2$ (in fact, the points of these part are the ones whose respective metric projections lie at such faces). Choosing $\delta > 0$ such that $C \subseteq \delta B^n$, we get from (6.2.4) that

$$\mathrm{vol}_n(R) \leq c\varepsilon^2$$

for $\varepsilon > 0$ small enough (we must have $\varepsilon\delta < \varepsilon_0$). We write (for $\varepsilon > 0$)

$$\frac{\mathrm{vol}_n(P + \varepsilon C) - \mathrm{vol}_n(P)}{\varepsilon} = \sum_{F \text{ facet of } P} h_C(u_F)\mathrm{vol}_{n-1}(F) + \frac{\mathrm{vol}_n(R)}{\varepsilon}.$$

From the inequality above, the latter goes to 0 as $\varepsilon \to 0$. Thus,

$$nV(C, P, \ldots, P) = \lim_{\varepsilon \to 0} \frac{\mathrm{vol}_n(P + \varepsilon C) - \mathrm{vol}_n(P)}{\varepsilon} = \sum_{F \text{ facet of } P} h_C(u_F)\mathrm{vol}_{n-1}(F),$$

and this concludes the proof.

□

Remark 6.2.2. As a consequence, we have

$$nV(B^n, P, \ldots, P) = \sum_{F \text{ facet of } P} \mathrm{vol}_{n-1}(F),$$

which is the usual surface area $\mathcal{H}^{n-1}(\partial P)$ of P (see equality (4.4.1)). Since $nV(B^n, P, \ldots, P)$ coincides with the Minkowski surface area of P (this follows from (6.1.2)), we get that these two surface area concepts coincide for polytopes. Because mixed volumes are continuous in the Hausdorff metric, it follows that for any convex body $K \subseteq \mathbb{R}^n$, the surface area $S_P(K)$ defined in Section 4.1 via approximation by polytopes coincides with the Minkowski surface area $S_M(K)$. The argument to prove that is the following: start with a sequence $(Q_j)_{j\in\mathbb{N}}$ of polytopes contained in K such that $\mathcal{H}^{n-1}(Q_j) \to S_P(K)$ as $j \to \infty$. In view of Proposition 3.4.1, for each $j \in \mathbb{N}$, we may let $\tilde{Q}_j \subseteq K$ be a polytope such that

$$d_{\mathcal{H}}(\tilde{Q}_j, K) < \frac{1}{2^j}.$$

For each $j \in \mathbb{N}$, define $P_j := \mathrm{conv}\big(\mathrm{ext}Q_j \cup \mathrm{ext}\tilde{Q}_j\big)$. Since it is clear that $Q_j \subseteq P_j \subseteq K$, we have

$$\mathcal{H}^{n-1}(Q_j) \leq \mathcal{H}^{n-1}(P_j) \leq S_P(K)$$

for any $j \in \mathbb{N}$ (see Corollary A.4.7). From this, we get that $\mathcal{H}^{n-1}(P_j) \to S_P(K)$ as $j \to \infty$. Once we also have that $\tilde{Q}_j \subseteq P_j \subseteq K$, it follows that

$$d_{\mathcal{H}}(P_j, K) \leq d_{\mathcal{H}}(\tilde{Q}_j, K) < \frac{1}{2^j}.$$

Hence, $P_j \to K$ in the Hausdorff metric as $j \to \infty$. Thus, we finally have

$$S_P(K) = \lim_{j\to\infty} \mathcal{H}^{n-1}(P_j) = \lim_{j\to\infty} \sum_{F_k \text{ facet of } P_j} \mathrm{vol}_{n-1}(F_k) =$$

$$= \lim_{j\to\infty} nV(B^n, P_j, \ldots, P_j) = nV(B^n, K, \ldots, K) = S_M(K),$$

where in the last equality we used (6.1.2) again.

Remark 6.2.3 (Surface area measure). Let $K \subseteq \mathbb{R}^n$ be a convex body. Denote by $\mathcal{B}_{\mathbb{S}^{n-1}}$ the Borel σ-algebra of $\mathbb{S}^{n-1}$. For each set $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$, let

$$\eta_K(E) := \{x \in \partial K : \exists\, u \in E \text{ such that } u \text{ is an outer normal of } K \text{ at } x\},$$

In other words, $\eta_K(E)$ is the set of boundary points of K having a vector of E as a unit outer normal vector. The set $\eta_K(E) \subseteq \partial K$ is known as the *reverse normal image* of E. We define a measure on $\mathbb{S}^{n-1}$ by

$$\mu_K(E) := \mathcal{H}^{n-1}(\eta_K(E)).$$

This is indeed a Borel measure on $\mathbb{S}^{n-1}$ (this is proved ahead, in Theorem 7.2.3), known as the *surface area measure of* K. We have the following version of (6.2.3):

$$V(L, K, \ldots, K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu_K(u) \tag{6.2.6}$$

for any convex bodies $K, L \subseteq \mathbb{R}^n$. This will be discussed in Section 7.2.

We actually have a result which is better than Lemma 6.2.1. Namely, we can even obtain a similar expression for the mixed volume when the polytopes are not all the same.

Proposition 6.2.2. *Let $P_2, \ldots, P_n$ be polytopes in $\mathbb{R}^n$. Letting $N \subseteq \mathbb{S}^{n-1}$ be a finite set containing the unit outer normal vectors of all combinations $t_2P_2 + \ldots + t_nP_n$, $t_2, \ldots, t_n \geq 0$ (see Proposition 6.1.1), we have*

$$V(C, P_2, \ldots, P_n) = \frac{1}{n} \sum_{u\in N} h_C(u) V^{(n-1)}\Big(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)\Big) \tag{6.2.7}$$

for every $C \in CC^n$, where $V^{(n-1)}$ denotes the mixed volume in $\mathbb{R}^{n-1}$.

Remark 6.2.4. All the sets $P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)$ lie in hyperplanes which are parallel to $u^\perp$. These hyperplanes can be naturally identified with $u^\perp$ by translations, and $u^\perp$ is naturally identified with $\mathbb{R}^{n-1}$ (by any isometric isomorphism). It follows that the sets $P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)$ are naturally identified with sets of $\mathbb{R}^{n-1}$, and this is why the mixed volume $V^{(n-1)}(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u))$ makes sense. This idea will be extensively used with no further comments from now on, even for subspaces of higher codimension (see, e.g., Theorem 6.2.4 ahead).

Proof. If $P = t_2P_2 + \ldots + t_nP_n$, then the linearity of the mixed volume gives

$$V(C, P, \ldots, P) = \sum_{j_2,\ldots,j_n=2}^{n} V(C, P_{j_2}, \ldots, P_{j_n}) t_{j_2} \ldots t_{j_n}.$$

Our strategy is to find another polynomial representation for $V(C, P, \dots, P)$ and then to compare the coefficients. From Lemma 6.2.1, we get

$$V(C, P, \dots, P) = \frac{1}{n} \sum_{F \text{ facet of } P} h_C(u_F)\text{vol}_{n-1}(F) = \frac{1}{n} \sum_{u \in N} h_C(u)\text{vol}_{n-1}(P \cap \mathbf{h}(P, u)),$$

where the last equality comes from the fact that N contains all unit outer normal vectors of the facets of F (and $P \cap \mathbf{h}(P, u_F) = F$, of course), and if $u \in N$ is not normal to a facet, then $\text{vol}_{n-1}(P \cap \mathbf{h}(P, u)) = 0$. Now we use Lemma 6.1.2 and the definition of mixed volumes (in $\mathbb{R}^{n-1}$) to write

$$\begin{aligned}\text{vol}_{n-1}(P \cap \mathbf{h}(P, u)) &= \text{vol}_{n-1}\Big(t_2(P_2 \cap \mathbf{h}(P_2, u)) + \dots + t_n(P_n \cap \mathbf{h}(P_n, u))\Big) = \\ &= \sum_{j_2, \dots, j_n = 2}^{n} V^{(n-1)}(P_{j_2} \cap \mathbf{h}(P_{j_2}, u), \dots, P_{j_n} \cap \mathbf{h}(P_{j_n}, u)) t_{j_2} \dots t_{j_n},\end{aligned}$$

for each $u \in N$. Then,

$$\begin{aligned}&V(C, P, \dots, P) = \\ &= \sum_{j_2, \dots, j_n = 2}^{n} \left(\frac{1}{n} \sum_{u \in N} h_C(u) V^{(n-1)}(P_{j_2} \cap \mathbf{h}(P_{j_2}, u), \dots, P_{j_n} \cap \mathbf{h}(P_{j_n}, u)) \right) t_{j_2} \dots t_{j_n}.\end{aligned}$$

Finally, comparing the coefficients of the term $t_2 \dots t_n$ in both polynomial representations obtained for $V(C, P, \dots, P)$, we get

$$V(C, P_2, \dots, P_n) = \frac{1}{n} \sum_{u \in N} h_C(u) V^{(n-1)}(P_2 \cap \mathbf{h}(P_2, u), \dots, P_n \cap \mathbf{h}(P_n, u)),$$

and the proof is complete.

□

Remark 6.2.5. Similarly to what was explained in Remark 6.2.3, there is a version of equality (6.2.7) for general convex bodies. This involves integrating the support measure of the first body with respect to the *mixed surface area measure* of the others. This concept will be defined and studied in Section 7.1 ahead.

Proof of Theorem 6.2.2: From the continuity of the mixed volume, it is sufficient to consider the case where $C_2 = P_2, \dots, C_n = P_n$ are polytopes. If $C \subseteq D$, then $h_C(u) \leq h_D(u)$ for every $u \in \mathbb{S}^{n-1}$. Hence

$$\begin{aligned}&V(D, P_2, \dots, P_n) - V(C, P_2, \dots, P_n) = \\ &\sum_{u \in N} \Big(h_D(u) - h_C(u)\Big) V^{(n-1)}\Big(P_2 \cap \mathbf{h}(P_2, u), \dots, P_n \cap \mathbf{h}(P_n, u)\Big) \geq 0,\end{aligned}$$

where the last inequality justifies because we already have that $V^{(n-1)}$ is non-negative.

□

Using monotonicity, we also can give an important description of the cases where a mixed volume is strictly positive.

Proposition 6.2.3. *Let $C_1, \dots, C_n \in CC^n$. Then $V(C_1, \dots, C_n) > 0$ if and only if there exist segments $s_i \subseteq C_i$, $i = 1, \dots, n$, with linearly independent directions.*

Proof. We start by investigating mixed volumes of segments. Let $s_1, \ldots, s_n \subseteq \mathbb{R}^n$ be segments whose directions are **not** linearly independent. Then $s_1 + \ldots + s_n$ is contained in some affine subspace whose dimension is not maximal, implying that $\mathrm{vol}_n(s_1 + \ldots + s_n) = 0$. It follows that

$$0 = \mathrm{vol}_n(s_1 + \ldots + s_n) = \sum_{1 \le i_1, \ldots, i_n \le n} V(s_{i_1}, \ldots, s_{i_n}),$$

and since all terms in the sum are non-negative, we get that $V(s_{i_1}, \ldots, s_{i_n}) = 0$ for any choice of $1 \le i_1, \ldots, i_n \le n$. This argument has the following consequence: if $s_1, \ldots, s_n$ **have** linearly independent directions, then

$$\mathrm{vol}_n(s_1 + \ldots + s_n) = \sum_{1 \le i_1, \ldots, i_n \le n} V(s_{i_1}, \ldots, s_{i_n}) = n! V(s_1, \ldots, s_n).$$

The last inequality justifies because if we repeat some segment, then the n-tuple $(s_{i_1}, \ldots, s_{i_n})$ will not have linearly independent directions, and hence their mixed volume vanishes.

Having this in mind, assume that there exist segments $s_i \subseteq C_i, i = 1, \ldots, n$, with linearly independent directions. Then, using monotonicity of mixed volumes, we get

$$V(C_1, \ldots, C_n) \ge V(s_1, \ldots, s_n) = \frac{\mathrm{vol}_n(s_1 + \ldots + s_n)}{n!} > 0.$$

The last inequality justifies because $s_1 + \ldots + s_n$ is clearly a convex body (see also Remark 6.2.6 below).

To prove the converse, we use induction on the dimension. For each $k \in \mathbb{N}$, denote by $V^{(k)}$ the mixed volume in $\mathbb{R}^k$. Hence, our induction hypothesis is the following: for any $C_1, \ldots, C_k \in CC^k$ such that $V^{(k)}(C_1, \ldots, C_k) > 0$, there exist segments $s_i \subseteq C_i, i = 1, \ldots, k$, having linearly independent directions. This is obviously true for $k = 1$, since in this case the non-empty compact convex sets are points and segments, and $V^{(1)}$ is just the usual length (one-dimensional volume).

Let $n > 1$ and assume that the induction hypothesis is true for every $k = n-1$. Let $C_1, \ldots, C_n \in CC^n$ be such that $V^{(n)}(C_1, \ldots, C_n) > 0$. Recalling that any convex body can be approximated (in the Hausdorff metric) by polytopes contained in it, and since mixed volumes are continuous in the Hausdorff metric, we may choose polytopes $P_i \subseteq C_i$, $i = 1, \ldots, n$, such that $V^{(n)}(P_1, \ldots, P_n) > 0$. Using (6.2.7), we may write

$$0 < V^{(n)}(P_1, \ldots, P_n) = \sum_{u \in N} h_{P_1}(u) V^{(n-1)}\big(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)\big),$$

where N is the finite subset of $\mathbb{S}^{n-1}$ containing the outer normal vectors of the facets of the polytopes $t_2 P_2 + \ldots + t_n P_n, t_2, \ldots, t_n \ge 0$. Since mixed volumes are translation-invariant, we may assume that, without loss of generality, $0_{\mathbb{R}^n} \in \mathrm{relint} P_1$. By doing so, we have that all of the terms in the sum above are non-negative. It follows that there exists $u_0 \in N$ such that $h_{P_1}(u_0) > 0$ and $V^{(n-1)}\big(P_2 \cap \mathbf{h}(P_2, u_0), \ldots, P_n \cap \mathbf{h}(P_n, u_0)\big) > 0$. By the induction hypothesis, for each $i = 2, \ldots, n$, one may choose a segment

$$s_i \subseteq P_i \cap \mathbf{h}(P_i, u_0) \subseteq P_i \subseteq C_i$$

in such a way that $s_2, \ldots, s_n$ have linearly independent directions. Moreover, the direction of each segment s_i is orthogonal to u_0, because $s_i \in \mathbf{h}(P_i, u_0)$. Finally, let $x \in P_1$ be such that $\langle x, u_0 \rangle = h_{P_1}(u_0)$. Then, since $h_{P_1}(u_0) > 0$, we get that $s_1 := \mathrm{seg}[0_{\mathbb{R}^n}, x]$ is a segment in P_1 (and hence in C_1) whose direction is not orthogonal to u_0, since, of course, $\langle x, u_0 \rangle > 0$. It follows that the directions of $s_1, \ldots, s_n$ are linearly independent, and this concludes the proof.

□

Remark 6.2.6. Actually, we can give a more precise description of the mixed volumes of segments. Let $s_1, \ldots, s_n$ be segments having linearly independent directions, and assume that $s_i = \text{seg}[0_{\mathbb{R}^n}, x_i]$ for each $i = 1, \ldots, n$. Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be the linear map such that $T(e_i) = x_i$ for each $i = 1, \ldots, n$, and let $C^n := [0,1] \times [0,1] \ldots \times [0,1] = \text{seg}[0_{\mathbb{R}^n}, e_1] + \ldots + \text{seg}[0_{\mathbb{R}^n}, e_n]$ be the standard n-dimensional unit cube. From the well-known behavior of the Lebesgue measure under a linear map (see Theorem A.3.3), we have

$$n!V(s_1, \ldots, s_n) = \text{vol}_n(s_1 + \ldots + s_n) = \text{vol}_n(TC^n) = |\det(T)|\text{vol}_n(C^n) = |\det(x_1, \ldots, x_n)|,$$

where the determinant is the determinant of the matrix whose columns are the vectors $x_1, \ldots, x_n$ (which clearly is the canonical matrix representation of T). Of course, equality also holds if $s_1, \ldots, s_n$ do not have linearly independent directions, since in this case both sides are zero.

The last property of mixed volumes that we will investigate in this section is known as the *valuation property*. We will follow a proof that, according to Peter Gruber [1070], was suggested by Károly Böröczky. This is based on the fact that the support function also has the valuation property.

Lemma 6.2.2. *Let $C, D \in CC^n$ be such that $C \cup D \in CC^n$. Then*

$$h_{C\cup D} + h_{C\cap D} = h_C + h_D.$$

Proof. First, observe that it is immediate that

$$h_{C\cup D}(u) = \max\{h_C(u), h_D(u)\} \tag{6.2.8}$$

for every $u \in \mathbb{R}^n$. Also, we claim that the hypothesis on $C \cup D$ being convex gives the equality

$$h_{C\cap D}(u) = \min\{h_C(u), h_D(u)\} \tag{6.2.9}$$

for any $u \in \mathbb{R}^n$. To prove this claim, first notice that if $C \cup D$ is convex, then $C \cap D \neq \emptyset$. In fact, if $C \cap D = \emptyset$, then compactness gives that $\text{dist}(C, D) > 0$, and hence this distance is attained in a segment $\text{seg}[x, y]$ with $x \in C$ and $y \in D$ such that $(C \cup D) \cap \text{seg}(x, y) = \emptyset$. This contradicts the convexity of $C \cup D$. Now let $u \in \mathbb{R}^n$ be a non-zero vector (the case for $u = 0_{\mathbb{R}^n}$ is obvious), and assume (without loss of generality) that $h_C(u) \leq h_D(u)$. Choose points $x \in C$ and $y \in D$ such that $\langle x, u\rangle = h_C(u)$ and $\langle y, u\rangle = h_D(u)$, and observe that $\text{seg}[x, y] \subseteq C \cup D$. Let

$$\lambda_0 = \max\{\lambda \in [0,1] : (1-\lambda)x + \lambda y \in C\},$$

where we notice that the maximum exists because C is compact. Put $z = (1-\lambda_0)x + \lambda_0 y$. By definition, we have that $z \in C$, and we claim that we also have $z \in D$. Indeed, put $\lambda_1 = \min\{\lambda \in [0,1] : (1-\lambda)x + \lambda y \in D\}$ and $w = (1-\lambda_1)x + \lambda_1 y$ (clearly, $w \in D$). If $z \notin D$, then $\lambda_0 < \lambda_1$, and hence $\text{seg}[z, w]$ is a non-degenerate segment connecting points of $C \cup D$ but which is not contained in $C \cup D$, contradicting the convexity of this set. Notice that

$$\begin{aligned}\langle z, u\rangle = \langle (1-\lambda_0)x + \lambda_0 y, u\rangle = (1-\lambda_0)h_C(u) + \lambda_0 h_D(u) \geq \\ \geq (1-\lambda_0)h_C(u) + \lambda_0 h_C(u) = h_C(u).\end{aligned}$$

This, together with the facts that $C \cap D \subseteq C$ and $z \in C \cap D$, yields the inequalities

$$h_{C\cap D}(u) \leq h_C(u) \leq \langle z, u\rangle \leq h_{C\cap D}(u),$$

and hence we have $h_{C\cap D}(u) = h_C(u) = \min\{h_C(u), h_D(u)\}$. Now we simply observe that (6.2.8) and (6.2.9) give immediately the desired statement.

□

Theorem 6.2.3 (Valuation property). *If $C, D \in CC^n$ are such that $C \cup D \in CC^n$, then*

$$V(C \cup D, C_2, \ldots, C_n) + V(C \cap D, C_2, \ldots, C_n) = V(C, C_2, \ldots, C_n) + V(D, C_2, \ldots, C_n)$$

for any $C_2, \ldots, C_n \in CC^n$.

Proof. By continuity, it is sufficient to consider that $C_2 = P_2, \ldots, C_n = P_n$ are polytopes. Hence, the result is an easy consequence of Proposition 6.2.2 and Lemma 6.2.2. Indeed, if $N \subseteq \mathbb{S}^{n-1}$ is such as in the statement of that proposition, then

$$\begin{aligned} &V(C \cup D, C_2, \ldots, C_n) + V(C \cap D, C_2, \ldots, C_n) = \\ &= \frac{1}{n} \sum_{u\in N} \Big(h_{C\cup D}(u) + h_{C\cap D}(u)\Big) V^{(n-1)}\Big(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)\Big) = \\ &= \frac{1}{n} \sum_{u\in N} \Big(h_C(u) + h_D(u)\Big) V^{(n-1)}\Big(P_2 \cap \mathbf{h}(P_2, u), \ldots, P_n \cap \mathbf{h}(P_n, u)\Big) = \\ &= V(C, C_2, \ldots, C_n) + V(D, C_2, \ldots, C_n), \end{aligned}$$

and the proof is complete.

□

Remark 6.2.7. A more general valuation property for mixed volumes will be given later, in Proposition 6.4.1.

While we defined mixed volumes as coefficients of certain polynomials and proved later that these coefficients are given by a symmetric and multilinear map $V\colon CC^n \times \ldots CC^n \to \mathbb{R}$ of n-tuples of sets in CC^n, in [1713], the authors adopted the somehow "reverse" approach. Namely, they defined the mixed volume of $\mathbb{R}^n$ as a symmetric and multilinear function $V\colon CC^n \times \ldots \times CC^n \to \mathbb{R}$ such that $V(C, \ldots, C) = \mathrm{vol}_n(C)$ for every $C \in CC^n$ and they proved that such a function must give the coefficients of the polynomial expansions of the n-dimensional volumes of Minkowski combinations of sets in CC^n. Their approach gives another "meaning" for the uniqueness of mixed volumes, which we explore next.

Proposition 6.2.4. *The mixed volume $V\colon CC^n \times \ldots CC^n \to \mathbb{R}$ is the unique symmetric and multilinear function defined for n-tuples of sets in CC^n such that*

$$V(C, C, \ldots, C) = \mathrm{vol}_n(C)$$

for any $C \in CC^n$.

Proof. Assume that V_1 is another symmetric and multilinear map with the property that

$$V_1(C, \ldots, C) = \mathrm{vol}_n(C)$$

for every $C \in CC^n$. Let $C_1, \ldots, C_n$ be non-empty compact convex sets given arbitrarily (not necessarily distinct). If we write

$$P = t_1 C_1 + \ldots + t_n C_n$$

for arbitrary $t_1, \ldots, t_n \geq 0$, then

$$V_1(P, P, \dots, P) = \mathrm{vol}_n(P) = V(P, P, \dots, P),$$

and hence the left and right-hand sides must be the same polynomial on the variables $t_1, \dots, t_n$. Indeed, the right-hand side is a polynomial because V is the mixed volume of $\mathbb{R}^n$, and the left-hand side must be a polynomial because V_1 is multilinear. Therefore, the coefficient of the term $t_1 t_2 \dots t_n$ must be the same in both sides. By symmetry, this gives that

$$n! V_1(C_1, \dots, C_n) = n! V(C_1, \dots, C_n),$$

and this concludes the proof.

□

Now we continue by stating and proving a reduction formula for mixed volumes, which allows us to express mixed volumes of bodies in $\mathbb{R}^n$ in terms of their projections onto vector subspaces. For any $m \in \mathbb{N}$, we denote the mixed volume of $\mathbb{R}^m$ by $V^{(m)}$ (and since we are working in $\mathbb{R}^n$, we still denote $V^{(n)}$ simply by V). If $\mathbf{g}$ is a k-dimensional vector (or affine) subspace of $\mathbb{R}^n$, $k \in \{1, \dots, n-1\}$, then, identifying $\mathbf{g}$ nauturally with $\mathbb{R}^k$, we are able to define $V^{(k)}$ for k-tuples of compact convex sets of $\mathbf{g}$, even though these sets are subsets of $\mathbb{R}^n$.

We also introduce a new notation: in what follows, for any $0 \leq j \leq n$, we write

$$V\big(K[j], L[n-j]\big) := V\big(\underbrace{K, \dots, K}_{j}, \underbrace{L, \dots, L}_{n-j}\big).$$

More generally, if $m_1, \dots, m_i$ are natural numbers such that $m_1 + \dots + m_i = n$ and $K_1, \dots, K_i \in CC^n$, then we write

$$V\big(K_1[m_1], K_2[m_2], \dots, K_i[m_i]\big) := V\big(\underbrace{K_1, \dots, K_1}_{m_1}, \underbrace{K_2, \dots, K_2}_{m_2}, \dots, \underbrace{K_i, \dots, K_i}_{m_i}\big).$$

Theorem 6.2.4. *Let $\mathbf{g}$ be a k-dimensional vector subspace of $\mathbb{R}^n$, $k \in \{1, \dots, n-1\}$, and let $\mathbf{g}^\perp$ be its orthogonal complement. For any $L_1, \dots, L_k, K_1, \dots, K_{n-k} \in CC^n$ such that $L_1, \dots, L_k \subseteq \mathbf{g}$, we have*

$$\binom{n}{k} V(L_1, \dots, L_k, K_1, \dots, K_{n-k}) = V^{(k)}(L_1, \dots, L_k) V^{(n-k)}(\mathrm{proj}_{\mathbf{g}^\perp}(K_1), \dots, \mathrm{proj}_{\mathbf{g}^\perp}(K_{n-k})). \tag{6.2.10}$$

In particular, if $K, L \in CC^n$ and $L \subseteq \mathbf{g}$, then

$$\binom{n}{k} V\big(K[n-k], L[k]\big) = \mathrm{vol}_k(L) \mathrm{vol}_{n-k}(\mathrm{proj}_{\mathbf{g}^\perp}(K)).$$

Proof. Actually, we will obtain the first equality as a consequence of the second one. Hence, we investigate first the case where $K, L \subseteq CC^n$ and $L \subseteq \mathbf{g}$. On the one hand, for any $\lambda > 0$, we have the equality

$$\mathrm{vol}_n(K + \lambda L) = \sum_{j=0}^{n} \binom{n}{j} \lambda^j V\big(K[n-j], L[j]\big), \tag{6.2.11}$$

and on the other hand, using Cavalieri's principle for k-dimensional slicing (Theorem 4.2.3), we get

$$\mathrm{vol}_n(K+\lambda L)=\int_{\mathbf{g}^\perp}\mathrm{vol}_k\big((K+\lambda L)\cap(z+\mathbf{g})\big)\,d\mathcal{H}^{n-k}(z), \tag{6.2.12}$$

also for any $\lambda>0$. Since $L\subseteq\mathbf{g}$, we have the equality

$$(K+\lambda L)\cap(z+\mathbf{g})=\big[K\cap(z+\mathbf{g})\big]+\lambda L.$$

In fact, assume that $x+\lambda y=z+w\in(K+\lambda L)\cap(z+\mathbf{g})$ for $x\in K$, $y\in L$ and $w\in\mathbf{g}$. Then $x-z=w-\lambda y\in\mathbf{g}$ (recall that $\mathbf{g}$ is a vector subspace), and hence $x\in z+\mathbf{g}$. It follows that $x+\lambda y\in[K\cap(z+\mathbf{g})]+\lambda L$, and this gives the "$\subseteq$" part. Now let $x+\lambda y\in[K\cap(z+\mathbf{g})]+\lambda L$ for $x\in K\cap(z+\mathbf{g})$ and $y\in L$. Once $x\in K$, we have $x+\lambda y\in K+\lambda L$. Because $x\in z+\mathbf{g}$, we may write $x=z+w$ for some $w\in\mathbf{g}$, and thus we have $x+\lambda y=z+(w+\lambda y)\in z+\mathbf{g}$, where the last inclusion comes from using the fact that $L\subseteq\mathbf{g}$ again. This gives the "$\supseteq$" part. Having this set equality in mind, we rewrite (6.2.12) as

$$\begin{aligned}\mathrm{vol}_n(K+\lambda L)&=\int_{\mathbf{g}^\perp}\mathrm{vol}_k\big([K\cap(z+\mathbf{g})]+\lambda L\big)\,d\mathcal{H}^{n-k}(z)=\\&=\int_{\mathrm{proj}_{\mathbf{g}^\perp}(K)}\mathrm{vol}_k\big([K\cap(z+\mathbf{g})]+\lambda L\big)\,d\mathcal{H}^{n-k}(z)=\\&=\int_{\mathrm{proj}_{\mathbf{g}^\perp}(K)}\sum_{j=0}^{k}\binom{k}{j}\lambda^jV^{(k)}\big(K\cap(z+\mathbf{g})[k-j],L[j]\big)\,d\mathcal{H}^{n-k}(z)=\\&=\sum_{j=0}^{k}\binom{k}{j}\lambda^j\int_{\mathrm{proj}_{\mathbf{g}^\perp}(K)}V^{(k)}\big(K\cap(z+\mathbf{g})[k-j],L[j]\big)\,d\mathcal{H}^{n-k}(z),\end{aligned}$$

where the second equality follows from the fact that if $z\notin\mathrm{proj}_{\mathbf{g}^\perp}(K)$, then $K\cap(z+\mathbf{g})=\emptyset$ (and $\emptyset+\lambda L=\emptyset$). This is a polynomial expression for $\mathrm{vol}_n(K+\lambda L)$, and hence this has to be the same polynomial expression as in (6.2.11). Comparing the coefficients of λ^k on both polynomials, we arrive at

$$\begin{aligned}\binom{n}{k}V\big(K[n-k],L[k]\big)&=\int_{\mathrm{proj}_{\mathbf{g}^\perp}(K)}V^{(k)}(L,\dots,L)\,d\mathcal{H}^{n-k}(z)=\\&=\mathrm{vol}_{n-k}\big(\mathrm{proj}_{\mathbf{g}^\perp}(K)\big)V^{(k)}(L,\dots,L)=\mathrm{vol}_{n-k}\big(\mathrm{proj}_{\mathbf{g}^\perp}(K)\big)\mathrm{vol}_k(L),\end{aligned}$$

and this is the second equality.

To prove the first equality, we simply let $K:=\sum_{j=1}^{n-k}\lambda_jK_j$ and $L:=\sum_{i=1}^{k}t_iL_i$ for coefficients $\lambda_1,\dots,\lambda_{n-k},t_1,\dots,t_k>0$. It is clear that $L\in\mathbf{g}$, provided $L_1,\dots,L_k\in\mathbf{g}$. Thus, from the equality that we have proved, we get

$$\begin{aligned}\binom{n}{k}V\left(\sum_{j=1}^{n-k}\lambda_jK_j[n-k],\sum_{i=1}^{k}t_iL_i[k]\right)&=\mathrm{vol}_{n-k}\left(\mathrm{proj}_{\mathbf{g}^\perp}\left(\sum_{j=1}^{n-k}\lambda_jK_j\right)\right)\mathrm{vol}_k\left(\sum_{i=1}^{k}t_iL_i\right)=\\&=\mathrm{vol}_{n-k}\left(\sum_{j=1}^{n-k}\lambda_j\mathrm{proj}_{\mathbf{g}^\perp}(K_j)\right)\mathrm{vol}_k\left(\sum_{i=1}^{k}t_iL_i\right),\end{aligned}$$

where in the last equality we used the linearity of the projection. From the linearity of mixed volumes the left-hand side of the above is a polynomial of degree n on $\lambda_1,\dots,\lambda_{n-k},t_1,\dots,t_k$. This also holds for the right-hand side when we expand the volumes as

$$\operatorname{vol}_{n-k}\left(\sum_{j=1}^{n-k}\lambda_j \operatorname{proj}_{\mathbf{g}^\perp}(K_j)\right) =$$

$$= \sum_{j_1,\dots,j_{n-k}=1}^{n-k} V^{(n-k)}\big(\operatorname{proj}_{\mathbf{g}^\perp}(K_{j_1}),\dots,\operatorname{proj}_{\mathbf{g}^\perp}(K_{j_{n-k}})\big)\lambda_{j_1}\dots\lambda_{j_{n-k}}$$

and

$$\operatorname{vol}_k\left(\sum_{i=1}^{k} t_i L_i\right) = \sum_{i_1,\dots,i_k=1}^{k} V^{(k)}(L_{i_1},\dots,L_{i_k})t_{i_1}\dots t_{i_k}.$$

Both polynomials are the same, and then the coefficient of the term $\lambda_1\dots\lambda_{n-k}t_1\dots t_k$ is the same on both sides. Therefore,

$$\binom{n}{k}\big((n-k)!k!V(K_1,\dots,K_{n-k},L_1,\dots,L_k)\big) =$$

$$= \big[(n-k)!V^{(n-k)}(\operatorname{proj}_{\mathbf{g}^\perp}(K_1),\dots\operatorname{proj}_{\mathbf{g}^\perp}(K_{n-k}))\big]\big[k!V^{(k)}(L_1,\dots,L_k)\big],$$

and this gives the desired.

□

We give an example to illustrate formula (6.2.10). In Figure 6.2.2, K is a ball and L is a segment. The body $K+L$ is the "bone" body. From equality (6.1.3), the area of the region (R, say) of the bone outside the ball K is equal to $2V(K,L)$. On the other hand, L is contained in a line $\mathbf{g}$, and the projection of K onto $\mathbf{g}^\perp$ is the dotted diameter. By formula (6.2.10), we have

$$2V(K,L) = \operatorname{length}(\operatorname{proj}_{g^\perp}(K))\operatorname{length}(L).$$

That is, $2V(K,L)$, which is the area of R, is equal to the area of a rectangle whose sides have the respective lengths of L and of the diameter of K. However, the fact that

$$\operatorname{area}(R) = \operatorname{length}(\operatorname{proj}_{g^\perp}(K))\operatorname{length}(L)$$

also follows immediately from Cavalieri's principle if we see R as a (parametrized) union of segments parallel to L, it is interesting to remember that Cavalieri's principle was used in the proof of (6.2.10). An even simpler way of obtaining the equality above is noticing that the "concavity" in the "bottom" of R can be "fullfiled" by the half-circle in the "top" of R (just "cut" the top and "fits" it in the bottom to obtain a rectangle).

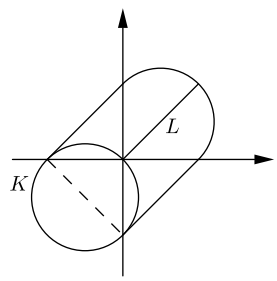


Fig. 6.2.2 An illustration of (6.2.10).

6.3 Minkowski inequalities

One of the main features of mixed volumes is that there are many of inequalities comparing them which lead to *geometric inequalities*. The objective of this section is to state and prove two of the most important of these inequalities, known as *first* and *second Minkowski inequalities*.

Theorem 6.3.1 (1st Minkowski inequality). *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Then*

$$V(K, L, \dots, L)^n \geq \mathrm{vol}_n(K)\mathrm{vol}_n(L)^{n-1},$$

with equality if and only if K and L are homothetical.

Remark 6.3.1. Of course, the inequality above is trivial if K or L do not have maximal affine dimension. On the other hand, for the characterization of the equality case, we need that K and L are assumed to be convex bodies.

Proof. The fourth version of the Brunn–Minkowski inequality (Theorem 5.1.2) gives that the function

$$f(\lambda) = \mathrm{vol}_n\big((1-\lambda)K + \lambda L\big)^{1/n}$$

is concave on $[0, 1]$, and that the equality

$$f(\lambda) = (1-\lambda)\mathrm{vol}_n(K)^{1/n} + \lambda \mathrm{vol}_n(L)^{1/n}$$

holds for every $\lambda \in [0, 1]$ if and only if K and L are homothetical. On the other hand, due to Theorem 6.1.1 we may write

$$f(\lambda)^n = \mathrm{vol}_n\big((1-\lambda)K + \lambda L\big) = \sum_{j=0}^{n} \binom{n}{j} V\big(K[j], L[n-j]\big)(1-\lambda)^j \lambda^{n-j},$$

and hence f is differentiable. Indeed, f is the composition of the strictly positive polynomial function f^n with the n-th root function $t \mapsto t^{1/n}$, which is differentiable if $t > 0$. Hence, we may differentiate both sides of the above to obtain

$$\begin{aligned} nf(\lambda)^{n-1} f'(\lambda) = \\ = \sum_{j=0}^{n} \binom{n}{j} V\big(K[j], L[n-j]\big)\Big((n-j)\lambda^{n-j-1}(1-\lambda)^j - j(1-\lambda)^{j-1}\lambda^{n-j}\Big). \end{aligned} \tag{6.3.1}$$

The unique terms without a $(1-\lambda)$ in the right-hand side polynomial appear for $j = 0$ and $j = 1$. Hence, for $\lambda = 1$, all the terms with $j > 1$ vanish, and consequently the equality above reads

$$\begin{aligned} nf(1)^{n-1} f'(1) = \binom{n}{0} V(L, \dots, L)n - \binom{n}{1} V(K, L, \dots, L) = \\ = n\big(\mathrm{vol}_n(L) - V(K, L, \dots, L)\big). \end{aligned} \tag{6.3.2}$$

Since $f(1) = \mathrm{vol}_n(L)^{1/n}$ and $V(L, \dots, L) = \mathrm{vol}_n(L)$, we get, rearranging the terms of the equality above,

$$V(K, L, \dots, L) = \mathrm{vol}_n(L) - f'(1)\mathrm{vol}_n(L)^{\frac{n-1}{n}}.$$

On the other hand, since f is concave, we may reverse inequality (1.2.6) to obtain

$$f'(1) \leq f(1) - f(0) = \mathrm{vol}_n(L)^{1/n} - \mathrm{vol}_n(K)^{1/n}.$$

Finally, plugging this inequality in the equality above gives

$$V(K, L, \dots, L) \geq \mathrm{vol}_n(L) - \left(\mathrm{vol}_n(L)^{1/n} - \mathrm{vol}_n(K)^{1/n}\right)\mathrm{vol}_n(L)^{\frac{n-1}{n}} = \mathrm{vol}_n(K)^{1/n}\mathrm{vol}_n(L),$$

which is the desired inequality. Of course, equality holds if and only if $f'(1) = f(1) - f(0)$, and we claim that this is equivalent to

$$f(\lambda) = (1-\lambda)f(0) + \lambda f(1) \ \ \forall\, \lambda \in [0,1]. \tag{6.3.3}$$

In fact, if there is some $c \in (0,1)$ for which this equality does not hold (observe that it holds trivially for $\lambda = 0$ and $\lambda = 1$), then we must have $f(c) > (1-c)f(0) + cf(1)$ because f is concave, and hence

$$f'(1) \leq \frac{f(1) - f(c)}{1-c} < \frac{f(1) - (1-c)f(0) - cf(1)}{1-c} = f(1) - f(0),$$

where we used again the version of (1.2.6) for concave functions. As it was mentioned in the beginning of the proof, from the Brunn–Minkowski inequality, we have that (6.3.3) holds if and only if K and L are homothetical. This concludes the proof.

□

As a consequence of the 1st Minkowski inequality, we get another proof of the classical isoperimetric inequality for convex bodies. For the sake of simplicity, we denote the surface area of a convex body K (which can be equivalently defined as the Minkowski surface area of K or as $\mathcal{H}^{n-1}(\partial K)$, see Section 4.1) by $S(K)$.

Corollary 6.3.1 (Isoperimetric inequality). *For any convex body $K \subseteq \mathbb{R}^n$, we have the inequality*

$$\left(\frac{S(K)}{S(B^n)}\right)^{\frac{1}{(n-1)}} \geq \left(\frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)}\right)^{\frac{1}{n}}.$$

Equality holds if and only if K is homothetical to B^n.

Proof. From (6.1.2), we have that $S(K) = nV(B^n, K, \dots, K)$. The first Minkowski inequality implies that

$$\frac{S(K)}{n} = V(B^n, K, \dots, K) \geq \mathrm{vol}_n(K)^{(n-1)/n}\mathrm{vol}_n(B^n)^{1/n}. \tag{6.3.4}$$

Using (6.1.2) for $K = B^n$, we have $S(B^n) = nV(B^n, \dots, B^n) = n\mathrm{vol}_n(B^n)$, and hence we rewrite the inequality above as

$$S(K) \geq n\mathrm{vol}_n(B^n)\frac{\mathrm{vol}_n(K)^{\frac{n-1}{n}}}{\mathrm{vol}_n(B^n)^{\frac{n-1}{n}}} = S(B^n)\frac{\mathrm{vol}_n(K)^{\frac{n-1}{n}}}{\mathrm{vol}_n(B^n)^{\frac{n-1}{n}}},$$

and this gives immediately the desired inequality. It is clear that equality holds in the above if and only if equality holds in the 1st Minkowski inequality, and this is equivalent to K and B^n being homothetical.

□

Remark 6.3.2. It is worth noticing that the Brunn–Minkowski inequality can also be obtained from the 1st Minkowski inequality (see Exercise 6.14).

Theorem 6.3.2 (2nd Minkowski inequality). *We have*

$$V(K, L, \dots, L)^2 \geq V(K, K, L, \dots, L)\mathrm{vol}_n(L)$$

for any convex bodies $K, L \subseteq \mathbb{R}^n$.

Proof. As in the proof of the first Minkowski inequality (Theorem 6.3.1), we let

$$f(\lambda) = \mathrm{vol}_n\big((1-\lambda)K + \lambda L\big)^{1/n}$$

for $\lambda \in [0, 1]$. By the polynomiality of f^n, we know that f is twice differentiable, and since f is concave, it follows from Corollary 1.2.2 that $f''(1) \leq 0$. Differentiating both sides of (6.3.1), we get

$$\begin{aligned} n(n-1)f(\lambda)^{n-2}f'(\lambda)^2 + nf(\lambda)^{n-1}f''(\lambda) = \\ = \sum_{j=0}^{n} \binom{n}{j} V\big(K[j], L[n-j]\big)\left[(n-j)(n-j-1)\lambda^{n-j-2}(1-\lambda)^j - \right. \\ \left. 2(n-j)j\lambda^{n-j-1}(1-\lambda)^{j-1} + j(j-1)\lambda^{n-j}(1-\lambda)^{j-2}\right]. \end{aligned}$$

Making $\lambda = 1$, the terms for $j > 2$ in the right-hand side of the above vanish, and we get

$$\begin{aligned} n(n-1)f(1)^{n-2}f'(1)^2 + nf(1)^{n-1}f''(1) = \binom{n}{0}V(L, \dots, L)n(n-1) + \\ -\binom{n}{1}V(K, L, \dots, L)2(n-1) + \binom{n}{2}V(K, K, L, \dots, L)2 = \\ = n(n-1)\mathrm{vol}_n(L) - 2n(n-1)V(K, L, \dots, L) + n(n-1)V(K, K, L, \dots, L). \end{aligned}$$

The definition of f gives that $f(1) = \mathrm{vol}_n(L)^{1/n}$, and from (6.3.2), we have an expression for $f'(1)$. This leads to

$$f(1)^{n-2}f'(1)^2 = \mathrm{vol}_n(L)^{\frac{n-2}{n}}\frac{\big(\mathrm{vol}_n(L) - V(K, L, \dots, L)\big)^2}{\mathrm{vol}_n(L)^{\frac{2(n-1)}{n}}} = \frac{\big(\mathrm{vol}_n(L) - V(K, L, \dots, L)\big)^2}{\mathrm{vol}_n(L)}.$$

Plugging this equality in the previous equality (and dividing by n), we get

$$\begin{aligned} (n-1)\frac{\big(\mathrm{vol}_n(L) - V(K, L, \dots, L)\big)^2}{\mathrm{vol}_n(L)} + \mathrm{vol}_n(L)^{\frac{n-1}{n}}f''(1) = \\ = (n-1)\mathrm{vol}_n(L) - 2(n-1)V(K, L, \dots, L) + (n-1)V(K, K, L, \dots, L). \end{aligned}$$

Multiplying both sides by $\mathrm{vol}_n(L)$, cancelling the equal terms on both sides, and rearranging the terms, we get

$$\mathrm{vol}_n(L)^{\frac{2n-1}{n}}f''(1) = (n-1)\Big(V(K, K, L, \dots, L)\mathrm{vol}_n(L) - V(K, L, \dots, L)^2\Big),$$

and we know that the left-hand side above is non-positive. Hence

$$V(K, K, L, \dots, L)\mathrm{vol}_n(L) - V(K, L, \dots, L)^2 \leq 0,$$

and this is the desired inequality.

□

6.4 Quermassintegrals and intrinsic volumes

For a given $C \in CC^n$, the polynomiality of the volume and the definition of mixed volumes yield the following formula for the n-dimensional volume of the set $C + tB^n$ (which consists of points whose distance to C is not greater than t; see Remark 2.5.1):

$$\begin{aligned}\mathrm{vol}_n(C + tB^n) = \\ = V(C,\dots,C) + \binom{n}{1} V(C,\dots,C,B^n)t + \binom{n}{2} V(C,\dots,C,B^n,B^n)t^2 + \dots \\ \dots + \binom{n}{n-1} V(C,B^n,\dots,B^n)t^{n-1} + V(B^n,\dots,B^n)t^n,\end{aligned}$$

for $t \geq 0$. This is known as the *Steiner formula* (or *Steiner polynomial*). For each $i = 0,\dots,n$, the number

$$W_i(C) := V\big(\underbrace{C,\dots,C}_{n-i},\underbrace{B^n,\dots,B^n}_{i}\big).$$

is the *i-th quermassintegral of C*. It is worth mentioning that some authors use the term *mean projection measure* instead of quermassintegral, but we will avoid this terminology (for the reasoning behind this name, see Remark 6.5.5). Under this notation, the Steiner formula becomes

$$\mathrm{vol}_n(C + tB^n) = \sum_{j=0}^{n} \binom{n}{j} W_j(C) t^j. \tag{6.4.1}$$

The quermassintegrals of a set $K \in CC^n$ carry a lot of geometric information about K. As immediate examples, we have that $W_0(K)$ is the *volume* of K and, if K is a convex body, then $nW_1(K)$ is its *surface area* (this follows from equality (6.1.2)). Even more, equality (7.1.11), that will be proved later, yields

$$\begin{aligned}W_{n-1}(K) = V(K,B^n,\dots,B^n) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\,d\mathcal{H}^{n-1}(u) = \\ = \frac{\mathrm{vol}_{n-1}(\mathbb{S}^{n-1})}{2n} w(K) = \frac{\mathrm{vol}_n(B^n)}{2} w(K),\end{aligned} \tag{6.4.2}$$

where $w(K)$ is the *mean width* of $K \in CC^n$ (introduced in Section 5.5). Summarizing, the $(n-1)$-th quermassintegral can be (uniformly) normalized to provide mean width of sets of CC^n. Observe that, if we restrict ourselves to convex bodies, then (6.4.2) together with the 1st Minkowski inequality (Theorem 6.3.1) give immediately Urysohn's inequality (5.5.1). Hence, we have a proof which does not use Steiner symmetrizations. We also have the advantage of characterizing the equality case, as announced in Remark 5.5.1: equality in Urysohn's inequality implies equality in the 1st Minkowski inequality, which gives that K must be a ball.

This geometric meaning justifies the investigation of quermassintegrals. Next, we state and prove some of their properties. Notice that, for each $i = 0,\dots,n$, we can regard the i-th quermassintegral as a function $W_i : CC^n \to [0,+\infty)$.

Theorem 6.4.1. *For any $i = 0,\dots,n$ the following properties hold:*

(i) *$W_i(\cdot)$ is continuous (with respect to the Hausdorff metric in CC^n), translation-invariant and $\mathrm{O}(n)$-invariant.*

(ii) *For every $C \in CC^n$ and every $t \geq 0$, we have $W_i(tC) = t^{n-i}W_i(C)$. In other words, $W_i(\cdot)$ is (positive) homogeneous of degree $n-i$.*

(iii) *If $C_1, C_2 \in CC^n$ are such that $C_1 \subseteq C_2$, then $W_i(C_1) \leq W_i(C_2)$.*

Proof. For **(i)**, we use Corollary 6.2.1. The continuity of $W_i(\cdot)$ comes immediately from the continuity of mixed volumes. The fact that $W_i(\cdot)$ is translation-invariant also comes from this property for mixed volumes. To prove that $W_i(\cdot)$ is $\mathrm{O}(n)$-invariant, simply notice that by item **(iii)** of the mentioned corollary, we have

$$\begin{aligned} W_i(TC) = V\big(TC[n-i], B^n[i]\big) = V\big(TC[n-i], TB^n[i]\big) = \\ = |\det T| V\big(C[n-i], B^n[i]\big) = W_i(C) \end{aligned}$$

whenever $T \in \mathrm{O}(n)$. Here, we used the facts that $TB^n = B^n$ and $|\det T| = 1$ if T is an orthogonal transformation.

The proof of **(ii)** is an easy calculation. From the multilinearity of mixed volumes, we get straightforwardly

$$W_i(tC) = V\big(tC[n-i], B^n[i]\big) = t^{n-i}V\big(C[n-i], B^n[i]\big) = t^{n-i}W_i(C).$$

The property **(iii)** is a consequence of the monotonicity property of mixed volumes (see Theorem 6.2.2) applied iteratively $n-i$ times:

$$\begin{aligned} W_i(C_1) = V(C_1, \ldots, C_1, B^n, \ldots, B^n) \leq V(C_2, C_1, \ldots, B^n, \ldots, B^n) \leq \\ V(C_2, C_2, C_1, \ldots, B^n, \ldots, B^n) \leq \ldots \leq V(C_2, \ldots, C_2, B^n, \ldots, B^n) = W_i(C_2), \end{aligned}$$

where at each step the C's appear in the first $n-i$ entries, and the i remaining entries are filled with B^n.

□

As a further property, we also have that the quermassintegrals have the valuation property. That is, for every $0 \leq j \leq n$, we have

$$W_j(K \cup L) + W_j(K \cap L) = W_j(K) + W_j(L), \tag{6.4.3}$$

whenever $K, L \in CC^n$ are such that $K \cup L \in CC^n$. The proof of this fact follows immediately as a particular case of the more general result below.

Proposition 6.4.1. *Let $K, L \in CC^n$ be such that $K \cup L \in CC^n$, and let $0 \leq j \leq n$. For any $C_{j+1}, \ldots, C_n \in CC^n$, we have*

$$\begin{aligned} V\big((K \cup L)[j], C_{j+1}, \ldots, C_n\big) + V\big((K \cap L)[j], C_{j+1}, \ldots, C_n\big) = \\ = V\big(K[j], C_{j+1}, \ldots, C_n\big) + V\big(L[j], C_{j+1}, \ldots, C_n\big). \end{aligned}$$

In other words, the mapping $CC^n \ni K \mapsto V\big(K[j], C_{j+1}, \ldots, C_n\big)$ has the valuation property.

Proof. We start by proving some set equalities that we will need. First, remember that if $K, L \in CC^n$ are such that $K \cup L \in CC^n$, then we obviously have that $K \cap L \neq \emptyset$. Let $D \subseteq \mathbb{R}^n$ be also non-empty, compact and convex. The equalities that we need are the following ones:

$$\begin{aligned} (K \cup L) + D = (K + D) \cup (L + D) \text{ and} \\ (K \cap L) + D = (K + D) \cap (L + D). \end{aligned}$$

The first equality is immediate, as well as the inclusion $(K \cap L) + D \subseteq (K + D) \cap (L + D)$. To prove the reverse inclusion, let $x \in (K + D) \cap (L + D)$. Then there exist $y \in K$, $z \in L$ and $a, b \in D$ such that $x = y + a = z + b$. Once $y, z \in K \cup L$, by convexity we get that $\mathrm{seg}[y, z] \subseteq K \cup L$. We claim that there exists a number $\lambda_0 \in [0, 1]$ such that $(1 - \lambda)y + \lambda z \in K \cap L$. To verify that, it is clear that it suffices to consider the case where $y \notin L$ and $z \notin K$. We let

$$\lambda_0 := \sup\big\{\lambda \in [0, 1] : (1 - \lambda)y + \lambda z \in K\big\}.$$

Because K is closed, it is clear that $y_0 := (1 - \lambda_0)y + \lambda_0 z \in K$, and by convexity it follows that $(1 - \lambda)y + \lambda z \in K$ whenever $0 \leq \lambda \leq \lambda_0$. Now we let

$$\lambda_1 := \inf\big\{\lambda \in [0, 1] : (1 - \lambda)y + \lambda z \in L\big\}.$$

Similarly to the previous argument, we get that $z_1 := (1 - \lambda_1)y + \lambda z \in L$, and that $(1 - \lambda)y + \lambda z \in L$ if and only if $\lambda_1 \leq \lambda \leq 1$. If $y_0 \notin L$, then $\lambda_0 < \lambda_1$, and the points $(1 - \lambda)y + \lambda z$ for $\lambda_0 < \lambda < \lambda_1$ are not in $K \cup L$. This contradicts the fact that $\mathrm{seg}[y, z] \subseteq K \cup L$.

Hence, we may fix a point $(1 - \lambda)y + \lambda z \in K \cap L$ for some $\lambda \in [0, 1]$. Since D is convex, we have that $(1 - \lambda)a + \lambda b \in D$. Thus, we write

$$x = (1 - \lambda)x + \lambda x = (1 - \lambda)(y + a) + \lambda(z + b) = \big[(1 - \lambda)y + \lambda z\big] + \big[(1 - \lambda)a + \lambda b\big],$$

and the latter is a sum of a point of $K \cap L$ with a point of D, and the second equality is proved (compare with Exercise 2.8).

Since we always have that $\mathrm{vol}_n(E \cup F) + \mathrm{vol}_n(E \cap F) = \mathrm{vol}_n(E) + \mathrm{vol}_n(F)$ for any measurable sets $E, F \subseteq \mathbb{R}^n$, the equalities above have the consequence

$$\mathrm{vol}_n\big((K \cup L) + D\big) + \mathrm{vol}_n\big((K \cap L) + D\big) = \mathrm{vol}_n(K + D) + \mathrm{vol}_n(L + D).$$

Even more, because $t(K \cup L) = tK \cup tL$ and $t(K \cap L) = tK \cap tL$ for any $t \geq 0$, we may write

$$\mathrm{vol}_n\big(t(K \cup L) + D\big) + \mathrm{vol}_n\big(t(K \cap L) + D\big) = \mathrm{vol}_n(tK + D) + \mathrm{vol}_n(tL + D), \tag{6.4.4}$$

for every $t \geq 0$.

Now let $C_2, \ldots, C_n \in CC^n$ be arbitrarily given. By the polynomiality of the volume, for every $t, t_2, \ldots, t_n \geq 0$, we may write

$$\mathrm{vol}_n\big(t(K \cup L) + t_2C_2 + \ldots + t_nC_n\big) =$$
$$= \sum_{j=0}^{n} \binom{n}{j} t^j \sum_{i_{j+1}, \ldots, i_n = 2}^{n} V\big((K \cup L)[j], C_{i_{j+1}}, \ldots, C_{i_n}\big) t_{i_{j+1}} \ldots t_{i_n}.$$

That is, we are organizing the polynomial $\mathrm{vol}_n\big(t(K \cup L) + t_2C_2 + \ldots + t_nC_n\big)$ by grouping the terms where t^j appears. Replacing $K \cup L$ by $K \cap L$, we have the equality

$$\mathrm{vol}_n\big(t(K \cap L) + t_2C_2 + \ldots + t_nC_n\big) =$$
$$= \sum_{j=0}^{n} \binom{n}{j} t^j \sum_{i_{j+1}, \ldots, i_n = 2}^{n} V\big((K \cap L)[j], C_{i_{j+1}}, \ldots, C_{i_n}\big) t_{i_{j+1}} \ldots t_{i_n},$$

and replacing $K \cup L$ by K and L, respectively, we derive the equalities

$$\mathrm{vol}_n\big(tK + t_2C_2 + \ldots + t_nC_n\big) = \sum_{j=0}^{n} \binom{n}{j} t^j \sum_{i_{j+1},\ldots,i_n=2}^{n} V\big(K[j], C_{i_{j+1}}, \ldots, C_{i_n}\big) t_{i_{j+1}} \ldots t_{i_n}$$

and

$$\mathrm{vol}_n\big(tL + t_2C_2 + \ldots + t_nC_n\big) = \sum_{j=0}^{n} \binom{n}{j} t^j \sum_{i_{j+1},\ldots,i_n=2}^{n} V\big(L[j], C_{i_{j+1}}, \ldots, C_{i_n}\big) t_{i_{j+1}} \ldots t_{i_n}.$$

Letting $t_2C_2 + \ldots + t_nC_n = E$ in (6.4.4), we arrive at an equality of polynomials, and comparing the coefficients, we obtain

$$\begin{aligned} V\big((K \cup L)[j], C_{i_{j+1}}, \ldots, C_{i_n}\big) + V\big((K \cap L)[j], C_{i_{j+1}}, \ldots, C_{i_n}\big) = \\ = V\big(K[j], C_{i_{j+1}}, \ldots, C_{i_n}\big) + V\big(L[j], C_{i_{j+1}}, \ldots, C_{i_n}\big), \end{aligned}$$

for any $0 \le j \le n$, and all indices $i_{j+1}, \ldots, i_n$. Since the bodies $C_2, \ldots, C_n$ were arbitrarily chosen in CC^n the proof is complete.

□

Remark 6.4.1. Notice that in the case $j = 1$ this gives another proof of Theorem 6.2.3.

The quermassintegrals have the "disadvantage" of being dimension-dependent in the following sense: if $C \subseteq \mathbb{R}^{n-1}$ is naturally identified as a subset of $\mathbb{R}^n$ (for example, identifying $\mathbb{R}^{n-1}$ with the subset $\{(x_1, \ldots, x_{n-1}, 0) : x_1, \ldots, x_{n-1} \in \mathbb{R}\}$ of $\mathbb{R}^n$), then the quermassintegrals of C calculated in $\mathbb{R}^n$ are not necessarily the same as the quermassintegrals calculated in $\mathbb{R}^{n-1}$. In fact, we have useful expressions for the quermassintegrals of $\mathbb{R}^n$ in terms of the quermassintegrals of $\mathbb{R}^{n-1}$. Recall that we use the following notation:

$$\kappa_n := \mathrm{vol}_n(B^n),$$

for each $n \in \mathbb{N}$. Also, we denote the j-th quermassintegral of $\mathbb{R}^{n-1}$ by w_j (for each $j = 0, \ldots, n-1$).

Proposition 6.4.2. *Let $C \in CC^{n-1}$, and let $T\colon \mathbb{R}^{n-1} \to \mathbb{R}^n$ be the natural identification*

$$T(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, 0),$$

as described above. Then

$$W_j(TC) = \frac{j\kappa_j}{n\kappa_{j-1}} w_{j-1}(C),$$

for $j = 1, \ldots, n$.

Proof. Abusing a little of the notation, we will denote TC simply by C, and also $T\mathbb{R}^{n-1}$ by $\mathbb{R}^{n-1}$ and TB^{n-1} by B^{n-1}. The key point of the proof is to calculate the n-dimensional volume of $C + \lambda B^n$ by Cavalieri's principle (see (4.2.1) and Theorem A.6.3). For each fixed $\lambda \ge 0$, we have

$$\mathrm{vol}_n(C + \lambda B^n) = \int_{-\lambda}^{\lambda} \mathrm{vol}_{n-1}\Big((C + \lambda B^n) \cap (\mathbb{R}^{n-1} + te_n)\Big)\, dt. \tag{6.4.5}$$

In fact, since C is contained in the hyperplane $\{(x_1, \ldots, x_{n-1}, 0)\}$, it is clear that $h_{C+\lambda B^n}(e_n) = \lambda = h_{C+\lambda B^n}(-e_n)$. Now we claim that

$$C + (\lambda^2 - t^2)^{1/2} B^{n-1} + te_n = (C + \lambda B^n) \cap (\mathbb{R}^{n-1} + te_n) \tag{6.4.6}$$

whenever $\lambda \geq 0$ and $-\lambda \leq t \leq \lambda$. To prove this, we assume that $\lambda > 0$, since the case where $t = \lambda = 0$ is trivial. Notice that a point of the set in the left-hand side has the form

$$\Big(x_1 + (\lambda^2 - t^2)^{1/2}u_1, \ldots, x_{n-1} + (\lambda^2 - t^2)^{1/2}u_{n-1}, t\Big),$$

where $(x_1, \ldots, x_{n-1}, 0) \in C$ and $(u_1, \ldots, u_{n-1}, 0) \in B^{n-1}$. It is clear that such a point belongs to $\mathbb{R}^{n-1} + te_n$ (the last coordinate is t). Now we write

$$\begin{aligned}\Big(x_1 + (\lambda^2 - t^2)^{1/2}u_1, \ldots, x_{n-1} + (\lambda^2 - t^2)^{1/2}u_{n-1}, t\Big) = \\ = (x_1, \ldots, x_{n-1}, 0) + \Big((\lambda^2 - t^2)^{1/2}u_1, \ldots, (\lambda^2 - t^2)^{1/2}u_{n-1}, t\Big),\end{aligned}$$

and we claim that the last summand in the right-hand side is a point of λB^n. Indeed

$$\Big|\big((\lambda^2 - t^2)^{1/2}u_1, \ldots, (\lambda^2 - t^2)^{1/2}u_{n-1}, t\big)\Big| = \sqrt{(\lambda^2 - t^2)(u_1^2 + \ldots + u_{n-1}^2) + t^2} \leq \lambda,$$

where the inequality justifies since $(u_1, \ldots, u_{n-1}, 0) \in B^{n-1}$. This gives the inclusion "$\subseteq$" in (6.4.6). Now we prove the reverse inclusion. Let

$$(x_1 + \lambda v_1, \ldots, x_{n-1} + \lambda v_{n-1}, \lambda v_n) \in (C + \lambda B^n) \cap (\mathbb{R}^{n-1} + te_n),$$

with $(x_1, \ldots, x_{n-1}, 0) \in C$, $(v_1, \ldots, v_n) \in B^n$ and $\lambda v_n = t$. We only have to prove that $(\lambda v_1, \ldots, \lambda v_{n-1}, 0) \in (\lambda^2 - t^2)^{1/2}B^{n-1}$. This comes immediately from the inequality

$$(\lambda v_1)^2 + \ldots + (\lambda v_{n-1})^2 = \lambda^2(v_1^2 + \ldots + v_{n-1}^2) \leq \lambda^2(1 - v_n^2) = \lambda^2\left(1 - \frac{t^2}{\lambda^2}\right) = \lambda^2 - t^2,$$

and hence equality (6.4.6) is proved. Now we observe that $C + (\lambda^2 - t^2)^{1/2}B^{n-1} + te_n$ is a translation of $C + (\lambda^2 - t^2)^{1/2}B^{n-1}$. Therefore,

$$\begin{aligned}\mathrm{vol}_{n-1}\big((C + \lambda B^n) \cap (\mathbb{R}^{n-1} + te_n)\big) = \mathrm{vol}_{n-1}\big(C + (\lambda^2 - t^2)^{1/2}B^{n-1} + te_n\big) = \\ = \mathrm{vol}_{n-1}\big(C + (\lambda^2 - t^2)^{1/2}B^{n-1}\big),\end{aligned}$$

for any $\lambda \geq 0$ and $-\lambda \leq t \leq \lambda$. Replacing this equality in (6.4.5), we get

$$\begin{aligned}\mathrm{vol}_n(C + \lambda B^n) = \int_{-\lambda}^{\lambda} \mathrm{vol}_{n-1}\Big((C + \lambda B^n) \cap (\mathbb{R}^{n-1} + te_n)\Big)\, dt = \\ = \int_{-\lambda}^{\lambda} \mathrm{vol}_{n-1}\big(C + (\lambda^2 - t^2)^{1/2}B^{n-1}\big)\, dt = \int_{-\lambda}^{\lambda} \sum_{j=0}^{n-1} \binom{n-1}{j} w_j(C)(\lambda^2 - t^2)^{j/2}\, dt = \\ = \sum_{j=0}^{n-1} \binom{n-1}{j} w_j(C) \int_{-\lambda}^{\lambda} (\lambda^2 - t^2)^{j/2}\, dt = \sum_{j=0}^{n-1} \binom{n-1}{j} w_j(C)\lambda^j \int_{-\lambda}^{\lambda} \left(1 - \left(\frac{t}{\lambda}\right)^2\right)^{\frac{j}{2}} dt.\end{aligned}$$

Making a change of variables ($s = t/\lambda$) in the last integral, we finally arrive at

$$\mathrm{vol}_n(C + \lambda B^n) = \sum_{j=0}^{n-1} \binom{n-1}{j} w_j(C)\lambda^{j+1} \int_{-1}^{1} \left(1 - s^2\right)^{\frac{j}{2}} ds.$$

Now we replace the right-hand side above by its Steiner polynomial expression to get

$$\sum_{j=0}^{n}\binom{n}{j}W_j(C)\lambda^j=\sum_{j=0}^{n-1}\binom{n-1}{j}w_j(C)\lambda^{j+1}\int_{-1}^{1}\left(1-s^2\right)^{\frac{j}{2}}ds,$$

and since this holds for any $\lambda \geq 0$, we get that the left and right-hand sides have to be the same polynomial. Then the coefficients of each power of λ are equal, and thus we obtain

$$\binom{n}{j}W_j(C)=\binom{n-1}{j-1}w_{j-1}(C)\int_{-1}^{1}(1-s^2)^{\frac{j-1}{2}}ds,$$

for each $j=1,\dots,n$. We rewrite this as

$$W_j(C)=\frac{j}{n}w_{j-1}(C)\int_{-1}^{1}(1-s^2)^{\frac{j-1}{2}}ds. \tag{6.4.7}$$

Observe that this already gives an expression for the quermassintegrals in $\mathbb{R}^n$ in terms of the quermassintegrals in $\mathbb{R}^{n-1}$. To get the final expression, first let $j=1$, and notice that

$$\int_{-1}^{1}(1-s^2)^{\frac{j-1}{2}}ds=\int_{-1}^{1}1\,ds=2=\frac{\mathrm{vol}_1(B^1)}{\mathrm{vol}_0(B^0)},$$

where we recall the reader that vol_0 is the counting measure (the 0-dimensional Hausdorff measure) and B^0 is a point, whereas $\mathrm{vol}_1(B^1)$ is the length of the segment $[-1,1]$. If $j>1$, then we observe that B^j can be parametrized as (u,s) where $u\in\sqrt{1-s^2}B^{j-1}$ and $-1\leq s\leq 1$. Hence, Cavalieri's principle (see (4.2.1) and Theorem A.6.3) gives

$$\begin{aligned}\mathrm{vol}_j(B^j)=\int_{-1}^{1}\mathrm{vol}_{j-1}\left((1-s^2)^{1/2}B^{j-1}\right)ds=\int_{-1}^{1}(1-s^2)^{\frac{j-1}{2}}\mathrm{vol}_{j-1}(B^{j-1})\,ds=\\ =\mathrm{vol}_{j-1}(B^{j-1})\int_{-1}^{1}(1-s^2)^{\frac{j-1}{2}}ds,\end{aligned}$$

and this yields

$$\int_{-1}^{1}(1-s^2)^{\frac{j-1}{2}}ds=\frac{\mathrm{vol}_j(B^j)}{\mathrm{vol}_{j-1}(B^{j-1})}=\frac{\kappa_j}{\kappa_{j-1}}.$$

Replacing this equality in (6.4.7), we finish the proof.

□

This proposition allows us to introduce new functionals defined for compact and convex sets which do not depend on the dimension of the ambient space. That is, these functionals produce the same value for a given set $C\subseteq CC^k$ if this is identified with a subset of a k-dimensional affine subspace of $\mathbb{R}^n$, $n>k$. These are the *intrinsic volumes* $V_j(\cdot)$ defined by

$$V_{n-i}(C)=\binom{n}{i}\frac{1}{\kappa_i}W_i(C), \tag{6.4.8}$$

for $i=0,\dots,n$ and $C\in CC^n$. In fact, if $C\in CC^{n-1}$, then for each $i=0,\dots,n-1$ we may calculate the intrinsic volume in $\mathbb{R}^{n-1}$ as

$$V_{n-1-i}(C)=\binom{n-1}{i}\frac{1}{\kappa_i}w_i(C).$$

If C is identified as a subset of an $(n-1)$-dimensional affine subspace of $\mathbb{R}^n$, then Proposition 6.4.2 gives

$$V_{n-1-i}(C) = \binom{n-1}{i}\frac{1}{\kappa_i}w_i(C) = \binom{n-1}{i}\frac{1}{\kappa_i}\frac{n\kappa_i}{(i+1)\kappa_{i+1}}W_{i+1}(C) = \\ = \binom{n}{i+1}\frac{1}{\kappa_{i+1}}W_{i+1}(C),$$

and the latter is precisely the intrinsic volume $V_{n-(i+1)}(C)$ calculated in $\mathbb{R}^n$ as defined in (6.4.8). If the dimension of C is lower than $n-1$, then we just have to repeat the argument indutively and it follows that the intrinsic volumes do not depend on the dimension of the ambient space, but only on the set. With intrinsic volumes, the Steiner polynomial (6.4.1) clearly reads

$$\mathrm{vol}_n(C+\lambda B^n) = \sum_{j=0}^{n}\kappa_j V_{n-j}(C)\lambda^j,$$

for $\lambda \geq 0$.

6.5 Formulas for quermassintegrals

In this section, we derive some formulas for quermassintegrals. First, we need a comment about the $(n-1)$-dimensional volume of the projection of a polytope onto a hyperplane. Let $P \subseteq \mathbb{R}^n$ be a polytope and $u \in \mathbb{S}^{n-1}$. Since the intersections of its facets have affine dimension smaller than $n-1$, we may write

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(P)\big) = \frac{1}{2}\sum_{F \text{ facet of } P}\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(F)\big),$$

where the multiplication by $1/2$ justifies since, up to a set of zero $(n-1)$-dimensional measure, all points of $\mathrm{proj}_{u^\perp}(P)$ have two pre-images on ∂P. Now, given a facet F of P, denote by u_F its outer normal vector. Since the volume is invariant under translations, we may assume that, without loss of generality, $F \subseteq u_F^\perp$. In this case, the projection of $u_F^\perp$ onto $u^\perp$ is a linear map whose determinant is $|\langle u, u_F\rangle|$ (the sign depends on the orientation of the respective idenfications of the hyperplanes with $\mathbb{R}^{n-1}$). Therefore,

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(F)\big) = |\langle u, u_F\rangle|\mathrm{vol}_{n-1}(F).$$

It follows that

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(P)\big) = \frac{1}{2}\sum_{F \text{ facet of } P}|\langle u, u_F\rangle|\mathrm{vol}_{n-1}(F). \tag{6.5.1}$$

Throughout this section, any hyperplane $u^\perp \subseteq \mathbb{R}^n$ is naturally identified with $\mathbb{R}^{n-1}$ by an isometric isomorphism. Hence, any projection onto $u^\perp$ is identified with a subset of $\mathbb{R}^{n-1}$, and its $(n-1)$-dimensional volume can be seen both as the Lebesgue volume in $\mathbb{R}^{n-1}$ or as the $(n-1)$-dimensional Hausdorff measure (in $\mathbb{R}^n$ or in $\mathbb{R}^{n-1}$).

Theorem 6.5.1 (Cauchy's formula). *For any $K \in CC^n$, we have the equality*

$$nW_1(K) = \frac{1}{\kappa_{n-1}} \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)\, d\mathcal{H}^{n-1}(u), \tag{6.5.2}$$

where $d\mathcal{H}^{n-1}(u)$ means that the integration is taken with respect to the Hausdorff $(n-1)$-dimensional measure on the variable $u \in \mathbb{S}^{n-1}$.

Remark 6.5.1. Recall that $nW_1(K)$ is the surface area of K.

Remark 6.5.2. This formula is also known as *Cauchy's surface area formula*, or *Cauchy's projection formula*. These names will be used throughout the text.

Proof. Notice that for any $v \in \mathbb{S}^{n-1}$, we may write

$$\int_{\mathbb{S}^{n-1}} |\langle v, u\rangle|\, d\mathcal{H}^{n-1}(u) = 2c_n$$

for some constant $c_n > 0$ which only depends on the dimension. Indeed, since $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ is rotation invariant, we get that the integral in the right-hand side of the above is constant for $v \in \mathbb{S}^{n-1}$. The correct expression for c_n will be obtained at the end of the proof.

The strategy of the proof is to consider first polytopes, and then to obtain the general case by approximations. Let $P \subseteq \mathbb{R}^n$ be a polytope. For any facet F of P, with outer normal u_F, we have

$$\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(F)\big)\, d\mathcal{H}^{n-1}(u) = \mathrm{vol}_{n-1}(F) \int_{\mathbb{S}^{n-1}} |\langle u_F, u\rangle|\, d\mathcal{H}^{n-1}(u) = 2c_n \mathrm{vol}_{n-1}(F).$$

Therefore, we get from (6.5.1) the equality

$$\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(P)\big)\, d\mathcal{H}^{n-1}(u) =$$

$$= \frac{1}{2} \cdot \sum_{F \text{ facet of } P} \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(F)\big)\, d\mathcal{H}^{n-1}(u) = c_n \sum_{F \text{ facet of } P} \mathrm{vol}_{n-1}(F),$$

and the last sum equals the surface area of P, that is, $nW_1(P)$. This shows that the equality (6.5.2) with κ_{n-1} replaced by c_n holds for polytopes.

Now let $K \in CC^n$. For each $j \in \mathbb{N}$, let P_j be a polytope such that $P_j \subseteq K \subseteq P_j + \frac{1}{j}B^n$ (see Theorem 2.6.2). It is clear that $P_j \to K$ in the Hausdorff metric as $j \to \infty$. Hence, we get that

$$W_1(P_j) \to W_1(K) \tag{6.5.3}$$

when $j \to \infty$. For any $u \in \mathbb{S}^{n-1}$, we clearly have

$$\mathrm{proj}_{u^\perp}(P_j) \subseteq \mathrm{proj}_{u^\perp}(K) \subseteq \mathrm{proj}_{u^\perp}\Big(P_j + \frac{1}{j}B^n\Big) = \mathrm{proj}_{u^\perp}(P_j) + \frac{1}{j}\mathrm{proj}_{u^\perp}(B^n),$$

where the last equality comes from the linearity of the projection. Consequently,

$$\begin{aligned} 0 \le \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) - \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(P_j)\big) \le \\ \le \mathrm{vol}_{n-1}\Big(\mathrm{proj}_{u^\perp}(P_j) + \frac{1}{j}\mathrm{proj}_{u^\perp}(B^n)\Big) - \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(P_j)\big). \end{aligned} \tag{6.5.4}$$

Under the canonical identification between $u^\perp$ and $\mathbb{R}^{n-1}$, we have that $\mathrm{proj}_{u^\perp}(B^n)$ is identified with B^{n-1}. Hence, recalling that we denote the i-th quermassintegral in $\mathbb{R}^{n-1}$ by w_i, we write

$$\begin{aligned}
&\operatorname{vol}_{n-1}\Big(\operatorname{proj}_{u^\perp}(P_j)+\frac{1}{j}\operatorname{proj}_{u^\perp}(B^n)\Big)-\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(P_j)\big)=\\
&=\sum_{i=1}^{n-1}\binom{n-1}{i}\Big(\frac{1}{j}\Big)^i w_i\big(\operatorname{proj}_{u^\perp}(P_j)\big)\le\frac{1}{j}\sum_{i=1}^{n-1}\binom{n-1}{i}\Big(\frac{1}{j}\Big)^{i-1} w_i\big(\operatorname{proj}_{u^\perp}(K)\big)\le\\
&\le\frac{1}{j}(n-1)!\sum_{i=1}^{n-1} w_i(\operatorname{proj}_{u^\perp}(K)),
\end{aligned}\tag{6.5.5}$$

where the last inequality justifies because we always have $\binom{n-1}{i}\le(n-1)!$ and $(1/j)^{i-1}\le 1$. To bound the last sum, let $\rho>0$ be such that $K\subseteq\rho B^n$. Then, for any $u\in\mathbb{S}^{n-1}$ and $i=1,\dots,n-1$, we have

$$w_i\big(\operatorname{proj}_{u^\perp}(K)\big)\le w_i\big(\operatorname{proj}_{u^\perp}(\rho B^n)\big)=w_i(\rho B^{n-1})=\rho^{n-1-i}w_i(B^{n-1}),$$

where the last equality holds because under the canonical identification between $u^\perp$ and $\mathbb{R}^{n-1}$, we have that $\operatorname{proj}_{u^\perp}(\rho B^n)$ is identified with ρB^{n-1}. Now we observe that $w_i(B^{n-1})=\kappa_{n-1}$ for every $i=0,\dots,n-1$. We get

$$\sum_{i=1}^{n-1} w_i\big(\operatorname{proj}_{u^\perp}(K)\big)\le\sum_{i=1}^{n-1}\rho^{n-1-i}\kappa_{n-1}=:c_0,\tag{6.5.6}$$

and c_0 is a constant which only depends on K and on the dimension n. Setting $c:=(n-1)!c_0$, we get a constant which also depends only on K and on n, and replacing inequality (6.5.6) in inequality (6.5.5), and also considering inequality (6.5.4), we get

$$\begin{aligned}
&\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(K)\big)-\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(P_j)\big)\le\\
&\le\operatorname{vol}_{n-1}\Big(\operatorname{proj}_{u^\perp}(P_j)+\frac{1}{j}\operatorname{proj}_{u^\perp}(B^n)\Big)-\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(P_j)\big)\le\\
&\le\frac{1}{j}(n-1)!\sum_{i=1}^{n-1} w_i(\operatorname{proj}_{u^\perp}(K))\le\frac{1}{j}(n-1)!c_0=\frac{c}{j},
\end{aligned}$$

were $c>0$ does not depend on u or j. We finally arrive at

$$\begin{aligned}
&0\le\int_{\mathbb{S}^{n-1}}\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(K)\big)\,d\mathcal{H}^{n-1}(u)-\int_{\mathbb{S}^{n-1}}\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(P_j)\big)\,d\mathcal{H}^{n-1}(u)=\\
&=\int_{\mathbb{S}^{n-1}}\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(K)\big)-\big(\operatorname{proj}_{u^\perp}(P_j)\big)\,d\mathcal{H}^{n-1}(u)\le\int_{\mathbb{S}^{n-1}}\frac{c}{j}\,d\mathcal{H}^{n-1}(u)=\frac{c}{j}\operatorname{vol}_{n-1}(\mathbb{S}^{n-1}).
\end{aligned}$$

Therefore, we have

$$\int_{\mathbb{S}^{n-1}}\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(P_j)\big)\,d\mathcal{H}^{n-1}(u)\longrightarrow\int_{\mathbb{S}^{n-1}}\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(K)\big)\,d\mathcal{H}^{n-1}(u)\tag{6.5.7}$$

as $j\to\infty$. Since we already have the Cauchy formula for polytopes (with κ_{n-1} replaced by c_n), the equality

$$nW_1(P_j)=\frac{1}{c_n}\int_{\mathbb{S}^{n-1}}\operatorname{vol}_{n-1}\big(\operatorname{proj}_{u^\perp}(P_j)\big)\,d\mathcal{H}^{n-1}(u)$$

holds for every $j\in\mathbb{N}$. Letting $j\to\infty$, we get from (6.5.3) and (6.5.7) that

$$nW_1(K) = \frac{1}{c_n} \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)\, d\mathcal{H}^{n-1}(u).$$

To give the correct value of c_n, notice that if we let $K = B^n$ in the above, then the left-hand side becomes the surface area of B^n. Also, the projection of B^n onto any hyperplane is clearly identified with B^{n-1}. Thus,

$$\mathrm{vol}_{n-1}(\mathbb{S}^{n-1}) = \frac{1}{c_n} \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}(B^{n-1})\, d\mathcal{H}^{n-1}(u) = \frac{\kappa_{n-1}\mathrm{vol}_{n-1}(\mathbb{S}^{n-1})}{c_n},$$

from where we get that $c_n = \kappa_{n-1}$. This concludes the proof.

□

Remark 6.5.3. The equality

$$\int_{\mathbb{S}^{n-1}} |\langle v, u\rangle|\, d\mathcal{H}^{n-1}(u) = 2\kappa_{n-1}, \tag{6.5.8}$$

which holds for any $v \in \mathbb{S}^{n-1}$, has interest in itself, since it will appear sometimes in the remaining text. Since it is clear that

$$h_{\mathrm{seg}[-v,v]}(\cdot) = |\langle v, \cdot\rangle|,$$

we get

$$\int_{\mathbb{S}^{n-1}} |\langle v, u\rangle|\, d\mathcal{H}^{n-1}(u) = \frac{\omega_{n-1}}{2} \int_{\mathbb{S}^{n-1}} 2h_{\mathrm{seg}[-v,v]}(u)\, d\sigma(u) = \frac{\omega_{n-1}}{2} w(\mathrm{seg}[-v, v]),$$

where we recall that σ is the standard probability measure on $\mathbb{S}^{n-1}$, which is just $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ normalized, and $\omega_{n-1} = \mathcal{H}^{n-1}(\mathbb{S}^{n-1})$. That is, the integral in the left-hand side of (6.5.8) is related to the mean width of a segment whose length is 2. From equality (6.4.2), which we stated momentaneously without a proof (this will be given in Chapter 7), the mean width is related to the $(n-1)$-th quermassintegral. Having this in mind, equality (6.5.8) can also be obtained with the help of Exercises 6.1 and 6.2.

It is also worth mentioning that (6.5.8) may be derived directly from a particular formula for integration over $\mathbb{S}^{n-1}$. This is made later, in Proposition 11.5.1.

Remark 6.5.4. For the sake of formalism, a comment is due on why are we able to integrate the function $\mathbb{S}^{n-1} \ni u \mapsto \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)$. This function is indeed continuous. This happens because the map

$$\mathbb{S}^{n-1} \ni u \mapsto \mathrm{proj}_{u^\perp}(K) + \{tu : t \in [0, 1]\} \in CC^n$$

is clearly continuous in the Hausdorff metric, and

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \mathrm{vol}_n\big(\mathrm{proj}_{u^\perp}(K) + \{tu : t \in [0, 1]\}\big).$$

The Lebesgue volume is continuous with respect to the Hausdorff metric in CC^n, and hence $\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)$ is a composition of continuous maps. Alternatively, after developing the theory of *mixed surface area measures*, we can use equality (7.2.5).

Actually, Cauchy's surface area formula is a particular case of the more general *formulas of Kubota*, which we will prove as a consequence of Cauchy's surface area formula itself. First, we need the following useful formula for the quermassintegrals of $C + \lambda B^n$.

Proposition 6.5.1. *For any $C \in CC^n$ and every $\lambda \geq 0$, we have*

$$W_i(C + \lambda B^n) = \sum_{j=0}^{n-i} \binom{n-i}{j} W_{i+j}(C)\lambda^j$$

for each $i = 0, \dots, n$.

Proof. We use the multilinearity and the symmetry of the mixed volume to write

$$W_i(C + \lambda B^n) = V\big((C + \lambda B^n)[n-i], B^n[i]\big) \sum_{j=0}^{n-i} \binom{n-i}{j} V\big(C[n-i-j], \lambda B^n[j], B^n[i]\big) =$$

$$= \sum_{j=0}^{n-i} \binom{n-i}{j} \lambda^j V\big(C[n-i-j], B^n[i+j]\big) = \sum_{j=0}^{n-i} \binom{n-i}{j} \lambda^j W_{i+j}(C),$$

where the second equality justifies because there are $\binom{n-i}{j}$ ways of filling j of the $n-i$ entries with λB^n. The proof is complete.

□

Notice that in the case where $i = 1$, we have a formula for the surface area of $C + \lambda B^n$:

$$S(C + \lambda B^n) = nW_1(C + \lambda B^n) = \sum_{j=0}^{n-1} n\binom{n-1}{j} W_{j+1}(C)\lambda^j. \tag{6.5.9}$$

Theorem 6.5.2 (Kubota's formulas). *For every* $K \in CC^n$ *and any* $i = 1, \dots, n$, *we have*

$$W_i(K) = \frac{1}{n\kappa_{n-1}} \int_{\mathbb{S}^{n-1}} w_{i-1}\big(\mathrm{proj}_{u^\perp}(K)\big)\, d\mathcal{H}^{n-1}(u),$$

where w_{i-1} *denotes the* $(i-1)$*-th quermassintegral of* $\mathbb{R}^{n-1}$.

Proof. Our first step is to discuss the continuity of the functions $\mathbb{S}^{n-1} \ni u \mapsto w_{i-1}\big(\mathrm{proj}_{u^\perp}(K)\big)$. It is not that easy to give a completely formal proof of this, but the argumentation can go as follows. Fix $u_0 \in \mathbb{S}^{n-1}$. Choose a neighborhood $U \subseteq \mathbb{S}^{n-1}$ of u_0 and a map $T\colon U \to \mathrm{O}(n)$ continuous with respect to the usual norm in $\mathrm{O}(n)$ and such that $T(u)u = e_n$ for each $u \in U$. Then, denoting $T(u) = T_u$, we have

$$T_u(\mathrm{proj}_{u^\perp}(K)) = \mathrm{proj}_{e_n^\perp}(T_u K)$$

for each $u \in U$. Since T_u is continuous in the usual norm of $\mathrm{O}(n)$, we get that the map $U \ni u \mapsto T_u(K)$ is continuous in the Hausdorff metric of CC^n. Now we choose any natural identification between $e_n^\perp$ and $\mathbb{R}^{n-1}$, and we get that $U \ni u \mapsto \mathrm{proj}_{e_n^\perp}(T_u K)$ is continuous in the Hausdorff metric of $\mathbb{R}^{n-1}$ (the projection onto a fixed hyperplane is clearly continuous with respect to the Hausdorff metric induced on this hyperplane). Since the quermassintegral w_{i-1} is continuous in the Hausdorff metric of $\mathbb{R}^{n-1}$, we get that the function $U \ni u \mapsto w_{i-1}\big(\mathrm{proj}_{e_n^\perp}(T_u K)\big) = w_{i-1}\big(\mathrm{proj}_{u^\perp}(K)\big)$ is continuous at U and, in particular, at u_0. Since u_0 was arbitrarily fixed, we get that the map $u \mapsto w_{i-1}\big(\mathrm{proj}_{u^\perp}(K)\big)$ is continuous on $\mathbb{S}^{n-1}$.

Now we recall that from the linearity of the projections, we have

$$\mathrm{proj}_{u^\perp}(K + \lambda B^n) = \mathrm{proj}_{u^\perp}(K) + \lambda \mathrm{proj}_{u^\perp}(B^n),$$

for any $\lambda \geq 0$ and $u \in \mathbb{S}^{n-1}$. Since any projection of B^n onto an $(n-1)$-dimensional subspace is identified with B^{n-1}, we get

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K+\lambda B^n)\big) = \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K) + \lambda \mathrm{proj}_{u^\perp}(B^n)\big) = \sum_{j=0}^{n-1} \binom{n-1}{j} w_i\big(\mathrm{proj}_{u^\perp}(K)\big)\lambda^i.$$

We integrate both sides in $u \in \mathbb{S}^{n-1}$ (and divide by κ_{n-1}), and we evaluate the integral of the left-hand side using Cauchy's surface area formula:

$$\begin{aligned} S(K+\lambda B^n) &= \frac{1}{\kappa_{n-1}} \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K+\lambda B^n)\big)\, d\mathcal{H}^{n-1}(u) = \\ &= \frac{1}{\kappa_{n-1}} \int_{\mathbb{S}^{n-1}} \sum_{j=0}^{n-1} \binom{n-1}{j} w_i\big(\mathrm{proj}_{u^\perp}(K)\big)\lambda^i = \\ &= \sum_{j=0}^{n-1} \left(\frac{1}{\kappa_{n-1}} \binom{n-1}{j} \int_{\mathbb{S}^{n-1}} w_i\big(\mathrm{proj}_{u^\perp}(K)\big)\, d\mathcal{H}^{n-1}(u) \right) \lambda^i. \end{aligned}$$

Finally, we only have to compare the coefficients of this polynomial expansion of $S(K+\lambda B^n)$ with the ones of the expression given in (6.5.9). Considering the term λ^{i-1}, we have

$$n\binom{n-1}{i-1} W_i(K) = \frac{1}{\kappa_{n-1}} \binom{n-1}{i-1} \int_{\mathbb{S}^{n-1}} w_{i-1}\big(\mathrm{proj}_{u^\perp}(K)\big)\, d\mathcal{H}^{n-1}(u),$$

for each $i = 1, \ldots, n$. This yields the desired, and the proof is complete.

□

Remark 6.5.5. In the book [133], more general versions of Kubota's formulas are presented which involve integrations over Grassmannians. For $K \in CC^n$ and $0 < i < n$, one may write

$$W_{n-i}(K) = \frac{\kappa_n}{\kappa_i} \int_{G(n,i)} \mathrm{vol}_i(\mathrm{proj}_E(K))\, d\nu_i(E), \tag{6.5.10}$$

where $G(n,i)$ is the Grassmannian of i-dimensional subspaces in $\mathbb{R}^n$ endowed with its usual probability Haar measure ν_i.

Providing detailed proofs for these formulas is beyond the scope of this introductory book, since we do not work with Grassmannians. However, it is worth noticing that (6.5.10) expresses $W_{n-i}(K)$ as the mean of the i-dimensional volumes of the projections of K onto the i-dimensional subspaces of $\mathbb{R}^n$ (up to the constant scalar multiplication by κ_n/κ_i, of course). This justifies why quermassintegrals are also called *mean projection measures*.

The Cauchy surface area formula (Theorem 6.5.1) is the case of (6.5.10) for $i = n-1$. Intuitively, the hyperplanes of $\mathbb{R}^n$ are indexed by their unit normal vectors, and hence the integration of $u \mapsto \mathrm{vol}_{n-1}(\mathrm{proj}_{u^\perp}(K))$ over $\mathbb{S}^{n-1}$ yields the mean of the $(n-1)$-dimensional volumes of the projections of K over the hyperplanes of $\mathbb{R}^n$ (up to the due normalization, of course). Similarly, the mean of the projections of K onto the 1-dimensional subspaces of $\mathbb{R}^n$ is just the mean width of K. Hence, the case of (6.5.10) for $i = 1$ gives us intuition about why the $(n-1)$-th quermassintegral is related to mean width (as it was mentioned in the beginning of Section 6.4, this will be proved in Section 7.1).

6.6 Exercises

Exercise 6.1. **(a)** *Let* $Q^2 := [-1,1]^2 \subseteq \mathbb{R}^2$*, and let* B^2 *be the unit ball of* $\mathbb{R}^2$*. Compute the mixed volume* $V(Q^2, B^2)$*.*

(b) *Compute the quermassintegrals of* $Q^3 := [-1,1]^3 \subseteq \mathbb{R}^3$*.*

(c) *Let* $u \in \mathbb{S}^{n-1}$*. Compute the quermassintegrals of the segment* $\mathrm{seg}[-u,u] \subseteq \mathbb{R}^n$*.*

Exercise 6.2. *Let* $u \in \mathbb{S}^{n-1}$ *be arbitrarily given. Compute the mean width of* $\mathrm{seg}[-u,u]$*.*

Hint: Use the relation between the mean width and the $(n-1)$-th quermassintegral given in the beginning of Section 6.4.

Exercise 6.3. *Let* $K \in CC^n$ *and* $i \in \{0,\dots,n\}$ *be arbitrarily given. Show that*

$$W_i(-K) = W_i(K).$$

Exercise 6.4. *Verify that*

$$V_j(B^n) = \binom{n}{n-j} \frac{\kappa_n}{\kappa_{n-j}}$$

for each $0 \le j \le n$*.*

Exercise 6.5. *Prove that for any* $K_1,\dots,K_n \in CC^n$ *we have*

$$V(K_1,\dots,K_n) = \frac{1}{n!} \frac{\partial^n}{\partial t_1 \partial t_2 \dots \partial t_n} \mathrm{vol}_n(t_1K_1 + \dots + t_nK_n)\Big|_{t_1=\dots=t_n=0}.$$

Exercise 6.6. *Let* $C \subseteq \mathbb{R}^n$ *be a compact convex set such that* $\dim(C) = n-1$*. Prove that*

$$nW_1(C) = 2\mathrm{vol}_{n-1}(C). \tag{6.6.1}$$

Hint: Use Proposition 6.4.2.

Remark 6.6.1. Recall that for convex bodies the functional $nW_1(\cdot)$ yields the surface area. According to the exercise above, in the $(n-1)$-dimensional case, we obtain twice the $(n-1)$-dimensional volume. One way to interpret that is the following: if we approximate C by orthogonal cylinders over C, then the surface area of each cylinder equals twice the area of C plus the area of the "lateral" part. The latter goes to 0 as the height of the cylinders goes to 0.

In item **(b)** of the exercise below, we provide an extension of equality (6.6.1).

Exercise 6.7. *Let* $0 \le i \le n$*, and assume that* $C \in CC^n$ *is such that* $\dim(C) = i$*.*

(a) *Verify that* $V_i(C) = \mathrm{vol}_i(C)$*.*

(b) *Prove that*

$$\binom{n}{i} W_{n-i}(C) = \kappa_{n-i}\mathrm{vol}_i(C).$$

Exercise 6.8. *Let* $K, L, M \in CC^n$ *be such that* $K + L = M$*. Prove that the equality*

$$\mathrm{vol}_n(K) = \sum_{i=0}^{n} \binom{n}{i} (-1)^i V(M[n-i], L[i])$$

holds.

Exercise 6.9. *Let $K, L \subseteq \mathbb{R}^2$ be convex bodies whose boundaries are C^2 curves, and parametrize their support functions as in Exercise 4.14. Prove the following expression for the mixed volume of K and L:*

$$V(K, L) = \frac{1}{2} \int_0^{2\pi} h_K(\theta) h_L(\theta) - h_K'(\theta) h_L'(\theta)\, d\theta.$$

Exercise 6.10. *Prove that if $K, L \in CC^n$ are such that $K \cup L \in CC^n$, then*

$$V(K, L, C_3, \ldots, C_n) = V(K \cup L, K \cap L, C_3, \ldots, C_n)$$

for any $C_3, \ldots, C_n \in CC^n$.

Remark 6.6.2. The statement may suggest that we must have $n \geq 3$, but the result also holds true in dimension 2. It reads

$$V(K, L) = V(K \cup L, K \cap L)$$

whenever $K, L \in CC^2$ are such that $K \cup L \in CC^2$.

Exercise 6.11 (Rademacher). *Let K be a planar convex body. Show that*

$$V(K, -K) \leq 2\mathrm{vol}_2(K).$$

Hint: Use the Rogers–Shephard inequality.

Exercise 6.12. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that*

$$V(-K, K, \ldots, K) \geq \mathrm{vol}_n(K).$$

Exercise 6.13. *Let $K_1, \ldots, K_n \in CC^n$. Prove that $V(K_1, \ldots, K_n) > 0$ if and only if for every $k \in \{1, \ldots, n\}$ and any choice of indices $1 \leq i_1 < \ldots < i_k \leq n$ the inequality $\dim(K_{i_1} + \ldots + K_{i_k}) \geq k$ holds.*

Exercise 6.14. *Show that the Brunn–Minkowski inequality (Theorem 5.1.2) can be obtained from the 1st Minkowski inequality (Theorem 6.3.1).*

For the next exercise, recall that $r_{\mathrm{in}}(K)$ and $r_{\mathrm{out}}(K)$ denote the inradius and circumradius of K, respectively. These concepts are defined and studied in Exercise 2.13.

Exercise 6.15. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that for any $i \in \{1, \ldots, n\}$ the inequalities*

$$r_{\mathrm{in}}(K) \leq \frac{W_{i-1}(K)}{W_i(K)} \leq r_{\mathrm{out}}(K)$$

hold.

Exercise 6.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body, and assume that $\mathbf{g}$ is a k-dimensional vector subspace of $\mathbb{R}^n$ for some $k \in \{1, \ldots, n-1\}$. Prove that for any $j \in \{0, \ldots, k\}$ the reduction formula*

$$V\big(K[n-j], B^k[j]\big) = \binom{n}{j}^{-1} \binom{k}{j} \int_{\mathrm{proj}_{\mathbf{g}^\perp}(K)} W_j^{(k)}\big(K \cap (\mathbf{g} + z)\big)\, d\mathcal{H}^{n-k}(z),$$

holds, where $W_j^{(k)}$ denotes the j-th quermassintegral of $\mathbb{R}^k$ and B^k is the unit ball of $\mathbb{R}^k$ identified with $B^n \cap \mathbf{g}$.

Exercise 6.17. *Let $1 \le j \le n$, and let $u_1, \dots, u_j \in \mathbb{S}^{n-1}$. Write $E := \operatorname{span}\{u_1, \dots, u_j\}$. Prove that for any $C_1, \dots, C_{n-j} \in CC^n$, we have*

$$\binom{n}{j} V(\operatorname{seg}[-u_1, u_1], \dots, \operatorname{seg}[-u_j, u_j], C_1, \dots, C_{n-j}) =$$
$$= \frac{2^j}{j!} D_j(u_1, \dots, u_j) V^{(n-j)}(\operatorname{proj}_{E^\perp}(C_1), \dots, \operatorname{proj}_{E^\perp}(C_{n-j})),$$

where

$$D_j(u_1, \dots, u_j) := \operatorname{vol}_j\big(\operatorname{seg}[0_{\mathbb{R}^n}, u_1] + \dots + \operatorname{seg}[0_{\mathbb{R}^n}, u_j]\big).$$

Exercise 6.18. *Verify that for any $K_1, \dots, K_{n-1} \in CC^n$ and every $u \in \mathbb{S}^{n-1}$ we have*

$$V^{(n-1)}\big(\operatorname{proj}_{u^\perp}(K_1), \dots, \operatorname{proj}_{u^\perp}(K_{n-1})\big) = \frac{n}{2} V(\operatorname{seg}[-u, u], K_1, \dots, K_{n-1}).$$

Remark 6.6.3. The particular case where $K_1 = \dots = K_{n-1} = K$ is derived later in the proof of Theorem 7.2.2 (see also Remark 7.2.1).

Exercise 6.19. *Let $K_1, \dots, K_{n-1} \in CC^n$, and define the function $f : \mathbb{R}^n \to \mathbb{R}$ by*

$$f(\lambda u) = \lambda V^{(n-1)}\big(\operatorname{proj}_{u^\perp}(K_1), \dots, \operatorname{proj}_{u^\perp}(K_{n-1})\big),$$

for any $\lambda \ge 0$ and every $u \in \mathbb{S}^{n-1}$. Prove that f is sublinear.

Hint: Use Exercise 6.18 and consider first the case where $K_1, \dots, K_{n-1}$ are polytopes (for this case, use Proposition 6.2.2).

Remark 6.6.4. Another approach for the sublinearity of f is given later in Exercise 10.17 (see also the definition of the *mixed projection body* above this exercise). There, we use *mixed surface area measures* to rewrite f. In the case where $K_1 = \dots = K_{n-1} = K$, this is related to the important definition of the *projection body* of K, which is studied in Section 10.3.

In the next exercises, we investigate some facts concerning mixed volumes (and quermassintegrals) involving polar bodies. For that sake, we remind the reader that $\mathcal{K}_0^n$ denotes the family of the convex bodies of $\mathbb{R}^n$ which cointain the origin as an interior point.

Exercise 6.20. *Let $K \in \mathcal{K}_0^n$, and let $C_2, \dots, C_n \in CC^n$ be arbitrarily given. Prove that*

$$V(K, C_2, \dots, C_n) V(K^\circ, C_2, \dots, C_n) \ge V(B^n, C_2, \dots, C_n)^2.$$

Hint: Consider first the case where $C_2, \dots, C_n$ are polytopes. For this case, use Proposition 6.2.2. Then, derive the general case by approximations.

Exercise 6.21. *Prove that for any $K \in \mathcal{K}_0^n$ the inequality*

$$W_{n-1}(K^\circ) W_{n-1}(K) \ge \kappa_n^2$$

holds. Conclude that

$$w(K) w(K^\circ) \ge 4,$$

where we remind the reader that $w(\cdot)$ stands for the mean width.

Remark 6.6.5. The latter inequality is obtained in a different way in Corollary 11.1.2.

Exercise 6.22. **(a)** *Let $K \in \mathcal{K}_0^n$. Show that*

$$B^n \subseteq \frac{K + K^\circ}{2}.$$

Verify that equality holds if and only if $K = B^n$.

(b) *If $K \in \mathcal{K}_0^2$, prove that*

$$V(K, K^\circ) \geq \kappa_2.$$

Check that this inequality is sharp.

Hint: For any $\lambda > 0$, let

$$M_\lambda := \frac{1}{2}\left(\lambda K + \frac{1}{\lambda} K^\circ\right).$$

Show that

$$\min\left\{W_0(M_\lambda) : \lambda > 0\right\} = \frac{1}{2}\Big(V(K, K^\circ) + V(K, K)^{\frac{1}{2}} V(K^\circ, K^\circ)^{\frac{1}{2}}\Big),$$

and use the 1st Minkowski inequality to bound the last term. Finally, apply item **(a)** to verify that

$$W_0(M_\lambda) \geq \kappa_2$$

for all $\lambda > 0$.

Exercise 6.23. *Show that in dimension $n > 3$ the unit ball B^n does **not** minimize the product*

$$W_1(K) W_1(K^\circ)$$

over $K \in \mathcal{K}_0^n$.

Hint: Recall that $nW_1(\cdot)$ is the surface area, and compare the cases where $K = B^n$ and where $K = P_\Delta^n$ (the regular n-dimensional cross-polytope). See Exercises 4.3 and 4.5.

The objective of the next exercise is to construct a multilinear (for Minkowski combinations) function $F\colon (CC^n)^n \to \mathbb{R}$ which has "good" properties but is not the same as the mixed volume (in this direction, see also Exercise 6.26 ahead). To do that, we first notice that since $\mathrm{GL}(n)$ has a topology endowed by the norm of $\mathrm{L}(n) \simeq \mathbb{R}^{n^2}$, we can endow it with a finite Borel measure.

Exercise 6.24. *Let μ be a finite Borel measure on $\mathrm{GL}(n)$ with the property that*

$$\int_{\mathrm{GL}(n)} ||T||_{\mathrm{L}(n)} \, d\mu(T) < +\infty.$$

(a) *For each $K \in CC^n$, $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$ prove that the map*

$$\mathrm{GL}(n) \ni T \mapsto h(TK, \lambda u)$$

is continuous, and define

$$h(K^\mu, \lambda u) := \int_{\mathrm{GL}(n)} h(TK, \lambda u)\, d\mu(T).$$

(b) *Verify that* $h(K^\mu, \lambda u) < +\infty$ *always.*

(c) *Prove that* $h(K^\mu, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$ *is a sublinear function. Conclude that* $h(K^\mu, \cdot)$ *is the support function of some set of* CC^n *(which we denote by* K^μ*, thus justifying the notation).*

Let $F_\mu\colon (CC^n)^n \to \mathbb{R}$ *be defined as*

$$F_\mu(K_1, \dots, K_n) = V(K_1^\mu, \dots, K_n^\mu), \quad K_1, \dots, K_n \in CC^n.$$

(d) *Show that* F_μ *is monotone for inclusions and linear for Minkowski combinations in each entry.*

(e) *Check that* F_μ *is symmetric in the entries.*

(f) *Prove that if* μ *is* even *(that is, if* $\mu(A) = \mu(-A)$ *for every Borel set* $A \subseteq \mathrm{GL}(n)$*), then* F^μ *is translation-invariant.*

(g) *Give an example of a finite even Borel measure* μ *on* $\mathrm{GL}(n)$ *such that* F_μ *is not a multiple of the mixed volume of* $\mathbb{R}^n$*.*

Of course, identifying each set of CC^n with its support function, it follows that the mixed volume can be seen as a functional defined for support functions (over $\mathbb{S}^{n-1}$, say). For the next exercise, we will use a theorem from [2167] which states, roughly speaking, that the two-dimensional mixed volume (defined for support functions) does not admit a bilinear continuous extension to $C(\mathbb{S}^1) \times C(\mathbb{S}^1)$. Here, "bilinear" has the usual linear algebra sense: we are considering also combinations with negative scalars.

Theorem 6.6.1. [2167, Theorem 5.2.2] *There is no bilinear function* $F\colon C(\mathbb{S}^1)^2 \to \mathbb{R}$ *which is continuous in each variable (with respect to the maximum norm) and satisfies* $F(h_K, h_L) = V(K, L)$ *for every* $K, L \in CC^n$*.*

Exercise 6.25. *Let* $n \geq 3$*. Prove that there is no multilinear (in the usual linear algebra sense) function* $F\colon C(\mathbb{S}^{n-1})^n \to \mathbb{R}$ *which is continuous in each variable (with respect to the maximum norm) and satisfies*

$$F(h_{K_1}, \dots, h_{K_n}) = V(K_1, \dots, K_n)$$

for every $K_1, \dots, K_n \in CC^n$*.*

Exercise 6.26. *Let* $f\colon (\mathbb{S}^{n-1})^n \to \mathbb{R}$ *be a function which is continuous, non-negative, symmetric, and even in each entry. Define* $H\colon (CC^n)^n \to \mathbb{R}$ *by*

$$H(K_1, \dots, K_n) :=$$
$$:= \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} h_{K_1}(u_1) \dots h_{K_n}(u_n) f(u_1, \dots, u_n)\, d\mathcal{H}^{n-1}(u_1) \dots d\mathcal{H}^{n-1}(u_n).$$

(a) *Verify that* H *is symmetric, Minkowski multilinear, and translation invariant.*

(b) *Show that* H *is non-decreasing for inclusions in each variable.*

(c) *Prove that H has the* valuation property *in each entry. That is, show that if $K, L \in CC^n$ are such that $K \cup L \in CC^n$, then*

$$H(K \cup L, K_2, \ldots, K_n) + H(K \cap L, K_2, \ldots, K_n) = H(K, K_2, \ldots, K_n) + H(L, K_2, \ldots, K_n)$$

for every $K_2, \ldots, K_n \in CC^n$.

(d) *Check that H is not a scalar multiple of the mixed volume of $\mathbb{R}^n$.*

The objective of the next exercise is to prove that the mixed volume of $\mathbb{R}^n$ cannot be "suitably" extended to a functional defined in a family of compact sets larger than CC^n. In what follows, for the convenience of the reader, we recall that $\mathcal{C}^n$ stands for the family of non-empty compact subsets of $\mathbb{R}^n$. Also, if $\mathcal{A}$ is a family of subsets of $\mathbb{R}^n$ which is closed under Minkowski addition, then we say that a functional $\varphi\colon \mathcal{A} \to \mathbb{R}$ is *Minkowski additive* if

$$\varphi(A + B) = \varphi(A) + \varphi(B)$$

for any $A, B \in \mathcal{A}$. Notice that the mixed volume (seen as a functional defined for n-tuples of sets from CC^n) is Minkowski additive in each entry.

Exercise 6.27. (a) *Let $A \subseteq \mathbb{R}^n$ be non-empty. Prove that*

$$A + n\mathrm{conv}(A) = (n+1)\mathrm{conv}(A).$$

(b) *Let $\varphi\colon \mathcal{A} \to \mathbb{R}$ be a Minkowski additive functional, where $\mathcal{A}$ is a family of subsets of $\mathbb{R}^n$ which is closed for Minkowski addition. Show that*

$$\varphi(A) = \varphi(\mathrm{conv}(A))$$

for any $A \in \mathcal{A}$.

(c) *Let $C \subseteq \mathcal{C}^n$ be full-dimensional and non-convex. Verify that*

$$\mathrm{vol}_n(C) < \mathrm{vol}_n(\mathrm{conv}(C)).$$

(d) *Assume that $\mathcal{D}$ is a family of subsets of $\mathbb{R}^n$ which is closed for Minkowski addition, and such that $CC^n \subseteq \mathcal{D} \subseteq \mathcal{C}^n$. Prove that if there exists a functional $\tilde{V}\colon \mathcal{D}^n \to \mathbb{R}$ which is Minkowski additive in each entry and such that*

$$\tilde{V}(D, \ldots, D) = \mathrm{vol}_n(D)$$

for each $D \in \mathcal{D}$, then $\mathcal{D} = CC^n$.

Hint: Compare the volumes of D and $\mathrm{conv}(D)$. Notice that if D is full-dimensional, then these sets have the same volume if and only if D is convex. In the case where D is not full-dimensional, let C be a non-empty, compact, and convex subset of an affine subspace complementary to $\mathrm{aff}(D)$, and compare the volumes of $C + D$ and $\mathrm{conv}(C + D)$.

Recall that a convex body K has *constant width* if $h_K(u) + h_K(-u)$ is a constant function of $u \in \mathbb{S}^{n-1}$. In the next exercise, we investigate the *Barbier theorem* for planar bodies of constant width. Observe that the perimeter of a planar convex body can be calculated in terms of mixed volumes by (6.1.2).

Exercise 6.28 (Barbier's theorem). *Let $K \subseteq \mathbb{R}^2$ be a convex body having constant width $c > 0$. Prove that the perimeter of K equals πc.*

Exercise 6.29 (Blaschke). *Let $K \subseteq \mathbb{R}^3$ be a convex body of constant width $c > 0$.*

(a) *Prove that*

$$3cS(K) - 6\mathrm{vol}_3(K) = 2\pi c^3.$$

(b) *Show that*

$$W_2(K) = \frac{2}{3}\pi c.$$

The *Wills functional* is defined as the functional $\mathcal{W}\colon CC^n \to \mathbb{R}$ which associates each set of CC^n to the sum of its intrinsic volumes. That is,

$$\mathcal{W}(K) := \sum_{j=0}^{n} V_j(K)$$

for each $K \in CC^n$. For motivations, further results and applications around this definition, we refer the reader to [83] and references therein.

Exercise 6.30. *Prove that the Wills functional has the following integral representation:*

$$\mathcal{W}(K) = 2\pi \int_0^{+\infty} te^{-\pi t^2} \mathrm{vol}_n(K + tB^n)\, dt.$$

Remark 6.6.6. Another integral representation of the Wills functional is given in Exercise 11.19. For that, we need *Hadwiger's theorem* on the characterization of *valuations* (see Section 11.3 and, in particular, Theorem 11.3.1), and this is why we do not enunciate it here.

Remember that (5.6.1) is a notion of mean width for log-concave functions introduced in [2064]. In the next exercises, we investigate a different notion of mean width for log-concave functions which is based on the characterization of the mean width as a multiple of the $(n-1)$-th quermassintegral. This was introduced by Klartag and Milman in [1411]. As it was the case in the exercises in which we deal with definition (5.6.1), here we will restrict ourselves to log-concave functions of the family

$$\mathrm{LC}_\infty(\mathbb{R}^n) := \left\{ f = e^{-\varphi} : \varphi \text{ is closed, convex, proper, and such that } \lim_{|x|\to+\infty} \varphi(x) = +\infty \right\}.$$

For our purposes, the family $\mathrm{LC}_\infty(\mathbb{R}^n)$ is the adequate family of log-concave functions to work with. We note that $1_K \in \mathrm{LC}_\infty(\mathbb{R}^n)$ for every $K \in CC^n$, and from Exercise 1.34, we get that

$$\int_{\mathbb{R}^n} f\, d\mathcal{L}^n < \infty.$$

Once the integral of a log-concave function plays the role of the "volume", we have that the functions of $\mathrm{LC}_\infty(\mathbb{R}^n)$ have "finite volume". In what follows, we remember that

$$G(x) := e^{-\frac{|x|^2}{2}}$$

is the standard Gaussian function defined for $x \in \mathbb{R}^n$. From Exercise 1.39, we get that this is the unique self-polar function of $\mathrm{LC}_\infty(\mathbb{R}^n)$, and hence it "plays the role" of the unit ball, in some sense. The integral of G is computed in (5.6.3).

In what follows, **all** integrations are taken with respect to the Lebesgue measure of $\mathbb{R}^n$. Recall that from (6.4.2) the mean width $w(K)$ of a convex body $K \subseteq \mathbb{R}^n$ can be computed in terms of mixed

volumes as

$$w(K) = \frac{2}{\kappa_n} W_{n-1}(K) = \frac{2}{\kappa_n} V(B^n[n-1], K) =$$
$$= \frac{2}{\kappa_n} \lim_{\varepsilon \to 0^+} \frac{\mathrm{vol}_n(B^n + \varepsilon K) - \mathrm{vol}_n(B^n)}{\varepsilon} = \frac{2}{\kappa_n} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} 1_{B^n + \varepsilon K} - \int_{\mathbb{R}^n} 1_{B^n} \right] =$$
$$= \frac{2}{\kappa_n} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} 1_{B^n} \star (\varepsilon \cdot 1_K) - \int_{\mathbb{R}^n} 1_{B^n} \right].$$

In the last equality, we remember that $\star$ denotes the *Asplund product* introduced in Exercise 1.26 and that the scalar multiplication $\varepsilon \cdot 1_K$ is as defined in Exercise 1.28. The equality $1_{B^n + \varepsilon K} = 1_{B^n} \star (\varepsilon \cdot 1_K)$ follows from these exercises (see item **(a)** of both).

To extend the definition of mean width for arbitrary log-concave functions, it is reasonable to assume that the standard Gaussian function $G(x)$ plays the role of the unit ball, since it is the unique self-polar function in $\mathrm{LC}_\infty(\mathbb{R}^n)$ (see item **(c)** of Exercise 1.39). Hence, we define the *mean width* of a function $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ as

$$w(f) := \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) - \int_{\mathbb{R}^n} G \right], \tag{6.6.2}$$

where all the integrations are taken with respect to the standard Lebesgue measure, as it was mentioned above. The normalization constant is chosen so that we have $w(G) = 2$. We use a normalization different to the one used in the references cited above (the reason is that, for us, the mean width of a convex body is really the mean of the width function over $\mathbb{S}^{n-1}$, whereas some authors take half of this mean). Notice that it is not *a priori* clear that the limit in the definition above exists. This was formally proved (in greater generality) in [638, Section 3.1], though we may have $w(f) = +\infty$.

It is important to mention that Rotem proved in [2064] that the notions of mean width for log-concave functions defined in (5.6.1) and (6.6.2) actually coincide (later we prove this for characteristic functions; see Exercise 11.22).

Exercise 6.31. **(a)** *Verify that indeed we have that $w(G) = 2$.*

(b) *Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an affine transformation having the form $T(\cdot) = x_0 + U(\cdot)$ for some $x_0 \in \mathbb{R}^n$ and some $U \in \mathrm{O}(n)$. Prove that*

$$w(f \circ T) = w(f).$$

In other words, w is translation-invariant and $\mathrm{O}(n)$-invariant, in some sense.

Exercise 6.32. *Let $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ be such that*

$$\int_{\mathbb{R}^n} G \star (\varepsilon \cdot f) \to \int_{\mathbb{R}^n} G \tag{6.6.3}$$

as $\varepsilon \to 0$. Prove that for every $a > 0$, we have

$$w(af) = \frac{4}{n} \log(a) + w(f),$$

*where af denotes the **usual** scalar multiplication. Conclude that $w(\cdot)$ is not necessarily positive.*

Remark 6.6.7. According to [2064], $f = e^{-\varphi}$ has the property (6.6.3) if and only if $\varphi^*(x) < +\infty$ for every $x \in \mathbb{R}^n$ (recall that φ^* is the conjugate function of φ; see Section 1.4).

Exercise 6.33. *Let $K \in CC^n$ and $\varepsilon > 0$. Prove that*

$$[G \star (\varepsilon \cdot 1_K)](x) = e^{-\frac{1}{2} d(\varepsilon K, x)^2}$$

for every $x \in \mathbb{R}^n$.

Remark 6.6.8. This equality will be used later, in Exercise 11.21, to prove that the function

$$\varepsilon \mapsto \int_{\mathbb{R}^n} G \star (\varepsilon \cdot 1_K) \tag{6.6.4}$$

is a polynomial on $\varepsilon > 0$. As a consequence, we will obtain that

$$w(1_K) = c_n w(K)$$

for any $K \in CC^n$, where c_n is a constant which only depends on the dimension. In some sense, this confirms that the definition (6.6.2) gives a reasonable extension of the concept of mean width to log-concave functions. Moreover, an equality which is very similar to (6.6.4) will be used to obtain a lower bound for the Wills functional (see Exercise 11.23).

Exercise 6.34 (Urysohn's inequality). *Let $f \in \mathrm{LC}_\infty(\mathbb{R}^n)$ be such that*

$$\int_{\mathbb{R}^n} f = \int_{\mathbb{R}^n} G \ \left(= (2\pi)^{n/2} \right).$$

Prove that $w(f) \geq w(G)$ $(= 2)$. In other words, G minimizes the mean width among all functions having the same volume as it.

Hint: Use the version of the Prékopa-Leindler inequality given in Exercise 1.28.

6.7 Notes

Chapter 6 is dedicated to the central and substantial notions of *mixed volumes* and *quermassintegrals*. Due to the importance of the latter ones, we discuss in this case even selected research papers.

6.7.1 Mixed volumes

The notion of *mixed volumes* goes back to Minkowski (cf. [1806]). In a letter to Hilbert, he wrote that his investigations of surface area and volume in 3-space are complete, and that the introduction of the idea of "mixed volume", connected with three convex bodies, was his most important improvement there (for this letter see pp. 133-136 in [1808]). Today we know that this notion and subcases of it (like, e.g., *quermassintegrals*, also discussed in these notes) belong to the central part of the theory of convex sets.

BOOKS AND BOOK PARTS

We start with books where mixed volumes are comprehensively presented. Bonnesen and Fenchel give definitions and also representations (in particular by *support functions*) in Section 29 and 37 of their basic monograph [446]. Chapter I and II of Busemann's book [549] on convex surfaces might be considered, in great parts, as continuation of the book [446]. In particular, Chapter II deals in a detailed manner with the Brunn–Minkowski theory; its second section (Section 6 of the whole book) particularly treats mixed volumes. In the following decades, various convexity books contained complete chapters or sections dealing with mixed volumes, related notions, and applications; examples are the books [1133], [1134], [803], and [273] of Hadwiger, Eggleston, and Benson, respectively. Leichtweiss presents this topic in [1531] in a very detailed manner; after definitions, basic properties, applications, and exercises for two dimensions in §15, he goes on with results on *mixed surface area and mixed quermassintegrals* in §24. Burago and Zalgaller continue with many aspects in [531]: definitions and fundamental properties of mixed volumes occur in §19; subsection 24.3 is dedicated to *dual mixed volumes*; §25 presents analogues of mixed volumes (e.g., vector analogues); in §26 mixed volumes for non-convex sets are studied; and §27 deals with *algebra and mixed volumes*.

A bridge to *algebraic geometry* is given in the two books [1909] and [826] of Oda and Ewald, respectively. In Chapter 2 of [1909], integral polytopes and projective toric varieties are treated. For example, the number of lattice points contained in a polytope is expressed as the dimension of the space of global sections of the corresponding line bundle, and the *mixed volume of several polytopes* can be expressed via the intersection indices of the corresponding line bundles. And Chapter IV of [826] is dedicated to Minkowski sum and mixed volumes. Besides basic definitions and properties of mixed volumes, the Alexandrov–Fenchel inequality is discussed, and in the given dictionary between both disciplines mixed volume is contrasted with intersection numbers of the Cartier divisors, and the Alexandrov–Fenchel inequality with the Hodge inequality. Here Section 1 of Ewald's handbook article [825] should be analogously taken into consideration.

The whole Chapter 6 of Webster's book [2437] refers to mixed volumes, related extremum problems and tools (like Steiner's symmetrization and the relations to the Brunn–Minkowski theorem). Groemer emphasizes in [1042] the following aspects: mixed volume and mean projection measures (Section 2.4), relations to Fourier series (Section 4.2 and 4.3), and relations to harmonic expansions (Section 5.1). Thompson sets in [2361] the following priorities: in Section 2.3, basic definitions of mixed volumes and related notions, *Minkowski's inequality* for them, in Chapter 5, their role in connection with *area and content* as well as *isoperimetrices in Minkowski spaces*, and in Chapters 6 and 7, special relations to the *Holmes–Thompson and the Busemann definition of area*.

Gardner covers the subject of mixed volumes in Appendix A of [930]; he studies there *mixed volumes and mixed area measures*, relations to *quermassintegrals and intrinsic volumes*, and finally also *dual mixed volumes*. In Gruber's monograph [1070], Section 6.1 is devoted to mixed volumes; subjects are their definition with *Minkowski's theorem on mixed volumes, Steiner's formula*, and a variety of basic properties of mixed volumes, including the *valuation property*.

The most comprehensive and structured representation of mixed volumes is contained in Schneider's book [2167]: Chapter 5 contains definitions and related notions like *mixed area measures, generalizations, moment vectors, curvature centroids, Minkowski tensors*, and *mixed discriminants* are discussed there; in Chapter 7, *inequalities for mixed volumes* (including isoperimetric inequalities and the Alexandrov–Fenchel inequality), cases of equality, stability questions, and *Wulff shapes* are studied; Section 6.3 deals with *mixed valuations and polynomiality*, Section 7.3 with *mixed quermassintegrals*; mixed bodies occur in the notes of Section 8.2, and in Section 9.4, one finds *mixed width integrals*; finally, Chapter 10 treats *mixed affine and mixed p-affine surface area, dual mixed volumes, mixed projection bodies, and mixed intersection bodies*. We mention also the book [496] of Brazitikos et al., where, in the first chapter, *mixed volumes and classical geometric inequalities* are discussed.

The monograph [1713] treating *bodies of constant width* as main subject contains a whole chapter (Chapter 12) dedicated to *mixed volumes as a tool*. It shows how they allow to prove several classical theorems on the volume of bodies constant width and several further results; in addition, the notes to this chapter should be taken into consideration. Also, Hug and Weil study mixed volumes in their book [1263]: Section 3.3 deals with basic properties of mixed volumes, Section 4.1 with *mixed (surface) area measures*, and Section 4.3 with the notion of *mixed discriminant* and a local Steiner formula. Finally, in this list, we mention the recent book [133] of Artstein-Avidan et al.; the first part B.1. of its Appendix B (Advanced Convexity) is devoted to mixed volumes, where the subjects of volume and surface area, comprehensively also polytopes and, coming from this, general convex bodies are studied in this setting.

We continue now with important surveys and expository papers, treating mixed volumes and related notions, and results.

SURVEYS AND EXPOSITIONS

An early survey in this interplay is [2351] of Teissier. In its first part, the author's deduction of the Alexandrov–Fenchel inequalities between mixed volumes of convex bodies from the *Hodge index theorem* is represented.

Chakerian [593] starts a discussion of how the *additivity and monotonicity of mixed volumes* can be used to study relations and estimates between geometric invariants of *sets of constant width*.

Goodey [991] gives a survey on several new results on the relationship between certain mixed volumes of convex bodies and the *mixed volumes of their projections*. For example, the following question is generalized, regarding the participating dimensions and mixed volumes: whether the n-dimensional volume of a convex body K must be larger than that of a convex body C if all the $(n-1)$-dimensional projections of K have volumes larger than those of the corresponding projections of C. It is emphasized that these results are related to properties of *zonoids*.

In the exposition [1788], Milman introduces applications of certain geometric inequalities and basic notions from convexity into the local theory of Banach spaces. For example, he introduces several *geometric inequalities involving mixed volumes*, and he includes proofs of special cases. He studies Brunn–Minkowski inequalities for (mixed) volumes, continuing with the inequalities of Urysohn and Blaschke–Santaló. Also, he estimates the dimensions of well complemented Euclidean sections of arbitrary n-dimensional spaces in terms of the mixed volumes of the unit ball of those spaces; thus a Levy mean approach is used to estimate mixed volumes. Basic information on mixed volumes and Euclidean sections is included as a survey part of the paper.

It is well known that mixed volumes give a powerful method for solving projection problems. In the famous paper [1630], Lutwak introduces the important notion of *dual mixed volumes* as a powerful tool for solving problems involving intersections, and also for extending the considerations of convex sets (with support functions) to starshaped sets (with radial functions). Therefore, mixed volumes in connection with *projection bodies and the Petty–Schneider result* are contrasted with dual mixed volumes in connection with *intersection bodies and section problems like that of Busemann–Petty*. Due to these cross-connections, this publication is an important milestone in convexity theory. It is continued by the same author with [1631]. He gives there a survey of the *mixed affine surface area* and the *dual mixed volume*. The *dual Brunn–Minkowski inequality* gets an alternate proof, and a new Brunn–Minkowski type inequality is derived. Besides the projection operator (yielding the *projection body*), the centroid operator (leading to the *centroid body*) is studied. Furthermore, the author succeeds in giving analogues to centroid bodies of the Petty–Schneider theorem (which is combined with projection bodies). It turns out that necessarily *polars of zonoids* have to be used.

Groemer's handbook survey [1040] presents recent results on the stability of various important geometric inequalities. The fourth section refers to *stability versions of inequalities for mixed volumes*. In this way, Minkowski inequalities as well as certain other cases of the Alexandrov–Fenchel inequalities are treated.

Regarding mixed volumes, the most important survey of the Handbook of Convexity is that of Sangwine-Yager [2098]. Comprehensively, it collects and describes *main properties of mixed volumes, quermassintegrals, and mixed surface area measures*, also describing their behaviour under basic symmetrization procedures. These are Steiner, central, and Blaschke symmetrization, Schwarz rounding and rotation averaging. The Alexandrov–Fenchel inequalities and some of their special cases are taken into consideration, and the equality cases are investigated, too. But also further inequalities for mixed volumes are studied, like those related to *Bonnesen's inequality*.

In the third section of Schneider's handbook survey [2152], *curvature and surface area measures* of a convex body K are studied. The author shows that these measures can be traced back to measures defined on the generalized normal bundle of K and discusses their relations to mixed volumes and gives respective characterizations using invariance properties.

Here we have also to mention the comprehensive paper [1027] of Gritzmann and Klee. It belongs to a series of expository papers dedicated to *computational convexity*, a field defined by interactions between convex geometry, mathematical programming, and computer science. This second part is concerned with *computing volumes and mixed volumes* of convex polytopes and more general convex bodies. The main emphasis of the authors is on problems involving the dimension as input part. However, the respective questions for fixed dimensions are treated, too. After defining the necessary notions, the authors present, among many other things, deterministic methods for computing mixed volumes (for polytopes and general convex bodies like also special types of convex bodies), for approximating mixed volumes, volumes of unions and intersections of special bodies, etc.; application fields are also presented. These are, e.g., counting integer points in lattice polytopes, zonotopes and mixture-management, integration over convex bodies, points on spheres, and so on.

It is well known that *zonoids* (i.e., limits of finite vector sums of line segments) and generalized zonoids were characterized as special centered convex bodies via *inequalities between mixed volumes*, and that analogous characterizations of *intersection bodies* (i.e., star bodies whose radial function is the Radon transform of a positive measure on the unit sphere) and *dual mixed volumes* were found. Analogues are the polars of projection bodies (where the characterization involves centroid bodies and dual mixed volumes). In the deep paper [996] of Goodey, Lutwak, and Weil such results are presented and extended (e.g., to the setting of generalized intersection bodies). The authors show where classes of sets form convex cones with respect to Minkowski and radial additions, and where not.

Peri [1964] nicely surveys analytic aspects of convexity, particularly also Steiner's formula for parallel bodies, mixed volumes, Minkowski's mixed volume theorem, and the inequalities of

Brunn–Minkowski and Alexandrov–Fenchel. In a similar style, Gruber [1067] surveys several problems from convexity; one of them treats connections between *algebraic geometry and mixed volume inequalities*.

Bertrand and Bihan [301] investigate intersection multiplicities for cells in the intersection of various tropical hypersurfaces. For this purpose, a sum of generalized mixed volumes and volumes of special polytopes are used.

Yu, Zhang, and Wang [2508] derive new results on *dual mixed volumes of star bodies*, e.g., cyclic Brunn–Minkowski inequalities for them. Similar inequalities are proven for radial and harmonic Blaschke addition.

Breiding et al. [501] say that in their paper three topics come together, namely *zonoids, (generalized) mixed volumes, and random determinants*. Starting with a tensor product of two (centered) zonoids, they construct vector spaces of zonoids. Using suitably a canonical multilinear map there, they are able to introduce an exterior product of zonoids. On this way, they develop a graded, associative and commutative real algebra, which is called the *zonoid algebra* and inherits a duality notion from the Hodge star operator. Unexpected relations to notions like *mixed volumes, projection bodies, mean width,* and *intrinsic volumes* occur. Applications of this structure go also in the following directions: applying the Alexandrov–Fenchel inequalities for mixed volumes, and presenting a new analogue of mixed volumes called the mixed J-volume. This leads also to special angular valuations on polytopes.

Izmestiev [1290] gives a nice introduction to the theory of mixed volumes of convex bodies studying important related inequalities, such as the *Brunn–Minkowski, the Alexandrov–Fenchel, and the two Minkowski inequalities*. Steiner's formula and integral-geometric formulas (also those of Cauchy and Crofton) are studied, and the interplay between smooth and polyhedral hypersurfaces is taken care of. The Blaschke–Santaló inequality, Mahler's conjecture, and the isoperimetric inequality are also discussed.

6.7.2 Minkowski's inequalities related to mixed volumes

It is clear that mixed volumes satisfy several types of inequalities, among them also the so-called *first and second inequality of Minkowski regarding mixed volumes*, where two n-dimensional convex bodies participate. In the first case, equality characterizes homothety, and for the second equality case, homothety and additional conditions have to be satisfied.

BOOK PARTS

These inequalities are comprehensively discussed already in §49 of the monograph [446] of Bonnesen and Fenchel. It is shown there how one can get further inequalities from them (also with more than two participating bodies), and in the following §50 of [446] interesting *sharpenings and subcases* are discussed, also using cross-section measures. Blaschke discusses in §15 of his booklet [356] the *planar case*, and in a detailed way Leichtweiss discusses and proves both inequalities in §22 of his monograph [1531] (Satz 22.3 and Satz 22.6), also giving various applications. And in §24 (Satz 24.3), he gives a common generalization of both Minkowski inequalities, again characterizing homothety in the first case, and homothety together with additional quadratic relations on mixed quermassintegrals in the second.

In the classical book [531] of Burago and Zalgaller, both inequalities occur in Section 20.1 as particular cases of the Alexandrov–Fenchel inequalities, and Santaló gives a discussion in the second note to Chapter 7 in [2103]. In Webster's book [2437], we refer to Section 6.5 (Theorem 6.5.5). Thompson [2361] shows the importance of the Minkowski inequalities (see Section 2.3) also in *Minkowski geometry*, to derive results for normed spaces. Namely, they occur in connection with the *Minkowskian isoperimetric theorem* and properties of the corresponding *isoperimetrices* (Sections 4.4

and 5.3), and with *Petty's projection inequality and Busemann's intersection inequality*, to obtain then *bounds for the surface area of the unit ball* (Sections 6.5 and 7.4).

Also, Gardner [930] needs these inequalities at several places. The *first Minkowski inequality* is presented as Theorem B.2.1. and obtained from that of Brunn–Minkowski, the *second one* at the end of Appendix B.2.; in Appendix B.3., they are discussed as special cases of the Alexandrov–Fenchel inequality. In Appendix B.4., the *dual Alexandrov–Fenchel inequality* and, following from it, the *dual Minkowski inequalities* are studied. Further on, the Minkowski inequalities are needed in the Sections 3.3 (*projection functions*), 4.2 (smaller bodies with larger projections), 6.1 (combined with *chord functions and chordal symmetrals*, see Theorem 6.1.4), and 9.2 (*Petty's projection inequality*).

In Gruber's book [1070], one can find Minkowski's inequalities in Section 6.3 (of Chapter 6 on mixed volumes and quermassintegrals), and in Schneider's monograph [2167] together in one theorem; namely, in Section 7.2 on Minkowski and isoperimetric inequalities. Hug and Weil [1263] treat both inequalities separately in their Section 3.4 on the Brunn–Minkowski theorem, together with a variety of related inequalities. Also, in the very recent book [134] by Artstein-Avidan on asymptotic geometric analysis these inequalities are presented, namely in the Appendix B.2.

We continue with some important surveys in which both inequalities are discussed.

SURVEYS AND EXPOSITIONS

In the handbook article [2098], Sangwine-Yager presents the Minkowski's inequalities in Theorem 6.3 and the text after it, and then she continues considering interesting subcases and restrictions of them (regarding the dimension and the degree of generality of the considered convex bodies).

Groemer studies the stability of Minkowski's inequalities in Section 4 of his handbook article [1040], giving interesting cross-connections to related inequalities and references.

Also in the handbook article of Lutwak [1633] Minkowski's inequalities occur: namely, in the relationships between the Busemann–Petty centroid and Petty projection inequalities (Section 5), Petty's surface area inequality and its relatives (Section 8), and in view of Petty's conjectured projection inequality (subsection 12.2). Finally, we mention that also in subsection 5.4.2 of Izmestiev's recent survey [1290] the two Minkowski inequalities are presented, the first one as consequence of the Alexandrov–Fenchel inequality.

6.7.3 Quermassintegrals

The precondition for defining quermassintegrals is the notion of *cross-section measure* (German: "quermass") of an n-dimensional convex body K at a given direction u; see §7 of Bonnesen and Fenchel [446], where the German name "quermassintegral" occurs for the first time.

THE PRECONDITION OF CROSS-SECTION MEASURES

The classical cases are two projection and two section measures (or, meant analogously, two "outer" and two "inner" cross-section measures). The first two are the projection measures of dimension $n-1$ (also called *brightness* and being the measure of the orthogonal projection of K onto a hyperplane with u as normal direction) and of dimension 1 (also called *width* and presenting the 1-measure of the orthogonal projection of K onto the 1-subspace of direction u). The inner or section measure of dimension $n-1$ (named *HA-measurement*) is the maximal measure of a hyperplane section of K normal to u, and that of dimension 1 measures the *maximum chord length* of K in direction u. In this order, the two projection measures present the *support functions of the projection body and of the difference body of* K, respectively, and the two section measures present the *radial function of the cross-section body*

and of the difference body of K, respectively. Further, basic notions (like central symmetral, Blaschke body, intersection body, chordal symmetral, covariogram, etc.) are strongly related to or obtained by modifications of these concepts. And it is clear that one can easily extend these projection and section settings to *intermediate dimensions*. Basic references directly devoted to cross-section measures are: the book [446] of Bonnesen and Fenchel (see Sections 30, 33, and 42 there), Hadwiger's monograph [1134] (cf. Sections 4.4.2 and 4.4.3, slightly different in the case of section measures), Gardner's book [930] (Chapters 3, 4, 6, 7, and 8), Schneider's monograph [2167] (Sections 4.1 and 4.2), and Martini's survey [1710]. In view of these and further basic references, we emphasize once more that, instead of quermassintegrals, in the literature often their normalized forms, the *intrinsic volumes*, occur. They were introduced by McMullen in [1738].

QUERMASSINTEGRALS IN BOOK PARTS

It is well known that the notion of *quermassintegrals* (or mean cross-section measures) can be introduced in various ways. One of these ways (justifying the name) uses the notion of outer cross-section measures (see above). Namely, quermassintegrals can be suitably defined as *integral means* of such measures, with the *surface area* (via *Cauchy's projection formula*) and the *mean width* as illustrative examples. More precisely, the kth quermassintegral of an n-dimensional convex body K is, up to a factor depending on n, the integral mean of the $(n-k)$th outer cross-section measures, $k = 1, ..., n$. For example, this yields with W_0 the *volume*, connected with W_1 the *surface area*, and connected with W_2 the *integral of the mean curvature* of the considered convex body K. Comprehensive representations of these connections and more related material can be found in Sections 30, 31, and 32 of the book [446] of Bonnesen–Fenchel (background material for mixed volumes and cross-section measures), and multiple times in the longest and integral-geometric chapter of Hadwiger's monograph [1134]; namely, in the Sections 6.1.5 (Cauchy's surface area formula), 6.1.6 (quermassintegrals via Kubota's recursion formula), 6.1.7 (mean width), 6.1.10 (the *famous characterization of quermassintegrals*, see also Klain's and Chen's new and short proofs in [1395] and [604]), 6.3.4 (referring to quermassintegrals in the convex ring, i.e., the set of finite unions of convex bodies called polyconvex sets), 6.4.8 (replacing the "arithmetic integral mean" by the harmonic one to get *harmonic quermassintegrals*), 6.5.7 (extending the classical isoperimetric problem), and 6.5.8 (mixed quermassintegrals).

The mentioned famous characterization theorem of quermassintegrals says: *the continuous, rigid motion invariant valuations on the space of n-dimensional convex bodies are precisely the linear combinations of the quermassintegrals*. The complete §16 of Leichtweiss' book [1531] is dedicated to quermassintegrals, where besides Steiner's formula, basic properties, Cauchy's projection formula, and Kubota's recursion formula also the dependence on the dimension of the space (which plays no role for intrinsic volumes) is comprehensively discussed. In §24 *mixed quermassintegrals* are used to generalize the Brunn–Minkowski theorem for quermassintegrals. Also, in Gardner's monograph [930], quermassintegrals are presented as an essential topic: in Section 9.3 and the corresponding notes 9.6 also *affine and harmonic quermassintegrals* are discussed, in the Appendix A.5 (also on intrinsic volumes) basic properties, Steiner's formula and further formulas, and the *valuation property* are presented, and in Appendix A.7 they are studied in the setting of *dual mixed volumes and star bodies*. Further on, quermassintegrals and related topics are discussed in Chapter 13 of Santaló's book [2103], Section 6.4 of Gruber's monograph [1070], Section 4.2 in Schneider's book [2167], Section 3.3 (on mixed volumes) of Hug and Weil [1263], and also Chapter 5 there. And in the book [133] of Artstein-Avidan et al., Appendix B.5. is devoted to quermassintegrals;see also subsection 9.1.4. in the second part [134] of that book.

SURVEYS

Wills [2461] gives a nice overview on results in the geometry of numbers which are strongly related to quermassintegrals and dual quermassintegrals, successive minima and integer-lattice problems. Further important surveys related to quermassintegrals can, of course, be found in the Handbook of Convexity. For example, the third section of Sanwine-Yager's survey [2098] is dedicated to *quermassintegrals and intrinsic volumes*, obtained as two forms of *Steiner's formula*. It is said that there are circumstances under which any of the two concepts can be preferred; e.g., the quermassintegrals for a convenient presentation of the Alexandrov–Fenchel inequality, and the intrinsic volumes since they depend on the dimension of the considered convex body, and not of the space where it lies. In addition, their index equals the degree of homogeneity. *Affine quermassintegrals* are introduced in Lutwak's survey [1633], and also interesting subcases are considered there. In Section 2.1 of his handbook article [1745], McMullen comprehensively discusses the *valuation property of quermassintegrals*. Section 5 of Groemer's survey [1040] treats quermassintegrals, intrinsic volumes, and higher-dimensional isoperimetric inequalities. Interesting inequalities in terms of quermassintegrals are given, and related stability versions are discussed, too. Bastero et al. [249] survey results on dual quermassintegrals connected with necessary and sufficient conditions for extremal positions of convex bodies and isotropic properties of measures. Also, results for the affine setting are presented.

We continue with selected, mainly recent research papers referring to quermassintegrals and not cited in the literature described above.

SELECTED RESEARCH PAPERS

Yu and Wu [2509] investigate *quermassintegrals and dual quermassintegrals* of convex bodies K and star bodies S, respectively. In particular, it is investigated what inclusions of *i-th projection bodies* of K and S and, in analogous relation, of *i-th intersection bodies* (here K also has to be starshaped) and *i-th centroid bodies* imply for their i-th quermassintegrals (or dual i-th quermassintegrals in case of i-th intersection bodies). For the equality cases, K and S have to be translates or even have to coincide.

Klain [1399] extends Hadwiger's characterization theorem to *hyperbolic space* via characterizing *simple valuations* (which vanish on lower-dimensional sets). This way the notion of *area in hyperbolic planes* is nicely characterized, and from this, the author derives a general integral formula for valuations in the hyperbolic plane, which extends the *fundamental kinematic formula of hyperbolic integral geometry*. For higher dimensions, partial results on characterizing hyperbolic volume are given. It is proved that continuous isometry-invariant simple valuations on polytopes in odd dimensions are uniquely determined by their values at ideal simplices. Also, Wang and Xia [2420] study analogues of quermassintegrals in hyperbolic space. Since in many ways this is natural in hyperbolic geometry, they treat *horospherically convex bodies*, establishing quite strong *Alexandrov–Fenchel type inequalities* there. Introducing suitably the k-th order quermassintegrals, they obtain several interesting related inequalities. Most of their results rely on a curvature flow under which one curvature integral is preserved, whereas the others vary monotonically. Also, Ge, Wang, and Wu [946] treat quermassintegrals in hyperbolic space. Establishing first a Sobolev-type inequality in hyperbolic n-space, the authors verify hyperbolic Alexandrov–Fenchel inequalities for curvature integrals and quermassintegrals.

Bastero et al. [249] confirm necessary and sufficient conditions for extrema of certain functions on i-th dual quermassintegrals (of convex bodies and also their polars), and the extremal positions are characterized in terms of *isotropic measures*. Earlier results, partly with new proofs, new background, and extensions to variants of the problem, are surveyed, where translations are allowed, or containment in the unit ball is assumed.

Gardner [931] extends results of the dual Brunn–Minkowski theory from starshaped sets to bounded Borel sets, among them new insights on *dual quermassintegrals*. And new results on radial sums and *dual affine quermassintegrals* in the Brunn–Minkowski sense are derived, like also an *extension of the Busemann intersection inequality*. Interesting new results on sections of bounded Borel sets are used to study successfully equality cases.

Lu and Leng [1605] confirm a family of *inequalities involving mixed intersection bodies*, starting with the Busemann intersection inequality. It turns out that all these inequalities are extensions of known inequalities between pairs of *dual quermassintegrals of star bodies*. Also, Xiong and Cheung [2482] refer to inequalities for dual quermassintegrals, coming from results on *chord power integrals*.

Guan and Li [1092] prove an isoperimetric inequality for quermassintegrals of non-convex star-shaped sets using as tool a certain curvature flow.

Hernández Cifre and Saorín [1189] introduce classes of convex bodies via differentiability properties of the quermassintegrals, where the parameters of inner parallel bodies are used. They consider outer parallel bodies of convex bodies K using a general convex body E instead of a ball, thus having the form $K + pE$, and analogously (for a negative parameter p) inner parallel bodies. Conditions to be in such a class are given, *tangential bodies* (which play an important role here) in every class are described, and sharp inequalities between *relative quermassintegrals*, involving convex bodies of a given class, are given, together with a complete characterization of the equality cases. The class obtained by $p = n - 1$ as the set of outer parallel bodies of lower-dimensional bodies is characterized, too. Strongly related is the paper [1190] of the same authors, continuing their study of inner parallel bodies and quermassintegrals and investigating certain differentiability properties originating in a question of Bol. Consequences of their results refer to the decomposability of convex bodies and to the behavior of the roots of the Steiner polynomial. In particular, it is shown that, for $(n \geq 3)$, there are many n-dimensional convex bodies not satisfying the inradius condition in Teissier's problem on the *roots of the Steiner polynomial*. Also, the paper [1188] of Hernández Cifre et al. goes into the direction of p-parallel bodies. Via the notion of L_p *mixed quermassintegrals* the authors study the *differentiability of the quermassintegrals* with respect to a parameter family of p-parallel bodies. They show that the volume case is always differentiable, and also a sharp lower bound for the derivative of the support function of the p-inner parallel bodies is given.

Makai and Martini prove, in a forerunning paper of [1675], that a convex body K is origin symmetric iff for each direction u the $(n-1)$-volume of the intersection of K and an arbitrary hyperplane, with normal direction u, attains its maximum if the hyperplane contains the origin. Further on, they proved that for an origin-symmetric convex body K and $0 \leq k \leq n-2$, the quermassintegral W_k of the intersection of K and an arbitrary hyperplane attains its maximum if the hyperplane passes through the origin. And they posed the question whether the converse implication holds. In [1675], they are concerned with this problem for k-dimensional sections of K, if $1 \leq k \leq n-2$. They prove an infinitesimal version of the converse, for small C^2 perturbations of the unit ball if $k = 1$, and for small C^3 perturbations of the unit ball if $2 \leq k \leq n-2$.

Dafnis and Paouris [701] study upper and lower bounds for *re-normalized versions of the affine quermassintegrals*, resp. for the *dual affine quermassintegrals* which describe the asymptotic behaviour of these functionals when the dimension increases. They obtain interesting bounds and show that the study of these bounds is related to the *Blaschke–Santaló inequality and the Mahler and hyperplane conjectures*. Furthermore, p-mean versions of affine and dual affine quermassintegrals are introduced and also used for getting sharp bounds. Finally, bounds for products of affine and dual affine quermass-integrals are derived, and as one important tool, the authors use more general functionals which arise as normalized p-means of projection and section functions of the convex body K.

Chang and Wang [601] study the *Alexandrov–Fenchel inequalities for quermassintegrals* on a class of *non-convex sets* relating curvature quantities of different orders defined on the boundary of these sets.

Bobkov, Colesanti, and Fragalà [376] extend the concept of quermassintegrals to a *functional setting*, in the class of quasi-concave functions defined on the Euclidean space, and with the hierarchy of their subclasses given by α-concave functions. So functional counterparts of classical results are studied, More precisely, it is proved that these extensions satisfy a Steiner-type formula and are *valuations*. Also, generalizations of the *inequalities of Prékopa–Leindler and Brascamp–Lieb*, and of Cauchy–Kubota formulae are derived, like also *functional variants of isoperimetric type and Urysohn inequalities*.

Feng, Wang, and Yuan [860] study *differences of quermassintegrals and dual quermassintegrals* of the so-called Blaschke–Minkowski and radial Blaschke–Minkowski homomorphisms. They establish Minkowski, Brunn–Minkowski, and Alexandrov–Fenchel type inequalities for them, and they get also some straightforward inequalities on quermassintegral differences involving the mixed intersection and mixed projection body operators.

A pair of convex bodies K, E determines a *sausage* if there exists a compact convex set D of dimension at most one such that either $K = D + E$ or $E = D + K$. It is well known that sausages occur in discrete geometry (in connection with extremal ball packings), regarding differentiability of quermassintegrals (combined with inner parallel bodies), and in the study of the equality case of linear refinements of the Brunn–Minkowski inequality (for convex bodies having a common/equal-volume hyperplane projection). Yepes Nicolás: [2501] shows that sausages are the family of extremal sets in combination with certain linear improvements of Minkowski's first inequality. There sausages yield the equality cases of the corresponding linear refinements of the isoperimetric inequality and Urysohn's inequality. Quermassintegrals (like mean width) and related functionals of convex bodies appear when the author characterizes sausages by algebraic properties of the roots of Steiner polynomials.

Alonso-Gutiérrez et al. [81] derive estimates for the *integrals of powered i-th mean curvatures* (with $i = 1, 2, \ldots, n-1$) of compact, convex hypersurfaces in terms of quermassintegrals of the corresponding C^2_+ convex bodies. Similar estimates for the integrals of any convex transformation of the elementary symmetric functions of the radii of curvature of C^2_+ convex bodies are also proved, again in terms of quermassintegrals. Finally, the authors switch to radial functions, and then the estimates of the corresponding integrals are obtained in terms of *dual quermassintegrals*.

As starting point, Alonso-Gutiérrez et al. [84] give a characterization of the instance when an ordered $(m + 1)$-tuple of numbers is the ordered sequence of r-th *dual quermassintegrals* of two n-dimensional *star bodies*. This can be seen as analogue to Shephard's classification of quermassintegrals of two convex bodies in the Brunn–Minkowski theory of convex bodies. Further on, it is shown that the characterization of dual quermassintegrals is related to the *moment problem* which, roughly speaking, deals with the existence of a measure with prescribed moments. Based on this, the authors get further results on dual quermassintegrals and structural properties of the set of roots of dual Steiner polynomials of star bodies.

Zhao [2544] extends the concept of *dual affine quermassintegrals* to the Orlicz space. By calculating the first-order variation of the dual affine quermassintegrals, this extension yields the notion of *Orlicz dual affine quermassintegral*. The fundamental results on the related Minkowski and Brunn–Minkowski inequalities are correspondingly generalized, extending a variety of previously known inequalities. Also included are the related concepts and inequalities on *Orlicz dual mixed volumes*.

Zhao and Cheung [2552] prove Minkowski and Brunn–Minkowski inequalities for *differences of quermassintegrals and dual quermassintegrals* of convex and star bodies, respectively. More precisely, they take volume difference, quermassintegral difference, and dual quermassintegral difference (having specific meanings in the (dual) Brunn–Minkowski theory) into consideration. Considering also the difference between quermassintegral and dual quermassintegral, they add in this sense some "mixed notion". As for analogous investigations in related papers, the Brunn–Minkowski inequality and the Minkowski inequality of mixed volume type are proved.

Yaskin [2493] defines a direction-dependent section function for convex bodies K and proves that it is a polynomial on its support for any direction iff K is an ellipsoid and the dimension is odd. Using dual mixed volumes as tool, the author derives an even more general analogous result; ellipsoids and even dimensions occur there analogously. He asks finally whether this result survives if dual volumes are replaced by dual quermassintegrals.

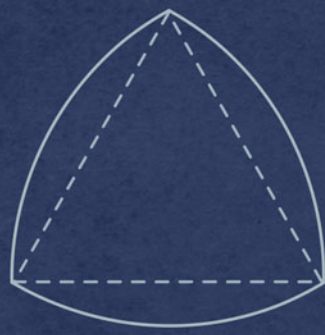
Besau, Hoehner, and Kur [304] give estimates in the framework of best asymptotic approximation of the Euclidean unit ball by polytopes; they use notions of distance induced by *intrinsic volumes* and also by the *Wills functional*. They succeed with asymptotically sharp bounds in high dimensions. Surprisingly, there is an almost optimal polytope doing this, at the same time, for all intrinsic volumes

(up to absolute constants). They extend their investigations also to asymptotic formulas for the *best approximation of smooth convex bodies by polytopes* with respect to a distance induced by dual volumes in the sense of Lutwak's dual Brunn–Minkowski theory.

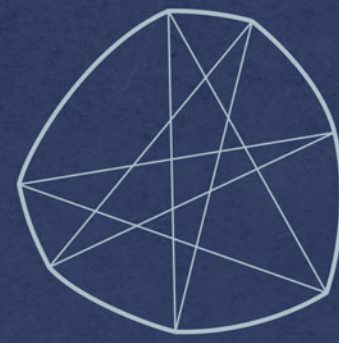
PLATE 7

Bodies of constant width

Reuleaux triangle



Reuleaux 7-gon



Figures 16 - 17

Rotationally symmetric body of constant width

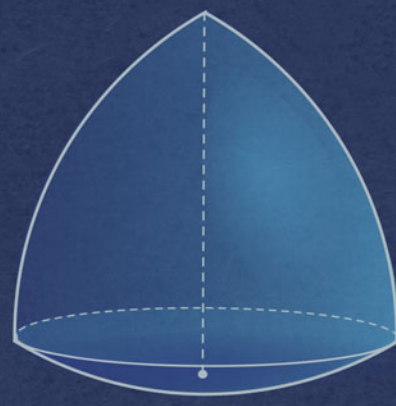


Figure 18

Bodies of constant width ► (see, e.g., 2.9.1.8)

Chapter 7
Mixed surface area measures

"The essence of mathematics lies in its freedom."

— Georg Cantor

To each $(n-1)$-tuple $(C_1, \ldots, C_{n-1})$ of compact convex sets, we may associate a Borel measure on $\mathbb{S}^{n-1}$ which "represents" (by integration) the functional $CC^n \ni L \mapsto V(L, C_1, \ldots, C_{n-1})$. This is the so-called *mixed surface area measure* of $C_1, \ldots, C_{n-1}$, and the formal definition is given right below. The existence of such a measure will be derived from the Riesz representation theorem for measures.

Together with introducing mixed surface area measure, one of the main tasks in this chapter is to prove that the construction of the surface area measure of a convex body of $\mathbb{R}^n$ described in Remark 6.2.3 is indeed a Borel measure on $\mathbb{S}^{n-1}$. We also show that this measure can be re-obtained as a mixed surface area measure, which makes it quite easier to work with it.

Of course, a very natural question arises: what are the Borel measures on $\mathbb{S}^{n-1}$ which are surface area measures of some convex body of $\mathbb{R}^n$? This question is known as *Minkowski's existence problem*. We give a complete solution for this problem in Section 7.3. The answer is surprisingly simple: a Borel measure on $\mathbb{S}^{n-1}$ is the surface area measure of a convex body of $\mathbb{R}^n$ if and only if it has centroid at the origin and it is not concentrated on a great subsphere.

From the differential geometric point of view, the Minkowski existence problem regards the existence of a hypersurface with a given prescribed curvature. It turns out that, in the smooth case, both problems are closely related. Indeed, if $K \subseteq \mathbb{R}^n$ is a convex body whose boundary is C^2_+ (the exact meaning of that will be clear later), then the inverse of the Gaussian curvature of ∂K is the density of the surface area measure of K with respect to the usual $(n-1)$-Hausdorff measure on $\mathbb{S}^{n-1}$. The relations between mixed surface area measures of convex bodies and their boundary curvatures are briefly investigated in Section 7.4. We do not go too deep in this direction, because this book does not assume any experience in differential geometry. The interested reader can find a much more complete treatment in [2167].

Plate 7 shows three examples of *sets of constant width*. Over the unit sphere, their width function is constant, and the third example is obtained by rotating the first one around one of its three axes of symmetry.

7.1 Definition and properties

Let $C_1, \ldots, C_{n-1}$ be non-empty, compact and convex subsets of $\mathbb{R}^n$ (that is, elements of the collection CC^n). The *mixed surface area measure* of $C_1, \ldots, C_{n-1}$ is the unique finite Borel measure

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_7

$S(C_1,\dots,C_{n-1},\cdot)$ on $\mathbb{S}^{n-1}$ with the property that

$$V(C_1,\dots,C_{n-1},L)=\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\,dS(C_1,\dots,C_{n-1},u) \tag{7.1.1}$$

for every $L\in CC^n$. It is not obvious that such measures exist and are unique, and proving this is our main objective in this section (see Theorem 7.1.2 below). We start by investigating the case where $C_1,\dots,C_{n-1}$ are polytopes. In view of (6.2.7), given polytopes $P_1,\dots,P_{n-1}\subseteq\mathbb{R}^n$, we write

$$S(P_1,\dots,P_{n-1},\cdot):=\sum_{u\in\mathbb{S}^{n-1}} V^{(n-1)}\big(P_1\cap\mathbf{h}(P_1,u),\dots,P_{n-1}\cap\mathbf{h}(P_{n-1},u)\big)\delta_u(\cdot), \tag{7.1.2}$$

where δ_u is the usual *Dirac delta measure* (concentrated) at u, defined as:

$$\delta_u(E)=\begin{cases}1, & \text{if } u\in E,\\ 0, & \text{if } u\notin E\end{cases}$$

for each $E\subseteq\mathcal{B}(\mathbb{S}^{n-1})$. Here we also remind the reader that $V^{(n-1)}$ denotes the mixed volume in $\mathbb{R}^{n-1}$, and that $\mathbf{h}(P,u)$ represents the supporting hyperplane of P which has u as outer normal.

However, this only makes sense if $V^{(n-1)}\big(P_1\cap\mathbf{h}(P_1,u),\dots,P_{n-1}\cap\mathbf{h}(P_{n-1},u)\big)\neq 0$ for at most finitely many vectors $u\in\mathbb{S}^{n-1}$. Our first objective is to prove that this is indeed the case.

Lemma 7.1.1. *If $V^{(n-1)}\big(P_1\cap\mathbf{h}(P_1,u)\dots,P_{n-1}\cap\mathbf{h}(P_{n-1},u)\big)\neq 0$ for $u\in\mathbb{S}^{n-1}$, then u is the outer normal of some facet of $P_1+\ldots+P_{n-1}$.*

Remark 7.1.1. In (6.2.7), we took the summation over the (finite) set N of outer normal vectors of the facets of all polytopes $t_2P_2+\ldots+t_nP_n$ for $t_2,\dots,t_n\geq 0$. By Lemma 7.1.1, it follows that N can be restricted to the set of outer normal vectors of the facets of $P_2+\ldots+P_n$ (this is clearly contained in N).

We will prove Lemma 7.1.1 as a consequence of the following result, which gives a sufficient condition for the disappearance of the mixed volume of an n-tuple of sets in CC^n (a necessary and sufficient condition was given in Proposition 6.2.3).

Lemma 7.1.2. *If $C_1,\dots,C_n$ are non-empty compact convex sets in $\mathbb{R}^n$ such that $\dim(C_1+\ldots+C_n)<n$, then $V(C_1,\dots,C_n)=0$.*

Proof. This is an easy consequence of the way we have defined mixed volumes. If $\dim(C_1+\ldots+C_n)<n$, then

$$0=\mathrm{vol}_n(C_1+\ldots+C_n)=\sum_{1\leq i_1,\dots,i_n\leq n} V(C_{i_1},\dots,C_{i_n}),$$

and the sum on the right-hand side contains only non-negative numbers (recall that mixed volumes are always non-negative). Hence, $V(C_{i_1},\dots,C_{i_n})=0$ for any $1\leq i_1,\dots,i_n\leq n$. In particular, $V(C_1,\dots,C_n)=0$.

□

Proof of Lemma 7.1.1. Let $P=P_1+\ldots+P_{n-1}$. From Lemma 6.1.2, we know that

$$P_1\cap\mathbf{h}(P_1,u)+\ldots+P_{n-1}\cap\mathbf{h}(P_{n-1},u)=P\cap\mathbf{h}(P,u).$$

Therefore, if u is not the outer normal of a facet of P, then

$$n-1>\dim(P\cap\mathbf{h}(P,u))=\dim\big(P_1\cap\mathbf{h}(P_1,u)+\ldots+P_{n-1}\cap\mathbf{h}(P_{n-1},u)\big),$$

and from Lemma 7.1.2 applied for $\mathbb{R}^{n-1}$, we get that

$$V^{(n-1)}\big(P_1 \cap \mathbf{h}(P_1, u), \ldots, P_{n-1} \cap \mathbf{h}(P_{n-1}, u)\big) = 0.$$

□

This proves that indeed (7.1.2) makes sense because the sum is taken over a finite subset of $\mathbb{S}^{n-1}$ (any polytope has at most finitely many facets). Thus, $S(P_1, \ldots, P_{n-1}, \cdot)$ is just a (non-negative) combination of Dirac delta measures. In view of that, and remembering that the integration of a function with respect to a (non-negative) combination of Dirac delta measures $\delta = \lambda_1 \delta_{u_1} + \ldots + \lambda_m \delta_{u_m}$ on $\mathbb{S}^{n-1}$ is given by

$$\int_{\mathbb{S}^{n-1}} f \, d\delta = \lambda_1 f(u_1) + \ldots + \lambda_m f(u_m),$$

we may rewrite (6.2.7) for each $C \in CC^n$ as

$$\begin{aligned} V(C, P_1, \ldots, P_{n-1}) &= \\ = \frac{1}{n} \sum_{u \in \mathbb{S}^{n-1}} \left[V^{(n-1)}\big(P_1 \cap \mathbf{h}(P_1, u) \ldots, P_{n-1} \cap \mathbf{h}(P_{n-1}, u)\big) \int_{\mathbb{S}^{n-1}} h_C(v) \, d\delta_u(v) \right] &= \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_C(v) \, dS(P_1, \ldots, P_{n-1}, v), & \end{aligned}$$

and this guarantees the existence (but not uniqueness) of mixed surface area measures for polytopes. Now we have to extend this (and to prove uniqueness, of course) for the case of general non-empty, convex, and compact sets. For doing this, we follow [1263] and have also to borrow some theory from functional analysis.

We endow the vector space $C(\mathbb{S}^{n-1})$ of continuous functions $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$ with the *maximum norm*, given by

$$||f||_{C(\mathbb{S}^{n-1})} := \max\{|f(x)| : x \in \mathbb{S}^{n-1}\}.$$

By doing so, $C(\mathbb{S}^{n-1})$ becomes a *Banach space*, meaning that every Cauchy sequence converges. A function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$ is said to be of C^2 *class* if its *radial extension* $\tilde{f}\colon \mathbb{R}^n \to \mathbb{R}$ given by

$$\tilde{f}(x) = \begin{cases} |x| f\left(\frac{x}{|x|}\right), & \text{if } x \neq 0_{\mathbb{R}^n} \text{ and} \\ 0, & \text{if } x = 0_{\mathbb{R}^n} \end{cases}$$

is twice continuously differentiable in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ (recall that this means that $\tilde{f}$ has continuous second-order partial derivatives in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$). Notice that $\tilde{f}$ is positively homogeneous (recall that we reserve this expression for functions which are positively homogeneous of degree one). The set of such functions is clearly a vector subspace of $C(\mathbb{S}^{n-1})$, and it is denoted by $C^2(\mathbb{S}^{n-1})$.

Proposition 7.1.1. *The subspace $C^2(\mathbb{S}^{n-1})$ is dense in $C(\mathbb{S}^{n-1})$ with respect to the maximum norm.*

Proof. We briefly describe the construction of a sequence of functions in $C^2(\mathbb{S}^{n-1})$ converging to a given function $f \in C(\mathbb{S}^{n-1})$, but without proving in detail why everything works. We use the technique of smoothing by *mollifiers* (see [821]). Define $F\colon \mathbb{R}^n \to \mathbb{R}$ by

$$F(x) := \begin{cases} \beta \exp\left(\frac{1}{|x|^2-1}\right), & \text{if } |x| < 1, \\ 0, & \text{if } |x| \geq 1 \end{cases},$$

where the constant $\beta \in \mathbb{R}$ is chosen such that $\int_{\mathbb{R}^n} F = 1$. It is not difficult to check that this function is C^2 (actually, it is C^∞, and it is known as the *standard mollifier*). For each $j \in \mathbb{N}$, define

$$F_j(x) = j^n F(jx).$$

If $f \in C(\mathbb{S}^{n-1})$, then its radial extension $\tilde{f}$ is clearly continuous in $\mathbb{R}^n$. For each $j \in \mathbb{N}$ and $x \in \mathbb{R}^n$, let

$$\tilde{f}_j(x) := F_j * \tilde{f}(x) := \int_{\mathbb{R}^n} F_j(x-y)\tilde{f}(y)\,dy,$$

where the integration is taken with respect to the usual Lebesgue measure of $\mathbb{R}^n$. In other words, $\tilde{f}_j$ is the *convolution* of F_j and $\tilde{f}$. Finally, let $f_j = \tilde{f}_j|_{\mathbb{S}^{n-1}}$. Item (i) of Theorem 7 of Appendix C.5 in [821] gives that $\tilde{f}_j$ is C^2 (in particular), and hence the radial extension of f_j is C^2 in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ (this is given by $|x|\tilde{f}_j(x/|x|)$, thus obtained as products and compositions of twice continuously differentiable functions). Even more, item (iii) of the same theorem guarantees that $f_j \to f$ uniformly, which is equivalent to convergence in the maximum norm.

□

Remark 7.1.2. The fact that $C^2(\mathbb{S}^{n-1})$ is dense in $C(\mathbb{S}^{n-1})$ can also be obtained by the Stone–Weierstrass Theorem (see [775, Theorem 4.6.16]). However, the method that we adopted has an advantage that will be useful later: if $f \in C(\mathbb{S}^{n-1})$ is non-negative, then the convolution of F_j and $\tilde{f}$ is also non-negative. Hence, we may approximate f by non-negative functions in $C^2(\mathbb{S}^{n-1})$.

Actually, we will work with another set of functions. We denote by $\mathcal{F}$ the set of continuous functions $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$ for which there exists $K, L \in CC^n$ such that $f = h_K - h_L$. From Proposition 2.3.2, it is clear that $\mathcal{F}$ is a vector subspace of $C(\mathbb{S}^{n-1})$.

Proposition 7.1.2. *We have $C^2(\mathbb{S}^{n-1}) \subseteq \mathcal{F}$. In particular, $\mathcal{F}$ is dense in $C(\mathbb{S}^{n-1})$.*

Proof. For simplicity of notation, we also denote by f the radial extension of f. What we have to prove is that, given $f \in C^2(\mathbb{S}^{n-1})$, there exists $C, L \in CC^n$ such that $f = h_C - h_L$, where h_C and h_L are defined in $\mathbb{R}^n$ (these are clearly the radial extensions of their restrictions to $\mathbb{S}^{n-1}$, since any support function is positively homogeneous). We will show that there exist $C \in CC^n$ and $r > 0$ such that $f = h_C - h_{rB^n}$, but for this, we need some preliminary steps.

(i) We need some facts about positively homogeneous C^2 functions. Let $g\colon \mathbb{R}^n \to \mathbb{R}$ be a positively homogeneous function which is twice continuously differentiable in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Then for each non-zero $x \in \mathbb{R}^n$, we have

$$g(x) = \langle \nabla g(x), x \rangle = \sum_{j=1}^{n} x_j \partial_j g(x). \tag{7.1.3}$$

The second equality comes just from writing $\nabla g(x)$ in coordinates, and for the first one we simply notice that

$$\langle \nabla g(x), x \rangle = \frac{d}{dt} g(x+tx)\Big|_{t=0} = g(x)\frac{d}{dt}(1+t)\Big|_{t=0} = g(x).$$

In possession of this fact, for each non-zero $x \in \mathbb{R}^n$ define the bilinear form $d^2 g_x\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$

$$d^2 g_x(u, v) := \big\langle \mathrm{hess}(g)_x(u), v \big\rangle,$$

where we remind the reader that $\mathrm{hess}(g)_x$ is the *Hessian matrix* of g at x, defined as $\mathrm{hess}(g)_x := \big[\partial_{ij} g(x)\big]_{i,j=1}^n$. We claim that for each $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ equality

$$d^2 g_x(x, v) = 0 \tag{7.1.4}$$

holds for every $v \in \mathbb{R}^n$. This is the same as proving that $\mathrm{hess}(g)_x(x) = 0$. To do so, we differentiate both sides of (7.1.3) with respect to e_i to obtain

$$\partial_i g(x) = \partial_i g(x) + \sum_{j=0}^{n} x_j \partial_{ij} g(x) = \partial_i g(x) + \langle \mathrm{hess}(g)_x(x), e_i \rangle,$$

and this equality holds for every $i \in \{1, \dots, n\}$. This proves that $\mathrm{hess}(g)_x(x) = 0$. Finally, we claim that

$$d^2 g_{\lambda x}(\cdot, \cdot) = \frac{1}{\lambda} d^2 g_x(\cdot, \cdot) \tag{7.1.5}$$

for any non-zero vector x and every $\lambda > 0$. Indeed, we have

$$\partial_i g(\lambda x) = \lim_{t \to 0} \frac{g(\lambda x + t e_i) - g(\lambda x)}{t} = \lim_{t \to 0} \lambda \frac{g\left(x + \frac{t}{\lambda} e_i\right) - g(x)}{\lambda \frac{t}{\lambda}} =$$
$$= \lim_{t \to 0} \frac{g\left(x + \frac{t}{\lambda} e_i\right) - g(x)}{\frac{t}{\lambda}} = \partial_i g(x)$$

for every $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, $\lambda > 0$ and $i \in \{1, \dots, n\}$. Thus, differentiating with respect to any e_j, we get

$$\partial_{ij} g(\lambda x) = \lim_{t \to 0} \frac{\partial_i g(\lambda x + t e_j) - \partial_i g(\lambda x)}{t} = \frac{1}{\lambda} \lim_{t \to 0} \frac{\partial_i g\left(x + \frac{t}{\lambda} e_j\right) - \partial_i g(x)}{\frac{t}{\lambda}} = \frac{1}{\lambda} \partial_{ij} g(x).$$

From this, we obtain that $\mathrm{hess}(g)_{\lambda x} = \frac{1}{\lambda} \mathrm{hess}(g)_x$, and (7.1.5) is proved.

(ii) Notice that $h_{B^n}(x) = |x|$ for all $x \in \mathbb{R}^n$. We claim that if $x, y \in \mathbb{S}^{n-1}$ are orthogonal vectors, then

$$\big\langle \mathrm{hess}(h_{B^n})_x(y), y \big\rangle = 1. \tag{7.1.6}$$

Indeed, a straightforward calculation for $x \in \mathbb{S}^{n-1}$ gives that

$$\mathrm{hess}(h_{B^n})_x = \big[\delta_{ij} - x_i x_j\big]_{i,j=1}^n,$$

where $x = (x_1, \dots, x_j)$, and δ_{ij} is the usual Kronecker delta. Then,

$$\mathrm{hess}(h_{B^n})_x(y) = y - x\langle x, y \rangle, \tag{7.1.7}$$

and from this we get that

$$\big\langle \mathrm{hess}(h_{B^n})_x(y), y \big\rangle = \langle y, y \rangle - \langle x, y \rangle^2 = 1,$$

because $\langle x, y \rangle = 0$ and $|y| = 1$.

Now we return to what we want to prove. Observe that the map

$$\mathbb{S}^{n-1} \times \mathbb{S}^{n-1} \ni (x, y) \mapsto d^2 f_x(y, y) = \big\langle \operatorname{hess}(f)_x(y), y \big\rangle$$

is continuous (because the second-order partial derivatives of f are continuous), and hence it attains a maximum. Thus, we may choose a number $r > 0$ such that

$$r > d^2 f_x(y, y)$$

for every $x, y \in \mathbb{S}^{n-1}$. It is clear that $h_{rB^n} = r h_{B^n}$ (see Proposition 2.3.2). Let $x \in \mathbb{S}^{n-1}$ and $u \in \mathbb{R}^n$ be arbitrary. Write $u = \alpha x + z$ for some $z \in x^\perp$ and some $\alpha \in \mathbb{R}$, assume first that $z \neq 0$, and let $y = z/|z|$. Using (7.1.4) for both f and h_{B^n}, and using also (7.1.7), we obtain

$$\begin{aligned} d^2(f + h_{rB^n})_x(u, u) &= d^2 f_x(u, u) + r d^2(h_{B^n})_x(u, u) = \\ = d^2 f_x(z, z) + r d^2(h_{B^n})_x(z, z) &= |z|^2 \big(d^2 f_x(y, y) + r d^2(h_{B^n})_x(y, y)\big) = \\ &= |z|^2 \big(d^2 f_x(y, y) + r\big) > 0. \end{aligned}$$

In case where $z = 0$, then we have $d^2(f + h_{rB^n})_x(u, u) = 0$ from (7.1.4). Consequently, $d^2(f + h_{rB^n})_x$ is positive semi-definite for every $x \in \mathbb{S}^{n-1}$. Since $f + h_{rB^n}$ is positively homogeneous (because it is the sum of positively homogeneous functions), we get from (7.1.5) that $d^2(f + h_{rB^n})_x$ is positive semi-definite for every $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. From Theorem 1.2.4, we get that $f + h_{rB^n}$ is convex when restricted to any convex subset of $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, and by continuity it is convex in $\mathbb{R}^n$.

Summarizing, we know that the function $f + h_{rB^n}$ is positively homogeneous and convex, implying that it is sublinear (see Remark 2.3.1). Since this function is clearly also finite, by Theorem 2.3.1, we get that there exists $C \in CC^n$ such that $f + h_{rB^n} = h_C$. Hence $f = h_C - h_{rB^n}$, and taking the restrictions of these functions to $\mathbb{S}^{n-1}$, we conclude the proof.

□

The last result we need from functional analysis is a type of representation theorem, known as *Riesz–Markov theorem*. Recall that a linear functional $\varphi\colon C(X) \to \mathbb{R}$ is said to be *positive* if $\varphi(f) \geq 0$ for every $f \in C(X)$ such that $f(x) \geq 0$ for all $x \in X$.

Theorem 7.1.1 (Riesz–Markov)**.** *Let X be a compact metric space, and let $\varphi\colon C(X) \to \mathbb{R}$ be a positive linear functional. Then there exists a unique finite Borel measure μ on X such that*

$$\varphi(f) = \int_X f \, d\mu$$

for every $f \in C(X)$.

Proof. See [2075, Theorem 2.14].

□

Before stating and proving the main theorem regarding the existence and uniqueness of mixed surface area measures, we need a technical lemma that bounds the difference of mixed volumes by the maximum norm of the difference of support functions.

Lemma 7.1.3. *Let $C_1, \ldots, C_{n-1}, C, L \in CC^n$. Then*

$$|V(C, C_1, \ldots, C_{n-1}) - V(L, C_1, \ldots, C_{n-1})| \leq ||h_C - h_L||_{C(\mathbb{S}^{n-1})} V(B^n, C_1, \ldots, C_{n-1}).$$

Proof. By the continuity of mixed volumes in the Hausdorff metric, it is enough to consider the case where $C_1, \ldots, C_{n-1}$ are polytopes $P_1, \ldots, P_{n-1}$, respectively. In this case, (6.2.7) gives

$$|V(C,P_1,\dots,P_{n-1})-V(L,P_1,\dots,P_{n-1})| =$$
$$= \frac{1}{n}\left|\sum_{u\in N}\big(h_C(u)-h_L(u)\big)V^{(n-1)}\big(P_1\cap\mathbf{h}(P_1,u),\dots,P_{n-1}\cap\mathbf{h}(P_{n-1},u)\big)\right| \leq$$
$$\leq \max_{u\in\mathbb{S}^{n-1}}|h_C(u)-h_L(u)|\frac{1}{n}\sum_{u\in N}V^{(n-1)}\big(P_1\cap\mathbf{h}(P_1,u),\dots,P_{n-1}\cap\mathbf{h}(P_{n-1},u)\big) =$$
$$= ||h_C-h_L||_{C(\mathbb{S}^{n-1})}V(B^n,P_1,\dots,P_{n-1}),$$

where in the last equality we used (6.2.7) recalling that $h_{B^n}(u)=1$ for every $u\in\mathbb{S}^{n-1}$.

□

Theorem 7.1.2. *Let $C_1,\dots,C_{n-1}\in CC^n$ be given. There exists a unique finite Borel measure $S(C_1,\dots,C_{n-1},\cdot)$ on $\mathbb{S}^{n-1}$ with the property that*

$$V(L,C_1,\dots,C_{n-1})=\frac{1}{n}\int_{\mathbb{S}^{n-1}}h_L(u)\,dS(C_1,\dots,C_{n-1},u) \tag{7.1.8}$$

for every $L\in CC^n$.

Remark 7.1.3. As mentioned before, the measure $S(C_1,\dots,C_{n-1},\cdot)$ is called the *mixed surface area measure* of $C_1,\dots,C_{n-1}$. Similar to what was discussed for mixed volumes, we may regard this as a map which associates to each $(n-1)$-tuple of non-empty compact convex sets a Borel measure on $\mathbb{S}^{n-1}$.

Proof. We define the functional $T(C_1,\dots,C_{n-1},\cdot)\colon\mathcal{F}\to\mathbb{R}$ as

$$T(C_1,\dots,C_{n-1},f):=nV(C,C_1,\dots,C_{n-1})-nV(L,C_1,\dots,C_{n-1}),$$

if $f=h_C-h_L$. This is well-defined because if $h_C-h_L=h_{C'}-h_{L'}$, then $h_C+h_{L'}=h_L+h_{C'}$, which gives that $C+L'=L+C'$ (see Proposition 2.3.2), and hence linearity of mixed volumes yields

$$V(C,C_1,\dots,C_{n-1})+V(L',C_1,\dots,C_{n-1})=V(C+L',C_1,\dots,C_{n-1})=$$
$$=V(L+C',C_1,\dots,C_{n-1})=V(L,C_1,\dots,C_{n-1})+V(C',C_1,\dots,C_{n-1}).$$

Also, it is easy to see that $T(C_1,\dots,C_{n-1},\cdot)$ is linear (if $\alpha<0$ and $f=h_C-h_L$, then we write $\alpha f=h_{-\alpha L}-h_{-\alpha C}$). Now we observe that $T(C_1,\dots,C_{n-1},\cdot)$ is continuous with respect to the maximum norm, because from Lemma 7.1.3, we have the inequality

$$|T(C_1,\dots,C_{n-1},f)|=n|V(C,C_1,\dots,C_{n-1})-V(L,C_1,\dots,C_{n-1})|\leq$$
$$\leq nV(B^n,C_1,\dots,C_{n-1})||h_C-h_L||_{C(\mathbb{S}^{n-1})}=nV(B^n,C_1,\dots,C_{n-1})||f||_{C(\mathbb{S}^{n-1})},$$

for any $f\in\mathcal{F}$, $f=h_C-h_L$. The constant $nV(B^n,C_1,\dots,C_{n-1})$ does not depend on f, but only on $C_1,\dots,C_{n-1}$. Since $\mathcal{F}$ is dense in $C(\mathbb{S}^{n-1})$, it follows that $T(C_1,\dots,C_{n-1},\cdot)$ admits a unique extension to a continuous linear function defined on $C(\mathbb{S}^{n-1})$, which is given as

$$T(C_1,\dots,C_{n-1},f):=\lim_{j\to\infty}T(C_1,\dots,C_{n-1},f_j),$$

for each $f\in C(\mathbb{S}^{n-1})$, where (f_j) is any sequence in $\mathcal{F}$ with $f_j\to f$ (in the maximum norm, of course). We are abusing a little of the notation still denoting this extension by $T(C_1,\dots,C_{n-1},\cdot)$.

We claim that $T(C_1,\dots,C_{n-1},\cdot)\colon C(\mathbb{S}^{n-1})\to\mathbb{R}$ is positive. It is clearly sufficient to prove that for functionals in $\mathcal{F}$ (see Remark 7.1.2). If $f=h_C-h_L\geq 0$, then $h_C\geq h_L$, and this gives that $L\subseteq C$.

Hence, monotonicity of mixed volumes yields that $V(C, C_1, \dots, C_{n-1}) \geq V(L, C_1, \dots, C_{n-1})$, from where we get that $T(C_1, \dots, C_{n-1}, f) \geq 0$.

As a consequence of $T(C_1, \dots, C_{n-1}, \cdot)$ being positive, Theorem 7.1.1 gives that there exists a unique finite Borel measure $S(C_1, \dots, C_{n-1}, \cdot)$ on $\mathbb{S}^{n-1}$ such that

$$T(C_1, \dots, C_{n-1}, f) = \int_{\mathbb{S}^{n-1}} f(u)\, dS(C_1, \dots, C_{n-1}, u).$$

for every $f \in C(\mathbb{S}^{n-1})$. Therefore, letting $f = h_L$ in the equality above for each $L \in CC^n$, we get

$$V(L, C_1, \dots, C_{n-1}) = \frac{1}{n} T(C_1, \dots, C_{n-1}, h_L) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS(C_1, \dots, C_{n-1}, u),$$

and hence the existence part is finished. For the uniqueness, suppose that ν is a finite Borel measure on $\mathbb{S}^{n-1}$ with the property that

$$V(L, C_1, \dots, C_{n-1}) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, d\nu(u),$$

for every $L \in CC^n$, and assume first that $\nu(\mathbb{S}^{n-1}) \neq 0$. Then, for any $f \in \mathcal{F}$ (with $f = h_C - h_L$, say), we have

$$\int_{\mathbb{S}^{n-1}} f(u)\, d\nu(u) = nV(L, C_1, \dots, C_{n-1}) - nV(C, C_1, \dots, C_{n-1}) = T(C_1, \dots, C_{n-1}, f).$$

We claim that this equality actually holds for every $f \in C(\mathbb{S}^{n-1})$. This is a consequence of $\mathcal{F}$ being dense in $C(\mathbb{S}^{n-1})$ in the maximum norm, and of ν being finite. In fact, if $f \in C(\mathbb{S}^{n-1})$ and (f_j) is a sequence in $\mathcal{F}$ with $f_j \to f$, then for any given $\varepsilon > 0$ we have $||f_j - f||_{C(\mathbb{S}^{n-1})} < \varepsilon/\nu(\mathbb{S}^{n-1})$ for $j \in \mathbb{N}$ large enough, and hence

$$\left| \int_{\mathbb{S}^{n-1}} f_j\, d\nu - \int_{\mathbb{S}^{n-1}} f\, d\nu \right| \leq \int_{\mathbb{S}^{n-1}} |f_j - f|\, d\nu \leq \int_{\mathbb{S}^{n-1}} \frac{\varepsilon}{\nu(\mathbb{S}^{n-1})}\, d\nu = \varepsilon.$$

Therefore, $\int_{\mathbb{S}^{n-1}} f_j\, d\nu \to \int_{\mathbb{S}^{n-1}} f\, d\nu$ as $j \to \infty$, and the claim is proved. Finally, the equality case in Riesz–Markov Theorem yields that $\nu(\cdot) = S(C_1, \dots, C_{n-1}, \cdot)$.

If $\nu(\mathbb{S}^{n-1}) = 0$, then $V(L, C_1, \dots, C_{n-1}) = 0$ for every $L \in CC^n$. As an immediate consequence, the functional $T(C_1, \dots, C_{n-1}, \cdot)$ vanishes in $\mathcal{F}$. Since this functional is continuous and $\mathcal{F}$ is dense, it follows that $T(C_1, \dots, C_{n-1}, \cdot)$ also vanishes in $C(\mathbb{S}^{n-1})$. Thus, $S(C_1, \dots, C_{n-1}, \cdot) = 0 = \nu(\cdot)$. The proof is complete.

□

Example 7.1.1. The mixed surface area measure $S(P_1, \dots, P_{n-1}, \cdot)$ of polytopes $P_1, \dots, P_{n-1} \subseteq \mathbb{R}^n$ is a (non-negative) combination of Dirac measures concentrated in the outer normal vectors of the facets of the polytope $P_1 + \dots + P_{n-1}$, as it is discussed in (7.1.2). In particular, if $P_1 = \dots = P_{n-1} = P$, then the facets of $P_1 + \dots + P_{n-1} = (n-1)P$ are respectively parallel to the facets of P, and thus

$$\begin{aligned} S(P, \dots, P, \cdot) &= \sum_{u \in \mathbb{S}^{n-1}} V^{(n-1)}(P \cap \mathbf{h}(P, u), \dots, P \cap \mathbf{h}(P, u)) \delta_u(\cdot) = \\ &= \sum_{F \text{ facet of } P} \mathrm{vol}_{n-1}(F) \delta_{u_F}(\cdot), \end{aligned} \tag{7.1.9}$$

where, as usual, u_F denotes the outer normal vector of the facet F. Then, for a given Borel set $A \subseteq \mathbb{S}^{n-1}$, the measure $S(P, \dots, P, A)$ equals the sum of the $(n-1)$-dimensional volumes of the facets of P whose outer normal vectors belong to A.

Now that we have the existence and uniqueness of the mixed surface area measures, it is time to study some of their properties. We remind the reader that sometimes we will naturally regard this concept as a map that associates each $(n-1)$-tuple of sets in CC^n to a finite Borel measure on $\mathbb{S}^{n-1}$. The space of finite Borel measures on $\mathbb{S}^{n-1}$ will be denoted by $\mathcal{M}(\mathbb{S}^{n-1})$. For the convenience of the reader, we also recall a concept from standard measure theory. We say that a sequence $(\mu_j)_{j\in\mathbb{N}}$ of finite Borel measures on $\mathbb{S}^{n-1}$ *converges weakly* to a finite Borel measure μ on $\mathbb{S}^{n-1}$ if

$$\int_{\mathbb{S}^{n-1}} f \, d\mu_j \to \int_{\mathbb{S}^{n-1}} f \, d\mu$$

for every $f \in C(\mathbb{S}^{n-1})$. From Proposition A.2.4, we obtain that the weak limit is unique, when it exists.

Theorem 7.1.3 (Properties of mixed surface area measures)**.** *The map*

$$S\colon \underbrace{CC^n \times \dots \times CC^n}_{n-1} \to \mathcal{M}(\mathbb{S}^{n-1}),$$

which associates each $(n-1)$-tuple of sets in CC^n to their mixed surface area measure has the properties below.

(i) *S is symmetric, meaning that $S(\mathrm{per}(C_1, \dots, C_{n-1}), \cdot) = S(C_1, \dots, C_{n-1}, \cdot)$ for any $(n-1)$-tuple $C_1, \dots, C_{n-1}$ of sets in CC^n and any permutation of them.*

(ii) *S is multilinear in the sense that for any $C, L, C_2, \dots, C_{n-1} \in CC^n$ and every $\alpha, \beta \geq 0$,*

$$S(\alpha C + \beta L, C_2, \dots, C_{n-1}, \cdot) = \alpha S(C, C_2, \dots, C_{n-1}, \cdot) + \beta S(L, C_2, \dots, C_{n-1}, \cdot).$$

(iii) *S is translation-invariant, that is,*

$$S(C_1 + x_1, \dots, C_{n-1} + x_{n-1}, \cdot) = S(C_1, \dots, C_{n-1}, \cdot),$$

for any $C_1, \dots, C_{n-1} \in CC^n$ and every $x_1, \dots, x_{n-1} \in \mathbb{R}^n$.

(iv) *S is $\mathrm{O}(n)$-invariant in the following sense: if $C_1, \dots, C_{n-1} \in CC^n$ and $T \in \mathrm{O}(n)$, then*

$$S(TC_1, \dots, TC_{n-1}, TA) = S(C_1, \dots, C_{n-1}, A)$$

for each Borel set $A \subseteq \mathbb{S}^{n-1}$.

(v) *S has the following continuity property: if $C_1, \dots, C_{n-1} \in CC^n$ and for each $k \in \{1, \dots, n-1\}$ we pick a sequence $(C_k^j)_{j\in\mathbb{N}}$ in CC^n converging to C_k in the Hausdorff metric, then*

$$S(C_1^j, \dots, C_{n-1}^j, \cdot) \to S(C_1, \dots, C_{n-1}, \cdot)$$

weakly as $j \to \infty$.

Proof. Let $C_1, \dots, C_{n-1} \in CC^n$. For any permutation $\mathrm{per}(C_1, \dots, C_{n-1})$ and every $C \in CC^n$, we have

$$\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_C(u)\,dS(\mathrm{per}(C_1,\dots,C_{n-1},u)) = V(\mathrm{per}(C_1,\dots,C_{n-1}),C) = \\ = V(C,C_1,\dots,C_{n-1}),$$

where we used the symmetry of mixed volumes. The uniqueness part of Theorem 7.1.2 gives that $S(\mathrm{per}(C_1,\dots,C_{n-1}),\cdot) = S(C_1,\dots,C_{n-1},\cdot)$. This settles **(i)**.

To prove **(ii)**, the argument is similar. Let $C, L, C_2, \dots, C_{n-1} \in CC^n$ and $\alpha,\beta \geq 0$, and put $\mu(\cdot) := \alpha S(C,C_2,\dots,C_{n-1},\cdot) + \beta S(L,C_2,\dots,C_{n-1})$. For any $K \in CC^n$, we have

$$\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\,d\mu(u) = \\ = \frac{\alpha}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\,dS(C,C_2,\dots,C_{n-1},u) + \frac{\beta}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\,dS(L,C_2,\dots,C_{n-1},u) = \\ = \alpha V(K,C,C_2,\dots,C_{n-1}) + \beta V(K,L,C_2,\dots,C_{n-1}) = V(K,\alpha C+\beta L,C_2,\dots,C_{n-1}),$$

and using again the uniqueness result in Theorem 7.1.2, we obtain the desired equality $\mu(\cdot) = S(\alpha C + \beta L, C_2, \dots, C_{n-1}, \cdot)$.

Property **(iii)** is proved in the same way, using the translation-invariance of mixed volumes and the uniqueness property of mixed surface area measures. We will not write down the details.

Now we prove **(iv)**. First observe that for any $C \in CC^n$, every $T \in \mathrm{O}(n)$ and each $u \in \mathbb{S}^{n-1}$ we have $h_{TC}(u) = h_C(T^{-1}u)$. This follows from (2.6.5) together with the fact that $T^* = T^{-1}$ whenever T is orthogonal (see Lemma 2.6.5).

Given $C_1,\dots,C_{n-1}$ and $T \in \mathrm{O}(n)$, let μ be the push-forward measure of the $S(TC_1,\dots,TC_{n-1},\cdot)$ under the restriction of T^{-1} to $\mathbb{S}^{n-1}$. That is, $\mu(\cdot) = S(TC_1,\dots,TC_{n-1},T(\cdot))$. For any $C \in CC^n$, standard change of variables yields

$$\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_C(u)\,d\mu(u) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_C(T^{-1}(v))\,dS(TC_1,\dots,TC_{n-1},v) = \\ = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{TC}(v)\,dS(TC_1,\dots,TC_{n-1},v) = V(TC,TC_1,\dots,TC_{n-1}) = V(C,C_1,\dots,C_{n-1}).$$

It follows that $\mu(\cdot) = S(C_1,\dots,C_{n-1},\cdot)$, and this gives that

$$S(TC_1,\dots,TC_{n-1},TA) = \mu(A) = S(C_1,\dots,C_{n-1},A)$$

for every Borel set $A \subseteq \mathbb{S}^{n-1}$.

It remains to prove **(v)**. If $f \in C(\mathbb{S}^{n-1})$, then let $(f_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{F}$ such that $f_m \to f$ in the maximum norm. For each $m \in \mathbb{N}$, writing $f_m = h_{K_m} - h_{L_m}$, we have the estimate

$$
\begin{aligned}
&\left|\int_{\mathbb{S}^{n-1}} f(u)\,dS(C_1^j,\dots,C_{n-1}^j,u)-\int_{\mathbb{S}^{n-1}} f(u)\,dS(C_1,\dots,C_{n-1},u)\right|\le\\
&\le\left|\int_{\mathbb{S}^{n-1}} f(u)\,dS(C_1^j,\dots,C_{n-1}^j,u)-\int_{\mathbb{S}^{n-1}} f_m(u)\,dS(C_1^j,\dots,C_{n-1}^j,u)\right|+\\
&+\left|\int_{\mathbb{S}^{n-1}} f_m(u)\,dS(C_1^j,\dots,C_{n-1}^j,u)-\int_{\mathbb{S}^{n-1}} f_m(u)\,dS(C_1,\dots,C_{n-1},u)\right|+\\
&+\left|\int_{\mathbb{S}^{n-1}} f_m(u)\,dS(C_1,\dots,C_{n-1},u)-\int_{\mathbb{S}^{n-1}} f(u)\,dS(C_1,\dots,C_{n-1},u)\right|\le\\
&\le\int_{\mathbb{S}^{n-1}}|f(u)-f_m(u)|\,dS(C_1^j,\dots,C_{n-1}^j,u)+\\
&+n\left|V(K_m,C_1^j,\dots,C_{n-1}^j)-V(K_m,C_1,\dots,C_{n-1})\right|+\\
&+n\left|V(L_m,C_1^j,\dots,C_{n-1}^j)-V(L_m,C_1,\dots,C_{n-1})\right|+\\
&+\int_{\mathbb{S}^{n-1}}|f(u)-f_m(u)|\,dS(C_1,\dots,C_{n-1},u)\le\\
&\le\|f-f_m\|_{C(\mathbb{S}^{n-1})}\big(V(B^n,C_1^j,\dots,C_{n-1}^j)+V(B^n,C_1,\dots,C_{n-1})\big)+\\
&+n\left|V(K_m,C_1^j,\dots,C_{n-1}^j)-V(K_m,C_1,\dots,C_{n-1})\right|+\\
&+n\left|V(L_m,C_1^j,\dots,C_{n-1}^j)-V(L_m,C_1,\dots,C_{n-1})\right|.
\end{aligned}
\tag{7.1.10}
$$

We bound these last terms in the following way: for an arbitrarily given $\varepsilon>0$, let $j_0\in\mathbb{N}$ be such that $j>j_0$ implies

$$V(B^n,C_1^j,\dots,C_{n-1}^j)<\varepsilon+V(B^n,C_1,\dots,C_{n-1}).$$

Now choose a number $m_0\in\mathbb{N}$ for which

$$\|f-f_{m_0}\|_{C(\mathbb{S}^{n-1})}<\frac{\varepsilon}{3(\varepsilon+2V(B^n,C_1,\dots,C_{n-1}))}.$$

Then, choose $j_1\in\mathbb{N}$, with $j_1>j_0$, such that both inequalities

$$\left|V(K_{m_0},C_1^{j_1},\dots,C_{n-1}^{j_1})-V(K_{m_0},C_1,\dots,C_{n-1})\right|<\frac{\varepsilon}{3n},\ \text{and}$$
$$\left|V(L_{m_0},C_1^{j_1},\dots,C_{n-1}^{j_1})-V(L_{m_0},C_1,\dots,C_{n-1})\right|<\frac{\varepsilon}{3n}$$

hold. Finally, inequality (7.1.10) for $j=j_1$ and $m=m_0$ reads

$$
\begin{aligned}
&\left|\int_{\mathbb{S}^{n-1}} f(u)\,dS(C_1^j,\dots,C_{n-1}^j,u)-\int_{\mathbb{S}^{n-1}} f(u)\,dS(C_1,\dots,C_{n-1},u)\right|\le\\
&\le\frac{\varepsilon}{3(\varepsilon+2V(B^n,C_1,\dots,C_{n-1}))}(\varepsilon+2V(B^n,C_1,\dots,C_{n-1}))+n\frac{\varepsilon}{3n}+n\frac{\varepsilon}{3n}=\varepsilon.
\end{aligned}
$$

This proves that $S(C_1^j,\dots,C_{n-1}^j,\cdot)$ converges weakly to $S(C_1,\dots,C_{n-1},\cdot)$.

□

Corollary 7.1.1. *The mixed surface area measure $S(B^n,\dots,B^n,\cdot)$ coincides with the restriction of the $(n-1)$-dimensional Hausdorff measure to $\mathbb{S}^{n-1}$. Consequently, we have the equality*

$$V(C, B^n, \dots, B^n) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_C(u)\, d\mathcal{H}^{n-1}(u) \tag{7.1.11}$$

for any $C \in CC^n$.

Proof. From item **(iv)** of Theorem 7.1.3, we get that $S(B^n, \dots, B^n, \cdot)$ is $\mathrm{O}(n)$-invariant, meaning that

$$S(B^n, \dots, B^n, TA) = S(B^n, \dots, B^n, A),$$

for any $T \in \mathrm{O}(n)$ and every Borel set $A \subseteq \mathbb{S}^{n-1}$. Hence this measure is uniformly distributed, and it follows from Theorem 4.5.1 that there exists a constant $c \in \mathbb{R}$ such that

$$S(B^n, \dots, B^n, \cdot) = c\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}(\cdot).$$

To prove that $c = 1$, we calculate

$$\begin{aligned} S(B^n, \dots, B^n, \mathbb{S}^{n-1}) = \int_{\mathbb{S}^{n-1}} dS(B^n, \dots, B^n, u) = \int_{\mathbb{S}^{n-1}} h_{B^n}(u)\, dS(B^n, \dots, B^n, u) = \\ = nV(B^n, \dots, B^n) = n\mathrm{vol}_n(B^n) = \mathcal{H}^{n-1}(\mathbb{S}^{n-1}), \end{aligned}$$

where we remind the reader that the last equality is discussed in Section 4.1. Now equality (7.1.11) follows immediately.

□

7.2 j-th order surface area measures

Let $C \in CC^n$ and $j \in \{0, \dots, n-1\}$. We define the *j-th order surface area measure of C* to be the mixed surface area measure

$$S_j(C, \cdot) := S\big(\underbrace{C, \dots, C}_{j}, \underbrace{B^n \dots, B^n}_{n-j-1}, \cdot\big).$$

Observe that from Corollary 7.1.1, we have that $S_0(C, \cdot) = \mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ for any $C \in CC^n$. Moreover,

$$S_{n-1}(C, \mathbb{S}^{n-1}) = \int_{\mathbb{S}^{n-1}} dS(C, \dots, C, u) = nV(C, \dots, C, B^n) = nW_1(C), \tag{7.2.1}$$

and hence $S_{n-1}(C, \mathbb{S}^{n-1})$ is the total surface area of C. More generally, we can express the j-th quermassintegral in terms of the j-th order total surface area measure as

$$\begin{aligned} W_j(C) = V\big(C[n-j], B^n[j]\big) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{B^n}(u)\, dS\big(\underbrace{C, \dots, C}_{n-j}, \underbrace{B^n, \dots, B^n}_{j-1}, u\big) = \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} dS_{n-j}(C, u) = \frac{1}{n} S_{n-j}(C, \mathbb{S}^{n-1}), \end{aligned} \tag{7.2.2}$$

for $j = 1, \dots, n$. It is also immediate to notice that

$$W_j(K) = V\big(K, K[n-j-1], B^n[j]\big) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(u)\, dS_{n-j-1}(K, u) \tag{7.2.3}$$

for $j = 0, \dots, n-1$. In particular, since $W_0(K)$ is the volume of K, we have the following formula:

$$\mathrm{vol}_n(K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(u)\, dS_{n-1}(K, u). \tag{7.2.4}$$

Moreover, we have the following Steiner formula for the $(n-1)$-th order surface area measure.

Proposition 7.2.1. *For any $C \in CC^n$ and every $\varepsilon \geq 0$, we have*

$$S_{n-1}(C + \varepsilon B^n, \cdot) = \sum_{j=0}^{n-1} \varepsilon^{n-j-1} \binom{n-1}{j} S_j(C, \cdot).$$

Proof. First observe that the linearity of mixed volumes gives

$$V(L, C + \varepsilon B^n, \dots, C + \varepsilon B^n) = \sum_{j=0}^{n-1} \binom{n-1}{j} V\big(L, \underbrace{C, \dots, C}_{j}, \underbrace{\varepsilon B^n, \dots, \varepsilon B^n}_{n-j-1}\big) =$$

$$= \sum_{j=0}^{n-1} \varepsilon^{n-j-1} \binom{n-1}{j} V\big(L, \underbrace{C, \dots, C}_{j}, \underbrace{B^n, \dots, B^n}_{n-j-1}\big),$$

for every $C, L \in CC^n$ and any $\varepsilon \geq 0$. Now let ν be the Borel measure on $\mathbb{S}^{n-1}$ given as

$$\nu(\cdot) = \sum_{j=0}^{n-1} \varepsilon^{n-j-1} \binom{n-1}{j} S_j(C, \cdot).$$

For any $L \in CC^n$, we have

$$\frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, d\nu(u) = \sum_{j=0}^{n} \varepsilon^{n-j-1} \binom{n-1}{j} \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS_j(C, u) =$$

$$= \sum_{j=0}^{n-1} \varepsilon^{n-j-1} \binom{n-1}{j} V\big(L, \underbrace{C, \dots, C}_{j}, \underbrace{B^n, \dots, B^n}_{n-j-1}\big) = V(L, C + \varepsilon B^n, \dots, C + \varepsilon B^n).$$

On the other hand, the measure $S_{n-1}(C + \varepsilon B^n, \cdot)$ is the unique finite Borel measure on $\mathbb{S}^{n-1}$ such that

$$V(L, C + \varepsilon B^n, \dots, C + \varepsilon B^n) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS_{n-1}(C + \varepsilon B^n, u)$$

for all $L \in CC^n$. It follows that $S_{n-1}(C + \varepsilon B^n, \cdot) = \nu(\cdot)$, and the proof is complete. □

A lot of interesting questions naturally arise from the definition of the j-th order surface areas of convex bodies. For example, one may ask what properties a given Borel measure on $\mathbb{S}^{n-1}$ should have to be equal to $S_j(C, \cdot)$ for some $C \in CC^n$ and some $j \in \{1, \dots, n\}$ (the case where $j = n-1$ is known as the *Minkowski existence problem*, and we investigate it in Section 7.3). Or, as another example, we may ask whether $S_j(K, \cdot) = S_j(L, \cdot)$ implies that K and L are equal (up to translation, of course). For this question, we deal with the case where $j = n-1$ below, and the general case is studied later in Corollary 8.4.3. The interested reader may find a very complete exposition about this topic in [2167, Chapter 8].

Theorem 7.2.1. *Let $K, L \in \mathcal{K}^n$. Then*

$$S_{n-1}(K, \cdot) = S_{n-1}(L, \cdot)$$

if and only if K and L are translates.

Proof. If K and L are translates, then the equality comes immediately from item **(iii)** of Theorem 7.1.3. Conversely, if K and L are convex bodies such that $S_{n-1}(K, \cdot) = S_{n-1}(L, \cdot)$, then

$$V(L, K, \dots, K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS_{n-1}(K, u) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS_{n-1}(L, u) = \\ = V(L, \dots, L) = \mathrm{vol}_n(L).$$

Analogously, we get that $V(K, L, \dots, L) = \mathrm{vol}_n(K)$. Thus, using the 1st Minkowski inequality (Theorem 6.3.1) twice, we get

$$\mathrm{vol}_n(L)^n = V(L, K, \dots, K)^n \geq \mathrm{vol}_n(K)^{n-1} \mathrm{vol}_n(L) \text{ and}$$
$$\mathrm{vol}_n(K)^n = V(K, L, \dots, L)^n \geq \mathrm{vol}_n(L)^{n-1} \mathrm{vol}_n(K).$$

These inequalities lead immediately to

$$\mathrm{vol}_n(K) = \mathrm{vol}_n(L) = V(K, L, \dots, L) = V(L, K, \dots, K),$$

and hence we have the equality case in the 1st Minkowski inequality. It follows that K and L are homothets. Since they have the same volume, we obtain that K and L are translates.

□

The surface area measure $S_{n-1}(K, \cdot)$ can be used to calculate the volume of the projections of K onto hyperplanes. In particular, this will lead to a new proof of the Cauchy's surface area formula (Theorem 6.5.1).

Theorem 7.2.2. *For any $K \in CC^n$ and every $u \in \mathbb{S}^{n-1}$, we have*

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle x, u \rangle|\, dS_{n-1}(K, x). \tag{7.2.5}$$

In particular, the projection function $\mathbb{S}^{n-1} \ni u \mapsto \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)$ *is continuous.*

Proof. Our first step is to calculate the volume of the body $K + \mathrm{seg}[-u, u]$. For each $z \in \mathrm{proj}_{u^\perp}(K)$, let $\lambda_1(z)$ and $\lambda_2(z)$ be the real numbers such that $z + \lambda_i(z)u \in \partial K$, and with $\lambda_1(z) \leq \lambda_2(z)$. Geometrically, these numbers give the intersection of the line $z + \mathbb{R}u$ with the boundary of K. Clearly, the intersections of $z + \mathbb{R}u$ with $K + \mathrm{seg}[-u, u]$ are represented by $\lambda_1(z) - 1$ and $\lambda_2(z) + 1$. Identifying $\mathbb{R}^{n-1} \times \mathbb{R}$ with $u^\perp \times \mathrm{span}\{u\}$, we have that $(z, t) \in K + \mathrm{seg}[-u, u]$ if and only if $z \in \mathrm{proj}_{u^\perp}(K)$ and $\lambda_1(z) - 1 \leq t \leq \lambda_2(z) + 1$. Therefore, using Cavalieri's principle (4.2.2), we get

$$\begin{aligned}\mathrm{vol}_n\big(K+\mathrm{seg}[-u,u]\big) &= \int_{\mathbb{R}^n} 1_{K+\mathrm{seg}[-u,u]}\,d\mathcal{L}^n = \int_{\mathrm{proj}_{u^\perp}(K)}\int_{\lambda_1(z)-1}^{\lambda_2(z)+1} d\mathcal{L}(t)\,d\mathcal{L}^{n-1}(z) = \\ &= \int_{\mathrm{proj}_{u^\perp}(K)}\left(2+\int_{\lambda_1(z)}^{\lambda_2(z)} d\mathcal{L}(t)\right)d\mathcal{L}^{n-1}(z) = \\ &= 2\int_{\mathrm{proj}_{u^\perp}(K)} d\mathcal{L}^{n-1}(z) + \int_{\mathrm{proj}_{u^\perp}(K)}\int_{\lambda_1(z)}^{\lambda_2(z)} d\mathcal{L}(t)\,d\mathcal{L}^{n-1}(z) = \\ &= 2\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) + \mathrm{vol}_n(K).\end{aligned}$$

On the other hand, our original definition of mixed volumes gives the equality

$$\mathrm{vol}_n\big(K+\mathrm{seg}[-u,u]\big) = \sum_{i=0}^{n}\binom{n}{i}V\big(\underbrace{K,\dots,K}_{n-i},\underbrace{\mathrm{seg}[-u,u],\dots,\mathrm{seg}[-u,u]}_{i}\big).$$

In view of Proposition 6.2.3, we have that

$$V\big(\underbrace{K,\dots,K}_{n-i},\underbrace{\mathrm{seg}[-u,u],\dots,\mathrm{seg}[-u,u]}_{i}\big) = 0$$

whenever $i = 2,\dots,n$. Then we may re-write the equality above as

$$\begin{aligned}\mathrm{vol}_n\big(K+\mathrm{seg}[-u,u]\big) &= V(K,\dots,K) + nV\big(K,\dots,K,\mathrm{seg}[-u,u]\big) = \\ &= \mathrm{vol}_n(K) + nV\big(K,\dots,K,\mathrm{seg}[-u,u]\big).\end{aligned}$$

Consequently,

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \frac{n}{2}V\big(\mathrm{seg}[-u,u],K,\dots,K\big) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} h_{\mathrm{seg}[-u,u]}(x)\,dS_{n-1}(K,x),$$

and the desired follows from the simple observation that the support function of $\mathrm{seg}[-u,u]$ is the function $|\langle u,\cdot\rangle|$ (see Corollary 2.3.2).

□

Remark 7.2.1. The equality

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \frac{n}{2}V\big(\mathrm{seg}[-u,u],K,\dots,K\big), \tag{7.2.6}$$

which we proved above, can also be obtained straightforwardly as a particular case of Theorem 6.2.4 (see Exercise 6.18). The general case of this theorem leads to an extension of (7.2.5). See Exercise 7.13.

Remark 7.2.2. As it was announced, formula (7.2.5) leads to an alternative proof of Cauchy's surface area formula. This is as follows:

$$\begin{aligned}&\int_{\mathbb{S}^{n-1}}\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)\,d\mathcal{H}^{n-1}(u) = \int_{\mathbb{S}^{n-1}}\frac{1}{2}\int_{\mathbb{S}^{n-1}}|\langle x,u\rangle|\,dS_{n-1}(K,x)\,d\mathcal{H}^{n-1}(u) = \\ &= \int_{\mathbb{S}^{n-1}}\left(\frac{1}{2}\int_{\mathbb{S}^{n-1}}|\langle x,u\rangle|\,d\mathcal{H}^{n-1}(u)\right)dS_{n-1}(K,x) = \int_{\mathbb{S}^{n-1}}\kappa_{n-1}\,dS_{n-1}(K,x) = \kappa_{n-1}nW_1(K),\end{aligned}$$

where we used equality (6.5.8) for $v = x$, of course.

Our next task is to formalize the claims that we have made earlier in Remark 6.2.3. For the convenience of the reader, we recall that the *reverse normal image* of a Borel set $E \subseteq \mathbb{S}^{n-1}$ on a given convex body K is the set

$$\eta_K(E) := \{x \in \partial K : \exists\, u \in E \text{ such that } u \text{ is an outer normal of } K \text{ at } x\},$$

and we also note that $\eta_K(\emptyset) = \emptyset$. Figure 7.2.1 illustrate this definition (in the two-dimensional case). We define the *surface area measure* of K to be the function μ_K defined as

$$\mu_K(E) = \mathcal{H}^{n-1}(\eta_K(E))$$

for $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$, where $\mathcal{B}_{\mathbb{S}^{n-1}}$ is the usual Borel σ-algebra of $\mathbb{S}^{n-1}$ (see Appendix A). Since $\mathcal{H}^{n-1}$ is an outer measure defined in $\mathbb{R}^n$, we have that μ_K is well-defined. This is actually a finite map (meaning that there is no set $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$ such that $\mu_K(E) = \infty$) because $0 < \mathcal{H}^{n-1}(\partial K) < \infty$ (this is proved in Theorem 4.4.1).

However, it is not immediate that μ_K is a measure on $\mathbb{S}^{n-1}$, because it is not obvious that it is σ-additive. We will prove this next, and we will also prove that the surface area measure μ_K of any convex body K coincides with its $(n-1)$-th surface area measure $S_{n-1}(K, \cdot)$. We emphasize that everything that we do in this section from now on relies heavily on measure-theoretic arguments, and the proofs can be omitted in a first reading (or in the case where the reader is not interested in technicalities). However, the results are very important and have to be kept in mind.

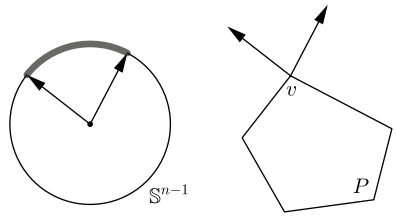


Fig. 7.2.1 The vertex v is the reverse normal image on ∂P of the arc highlighted in the boundary of $\mathbb{S}^{n-1}$.

Theorem 7.2.3. *For any $K \in \mathcal{K}^n$, the surface area measure μ_K is a Borel measure on $\mathbb{S}^{n-1}$ and*

$$\mu_K(\cdot) = S_{n-1}(K, \cdot) \tag{7.2.7}$$

over $\mathcal{B}_{\mathbb{S}^{n-1}}$.

Remark 7.2.3. In view of the definition of $S_{n-1}(K, \cdot)$, we have

$$V(L, K, \ldots, K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS_{n-1}(K, u) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu_K(u),$$

and hence this gives the announced proof of (6.2.6).

Remark 7.2.4. The case where $K = P$ is a proper polytope is easy to prove. Indeed, since every face of a polytope is exposed, and because any polytope has finitely many faces, we have that for every $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$ the reverse normal image $\eta_P(E)$ is a finite union of faces of P. Being a (finite) union of compact sets, $\eta_P(E)$ is a Borel set. If $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$, then recalling that the only proper faces of P which have non-zero $(n-1)$-dimensional measure are the facets, we have

$$\mu_P(E) = \mathcal{H}^{n-1}(\eta_P(E)) = \sum_{u_F \in E} \mathrm{vol}_{n-1}(F). \tag{7.2.8}$$

If $E = \bigcup_{j\in\mathbb{N}} E_j$ is a disjoint union, then the last sum can also be seen as taken over the facets whose respective outer normal vectors belong to some E_j. Since these sets are mutually disjoint, the outer normal vector of a given facet F can belong to at most one of the E_j's, and hence we also have

$$\sum_{j=1}^{\infty} \mu_P(E_j) = \sum_{u_F \in E_j \text{ for some } j} \mathrm{vol}_{n-1}(F) = \sum_{u_F \in E} \mathrm{vol}_{n-1}(F) = \mu_P(E).$$

This shows that μ_P is σ-additive. Therefore, μ_P is indeed a Borel measure on $\mathbb{S}^{n-1}$. Also, from (7.1.9) and (7.2.8), we get

$$S_{n-1}(P, E) = \sum_{u_F \in E} \mathrm{vol}_{n-1}(F) = \mu_P(E)$$

for any $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$. This settles (7.2.7) for proper polytopes.

Now that we have Theorem 7.2.3 for polytopes, we jump to the general case. For that sake, we will need two other results, which are Propositions 7.2.2 and 7.2.3 (we give a complete proof for the first one and a reference for the second one). These propositions are fairly technical and heavily use measure theory, so the reader which is not interested in measure theoretical details may only read their enunciates and skip to the proof of the theorem on page 453.

Proposition 7.2.2. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For any $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$, the reverse normal image $\eta_K(E)$ is $\mathcal{H}^{n-1}$-measurable. As a consequence, μ_K is a Borel measure on $\mathbb{S}^{n-1}$.*

Proof. Define the collection

$$\Sigma := \left\{E \subseteq \mathbb{S}^{n-1} : \eta_K(E) \text{ is } \mathcal{H}^{n-1} - \text{measurable}\right\}.$$

We want to prove that Σ contains the Borel σ-algebra of $\mathbb{S}^{n-1}$. Our strategy is to show that Σ is a σ-algebra of $\mathbb{S}^{n-1}$ which contains all open sets of $\mathbb{S}^{n-1}$.

We prove that Σ contains all compact subsets of $\mathbb{S}^{n-1}$. For that sake, let $E \subseteq \mathbb{S}^{n-1}$ be compact and let $(x_j)_{j\in\mathbb{N}}$ be a sequence in $\eta_K(E)$ converging to a point $x \in \partial K$ (recall that ∂K is compact). For each $j \in \mathbb{N}$, choose $u_j \in E$ such that u_j is an outer normal of K at x_j. Since E is compact, passing to a subsequence if necessary we may assume that $u_j \to u \in E$ as $j \to \infty$. Notice that the inequality

$$\langle z, u_j\rangle \leq \langle u_j, x_j\rangle$$

holds for each $j \in \mathbb{N}$ and every $z \in K$, because of the definition of outer normals. Fixing an arbitrary $z \in K$ and making $j \to \infty$ in the inequality above, we get

$$\langle z, u\rangle \leq \langle u, x\rangle,$$

and hence u is an outer normal of K at x. Since $u \in E$, we get that $x \in \eta_K(E)$, and this proves that $\eta_K(E)$ is closed, and hence compact. We know from Proposition A.4.2 that $\mathcal{H}^{n-1}$ is a Borel regular outer measure, and this gives that the Borel sets of $\mathbb{R}^n$ are $\mathcal{H}^{n-1}$-measurable. Hence, $\eta_K(E)$ is $\mathcal{H}^{n-1}$-measurable whenever $E \subseteq \mathbb{S}^{n-1}$ is compact, and this shows that Σ contains all compact subsets of $\mathbb{S}^{n-1}$.

Now we show that Σ is indeed a σ-algebra. First, it is clear that $\emptyset, \mathbb{S}^{n-1} \in \Sigma$, because $\eta_K(\emptyset) = \emptyset$ and $\eta_K(\mathbb{S}^{n-1}) = \partial K$ are $\mathcal{H}^{n-1}$-measurable. To prove that Σ is closed under complementation, notice that for any non-empty proper subset $E \in \Sigma$, we have the equality

$$\eta_K(\mathbb{S}^{n-1} \setminus E) = \big(\partial K \setminus \eta_K(E)\big) \cup \big(\eta_K(E) \cap \eta_K(\mathbb{S}^{n-1} \setminus E)\big).$$

The set $\partial K \setminus \eta_K(E)$ is $\mathcal{H}^{n-1}$-measurable, because this is the difference of $\mathcal{H}^{n-1}$-measurable sets (recall that for any outer measure μ the collection of μ-measurable sets is a σ-algebra). On the other hand, we claim that

$$\mathcal{H}^{n-1}\big(\eta_K(E) \cap \eta_K(\mathbb{S}^{n-1} \setminus E)\big) = 0.$$

In fact, any point of $\eta_K(E) \cap \eta_K(\mathbb{S}^{n-1} \setminus E)$ is singular, since it has at least two distinct outer normal vectors (one in E and another in $\mathbb{S}^{n-1} \setminus E$). From Corollary 2.4.2 we get the claim. From Remark A.1.2, it follows that $\eta_K(E) \cap \eta_K(\mathbb{S}^{n-1} \setminus E)$ is $\mathcal{H}^{n-1}$-measurable. Therefore, $\eta_K(\mathbb{S}^{n-1} \setminus E)$ is the union of $\mathcal{H}^{n-1}$-measurable sets, and hence it is measurable. The definition of Σ then gives $\mathbb{S}^{n-1} \setminus E \in \Sigma$.

It remains to prove that Σ is closed under countable union. This follows immediately from the trivial equality

$$\eta_K\left(\bigcup_{j=1}^{\infty} E_j\right) = \bigcup_{j=1}^{\infty} \eta_K(E_j),$$

and if each E_j is a set of Σ, then the right-hand side is a countable union of $\mathcal{H}^{n-1}$-measurable sets, and hence it is also $\mathcal{H}^{n-1}$-measurable.

Hence, Σ is indeed a σ-algebra. Since Σ contains all compact subsets of $\mathbb{S}^{n-1}$, we get that Σ also contains all open subsets of $\mathbb{S}^{n-1}$. In fact, if $A \subseteq \mathbb{S}^{n-1}$ is open, then $\mathbb{S}^{n-1} \setminus A$ is compact, and hence it lies in Σ. Since Σ is closed under complementation, we get that $A \in \Sigma$.

By definition, $\mathcal{B}_{\mathbb{S}^{n-1}}$ is the smallest σ-algebra of $\mathbb{S}^{n-1}$ which contains all open sets of $\mathbb{S}^{n-1}$, and this gives that $\Sigma \supseteq \mathcal{B}_{\mathbb{S}^{n-1}}$. To explain why this gives that μ_K is a Borel measure (in the classical sense) on $\mathbb{S}^{n-1}$, we only need to prove that μ_K is σ-additive when restricted to $\mathcal{B}_{\mathbb{S}^{n-1}}$. For that sake, let $(E_j)_{j\in\mathbb{N}}$ be a countable disjoint sequence of sets in $\mathcal{B}_{\mathbb{S}^{n-1}}$. Their reverse normal images are $\mathcal{H}^{n-1}$-measurable, but may not be mutually disjoint. This is not a problem, as we explain now. Writing, for each $j \in \mathbb{N}$,

$$F_j = \eta_K(E_j) \setminus \bigcup_{m\neq j} \eta_K(E_m),$$

we have that $(F_j)_{j\in\mathbb{N}}$ is a disjoint sequence of sets, and since $\eta_K(E_j)$ is $\mathcal{H}^{n-1}$-measurable, we get

$$\begin{aligned}
\mathcal{H}^{n-1}(\eta_K(E_j)) &= \mathcal{H}^{n-1}(F_j) + \mathcal{H}^{n-1}\left(\eta_K(E_j) \cap \left(\bigcup_{m\neq j} \eta_K(E_m)\right)\right) = \\
&= \mathcal{H}^{n-1}(F_j) + \mathcal{H}^{n-1}\left(\bigcup_{m\neq j} \big(\eta_K(E_j) \cap \eta_K(E_m)\big)\right) \leq \\
&\leq \mathcal{H}^{n-1}(F_j) + \sum_{j\neq m} \mathcal{H}^{n-1}\big(\eta_K(E_j) \cap \eta_K(E_m)\big) = \mathcal{H}^{n-1}(F_j) \leq \mathcal{H}^{n-1}(\eta_K(E_j))
\end{aligned}$$

for each $j \in \mathbb{N}$. Here the first equality comes from (A.1.2), the inequality follows from $\mathcal{H}^{n-1}$ being an outer measure, and the last equality comes from the fact that each intersection between the sets $\eta_K(E_j)$ consists only of singular points, and then has zero measure (as it was explained before). Moreover, it is easy to see that

$$\bigcup_{j=1}^{\infty} F_j = \bigcup_{j=1}^{\infty} \eta_K(E_j).$$

Hence, since any outer measure μ is σ-additive when restricted to (disjoint) μ-measurable sets, we obtain that

$$\mu_K\left(\bigcup_{j=1}^{\infty} E_j\right) = \mathcal{H}^{n-1}\left(\eta_K\left(\bigcup_{j=1}^{\infty} E_j\right)\right) = \mathcal{H}^{n-1}\left(\bigcup_{j=1}^{\infty} \eta_K(E_j)\right) = \mathcal{H}^{n-1}\left(\bigcup_{j=1}^{\infty} F_j\right) =$$
$$= \sum_{j=1}^{\infty} \mathcal{H}^{n-1}(F_j) = \sum_{j=1}^{\infty} \mathcal{H}^{n-1}(\eta_K(E_j)) = \sum_{j=1}^{\infty} \mu_K(E_j),$$

and this concludes the proof.

□

Proposition 7.2.3. *Let $(K_j)_{j\in\mathbb{N}}$ be a sequence of convex bodies converging in the Hausdorff metric to a convex body K. Then the sequence $(\mu_{K_j})_{j\in\mathbb{N}}$ of measures converges weakly to μ_K.*

For a proof, see [1070].

Proof of Theorem 7.2.3. From Proposition 7.2.2, we already know that μ_K is a Borel measure. For a given convex body K, let $(P_j)_{j\in\mathbb{N}}$ be a sequence of proper polytopes converging to K in the Hausdorff metric. Then, since the theorem is true for proper polytopes, we have

$$V(L, P_j, \dots, P_j) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\, dS_{n-1}(P_j, u) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu_{P_j}(u)$$

for any $L \in CC^n$. As $j \to \infty$, the left-hand side above converges to $V(L, K, \dots, K)$ because of the continuity of mixed volumes, whereas Proposition 7.2.3 gives that

$$\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu_{P_j}(u) \to \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu_K(u).$$

It follows that

$$V(L, K, \dots, K) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu_K(u)$$

for every $L \in CC^n$. Since $S_{n-1}(K, \cdot)$ is the unique Borel measure on $\mathbb{S}^{n-1}$ with this property, we get that $\mu_K(\cdot) = S_{n-1}(K, \cdot)$.

□

Remark 7.2.5. Now we know that the surface area μ_K of a convex body K coincides with its $(n-1)$-th mixed surface area. Thus, from now on the reader may keep in mind that every result proved for one of these measures also holds for the other one.

Remark 7.2.6. If $K \subseteq \mathbb{R}^n$ is a smooth convex body, then we may define a map $\xi_K \colon \partial K \to \mathbb{S}^{n-1}$ associating each point $x \in \partial K$ to the (unique) unit outer normal of K at x. This map is known as the *Gauss map* of K. From Theorem 2.3.2 and Corollary 2.4.2, we have that $\mathcal{H}^{n-1}$-almost every boundary point of any given convex body K is smooth. Therefore, in the non-smooth case this definition can be extended to a function defined $\mathcal{H}^{n-1}$-almost everywhere in ∂K: the *Gauss map* of K is the map $u_K \colon \partial K \to \mathbb{S}^{n-1}$ which associates each smooth point $x \in \partial K$ to the unit outer normal of K at x. Notice that by writing $u_K \colon \partial K \to \mathbb{S}^{n-1}$, we are abusing of the notation, since u_K is defined only for the

smooth points of ∂K. As it will be clear below, there is no difference for our purposes in working with maps which are defined only almost everywhere, and this is why we stick to this notation. The reader may notice, however, that we use ξ_K for the Gauss map of a smooth body and u_K for the non-smooth case. This is only a stylistic option. Having said that, for each $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$, we get

$$u_K^{-1}(E) = \eta_K(E) \cap \big\{\text{smooth points of } \partial K\big\}.$$

The set of smooth points of ∂K is $\mathcal{H}^{n-1}$-measurable, because its complement has zero measure. Therefore, since we already proved that $\eta_K(E)$ is also $\mathcal{H}^{n-1}$-measurable, it follows that u_K is a measurable map with respect to the σ-algebra of $\mathcal{H}^{n-1}$-measurable subsets of ∂K and to the Borel σ-algebra of $\mathbb{S}^{n-1}$. Even more, momentaneously representing $\mathrm{Sm}(\partial K) := \big\{\text{smooth points of } \partial K\big\}$, we have

$$\begin{aligned} \mu_K(E) &= \mathcal{H}^{n-1}(\eta_K(E)) = \\ &= \mathcal{H}^{n-1}\big(\eta_K(E) \cap \mathrm{Sm}(\partial K)\big) + \mathcal{H}^{n-1}\big(\eta_K(E) \setminus \mathrm{Sm}(\partial K)\big) = \mathcal{H}^{n-1}(u_K^{-1}(E)), \end{aligned}$$

for every $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$. From this, we get that

$$\mu_K = (u_K)_* \mathcal{H}^{n-1}\big|_{\partial K},$$

where we remind the reader that the terminology $f_*(\mu)$ stands for the usual push-forward measure of μ with respect to f (see Appendix A). Summarizing, we have the following.

Theorem 7.2.4. *The surface area measure μ_K of a convex body $K \subseteq \mathbb{R}^n$ is the push-forward measure of $\mathcal{H}^{n-1}\big|_{\partial K}$ under the Gauss map of K.*

In case of any doubt regarding push-forward measures given by maps defined only almost everywhere, we refer to the discussion at the end of Section A.2. Moving forward, from (A.2.4), we have

$$\int_{\mathbb{S}^{n-1}} f(u)\, d\mu_K(u) = \int_{\partial K} f(u_K(z))\, d\mathcal{H}^{n-1}(z) \tag{7.2.9}$$

for every measurable function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$. To be completely formal, we have to notice that the integration in the right-hand side is defined with respect to the σ-algebra of $\mathcal{H}^{n-1}$-measurable subsets of ∂K (denote it by Σ, say). However, since $\mathcal{H}^{n-1}$ is a Borel regular outer measure when restricted to the Borel σ-algebra of ∂K, it comes from Proposition A.2.6 that we can replace Σ by the Borel σ-algebra of ∂K, and the mentioned integral can be seen as an integral with respect to a Borel σ-algebra.

This, together with the fact that $\mu_K(\cdot)$ coincides with $S_{n-1}(K, \cdot)$, provides new formulas for the volume of a convex body. We state and prove them now.

Theorem 7.2.5. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then*

$$\mathrm{vol}_n(K) = \frac{1}{n} \int_{\partial K} h_K(u_K(z))\, d\mathcal{H}^{n-1}(z).$$

Even more, if K has the origin as an interior point, then we can derive a sort of analogue of the Gaussian integral in which B^n is replaced by K:

$$\mathrm{vol}_n(K) = \frac{1}{n!} \int_{\mathbb{R}^n} e^{-||x||_K}\, dx,$$

where dx shows that the integration is meant with respect to the Lebesgue measure, of course, and we also recall that $||\cdot||_K$ stands for the gauge function of K.

Proof. For the first equality, we just have to use (7.2.9), the definition of $S_{n-1}(K,\cdot)$ and the fact that this measure coincides with μ_K. We compute

$$\frac{1}{n}\int_{\partial K} h_K(u_K(z))\,d\mathcal{H}^{n-1}(z) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\,d\mu_K(u) =$$
$$= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\,dS_{n-1}(K,u) = V(K,\dots,K) = \mathrm{vol}_n(K),$$

where we also observe that the third equality (which follows immediately from the definition of S_{n-1}) is the same as in (7.2.4).

To prove the other equality, we have to use the equality that we just have proved and Proposition A.6.1, which is a consequence of the co-area formula. We also need to have in mind the definition of the Gamma function, as well as the fact that $\Gamma(n) = (n-1)!$ for every $n \in \mathbb{N}$. We compute

$$\frac{1}{n!}\int_{\mathbb{R}^n} e^{-||x||_K}\,dx = \int_0^{+\infty}\int_{\partial K} e^{-||tz||_K} t^{n-1} h_K(u_K(z))\,d\mathcal{H}^{n-1}(z)dt =$$
$$= \frac{1}{n!}\int_0^{+\infty} e^{-t}t^{n-1}\,dt \int_{\partial K} h_K(u_K(z))\,d\mathcal{H}^{n-1}(z) =$$
$$= \frac{\Gamma(n)}{n!}\int_{\partial K} h_K(u_K(z))\,d\mathcal{H}^{n-1}(z) = \frac{1}{n}\int_{\partial K} h_K(u_K(z))\,d\mathcal{H}^{n-1}(z) = \mathrm{vol}_n(K),$$

and the proof is complete. The separation of the integrals in the second equality is justified by Fubini's theorem (Theorem A.5.3).

□

We introduce some terminology. A Borel measure μ on $\mathbb{S}^{n-1}$ is said to be *even* if $\mu(E) = \mu(-E)$ for every set $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$. In other words, a measure μ on $\mathbb{S}^{n-1}$ is even when

$$(-\mathrm{Id}_{\mathbb{R}^n})_*\mu = \mu,$$

where the identity map $\mathrm{Id}_{\mathbb{R}^n}$ is considered as restricted to $\mathbb{S}^{n-1}$, of course. It is not difficult to check that the weak limit of a sequence of even measures is an even measure (this follows from Proposition A.2.4).

Let $K \subseteq \mathbb{R}^n$ be a convex body. From the fact that $S_{n-1}(K,\cdot) = \mu_K(\cdot)$, we notice that $S_{n-1}(K,\cdot)$ is even if and only if the $(n-1)$-dimensional Hausdorff measure is the same for measurable subsets of ∂K having respectively "opposite" unit outer normal vectors. Hence, we suspect that the evenness of surface area measures might be somehow related to central symmetry. Next, we elucidate this relation as a consequence of the Theorem 7.2.1.

Proposition 7.2.4. *Let $K \subseteq \mathbb{R}^n$ be a convex body. The measure $S_{n-1}(K,\cdot)$ is even if and only if K is centrally symmetric.*

Proof. Let $K \subseteq \mathbb{R}^n$ be any convex body. Observe that $S_{n-1}(K,E) = S_{n-1}(-K,-E)$ for every Borel subset $E \subseteq \mathbb{S}^{n-1}$. There are two ways of seeing this: first, we may use the equality $S_{n-1}(K,\cdot) = \mu_K$ together with the fact that u is an outer normal of K at $x \in \partial K$ if and only if $-u$ is an outer normal of $-K$ at $-x \in \partial(-K)$. Thus, for any Borel subset $E \subseteq \mathbb{S}^{n-1}$, we have that

$$-\eta_K(E) = \eta_{-K}(-E).$$

Thus, the sets $\eta_K(E)$ and $\eta_{-K}(-E)$ have the same $(n-1)$-dimensional Hausdorff measure. It follows that $\mu_K(E) = \mu_{-K}(-E)$.

The second way of deriving equality does not rely on μ_K. For every $L \in CC^n$, we may write

$$\int_{\mathbb{S}^{n-1}} h_L(u)\, dS_{n-1}(-K,u) = nV(-K[n-1],L) = nV(K[n-1],-L) =$$
$$= \int_{\mathbb{S}^{n-1}} h_{-L}(u)\, dS_{n-1}(K,u) = \int_{\mathbb{S}^{n-1}} h_L(-u)\, dS_{n-1}(K,u) =$$
$$= \int_{\mathbb{S}^{n-1}} h_L(v)\, d(-\mathrm{Id}_* S_{n-1}(K,v)),$$

where in the last equality we did a simple change of variables; see, e.g., (A.2.4). By linearity, the equality above also holds if we replace h_L by any function having the form $h_L - h_M$, for $L, M \in CC^n$, and by approximations, we can go further and replace h_L by any continuous functions. By Proposition A.2.4 this gives that

$$S_{n-1}(-K,\cdot) = -\mathrm{Id}_* S_{n-1}(K,\cdot).$$

This means that $S_{n-1}(K,E) = S_{n-1}(-K,-E)$ for any Borel subset $E \subseteq \mathbb{S}^{n-1}$.

If K is centrally symmetric, then translation-invariance gives that $S_{n-1}(K,\cdot) = S_{n-1}(-K,\cdot)$. Thus, for any Borel subset $E \subseteq \mathbb{S}^{n-1}$, we have that

$$S_{n-1}(K,-E) = S_{n-1}(-K,E) = S_{n-1}(K,E),$$

and hence $S_{n-1}(K,\cdot)$ is even.

Conversely, if $S_{n-1}(K,\cdot)$ is even, then $S_{n-1}(K,\cdot) = S_{n-1}(-K,\cdot)$, and from Theorem 7.2.1, we get that K and $-K$ are translates. This gives that $K = -K + x$ for some $x \in \mathbb{R}^n$, and hence

$$K - \frac{x}{2} = -K + \frac{x}{2} = -\left(K - \frac{x}{2}\right).$$

In other words, K is centrally symmetric with respect to $x/2$.

□

7.3 Minkowski's existence theorem

The objective of this section is to answer the following question, known as the *Minkowski problem*:

Question 1. Let μ be a Borel measure on $\mathbb{S}^{n-1}$. What properties must μ have that we may guarantee that it is the surface area measure of some convex body of $\mathbb{R}^n$?

We remind the reader that the surface area measure μ_K of a convex body $K \subseteq \mathbb{R}^n$ was defined in Remark 6.2.3 and investigated in Section 7.2, and that we always have $\mu_K = S_{n-1}(K,\cdot)$; see Theorem 7.2.3 for proof of this equality.

The answer to this question is known as the *Minkowski existence theorem*. In rough terms, a Borel measure μ on $\mathbb{S}^{n-1}$ is the surface area measure of a convex body if and only if it is finite, has its *centroid at the origin*, and is not concentrated on any *great subsphere* of $\mathbb{S}^{n-1}$ (the precise formal meanings of these concepts will be given later). Of course, it is natural to start by investigating some properties that any measure of the form $S_{n-1}(K,\cdot)$ has.

Lemma 7.3.1. *For any convex body $K \subseteq \mathbb{R}^n$, we have*

$$\int_{\mathbb{S}^{n-1}} u\, dS_{n-1}(K,u) = 0_{\mathbb{R}^n},$$

where the integration is made in each coordinate, as usual.

Remark 7.3.1. For a given finite Borel measure μ on the sphere, the point

$$c(\mu) := \int_{\mathbb{S}^{n-1}} u \, d\mu$$

is the *centroid* of μ.

Proof. Letting $\{e_1, \ldots, e_n\}$ be the canonical basis of $\mathbb{R}^n$ and writing $u = \langle u, e_1\rangle e_1 + \ldots + \langle u, e_n\rangle e_n$, we have

$$\int_{\mathbb{S}^{n-1}} u \, dS_{n-1}(K, u) = \sum_{j=1}^{n} \left(\int_{\mathbb{S}^{n-1}} \langle u, e_j\rangle \, dS_{n-1}(K, u) \right) e_j =$$

$$= \sum_{j=1}^{n} \left(\int_{\mathbb{S}^{n-1}} h_{\{e_j\}}(u) \, dS_{n-1}(K, u) \right) e_j = \sum_{j=1}^{n} nV(\{e_j\}, K, \ldots, K)e_j = 0_{\mathbb{R}^n},$$

where the last equality follows directly from Proposition 6.2.3.

□

A *great subsphere* of $\mathbb{S}^{n-1}$ is an intersection of $\mathbb{S}^{n-1}$ with a hyperplane passing through the origin, that is, a set having the form $\mathbb{S}^{n-1} \cap v^{\perp}$ for some $v \in \mathbb{S}^{n-1}$. We say that a finite and non-zero Borel measure μ on $\mathbb{S}^{n-1}$ is *concentrated on a great subsphere* if there exists a great subsphere $S = \mathbb{S}^{n-1} \cap v^{\perp}$ such that $\mu(\mathbb{S}^{n-1} \setminus S) = 0$. Next, we give a useful characterization of measures concentrated on great subspheres.

Lemma 7.3.2. *Let μ be a finite and non-zero Borel measure on $\mathbb{S}^{n-1}$. Then μ is concentrated on the great subsphere $S = \mathbb{S}^{n-1} \cap v^{\perp}$ if and only if*

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, d\mu(u) = 0.$$

Proof. Assume that μ is concentrated on $S = \mathbb{S}^{n-1} \cap v^{\perp}$. Thus, $\mu(\mathbb{S}^{n-1} \setminus S) = 0$. Writing $\mathbb{S}^{n-1}$ as the disjoint union of S and $\mathbb{S}^{n-1} \setminus S$, we arrive at

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, d\mu(u) = \int_{\mathbb{S}^{n-1}\setminus S} |\langle u, v\rangle| \, d\mu(u) + \int_{S} |\langle u, v\rangle| \, d\mu(u) = \int_{S} |\langle u, v\rangle| \, d\mu(u) = 0,$$

where the second equality comes directly from the fact that $\mathbb{S}^{n-1} \setminus S$ has zero measure, and the last equality follows from the fact that $S \subseteq v^{\perp}$, and hence $\langle u, v\rangle = 0$ for every $u \in S$.

For the converse, assume that the equality

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, d\mu(u) = 0$$

holds for some $v \in \mathbb{S}^{n-1}$. Let $S := \mathbb{S}^{n-1} \cap v^{\perp}$ and, for any number $m \in \mathbb{N}$, consider the "slab"

$$S(v, m) := \mathbb{S}^{n-1} \cap \operatorname{int}(\mathbf{h}^{-}(v, 1/m)) \cap \operatorname{int}(\mathbf{h}^{+}(v, -1/m)).$$

This can be described as the set of points $u \in \mathbb{S}^{n-1}$ such that $-1/m < \langle u, v\rangle < 1/m$. It follows that $S(v, m)$ is a sequence of open subsets of $\mathbb{S}^{n-1}$ such that $S(v, m+1) \subseteq S(v, m)$ for each $m \in \mathbb{N}$, and it is clear that

$$\mathbb{S}^{n-1} \setminus S = \bigcup_{m=1}^{\infty} \big(\mathbb{S}^{n-1} \setminus S(v,m)\big).$$

Since μ is a finite Borel measure, item **(i)** of Lemma A.1.1 gives that

$$\mu(\mathbb{S}^{n-1} \setminus S) = \lim_{m\to\infty} \mu\big(\mathbb{S}^{n-1} \setminus S(v,m)\big).$$

If μ is not concentrated on S, then the left-hand side of the equality above is positive, implying that there exists $m_0 \in \mathbb{N}$ such that

$$\mu\big(\mathbb{S}^{n-1} \setminus S(v,m_0)\big) > 0.$$

On the other hand, from the definition of $S(v,m_0)$, it follows that $|\langle u,v\rangle| \geq 1/m_0$ for every $u \in \mathbb{S}^{n-1} \setminus S(v,m_0)$. Since we clearly have that $\mathbb{S}^{n-1} \setminus S(v,m_0) \subseteq \mathbb{S}^{n-1} \setminus S$, and remembering that $\langle u,v\rangle = 0$ whenever $u \in S$, we get

$$\begin{aligned} 0 = \int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\, d\mu(u) = \int_{\mathbb{S}^{n-1}\setminus S} |\langle u,v\rangle|\, d\mu(u) \geq \int_{\mathbb{S}^{n-1}\setminus S(v,m_0)} |\langle u,v\rangle|\, d\mu(u) \geq \\ \geq \int_{\mathbb{S}^{n-1}\setminus S(v,m_0)} \frac{1}{m_0}\, d\mu(u) = \frac{\mu\big(\mathbb{S}^{n-1} \setminus S(v,m_0)\big)}{m_0} > 0, \end{aligned}$$

and this contradiction gives that $\mu(\mathbb{S}^{n-1} \setminus S) = 0$. Hence, μ is concentrated on S, and the proof is complete.

□

Remark 7.3.2. An equivalent definition follows immediately: a finite non-zero Borel measure on $\mathbb{S}^{n-1}$ is *not* concentrated on a great subsphere if and only if

$$\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\, d\mu(u) > 0$$

for every $v \in \mathbb{S}^{n-1}$ (or, equivalently, for every non-zero vector $v \in \mathbb{R}^n$).

Proposition 7.3.1. *The surface area measure of a convex body $K \subseteq \mathbb{R}^n$ is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$.*

Proof. Let $K \subseteq \mathbb{R}^n$ be an arbitrarily given convex body. All we have to prove is that

$$\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\, dS_{n-1}(K,u) > 0$$

for every $v \in \mathbb{S}^{n-1}$. To do that, we simply recall that $|\langle u,v\rangle| = h_{\mathrm{seg}[-v,v]}(u)$ and use the of definition $S_{n-1}(K,\cdot)$ to write

$$\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\, dS_{n-1}(K,u) = \int_{\mathbb{S}^{n-1}} h_{\mathrm{seg}[-v,v]}(u)\, dS_{n-1}(K,u) = nV\big(\mathrm{seg}[-v,v],K,\ldots,K\big),$$

and from Proposition 6.2.3, we get that the latter is a positive number.

□

Summarizing from Lemma 7.3.1 and Proposition 7.3.1, we have that the surface area measure of any convex body has its centroid at the origin and is not concentrated on a great subsphere. The

Minkowski existence theorem states that these two properties are sufficient to guarantee that a finite Borel measure on $\mathbb{S}^{n-1}$ is the surface area measure of some convex body.

Theorem 7.3.1 (Minkowski's existence theorem)**.** *A finite non-zero Borel measure on $\mathbb{S}^{n-1}$ is the surface area measure of a convex body of $\mathbb{R}^n$ if and only if it has its centroid at the origin and it is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$.*

Remark 7.3.3 (Uniqueness)**.** The body is uniquely determined in the following sense: if $\mu = S_{n-1}(K, \cdot)$ for some convex body $K \subseteq \mathbb{R}^n$, then $\mu = S_{n-1}(L, \cdot)$ for another convex body $L \subseteq \mathbb{R}^n$ if and only if L is a translate of K. This is straightforward due to Theorem 7.2.1. It is also worth having in mind that, somewhat conversely, if μ is the surface area measure of a convex body K, then μ is also the surface area measure of any translate of K.

As it was mentioned above, we already know that the surface area measure of any convex body has centroid at the origin and is not concentrated on a great subsphere. Therefore, the novelty here is that for any finite non-zero Borel measure μ on $\mathbb{S}^{n-1}$ such that

$$\int_{\mathbb{S}^{n-1}} u \, d\mu(u) = 0_{\mathbb{R}^n}$$

and

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, d\mu(u) > 0$$

for every $v \in \mathbb{S}^{n-1}$ there exists a convex body $K \subseteq \mathbb{R}^n$ such that $\mu = S_{n-1}(K, \cdot)$. To prove Theorem 7.3.1, we follow [1263] and [2167], and consider first polytopes. Then we obtain the general case by approximations.

Let $P \subseteq \mathbb{R}^n$ be a proper polytope whose facets and corresponding unit outer normal vectors are $F_1, \ldots, F_m$ and $u_1, \ldots, u_m$, respectively. From (7.1.2), we have that the surface area measure of P is given as

$$S_{n-1}(P, \cdot) = \sum_{j=1}^{m} \mathrm{vol}_{n-1}(F_j)\delta_{u_j}. \tag{7.3.1}$$

That is, the surface area measure of a polytope is given by a positive combination of the Dirac delta measures concentrated in the unit outer normal vectors of its facets. The "weights" are the respective $(n-1)$-dimensional volumes of the facets. From now on, a Borel measure μ on $\mathbb{S}^{n-1}$ is said to be *discrete* if we may write

$$\mu = \sum_{j=1}^{m} \alpha_j \delta_{u_j}$$

for some natural number m, some pairwise distinct $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ and some real numbers $\alpha_1, \ldots, \alpha_m > 0$. It is clear that any discrete measure is finite (in fact, $\mu(\mathbb{S}^{n-1})$ is just the sum of the α_j's).

We may expect that Minkowski's problem is easier to solve when restricted to polytopes, since, in this case, one only has to look at the discrete measures of $\mathbb{S}^{n-1}$. For discrete measures, we may express the centroid and concentration properties described above as follows.

Lemma 7.3.3. *Assume that $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ are pairwise distinct vectors and let $\alpha_1, \ldots, \alpha_m > 0$. Let μ be the discrete measure on $\mathbb{S}^{n-1}$ given as*

$$\mu = \sum_{j=1}^{m} \alpha_j \delta_{u_j}.$$

We have:

(i) *μ has centroid at the origin if and only if*

$$\sum_{j=1}^{m} \alpha_j u_j = 0_{\mathbb{R}^n}.$$

(ii) *μ is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$ if and only if* $\operatorname{span}\{u_1, \ldots, u_m\} = \mathbb{R}^n$.

Proof. Item **(i)** comes immediately from the equality

$$\int_{\mathbb{S}^{n-1}} u \, d\mu(u) = \sum_{j=1}^{m} \alpha_j u_j.$$

For **(ii)**, first suppose that μ is concentrated on a great subsphere $\mathbb{S}^{n-1} \cap v^{\perp}$. Then we obviously have that $u_1, \ldots, u_m \in v^{\perp}$, and hence $\operatorname{span}\{u_1, \ldots, u_m\} \subseteq v^{\perp}$. This gives that $\operatorname{span}\{u_1, \ldots, u_m\} \neq \mathbb{R}^n$. On the other hand, if $\operatorname{span}\{u_1, \ldots, u_m\} \neq \mathbb{R}^n$, then $\operatorname{span}\{u_1, \ldots, u_m\}$ is contained in some $(n-1)$-dimensional vector subspace of $\mathbb{R}^n$ (**h**, say). Then μ is concentrated on $\mathbb{S}^{n-1} \cap \mathbf{h}$.

□

Remark 7.3.4. For a proper polytope $P \subseteq \mathbb{R}^n$ with faces $F_1, \ldots, F_m$ and corresponding outer unit normal vectors $u_1, \ldots, u_m$, we have from Lemma 7.3.1 that

$$0_{\mathbb{R}^n} = \int_{\mathbb{S}^{n-1}} u \, dS_{n-1}(P, u) = \sum_{j=1}^{m} \operatorname{vol}_{n-1}(F_j) u_j.$$

On the other hand, since $S_{n-1}(P, \cdot)$ is not concentrated on a great subsphere (this follows from Proposition 7.3.1), we must have that $\operatorname{span}\{u_1, \ldots, u_m\} = \mathbb{R}^n$, since otherwise all u_j's would lie in the same hyperplane (actually, equality (3.1.3) gives an even stronger condition for the outer normal vectors of the facets of a proper polytope).

Theorem 7.3.2 (Minkowski's existence theorem for polytopes)**.** *Let $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ are pairwise distinct vectors and let $\alpha_1, \ldots, \alpha_m > 0$. Assume that*

$$\begin{aligned} &\operatorname{span}\{u_1, \ldots, u_m\} = \mathbb{R}^n \quad \text{and} \\ &\sum_{j=1}^{m} \alpha_j u_j = 0_{\mathbb{R}^n}. \end{aligned} \tag{7.3.2}$$

Then there exists a (unique up to translations) proper polytope $P \subseteq \mathbb{R}^n$ having (precisely) $u_1, \ldots, u_m$ as the normal vectors of its facets, and such that

$$\alpha_j = \operatorname{vol}_{n-1}(F(P, u_j)) \tag{7.3.3}$$

for each $j \in \{1, \ldots, m\}$.

Remark 7.3.5. Notice very carefully that the statement of this theorem does not involve any measure-theoretic concept. By stating it in this way, we have given conditions under which one can construct a proper polytope with prescribed outer unit normal vectors and areas for the facets.

In view of Lemma 7.3.3, the conditions given in (7.3.2) are simply another way to state that the measure $\mu := \sum_{j=1}^m \alpha_j \delta_{u_j}$ has centroid at the origin and is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. If this is the case, and if P is a proper polytope as in the statement, then from (7.3.1) and (7.3.3), we get that $\mu = S_{n-1}(P, \cdot)$. Thus, it follows that this theorem is indeed the particular case of Theorem 7.3.1 for polytopes.

Proof. We let $\mathbb{R}^m_+ := \{(\beta_1, \dots, \beta_m) : \beta_1, \dots, \beta_m \geq 0\}$. For each $\beta \in \mathbb{R}^m_+$, we define

$$p(\beta) := \bigcap_{j=1}^m \mathbf{h}^-(u_j, \beta_j).$$

We have immediately that for any $\beta \in \mathbb{R}^m_+$ the set $p(\beta)$ is closed (because it is the intersection of finitely many closed sets), convex (since it is the intersection of convex sets) and non-empty (this follows from the fact that $\beta_j \geq 0$ for any $j = 1, \dots, m$). We claim that $p(\beta)$ is also bounded. If this is not true, then we may let $(x_k)_{k\in\mathbb{N}}$ be a sequence in $p(\beta)$ such that $|x_k| \to \infty$ as $k \to \infty$. Thus, the points $x_k/|x_k|$ form a sequence of points in $\mathbb{S}^{n-1}$, and passing to a subsequence if necessary, we may assume that $x_k/|x_k| \to u$ for some $u \in \mathbb{S}^{n-1}$. Since $x_k \in p(\beta)$ for any $k \in \mathbb{N}$, we have

$$\langle x_k, u_j \rangle \leq \beta_j$$

for any $k \in \mathbb{N}$ and every $j \in \{1, \dots, m\}$. Consequently,

$$\left\langle \frac{x_k}{|x_k|}, u_j \right\rangle \leq \frac{\beta_j}{|x_k|},$$

and letting $k \to \infty$, we obtain

$$\langle u, u_j \rangle \leq 0$$

for every $j = 1, \dots, m$. On the other hand, the centroid condition gives

$$0 = \langle u, 0_{\mathbb{R}^n} \rangle = \left\langle u, \sum_{j=1}^m \alpha_j u_j \right\rangle = \sum_{j=1}^m \alpha_j \langle u, u_j \rangle,$$

and because all α_j's are positive, we must have $\langle u, u_j \rangle = 0$ for every $j = 1, \dots, m$. This is a contradiction, because $u \in \mathbb{S}^{n-1}$ and $\{u_1, \dots, u_m\}$ span $\mathbb{R}^n$.

By a similar argument, we may prove that

$$p(0, \dots, 0) = 0_{\mathbb{R}^n}.$$

Indeed, if $z \in p(0, \dots, 0)$, then $\langle z, u_j \rangle \leq 0$ for every $j = 1, \dots, m$. Hence

$$0 = \left\langle z, \sum_{j=1}^m \alpha_j u_j \right\rangle = \sum_{j=1}^m \alpha_j \langle z, u_j \rangle,$$

and if $\langle z, u_j \rangle < 0$ for some j, then the summation in the right-hand side is strictly negative (again, because all α_j's are positive). As a consequence, we get that $\langle z, u_j \rangle = 0$ for every $j = 1, \dots, m$, and since $\operatorname{span}\{u_1, \dots, u_m\} = \mathbb{R}^n$, we get that $z = 0_{\mathbb{R}^n}$.

Summarizing, what we have proved is that $p(\beta)$ is a set of CC^n for each $\beta \in \mathbb{R}^m_+$, and that $p(0_{\mathbb{R}^m}) = 0_{\mathbb{R}^n}$. Thus, p can be seen as a map from $\mathbb{R}^m_+$ to CC^n which fixes the neutral element of the sum (in CC^n we consider the Minkowski sum, of course). We also have that p is *concave* in the

following sense: for any $\beta, \gamma \in \mathbb{R}^m_+$ and every $\lambda \in [0,1]$, we have that

$$\lambda p(\beta) + (1-\lambda)p(\gamma) \subseteq p(\lambda\beta + (1-\lambda)\gamma).$$

Indeed, if $z \in \lambda p(\beta) + (1-\lambda)p(\gamma)$, then $z = \lambda y + (1-\lambda)w$ for some $y \in p(\beta)$ and some $w \in p(\gamma)$. Writing in coordinates $\beta = (\beta_1, \ldots, \beta_m)$ and $\gamma = (\gamma_1, \ldots, \gamma_m)$, we get that

$$\langle z, u_j\rangle = \lambda\langle y, u_j\rangle + (1-\lambda)\langle w, u_j\rangle \leq \lambda\beta_j + (1-\lambda)\gamma_j$$

for each $j = 1, \ldots, m$. Once $\lambda\beta + (1-\lambda)\gamma = (\lambda\beta_1 + (1-\lambda)\gamma_1, \ldots, \lambda\beta_m + (1-\lambda)\gamma_m)$ we have that $z \in p(\lambda\beta + (1-\lambda)\gamma)$, and this proves the inclusion.

Moving forward, we prove that the map $p\colon \mathbb{R}^m_+ \to CC^n$ is continuous (with respect to the usual topology in $\mathbb{R}^m_+$ and to the Hausdorff metric in CC^n). Let $(\beta_1, \ldots, \beta_m) \in \mathbb{R}^m_+$, $\varepsilon > 0$, and $(t_1, \ldots, t_m) \in (-\varepsilon, \varepsilon)^m$. Let $p(\beta + t)$ denote the polytope $p(\beta_1 + t_1, \ldots, \beta_m + t_m)$. Writing the Hausdorff distance in terms of support functions, we want to estimate

$$\max_{u \in \mathbb{S}^{n-1}} \left| h_{p(\beta+t)}(u) - h_{p(\beta)}(u) \right|$$

For $\varepsilon > 0$ small enough, we may assume that $p(\beta + t)$ has parallel facets to $p(\beta)$, that is, the "structure" of both polytopes are the same (formally, they are strongly isomorphic). Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. Then, u is the outer normal of $p(\beta)$ at a vertex x, and we may assume that x is the intersection of the facets of $p(\beta)$ whose outer normal vectors are $u_{i_1}, \ldots, u_{i_k}$. Of course, u is the outer normal of $p(\beta + t)$ in the corresponding vertex $\bar{x}$ which is the intersection of the facets of $p(\beta + t)$ whose outer normal vectors are the same. From Theorem 3.2.1, there exist $c_1, \ldots, c_k \geq 0$ such that

$$u = c_1 u_{i_1} + \ldots + c_k u_{i_k}.$$

It is easy to see from Proposition 3.2.1 that this decomposition is unique. In fact, otherwise we would have a subset $\{v_1, \ldots, v_s\}$ of $\{u_{i_1}, \ldots, u_{i_k}\}$ and numbers $a_1, \ldots, a_s > 0$ and $1 \leq r \leq s$ such that

$$a_1 v_1 + \ldots + a_r v_r = a_{r+1} v_{r+1} + \ldots + a_s v_s.$$

Thus, from the mentioned proposition (and having also Lemma 3.2.5 in mind), we would have that the faces F and G (of $p(\beta)$, say) obtained by the intersections of the facets whose outer normal vectors are $v_1, \ldots, v_r$ and $v_{r+1}, \ldots, v_s$, respectively, are the same. This is clearly a contradiction.

Once $u, u_{i_1}, \ldots, u_{i_k}$ are outer normal vectors of $p(\beta)$ at x, we have

$$h_{p(\beta)}(u) = \langle u, x\rangle = c_1\langle u_{i_1}, x\rangle + \ldots + c_k\langle u_{i_k}, x\rangle = c_1\beta_{i_1} + \ldots + c_k\beta_{i_k}.$$

Similarly,

$$h_{p(\beta+t)}(u) = \langle u, \bar{x}\rangle = c_1\langle u_{i_1}, \bar{x}\rangle + \ldots + c_k\langle u_{i_k}, \bar{x}\rangle = c_1(\beta_{i_1} + t_1) + \ldots + c_k(\beta_{i_k} + t_k).$$

Thus,

$$\left|h_{p(\beta+t)}(u) - h_{p(\beta)}(u)\right| = \left|\sum_{j=1}^k c_j(\beta_{i_j} + t_j) - \sum_{j=1}^k c_j\beta_{i_j}\right| = \left|\sum_{j=1}^k c_j t_j\right| \leq \sum_{j=1}^k c_j|t_j| \leq \varepsilon\sum_{j=1}^k c_j.$$

The coefficients c_j are clearly continuous functions on the compact set

$$\mathbb{S}^{n-1} \cap \operatorname{pos}\{u_{i_1}, \ldots, u_{i_k}\} = \mathbb{S}^{n-1} \cap \mathrm{NC}_{p(\beta)}(x) = \mathbb{S}^{n-1} \cap \mathrm{NC}_{p(\beta+t)}(\bar{x}).$$

From compactness, we get that $\sum_{j=1}^{k} c_j$ has an uniform upper bound as u ranges in the set above. Once there are finitely many vertices, we actually have an uniform upper bound for $u \in \mathbb{S}^{n-1}$ (taking the greatest upper bound among all vertices, of course). Considering this upper bound in the inequality above, we get the continuity of p.

As a consequence of the continuity of p, we have that the function $v\colon \mathbb{R}^m_+ \to \mathbb{R}$ given as

$$v(\beta) := \mathrm{vol}_n(p(\beta))$$

is continuous. Notice that $v(0,\dots,0) = 0$, and if $\beta \in \mathbb{R}^m_+$ is such that $\beta_j > \rho$ for each $j \in \{1,\dots,m\}$, then $\rho B^n \subseteq p(\beta)$. It follows immediately from the classical intermediate value theorem that $v^{-1}(1)$ is non-empty. Moreover, by continuity, we obtain that $v^{-1}(1)$ is closed. It follows that $v^{-1}(1) \cap [0,1]^m$ is a non-empty compact set.

Since the numbers $\alpha_1,\dots,\alpha_m$ are positive, we have that $\alpha := (\alpha_1,\dots,\alpha_m) \in \mathbb{R}^m_+$. Consider the function $f\colon v^{-1}(1) \to \mathbb{R}$ given as

$$f(\cdot) := \frac{\langle \cdot, \alpha\rangle}{n},$$

where $\langle\cdot,\cdot\rangle$ denotes the usual inner product of $\mathbb{R}^m$. It is clear that f is continuous, and by the compactness of $v^{-1}(1) \cap [0,1]^m$ there exists $\bar\beta \in v^{-1}(1) \cap [0,1]^m$ such that

$$f\big(\bar\beta\big) = \min\big\{f(\beta) : \beta \in v^{-1}(1) \cap [0,1]^m\big\} =: c.$$

Notice that we actually have

$$c = \min\big\{f(\beta) : \beta \in v^{-1}(1)\big\},$$

that is, c is the minimum of f over $v^{-1}(1)$, and not only over $v^{-1}(1) \cap [0,1]^m$. One way to see this is observing that for any $\beta_0 \in v^{-1}(1) \setminus [0,1]^m$, we have that $f(\beta_0) > f(1,\dots,1)$.

We claim that $c > 0$. Indeed, since $\bar\beta := \big(\bar\beta_1,\dots,\bar\beta_m\big) \in v^{-1}(1)$, it follows that $1 = v\big(\bar\beta\big)$. Hence, we must have $\bar\beta_k > 0$ for some $k \in \{1,\dots,m\}$, since $p(0,\dots,0) = 0_{\mathbb{R}^n}$. Thus, we write

$$c = f\big(\bar\beta\big) = \frac{1}{n}\sum_{j=1}^{m} \alpha_j\bar\beta_j \geq \frac{\alpha_k\bar\beta_k}{n} > 0,$$

where the first inequality justifies because all α_j's and all $\bar\beta_j$'s are non-negative. Once $p(\bar\beta) = 1$, we get that $p(\bar\beta)$ is a set of CC^n having positive volume. Therefore, $p(\bar\beta)$ is a convex body.

It is clear that $0_{\mathbb{R}^n} \in p(\bar\beta)$, and now we show that the minimizer $\bar\beta \in v^{-1}(1)$ can be chosen so that $0_{\mathbb{R}^n} \in \mathrm{int}(p(\bar\beta))$. If $0_{\mathbb{R}^n} \in \partial p(\bar\beta)$, then there exists a non-zero vector $z = (z_1,\dots,z_n) \in \mathbb{R}^n$ such that $0_{\mathbb{R}^n} \in \mathrm{int}(p(\bar\beta) + z)$. For each $j = 1,\dots,m$, we have

$$\langle y, u_j\rangle \leq \bar\beta_j \iff \langle y + z, u_j\rangle \leq \bar\beta_j + \langle z, u_j\rangle,$$

and hence

$$y \in p\big(\bar\beta\big) \iff y + z \in \bigcap_{j=1}^{m} \mathbf{h}^-\big(u_j, \bar\beta_j + \langle z, u_j\rangle\big),$$

and this gives

$$p(\bar{\beta}) + z = \bigcap_{j=1}^{m} \mathbf{h}^{-}\big(u_j, \bar{\beta}_j + \langle z, u_j\rangle\big). \tag{7.3.4}$$

Since $0_{\mathbb{R}^n} \in \operatorname{int}(p(\bar{\beta}) + z)$, and with arguments from standard point-set topology, we get that

$$0_{\mathbb{R}^n} \in \operatorname{int}\left(\bigcap_{j=1}^{m} \mathbf{h}^{-}\big(u_j, \bar{\beta}_j + \langle z, u_j\rangle\big)\right) \subseteq \bigcap_{j=1}^{m} \operatorname{int}\big(\mathbf{h}^{-}\big(u_j, \bar{\beta}_j + \langle z, u_j\rangle\big)\big),$$

and this yields

$$\bar{\beta}_j + \langle z, u_j\rangle > \langle 0_{\mathbb{R}^n}, u_j\rangle = 0$$

for every $j \in \{1, \dots, m\}$. For simplicity, let $\gamma_j := \bar{\beta}_j + \langle z, u_j\rangle$ for each j. From the inequalities above, we have that $\gamma := (\gamma_1, \dots, \gamma_m) \in \mathbb{R}^m_+$, and hence (7.3.4) reads

$$p(\bar{\beta}) + z = p(\gamma).$$

Because the volume is translation-invariant, we get

$$\operatorname{vol}_n(p(\gamma)) = \operatorname{vol}_n\big(p(\bar{\beta}) + z\big) = \operatorname{vol}_n\big(p(\bar{\beta})\big) = 1,$$

and then $\gamma \in v^{-1}(1)$. Now we calculate

$$f(\gamma) = \frac{\langle \gamma, \alpha\rangle}{n} = \frac{\langle \bar{\beta}, \alpha\rangle}{n} + \frac{1}{n}\sum_{j=1}^{m} \alpha_j \langle z, u_j\rangle = f(\bar{\beta}) + \left\langle z, \sum_{j=1}^{m} \alpha_j u_j \right\rangle = f(\bar{\beta}),$$

where the last equality follows from the centroid condition. It follows that γ is also a minimizer of f over $v^{-1}(1)$, and $0_{\mathbb{R}^n} \in \operatorname{int}(p(\gamma))$. Hence, we may replace $\bar{\beta}$ by γ and return to the original terminology to assume that $\bar{\beta}$ is a minimizer of f over $v^{-1}(1)$ such that $p(\bar{\beta})$ contains the origin as an interior point. Notice carefully that, by doing this, we also have that $\bar{\beta}_j > 0$ for each $j \in \{1, \dots, m\}$.

For simplicity, let $Q = p(\bar{\beta})$. For each $j \in \{1, \dots, m\}$, let $\xi_j := \operatorname{vol}_{n-1}(F(Q, u_j))$, where we recall that $F(Q, u_j)$ denotes the face of Q whose outer normal is u_j (this is the intersection of Q with the supporting hyperplane having u_j as outer normal). We obviously have $\xi := (\xi_1, \dots, \xi_m) \in \mathbb{R}^m_+$. Since Q is the intersection of half-spaces bounded by hyperplanes orthogonal to $u_1, \dots, u_m$, we get that the outer normal of the facets of Q are among $u_1, \dots, u_m$. In fact, the boundary of Q is contained in the union of these hyperplanes, and hence any $(n-1)$-dimensional subset of ∂Q must be contained in some of them (recall that the u_j's are pairwise distinct). If $F(Q, u_j)$ is a facet, then $\xi_j > 0$ and $h_Q(u_j) = \bar{\beta}_j$. On the other hand, if $F(Q, u_i)$ is not a facet, then $\xi_i = 0$. Hence, we may use Theorem 4.4.2 to write

$$1 = \operatorname{vol}_n(Q) = \frac{1}{n}\sum_{j=1}^{m} h_Q(u_j)\operatorname{vol}_{n-1}(F(Q, u_j)) = \frac{1}{n}\sum_{j=1}^{m} \bar{\beta}_j \xi_j = \frac{\langle \bar{\beta}, \xi\rangle}{n}.$$

Thus, we have

$$\frac{\langle \bar{\beta}, \xi\rangle}{n} = 1 = \frac{f(\bar{\beta})}{c} = \frac{\langle \bar{\beta}, \alpha\rangle}{cn}.$$

These equalities give that

$$\bar{\beta} \in \mathbf{h}_m(\xi, n) \cap \mathbf{h}_m(\alpha/c, n),$$

where $\mathbf{h}_m$ denotes hyperplanes of $\mathbb{R}^m$. Since it is clear that small "shifts" in the facets of a polytope do not change its "structure", we have that there exists a number $\varepsilon > 0$ small enough such that for each

$$\beta = (\beta_1, \ldots, \beta_m) \in \bar{\beta} + \underbrace{[-\varepsilon, \varepsilon] \times \ldots \times [-\varepsilon, \varepsilon]}_{m \text{ times}} =: C \ \left(\subseteq \mathbb{R}^m \right)$$

we have that

(i) $\beta_j > 0$ for each $j = 1, \ldots, m$ and

(ii) the outer normal vectors of the facets of $p(\beta)$ are exactly the same as the outer normal vectors of the facets of $p(\bar{\beta}) = Q$.

We claim that $v(\beta) \leq 1$ for every $\beta \in C \cap \mathbf{h}_m(\alpha/c, n)$. To verify this, assume that $\beta = (\beta_1, \ldots, \beta_m) \in C \cap \mathbf{h}_m(\alpha/c, n)$ is such that $v(\beta) > 1$. Then, by the continuity of v and from the fact that $v(0_{\mathbb{R}^m}) = 0$, it follows that for some number $0 < t < 1$, we have that $v(t\beta) = 1$ (we just use the intermediate value theorem for $t \mapsto v(t\beta)$). Then, $t\beta \in v^{-1}(1)$, and since c is the minimum of f over $v^{-1}(1)$, we get

$$c \leq f(t\beta) = \frac{\langle t\beta, \alpha \rangle}{n} = tc \frac{\langle \beta, \alpha/c \rangle}{n} = tc < c,$$

which is a contradiction. Hence, we indeed have that $v \leq 1$ in $C \cap \mathbf{h}_m(\alpha/c, n)$.

It is clear that C is a convex body of $\mathbb{R}^m$, and hence $C \cap \mathbf{h}_m(\alpha/c, n)$ is a convex set. Also, we already know that $\bar{\beta} \in C \cap \mathbf{h}_m(\alpha/c, n)$. Thus, for any $\beta \in C \cap \mathbf{h}_m(\alpha/c, n)$ and $\lambda \in [0, 1]$, we have that $\lambda\beta + (1 - \lambda)\bar{\beta} \in C \cap \mathbf{h}_m(\alpha/c, n)$. From the concavity of p and from the monotonicity of the volume for inclusion, we obtain

$$\begin{aligned} 1 \geq v\big(\lambda\beta + (1-\lambda)\bar{\beta}\big) = \mathrm{vol}_n\big(p\big(\lambda\beta + (1-\lambda)\bar{\beta}\big)\big) \geq \\ \geq \mathrm{vol}_n\big(\lambda p(\beta) + (1-\lambda)p\big(\bar{\beta}\big)\big) = \mathrm{vol}_n\big(\lambda p(\beta) + (1-\lambda)Q\big). \end{aligned} \tag{7.3.5}$$

Using the continuity of the volume (in the Hausdorff metric) and a change of variables, we also have

$$\begin{aligned} \lim_{\lambda \to 0^+} \frac{\mathrm{vol}_n((1-\lambda)Q + \lambda p(\beta)) - \mathrm{vol}_n((1-\lambda)Q)}{\lambda} = \\ = \lim_{\lambda \to 0^+} \frac{(1-\lambda)^n \mathrm{vol}_n\left(Q + \frac{\lambda}{1-\lambda} p(\beta)\right) - (1-\lambda)^n \mathrm{vol}_n(Q)}{\lambda} = \\ = \lim_{\lambda \to 0^+} \left[\frac{\mathrm{vol}_n\left(Q + \frac{\lambda}{1-\lambda} p(\beta)\right) - \mathrm{vol}_n(Q)}{\lambda} - n \left(\mathrm{vol}_n\left(Q + \frac{\lambda}{1-\lambda} p(\beta)\right) - \mathrm{vol}_n(Q) \right) \right] = \\ = \lim_{t \to 0^+} \frac{\mathrm{vol}_n(Q + tp(\beta)) - \mathrm{vol}_n(Q)}{\frac{t}{t+1}} = \lim_{t \to 0^+} \frac{\mathrm{vol}_n(Q + tp(\beta)) - \mathrm{vol}_n(Q)}{t}. \end{aligned} \tag{7.3.6}$$

From (7.3.5) and (7.3.6), we get that

$$V(Q,\dots,Q,p(\beta)) = \frac{1}{n}\lim_{t\to 0^+}\frac{\mathrm{vol}_n(Q+tp(\beta))-\mathrm{vol}_n(Q)}{t} =$$
$$= \frac{1}{n}\lim_{\lambda\to 0^+}\frac{\mathrm{vol}_n((1-\lambda)Q+\lambda p(\beta))-\mathrm{vol}_n((1-\lambda)Q)}{\lambda} \leq$$
$$\leq \frac{1}{n}\lim_{\lambda\to 0^+}\frac{1-(1-\lambda)^n\mathrm{vol}_n(Q)}{\lambda} = \frac{1}{n}\lim_{\lambda\to 0^+}\frac{1-(1-\lambda)^n}{\lambda} = 1$$

On the other hand, from Lemma 6.2.1, we have

$$V(Q,\dots,Q,p(\beta)) = \frac{1}{n}\sum_{F \text{ facet of } Q} h_{p(\beta)}(u_F)\mathrm{vol}_{n-1}(F) = \frac{1}{n}\sum_{j=1}^{m} h_{p(\beta)}(u_j)\xi_j =$$
$$= \frac{1}{n}\sum_{j=1}^{m}\beta_j\xi_j = \frac{\langle\beta,\xi\rangle}{n}.$$

Here, the second equality is justified because all outer normal vectors of the facets of Q are among the u_j's, and whenever $F(Q,u_i)$ is not a facet of Q, we have that $\xi_i = 0$. The third equality comes from the fact that the facets of $p(\beta)$ have the same outer normal vectors as the facets of Q. It follows that

$$\langle\xi,\beta\rangle \leq n$$

for every $\beta \in C \cap \mathbf{h}_m(\alpha/c,n)$, and this leads to the inclusion

$$C \cap \mathbf{h}_m(\alpha/c,n) \subseteq \mathbf{h}_m^-(\xi,n).$$

Because $\bar{\beta}$ is clearly an interior point of C, and since $\bar{\beta} \in \mathbf{h}_m(\xi,n)$, we must have $\mathbf{h}_m(\alpha/c,n) = \mathbf{h}_m(\xi,n)$. As a consequence, we get

$$\xi = \frac{\alpha}{c},$$

and hence

$$\mathrm{vol}_{n-1}(F(Q,u_j)) = \xi_j = \frac{\alpha_j}{c} > 0$$

for each $j \in \{1,\dots,m\}$. Therefore, $u_1,\dots,u_m$ are precisely the outer normal vectors of the facets of Q. Letting $P := c^{\frac{1}{n-1}}Q$, we get immediately that $u_1,\dots,u_m$ are also exactly the outer normal vectors of the facets of P, and

$$\mathrm{vol}_{n-1}(F(P,u_j)) = \alpha_j$$

for every $j \in \{1,\dots,m\}$. Once Q is a proper polytope, we also have that P is a proper polytope, and this concludes the proof of the existence. Once $\sum_{j=1}^{m}\alpha_j\delta_{u_j}$ is the surface area measure of P, the uniqueness follows from Remark 7.3.3.

□

The strategy for the general case is as follows: given a finite Borel measure μ on $\mathbb{S}^{n-1}$ which has its centroid at $0_{\mathbb{R}^n}$ and is not concentrated on a great subsphere, we find a sequence of discrete Borel measures having these same properties and converging to μ. Then we prove that, up to a subsequence, the polytopes associated with these discrete measures converge to a convex body whose surface area measure is μ. We prove the existence of such a sequence of discrete measures in the next proposition. In what follows, recall that the *positive part* of a function $f\colon X \to \mathbb{R}$ is the function $f_+\colon X \to [0,+\infty)$

given for each $x \in X$ as

$$f_+(x) := \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ 0 & \text{if } f(x) < 0 \end{cases}$$

or, equivalently, as $f_+(x) := \max\{f(x), 0\}$ for each $x \in X$. Similarly, the *negative part* of f is defined as $f_-(x) := \min\{f(x), 0\}$ for all $x \in X$.

Proposition 7.3.2. *Let μ be a finite Borel measure on $\mathbb{S}^{n-1}$ which has its centroid at $0_{\mathbb{R}^n}$ and is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. Then there exists a sequence $(\mu_k)_{k\in\mathbb{N}}$ of discrete Borel measures of $\mathbb{S}^{n-1}$ such that the following statements hold:*

(i) *For each $k \in \mathbb{N}$, the measure μ_k has centroid at $0_{\mathbb{R}^n}$.*

(ii) *μ_k converges weakly to μ.*

(iii) *For every $\varepsilon > 0$, there exists $k_0 \in \mathbb{N}$ with the property that the inequality*

$$\left| \int_{\mathbb{S}^{n-1}} \langle v, u\rangle_+ \, d\mu_k(u) - \int_{\mathbb{S}^{n-1}} \langle v, u\rangle_+ \, d\mu(u) \right| < \varepsilon$$

holds for any $k > k_0$ and every $v \in \mathbb{S}^{n-1}$. As a consequence, for k large enough we have that μ_k is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$.

Proof. We divide the proof into several steps:

Step 1. [Covering $\mathbb{S}^{n-1}$ by suitable sets] For each $k \in \mathbb{N}$ and $z \in \mathbb{S}^{n-1}$, consider the *spherical cap*

$$C_k(z) := \mathbb{S}^{n-1} \cap \mathbf{h}^+\left(z, 1 - \frac{1}{2k^2}\right).$$

It is easy to see that $C_k(z)$ is contained in the ball having the unique point of the intersection $\operatorname{seg}[0_{\mathbb{R}^n}, z] \cap \mathbf{h}(z, 1 - 1/2k^2)$ as center, and whose radius is the same as the radius (r, say) of the circle $\mathbb{S}^{n-1} \cap \mathbf{h}(z, 1 - 1/2k^2)$. By the Pythagorean theorem, we get

$$r^2 = 1 - \left(1 - \frac{1}{2k^2}\right)^2 = \frac{1}{k^2} - \frac{1}{4k^4} < \frac{1}{k^2},$$

and hence $r < 1/k$. Since $C_k(z)$ is contained in a ball of radius r, we get

$$\operatorname{diam}(C_k(z)) \leq 2r < \frac{2}{k},$$

where the diameter is taken with respect to the metric of $\mathbb{R}^n$, of course. Moreover, notice carefully that for any $u \in C_k(z)$, we have

$$\langle z, u\rangle \geq 1 - \frac{1}{2k^2}, \tag{7.3.7}$$

since $u \in \mathbf{h}^+(z, 1 - 1/2k^2)$. In particular, $\langle z, u\rangle > 0$ for every $u \in C_k(z)$.

Now fix $k \in \mathbb{N}$. By compactness, we have that $\mathbb{S}^{n-1}$ can be covered by finitely many spherical caps $C_k(z_1), \ldots, C_k(z_{M(k)})$. From that covering, we claim that we may extract a finite family

$E_1^k, \dots, E_{N(k)}^k$ of pairwise disjoint Borel subsets of $\mathbb{S}^{n-1}$ such that

(a) each E_i^k is contained in some of the spherical caps $C_k(z_j)$,

(b) $\mu(E_i^k) > 0$ for every $i = 1, \dots, N(k)$, and

(c) up to a set of μ-measure zero, $\mathbb{S}^{n-1}$ equals the union of the E_i^k's

The construction of this family follows from a standard argument: first we consider the intersection of all the spherical caps and let this intersection be E_1^k. Then we take all the intersections of $M(k) - 1$ spherical caps and consider the set difference of each of these intersections with E_1^k, and we let these sets be $E_2^k, \dots, E_{M(k)+1}^k$ (notice that there are exactly $M(k)$ intersections of $M(k) - 1$ caps). Next, we take all the intersections between $M(k) - 2$ spherical caps and consider the set difference of each of these intersections with the union of all E_i^k's constructed before. These sets will be the next E_i^k's. Proceeding inductively, after a finite time of steps, we get a disjoint cover of $\mathbb{S}^{n-1}$ by Borel sets, each of which is contained in some spherical cap $C_k(z_j)$. Finally, we discard all of these sets which have zero μ-measure, and the remaining ones are (after renumbering) the desired family $E_1^k, \dots, E_{N(k)}^k$.

Step 2. [Constructing the μ_k's] For each $k \in \mathbb{N}$ and every $1 \le i \le N(k)$, consider the vector

$$v_i^k := \frac{1}{\mu(E_i^k)} \int_{E_i^k} u \, d\mu(u).$$

We bound the norm of v_i^k from above and from below. First, notice that

$$|v_i^k| \le \frac{1}{\mu(E_i^k)} \int_{E_i^k} |u| \, d\mu(u) = 1,$$

because $E_i^k \subseteq \mathbb{S}^{n-1}$, and hence $|u| = 1$ for every $u \in E_i^k$. Now recall that E_i^k is contained in some spherical cap, which for simplicity we denote by $C_k(z)$. In view of (7.3.7) and using Cauchy–Schwarz inequality, we obtain

$$|v_i^k| = |v_i^k||z| \ge \langle v_i^k, z\rangle = \frac{1}{\mu(E_i^k)} \int_{E_i^k} \langle u, z\rangle \, d\mu(u) \ge 1 - \frac{1}{2k^2}. \tag{7.3.8}$$

Summarizing, we have

$$0 < 1 - \frac{1}{2k^2} \le |v_i^k| \le 1$$

for each $k \in \mathbb{N}$ and every $1 \le i \le N(k)$. Now we let

$$u_i^k := \frac{v_i^k}{|v_i^k|} \in \mathbb{S}^{n-1}$$

and define

$$\mu_k := \sum_{i=1}^{N(k)} \mu(E_i^k)|v_i^k| \delta_{u_i^k},$$

for every $k \in \mathbb{N}$. This defines, for each $k \in \mathbb{N}$, a discrete Borel measure on $\mathbb{S}^{n-1}$, and we claim that each of these measures has its centroid at the origin. Indeed,

$$\int_{\mathbb{S}^{n-1}} u\, d\mu_k(u) = \sum_{i=1}^{N(k)} \mu(E_i^k)|v_i^k|u_i^k = \sum_{i=1}^{N(k)} \mu(E_i^k)v_i^k = \sum_{i=1}^{N(k)} \int_{E_i^k} u\, d\mu(u) = \\ = \int_{\mathbb{S}^{n-1}} u\, d\mu(u) = 0_{\mathbb{R}^n},$$

where we used the fact that $\mathbb{S}^{n-1}$ is the union of the E_i^k's up to a set of zero μ-measure. The last equality follows from the hypothesis on μ having its centroid at $0_{\mathbb{R}^n}$.

Step 3. [μ_k converges weakly to μ] We first prove that

$$|u_i^k - u| < \frac{2}{k}$$

for any $k \in \mathbb{N}$, $1 \leq i \leq N(k)$ and $u \in E_i^k$. To verify this, assume again that E_i^k is contained in the cap $C_k(z)$. In this case, we already know from (7.3.8) that $\langle v_i^k, z\rangle \geq 1 - 1/2k^2$. Recalling that $|v_i^k| \leq 1$, this leads to the inequality

$$\langle u_i^k, z\rangle = \frac{\langle v_i^k, z\rangle}{|v_i^k|} \geq \langle v_i^k, z\rangle \geq 1 - \frac{1}{2k^2}.$$

Thus, $u_i^k \in \mathbf{h}^+(z, 1 - 1/2k^2)$. Since we already know that $u_i^k \in \mathbb{S}^{n-1}$, we obtain that $u_i^k \in C_k(z)$. Since any $u \in E_i^k$ lies in $C_k(z)$, we get

$$|u_i^k - u| \leq \operatorname{diam}(C_k(z)) < \frac{2}{k}.$$

Now let $f \in C(\mathbb{S}^{n-1})$ be arbitrarily given. Since f is continuous and $\mathbb{S}^{n-1}$ is compact, we have that f is uniformly continuous. For an arbitrarily given $\varepsilon > 0$, let $k_0 \in \mathbb{N}$ be such that $|f(u) - f(v)| < \varepsilon/2\mu(\mathbb{S}^{n-1})$ whenever $u, v \in \mathbb{S}^{n-1}$ are such that $|u - v| < 2/k_0$. It follows that, if $k > k_0$, then

$$|f(u_i^k) - f(u)| < \frac{\varepsilon}{2\mu(\mathbb{S}^{n-1})}$$

for any $i = 1, \ldots, N(k)$ and every $u \in E_i^k$. Since f is bounded (because it is a continuous function over a compact set), we may also assume (by enlarging k_0 if necessary) that

$$\frac{1}{2k^2} \max\left\{|f(u)| : u \in \mathbb{S}^{n-1}\right\} < \frac{\varepsilon}{2\mu(\mathbb{S}^{n-1})}$$

for every $k > k_0$. From the bounds for $|v_k^i|$, we get as a consequence that

$$\begin{aligned} \left||v_i^k|f(u_i^k) - f(u)\right| &\leq \left||v_i^k|f(u_i^k) - |v_i^k|f(u)\right| + \left||v_i^k|f(u) - f(u)\right| = \\ = |v_i^k||f(u_i^k) - f(u)| + (1 - |v_i^k|)|f(u)| &\leq |f(u_i^k) - f(u)| + \frac{|f(u)|}{2k^2} \leq \\ &\leq \frac{\varepsilon}{2\mu(\mathbb{S}^{n-1})} + \frac{\varepsilon}{2\mu(\mathbb{S}^{n-1})} = \frac{\varepsilon}{\mu(\mathbb{S}^{n-1})}, \end{aligned} \tag{7.3.9}$$

for every $k > k_0$, all $u \in E_i^k$ and any $i = 1, \dots, N(k)$. With these bounds in mind, for $k > k_0$, we estimate

$$\begin{aligned} &\left| \int_{\mathbb{S}^{n-1}} f(u)\, d\mu_k(u) - \int_{\mathbb{S}^{n-1}} f(u)\, d\mu(u) \right| = \\ &= \left| \sum_{i=1}^{N(k)} \mu(E_i^k) |v_i^k| f(u_i^k) - \sum_{i=1}^{N(k)} \int_{E_i^k} f(u)\, d\mu(u) \right| = \\ &= \left| \sum_{i=1}^{N(k)} \int_{E_i^k} |v_i^k| f(u_i^k) - f(u)\, d\mu(u) \right| \leq \sum_{i=1}^{N(k)} \int_{\mathbb{S}^{n-1}} \left| |v_i^k| f(u_i^k) - f(u) \right| d\mu(u) \leq \\ &\leq \sum_{i=1}^{N(k)} \int_{E_i^k} \frac{\varepsilon}{\mu(\mathbb{S}^{n-1})}\, d\mu(u) = \frac{\varepsilon}{\mu(\mathbb{S}^{n-1})} \sum_{i=1}^{N(k)} \mu(E_i^k) = \varepsilon. \end{aligned} \tag{7.3.10}$$

This proves that μ_k converges weakly to μ as $k \to \infty$.

Step 4. [Proving that **(iii)** holds] For any $v, u \in \mathbb{S}^{n-1}$, we have from the Cauchy–Schwarz inequality that $\langle v, u \rangle_+ \leq |\langle v, u \rangle_+| \leq |\langle v, u \rangle| \leq 1$. Also, recall that we always have $1 - 1/2k^2 \leq |v_i^k| \leq 1$. Thus, for any $k \in \mathbb{N}$, $1 \leq i \leq N(k)$, $u \in E_i^k$ and $v \in \mathbb{S}^{n-1}$, we have the bound

$$\begin{aligned} &\left| |v_i^k| \langle v, u_i^k \rangle_+ - \langle v, u \rangle_+ \right| \leq \left| |v_i^k| \langle v, u_i^k \rangle_+ - |v_i^k| \langle v, u \rangle_+ \right| + \left| |v_i^k| \langle v, u \rangle_+ - \langle v, u \rangle_+ \right| = \\ &= |v_i^k| \left| \langle v, u_i^k \rangle_+ - \langle v, u \rangle_+ \right| + \langle v, u \rangle_+ (1 - |v_i^k|) \leq \left| \langle v, u_i^k \rangle_+ - \langle v, u \rangle_+ \right| + \frac{1}{2k^2} \langle v, u \rangle_+ \leq \\ &\leq |\langle v, u_i^k - u \rangle| + \frac{1}{2k^2} \leq |u - u_i^k| + \frac{1}{2k^2} \leq \frac{2}{k} + \frac{1}{2k^2}, \end{aligned}$$

where we used the fact that $|\langle v, u_i^k \rangle_+ - \langle v, u \rangle_+| \leq |\langle v, u_i^k \rangle - \langle v, u \rangle|$ (to verify this, simply separate in cases). Hence, for $\varepsilon > 0$ arbitrarily given, we may let $k_0 \in \mathbb{N}$ be large enough such that $k > k_0$ implies

$$\left| |v_i^k| \langle v, u_i^k \rangle_+ - \langle v, u \rangle_+ \right| < \frac{\varepsilon}{\mu(\mathbb{S}^{n-1})}$$

for every $1 \leq i \leq N(k)$, $u \in E_i^k$ and $v \in \mathbb{S}^{n-1}$. This is the same estimate that we derived for f in (7.3.9). Therefore, replacing $f(\cdot)$ by $\langle v, \cdot \rangle_+$ (for each fixed $v \in \mathbb{S}^{n-1}$) in (7.3.10), we get that for any $k > k_0$, the inequality

$$\left| \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+\, d\mu_k(u) - \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+\, d\mu(u) \right| < \varepsilon$$

holds for all $v \in \mathbb{S}^{n-1}$. This gives the "uniform" convergence described in **(iii)**.

To conclude the proof, it only remains to show that for k large enough, we have that μ_k is not concentrated in a great subsphere of $\mathbb{S}^{n-1}$. We claim that the hypotheses on μ imply that

$$\int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+\, d\mu(u) > 0 \tag{7.3.11}$$

for any $v \in \mathbb{S}^{n-1}$. Indeed, since μ has its centroid at the origin, we have

$$0 = \langle v, 0_{\mathbb{R}^n} \rangle = \left\langle v, \int_{\mathbb{S}^{n-1}} u \, d\mu(u) \right\rangle = \int_{\mathbb{S}^{n-1}} \langle v, u \rangle \, d\mu(u) =$$
$$= \int_{\mathbb{S}^{n-1} \cap \mathbf{h}^+(v,0)} \langle v, u \rangle \, d\mu(u) + \int_{\mathbb{S}^{n-1} \cap \mathbf{h}^-(v,0)} \langle v, u \rangle \, d\mu(u).$$

Notice carefully that the sets $\mathbb{S}^{n-1} \cap \mathbf{h}^+(v,0)$ and $\mathbb{S}^{n-1} \cap \mathbf{h}^-(v,0)$ overlap on the great subsphere $\mathbb{S}^{n-1} \cap v^\perp$, and μ may not be zero on this great subsphere. However, $\langle v, u \rangle = 0$ over this subsphere, and hence the last equality above in fact holds. From the definition of the positive part of a function, it is clear that the integral of $\langle v, u \rangle_+$ over $\mathbb{S}^{n-1}$ equals the integral of $\langle v, u \rangle$ over $\mathbb{S}^{n-1} \cap \mathbf{h}^+(v,0)$. Thus,

$$\int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu(u) = \int_{\mathbb{S}^{n-1} \cap \mathbf{h}^+(v,0)} \langle v, u \rangle \, d\mu(u) = \int_{\mathbb{S}^{n-1} \cap \mathbf{h}^-(v,0)} \langle -v, u \rangle \, d\mu(u) =$$
$$= \int_{\mathbb{S}^{n-1}} \langle -v, u \rangle_+ \, d\mu(u).$$

It is also clear from the definition of the positive part that $|\langle v, u \rangle| = \langle v, u \rangle_+ + \langle -v, u \rangle_+$ for every $u \in \mathbb{S}^{n-1}$. Therefore, since μ is not centered at any great subsphere, we get

$$0 < \int_{\mathbb{S}^{n-1}} |\langle v, u \rangle| \, d\mu(u) = \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu(u) + \int_{\mathbb{S}^{n-1}} \langle -v, u \rangle_+ \, d\mu(u) = 2 \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu(u),$$

and this gives (7.3.11). It follows that the function

$$\mathbb{S}^{n-1} \ni v \mapsto \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu(u)$$

is strictly positive. Because the function is clearly continuous, and since $\mathbb{S}^{n-1}$ is compact, we have that it attains a positive minimum (c_0, say). Fixing any $0 < \varepsilon < c_0$, we may let $k_0 \in \mathbb{N}$ be such that $k > k_0$ implies

$$\left| \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu_k(u) - \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu(u) \right| < \varepsilon.$$

Hence, we get

$$\int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu_k(u) \geq \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu(u) - \varepsilon \geq c_0 - \varepsilon > 0,$$

and since it is clear that

$$\int_{\mathbb{S}^{n-1}} |\langle v, u \rangle| \, d\mu_k(u) \geq \int_{\mathbb{S}^{n-1}} \langle v, u \rangle_+ \, d\mu_k(u),$$

we finally arrive at

$$\int_{\mathbb{S}^{n-1}} |\langle v, u \rangle| \, d\mu_k(u) > 0$$

for every $k > k_0$ and any $v \in \mathbb{S}^{n-1}$. It follows that, for each $k > k_0$, the measure μ_k is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$.

□

Now we can prove the general case of Minkowski's existence theorem.

Proof of Theorem 7.3.1. By Proposition 7.3.2, we may let $(\mu_k)_{k\in\mathbb{N}}$ be a sequence of discrete measures on $\mathbb{S}^{n-1}$, each of which having its centroid at the origin and converging weakly to μ. By discarding finitely many initial terms, we may also assume that each μ_k is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. By Theorem 7.3.2, for each $k \in \mathbb{N}$ we may choose a polytope $P_k \subseteq \mathbb{R}^n$ such that $\mu_k = S_{n-1}(P_k, \cdot)$. Since μ_k is the surface area measure of any translate of P_k (see Remark 7.3.3), we may assume that $0_{\mathbb{R}^n} \in \operatorname{int}(P_k)$ for each $k \in \mathbb{N}$.

Our strategy is to prove that $(P_k)_{k\in\mathbb{N}}$ is a uniformly bounded family of sets, and hence to use Blaschke's selection theorem. Since μ_k converges weakly to μ, we have that $\mu_k(\mathbb{S}^{n-1}) \to \mu(\mathbb{S}^{n-1})$ as $k \to \infty$. Hence, $\mu_k(\mathbb{S}^{n-1})$ is a bounded sequence of numbers, meaning that there exists a number $C_1 > 0$ such that

$$\mu_k(\mathbb{S}^{n-1}) < C_1$$

for every $k \in \mathbb{N}$. Because μ_k is the surface area measure of P_k, we have that $\mu_k(\mathbb{S}^{n-1})$ is the total surface area of P_k (see, e.g., (7.2.1)). Denote the total surface area of P_k by $S(P_k)$. From the isoperimetric inequality (Theorem 5.2.1), we have a uniform upper bound for the volume of P_k:

$$\operatorname{vol}_n(P_k) \leq \operatorname{vol}_n(B^n)\left(\frac{S(P_k)}{S(B^n)}\right)^{\frac{n}{n-1}} \leq \kappa_n\left(\frac{C_1}{\omega_n}\right)^{\frac{n}{n-1}} =: C_2,$$

for each $k \in \mathbb{N}$, where we remind the reader that κ_n and ω_n denote the volume and the total surface area of the unit ball of $\mathbb{R}^n$, respectively.

Now let $k \in \mathbb{N}$, $v \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$ such that $\lambda v \in P_k$ be arbitrarily given. Then it is clear that $h_{\operatorname{seg}[0_{\mathbb{R}^n},\lambda v]}(\cdot) = \lambda\langle v, \cdot\rangle_+$, and since $\operatorname{seg}[0_{\mathbb{R}^n}, \lambda v] \subseteq P_k$, we get the inequality

$$h_{\operatorname{seg}[0_{\mathbb{R}^n},\lambda v]} \leq h_{P_k}.$$

Thus,

$$C_2 \geq \operatorname{vol}_n(P_k) = V(P_k, \ldots, P_k) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{P_k}(u)\, dS_{n-1}(P_k, u) =$$
$$= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{P_k}(u)\, d\mu_k(u) \geq \frac{1}{n}\int_{\mathbb{S}^{n-1}} \lambda\langle v, u\rangle_+\, d\mu_k(u) = \frac{\lambda}{n}\int_{\mathbb{S}^{n-1}} \langle v, u\rangle_+\, d\mu_k(u).$$

From the proof of item **(iii)** of Proposition 7.3.2, we have that

$$c_0 := \min\left\{\int_{\mathbb{S}^{n-1}} \langle v, u\rangle_+\, d\mu(u) : v \in \mathbb{S}^{n-1}\right\} > 0.$$

Hence, choosing $0 < \varepsilon < c_0/2$, we have from the mentioned result that there exists a number $k_0 \in \mathbb{N}$ such that $k > k_0$ implies

$$\int_{\mathbb{S}^{n-1}} \langle v, u\rangle_+\, d\mu_k(u) \geq \int_{\mathbb{S}^{n-1}} \langle v, u\rangle_+\, d\mu(u) - \varepsilon \geq c_0 - \varepsilon > \frac{c_0}{2}$$

for any $v \in \mathbb{S}^{n-1}$. Replacing this in the previous inequality and rearranging the terms, we arrive at

$$\lambda \leq \frac{2nC_2}{c_0} =: r.$$

That is, $k > k_0$, $v \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$ is such that $\lambda v \in P_k$, then $\lambda \leq r$. As a consequence, we obtain immediately that

$$P_k \subseteq rB^n$$

for every $k > k_0$. Enlarging r if necessary, we may assume that all P_k's are contained in rB^n (since there are at most finitely many remaining polytopes $P_1, \ldots, P_k$).

Now we have immediately from the Blaschke Selection Theorem (see Theorem 2.5.4) that, passing to a subsequence if necessary, we may assume that P_k converges in the Hausdorff metric to a set $K \in CC^n$. Because $\mu_k = S_{n-1}(P_k, \cdot)$ for any $k \in \mathbb{N}$, we have

$$V(P_k, \ldots, P_k, L) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu_k(u)$$

for any $L \in CC^n$ and every $k \in \mathbb{N}$. Letting $k \to \infty$, using the continuity of the mixed volume in the Hausdorff metric (in the left-hand side) and using the fact that μ_k converges weakly to μ (in the right-hand side), we finally arrive at

$$V(K, \ldots, K, L) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, d\mu(u),$$

for every $L \in CC^n$. Hence, it follows that $\mu = S_{n-1}(K, \cdot)$. The fact that K is a convex body (so far we only have that $K \in CC^n$; that is, we still do not know whether K has maximal affine dimension) comes straightforwardly from the fact that μ is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. Indeed, if K does not have maximal dimension, then up to a translation we may assume that $K \subseteq v^\perp$ for some $v \in \mathbb{S}^{n-1}$. Thus, for any $w \in v^\perp \cap \mathbb{S}^{n-1}$, we have from Proposition 6.2.3 that

$$0 = V\big(K, \ldots, K, \mathrm{seg}[-w, w]\big) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{\mathrm{seg}[-w,w]}(u)\, dS_{n-1}(K, u) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} |\langle w, u\rangle|\, d\mu(u),$$

and this contradicts the fact that μ is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. The proof is complete.

□

The Minkowski existence theorem gives a complete characterization of the measures $S_{n-1}(K, \cdot)$ for convex bodies $K \subseteq \mathbb{R}^n$. However, we have not mentioned the case where $K \in CC^n$ does not have a maximal affine dimension. It is not difficult to get the **intuition** about what happens in this case. If $\dim(K) \leq n-2$, then $K = \partial K$, and the $(n-1)$-dimensional Hausdorff measure of ∂K equals 0. Hence we suspect that $S_{n-1}(K, \cdot)$ is the zero measure. In the case where $\dim(K) = n-1$, K is contained in a hyperplane ($u^\perp$, with $u \in \mathbb{S}^{n-1}$, say), and the reverse normal image of a Borel set $E \subseteq \mathbb{S}^{n-1}$ has positive $(n-1)$-dimensional Hausdorff measure if and only if $u \in E$ or $-u \in E$. Thus, our intuition says that $S_{n-1}(K, \cdot)$ must be a combination of the Dirac delta measures δ_u and δ_{-u}. Because u and $-u$ support K at each of its points, the weight of each delta measure would be $\mathrm{vol}_{n-1}(K)$.

By now, these ideas are not formal because the characterization of $S_{n-1}(K, \cdot)$ as the push-forward measure of $\mathcal{H}^{n-1}|_{\partial K}$ under the Gauss map (that is, $S_{n-1}(K, \cdot) = \mu_K$) was only proved for the case where K is a convex body. In the case where $K \in CC^n$ is such that $\dim(K) < n$, we only have the definition of $S_{n-1}(K, \cdot)$ via mixed volumes (and the Riesz representation theorem) to work with (μ_K was not even defined for this case). Nevertheless, it is not difficult to give formal proofs for our intuitive claims, and the reader is invited to do that in Exercises 7.6 and 7.27. These exercises give complete descriptions of the measures $S_{n-1}(K, \cdot)$ for sets $K \in CC^n$ with non-maximal affine dimension.

7.4 Surface area measures and curvature

At this point, it is useful to investigate the relation between μ_K and the restriction of the $(n-1)$-dimensional Hausdorff measure to $\mathbb{S}^{n-1}$, since both are measures on $\mathbb{S}^{n-1}$. The reader should take into consideration that, to do so, we will assume some familiarity with standard concepts and terminology from smooth manifolds theory (also, some differential geometry might be helpful, but it is not necessary). In this direction, we refer the reader to Lee's book [1526].

First, we restrict ourselves to the particular case where K is a smooth convex body, and we remember that this means that each point of ∂K has a unique unit outer normal vector (that is, the map u_K introduced in Remark 7.2.6 is defined everywhere). In this case, we recall that the *Gauss map* of K is the map $\xi_K : \partial K \to \mathbb{S}^{n-1}$ which associates each point $x \in \partial K$ to the unit outer normal vector of K at x (now defined for every point of K; see Remark 7.2.6). It is clear that the reverse normal image $\eta_K(E)$ of any Borel set $E \subseteq \mathbb{S}^{n-1}$ is just $\xi_K^{-1}(E)$, and hence the measure μ_K is given as

$$\mu_K(E) = \mathcal{H}^{n-1}(\xi_K^{-1}(E)),$$

for every Borel set $E \subseteq \mathbb{S}^{n-1}$. Once ξ_K is clearly continuous (hence measurable), it follows that

$$\mu_K = (\xi_K)_* \mathcal{H}^{n-1}\big|_{\partial K},$$

that is, μ_K is the push-forward measure of the $(n-1)$-dimensional Hausdorff measure (restricted to ∂K, of course) to $\mathbb{S}^{n-1}$ under ξ_K. Notice that this is just Theorem 7.2.4 for the case where K is smooth. We emphasize that, in this particular case, the fact that ξ_K is defined everywhere and continuous allows us to regard directly the push-forward measure of $\mathcal{H}^{n-1}|_{\partial K}$ under the Gauss map of K avoiding the machinery used in the previous section to tackle the non-smooth case. From formula (A.2.4) of change of variables, we obtain the equality

$$\int_{\mathbb{S}^{n-1}} f \, d\mu_K = \int_{\partial K} f \circ \xi_K \, d\mathcal{H}^{n-1} \tag{7.4.1}$$

for any measurable function $f : \mathbb{S}^{n-1} \to \mathbb{R}$. This is, of course, the version of (7.2.9) for the case where u_K is defined everywhere.

This equality might be useful, but it does not solve our problem of establishing a relation between μ_K and the restriction of $\mathcal{H}^{n-1}$ to $\mathbb{S}^{n-1}$. To find a suitable relation, also involving objects with geometrical meaning, we restrict ourselves once again and consider the so-called C^2_+ *convex bodies*. We say that a convex body K is C^2_+ if:

(i) its boundary is a C^2 hypersurface of $\mathbb{R}^n$, meaning it is locally parametrizable by C^2 charts (in particular, this gives that K is smooth in the sense described above) and

(ii) $\det(d\xi_K)_x > 0$ for every $x \in \partial K$, where $(d\xi_K)_x$ denotes the differential map of ξ_K at x, and the determinant is taken by considering the matrix representation of $(d\xi_K)_x$ obtained when we fix a same basis on the tangent spaces $T_x\partial K$ and $T_{\xi_K(x)}\mathbb{S}^{n-1}$ (it is clear that $T_x\partial K = T_{\xi_K(x)}\mathbb{S}^{n-1} = \xi_K(x)^\perp$).

It is a standard result of differential geometry that the Gauss map of a C^2 hypersurface of $\mathbb{R}^n$ is differentiable; this is easily verified working with local parametrizations. Thus, we have that $(d\xi_K)_x$ is indeed well-defined for each $x \in \partial K$. The number

$$\kappa(x) := \det(d\xi_K)_x$$

is the *Gauss curvature* of ∂K at x. This is a widely studied concept of differential geometry having a lot of equivalent definitions and geometric interpretations. For us, it is sufficient to work with the definition above; we will show that, roughly speaking, for a C^2_+ convex body K the relation between μ_K and $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ can be described in terms of the Gauss curvature of ∂K. It is also useful to define the function $\bar{\kappa}\colon \mathbb{S}^{n-1} \to \mathbb{R}$ as

$$\bar{\kappa}(u) = \kappa\big(\xi_K^{-1}(u)\big),$$

for each $u \in \mathbb{S}^{n-1}$. That is, $\bar{\kappa}(u)$ is the Gauss curvature of ∂K at the point where the unit outer normal vector is u.

Condition **(ii)** yields that K is strictly convex: in fact, if this is not the case, then ∂K contains a segment s, and all the points of s have the same unit outer normal. Consequently, if v is a vector in the direction of s, then it is easy to see that $(d\xi_K)_x(v) = 0$ for any $x \in \operatorname{relint}(s)$. Hence, if K is a C^2_+ convex body, then it is smooth and strictly convex, from which it follows that its Gauss map ξ_K is a bijection. Using **(ii)** and the classical inverse function theorem (see, e.g., [1526]), we obtain that ξ_K is a diffeomorphism.

Proposition 7.4.1. *Let $K \subseteq \mathbb{R}^n$ be a C^2_+ convex body. Then for every measurable function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$, we have*

$$\int_{\mathbb{S}^{n-1}} f(u)\, dS_{n-1}(K,u) = \int_{\mathbb{S}^{n-1}} f(u)\, d\mu_K(u) = \int_{\mathbb{S}^{n-1}} f(u) \frac{1}{\bar{\kappa}(u)}\, d\mathcal{H}^{n-1}(u).$$

Remark 7.4.1. Of course, we already have the first equality, and we put it here only for the sake of summarization.

Proof. In view of the well-known fact that the differential of the Gauss map is always self-adjoint (and hence so it is the inverse), this result is a consequence of the area formula (A.6.1). Notice that

$$(d\xi_K^{-1})_u = \big((d\xi_K)_{\xi_K^{-1}(u)}\big)^{-1}$$

for each $u \in \mathbb{S}^{n-1}$, which gives

$$\det\big((d\xi_K^{-1})_u\big) = \big[\det\big((d\xi_K)_{\xi_K^{-1}(u)}\big)\big]^{-1}.$$

Consequently,

$$\sqrt{\det\big((d\xi_K^{-1})_u (d\xi_K^{-1})_u^*\big)} = \det\big((d\xi_K^{-1})_u\big) = \big[\det\big((d\xi_K)_{\xi_K^{-1}(u)}\big)\big]^{-1}.$$

Applying the area formula for $F = \xi_K^{-1}$, which is a diffeomorphism, observing that the equality above gives the term $\sqrt{\det\big((dF_u)(dF_u)^*\big)}$ of the area formula, and also using (7.4.1), we get

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} f(u)\, d\mu_K &= \int_{\partial K} f\circ\xi_K(x)\, d\mathcal{H}^{n-1}(x) = \int_{\xi_K^{-1}(\mathbb{S}^{n-1})} f\circ\xi_K(x)\, d\mathcal{H}^{n-1}(x) = \\ = \int_{\mathbb{S}^{n-1}} f(u)\det\big((d\xi_K^{-1})_u\big)\, d\mathcal{H}^{n-1}(u) &= \int_{\mathbb{S}^{n-1}} f(u)\big[\det\big((d\xi_K)_{\xi_K^{-1}(u)}\big)\big]^{-1}\, d\mathcal{H}^{n-1}(u) = \\ &= \int_{\mathbb{S}^{n-1}} f(u)\frac{1}{\bar{\kappa}(u)}\, d\mathcal{H}^{n-1}(u),\end{aligned}$$

and this concludes the proof. □

Remark 7.4.2. In the language of measure theory, we say that $\bar{\kappa}^{-1}$ is the *density* of the measure $S_{n-1}(K,\cdot)$ with respect to the measure $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ because, for any Borel set $E \subseteq \mathbb{S}^{n-1}$, setting $f = 1_E$, we have

$$S_{n-1}(K,E) = \int_{\mathbb{S}^{n-1}} 1_E(u)\, dS_{n-1}(K,u) = \int_{\mathbb{S}^{n-1}} 1_E(u)\bar{\kappa}(u)^{-1}\, d\mathcal{H}^{n-1}(u) = \\ = \int_E \bar{\kappa}(u)^{-1}\, d\mathcal{H}^{n-1}(u). \tag{7.4.2}$$

In particular, this leads to a formula for the surface area $S(K)$ of a C^2_+ body K in terms of its Gaussian curvature. We simply let $E = \mathbb{S}^{n-1}$ in the equality above and use (7.2.1) to obtain

$$S(K) = S_{n-1}(K,\mathbb{S}^{n-1}) = \int_{\mathbb{S}^{n-1}} \bar{\kappa}(u)^{-1} d\mathcal{H}^{n-1}(u). \tag{7.4.3}$$

We remind the reader that the surface area of a convex body K can be equivalently defined as the total $(n-1)$-dimensional Hausdorff measure of K, as the Minkowski surface area of K or via approximations by polytopes.

In view of Proposition 7.4.1, a terminology which is often used in the literature is

$$\frac{dS_{n-1}(K,\cdot)}{d\mathcal{H}^{n-1}} = \bar{\kappa}^{-1},$$

or else $dS_{n-1}(K,\cdot) = \bar{\kappa}^{-1} d\mathcal{H}^{n-1}$. Again, of course the same holds if we replace $S_{n-1}(K,\cdot)$ by μ_K, since we have proved that these measures coincide.

A consequence of Proposition 7.4.1 is that the *total curvature* of any C^2_+ convex body, which is defined as the integral of the Gauss curvature of ∂K over ∂K with respect to the $(n-1)$-dimensional Hausdorff measure, equals $\mathrm{vol}_{n-1}(\mathbb{S}^{n-1})$. In fact, using also (7.4.1), we get

$$\int_{\partial K} \kappa(x)\, d\mathcal{H}^{n-1}(x) = \int_{\mathbb{S}^{n-1}} \kappa\big(\xi_K^{-1}(u)\big)\, d\mu_K(u) = \int_{\mathbb{S}^{n-1}} \bar{\kappa}(u)\bar{\kappa}(u)^{-1}\, d\mathcal{H}^{n-1}(u) = \\ = \int_{\mathbb{S}^{n-1}} d\mathcal{H}^{n-1}(u) = \mathrm{vol}_{n-1}(\mathbb{S}^{n-1}).$$

This is related to a very important topological invariant called the *Euler characteristic*. Seen as a topological manifold, the boundary of a convex body and the sphere are homeomorphic, and so they have the same Euler characteristic.

We can go even further and find densities for the other j-th surface area measures of a C^2_+ body K (with respect to $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$) in terms of its *principal curvatures*. We explain how this works, but giving the formal proofs is beyond the scope of this book. For the interested reader, we refer to [2167].

At each point of ∂K, the differential $d(\xi_K)_x$ of the Gauss map has $n-1$ positive eigenvalues which we denote by $\lambda_1(x),\dots,\lambda_{n-1}(x)$. For each $j = 1,\dots,n-1$, the *j-th curvature* of ∂K at x is the number

$$H_j(x) := \binom{n-1}{j}^{-1} \sum_{1\le i_1<\ldots<i_j\le n-1} \lambda_{i_1}(x)\dots\lambda_{i_j}(x).$$

We also make the convention that $H_0(x) = 1$ for every $x \in \partial K$. Notice that H_{n-1} agrees with the Gauss curvature of ∂K, and $H_1(x)$ is the usual *mean curvature* of ∂K at x. Now for each $j = 0,\dots,n-1$ and any $u \in \mathbb{S}^{n-1}$, we define

$$s_j(u) := \frac{H_{n-1-j}(\xi_K^{-1}(u))}{H_{n-1}(\xi_K^{-1}(u))} = \frac{H_{n-1-j}(\xi_K^{-1}(u))}{\bar{\kappa}(u)}.$$

Here we observe that s_{n-1} is the same as $\bar{\kappa}^{-1}$ (and $s_0 = 1$, of course). It turns out that, for each $j = 0, \dots, n-1$, the function s_j is the density of $S_j(K, \cdot)$ with respect to $\mathcal{H}^{n-1}$, that is,

$$\frac{dS_j(K, \cdot)}{d\mathcal{H}^{n-1}} = s_j,$$

meaning that for every Borel set $E \subseteq \mathbb{S}^{n-1}$, we have

$$S_j(K, E) = \int_E s_j(u)\, d\mathcal{H}^{n-1}(u).$$

Since the density of $S_{n-1}(K, \cdot)$ with respect to $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ is $\bar{\kappa}(u)^{-1}$, and by the change of variables (7.4.1), we also have

$$S_j(K, E) = \int_E s_j(u)\, d\mathcal{H}^{n-1}(u) = \int_E \frac{H_{n-1-j}(\xi_K^{-1}(u))}{\bar{\kappa}(u)}\, d\mathcal{H}^{n-1}(u) =$$
$$= \int_E H_{n-1-j}(\xi_K^{-1}(u))\, dS_{n-1}(K, u) = \int_{\xi_K^{-1}(E)} H_{n-1-j}(x)\, d\mathcal{H}^{n-1}(x).$$

That is, the j-th surface area measures of K can be obtained by integrating the j-th curvatures of ∂K. As a particular consequence, we have an expression for the quermassintegrals of K in terms of these curvatures. Using (7.2.2), we obtain that, for each $j = 1, \dots, n$,

$$W_j(K) = \frac{1}{n} S_{n-j}(K, \mathbb{S}^{n-1}) = \frac{1}{n} \int_{\partial K} H_{j-1}(x)\, d\mathcal{H}^{n-1}(x).$$

Actually, there is a way of getting these equalities related to quermassintegrals and curvature without using mixed surface area measures. To do so, we compute the volume of the "tube" between K and $K + \varepsilon B^n$ in two different ways. On the one hand, we use mixed volumes to obtain a polynomial on ε whose coefficients are given in terms of the quermassintegrals of K. On the other hand, we consider a parametrization of the tube and compute its volume in the "differential geometric way". This will lead to a polynomial in ε whose coefficients are given in terms of the integrals of the j-th curvatures of ∂K. We obtain the equalities above by comparing the coefficients of these polynomials.

7.5 Exercises

Exercise 7.1. *Prove that for any $K_1, \dots, K_n \in CC^n$ the mixed surface area measure $S(K_1, \dots, K_n, \cdot)$ has its centroid at the origin. That is,*

$$\int_{\mathbb{S}^{n-1}} v\, dS(K_1, \dots, K_n, v) = 0_{\mathbb{R}^n}.$$

Remark 7.5.1. In particular, this shows that $S_{n-1}(K, \cdot)$ has its centroid at the origin also in the case when $K \in CC^n$ is not a convex body. Notice that in Section 7.3 only convex bodies were considered. See also Exercise 7.27 ahead.

Exercise 7.2. *Let $K_1, \dots, K_{n-1} \subseteq \mathbb{R}^n$ be convex bodies. Prove that the mixed surface area measure $S(K_1, \dots, K_{n-1}, \cdot)$ is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$.*

We introduce some terminology for the next exercises. A *translation family of convex bodies* is the family of translates of a given convex body $K \subseteq \mathbb{R}^n$. Notice carefully that, as a consequence of Theorem 7.2.1, we have that $S_{n-1}(K, \cdot) = S_{n-1}(L, \cdot)$ if and only if K and L belong to a same translation family of convex bodies.

From now on, we denote the *translation family* of K by $[K]$. Of course, it consists of all translates of K. The concepts of Minkowski sum and scalar multiplication extend naturally for translation families of convex bodies: we just let $[K]+[L] = [K+L]$ and $\lambda[K] = [\lambda K]$ for any $K, L \in \mathcal{K}^n$ and every $\lambda > 0$. In view of the comment above, we also write $S_{n-1}([K], \cdot) := S_{n-1}(K, \cdot)$ for any convex body $K \subseteq \mathbb{R}^n$. In the same way, volumes and mixed volumes can be seen as functionals defined for translation families of convex bodies.

Exercise 7.3. *From the previous exercises, conclude that for any $K_1, \dots, K_{n-1} \in \mathcal{K}^n$ the mixed surface area measure $S(K_1, \dots, K_{n-1}, \cdot)$ is the surface area measure of some convex body, and hence of a unique translation family of convex bodies.*

Remark 7.5.2. For given $K_1, \dots, K_{n-1} \in \mathcal{K}^n$, denote by $[K_1, \dots, K_{n-1}]$ the translation family of convex bodies satisfying

$$S_{n-1}([K_1, \dots, K_{n-1}], \cdot) = S(K_1, \dots, K_{n-1}, \cdot).$$

Then $[K_1, \dots, K_{n-1}]$ is called the *mixed body* of $K_1, \dots, K_{n-1}$. Next, we investigate some properties of mixed bodies. These are mostly taken from [1629].

Exercise 7.4. *Let $L_1, K_1, \dots, K_{n-1} \in \mathcal{K}^n$ and $\lambda_1, \dots, \lambda_{n-1} > 0$ be arbitrarily given. Verify that*

(a) $[\lambda_1 K_1, \dots, \lambda_{n-1} K_{n-1}] = (\lambda_1 \dots \lambda_{n-1})^{1/(n-1)} [K_1, \dots, K_{n-1}]$.

(b) $V\big([K_1, \dots, K_{n-1}][n-1], L_1\big) = V(K_1, \dots, K_{n-1}, L_1)$.

Exercise 7.5. *Prove that for any $K_1, \dots, K_{n-1} \in \mathcal{K}^n$, the inequality*

$$\mathrm{vol}_n([K_1, \dots, K_{n-1}])^{n-1} \leq \frac{1}{\kappa_n} V(B^n, K_1, \dots, K_{n-1})^n$$

holds, with equality if and only if $[K_1, \dots, K_{n-1}]$ is homothetic to a ball.

Hint: Use the 1st Minkowski inequality (Theorem 6.3.1).

Exercise 7.6. *Let $n \geq 2$ and $K \in CC^n$ be such that $\dim(K) \leq n-2$. Verify that $S_{n-1}(K, \cdot)$ is the zero measure on $\mathbb{S}^{n-1}$.*

Exercise 7.7. **(a)** *Prove that in the case where $n = 2$, we may replace $\mathcal{K}^n$ by CC^n in Theorem 7.2.1.*

(b) *Explain why for $n \geq 3$ we cannot replace $\mathcal{K}^n$ by CC^n in Theorem 7.2.1, even if we assume the additional hypothesis that the involved mixed surface area measures are non-zero.*

Exercise 7.8. *Let $K \subseteq \mathbb{R}^n$ be a convex body with the property that*

$$V(K[n-1], L) = V(K[n-1], -L)$$

for every $L \in CC^n$. Prove that K is centrally symmetric.

Exercise 7.9. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope. Prove that P is centrally symmetric if and only if for each facet F of P there is a facet $\tilde{F}$ of P having opposite unit outer normal vector as F, and such that* $\mathrm{vol}_{n-1}(F) = \mathrm{vol}_{n-1}(\tilde{F})$.

Exercise 7.10. **(a)** *Show that for any $L, C_1, \ldots, C_{n-1} \in CC^n$ and every $T \in \mathrm{GL}(n)$, we have*

$$\int_{\mathbb{S}^{n-1}} h_{TL}(u)\, dS(TC_1, \ldots, TC_{n-1}, u) = |\det(T)| \int_{\mathbb{S}^{n-1}} h_L(u)\, dS(C_1, \ldots, C_{n-1}, u).$$

(b) *Conclude that for any $f \in C(\mathbb{S}^{n-1})$, all $T \in \mathrm{GL}(n)$ and every $C_1, \ldots, C_{n-1} \in CC^n$, we have the equality*

$$\int_{\mathbb{S}^{n-1}} f(u)\, dS(TC_1, \ldots, TC_{n-1}, u) =$$
$$= |\det(T)| \int_{\mathbb{S}^{n-1}} f \circ (T^*)^{-1}(u)\, dS(C_1, \ldots, C_{n-1}, u).$$

Exercise 7.11. *Verify that the following expression for the surface area of a convex body $K \subseteq \mathbb{R}^n$ holds:*

$$S(K) = \frac{1}{2\kappa_{n-1}} \int_{\mathbb{S}^{n-1}} \int_{\partial K} |\langle u_K(x), v\rangle|\, d\mathcal{H}^{n-1}(x) d\mathcal{H}^{n-1}(v),$$

where u_K stands for the Gauss map of ∂K (defined almost everywhere; see Remark 7.2.6).

Exercise 7.12. *Recall that we denote by $M_u(K)$ the Minkowski symmetrization of $K \in CC^n$ in the direction of $u \in \mathbb{S}^{n-1}$, introduced in Exercise 2.58. Prove that*

$$W_i(M_u(K)) \geq W_i(K) \tag{7.5.1}$$

for any $K \in CC^n$, $u \in \mathbb{S}^{n-1}$ and $0 \leq i \leq n-1$.

Remark 7.5.3. From Exercise 5.4, we have that Minkowski symmetrization preserves mean width, and hence equality in (7.5.1) always holds in the case where $i = n-1$. In Exercise 8.12, we characterize the equality case for $i \in \{0, \ldots, n-2\}$.

Exercise 7.13. *Prove the following extension of Theorem 7.2.2: for any $K_1, \ldots, K_{n-1} \in CC^n$ and every $u \in \mathbb{S}^{n-1}$, show that the equality*

$$V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(K_1, \ldots, K_{n-1}, v)$$

holds. Conclude that

$$\int_{\mathbb{S}^{n-1}} V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big)\, d\mathcal{H}^{n-1}(u) =$$
$$= n\kappa_{n-1} V(K_1, \ldots, K_{n-1}, B^n).$$

Remark 7.5.4. Notice that we may reobtain Theorem 6.5.2 directly from the equality above.

Exercise 7.14. **(a)** *Let $i \in \{1, \ldots, n-1\}$, $K \in CC^n$, and $u \in \mathbb{S}^{n-1}$. Show that*

$$\frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS_i(K, v) = W_i^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K)\big),$$

where $W_i^{(n-1)}$ denotes the i-th quermassintegral of $\mathbb{R}^{n-1}$.

Fix $K \in CC^n$ and $i \in \{1, \dots, n-1\}$. For each $\lambda \geq 0$ and $u \in \mathbb{S}^{n-1}$, let

$$h(\Pi_i K, \lambda u) := \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle \lambda u, v \rangle| \, dS_i(K, v).$$

Of course, this defines a function $h(\Pi_i K, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$.

(b) *Prove that the function $h(\Pi_i K, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$ is sublinear. Conclude that it is the support function of a set from CC^n (which we denote by $\Pi_i K$, hence justifying the notation that we introduced above).*

(c) *Show that if K is a convex body and $i \in \{1, \dots, n-1\}$, then $\Pi_i K$ is a convex body symmetric with respect to the origin.*

Remark 7.5.5. In the case $i = n-1$, we obtain the so-called *projection body* of K. This construction is investigated in Section 10.3.

Exercise 7.15. *Let $K \in CC^n$, where $n \geq 2$.*

(a) *Verify that for any $u \in \mathbb{S}^{n-1}$ and every $v \in u^\perp$, one has that*

$$h\big(\mathrm{proj}_{u^\perp}(K), v\big) = h(K, v).$$

(b) *Prove that*

$$h(\Pi_1 K, u) = \frac{1}{n-1} \int_{\mathbb{S}^{n-1} \cap u^\perp} h_K(v) \, d\mathcal{H}^{n-2}(v)$$

for any $u \in \mathbb{S}^{n-1}$.

(c) *Conclude that $\Pi_1(K + L) = \Pi_1(K) + \Pi_1(L)$ for any $K, L \in CC^n$.*

The next exercise yields a proof of the equality $\Pi_1(K + L) = \Pi_1(K) + \Pi_1(L)$ which does not rely on the integral expression given in item **(b)** above.

Exercise 7.16. *Show that*

$$S_1(\lambda K + L, \cdot) = \lambda S_1(K, \cdot) + S_1(L, \cdot)$$

for any $K, L \in CC^n$ and $\lambda > 0$. In other words, the 1st order surface area measure can be seen as a Minkowski linear map from CC^n to the space of finite Borel measures of $\mathbb{S}^{n-1}$.

Exercise 7.17. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Prove that the measure*

$$\nu(\cdot) := S_{n-1}(K, \cdot) + S_{n-1}(L, \cdot)$$

has its centroid at the origin, and is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$. Conclude that ν is the surface area measure of some convex body, and hence of a unique translation family of convex bodies.

Remark 7.5.6. This observation gives a new notion of addition in the space of convex bodies, which is called *Blaschke addition*. For $K, L \in \mathcal{K}^n$, we define the *Blaschke sum* of K and L as the unique convex body $K\#L$ having its centroid at the origin and such that

$$S_{n-1}(K\#L,\cdot) = S_{n-1}(K,\cdot) + S_{n-1}(L,\cdot).$$

This equality is satisfied for any body in the translation family of $K\#L$, and the condition on the centroid lying in the origin appears because we want to choose a just one of these bodies.

Recall that to deal with mixed bodies, we worked with translation families rather than with convex bodies (the condition on the centroid was not adopted). Of course, this can be done here as well. The translation family $[K\#L]$ of $K\#L$ satisfies

$$S_{n-1}([K\#L],\cdot) = S_{n-1}([K],\cdot) + S_{n-1}([L],\cdot).$$

Hence, we may write $[K]\#[L] = [K\#L]$, thus regarding the Blaschke addition as an operation between translation families of convex bodies. It is clear that both approaches are equivalent.

Exercise 7.18. *Let $L_1, K_1, K_2, \ldots, K_{n-1}$ be convex bodies in $\mathbb{R}^n$, and recall the notation introduced in Remark 7.5.2. Verify that*

$$[K_1 + L_1, K_2, \ldots, K_{n-1}] = [K_1, K_2, \ldots, K_{n-1}]\#[L_1, K_2, \ldots, K_{n-1}].$$

Exercise 7.19. *Let $K, L \in \mathcal{K}^n$ and $T \in \mathrm{SL}(n)$. Prove that $TK\#TL = T(K\#L)$.*

Exercise 7.20 (Kneser-Süss inequality)**.** *Prove that for any $K, L \in \mathcal{K}^n$, we have the inequality*

$$\mathrm{vol}_n(K\#L)^{(n-1)/n} \geq \mathrm{vol}_n(K)^{(n-1)/n} + \mathrm{vol}_n(L)^{(n-1)/n}.$$

Show that equality holds if and only if K and L are homothetic.

Exercise 7.21. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Verify that*

$$\Pi_{n-1}(K\#L) = \Pi_{n-1}K + \Pi_{n-1}L,$$

where Π_{n-1} is as defined in Exercise 7.14.

Exercise 7.22. *Let $K, L \subseteq \mathbb{R}^n$ be arbitrarily given convex bodies. Recall that $[K_1, \ldots, K_{n-1}]$ denotes the mixed body of $K_1, \ldots, K_{n-1} \in \mathcal{K}^n$, which, as we remember, is a translation family of convex bodies. Letting $[K + L]$ be the translation family of $K + L$, show that*

$$[K + L] =$$
$$= [K, \ldots, K]\#\binom{n-1}{1}^{\frac{1}{n-1}} \big[K[n-2], L\big]\#\ldots\#\binom{n-1}{n-2}^{\frac{1}{n-1}} \big[K, L[n-2]\big]\#[L, \ldots, L],$$

where $[K[n-i-1], L[i]]$ denotes the mixed body with $n-i-1$ copies of K and i copies of L, of course.

Let $K \subseteq \mathbb{R}^n$ be a convex body whose boundary is C^2_+. Denote by $f(K,\cdot)\colon \mathbb{S}^{n-1} \to \mathbb{R}$ the function which associates each unit vector $u \in \mathbb{S}^{n-1}$ to the inverse of the Gauss curvature of ∂K at the point where u is the unit outer normal vector. The *affine surface area* of K is defined as

$$\Omega(K) := \int_{\mathbb{S}^{n-1}} f(K,u)^{\frac{n}{n+1}}\, d\mathcal{H}^{n-1}(u).$$

For more on this important concept, we refer the reader to the recent survey [2200] by Schütt and Werner.

Exercise 7.23. *Show that for any convex body with C^2_+ boundary, we have the equality*

$$\Omega(K) = \int_{\partial K} \kappa(x)^{\frac{1}{n+1}}\, d\mathcal{H}^{n-1}(x),$$

where $\kappa(x)$ denotes the Gauss curvature of ∂K at x, as usual.

Let μ be a finite Borel measure on $\mathbb{S}^{n-1}$. We say that μ is *concentrated on a closed hemisphere* if there exists a vector $u_0 \in \mathbb{S}^{n-1}$ for which the hemisphere $H := \mathbb{S}^{n-1} \cap \mathbf{h}^+(u_0, 0)$ has the property that $\mu(\mathbb{S}^{n-1} \setminus H) = 0$.

Exercise 7.24. *Let μ be a finite Borel measure that is not concentrated on a great subsphere and has its centroid at the origin. Prove that μ is not concentrated on a closed hemisphere. Conclude that the surface area measure of any given convex body is not concentrated on a closed hemisphere.*

For the next exercise, we recall that for a given fixed $u \in \mathbb{S}^{n-1}$, we define and denote the *positive part* and the *negative part* of the function $x \mapsto \langle u, x\rangle$ as

$$\langle u, x\rangle_+ := \max\{\langle u, x\rangle, 0\} \quad \text{and}$$
$$\langle u, x\rangle_- := \min\{\langle u, x\rangle, 0\},$$

respectively.

Exercise 7.25. *Let $K \in CC^n$ and $u \in \mathbb{S}^{n-1}$ be arbitrarily given.*

(a) *Verify that*

$$h(\mathrm{seg}[0_{\mathbb{R}^n}, u], x) = \langle u, x\rangle_+$$

for any $x \in \mathbb{R}^n$.

(b) *Show that*

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+\, dS_{n-1}(K, v) = \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big),$$

and conclude that

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_-\, dS_{n-1}(K, v) = -\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big).$$

(c) *Prove the following estimate:*

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+\, dS_{n-1}(K, v) \geq \frac{\mathrm{vol}_n(K)}{2\max_{\mathbb{S}^{n-1}} h_K}.$$

Exercise 7.26. *Extend item* **(b)** *of the previous exercise showing that for any $K_1, \ldots, K_{n-1} \in CC^n$, and every $u \in \mathbb{S}^{n-1}$, one has*

$$\int_{\mathbb{S}^{n-1}} \langle u, v\rangle_+\, dS(K_1, \ldots, K_{n-1}, v) = V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \ldots, \mathrm{proj}_{u^\perp}(K_{n-1})\big).$$

As it was announced at the end of Section 7.3, in the next exercise, we give a complete characterization of the area measure $S_{n-1}(K, \cdot)$ in the case where K is an $(n-1)$-dimensional compact convex set.

Exercise 7.27. *Let $K \in CC^n$ and $u \in \mathbb{S}^{n-1}$. Assume that $S_{n-1}(K, \cdot)$ is non-zero. Prove that the following statements are equivalent:*

(a) $S_{n-1}(K,\cdot)$ *is* concentrated on $\{-u,u\}$, *that is*

$$S_{n-1}(K,\mathbb{S}^{n-1}\setminus\{-u,u\})=0.$$

(b) $\dim(K)=n-1$ *and* $K\subseteq u^{\perp}$, *up to a translation.*

(c) *The equality*

$$\int_{\mathbb{S}^{n-1}}|\langle u,v\rangle|\,dS_{n-1}(K,v)=S_{n-1}(K,\mathbb{S}^{n-1})$$

holds.

(d) *There exist numbers* $\alpha,\beta\geq 0$ *such that*

$$S_{n-1}(K,\cdot)=\alpha\delta_u+\beta\delta_{-u}.$$

In this case, we have that $\alpha=\beta=\mathrm{vol}_{n-1}(K)$.

Exercise 7.28. *Let* $f,h\in C(\mathbb{S}^{n-1})$. *Prove that*

$$\int_{\mathbb{S}^{n-1}}f(v)\,dS_{n-1}(K,v)=\int_{\mathbb{S}^{n-1}}h(v)\,dS_{n-1}(K,v)$$

for every $K\in CC^n$ *if and only if* $f-h$ *is the restriction of a linear function to* $\mathbb{S}^{n-1}$.

In the next exercise, we give a reverse Minkowski inequality which provides an upper bound for the mixed volume $V(K,L,\dots,L)$ in terms of intrinsic volumes of K and L. We remind the reader that V_j stands for the j-th intrinsic volume (see Section 6.4 for definitions). The reader may use the following theorem.

Theorem 7.5.1. [1263, Theorem 4.6] *If* $K\in CC^n$ *has positive diameter, then*

$$V_1(K)\geq 2r_{\mathrm{out}}(K).$$

Equality holds if and only if K *is a segment.*

Exercise 7.29. *Let* $K,L\in CC^n$. *Prove that*

$$V(K,L[n-1])\leq\frac{1}{n}V_1(K)V_{n-1}(L).$$

Show that if $\dim(K)\geq 1$ *and* $\dim(L)\geq n-1$, *then equality holds if and only if* K *is a segment and* L *is contained in a hyperplane orthogonal to* K.

7.6 Notes

Chapter 7 mainly refers to (j-th) surface area measures, curvature, and Minkowski's existence theorem. We decided to order the notes in such a way that these themes are treated in two subsections.

7.6.1 Minkowski's existence theorem and related topics

The notions *Minkowski's existence theorem* or also *Minkowski's problem* are used in connection with a fundamental problem from the Brunn–Minkowski theory which concerns the existence and uniqueness of closed convex hypersurfaces whose curvatures, as functions of the outer normals, are prescribed. Thus, Minkowski's problem asks for necessary and sufficient conditions on a finite Borel measure on the unit sphere that guarantees that it is the surface area measure of a convex body. Furthermore, the *Alexandrov problem*, or the Minkowski problem for Alexandrov's integral curvature, asks for analogous conditions for such Borel measures, to be the Alexandrov integral curvature of some convex body. Similarly, the L_p *Minkowski's problem* asks for existence and uniqueness of a convex body such that its L_p area measure is a given Borel measure on the unit sphere. Of course, further generalizations exist, such as dual settings or concepts referring to Orlicz spaces.

BOOK PARTS, SURVEYS, AND BASIC ARTICLES

Starting with the three-dimensional case, *Minkowski's problem* (see Section 7.3 above) was introduced and first discussed by Minkowski himself (see Vol. II of [1807]); cf. there pp. 103-121 and pp. 230-276, where the polytope case is also extended to general convex bodies. Further contributions and extensions are given in the monographs of Bonnesen and Fenchel [446] (see §13), A. D. Alexandrov [39] (cf. Chapter VII), and Pogorelov [1980] (see §2 there). Important for deducing the general case via approximation from the polytopal one is also the article [857] of Fenchel and Jessen. A different approach is due to McMullen [1737], and a modified treatment for polytopes, referring simultaneously to the existence problem and the Minkowski's inequality, is given by Klain [1398]. Chapter 8 of Schneider's book [2167] and, particularly, Section 8.2 and its notes there contain the most comprehensive representation of this topic, giving outlooks to many related notions and topics. The aspects and viewpoints discussed there in detail refer to the following themes (which we give now with additional important references): *differential geometry* (see, e.g., Pogorelov [1980] and Gluck [981]), *analoguous subjects* (e.g., in physics; cf. Colesanti [635]), *algorithmic aspects* (see, e.g., Gritzmann and Hufnagel [1023]), *applications* (e.g., for random hyperplane systems in stochastic geometry and Blaschke bodies; see the book [2179] of Schneider and Weil), *Blaschke addition and Blaschke bodies* (cf. Section 3.3, the Notes 3.4, Section 6.1, and the Appendix 3 in Gardners monograph [930], as well as Section 6.2 and Notes 6.7 of Thompson's book [2361]), and *mixed bodies* (see Lutwak [1629]). Further books containing material on Minkowski's existence theorem or problem are Leichtweiss [1537] (Section 1.2), Artstein-Avidian et al. [134] (Section 1.6), Hug and Weil [1263] (Section 4.2), and Schneider [2174] (Sections 7.4 and 7.7).

We continue with (mostly more recent) research papers on the Minkowski problem which are not cited in the references above.

RECENT RESEARCH ARTICLES

Zhu [2557] solves the L_p *Minkowski problem* for an atomic measure whose support consists of vectors in general position, not all contained in a hemisphere. He proves that, under these conditions,

a polytopal solution always exists. This is the first step in the direction of the full solution to the problem of atomic measures. In the related paper [2558], Zhu discusses the L_p Minkowski problem in its relation to certain topics like equations of Monge-Ampère type, Sobolev inequalities, and flows. He proves that for $0 < p < 1$ and a discrete measure μ on the unit sphere, this measure μ is the L_p surface area measure of a convex polytope iff the support of μ is not concentrated on a closed hemisphere.

Klyachin [1424] shows that the existence of a convex polytope in n-space with prescribed values of curvatures at its vertices can be reduced to the existence of a particular *simplicial triangulation* of the unit sphere with prescribed measures of simplices. This existence is also verified with the given measures of simplices, but only for $n = 3$.

In [2559], Zhu answers the following two questions related to the L_p Minkowski problem: Suppose that the L_p surface area measure of convex bodies K_i converges weakly to that of a convex body K. Does the sequence K_i then converge to K in the Hausdorff metric? And if K_p and K_q are solutions of the L_p and L_q, Minkowski problems associated to the measure μ, respectively, does then, if $q \to p$, also K_q converge to K_p in the Hausdorff metric? In the first case, the answer is positive for all $p > 1$ and $p \neq n$ (completing earlier results), and the second problem has an affirmative answer, too. K. J. Böröczky and Trinh [414] give the complete solution of the planar L_p Minkowski problem by verifying the following statement: For $p \in (0, 1)$ and a non-trivial bounded Borel measure μ on the unit circle, μ is the L_p-surface area measure of a planar convex body, containing the origin, if and only if the support of μ does not consist of a pair of opposite vectors. Bianchi et al. [328] assume that the used measure has a density f with respect to the Hausdorff measure on the unit sphere; their task is to solve a differential equation on that sphere using also the support function of convex bodies. In this way, an *Orlicz extension* of the L_p version of the Minkowski problem is derived. The same authors deal in [329] with the L_p Minkowski problem. They study *smoothness of solutions* of the L_p Minkowski problem when $p < 1$ and the used measure has a positive and bounded density with respect to the Hausdorff measure on the sphere. Improving older results and using an associated Monge-Ampère equation, they give sufficient conditions under which the solutions of the L_p Minkowski problem with $p < 1$ are C^1-regular.

Based on the deep paper [1245] of Huang et al. which treats L_p Alexandrov integral curvature for the given solution of the respective L_p *Alexandrov problem* for $p > 0$, Zhao [2553] investigates the case $-1 < p < 0$ and presents the solution for polytopes centered at the origin. Mui [1863] directly continues this work and extends the results above by finding a necessary and sufficient condition for origin-symmetric convex bodies, also for the L_p-Alexandrov problem in case of $-1 < p < 0$. Introducing an L_p*-Gaussian surface area measure*, Liu [1588] succeeds in extending the Minkowski problem from Gaussian probability spaces to a so-called L_p-Gaussian problem, and he proves respective existence and uniqueness results for $p \geq 1$. The planar L_p-Minkowski problem is, for arbitrary reals p, comprehensively studied by Du in the recent paper [767]. This paper improves and completes many older related results; in particular, some interesting new uniqueness results are derived.

Although it is cited elsewhere, we have to bring the deep paper [1245] of Huang et al. also here. As a new family of geometric measures, the authors present the so-called *dual curvature measures* and study their associated Minkowski problems, which were both elusive in the dual Brunn–Minkowski theory. They treat the important subcase when the measure μ is even and the studied bodies are origin-symmetric.

Dual quermassintegrals and the fundamental geometric functionals in the *dual Brunn–Minkowski theory* are also the topic of Zhao's paper [2551]. There the corresponding *dual Minkowski problem* (which has the Alexandrov problem and the logarithmic Minkowski problem as special cases) is completely solved for the case $q < 0$, including the existence and the uniqueness part. The used methods work also in the case when the studied measure is not only even. These investigations are continued by the same author in [2552]. He introduces a new set of subspace mass inequalities and solves, under new sufficiency assumptions, the existence part of the even dual Minkowski problem. It should be noticed that, together with results obtained by K. J. Böröczky, Henk, and Pollehn [402],

the existence part of the *dual even Minkowski problem* is completely solved. Chen and Li [611] deal with the planar dual Minkowski problem, presenting the first existence result for this type of problem (in the case $q > 0$), without any assumptions regarding the evenness of the used measure μ. The authors emphasize that their methods cannot be applied in higher dimensions. Also, by K. J. Böröczky et al. [407], the dual Minkowski problem is deeply studied for symmetric bodies. The authors go on the lines of a variational method, where the used functional is representable as the sum of a dual quermassintegral and an entropy integral. Besides other results, Li, Sheng, and Wang [1566] succeed in giving a new proof for the classical Alexandrov problem (smooth case), and for the dual q-Minkowski problem coming from [1245] if $q < 0$. Other cases are derived also for even functions, and for the non-symmetric case an interesting counterexample is constructed.

Huang and Zhao [1248] derive results on the L_p *dual Minkowski problem* (using L_p dual curvature measure) with existence results in a weak sense. They also present existence and uniqueness results for the smooth case. This is continued by Chen, Huang, and Zhao in [605], where geometric variational methods are used to investigate the existence of smooth solutions of the L_p dual Minkowski problem (even case). The authors also look for conditions implying that the only even, smooth, and uniformly convex solution is the unit ball. Lu and Pu [1606] give the solution to the existence part of the L_p dual Minkowski problem for $0 < p < 1$, $q > 0$, and $p \neq q$. Also K. J. Böröczky and Fodor [395] get existence results on the L_p dual Minkowski problem; for $p > 1$ and the second parameter $q > 0$ also regularity results for the solutions of such problems are obtained.

Yijing [2504] investigates the existence and uniqueness of positive solutions of some *planar Orlicz Minkowski problems* for $0 < p < 1$. He obtains existence and uniqueness results on plane convex bodies that are not origin-symmetric. On the other hand, under certain circumstances, the non-existence of smooth convex bodies having the origin as interior point can occur. As the title says, Yijing and Duanzhi [2505] investigate the planar Orlicz Minkowski problem involving the log Minkowski problem. The Orlicz Minkowski problem was posed and largely solved by Haberl, Lutwak, Yang, and Zhang in [1124], where no evenness assumptions on the measure are made, and the authors restrict to the case $p = 0$. A further related paper is [2342] by Tan and Sun. Feng and He [858] introduce the *Orlicz integral curvature* and give basic properties and related references. They also present the *Orlicz Alexandrov problem* on convex bodies and give a complete, well-structured solution of it for even measures. As a tool, variational formulas for the entropy of convex bodies are used which are motivated by Huang et al. [1246]. Xie [2481] solves a (not necessarily even) end-point case of the volume-normalized Orlicz Minkowski problem considered by Haberl et al. in [1124]. There are also recent results on the *discrete Orlicz Minkowski problem* based on Zhu's paper [2557]. Namely, in [2557], it is shown that if $0 < p < 1$ and μ is a discrete, not necessarily even measure on the unit sphere, then μ is the L_p surface area measure of a polytope if and only if the support of μ is not concentrated on a closed hemisphere (offering also the extension to general convex bodies via approximation arguments). Wu et al. (see [2473] and [2474]) generalize this to the discrete Orlicz Minkowski problem: In [2473], the existence of the discrete Orlicz Minkowski problem for a continuously differentiable, strictly increasing Orlicz function satisfying some further criteria is considered. And in [2474], the same authors deal further with the discrete Orlicz Minkowski problem based on the concepts given by Haberl et al. [1124] for suitable functions. For example, they solve a case of the discrete Orlicz-Minkowski problem which corresponds to the case $p < 0$ of the L_p Minkowski problem.

Gardner et al. (cf. [936] and [937]) derive deep results treating *dual Orlicz-Minkowski-type problems*. In the first paper, the variational formulas of the general volume functional with respect to two different types of Orlicz linear additions are introduced, leading to a new *general dual Orlicz curvature measure*. Then the authors pose a new dual Orlicz-Minkowski problem fitting to this measure, and some special cases from this framework are solved. In addition, some interesting *dual Orlicz-Minkowski-type inequalities* are derived. In the second paper [937], also *general dual Orlicz curvature measures* and their associated *general dual Orlicz-Minkowski problems* are studied, where general means that the assumptions are broader (e.g., the used convex bodies can have the origin as boundary point). So some

existence results for the general dual Orlicz Minkowski problem are obtained. Also in [610], Chen et al. study the dual Orlicz-Minkowski problem. The authors use a geometric flow involving Gauss curvature and functions of normal vectors and radial vectors. On this way, they obtain a new existence result of smooth and even solutions.

Using the support functions of all convex bodies with the origin as an interior point and with the volume of their polars equal to that of the standard Euclidean ball, Luo, Ye, and Zhu [1624] introduce the so-called *polar Orlicz-Minkowski problem*. They give conditions for its solvability and the uniqueness of solutions. In addition, questions of existence, uniqueness, and continuity of the so-called p-capacitary Orlicz-Petty bodies are given, a notion coming from a special functional on the set of convex bodies (the Orlicz mixed p-capacity). Mou and Zhu [1860] study the Orlicz-Minkowski problem for a measure with density that has a positive degree of concavity, and the existence of the even case is clarified. Showing the existence of *Orlicz-Petty bodies*, the authors verify also the continuity of the corresponding Orlicz geominimal measure.

Livshyts [1591] generalizes the Minkowski problem and Alexandrov's theorem on centered measures on the sphere in the following way: Given a measure μ on Euclidean n-space, with positive degrees of concavity and homogeneity, a convex body is uniquely determined, up to μ-measure zero, by a *surface area measure of a convex set* weighted regarding μ. Under certain assumptions, a corresponding existence result is confirmed, and interesting applications in convexity are given. Using the standard Gaussian probability measure, Huang, Xi, and Zhao [1247] introduce the Gaussian surface area measure of a convex body and investigate the corresponding *Gaussian Minkowski problem* on Borel measures on the unit sphere. They prove a respective uniqueness result, for the symmetric case also an existence result, and an existence and uniqueness result for the smooth case.

Developing a very interesting *Brunn–Minkowski theory for coconvex sets* (see also our next subsection) and unbounded sets, analogues of notions like *mixed volume, surface area measure, mixed surface area measure, and cone volume measure* are introduced in Schneider's papers [2171] and [2173]. These new concepts are promising regarding extensions of the *Minkowski problem to coconvex sets and unbounded convex bodies*, cf. Section 7.7 of Schneider's very recent book [2174].

Finally, we mention here also the interesting expository paper [1528] of Lehmann and Xiao, discussing *relations between convex geometry, algebraic geometry, and complex geometry*. Among various other topics, complex-geometric analogues to Minkowski's existence theorem and also to Blaschke addition are presented.

7.6.2 Surface area measures and curvature measures

The notion of *surface area measures* was independently introduced by A. D. Alexandrov in 1937 and by Fenchel and Jessen in 1938 (see [37] and [857], respectively). A well-known curvature measure is *Alexandrov's integral curvature* (also called integral Gauss curvature). In a more general setting, curvature measures were introduced by Federer in 1959 (cf. [833]), namely treating families of sets being more general than convex bodies, and Schneider [2145] developed the corresponding theory for general compact, convex sets (see also his handbook survey [2152] and Chapter 8 of his monograph [2152]).

BOOK PARTS

We start with representations of results on convex surfaces represented in *monographs*, seeing them as boundaries of convex bodies K mainly without smoothness assumptions. Thus, the subcases interesting from the viewpoint of differential geometry are not in the focus, and also properties like strict convexity of K are not emphasized. The same holds for the case of convex polytopes (see, e.g., Alexandrov's monograph [39]).

Early books, treating that philosophy regarding curvature properties, are the following ones: A. D. Alexandrov [40] (treating, e.g., *approximation of two-dimensional manifolds of restricted curvature by polyhedral metrics* in 3-space, generalizing the Gaussian curvature and convergent metrics) and also the collection [38] (basic notions like *mixed volumes, surface and area measures, and the Alexandrov–Fenchel inequality*), Busemann [549] (for example, several types of curvature and their influence on the local shape of a surface, *intrinsic metrics and curvature*, local realization of metrics with non-negative curvature, etc.; cf., e.g., Chapters II, III, and Section 21), Pogorelov [1979] (summarizing his and Alexandrov's former monographs and articles on the *intrinsic geometry of convex surfaces*, regularity of convex surfaces with regular metric, related uniqueness results, *infinitesimal bending of convex surfaces*, convex surfaces with given *spherical map*, etc.; see mainly Chapters I, II, III, IV, V, and VII), Hadwiger [1134] (Minkowski's relative surface area, the Brunn–Minkowski inequality, the isoperimetric theorem as an inequality for *relative surface areas and measures of two arbitrary compact sets*; see Chapters 5 and 6), Burago and Zalgaller [531] (particularly interesting: *closed surfaces of bounded curvature*, various *definitions of area* and isoperimetric inequalities, also for sets more general than compacta; see Chapters 1 and 3), Leichtweiss [1531] (surface area and Cauchy's projection formula in §16, and *curvature properties* in §24), and Groemer [1042] (various approaches to the *notion of surface area*, Cauchy's surface area formula and several related inequalities, see especially Sections 2.4, 5.2, and 5.3).

Surface area is also a central theme of Thompson's monograph [2361], treating *the geometry of finite dimensional real Banach spaces* (= Minkowski geometry). Various definitions of surface area were explored in mainly three-dimensional Minkowski spaces. Using the cases of *Choquet–Busemann area* and *Holmes–Thompson area* (further concepts, like Benson area, are also treated, but shorter), the author broadly discusses the respective variants of the isoperimetric problem in such spaces. It turns out that in the first two cases, for constructing the discovered isoperimetrices, the notions of *intersection body, projection body*, and *polar body* applied to the corresponding unit balls play the key role; see also the interesting paper [2158] of Schneider, where the importance of the Holmes–Thompson surface area is underlined. (Also referring to convex surfaces, other parts of Thompson's book [2361] treat the *inner metric of such surfaces* bounding the unit ball. Thus, related geometric properties of unit spheres of normed spaces should also be mentioned here; see, e.g., Schäffer's monograph [2126] and Chapter 2 of Diestel's book [750].) In [2361], the greatest part of the material on surface area is concentrated on Chapters 5, 6, and 7. It is well known that, although the definition of convex bodies is based on affine concepts only, convexity topics (like also *curvature measures*) are mainly studied in the Euclidean setting.

However, there is an interesting affine geometry of convex sets, which is the theory of invariant entities and equivariant associated bodies with respect to the group of *volume-preserving affinities* in n-dimensional affine space. Leichtweiss took this as topic of his second book [1537], and the most important notion there is that of *affine surface area*. He presents the approach to this notion which is due to Petty and Lutwak, and (on the other hand) that due to himself as well as Schütt and Werner, and he shows their equivalence. *Affine characterizations of special types of convex bodies* (simplices, ellipsoids, affine cubes etc.; see also our subsection 2.9.1) are given, such as the *affine isoperimetric problem, the Blaschke-Santaló inequality*, and also *inequalities for difference, centroid, intersection, and projection bodies*.

Gardner approaches area measures via outer parallel sets in the complete appendix A.2 of his monograph [930], treating also *jth area measures obtained via approximation by polytopes*. And *mixed area measures* are introduced in appendix A.3. Various results in Chapters 3 and 9 need these and related notions and are based on theorems of A. D. Alexandrov, Blaschke, and others. More precisely, here we mean results on the *Blaschke body* (Section 3.3 of [930] and its Notes 3.4), *affine surface area* (Notes 9.3) and *affine isoperimetric inequalities* (Notes 9.4). We also refer to the appendix A.6, where, e.g., Cauchy's surface area formula is discussed. In the second part of her book [1857], Moszyńska particularly deals with curvature and surface area measures (the whole Chapter 10 is

dedicated to this theme). For example, she shows, using curvature measures, the relationship between the *mean curvature* considered in Section 9.1 and the *integral of the mean curvatures* in view of differential geometry. Although Morvan's monograph [1853] is mainly devoted to differential geometry, it treats also several further branches of mathematics which are close to convexity (among them polytope geometry, classical convexity, and computational themes like Delaunay triangulations), thus bringing together also several concepts and various aspects of curvature measures. Parts VI, VII, and VIII of [1853] contain interesting viewpoints on *curvatures and related approximation results*. For example, Part VI refers to the *Steiner formula*, for convex bodies and more general, and also *quermassintegrals* occur there.

In Section 4.2 of his monograph [2167], Schneider presents the most comprehensive representation of *curvature measures and area measures of order* j ($j = 0, 1, \ldots, n-1$) *for general n-dimensional convex bodies*. He shows how they are derived from the corresponding *support measures* (or generalized curvature measures, concentrated on the corresponding support elements) by using also the *local Steiner formula* and the notion of *intrinsic volumes*. For $j = n-1$, the area measures become the so-called *surface area measures*; in the smooth and in the strictly convex case, the more general support measures can be replaced by the *curvature measures and area measures*. Basic properties of these measures and their relations to each other are also studied in Section 4.2, and in the notes to this section various aspects and generalizations of used notions are discussed, e.g.: (local) *Steiner-type formulae* (e.g., also in *non-Euclidean spaces* or for *non-convex sets*), *sets of positive reach* (defined via unique nearest points), the *Steiner polynomial* of degree n (expressing the volume of outer parallel bodies) and its roots, inner parallel bodies, *axiomatic characterizations of curvature and area measures*, and also of support measures, *Hermitian curvature and area measures*, as well as *Hessian measures of convex functions*. Chapter 8 of Schneider's book [2167] is inspired by (uniqueness) results on the determination (i.e., on questions of existence, uniqueness and stability) of smooth closed convex hypersurfaces by curvature functions, such as Gauss curvature, mean curvature and respective generalizations. Neglecting differentiability assumptions leads to general curvature measures and area measures. Here also Minkowski's existence theorem (for given surface area measure of order $n-1$), Christoffel's problem, and area measures of intermediate order are studied, and related stability estimates are given. The notes in this chapter impressively present the knowledge around these topics. In Section 9.1 of [2167], also the L_p *surface area measure* is discussed (as part of the L_p Brunn–Minkowski theory), and Section 10.5 is devoted to (generalizations of) *curvature functions and affine surface areas*, inspired by affine differential geometry (see, e.g., also the book [1557] of Li, Simon, and Zhao, and Section 1.4 in the monograph [1537] of Leichtweiss). Various related *affine inequalities* are collected there, and in the notes to this section of [1537] further related or more general notions are surveyed.

From the viewpoint of functional analysis, *minimal surface area positions* are treated in Section 1.5 of the book [496] of Brazitikos et al., and also Chapter 2 of the monograph [133] of Artstein-Avidan et al. is devoted to classical positions of convex bodies, treating the *minimal positions for surface area and mean width*, and also *Ball's reverse isoperimetric inequality*.

Generalizing also topics from convexity, the monograph [2023] of Rataj and Zähle is dedicated to *curvature measures of sets of positive reach and fractal sets*, where the invariance under certain transformation groups (in the sense of Klein's Erlangen program) is a central aspect. In the first three chapters an appropriate introduction to geometric measure theory, convex geometry, and differential geometry is given, and Chapter 7 offers approximation procedures and characterization theorems referring to curvature measures.

Although mainly devoted to bodies of constant width, the book [1713] of Martini, Montejano, and Oliveros contains also passages which refer to surface area and curvature results of convex bodies in general. And it turns out that the results on constant width bodies are somehow also interesting for the general case. In this sense, Section 11.2 of [1713] is especially interesting (it is the part where aspects of *differential geometry* are treated), but also Sections 10.3 and 12.2 contain related material.

The monograph [1263] of Hug and Weil refers to surface area in Section 3.2, and to *mixed area measures* in Section 4.1 and (in other sections of Chapter 4) to further related topics, like the (local) Steiner formula, Minkowski's existence and uniqueness result, and (at the beginning of Chapter 5) Cauchy's projection formula.

Finally, we mention here Schneider's book [2174] treating the *theory of convex cones*. Curvature measures are presented in Section 2.5, surface area measures of coconvex sets are introduced in Section 7.1, and Minkowski's problem (see also our previous subsection) is covered in Section 7.7.

We continue with related *surveys*.

SURVEYS

Schneider's survey [2147] collects results on the boundary or face structure of convex bodies and related results on their curvature measures. With the latter the *area functions* (in the sense of Alexandrov and Fenchel–Jessen, see Alexandrov [37] and Fenchel–Jessen [857]) as well as *curvature measures* (in the sense of Federer [833]) are meant. All these are defined, various (also local) *properties and characterization theorems* are given, and related *integral geometric formulas* are described. Open problems are collected at the end of this very informative article.

A natural and comprehensive continuation is Schneider's handbook article [2152], clearly presenting the knowledge about properties of the boundary and on curvature and surface area measures of convex bodies. The first section discusses *degrees of deviation from strict convexity or smoothness*. The author collects results like this: to what extent boundary parts or points of convex bodies can fail to be strictly convex or smooth? For example, this yields degrees of smoothness or the clarification of the question how "large" the set of directions of non-degenerate line segments in the boundary of convex bodies can be. Then the notion of pointwise curvature (for the nonsmooth case) and its insufficiency for local descriptions of convex hypersurfaces is presented, and so, based on the more appropriate notions of curvature measures and surface area measures, some *generalized differential geometry for convex hypersurfaces* is developed. Further on, the author collects related *existence, uniqueness and stability results*, including nonsmooth versions of the related theorems of Minkowski and Alexandrov–Fenchel–Jessen (see above).

Here we mention also the handbook article [2182] of Schneider and Wieacker, where the authors use a *measure-theoretic approach* (in contrast to monographs existing in those days) and discuss convexity-related parts of integral geometry. E.g., after the introductory part in a section on intersection formulas, the relations of *curvature measures* of Federer [833] to *kinematic formulas in the integral geometry of convex bodies* are presented.

Schneider's exposition [2159] nicely demonstrates where and in which way measures occur in geometric convexity, particularly in view of measures associated with the boundary structure of a convex body. In particular, the (main) second part is devoted to *curvature measures*, suitably extending the differential geometric notions of mean curvature and Gauss curvature. Also, the classical *uniqueness and existence results referring to surface area measures* are presented (see also our previous subsection).

In [2162], Schneider surveys *techniques from integral geometry needed in stochastic geometry* which refer to convex bodies. In the second chapter, he introduces *curvature measures as local versions of the intrinsic volumes*, and extensions of classical formulae (like that of Steiner or Crofton) are presented in the context of *generalized curvature measures*. And in the fourth part *Holmes–Thompson areas* in smooth or hypergeometric Minkowski spaces play a role.

In the exposition [1872], Nagy, Schütt, and Werner present and derive relations between the concept of *halfspace depth* (one of the most promising tools for the efficient classifiers design for multivariate and functional data) and basic notions from convex geometry, e.g. *polytope approximation of general convex bodies* (therefore the link). Maximum halfspace depth can be seen as a *measure of symmetry* for random vectors, showing a further particular bridge to convexity. Another bridge is given by the strong

relation to the *convex floating body problem*. First a comprehensive historical survey on the notion of halfspace depth is given, and then the approximation of convex bodies by polytopes is discussed. The authors prove that the concept of convex floating bodies corresponds to the *upper level sets of the halfspace depth*. Summarizing this expository paper shows nice *deep connections between statistics and convexity*.

And now we present some *research articles*; most of them cannot be found in the references given above.

RECENT RESEARCH ARTICLES

Lamberg's dissertation [1490] deals with modelling asteroids simplified to be three-dimensional convex bodies. Measurements of reflected light intensity (i.e., *brightness* in terms of convexity) allow the determination of their *area functions* from which, via Minkowski's celebrated theorem, the convex body can be constructed; finally, this procedure determines the modelled shape of the asteroid. *Polytope approximation* is used, and inverting the procedure yields a description of the *reciprocal Gauss curvature* or, more generally, the area function of the convex surface bounding the asteroid.

In [2168], Schneider studies the concept of typical n-dimensional convex bodies in the sense of *Baire category*. He shows that such a typical body has the following properties: the spherical image of the set of umbilics of zero curvature has measure zero; the set of umbilics of infinite curvature is dense in the boundary and uncountable, and its spherical image has full measure in the unit sphere. For a typical convex set which is smooth and strictly convex, the author gives a complete picture of *curvatures of typical convex bodies*.

The strong influence of Minkowski problems on the L_p Brunn–Minkowski theory motivated Huang et al. [1245] to introduce a new family of geometric j-measures which they call *dual curvature measures*; they investigate also their respective Minkowski problems (cf. also our related subsection). To establish such "conceptually duals" of the usual curvature measures, also a transformation to the unit sphere (using the inverse radial mapping) is necessary. The *dual curvature measures and dual area measures* are then obtained via local Steiner-type formulas. For example, the dual area measures occur as (spherical) integrals of powers of radial functions. The authors of [1245] verify continuity and *valuation properties*, and they show that the dual curvature measures can be interpreted as differentials of the *dual quermassintegrals*. It turns out that this new family of j-measures connects well-known measures like Alexandrov's integral curvature and the cone volume measure (for $j = 0$ giving the Alexandrov problem, and for $j = n$ yielding the logarithmic Minkowski problem). The authors derive also a *Minkowski-type existence theorem*; they give a sufficient condition on an even Borel measure on the unit sphere to be the jth dual curvature measure of a convex body centered at the origin.

Zhao [2551] gives a complete solution for the case when $j < 0$, including existence and uniqueness, and his approach functions not only for even measures. Continuing these developments, Lutwak et al. [1642] construct L_p *dual curvature measures*. This yields a unified concept of mixed volume, e.g., including, as special cases of Minkowsk's classical first mixed volume, L_p *mixed volumes*, and *dual mixed volumes*. Establishing geometric inequalities and variational integral formulas for the unified mixed volumes and for the new L_p dual curvature measures, they succeed in showing surprising *cross-connections between the L_p Brunn–Minkowski theory and the dual Brunn–Minkowski theory*. Also the L_p dual Minkowski problem for L_p dual curvature measure is posed. Strongly related to this, Huang et al. [1246] solve the L_p-Alexandrov problem in the case $p > 0$; their proof is quite different to that given by Alexandrov for $p = 0$. Further on, for $p < 0$, they give a sufficient condition for the existence of solutions of the L_p-Alexandrov problem, for finite even Borel measures. And if the Borel measure has a density, then this condition is shown to be also necessary.

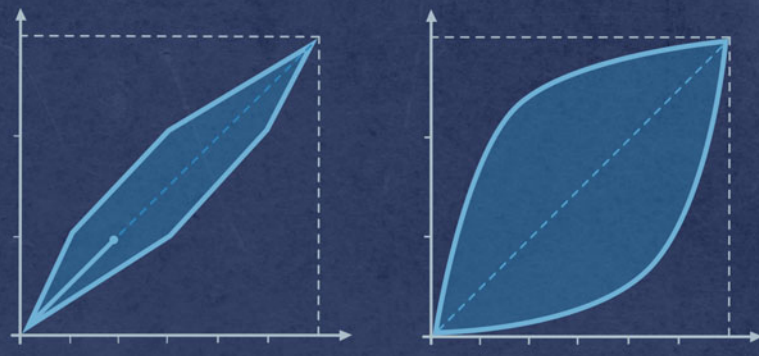
Schneider [2171] develops a *Brunn–Minkowski theory for coconvex sets*. Namely, if C is a fixed (but arbitrary) pointed closed convex cone with interior points and apex at the origin, a closed convex set $K \subset C$ is called C-close if $C \setminus K$ has finite volume. In this case, the set $C \setminus K$ is called

C-coconvex. Defining also the *cosum* of two C-coconvex sets (as analogue of Minkowski sum), this theory can be established, and the corresponding concepts of *mixed volume, surface area measure, mixed surface area measure*, and *cone volume measure* are introduced. Also, the *Minkowski problem and the logarithmic Minkowski problem for coconvex sets* are formulated, and various results in this framework are obtained. Further related material can be found in Schneider's recent book [2174], see also above.

Further essential papers by K. J. Böröczky et al., in which *jth dual curvature measures* are treated, are [407] and [408]. The first one contains a survey on the Minkowski problem, and the dual Minkowski problem (regarding necessary and sufficient conditions on a prescribed measure on the unit sphere, such that it is the jth dual curvature measure of a 0-symmetric convex body) is then introduced and solved for $1 < j < n$. The proof uses also dual quermassintegrals. In the second paper [408], the authors want to generalize Alexandrov's characterization of curvature measures of convex bodies. For this purpose they introduce, motivated by Huang et al. [1245], a new concept called the *Gauss image measure*. Alexandrov's integral curvature, Alexandrov–Fenchel–Jessen surface area measures, and the dual curvature measures discovered in [1245] are all Gauss image measures. Böröczky et al. [408] pose the so-called *Gauss image problem* (also for necessary and sufficient conditions to be curvature measures) and give a solution. They show that there is a close relation between the Gauss image problem and the log-volume product maximization problem. The L_p *dual curvature measure*, introduced in [1642], has the L_p surface area measure, the L_p integral curvatures, and the dual curvature measures as subcases. Following this, Li, Liu, and Lu [1565] deal with various cases of the corresponding L_p *dual Minkowski problem*' which asks for necessary and sufficient conditions on non-negative Borel measures on the unit sphere to be the L_p dual curvature measure of a convex body. They prove a new existence result of solutions and succeed, based on it, with constructing an example from which they get non-uniqueness of the solution. More related material on Minkowski problems of various types are summarized in our previous subsection.

PLATE 8

Zonoids



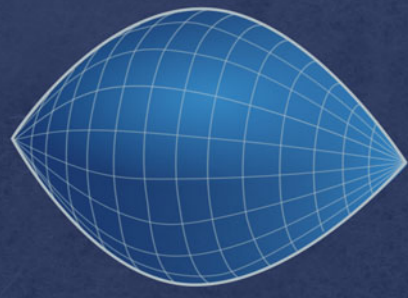
Lorenz zonoids

(statistical distributions)

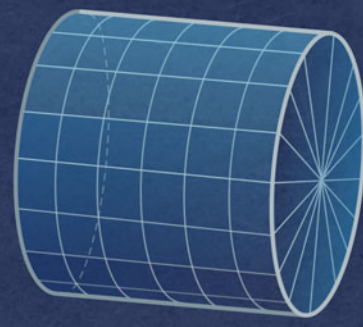
Figures 19 - 20

Figures 21 - 23

Spindle-shaped zonoids



Cylindrical zonoids



Ellipsoids



Zonoids ► (see, e.g., 2.9.11, 10.2, and 10.6.2)

Chapter 8
The Alexandrov–Fenchel inequality

"Mathematics knows no races or geographical boundaries; for mathematics, the cultural world is one country."

— David Hilbert

The so-called *Alexandrov–Fenchel inequality* is one of the deepest and strongest results in the theory of mixed volumes. Formulating it is quite simple. Namely, it says that

$$V(K, L, K_3, \dots, K_n)^2 \geq V(K, K, K_3, \dots, K_n)V(L, L, K_3, \dots, K_n)$$

for any $K, L, K_3, \dots, K_n \in CC^n$. The proof of that is difficult, and we dedicate an entire section for this. It is worth saying that several consequences and equivalent forms will also be derived in this chapter.

To illustrate the strength of this inequality, we observe that it yields the 1st Minkowski inequality and the 2nd Minkowski inequality as particular cases. This can be seen from the equivalent formulation (8.3.1) given below. Consequently, the Alexandrov–Fenchel inequality also yields the Urysohn inequality and the classical isoperimetric inequality (both studied in Chapter 5).

The Brunn–Minkowski inequality can also be obtained from the Alexandrov–Fenchel inequality (with equality case), because it can be obtained from the 1st Minkowski inequality (see Exercise 6.14). Since the proof of the Alexandrov–Fenchel inequality that we give here does not rely on the Brunn–Minkowski inequality, we get that the 1st Minkowski inequality and the Brunn–Minkowski inequality are equivalent. In Exercise 8.6, we actually obtain Brunn–Minkowski inequalities for general quermassintegrals, and not only for the volume. The equality cases are studied in Exercise 8.11.

A complete classification of the equality case in the Alexandrov–Fenchel inequality is still an open problem. However, in view of the formulation given above, we know that equality implies that K and L are homothetical provided $K_3, \dots, K_n$ are smooth convex bodies. We prove this in Section 8.4, but we do not go much further in this direction. We refer the reader to Schneider [2167] for a more complete treatment.

We start this chapter by studying *Wulff shapes*, which are important in the proof of the Alexandrov–Fenchel inequality. Roughly speaking, we are interested in seeing with more attention convex bodies as the intersections of their respective supporting half-spaces.

Plate 8 shows examples of *zonoids* which are the limits of zonotopes. Zonoids are mainly discussed in Chapters 2 and 10.

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_8

8.1 Wulff shapes

From Theorem 2.3.1 (and comments below), we know that a function $h\colon \mathbb{S}^{n-1} \to \mathbb{R}$ is the support function of a compact, convex and non-empty subset of $\mathbb{R}^n$ if and only if its radial extension is sublinear. However, there is a somewhat natural way to associate a convex body to a given (continuous and positive) function defined over $\mathbb{S}^{n-1}$ (or even over a proper subset of $\mathbb{S}^{n-1}$ having certain properties) whose radial extension is not necessarily sublinear. We describe this construction next.

Throughout this section, we denote by $C^+(\mathbb{S}^{n-1})$ the space of continuous positive functions $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$. Let $E \subseteq \mathbb{S}^{n-1}$ be a closed subset such that $\operatorname{pos}(E) = \mathbb{R}^n$ (this is clearly equivalent to E not being contained in any closed hemisphere of $\mathbb{S}^{n-1}$; see Exercise 3.15), and let $f \in C^+(\mathbb{S}^{n-1})$. The *Wulff shape* of the pair (E, f) is the set

$$W(E, f) := \bigcap_{u\in E} \mathbf{h}^-(u, f(u)) = \Big\{x \in \mathbb{R}^n : \langle x, u\rangle \le f(u) \text{ for every } u \in E\Big\}.$$

In the particular case where $E = \mathbb{S}^{n-1}$, we simply write $W(f) := W(\mathbb{S}^{n-1}, f)$ and call $W(f)$ the *Wulff shape of* f. This section is devoted to investigate Wulff shapes, presenting a topic of interest in itself, but which will be also used in the proof of the Alexandrov–Fenchel inequality, and later in the proof of the so-called L_p *Minkowski existence problem* (see Section 11.2). The name justifies since this construction appeared first in Wulff's work on crystallography. The Wulff shapes are, roughly speaking, the shapes which minimize a certain type of "surface energy" among shapes having prescribed volume (in this direction, we refer the reader to Gardner [929, Section 6] and references therein). This "variational" (classical) approach will be overlooked for now, and considered later in Theorem 8.1.2; first we investigate early properties of Wulff shapes which follow from the explicit construction that we introduced above. We start by proving that any Wulff shape is a convex body.

Lemma 8.1.1. *For any closed subset $E \subseteq \mathbb{S}^{n-1}$ such that $\operatorname{pos}(E) = \mathbb{R}^n$ and every function $f \in C^+(\mathbb{S}^{n-1})$, the Wulff shape $W(E, f)$ is a convex body containing the origin as an interior point.*

Proof. We first notice that $W(E, f)$ is convex because it is defined as an intersection of convex sets. It is also easy to see that $0_{\mathbb{R}^n} \in \operatorname{int}(W(E, f))$. In fact, let

$$c := \min\big\{f(u) : u \in \mathbb{S}^{n-1}\big\}.$$

Once f is positive and $\mathbb{S}^{n-1}$ is compact, we have that $c > 0$. We claim that $cB^n \subseteq W(E, f)$. Indeed, let $z \in cB^n$ be arbitrarily given. For every $u \in E$, we have

$$\langle z, u\rangle \le |z||u| \le c \le f(u),$$

and hence $z \in W(E, f)$. This shows that $cB^n \subseteq W(E, f)$, and we get that $W(E, f)$ has the origin as an interior point.

Since it is also clear that $W(E, f)$ is closed (it is an intersection of closed sets), it only remains to show that $W(E, f)$ is bounded. For that sake, we use essentially the same argument as in the proof of Lemma 8.2.5. From the equality $\operatorname{pos}(E) = \mathbb{R}^n$, it is not difficult to see that E contains a finite subset $\{u_1, \dots, u_m\}$ such that $\operatorname{pos}\{u_1, \dots, u_m\} = \mathbb{R}^n$. We may write

$$W(E, f) = \bigcap_{u\in E} \mathbf{h}^-(u, f(u)) \subseteq \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, f(u_j)) =: P,$$

and hence it suffices to verify that P is bounded. Assume, by contradiction, that there exists a sequence of points $(x_k)_{k\in\mathbb{N}}$ in P such that $|x_k| \to +\infty$. For each $k \in \mathbb{N}$, let

$$v_k := \frac{x_k}{|x_k|}.$$

Each v_k is a point of $\mathbb{S}^{n-1}$, and hence passing to a subsequence if necessary we may assume that (v_k) converges to some point $v \in \mathbb{S}^{n-1}$. Let $M := \max_{u\in\mathbb{S}^{n-1}} f(u)$. From the definition of P, we have that $\langle x_k, u_j\rangle \le f(u_j) \le M$ for each $k \in \mathbb{N}$ and $j \in \{1,\dots,m\}$, and this leads to

$$\left\langle \frac{x_k}{|x_k|}, u_j \right\rangle \le \frac{M}{|x_k|}.$$

Letting $k \to \infty$, we get that

$$\langle v, u_j\rangle \le 0$$

for every $j \in \{1,\dots,m\}$. Recalling that v is non-zero (because it is a vector of $\mathbb{S}^{n-1}$), this means that $u_j \in \mathbf{h}^-(v,0)$ for each $j \in \{1,\dots,m\}$, and this gives that $\operatorname{pos}\{u_1,\dots,u_m\} \subseteq \mathbf{h}^-(v,0)$. This is a contradiction to the fact that $\operatorname{pos}\{u_1,\dots,u_m\} = \mathbb{R}^n$, and the proof is complete.

□

Notice that if E is a finite subset of $\mathbb{S}^{n-1}$ such that $\operatorname{pos}(E) = \mathbb{R}^n$, then $W(E,f)$ is a polytope (because it is a convex body which can be written as the intersection of finitely many half-spaces). This particular case was briefly touched in the proof of the Alexandrov–Fenchel inequality (see Section 8.2).

In the case where $E = \mathbb{S}^{n-1}$, we have a useful characterization of Wulff shapes in terms of support functions. This is given next.

Lemma 8.1.2. *Let $f \in C^+(\mathbb{S}^{n-1})$, and consider the family*

$$\mathcal{K}(f) := \big\{K \in \mathcal{K}_0^n : h_K \le f\big\}.$$

Then $W(f) \in \mathcal{K}(f)$, and $K \subseteq W(f)$ whenever $K \in \mathcal{K}(f)$. In other words, $W(f)$ is the maximal element of $\mathcal{K}(f)$.

Proof. Let $u \in \mathbb{S}^{n-1}$ be arbitrarily given. By definition, we have that

$$W(f) \subseteq \mathbf{h}^-(u, f(u)).$$

Thus, $h_{W(f)}(u) \le f(u)$. Since this holds for any $u \in \mathbb{S}^{n-1}$, it follows that $W(f) \in \mathcal{K}(f)$.

Now let $K \in \mathcal{K}(f)$. For any given $u \in \mathbb{S}^{n-1}$, we have $h_K(u) \le f(u)$, and this leads to the inclusion $K \subseteq \mathbf{h}^-(u, f(u))$. Therefore,

$$K \subseteq \bigcap_{u\in\mathbb{S}^{n-1}} \mathbf{h}^-(u, f(u)) = W(f).$$

This shows that $W(f)$ is the maximal element of $\mathcal{K}(f)$.

□

Notice that if $f = h_L$ for some $L \in \mathcal{K}_0^n$, then $W(f) = L$. This can be verified directly from the definition of $W(f)$, or by noticing that L is the maximal element of $\mathcal{K}(h_L)$.

The Wulff shape $W(E,f)$ may be seen as the convex body whose bounding half-spaces are indexed by the restriction of f to E. One has to be careful here, since some of these half-spaces might be "superfluous"; see Figure 8.1.1. This means that for $u \in E$ the hyperplane $\mathbf{h}(u, f(u))$ may **not** be a supporting hyperplane of $W(E,f)$. As we will see next, $W(E,f)$ is supported at each fixed boundary point by some of the hyperplanes $\big\{\mathbf{h}(u, f(u)) : u \in E\big\}$.

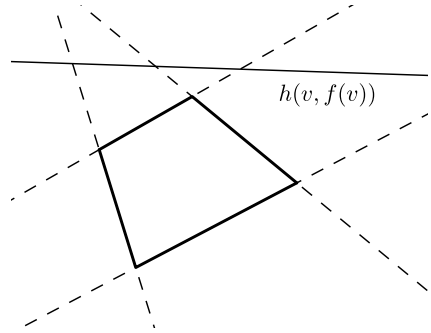


Fig. 8.1.1 Removing the half-space $\mathbf{h}^-(v, f(v))$ leads to the same Wulff shape.

Lemma 8.1.3. *Let $f \in C^+(\mathbb{S}^{n-1})$, and let $E \subseteq \mathbb{S}^{n-1}$ be a closed set such that* $\mathrm{pos}(E) = \mathbb{R}^n$. *For simplicity, write $K = W(E, f)$. We have the following statements:*

(i) *If $x \in \partial K$, then $x \in \mathbf{h}(u, f(u))$ for some $u \in E$. In particular, K has an outer normal belonging to E in each of its boundary points.*

(ii) $\left\{x \in \mathbb{R}^n : \langle x, u\rangle < f(u) \text{ for every } u \in E\right\} = \mathrm{int}(K)$.

Proof. Recall that

$$K = \bigcap_{u \in E} \mathbf{h}^-(u, f(u)),$$

and that we already know that $K \in \mathcal{K}_0^n$. If $x \in \partial K$, then we have that

$$\langle x, u\rangle \leq f(u)$$

for every $u \in E$. Assume, by contradiction, that there is no $u \in E$ such that $x \in \mathbf{h}(u, f(u))$. Then we have that $\langle x, u\rangle < f(u)$ for each $u \in E$, and hence the continuous mapping

$$u \mapsto f(u) - \langle x, u\rangle$$

is positive over the compact set E. Thus, it attains a positive global minimum, and then there exists a number $\varepsilon > 0$ such that

$$f(u) - \langle x, u\rangle > 2\varepsilon$$

for every $u \in E$. We claim that this gives the inclusion $x + \varepsilon B^n \subseteq K$. In fact, for any $v \in B^n$ and each $u \in E$, the Cauchy–Schwarz inequality leads to

$$\langle x + \varepsilon v, u\rangle = \langle x, u\rangle + \varepsilon\langle v, u\rangle < f(u) - 2\varepsilon + \varepsilon|v||u| \leq f(u) - \varepsilon,$$

from which

$$x + \varepsilon v \in \bigcap_{u \in E} \mathbf{h}^-(u, f(u)) = K$$

follows, and hence we have the desired inclusion $x + \varepsilon B^n \subseteq K$. This is a contradiction to the fact that $x \in \partial K$. It follows that $x \in \mathbf{h}(u, f(u))$ for some $u \in E$. In this case, it is clear that $\mathbf{h}(u, f(u))$ is a supporting hyperplane of K at x, and then u is an outer normal of K at x. This proves **(i)**.

Now we prove **(ii)**. If $x \in \mathbb{R}^n$ is such that $\langle x, u\rangle < f(u)$ for any $u \in E$, then we have from the definition that $x \in K$. Once the inequality is strict for every $u \in E$, it follows from item **(i)** that $x \notin \partial K$. Hence $x \in \operatorname{int}(K)$, and this gives the inclusion "$\subseteq$".

For the reverse inclusion, let $x \in \operatorname{int}(K)$. Then we may fix $\varepsilon > 0$ such that $x + \varepsilon B^n \subseteq K$. Since we have in particular that $x \in K$, it follows that $\langle x, u\rangle \leq f(u)$ for every $u \in E$. If $\langle x, u\rangle = f(u)$ for some $u \in E$, then

$$\langle x + \varepsilon u, u\rangle = f(u) + \varepsilon > f(u),$$

a contradiction. As a consequence, we have $\langle x, u\rangle < f(u)$ for every $u \in E$, and the proof is complete. □

At this point, we know that Wulff shapes are convex bodies given by functions of $C^+(\mathbb{S}^{n-1})$. Hence, it is natural to ask whether convergence of functions implies convergence of the respective Wulff shapes (in the Hausdorff metric, of course). We investigate this next.

Proposition 8.1.1. *Let $(f_j)_{j\in\mathbb{N}}$ be a sequence of functions in $C^+(\mathbb{S}^{n-1})$ converging uniformly to a function $f \in C^+(\mathbb{S}^{n-1})$, and let $E \subseteq \mathbb{S}^{n-1}$ be a closed set having the property that $\operatorname{pos}(E) = \mathbb{R}^n$. Then the sequence $(W(E, f_j))_{j\in\mathbb{N}}$ converges to $W(E, f)$ in the Hausdorff metric.*

Proof. Our strategy is to use Theorem 2.6.1. For simplicity, we write $K_j := W(E, f_j)$ for each $j \in \mathbb{N}$, and $K := W(E, f)$. Fix a point $x \in \operatorname{int}(K)$. Then we know from Lemma 8.1.3 that $\langle x, u\rangle < f(u)$ for any $u \in E$. By compactness, the minimum of the function $E \ni u \mapsto f(u) - \langle x, u\rangle$ is positive, and hence we may let $\varepsilon > 0$ be such that $\langle x, u\rangle < f(u) - \varepsilon$ for all $u \in E$ (as in the proof of Lemma 8.1.3). Since $f_j \to f$ uniformly, there exists $j_0 \in \mathbb{N}$ such that $f(u) < f_j(u) + \varepsilon$ for every $j > j_0$ and any $u \in E$. Thus,

$$\langle x, u\rangle < f(u) - \varepsilon < f_j(u)$$

for each $j > j_0$ and every $u \in E$. This means that $x \in K_j$ for all $j > j_0$, and this gives that x is the limit of a sequence of points $(x_j)_{j\in\mathbb{N}}$ such that $x_j \in K_j$ for each $j \in \mathbb{N}$ (simply set $x_j = x$ for $j > j_0$ and pick any point of K_j in the case where $j \leq j_0$).

Summarizing, any interior point of K is the limit of a sequence of points $(x_j)_{j\in\mathbb{N}}$ such that x_j is a point K_j for each $j \in \mathbb{N}$. But this actually holds for any point of K (not only for the interior points), because any boundary point of K can be approximated by interior points of K. Hence, we have condition **(a)** of Theorem 2.6.1.

It remains to verify that also condition **(b)** of the mentioned theorem holds true. Let $(x_{i_j})_{j\in\mathbb{N}}$ be a sequence such that $x_{i_j} \in K_{i_j}$ for every $j \in \mathbb{N}$, and assume that (x_{i_j}) converges to a point $x \in \mathbb{R}^n$. For each $u \in E$, we have

$$\langle x_{i_j}, u\rangle \leq f_{i_j}(u).$$

The uniform convergence $f_j \to f$ implies, in particular, that f_{i_j} converges pointwise to f. Hence, letting $j \to \infty$ in the inequality above, we get that

$$\langle x, u\rangle \leq f(u)$$

for every $u \in E$. This shows that $x \in K$, and the proof is complete.

□

We move forward to derive a formula to describe the volume of a Wulff shape. Our first task is to prove that the surface area measure of a Wulff shape $W(E, f)$ is concentrated at E.

Corollary 8.1.1 (of Lemma 8.1.3)**.** *In the same notation and conditions of Lemma 8.1.3, we have*

$$S_{n-1}(K, \mathbb{S}^{n-1} \setminus E) = 0. \tag{8.1.1}$$

In other words, the surface area measure of $W(E, f)$ is concentrated at E.

Proof. We recall that

$$S_{n-1}(K, \mathbb{S}^{n-1} \setminus E) = \mu_K(\mathbb{S}^{n-1} \setminus E) = \mathcal{H}^{n-1}(\eta_K(\mathbb{S}^{n-1} \setminus E)), \tag{8.1.2}$$

where $\eta_K(\mathbb{S}^{n-1} \setminus E)$ denotes the *reverse normal image* of $\mathbb{S}^{n-1} \setminus E$, defined as

$$\eta_K(\mathbb{S}^{n-1} \setminus E) :=$$
$$:= \big\{x \in \partial K : \exists\, v \in \mathbb{S}^{n-1} \setminus E \text{ such that } v \text{ is an outer normal of } K \text{ at } x\big\}.$$

For the proper definitions and proofs, we refer the reader to Section 7.2 (and, in particular, to Theorem 7.2.3).

We may guarantee that K has at least two distinct unit outer normal vectors at any given point of $\eta_K(\mathbb{S}^{n-1} \setminus E)$: one in $\mathbb{S}^{n-1} \setminus E$ (by the definition of η_K) and the other belonging to E (from Lemma 8.1.3 above). It follows that all points of $\eta_K(\mathbb{S}^{n-1} \setminus E)$ are singular points of ∂K. Now we remember that Corollary 2.4.2 states that the set of singular points of ∂K has zero $(n-1)$-dimensional Hausdorff measure. This leads to the equality

$$\mathcal{H}^{n-1}(\eta_K(\mathbb{S}^{n-1} \setminus E)) = 0,$$

and, together with (8.1.2), this yields (8.1.1). The proof is complete. □

Now we are ready to state and prove the announced volume formula for $W(E, f)$.

Theorem 8.1.1. *Let $f \in C^+(\mathbb{S}^{n-1})$ and let $E \subseteq \mathbb{S}^{n-1}$ be a closed set such that* $\operatorname{pos}(E) = \mathbb{R}^n$*. Writing again $K = W(E, f)$, we have the equality*

$$\operatorname{vol}_n(K) = \frac{1}{n} \int_E f(u)\, dS_{n-1}(K, u). \tag{8.1.3}$$

Proof. From the definition of $S_{n-1}(K, \cdot)$ and using (8.1.1), we get

$$\operatorname{vol}_n(K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(u)\, dS_{n-1}(K, u) = \frac{1}{n} \int_E h_K(u)\, dS_{n-1}(K, u). \tag{8.1.4}$$

Now we prove that $h_K(v) = f(v)$ for $S_{n-1}(K, \cdot)$-almost all $v \in E$. Assume that

$$x \in \eta_K\big(\{v \in E : h_K(v) \neq f(v)\}\big),$$

that is, x is a boundary point where K has an outer normal $v \in E$ with the property that $f(v) \neq h_K(v)$. By Lemma 8.1.3, there exists $u \in E$ such that $x \in \mathbf{h}(u, f(u))$, and we also know that in this case u is an outer normal of K at x. In particular, we have $h_K(u) = \langle x, u\rangle = f(u)$, and this gives that $u \neq v$. Then K has at least two distinct unit outer normal vectors at x, and this means that x is a singular point of ∂K. It follows that all points of $\eta_K\big(\{v \in E : h_K(v) \neq f(v)\}\big)$ are singular, and from Corollary 2.4.2, we get

$$S_{n-1}\big(K, \{v \in E : h_K(v) \neq f(v)\}\big) = \mathcal{H}^{n-1}\big(\eta_K\big(\{v \in E : h_K(v) \neq f(v)\}\big)\big) = 0.$$

This shows that $h_K(v) = f(v)$ for $S_{n-1}(K,\cdot)$-almost all points of E. Then we may write

$$\int_E h_K(v)\,dS_{n-1}(K,v) = \int_E f(v)\,dS_{n-1}(K,v),$$

and together with (8.1.4) this gives the desired.

□

Now we discuss the energy minimizing property of Wulff shapes, which originally inspired their definition. This appears in many ways througout the literature; see, e.g., Fonseca [889] and Taylor [2347]. A version for polytopes is stated and proved by Gruber [1070, Theorem 8.13]. Because of our purposes in this book, we choose to present a version which relies solely on the Brunn–Minkowski theory.

Theorem 8.1.2. *Let $f \in C^+(\mathbb{S}^{n-1})$, and let $E \subseteq \mathbb{S}^{n-1}$ be a closed set for which $\operatorname{pos}(E) = \mathbb{R}^n$ holds. We write $K = W(E,f)$. Among all convex bodies $L \subseteq \mathbb{R}^n$ with surface area measure $S_{n-1}(L,\cdot)$ concentrated at E and such that $\operatorname{vol}_n(L) = \operatorname{vol}_n(K)$, the ones which minimize the functional*

$$L \mapsto \int_{\mathbb{S}^{n-1}} f(u)\,dS_{n-1}(L,u) \tag{8.1.5}$$

are precisely the translates of K.

Remark 8.1.1. The functional defined by (8.1.5) models the energy of the boundary of L subject to a "tension" given by $f|_E$. By the so-called *Gibbs–Curie principle*, the minimization of surface energy models the shape of crystals, and this is why the Wulff shapes appear in crystallography.

Proof. Let $L \subseteq \mathbb{R}^n$ be any convex body whose surface area measure is concentrated at E and whose volume is the same as the volume of K. We know from Lemma 8.1.2 that $f \geq h_K$. Together with the 1st Minkowski inequality (Theorem 6.3.1) and with formula (8.1.3), this yields

$$\int_E f(u)\,dS_{n-1}(L,u) \geq \int_E h_K(u)\,dS_{n-1}(L,u) =$$
$$= \int_{\mathbb{S}^{n-1}} h_K(u)\,dS_{n-1}(L,u) = nV(K,L,\dots,L) \geq n\operatorname{vol}_n(K) = \int_E f(u)\,dS_{n-1}(K,u).$$

This proves that K minimizes the functional (8.1.5). Since $S_{n-1}(K,\cdot)$ is translation invariant, we have that any translate of K also minimizes this functional. For the uniqueness, observe that if equality holds in the inequality above, then we have equality in the 1st Minkowski inequality. This gives that K and L are homothetical, and since they have the same volume we get that they are translates.

□

Remark 8.1.2. The particular case where E is finite is precisely the result for polytopes given in [1070, Theorem 8.13].

Next, we study the first variation of the volume of a Wulff shape. More concretely, we try to understand how the volume of a Wulff shape $W(E,f)$ varies if we perform a small perturbation on f. First, we need a technical lemma from measure theory that we state and prove for the sake of completeness. In what follows, recall that $\mathcal{M}(\mathbb{S}^{n-1})$ denotes the space of finite Borel measures on $\mathbb{S}^{n-1}$.

Lemma 8.1.4 (Simultaneous convergence)**.** *Let $(f_m)_{m\in\mathbb{N}}$ be a sequence of functions in $C(\mathbb{S}^{n-1})$ converging in the maximum norm to a function $f \in C(\mathbb{S}^{n-1})$, and let $(\mu_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{M}(\mathbb{S}^{n-1})$ converging weakly to a measure $\mu \in \mathcal{M}(\mathbb{S}^{n-1})$. Then we have that*

$$\int_{\mathbb{S}^{n-1}} f_m \, d\mu_m \to \int_{\mathbb{S}^{n-1}} f \, d\mu$$

as $m \to \infty$.

Proof. For simplicity, denote the maximum norm of $C(\mathbb{S}^{n-1})$ by $||\cdot||$. Also, we omit the domain of integrations, since all integrations are considered over the sphere. Because $\mu_m \to \mu$ weakly, we have that $\mu_m(\mathbb{S}^{n-1}) \to \mu(\mathbb{S}^{n-1})$. It follows immediately that the sequence (μ_m) is uniformly bounded, meaning that there exists a number $c > 0$ such that

$$\mu_m(\mathbb{S}^{n-1}) \leq c$$

for every $m \in \mathbb{N}$. The proof is based on the following estimate:

$$\begin{aligned}
\left| \int f_m \, d\mu_m - \int f \, d\mu \right| \leq \\
\leq \left| \int f_m \, d\mu_m - \int f \, d\mu_m \right| + \left| \int f \, d\mu_m - \int f \, d\mu \right| \leq \\
\leq ||f_m - f|| \mu_m(\mathbb{S}^{n-1}) + \left| \int f \, d\mu_m - \int f \, d\mu \right| \leq \\
\leq ||f_m - f|| c + \left| \int f \, d\mu_m - \int f \, d\mu \right|.
\end{aligned}$$

The first term in the last line goes to 0 as $m \to \infty$ because $f_m \to f$ in the maximum norm of $C(\mathbb{S}^{n-1})$. The same happens to the second term as a consequence of the weak convergence $\mu_m \to \mu$. The proof is complete.

□

First variation of the volume of a Wulff shape. Let $E \subseteq \mathbb{S}^{n-1}$ be a closed set having the property that $\operatorname{pos}(E) = \mathbb{R}^n$. Assume that $G\colon [-\varepsilon, \varepsilon] \times E \to \mathbb{R}$ and $g\colon E \to \mathbb{R}$ be continuous functions such that

$$\lim_{t \to 0} \frac{G(t, \cdot)}{t} = g(\cdot)$$

uniformly on E. This means that for every $\beta > 0$, there exists a number $\delta > 0$ such that $t \neq 0$ and $|t| < \delta$ imply

$$\left| \frac{G(t, u)}{t} - g(u) \right| < \beta$$

for every $u \in E$. Notice that, in this case, for each $u \in E$, we have

$$G(0, u) = \lim_{t \to 0} G(t, u) = \lim_{t \to 0} t \frac{G(t, u)}{t} = g(u) \lim_{t \to 0} t = 0.$$

Moreover, we have that $G(t, \cdot)$ converges uniformly on E to the zero function as $t \to 0$. This follows immediately from the inequality

$$|G(t, u)| \leq |G(t, u) - t g(u)| + |t g(u)| = |t| \left(\left| \frac{G(t, u)}{t} - g(u) \right| + |g(u)| \right),$$

since the right-hand side can be made arbitrarily small if t is small enough. For each $t \in [-\varepsilon, \varepsilon]$, we let $m_t := \min_{u \in E} G(t, u)$. We claim that

$$\lim_{t\to 0} m_t = 0.$$

If this is not true, then there exist a number $c > 0$ and a sequence $(t_j)_{j\in\mathbb{N}}$ in $[-\varepsilon, \varepsilon]$ such that $t_j \to 0$ when $j \to +\infty$ and with $|m_{t_j}| > c$ for every $j \in \mathbb{N}$. For any given $j \in \mathbb{N}$, the map $G(t_j, \cdot)$ is a continuous function defined over the compact set E, and hence we may choose a point $u_j \in E$ such that

$$G(t_j, u_j) = m_{t_j}.$$

Up to a subsequence, we may assume that the sequence $(u_j)_{j\in\mathbb{N}}$ converges to a point $u_0 \in E$. Thus, from the continuity of G, we get

$$0 = |G(0, u_0)| = \lim_{j\to+\infty} |G(t_j, u_j)| = \lim_{j\to+\infty} |m_{t_j}| > c > 0,$$

and this contradiction gives the desired statement.

Now let $f \in C^+(\mathbb{S}^{n-1})$. Since $m_t \to 0$ as $t \to 0$, and because f attains a positive minimum, we have that there exists a number $0 < \delta < \varepsilon$ such that

$$-m_t < \min_{u\in E} f(u)$$

whenever $|t| < \delta$. For simplicity, from now on, we simply say that if $|t| < \delta$, then t is small. This gives that

$$f(u) + G(t, u) \geq f(u) + m_t \geq \min_{u\in E} f(u) + m_t > 0$$

for every $u \in E$ and for t small enough. This means that the (continuous) function $E \ni u \mapsto f + G(t, \cdot)$ is positive for t sufficiently small. It follows that the Wulff shape $K_t := W(E, f + G(t, \cdot))$ is a convex body, and by formula (8.1.3), we have

$$\mathrm{vol}_n(K_t) = \frac{1}{n}\int_E f(u) + G(t, u)\, dS_{n-1}(K_t, u) \tag{8.1.6}$$

for t small.

Proposition 8.1.2. *In the same hypothesis and notation as above, denote $K := W(E, f)$ and let $\mathrm{v}\colon [-\varepsilon, \varepsilon] \to \mathbb{R}$ be defined as*

$$\mathrm{v}(t) := \mathrm{vol}_n(K_t).$$

Then v is differentiable at $t = 0$, and

$$\mathrm{v}'(0) = \int_E g(u)\, dS_{n-1}(K, u).$$

Proof. Since $G(0, \cdot) = 0$, we have that $K_0 = K$, and hence $\mathrm{v}(0) = \mathrm{vol}_n(K)$. Thus,

$$\mathrm{v}'(0) = \lim_{t\to 0} \frac{\mathrm{vol}_n(K_t) - \mathrm{vol}_n(K)}{t}.$$

To prove the desired equality, we start by considering the right-hand sided derivative. That is, we will show that

$$\mathrm{v}'_+(0) = \lim_{t\to 0^+} \frac{\mathrm{vol}_n(K_t) - \mathrm{vol}_n(K)}{t} = \int_E g(u)\, dS_{n-1}(K,u).$$

Write

$$V_j(t) := V(K_t[j], K[n-j])$$

for any small t and each $j = 0, \dots, n$. In particular,

$$V_0(t) = V(K, \dots, K) = \mathrm{vol}_n(K)$$

for all t, and

$$V_j(0) = V(K_0[j], K[n-j]) = V(K, \dots, K) = \mathrm{vol}_n(K)$$

for each $j = 0, \dots, n$. For simplicity, write $V_0 := \mathrm{vol}_n(K)$ ($= V_0(t)$ for any t). From Lemma 8.1.2, we have

$$h_{K_t}(u) \le f(u) + G(t,u)$$

for any small t and every $u \in E$. Since the measure $S_{n-1}(K,\cdot)$ is concentrated at E (see Corollary 8.1.1), we obtain from this inequality and from formula (8.1.3) that

$$\begin{aligned} V_1(t) = V(K_t, K[n-1]) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{K_t}(u)\, dS_{n-1}(K,u) = \\ = \frac{1}{n}\int_E h_{K_t}(u)\, dS_{n-1}(K,u) \le \frac{1}{n}\int_E f(u) + G(t,u)\, dS_{n-1}(K,u) = \\ = \mathrm{vol}_n(K) + \frac{1}{n}\int_E G(t,u)\, dS_{n-1}(K,u). \end{aligned}$$

Dividing both sides by (small) $t > 0$ and rearranging the terms, we arrive at

$$\frac{V_1(t) - V_0}{t} \le \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K,u). \tag{8.1.7}$$

Because $G(t,\cdot)/t$ converges uniformly to g (as $t \to 0$, and in particular as $t \to 0^+$) and $S_{n-1}(K,\cdot)$ is a finite measure, it is immediate to check that the integral in the right-hand side of the above inequality converges to the integral of g as $t \to 0^+$. This leads to

$$\begin{aligned} \limsup_{t\to 0^+} \frac{V_1(t) - V_0}{t} \le \limsup_{t\to 0^+} \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K,u) = \\ = \frac{1}{n}\lim_{t\to 0^+}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K,u) = \frac{1}{n}\int_E g(u)\, dS_{n-1}(K,u). \end{aligned} \tag{8.1.8}$$

Now we observe that each measure $S_{n-1}(K_t,\cdot)$ is concentrated at E, and using (8.1.3) for K_t together with the inequality $h_K \le f$, we write

$$V_n(t) = \mathrm{vol}_n(K_t) = \frac{1}{n}\int_E f(u) + G(t,u)\, dS_{n-1}(K_t,u) \geq$$
$$\geq \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\, dS_{n-1}(K_t,u) + \frac{1}{n}\int_E G(t,u)\, dS_{n-1}(K_t,u) =$$
$$= V(K_t[n-1],K) + \frac{1}{n}\int_E G(t,u)\, dS_{n-1}(K_t,u) =$$
$$= V_{n-1}(t) + \frac{1}{n}\int_E G(t,u)\, dS_{n-1}(K_t,u).$$

Dividing both sides by $t > 0$, we may rewrite this inequality as

$$\frac{V_n(t) - V_{n-1}(t)}{t} \geq \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K_t,u),$$

which leads to

$$\liminf_{t\to 0^+} \frac{V_n(t) - V_{n-1}(t)}{t} \geq \liminf_{t\to 0^+} \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K_t,u). \tag{8.1.9}$$

Since we know that $G(t,\cdot)$ converges uniformly on E to the zero function as $t \to 0$, we get that $f(\cdot) + G(t,\cdot) \to f(\cdot)$ uniformly on E as $t \to 0$. Thus, Proposition 8.1.1 gives that $K_t \to K$ in the Hausdorff metric as $t \to 0$. It follows from Theorem 7.1.3 that $S_{n-1}(K_t,\cdot)$ converges weakly to $S_{n-1}(K,\cdot)$. Using again the fact that $G(t,\cdot)/t$ converges uniformly on E to g, we obtain as an immediate consequence of Lemma 8.1.4 that

$$\lim_{t\to 0} \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K_t,u) = \frac{1}{n}\int_E g(u)\, dS_{n-1}(K,u).$$

In particular, the limit in the left-hand side of the above equality exists (it was not clear *a priori*). This yields

$$\frac{1}{n}\int_E g(u)\, dS_{n-1}(K,u) = \lim_{t\to 0} \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K_t,u) =$$
$$= \lim_{t\to 0^+} \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K_t,u) = \liminf_{t\to 0^+} \frac{1}{n}\int_E \frac{G(t,u)}{t}\, dS_{n-1}(K_t,u),$$

and thus, inequality (8.1.9) reads

$$\liminf_{t\to 0^+} \frac{V_n(t) - V_{n-1}(t)}{t} \geq \frac{1}{n}\int_E g(u)\, dS_{n-1}(K,u). \tag{8.1.10}$$

Now we notice that a telescope sum yields the equality

$$(V_1(t) - V_0)\sum_{j=0}^{n-1}\left(\frac{V_1(t)}{V_0}\right)^j = \sum_{j=0}^{n-1}\left[\frac{V_1(t)^{j+1}}{V_0^j} - \frac{V_1(t)^j}{V_0^{j-1}}\right] = \left(V_1(t)^n - V_0^n\right)V_0^{1-n},$$

and the 1st Minkowski inequality (Theorem 6.3.1) gives

$$V_1(t)^n = V(K_t, K[n-1])^n \geq \mathrm{vol}_n(K_t)\mathrm{vol}_n(K)^{n-1} = V_n(t)V_0^{n-1}.$$

Replacing this inequality in the equality above yields

$$(V_1(t)-V_0)\sum_{j=0}^{n-1}\left(\frac{V_1(t)}{V_0}\right)^j = \big(V_1(t)^n - V_0^n\big)V_0^{1-n} \geq \big(V_n(t)V_0^{n-1} - V_0^n\big)V_0^{1-n} = \\ = V_n(t) - V_0,$$

and dividing by $t > 0$, we arrive at

$$\frac{V_1(t)-V_0}{t}\sum_{j=0}^{n-1}\left(\frac{V_1(t)}{V_0}\right)^j \geq \frac{V_n(t)-V_0}{t}.$$

Recall that K_t converges to K in the Hausdorff metric as $t \to 0$. Then the continuity of mixed volumes gives that $V_1(t) \to V_0$ as $t \to 0$ (and, in particular, as $t \to 0^+$). Consequently, $V_1(t)/V_0$ converges to 1 (keep in mind that $V_0 > 0$), and we get that the summation in the inequality above goes to n. Therefore,

$$\limsup_{t\to 0^+}\frac{V_1(t)-V_0}{t} \geq \frac{1}{n}\limsup_{t\to 0^+}\frac{V_n(t)-V_0}{t}. \tag{8.1.11}$$

Now we use the same strategy again (that is, the telescope sum and the 1st Minkowski inequality) and write

$$(V_{n-1}(t)-V_n(t))\sum_{j=0}^{n-1}\left(\frac{V_{n-1}(t)}{V_n(t)}\right)^j = \big(V_{n-1}(t)^n - V_n(t)^n\big)V_n(t)^{1-n} \geq \\ \geq \big(V_0V_n(t)^{n-1} - V_n(t)^n\big)V_n(t)^{1-n} = V_0 - V_n(t).$$

Dividing both sides by $-t$, we arrive at

$$\frac{V_n(t)-V_{n-1}(t)}{t}\sum_{j=0}^{n-1}\left(\frac{V_{n-1}(t)}{V_n(t)}\right)^j \leq \frac{V_n(t)-V_0}{t}.$$

Letting $t \to 0^+$, we have that both $V_{n-1}(t)$ and $V_n(t)$ converge to V_0, and hence we obtain the inequality

$$\liminf_{t\to 0^+}\frac{V_n(t)-V_{n-1}(t)}{t} \leq \frac{1}{n}\liminf_{t\to 0^+}\frac{V_n(t)-V_0}{t}. \tag{8.1.12}$$

Now we use, in this order, the inequalities (8.1.8) and (8.1.11), the fact that $\liminf \leq \limsup$, and the inequalities (8.1.12) and (8.1.10) to get

$$\frac{1}{n}\int_E g(u)\,dS_{n-1}(K,u) \geq \limsup_{t\to 0^+}\frac{V_1(t)-V_0}{t} \geq \frac{1}{n}\limsup_{t\to 0^+}\frac{V_n(t)-V_0}{t} \geq \\ \geq \frac{1}{n}\liminf_{t\to 0^+}\frac{V_n(t)-V_0}{t} \geq \liminf_{t\to 0^+}\frac{V_n(t)-V_{n-1}(t)}{t} \geq \frac{1}{n}\int_E g(u)\,dS_{n-1}(K,u).$$

Thus, all quantities above are the same. In particular, this leads to

$$\liminf_{t\to 0^+}\frac{V_n(t)-V_0}{t} = \limsup_{t\to 0^+}\frac{V_n(t)-V_0}{t} = \int_E g(u)\,dS_{n-1}(K,u),$$

and hence

$$\mathrm{v}'_+(0) = \lim_{t\to 0^+}\frac{\mathrm{vol}_n(K_t)-\mathrm{vol}_n(t)}{t} = \lim_{t\to 0^+}\frac{V_n(t)-V_0}{t} = \int_E g(u)\,dS_{n-1}(K,u).$$

This settles the case of the right-hand sided derivative. It remains to consider the left-hand sided derivative. That is, we have to prove that

$$\mathrm{v}'_-(0) = \lim_{t\to 0^-} \frac{\mathrm{vol}_n(K_t) - \mathrm{vol}_n(K)}{t} = \int_E g(u)\, dS_{n-1}(K,u). \tag{8.1.13}$$

We may obtain this from a slight modification of the previous argument: we simply divide by $t < 0$ the inequalities which we divided by $t > 0$ previously (hence changing the sign). For example, inequality (8.1.7) becomes

$$\frac{V_1(t) - V_0}{t} \geq \frac{1}{n} \int_E \frac{G(t,u)}{t}\, dS_{n-1}(K,u)$$

for (small) $t < 0$. Since $G(t,u)/t$ also converges uniformly to g as $t \to 0^-$, this leads to

$$\liminf_{t\to 0^-} \frac{V_1(t) - V_0}{t} \geq \frac{1}{n} \int_E g(u)\, dS_{n-1}(K,u). \tag{8.1.14}$$

Similarly, dividing by $t < 0$ instead of by $t > 0$, inequalities (8.1.10), (8.1.11), and (8.1.12) are "replaced" by

$$\begin{aligned}
\limsup_{t\to 0^-} \frac{V_n(t) - V_{n-1}(t)}{t} &\leq \frac{1}{n} \int_E g(u)\, dS_{n-1}(K,u),\\
\liminf_{t\to 0^-} \frac{V_1(t) - V_0}{t} &\leq \frac{1}{n} \liminf_{t\to 0^-} \frac{V_n(t) - V_0}{t} \quad \text{and}\\
\limsup_{t\to 0^-} \frac{V_n(t) - V_{n-1}(t)}{t} &\geq \frac{1}{n} \limsup_{t\to 0^-} \frac{V_n(t) - V_0}{t},
\end{aligned}$$

respectively. As before, these inequalities, together with (8.1.14), yield

$$\begin{aligned}
&\frac{1}{n} \int_E g(u)\, dS_{n-1}(K,u) \leq \liminf_{t\to 0^-} \frac{V_1(t) - V_0}{t} \leq \frac{1}{n} \liminf_{t\to 0^-} \frac{V_n(t) - V_0}{t} \leq\\
&\leq \frac{1}{n} \limsup_{t\to 0^-} \frac{V_n(t) - V_0}{t} \leq \limsup_{t\to 0^-} \frac{V_n(t) - V_{n-1}(t)}{t} \leq \frac{1}{n} \int_E g(u)\, dS_{n-1}(K,u).
\end{aligned}$$

Hence, these numbers are all the same. Thus, as before, it follows that

$$\begin{aligned}
\int_E g(u)\, dS_{n-1}(K,u) = \liminf_{t\to 0^-} \frac{V_n(t) - V_0}{t} = \limsup_{t\to 0^-} \frac{V_n(t) - V_0}{t} = \lim_{t\to 0^-} \frac{V_n(t) - V_0}{t} =\\
= \lim_{t\to 0^-} \frac{\mathrm{vol}_n(K_t) - \mathrm{vol}_n(K)}{t} = \mathrm{v}'_-(0),
\end{aligned}$$

and the proof is complete.

□

8.2 A proof of the inequality

As it was mentioned in the beginning of this chapter, the statement of the Alexandrov–Fenchel inequality is as follows (there are other equivalent enunciates in the literature; we give some of them later).

Theorem 8.2.1 (Alexandrov–Fenchel inequality). *For any $K_1, K_2, K_3, \ldots, K_n \in CC^n$, we have*

$$V(K_1, K_2, K_3, \dots, K_n)^2 \geq V(K_1, K_1, K_3, \dots, K_n)V(K_2, K_2, K_3, \dots, K_n). \tag{8.2.1}$$

Notice that the case where $n = 2$ is simply the 1st Minkowski inequality. We will follow the proof of the book Hug and Weil [1263] (the argument was originally given in Cordero-Erausquin et al. [659]). This goes by approximating the involved bodies by strongly isomorphic polytopes.

We prove the inequality for convex bodies $K_1, \dots, K_n \subseteq \mathbb{R}^n$, since the general case (for sets in CC^n) follows immediately by approximations. This approach has the advantage that all involved mixed volumes become strictly positive (this is a consequence of Proposition 6.2.3; this fact will be used many times in this section).

Because we aim to provide a detailed (and self-contained) proof, we start by proving several auxiliary results which will be combined to obtain the Alexandrov–Fenchel inequality. The reader which is not interested in all these technical details can move directly to the main proof starting in page 527 and return to read the statement of each auxiliary result when it is needed (we will always refer to them when they are used).

The first result we need is a weaker version of the classical Perron–Frobenius theorem from linear algebra. In what follows, recall that a square $m \times m$ matrix $M = (a_{ij})$ is said to be *irreducible* if for any proper subset $\{e_{i_1}, \dots, e_{i_k}\}$ $(0 < k < m)$ of the canonical basis of $\mathbb{R}^m$ it holds that

$$M\big(\operatorname{span}\{e_{i_1}, \dots, e_{i_k}\}\big) \not\subseteq \operatorname{span}\{e_{i_1}, \dots, e_{i_k}\}.$$

In other words, M is irreducible when it does not admit an invariant proper coordinate subspace. Next, we present a useful criterion for the irreducibility of a matrix.

Lemma 8.2.1. *Let $M = (m_{ij})$ be an $m \times m$ matrix. Then M is irreducible if for any pair $i, j \in \{1, \dots, m\}$ for which $m_{ji} = 0$ there exists a chain $j, j_1, \dots, j_r, i$ of elements of $\{1, \dots, m\}$ such that the entries $m_{jj_1}, m_{j_1j_2}, \dots, m_{j_ri}$ are non-zero.*

Proof. Assume that M is not irreducible, and let $\{i_1, \dots, i_k\}$ be a non-empty proper subset of $\{1, \dots, m\}$ for which

$$M\big(\operatorname{span}\{e_{i_1}, \dots, e_{i_k}\}\big) \subseteq \operatorname{span}\{e_{i_1}, \dots, e_{i_k}\}.$$

For simplicity, write $A := \{i_1, \dots, i_k\}$ and $B := \{1, \dots, m\} \setminus \{i_1, \dots, i_k\}$. The inclusion above means that for any $i \in A$ and $j \in B$ the j-th coordinate of Me_i is zero. This leads to

$$0 = \langle Me_i, e_j \rangle = m_{ji}$$

whenever $i \in A$ and $j \in B$. Since both sets A and B are non-empty, we may fix $i_0 \in A$ and $j_0 \in B$. We claim that there is no chain $j_0, j_1, \dots, j_r, i_0$ of indices in $\{1, \dots, m\}$ such that $m_{j_0j_1}, \dots, m_{j_ri_0} \neq 0$. Indeed, if such a chain exists, then $m_{j_0j_1} \neq 0$ implies that $j_1 \in B$. As a consequence, $m_{j_1j_2} \neq 0$ implies that we also have $j_2 \in B$. Proceeding inductively, we get that $j_r \in B$, and hence $m_{j_ri_0} = 0$, a contradiction.

□

Before moving forward, we introduce some terminology. From now on, we say that a vector is *positive* (resp. *negative*) if all its entries are positive (resp. negative).

Proposition 8.2.1. *Let M be an $m \times m$ real matrix which is symmetric, irreducible, and whose entries are all non-negative. Assume also that M has a positive eigenvector v^1 with corresponding eigenvalue $\lambda_1 > 0$. Then λ_1 is a simple eigenvalue, and one has that*

$$\lambda \leq \lambda_1$$

for any eigenvalue λ of M. Moreover, any positive eigenvector of M is a multiple of v^1.

Proof. Let u be an eigenvector which is linearly independent of v^1 and is associated to an eigenvalue $\lambda \geq \lambda_1$. Up to replacing u by $-u$, we may assume that at least one entry of u is negative. For $t \in [0,1]$, let $w(t) = (1-t)v^1 + tu$. Since all coordinates of $w(0) = v^1$ are positive and $w(1) = u$ has at least a negative coordinate, by continuity, we may let $t_0 \in (0,1)$ be the smallest value for which $w = w(t_0)$ has at least one zero coordinate (meaning that all non-zero coordinates are positive). Assume that w has precisely $k-1$ zero coordinates ($k \geq 2$). For the sake of simplicity, up to rearranging the order of the coordinates, we may write

$$w = (0, \ldots, 0, w_k, \ldots, w_m),$$

where $0 < w_k \leq w_{k+1} \leq \ldots \leq w_m$. Now notice that

$$Mw = M\big((1-t_0)v^1 + t_0 u\big) = (1-t_0)\lambda_1 v^1 + t_0 \lambda u = \lambda w + (1-t_0)(\lambda_1 - \lambda)v^1.$$

Thus, the first $k-1$ coordinates of Mw are respectively the same as the first $k-1$ coordinates of $(1-t_0)(\lambda_1 - \lambda)v^1$, which are non-positive because $\lambda \geq \lambda_1$ and v^1 is a positive vector. Hence, for each $i = 1, \ldots, k-1$, we have

$$0 \geq (Mw)_i = \sum_{j=1}^{m} a_{ij} w_j = \sum_{j=k}^{m} a_{ij} w_j \geq 0,$$

where the last inequality justifies because $a_{ij} \geq 0$ for all $i, j = 1, \ldots, m$ and $w_j > 0$ for every $j = k, \ldots, m$. It follows that $a_{ij} = 0$ for all $i = 1, \ldots, k-1$ and $j = k, \ldots, m$. It is immediate to notice that this gives

$$M\big(\operatorname{span}\{e_k, \ldots, e_m\}\big) \subseteq \operatorname{span}\{e_k, \ldots, e_m\},$$

and this contradicts the irreducibility of M. Consequently, there is no eigenvector linearly independent of v^1 associated to an eigenvalue $\lambda \geq \lambda_1$. This shows that λ_1 is a simple eigenvalue and that no eigenvalue of M is greater than λ_1.

It remains to prove that v^1 is the unique positive eigenvector of M up to scalar multiplication. Assume that u is a positive eigenvector which is linearly independent of v^1, and let λ be the eigenvalue associated to u. Using the symmetry of M, we get

$$\langle Mv^1, u\rangle = \langle v^1, Mu\rangle,$$

and this gives

$$\lambda_1 \langle v^1, u\rangle = \lambda \langle v^1, u\rangle.$$

Since v_1 and u are positive vectors, we have $\langle v^1, u\rangle \neq 0$, and this gives $\lambda_1 = \lambda$. We already know that λ_1 is a simple eigenvalue, and hence u must be a multiple of v^1. The proof is complete.

□

Corollary 8.2.1. *Let $M = (a_{ij})$ be an $m \times m$ matrix which is symmetric, irreducible, and such that $m_{ij} \geq 0$ whenever $i \neq j$. Assume that M has a positive eigenvector $v^1 \in \mathbb{S}^{m-1}$, and that the corresponding eigenvalue λ_1 is positive. If M does not have an eigenvalue in the interval $(0, \lambda_1)$, then*

$$\langle Mx, x\rangle \leq \lambda_1 \langle v^1, x\rangle^2 \tag{8.2.2}$$

for every $x \in \mathbb{R}^m$.

Proof. Since M is symmetric, it admits an orthonormal basis $\{v^1, \dots, v^m\}$ of eigenvectors, with corresponding eigenvalues $\lambda_1, \dots, \lambda_m$. We already know that v^1 is a positive vector, and that $\lambda_1 > 0$. Let $c := \max\{|a_{ii}| : i = 1, \dots, m\}$, and consider the matrix $M_c := M + c\mathrm{Id}_{\mathbb{R}^m}$. This matrix is clearly symmetric and has non-negative entries. Also, it is immediate to see that $v^1, \dots, v^m$ are eigenvectors of M_c with corresponding eigenvalues $\lambda_1 + c, \dots, \lambda_m + c$. It is clear from the definition that M_c is irreducible, because any invariant subspace of M_c is also invariant under M. Since the (positive) eigenvalue $\lambda_1 + c$ of M_c corresponds to the positive vector v^1, from Proposition 8.2.1 (applied for M_c, of course) we get that

$$\lambda_j + c \leq \lambda_1 + c$$

for any $j = 1, \dots, m$. This gives that $\lambda_j \leq \lambda_1$ for all $j = 1, \dots, m$. Since λ_1 is a simple eigenvalue, none of the vectors $v^2, \dots, v^m$ can correspond to λ_1, and hence we get that $\lambda_j \neq \lambda_1$ whenever $j = 2, \dots, m$. Once M does not have any eigenvalue in the interval $(0, \lambda_1)$, we conclude that $\lambda_j \leq 0$ for $j = 2, \dots, m$.

Finally, letting $x \in \mathbb{R}^m$ be arbitrarily given, recalling that $\{v^1, \dots, v^m\}$ is an orthonormal basis of $\mathbb{R}^m$, and writing $x = \sum_{j=1}^m \alpha_j v^j$, we arrive at

$$\langle Mx, x\rangle - \lambda_1 \langle v^1, x\rangle^2 = \sum_{j=1}^m \lambda_j \alpha_j^2 - \lambda_1 \alpha_1^2 = \sum_{j=2}^m \lambda_j \alpha_j^2 \leq 0,$$

and the proof is complete.

□

Here and throughout the text, we do not make distinctions between matrices and linear transformations: any matrix defines a linear transformation by standard matrix multiplication, and any linear transformation is naturally represented as a matrix by fixing the canonical basis. For a given fixed vector $v \in \mathbb{R}^m$, we let $v \otimes v \colon \mathbb{R}^m \to \mathrm{span}\{v\}$ be the linear transformation defined as

$$v \otimes v(x) = \langle v, x\rangle v$$

for each $x \in \mathbb{R}^m$. Write $v = (v_1, \dots, v_m)$ in canonical coordinates. In the canonical matrix form, the ij-th entry of $v \otimes v$ is given as

$$\langle v \otimes v(e_j), e_i\rangle = \langle v, e_j\rangle\langle v, e_i\rangle = v_i v_j.$$

Thus, the canonical matrix representation of $v \otimes v$ is

$$v \otimes v = \begin{bmatrix} v_1v_1 & v_1v_2 & \dots & v_1v_m \\ v_2v_1 & v_2v_2 & \dots & v_2v_m \\ \vdots & \vdots & \dots & \vdots \\ v_mv_1 & v_mv_2 & \dots & v_mv_m \end{bmatrix}. \tag{8.2.3}$$

For the sake of simplicity, we write $M \leq 0$ when the matrix M is negative semi-definite. We also write $M_1 \leq M_2$ for the case where the matrix $M_1 - M_2$ is negative semi-definite. Of course, we can also consider analogous notations for positive semi-definite and (positive and negative) definite matrices. Observe that under this notation inequality (8.2.2) reads

$$M \leq \lambda_1 v^1 \otimes v^1. \tag{8.2.4}$$

For what follows, we also remind the reader that a set $C \subseteq \mathbb{R}^m$ is an *open convex cone* if it is open, convex, and has the property that $\lambda x \in C$ whenever $x \in C$ and $\lambda > 0$ (see Section 2.4). If $C \subseteq (0, +\infty)^m$ is an open convex cone and $p \geq 0$, then we say that a function $g\colon C \to \mathbb{R}$ is *positively homogeneous of degree* p if $g(tx) = t^p g(x)$ for any $x \in C$ and $t > 0$. We also remind the reader that, in view of this definition, the simpler expression *positively homogeneous* is reserved for the functions which are positively homogeneous of degree one.

Lemma 8.2.2. *If $C \subseteq (0, +\infty)^m$ is an open convex cone and $g\colon C \to \mathbb{R}$ is a C^1 function which is positively homogeneous of degree $p \geq 0$, then*

$$\big\langle x, \nabla g(x)\big\rangle = \sum_{i=1}^{m} x_i \partial_i g(x) = p g(x)$$

for any $x = (x_1, \ldots, x_m) \in C$.

Proof. The first equality is immediate by the definition of $\nabla g(x)$. To prove the other equality, we just compute

$$\begin{aligned}\langle x, \nabla g(x)\rangle = \frac{\partial g}{\partial x}(x) = \lim_{t\to 0} \frac{g(x+tx) - g(x)}{t} = \lim_{t\to 0} \frac{(1+t)^p g(x) - g(x)}{t} = \\ = g(x) \lim_{t\to 0} \frac{(1+t)^p - 1}{t} = g(x) \frac{d}{dt}(1+t)^p\Big|_{t=0} = p g(x).\end{aligned}$$

This gives the desired.

□

Proposition 8.2.2. *Let $C \subseteq (0, +\infty)^m$ be an open convex cone, and let $f\colon C \to (0, +\infty)$ be a C^3 function having the following properties:*

(a) *f is positively homogeneous of degree* 3,

(b) *for each $i = 1, \ldots, m$ and $x \in C$ we have $\partial_i f(x) > 0$, and the function $x \mapsto \partial_i f(x)^{\frac{1}{2}}$ is concave in C,*

(c) *for every $x \in C$ the Hessian matrix of f at x is irreducible, and $\partial_{ij} f(x) \geq 0$ for $i \neq j$.*

Then the function $x \mapsto f(x)^{\frac{1}{3}}$ is concave in C.

Proof. We write $\mathrm{hess}(g)(x)$ to represent the Hessian matrix of a C^2 function g at a point x. For each $i = 1, \ldots, m$ the function $(\partial_i f)^{\frac{1}{2}}$ is concave, and hence its Hessian matrix is negative semi-definite in each point $x \in C$. That is, in view of the notation introduced above for negative semi-definite matrices, we may write

$$\mathrm{hess}\big((\partial_i f)^{\frac{1}{2}}\big)(x) \leq 0$$

for each $x \in C$. We use the chain rule to rewrite the entries of the Hessian matrix of $(\partial_i f)^{\frac{1}{2}}$. A straightforward calculation yields

$$\partial_{kj}\big((\partial_i f)^{\frac{1}{2}}\big) = -\frac{1}{4}\frac{1}{(\partial_i f)^{\frac{3}{2}}} \partial_k(\partial_i f)\partial_j(\partial_i f) + \frac{1}{2}\frac{1}{(\partial_i f)^{\frac{1}{2}}} \partial_{kj}(\partial_i f),$$

where all derivatives and functions are evaluated at a given point $x \in C$. Hence, having in mind the matrix representation of $\nabla(\partial_i f)(x) \otimes \nabla(\partial_i f)(x)$ given as in (8.2.3), we get

$$\operatorname{hess}\big((\partial_i f)^{\frac{1}{2}}\big)(x) = \frac{(\partial_i f(x))^{-\frac{1}{2}}}{2}\left(\operatorname{hess}\big(\partial_i f\big)(x) - \frac{\nabla(\partial_i f)(x)\otimes\nabla(\partial_i f)(x)}{2\partial_i f(x)}\right)$$

for any $x \in C$. Since $(\partial_i f(x))^{\frac{1}{2}} > 0$ and $\operatorname{hess}\big((\partial_i f)^{\frac{1}{2}}\big)(x)$ is negative semi-definite, it follows that the matrix between the parentheses in the right-hand side is also negative semi-definite. Thus, we may write

$$\operatorname{hess}(\partial_i f)(x) \leq \frac{\nabla(\partial_i f)(x)\otimes\nabla(\partial_i f)(x)}{2\partial_i f(x)} \tag{8.2.5}$$

for all $x \in C$. For each $i = 1, \ldots, m$ and $x \in C$, denote by $\partial_i\big(\operatorname{hess}(f)\big)(x)$ the matrix whose entries are the respective i-th derivatives (evaluated at x) of the entries of $\operatorname{hess}(f)(x)$. That is, the element in the position jk is simply $\partial_i(\partial_{jk} f)(x)$. Since f is C^3, the partial derivatives up to the third order commute, and then we have

$$\operatorname{hess}(\partial_i f)(x) = \partial_i\big(\operatorname{hess} f\big)(x). \tag{8.2.6}$$

Since f is positively homogeneous of degree 3, we have that each partial derivative $\partial_i f$ is positively homogeneous of degree 2, and each second order derivative $\partial_{jk} f$ is positively homogeneous of degree 1. From Lemma 8.2.2, we get

$$\partial_{jk} f(x) = \sum_{i=1}^{m} x_i \partial_i(\partial_{jk} f)(x) \tag{8.2.7}$$

for each $x = (x_1, \ldots, x_m) \in C$ and any $j, k \in \{1, \ldots, m\}$. Thus, as a consequence of (8.2.6) we obtain

$$\sum_{i=1}^{m} x_i \operatorname{hess}(\partial_i f)(x) = \sum_{i=1}^{m} x_i \partial_i(\operatorname{hess} f)(x) = \operatorname{hess}(f)(x),$$

where the last equality justifies since the (jk)-th element of the matrix in the middle is precisely the right-hand side of (8.2.7). In view of (8.2.5), and since each x_i is positive, this leads directly to

$$\operatorname{hess}(f)(x) = \sum_{i=1}^{m} x_i \operatorname{hess}(\partial_i f)(x) \leq \frac{1}{2}\sum_{i=1}^{m} x_i \frac{\nabla(\partial_i f)(x)\otimes\nabla(\partial_i f)(x)}{\partial_i f(x)}. \tag{8.2.8}$$

For each $x = (x_1, \ldots, x_m) \in C$, consider the matrix

$$D(x) := \operatorname{diag}\left(\frac{x_1}{\partial_1 f(x)}, \ldots, \frac{x_m}{\partial_m f(x)}\right),$$

where we recall that, by hypothesis, $\partial_j f(x) > 0$ for any $x \in C$ and every $j = 1, \ldots, m$. We claim that

$$\sum_{i=1}^{m} x_i \frac{\nabla(\partial_i f)(x)\otimes\nabla(\partial_i f)(x)}{\partial_i f(x)} = \operatorname{hess}(f)(x)\cdot D(x)\cdot\operatorname{hess}(f)(x) \tag{8.2.9}$$

for any $x \in C$, and where the dots stand for matrix multiplication. To verify this equality between matrices, it is clearly enough to prove that

$$\left\langle\left[\sum_{i=1}^{m} x_i \frac{\nabla(\partial_i f)(x)\otimes\nabla(\partial_i f)(x)}{\partial_i f(x)}\right] e_j, e_k\right\rangle = \Big\langle\big[\operatorname{hess}(f)(x)\cdot D(x)\cdot\operatorname{hess}(f)(x)\big] e_j, e_k\Big\rangle$$

for any $j, k \in \{1, \dots, m\}$. We fix any such pair j, k and compute each side separately:

$$\left\langle \left[\sum_{i=1}^{m} x_i \frac{\nabla(\partial_i f)(x) \otimes \nabla(\partial_i f)(x)}{\partial_i f(x)} \right] e_j, e_k \right\rangle = \left\langle \sum_{i=1}^{m} x_i \frac{\langle \nabla(\partial_i f)(x), e_j \rangle \nabla(\partial_i f)(x)}{\partial_i f(x)}, e_k \right\rangle =$$
$$= \sum_{i=1}^{m} x_i \frac{\langle \nabla(\partial_i f)(x), e_j \rangle}{\partial_i f(x)} \langle \nabla(\partial_i f)(x), e_k \rangle = \sum_{i=1}^{m} x_i \frac{\partial_{ji} f(x) \partial_{ki} f(x)}{\partial_i f(x)},$$

and

$$\left\langle \left[\mathrm{hess}(f)(x) \cdot D(x) \cdot \mathrm{hess}(f)(x) \right] e_j, e_k \right\rangle =$$
$$= \left\langle \mathrm{hess}(f)(x) \cdot D(x) \cdot (\partial_{1j} f(x), \dots, \partial_{mj} f(x))^T, e_k \right\rangle =$$
$$= \left\langle \mathrm{hess}(f)(x) \cdot \left(\frac{x_1 \partial_{1j} f(x)}{\partial_1 f(x)}, \dots, \frac{x_m \partial_{mj} f(x)}{\partial_m f(x)} \right)^T, e_k \right\rangle = \sum_{i=1}^{m} x_i \frac{\partial_{ij} f(x) \partial_{ki} f(x)}{\partial_i f(x)},$$

and hence we have (8.2.9). Thus, we may rewrite (8.2.8) as

$$\mathrm{hess}(f)(x) \leq \frac{1}{2} \mathrm{hess}(f)(x) \cdot D(x) \cdot \mathrm{hess}(f)(x). \tag{8.2.10}$$

For each $x \in C$, we let

$$D(x)^{\frac{1}{2}} := \mathrm{diag} \left(\left(\frac{x_1}{\partial_1 f(x)} \right)^{\frac{1}{2}}, \dots, \left(\frac{x_m}{\partial_m f(x)} \right)^{\frac{1}{2}} \right).$$

The notation justifies because it is clear that $D(x)^{\frac{1}{2}} D(x)^{\frac{1}{2}} = D(x)$. We recall again that all coordinates of any point $x \in C$ are positive, as well as all partial derivatives $\partial_j f(x)$ (by hypothesis), and then $D(x)^{\frac{1}{2}}$ is indeed well-defined. Now we let

$$M(x) := D(x)^{\frac{1}{2}} \cdot \mathrm{hess}(f)(x) \cdot D(x)^{\frac{1}{2}}$$

for each $x \in C$. The (jk)-th entry of $M(x)$ is clearly given as

$$\big(M(x)\big)_{jk} = \left(\frac{x_j}{\partial_j f(x)} \right)^{\frac{1}{2}} \left(\frac{x_k}{\partial_k f(x)} \right)^{\frac{1}{2}} \partial_{jk} f(x),$$

and this yields immediately that $M(x)$ is symmetric (because the partial derivatives commute) and $\big(M(x)\big)_{jk} \geq 0$ whenever $j \neq k$ (here we used the second hypothesis in **(c)**). Once $D(x)^{1/2}$ is a diagonal matrix whose diagonal entries are positive, we have that $D(x)^{1/2}$ admits an inverse matrix $D(x)^{-1/2}$. Because $\mathrm{hess}(f)(x) = D(x)^{-1/2} M(x) D(x)^{-1/2}$ is irreducible and $D(x)^{-1/2}$ is diagonal, it follows straightforwardly that $M(x)$ is irreducible. This happens because any coordinate subspace is invariant under a diagonal matrix, and hence any proper coordinate subspace invariant under $M(x)$ would also be invariant under $\mathrm{hess}(f)(x)$, which is irreducible by the hypothesis.

Now we claim that if N is a negative semi-definite matrix and D is a diagonal matrix (both being $m \times m$ matrices), then the matrix DND is also negative semi-definite. In fact, for any $z \in \mathbb{R}^m$, we have

$$0 \geq \langle N(Dz), Dz \rangle = \langle D^T N D z, z \rangle = \langle DNDz, z \rangle.$$

From (8.2.10), the matrix

$$A := \mathrm{hess}(f)(x) - \frac{1}{2}\mathrm{hess}(f)(x) \cdot D(x) \cdot \mathrm{hess}(f)(x)$$

is negative semi-definite. Hence, we have that the matrix $D(x)^{\frac{1}{2}} A D(x)^{\frac{1}{2}}$ is negative semi-definite. Thus, we may write

$$\begin{aligned} 0 \geq D(x)^{\frac{1}{2}} \left(\mathrm{hess}(f)(x) - \frac{1}{2}\mathrm{hess}(f)(x) D(x) \mathrm{hess}(f)(x) \right) D(x)^{\frac{1}{2}} = \\ = M(x) - \frac{1}{2} D(x)^{\frac{1}{2}} \mathrm{hess}(f)(x) D(x) \mathrm{hess}(f)(x) D(x)^{\frac{1}{2}} = \\ = M(x) - \left(D(x)^{\frac{1}{2}} \mathrm{hess}(f)(x) D(x)^{\frac{1}{2}} \right) \left(D(x)^{\frac{1}{2}} \mathrm{hess}(f)(x) D(x)^{\frac{1}{2}} \right) = M(x) - \frac{1}{2} M(x)^2. \end{aligned}$$

Let λ be any eigenvalue of $M(x)$, and let v be an associated eigenvector. Then, we have

$$0 \geq \left\langle \left(M(x) - \frac{1}{2} M(x)^2 \right) v, v \right\rangle = \left\langle \lambda v - \frac{\lambda^2}{2} v, v \right\rangle = |v|^2 \left(\lambda - \frac{\lambda^2}{2} \right),$$

and this gives that $\lambda \leq 0$ or $\lambda \geq 2$. Equivalently, this means that $M(x)$ does not have any eigenvalue in the interval $(0, 2)$.

Now observe that

$$D(x)^{-\frac{1}{2}} := \mathrm{diag}\left(\left(\frac{\partial_1 f(x)}{x_1} \right)^{\frac{1}{2}}, \ldots, \left(\frac{\partial_m f(x)}{x_m} \right)^{\frac{1}{2}} \right)$$

for each $x \in C$. We claim that, for each $x \in C$, the vector

$$v(x) := D(x)^{-\frac{1}{2}} x = \begin{bmatrix} (x_1 \partial_1 f(x))^{\frac{1}{2}} \\ (x_2 \partial_2 f(x))^{\frac{1}{2}} \\ \vdots \\ (x_m \partial_m f(x))^{\frac{1}{2}} \end{bmatrix}$$

is an eigenvector of $M(x)$ with associated eigenvalue $\lambda_1 = 2$ (notice carefully that $v(x)$ is a positive vector). To verify this, notice first that since each partial derivative $\partial_k f$ is positively homogeneous of degree 2, Lemma 8.2.2 yields

$$\sum_{i=1}^{m} x_i \partial_{ki} f(x) = 2 \partial_k f(x) \tag{8.2.11}$$

for any $k = 1, \ldots, m$ and each $x \in C$. Using these equalities we compute:

$$M(x)v(x) = M(x)D(x)^{-\frac{1}{2}}x = D(x)^{\frac{1}{2}}\mathrm{hess}(f)(x)x =$$

$$= D(x)^{\frac{1}{2}}\begin{bmatrix} \partial_{11}f(x) & \partial_{12}f(x) & \dots & \partial_{1m}f(x) \\ \partial_{21}f(x) & \partial_{22}f(x) & \dots & \partial_{2m}f(x) \\ \vdots & \vdots & \dots & \vdots \\ \partial_{m1}f(x) & \partial_{m2}f(x) & \dots & \partial_{mm}f(x) \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = D(x)^{\frac{1}{2}}\begin{bmatrix} \sum_{i=1}^m x_i\partial_{1i}f(x) \\ \sum_{i=1}^m x_i\partial_{2i}f(x) \\ \vdots \\ \sum_{i=1}^m x_i\partial_{mi}f(x) \end{bmatrix} =$$

$$= 2D(x)^{\frac{1}{2}}\begin{bmatrix} \partial_1 f(x) \\ \partial_2 f(x) \\ \vdots \\ \partial_m f(x) \end{bmatrix} = 2\begin{bmatrix} (x_1\partial_1 f(x))^{\frac{1}{2}} \\ (x_2\partial_2 f(x))^{\frac{1}{2}} \\ \vdots \\ (x_m\partial_m f(x))^{\frac{1}{2}} \end{bmatrix} = 2v(x),$$

and hence $v(x)$ is indeed an eigenvector of $M(x)$ associated to the eigenvalue 2. Summarizing, for each $x \in C$, we have the following:

(i) $M(x)$ is symmetric, irreducible, and all of its non-diagonal entries are non-negative,

(ii) $M(x)$ has the positive vector $v^1(x) := v(x)/|v(x)| \in \mathbb{S}^{m-1}$ as an eigenvector associated to the eigenvalue $\lambda_1 = 2$, and

(iii) $M(x)$ has no eigenvalue in the interval $(0, 2)$.

It follows that we may apply Corollary 8.2.1 to $M(x)$. This leads to

$$M(x) \leq 2v^1(x) \otimes v^1(x), \tag{8.2.12}$$

where we used the rewriting of (8.2.2) given in (8.2.4), of course. Now observe that

$$|v(x)|^2 = \left|D(x)^{-\frac{1}{2}}x\right|^2 = \left\langle D(x)^{-\frac{1}{2}}x, D(x)^{-\frac{1}{2}}x\right\rangle = \left\langle x, D(x)^{-1}x\right\rangle = \left\langle x, \nabla f(x)\right\rangle =$$

$$= \sum_{i=1}^m x_i\partial_i f(x) = 3f(x),$$

where the last equality follows immediately when we apply Lemma 8.2.2 for f (which is positively homogeneous of degree 3). It is clear from the definition that

$$v^1(x) \otimes v^1(x) = \frac{v(x)}{|v(x)|} \otimes \frac{v(x)}{|v(x)|} = \frac{v(x) \otimes v(x)}{|v(x)|^2} = \frac{D(x)^{-\frac{1}{2}}x \otimes D(x)^{-\frac{1}{2}}x}{3f(x)},$$

from which we may rewrite (8.2.12) as

$$D(x)^{\frac{1}{2}}\mathrm{hess}(f)(x)D(x)^{\frac{1}{2}} \leq \frac{2}{3}\frac{D(x)^{-\frac{1}{2}}x \otimes D(x)^{-\frac{1}{2}}x}{f(x)}.$$

We proved above that if N is negative semi-definite and D is diagonal, then DND is negative semi-definite. Hence, we may multiply both sides (on the left and on the right) of the inequality above by $D(x)^{-\frac{1}{2}}$. We arrive at

$$\mathrm{hess}(f)(x) \leq \frac{2}{3f(x)}D(x)^{-\frac{1}{2}}\left(D(x)^{-\frac{1}{2}}x \otimes D(x)^{-\frac{1}{2}}x\right)D(x)^{-\frac{1}{2}}. \tag{8.2.13}$$

To understand the linear map in the right-hand side, we apply it to a given vector $z \in \mathbb{R}^m$:

$$\begin{aligned} D(x)^{-\frac{1}{2}}\Big(D(x)^{-\frac{1}{2}}x \otimes D(x)^{-\frac{1}{2}}x\Big)D(x)^{-\frac{1}{2}}(z) = \\ = D(x)^{-\frac{1}{2}}\Big(\big\langle D(x)^{-\frac{1}{2}}x, D(x)^{-\frac{1}{2}}z\big\rangle D(x)^{-\frac{1}{2}}x\Big) = \\ = \big\langle D(x)^{-\frac{1}{2}}x, D(x)^{-\frac{1}{2}}z\big\rangle D(x)^{-\frac{1}{2}}D(x)^{-\frac{1}{2}}x = \big\langle D(x)^{-1}x, z\big\rangle D(x)^{-1}x = \\ = D(x)^{-1}x \otimes D(x)^{-1}x(z). \end{aligned}$$

Thus, we have the equality

$$D(x)^{-\frac{1}{2}}\Big(D(x)^{-\frac{1}{2}}x \otimes D(x)^{-\frac{1}{2}}x\Big)D(x)^{-\frac{1}{2}} = D(x)^{-1}x \otimes D(x)^{-1}x,$$

and a direct computation gives

$$D(x)^{-1}x = \nabla f(x).$$

Now we use these equalities to rewrite the right-hand side of (8.2.13). We get

$$\operatorname{hess}(f)(x) \le \frac{2}{3}\frac{\nabla f(x) \otimes \nabla f(x)}{f(x)},$$

which means that the matrix

$$H(x) := \operatorname{hess}(f)(x) - \frac{2}{3}\frac{\nabla f(x) \otimes \nabla f(x)}{f(x)}$$

is negative semi-definite. Now observe that for any $i, j \in \{1, \dots, m\}$, we have

$$\partial_{ij}\big(f^{\frac{1}{3}}\big)(x) = \frac{1}{3}f(x)^{-\frac{2}{3}}\left[\partial_{ij}f(x) - \frac{2}{3}\frac{\partial_i f(x)\partial_j f(x)}{f(x)}\right]$$

for each $x \in C$. Thus, having in mind the canonical matrix representation of $\nabla f(x) \otimes \nabla f(x)$ given as in (8.2.3), the Hessian matrix of $f^{\frac{1}{3}}$ at a point $x \in C$ can be written as

$$\operatorname{hess}\big(f^{\frac{1}{3}}\big)(x) = \frac{1}{3}f(x)^{-\frac{2}{3}}\left(\operatorname{hess}(f)(x) - \frac{2}{3}\frac{\nabla f(x) \otimes \nabla f(x)}{f(x)}\right) = \frac{1}{3}f(x)^{-\frac{2}{3}}H(x).$$

Since $H(x)$ is negative semi-definite and $\frac{1}{3}f(x)^{-\frac{2}{3}} > 0$, we get that the Hessian matrix of the function $f^{\frac{1}{3}}$ is negative semi-definite at each point of C. This shows that $f^{\frac{1}{3}}$ is concave and the proof is complete. □

Our next step is to understand how the Alexandrov–Fenchel inequality is related to the concavity of certain functions given by mixed volumes and Minkowski combinations.

Lemma 8.2.3. *Assume that $n \ge 3$. For given convex bodies $K_1, K_2, K_3, \dots, K_n \in \mathcal{K}^n$, consider the functions*

$$f_2(K_1, K_2, K_3, \dots, K_n, t) := V(K_1 + tK_2[2], K_3, \dots, K_n)^{\frac{1}{2}}$$

and

$$f_3(K_1, K_2, K_4, \dots, K_n, t) := V(K_1 + tK_2[3], K_4, \dots, K_n)^{\frac{1}{3}}.$$

defined for $t \geq 0$. The following statements are equivalent:

(a) *Inequality (8.2.1) holds for any convex bodies $K_1, \ldots, K_n \in \mathcal{K}^n$.*

(b) *For any fixed n-tuple $K_1, K_2, K_3, \ldots, K_n \in \mathcal{K}^n$ the function f_2 defined as above is concave.*

(c) *For every fixed $(n-1)$-tuple $K_1, K_2, K_4, \ldots, K_n \in \mathcal{K}^n$ the function f_3 defined as above is concave.*

Remark 8.2.1. Actually, the Alexandrov–Fenchel inequality implies the concavity of a larger class of functions, as it will be proved later in Theorem 8.3.1.

Proof. We start by proving the equivalence between **(a)** and **(b)**, which follows from an easy calculation. For the sake of simplicity, we write

$$V_2(t) := V(K_1 + tK_2[2], K_3, \ldots, K_n), \quad \text{and}$$
$$f_2(t) := f_2(K_1, \ldots, K_n, t) = V_2(t)^{\frac{1}{2}}.$$

From the linearity of mixed volumes (recall that we are assuming $t \geq 0$), we may write $V_2(t)$ as

$$V_2(t) = t^2 V_{22} + 2tV_{12} + V_{11},$$

where $V_{ij} = V(K_i, K_j, K_3, \ldots, K_n)$ for $i, j \in \{1, 2\}$. Hence, $V_2(t)$ is differentiable, and consequently $f_2(t)$ is also differentiable (observe that always $V_2(t) > 0$, since the involved sets are all convex bodies). A straightforward computation yields

$$f_2''(t) = -\frac{1}{4} f_2(t)^{-3} V_2'(t)^2 + \frac{1}{2} f_2(t) V_2''(t) = -f_2(t)^{-3} \big(tV_{22} + V_{12}\big)^2 + f_2(t)^{-1} V_{22} =$$
$$= -f_2(t)^{-3} \big[t^2 V_{22}^2 + 2tV_{22}V_{12} + V_{12}^2 - f_2(t)^2 V_{22}\big] = -f_2(t)^{-3} \big(V_{12}^2 - V_{11}V_{22}\big).$$

Since $f_2(t) > 0$, it follows that $f_2''(t) \leq 0$ for every $t \geq 0$ if and only if $V_{12}^2 - V_{11}V_{22} \geq 0$. This means precisely that the function $f_2(K_1, \ldots, K_n, \cdot)$ is concave if and only if inequality (8.2.1) holds for the bodies $K_1, \ldots, K_n$, and this settles the equivalence between **(a)** and **(b)**.

Now we prove that **(a)** implies **(c)**. To do that, we first write

$$V_3(t) := V(K_1 + tK_2[3], K_4, \ldots, K_n) \quad \text{and}$$
$$f_3(t) := f_3(K_1, K_2, K_4, \ldots, K_n, t) = V_3(t)^{\frac{1}{3}}.$$

For $t \geq 0$, the linearity of mixed volumes gives that $V_3(t)$ is a (positive) polynomial of third degree. Thus, as it was the case for V_2 and f_2, the functions V_3 and f_3 are differentiable. We have

$$V_3(t) = t^3 V_{222} + 3t^2 V_{122} + 3tV_{112} + V_{111},$$

where $V_{ijl} = V(K_i, K_j, K_l, K_4, \ldots, K_n)$ for $i, j, l \in \{1, 2\}$. For simplicity, we introduce $\mathbf{K} := (K_4, \ldots, K_n)$ and $K_t := K_1 + tK_2$. Differentiating $f_3(t)$ twice, and using the expression above to get the first and second derivatives of $V_3(t)$, we obtain

$$f_3''(t) = -\frac{2}{9} f_3(t)^{-5} V_3'(t)^2 + \frac{1}{3} f_3(t)^{-2} V_3''(t) =$$
$$= -f_3(t)^{-5} \left[\frac{2}{9} V_3'(t)^2 - \frac{1}{3} V_3(t) V_3''(t) \right] = -f_3(t)^{-5} \big[2(t^2 V_{222} + 2t V_{221} + V_{112})^2 -$$
$$-(2t V_{222} + 2V_{122})(t^3 V_{222} + 3t^2 V_{122} + 3t V_{112} + V_{111}) \big] =$$
$$= -2 f_3(t)^{-5} \big(- t^2 V_{222} V_{112} + t^2 V_{221}^2 - t V_{222} V_{111} + t V_{112} V_{221} + V_{112}^2 - V_{122} V_{111} \big) =$$
$$= -2 f_3(t)^{-5} \Big[(V_{112} + t V_{122})^2 - (V_{111} + t V_{112})(V_{221} + t V_{222}) \Big] =$$
$$= -2 f_3(t)^{-5} \Big[V(K_1, K_2, K_t, \mathbf{K})^2 - V(K_1, K_1, K_t, \mathbf{K}) V(K_2, K_2, K_t, \mathbf{K}) \Big].$$

Since our hypothesis is that inequality (8.2.1) holds for every n-tuple of convex bodies of $\mathbb{R}^n$, we may apply it for the bodies $K_1, K_2, K_t, K_4, \ldots, K_n$ to obtain

$$V(K_1, K_2, K_t, \mathbf{K})^2 - V(K_1, K_1, K_t, \mathbf{K}) V(K_2, K_2, K_t, \mathbf{K}) \geq 0,$$

and once f_3 is positive, this gives that $f_3''(t) \leq 0$ for every $t \geq 0$. This means that f_3 is concave, as we wanted to show.

Last, we prove that **(c)** implies **(b)**. Let $K_1, K_2, K_3, K_4, \ldots, K_n \in \mathcal{K}^n$ be arbitrarily given convex bodies. We write a point of $\mathbb{R}^3$ in coordinates as $x = (x_1, x_2, x_3)$, as usual, and we extend this notation for vectors of the subsets $[0, +\infty)^3$ and $(0, +\infty)^3$ of $\mathbb{R}^3$. Define the function $G \colon \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$ by

$$G(x, y, z) := \sum_{i,j,l=1}^{3} x_i y_j z_l V(K_i, K_j, K_l, K_4, \ldots, K_n).$$

This function is clearly symmetric and linear in each entry. From the linearity of mixed volumes (for non-negative scalars), if $x, y, z \in [0, +\infty)^3$, we have

$$G(x, y, z) =$$
$$= V\big(x_1 K_1 + x_2 K_2 + x_3 K_3, y_1 K_1 + y_2 K_2 + y_3 K_3, z_1 K_1 + z_2 K_2 + z_3 K_3, K_4, \ldots, K_n\big).$$

Of course, if in addition, we have that $x, y, z \neq 0_{\mathbb{R}^3}$, then we have that $G(x, y, z) > 0$ (since in this case all entries of the mixed volume which defines G are convex bodies). Now we let $F \colon \mathbb{R}^3 \to \mathbb{R}$ be given as

$$F(x) := G(x, x, x),$$

and $f_{x,y} \colon [0, +\infty) \to \mathbb{R}$ be given as

$$f_{x,y}(t) = F(x + ty)^{\frac{1}{3}},$$

where $x, y \in [0, +\infty)^3$ are fixed. We claim that $f_{x,y}(t)$ is a concave function for any fixed $x, y \in [0, +\infty)^3$. This is obvious if $x = 0_{\mathbb{R}^3}$ or $y = 0_{\mathbb{R}^3}$. Then we assume that both x and y are non-zero, and we write

$$\begin{aligned} f_{x,y}(t) = F(x+ty)^{\frac{1}{3}} = \\ = V\big((x_1+ty_1)K_1 + (x_2+ty_2)K_2 + (x_3+ty_3)K_3[3], K_4, \dots, K_n\big)^{\frac{1}{3}} = \\ = V\big((x_1K_1 + x_2K_2 + x_3K_3) + t(y_1K_1 + y_2K_2 + y_3K_3)[3], K_4, \dots, K_n\big)^{\frac{1}{3}} = \\ = f_3(x_1K_1 + x_2K_2 + x_3K_3, y_1K_1 + y_2K_2 + y_3K_3, K_4, \dots, K_n, t). \end{aligned}$$

Since x and y are non-zero, we get that $x_1K_1 + x_2K_2 + x_3K_3$ and $y_1K_1 + y_2K_2 + y_3K_3$ are convex bodies. Since we are assuming that **(c)** holds, this equality gives immediately that $f_{x,y}$ is concave.

Now we claim that the fact that $f_{x,y}$ is always concave yields the inequality

$$G(x,y,z)^2 \geq G(x,x,z)G(y,y,z) \tag{8.2.14}$$

for any $x, y, z \in [0,+\infty)^3$. Since G is continuous it suffices to consider the case where all entries of $x, y, z \in (0,+\infty)^3$ (the case for $[0,+\infty)^3$ follows from approximations). In this case, we already know $G(x,x,z)$, $G(y,y,z)$ and $G(x,y,z)$ are strictly positive. Because G is linear (in each entry) and symmetric, we have

$$\frac{d}{dt}F(x+ty) = \frac{d}{dt}G(x+ty, x+ty, x+ty) = 3G(y, x+ty, x+ty)$$

and

$$\frac{d^2}{dt^2}F(x+ty) = 6G(y,y,x+ty).$$

Thus, we have the following expression to the second derivative of $f_{x,y}$:

$$\begin{aligned} f''_{x,y}(t) = -\frac{2}{9}F(x+ty)^{-\frac{5}{3}}\left(\frac{d}{dt}F(x+ty)\right)^2 + \frac{1}{3}F(x+ty)^{-\frac{2}{3}}\frac{d^2}{dt^2}F(x+ty) = \\ = -2F(x+ty)^{-\frac{5}{3}}G(y,x+ty,x+ty)^2 + 2F(x+ty)^{-\frac{2}{3}}G(y,y,x+ty). \end{aligned}$$

Letting $t = 0$ (and, for the sake of the formalism, considering the right-sided derivatives, once we are assuming that $t \geq 0$), we get the equality

$$\begin{aligned} f''_{x,y}(0) = -2F(x)^{-\frac{5}{3}}G(y,x,x)^2 + 2F(x)^{-\frac{2}{3}}G(y,y,x) = \\ = -2F(x)^{-\frac{5}{3}}\big[G(y,x,x)^2 - F(x)G(y,y,x)\big]. \end{aligned}$$

Since we are assuming that the entries of x are non-zero, we have that $F(x) > 0$. Because $f_{x,y}$ is concave, we have that $f''_{x,y}(0) \leq 0$. Hence, recalling also that $F(x) = G(x,x,x)$, we get

$$G(y,x,x)^2 - G(x,x,x)G(y,y,x) \geq 0 \tag{8.2.15}$$

for any $x, y \in (0,+\infty)^3$. This can be easily extended to the case where $x \in (0,+\infty)^3$ and y is any vector of $\mathbb{R}^3$. If this is the case, we let $t_0 > 0$ be large enough such that $y + t_0x \in (0,+\infty)^3$. Then the inequality above holds if we replace y by $y + t_0x$, and using the linearity of G we arrive at

$$\begin{aligned} 0 \leq G(y+t_0x, x, x)^2 - G(x,x,x)G(y+t_0x, y+t_0x, x) = \\ = G(y,x,x)^2 - G(x,x,x)G(y,y,x). \end{aligned}$$

Now let $x, y, z \in (0,+\infty)^3$ be arbitrarily fixed, and let $h\colon \mathbb{R} \to \mathbb{R}$ be the quadratic function given as

$$h(t) := G(y + tz, y + tz, x) = t^2 G(z, z, x) + 2tG(y, z, x) + G(y, y, x).$$

Recalling that $G(z, x, x) > 0$, we may write

$$t_1 := -\frac{G(y, x, x)}{G(z, x, x)}.$$

Notice carefully that $t_1 < 0$. We have that $G(y+t_1 z, x, x) = 0$. Since we already know that (8.2.15) holds if $x \in (0, +\infty)^3$ and $y \in \mathbb{R}^3$, we may apply it here for the vectors x and $y + t_1 z$. By doing this, we get

$$0 \leq G(y + t_1 z, x, x)^2 - G(x, x, x)G(y + t_1 z, y + t_1 z, x) = -G(x, x, x)h(t_1).$$

Because $G(x, x, x) > 0$, we arrive at $h(t_1) \leq 0$. On the other hand, $h(0) = G(y, y, x) > 0$. It follows that h has a root in the interval $[t_1, 0)$, and this means that its discriminant is non-negative. This yields inequality (8.2.14) for $x, y, z \in (0, +\infty)^3$ and, as we have mentioned earlier, from the continuity of G, this naturally extends to the case where $x, y, z \in [0, +\infty)^3$.

Finally, inequality (8.2.1) is precisely the case of (8.2.14), where $x = e_1$, $y = e_2$, and $z = e_3$. This shows that **(c)** implies **(a)**, and the proof is complete.

□

Recall that a function $f : C^n \to \mathbb{R}$ is *concave* if

$$f((1 - t)K + tL) \geq (1 - t)f(K) + tf(L)$$

for every $K, L \in C^n$ and any $t \in [0, 1]$. Since $\mathcal{K}^n$ is closed for convex combinations, the same definition works if we replace C^n by $\mathcal{K}^n$.

Lemma 8.2.4. *Let $m \in \{2, \dots, n\}$ and let $K_1, K_2, K_{m+1}, \dots, K_n \in \mathcal{K}^n$. Define $f_m : [0, +\infty) \to \mathbb{R}$ by*

$$f_m(t) = V(K_1 + tK_2[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}}, \quad t \geq 0,$$

and $g_m : [0, 1] \to \mathbb{R}$ by

$$g_m(t) = V((1 - t)K_1 + tK_2[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}}, \quad 0 \leq t \leq 1.$$

For fixed $K_{m+1}, \dots, K_n \in \mathcal{K}^n$, we also let $v_m : \mathcal{K}^n \to \mathbb{R}$ be given as

$$v_m(K) := V(K[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}}, \quad K \in \mathcal{K}^n.$$

We have:

(a) *For fixed $K_1, K_2, K_{m+1}, \dots, K_n$, we have that f_m is concave if and only if g_m is concave.*

(b) *Fixing $K_{m+1}, \dots, K_n \in \mathcal{K}^n$, we have that v_m is concave if and only if g_m is concave for any bodies in place of K_1 and K_2.*

Proof. We start by proving **(a)**. Let $s : [0, 1) \to [0, +\infty)$ be given as

$$s(t) = \frac{t}{1 - t},$$

and notice that this is clearly a bijection. Write

$$g_m(t) = V((1-t)K_1 + tK_2[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}} =$$
$$= (1-t)V(K_1 + s(t)K_2[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}} = (1-t)f_m(s(t)).$$

By the chain rule, derivating twice, we get

$$\frac{d^2 g_m}{dt^2} = \frac{df_m}{ds}\left(-2\frac{ds}{dt} + (1-t)\frac{d^2 s}{dt^2}\right) + (1-t)\left(\frac{df_m}{ds}\right)^2 \frac{d^2 f_m}{ds^2}.$$

A straightforward calculation gives that

$$-2\frac{ds}{dt} + (1-t)\frac{d^2 s}{dt^2} = 0,$$

and hence we arrive at

$$\frac{d^2 g_m}{dt^2} = (1-t)\left(\frac{df_m}{ds}\right)^2 \frac{d^2 f_m}{ds^2}.$$

Hence, we have that $d^2 g_m/dt^2 \le 0$ for every $t \in [0,1)$ if and only if $d^2 f_m/ds^2 \le 0$ for any $s \ge 0$. This means that g_m is concave in $[0,1)$ if and only if f_m is concave in $[0,+\infty)$. Of course, continuity gives that in this case g_m is concave in $[0,1]$.

Now we prove **(b)**. Assume first that g_m is concave for any bodies in place of K_1 and K_2. Let $K, L \in \mathcal{K}^n$, and $\lambda \in [0,1]$ be arbitrarily given. Defining g_m with K and L in place of K_1 and K_2, respectively, we have that

$$g_m(\lambda) = g_m((1-\lambda)0 + \lambda 1) \ge (1-\lambda)g_m(0) + \lambda g_m(1),$$

for any $\lambda \in [0,1]$. Letting $\mathbf{K} := (K_{m+1}, \dots, K_n)$, this reads

$$V\big((1-\lambda)K + \lambda L[m], \mathbf{K}\big)^{\frac{1}{m}} \ge (1-\lambda)V\big(K[m], \mathbf{K}\big)^{\frac{1}{m}} + V\big(L[m], \mathbf{K}\big)^{\frac{1}{m}}.$$

This is the same as

$$v_m\big((1-\lambda)K + \lambda L\big) \ge (1-\lambda)v_m(K) + \lambda v_m(L),$$

and hence v_m is concave. For the converse, assume that v_m is concave, let K_1 and K_2 be arbitrarily given convex bodies, and set

$$g_m(t) = V((1-t)K_1 + tK_2, \mathbf{K})^{\frac{1}{m}}, \quad 0 \le t \le 1.$$

We have to prove that g_m is concave. For that sake, let $\lambda, t, s \in [0,1]$ be arbitrarily given. Notice that we have

$$\begin{aligned} \big(1 - \big[(1-\lambda)t + \lambda s\big]\big)K_1 + \big((1-\lambda)t + \lambda s\big)K_2 = \\ = (1-\lambda)\big[(1-t)K_1 + tK_2\big] + \lambda\big[(1-s)K_1 + sK_2\big]. \end{aligned} \tag{8.2.16}$$

Applying the concavity of v_m for the bodies $(1-t)K_1 + tK_2$ and $(1-s)K_1 + sK_2$, and for $\lambda \in [0,1]$, we get

$$V\big((1-\lambda)((1-t)K_1+tK_2)+\lambda((1-s)K_1+sK_2)[m],\mathbf{K}\big)^{\frac{1}{m}} \geq$$
$$\geq (1-\lambda)V\big((1-t)K_1+tK_2[m],\mathbf{K}\big)^{\frac{1}{m}}+\lambda V\big((1-s)K_1+sK_2[m],\mathbf{K}\big)^{\frac{1}{m}} =$$
$$= (1-\lambda)g_m(t)+\lambda g_m(s).$$

Using (8.2.16) to rewrite the left-hand side of the above, we arrive at

$$V\big((1-\lambda)((1-t)K_1+tK_2)+\lambda((1-s)K_1+sK_2)[m],\mathbf{K}\big)^{\frac{1}{m}} =$$
$$= V\big(\big(1-\big[(1-\lambda)t+\lambda s\big]\big)K_1+\big((1-\lambda)t+\lambda s\big)K_2[m],\mathbf{K}\big)^{\frac{1}{m}} =$$
$$= g_m\big((1-\lambda)t+\lambda s\big),$$

and hence we get

$$g_m\big((1-\lambda)t+\lambda s\big) \geq (1-\lambda)g_m(t)+\lambda g_m(s).$$

Since this holds for any $t,s,\lambda \in [0,1]$, we conclude that g_m is concave, and the proof is complete (recall that K_1 and K_2 were arbitrarily taken).

□

Remark 8.2.2. Later, as a consequence of the Alexandrov–Fenchel inequality, we will prove that all functions f_m, g_m and v_m defined for $m \in \{2,\dots,n\}$ in Lemma 8.2.4 are indeed concave.

Corollary 8.2.2. *The Alexandrov–Fenchel inequality (8.2.1) holds for any bodies $K_1,K_2,K_4,\dots,K_n \in \mathcal{K}^n$ if and only if the function*

$$K \mapsto V(K[3],K_4,\dots,K_n)^{\frac{1}{3}} \tag{8.2.17}$$

is concave in $\mathcal{K}^n$ for any $K_4,\dots,K_n \in \mathcal{K}^n$.

Proof. By Lemma 8.2.4, the function given in (8.2.17) is concave for any $K_4,\dots,K_n \in \mathcal{K}^n$ if and only if $f_3(t) := V(K_1+tK_2[3],K_4,\dots,K_n)^{\frac{1}{3}}$ is concave in $[0,+\infty)$ for every $K_1,K_2,K_4,\dots,K_n \in \mathcal{K}^n$. By Lemma 8.2.3, this is equivalent to the Alexandrov–Fenchel inequality.

□

Next, we investigate mixed volumes of strongly isomorphic polytopes. The reader should be aware of the fact that we will use heavily the notation and results presented in Section 3.3. For the sake of convenience, we recall that for a polytope $Q \subseteq \mathbb{R}^n$ and a vector $u \in \mathbb{S}^{n-1}$ we denote by $\mathbf{h}(Q,u)$ the supporting hyperplane of Q having u as unit outer normal vector. Also, we write $F(Q,u) := Q \cap \mathbf{h}(Q,u)$, which is the face of Q having u as unit outer normal.

Lemma 8.2.5. *Let $u_1,\dots,u_m \in \mathbb{S}^{n-1}$ be distinct vectors such that*

$$\operatorname{pos}\{u_1,\dots,u_m\} = \mathbb{R}^n,$$

and let $h = (h_1,\dots,h_m) \in (0,+\infty)^m$ be any m-tuple of positive numbers. Then the set

$$P_{[h]} := \bigcap_{j=1}^{m} \mathbf{h}^-(u_j,h_j) \tag{8.2.18}$$

is a proper polytope having the origin as an interior point. Moreover, we have that

$$\partial P_{[h]} \subseteq \bigcup_{j=1}^{m} \mathbf{h}(u_j, h_j). \tag{8.2.19}$$

Remark 8.2.3. Notice that this construction is a particular case of the Wulff shape.

Proof. From Corollary 3.1.6, we know that any bounded finite intersection of half-spaces is a polytope. The fact that $P_{[h]}$ is a convex body containing the origin as an interior point follows straightforwardly from Lemma 8.1.1. The inclusion in (8.2.19) follows immediately from Lemma 8.1.3.

□

Having fixed (distinct) unit outer normal vectors $u_1, \ldots, u_m$ which span $\mathbb{R}^n$ by positive combinations, $P_{[h]}$ will always denote the construction introduced in (8.2.18).

Now we restrict ourselves to the case where the fixed (distinct) vectors $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ are the unit outer normal vectors of the facets of a given proper polytope $P \subseteq \mathbb{R}^n$. Letting $h_0 = (h_P(u_1), \ldots, h_P(u_m))$, we have from Corollary 3.1.4 that $P = P_{[h_0]}$. In this case, we say that h_0 is the *vector of support numbers* of P, and we have the following uniqueness result: if $h \in (0, +\infty)^m$ is such that $P_{[h]} = P$, then $h = h_0$. Indeed, fix any $k \in \{1, \ldots, m\}$ and let F_k be the facet of P whose unit outer normal vector is u_k. Recalling that $F_k = \partial P \cap \mathbf{h}(u_k, h_P(u_k))$ and using (8.2.19), we get

$$F_k = \partial P_{[h]} \cap \mathbf{h}(u_k, h_P(u_k)) \subseteq \mathbf{h}(u_k, h_P(u_k)) \cap \left(\bigcup_{j=1}^{m} \mathbf{h}(u_j, h_j) \right) =$$
$$= \bigcup_{j=1}^{m} \left[\mathbf{h}(u_k, h_P(u_k)) \cap \mathbf{h}(u_j, h_j) \right],$$

and then the latter must be $(n-1)$-dimensional because F_k is $(n-1)$-dimensional. Of course, this is only possible if $h_k = h_P(u_k)$, and since this holds for every $k = 1, \ldots, m$, we arrive at the equality $h = h_0$.

Notice very carefully that this uniqueness result has the following immediate consequence: if $h = (h_1, \ldots, h_m) \in (0, +\infty)^m$ is such that $P_{[h]}$ is a polytope having precisely the vectors $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ as unit outer normal vectors of the facets, then, representing the support function of $P_{[h]}$ as $h(P_{[h]}, \cdot)$, we have

$$h(P_{[h]}, u_j) = h_j \tag{8.2.20}$$

for each $j \in \{1, \ldots, m\}$. As in other parts of the book, this alternative notation for support functions is adopted for the sake of clarity (by doing that we avoid too complicated subindices).

From now on, we work only with proper simple polytopes. By doing that, we have the advantage that each k-face lies in exactly $n-k$ facets (see Section 3.3 again). The reader must keep this fact in mind. Also, recall that we denote the equivalence class of polytopes which are strongly isomorphic to a given polytope P by $[P]_{\mathrm{si}}$.

Proposition 8.2.3. *Let $P \subseteq \mathbb{R}^n$ be a proper simple polytope containing the origin as an interior point, and having facets $F_1, \ldots, F_m$ with corresponding unit outer normal vectors $u_1, \ldots, u_m$. Define*

$$\mathcal{A}(P) := \left\{ h \in (0, +\infty)^m : P_{[h]} \in [P]_{\mathrm{si}} \right\}.$$

Then $\mathcal{A}(P)$ is an open convex cone, and

$$(h_P(u_1), \ldots, h_P(u_m)) \in \mathcal{A}(P).$$

Proof. The fact that $(h_P(u_1), \dots, h_P(u_m)) \in \mathcal{A}(P)$ follows immediately from the comment above the statement. Also, since homothetical polytopes are strongly isomorphic (this comes from the definition of strongly isomorphic polytopes), we get that $\lambda h \in \mathcal{A}(P)$ whenever $h \in \mathcal{A}(P)$ and $\lambda > 0$.

Now, let $h = (h_1, \dots, h_m)$ and $h' = (h'_1, \dots, h'_m)$ be arbitrary elements of $\mathcal{A}(P)$. We will prove that $P_{[h+h']}$ is strongly isomorphic to P, which gives that $h + h' \in \mathcal{A}(P)$. Since $P_{[h]}$ and $P_{[h']}$ are strongly isomorphic to P, it follows from Theorem 3.3.1 that $P_{[h]} + P_{[h']}$ is also strongly isomorphic to P. Then $u_1, \dots, u_m$ are precisely the unit normal vectors of the facets of $P_{[h]} + P_{[h']}$, and from Corollary 3.1.4, it follows that we may write

$$P_{[h]} + P_{[h']} = \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, \alpha_j),$$

where $\alpha_j = h(P_{[h]} + P_{[h']}, u_j)$ for each $j = 1, \dots, m$. Once $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ are the unit outer normal vectors of the facets of $P_{[h]}$ and of $P_{[h']}$ (since both are strongly isomorphic to P), we may use (8.2.20) to obtain

$$\alpha_j = h(P_{[h]} + P_{[h']}, u_j) = h(P_{[h]}, u_j) + h(P_{[h']}, u_j) = h_j + h'_j,$$

for any $j = 1, \dots, m$. Therefore, we have

$$P_{[h]} + P_{[h']} = \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, \alpha_j) = \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_j + h'_j) = P_{[h+h']}, \tag{8.2.21}$$

and once we already know that $P_{[h]} + P_{[h']}$ is strongly isomorphic to P, we get from this equality that $P_{[h+h']} \in [P]_{\mathrm{si}}$. This shows that $h + h' \in \mathcal{A}(P)$.

Finally, we have that $\mathcal{A}(P)$ is open as a straightforward consequence of Proposition 3.3.2. Thus, $\mathcal{A}(P)$ is an open set which is closed for sums and positive scalar multiplications. This shows that $\mathcal{A}(P)$ is an open convex cone.

□

Of course, the definition of $P_{[h]}$ can be naturally extended to the case where $h = (h_1, \dots, h_m)$ do not necessarily have only positive entries. That is, we may let

$$P_{[h]} := \bigcap_{j=1}^{m} \mathbf{h}^-(u_j, h_j)$$

for any $h = (h_1, \dots, h_m) \in \mathbb{R}^m$. By doing this, we can describe also polytopes which do not have the origin as an interior point. It is important to mention that, however, if $h \notin (0, +\infty)^m$, then $P_{[h]}$ may not be a proper polytope. In the case where $P_{[h]}$ is a proper polytope whose facets have unit outer normal vectors $u_1, \dots, u_m$, we still call h the *vector of support numbers of* P.

Letting $P \subseteq \mathbb{R}^n$ be any proper polytope having $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ as unit outer normal vectors of the facets, we define

$$\mathcal{E}(P) := \left\{ h \in \mathbb{R}^m : P_{[h]} \in [P]_{\mathrm{si}} \right\}.$$

Hence, we are considering also polytopes in $[P]_{\mathrm{si}}$ which do not have the origin as an interior point. This is important for us because we want to describe a facet $F(P, u)$ of a given polytope P (projected onto $u^\perp \simeq \mathbb{R}^{n-1}$) in terms of support numbers. The projection of $F(P, u)$ onto $u^\perp$ may not contain the origin, even if P does. Notice that this projection is simply given by the translation of the supporting hyperplane

$\mathbf{h}(P,u)$ to $u^\perp$, and from now on, we will often regard a facet $F(P,u)$ as a subset of $u^\perp$ (identified with $\mathbb{R}^{n-1}$) without further comments. We write these natural identifications as $\mathbf{h}(P,u) \simeq u^\perp \simeq \mathbb{R}^{n-1}$.

As before, let $u_1,\dots,u_m \in \mathbb{S}^{n-1}$ be the unit outer normal vectors of the facets of a proper simple polytope $P \subseteq \mathbb{R}^n$, and let $F_i := F(P,u_i)$ be a fixed facet. Of course, the facets of F_i are the $(n-2)$-dimensional faces of P which are contained in F_i (see Proposition 3.1.1 if any clarification is needed). An $(n-2)$-dimensional face of P is always given by an intersection of two facets of P (see Corollary 3.2.3), and hence the facets of F_i are precisely the $(n-2)$-dimensional intersections of F_i with the other facets of P. Letting

$$J := \Big\{(i,j) \in \{1,\dots,m\}^2 : \dim\big(F(P,u_i) \cap F(P,u_j)\big) = n-2\Big\}, \tag{8.2.22}$$

we get that the facets of F_i (as a proper polytope of $\mathbf{h}(P,u_i) \simeq u_i^\perp \simeq \mathbb{R}^{n-1}$) are indexed by the indices $j \in \{1,\dots,m\}$ such that $(i,j) \in J$. For a pair $(i,j) \in J$, we have that $F_{ij} := F(P,u_i) \cap F(P,u_j)$ is a facet of both F_i and $F_j := F(P,u_j)$. Once we are interested in describing F_i (and not F_j), we see F_{ij} as a facet of F_i and denote by u_{ij} the unit outer normal vector of this facet (notice that this is a vector of $u_i^\perp$). By doing this, we have that the set of unit outer normal vectors of the facets of F_i is the set

$$U_i := \{u_{ij} : (i,j) \in J\}. \tag{8.2.23}$$

Therefore, if we let $h_{ij} := h_{u^\perp}(F_i,u_{ij})$, where $h_{u^\perp}(F_i,\cdot)$ denotes the support function of F_i as a subset of $u_i^\perp$, we get that

$$F_i \simeq \mathrm{proj}_{u_i^\perp}(F_i) = \bigcap_{(i,j)\in J} \mathbf{h}^-_{u_i^\perp}(u_{ij},h_{ij}), \tag{8.2.24}$$

where $\mathbf{h}^-_{u_i^\perp}$ stands for half-spaces in $u_i^\perp$, of course. Then the numbers h_{ij} (with $(i,j) \in J$) are the support numbers of F_i (as a proper polytope of $u_i^\perp \simeq \mathbb{R}^{n-1}$).

We want to describe the unit outer normal vector u_{ij} of the facet F_{ij} (of F_i as a proper polytope of $\mathbf{h}(P,u_i) \simeq u_i^\perp \simeq \mathbb{R}^{n-1}$, as we emphasize again) in terms of the vectors u_i and u_j. We are looking for a vector of $u_i^\perp$ which is orthogonal to the intersection of the supporting hyperplanes $\mathbf{h}(P,u_i)$ and $\mathbf{h}(P,u_j)$ of F_i and F_j, respectively. It is not difficult to see that u_{ij} must point in the direction of $\mathrm{proj}_{u_i^\perp}(u_j)$. Indeed, this projection is given as

$$\mathrm{proj}_{u_i^\perp}(u_j) = u_j - \langle u_i,u_j\rangle u_i,$$

and if $y,z \in \mathbf{h}(P,u_i) \cap \mathbf{h}(P,u_j)$, then $\langle y,u_i\rangle = \langle z,u_i\rangle = h_i$ and $\langle y,u_j\rangle = \langle z,u_j\rangle = h_j$. Thus, we have

$$\langle y-z, \mathrm{proj}_{u_i^\perp}(u_j)\rangle = \langle y-z, u_j - \langle u_i,u_j\rangle u_i\rangle = h_j - \langle u_i,u_j\rangle h_i - h_j + \langle u_i,u_j\rangle h_i = 0.$$

In other words, the direction of $\mathrm{proj}_{u_i^\perp}(u_j)$ is the (unique) direction in $u_i^\perp$ which is orthogonal to the facet F_{ij} of F_i. It follows that

$$u_{ij} = \frac{\mathrm{proj}_{u_i^\perp}(u_j)}{|\mathrm{proj}_{u_i^\perp}(u_j)|} = \frac{u_j - \langle u_i,u_j\rangle u_i}{|\mathrm{proj}_{u_i^\perp}(u_j)|}. \tag{8.2.25}$$

Some confusion may occur regarding the question whether this is the correct orientation of u_{ij} (that is, pointing outwards F_i). However, it is easy to see that the orientation above is indeed correct: let $z \in F_{ij}$ and $y \in F_i$. Then $\langle z,u_i\rangle = \langle y,u_i\rangle = h_i$, $\langle z,u_j\rangle = h_j$ and $\langle y,u_j\rangle \le h_j$ (because y is, in particular, a point of P, and P is contained in the half-space $\mathbf{h}^-(u_j,h_j)$). We have

$$\langle z - y, u_j - \langle u_i, u_j\rangle u_i\rangle = \langle z, u_j\rangle - \langle y, u_j\rangle + \langle u_i, u_j\rangle\big(\langle y, u_i\rangle - h_i\big) = h_j - \langle y, u_j\rangle \geq 0,$$

and this gives that u_{ij} is in fact given as in (8.2.25).

Now we describe the numbers h_{ij}. For that sake, let $z \in F_{ij}$ be given. We have that $\langle z, u_i\rangle = h_i$ and $\langle z, u_j\rangle = h_j$. Thus,

$$\begin{aligned} h_{ij} = h_{u_i^\perp}(F_i, u_{ij}) = \langle z, u_{ij}\rangle &= \frac{1}{|\mathrm{proj}_{u_i^\perp}(u_j)|}\langle z, u_j - \langle u_i, u_j\rangle u_i\rangle = \\ &= \frac{1}{|\mathrm{proj}_{u_i^\perp}(u_j)|}h_j - \frac{\langle u_i, u_j\rangle}{|\mathrm{proj}_{u_i^\perp}(u_j)|}h_i. \end{aligned} \tag{8.2.26}$$

Observe carefully that z is not really a point of $u_i^\perp$, but of $\mathbf{h}(P, u_i)$. This is not a problem since these hyperplanes are identified by a translation in the direction of u_i (hence orthogonal to u_{ij}). In other words, the calculation above does not change in case we are strictly formal and replace $z \in F_{ij}$ by its projection onto $u_i^\perp$.

Now assume that $\bar{P}$ is a polytope which is strongly isomorphic to P. Then $\bar{P}$ has a facet $\bar{F}_i$ whose unit outer normal vector is u_i, and from Lemma 3.3.1, we have that F_i and $\bar{F}_i$ are strongly isomorphic (as polytopes of $\mathbb{R}^n$, and consequently also as proper polytopes of $u_i^\perp$; every hyperplane of $u_i^\perp$ can be realized as a hyperplane of $\mathbb{R}^n$ by a direct sum with u_i). It follows that the unit outer normal vectors of the facets of $\bar{F}_i$ are the same as of the facets of F_i (see (8.2.25)), and hence they rely only on the strongly isomorphic class of P (and not on P itself). As a consequence, letting $\bar{h} = (\bar{h}_1, \ldots, \bar{h}_m)$ be the vector of supporting numbers of $\bar{P}$ and letting $\bar{h}_{ij}$ denote the supporting numbers of $\bar{F}_i$, we may repeat the previous arguments and write

$$\bar{h}_{ij} = \frac{1}{|\mathrm{proj}_{u_i^\perp}(u_j)|}\bar{h}_j - \frac{\langle u_i, u_j\rangle}{|\mathrm{proj}_{u_i^\perp}(u_j)|}\bar{h}_i.$$

Hence, the coefficients of the expression (8.2.26) depend only on the strongly isomorphic class $[P]_{\mathrm{si}}$, and not on P itself. In other words, (8.2.26) defines a linear map which carries each supporting vector $h = (h_1, \ldots, h_m) \in \mathcal{E}(P)$ to the supporting numbers h_{ij} of the facet of $P_{[h]}$ which is orthogonal to u_i. In the literature, this map is often denoted by $\Lambda_i \colon \mathcal{E}(P) \to \mathcal{E}(F_i)$.

Proposition 8.2.4. *Let $P \subseteq \mathbb{R}^n$ be a simple proper polytope having the origin as an interior point, and let $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ be the unit outer normal vectors of its facets. Let $f \colon \mathcal{E}(P)^n \to \mathbb{R}$ be the function given as*

$$f(h^1, \ldots, h^n) := V(P_{[h^1]}, \ldots, P_{[h^n]})$$

for $(h^1, \ldots, h^n) \in \mathcal{E}(P)^n$, and write each h^k in coordinates as $h^k := (h^k_1, \ldots, h^k_m)$. Then there exist numbers $c_{j_1,\ldots,j_n}$, $1 \leq j_1, \ldots, j_n \leq m$, dependending only on $[P]_{\mathrm{si}}$ and symmetric in the indices $j_1, \ldots, j_n$, such that

$$f(h^1, \ldots, h^n) = \sum_{j_1,\ldots,j_n=1}^{m} c_{j_1,\ldots,j_n} h^1_{j_1} h^2_{j_2} \ldots h^n_{j_n}. \tag{8.2.27}$$

As a particular consequence, we have that the restriction of f to $\mathcal{A}(P)^n$ is of class C^∞.

Proof. We proceed by induction on the dimension n. In the case where $n = 1$, any proper polytope is a non-denegerate interval whose facets are the endpoints. Also, the unit outer normal vectors of the facets are $-e_1$ and e_1 (and we consider them in this order). In particular, all proper polytopes of $\mathbb{R}$ are strongly isomorphic. Letting P be such a proper polytope and $h = (h_1, h_2) \in \mathcal{E}(P)$, we have

$$P_{[h]} = \mathbf{h}^-(-e_1, h_1) \cap \mathbf{h}^-(e_1, h_2) = [-h_1, h_2],$$

and then

$$V(P_{[h]}) = \mathrm{vol}_1(P_{[h]}) = h_1 + h_2.$$

This settles the case where $n = 1$ (with $c_1 = c_2 = 1$, of course).

Now we assume that the result holds in $\mathbb{R}^{n-1}$, and we prove that it holds in $\mathbb{R}^n$. Let $(h^1, \dots, h^n) \in \mathcal{E}(P)^n$. The polytopes $P_{[h^1]}, \dots, P_{[h^n]}$ are strongly isomorphic, and thus each of them has $N := \{u_1, \dots, u_m\}$ as unit outer normal vectors of the facets. As a consequence of Theorem 3.3.1, we have that N contains the unit outer normal vectors of the facets of any combination of $t_1 P_{[h^1]} + \dots + t_n P_{[h^n]}$ for $t_1, \dots, t_n \geq 0$ (provided they are not all zero). Therefore, in view of Proposition 6.2.2, we may write

$$\begin{aligned} V(P_{[h^1]}, \dots, P_{[h^n]}) &= \frac{1}{n} \sum_{i=1}^{m} h(P_{[h^n]}, u_i) V^{(n-1)}\big(F(P_{[h^1]}, u_i), \dots, F(P_{[h^{n-1}]}, u_i)\big) = \\ &= \frac{1}{n} \sum_{i=1}^{m} h_i^n V^{(n-1)}\big(F(P_{[h^1]}, u_i), \dots, F(P_{[h^{n-1}]}, u_i)\big). \end{aligned} \tag{8.2.28}$$

From Lemma 3.3.1, we have that for each fixed i, the facets $F(P_{[h^k]}, u_i)$, $k = 1, \dots, n-1$, are strongly isomorphic, and hence also strongly isomorphic as polytopes of $u_i^\perp \simeq \mathbb{R}^{n-1}$. Then we may use the induction hypothesis. We let J be given as in (8.2.22) and write

$$V^{(n-1)}\big(F(P_{[h^1]}, u_i), \dots, F(P_{[h^{n-1}]}, u_i)\big) = \sum_{(i, j_k) \in J} c^i_{j_1, \dots, j_{n-1}} h^1_{ij_1} \dots h^{n-1}_{ij_{n-1}} \tag{8.2.29}$$

for each $i \in \{1, \dots, m\}$, where the coefficients $c^i_{j_1, \dots, j_{n-1}}$ depend only on the class of the facets $F(P_{[h^1]}, u_i), \dots, F(P_{[h^{n-1}]}, u_i)$, and hence depend only on $[P]_{\mathrm{si}}$. Now we use (8.2.26) to write each $h^k_{ij_k}$ in terms of h^k_i and $h^k_{j_k}$ as

$$h^k_{ij_k} = \frac{1}{|\mathrm{proj}_{u_i^\perp}(u_{j_k})|} h^k_{j_k} - \frac{\langle u_i, u_{j_k} \rangle}{|\mathrm{proj}_{u_i^\perp}(u_{j_k})|} h^k_i =: \alpha_{ij_k} h^k_{j_k} - \beta_{ij_k} h^k_i,$$

where we adopt easier notations for the coefficients. Substituting this in (8.2.29), and then in (8.2.28), we arrive at

$$\begin{aligned} &V(P_{[h^1]}, \dots, P_{[h^n]}) = \\ &= \frac{1}{n} \sum_{i=1}^{m} h_i^n \left(\sum_{(i, j_k) \in J} c^i_{j_1, \dots, j_{n-1}} (\alpha_{ij_1} h^1_{j_1} - \beta_{ij_1} h^1_i) \dots (\alpha_{ij_{n-1}} h^{n-1}_{j_{n-1}} - \beta_{ij_{n-1}} h^{n-1}_i) \right). \end{aligned}$$

Rearranging the terms, inserting zero coefficients for the missing terms (if needed), and doing a trivial symmetrization process, we get that this expression has the same format as (8.2.27). Since the $c^i_{j_1, \dots, j_{n-1}}$, the α_{ij_k}'s and the β_{ij_k}'s only depend on $[P]_{\mathrm{si}}$, the proof is complete.

□

Proof of Theorem 8.2.1. We use induction on the dimension. We already know that the inequality is true if the dimension is $n = 2$. Thus, let $n \geq 3$ and assume that (8.2.1) holds in any dimension smaller than n. Let $P \subseteq \mathbb{R}^n$ be any fixed simple proper polytope having the origin as an interior point, and having $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ as unit outer normal vectors of the facets. Let $P_4, \dots, P_n \in [P]_{\mathrm{si}}$ be arbitrarily given. Consider the function $f \colon \mathcal{A}(P) \to \mathbb{R}$ given as

$$f(h) := V\big(P_{[h]}[3], P_4, \dots, P_n\big), \quad h = (h_1, \dots, h_m) \in \mathcal{A}(P). \tag{8.2.30}$$

Using the induction hypothesis, we will show that f has the conditions of Proposition 8.2.2. First, from Proposition 8.2.4, we have that f is a (positive) homogeneous polynomial of degree 3 whose variables are the entries of h. Thus, in particular, we get that f is of class C^∞ and positively homogeneous of degree 3.

We proceed to compute the partial derivatives of f. For simplicity, for any $j \in \{1, \dots, m\}$, we write

$$(F(P_4, u_j), \dots, F(P_n, u_j)) = \mathbf{P}(u_j).$$

Thus, using Proposition 6.2.2 as we did in (8.2.28), it follows that

$$f(h) = \frac{1}{n} \sum_{j=1}^{m} h_j V^{(n-1)}(F(P_{[h]}, u_j)[2], \mathbf{P}(u_j)),$$

and from Propositon 8.2.4 we get that each mixed volume in the sum is a homogeneous polynomial of degree 2. Indeed, $F(P_{[h]}, u_j), F(P_4, u_j), \dots, F(P_n, u_j)$ are strongly isomorphic polytopes in $u_j^\perp$. Hence, for each $j = 1, \dots, m$, we may write

$$V^{(n-1)}\big(F(P_{[h]}, u_j)[2], \mathbf{P}(u_j)\big) = \sum_{a,s=1}^{m} c_{as}^j h_a h_s,$$

for some coefficients c_{as}^j, which are symmetric in a, s and j, and consequently, we have

$$f(h) = \frac{1}{n} \sum_{j=1}^{m} h_j \left(\sum_{a,s=1}^{m} c_{as}^j h_a h_s \right)$$

Thus, for each fixed $i \in \{1, \dots, m\}$, we may write

$$f(h) = \frac{1}{n} \left[c_{ii}^i h_i^3 + 3 \sum_{k=1}^{m} c_{ii}^k h_i^2 h_k + 3 \sum_{\substack{a,s=1 \\ a,s \neq i}}^{m} c_{as}^i h_i h_a h_s + \sum_{\substack{a,s,j=1 \\ a,s,j \neq i}}^{m} c_{as}^j h_j h_a h_s \right],$$

from where we compute

$$\partial_i f(h) = \frac{1}{n} \left(3c_{ii}^i h_i^2 + 6 \sum_{k=1}^{m} c_{ii}^k h_i h_k + 3 \sum_{\substack{a,s=1 \\ a,s \neq i}}^{m} c_{as}^i h_a h_s \right) = \frac{3}{n} \sum_{a,s=1}^{m} c_{as}^i h_a h_s =$$
$$= \frac{3}{n} V^{(n-1)}\big(F(P_{[h]}, u_i)[2], \mathbf{P}(u_i)\big).$$

Once all polytopes in the mixed volume of the right-hand side are always $(n-1)$-dimensional, we get that $\partial_i f(h) > 0$ for any $h \in \mathcal{A}(P)$.

Now we investigate the second derivatives. For that sake, first observe that, for each fixed i, the mixed volume $V^{(n-1)}(F(P_{[h]}, u_i)[2], \mathbf{P}(u_i))$ is a homogeneous polynomial of degree 2 on the supporting numbers of the facets of $F(P_{[h]}, u_i)$. These are precisely the numbers h_{ij} for $j \in J_i := \{j : (i, j) \in J\}$ (pay attention to the new notation that we are introducing). From (8.2.26), we get that each h_{ij} depends only on h_i and h_j (linearly). Thus, $V^{(n-1)}(F(P_{[h]}, u_i)[2], \mathbf{P}(u_i))$ is a homogeneous polynomial of degree

2 on h_k for $k \in \{i\} \cup J_i$. The first immediate consequence of this fact is that

$$\partial_{ki} f = 0$$

whenever $k \notin \{i\} \cup J_i$. For the case where $j \in J_i$, using (8.2.28) again we arrive at the equality

$$V^{(n-1)}(F(P_{[h]}, u_i)[2], \mathbf{P}(u_i)) =$$
$$= \frac{1}{n-1} \sum_{j \in J_i} h_{ij} V^{(n-2)}(F(P_{[h]}, u_i) \cap F(P_{[h]}, u_j), \mathbf{P}(u_i)(u_j)),$$

where

$$\mathbf{P}(u_i)(u_j) := (F(P_4, u_i) \cap F(P_4, u_j), \ldots, F(P_n, u_i) \cap F(P_n, u_j)).$$

The mixed volume $V^{(n-2)}(F(P_{[h]}, u_i) \cap F(P_{[h]}, u_j), \mathbf{P}(u_i)(u_j))$ is an homogeneous polynomial of degree 1 on the h_{ij}'s, and hence we may write

$$V^{(n-2)}(F(P_{[h]}, u_i) \cap F(P_{[h]}, u_j), \mathbf{P}(u_i)(u_j)) = \sum_{a \in J_i} b^j_{ia} h_{ia} \tag{8.2.31}$$

for some coefficients b^j_{ia} which are symmetric in i, a, and j. Thus,

$$V^{(n-1)}(F(P_{[h]}, u_i)[2], \mathbf{P}(u_i)) = \frac{1}{n-1} \sum_{j \in J_i} h_{ij} \left(\sum_{a \in J_i} b^j_{ia} h_{ia} \right),$$

and

$$\partial_i f(h) = \frac{3}{n} V^{(n-1)}\big(F(P_{[h]}, u_i)[2], \mathbf{P}(u_i)\big) = \frac{3}{n} \frac{1}{n-1} \sum_{j \in J_i} h_{ij} \left(\sum_{a \in J_i} b^j_{ia} h_{ia} \right) =$$
$$= \frac{3}{n} \frac{1}{n-1} \left[\sum_{j \in J_i} b^j_{ij} h^2_{ij} + \sum_{\substack{j,a \in J_i \\ j \neq a}} b^j_{ia} h_{ij} h_{ia} \right].$$

Derivating with respect to some $k \in J_i$, we get

$$\partial_{ki} f(h) = \frac{3}{n} \frac{1}{n-1} \left[2 h_{ik} b^k_{ik} (\partial_k h_{ik}) + 2 \sum_{\substack{j \in J_i \\ j \neq k}} b^k_{ij} h_{ij} (\partial_k h_{ik}) \right] = \frac{3}{n} \frac{2 \partial_k (h_{ik})}{n-1} \sum_{j \in J_i} b^k_{ij} h_{ij}.$$

The derivative $\partial_k (h_{ik})$ can be easily computed using (8.2.26). Taking also (8.2.31) into consideration, we finally get

$$\partial_{ki} f(h) = \frac{3}{n} \frac{2}{n-1} \frac{1}{|\mathrm{proj}_{u_i^\perp}(u_k)|} V^{(n-2)}(F(P_{[h]}, u_i) \cap F(P_{[h]}, u_k), \mathbf{P}(u_i)(u_k)).$$

Hence, $\partial_{ki} f > 0$ for any $k \in J_i$. Indeed, since $k \in J_i$, we have that all polytopes inside the mixed volume in the right-hand side are $(n-2)$-dimensional.

Summarizing, if $i \neq j$, then we have $\partial_{ji} f > 0$ if $(i,j) \in J$ and $\partial_{ji} f = 0$ if $(i,j) \notin J$. It follows that $\partial_{ji} f \geq 0$ whenever $i \neq j$. We also claim that $\mathrm{hess}(f)(h)$ is irreducible for any $h \in \mathcal{A}(P)$. Indeed, for any $i, j \in \{1, \dots, m\}$, the facets $F(P, u_i)$ and $F(P, u_j)$ are connected by a sequence of $(n-2)$-dimensional faces (and the same holds for any $Q \in [P]_{\mathrm{si}}$, of course); see Exercise 3.28. Hence, we may find $j_1, \dots, j_r \in \{1, \dots, m\}$ such that $\partial_{jj_1} f(h), \dots, \partial_{j_r i} f(h) > 0$ for any $h \in \mathcal{A}(P)$.

Since it is clear that $P_{[th]} = tP_{[h]}$ for every $t > 0$, and using (8.2.21), we get that

$$P_{[(1-\lambda)h_1 + \lambda h_2]} = (1-\lambda)P_{[h_1]} + \lambda P_{[h_2]} \tag{8.2.32}$$

for any $h_1, h_2 \in \mathcal{A}(P)$ and all $\lambda \in [0,1]$. Moreover, using Lemma 3.3.2, we get that

$$\begin{aligned} F(P_{[(1-\lambda)h_1+\lambda h_2]}, u_i) &= F((1-\lambda)P_{[h_1]} + \lambda P_{[h_2]}, u_i) = \\ &= (1-\lambda)F(P_{[h_1]}, u_i) + \lambda F(P_{[h_2]}, u_i). \end{aligned}$$

By our induction hypothesis, the Alexandrov–Fenchel inequality holds in $u_i^{\perp} \simeq \mathbb{R}^{n-1}$. Applying implication **(a)⇒(b)** of Lemma 8.2.3 together with Lemma 8.2.4 for $m = 2$, we obtain that the function

$$\mathcal{K}^{n-1} \ni K \mapsto V^{(n-1)}\big(K[2], \mathbf{P}(u_i)\big)^{\frac{1}{2}}$$

is concave. In view of these facts, for every $i \in \{1, \dots, n\}$, any $h_1, h_2 \in \mathcal{A}(P)$, and all $\lambda \in [0,1]$, we get that

$$\begin{aligned} \partial_i f\big((1-\lambda)h_1 + \lambda h_2\big)^{\frac{1}{2}} &= \left(\frac{3}{n}\right)^{\frac{1}{2}} V^{(n-1)}\big(F(P_{[(1-\lambda)h_1+\lambda h_2]}, u_i)[2], \mathbf{P}(u_i)\big)^{\frac{1}{2}} = \\ &= \left(\frac{3}{n}\right)^{\frac{1}{2}} V^{(n-1)}\big((1-\lambda)F(P_{[h_1]}, u_i) + \lambda F(P_{[h_2]}, u_i)[2], \mathbf{P}(u_i)\big)^{\frac{1}{2}} \geq \\ &\geq (1-\lambda)\left(\frac{3}{n}\right)^{\frac{1}{2}} V^{(n-1)}\big(F(P_{[h_1]}, u_i)[2], \mathbf{P}(u_i)\big)^{\frac{1}{2}} + \\ &\quad + \lambda\left(\frac{3}{n}\right)^{\frac{1}{2}} V^{(n-1)}\big(F(P_{[h_2]}, u_i)[2], \mathbf{P}(u_i)\big)^{\frac{1}{2}} = \\ &= (1-\lambda)\partial_i f(h_1)^{\frac{1}{2}} + \lambda \partial_i f(h_2)^{\frac{1}{2}}. \end{aligned}$$

This shows that the functions $h \mapsto (\partial_i f)^{\frac{1}{2}}$ are concave in $\mathcal{A}(P)$.

It follows that f has all the conditions of Proposition 8.2.2, and hence $f^{\frac{1}{3}}$ is concave. Now let $K_1, K_2, K_4, \dots, K_n \in \mathcal{K}^n$ and $\lambda \in [0,1]$ be arbitrarily given. Let $(P_1^r, P_2^r, P_4^r, \dots, P_n^r)_{r \in \mathbb{N}}$ represent sequences of simple polytopes converging simultaneously to $(K_1, K_2, K_4, \dots, K_n)$ as $r \to \infty$, and having the property that for each $r \in \mathbb{N}$ the polytopes $P_1^r, \dots, P_n^r$ are strongly isomorphic (to a fixed polytope P^r, say); the existence of such sequences is guaranteed by Theorem 3.4.2. For each $r \in \mathbb{N}$, we may write P_1^r and P_2^r in terms of their support numbers as $P_1^r = P_{[h_1^r]}$ and $P_2^r = P_{[h_2^r]}$, and define f_r over $\mathcal{A}(P^r)$ as in (8.2.30). Therefore, continuity of mixed volumes, equality (8.2.32) and the concavity of $f_r^{\frac{1}{3}}$ imply

$$
\begin{aligned}
V((1-\lambda)K_1 + \lambda K_2[3], K_4, \dots, K_n)^{\frac{1}{3}} &= \\
= \lim_{r\to\infty} V((1-\lambda)P_{[h_1^r]} + \lambda P_{[h_2^r]}[3], P_4^r, \dots, P_n^r)^{\frac{1}{3}} &= \\
= \lim_{r\to\infty} V\big(P_{[(1-\lambda)h_1^r+\lambda h_2^r]}[3], P_4^r, \dots, P_n^r\big)^{\frac{1}{3}} = \lim_{r\to\infty} f_r((1-\lambda)h_1^r + \lambda h_2^r)^{\frac{1}{3}} &\geq \\
\geq \lim_{r\to\infty} \left[(1-\lambda) f_r(h_1^r)^{\frac{1}{3}} + \lambda f_r(h_2^r)^{\frac{1}{3}}\right] &= \\
= \lim_{r\to\infty} \left[(1-\lambda) V(P_{[h_1^r]}[3], P_4^r, \dots, P_n^r)^{\frac{1}{3}} + \lambda V(P_{[h_2^r]}[3], P_4^r, \dots, P_n^r)\right] &= \\
= (1-\lambda) V(K_1[3], K_4, \dots, K_n)^{\frac{1}{3}} + \lambda V(K_2[3], K_4, \dots, K_n)^{\frac{1}{3}}.
\end{aligned}
$$

This gives that the function $K \mapsto V(K[3], K_4, \dots, K_n)^{\frac{1}{3}}$ is concave in $\mathcal{K}^n$. Once this holds for any $K_4, \dots, K_n \in \mathcal{K}^n$, it follows from Corollary 8.2.2 that (8.2.1) holds for any convex bodies $K_1, K_2, K_4, \dots, K_n \in \mathcal{K}^n$. As it was mentioned in the beginning of this section, the case for sets from CC^n follows immediately by approximations (recall that $\mathcal{K}^n$ is dense in CC^n with respect to the Hausdorff metric).

□

8.3 Other versions and consequences

The Alexandrov–Fenchel inequality implies (or even is equivalent to) a lot of other inequalities, including a Brunn–Minkowski inequality for mixed volumes. We discuss some of these consequences now.

Proposition 8.3.1. *For any natural number $1 \leq m \leq n$ and every $K_1, \dots, K_n \in CC^n$, we have the inequality*

$$
V(K_1, \dots, K_n)^m \geq \prod_{j=1}^{m} V\big(K_j[m], K_{m+1}, \dots, K_n\big). \tag{8.3.1}
$$

Remark 8.3.1. In some places, (8.3.1) is enunciated as the Alexandrov–Fenchel inequality. Both formulations are equivalent. Indeed, as we will see below, (8.3.1) is proved as a consequence of (8.2.1), and conversely, it is immediate to see that (8.2.1) is the case of (8.3.1) for $m = 2$.

Proof. First, observe that it suffices to prove the inequality for the particular case where $K_1, \dots, K_n$ are convex bodies. Indeed, since $\mathcal{K}^n$ is clearly dense in CC^n (with respect to the Hausdorff metric, see Lemma 2.5.1), we derive the general case for $K_1, \dots, K_n \in CC^n$ from this particular case by the continuity of mixed volumes. We see that $K_1, \dots, K_n \in \mathcal{K}^n$ brings the advantage that all involved mixed volumes are positive (see Proposition 6.2.3).

We say that a finite sequence $(c_0, \dots, c_m)$ of numbers is *concave* if $c_{i-1} - 2c_i + c_{i+1} \leq 0$ for every $i \in \{1, \dots, m-1\}$ or, equivalently, if

$$
c_0 - c_1 \leq c_1 - c_2 \leq \dots \leq c_{m-1} - c_m.
$$

In this case, we claim that for any $0 \leq i < j < k \leq m$ the inequality

$$
\frac{(c_i - c_{i+1}) + \dots + (c_{j-1} - c_j)}{j-i} \leq \frac{(c_j - c_{j+1}) + \dots + (c_{k-1} - c_k)}{k-j}
$$

holds. In fact,

$$\frac{(c_i - c_{i+1}) + \ldots + (c_{j-1} - c_j)}{j-i} \leq \frac{(j-i)(c_{j-1} - c_j)}{j-i} = c_{j-1} - c_j = \\ = \frac{(k-j)(c_{j-1} - c_j)}{k-j} \leq \frac{(c_j - c_{j+1}) + \ldots + (c_{k-1} - c_k)}{k-j}.$$

Moreover, it is clear that equality holds if and only if

$$c_i - c_{i+1} = c_{i+1} - c_{i+2} = \ldots = c_{j-1} - c_j = c_j - c_{j+1} = \ldots = c_{k-1} - c_k. \tag{8.3.2}$$

As a consequence, we have that

$$\frac{c_i - c_j}{j-i} \leq \frac{c_j - c_k}{k-j},$$

which can be rewritten as

$$(k-j)c_i + (i-k)c_j + (j-i)c_k \leq 0, \tag{8.3.3}$$

and equality holds if and only if (8.3.2) holds or, equivalently, if $c_{p-1} - 2c_p + c_{p+1} = 0$ for every $p = i+1, \ldots, k-1$.

A finite sequence $(c_0, \ldots, c_m)$ of positive numbers is said to be *log-concave* if the sequence $(\log c_0, \ldots, \log c_m)$ is concave or, equivalently, if

$$c_i^2 \geq c_{i-1}c_{i+1}$$

for $i = 1, \ldots, m-1$. In this case, from (8.3.3), we get immediately that

$$c_i^{k-j} c_j^{i-k} c_k^{j-i} \leq 1$$

whenever $0 \leq i < j < k \leq m$. Of course, this can be written (with positive exponents) as

$$c_j^{k-i} \geq c_i^{k-j} c_k^{j-i}. \tag{8.3.4}$$

It is also clear that equality holds if and only if $c_p^2 = c_{p-1}c_{p+1}$ for every $p \in \{i+1, \ldots, k-1\}$.

Let $0 \leq m \leq n$, and $K, L, K_{m+1}, \ldots, K_n \in \mathcal{K}^n$. For every $0 \leq i \leq m$, let

$$c_i := V\big(K[m-i], L[i], K_{m+1}, \ldots, K_n\big).$$

From the Alexandrov–Fenchel inequality, we have

$$\begin{aligned} c_i^2 = V\big(K[m-i], L[i], K_{m+1}, \ldots, K_n\big)^2 \geq \\ \geq V\big(K[m-i+1], L[i-1], K_{m+1}, \ldots, K_n\big) V\big(K[m-i-1], L[i+1], \ldots, K_n\big) = \\ = c_{i-1}c_{i+1}, \end{aligned} \tag{8.3.5}$$

and hence the sequence $(c_0, \ldots, c_m)$ is log-concave if $c_i > 0$ for every $i = 0, \ldots, m$.

Using this fact, we prove (8.3.1) by induction on m. The case $m = 2$ is just inequality (8.2.1), and hence we assume that the result holds for some $m \geq 2$. We have to prove that

$$V(K_1,\dots,K_n)^{m+1} \geq \prod_{j=1}^{m+1} V\big(K_j[m+1],K_{m+2},\dots,K_n\big). \tag{8.3.6}$$

For that sake, we fix any $p \in \{1,\dots,m\}$ and consider the sequence

$$c_i := V\big(K_p[m-i+1],K_{m+1}[i],K_{m+2},\dots,K_n\big),$$

$i = 0,\dots,m+1$. Since we are considering that all involved mixed volumes are non-zero, it follows that the sequence $(c_0,\dots,c_{m+1})$ is log-concave. Hence, we may apply (8.3.4) for $i = 0$, $j = 1$ and $k = m+1$ to get

$$\begin{aligned} V\big(K_p[m],K_{m+1},K_{m+2},\dots,K_m\big)^{m+1} \geq \\ \geq V\big(K_p[m+1],K_{m+2},\dots,K_n\big)^m V\big(K_{m+1}[m+1],K_{m+2},\dots,K_n\big). \end{aligned}$$

Multiplying on $p = 1,\dots,m$ and rearranging the terms, we arrive at

$$\prod_{p=1}^{m+1} V\big(K_p[m+1],K_{m+2},\dots,K_n\big)^m \leq \prod_{p=1}^{m} V\big(K_p[m],K_{m+1},K_{m+2},\dots,K_n\big)^{m+1},$$

which gives

$$\prod_{p=1}^{m+1} V\big(K_p[m+1],K_{m+2},\dots,K_n\big) \leq \left[\prod_{p=1}^{m} V\big(K_p[m],K_{m+1},K_{m+2},\dots,K_n\big)\right]^{(m+1)/m}.$$

Using the induction hypothesis to bound the right-hand side, we finally get

$$\begin{aligned} \left[\prod_{p=1}^{m} V\big(K_p[m],K_{m+1},\dots,K_{m+2}\big)\right]^{(m+1)/m} &\leq [V(K_1,\dots,K_n)^m]^{(m+1)/m} = \\ &= V(K_1,\dots,K_n)^{m+1}. \end{aligned}$$

This gives that (8.3.1) holds for $m+1$, and the proof is complete.

□

Corollary 8.3.1. *For any $K_1,\dots,K_n \in CC^n$, we have that*

$$V(K_1,\dots,K_n)^n \geq \mathrm{vol}_n(K_1)\dots\mathrm{vol}_n(K_n).$$

Proof. This is just the case $m = n$ of inequality (8.3.1).

□

Now we move forward to justify the claim of Remark 8.2.2, which states that the functions f_m, g_m and v_m defined for $m \in \{2,\dots,n\}$ in Lemma 8.2.4 are concave. From the equivalences given in this lemma, it is clear that we may choose one of these families of functions to work with.

Considering the functions f_m and having that the Alexandrov–Fenchel inequality holds, the cases where $m = 2$ and $m = 3$ are essentially the implications **(a)⇒(b)** and **(a)⇒(c)** of Lemma 8.2.3. For the proof of the general case presented next, we work with the functions g_m, which from now on we denote simply by f.

Of course, by approximations, we may notice that all involved sets can be taken from the larger family CC^n, instead of from $\mathcal{K}^n$. In what follows, we take that into consideration without further comments.

From now on, a given one-variable function $f\colon A \subseteq \mathbb{R} \to \mathbb{R}$ having the form $f(t) = at + b$ for some $a, b \in \mathbb{R}$ will be called *linear*. Of course, the term *affine* would be more accurate, but we use the term *linear* to be more consistent with the existing literature.

Theorem 8.3.1. *Let $m \in \{1, \dots, n\}$ and $K, L, K_{m+1}, \dots, K_n \in CC^n$ be arbitrarily given, and define $K_t := (1-t)K + tL$ for $t \in [0,1]$. Then the function $f\colon [0,1] \to \mathbb{R}$ given as*

$$f(t) := V\big(K_t[m], K_{m+1}, \dots, K_n\big)^{1/m}$$

is concave.

Remark 8.3.2. It is possible to prove this result by adapting the argument used to prove the implication **(a)⇒(c)** of Lemma 8.2.3 (this was done by Hug and Weil in [1263, Lemma 3.4], and we propose this approach in Exercise 8.8). However, here we follow the proof given by Schneider [2167, Theorem 7.4.5]. Although this proof being slightly more complicated, it gives us a glimpse of what happens if f is linear (see Theorem 8.3.2 below). This is (in some cases) related to the equality case in the Alexandrov–Fenchel inequality, as it will be investigated in Section 8.4.

Proof. Again, we may consider that $K, L, K_{m+1}, \dots, K_n$ are convex bodies, since the general case for sets in CC^n follows from this particular one by approximations. In fact, if we prove that the inequality

$$\begin{aligned} V\big(K_{(1-\lambda)s+\lambda t}[m], K_{m+1}, \dots, K_n\big)^{1/m} \geq \\ \geq (1-\lambda)V\big(K_s[m], K_{m+1}, \dots, K_n\big)^{1/m} + \lambda V\big(K_t[m], K_{m+1}, \dots, K_n\big)^{1/m} \end{aligned}$$

holds for any $K, L, K_{m+1}, \dots, K_n \in \mathcal{K}^n$ and every $\lambda, s, t \in [0,1]$, then the continuity of mixed volumes and the fact that $\mathcal{K}^n$ is dense in CC^n yield the general case.

Having said this, we assume from now on that $K, L, K_{m+1}, \dots, K_n$ are convex bodies. We also assume that $m \geq 2$, since if $m = 1$, then f is clearly linear (and hence concave). It follows from Proposition 6.2.3 that

$$V\big(K[m-i], L[i], K_{m+1}, \dots, K_n\big) > 0$$

for every $i = 0, \dots, m$. Also, from the linearity of mixed volumes we get that f is the m-th root of a polynomial whose coefficients are given in terms of the mixed volumes above. Since these coefficients are positive (in particular, the independent term is positive), we get that f is differentiable in $(0,1)$ and has the corresponding one-sided derivatives at $t = 0$ and $t = 1$.

For simplicity, write

$$V_{(i)} := V\big(K[m-i], L[i], K_{m+1}, \dots, K_n\big)$$

for each $i = 0, \dots, m$. Then, from the linearity of mixed volumes, we derive the polynomial expression

$$V\big(K_t[m], K_{m+1}, \dots, K_n\big) = \sum_{i=0}^{m} \binom{m}{i} (1-t)^{m-i} t^i V_{(i)}.$$

We will compute the first and the second derivatives of this polynomial expression in order to obtain the second (right-sided) derivative of f at $t = 0$. This derivative will be calculated mainly to fix ideas and notation, since we will actually need the derivative of another (auxiliary) function, as the reader will notice later. For simplicity, we write

$$(1-t)^{m-i}t^i = \sum_{j=0}^{m-i}\binom{m-i}{j}(-1)^j t^{j+i}.$$

We have

$$\left[\sum_{i=0}^{m}\binom{m}{i}(1-t)^{m-i}t^i V_{(i)}\right]' = \left[\sum_{i=0}^{m}\binom{m}{i}V_{(i)}\left(\sum_{j=0}^{m-i}\binom{m-i}{j}(-1)^j t^{i+j}\right)\right]' =$$
$$= \sum_{i=0}^{m}\binom{m}{i}V_{(i)}\left(\sum_{j=0}^{m-i}\binom{m-i}{j}(-1)^j (i+j)t^{i+j-1}\right)$$

and

$$\left[\sum_{i=0}^{m}\binom{m}{i}(1-t)^{m-i}t^i V_{(i)}\right]'' = \left[\sum_{i=0}^{m}\binom{m}{i}V_{(i)}\left(\sum_{j=0}^{m-i}\binom{m-i}{j}(-1)^j (i+j)t^{i+j-1}\right)\right]' =$$
$$= \sum_{i=0}^{m}\binom{m}{i}V_{(i)}\left(\sum_{j=0}^{m-i}\binom{m-i}{j}(-1)^j (i+j)(i+j-1)t^{i+j-2}\right).$$

For $t=0$, and writing $V(t) := V\big(K_t[m], K_{m+1}, \dots, K_n\big)$, these expressions give straightforwardly that

$$V'(0) = m\big(V_{(1)} - V_{(0)}\big)$$

and

$$V''(0) = m(m-1)\big(V_{(0)} - 2V_{(1)} + V_{(2)}\big),$$

where we are considering only the right-sided derivatives (and abusing a little of the notation), of course. Now, since $f(t) = V(t)^{1/m}$, we have

$$f''(t) = -\frac{m-1}{m^2}V(t)^{1/m-2}V'(t)^2 + \frac{1}{m}V(t)^{1/m-1}V''(t),$$

and hence, once it is clear that $V(0) = V_{(0)}$, we arrive at

$$f''(0) =$$
$$= -\frac{m-1}{m^2}V_{(0)}^{1/m-2}m^2\big(V_{(1)} - V_{(0)}\big)^2 + \frac{1}{m}V_{(0)}^{1/m-1}m(m-1)\big(V_{(0)} - 2V_{(1)} + V_{(2)}\big) =$$
$$= (m-1)V_{(0)}^{1/m-2}\big[-\big(V_{(1)} - V_{(0)}\big)^2 + V_{(0)}\big(V_{(0)} - 2V_{(1)} + V_{(2)}\big)\big] =$$
$$= (m-1)V_{(0)}^{1/m-2}\big(V_{(0)}V_{(2)} - V_{(1)}^2\big).$$

Finally, the Alexandrov–Fenchel inequality for the bodies $K, L, K_{m+1}, \dots, K_n$ reads

$$V_{(1)}^2 = V\big(K[m-1], L, K_{m+1}, \dots, K_n\big)^2 \geq$$
$$\geq V\big(K[m], K_{m+1}, \dots, K_n\big)V\big(K[m-2], L[2], K_{m+1}, \dots, K_n\big) = V_{(0)}V_{(2)},$$

and it follows that $V_{(0)}V_{(2)} - V_{(1)}^2 \leq 0$. Since $V_{(0)} > 0$, we get that $f''(0) \leq 0$.

Now fix any number $t \in (0,1)$, and let $K(s) = (1-s)K_t + sL$ for $s \in [0,1]$. Define the function $g\colon [0,1] \to \mathbb{R}$ by

$$g(s) = V\big(K(s)[m], K_{m+1}, \dots, K_n\big)^{1/m}.$$

Replacing K by K_s and repeating the argument that we used to prove that $f''(0) \le 0$, we get straightforwardly that $g''(0) \le 0$ (where we are also considering the right-sided derivative, of course). For small $\varepsilon > 0$, we may write

$$K\left(\frac{\varepsilon}{1-t}\right) = \left(1 - \frac{\varepsilon}{1-t}\right) K_t + \frac{\varepsilon}{1-t} L = \left(1 - \frac{\varepsilon}{1-t}\right)\big((1-t)K + tL\big) + \frac{\varepsilon}{1-t} L = \\ = (1 - t - \varepsilon)K + (t+\varepsilon)L = K_{t+\varepsilon}.$$

Thus,

$$f(t+\varepsilon) = g\left(\frac{\varepsilon}{1-t}\right)$$

for every $\varepsilon > 0$ small enough (we actually need to have $t + \varepsilon < 1$). It follows that

$$f''_+(t) = \frac{1}{(1-t)^2} g''_+(0) \le 0.$$

At first glance, this holds only for the right-sided derivative, but since f is the m-th root of a non-vanishing polynomial in a (two-sided) neighborhood of t, we get that the inequality $f''(t) \le 0$ holds for the usual derivative. Because $t \in (0,1)$ was arbitrarily fixed, we obtain that $f''(t) \le 0$ for every $t \in (0,1)$. This shows that f is concave (continuity, say, guarantees that there is no problem for $t = 1$). □

As it was mentioned before, by Lemma 8.2.4 the concavity of f yields that the function $v_m\colon CC^n \to \mathbb{R}$ given as $v_m(K) = V(K[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}}$ is concave. This leads to

$$V\big((1-t)K + tL[m], K_{m+1}, \dots, K_n\big)^{1/m} \ge \\ \ge (1-t)V(K[m], K_{m+1}, \dots, K_n)^{1/m} + tV(L[m], K_{m+1}, \dots, K_n)^{1/m}, \tag{8.3.7}$$

for any $K, L \in CC^n$ and every $t \in [0,1]$. The case where $m = n$ is obviously the classical Brunn–Minkowski inequality, and hence (8.3.7) can be seen as an extension of this inequality for mixed volumes. In the case $m = n$, as an immediate consequence of Theorem 5.1.2, we have that if K and L are convex bodies, then f is linear if and only if K and L are homothetical. There is no analogous extension of this equivalence for the general case, but we still can find useful equivalences for the linearity of f. This is made next. In what follows, we keep writing

$$V_{(i)} := V\big(K[m-i], L[i], K_{m+1}, \dots, K_n\big)$$

for each $i = 0, \dots, m$.

Theorem 8.3.2. *Under the hypotheses and notation of Theorem 8.3.1, and if we additionaly have that $V_{(i)} > 0$ for every $i \in \{0, \dots, m\}$, the following items are equivalent:*

(i) *f is linear,*

(ii) $V_{(i)}^2 = V_{(i-1)}V_{(i+1)}$ *for every* $i = 1, \dots, m-1$,

(iii) $V_{(j)}^m = V_{(0)}^{m-j} V_{(m)}^j$ *for some* $0 < j < m$,

(iv) $V_{(i)}^m = V_{(0)}^{m-i} V_{(m)}^i$ *for every* $0 \le i \le m$.

Remark 8.3.3. There is yet another (easy) equivalence: notice that, since f is concave, we have that f is linear if and only if in (8.3.7) equality holds for **some** $t \in (0,1)$.

Proof. Assume that f is linear. Then we must have

$$f(t) = (1-t)f(0) + tf(1) = (1-t)V_{(0)}^{1/m} + tV_{(1)}^{1/m},$$

from which it follows that

$$V\big(K_t[m], K_{m+1}, \dots, K_n\big) = \Big[(1-t)V_{(0)}^{1/m} + tV_{(1)}^{1/m}\Big]^m \tag{8.3.8}$$

for each $t \in [0,1]$. From the polynomial expression in the left-hand side of the above, derived in the proof of Theorem 8.3.1 (which follows from the linearity of mixed volumes), we arrive at

$$\sum_{i=0}^{m} \binom{m}{i} V_{(i)}(1-t)^{m-i}t^i = \sum_{i=0}^{m} \binom{m}{i} V_{(0)}^{(m-i)/m} V_{(m)}^{i/m} (1-t)^{m-i}t^i. \tag{8.3.9}$$

Comparing the coefficients, we get the equalities

$$V_{(i)} = V_{(0)}^{(m-i)/m} V_{(m)}^{i/m},$$

for $i = 0, \dots, m$. By taking the appropriate powers on both sides, these equalities lead immediately to

$$\begin{aligned} V_{(i)}^{2m} &= V_{(0)}^{2(m-i)} V_{(m)}^{2i}, \\ V_{(i-1)}^{m} = V_{(0)}^{m-i+1} V_{(m)}^{i-1} &\quad \text{and} \\ V_{(i+1)}^{m} &= V_{(0)}^{m-i-1} V_{(m)}^{i+1} \end{aligned}$$

for every $i \in \{1, \dots, m-1\}$. Therefore,

$$V_{(i-1)}^m V_{(i+1)}^m = V_{(0)}^{m-i+1} V_{(m)}^{i-1} V_{(0)}^{m-i-1} V_{(m)}^{i+1} = V_{(0)}^{2(m-i)} V_{(m)}^{2i} = V_{(i)}^{2m}.$$

Thus, taking the m-th root on both sides we get **(ii)**. On the other hand, if **(ii)** holds, then we may write

$$\frac{V_{(i+1)}}{V_{(i)}} = \frac{V_{(i)}}{V_{(i-1)}}$$

for each $i \in \{1, \dots .m-1\}$. This gives

$$\frac{V_{(i)}}{V_{(0)}} = \frac{V_{(1)}}{V_{(0)}} \frac{V_{(2)}}{V_{(1)}} \cdots \frac{V_{(i)}}{V_{(i-1)}} = \left(\frac{V_{(1)}}{V_{(0)}}\right)^i$$

for each $i = 0, \dots, m$. Moreover, the case for $i = m$ reads

$$\frac{V_{(m)}}{V_{(0)}} = \frac{V_{(1)}}{V_{(0)}} \cdots \frac{V_{(m)}}{V_{(m-1)}} = \left(\frac{V_{(1)}}{V_{(0)}}\right)^m. \tag{8.3.10}$$

Consequently,

$$\left(\frac{V_{(i)}}{V_{(0)}}\right)^m = \left(\frac{V_{(1)}}{V_{(0)}}\right)^{mi} = \left(\frac{V_{(m)}}{V_{(0)}}\right)^i,$$

and this can be finally rewritten as

$$V_{(i)} = V_{(0)}^{(m-i)/m} V_{(m)}^{i/m}, \tag{8.3.11}$$

and this holds for each $i = 0, \dots, m$. It follows that equality holds in (8.3.9), and hence we have (8.3.8). Thus, f is linear. The equivalence of **(i)** and **(ii)** is proved. Notice that the equalities (8.3.11) also yield that **(ii)** implies **(iv)**.

Since the implication **(iv)⇒(iii)** is obvious, it just remains to prove that **(iii)⇒(ii)**. This follows from the equality case in (8.3.4), but for the convenience of the reader we write a detailed proof below. We may rewrite the equality $V_{(j)}^m = V_{(0)}^{m-j} V_{(m)}^j$ as

$$\left(\frac{V_{(m)}}{V_{(j)}}\right)^{1/(m-j)} = \left(\frac{V_{(j)}}{V_{(0)}}\right)^{1/j},$$

and we already know from the Alexandrov–Fenchel inequality that

$$\frac{V_{(1)}}{V_{(0)}} \geq \frac{V_{(2)}}{V_{(1)}} \geq \dots \geq \frac{V_{(j)}}{V_{(j-1)}} \geq \dots \geq \frac{V_{(m)}}{V_{(m-1)}}, \tag{8.3.12}$$

see the calculation in (8.3.5), in case of any doubt. This leads to

$$\begin{aligned} \left(\frac{V_{(j)}}{V_{(0)}}\right)^{1/j} = \left(\frac{V_{(1)}}{V_{(0)}} \cdots \frac{V_{(j)}}{V_{(j-1)}}\right)^{1/j} \geq \frac{V_{(j)}}{V_{(j-1)}} \geq \left(\frac{V_{(j+1)}}{V_{(j)}} \cdots \frac{V_{(m)}}{V_{(m-1)}}\right)^{1/(m-j)} = \\ = \left(\frac{V_{(m)}}{V_{(j)}}\right)^{1/(m-j)} = \left(\frac{V_{(j)}}{V_{(0)}}\right)^{1/j}, \end{aligned} \tag{8.3.13}$$

from which we get that

$$\left(\frac{V_{(j)}}{V_{(j-1)}}\right)^j = \frac{V_{(j)}}{V_{(0)}} = \frac{V_{(1)}}{V_{(0)}} \cdots \frac{V_{(j)}}{V_{(j-1)}}.$$

Observe that the right-hand side would be strictly greater than the left-hand side, if one of the first j inequalities in (8.3.12) should be strict. Consequently, we get that

$$\frac{V_{(1)}}{V_{(0)}} = \dots = \frac{V_{(j)}}{V_{(j-1)}}.$$

On the other hand, (8.3.13) also gives

$$\left(\frac{V_{(j)}}{V_{(j-1)}}\right)^{m-j} = \frac{V_{(m)}}{V_{(j)}} = \frac{V_{(j+1)}}{V_{(j)}} \cdots \frac{V_{(m)}}{V_{(m-1)}},$$

and then the last $m - j$ inequalities in (8.3.12) yield

$$\frac{V_{(j)}}{V_{(j-1)}} = \frac{V_{(j+1)}}{V_{(j)}} = \dots = \frac{V_{(m)}}{V_{(m-1)}}.$$

Summarizing, we have equality in all inequalities of (8.3.12), and this is precisely condition **(ii)**.

□

Remark 8.3.4. It is worth noticing that equality in **(iii)** for $j = 1$ is related to the linearity of f in the following way: using again the notation

$$V(t) = V\big(K_t[m], K_{m+1}, \dots, K_n\big),$$

we know from the proof of Theorem 8.3.1 that

$$V'(0) = m\big(V_{(1)} - V_{(0)}\big).$$

Once $f(t) = V(t)^{1/m}$, and since it is clear that $V(0) = V_{(0)}$, we get that the first (right-sided) derivative of f at $t = 0$ is given as

$$f'(0) = \frac{1}{m} V(0)^{1/m-1} V'(0) = V_{(0)}^{1/m-1}\big(V_{(1)} - V_{(0)}\big) = V_{(0)}^{1/m-1} V_{(1)} - V_{(0)}^{1/m}.$$

On the other hand, since we obviously have that $V(1) = V_{(m)}$, and can use the equality $V_{(1)}^m = V_{(0)}^{m-1} V_{(m)}$ to rewrite $V_{(m)}$, we get

$$f(1) - f(0) = V_{(m)}^{1/m} - V_{(0)}^{1/m} = \left(\frac{V_{(1)}^m}{V_{(0)}^{m-1}}\right)^{1/m} - V_{(0)}^{1/m} = V_{(0)}^{1/m-1} V_{(1)} - V_{(0)}^{1/m} = f'(0).$$

Therefore, we have the equality $f'(0) = f(1) - f(0)$, and since f is concave this gives that f is linear.

Equality in the Alexandrov–Fenchel inequality (8.2.1) clearly holds if K and L are translates. However, in dimension $n \geq 3$, this is not the unique case for equality, and the complete characterization of the equality case in (8.2.1) is still open. We briefly discuss this problem in the next section.

8.4 A brief discussion of the equality case

As it was mentioned in Section 8.2, the complete classification of the equality case in the Alexandrov–Fenchel inequality (8.2.1) is still an open problem. This section is devoted to provide a brief glimpse of what is already known about this problem. We do not survey the existing results in the literature; our aim is to give formal and comprehensive proofs of some particular cases where the equality case in (8.2.1) is already completely understood. Roughly speaking, we guide ourselves by the following question.

Question 2. Are there any particular assumptions on $K, L, K_3, \dots, K_n \in CC^n$ under which the equality

$$V(K, L, K_3, \dots, K_n)^2 = V(K, K, K_3, \dots, K_n) V(L, L, K_3, \dots, K_n) \tag{8.4.1}$$

implies that K and L are homothetical?

We start by proving a useful inequality derived from the Alexandrov–Fenchel inequality which has some important consequences for us.

Proposition 8.4.1. *Let $K_0, K_1, K_2, K_3, \dots, K_n \in CC^n$, and write*

$$V_{ij} := V(K_i, K_j, K_3, \dots, K_n)$$

for $i, j = 0, 1, 2$. Then we have the inequality

$$\big(V_{00}V_{12} - V_{01}V_{02}\big)^2 \le \big(V_{01}^2 - V_{00}V_{11}\big)\big(V_{02}^2 - V_{00}V_{22}\big). \tag{8.4.2}$$

Proof. Notice that for every $t_1, t_2 \ge 0$ the linearity of mixed volumes and the Alexandrov–Fenchel inequality yield

$$\begin{aligned}
0 \le V(K_1 + t_2K_0, K_2 + t_1K_0, K_3, \dots, K_n)^2 - \\
-V(K_1 + t_2K_0, K_1 + t_2K_0, K_3, \dots, K_n)V(K_2 + t_1K_0, K_2 + t_1K_0, K_3, \dots, K_n) = \\
= \big(V_{12} + t_2V_{02} + t_1V_{01} + t_1t_2V_{00}\big)^2 - \big(V_{11} + 2t_2V_{01} + t_2^2V_{00}\big)\big(V_{22} + 2t_1V_{02} + t_1^2V_{00}\big) = \\
= t_1^2\big(V_{01}^2 - V_{11}V_{00}\big) + 2t_1t_2\big(V_{12}V_{00} - V_{01}V_{02}\big) + t_2^2\big(V_{02}^2 - V_{00}V_{22}\big) + \\
+2t_1\big(V_{12}V_{01} - V_{11}V_{02}\big) + 2t_2\big(V_{12}V_{02} - V_{22}V_{01}\big) - V_{11}V_{22} + V_{12}^2.
\end{aligned}$$

Denoting the coefficients of the last expression by $A := V_{01}^2 - V_{11}V_{00}$, $B := V_{12}V_{00} - V_{01}V_{02}$, $C := V_{02}^2 - V_{00}V_{22}$, $D := V_{12}V_{01} - V_{11}V_{02}$, $E := V_{12}V_{02} - V_{22}V_{01}$, and $F := -V_{11}V_{22} + V_{12}^2$, we may write

$$At_1^2 + 2Bt_1t_2 + Ct_2^2 + 2Dt_1 + 2Et_2 + F \ge 0$$

for any $t_1, t_2 \ge 0$. Replacing t_1 and t_2 by λt_1 and λt_2 (for $\lambda \ge 0$), respectively, we obtain immediately that

$$\lambda^2\big(At_1^2 + 2Bt_1t_2 + Ct_2^2\big) + \lambda(2Dt_1 + 2Et_2) + F \ge 0$$

for every $\lambda, t_1, t_2 \ge 0$. For any fixed $t_1, t_2 \ge 0$, the expression in the left-hand side becomes a polynomial of second degree in the variable λ. Hence, the coefficient of λ^2 must be non-negative, since otherwise this polynomial would attain negative values for λ large enough. It follows that

$$At_1^2 + 2Bt_1t_2 + Ct_2^2 \ge 0$$

for any $t_1, t_2 \ge 0$. Letting $t_1 = x \ge 0$ and $t_2 = 1$, we have that $Ax^2 + 2Bx + C \ge 0$ for every $x \ge 0$. Using again the Alexandrov–Fenchel inequality and the linearity of mixed volumes, we also get the inequality

$$\begin{aligned}
0 \le V(t_1K_1 + t_2K_2, K_0, K_3, \dots, K_n)^2 - \\
-V(t_1K_1 + t_2K_2, t_1K_1 + t_2K_2, K_3, \dots, K_n)V(K_0, K_0, K_3, \dots, K_n) = \\
= \big(t_1V_{01} + t_2V_{02}\big)^2 - \big(t_1^2V_{11} + 2t_1t_2V_{12} + t_2^2V_{22}\big)V_{00} = \\
= t_1^2\big(V_{01}^2 - V_{11}V_{00}\big) - 2t_1t_2\big(V_{12}V_{00} - V_{01}V_{02}\big) + t_2^2\big(V_{02}^2 - V_{22}V_{00}\big) = \\
= At_1^2 - 2Bt_1t_2 + Ct_2^2
\end{aligned}$$

for every $t_1, t_2 \ge 0$. If $x < 0$, then with $t_1 = -x$ and $t_2 = 1$ we obtain that $Ax^2 + 2Bx + C \ge 0$. Therefore, the inequality

$$Ax^2 + 2Bx + C \ge 0$$

holds for every $x \in \mathbb{R}$. This gives that $B^2 - AC \le 0$, which is precisely the inequality (8.4.2).

□

An important consequence of inequality (8.4.2) is that, under slightly more restrictive assumptions on the involved sets, it allows us to rewrite the Alexandrov–Fenchel inequality with an additional "independent body", in some sense. We see this next.

Proposition 8.4.2. *Let $K_0, K_1, \dots, K_n \in CC^n$, and still denote*

$$V_{ij} := V(K_i, K_j, K_3, \dots, K_n)$$

for $i, j = 0, 1, 2$. Assume that all V_{00}, V_{01} and V_{02} are positive. Then we have the inequality

$$\frac{V_{11}}{V_{01}^2} - \frac{2V_{12}}{V_{01}V_{02}} + \frac{V_{22}}{V_{02}^2} \leq 0. \tag{8.4.3}$$

Now fix any $K_1, K_2, K_3, \dots, K_n \in CC^n$. Then equality in (8.4.3) holds for **some** *$K_0 \in CC^n$ such that V_{00}, V_{01} and V_{02} are positive if and only if the equality*

$$V_{12}^2 = V_{11}V_{22} \tag{8.4.4}$$

holds. In this case, equality in (8.4.3) holds for **every** *$K_0 \in CC^n$ such that V_{00}, V_{01} and V_{02} are positive.*

Proof. We rewrite inequality (8.4.2) as

$$\left(V_{00}V_{12} - V_{01}V_{02}\right)^2 \leq V_{01}^2 V_{02}^2 \left(1 - \frac{V_{00}V_{11}}{V_{01}^2}\right)\left(1 - \frac{V_{00}V_{22}}{V_{02}^2}\right).$$

From the Alexandrov–Fenchel inequality, both expressions between parentheses in the right-hand side are non-negative. Hence, we may use the arithmetic-geometric inequality $ab \leq \frac{1}{4}(a+b)^2$ for these expressions to obtain

$$\left(V_{00}V_{12} - V_{01}V_{02}\right)^2 \leq \frac{V_{01}^2 V_{02}^2}{4}\left(2 - \frac{V_{00}V_{11}}{V_{01}^2} - \frac{V_{00}V_{22}}{V_{02}^2}\right)^2.$$

The expression between the parentheses in the right-hand side is clearly non-negative (because it is the sum of non-negative expressions), as well as V_{01} and V_{02}. Thus, observing that $V_{01}V_{02} - V_{00}V_{12} \leq |V_{00}V_{12} - V_{01}V_{02}|$ and taking the square root on both sides, we arrive at

$$V_{01}V_{02} - V_{00}V_{12} \leq \frac{V_{01}V_{02}}{2}\left(2 - \frac{V_{00}V_{11}}{V_{01}^2} - \frac{V_{00}V_{22}}{V_{02}^2}\right).$$

which reads

$$2 - \frac{2V_{00}V_{12}}{V_{01}V_{02}} \leq 2 - \frac{V_{00}V_{11}}{V_{01}^2} - \frac{V_{00}V_{22}}{V_{02}^2},$$

or else

$$V_{00}\left(\frac{V_{11}}{V_{01}^2} - \frac{2V_{12}}{V_{01}V_{02}} + \frac{V_{22}}{V_{02}^2}\right) \leq 0.$$

Since $V_{00} > 0$, this gives inequality (8.4.3).

Now assume that equality holds in this inequality for some $K_0, K_1, \dots, K_n \in CC^n$ such that V_{00}, V_{01} and V_{02} are positive. Then we have equality in all previous steps and, in particular, we get $ab = \frac{1}{4}(a+b)^2$ for

$$a = 1 - \frac{V_{00}V_{11}}{V_{01}^2} \text{ and}$$
$$b = 1 - \frac{V_{00}V_{22}}{V_{02}^2}.$$

Since $ab = \frac{1}{4}(a+b)^2$ is equivalent to $(a-b)^2 = 0$, it follows that $a = b$, and this leads to

$$\frac{V_{11}}{V_{01}^2} = \frac{V_{22}}{V_{02}^2}.$$

Using this twice in equality in (8.4.3), we get

$$2\frac{V_{11}}{V_{01}^2} - \frac{2V_{12}}{V_{01}V_{02}} = 0 \text{ and}$$
$$2\frac{V_{22}}{V_{02}^2} - \frac{2V_{12}}{V_{01}V_{02}} = 0.$$

These can be written as

$$\frac{2}{V_{01}}\left(\frac{V_{11}}{V_{01}} - \frac{V_{12}}{V_{02}}\right) = 0 \text{ and}$$
$$\frac{2}{V_{02}}\left(\frac{V_{22}}{V_{02}} - \frac{V_{12}}{V_{01}}\right) = 0,$$

and hence we have

$$V_{11} = V_{12}\frac{V_{01}}{V_{02}} \text{ and}$$
$$V_{22} = V_{12}\frac{V_{02}}{V_{01}}.$$

Multiplying these equalities, we arrive at (8.4.4).

For the converse, assume that equality holds in (8.4.4) for the sets $K_1, K_2, \ldots, K_n \in CC^n$. We first consider that $V_{11} > 0$. In this case, it is immediate to see that the quadratic function

$$q(x) = V_{11}x^2 - 2V_{12}x + V_{22}$$

is non-negative for every $x \in \mathbb{R}$. Let $K_0 \in CC^n$ be any set such that V_{00}, V_{01} and V_{02} are positive. Letting $x = V_{02}/V_{01}$, we arrive at

$$V_{11}\frac{V_{02}^2}{V_{01}^2} - 2V_{12}\frac{V_{02}}{V_{01}} + V_{22} \geq 0,$$

and dividing both sides by V_{02}^2, we finally get

$$\frac{V_{11}}{V_{01}^2} - \frac{2V_{12}}{V_{01}V_{02}} + \frac{V_{22}}{V_{02}^2} \geq 0.$$

Since we already know that the reverse inequality also holds, we have equality in (8.4.3). If $V_{11} = 0$, then (8.4.4) gives that $V_{12} = 0$, and hence (8.4.3) becomes $V_{22} \leq 0$. Because V_{22} is a mixed volume,

the reverse inequality also holds, and hence $V_{22} = 0$. Thus, we have equality in (8.4.3). The proof is complete.

□

As another consequence of inequality (8.4.2), we show that equality (8.4.1) can be characterized by certain mixed surface area measures.

Theorem 8.4.1. *Let $K, L, K_3, \dots, K_n \in CC^n$ be such that $V(K, L, K_3, \dots, K_n) > 0$. Then the equality (8.4.1) holds if and only if there exists a number $\alpha > 0$ such that*

$$S(K, K_3, \dots, K_n, \cdot) = \alpha S(L, K_3, \dots, K_n, \cdot).$$

Proof. Assume first that equality (8.4.1) holds for sets $K, L \in CC^n$ such that $V(K, L, K_3, \dots, K_n) > 0$. Writing $K = K_0$ and $L = K_1$, and using the notation introduced in Proposition 8.4.1, we get that $V_{01}^2 = V_{00}V_{11}$. Thus, from (8.4.2), it follows that

$$V_{00}V_{12} = V_{01}V_{02}$$

for every $K_2 \in CC^n$. From the definition of mixed surface area measures, it follows that this equality can be written as

$$V_{00} \int_{\mathbb{S}^{n-1}} h_{K_2}(u)\, dS(L, K_3, \dots, K_n, u) = V_{01} \int_{\mathbb{S}^{n-1}} h_{K_2}(u)\, dS(K, K_3, \dots, K_n, u).$$

Observe also that since $V_{01} = V(K, L, K_3, \dots, K_n) > 0$ and $V_{01}^2 = V_{11}V_{00}$, we get that $V_{00} > 0$. Hence, we may let $\alpha := V_{00}/V_{01}$, and by the equality above (replacing successively K_2 by arbitrary sets C and D of CC^n), we have

$$\begin{aligned} &\int_{\mathbb{S}^{n-1}} h_C(u) - h_D(u)\, d(\alpha S(L, K_3, \dots, K_n, u)) = \\ &\quad = \int_{\mathbb{S}^{n-1}} h_C(u) - h_D(u)\, dS(K, K_3, \dots, K_n, u) \end{aligned}$$

for any $C, D \in CC^n$. We know from Proposition 7.1.2 that the space of functions $g\colon \mathbb{S}^{n-1} \to \mathbb{R}$ which can be written as $g = h_C - h_D$ for some $C, D \in CC^n$ is dense in $C(\mathbb{S}^{n-1})$. Thus, we obtain that

$$\int_{\mathbb{S}^{n-1}} f(u)\, dS(K, K_3, \dots, K_n, u) = \int_{\mathbb{S}^{n-1}} f(u)\, d(\alpha S(L, K_3, \dots, K_n, u))$$

for every $f \in C(\mathbb{S}^{n-1})$. From Proposition A.2.4, we get that

$$S(K, K_3, \dots, K_n, \cdot) = \alpha S(L, K_3, \dots, K_n, \cdot).$$

For the converse, just notice that if the equality above holds, then, representing again $K = K_0$ and $L = K_1$, we have by integration that

$$\begin{aligned} V_{01} = V(K_0, K_1, K_3, \dots, K_n) &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{K_1}(u)\, dS(K_0, K_3, \dots, K_n, u) = \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{K_1}(u)\, d(\alpha S(K_1, K_3, \dots, K_n, u)) &= \alpha V(K_1, K_1, K_3, \dots, K_n) = \alpha V_{11}, \end{aligned}$$

and

$$V_{00} = V(K_0, K_0, K_3, \dots, K_n) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{K_0}(u)\, dS(K_0, K_3, \dots, K_n, u) =$$
$$= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{K_0}(u)\, d(\alpha S(K_1, K_3, \dots, K_n, u)) = \alpha V(K_0, K_1, K_3, \dots, K_n) = \alpha V_{01}.$$

Finally, this leads to

$$V_{01}^2 - V_{00}V_{11} = (\alpha V_{11})^2 - \alpha V_{01}V_{11} = \alpha^2 V_{11}^2 - \alpha(\alpha V_{11})V_{11} = 0,$$

and this is (8.4.1). The proof is complete.

□

We move forward to provide a result concerning the equality case in the Alexandrov–Fenchel inequality for projections onto hyperplanes. Again, we relate it to the equality between certain mixed surface area measures. In what follows, for the sake of simplicity, we write

$$C^u := \mathrm{proj}_{u^\perp}(C)$$

for any $u \in \mathbb{S}^{n-1}$ and every $C \subseteq \mathbb{R}^n$. Also, we denote the unit ball B^n of $\mathbb{R}^n$ by B (so we can write B^u for the projection of B^n onto $u^\perp$). First, we need a technical lemma.

Lemma 8.4.1. *For any $K_1, \dots, K_{n-1} \in \mathcal{CC}^n$ and every $u \in \mathbb{S}^{n-1}$, we have the equalities*

$$V^{(n-1)}(K_1^u, \dots, K_{n-1}^u) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(K_1, \dots, K_{n-1}, v) \tag{8.4.5}$$

and

$$\int_{\mathbb{S}^{n-1}} V^{(n-1)}(K_1^u, \dots, K_{n-1}^u)\, d\mathcal{H}^{n-1}(u) = n\kappa_{n-1} V(K_1, \dots, K_{n-1}, B^n). \tag{8.4.6}$$

Proof. First of all, from the definition of $S(K_1, \dots, K_{n-1}, \cdot)$ and since the support function of $\mathrm{seg}[-u, u]$ is given as $|\langle u, \cdot\rangle|$, we get the equality

$$V\big(K_1, \dots, K_{n-1}, \mathrm{seg}[-u, u]\big) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, dS(K_1, \dots, K_{n-1}, v). \tag{8.4.7}$$

Now we apply Theorem 6.2.4 for $\mathbf{g} = \mathrm{span}\{u\}$ (and, consequently, $\mathbf{g}^\perp = u^\perp$) to write

$$\binom{n}{1} V\big(K_1, \dots, K_{n-1}, \mathrm{seg}[-u, u]\big) = V^{(1)}\big(\mathrm{seg}[-u, u]\big) V^{(n-1)}(K_1^u, \dots, K_{n-1}^u).$$

Because $V^{(1)}$ is just the (Euclidean) length, we arrive at

$$V\big(K_1, \dots, K_{n-1}, \mathrm{seg}[-u, u]\big) = \frac{2}{n} V^{(n-1)}(K_1^u, \dots, K_{n-1}^u).$$

Replacing this equality in (8.4.7), we get the first desired equality. For the other equality, just integrate both sides of (8.4.5), and use Fubini's theorem and Cauchy's surface area formula (6.5.2):

$$\int_{\mathbb{S}^{n-1}} V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1),\dots,\mathrm{proj}_{u^\perp}(K_{n-1})\big)\,d\mathcal{H}^{n-1}(u) =$$
$$= \frac{1}{2}\int_{\mathbb{S}^{n-1}}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS(K_1,\dots,K_{n-1},v)\,d\mathcal{H}^{n-1}(u) =$$
$$= \frac{1}{2}\int_{\mathbb{S}^{n-1}}\left(\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,d\mathcal{H}^{n-1}(u)\right) dS(K_1,\dots,K_{n-1},v) =$$
$$= \kappa_{n-1}\int_{\mathbb{S}^{n-1}} dS(K_1,\dots,K_{n-1},v) = n\kappa_{n-1}V(K_1,\dots,K_{n-1},B^n),$$

where in the last equality we used the definition of $S(K_1,\dots,K_{n-1},\cdot)$ and the fact that $h_{B^n}=1$. The proof is complete.

□

Remark 8.4.1. Compare this with Exercise 7.13.

Now we are ready to state and prove the aforementioned result.

Proposition 8.4.3. *Let $K,L,K_3,\dots,K_{n-1}\in CC^n$, where $n\geq 3$, and assume that*

$$V^{(n-1)}(K^u,B^u,K_3^u,\dots,K_{n-1}^u)>0 \text{ and}$$
$$V^{(n-1)}(L^u,B^u,K_3^u,\dots,K_{n-1}^u)>0$$

for any $u\in\mathbb{S}^{n-1}$ (this happens, e.g., if $K,L,K_3,\dots,K_{n-1}$ are convex bodies). If

$$S(B,K,K_3,\dots,K_{n-1},\cdot)=S(B,L,K_3,\dots,K_{n-1},\cdot),$$

then the equality

$$V^{(n-1)}(K^u,L^u,K_3^u,\dots,K_{n-1}^u)^2 =$$
$$= V^{(n-1)}(K^u,K^u,K_3^u,\dots,K_{n-1}^u)V^{(n-1)}(L^u,L^u,K_3^u,\dots,K_{n-1}^u)$$

holds for every $u\in\mathbb{S}^{n-1}$.

Proof. Let $K_0^u:=B^u$, $K_1^u:=K^u$ and $K_2^u:=L^u$. Also, for simplicity, we fix the notation

$$V_{ij}^u = V^{(n-1)}(K_i^u,K_j^u,K_3^u,\dots,K_{n-1}^u),$$

for $i,j=0,1,2$. Even more, write

$$S_1 := S(B,K,K_3,\dots,K_{n-1},\cdot) \text{ and}$$
$$S_2 := S(B,L,K_3,\dots,K_{n-1},\cdot).$$

From (8.4.5), the equality $S_1=S_2$ gives

$$V_{01}^u = V^{(n-1)}(B^u,K^u,K_3^u,\dots,K_{n-1}^u) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS(B,K,K_3,\dots,K_{n-1},v) =$$
$$= \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,dS(B,L,K_3,\dots,K_{n-1},v) = V^{(n-1)}(B^u,L^u,K_3^u,\dots,K_{n-1}^u) = V_{02}^u.$$

Further, we have by the hypothesis that $V_{01}^u>0$ and $V_{02}^u>0$. Hence, one can certainly find $n-3$ linearly independent segments each of which lies in one of the K_j^u's (for $j=3,\dots,n-1$), and this gives immediately $V_{00}^u>0$. Thus, identifying $u^\perp$ with $\mathbb{R}^{n-1}$ we may apply inequality (8.4.3) in dimension

$n-1$ for the sets $K_0 = B^u$, $K_1 = K^u$, $K_2 = L^u$ and $K_j = K_j^u$ for $j = 3, \dots, n-1$. In view of the equality $V_{01}^u = V_{02}^u$ obtained above, inequality (8.4.3) reads

$$V_{11}^u - 2V_{21}^u + V_{22}^u \leq 0,$$

or else

$$g(u) := V^{(n-1)}(K^u, K^u, K_3^u, \dots, K_{n-1}^u) - 2V^{(n-1)}(K^u, L^u, K_3^u, \dots, K_{n-1}^u) + \\ + V^{(n-1)}(L^u, L^u, K_3^u, \dots, K_{n-1}^u) \leq 0$$

for every $u \in \mathbb{S}^{n-1}$. It is immediate by (8.4.5) that g is continuous. Integrating on $u \in \mathbb{S}^{n-1}$ with respect to $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ and using (8.4.6), we arrive at

$$\frac{1}{n\kappa_{n-1}} \int_{\mathbb{S}^{n-1}} g(u)\, d\mathcal{H}^{n-1}(u) = \\ = V(B, K, K, K_3, \dots, K_{n-1}) - 2V(B, K, L, K_3, \dots, K_{n-1}) + V(B, L, L, \dots, K_{n-1}) = \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K\, dS_1 - \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K\, dS_2 - \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L\, dS_1 + \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L\, dS_2 = \\ = 0,$$

where in the last equality, we used again the fact that $S_1 = S_2$. Since g is a non-positive continuous function whose integral vanishes, we get that $g(u) = 0$ for every $u \in \mathbb{S}^{n-1}$. This gives equality in (8.4.3), and from that, we finally obtain

$$\left(V_{12}^u\right)^2 = V_{11}^u V_{22}^u$$

for any $u \in \mathbb{S}^{n-1}$. This is precisely the equality that we wanted to show, and hence the proof is complete. □

We will use this result to prove the following theorem, which is very central for this section. This states that equality (8.4.1) implies that K and L are homothetical in the case where $K_3 = \dots = K_n = B^n$.

Theorem 8.4.2. *If $K, L \in CC^n$ are such that*

$$V(K, L, B^n, \dots, B^n)^2 = V(K, K, B^n, \dots, B^n)V(L, L, B^n, \dots, B^n), \tag{8.4.8}$$

then K and L are homothetical.

Before proving this theorem, we need a technical result which has independent interest. It states that two sets of CC^n are translates if all of their projections onto hyperplanes are translates. We keep denoting the orthogonal projection of a set $C \subseteq \mathbb{R}^n$ onto $u^\perp$ by C^u.

Proposition 8.4.4. *Let $n \geq 3$. If $K, L \in CC^n$ are such that K^u and L^u are translates for every $u \in \mathbb{S}^{n-1}$, then K and L are translates.*

Proof. The bodies K and L have hyperplane projections which are translates of each other, and this gives that for each $u \in \mathbb{S}^{n-1}$ there exists a (unique) $x^u \in \mathbb{R}^n$ such that

$$K^u = L^u + x^u.$$

For any $t \in \mathbb{R}$, if we set $z = x^u + tu$, then

$$(L + z)^u = L^u + x^u = K^u,$$

and by the uniqueness of x^u the points of the line $x^u + \mathbb{R}u$ are the unique points having this property. Hence, we may write

$$R(u) := \left\{z \in \mathbb{R}^n : K^u = (L+z)^u\right\} = x^u + \mathbb{R}u.$$

Now let $u, v \in \mathbb{S}^{n-1}$ be linearly independent vectors, and let $z \in x^u + \mathbb{R}u$. We observe that $(L+z)^v$ is a translate of L^v, and since L^v and K^v are translates, it follows that K^v and $(L+z)^v$ are translates. Hence, we may write

$$K^v = (L + z + y)^v$$

for some $y \in \mathbb{R}^n$.

It is easy to see that for any non-empty subset $A \subseteq \mathbb{R}^n$ and every vector $v_0 \in \mathbb{S}^{n-1}$ we have that $h_A(w) = h_{A^{v_0}}(w)$ whenever $w \in v_0^\perp$. Thus, if $w \in u^\perp \cap v^\perp$ (that is, if w is simultaneously orthogonal to u and v), then we may write

$$\begin{aligned} h_K(w) = h_{K^u}(w) = h_{(L+z)^u}(w) = h_{L+z}(w) \ \text{ and} \\ h_K(w) = h_{K^v}(w) = h_{(L+z+y)^v}(w) = h_{L+z+y}(w) = h_{L+z}(w) + \langle y, w\rangle. \end{aligned}$$

It follows that $\langle y, w\rangle = 0$. Because this holds for every $w \in u^\perp \cap v^\perp$, we must have $y = \alpha u + \beta v$ for some $\alpha, \beta \in \mathbb{R}$. Hence, the vector $x := z + y - \beta v$ is clearly in the line $R(u)$, since it is the sum of the point z of this line with a vector in the direction of u (namely, $y - \beta v = \alpha u$). It is also clear that $K^v = (L+x)^v$, and hence

$$x \in R(u) \cap R(v).$$

Even more, since $R(u)$ and $R(v)$ are lines in linearly independent directions, their intersection may have at most one point. Thus, we actually have that $R(u) \cap R(v) = \{x\}$.

Now let $u_1, u_2 \in \mathbb{S}^{n-1}$ be linearly independent vectors. We already know that the lines $R(u_1)$ and $R(u_2)$ intersect in a point (z_0, say). Let $v \in \mathbb{S}^{n-1}$ be such that v is not in the vector subspace spanned by u_1 and u_2 (remember that the dimension n is greater than or equal to 3). The line $R(v)$ intersects both $R(u_1)$ and $R(u_2)$, and since v is transversal to the plane which contains these lines, we necessarily have $R(v) \cap R(u_1) = R(v) \cap R(u_2) = \{z_0\}$. It follows that $K^v = (L + z_0)^v$ for any $v \in \mathrm{S}^{n-1} \setminus \operatorname{span}\{u_1, u_2\}$. Fix such a vector v and let $w \in \mathbb{S}^{n-1} \cap v^\perp$. We have

$$h_K(w) = h_{K^v}(w) = h_{(L+z_0)^v}(w) = h_{L+z_0}(w).$$

Notice that this equality holds for every w in the set

$$E := \left\{w \in \mathbb{S}^{n-1} : w \in v^\perp \text{ for some } v \in \mathbb{S}^{n-1} \setminus \operatorname{span}\{u_1, u_2\}\right\}.$$

We claim that $\mathbb{S}^{n-1} \setminus \operatorname{span}\{u_1, u_2\}^\perp \subseteq E$. To verify that, let $w \in \mathbb{S}^{n-1} \setminus \operatorname{span}\{u_1, u_2\}^\perp$ be arbitrarily given. Then, $w^\perp$ is an $(n-1)$-dimensional vector subspace whose intersection with $\mathbb{S}^{n-1}$ cannot be contained in $\operatorname{span}\{u_1, u_2\}$. Indeed, if $n \geq 4$ this follows immediately from the dimension of $w^\perp$ being greater than 2. In the case $n = 3$, we have that if $w^\perp = \operatorname{span}\{u_1, u_2\}$, then $w \in \operatorname{span}\{u_1, u_2\}^\perp$, a contradiction. Hence, there exists $v \in w^\perp \cap (\mathbb{S}^{n-1} \setminus \operatorname{span}\{u_1, u_2\})$, and this gives that $w \in v^\perp$ for some $v \in \mathbb{S}^{n-1} \setminus \operatorname{span}\{u_1, u_2\}$. In other words, $w \in E$, and the claim is proved. From this inclusion, we get immediately that E is dense in $\mathbb{S}^{n-1}$. Hence, by continuity, we finally obtain that

$$h_K(w) = h_{L+z_0}(w)$$

for any $w \in \mathbb{S}^{n-1}$. It follows that $K = L + z_0$, and this concludes the proof.

□

Proof of Theorem 8.4.2. We proceed by induction on the dimension. We start by proving that the result holds in dimension $n = 2$. In this case, equality (8.4.8) is the equality case in the 1st Minkowski inequality. Hence, if K and L are convex bodies, Theorem 6.3.1 gives that they are homothetical. If K and L are segments, then $\mathrm{vol}_n(K) = \mathrm{vol}_n(L) = 0$, and $V(K, L) \neq 0$ if and only if they are linearly independent (see Proposition 6.2.3). This gives that equality (8.4.8) holds if and only if K and L are in the same direction, and hence if and only if K and L are homothetical. Last, if either K or L is a point, then equality holds trivially and K and L are trivially homothetical (every non-empty set is homothetical to any point by considering 0 as the scaling factor).

Now let $n \geq 3$ and assume that in dimension $n - 1$ the equality (8.4.8) implies that K and L are homothetical. Let $K, L \in CC^n$ be such that (8.4.8) holds.

We first observe that the result is easy to prove in the case where both sides of (8.4.8) are zero. Indeed, if $V(K, K, B^n, \dots, B^n) = 0$, then from Proposition 6.2.3, we must have that $\dim(K) < 2$. Hence, K must be a point or a segment. If K is a point, then K and L are trivially homothetical. On the other hand, if K is a segment, then, since we also have that $V(K, L, B^n, \dots, B^n) = 0$ and using Proposition 6.2.3 again, we necessarily have that L is either a point or a segment in the direction of K. It follows that K and L are homothetical. It is immediate that the same holds if $V(L, L, B^n, \dots, B^n) = 0$.

Thus, it suffices to consider that $V(K, K, B^n, \dots, B^n)$ and $V(L, L, B^n, \dots, B^n)$ are positive. This yields that $\dim(K) \geq 2$ and $\dim(L) \geq 2$. In view of (8.4.8), this also gives that $V(K, L, B^n, \dots, B^n)$ is positive as well. Therefore, it follows from Theorem 8.4.1 that $S(K, B^n[n-2], \cdot) = \alpha S(L, B^n[n-2], \cdot)$. Then, setting $\bar{L} := \alpha L$, we get that

$$S(K, B^n[n-2], \cdot) = S(\bar{L}, B^n[n-2], \cdot).$$

Even more, since K and $\bar{L}$ have dimension at least 2, we get that

$$V^{(n-1)}\big(K^u, B^u[n-2]\big) > 0 \quad \text{and} \quad V^{(n-1)}\big(\bar{L}^u, B^u[n-2]\big) > 0$$

for every $u \in \mathbb{S}^{n-1}$ (indeed, the kernel of $\mathrm{proj}_{u^\perp}$ is always one-dimensional). From Proposition 8.4.3, we obtain that

$$V^{(n-1)}\big(K^u, \bar{L}^u, B^u[n-3]\big)^2 = V^{(n-1)}\big(K^u, K^u, B^u[n-3]\big) V^{(n-1)}\big(\bar{L}^u, \bar{L}^u, B^u[n-3]\big)$$

for every $u \in \mathbb{S}^{n-1}$. By the induction hypothesis, it follows that K^u and $\bar{L}^u$ are translates for each $u \in \mathbb{S}^{n-1}$, and hence Proposition 8.4.4 gives that K and $\bar{L}$ are translates. This means that K and L are homothetical, and the proof is complete.

□

We head now to prove the main result of this section, which states that if equality (8.4.1) holds for smooth convex bodies $K_3, \dots, K_n$, then we still ensure that K and L are homothetical. We follow the original proof given in [2151].

Let us fix some concepts and ideas. The *Minkowski difference* between two sets $C, D \in CC^n$ is defined as

$$C \div D := \bigcap_{x \in D} (C - x) = \big\{ z \in \mathbb{R}^n : z + D \subseteq C \big\}.$$

Notice that this set can be empty; it is clear that this happens if and only if C does not contain a translate of D. Moreover, if $C \div D$ is non-empty, then we have $C \div D \in CC^n$ (because any intersection of compact convex sets is compact and convex).

If $K \subseteq \mathbb{R}^n$ is a convex body and $r_{\text{in}}(K)$ denotes the inradius of K (see Exercise 2.13 for the definition), then for $0 \leq t \leq r_{\text{in}}(K)$, we have that $K \div tB^n \neq \emptyset$. Indeed, in this case K contains a translate of tB^n. It follows that for each $t \in [0, r_{\text{in}}(K)]$ the set

$$K_t := (K \div tB^n) + tB^n \tag{8.4.9}$$

is a convex body. In fact, it is clear that if $t = 0$, then $K_t = K$. If $0 < t \leq r_{\text{in}}(K)$, then $K \div tB^n \in CC^n$ and tB^n is a convex body, and hence their sum is a convex body. We also have clearly the inclusion

$$K_t \subseteq K \tag{8.4.10}$$

for every $0 \leq t \leq r_{\text{in}}(K)$. Indeed, if $y \in K_t$, then $y = z + v$ for some $z \in K \div tB^n$ and some $v \in tB^n$. Thus, $z \in K - x$ for every $x \in tB^n$, from where we get that $z \in K - v$. Hence, $z + v \in K$. We will need the following technical result.

Proposition 8.4.5. *Let $K \subseteq \mathbb{R}^n$ be a smooth convex body, and let K_t be defined for each $0 \leq t < r_{\text{in}}(K)$ as above. For any fixed $u \in \mathbb{S}^{n-1}$, define $\varphi_u : [0, r_{\text{in}}(K)] \to \mathbb{R}$ by*

$$\varphi_u(t) := h_{K_t}(u).$$

Then $(\varphi_u)'_+(0) = 0$.

Proof. Fix $u \in \mathbb{S}^{n-1}$ and let $0 < t < r_{\text{in}}(K)$. Let $z_0 \in K \div tB^n$ be such that

$$\langle z_0, u \rangle = h_{K \div tB^n}(u).$$

By the definition of $K \div tB^n$, we have that

$$\langle z_0, u \rangle = \max \left\{ \langle z, u \rangle : z + tB^n \subseteq K \right\}.$$

It follows that $\partial K \cap \partial(z_0 + tB^n)$ must be non-empty, since $z_0 + tB^n$ being contained in the interior of K would contradict the maximality of z_0 (indeed, we would be able to translate the ball $z_0 + tB^n$ slightly within K and in the direction of u).

If $v \in \mathbb{S}^{n-1}$ is such that $z_0 + tv \in \partial K \cap \partial(z_0 + tB^n)$, then v is clearly an outer normal of $z_0 + tB^n$ at $z_0 + tv$. Since $z_0 + tB^n \subseteq K$, we also have that v is an outer normal of K at $z_0 + tB^n$ (see, e.g., Exercise 2.41). Therefore, we have that

$$h_{z_0 + tB^n}(v) = \langle z_0 + tv, v \rangle = h_K(v).$$

Conversely, if $v \in \mathbb{S}^{n-1}$ is such that the equality $h_{z_0+tB^n}(v) = h_K(v)$ holds, then $z_0 + tv \in \partial K$. In fact, $z_0 + tv$ is the point of $z_0 + t\mathbb{S}^{n-1}$ where v is the outer normal. If this point lies in the interior of K, then $h_K(v) > h_{z_0+tB^n}(v)$. Summarizing, we have

$$\begin{aligned} N_t &:= \left\{ v \in \mathbb{S}^{n-1} : z_0 + tv \in \partial K \cap (z_0 + t\mathbb{S}^{n-1}) \right\} = \\ &= \left\{ v \in \mathbb{S}^{n-1} : h_{z_0+tB^n}(v) = h_K(v) \right\}. \end{aligned}$$

See Figure 8.4.1 for an illustration of N_t. It is clear that N_t is a non-empty compact set (it is bounded because it is a subset of $\mathbb{S}^{n-1}$, and it is closed since it is the pre-image of 0 under the continuous function $h_{z_0+tB^n} - h_K$).

We claim that there exists a non-zero vector $w_0 \in \mathbb{R}^n$ such that $N_t \subseteq \text{int}(\mathbf{h}^-(w_0, 0))$. In other words, we have that $\langle v, w_0 \rangle < 0$ for every $v \in N_t$. To prove that, denote $r := r_{\text{in}}(K)$ and let $z_1 + rB^n$ be a ball of maximal radius contained in K. Since the boundary of $z_0 + tB^n$ intersects the boundary of K and $t < r$, we necessarily have that $z_1 \neq z_0$. Thus, letting $w_0 = z_1 - z_0$ we have that $w_0 \neq 0_{\mathbb{R}^n}$.

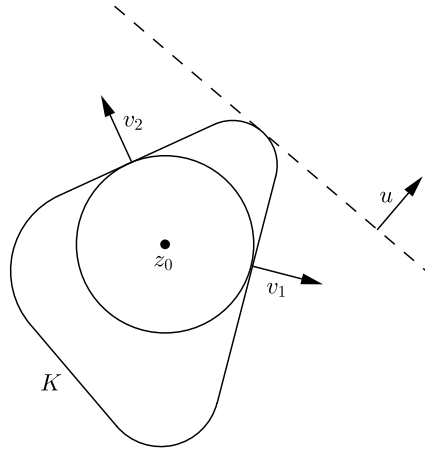


Fig. 8.4.1 In this illustration, we have that $N_t = \{v_1, v_2\}$.

Assume by contradiction that $\langle v, w_0\rangle \geq 0$ for some $v \in N_t$. Since $z_1 + rB^n \subseteq K$, we have the inequality $h_K(v) \geq h_{z_1+rB^n}(v)$. This gives

$$\begin{aligned}\langle z_0 + tv, v\rangle = h_{z_0+tB^n}(v) = h_K(v) \geq h_{z_1+rB^n}(v) = \langle z_1 + rv, v\rangle = \\ = \langle z_0 + w_0 + rv, v\rangle = \langle w_0, v\rangle + \langle z_0 + tv, v\rangle + \langle (r-t)v, v\rangle \geq \\ \geq \langle z_0 + tv, v\rangle + (r-t) > \langle z_0 + tv, v\rangle,\end{aligned}$$

which is a contradiction. Therefore, we have $\langle v, w_0\rangle < 0$ for any $v \in N_t$, and the claim is proved. Notice that the inclusion $N_t \subseteq \operatorname{int}(\mathbf{h}^-(w_0, 0))$ yields that $0_{\mathbb{R}^n} \notin \operatorname{conv}(N_t)$.

Because N_t is a non-empty compact set whose convex hull does not contain the origin, it follows from Exercise 2.47 that $\operatorname{pos}(N_t)$ is a closed convex cone. We obviously have that

$$\operatorname{pos}(N_t) \subseteq \operatorname{int}(\mathbf{h}^-(w_0, 0)) \cup \{0_{\mathbb{R}^n}\}, \tag{8.4.11}$$

and in particular, this implies that $\operatorname{pos}(N_t)$ does not contain a line (that is, $\operatorname{pos}(N_t)$ is a *pointed cone*).

Now we prove that $u \in \operatorname{pos}(N_t)$. To do that, assume by contradiction that $u \notin \operatorname{pos}(N_t)$. Since $\operatorname{pos}(N_t)$ is a closed convex set which is not $\mathbb{R}^n$, we may let $a \in \partial(\operatorname{pos}(N_t))$ be the metric projection of u onto $\operatorname{pos}(N_t)$, and $w_1 := u - a$ is an outer normal of $\operatorname{pos}(N_t)$ at a. We know from Section 2.2 that the hyperplane $\mathbf{h}(w_1, \langle w_1, a\rangle)$ supports $\operatorname{pos}(N_t)$ at a with $\langle w_1, u\rangle > \langle w_1, a\rangle$ and $\langle w_1, y\rangle \leq \langle w_1, a\rangle$ for every $y \in \operatorname{pos}(N_t)$. Since $0_{\mathbb{R}^n} \in \operatorname{pos}(N_t)$, we have that

$$0 = \langle w_1, 0_{\mathbb{R}^n}\rangle \leq \langle w_1, a\rangle,$$

and because $\lambda a \in \operatorname{pos}(N_t)$ for every $\lambda \geq 0$, we get that

$$\lambda\langle w_1, a\rangle = \langle w_1, \lambda a\rangle \leq \langle w_1, a\rangle$$

for any $\lambda \geq 0$. If $\langle w_1, a\rangle > 0$, then this is a contradiction, and hence we obtain that $\langle w_1, a\rangle = 0$. It follows that

$$\langle w_1, u\rangle > 0 \text{ and } \langle w_1, y\rangle \leq 0 \text{ for every } y \in \operatorname{pos}(N_t).$$

Since $\langle w_1, u\rangle > 0$, we may fix $\alpha_0 > 0$ small enough such that for the vector $w_2 := w_1 + \alpha_0 w_0$, we have

$$\langle w_2, u\rangle = \langle w_1, u\rangle + \alpha_0\langle w_0, u\rangle > 0.$$

We know from (8.4.11) that if $y \in \mathrm{pos}(N_t) \setminus \{0_{\mathbb{R}^n}\}$, then $\langle w_0, y\rangle < 0$. Thus, since $\langle w_1, y\rangle \leq 0$ for any $y \in \mathrm{pos}(N_t)$, we arrive at

$$\langle w_2, y\rangle = \langle w_1, y\rangle + \alpha_0\langle w_0, y\rangle \leq \alpha_0\langle w_0, y\rangle < 0$$

for each $y \in \mathrm{pos}(N_t) \setminus \{0_{\mathbb{R}^n}\}$. Because it is clear that w_2 is non-zero, up to rescaling we may assume that $|w_2| = 1$. Since $N_t \subseteq \mathrm{pos}(N_t) \setminus \{0_{\mathbb{R}^n}\}$, we have the inclusion

$$N_t \subseteq \mathrm{int}(\mathbf{h}^-(w_2, 0)).$$

Since $z_0 + tB^n \subseteq K$, we have that $h_K \geq h_{z_0+tB^n}$. The unit vectors for which the equality holds are precisely the elements of the set N_t, and from the inclusion above we get that none of them lie in the closed half-space $\mathbf{h}^+(w_2, 0)$. It follows that the function

$$\mathbb{S}^{n-1} \cap \mathbf{h}^+(w_2, 0) \ni v \mapsto h_K(v) - h_{z_0+tB^n}(v)$$

is strictly positive. By continuity, and since $\mathbb{S}^{n-1} \cap \mathbf{h}^+(w_2, 0)$ is clearly compact, we get that this function attains a positive minimum (β, say). Thus, remembering that $w_2 \in \mathbb{S}^{n-1}$, the Cauchy–Schwarz inequality gives

$$\begin{aligned} h_K(v) \geq h_{z_0+tB^n}(v) + \beta = \langle z_0 + tv, v\rangle + \beta|w_2||v| \geq \langle z_0 + tv, v\rangle + \beta\langle w_2, v\rangle = \\ = \langle z_0 + tv + \beta w_2, v\rangle = h_{z_0+\beta w_2+tB^n}(v) \end{aligned}$$

for every $v \in \mathbb{S}^{n-1}$. This leads to the inclusion

$$z_0 + \beta w_2 + tB^n \subseteq K,$$

and we claim that this contradicts the maximality of z_0. Indeed, since $\beta > 0$ and $\langle w_2, u\rangle > 0$, we have

$$\langle z_0 + \beta w_2, u\rangle = \langle z_0, u\rangle + \beta\langle w_2, u\rangle > \langle z_0, u\rangle.$$

This contradiction gives that $u \in \mathrm{pos}(N_t)$. We claim that, as a consequence, there exist $v_1, \ldots, v_{n+1} \in N_t$ (not necessarily distinct) and $\lambda_1, \ldots, \lambda_{n+1} \geq 0$ such that

$$u = \lambda_1 v_1 + \ldots + \lambda_{n+1} v_{n+1}. \tag{8.4.12}$$

Indeed, since $u \in \mathrm{pos}(N_t)$, we may write

$$u = \alpha_1 u_1 + \ldots + \alpha_k u_k$$

for some $u_1, \ldots, u_k \in N_t$ and some $\alpha_1, \ldots, \alpha_k \geq 0$ (which are not all zero, since $u \in \mathbb{S}^{n-1}$). Let $\alpha := \alpha_1 + \ldots + \alpha_k$. It follows that $u/\alpha \in \mathrm{conv}(N_t)$, and hence Carathéodory's theorem (Theorem 2.1.2) gives that u/α can be written as a convex combination of affinely independent points of N_t, which we may denote by $v_1, \ldots, v_{n+1}$ because the affine dimension of N_t is at most $n + 1$, of course. We write

$$\frac{u}{\alpha} = \beta_1 v_1 + \ldots + \beta_{n+1} v_{n+1}.$$

Hence, we simply set $\lambda_j := \alpha\beta_j$ for each $j = 1, \ldots, n+1$ and we have (8.4.12). For each $j = 1, \ldots, n+1$ and every $x \in K$, we have that

$$\langle x, v_j \rangle \leq h_K(v_j) = h_{z_0+tB^n}(v_j),$$

and hence it follows immediately that

$$\langle x, u \rangle \leq \sum_{j=1}^{n+1} \lambda_j h_{z_0+tB^n}(v_j).$$

Since this holds for every $x \in K$, we obtain, passing to the supremum, the inequality

$$h_K(u) \leq \sum_{j=1}^{n+1} \lambda_j h_{z_0+tB^n}(v_j).$$

Now observe that it is clear that $K_0 = K$, and then from (8.4.10), we get that $\varphi_u(t) \leq \varphi_u(0)$ for every $0 < t < r_{\mathrm{in}}(K)$. We also have that

$$z_0 + tB^n \subseteq K_t,$$

because $z_0 \in K \div tB^n$. It follows that

$$\begin{aligned} 0 \leq \varphi_u(0) - \varphi_u(t) = h_K(u) - h_{K_t}(u) \leq h_K(u) - h_{z_0+tB^n}(u) \leq \\ \leq \sum_{j=1}^{n+1} \lambda_j h_{z_0+tB^n}(v_j) - h_{z_0+tB^n}(u) = \\ = \left\langle z_0, \sum_{j=1}^{n+1} \lambda_j v_j \right\rangle + \sum_{j=1}^{n+1} t\lambda_j h_{B^n}(v_j) - \langle z_0, u \rangle - t h_{B^n}(u) = t\left(-1 + \sum_{j=1}^{n+1} \lambda_j\right), \end{aligned}$$

where we used (8.4.12) and the fact that $h_{B^n}(v_j) = h_{B^n}(u) = 1$ for every $j = 1, \ldots, n+1$. Dividing both sides by t, we arrive at

$$0 \leq \frac{\varphi_u(0) - \varphi_u(t)}{t} \leq -1 + \sum_{j=1}^{n+1} \lambda_j. \tag{8.4.13}$$

Since $t \in (0, r_{\mathrm{in}}(K))$ was arbitrarily fixed, we may write $z_0 = z_0(t)$, and $\lambda_j = \lambda_j(t)$ and $v_j = v_j(t)$ for each $j = 1, \ldots, n+1$. Notice carefully that $z_0(t)$, the $\lambda_j(t)$'s and the $v_j(t)$'s are not necessarily unique, and hence we have to fix them for each t. It is not difficult to check that for each fixed j any converging subsequence of the "touching points" $z_0(t) + tv_j(t)$ must converge to a point of $F(K, u)$, which is the face of K consisting of boundary points where the unit outer normal is u (an argument by contradiction gives a formal proof of this fact). The Gauss map of K is continuous (because K is smooth), and because $v_j(t)$ is the unit outer normal of K at $z_0(t) + tv_j(t)$, this gives that $v_j(t) \to u$. Since

$$u = \lambda_1(t)v_1(t) + \ldots + \lambda_{n+1}(t)v_{n+1}(t),$$

we get that

$$\sum_{j=1}^{n+1} \lambda_j(t) \to 1$$

as $t \to 0$. Therefore, letting $t \to 0$ in (8.4.13) gives the desired, and the proof is complete.

□

Corollary 8.4.1. *Let K be a smooth convex body, and let $K_2, \dots, K_n \in CC^n$. Define*

$$\phi(t) := V(K_t, K_2, \dots, K_n)$$

for $0 \le t \le r_{\text{in}}(K)$. Then, $\phi'_+(0) = 0$.

Proof. Using the notation $\varphi_u(t) := h_{K_t}(u)$ for $t \in [0, r_{\text{in}}(K)]$ and $u \in \mathbb{S}^{n-1}$, and using also the definition of mixed surface area measures, we write

$$\begin{aligned}\frac{\phi(t) - \phi(0)}{t} &= \frac{V(K_t, K_2, \dots, K_n) - V(K_0, K_2, \dots, K_n)}{t} = \\ &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \frac{h_{K_t}(u) - h_{K_0}(u)}{t}\, dS(K_2, \dots, K_n, u) = \\ &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \frac{\varphi_u(t) - \varphi_u(0)}{t}\, dS(K_2, \dots, K_n, u).\end{aligned}$$

Hence

$$\phi'_+(0) = \frac{1}{n} \lim_{t \to 0^+} \int_{\mathbb{S}^{n-1}} \frac{\varphi_u(t) - \varphi_u(0)}{t}\, dS(K_2, \dots, K_n, u). \tag{8.4.14}$$

We will show that the integrand remains bounded as $t \to 0$ (uniformly on $u \in \mathbb{S}^{n-1}$), so we can use the dominated convergence theorem. For that sake, choose $z \in K$ and $\lambda > 0$ such that $z + \lambda B^n \subseteq \operatorname{int}(K)$. Fix any point $x \in F(K, u)$. For any given $0 < t < \lambda$, consider the dilation $D\colon \mathbb{R}^n \to \mathbb{R}^n$ given as

$$D(y) = x + \frac{t}{\lambda}(y - x).$$

We observe that $D(z + \lambda B^n) \subseteq K$. Indeed, for any $v \in \mathbb{S}^{n-1}$, we have

$$D(z + \lambda v) = x + \frac{t}{\lambda}(z + \lambda v - x) = \left(1 - \frac{t}{\lambda}\right) x + \frac{t}{\lambda}(z + \lambda v),$$

and hence $D(z + \lambda v)$ is a convex combination of the points x and $z + \lambda v$. Because these points are points of K, we get from convexity that $D(z + \lambda v) \in K$. Also, letting

$$z_t := x - \frac{t}{\lambda} x + \frac{t}{\lambda} z,$$

we get immediately that $D(z + \lambda B^n) = z_t + tB^n$. Thus, $z_t + tB^n \subseteq K$, and this leads to $z_t \in K \div tB^n$. Therefore,

$$\langle z_t, u \rangle \le h_{K \div tB^n}(u) = h_{K_t}(u) - h_{tB^n}(u) = h_{K_t}(u) - t,$$

and we may rewrite this inequality as

$$h_{K_t}(u) \ge \langle z_t, u \rangle + t = \langle z_t + tu, u \rangle = h_{z_t + tB^n}(u), \tag{8.4.15}$$

where the last equality justifies since it is clear that $z_t + tu$ is the point of $z_t + t\mathbb{S}^{n-1}$ where u is the unit outer normal of $z_t + tB^n$. Now remember that $x \in F(K, u)$ leads to $\langle x, u \rangle = h_K(u)$. Let

$$\alpha := h_K(u) - h_{z_t+tB^n}(u) = \langle x, u\rangle - \langle z_t + tu, u\rangle =$$
$$= \langle x, u\rangle - \left\langle x - \frac{t}{\lambda}x + \frac{t}{\lambda}z + tu, u\right\rangle = \frac{t}{\lambda}\langle x - z, u\rangle - t,$$

and

$$\beta := h_K(u) - h_{z+\lambda B^n}(u) = \langle x, u\rangle - \langle z + \lambda u, u\rangle = \langle x - z, u\rangle - \lambda.$$

Hence, we have that

$$\alpha = \frac{t}{\lambda}\beta.$$

This, together with inequality (8.4.15), yields

$$0 \le \varphi_u(0) - \varphi_u(t) = h_K(u) - h_{K_t}(u) \le h_K(u) - h_{z_t+tB^n}(u) = \alpha = \frac{\beta}{\lambda}t,$$

where the first inequality comes from the inclusion $K_t \subseteq K$ given in (8.4.10). Finally, letting

$$\beta_0 := \max\left\{h_K(v) - h_{z+\lambda B^n}(v) : v \in \mathbb{S}^{n-1}\right\},$$

we get

$$\left|\frac{\varphi_u(t) - \varphi_u(0)}{t}\right| \le \frac{\beta_0}{\lambda}$$

for any $u \in \mathbb{S}^{n-1}$ and every $0 < t < \lambda$. Since the constant β_0/λ only depends on the fixed body K, we get the desired.

Finally, since the integrand in (8.4.14) is uniformly bounded on $u \in \mathbb{S}^{n-1}$ as $t \to 0$, we may use the dominated convergence theorem (Theorem A.2.4) and Proposition 8.4.5 to get

$$\phi'_+(0) = \frac{1}{n}\int_{\mathbb{S}^{n-1}}\left[\lim_{t\to 0^+}\frac{\varphi_u(t) - \varphi_u(0)}{t}\right] dS(K_2, \dots, K_n, u) =$$
$$= \frac{1}{n}\int_{\mathbb{S}^{n-1}}(\varphi_u)'_+(0)\, dS(K_2, \dots, K_n, u) = 0,$$

and the proof is complete.

□

Now we are in position to state and prove the main result of this section.

Theorem 8.4.3. *Let $L_3, \dots, L_n \subseteq \mathbb{R}^n$ be smooth convex bodies. If $K, L \in CC^n$ are such that*

$$V(K, L, L_3, \dots, L_n)^2 = V(K, K, L_3, \dots, L_n)V(L, L, L_3, \dots, L_n), \tag{8.4.16}$$

then K and L are homothetical.

Proof. The case where $n = 2$ is simply the equality case of the 1st Minkowski inequality (Theorem 6.3.1), and hence we may assume that $n \ge 3$. We also may assume that K and L are not points, since if one of them is a point, then they are trivially homothetical. If K (or L) is a segment, then $V(K, K, K_3, \dots, K_n) = 0$, and hence $V(K, L, K_3, \dots, K_n) = 0$. It follows that L is a point or a segment in the direction of L, and this gives that K and L are homothetical.

Thus, the only non-trivial case is the case where $\dim K \ge 2$ and $\dim L \ge 2$, and from now on, we assume that this holds. In this case, all mixed volumes appearing in (8.4.16) are positive. Rescaling L

if necessary, we have from Theorem 8.4.1 that

$$S(K, L_3, L_4, \dots, L_n, \cdot) = S(L, L_3, L_4, \dots, L_n, \cdot).$$

From the definition of mixed surface area measures, this leads to the equalities

$$\begin{aligned} V(K, K, L_3, \dots, L_n) &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(v)\, dS(K, L_3, \dots, L_n, v) = \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(v)\, dS(L, L_3, \dots, L_n, v) &= V(K, L, L_3, \dots, L_n) = \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(v)\, dS(K, L_3, \dots, L_n) &= \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(v)\, dS(L, L_3, \dots, L_n, v) &= V(L, L, L_3, \dots, L_n). \end{aligned} \tag{8.4.17}$$

For simplicity, from now on, we write $L_3 = M$. Let $C \subseteq \mathbb{R}^n$ be a convex body. Repeating the argument as above, we obtain

$$\begin{aligned} V(C, K, M, L_4, \dots, L_n) &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_C(v)\, dS(K, M, L_4, \dots, L_n, v) = \\ = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_C(v)\, dS(L, M, L_4, \dots, L_n, \cdot, v) &= V(C, L, M, L_4, \dots, L_n). \end{aligned} \tag{8.4.18}$$

Now we will use inequality (8.4.3) with K_3 replaced by C, $K_j = L_j$ for each $j = 4, \dots, n$, and for $K_1 = K$ and $K_2 = L$, in such a way that

$$\begin{aligned} V_{11} &= V(K, K, C, L_4, \dots, L_n), \\ V_{12} &= V(K, L, C, L_4, \dots, L_n) \quad \text{and} \\ V_{22} &= V(L, L, C, L_4, \dots, L_n). \end{aligned} \tag{8.4.19}$$

In addition, letting K_0 in inequality (8.4.3) be the body M, and writing for simplicity $\mathbf{L} := (L_4, \dots, L_n)$, we finally arrive at

$$\frac{V(K, K, C, \mathbf{L})}{V(M, K, C, \mathbf{L})^2} - \frac{2V(K, L, C, \mathbf{L})}{V(M, K, C, \mathbf{L})V(M, L, C, \mathbf{L})} + \frac{V(L, L, C, \mathbf{L})}{V(M, L, C, \mathbf{L})^2} \leq 0. \tag{8.4.20}$$

Because (8.4.18) reads $V(M, K, C, \mathbf{L}) = V(M, L, C, \mathbf{L})$, this inequality is equivalent to

$$V(K, K, C, \mathbf{L}) - 2V(K, L, C, \mathbf{L}) + V(L, L, C, \mathbf{L}) \leq 0. \tag{8.4.21}$$

So, if we let

$$\Phi(C) = V(K, K, C, \mathbf{L}) - 2V(K, L, C, \mathbf{L}) + V(L, L, C, \mathbf{L}),$$

then we have $\Phi(C) \leq 0$ for every convex body $C \subseteq \mathbb{R}^n$. Recall that we are writing $L_3 = M$ and let M_t be as in (8.4.9) for $t \geq 0$ small enough such that M_t is a convex body. We define

$$f(t) := \Phi(M_t) = V(K, K, M_t, \mathbf{L}) - 2V(K, L, M_t, \mathbf{L}) + V(L, L, M_t, \mathbf{L}),$$

and we notice that, since $M_0 = M = L_3$, from the equalities in (8.4.17), it follows that

$$f(0) = V(K, K, L_3, \mathbf{L}) - 2V(K, L, L_3, \mathbf{L}) + V(L, L, L_3, \mathbf{L}) = 0.$$

Moreover, from Corollary 8.4.1, we have immediately that $f'_+(0) = 0$. On the other hand, by the linearity of mixed volumes, we may write

$$f(t) = \Phi(M_t) = \Phi\big((M \dot{-} tB^n) + tB^n\big) = \Phi(M \dot{-} tB^n) + t\Phi(B^n),$$

and hence, letting $g(t) := \Phi(M \dot{-} tB^n)$, we get that

$$g'_+(0) = f'_+(0) - \Phi(B^n) = -\Phi(B^n) \geq 0,$$

where the last inequality justifies since we already know that $\Phi(C) \leq 0$ for every convex body $C \subseteq \mathbb{R}^n$. On the other hand, $\Phi(C) \leq 0$ also implies that $g(t) \leq 0$ for every $t > 0$ small enough such that $M \dot{-} tB^n$ is a convex body. Since it is also clear that $g(0) = 0$, this leads to $g'_+(0) = 0$. It follows from the equality above that $\Phi(B^n) = 0$, and this equality reads

$$V(K, K, B^n, \mathbf{L}) - 2V(K, L, B^n, \mathbf{L}) + V(L, L, B^n, \mathbf{L}) = 0.$$

Therefore, equality holds for $C = B^n$ in (8.4.21), and thus also in (8.4.20). In view of Proposition 8.4.2 and (8.4.19), we get that

$$V(K, L, B^n, L_4, \dots, L_n)^2 = V(K, K, B^n, L_4, \dots, L_n)V(L, L, B^n, L_4, \dots, L_n).$$

Repeating the argument with L_4 in the role of (the former) L_3 and with B^n in place of (the former) L_4, we arrive at

$$\begin{aligned} V(K, L, B^n, B^n, L_5, \dots, L_n)^2 = \\ = V(K, K, B^n, B^n, L_5, \dots, L_n)V(L, L, B^n, B^n, L_5, \dots, L_n). \end{aligned}$$

Next, we repeat again the original argument for L_5 in the role of L_3, and letting the original L_4 and L_5 be replaced by B^n. Proceeding iteratively, after a finite number of steps we obtain

$$V(K, L, B^n, \dots, B^n)^2 = V(K, K, B^n, \dots, B^n)V(L, L, B^n, \dots, B^n),$$

and from Theorem 8.4.2, we get that K and L are homothetical.

□

We finish this section by investigating another equality case which will be useful to study the so-called *mixed quermassintegrals* later. Roughly speaking, it states that for smooth convex bodies $K_{m+1}, \dots, K_n \subseteq \mathbb{R}^n$, equality in the generalized Brunn–Minkowski theorem (see Theorem 8.3.1) implies that K and L are homothetical.

Theorem 8.4.4. *Let $m \in \{2, \dots, n\}$ and $K_{m+1}, \dots, K_n \subseteq \mathbb{R}^n$ be smooth convex bodies. Given $K, L \in CC^n$ such that $\dim(K) \geq m$ and $\dim(L) \geq m$, we let $K_t := (1-t)K + tL$ for $t \in [0,1]$, and define $f\colon [0,1] \to \mathbb{R}$ by*

$$f(t) = V\big(K_t[m], K_{m+1}, \dots, K_n\big)^{1/m}.$$

Then f is linear if and only if K and L are homothetical.

Proof. The fact that f is linear if K and L are homothetical is immediate. Thus, all we have to prove is the converse implication. Hence, we assume that f is linear, and for each $i = 0, \dots, m$ we let

$$V_{(i)} = V(K[m-i], L[i], K_{m+1}, \dots, K_n),$$

and we observe that the condition on the dimensions of K and L (together with the fact that $K_{m+1}, \dots, K_n$ have maximal dimension) gives that $V_{(i)} > 0$ for each $i = 0, \dots, m$. From Theorem 8.3.2, we have

$$V_{(i)}^2 = V_{(i-1)} V_{(i+1)}$$

for each $i = 1, \dots, m-1$. In this case, if for simplicity, we denote

$$\mathbf{K}_i := (K[m-i-1], L[i-1], K_{m+1}, \dots, K_n),$$

then the equality above reads

$$V(K, L, \mathbf{K}_i)^2 = V(K, K, \mathbf{K}_i) V(L, L, \mathbf{K}_i).$$

From Theorem 8.4.1, it follows immediately that

$$S(K, \mathbf{K}_i, \cdot) = \alpha_i S(L, \mathbf{K}_i, \cdot)$$

for some constant $\alpha_i > 0$. To find this constant, we simply write

$$V(K, K, \mathbf{K}_i) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(v)\, dS(K, \mathbf{K}_i, v) = \frac{\alpha_i}{n} \int_{\mathbb{S}^{n-1}} h_K(v)\, dS(L, \mathbf{K}_i, v) = \\ = \alpha_i V(K, L, \mathbf{K}_i),$$

which yields

$$\alpha_i = \frac{V(K, K, \mathbf{K}_i)}{V(K, L, \mathbf{K}_i)} = \frac{V(K[m-i+1], L[i-1], K_{m+1}, \dots, K_n)}{V(K[m-i], L[i], K_{m+1}, \dots, K_n)} = \frac{V_{(i-1)}}{V_{(i)}}.$$

Now we observe that

$$V(B^n, K, \mathbf{K}_i) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_{B^n}(v),\ dS(K, \mathbf{K}_i, v) = \frac{\alpha_i}{n} \int_{\mathbb{S}^{n-1}} h_{B^n}(v),\ dS(L, \mathbf{K}_i, v) = \\ = \alpha_i V(B^n, L, \mathbf{K}_i),$$

and in view of the expression for α_i derived above and of the definition of $\mathbf{K}_i$, this can be rewritten as

$$V_{(i)} V(K[m-i], L[i-1], B^n, K_{m+1}, \dots, K_n) = \\ = V_{(i-1)} V(K[m-i-1], L[i], B^n, K_{m+1}, \dots, K_n),$$

and this holds for every $i = 1, \dots, m-1$. If we write

$$\bar{V}_{(i)} := V\big(K[m-i-1], L[i], B^n, K_{m+1}, \dots, K_n\big)$$

for each $i = 0, \dots, m-1$, then the equality above reads

$$V_{(i)} \bar{V}_{(i-1)} = V_{(i-1)} \bar{V}_{(i)}, \tag{8.4.22}$$

for every $i = 1, \dots, m-1$. If $i \in \{0, \dots, m-2\}$, then the equality above for $i+1$ yields

$$V_{(i+1)} \bar{V}_{(i)} = V_{(i)} \bar{V}_{(i+1)}. \tag{8.4.23}$$

Multiplying the left-hand side of (8.4.22) by the right-hand side of (8.4.23) and vice-versa, we arrive at

$$V_{(i)}^2 \bar{V}_{(i-1)} \bar{V}_{(i+1)} = \bar{V}_{(i)}^2 V_{(i-1)} V_{(i+1)},$$

and since $V_{(i)}^2 = V_{(i-1)} V_{(i+1)}$ for $i = 1, \dots, m-1$, we get

$$\bar{V}_{(i)}^2 = \bar{V}_{(i-1)} \bar{V}_{(i+1)}$$

for every $i = 1, \dots, m-2$. This is the same as

$$V(K[m-i-1], L[i], B^n, K_{m+1}, \dots, K_n)^2 = \\ = V(K[m-i], L[i-1], B^n, K_{m+1}, \dots, K_n) V(K[m-i-2], L[i+1], B^n, \dots, K_n).$$

Now writing

$$\mathbf{K}_i' := (K[m-i-2], L[i-1], B^n, K_{m+1}, \dots, K_n),$$

for each $i = 1, \dots, m-2$, the equality above reads

$$V(K, L, \mathbf{K}_i')^2 = V(K, K, \mathbf{K}_i') V(L, L, \mathbf{K}_i'),$$

and using Theorem 8.4.1, we get again $S(K, \mathbf{K}_i', \cdot) = \alpha_i' S(L, \mathbf{K}_i', \cdot)$ for some $\alpha_i' > 0$. Repeating exactly the previous argument, we arrive at

$$V(K[m-i-2], L[i], B^n[2], K_{m+1}, \dots, K_n)^2 = \\ = V(K[m-i-1], L[i-1], B^n[2], \dots, K_n) V(K[m-i-3], L[i+1], B^n[2], \dots, K_n)$$

for every $i = 1, \dots, m-3$ (in both mixed volumes of the last line, the body which comes after $B^n[2]$ is K_{m+1}, of course; we omitted that so the equation fits the margins). Proceeding iteratively, after finitely many steps we must have $i = m - (m-1) = 1$, and

$$V(K, L, B^n[m-2], K_{m+1}, \dots, K_n)^2 = \\ = V(K, K, B^n[m-2], K_{m+1}, \dots, K_n) V(L, L, B^n[m-2], K_{m+1}, \dots, K_n).$$

As a consequence, it follows immediately from Theorem 8.4.3 that K and L are homothetical. The proof is complete.

□

Corollary 8.4.2. *Let $m \in \{2, \dots, n\}$, and let $\mathbf{K} := (K_{m+1}, \dots, K_n)$ be an $(n-m)$-tuple of smooth convex bodies. If $K, L \in CC^n$ have dimension greater than or equal to m, then the equality*

$$V(K[m-1], L, \mathbf{K})^m = V(K[m], \mathbf{K})^{m-1} V(L[m], \mathbf{K}) \tag{8.4.24}$$

holds if and only if K and L are homothetical.

Proof. The fact that the (8.4.24) holds if K and L are homothetical is straightforward. We prove the converse. Assume that (8.4.24) holds, take again $K_t := (1-t)K + tL$ for $t \in [0, 1]$, and

$$f(t) = V(K_t[m], K_{m+1}, \dots, K_n)^{1/m}.$$

Writing again

$$V_{(i)} := V(K[m-i], L[i], K_{m+1}, \dots, K_n)$$

for each $i = 0, \dots, m$, it follows from (8.4.24) that

$$V_{(1)}^m = V_{(0)}^{m-1} V_{(m)}.$$

From Theorem 8.3.2, we get that f is linear. Therefore, using Theorem 8.4.4, we obtain that K and L are homothetical.

□

As another consequence of Theorem 8.4.4, we have an extension of the determination by surface area measures given in Theorem 7.2.1.

Corollary 8.4.3. *Let $m \in \{2, \dots, n\}$, and assume that $\mathbf{K} := (K_{m+1}, \dots, K_n)$ is an $(n-m)$-tuple of smooth convex bodies. If $K, L \in CC^n$ have dimension at least m and*

$$S(K[m-1], \mathbf{K}, \cdot) = S(L[m-1], \mathbf{K}, \cdot), \tag{8.4.25}$$

then K and L are translates. In particular, it follows that if

$$S_{m-1}(K, \cdot) = S_{m-1}(L, \cdot) \tag{8.4.26}$$

for some $m \in \{2, \dots, n\}$, then K and L are translates.

Remark 8.4.2. In other words, we have that for any $j \in \{1, \dots, n-1\}$ the j-th surface area measure $S_j(K, \cdot)$ of a convex body K determines K up to translation. This is known as the *Alexandrov–Fenchel–Jessen theorem*.

Proof. As usual, write

$$V_{(i)} := V\big(K[m-i], L[i], \mathbf{K}\big),$$

for $i \in \{0, \dots, m\}$, and

$$f(t) = V\big((1-t)K + tL[m], \mathbf{K}\big)^{1/m}$$

for $t \in [0, 1]$. From Theorem 8.3.1, we know that f is concave, which leads to the inequality $f'(0) \geq f(1) - f(0)$. By the calculation of $f'(0)$ given in Remark 8.3.4, this leads to

$$V_{(0)}^{1/m-1} V_{(1)} - V_{(0)}^{1/m} = f'(0) \geq f(1) - f(0) = V_{(m)}^{1/m} - V_{(0)}^{1/m}.$$

On the other hand, equality (8.4.25) yields

$$\begin{aligned} V_{(1)} = V(K[m-1], L, \mathbf{K}) &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS(K[m-1], \mathbf{K}, u) = \\ &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)\, dS(L[m-1], \mathbf{K}, u) = V(L[m], \mathbf{K}) = V_{(m)}. \end{aligned}$$

Replacing this equality in the previous inequality, we arrive immediately at

$$V_{(0)} \geq V_{(m)},$$

and interchanging the roles of K and L, we also get the reverse inequality. Hence, $V_{(0)} = V_{(m)}$. Since we already know from the above that $V_{(m)} = V_{(1)}$, the equality in item **(iii)** of Theorem 8.3.2 holds for

$j = 1$, and this gives that f is linear. From Theorem 8.4.4, we get that K and L are homothetical. Since the measures in (8.4.25) are non-zero (because of the hypothesis on the dimensions of K and L) and homogeneous of degree $m-1$ on K and L, respectively, it follows that the scaling factor must be 1, and hence K and L are translates.

Finally, equality (8.4.26) also implies that K and L are translates, since it is simply the particular case of (8.4.25) where $K_{m+1} = \ldots = K_n = B^n$.

□

8.5 Exercises

For the exercise below, recall from Remark 7.5.2 that $[K_1, \ldots, K_{n-1}]$ denotes the mixed body of an $(n-1)$-tuple $K_1, \ldots, K_{n-1} \in \mathcal{K}^n$.

Exercise 8.1. *Let $K_1, \ldots, K_{n-1} \in \mathcal{K}^n$. Prove that*

$$\mathrm{vol}_n([K_1, \ldots, K_{n-1}])^{n-1} \geq \mathrm{vol}_n(K_1) \ldots \mathrm{vol}_n(K_{n-1}).$$

Hint: Use Corollary 8.3.1.

Exercise 8.2. *Let $0 \leq i < n-1$ and $K \in CC^n$. Prove that*

$$W_{i+1}(K)^{n-i} \geq \kappa_n W_i(K)^{n-i-1}. \tag{8.5.1}$$

Hint: Use inequality (8.3.1).

Inequality (8.5.1) relates consecutive quermassintegrals of a set $K \in CC^n$. We can actually do better, and relate any pair of quermassintegrals of K. This is done in the next exercise.

Exercise 8.3. *Let $0 \leq i < j < n$. Prove that*

$$W_j(K)^{n-i} \geq \kappa_n^{j-i} W_i(K)^{n-j} \tag{8.5.2}$$

for every $K \in CC^n$.

Of course, (8.5.1) is just (8.5.2) for $j = i+1$. In the next exercise, we discuss the equality case in the more general inequality and, of course, the case where $j = i+1$ provides a version for the more restricted one.

Exercise 8.4. *Fix $0 \leq i < j < n$, and assume that $K \in CC^n$ is such that $\dim(K) \geq n-i$. Prove that in (8.5.2), equality holds if and only if the function $f\colon [0,1] \to \mathbb{R}$ given as*

$$f(t) = V((1-t)K + tB^n[n-i], B^n[i])^{1/(n-i)}$$

is linear.

Hint: Use Theorem 8.3.2.

Remark 8.5.1. For fixed $0 \leq i < j < n$, if $\dim(K) \geq n-i$, then the equality case in inequality (8.5.2) holds if and only if K is a ball. The reader is invited to prove it in Exercise 8.13 below.

Exercise 8.5. *Verify that the case $i = 0$ of inequality (8.5.1) is the isoperimetric inequality (Theorem 5.2.1).*

Exercise 8.6 (Brunn–Minkowski inequality for quermassintegrals)**.** *Let $1 \le m \le n$. Verify that for any $K, L, K_{m+1}, \dots, K_n \in CC^n$, we have*

$$V(K + L[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}} \ge \\ \ge V(K[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}} + V(L[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}}.$$

Conclude that for all $0 \le i \le n-1$ and every $K, L \in CC^n$, we have

$$W_i(K + L)^{\frac{1}{n-i}} \ge W_i(K)^{\frac{1}{n-i}} + W_i(L)^{\frac{1}{n-i}}.$$

Remark 8.5.2. The equality cases in these inequalities are discussed later in Exercise 8.11.

Exercise 8.7. *Let $K \in \mathcal{K}_0^3$. Prove that*

$$V(K, K^\circ, B^3) \ge \kappa_3.$$

Check that this inequality is sharp.

Hint: Use the same strategy as in Exercise 6.22, but taking the Alexandrov–Fenchel inequality instead of the 1st Minkowski inequality.

The exercise below gives an alternative proof of Theorem 8.3.1 (see also Remark 8.3.2).

Exercise 8.8. *Let $m \in \{2, \dots, n\}$, and let $K_1, K_2, K_{m+1}, \dots, K_n \in \mathcal{K}^n$. Define $g_m \colon [0, +\infty) \to \mathbb{R}$ by*

$$g_m(t) = V(K_1 + tK_2[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}}, \quad t \ge 0.$$

Denote $\mathbf{K}(t) := (K_1 + tK_2[m-2], K_{m+1}, \dots, K_n)$. Prove that

$$g_m''(t) = -(m-1)g_m(t)^{1-2m}\Big[V(K_1, K_2, \mathbf{K}(t))^2 - V(K_1, K_1, \mathbf{K}(t))V(K_2, K_2, \mathbf{K}(t))\Big]$$

for any $t \ge 0$. Conclude that the Alexandrov–Fenchel inequality implies the concavity of g_m, and show how this implies Theorem 8.3.1.

The next exercise can be seen as an extension of the Minkowski inequalities (see Section 6.3).

Exercise 8.9. *Let $m \in \{2, \dots, n\}$, and let $K_1, K_2, K_{m+1}, \dots, K_n \in \mathcal{K}^n$. Prove that*

$$V(K_1[m-1], K_2, K_{m+1}, \dots, K_n)^m \ge \\ \ge V(K_1[m], K_{m+1}, \dots, K_n)^{m-1} V(K_2[m], K_{m+1}, \dots, K_n).$$

Exercise 8.10. *Give an example to show that Proposition 8.4.4 does not hold in the case where the dimension is* 2.

Remember that in Exercise 8.6 above, the inequality

$$V(K + L[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}} \ge \\ \ge V(K[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}} + V(L[m], K_{m+1}, \dots, K_n)^{\frac{1}{m}} \tag{8.5.3}$$

has been stated for every $1 \le m \le n$ and any $K, L, K_{m+1}, \dots, K_n \in CC^n$. In the case $m = 1$, the linearity of mixed volumes gives equality. Below, we discuss the case where $m > 1$.

Exercise 8.11. (a) *Let $m \in \{2, \dots, n\}$. Show that if $K_{m+1}, \dots, K_n$ are smooth convex bodies, and if K, L are convex bodies, then we have equality in (8.5.3) if and only if K and L are homothetical.*

(b) *Conclude that for any convex bodies $K, L \subseteq \mathcal{K}^n$ and every $i \in \{0, \dots, n-2\}$, the equality*

$$W_i(K+L)^{\frac{1}{n-i}} = W_i(K)^{\frac{1}{n-i}} + W_i(L)^{\frac{1}{n-i}}$$

holds if and only if K and L are homothetical.

Exercise 8.12. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that equality holds in (7.5.1) for some $0 \leq i \leq n-2$ and $u \in \mathbb{S}^{n-1}$ if and only if $\mathrm{ref}_{u^\perp}(K)$ is a translate of K.*

Exercise 8.13. *Fix $0 \leq i < j < n$, and let $K \in CC^n$ be such that $\dim(K) \geq n - i$. Prove that equality in (8.5.2) holds for K if and only if K is a ball.*

Exercise 8.14. *Where in the proof of Theorem 8.4.3 do we use the smoothness hypothesis?*

8.6 Notes

The two main themes of Chapter 8 are clearly the Wulff shapes and the Alexandrov–Fenchel inequality and its variants and generalizations. Our notes below follow precisely that separation.

8.6.1 Wulff shapes

Wulff shapes are obtained as the solutions of an anisotropic isoperimetric problem under direction-dependent surface tensions. Thus, they are natural *generalizations of spheres in the isotropic setting*, and a special case of them, studied by the Russian crystallographer Wulff [2475], appeared for the first time in 1901. It turns out that Wulff shapes are essential for the equilibrium of (small) elastic crystals. Alternatively, they can be obtained by using a minimum volume principle and the search of packings having maximal density based on the *parametric density from the theory of packings*. In crystallography, Wulff shapes describe the shape of ideal crystals.

BOOK PARTS AND SURVEYS

As starting points for discussing Wulff shapes and their applications, one can take the following surveys: Böröczky et al. [417] (crystals and quasicrystals and their mathematical theory from discrete geometry), Shlosman [2238] (applications in statistical mechanics and combinatorics), Section 6 of Gardner's nice exposition [929] on related inequalities, and Han and Nishimura [1153] (presentation of the so-called spherical method for studying Wulff shapes), as well as the following monographs: the whole book [757] of Dobrushin et al. (giving also a rigorous study of the original Wulff construction), Chapter 10 of Böröczky [391] on *finite packings and coverings*, Section 8.4 of Gruber's monograph [1070], Section 7.5 of Schneider's book [2167], and Appendix B.2 of Artstein-Avidan et al. [133]. Taylor's nice review [2349] of the monograph Dobrushin et al. [757] is recommendable, too. We also mention the exposition [889] of Fonseca and the first chapter of the booklet [911] of Funaki, where (more applied) variational problems are used to describe shapes of crystals (among them also the Wulff shape) and the convergence of the corresponding discrete (microscopic) interfaces to solutions of such variational problems.

In the following, we try to concentrate on research references that appeared after Schneider's monograph [2167] or cannot be found in the above references.

RECENT RESEARCH CONTRIBUTIONS

We start with some papers related to Wulff shapes in *Minkowski geometry*, i.e., in the geometry of real, finite dimensional Banach spaces. A first basic reference is [915] of Gage. Under certain technical assumptions, every anisotropic curvature flow arises as a "gradient" flow for the length functional of a unique Minkowski geometry. Then each geometry has a unique curve generated by a shrinking procedure and attracting all other embedded closed curves. Every closed smooth embedded curve shrinks to a point and becomes asymptotic to the self-similar flow which is the *isoperimetrix* of the respective Minkowski geometry (cf. Thompson [2361]). Thus, the isoperimetrix shrinks in a self-similar way. Using similar techniques, Stancu [2287] applies the analogy between the *motion of convex, polygonal curves by crystalline curvature and the motion of smooth, convex curves by curvature in Minkowski planes* to verify the uniqueness of self-similar solutions for a crystalline flow; the only self-similar solution is a homothet of the Wulff shape. There are interesting *applications in material science models*. Morgan [1851] shows that for any normed plane an immersed closed rectifiable curve

in equilibrium for fixed area must be a Wulff shape (possibly with multiplicity). He and Li [1174] obtain several new characterizations of the Wulff shape via a *new Minkowski-type integral formula for compact hypersurfaces* with given anisotropic mean curvature. Xia [2477] studies the *anisotropic Minkowski problem* of prescribing the anisotropic Gauss–Kronecker curvature for a closed, strictly convex hypersurface as a function on its anisotropic normals in Minkowski geometry. The solution guarantees the existence and uniqueness of the required surface.

We continue with *Wulff shapes in discrete geometry*, more precisely related to *packings and coverings*; see again the survey [417] and the book [391] (but also the introductary part of Section 1.11 of the problem book [492] of Brass et al.). Wills shows in [2463] that the shape of large densest sphere packings in a lattice, measured by the *parametric density*, does asymptotically not tend to a sphere but to the *polytopal Wulff shape*, depending only on the lattice and the parameter. Böröczky and Schnell [415] introduce an analogue of the known Gibbs–Curie energy for *quasicrystals* and verify that there exists a shape of Wulff type (always being a polytope) optimizing this energy. Based on this result, they model the triacontahedron and the dodecahedron as shapes of quasicrystals. Continuing this, in [416] Böröczky and Schnell show that *dense finite quasiperiodic circle packings in the plane* also yield a Wulff shape. Thus, shapes of quasicrystals might be explained via finite packing densities. Their analogue to the Gibbs–Curie surface energy is minimized by the Wulff shape, i.e., a polygon. Due to the five-fold symmetry of quasicrystals, this might be a regular decagon. The asymptotic Wulff-shape behaviour of large densest sphere packings in a lattice also holds for *periodic packings with spheres of different size*, see [2185] of Schnell, where he extends the result of Wills (see [2463] above) from lattice packings to periodic sphere packings, and for the further extension to the *asymptotic shapes of periodic packings of spheres of different radii*, we refer again to Wills [2466]. Continuing this sequence of papers, Schnell gives in [2186] rigorous proofs of certain properties of the used tools for periodic structures, and even *stability results* for increasing parameters are derived.

Using further on the parametric density, Betke and Böröczky [305] extend the spheres in finite lattice packings to centrally symmetric convex bodies, getting again approximately the Wulff condition. And also regarding the parametric density, Meyer and Schnell [1776] show for *finite lattice coverings of strictly convex bodies in the plane* that the optimal arrangement is either a sausage, so-called double sausage, or tending to a Wulff shape.

Li, Huang, and Xi [1554] obtain upper estimates and lower bounds for k-volumes of sections and projections of Wulff shapes, respectively. These shapes are determined by a discrete and isotropic measure, and implicitly by a positive function; for example, the regular simplex and the cube are examples defined by a discrete measure. Pan and Zhang [1934] study crystalline flows on a class of admissible convex polygons in the plane which can be interpreted as *discrete versions of smooth curvature flows of convex curves of constant perimeter*. The authors verify that for any such initial polygon the flow converges to a Wulff shape. For the centrally symmetric case, the convergence to the asymptotic shape is exponential, and in the non-symmetric case a method via generalized eigenvectors suffices.

Now we come to *stability results on Wulff shapes* (already mentioned regarding [2186]). Palmer [1932] studies the functional of a hypersurface given by a convex elliptic integrand with a volume constraint and verifies that, up to homothety and translation, the only closed, oriented, and stable critical point is the Wulff shape. His result is reproved by Winklmann [2469]. Novaga and Paolini [1904] consider crystalline mean curvature flow and facets of special polytopes shifted with a certain velocity such that the resulting flow satisfies fitting conditions. Systems of ordinary differential equations control the progress as long as the polytopes have facets parallel to those of the corresponding Wulff shape. In view of the stability of the Wulff shape, it is shown that, after rescaling and translating (which can also be avoided), a limiting shape, if existent, is essentially unique. Via numerical computations, examples of *stable dodecahedral Wulff shapes* are given, and various unstable examples are presented, too. Mainly generalizing Alexandrov-type theorems, Roth [2065] offers several interesting stability results for closed convex hypersurfaces with almost constant anisotropic first and second mean curvatures which are close to Wulff shapes.

The remaining part is dedicated to *results on Wulff shapes having different character*.

Peng et al. [1957] develop a theoretical connection between shapes of crystalline materials and the dynamics of nonlinear shock waves in gases. Using this like also the geometric interpretation of ideal crystal growth, the authors establish new formulas for the Wulff shape, also for higher dimensions.

Nishimura and Sakemi [1900] investigate Wulff shapes from the *topological and geometric viewpoint*. They focus on tools like spherical polar sets, perspective projections, and pedal sets, and they give conditions when Wulff shapes cannot be polytopes.

Continuing earlier work of Schuster and Weberndorfer [2209] referring to Lebesgue measure, Huang and He [1244] prove exact inequalities for the Gaussian measure of Wulff shapes and their polars. For example, Barthe's *mean width inequality* for continuous isotropic measures and its dual are subcases of these new inequalities, and also two new Gaussian inequalities related to the *LYZ ellipsoid* are derived. Guo and Leng [1109] establish *mean width inequalities for symmetric Wulff shapes and duals*, discussing also equality conditions. Again, Barthe's mean width inequality for even isotropic measures and its dual form are special cases of these results.

Neumayer [1886] proves *quantitative isoperimetric inequalities for anisotropic surface energies*, where the corresponding *isoperimetric deficit* controls both the Fraenkel asymmetry (a special distance measuring how far a set is from achieving equality in the Wulff inequality) and an oscillation measure between the boundary of the set and the boundary of the respective Wulff shape.

Xia [2478] considers the inverse anisotropic curvature flow from strictly convex hypersurfaces and studies the long-time convergence to the Wulff shape (suitably rescaling and shaping the involved speed functions).

Using support and radial functions, Han and Nishimura [1150] define *duality and self-duality of Wulff shapes*, coming up then with a nice characterization of spherical bodies of constant width. They continue in [1151] by giving a necessary and sufficient condition for the *strict convexity of Wulff shapes*. The survey [1153] of the same authors gives a nice overview on the so-called *spherical method to construct Wulff shapes*, and on related applications. The second section then collects the tools grouped together under the notion of "spherical method", e.g.: spherical duals and spherical pedals, spherical polar sets, spherical Wulff shapes (of constant width), spherical caustics, and spherical symmetry sets. Various open questions are posed, mainly treating the space consisting of convex integrands and the space consisting of Wulff shapes. A comprehensive list of related references is given. The same authors investigate in [1152] the *limit of the Hausdorff distance for one-parameter families of Wulff shapes* constructed by affine perturbations of dual Wulff shapes, and in [1154], they study *spherical Wulff shapes* and their duality (see also Lassak [1503]). They show that the spherical dual transform for spherical Wulff shapes is an isometry with respect to the Pompeiu–Hausdorff metric.

Saroglou, Soprunov, and Zvavitch [2117] show that if K is a convex body satisfying the *Bézout type inequality*

$$V(L_1, \ldots, L_n)V_n(K) \leq V(L_1, K[n-1])V(L_2, \ldots, L_n, K)$$

for mixed volumes of all convex bodies L_i, $i = 1, 2, ..., n$, then K has to be a simplex. For their tool, which is interesting for itself, they use a new result on *differentiability of support functions of Wulff shapes*.

In the very recent paper [439] of Bonacini et al., the authors minimize, under volume constraint, an energy functional given by the sum of a crystalline perimeter and a non-local interaction of Riesz type. They observe that the following holds in the small mass regime under consideration: if the Wulff shape has *certain symmetry properties*, then it is the unique global minimizer of the total energy. In the planar situation, this applies to convex polygons which are reflection symmetric with respect to their angle bisectors.

8.6.2 The Alexandrov–Fenchel inequality

Meaning mixed volumes of n-dimensional convex bodies K_i, the famous Alexandrov–Fenchel inequality from the 1930s states that

$$V(K_1, K_2, K_3, ..., K_n)^2 \geq V(K_1, K_1, K_3, ..., K_n)V(K_2, K_2, K_3, ..., K_n).$$

This famous inequality and variants of it express inequalities of Brunn–Minkowski type for mixed volumes. Its history until the 1980s can be found in Section 20.3 of the book [531] of Burago and Zalgaller. First, we present shortly book parts and surveys dedicated to this famous type of inequalities, and later, we give a selection of recent research papers on this topic.

BOOK PARTS AND SURVEYS

The *Alexandrov–Fenchel inequality*, its origins (see Alexandrov [37] and Fenchel [854]), *its dual, various extensions, sharpenings and consequences* are comprehensively discussed in the Appendices B.3, B.4, and B.5 of Gardner's book [930] and in Sections 7.3, 7.4, and 7.6 of Schneider's monograph [2167]. Alexandrov's two known proofs are also printed in several convexity books cited there, namely those of Busemann [549], Leichtweiss [1531], Hörmander [1209], and Martini, Montejano, and Oliveros [1713]; see also §27 of the monograph [531] regarding an algebraic approach. Artstein-Avidan et al. [134] present in Chapter 9 (besides many other deep inequalities inspired by convex geometry) various inequalities for functions, among them Santaló-type, Alexandrov-type, and Alexandrov–Fenchel-type ones. Here we take the representations in the books of Gardner and Schneider as starting points.

Survey-like publications which appeared before [930] and [2167] are [2148] by Schneider (surveying known results and formulating a conjecture on the general case of equality, with new supporting evidence), Sangwine-Yager [2098] (detailed discussion of the unsettled equality case), Groemer [1040] (stability aspects), and [2154] (update with three conjectures on the equality case). The survey [1404] of Klartag should also be mentioned here, presenting a rich collection of functional inequalities coming from the Brunn–Minkowski inequality and deriveable by interesting, mainly geometric techniques. He discusses and even proves functional versions of Alexandrov–Fenchel inequalities. Ge et al. [947] present a survey about a new higher-order mass, more precisely the Gauss–Bonnet–Chern mass for asymptotically flat and asymptotically hyperbolic manifolds. To prove the related Penrose inequality, the crucial step is the use of the Alexandrov–Fenchel inequality established for the hyperbolic space.

RECENT RESEARCH CONTRIBUTIONS

If the complement of a closed convex set in a closed convex pointed cone is bounded, then this complement minus the apex of the cone is called a *coconvex set* (The main motivation of the authors to study coconvex bodies comes from the local study of singularities of polynomials; there conconvex sets occur and are strongly related to Newton diagrams.) Khovanskii and Timorin [1375] extend the Alexandrov–Fenchel inequality to the setting of coconvex bodies. This yields coconvex versions of the Brunn–Minkowski and the Minkowski inequalities; see also Schneider's paper [2171] where the foundations for a Brunn–Minkowski theory treating coconvex sets are laid.

Abardia and Wannerer [4] extend the Alexandrov–Fenchel inequality for mixed volumes to functionals arising naturally in *hermitian integral geometry*. Martinez-Maure [1702] enriches his investigations on *hedgehogs* (i.e., envelopes parametrized by their Gauss map) by giving, under certain regularity assumptions, a stability estimate of their Alexandrov–Fenchel inequality. The proof of the main result of [1702] is based on the knowledge on equality cases in partial extensions of the Alexandrov–Fenchel inequality to hedgehogs.

The papers [2542] and [2543] combine *volume sums and differences* with (dual) versions of the Alexandrov–Fenchel inequality. For example, in [2542], an Alexandrov–Fenchel type inequality for the volume difference function of a Blaschke–Minkowski and a radial Blaschke–Minkowski homomorphism is derived. These latter notions, based on *Minkowski, Blaschke, radial, and radial Blaschke addition*, are due to Schuster [2206], who also proved Alexandrov–Fenchel inequalities within this setting. And a dual Alexandrov–Fenchel inequality for *dual mixed volume differences* is obtained in [2543].

We continue with L_p-type and Orlicz-type results. Ye [2498] considers the mixed notions for the Orlicz L_φ affine and geominimal surface areas, and he proves Alexandrov–Fenchel inequalities for this concept. Introducing a new affine geometric quantity by calculating the Orlicz first-order variation of the mixed volume (calling the quantity *Orlicz multiple mixed volume*), Zhao extends in [2544] the Alexandrov–Fenchel inequality to an Orlicz setting. It turns out that his Orlicz-Alexandrov–Fenchel inequality yields in special cases an L_p-Alexandrov–Fenchel inequality. In [2546], Zhao establishes a new Alexandrov–Fenchel type inequality involving logarithms and based on new concepts of mixed volume measure and L_p-multiple mixed volume measure; the L_p-Alexandrov–Fenchel inequality is used for this purpose. It leads also to four corresponding logarithmic Minkowski inequalities. Zhang [2528] introduces the notion of L_p-mixed intersection body for any $p \neq 0$, and he establishes a dual Alexandrov–Fenchel inequality for L_p-mixed intersection bodies. In [2547] the authors use their concept of q-dual mixed volume and that of L_p-mixed intersection body to obtain an L_p-Alexandrov–Fenchel inequality for the so-called q-dual volume sum function.

Also, *Alexandrov–Fenchel inequalities in hyperbolic geometry* were derived. Ge, Wang, and Wu [946] extend such inequalities, referring to closed hypersurfaces and their principal curvatures in the Euclidean setting, to *horospherically convex hypersurfaces in hyperbolic n-space* (whose principal curvatures are greater than or equal to 1). Equality characterizes geodesic spheres. Based on a curvature flow under which one curvature integral is preserved while the other vary monotonically, Wang and Xia [2420] show that, for horospherically convex bodies, this flow has long-term existence and converges exponentially to *geodesic spheres*. This yields *strong Alexandrov–Fenchel type inequalities in hyperbolic n-space*. Wei and Xiong [2438] apply the unit normal flow of closed convex hypersurfaces to get new inequalities of Alexandrov–Fenchel type in hyperbolic and spherical n-space. The authors of [1572] obtain a sharp Alexandrov–Fenchel type inequality for star-shaped and convex hypersurfaces in hyperbolic n-space, using two new monotone quantities along the inverse mean curvature flow. This is applied to a certain *Penrose inequality*. In [976], the authors prove an inequality which is similar to one conjectured by Ge, Wang, and Wu and treating *weighted Alexandrov–Fenchel inequalities for horospherically convex hypersurfaces* (referring to a physically motivated, weighted integral of the mean curvature in terms of the area of the hypersurface). For a certain case, counterexamples to the conjectured inequality are presented. *Fuchsian convex bodies* are defined as the closed convex sets in Minkowski space globally invariant under the action of a Fuchsian group. In [873], Fillastre starts to develop a Brunn–Minkowski theory for Fuchsian convex bodies in hyperbolic space. For example, as functions on a compact hyperbolic manifold instead of the sphere, *support functions* can be defined. It is verified that the associated volume is convex (it is log-concave in the classical setting). Based on this, analogues of Alexandrov–Fenchel and Brunn–Minkowski inequalities are obtained, and the inequalities are reversed.

There are also results about *Alexandrov–Fenchel inequalities in spherical spaces* (see also [2438] cited above) or related to spheres. For example, Makowski and Scheuer [1685] prove a rigidity result in the sphere, which yields a generalization of results on smooth convex hypersurfaces in spheres related to certain inverse curvature flows in the sphere. This way they obtain some Alexandrov–Fenchel type inequalities. Continuing [1685] and also [2438], in [975], an Alexandrov–Fenchel-type inequality for strictly convex hypersurfaces in the n-dimensional sphere is proved.

We mention some further papers related to Alexandrov–Fenchel inequalities. Ewald and Tondorf [827] prove partial results on the equality case of the Alexandrov–Fenchel inequality by modifying a conjecture of Schneider. The Alexandrov–Fenchel inequality can be reformulated by stating that the

logarithmized mixed volume f (seen as a function on the discrete simplex) is concave on any segment in the discrete simplex that is parallel to one of the sides. Gromov asked whether it is true that f is concave on the discrete simplex. Burda [536] gives a negative answer to this question about mixed volumes. A complex-analytic new proof of the Alexandrov–Fenchel inequality is given by Wang in [2431]. He basically uses the Legendre transform to develop the Brascamp–Lieb proof of the Prékopa theorem. Chang and Wang [600] discuss how to extend Alexandrov–Fenchel inequalities to non-convex domains or generalized convexity notions. They give a survey of known results in this direction and establish Alexandrov–Fenchel inequalities for the notion of k-convex domains.

PLATE 9

Ellipsoids

General ellipsoid

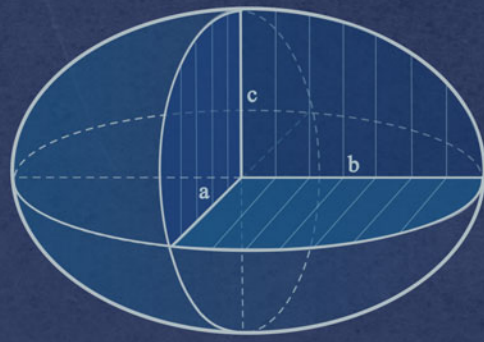
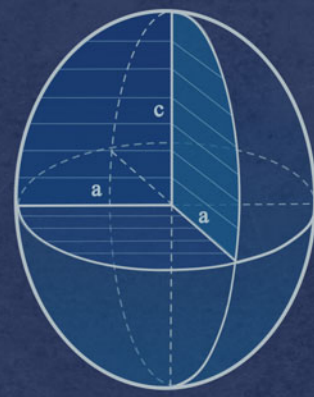


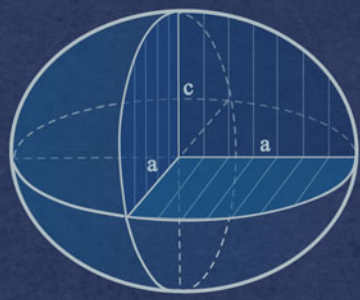
Figure 24

Figures 25 - 26

Prolate spheroid (a < c)



Oblate spheroid (a > c)



Ellipsoids ► (see, e.g., 2.9.1.6 and Chapter 9)

Chapter 9
Affine convex geometry – Part 1

"Mathematics consists of proving the most obvious thing in the least obvious way."

— George Pólya

This chapter is dedicated to the affine geometry of convex bodies (meant in view of their affine positions). Roughly speaking, we want to discuss how quantities and geometric objects related to convex bodies behave under (certain types of) affine transformations. We start by investigating the *John ellipsoid* and the *Löwner ellipsoid*, both associated with a given convex body $K \subseteq \mathbb{R}^n$. They are the ellipsoid of maximal volume inscribed in K and the ellipsoid of minimum volume circumscribed to K, respectively. By definition, any ellipsoid can be transformed into the unit ball B^n by an affine transformation. *John's theorem*, stated and proved in Section 9.2, provides an important characterization of centrally symmetric convex bodies having the unit ball as John ellipsoid. This theorem has far-reaching consequences and applications, and we discuss some of them in the Notes of this chapter. In Plate 9 three types of ellipsoids are presented; the second and the third one are bodies of revolution.

Of course, the volume of a convex body K is invariant under linear transformations whose determinants have absolute value 1. However, this is not the case for surface area or mean width. A natural question arises: is there a lower bound for these geometric quantities when we range through all such transformations? And if the answer is positive, are there characterizations or special properties of the limiting cases? We investigate these questions in Section 9.3. It is interesting to notice that we can perform these investigations without using any theory of mixed volumes.

Since surface area and mean width are, up to rescaling, quermassintegrals, the natural next step is to extend the results obtained on these quantities to arbitrary quermassintegrals. This is made in Section 9.4, but then heavily using the theory of mixed volumes. It is worth mentioning that some results of Section 9.3 (for surface area and mean width) can be re-obtained as particular cases of the results obtained in Section 9.4 (treating general quermassintegrals). This does not mean that Section 9.3 is redundant, because some methods introduced in this section are used in Section 9.4. However, the presentations of these topics follow more a didactic logic rather than an "optimal" one. Namely, we might have written only Section 9.4, introducing all of the necessary methods directly when dealing with quermassintegrals. But we decided to introduce, as an alternative way, some methods while working with notions that have a very clear geometric intuition (surface area and mean width), and so we can use these methods later when working with more abstract quantities (quermassintegrals).

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9.1 John and Löwner ellipsoids

We start this chapter by recalling some terminology regarding concepts from standard linear algebra. We denote by $\mathrm{L}(n)$ the collection of all linear maps $T\colon \mathbb{R}^n \to \mathbb{R}^n$. This is clearly a group under composition and a (finite-dimensional) normed space with the norm

$$||T||_{\mathrm{L}(n)} := \max\{|Tx| : x \in B^n\},$$

where we remind the reader that $B^n = \{x \in \mathbb{R}^n : |x| \leq 1\}$ is the Euclidean unit ball of $\mathbb{R}^n$ (this norm was introduced in Section 2.6). Below, we recall some important subgroups of $\mathrm{L}(n)$

(a) We denote by $\mathrm{GL}(n)$ the *general linear group of* $\mathbb{R}^n$, which is the group (under composition) of invertible linear maps $T\colon \mathbb{R}^n \to \mathbb{R}^n$ (that is, isomorphisms of $\mathbb{R}^n$).

(b) The *special linear group of* $\mathbb{R}^n$ is the subgroup of $\mathrm{GL}(n)$ consisting of all linear transformations whose determinant equals 1. We denote the special linear group of $\mathbb{R}^n$ by $\mathrm{SL}(n)$.

(c) The *orthogonal group* of $\mathbb{R}^n$ is the subgroup $\mathrm{O}(n)$ of $\mathrm{GL}(n)$ consisting of the orthogonal transformations of $\mathbb{R}^n$. For properties of orthogonal transformations, which will be intensively used, we refer the reader to Lemma 2.6.5.

Remark 9.1.1. The reader may notice carefully that the maps in $\mathrm{SL}(n)$ preserve both orientation and Lebesgue volume. Moreover, any orthogonal transformation is clearly an isometry whose determinant equals 1 or -1. Hence, orthogonal transformations preserve the Hausdorff measures of $\mathbb{R}^n$ (and also preserve Lebesgue volume, in particular).

Remark 9.1.2. It is not difficult to understand the topology of these spaces. The space $\mathrm{L}(n)$ is a normed space with dimension n^2, and hence it is homeomorphic to $\mathbb{R}^{n^2}$ endowed with the Euclidean norm. The general linear group $\mathrm{GL}(n)$ is an open subset of $\mathrm{L}(n)$ because the determinant is a continuous function from $\mathrm{L}(n)$ onto $\mathbb{R}$, and $\mathrm{GL}(n) = \det^{-1}(\mathbb{R} \setminus \{0\})$. Since $\mathrm{SL}(n) = \det^{-1}(1)$, we get that the special linear group is a closed subset of $\mathrm{L}(n)$. From the continuity of the inner product, it is also clear that the orthogonal group is closed.

Recall that an *affine transformation* (or *affine map*) of $\mathbb{R}^n$ is a map $A\colon \mathbb{R}^n \to \mathbb{R}^n$ of the form

$$A(\cdot) = T(\cdot) + x_0,$$

where $T \in \mathrm{GL}(n)$ and $x_0 \in \mathbb{R}^n$. In other words, an affine transformation is a composition of an isomorphism with a translation. The image of a set $U \subseteq \mathbb{R}^n$ under an affine transformation is called an *affine image* of U. An affine image of a convex body $K \subseteq \mathbb{R}^n$ is often called a *position* of K, and we will adopt this terminology. A *linear image* of U is the image of U under a linear map. An affine image of the Euclidean unit ball $B^n = \{x \in \mathbb{R}^n : |x| \leq 1\}$ is called an *ellipsoid*.

A standard result in linear algebra (known as *polar decomposition*) gives that we need not consider all maps in $\mathrm{GL}(n)$ to obtain the ellipsoids of $\mathbb{R}^n$, but only the symmetric and positive-definite ones. For the convenience of the reader, we explain this next.

Lemma 9.1.1 (Polar decomposition)**.** *For every $T \in \mathrm{GL}(n)$ there exists some $U \in \mathrm{O}(n)$ such that TU is symmetric (self-adjoint) and positive-definite.*

Remark 9.1.3. Equivalently, every $T \in \mathrm{GL}(n)$ can be written as $T = PV$, where V is orthogonal and P is symmetric and positive-definite. Indeed, if $U \in \mathrm{O}(n)$, then $U^{-1} \in \mathrm{O}(n)$ and $TU = P$ if and only if $T = PU^{-1}$.

Proof. First observe that the linear map TT^* is clearly symmetric and positive-definite. Indeed, for any $x, y \in \mathbb{R}^n$ we have

$$\langle (TT^*)^* x, y\rangle = \langle x, TT^* y\rangle = \langle T^* x, T^* y\rangle = \langle TT^* x, y\rangle \text{ and}$$
$$\langle TT^* x, x\rangle = \langle T^* x, T^* x\rangle = |T^* x|^2.$$

The first equality gives that $(TT^*)^* = TT^*$, from where we get that TT^* is symmetric. In the second equality, we observe that since T^* is an isomorphism, we have that $|T^* x|^2 \geq 0$ with equality if and only if $x = 0_{\mathbb{R}^n}$, and this yields that TT^* is positive-definite.

The spectral theorem (which is Theorem 9.1.1 stated below) gives that there exists $V \in \mathrm{O}(n)$ such that $TT^* = V^{-1}DV$, where D is a diagonal matrix with positive entries in the principal diagonal. Hence we may define the *square root* $(TT^*)^{1/2}$ of TT^* to be $(TT^*)^{1/2} := V^{-1}D^{1/2}V$, where $D^{1/2}$ is the diagonal matrix whose entries in the principal diagonal are the respective positive square roots of the entries of the principal diagonal of D. It is readily seen that

$$(TT^*)^{1/2}(TT^*)^{1/2} = TT^*.$$

It is also immediate that $(TT^*)^{1/2}$ is a positive-definite isomorphism (because all the entries in the principal diagonal of $D^{1/2}$ are strictly positive). We also have that $(TT^*)^{1/2}$ is symmetric. Indeed, since $D^{1/2}$ is symmetric, V^{-1} is also an orthogonal transformation, and $V^* = V^{-1}$ (this is true for any orthogonal transformation), we get

$$\langle (TT^*)^{1/2} x, y\rangle = \langle V^{-1}D^{1/2}Vx, V^{-1}(Vy)\rangle = \langle D^{1/2}Vx, Vy\rangle = \langle Vx, D^{1/2}Vy\rangle =$$
$$= \langle x, V^{-1}D^{1/2}Vy\rangle = \langle x, (TT^*)^{1/2} y\rangle,$$

for every $x, y \in \mathbb{R}^n$. Now we define $U := T^{-1}(TT^*)^{1/2}$. By doing so, we have $TU = (TT^*)^{1/2}$, which is symmetric and positive-definite. Therefore, if we prove that U is orthogonal, then we are done. To do so, we need some ingredients. First, we have to verify that $(T^*)^{-1} = (T^{-1})^*$. In fact,

$$\langle x, (T^{-1})^* y\rangle = \langle T^{-1}x, y\rangle = \langle T^{-1}x, T^*(T^*)^{-1}y\rangle = \langle TT^{-1}x, (T^*)^{-1}y\rangle = \langle x, (T^*)^{-1}y\rangle,$$

for every $x, y \in \mathbb{R}^n$. It is clear that $(TT^*)^{-1}$ is also symmetric and positive-definite, and hence it has a square root $((TT^*)^{-1})^{1/2}$, as above. We prove that $((TT^*)^{-1})^{1/2} = ((TT^*)^{1/2})^{-1}$, which we will denote simply by $(TT^*)^{-1/2}$. Since the inverse of a diagonal matrix is obtained simply by taking the multiplicative inverses of the entries of the principal diagonal, it is obvious that $(D^{1/2})^{-1} = (D^{-1})^{1/2}$. Thus,

$$((TT^*)^{-1})^{1/2} = (VD^{-1}V^{-1})^{1/2} = V^{-1}(D^{-1})^{1/2}V = V^{-1}(D^{1/2})^{-1}V =$$
$$= (V^{-1}D^{1/2}V)^{-1} = ((TT^*)^{1/2})^{-1}.$$

In other words,

$$(TT^*)^{-1} = (TT^*)^{-1/2}(TT^*)^{-1/2},$$

and

$$(TT^*)^{-1/2}(TT^*)^{1/2} = (TT^*)^{1/2}(TT^*)^{-1/2} = \mathrm{Id}_{\mathbb{R}^n}.$$

Finally,

$$\begin{aligned}\langle Ux, Uy\rangle &= \langle T^{-1}(TT^*)^{1/2}x, T^{-1}(TT^*)^{1/2}y\rangle = \langle (TT^*)^{1/2}x, (T^{-1})^*T^{-1}(TT^*)^{1/2}y\rangle = \\ &= \langle (TT^*)^{1/2}x, (T^*)^{-1}T^{-1}(TT^*)^{1/2}y\rangle = \langle (TT^*)^{1/2}x, (TT^*)^{-1}(TT^*)^{1/2}y\rangle = \\ &= \langle (TT^*)^{1/2}x, (TT^*)^{-1/2}(TT^*)^{-1/2}(TT^*)^{1/2}y\rangle = \\ &= \langle (TT^*)^{1/2}x, (TT^*)^{-1/2}y\rangle = \langle x, (TT^*)^{1/2}(TT^*)^{-1/2}y\rangle = \langle x, y\rangle,\end{aligned}$$

for every $x, y \in \mathbb{R}^n$. This concludes the proof.

□

Remark 9.1.4. Similarly, for any $T \in \mathrm{GL}(n)$, there exists an orthogonal transformation U such that UT is symmetric and positive-definite (or, equivalently, every isomorphism T can be written as $T = UP$ for U orthogonal and P symmetric and positive-definite). To verify that, we work with T^*T instead of TT^*.

Remark 9.1.5. It is also worth noticing that if $T \in \mathrm{SL}(n)$, and when we write $T = UP$ (or $T = PU$) for $U \in \mathrm{O}(n)$ and P is symmetric and positive-definite, then we have that $P \in \mathrm{SL}(n)$. In fact, the determinant of P is positive because P is positive-definite, and once $1 = \det(T) = |\det(U)|$ we obtain that $|\det(P)| = 1$.

The advantage of working with positive-definite symmetric maps is that they are *diagonalizable* by orthonormal vectors. This is a consequence of the widely known *spectral theorem* of linear algebra. Since this theorem is largely used in this chapter, we state it now for the convenience of the reader. A proof can be found in most undergraduate linear algebra books.

Theorem 9.1.1 (Spectral theorem)**.** *Let $T \in \mathrm{L}(n)$ be symmetric. There exist an orthonormal basis $\{v_1, \ldots, v_n\}$ of $\mathbb{R}^n$ and numbers $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, possibly repeated, such that $Tv_j = \lambda_j v_j$ for each $j = 1, \ldots, n$. In other words, $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of T. If we identify linear maps with their respective canonical matrix representations, this is the same as saying that we can write*

$$T = U\big[\mathrm{diag}(\lambda_1, \ldots, \lambda_n)\big]U^{-1},$$

where $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ *stands for the diagonal $n \times n$ matrix whose principal diagonal has entries $\lambda_1, \ldots, \lambda_n$ and U is the orthogonal matrix whose columns are $v_1, \ldots, v_n$. If, more than being symmetric, we also have that T is positive-definite, then $\lambda_1, \ldots, \lambda_n > 0$ and, in particular,* $\det(T) = \lambda_1 \ldots \lambda_n > 0$.

For a given $T \in \mathrm{GL}(n)$, let $U \in \mathrm{O}(n)$ be such that TU is self-adjoint and positive-definite. In this case, of course we have $T(B^n) = TU(B^n)$, because U is an isometry. Hence, if $E = x_0 + T(B^n)$ is an ellipsoid, then there exists a symmetric and positive-definite linear transformation $S \in \mathrm{GL}(n)$ such that $E = x_0 + S(B^n)$. Using the spectral theorem, we get that there exists an orthonormal basis $\{v_1, \ldots, v_n\}$ of $\mathbb{R}^n$ which diagonalizes S, meaning that for each $1 \le j \le n$ there exists a number λ_j such that $Sv_j = \lambda_j v_j$ (and since S is positive-definite, each eigenvalue λ_j is positive). Thus, we can express E as follows:

$$E = \left\{ x \in \mathbb{R}^n : \sum_{j=1}^{n} \frac{\langle x - x_0, v_j\rangle^2}{\lambda_j^2} \le 1 \right\}.$$

Since every ellipsoid can be obtained by regarding a symmetric and positive-definite linear map, we get that any ellipsoid may be described as above. Each v_j is called a *principal axis* of E. Moreover, in this case we notice that

$$\mathrm{vol}_n(E) = \mathrm{vol}_n(SB^n) = \det(S)\mathrm{vol}_n(B^n) = (\lambda_1\lambda_2 \ldots \lambda_n)\mathrm{vol}_n(B^n). \tag{9.1.1}$$

One of the key results of this section, that we will prove next, states that any convex body $K \subseteq \mathbb{R}^n$ contains a unique ellipsoid of maximal volume. We start by proving a linear algebra auxiliary result.

Lemma 9.1.2. *Let $T_1, T_2 \in \mathrm{GL}(n)$ be symmetric and positive-definite linear maps, and let $\lambda \in [0,1]$. Then*

$$\det((1-\lambda)T_1 + \lambda T_2) \geq \det(T_1)^{1-\lambda}\det(T_2)^{\lambda},$$

and equality holds for some $\lambda \in (0,1)$ if and only if $T_1 = T_2$.

Proof. This is a consequence of the formula

$$\int_{\mathbb{R}^n} \exp(-\langle Tx, x\rangle)\, dx = \frac{\pi^{n/2}}{\det(T)^{1/2}},$$

which holds for any symmetric and positive-definite isomorphism $T \in \mathrm{GL}(n)$. The proof is easy: just diagonalize T by an orthogonal map U, and use change of variables, Fubini's theorem, and the Gaussian integral (see Section 4.3). From this formula and Hölder's inequality, we obtain

$$\begin{aligned}
\left(\frac{\pi^n}{\det((1-\lambda)T_1 + \lambda T_2)}\right)^{1/2} &= \int_{\mathbb{R}^n} \exp(-\langle (1-\lambda)T_1x + \lambda T_2 x, x\rangle)\, dx = \\
&= \int_{\mathbb{R}^n} (\exp(-\langle T_1x, x\rangle))^{1-\lambda}(\exp(-\langle T_2x, x\rangle))^{\lambda}\, dx \leq \\
&\leq \left(\int_{\mathbb{R}^n} \exp(-\langle T_1x, x\rangle)\, dx\right)^{1-\lambda}\left(\int_{\mathbb{R}^n} \exp(-\langle T_2x, x\rangle)\, dx\right)^{\lambda} = \\
&= \left(\frac{\pi^n}{\det(T_1)^{1-\lambda}\det(T_2)^{\lambda}}\right)^{1/2},
\end{aligned}$$

and hence the desired inequality follows. Now notice that equality holds if and only if we have equality in Hölder's inequality above. The equality condition for Hölder's inequality gives immediately that there exists $\alpha > 0$ such that

$$\alpha \exp(-\langle T_1x, x\rangle) = \exp(-\langle T_2x, x\rangle)$$

for every $x \in \mathbb{R}^n$. Letting $x = 0_{\mathbb{R}^n}$, we get that $\alpha = 1$. Thus,

$$\langle T_1x, x\rangle = \langle T_2x, x\rangle$$

for any $x \in \mathbb{R}^n$ or, equivalently,

$$\langle (T_1 - T_2)x, x\rangle = 0$$

for all $x \in \mathbb{R}^n$. Since T_1 and T_2 are symmetric, we have that $T_1 - T_2$ is symmetric, and then it follows from the Spectral theorem above that it has an orthonormal basis of eigenvectors. Replacing each eigenvector in the equality above, we get that the associated eigenvalue is 0. Hence $T_1 = T_2$. □

Theorem 9.1.2 (John ellipsoid)**.** *If K is a convex body in $\mathbb{R}^n$, then there exists a unique ellipsoid $E_0 \subseteq K$ such that*

$$\mathrm{vol}_n(E_0) = \sup\left\{\mathrm{vol}_n(E) : E \subseteq K \text{ and } E \text{ is an ellipsoid}\right\}.$$

We call E_0 the John ellipsoid *of K.*

Proof. First observe that $E \subseteq K$ implies $\mathrm{vol}_n(E) \leq \mathrm{vol}_n(K)$, and hence we have

$$c_0 := \sup\left\{\mathrm{vol}_n(E) : E \subseteq K \text{ and } E \text{ is an ellipsoid}\right\} < \infty.$$

This supremum is also obviously positive, because in particular K contains a ball. Let $(E_j)_{j\in\mathbb{N}}$ be a sequence of ellipsoids contained in K such that $\mathrm{vol}_n(E_j) \to c_0$ as $j \to \infty$. Since $(E_j)_{j\in\mathbb{N}}$ is a bounded family of convex bodies, passing to a subsequence (if necessary), we may write that $E_j \to C$ for some $C \in CC^n$ in the Hausdorff metric (from Blaschke's selection theorem). Further, observing that $\mathrm{vol}_n(C) = \lim_{j\to\infty} \mathrm{vol}_n(E_j) = c_0 > 0$ we obtain that C is a convex body. Of course, it is also readily seen that $C \subseteq K$ (because all E_j's are contained in K).

We will show that C is an ellipsoid, thus proving the existence part of the theorem. To do so, notice that, for each $j \in \mathbb{N}$, we may write $E_j = x_j + T_j(B^n)$ for some $x_j \in K$ and some $T_j \in \mathrm{GL}(n)$ (we get $x_j \in K$ from $0_{\mathbb{R}^n} \in T_j(B^n)$ and $E_j \subseteq K$). We claim that $(T_j)_{j\in\mathbb{N}}$ is bounded in the norm of $\mathrm{L}(n)$. Indeed, since we always have $x_j \in K$ and $x_j + T_j(B^n) \subseteq K$, we obtain that $T_j(B^n) \subseteq K - K$ for each $j \in \mathbb{N}$. Thus,

$$||T_j||_{\mathrm{L}(n)} = \max\{|T_j(x)| : x \in B^n\} \leq \max\{|z| : z \in K - K\} < \infty,$$

where the last inequality comes from the compactness of $K - K$. It follows from $\mathrm{L}(n)$ being a finite-dimensional normed space that $(T_j)_{j\in\mathbb{N}}$ admits a subsequence $(T_{j_k})_{k\in\mathbb{N}}$ which converges to some $T \in \mathrm{L}(n)$. The sequence $(x_{j_k})_{k\in\mathbb{N}}$ is bounded in $\mathbb{R}^n$, and hence it admits also a subsequence which converges to some point $x \in \mathbb{R}^n$. To simplify the notation, we just write that $T_m \to T$ and $x_m \to x$ as $m \to \infty$, and that $(E_m)_{m\in\mathbb{N}}$ is a subsequence of $(E_j)_{j\in\mathbb{N}}$. We claim that $E_m \to x + T(B^n)$ in the Hausdorff metric as $m \to \infty$. In fact, for an arbitrarily given $\varepsilon > 0$, if $m \in \mathbb{N}$ is large enough such that $||T_m - T||_{\mathrm{L}(n)} < \varepsilon$, we have that

$$|T_m(z) - T(z)| < \varepsilon,$$

for all $z \in B^n$. This means that each point of $T(B^n)$ is ε-close to some point of $T_m(B^n)$, and vice versa. Consequently, we have

$$\begin{aligned} T_m(B^n) &\subseteq T(B^n) + \varepsilon B^n \text{ and} \\ T(B^n) &\subseteq T_m(B^n) + \varepsilon B^n. \end{aligned}$$

If $m \in \mathbb{N}$ is also large enough such that $|x_m - x| < \varepsilon$, then $x_m \in x + \varepsilon B^n$ and $x \in x_m + \varepsilon B^n$. From that we obtain

$$\begin{aligned} x_m + T_m(B^n) \subseteq x_m + T(B^n) + \varepsilon B^n \subseteq x + T(B^n) + 2\varepsilon B^n \text{ and} \\ x + T(B^n) \subseteq x + T_m(B^n) + \varepsilon B^n \subseteq x_m + T_m(B^n) + 2\varepsilon B^n, \end{aligned}$$

and because $E_m = x_m + T_m(B^n)$ we finally arrive at $d_{\mathcal{H}}(E_m, x + T(B^n)) \leq 2\varepsilon$. Now we recall that $(E_j)_{j\in\mathbb{N}}$ converges to C, which means that any of its subsequences must converge to C. Therefore, $C = x + T(B^n)$. Since $\mathrm{vol}_n(C) > 0$, we have that $T \in \mathrm{GL}(n)$, and this proves that C is an ellipsoid.

Now we prove the uniqueness. Assume that K contains two distinct ellipsoids of maximal volume. Up to an affine transformation, we may assume that one of them is precisely the Euclidean unit ball B^n (any affine transformation multiplies all volumes by the same constant, and hence the maximality of the volumes of the ellipsoids is maintained). By the comment below the proof of Lemma 9.1.1, the other maximal ellipsoid can be written as $x_0 + T(B^n)$ where $T \in \mathrm{GL}(n)$ is symmetric and positive-definite (and $x_0 \in K$, of course). Also, since $x_0 + T(B^n)$ has the same volume as B^n, we must have that

$\det(T) = 1$. Further, we claim that

$$E := \frac{x_0}{2} + \frac{T + \mathrm{Id}_{\mathbb{R}^n}}{2} B^n \subseteq K.$$

Indeed, if $z \in B^n$, then $x_0 + Tz \in x_0 + T(B^n) \subseteq K$, and therefore the point

$$\frac{x_0}{2} + \frac{Tz + z}{2} = \frac{1}{2}(x_0 + Tz) + \frac{1}{2}z$$

is a convex combination of points of K, from which it lies in K. Hence E is an ellipsoid contained in K, from where we get

$$\mathrm{vol}_n(B^n) \geq \mathrm{vol}_n(E) = \mathrm{vol}_n\left(\frac{T + \mathrm{Id}_{\mathbb{R}^n}}{2} B^n\right) = \det\left(\frac{T + \mathrm{Id}_{\mathbb{R}^n}}{2}\right) \mathrm{vol}_n(B^n).$$

Consequently, from Lemma 9.1.2, we obtain

$$1 \geq \det\left(\frac{T + \mathrm{Id}_{\mathbb{R}^n}}{2}\right) \geq \det(T)^{1/2} \det(\mathrm{Id}_{\mathbb{R}^n})^{1/2} = 1,$$

and hence the equality case in the mentioned lemma holds. This gives $T = \mathrm{Id}_{\mathbb{R}^n}$. It follows that B^n and $x_0 + B^n$ are ellipsoids of maximal volume contained in K, and we have that $x_0 \neq 0_{\mathbb{R}^n}$ because we are assuming that they are distinct. Thus,

$$L := \mathrm{conv}(B^n \cup (x_0 + B^n)) \subseteq K,$$

and we claim that L contains an ellipsoid with volume greater than $\mathrm{vol}_n(B^n)$. This is illustrated in Figure 9.1.1. Indeed, assume that

$$\left\{\frac{x_0}{|x_0|}, v_2, \ldots, v_n\right\}$$

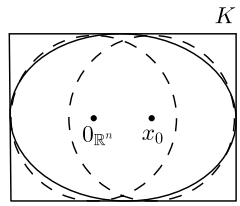


Fig. 9.1.1 The convex hull of the balls contains an ellipsoid of greater volume.

is an orthonormal basis of $\mathbb{R}^n$, and let $R \in \mathrm{GL}(n)$ be the linear transformation such that $R(v_j) = v_j$ for each $j = 2, \ldots, n$, and

$$R\left(\frac{x_0}{|x_0|}\right) = \left(\frac{|x_0|}{2} + 1\right) \frac{x_0}{|x_0|}.$$

Then the ellipsoid $F := x_0/2 + R(B^n)$ is such that $F \subseteq L$, and we explain how to prove this inclusion. A point of F can be written as $x_0/2 + R(z)$, for some $z \in B^n$. We can decompose z in the fixed basis above as

$$z = \alpha_1 \frac{x_0}{|x_0|} + \alpha_2 v_2 + \ldots + \alpha_n v_n,$$

for numbers $-1 \leq \alpha_j \leq 1$ such that $\alpha_1^2 + \ldots + \alpha_n^2 \leq 1$. If $\alpha_1 \geq 0$, then we write

$$\frac{x_0}{2} + R(z) = (1-\lambda)x + \lambda y,$$

where

$$x = -\alpha_1 \frac{x_0}{|x_0|} + \alpha_2 v_2 + \ldots + \alpha_n v_n \in B^n \text{ and}$$
$$y = x_0 + \alpha_1 \frac{x_0}{|x_0|} + \alpha_2 v_2 + \ldots + \alpha_n v_n \in x_0 + B^n,$$

and then

$$\lambda = \frac{\frac{|x_0|}{2}(\alpha_1 + 1) + 2\alpha_1}{|x_0| + 2\alpha_1}.$$

The fact that $0 \leq \lambda \leq 1$ comes from $0 \leq \alpha_1 \leq 1$. Hence $x_0/2 + R(z)$ is a convex combination of points of $B^n \cup (x_0 + B^n)$. If $\alpha_1 < 0$, we replace x, y, and λ by

$$x = \alpha_1 \frac{x_0}{|x_0|} + \alpha_2 v_2 + \ldots + \alpha_n v_n \in B^n,$$
$$y = x_0 - \alpha_1 \frac{x_0}{|x_0|} + \alpha_2 v_2 + \ldots + \alpha_n v_n \in x_0 + B^n \text{ and}$$
$$\lambda = \frac{\frac{|x_0|}{2}(\alpha_1 + 1)}{|x_0| - 2\alpha_1},$$

and the fact that $0 \leq \lambda \leq 1$ comes from $-1 \leq \alpha_1 < 0$. Thus, we have indeed that $F \subseteq L$, and consequently we have that F is an ellipsoid contained in K. Last, observe that

$$\mathrm{vol}_n(F) = \det(R)\mathrm{vol}_n(B^n) = \left(\frac{|x_0|}{2} + 1\right)\mathrm{vol}_n(B) > \mathrm{vol}_n(B).$$

This contradicts the maximality of the volume of B^n among the volumes of the ellipsoids contained in K, and hence the proof is complete.

□

Remark 9.1.6. Notice carefully that to prove the existence part we showed that the limit of any sequence of ellipsoid converging in the Hausdorff metric is a set of the form $C = x_0 + T(B^n)$, with $x_0 \in \mathbb{R}^n$ and $T \in \mathrm{L}(n)$. In the case where T is not invertible we say that C is a *degenerate ellipsoid*.

Now we prove that any convex body $K \subseteq \mathbb{R}^n$ is contained in a unique ellipsoid of minimal volume. This is known as the *Löwner ellipsoid* of K. First, we need a geometrical lemma.

Lemma 9.1.3. *Let $E \subseteq \mathbb{R}^n$ be an ellipsoid which contains a ball of radius $r > 0$ and a segment of length d. Then,*

$$\mathrm{vol}_n(E) \geq \frac{d}{2} r^{n-1} \mathrm{vol}_n(B^n).$$

Proof. Without loss of generality, assume that E is centered at the origin and consider the representation

$$E = \left\{x \in \mathbb{R}^n : \langle x, v_j\rangle^2 \leq \lambda_j^2 \text{ for } j = 1, \ldots, n\right\}$$

as discussed above. For any j, it is readily seen that E is contained in the "slab" of the parallel supporting hyperplanes of E which are normal to v_j. Clearly, the width of this slab is $2\lambda_j$. Since this slab also contains a ball of radius r (which is contained in E), we get that $\lambda_j \geq r$.

Now let $\lambda_M = \max\{\lambda_1, \ldots, \lambda_M\}$. It is easy to see from the representation above that $E \subseteq \lambda_M B^n$. Hence, if E contains a segment of length d, we get that $\lambda_M B^n$ contains a segment of length d. Thus, we have the inequality $\lambda_M \geq d/2$. Recalling the formula for the volume of an ellipsoid as a function of $\lambda_1, \ldots, \lambda_n$, we finally arrive at

$$\mathrm{vol}_n(E) = \lambda_1\lambda_2 \ldots \lambda_M \mathrm{vol}_n(B^n) \geq \frac{d}{2} r^{n-1} \mathrm{vol}_n(B^n)$$

and the proof is complete.

□

Theorem 9.1.3 (Löwner ellipsoid)**.** *Let K be a convex body. There exists a unique ellipsoid $F_0 \supseteq K$ such that*

$$\mathrm{vol}_n(F_0) = \inf\left\{\mathrm{vol}_n(F) \colon K \subseteq F \textit{ and } F \textit{ is an ellipsoid}\right\}.$$

Proof. For the existence, we let (E_j) be a sequence of ellipsoids whose volumes converge to the infimum above, and we choose a ball $x_0 + r_1 B^n$ contained in K (with $r > 0$). Let $c > 0$ be the volume of some ellipsoid which contains K, and fix any $\varepsilon > 0$. Now we choose $r_2 > 0$ such that $K \subseteq x_0 + r_2 B^n$ and

$$\frac{r_2}{2} r_1^{n-1} \mathrm{vol}_n(B^n) > c + \varepsilon.$$

Let E be an ellipsoid containing K which is not contained in $x_0 + r_2 B^n$. Because E is not contained in $x_0 + r_2 B^n$ but contains x_0 (in its interior), we get that E has a segment whose length is greater than r_2 (just connect x_0 to a point of E outside the ball). Since E contains K, it also contains $x_0 + r_1 B^n$. From the previous lemma, we have

$$\mathrm{vol}_n(E) \geq \frac{r_2}{2} r_1^{n-1} \mathrm{vol}_n(B^n),$$

from where we get that

$$\mathrm{vol}_n(E) \geq c + \varepsilon \geq \inf\{\mathrm{vol}_n(F) \colon K \subseteq F \text{ and } F \text{ is an ellipsoid}\} + \varepsilon.$$

Since this holds for any ellipsoid containing K which is not contained in $x_0 + r_2 B^n$, we get that there exists $n_0 \in \mathbb{N}$ such that $E_j \subseteq x_0 + r_2 B^n$ for every $j > n_0$. From Blaschke's selection theorem we guarantee that a subsequence of (E_j) converges, and from Remark 9.1.6, we have that its limit is some ellipsoid F_0. Finally, we notice that F_0 is not degenerate because $K \subseteq F_0$ (this is easy to prove) and this gives $\mathrm{vol}_n(F_0) > 0$.

The proof of the uniqueness of the minimal enclosing ellipsoid is quite different from the proof of the uniqueness of the maximal enclosed ellipsoid, thus we will show it in detail. Assume that there are two distinct ellipsoids F_0 and F_1, both containing K, such that

$$\mathrm{vol}_n(F_1) = \mathrm{vol}_n(F_0) = \inf\left\{\mathrm{vol}_n(F) : K \subseteq F \text{ and } F \text{ is an ellipsoid}\right\}.$$

Now we apply iteratively some affine transformations on $\mathbb{R}^n$ to make calculations easier. Since any affine transformation takes ellipsoids to ellipsoids, preserves inclusion, and multiplies every volume by

the same constant, we stay with two distinct enclosing ellipsoids of minimal volume by doing so. First we consider an affine transformation which carries F_0 to B^n. Next, up to an orthogonal transformation we may consider that the principal axes of F_1 are the vectors of the canonical basis $\{e_1, \ldots, e_n\}$, and we observe carefully that this transformation preserves the ball. Finally, we apply a translation such that F_1 is centered at the origin. Summarizing, now F_0 is a ball centered at some point $x_0 = (\beta_1, \ldots, \beta_n) \in \mathbb{R}^n$, and F_1 is an ellipsoid centered at the origin with canonical principal axes. Then we write in coordinates

$$F_0 = \left\{ (\alpha_1 \ldots, \alpha_n) \in \mathbb{R}^n : \sum_{j=1}^{n} (\alpha_j - \beta_j)^2 \leq 1 \right\} \text{ and}$$

$$F_1 = \left\{ (\alpha_1, \ldots, \alpha_n) \in \mathbb{R}^n : \sum_{j=1}^{n} \frac{\alpha_j^2}{\lambda_j^2} \leq 1 \right\}.$$

Here we get from (9.1.1) that $\lambda_1 \ldots \lambda_n = 1$, because $\mathrm{vol}_n(F_1) = \mathrm{vol}_n(F_0) = \mathrm{vol}_n(B^n)$. We also may assume that each λ_j is positive, of course. Our strategy is to define an "average" ellipsoid which still contains K, but having smaller volume than F_0 and F_1. We do so by defining

$$F := \left\{ (\alpha_1, \ldots, \alpha_n) \in \mathbb{R}^n : \frac{1}{2} \sum_{j=1}^{n} \left[(\alpha_j - \beta_j)^2 + \frac{\alpha_j^2}{\lambda_j^2} \right] \leq 1 \right\}.$$

The fact that $K \subseteq F$ is immediate. Indeed, since $K \subseteq F_0$ and $K \subseteq F_1$, the inequalities defining F_0 and F_1 hold true for any point $(\alpha_1, \ldots, \alpha_n) \in K$, and then the inequality defining F is also verified for this point. To check that F is an ellipsoid, we perform a simple calculation to obtain

$$(\alpha - \beta)^2 + \frac{\alpha^2}{\lambda^2} = \frac{\lambda^2 + 1}{\lambda^2} \left(\alpha - \frac{\beta \lambda^2}{\lambda^2 + 1} \right)^2 + \frac{\beta^2}{\lambda^2 + 1},$$

and we replace it in the inequality which defines F (for each $j = 1, \ldots, n$), getting that $(\alpha_1, \ldots, \alpha_n) \in F$ if and only if

$$\sum_{j=1}^{n} \frac{\lambda_j^2 + 1}{2\lambda_j^2} \left(\alpha_j - \frac{\beta_j \lambda_j^2}{\lambda_j^2 + 1} \right)^2 \leq 1 - \sum_{j=1}^{n} \frac{\beta_j^2}{2(\lambda_j^2 + 1)}.$$

It is clear that the number on the right-hand side of the inequality above is smaller than or equal to 1. Additionally, we have that it is strictly positive, because otherwise F would contain at most one point and we already know that F contains a convex body. Hence, dividing both sides of the inequality above by the number on the right-hand side, we write F as

$$F = \left\{ (\alpha_1, \ldots, \alpha_n) \in \mathbb{R}^n : \sum_{j=1}^{n} \frac{(\alpha_j - \gamma_j)^2}{\xi_j^2} \leq 1 \right\},$$

where

$$\gamma_j := \frac{\beta_j \lambda_j^2}{\lambda_j^2 + 1},$$

$$\xi_j := \frac{\lambda_j \sqrt{2}}{\sqrt{\lambda_j^2 + 1}} \sqrt{\eta} \text{ and}$$

$$\eta := 1 - \sum_{j=1}^{n} \frac{\beta_j^2}{2(\lambda_j + 1)}.$$

The inequality $(\lambda_j - 1)^2 \geq 0$ leads immediately to $\lambda_j^2 + 1 \geq 2\lambda_j$, with equality if and only if $\lambda_j = 1$. This can be written as

$$\frac{\lambda_j \sqrt{2}}{\sqrt{\lambda_j^2 + 1}} \leq \sqrt{\lambda_j}, \tag{9.1.2}$$

with equality if and only if $\lambda_j = 1$, and this holds for each $j = 1, \ldots, n$. Consequently, recalling also that $\eta \leq 1$, we have the inequality

$$\text{vol}_n(F) = \text{vol}_n(B^n) \prod_{j=1}^{n} \xi_j \leq \text{vol}_n(B^n) \eta^{n/2} \prod_{j=1}^{n} \frac{\lambda_j \sqrt{2}}{\sqrt{\lambda_j^2 + 1}} \leq$$

$$\leq \text{vol}_n(B^n) \prod_{j=1}^{n} \sqrt{\lambda_j} = \text{vol}_n(B^n) = \text{vol}_n(F_0),$$

where we also remember that $\lambda_1 \ldots \lambda_n = 1$. Since F_0 has minimal volume among the ellipsoids which enclose K, we must have equality in the above. This means that

(i) $\eta = 1$, since otherwise we would have $\eta^{n/2} < 1$, and the inequality would be strict. This leads to $\beta_j = 0$ for each $j = 1, \ldots, n$ and, returning to the definition of F_0 in coordinates, we get that F_0 is centered at the origin. Hence $F_0 = B^n$.

(ii) We must have

$$\prod_{j=1}^{n} \frac{\lambda_j \sqrt{2}}{\sqrt{\lambda_j^2 + 1}} = \prod_{j=1}^{n} \sqrt{\lambda_j}.$$

From (9.1.2), the j-th term of the left-hand side product is smaller than or equal to the j-th term of the right-hand sided product, and hence we get that

$$\frac{\lambda_j \sqrt{2}}{\sqrt{\lambda_j^2 + 1}} = \sqrt{\lambda_j}$$

for each $j = 1, \ldots, n$. Consequently, we get that $\lambda_1 = \lambda_2 = \ldots = \lambda_n = 1$, and it follows from the expression of F_1 in coordinates that $F_1 = B^n$. This contradicts the fact that $F_0 \neq F_1$, and so we are done.

□

9.2 John's theorems for the centrally symmetric case

Our main objective in this section is to characterize the John ellipsoid of a centrally symmetric convex body, and we do that by the so-called John's theorems. We say that a convex body is in *John position* if its John ellipsoid is the Euclidean unit ball B^n, and in *Löwner position* if its Löwner ellipsoid is a multiple of B^n.

These definitions make sense because the image $A(E)$ of the John (Löwner) ellipsoid E of K by an affine transformation A is the John (Löwner) ellipsoid of $A(K)$. We recall again that an affine transformation preserves inclusion, takes ellipsoids onto ellipsoids, and multiplies every volume by the same constant. In other words, if $E = x_0 + T(B^n)$ is the John ellipsoid of K, then we have

$$B^n = T^{-1}(E) - T^{-1}(x_0) \subseteq T^{-1}(K) - T^{-1}(x_0),$$

and B^n is the John ellipsoid of the position $T^{-1}(K) - T^{-1}(x_0)$ of K. Moreover, notice that the John position of a convex body is unique up to orthogonal transformations: if K is in John position, then $T(K) + x_1$ is in John position if and only if $x_1 = 0_{\mathbb{R}^n}$ and $T \in \mathrm{O}(n)$. Similarly, the Löwner position is unique up to orthogonal transformations and scalar multiplications.

Lemma 9.2.1. *The John ellipsoid of a centrally symmetric convex body $K \subseteq \mathbb{R}^n$ has the same center as K.*

Proof. It is clearly enough to assume that K is centered at the origin, because if E is the John ellipsoid of $K - x_0$, then $E + x_0$ is the John ellipsoid of K. Let $E = x_0 + T(B^n)$ be any ellipsoid contained in K (and not necessarily centered at the origin). The convex body

$$F := \frac{1}{2}(E - E) = \frac{1}{2}(x_0 + T(B^n) - x_0 - T(B^n)) = \frac{1}{2}T(B^n - B^n) = T(B^n)$$

is clearly an ellipsoid centered at the origin having the same volume as E. Since K is centered at the origin and $E \subseteq K$, we get that $-E \subseteq K$, and this leads to $F \subseteq K$. This construction shows that each ellipsoid $E \subseteq K$ corresponds to an ellipsoid $F \subseteq K$ centered at the origin having the same volume as E. It follows that the ellipsoid of maximal volume contained in K (which we know to be unique) is centered at the origin.

□

Lemma 9.2.2. *If $K \subseteq \mathbb{R}^n$ is a convex body centered at the origin, then the Löwner ellipsoid of K is centered at the origin.*

Proof. Assume that the Löwner ellipsoid F of K is not centered at the origin. Then $F \neq -F$, and we claim that $-F$ is also an ellipsoid containing K. Indeed, since $K \subseteq F$, we have that $-K \subseteq -F$. But since K is centered at the origin, we have that $K = -K$, from where it follows that $K \subseteq -F$. Since $\mathrm{vol}_n(F) = \mathrm{vol}_n(-F)$ this contradicts the uniqueness of the enclosing ellipsoid of minimal volume.

□

Next we show that for bodies centered at the origin the John and Löwner ellipsoids are dual to each other, in some sense.

Proposition 9.2.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body centered at the origin. Then $E \subseteq K$ is the John ellipsoid of K if and only if E° is the Löwner ellipsoid of K°.*

Proof. Let E be the John ellipsoid of K. Since polarity reverses inclusion, we have that $K^\circ \subseteq E^\circ$. From Lemma 9.2.1 we have that E is centered at the origin, and hence we can write $E = T(B^n)$ for

some $T \in \mathrm{GL}(n)$. Now Lemma 5.4.1 gives that E° is also an ellipsoid centered at the origin, because $E^\circ = (T(B^n))^\circ = (T^*)^{-1}(B^n)$.

Summarizing, we already know that E° is an ellipsoid centered at the origin which contains K°. It remains to prove that it has minimal volume. Assume that F is the Löwner ellipsoid of K°, and notice that F is centered at the origin because K° is centered at the origin. If $F \neq E^\circ$, then uniqueness of the Löwner ellipsoid gives that $\mathrm{vol}_n(F) < \mathrm{vol}_n(E^\circ)$. On the other hand, using Lemma 5.4.1 again we get

$$\mathrm{vol}_n(F)\mathrm{vol}_n(F^\circ) = \mathrm{vol}_n(B^n)^2 = \mathrm{vol}_n(E)\mathrm{vol}_n(E^\circ),$$

from where it follows that $\mathrm{vol}_n(F^\circ) > \mathrm{vol}_n(E)$. But F° is an ellipsoid contained in K (because $F^\circ \subseteq (K^\circ)^\circ = K$), and this contradicts the fact that E is the John ellipsoid of K. The proof of the reverse implication is completely analogous, and we will omit it.

□

A *contact point* (or *touching point*) of two convex bodies $K, L \subseteq \mathbb{R}^n$ is a point $x \in \partial K \cap \partial L$ with the property that K and L are supported by the same hyperplane at x. If $K \subseteq L$, then any point $x \in \partial K \cap \partial L$ is a contact point between K and L, because any hyperplane which supports L at x also supports K at x.

Let $K \subseteq \mathbb{R}^n$ be a convex body centered at the origin which is in John position. Then the intersection $\mathbb{S}^{n-1} \cap \partial K$ must be non-empty, otherwise we would have that $0 < \mathrm{dist}(\partial K, \mathbb{S}^{n-1}) =: 2\delta$ from where

$$(1+\delta)B^n \subseteq K,$$

and this would contradict the fact that B^n is the ellipsoid of maximum volume contained in K. By the previous comment, the points in $\mathbb{S}^{n-1} \cap \partial K$ are contact points between K and B^n. Even more, we can describe the common supporting hyperplane of K and B^n at a contact point: if $x \in \mathbb{S}^{n-1} \cap \partial K$ and $\mathbf{h}$ is a supporting hyperplane of K at x, then since $B^n \subseteq K$ we have that $\mathbf{h}$ is also a supporting hyperplane of B^n at x, from where it follows that $\mathbf{h} = \mathbf{h}(x, 1) = x^\perp + x$ (observe that this implies, in particular, that the John ellipsoid of a symmetric convex body only touches the boundary at smooth points).

A consequence, which comes from Proposition 2.3.1, is that $h_K(x) = 1$. By Proposition 2.3.6, we have $\|\cdot\|_{K^\circ} = h_K(\cdot)$, and hence $\|x\|_{K^\circ} = 1$. This means that if x is a contact point of K and B^n, then $x \in \partial K^\circ$. It follows that x is also a contact point between K° and its Löwner ellipsoid.

In what follows we state and prove a theorem due to Fritz John which characterizes how the contact points of a convex body in John position and its John ellipsoid are distributed. We give a proof which involves some machinery from the theory of Lie groups (namely, regarding certain subgroups of $\mathrm{L}(n)$ as differentiable manifolds). To do so, we introduce now several new concepts. For the convenience of the reader, we will also provide an alternative proof of one of the implications of the theorem where the tools from Lie group theory are not necessary. However, the other concepts we introduce now then are still needed.

From now on, always assume that each linear map in $\mathrm{L}(n)$ is naturally identified with its matrix representation in the canonical basis of $\mathbb{R}^n$, yielding a natural linear isomorphism between $\mathrm{L}(n)$ and the set $\mathrm{M}(n)$ of $n \times n$ matrices with entries from $\mathbb{R}$. By doing so, the vector subspace $\mathrm{Adj}(n) \subseteq \mathrm{L}(n)$ of self-adjoint linear maps is naturally identified with the set $\mathrm{Sym}(n)$ of symmetric $n \times n$ matrices. Since we can "choose" $\frac{1}{2}n(n+1)$ entries for a symmetric matrix, it follows that

$$\dim(\mathrm{Sym}(n)) = \frac{1}{2}n(n+1).$$

Thus, from now on we always keep in mind that $\mathrm{Sym}(n)$ is a vector space with dimension $\frac{1}{2}n(n+1)$. We consider that the n^2-dimensional vector space $\mathrm{M}(n)$ is endowed with the *matrix inner product*

$$\left\langle [a_{ij}]_{i,j=1}^n, [b_{ij}]_{i,j=1}^n \right\rangle_M := \sum_{i,j=1}^n a_{ij} b_{ij},$$

which is also naturally induced on $\mathrm{Sym}(n)$. The topology defined on $\mathrm{M}(n)$ by the matrix inner product is the same topology induced by the norm of $\mathrm{L}(n)$ under the isomorphism $\mathrm{L}(n) \simeq \mathrm{M}(n)$. In fact, to verify this one just needs to remember that any norm in a finite-dimensional normed space gives rise to the same topology. Notice also that the trace of a matrix M can be simply described as the matrix inner product of M with $\mathrm{Id}_{\mathbb{R}^n}$.

A matrix $M \in \mathrm{M}(n)$ is *positive-definite* if it is identified with a positive-definite linear map in $\mathrm{L}(n)$. We denote the set of symmetric and positive-definite linear matrices by $\mathrm{Sym}_+(n)$.

Lemma 9.2.3. *The set* $\mathrm{Sym}_+(n)$ *is an open convex cone in* $\mathrm{Sym}(n)$ *whose apex is the origin.*

Proof. If S, T are positive-definite matrices and $\lambda \in (0,1)$, then

$$\langle ((1-\lambda)S + \lambda T)x, x\rangle = (1-\lambda)\langle Sx, x\rangle + \lambda \langle Tx, x\rangle > 0$$

for every non-zero vector $x \in \mathbb{R}^n$. Hence the set of positive-definite matrices is convex. On the other hand, $\mathrm{Sym}(n)$ is also convex, because it is a vector subspace. It follows that $\mathrm{Sym}_+(n)$ is the intersection of convex subsets of $\mathrm{M}(n)$, and hence it is convex.

To prove that $\mathrm{Sym}_+(n)$ is open, it is sufficient to prove that the set of positive-definite matrices is open in $\mathrm{M}(n)$. For that, it is enough to prove that the set of positive-definite linear maps is open in $\mathrm{L}(n)$. Let $P \in \mathrm{L}(n)$ be positive-definite, and let

$$c_0 := \inf \left\{ \langle Px, x\rangle : x \in \mathbb{S}^{n-1} \right\}.$$

Observe that $c_0 > 0$ because $\mathbb{S}^{n-1}$ is compact and P is positive-definite. For any $0 < \varepsilon < c_0$, if $||T - P||_{\mathrm{L}(n)} < c_0 - \varepsilon$, then

$$\langle Tx, x\rangle = \langle Px, x\rangle - \langle (T-P)x, x\rangle \geq c_0 - |(T-P)x|\,|x| \geq c_0 - ||T-P||_{\mathrm{L}(n)} > \\ > \varepsilon > 0$$

for each $x \in \mathbb{S}^{n-1}$. Notice that we have used the Cauchy–Schwarz inequality to write $\langle (T-P)x, x\rangle \geq -|(T-P)x|\,|x|$. It follows that all linear maps in the open ball

$$\left\{ T \in \mathrm{L}(n) : ||T-P||_{\mathrm{L}(n)} < c_0 - \varepsilon \right\}$$

are positive-definite.

The fact that $\mathrm{Sym}_+(n)$ is a cone with apex in the origin comes immediately from the fact that if $P \in \mathrm{Sym}_+(n)$ and $\lambda > 0$, then $\lambda P \in \mathrm{Sym}_+(n)$.

□

Now we look at the set of positive-definite symmetric matrices with determinant equal to or greater than 1.

Proposition 9.2.2. *The set*

$$\mathcal{D} := \{T \in \mathrm{Sym}_+(n) : \det(T) \geq 1\}$$

is a smooth and strictly convex closed set with non-empty interior.

Proof. Strict convexity of $\mathcal{D}$ comes immediately from Lemma 9.1.2. To prove that $\mathcal{D}$ is smooth, observe that its boundary is $\partial\mathcal{D} = \det^{-1}(1)$, and 1 is a regular value of the differentiable map

$\det\colon \mathrm{Sym}_+(n) \to \mathbb{R}$. Consequently, $\partial\mathcal{D}$ is (locally) a parametrized hypersurface, thus smooth. The continuity of the determinant gives that $\mathcal{D}$ is closed and has non-empty interior.

□

Remark 9.2.1. The reader who is not familiar with differential geometry may find the proof of smoothness of $\mathcal{D}$ quite confusing. What we use is a consequence of the classical implicit function theorem, and we clarify the ideas now. If $f\colon X \to Y$ is a differentiable map between smooth manifolds, then we say that $y \in Y$ is a *regular value* of f if the differential df_x is surjective for every $x \in f^{-1}(y)$. For any regular value $y \in Y$ of f, the pre-image $f^{-1}(y)$ is a submanifold of X whose codimension is the same as the dimension of Y. Being an open subset of a vector space, the set $\mathrm{Sym}_+(n)$ is a smooth manifold, and the map $\det\colon \mathrm{Sym}_+(n) \to \mathbb{R}$ is a differentiable map between $\mathrm{Sym}_+(n)$ and $\mathbb{R}$ (because it is polynomial in the variables). Since it is easy to check that 1 is a regular value of the determinant, we get that $\det^{-1}(1)$ is a smooth hypersurface. The "differential geometry" smoothness implies the "convex geometry" smoothness: a convex region bounded by a smooth hypersurface is supported by a unique hyperplane at each boundary point (which is the *tangent space* of the hypersurface at the point, in the language of differential geometry). There are several ways to prove this claim. One of them is to identify (locally) the bounding hypersurface as the graph of a convex differentiable function and to use the theory developed in Section 1.3.

Remark 9.2.2. The tangent space of $\det^{-1}(1)$ at $\mathrm{Id}_{\mathbb{R}^n}$ is the so-called *Lie algebra* of the special linear group $\mathrm{SL}(n)$. It is a well-known fact that every element of the Lie algebra of $\mathrm{SL}(n)$ has trace equal to 0, and hence we get that $\mathrm{Id}_{\mathbb{R}^n}$ is an (inner) normal of $\mathcal{D}$ at $\mathrm{Id}_{\mathbb{R}^n}$. This comes immediately from the fact that $\mathrm{tr}(T) = \langle \mathrm{Id}_{\mathbb{R}^n}, T\rangle_M$.

For any vectors $u, v \in \mathbb{R}^n$ we define the linear map $u \otimes v\colon \mathbb{R}^n \to \mathrm{span}\{u\}$ by

$$u \otimes v(x) = \langle x, v\rangle u$$

for each $x \in \mathbb{R}^n$. Under the adopted canonical isomorphism between $\mathrm{L}(n)$ and $\mathrm{M}(n)$, the map $u \otimes v$ is clearly identified with the matrix $u \cdot v^T$, where the vectors are represented as column matrices, as usual.

Observe that if $u \in \mathbb{S}^{n-1}$, then $u \otimes u\colon \mathbb{R}^n \to \mathrm{span}\{u\}$ is just the projection onto u:

$$u \otimes u(z) = \langle z, u\rangle u,$$

for all $z \in \mathbb{R}^n$. It is easy to see that $u \otimes u$ is simply the linear transformation which carries u to $|u|^2 u$, and whose kernel is $u^\perp$.

Lemma 9.2.4. *Let $u, v \in \mathbb{R}^n$ and $T \in \mathrm{M}(n)$. Then*

$$\langle Tu, v\rangle = \langle T, v \otimes u\rangle_M.$$

As a consequence, if $T \in \mathrm{Sym}(n)$, then we have

$$\langle Tu, v\rangle = \langle T, u \otimes v\rangle_M.$$

Proof. This can be proved simply by writing everything in coordinates, but we can avoid some calculations by observing that, since both sides in the equality are linear in u and v (and also in T), it is sufficient to consider the case where $u = e_i$ and $v = e_k$ for some $i, k = 1, \ldots, n$. If $T = [t_{ij}]_{i,j=1}^n$, then

$$\langle Tu, v\rangle = \langle (t_{1i}, t_{2i}, \ldots, t_{ni}), e_k\rangle = t_{ki}.$$

On the other hand, a direct calculation shows that the matrix $e_k \cdot e_i^T$ which represents $v \otimes u$ has 0 in all entries, except for the entry in the position ki, which is 1. Hence, by the definition of matrix inner

product, we have

$$\langle T, v \otimes v \rangle_M = t_{ki} = \langle Tu, v \rangle.$$

To prove the other equality, simply notice that if T is symmetric, then

$$\langle Tu, v \rangle = \langle Tv, u \rangle = \langle T, u \otimes v \rangle_M,$$

for any $u, v \in \mathbb{R}^n$.

□

Theorem 9.2.1 (John). *A convex body $K \subseteq \mathbb{R}^n$ centered at the origin which contains B^n is in John position if and only if there exist an integer $n \leq m \leq \frac{1}{2}n(n+1)$, points $x_1, \ldots, x_m \in \mathbb{S}^{n-1} \cap \partial K$, and positive numbers $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$ such that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} \lambda_j x_j \otimes x_j. \tag{9.2.1}$$

Remark 9.2.3. The equality in (9.2.1) is called a *decomposition of the identity*. In particular, in this case we have

$$\sum_{j=1}^{m} \lambda_j = n. \tag{9.2.2}$$

Indeed, recalling that $\{e_1, \ldots, e_n\}$ denotes the canonical basis of $\mathbb{R}^n$, we write

$$n = \sum_{i=1}^{n} |e_i|^2 = \sum_{i=1}^{n} \langle \mathrm{Id}_{\mathbb{R}^n}(e_i), e_i \rangle = \sum_{i=1}^{n} \left\langle \sum_{j=1}^{m} \lambda_j x_j \otimes x_j(e_i), e_i \right\rangle =$$
$$= \sum_{i=1}^{n} \sum_{j=1}^{m} \lambda_j \langle x_j, e_i \rangle^2 = \sum_{j=1}^{m} \lambda_j \sum_{i=1}^{n} \langle x_j, e_i \rangle^2 = \sum_{j=1}^{m} \lambda_j |x_j|^2 = \sum_{j=1}^{m} \lambda_j,$$

where the last equality comes from the fact that $x_j \in \mathbb{S}^{n-1}$ for each $j = 1, \ldots, m$.

Proof. We first assume that K is in John position. Any ellipsoid centered at the origin is the image of the Euclidean unit ball under some symmetric and positive-definite linear map. We will describe the ellipsoids centered at the origin which are contained in K in terms of these linear maps. Let T be a symmetric and positive-definite linear map, and $E = T(B^n)$. Then, from Proposition 2.3.2, we get that $E \subseteq K$ if and only if

$$h_E(v) \leq h_K(v),$$

for all $v \in \mathbb{S}^{n-1}$. On the other hand, we have the equality

$$h_E(v) = h_{T(B^n)}(v) = \max\left\{ \langle Tu, v \rangle : u \in \mathbb{S}^{n-1} \right\},$$

for every $v \in \mathbb{S}^{n-1}$. It follows that

$$h_E(v) \leq h_K(v) \text{ for all } v \in \mathbb{S}^{n-1} \iff \langle Tu, v \rangle \leq h_K(v) \text{ for all } u, v \in \mathbb{S}^{n-1}.$$

Finally, from Lemma 9.2.4 we get that $E \subseteq K$ if and only if

$$\langle T, u \otimes v \rangle_M \leq h_K(v)$$

for any $u, v \in \mathbb{S}^{n-1}$. Consequently, the collection of ellipsoids centered at the origin and contained in K can be described by the set

$$\mathcal{E} := \left\{ T \in \mathrm{Sym}_+(n) : \langle T, u \otimes v \rangle_M \leq h_K(v) \text{ for any } u, v \in \mathbb{S}^{n-1} \right\}$$

in the following sense: if $T \in \mathrm{Sym}_+(n)$ is such that $T(B^n) \subseteq K$, then $T \in \mathcal{E}$ and, conversely, if $E \subseteq K$ is an ellipsoid centered at the origin, then there exists $T \in \mathcal{E}$ such that $E = T(B^n)$. Next we collect some information about $\mathcal{E}$.

(1) $\mathcal{E}$ is non-empty and convex. Indeed, since K is in John position we get that $\mathrm{Id}_{\mathbb{R}^n} \in \mathcal{E}$. If $S, T \in \mathcal{E}$ and $\lambda \in (0,1)$, then for each $z \in B^n$ we have $Tz \in K$ and $Sz \in K$. By the convexity of K, we obtain that $(1-\lambda)Tz + \lambda Sz \in K$. Thus $((1-\lambda)T + \lambda S)(B^n) \subseteq K$, and this leads to $(1-\lambda)T + \lambda S \in \mathcal{E}$ (here we are also using the fact that $\mathrm{Sym}_+(n)$ is convex).

(2) $\det(T) \leq 1$ for every $T \in \mathcal{E}$, with equality if and only if $T = \mathrm{Id}_{\mathbb{R}^n}$. This comes immediately from K being in John position: B^n is the unique ellipsoid of maximal volume contained in K. Any map $T \in \mathcal{E}$ with determinant greater than 1 would yield an ellipsoid $T(B^n)$ contained in K with volume greater than $\mathrm{vol}_n(B^n)$, contradicting maximality. Further, the existence of a map $T \in \mathcal{E}$ different from $\mathrm{Id}_{\mathbb{R}^n}$ and with determinant 1 would contradict uniqueness of the John ellipsoid.

(3) $\mathcal{E}$ is an intersection of half-spaces in $\mathrm{M}(n)$, up to intersection with $\mathrm{Sym}_+(n)$. Indeed, we may write

$$\mathcal{E} = \mathrm{Sym}_+(n) \cap \left(\bigcap_{u,v \in \mathbb{S}^{n-1}} \mathbf{h}_M^-(u \otimes v, h_K(v)) \right), \tag{9.2.3}$$

where we are denoting by $\mathbf{h}_M$ the hyperplanes of $\mathrm{M}(n)$. Notice that this gives another proof of the convexity of $\mathcal{E}$.

It follows from **(2)** that $\mathcal{D} \cap \mathcal{E} = \{\mathrm{Id}_{\mathbb{R}^n}\}$. Since both sets are closed and convex subsets of $\mathrm{M}(n)$, they have a common supporting hyperplane at $\mathrm{Id}_{\mathbb{R}^n}$ (see Remark 2.2.5), which is precisely the Lie algebra of $\mathrm{SL}(n)$. This gives that $\mathrm{Id}_{\mathbb{R}^n}$ lies in the normal cone of $\mathcal{E}$ at $\mathrm{Id}_{\mathbb{R}^n}$ (see Remark 9.2.2). Now we proceed to describe this normal cone, which we will denote by $\mathrm{NC}_{\mathcal{E}}(\mathrm{Id}_{\mathbb{R}^n})$, and we divide this task into steps.

(i) We prove that $\mathrm{Id}_{\mathbb{R}^n} \in \mathbf{h}_M(u \otimes v, h_K(v))$ for some $u, v \in \mathbb{S}^{n-1}$ if and only if $u = v$ and $u \in \mathbb{S}^{n-1} \cap \partial K$. First, if $\mathrm{Id}_{\mathbb{R}^n} \in \mathbf{h}_M(u \otimes v, h_K(v))$ for some $u, v \in \mathbb{S}^{n-1}$, then recalling that $\mathbb{S}^{n-1} \subseteq K$ we have

$$h_K(v) \geq \langle u, v \rangle = \langle \mathrm{Id}_{\mathbb{R}^n} u, v \rangle = \langle \mathrm{Id}_{\mathbb{R}^n}, u \otimes v \rangle_M = h_K(v),$$

from where we get that $h_K(v) = \langle u, v \rangle$. This gives, in particular, that $u \in \partial K$. Further, since $B^n \subseteq K$, we have that $\langle u, v \rangle = h_K(v) \geq h_{B^n}(v) = 1$. Hence, we get from the Cauchy–Schwarz inequality that $u = v$.

Now let $u \in \mathbb{S}^{n-1} \cap \partial K$. Since $u \in K$, we have that $h_K(u) \geq 1$. Then

$$\langle \mathrm{Id}_{\mathbb{R}^n}, u \otimes u \rangle_M = \langle u, u \rangle = 1 \leq h_K(u),$$

from where it follows that $u \in \mathbf{h}_M(u \otimes u, h_K(u))$.

(ii) The support cone of $\mathcal{E}$ at $\mathrm{Id}_{\mathbb{R}^n}$ is given by

$$\mathrm{SC}_{\mathcal{E}}(\mathrm{Id}_{\mathbb{R}^n}) = \bigcap_{u\in\mathbb{S}^{n-1}\cap\partial K} \mathbf{h}^-_M(u\otimes u, h_K(u)).$$

To prove the inclusion "$\subseteq$", let $\lambda \geq 0$ and $T \in \mathcal{E}$. For any $u \in \mathbb{S}^{n-1}\cap\partial K$ we have that $\langle \mathrm{Id}_{\mathbb{R}^n}, u\otimes u\rangle_M = 1 = h_K(u)$, and $\langle T, u\otimes u\rangle_M \leq h_K(u)$. Thus,

$$\langle \lambda(T-\mathrm{Id}_{\mathbb{R}^n}), u\otimes u\rangle_M = \lambda\Big(\langle T, u\otimes u\rangle_M - h_K(u)\Big) \leq 0,$$

from where $\lambda(T-\mathrm{Id}_{\mathbb{R}^n}) \in \mathbf{h}^-_M(u\otimes u, h_K(u))$. The reverse inclusion comes from (9.2.3): should $\mathcal{E}$ be supported at $\mathrm{Id}_{\mathbb{R}^n}$ by half-spaces other than that described in (9.2.3), then these half-spaces would have to appear in the intersection. This, together with the fact that $\mathrm{Sym}_+(n)$ is open (hence intersecting with $\mathrm{Sym}_+(n)$ does not "produce new supporting half-spaces"), gives the desired.

(iii) Step **(ii)** and Proposition 2.4.5 give immediately that

$$\mathrm{NC}_{\mathcal{E}}(\mathrm{Id}_{\mathbb{R}^n}) = \mathrm{pos}\{x\otimes x : x\in\mathbb{S}^{n-1}\cap\partial K\},$$

where we observe carefully that the set on the right-hand side is closed, since $\mathbb{S}^{n-1}\cap\partial K$ is compact. Consequently, since $\mathrm{Id}_{\mathbb{R}^n}$ lies in $\mathrm{NC}_{\mathcal{E}}(\mathrm{Id}_{\mathbb{R}^n})$, we get that it can be written as a combination

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^m \lambda_j x_j\otimes x_j,$$

with $\lambda_j > 0$ and $x_j \in \mathbb{S}^{n-1}\cap\partial K$ for each $j = 1,\dots,m$. The fact that we can take $m \leq \frac{1}{2}n(n+1)$ comes from Theorem 2.1.2. Indeed, since we are working in a vector space whose dimension is $\frac{1}{2}n(n+1)$, there are at most $\frac{1}{2}n(n+1)+1$ affinely independent points, and hence any point in $\mathrm{NC}_{\mathcal{E}}(\mathrm{Id}_{\mathbb{R}^n})$ can be written as a convex combination of $\frac{1}{2}n(n+1)+1$ points. Finally, due to this set being a convex cone one of the points can be discarded.

It remains to show that $m \geq n$. To do so, it is enough to prove that $\{x_1,\dots,x_m\}$ spans $\mathbb{R}^n$. If this is not the case, then we can choose a vector $x \in \mathbb{S}^{n-1}$ which is orthogonal to each x_j. This yields

$$1 = |x|^2 = \langle x, x\rangle = \left\langle \sum_{j=1}^m \lambda_j x_j\otimes x_j(x), x\right\rangle = \langle 0_{\mathbb{R}^n}, x\rangle = 0,$$

a contradiction. Notice very carefully that, in particular, this gives that, regrading boundaries, the intersection of a symmetric convex body in $\mathbb{R}^n$ with its John ellipsoid contains at least $2n$ points (that is, at least n linearly independent points and their respective opposite points).

This settles one of the directions of the theorem. To prove the converse, our first observation is: if there exists a decomposition of the identity as in (9.2.1), then each point x_j is a contact point between B^n and K (see the comment below Proposition 9.2.1). This happens because $x_j \in \mathbb{S}^{n-1}\cap\partial K$ and $B^n \subseteq K$. In particular, we have that K is supported by $x_j^\perp$ at x_j for each $j = 1,\dots,m$. Having this in mind, we prove that $\mathrm{Id}_{\mathbb{R}^n}$ is a boundary point of $\mathcal{E}$. Indeed, given an arbitrary $\varepsilon > 0$, let

$$T_\varepsilon := (\lambda_1+\varepsilon)x_1\otimes x_1 + \sum_{j=2}^m \lambda_j x_j\otimes x_j = \mathrm{Id}_{\mathbb{R}^n} + \varepsilon x_1\otimes x_1.$$

Then T_ε is clearly symmetric and positive-definite, and

$$T_\varepsilon(x_1) = x_1 + \varepsilon\langle x_1, x_1\rangle x_1 = (1+\varepsilon)x_1,$$

where we recall that $x_1 \in \mathbb{S}^{n-1}$. Now observe that $(1+\varepsilon)x_1 \notin K$, because x_1 is a boundary point where K is supported by $x_1^\perp$. This gives that $T_\varepsilon \notin \mathcal{E}$. On the other hand,

$$||T_\varepsilon - \mathrm{Id}_{\mathbb{R}^n}||_{\mathrm{L}(n)} = \varepsilon||x_1 \otimes x_1||_{\mathrm{L}(n)} = \varepsilon \max\{|\langle z, x_1\rangle x_1| : z \in B^n\} = \varepsilon,$$

from where we get that every neighborhood of $\mathrm{Id}_{\mathbb{R}^n}$ in $\mathrm{Sym}_+(n)$ has points outside $\mathcal{E}$. Since $\mathrm{Id}_{\mathbb{R}^n} \in \mathcal{E}$, we get that $\mathrm{Id}_{\mathbb{R}^n} \in \partial\mathcal{E}$.

Now we claim that $\mathrm{Id}_{\mathbb{R}^n} \in \mathrm{NC}_{\mathcal{E}}(\mathrm{Id}_{\mathbb{R}^n})$. In fact, the existence of the decomposition (9.2.1) gives

$$\mathrm{Id}_{\mathbb{R}^n} \in \mathrm{pos}\big\{x \otimes x : x \in \mathbb{S}^{n-1} \cap \partial K\big\},$$

and $\mathcal{E}$ is supported at $\mathrm{Id}_{\mathbb{R}^n}$ by hyperplanes of the form $\mathbf{h}_M(x \otimes x, h_K(x))$ with $x \in \mathbb{S}^{n-1} \cap \partial K$ (we can simply repeat the previous proof of this fact; we did not need to regard B^n as the John ellipsoid of K). Hence we have our claim (Figure 9.2.1).

It follows that the hyperplane through $\mathrm{Id}_{\mathbb{R}^n}$ and normal to $\mathrm{Id}_{\mathbb{R}^n}$ (which is the Lie algebra of $\mathrm{SL}(n)$ at $\mathrm{Id}_{\mathbb{R}^n}$) separates $\mathcal{E}$ and $\mathcal{D}$. Consequently, $\det(T) \leq 1$ for every $T \in \mathcal{E}$, and hence B^n is the ellipsoid of maximal volume in K.

□

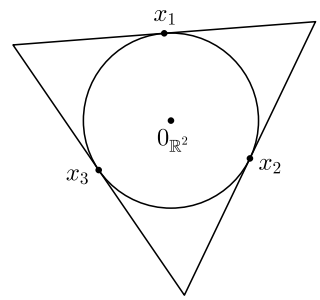


Fig. 9.2.1 Using Theorem 9.2.1 it is not difficult to prove that an equilateral triangle whose barycenter is the origin (and which is properly scaled) is in John position.

As it was previously announced, we give now a proof of the first implication of Theorem 9.2.1 (**up to admitting a weaker upper bound for the maximum *m***) which does not involve the Lie groups machinery (that is, in this proof we do not need to regard $\mathrm{SL}(n)$ as a hypersurface of $\mathrm{M}(n)$). This is heavily based on the proof given in [133].

Alternative proof of the first implication of Theorem 9.2.1. Our hypothesis is that K is a convex body centered at the origin and in John position. We already know that the set of points in the intersection $\mathbb{S}^{n-1} \cap \partial K$ is non-empty (the argument is given below Proposition 9.2.1). We claim that the set

$$C := \{x \otimes x : x \in \mathbb{S}^{n-1} \cap \partial K\} \subseteq \mathrm{Sym}(n)$$

is compact (with the topology of $\mathrm{L}(n)$, of course). Notice that C is the image of the (compact) set $\mathbb{S}^{n-1} \cap \partial K$ under the map

$$\mathbb{R}^n \ni x \mapsto x \otimes x \in \mathrm{L}(n),$$

and hence it is sufficient to prove that this map is continuous. For that, we first write

$$||x \otimes x - y \otimes y||_{\mathrm{L}(n)} = \max\{|x \otimes x(u) - y \otimes y(u)| : u \in \mathbb{S}^{n-1}\} = \\ = \max\{|\langle u, x\rangle x - \langle u, y\rangle y| : u \in \mathbb{S}^{n-1}\},$$

for any $x, y \in \mathbb{R}^n$. Now for any $u \in \mathbb{S}^{n-1}$ we have the estimate

$$|\langle u, x\rangle x - \langle u, y\rangle y| \leq |(\langle u, x\rangle - \langle u, y\rangle)x| + |\langle u, y\rangle(x - y)| = \\ = |\langle u, x - y\rangle||x| + |\langle u, y\rangle||x - y| \leq |u||x - y||x| + |u||y||x - y| = |x - y|(|x| + |y|),$$

and hence

$$||x \otimes x - y \otimes y||_{\mathrm{L}(n)} \leq |x - y|(|x| + |y|).$$

This gives immediately the continuity at $x = 0_{\mathbb{R}^n}$, because in this case the inequality above reads $||y \otimes y||_{\mathrm{L}(n)} \leq |y|^2$. If $x \neq 0_{\mathbb{R}^n}$, then for an arbitrarily given number $\varepsilon > 0$ we choose a number $0 < \varepsilon_1 < \min\{\varepsilon, 3|x|^2\}$. If $|x - y| < \varepsilon_1/3|x|$, then

$$|x| + |y| \leq |x| + |x| + \frac{\varepsilon_1}{3|x|} \leq 3|x| \text{ and, consequently,}$$
$$||x \otimes x - y \otimes y||_{\mathrm{L}(n)} \leq |x - y|(|x| + |y|) \leq \frac{\varepsilon_1}{3|x|}(3|x|) = \varepsilon_1 < \varepsilon,$$

and then we have continuity at x.

Now that we have settled the convexity of C, it is clear that $\mathrm{conv}(C)$ is also compact (see Theorem 2.1.3). In view of Remark 9.2.3, proving that there exists a decomposition of the identity as in (9.2.1) is the same as proving that

$$\frac{\mathrm{Id}_{\mathbb{R}^n}}{n} \in \mathrm{conv}(C).$$

We proceed by contradiction, assuming that $\mathrm{Id}/n \notin \mathrm{conv}(C)$, where we adopted the easier notation $\mathrm{Id} := \mathrm{Id}_{\mathbb{R}^n}$. In this case, Id/n and $\mathrm{conv}(C)$ can be strongly separated in $\mathrm{Sym}(n)$ by a hyperplane $\mathbf{h}_{\mathrm{Sym}}(T_0, \alpha)$, say. Observe that $T_0 \in \mathrm{Sym}(n)$ can, for example, be taken to be the difference vector between the metric projection of Id/n onto $\mathrm{conv}(C)$ and Id/n (see the proof of Corollary 2.2.1). By changing the orientation of T_0 if necessary, we may assume that

$$\left\langle T_0, \frac{\mathrm{Id}}{n} \right\rangle_M < \alpha < \langle T_0, x \otimes x\rangle_M \tag{9.2.4}$$

for every $x \in \mathbb{S}^{n-1} \cap \partial K$. These inequalities simply state that $\mathrm{conv}(C)$ and Id/n lie in distinct (open) half-spaces determined by the separating hyperplane. Now let

$$T := T_0 - \frac{\mathrm{tr}(T_0)}{n}\mathrm{Id},$$

where tr denotes the trace of the matrix. It is clear that T is symmetric (because T_0 and the identity are symmetric). We recall that $\mathrm{tr}(\cdot) = \langle \mathrm{Id}, \cdot\rangle$ (in other words, Id is the Riesz identification of the trace, which is a linear functional, with respect to the matrix inner product). We also have immediately that $\mathrm{tr}(x \otimes x) = 1$ whenever $x \in \mathbb{S}^{n-1}$, because the matrix representation of $x \otimes x$ is $x \cdot x^T$, and the entries of the principal diagonal of this matrix are the squares of the coordinates of x. Therefore, in view of (9.2.4), for each $x \in \mathbb{S}^{n-1} \cap \partial K$ we have

$$\langle T, x\otimes x\rangle_M = \langle T_0, x\otimes x\rangle_M - \left\langle \frac{\mathrm{tr}(T_0)}{n}\mathrm{Id}, x\otimes x\right\rangle_M > \alpha - \frac{\mathrm{tr}(T_0)}{n}\langle \mathrm{Id}, x\otimes x\rangle_M =$$
$$= \alpha - \frac{\mathrm{tr}(T_0)}{n} > \left\langle T_0, \frac{\mathrm{Id}}{n}\right\rangle_M - \frac{\mathrm{tr}(T_0)}{n} = \frac{1}{n}\langle T_0, \mathrm{Id}\rangle_M - \frac{\mathrm{tr}(T_0)}{n} = 0,$$

where the last equality justifies since $\mathrm{tr}(T_0) = \langle T_0, \mathrm{Id}\rangle_M$. Summarizing, if we put $\beta := \alpha - \mathrm{tr}(T_0)/n$, then

$$\langle T, x\otimes x\rangle_M > \beta > 0 \tag{9.2.5}$$

for every $x \in \mathbb{S}^{n-1}\cap \partial K$, where T is a symmetric matrix. Moreover, observe that the trace of T is 0. In fact,

$$\mathrm{tr}(T) = \mathrm{tr}(T_0) - \frac{\mathrm{tr}(T_0)}{n}\mathrm{tr}(\mathrm{Id}) = 0.$$

Now we put $T_\delta := \mathrm{Id} + \delta T$ for $\delta > 0$. It is clear that T_δ is symmetric, and we know that if δ is small enough, then T_δ is positive-definite, since the set of positive-definite matrices is open (see the proof of Lemma 9.2.3) and Id is positive-definite. Though not necessary here, we provide an explicit bound: if $\delta < 1/||T||_{\mathrm{L}(n)}$, then, using the Cauchy–Schwarz inequality and the definition of the norm in $\mathrm{L}(n)$, we get

$$\langle T_\delta x, x\rangle = \langle (\mathrm{Id}+\delta T)x, x\rangle = |x|^2 + \delta\langle Tx, x\rangle \geq |x|^2 - \delta|Tx|\,|x| \geq$$
$$\geq |x|^2 - \delta||Tx||_{\mathrm{L}(n)}|x|^2 = |x|^2\Big(1-\delta||T||_{\mathrm{L}(n)}\Big) > 0$$

whenever $x \neq 0_{\mathbb{R}^n}$.

We want to construct an ellipsoid which is different from B^n but is contained in K and also has maximal volume, thus obtaining a contradiction. To do so, the "correct" δ will be chosen later, but from now on we always assume that $\delta < 1/||T||_{\mathrm{L}(n)}$. Since this assumption implies that T_δ is symmetric and positive-definite, it follows that T_δ has a square root S_δ, say. Then S_δ is also symmetric and positive-definite (see the proof of Lemma 9.1.1), and hence invertible. We define the ellipsoid

$$E_\delta := S_\delta^{-1}(B^n) = \{x\in\mathbb{R}^n : \langle T_\delta x, x\rangle \leq 1\},$$

where the inequality justifies since $x\in E_\delta$ if and only if $S_\delta x\in B^n$, which is equivalent to $\langle S_\delta x, S_\delta x\rangle \leq 1$. Since S_δ is symmetric, we finally get that $\langle S_\delta x, S_\delta x\rangle = \langle (S_\delta)^2 x, x\rangle = \langle T_\delta x, x\rangle$. Now we define the sets

$$D_1 := \left\{ z\in\mathbb{S}^{n-1} : \mathrm{dist}(z, \mathbb{S}^{n-1}\cap\partial K) \leq \frac{\beta}{2||T||_{\mathrm{L}(n)}}\right\} \text{ and}$$
$$D_2 := \left\{ z\in\mathbb{S}^{n-1} : \mathrm{dist}(z, \mathbb{S}^{n-1}\cap\partial K) \geq \frac{\beta}{2||T||_{\mathrm{L}(n)}}\right\},$$

where $\beta > 0$ is as in (9.2.5). These sets are clearly compact, and of course we have that $D_2\cap\partial K = \emptyset$. Hence $\mathrm{dist}(D_2, \partial K) > 0$, and in particular $\gamma := \max\{||x||_K : x\in D_2\} < 1$ (recall that $D_2 \subseteq \mathbb{S}^{n-1}\subseteq K$). We claim that for $\delta > 0$ small enough we have

$$\max\{\lambda > 0 : \lambda x\in E_\delta\} < \frac{1}{||x||_K} \tag{9.2.6}$$

for every $x \in D_2$. Of course, the maximum on the left-hand side is attained when $\lambda x \in \partial E_\delta$. In other words, the ray starting at $0_{\mathbb{R}^n}$ and going in the direction of $x \in D_2$ reaches the boundary of E_δ before touching the boundary of K. Since T is a symmetric non-zero matrix which has trace equal to 0, it follows that T has some negative eigenvalue (to see this, just diagonalize T and recall that the trace is invariant under matrix similarity). Therefore, there exists some $u_0 \in \mathbb{S}^{n-1}$ such that $\langle Tu_0, u_0\rangle < 0$. We claim that $u_0 \in D_2$. Indeed, if $u_0 \in \mathbb{S}^{n-1} \setminus D_2$, then there exists $u \in \mathbb{S}^{n-1} \cap \partial K$ such that $|u - u_0| < \beta/2||T||_{\mathrm{L}(n)}$, and (9.2.5) gives

$$\begin{aligned}\langle Tu_0, u_0\rangle &= \langle Tu, u\rangle + \langle Tu, u_0 - u\rangle + \langle T(u_0 - u), u_0\rangle > \\ > \beta + \langle Tu, u_0 - u\rangle + \langle T(u_0 - u), u_0\rangle &\geq \beta - |Tu|\,|u_0 - u| - |T(u_0 - u)|\,|u_0| \geq \\ &\geq \beta - 2||T||_{\mathrm{L}(n)}|u_0 - u| > \beta - \beta = 0,\end{aligned}$$

a contradiction. This proves that we must have $u_0 \in D_2$. Hence, setting $\xi := \min_{x\in D_2}\langle Tx, x\rangle$ we have $\xi \leq \langle Tx, x\rangle < 0$ for every $x \in D_2$.

Now we observe that $(\gamma^2 - 1)/\xi > 0$ (because numerator and denominator are negative) and prove that if $0 < \delta < (\gamma^2 - 1)/\xi$, then (9.2.6) holds for every $x \in D_2$. Indeed, if $x \in D_2$ and $\lambda x \in \partial E_\delta$ (for $\lambda > 0$), then $S_\delta(\lambda x) \in \mathbb{S}^{n-1}$, and hence

$$\begin{aligned}1 = \langle S_\delta(\lambda x), S_\delta(\lambda x)\rangle = \lambda^2\langle(\mathrm{Id} + \delta T)x, x\rangle = \lambda^2(1 + \delta\langle Tx, x\rangle) \geq \lambda^2(1 + \delta\xi) > \\ > \lambda^2\left(1 + \frac{\gamma^2 - 1}{\xi}\xi\right) = (\lambda\gamma)^2,\end{aligned}$$

from where we get that

$$\lambda < \frac{1}{\gamma} \leq \frac{1}{||x||_K},$$

as we wanted to prove. We claim that (9.2.6) also holds for $x \in D_1$. In fact, if $x \in D_1$, then there exists $u \in \mathbb{S}^{n-1} \cap \partial K$ such that $|u - x| \leq \beta/2||T||_{\mathrm{L}(n)}$. Then, letting $\lambda > 0$ be such that $\lambda x \in \partial E_\delta$ we have (as before) that $S_\delta(\lambda x) \in \mathbb{S}^{n-1}$, and

$$\begin{aligned}1 = \langle S_\delta(\lambda x), S_\delta(\lambda x)\rangle = \lambda^2\langle(\mathrm{Id} + \delta T)x, x\rangle = \lambda^2(1 + \delta\langle Tx, x\rangle) = \\ = \lambda^2\Big(1 + \delta(\langle Tu, u\rangle) + \langle Tu, x - u\rangle + \langle T(x - u), x\rangle)\Big) > \\ > \lambda^2\Big(1 + \delta(\beta - |Tu|\,|x - u| - |T(x - u)|\,|x|)\Big) \geq \lambda^2\Big(1 + \delta\Big(\beta - 2||T||_{\mathrm{L}(n)}|u - x|\Big)\Big) > \lambda^2,\end{aligned}$$

where the last inequality justifies because $\beta - 2||T||_{\mathrm{L}(n)}|u - x| > 0$. It follows that

$$\lambda < 1 < \frac{1}{\gamma} \leq \frac{1}{||x||_K},$$

since $\gamma < 1$. Because $\mathbb{S}^{n-1} = D_1 \cup D_2$, we have proved that for every $v \in \mathbb{S}^{n-1}$, the ray $\{\lambda v : \lambda > 0\}$ intersects the boundary of E_δ before it intersects the boundary of K. It follows that $E_\delta \subseteq K$.

Now we use the classical inequality $\det(A)^{1/n} \leq \mathrm{tr}(A)/n$, and also the fact that $\mathrm{tr}(T) = 0$ to obtain

$$\det(T_\delta)^{1/n} \leq \frac{\mathrm{tr}(T_\delta)}{n} = \frac{\mathrm{tr}(\mathrm{Id}) + \delta\mathrm{tr}(T)}{n} = 1,$$

and hence we also have $\det(T_\delta)^{1/2} \leq 1$. This leads to

$$\mathrm{vol}_n(E_\delta) = \det(S_\delta^{-1})\mathrm{vol}_n(B^n) = \frac{1}{\det(T_\delta)^{1/2}}\mathrm{vol}_n(B^n) \geq \mathrm{vol}_n(B^n),$$

and since K is in John position, we must have $\mathrm{vol}_n(E_\delta) = \mathrm{vol}_n(B^n)$, and $E_\delta = B^n$ (because of uniqueness). This is a contradiction since

$$E_\delta = \{x \in \mathbb{R}^n : \langle(\mathrm{Id} + \delta T)x, x\rangle \leq 1\}$$

and $T \neq 0$.

Finally, this contradiction shows that $\mathrm{Id}/n \in \mathrm{conv}(C)$, and the decomposition of the identity follows. The fact that we can write this decomposition with at most $\frac{1}{2}n(n+1)+1$ points of $\mathbb{S}^{n-1} \cap \partial K$ comes from Theorem 2.1.2, because $\mathrm{conv}(C)$ is contained in the space $\mathrm{Sym}(n)$, which has dimension $\frac{1}{2}n(n+1)$, and hence the maximal possible affine dimension of $\mathrm{conv}(C)$ is $\frac{1}{2}n(n+1)+1$.

□

The next theorem is also attributed to John. Some authors prove it independently (see, e.g., [133] and [2361]), but we will prove it as a consequence of Theorem 9.2.1.

Theorem 9.2.2 (John). *If $K \subseteq \mathbb{R}^n$ is a convex body centered at the origin which is in John position, then $K \subseteq \sqrt{n}B^n$.*

Proof. Let $x_1, \ldots, x_m \in \partial K \cap \mathbb{S}^{n-1}$ and $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$ be as in Theorem 9.2.1. Then we have that $K \subseteq \mathbf{h}^-(x_j, 1)$ for each $j = 1, \ldots, m$ (see the comment right before the mentioned theorem). Therefore, inequality

$$\langle x, x_j\rangle \leq 1$$

holds for every $x \in K$ and any $j = 1, \ldots, m$. It follows that

$$|x|^2 = \langle \mathrm{Id}_{\mathbb{R}^n}(x), x\rangle = \left\langle \sum_{j=1}^m \lambda_j x_j \otimes x_j(x), x \right\rangle = \sum_{j=1}^m \lambda_j \langle x, x_j\rangle^2 \leq \sum_{j=1}^m \lambda_j = n$$

for any $x \in K$, where the last equality comes from (9.2.2). This gives immediately the inclusion $K \subseteq \sqrt{n}B^n$.

□

We describe now a consequence of this theorem that will be important later. The *Banach–Mazur distance* between convex bodies K and L centered at the origin is defined as

$$d_{BM}(K, L) =$$
$$= \inf\{c \geq 1 : \exists\, T \in \mathrm{GL}(n) \text{ such that } ||x||_K \leq ||Tx||_L \leq c||x||_K \ \forall\, x \in \mathbb{R}^n\}.$$

There is also a useful equivalent definition that we introduce now.

Proposition 9.2.3. *For any convex bodies $K, L \subseteq \mathbb{R}^n$ centered at the origin we have the equality*

$$d_{BM}(K, L) = \inf\{c \geq 1 : \exists\, T \in \mathrm{GL}(n) \text{ such that } L \subseteq T(K) \subseteq cL\}.$$

Proof. To organize the ideas, define the sets

$$C = \{c \geq 1 : \exists\, T \in \mathrm{GL}(n) \text{ such that } ||x||_K \leq ||Tx||_L \leq c||x||_K \ \forall\, x \in \mathbb{R}^n\} \text{ and}$$
$$R = \{c \geq 1 : \exists\, T \in \mathrm{GL}(n) \text{ such that } L \subseteq T(K) \subseteq cL\}.$$

Hence we have to prove that $\inf C = \inf R$. Let $c \in C$. Then there exists $T \in \mathrm{GL}(n)$ such that $||x||_K \leq ||Tx||_L \leq c||x||_K$ for every $x \in \mathbb{R}^n$. If $x \in T^{-1}(L)$, then $Tx \in L$ and $||x||_K \leq ||Tx||_L \leq 1$, from where we get that $x \in K$. This gives the inclusion $T^{-1}(L) \subseteq K$, which is equivalent to $L \subseteq T(K)$. On the other hand, if $z \in K$, then $||Tz||_L \leq c||z||_K \leq c$, and this gives that $Tz \in cL$. Thus, $T(K) \subseteq cL$. It follows that $c \in R$, and hence $\inf R \leq c$. Since this holds for every $c \in C$, we get

$$\inf R \leq \inf C.$$

Now we prove the reverse inequality. Let $r \in R$. Then, there exists $T \in \mathrm{GL}(n)$ such that $L \subseteq T(K) \subseteq rL$. If $x \in \partial K$, then $Tx \in rL$, and hence $||Tx||_L \leq r = r||x||_K$. Since both sides are homogeneous (of degree 1) and K is a convex body with the origin as interior point, we get that this inequality holds for every $x \in \mathbb{R}^n$. For the other inequality, observe that for each $x \neq 0_{\mathbb{R}^n}$ the point $Tx/||x||_K$ lies in the boundary of $T(K)$. Thus, since $L \subseteq T(K)$ we have

$$\left\| \frac{Tx}{||x||_K} \right\|_L \geq \left\| \frac{Tx}{||x||_K} \right\|_{T(K)} = 1,$$

and then we get $||Tx||_L \geq ||x||_K$. It follows that $r \in C$. By the same argument as previously, we get that $\inf C \leq \inf R$, and this concludes the proof.

□

Corollary 9.2.1. *The Banach–Mazur distance is* $\mathrm{GL}(n)$*-invariant in the following sense: if* $K, L \subseteq \mathbb{R}^n$ *are convex bodies centered at the origin, then*

$$d_{BM}(S_1K, S_2L) = d_{BM}(K, L)$$

for every $S_1, S_2 \in \mathrm{GL}(n)$.

Proof. Indeed, $L \subseteq T(K) \subseteq cL$ if and only if $S_2L \subseteq S_2TS_1^{-1}(S_1K) \subseteq cS_2L$.

□

Theorem 9.2.3. *For any convex body* $K \subseteq \mathbb{R}^n$ *centered at the origin we have*

$$d_{BM}(K, B^n) \leq \sqrt{n}.$$

Proof. From Corollary 9.2.1, we may assume that K is in John position. Then the result is straightforward from Theorem 9.2.2.

□

9.3 Minimal positions I: surface area and mean width

Let $K \subseteq \mathbb{R}^n$ be a convex body. From the Urysohn inequality (Theorem 5.5.1) there is a positive uniform lower bound for the mean width $w(TK)$ of the body TK when T ranges through $\mathrm{SL}(n)$. In fact, since the transformations of $\mathrm{SL}(n)$ preserve the Lebesgue volume, it follows that

$$w(TK) \geq 2\left(\frac{\mathrm{vol}_n(TK)}{\mathrm{vol}_n(B^n)}\right)^{1/n} = 2\left(\frac{\mathrm{vol}_n(K)}{\mathrm{vol}_n(B^n)}\right)^{1/n}$$

for every $T \in \mathrm{SL}(n)$. Hence a natural question arises: for a given convex body $K \subseteq \mathbb{R}^n$, is there a map $T_0 \in \mathrm{SL}(n)$ such that $w(T_0K) = \inf\{w(TK) : T \in \mathrm{SL}(n)\}$? Or else, how can one characterize a convex

body $K \subseteq \mathbb{R}^n$ such that $w(K) \leq w(TK)$ for every $T \in \mathrm{SL}(n)$? If K has this minimizing property, then it is clear that $w(K) = \inf\{w(TK) : T \in \mathrm{SL}(n)\}$, and we say that K is in *minimal mean width position*.

Notice that the minimal mean width position is invariant under orthogonal transformations in the following sense: if K is in minimal mean width position, then UK is also in minimal mean width position for every $U \in \mathrm{O}(n)$. This happens, of course, because the mean width of a convex body is invariant under orthogonal maps, as it is explained in (5.5.1).

Similarly, the isoperimetric inequality (Corollary 6.3.1) gives a positive lower bound for the surface area of the bodies TK as T ranges through $\mathrm{SL}(n)$:

$$S(TK) \geq S(B^n)\left(\frac{\mathrm{vol}_n(TK)}{\mathrm{vol}_n(B^n)}\right)^{\frac{n-1}{n}} = \frac{\omega_n}{\kappa_n^{\frac{n-1}{n}}}\mathrm{vol}_n(K)^{\frac{n-1}{n}},$$

where $S(\cdot)$ denotes the total surface area measure of a convex body, which, as we know, can be equivalently defined as the Minkowski surface area measure, as the $(n-1)$-dimensional Hausdorff measure of the boundary, or via approximations by polytopes (see Section 4.1). As we will see, there exists a map $T_0 \in \mathrm{SL}(n)$ such that

$$S(T_0K) = \inf\left\{S(TK) : T \in \mathrm{SL}(n)\right\}.$$

In this case, we say that T_0K is a *minimal surface area position* of K, and if the infimum above is attained for K itself, then we say that K is in *minimal surface area position*. As the mean width, the surface area of a convex body is invariant under orthogonal transformations. Hence if K is in minimal surface area position, then so is UK for any $U \in \mathrm{O}(n)$.

Remark 9.3.1. The reader might wonder why we do not consider linear maps whose determinant has absolute value 1 instead of restricting ourselves to transformations of $\mathrm{SL}(n)$, once maps with determinant -1 also preserve volume. The reason is that it is unnecessary: if $\det(R) = -1$, then choosing any transformation $U \in \mathrm{O}(n)$ such that $\det U = -1$ we have

$$\det(UR) = 1.$$

It follows that $UR \in \mathrm{SL}(n)$, and because U is orthogonal we have $w(URK) = w(RK)$ and $S(URK) = S(RK)$. This shows that for any linear map R whose determinant is -1, there exists a map $\bar{R} \in \mathrm{SL}(n)$ such that $w(RK) = w(\bar{R}K)$ and $S(RK) = S(\bar{R}K)$. Analogously, for every $T \in \mathrm{SL}(n)$ there exists a transformation $\bar{T}$ having determinant -1 such that TK and $\bar{T}K$ have the same mean width and the same surface area.

The minimal mean width and the minimal surface area are examples of the general problem of characterizing positions of a convex body which minimize a given functional $f : \mathcal{K}^n \to \mathbb{R}$. Our objective in this section is to investigate the existence and uniqueness (up to orthogonal transformations) of these minimal positions, and also to derive characterizations of them. For this sake, we follow the approach presented in [133].

The surface area and mean width of K are, up to constant multiplications, the quermassintegrals $W_1(K)$ and $W_{n-1}(K)$. Thus, we are dealing here with two particular cases of the more general problem of minimizing the i-th quermassintegral $W_i(TK)$ as T ranges through $\mathrm{SL}(n)$. We investigate this more general situation in the next section. As a consequence, some of the results proved in this present section for minimal mean width and minimal surface area (namely, Theorems 9.3.1, 9.3.2, and 9.3.3) can also be derived as particular cases of the theorems regarding minimal positions of quermassintegrals given in the next section. However, it is illustrative that we barely need any "mixed volumes machinery" here (this is used only in Lemma 9.3.1, and we even give an alternative proof which does not rely on mixed volumes), and this is why we prove these results in this section even if they can also be obtained as particular cases of results which will come later. It is also worth mentioning that some techniques

established here will be applied for the more general case of arbitrary quermassintegrals in the next section.

We start with the existence problem for the minimal surface area position. Throughout this section, we will always assume for simplicity that the considered convex bodies have the origin in their respective interiors, as the reader will notice. This is because both mean width and surface area are clearly invariant under translations.

Theorem 9.3.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as interior point. There exists a linear transformation $T_0 \in \mathrm{SL}(n)$ such that*

$$S(T_0K) = \inf\big\{S(TK) : T \in \mathrm{SL}(n)\big\}.$$

Proof. First observe that if $T \in \mathrm{SL}(n)$, then letting $U \in \mathrm{O}(n)$ be such that UT is symmetric and positive-definite (see Remark 9.1.4) we get that $UT \in \mathrm{SL}(n)$ and

$$S((UT)K) = \mathcal{H}^{n-1}(U(TK)) = \mathcal{H}^{n-1}(TK) = S(TK),$$

because U is an isometry and any Hausdorff measure is invariant under isometries. This shows that for every $T \in \mathrm{SL}(n)$ there exists a matrix $P \in \mathrm{SL}(n)$ which is symmetric and positive-definite such that $S(PK) = S(TK)$. Consequently,

$$\inf\big\{S(TK) : T \in \mathrm{SL}(n)\big\} = \inf\big\{S(PK) : P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)\big\}, \tag{9.3.1}$$

where we recall the reader that $\mathrm{Sym}_+(n)$ denotes the space of symmetric and positive-definite linear maps of $\mathbb{R}^n$. Let $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$. Then P has positive eigenvalues $\lambda_1, \ldots, \lambda_n$ corresponding to eigenvectors $v_1, \ldots, v_n$ which we may assume to form an orthonormal basis of $\mathbb{R}^n$. Moreover, since $\det(P) = 1$, we have that $\lambda_1 \ldots \lambda_n = 1$. Let

$$\lambda_M := \max\big\{\lambda_j : 1 \leq j \leq n\big\},$$

where $1 \leq M \leq n$. We claim that

$$||P||_\infty = \lambda_M. \tag{9.3.2}$$

Indeed, if $u \in \mathbb{S}^{n-1}$ is written as $u = \alpha_1 v_1 + \ldots + \alpha_n v_n$, then $\alpha_1^2 + \ldots + \alpha_n^2 = 1$, and then

$$\begin{aligned} |Pu| = |\lambda_1\alpha_1 v_1 + \ldots + \lambda_n\alpha_n v_n| &= \sqrt{(\lambda_1\alpha_1)^2 + \ldots + (\lambda_n\alpha_n)^2} \leq \\ &\leq \sqrt{\lambda_M^2(\alpha_1^2 + \ldots + \alpha_n^2)} = \lambda_M. \end{aligned}$$

Passing to the supremum as u ranges through $\mathbb{S}^{n-1}$ we get that $||P||_\infty \leq \lambda_M$. Once $v_M \in \mathbb{S}^{n-1}$ and $Pv_M = \lambda_M v_M$ we get that $|Pv_M| = \lambda_M$, and hence the equality holds.

Now notice that since K has the origin as an interior point, we have that K contains a closed ball centered at the origin. Then, K also contains the simplex

$$V = \mathrm{conv}\{0_{\mathbb{R}^n}, rv_1, \ldots, rv_n\}$$

for some $r > 0$ small enough. Up to the orthogonal transformation which carries the canonical basis of $\mathbb{R}^n$ onto $\{v_1, \ldots, v_n\}$, this is a dilation (by the factor r) of the standard orthogonal n-simplex Δ_n defined in (4.4.4). It is clear that

$$PV = \mathrm{conv}\{0_{\mathbb{R}^n}, r\lambda_1 v_1, \ldots, r\lambda_n v_n\},$$

and since $V \subseteq K$ we have that $PV \subseteq PK$. For each $1 \leq j \leq n$, the set

$$F_j := \operatorname{conv}\{0_{\mathbb{R}^n}, r\lambda_1 v_1, \ldots, r\lambda_{j-1} v_{j-1}, r\lambda_{j+1} v_{j+1}, \ldots, r\lambda_n v_n\}$$

is a facet of TV whose area we claim to be

$$\operatorname{vol}_{n-1}(F_j) = \frac{r^{n-1}}{(n-1)!}\frac{1}{\lambda_j}.$$

Indeed, the volume of F_j can be computed by identifying the hyperplane

$$\operatorname{span}\{v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n\} \supseteq F_j$$

with $\mathbb{R}^{n-1}$ by the linear map which associates $\{v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_n\}$ with the canonical basis of $\mathbb{R}^{n-1}$, which we denote by $\{e_1, \ldots, e_{j-1}, e_{j+1}, \ldots, e_n\}$. By doing that, F_j is identified with the set

$$\operatorname{conv}\{0_{\mathbb{R}^n}, r\lambda_1 e_1, \ldots, r\lambda_{j-1} e_{j-1}, r\lambda_{j+1} e_{j+1}, \ldots, r\lambda_n e_n\} \subseteq \mathbb{R}^{n-1}.$$

This is clearly the image of the standard orthogonal $(n-1)$-dimensional simplex Δ_{n-1} of $\mathbb{R}^{n-1}$ under the linear D map of $\mathbb{R}^{n-1}$ whose matrix representation in the canonical basis $\{e_1, \ldots, e_{j-1}, e_{j+1}, \ldots, e_n\}$ is

$$D \simeq \operatorname{diag}(r\lambda_1, \ldots, r\lambda_{j-1}, r\lambda_{j+1}, \ldots, r\lambda_n),$$

where we use the symbol "$\simeq$" because on the left-hand side we have a linear map, and on the right-hand side we have a matrix. The determinant of D is given as

$$\det(D) = r^{n-1}\lambda_1 \ldots \lambda_{j-1}\lambda_{j+1} \ldots \lambda_n = \frac{r^{n-1}}{\lambda_j},$$

where in the last equality we used the fact that the product of the eigenvalues of P is 1. Once by (4.4.5) we have the equality $\operatorname{vol}_{n-1}(\Delta_{n-1}) = 1/(n-1)!$, we finally get

$$\operatorname{vol}_{n-1}(F_j) = \operatorname{vol}_{n-1}(D\Delta_{n-1}) = \det(D)\operatorname{vol}_{n-1}(\Delta_{n-1}) = \frac{r^{n-1}}{(n-1)!}\frac{1}{\lambda_j},$$

as we have claimed. Since $PV \subseteq PK$, we have that the surface area of PK is not smaller than the surface area of PV (see Corollary A.4.7). On the other hand, the surface area of PV is greater than the sum of the areas of the facets F_j, $1 \leq j \leq n$. Let $\lambda_m := \min\{\lambda_j : 1 \leq j \leq n\}$, where $1 \leq m \leq j$. Then

$$S(PK) \geq S(PV) > \operatorname{vol}_{n-1}(F_m) = \frac{r^{n-1}}{(n-1)!}\frac{1}{\lambda_m}. \tag{9.3.3}$$

Now we bound $1/\lambda_m$ in terms of λ_M. We write

$$1 = \lambda_1 \ldots \lambda_n \geq \lambda_M(\lambda_m)^{n-1},$$

and this leads to

$$\frac{1}{\lambda_m} \geq (\lambda_M)^{\frac{1}{n-1}}. \tag{9.3.4}$$

Plugging this inequality in (9.3.3) and using (9.3.2) we finally arrive at

$$S(PK) \geq \frac{r^{n-1}}{(n-1)!}(\lambda_M)^{\frac{1}{n-1}} \geq \frac{r^{n-1}}{(n-1)!}||P||_\infty^{\frac{1}{n-1}}.$$

This inequality shows that $S(PK) \to \infty$ if $||P||_\infty \to \infty$ (notice that the constant on the right-hand side of the above only depends on K and on the dimension). Therefore, if we choose a sequence $(P_m)_{m\in\mathbb{N}}$ of transformations in $\mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$ such that

$$S(P_m K) \to \inf\left\{S(PK) : P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)\right\}$$

as $m \to \infty$, then the sequence (P_m) is bounded in the norm of $\mathrm{L}(n)$. Thus, passing to a subsequence if necessary, we may assume that P_m converges in $\mathrm{L}(n)$ to some linear transformation T_0, and since $\mathrm{SL}(n)$ is closed in $\mathrm{L}(n)$ we get that $T_0 \in \mathrm{SL}(n)$. From Proposition 2.6.2, it follows immediately that $P_m K \to T_0 K$ in the Hausdorff metric. Finally, expressing the surface area as a (multiple of a) quermassintegral and recalling that quermassintegrals are continuous in the Hausdorff metric, we get

$$\begin{aligned} S(T_0 K) = \lim_{m\to\infty} S(P_m K) = \inf\left\{S(PK) : P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)\right\} = \\ = \inf\left\{S(TK) : T \in \mathrm{SL}(n)\right\}, \end{aligned}$$

and this concludes the proof.

□

Remark 9.3.2. The idea of the proof is that, unlike in the case of the volume, the surface area of a convex body becomes "too large" if we "stretch it too much" in a given direction, even if we "shrink" proportionally some other direction. Thus, it is not strictly necessary to work with orthogonal simplices. What we really needed was to find some convex body contained in K which we can "stretch" in some direction to compute the surface area (or, at least, of a "piece") of the resulting body to show that it becomes too large. One can also work with parallelotopes, for example.

We move forward to characterize convex bodies which are in minimal surface area position. To do so, we recall that μ_K denotes the *surface area measure* of a convex body $K \subseteq \mathbb{R}^n$. Roughly speaking, μ_K is the push-forward measure of $\mathcal{H}^{n-1}|_{\partial K}$ under the Gauss map of K. The precise definition of μ_K is given in Remark 6.2.3 (we have to be careful because the Gauss map of a nonsmooth convex body is defined only $\mathcal{H}^{n-1}$-almost everywhere). In Theorem 7.2.3 it is proved that this is indeed a Borel measure in $\mathbb{S}^{n-1}$ and that it coincides with the $(n-1)$-th mixed surface area measure of K, which we denote by $S_{n-1}(K,\cdot)$.

We start by stating and proving a useful formula to compute the surface area of the image of a convex body under a linear map.

Lemma 9.3.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then the equality*

$$S((T^*)^{-1}K) = \int_{\mathbb{S}^{n-1}} |Tu|\, d\mu_K(u)$$

holds for any $T \in \mathrm{SL}(n)$.

Proof. From (6.1.2), the surface area measure of $(T^*)^{-1}K$ can be computed in terms of mixed volumes as

$$S((T^*)^{-1}K) = nV\left((T^*)^{-1}K, \dots, (T^*)^{-1}K, B^n\right).$$

Thus, writing $B^n = (T^*)^{-1}T^*B^n$ and using Item **(iii)** of Corollary 6.2.1 we get

$$S((T^*)^{-1}K) = n|\det(T^*)^{-1}|V(K,\dots,K,T^*B^n) = nV(K,\dots,K,T^*B^n),$$

where the last equality justifies because, since $T \in \mathrm{SL}(n)$, we have $\det(T^*)^{-1} = 1$. By the definition of mixed surface area measures given in (7.1.1), we have

$$V(K,\dots,K,T^*B^n) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{T^*B^n}(u)\,dS_{n-1}(K,u) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{T^*B^n}(u)\,d\mu_K(u),$$

and hence

$$S((T^*)^{-1}K) = \int_{\mathbb{S}^{n-1}} h_{T^*B^n}(u)\,d\mu_K(u).$$

Therefore, to conclude the proof all we have to do is to show that $h_{T^*B^n}(u) = |Tu|$ for any $u \in \mathbb{S}^{n-1}$. To verify that, we recall that h_{B^n} is simply the Euclidean norm of $\mathbb{R}^n$ and write

$$\begin{aligned} h_{T^*B^n}(u) = \sup\left\{\langle u, y\rangle : y \in T^*B^n\right\} = \sup\left\{\langle u, T^*z\rangle : z \in B^n\right\} = \\ = \sup\left\{\langle Tu, z\rangle : z \in B^n\right\} = h_{B^n}(Tu) = |Tu|, \end{aligned}$$

and the proof is complete.

□

Now we give an alternative proof which does not rely on mixed volumes, but only on approximations by polytopes.

Alternative proof. Let $P \subseteq \mathbb{R}^n$ be a proper polytope, and let $T \in \mathrm{SL}(n)$. If F is a facet of P with u as unit outer normal vector, then $(T^*)^{-1}F$ is a facet of $(T^*)^{-1}P$ having unit outer normal vector $Tu/|Tu|$ (this is easy to verify; see, e.g., Exercise 3.25). The area of F equals the volume of the orthogonal cylinder $F + \mathrm{seg}[0_{\mathbb{R}^n}, u]$. Also, the height of the (non-orthogonal) cylinder $(T^*)^{-1}F + \mathrm{seg}[0_{\mathbb{R}^n}, (T^*)^{-1}u]$ equals the projection of $(T^*)^{-1}u$ onto the normal vector $Tu/|Tu|$. Thus, once T is volume-preserving (and hence also it is $(T^*)^{-1}$), we arrive at

$$\begin{aligned} \mathrm{vol}_{n-1}(F) = \mathrm{vol}_n(F+u) = \mathrm{vol}_n((T^*)^{-1}F + (T^*)^{-1}u) = \\ = \mathrm{vol}_{n-1}((T^*)^{-1}F)\frac{\langle (T^*)^{-1}u, Tu\rangle}{|Tu|} = \mathrm{vol}_{n-1}((T^*)^{-1}F)\frac{1}{|Tu|}, \end{aligned}$$

where the third inequality is obtained by expressing the volume of the non-orthogonal cylinder $(T^*)^{-1}F + \mathrm{seg}[0_{\mathbb{R}^n}, (T^*)^{-1}u]$ as the product of the area of its basis and its height (compare with Exercise 4.9). Therefore,

$$S((T^*)^{-1}P) = \sum_F \mathrm{vol}_{n-1}((T^*)^{-1}F) = \sum_F |Tu|\mathrm{vol}_{n-1}(F) = \int_{\mathbb{S}^{n-1}} |Tu|\,d\mu_P,$$

where μ_P denotes the surface area measure of P, of course. From Proposition 7.2.3 (which was proved independently of mixed volumes and mixed surface area measures) we have that μ_{P_j} converges weakly to μ_K if (P_j) is a sequence of proper polytopes converging to a convex body K in the Hausdorff metric. The desired equality follows.

□

Before moving forward, we also need some results about determinants which we collect now. Sometimes we restrict ourselves to the case of linear maps in $\mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, once we can study the minimal surface area position looking only to these maps (this is also true for the minimal mean width position).

Proposition 9.3.1. **(i)** *For any* $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$ *the inequality*

$$\mathrm{tr}(P) \geq n$$

holds, with equality if and only if $P = \mathrm{Id}_{\mathbb{R}^n}$.

(ii) *For every* $A \in \mathrm{L}(n)$ *the function* $g(t) = \det(\mathrm{Id}_{\mathbb{R}^n} + tA)^{1/n}$ *is of class* C^∞ *in* $(-\varepsilon, \varepsilon)$ *for some small* $\varepsilon > 0$*, and at* $t = 0$ *we have*

$$g'(0) = \frac{\mathrm{tr}(A)}{n}.$$

Proof. We prove **(i)**. By the spectral theorem, we may let $U \in \mathrm{O}(n)$ be such that

$$U^{-1}PU = \mathrm{diag}(\lambda_1, \ldots, \lambda_n),$$

where $\lambda_1, \ldots, \lambda_n > 0$ are the eigenvalues of P. Once we always have that $\mathrm{tr}(AB) = \mathrm{tr}(BA)$, we get

$$\begin{aligned}\mathrm{tr}(P) = \mathrm{tr}(PUU^{-1}) = \mathrm{tr}(U^{-1}PU) = \lambda_1 + \ldots + \lambda_n \geq n(\lambda_1 \ldots \lambda_n)^{1/n} = \\ = n\det(U^{-1}PU)^{1/n} = n\det(P)^{1/n} = n,\end{aligned}$$

where the reader may notice that we used the arithmetic–geometric means inequality and the fact that $\det(P) = 1$. If the equality $\mathrm{tr}(P) = n$ holds, then dividing the above by n we have

$$1 = \frac{\lambda_1 + \ldots + \lambda_n}{n} \geq (\lambda_1 \ldots \lambda_n)^{1/n} = 1,$$

and this leads to the equality case of the arithmetic–geometric means inequality. It follows that $\lambda_1 = \ldots = \lambda_n = 1$. Hence

$$U^{-1}PU = \mathrm{diag}(\lambda_1, \ldots, \lambda_n) = \mathrm{Id}_{\mathbb{R}^n},$$

and this gives $P = U(\mathrm{Id}_{\mathbb{R}^n})U^{-1} = UU^{-1} = \mathrm{Id}_{\mathbb{R}^n}$.

For **(ii)**, we first warn the reader that we identify all involved linear maps with their canonical matrix representations. The function $t \mapsto \det(\mathrm{Id}_{\mathbb{R}^n} + tA)$ is polynomial in t, and hence it is of class C^∞ over $\mathbb{R}$. On the other hand, by continuity we may assume that there exists $\varepsilon > 0$ such that $\det(\mathrm{Id}_{\mathbb{R}^n} + tA) > 0$ for every $t \in (-\varepsilon, \varepsilon)$. Once the function $\lambda \to \lambda^{1/n}$ is C^∞ in $(0, +\infty)$ we get the first claim. To get the desired expression for $g'(0)$ we use the *Jacobi formula*, which states that if $t \mapsto M(t)$ is a matrix-valued differentiable map on one variable, then

$$\frac{d}{dt}\det(M(t)) = \mathrm{tr}\big(\mathrm{adj}(M(t))M'(t)\big),$$

where $\mathrm{adj}(M(t))$ stands for the *adjugate matrix* of $M(t)$, which is the transpose of the co-factor matrix of $M(t)$. For $M(t) = \mathrm{Id}_{\mathbb{R}^n} + tA$, we have $M(0) = \mathrm{Id}_{\mathbb{R}^n}$ and $M'(0) = A$. Hence

$$\begin{aligned}g'(0) = \frac{d}{dt}\Big[\det(\mathrm{Id}_{\mathbb{R}^n} + tA)^{1/n}\Big]\Big|_{t=0} = \frac{1}{n}\det(\mathrm{Id}_{\mathbb{R}^n})^{1/n-1}\frac{d}{dt}\det(\mathrm{Id}_{\mathbb{R}^n} + tA)\Big|_{t=0} = \\ = \frac{1}{n}\mathrm{tr}\big(\mathrm{adj}(\mathrm{Id}_{\mathbb{R}^n})A\big) = \frac{\mathrm{tr}(A)}{n},\end{aligned}$$

where the last equality justifies because it is clear that $\mathrm{adj}(\mathrm{Id}_{\mathbb{R}^n}) = \mathrm{Id}_{\mathbb{R}^n}$.

□

Theorem 9.3.2. *A convex body $K \subseteq \mathbb{R}^n$ is in minimal surface area position if and only if*

$$\int_{\mathbb{S}^{n-1}} \langle u, Tu \rangle \, d\mu_K(u) = S(K) \frac{\operatorname{tr}(T)}{n} \tag{9.3.5}$$

for every $T \in \mathrm{L}(n)$.

Proof. First, assume that K is in minimal surface area position, and fix an arbitrary $T \in \mathrm{L}(n)$. Once the determinant is continuous, we have that for $\varepsilon > 0$ small the linear map $\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T$ is invertible, and hence the linear map

$$T_\varepsilon := \frac{\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T}{\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}}$$

is in $\mathrm{SL}(n)$. Consequently, we also have that $(T_\varepsilon^*)^{-1} \in \mathrm{SL}(n)$. Because K is in minimal surface area position, it holds that $S((T_\varepsilon^*)^{-1}K) \geq S(K)$, and in view of Lemma 9.3.1 this inequality can be written as

$$S(K) \leq \int_{\mathbb{S}^{n-1}} |T_\varepsilon u| \, d\mu_K(u) = \frac{1}{\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}} \int_{\mathbb{S}^{n-1}} |u + \varepsilon T u| \, d\mu_K(u). \tag{9.3.6}$$

Now observe that for each fixed $u \in \mathbb{S}^{n-1}$, the function $f(\varepsilon) = |u + \varepsilon T u|$ is clearly differentiable at $\varepsilon = 0$ (because the Euclidean norm is differentiable at each point of $\mathbb{S}^{n-1}$). Therefore, the standard Taylor polynomial up to order 1 can be written as

$$|u + \varepsilon T u| = f(\varepsilon) = f(0) + \varepsilon f'(0) + r_u(\varepsilon),$$

where

$$\lim_{\varepsilon \to 0} \frac{r_u(\varepsilon)}{\varepsilon} = 0. \tag{9.3.7}$$

It is obvious that $f(0) = 1$, and a straightforward calculation gives that $f'(0) = \langle u, Tu \rangle$. Then we write the equality above as

$$|u + \varepsilon T u| = 1 + \varepsilon \langle u, Tu \rangle + r_u(\varepsilon). \tag{9.3.8}$$

Similarly, since for each fixed $u \in \mathbb{S}^{n-1}$ the function $\varepsilon \mapsto \det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}$ is also clearly differentiable, we use Item **(ii)** of Proposition 9.3.1 to write

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n} = 1 + \varepsilon \frac{\operatorname{tr}(T)}{n} + s(\varepsilon),$$

where

$$\lim_{\varepsilon \to 0} \frac{s(\varepsilon)}{\varepsilon} = 0.$$

Now we rewrite (9.3.6) as

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n} S(K) \leq \int_{\mathbb{S}^{n-1}} |u + \varepsilon T u| \, d\mu_K(u),$$

and replace $|u + \varepsilon T u|$ and $\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}$ by the expressions obtained above to get

$$S(K) + S(K)\varepsilon\frac{\operatorname{tr}(T)}{n} + S(K)s(\varepsilon) \leq$$
$$\leq \int_{\mathbb{S}^{n-1}} d\mu_K + \int_{\mathbb{S}^{n-1}} \varepsilon\langle u, Tu\rangle \, d\mu_K(u) + \int_{\mathbb{S}^{n-1}} r_u(\varepsilon)\, d\mu_K(u).$$

The first integral on the right-hand side clearly equals $S(K)$. Hence cancelling it on both sides and dividing by $\varepsilon > 0$, we arrive at

$$S(K)\frac{\operatorname{tr}(T)}{n} + S(K)\frac{s(\varepsilon)}{\varepsilon} \leq \int_{\mathbb{S}^{n-1}} \langle u, Tu\rangle \, d\mu_K(u) + \int_{\mathbb{S}^{n-1}} \frac{r_u(\varepsilon)}{\varepsilon}\, d\mu_K(u).$$

Taking the limit on both sides as $\varepsilon \to 0^+$, we obtain

$$S(K)\frac{\operatorname{tr}(T)}{n} \leq \int_{\mathbb{S}^{n-1}} \langle u, Tu\rangle \, d\mu_K(u) + \lim_{\varepsilon\to 0^+} \int_{\mathbb{S}^{n-1}} \frac{r_u(\varepsilon)}{\varepsilon}\, d\mu_K(u), \tag{9.3.9}$$

and we have to deal with the limit on the right-hand side of the above. The strategy is to use the dominated convergence theorem (Theorem A.2.4). First notice that for each fixed (small) $\varepsilon > 0$ the function $\mathbb{S}^{n-1} \ni u \mapsto r_u(\varepsilon)/\varepsilon = \big(|u + \varepsilon Tu| - 1\big)/\varepsilon - \langle u, Tu\rangle$ is continuous. Once $\mathbb{S}^{n-1}$ is compact and μ_K is a finite measure, it follows that each of these functions is μ_K-integrable. Hence the function

$$h(\varepsilon) := \int_{\mathbb{S}^{n-1}} \frac{r_u(\varepsilon)}{\varepsilon}\, d\mu_K(u)$$

is well-defined for $\varepsilon > 0$, and we want to evaluate $\lim_{\varepsilon\to 0^+} h(\varepsilon)$. For that sake, let $(\varepsilon_m)_{m\in\mathbb{N}}$ be any sequence of positive numbers such that $\varepsilon_m \to 0$ as $m \to \infty$, and for each $m \in \mathbb{N}$ define the function $g_m \colon \mathbb{S}^{n-1} \to \mathbb{R}$ as

$$g_m(u) = \frac{r_u(\varepsilon_m)}{\varepsilon_m}.$$

Then (g_m) is a sequence of (μ_K-integrable) functions defined over $\mathbb{S}^{n-1}$ which converges pointwise to 0 as $m \to \infty$ by (9.3.7). Now we notice that

$$|g_m(u)| = \left|\frac{1}{\varepsilon_m}|u + \varepsilon_m Tu| - \frac{1}{\varepsilon_m} - \langle u, Tu\rangle\right| \leq \left|\left|\frac{1}{\varepsilon_m}u + Tu\right| - \frac{1}{\varepsilon_m}\right| + |\langle u, Tu\rangle| \leq$$
$$\leq \left|\left|\frac{1}{\varepsilon_m}u + Tu\right| - \left|\frac{1}{\varepsilon_m}u\right|\right| + |Tu||u| \leq \left|\frac{1}{\varepsilon_m}u + Tu - \frac{1}{\varepsilon_m}u\right| + |Tu| = 2|Tu| \leq 2||T||_\infty,$$

for any $m \in \mathbb{N}$ and $u \in \mathbb{S}^{n-1}$. It follows that the (pointwise) convergence $g_m \to 0$ is dominated. Therefore, from the dominated convergence theorem we get that

$$\lim_{m\to\infty} \int_{\mathbb{S}^{n-1}} g_m(u)\, d\mu_K(u) = \int_{\mathbb{S}^{n-1}} \lim_{m\to\infty} g_m(u)\, d\mu_K(u) = \int_{\mathbb{S}^{n-1}} 0\, d\mu_K(u) = 0.$$

Consequently,

$$0 = \lim_{m\to\infty} \int_{\mathbb{S}^{n-1}} g_m(u)\, d\mu_K(u) = \lim_{m\to\infty} \int_{\mathbb{S}^{n-1}} \frac{r_u(\varepsilon_m)}{\varepsilon_m}\, d\mu_K(u) = \lim_{m\to\infty} h(\varepsilon_m),$$

and this holds for any sequence (ε_m) of positive numbers converging to 0. From standard analysis arguments, we obtain that

$$\lim_{\varepsilon\to 0^+}\int_{\mathbb{S}^{n-1}}\frac{r_u(\varepsilon)}{\varepsilon}\,d\mu_K(u)=\lim_{\varepsilon\to 0^+}h(\varepsilon)=0.$$

Returning to (9.3.9) with this equality in mind we finally get that

$$S(K)\frac{\operatorname{tr}(T)}{n}\le\int_{\mathbb{S}^{n-1}}\langle u,Tu\rangle\,d\mu_K(u),$$

and this holds for any $T\in \mathrm{L}(n)$. In particular, replacing T by $-T$ in the above we arrive at the reverse inequality, and hence

$$S(K)\frac{\operatorname{tr}(T)}{n}=\int_{\mathbb{S}^{n-1}}\langle u,Tu\rangle\,d\mu_K(u)$$

for any $T\in \mathrm{L}(n)$.

Now we assume that (9.3.5) holds, and we have to prove that $S(K)\le S(TK)$ for any $T\in \mathrm{SL}(n)$. From equality (9.3.1), it suffices to consider that $T\in \mathrm{SL}(n)\cap \mathrm{Sym}_+(n)$. In this case, we also have that $T^{-1}\in \mathrm{SL}(n)\cap \mathrm{Sym}_+(n)$ and $(T^{-1})^*=(T^*)^{-1}$. Using Lemma 9.3.1 (replacing $(T^*)^{-1}$ by T), the Cauchy–Schwarz inequality, and Item **(i)** of Proposition 9.3.1 (for T^{-1}) we get

$$\begin{aligned}S(TK)&=\int_{\mathbb{S}^{n-1}}|(T^*)^{-1}u|\,d\mu_K(u)\ge\int_{\mathbb{S}^{n-1}}\langle u,(T^{-1})^*u\rangle\,d\mu_K(u)=\\&=\int_{\mathbb{S}^{n-1}}\langle T^{-1}u,u\rangle\,d\mu_K(u)=\frac{\operatorname{tr}(T^{-1})}{n}S(K)\ge S(K),\end{aligned}\tag{9.3.10}$$

and this concludes the proof.

□

Remark 9.3.3. A Borel measure μ on $\mathbb{S}^{n-1}$ for which it holds that

$$\int_{\mathbb{S}^{n-1}}\langle u,Tu\rangle\,d\mu=\mu(\mathbb{S}^{n-1})\frac{\operatorname{tr}(T)}{n}\tag{9.3.11}$$

for every $T\in \mathrm{L}(n)$ is said to be *isotropic*. Once for any convex body $K\subseteq\mathbb{R}^n$ it holds that $\mu_K(\mathbb{S}^{n-1})=S(K)$, what we have proved is that K is in minimal surface area position if and only if μ_K is isotropic. Below we give another useful characterization of isotropic measures.

Proposition 9.3.2. *A Borel measure μ on $\mathbb{S}^{n-1}$ is isotropic if and only if*

$$\frac{\mu(\mathbb{S}^{n-1})}{n}=\int_{\mathbb{S}^{n-1}}\langle u,u_0\rangle^2\,d\mu(u)\tag{9.3.12}$$

for every $u_0\in\mathbb{S}^{n-1}$.

Proof. Assume that μ is isotropic and let $u_0\in\mathbb{S}^{n-1}$ be arbitrarily given. Put $T_0:=u_0\otimes u_0$, that is

$$T_0(z)=\langle z,u_0\rangle u_0$$

for each $z\in\mathbb{R}^n$. Then $T_0\in \mathrm{L}(n)$ and fixing any orthonormal basis of $\mathbb{R}^n$ such that one of the vectors is u_0, we conclude that $\operatorname{tr}(T_0)=1$. Thus,

$$\frac{\mu(\mathbb{S}^{n-1})}{n}=\mu(\mathbb{S}^{n-1})\frac{\operatorname{tr}(T_0)}{n}=\int_{\mathbb{S}^{n-1}}\langle u,T_0u\rangle\,d\mu(u)=\int_{\mathbb{S}^{n-1}}\langle u,u_0\rangle^2\,d\mu(u),$$

and this gives (9.3.12).

Conversely, assume that (9.3.12) holds true and let $\{e_1, \dots, e_n\}$ be the canonical basis of $\mathbb{R}^n$. For each $j = 1, \dots, n$, we have

$$\frac{\mu(\mathbb{S}^{n-1})}{n} = \int_{\mathbb{S}^{n-1}} \langle u, e_j \rangle^2 \, d\mu = \int_{\mathbb{S}^{n-1}} \langle u, e_j \otimes e_j(u) \rangle \, d\mu, \tag{9.3.13}$$

and we claim that

$$\int_{\mathbb{S}^{n-1}} \langle u, e_j \otimes e_k(u) \rangle \, d\mu = 0 \tag{9.3.14}$$

whenever $j \neq k$, where we remind the reader that the linear map $e_j \otimes e_k$ is defined as $e_j \otimes e_k(x) = \langle x, e_k \rangle e_j$ for each $x \in \mathbb{R}^n$. To verify this equality, observe that if $j \neq k$, then $|e_j + e_k| = \sqrt{2}$, and since $(e_j + e_k)/|e_j + e_k|$ is unit, it follows that

$$\begin{aligned} \frac{\mu(\mathbb{S}^{n-1})}{n} &= \int_{\mathbb{S}^{n-1}} \left\langle u, \frac{e_j + e_k}{|e_j + e_k|} \right\rangle^2 d\mu(u) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle u, e_j + e_k \rangle^2 \, d\mu = \\ &= \frac{1}{2} \int_{\mathbb{S}^{n-1}} \big(\langle u, e_j \rangle^2 + \langle u, e_k \rangle^2 + 2 \langle u, e_j \rangle \langle u, e_k \rangle \big) \, d\mu(u) = \\ &= \frac{\mu(\mathbb{S}^{n-1})}{n} + \int_{\mathbb{S}^{n-1}} \langle u, e_j \rangle \langle u, e_k \rangle \, d\mu(u) = \frac{\mu(\mathbb{S}^{n-1})}{n} + \int_{\mathbb{S}^{n-1}} \langle u, e_j \otimes e_k(u) \rangle \, d\mu(u). \end{aligned}$$

This gives (9.3.14). Now let $T \in \mathrm{L}(n)$ be arbitrarily given. We clearly can write

$$T = \sum_{j,k=1}^{n} \langle T e_k, e_j \rangle e_j \otimes e_k,$$

and hence for each $u \in \mathbb{S}^{n-1}$ we have

$$\langle u, Tu \rangle = \sum_{j,k=1}^{n} \langle T e_k, e_j \rangle \langle u, e_j \otimes e_k(u) \rangle.$$

Integrating over $\mathbb{S}^{n-1}$ (with respect to μ, of course) and using the equalities (9.3.13) and (9.3.14) we get

$$\begin{aligned} \int_{\mathbb{S}^{n-1}} \langle u, Tu \rangle \, d\mu(u) &= \int_{\mathbb{S}^{n-1}} \left(\sum_{j,k=1}^{n} \langle T e_k, e_j \rangle \langle u, e_j \otimes e_k(u) \rangle \right) d\mu(u) = \\ = \sum_{j,k=1}^{n} \left(\langle T e_k, e_j \rangle \int_{\mathbb{S}^{n-1}} \langle u, e_j \otimes e_k(u) \rangle \, d\mu(u) \right) &= \sum_{j=1}^{n} \left(\langle T e_j, e_j \rangle \int_{\mathbb{S}^{n-1}} \langle u, e_j \otimes e_j(u) \rangle \, d\mu(u) \right) = \\ &= \frac{\mu(\mathbb{S}^{n-1})}{n} \sum_{j=1}^{n} \langle T e_j, e_j \rangle = \frac{\mu(\mathbb{S}^{n-1})}{n} \mathrm{tr}(T), \end{aligned}$$

and the proof is complete.

□

Now we discuss uniqueness for the minimal surface area measure position. It turns out that this position is unique up to orthogonal transformations.

Proposition 9.3.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body which is in minimal surface area position. If $T \in \mathrm{SL}(n)$, then TK is in minimal surface area position if and only if T is orthogonal.*

Proof. If T is orthogonal, then we already know that $S(TK) = S(K)$, and hence TK is also in minimal surface area position.

Now let $T \in \mathrm{SL}(n)$ and assume that TK is in minimal surface area position. Write $T = UP$ for some $U \in \mathrm{O}(n)$ and some P symmetric and positive-definite (see Remark 9.1.4). Observe that we also have $P \in \mathrm{SL}(n)$, once the determinant of P is necessarily positive (see Remark 9.1.5). Thus,

$$S(K) = S(TK) = S(UPK) = S(PK) \geq \frac{\mathrm{tr}(P^{-1})}{n} S(K),$$

where the last inequality follows from (9.3.10) for T replaced by P. This leads to $\mathrm{tr}(P^{-1}) \leq n$. Once $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, we get that $P^{-1} \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, and from Item **(i)** of Proposition 9.3.1 we obtain that $\mathrm{tr}(P^{-1}) \geq n$. Hence we have the equality

$$\mathrm{tr}(P^{-1}) = n,$$

and then $P^{-1} = \mathrm{Id}_{\mathbb{R}^n}$ (the fact that the identity is the unique transformation in $\mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$ whose trace equals n is also proved in Item **(i)** of Proposition 9.3.1). Therefore, P is also the identity, and $T = UP = U \in \mathrm{O}(n)$. This is what we wanted to prove.

□

Now we study convex bodies which are in minimal mean width position. We start investigating the existence, and after that we give a characterization theorem which holds under the geometric hypothesis of strict convexity. In what follows, we remind the reader that we denote by σ the spherical probability measure, that is,

$$\sigma(\cdot) = \frac{1}{\mathcal{H}^{n-1}(\mathbb{S}^{n-1})} \mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}(\cdot).$$

Theorem 9.3.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body. There exists a linear map $T_0 \in \mathrm{SL}(n)$ such that $T_0 K$ is in minimal mean width position.*

Proof. We may assume that K has the origin as an interior point. Let $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, and assume that P has an orthonormal basis of eigenvectors $\{v_1, \ldots, v_n\}$ corresponding to eigenvalues $\lambda_1, \ldots, \lambda_n > 0$, respectively. Let $M \in \{1, \ldots, n\}$ be such that

$$\lambda_M = \max\{\lambda_j : j = 1, \ldots, n\}.$$

Let $\rho > 0$ be such that $\rho B^n \subseteq K$. Then, the segment $\mathrm{seg}[-\rho v_M, \rho v_M]$ is contained in K. Since the mean width is monotone under inclusion (because $h_K \geq h_L$ whenever $L \subseteq K$), we get

$$\begin{aligned} w(PK) &\geq w\big(P(\mathrm{seg}[-\rho v_M, \rho v_M])\big) = w\big(\mathrm{seg}[-\rho\lambda_M v_M, \rho\lambda_M v_M]\big) = \\ &= w\big(\rho\lambda_M \mathrm{seg}[-v_M, v_M]\big) = \rho\lambda_M w\big(\mathrm{seg}[-v_M, v_M]\big) = \rho\lambda_M w\big(\mathrm{seg}[-e_1, e_1]\big), \end{aligned} \tag{9.3.15}$$

where the third equality comes from the fact that $h_{\alpha K} = \alpha h_K$ for any $\alpha > 0$ (geometrically, if we dilate a body by a factor α, then its mean width is also dilated by α, of course). Also, we obtain the last equality by considering any orthogonal transformation which carries v_M onto e_1 (mean width is invariant under orthogonal transformations). To calculate the mean width of the segment $\mathrm{seg}[-e_1, e_1]$ we first compute its support function:

$$h_{\mathrm{seg}[-e_1,e_1]}(x) = \max\left\{\langle x, v\rangle : v \in \mathrm{seg}[-e_1, e_1]\right\} = |\langle x, e_1\rangle| = |x_1|,$$

where in the last equality we are writing $x = (x_1, \ldots, x_n)$. Thus,

$$w\big(\mathrm{seg}[-e_1, e_1]\big) = 2\int_{\mathbb{S}^{n-1}} |x_1|\, d\sigma(x) := c. \tag{9.3.16}$$

We actually know the precise value of c (this is calculated in different ways in Proposition 11.5.1 and Remark 6.5.3), but we do not need it here. We only have to observe that $0 < c < +\infty$ (this is obvious) and that c does not depend on P and K. Replacing equality (9.3.16) in inequality (9.3.15) above yields

$$w(PK) \geq \rho c \lambda_M = \rho c ||P||_\infty, \tag{9.3.17}$$

where the last equality comes from the fact that $||P||_\infty = \lambda_M$ (this was shown in the proof of Theorem 9.3.1).

As in the case of the surface area, if we write $T \in \mathrm{SL}(n)$ as $T = UP$ for some $U \in \mathrm{O}(n)$ and $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, we have that $w(TK) = w(PK)$. Hence

$$\inf\big\{w(TK) : T \in \mathrm{SL}(n)\big\} = \inf\big\{w(PK) : P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)\big\}.$$

Now the proof goes as the proof of Theorem 9.3.1. Letting $(P_m)_{m\in\mathbb{N}}$ be a sequence in $\mathrm{SL}(n)\cap\mathrm{Sym}_+(n)$ such that $w(P_mK)$ converges to this infimum as $m \to \infty$, from (9.3.17) we must have that $||P_m||_\infty$ is bounded. Passing to a subsequence if necessary, we may assume that P_m converges in $\mathrm{L}(n)$. Letting T_0 be the limit of P_m, we have that $T_0 \in \mathrm{SL}(n)$, once the special linear group is closed. From Proposition 2.6.2, we have that $P_mK \to T_0K$ in the Hausdorff metric as $m \to \infty$. Also, it is clear that the mean width is continuous in the Hausdorff metric (this is proved in the beginning of Section 5.5; alternatively, simply observe that the mean width is a constant multiple of the quermassintegral W_{n-1}). Therefore,

$$w(T_0K) = \lim_{m\to\infty} w(P_mK) = \inf\big\{w(TK) : T \in \mathrm{SL}(n)\big\},$$

and the proof is complete.

□

Theorem 9.3.4. *A convex body $K \subseteq \mathbb{R}^n$ which is strictly convex is in minimal mean width position if and only if*

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu\rangle\, d\sigma(u) = \mathrm{tr}(T)\frac{w(K)}{2n} \tag{9.3.18}$$

for every $T \in \mathrm{L}(n)$.

Proof. The proof is based in the same variational argument as in the proof of Theorem 9.3.2. Assume that K is in minimal mean width position, and let $T \in \mathrm{L}(n)$ be arbitrarily given. Then, for $\varepsilon > 0$ small enough we have that

$$T_\varepsilon := \frac{\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T}{\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}} \in \mathrm{SL}(n).$$

Of course, we also have that $(T_\varepsilon)^* \in \mathrm{SL}(n)$, and since K is in minimal mean width position, this leads to $w(K) \leq w((T_\varepsilon)^*K)$. We write this inequality as

$$w(K) \le w((T_\varepsilon)^* K) = 2\int_{\mathbb{S}^{n-1}} h_{(T_\varepsilon)^* K}(u)\, d\sigma(u) = 2\int_{\mathbb{S}^{n-1}} h_K(T_\varepsilon u)\, d\sigma(u) =$$
$$= \frac{2}{\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}} \int_{\mathbb{S}^{n-1}} h_K(u + \varepsilon T u)\, d\sigma(u),$$

and this leads to

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n} w(K) \le 2\int_{\mathbb{S}^{n-1}} h_K(u + \varepsilon T u)\, d\sigma(u). \tag{9.3.19}$$

Again, we write

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n} = 1 + \varepsilon \frac{\mathrm{tr}(T)}{n} + s(\varepsilon), \tag{9.3.20}$$

where

$$\lim_{\varepsilon \to 0} \frac{s(\varepsilon)}{\varepsilon} = 0.$$

Once we are assuming that K is strictly convex, we have from Proposition 2.3.9 that h_K is C^1 in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Therefore, for each fixed $u \in \mathbb{S}^{n-1}$ the function $f(\varepsilon) = h_K(u + \varepsilon T u)$ is differentiable at $\varepsilon = 0$. The derivative of this function at $\varepsilon = 0$ is the directional derivative of h_K at u in the direction Tu, and hence the Taylor expansion up to the first order reads

$$h_K(u + \varepsilon T u) = f(\varepsilon) = f(0) + \varepsilon f'(0) + \rho_u(\varepsilon) = h_K(u) + \varepsilon \langle \nabla h_K(u), Tu \rangle + \rho_u(\varepsilon), \tag{9.3.21}$$

where

$$\lim_{\varepsilon \to 0^+} \frac{\rho_u(\varepsilon)}{\varepsilon} = 0. \tag{9.3.22}$$

Replacing (9.3.20) and (9.3.21) in (9.3.19) we arrive at

$$w(K) + \varepsilon w(K) \frac{\mathrm{tr}(T)}{n} + s(\varepsilon) w(K) \le$$
$$\le 2\int_{\mathbb{S}^{n-1}} h_K(u) + \varepsilon \langle \nabla h_K(u), Tu \rangle + \rho_u(\varepsilon)\, d\sigma(u) =$$
$$= w(K) + 2\varepsilon \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu \rangle\, d\sigma(u) + 2\int_{\mathbb{S}^{n-1}} \rho_u(\varepsilon)\, d\sigma(u).$$

Cancelling the term $w(K)$ on both sides, dividing by $\varepsilon > 0$, and making $\varepsilon \to 0^+$, we get

$$w(K) \frac{\mathrm{tr}(T)}{2n} \le \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu \rangle\, d\sigma(u) + \lim_{\varepsilon \to 0^+} \int_{\mathbb{S}^{n-1}} \frac{\rho_u(\varepsilon)}{\varepsilon}\, d\sigma(u). \tag{9.3.23}$$

To deal with the last term on the right-hand side, we do the same as we did for $r_u(\varepsilon)/\varepsilon$ in the proof of Theorem 9.3.2. First notice that for each (small) $\varepsilon > 0$, we have

$$\frac{\rho_u(\varepsilon)}{\varepsilon} = \frac{h_K(u + \varepsilon T u) - h_K(u)}{\varepsilon} - \langle \nabla h_K(u), Tu \rangle$$

for any $u \in \mathbb{S}^{n-1}$. Since h_K is C^1 over $\mathbb{S}^{n-1}$, we have that $\mathbb{S}^{n-1} \ni u \mapsto \nabla h_K(u)$ is continuous, and hence the function

$$\mathbb{S}^{n-1} \ni u \mapsto \frac{\rho_u(\varepsilon)}{\varepsilon}$$

is continuous for each (small) $\varepsilon > 0$. Once σ is a finite measure, each of these maps is σ-integrable. Because h_K is a sublinear function and $\varepsilon > 0$, we have the inequalities

$$h(u + \varepsilon Tu) - h_K(u) \leq h_K(\varepsilon Tu) = \varepsilon h_K(Tu) \leq \varepsilon \max\left\{h_K(Tu) : u \in \mathbb{S}^{n-1}\right\} \text{ and}$$
$$h(u) - h_K(u + \varepsilon Tu) \leq h_K(-\varepsilon Tu) = \varepsilon h_K(T(-u)) \leq \varepsilon \max\left\{h_K(Tu) : u \in \mathbb{S}^{n-1}\right\}$$

for any $\varepsilon > 0$ and every $u \in \mathbb{S}^{n-1}$. We note that the maximum $m_0 := \max\{h_K(Tu) : u \in \mathbb{S}^{n-1}\}$ is finite because $h_K \circ T$ is continuous and $\mathbb{S}^{n-1}$ is compact. Thus, we arrive at

$$|h_K(u + \varepsilon Tu) - h_K(Tu)| \leq \varepsilon m_0,$$

and since ∇h_K is continuous on $\mathbb{S}^{n-1}$, we also have that $m_1 := \max\{|\nabla h_K(u)| : u \in \mathbb{S}^{n-1}\}$ is finite. Thus, we estimate

$$\left|\frac{\rho_u(\varepsilon)}{\varepsilon}\right| = \left|\frac{h_K(u + \varepsilon Tu) - h_K(u)}{\varepsilon} - \langle \nabla h_K(u), Tu\rangle\right| \leq$$
$$\leq \frac{1}{\varepsilon}|h_K(u + \varepsilon Tu) - h_K(u)| + |\langle \nabla h_K(u), Tu\rangle| \leq m_0 + |\nabla h_K(u)||Tu| \leq$$
$$\leq m_0 + m_1||T||_\infty,$$

and this upper bound does not depend on u and ε (but only on K and T). Therefore, we may use the dominated convergence theorem in the same way as we did in the proof of Theorem 9.3.2 to conclude that

$$\lim_{\varepsilon \to 0^+} \int_{\mathbb{S}^{n-1}} \frac{\rho_u(\varepsilon)}{\varepsilon}\, d\sigma(u) = 0.$$

Returning to (9.3.23), we get

$$w(K)\frac{\operatorname{tr}(T)}{2n} \leq \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu\rangle\, d\sigma(u),$$

and this holds for every $T \in \mathrm{L}(n)$. Replacing T by $-T$ we obtain the reverse inequality, and this shows (9.3.18).

Now we prove the other direction: we assume that (9.3.18) holds for any $T \in \mathrm{L}(n)$ and show that K is in minimal mean width position. For that sake, we need to show that $w(K) \leq w(TK)$ for any $T \in \mathrm{SL}(n)$.

As in the case of the surface area, the minimal mean width is clearly invariant under orthogonal transformations. Then, for any $T \in \mathrm{SL}(n)$, using again the polar decomposition to write $T = UP$ for $U \in \mathrm{O}(n)$ and $P \in \mathrm{Sym}_+(n)$, we have that $P \in \mathrm{SL}(n)$, as it is explained in Remark 9.1.5, and $w(TK) = w(U(PK)) = w(PK)$. Consequently, it is enough to prove that $w(K) \leq w(PK)$ for every $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$. To do so, recall that Proposition 2.3.9 states that $\nabla h_K(u)$ is a point of ∂K for each $u \in \mathbb{S}^{n-1}$, and hence

$$h_K(x) = \sup\left\{\langle x, z\rangle : z \in K\right\} \geq \langle \nabla h_K(u), x\rangle$$

for any $x \in \mathbb{R}^n$ and every $u \in \mathbb{S}^{n-1}$. Because P is symmetric, we have that $h_{PK}(x) = h_K(P^*x) = h_K(Px)$ for every $x \in \mathbb{R}^n$. This, together with the inequality above (for $z = Pu$), leads to

$$\begin{aligned} w(PK) &= 2\int_{\mathbb{S}^{n-1}} h_{PK}(u)\,d\sigma(u) = 2\int_{\mathbb{S}^{n-1}} h_K(Pu)\,d\sigma(u) \geq \\ &\geq 2\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Pu\rangle\,d\sigma(u) = w(K)\frac{\mathrm{tr}(P)}{n} \geq w(K), \end{aligned} \tag{9.3.24}$$

where the last inequality comes from Item **(i)** of Proposition 9.3.1. This concludes the proof.

□

Remark 9.3.4. Notice carefully that we only need that (9.3.18) holds for transformations $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$ to have that K is in minimal mean width position.

Scholium 9.3.1. The minimal mean width position is $\mathrm{O}(n)$-invariant in the following sense: if K is a convex body in minimal mean width position and $T \in \mathrm{SL}(n)$, then TK is in minimal mean width position if and only if $T \in \mathrm{O}(n)$.

Proof. If K is in minimal mean width position and $U \in \mathrm{SL}(n) \cap \mathrm{O}(n)$, then UK is in minimal mean width position because the mean width is $\mathrm{O}(n)$-invariant, and hence $w(UK) = w(K) \leq w(TK)$ for every $T \in \mathrm{SL}(n)$.

Now assume that K and TK are both in minimal mean width position for some $T \in \mathrm{SL}(n)$. Then of course we have $w(K) = w(TK)$, and writing $T = UP$ for $U \in \mathrm{O}(n)$ and $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, we have $w(K) = w(PK)$. This leads to the equality case in (9.3.24), and in particular we get that $\mathrm{tr}(P) = n$. From Item **(i)** of Proposition 9.3.1 we obtain that $P = \mathrm{Id}_{\mathbb{R}^n}$, and hence $T = U \in \mathrm{O}(n)$.

□

As we have seen, the minimal surface area position of a convex body is characterized by an isotropic condition to the surface area measure of the body. The natural question that arises is whether there is an isotropic characterization of the minimal mean width position. The answer is positive, as we will see now: if $K \subseteq \mathbb{R}^n$ is a convex body having the origin as an interior point, then the function

$$\sigma_K(E) := \int_E h_K(u)\,d\sigma(u) \tag{9.3.25}$$

defined for any $E \in \mathcal{B}(\mathbb{S}^{n-1})$ is a Borel measure in $\mathbb{S}^{n-1}$, and we will prove that it is isotropic if and only if K is in minimal mean width position. Before this, we need to derive an equality which follows from the classical Green formula. Recall that the *Laplacian* of a function $f\colon U \to \mathbb{R}$ defined over an open set $U \subseteq \mathbb{R}^n$ is the function $\Delta f\colon U \to \mathbb{R}$ defined as

$$\Delta f(x) = \frac{\partial^2 f}{\partial x_1^2}(x) + \ldots + \frac{\partial^2 f}{\partial x_n^2}(x)$$

for each $x \in U$.

Lemma 9.3.2. *Let $K \subseteq \mathbb{R}^n$ be a strictly convex body having the origin as interior point. Then*

$$\frac{w(K)}{2} + \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), u_0\rangle\langle u, u_0\rangle\,d\sigma(u) = (n+1)\int_{\mathbb{S}^{n-1}} h_K(u)\langle u, u_0\rangle^2\,d\sigma(u)$$

for each fixed $u_0 \in \mathbb{S}^{n-1}$.

Proof. Fix $u_0 \in \mathbb{S}^{n-1}$ and consider the function $f\colon \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ given as

$$f(x) = \frac{1}{2|x|^2}\langle x, u_0\rangle^2.$$

It is clear that f is twice continuously differentiable and $f(tu) = f(u)$ for any $t > 0$ and $u \in \mathbb{S}^{n-1}$. A straightforward calculation shows that

$$\nabla f(x) = \frac{\langle x, u_0\rangle}{|x|^2} u_0 - \frac{\langle x, u_0\rangle^2}{|x|^4} x, \tag{9.3.26}$$

and in particular we have that

$$\nabla f(tu) = \frac{1}{t} \nabla f(u) \tag{9.3.27}$$

for every $t > 0$ and each $u \in \mathbb{S}^{n-1}$. Moreover, writing $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ as $x = tu$ for $t > 0$ and $u \in \mathbb{S}^{n-1}$ we derive the equality

$$\langle \nabla f(x), x\rangle = \langle \nabla f(tu), tu\rangle = \langle \nabla f(u), u\rangle = \langle \langle u, u_0\rangle u_0 - \langle u, u_0\rangle^2 u, u\rangle = 0, \tag{9.3.28}$$

which can be interpreted as follows: the number $\langle \nabla f(x), x\rangle$ is the directional derivative of f at x in the direction of x. This number is 0 because f is constant along the line $\mathbb{R}_+ \ni \lambda \mapsto \lambda x$. Equality (9.3.28) can also be obtained as a direct consequence of Lemma 8.2.2. Another direct computation leads to the equality

$$\Delta f(x) = \frac{1}{|x|^2} - \frac{n\langle x, u_0\rangle^2}{|x|^4} \tag{9.3.29}$$

for each $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Here we observe that

$$\Delta f(tu) = \frac{1}{t^2} \Delta f(u) \tag{9.3.30}$$

for any $t > 0$ and $u \in \mathbb{S}^{n-1}$. We will also need to use the fact that

$$\nabla h_K(tu) = \nabla h_K(u)$$

for each $u \in \mathbb{S}^{n-1}$ and every $t > 0$. This is a consequence of h_K being positively homogeneous, but this can also be seen, e.g., by recalling that $h_K = \gamma_{K^\circ}$ and using (2.3.2).

Let $0 < r_0 < r$ and notice that f is twice continuously differentiable (and h_K is continuously differentiable) in an open set containing the annulus $rB^n \setminus r_0 B^n$. Then the classical Green formula states that

$$\begin{aligned} &\int_{\partial(rB^n \setminus r_0 B^n)} h_K(x) \langle \nabla f(x), x\rangle \, d\mathcal{H}^{n-1}(x) = \\ &= \int_{rB^n \setminus r_0 B^n} \langle \nabla h_K(x), \nabla f(x)\rangle + h_K(x) \Delta f(x) \, dx. \end{aligned}$$

From (9.3.28) we have that the left-hand side vanishes. We rewrite the integral of the right-hand side in polar coordinates and use (9.3.27) and (9.3.30) to obtain

$$\begin{aligned} 0 &= \int_{r_0}^{r} \int_{\mathbb{S}^{n-1}} \big(\langle \nabla h_K(tu), \nabla f(tu)\rangle + h_K(tu) \Delta f(tu)\big) t^{n-1} \, d\mathcal{H}^{n-1}(u) dt = \\ &= \int_{r_0}^{r} \int_{\mathbb{S}^{n-1}} \big(\langle \nabla h_K(u), \nabla f(u)\rangle + h_K(u) \Delta f(u)\big) t^{n-2} \, d\mathcal{H}^{n-1}(u) dt. \end{aligned}$$

Differentiating both sides with respect to r at $r = 1$ we arrive at

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), \nabla f(u) \rangle \, d\mathcal{H}^{n-1}(u) = -\int_{\mathbb{S}^{n-1}} h_K(u) \Delta f(u) \, d\mathcal{H}^{n-1}(u). \tag{9.3.31}$$

Dividing both sides by the Hausdorff area ω_n of $\mathbb{S}^{n-1}$ we may replace $\mathcal{H}^{n-1}$ by the standard spherical probability measure σ. We use the expressions for ∇f and Δf given in (9.3.26) and (9.3.29), respectively, and rewrite the equality above as

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), u_0 \rangle \langle u, u_0 \rangle - \langle \nabla h_K(u), u \rangle \langle u, u_0 \rangle^2 \, d\sigma(u) =$$
$$= -\int_{\mathbb{S}^{n-1}} h_K(u) \big(1 - n \langle u, u_0 \rangle^2\big) \, d\sigma(u).$$

From Remark 2.3.16, we have the equality $\langle \nabla h_K(u), u \rangle = h_K(u)$. Using this fact and rearranging the terms we can rewrite the above as

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), u_0 \rangle \langle u, u_0 \rangle \, d\sigma(u) =$$
$$= -\int_{\mathbb{S}^{n-1}} h_K(u) \, d\sigma(u) + (n+1) \int_{\mathbb{S}^{n-1}} h_K(u) \langle u, u_0 \rangle^2 \, d\sigma(u) =$$
$$= -\frac{w(K)}{2} + (n+1) \int_{\mathbb{S}^{n-1}} h_K(u) \langle u, u_0 \rangle^2 \, d\sigma(u),$$

and the proof is complete.

□

Remark 9.3.5. The interested reader can find more information about the Green formula for integrations of "radial" functions over the sphere in the first chapter of Groemer's book [1042]. The argument to derive equality (9.3.31) was taken from there.

Theorem 9.3.5. *Let $K \subseteq \mathbb{R}^n$ be a strictly convex body having the origin as an interior point. Then, K is in minimal mean width position if and only if the equality*

$$\int_{\mathbb{S}^{n-1}} \langle u, u_0 \rangle^2 h_K(u) \, d\sigma(u) = \frac{w(K)}{2n} \tag{9.3.32}$$

holds for every $u_0 \in \mathbb{S}^{n-1}$.

Remark 9.3.6. In view of Proposition 9.3.2, and since it is clear that $\sigma_K(\mathbb{S}^{n-1}) = w(K)/2$, equality (9.3.32) holds for any $u_0 \in \mathbb{S}^{n-1}$ if and only if the measure σ_K defined in (9.3.25) is isotropic.

Proof. If K is in minimal mean width position and $u_0 \in \mathbb{S}^{n-1}$, then applying (9.3.18) for $T_0 := u_0 \otimes u_0$ (and recalling that $\mathrm{tr}(T_0) = 1$) we arrive at the equality

$$\frac{w(K)}{2n} = \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), T_0 u \rangle \, d\sigma(u) = \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), u_0 \rangle \langle u, u_0 \rangle \, d\sigma(u).$$

By Lemma 9.3.2, this leads to

$$\frac{w(K)}{2n} = -\frac{w(K)}{2} + (n+1) \int_{\mathbb{S}^{n-1}} h_K(u) \langle u, u_0 \rangle^2 \, d\sigma(u),$$

and hence equality (9.3.32) follows.

Now we prove the other claim. Assume that (9.3.32) holds for any $u_0 \in \mathbb{S}^{n-1}$. Then from Lemma 9.3.2 we get immediately that

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), u_0 \rangle \langle u, u_0 \rangle \, d\sigma(u) = \frac{w(K)}{2n}$$

for every $u_0 \in \mathbb{S}^{n-1}$. Let $\{e_1, \dots, e_n\}$ be the canonical basis of $\mathbb{R}^n$. The equality above leads to

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \rangle \langle u, e_j \rangle \, d\sigma(u) = \frac{w(K)}{2n} \tag{9.3.33}$$

for each $j = 1, \dots, n$. On the other hand, if $j \neq k$ we have that $|e_j + e_k| = \sqrt{2}$, and hence

$$\begin{aligned}
\frac{w(K)}{2n} &= \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j + e_k \rangle \langle u, e_j + e_k \rangle \, d\sigma(u) = \\
&= \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \rangle \langle u, e_j \rangle \, d\sigma(u) + \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \rangle \langle u, e_k \rangle \, d\sigma(u) + \\
&+ \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_k \rangle \langle u, e_j \rangle \, d\sigma(u) + \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_k \rangle \langle u, e_k \rangle \, d\sigma(u) = \\
&= \frac{w(K)}{2n} + \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \rangle \langle u, e_k \rangle \, d\sigma(u) + \frac{1}{2} \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_k \rangle \langle u, e_j \rangle \, d\sigma(u),
\end{aligned}$$

from where we get that

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \rangle \langle u, e_k \rangle \, d\sigma(u) = - \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_k \rangle \langle u, e_j \rangle \, d\sigma(u) \tag{9.3.34}$$

whenever $j \neq k$. Now let $S \in \mathrm{L}(n)$ be symmetric. Then we may write

$$S = \sum_{j,k=1}^{n} \langle Se_k, e_j \rangle e_j \otimes e_k,$$

and since S is symmetric we have that $\langle Se_k, e_j \rangle = \langle Se_j, e_k \rangle$ for any $k, j = 1, \dots, n$. Thus, from (9.3.33) and (9.3.34) we get

$$\begin{aligned}
\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Su \rangle \, d\sigma(u) &= \sum_{j,k=1}^{n} \langle Se_k, e_j \rangle \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \otimes e_k(u) \rangle \, d\sigma(u) = \\
= \sum_{i=1}^{n} \langle Se_i, e_i \rangle \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_i \rangle \langle u, e_i \rangle \, d\sigma(u) &+ \sum_{\substack{j,k=1 \\ j \neq k}}^{n} \langle Se_k, e_j \rangle \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \rangle \langle u, e_k \rangle \, d\sigma(u) = \\
= \frac{w(K)}{2n} \sum_{i=1}^{n} \langle Se_i, e_i \rangle &+ \sum_{\substack{j,k=1 \\ j<k}}^{n} \left[\left(\langle Se_k, e_j \rangle - \langle Se_j, e_k \rangle \right) \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), e_j \rangle \langle u, e_k \rangle \, d\sigma(u) \right] = \\
&= \mathrm{tr}(S) \frac{w(K)}{2n}.
\end{aligned}$$

This shows that equality if (9.3.32) holds for any $u_0 \in \mathbb{S}^{n-1}$, then (9.3.18) holds for any symmetric transformation $S \in \mathrm{L}(n)$ and, in particular, for any transformation of $\mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$. In view of Remark 9.3.4, this gives that K is in minimal mean width position.

□

Remark 9.3.7. By an approximation argument, one can show that for any convex body in minimal mean width position the measure σ_K is isotropic (that is, we do not need to assume that K is strictly convex). See [959] for more details.

9.4 Minimal positions II: quermassintegrals

In the previous section, we studied the problems of minimizing (for a fixed convex body $K \subseteq \mathbb{R}^n$) the quermassintegrals $W_1(TK)$ and $W_{n-1}(TK)$ as T ranges through $\mathrm{SL}(n)$. Indeed, these quermassintegrals are, up to constant multiplications, the surface area and the mean width of TK. Now we consider the other i-th quermassintegrals. Once $W_i(K) > 0$ for each convex body $K \subseteq \mathbb{R}^n$ and any $i = 1, \ldots, n-1$, we get that the infimum

$$\inf \big\{ W_i(TK) : T \in \mathrm{SL}(n) \big\}$$

exists and is non-negative. We prove now the existence of a minimal position for each quermassintegral, that is, for each $i = 1, \ldots, n-1$ there exists a map in $\mathrm{SL}(n)$ for which the infimum above is attained. The proof is somewhat inspired by the proof of the existence of a minimal mean width position.

Theorem 9.4.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For each $i = 1, \ldots, n-1$ there exists a map $T^i \in \mathrm{SL}(n)$ such that*

$$W_i(T^i K) = \inf \big\{ W_i(TK) : T \in \mathrm{SL}(n) \big\},$$

and consequently this infimum is strictly positive.

Proof. First of all, from Theorem 6.4.1 we have that each quermassintegral W_i is translation-invariant and $\mathrm{O}(n)$-invariant. Hence we may assume that K has the origin as an interior point. Moreover, if $T \in \mathrm{SL}(n)$, then using the polar decomposition to write $T = UP$ for $U \in \mathrm{O}(n)$ and $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, we get that $W_i(TK) = W_i(PK)$. Consequently,

$$\inf \big\{ W_i(TK) : T \in \mathrm{SL}(n) \big\} = \inf \big\{ W_i(PK) : P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n) \big\}$$

for each $i = 1, \ldots, n-1$.

Let $P \in \mathrm{SL}(n) \cap \mathrm{Sym}_+(n)$, and assume that the eigenvalues of P are $0 < \lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n$, and that they correspond to an orthonormal basis $\{v_1, \ldots, v_n\}$ of eigenvectors. Observe that $\lambda_1 \ldots \lambda_n = 1$, since $P \in \mathrm{SL}(n)$. Let U be the orthogonal transformation such that $Ue_j = v_j$ for each $j = 1, \ldots, n$, where $\{e_1, \ldots, e_n\}$ is the canonical basis of $\mathbb{R}^n$. Then for each $i = 1, \ldots, n-1$ we have

$$\begin{aligned} &V\big(\mathrm{seg}[0, v_1], \ldots, \mathrm{seg}[0, v_i], B^n, \ldots, B^n\big) = \\ &= V\big(U(\mathrm{seg}[0, e_1]), \ldots, U(\mathrm{seg}[0, e_i]), UB^n, \ldots, UB^n\big) = \\ &= V\big(\mathrm{seg}[0, e_1], \ldots, \mathrm{seg}[0, e_i], B^n, \ldots, B^n\big) =: c_i. \end{aligned}$$

Each constant c_i is strictly positive (from Proposition 6.2.3) and does not depend on P or K. Let $\rho > 0$ be such that $\rho B^n \subseteq K$. Then $\mathrm{seg}[0, \rho v_j] \subseteq K$ for each $j = 1, \ldots, n$. Consequently, $P(\mathrm{seg}[0, \rho v_j]) \subseteq PK$ for each j. Fixing any $i = 1, \ldots, n-1$, we have from the monotonicity and linearity of mixed volumes that

$$W_{n-i}(PK) = V\big(PK[i], B^n[n-i]\big) \geq V\big(P(\mathrm{seg}[0,\rho v_1]),\dots,P(\mathrm{seg}[0,\rho v_i]),B^n,\dots,B^n\big) =$$
$$= V\big(\mathrm{seg}[0,\lambda_1\rho v_1],\dots,\mathrm{seg}[0,\lambda_i\rho v_i],B^n,\dots,B^n\big) =$$
$$= \rho^i\lambda_1\dots\lambda_i V\big(\mathrm{seg}[0,v_1],\dots,\mathrm{seg}[0,v_i],B^n,\dots,B^n\big) = \rho^i c_i(\lambda_1\dots\lambda_i) = \rho^i c_i\frac{1}{\lambda_{i+1}\dots\lambda_n}.$$

Now recall that λ_1 and λ_n are the smallest and the largest eigenvalues of P, respectively. From the estimate (9.3.4) we have

$$\frac{1}{\lambda_1} \geq (\lambda_n)^{\frac{1}{n-1}},$$

and this can be rewritten as

$$\frac{1}{\lambda_n} \geq (\lambda_1)^{n-1}.$$

Therefore, we arrive at

$$\frac{1}{\lambda_{i+1}\dots\lambda_n} \geq \frac{1}{(\lambda_n)^{n-i}} \geq \lambda_1^{(n-i)(n-1)},$$

and we notice carefully that the exponent on the right-hand side is non-zero since $1 \leq i \leq n-1$. Once $||P||_\infty = \lambda_1$ (this is clear from standard linear algebra, and it was shown in the proof of Theorem 9.3.1), we obtain

$$W_{n-i}(PK) \geq \rho^i c_i ||P||_\infty^{(n-i)(n-1)},$$

with a positive exponent on $||P||_\infty$. Now the proof goes as in Theorems 9.3.1 and 9.3.3. Let $(P_m)_{m\in\mathbb{N}}$ be a sequence in $\mathrm{SL}(n)\cap\mathrm{Sym}_+(n)$ such that

$$W_{n-i}(P_mK) \to \inf\big\{W_{n-i}(TK) : T\in\mathrm{SL}(n)\big\}.$$

By the previous inequality relating $W_{n-i}(PK)$ with a positive power of $||P||_\infty$, we get that the sequence (P_m) is bounded in $\mathrm{L}(n)$. Thus, passing to a subsequence if necessary, we may assume that P_m converges in $\mathrm{L}(n)$ to some $T^{n-i}\in\mathrm{L}(n)$. We actually have that $T^{n-i}\in\mathrm{SL}(n)$, because $\mathrm{SL}(n)$ is closed. Once $P_mK\to T^{n-i}K$ in the Hausdorff metric, and since W_{n-i} is continuous in the Hausdorff metric we get

$$W_{n-i}(T^{n-i}K) = \lim_{m\to\infty} W_{n-i}(P_mK) = \inf\{W_{n-i}(TK) : T\in\mathrm{SL}(n)\big\},$$

and hence the position $T^{n-i}K$ minimizes $W_{n-i}(TK)$ as T ranges through $\mathrm{SL}(n)$. Notice that since $T^{n-i}\in\mathrm{SL}(n)$, we have that $T^{n-i}K$ is a convex body, and this gives that $W_{n-i}(T^{n-i}K)>0$. Once we have fixed $i=1,\dots,n-1$ arbitrarily the proof is complete.

□

For a given $i\in\{1,\dots,n-1\}$, we say that K *minimizes the i-th quermassintegral* if $W_i(K) = \inf\{W_i(TK) : T\in\mathrm{SL}(n)\}$ or, equivalently, if $W_i(K)\leq W_i(TK)$ for every $T\in\mathrm{SL}(n)$. Now we follow [959] showing that if K minimizes the i-th quermassintegral, then the measure $S_{n-i}(K,\cdot)$ is isotropic.

Proposition 9.4.1. *Let K be a strictly convex body having the origin as an interior point, and let $i\in\{1,\dots,n-1\}$. If K minimizes W_i, then*

$$\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu \rangle \, dS_{n-i-1}(K,u) = \mathrm{tr}(T) W_i(K)$$

for every $T \in \mathrm{L}(n)$.

Proof. The proof is based on the same variational argument used in the proof of Theorem 9.3.4, and we recall that the hypothesis on K being strictly convex implies that its support function h_K is C^1. For a given $T \in \mathrm{L}(n)$, we let $\varepsilon > 0$ be small enough such that $\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T$ is an isomorphism, and define

$$T_\varepsilon := \frac{(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^*}{\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}}.$$

Thus, $T_\varepsilon \in \mathrm{SL}(n)$, and since we are assuming that K minimizes the i-th quermassintegral (for some fixed $i \in \{1, \dots, n-1\}$), we have that $W_i(T_\varepsilon K) \geq W_i(K)$. Because W_i is homogeneous of degree $n-i$ (see Theorem 6.4.1 again) we may write this inequality as

$$W_i((\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^* K) \geq \det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{(n-i)/n} W_i(K). \tag{9.4.1}$$

Since $(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^* K = (\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T^*)K \subseteq K + \varepsilon T^* K$, and using the linearity of mixed volumes, we may write

$$\begin{aligned} W_i((\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^* K) &= V\big((\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^* K[n-i], B^n[i]\big) \leq \\ \leq V\big(K + \varepsilon T^* K[n-i], B^n[i]\big) &= W_i(K) + \varepsilon(n-i) V\big(T^* K, K[n-i-1], B^n[i]\big) + \\ + \varepsilon^2 \binom{n-i}{2} V\big(T^* K[2], K[n-i-2], B^n[i]\big) &+ \ldots + \varepsilon^{n-i} V\big(T^* K[n-i], B^n[i]\big), \end{aligned}$$

and for the sake of simplicity we write

$$\begin{aligned} O(\varepsilon^2) &:= \\ := \varepsilon^2 \binom{n-i}{2} V\big(T^* K[2], K[n-i-2], B^n[i]\big) &+ \ldots + \varepsilon^{n-i} V\big(T^* K[n-i], B^n[i]\big). \end{aligned}$$

Observe that $\lim_{\varepsilon \to 0} O(\varepsilon^2)/\varepsilon = 0$. Having this notation in mind, and using formula (7.2.3) we get

$$W_i((\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^* K) \leq W_i(K) + \frac{\varepsilon(n-i)}{n} \int_{\mathbb{S}^{n-1}} h_{T^* K}(u) \, dS_{n-i-1}(K,u) + O(\varepsilon^2).$$

This, together with (9.4.1), and because $h_{T^* K}(u) = h_K(Tu)$ for any $T \in \mathrm{L}(n)$ and $u \in \mathbb{S}^{n-1}$, finally leads to

$$\begin{aligned} \det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{(n-i)/n} W_i(K) &\leq \\ \leq W_i(K) + \frac{\varepsilon(n-i)}{n} \int_{\mathbb{S}^{n-1}} h_K(Tu) \, dS_{n-i-1}(K,u) &+ O(\varepsilon^2). \end{aligned}$$

Moving forward, the chain rule together with Item **(ii)** of Proposition 9.3.1 gives the Taylor expansion of $\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{(n-i)/n}$ up to the first order:

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{(n-i)/n} = 1 + \varepsilon \frac{(n-i)\mathrm{tr}(T)}{n} + s(\varepsilon),$$

where $\lim_{\varepsilon \to 0} s(\varepsilon)/\varepsilon = 0$. Therefore, for small $\varepsilon > 0$ we have the inequality

$$W_i(K) + \varepsilon W_i(K)\frac{(n-i)\mathrm{tr}(T)}{n} + s(\varepsilon)W_i(K) \leq$$
$$\leq W_i(K) + \frac{\varepsilon(n-i)}{n}\int_{\mathbb{S}^{n-1}} h_K(Tu)\, dS_{n-i-1}(K,u) + O(\varepsilon^2),$$

and cancelling the term $W_i(K)$ on both sides, dividing everything by $\varepsilon > 0$, and letting $\varepsilon \to 0^+$ lead to

$$W_i(K)\mathrm{tr}(T) \leq \int_{\mathbb{S}^{n-1}} h_K(Tu)\, dS_{n-i-1}(K,u).$$

Once this holds for every $T \in \mathrm{L}(n)$, we may replace T by $\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T$. Using the linearity of the trace, we get $\mathrm{tr}(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T) = n + \varepsilon\mathrm{tr}(T)$, and hence the inequality above for T replaced by $\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T$ reads

$$nW_i(K) + \varepsilon W_i(K)\mathrm{tr}(T) \leq \int_{\mathbb{S}^{n-1}} h_K(u + \varepsilon Tu)\, dS_{n-i-1}(K,u).$$

Recall that (9.3.21) gives the Taylor expansion of $h_K(u + \varepsilon Tu)$ up to the first order:

$$h_K(u + \varepsilon Tu) = h_K(u) + \varepsilon\langle \nabla h_K(u), Tu\rangle + \rho_u(\varepsilon),$$

where $\lim_{\varepsilon\to 0^+} \rho_u(\varepsilon)/\varepsilon = 0$. Replacing this expression in the inequality above and using (7.2.3) we obtain

$$nW_i(K) + \varepsilon W_i(K)\mathrm{tr}(T) \leq \int_{\mathbb{S}^{n-1}} h_K(u) + \varepsilon\langle \nabla h_K(u), Tu\rangle + \rho_u(\varepsilon)\, dS_{n-i-1}(K,u) =$$
$$= nW_i(K) + \varepsilon\int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu\rangle\, dS_{n-i-1}(K,u) + \int_{\mathbb{S}^{n-1}} \rho_u(\varepsilon)\, dS_{n-i-1}(K,u).$$

Cancelling the term $nW_i(K)$ on both sides, dividing by $\varepsilon > 0$, and letting $\varepsilon \to 0^+$ we arrive at

$$W_i(K)\mathrm{tr}(T) \leq \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu\rangle\, dS_{n-i-1}(K,u) + \lim_{\varepsilon\to 0^+}\int_{\mathbb{S}^{n-1}} \frac{\rho_u(\varepsilon)}{\varepsilon}\, dS_{n-i-1}(K,u).$$

Finally, since the measure $S_{n-i-1}(K,\cdot)$ is a finite Borel measure on $\mathbb{S}^{n-1}$, we may use the dominated convergence theorem in exactly the same way as we did in the proof of Theorem 9.3.4 to conclude that

$$\lim_{\varepsilon\to 0^+}\int_{\mathbb{S}^{n-1}} \frac{\rho_u(\varepsilon)}{\varepsilon}\, dS_{n-i-1}(K,u) = 0.$$

Indeed, $\rho_u(\varepsilon)$ is exactly the same here and there, and the argument presented there clearly holds for any finite Borel measure on $\mathbb{S}^{n-1}$. Consequently, we have

$$W_i(K)\mathrm{tr}(T) \leq \int_{\mathbb{S}^{n-1}} \langle \nabla h_K(u), Tu\rangle\, dS_{n-i-1}(K,u)$$

for any $T \in \mathrm{L}(n)$. Replacing T by $-T$ we get the reverse inequality, and hence the proof is complete. □

Theorem 9.4.2. *If $i \in \{1,\ldots,n-1\}$ and $K \subseteq \mathbb{R}^n$ is a convex body which minimizes the i-th quermassintegral, then the mixed surface area measure $S_{n-i}(K,\cdot)$ is isotropic.*

Proof. For any $R \in \mathrm{SL}(n)$ we have that $R^{-1} \in \mathrm{SL}(n)$, and consequently $W_i(R^{-1}K) \geq W_i(K)$. Thus, using the $\mathrm{SL}(n)$-invariance of mixed volumes, we get

$$W_i(K) \le W_i(R^{-1}K) = V\big(R^{-1}K[n-i], B^n[i]\big) = V\big(K[n-i], RB^n[i]\big).$$

Let $T \in \mathrm{L}(n)$ be arbitrarily given. For $\varepsilon > 0$ small enough the linear map

$$T_\varepsilon := \frac{\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T}{\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{1/n}}$$

is in $\mathrm{SL}(n)$, with $\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T) > 0$. Letting $R = T_\varepsilon$ in the inequality above and using the linearity of mixed volumes we arrive at

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{i/n} W_i(K) \le V\big(K[n-i], (\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)B^n[i]\big).$$

Once it is clear that $(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)B^n \subseteq B^n + \varepsilon T B^n$, from monotonicity and linearity of mixed volumes we obtain

$$\begin{aligned}\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{i/n} W_i(K) &\le V\big(K[n-i], B^n + \varepsilon T B^n[i]\big) = V\big(K[n-i], B^n[i]\big) + \\ &+ \varepsilon i V\big(K[n-i], TB^n, B^n[i-1]\big) + \varepsilon^2 \binom{i}{2} V\big(K[n-i], TB^n[2], B^n[i-2]\big) + \dots \\ &+ \dots + \varepsilon^i V\big(K[n-i], TB^n[i]\big).\end{aligned}$$

Here we observe that the first term on the right-hand side is $W_i(K)$. For the sake of simplicity we write

$$O(\varepsilon^2) := \varepsilon^2 \binom{i}{2} V\big(K[n-i], TB^n[2], B^n[i-2]\big) + \dots + \varepsilon^i V\big(K[n-i], TB^n[i]\big),$$

and observe that $\lim_{\varepsilon \to 0} O(\varepsilon^2)/\varepsilon = 0$. Having in mind the definition of $S_{n-i}(K, \cdot)$, we rewrite the inequality above as

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{i/n} W_i(K) \le W_i(K) + \frac{\varepsilon i}{n} \int_{\mathbb{S}^{n-1}} h_{TB^n}(u)\, dS_{n-i}(K, u) + O(\varepsilon^2).$$

As in the proof of Proposition 9.4.1, we write the Taylor expansion up to order 1 of the function $\varepsilon \mapsto \det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{i/n}$:

$$\det(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^{i/n} = 1 + \frac{\varepsilon i}{n} \mathrm{tr}(T) + s(\varepsilon),$$

with $\lim_{\varepsilon \to 0^+} s(\varepsilon)/\varepsilon = 0$. Substituting this expression in the inequality above (and subsequently cancelling the term $W_i(K)$ on both sides) yields

$$\frac{\varepsilon i}{n} \mathrm{tr}(T) W_i(K) + W_i(K) s(\varepsilon) \le \frac{\varepsilon i}{n} \int_{\mathbb{S}^{n-1}} h_{TB^n}(u)\, dS_{n-i}(K, u) + O(\varepsilon^2).$$

Dividing by $\varepsilon > 0$ and letting $\varepsilon \to 0^+$ we arrive at

$$\mathrm{tr}(T) W_i(K) \le \int_{\mathbb{S}^{n-1}} h_{TB^n}(u)\, dS_{n-i}(K, u).$$

Because this holds for any $T \in \mathrm{L}(n)$, we may replace T by $(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^*$. Recalling that we always have $h_{A^*K}(u) = h_K(Au)$ and also that h_{B^n} is simply the Euclidean norm, this leads to

$$\mathrm{tr}((\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^*) W_i(K) \le \int_{\mathbb{S}^{n-1}} |u + \varepsilon T u|\, dS_{n-i}(K, u).$$

It is clear that $\mathrm{tr}((\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T)^*) = \mathrm{tr}(\mathrm{Id}_{\mathbb{R}^n} + \varepsilon T) = n + \varepsilon \mathrm{tr}(T)$. Thus, we may rewrite the above as

$$nW_i(K) + \varepsilon W_i(K)\mathrm{tr}(T) \leq \int_{\mathbb{S}^{n-1}} |u + \varepsilon Tu| \, dS_{n-i}(K, u).$$

Now we write the Taylor expansion up to order 1 of the function $\varepsilon \mapsto |u + \varepsilon Tu|$ (for $u \in \mathbb{S}^{n-1}$ fixed):

$$h_{B^n}(u + \varepsilon Tu) = |u + \varepsilon Tu| = 1 + \varepsilon \langle u, Tu \rangle + \rho_u(\varepsilon),$$

where $\lim_{\varepsilon \to 0^+} \rho_u(\varepsilon)/\varepsilon = 0$. Replacing $|u + \varepsilon Tu|$ by this expression in the inequality above, we get

$$nW_i(K) + \varepsilon W_i(K)\mathrm{tr}(T) \leq \int_{\mathbb{S}^{n-1}} 1 + \varepsilon \langle u, Tu \rangle + \rho_u(\varepsilon) \, dS_{n-i}(K, u) =$$

$$= S_{n-i}(K, \mathbb{S}^{n-1}) + \varepsilon \int_{\mathbb{S}^{n-1}} \langle u, Tu \rangle \, dS_{n-i}(K, u) + \int_{\mathbb{S}^{n-1}} \rho_u(\varepsilon) \, dS_{n-i}(K, u).$$

From (7.2.2), it follows that $S_{n-i}(K, \mathbb{S}^{n-1}) = nW_i(K)$. Cancelling this term on both sides, dividing everything by $\varepsilon > 0$, and letting $\varepsilon \to 0^+$ yield

$$\mathrm{tr}(T)W_i(K) \leq \int_{\mathbb{S}^{n-1}} \langle u, Tu \rangle \, dS_{n-i}(K, u) + \lim_{\varepsilon \to 0^+} \int_{\mathbb{S}^{n-1}} \frac{\rho_u(\varepsilon)}{\varepsilon} \, dS_{n-i}(K, u).$$

Since here $\rho_u(\varepsilon)$ is the remainder of the Taylor expansion up to order 1 of the support function of B^n (which is a strictly convex body), by exactly the same argument as in Theorem 9.3.4 (i.e., using the dominated convergence theorem), we have that the limit appearing on the right-hand side of the above is 0. Therefore,

$$\mathrm{tr}(T)W_i(K) \leq \int_{\mathbb{S}^{n-1}} \langle u, Tu \rangle \, dS_{n-i}(K, u)$$

for every $T \in \mathrm{L}(n)$. Replacing T by $-T$ we obtain the reverse inequality, and this concludes the proof. □

9.5 Exercises

Let $K \subseteq \mathbb{R}^n$ be a convex body. A *chord* of K is a segment joining two boundary points of K. A *metric diameter* of K is a chord whose length equals $\mathrm{diam}(K)$ (by a compactness argument, the diameter of K is indeed attained for a pair of points, and it is obvious that these points must be boundary points).

A chord $\mathrm{seg}[x, y] \subseteq K$ is an *affine diameter* of K if K is supported by parallel hyperplanes at x and y. In the exercise below, we explore some properties of affine diameters. These were mostly taken from Soltan's survey [2273].

Exercise 9.1. *Let $K \subseteq \mathbb{R}^n$ be any convex body.*

(a) *Assume that* $\mathrm{seg}[x, y]$ *is a metric diameter of K. Let $\mathbf{h}_1$ be the hyperplane through x and orthogonal to $v := y - x$, and let $\mathbf{h}_2$ be the hyperplane through y and orthogonal to v. Prove that $\mathbf{h}_1$ and $\mathbf{h}_2$ are supporting hyperplanes of K.*

(b) *Prove that a chord* $\operatorname{seg}[x, y] \subseteq K$ *is an affine diameter of* K *if and only if it is the longest chord in a given direction of* $\mathbb{R}^n$.

(c) *Use Item* **(b)** *to show that* K *has an affine diameter in each direction of* $\mathbb{R}^n$.

(d) *Prove that affine diameters are affine invariants in the following sense: if* $\operatorname{seg}[x, y]$ *is an affine diameter of* K *and* $A(\,) = x_0 + T(\cdot)\colon \mathbb{R}^n \to \mathbb{R}^n$ *is an affine transformation of* $\mathbb{R}^n$, *then* $\operatorname{seg}[Ax, Ay]$ *is an affine diameter of* AK.

For given convex bodies $K, L \subseteq \mathbb{R}^n$, we define the *inner radius of* K *with respect to* L as the number

$$r_{\text{in}}(K, L) = \max\left\{\lambda \geq 0 : \lambda L + x \subseteq K \text{ for some } x \in \mathbb{R}^n\right\}.$$

Thus, r_{in} gives the largest homothetical copy of L which can be translated to lie within K. The number $r_{\text{in}}(K, -K)$ is called the *Minkowski asymmetry measure* of K. Recently, Brandenberg, Dichter, and González Merino investigated this concept, finding interesting results. See, e.g., [485], [483], [484], and [749].

Exercise 9.2. **(a)** *Prove that* $r_{\text{in}}(K, L)$ *is an affine invariant in the following sense: if* $A(\cdot) := x_0 + T(\cdot)\colon \mathbb{R}^n \to \mathbb{R}^n$ *is any affine transformation of* $\mathbb{R}^n$, *then*

$$r_{\text{in}}(AK, AL) = r_{\text{in}}(K, L).$$

(b) *Show that* $r_{\text{in}}(K, -K) \leq 1$ *for any* $K \in \mathcal{K}^n$.

(c) *Show that* $r_{\text{in}}(K, -K) = 1$ *if and only if* K *is symmetric with respect to an interior point (hence the Minkowski asymmetry measure is indeed an asymmetry measure).*

(d) *Compute the Minkowski asymmetry of a triangle in* $\mathbb{R}^2$.

(e) *Extend Item* **(d)** *for a (full-dimensional) simplex in arbitrary dimension* $n \geq 2$.

Exercise 9.3. *Prove that the Banach–Mazur distance is symmetric in its entries.*

Exercise 9.4. *Let* $K, L \subseteq \mathbb{R}^n$ *be convex bodies.*

(a) *Verify that if* K *and* L *are in minimal mean width position, then* $K + L$ *is also in minimal mean width position.*

(b) *Show that if* $K + L$ *and* K *are in minimal mean width position, then* L *is also in minimal mean width position.*

The next exercise was taken from the paper [962] of Giannopoulos and Papadimitrakis.

Exercise 9.5. *Let* $K \subseteq \mathbb{R}^n$ *be a convex body in minimal surface area position. Prove that*

$$\frac{S(K)}{2n} \leq \operatorname{vol}_{n-1}\left(\operatorname{proj}_{u^\perp}(K)\right) \leq \frac{S(K)}{2\sqrt{n}}$$

for every $u \in \mathbb{S}^{n-1}$.

Hint: Use the integral representation for the volume of a projection given in the beginning of Section 10.3. For the upper bound, use Hölder's inequality, and for the lower bound simply observe that $\langle u, v\rangle^2 \leq |\langle u, v\rangle|$ for any $u, v \in \mathbb{S}^{n-1}$.

Exercise 9.6. *Let $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ and $c_1, \dots, c_m > 0$ be such that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j.$$

Prove that the Borel measure μ on $\mathbb{S}^{n-1}$ given as

$$\mu = \sum_{j=1}^{m} c_j \delta_{u_j}$$

is isotropic. Conclude that if $K \subseteq \mathbb{R}^n$ is a convex body symmetric with respect to the origin and in John position, then there is an isotropic Borel measure on the sphere which is concentrated in $\partial K \cap \mathbb{S}^{n-1}$.

Remark 9.5.1. A converse of this result will be given later in Exercise 9.12: we show that if $K \subseteq \mathbb{R}^n$ is an $0_{\mathbb{R}^n}$-symmetric convex body containing B^n, then the existence of a non-zero isotropic Borel measure concentrated in $\partial K \cap \mathbb{S}^{n-1}$ guarantees that K is in John position. This means that, for origin-symmetric convex bodies, the John position is an isotropic position, in some sense.

Exercise 9.7. *Let $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ and $c_1, \dots, c_m > 0$. Prove that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j \tag{9.5.1}$$

if and only if

$$\sum_{j=1}^{m} c_j \langle u, u_j \rangle^2 = 1 \quad \forall \ u \in \mathbb{S}^{n-1}. \tag{9.5.2}$$

Exercise 9.8. *Let $P \subseteq \mathbb{R}^n$ be a proper polytope with unit outer normal vectors $u_1, \dots, u_m$ and corresponding facets $F_1, \dots, F_m$. Prove that P is in minimal surface area position if and only if*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j,$$

where

$$c_j = \frac{n \mathrm{vol}_{n-1}(F_j)}{S(P)}$$

for each $1 \le j \le m$.

For the next exercise, we need an inequality known as the *geometric Brascamp–Lieb inequality*. This is stated below. In [200], Ball presented a proof which, according to him, is due to Barthe. For Exercises 9.10 and 9.11, we refer the reader to [197] and [194], respectively (also, we thank Julian Haddad for showing us the result in Exercise 9.11).

Theorem 9.5.1 (Geometric Brascamp–Lieb inequality)**.** *Let $u_1, \dots, u_m \in \mathbb{S}^{n-1}$ and $c_1, \dots, c_m > 0$ be such that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j.$$

For any integrable functions $f_1, \dots, f_m : \mathbb{R} \to [0, +\infty)$ it holds that

$$\int_{\mathbb{R}^n} \prod_{j=1}^m f(\langle u_j, x\rangle)^{c_j}\, dx \le \prod_{j=1}^m \left(\int_{\mathbb{R}} f_j(t)\, dt \right)^{c_j}.$$

Exercise 9.9. **(a)** *Verify that the cube $Q^n = [-1,1]^n$ is in John position.*

(b) *Prove that Q^n has the maximal volume among all origin-symmetric convex bodies in John position.*

Hint: If K is origin-symmetric and in John position, then we may consider touching points $u_1, \dots, u_m$ of $\mathbb{S}^{n-1} \cap \partial K$ as in Theorem 9.2.1. The points $-u_1, \dots, -u_m$ are also touching points of $\mathbb{S}^{n-1} \cap \partial K$, and hence K is contained in the "strips" bounded by its (parallel) supporting hyperplanes at u_j and $-u_j$, for $j = 1, \dots, m$. Use the geometric Brascamp–Lieb inequality to bound the volume of the intersection of these strips. It is also useful to notice that for any characteristic function 1_X and every $c > 0$ it holds that $(1_X)^c = 1_X$.

Exercise 9.10 (Reverse isoperimetric inequality)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body. Prove that there is a position $\tilde{K}$ of K such that*

$$\operatorname{vol}_n(\tilde{K}) = \operatorname{vol}_n(Q^n) \text{ and}$$
$$S(\tilde{K}) \le S(Q^n).$$

Hint: Consider the scalar multiple of the John position of K which satisfies the volume condition.

Exercise 9.11. *Let $Q_n = \left[-\frac{1}{2}, \frac{1}{2}\right]^n \subseteq \mathbb{R}^n$, and let $\mathbf{g}$ be a k-dimensional subspace of $\mathbb{R}^n$, where $k < n$. Prove that*

$$\operatorname{vol}_k(\mathbf{g} \cap Q_n) \le \left(\frac{n}{k}\right)^{\frac{k}{2}}.$$

Hint: Let $P : \mathbb{R}^n \to \mathbf{g}$ be the orthogonal projection onto $\mathbf{g}$ and, up to reordering the terms, assume that $Pe_i \neq 0_{\mathbf{g}}$ for $i = 1, \dots, m$ (for some $m \le n$). Letting $u_i = Pe_i/|Pe_i|$ and $c_i = |Pe_i|^2$ for each $i = 1, \dots, m$, show that

$$\mathbf{g} \cap Q_n = \left\{ x \in \mathbf{g} : |\langle x, u_i\rangle| \le 1/2\sqrt{c_i} \text{ for all } i = 1, \dots, m \right\}.$$

Letting f_i be the characteristic function of $[-1/2\sqrt{c_i}, 1/2\sqrt{c_i}]$ for each $i = 1, \dots, m$, notice that we always have $f_i = f_i^{c_i}$ and conclude that

$$\operatorname{vol}_k(\mathbf{g} \cap Q_n) = \int_{\mathbf{g}} \prod_{i=1}^m f_i(\langle x, u_i\rangle)^{c_1}\, dx.$$

Further, prove that

$$\mathrm{Id}_{\mathbf{g}} = \sum_{j=1}^m c_i u_i \otimes u_i,$$

and use the Brascamp–Lieb inequality. Finally, use the well-known fact that the product $\prod c_i^{c_i}$ is minimized (for positive c_i's) when all the c_i's are the same.

The objective of the next exercise is to prove a sort of converse of Exercise 9.6. We want to prove that if $K \subseteq \mathbb{R}^n$ is an origin-symmetric convex body such that $B^n \subseteq K$, then the existence of an isotropic

Borel measure on $\mathbb{S}^{n-1}$ concentrated on $\partial K \cap \mathbb{S}^{n-1}$ guarantees that K is in John position. This result appeared originally in [959].

Exercise 9.12. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin such that $B^n \subseteq K$. Assume that there exists a non-zero isotropic Borel measure μ on $\mathbb{S}^{n-1}$ concentrated in $\partial K \cap \mathbb{S}^{n-1}$.*

(a) *For simplicity, write $A = \partial K \cap \mathbb{S}^{n-1}$. Show that* $\mathrm{pos}(A) = \mathbb{R}^n$.

(b) *Let L be the Wulff shape*

$$L = W(A, 1) = \bigcap_{u \in A} \mathbf{h}^-(u, 1).$$

Show that

$$L = \big\{ z \in \mathbb{R}^n : |\langle z, u \rangle| \leq 1 \ \ \forall\, u \in A \big\}.$$

Check that L is a convex body. Verify also that $K \subseteq L$.

(c) *Assume that for an orthonormal basis $\{v_1, \ldots, v_n\}$ of $\mathbb{R}^n$ and for numbers $\alpha_1, \ldots, \alpha_n > 0$ the ellipsoid*

$$E = \left\{ y \in \mathbb{R}^n : \sum_{i=1}^n \frac{\langle y, v_i \rangle^2}{\alpha_i^2} \leq 1 \right\}$$

is contained in L. Prove that

$$\sum_{j=1}^n \alpha_j \leq n.$$

(d) *Using the inequality of arithmetic and geometric means, verify that*

$$\mathrm{vol}_n(E) \leq \mathrm{vol}_n(B^n).$$

(e) *Conclude that B^n is the ellipsoid of maximal volume contained in K, that is, K is in John position.*

Hint for **(c)**: For an arbitrary $u \in A$, let

$$z(u) = \left(\sum_{j=1}^n \alpha_j^2 \langle u, v_j \rangle^2 \right)^{-\frac{1}{2}} \sum_{j=1}^n \alpha_j^2 \langle u, v_j \rangle v_j .$$

Show that $z(u) \in E$. From $E \subseteq L$, conclude that $|\langle u, z(u) \rangle| \leq 1$. Use this inequality to get

$$\sum_{j=1}^n \alpha_j^2 \langle u, v_j \rangle^2 \leq 1.$$

Using the classical Cauchy–Schwarz inequality for the vectors

$$(\alpha_1 \langle u, v_1 \rangle, \ldots, \alpha_n \langle u, v_n \rangle) \ \text{ and } \ (\langle u, v_1 \rangle, \ldots, \langle u, v_n \rangle),$$

conclude that

$$\sum_{j=1}^{n} \alpha_j \langle u, v_j \rangle^2 \leq 1.$$

Finally, recalling that μ is isotropic and concentrated in A, write

$$\sum_{j=1}^{n} \alpha_j = \frac{n}{\mu(\mathbb{S}^{n-1})} \int_A \sum_{j=1}^{n} \alpha_j \langle u, v_j \rangle^2 \, d\mu(x)$$

and notice that the right-hand side is bounded by the inequality above.

Remark 9.5.2. Recently, Baêta and Haddad considered the problem of constructing a non-negative centered isotropic measure supported on the contact points of a convex body in Löwner position; see [178].

Exercise 9.13. *Let $\phi\colon \mathbb{R} \to [0, +\infty)$ be a convex function such that $\phi(0) = 0$.*

(a) *Verify that ϕ is non-increasing in $(-\infty, 0]$ and non-decreasing in $[0, +\infty)$.*

From now on, we also assume that ϕ is strictly decreasing in $(-\infty, 0]$ or strictly increasing in $[0, +\infty)$. For a given convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point, we define

$$h(\Pi_\phi K, x) = \inf \left\{ \lambda > 0 : \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v \rangle}{\lambda h_K(v)} \right) h_K(v) \, dS_{n-1}(K, v) \leq 1 \right\}$$

for each $x \in \mathbb{R}^n$.

(b) *Show that if ϕ is strictly decreasing in $(-\infty, 0]$, then we have $\lim_{t \to -\infty} \phi(t) = +\infty$, and if ϕ is strictly increasing in $[0, +\infty)$, then it holds that $\lim_{t \to +\infty} \phi(t) = +\infty$.*

(c) *Check that for any given $x \in \mathbb{R}^n$ there exists a number $\lambda > 0$ for which we indeed have*

$$\frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v \rangle}{\lambda h_K(v)} \right) h_K(v) \, dS_{n-1}(K, v) \leq 1.$$

Conclude that $h(\Pi_\phi K, \cdot)$ defines a real-valued function.

(d) *Prove that if $x \neq 0_{\mathbb{R}^n}$, then the function*

$$\lambda \mapsto \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v \rangle}{\lambda h_K(v)} \right) h_K(v) \, dS_{n-1}(K, v)$$

is continuous and strictly decreasing in $(0, +\infty)$. In particular, show that if $x \neq 0_{\mathbb{R}^n}$, then $h(\Pi_\phi K, x) > 0$. Conclude that in this case one has that

$$\frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \phi\left(\frac{\langle x, v \rangle}{\lambda_0 h_K(v)} \right) h_K(v) \, dS_{n-1}(K, v) = 1$$

if and only if $\lambda_0 = h(\Pi_\phi K, x)$.

(e) *Prove that the function $h(\Pi_\phi K, \cdot)\colon \mathbb{R}^n \to \mathbb{R}$ is sublinear.*

It follows that $h(\Pi_\phi K, \cdot)$ is the support function of a set of CC^n, which we denote by $\Pi_\phi K$, of course. This is known as the Orlicz projection body *of K with respect to ϕ. Notice that from Item* **(c)**

we have that $\Pi_\phi K \in \mathcal{K}_0^n$.

(f) *Let*

$$c_\phi := \max\left\{c > 0 : \max\{\phi(-c), \phi(c)\} \le 1\right\}.$$

Verify that $c_\phi > 0$ *and prove that*

$$\frac{1}{2nc_\phi \max_{v\in\mathbb{S}^{n-1}} h_K(v)} \le h(\Pi_\phi K, u) \le \frac{1}{c_\phi \min_{v\in\mathbb{S}^{n-1}} h_K(v)}$$

for any $u \in \mathbb{S}^{n-1}$.

Hint for **(f)**: For the lower bound, use Jensen's inequality and the estimate derived in Item **(c)** of Exercise 7.25.

In the same notation and under analogous conditions as in the previous exercise, we define the *Orlicz projection map* (associated with ϕ) as the map $\Pi_\phi : \mathcal{K}_0^n \to \mathcal{K}_0^n$ which relates each convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point to its Orlicz projection body $\Pi_\phi K$.

Exercise 9.14. *Prove that the Orlicz projection map* $\Pi_\phi : \mathcal{K}_0^n \to \mathcal{K}_0^n$ *is continuous in the Hausdorff metric.*

The next exercise is dedicated to the behaviour of the Orlicz projection body under volume-preserving linear transformations of $\mathbb{R}^n$. Hence we still consider the same conditions and notation as in Exercise 9.13 above.

Exercise 9.15. (a) *Prove that for any* $K \in \mathcal{K}_0^n$ *and every* $T \in \mathrm{GL}(n)$ *such that* $|\det(T)| = 1$ *one has*

$$\Pi_\phi(TK) = (T^*)^{-1}\Pi_\phi K. \tag{9.5.3}$$

Hint: Consider first the particular case where K is a polytope, and then settle the general case by approximation arguments.

(b) *Show that for every* $c > 0$ *and any* $K \in \mathcal{K}_0^n$ *we have*

$$\Pi_\phi(cK) = \frac{1}{c}\Pi_\phi K.$$

(c) *Verify that (9.5.3) holds for any* $T \in \mathrm{GL}(n)$. *Denoting by* $\Pi_\phi^\circ K$ *the polar body of* $\Pi_\phi K$, *conclude that*

$$\Pi_\phi^\circ(TK) = T\Pi_\phi^\circ K$$

for every $T \in \mathrm{GL}(n)$.

(d) *Prove that if* ϕ *is even, then* $\Pi_\phi(-K) = \Pi_\phi K$ *for every* $K \in \mathcal{K}_0^n$.

Remark 9.5.3. We deal with some examples of Orlicz projection bodies in Exercise 11.14 below. In particular, we verify that the usual projection body can be re-obtained (up to a normalization) if we use a suitable function ϕ.

The exercises about Orlicz projection bodies are taken from [1641], which is the main reference for this subject. An important result concerning these bodies states that, denoting by $\Pi_\phi^\circ K$ the polar body of $\Pi_\phi K$, the ($\mathrm{GL}(n)$-invariant) ratio

$$\frac{\mathrm{vol}_n(\Pi^\circ_\phi K)}{\mathrm{vol}_n(K)}$$

is maximal (among all $K \in \mathcal{K}^n_0$) if and only if K is an ellipsoid symmetric with respect to the origin (this was proved in the mentioned reference only for the case where ϕ is strictly convex; the general case was proved in [393]). Later, in Exercise 11.15, the reader is invited to use this result to give an alternative proof of the so-called *Petty projection inequality*, which we state and prove in Theorem 10.4.2. Even more, in Exercise 11.17 this result is applied to establish an L_p *extension* of the Petty projection inequality.

In Section 9.3, we dealt with *isotropic measures*, but in the exercises section of Chapter 4 we introduced the (related) notion of *isotropic body*. For the next exercises, which are basically results taken from [133], we recall that a convex body $K \subseteq \mathbb{R}^n$ is said to be *isotropic* if it has its centroid at the origin and if there exists a constant $\rho \in \mathbb{R}$ such that

$$\int_K \langle x, u\rangle^2 \, dx = \rho^2$$

for every $u \in \mathbb{S}^{n-1}$.

As we have said above in Exercise 4.34, most authors also demand that $\mathrm{vol}_n(K) = 1$, but we do not do it here. We do it in this way because the interesting features of isotropic bodies are invariant under rescaling. It is useful, however, to work with normalized volume when dealing with *isotropic constants* (which will be defined below), but in this case we may simply restrict ourselves to bodies of unit volume.

If $K \subseteq \mathbb{R}^n$ is a convex body, and $T \in \mathrm{GL}(n)$ and $x_0 \in \mathbb{R}^n$ are such that $x_0 + TK$ is isotropic, then we say that $x_0 + TK$ is an *isotropic position* of K.

Exercise 9.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Define the linear map $M : \mathbb{R}^n \to \mathbb{R}^n$ as*

$$My = \int_K \langle x, y\rangle x \, dx$$

for $y \in \mathbb{R}^n$.

(a) *Prove that M is symmetric and positive-definite. Conclude that M has a symmetric and positive-definite square root (as it is defined in the proof of Lemma 9.1.1).*

(b) *Let $T \in \mathrm{SL}(n)$. For a moment, denote by M_L the inertia map of the convex body L. Prove that $M_{TK} = TM_K T^*$.*

Remark 9.5.4. The linear map M is called the *inertia map* (or *moment map*) of K; of course, in matrix form, we may write $M = (m_{ij})_{i,j=1,\ldots,n}$ as

$$m_{ij} = \int_K x_i x_j \, dx.$$

This matrix is called the *inertia matrix* (or *moment matrix*) of K.

Exercise 9.17. *Let $K \subseteq \mathbb{R}^n$ be an arbitrarily given convex body. Prove that K has an isotropic position.*

Hint: Let $x_0 \in \mathbb{R}^n$ be such that $K_0 = x_0 + K$ has its centroid at the origin. Define the linear map $M : \mathbb{R}^n \to \mathbb{R}^n$ as

$$M(y) = \int_{K_0} \langle x, y\rangle x \, dx.$$

Use Exercise 9.16 and denote the square root of M by S. Then, prove that $K_1 = S^{-1}(K_0)$ is isotropic.

Exercise 9.18. *Let $K \subseteq \mathbb{R}^n$ be a convex body having its centroid at the origin. Prove that the following assertives are equivalent:*

(i) *K is isotropic with*

$$\int_K \langle x, u\rangle^2 \, dx = \rho^2$$

for every $u \in \mathbb{S}^{n-1}$.

(ii) *Writing in coordinates $x = (x_1, \ldots, x_n)$, we have*

$$\int_K x_i x_j \, dx = \rho^2 \delta_{i,j}$$

for any $i, j = 1, \ldots, n$, where $\delta_{i,j}$ denotes the usual Kronecker delta.

(iii) *For every $T \in \mathrm{L}(n)$ we have the equality*

$$\int_K \langle x, Tx\rangle \, dx = \rho^2 \mathrm{tr}(T).$$

Remark 9.5.5. Notice that the equivalence between **(i)** and **(ii)** means that K is isotropic if and only if its inertia matrix is a multiple of the identity matrix.

It is easy to see that isotropic convex bodies remain isotropic under rescaling. Thus, in the next exercise we work with transformations of $\mathrm{SL}(n)$ instead of $\mathrm{GL}(n)$.

Exercise 9.19. *Let $K \subseteq \mathbb{R}^n$ be a convex body having its centroid at the origin. Define*

$$B(K) := \inf\left\{ \int_{TK} |x|^2 \, dx : T \in \mathrm{SL}(n) \right\}. \tag{9.5.4}$$

(a) *Prove that $T_0 K$ is isotropic for a given transformation $T_0 \in \mathrm{SL}(n)$ if and only if*

$$\int_{T_0 K} |x|^2 \, dx = B(K).$$

(b) *Conclude that if $T_0, T_1 \in \mathrm{SL}(n)$ are such that $K_0 = T_0 K$ and $K_1 = T_1 K$ are isotropic, then $T_1 = U T_0$ for some $U \in \mathrm{O}(n)$.*

Hint. If $T \in \mathrm{SL}(n)$, then we may use Item **(i)** of Proposition 9.3.1 for $P = T^* T$.

Remark 9.5.6. Notice that this exercise gives that the isotropic position of a convex body is unique up to orthogonal transformations and rescalings.

Since we have existence of an isotropic position for any convex body $K \subseteq \mathbb{R}^n$ having its centroid at the origin, Item **(a)** of the previous exercise yields that the infimum in the definition of $B(K)$ given in (9.5.4) is actually a (positive) minimum. The number $L_K > 0$ such that

$$L_K^2 = \frac{1}{n} B(K)$$

is called the *isotropic constant* of K.

Exercise 9.20. **(a)** *In the same assumptions and notation as in Exercise 9.19, check that*

$$B(K) = \min\left\{ \frac{1}{|\det(T)|^{1+\frac{2}{n}}} \int_{TK} |x|^2 \, dx : T \in \mathrm{GL}(n) \right\}.$$

(b) *Verify that if K is isotropic, then*

$$\int_K \langle x, u\rangle^2 \, dx = L_K^2$$

for every $u \in \mathbb{S}^{n-1}$.

(c) *Show that for any $\alpha \neq 0$ and every convex body $K \subseteq \mathbb{R}^n$ having its centroid at the origin we have*

$$L_{\alpha K} = |\alpha|^{1+\frac{n}{2}} L_K.$$

Remark 9.5.7. Item **(c)** of the previous exercise gives that our definition of the isotropic constant is not "invariant under rescaling" in a good way. In other words, rescaling K can make L_K become arbitrarily large or arbitrarily small. However, if we look at the class of convex bodies whose volume is unit, then we can ask about bounds for the isotropic constant. Alternatively, we may define the *normalized isotropic constant* as the number $\bar{L}_K > 0$ such that

$$\bar{L}_K^2 = \frac{1}{n} \min\left\{ \frac{1}{\mathrm{vol}_n(TK)^{1+\frac{2}{n}}} \int_{TK} |x|^2 \, dx : T \in \mathrm{GL}(n) \right\}.$$

Of course, we still work with convex bodies having their respective centroids at the origin. Also, it is easy to see that $\bar{L}_K$ is invariant in the linear class of K, and that $\bar{L}_K = L_K$ if K has unit volume.

Exercise 9.21. **(a)** *Let $K \subseteq \mathbb{R}^n$ be a convex body having its centroid at the origin. Prove that*

$$\bar{L}_K \geq \bar{L}_{B^n}.$$

(b) *Prove that there exists a universal constant $c > 0$ such that*

$$\bar{L}_{B^n} \geq c$$

for every natural number $n \geq 1$.

Hint for **(b)**: Write $\bar{L}_{B^n}$ in terms of the volume κ_n of B^n and use Stirling's approximation (11.5.2) for the Gamma function.

In the next exercise, we estimate $\bar{L}_K$ by the above in the case where K is symmetric with respect to the origin.

Exercise 9.22. *Prove that there exists a universal constant $c > 0$ such that*

$$\bar{L}_K \leq c\sqrt{n}$$

for every convex body $K \subseteq \mathbb{R}^n$ which is symmetric with respect to the origin.

Hint: Up to a linear transformation, we may assume that K is in John position, and up to a further rescaling, we may assume that K has volume 1 (the maximal ellipsoid inscribed in K becomes rB^n for some $r > 0$). In this case, check that $r \leq \kappa_n^{-1/n}$. Using Proposition 4.3.1 (and 11.5.2) show that $r \leq c\sqrt{n}$ for some universal constant c. Then, use Theorem 9.2.2.

Exercise 9.23. *Let $K \subseteq \mathbb{R}^n$ be a convex body which is symmetric with respect to the origin and such that $\mathrm{vol}_n(K) = 1$. Let $M\colon \mathbb{R}^n \to \mathbb{R}^n$ be the inertia map of K, and let $S\colon \mathbb{R}^n \to \mathbb{R}^n$ be the square root of M (see Exercise 9.16 for the existence of this square root). Define the ellipsoid*

$$\mathcal{E}_{\mathcal{B}}(K) = S^{-1}B^n.$$

This is called the Binet ellipsoid *of K.*

(a) *Prove that K is isotropic if and only if*

$$\mathcal{E}_{\mathcal{B}}(K) = L_K^{-1}B^n.$$

(b) *Still assuming that the convex body K is symmetric with respect to the origin and has unit volume, but not necessarily isotropic, show that*

$$\mathrm{vol}_n(\mathcal{E}_{\mathcal{B}}(K)) = \kappa_n L_K^{-n}.$$

Exercise 9.24. *Let $K \subseteq \mathbb{R}^n$ be a convex body symmetric with respect to the origin, and assume that $\mathrm{vol}_n(K) = 1$.*

(a) *Show that*

$$||y||_{\mathcal{E}_{\mathcal{B}}(K)} = \left(\int_K \langle x, y\rangle^2\, dx\right)^{1/2}$$

for every $y \in \mathbb{R}^n$.

(b) *Prove that there exists $u_0 \in \mathbb{S}^{n-1}$ such that*

$$\int_K \langle x, u_0\rangle^2\, dx \leq L_K^2.$$

Hint: Use the equality in Item **(b)** of Exercise 9.23 and write the volume of $\mathcal{E}_{\mathcal{B}}(K)$ using formula (4.3.6).

9.6 Notes

The whole Chapter 9 deals with two main topics: the John–Löwner ellipsoids and (minimal) positions of convex bodies. Therefore we have chosen both these themes for the two corresponding parts of our notes to this chapter here.

9.6.1 Löwner–John ellipsoids and related topics

Is is well known that for each n-dimensional convex body K there is a unique ellipsoid of minimal volume containing K (called the *Löwner ellipsoid* or *Loewner ellipsoid of* K), and a unique ellipsoid of maximal volume contained in K (named *John's ellipsoid of* K). As the monographs and handbook surveys cited in the following will demonstrate, the *Löwner–John ellipsoids* (as they are usually called together) as well as other types (and properties) of ellipsoids are, as well-studied objects with many properties, important in different mathematical fields, among them being convex geometry. See also subsection 2.9.1.6 in this book here.

THE NOTION IN BASIC EXPOSITIONS AND BOOKS

For earlier contributions to Löwner–John ellipsoids in the field of *convexity* we refer to Section 70 of Bonnesen–Fenchel [446], the survey [1074] of Gruber and Höbinger, Petty's survey [1969], and subsection 1.2 as well as Section 3 of the handbook survey [1177] of Heil and Martini.

In several book parts, this topic was later comprehensively discussed: We refer to Section 10.12 of Schneider's monograph [2167], Chapter 9 of Gardner's book [930], and Section 2.12 of the book [1713] of Martini, Montejano, and Oliveros. With respect to *Banach-space theory and Minkowski geometry* we refer to Chapter 1 and Section 2.1 of Artstein-Avidan et al. [133], Chapter 3 in Thompson's book [2361], and Section 4 of the survey [1579] of Lindenstrauss and V. Milman.

Regarding aspects of *approximation theory* we note that Gruber discusses this topic in Chapter 11, which is dedicated to approximation themes, of his monograph [1070]. (For more explicit approximation aspects the paper [379] of Bochi is a good example.)

And for *mathematical programming* we mention Chapter 3 of the book [1047] of Grötschel et al., Section 9 of the handbook article [1025] of Gritzmann and Klee, and Section 27.5 of the handbook article [1028] of the same authors; see also below regarding applications in optimization.

There are several notions in these and further fields which are closely related to the Löwner–John ellipsoids, such as the notion of *Banach–Mazur distance* (see here also Tomczak-Jaegermann [2375]), *positions of convex bodies* (cf. the first chapter in [496] of Brazitikos et al., and Chapter 2 of Artstein-Avidan et al. [133]), and the *ellipsoid method in optimization* (see again Chapter 3 of Grötschel et al. [1047]).

FIRST FUNDAMENTAL RESULTS ON LÖWNER–JOHN ELLIPSOIDS

It is not hard to see that for a given n-dimensional convex body K such extremal ellipsoids exist, but to show their uniqueness requires some efforts. The planar case was solved in the early paper [268] by Behrend, and perhaps Danzer, Laugwitz, and Lenz [709] derived first the general case, independently followed by Zaguskin [2517].

John came into the game with his well-known paper [1319], where (with other motivations) it is shown that a convex body K can be positioned "between" two concentric ellipsoids of ratio n, and that n can be replaced by the factor $\sqrt{n}$ when K is centrally symmetric. It turned out that the (number

of) *contact points* between the boundaries of K and the minimal volume ellipsoid are essential (for the latter, also the papers [198] of Ball and [1956] of Pelczyński are important). John showed that the upper bound on their number is $n(n+3)/2$, for the symmetric case replaceable by $n(n+1)/2$.

The maximum volume ellipsoid contained in K have analogous properties, and with this observation (in unpublished form, but see p. 90 in [548] of Busemann) Löwner joined this field. So it was natural to call the minimal volume ellipsoid *John's ellipsoid*, and the maximal volume one *Löwner ellipsoid*, or meaning both with the name *Löwner–John ellipsoids*.

We recommend to read Schneider's Section 10.2 (and the notes thereof) for a fast and direct approach to further types of ellipsoids associated with unit balls of normed spaces or convex bodies K (or more general sets). These are the *fundamental ellipsoids of inertia*, also called *Binet ellipsoids* and related to the 2-centroid body of K, the *Legendre ellipsoids* having the same moments of inertia with respect to all axes of K through the origin, and the Lutwak-Yang-Zhang (LYZ) ellipsoids as a "dual" counterpart of the 2-centroid body. Further generalizations are the L_p *John ellipsoids* and their duals; see Lutwak et al. [1639]. We refer also to our representation of ellipsoids here in this book, namely in subsection 2.9.1.6.

FURTHER GEOMETRIC PROPERTIES OF LÖWNER–JOHN ELLIPSOIDS

Let us continue with some *geometric results on and properties of Löwner–John ellipsoids*. A very nice survey on these ellipsoids is Henk [1179], containing also short, interesting biographies of Karel Löwner and Fritz John. And also in the beautiful paper [284] on convexity in general, Berger nicely discusses Löwner–John ellipsoids.

Nüesch [1907] shows that the Löwner–John ellipsoids of a simplex S are *homothets* with centers at the centroid of S, and their equations are calculated from the vertex coordinates of S. Vandev [2393] gives a formula for the volume of the Löwner ellipsoid of an arbitrary simplex. Following further the idea of restricting the class of convex bodies approximated by Löwner–John ellipsoids, it turns out that in the case of convex polyhedra characterizations of affine regular polyhedra are obtained; see L. Fejes Tóth [844].

Continuing John, Gruber [1057] proves that the boundaries of most (in the sense of *Baire category*) convex bodies have exactly $n(n+3)/2$ points in common with the boundaries of both ellipsoids.

Analogues of Löwner–John ellipsoids defined via other quermassintegrals (instead of volume) are considered by Firey [879], but there are various open questions (e.g., regarding uniqueness). Guderlay and Keller [1094] investigate *Minkowski sums of ellipsoids and balls*, getting upper bounds for Löwner ellipsoids of parallel bodies if the Löwner ellipsoids of the original bodies are given.

In the two papers [440] by Bonan and [1332] by Juhnke, the authors deal with *inequalities for extremal width values* of convex bodies in terms of such values of their Löwner–John ellipsoids and polarity properties; e.g., it is shown that the Löwner ellipsoid of a convex body K equals the John ellipsoid of the *polar body* of K.

Klartag [1402] introduces a mapping depending on a given centered convex body which associates ellipsoids with ellipsoids; he characterizes this mapping in terms of *isotropic measures* and shows that the unique fixed point of it is the Löwner–John ellipsoid.

The role that Löwner–John ellipsoids can also play in *volume computation of convex bodies* is discussed in the survey [2253] of Simonovits.

Gruber [1072] studies *local and global maximum properties* of the volume on the space of all ellipsoids in a convex body K, taking also ultra maximality (when the volume is nowhere stationary) into consideration. He characterizes those ellipsoids for which the volume is locally maximum, resp. locally ultra maximum; the global maximum is the only local maximum, and for generic K it is an ultra maximum. Extensions to affine copies of a convex body (instead of ellipsoids) are given, dual counterparts are studied, and also for surface area and, more generally, intrinsic volumes similar results are obtained.

Ivanov [1286] derives a tight upper bound on the volume of the John ellipsoid of a k-dimensional section of the n-cube, and the optimal lower bound on the volume of the Löwner ellipsoid of a k-dimensional projection of the n-dimensional cross-polytope. The author also discusses possible *applications in geometric inequalities*, including Ball's version of the Brascamb–Lieb inequality.

Xiong and Hu [2484] investigate relations between the classical *John ellipsoid* and the *Lutwak–Yang–Zhang (LYZ) ellipsoid*. The authors succeed to show differences via interesting examples, and they give exact conditions for their coincidence.

Bochi [379] shows the following nice result: Let K be a centrally symmetric convex body, and λ be a number between the volumes of the John and the Löwner ellipsoid. Let E be an ellipsoid of volume λ such that the intersection of K and E has maximal volume. It is proved that, in two dimensions, this condition of maximality determines E, and that it leads to a continuous 1-parameter family of ellipsoids interpolating the John and Löwner ellipsoids.

Here we should mention also the very recent survey [2278] on *geometric characterizations of ellipsoids* of V. Soltan. With geometric flavour and skill, the author painstakingly collects results on the following themes around ellipsoids: *hyperplanarity conditions for sections, projections, mid-surfaces and shadow boundaries, intersections of homothets, projective centers*, and *invariance under affine or projective transformations*. The paper shows in great detail a large variety of characteristic properties of ellipsoids with interesting *extensions to convex quadrics*, and the bibliography (more than 100 references) is very useful.

There are many *generalizations and extensions of the notions of Löwner–John ellipsoids*. We present now many of them.

GENERALIZATIONS

We start with a result on *normed spaces*, called *Lewis' theorem*. It generalizes John's theorem so that the contained ellipsoid is replaced by any affine image of a convex body; see [1551]. For example, the result is also nicely presented in Chapter 2 of the book [2375] of Tomczak-Jaegermann.

Lutwak et al. [1639] *generalize three types of ellipsoids* associated with a convex body, namely John's ellipsoid, *Petty's ellipsoid* (being the image of the unit ball under volume-preserving linear transformation with position having minimal surface area), and the *dual of the known Legendre ellipsoid*. Using notions like L_p-curvature and L_p-mixed volume, they introduce an ellipsoid class having the mentioned types as subcases. Uniqueness results are derived, and it is verified that these ellipsoids satisfy Ball's volume ratio inequality (see [197]).

Using the tool of Orlicz mixed volume, Zou and Xiong [2576] introduce (in analogy to the L_p-ellipsoids from [1639]) *unique Orlicz–John ellipsoids* and prove various related results, e.g., treating *Orlicz versions of the isoperimetric and Ball's volume ratio inequality*, and an exact condition when this ellipsoid is a ball.

Also generalizing John's theorem, Giannopoulos et al. [963] prove that if a smooth convex body K is in maximal volume position inside a smooth convex body L, then there is a decomposition of the identity using only contact points of the boundaries of K, L, and their polar bodies. (For K itself being an ellipsoid, this is John's result.) From this result they obtain more direct proofs of known results on the Banach–Mazur distance. Bastero and Romance [251] extend this to the case that the inner body is not convex. Lasserre [1510] replaces the given convex bodies by non-convex sets, considering then outer approximations of them by suitable sublevel sets and proving also analogues to John's results on the existence of the solution and on contact points. In general, the uniqueness remains open.

Continuing Gruber [1071], Schröcker [2193] shows that the uniqueness of the Löwner ellipsoid still holds with respect to an arbitrary, strictly eigenvalue-convex-size function; this generalization refers also to *other quermassintegrals than volume* (see also Weber and Schröcker [2435]). John's ellipsoid characterization says that if the standard Euclidean ball is the John ellipsoid, then there are many points of contact between the boundaries of the ball and the convex body under consideration,

in a strong geometric sense. Giannopoulos et al. [963] give an extension of this (obtained for pairs of bodies without any symmetry assumptions), further extended to the case in which the inner body is even not convex.

Generalizing the concept of L_p-John ellipsoids (cf. Lutwak et al. [1639]), Ma, Wu, and Feng [1655] introduce so-called (p, q)-*John ellipsoids* centred at the origin. These are associated with the problem of minimizing (for $0 < p \leq q$) the total (p, q)-dual curvature measure which was introduced by Lutwak et al. as a natural extension of the L_p-surface area measure, the qth dual curvature measure, and the L_p-integral curvature. Analogues of John's theorem and Ball's volume ratio inequality are verified for this new type of ellipsoids, a connection between the isotropy of measures and a characterization of (p, q)-John ellipsoids is shown, and basic properties of L_p-John ellipsoids (like continuity, monotonicity, and John's inclusion) are extended to (p, q)-John ellipsoids.

Zou and Xiong [2578] obtain conditions for the coincidence of John and LYZ ellipsoids. They yield a characterization of convex bodies for which these two ellipsoids coincide. Conclusions and consequences (e.g., regarding sharp *affine isoperimetric inequalities*) are presented.

Seeger and Torki [2214] derive extended analogues of John's and Löwner's ellipsoids for proper (pointed) convex cones whose apex is the origin. Namely, it is shown that a proper convex cone is contained in a unique circumscribed ellipsoidal cone of minimum conical volume and contains a unique inscribed ellipsoidal cone of maximum conical volume. (The conical volume of a cone is defined as the common volume of a certain truncation of the cone.)

Arocha, Bracho, and Montejano show in [126] and [127] that if a convex set in complex n-space contains two inscribed *complex ellipsoids of maximal volume*, then they are translates of each other. And that, surprisingly, the *circumscribed complex ellipsoid of minimal volume is unique*. Further on, this yields a complex analogue of Brunn's classical characterization of ellipsoids. In [127] the same authors derive an unexpected characterization of complex ellipsoids which has no real-number analogue. Namely, they confirm that if every complex line intersects a complex n-dimensional convex body in a disk, then the convex body is a complex ellipsoid.

There is a large variety of *applications of Löwner–John ellipsoids*, and we cite here some interesting examples.

APPLICATIONS IN CONVEXITY: INEQUALITIES AND EXTREMAL PROBLEMS

So-called *reverse inequalities in convexity* (which are affine invariant inequalities) were already considered by Behrend [268] and John [1319]. Recall that a convex body K is said to be in *John's position* if the ellipsoid of maximal volume contained in K is the Euclidean unit ball. A deep paper in this direction is due to Ball [197], in which he proves *reverse isoperimetric inequalities* (see also the presentation in the first chapter of Artstein-Avidan et al. [133]). Ball shows that the regular simplex has maximal volume among all convex bodies in John's position, and the n-cube has maximal volume among the centrally symmetric convex bodies in John's position. Thus, if S is a regular n-simplex and K an n-dimensional convex body, there is an affine image of K, of the same volume as S, whose surface area is not larger than that of S. And for K being centrally symmetric, there is an affine image of K with the same volume as a cube whose surface area is not larger than that of K. The necessary upper bounds for surface area depend on the *volume ratio* of K, which is the n-th root of the quotient of the volume of K divided by that of the volume-maximal ellipsoid in K. These and further related results are nicely presented in Section 10.13 (and its notes) from Schneider's book [2167].

Schmuckenschläger [2138] shows with a closely related proof that the *regular simplex* inscribed to the Euclidean unit ball has *minimal mean width* among all convex bodies whose Löwner ellipsoid is that unit ball. And Barthe [232] proves that if K is a convex body such that the Euclidean unit ball is the Löwner ellipsoid contained in K, then the mean width of K is less than the mean width of any regular simplex circumscribed to that ellipsoid.

Li and Leng [1556] extend this mean width inequality to the context of *isotropic measures on the unit sphere*, where the contact points of Löwner ellipsoids play again a role. They show that among the convex hulls of the support of isotropic measures on the unit sphere, the regular simplex inscribed in the Euclidean unit ball has maximal l-norm, and that in the dual situation there is a reverse result for their polar bodies. In this sense, again *isotropic measures* are investigated. The paper contains an interesting collection of further related references.

APPLICATIONS IN CONVEXITY: MEASURES OF SYMMETRY

In Chapter 3 of Toth's monograph [2381], John's theorem is discussed in connection with *stability problems, measures of symmetry*, and the *Banach–Mazur distance* (for the latter see also Tomczak-Jaegermann [2375]). Meyer et al. [1777] introduce *affine measures of symmetry* based on pairs of affine covariant points; the appropriately normalized Euclidean distance between these points presents the measure. For convex bodies with "enough symmetries" (i.e., whose group of symmetries has exactly one fixed point), this distance will be zero, and simplices take the value 1.

As such *affine covariant points* one can take the centroid and the Santaló point, but also the *centers of the John and the Löwner ellipsoids* are used in [1777]. These points are also studied by Safronenko [2086] who continues the investigations of Meyer et al. [1777] by giving new bounds for the measure under consideration in the above-mentioned cases, therefore including John centers and Löwner centers.

Staying with centers, Huang [1242] investigates the natural question whether the center of mass c of a convex body lies in its John ellipsoid; this is important for the efficiency of many algorithms treating convex bodies. With two counterexamples, where c does not belong to a certain homothet of John's ellipsoid, he derives an unexpected negative result.

APPLICATIONS IN CONVEXITY: CONVEX POLYTOPES

An unexpected application of Löwner–John ellipsoids in *polytope theory* due to Figiel, Lindenstrauss, and Milman [872] (namely bounding the product of the facet number and the vertex number of centrally symmetric polytopes and inspired in this survey by the famous "3^n-conjecture" of Kalai) is shortly discussed by Henk in [1179]; see also Proposition (8.1) and pages 274–276 in Barvinok's monograph [242].

For an interesting *discrete analogue* of John's theorem we refer to Berg and Henk [281]. Clearly, a natural goal of approximation is counting the set of lattice points in ellipsoids, but these are hard to compute. However, by a so-called generalized arithmetic progression method one succeeds to cover (many of) the lattice points inside a convex body, using a simple geometric structure. Berg and Henk [281] improve former results on this concept. It is a nice consequence of John's theorem that every n-dimensional symmetric convex body K in John's position can be approximated by a convex polytope P with a polynomial number of facets in n such that $P \subset K \subset \sqrt{n}P$. With homothety ratio n instead of $\sqrt{n}$, this extends also to the non-symmetric case.

Huang [1243] studies also the non-symmetric case, but for homothety ratios below n. He shows that for $r = o(n)$ and a sufficiently large dimension n there exists a convex body K in John's position for which there is no n-polytope P with facet number polynomial in n such that $K \subset P \subset rK$. In addition he shows that for $r = O(\sqrt{n})$ there is a convex body for which a polytope with exponential facet number is needed.

APPLICATIONS IN CONVEXITY: POSITIONS

Artstein-Avidan and Katzin [135] also treat isotropic measures. They define a one-parametric family of positions of a centrally symmetric convex body K which interpolates between the John position and the Löwner position. For K, the ellipsoid E_r of volume $r^n\kappa_n$ is a *maximum intersection ellipsoid*

of volume r if the volume of the intersection of K and all ellipsoids of volume $r^n\kappa_n$ is not larger than the volume of the intersection of K and E_r. Based on this, K is in *maximal intersection position of radius r* if the r-fold homothet of the unit ball is a maximum intersection ellipsoid of (volume)-radius r (here κ_n denotes the volume of the unit ball). It turns out that the set of maximal intersection positions interpolates between the John and the Löwner position, and that a centrally symmetric convex body in maximal intersection position of radius r, satisfying simple additional properties, has the property that the restriction of the surface area measure on the sphere to the intersection with $r^{-1}K$ is an isotropic measure. It follows that when K is in John position (and analogously for the Löwner position), the isotropic measure, guaranteed to exist by John's theorem, can be obtained as a limit of isotropic measures.

Continuing Ball [197] and Barthe [232], Li [1552] uses maximum volume ellipsoids contained in convex bodies to derive *isomorphic versions of the simplex inequalities of Ball and Barthe*.

MORE APPLICATIONS IN THE LOCAL THEORY OF BANACH SPACES

More known applications of Löwner–John ellipsoids in the local theory of Banach spaces can be found in the two important handbook surveys [1579] by Lindenstrauss and V. Milman, and [960] by Giannopoulos and Milman, discussed also in our subsection 12.4.5. In Section 4 of [1579] the authors discuss, besides John's theorem and related ellipsoids in this framework, the role of *inertia ellipsoids* and *M-ellipsoids*.

And in [960] Giannopoulos and V. Milman emphasize that an important step to study a finite-dimensional normed space (or a convex body) is the introduction of a proper Euclidean structure on it. This can be represented by a maximum volume ellipsoid inside (or by a minimum volume ellipsoid containing) the respective unit ball (or convex body). The authors describe also various further Euclidean structures.

APPLICATIONS: OPTIMIZATION

Also in *optimization* Löwner–John ellipsoids play an essential role; e.g., they are used to bring convex bodies K by an affine transformation into a needed optimal position, e.g., the *Löwner–John-position* (with the unit ball as minimal volume ellipsoid). It is clearly useful to bring convex bodies into Löwner–John-positions, since then, e.g., almost degenerate, flat bodies are avoided.

John's result on sandwiching a convex body between two concentric ellipsoids (best possible for the simplex) is basic for *computational approximations of Löwner–John ellipsoids* via the ellipsoid method (see Section 3.1 of Grötschel et al. [1047] and, for asymptotical improvement, [1431]), and also via the Löwner–John-section approach due to Tarasov, Khachiyan, and Erlikh [2345]. Here we also cite once more Section 9 of the handbook article [1025] of Gritzmann and Klee, and Section 27.5 of the handbook article [1028], by the same authors.

For further related aspects, referring to Khachiyan's polynomial time algorithm for solving linear programming tasks (based on sequences of ellipsoids of strictly decreasing volume and called *ellipsoid algorithm*) and to Karmarkar's projective algorithm, we refer to the surveys [2372] of Todd and Yildirim, [1179] of Henk, and [1025] of Gritzmann and Klee (Section 9 there).

Juhnke (see [1330] and [1331]) studies Löwner–John ellipsoids from the viewpoint of *nonlinear semi-infinite optimization* using support functions, and he derives also uniqueness conditions via "extended convexity properties" of both the objective functions and the feasible regions.

Dwyer and Eddy [784] deal with so-called empty sets, i.e., with sets from the Euclidean unit ball B whose interior contains no point of a finite point set also lying in B. Asking for *volume-maximal empty sets*, they show that the question for empty ellipsoids helps to approach the general case (which, except for finite sets of small cardinality, seems to be very hard).

Gruber [1071] studies *minimum distance ellipsoids* of convex bodies in the sense of optimal sandwiching of convex bodies between two ellipsoids, and he uses John's criteria to handle the symmetric case in the Baire category sense. The paper [1475] of Kumar and Yildirim is dedicated to the ε-approximation of Löwner ellipsoids of given point sets by a modification of Khachyian's method, using a suitable sequence of ellipsoids.

Finally we mention here Todd's recent booklet [2371] on *minimum volume ellipsoids*, interesting for several modern disciplines like *(matrix) optimization, computational geometry, convex analysis, data representations*, and *statistics*. In the last chapter, *polytope approximation* plays an essential role.

FURTHER APPLICATIONS

Dwyer and Eddy [784] study empty sets in given point sets X (so named if their interior contains no point from X), and in particular they investigate *volume-maximal empty ellipsoids* from the viewpoint of computational geometry.

Lin and Bioucas-Dias [1574] study non-negative *blind source separation* which is often an inverse problem, e.g., when the mixing system is ill-conditioned. In particular, the authors deal with hyperspectral unmixing (HU) in remote sensing. Here HU is a *matrix factorization problem* factoring the so-called endmember matrix (which holds the material hyperspectral signatures) and the abundance matrix (which holds the material fractions at each image pixel). To create a new way for attacking such problems, the authors use *John's maximum volume ellipsoid* E inscribed in the convex hull of the data set, then affinely mapping E into a Euclidean ball. Applying the same affinity to the data mixtures, the authors study the endmember matrix associated with the mapped data; it turns out that the (preconditioned) endmembers form a regular structure that can be algorithmically used, and the authors substantiate their new method by numerical simulations and experiments with real data.

9.6.2 Positions

A *position* of a convex body K is the image of K under a non-degenerate affine transformation. In many cases, the view on this notion is restricted only to volume-preserving affinities. A good example of a position (also related to our theme above) is the *John position* that a convex body has if its John ellipsoid is the Euclidean unit ball. Further on, a convex body K is said to be in *isotropic position* (or shortly *isotropic*) if it has volume 1, its center of gravity is the origin, and the covariance matrix of a random vector uniformly distributed in K is the identity matrix times a constant, which is called the *isotropic constant* of K.

THE NOTION IN BOOKS AND SURVEYS

Up to orthogonal transformations, each convex body has a unique affine image being *isotropic*, and the Euclidean ball has the smallest isotropic constant, thus yielding a lower bound for the isotropic constant by an absolute positive constant. Schneider nicely summarizes the relations between the notions of *positions, isotropic measures*, and *reverse inequalities* in convexity in Section 10.13 (and its notes) of his book [2167]. Various of our references above, e.g., referring to *John and Löwner ellipsoids* and to *mean width*, are also discussed there, and basic results (like *Ball's reverse isoperimetric inequality* and the *Ball–Barthe inequality for isotropic measures*) are presented with basic proof ideas.

Shortly after (the second edition of) [2167] the book [496] of Brazitikos et al. appeared, which is dedicated to *isotropic positions and isotropic convex bodies*. In Chapter 1 classical positions (like *John, minimal mean width, and minimal surface area position*) and reverse inequalities are comprehensively discussed, and in Chapter 2 *isotropic log-concave measures*, their concentration inequalities, and the

convex bodies associated with them are studied. The *isotropic constant conjecture* follows, like also *bodies with maximal isotropic constants* and the *M-position* (defined here below). The next monograph containing a lot of material on positions is [133] by Artstein-Avidan et al.; here the classical positions (as mentioned also for [496]) and *reverse inequalities* fill Chapter 2. In Chapter 8 the M-position and in Chapter 10 the isotropic position and isotropic log-concave measures are treated. And in the second part [134] *isotropic log-concave probability measures* are discussed (see Section 2.1 there), whereas isotropic measure and position are also treated in subsection 7.3.1, in connection with Steiner symmetrizations.

Two surveys, in which isotropic positions, ellipsoids, and further related notions from convex geometry play an important role, are [1794] by Milman and Pajor as well as [249] of Bastero, Bernués, and Romance.

We continue with selected results on (mainly isotropic) positions which are not discussed in these three books.

SELECTED RESULTS: INEQUALITIES AND EXTREMAL PROBLEMS

In Euclidean n-space, a convex body K of unit volume is said to be in *M-position* with positive constant c if the volume of $(K + B_n)^{1/n}$ is not larger than c, B_n denoting a ball of unit volume. And a convex body K is in *minimal surface area position* if its surface area is minimum among the affine images of K of the same volume. Giannopoulos and Milman asked whether every convex body of unit volume in minimal surface area position is also in M-position. Saroglou [2113] answers this question in the negative, even already for 1-unconditional convex bodies (which are symmetric to all coordinate hyperplanes). Further related results (also on isotropic constants) are given.

Alonso-Gutiérrez [72] studies convex bodies in isotropic position and gives elegant geometric proofs of results on the isotropy constants of convex bodies which were known before only for the centrally symmetric subcase. These results refer to convex polytopes of given vertex and given facet numbers, and the used tools are, among other things, John and Löwner positions.

Petty showed that a convex body has minimal surface area among all its SL(n) images iff its surface area measure is isotropic. Introducing *minimal Orlicz surface area*, Zou and Xiong [2577] extend this fact to the *Orlicz setting*, establishing also the analogue of Ball's reverse isoperimetric inequality. And analogously introducing *minimal dual Orlicz surface area*, Ma [1649] derives the dual results and the dual form of Ball's isoperimetric inequality.

For K an n-dimensional origin-symmetric convex body in *isotropic position*, E. Milman [1785] gives an upper bound for the (half) mean width in terms of the isotropic constant and a universal constant. This improves former related results and implies that the isotropic position is (up to the isotropic constant term of K) an almost 2-regular M-position. It extends to arbitrary positions and also the mean width of L_p-centroid bodies. The question of whether it is possible to remove the isotropic constant term is essentially equivalent to the *slicing problem*.

Giannopoulos et al. [956] treat symmetric convex bodies K in isotropic position. It is well known that for the circumradius $R(K)$ and the isotropic constant L_K of K the following holds: there is an absolute constant c such that $cL_K \leq R(C) \leq (n+1)L_K$. The authors of [956] extend this to lower dimensional sections of K and derive also further results, e.g., on centroid bodies of log-concave measures.

Giannopoulos et al. [958] establish inequalities comparing the surface area of a convex body K with the minimal, average, or maximal surface area of its hyperplane or lower dimensional projections. Analogous problems for quermassintegrals of K are discussed, too. Needed tools are, e.g., the *minimal surface area position* and the volume of the projection body. The obtained results are related to the famous *hyperplane conjecture*, with sections replaced by projections, and volume replaced by surface area.

As we said above, there is a lower bound for the isotropic constant by an absolute positive constant. It is interesting to ask whether there exists an upper bound being also an absolute constant. For example, this question is again related to the hyperplane conjecture. In [86], Alonso-Gutiérrez et al. use *random polytopes* as a tool for getting new results in this direction. The authors estimate the isotropic constant of a random polytope in an isotropic convex body K from above, using the symmetric convex hull of independent random vectors uniformly distributed in K. They get an upper bound that is very interesting in this environment. The papers [1210] of Hörrmann et al. and [1998] of Prochno et al. go into the same direction; for unconditional bodies the authors argue that even an absolute constant c forms the upper bound, with overwhelming probability, and they verify the hyperplane conjecture for this model.

For a convex body K, González Merino and Schymura [990] study sharp estimates and asymptotics on the *isodiametric and isominwidth quotients* of K (i.e., the volume of K divided by the n-th power of the diameter, and the nth power of the minimum width, respectively). They succeed with the isodiametric case (and for the isodiametric position) with nice lower bounds and derive, for the symmetric case, a nice characterization of the regular cross-polytope. Regarding the isominwidth case the cube occurs as an extremal case. They characterize completely the *Behrend position* of convex bodies and discuss the relation to John's and other positions. In the introductory part, the authors survey related problems and results in a clear manner, thus giving a refreshing new view on the field.

APPLICATIONS

The paper [1199] of Hinrichs et al. refers to information-based complexity (IBC); the authors prove the curse of dimensionality in the worst-case setting for multivariate numerical integration with respect to various classes of smooth functions. (It should be noted that in (IBC) it is essential to know when problems there suffer from the curse of dimensionality.) The domains they use are *isotropic convex bodies* with small diameter, and sequences of symmetric, isotropic convex bodies are important. The authors say that this is the first paper in which results from asymptotic geometric analysis have been used to study a problem in (IBC).

PLATE 10

Zonotopes and related polytope types

Translative space-fillers

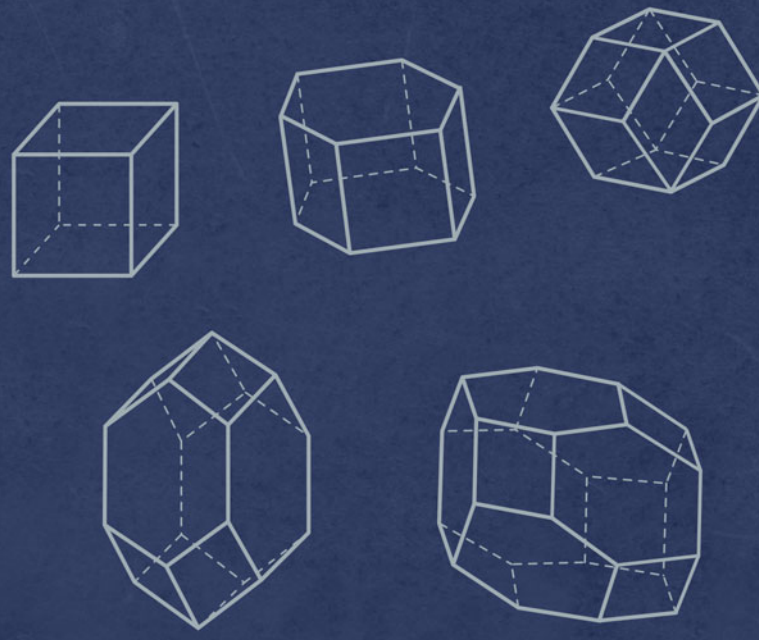
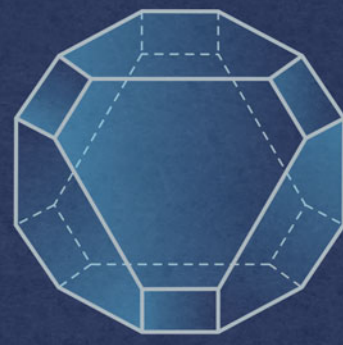


Figure 27

A belt polytope

Figure 28



Zonotopes and related polytope types ► (see, e.g., 2.9.1.11, 10.2, and 10.6.1)

Chapter 10
Affine convex geometry – Part 2

"The only way to learn mathematics is to do mathematics."
— Paul Halmos

The main objective of this chapter is to study some important (starlike or convex) bodies associated to a given convex body. These are the so-called *associated bodies*, and they play a central role in the affine geometry of convex bodies. For example, we state and prove the *Petty projection inequality*, which is stronger than the classical isoperimetric inequality.

Recall that we regard the space CC^n (of non-empty compact convex subsets of $\mathbb{R}^n$) as a metric space (with the Hausdorff distance) endowed with Minkowski addition and scalar multiplication operations. Subsets of CC^n which, in a certain sense, "preserve" these structures are known as *Minkowski classes*. We briefly investigate this concept in Section 10.1.

In Section 10.2, we introduce *zonotopes*, which are Minkowski sums of finitely many segments. Further on, *zonoids* are the Hausdorff limits of the zonotopes. As we will see, in some sense zonoids form the simplest Minkowski class of CC^n. It is worth mentioning that zonoids appear in various contexts in convexity and also in other areas of mathematics (in this direction, see Schneider and Weil [2176] and references therein). Plate 10 shows the five types of zonotopes which are the *translative space-fillers* in three dimensions. And Figure 28 there presents a generalization of the notion of zonotopes, a so-called belt polytope. It has a segment summand parallel to each of its edges.

The central concepts of *projection*, *centroid*, and *intersection* bodies are studied in Section 10.3. Roughly speaking, the projection and the intersection bodies of a given convex body K are defined via their support and radial functions, respectively; in a direction-dependent way these represent the $(n-1)$-dimensional volumes of projections and sections of K. These notions have been extensively studied in the literature, and this is the reason why we present them here. A particular aspect that we emphasize is that of geometric inequalities involving these bodies. We discuss this topic in Section 10.4, where, among other inequalities, we prove the classical *Petty projection inequality*.

10.1 Minkowski classes

A subset $\mathcal{M} \subseteq CC^n$ which is closed for Minkowski sums and positive scalar multiplications (that is, closed for Minkowski combinations), and also closed in the Hausdorff metric, is called a *Minkowski class* (notice that it is clear that CC^n is a Minkowski class itself). To illustrate this concept, we start with an easy example. Recall that a set $A \subseteq \mathbb{R}^n$ is said to be *centrally symmetric* if there exists a point $x_0 \in A$ such that $(A - x_0) = -(A - x_0)$. In this case, we say that A is *symmetric with respect to* x_0.

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_10

Lemma 10.1.1. *The subset of CC^n consisting of the centrally symmetric sets is a Minkowski class.*

Proof. Let $C, D \in CC^n$ be centrally symmetric, and let $x_0 \in C$ and $z_0 \in D$ be such that $C - x_0 = -(C - x_0)$ and $D - z_0 = -(D - z_0)$. Then

$$C + D - x_0 - z_0 = (C - x_0) + (D - z_0) = -(C - x_0) - (D - z_0) = \\ = -(C + D - x_0 - z_0),$$

and hence $C + D$ is symmetric with respect to $x_0 + z_0$. Also, if $\lambda > 0$, then it is straightforward to see that λC is symmetric with respect to λx_0. This proves that the set of centrally symmetric sets of CC^n is closed for Minkowski combinations.

It remains to prove that the Hausdorff limit of a converging sequence of centrally symmetric sets is centrally symmetric. For that sake, let $(C_m)_{m\in\mathbb{N}}$ be a sequence in CC^n such that $C_m \to C$ for some $C \in CC^n$, and assume that, for each $m \in \mathbb{N}$, the set C_m is symmetric with respect to $x_m \in C_m$. Once (C_m) is a converging sequence, we have that (C_m) is bounded in the Hausdorff metric, and by Proposition 2.5.3 we get that (C_m) is also bounded as a family of subsets of $\mathbb{R}^n$. In particular, (x_m) is a bounded sequence of points of $\mathbb{R}^n$, and passing to a subsequence if necessary we may assume that $x_m \to x_0$ for some $x_0 \in \mathbb{R}^n$. We claim that C is symmetric with respect to x_0. To verify that, simply recall that $C_m - x_m = -(C_m - x_m)$ for every $m \in \mathbb{N}$ and use Lemma 2.6.1. Indeed, since the Minkowski addition is continuous in the Hausdorff metric, and translations are, in particular, Minkowski additions in CC^n, we get that $C - x_0 = -(C - x_0)$. Compare this argument with Exercise 2.51.

□

Remark 10.1.1. We proved that the Hausdorff limit C of (C_m) is centered at a point which is the limit of a subsequence of the centers of the sets C_m. As a consequence, we have that if all the sets C_m are centered at the same point $z \in \mathbb{R}^n$, then C is also centered at z.

When we identify sets of CC^n with their respective support functions (restricted to $\mathbb{S}^{n-1}$, say), the centrally symmetric ones correspond, up to translations, to the even functions. From the functional viewpoint, the fact that the class of centrally symmetric sets of CC^n is closed with respect to the Hausdorff metric is equivalent to the fact that the even functions of $\mathbb{S}^{n-1}$ are also closed in $C(\mathbb{S}^{n-1})$ in the maximum norm. Similarly, the constant functions correspond to the *bodies of constant width* (up to translations, again). Recall that a set $K \in CC^n$ is said to have *constant width* if its width map $w_K : \mathbb{S}^{n-1} \to \mathbb{R}$ defined for each $u \in \mathbb{S}^{n-1}$ by

$$w_K(u) = h_K(u) + h_K(-u)$$

is constant. Below we prove that these bodies also form a Minkowski class. Of course, this is related to the fact that the constant functions in $\mathbb{S}^{n-1}$ are a closed family in $C(\mathbb{S}^{n-1})$ with respect to the maximum norm.

Proposition 10.1.1. *The family of all bodies of constant width of $\mathbb{R}^n$ is a Minkowski class.*

Proof. For simplicity, let us denote the family of bodies of constant width of $\mathbb{R}^n$ by $C\mathcal{W}^n$. Since $h_{K+\lambda L} = h_K + \lambda h_L$ for any $K, L \in CC^n$ and all $\lambda \geq 0$, we have immediately that $C\mathcal{W}^n$ is closed for Minkowski combinations. Thus, it remains to prove that $C\mathcal{W}^n$ is closed in CC^n with respect to the Hausdorff metric. For that sake, let $(K_m)_{m\in\mathbb{N}}$ be a sequence of bodies of constant width converging to a set $C \in CC^n$. It follows from Proposition 2.5.2 (see also Remark 2.5.3) that the width functions $(w_{K_m})_{m\in\mathbb{N}}$ (restricted to $\mathbb{S}^{n-1}$) form a sequence of constant functions converging to w_C in the maximum norm of $C(\mathbb{S}^{n-1})$. It is immediate to check that this gives that w_C is constant in $\mathbb{S}^{n-1}$. Hence, $C \in C\mathcal{W}^n$.

□

We move forward investigating Minkowski classes which are closed under certain groups of transformations of $\mathbb{R}^n$. First, for any given $K \in CC^n$ consider the family of positions of K:

$$L(K) := \Big\{x + T(K) : x \in \mathbb{R}^n \text{ and } T \in \mathrm{L}(n)\Big\}.$$

The set

$$\mathcal{L}(K) := \mathrm{cl}_{\mathcal{H}}\big\{\lambda_1 L_1 + \ldots + \lambda_m L_m : m \in \mathbb{N}, \lambda_j > 0 \text{ and } L_j \in L(K) \text{ for all } 1 \le j \le m\big\},$$

where the closure is taken in the Hausdorff metric, of course, is a Minkowski class (generated by the positions of K). The fact that $\mathcal{L}(K)$ is closed for Minkowski combinations is immediate. To prove that $\mathcal{L}(K)$ is closed in CC^n, one just has to take the same argument as used in Theorem 9.1.2 to prove that the (non-degenerate) Hausdorff limit of a sequence of ellipsoids is an ellipsoid.

If $\mathcal{G}$ is a subgroup of $\mathrm{L}(n)$, then we say that a given Minkowski class $\mathcal{M} \subseteq CC^n$ is *$\mathcal{G}$-invariant* if $T(K) \in \mathcal{M}$ for any $T \in \mathcal{G}$ and every $K \in \mathcal{M}$. Notice that for every fixed $K \in CC^n$ the class $\mathcal{L}(K)$ is $\mathrm{L}(n)$-invariant. The family of centrally symmetric compact convex sets of $\mathbb{R}^n$ is also clearly $\mathrm{L}(n)$-invariant. Observe also that the class of constant width bodies is not $\mathrm{L}(n)$-invariant, but it is obviously $\mathrm{O}(n)$-invariant.

The class $\mathcal{L}(K)$ is a particular case of a much more general construction: for any non-empty subset $\mathcal{M} \subseteq CC^n$ and every subgroup $\mathcal{G} \subseteq \mathrm{L}(n)$, it is not difficult to see that the family

$$\mathrm{cl}_{\mathcal{H}}\big\{\lambda_1 T_1(K_1) + \ldots + \lambda_m T_m(K_m) : m \in \mathbb{N},\ \lambda_j > 0,\ T_j \in \mathcal{G},\ K_j \in \mathcal{M} \text{ for all } 1 \le j \le m\big\} \quad (10.1.1)$$

is a $\mathcal{G}$-invariant Minkowski class. In Exercise 10.1, the reader is invited to give a formal proof of this fact. The natural question that arises is whether we can re-obtain CC^n in this way from a smaller family of sets, but this is not the case. The proof of this fact is not in the scope of this book, but can be found in [2167, Section 3.5].

For us, the most important Minkowski class is that of *zonoids*, which are $\mathcal{L}(K)$ for any non-degenerate segment K. That is, zonoids are a Minkowski class generated by segments. We investigate it in the next section.

10.2 Zonotopes and zonoids

A *zonotope* of $\mathbb{R}^n$ is a Minkowski sum of finitely many segments of $\mathbb{R}^n$ (the sum having at least one segment). That is, a zonotope is a set having the form

$$Z = \mathrm{seg}[a_1, b_1] + \ldots + \mathrm{seg}[a_k, b_k],$$

where $a_1, \ldots, a_k, b_1, \ldots, b_k \in \mathbb{R}^n$. See Figure 10.2.1. Notice carefully that the segments may be degenerate (meaning their endpoints are the same), and hence we assume that points are zonotopes. It is readily seen that any zonotope is a non-empty compact and convex set (because the Minkowski sum of convex sets is always convex, and any segment is a convex set). We say that a zonotope is *proper* if it is a convex body (that is, if it has maximal affine dimension). Even more, every zonotope is a polytope, because it is a sum of polytopes (and every segment is a polytope, of course). The fact that a Minkowski sum of polytopes is a polytope is given in Lemma 6.1.1. Next we show that, up to translations, zonotopes are Minkowski sums of centrally symmetric segments.

Lemma 10.2.1. *Let $Z \subseteq \mathbb{R}^n$ be a zonotope. Up to a translation, Z is the sum of finitely many segments centered at the origin. In particular, any zonotope is centrally symmetric.*

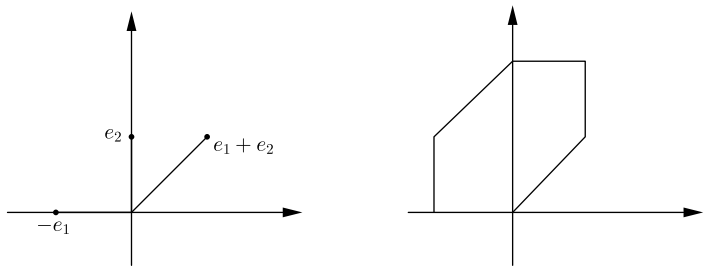


Fig. 10.2.1 The Minkowski sum of the segments $\mathrm{seg}[0_{\mathbb{R}^n}, -e_1]$, $\mathrm{seg}[0_{\mathbb{R}^n}, e_2]$, and $\mathrm{seg}[0_{\mathbb{R}^n}, e_1 + e_2]$.

Proof. Write

$$Z = \mathrm{seg}[a_1, b_1] + \ldots + \mathrm{seg}[a_k, b_k].$$

For each $j = 1, \ldots, k$, we may write

$$\mathrm{seg}[a_j, b_j] = \frac{a_j + b_j}{2} + \mathrm{seg}\left[\frac{a_j - b_j}{2}, \frac{b_j - a_j}{2}\right],$$

and the segment in the right-hand side is clearly symmetric with respect to the origin (geometrically, we are simply translating the midpoint of the segment so that it lies in the origin). For simplicity, we denote $\mathrm{seg}[a_j, b_j] = z_j + \mathrm{seg}[-v_j, v_j]$. Thus,

$$\begin{aligned} Z &= z_1 + \mathrm{seg}[-v_1, v_1] + \ldots + z_k + \mathrm{seg}[-v_k, v_k] = \\ &= (z_1 + \ldots + z_k) + \mathrm{seg}[-v_1, v_1] + \ldots + \mathrm{seg}[-v_k, v_k], \end{aligned}$$

and the right-hand side is a translation (by the factor $z_1 + \ldots + z_k$) of a Minkowski sum of segments centered at the origin.

To prove the remaining claim, notice that since we may write any zonotope as $Z = z + \mathrm{seg}[-v_1, v_1] + \ldots + \mathrm{seg}[-v_k, v_k]$ it is clear that $Z - z = -(Z - z)$, and hence Z is symmetric with respect to $z \in Z$. □

Remark 10.2.1. Writing a zonotope as a translation of a Minkowski sum of origin-symmetric segments has also another useful consequence: it is very easy to obtain an expression for the support function of a zonotope. Indeed, for a zonotope $Z = z + \mathrm{seg}[-v_1, v_1] + \ldots + \mathrm{seg}[-v_k, v_k]$ we have that

$$h_Z(x) = h_{\{z\}}(x) + h_{\mathrm{seg}[-v_1, v_1]}(x) + \ldots + h_{\mathrm{seg}[-v_k, v_k]}(x) = \langle z, x\rangle + \sum_{j=1}^{k} |\langle x, v_j\rangle| \tag{10.2.1}$$

for any $x \in \mathbb{R}^n$. Of course, we may always assume that $v_j \neq 0$ for each $j \in \{1, \ldots, k\}$, since otherwise the corresponding segment would be simply $0_{\mathbb{R}^n}$ (which is the neutral element for Minkowski sums). In this case, we can write $v_j = \beta_j u_j$ for some $\beta_j > 0$ and $u_j \in \mathbb{S}^{n-1}$. By doing that, the support function of Z reads

$$h_Z(x) = \langle z, x\rangle + \sum_{j=1}^{k} \beta_j |\langle x, u_j\rangle|, \tag{10.2.2}$$

for every $x \in \mathbb{R}^n$. Conversely, any set of CC^n having the support function in this form is a zonotope (because sets of CC^n which have the same support function are equal).

Observe that the expressions (10.2.1) and (10.2.2) make explicit that the center of Z is z. In particular, the support function of any zonotope Z_0 centered at the origin can be written as

$$h_{Z_0}(x) = \sum_{j=1}^{k} \beta_j |\langle x, u_j \rangle|, \tag{10.2.3}$$

for some numbers $\beta_1, \ldots, \beta_k > 0$ and some $u_1, \ldots, u_k \in \mathbb{S}^{n-1}$.

Recall that $\mathbf{h}(P, u)$ represents the unique supporting hyperplane of a polytope P having unit outer normal $u \in \mathbb{S}^{n-1}$. We also denote $F(P, u) := P \cap \mathbf{h}(P, u)$, and by definition $F(P, u)$ is an exposed face of P. Once every proper face of a polytope is exposed (see Corollary 3.1.1), we get that every proper face of P is of the form $F(P, u)$ for some $u \in \mathbb{S}^{n-1}$. Letting $Z = \operatorname{seg}[a_1, b_1] + \ldots + \operatorname{seg}[a_k, b_k]$ be a zonotope, we get from Lemma 6.1.2 that

$$F(Z, u) = F(\operatorname{seg}[a_1, b_1], u) + \ldots + F(\operatorname{seg}[a_k, b_k], u),$$

for each $u \in \mathbb{S}^{n-1}$ (for an illustration of this equality, let, e.g., $u = -e_1$ in Figure 10.2.1). Once the intersection of a segment with a supporting hyperplane is either the segment itself or a point, the right-hand side of the equality above is a Minkowski sum of finitely many segments. Thus, every face of a zonotope is also a zonotope. In particular, every face of a zonotope is centrally symmetric. One can also show that a convex polytope all whose two-dimensional faces are centrally symmetric is a zonotope (see [2167, Theorem 3.5.2]).

A set $Z \in CC^n$ is a *zonoid* if it is the limit in the Hausdorff metric of a sequence of zonotopes. A *proper zonoid* is a zonoid which is a convex body. As we will see next, the family of zonoids is a Minkowski class, and since it is generated by segments, it can be seen as the simplest non-trivial Minkowski class of CC^n.

Proposition 10.2.1. *The subset $\mathcal{Z}^n \subseteq CC^n$ consisting of the zonoids is a Minkowski class.*

Proof. Let Z and W be zonoids which are the limits in the Hausdorff metric of sequences $(C_m)_{m\in\mathbb{N}}$ and $(D_m)_{m\in\mathbb{N}}$ of zonotopes, respectively. For any $\lambda \geq 0$, $C_m + \lambda D_m \to Z + \lambda W$ by Lemma 2.6.1. Because any sum and scalar multiplication of zonotopes is a zonotope, we obtain that $Z + \lambda W$ is a zonoid, and this shows that the family of zonoids of $\mathbb{R}^n$ is closed for Minkowski combinations.

It remains to prove that $\mathcal{Z}^n$ is closed under Hausdorff limits. Let $(Z_m)_{m\in\mathbb{N}}$ be a sequence of zonoids converging in the Hausdorff metric to a set $Z \in CC^n$. For each $m \in \mathbb{N}$, we may choose a zonotope $C_m \in CC^n$ such that $d_{\mathcal{H}}(C_m, Z_m) < 1/m$. It is immediate to see that $C_m \to Z$, and hence Z is a zonoid. This concludes the proof.

□

It turns out that, as in the case of zonotopes, we may characterize zonoids in terms of their support functions. To do that, we need an auxiliary result on weak convergence of measures which we state now. Remember that a sequence $(\mu_j)_{j\in\mathbb{N}}$ of (finite, say) Borel measures on a compact topological space X *converges weakly* to a Borel measure μ on X if

$$\int_X f \, d\mu_j \to \int_X f \, d\mu$$

for every function $f \in C(X)$, where we recall that $C(X)$ denote the space of all continuous functions $f\colon X \to \mathbb{R}$.

Theorem 10.2.1. *Let $(\mu_j)_{j\in\mathbb{N}}$ be a sequence of Borel measures on $\mathbb{S}^{n-1}$ which is* bounded *in the sense that there exists a number $c > 0$ such that $\mu_j(\mathbb{S}^{n-1}) < c$ for every $j \in \mathbb{N}$. Then there is a subsequence of (μ_j) which converges weakly to a finite Borel measure μ on $\mathbb{S}^{n-1}$.*

Proof. See, e.g., [775].

□

In what follows, we remind the reader that a Borel measure μ on $\mathbb{S}^{n-1}$ is said to be *even* if $\mu(E) = \mu(-E)$ for every set $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$ (this can also be expressed by writing $-\mathrm{Id}_*\mu = \mu$). As it was commented in Section 7.2, the weak limit of a sequence of even measures is even.

Theorem 10.2.2. *A set $K \in CC^n$ is a zonoid centered at the origin if and only if there exists a finite even Borel measure μ on $\mathbb{S}^{n-1}$ such that*

$$h_K(x) = \int_{\mathbb{S}^{n-1}} |\langle x, u\rangle| \, d\mu(u) \tag{10.2.4}$$

for any $x \in \mathbb{R}^n$. In this case, μ is said to be a generating measure *of K.*

Remark 10.2.2. If Z_0 is a zonotope centered at the origin written as

$$Z_0 = \beta_1 \mathrm{seg}[-u_1, u_1] + \ldots + \beta_k \mathrm{seg}[-u_k, u_k]$$

for $\beta_1, \ldots, \beta_k > 0$ and $u_1, \ldots, u_k \in \mathbb{S}^{n-1}$, then the support function h_{Z_0} is given as in (10.2.3), and we get that

$$h_{Z_0}(x) = \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \, d\mu(v)$$

where μ is the combination of Dirac delta measures given as

$$\mu = \sum_{j=1}^{k} \frac{\beta_j}{2} \left(\delta_{u_j} + \delta_{-u_j}\right),$$

which is clearly a finite even Borel measure on $\mathbb{S}^{n-1}$. Here, δ_{u_j} is the usual Dirac delta measure. This happens because of the equality

$$\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \, d\delta_{u_j}(v) = |\langle u_j, x\rangle|,$$

that is immediate from the definition of the Dirac delta measure (the integral of any function with respect to a Dirac delta measure is simply its value in the point where the mass is concentrated; see the definition of the Lebesgue integral in Appendix A).

Proof. Assume first that the support function of K is given as in (10.2.4) for some even finite Borel measure μ on $\mathbb{S}^{n-1}$. We may assume that μ is non-zero (i.e., that there exists a Borel subset $E \subseteq \mathbb{S}^{n-1}$ such that $\mu(E) > 0$), because if μ is the zero measure, then it is clear that $K = \{0_{\mathbb{R}^n}\}$. For an arbitrarily given $\varepsilon > 0$, choose a partition $\{E_1, \ldots, E_k\}$ of $\mathbb{S}^{n-1}$ consisting of non-empty disjoint Borel sets, each of which having diameter (calculated in the metric of $\mathbb{R}^n$, say) less than $\varepsilon/\mu(\mathbb{S}^{n-1})$. For each $j \in \{1, \ldots, k\}$, fix a point $u_j \in E_j$. For any $u \in \mathbb{S}^{n-1}$, we estimate

$$\left|\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,d\mu(v) - \sum_{j=1}^{k}\mu(E_j)|\langle u,u_j\rangle|\right| = \left|\int_{\mathbb{S}^{n-1}} |\langle u,v\rangle|\,d\mu(v) - \sum_{j=1}^{k}\int_{E_j}|\langle u,u_j\rangle|\,d\mu(v)\right| =$$

$$= \left|\sum_{j=1}^{k}\int_{E_j}|\langle u,v\rangle|\,d\mu(v) - \sum_{j=1}^{k}\int_{E_j}|\langle u,u_j\rangle|\,d\mu(v)\right| = \left|\sum_{j=1}^{k}\int_{E_j}|\langle u,v\rangle| - |\langle u,u_j\rangle|\,d\mu(v)\right| \leq$$

$$\leq \sum_{j=1}^{k}\int_{E_j}\big||\langle u,v\rangle| - |\langle u,u_j\rangle|\big|\,d\mu(v) \leq \sum_{j=1}^{k}\int_{E_j}|\langle u,v-u_j\rangle|\,d\mu(v) \leq \sum_{j=1}^{k}\int_{E_j}|u||v-u_j|\,d\mu(v) \leq$$

$$\leq \sum_{j=1}^{k}\int_{E_j}\operatorname{diam}(E_j)\,d\mu(v) \leq \sum_{j=1}^{k}\frac{\varepsilon}{\mu(\mathbb{S}^{n-1})}\int_{E_j}d\mu(v) = \frac{\varepsilon}{\mu(\mathbb{S}^{n-1})}\sum_{j=1}^{k}\mu(E_j) = \varepsilon.$$

Thus, if Z is the zonotope whose support function is given as

$$h_Z(x) = \sum_{j=1}^{k}\mu(E_j)|\langle x,u_j\rangle|,$$

we get from the estimate above that

$$|h_K(u) - h_Z(u)| \leq \varepsilon$$

for every $u \in \mathbb{S}^{n-1}$. From Proposition 2.5.2, it follows that

$$d_{\mathcal{H}}(K,Z) = \sup\left\{|h_K(u) - h_Z(u)| : u \in \mathbb{S}^{n-1}\right\} \leq \varepsilon.$$

This shows that K can be approximated (in the Hausdorff metric) by zonotopes, and then K is a zonoid.

Now let K be a zonoid centered at the origin, and assume that $(Z_m)_{m\in\mathbb{N}}$ be a sequence of zonotopes such that $K = \lim_{m\to\infty} Z_m$, the limit taken in the Hausdorff metric. Thus,

$$K = \frac{K-K}{2} = \lim_{m\to\infty}\frac{Z_m - Z_m}{2},$$

where the last equality comes from Exercise 2.50 (or from the arguments given in the proof of Proposition 10.2.1), and it is clear that $(Z_m - Z_m)/2$ is a zonotope centered at the origin. Therefore, K is the limit of a sequence of zonotopes centered at the origin, and for the sake of simplicity we may assume that each Z_m is centered at the origin. Hence, for each $m \in \mathbb{N}$, in view of Remark 10.2.2 we may write the support function of Z_m as

$$h_{Z_m}(x) = \int_{\mathbb{S}^{n-1}}|\langle x,v\rangle|\,d\mu_m(v),$$

for some even measure μ_m. We claim that $(\mu_m)_{m\in\mathbb{N}}$ is a bounded sequence of measures in the sense of Theorem 10.2.1. To prove this, recall that for every $u \in \mathbb{S}^{n-1}$ we have the equality

$$2\kappa_{n-1} = \int_{\mathbb{S}^{n-1}}|\langle u,v\rangle|\,d\mathcal{H}^{n-1}(v),$$

as it is discussed in Remark 6.5.3, and where κ_{n-1} denotes the (Lebesgue) volume of the unit ball of $\mathbb{R}^{n-1}$. Integrating both sides over $\mathbb{S}^{n-1}$ with respect to μ_m we get

$$\int_{\mathbb{S}^{n-1}} 2\kappa_{n-1}\, d\mu_m(u) = \int_{\mathbb{S}^{n-1}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mathcal{H}^{n-1}(v) d\mu_m(u).$$

Now we rewrite the left-hand side in the obvious way (it is the integral of a constant) and use Fubini's theorem on the right-hand side to arrive at

$$2\kappa_{n-1}\mu_m(\mathbb{S}^{n-1}) = \int_{\mathbb{S}^{n-1}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mu_m(u) d\mathcal{H}^{n-1}(v) = \int_{\mathbb{S}^{n-1}} h_{Z_m}(v)\, d\mathcal{H}^{n-1}(v).$$

We warn the interested reader that the formal reason why we can use Fubini's theorem in this context (and in other parts of this chapter) is explained in the comment below Proposition A.5.4. Once (Z_m) converges in the Hausdorff metric, we get that it is a bounded sequence. By Proposition 2.5.3, this gives that there exists a number $\rho > 0$ such that $Z_m \subseteq \rho B^n$ for each $m \in \mathbb{N}$. It follows that $h_{Z_m}(v) \leq h_{\rho B^n}(v) = \rho$ for every $v \in \mathbb{S}^{n-1}$ and any $m \in \mathbb{N}$. Applying this estimate to the equality above yields

$$2\kappa_{n-1}\mu_m(\mathbb{S}^{n-1}) \leq \int_{\mathbb{S}^{n-1}} \rho\, d\mathcal{H}^{n-1}(v) = \rho\omega_n,$$

where we recall that ω_n denotes the usual area of the unit sphere, that is, $\omega_n = \mathcal{H}^{n-1}(\mathbb{S}^{n-1})$. Finally, this leads to

$$\mu_m(\mathbb{S}^{n-1}) \leq \frac{\rho\omega_n}{2\kappa_{n-1}},$$

and the right-hand side is a constant which does not depend on m. It follows that (μ_m) is indeed bounded. Therefore, passing to a subsequence if necessary, we may assume that μ_m converges weakly to a finite even measure μ on $\mathbb{S}^{n-1}$ (the weak limit is even because each μ_m is even; see the comment after the definition of even measures). For each $u \in \mathbb{S}^{n-1}$ and every $m \in \mathbb{N}$, we may write

$$\begin{aligned}
&\left| \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mu(v) - h_K(u) \right| \leq \\
&\leq \left| \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mu(v) - h_{Z_m}(u) \right| + \left| h_{Z_m}(u) - h_K(u) \right| \leq \\
&\leq \left| \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mu(v) - \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mu_m(v) \right| + d_{\mathcal{H}}(K, Z_m),
\end{aligned}$$

and, for any fixed $u \in \mathbb{S}^{n-1}$, the latter goes to 0 as $m \to \infty$. This happens because $Z_m \to K$ in the Hausdorff metric, and μ_m converges weakly to μ. This shows that

$$h_K(u) = \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mu(v)$$

for each $u \in \mathbb{S}^{n-1}$, and hence it follows immediately that (10.2.4) holds for every $x \in \mathbb{R}^n$ (since both sides of the equality above are positively homogeneous). The proof is complete.

□

Remark 10.2.3. It turns out that generating measures are unique, that is, for a zonoid $K \subseteq \mathbb{R}^n$ centered at the origin there is a unique finite even measure μ on $\mathbb{S}^{n-1}$ satisfying (10.2.4). This is an immediate consequence of Theorem 10.3.1 ahead. However, if one drops the hypothesis of μ being even, then uniqueness does not hold anymore.

Scholium 10.2.1. Let $(Z_m)_{m\in\mathbb{N}}$ be a sequence of zonotopes centered at the origin converging in the Hausdorff metric to a zonoid Z (also centered at the origin; see Remark 10.1.1). Assuming that μ_m is the generating measure of Z_m (for each $m \in \mathbb{N}$) and μ be the generating measure of Z, we have that (μ_m) converges weakly to μ.

As it will be clear later, the knowledge on zonoids is very useful to obtain some important affine geometric inequalities. For these later applications, now we derive explicit formulas for volumes and mixed volumes of zonoids. In what follows, we write

$$D_n(x_1,\ldots,x_n) := \mathrm{vol}_n\big(\mathrm{seg}[0_{\mathbb{R}^n},x_1]+\ldots+\mathrm{seg}[0_{\mathbb{R}^n},x_n]\big) \tag{10.2.5}$$

for any $x_1,\ldots,x_n \in \mathbb{R}^n$. The zonotope $\mathrm{seg}[0_{\mathbb{R}^n},x_1]+\ldots+\mathrm{seg}[0_{\mathbb{R}^n},x_n]$ is often called the *parallelotope* (or *affine cube*) *spanned by* $x_1,\ldots,x_n$. From Remark 6.2.6 we have the expressions

$$D_n(x_1,\ldots,x_n) = n!V\big(\mathrm{seg}[0_{\mathbb{R}^n},x_1],\ldots,\mathrm{seg}[0_{\mathbb{R}^n},x_n]\big) = |\det(x_1,\ldots,x_n)|,$$

where the determinant is the determinant of the matrix whose columns are the vectors $x_1,\ldots,x_n$. Notice carefully that, as a consequence of expressing $D_n(x_1,\ldots,x_n)$ as the determinant above, we get that D_n is symmetric and invariant under changing the sign of any entry. Even more, from the translation-invariance (and linearity) of mixed volumes we get the expression

$$\begin{aligned} V\big(\mathrm{seg}[-x_1,x_1],\ldots,\mathrm{seg}[-x_n,x_n]\big) &= V(-x_1+\mathrm{seg}[0_{\mathbb{R}^n},2x_1],\ldots,-x_n+\mathrm{seg}[0_{\mathbb{R}^n},2x_n]) = \\ &= V\big(2\mathrm{seg}[0_{\mathbb{R}^n},x_1],\ldots,2\mathrm{seg}[0_{\mathbb{R}^n},x_n]\big) = 2^nV(\mathrm{seg}[0_{\mathbb{R}^n},x_1],\ldots,\mathrm{seg}[0_{\mathbb{R}^n},x_n]) = \\ &= \frac{2^n}{n!}D_n(x_1,\ldots,x_n). \end{aligned} \tag{10.2.6}$$

Before moving forward, we will need to borrow another result from measure theory. This regards weak convergence of product measures. Once the proof is simple, we give it explicitly here.

Proposition 10.2.2. *Let $\mu_1,\ldots,\mu_m$ be finite Borel measures on compact spaces $X_1,\ldots,X_m$ (also Hausdorff and second countable), respectively, and assume that for each $k = 1,\ldots,m$ the sequence $(\mu_k^j)_{j\in\mathbb{N}}$ consists of finite Borel measures on X_k and converges weakly to μ_k as $j\to\infty$. Then*

$$\int_{X_m}\ldots\int_{X_1} f\,d\mu_1^j\ldots d\mu_m^j \to \int_{X_m}\ldots\int_{X_1} f\,d\mu_1\ldots d\mu_m$$

for any $f \in C(X_1\times\ldots\times X_m)$ as $j\to\infty$.

Proof. From Fubini's theorem, this is the same as saying that the sequence $(\mu_1^j\otimes\ldots\otimes\mu_m^j)_{j\in\mathbb{N}}$ of product measures converges weakly to the product measure $\mu_1\otimes\ldots\otimes\mu_m$ as $j\to\infty$. Thus, it is sufficient to prove the case where $m = 2$, once the general case comes from applying iteratively this particular one. Let $f \in C(X_1\times X_2)$. The function $g\colon X_2\to\mathbb{R}$ given as

$$g(x_2) = \int_{X_1} f(x_1,x_2)\,d\mu_1(x_1)$$

is clearly continuous (because f is continuous and μ_1 is finite). Therefore, the weak convergence $\mu_2^j\to\mu_2$ gives

$$\int_{X_2}\int_{X_1} f(x_1,x_2)\,d\mu_1 d\mu_2^j \to \int_{X_2}\int_{X_1} f(x_1,x_2)\,d\mu_1 d\mu_2$$

as $j\to\infty$. It follows that for an arbitrarily given $\varepsilon > 0$ there exists a number $j_0\in\mathbb{N}$ such that

$$\left|\int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2 - \int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2^j\right| < \frac{\varepsilon}{2},$$

whenever $j > j_0$. On the other hand, for each $k \in \mathbb{N}$, let $g_k \colon X_2 \to \mathbb{R}$ be given as

$$g_k(x_2) = \int_{X_1} f(x_1, x_2)\, d\mu_1^k(x_1).$$

From the weak convergence of μ_1^k to μ_1 (and because each function g_k is clearly continuous), we get that $g_k \to g$ pointwise. By standard analysis arguments, once X_2 is compact we have that the convergence is uniform. Because (μ_2^j) converges weakly to μ_2 (and since μ_2 is finite), we may let $c > 0$ be a constant such that $\mu_2^j(X_2) \leq c$ for every $j \in \mathbb{N}$. Once the convergence $g_k \to g$ is uniform, there exists $k_0 \in \mathbb{N}$ such that if $k > k_0$, then

$$|g_k(x_2) - g(x_2)| < \frac{\varepsilon}{2c}$$

for every $x_2 \in X_2$. Now, for any $j, k \in \mathbb{N}$ such that $j > j_0$ and $k > k_0$, we write

$$\begin{aligned}\left|\int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2 - \int_{X_2}\int_{X_1} f\,d\mu_1^k d\mu_2^j\right| &\leq \left|\int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2 - \int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2^j\right| + \\ + \left|\int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2^j - \int_{X_2}\int_{X_1} f\,d\mu_1^k d\mu_2^j\right| &\leq \left|\int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2 - \int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2^j\right| + \\ + \int_{X_2} |g(x_2) - g_k(x_2)| d\mu_2^j(x_2) &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2c}\mu_2^j(X_2) \leq \varepsilon.\end{aligned}$$

Hence, for any $j > \max\{j_0, k_0\}$, we have that

$$\left|\int_{X_2}\int_{X_1} f\,d\mu_1 d\mu_2 - \int_{X_2}\int_{X_1} f\,d\mu_1^j d\mu_2^j\right| < \varepsilon,$$

and this concludes the proof.

□

Now we are in the position of stating and proving the expression for volumes and mixed volumes of zonoids. Since in the proof we will work also with zonotopes, we reserve the letter Z for zonotopes and take W for zonoids.

Theorem 10.2.3. *Let $W \subseteq \mathbb{R}^n$ be a zonoid centered at the origin and having generating measure μ. We have*

$$\mathrm{vol}_n(W) = \frac{2^n}{n!}\int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, d\mu(v_1) \ldots d\mu(v_n). \tag{10.2.7}$$

More generally, if $W_1, \ldots, W_n$ are zonoids centered at the origin whose generating measures are $\mu_1, \ldots, \mu_n$, respectively, then

$$V(W_1, \ldots, W_n) = \frac{2^n}{n!}\int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, d\mu_1(v_1) \ldots d\mu_n(v_n). \tag{10.2.8}$$

Proof. Consider first that Z is a zonotope centered at the origin given as

$$Z = \beta_1 \mathrm{seg}[-u_1, u_1] + \ldots + \beta_k \mathrm{seg}[-u_k, u_k],$$

where $\beta_1, \ldots, \beta_k > 0$ and $u_1, \ldots, u_k \in \mathbb{S}^{n-1}$. Then, from linearity of mixed volumes, we can compute the volume of Z as

$$\begin{aligned} \operatorname{vol}_n(Z) = V(Z, \ldots, Z) = \sum_{i_1, \ldots, i_n = 1}^{k} \beta_{i_1} \ldots \beta_{i_n} V\big(\operatorname{seg}[-u_{i_1}, u_{i_1}], \ldots, \operatorname{seg}[-u_{i_n}, u_{i_n}]\big) = \\ = \frac{2^n}{n!} \sum_{i_1, \ldots, i_n = 1}^{k} \beta_{i_1} \ldots \beta_{i_n} D_n(u_{i_n}, \ldots, u_{i_n}), \end{aligned} \tag{10.2.9}$$

where the last equality comes from (10.2.6). On the other hand, we know from Remark 10.2.2 that the generating measure μ of Z is given as

$$\mu = \sum_{j=1}^{k} \frac{\beta_j}{2} \big(\delta_{u_j} + \delta_{-u_j}\big).$$

Recall that D_n is invariant under changing the sign of any entry, and that the integral of a function with respect to a Dirac delta measure is simply its value in the point where the mass is concentrated. We integrate iteratively $D_n(v_1, \ldots, v_n)$ with respect to μ in each variable to obtain

$$\begin{aligned} \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, d\mu(v_1) \ldots d\mu(v_n) = \\ = \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} \sum_{i_1=1}^{k} \frac{\beta_{i_1}}{2} \big(D_n(u_{i_1}, \ldots, v_n) + D_n(-u_{i_1}, \ldots, v_n)\big)\, d\mu(v_2) \ldots d\mu(v_n) = \\ = \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} \sum_{i_1=1}^{k} \beta_{i_1} D_n(u_{i_1}, v_2, \ldots, v_n)\, d\mu(v_2) \ldots d\mu(v_n) = \\ = \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} \sum_{i_1, i_2=1}^{k} \beta_{i_1} \beta_{i_2} D_n(u_{i_1}, u_{i_2}, \ldots, v_n)\, d\mu(v_3) \ldots d\mu(v_n) = \ldots \\ \ldots = \sum_{i_1, \ldots, i_n = 1}^{k} \beta_{i_1} \ldots \beta_{i_n} D_n(u_{i_n}, \ldots, u_{i_n}). \end{aligned}$$

Substituting this equality in (10.2.9) settles (10.2.7). We move forward to prove (10.2.8) for zonotopes $Z_1, \ldots, Z_n \subseteq \mathbb{R}^n$ centered at the origin (whose corresponding generating measures are $\mu_1, \ldots, \mu_n$, respectively, say). It is clear that for any numbers $\alpha_1, \ldots, \alpha_n > 0$ the set $\alpha_1 Z_1 + \ldots + \alpha_n Z_n$ is a zonotope centered at the origin whose generating measure is $\alpha_1 \mu_1 + \ldots + \alpha_n \mu_n$. Once the integral is linear for combinations of measures, we get from the formula that we just have proved the equality

$$\begin{aligned} \operatorname{vol}_n(\alpha_1 Z_1 + \ldots + \alpha_n Z_n) = \\ = \frac{2^n}{n!} \sum_{j_1, \ldots, j_n = 1}^{n} \alpha_{j_1} \ldots \alpha_{j_n} \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, d\mu_{j_1}(v_1) \ldots d\mu_{j_n}(v_n). \end{aligned}$$

On the other hand, Minkowski's theorem on the polynomiality of the volume (Theorem 6.1.1) gives

$$\operatorname{vol}_n(\alpha_1 Z_1 + \ldots + \alpha_n Z_n) = \sum_{j_1, \ldots, j_n = 1}^{n} \alpha_{j_1} \ldots \alpha_{j_n} V(Z_{j_1}, \ldots, Z_{j_n}).$$

Hence we have two polynomial expressions for the volume of $\alpha_1 Z_1 + \ldots + \alpha_n Z_n$. These polynomials must be the same, and then they have the same coefficient for the term $\alpha_1 \ldots \alpha_n$. In both polynomials, this term appears once for each permutation of $\{1, \ldots, n\}$. From the symmetry of mixed volumes, and from Fubini's theorem (which gives that the integral is invariant under permutating the order of integration), we get

$$n!V(Z_1, \ldots, Z_n) = n!\frac{2^n}{n!} \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, d\mu_1(v_1) \ldots d\mu_n(v_n),$$

and this proves (10.2.8) for zonotopes.

In the case of zonoids, we simply consider approximations by zonotopes. Thus, both equalities come immediately from Scholium 10.2.1 and Proposition 10.2.2, and from the fact that mixed volumes are continuous in the Hausdorff metric.

□

We finish this section by briefly mentioning the class of *generalized zonoids*. A set $K \in CC^n$ is said to be a *generalized zonoid* if there exist zonoids $Z_1, Z_2 \subseteq \mathbb{R}^n$ such that $K + Z_1 = Z_2$. This name justifies since these sets are, up to translations, precisely the ones whose support functions can be expressed by (10.2.4) allowing μ to be a signed (even) measure. This follows from the classical Hanh–Jordan decomposition of signed measures (see, e.g., [239]). It is immediate that the family of generalized zonoids is a Minkowski class in CC^n, and it is also clear that every zonoid is a generalized zonoid. To give a complete and thorough account on generalized zonoids we would need some techniques which are not in the scope of this book (such as properties of the *cosine transform*, for example). However, later we will need the following result, which we state without a proof.

Theorem 10.2.4. *The class of generalized zonoids is dense in the class of centrally symmetric bodies with respect to the Hausdorff metric.*

In other words, every set $K \in CC^n$ which is symmetric with respect to some point can be approximated (in the Hausdorff metric) by a sequence of generalized zonoids. For more details, we refer the reader to [930], [1263], and [2167].

10.3 Projection, centroid, and intersection bodies

Now that we have introduced zonotopes and zonoids, we move forward to define and investigate the concept of *projection body* of a convex body. Let $K \subseteq \mathbb{R}^n$ be a convex body, and consider the function $g_K : \mathbb{S}^{n-1} \to \mathbb{R}$ given as

$$g_K(u) := \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle v, u\rangle|\, dS_{n-1}(K, v),$$

where the last equality is given by formula (7.2.5). We recall that $S_{n-1}(K, \cdot)$ is the $(n-1)$-th mixed surface area measure of K. That is, $g_K(u)$ is the $(n-1)$-dimensional volume of the orthogonal projection of K onto the hyperplane $u^\perp$. We extend g_K to a function $f_K : \mathbb{R}^n \to \mathbb{R}$ letting $f_K(\lambda u) = \lambda g_K(u)$ for each $\lambda \geq 0$ and $u \in \mathbb{S}^{n-1}$. That is, letting $x \in \mathbb{R}^n$ and writing $x = \lambda u$ for some $\lambda \geq 0$ and some $u \in \mathbb{S}^{n-1}$, we deduce the expression

$$f_K(x) = \lambda g_K(u) = \frac{\lambda}{2}\int_{\mathbb{S}^{n-1}} |\langle v, u\rangle|\, dS_{n-1}(K, v) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, \lambda u\rangle|\, dS_{n-1}(K, v) =$$
$$= \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, x\rangle|\, dS_{n-1}(K, v).$$

This function has the following properties.

(i) f_K is sublinear. Indeed, f_K is positively homogeneous by definition. To prove subadditivity, let $x, y \in \mathbb{R}^n$ be arbitrarily given. From the expression above, we get

$$f_K(x+y) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, x+y\rangle|\, dS_{n-1}(K, v) \leq$$
$$\leq \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, x\rangle|\, dS_{n-1}(K, v) + \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, y\rangle|\, dS_{n-1}(K, v) = f_K(x) + f_K(y).$$

(ii) f_K is even, that is $f_K(x) = f_K(-x)$ for each $x \in \mathbb{R}^n$. This follows straightforwardly from the expression for f_K given above.

(iii) $f_K(u) > 0$ for each $u \in \mathbb{S}^{n-1}$. This follows from the immediate observation that the $(n-1)$-dimensional volume of the orthogonal projection of K onto any hyperplane is positive (because K has maximal affine dimension).

From Theorem 2.3.1, the sublinearity gives that f_K is the support function of a unique set of $\Pi K \in CC^n$. Once f_K is even, we get that ΠK is centered at the origin (this is straightforward to prove; see, e.g., Exercise 2.31). We also claim that ΠK is a convex body. Indeed, if we let

$$c := \min\left\{f_K(u) : u \in \mathbb{S}^{n-1}\right\},$$

then from continuity (any sublinear function is continuous) and from the fact that $f_K > 0$ in $\mathbb{S}^{n-1}$, we obtain that $c > 0$. Because $h_{\Pi K} = f_K$, it follows that $h_{\Pi K} \geq c$ in $\mathbb{S}^{n-1}$, and this means that $cB^n \subseteq \Pi K$. Thus, ΠK has maximal affine dimension.

The convex body ΠK constructed in this way is called the *projection body* of K. In other words, the projection body of K is the (unique) convex body ΠK whose support function is given as

$$h_{\Pi K}(x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, x\rangle|\, dS_{n-1}(K, v) \tag{10.3.1}$$

for each $x \in \mathbb{R}^n$. Of course, we could have defined the projection body directly by this expression, but then we would have lost the geometric meaning of the definition. Observe also that $h_{\Pi K}$ can be defined in terms of mixed volumes as a consequence of the definition of $S_{n-1}(K, \cdot)$:

$$h_{\Pi K}(x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, x\rangle|\, dS_{n-1}(K, v) = \frac{n}{2}\left(\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{\mathrm{seg}[-x,x]}(v)\, dS_{n-1}(K, v)\right) =$$
$$= \frac{n}{2} V\big(K, \ldots, K, \mathrm{seg}[-x, x]\big).$$

Example 10.3.1. (Projection bodies of proper polytopes) Let $P \subseteq \mathbb{R}^n$ be a proper polytope with unit outer normal vectors $u_1, \ldots, u_m$ and corresponding facets $F_1, \ldots, F_m$. Then, the projection body of P is the zonotope

$$\Pi P = \frac{1}{2}\sum_{j=1}^{m} \mathrm{vol}_{n-1}(F_j)\mathrm{seg}[-u_j, u_j]. \tag{10.3.2}$$

Indeed, recalling that from (7.1.9) we have $S_{n-1}(P,\cdot) = \sum_{j=1}^{n} \mathrm{vol}_{n-1}(F_j)\delta_{u_j}$, we get

$$h(\Pi P, x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle|\, dS_{n-1}(P, v) = \frac{1}{2}\sum_{j=1}^{m} \mathrm{vol}_{n-1}(F_j)|\langle x, u_j\rangle| =$$

$$= \frac{1}{2}\sum_{j=1}^{m} \mathrm{vol}_{n-1}(F_j)h(\mathrm{seg}[-u_j, u_j], x) = h\left(\frac{1}{2}\sum_{j=1}^{m} \mathrm{vol}_{n-1}(F_j)\mathrm{seg}[-u_j, u_j], x\right)$$

for any $x \in \mathbb{R}^n$. This gives (10.3.2).

Figure 10.3.1 illustrates the case where P is an equilateral triangle in $\mathbb{R}^2$ (whose side length is 2, say). In this case, letting u_1, u_2, and u_3 be the unit outer normal vector of the sides, equality above yields

$$h(\Pi P, \cdot) = h(\mathrm{seg}[-u_1, u_1] + \mathrm{seg}[-u_2, u_2] + \mathrm{seg}[-u_3, u_3], \cdot).$$

Hence, ΠP is the zonotope $\mathrm{seg}[-u_1, u_1] + \mathrm{seg}[-u_2, u_2] + \mathrm{seg}[-u_3, u_3]$, as it is shown in the figure.

An even simpler expression for the projection body of a polytope is given in Exercise 10.3. See also Figure 10.5.1.

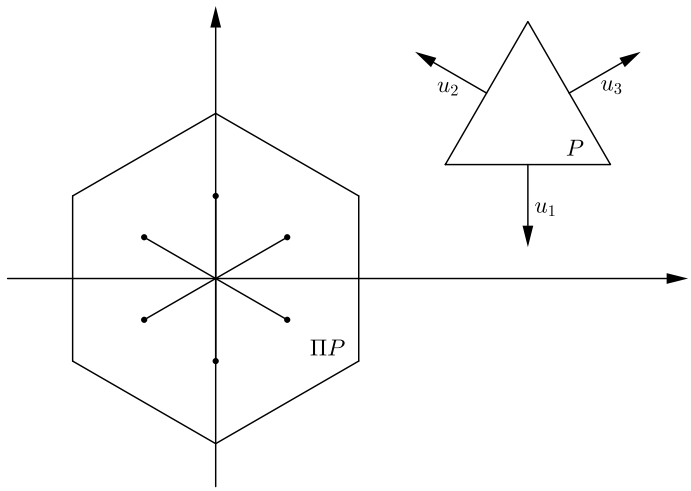


Fig. 10.3.1 The projection body ΠP of the triangle P is the hexagon obtained as the Minkowski sum of the segments drawn through the origin.

From now on, we explore some properties of projection bodies. We start by understanding the projection bodies of different positions of the same convex body. In what follows, we remember that for any $T \in \mathrm{GL}(n)$ we have $(T^{-1})^* = (T^*)^{-1}$ (this is a standard fact from linear algebra, and it is shown in the proof of Theorem 9.1.1). This fact will be used sometimes below without any further comments.

Proposition 10.3.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then*

(i) $\Pi(x+K) = \Pi K$ *for any* $x \in \mathbb{R}^n$ *and*

(ii) $\Pi(TK) = |\det(T)| \cdot (T^{-1})^* \Pi K$ *for every* $T \in \mathrm{GL}(n)$.

Proof. For **(i)**, fix $x \in \mathbb{R}^n$ and simply notice that

$$\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(x+K)\big) = \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)$$

for any $u \in \mathbb{S}^{n-1}$. Therefore, $\Pi(x+K)$ and ΠK are given by the same sublinear function, and hence they are the same convex body.

To prove **(ii)**, we recall that $h_{\mathrm{seg}[-x,x]}(v) = |\langle x, v\rangle|$ for any $x, v \in \mathbb{R}^n$, and that $h_{TA}(x) = h_A(T^* x)$ for every $T \in \mathrm{GL}(n)$, $A \subseteq \mathbb{R}^n$, and $x \in \mathbb{R}^n$. We use the definition of S_{n-1} and item **(iii)** of Corollary 6.2.1 to compute, for any fixed $x \in \mathbb{R}^n$, the support function of $\Pi(TK)$ at x as

$$\begin{aligned}
h_{\Pi(TK)}(x) &= \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle|\, dS_{n-1}(TK, v) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} h_{\mathrm{seg}[-x.x]}(v)\, dS_{n-1}(TK, v) = \\
&= \frac{n}{2} V\big(\mathrm{seg}[-x,x], TK, \dots, TK\big) = \frac{|\det(T)|}{2} n V\big(T^{-1}\mathrm{seg}[-x,x], K, \dots, K\big) = \\
&= \frac{|\det(T)|}{2}\int_{\mathbb{S}^{n-1}} h_{T^{-1}\mathrm{seg}[-x,x]}(v)\, dS_{n-1}(K, v) = \frac{|\det(T)|}{2}\int_{\mathbb{S}^{n-1}} h_{\mathrm{seg}[-x,x]}((T^{-1})^* v)\, dS_{n-1}(K, v) = \\
&= \frac{|\det(T)|}{2}\int_{\mathbb{S}^{n-1}} |\langle x, (T^{-1})^* v\rangle|\, dS_{n-1}(K, v) = \frac{|\det(T)|}{2}\int_{\mathbb{S}^{n-1}} |\langle T^{-1}x, v\rangle|\, dS_{n-1}(K, v) = \\
&= |\det(T)| h_{\Pi K}(T^{-1}x) = |\det(T)| h_{(T^{-1})^*\Pi K}(x) = h_{|\det(T)|\cdot(T^{-1})^*\Pi K}(x).
\end{aligned}$$

Therefore, the convex bodies $\Pi(TK)$ and $|\det(T)| \cdot (T^*)^{-1}\Pi K$ have the same support function, and hence they are the same.

□

This proposition has some consequences which we explore now. In what follows, we denote the polarity of ΠK by $\Pi^\circ K$. That is, $\Pi^\circ K := (\Pi K)^\circ$. We also write $\Pi^2 K := \Pi(\Pi K)$.

Corollary 10.3.1. *For any convex body* $K \subseteq \mathbb{R}^n$ *it holds that* $\Pi K = \Pi(-K)$.

Proof. From item **(ii)** of Proposition 10.3.1, we have that $\Pi(-K) = \Pi(-\mathrm{Id}_{\mathbb{R}^n} K) = -\Pi K$. Because ΠK is centered at the origin, we have that $-\Pi K = \Pi K$, and hence $\Pi(-K) = \Pi K$. Notice that this equality also follows from the immediate observation that the $(n-1)$-dimensional volumes of the orthogonal projections of K and $-K$ onto hyperplanes are always the same.

□

Corollary 10.3.2. *For any convex body* $K \subseteq \mathbb{R}^n$ *and every* $T \in \mathrm{SL}(n)$, *we have* $\Pi^\circ(TK) = T(\Pi^\circ K)$ *and* $\Pi^2(TK) = T(\Pi^2 K)$.

Proof. Item **(ii)** of Proposition 10.3.1 for $T \in \mathrm{SL}(n)$ reads $\Pi(TK) = (T^{-1})^* \Pi K$. Since we clearly have that $(((T^{-1})^*)^{-1})^* = T$, we get

$$\Pi^2(TK) = \Pi(\Pi(TK)) = \Pi((T^{-1})^*\Pi K) = T(\Pi(\Pi K)) = T(\Pi^2 K).$$

On the other hand, from Lemma 5.4.1 we know that $((T^{-1})^* K)^\circ = (((T^{-1})^*)^{-1})^* K^\circ = TK^\circ$. Hence

$$\Pi^\circ(TK) = (\Pi(TK))^\circ = ((T^{-1})^*\Pi K)^\circ = T(\Pi K)^\circ = T(\Pi^\circ K),$$

and the proof is complete.

□

Remark 10.3.1. If $T \in \mathrm{GL}(n)$, then it is not difficult to see that

$$\Pi^\circ(TK) = \frac{1}{|\det(T)|} T(\Pi^\circ K). \tag{10.3.3}$$

Indeed, if $\alpha > 0$, then $\det(\alpha \mathrm{Id}_{\mathbb{R}^n}) = \alpha^n$ and $((\alpha \mathrm{Id}_{\mathbb{R}^n})^{-1})^* = \frac{1}{\alpha}\mathrm{Id}_{\mathbb{R}^n}$. Hence, applying item **(ii)** of Proposition 10.3.1 for $T = \alpha \mathrm{Id}_{\mathbb{R}^n}$, we get

$$\Pi^\circ(\alpha K) = (\Pi(\alpha K))^\circ = (\Pi(\alpha \mathrm{Id}_{\mathbb{R}^n}(K)))^\circ = (\alpha^{n-1}\Pi K)^\circ = \frac{1}{\alpha^{n-1}}(\Pi K)^\circ = \\ = \frac{1}{\alpha^{n-1}}\Pi^\circ K.$$

Notice that the case $\alpha < 0$ can be understood without any further calculations because we always have that $\Pi(-K) = \Pi K$. If $T \in \mathrm{GL}(n)$, then letting $\alpha := |\det(T)|^{1/n}$, we have that $T_0 := T/\alpha \in \mathrm{SL}(n)$. Thus, T_0 commutes with Π°, and hence

$$\Pi^\circ(TK) = \Pi^\circ(\alpha T_0(K)) = \Pi^\circ(T_0(\alpha K)) = T_0(\Pi^\circ(\alpha K)) = \\ = T_0\left(\frac{1}{\alpha^{n-1}}\Pi^\circ K\right) = \frac{1}{\alpha^n} T(\Pi^\circ K) = \frac{1}{|\det(T)|} T(\Pi^\circ K).$$

As an immediate consequence of this equality, we have that the quantity

$$\mathrm{vol}_n(K)^{n-1}\mathrm{vol}_n(\Pi^\circ K) \tag{10.3.4}$$

is $\mathrm{GL}(n)$-invariant. From item **(i)** of Proposition 10.3.1, it is also translation-invariant, and hence it is an affine invariant of K. The *Petty projection inequality* states that the value for $K = B^n$ is the optimal upper bound for this affine invariant as K ranges through $\mathcal{K}^n$, and that this upper bound is attained only for ellipsoids. We will prove this later.

Now we explain the relation between zonoids and projection bodies. It turns out that the projection bodies are precisely the proper zonoids (cf. Plate 8 at the beginning of Chapter 8). This we will see next.

Proposition 10.3.2. *Let $K \subseteq \mathbb{R}^n$ be a convex body. Then the projection body ΠK of K is a proper zonoid centered at the origin. Conversely, any proper zonoid centered at the origin is the projection body of some convex body.*

Proof. Recall that $S_{n-1}(K, \cdot)$ coincides with the usual surface area measure μ_K of K (see Theorem 7.2.3). Since the reverse normal image of a subset of $\mathbb{S}^{n-1}$ on K is clearly the same as the reverse normal image of $-E$ on $-K$, we get that $S_{n-1}(K, E) = S_{n-1}(-K, -E)$ for every Borel subset $E \subseteq \mathbb{S}^{n-1}$ (see also item **(iv)** of Theorem 7.1.3). Thus, the Borel measure on $\mathbb{S}^{n-1}$ defined for each $E \in \mathcal{B}_{\mathbb{S}^{n-1}}$ as

$$\nu_K(E) := \frac{1}{4}\big(S_{n-1}(K, E) + S_{n-1}(-K, E)\big) \tag{10.3.5}$$

is even. For any $x \in \mathbb{R}^n$, we have

$$\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \, d\nu_K(v) = \frac{1}{2}\left(\frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, x\rangle| \, dS_{n-1}(K, v) + \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle v, x\rangle| \, dS_{n-1}(-K, v)\right) = \\ = \frac{1}{2}\big(h_{\Pi K}(x) + h_{\Pi(-K)}(x)\big) = h_{\Pi K}(x),$$

where in the last equality we used the fact that $\Pi K = \Pi(-K)$, proved in Corollary 10.3.1. Then it follows from Theorem 10.2.2 that ΠK is a zonoid centered at the origin. Once we already know that ΠK is a convex body, it follows that it is a proper zonoid.

Now we prove the converse. Let $Z \subseteq \mathbb{R}^n$ be a proper zonoid centered at the origin. Then with Theorem 10.2.2 we may write the support function of Z as

$$h_Z(x) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle x, u\rangle| \, d\mu(u)$$

for some even finite Borel measure μ on $\mathbb{S}^{n-1}$ (notice that we modified slightly the statement of the mentioned theorem dividing the integral by $1/2$; to do that one just has to rescale μ). Once Z is proper, we get that $h_Z(x) > 0$ for any $x \in \mathbb{S}^{n-1}$, and this yields that μ is not concentrated on a great subsphere (see Lemma 7.3.2). Moreover, since μ is even, we get immediately that μ has its centroid at the origin. It follows from Minkowski's existence theorem (Theorem 7.3.1) that $\mu = S_{n-1}(K, \cdot)$ for some convex body $K \subseteq \mathbb{R}^n$. Thus,

$$h_Z(x) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle x, u\rangle| \, d\mu(u) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle x, u\rangle| \, dS_{n-1}(K, u) = h_{\Pi K}(x),$$

and this gives that $Z = \Pi K$.

□

Remark 10.3.2. Notice that we proved a little more than that every projection body ΠK is a zonoid: we also described how its generating measure is constructed from K. We proved that any given projection body ΠK is the zonoid whose generating measure is the measure ν_K defined in (10.3.5).

As a corollary, we have the following useful formula for the volume of a projection body.

Corollary 10.3.3. *For any convex body $K \subseteq \mathbb{R}^n$ we have that*

$$\mathrm{vol}_n(\Pi K) = \frac{1}{n!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n) \, dS_{n-1}(K, v_1) \dots dS_{n-1}(K, v_n).$$

Proof. Since ΠK is the zonoid whose generating measure is the measure ν_K defined in (10.3.5), it follows from Theorem 10.2.3 that

$$\mathrm{vol}_n(\Pi K) = \frac{2^n}{n!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n) \, d\nu_K(v_1) \dots d\nu_K(v_n). \tag{10.3.6}$$

The equality $S_{n-1}(K, -E) = S_{n-1}(-K, E)$, which holds for any Borel set $E \subseteq \mathbb{S}^{n-1}$ (see the proof of Proposition 7.2.4), means that the push-forward of the measure $S_{n-1}(K, \cdot)$ under the map $-\mathrm{Id}$ is precisely the measure $S_{n-1}(-K, \cdot)$. Indeed, for any Borel set $E \in \mathbb{S}^{n-1}$,

$$-\mathrm{Id}_* S_{n-1}(K, \cdot)(E) = S_{n-1}(K, -E) = S_{n-1}(-K, E).$$

From the formula (A.2.4) for changes of variables, we get that if $f : \mathbb{S}^{n-1} \to \mathbb{R}$ is even (and integrable), then

$$\int_{\mathbb{S}^{n-1}} f(v) \, dS_{n-1}(-K, v) = \int_{\mathbb{S}^{n-1}} f(v) \, d(-\mathrm{Id}_{\mathbb{R}^n})_* S_{n-1}(K, v) =$$
$$= \int_{\mathbb{S}^{n-1}} f(-v) \, dS_{n-1}(K, v) = \int_{\mathbb{S}^{n-1}} f(v) \, dS_{n-1}(K, v).$$

In particular, we have that

$$\int_{\mathbb{S}^{n-1}} f(v)\, d\nu_K(v) = \frac{1}{4}\left(\int_{\mathbb{S}^{n-1}} f(v)\, dS_{n-1}(K,v) + \int_{\mathbb{S}^{n-1}} f(v)\, dS_{n-1}(-K,v)\right) = \\ = \frac{1}{2}\int_{\mathbb{S}^{n-1}} f(v)\, dS_{n-1}(K,v)$$

for any even measurable function $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$. Once we know that $D_n(v_1,\dots,v_n)$ is even in each entry, this allows us to rewrite (10.3.6) as

$$\begin{aligned}\operatorname{vol}_n(\Pi K) &= \\ &= \frac{2^{n-1}}{n!}\int_{\mathbb{S}^{n-1}}\dots\int_{\mathbb{S}^{n-1}} D_n(v_1,\dots,v_n)\, dS_{n-1}(K,v_1) d\nu_K(v_2)\dots d\nu_K(v_n) = \\ &= \frac{2^{n-2}}{n!}\int_{\mathbb{S}^{n-1}}\dots\int_{\mathbb{S}^{n-1}} D_n(v_1,\dots,v_n)\, dS_{n-1}(K,v_1) dS_{n-1}(K,v_2) d\nu_K(v_3)\dots d\nu_K(v_n) = \\ &\dots = \frac{1}{n!}\int_{\mathbb{S}^{n-1}}\dots\int_{\mathbb{S}^{n-1}} D_n(v_1,\dots,v_n)\, dS_{n-1}(K,v_1)\dots dS_{n-1}(K,v_n),\end{aligned}$$

where the reader may notice that repeatedly we are using Fubini's theorem to change the order of the integrations conveniently. This concludes the proof.

□

Next we discuss a sort of uniqueness result for projection bodies. Roughly speaking, it states that any given projection body is the projection body of a unique body symmetric with respect to the origin.

Theorem 10.3.1. *Let μ, ν be finite even measures on $\mathbb{S}^{n-1}$. If*

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\mu(v) = \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|\, d\nu(v)$$

for each $u \in \mathbb{S}^{n-1}$, then $\mu = \nu$.

Remark 10.3.3. A consequence of this result is that each zonoid has a unique generating measure (notice that, by definition, we assume that generating measures are always even).

Comment on the proof. This is a consequence of Theorem 3.5.4 in [2167], whose proof involves the theory of *spherical harmonics*, which is beyond the scope of this book. Alternatively, the reader may read Theorem C.2.1 in [930]. There, the proof is obtained by an approximation argument for continuous even functions defined on the sphere.

□

Before moving forward, a quick remark about terminology is due to avoid any confusion. Remember that we say that a body $K \subseteq \mathbb{R}^n$ *has its centroid at the origin* when the *centroid* of K, defined as

$$\frac{1}{\operatorname{vol}_n(K)}\int_K x\, dx,$$

is $0_{\mathbb{R}^n}$ (see Exercises 4.10 and 4.11 for more on the centroid of a convex body). Thus, the fact that K has its centroid at the origin does not bring any information about symmetry. This shall not be confused with K being *centered at the origin*, which means that K is symmetric with respect to the origin (as it was extensively used in the above). Recall also that K is said to be *centrally symmetric* if some translate of K is centered at the origin. In this case, we have that $K - x_0 = -(K - x_0)$ for some $x_0 \in K$, and we also say that K is *centered at* x_0. Again, this is not the same as saying that K *has centroid at* x_0.

In Proposition 10.3.2, we proved that any projection body ΠK, $K \in \mathcal{K}^n$, is a proper zonoid by writing

$$h_{\Pi K}(x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \, d\left(\frac{1}{2}S_{n-1}(K, v) + \frac{1}{2}S_{n-1}(-K, v)\right).$$

The measure $\frac{1}{2}S_{n-1}(K, \cdot) + \frac{1}{2}S_{n-1}(-K, \cdot)$ is just the measure ν_K introduced (10.3.5) multiplied by 2 (we do this multiplication for our convenience). Let us take a careful look at this measure. We already know that this measure is even, and hence it has centroid at the origin. Moreover, this measure is not concentrated on a great subsphere of $\mathbb{S}^{n-1}$, since K is a convex body. It follows from the Minkowski existence theorem (Theorem 7.3.1) that $2\nu_K$ is the surface area measure of a convex body (which is unique up to translations; keep Theorem 7.2.1 in mind). We denote by ∇K the unique convex body whose centroid is the origin, and such that

$$S_{n-1}(\nabla K, \cdot) = \frac{1}{2}S_{n-1}(K, \cdot) + \frac{1}{2}S_{n-1}(-K, \cdot).$$

The body ∇K is called the *Blaschke body* of K. This name justifies because of its clear relation with the concept of Blaschke addition, introduced in Remark 7.5.6. Indeed, it is clear that

$$\nabla K = \frac{1}{2^{1/(n-1)}} K\#(-K).$$

Notice that $S_{n-1}(\nabla K, \cdot)$ is even, and then ∇K is centrally symmetric (see Proposition 7.2.4). Once ∇K has the centroid at the origin by definition, we get that ∇K is symmetric with respect to the origin (see Exercise 4.11). Also, it is straightforward to check that if K is centrally symmetric, then ∇K is just a translation of K. Indeed, in the case where K is centrally symmetric it holds that $S_{n-1}(K, \cdot) = S_{n-1}(-K, \cdot)$, and then it follows that $S_{n-1}(\nabla K, \cdot) = S_{n-1}(K, \cdot)$. Next we prove that the Blaschke body of K is the unique $0_{\mathbb{R}^n}$-symmetric body among all bodies whose projection body is ΠK.

Theorem 10.3.2. *For any convex body $K \subseteq \mathbb{R}^n$, it holds that $\Pi K = \Pi(\nabla K)$. Moreover, if $K, L \subseteq \mathbb{R}^n$ are convex bodies, then $\Pi K = \Pi L$ if and only if $\nabla K = \nabla L$.*

Proof. Once $\Pi K = \Pi(-K)$, we have

$$2h(\Pi K, u) = h(\Pi(K) + \Pi(-K), u) =$$
$$= \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, d(S_{n-1}(K, v) + S_{n-1}(-K, v)) = 2h(\Pi(\nabla K), u)$$

for each $u \in \mathbb{S}^{n-1}$. This shows that $\Pi K = \Pi(\nabla K)$.

Moving forward, if $\nabla K = \nabla L$, then $\Pi K = \Pi(\nabla K) = \Pi(\nabla L) = \Pi L$. For the converse, if $\Pi K = \Pi L$, then $\Pi(\nabla K) = \Pi(\nabla L)$, and this leads to

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, dS_{n-1}(\nabla K, v) = \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle| \, dS_{n-1}(\nabla L, v)$$

for all $u \in \mathbb{S}^{n-1}$. From Theorem 10.3.1 we get that ∇K and ∇L have the same surface area measure. Using Theorem 7.2.1, it follows that ∇K and ∇L are translates. Since both bodies are symmetric with respect to the origin, it follows that $\nabla K = \nabla L$ (indeed, it is not difficult to prove that a convex body cannot be centered at two different points; see Exercise 2.32).

□

As a consequence, if L is centered at the origin, and $\Pi L = \Pi K$, then $\nabla L = \nabla K$ and $L = \nabla L$. It follows that $L = \nabla K$. This shows that ∇K is the unique $0_{\mathbb{R}^n}$-symmetric body whose projection body is ΠK. Yet another consequence is given below.

Corollary 10.3.4. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies centered at the origin. Then $\Pi K = \Pi L$ if and only if $K = L$.*

Proof. We already know that $\Pi K = \Pi L$ if and only if $\nabla K = \nabla L$. Once K and L are centered at the origin, we have that $\nabla K = K$ and $\nabla L = L$, and this gives the desired statement. □

We originally defined the projection body *of a convex body*, but it is worth mentioning that we can also define the projection body *of a non-empty compact convex set* (thus dropping the maximal affine dimension hypothesis). We do this in the natural way: if $C \in CC^n$, then the function which associates $\lambda u \in \mathbb{R}^n$ (with $\lambda \geq 0$ and $u \in \mathbb{S}^{n-1}$) to the number $\lambda \mathrm{vol}_n(\mathrm{proj}_{u^\perp}(C))$ is sublinear (for exactly the same reason why this holds for $K \in \mathcal{K}^n$), and hence it is the support function of some set $\Pi C \in CC^n$. We define ΠC to be the *projection body* of C. Alternatively (and equivalently), one may straightforwardly extend the expression (10.3.1) to sets of CC^n. For more about this extension, we refer the reader to Exercise 10.20. Most of the results about projection bodies of convex bodies that we proved above can be directly extended for the case of projection bodies of sets of CC^n (see, e.g., Exercise 10.21), but we leave the details for the interested reader. We originally restricted ourselves to convex bodies because this is sufficient for the applications in which we are interested (namely, affine isoperimetric inequalities for convex bodies).

We move forward to study other bodies associated with a given convex body and defined via their support functions. Remember that we denote by $\mathcal{K}_0^n$ the family of convex bodies of $\mathbb{R}^n$ which contain the origin as an interior point. For a given convex body $K \in \mathcal{K}_0^n$, we let $m_K : \mathbb{R}^n \to \mathbb{R}$ be the function given as

$$m_K(x) = \int_K |\langle x, z\rangle| \, dz$$

for each $x \in \mathbb{R}^n$, where we recall the reader that "dz" stands for integration with respect to the usual Lebesgue measure of $\mathbb{R}^n$. It is immediate to see that m_K is a sublinear even function. Also, we have that $m_K(u) > 0$ for every $u \in \mathbb{S}^{n-1}$. Indeed, since K is a convex body having the origin as an interior point, then K contains a ball rB^n, and hence

$$m_K(u) = \int_K |\langle u, z\rangle| \, dz \geq \int_{rB^n} |\langle u, z\rangle| \, dz, \tag{10.3.7}$$

and the last integral is clearly positive (we can actually compute it using polar coordinates and the equality given in Remark 6.5.3, but this is not really necessary).

It follows that m_K is the support function of a convex body centered at the origin, which we call the *moment body* of K, and denote by MK. Summarizing, the moment body of a convex body $K \in \mathcal{K}_0^n$ is the unique convex body $MK \subseteq \mathbb{R}^n$ whose support function is given as

$$h_{MK}(x) = \int_K |\langle x, z\rangle| \, dz \tag{10.3.8}$$

for any $x \in \mathbb{R}^n$. As a consequence of this definition, we have that MK is centered at the origin.

We are mainly interested in a different normalization of the moment body. The *centroid body* of a convex body $K \in \mathcal{K}_0^n$ is the convex body

$$\Gamma K := \frac{1}{\mathrm{vol}_n(K)} MK.$$

Equivalently, we may define the centroid body of $K \in \mathcal{K}_0^n$ as the unique convex body $\Gamma K \subseteq \mathbb{R}^n$ whose support function is given as

$$h_{\Gamma K}(x) = \frac{1}{\mathrm{vol}_n(K)} \int_K |\langle x, z\rangle| \, dz \tag{10.3.9}$$

for each $x \in \mathbb{R}^n$. Of course, the centroid body is also always centered at the origin. As an immediate consequence of this expression, we can understand how centroid bodies of linearly equivalent bodies behave.

Lemma 10.3.1. *For any convex body $K \subseteq \mathbb{R}^n$ and every $T \in \mathrm{GL}(n)$ we have that*

$$\Gamma(TK) = T(\Gamma K).$$

Proof. We simply use change of variables in the integration to calculate

$$h_{\Gamma(TK)}(x) = \frac{1}{\mathrm{vol}_n(TK)} \int_{TK} |\langle x, z\rangle| \, dz = \frac{1}{|\det(T)|\mathrm{vol}_n(K)} \int_K |\langle x, Ty\rangle||\det(T)| \, dy =$$
$$= \frac{1}{\mathrm{vol}_n(K)} \int_K |\langle T^* x, y\rangle| \, dy = h_{\Gamma K}(T^* x).$$

From (2.6.5), it follows that $h_{\Gamma K}(T^* x) = h_{T(\Gamma K)}(x)$. This concludes the proof. □

Corollary 10.3.5. *Let $E \subseteq \mathbb{R}^n$ be an ellipsoid centered at the origin. Then*

$$\Gamma E = \frac{2\kappa_{n-1}}{(n+1)\kappa_n} E,$$

where we recall that $\kappa_d := \mathrm{vol}_d(B^d)$.

Proof. Once E is an ellipsoid centered at the origin, there exists $T \in \mathrm{GL}(n)$ such that $E = TB^n$. Hence

$$\Gamma E = \Gamma(TB^n) = T(\Gamma B^n), \tag{10.3.10}$$

and then we have to investigate the centroid body of the unit ball. From expression (10.3.9), and using polar coordinates and the equality described in Remark 6.5.3, we get

$$h_{\Gamma B^n}(x) = \frac{1}{\mathrm{vol}_n(B^n)} \int_{B^n} |\langle x, z\rangle| \, dz =$$
$$= \frac{1}{\kappa_n} \int_0^1 \int_{\mathbb{S}^{n-1}} |\langle x, u\rangle| t^n \, d\mathcal{H}^{n-1}(u) dt = \frac{1}{\kappa_n} \int_0^1 2\kappa_{n-1} t^n \, dt = \frac{2\kappa_{n-1}}{(n+1)\kappa_n},$$

for any $x \in \mathbb{R}^n$. Therefore, the support function of ΓB^n is the constant $2\kappa_{n-1}/(n+1)\kappa_n$. It follows that

$$\Gamma B^n = \frac{2\kappa_{n-1}}{(n+1)\kappa_n} B^n.$$

Using this equality in (10.3.10) above we get

$$\Gamma E = T\left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n} B^n\right) = \frac{2\kappa_{n-1}}{(n+1)\kappa_n} TB^n = \frac{2\kappa_{n-1}}{(n+1)\kappa_n} E,$$

and the proof is complete. □

Remark 10.3.4. As in the case of the projection bodies, it is common in the literature to define moment and centroid bodies for a larger class of sets. While in the case of the projection bodies the definition can be (reasonably) extended to CC^n, the moment and centroid bodies are often defined for any *star body*, what we do also now.

Let $C \subseteq \mathbb{R}^n$ be a compact set such that $0_{\mathbb{R}^n} \in C$. Then, for any $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, we have that

$$\rho_C(x) := \sup\left\{\lambda \geq 0 : \lambda x \in C\right\}$$

is a real number. This defines a function $\rho_C : \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ known as the *radial function* of C (this is only an extension of the radial function defined in Section 2.3 for convex bodies in $\mathcal{K}_0^n$). Notice carefully that it is immediate from this definition that

$$\rho_C(\lambda x) = \frac{1}{\lambda}\rho_C(x)$$

for any $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ and $\lambda > 0$.

We say that a non-empty compact set $S \subseteq \mathbb{R}^n$ is a *star body* if $\mathrm{seg}[0_{\mathbb{R}^n}, x] \subseteq S$ for every $x \in S$, and if its radial function $\rho_S : \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ is positive and continuous. See Figure 10.3.2.

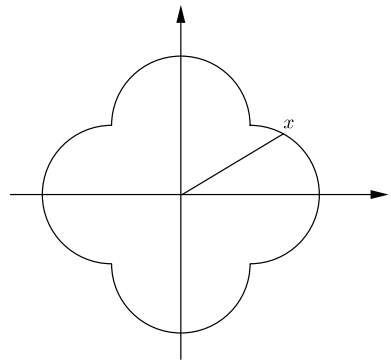


Fig. 10.3.2 A star body (which is origin-symmetric).

The condition on the radial function being positive and continuous implies that any star body contains a ball centered at the origin. In fact, if $r := \min\{\rho_S(u) : u \in \mathbb{S}^{n-1}\}$, then $r > 0$ and it is immediate that $rB^n \subseteq S$. As a consequence, we get that any star body has positive volume. Then, we may define the *moment* and *centroid bodies* of a star body $S \subseteq \mathbb{R}^n$ straightforwardly replacing K by S in the expressions (10.3.8) and (10.3.9), respectively. The functions h_{MS} and $h_{\Gamma S}$ obtained in this way clearly remain sublinear and even. Even more, since S contains a ball we may repeat the calculation in (10.3.7) and conclude that h_{MS} and $h_{\Gamma S}$ are positive in $\mathbb{S}^{n-1}$. Hence h_{MS} and $h_{\Gamma S}$ are support functions of (unique) convex bodies MS and ΓS centered at the origin, respectively. The bodies MS and ΓS are called the *moment body* and the *centroid body* of the star body S, respectively.

However, from now on we return to restrict ourselves to the case of convex bodies in $\mathcal{K}_0^n$. Radial functions for these bodies were studied in Section 2.3. We know from Proposition 2.3.6 that

$$\rho_K(x) = \frac{1}{h_{K^\circ}(x)}$$

for every $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, and this gives that ρ_K is finite, positive, and continuous. In particular, any convex body $K \in \mathcal{K}_0^n$ is a star body.

From now on, we will work mainly with centroid bodies. Once the moment body of a convex body $K \in \mathcal{K}_0^n$ is a scalar multiple of the centroid body, most of the results for centroid bodies which we will derive admit immediate analogues for moment bodies.

Proposition 10.3.3. *Let $K \in \mathcal{K}_0^n$. Then, the support function of the centroid body ΓK of K admits the following expression:*

$$h_{\Gamma K}(x) = \frac{1}{(n+1)\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \rho_K(v)^{n+1}\, d\mathcal{H}^{n-1}(v), \tag{10.3.11}$$

for each $x \in \mathbb{R}^n$. As a consequence, we have that ΓK is a proper zonoid.

Proof. From integration by polar coordinates (see Corollary A.6.1) we get

$$h_{\Gamma K}(x) = \frac{1}{\mathrm{vol}_n(K)} \int_K |\langle x, z\rangle|\, dz = \frac{1}{\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \int_0^{\rho_K(v)} |\langle x, tv\rangle| t^{n-1}\, dt d\mathcal{H}^{n-1}(v) =$$
$$= \frac{1}{\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} \left(|\langle x, v\rangle| \int_0^{\rho_K(v)} t^n\, dt \right) d\mathcal{H}^{n-1}(v) = \frac{1}{\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \frac{\rho_K(v)^{n+1}}{n+1}\, d\mathcal{H}^{n-1}(v) =$$
$$= \frac{1}{(n+1)\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \rho_K(v)^{n+1}\, d\mathcal{H}^{n-1}(v).$$

This settles (10.3.11). For the other claim, first observe that once we obviously have that $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ is an even Borel measure, we may change the variable in the last integral to write

$$h_{\Gamma K}(x) = \frac{1}{(n+1)\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle x, -v\rangle| \rho_K(-v)^{n+1}\, d\mathcal{H}^{n-1}(v) =$$
$$= \frac{1}{(n+1)\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \rho_K(-v)^{n+1}\, d\mathcal{H}^{n-1}(v).$$

This leads to

$$h_{\Gamma K}(x) = \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \frac{\rho_K(v)^{n+1} + \rho_K(-v)^{n+1}}{2(n+1)\mathrm{vol}_n(K)}\, d\mathcal{H}^{n-1}(v). \tag{10.3.12}$$

This equality reads

$$h_{\Gamma K}(x) = \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle|\, d\nu_{\Gamma K}(v), \tag{10.3.13}$$

where $\nu_{\Gamma K}$ is the Borel measure on $\mathbb{S}^{n-1}$ defined as

$$\nu_{\Gamma K}(E) := \int_E \frac{\rho_K(v)^{n+1} + \rho_K(-v)^{n+1}}{2(n+1)\mathrm{vol}_n(K)}\, d\mathcal{H}^{n-1}(v) \tag{10.3.14}$$

for each Borel set $E \subseteq \mathbb{S}^{n-1}$. That is, $\nu_{\Gamma K}$ is the Borel measure on $\mathbb{S}^{n-1}$ which has density

$$\delta_{\Gamma K}(v) := \frac{\rho_K(v)^{n+1} + \rho_K(-v)^{n+1}}{2(n+1)\mathrm{vol}_n(K)} \tag{10.3.15}$$

with respect to $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$. Since $\nu_{\Gamma K}$ is clearly even and finite, it follows from Theorem 10.2.2 that ΓK is a zonoid. Once ΓK has maximal affine dimension, we get that it is a proper zonoid.

□

Corollary 10.3.6. *We have the following formula for the volume of the centroid body of a convex body $K \in \mathcal{K}_0^n$:*

$$\mathrm{vol}_n(\Gamma K) = \frac{2^n}{n!\mathrm{vol}_n(K)^n} \int_K \dots \int_K D_n(x_1, \dots, x_n)\, dx_1 \dots dx_n.$$

Proof. Once (10.3.13) gives that $\nu_{\Gamma K}$ is the generating measure of (the proper zonoid) ΓK, we have from equality 10.2.7 that

$$\mathrm{vol}_n(\Gamma K) = \frac{2^n}{n!} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n)\, d\nu_{\Gamma K}(v_1) \dots d\nu_{\Gamma K}(v_n). \tag{10.3.16}$$

Now we remember that the function D_n is even in each entry and that $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ is an even measure. Consequently, fixing $v_2, \dots, v_n \in \mathbb{S}^{n-1}$, we get that

$$\int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n)\, \rho_K(v_1)^{n+1} d\mathcal{H}^{n-1}(v_1) = \int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n)\, \rho_K(-v_1)^{n+1} d\mathcal{H}^{n-1}(v_1).$$

Then, since $\nu_{\Gamma K}$ has density given as in (10.3.15), we get that

$$\begin{aligned}\int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n)\, d\nu_{\Gamma K}(v_1) &= \int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n) \frac{\rho_K(v_1)^{n+1} + \rho_K(-v_1)^{n+1}}{2(n+1)\mathrm{vol}_n(K)} d\mathcal{H}^{n-1}(v_1) = \\ = \frac{1}{\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} D_n(v_1, \dots, v_n) \frac{\rho_K(v_1)^{n+1}}{n+1} d\mathcal{H}^{n-1}(v_1) &= \frac{1}{\mathrm{vol}_n(K)} \int_K D_n(x_1, v_2, \dots, v_n)\, dx_1,\end{aligned}$$

where in the last equality we simply "reverted" the integration in polar coordinates used in the proof of (10.3.11). Notice carefully that we can do this because D_n is positively homogeneous in each entry. Now we just repeat this process iteratively to rewrite the integral in the right-hand side of (10.3.16). This gives the desired.

□

Remark 10.3.5. If we do not "revert" the integration by polar coordinates, then we arrive at the formula

$$\begin{aligned}&\mathrm{vol}_n(\Gamma K) = \\ &= \frac{\gamma_n}{\mathrm{vol}_n(K)^n} \int_{\mathbb{S}^{n-1}} \dots \int_{\mathbb{S}^{n-1}} \frac{D_n(v_1, \dots, v_n)}{\left[\rho_K(v_1) \dots \rho_K(v_n)\right]^{-(n+1)}} d\mathcal{H}^{n-1}(v_1) \dots d\mathcal{H}^{n-1}(v_n),\end{aligned}$$

where γ_n is the constant given as

$$\gamma_n := \frac{2^n}{n!(n+1)^n}.$$

As in the case of projection bodies, we have a uniqueness result for centroid bodies of bodies which are symmetric with respect to the origin.

Lemma 10.3.2. *Let K and L be convex bodies in $\mathbb{R}^n$ which are symmetric with respect to the origin. If $\Gamma K = \Gamma L$, then $K = L$.*

Proof. From (10.3.13), the equality

$$\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle|\, d\nu_{\Gamma K}(v) = \int_{\mathbb{S}^{n-1}} |\langle x, v\rangle|\, d\nu_{\Gamma L}(v)$$

holds for every $x \in \mathbb{R}^n$, where $\nu_{\Gamma K}$ and $\nu_{\Gamma L}$ are the generating measures of ΓK and ΓL, respectively (as given in (10.3.14)). It follows from Theorem 10.3.1 that $\nu_{\Gamma K} = \nu_{\Gamma L}$. Since K is symmetric with respect to the origin, the density $\delta_{\Gamma K}$ of $\nu_{\Gamma K}$ with respect to $\mathcal{H}^{n-1}$ given in (10.3.15) reads

$$\delta_{\Gamma K}(v) = \frac{\rho_K(v)^{n+1}}{(n+1)\mathrm{vol}_n(K)},$$

and we have an analogous expression for $\delta_{\Gamma L}$. We claim that $\delta_{\Gamma K}(v) = \delta_{\Gamma L}(v)$ for every $v \in \mathbb{S}^{n-1}$. In fact, if there is a point v_0 where $\delta_{\Gamma K}(v_0) > \delta_{\Gamma L}(v_0)$, say, then by continuity there exist numbers $\rho, \varepsilon > 0$ such that $\delta_{\Gamma K}(v) > \delta_{\Gamma L}(v)$ for every $v \in B_{\mathbb{S}^{n-1}}(v_0, \rho)$. Since $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ attributes a positive value for any open ball of $\mathbb{S}^{n-1}$ (see Proposition 4.5.3), we get

$$\nu_{\Gamma K}(B_{\mathbb{S}^{n-1}}(v_0,\rho)) = \int_{B_{\mathbb{S}^{n-1}}(v_0,\rho)} \delta_{\Gamma K}\, d\mathcal{H}^{n-1} > \int_{B_{\mathbb{S}^{n-1}}(v_0,\rho)} \delta_{\Gamma L}\, d\mathcal{H}^{n-1} = \\ = \nu_{\Gamma L}(B_{\mathbb{S}^{n-1}}(v_0,\rho)),$$

and this contradicts the fact that $\nu_{\Gamma K} = \nu_{\Gamma L}$. Hence we indeed have that $\delta_{\Gamma K} = \delta_{\Gamma L}$, and this leads to

$$\rho_L(v) = \left(\frac{\mathrm{vol}_n(L)}{\mathrm{vol}_n(K)}\right)^{\frac{1}{n+1}} \rho_K(v)$$

for each $v \in \mathbb{S}^{n-1}$. Consequently

$$L = \left(\frac{\mathrm{vol}_n(L)}{\mathrm{vol}_n(K)}\right)^{\frac{1}{n+1}} K.$$

Now we only have to show that $\mathrm{vol}_n(K) = \mathrm{vol}_n(L)$. For that sake, simply notice that

$$\mathrm{vol}_n(L) = \mathrm{vol}_n\left(\left(\frac{\mathrm{vol}_n(L)}{\mathrm{vol}_n(K)}\right)^{\frac{1}{n+1}} K\right) = \left(\frac{\mathrm{vol}_n(L)}{\mathrm{vol}_n(K)}\right)^{\frac{n}{n+1}} \mathrm{vol}_n(K),$$

and this gives immediately the desired.

□

Another important body associated to a given convex body is the so-called *intersection body*. To provide a complete introduction to this topic we would need some background material which is out of the scope of this book. Nevertheless, we give at least a superficial overview of this concept. First we need a result about radial functions of star bodies. By definition, the radial function of a star body is continuous, positive, and "positively homogeneous of degree -1" (it will be clear below what this means). We show that any function having these properties is the radial function of a unique star body.

Proposition 10.3.4. *Let $\rho\colon \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ be a continuous function which is positive, and such that $\rho(\lambda x) = \lambda^{-1}\rho(x)$ for any $\lambda > 0$ and every $x \in \mathbb{R}^n$. Then ρ is the radial function of a unique star body of $\mathbb{R}^n$.*

Proof. We let

$$S := \{\lambda x : \lambda \in [0,1] \text{ and } x \in \rho^{-1}(1)\}.$$

In other words, S is the union of the segments connecting the points of $\rho^{-1}(1)$ to $0_{\mathbb{R}^n}$. To verify that S is bounded, observe that for any $x \in \rho^{-1}(1)$ we have

$$1 = \rho(x) = |x|^{-1}\rho\left(\frac{x}{|x|}\right),$$

and hence

$$|x| = \rho\left(\frac{x}{|x|}\right) \leq \max\left\{\rho(u) : u \in \mathbb{S}^{n-1}\right\}.$$

Since ρ is continuous and $\mathbb{S}^{n-1}$ is compact we get that the maximum in the right-hand side is finite. Once for any $\lambda \in [0,1]$ we have $|\lambda x| \leq |x|$ it follows that S is bounded.

Now we prove that S is closed. First observe that $\rho^{-1}(1)$ is clearly closed, and hence compact (above we proved that it is bounded). If $(\lambda_n x_n)$ is a sequence in S converging to a point $z \in \mathbb{R}^n$, then (x_n) is a sequence of points in the compact set $\rho^{-1}(1)$, and hence it has a subsequence (x_{n_k}) converging to a point $x \in \rho^{-1}(1)$. On the other hand, (λ_{n_k}) is a bounded sequence of numbers, and then it has a converging subsequence $(\lambda_{n_{k_j}})$ converging to a number $\lambda \in [0,1]$. It follows that $(\lambda_{n_{k_j}} x_{n_{k_j}})$ converges to λx, and then $z = \lambda x \in S$.

It is also immediate that $\operatorname{seg}[0_{\mathbb{R}^n}, z] \subseteq S$ for any $z \in S$. Indeed, if $z \in S$, then $z = \lambda x$ for some $\lambda \in [0,1]$ and some $x \in \rho^{-1}(1)$, and any point of $\operatorname{seg}[0_{\mathbb{R}^n}, z]$ can be written as αz for some $\alpha \in [0,1]$. Thus, $\alpha z = (\alpha\lambda)x$, and $\alpha\lambda \in [0,1]$. Hence $\alpha z \in S$.

It only remains to prove that

$$\rho_S(z) := \sup\{\alpha \geq 0 : \alpha z \in S\} = \rho(z)$$

for every $z \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. For any of these points, we have that $\rho(z) > 0$, from where we get that $x := \rho(z)z \in \rho^{-1}(1) \subseteq S$. Thus,

$$\sup\{\alpha \geq 0 : \alpha z \in S\} \geq \rho(z).$$

If there exists $\alpha_0 > \rho(z)$ such that $\alpha_0 z \in S$, then writing

$$\alpha_0 z = \frac{\alpha_0}{\rho(z)}\rho(z)z = \frac{\alpha_0}{\rho(z)}x \in S$$

we have $\beta x \in S$ for

$$\beta := \frac{\alpha_0}{\rho(z)} > 1.$$

However, since $x \in \rho^{-1}(1)$, by the definition of S it is reasonable to expect that S only contains the points of the form λx for $0 \leq \lambda \leq 1$. We prove that this is indeed true. Because $\beta x \in S$, there exist $\lambda \in [0,1]$ and $y \in \rho^{-1}(1)$ such that $\lambda y = \beta x$. Hence

$$1 = \rho(y) = \rho\left(\frac{\beta x}{\lambda}\right) = \frac{\lambda}{\beta}\rho(x) = \frac{\lambda}{\beta} < 1,$$

where the last inequality comes from the facts that $\beta > 1$ and $0 \leq \lambda \leq 1$. This contradiction concludes the proof of the fact that S is a star body whose radial function is ρ.

For the uniqueness, we observe that for a given star body $S \subseteq \mathbb{R}^n$ with radial function ρ_S, for a point $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$

$$x \in S \iff \rho_S(x) \geq 1. \tag{10.3.17}$$

Indeed, if $x \in S$, then $\sup\{\lambda \geq 0 : \lambda x \in S\} \geq 1$. On the other hand, if $\rho_S(x) \geq 1$, then there exists a number $\lambda \geq 1$ such that $\lambda x \in S$. Once S contains the segment $\operatorname{seg}[0_{\mathbb{R}^n}, \lambda x]$, we get that $x \in S$.

As a consequence, we prove that two different star bodies have different radial functions. Indeed, if S_1 and S_2 are distinct star bodies, then we may assume, without loss of generality, that there exists a point $x \neq 0_{\mathbb{R}^n}$ such that $x \in S_1$ and $x \notin S_2$. Thus, $\rho_{S_1}(x) \geq 1$ and $\rho_{S_2}(x) < 1$. It follows that ρ_{S_1} and ρ_{S_2} do not represent the same function.

□

At this point, we notice that it is sufficient to consider the radial function of a star body as a function defined over $\mathbb{S}^{n-1}$. In fact, if $\tilde{\rho}\colon \mathbb{S}^{n-1} \to \mathbb{R}$ is positive and continuous, then we extend $\tilde{\rho}$ to a function $\rho\colon \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ by setting $\rho(\lambda u) := \lambda^{-1}\tilde{\rho}(u)$ for each $u \in \mathbb{S}^{n-1}$ and $\lambda > 0$. By doing that, ρ is the radial function of a unique star body of $\mathbb{R}^n$. Next we collect a few properties of radial functions of star bodies which will be useful later.

Lemma 10.3.3. *For any map $A \in \mathrm{GL}(n)$, every star body $S \subseteq \mathbb{R}^n$ and each number $\alpha > 0$ we have*

(i) *AS is a star body and $\rho_{AS}(x) = \rho_S(A^{-1}x)$ for each $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$.*

(ii) *αS is a star body and $\rho_{\alpha S}(x) = \alpha\rho_S(x)$ for every $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$.*

Proof. For **(i)**, first observe that it is immediate that AS is compact. Now let $z \in AS$. Then $\operatorname{seg}[0_{\mathbb{R}^n}, z] = A\big(\operatorname{seg}[0_{\mathbb{R}^n}, A^{-1}z]\big)$ and $A^{-1}z \in S$. Since $\operatorname{seg}[0_{\mathbb{R}^n}, A^{-1}z] \subseteq S$, we get that $\operatorname{seg}[0_{\mathbb{R}^n}, z] \subseteq AS$. Moving forward, notice that for every $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ and each $\lambda > 0$ it holds that $\lambda A^{-1}x \in S$ if and only if $\lambda x \in AS$. As a consequence, we obtain that $\rho_S(A^{-1}x) = \rho_{AS}(x)$ for every $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Hence ρ_{AS} is continuous and positive in $\mathbb{R}^{n-1}$ and AS is a star body.

To prove **(ii)**, we use **(i)** for $A = \alpha \mathrm{Id}_{\mathbb{R}^n}$. Indeed, $\alpha\rho_S(x) = \rho_S(\alpha^{-1}x) = \rho_S(A^{-1}x) = \rho_{AS}(x) = \rho_{\alpha S}(x)$.

□

Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point. The function $\mathbb{S}^{n-1} \ni u \mapsto \mathrm{vol}_{n-1}(K \cap u^{\perp})$ is positive and continuous (for the continuity, see Remark 10.3.6 below). Hence we define the *intersection body* of K as the unique star body $IK \subseteq \mathbb{R}^n$ whose radial function (over the unit sphere) is given as

$$\rho_{IK}(u) := \mathrm{vol}_{n-1}(K \cap u^{\perp}),$$

for every $u \in \mathbb{S}^{n-1}$.

Remark 10.3.6. A formal argument to prove the continuity of the function $u \mapsto \mathrm{vol}_{n-1}(K \cap u^{\perp})$ would be as follows: it is clear that we have that $\mathrm{vol}_{n-1}(K \cap u^{\perp})$ equals the (n-dimensional) volume of the "orthogonal" cylinder $c(K, u) := K \cap u^{\perp} + \operatorname{seg}[0_{\mathbb{R}^n}, u]$ (see, e.g., Exercise 4.9). One can check that if $u_n \to u$ in $\mathbb{S}^{n-1}$, then $c(K, u_n) \to c(K, u)$ in the Hausdorff metric of $\mathbb{R}^n$. Hence the continuity of the n-dimensional volume with respect to the Hausdorff metric gives the desired.

Notice that formula (4.3.7) clearly extends for star bodies. In fact, a star body S can be parametrized in polar coordinates by $x = \lambda u$ for $u \in \mathbb{S}^{n-1}$ and $0 \leq \lambda \leq \rho_S(u)$. Thus,

$$\mathrm{vol}_n(S) = \int_{\mathbb{S}^{n-1}} \int_0^{\rho_S(u)} t^{n-1}\, dt\, d\mathcal{H}^{n-1}(u) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} \rho_S(u)^n\, d\mathcal{H}^{n-1}(u).$$

As a consequence, the volume of the intersection body of K is given by the formula

$$\mathrm{vol}_n(IK) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}(K \cap u^{\perp})^n\, d\mathcal{H}^{n-1}(u). \tag{10.3.18}$$

This equality is used later to derive the so-called *Busemann intersection inequality*, which gives an upper bound for the volume of IK in terms of the volume of K. Now we move forward to investigate the behavior of the intersection body operator (i.e., the operator which associates each $K \in \mathcal{K}_0^n$ to its intersection body) under linear isomorphisms. To do that, we need to understand how the volumes of subsets of hyperplanes are transformed under maps of $\mathrm{GL}(n)$. We explain this in the next technical lemma, which has independent interest.

Lemma 10.3.4. *Let $u \in \mathbb{S}^{n-1}$ and let $E \subseteq u^\perp$ be a measurable set. For any $T \in \mathrm{GL}(n)$, we have*

$$\mathrm{vol}_{n-1}(TE) = |\det(T)||(T^{-1})^* u|\mathrm{vol}_{n-1}(E). \tag{10.3.19}$$

Proof. We first show that $T(u^\perp) = \left((T^{-1})^* u\right)^\perp$. Indeed, $T(u^\perp)$ if and only if $x \in T^{-1}x \in u^\perp$. This is equivalent to $\langle T^{-1}x, u\rangle = 0$, and this happens if and only if $\langle x, (T^{-1})^* u\rangle = 0$. The latter is the same as saying that $x \in \left((T^{-1})^* u\right)^\perp$.

As a consequence, if $E \subseteq u^\perp$, then since $TE \subseteq T(u^\perp)$ we get that $(T^{-1})^* u$ is orthogonal to the hyperplane which contains TE. Thus, in view of Exercise 4.9, we get that $\mathrm{vol}_{n-1}(TE)$ is the same as the n-dimensional volume of the "orthogonal" cylinder

$$TE + \mathrm{seg}\left[0_{\mathbb{R}^n}, \frac{(T^{-1})^* u}{|(T^{-1})^* u|}\right].$$

Therefore, using Exercise 4.9 again (but now for a "non-orthogonal" cylinder), we get

$$\begin{aligned}\mathrm{vol}_{n-1}(TE) = \mathrm{vol}_n\left(TE + \mathrm{seg}\left[0_{\mathbb{R}^n}, \frac{(T^{-1})^* u}{|(T^{-1})^* u|}\right]\right) = |\det(T)|\mathrm{vol}_n\left(E + \mathrm{seg}\left[0_{\mathbb{R}^n}, \frac{T^{-1}(T^{-1})^* u}{|(T^{-1})^* u|}\right]\right) = \\ = |\det(T)|\left|\left\langle u, \frac{T^{-1}(T^{-1})^* u}{|(T^{-1})^* u|}\right\rangle\right|\mathrm{vol}_{n-1}(E) = \\ = |\det(T)|\left|\left\langle (T^{-1})^* u, \frac{(T^{-1})^* u}{|(T^{-1})^* u|}\right\rangle\right|\mathrm{vol}_{n-1}(E) = |\det(T)||(T^{-1})^* u|\mathrm{vol}_{n-1}(E),\end{aligned}$$

and this concludes the proof.

□

Proposition 10.3.5. *For any convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point, and for any $T \in \mathrm{GL}(n)$, we have the equality*

$$I(TK) = |\det(T)|(T^*)^{-1} IK.$$

In particular, $I(OK) = O(IK)$ for any $O \in \mathrm{O}(n)$.

Proof. For any $u \in \mathbb{S}^{n-1}$, we have

$$\rho_{I(TK)}(u) = \mathrm{vol}_{n-1}(TK \cap u^\perp) = \mathrm{vol}_{n-1}(T(K \cap T^{-1}u^\perp)).$$

The set $K \cap T^{-1}u^\perp$ is contained in the hyperplane $T^{-1}u^\perp$. By the argument in the beginning of the proof of Lemma 10.3.4, we have $T^{-1}u^\perp = (T^* u)^\perp$. This hyperplane is obviously the orthogonal hyperplane of the unit vector $v := T^* u/|T^* u|$, that is, $v^\perp = (T^* u)^\perp$. Hence, applying (10.3.19) for $v \in \mathbb{S}^{n-1}$ and $E = K \cap T^{-1}u^\perp = K \cap v^\perp \subseteq v^\perp$, we have

$$\begin{aligned}\rho_{I(TK)}(u) = \mathrm{vol}_{n-1}(T(K \cap T^{-1}u^\perp)) = \mathrm{vol}_{n-1}(T(K \cap v^\perp) = \\ = |\det(T)||(T^{-1})^* v|\mathrm{vol}_{n-1}(K \cap v^\perp).\end{aligned}$$

Since $(T^{-1})^* = (T^*)^{-1}$, we have from the definition of v that $(T^{-1})^* v = u/|T^* u|$. Also, it is immediate that $\mathrm{vol}_{n-1}(K \cap v^\perp) = \rho_{IK}(v)$. Then,

$$\rho_{I(TK)}(u) = \frac{|\det(T)|}{|T^* u|}\rho_{IK}(v) = \frac{|\det(T)|}{|T^* u|}\rho_{IK}\left(\frac{T^* u}{|T^* u|}\right) = |\det(T)|\rho_{IK}(T^* u).$$

Using Lemma 10.3.3, we get that

$$\rho_{I(TK)}(u) = \rho_{|\det(T)|(T^*)^{-1}IK}(u),$$

and since this holds for every $u \in \mathbb{S}^{n-1}$, it follows that $I(TK) = |\det(T)|(T^*)^{-1}IK$. The claim for orthogonal transformations comes from the facts that $(O^*)^{-1} = O$ and $|\det(O)| = 1$.

□

It is worth mentioning that the intersection body of a convex body symmetric with respect to the origin is a convex body. A proof of that can be found in Gardner's book [930].

10.4 Some related geometric inequalities

In this section we introduce the main inequalities related to the projection and centroid bodies, namely, the *Petty projection inequality* and the *Busemann–Petty centroid inequality*. We follow the proofs given in [930]. For the sake of simplicity, we introduce the notation

$$F(K) := \int_K \dots \int_K D_n(x_1, \dots, x_n)\, dx_1 \dots dx_n, \tag{10.4.1}$$

where D_n is as defined in (10.2.5). Of course, this can be seen as a functional defined over the space $\mathcal{K}^n$ of convex bodies of $\mathbb{R}^n$. Since D_n is non-negative, we have that $F(\cdot)$ is non-decreasing for inclusion, that is, if $K \subseteq L$, then $F(K) \leq F(L)$. It is worth understanding the behavior of $F(\cdot)$ for linear images of K. For any $T \in \mathrm{GL}(n)$, we perform n changes of variables to obtain

$$\begin{aligned} F(TK) &= \int_{TK} \dots \int_{TK} D_n(x_1, \dots, x_n)\, dx_1 \dots dx_n = \\ &= \int_K \dots \int_K D_n(Tz_1, \dots, Tz_n)|\det(T)|^n\, dz_1 \dots dz_n = \\ &= \int_K \dots \int_K |\det(Tz_1, \dots, Tz_n)||\det(T)|^n\, dz_1 \dots dz_n = \\ &= \int_K \dots \int_K |\det(z_1, \dots, z_n)||\det(T)|^{n+1}\, dz_1 \dots dz_n = \\ &= |\det(T)|^{n+1} \int_K \dots \int_K D_n(z_1, \dots, z_n)\, dz_1 \dots dz_n = |\det(T)|^{n+1} F(K). \end{aligned} \tag{10.4.2}$$

In particular, $F(\cdot)$ is $\mathrm{SL}(n)$-invariant and, consequently, $\mathrm{O}(n)$-invariant. In the next two technical lemmas, we prove that $F(\cdot)$ is continuous in the Hausdorff metric and that it is non-increasing for Steiner symmetrizations.

Lemma 10.4.1. *The functional $F\colon \mathcal{K}^n \to \mathbb{R}$ is continuous with respect to the Hausdorff metric.*

Proof. Consider that $\mathbb{R}^{n^2}$ is endowed with the usual Euclidean structure and with the corresponding Hausdorff distance, which we will also denote by $d_{\mathcal{H}}$ (abusing a little of the notation). The unit ball of

$\mathbb{R}^{n^2}$ will be denoted by B^{n^2}, and for a given convex body $L \subseteq \mathbb{R}^n$ we denote the cartesian product of n copies of L as $L^n := L \times L \ldots \times L \subseteq \mathbb{R}^{n^2}$. We claim that for each convex body $L \subseteq \mathbb{R}^n$ and every number $\delta > 0$, the inclusion

$$(L + \delta B^n)^n \subseteq L^n + n\delta B^{n^2} \tag{10.4.3}$$

holds. Indeed, any point of the left-hand side set can be written as $(x_1 + \delta_1 u_1, \ldots, x_n + \delta_n u_n)$, where $x_1, \ldots, x_n \in L$, $0 \leq \delta_1 < \ldots < \delta_n \leq \delta$, and $u_1, \ldots, u_n \in B^n$. Thus $(x_1, \ldots, x_n) \in L^n$ and

$$|(x_1 + \delta_1 u_1, \ldots, x_n + \delta_n u_n) - (x_1, \ldots, x_n)| = |(\delta_1 u_1, \ldots, \delta_n u_n)| \leq \sum_{j=1}^{n} |\delta_j| \leq n\delta.$$

Since $L^n + n\delta B^{n^2}$ are the set of points of $\mathbb{R}^{n^2}$ having distance at most $n\delta$ to L^n, this inequality leads to $(x_1 + \delta_1 u_1, \ldots, x_n + \delta_n u_n) \in L^n + n\delta B^{n^2}$.

As a consequence, if $K \subseteq L + \delta B^n$, then

$$K^n \subseteq (L + \delta B^n)^n \subseteq L^n + n\delta B^{n^2}.$$

It follows that if K and L are convex bodies of $\mathbb{R}^n$ such that $d_{\mathcal{H}}(K, L) < \delta$, then $d_{\mathcal{H}}(K^n, L^n) < n\delta$. Hence, if $(K_m)_{m\in\mathbb{N}}$ is a sequence in $\mathcal{K}^n$ converging in the Hausdorff metric (of $\mathbb{R}^n$) to $K \in \mathcal{K}^n$, then K_m^n converges in the Hausdorff metric of $\mathbb{R}^{n^2}$ to K^n.

Let $(K_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{K}^n$ converging in the Hausdorff metric to $K \in \mathcal{K}^n$. If $d_{\mathcal{H}}(K_m, K) \leq \delta$, then $K \subseteq K_m + \delta B^n$ and $K_m \subseteq K + \delta B^n$. Because (K_m) is a converging sequence, we get that it is bounded, and by Proposition 2.5.3 it follows that there exists $\rho > 0$ such that $K_m \subseteq \rho B^n$ for every $m \in \mathbb{N}$. Of course, we also have that $K \subseteq \rho B^n$ (alternatively, one may simply enlarge ρ enough such that the ball ρB^n also contains K). Because of continuity, we may set

$$c := \max\left\{D_n(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \rho B^n\right\}.$$

Using Fubini's theorem and (10.4.3), and because F is non-decreasing under inclusion, we may estimate

$$\begin{aligned}
F(K_m) - F(K) \leq F(K + \delta B^n) - F(K) = \\
= \int_{K+\delta B^n} \ldots \int_{K+\delta B^n} D_n \, dx_1 \ldots dx_n - \int_K \ldots \int_K D_n \, dx_1 \ldots dx_n = \\
= \int_{(K+\delta B^n)^n} D_n \, d\mathcal{L}^{n^2} - \int_{K^n} D_n \, d\mathcal{L}^{n^2} = \int_{(K+\delta B^n)^n \setminus K^n} D_n \, d\mathcal{L}^{n^2} \leq c\mathrm{vol}_{n^2}\left((K + \delta B^n)^n \setminus K^n\right) = \\
= c\Big(\mathrm{vol}_{n^2}\left((K + \delta B^n)^n\right) - \mathrm{vol}_{n^2}(K^n)\Big) \leq c\Big(\mathrm{vol}_{n^2}\left(K^n + n\delta B^{n^2}\right) - \mathrm{vol}_{n^2}(K^n)\Big).
\end{aligned}$$

Interchanging the roles of K and K_m we get immediately

$$F(K) - F(K_m) \leq c\Big(\mathrm{vol}_{n^2}\left(K_m^n + n\delta B^{n^2}\right) - \mathrm{vol}_{n^2}(K_m^n)\Big),$$

and since $K_m \subseteq K + \delta B^n$, (10.4.3) gives that $K_m^n + n\delta B^{n^2} \subseteq K^n + 2n\delta B^{n^2}$. Hence

$$\begin{aligned}
F(K) - F(K_m) \leq c\Big(\mathrm{vol}_{n^2}\left(K_m^n + n\delta B^{n^2}\right) - \mathrm{vol}_{n^2}(K_m^n)\Big) \leq c\Big(\mathrm{vol}_{n^2}\left(K^n + 2n\delta B^{n^2}\right) - \mathrm{vol}_{n^2}(K_m^n)\Big) = \\
= c\Big(\mathrm{vol}_{n^2}\left(K^n + 2n\delta B^{n^2}\right) - \mathrm{vol}_{n^2}(K^n)\Big) + c\Big(\mathrm{vol}_{n^2}(K^n) - \mathrm{vol}_{n^2}(K_m^n)\Big).
\end{aligned}$$

Now we use the continuity of the Lebesgue volume in the Hausdorff metric (Theorem 2.6.3). For $\delta_0 > 0$ small enough, we have simultaneously

$$\mathrm{vol}_{n^2}\left(K^n + n\delta_0 B^{n^2}\right) - \mathrm{vol}_{n^2}(K^n) < \varepsilon/c \quad \text{and}$$
$$\mathrm{vol}_{n^2}\left(K^n + 2n\delta_0 B^{n^2}\right) - \mathrm{vol}_{n^2}(K^n) < \varepsilon/2c.$$

We let $m_0 \in \mathbb{N}$ be such that $m > m_0$ implies that $d_{\mathcal{H}}(K, K_m) < \delta_0$, and that

$$\mathrm{vol}_{n^2}(K^n) - \mathrm{vol}_{n^2}(K_m^n) < \varepsilon/2c,$$

where we are using the fact that (K_m^n) converges in the Hausdorff metric of $\mathbb{R}^{n^2}$ to K^n. From the estimates above, it follows that $m > m_0$ implies

$$|F(K) - F(K_m)| < \varepsilon,$$

and this gives the continuity of F in the Hausdorff metric.

□

In what follows, we recall that for each $K \in \mathcal{K}^n$ and every $u \in \mathbb{S}^{n-1}$ we denote by $S_u(K)$ the Steiner symmetrization of K with respect to u (see Section 2.7). We also introduce a new terminology: any segment connecting two boundary points of K is called a *chord* of K.

Proposition 10.4.1. *For every $K \in \mathcal{K}^n$ and any $u \in \mathbb{S}^{n-1}$ we have that*

$$F(K) \geq F(S_u(K)).$$

In other words, the functional F is non-increasing under Steiner symmetrizations. Equality holds if and only if the midpoints of all chords of K parallel to u lie in a hyperplane containing the origin.

Proof. We first claim that for the inequality it suffices to prove the case where $u = e_n$. Indeed, if this case is settled (for any convex body), then for a given $u \in \mathbb{S}^{n-1}$ we let $O \in \mathrm{O}(n)$ be such that $Oe_n = u$, and the inequality $F(S_{e_n}(O^{-1}K)) \leq F(O^{-1}K)$ leads to

$$F(K) = F(O^{-1}K) \geq F(S_{e_n}(O^{-1}K)) = F(O(S_{e_n}(O^{-1}K)) = F(S_u K).$$

Assuming the case where $u = e_n$, it follows that $u^\perp = \{(t_1, \ldots, t_{n-1}, 0) : t_1, \ldots, t_{n-1} \in \mathbb{R}\}$. Hence we will represent a point $x \in \mathbb{R}^n$ as $x = (z, t)$, where $z \in u^\perp$ and $t \in \mathbb{R}$.

To make the notation less lengthy, we write $\mathrm{proj}_{u^\perp}(K) = K|_{u^\perp}$. Let $z \in \mathrm{proj}_{u^\perp}(K)$. Then we may write

$$K \cap (z + \mathbb{R}u) = \mathrm{seg}[z + \alpha u, z + \beta u],$$

with $\beta \geq \alpha$. Letting $s = (\beta - \alpha)/2$, it follows from the definition of the Steiner symmetrization process that

$$S_u(K) \cap (z + \mathbb{R}u) = \mathrm{seg}[z - su, z + su].$$

For simplicity, write $\gamma = (\alpha + \beta)/2$, and observe that (z, γ) is the midpoint of $K \cap (z + \mathbb{R}u)$. Now let points $(z_1, 0), \ldots, (z_n, 0) \in \mathrm{proj}_{u^\perp}(K)$. For each j, we let $\alpha_j, \beta_j, \gamma_j$, and s_j be the numbers α, β, s, and γ, defined as above, associated to z_j. We write $x_j = (z_j, t_j)$ and use Fubini's theorem to get

$$F(S_u(K)) = \int_{S_u(K)} \dots \int_{S_u(K)} D_n(x_1, \dots, x_n)\, dx_1 \dots dx_n =$$
$$= \int_{K|_{u^\perp}} \int_{-s_n}^{s_n} \dots \int_{K|_{u^\perp}} \int_{-s_1}^{s_1} D_n((z_1, t_1), \dots, (z_n, t_n))\, dt_1 d\mathcal{L}^{n-1}(z_1) \dots dt_n d\mathcal{L}^{n-1}(z_n).$$

Remember that the function $D_n(x_1, \dots, x_n)$ gives the absolute value of the determinant of the matrix whose columns are $x_1, \dots, x_n$, which we denote by $(x_1, \dots, x_n)$, abusing a little of the notation. Since the matrix $((z_1, t_1), \dots, (z_n, t_n))$ can be obtained by the matrix $((z_1, t_1 + \gamma_1), \dots, (z_n, t_n + \gamma_n))$ by adding the vector $(t_1 - \gamma_1, \dots, t_n - \gamma_n)$ to the last row and dividing this last row by 2, it follows from the linearity (for vector addition and scalar multiplication in each row) of determinants that

$$\begin{aligned} D_n((z_1, t_1), \dots, (z_n, t_n)) &= |\det((z_1, t_1), \dots, (z_n, t_n))| = \\ &= \frac{1}{2}|\det((z_1, t_1 - \gamma_1), \dots, (z_n, t_n - \gamma_n)) + \det((z_1, t_1 + \gamma_1), \dots, (z_n, t_n + \gamma_n))| \leq \\ &\leq \frac{1}{2}|\det((z_1, t_1 - \gamma_1), \dots, (z_n, t_n - \gamma_n))| + \frac{1}{2}|\det((z_1, t_1 + \gamma_1), \dots, (z_n, t_n + \gamma_n))| = \\ &= \frac{1}{2} D_n((z_1, t_1 - \gamma_1), \dots, (z_n, t_n - \gamma_n)) + \frac{1}{2} D_n((z_1, t_1 + \gamma_1), \dots, (z_n, t_n + \gamma_n)). \end{aligned} \tag{10.4.4}$$

Also, it is clear that $D_n((z_1, -t_1), \dots, (z_n, -t_n)) = D_n((z_1, t_1), \dots, (z_n, t_n))$. Hence, by the equality above, we may use Fubini's theorem (in the reverse direction as used above) and some simple changes of variables to obtain an estimate for $F(S_u(K))$. For the sake of simplicity of the notation, we denote the multiple integrations with respect to $dt_1 d\mathcal{L}^{n-1}(z_1) \dots dt_n d\mathcal{L}^{n-1}(z_n)$ by the symbol $d(t, z)$.

$$\begin{aligned} F(S_u(K)) &\leq \\ &\leq \frac{1}{2} \int_{K|_{u^\perp}} \int_{-s_n}^{s_n} \dots \int_{K|_{u^\perp}} \int_{-s_1}^{s_1} D_n((z_1, t_1 - \gamma_1), \dots, (z_n, t_n - \gamma_n))\, d(t, z) + \\ &+ \frac{1}{2} \int_{K|_{u^\perp}} \int_{-s_n}^{s_n} \dots \int_{K|_{u^\perp}} \int_{-s_1}^{s_1} D_n((z_1, t_1 + \gamma_1), \dots, (z_n, t_n + \gamma_n))\, d(t, z) = \\ &= \frac{1}{2} \int_{K|_{u^\perp}} \int_{-\beta_n}^{-\alpha_n} \dots \int_{K|_{u^\perp}} \int_{-\beta_1}^{-\alpha_1} D_n((z_1, t_1), \dots, (z_n, t_n))\, d(t, z) + \\ &+ \frac{1}{2} \int_{K|_{u^\perp}} \int_{\alpha_n}^{\beta_n} \dots \int_{K|_{u^\perp}} \int_{\alpha_1}^{\beta_1} D_n((z_1, t_1), \dots, (z_n, t_n))\, d(t, z) = \\ &= \frac{1}{2} \int_{K|_{u^\perp}} \int_{\alpha_n}^{\beta_n} \dots \int_{K|_{u^\perp}} \int_{\alpha_1}^{\beta_1} D_n((z_1, -t_1), \dots, (z_n, -t_n))\, d(t, z) + \\ &+ \frac{1}{2} \int_K \dots \int_K D_n(x_1, \dots, x_n)\, dx_1 \dots dx_n = \\ &= \frac{1}{2} \int_{K|_{u^\perp}} \int_{\alpha_n}^{\beta_n} \dots \int_{K|_{u^\perp}} \int_{\alpha_1}^{\beta_1} D_n((z_1, t_1), \dots, (z_n, t_n))\, d(t, z) + \frac{1}{2} F(K) = \\ &= F(K). \end{aligned}$$

This settles the inequality.

Now we deal with the equality. We keep considering the case $u = e_n$, and in the ending of the proof we explain how this yields the general case. Regard the three assertives below:

(i) $F(K) = F(S_u(K))$,

(ii) we have equality in (10.4.4) for any $(z_1, t_1), \ldots, (z_n, t_n) \in S_u(K)$, and

(iii) $\det((z_1, \gamma_1), \ldots, (z_n, \gamma_n)) = 0$ for every $(z_1, 0), \ldots, (z_n, 0) \in \mathrm{proj}_{u^\perp}(K)$.

From continuity, it is immediate that **(i)** holds if and only if **(ii)** holds. We will prove that **(ii)** and **(iii)** are also equivalent. Assume that **(ii)** holds, and let $(z_1, 0), \ldots, (z_n, 0) \in \mathrm{proj}_{u^\perp}(K)$. Then it is clear that $(z_1, 0), \ldots, (z_n, 0) \in S_u(K)$, and equality in (10.4.4) for $t_1 = \ldots = t_n = 0$ yields

$$0 = D_n((z_1, 0), \ldots, (z_n, 0)) =$$
$$= \frac{1}{2} D_n((z_1, -\gamma_1), \ldots, (z_n, -\gamma_n)) + \frac{1}{2} D_n((z_1, \gamma_1), \ldots, (z_n, \gamma_n)) = D_n((z_1, \gamma_1), \ldots, (z_n, \gamma_n)),$$

where the first equality justifies since $D_n((z_1, 0), \ldots, (z_n, 0))$ is the absolute value of the determinant of a matrix whose last row has 0 in all of its entries. On the other hand, if **(iii)** holds, then for any points $(z_1, t_1), \ldots, (z_n, t_n) \in S_u(K)$ we have from the linearity of the determinant (again, for vector addition in the last row)

$$D_n((z_1, t_1 + \gamma_1), \ldots, (z_n, t_n + \gamma_n)) =$$
$$= |\det((z_1, t_1), \ldots, (z_n, t_n)) + \det((z_1, \gamma_1), \ldots, (z_n, \gamma_n))| = D_n((z_1, t_1), \ldots, (z_n, t_n)),$$

and by the same argument it also holds that

$$D_n((z_1, t_1 - \gamma_1), \ldots, (z_n, t_n - \gamma_n)) = D_n((z_1, t_1), \ldots, (z_n, t_n)).$$

This yields equality in (10.4.4), once the latter term becomes the same as the first term.

Now we simply observe that for any $(z, 0) \in \mathrm{proj}_{u^\perp}(K)$, the point (z, γ) is the midpoint of the chord of K parallel to u whose projection onto $u^\perp$ is $(z, 0)$. Moreover, the equality

$$\det(x_1, \ldots, x_n) = 0$$

holds if and only if one of the vectors $x_1, \ldots, x_n$ lies in the hyperplane spanned by the other vectors. This is clearly equivalent to the points $x_1, \ldots, x_n$ belonging to the same hyperplane containing the origin. Hence the equivalence between **(i)** and **(iii)** yields that $F(K) = F(S_u(K))$ if and only if the midpoints of any n chords of K parallel to u lie in the same hyperplane containing the origin. It is immediate to check that this is equivalent to the property that the midpoints of all chords of K are lying in the same hyperplane (fix any $n - 1$ chords parallel to u; the midpoint of any other chord parallel to u will have to be in the hyperplane spanned by the midpoints of these fixed chords seen as vectors).

Last, we investigate the general case of equality. Assume that $F(K) = F(S_u(K))$ for some $u \in \mathbb{S}^{n-1}$. Letting $O \in \mathrm{O}(n)$ be such that $e_n = Ou$, we have $F(S_{e_n}(OK)) = F(O(S_u(K)) = F(S_u(K)) = F(K) = F(OK)$. Since we already settled the case where the Steiner symmetrization is taken with respect to e_n, it follows that the midpoints of all the chords of OK parallel to e_n lie in the same hyperplane $\mathbf{h}$ passing through the origin. It follows straightforwardly that the midponts of the chords of $K = O^{-1}OK$ parallel to $u = O^{-1}e_n$ lie in the hyperplane $O^{-1}\mathbf{h}$.

Conversely, assume that the midpoints of all chords of K parallel to a given direction $u \in \mathbb{S}^{n-1}$ lie in the hyperplane $\mathbf{h}$ (containing the origin). Then if $O \in \mathrm{O}(n)$ is such that $Ou = e_n$, we get that the midpoints of the chords of OK parallel to e_n lie in the hyperplane $O\mathbf{h}$. It follows that $F(OK) = F(S_{e_n}(OK))$. Hence

$$F(K) = F(OK) = F(S_{Ou}(OK)) = F(O(S_u(K)) = F(S_u(K)),$$

and this concludes the proof.

□

We can also give another (equivalent) interpretation for the equality condition, which has less geometric flavor but is easier to deal with. We do it next.

Corollary 10.4.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body and let $u \in \mathbb{S}^{n-1}$. Then $F(K) = F(S_u(K))$ if and only if there exists a transformation $T \in \mathrm{SL}(n)$ such that $S_u(K) = TK$.*

Proof. First assume that $S_u(K) = TK$ for some $T \in \mathrm{SL}(n)$. Then, since F is $\mathrm{SL}(n)$-invariant, we get that $F(K) = F(TK) = F(S_u(K))$.

For the converse, if $F(K) = F(S_u(K))$, then there exists a hyperplane $\mathbf{h}$ containing the origin such that the midpoints of all chords of K parallel to u lie in $\mathbf{h}$. In particular, it follows that $u \notin \mathbf{h}$, and hence we may write $\mathbb{R}^n = \mathbf{h} \oplus \mathrm{span}\{u\}$. Let T be defined as

$$T|_{\mathbf{h}} = \mathrm{proj}_{u^\perp}|_{\mathbf{h}} \text{ and}$$
$$Tu = u.$$

That is, the restriction of T to $\mathbf{h}$ coincides with $\mathrm{proj}_{u^\perp}$ and T fixes u. Having in mind the direct sum $\mathbb{R}^n = \mathbf{h} \oplus \mathrm{span}\{u\}$, if $x = v + \alpha u$ for $v \in \mathbf{h}$ and $\alpha \in \mathbb{R}$, we have

$$T(x) = T(v + \alpha u) = \mathrm{proj}_{u^\perp}(v) + \alpha u = v - \langle u, v\rangle u + \alpha u.$$

This gives immediately that $T \in \mathrm{GL}(n)$. We claim that $S_u(K) = TK$. Indeed, let $\mathrm{seg}[z - \beta u, z + \beta u]$ be any chord of K parallel to u and having z as midpoint. Then $z \in \mathbf{h}$, and the Steiner symmetrization process takes this chord to the segment

$$\mathrm{seg}\big[\mathrm{proj}_{u^\perp}(z) - \beta u, \mathrm{proj}_{u^\perp}(z) + \beta u\big] \subseteq S_u(K).$$

On the other hand, it is clear from the definition of T that

$$T\big(\mathrm{seg}[z - \beta u, z + \beta u]\big) = \mathrm{seg}\big[\mathrm{proj}_{u^\perp}(z) - \beta u, \mathrm{proj}_{u^\perp}(z) + \beta u\big].$$

This shows that each chord of K parallel to u has the same image by the Steiner symmetrization process and by T. It follows that $S_u(K) = TK$. To verify that $T \in \mathrm{SL}(n)$, we first let $\{z_1, \ldots, z_{n-1}\}$ be a basis of $\mathbf{h}$. Then $\{z_1, \ldots, z_{n-1}, u\}$ is a basis of $\mathbb{R}^n$, and we may calculate the determinant of T using its matrix representation with respect to this basis (fixed in both domain and co-domain). Since $T(z_j) = z_j - \langle u, z_j\rangle u$ for each $j = 1, \ldots, n-1$ and $Tu = u$, this matrix representation is

$$T = \begin{bmatrix} 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & 0 \\ -\langle u, z_1\rangle & -\langle u, z_2\rangle & \ldots & -\langle u, z_{n-1}\rangle & 1 \end{bmatrix}.$$

Calculating the determinant by the Laplace formula using the last column we get immediately that $\det(T) = 1$. This concludes the proof.

□

Theorem 10.4.1 (Busemann random simplex inequality)**.** *For every convex body $K \subseteq \mathbb{R}^n$ we have that*

$$F(K) \geq n! \left(\frac{\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(K)^{n+1}. \tag{10.4.5}$$

Equality holds if and only if K is an ellipsoid centered at the origin.

Proof. First observe that if $E \subseteq \mathbb{R}^n$ is an ellipsoid centered at the origin. Then Corollaries 10.3.5 and 10.3.6 give that

$$\begin{aligned} F(E) &= \frac{n!\mathrm{vol}_n(E)^n}{2^n}\mathrm{vol}_n(\Gamma E) = \frac{n!\mathrm{vol}_n(E)^n}{2^n}\mathrm{vol}_n\left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}E\right) = \\ &= \frac{n!\mathrm{vol}_n(E)^n}{2^n}\left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(E) = n!\left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(E)^{n+1}. \end{aligned} \tag{10.4.6}$$

This shows that equality in (10.4.5) holds for ellipsoids centered at the origin.

To prove the inequality, let $K \subseteq \mathbb{R}^n$ be any convex body and let $(u_m)_{m\in\mathbb{N}}$ be a sequence in $\mathbb{S}^{n-1}$ such that sequence $(K_m)_{m\in\mathbb{N}}$ of bodies defined recursively as

$$\begin{aligned} K_1 &= S_{u_1}(K) \text{ and} \\ K_m &= S_{u_m}(K_{m-1}) \text{ for } m \geq 2 \end{aligned}$$

converges in the Hausdorff metric to a ball ρB^n (see Theorem 2.7.1). Since the volume is invariant under Steiner symmetrizations, we have $\mathrm{vol}_n(K_j) = \mathrm{vol}_n(K)$ for every $j \in \mathbb{N}$, and since the volume is continuous in the Hausdorff metric, this leads to $\mathrm{vol}_n(\rho B^n) = \mathrm{vol}_n(K)$. Because F is non-increasing under Steiner symmetrizations, we get that $F(K) \geq F(K_1)$ and $F(K_m) \geq F(K_{m+1})$ for every $m \in \mathbb{N}$. In particular, $F(K) \geq F(K_m)$ for each $m \in \mathbb{N}$, and since F is continuous in the Hausdorff metric, it follows that

$$F(K) \geq \lim_{m\to\infty} F(K_m) = F(\rho B^n).$$

Recalling that ρB^n is an ellipsoid centered at the origin, we may use (10.4.6) and the fact that K and ρB^n have the same volume to obtain

$$F(K) \geq F(\rho B^n) = n!\left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(\rho B^n)^{n+1} = n!\left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(K)^{n+1},$$

and this settles the inequality.

Now assume that equality holds in (10.4.5) for a convex body $K \subseteq \mathbb{R}^n$. Then, setting $K = K_0$ for simplicity, by the above we must have

$$F(K_0) = F(K_1) = \ldots = F(K_m) = \ldots = F(\rho B^n).$$

Since $K_m = S_{u_m}(K_{m-1})$ for each $m \in \mathbb{N}$, Corollary 10.4.1 gives that for each $m \in \mathbb{N}$ there exists a map $\tilde{T}_m \in \mathrm{SL}(n)$ with the property that

$$(K_m =)\ S_{u_m}(K_{m-1}) = \tilde{T}_m(K_{m-1}).$$

Hence for each $m \in \mathbb{N}$, we have that

$$K_m = \tilde{T}_m(\tilde{T}_{m-1}(\ldots(\tilde{T}_1 K)\ldots)),$$

and setting $T_m := \tilde{T}_m \circ \tilde{T}_{m-1} \circ \ldots \circ \tilde{T}_1$ we have that $T_m \in \mathrm{SL}(n)$ and $K_m = T_m K$. In other words, each K_m is the linear image of K under a map of $\mathrm{SL}(n)$. We claim that $(T_m)_{m\in\mathbb{N}}$ is bounded in the norm of $\mathrm{L}(n)$. To prove this, let first $x_0 \in \mathbb{R}^n$ and $r_1 > 0$ be such that $x_0 + r_1 B^n \subseteq K$. Since (K_m) converges to K in the Hausdorff metric, we have from Proposition 2.5.3 that there exists $r_2 > 0$ such that $K_m \subseteq r_2 B^n$ for every $m \in \mathbb{N}$. Hence

$$T_m(x_0 + r_1 B^n) \subseteq T_m(K) = K_m \subseteq r_2 B^n$$

for each $m \in \mathbb{N}$. Observe that since $x_0 \in K$, we get that $T_m(x_0) \in T_m(K) \subseteq r_2 B^n$ for any $m \in \mathbb{N}$. This, together with linearity, leads to

$$r_1 T_m(B^n) \subseteq r_2 B^n - T_m(x_0) \subseteq 2r_2 B^n,$$

for every $m \in \mathbb{N}$. Rewriting this as

$$T_m(B^n) \subseteq \frac{2r_2}{r_1} B^n,$$

we obtain that $||T_m||_{\mathrm{L}(n)} \leq 2r_2/r_1$ for each $m \in \mathbb{N}$. As a consequence, passing to a subsequence if necessary we may assume that $T_m \to T$ in $\mathrm{L}(n)$, and from the continuity of the determinant we get that $T \in \mathrm{SL}(n)$. From Proposition 2.6.2, it follows that $T_m(K) \to TK$ in the Hausdorff metric, and since we already have that $T_m(K)$ converges to ρB^n, we get $TK = \rho B^n$. Thus, $K = \rho T^{-1}(B^n)$, and this shows that K is an ellipsoid centered at the origin.

□

Remark 10.4.1. It is worth explaining why the name of inequality (10.4.5) is *Busemann random simplex inequality*. We first claim that for any $x_1, \ldots, x_n \in \mathbb{R}^n$ the number $D_n(x_1, \ldots, x_n)$ is $n!$ times the volume of the simplex whose vertices are $0_{\mathbb{R}^n}, x_1, \ldots, x_n$, that is,

$$\frac{1}{n!} D_n(x_1, \ldots, x_n) = \mathrm{vol}_n\big(\mathrm{conv}\{0_{\mathbb{R}^n}, x_1, \ldots, x_n\}\big). \tag{10.4.7}$$

To prove that, consider first the case where $x_1 = e_1, \ldots, x_n = e_n$. Then from (4.4.5) we get that

$$\frac{1}{n!} D_n(e_1, \ldots, e_n) = \frac{1}{n!} |\det(e_1, \ldots, e_n)| = \frac{1}{n!} = \mathrm{vol}_n\big(\mathrm{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\}\big).$$

Now, for $x_1, \ldots, x_n \in \mathbb{R}^n$ arbitrarily given, we let T be the linear map such that $Te_j = x_j$ for each $j = 1, \ldots, n$. Hence

$$\begin{aligned} \frac{1}{n!} D_n(x_1, \ldots, x_n) = \frac{1}{n!} |\det(Te_1, \ldots, Te_n)| = \frac{|\det(T)|}{n!} D_n(e_1, \ldots, e_n) = \\ = |\det(T)| \mathrm{vol}_n\big(\mathrm{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\}\big) = \mathrm{vol}_n\big(T\big(\mathrm{conv}\{0_{\mathbb{R}^n}, e_1, \ldots, e_n\}\big)\big) = \\ = \mathrm{vol}_n\big(\mathrm{conv}\{0_{\mathbb{R}^n}, x_1, \ldots, x_n\}\big), \end{aligned}$$

and this settles (10.4.7). Having this equality in mind, we divide both sides of inequality (10.4.5) by $n!\mathrm{vol}_n(K)^n$ and rewrite it as

$$\frac{1}{\mathrm{vol}_n(K)^n} \int_K \ldots \int_K \mathrm{vol}_n\big(\mathrm{conv}\{0_{\mathbb{R}^n}, x_1, \ldots, x_n\}\big)\, dx_1 \ldots dx_n \geq \left(\frac{\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(K). \tag{10.4.8}$$

In the language of probability theory, the left-hand side is the *expected value* of the volume of a simplex whose vertices are $0_{\mathbb{R}^n}$ and n randomly chosen points in K (with uniform probability). Hence, inequality (10.4.5) states that if one randomly chooses n points $x_1, \ldots, x_n$ in K, then the expected value of the volume of the simplex $\mathrm{conv}\{0_{\mathbb{R}^n}, x_1, \ldots, x_n\}$ is, up to multiplication by a constant which only depends on the dimension, greater than or equal to $\mathrm{vol}_n(K)$. Moreover, equality holds if and only if K is an ellipsoid centered at the origin.

Remark 10.4.2. The constant in inequality (10.4.5) is sometimes presented in a different way. Using the identities given in items **(a)** and **(b)** of Exercise 4.7 (which follow from the identity $\Gamma(x+1) = x\Gamma(x)$ of the Gamma function), we can rewrite

$$n!\left(\frac{\kappa_{n-1}}{(n+1)\kappa_n}\right)^n = n!\left(\frac{\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \frac{2^n\pi^{n-1}}{n!\kappa_n\kappa_{n-1}} = \frac{2}{n+1}\left(\frac{2\pi\kappa_{n-1}}{n+1}\right)^{n-1}\frac{1}{\kappa_n^{n+1}} = \frac{2\kappa_{n+1}^{n-1}}{(n+1)\kappa_n^{n+1}}.$$

As a corollary of the Busemann random simplex inequality we get a lower bound for the volume of the centroid body of a body $K \in \mathcal{K}_0^n$ in terms of the volume of K.

Corollary 10.4.2. (Busemann–Petty centroid inequality) *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point. We have the inequality*

$$\mathrm{vol}_n(\Gamma K) \geq \left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(K), \tag{10.4.9}$$

and equality holds if and only if K is an ellipsoid centered at the origin.

Proof. This follows immediately from the expression for the volume $\mathrm{vol}_n(\Gamma K)$ in terms of $F(K)$ and from the Busemann random simplex inequality:

$$\mathrm{vol}_n(\Gamma K) = \frac{2^n}{n!\mathrm{vol}_n(K)^n}F(K) \geq \frac{2^n}{n!\mathrm{vol}_n(K)^n}n!\left(\frac{\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(K)^{n+1} = \left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(K).$$

Equality holds above if and only if we have equality in the Busemann random simplex inequality, and this happens if and only if K is an ellipsoid centered at the origin.

□

We move forward to present the so-called *Petty projection inequality*, which gives an upper bound for the affine invariant (10.3.4) discussed in Remark 10.3.1. In what follows, recall that we denote by $\Pi^\circ K$ the polar of the projection body of K, that is, $\Pi^\circ K = (\Pi K)^\circ$.

Lemma 10.4.2. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies, and assume that L has the origin as an interior point. Then*

$$V(K,\dots,K,\Gamma L) = \frac{2}{n(n+1)\mathrm{vol}_n(L)}\int_{\mathbb{S}^{n-1}} \rho_L(u)^{n+1}\rho_{\Pi^\circ K}(u)^{-1}\,d\mathcal{H}^{n-1}(u).$$

Proof. By the definition of the mixed surface area measure $S_{n-1}(K,\cdot)$, expressions (10.3.11) and (10.3.1) for the support functions of the centroid and projection bodies, and Fubini's theorem we have

$$\begin{aligned} V(K,\dots,K,\Gamma L) &= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{\Gamma L}(v)\,dS_{n-1}(K,v) = \\ &= \frac{1}{n(n+1)\mathrm{vol}_n(L)}\int_{\mathbb{S}^{n-1}}\int_{\mathbb{S}^{n-1}} |\langle v,u\rangle|\rho_L(u)^{n+1}\,d\mathcal{H}^{n-1}(u)dS_{n-1}(K,v) = \\ &= \frac{1}{n(n+1)\mathrm{vol}_n(L)}\int_{\mathbb{S}^{n-1}} \rho_L(u)^{n+1}\left(\int_{\mathbb{S}^{n-1}} |\langle v,u\rangle|\,dS_{n-1}(K,v)\right)d\mathcal{H}^{n-1}(u) = \\ &= \frac{2}{n(n+1)\mathrm{vol}_n(L)}\int_{\mathbb{S}^{n-1}} \rho_L(u)^{n+1}h_{\Pi K}(u)\,d\mathcal{H}^{n-1}(u). \end{aligned}$$

Since ΠK is a convex body having the origin as an interior point, we get that $h_{\Pi K}(u) = \rho_{\Pi^\circ K}(u)^{-1}$ for each $u \in \mathbb{S}^{n-1}$. This concludes the proof.

□

Theorem 10.4.2 (Petty projection inequality)**.** *For any convex body $K \subseteq \mathbb{R}^n$ we have the inequality*

$$\mathrm{vol}_n(K)^{n-1}\mathrm{vol}_n(\Pi^\circ K) \le \left(\frac{\kappa_n}{\kappa_{n-1}}\right)^n. \tag{10.4.10}$$

Equality holds if and only if K is an ellipsoid.

Proof. Letting $L = \Pi^\circ K$ in Lemma 10.4.2, we obtain from the first equality in (4.3.7) that

$$V(K,\dots,K,\Gamma(\Pi^\circ K)) = \frac{2}{n(n+1)\mathrm{vol}_n(\Pi^\circ K)}\int_{\mathbb{S}^{n-1}} \rho_{\Pi^\circ K}(u)^n\, d\mathcal{H}^{n-1}(u) = \\ = \frac{2}{(n+1)\mathrm{vol}_n(\Pi^\circ K)}\mathrm{vol}_n(\Pi^\circ K) = \frac{2}{n+1}.$$

By the 1st Minkowski inequality (Theorem 6.3.1), this gives

$$\left(\frac{2}{n+1}\right)^n = V(K,\dots,K,\Gamma(\Pi^\circ K))^n \ge \mathrm{vol}_n(\Gamma(\Pi^\circ K))\mathrm{vol}_n(K)^{n-1},$$

and hence, using Busemann–Petty centroid inequality (10.4.9) for $\Pi^\circ K$, we get

$$\left(\frac{2}{n+1}\right)^n \ge \mathrm{vol}_n(\Gamma(\Pi^\circ K))\mathrm{vol}_n(K)^{n-1} \ge \left(\frac{2\kappa_{n-1}}{(n+1)\kappa_n}\right)^n \mathrm{vol}_n(\Pi^\circ K)\mathrm{vol}_n(K)^{n-1},$$

and this gives (10.4.10).

Notice that if equality holds, then we have equality in the Busemann–Petty centroid inequality (for $\Pi^\circ K$) and in the 1st Minkowski inequality (for K and $\Gamma(\Pi^\circ K)$). The equality case in the Busemannn–Petty centroid inequality gives that $\Pi^\circ K$ is an ellipsoid centered at the origin, and hence $\Gamma(\Pi^\circ K)$ is an ellipsoid centered at the origin (see Corollary 10.3.5). The equality case in the 1st Minkowski inequality yields that K and $\Gamma(\Pi^\circ K)$ are homothetical. Since $\Gamma(\Pi^\circ K)$ is an ellipsoid, it follows that K is an ellipsoid.

On the other hand, it is not difficult to verify that we have equality in (10.4.10) whenever K is an ellipsoid. Since $\mathrm{vol}_n(K)^{n-1}\mathrm{vol}_n(\Pi^\circ K)$ is affine invariant, we may assume that $K = B^n$. Since

$$h_{\Pi B^n}(u) = \mathrm{vol}_n\big(\mathrm{proj}_{u^\perp}(B^n)\big) = \mathrm{vol}_{n-1}(B^{n-1}) = \kappa_{n-1}$$

for each $u \in \mathbb{S}^{n-1}$, it follows that $\Pi B^n = \kappa_{n-1}B^n$ (compare with Exercise 10.11). Hence

$$\Pi^\circ B^n = (\kappa_{n-1}B^n)^\circ = \frac{1}{\kappa_{n-1}}B^n.$$

Thus,

$$\mathrm{vol}_n(B^n)^{n-1}\mathrm{vol}_n(\Pi^\circ B^n) = \kappa_n^{n-1}\mathrm{vol}_n\left(\frac{1}{\kappa_{n-1}}B^n\right) = \left(\frac{\kappa_n}{\kappa_{n-1}}\right)^n,$$

and the proof is complete. □

Remark 10.4.3. One of the important features of the Petty projection inequality is that it is, in some sense, stronger than the classical isoperimetric inequality. This is discussed later in Exercise 11.1.

One may ask whether a lower bound for the affine invariant $\mathrm{vol}_n(K)^{n-1}\mathrm{vol}_n(\Pi^\circ K)$ is known. The answer is positive: the optimal lower bound was obtained by Zhang in the early 90s. He also characterized the equality case. His result is now known as the *Zhang projection inequality*. Despite being a recent result, we can present a proof (without the characterization of the equality case; this

will be only briefly mentioned in Remark 10.4.6) using the theory that we developed thus far. To do so, we just need some preparation material.

Recall that in the proof of the Rogers–Shephard inequality (Theorem 5.3.1) we have worked with the function $g_K : \mathbb{R}^n \to \mathbb{R}$ defined as

$$g_K(x) = \mathrm{vol}_n\big(K \cap (K + x)\big),$$

for each $x \in \mathbb{R}^n$. This is known as the *covariogram* of the convex body $K \subseteq \mathbb{R}^n$. In the following lemma we collect some early properties of the covariogram.

Lemma 10.4.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body, and let $g_K : \mathbb{R}^n \to \mathbb{R}$ be its covariogram. Then,*

(i) *g_K is an even function and*

(ii) *we have the equality $g_K(x) = 1_K * 1_{-K}(x)$ for every $x \in \mathbb{R}^n$, where $*$ denotes the usual convolution (see Section 1.5).*

Proof. For **(i)** we have to prove that $g_K(x) = g_K(-x)$ for every $x \in X$. This is a consequence of the fact that

$$K \cap (K + x) = x + K \cap (K - x),$$

that is, the set $K \cap (K + x)$ is a translation of $K \cap (K - x)$ (and hence both have the same volume). Indeed, if $z \in K \cap (K - x)$, then $z + x \in K + x$ and $z + x \in K - x + x = K$, and this shows that $x + K \cap (K - x) \subseteq K \cap (K + x)$. The reverse inclusion follows analogously.

To prove **(ii)**, just notice that for any fixed $x \in \mathbb{R}^n$ we have that $x - y \in -K$ if and only if $y \in K + x$. Thus, $1_{-K}(x - y) = 1_{K+x}(y)$. It follows that

$$\begin{aligned} 1_K * 1_{-K}(x) &= \int_{\mathbb{R}^n} 1_K(y) 1_{-K}(x - y)\, dy = \int_{\mathbb{R}^n} 1_K(y) 1_{K+x}(y)\, dy = \\ &= \int_{\mathbb{R}^n} 1_{K\cap(K+x)}(y)\, dy = \mathrm{vol}_n\big(K \cap (K + x)\big) = g_K(x). \end{aligned}$$

□

In the proof of Theorem 5.3.1, we proved that

$$K - K = \big\{x \in \mathbb{R}^n : K \cap (K + x) \neq \emptyset\big\},$$

and it follows that $g_K(x) = 0$ whenever $x \notin K - K$. Hence we may restrict g_K to the difference body $K - K$. It will be useful later to know the integral of g_K over $\mathbb{R}^n$ (or, equivalently, over $K - K$). This calculation was actually done in (5.3.3), where we implicitly used item **(ii)** of Lemma 10.4.3. For the convenience of the reader, we repeat this calculation:

$$\begin{aligned} \int_{\mathbb{R}^n} g_K(x)\, dx &= \int_{\mathbb{R}^n} 1_K * 1_{-K}(x)\, dx = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} 1_K(y) 1_{-K}(x - y)\, dy dx = \\ &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} 1_K(y) 1_{y-K}(x)\, dx dy = \int_{\mathbb{R}^n} 1_K(y) \mathrm{vol}_n(y - K)\, dy = \\ &= \mathrm{vol}_n(K) \int_{\mathbb{R}^n} 1_K(y)\, dy = \mathrm{vol}_n(K)^2. \end{aligned} \tag{10.4.11}$$

Next we move forward to calculate the directional derivatives of g_K at $0_{\mathbb{R}^n}$.

Proposition 10.4.2. *For any convex body $K \subseteq \mathbb{R}^n$ and every $u \in \mathbb{S}^{n-1}$, we have that*

$$(g_K)'_+(0_{\mathbb{R}^n}, u) = -\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big). \tag{10.4.12}$$

Remark 10.4.4. Since we have that

$$(g_K)'_+(0_{\mathbb{R}^n}, u) := \lim_{t \to 0^+} \frac{g_K(tu) - g_K(0_{\mathbb{R}^n})}{t},$$

and because g_K is an even function, it follows from (10.4.12) that

$$(g_K)'_-(0_{\mathbb{R}^n}, u) := \lim_{t \to 0^-} \frac{g_K(tu) - g_K(0_{\mathbb{R}^n})}{t} = \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big).$$

Proof. We first claim that for t small enough $K \cap (K + tu)$ is a convex body. Indeed, since K is a convex body, we may fix a ball $B \subseteq K$. For small t the intersection $B \cap (B + tu)$ clearly contains a (non-degenerate) ball, and $B \cap (B + tu) \subseteq K \cap (K + tu)$. The convexity and compactness of $K \cap (K + tu)$ are immediate.

For the sake of simplicity, throughout this proof we always assume that $t > 0$ is small enough such that $K \cap (K + tu)$ is a convex body. We prove (10.4.12) as a consequence of the inequalities

$$t\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K \cap (K + tu))\big) \leq \mathrm{vol}_n\big(K \setminus (K + tu)\big) \leq t\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big). \tag{10.4.13}$$

We start with the second inequality. Since the volume is translation-invariant, we may assume without loss of generality that K and $K + tu$ are contained in $\mathrm{int}(\mathbf{h}^+(u, 0))$, which is the "positive" half-space determined by $u^\perp$. It is clear that

$$\mathrm{proj}_{u^\perp}(K) = \mathrm{proj}_{u^\perp}(K + tu).$$

We describe the points of $K \setminus (K + tu)$ in terms of their intersections with the half-lines starting at $u^\perp$ in the direction of u (see Figure 10.4.1). Let $x \in K \setminus (K + tu)$. Then we may write $x = z + \lambda u$ for some $\lambda > 0$, where $z = \mathrm{proj}_{u^\perp}(x) \in \mathrm{proj}_{u^\perp}(K)$. We obviously have the inequality

$$\lambda \geq \min\{\alpha > 0 : z + \alpha u \in K\} := \alpha_0(z),$$

and we claim that we also have

$$\lambda \leq \alpha_0(z) + t.$$

To verify this inequality, observe that since $z + \alpha_0(z)u$ and $z + \lambda u$ are points of K, we have that $z + (\alpha_0(z) + t)u$ and $z + (\lambda + t)u \in K + tu$, and by convexity $K + tu$ contains the segment connecting these points. If $\lambda > \alpha_0(z) + t$, we obtain that $\alpha_0(z) + t \leq \lambda \leq \lambda + t$, from which we have $z + \lambda u \in K + tu$. This contradiction gives that $\lambda \leq \alpha_0(z) + t$. As a consequence of these bounds for λ, we get that

$$\big(K \setminus (K + tu)\big) \cap (z + \mathbb{R}_+ u) \subseteq \mathrm{seg}\big[z + \alpha_0(z)u, z + (\alpha_0(z) + t)u\big].$$

Hence by calculating the volume of $K \setminus (K + tu)$ by Cavalieri's principle we obtain the inequality

$$\mathrm{vol}_n\big(K \setminus (K + tu)\big) \leq \int_{\mathrm{proj}_{u^\perp}(K)} \int_{\alpha_0(z)}^{\alpha_0(z)+t} ds d\mathcal{H}^{n-1} = t\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big),$$

which is the right-hand inequality in (10.4.13).

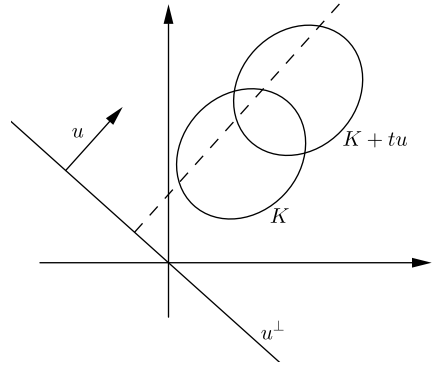


Fig. 10.4.1 The points of $K \setminus (K + tu)$ have the form $z + \lambda u$, where z is a point of the orthogonal projection of K onto $u^{\perp}$.

Now we prove the other inequality. Let $z \in \mathrm{proj}_{u^{\perp}}(K \cap (K + tu))$, and still write $\alpha_0(z) := \min\{\alpha > 0 : z + \alpha u \in K\}$. We claim that

$$\mathrm{seg}\big[z + \alpha_0(z)u, z + (\alpha_0(z) + t)u\big) \subseteq K \setminus (K + tu). \tag{10.4.14}$$

Indeed, assume that $z + \beta u \in K \cap (K + tu)$. Then $z + (\beta - t)u \in K$, from what we get that $\beta - t \geq \alpha_0(z)$, and this leads to $\beta \geq \alpha_0(z) + t$. This has two consequences. First, we have that $z + (\alpha_0(z) + t)u \in K$, because

$$z + (\alpha_0(z) + t)u \in \mathrm{seg}\big[z + \alpha_0(z)u, z + \beta u\big],$$

and this segment connects two points of K. In particular, this leads to

$$\mathrm{seg}\big[z + \alpha_0(z)u, z + (\alpha_0(z) + t)u\big] \subseteq K. \tag{10.4.15}$$

Second, we have

$$(K + tu) \cap \mathrm{seg}\big[z + \alpha_0(z)u, z + (\alpha_0(z) + t)u\big) = \emptyset. \tag{10.4.16}$$

In fact, let $z + \beta u$ be any point of the segment $\mathrm{seg}[z + \alpha_0(z)u, z + (\alpha_0(z) + t)u)$. Then $\alpha_0(z) \leq \beta < \alpha_0(z) + t$, and $z + \beta u \in K$ by (10.4.15) above. If we also have that $z + \beta u \in K + tu$, then it holds that $z + \beta u \in K \cap (K + tu)$, and by what we have proved, this leads to $\beta \geq \alpha_0(z) + t$. This contradiction gives (10.4.16). The inclusion (10.4.15) and the equality (10.4.16) give (10.4.14).

It follows that, for every point $z \in \mathrm{proj}_{u^{\perp}}(K \cap (K + tu))$, the set $K \setminus (K + tu)$ contains the segment $\mathrm{seg}[z + \alpha_0(z)u, z + (\alpha_0(z) + t)u)$. Using Cavalieri's principle again to compute the volume of $K \setminus (K + tu)$, we get

$$\mathrm{vol}_n\big(K \setminus (K + tu)\big) \geq \int_{\mathrm{proj}_{u^{\perp}}(K \cap (K + tu))} \int_{\alpha_0(z)}^{\alpha_0(z)+t} ds d\mathcal{H}^{n-1} = t\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^{\perp}}(K \cap (K + tu))\big),$$

and this settles (10.4.13).

Dividing (10.4.13) by $t > 0$, and taking the limit as $t \mapsto 0^+$, we obtain

$$\lim_{t \to 0^+} \big[\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^{\perp}}(K \cap (K + tu))\big)\big] \leq \lim_{t \to 0^+} \frac{\mathrm{vol}_n\big(K \setminus (K + tu)\big)}{t} \leq \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^{\perp}}(K)\big).$$

Since K can be written as the disjoint union

$$K = \big(K \setminus (K + tu)\big) \cup \big(K \cap (K + tu)\big),$$

we have that

$$\mathrm{vol}_n\big(K \setminus (K + tu)\big) = \mathrm{vol}_n(K) - \mathrm{vol}_n\big(K \cap (K + tu)\big) = g_K(0_{\mathbb{R}^n}) - g_K(tu),$$

and hence we may rewrite the inequality above (multiplying by -1) as

$$\begin{aligned} -\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big) \leq \lim_{t\to 0^+} \frac{g_K(tu) - g_K(0_{\mathbb{R}^n})}{t} = (g_K)'_+(0_{\mathbb{R}^n}, u) \leq \\ \leq -\lim_{t\to 0^+} \Big[\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K \cap (K + tu))\big)\Big]. \end{aligned} \tag{10.4.17}$$

It is clear that $K \cap (K + tu)$ converges in the Hausdorff metric to K as $t \to 0$ (this can be obtained, e.g., as a consequence of Theorem 2.6.1). Therefore, we have that

$$\mathrm{proj}_{u^\perp}\big(K \cap (K + tu)\big) \to \mathrm{proj}_{u^\perp}(K)$$

in the Hausdorff metric of $u^\perp$ as $t \to 0$. Indeed, the unit ball $B_{u^\perp}$ of $u^\perp$ is just the projection of B^n onto $u^\perp$, and hence an inclusion $C \subseteq L + \varepsilon B^n$ always implies the inclusion $\mathrm{proj}_{u^\perp}(C) \subseteq \mathrm{proj}_{u^\perp}(L) + \varepsilon B_{u^\perp}$ (see Exercise 2.52 and the comment above it for more details). Now we recall that vol_{n-1} denotes the Lebesgue measure of $\mathbb{R}^{n-1}$ induced on $u^\perp$ by a natural identification (i.e., an isometric isomorphism between $u^\perp$ and $\mathbb{R}^{n-1}$). Since it is clear that the Hausdorff metric of $u^\perp$ coincides with the Hausdorff metric of $\mathbb{R}^{n-1}$ under such an identification, and because the Hausdorff metric of $\mathbb{R}^{n-1}$ is continuous in the Lebesgue measure of $\mathbb{R}^{n-1}$, we finally arrive at

$$\lim_{t\to 0^+} \Big[\mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K \cap (K + tu))\big)\Big] = \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big).$$

Replacing this equality in the right-hand side of (10.4.17) gives (10.4.12), and the proof is complete. □

Corollary 10.4.3. *For any fixed $u \in \mathbb{S}^{n-1}$, the inequality*

$$g_K(tu) \leq \mathrm{vol}_n(K)\left(1 - \frac{t h_{\Pi K}(u)}{n\mathrm{vol}_n(K)}\right)^n \tag{10.4.18}$$

holds whenever $0 \leq t \leq \rho_{K-K}(u)$.

Proof. In the proof of Theorem 5.3.1, we showed that the function

$$x \mapsto g_K(x)^{1/n} = \mathrm{vol}_n\big(K \cap (K + x)\big)^{1/n}$$

is concave on $K - K$. Fix any $u \in \mathbb{S}^{n-1}$. Since $0 \leq tu \leq \rho_{K-K}(u)$ implies that $tu \in K - K$, we get that the function $f\colon [0, \rho_{K-K}(u)] \to \mathbb{R}$ given as

$$f(t) := g_K(tu)^{1/n} = \mathrm{vol}_n\big(K \cap (K + tu)\big)^{1/n}$$

is concave. From (10.4.12) and the chain rule we get that

$$f'_+(0) = \frac{1}{n} g_K(0_{\mathbb{R}^n})^{(1-n)/n} (g_K)'_+(0_{\mathbb{R}^n}, u) = -\frac{1}{n\mathrm{vol}_n(K)^{(n-1)/n}} \mathrm{vol}_{n-1}(\mathrm{proj}_{u^\perp}(K)) = \\ = -\frac{h_{\Pi K}(u)}{n\mathrm{vol}_n(K)^{(n-1)/n}}.$$

The concavity of f gives that its graph lies below the line passing through $(0, f(0))$ and with inclination $f'_+(0)$. Then

$$g_K(tu)^{1/n} = f(t) \leq f(0) + tf'_+(0) = \mathrm{vol}_n(K)^{1/n} - \frac{th_{\Pi K}(u)}{n\mathrm{vol}_n(K)^{(n-1)/n}} = \\ = \mathrm{vol}_n(K)^{1/n} \left(1 - \frac{th_{\Pi K}(u)}{n\mathrm{vol}_n(K)}\right), \tag{10.4.19}$$

and taking the n-th power on both sides we get the desired.

□

Remark 10.4.5. In particular, for $t = \rho_{K-K}(u)$ we have that

$$1 - \frac{\rho_{K-K}(u) h_{\Pi K}(u)}{n\mathrm{vol}_n(K)} \geq \left(\frac{g_K(\rho_{K-K}(u)u)}{\mathrm{vol}_n(K)}\right)^{1/n} \geq 0,$$

and hence

$$\rho_{K-K}(u) \frac{h_{\Pi K}(u)}{n\mathrm{vol}_n(K)} \leq 1. \tag{10.4.20}$$

This will be used ahead.

Now we are in the position to state and prove the Zhang projection inequality.

Theorem 10.4.3 (Zhang projection inequality)**.** *For every convex body $K \subseteq \mathbb{R}^n$ the inequality*

$$\mathrm{vol}_n(K)^{n-1} \mathrm{vol}_n(\Pi^\circ K) \geq \frac{1}{n^n} \binom{2n}{n}$$

holds.

Proof. As in the proof of Theorem 5.3.1, we will use the equality

$$\int_0^1 (1-r)^n r^{n-1}\, dr = \frac{1}{n} \binom{2n}{n}^{-1},$$

which comes easily by multiple successive integrations by parts. The points of $K - K$ can be written in polar coordinates as $x = tu$ for $u \in \mathbb{S}^{n-1}$ and $0 \leq t \leq \rho_{K-K}(u)$. Hence, using equality (10.4.11) and inequality (10.4.18) we write

$$\mathrm{vol}_n(K)^2 = \int_{\mathbb{R}^n} g_K(x)\, dx = \int_{K-K} g_K(x)\, dx = \int_{\mathbb{S}^{n-1}} \int_0^{\rho_{K-K}(u)} g_K(tu) t^{n-1}\, dt d\mathcal{H}^{n-1}(u) \leq \\ \leq \int_{\mathbb{S}^{n-1}} \int_0^{\rho_{K-K}(u)} \mathrm{vol}_n(K) \left(1 - \frac{th_{\Pi K}(u)}{n\mathrm{vol}_n(K)}\right)^n t^{n-1}\, dt d\mathcal{H}^{n-1}(u).$$

To deal with this last integral (with respect to t), we make the change of variables:

$$r = t\frac{h_{\Pi K}(u)}{n\mathrm{vol}_n(K)},$$

and because $0 \le t \le \rho_{K-K}(u)$, we have that

$$0 \le r \le \rho_{K-K}(u)\frac{h_{\Pi K}(u)}{n\mathrm{vol}_n(K)} =: \beta.$$

This leads to

$$\mathrm{vol}_n(K)^2 \le \mathrm{vol}_n(K)\int_{\mathbb{S}^{n-1}}\int_0^{\beta}(1-r)^n\left(\frac{n\mathrm{vol}_n(K)}{h_{\Pi K}(u)}\right)^n r^{n-1}\,dr\,d\mathcal{H}^{n-1}(u) =$$
$$= n^n\mathrm{vol}_n(K)^{n+1}\int_{\mathbb{S}^{n-1}} h_{\Pi K}(u)^{-n}\int_0^{\beta}(1-r)^n r^{n-1}\,dr\,d\mathcal{H}^{n-1}(u).$$

Because by (10.4.20) we have that $\beta \le 1$, and since $(1-r)^n r^{n-1}$ is non-negative for $r \in [\beta, 1]$, we get

$$\int_0^{\beta}(1-r)^n r^{n-1}\,dr \le \int_0^1 (1-r)^n r^{n-1}\,dr = \frac{1}{n}\binom{2n}{n}^{-1}.$$

Because $h_{\Pi K}(u)^{-n}$ is always positive for $u \in \mathbb{S}^{n-1}$, we finally arrive at

$$\mathrm{vol}_n(K)^2 \le n^n\mathrm{vol}_n(K)^{n+1}\binom{2n}{n}^{-1}\left(\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{\Pi K}(u)^{-n}\,d\mathcal{H}^{n-1}(u)\right) =$$
$$= n^n\mathrm{vol}_n(K)^{n+1}\binom{2n}{n}^{-1}\mathrm{vol}_n(\Pi^\circ K),$$

where the last equality follows from (4.3.8). Rearranging the terms yields the desired.

□

Remark 10.4.6. As in the Rogers–Shephard inequality, the equality case in the Zhang projection inequality implies that $K \cap (K + x)$ is homothetical to K for each $x \in K - K$. Indeed, if equality holds in the Zhang projection inequality, then equality that holds in (10.4.18) holds for every $u \in \mathbb{S}^{n-1}$ and any $0 \le t \le \rho_{K-K}(u)$. Hence we also have equality in (10.4.19), and in particular this gives that

$$f(t) = f(0) + tf'_+(0).$$

Thus, f is a polynomial of first degree. It follows that for any $0 \le t, s \le \rho_{K-K}(u)$ and every $\lambda \in [0, 1]$ we have

$$f\big((1-\lambda)s + \lambda t\big) = (1-\lambda)f(s) + \lambda f(t).$$

For any $x \in K - K$, we write $x = tu$ for some $u \in \mathbb{S}^{n-1}$ and $0 \le t \le \rho_{K-K}(u)$. Applying the above for $s = 0$, recalling the definition of f (for any fixed $u \in \mathbb{S}^{n-1}$), and using the Brunn–Minkowski inequality (Theorem 5.1.2), we get

$$\mathrm{vol}_n(K \cap (K + \lambda x))^{1/n} = (1-\lambda)\mathrm{vol}_n(K)^{1/n} + \lambda\mathrm{vol}_n(K \cap (K + x))^{1/n} \le$$
$$\le \mathrm{vol}_n\big((1-\lambda)K + \lambda(K \cap (K + x))\big)^{1/n} \le \mathrm{vol}_n(K \cap (K + \lambda x))^{1/n},$$

where the last inequality justifies since (5.3.2) gives

$$(1-\lambda)K + \lambda(K \cap (K+x)) \subseteq K \cap (K+\lambda x). \tag{10.4.21}$$

Hence we have equality in the Brunn–Minkowski inequality for the bodies K and $K \cap (K+x)$, and then they are homothetical.

As it was mentioned (without a proof) in Remark 5.3.1, the property of K being homothetical to $K \cap (K+x)$ for every $x \in K - K$ characterizes the simplex. It follows that the equality case in the Zhang projection inequality characterizes the simplex.

We finish this section by briefly mentioning another consequence of the Busemann random simplex inequality, namely, the *Busemann intersection inequality*. This inequality gives an upper bound for the volume of the intersection body of a convex body. We need a formula for the volume of a convex body which we give without a proof.

Proposition 10.4.3. *For any bounded measurable set $C \subseteq \mathbb{R}^n$ it holds that*

$$\mathrm{vol}_n(C)^{n-1} = \frac{1}{2}\int_{\mathbb{S}^{n-1}} \int_{C\cap u^\perp} \dots \int_{C\cap u^\perp} D_{n-1}(x_1,\dots,x_{n-1})\, dx_1 \dots dx_{n-1} d\mathcal{H}^{n-1}(u),$$

where each dx_j stands for the integration with respect to the Lebesgue $(n-1)$-dimensional measure.

Proof. This is formula (10.102) in [2167].

□

Theorem 10.4.4 (Busemann intersection inequality)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point, and recall that IK denotes the intersection body of K. We have*

$$\mathrm{vol}_n(IK) \le \frac{\kappa_{n-1}^n}{\kappa_n^{n-2}} \mathrm{vol}_n(K)^{n-1}. \tag{10.4.22}$$

Proof. This follows immediately from the formula above by applying the Busemann random simplex inequality in $\mathbb{R}^{n-1}$ for $K \cap u^\perp$:

$$\begin{aligned}
\mathrm{vol}_n(K)^{n-1} &= \frac{1}{2}\int_{\mathbb{S}^{n-1}} \int_{K\cap u^\perp} \dots \int_{K\cap u^\perp} D_{n-1}(x_1,\dots,x_{n-1})\, dx_1 \dots dx_{n-1} d\mathcal{H}^{n-1}(u) = \\
&= \frac{1}{2}\int_{\mathbb{S}^{n-1}} F(K\cap u^\perp)\, d\mathcal{H}^{n-1}(u) \ge \frac{1}{2}\int_{\mathbb{S}^{n-1}} (n-1)! \left(\frac{\kappa_{n-2}}{n\kappa_{n-1}}\right)^{n-1} \mathrm{vol}_{n-1}(K\cap u^\perp)^n\, d\mathcal{H}^{n-1}(u) = \\
&= \frac{n!}{2}\left(\frac{\kappa_{n-2}}{n\kappa_{n-1}}\right)^{n-1} \frac{1}{n}\int_{\mathbb{S}^{n-1}} \rho_{IK}(u)^n\, d\mathcal{H}^{n-1}(u) = \frac{n!}{2}\left(\frac{\kappa_{n-2}}{n\kappa_{n-1}}\right)^{n-1} \mathrm{vol}_n(IK),
\end{aligned}$$

where the last equality comes from (10.3.18). This leads to

$$\mathrm{vol}_n(IK) \le \frac{2}{n!}\left(\frac{n\kappa_{n-1}}{\kappa_{n-2}}\right)^{n-1} \mathrm{vol}_n(K)^{n-1},$$

and the constant in the right-hand side coincides with the constant in the right-hand side of (10.4.22). This is a consequence of the identities given in items **(a)** and **(b)** of Exercise 4.7:

$$\frac{2}{n!}\left(\frac{n\kappa_{n-1}}{\kappa_{n-2}}\right)^{n-1} = \frac{2}{n!}\left(\frac{2\pi\kappa_{n-1}}{\kappa_n}\right)^{n-1} = \frac{2^n\pi^{n-1}}{n!}\left(\frac{\kappa_{n-1}}{\kappa_n}\right)^{n-1} = \kappa_{n-1}\kappa_n\left(\frac{\kappa_{n-1}}{\kappa_n}\right)^{n-1} = \frac{\kappa_{n-1}^n}{\kappa_n^{n-2}},$$

and the proof is complete.

□

Remark 10.4.7. Notice that equality in (10.4.22) holds if and only if we have equality in the Busemann random simplex inequality for all hyperplane sections $K \cap u^{\perp}$ of K. Hence, in this case, all hyperplane sections of K (passing through the origin) are $(n-1)$-dimensional centered ellipsoids. For $n \geq 3$ this property characterizes ellipsoids (see, e.g., Lemma 10.2.4 in [2167]). Also, it is clear that for $n = 2$ this characterizes bodies which are centered at the origin.

10.5 Exercises

Exercise 10.1. *Give a formal proof of the fact that the family described in (10.1.1) is a Minkowski class.*

Exercise 10.2. *Prove that a set $Z \subseteq \mathbb{R}^n$ is a zonotope centered at the origin if and only if there exist a number $k \in \mathbb{N}$ and a linear map $T\colon \mathbb{R}^k \to \mathbb{R}^n$ such that $T[-1,1]^k = Z$.*

Remark 10.5.1. This exercise provides a classical characterization of zonotopes (see also Plate 10 above) as projections of cubes. It is enough to work with zonotopes which are centered at the origin, of course. Indeed, we already know that every zonotope can be translated to be centered at the origin.

Exercise 10.3. *Let $P \subseteq \mathbb{R}^n$ be a polytope whose facets are $F_1, \ldots, F_m$ with corresponding unit outer normal vectors $u_1, \ldots, u_m$. Prove that*

$$\Pi P = \sum_{j=1}^{m} \operatorname{seg}\bigl[0, \operatorname{vol}_{n-1}(F_j)u_j\bigr].$$

That is, the projection body of a polytope is the sum of the outer normal vectors of the facets respectively re-scaled to have their lengths equal to the areas of the corresponding facets. See Figure 10.5.1.

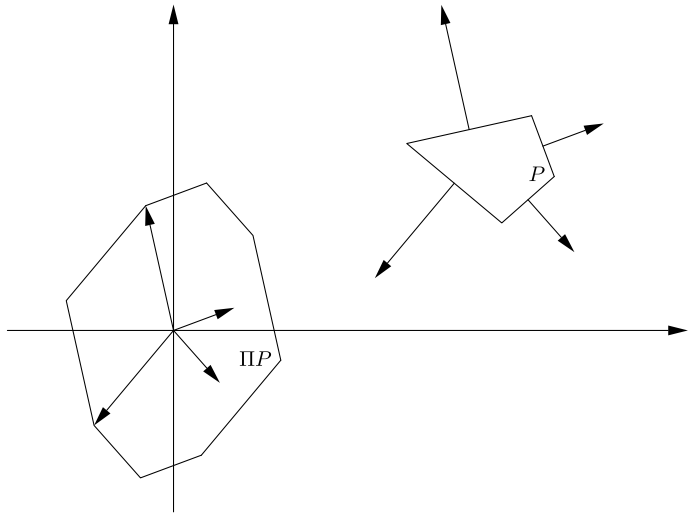


Fig. 10.5.1 The outer normal vectors are $\operatorname{vol}_{n-1}(F_i)u_i$. The Minkowski sum of the corresponding segments (starting at the origin) is ΠP.

Exercise 10.4. *Let $Z \subseteq \mathbb{R}^n$ be a zonotope (centered at the origin) given as*

$$Z = \sum_{k=1}^{m} \beta_k \operatorname{seg}[-u_k, u_k],$$

for $\beta_1, \ldots, \beta_k > 0$ *and* $u_1, \ldots, u_k \in \mathbb{S}^{n-1}$*, and denote by*

$$\mu := \sum_{k=1}^{m} \frac{\beta_k}{2} \left(\delta_{u_k} + \delta_{-u_k}\right)$$

its generating measure. Given $C_1, \ldots, C_{n-j} \in CC^n$*, we denote for simplicity*

$$\mathbf{C}|\{v_1, \ldots, v_j\}^{\perp} := \left(C_1|\{v_1, \ldots, v_j\}^{\perp}, \ldots, C_{n-j}|\{v_1, \ldots, v_j\}^{\perp}\right),$$

where $A|\{v_1, \ldots, v_j\}^{\perp}$ *denotes the orthogonal projection of the set* $A \subseteq \mathbb{R}^n$ *onto the orthogonal complement of the vector subspace* $\operatorname{span}\{v_1, \ldots, v_j\}$*, that is,*

$$A|\{v_1, \ldots, v_j\}^{\perp} := \operatorname{proj}_{(\operatorname{span}\{v_1, \ldots, v_j\})^{\perp}}(A). \tag{10.5.1}$$

(a) *Show that for any* $1 \leq j < n$ *and* $C_1, \ldots, C_{n-j} \in CC^n$ *it holds that*

$$\frac{n!}{2^j (n-j)!} V(Z[j], C_1, \ldots, C_{n-j}) =$$
$$= \int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_j(v_1, \ldots, v_j) V^{(n-j)}\left(\mathbf{C}|\{v_1, \ldots, v_j\}^{\perp}\right) d\mu(v_1) \ldots d\mu(v_j).$$

(b) *For a given* $1 \leq j < n$*, let* $Z_1, \ldots, Z_j$ *be zonotopes as above with generating measures* $\mu_1, \ldots, \mu_j$*, respectively. Prove that*

$$\frac{n!}{2^j (n-j)!} V(Z_1, \ldots, Z_j, C_1, \ldots, C_{n-j}) =$$
$$= \int_{\mathbb{S}^{n-1}} \cdots \int_{\mathbb{S}^{n-1}} D_j(v_1, \ldots, v_j) V^{(n-j)}\left(\mathbf{C}|\{v_1, \ldots, v_j\}^{\perp}\right) d\mu_1(v_1) \ldots d\mu_j(v_j)$$

for any $C_1, \ldots, C_{n-j} \in CC^n$*.*

(c) *Prove that the formula in item* **(b)** *still holds true if* $Z_1, \ldots, Z_j$ *are zonoids with generating measures* $\mu_1, \ldots, \mu_j$*, respectively.*

Exercise 10.5. *Let* $Z \subseteq \mathbb{R}^n$ *be a proper zonotope given as*

$$Z = \sum_{j=1}^{m} \alpha_j \operatorname{seg}[-u_j, u_j],$$

for $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ *and* $\alpha_1, \ldots, \alpha_m > 0$*. Prove that*

$$\operatorname{vol}_n(Z) = \frac{2}{n} \sum_{j=1}^{m} \alpha_j \operatorname{vol}_{n-1}\left(\operatorname{proj}_{u_j^{\perp}}(Z)\right).$$

Exercise 10.6. *Let* $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ *and* $c_1, \ldots, c_m > 0$ *be such that*

$$\mathrm{Id}_{\mathbb{R}^n} = \sum_{j=1}^{m} c_j u_j \otimes u_j.$$

For any $\alpha_1, \ldots, \alpha_m > 0$, consider the zonotope

$$Z = \sum_{j=1}^{m} \alpha_j \mathrm{seg}[-u_j, u_j].$$

Prove that

$$\mathrm{vol}_n(Z) \geq 2^n \prod_{j=1}^{m} \left(\frac{\alpha_j}{c_j}\right)^{c_j}.$$

Hint: Use Exercise 10.5 and induction on the dimension. A formal and detailed proof can be found in [196].

Exercise 10.7. *Let $K_1, \ldots, K_n \in \mathcal{K}^n$ be arbitrarily given. Prove that*

$$V(\Pi K_1, \ldots, \Pi K_n) =$$
$$= \frac{1}{n!} \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} D_n(v_1, \ldots, v_n)\, dS_{n-1}(K_1, v_1) \ldots dS_{n-1}(K_n, v_n).$$

The objective of the next exercise is to derive a formula for mixed volumes of projections of zonoids. We still denote the orthogonal complement of $\mathrm{span}\{v_1, \ldots, v_j\}$ by $\{v_1, \ldots, v_j\}^{\perp}$.

Let $\mathbf{g}_1, \mathbf{g}_2 \subseteq \mathbb{R}^n$ be k-dimensional vector subspaces, where $k \in \{1, \ldots, n-1\}$. We denote by $[\mathbf{g}_1, \mathbf{g}_2]$ the factor by which the k-dimensional volume of a measurable subset of $\mathbf{g}_1$ is multiplied when orthogonally projected onto $\mathbf{g}_2$ (for example, if $\mathbf{g}_1$ and $\mathbf{g}_2$ are lines, then $[\mathbf{g}_1, \mathbf{g}_2]$ is the cosine of the acute angle between them).

Exercise 10.8. *Let $1 \leq j \leq n-1$, and let $\mathbf{g} \subseteq \mathbb{R}^n$ be a j-dimensional vector subspace. Prove that if $Z_1, \ldots, Z_j$ are zonoids with respective generating measures $\mu_1, \ldots, \mu_j$, then*

$$V^{(j)}\big(\mathrm{proj}_{\mathbf{g}}(Z_1), \ldots, \mathrm{proj}_{\mathbf{g}}(Z_j)\big) =$$
$$= \frac{2^j}{j!} \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} D_j(v_1, \ldots, v_j)[\mathbf{g}^{\perp}, \{v_1, \ldots, v_j\}^{\perp}]\, d\mu_1(v_1) \ldots d\mu_j(v_j).$$

Exercise 10.9. *Let $1 \leq j \leq n-1$, and let $Z_1, \ldots, Z_j \subseteq \mathbb{R}^n$ be zonoids whose generating measures are $\mu_1, \ldots, \mu_j$, respectively. Prove that*

$$\int_{\mathbb{S}^{n-1}} f(v)\, dS(Z_1, \ldots, Z_j, B^n[n-j-1], v) =$$
$$= \frac{2^j (n-j-1)!}{(n-1)!} \int_{\mathbb{S}^{n-1}} \ldots \int_{\mathbb{S}^{n-1}} \left[\int_{\mathbb{S}^{n-1} \cap \{v_1, \ldots, v_j\}^{\perp}} f(v)\, d\mathcal{H}^{n-j-1}(v) \right] \times$$
$$\times D_j(v_1, \ldots, v_j)\, d\mu_1(v_1) \ldots d\mu_j(v_j)$$

for any $f \in C(\mathbb{S}^{n-1})$.

Let C be a closed convex cone of some (real, finite- or infinite-dimensional, topological) vector space. A subset having the form $R := \{\lambda v : \lambda \geq 0\}$ for some non-zero vector $v \in C$ is called an *extreme ray* of C if $x, y \in C$ and $x + y \in R$ imply $x, y \in R$.

Exercise 10.10. *Consider the family* $\mathcal{Z} \subseteq C(\mathbb{S}^{n-1})$ *given as*

$$\mathcal{Z} := \Big\{\text{support functions of zonoids centered at } 0_{\mathbb{R}^n}\Big\}.$$

(a) *Verify that* $\mathcal{Z}$ *is a convex cone.*

(b) *Show that* $\mathcal{Z}$ *is closed in the maximum norm of* $C(\mathbb{S}^{n-1})$.

(c) *Let* $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. *Prove that the subset*

$$\big\{|\langle \cdot, \lambda x\rangle| \colon \mathbb{S}^{n-1} \to \mathbb{R} : \lambda \geq 0\big\} \subseteq \mathcal{Z}$$

is an extreme ray of $\mathcal{Z}$.

Exercise 10.11. (a) *Show that*

$$\Pi B^n = \kappa_{n-1} B^n.$$

(b) *Verify that the projection body of any ellipsoid is an ellipsoid.*

(c) *Let* $E \subseteq \mathbb{R}^n$ *be an ellipsoid centered at the origin. Prove that*

$$\Pi E = \mathrm{vol}_n(E) \frac{\kappa_{n-1}}{\kappa_n} E^\circ.$$

Exercise 10.12. *Let* $\Pi \colon \mathcal{K}^n \to \mathcal{K}^n$ *be the* projection map, *which associates each* $K \in \mathcal{K}^n$ *to its projection body* ΠK. *Prove that this map is continuous in the Hausdorff metric.*

Exercise 10.13. *Let* $K \subseteq \mathbb{R}^n$ *be a convex body in minimal surface area position. Prove that*

$$S(K) \leq n \mathrm{vol}_n(\Pi K)^{\frac{1}{n}}. \tag{10.5.2}$$

Hint: Consider first polytopes. For this case, use (10.3.2) and Exercises 9.8 and 10.6. Approach the general case using approximations by polytopes.

Remark 10.5.2. Inequality (10.5.2) appears in [966] and [962].

Exercise 10.14. *Consider the functional* $\psi : \mathcal{K}^n \to \mathbb{R}$ *given by*

$$\psi(K) = \mathrm{vol}_n(\Pi^\circ K)^{-1/n}, \quad K \in \mathcal{K}^n.$$

(a) *Verify that the expression*

$$\psi(K) = \left[\frac{1}{n} \int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)^{-n} \, d\mathcal{H}^{n-1}(u)\right]^{-1/n}$$

holds.

(b) *Show that* ψ *is (strictly) monotone for inclusion, that is,* $\psi(K) \leq \psi(L)$ *whenever* $K \subseteq L$.

(c) *Prove that* ψ *is homogeneous of degree* $n-1$ *and invariant under volume-preserving transformations.*

Remark 10.5.3. Notice that ψ "behaves" as an affine surface area concept, since it is $(n-1)$-homogeneous, invariant under volume-preserving transformations and strictly monotone for inclusion. Recall that bounds for the volume of the polar projection body were derived in Section 10.4.

Exercise 10.15. *Show that for any convex bodies $K, L \subseteq \mathbb{R}^n$ the equality*

$$V(\Pi K, L, \dots, L) = V(\Pi L, K, \dots, K)$$

holds.

Exercise 10.16. *Let $K \subseteq \mathbb{R}^n$ be a convex body, and recall that ∇K denotes the Blaschke body of K. Prove that*

$$\mathrm{vol}_n(\nabla K) \geq \mathrm{vol}_n(K).$$

Show that equality holds if and only if K is centrally symmetric.

Hint: Use Exercise 7.20.

For the next exercise we introduce an extension of the notion of projection body. Let $K_1, \dots, K_{n-1} \in \mathcal{K}^n$. The *mixed projection body* of $(K_1, \dots, K_{n-1})$ is the (unique) convex body $\Pi(K_1, \dots, K_{n-1})$ whose support function (over $\mathbb{S}^{n-1}$) is given as

$$h(\Pi(K_1, \dots, K_{n-1}), u) := V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(K_1), \dots, \mathrm{proj}_{u^\perp}(K_{n-1})\big)$$

for any $u \in \mathbb{S}^{n-1}$. Of course, we extend this function to $\mathbb{R}^n$ by letting

$$h(\Pi(K_1, \dots, K_{n-1}), \lambda u) = \lambda h(\Pi(K_1, \dots, K_{n-1}), u)$$

for every $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$. Notice that it is not clear *a priori* that this function is sublinear, and hence it is not clear that $\Pi(K_1, \dots, K_{n-1})$ is a convex body; proving that is one of the objectives of the next exercise. It is immediate to check that in the case where $K_1 = \dots = K_{n-1}$ we just re-obtain the usual projection body. For more on mixed projection bodies, we refer the reader to [1632].

Exercise 10.17. **(a)** *Prove that*

$$h(\Pi(K_1, \dots, K_{n-1}), x) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\langle x, v \rangle| \, dS(K_1, \dots, K_{n-1}, v)$$

for every $x \in \mathbb{R}^n$.

(b) *Show that the mixed projection body $\Pi(K_1, \dots, K_{n-1})$ is a convex body.*

(c) *Let $K_1, \dots, K_{n-1}, L_1, \dots, L_{n-1} \subseteq \mathbb{R}^n$ be convex bodies. Prove that*

$$V(\Pi(K_1, \dots, K_{n-1}), L_1, \dots, L_{n-1}) = V(\Pi(L_1, \dots, L_{n-1}), K_1, \dots, K_{n-1}).$$

(d) *Show that the* mixed projection map

$$\mathcal{K}^n \times \dots \times \mathcal{K}^n \ni (K_1, \dots, K_{n-1}) \mapsto \Pi(K_1, \dots, K_{n-1})$$

is continuous in each entry with respect to the Hausdorff metric.

Exercise 10.18. *Prove that for any convex bodies $K, L \subseteq \mathbb{R}^n$ the Brunn–Minkowski-type inequality*

$$\mathrm{vol}_n(\Pi(K + L))^{\frac{1}{n(n-1)}} \geq \mathrm{vol}_n(\Pi K)^{\frac{1}{n(n-1)}} + \mathrm{vol}_n(\Pi L)^{\frac{1}{n(n-1)}} \tag{10.5.3}$$

holds. Show that equality holds if and only if $\Pi(K + L)$, ΠK, and ΠL are homothetical.

Exercise 10.19. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies. Prove that if ΠK, ΠL, and $\Pi(K+L)$ are homothetical, then K and L are homothetical. Conclude that the equality case in (10.5.3) holds if and only if K and L are homothetical.*

Hint. Using the definition of projection body, compare the volumes of the projections of $K + L$, K, and L onto any given hyperplane $u^\perp$. Conclude from the equality case in the classical Brunn–Minkowski inequality that $\mathrm{proj}_{u^\perp}(K)$ and $\mathrm{proj}_{u^\perp}(L)$ are always homothetical. Prove that, up to dilating one of the bodies, one may assume that these projections are actually translates. Last, use Proposition 8.4.4.

For the next exercise, we extend the definition of *projection body* for a set $C \in CC^n$ naturally by letting

$$h(\Pi C, \lambda u) = \lambda \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(C)\big)$$

for every $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$.

In the same manner, we extend the notion of *mixed projection body* for each $(n-1)$-tuple $C_1, \ldots, C_{n-1}$ in CC^n by letting

$$h(\Pi(C_1, \ldots, C_{n-1}), \lambda u) = \lambda V^{(n-1)}\big(\mathrm{proj}_{u^\perp}(C_1), \ldots, \mathrm{proj}_{u^\perp}(C_{n-1})\big)$$

for any $u \in \mathbb{S}^{n-1}$ and $\lambda \geq 0$. By these definitions, it is immediate to check that $\Pi(C, \ldots, C) = \Pi C$.

Exercise 10.20. **(a)** *Prove that the functions $h(\Pi C, \cdot)$ and $h(\Pi(C_1, \ldots, C_{n-1}), \cdot)$ defined above are indeed support functions of sets in CC^n.*

(b) *Verify that*

$$h(\Pi C, x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \, dS_{n-1}(C, v) = \frac{n}{2} V\big(C[n-1], \mathrm{seg}[-x, x]\big)$$

and that

$$h(\Pi(C_1, \ldots, C_{n-1}), x) = \frac{1}{2}\int_{\mathbb{S}^{n-1}} |\langle x, v\rangle| \, dS(C_1, \ldots, C_{n-1}, v) = \frac{n}{2} V\big(C_1, \ldots, C_{n-1}, \mathrm{seg}[-x, x]\big)$$

for any $x \in \mathbb{R}^n$.

(c) *Assume that $C \in CC^n$ is such that $\dim(C) \leq n-2$. Show that $\Pi C = \{0_{\mathbb{R}^n}\}$.*

(d) *Prove that if $C \in CC^n$ is such that $\dim(C) = n-1$, then ΠC is a segment.*

Now we may see the mixed projection map as a map which associates each $(n-1)$-tuple $(C_1, \ldots, C_{n-1})$ of sets of CC^n (instead of $\mathcal{K}^n$) to the mixed projection body $\Pi(C_1, \ldots, C_{n-1})$. Below we investigate some properties of this map. As a first step, the reader may verify that items **(c)** and **(d)** of Exercise 10.17 still hold true when we consider the larger family CC^n.

Exercise 10.21. **(a)** *Let $C_1, \ldots, C_{n-1} \in CC^n$ and $x_1, \ldots, x_{n-1} \in \mathbb{R}^n$. Show that*

$$\Pi(x_1 + C_1, \ldots, x_{n-1} + C_{n-1}) = \Pi(C_1, \ldots, C_{n-1}).$$

(b) *Prove that for any $T \in \mathrm{GL}(n)$ and all $C_1, \ldots, C_{n-1} \in CC^n$ we have*

$$\Pi(TC_1, \ldots, TC_{n-1}) = |\det(T)|(T^{-1})^* \Pi(C_1, \ldots, C_{n-1}).$$

Exercise 10.22. **(a)** *Let $C_1, \dots, C_{n-1}, L_1, \dots, L_{n-1} \in CC^n$ be such that $C_j \subseteq L_j$ for each $j = 1, \dots, n-1$. Verify that*

$$\Pi(C_1, \dots, C_{n-1}) \subseteq \Pi(L_1, \dots, L_{n-1}).$$

(b) *Prove that if $C, L \in CC^n$ are such that $C \cup L \in CC^n$ and $C \cap L \neq \emptyset$, then for every $C_2, \dots, C_{n-1} \in CC^n$ we have*

$$\Pi(C \cup L, C_2, \dots, C_{n-1}) + \Pi(C \cap L, C_2, \dots, C_{n-1}) = \Pi(C, C_2, \dots, C_{n-1}) + \Pi(L, C_2, \dots, C_{n-1}).$$

In other words, the mixed projection map has the valuation property.

The objective of the next exercise is to prove that the mixed projection map is polynomial for Minkowski combinations in a similar way as the mixed volume.

Exercise 10.23. *Let $C_1, \dots, C_m \in CC^n$ be arbitrarily given. Prove that for any $\lambda_1, \dots, \lambda_m \geq 0$ we have the equality*

$$\Pi(\lambda_1 C_1 + \dots + \lambda_m C_m) = \sum_{1 \leq i_1, \dots, i_{n-1} \leq m} \lambda_{i_1} \dots \lambda_{i_m} \Pi(C_{i_1}, \dots, C_{i_m}).$$

Exercise 10.24. **(a)** *Prove that the functional $\phi \colon \mathcal{K}^n \to \mathbb{R}$ defined as*

$$\phi(K) = \mathrm{vol}_n(K)^{1-n} \mathrm{vol}_n(\Pi K)$$

is invariant under affine transformations.

(b) *Show that ϕ attains a minimum over $\mathcal{K}^n$.*

(c) *Show that for any convex body $K \subseteq \mathbb{R}^n$ we have the inequality*

$$\mathrm{vol}_n(\Pi K)\mathrm{vol}_n(K)^{1-n} \geq \mathrm{vol}_n(\Pi^2 K)\mathrm{vol}_n(\Pi K)^{1-n}.$$

(d) *Prove that if ϕ attains a minimum at $K_0 \in \mathcal{K}^n$, then K_0 and ΠK_0 are homothetical. Conclude that any minimizer of ϕ is a proper zonoid.*

From Exercise 10.14 (see also Remark 10.5.3), we have that the functional $\psi \colon \mathcal{K}^n \to \mathbb{R}$ given as $\psi(K) := \mathrm{vol}_n(\Pi^\circ K)^{-1/n}$ behaves as an affine surface area concept for convex bodies. The computations in the next exercise lead to bounds for $\psi(K)$ in terms of $S(K)$ and $\mathrm{vol}_n(K)$.

Exercise 10.25. **(a)** *Verify that*

$$\frac{\kappa_{n-1}}{\kappa_n} \mathrm{vol}_n(K)^{\frac{n-1}{n}} \leq \psi(K) \leq \frac{\kappa_{n-1}}{n\kappa_n^{(n+1)/n}} S(K)$$

for every convex body $K \subseteq \mathbb{R}^n$.

(b) *Conclude that*

$$\inf \left\{ S(TK) : T \in \mathrm{SL}(n) \right\} \geq \frac{n\kappa_n^{(n+1)/n}}{\kappa_{n-1}} \psi(K).$$

10.6 Notes

The notes to Chapter 10 are splitted into three parts, taking the corresponding literature comprehensively into consideration. These three parts are: zonotopes, zonoids, and (as third part) three types of convex or star-shaped bodies associated with a given convex body, namely its projection, centroid, and intersection body. Especially with the first two themes we try to complete and update existing basic surveys and expositions.

10.6.1 Zonotopes

Since zonotopes are special polytopes, many references cited here would fit also to the subsections on polytopes and/or discrete geometry. But we decided to leave them here, close to the subsection treating zonoids. Zonotopes are special polytopes that arise mainly in the intersection of combinatorics, discrete and convex geometry, computational geometry, and linear optimization. But one meets them also in many other fields of (applied) mathematics, and also as useful tools in non-mathematical fields. They occur already in our subsection 2.9.1.11, but their versatile applications ensure that they occur sometimes also in other chapters, e.g., as projection bodies of polytopes.

DEFINITION, EXAMPLES, AND SOME INTERESTING FACTS

Zonotopes of dimension n are precisely those n-dimensional convex polytopes which can be represented as Minkowski (or vector) sums of finitely many segments or, equivalently, as *totally symmetric n-polytopes*. With this we mean that each k-dimensional face of them has a center of symmetry, $0 \leq k \leq n$; it suffices to have this central symmetry only for all 2-faces (see Plate 10 above).

Zonotopes are also representable as *parallel projections of higher dimensional cubes*. *Zonoids* (see below, our next subsection) are precisely the limits of zonotopes in the Hausdorff metric. For $n = 2$ one sees that any centrally symmetric polygon is a zonotope (usually also called *zonogon*). Three-dimensional zonotopes are also called *zonohedra*, and special representatives of them can be found in the class of *Catalan solids* (namely the *rhombic dodecahedron* and the *rhombic triacontahedron*) or as translative space-fillers like the *affine cube*, the *Archimedean truncated octahedron*, and the *prolonged rhombic dodecahedron*.

There are also extensive families of polytopes that are zonotopes, like, e.g., a large variety of *suitable k-fold prisms* (see Section 4.4 of Grünbaum's book [1085]) or the class of *permutahedra* (see Example 0.10 in Ziegler's monograph [2560]).

It is well-known that the *projection body of any n-dimensional convex polytope* is an n-dimensional zonotope centered at the origin. This yields an important metrical application of zonotopes going back to Minkowski and is particularly discussed in our third part here.

And in view of spherical images, normal fans, and combinatorial properties, zonotopes are combinatorially combined with notions like *arrangements of hyperplanes* and *oriented matroids*. There are many more properties and applications of zonotopes.

Already here we would like to draw attention to a homonymous use of the word *parallelotope*. On the one hand, this notion is often used for zonotopes which are vector sums of line segments whose direction vectors (one for each segment summand) form a linearly independent system. Here equivalent notions are then "affine cubes" or "parallelepipeds" (the latter, in linear algebra, preferably used for the three-dimensional situation). On the other hand, many authors use the word "parallelotope" in the meaning of "translative space-filler". This second meaning is not perfect since (already) in dimension three also space-fillers with non-parallelogramic facets exist, and in dimensions higher than three the

situation is even worse (the translative space-fillers are even not necessarily zonotopes). However, in concordance with many references, we will use both meanings suitably and hope that the reader will always recognize the meaning in the respective context.

ZONOTOPES IN BOOK PARTS

As starting points, we take the occurrence of zonotopes in some *monographs* that are not too old (latest edition), and also in basic surveys. From the viewpoint of *oriented matroids*, Section 2.2 (and page 95 ff., in view of problems and exercises) in the book [354] of Björner et al. is dedicated to zonotopes, followed by Lecture 7 in Ziegler's monograph [2560], which collects many aspects of *zonotopal geometry and combinatorics*, also including inspiring problems and exercises.

In Thompson's book [2361] on the geometry of Minkowski spaces zonotopes appear surprisingly often: in Sections 5.4 and 6.2 as construction steps for *isoperimetrices based on the Holmes–Thompson definition of Minkowski surface area*, in Section 5.6 regarding *Wulff shapes*, in Section 6.1 with respect to *approximation aspects*, and in Section 9.5 with respect to *numerical parameters in normed spaces with zonotopal unit balls*; a little survey on zonotopes is presented in Section 6.7.

Deza and Laurent [746] treat zonotopes and zonoids in Section 8.3, with respect to L_1-embeddable norms. In Section 6.4 of Felsner's book [852] *zonotopal tilings, pseudoline arrangements* (see here also Grünbaum's booklet [1083]), and applications (like k-sets) are discussed. Ewald [826] introduces zonotopes in Section 7 of Chapter VI, and Barvinok [242] poses problems on zonotopes in Section 2 of Chapter VI and Section 2 of Chapter VII. The second edition [1085] of Grünbaum's famous monograph contains several comments and references on zonotopes; in particular, we refer to Section 18.4.

Gardner (see [930]) discusses zonotopes in Sections 4.1 and 4.2 (mainly as projection bodies, and in certain inequalities), in the Notes of Chapter 4, and in Chapters 8 and 9 (in connection with projection functions). In Schneider's book [2167] Section 3.5 and its Notes refer comprehensively to them. Also at the end of Section 10.14 on L_p *zonoids* zonotopes are treated.

The strong relation between *zonotopes and hyperplane arrangements* gave de Concini and Procesi the motivation to dedicate the whole Section 2.3 of their monograph [645] to zonotopes, discussing them as *shadows of cubes, regarding tilings, lattices*, and the support of an interesting class of piecewise polynomial functions called *box splines*. For here and the following references we say that a *zonotopal tiling* of a zonotope Z is a subdivision S of Z such that the maximal cells (and therefore all faces) of S are translates of zonotopes determined by subsets from the family of segment summands of Z. In their book [264] devoted to *geometric combinatorics*, Beck and Sanyal treat zonotopes in Section 7.5; they present *zonotopal tilings, zonotopes as projections of cubes, their facial structure*, and theorems on them related to *lattice theory*. Also the monograph [263] (on counting integer points of integer polytopes) contains related material, namely nice examples which are zonotopes and, in particular, *permutahedra*. We repeat here the two important books [2560] and [852] of Ziegler and Felsner, studying zonotopal tilings in detail.

In Section 4.4 of the recent book [1263] Hug and Weil treat various properties of zonotopes and zonoids, especially their occurrence as *projection bodies*. In addition, *generalized zonoids* are discussed there. And also in Section 14.3 of the book [2179] on stochastic and integral geometry basic properties of zonotopes and zonoids are presented.

We should mention that, due to their relations to hyperplane arrangements, zonotopes are also discussed in several books and proceedings from *computational geometry*. We give here only one example, namely Edelsbrunner's monograph [793], where zonotopes are introduced in Section 1.7, and several properties of them occur directly or indirectly in several further sections.

The book [1522] of Le et al. can be located "between" state estimation and robust control, applied to uncertain *discrete-time linear-time invariant systems*. Using properties of zonotopes, the authors study disturbances, measurement noises, and constraints with the aim to guarantee feasibility and

stability. To model possible uncertainties, disturbances, and measurement noises, they have chosen the deterministic way. In subsection 1.1.4 of [1522] zonotopes are introduced, and in Section 1.2 various properties and applications of them (e.g., certain approximation procedures suitable for later use) are presented. And in later sections, topics like volume minimization and set membership within the framework of zonotopes are discussed.

SURVEYS TREATING ZONOTOPES

The most important *surveys* discussing many aspects of the geometry of zonotopes (and also of zonoids) are [422] of Bolker, [2176] of Schneider and Weil, and [997] of Goodey and Weil (see also Martini's exposition [1705]). It seems that a comprehensive *recent* survey on their combinatorics and further topics is missing, although there are many papers in that direction. However, in the Handbook of Convex Geometry [1077], the following articles refer (besides Goodey and Weil [997]) explicitly to results on zonotopes.

Approximation problems treating zonoids and zonotopes are collected in Section 7 of Gruber's exposition [1059], and also in the handbook article [1447] of Koldobsky and König approximation aspects are discussed. In its fifth part, results on the *approximation of n-dimensional zonoids by zonotopes* are presented (namely, to estimate the minimal number of segment summands needed to place a zonoid suitably between two homothetical zonotopes).

In subsection 4.3 of [1177] Heil and Martini discuss how *degrees of symmetry and central symmetry of faces* lead to zonoids and zonotopes, respectively. In Section 5 of the survey [260] of Bayer and Lee *graphs of zonotopes* are discussed, and in Bayer's related survey [257] *polyhedral subdivisions* of convex polytopes and their effect on the h-vectors are studied. In particular, *fiber zonotopes, zonotopal subdivisions*, and *associated hyperplane arrangements* are discussed. In view of *computational geometry*, Edelsbrunner [794] shows applications of *arrangements of hyperplanes*.

In subsection 3.1 of [1034] Gritzmann and Wills study inequalities for *lattice zonotopes in connection with intrinsic volumes*, and Engel treats *translative space-fillers* in subsection 3.4 of [812].

Davydov and Paulauskas [715] work on limit theorems for vector sums of random sets and survey results on stable compact convex sets in Banach spaces. They take special care of *stable random zonotopes* and study also an application in statistics. Ostrovskii [1928] surveys enlargements of unit balls of Banach spaces in the study of projections there, and connections to *space-tiling zonotopes and zonoid theory* are particularly described.

The survey [706] of Danilov et al. refers to *cyclic zonotopes* (which are n-zonotopes generated by any m distinct vectors of the form $(1, t, ..., t^{n-1})$). The authors treat geometric and combinatorial aspects of *tilings* of such zonotopes by affine cubes, called *cubillages*. This theory is interesting since it is related to the theory of *higher Bruhat orders* (see, e.g., [853] by Felsner and Ziegler), and also to the theory of *triangulations of cyclic polytopes*.

Braun and Vindas [493] present a brief survey on *lattice zonotopes* (see also below); they lay emphasis on enumeration aspects (e.g., Ehrhart theory) and combinatorial structures (for example, regarding graphs and permutations).

We now start to present research publications on zonotopes that are (in almost all cases) not mentioned in the references above. Our approach is structured by certain aspects (like, e.g., "special types of zonotopes"), and within the corresponding subsections the references are mainly (but not only) ordered in historical sequence.

GEOMETRIC PROPERTIES AND GENERALIZATIONS OF ZONOTOPES

Comparing definitions of Fedorov and Coxeter, Taylor [2348] compares generalized zonohedra and usual zonohedra, respectively. As a polytope, a *generalized zonohedron* has a non-degenerate *segment summand parallel to each of its edges* and is, by other authors, also called a *belt polytope*.

The spherical images of belt polytopes (coinciding with those of the zonotopes obtained as sums of the mentioned segments summands) and relations to *Wulff shapes* were considered by Taylor [2348], Bolker [422], and, independently, by Martini [1710]. Later investigations showed that these polytopes and their non-polytopal analogues, called *belt bodies*, play an important role in the combinatorial geometry of convex bodies (see [424] and [186] by Boltyanski and Baladze, [427] by Boltyanski and Martini, and Chapters VII and VIII of the monograph [431] by Boltyanski, Martini, and P. Soltan). In this field, generally unsolved problems are solved at least for these classes. For example, one finds there the solution of Hadwiger's famous covering problem for belt bodies which are, regarding the Hausdorff metric, dense within the set of all convex bodies (this holds even already for belt polytopes). In the sense of their spherical images, belt polytopes are the most general family of polytopes whose spherical images are projective arrangements of $(n-2)$-planes in the hyperplane at infinity.

Schnell [2154] generalizes a result of McMullen (for unit cubes) in the following way: Let P be the parallelotope spanned by the linearly independent vectors $a_1, ..., a_n$ in n-space, and P^* be the parallelotope obtained from the uniquely determined vectors $b_1, ..., b_n$ defined by $b_i \cdot a_j = \delta_{ij}$ (Kronecker delta) with $1 \leq i, j \leq n$. Let $P|E_i$ be the image of P under orthogonal projection onto the i-dimensional subspace E_i, and let V_i denote the i-dimensional volume. The author proves that $V_i(P_i) = V_n(P) \cdot V_{n-i}(P^*|E_{n-i})$ holds, for $i = 1, ..., n$ and all i-dimensional subspaces E_i and their orthocomplements E_{n-i}. He poses the question of whether there is a corresponding statement for zonotopes. A further generalization is given by Schneider [2157]: A pair (K, K') of convex bodies in Euclidean n-space is called a *(VP)-pair* if, for any pair of orthogonal complementary subspaces, the orthogonal projections of K onto the first and of K' onto the second of these subspaces have numerically the same volumes. It is shown that (K, K') is a *(VP)-pair* iff K is a direct sum of centrally symmetric polygons and segments, the n-dimensional volume of K is 1, and $K' = 1/2\Pi K$, where ΠK is the projection body of K. An application to integral geometry in Minkowski spaces is given.

As shown by Billera, Ehrenborg, and Readdy [345], the *affine hull of the flag vectors of all n-dimensional zonotopes* or, analogously, of all essential central hyperplane arrangements in n-space coincides with the affine hull of the flag vectors of all n-polytopes. The authors give a characterization of the lattice spanned by these flag vectors of zonotopes and show that this is equivalent to the property that the integral lattice spanned by the cd-indices (i.e., non-commutative polynomials in the variables c and d which efficiently encode the flag vectors) of all zonotopes coincides with the ring of all non-commutative polynomials with integral coefficients in the variables c and $2d$.

Dyer, Gritzmann, and Hufnagel [785] deal with the complexity of *computing mixed volumes of polytopes*, showing also that *computing the volume of zonotopes is #P-hard* (while each corresponding mixed volume can be computed easily). On the other hand, they give general examples which show that computing mixed volumes is hard, even when volume computation is easy. Cifuentes and Parrilo [627] study the time complexity of mixed volumes of m-tuples of zonotopes via determinants and properties of edge graphs and their tree decompositions.

Fern et al. [863] examine the linear dependence matroid M_W on the column vectors corresponding to reflections in a subgroup W of the *symmetry group of the n-dimensional hypercube*. They succeed in determining all possible automorphism groups of M_W. Going further, the authors analogously study reflection groups of more general polytopes and, in particular, of related zonotopes. So for the Platonic solids connections between a given reflection group W and the automorphism group of a corresponding zonotope are studied. E.g., it is shown that if W is the automorphism group of the hypercube, then the symmetry group of the corresponding zonotope is isomorphic to W.

Gover and Krikorian [1011] give volume formulas of affine cubes in terms of their defining matrix A, the maximal minors, the edge lengths, and cosines of angles between edge pairs. They continue using this for real-case versions of several well-known determinantal inequalities involving principal minors of a positive-definite Hermitian matrix. Finally, zonotopes are studied. E.g., *decomposition procedures of zonotopes into generating affine cubes* are given, and a volume formula for zonotopes is derived, too.

Among other results, Donoho and Tanner [760] obtain results on the number of k-faces of n-dimensional zonotopes considered as *projections of N-dimensional hypercubes*. These projections are considered for $N, n, k \to \infty$ with ratios n/N and k/n that approach limits in $(0, 1)$. Finally, applications of these results in *signal processing, information theory, and probability theory* are presented.

Here we mention also the paper [3] of Abardia and Saorin on *complex difference bodies*, i.e., difference bodies in complex vector spaces (see also subsection 5.7.3). The authors present (among many other results) also the case that a complex difference body is a *zonotope*.

Continuing investigations of Hensley, Vaaler, and Ball, Anantharaman [106] studies central cube slices in 4-space. Examples of non-zonotopal such slices are given, which is in strict contrast to the three-dimensional situation (see the related classical theorem of Herz and Lindenstrauss). The author constructs also non-trivial zonotopal slices.

APPROXIMATION AND INTERPOLATION

Recall that approximation aspects are already mentioned in the part above devoted to surveys.

Linhart [1583] deals with the approximation of the shape of the ball by zonotopes in terms of the *variance of the width*. More precisely, among all n-dimensional zonotopes created by the same number of segment summands of equal length he finds those with minimal variance of the width.

In [572] the following statement is proved in two ways (one of which is non-constructive and due to Bourgain): Given an n-dimensional zonoid Z and unit vectors $u_1, ..., u_k$, there is always a zonotope Z' being the sum of at most k segments and having the same supporting hyperplanes as Z in k parallel pairs, namely perpendicular to $u_1, ..., u_k$, respectively. The second and constructive proof uses linear programming and gives an approach to an application in stochastic geometry. A further application refers to sizes of projections of convex bodies, using the notion of *projection body*. In [1897] the approximation of a compact, convex set in the Hausdorff metric by linear combinations of convex compacta is studied, where the subcases of *approximating parallelotopes and zonotopes* are particularly taken care of.

Wagner [2412] studies the problem to distribute points $\{x_1, ..., x_m\}$, where m is large, as uniformly as possible over the unit sphere of Euclidean n-space. These results yield almost best approximations of balls, within a chosen Hausdorff distance, by zonotopes for dimensions n with $3 \leq n \leq 6$. Bourgain and Lindenstrauss [475] extend this approximation result with different methods to all $n \geq 3$.

Matoušek [1731] says that *a zonotope Y approximates a zonoid Z up to ε* if the inclusion $Z \subset Y \subset (1 + \varepsilon)Z$ holds. He shows that, for dimension $n > 4$, there is a constant C which depends only on n such that an arbitrary centred zonoid can be approximated up to ε by a centred zonotope whose number of segment summands of equal lengths is at most $C\varepsilon^{-2+6/(n+2)}$. In various directions this improves well-known results of Bourgain, Lindenstrauss, and Wagner. In addition, it is shown that, for any $n > 2$, a zonotope with m segment summands approximating a (suitably defined) zonoid up to ε can be computed in time polynomial in m. Here the upper bound for m differs from that given above only by a logarithmic factor. The author formulates these results also in terms of probability measures on the sphere.

In Section 5 of the handbook survey [1447] the approximation of zonoids by zonotopes in the following sense is discussed: estimate the minimal number $N(F, \varepsilon)$ of needed segment summands such that the corresponding zonotope Z and a homothet $(1 + \varepsilon)Z$ of it enclose a zonoid F between them.

Gluskin [983] investigates the problem of approximating a Euclidean n-ball by a zonotope with N segment summands. He estimates the minimal number $d(n, N)$ such that it is possible to find such a zonotope F and for which the inclusions $B \subset F \subset d(n, N)B$ hold, with B as standard Euclidean n-ball. For all values of n and N, the author gives precise estimates.

Markov [1698] studies *quasi-vector spaces* (for which spaces of convex bodies are known examples) and discusses, with this background, an approximation problem for zonogons in the two-dimensional case.

Voronoi splines are B-spline-like elements which inherit the geometry of a sampling lattice from its Voronoi cell generating a lattice-shift-invariant spline space of approximation. Mirzargar and Entezari [1809] establish the relations between Voronoi splines and box splines (see the monograph [645] of de Concini and Procesi) by using geometric properties of Voronoi polytopes and zonotopes.

We note that the *zonotope algebra* developed by Holtz and Ron [1226] (and in further papers, see below) has applications in and motivations from approximation theory and other fields.

Hyperplane arrangements, zonotope properties, and box splines are comprehensively discussed in the monograph [645] (see, particularly, Section 2.3). One should mention here also the forerunner exposition [644] and the research paper [1542], nicely connecting algebraic and combinatorial structures with matroid, interpolation, and approximation theory.

For a set X of unimodular vectors and B_X as corresponding box spline, Lenz [1543] discusses the proof of the following interpolation problem of Holtz and Ron: every real-valued function on the set of *interior lattice points of the zonotope* obtained from X can be extended in a unique way to a function on the whole zonotope having the form $p(D)B_X$, with $p(D)$ as certain differential operator.

Rahmani, Pinoli, and Debayle [2014] study the *approximation of two-dimensional convex bodies by zonotopes regarding the width function*. Their issue is first described in the deterministic case and then extended to random closed sets (thus, random symmetric convex bodies are approximated). The used zonotopes are of special type: they have a fixed upper bound regarding the edge number and are equiangular (between the normals).

Alonso-Gutiérrez and Prochno [88] study a certain rich class of random convex sets which suitably "interpolate" between polytopes and zonotopes and can occur as special unit balls of norms (e.g., also polars of zonotopes are studied as such bodies). Geometric properties of such bodies are discussed in terms of their *support function and mean width*.

The paper [2350] of Taylor is related to *shape reconstruction of (two-dimensional) convex bodies* when a certain volume function is given. The discretization procedure of the author yields a method in which it is the aim to *approximate convex bodies by zonotopes* with a fixed number of facets.

We continue with zonotopes occurring regarding *inequalities and extremal problems*.

INEQUALITIES AND EXTREMUM PROBLEMS WITH RESPECT TO ZONOTOPES

In [1581], Linhart shows that among all 3-zonotopes having 3, 4, or 6 congruent segment summands, the regular ones have the *maximal total surface and the maximal inradius*. And in terms of n and m, he gives in [1582] an *upper bound on kth intrinsic volumes of equilateral n-zonotopes* which are sums of m segments of equal lengths. The equality cases are also discussed, and the case $k = 2$ is particularly interesting since the segment configuration then forms a *regular eutactic star*.

Filliman [874] uses decompositions of zonotopes into affine cubes to derive a *volume formula for zonotopes*. This is used to study the problem of *maximizing the volume of zonotopes* restricted by various conditions, e.g., that the sum of the squared lengths of the participating segment summands is constant, or that the considered zonotopes are *orthogonal* projections of a unit n-cube. The author gives various upper bounds for the volume and other specific results with regard to zonotopes of the described type. A nice specific result says that the Catalan triacontahedron has the largest volume among all orthogonal projections of a 6-cube into a 3-space.

K. Bezdek [314] studies n-zonotopes which are sums of $n+1$ segments, calling them *regular* if these segments go from the centroid of a regular n-simplex to its vertices. He shows that if such a zonotope Z contains the n-dimensional unit ball, then its ith intrinsic volume $V_i(Z)$ (for every $i = 1, \ldots, n$) is minimized iff Z is regular having inradius 1. This result has applications in the subfield of *sphere packings*, and also the paper [711] of Daróczy-Kiss is strongly related.

An *f-monotone path of an n-dimensional polytope P* (with f being a linear functional) admits a *flip* if a part of the path can be moved across some 2-face to get another f-monotone path. For P a zonotope with m generating segments and f a generic linear functional on it, Athanasiadis and Santos [147] give a lower bound on the number of flips in terms of n and m. They conjecture that this bound can be reached for all n and m, and it is shown that for each fixed difference $m - n$ it is sharp for infinitely many values of m. *Monotone path polytopes* arise as a special case of the construction of fiber polytopes, and a simple example is the *permutahedron* which is a zonotope being polar to the so-called braid arrangement. Athanasiadis [144] shows how the zonotopes polar to the cones of certain deformations of braid arrangements can be realized as monotone path polytopes.

Gordon et al. [1007] study special *Banach spaces whose unit balls are zonotopes*, and their duals. They derive upper and lower estimates for the volumes of the unit balls and their lower dimensional sections, and the dimension of almost Euclidean sections is determined, yielding a Dvoretzky-type theorem. Further on, asymptotically sharp estimates for important geometric parameters are obtained, such as type and cotype constants, p-summing norms, projection constants, volume ratios, and so on.

Ostrovskii [1927] selects, among all linear projections of the n-cube onto linear subspaces, those which *minimize the volume of the image*. The shape of such images is described, and it is characterized in terms of zonotopes. Further interesting contributions in this direction (mainly due to the author) are summarized in his exposition [1928], where a survey on sufficient enlargements of unit balls of Banach spaces in the study of such projections is given, with open problems. Connections to zonotopes and zonoids are discussed.

Zhao and Leng [2550] prove an extension of a lower bound on the volume of zonotopes due to Ball, and they give also a related upper bound. In addition, they deduce inequalities for the in- and circumradius of a certain type of zonotopes.

Ehrenborg [804] studies the cd-index (which is an encoding of the numbers of chains, specified by ranks, in a given poset) of zonotopes which span the space of cd-indices of all n-dimensional polytopes. He applies his lifting techniques and derives two classes of *linear inequalities for the flag f-vectors of zonotopes*. Also he shows that certain linear combinations of cd-coefficients are, among zonotopes, minimized by the n-cube. Also the paper [2297] of Stenson treats inequalities for flag f-vectors of zonotopes. In particular, cubical zonotopes are studied in this framework. An upper bound on the dimension of flag vectors of them is given.

Rörig, Witte, and Ziegler [2052] ask for the maximal number of vertices for a two-dimensional *central section of an n-dimensional zonotope* having m zones. They show that for any $n \leq 2$ the maximal complexity (number of vertices) for such a central 2-cut is $\Theta(m^{n-1})$. Considering polars of zonotopes, they obtain also lower bounds and prove that the maximal complexity for a two-dimensional projection of the polar of an n-dimensional zonotope with m zones is $\Theta(m^{n-1})$.

Inspired by lower bounds of *successive minima for zonotopes*, Henk, Linke, and Wills [1182] determine the minimal volume of a zonotope that contains the classical crosspolytope. In every dimension the class of such "minimal zonotopes" contains also an affine cube, and in three dimensions all such zonotopes are obtained.

Campi and Gronchi [568] present *volume inequalities for L_p-zonotopes* estimating the volume ratios of such zonotopes Z_p and Z_1, and Z_1 and Z_p^* (with Z^* meaning the polar). Both ratios are minimized if Z_1 is a parallelotope. Generalizing these results and continuing also the work of Weberndorfer [2436], Wang, Leng, and Huang [2419] define *Orlicz zonotopes* and give, using the notion of shadow systems, a sharp upper estimate for their volume and a sharp lower estimate for the volume of their polars, in terms of the volume of Z_1. Alonso-Gutiérez [71] derives lower bounds on the volumes of *L_p-zonotopes* which are taken as polars of unit balls. Based on the concept of shadow systems of convex bodies or zonotopes, Weberndorfer [2436] gives sharp *reverse isoperimetric inequalities* for the asymmetric L_p volume product and the asymmetric L_p volume ratio within the framework of asymptotic L_p zonotopes. Reisner's well-known volume product inequality (for L_1-zonotopes) presents a subcase of these results. Related to this, Ryabogin and Zvavitch devoted Section 4 of their exposition [2082]

to zonotopes and zonoids, and the validity of Mahler's conjecture for them. The work [2436] of Weberndorfer is extended by Yang and Chen [2488] to so-called *asymmetric Orlicz zonotopes* in the Orlicz–Brunn–Minkowski setting.

Blichfeldt's theorem establishes a sharp lower bound on the volume of a convex body K whose lattice points span the whole space, in terms of the amount of integer points contained in K. Henk, Henze, and Wills [1180] give a generalization for centrally symmetric convex bodies and obtain sharp Blichfeldt-type inequalities for 0-symmetric lattice polygons, lattice crosspolytopes, and lattice zonotopes.

Barber and Ben [225] study the *edge isoperimetric problem for graphs* (namely, to determine, for each m, the minimum number of edges leaving any set of m vertices). In general, this problem is NP-hard, but for special graph classes exact results exist. Here the problem is asymptotically solved for every Cayley graph over the integer lattice; the derived near-optimal shapes are zonotopes generated by line segments corresponding to the generators of the Cayley graph.

COMBINATORICS AND RELATIONS OF ZONOTOPES TO ALGEBRAIC GEOMETRY

Greene and Zaslavsky [1019] give various examples to indicate the role that Whitney numbers play in several branches of *combinatorial geometry*. They study Euclidean and projective arrangements of hyperplanes, where the basic poset is given by the family of intersections of the hyperplanes, ordered by reverse inclusion. In particular, Section 5 shows how to attach geometric meanings to the *Whitney numbers in the context of zonotopes*.

Also in the *combinatorial geometry of convex bodies* zonotopes play an important role. The *Helly dimension* of a well-defined family F of convex bodies is the smallest integer m such that for every collection of s members of F the following holds: if each $m+1$ sets from F have a point in common, then all s sets have non-empty intersection. Baladze and Boltyanski (see [187], [186], and the book [424]) derived several related results referring to (Minkowski sums of) zonotopes, zonoids, and belt bodies, see also [427] and Chapters IV and VII of the monograph [431]. E.g., in [308] Beutelspacher and K. Bezdek combine the notion of Helly dimension with integral polytopes. Namely, if any two homothetic images of a lattice-dependent affine cube in the given subfamily intersect on the considered lattice, then all the homothetic images in the subfamily have a lattice point in common, and these are the only zonotopes for which this Helly-type theorem holds. And regarding *Hadwiger's covering problem* we mention (besides [427] and [431]) the paper [430] in which the best known upper bounds on the covering (or illumination) number for zonotopes and belt bodies are given. These numbers result from considerations about a type of indecomposability of zonotopes and approximation procedures of belt bodies by zonotopes. Also problems on (primitive) *fixing systems* belong to the combinatorial geometry of convex bodies (see §44 of [431]). In [428] counterexamples to a related conjecture of Danzer are constructed, and these are zonotopes and zonoids.

The *Bohne–Dress theorem* confirms that the tilings of a zonotope Z by zonotopes are in bijection with the single-element liftings of the associated oriented matroid. Richter-Gebert and Ziegler [2032] prove a natural bijection between the polytopal tilings of a zonotope Z by zonotopes and the one-element liftings of the oriented matroid associated with Z. From this a simple proof and an extension of the *Bohne–Dress theorem on zonotopal tilings* follow. In addition the authors prove that not every oriented matroid is representable by a suitable zonotopal tiling. They construct a rank 4 oriented matroid having 40 elements which cannot be derived as substructure of a lifting of a realizable oriented matroid.

Brown and Diaconis [506] study random walks on the system of chambers of real hyperplane arrangements. Their investigations contain also *random walks on zonotopes and zonotopal tilings*. Bounds on the rate of convergence to stationarity are derived, and the obtained results are extended to oriented matroids.

Let Q be a configuration of points in n-space. Fixing a generic direction u, one gets a linear ordering of the points of Q. Edelman [791] establishes upper and lower bounds on the number $f(Q)$ of such

orderings, using results from matroid theory. Among many other results (also treating Whitney and Stirling numbers), the author gets the following result on *coherent monotone paths of zonotopes* (a *coherent monotone path on a polytope* is an edge-path which is monotonic increasing relative to some direction): he shows how to construct from a given zonotope Z a point configuration $Q(Z)$ such that $f(Q(Z))$ is exactly the number of coherent monotone paths on Z.

Generalizing the Tutte polynomial, Moci [1817] introduces a multiplicity Tutte polynomial $M(x, y)$ which has applications to zonotopes and toric arrangements. Interesting subcases are presented, and it is shown how $M(x, 1)$ yields the volume and the number of integral points of the respective zonotope.

Related to box splines, Adila and Postnikov [124] investigate ideals in a polynomial ring generated by powers of linear forms. In particular, they lay emphasis on power ideals which arise from an *arrangement of hyperplanes* whose combinatorics and Tutte polynomial are studied and used. Among further unifying and deep results, they confirm a conjecture of Holtz and Ron on the *spline interpolation of functions on the lattice points of a zonotope*. Motivated by approximation theory, Holtz and Ron [1226] develop an algebraic concept with the aim to study hyperplane arrangements and their associated zonotopes; it is called *zonotopal algebra* and consists of three nested pairs of ideals in the ring of polynomials of the space of the arrangement and of its dual, using their geometric and combinatorial properties. This algebra has applications and relations to many mathematical fields, e.g., *approximation theory, multivariate polynomial interpolation, group representations, matroid theory, and algebraic graph theory*. Extensions of these results are given in Holtz et al. [1227], where the mentioned nested triple is extended to a more general structure. For example, analogues of various results from [1226] are obtained when the number of pairs of ideals grows with the hyperplane arrangement. Li and Ron [1564] present a unified framework for the so-called external chapter of zonotopal algebra invoking the underlying matroidal and geometrical structures. In the second part of [1544], Lenz constructs a family of zonotopal spaces much more general than those in [124], [1226], [1227], and [1564]. The combinatorial structure on which it is based is called *forward exchange matroid*, which is an ordered matroid together with a subset of its bases set satisfying a weak version of the basis exchange axiom.

The deep paper [876] of Finis and Lapid deals with two related topics. The first one is the study of the relation complexes of hyperplane arrangements and of several related constructions. E.g., complexes graded by geometric lattices and inductive constructions of such complexes are considered. The second topic is given by certain algebras over polynomial rings defined by the labeled graphs of *fiber zonotopes*. A series of interesting examples is given.

In [2246], da Silva studies Euclidean *representations of matroids* as geometric simplicial complexes; these are defined by symmetry properties abstracting such properties of zonotopes.

We mention here also the paper [122] of Arbo and Proudfoot whose content we describe under "zonotopal tilings".

Let M be a *regular matroid* which is representable by a matrix A of rank r with integer entries such that every quadratic k-submatrix has determinant in $\{-1, 0, 1\}$ for all $1 \leq k \leq r$. The *Jacobian group* $Jac(M)$ of M is a finite abelian group whose cardinality equals the number of bases of M. There are many explicit combinatorial bijections in the literature. Backman et al. [170] study a family of explicit bijections between $Jac(M)$ and bases of M. This yields many examples, new even in the more general case of studying graphs. The authors give a purely combinatorial description and consider also the Tutte polynomial associated to M and the *Ehrhart polynomial* associated to the Minkowski sum of the columns of A. This yields interesting connections to *zonotopal subdivisions and lattice points in zonotopes*, and so a new approach to Stanley's formula combining Ehrhart polynomials of the studied zonotopes and the Tutte polynomial of M is obtained.

ZONOTOPAL TILINGS

A *zonotopal tiling of a zonotope* Z is a collection of zonotopes forming a polyhedral complex with support Z. The minimal elements of the poset P of zonotopal tilings of a zonotope Z, ordered by

refinement, are the *tight zonotopal tilings of* Z. Further on, this notion and closely related notions are discussed.

Bailey [181] defines a *rhombohedral tiling of an n-zonotope* Z as decomposition of Z into a non-disjoint union of cells, where each of these cells is a translate of the vector sum of an independent n-subset of the system of generating segments. Such a tiling is called *coherent* if it can be obtained by projecting the "top faces" of some $(n+1)$-zonotope onto Z. In the three-dimensional setting, the author studies the enumeration and structure of the set of all tilings of the zonotope Z and the subset of all coherent tilings of it. His methods come from the theory of hyperplane arrangements and oriented matroids.

Latapy [1513] investigates two types of combinatorial objects: *integer partitions and tilings*. He studies structures of solution sets to partition problems and uses then the obtained results and their correspondence with partitions to study *tilings of zonotopes with flips*. The used methods come from dynamical systems and order theory.

Chavanon and Rémila [602] study *rhombic tilings* which are graphs whose vertices are tilings of a given zonotope. They use ideas from matroid theory to develop a decomposition method for such tilings, to encode them by a sequence of small tilings, containing $n+1$ tiles. Also the relations to higher Bruhat orders (see [853]) and the Bohne–Dress theorem (cf. [2032]) in this framework are discussed.

A *multiple k-tiling* of Euclidean n-space by a convex body K is obtained when translates of K with respect to a discrete multiset exist in such a way that each point of the space is covered exactly k times (except perhaps the translated copies of the boundary of K); all possible multiple tilers in 3-space are zonotopes. Extending results on planar k-tilings, Gravin et al. [1016] verify the conjecture that in 3-space the corresponding multiset of translation vectors has to be a finite union of translated lattices (also known as quasi-periodic sets), unless the multitiler belongs to a special class of zonotopes. The members of this class are Minkowski sums of two two-dimensional symmetric polygons in 3-space, one of which may be a single line segment. They show that *rational zonotopes* of this type admit a multiple tiling with a non-quasi-periodic family of translation vectors. We should mention here the strongly related older paper [1035] by Groemer, which is not cited in [1016]. Groemer discusses multiple space subdivisions by zonotopes.

Using an abstract definition, Arbo and Proudfoot [122] investigate *hypertoric varieties* (which are quaternionic analogues of toric varieties). They take the natural combinatorial structure dual to a central arrangement (called a zonotope) and a natural structure dual to an affine arrangement (called a zonotopal tiling). One can construct the tiling from sign vectors and oriented matroids. Furthermore, hypertoric varieties are classified with the help of zonotopal tilings. It is shown that hypertoric varieties corresponding to so-called irregular zonotopal tilings (presented also by examples not known before) are particularly interesting and impressive.

Two minimal elements of the poset P of zonotopal tilings of a zonotope Z are connected by a flip if P has an element whose only proper refinements are these two minimal elements. The flip graph of Z is the graph whose vertices are minimal elements of P and whose edges are flips. Liu [1586] constructs a three-dimensional zonotope whose set of tight zonotopal tilings is not connected by flips. Interestingly, he uses this to prove that the flip graph of a product of two simplices (one 4- and one n-dimensional) is not connected for large n.

SPECIAL TYPES OF ZONOTOPES: TRANSLATIVE SPACE-FILLERS

Here we start with results on *translative space-fillers* (see Fig. 27 in Plate 10 above).

Using regular chain groups, Jaeger [1291] gives a strong characterization of those zonotopes which are *translative space-fillers* (also called *parallelotopes* or *parallelohedra*). A nice paper discussing the related discoveries of Fedorov (see his famous book [836]) is [2222] of Senechal and Galiulin. Meaning the translative space-fillers in 3-space, the authors cite Delone: "Tradition ascribes to Plato the discovery of the five regular convex polyhedra....and Fedorov found the five parallelohedra". They underline that Fedorov was also the first who posed the task to find them, and that he recognized them

as representatives (with parallelograms and hexagons as faces, and 4 or 6 faces in each zone) of the *infinite family of zonohedra*, i.e., of those convex polyhedra which have only centrally symmetric faces. In addition, Fedorov was interested in *isohedra* (convex polyhedra with congruent faces), and he found four of the five *isozonohedra*. The fifth, which can be obtained from the rhombic icosahedron (also in this class) by collapsing one zone, is the *second isohedral rhombic dodecahedron*; it was found only in 1960 by Bilinski (see [341]). Its segment summands have the directions of four (of the six) joins of pairs of opposite vertices of a regular icosahedron.

Voronoi (1909) conjectured that if a set of translates of a polytope P forms a facet-to-facet tiling of Euclidean space, then P is an affine image of a *Voronoi cell* for a lattice. Erdahl [813] verifies this conjecture at least for the family of zonotopes, and he also confirms that the Voronoi cell for a lattice is a zonotope iff the corresponding Delaunay tiling is a dicing (which is an arrangement of hyperplanes such that the vertices of the corresponding partition form a lattice). Voronoi's conjecture was in general confirmed for $n = 4$ by Delaunay [725]. All higher dimensional general cases remained open until a proof was announced in 2019 by Garber and Magazinov for $n = 5$; we refer to their most recent paper [927] where they explain how the arguments of their proof for $n = 5$ (to which they give an outline) can be alternatively adapted into Delaunay's proof for $n = 4$. A further interesting paper in this direction is [1022] by Grishukhin. On the lines of Voronoi, the author studies polytopes defined with the help of lattices and quadratic forms which are parallelotopes. He presents the related results in modern terms and reflects also the known modern development, including Venkov's and McMullen's characterization of translative space-fillers with the help of 4- and 6-belts of centrally symmetric facets, and the zonotopal subcases; see McMullen's paper [1740]. Deza and Grishukhin [745] show that Voronoi's conjecture is equivalent to the following: For each n-dimensional parallelotope P, there exist scalars γ_i and a positive-definite n-quadratic matrix Q such that $\gamma_i p_i = Q t_i$ for every i, where p_i denotes the normal vectors of the facets of P and t_i are the associated translation vectors. Based on these cases the authors verify that every zonotopal translative space-filler is affinely equivalent to a Voronoi polytope, giving then also explicitly the corresponding matrix Q.

A second conjecture of Voronoi says that the (unique) lattice L creating a facet-to-facet tiling of an n-parallelotope P has a basis $\{b_1, ..., b_n\}$ of vectors, with the property that for each b_i the polytopes P and $P + b_i$ have a complete facet in common. Garber [924] confirms this conjecture for translative space-fillers that are zonotopes.

The same author (see [925]) derives results on belt distances of parallelotopes. Namely, a *belt of a polytope* P is the set of all facets parallel to a given $(n-2)$-face of P, and a sequence of facets is called a *belt path* if any two consecutive facets in that sequence belong to the same belt. The number of different belts that the facets in a belt path belong to is called the *length of the path*, and the *belt distance* of two facets is the length of the shortest belt path between them. Finally, the *belt diameter* of P is the maximal belt distance between any two facets of P. It is shown in [925] that belt diameters of zonotopes that are parallelotopes are bounded by belt diameters of certain symmetric zonotopes, and the precise upper bound of $\log_2(4/5n)$ for n-dimensional space-filling zonotopes is confirmed. Garber continues this in [926], where related results on belt diameters of specific space-filling zonotopes (constructed with the help of permutahedra) are derived.

The notion of *principal domain* for a domain of quadratic forms is due to Voronoi, and positive quadratic functions taken on incidence vectors of suitable subsets are called *set functions* in this framework. Dutour-Sikirić and Grishukhin [779] prove that a set function is submodular iff the quadratic function belongs to a principal domain. Therefore, closures of principal domains can be identified with cones of so-called cut submodular set functions. And parallelotopes of a closed principal domain are zonotopes that are generalized polymatroids related to cut submodular set functions.

In his survey [1928], Ostrovskii nicely discusses open problems from Banach space theory which can be made easier by sufficient enlargements of unit balls. It turns out that also space-tiling zonotopes play a role there.

We continue with results on *lattice and integral zonotopes*.

SPECIAL TYPES OF ZONOTOPES: LATTICE AND INTEGRAL ZONOTOPES

Dealing with upper bounds on *lattice point enumerators* of convex bodies, Gritzmann and Wills [1032] prove an important conjecture in terms of intrinsic volumes for lattice zonotopes. Also Betke and Gritzmann [306] prove an upper bound for lattice zonotopes (being not true for convex bodies in general), and they use valuation theory. Inspired by *Minkowski's second theorem on successive minima and volumes*, Bey et al. [309] study the role of volume and the lattice point enumerator, and they consider the bounds on the coefficients of Ehrhart polynomials of lattice polytopes via the successive minima. Their results on lattice zonotopes and lattice-face polytopes show that for 0-symmetric lattice-face polytopes and lattice parallelepipeds the volume can be replaced by the lattice point enumerator.

D'Addero and Moci [17] investigate the relations between two *polynomials of integral zonotopes* (the Ehrhart polynomial and the arithmetic Tutte polynomial). They use suitable decompositions of integral zonotopes into sub-zonotopes. Their method yields some combinatorial understanding of the coefficients of Ehrhart polynomials of integral zonotopes.

Crapo and Senechal [683] study properties of *zonotopes whose vertices have integer coordinates*. Zonotopes are said to belong to the *same family* if each of them is the sum of a subset of the given family of segment summands. Tiling problems for zonotopes with integer coordinates related in this way are studied, and it is shown that any such zonotope yields a periodic tiling (formed by it and other family members).

Henze and Malikiosis [1187] discuss the equivalence of the problem of estimating *covering radii of zonotopes* to the problems of billiard ball motions inside a cube avoiding an inner cube, lines in a multidimensional torus avoiding an inner cube, and views unobstructed by a lattice arrangement of cubes. They use *zonotopal geometry* to present a refined view on these equivalences. Additionally, in terms of zonotopes they show connections to the *lonely runner problem*.

Given a convex cone C in n-space, let $T(C, x)$ be the family of all integral zonotopes in C whose endpoint x (i.e., the sum vector of all generating vectors) is from C. Bárány, Bureaux, and Lund [221] show that, "for large x" and after suitable scaling, the majority of the appearing zonotopes in $T(C, x)$ is (regarding shape) very close to a fixed convex set. Also combinatorial properties of typical such zonotopes in $T(C, x)$ are presented.

For the sake of completeness, we cite here the brief survey [493] of Braun and Vindas-Meléndezon on *lattice zonotopes*.

Based on the "paving technique" that Shephard considered in his paper [2227], Galashin et al. [918] slightly generalize a concept coming from there. Namely, they prove that if P is *any fixed lattice polytope* and Z a *lattice zonotope*, then for $k \geq 1$ the number of lattice points in the set $P + kZ$ is a polynomial with non-negative integer coefficients in k. They conjecture that the *Ehrhart-like polynomials* have non-negative and integer coefficients, saying also that this might have deeper relations to *representation theory* and *algebraic geometry*. Beck, Jochemko, and McCollough [262] study the h^*-polynomials of lattice polytopes (i.e., the numerator polynomials of the generating functions of their Ehrhart polynomials). In particular, they obtain several deep results on *h^*-polynomials of lattice zonotopes*. E.g., they are able to describe the convex hull of the h^*-polynomials of all lattice n-zonotopes as an n-dimensional simplicial cone whose extreme rays correspond to certain polynomials which encode the distribution of refined descent statistics on permutations. A closed formula for the h^*-polynomial of a lattice zonotope is given, too.

Deza, Pournin, and Sukegawa [742] deal with asymptotic upper bounds on the graph diameter (= number of generating segments) of *lattice zonotopes* and, in particular, *primitive lattice zonotopes* contained in the cube $[0, k]^n$. The latter are sums of primitive lattice vectors having lattice points as endpoints, but no interior lattice points. They describe cases where a primitive lattice zonotope is the unique lattice zonotope of maximum diameter contained in $[0, k]^n$.

McCormick et al. [1735] study the *closest vector problem*: to find, for a lattice L and a given vector v not necessarily from L, the vector from L which is closest to v. For special classes of lattices this can be solved in polynomial time. Using combinatorial properties of zonotopal lattices, the authors present a polynomial-time solution for the closest vector problem in this type of lattices.

We continue with results on so-called *primitive zonotopes*.

SPECIAL TYPES OF ZONOTOPES: PRIMITIVE ZONOTOPES

For positive integers n, p, q or $q = \infty$, the zonotopes generated by the primitive integer vectors v satisfying $\|v\|_q \leq p$ (i.e., zonotopes generated by all primitive integer vectors whose q-norm is at most p) are called *primitive zonotopes*. These zonotopes are interesting since they have a large symmetry group, a large diameter, and, in view of their grid embedding size, many vertices. A. Deza et al. (see [737] and [738]) study basic properties of them, e.g., lower bounds on their diameters and connections to the *computational complexity of multicriteria matroid optimization*. Also *linear optimization over primitive zonotopes* is taken into consideration. *Primitive lattice zonotopes* are discussed by A. Deza et al. in [742], and primitive zonotopes whose generators have q-norm bounded by a constant are considered by A. Deza et al. in [741], e.g., giving bounds and asymptotic estimates for the number of vertices of them. It is shown that there are also connections to *permutahedra and generalized retarded functions in quantum field theory*.

FURTHER SPECIAL TYPES OF ZONOTOPES

A zonotope is called *graphical* if it is the vector sum of segments $[-e_i + e_j, e_i - e_j]$ whose indices (i, j) run over the set of edges of a given graph G. For a complete graph and a tree on $n+1$ vertices, one has as examples a permutahedron and a cube, respectively. It is known that the number of vertices of the graphical zonotope Z_G equals the number of acyclic orientations of a graph G. There are so-called q-analogues of the chromatic symmetric function of the graph G, and Grujić [1089] proves results in this direction. Let Z_n denote the n-dimensional convex hull of all (ordered) degree sequences of simple n-vertex graphs. Using the fact that Z_n is a zonotope, Stanley [2289] finds an explicit generating function for the number of these degree sequences. With the help of signed graph colorings, also the f-vector of Z_n is found.

Athanasiadis [145] investigates zonotopal subdivisions of *cyclic zonotopes*, generated by any m distinct vectors of the form $(1, t, \ldots, t^{n-1})$. It is shown that all zonotopal subdivisions of such zonotopes are shellable, and that all such subdivisions are also stackable in a suitable direction (i.e., there is a height function characterizing the tiles of the subdivision). Relations to Euclidean matroid programs are discussed. Davidov, Karzanov, and Koshevoy [706] give a comprehensive survey on the theory of *tilings of cyclic zonotopes by parallelohedra*, divided into a geometric and a combinatorial part. It turns out that, for example, this is related to the theory of higher Bruhat orders (see also [853] below) and to the parallel theory of triangulations of cyclic polytopes. Felsner and Ziegler [853] investigate graphs called *higher Bruhat orders* $B(m, k)$. These are combinatorially defined partial orders which "look like" the graphs of $(m-k)$-dimensional zonotopes. E.g., arrangements of m pseudolines connected by flips represent the case $B(m, 2)$. Higher Bruhat orders contain the graphs of such zonotopes, but are in general not covered by them; and in general they are even not polytopal. But they contain large zonotopal subgraphs, for some parameters even covering the whole vertex sets of $B(m, k)$.

If P and Q are convex polytopes such that $Q = \pi(P)$ is a projection of P, and the fibers $\pi^{-1}(x)$ with $x \in Q$ are also convex polytopes. The *fiber polytope* $\Sigma(P, Q)$ is, in a well-defined sense, the average over all fibers $\pi^{-1}(x)$, see Billera and Sturmfels [346]. There are also closely related *fiber zonotopes*. Fiber zonotopes play also a role in [346] and [1752]: in the latter paper McMullen shows that if P is an n-dimensional polytope all whose fiber polytopes are known, with regard to each r-dimensional

linear subspace with $2 \leq r < n$, then P is uniquely determined. He also shows that if $r \geq 4$ and the r-dimensional fiber polytopes of P are zonotopes, then P is itself a zonotope.

Postnikov et al. [1990] study the f-, h-, and γ-vectors of *simple generalized permutahedra* (this family including *permutahedra, associahedra, graph-associahedra, simple graphical zonotopes*, and other types). Besides many further interesting results (among them a combinatorial interpretation for γ-vectors for a large family of generalized permutahedra), they derive or conjecture upper and lower bounds for the f-, h-, and γ-vectors within several such polytope classes.

Consider a simple Lie algebra with a Cartan subalgebra h and $\Phi \subset h^*$ as set of roots. Then let P_Φ be the *root polytope* of a finite irreducible crystallographic root system Φ, i.e., the convex hull of all roots in Φ. Cellini and Marietti [586] find out for which Lie algebra the polars of root polytopes are zonotopes. In these cases the zonotopes are determined explicitly, with a characteristic isomorphism property.

ALGORITHMICAL ASPECTS AND APPLICATIONS

Kühn [1469] studies *discrete dynamical systems* and the validated inclusion of occurring n-dimensional sets by special sets which are computationally easy to represent (like, e.g., parallelotopes). Successive computation then yields the inconvenient *wrapping effect*, i.e., an exponential growing overestimate of solution sets. This effect is kept arbitrarily small by a new method developed by the author and using k-zonotopes (having $k \geq n$ segment summands) as inclusion sets. The steps of the approach use the geometry of zonotopes in a detailed manner. Alamo, Bravo, and Camacho [35] show a new approach to *guaranteed state estimation for nonlinear discrete-time systems* with a bounded description of noise and parameters. For their purpose, they compute a set represented by a zonotope whose size is minimized at each sample time. Also Wan, Vehi, and Luo [2417] give a numerical approach to control invariant sets for constrained nonlinear discrete-time systems, and *zonotope geometry is used to reduce the wrapping effect*. The investigations are based on the fact that, for representing sets, zonotopes are more flexible than boxes in interval analysis. And similarly Chai et al. [587] use an improved use of zonotope geometry to *set-membership identification for linear discrete-time systems*. The authors introduce a new family of zonotopes containing the intersection of a zonotope with a strip, and the minimal volume zonotope in this family is taken as the outer bound of the feasible solution set. These investigations are continued by Chai et al. in [588].

In the spirit of computational convexity, Burger and Gritzmann [539] investigate the problem of maximal and minimal (regarding respective volumes) orthogonal projections of convex polytopes onto subspaces. Taking the *projection bodies* of these polytopes into consideration (which are centred zonotopes), this means to ask for the diameter and the minimal width of zonotopes (see also [1723]). They show that to find maximal volume projections on hyperplanes is already NP-hard for simplices, and that getting the minimal volume projections is easy for simplices but NP-hard for bipyramids over parallelotopes. The authors prove also interesting results for lower dimensional projections, and also the inradius computation of zonotopes is dealt with. Martini (see Section 7 of [1705] and §21 in [432]) shows that the *median-hyperplane problem* from location science can be studied in the same way like the function of $(n-1)$-dimensional outer cross-section measures of n-polytopes, namely via support functions of zonotopes.

Bern and Eppstein [293] combine properties of zonotopes and so-called *support vector machine classifiers* (computer algorithms that learn by example to assign labels to objects, e.g., successfully applied to an increasingly wide variety of biological applications). To get also new theoretical results on support vector machine classifiers, they study relations to the ellipsoid method.

In their note [1105] Guibas, Nguyen, and Zhang show how to construct (approximately) *optimal zonotopes enclosing given point sets*. Using an implicit representation of them, their approach has applications in *collision detection*. It suffices to study the generating segment system; their explicit description as whole polytopes is not needed.

Replacing line segments by convex n-polytopes in vertex-representation, Fukuda [909] considers the construction of the Minkowski sum of k polytopes as *generalization of the zonotope construction problem*. Gritzmann and Sturmfels [1031] study various aspects of this and present polynomial algorithms for the problem when one of the parameters k or n is fixed. Extending a known algorithm for the zonotope construction, Fukuda creates here an efficient algorithm for variable n and k, more precisely an algorithm that runs in time bounded by a polynomial function of both the input and the output size. The algorithm is based on linear programming and reverse search. The motivation comes from the use of polyhedral computation for *optimal tolerance determination in mechanical engineering*.

Ferrez et al. [864] refer to a subcase of the problem of *maximizing quadratic forms* in n binary variables, namely when the underlying (symmetric) matrix is positive semi-definite of fixed rank k. This can be reduced to *searching the extreme points of a zonotope*, yielding polynomial complexity in $O(n^{k-1})$. Reverse search leads to an efficient enumeration of the cells of *regular hyperplane arrangements*. Affine forms are used to represent convex sets via bases of error terms $\varepsilon \in [-1, 1]^m$, and quadratic forms are an extension of affine forms allowing the use of quadratic error terms. In *static analysis, set-based simulation for hybrid systems, and floating point analysis*, the zonotope domain based on affine forms has been widely used providing abstraction of functions with a cost linear in the number of error terms. Adjé et al. [20] study *quadratic versions of zonotopes* and present also a new algorithm (in the setting of semi-definite programming) to project quadratic zonotopes and forms to intervals. Instructive examples are represented. Coming from *applied stabilization problems*, Il'in et al. [1266] study stability of affinely parameterized families of polynomials leading to a linear transformation of a parallelotope of parameters into a zonotope of polynomial coefficients. An *inner approximation of the zonotope by a union of parallelotopes* is constructed, whose stability is checked via further subdivisions.

Onn and Rothblum [1916] present a *vertex enumeration algorithm* for n-dimensional polytopes P with $n > 2$ and, in particular, for zonotopes (the latter needed as a step). Their input is a vector set containing a direction for every edge of P and an oracle solving linear programming tasks over P. With this input they obtain a list of vertices of the Minkowski sum of the respective segments and a list of linear functionals, such that each of them is maximized precisely at the associated vertex over the whole zonotope. Further steps refer to P itself, with a resulting list that contains all vertices of P.

In view of linear programming problems, also Fujishige et al. [908] propose a method called the *LP-Newton method* that is based on geometric properties of zonotopes. Using them, also here linear optimization is made easy. To show the behavior of the LP-Newton method, some concrete computational results are obtained.

Althoff, Stursberg, and Buss [91] compute reachable sets for *hybrid systems with linear continuous dynamics*, which are represented with the help of zonotopes. This yields low computational complexity regarding the dimension of the studied system. Since zonotopes have also disadvantages in this framework, polytopes as an intermediate step are used. Afterwards, these polytopes are enclosed by zonotopes.

In [699] an enhanced version of reverse search of significantly reduced computational complexity is presented. The starting point (see also [700]) is Goffin's algorithm which in polynomial time constructs a tight approximation of the *Löwner–John ellipsoid of a polytope* given by its facet description. The author adapts this algorithm for zonotopes given by their generators and shows that this approach works in polynomial time, in terms of the system of generators of the zonotopes. And their description may be superpolynomially shorter than the facet description. Given an n-vector p of processing times of jobs, the *single machine scheduling polytope* S occurs as the convex hull of completion times of jobs when these are scheduled without waiting time on a single machine. Carathéodory's theorem secures that a given point $x \in S$ can be obtained as convex combination of at most n vertices of S. Hoeksma et al. [1207] confirm that such a convex combination can be computed from x and p in $O(n^2)$ time, being linear in the naive encoding of the output. This result can be derived based on the fact that the

scheduling polytope S is a zonotope. These investigations are continued in Hoeksma et al. [1208]; besides further applications for scheduling polytopes the presented algorithm may serve as a blueprint for similar decomposition problems referring to more general zonotopes.

Moser [1854] deals with Weyl's discrepancy misalignment measure which induces a certain norm that, when applied to differences of index-shifted sequences, shows a monotonicity and a Lipschitz property. The corresponding unit ball is an n-dimensional zonotope resulting from a multiple projection from the $(n + 1)$-dimensional hypercube which can be interpreted as a discrete differentiation. Thus, the discrepancy norm is related to *measuring the distance between index-shifted sequences*.

Montina [1840] develops an output-sensitive algorithm for generating the whole set of flats of a finite matroid. Also a special algorithm for the case of vector matroids is given, and in some cases this one yields an effective method for computing zonotopes in H-representation, when their generating system of segments is given. This work is motivated by *quantum communication*.

Stenson [2298] describes the relations between certain types of functions and zonotopes. These functions occur in *voting systems of players who can form coalitions*, where a winning coalition is passing a certain measure. In this setting, the author treats *switching functions, threshold functions, weighted voting systems, and simple games*, and a zonotope suitably related to them is defined. Coordinates of these zonotopes are combined with "voting interpretations" in terms of a measure of voting power.

Montúfar and Morton [1841] study successfully relations between *theoretical properties of restricted Boltzmann machines, models of machine learning forming partial blocks of models for deep learning*, and certain notions from discrete mathematics and convex geometry. Among the latter are *probability distributions, Hamming modes, zonotopes, hyperplane arrangements, linear threshold codes*, and also *multicovering numbers of hypercubes*. Based on these relations they prove results on mixtures of product distributions and products of mixtures of pairs of product distributions

Ryabogin, Yaskin, and Zhang [2080] consider a typical question from *discrete geometric tomography*, namely a discrete version of *Alexandrov's projection theorem for surface areas*. It is asked whether two 0-symmetric convex lattice sets K and L coincide when there are coinciding discrete surface areas of projections of them in the same projection directions. (The discrete surface area is measured by numbers of projected lattice points, suitably contained in convex hulls of analogous projections.) The authors answer this question affirmatively for the three-dimensional case, and for higher dimensions they confirm the same at least for the case that the convex hulls of K and L are zonotopes. The proof methods mainly come from the geometry of numbers and from discrete geometry.

Wang et al. [2418] present two *zonotope-based algorithms on set-membership estimation* for identification of time-varying parameters in linear static models, with additive and also multiplicative uncertainties. They are called "Cone And Zonotope Intersection" and "Polyhedron And Zonotope Intersection", which involve linear programming and linear matrix inequalities and can be successfully applied in *health monitoring*, for studying *mini-batches of measurement data and the identification of time-varying parameters*.

Kulmburg and Althoff [1474] define a new norm type for zonotopes, to solve the problem of whether a point lies within a zonotope. Based on this, they prove the co-NP-completeness of the *zonotope containment problem* (whether a zonotope is contained in another one). Based on new algorithms, they are able to solve the zonotope containment problem in polynomial time when the dimension or the number of generating segments of one of both zonotopes is fixed.

In the nice paper [740] of A. Deza and Pournin several problems around the *geometry of zonotopes* are attacked. The first one is the question of whether a polytope P, given by its vertex set, is a zonotope. Without using any convex hull computation, the authors present an algorithm solving this problem and delivering, in the positive case, the configuration of generating line segments. The second presented algorithm decomposes a polytope into a Minkowski sum of segments and other polytopes, where the latter cannot be decomposed into zonotopes. And a third algorithm yields the generating segments from the given vertex set of a given zonotope.

For a given point configuration A, Barucq et al. [241] study relations between polynomial-reproducing spline spaces over certain subsets of the convex hull of A and zonotopal tilings of the zonotope associated with the respective vector configuration. This yields a new result on *Delaunay configurations*, and a related iterative construction of such spaces is verified. This leads also to interesting algorithms regarding the related *spline functions*.

Motivated by applications in *smart grids, autonomous diving, and human–robot collaboration*, Chalkis et al. [599] deal with estimating the volume of zonotopes. To reduce the related complexity, they use a second zonotope of smaller order whose volume is easier obtained by a closed formula. The approach is based on Multiphase Monte Carlo Methods. It is shown that the developed algorithm for zonotopes requires a sublinear number of oracle calls in the dimension, and that the best theoretical bound is cubic. This can also be easily extended to arbitrary convex bodies.

10.6.2 Zonoids

Zonoids are limits (in the Hausdorff metric) of convergent sequences of zonotopes, and limits of convergent sequences of zonoids are again zonoids. These interesting convex bodies occur as the *ranges of non-atomic vector-valued measures* in measure theory and related fields, as *projection bodies and centroid bodies* in convexity, and as *lift zonoids* in statistics (see Plate 10 above). They have many further properties and applications. We refer the reader also to our subsections dedicated to projection and centroid bodies and to the theory of Banach spaces. Special types of zonoids are ellipsoids and the spindle-shaped convex body whose boundary is obtained by rotating a suitable piece (between $-\pi/2$ and $\pi/2$) of the cosine curve around the x-axis. We note that the introductory part of 10.6.1 above contains already references on various monographs and surveys treating, besides zonotopes, also zonoids.

MINKOWSKI CLASSES

Fitting to Chapter 10 here, a few words on Minkowski classes of convex bodies should make sense—also since this notion is connected with that of zonoids. A non-empty subset of the family of all convex bodies which is closed in the Hausdorff metric and closed also under Minkowski addition and dilatation is said to be a *Minkowski class* in Euclidean n-space. There are subcases like, e.g., rotation invariant, motion invariant, and translation-invariant Minkowski classes obtained by the property that the images of their members under the corresponding transformations still belong to the same class. Examples of motion invariant Minkowski classes are given by the *bodies of constant width* and by the *centrally symmetric convex bodies*, and the class of *rotors* in this framework is nicely discussed by Schneider in Section 3.5 of his monograph [2167] (see also Sections 4.5 and 5.7 in Groemer's book [1042], and Section 17.1 in [1713]). In [2167] it is also emphasized why *zonoids* form the smallest non-trivial affine-invariant Minkowski class. This Section 3.5 from [2167] is also the main reference for Minkowski classes in general, and the particularly comprehensive notes to it deserve special mention. As example of a more recent contribution we cite [1823] of Molchanov and Nagel, where a new Minkowski class is introduced and zonoids also play an important role (see below for more precise informations on this paper).

ZONOIDS IN BOOK PARTS

We start with *books treating zonoids*.

In Thompson's book [2361] on *Minkowski geometry* they occur in Section 2.3 (Mahler–Reisner inequality), in the Notes 3.5 (zonoids whose polars are zonoids), in Sections 5.4 and 6.2 (as tool to

get isoperimetrices for the Holmes–Thompson definition of surface area in normed spaces), in Section 6.1 (approximation), in Section 6.4 (integral geometry in normed spaces), in Section 6.5 (certain inequalities), several times in the Notes 6.7 and in the Notes 7.5 (e.g., in relation to associated bodies like intersection bodies), and in Section 9.5 (numerical parameters in normed spaces).

In Gardner's book [930] on *geometric tomography* zonoids are introduced and discussed in Section 4.1 (with many properties and also as projection bodies), in Section 4.2 (in connection with smaller bodies having larger projections), in the Open Problems and Notes to Chapter 4 (also their history and notions like generalized zonoids, projection bodies of order i, polars of projection bodies etc.), in the Notes to Chapter 8 (generalized zonoids), in Section 9.1 (as centroid bodies), in Section 9.3 (certain inequalities), and in the Notes to Chapter 9 (affine isoperimetric inequalities).

In Schneider's monograph [2167], Section 3.5 and its Notes comprehensively refer to zonoids, like also Section 10.14 (treating L_p *zonoids*). At the beginning of Section 3.5 the family of zonoids is discussed as the simplest non-trivial Minkowski class, generated by segments, and also the *smallest non-trivial affine-invariant Minkowski class* (see also above). Section 4 of the Lecture Notes [2082] of Ryabogin and Zvavitch is dedicated to zonotopes and zonoids, and to the proof of Mahler's conjecture for them. From the viewpoint of *functional analysis*, zonoids are treated in Chapter 4 (dedicated to partial answers of the hyperplane conjecture) and Chapter 7 (discussing centroid bodies) of the monograph [496] of Brazitikos et al., in Chapter 12 (discussing the relations between positive operated-valued measures and zonoids) of [151] by Aubrun and Szarek, in Section 2.5 of [133] by Artstein-Avidan et al. (as projection bodies), and in subsection 4.6.2 of the second volume [134] (regarding Shephard's problem on projections of convex bodies).

Leichtweiss [1537] discusses zonoids in Section 1.2, and Deza and Laurent [746] present them in Section 8.3. And in the recent book [1263] of Hug and Weil especially Section 4.4 is dedicated to zonoids, mainly in form of projection bodies. Further monographs treating also zonoids are: Groemer's book [1042] (at the end of Sections 2.2 and 5.5), and the book [2179] of Schneider and Weil treating stochastic and integral geometry (Section 14.3).

Finally here, we mention also Mosler's monograph [1855] dedicated to the notion of lift zonoids. This book gives an informative description of sets of multivariate data in *statistics* in geometric language. Namely, given a set M of m pairs $(a_i, p_i), i = 1, ..., m$, where the a_i are n-dimensional vectors with probability p_i, the *lift zonoid* Z_n is defined as the convex hull spanned over all segments starting from the origin to points (P, M') in $(n + 1)$-space formed by all subsets $M' \in M$ with accumulated probabilities $P = P(M')$. The lift zonoid for a distribution is calculated as a limit of Z_n when $m \to \infty$. Thus, by the lift zonoid the space of n-variate measures having finite first moments is mapped into the space of $(n + 1)$-dimensional, centrally symmetric, convex bodies, and if the n-variate measure has finite support, then *lift zonotopes* are obtained. In contrast to the usual notion of zonoid, that of lift zonoid characterizes the underlying measure uniquely. Especially Sections 1.3 and 1.4, the whole Chapter 2, and Appendix A of Mosler's book [1855] contain a lot of material about zonoids and lift zonoids; some further related and useful notions (like zonoid trimmed regions, zonoid depth, lift zonoid volume, etc.) can be found in further sections of that book. *Lorenz zonoids* as special type of lift zonoids are also introduced and discussed in this monograph, namely in Sections 7.2 and 9.2 there.

We continue now with *related surveys*.

SURVEYS TREATING ZONOIDS

As we said already above, the two most *important surveys on zonoids* (mainly from the viewpoint of convexity) are [2176] of Schneider and Weil as well as [997] of Goodey and Weil. The paper [422] of Bolker is an essential forerunner. The most important topics studied in these surveys (and treating zonoids) are: geometric properties, geometric and analytic characterizations, zonoids as ranges of vector measures, subspaces of L_1-spaces, related norm inequalities, zonoids as auxiliary bodies (also

projection bodies), generalized zonoids (which are dense in the family of centrally symmetric convex bodies), approximation problems, relations to the spherical Radon transform, projections and projection functions, degrees of central symmetry, and zonoids in integral and stochastic geometry.

In Section 6 of the handbook article [2182] of Schneider and Wieacker *translative integral geometry* and associated auxiliary zonoids are discussed. In the research paper [2183], integral geometry is extended to Minkowski spaces. It turns out that for the Holmes–Thompson definition of area, which is more tractable than that of Busemann, the authors derive a translative intersection formula, and for hypermetric spaces (whose unit balls are polars of zonoids) they get also full *analogues of the Crofton formulae* for rectifiable sets and for convex bodies. We mention here also the paper [2158], where similarly the *Holmes–Thompson definition and polars of zonoids* play a role, and mainly *intrinsic volumes in Minkowski spaces* are investigated.

The survey [1710] contains extensive material about convex or star-shaped bodies associated with given convex bodies via *outer and inner cross-section measures*, thus yielding notions like *difference body, projection body, intersection body*, and *cross-section body*. The relationships between these bodies and the consequences of combining them suitably are also discussed, mainly within the framework of *geometric tomography*. The central role of projection bodies brings zonoids into the game.

In his survey [1059], Gruber discusses *approximation results* regarding special classes of convex bodies, among them also zonoids. In the handbook article [1579] of Lindenstrauss and Milman *approximation and symmetrization problems around zonoids* are discussed (see subsection 6.2). And also Section 5 of the handbook article [1447] of Koldobsky and König treats approximation results, namely of zonoids by zonotopes.

Schneider's survey [2159] comprehensively presents where in convexity *measures* occur, and in which way they do this. E.g., this refers to Hausdorff measures, measures associated with boundary structures, and classical uniqueness and existence results with respect to surface area measures. In the final section zonoids are discussed, mainly in their role as *ranges of non-atomic vector measures*. Variations of this property are discussed, and finally an application to *Poisson processes of hyperplanes* is shown, with zonoids as auxiliary bodies.

Schneider [2161] gives a complete and deep survey on the developments around generalizations of the classical Crofton formula from integral geometry in Euclidean space to *finite-dimensional normed spaces and projective Finsler spaces*. He also clarifies the relation to Hilbert's fourth problem, shows different ways to this generalization, and emphasizes the relations to geometric convexity and, in particular, to the theory of zonoids.

Ostrovskii [1928] surveys sufficient enlargements of unit balls in Banach spaces in the study of projections, which yields natural possibilities to approach some well-known open problems of Banach space theory. The connections to zonotopes and zonoids are also discussed, and open problems are posed.

We continue with surveying research papers related to zonoids which (in most cases) cannot be found in the books and surveys above.

GEOMETRIC PROPERTIES AND CHARACTERIZATIONS OF ZONOIDS

Rodríguez-Piazza [2049] studies zonoids as *ranges of vector measures*. He shows that if the ranges of two finitely additive measures with values in a normed space have the same closed convex hull, then these measures have equal total variation. This implies that in the Euclidean space the total variation is continuous with respect to the range when the Hausdorff topology is considered. Further related results are derived.

Campi, Haas, and Weil [572] prove that if Z is an n-dimensional zonoid and $u_1, \ldots, u_k$ are fixed unit vectors, then there exists a zonotope Z' being the sum of at most k line segments and having the same supporting hyperplanes as Z in the directions $u_1, \ldots, u_k$. They prove this twice: the first proof is due to Bourgain (using Carathéodory's theorem), and the second is based on linear programming. Using the

projection body property of full-dimensional zonoids, also results on sizes of projections of general convex bodies are obtained.

Sekerin [2217] presents a *characterization of zonoids via their support functions*. Namely, he characterizes these functions by an integral condition within the class of subharmonic functions on the n-dimensional space which are positively homogeneous of degree one (in particular, the support functions of general convex bodies form a subclass of these functions).

Lonke [1600] studies *zonoids whose polars are zonoids*. Bolker conjectured that they have to be ellipsoids, and Schneider disproved this conjecture by constructing smooth perturbations of balls as counterexamples. The question remained whether there are also nonsmooth zonoids whose polars are zonoids, and Lonke constructs concrete three- and four-dimensional examples. It is likely that they exist also in higher dimensions. He also shows that if a convex body K has a summand whose dimension is at least 1 and at most $n-2$, then the polar of the body cannot be a zonoid. Hence the property of K to be together with its polar zonoid implies that the boundary of K contains only faces of dimensions 0 and $n-1$.

It is known that for $n > 2$ an n-dimensional 0-symmetric convex body Z is a zonoid if and only if each piecewise linear inequality holds in Euclidean n-space, with the absolute value replaced by the norm given by the support function of Z. It was an open problem of whether a single piecewise linear inequality is sufficient to characterize the centered zonoids among all centered convex bodies. Burger [537] shows that this is not true and proves even that no finite set of piecewise linear inequalities suffices to characterize the class of zonoids. Consequently, even the *Hlawka inequality* in 3-space does not characterize zonoids.

Gordon and Junge [1005] generalize the *classical volume ratio* (with respect to ellipsoids) to the L_p case. This yields an essential connection between such quantities and norms in classical ideal spaces of operators. The authors can show that, although there are many more zonoids than ellipsoids, every n-dimensional unit ball B has a slice of dimension at most $n/2$ whose volume ratio with respect to zonoids is, up to a universal factor, at least as big as the classical volume ratio of B. Thus, the volume ratio with respect to ellipsoids is controlled by the volume ratio with respect to zonoids in a section of "small" dimension.

In view of two-dimensional zonoids and *Chebyshev measures*, Bianchi et al. [336] prove that a strictly convex, centrally symmetric, and compact set in the plane whose boundary contains the origin is the range of a Chebyshev measure.

Goodey and Jiang [993] are interested in the question of whether a convex body is determined by the average of its k-dimensional projections. They denote with P_k the average (in the sense of Minkowski sums) of projections of convex bodies onto k-dimensional subspaces of the n-space ($k \in \{1, ..., n-1\}$), and the average is formed by integration of support functions with respect to the rotation invariant probability measure on the corresponding Grassmannian manifold of k-dimensional subspaces. Among other results, it is proved that P_k is injective if $k \geq n/2$, and that each n-dimensional convex body is determined by the Minkowski sum of its three-dimensional projections. The authors study also the range of P_k, getting a closed proper subset of the class of all convex bodies, and a proper subfamily of the class of zonoids when restricted to all centrally symmetric convex bodies.

Rodíguez-Piazza and Romero-Moreno [2051] extend results on zonoids (as closed convex hulls of ranges of vector measures) due to Neyman from Euclidean space to Banach spaces. Namely, the equivalence of the following statements on zonoids Z in Banach spaces are proved: (1) Z uniquely determines the associated conical measure. (2) There is a vector measure such that every decomposition of Z into the sum of zonoids can be obtained by decomposition. (3) The set $Z \cap (\alpha Z)$ is a parallelepiped for any $\alpha > 0$. Further results about *decompositions of zonoids* are derived. And also in [2050] results on zonoids as ranges of vector measures are extended. The investigations there are also using a decomposition result for zonoids regarding vector sum and based on a measurable function, which is then extended to countable sums.

Burger [538] gives a new proof of the Bolker's theorem saying that a centrally symmetric convex body K in Euclidean n-space is a zonoid iff the function $-h$ is conditionally positive-definite (where h is the support function of K).

The authors of [966] consider the *minimum of the mean width* under volume-preserving linear maps. More precisely, they give three applications of such an isotropic characterization of the minimal mean width position. The first one is the reverse of an Urysohn-type inequality, and it is also proved that for zonoids this minimum is smaller than that for the cube.

Aramyan [118] studies *radii of curvature of plane projections* of n-dimensional centrally symmetric bodies which are smooth. The obtained results on support functions and generating density of convex bodies are presented in the context of Weil's conjecture on the *zonal characterization of zonoids* in even-dimensional spaces. The same author, inspired by the question of whether there is at least a describable subclass of zonoids admitting a local equatorial characterization of zonoids, derives in [116] a sufficient condition, based on such a local equatorial condition, that a centrally symmetric body is a zonoid. This yields indeed a definition of a subclass of zonoids admitting a *local equatorial characterization* (e.g., ellipsoids belong to this subclass). In [117], Aramyan studies a new representation for the support function of an origin-symmetric n-dimensional convex body K in terms of surface curvature functions of K. Using it and a local equatorial property, he finds a sufficient condition for an origin-symmetric n-dimensional convex body to be a zonoid.

Alesker [46] studies a generalization of the fact that the class of *generalized zonoids* is dense in the class of centrally symmetric convex bodies.

Klebanov et al. [1413] prove that the *support function of an n-dimensional zonoid* is a negative-definite first degree homogeneous function, and that the converse holds, too. Furthermore, if the support of a generating measure of a zonoid coincides with the unit sphere, then the support function is strongly negative-definite, and due to this it generates a distance on the space of Borel probability measures.

Anantharaman [105] studies a special case of a result of Bolker: if M is the range of a non-atomic vector measure on a measurable space (S, Σ) and not a line segment, then M is the sum of two sets, neither of which is a homothetic copy of M). In [105], M is assumed to be the unit disk in a 2-space, an explicit decomposition of which in the sense of Bolker's conditions is given. On the other hand, an explicit decomposition of M not following Bolker's conditions is presented, where the sets are the ranges of the restrictions of the considered vector measure to certain subsets of S.

Koldobsky [1441] found that the family of *intersection bodies* contains the class of polars of zonoids as a subfamily, and Schneider and Wieacker [2183] conjectured that for most convex bodies their intersection body is not the *polar of a zonoid*. In this framework (and only for dimensions four and six), Alfonseca [54] presents geometric criteria for a convex body of revolution whose intersection body is not the polar of some zonoid. Various examples of such intersection bodies are given.

Aramyan [120] introduces asymmetric versions of zonoids and of generalized zonoids via their *support functions*. He then proves some results on these notions.

Davydov and Paulauskas [716] study sets which are essentially countable Minkowski sums of sums of k independent, identically distributed, exponential random variables, suitably combined with independent random vectors, identically distributed according to some probability measure on the Euclidean unit sphere. They call them *random stable countable zonotopes*. For the planar case, the Hausdorff dimension of the set of extremal points of the boundary of such a set is obtained, and the authors give also a (non-random) characterization for zonoids to be a countable zonotope (i.e., a countable sum of zonotopes) in terms of the Hausdorff measure of its boundary. The main condition therein says that a zonoid is a countable zonotope if and only if all its facets are themselves countable zonotopes.

Molchanov and Nagel [1823] study the family of convex bodies obtained from 0-symmetric convex bodies K via multiplication with diagonal matrices, by forming Minkowski sums of the transformed sets, and by taking limits in the Hausdorff metric. This yields the natural notion of *diagonal Minkowski*

classes which are one of the main topics of investigation in this paper. The obtained support functions arise by an integral transform of measures on the family of diagonal matrices, called K-transforms. If, in particular, K is a line segment not from a coordinate hyperplane, then the class of *zonoids and the cosine transform* are obtained. The authors show that the following two statements are equivalent: the vector space generated by support functions of zonoids is dense in the family of support functions of 0-symmetric convex bodies; the cosine transform is injective. If K is a generalized zonoid, conditions are clarified that guarantee the injectivity of the K-transform. Also relations to *mixed volumes* and a geometric description of one-sided stable laws are discussed.

In the interesting paper [501] Breiding et al. combine three topics: *zonoids, mixed volumes, and random determinants*. The authors introduce "differences of zonoids" to obtain vector spaces of zonoids, and considering a canonical multilinear map of such vector spaces of zonoids, they are able to introduce an exterior product of zonoids which itself is a zonoid. This yields a graded, associative, and commutative real algebra, called *zonoid algebra*. Interesting relations to mixed volumes can be confirmed, and notions like *projection body, mean width , and intrinsic volume* occur. Applications like explicit formulas for the expectation of the determinant of random matrices with independent columns are obtained, and an extension of Vitale's results from [2407] can be obtained. With the help of Alexandrov–Fenchel inequalities for mixed volumes, further consequences are derived, and also valuations come in. A new proof of a *reverse form of the Alexandrov–Fenchel inequality for zonoids* due to Böröczky and Hug (see [404]) is given.

The paper [1549] of Lerario and Mathis is motivated by a question of Sturmfels: given a *semialgebraic convex body*, is there an algorithm to decide whether it is a zonoid? Indeed, for a given semialgebraic family of convex bodies it is not known whether the subfamily of zonoids of the family is semialgebraic or not. The authors prove that if a family of convex bodies is definable in the structure generated by globally subanalytic sets, then the set of zonoids in this family is definable in the structure generated by globally subanalytic sets and the graph of the exponential function. This shows the "tameness" of the subfamily of zonoids.

INEQUALITIES AND EXTREMUM PROBLEMS INVOLVING ZONOIDS

Hadwiger's covering problem asks for the upper bound on the minimum number of smaller homothets of a convex body K needed to cover K, and the conjecture (verified only in the planar case) is 2^n with equality if, and only if, K is an affine cube. There are many partial results around this conjecture, also treating zonoids. Boltyanski and P. S. Soltan [435] succeeded in confirming the conjecture for zonoids (see also [434]); for the more general class of belt bodies it was verified in [427], see also Chapter VII of the book [431]. For the class belt polytopes, which are already dense in the family of convex bodies, the confirmation was given already in [1705]. This covering problem is equivalent to a famous illumination problem. In [322] a related illumination problem is solved for zonoids. These problems belong to the so-called combinatorial geometry of convex bodies. Another problem from that field is called the Szökefalvi-Nagy problem and treats sharpenings of the Helly dimension for subclasses of all convex bodies. Baladze [185] succeeded in giving a complete solution of this problem for zonoids.

Leng [1541] derives some inequalities for the *affine surface area* of the curvature images of simplices, zonoids, and centrally symmetric convex bodies. He also verifies an inequality for the mixed volumes and surface areas of projection bodies.

Xi et al. [2476] study the L_p *mean zonoid*, ($p \geq 1$), of a convex body K. It is shown that, up to homothety, this mean zonoid is the L_p centroid body of the radial $(n + p)$th mean body of K. Related affine inequalities and geometric properties (e.g., observed via Steiner symmetrization) are derived.

Du, Leng, and Xi [766] introduce the notion of *Orlicz mean zonoid* of a convex body K, and they derive a sharp lower bound for its volume normalized by the volume of K. It turns out that the minimum characterizes the ellipsoids. Generalizingvadjust a concept of Zhang and continuing these

considerations, Guo, Leng, and Du [1110] introduce an *Orlicz mean zonoid operator* and prove that it is commutative, GL(n)-covariant, and continuous. Also for this operator, an affine isoperimetric inequality is established. And Ma [1650] gives, based on shadow systems, a new proof of an affine isoperimetric inequality for Orlicz mean zonoids.

Generalizing the notion of mean zonoids given by Zhang, and the notion of *L_p-mean zonoids* introduced by Xi, Guo, and Leng, the authors of [1654] introduce and investigate L_p-mixed mean zonoids of pairs of convex bodies, confirming some essential properties (such as monotonicity and GL(n) covariance). Also *affine isoperimetric inequalities* for those zonoids are proved.

Li and Huang [1553] establish the *L_p Loomis–Whitney inequality* for even isotropic measures in terms of the support function of L_p projection bodies and thus extend Ball's Loomis–Whitney inequality to the L_p setting. In addition, the authors obtain a sharp upper bound of the minimal p-mean width of L_p-zonoids. Tang and Si [2343] establish a Loomis–Whitney-type inequality for a volume normalized L_p projection body when $p \geq 1$. The equality conditions are given for $p \neq 2$. The *dual Loomis–Whitney inequality* establishes a lower bound for the volume of a convex body in terms of its coordinate sections. Similar results (thus extending the Loomis–Whitney inequality and its dual further onto Grassmann manifolds) are obtained by Li et al. [1558], and they are obtained via some operators meant to replace usual projections and sections. This concept yields a characterization of weighted balls formed by an isotropic measure on a Grassmann manifold.

Li et al. [1555] and further authors below study *L_p-zonoids and their polars*. Continuing a series of volume inequalities of Ball, Barthe, Meyer, Lutwak, Yang, and Zhang, which refer to L_p-zonoids and their polars with even isotropic generating measures, in [1555] Li et al. give *sharp upper and lower bounds for the volumes of k-dimensional projections of L_p-zonoids*, and of k-dimensional sections of their polars. The necessary conditions for equality cases are completely new. Also Böröczky et al. [397] give volume inequalities for L_p zonoids of even isotropic measures and for their polars.

Hereby the case $p = \infty$ gives a stability version of the *reverse isoperimetric inequality for centrally symmetric convex bodies*. The authors state an interesting conjecture on volume inequalities for L_p zonoids of general centered (non-symmetric) isotropic measures. They use strengthenings of notions used by Barthe for results around the Brascamp–Lieb and reverse Brascamp–Lieb inequalities, and they phrase an interesting conjecture on a possible stability version of the Brascamp-Lieb and the reverse Brascamp–Lieb inequalities.

In [1568] Li and Wang study *asymmetric L_p-mean zonoids*, a typical notion from the *L_p-Brunn–Minkowski theory*. They present an affine isoperimetric inequality with a lower bound on the volume of the asymmetric L_p-mean zonoids of any convex body K, having the origin as interior point, by the volume of K. Further on, *extremal values for volume and quermassintegrals* are given, as well as *dual quermassintegrals of the polar*. And also certain monotonic inequalities for this type of zonoids are proved.

It is known that *Hlawka's classical inequality* has interesting connections to properties of zonotopes and zonoids. Luo [1623] introduces so-called Hlawka type-1 and type-2 quantities and investigates how "Hlawka-like" they are related to each other. Connections between several variants of Hlawka inequalities become clear from this viewpoint, such as Serre's reverse Hlawka inequality in the future cone of the Minkowski space. A complete study of Hlawka's inequality for quadratic forms is given, too.

Böröczky and Hug [404] bring together and extend results in the spirit of a *reverse Alexandrov–Fenchel inequality*. Particularly, they give generalizations when most of the participating convex bodies are zonoids. This supports further evidence that the *conjectured reverse Alexandrov-Fenchel inequality* is true. They also characterize the equality cases in all established inequalities and stronger stability results.

APPLICATIONS OF ZONOIDS

One of the main applications of n-dimensional zonoids and, in the polytopal case, of n-dimensional zonotopes is the fact that in the origin-symmetric case they occur as *projection bodies of general convex bodies and polytopes*, respectively. Likewise their appearance as *centroid bodies* is essential. For related results and references in this direction we refer to our next subsection below.

Hanegraaf [1156] offers an interesting series of problems which are related to the geometric design of pieces of surfaces, e.g., as building elements of global surfaces by mitigating also a local convexity condition. In particular, he studies zonogonal, zonohedral, and zonoidal pieces whose global union forms then interesting surfaces. The posed problems mainly concern morphological aspects. Special attention is paid to the occurrence of (also globally) non-convex regions.

Martini [1707] poses the question for which convex bodies their *projection bodies and difference bodies are polars*. He clarifies this for polytopes (whose projection bodies are centred zonotopes): only the simplices have this property. So the problem is still open for all convex bodies which are not polytopes and also not ellipsoids, i.e., for convex bodies having non-polytopal and non-ellipsoidal projection bodies.

Vitale [2407] investigates expectations of absolute values of determinants of random matrices, whose columns present random vectors, and random convex bodies. Among them, he concentrates on zonotopes and zonoids, giving also a nice characterization of zonoids.

Rataj [2022] discusses integrals needed for translation versions of principal kinematic formulae to get a general Poincaré formula for a finite number of Hausdorff rectifiable sets. He generalizes former results in this direction expressing such integrals in terms of *mixed volumes of certain auxiliary zonoids*. Further results in this direction are discussed in surveys on zonoids (see above and also Section 6 of the handbook exposition [2182] of Schneider and Wieacker).

Daoudi et al. [710] study local regularity properties of the boundary of some special types of zonoids. These considerations are motivated by an application in *chemical engineering*.

One should also mention here that Klain's new and shorter proof of Hadwiger's famous *characterization theorem for quermassintegrals* of convex bodies (see [1395]) is also based on the representation of a convex body with C^∞ support function as the difference of two zonoids.

Koshlevoy and Mosler [1457] study so-called *lift zonoids* (see also the book [1855], already mentioned above). With the help of a measure M in n-space with finite absolute moment, the usual support function of a zonoid Z can be introduced, and Z uniquely determines M when M is an even measure concentrated on the unit sphere. If M is suitably lifted to a measure M', then the zonoid Z' generated by M' is the *lift zonoid* of M. The authors derive structural results for lift zonoids (e.g., continuity and behaviour under projections) and give applications. The lift zonoid is related to *random convex sets* and to the *convex hull of a multivariate random sample*. For an arbitrary sampling distribution, the authors give bounds on the expected volume of the random convex hull. And the set inclusion of lift zonoids defines an ordering of random vectors which reflects their variability.

The subcase of *Lorenz zonoids* (see also Plate 10 above) was introduced by the same authors and earlier, see Koshlevoy and Mosler [1456]. Continuing these investigations, Dall'Aglio and Scarsini [702] characterize these notions (lift zonoids and Lorenz zonoids of a random vector). Such a characterization allows a direct use of Lyapunov's theorem on the range of non-atomic vector measures and helps to simplify proofs of further related results. In the later paper [703], these authors consider a zonoid of an n-dimensional random vector X as tool to measure linear dependence among its components. As opposed to the lift zonoid, it does not determine the distribution of X, but describes the size-biased distribution of its compositional variables. A class of *measures of linear dependence* is proposed, and special results for $n = 2$ are also obtained.

Borell [455] considers lift zonoids in view of *expectations of random sets in real separable Banach spaces*. He expresses the support function of the lift zonoid in terms of the Gaussian isoperimetric function, the unit ball in the corresponding reproducing kernel Hilbert space, and affine transformations of that zonoid. Finally he shows an interesting application of lift zonoids to derive bounds on the average excess return in the Markowitz mean-variance portfolio selection setting.

Koshlevoy and Mosler [1459] survey generalizations of the Lorenz curve and Lorenz dominance for multidimensional data. Clear definitions of Lorenz zonoids and their relation to lift zonoids are given. Three multivariate extensions of univariate Lorenz dominance are exposed, their differences and the set inclusion of lift zonoids, the scaled convex order, and the price Lorenz order are discussed. The latter is based on the *set inclusion of extended Lorenz zonoids*.

Raffinetti and Giudici [2013] propose a new *relative dependence measure in statistics* which presents an alternative to the partial correlation coefficient to establish categorical covariate contributions in *multiple linear regression models*. This new measure is *based on Lorenz zonoids*. Similarly, the same authors study in [977] new *model selection measures on Lorenz zonoids*; these are robust and based on a mutual notion of variability. Using these Lorenz zonoids, the contribution of each explanatory variable can be measured more accurately.

Koshlevoy et al. [1458] introduce a new scatter matrix functional which is a *multivariate affine equivariant extension of the so-called mean deviation*. It is constructed from the data vectors and their angular distances. Since it is the regular covariance matrix of the centers of the facets of the zonotope based on the data set, the new estimate is called the *zonoid covariance matrix*. It is shown that between the zonoid covariance matrix and the affine equivariant sign covariance matrix the following connection holds: interchanging the roles of data vectors and hyperplanes yields the sign covariance matrix as the zonoid covariance matrix.

Hoffmann [1212] investigates special *Poisson processes of hypersurfaces* extending results of Wieacker (where the projection body plays already a role) to the non-stationary case and to cylinders. He introduces so-called *associated zonoids* whose jth intrinsic volume is determined by some intersection intensity. Deleting the Poisson assumption, the intensity function of the intersection with an affine subspace is determined; for linear sections this is expressed in terms of the support function of the associated zonoid (having an appropriate location).

Mosler et al. [1856] discuss how probability distributions on Euclidean n-space can be described by their *zonoid regions* forming a nested family of compact, convex sets around the expectation. This has many applications in *multivariate statistical analysis*, i.e., for example in cluster analysis, multivariate location, and risk analysis. For constructing such *nested zonoid regions* of an n-variate empirical distribution, $n \geq 3$, here an exact algorithm is given.

Using projection bodies, Schneider [2165] investigates the *tomographical problem* "how small" a direction set with given corresponding brightness values of an n-dimensional convex polytope or convex body ($n > 2$) has to be such that it is uniquely determined in the family of origin-symmetric convex bodies. Denoting the direction set yielding such a result by A, the aim here is to reduce it as far as possible. It is shown that for polytopes the set A consists of a finite union of great subspheres of the unit sphere. Thus, a set of spherical measure zero is possible (former investigations delivered only sets of positive spherical measure). As tools, the author uses surface area measures, zonoid properties, and inequalities of Alexandrov–Fenchel type.

Each symmetric stable distribution in n-space is related to a norm that, in case of a multivariate Cauchy distribution, has *the polar of a zonoid as unit ball*. In [1821], Molchanov interprets symmetric stable laws by using convex or star-shaped sets, and he presents concepts from convex geometry useful for new probabilistic results regarding *multivariate symmetric stable distributions*. E.g., he discusses expressions for moments of the Euclidean norm of a stable vector, mixed moments, and various integrals of the density function; geometric inequalities are used for bounding important parameters of stable laws.

Kulik and Tymoshkevych [1472] prove that the lift zonoid concept for a probability measure on Euclidean n-space yields a one-to-one representation of any interior point of the convex hull of the support of a continuous measure as the barycenter with respect to this measure, of either a half-space or the whole space. An infinite-dimensional generalization is given, based on the extended concept of lift zonoid for a cylindrical probability measure. In [1473] these authors introduce the concept of *weighted lift zonoids*, and they confirm that the ordering condition on a measure, formulated in terms

of the weighted lift zonoids of this measure, yields certain functional inequalities for this measure, such as the weighted inverse log-*Sobolev inequality*.

Two integrable random vectors are said to be *zonoid equivalent* if both their scalar products with any vector have the same first absolute moments. This notion, comprehensively discussed and applied by Molchanov et al. [1824], is clearly based on the geometric concept of zonoids. With the help of this background, one gets a nice *geometric description of multivariate random variables* useful in multivariate statistics (for trimming and data depth). One important consequence of zonoid equivalence of x and x' is the equality of the expected values of $f(x)$ and $f(x')$ for any even, one-homogeneous function f. The authors of [1824] mainly investigate stochastic processes whose finite-dimensional distributions are zonoid equivalent with respect to time shift (zonoid stationarity) and permutation of its components (swap invariance).

Hug, Thäle, and Weil [1262] study *weakly stationary random processes of k-dimensional flats*. In terms of parameters of the underlying k-flat process, the corresponding intensity measures are derived. By a translation into geometric parameters of associated zonoids, several new related uniqueness and stability results are obtained. These are based on partially new estimates for area measures of zonoids.

Aubrun and Lancien [150] find an interesting correspondence between convex geometry (zonoids) and the distinguishability of norms associated to *quantum measurements or positive operator-valued measures (POVM)*. The latter, being special vector measures, were introduced in *quantum information theory*. The authors obtain the POVM version of classical results from the local theory of Banach spaces, namely results about *approximation of zonoids by zonotopes*. (At the beginning of the paper, a helpful table is given which can be seen as "dictionary" between zonoids and POVMs.) So they show how the most symmetric POVM can be approximated by a discrete POVM, and that for *any* POVM similar (but weaker) approximation results hold.

10.6.3 Projection, centroid, and intersection bodies

We now discuss important references and results referring to these three types of convex or star-shaped bodies associated with a given convex or star-shaped body.

PROJECTION BODIES: THE DEFINITION

The *projection body* ΠK *of a convex body* K is the convex body whose support function value in direction u equals the $(n-1)$-area of the orthogonal projection of K onto a hyperplane with normal vector u (also called the outer $(n-1)$-dimensional cross-section measure of K at u). This notion goes back to Minkowski, and it is one of the nicest applications of *zonotopes* [and of *zonoids*] that the *projection body* of any n-dimensional convex polytope [of any general n-dimensional convex body] is a zonotope [zonoid] centered at the origin. Thus, e.g., the maximal and minimal content of orthogonal hyperplanar shadows of K are given by the halved diameter and the halved minimal width of ΠK, respectively.

PROJECTION BODIES: BOOK PARTS AND SURVEYS

In several monographs one finds important contributions devoted to *projection bodies*. These are §7 of Bonnesen–Fenchel [446], Sections 2.6 and 5.5 of Groemer's book [1042], Chapters 5 and 6 of Thompson's monograph [2361], Sections 3.3, 9.3 and the whole Chapter 4 of Gardner's book [930], Section 10.9 of Schneider's book [2167], and Section 4.4 of the book [1263] of Hug and Weil. Surveys containing important material in this direction and taken into consideration here are [2176] by Schneider and Weil, [997] by Goodey and Weil, and [1710] by Martini. Looking at the appearance

dates of these publications, it is our goal to cite (with some exceptions) results that came to be after the second editions of [930] and [2167] (or cannot be found in these publications). Of course, we take also the later books [1263] and [1713] into consideration.

First we shortly mention *important topics* that might be also useful for projection bodies and related topics, and which are discussed in the surveys and books mentioned above, like also in the expositions [474] by Bourgain and Lindenstrauss, [1634] by Lutwak, and Section 7.2 in Groemer's handbook article [1040], [1710], and [5]):

Referring to *zonotopes*, such topics are: (projective) *arrangements of hyperplanes, translative space-fillers, cube projections, totally symmetric polytopes, approximations of and by zonotopes, and oriented matroids* (here we additionally cite Chapter 2 of Björner et al. [354], Chapters 6 and 7 of Ziegler's important monograph [2560], and the expository paper [1705]).

Regarding *zonoids* we mention: *ranges of vector measures, subspaces of L_1-spaces, (analytic) characterization theorems and generalized zonoids (also belt bodies), projection functions (here also themes like Blaschke bodies, bodies of constant brightness, and tomographic problems), isoperimetrices in normed spaces, zonoids in integral and stochastic geometry, affine (isoperimetric) inequalities, and related stability results*.

Now let us collect results on *generalizations of the notion of projection body*, where also other viewpoints (like valuations) play a role.

GENERALIZATIONS OF THE NOTION OF PROJECTION BODIES

For a given an n-dimensional convex body K, the support function of its *projection body of order k*, denoted by $\Pi_k K$, is a scalar multiple of the kth intrinsic volume of the corresponding orthogonal projections of K onto $(n-1)$-subspaces. Every projection body of order k is a zonoid, but for $k < n-1$ no characterization of these bodies is known (the usual projection bodies present the case $k = n-1$). Campi and Gronchi [570] give a necessary condition for a function to be the support function of a projection body of order $k = 1$, and they also study a reconstruction problem of a convex body from the kth intrinsic volumes of its orthogonal projections onto all (or some) hyperplanes. Comparing projection bodies of order 1 with those of order $n-1$, they observe existence results which only hold for the case $n-1$.

As duals of j-intersection bodies, Dorrek and Schuster [762] generalize the notion of projection body by considering pairs K, L of origin-symmetric convex bodies whose codimensional projections onto respective subspaces have equal codimensional volumes, then considering K as *j-projection body* of L (the classical case is $j = 1$.) Various characterizations of j-projection bodies are derived, certain properties of them are studied, and several examples are given in [762].

The notion of *mixed projection bodies*, going back to Lutwak [1632], is defined as follows: Let $K_1, \ldots, K_{n-1}$ be $n-1$ n-dimensional convex bodies, and $K_1^u, \ldots, K_{n-1}^u$ be their projections onto the $(n-1)$-subspace orthogonal to an arbitrary unit vector u and containing the origin. The mixed projection body $\Pi(K_1, \ldots, K_{n-1})$ of $K_1, \ldots, K_{n-1}$ is the convex body whose support function is given by the mixed volume of $K_1^u, \ldots, K_{n-1}^u$. Continuing Lutwak [1632], Zhao and Leng [2549] establish *polar forms of the general Minkowski, Alexandrov–Fenchel, and Brunn–Minkowski inequality for mixed projection bodies*, thus in terms of dual mixed volumes of polars of mixed projection bodies. Equality cases and applications are also treated. And based on [2], in [2537] and [1589] the polars of *mixed complex projection bodies* are studied, and also the polar forms of the three types of inequalities mentioned above are established.

Lutwak et al. [1638] introduce *L_p-projection bodies*, and the exact L_p-analogues of the Busemann–Petty centroid inequality and the Petty projection inequality are verified; also equality cases are obtained. The authors use basically *Steiner symmetrization* on smooth L_p-projection bodies. Ludwig [1611] (see also Haberl and Schuster [1126]) extends the notion of L_p-projection bodies to that of *general L_p-projection bodies*. Results and methods analogous to those in [1638] occur for this more

general setting in [1126], and smooth L_p-projection bodies play again an important role; as applications this yields new sharp affine L_p-Sobolev inequalities, see also Haberl and Schuster [1125]. Extending Ball's Loomis–Whitney inequality to the L_p setting and using support functions of L_p projection bodies, Li and Huang [1553] establish the L_p *Loomis–Whitney inequality* for even isotropic measures, with equality cases. They also derive the sharp upper bound of the minimal p-mean width of L_p zonoids. In addition we mention here the paper [862], in which the further extension of (p, q)-*mixed projection bodies* is introduced; it is related to L_p dual curvature measures.

Another way of generalizing projection bodies is presented by Lutwak et al. in [1641]. The authors introduce the notion of *Orlicz projection body*, and the main result of the paper is the Orlicz–Petty projection inequality: for a convex body K with the origin as interior point, the volume quotient between the polar Orlicz projection body of K and K itself is maximal when K is an origin-symmetric ellipsoid. It is also proved that if the used convex function is strictly convex, then these ellipsoids are the only extremal candidates. K. J. Böröczky [393] succeeded in proving this without the strict convexity assumption, adding also corresponding stability results. See also Li and Leng [1561] for Orlicz extensions.

We switch now to *affine (isoperimetric) inequalities*, also closely related to *valuations and generalizations of the notion of projection body*. Saroglou (see [2111] and [2115]) deals with sharp bounds for the affine volume invariant $V(\Pi K)/V^{n-1}(K)$. Petty conjectured that the minimum is attained (only) for ellipsoids, and Schneider conjectured the maximum to be 2^n, holding iff the convex body K is the affine image of Cartesian products of line segments or centrally symmetric two-dimensional convex bodies. These *conjectured projection inequalities of Petty and Schneider* are also presented in Section 12 of Lutwak's survey [1633]. A minimizer must have the property that $\Pi^2 K$ is a suitable homothet of K, motivating to study fixed points of iterates of the projection body operator. Brannen constructed counterexamples to the second, and he conjectured that the upper bound remains true in the class of zonoids, in 3-space confirmed by Saroglou (see [2111]); for all dimensions > 2 he shows that there is a convex body K and a direction such that the volume of the projection body of the Steiner symmetrization of K along this direction is strictly larger than the volume of ΠK. The second paper [2115] completely refers to the three-dimensional situation. Saroglou shows that if K is a three-dimensional zonoid of volume 1, then its second projection body $\Pi^2 K$ is contained in $8K$ with equality iff K is a cylinder; and second, if K is a centrally symmetric convex body of volume 1, then $\Pi^2 K$ contains $6K$, this bound being attained iff K is a double cone with circular base. One consequence is a strengthening of a reverse isoperimetric inequality for three-dimensional zonoids, previously established by the author.

Also Saroglou and Zvavitch [2118] deal with the affine invariant $V(\Pi K)/V^{d-1}(K)$. There it is proved that if a convex body K has absolutely continuous surface area measure, whose density is sufficiently close to the constant, then the iteration of the projection body operator of convex bodies converges to the ball with respect to the Banach–Mazur distance. Based on this, the ellipsoid is a local solution to the conjectured inequality of Petty on the volume product $V(\Pi K)/V^{n-1}(K)$, thus supporting it. Continuing [2118], Ivaki [1285] studies the equation $\Pi(\Pi K) = cK$ for some $c > 0$ and K a convex body. He succeeds to show that in a neighborhood of the unit ball, the only solutions of this equation are the *origin-symmetric ellipsoids*. He also generalizes the stability result of the paper to the kth projection body and applies it to further related problems, also treating iterates of the projection and centroid operators.

Replacing in the conjectured Petty projection inequality the volume of the projection body by that of its dual, the *Petty projection inequality* is obtained, which was extended by Haberl and Schuster [1126] for certain Minkowski valuations. Haberl [1122] gives *complex extensions of the Petty projection inequality*, and Wang and Liu [2426] extend this to the L_p setting. Berg and Schuster [279] derive an inequality for polarizations of these Minkowski valuations. It is also shown that the volume of the (generalized) polar projection body of order i, $1 \leq i \leq n-1$, can be bounded in terms of the corresponding affine quermassintegrals. Also the dual framework is investigated, yielding inequalities for generalizations of the intersection body and the *dual affine quermassintegrals*. Finally, a general-

ization of the Busemann–Petty centroid inequality for generalized centroid bodies is established and used as important proof tool.

For the literature on *generalized zonoids* (and related extensions) we refer to the surveys [2176] and [997], and a further and very natural generalization is that of belt polytopes: a convex n-polytope is said to be a *belt polytope* if it has a non-degenerate segment summand parallel to each of its edges. This notion was introduced by Fedorov (see Taylor [2347]). Projection bodies of convex n-polytopes (i.e., zonotopes) and belt polytopes with the same system of the mentioned segment summands cannot be combinatorially distinguished. Namely, their spherical images are the intersections of the projective arrangements of facet hyperplanes with the hyperplane at infinity, consisting there of projective arrangements of $(n-2)$-planes. *More combinatorial informations regarding projections of convex polytopes* can be obtained from these arrangements. E.g., these informations can refer to (sharp) shadow boundaries for central or parallel projections, and they create also interesting research problems; see the survey [1712] of Martini and Chapter VIII (pp. 384–388) in the monograph [431].

PROJECTION BODIES AND VALUATIONS

We continue with results treating *valuations* (see also Section 11.3 and Subsection 11.7.2 below). Let us first shortly mention Alesker's recent monograph [49] on valuations, honouring also Hadwiger's monograph [1134] as a classic in the field (due to the famous characterization of quermassintegrals as continuous, valuations invariant under rigid motions).

We take now Ludwig's results on *Minkowski valuations* from [1611] as (one) starting point. Moment bodies, difference bodies, and projection bodies of n-dimensional convex bodies have the property that their corresponding operators Z are *valuations* regarding Minkowski addition $+$. This means that they satisfy $Z(K_1 \cup K_2) + Z(K_1 \cap K_2) = ZK_1 + ZK_2$, being homogeneous, compatible with the action of $\mathrm{SL}(n)$, and with $K_1, K_2, K_1 \cup K_2$ as convex bodies. In [1611] it is shown that the operators Z can, up to minor modifications, be characterized by these properties. For $p > 1$, the results are also extended to L_p Minkowski addition. The author classifies $\mathrm{SL}(n)$ contravariant Minkowski valuations which, for $n \geq 3$, are closely related to the *projection body operator* Π. Several investigated problems are completely solved for the planar case. Abardia and Bernig [2] study such valuations on a k-dimensional complex vector space, where $k > 2$. Particularly, they introduce the concept of *mixed complex projection body*, and they establish an analogue of Ludwig's characterization of the projection body operator Π by its $\mathrm{SL}(n)$ contravariance given in [1611]. The discussed valuations are shown to satisfy geometric inequalities of Brunn–Minkowski, Alexandrov–Fenchel, and Minkowski type (later papers related to this are [2537] and [1589]).

Also motivated by Ludwig's characterization of the projection body operator Π, Schneider and Schuster [2175] replace the mentioned contravariance with Euclidean rotation equivariance. Then the projection body operator is no longer characterized in the mentioned way, but the assumption that polytopes are mapped to polytopes suffices to characterize combinations of the operator Π; other characterizations of Π are also derived. Among other related results (also involving centroid bodies), Haberl [1121] shows that a map Z from the class of convex bodies containing the origin onto itself is a continuous $\mathrm{SL}(n)$-*contravariant Minkowski valuation* if and only if there is a real non-negative constant c such that $ZK = c\Pi K$.

Related extensions to Sobolev spaces are given by Ludwig in [1616]. Further on, Li and Leng [1561] investigate *Orlicz projection bodies* and establish the following: Unlike their L_p analogues, the *Orlicz projection body operator* is not an Orlicz valuation for any convex function. In addition, one cannot modify the definitions of the Orlicz projection operator to get from it a valuation. These results are obtained by *classifying the* $\mathrm{SL}(n)$-*compatible Orlicz valuations*. In the setting of log-concave functions, Fang and Zhou [830] establish *functional analogues for the projection bodies*. The (functional) projection body of a given log-concave function f is defined in terms of its support function, using an integral expression similar to that of the geometric setting.

In [406] the authors give a complete classification of *Minkowski valuations on lattice polytopes* which intertwine with the special linear group over the integers and are translation-invariant. It turns out that, in the contravariant case, *the only such valuations are multiples of projection bodies*.

Lutwak's observation that the kth intrinsic volume of the ith projection body is log-concave (see [1632]) is extended to translation-invariant, $SO(n)$ equivariant, homogeneous, and continuous Minkowski valuations by Berg et al. [278]. It is shown that besides the projection body operators a large class of Minkowski valuations has the log-concave property. A similar property for *commutative Orlicz Minkowski additions* including the L_p Minkowski addition is also obtained. The main inequality is extended further, replacing Minkowski addition by any commutative Orlicz Minkowski addition.

We continue with some further results (of different types) related to projection bodies.

SOME FURTHER RESULTS ON PROJECTION BODIES

Schneider [2165] studies tomographic questions of the following type: What can be said about two convex bodies C, L which have *equal brightness values in a direction set* U which is a proper subset of the unit sphere? It is shown that C and L are translates of each other when U is chosen to reflect the structure of the set of extreme normal vectors of one of the bodies, e.g., C. For C a polytope, U can be a finite union of great $(n-2)$-subspheres. Different from known results, this set has spherical measure zero.

Gardner et al. [939] study *Blaschke addition* # of origin-symmetric convex bodies. They show that a binary operation between origin-symmetric convex bodies, uniformly continuous in an appropriate metric and $\mathrm{GL}(n)$-covariant, corresponds to a linear combination of convex bodies in the sense of Blaschke addition. Among other results this shows that any *characterization of Blaschke addition* is equivalent to a characterization of Minkowski addition, restricted to the class of origin-symmetric *zonoids*, where the relation $\Pi(K\#L) = \Pi K + \Pi L$ between Blaschke addition and Minkowski addition via the projection operator is essential.

Based on properties of projection bodies, Schneider [2165] asks "how small" a direction set with given corresponding brightness values of a convex polytope or convex body can be such that it is uniquely determined in the family of origin-symmetric convex bodies. This paper is already described above, in the part on zonoids.

Referring to an old question of Blaschke and treating relations between projection bodies and intersection bodies, Nazarov, Ryabogin, and Zvavitch [1882] prove that there is no *local equatorial characterization of zonoids* in odd dimensions. This gives a negative answer to the corresponding conjecture posed by Weil in 1977, showing that the local equatorial characterization of zonoids may be given only in even dimensions (the latter was confirmed by several authors; see, e.g., [1936]). An analogously negative result for intersection bodies is proved. This is continued in [998] where, not depending on Radon and cosine transforms, the authors present a general principle which shows that such local or equatorial characterization problems are equivalent to corresponding *support properties of the spherical operators*. This yields further geometric constructions (also related to *L_p-centroid bodies and to k-intersection bodies*). Aramyan [120] defines asymmetric versions of zonoids and generalizes zonoids via their support function, with the origin as centroid, and he describes several properties of them. In [116] the same author asks: is there a subclass of zonoids admitting a local equatorial characterization? He finds a sufficient condition for centrally symmetric convex bodies to be zonoids, and this yields such a subclass (e.g., the ellipsoids belong to it). Here we mention also the paper [710], in which (from the viewpoint of geometric design and applications in chemical engineering) local regularity properties of the *boundaries of special types of zonoids* are presented.

Finally we cite the paper [2292] of Steineder showing an interesting *connection between symbolic dynamics and convex geometry via polytopal projection bodies*.

CENTROID BODIES: THE DEFINITION

The notion of *centroid body* of a convex body C goes back to Petty [1967], although the basic ideas are older (see [776] and [355]). It can be presented as the integral arithmetic mean of the support functions $|\langle x, u\rangle$ of the line segments joining $-x$ and x in the centred convex body K (i.e., it is built from the centroids of the intersections of K with half-spaces containing the origin in their boundary).

CENTROID BODIES: SURVEYS AND BOOK PARTS

In [1631] Lutwak starts with an exposition of results on *mixed affine surface area and dual mixed volumes*, and with related *connections between the projection operator and the centroid operator*. Based on also new inequalities for affine surface area, an alternative proof of Petty's generalization of the Winternitz result on the monotonicity of affine surface area is given. Further on, a *centroid-body analogue to the Petty–Schneider result on projection bodies* is derived, namely: if K is star-shaped (with positive continuous radial function) and L is the dual of a zonoid such that the centroid body of K contains that of L, then the volume of K is not larger than that of L, with equality if and only if $K = L$. After [1631] the important Handbook article [1633] appeared, reflecting the state of knowledge regarding centroid bodies.

Although, like also every centroid body, each projection body is a zonoid, the *projection operator* (e.g., being translation invariant) and the *centroid operator* (for instance, commuting with linear maps) are different. In addition to the expository paper [1633] of Lutwak, V. Milman and Pajor [1794] and Groemer, Lindenstrauss, and V. Milman (see their handbook surveys [1039] and [1579]) discuss centroid bodies. Collected in monographs, the basic knowledge on centroid bodies can be found in Section 5.6 of Groemer's book [1042], Sections 9.1 and 9.2 of Gardner's monograph [930], Section 10.8 of Schneider's book [2167], Chapter 5 of [496], and Chapter 10 of [133]. More precisely regarding the last two references, the essential role of *L_p-centroid bodies* in the *geometry of isotropic convex bodies* should be described here. We start with Chapters 5 and 7 of the book Brazitikos et al. [496]. In Chapter 5 *fundamental properties of L_p-centroid bodies* are presented, e.g., together with sharp moment inequalities for the Euclidean norm and a proof of the fundamental deviation inequality of Paouris. A combination of the theory of L_p-centroid bodies and the logarithmic Laplace transform in Chapter 7 yields a new approach to the hyperplane conjecture by Klartag and E. Milman. In Chapter 10 of the monograph [133] of Artstein-Avidan et al., the authors also discuss the deviation inequality of Paouris, proved with the help of L_p-centroid bodies, and the connection between isotropic log-concave measures and convex bodies developed by Ball.

CENTROID BODIES: INEQUALITIES

We continue with some *basic inequalities* treating also centroid bodies. A first important result on centroid bodies is the *Busemann–Petty centroid inequality* giving a lower bound on the ratio of the volume of the centroid body of a convex body K and that of K itself, with equality iff K is an ellipsoid centered at the origin (see [1967]). Then in [1638] Lutwak et al. study centroid bodies (and projection bodies), extending these notions to the *L_p setting*. First the L_p version of the Busemann–Petty centroid inequality is proved, and then also the L_p analogue of Petty's result is derived, namely that among bodies star-shaped with respect to the origin, only the ellipsoids have *centroid bodies of minimal volume*. The corresponding result for the *polars of projection bodies* (having then maximal volume) is also obtained in this setting, where the authors carefully use the tool of *Steiner symmetrization* (on smooth L_p-projection bodies).

Campi and Gronchi [567] generalize the L_p version of the Busemann–Petty centroid inequality to the *intrinsic volumes of the L_p centroid body*, see also their paper [569]. In view of results on the *reverse Busemann–Petty centroid inequality* (meaning an upper bound) in the planar case we refer to Bisztriczky and K. J. Böröczky [349], and for the L_p version to Campi and Gronchi [566], where triangles and (in the symmetric case) parallelograms are the extremal convex figures.

The authors of [2509] derive inequalities for dual quermassintegrals from inclusions of centroid body variants of star bodies and polars of projection bodies. And the paper [2495] deals with the question of whether the inclusion of polar p-centroid bodies of two star bodies implies the respective volume inequality. It is verified that if $0 < p < 1$, then the implication is true in the planar case, but not for $n > 2$. And if $-1 < p \leq 0$, then the implication is true if and only if n is at most 3. Spencer [2284] gives a quantitative version of the main result from [2495]; corresponding to this volume comparison result, he proves a separation inequality.

Volume inequalities and a Brunn–Minkowski-type inequality involving *Blaschke sums* for analogues of centroid bodies in the p-dual Brunn–Minkowski theory are studied by Si, Xiong, and Yu [2242]. Lutwak, Yang, and Zhang [1640] present the *extension of the concept of L_p-centroid bodies to Orlicz spaces*; the authors prove a corresponding analogue of the Busemann–Petty centroid inequality, also characterizing centred ellipsoids and conjecturing that this ratio of volumes is also minimized by ellipsoids among all bodies star-shaped with respect to the origin. This was confirmed by Zhu [2556]. Also based on the concept of Orlicz centroid body, Chen et al. [609] prove the *reverse Orlicz Busemann–Petty centroid inequality* and study extrema of further affine-invariant functionals. E.g., in the class of all planar convex bodies the maximum is attained by triangles. Also by Xiong and Xu [2483] Orlicz extensions are studied; it is shown that the *Orlicz centroid bodies of ellipsoids* are again ellipsoids (explicit examples are given, too).

Klartag and Milman [1411] unify and slightly improve several bounds on the isotropic constant of high-dimensional convex bodies, and some new bounds on the *volume of L_p-centroid bodies* are also presented. Ivaki [1283] gives a nice proof of the *stability of the Busemann–Petty centroid inequality* in the plane, using a version of a centro-affine flow introduced by Stancu and applied first to smooth planar convex bodies, and then extended to general convex bodies via approximation arguments. Another stability result is obtained by Guo et al. in [1111]: the *Hausdorff distance of the p-centroid bodies* of two convex bodies K, L having equal volumes is used to bound the Hausdorff distance of K and L.

Paouris and Werner [1940] introduce a new affine invariant which can be derived in three ways: as the limit of normalized L_p-affine surface areas, as the relative entropy of the cone measure of a convex body K and the cone measure of its polar body, and as the limit of the *volume difference of K and associated L_p-centroid bodies*. Based on this, they obtain new affine isoperimetric inequalities for convex bodies and show relations to quantum information theory.

Lutwak and Zhang [1643] establish an *upper bound on the volume product of a convex body and the dual of its p-centroid body*; Chai and Lee [590] give a respective lower bound for convex bodies K with suitable centered homothetic ellipsoids which are contained in, and do contain K, respectively. The subcase of *bodies of constant width* is particularly studied.

Using the notions of L_p-dual affine surface area and L_p-centroid body, *Shephard-type inequalities* are established by Feng and Weidong in [861]. The equality cases are reached when the L_p-centroid bodies of the participating star bodies coincide. We mention also the paper [2425] of Wang and Li, in which different types of inequalities on *quermassintegrals of variants of L_p-centroid bodies* are proved, also discussing the equality cases. Improving former results, E. Milman [1785] derives an *upper bound on the mean width* of origin-symmetric convex bodies in isotropic position, depending on the isotropic constant and a universal constant. He uses L_p zonoids, a generalization of L_p-centroid bodies.

We continue with *further generalizations of Orlicz centroid bodies*. In [2234] the authors extend this concept for probability measures to the multivariate case and derive the *affine isoperimetric-type*

inequality for this generalization. This way they bring together the settings of (thus extended) *Orlicz centroid bodies* and so-called *mean zonoids* (a setting going back to [2531]).

The extension of the notion of Orlicz centroid bodies to the Orlicz–Lorentz concept on star bodies is obtained by Nguyen in [1892]. The author proves the *Orlicz–Lorentz Busemann–Petty centroid inequality*, showing that the volume ratio for the Orlicz–Lorentz centroid body and the original body is minimal iff this body is a centered ellipsoid; see also [608] of Chen and Yang, where additionally a related result on *shadow systems* (generated with the help of Orlicz–Lorentz centroid bodies) is proved. An alternative proof of Nguyen's main result is obtained by Zhang and Fang [2534], again suitably using shadow systems. Further variants of the Orlicz–Lorentz Busemann–Petty centroid inequality are derived by Feng and Ma [859], together with a reverse of it in the planar case.

CENTROID BODIES AND VALUATIONS

Centroid bodies occur also in recent investigations on *valuations*; cf. also Section 11.3 and Subsection 11.7.2. Haberl [1119] studies a mapping of a convex polytope to its *polar L_p-centroid body* with values in the semigroup of centered star bodies and with respect to a certain L_p harmonic addition with an adjoined identity element as valuation, and similar results are also obtained for intersection bodies. In [1121] he can use the (asymmetric) centroid body of a convex body K and of $-K$ to characterize *continuous* SL(n)*-covariant Minkowski valuations* (with respect to Minkowski addition) on the set of convex bodies with the origin as interior point (the analogous result for contravariant Minkowski valuations uses projection bodies).

Li and Leng [1556] deals with *Orlicz extensions of valuations*, showing that the Orlicz operators coming from the projection body, centroid body, and difference body yield no Orlicz valuations.

APPLICATIONS OF AND FURTHER RESULTS ON CENTROID BODIES

We finish with *applications of and further results on (generalized) concepts of centroid bodies*.

Presenting a new proof of Klartag's result on the concentration hypothesis regarding the volume of a convex body within a thin Euclidean shell, Fleury et al. [881] prove an almost isometric reverse Hölder inquality for the *mean width of L_p-centroid bodies* and a related stability result. Paouris and Werner [1941] treat the problem of *approximating symmetric polytopes by the duals of their L_p-centroid bodies*. The rate of convergence is shown to be independent of the geometry of the polytope; this and further results have applications in Banach spaces (approximation of polytopal norms by uniformly convex ones). It is noted that with the same methods the extension to the non-symmetric case is possible.

Haberl and Schuster [1126] prove *sharp L_p affine isoperimetric inequalities for all L_p projection bodies and all L_p centroid bodies*. They show that these new inequalities for the asymmetric L_p projection bodies and asymmetric L_p centroid bodies are the strongest inequalities within these families of inequalities. The results on the L_p-Busemann–Petty centroid inequality are applied to obtain an L_p-analogue of the Blaschke–Santaló inequality. Also regarding generalizations of the Blaschke–Santaló inequality, we cite the paper [1128] of Haddad, Jiménez and Montenegro. In this paper the authors give, after a survey of extensions of this inequality, (asymmetric) functional versions of the *L_p Blaschke–Santaló inequality for L_p-centroid bodies*. These are sharp, and in one case also the symmetric version is presented.

Ma [1652] studies the extrema of affine-invariant functionals treating the *volume of the Orlicz centroid body* and the reverse Orlicz Blaschke–Santaló inequality. He establishes such an inequality in the planar case, confirming also the extremality of triangles and, in the centrally symmetric case, of parallelograms. Also a nice characterization of ellipsoids in this framework is obtained.

Guerrero-Zarazua and Jerónimo-Castro [1098] derive geometric results on *planar floating and centroid bodies*. In the latter case, they give an isoperimetric-type inequality for the perimeter of the centroid body of a convex body in the plane; equality holds iff this body is a Euclidean disc.

Ivaki [1284] provides local and partial answers to so-called *projection centroid conjectures*, combining and/or iterating ith projection operators and centroid operators, and yielding with corresponding homotheties to the original convex body K necessary or exact conditions for K being a ball or an ellipsoid. Bernig [298] introduces a *new definition of volume in normed vector spaces* which is based on a version of the centroid body and uses the associated affine invariant of that volume; it has strong convexity properties, and also the projection body plays a role there.

The role of L_p-centroid bodies in the *geometry of isotropic convex bodies* is described above, under book parts and surveys (see Chapters 5 and 7 of [496] and Chapter 10 of [133]).

Giannopoulos et al. [964] study *geometric properties of the L_p-centroid bodies* of an isotropic log-concave measure, like the inradius random projections of them, yielding then estimates for related covering numbers.

Yang [2489] discusses *connections between convex geometry and information science*. More precisely, it is shown how well-known inequalities from information science (like, e.g., the moment-entropy inequality) are related to inequalities for volumes of centroid bodies.

INTERSECTION BODIES: DEFINITION AND SOME HISTORY

As we see in Section 10.3, the radial function value of the *intersection body IK of a star body K* in direction u is the $(n-1)$-volume of the intersection of this star body with an $(n-1)$-subspace normal to u. Equivalently, the radial function of IK is the Funk transform of the radial function of the original body K. Koldobsky [1441] shows that a centrally symmetric star body is an intersection body iff the function $1/\|x\|$ is a positive-definite distribution, where $\|x\|$ is the homogeneous function of degree 1 being 1 on the boundary of the body. The notion of intersection body, basically important in convexity and in the theory of star-shaped sets, goes back to Lutwak [1630], who proved an analogue (namely for sections instead of projections) of *Shephard's projection problem*, which was independently solved by Schneider and Petty. However, there are older contributions related to intersection bodies. For instance, we cite here Busemann [545] who proved that the intersection body of a convex body centered at the origin is convex.

INTERSECTION BODIES IN BOOK PARTS AND SURVEYS

The Busemann definition of surface area in Minkowski spaces (with a centered convex body as unit ball B) yields as solution of the isoperimetric problem a homothetical copy of the polar of the intersection body IB of B (see, e.g., Busemann's work [546] and Chapter 5 of Thompson's monograph [2361], the latter with many nice figures). The knowledge on and applications of intersection bodies until 2005 are perfectly presented in Chapter 8 of Gardner's monograph [930], besides which we also refer to Section 10.10 of Schneider's book [2167], the surveys [1633] (Section 12) and [1710] (Section 2.3). Due to the comprehensive representation in [930] we give below a survey on intersection bodies concentrating only on important results that appeared after (this second edition of) [930]. Here we also cite Moszyńska's monograph [1857]. In its Part III, also star bodies and intersection bodies are discussed as important topics in geometric tomography. E.g., *dual intrinsic volumes, the Shephard problem, the Busemann–Petty problem, and the notion of star duality* are discussed there. And the comprehensive survey [1157] of Hansen et al. contains in its Sections 14 and 17 related material, like the *dual Brunn–Minkowski inequality, the Orlicz-Brunn–Minkowski theory and its dual, intersection bodies, and valuations on star bodies*.

INTERSECTION BODIES: INEQUALITIES AND COMPARISON RESULTS

We start now to discuss *volume inqualities and further inequalities related to intersection bodies*. Based on the role that Blaschke addition and *radial Blaschke addition* play for the projection

body operator and the intersection body operator, respectively, Schuster [2206] generalizes Lutwak's Brunn–Minkowski theory for mixed projection operators resp. the dual Brunn–Minkowski theory treating the *intersection body analogue* and hence referring again to *star bodies* (see the basic survey [1157] by Hansen et al. on star-shaped sets). In the first case, he obtains results on mixed Blaschke–Minkowski homomorphisms (which are continuous, rotation intertwining and Blaschke–Minkowski additive maps) and proves that the volume of a Blaschke–Minkowski homomorphism satisfies a Brunn–Minkowski inequality. He derives volume inequalities for the polars of mixed Blaschke–Minkowski homomorphisms restricting to the case of even operators and presents *analogous results for intersection bodies*.

Rubin [2068] deals with volume comparison in the sense of *lower dimensional Busemann–Petty problems*, asking whether n-dimensional centrally symmetric convex bodies with smaller k-dimensional central sections have necessarily smaller volumes. For $k = 1$ this is obvious, and for $k > 3$ non-true conclusions exist. The problem is still open for $k = 2$ or $k = 3$ ($n > 4$), but if the body with smaller sections is a body of revolution, the answer is positive. Rubin solves the problem for the case that the body with smaller sections is invariant under rotations, preserving mutually orthogonal subspaces of dimensions l and $n - l$, respectively, with $k + l \leq n$, where the answer depends on l. For the proof he uses *spherical Radon transforms, generalized cosine transforms, and properties of intersection bodies* (see also [2067]).

Using the generalized notion of k-intersection bodies, also the paper [1448] of Koldobsky and Ma deals with lower dimensional variants of the Busemann–Petty problem proving a *stability version*. Strongly related is Koldobsky's paper [1445], where also stability results for generalized k-intersection bodies and extensions to *complex intersection bodies* are obtained (these extended notions are discussed below). Based on the concept of *star duality* introduced by Moszńska (and different to the common polar duality concept), Yuan and Zhu [2511] study *star duals of intersection bodies*. They prove analogues of well-known inequalities for star duals of intersection bodies, e.g. of the Busemann intersection inequality and the dual Brunn–Minkowski inequalities for the *radial Blaschke sum and the harmonic Blaschke sum*. We mention here also the paper [2509] by Yu and Wu, shortly discussed also in the part on centroid bodies above.

GENERALIZATIONS OF INTERSECTION BODIES

We continue with concepts that present *generalizations of the notion of intersection body*. Some results on different extensions of this notion are also discussed in other parts of these notes; the decision where to consider them depends on the leading aspects of the respective papers.

We start here with the setting of *mixed intersection bodies* going back to Lutwak's basic paper [1630]. For star bodies K and L, the mixed intersection bodies $I_i(K, L)$ and the ith intersection bodies $I_i(K)$ (when L is taken as ball) are introduced by Zhang in [2532] as the analogues, in the dual Brunn–Minkowski theory, of mixed projection bodies and ith projection bodies investigated by Lutwak in [1632] (where the considered bodies have to be convex). This dual theory is naturally based on notions like *radial addition, dual mixed volumes, and dual quermassintegrals* (see again [1630]), and it refers of course to star bodies.

Yuan and Leng [2514], Zhao and Leng [2548], and Zhao [2536] establish inequalities of Brunn–Minkowski type, dual to those of Lutwak on mixed projection bodies; e.g., they refer to dual quermassintegrals of mixed intersection bodies and present *dual analogues of Alexandrov–Fenchel inequalities and Brunn–Minkowski inequalities treating mixed intersection bodies*. Similarly, Lu, Weihomg, and Leng (see [1607] and [1605]) analogues of Busemann's intersection inequality and an inequality of Lutwak between the volumes of convex bodies and their associated mixed projection integrals are obtained. For *polars of mixed intersection bodies*, Ma [1647] gives a stability version of the Alexandrov–Fenchel inequality. Following the ideas from Koldobsky et al. [1450], the concept of mixed intersection bodies was also extended to the complex case. Namely, Wang et al. [2422] derive

inequalities for *dual quermassintegrals of the mixed complex intersection body* (when two star bodies are involved).

Haberl [1118] obtains various nice results on *generalizations of intersection bodies within the dual L_p Brunn–Minkowski theory*. The notion of non-symmetric L_p *intersection body* I_p^+K of star bodies K is introduced, and the author shows that if K is a non-symmetric L_p intersection body, then the analogue of the implication in the Busemann–Petty problem holds, i.e., then $I_p^+K \subset I_p^+L$ implies that K has smaller volume than L. This yields a sufficient condition for comparing volumes of non-symmetric bodies, whereas the original Busemann–Petty problem refers only to symmetric bodies. Furthermore, stability results involving L_p intersection bodies are proved, and also ellipsoid formulas are extended to L_p intersection bodies.

Berck [274] studies properties of L_p intersection bodies I_pK which coincide, if K is centered, with the polar p-centroid bodies considered by Gardner and Giannopoulos in [932]. It is shown that the L_p-intersection body of a centered convex body is convex. For proving this, the author extends the Brunn–Minkowski theorem and Brunn's theorem to pth moments. For further results on L_p intersection bodies in view of Brunn–Minkowski inequalities and Busemann–Petty-type problems, and on generalizations of L_p intersection bodies, we refer to Wang and Li (cf. [2423] and [2424]) and Shen et al. [2226].

To extend the notion of L_p-intersection body to that of L_p-*mixed intersection body*, the concept of L_p-dual mixed volume for a star body is introduced by Zhao [2535] as well as Ma [1648]; *Alexandrov–Fenchel type, Brunn–Minkowski type, and further inequalities for L_p-mixed intersection bodies* are confirmed there. Particularly by Ma [1648], some inequalities for dual L_p-mixed integrals and monotonicity properties of mixed L_p intersection bodies are proved. Zhang [2528] establishes *a Minkowski-type inequality and a dual Alexandrov–Fenchel inequality for L_p-mixed intersection bodies*, and using operations like L_p-radial addition and the log-Minkowski radial addition of star bodies, *three Brunn–Minkowski-type inequalities* (also for L_p-mixed intersection bodies) are obtained. *Busemann–Petty-type problems for L_p-mixed intersection bodies* are presented by Zhang, Wu, and Feng [2533]. A notion introduced Zhou and Wu [2554] is called *quasi-L_p intersection body*; it is also defined with the help of L_p-dual mixed volumes, and the authors discuss *Busemann–Petty-type problems for quasi-L_p intersection bodies*.

Continuing Koldobsky [1443], Kalton and Koldobsky [1344] study the extended notion of k-*intersection bodies*; namely, a star body K is the k-intersection body of L if for every $(n-k)$-dimensional subspace H the k-volume of $K \cap H^\perp$ equals the $(n-k)$-volume of $L \cap H$. They show the following: For $0 < q < 1$, to every symmetric convex and n-dimensional k-intersection body K there exists an n-dimensional subspace of $L_q([0,1])$ whose unit ball B satisfies $B \subset K \subset c \cdot B$, where the constant c depends only on k and q. Yaskina [2497] constructs a body all whose $(n-1)$-dimensional central cross-sections are 1-intersection bodies, but which itself is not an intersection body. Yaskin [2492] shows that there is an n-dimensional body all of whose m-dimensional sections are k-intersection bodies, but not all $(m+1)$-dimensional sections are of this type. He shows also that the family of j-intersection bodies is not contained in that of k-intersection bodies for $j > k$. Yaskin [2493] studies k-intersection bodies in view of Busemann's convexity result from [545]. He considers the cases $4 \le k \le n-1$, $k = 2$, and $k = 3$, and he verifies via corresponding counterexamples that k-intersection bodies of convex bodies centered at the origin are not necessarily convex for $k > 1$.

Also Koldobsky et al. [1449] treat generalizations of usual intersection bodies. The authors also deal with problems on the shape and the volume of a k-intersection body of a symmetric star body. If the k-intersection body exists and is convex, then its *Banach–Mazur distance to the n-dimensional Euclidean ball* is less than a constant depending only on k. Further recent papers strongly based on (generalized) k-intersection bodies and treating the slicing problem and related inequalities from high-dimensional convex geometry are Koldobsky [1446], Brazitikoz et al. [495], Giannopoulos and Koldobsky [957], Koldobsky et al. [1451], and Yu [2510].

Kim et al. [1380] extend an analogue of Busemann's theorem on the convexity of intersection bodies of centered bodies to a *generalized convexity notion*, the so-called p-convexity (p giving the power in a generalized triangle inequality, yielding with $p = 1$ the usual convexity notion): the intersection body of a centered p-convex body is *q-convex* for all q smaller than or equal to a suitable constant $q_0 = q_0(p, n)$ which is asymptotically best possible. The authors extend their machinery to more general results in measure spaces with log-concave and s-concave measures. In addition, high dimensional p-convex bodies K are constructed whose intersection bodies are much farther away from the Euclidean ball than K.

Rubin [2067] studies relations between geometric properties of intersection bodies in convex geometry and generalized *cosine transforms in harmonic analysis*. The author introduces the new concept of *λ-intersection body* (extending that of k-intersection body by letting k a real λ). This notion unifies some known types of geometric objects. Parallels between trace theorems in function theory, restrictions of the spherical Radon transforms and the generalized cosine transforms to lower dimensional subspaces, and sections of λ-intersection bodies are studied, and also explicit examples of λ-intersection bodies are given. Also Wu et al. [2472] treat λ-intersection bodies, namely for *volume comparisons in the spirit of the Busemann–Petty problem*.

Finally we mention here the paper [1450] of Koldobsky et al., in which *complex intersection bodies* are introduced (see also [1445]). The authors confirm that various properties of complex intersection bodies are similar to those of real intersection bodies. E.g., they prove that complex intersection bodies of symmetric complex convex bodies are also convex, an analogue of Busemann's theorem from [545].

INTERSECTION BODIES AND VALUATIONS

Now we want to discuss the relations between intersection bodies and the notion of *valuation* which is broadly discussed in Section 11.3 and Subsection 11.7.2 below.

The two papers [1612] of Ludwig and [1123] of Ludwig and Haberl are important for this topic. In the first one (discussed also in Note 8.3, p. 340, of Gardner's book [930]) it is shown that, with the operation of *radial addition* and for the set of n-dimensional star bodies having continuous radial functions, *the intersection body operator is a valuation*. Further on, an operator which maps the family of convex polytopes having the origin as interior point to the mentioned type of star bodies is a non-trivial covariant valuation iff the image of such a polytope is a homothet of the *intersection body of its polar*. This is continued in the second paper [1123], where $\mathrm{GL}(n)$-covariant L_p radial valuations (with $p > 0$) on n-dimensional convex polytopes, with $n > 2$, are classified, finally giving a characterization of L_p-intersection bodies analogous to that described above. A related survey is Ludwig [1613], discussing valuations invariant or covariant under $\mathrm{SL}(n)$ and presenting the characterizations known in this direction.

FURTHER RESULTS ON INTERSECTION BODIES

We continue with a bunch of *further results on intersection bodies*.

We start with *local characterizations*. Inspired by problems and results around local and equatorial characterizations of zonoids (to which the important result belongs that in odd dimensions there is no local equatorial characterization of zonoids), Nazarov et al. [1882] derive the following results concerning intersection bodies: For odd $n > 4$, there is no local equatorial characterization of intersection bodies among all the symmetric convex bodies, for even n there is a local equatorial characterization of the intersection bodies among all the centered convex bodies, and for $n > 4$ there is no local characterization of intersection bodies among all convex bodies.

Alfonseca [53] derives conditions for or against the fact that a centered convex body is an *intersection body of a star body* and studies (as generalization of the usual notion of intersection body) also star bodies whose radial function is the spherical Radon transform of an even (positive) measure. It is

shown that, for $n > 4$, a direct sum of convex bodies whose first has dimension at least one, and second at least four, cannot be an intersection body. For the case of *bodies of revolution*, in terms of differentiability properties and convexity conditions also necessary criteria for intersection bodies are derived. The author strongly uses Fourier analytic methods.

Alfonseca continues in [54], studying geometric conditions and certain bodies of revolution (in dimensions four and six) whose intersection bodies *are not polars of zonoids*. Also related to (intersection) bodies of revolution, Alfonseca and Kim [56] investigate the following question: Does the *iterated intersection body operation* actually improve convexity? It is proved that the intersection body of a symmetric convex body of revolution has the same asymptotic behavior near the equator as the Euclidean ball. This implies that in sufficiently high dimensions *the "second" intersection body of a symmetric convex body of revolution is very close to an ellipsoid in the Banach–Mazur distance*. And also for the class of star bodies of revolution, local convexity properties at the equator of corresponding intersection bodies are studied.

Koldobsky [1441] verifies that *the class of intersection bodies contains the class of polars of zonoids*, and Schneider and Wieacker [2183] show that *for most convex bodies the intersection body is not the polar of some zonoid*. Based on [1441], Koldobsky shows in [1442] that the unit balls of l_p^n, $2 < p \leq \infty$, are intersection bodies for $n = 4$, but are not intersection bodies for $n \geq 5$. Meyer and Reisner [1774] introduce a new convex body associated with a given convex body K and some interior point x of it; *it generalizes both the intersection body and the cross-section body of* K (for the latter see also below). They call it the *convex intersection body* of K. It is a subset of the intersection body of K with respect to x, with equality iff K is symmetric about x. The authors pose interesting problems, e.g., the question of whether for an isotropic convex body K the convex intersection body is equivalent to the Euclidean ball. The paper [2300] of Stephen is also dedicated to the notion introduced by Meyer and Reisner. It is shown there that the *convex intersection body map* is not injective; it satisfies the condition that there is a sequence of non-symmetric smooth convex bodies whose images approximate a ball in a variety of different natural metrics. And also Fradelizi et al. [895] study convex intersection bodies in the sense of Meyer and Reisner [1774]: It is shown that for every convex body K with centroid at the origin the intersection body IK and the convex intersection body $C(IK)$ satisfy $c \cdot IK \subset C(IK) (\subset IK)$ for some absolute constant $c > 0$, thus solving a problem positively which was posed in [1774].

Yaskin and Yaskina [2496] consider problems of the following type: Let K and L be convex bodies centered at the origin. If the intersections of every slab (thick section) of fixed width and symmetric about the origin with K resp. L have equal volumes, is then necessarily $K = L$? This is verified within the class of centered convex polytopes. A related problem with section hyperplanes at the same distance from the origin is also discussed.

We finish this part on intersection bodies by discussing the closely related notion of *cross-section body*.

THE CROSS-SECTION BODY

This concept (see [1710], where the term occurs for the first time, and Chapter 8 of Gardner's monograph [930]) is closely related to that of intersection bodies, but refers to convex bodies (instead of star-shaped sets). For K an n-dimensional convex body, the $(n-1)$-dimensional outer cross-section measure (or brightness) in direction u is the radial function value of its *cross-section body* CK in the same direction u. Equivalently, CK is identical with the union of all intersection bodies of K (each depending on the position of the origin). We start with two results which were still open problems in the first edition of Gardner's book [930].

The first, being a typical tomographic problem, is due to Klee [1416]: Is an n-dimensional convex body determined (up to translation and reflection in the origin) by its cross-section body, i.e., by the *union of all its intersection bodies*? Gardner et al. [940] answer this with a nice *mixture of geometric*

and analytic methods; namely, their main proof idea is based on the construction of two convex bodies, one being centrally symmetric and the other one not, which have the same cross-section body. And the other problem asks: Assume there is an intersection body of a convex body K coinciding with the cross-section body of K. Is K then centered at the origin? As shown by Makai et al. [1677] via a *generalized spherical Radon transformation* (see also [1676] for properties), the answer is positive. There also the following is proved: for a convex body K centered at the origin, the *kth quermassintegral* ($0 \le k \le n-2$) of the intersection of K and an arbitrary hyperplane attains its maximum if the hyperplane contains the origin. Inspired by this, Makai et al. [1675] prove an infinitesimal variant of the converse implication, namely for small C^2-perturbations of the unit ball if $k = 1$, and for small C^3-perturbations of the unit ball if $2 \le k \le n-2$.

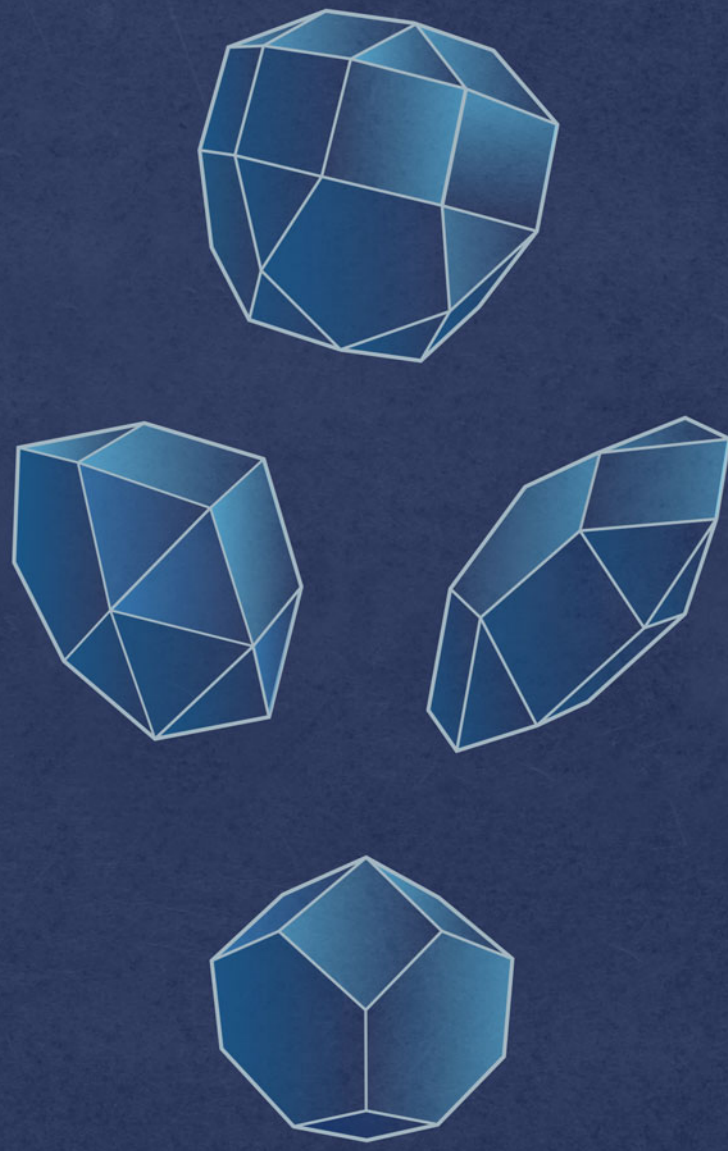
For related tomographic questions, also using *intrinsic volumes of sections of convex bodies* and giving counterexamples for non-symmetric convex bodies, the reader is referred to Ryabogin and Yaskin [2079]. And one should cite here the paper [2301] of Stephen and Yaskin. It contains a stability version of the result from [1677] on the coincidence of an intersection body with the corresponding cross-section body, quantifying it by the *radial distance* of both bodies. It follows that if for any direction the maximal hyperplane section is close to the central hyperplane section, then K and $-K$ are close. Gardner and Giannopoulos [932] define the notion of *p-cross-section body* C_pK of a convex body K (with $p > -1$) as the set whose radial function in any direction is the pth mean of the volumes of intersections of K with hyperplanes through points of K and orthogonal to that direction. Many interesting properties of C_pK are established, as well as relations of this concept to other well-known and important settings (for instance, that of the projection body or the intersection body of K). E.g., related inclusion properties (including also the projection body of K) are derived.

Wang and Zhou [2430] continue with the investigations of p-cross-section bodies. Using *L_p-dual mixed volumes*, they establish some inequalities for such bodies, like for instance certain volume inequalities (between K, C_pK, and the intersection body of K). The question of whether p-cross-section bodies are in general convex seems to be open, yet. Wang and Yan [2428] establish some new inequalities for dual quermassintegrals of p-cross-section bodies, extending corresponding inequalities for volumes given in [2430]. In addition, some Busemann–Petty-type problems for p-cross-section bodies are given there.

PLATE 11

Figure 29

Regular-faced polyhedra



Four of the 92 Johnson-Zalgaller polyhedra ► (see, e.g., 2.9.1.4)

Chapter 11
Further selected topics

"In mathematics, you don't understand things. You just get used to them."

— John von Neumann

This chapter is devoted to more topics and developments in convex geometry, some of them being very modern. The corresponding content selection depends also on the expertise and the particular interests of the authors, and this selection has two objectives: first, to show how the "common core" of the theory studied in this book is applied to obtain more profound results; and second, to bring recent research developments close to the reader.

We start with some extensions of the Brunn–Minkowski theory. In Section 11.1, we deal with *dual mixed volumes*, which are functionals defined for convex bodies (having the origin as an interior point) via their respective radial functions. These functionals are useful, e.g., to derive geometric inequalities. In Section 11.2, we introduce the so-called L_p *Brunn–Minkowski theory*. Roughly speaking, we develop a Brunn–Minkowski theory where Minkowski addition and scalar multiplication are replaced by more general operations. This leads to new definitions of mixed volumes, mixed surface area measures, quermassintegrals, and so on.

In Section 11.3, we introduce *valuations*, which can be seen as abstract ways of "measuring the size" of sets. The theory of valuations has been intensively studied recently, and there is a vast and deep literature about it. We restrict ourselves to a complete and thorough treatment of the classical *Hadwiger theorem*, which, roughly speaking, states that any valuation defined for convex bodies and having "good" geometric-flavored properties is a linear combination of quermassintegrals.

The last topic we deal with is the *Bourgain–Milman inequality*, which was stated previously in Section 5.4. This inequality gives an "asymptotic" lower bound for the Mahler product of $0_{\mathbb{R}^n}$-symmetric convex bodies. One can argue that this is a result from asymptotic geometric analysis, rather than from convex geometry, but we give here an almost self-contained proof. The only result that we use without a proof is the MM^*-estimate of Pisier. The proof presented here is taken from the book [133] of Artstein-Avidan et al., which is our main reference for asymptotic geometric analysis as a whole. For that sake, we need some inequalities involving covering numbers (namely, the *Sudakov inequalities*), and we state and prove them in Section 11.4.

Plate 11 shows four of the 92 regular-faced 3-polytopes that are usually called *Johnson–Zalgaller polyhedra*.

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V. Balestro et al., *Convexity from the Geometric Point of View*, Cornerstones,
https://doi.org/10.1007/978-3-031-50507-2_11

11.1 Dual mixed volumes

In this section, we study the theory of *dual mixed volumes*, which is a dual concept of mixed volumes introduced by Lutwak in [1625]. We start by recalling that $\mathcal{K}_0^n$ denotes the family of convex bodies of $\mathbb{R}^n$ which contain the origin as an interior point, and the *radial function* of a body $K \in \mathcal{K}_0^n$ is the function $\rho_K : \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ given as

$$\rho_K(x) := \sup\left\{\lambda \geq 0 : \lambda x \in K\right\}$$

for each $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. The radial function was introduced in Section 2.3 for bodies of $\mathcal{K}_0^n$ and extended in Remark 10.3.4 to a larger class of sets (namely, compact sets containing the origin). For the convenience of the reader, we describe now some important features that radial functions have.

Remember that, from Proposition 2.3.6, the equalities

$$\rho_K(x) = \frac{1}{h_{K^\circ}(x)} = \frac{1}{||x||_K}$$

hold for any $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$. Thus, from the properties of the support function and of polarity (see Section 2.3 again), we get straightforwardly the following facts about ρ_K:

(i) ρ_K is positive, continuous,

(ii) $\rho_K(\lambda x) = \lambda^{-1}\rho_K(x)$ for any $x \in \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$ and every $\lambda > 0$, and

(iii) $\rho_{\lambda K} = \lambda \rho_K$ for any $\lambda > 0$.

Moreover, if $K, L \in \mathcal{K}_0^n$ are such that $\rho_K = \rho_L$, then $h_{K^\circ} = h_{L^\circ}$. This gives that $K^\circ = L^\circ$, and hence $K = L$. In other words, a body from $\mathcal{K}_0^n$ is uniquely determined by its radial function (this actually holds for the larger class of *star bodies*; see Proposition 10.3.4).

By property **(ii)** above, we get that, for any $K \in \mathcal{K}_0^n$, the radial function ρ_K is completely determined by its restriction to $\mathbb{S}^{n-1}$. By item **(i)**, we have that $\rho_K|_{\mathbb{S}^{n-1}} \in C(\mathbb{S}^{n-1})$, where we remind the reader that $C(\mathbb{S}^{n-1})$ stands for the space of continuous functions $f : \mathbb{S}^{n-1} \to \mathbb{R}$. We recall also that this space is always endowed with the *maximum norm*:

$$||f||_\infty := \max\left\{|f(x)| : x \in \mathbb{S}^{n-1}\right\}.$$

A natural question that arises is whether convergence of bodies in the Hausdorff metric implies convergence of their respective radial functions in $C(\mathbb{S}^{n-1})$ (this is true for support functions, as an immediate consequence of item **(ii)** of Proposition 2.5.2). We answer this question positively.

Proposition 11.1.1. *Let $(K_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{K}_0^n$ converging in the Hausdorff metric to a body $K \in \mathcal{K}_0^n$. Then the sequence $(\rho_{K_m})_{m\in\mathbb{N}}$ converges to ρ_K in $C(\mathbb{S}^{n-1})$.*

Proof. Since K_m converges to K in the Hausdorff metric, it follows from Proposition 2.6.1 that K_m° converges to K° in the Hausdorff metric. Hence, using item **(ii)** of Proposition 2.5.2 we obtain that $h_{K_m^\circ}$ converges to h_{K° in $C(\mathbb{S}^{n-1})$. This means that for every $\varepsilon_1 > 0$, there exists $m_0 \in \mathbb{N}$ such that $m > m_0$ implies that

$$\left|h_{K_m^\circ}(u) - h_{K^\circ}(u)\right| < \varepsilon_1$$

for every $u \in \mathbb{S}^{n-1}$. This gives that

$$h_{K^\circ}(u) - \varepsilon_1 \le h_{K_m^\circ}(u) \le h_{K^\circ}(u) + \varepsilon_1$$

for every $u \in \mathbb{S}^{n-1}$, and in particular, we have

$$\min\left\{h_{K_m^\circ}(u) : u \in \mathbb{S}^{n-1}\right\} \ge \min\left\{h_{K^\circ}(u) : u \in \mathbb{S}^{n-1}\right\} - \varepsilon_1,$$

and of course we may assume that ε_1 is small enough in order that the right-hand side of the above is positive. For simplicity, let $c := \min\left\{h_{K^\circ}(u) : u \in \mathbb{S}^{n-1}\right\}$. For each $u \in \mathbb{S}^{n-1}$, we have

$$\left|\rho_K(u) - \rho_{K_m}(u)\right| = \frac{\left|h_{K_m^\circ}(u) - h_{K^\circ}(u)\right|}{h_{K^\circ}(u) h_{K_m^\circ}(u)} < \frac{\varepsilon_1}{(\min_{\mathbb{S}^{n-1}} h_{K^\circ})(\min_{\mathbb{S}^{n-1}} h_{K_m^\circ})} \le \frac{\varepsilon_1}{c(c-\varepsilon_1)}.$$

For small $\varepsilon > 0$ given, we let

$$\varepsilon_1 := \frac{c^2\varepsilon}{1 - c\varepsilon}.$$

Then the inequality above reads

$$\left|\rho_K(u) - \rho_{K_m}(u)\right| < \varepsilon,$$

and we already know that there exists a number $m_0 \in \mathbb{N}$ such that this holds for every $u \in \mathbb{S}^{n-1}$. This proves that ρ_{K_m} converges to ρ_K in $C(\mathbb{S}^{n-1})$.

□

The *dual mixed volume* of an n-tuple of bodies $K_1, \ldots, K_n \in \mathcal{K}_0^n$ is the number

$$\tilde{V}(K_1, \ldots, K_n) := \frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_{K_1}(u) \ldots \rho_{K_n}(u)\, d\mathcal{H}^{n-1}(u). \tag{11.1.1}$$

Of course, similarly to the case of mixed volumes, this yields a functional $\tilde{V} : \mathcal{K}_0^n \times \ldots \times \mathcal{K}_0^n \to [0, +\infty)$ defined for n-tuples of convex bodies (each of them having the origin as an interior point). As in the case of mixed volumes, we will use the terminology "dual mixed volume" to refer both to this functional and to the number attributed by a given n-tuple of bodies of $\mathcal{K}_0^n$ defined in (11.1.1). Next we establish some basic properties of dual mixed volumes.

Proposition 11.1.2. *The dual mixed volume has the following properties:*

(i) *As a functional defined over $\mathcal{K}_0^n \times \ldots \times \mathcal{K}_0^n$, the dual mixed volume $\tilde{V}$ is positive, continuous in the Hausdorff metric, and symmetric in all entries.*

(ii) *$\tilde{V}$ is positively homogeneous in each entry. In view of the symmetry, this is the same as saying that*

$$\tilde{V}(\lambda K_1, K_2, \ldots, K_n) = \lambda \tilde{V}(K_1, K_2, \ldots, K_n)$$

for any $\lambda > 0$ and every $K_1, \ldots, K_n \in \mathcal{K}_0^n$.

(iii) *For any $K, L, K_2, \ldots, K_n \in \mathcal{K}_0^n$ such that $K \subseteq L$, we have that*

$$\tilde{V}(K, K_2, \ldots, K_n) \le \tilde{V}(L, K_2, \ldots, K_n).$$

Again due to symmetry, this holds for all entries. In other words, $\tilde{V}$ is non-increasing for inclusion in each entry.

(iv) *$\tilde{V}$ is $\mathrm{O}(n)$-invariant in the following sense: for any $U \in \mathrm{O}(n)$ and every $K_1, \dots, K_n \in \mathcal{K}_0^n$, the equality*

$$\tilde{V}(UK_1, \dots, UK_n) = \tilde{V}(K_1, \dots, K_n)$$

holds.

(v) *For every $K \in \mathcal{K}_0^n$, we have*

$$\tilde{V}(K, \dots, K) = \mathrm{vol}_n(K).$$

Proof. The facts that $\tilde{V}$ is positive and symmetric are obvious. We show that $\tilde{V}$ is continuous in the Hausdorff metric in the first entry (by symmetry, this implies that the same holds for all entries). Let $K, L, K_2, \dots, K_n \in \mathcal{K}_0^n$. Then

$$\begin{aligned}
\left|\tilde{V}(K, K_2, \dots, K_n) - \tilde{V}(L, K_2, \dots, K_n)\right| \leq \\
\leq \frac{1}{n} \int_{\mathbb{S}^{n-1}} |\rho_K(u) - \rho_L(u)| \rho_{K_2}(u) \dots \rho_{K_n}(u)\, d\mathcal{H}^{n-1}(u) \leq \\
\leq \frac{\max\left\{\rho_{K_2}(u) \dots \rho_{K_n}(u) : u \in \mathbb{S}^{n-1}\right\}}{n} \int_{\mathbb{S}^{n-1}} |\rho_K(u) - \rho_L(u)|\, d\mathcal{H}^{n-1}(u) \leq \\
\leq \frac{\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) \max\left\{\rho_{K_2}(u) \dots \rho_{K_n}(u) : u \in \mathbb{S}^{n-1}\right\}}{n} ||\rho_K - \rho_L||_\infty,
\end{aligned}$$

and hence the continuity of $\tilde{V}$ in the first entry follows from Proposition 11.1.1.

Property **(ii)** follows immediately from the fact that $\rho_{\lambda K} = \lambda \rho_K$ for any $K \in \mathcal{K}_0^n$ and every $\lambda > 0$. For **(iii)**, simply notice that it is immediate from the definition of the radial function that $\rho_K \leq \rho_L$ whenever $K \subseteq L$. Since all radial functions involved are always positive, the inequality follows.

To prove **(iv)**, we use item **(i)** of Lemma 10.3.3 and the fact that $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ is invariant under orthogonal maps (see Proposition 4.5.1 and Remark 4.5.2). We calculate

$$\begin{aligned}
\tilde{V}(UK_1, \dots, UK_n) &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_{UK_1}(u) \dots \rho_{UK_n}(u)\, d\mathcal{H}^{n-1}(u) = \\
&= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_{K_1}(U^{-1}u) \dots \rho_{K_n}(U^{-1}u)\, d\mathcal{H}^{n-1}(u) = \\
&= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_{K_1}(u) \dots \rho_{K_n}(u)\, d\mathcal{H}^{n-1}(u) = \tilde{V}(K_1, \dots, K_n).
\end{aligned}$$

Finally, item **(v)** comes straightforwardly from formula (4.3.7), and this concludes the proof. □

We move forward to derive several geometric inequalities for dual mixed volumes. These are mostly counterparts to inequalities for (classical) mixed volumes (such as Minkowski inequalities, for example). First, we need an extension of Hölder's inequality. For the convenience of the reader, we state the classical Hölder inequality, and then prove the needed extension as a corollary. The proof is technical and can be skipped, since this is a result from classical analysis (rather than from convexity) and the details will not be used throughout the remainder of the text.

Theorem 11.1.1. (Classical Hölder inequality)**.** *Let X be a topological space endowed with a finite Borel measure μ, and let $f, g\colon X \to \mathbb{R}$ be continuous, positive and integrable functions. For any numbers $p, q \in (1, +\infty)$ with $1/p + 1/q = 1$, we have that*

$$\int_X fg\, d\mu \leq \left(\int_X f^p\, d\mu\right)^{\frac{1}{p}} \left(\int_X g^q\, d\mu\right)^{\frac{1}{q}},$$

and equality holds if and only if there exists a number $\lambda > 0$ such that $f(x)^p = \lambda g(x)^q$ almost everywhere in X.

Proof. See, e.g., [239].

□

Corollary 11.1.1. *Still assuming that X is a topological space endowed with a finite Borel measure μ, let $f_0, f_1, \ldots, f_m\colon X \to \mathbb{R}$ be continuous, positive, and integrable functions. If $p_1, \ldots, p_m \in (1, +\infty)$ are such that*

$$\frac{1}{p_1} + \ldots + \frac{1}{p_m} = 1,$$

then

$$\int_X f_0(x) f_1(x) \ldots f_m(x)\, d\mu(x) \leq \prod_{j=1}^{m} \left(\int_X f_0(x) f_j(x)^{p_j}\, d\mu(x)\right)^{\frac{1}{p_j}}.$$

Equality holds if and only if there exist numbers $\lambda_1, \ldots, \lambda_m > 0$ such that $\lambda_1 f_1(x)^{p_1} = \ldots = \lambda_m f_m(x)^{p_m}$ almost everywhere in X.

Proof. We proceed with induction on $m \in \mathbb{N}$. The case $m = 1$ is obvious. Now let $m \geq 2$, and assume that the result holds for $m - 1$. Let $f_0, f_1, \ldots, f_m\colon X \to \mathbb{R}$ and $p_1, \ldots, p_m$ be as in the statement, and define

$$\frac{1}{q_1} := \frac{1}{p_2} + \ldots + \frac{1}{p_m}.$$

We use Hölder's inequality for the functions $f_0^{1/p_1} f_1$ and $f_0^{1/q_1} f_2 \ldots f_m$, and the exponents p_1 and q_1, respectively, to write

$$\int_X f_0 f_1 \ldots f_m\, d\mu = \int_X \left(f_0^{\frac{1}{p_1}} f_1\right)\left(f_0^{\frac{1}{q_1}} f_2 \ldots f_m\right) d\mu \leq \left(\int_X f_0 f_1^{p_1}\, d\mu\right)^{\frac{1}{p_1}} \left(\int_X f_0 f_2^{q_1} \ldots f_m^{q_1}\, d\mu\right)^{\frac{1}{q_1}}.$$

Now we bound the last integral applying the induction hypothesis for the $m-1$ functions $f_2^{q_1}, \ldots, f_m^{q_1}$, and with exponents

$$\beta_j := \frac{p_j}{q_1},$$

for $j = 2, \ldots, m$. However, we only can do that if the reciprocals of these numbers sum up to 1. This is indeed true, because

$$\frac{1}{\beta_2} + \ldots + \frac{1}{\beta_m} = q_1 \left(\frac{1}{p_2} + \ldots + \frac{1}{p_m}\right) = 1.$$

We arrive at

$$\int_X f_0 f_1 \dots f_m \, d\mu \le \left(\int_X f_0 f_1^{p_1} \, d\mu \right)^{\frac{1}{p_1}} \left(\prod_{j=2}^{m} \left(\int_X f_0 f_j^{\beta_j q_1} \, d\mu \right)^{\frac{1}{\beta_j}} \right)^{\frac{1}{q_1}} =$$

$$= \left(\int_X f_0 f_1^{p_1} \, d\mu \right)^{\frac{1}{p_1}} \prod_{j=2}^{m} \left(\int_X f_0 f_j^{p_j} \, d\mu \right)^{\frac{1}{p_j}} = \prod_{j=1}^{m} \left(\int_X f_0 f_j^{p_j} \, d\mu \right)^{\frac{1}{p_j}},$$

and this settles the inequality.

Equality holds in the above if and only if equality holds in the case of $m-1$ functions and there exists a number $\lambda > 0$ such that

$$\left(f_0^{\frac{1}{p_1}} f_1 \right)^{p_1} = \lambda \left(f_0^{\frac{1}{q_1}} f_2 \dots f_m \right)^{q_1}. \tag{11.1.2}$$

In the case for $m-1$ functions, the induction hypothesis gives that equality holds if and only if there exist numbers $\lambda_2, \dots, \lambda_m > 0$ such that

$$\lambda_2 f_2^{q_1 \beta_2} = \dots = \lambda_m f_m^{q_1 \beta_m},$$

almost everywhere, that is,

$$\lambda_2 f_2^{p_2} = \dots = \lambda_m f_m^{p_m} \tag{11.1.3}$$

almost everywhere. In this case, equality (11.1.2) can be rewritten as

$$f_1^{p_1} = \lambda \left(\left(\frac{\lambda_m}{\lambda_2} \right)^{\frac{1}{p_2}} f_m^{\frac{p_m}{p_2}} \left(\frac{\lambda_m}{\lambda_3} \right)^{\frac{1}{p_3}} f_m^{\frac{p_m}{p_3}} \dots f_m \right)^{q_1} = \lambda \left(\frac{\lambda_m}{\lambda_2} \right)^{\frac{q_1}{p_2}} \dots \left(\frac{\lambda_m}{\lambda_{m-1}} \right)^{\frac{q_1}{p_{m-1}}} f_m^{\frac{q_1 p_m}{p_2} + \dots + \frac{q_1 p_m}{p_{m-1}} + q_1},$$

and because a simple calculation gives that $q_1 = p_1/(p_1 - 1)$, we have that

$$\frac{q_1 p_m}{p_2} + \dots + \frac{q_1 p_m}{p_{m-1}} + q_1 = q_1 p_m \left(\frac{1}{p_2} + \dots + \frac{1}{p_{m-1}} \right) + q_1 =$$

$$= \frac{p_1 p_m}{p_1 - 1} \left(1 - \frac{1}{p_1} - \frac{1}{p_m} \right) + \frac{p_1}{p_1 - 1} = \frac{p_1 p_m - p_m - p_1 + p_1}{p_1 - 1} = p_m.$$

Thus, writing

$$\lambda_1 := \left[\lambda \left(\frac{\lambda_m}{\lambda_2} \right)^{\frac{q_1}{p_2}} \dots \left(\frac{\lambda_m}{\lambda_{m-1}} \right)^{\frac{q_1}{p_{m-1}}} \right]^{-1},$$

we get that (11.1.2) and (11.1.3) hold if and only if there exist numbers $\lambda_1, \dots, \lambda_m > 0$ such that $\lambda_1 f_1^{p_1} = \dots = \lambda_m f_m^{p_m}$ almost everywhere. The proof is complete.

□

Remark 11.1.1. We are mainly interested in the case where $X = \mathbb{S}^{n-1}$ and $\mu = \mathcal{H}^{n-1}$. In this case, since $\mathcal{H}^{n-1}(A) > 0$ for any open non-empty set $A \subseteq \mathbb{S}^{n-1}$, if two continuous functions are equal almost everywhere, then they are equal everywhere. This will be used throughout this section without further comments.

From now on, as in the case of the classical mixed volumes, we will use square brackets to represent the number of times that a given body appears in the entries of the dual mixed volume. Formally, if $m_1, \dots, m_i$ are natural numbers such that $m_1 + \dots + m_i = n$ and $K_1, \dots, K_i \in \mathcal{K}_0^n$, then we write

$$\tilde{V}\big(K_1[m_1], K_2[m_2], \dots, K_i[m_i]\big) := \tilde{V}\big(\underbrace{K_1, \dots, K_1}_{m_1}, \underbrace{K_2, \dots, K_2}_{m_2}, \dots, \underbrace{K_i, \dots, K_i}_{m_i}\big).$$

Using the extension of Hölder's inequality obtained in Corollary 11.1.1, we can derive an inequality that can be seen as a reverse Alexandrov–Fenchel inequality (Theorem 8.2.1) for dual mixed volumes.

Theorem 11.1.2 (Dual Alexandrov–Fenchel inequality)**.** *For any $K_1, \dots, K_n \in \mathcal{K}_0^n$ and every natural number $1 \leq m \leq n$, we have the inequality*

$$\tilde{V}(K_1, \dots, K_n)^m \leq \prod_{j=1}^{m} \tilde{V}\big(K_j[m], K_{m+1}, \dots, K_n\big). \tag{11.1.4}$$

Equality holds if and only if $K_1, \dots, K_m$ represent all the same body up to positive scalar multiplication.

Proof. We use the inequality proved in Corollary 11.1.1 for the functions $f_1 = \rho_{K_1}$, $f_2 = \rho_{K_2}$,..., $f_m = \rho_{K_m}$ and $f_0 = \rho_{K_{m+1}} \dots \rho_{K_n}$, with exponents $p_j = m$ for all $j = 1, \dots, m$. Thus,

$$\begin{aligned}
\tilde{V}(K_1, \dots, K_n) &= \frac{1}{n} \int_{\mathbb{S}^{n-1}} \big(\rho_{K_{m+1}} \dots \rho_{K_n}\big) \rho_{K_1} \dots \rho_{K_m} \, d\mathcal{H}^{n-1} \leq \\
&\leq \frac{1}{n} \prod_{j=1}^{m} \left(\int_{\mathbb{S}^{n-1}} \rho_{K_{m+1}} \dots \rho_{K_n} (\rho_{K_j})^m \, d\mathcal{H}^{n-1} \right)^{\frac{1}{m}} = \\
= \prod_{j=1}^{m} \left(\frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_{K_{m+1}} \dots \rho_{K_n} (\rho_{K_j})^m \, d\mathcal{H}^{n-1} \right)^{\frac{1}{m}} &= \prod_{j=1}^{m} \tilde{V}\big(K_j[m], K_{m+1}, \dots, K_n\big),
\end{aligned}$$

and this proves the inequality. By the equality case in Corollary 11.1.1, we get that equality in (11.1.4) holds if and only if there exist numbers $\lambda_1, \dots, \lambda_m > 0$ such that

$$\lambda_1 (\rho_{K_1})^m = \lambda_2 (\rho_{K_2})^m = \dots = \lambda_m (\rho_{K_m})^m,$$

and this can be rewritten as

$$\rho_{\alpha_1 K_1} = \dots = \rho_{\alpha_m K_m},$$

with $\alpha_j = (\lambda_j)^{\frac{1}{m}}$ for each $j = 1, \dots, m$. It follows that equality holds in (11.1.4) if and only if there exist numbers $\alpha_1, \dots, \alpha_m > 0$ such that $\alpha_1 K_1 = \alpha_2 K_2 = \dots = \alpha_m K_m$.

□

As we know, particular cases of the classical Alexandrov–Fenchel inequality imply several important inequalities. For example, the 1st and 2nd Minkowski inequalities (see Section 6.3) can be obtained as immediate consequences of the Alexandrov–Fenchel inequality (but without the equality case), despite we have proved them independently. In the same way, particular cases of the dual Alexandrov–Fenchel inequality (11.1.4) lead to corresponding reverse inequalities for dual mixed volumes, as we discuss now. First, applying (11.1.4) for $m = n$, we get the inequality

$$\tilde{V}(K_1, \dots, K_n)^n \leq \mathrm{vol}_n(K_1) \mathrm{vol}_n(K_2) \dots \mathrm{vol}_n(K_n), \tag{11.1.5}$$

for any convex bodies $K_1, \dots, K_n \in \mathcal{K}_0^n$. Equality holds if and only if $K_1, \dots, K_n$ are equal up to positive scalar multiplications.

The particular case of (11.1.5) where $K_1 = K$ and $K_2 = \dots = K_n = L$ for some $K, L \in \mathcal{K}_0^n$ leads to a reverse version of the 1st Minkowski inequality (Theorem 6.3.1) for dual mixed volumes:

$$\tilde{V}(K, L, \dots, L)^n \leq \mathrm{vol}_n(K)\mathrm{vol}_n(L)^{n-1}. \tag{11.1.6}$$

A reverse 2nd Minkowski inequality (Theorem 6.3.2) for dual mixed volumes can be derived from (11.1.4) by letting $m = 2$, $K_1 = K$ and $K_2 = \dots = K_n = L$:

$$\tilde{V}(K, L, \dots, L)^2 \leq \tilde{V}(K, K, L, \dots, L)\tilde{V}(L, \dots, L) = \tilde{V}(K, K, L, \dots, L)\mathrm{vol}_{n-1}(L). \tag{11.1.7}$$

In both cases, equality holds if and only if $K = \lambda L$ for some $\lambda > 0$.

We move forward to derive an inequality which relates the dual mixed volumes with the classical mixed volumes. For that sake, recall that for classical mixed volumes, as a consequence of the Alexandrov–Fenchel inequality, we have the reverse inequality

$$V(K_1, \dots, K_n)^n \geq \mathrm{vol}_n(K_1)\mathrm{vol}_n(K_2) \dots \mathrm{vol}_n(K_n) \tag{11.1.8}$$

for any $K_1, \dots, K_n \in CC^n$ (this inequality is given in Corollary 8.3.1). As a consequence, we have that the dual mixed volume of any given n-tuple of bodies of $\mathcal{K}_0^n$ is not greater than their mixed volume.

Proposition 11.1.3. *For any $K_1, \dots, K_n \in \mathcal{K}_0^n$, we have*

$$\tilde{V}(K_1, \dots, K_n) \leq V(K_1, \dots, K_n), \tag{11.1.9}$$

and equality holds if and only if $K_1, \dots, K_n$ are equal up to positive scalar multiplications.

Proof. The inequality follows immediately from (11.1.5) and (11.1.8), because

$$\tilde{V}(K_1, \dots, K_n)^n \leq \mathrm{vol}_n(K_1)\mathrm{vol}_n(K_2) \dots \mathrm{vol}_n(K_n) \leq V(K_1, \dots, K_n)^n.$$

If equality holds, then we have equality in (11.1.5), and hence all K_j's are positive dilations of each other. On the other hand, if all the K_j's are positive dilations of each other, then we already know that equality in (11.1.5) holds. To verify that we also have equality in (11.1.8), notice that since all K_j's are equal up to positive scalar multiplication, they are all positive dilates of the same body $K \in \mathcal{K}_0^n$, say. Let $\lambda_1, \dots, \lambda_n > 0$ be numbers such that $K_j = \lambda_j K$ for each $j = 1, \dots, n$. Thus,

$$\begin{aligned}\mathrm{vol}_n(K_1) \dots \mathrm{vol}_n(K_n) = (\lambda_1 \dots \lambda_n)^n \mathrm{vol}_n(K)^n = (\lambda_1 \dots \lambda_n)^n V(K, \dots, K)^n = \\ = V(\lambda_1 K, \dots, \lambda_n K)^n = V(K_1, \dots, K_n)^n,\end{aligned}$$

and this concludes the proof. □

The radial function ρ_K of a body $K \in \mathcal{K}_0^n$ is strictly positive in $\mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\}$, and hence the function $(\rho_K)^i$ is well-defined, continuous and positive in this domain (and, in particular, in $\mathbb{S}^{n-1}$) for any $i \in \mathbb{R}$. Therefore, Hölder's inequality proved in Corollary 11.1.1 can also be used for these functions. For that reason, it is convenient to extend the definition of dual mixed volumes by letting

$$\tilde{V}\big(K[n-i], L[i]\big) := \frac{1}{n} \int_{\mathbb{S}^{n-1}} \rho_K(u)^{n-i} \rho_L(u)^i \, d\mathcal{H}^{n-1}(u)$$

for every $K, L \in \mathcal{K}_0^n$ and any $i \in \mathbb{R}$. We will also use the notation

$$\tilde{V}_i(K,L) = \tilde{V}\big(K[n-i], L[i]\big).$$

We warn the reader to be very careful about the range of the indices considered in each of the next results because, despite of this extension, some of them only hold for natural indices (this is the case for inequalities involving classical mixed volumes, for example). For the general case, we first have the following inequality.

Proposition 11.1.4. *For any $K, L \in \mathcal{K}_0^n$ and every $i, j, k \in \mathbb{R}$ such that $i < j < k$, we have the inequality*

$$\tilde{V}_j(K,L)^{k-i} \leq \tilde{V}_i(K,L)^{k-j}\tilde{V}_k(K,L)^{j-i}, \tag{11.1.10}$$

with equality if and only if $K = \lambda L$ for some $\lambda > 0$.

Proof. We apply Corollary 11.1.1 to the functions $f_0 = \rho_K^{n-i}\rho_L^i$, $f_1 = \rho_K^{i-j}\rho_L^{j-i}$ and $f_2 = 1$, with scalars $p_1 = (k-i)/(j-i)$ and $p_2 = (k-i)/(k-j)$, to get

$$\begin{aligned} n\tilde{V}_j(K,L) &= \int_{\mathbb{S}^{n-1}} \rho_K^{n-j}\rho_L^j \, d\mathcal{H}^{n-1} = \int_{\mathbb{S}^{n-1}} (\rho_K^{n-i}\rho_L^i)(\rho_K^{i-j}\rho_L^{j-i}) \, d\mathcal{H}^{n-1} \leq \\ &\leq \left(\int_{\mathbb{S}^{n-1}} \rho_K^{n-i}\rho_L^i(\rho_K^{i-j}\rho_L^{j-i})^{\frac{k-i}{j-i}} \, d\mathcal{H}^{n-1}\right)^{\frac{j-i}{k-i}} \left(\int_{\mathbb{S}^{n-1}} \rho_K^{n-i}\rho_L^i \, d\mathcal{H}^{n-1}\right)^{\frac{k-j}{k-i}} = \\ &= \left(\int_{\mathbb{S}^{n-1}} \rho_K^{n-k}\rho_L^k \, d\mathcal{H}^{n-1}\right)^{\frac{j-i}{k-i}} \big(n\tilde{V}_i(K,L)\big)^{\frac{k-j}{k-i}} = \big(n\tilde{V}_k(K,L)\big)^{\frac{j-i}{k-i}} \big(n\tilde{V}_i(K,L)\big)^{\frac{k-j}{k-i}} = \\ &= n\big(\tilde{V}_k(K,L)\big)^{\frac{j-i}{k-i}} \big(\tilde{V}_i(K,L)\big)^{\frac{k-j}{k-i}}. \end{aligned}$$

Dividing both sides by n and taking the $(k-i)$-th power we get the desired inequality. The equality case follows immediately from the equality case of Corollary 11.1.1.

□

Continuing, we may follow the case of classical mixed volumes to define the *dual quermassintegrals* of bodies from $\mathcal{K}_0^n$. For $i \in \mathbb{R}$, the *i-th quermassintegral* of a body $K \in \mathcal{K}_0^n$ is the number

$$\tilde{W}_i(K) := \tilde{V}\big(K[n-i], B^n[i]\big) \ \ \big(= \tilde{V}_i(K, B^n)\big).$$

Of course, each $\tilde{W}_i$ can be seen as a functional defined over $\mathcal{K}_0^n$. As in the case of the classical mixed volumes, we have that $\tilde{W}_0(K) = \mathrm{vol}_n(K)$ and $\tilde{W}_n(K) = \kappa_n$ for every $K \in \mathcal{K}_0^n$, where we remind the reader that κ_n denotes the volume of the Euclidean unit ball of $\mathbb{R}^n$. This follows immediately from item **(v)** of Proposition 11.1.2. Also, letting $L = B^n$, $i = 0$, $j = i$ and $k = n$ in inequality (11.1.10), it follows that

$$\tilde{W}_i(K) \leq \mathrm{vol}_n(K)^{\frac{n-i}{n}} \kappa_n^{\frac{i}{n}} \tag{11.1.11}$$

for every $K \in \mathcal{K}_0^n$ and any $i \in \mathbb{R}$ such that $0 < i < n$. From the characterization of the equality case of (11.1.10), we get that equality in the above holds if and only if K is a multiple of B^n (that is, if K is a ball centered at the origin).

An interesting application of the theory of dual mixed volumes is an inequality to bound from below the classical quermassintegrals of polar bodies. This is derived next.

Theorem 11.1.3. *For any natural number $0 \leq i < n$ and $K \in \mathcal{K}_0^n$, we have the inequality*

$$W_i(K^\circ) \geq \kappa_n^{n-i+1} W_{n-1}(K)^{i-n}. \tag{11.1.12}$$

Equality holds if and only if K is a ball centered at the origin.

Remark 11.1.2. In view of the expression of the mean width in terms of the $(n-1)$-th quermassintegral given in (6.4.2), we may rewrite (11.1.12) as

$$W_i(K^\circ) \geq 2^{n-i}\kappa_n w(K)^{i-n},$$

where we recall the reader that $w(K)$ denotes the mean width of K. This inequality can be interpreted as follows: among all bodies of $\mathcal{K}_0^n$ having the same prescribed mean width, the least value for the i-th quermassintegral ($0 \leq i < n$) of the polar body is attained for (and only for) the ball. Notice also that in the case $i = 0$ we just re-obtain inequality (5.5.3).

Proof. First consider inequality (11.1.10) letting K° in place of K, B^n in place of L, $i = i$, $j = n$ and $k = n+1$. We get

$$\tilde{W}_n(K^\circ)^{n-i+1} \leq \tilde{W}_i(K^\circ)\tilde{W}_{n+1}(K^\circ)^{n-i}.$$

Notice that

$$\tilde{W}_{n+1}(K^\circ) = \tilde{V}\big(K^\circ[-1], B^n[n+1]\big) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} \rho_{K^\circ}(u)^{-1}\rho_{B^n}(u)^{n+1}\, d\mathcal{H}^{n-1}(u) =$$
$$= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K(u)\, d\mathcal{H}^{n-1}(u) = \frac{\kappa_n}{2}\int_{\mathbb{S}^{n-1}} 2h_K(u)\, d\sigma(u) = \frac{\kappa_n}{2}w(K),$$

where we recall the reader that σ is the standard spherical probability measure, which is just $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ normalized, and we also remember that $\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = n\kappa_n$. Replacing this equality in the inequality above, and in view of the equality $\tilde{W}_n(K^\circ) = \kappa_n$, we arrive at

$$\kappa_n^{n-i+1} \leq \tilde{W}_i(K^\circ)\left(\frac{\kappa_n}{2}w(K)\right)^{n-i}.$$

Since i is a natural number such that $0 \leq i < n$, we may use inequality (11.1.9) for $K_1 = \ldots = K_{n-i} = K^\circ$ and $K_{n-i+1} = \ldots = K_n = B^n$ to get that $\tilde{W}_i(K^\circ) \leq W_i(K^\circ)$. Hence

$$\kappa_n^{n-i+1} \leq W_i(K^\circ)\left(\frac{\kappa_n}{2}w(K)\right)^{n-i},$$

and reorganizing the terms we get the desired. Equality holds if and only if we have equality in (11.1.10) for the bodies K° and B^n and in (11.1.9) for $K_1 = \ldots = K_{n-i} = K^\circ$ and $K_{n-i+1} = \ldots = K_n = B^n$. Equality in (11.1.10) holds for K° and B^n if and only if K° is a ball centered at the origin, which is equivalent to K being a ball centered at the origin. In this case, it is obvious that we also have equality in (11.1.9), since all the K_j's are dilations of each other.

□

As a corollary, we have a lower bound for the product of the mean width of a body with the mean width of its polar.

Corollary 11.1.2. *For any $K \in \mathcal{K}^n$ the inequality*

$$w(K)w(K^\circ) \geq 4 \tag{11.1.13}$$

holds, with equality if and only if K is a ball centered at the origin.

Proof. Letting $i = n-1$ in the inequality (11.1.12) and observing that

$$W_{n-1}(K^\circ) = \frac{\kappa_n}{2} w(K^\circ),$$

we have the desired.

□

Remark 11.1.3. In Theorem 11.5.2, inequality (11.1.13) will be obtained as a direct consequence of Hölder's inequality.

11.2 The L_p Brunn–Minkowski theory

In Chapters 5, 6, and 7, we have introduced the "main core" of the so-called *Brunn–Minkowski theory*. The reader may notice that all this theory relies fundamentally on two concepts: the Minkowski addition of sets and the notion of volume (given by the Lebesgue and Hausdorff measures). As a rough overview, we may say that the so-called L_p *Brunn–Minkowski theory* (or *Brunn-Minkowski-Firey theory*) is an extension of the (classical) Brunn–Minkowski theory where the Minkowski addition is replaced by a more general addition concept.

For the convenience of the reader, we recall that CC^n is the family of non-empty compact convex subsets of $\mathbb{R}^n$. As it was mentioned in the beginning of Section 6.1, when endowed with the Minkowski addition this family becomes a commutative semigroup with cancelation law. Recall that $C(\mathbb{S}^{n-1})$ denotes the space of continuous functions $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$. Let $H\colon CC^n \to C(\mathbb{S}^{n-1})$ be the map which associates each $C \in CC^n$ to its support function h_C (restricted to $\mathbb{S}^{n-1}$, say). From Remark 2.5.3, we have that this map is an isometry when CC^n is endowed with the Hausdorff metric and $C(\mathbb{S}^{n-1})$ is endowed with the maximum norm. Even more, it is clear that $C(\mathbb{S}^{n-1})$ is an additive group (under the usual addition of functions), and from item **(a)** of Proposition 2.3.2, we get that H preserves addition (and positive scalar multiplication).

The support function of $C \in CC^n$ seen as a map of $\mathbb{R}^n$ is simply the usual radial extension of its restriction to $\mathbb{S}^{n-1}$ (for this reason, there is no essential difference in regarding support functions as defined over $\mathbb{S}^{n-1}$ or $\mathbb{R}^n$). By Theorem 2.3.1, any sublinear function $h\colon \mathbb{R}^n \to \mathbb{R}$ is the support function of a (unique) set of CC^n. Therefore, this approach yields an equivalent way of defining the Minkowski addition in CC^n: for any $C, D \in CC^n$, it is clear that the function $h\colon \mathbb{R}^n \to \mathbb{R}$ given as

$$h = h_C + h_D$$

is sublinear. Then, h is the support function of a non-empty compact and convex set, and this set can be defined as the Minkowski sum $C + D$. It is clear that this definition agrees with the usual one (just use Proposition 2.3.2 again).

The importance of this approach is given by the fact that it leads to the mentioned extension of the Minkowski addition which gives rise to the L_p Brunn–Minkowski theory, as we will see next. From now on, we restrict ourselves to the family

$$CC_0^n := \left\{C \in CC^n : 0_{\mathbb{R}^n} \in C\right\}.$$

It is immediate to check that $h_C \geq 0$ for any $C \in CC_0^n$. This fact will be extensively used without further comments. We also recall that $\mathcal{K}_0^n$ denotes the family of convex bodies having the origin as an interior point.

Lemma 11.2.1. *Let $K, L \in CC_0^n$ and $p \geq 1$. Then the function $M_p(K, L)\colon \mathbb{R}^n \to \mathbb{R}$ given as*

$$M_p(K,L)(x) := \left(h_K(x)^p + h_L(x)^p\right)^{1/p}$$

for each $x \in \mathbb{R}^n$ is sublinear.

Proof. The positive homogeneity is obvious. For the subadditivity, we use Minkowski's classical inequality and the sublinearity of h_K and h_L to write

$$\begin{aligned} &M_p(K,L)(x+y) = \\ &= \left(h_K(x+y)^p + h_L(x+y)^p\right)^{1/p} \leq \left(\left(h_K(x)+h_K(y)\right)^p + \left(h_L(x)+h_L(y)\right)^p\right)^{1/p} \leq \\ &\leq \left(h_K(x)^p + h_L(x)^p\right)^{1/p} + \left(h_K(y)^p + h_L(y)^p\right)^{1/p} = M_p(K,L)(x) + M_p(K,L)(y). \end{aligned}$$

The proof is complete. □

As a consequence of Theorem 2.3.1, it follows that for any $K, L \in CC_0^n$, there exists a unique non-empty compact and convex set whose support function is $M_p(K,L)$. This set is called the *p-sum* (or *Firey sum*) of K and L, and it is denoted by $K +_p L$. Of course, the case $p = 1$ is the usual Minkowski addition. It is also immediate to notice from this definition that the p-sum is commutative. Moreover, we have that if K (say) belongs to $\mathcal{K}_0^n$, then $K +_p L$ is a convex body having the origin as an interior point. We show this next.

Lemma 11.2.2. *Let $K \in \mathcal{K}_0^n$, $L \in CC_0^n$, and $p \geq 1$. The p-sum $K +_p L$ is a convex body having the origin as an interior point.*

Proof. We already know that $K +_p L$ is non-empty compact and convex. Hence, it suffices to show that $0_{\mathbb{R}^n} \in \operatorname{int}(K +_p L)$. Since $K \in \mathcal{K}_0^n$, we have that h_K is strictly positive in $\mathbb{S}^{n-1}$, and consequently it follows that $M_p(K,L)$ is also strictly positive in $\mathbb{S}^{n-1}$. From continuity and compactness, we obtain that

$$c := \min\left\{M_p(K,L)(u) : u \in \mathbb{S}^{n-1}\right\} > 0,$$

and since $M_p(K,L)(u) \geq c$ for every $u \in \mathbb{S}^{n-1}$, we get from item **(iv)** of Proposition 2.3.2 that

$$cB^n \subseteq K +_p L.$$

Thus, $K +_p L$ contains a ball centered at the origin, and the argument is complete. □

It is also useful to define a "coherent" (in a sense that will become clear later) scalar multplication operation. We introduce the *p-scalar multiplication*: for any $K \in CC_0^n$, $p \geq 1$, and $\lambda \geq 0$, we write

$$\lambda \cdot_p K := \lambda^{1/p} K.$$

It is clear that $\lambda \cdot_p K \in CC_0^n$. Throughout this section, *for the sake of simplicity we will denote the p-scalar multplication $\alpha \cdot_p K$ simply by $\alpha \cdot K$.*

For $p \geq 1$, $K, L \in CC_0^n$, and $\alpha, \beta \geq 0$, we write

$$\alpha \cdot K +_p \beta \cdot L$$

to represent the p-sum of the sets $\alpha \cdot K$ and $\beta \cdot L$. That is, $\alpha \cdot K +_p \beta \cdot L$ is the (unique) set of CC^n whose support function is $M_p(\alpha^{1/p}K, \beta^{1/p}L)(x)$. Such a set is a *p-linear combination* of K and L.

From now on, we will write $h_K(\cdot) := h(K, \cdot)$, just in order to avoid big expressions in the subindex. Observe that

$$h(\alpha \cdot K +_p \beta \cdot L, x) = M_p(\alpha^{1/p}K, \beta^{1/p}L)(x) = \left(h_{\alpha^{1/p}K}(x)^p + h_{\beta^{1/p}L}(x)^p\right)^{1/p} = \\ = \left(\alpha h_K(x)^p + \beta h_L(x)^p\right)^{1/p},$$

for any $x \in \mathbb{R}$. Elevating both sides of this equality to the p-th power, we may write

$$h(\alpha \cdot K +_p \beta \cdot L, x)^p = \alpha h_K(x)^p + \beta h_L(x)^p, \tag{11.2.1}$$

and this is why the p-scalar multiplication is the "reasonable" concept of scalar multiplication to be considered.

To develop the L_p Brunn–Minkowski theory, we follow the approach of [1634]. Returning to the classical case for a moment, if $K, L \in CC^n$ and $i \in \{0, \dots, n-1\}$, then the *i-th mixed quermassintegral of K and L* is the number defined as

$$W_i(K, L) := \frac{1}{n-i} \lim_{\varepsilon \to 0^+} \frac{W_i(K + \varepsilon L) - W_i(K)}{\varepsilon}, \tag{11.2.2}$$

where $W_i(\cdot)$ denotes the usual i-th quermassintegral, of course.

Lemma 11.2.3. *For any $i \in \{0, \dots, n-1\}$ and every $K, L \in CC^n$, we have*

$$W_i(K, L) = V\big(K[n-i-1], L, B^n[i]\big).$$

In particular, we always have that $W_i(K, K) = W_i(K)$.

Proof. Due to the linearity of mixed volumes, we may write

$$W_i(K + \varepsilon L) = V(K + \varepsilon L[n-i], B^n[i]) = \sum_{j=0}^{n-i} \binom{n-i}{j} V(K[n-i-j], \varepsilon L[j], B^n) = \\ = \sum_{j=0}^{n-i} \binom{n-i}{j} \varepsilon^j V(K[n-i-j], L[j], B^n[i]).$$

The term corresponding to $j = 0$ is $V(K[n-i], B^n[i]) = W_i(K)$. Thus,

$$W_i(K, L) = \frac{1}{n-i} \lim_{\varepsilon \to 0^+} \frac{W_i(K, L) - W_i(K)}{\varepsilon} = \\ = \frac{1}{n-i} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \sum_{j=1}^{n-i} \binom{n-i}{j} \varepsilon^j V(K[n-i-j], L[j], B^n[i]) = \\ = \frac{1}{n-i} \binom{n-i}{1} V(K[n-i-1], L, B^n[i]) = V(K[n-i-1], L, B^n[i]).$$

This is the equality we wanted to prove. □

Remark 11.2.1. From the definition of mixed surface area measures, we also have the integral representation

$$W_i(K,L)=\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\,dS(K[n-i-1],B^n[i],\cdot)=\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)\,dS_{n-i-1}(K,\cdot),$$

where $S_{n-i-1}(K,\cdot)$ is the mixed surface area measure

$$S_{n-i-1}(K,\cdot)=S(K[n-i-1],B^n[i],\cdot),$$

as it was defined in Section 7.2.

Further, as an immediate consequence of (8.3.1), we have the inequality

$$W_i(K,L)^{n-i}\geq W_i(K)^{n-i-1}W_i(L) \tag{11.2.3}$$

for any $K,L\in CC^n$ and every $0\leq i\leq n-2$. It is clear that we have equality in the case where K and L are homothetical. In the case of convex bodies, this characterizes completely the equality case, as we will see next.

Proposition 11.2.1. *If $K,L\in\mathcal{K}^n$ are such that*

$$W_i(K,L)^{n-i}=W_i(K)^{n-i-1}W_i(L)$$

for some $0\leq i\leq n-2$, then K and L are homothetical.

Proof. Let $m:=n-i$. Then $m\in\{2,\dots,n\}$, and the equality in the statement is the same as

$$V(K[m-1],L,B^n[n-m])^m=V(K[m],B^n[n-m])^{m-1}V(L[m],B^n[n-m]).$$

It follows immediately from Corollary 8.4.2 applied for $K_{m+1}=\dots=K_n=B^n$ that K and L are homothetical.

□

Remark 11.2.2. It is not really necessary that K and L have dimension n. It suffices to require that K and L have dimension greater than or equal to $n-i$ (in the case where equality holds for a given $0\leq i\leq n-2$, of course). We have stated and proved this simpler version because in this section we are mainly interested in convex bodies.

In view of the definition (11.2.2), the i-th mixed quermassintegral $W_i(K,L)$ can be seen as the first variation of W_i with respect to the usual Minkowski sum (and scalar multiplication). Now we extend this definition to consider the first variation of W_i with respect to p-sums and p-scalar multiplications. For any $i=0,\dots,n-1$, $p\geq 1$ and $K,L\in\mathcal{K}_0^n$, we define the *p-th mixed quermassintegral* of K and L as

$$W_{p,i}(K,L):=\frac{p}{n-i}\lim_{\varepsilon\to 0^+}\frac{W_i(K+_p\,\varepsilon\cdot L)-W_i(K)}{\varepsilon}, \tag{11.2.4}$$

where we remind the reader that $\varepsilon\cdot L$ stands for the p-th scalar multiplication of L by ε, that is, $\varepsilon\cdot L:=\varepsilon^{1/p}L$. The case for $p=1$ gives the usual mixed quermassintegrals defined previously. For $p>1$, it is, at first glance, even not completely clear whether this limit exists. Thus, in order to ensure formally the existence of the p-th mixed quermassintegrals, and to have an equivalent definition which is easier to deal with, we provide next an integral representation of (11.2.4). This extends the integral representation of the usual mixed quermassintegrals given in Remark 11.2.1.

Theorem 11.2.1. *Let $p\geq 1$, $i\in\{0,\dots,n-1\}$ and $K,L\in\mathcal{K}_0^n$. Then we have the equality*

$$W_{p,i}(K,L)=\frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)^p h_K(u)^{1-p}\,dS_{n-i-1}(K,u).$$

Proof. For simplicity, we write $K_\varepsilon := K +_p \varepsilon \cdot L$ for each $\varepsilon > 0$. Our first task is to prove that $K_\varepsilon \to K$ in the Hausdorff metric as $\varepsilon \to 0$. Since $K \in \mathcal{K}_0^n$, there exist numbers $c_0, c_1 \in \mathbb{R}$ such that

$$0 < c_0 < h_K(u) < c_1$$

for every $u \in \mathbb{S}^{n-1}$. Since the function $t \mapsto t^{1/p}$ is continuous in $(0, +\infty)$, we have that it is uniformly continuous in the compact interval $[c_0, c_1]$. Let $\gamma > 0$ be arbitrarily given. Then there exists a number $\delta > 0$ such that

$$|t^{1/p} - s^{1/p}| < \gamma$$

whenever $t, s \in [c_0, c_1]$ are such that $|t - s| \leq \delta$. Now let $\varepsilon > 0$ be small enough such that

$$\varepsilon \max\{h(L, u) : u \in \mathbb{S}^{n-1}\}^p < \delta.$$

It follows that

$$\begin{aligned} \left|h_{K_\varepsilon}(u) - h_K(u)\right| = \left|(h(K +_p \varepsilon \cdot L, u) - h(K, u)\right| = \left|\left(h(K, u)^p + \varepsilon h(L, u)^p\right)^{1/p} - h(K, u)\right| = \\ = \left(h(K, u)^p + \varepsilon h(L, u)^p\right)^{1/p} - h(K, u) < \left(h(K, u)^p + \delta\right)^{1/p} - h(K, u) = \\ = \left|(h(K, u)^p + \delta)^{1/p} - (h(K, u)^p)^{1/p}\right| < \gamma \end{aligned}$$

for every $u \in \mathbb{S}^{n-1}$. Therefore, we have that

$$\max\left\{\left|h_{K_\varepsilon}(u) - h_K(u)\right| : u \in \mathbb{S}^{n-1}\right\} < \gamma,$$

and this proves that $h_{K_\varepsilon} \to h_K$ in the maximum norm of $C(\mathbb{S}^{n-1})$ as $\varepsilon \to 0$. From Proposition 2.5.2, we get that $K_\varepsilon \to K$ in the Hausdorff metric, as we wanted to show.

For the sake of simplicity, write

$$g(\varepsilon) := \frac{W_i(K_\varepsilon) - W_i(K)}{\varepsilon}$$

for each $\varepsilon > 0$. Notice that, using a telescope sum and the definition of mixed surface area measures, this can be written as

$$\begin{aligned} g(\varepsilon) = \frac{V(K_\varepsilon[n-i], B^n[i]) - V(K[n-i], B^n[i])}{\varepsilon} = \\ = \sum_{j=0}^{n-i-1} \frac{V(K_\varepsilon[j+1], K[n-i-j-1], B^n[i]) - V(K_\varepsilon[j], K[n-i-j], B^n[i])}{\varepsilon} = \\ = \sum_{j=0}^{n-i-1} \frac{1}{n} \int_{\mathbb{S}^{n-1}} \frac{h_{K_\varepsilon}(u) - h_K(u)}{\varepsilon}\, dS(K_\varepsilon[j], K[n-i-j-1], B^n[i], u). \end{aligned} \tag{11.2.5}$$

For each $\varepsilon > 0$, write

$$f_\varepsilon(u) := \frac{h_{K_\varepsilon}(u) - h_K(u)}{\varepsilon} = \frac{(h_K(u)^p + \varepsilon h_L(u)^p)^{1/p} - h_K(u)}{\varepsilon}, \quad u \in \mathbb{S}^{n-1}.$$

If we fix any $u \in \mathbb{S}^{n-1}$, then we clearly have that

$$\lim_{\varepsilon\to 0^+} f_\varepsilon(u) = \lim_{\varepsilon\to 0^+} \frac{(h_K(u)^p + \varepsilon h_L(u)^p)^{1/p} - h_K(u)}{\varepsilon} =$$
$$= \frac{d}{d\varepsilon}\Big[\big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{1/p}\Big]\Big|_{\varepsilon=0} = \frac{1}{p} h_L(u)^p h_K(u)^{1-p}.$$

We also claim that the convergence is monotone for every $u \in \mathbb{S}^{n-1}$. Indeed, fixing any $u \in \mathbb{S}^{n-1}$, we may write

$$\frac{d}{d\varepsilon}(f_\varepsilon(u)) = \frac{d}{d\varepsilon}\left(\frac{h_{K_\varepsilon}(u) - h_K(u)}{\varepsilon}\right) = \frac{d}{d\varepsilon}\left(\frac{\big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}} - h_K(u)}{\varepsilon}\right) =$$
$$= \frac{\frac{1}{p}\big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}-1} h_L(u)^p \varepsilon - \big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}} + h_K(u)}{\varepsilon^2}.$$

The expression in the numerator equals 0 for $\varepsilon = 0$, and we show that it is non-increasing for $\varepsilon > 0$: in fact, estimating the derivative, we get

$$\frac{d}{d\varepsilon}\left(\frac{1}{p}\big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}-1} h_L(u)^p \varepsilon - \big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}} + h_K(u)\right) =$$
$$= \frac{1}{p}\left(\frac{1}{p} - 1\right)\big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}-2} h_L(u)^{2p}\varepsilon \le 0,$$

since $p \ge 1$. Consequently,

$$\frac{1}{p}\big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}-1} h_L(u)^p \varepsilon - \big(h_K(u)^p + \varepsilon h_L(u)^p\big)^{\frac{1}{p}} + h_K(u) \le 0$$

whenever $\varepsilon > 0$. Returning to the expression for the derivative of the function $\varepsilon \mapsto f_\varepsilon(u)$, we finally get that

$$\frac{d}{d\varepsilon}(f_\varepsilon(u)) \le 0$$

for $\varepsilon > 0$, and this yields that f_ε converges monotonically (pointwise) as $\varepsilon \to 0^+$.

Let $(\varepsilon_m)_{m\in\mathbb{N}}$ be a sequence of positive numbers such that $\varepsilon_m \to 0$ as $m \to \infty$. For simplicity, denote f_{ε_m} by f_m. From the above, we have that f_m is a sequence of (continuous) functions defined in $\mathbb{S}^{n-1}$ which converge pointwise monotonically to the (continuous) function

$$f(u) = \frac{1}{p} h_L(u)^p h_K(u)^{1-p}.$$

Since $\mathbb{S}^{n-1}$ is compact, we get from Dini's classical theorem (see, e.g., Rudin [2074]) that f_m converges to f in the maximum norm of $C(\mathbb{S}^{n-1})$.

We have to prove that

$$\lim_{\varepsilon\to 0^+} g(\varepsilon) = \frac{n-i}{pn} \int_{\mathbb{S}^{n-1}} h_L(u)^p h_K(u)^{1-p}\, dS_{n-i-1}(K,u),$$

and from standard analysis it is suffices to prove that $g(\varepsilon_m)$ converges to the number in the right-hand side for any sequence $(\varepsilon_m)_{m\in\mathbb{N}}$ of positive numbers such that $\varepsilon_m \to 0$ as $m \to \infty$. For such a sequence, we use again the notation $f_m := f_{\varepsilon_m}$, and we also write

$$\mu_m^j := S(K_{\varepsilon_m}[j], K[n-i-j-1], B^n[i], \cdot)$$

for each $j = 0, \dots, n-i-1$ and every $m \in \mathbb{N}$. Thus, in view of (11.2.5), we have

$$g(\varepsilon_m) = \sum_{j=0}^{n-i-1} \frac{1}{n} \int_{\mathbb{S}^{n-1}} f_m(u)\, d\mu_m^j(u).$$

Since $K_\varepsilon \to K$ in the Hausdorff metric, it follows from item **(v)** of Theorem 7.1.3 that, for each $j = 0, \dots, n-i-1$, the sequence $(\mu_m^j)_{m\in\mathbb{N}}$ converges weakly to the measure

$$S(K[n-i-1], B^n[i], \cdot) = S_{n-i-1}(K, \cdot) =: \mu$$

as $m \to \infty$. Therefore, applying Lemma 8.1.4, we arrive immediately at

$$\lim_{m\to\infty} g(\varepsilon_m) = \sum_{j=0}^{n-i-1} \frac{1}{n} \lim_{m\to\infty} \int_{\mathbb{S}^{n-1}} f_m\, d\mu_m^j = \sum_{j=0}^{n-i-1} \frac{1}{n} \int_{\mathbb{S}^{n-1}} f\, d\mu =$$
$$= \frac{n-i}{n} \int_{\mathbb{S}^{n-1}} f\, d\mu = \frac{n-i}{pn} \int_{\mathbb{S}^{n-1}} h_L(u)^p\, h_K(u)^{1-p}\, dS_{n-i-1}(K, u),$$

and the proof is complete.

□

Corollary 11.2.1. *For any $p \geq 1$, every $i \in \{0, \dots, n-1\}$ and each $K \in \mathcal{K}_0^n$, we have that*

$$W_{p,i}(K, K) = W_i(K).$$

Proof. We simply notice that

$$W_{p,i}(K, K) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_K(u)\, dS_{n-i-1}(K, u) = V\big(K[n-i], B^n[i]\big) = W_i(K),$$

where we used the definition of the measure $S_{n-i-1}(K, \cdot)$. The proof is complete.

□

This integral representation leads to a version of inequality (11.2.3) for the p-th mixed quermass-integrals.

Theorem 11.2.2. *Let $p > 1$ and $0 \leq i < n$. For any $K, L \in \mathcal{K}_0^n$, we have*

$$W_{p,i}(K, L)^{n-i} \geq W_i(K)^{n-i-p} W_i(L)^p. \tag{11.2.6}$$

Equality holds if and only if K and L are dilates.

Proof. For simplicity, we write $\mu := S_{n-i-1}(K, \cdot)$, and we let $q \geq 1$ be such that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

We use Hölder's inequality in Corollary 11.1.1 with $f_0 = h_K$, $f_1 = h_L/h_K$, $f_2 = 1$, $p_1 = p$ and $p_2 = q$. This yields

$$W_{p,i}(K,L) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L^p h_K^{1-p}\,d\mu = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_K\left(\frac{h_L}{h_K}\right)^p d\mu = \frac{1}{n}\left[\left(\int_{\mathbb{S}^{n-1}} h_K\left(\frac{h_L}{h_K}\right)^p d\mu\right)^{\frac{1}{p}}\right]^p =$$

$$= \frac{1}{n}\left[\left(\int_{\mathbb{S}^{n-1}} h_K\left(\frac{h_L}{h_K}\right)^p d\mu\right)^{\frac{1}{p}}\right]^p \left(\int_{\mathbb{S}^{n-1}} h_K\,d\mu\right)^{\frac{p}{q}} \left(\int_{\mathbb{S}^{n-1}} h_K\,d\mu\right)^{-\frac{p}{q}} =$$

$$= \frac{1}{n}\left[\left(\int_{\mathbb{S}^{n-1}} h_K\left(\frac{h_L}{h_K}\right)^p d\mu\right)^{\frac{1}{p}} \left(\int_{\mathbb{S}^{n-1}} h_K 1^q\,d\mu\right)^{\frac{1}{q}}\right]^p \big(nW_i(K)\big)^{-\frac{p}{q}} \geq$$

$$\geq \frac{1}{n}\left(\int_{\mathbb{S}^{n-1}} h_L\,d\mu\right)^p \big(nW_i(K)\big)^{-\frac{p}{q}} = \frac{1}{n}\big(nV(L,K[n-i-1],B^n[i])\big)^p \big(nW_i(K)\big)^{-\frac{p}{q}} =$$

$$= \frac{n^p}{n} W_i(K,L)^p n^{-\frac{p}{q}} W_i(K)^{-\frac{p}{q}} = n^{p-1-\frac{p}{q}} W_i(K,L)^p W_i(K)^{-\frac{p}{q}}.$$

Since it is clear that $-p/q = 1-p$, we arrive at

$$W_{p,i}(K,L) \geq W_i(K,L)^p W_i(K)^{1-p}. \tag{11.2.7}$$

Let us consider first the case $i < n-1$, such that we can use inequality (11.2.3). We use this inequality after elevating both sides of (11.2.7) to the $(n-i)$-th power, to obtain

$$W_{p,i}(K,L)^{n-i} \geq \big(W_i(K,L)^{n-i}\big)^p W_i(K)^{(1-p)(n-i)} \geq$$
$$\geq \big(W_i(K)^{n-i-1}W_i(L)\big)^p W_i(K)^{(1-p)(n-i)} = W_i(K)^{n-i-p}W_i(L)^p,$$

and this is the inequality that we wanted to prove.

Now assume that equality holds. Then we have equality in (11.2.3), and hence K and L are homothetical (see Proposition 11.2.1). Thus, we may write $L = x + \beta K$ for some $x \in \mathbb{R}^n$ and $\beta > 0$. We also have equality in the Hölder inequality that we used, and the equality case of this inequality is described in Corollary 11.1.1. We obtain that there exist numbers $\lambda_1, \lambda_2 > 0$ such that

$$\lambda_1\left(\frac{h_L}{h_K}\right)^p = \lambda_2 1^q,$$

$S_{n-i-1}(K,\cdot)$-almost everywhere in $\mathbb{S}^{n-1}$, and setting $\lambda := (\lambda_2/\lambda_1)^{1/p}$ we get

$$h_L = \lambda h_K$$

$S_{n-i-1}(K,\cdot)$-almost everywhere. Thus,

$$h_{x+\beta K}(u) = \langle x,u\rangle + h_{\beta K}(u) = h_{\lambda K}(u)$$

for $S_{n-i-1}(K,\cdot)$-almost all $u \in \mathbb{S}^{n-1}$. Integrating with respect to $S_{n-i-1}(K,\cdot)$, and using the linearity of mixed volumes, we obtain

$$\beta V(K[n-i],B^n) = V(x+\beta K,K[n-i-1],B^n[i]) =$$
$$= \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{x+\beta K}(u)\,dS_{n-i-1}(K,u) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{\lambda K}(u)\,dS_{n-i-1}(K,u) =$$
$$= V(\lambda K,K[n-i-1],B^n[i]) = \lambda V(K[n-i],B^n[i]),$$

from what it follows that $\beta = \lambda$. Hence, we arrive at

$$\langle x, u\rangle = 0$$

for $S_{n-i-1}(K,\cdot)$-almost all $u \in \mathbb{S}^{n-1}$. Therefore,

$$V(\mathrm{seg}[-x,x], K[n-i-1], B^n[i]) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_{\mathrm{seg}[-x,x]}(u)\, dS_{n-i-1}(K,u) = \\ = \frac{1}{n}\int_{\mathbb{S}^{n-1}} |\langle x, u\rangle|\, dS_{n-i-1}(K,u) = 0,$$

and since K is a convex body this is only possible if $x = 0_{\mathbb{R}^n}$. It follows that $L = \beta K$, and this shows that K and L are dilates. The proof in the case $i < n-1$ is complete.

Now we deal with the case where $i = n-1$. It is clear that $W_{n-1}(K,L) = V(L, B^n[n-1]) = W_{n-1}(L)$, and hence, returning to (11.2.7), we get

$$W_{p,n-1}(K,L) \geq W_{n-1}(K,L)^p W_{n-1}(K)^{1-p} = W_{n-1}(L)^p W_{n-1}(K)^{1-p}.$$

This is already the desired inequality. If equality holds, then we have the equality case in Hölder's inequality, and as before we get that $h_L = \lambda h_K$ μ-almost everywhere for some $\lambda > 0$. However, in this case, we have that $\mu = S_0(K,\cdot) = \mathcal{H}^{n-1}$, and it follows from Remark 11.1.1 that actually we have $h_L = \lambda h_K$ everywhere. Hence $L = \lambda K$, and the proof is complete (the equality is easy to prove if K and L are dilates).

□

From (11.2.1) and from the integral representation of the p-th mixed quermassintegrals given in Theorem 11.2.1, we have immediately that $W_{p,i}$ is linear in the second entry with respect to p-sums and p-scalar multiplications (we call it *p-linearity*). That is,

$$W_{p,i}(M, K +_p \varepsilon \cdot L) = W_{p,i}(M,K) + \varepsilon W_{p,i}(M,L) \tag{11.2.8}$$

for any $K, L, M \in \mathcal{K}_0^n$, every $p \geq 1$, each $\varepsilon > 0$ and all $i \in \{0,\dots,n-1\}$. As we will see next, this leads to a Brunn–Minkowski type inequality for p-sums.

Proposition 11.2.2. *Let $K, L \in \mathcal{K}_0^n$, $p > 1$ and $0 \leq i < n$. Then we have*

$$W_i(K +_p L)^{\frac{p}{n-i}} \geq W_i(K)^{\frac{p}{n-i}} + W_i(L)^{\frac{p}{n-i}},$$

and equality holds if and only if K and L are dilates.

Proof. Let $M \in \mathcal{K}_0^n$. We use the p-linearity given in (11.2.8) and the inequality (11.2.6) to get

$$W_{p,i}(M, K +_p L) = W_{p,i}(M,K) + W_{p,i}(M,L) \geq W_i(M)^{\frac{n-i-p}{n-i}} W_i(K)^{\frac{p}{n-i}} + W_i(M)^{\frac{n-i-p}{n-1}} W_i(L)^{\frac{p}{n-i}} = \\ = W_i(M)^{\frac{n-i-p}{n-i}} \left[W_i(K)^{\frac{p}{n-i}} + W_i(L)^{\frac{p}{n-i}}\right].$$

Once this holds for any $M \in \mathcal{K}_0^n$ (K, L, p and i are given as in the statement, of course), we may let $M = K +_p L$. Using Corollary 11.2.1, we arrive at

$$W_i(K +_p L) = W_{p,i}(K +_p L, K +_p L) \geq W_i(K +_p L)^{\frac{n-i-p}{n-i}} \left[W_i(K)^{\frac{p}{n-i}} + W_i(L)^{\frac{p}{n-i}}\right],$$

and dividing both sides by $W_i(K +_p L)^{\frac{n-i-p}{n-i}}$ we arrive at the desired inequality. The equality case leads to equality in (11.2.6) for the bodies $K +_p L$ and K, and for the bodies $K +_p L$ and L. Hence K and $K +_p L$ are dilates, and L and $K +_p L$ are dilates. It follows that K and L are dilates. The verification that equality holds if K and L are dilates is immediate, and hence we are done.

□

Yet another consequence of the integral representation given in Theorem 11.2.1 is that we can find L_p analogues for the measures $S_{n-i-1}(K,\cdot)$, where $K \in \mathcal{K}_0^n$. Indeed, given $p \geq 1$ and $i \in \{0,\dots,n-1\}$, we define

$$S_{p,i}(K,E) := \int_E h_K(u)^{1-p}\, dS_{n-i-1}(K,u)$$

for each $E \in \mathcal{B}(\mathbb{S}^{n-1})$. Since h_K^{1-p} is continuous and positive in $\mathbb{S}^{n-1}$, it follows immediately that $S_{p,i}(K,\cdot)$ is a non-zero finite Borel measure on $\mathbb{S}^{n-1}$. This measure is known as the *i-th L_p surface area measure* of K. Of course, we may see $S_{p,i}(K,\cdot)$ as the measure having density h_K^{1-p} with respect to the measure $S_{n-i-1}(K,\cdot)$, and hence (using classical measure theory), we may write

$$dS_{p,i}(K,\cdot) = h_K^{1-p}\, dS_{n-i-1}(K,\cdot).$$

Moreover, we clearly have that $S_{1,i}(K,\cdot) = S_{n-i-1}(K,\cdot)$, and the integral representation of the p-th mixed quermassintegrals given in Theorem 11.2.1 yields

$$W_{p,i}(K,L) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)^p\, dS_{p,i}(K,u).$$

This can be seen as an L_p generalization of the integral representation of $W_i(K,L)$ given in Remark 11.2.1 (which is the case $p=1$, of course).

Since $S_{1,0}(K,\cdot) = S_{n-1}(K,\cdot)$ for any $K \in \mathcal{K}_0^n$, the measures $S_{p,0}(K,\cdot)$ for $p > 1$ are the natural L_p generalizations of the surface area measures $S_{n-1}(K,\cdot)$. Hence, we may call $S_{p,0}(K,\cdot)$ simply the *L_p surface area measure* of K. Next we state and prove a determination result for convex bodies in terms of their i-th L_p surface area measures which can be seen as the L_p extension of the Alexandrov–Fenchel–Jessen theorem given in Corollary 8.4.3.

Theorem 11.2.3. **(a)** *Let $p > 1$ and $i \in \{0,\dots,n-1\}$, and let $K, L \in \mathcal{K}_0^n$. If $p \neq n-i$ and*

$$S_{p,i}(K,\cdot) = S_{p,i}(L,\cdot),$$

then $K = L$.

(b) *For $i \in \{0,\dots,n-2\}$ and $K, L \in \mathcal{K}_0^n$, if*

$$S_{n-i,i}(K,\cdot) \leq S_{n-i,i}(L,\cdot),$$

then K and L are dilates. Consequently, the equality $S_{n-i,i}(K,\cdot) = S_{n-i,i}(L,\cdot)$ holds.

Proof. We start by **(a)**. The equality $S_{p,i}(K,\cdot) = S_{p,i}(L,\cdot)$ gives

$$W_{p,i}(K,L) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)^p\, dS_{p,i}(K,u) = \frac{1}{n}\int_{\mathbb{S}^{n-1}} h_L(u)^p\, dS_{p,i}(L,u) = W_i(L),$$

and hence inequality (11.2.6) yields

$$W_i(L)^{n-i} \geq W_i(K)^{n-i-p} W_i(L)^p.$$

Since $p \neq n-i$, this leads immediately to

$$W_i(L) \geq W_i(K).$$

By symmetry, the reverse inequality also holds, and then we have that $W_i(L) = W_i(K)$. Since we already have that $W_{p,i}(K, L) = W_i(L)$, we obtain that (11.2.6) holds with equality. Hence, K and L are dilates. To prove that the scaling factor is 1, we write $L = \alpha K$ for some $\alpha > 0$. Then, for any $E \in \mathcal{B}(\mathbb{S}^{n-1})$, we have

$$S_{p,i}(L, E) = S_{p,i}(\alpha K, E) = \int_E h_{\alpha K}(u)^{1-p}\, dS_{n-i-1}(\alpha K, u) =$$
$$= \int_E \alpha^{1-p} h_K(u)^{1-p} \alpha^{n-i-1}\, dS_{n-i-1}(K, u) = \alpha^{n-i-p} \int_E h_K(u)^{1-p}\, dS_{n-i-1}(K, u) =$$
$$= \alpha^{n-i-p} S_{p,i}(K, u).$$

In other words, $S_{p,i}(K, \cdot)$ is homogeneous of degree $n - i - p$ with respect to scalar multiplication of K. Once $S_{p,i}(K, \cdot) = S_{p,i}(L, \cdot)$ and $n - i - p \neq 0$, we get from the above that $\alpha = 1$, and this gives that $K = L$.

Now we prove **(b)**. Notice carefully that the fact that $i < n-1$ yields $n-i > 1$, and hence we may use inequality (11.2.6) for $p = n - i$. Since h_L^{n-i} is positive in $\mathbb{S}^{n-1}$, the inequality $S_{n-i,i}(K, \cdot) \leq S_{n-i,i}(L, \cdot)$ leads to

$$W_{n-i,i}(K, L) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)^{n-i}\, dS_{n-i,i}(K, u) \leq \frac{1}{n} \int_{\mathbb{S}^{n-1}} h_L(u)^{n-i}\, dS_{n-i,i}(L, u) = W_i(L).$$

This, together with inequality (11.2.6) for $p = n - i$, gives

$$W_i(L)^{n-i} \geq W_{n-i,i}(K, L)^{n-i} \geq W_i(K)^0 W_i(L)^{n-i} = W_i(L)^{n-i},$$

and then we have again equality in (11.2.6). It follows that K and L are dilates. Therefore, we may write $L = \alpha K$ for some $\alpha > 0$. We already proved that $S_{p,i}(K, \cdot)$ is homogeneous of degree $n - i - p$ with respect to scalar multiplication of K, and hence $S_{n-i,i}(K, \cdot)$ is homogeneous of degree 0 on K. Then

$$S_{n-i,i}(L, \cdot) = S_{n-i,i}(\alpha K, \cdot) = \alpha^0 S_{n-i,i}(K, \cdot) = S_{n-i,i}(K, \cdot),$$

and this concludes the proof.

□

The *Minkowski problem*, which we investigated in Section 7.3, asks what properties a finite Borel measure μ on $\mathbb{S}^{n-1}$ must have in order that there exists a convex body $K \subseteq \mathbb{R}^n$ such that $\mu = S_{n-1}(K, \cdot)$. This question was answered in Theorem 7.3.1, known as the *Minkowski existence theorem*. We head now to study the L_p extension of the Minkowski problem, which is called the *L_p Minkowski problem*. Roughly speaking, we want to find conditions that guarantee that a given finite Borel measure on $\mathbb{S}^{n-1}$ is the L_p surface area measure of some body $K \in \mathcal{K}_0^n$. For the sake of simplicity of the exposition, we deal only with the case where the given measure is even. This particular case is less complicated than the general case, and it is sufficient for the objective of this introductory exposition. We prove the following result.

Theorem 11.2.4. *Let $p > 1$, and let μ be an even, finite and non-zero Borel measure on $\mathbb{S}^{n-1}$ which is not concentrated on a great subsphere. Then there exist a number $\lambda > 0$ and a convex body $K_0 \subseteq \mathbb{R}^n$ symmetric with respect to the origin such that*

$$\mu = \lambda S_{p,0}(K_0, \cdot).$$

If in addition we have that $p \neq n$, then there exists a unique convex body $\bar{K}_0 \subseteq \mathbb{R}^n$ symmetric with respect to the origin such that

$$S_{p,0}(\bar{K}_0, \cdot) = \mu.$$

Proof. Recall that we denote by $C^+(\mathbb{S}^{n-1})$ the space of continuous and positive functions $f\colon \mathbb{S}^{n-1} \to \mathbb{R}$, and remember also that $W(f)$ stands for the Wulff shape of a function $f \in C^+(\mathbb{S}^{n-1})$, that is,

$$W(f) := \bigcap_{u \in \mathbb{S}^{n-1}} \mathbf{h}^-(u, f(u)).$$

We denote by $C_e^+(\mathbb{S}^{n-1})$ the subspace of $C^+(\mathbb{S}^{n-1})$ consisting of the even functions. We know from Lemma 8.1.1 that the Wulff shape of a function $f \in C^+(\mathbb{S}^{n-1})$ is a convex body, and it is easy to see that if $f \in C_e^+(\mathbb{S}^{n-1})$, then the Wulff shape is symmetric with respect to the origin. Indeed, if $x \in W(f)$, then $\langle x, u\rangle \leq f(u)$ for every $u \in \mathbb{S}^{n-1}$, and hence

$$\langle -x, u\rangle = \langle x, -u\rangle \leq f(-u) = f(u)$$

for any $u \in \mathbb{S}^{n-1}$. It follows that $-x \in W(f)$. The volume of the Wulff shape of a function $f \in C^+(\mathbb{S}^{n-1})$ will be denoted by $V(f)$. Define the functional $\varphi\colon C^+(\mathbb{S}^{n-1}) \to \mathbb{R}$ by

$$\varphi(f) := V(f)^{-p/n} \int_{\mathbb{S}^{n-1}} f(u)^p \, d\mu(u).$$

We want to minimize φ over $C^+(\mathbb{S}^{n-1})$. For that sake, it is easier to consider first functions of $C_e^+(\mathbb{S}^{n-1})$. We have the following properties.

(i) φ is homogeneous of degree 0, in the sense that $\varphi(\alpha f) = \varphi(f)$ for any $\alpha > 0$ and every $f \in C_e^+(\mathbb{S}^{n-1})$. This is immediate by the definition.

(ii) If φ attains a minimum over the family

$$\mathcal{F}_0 := \big\{\text{support functions of convex bodies centered at } 0_{\mathbb{R}^n}\big\},$$

then this is also a minimum of φ over $C_e^+(\mathbb{S}^{n-1})$. Indeed, assume that $h_L \in \mathcal{F}_0$ is such that $\varphi(h_L) = \min_{\mathcal{F}_0} \varphi$. Let $f \in C_e^+(\mathbb{S}^{n-1})$, and write $K := W(f)$. We know from Lemma 8.1.2 that $h_K \leq f$, and from the comment below this lemma we also have that $W(h_K) = K$. Thus, $V(h_K) = V(f)$. Therefore,

$$\begin{aligned}\varphi(f) = V(f)^{-p/n} \int_{\mathbb{S}^{n-1}} f(u)^p \, d\mu(u) &= V(h_K)^{-p/n} \int_{\mathbb{S}^{n-1}} f(u)^p \, d\mu(u) \geq \\ &\geq V(h_K)^{-p/n} \int_{\mathbb{S}^{n-1}} h_K(u)^p \, d\mu(u) = \varphi(h_K) \geq \varphi(h_L).\end{aligned}$$

(iii) If φ attains a minimum over the subfamily

$$\mathcal{F}_1 := \big\{h_L \in \mathcal{F}_0 : \ \mathrm{vol}_n(L) = 1 \text{ and } \varphi(h_L) \leq \beta\big\},$$

where $\beta := \varphi(h_{B^n})$, then this is also a minimum of φ over $\mathcal{F}_0$. Notice that this subfamily is non-empty since it contains the multiple of B^n which has unit volume (by homogeneity, φ attains the value β in this body). To verify the claim, let $h_L \in \mathcal{F}_1$ be such that $\varphi(h_L) = \min_{\mathcal{F}_1} \varphi$. If $h_K \in \mathcal{F}_0$, then there exists a number $c > 0$ such that $\varphi(h_{cK}) = 1$. If $\varphi(h_K) < \varphi(h_L)$, then, in particular, we have $\varphi(h_{cK}) = \varphi(h_K) \leq \beta$, and hence $h_{cK} \in \mathcal{F}_1$. This contradicts the fact that h_L minimizes φ over $\mathcal{F}_1$, and hence $\varphi(h_K) \geq \varphi(h_L)$.

(iv) φ attains a minimum over $\mathcal{F}_1$. To prove that, we first observe that since φ is obviously positive, its infimum over $\mathcal{F}_1$ is a non-negative number. We may let $(K_j)_{j\in\mathbb{N}}$ be a sequence of convex bodies centered at the origin such that the corresponding support functions form a sequence $(h_{K_j})_{j\in\mathbb{N}}$ in $\mathcal{F}_1$ with

$$\lim_{j\to\infty} \varphi(h_{K_j}) = \inf\left\{\varphi(f) : f \in \mathcal{F}_1\right\}.$$

We will use the Blaschke selection theorem (Theorem 2.5.4) to prove that the sequence (K_j) has a converging subsequence. First, observe that the function

$$v \mapsto \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, d\mu(u)$$

defined for $v \in \mathbb{S}^{n-1}$ is clearly continuous. Moreover, since μ is not concentrated on a great subsphere, we get, adapting trivially the proof of Lemma 7.3.2, that this function is positive for every $v \in \mathbb{S}^{n-1}$. Thus, from the compactness of $\mathbb{S}^{n-1}$, it follows that there exists a number $\alpha > 0$ such that

$$\int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, d\mu(u) \geq \alpha \tag{11.2.9}$$

for any $v \in \mathbb{S}^{n-1}$. Now let $j \in \mathbb{N}$ and $x \in K_j$ be arbitrarily given. Since K_j is symmetric with respect to the origin, we have that $-x$ is also a point of K_j, and thus

$$h_{K_j}(u) \geq \max\left\{\langle x, u\rangle, \langle -x, u\rangle\right\} = |\langle x, u\rangle|$$

for every $u \in \mathbb{S}^{n-1}$. Writing $x = |x|v$ for some $v \in \mathbb{S}^{n-1}$, we may write the inequality above as $h_{K_j}(u) \geq |x||\langle u, v\rangle|$ and, together with the bound (11.2.9), this yields

$$\alpha|x|^p \leq |x|^p \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, d\mu(u) \leq \int_{\mathbb{S}^{n-1}} h_{K_j}(u)^p \, d\mu(u) = \varphi(h_{K_j}) \leq \beta.$$

Notice that we also used the equality $1 = \mathrm{vol}_n(K_j) = V(h_{K_j})$, which comes from the fact that $h_{K_j} \in \mathcal{F}_1$ (the last inequality also follows from this inclusion). As a consequence, we have the bound

$$|x| \leq \left(\frac{\beta}{\alpha}\right)^{1/p} =: \gamma$$

for every $x \in K_j$, and it follows that $K_j \subseteq \gamma B^n$ for any $j \in \mathbb{N}$. Observing carefully that γ does not depend on j, this gives that $(K_j)_{j\in\mathbb{N}}$ is a bounded sequence of convex bodies of $\mathbb{R}^n$. By the Blaschke selection theorem, up to a subsequence, we may assume that (K_j) converges in the Hausdorff metric to a set $K_0 \in CC^n$. It is clear that $\mathrm{vol}_n(K_0) = 1$, and hence K_0 is a convex body. Also, from Remark 10.1.1, we have that K_0 is centered at the origin. Once φ is clearly continuous in the maximum norm of $C_e^+(\mathbb{S}^{n-1})$, and since the convergence of support functions in this norm is equivalent to the convergence of the corresponding bodies in the Hausdorff metric, we obtain that

$$\varphi(h_{K_0}) = \lim_{j\to\infty} \varphi(h_{K_j}) = \inf\left\{\varphi(f) : f \in \mathcal{F}_1\right\} \leq \beta,$$

and hence $K_0 \in \mathcal{F}_1$ (more generally, this argument shows that $\mathcal{F}_1$ is closed in $C^+(\mathbb{S}^{n-1})$ with respect to the maximum norm). From the above, we also have that φ attains its infimum over $\mathcal{F}_1$ at h_{K_0}.

From **(iii)** and **(ii)**, in this order, we have that h_{K_0} minimizes φ over $C_e^+(\mathbb{S}^{n-1})$. We claim that, even better, h_{K_0} minimizes φ over $C^+(\mathbb{S}^{n-1})$. To verify that, let $f \in C^+(\mathbb{S}^{n-1})$ be arbitrarily given, and let $K = W(f)$. The argument in **(ii)** can be trivially extended to the case where f is not necessarily even, and by doing this, we obtain the inequality $\varphi(f) \geq \varphi(h_K)$. Now let $\Delta_p K$ be defined as

$$2 \cdot \Delta_p K := K +_p -K,$$

and notice carefully that here we are considering the p-scalar multiplication. Then we have

$$2h_{\Delta_p K}(u)^p = h_K(u)^p + h_{-K}(u)^p = h_K(u)^p + h_K(-u)^p,$$

and, in particular, $h_{\Delta_p K} \in C_e^+(\mathbb{S}^{n-1})$. From the case $i = 0$ of the L_p Brunn–Minkowski inequality given in Proposition 11.2.2, we get straightforwardly that $\mathrm{vol}_n(\Delta_p K) \geq \mathrm{vol}_n(K)$, and hence $V(h_{\Delta_p K}) \geq V(h_K)$. This, together with the fact that μ is even, leads to

$$\begin{aligned}
\varphi(f) \geq \varphi(h_K) = V(h_K)^{-p/n} \int_{\mathbb{S}^{n-1}} h_K(u)^p \, d\mu(u) \geq \\
\geq V(h_{\Delta_p K})^{-p/n} \frac{1}{2} \int_{\mathbb{S}^{n-1}} h_K(u)^p + h_K(-u)^p \, d\mu(u) = \\
= V(h_{\Delta_p K})^{-p/n} \int_{\mathbb{S}^{n-1}} h_{\Delta_p K}(u)^p \, d\mu(u) = \varphi(h_{\Delta_p K}) \geq \varphi(h_{K_0}),
\end{aligned}$$

where the last inequality follows from the fact that h_{K_0} minimizes φ over $C_e^+(\mathbb{S}^{n-1})$, of course.

Let $h \in C^+(\mathbb{S}^{n-1})$ be an arbitrarily given function. For any $t \in \mathbb{R}$, the function $(h_{K_0}^p + th)^{1/p}$ defined over $\mathbb{S}^{n-1}$ is continuous, and for t small enough, it is strictly positive (this follows from a simple compactness argument). Hence, we may fix $\varepsilon > 0$ small enough such that $(h_{K_0}^p + th)^{1/p} \in C^+(\mathbb{S}^{n-1})$ whenever $|t| \leq \varepsilon$, and then the function $\phi \colon [-\varepsilon, \varepsilon] \to \mathbb{R}$ given by

$$\begin{aligned}
\phi(t) := \varphi\big((h_{K_0}^p + th)^{1/p}\big) = \\
= V\big((h_{K_0}^p + th)^{1/p}\big)^{-p/n} \left[\int_{\mathbb{S}^{n-1}} h_{K_0}(u)^p \, d\mu(u) + t \int_{\mathbb{S}^{n-1}} h(u) \, d\mu(u) \right]
\end{aligned}$$

is well-defined. Because h_{K_0} minimizes φ over $C^+(\mathbb{S}^{n-1})$, we have that ϕ attains a minimum at $t = 0$. Now we use Proposition 8.1.2 for $E = \mathbb{S}^{n-1}$, $f = h_{K_0}$, and $G(t, \cdot) = \big(h_{K_0}^p + th\big)^{1/p} - h_{K_0}$. Following the notation as established in the mentioned proposition (for K, K_t and v), we have

$$K = W(\mathbb{S}^{n-1}, h_{K_0}) = K_0,$$

$$K_t = W(\mathbb{S}^{n-1}, f + G(t, \cdot)) = W\big((h_{K_0}^p + th)^{1/p}\big), \quad t \in [-\varepsilon, \varepsilon],$$

and

$$\mathrm{v}(t) = V\big((h_{K_0}^p + th)^{1/p}\big), \quad t \in [-\varepsilon, \varepsilon].$$

Moving forward, notice that for each $u \in \mathbb{S}^{n-1}$, one has that

$$\lim_{t\to 0}\frac{G(t,u)}{t}=\lim_{t\to 0}\frac{\left(h_{K_0}(u)^p+th(u)\right)^{1/p}-h_{K_0}(u)}{t}=$$
$$=\frac{d}{dt}\left[\left(h_{K_0}(u)^p+th(u)\right)^{1/p}-h_{K_0}(u)\right]\Big|_{t=0}=\frac{1}{p}h(u)h_{K_0}(u)^{1-p},$$

and it is not difficult to check that this convergence is uniform on $\mathbb{S}^{n-1}$ (we may use exactly the same argument as applied in the proof of Theorem 11.2.1 to verify that the convergence of the f_ε's is uniform). Thus, the function in the right-hand side plays the role of the function g of Proposition 8.1.2. Finally, the formula for $\mathrm{v}'(0)$ given in this proposition yields

$$\frac{d}{dt}V\left((h_{K_0}^p+th)^{1/p}\right)\Big|_{t=0}=\mathrm{v}'(0)=\int_{\mathbb{S}^{n-1}}\frac{1}{p}h(u)h_{K_0}(u)^{1-p}\,dS_{n-1}(K_0,u).$$

Using this expression, we may differentiate $\phi(t)$ at $t=0$. Recalling also that $V(h_{K_0})=1$, we obtain

$$\phi'(0)=$$
$$=-\frac{1}{n}\int_{\mathbb{S}^{n-1}}h(u)h_{K_0}(u)^{1-p}\,dS_{n-1}(K_0,u)\int_{\mathbb{S}^{n-1}}h_{K_0}(u)^p\,d\mu(u)+\int_{\mathbb{S}^{n-1}}h(u)\,d\mu(u).$$

Since ϕ is minimized for $t=0$, the expression above equals 0. Then, letting

$$\lambda:=\frac{1}{n}\int_{\mathbb{S}^{n-1}}h_{K_0}(u)^p\,d\mu(u),$$

we clearly have that $\lambda>0$, and we may write

$$\int_{\mathbb{S}^{n-1}}h(u)\,d\mu(u)=\lambda\int_{\mathbb{S}^{n-1}}h(u)h_{K_0}(u)^{1-p}\,dS_{n-1}(K_0,u)=\lambda\int_{\mathbb{S}^{n-1}}h(u)\,dS_{p,0}(K_0,u).$$

This holds for any $h\in C^+(\mathbb{S}^{n-1})$. By Proposition A.2.4, this gives that $\mu=\lambda S_{p,0}(K_0,\cdot)$. If $p\neq n$, then letting $c>0$ be such that $c^{n-p}=\lambda$, we have immediately $\mu=\lambda S_{p,0}(K_0,\cdot)=S_{p,0}(cK_0,\cdot)$. Hence, we simply let $\bar{K}_0=cK_0$. The uniqueness follows directly from item **(a)** of Theorem 11.2.3.

□

Remark 11.2.3. Notice that this proof also works for the case $p=1$, since, in this case, we may use the classical Brunn–Minkowski inequality (instead of Proposition 11.2.2) to obtain the bound $\mathrm{vol}_n(\Delta_p K)\geq\mathrm{vol}_n(K)$. However, this particular case was studied in greater generality in Theorem 7.3.1 (namely, there we do not demand μ to be even), and this is why we only work with $p>1$ here.

11.3 Valuations on convex bodies

Let $A,B\subseteq\mathbb{R}^n$ be non-empty Lebesgue measurable sets (with finite measure, say). Since we may write the union $A\cup B$ as the disjoint union of the (measurable) sets $A\setminus(A\cap B)$, $B\setminus(A\cap B)$ and $A\cap B$, we get the equality

$$\mathrm{vol}_n(A\cup B)=\mathrm{vol}_n(A\setminus(A\cap B))+\mathrm{vol}_n(B\setminus(A\cap B))+\mathrm{vol}_n(A\cap B)=$$
$$=\mathrm{vol}_n(A)+\mathrm{vol}_n(B)-\mathrm{vol}_n(A\cap B).$$

That is, the volume of $A \cup B$ plus the volume of $A \cap B$ equals the sum of the volumes of A and B. There are many other examples of applications appearing in convexity which have this same (or an analogous) property, known as *valuation property*. For example, in Theorem 6.2.3, we have shown that for any fixed $C_2, \dots, C_n \in CC^n$, the equality

$$V(C \cup D, C_2, \dots, C_n) + V(C \cap D, C_2, \dots, C_n) = V(C, C_2, \dots, C_n) + V(D, C_2, \dots, C_n)$$

holds whenever $C, D \in CC^n$ are such that $C \cup D \in CC^n$ (convexity may be lost under unions). In the proof of this result, we used the fact that the application $H\colon CC^n \to C(\mathbb{S}^{n-1})$ which associates each $C \in CC^n$ to its support function $h_C\colon \mathbb{S}^{n-1} \to \mathbb{R}$ has the property that

$$H(C \cup D) + H(C \cap D) = H(C) + H(D),$$

whenever $C \cup D \in CC^n$, where the considered sum is the usual sum in $C(\mathbb{S}^{n-1})$, of course. This is proved in Lemma 6.2.2. Even more, equality (6.4.3) states that quermassintegrals (and, consequently, also intrinsic volumes) also have the valuation property.

Let $\mathcal{S}$ be a family of sets which is *intersectional*, meaning that $A \cap B \in \mathcal{S}$ whenever $A, B \in \mathcal{S}$, and that $\emptyset \in \mathcal{S}$. If $\mathbb{A}$ is an (additive) abelian group, then a map $\varphi\colon \mathcal{S} \to \mathbb{A}$ is a *valuation* if $\varphi(\emptyset) = 0$ and

$$\varphi(A \cup B) + \varphi(A \cap B) = \varphi(A) + \varphi(B) \tag{11.3.1}$$

whenever $A, B \in \mathcal{S}$ are such that $A \cup B \in \mathcal{S}$. Observe that this definition remains unaltered if we require that $A, B \neq \emptyset$.

We may replace the group $\mathbb{A}$ by an additive abelian semigroup with cancelation law, such as CC^n with the Minkowski addition or the space of finite Borel measures on $\mathbb{S}^{n-1}$, for example. The reason for our original definition with abelian groups is that in this introductory text we are mainly interested in valuations which take values on $\mathbb{R}$. Actually, from now on, we will work exclusively with functionals over $\mathbb{R}$, but the most part of the results that we present below can be extended to the more general case of maps over arbitrary abelian groups without any modifications in the proofs.

Observe carefully that we reserve the term *valuation* for maps which

(a) are defined on intersectional families,

(b) have the valuation property (11.3.1), and

(c) attain the value 0 for the empty set.

Hence, the maps that we mentioned above are not valuations by our definition, despite having the valuation property. In fact, the family of non-empty compact and convex sets CC^n is not intersectional, because $\emptyset \notin CC^n$. The reader may notice that this is not a problem if we work solely with functionals in CC^n which have the valuation property, since $K \cap L \in CC^n$ whenever $K, L \in CC^n$ are such that $K \cup L$ is convex. However, to develop the theory, we also regard other types of functionals (such as the ones which are *fully* or *weakly additive*, for example), and it is much more natural to work with the family $CC^n \cup \{\emptyset\}$. In general, the considered functionals defined in CC^n will be extended to attain the value 0 in the empty set.

That being said, according to our definition we have immediately that the functionals $\mathrm{vol}_n\colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ and $V(\cdot, C_2, \dots, C_n)\colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ mentioned above are valuations if we agree that $\mathrm{vol}_n(\emptyset) = 0$ and $V(\emptyset, C_2, \dots, C_n) = 0$. Of course, the same holds for the quermassintegrals (and intrinsic volumes). Since $C(\mathbb{S}^{n-1})$ is an additive abelian group, we see that the map $H\colon CC^n \cup \{\emptyset\} \to C(\mathbb{S}^{n-1})$ is also a valuation (similarly, here we agree that $h_\emptyset$ is the zero function).

Intuitively, one can think of valuations as the "reasonable" ways of "measuring the size" of sets, since when measuring sizes one may expect that (11.3.1) holds. Hence, it is natural to study valuations in their full generality, from the "abstract" viewpoint. This approach has been intensively studied in the past years, and this is the reason why we dedicate a section to present the basic results on the theory of valuations on convex bodies. More specifically, our principal aim in this section is to prove a key characterization result known as *Hadwiger's theorem*. We also discuss its meaning and some consequences at the end of the section. Before stating this theorem, we need to introduce some terminology. The *special orthogonal group of* $\mathbb{R}^n$ (or *rotation group of* $\mathbb{R}^n$) is the group consisting of orthogonal transformations of $\mathbb{R}^n$ whose determinant is 1. This is denoted by $\mathrm{SO}(n)$. The maps of $\mathrm{SO}(n)$ are also called the *rotations* of $\mathbb{R}^n$. Moreover, from now on, for each $0 \leq i \leq n$ we consider that the i-th intrinsic volume $V_i \colon CC^n \to \mathbb{R}$ defined in (6.4.8) is extended in such a way that $V_i(\emptyset) = 0$. By doing this, it follows from (6.4.3) that these maps are valuations.

Theorem 11.3.1 (Hadwiger's theorem)**.** *Let $\varphi \colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ be a valuation satisfying the following items:*

(i) *the restriction of φ to CC^n is continuous in the Hausdorff metric and*

(ii) *φ is $\mathrm{SO}(n)$-invariant and translation-invariant, meaning that $\varphi(x + TK) = \varphi(K)$ for any $x \in \mathbb{R}^n$, $T \in \mathrm{SO}(n)$ and $K \in CC^n$.*

Then, there are constants $c_0, \ldots, c_n \in \mathbb{R}$ such that

$$\varphi = \sum_{i=0}^{n} c_i V_i.$$

Remark 11.3.1. Of course, we may replace the intrinsic volumes by quermassintegrals in the statement above. The advantage of working with intrinsic volumes is their dimensional invariance explained at the end of Section 6.4.

Remark 11.3.2. The composition of a translation with a map of $\mathrm{O}(n)$ is commonly called a *rigid motion*, and the composition of a translation with a map of $\mathrm{SO}(n)$ is a *proper rigid motion*. By this terminology, property **(ii)** above means that φ is *invariant under proper rigid motions*.

To prove this theorem, we follow Hug and Weil [1263]. Before that, we have to study several properties of valuations. The reader which is not interested in exploring all technicalities in full detail may just read the statement of Proposition 11.3.2 and jump to the proof of the theorem on page 776. After the proof, some consequences of Hadwiger's theorem are briefly discussed.

Notice that valuations do not give information about unions of more than 2 sets belonging to the domain. Hence, it is useful to work with the more restrictive concept of *fully additive functional*. For each $m \in \mathbb{N}$, let $S(m)$ be the family of all non-empty subsets of $\{1, \ldots, m\}$. For each $I \in S(m)$, we denote by $|I|$ the cardinality of I and we also write $A_I := \bigcap_{i \in I} A_i$. A functional $\varphi \colon \mathcal{S} \to \mathbb{R}$ is *fully additive* if $\varphi(\emptyset) = 0$ and

$$\varphi(A_1 \cup \ldots \cup A_m) = \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(A_I) \tag{11.3.2}$$

for any $m \in \mathbb{N}$ and every $A_1, \ldots, A_m \in \mathcal{S}$ such that $A_1 \cup \ldots \cup A_m \in \mathcal{S}$. Observe that the definition remains unaltered if we require that each A_j is non-empty. Of course, the case $m = 2$ gives that any fully additive functional is a valuation.

Remark 11.3.3. Given a valuation $\varphi \colon \mathcal{S} \to \mathbb{R}$, one can think of naively defining

$$\varphi(A \cup B) = \varphi(A) + \varphi(B) - \varphi(A \cap B),$$

and extending indutively to unions of more than 2 sets. The problem is that a given union of three (say) sets might be already in $\mathcal{S}$, and φ may have *a priori* a different value than the value defined by this extension. The definition of *fully additive functional* automatically avoids this complication (see Exercise 11.2).

Even if Exercise 11.2 establishes that valuations and fully additive functionals are not the same in the general case, they do coincide if the considered domain is closed under finite unions. Intuitively, this happens because all "intermediate" unions are in the domain, and hence the functional is *a priori* defined for them. To give a formal explanation, we need first a new definition. For any given intersectional family $\mathcal{S}$, we define

$$U(\mathcal{S}) := \Big\{\text{finite unions of sets of } \mathcal{S}\Big\}.$$

That is, the elements of $U(\mathcal{S})$ are the unions $A_1 \cup \ldots \cup A_m$ of finitely many sets $A_1, \ldots, A_m \in \mathcal{S}$. We also made the convention that $\emptyset \in U(\mathcal{S})$ (as the union of no sets, say). Notice that it is clear that $U(\mathcal{S})$ is also intersectional.

Lemma 11.3.1. *Let $\varphi\colon U(\mathcal{S}) \to \mathbb{R}$ be a valuation. Then φ is fully additive.*

Proof. More than being intersectional, it is immediate that $U(\mathcal{S})$ is closed for finite unions. Thus, since φ is a valuation defined in $U(\mathcal{S})$, we have

$$\varphi(A \cup B) = \varphi(A) + \varphi(B) - \varphi(A \cap B)$$

for any $A, B \in U(\mathcal{S})$. In other words, we do not need to demand that $A \cup B \in U(\mathcal{S})$, because this holds trivially for any $A, B \in U(\mathcal{S})$.

To prove that (11.3.2) holds, we proceed by induction. The case $m = 2$ comes from the definition of valuations, and we assume that the case for $m - 1$ holds for some $m > 2$. Let $A_1, \ldots, A_m \in U(\mathcal{S})$. It is easy to see that

$$S(m) = S(m-1) \cup \{m\} \cup \Big\{I \cup \{m\} : I \in S(m-1)\Big\}.$$

Letting $A := A_1 \cup \ldots \cup A_{m-1}$, from the valuation property of φ and using the induction hypothesis, we get

$$\begin{aligned}
\varphi(A_1 \cup \ldots \cup A_m) = \varphi(A \cup A_m) = \varphi(A) + \varphi(A_m) - \varphi(A \cap A_m) = \\
= \varphi(A_1 \cup \ldots \cup A_{m-1}) + \varphi(A_m) - \varphi\big((A_1 \cap A_m) \cup \ldots \cup (A_{m-1} \cap A_m)\big) = \\
= \sum_{I \in S(m-1)} (-1)^{|I|-1} \varphi(A_I) + \varphi(A_m) + \sum_{I \in S(m-1)} (-1)^{|I|} \varphi(A_I \cap A_m) = \sum_{J \in S(m)} (-1)^{|J|-1} \varphi(A_J),
\end{aligned}$$

and this concludes the proof. □

Even more, any fully additive functional on an intersectional family $\mathcal{S}$ can be naturally extended to a valuation on $U(\mathcal{S})$, as we will show below.

Proposition 11.3.1. *Let $\varphi\colon \mathcal{S} \to \mathbb{R}$ be a fully additive functional. Then φ can be extended to a valuation on $U(\mathcal{S})$.*

Remark 11.3.4. Any such extension is fully additive, since any valuation on $U(\mathcal{S})$ is fully additive.

Proof. Once φ is fully additive, our strategy is to define

$$\tilde{\varphi}(A_1 \cup \ldots \cup A_m) := \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(A_I)$$

for any $A_1, \ldots, A_m \in \mathcal{S}$. Indeed, in this case we have $\tilde{\varphi}(A_1 \cup \ldots \cup A_m) = \varphi(A_1 \cup \ldots \cup A_m)$ whenever $A_1 \cup \ldots \cup A_m \in \mathcal{S}$. The first thing that we have to do is to prove that this definition is coherent with different representations of a set of $U(\mathcal{S})$ as two different unions of sets of $\mathcal{S}$. That is, we let $A \in U(\mathcal{S})$ be given as

$$A = A_1 \cup \ldots \cup A_m = E_1 \cup \ldots \cup E_k$$

for $A_1, \ldots, A_m, E_1, \ldots, E_k \in \mathcal{S}$. We have to show that

$$\sum_{I \in S(m)} (-1)^{|I|-1} \varphi(A_I) = \sum_{J \in S(k)} (-1)^{|J|-1} \varphi(E_J). \tag{11.3.3}$$

Notice that $A_I \subseteq A = E_1 \cup \ldots \cup E_k$ for any $I \in S(m)$. Then we may write

$$A_I = A_I \cap A = A_I \cap \left(\bigcup_{j=1}^{k} E_j \right) = \bigcup_{j=1}^{k} (A_I \cap E_j),$$

and $A_I \cap E_j \in \mathcal{S}$ for any $I \in S(m)$ and every $j \in \{1, \ldots, k\}$, since $A_I \cap E_j$ is an intersection of sets of $\mathcal{S}$. Thus, the sets $A_I \cap E_1, \ldots A_I \cap E_k$ are sets of $\mathcal{S}$ whose union also belongs to $\mathcal{S}$ (from the above, their union is A_I, which is an intersection of sets of $\mathcal{S}$). Using the fact that φ is fully additive on $\mathcal{S}$, we get that

$$\varphi(A_I) = \varphi \left(\bigcup_{j=1}^{k} (A_I \cap E_j) \right) = \sum_{J \in S(k)} (-1)^{|J|-1} \varphi(A_I \cap E_J)$$

for each $I \in S(m)$. Consequently,

$$\sum_{I \in S(m)} (-1)^{|I|-1} \varphi(A_I) = \sum_{I \in S(m)} \sum_{J \in S(k)} (-1)^{|I|+|J|} \varphi(A_I \cap E_J).$$

Now we repeat the argument writing $E_J = \bigcup_{i=1}^{m} E_J \cap A_i$ for each $J \in S(k)$. We arrive at

$$\sum_{J \in S(k)} (-1)^{|J|-1} \varphi(E_J) = \sum_{J \in S(k)} \sum_{I \in S(m)} (-1)^{|J|+|I|} \varphi(E_J \cap A_I),$$

and by symmetry this gives (11.3.3). Therefore, we have, in fact, that $\tilde{\varphi}$ is well-defined, meaning that it attains the same value for any representation of a set of $U(\mathcal{S})$ as a union of sets of $\mathcal{S}$.

It remains to prove that $\tilde{\varphi} \colon U(\mathcal{S}) \to \mathbb{R}$ is a valuation. For any given $A, B \in U(\mathcal{S})$, fix representations

$$A = A_1 \cup \ldots \cup A_m \quad \text{and}$$
$$B = B_1 \cup \ldots \cup B_k,$$

for some $A_1, \ldots, A_m, B_1, \ldots, B_k \in \mathcal{S}$. Hence, by the definition of $\tilde{\varphi}$, we may write

$$\tilde{\varphi}(A \cup B) = \tilde{\varphi}(A_1 \cup \ldots \cup A_m \cup B_1 \cup \ldots \cup B_k) = \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(A_I) + \sum_{J \in S(k)} (-1)^{|J|-1} \varphi(B_J) +$$
$$+ \sum_{I \in S(m),\, J \in S(k)} (-1)^{|I|+|J|-1} \varphi(A_I \cap B_J) = \tilde{\varphi}(A) + \tilde{\varphi}(B) - \sum_{I \in S(m),\, J \in S(k)} (-1)^{|I|+|J|} \varphi(A_I \cap B_J). \tag{11.3.4}$$

The reader may notice that in the last summation we gather all "mixed" intersections between A_i's and B_j's. Now let

$$S(m,k) = \Big\{\text{finite non-empty subsets of } \{1,\ldots,m\} \times \{1,\ldots,k\}\Big\}.$$

That is, each element $L \in S(m,k)$ is a non-empty set consisting of pairs (i,j) where $i \in \{1,\ldots,m\}$ and $j \in \{1,\ldots,k\}$. We explain the reason why we introduced this notation. Notice that

$$\left(\bigcup_{i=1}^{m} A_i\right) \cap \left(\bigcup_{j=1}^{k} B_j\right) = \bigcup_{i=1}^{m} \bigcup_{j=1}^{k} A_i \cap B_j.$$

Any $L \in S(m,k)$ can be written as

$$L = \big\{(i_1, j_1), \ldots, (i_{|L|}, j_{|L|})\big\}$$

for some natural numbers $1 \le i_1, \ldots, i_{|L|} \le m$ and $1 \le j_1, \ldots, j_{|L|} \le k$. Since each pair (i,j) corresponds to the intersection $A_i \cap B_j$, we have that L corresponds to the intersection $(A_{i_1} \cap B_{j_1}) \cap (A_{i_2} \cap B_{j_2}) \cap \ldots \cap (A_{i_{|L|}} \cap B_{j_{|L|}})$, which we denote by E_L. By the equality above and the definition of $\tilde{\varphi}$, the value of $\tilde{\varphi}(A \cap B)$ can be expressed by the values of φ in these intersections as

$$\tilde{\varphi}(A \cap B) = \tilde{\varphi}\left(\bigcup_{i=1}^{m} \bigcup_{j=1}^{k} A_i \cap B_j\right) = \sum_{L \in S(m,k)} (-1)^{|L|-1} \varphi(E_L). \tag{11.3.5}$$

Of course, we may modify the order of the intersection to aggregate the A_i's and the B_j's, and write

$$E_L = (A_{i_1} \cap \ldots \cap A_{i_{|L|}}) \cap (B_{j_1} \cap \ldots \cap B_{j_{|L|}}).$$

Let $\pi_1(L)$ be the subset of $S(m)$ consisting of $\{i_1, \ldots, i_{|L|}\}$, and let $\pi_2(L)$ be the subset of $S(k)$ given by $\{j_1, \ldots, j_{|L|}\}$. It is immediate that

$$E_L = A_{\pi_1(L)} \cap B_{\pi_2(L)}.$$

This equality has the following consequence: let $I \in S(m)$ and $J \in S(k)$. Then $E_L = A_I \cap B_J$ for every $L \in S(m,k)$ such that $\pi_1(L) = I$ and $\pi_2(L) = J$. Hence, we may write (11.3.5) as

$$\tilde{\varphi}(A \cap B) = \sum_{I \in S(m),\, J \in S(k)} \sum_{\substack{L \in S(m,k) \\ \pi_1(L)=I,\, \pi_2(L)=J}} (-1)^{|L|-1} \varphi(A_I \cap B_J) =$$
$$= \sum_{I \in S(m),\, J \in S(k)} \left[\varphi(A_I \cap B_J) \sum_{\substack{L \in S(m,k) \\ \pi_1(L)=I,\, \pi_2(L)=J}} (-1)^{|L|-1} \right].$$

Comparing this equality with the last summation in (11.3.4), it follows that all we have to prove is that the equality

$$\sum_{\substack{L\in S(m,k)\\ \pi_1(L)=I,\,\pi_2(L)=J}} (-1)^{|L|-1} = (-1)^{|I|+|J|}$$

holds for any $I \in S(m)$ and $J \in S(k)$. To do this, we fix $I = \{i_1, \dots, i_{|I|}\} \in S(m)$ and proceed by induction on $|J|$. If $|J| = 1$, then we may write $J = \{j_1\}$, and the unique $L \in S(m,k)$ such that $\pi_1(L) = I$ and $\pi_2(L) = J$ is $L = \{(i_p, j_1) : 1 \le p \le |I|\}$. In this case, we clearly have that $|L| = |I|$. Therefore,

$$\sum_{\substack{L\in S(m,k)\\ \pi_1(L)=I,\,\pi_2(L)=J}} (-1)^{|L|-1} = (-1)^{|I|-1} = (-1)^{|I|+1} = (-1)^{|I|+|J|}.$$

Now let $J = \{j_1, \dots, j_{|J|}\} \in S(k)$ be such that $|J| \ge 2$ and assume that the equality is true for any $J' \in S(k)$ with $|J'| < |J|$ and any $I \in S(m)$. For simplicity, write $J_1 := \{j_1, \dots, j_{|J|-1}\}$. Each $L \in S(m,k)$ for which $\pi_1(L) = I$ and $\pi_2(L) = J$ can be written by separating all the terms whose second coordinate is $j_{|J|}$ as the disjoint union

$$L = M \cup \big\{(i, j_{|J|}) : (i, j_{|J|}) \in L\big\},$$

where $\pi_2(M) = J_1$ and

$$\pi_1(M) := I_1 := \big\{i \in \{i_1, \dots, i_{|I|}\} : (i,j) \in L \text{ for some } j \in J_1\big\}.$$

For simplicity, write

$$\mathcal{L} := \Big\{L \in S(m,k) : \pi_1(L) = I \text{ and } \pi_2(L) = J\Big\} \text{ and}$$
$$\mathcal{L}_p := \Big\{L_p \in S(m,k) : \pi_2(L_1) = J_1,\ \pi_1(L_p) \subseteq I \text{ and } |\pi_1(L_p)| = p\Big\},$$

for $1 \le p \le |I|$. We already know that any set of the family $\mathcal{L}$ can be obtained by a set of some family $\mathcal{L}_p$ by adding pairs whose second coordinate is $j_{|J|}$. Now we investigate how many of these sets of $\mathcal{L}$ are associated to a given $L_p \in \mathcal{L}_p$. Notice that for the sets of any $\mathcal{L}_p$, we can use the induction hypothesis, since $|J_1| < |J|$.

Assume first that $1 \le p \le |I| - 1$, and let $L_p \in \mathcal{L}_p$. Then, to construct a set $L \in S(m,k)$ from L_p, we must add terms with second coordinate $j_{|J|}$, and whose respective first coordinates are the $|I| - p$ terms of I which are missing in $\pi_1(L_p)$. Now we guarantee that the obtained set is in $\mathcal{L}$ (because all terms of I appear in the first coordinate of some pair, and all terms of J appear in the second coordinate), we can freely add points having the form $(i, j_{|J|})$ for $i \in I_p$. Thus, for each $k = 0, \dots, p$ there exists $\binom{p}{k}$ sets $L \in \mathcal{L}$ associated to L_p, each of which having $|L_p| + |I| - p + k$ terms. In the case $p = |I|$, we have $\pi_1(L_p) = I$, and hence we add freely pairs $(i, j_{|J|})$ to obtain a set of $\mathcal{L}$ (but now we must add at least one term). It follows that for each $k = 1, \dots, |I|$, we may construct $\binom{|I|}{k}$ sets of $\mathcal{L}$ from every $L_I \in \mathcal{L}_{|I|}$. Each of these sets has $|L_p| + k$ elements. Finally, since every term of the summation that we want to calculate depends only on the number of elements of the associated set, we get

$$\sum_{L\in\mathcal{L}}(-1)^{|L|-1} =$$

$$= \sum_{p=1}^{|I|-1}\sum_{L_p\in\mathcal{L}_p}\sum_{k=0}^{p}\binom{p}{k}(-1)^{|L_p|+|I|-p+k-1} + \sum_{L_p\in\mathcal{L}_{|I|}}\sum_{k=1}^{|I|}\binom{|I|}{k}(-1)^{|L_p|+k-1} =$$

$$= \sum_{p=1}^{|I|-1}\sum_{L_p\in\mathcal{L}_p}(-1)^{|L_p|+|I|-p-1}\sum_{k=0}^{p}\binom{p}{k}(-1)^{k} + \sum_{L_p\in\mathcal{L}_{|I|}}(-1)^{|L_p|-1}\sum_{k=1}^{|I|}\binom{|I|}{k}(-1)^{k}.$$

The first summation in the right-hand side vanishes, because for every natural number $p \geq 1$, we have

$$\sum_{k=0}^{p}\binom{p}{k}(-1)^{k} = \sum_{k=0}^{p}\binom{p}{k}(-1)^{k}1^{p-k} = (1-1)^{p} = 0,$$

and by the same reason, we also have

$$\sum_{k=1}^{|I|}\binom{|I|}{k}(-1)^{k} = -\binom{|I|}{0}(-1)^{0} + \sum_{k=0}^{|I|}\binom{|I|}{k}(-1)^{k} = -1.$$

Therefore, using the induction hypothesis for the sets of $\mathcal{L}_{|I|}$, we arrive at

$$\sum_{L\in\mathcal{L}}(-1)^{|L|-1} = -\sum_{L_p\in\mathcal{L}_{|I|}}(-1)^{|L_p|-1} = -(-1)^{|I|+|J_1|} = (-1)^{|I|+|J_1|+1} = (-1)^{|I|+|J|},$$

and the proof is complete.

□

Recall that we denote the family of polytopes of $\mathbb{R}^n$ by $\mathcal{P}^n$. The family $\mathcal{P}^n \cup \{\emptyset\}$ is clearly intersectional, since the intersection of polytopes is either a polytope or the empty set. Indeed, if P_1 and P_2 are polytopes, then $P_1 \cap P_2$ is the intersection of all the bounding half-spaces of P_1 and of P_2. Thus, $P_1 \cap P_2$ is a bounded intersection of finitely many closed half-spaces of $\mathbb{R}^n$, and this means that $P_1 \cap P_2$ is either a polytope or the empty set. As a consequence, we may define valuations on $\mathcal{P}^n \cup \{\emptyset\}$. We will call these simply *valuations on polytopes*.

As an important step to prove Hadwiger's theorem, we show that any valuation on polytopes is fully additive. We actually prove an even stronger result, but for that we need to introduce some terminology. We say that a functional $\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R}$ is *weakly additive* if $\varphi(\emptyset) = 0$ and for any $P \in \mathcal{P}^n \cup \{\emptyset\}$ and every hyperplane $\mathbf{h} \subseteq \mathbb{R}^n$ the equality

$$\varphi(P) = \varphi(P \cap \mathbf{h}^+) + \varphi(P \cap \mathbf{h}^-) - \varphi(P \cap \mathbf{h})$$

holds, where $\mathbf{h}^+$ and $\mathbf{h}^-$ denote the closed half-spaces determined by $\mathbf{h}$, as usual. Our first observation is that any valuation defined for polytopes is weakly additive. This is discussed next.

Lemma 11.3.2. *Let $\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R}$ be a valuation. Then φ is weakly additive.*

Proof. Let $P \in \mathcal{P}^n$ be a polytope and let $\mathbf{h} \subseteq \mathbb{R}^n$ be a hyperplane. Then we claim that the sets $P \cap \mathbf{h}$, $P \cap \mathbf{h}^+$ and $P \cap \mathbf{h}^-$ are either polytopes or the empty set. Indeed, $P \cap \mathbf{h}^+$ and $P \cap \mathbf{h}^-$ are the intersections of the bounding half-spaces of P with $\mathbf{h}^+$ and with $\mathbf{h}^-$, respectively. Thus, $P \cap \mathbf{h}^+$ and $P \cap \mathbf{h}^-$ are bounded (or empty) intersections of finitely many closed half-spaces of $\mathbb{R}^n$, and this gives that each of these sets is either a polytope or the empty set. The set $P \cap \mathbf{h}$ can be written as the intersection of polytopes $(P \cap \mathbf{h}^+) \cap (P \cap \mathbf{h}^-)$, and hence it is either a polytope or the empty set (as it was commented above).

Let $P \in \mathcal{P}^n \cup \{\emptyset\}$ and let $\mathbf{h} \subseteq \mathbb{R}^n$ be a hyperplane. Since $P \cap \mathbf{h}^+, P \cap \mathbf{h}^- \in \mathcal{P}^n \cup \{\emptyset\}$, if $\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R}$ is a valuation, then we may write

$$\varphi\big((P \cap \mathbf{h}^+) \cup (P \cap \mathbf{h}^-)\big) + \varphi\big((P \cap \mathbf{h}^+) \cap (P \cap \mathbf{h}^-)\big) = \varphi(P \cap \mathbf{h}^+) + \varphi(P \cap \mathbf{h}^-).$$

Since $(P \cap \mathbf{h}^+) \cup (P \cap \mathbf{h}^-) = P$ and $(P \cap \mathbf{h}^+) \cap (P \cap \mathbf{h}^-) = P \cap \mathbf{h}$, the desired follows.

□

Theorem 11.3.2. *Any weakly additive functional $\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R}$ is fully additive.*

Proof. From the definition of weak additivity, we already have that $\varphi(\emptyset) = 0$. Thus, all we have to prove is that for any $P_1, \dots, P_m \in \mathcal{P}^n \cup \{\emptyset\}$ such that $P_1 \cup \dots \cup P_m \in \mathcal{P}^n \cup \{\emptyset\}$ the equality

$$\varphi(P_1 \cup \dots \cup P_m) = \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I) \tag{11.3.6}$$

holds. We proceed by induction on the dimension n. If $n = 0$, then we may write $\mathbb{R}^0 = \{0\}$, and hence any polytope is $\{0\}$. It follows that for any $P_1, \dots, P_m \in \mathcal{P}^0$, we have $P_I = \{0\}$ for each $I \in S(m)$, and hence

$$\sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I) = \varphi(\{0\}) \sum_{I \in S(m)} (-1)^{|I|-1} = \varphi(\{0\}) \sum_{k=1}^{m} \binom{m}{k} (-1)^{k-1} =$$

$$= -\varphi(\{0\}) \left[\sum_{k=0}^{m} \binom{m}{k} (-1)^k - \binom{m}{0} (-1)^0 \right] = \varphi(\{0\}) = \varphi(P_1 \cup \dots \cup P_m).$$

This settles (11.3.6) for dimension $n = 0$.

Now we let $n \geq 1$ and assume that equality (11.3.6) holds for every weakly additive functional over $\mathcal{P}^k \cup \{\emptyset\}$ in any dimension $k < n$. To prove that this also holds for any weakly additive functional φ defined over $\mathcal{P}^n \cup \{\emptyset\}$, we first claim that it is enough to consider the case where $\dim(P_1 \cup \dots \cup P_m) = n$. Indeed, if $\dim(P_1 \cup \dots \cup P_m) = k < n$, then there exists a k-dimensional affine subspace $\mathbf{g}$ such that $P_1, \dots, P_m \subseteq \mathbf{g}$. The restriction of φ to $\mathbf{g}$ is naturally identified with a weakly additive functional defined over $\mathbb{R}^k$, and hence we have (11.3.6) by the induction hypothesis.

Having said this, let $P_1, \dots, P_m \in \mathcal{P}^n \cup \{\emptyset\}$ be such that $P_1 \cup \dots \cup P_m \in \mathcal{P}^n \cup \{\emptyset\}$ and $\dim(P_1 \cup \dots \cup P_m) = n$. For simplicity, write $P = P_1 \cup \dots \cup P_m$. As a first special case, assume that $P = P_j$ for some $j \in \{1, \dots, m\}$. Up to renumbering the polytopes if necessary, we may let $P = P_1$. Thus, $P_2, \dots, P_m \subseteq P_1$, and this leads to

$$P_1 \cap P_I = P_I \tag{11.3.7}$$

for every $I \in S(m)$. Denote by $S_1(m)$ the family of non-empty subsets of $S(m)$ which contain 1, and by $S_{\neq 1}(m)$ the family of non-empty subsets of $S(m)$ which do not contain 1. Then it is clear that $S_1(m)$ and $S_{\neq 1}(m)$ are disjoint and that $S(m) = S_1(m) \cup S_{\neq 1}(m)$. For each $I \in S_{\neq 1}(m)$, we have that $J = I \cup \{1\} \in S_1(m)$, and by (11.3.7), we get that $P_I = P_J$. Conversely, any $J \in S_1(m)$ which is not $\{1\}$ can be written as $J = I \cup \{1\}$ for a unique $I \in S_{\neq 1}(m)$, and again we have $P_I = P_J$. Thus, we write

$$\sum_{I\in S(m)}(-1)^{|I|-1}\varphi(P_I)=\sum_{J\in S_1(m)}(-1)^{|J|-1}\varphi(P_J)+\sum_{I\in S_{\neq 1}(m)}(-1)^{|I|-1}\varphi(P_I)=$$
$$=(-1)^{1-1}\varphi(P_1)+\sum_{I\in S_{\neq 1}(m)}(-1)^{|I\cup\{1\}|-1}\varphi(P_I)+\sum_{I\in S_{\neq 1}(m)}(-1)^{|I|-1}\varphi(P_I)=$$
$$=\varphi(P_1)+\sum_{I\in S_{\neq 1}(m)}\left[(-1)^{|I|+1-1}+(-1)^{|I|-1}\right]\varphi(P_I)=\varphi(P_1)=\varphi(P),$$

and this settles (11.3.6) for the particular case where some P_j equals P.

Now we consider induction on m. If $m=1$, then (11.3.6) clearly holds. We let $m\geq 2$ and assume that if $k<m$, then

$$\varphi(Q_1\cup\ldots\cup Q_k)=\sum_{I\in S(k)}(-1)^{|I|-1}\varphi(Q_I)$$

holds for any $Q_1,\ldots,Q_k\in\mathcal{P}^n\cup\{\emptyset\}$ such that $Q_1\cup\ldots\cup Q_k\in\mathcal{P}^n\cup\{\emptyset\}$. Let $P_1,\ldots,P_m\in\mathcal{P}^n$ (we obviously may assume that all sets are non-empty), and assume that $P=P_1\cup\ldots\cup P_m\in\mathcal{P}^n$ and that $\dim(P)=n$. As another special case, we first assume that one of the P_j's is not proper (that is, its affine dimension is smaller than n). Up to renumbering the polytopes, we may assume that $\dim(P_m)=k<n$, and then P_m is contained in an affine subspace of dimension k (**g**, say).

We claim that, in this case, $P_m\subseteq P_1\cup\ldots\cup P_{m-1}$. Indeed, since $\dim(P)=n$, there exists a point $z_0\in P_1\cup\ldots\cup P_{m-1}$ such that $z_0\notin\mathbf{g}$. If P_m is not contained in $P_1\cup\ldots\cup P_{m-1}$, then we may fix a point $x_0\in P_m$ (hence $x_0\in\mathbf{g}$) with $x_0\notin P_1\cup\ldots\cup P_{m-1}$. Since z_0 and x_0 are points of the polytope P, we get that $\mathrm{seg}[z_0,x_0]\subseteq P$. Let

$$\lambda_0:=\sup\left\{\lambda\in[0,1]:\lambda z_0+(1-\lambda)x_0\in P_1\cup\ldots\cup P_{m-1}\right\}.$$

Once $x_0\notin P_1\cup\ldots\cup P_{m-1}$, we get that $\lambda_0<1$. Thus, fixing any $\lambda_1\in(\lambda_0,1)$, we have that $(1-\lambda_1)z_0+\lambda_1x_0\in P_m$ (because this is a point of P which is not in $P_1\cup\ldots\cup P_{m-1}$). This is a contradiction to the facts that $z_0\notin\mathbf{g}$ and $x_0\in\mathbf{g}$, and then our claim is indeed true.

Since $P_m\subseteq P_1\cup\ldots\cup P_{m-1}$, we may write $P_m=(P_1\cap P_m)\cup\ldots\cup(P_{m-1}\cap P_m)$. This is the union of $m-1$ sets of $\mathcal{P}^n\cup\{\emptyset\}$, and then the induction hypothesis gives

$$\varphi(P_m)=\varphi\big((P_1\cap P_m)\cup\ldots\cup(P_{m-1}\cap P_m)\big)=\sum_{J\in S(m-1)}(-1)^{|J|-1}\varphi(P_m\cap P_J).$$

Let $S_m(m)$ be the family of non-empty subsets of $\{1,\ldots,m\}$ containing m. Then for any $J\in S(m-1)$, the term $P_m\cap P_J$ corresponds to $I=J\cup\{m\}$, which is an element of $S_m(m)$ distinct from $\{m\}$. Conversely, any $I\in S_m(m)$ which is not $\{m\}$ is the union of $\{m\}$ with some $J\in S(m-1)$. Thus, we may rewrite the above as

$$\varphi(P_m)=\sum_{\substack{I\in S_m(m)\\ I\neq\{m\}}}(-1)^{|I-1|-1}\varphi(P_I)=-\sum_{\substack{I\in S_m(m)\\ I\neq\{m\}}}(-1)^{|I|-1}\varphi(P_I).$$

Moreover, since the inclusion $P_m\subseteq P_1\cup\ldots\cup P_{m-1}$ clearly yields the equality $P=P_1\cup\ldots\cup P_{m-1}$, it also follows from the induction hypothesis that

$$\varphi(P)=\sum_{I\in S(m-1)}(-1)^{|I|-1}\varphi(P_I).$$

Because it is clear that $S(m)$ is the disjoint union $S(m-1) \cup S_m(m) \cup \{m\}$, and using the equalities above, we arrive at

$$\sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I) = \sum_{I \in S(m-1)} (-1)^{|I|-1} \varphi(P_I) + \sum_{\substack{I \in S_m(m) \\ I \neq \{m\}}} (-1)^{|I|-1} \varphi(P_I) + \varphi(P_m) =$$
$$= \varphi(P) - \varphi(P_m) + \varphi(P_m) = \varphi(P),$$

and this gives (11.3.6) for the case where one of the P_j's is not full-dimensional.

Now we exclude the two particular cases that we already dealt with, and hence we let $P_1, \ldots, P_m \in \mathcal{P}^n$ be such that $P = P_1 \cup \ldots \cup P_m \in \mathcal{P}^n$, each P_j is full-dimensional, and $P \neq P_j$ for each $j \in \{1, \ldots, m\}$. Recall that we are also assuming two induction hypotheses: first, equality (11.3.6) holds in any dimension less than n; second, (11.3.6) holds in $\mathbb{R}^n$ for $k < m$ polytopes. In particular, $P_1 \subseteq P$ is an n-dimensional polytope properly contained in P. Fixing points $x_0 \in \operatorname{int}(P_1)$ and $x_1 \in \operatorname{int}(P \setminus P_1)$, we have immediately that the segment $\operatorname{seg}[x_0, x_1]$ intersects the boundary of P_1 in an interior point of P (z_0, say). Let F be a facet of P_1 containing z_0 (remember that the boundary of a polytope is the union of its facets), and denote by $\mathbf{h} := \mathbf{h}(u, \alpha)$ the affine hull of F. Thus, $\mathbf{h} \cap \operatorname{int}(P) \neq \emptyset$, and we also have that $F = P_1 \cap \mathbf{h}$ (because $\mathbf{h}$ is the supporting hyperplane of P_1 containing the facet F). Moreover, up to replacing u by $-u$ if necessary, we may assume that $P_1 \subseteq \mathbf{h}^+$. By doing this, we also have that $P_1 \cap \mathbf{h}^- = F$. Since φ is weakly additive, we may write

$$\varphi(P) = \varphi(P \cap \mathbf{h}^+) + \varphi(P \cap \mathbf{h}^-) - \varphi(P \cap \mathbf{h}). \tag{11.3.8}$$

Notice that we may write $P \cap \mathbf{h}^- = (P_1 \cup \ldots \cup P_m) \cap \mathbf{h}^- = (P_1 \cap \mathbf{h}^-) \cup \ldots \cup (P_m \cap \mathbf{h}^-)$, and each set in this latter union is either a polytope or the empty set. The polytope $P_1 \cap \mathbf{h}^-$ is the facet F of P_1, which is an $(n-1)$-dimensional polytope. Using one of the particular cases already presented (namely, the case where one of the polytopes of the union is not full-dimensional), we may write

$$\varphi(P \cap \mathbf{h}^-) = \varphi\big((P_1 \cap \mathbf{h}^-) \cup \ldots \cup (P_m \cap \mathbf{h}^-)\big) = \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I \cap \mathbf{h}^-).$$

On the other hand, writing $P \cap \mathbf{h} = (P_1 \cap \mathbf{h}) \cup \ldots \cup (P_m \cap \mathbf{h})$, we get that the polytope $P \cap \mathbf{h}$ is written as a union of polytopes with dimension less than n. Because we already know that (11.3.6) holds in this case (as a consequence of our first induction hypothesis), we get

$$\varphi(P \cap \mathbf{h}) = \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I \cap \mathbf{h}).$$

Replacing the expressions derived above in (11.3.8) and rearranging the terms, we arrive at

$$\varphi(P) = \varphi(P \cap \mathbf{h}^+) + \sum_{I \in S(m)} (-1)^{|I|-1} \big(\varphi(P_I \cap \mathbf{h}^-) - \varphi(P_I \cap \mathbf{h})\big).$$

Because each P_I is either a polytope of the empty set, we may use again the weak additivity of φ to write

$$\varphi(P_I \cap \mathbf{h}^-) - \varphi(P_I \cap \mathbf{h}) = \varphi(P_I) - \varphi(P_I \cap \mathbf{h}^+),$$

and hence

$$\varphi(P) = \varphi(P \cap \mathbf{h}^+) + \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I) - \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I \cap \mathbf{h}^+).$$

Therefore, it suffices to prove that

$$\varphi(P \cap \mathbf{h}^+) = \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I \cap \mathbf{h}^+).$$

If $P \cap \mathbf{h}^+ = P_1$, then since $P_1 \subseteq \mathbf{h}^+$ we get that $P \cap \mathbf{h}^+ = P_1 \cap \mathbf{h}^+$. Thus, writing $P \cap \mathbf{h}^+ = (P_1 \cap \mathbf{h}^+) \cup \ldots \cup (P_m \cap \mathbf{h}^+)$ we arrive at the first special case that we dealt with (namely, the case where one of the polytopes of the union equals the union itself), and the equality above holds true. If $P \cap \mathbf{h}^+ \neq P_1$, then P_1 is properly contained in $P \cap \mathbf{h}^+$. In this case, we repeat the argument replacing P by $P^2 := P \cap \mathbf{h}^+$ and each P_j by $P_j^2 := P_j \cap \mathbf{h}^+$ (the new hyperplane will be denoted by $\mathbf{h}_s$, say). Since P_1 has finitely many facets (and can be written as the intersection of the supporting half-spaces of these facets), after finitely many steps (s, say) we will have that $P^s \cap \mathbf{h}_s^+ = P_1^s$, and hence the desired equality will follow. This concludes the proof.

□

This result, together with the facts that every valuation is weakly additive and that every fully additive functional is a valuation, yields the following equalities:

$$\begin{aligned} &\left\{\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R} : \varphi \text{ is a valuation}\right\} = \\ &= \left\{\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R} : \varphi \text{ is weakly additive}\right\} = \\ &= \left\{\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R} : \varphi \text{ is fully additive}\right\}. \end{aligned}$$

Moreover, from Proposition 11.3.1, we also get that any weakly additive functional $\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R}$ can be extended to a valuation on $U(\mathcal{P}^n)$. We recall that, by Remark 11.3.4, any such extension is fully additive on $U(\mathcal{P}^n)$.

Moving forward, we have the following theorem which relates continuity in the Hausdorff metric and full additivity. This will also be important for proving Hadwiger's theorem.

Theorem 11.3.3. *Let $\varphi\colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ be a functional whose restriction to CC^n is continuous with respect to the Hausdorff metric, and whose restriction to $\mathcal{P}^n \cup \{\emptyset\}$ is weakly additive. Then φ is fully additive in $CC^n \cup \{\emptyset\}$.*

Remark 11.3.5. In this case, Proposition 11.3.1 yields that φ can be extended to a valuation on $U(CC^n)$.

Proof. Let $C_1, \ldots, C_m \in CC^n$ be such that $C_1 \cup \ldots \cup C_m \in CC^n$. We have to prove that

$$\varphi(C_1 \cup \ldots \cup C_m) = \sum_{J \in S(m)} (-1)^{|J|-1} \varphi(C_J). \tag{11.3.9}$$

For any $j \in \{1, \ldots, m\}$ and every $i \in \mathbb{N}$, consider the set $C_j(i) := C_j + 2^{-i}B^n$. For each fixed $i \in \mathbb{N}$, the equality

$$C_1(i) \cup \ldots \cup C_m(i) = \bigcup_{j=1}^{m} (C_j + 2^{-i}B^n) = (C_1 \cup \ldots \cup C_m) + 2^{-1}B^n$$

holds, and hence $C_1(i) \cup \ldots \cup C_m(i) \in CC^n$. Applying Theorem 3.4.1 to the sets $C_1(i), \ldots, C_m(i)$ and $\varepsilon = 2^{-i}$, we get that for each $i \in \mathbb{R}$ there exist polytopes $P_1^i, \ldots, P_m^i$ such that

$$C_j(i) \subseteq P_j^i \subseteq C_j(i) + 2^{-i}B^n$$

for any $j \in \{1, \ldots, m\}$, and $P_1^i \cup \ldots \cup P_m^i \in \mathcal{P}^n$. Of course, the inclusions above can be rewritten as

$$C_j + 2^{-i}B^n \subseteq P_j^i \subseteq C_j + 2^{1-i}B^n, \tag{11.3.10}$$

and since $C_j + 2^{1-i}B^n = C_j + 2^{-(i-1)}B^n \subseteq P_j^{i-1}$, we have that for each fixed $j \in \{1, \ldots, m\}$ the sequence $(P_j^i)_{i\in\mathbb{N}}$ is decreasing (in the sense that $P_j^{i+1} \subseteq P_j^i$ for every $i \in \mathbb{N}$). Moreover, it is immediate that $P_j^i \to C_j$ in the Hausdorff metric for each $j \in \{1, \ldots, m\}$. As a consequence of Lemma 2.6.2 (applied iteratively), we have that $P_1^i \cup \ldots \cup P_m^i \to C_1 \cup \ldots \cup C_m$ in the Hausdorff metric. Therefore, from the continuity of φ, we get

$$\lim_{i\to\infty} \varphi(P_1^i \cup \ldots \cup P_m^i) = \varphi(C_1 \cup \ldots \cup C_m). \tag{11.3.11}$$

Also from (11.3.10), it follows that for any $J = \{j_1, \ldots, j_k\} \in S(m)$, and each $i \in \mathbb{N}$, we have the inclusions

$$C_{j_1} \cap \ldots \cap C_{j_k} \subseteq (C_{j_1} + 2^{-i}B^n) \cap \ldots \cap (C_{j_k} + 2^{-i}B^n) \subseteq P_{j_1}^i \cap \ldots \cap P_{j_k}^i \subseteq$$
$$\subseteq (C_{j_1} + 2^{1-i}B^n) \cap \ldots \cap (C_{j_k} + 2^{1-i}B^n).$$

Thus,

$$C_{j_1} \cap \ldots \cap C_{j_k} \subseteq \bigcap_{i=1}^{\infty} P_{j_1}^i \cap \ldots \cap P_{j_k}^i \subseteq$$
$$\subseteq \bigcap_{i=1}^{\infty} \left((C_{j_1} + 2^{1-i}B^n) \cap \ldots \cap (C_{j_k} + 2^{1-i}B^n)\right) = C_{j_1} \cap \ldots \cap C_{j_k}.$$

That is, writing $P_{j_1}^i \cap \ldots \cap P_{j_k}^i = P_J^i$, we have that

$$\bigcap_{i=1}^{\infty} P_J^i = C_J \tag{11.3.12}$$

for every $J \in S(m)$. Since the sequence $(P_J^i)_{i\in\mathbb{N}}$ is clearly decreasing (because each sequence (P_j^i) is decreasing), from Proposition 2.5.4 and using the continuity of φ in th Hausdorff metric, it follows that

$$\varphi(C_J) = \varphi\left(\bigcap_{i=1}^{\infty} P_J^i\right) = \lim_{i\to\infty} \varphi(P_J^i), \tag{11.3.13}$$

for any $J \in S(m)$ such that $C_J \neq \emptyset$. This also holds in the case $C_J = \emptyset$, but for a different reason: if $C_J = \emptyset$, then equality (11.3.12) forces P_J^i to be empty for sufficiently large i (the left-hand side is the intersection of a decreasing family of compact sets). Hence, both sides of the equality above are 0 in this case.

Now notice that since φ is weakly additive on $\mathcal{P}^n \cup \{\emptyset\}$, we get from Theorem 11.3.2 that φ is fully additive on $\mathcal{P}^n \cup \{\emptyset\}$, and hence

$$\varphi(P_1^i \cup \ldots \cup P_m^i) = \sum_{J\in S(m)} (-1)^{|J|-1} \varphi(P_J^i)$$

for each $i \in \mathbb{N}$. Letting $i \to \infty$, from (11.3.11) and (11.3.13), we get (11.3.9). This concludes the proof. □

Remark 11.3.6. Actually, this theorem holds in more generality, as it was observed in [1263]. We say that a functional $\psi : CC^n \to \mathbb{R}$ is σ *-continuous* if

$$\lim_{i\to\infty} \psi(D_i) = \psi\left(\bigcap_{i=1}^{\infty} D_i\right)$$

for any decreasing sequence $(D_i)_{i\in\mathbb{N}}$ in CC^n. It is easy to see that if we replace the continuity of $\varphi|_{CC^n}$ by σ-continuity, then Theorem 11.3.3 still holds true. Indeed, it is immediate that equality (11.3.13) is still true if we only assume that $\varphi|_{CC^n}$ is σ-continuous, and all we have to do is to verify that equality (11.3.11) also remains true if we assume this weaker continuity hypothesis. In fact, using (11.3.10), we obtain that

$$C_1 \cup \ldots \cup C_m \subseteq (C_1 + 2^{-i}B^n) \cup \ldots \cup (C_m + 2^{-i}B_n) \subseteq P_1^i \cup \ldots \cup P_m^i \subseteq$$
$$\subseteq (C_1 + 2^{1-i}B^n) \cup \ldots \cup (C_m + 2^{1-i}B_n)$$

for each $i \in \mathbb{N}$. Thus,

$$C_1 \cup \ldots \cup C_m \subseteq \bigcap_{i=1}^{\infty}(P_1^i \cup \ldots \cup P_m^i) \subseteq$$
$$\subseteq \bigcap_{i=1}^{\infty}\left((C_1 + 2^{1-i}B^n) \cup \ldots \cup (C_m + 2^{1-i}B_n)\right) = C_1 \cup \ldots \cup C_m,$$

that is,

$$C_1 \cup \ldots \cup C_m = \bigcap_{i=1}^{\infty}(P_1^i \cup \ldots \cup P_m^i).$$

This equality gives immediately that (11.3.11) holds if $\varphi|_{CC^n}$ is σ-continuous.

As an immediate consequence of Theorem 11.3.3 we have that, roughly speaking, continuous valuations on $CC^n \cup \{\emptyset\}$ must be fully additive. We explain it below.

Corollary 11.3.1. *Let $\varphi : CC^n \cup \{\emptyset\} \to \mathbb{R}$ be a valuation whose restriction to CC^n is continuous in the Hausdorff metric. Then φ is fully additive.*

Proof. The restriction of φ to $\mathcal{P}^n \cup \{\emptyset\}$ is a valuation, and hence it is weakly additive (see Lemma 11.3.2). Thus, φ satisfies the conditions of Theorem 11.3.3, and then it is fully additive. □

Now we are in the position to start the proof of Hadwiger's theorem. First we deal with a particular case, which will be used to prove the general case. In what follows, we say that a valuation $\varphi : CC^n \cup \{\emptyset\} \to \mathbb{R}$ is *simple* if $\varphi(C) = 0$ for every $C \in CC^n$ such that $\dim(C) < n$.

Proposition 11.3.2. *Let $\varphi : CC^n \cup \{\emptyset\} \to \mathbb{R}$ be a simple valuation which is invariant under proper rigid motions, and whose restriction to CC^n is continuous in the Hausdorff metric. If $\varphi([0,1]^n) = 0$, then φ vanishes identically.*

Proof. We proceed by induction on the dimension of $\mathbb{R}^n$. The case where $n = 0$ is trivial, and if $n = 1$, then, since φ is simple, we have that $\varphi(\{p\}) = 0$ for every $p \in \mathbb{R}$. Moreover, we have that $\varphi([0,1]) = 0$, and translation invariance gives that φ agrees on compact intervals of the same length. Hence, we may define a function $g : [0, +\infty) \to \mathbb{R}$ letting $g(\ell) := \varphi(I_\ell)$, where $I_\ell \subseteq \mathbb{R}$ is any compact interval whose

length is ℓ. At first glance, we already have that $g(0) = g(1) = 0$. We claim that g is additive, in the sense that

$$g(\ell_1 + \ell_2) = g(\ell_1) + g(\ell_2)$$

for any $\ell_1, \ell_2 \in [0, +\infty)$. Indeed, any compact interval J whose length is $\ell_1 + \ell_2$ can be naturally decomposed as the union of compact intervals I_1 and I_2 whose lengths are ℓ_1 and ℓ_2, respectively, and whose intersection is a singleton. Thus, since φ is a valuation, and because φ vanishes on singletons, we get

$$g(\ell_1 + \ell_2) = \varphi(J) = \varphi(I_1 \cup I_2) = \varphi(I_1) + \varphi(I_2) - \varphi(I_1 \cap I_2) = g(\ell_1) + g(\ell_2).$$

This additivity property yields that $g(r\ell) = rg(\ell)$ for any non-negative $r \in \mathbb{Q}$. We already have this equality if $r = 0$, and otherwise, we may write $r = p/q$ for some positive integers p, q. Then we obtain

$$pg(\lambda) = g\big(\underbrace{\lambda + \ldots + \lambda}_{p \text{ times}}\big) = g(p\lambda) = g\Big(\underbrace{(p/q)\lambda + \ldots + (p/q)\lambda}_{q \text{ times}}\Big) = qg\left(\frac{p}{q}\lambda\right),$$

and this gives the desired. Thus, since $g(1) = 0$, we have that $g(r) = rg(1) = 0$ for any non-negative number rational number r. Finally, we notice that the continuity of φ in the Hausdorff metric of CC^1 yields straightforwardly that g is continuous. Indeed, for any $\ell_0 \geq 0$, the intervals $[0, \ell]$ converge in the Hausdorff metric to $[0, \ell_0]$ as $\ell \to \ell_0$. Thus,

$$g(\ell_0) = \varphi([0, \ell_0]) = \lim_{\ell \to \ell_0} \varphi([0, \ell]) = \lim_{\ell \to \ell_0} g(\ell).$$

As a consequence, recalling that $\mathbb{Q}$ is dense in $\mathbb{R}$ we get that $g = 0$. Therefore, it follows that $\varphi(J) = 0$ for any compact interval $J \subseteq \mathbb{R}$. Since CC^1 is precisely the family of singletons and compact intervals of $\mathbb{R}$, we have that φ vanishes identically. This settles the one-dimensional case.

Now we fix $n \geq 2$, and assume as our induction hypothesis that the result to be proved holds true in any dimension $k < n$. In what follows, we naturally identify the hyperplane $\mathbb{R}^{n-1} \times \{0\}$ with $\mathbb{R}^{n-1}$. Let S be the segment $S = \text{seg}[0_{\mathbb{R}^n}, e_n]$ (hence orthogonal to $\mathbb{R}^{n-1} \times \{0\}$). We define the functional $\psi_S \colon CC^{n-1} \cup \{\emptyset\} \to \mathbb{R}$ by

$$\psi_S(C) := \varphi(C + S)$$

for each $C \in CC^{n-1} \cup \{\emptyset\}$ (notice that we are naturally identifying CC^{n-1} with the family of non-empty compact and convex subsets of $\mathbb{R}^{n-1} \times \{0\}$ to make sense of the Minkowski sum $C + S$; the reader should keep this in mind for what follows). We have the following properties.

(i) The restriction of ψ_S to CC^{n-1} is continuous in the Hausdorff metric. In fact, if $C \in CC^{n-1}$ and $(C_j)_{j \in \mathbb{N}}$ is a sequence in CC^{n-1} such that $C_j \to C$ in the Hausdorff metric of $\mathbb{R}^{n-1}$, then it is not difficult to check that $C_j + I \to C + I$ in the Hausdorff metric of $\mathbb{R}^n$. Thus, from the continuity of φ, we obtain the continuity of ψ_S.

(ii) ψ_S is a simple valuation. The fact that $\psi_S(\emptyset) = 0$ follows immediately from the equality $\emptyset + S = \emptyset$. Let $C, D \in CC^{n-1}$ be such that $C \cup D \in CC^{n-1}$. Then it is immediate that $(C \cup D) + S \in CC^n$, and that the equality $(C \cup D) + S = (C + S) \cup (D + S)$ holds. We clearly have the inclusion $(C \cap D) + S \subseteq (C + S) \cap (D + S)$, and we claim that the reverse inclusion also holds. In fact, if $z \in (C + S) \cap (D + S)$, then we may write $z = c + s_1 = d + s_2$ for some $c \in C$, $d \in D$ and $s_1, s_2 \in S$. It follows that $s_1 - s_2 = d - c \in \mathbb{R}^{n-1}$. Once s_1 and s_2 are points of a segment which is orthogonal to the hyperplane $\mathbb{R}^{n-1} \times \{0\}$, we must have

$s_1 = s_2$, from which we get that $c = d$. Thus, $c \in C \cap D$, and $z = c + s_1 \in (C \cap D) + S$. Having these set equalities in mind, we write

$$\psi_S(C \cup D) + \psi_S(C \cap D) = \varphi((C \cup D) + S) + \varphi((C \cap D) + S) =$$
$$= \varphi((C + S) \cup (D + S)) + \varphi((C + S) \cap (D + S)) = \varphi(C + S) + \varphi(D + S) = \psi_S(C) + \psi_S(D),$$

and hence ψ_S is indeed a valuation on $CC^{n-1} \cup \{\emptyset\}$. To check that ψ_S is simple, just observe that if $C \in CC^{n-1}$ is such that $\dim(C) < n - 1$, then $\dim(C + S) < n$. Therefore, using the fact that φ is simple, we get $\psi_S(C) = \varphi(C + S) = 0$.

(iii) ψ_S is invariant under proper rigid motions of $\mathbb{R}^{n-1}$. The translation invariance follows immediately from the translation invariance of φ: for any $C \in CC^{n-1}$ and $x \in \mathbb{R}^n$, we have

$$\psi_S(C + x) = \varphi(C + x + S) = \varphi(C + S) = \psi_S(C).$$

For any given $T \in \mathrm{SO}(n-1)$, we let $\tilde{T}$ be the unique linear isomorphism of $\mathbb{R}^n$ which agrees with T on $\mathbb{R}^{n-1}$ (that is, $\tilde{T}(v, 0) = (Tv, 0)$ for every $v \in \mathbb{R}^{n-1}$), and $\tilde{T}(e_n) = e_n$. It is easy to see that $\tilde{T} \in \mathrm{SO}(n)$, and the direction of S is e_n, we also have that $\tilde{T}(C + S) = TC + S$ (where TC is identified with $TC \times \{0\}$, of course) for every $C \in CC^{n-1}$. Thus

$$\psi_S(TC) = \varphi(TC + S) = \varphi(\tilde{T}(C + S)) = \varphi(C + S) = \psi_S(C),$$

and this shows that ψ_S is $\mathrm{SO}(n-1)$-invariant.

(iv) We have the equality $\psi_S([0,1]^{n-1}) = 0$. Indeed, since it is clear that $S + [0,1]^{n-1} \times \{0\} = \mathrm{seg}[0_{\mathbb{R}^n}, e_n] + [0,1]^{n-1} \times \{0\} = [0,1]^n$, we may write

$$\psi_S\big([0,1]^{n-1}\big) = \varphi\big(S + [0,1]^{n-1}\big) = \varphi\big(\mathrm{seg}[0_{\mathbb{R}^n}, e_n] + [0,1]^{n-1}\big) = \varphi\big([0,1]^n\big) = 0,$$

where in the last equality we used the fact that φ vanishes for the unit cube of $\mathbb{R}^n$.

Summarizing, ψ_S is a simple valuation on $CC^{n-1} \cup \{\emptyset\}$ which vanishes in the unit cube of $\mathbb{R}^{n-1}$, is invariant under proper rigid motions, and whose restriction to CC^{n-1} is continuous. Hence, it follows from our induction hypothesis that $\psi_S = 0$, and this yields that

$$\varphi(C + S) = 0$$

holds for any $C \in CC^{n-1} \times \{0\}$. Next, we argue that this can be extended for positive multiples of S, that is,

$$\varphi(C + \lambda S) = 0$$

for any $\lambda \geq 0$ and every $C \in CC^{n-1} \times \{0\}$. To verify this, fix any $C \in CC^{n-1} \times \{0\}$, and let $g_C \colon [0, +\infty) \to \mathbb{R}$ be given as

$$g_C(\lambda) = \varphi(C + \lambda S)$$

for each $\lambda \geq 0$. We have that $g_C(0) = \varphi(C) = 0$ (because φ is a simple valuation), and that $g_C(1) = \varphi(C + S) = 0$. We claim that g_C is additive (in the same sense as the function g defined above to deal with the case $n = 1$). In fact, for any $\lambda_1, \lambda_2 \geq 0$, we clearly may write

$$C + (\lambda_1 + \lambda_2)S = (C + \lambda_1 S) \cup (C + \lambda_1 e_n + \lambda_2 S),$$

and it is also immediate that the intersection between the sets $C + \lambda_1 S$ and $C + \lambda_1 e_n + \lambda_2 S$ is the set $C + \lambda_1 e_n$, which is not full-dimensional (because it is a translate of C). Hence, using the fact that φ is a simple translation-invariant valuation, we get

$$\begin{aligned} g_C(\lambda_1 + \lambda_2) = \varphi(C + (\lambda_1 + \lambda_2)S) = \varphi((C + \lambda_1 S) \cup (C + \lambda_1 e_n + \lambda_2 S)) = \\ = \varphi((C + \lambda_1 S) \cup (C + \lambda_1 e_n + \lambda_2 S)) + \varphi((C + \lambda_1 S) \cap (C + \lambda_1 e_n + \lambda_2 S)) = \\ = \varphi(C + \lambda_1 S) + \varphi(C + \lambda_1 e_n + \lambda_2 S) = g_C(\lambda_1) + \varphi(C + \lambda_2 S) = g_C(\lambda_1) + g_C(\lambda_2). \end{aligned}$$

Since g_C is continuous (again, this follows straightforwardly from the continuity of φ in the Hausdorff metric), we repeat the argument of the case $n = 1$ to get that g_C vanishes identically. Since $C \in CC^{n-1}$ was arbitrarily given, we get that the equality

$$\varphi(C + \lambda S) = 0$$

holds for any $C \in CC^{n-1}$ and every $\lambda \geq 0$. Due to the translation-invariance of φ, this extends for every $\lambda \in \mathbb{R}$ (in fact, simply write $C - \lambda S = -\lambda e_n + C + \lambda S$ for $\lambda \geq 0$).

In other words, φ vanishes on any orthogonal cylinder whose base is a compact and convex set of $\mathbb{R}^{n-1} \times \{0\}$. From the invariance of φ under proper rigid motions, it follows immediately that, more generally, φ vanishes on any orthogonal cylinder of $\mathbb{R}^n$ having convex base. Again in other words, if $\mathbf{h}$ is any (affine) hyperplane orthogonal to a vector $u \in \mathbb{S}^{n-1}$, then

$$\varphi\big(C + \lambda \mathrm{seg}[0_{\mathbb{R}^n}, u]\big) = 0$$

for any compact convex set $C \subseteq \mathbf{h}$ and every $\lambda \in \mathbb{R}$.

Now we deal with non-orthogonal cylinders (but still having convex base). Let $\mathbf{h}$ be any hyperplane, and let $v \in \mathbb{S}^{n-1} \setminus \mathbf{h}$ be a vector which is not orthogonal to $\mathbf{h}$. Write $S := \mathrm{seg}[0_{\mathbb{R}^n}, v]$, and let $C \subseteq \mathbf{h}$ be a non-empty compact convex set. We first argue that, in this case, we still have that the map $g_C \colon [0, +\infty) \to \mathbb{R}$ given by $g_C(\lambda) = \varphi(C + \lambda S)$ is additive. Indeed, for any $\lambda_1, \lambda_2 \geq 0$, we still can decompose

$$C + (\lambda_1 + \lambda_2)S = (C + \lambda_1 S) \cup (C + \lambda_1 v + \lambda_2 S),$$

and the intersection $(C + \lambda_1 S) \cap (C + \lambda_1 v + \lambda_2 S)$ equals $C + \lambda_1 v$, which is not full-dimensional. Hence, the additivity of g_C comes by repeating the argument used for the orthogonal case.

It is immediate to see that if $m \in \mathbb{N}$ is large enough, then we may consider a hyperplane $\mathbf{g}$ orthogonal to v cutting the cylinder $C + mS$ in such a way that $C \subseteq \mathbf{g}^-$ and $C + mv \subseteq \mathbf{g}^+$. For simplicity, write $A := C + mS$. Then we obviously may write $A = (A \cap \mathbf{g}^-) \cup (A \cap \mathbf{g}^+)$, and φ vanishes in the intersection $(A \cap \mathbf{g}^-) \cap (A \cap \mathbf{g}^+)$, since it lies in $\mathbf{g}$. Thus, we may write

$$\varphi(A) = \varphi\big((A \cap \mathbf{g}^-) \cup (A \cap \mathbf{g}^+)\big) = \varphi(A \cap \mathbf{g}^-) + \varphi(A \cap \mathbf{g}^+).$$

Let $E := (A \cap \mathbf{g}^-) + mv$. It is readily seen that the union $(A \cap \mathbf{g}^+) \cup E$ is the orthogonal cylinder $(A \cap \mathbf{g}) + mS$, and that the intersection $(A \cap \mathbf{g}^+) \cap E$ lies in a hyperplane (see Figure 11.3.1). Therefore, recalling that φ is simple, translation invariant and vanishes on orthogonal cylinders, and using the equality above, we arrive at

$$\begin{aligned} 0 = \varphi\big((A \cap \mathbf{g}) + mS\big) = \varphi\big((A \cap \mathbf{g}^+) \cup E\big) = \varphi(A \cap \mathbf{g}^+) + \varphi(E) - \varphi\big((A \cap \mathbf{g}^+) \cap E\big) = \\ = \varphi(A \cap \mathbf{g}^+) + \varphi\big((A \cap \mathbf{g}^-) + mv\big) = \varphi(A \cap \mathbf{g}^+) + \varphi(A \cap \mathbf{g}^-) = \varphi(A). \end{aligned}$$

Thus, from the additivity of g_C, it follows that

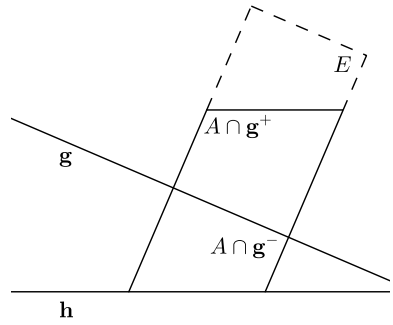


Fig. 11.3.1 Replacing a "piece" from the bottom to the top yields an orthogonal cylinder.

$$0 = \varphi(A) = \varphi(C + mS) = m\varphi(C + S) = mg_C(1),$$

and we get that $g_C(1) = 0$. Since g_C is clearly also continuous, from the same argument used before, we get that $g_C(\lambda) = 0$ for every $\lambda \geq 0$, that is,

$$\varphi(C + \lambda S) = 0$$

for any $\lambda \geq 0$. Moreover, writing $C - \lambda S = -2\lambda v + C + \lambda S$ (for $\lambda \geq 0$), from the translation-invariance of φ we obtain that the equality above holds for any $\lambda \in \mathbb{R}$.

Since this argument holds for any hyperplane $\mathbf{h}$ of $\mathbb{R}^n$, every non-empty set $C \subseteq \mathbf{h}$ which is compact and convex, and each vector $v \in \mathbb{S}^{n-1} \setminus \mathbf{h}$, we get that φ vanishes for any cylinder with convex base (not only for the orthogonal ones). Of course, the segment S need not to be in the form $S = \mathrm{seg}[0_{\mathbb{R}^n}, v]$. In fact, to give a formal explanation, notice that for any $p, q \in \mathbb{R}^n$ we may write $\mathrm{seg}[p, q] = p + \mathrm{seg}[0_{\mathbb{R}^n}, q - p]$. Thus, the translation-invariance of φ gives that

$$\varphi(C + \mathrm{seg}[p, q]) = \varphi(C + \mathrm{seg}[0_{\mathbb{R}^n}, q - p]) = 0,$$

whenever $C \subseteq \mathbb{R}^n$ is a non-empty compact convex set whose dimension is strictly smaller than n. Notice very carefully that this equality still holds true if the segment $\mathrm{seg}[p, q]$ is contained in the affine hull of C (this case was not considered previously, since we fixed $v \in \mathbb{S}^{n-1} \setminus \mathbf{h}$), since in this case $C + \mathrm{seg}[p, q]$ is not full-dimensional (and φ is simple).

Now let $P \subseteq \mathbb{R}^n$ be a polytope, and S be any segment. If $\dim(P) \leq n - 1$, then $P + S$ is a cylinder with convex base, and hence we have that $\varphi(P + S) = 0$. We want to understand what happens if P is proper. For that sake, first assume that $S = \mathrm{seg}[-v, v]$ for some $v \in \mathbb{R}^n$. If v is transversal to the affine hull of a given facet F of P, then either $F + \mathrm{seg}[0_{\mathbb{R}^n}, v]$ or $F + \mathrm{seg}[0_{\mathbb{R}^n}, -v]$ is a cylinder with convex base which does not intersect $\mathrm{int}(P)$. The intersection of any pair of such cylinders is not full-dimensional, as well as the intersection of any of these cylinders with P (in fact, this intersection is clearly the corresponding facet). Moreover, each of these cylinders is clearly a polytope (because it is the Minkowski sum of polytopes). Since the boundary of P is the union of its facets, the set $P + S$ can be written as the union of P with these cylinders. To be completely formal, we actually need to verify that any point of $(P + S) \setminus P$ lies in some of these "external" cylinders. We outline an argument now, and this is illustrated in Figure 11.3.2. Up to translation, we may assume that $P + S$ lies in the interior of one of the half-spaces determined by $v^\perp$. The orthogonal projections P and $P + S$ onto $v^\perp$ are the same, and for each point of $z \in (P + S) \setminus P$, we consider the half-line ℓ_z starting at $\mathrm{proj}_{v^\perp}(z)$ and passing through z. If z lies "after" the last intersection of ℓ_z with P, then z is in the external cylinder determined by some facet containing this last intersection, and if z lies "before" the first intersection

of ℓ_z with P, then z belongs to the external cylinder given by some facet containing this intersection. If z lies between the first and last intersections of ℓ_z with P, then z is a point of P.

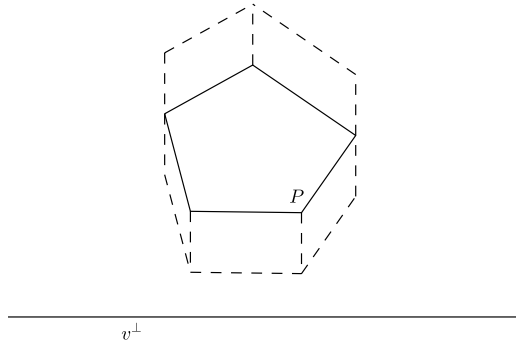


Fig. 11.3.2 The decomposition of $P + S$ as the union of P with "external" cylinders.

It follows that letting $P = P_1$ and denoting the "external" cylinders by $P_2, \ldots, P_m$, we have

$$P + S = P_1 \cup P_2 \cup \ldots \cup P_m.$$

Since φ is a valuation on $CC^n \cup \{\emptyset\}$, we have that the restriction of φ to $\mathcal{P}^n \cup \{\emptyset\}$ is also a valuation, and hence it is weakly additive. It follows from Theorem 11.3.2 that

$$\varphi(P + S) = \varphi(P_1 \cup \ldots \cup P_m) = \sum_{I \in S(m)} (-1)^{|I|-1} \varphi(P_I).$$

Since φ is simple, because any intersection different P_j's is either empty or has dimension smaller than n, and since P_j is a cylinder with convex base whenever $2 \leq j \leq n$, we get from the equality above that

$$\varphi(P + S) = \varphi(P_1) + \varphi(P_2) + \ldots + \varphi(P_m) = \varphi(P_1) = \varphi(P).$$

Since any segment of $\mathbb{R}^n$ can be written as a translation of some segment having the form $\mathrm{seg}[-v, v]$, and because φ is translation-invariant, we get that the equality above holds for any segment S.

If $K \in CC^n$, then we may let $(P_j)_{j \in \mathbb{N}}$ be a sequence of polytopes such that $P_j \to K$ in the Hausdorff metric. In this case, it is readily seen that for any segment S, we also have the convergence $P_j + S \to K + S$. Thus, from the continuity of φ, we obtain

$$\varphi(K + S) = \lim_{j \to \infty} \varphi(P_j + S) = \lim_{j \to \infty} \varphi(P_j) = \varphi(K).$$

Applying this equality iteratively, we get that $\varphi(K + W) = \varphi(W)$ for any $K \in CC^n$ and every zonotope $W \subseteq \mathbb{R}^n$ (recall that zonotopes are Minkowski sums of segments). This also yields that $\varphi(W) = 0$ for every zonotope $W \subseteq \mathbb{R}^n$. Indeed, if we write $W = S_1 + \ldots + S_m$, where each S_j is a segment, then since adding a segment does not modify the value of φ, we get

$$\varphi(W) = \varphi(S_1 + \ldots + S_m) = \varphi(S_1 + \ldots + S_{m-1}) = \ldots = \varphi(S_1) = 0,$$

where in the last equality we used the fact that S_1 is not full-dimensional (we are assuming that $n \geq 2$).

From the continuity of φ in the Hausdorff metric, it follows that $\varphi(K+Z) = \varphi(K)$ for every $K \in CC^n$ and each zonoid $Z \subseteq \mathbb{R}^n$ (remember that zonoids are limits of sequences of zonotopes in the Hausdorff metric). Even more, because φ vanishes for zonotopes, we get (using again continuity) that $\varphi(Z) = 0$ for every zonoid $Z \subseteq \mathbb{R}^n$. If $K \in CC^n$ is a generalized zonoid, then writing $K + Z_1 = Z_2$ for zonoids $Z_1, Z_2 \subseteq \mathbb{R}^n$, we get that

$$\varphi(K) = \varphi(K + Z_1) = \varphi(Z_2) = 0.$$

Thus, from Theorem 10.2.4, it follows that $\varphi(K) = 0$ for every centrally symmetric convex body $K \subseteq \mathbb{R}^n$ (this is also true if K is not full-dimensional, as a consequence of the fact that φ is simple, of course).

Now we prove that φ vanishes for simplices. Let $X := \operatorname{conv}\{0_{\mathbb{R}^n}, x_1, \ldots, x_n\}$ be an n-dimensional simplex, and let $x := x_1 + \ldots + x_n$. Consider the simplex $Y := x - X = \operatorname{conv}\{x, x - x_1, \ldots, x - x_n\}$. Let P be the parallelotope

$$P = \operatorname{seg}[0_{\mathbb{R}^n}, x_1] + \ldots + \operatorname{seg}[0_{\mathbb{R}^n}, x_n].$$

Hence it is clear that $X, Y \subseteq P$. Moreover, it is easy to see that P is the set of points $z \in \mathbb{R}^n$ which can be written as

$$z = \lambda_1 x_1 + \ldots + \lambda_n x_n,$$

for $0 \leq \lambda_1, \ldots, \lambda_n \leq 1$. In this case, we have immediately that $0 \leq \lambda_1 + \ldots + \lambda_n \leq n$. For such a point $z \in P$ written in this way, we have the following possibilities:

(i) $0 \leq \lambda_1 + \ldots + \lambda_n \leq 1$. It follows straightforwardly from the definition of X that this happens if and only if $z \in X$.

(ii) $n - 1 \leq \lambda_1 + \ldots + \lambda_n \leq n$. We claim that this holds true if and only if $z \in Y$. To verify that, first let $z \in Y$. Then we may write

$$\lambda_1 x_1 + \ldots + \lambda_n x_n = z = \alpha_0 x + \alpha_1 (x - x_1) + \ldots + \alpha_n (x - x_n)$$

for some $0 \leq \alpha_0, \ldots, \alpha_n \leq 1$ such that $\alpha_0 + \ldots + \alpha_n = 1$. Thus,

$$\lambda_1 x_1 + \ldots + \lambda_n x_n = x - \alpha_1 x_1 - \ldots - \alpha_n x_n = (1 - \alpha_1) x_1 + \ldots + (1 - \alpha_n) x_n,$$

and since $x_1, \ldots, x_n$ are affinely independent points (because X is an n-dimensional simplex), we have that $\lambda_j = 1 - \alpha_j$ for each $j = 1, \ldots, n$. Then

$$\lambda_1 + \ldots + \lambda_n = n - (\alpha_1 + \ldots + \alpha_n) = n - (1 - \alpha_0) = n - 1 + \alpha_0,$$

and since $0 \leq \alpha_0 \leq 1$, it follows that $n - 1 \leq \lambda_1 + \ldots + \lambda_n \leq n$. Conversely, if these inequalities hold, then defining $\alpha_j := 1 - \lambda_j$ for each $j \in \{1, \ldots, n\}$ we have that $0 \leq \alpha_j \leq 1$ for each j, and

$$\alpha_1 + \ldots + \alpha_n = n - (\lambda_1 + \ldots + \lambda_n),$$

from which, we get that

$$0 \leq \alpha_1 + \ldots + \alpha_n \leq 1.$$

Therefore, letting $\alpha_0 := 1 - (\alpha_1 + \ldots + \alpha_n)$, we have that $0 \leq \alpha_0 \leq \alpha_1$. It follows that

$$\alpha_0 x + \alpha_1(x - x_1) + \ldots + \alpha_n(x - x_n) \in Y,$$

but

$$\alpha_0 x + \alpha_1(x - x_1) + \ldots + \alpha_n(x - x_n) = (\alpha_0 + \alpha_1 + \ldots + \alpha_n)x - \alpha_1 x_1 - \ldots - \alpha_n x_n =$$
$$= x - \alpha_1 x_1 - \ldots - \alpha_n x_n = (1 - \alpha_1)x_1 + \ldots + (1 - \alpha_n)x_n = \lambda_1 x_1 + \ldots + \lambda_n x_n = z,$$

yielding that $z \in Y$.

(iii) $1 \leq \lambda_1 + \ldots + \lambda_n \leq n - 1$. We denote by Q the set of points $z = \lambda_1 x_1 + \ldots + \lambda_n x_n \in P$ for which these inequalities hold.

It follows that $P = X \cup Q \cup Y$, and that the intersections $X \cap Q$ and $Y \cap Q$ are determined by the equalities $\lambda_1 + \ldots + \lambda_n = 1$ and $\lambda_1 + \ldots + \lambda_n = n - 1$, respectively. The hyperplanes spanned by the points $x_1, \ldots, x_n$ and $x - x_1, \ldots, x - x_n$, respectively, are obviously parellel. Let v be a vector which is orthogonal to these hyperplanes, and up to rescaling if necessary, assume that

$$\langle x_j, v \rangle = 1$$

for every $j \in \{1, \ldots, n\}$. Then the hyperplane containing the points $x_1, \ldots, x_n$ is the hyperplane $\mathbf{h}_1 := \mathbf{h}(v, 1)$, and the hyperplane containing the points $x - x_1, \ldots, x - x_n$ is $\mathbf{h}_2 := \mathbf{h}(v, \langle x, v \rangle - 1)$. From **(i)** and **(ii)**, we obtain that

$$X = P \cap \mathbf{h}_1^- \text{ and } Y = P \cap \mathbf{h}_2^+.$$

Of course, we also have that $Q = P \cap \mathbf{h}_1^+ \cap \mathbf{h}_2^-$, and this gives that Q is a polytope. Even more, the points of $X \cap Q$ are characterized by the equality $\lambda_1 + \ldots + \lambda_n = 1$, and hence we have that $X \cap Q \subseteq \mathbf{h}_1$. Similarly, we get that $Y \cap Q \subseteq \mathbf{h}_2$. Because $P = X \cup Q \cup Y$, and since these sets are polytopes, we get that

$$\varphi(P) = \varphi(X) + \varphi(Y) + \varphi(Q) - \varphi(X \cap Q) - \varphi(Y \cap Q) - \varphi(X \cap Y).$$

Since P is centrally symmetric, we get that $\varphi(P) = 0$, and because $X \cap Y$ is either the empty set (for $n \geq 3$) or a segment (if $n = 2$), we also have that $\varphi(X \cap Y) = 0$. Since $X \cap Q$ and $Y \cap Q$ are contained in hyperplanenes, they are not full-dimensional, and then it follows that $\varphi(X \cap Q) = \varphi(Y \cap Q) = 0$. It is not difficult to see that Q is symmetric with respect to the point $x/2$, and hence $\varphi(Q) = 0$. Finally, using the fact that $Y = x - X$ and recalling that φ is translation-invariant, we arrive at

$$\varphi(X) + \varphi(-X) = 0.$$

If n is even, then $-\mathrm{Id}_{\mathbb{R}^n} \in \mathrm{SO}(n)$, and then we get $\varphi(X) = \varphi(-X)$. This, together with the equality above, leads to $\varphi(X) = 0$.

The same holds if n is odd, but to prove that, we need another strategy (since in this case the map $-\mathrm{Id}_{\mathbb{R}^n}$ is not a proper rigid motion). Still considering that $X = \mathrm{conv}\{0_{\mathbb{R}^n}, x_1, \ldots, x_n\}$ is an n-simplex, let z be the center of the ball (B, say) inscribed in X (see Exercise 3.23 for the existence of such a ball). Letting $F_1, \ldots, F_{n+1}$ be the facets of X, for each $j \in \{1, \ldots, n+1\}$ we denote by z_j the touching point of B and F_j. For any $i \neq j$ such that $F_i \cap F_j$ is non-empty, we consider the convex hull $E_{i,j} := \mathrm{conv}((F_i \cap F_j) \cup \{z, z_i, z_j\})$. It is readily seen that $E_{i,j}$ is invariant under reflection around the hyperplane spanned by $F_i \cap F_j$ and z. Hence, we get that $-E_{i,j}$ can be written as the image of $E_{i,j}$ under a composition of a reflection through a point with the reflection around a hyperplane (indeed, the reflection with respect to the hyperplane keeps $E_{i,j}$ unaltered, and the reflection through the point

leads to $-E_{i,j}$). Since we are in odd dimension, both reflections have determinant -1, and hence their composition is a proper rigid motion. Consequently, $-E_{i,j}$ is the image of $E_{i,j}$ under a proper rigid motion, and this gives that $\varphi(E_{i,j}) = \varphi(-E_{i,j})$. We may renumber the sets $E_{i,j}$ as $E_1, \ldots, E_m$, and it is clear that $X = E_1 \cup \ldots \cup E_m$, and that any intersection $E_p \cap E_q$ is not full-dimensional (for $p \neq q$, of course). It follows that

$$\varphi(X) = \varphi(E_1 \cup \ldots \cup E_m) = \varphi(E_1) + \ldots + \varphi(E_m) = \varphi(-E_1) + \ldots + \varphi(-E_m) = \\ = \varphi\big((-E_1) \cup \ldots \cup (-E_m)\big) = \varphi(-X),$$

and hence we also have that $\varphi(X) = 0$ in the case where the dimension n is odd.

Finally, since every proper polytope can be decomposed as a union of n-dimensional simplices $X_1, \ldots, X_m$ such that $X_i \cap X_j$ has dimension smaller than n whenever $i \neq j$ (see Exercise 3.24), and using again the fact that φ is fully additive on polytopes, we get that

$$\varphi(P) = \varphi(X_1) + \ldots + \varphi(X_m) - \sum_{i \neq j} \varphi(X_i \cap X_j) = 0.$$

That is, φ vanishes on polytopes. Since polytopes are dense in $\mathcal{K}^n$, it follows from the continuity of φ that $\varphi(K) = 0$ for any convex body $K \subseteq \mathbb{R}^n$. Once we already know that φ vanishes for compact convex sets which are not full-dimensional (because it is simple), we get that φ vanishes in CC^n, and the proof is complete.

□

Proof of Theorem 11.3.1. Recall that the intrinsic volume $V_0 \colon CC^n \to \mathbb{R}$ is identically 1 for any dimension $n \geq 1$ (see (6.4.8)). Hence, we extend it to be 1 also for dimension $n = 0$ (remember also that intrinsic volumes are invariant under natural embeddings into higher dimensional spaces). By doing that, the result becomes trivial in the zero-dimensional case. Indeed, if $\varphi(\{0\}) = c$, then $\varphi = cV_0$ (once $CC^0 = \{0\}$, and we get that φ is completely determined by its value in $\{0\}$).

We proceed by induction on the dimension. Fix $n \geq 1$ and assume that the theorem holds true for any dimension smaller than n. As in the proof of Proposition 11.3.2, we naturally identify the hyperplane $\mathbf{h} := \mathbb{R}^{n-1} \times \{0\}$ with $\mathbb{R}^{n-1}$, and recall that by doing that we also have a natural identification between CC^{n-1} and the family of non-empty compact convex subsets of $\mathbf{h}$. To avoid any confusion, we will denote this family by $CC(\mathbf{h})$. Under this identifications, it is clear that the restriction of φ to $CC(\mathbf{h}) \cup \{\emptyset\}$ leads to a valuation of $CC^{n-1} \cup \{\emptyset\}$ which is continuous in the Hausdorff metric and invariant under proper rigid motions (to verify this, simply adjust the arguments used for the mapping ψ_S in the proof of Proposition 11.3.2). Therefore, from the induction hypothesis, we get that there exist constants $c_0, \ldots, c_{n-1} \in \mathbb{R}$ such that

$$\varphi(C) = \sum_{i=0}^{n-1} c_i V_i(C), \tag{11.3.14}$$

for every $C \in CC(\mathbf{h})$. Notice that we are also using the fact that any intrinsic volume of C is the same when calculated regarding C as a subset of $\mathbb{R}^{n-1}$ or of $\mathbf{h}$ (that is, of $\mathbb{R}^n$).

Since φ and the V_i's are all invariant under proper rigid motions, and since any hyperplane of $\mathbb{R}^n$ is the image of $\mathbf{h}$ by some proper rigid motion, it follows that (11.3.14) holds for any $C \in CC^n$ such that $\dim(C) < n$. To deal with convex bodies, we define the function

$$\psi(C) = \varphi(C) - \sum_{i=0}^{n-1} c_i V_i(C) - c_n V_n(C),$$

where the constant c_n is chosen as

$$c_n := \frac{1}{V_n([0,1]^n)}\left(\varphi([0,1]^n) - \sum_{i=0}^{n-1} V_i([0,1]^n)\right).$$

This choice gives immediately that $\psi([0,1]^n) = 0$. It is also clear that ψ is a continuous valuation which is invariant under proper rigid motions. Indeed, ψ is a linear combination of valuations having these properties. Moreover, from (11.3.14) and because $V_n(C) = 0$ whenever C is not full-dimensional (in fact, V_n is just the Lebesgue volume), we have that ψ is simple. Thus, it follows from Proposition 11.3.2 that $\psi = 0$, and this concludes the proof.

□

As it was mentioned, Hadwiger's theorem is the most central theorem in the theory of valuations. To illustrate its strength, notice that the space $\mathbf{Val}_n$ consisting of the valuations $\varphi\colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ which are continuous and invariant under proper rigid motions is clearly a vector space over $\mathbb{R}$ (since all of these properties are maintained under linear combinations). It is *a priori* not clear whether $\mathbf{Val}_n$ is finite-dimensional, but it turns out that Hadwiger's theorem gives a complete description of this vector space. Indeed, we have

$$\mathbf{Val}_n = \operatorname{span}\{V_0, \ldots, V_n\},$$

and as a consequence, we get that $\mathbf{Val}_n$ is isomorphic to $\mathbb{R}^{n+1}$. From the heuristic viewpoint, Hadwiger's theorem states that the intrinsic volumes are essentially the unique ways of "measuring size" of sets from CC^n (if we agree that continuity, invariance under proper rigid motions, and property (11.3.1) are "natural" requirements for any functional which "measures size" of compact convex sets).

Remember that the i-th intrinsic volume is a positively homogeneous of degree i functional, meaning that $V_i(tC) = t^i V_i(C)$ for each $C \in CC^n$ and any $t > 0$. Hence, the expression of a valuation as a linear combination of intrinsic volumes can also be seen as a combination of homogeneous functionals of degree $0, \ldots, n$. In particular, we have the following corollary (of Hadwiger's theorem).

Corollary 11.3.2. *Let $\varphi\colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ be a valuation which is continuous, invariant under proper rigid motions and positively homogeneous of degree i for some $i \in \{0, \ldots, n\}$. Then there exists a constant $c \in \mathbb{R}$ such that*

$$\varphi = cV_i.$$

Proof. First observe that if $\varphi \in \mathbf{Val}_n$ is such that $\varphi(K) = 0$ for every convex body, then $\varphi(C) = 0$ for any $C \in CC^n$. Indeed, fixing $K \in \mathcal{K}^n$, if φ vanishes for convex bodies then $\varphi(tK) = 0$ for every $t > 0$. Hence, the expression for φ given in Hadwiger's theorem and the homogeneity of the intrinsic volumes give that the equality

$$0 = \sum_{j=0}^{n} t^j c_j V_j(K)$$

holds for every $t > 0$. Thus, the polynomial in the right-hand side must be zero. Since K is a convex body, we have $V_j(K) \neq 0$ for every j, which leads to $c_j = 0$ for every j. This means that $\varphi = 0$.

We may assume that φ does not vanish identically, since, in this case, there is nothing to prove. By doing this, we let $K \in \mathcal{K}^n$ be such that $\varphi(K) \neq 0$. From Hadwiger's theorem, we may write

$$\varphi = \sum_{j=0}^{n} c_j V_j$$

for some coefficients $c_0, \ldots, c_n \in \mathbb{R}$. Because $\varphi(tK) = t^i \varphi(K)$ for any $t > 0$, and using the homogeneity of the intrinsic volumes, we arrive at the equality

$$\sum_{j=0}^{n} t^j c_j V_j(K) = t^i \sum_{j=0}^{n} c_j V_j(K).$$

Because this equality holds for every $t > 0$, we get that the polynomials on the left and right-hand sides must be the same (i.e., they must have the same degree and the same coefficients). The polynomial on the right-hand side has the form αt^i, and hence the coefficients of the terms of degree different of i must vanish in the left-hand side polynomial. That is, it holds that

$$c_j V_j(K) = 0$$

for every $j \neq i$. Because K is a convex body, we have $V_j(K) \neq 0$ for every j. Thus, $c_j = 0$ for every $j \neq i$. Returning to the expression of φ, this yields

$$\varphi = c_i V_i,$$

and the proof is complete.

□

Hadwiger's theorem has a lot of important applications (it can be used, e.g., to derive Crofton's formulas of integral geometry), different versions and extensions. For example, the so-called *second characterization theorem of Hadwiger* states that the continuity hypothesis can be replaced by *monotonicity* (a valuation φ is *monotone* if $\varphi(C) \leq \varphi(D)$ whenever $C \subseteq D$). As an illustration, we finish this section by explaining how we may obtain the equality

$$w(C) = \frac{2}{\kappa_n} W_{n-1}(C)$$

for the mean width $w(C)$ of a set $C \in CC^n$ (recall that this equality was obtained as a consequence of Corollary 7.1.1, which relies on the theory of mixed surface area measures). Indeed, we know that the functional $w \colon CC^n \to \mathbb{R}$ which associates each set $C \in CC^n$ to its mean width is given as

$$w(C) = 2 \int_{\mathbb{S}^{n-1}} h_C(u)\, d\sigma(u).$$

Therefore, we have the following items:

(i) w is continuous in the Hausdorff metric because if $C_i \to C$, then $h_{C_i} \to h_C$ in the maximum norm of $C(\mathbb{S}^{n-1})$.

(ii) w is invariant under proper rigid motions, since for any $x_0 \in \mathbb{R}^n$ and $T \in \mathrm{SO}(n)$ we have $h_{x_0+TC}(u) = \langle x_0, u\rangle + h_C(T^*u)$. In fact, from this equality we have

$$w(x + TC) = 2 \int_{\mathbb{S}^{n-1}} \langle x_0, u\rangle + h_C(T^*u)\, d\sigma(u) = 2 \int_{\mathbb{S}^{n-1}} h_C(v)\, d\sigma(v) = w(C),$$

where we used the fact that σ is invariant under orthogonal transformations.

(iii) w is positively homogeneous of degree 1, since $h_{tC} = t h_C$ for any $C \in CC^n$ and every $t > 0$.

(iv) If we assume that $w(\emptyset) = 0$, then w is a valuation. This follows straightforwardly from Lemma 6.2.2.

From Corollary 11.3.2, we get immediately that

$$w = cV_1$$

for some $c \in \mathbb{R}$. To find the correct value of c, we consider the case for the unit ball B^n. We obviously have that $w(B^n) = 2$, and from the definition of V_1, we may also calculate

$$V_1(B^n) = \binom{n}{n-1} \frac{1}{\kappa_{n-1}} W_{n-1}(B^n) = \frac{n}{\kappa_{n-1}} V(B^n, \dots, B^n) = \frac{n\kappa_n}{\kappa_{n-1}}.$$

Then

$$c = \frac{w(B^n)}{V_1(B^n)} = \frac{2\kappa_{n-1}}{n\kappa_n}.$$

Finally, using again the definition of V_1, we arrive at

$$w = \frac{2\kappa_{n-1}}{n\kappa_n} V_1 = \frac{2\kappa_{n-1}}{n\kappa_n} \binom{n}{n-1} \frac{1}{\kappa_{n-1}} W_{n-1} = \frac{2}{\kappa_n} W_{n-1},$$

as we wanted to prove.

11.4 Inequalities for covering numbers

Roughly speaking, here we are interested in investigating how many copies of a given convex body are necessary to cover another given convex body. More specifically,

Definition 11.4.1. Let K and L be convex bodies in $\mathbb{R}^n$. The *covering number* of K by L is the number

$$N(K, L) = \min \left\{ m \in \mathbb{N} : \exists\, x_1, \dots, x_m \in \mathbb{R}^n \text{ such that } K \subseteq \bigcup_{j=1}^{m} (x_j + L) \right\}.$$

In other words, the covering number of K by L is the minimal number of translates of L needed to cover K. See Figure 11.4.1.

There is also a variant which demands the points x_j to be in K:

$$\bar{N}(K, L) = \min \left\{ m \in \mathbb{N} : \exists\, x_1, \dots, x_m \in K \text{ such that } K \subseteq \bigcup_{j=1}^{m} (x_j + L) \right\}.$$

The next auxiliary lemmas will be useful later.

Lemma 11.4.1. *For any convex bodies $K, L \subseteq \mathbb{R}^n$ we have*

$$N(K, (K - K) \cap (L - L)) \leq N(K, L).$$

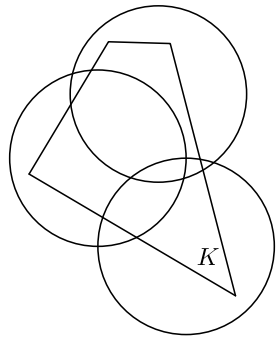


Fig. 11.4.1 Three homothets of the ball B^2 covering the quadrilateral K.

In particular, if K and L are centered at the origin, then $N(K, 2(K \cap L)) \leq N(K, L)$.

Proof. Let $N(K, L) = N$ and let $\{x_1, \ldots, x_N\} \subseteq \mathbb{R}^n$ be a (minimal) set such that

$$K \subseteq \bigcup_{j=1}^{N} (x_j + L) \tag{11.4.1}$$

Since N is minimal, it follows that for each $1 \leq j \leq N$ there exists a point $y_j \in (x_j + L) \cap A$. We claim that $(x_j + L) \cap K \subseteq y_j + (K - K) \cap (L - L)$. In fact, if $z \in (x_j + L) \cap K$, then in particular $z \in K$. Since $y_j \in K$, we get that $z - y_j \in K - K$. Also, from $z \in x_j + L$ we get $z - x_j \in L$, and from $y_j \in x_j + L$ we have $y_j - x_j \in L$. Hence $z - y_j = (z - x_j) - (y_j - x_j) \in L - L$. Finally,

$$K \subseteq \bigcup_{j=1}^{N} (x_j + L) \cap K \subseteq \bigcup_{j=1}^{N} (y_j + (K - K) \cap (L - L)),$$

where the first inclusion comes immediately by intersecting both sides of (11.4.1) with K. This shows that K can be covered by N copies of $(K - K) \cap (L - L)$, and hence the desired inequality follows. If K and L are centered at the origin, then $K - K = 2K$, $L - L = 2L$, and $(2K) \cap (2L) = 2(K \cap L)$. This concludes the proof.

□

Lemma 11.4.2. *For any convex bodies K, L and J, it holds that*

$$N(K, L) \leq N(K, J)N(J, L).$$

Proof. This follows straightforwardly from the fact that if $K \subseteq \bigcup_{j=1}^{N} (x_j + J)$ and $J \subseteq \bigcup_{k=1}^{M} (y_k + L)$, then

$$K \subseteq \bigcup_{j=1}^{N} \bigcup_{k=1}^{M} ((x_j + y_k) + L)).$$

□

We also need some volume bounds for the covering number.

Lemma 11.4.3. *If $K \subseteq \mathbb{R}^n$ is a convex body centered at the origin, $C \subseteq \mathbb{R}^n$ is a convex compact set also centered at the origin, and $s > 0$, then*

$$N(K, sB^n) \geq \frac{\mathrm{vol}_n(K + C)}{\mathrm{vol}_n((sB^n \cap K) + C)}.$$

Proof. First we will prove that if $K, D \subseteq \mathbb{R}^n$ are symmetric convex bodies and $C \subseteq \mathbb{R}^n$ is a compact convex set, then

$$\max_{x \in \mathbb{R}^n} \mathrm{vol}_n(((x + D) \cap K) + C) = \mathrm{vol}_n((D \cap K) + C). \tag{11.4.2}$$

This is a consequence of the Brunn–Minkowski inequality, as we will show. Let $L_x := (((x+D)\cap K)+C)$. It is easy to see that

$$\frac{L_x + L_{-x}}{2} \subseteq L_0 = (D \cap K) + C.$$

Also, since K, D and C are centered at the origin, we have that $\mathrm{vol}_n(L_x) = \mathrm{vol}_n(L_{-x})$. Therefore, the Brunn–Minkowski inequality gives

$$\mathrm{vol}_n(L_x)^{1/n} = \frac{1}{2}\mathrm{vol}_n(L_x)^{1/n} + \frac{1}{2}\mathrm{vol}_n(L_{-x})^{1/n} \leq \mathrm{vol}_n\left(\frac{L_x + L_{-x}}{2}\right)^{1/n} \leq$$
$$\leq \mathrm{vol}_n(L_0)^{1/n},$$

from where (11.4.2) follows. Now let $m = N(K, sB^n)$. There exist points $x_1, \ldots, x_m \in \mathbb{R}^n$ such that

$$K \subseteq \bigcup_{j=1}^{m}((x_j + sB^n) \cap K).$$

Thus,

$$K + C \subseteq \bigcup_{j=1}^{m}((x_j + sB^n) \cap K) + C.$$

This inclusion, together with (11.4.2), yields immediately

$$\mathrm{vol}_n(K + C) \leq \sum_{j=1}^{m} \mathrm{vol}_n(((x_j + sB^n) \cap K) + C) \leq$$
$$\leq N(K, sB^n) \max_{1 \leq j \leq m} \mathrm{vol}_n(((x_j + sB^n) \cap K) + C)) \leq$$
$$\leq N(K, sB^n)\mathrm{vol}_n((K \cap sB^n) + C),$$

and this concludes the proof.

□

Lemma 11.4.4. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies centered at the origin and assume that $L \subseteq bK$ for some $b \geq 1$. Then,*

$$N(\mathrm{conv}(K \cup L), (1 + 1/n)K) \leq 2bnN(L, K).$$

Proof. Let $N = N(L, K)$. Then there exist points $x_1, \ldots, x_N \in \mathbb{R}^n$ such that $L \subseteq \bigcup_{j=1}^{N}(x_j + K)$. Of course, we may assume that $L \cap (x_j + K) \neq \emptyset$ for each $1 \leq j \leq N$, and hence $x_j \in L + K \subseteq (b + 1)K$. For $\lambda \in [0, 1]$ we have

$$\lambda L + (1-\lambda)K \subseteq \bigcup_{j=1}^{N}(\lambda x_j + \lambda K) + (1-\lambda)K = \bigcup_{j=1}^{N} \lambda x_j + K.$$

This yields

$$\operatorname{conv}(L \cup K) \subseteq \bigcup_{j=1}^{N} \bigcup_{0\le\lambda\le 1} (\lambda x_j + K), \tag{11.4.3}$$

because it is easy to see that $\operatorname{conv}(L \cup K) = \bigcup_{0\le\lambda\le 1}(\lambda L + (1-\lambda)K)$. In fact, if $v = \alpha_1 y_1 + \ldots \alpha_s y_s + \beta_1 z_1 + \beta_k z_k$ is a point of $\operatorname{conv}(L \cup K)$, where $y_j \in L$ for each $1 \le j \le s$, $z_i \in K$ for each $1 \le s \le k$, and the α_j's and β_i's are non-negative numbers such that $\alpha_1 + \ldots + \alpha_s + \beta_1 + \ldots \beta_k = 1$. We may also assume that the α_j's are not all zero (as well as the β_i's), since otherwise there is nothing to prove. Writing

$$v = (\alpha_1 + \ldots + \alpha_s)\left(\frac{\alpha_1}{\alpha_1 + \ldots + \alpha_s} y_1 + \ldots + \frac{\alpha_s}{\alpha_1 + \ldots + \alpha_s} y_s\right) +$$
$$+(\beta_1 + \ldots + \beta_k)\left(\frac{\beta_1}{\beta_1 + \ldots + \beta_k} z_1 + \ldots \frac{\beta_k}{\beta_1 + \ldots + \beta_k} z_k\right).$$

Letting $\lambda := \alpha_1 + \ldots + \alpha_s$ we have that $0 \le \lambda \le 1$ and $\beta_1 + \ldots + \beta_k = 1 - \lambda$. It follows immediately that $v \in \lambda L + (1-\lambda)K$. The reverse inclusion is trivial.

For each $i = 1, 2, \ldots, \lfloor 2bn \rfloor$, define $\lambda_i = \frac{i}{2bn}$, and notice that $\lambda_i \in [0, 1]$. From (11.4.3), we have that for each $z \in \operatorname{conv}(L \cup K)$ there exists $1 \le j \le N$ and $\lambda \in [0, 1]$ such that $z \in \lambda x_j + K$. Taking λ_i such that $|\lambda - \lambda_i| \le \frac{1}{2bn}$, we get that

$$z \in \lambda_i x_j + \left(\frac{b+1}{2bn} + 1\right) K.$$

In fact, if $z = \lambda x_j + y$ for some $y \in K$, then we may write

$$z = \lambda_i x_j + (\lambda - \lambda_i) x_j + y,$$

and since $x_j \in (b+1)K$, we have that

$$(\lambda - \lambda_i) x_j \in \frac{b+1}{2bn} K,$$

where we are also using the fact that K is centered at the origin. It follows that $(\lambda - \lambda_i)x_j + y \in \left(\frac{b+1}{2bn} + 1\right) K$. Finally, this argument yields

$$\operatorname{conv}(L \cup K) \subseteq \bigcup_{i=1}^{\lfloor 2bn \rfloor} \bigcup_{j=1}^{N} \left(\lambda_i x_j + \left(\frac{b+1}{2bn} + 1\right) K\right) \subseteq \bigcup_{i=1}^{\lfloor 2bn \rfloor} \bigcup_{j=1}^{N} \left(\lambda_i x_j + \left(\frac{1}{n} + 1\right) K\right),$$

where the last inclusion holds because the condition $b \ge 1$ gives $\frac{b+1}{2b} \le 1$. It follows that $\operatorname{conv}(L \cup K)$ can be covered by $N\lfloor 2bn \rfloor$ translates of $\left(\frac{1}{n} + 1\right) K$. Therefore,

$$N(\operatorname{conv}(L \cup K), (1 + 1/n)K) \le N(L, K)\lfloor 2bn \rfloor \le 2bnN(L, K),$$

as we wanted to prove.

□

Corollary 11.4.1. *Let L be a convex body, and K be a convex body centered at the origin. If $L \subseteq bK$ for some $b \geq 1$, then*

$$\mathrm{vol}_n(\mathrm{conv}(K \cup L)) \leq 2ebnN(L, K)\mathrm{vol}_n(K).$$

Proof. Let $N = N(\mathrm{conv}(K \cup L), (1 + 1/n)K)$. From the definition of covering numbers, we may write

$$\mathrm{conv}(K \cup L) \subseteq \bigcup_{j=1}^{N}(x_j + (1 + 1/n)K),$$

from where we get

$$\mathrm{vol}_n(\mathrm{conv}(K \cup L)) \leq N\mathrm{vol}_n((1 + 1/n)K) = N(1 + 1/n)^n\mathrm{vol}_n(K)$$

Now we use the previous lemma to bound N. We get

$$\mathrm{vol}_n(\mathrm{conv}(K \cup L)) \leq N(1 + 1/n)^n\mathrm{vol}_n(K) \leq 2bnN(L, K)(1 + 1/n)^n\mathrm{vol}_n(K) \leq \\ \leq 2ebnN(L, K)\mathrm{vol}_n(K),$$

where we use the fact that the function $n \mapsto (1 + 1/n)^n$ is increasing and converges to e.

□

Corollary 11.4.2. *If $B^n \subseteq sbK$ for some $s > 0$ and some $b \geq 1$, then*

$$\mathrm{vol}_n\left(\mathrm{conv}\left(K \cup \frac{1}{s}B^n\right) + L\right) \leq 2ebnN(B^n, sK)\mathrm{vol}_n(K + L)$$

for every convex body $L \subseteq \mathbb{R}^n$ centered at the origin.

Proof. Put $B := \frac{1}{s}B^n$ and $N := N(B, K) = N(B^n, sK)$. By the proof of the previous lemma, we can take $M := N\lfloor 2bn \rfloor$ points $x_1, \ldots, x_M \in \mathbb{R}^n$ such that

$$\mathrm{conv}(K \cup B) \subseteq \bigcup_{j=1}^{M}\left(x_j + \left(1 + \frac{1}{n}\right)K\right).$$

Consequently,

$$\mathrm{conv}(K \cup B) + L \subseteq \bigcup_{j=1}^{M}\left(x_j + \left(1 + \frac{1}{n}\right)K\right) + L \subseteq \bigcup_{j=1}^{M}\left(x_j + \left(1 + \frac{1}{n}\right)(K + L)\right).$$

Therefore, since the volume is translation-invariant we get

$$\mathrm{vol}_n\big(\mathrm{conv}(K \cup B) + L\big) \leq M\mathrm{vol}_n\left(\left(1 + \frac{1}{n}\right)(K + L)\right) = \\ = \lfloor 2bn \rfloor N(B^n, sK)\left(1 + \frac{1}{n}\right)^n \mathrm{vol}_n(K + L) \leq 2ebnN(B^n, sK)\mathrm{vol}_n(K + L),$$

and this is the inequality we wanted to prove.

□

Definition 11.4.2. Let A be a convex body, and B be a convex body centered at the origin. The *separation number* of A by B is the number

$$M(A,B) := \max\left\{N \in \mathbb{N} : \exists\, x_1,\dots,x_N \in A \text{ such that } \forall j \neq i,\ x_j \notin (x_i + B)\right\},$$

or equivalently,

$$M(A,B) = \max\left\{N \in \mathbb{N} : \exists\, x_1,\dots,x_N \in A \text{ such that } ||x_i - x_j||_B > 1 \text{ if } i \neq j\right\},$$

where $||\cdot||_B$ is the Minkowski functional of B. See Figure 11.4.2. To check that this maximum indeed exists, see Exercise 11.24.

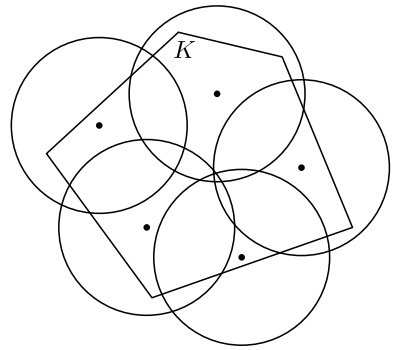


Fig. 11.4.2 Points of K separated by the ball.

In the next lemma we clarify some relations between covering and separation numbers.

Lemma 11.4.5. *Let $K, T \subseteq \mathbb{R}^n$ be convex bodies, and assume that T is centered at the origin. Then*

$$N(K,T) \leq \bar{N}(K,T) \leq M(K,T).$$

Proof. The inequality $N(K,T) \leq \bar{N}(K,T)$ comes immediately from the definitions, so all we have to prove is that $\bar{N}(K,T) \leq M(K,T)$. Let $\{x_1,\dots,x_M\} \subseteq K$ be a maximal subset of K with the property that $x_i \notin x_j + T$ whenever $i \neq j$. From the maximality, it follows that no point $x \in K$ is such that $x \notin x_j + T$ for every $1 \leq j \leq M$, which gives

$$K \subseteq \bigcup_{j=1}^{M} (x_j + T).$$

Hence $M(K,T) = M \geq \bar{N}(K,T)$.

□

Now we investigate the important *Sudakov inequality*, which estimates the covering number $N(K, tB^n)$ of a convex body K centered at the origin by a multiple of the Euclidean ball in terms of the *mean width* $w(K)$ of K (defined below). More precisely, it states that

$$\log N(K, tB^n) \leq cn\left(\frac{M(K^\circ)}{t}\right)^2 \tag{11.4.4}$$

where $c > 0$ is a universal constant. For the correct geometric interpretation, recall that the parameter $M(K^\circ)$ is just half of the *mean width* of K. For the convenience of the reader, we remember that the

width of K in the direction of $u \in \mathbb{S}^{n-1}$ is the distance between the supporting hyperplanes of K whose normal is u. This is the number $w_K(u) := h_K(u) + h_K(-u)$, as it is seen in Lemma 2.3.3. We recall that the *mean width* of K is the number

$$w(K) := \int_{\mathbb{S}^{n-1}} w_K(u)\, d\sigma(u) = 2\int_{\mathbb{S}^{n-1}} h_K(u)\, d\sigma(u),$$

where σ denotes the usual spherical probability measure on $\mathbb{S}^{n-1}$. That is, $w(K)$ is the mean of the width function of K over $\mathbb{S}^{n-1}$.

Recall that he *Gaussian measure* of $\mathbb{R}^n$ is the Borel measure γ_n with density

$$G(x) = (2\pi)^{-n/2} e^{-|x|^2/2},$$

where we also recall that $|\cdot|$ is the Euclidean norm in $\mathbb{R}^n$. In other words, the Gaussian measure of a Borel set $A \subseteq \mathbb{R}^n$ is given by

$$\gamma_n(A) = \frac{1}{(2\pi)^{n/2}} \int_A e^{-|x|^2/2}\, dx.$$

The Gaussian measure has a lot of "good" properties, as being invariant under orthogonal transformations, for example. Some of these properties were presented as exercises in Chapter 4. Here we will restrict ourselves to the properties that will be necessary later.

Lemma 11.4.6. *Let $K \subseteq \mathbb{R}^n$ be a convex body centered at the origin. For every $y \in \mathbb{R}^n$ we have the inequality*

$$\gamma_n(y + K) \geq e^{-|y|^2/2}\gamma_n(K).$$

Proof. First observe that the fact that K is centered at the origin gives $K - y = -(K + y)$, from where we have $\gamma_n(K + y) = \gamma_n(K - y)$. Thus,

$$\gamma_n(K+y) = \frac{1}{(2\pi)^{n/2}} \int_{K+y} e^{-|x|^2/2}\, dx = \frac{1}{(2\pi)^{n/2}} \int_K e^{-|x+y|^2/2}\, dx = \gamma_n(K-y) =$$
$$= \frac{1}{(2\pi)^{n/2}} \int_{K-y} e^{-|x|^2/2}\, dx = \frac{1}{(2\pi)^{n/2}} \int_K e^{-|x-y|^2/2}\, dx.$$

As a consequence, we may write

$$\gamma_n(K+y) = \frac{1}{(2\pi)^{n/2}} \int_K \frac{e^{-|x+y|^2/2} + e^{-|x-y|^2/2}}{2}\, dx.$$

From now on, we also denote the exponential function e^x by $\exp(x)$, so the notation is clearer. Since the exponential function is convex, we get immediately the inequality

$$\gamma_n(K+y) \geq \frac{1}{(2\pi)^{n/2}} \int_K \exp\left(\frac{-|x+y|^2 - |x-y|^2}{4}\right) dx =$$
$$= \frac{1}{(2\pi)^{n/2}} \int_K \exp\left(\frac{-|x|^2 - |y|^2}{2}\right) dx = \frac{e^{-|y|^2/2}}{(2\pi)^{n/2}} \int_K e^{-|x|^2/2}\, dx = e^{-|y|^2/2}\gamma_n(K).$$

□

We also need a technical lemma regarding the logarithm function.

Lemma 11.4.7. *Let $1 < \delta < 2$. There exists a constant $c_0 > 0$ such that*

$$\log(k) \leq c_0 \log(k/\delta)$$

for every $k \geq 2$.

Proof. Simply notice that the inequality in the statement is true if and only if

$$c_0 \geq \frac{\log(k)}{\log(k) - \log(\delta)}$$

for every $k \geq 2$. Since the function

$$t \mapsto \frac{\log(t)}{\log(t) - \log(\delta)}$$

is clearly continuous and decreasing on $[2, +\infty)$, the desired follows.

□

We are now in position to state and prove a "dual" version of Sudakov's inequality, that will be used to prove the "original one". For the convenience of the reader, we recall some concepts and results studied in Section 2.3. The *polar body* of a convex body $K \subseteq \mathbb{R}^n$ is the set $K^\circ := \{y \in \mathbb{R}^n : \langle x, y\rangle \leq 1 \; \forall \, x \in K\}$. If K contains the origin as an interior point, then K° is also a convex body. Even more, if K is centered at the origin, then so is K°. In this case, the norm whose unit ball is K and the support function of K° are equal, that is, $||\cdot||_K = h_{K^\circ}(\cdot)$ on $\mathbb{S}^{n-1}$ (this last fact is proved in Proposition 2.3.6).

Theorem 11.4.1. *There exists a constant $c > 0$ such that*

$$\log \bar{N}(B^n, tK) \leq cn \left(\frac{M(K)}{t}\right)^2$$

for any convex body $K \subseteq \mathbb{R}^n$ centered at the origin and every $t > 0$.

Proof. Denote by $||\cdot||_K$ the norm in $\mathbb{R}^n$ whose unit ball is K, and let $x_1, \ldots, x_N$ form a set of points in B^n which is maximal for the condition $i \neq j \Rightarrow ||x_i - x_j||_K > t$. It follows immediately that $N = M(B^n, tK)$, and that the sets $x_j + \frac{t}{2}K$ have disjoint interiors. Consequently, for any $\lambda > 0$ the sets $\lambda x_j + \frac{\lambda t}{2}K$ have disjoint interiors, and this leads to

$$1 \geq \gamma_n\left(\bigcup_{j=1}^N \left(\lambda x_j + \frac{\lambda t}{2}K\right)\right) = \sum_{j=1}^N \gamma_n\left(\lambda x_j + \frac{\lambda t}{2}K\right),$$

where we also recall that γ_n is a probability measure. Now we recall that each x_j is a point of B^n, and this gives that $|\lambda x_j| \leq \lambda$ for $1 \leq j \leq N$. Hence Lemma 11.4.6 gives

$$\gamma_n\left(\lambda x_j + \frac{\lambda t}{2}K\right) \geq e^{-\lambda^2/2}\gamma_n\left(\frac{\lambda t}{2}K\right).$$

With the previous inequality, this leads to

$$1 \geq N e^{-\lambda^2/2}\gamma_n\left(\frac{\lambda t}{2}K\right) \tag{11.4.5}$$

for every $\lambda > 0$. Notice that this gives an λ-dependent upper bound for N. To find the correct λ, we use polar coordinates $x = tu$, where $t \geq 0$ and $u \in \mathbb{S}^{n-1}$, to write

$$\begin{aligned}
\int_{\mathbb{R}^n} ||x||_K \, \gamma_n(dx) &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} ||x||_K e^{-|x|^2/2} \, dx = \\
&= \frac{1}{(2\pi)^{n/2}} \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} t^n ||u||_K e^{-t^2/2} \, d\mathcal{H}^{n-1}(u) \, dt = \\
&= \frac{n\mathrm{vol}(B^n)}{(2\pi)^{n/2}} \int_{\mathbb{S}^{n-1}} ||u||_K \, d\sigma(u) \int_0^{+\infty} t^n e^{-t^2/2} \, dt = \\
&= \frac{n}{2^{n/2}\Gamma\left(\frac{n}{2}+1\right)} \int_{\mathbb{S}^{n-1}} ||u||_K \, d\sigma(u) \int_0^{+\infty} t^n e^{-t^2/2} \, dt = \\
&= \frac{nM(K)}{2^{n/2}\Gamma\left(\frac{n}{2}+1\right)} \int_0^{+\infty} t^n e^{-t^2/2} \, dt = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{2}\Gamma\left(\frac{n}{2}+1\right)} M(K) \leq c\sqrt{n} M(K),
\end{aligned}$$

for some universal constant $c > 0$. We are using some well-known facts about the Gamma function without further comments (however, see Remark 11.4.1 below and (11.5.2) in the next section). Also, observe that we are denoting the integration with respect to the Gaussian measure by $\gamma_n(dx)$, meaning that we are integrating with respect to the Lebesgue measure dx with the density g_n that defines γ_n. Now let $\phi\colon \mathbb{R}^n \to \mathbb{R}$ be the simple function given as

$$\phi(x) := \begin{cases} \lambda t/2, & \text{if } ||x||_K \geq \lambda t/2 \text{ and} \\ 0, & \text{if } ||x||_K < \lambda t/2 \end{cases}.$$

Since $\phi(x) \leq ||x||_K$ for any $x \in \mathbb{R}^n$, we get from the definition of the Lebesgue integral that

$$\int_{\mathbb{R}^n} ||x||_K \, \gamma_n(dx) \geq \int_{\mathbb{R}^n} \phi(x) \, \gamma_n(dx) = \int_{\{||x||_K \geq \lambda t/2\}} \frac{\lambda t}{2} \, \gamma_n(dx) = \frac{\lambda t}{2} \gamma_n\Big(\{||x||_K \geq \lambda t/2\}\Big),$$

that is,

$$\gamma_n\Big(\{||x||_K \geq \lambda t/2\}\Big) \leq \frac{2}{\lambda t} \int_{\mathbb{R}^n} ||x||_K \gamma_n(dx) \leq \frac{2c\sqrt{n}}{\lambda t} M(K),$$

where we also used the previous inequality. The reader familiar with probability theory will notice that the first inequality above is simply the classical Markov inequality (see Exercise 4.19). On the other hand, the set $\{||x||_K \geq \lambda t/2\}$ is simply the complement of the body $\frac{\lambda t}{2} K$. Thus,

$$1 - \gamma_n\left(\frac{\lambda t}{2} K\right) \leq \frac{2c\sqrt{n}}{\lambda t} M(K),$$

and we recall that this holds for any $\lambda > 0$. Now choose a number $1/2 < \varepsilon < 1$. Setting $\lambda = c\sqrt{n} w(K^\circ)/(1-\varepsilon)t$ yields immediately

$$\gamma_n\left(\frac{\lambda t}{2} K\right) = \gamma_n\Big(\frac{c}{1-\varepsilon}\sqrt{n} M(K) K\Big) \geq \varepsilon.$$

This inequality together with (11.4.5) leads to

$$1 \geq \frac{N e^{-\lambda^2/2}}{\varepsilon^{-1}},$$

and replacing λ and rearranging the terms gives

$$\frac{N}{\varepsilon^{-1}} \leq \exp\left(\frac{2c^2 n M(K)^2}{(1-\varepsilon)^2 t^2}\right).$$

Now we will settle the case where $M(B^n, tK) \geq 2$. In this case, since $N = M(B^n, tK)$, it follows from Lemma 11.4.5 that

$$\log \bar{N}(B^n, tK) \leq \log N \leq c_0 \log\left(\frac{N}{\varepsilon^{-1}}\right) \leq \frac{2c_0 c^2 n}{(1-\varepsilon)^2}\left(\frac{M(K)}{t}\right)^2,$$

where $c_0 > 0$ is some constant given as in Lemma 11.4.7 (observe that $1 < \varepsilon^{-1} < 2$). It is clear that the constant $2c_0 c^2/(1-\varepsilon)^2$ does not depend on K or n. Finally, if $M(B^n, tK) = 1$, then $\bar{N}(B^n, tK) = 1$ and consequently $\log \bar{N}(B^n, tK) = 0$. Hence the desired inequality is trivial.

□

Remark 11.4.1. Using standard change of variables, and from the definition of the Gamma function for real numbers, we have

$$\int_0^{+\infty} u^n e^{-u^2/2}\, du = 2^{n/2} \int_0^{+\infty} \left(\frac{u^2}{2}\right)^{n/2} e^{-u^2/2}\, du = 2^{n/2} \int_0^{+\infty} t^{n/2} e^{-t} \frac{2^{1/2}}{t^{1/2}}\, dt =$$
$$= 2^{\frac{n+1}{2}} \int_0^{+\infty} t^{\frac{n-1}{2}} e^{-t}\, dt = 2^{\frac{n+1}{2}} \int_0^{+\infty} t^{\frac{n+1}{2}-1} e^{-t}\, dt = 2^{\frac{n+1}{2}} \Gamma\left(\frac{n+1}{2}\right).$$

The proved inequality can be equivalently re-stated as follows.

Corollary 11.4.3. *For any convex body $K \subseteq \mathbb{R}^n$ centered at the origin we have*

$$\sup_{t>0} t^2 \log \bar{N}(B^n, tK) \leq cnM(K)^2,$$

where $c > 0$ is a universal constant which does not depend on K nor n.

This leads to a certain duality between $N(K, tB^n)$ and $N(B^n, tK^\circ)$ which is described and proved next.

Lemma 11.4.8. *If $K \subseteq \mathbb{R}^n$ is a convex body centered at the origin, then*

$$N(K, tB^n) \leq N(K, 2tB^n) N\left(B^n, \frac{t}{8} K^\circ\right)$$

for any $t > 0$.

Proof. First notice that

$$(2K) \cap \frac{t^2}{2} K^\circ \subseteq tB^n.$$

Indeed, from the inequality $|x|^2 = \langle x, x\rangle \leq ||x||_K ||x||_{K^\circ}$ (proved in Proposition 2.3.6) we get that if $x \in (2K) \cap \left(\frac{t^2}{2} K^\circ\right)$, then

$$|x|^2 \leq 2\frac{t^2}{2} = t^2,$$

from where $x \in tB^n$. This inclusion gives immediately that

$$N(K, tB^n) \le N\left(K, (2K) \cap \left(\frac{t^2}{2}K^\circ\right)\right).$$

Now, since $(2K) \cap \left(\frac{t^2}{2}K^\circ\right) = 2\left(K \cap \left(\frac{t^2}{4}K^\circ\right)\right)$, we have from Lemma 11.4.1 that

$$N(K, tB^n) \le N\left(K, 2\left(K \cap \left(\frac{t^2}{4}K^\circ\right)\right)\right) \le N\left(K, \frac{t^2}{4}K^\circ\right).$$

Finally, from Lemma 11.4.2, we have

$$N(K, tB^n) \le N\left(K, \frac{t^2}{4}K^\circ\right) \le N(K, 2tB^n)\, N\left(2tB^n, \frac{t^2}{4}K^\circ\right) =$$
$$= N(K, 2tB^n) N\left(B^n, \frac{t}{8}K^\circ\right),$$

where we also used the obvious fact that $N(K, T) = N(\lambda K, \lambda T)$ for any $\lambda > 0$.

□

Theorem 11.4.2. *For every convex body $K \subseteq \mathbb{R}^n$ centered at the origin, we have*

$$\sup_{t>0} t^2 \log(N(K, tB^n) \le 100 \sup_{t>0} t^2 \log(\bar{N}(B^n, tK^\circ)).$$

As a consequence, it follows that there exists a universal constant $c > 0$ such that

$$\log N(K, tB^n) \le cn\left(\frac{M(K^\circ)}{t}\right)^2$$

for any convex body $K \subseteq \mathbb{R}^n$ centered at the origin and every $t > 0$. This is known as Sudakov's inequality.

Proof. From Lemma 11.4.8 we get, for every $t > 0$, that

$$t^2 \log N(K, tB^n) \le t^2 \log N(K, 2tB^n) + t^2 \log N\left(B^n, \frac{t}{8}K^\circ\right) =$$
$$= \frac{1}{4}(2t)^2 \log N(K, 2tB^n) + 64\left(\frac{t}{8}\right)^2 \log N\left(B^n, \frac{t}{8}K^\circ\right)$$

Taking the supremum over $t > 0$ on both sides we get

$$\sup_{t>0} t^2 \log N(K, tB^n) \le \frac{1}{4} \sup_{t>0} t^2 \log N(K, tB^n) + 64 \sup_{t>0} t^2 \log N(B^n, tK^\circ),$$

from where it follows that

$$\sup_{t>0} t^2 \log N(K, tB^n) \le \frac{256}{3} \sup_{t>0} t^2 \log N(B^n, tK^\circ) \le 100 \sup_{t>0} t^2 \log N(B^n, tK^\circ),$$

and this gives the first stated inequality. Now we recall that $N \le \bar{N}$ and use Corollary 11.4.3 for K° to obtain

$$\sup_{t>0} t^2 \log N(K, tB^n) \le 100 \sup_{t>0} t^2 \log \bar{N}(B^n, tK^\circ) \le 100cnM(K^\circ)^2,$$

and this gives Sudakov's inequality (replacing $100c$ by c).

□

11.5 The Bourgain–Milman inequality

In this section, we will prove the Bourgain–Milman inequality as an introduction to the method known as *isomorphic symmetrization*. The Bourgain–Milman inequality can be seen as a reverse Blaschke–Santaló inequality. Our reference for the proof of Bourgain–Milman inequality, as well as for asymptotic geometric analysis as a whole, is the outstanding recent book [133] by Shiri Artstein-Avidan, Apostolos Giannopoulos and Vitali Milman. For any convex body $K \subseteq \mathbb{R}^n$ centered at the origin, define

$$\bar{s}(K) := \left(\frac{\mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ)}{\mathrm{vol}_n(B^n)^2} \right)^{1/n}.$$

This is a variant of the quantity $s(K) := \mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ)$ (known as *Mahler product*). It is clear that both quantities are invariant under isomorphisms, that is,

$$s(TK) = \mathrm{vol}_n(TK)\mathrm{vol}_n((TK)^\circ) = \mathrm{vol}_n(K)\mathrm{vol}_n(K^\circ) = s(K),$$

and the same holds for $\bar{s}(K)$. This happens because $(TK)^\circ = (T^*)^{-1}K^\circ$, where T^* denotes the adjoint of T, and clearly we have $\det((T^*)^{-1}) = 1/\det(T)$. The Blaschke–Santaló inequality states that

$$s(K) \leq (\mathrm{vol}_n(B^n))^2$$

for every convex body $K \subseteq \mathbb{R}^n$ centered at the origin, with equality if and only if K is an ellipsoid. Notice that this can be also stated as $\bar{s}(K) \leq 1$. Before proving the Bourgain–Milman inequality, we need to introduce and investigate some quantities associated to convex bodies.

Let $K \subseteq \mathbb{R}^n$ be a convex body having the origin as an interior point, and recall that we denote the gauge function, radial function and the support function of K by $||\cdot||_K$, ρ_K and h_K, respectively (see Section 2.3). We define

$$b(K) := \max\{||x||_K : x \in B^n\} \text{ and}$$
$$M(K) = \int_{\mathbb{S}^{n-1}} ||x||_K \, d\sigma \left(= \frac{w(K^\circ)}{2} \right).$$

Observe that $M(K)$ is the mean of the gauge function $||\cdot||_K$ over $\mathbb{S}^{n-1}$, and recall that $w(K^\circ)$ denotes the mean width of K°. Moreover, it is easy to see that $K \subseteq K'$ implies $b(K) \geq b(K')$ and $M(K) \geq M(K')$. It is also clear that both numbers depend on the fixed Euclidean structure. Even more, we have the trivial bound $M \leq b$. Next we collect some important facts between these parameters.

Lemma 11.5.1. *For any $K \in \mathcal{K}_0^n$ we have the equality*

$$b := b(K) = \min\{c > 0 : ||x||_K \leq c|x| \ \forall \ x \in \mathbb{R}^n\}.$$

Moreover, b is the inverse of the radius of the greatest ball contained in K. In particular, there exists a point $x \in \partial K \cap \frac{1}{b}\mathbb{S}^{n-1}$, and this point has the property that $h_K(bx) = 1/b$.

Proof. Let us prove the equality. For any non-zero vector $x \in \mathbb{R}^n$, we have that $x/|x| \in B^n$. Hence

$$\left\| \frac{x}{|x|} \right\|_K \leq b,$$

and hence $||x||_K \leq b|x|$ for every $x \in \mathbb{R}^n$. Then we have the inequality

$$\min\{c > 0 : ||x||_K \leq c|x| \ \forall\ x \in \mathbb{R}^n\} \leq b.$$

Now assume that $c > 0$ is such that $||x||_K \leq c|x|$ for every $x \in \mathbb{R}^n$. This means that

$$\left\| \frac{x}{|x|} \right\|_K \leq c$$

for every non-zero $x \in \mathbb{R}^n$, which implies that $||x||_K \leq c$ for any $x \in B^n$. Hence

$$b = \max\{||x||_K : x \in B^n\} \leq c.$$

Since this holds for any $c > 0$ with the property that $||x||_K \leq c|x|$ for every $x \in \mathbb{R}^n$, it follows that

$$\min\{c > 0 : ||x||_K \leq c|x| \ \forall\ x \in \mathbb{R}^n\} \geq b,$$

and this concludes the proof of the equality.

Now assume that $r > 0$ is the greatest number such that $rB^n \subseteq K$. Then, for any non-zero vector $x \in \mathbb{R}^n$, we have that $rx/|x| \in K$. Thus,

$$\left\| \frac{rx}{|x|} \right\|_K \leq 1,$$

from which we get that $||x||_K \leq \frac{1}{r}|x|$ for every $x \in \mathbb{R}^n$. As a consequence, we have

$$b = \min\{c > 0 : ||x||_K \leq c|x| \ \forall\ x \in \mathbb{R}^n\} \leq \frac{1}{r}.$$

For the other inclusion, notice that if $x \in B^n$, then

$$\left\| \frac{x}{b} \right\|_K = \frac{||x||_K}{b} \leq 1,$$

from which it follows that $x/b \in K$ (recall that K is precisely the set of points $x \in \mathbb{R}^n$ such that $||x||_K \leq 1$). Consequently, we have that $\frac{1}{b}B^n \subseteq K$, and from the definition of r this gives $r \geq 1/b$. This leads to $b \geq 1/r$.

At this point, we know that $\frac{1}{b}\mathbb{S}^{n-1}$ is the greatest sphere contained in K. It is clear that there exists a point $x \in \partial K \cap \frac{1}{b}\mathbb{S}^{n-1}$, because otherwise the distance between these sets would be positive (both are compact). Hence an easy topological argument gives that we would be able to "enlarge" the sphere before it touches ∂K. It follows that $bx \in \mathbb{S}^{n-1}$. Let $\mathbf{h}$ be a supporting hyperplane of K at x. Then $\mathbf{h}$ is also a supporting hyperplane of $\frac{1}{b}\mathbb{S}^{n-1}$ at x. The (unique) outer normal of $\frac{1}{b}\mathbb{S}^{n-1}$ at x points in the direction of x itself. In other words, bx is the unique unit outer normal of $\frac{1}{b}\mathbb{S}^{n-1}$ at x, and hence bx is also an outer normal of K at x (it is actually the unique one). Thus,

$$h_K(bx) = \langle x, bx \rangle = b|x|^2 = \frac{1}{b},$$

and this concludes the proof.

□

Remark 11.5.1. Compare this lemma with Exercise 2.13.

Lemma 11.5.2. *For any $T \in \mathrm{O}(n)$ and every convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point we have the equalities $M(K) = M(TK)$ and $b(K) = b(TK)$.*

Proof. Since $T \in \mathrm{O}(n)$, we have that $TB^n = B^n$. Hence

$$b(K) = \max\{||x||_K : x \in B^n\} = \max\{||x||_K : x \in TB^n\} = \max\{||Ty||_K : y \in B^n\} = b(TK).$$

For the other equality, first observe that $x \in \lambda TK$ if and only if $T^{-1}x \in \lambda K$ for any $\lambda \geq 0$. Hence we get that $||x||_{TK} = ||T^{-1}x||_K$. Using standard change of variables, and since it is clear that the push-forward of σ under any orthogonal transformation is σ itself, we get

$$M(TK) = \int_{\mathbb{S}^{n-1}} ||x||_{TK}\, d\sigma(x) = \int_{\mathbb{S}^{n-1}} ||T^{-1}x||_K\, d\sigma(x) =$$
$$= \int_{T^{-1}(\mathbb{S}^{n-1})} ||y||_K\, d\sigma(Ty) = \int_{\mathbb{S}^{n-1}} ||y||_K |\det(T)|\, d\sigma(y) = \int_{\mathbb{S}^{n-1}} ||y||_K\, d\sigma(y) = M(K).$$

□

Lemma 11.5.3. *For any $\lambda > 0$ and every convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point we have*

$$M(\lambda K) = \frac{1}{\lambda} M(K).$$

As a consequence, the product $M(K)M(K^\circ)$ is invariant under rescaling, that is,

$$M(K)M(K^\circ) = M(\lambda K)M((\lambda K)^\circ)$$

for any $\lambda > 0$ and every convex body K containing the origin as an interior point.

Proof. This is a consequence of the fact that $||\cdot||_{\lambda K} = \frac{1}{\lambda}||\cdot||_K$. We have

$$M(\lambda K) = \int_{\mathbb{S}^{n-1}} ||x||_{\lambda K}\, d\sigma = \frac{1}{\lambda} \int_{\mathbb{S}^{n-1}} ||x||_K\, d\sigma = \frac{1}{\lambda} M(K).$$

Now, since $(\lambda K)^\circ = \frac{1}{\lambda} K^\circ$, we may write

$$M((\lambda K)^\circ) = M\left(\frac{1}{\lambda} K^\circ\right) = \lambda M(K^\circ),$$

and the second equality follows immediately.

□

Remark 11.5.2. The second inequality in the lemma above has a consequence that will be important for us: we can normalize the volume of K without modifying the value of the product $M(K)M(K^\circ)$.

Before moving forward, we need some technical results. Parametrizing $\mathbb{S}^{n-1}$ by $(x, \sqrt{1-x^2}u)$ for $u \in \mathbb{S}^{n-2}$ and $-1 \leq x \leq 1$, we obtain the following formula for integration over $\mathbb{S}^{n-1}$:

$$\int_{\mathbb{S}^{n-1}} f\, d\sigma = \frac{(n-1)\mathrm{vol}_{n-1}(B^{n-1})}{n\mathrm{vol}_n(B^n)} \int_{-1}^{1} (1-x^2)^{\frac{n-3}{2}} \int_{\mathbb{S}^{n-2}} f(x, \sqrt{1-x^2}u)\, d\sigma_{n-2}(u)\, dx. \qquad (11.5.1)$$

For a proof we refer the reader to Appendix A of the book [166] of Axler et al. Notice also that the fraction multiplying the integral in the right-hand side is the ratio of the surface areas of $\mathbb{S}^{n-2}$ and $\mathbb{S}^{n-1}$. As a consequence of this formula we have the following proposition.

Proposition 11.5.1. *Let $(x_1, \dots, x_n)$ be coordinates in $\mathbb{R}^n$. Then*

$$\int_{\mathbb{S}^{n-1}} |x_1| \, d\sigma(x) = \frac{2}{n} \frac{\mathrm{vol}_{n-1}(B^{n-1})}{\mathrm{vol}_n(B^n)} = \frac{2}{n\sqrt{\pi}} \frac{\Gamma\left(\frac{n}{2}+1\right)}{\Gamma\left(\frac{n+1}{2}\right)},$$

and the same holds for any coordinate other than x_1.

Remark 11.5.3. It is illustrative to prove this equality using (11.5.1), as we do it below, but this integration formula is not really necessary. Since $|x_1| = |\langle x, e_1 \rangle|$, we get that the desired equality is already given in Remark 6.5.3.

Proof. We apply (11.5.1) for $f = |x_1|$. Since f only depends on the first coordinate, we calculate

$$\begin{aligned} \int_{-1}^{1} (1-x_1^2)^{\frac{n-3}{2}} \int_{\mathbb{S}^{n-2}} |x_1| \, d\sigma_{n-2}(u) \, dx_1 &= \int_{-1}^{1} (1-x_1^2)^{\frac{n-3}{2}} |x_1| \, dx_1 = \\ = 2 \int_0^1 x_1 (1-x_1^2)^{\frac{n-3}{2}} \, dx_1 &= 2 \left[\frac{(1-x_1^2)^{\frac{n-1}{2}}}{-(n-1)} \right]_0^1 = \frac{2}{n-1}. \end{aligned}$$

Therefore,

$$\begin{aligned} \int_{\mathbb{S}^{n-1}} |x_1| \, d\sigma = \frac{(n-1)\mathrm{vol}_{n-1}(B^{n-1})}{n\mathrm{vol}_n(B^n)} \int_{-1}^{1} (1-x_1^2)^{\frac{n-3}{2}} \int_{\mathbb{S}^{n-2}} |x_1| \, d\sigma_{n-2}(u) \, dx_1 = \\ = \frac{(n-1)\mathrm{vol}_{n-1}(B^{n-1})}{n\mathrm{vol}_n(B^n)} \frac{2}{n-1} = \frac{2}{n} \frac{\mathrm{vol}_{n-1}(B^{n-1})}{\mathrm{vol}_n(B^n)}, \end{aligned}$$

and the remaining equality comes immediately from the expression for the volume of the n-dimensional unit ball in terms of the Gamma function.

□

Stirling's approximation states that

$$\Gamma\left(\frac{n}{2}+1\right) \sim \sqrt{2\pi} e^{-n/2} \left(\frac{n}{2}\right)^{(n+1)/2}, \tag{11.5.2}$$

where the symbol $\sim$ means that the ratio between the two numbers converges to 1 as $n \to +\infty$. Consequently, we have

$$\frac{\Gamma\left(\frac{n}{2}+1\right)}{\Gamma\left(\frac{n+1}{2}\right)} \sim \frac{e^{-n/2}\left(\frac{n}{2}\right)^{(n+1)/2}}{e^{-(n-1)/2}\left(\frac{n-1}{2}\right)^{n/2}} = \frac{\sqrt{n}}{\sqrt{2e}} \left(\frac{n}{n-1}\right)^{n/2} \sim \sqrt{\frac{n}{2}},$$

since it is easy to see from standard calculus that

$$\lim_{n \to +\infty} \left(\frac{n}{n-1}\right)^{n/2} = \sqrt{e}.$$

It follows that

$$\frac{\mathrm{vol}_{n-1}(B^{n-1})}{\mathrm{vol}_n(B^n)} = \frac{1}{\sqrt{\pi}} \frac{\Gamma\left(\frac{n}{2}+1\right)}{\Gamma\left(\frac{n+1}{2}\right)} \sim \frac{\sqrt{n}}{\sqrt{2\pi}}.$$

From Proposition 11.5.1, we get that

$$\int_{\mathbb{S}^{n-1}} |x_1| \, d\sigma = \frac{2}{n} \frac{\mathrm{vol}_{n-1}(B^{n-1})}{\mathrm{vol}_n(B^n)} \sim \frac{\sqrt{2}}{\sqrt{\pi}} \frac{1}{\sqrt{n}}. \tag{11.5.3}$$

Due to the asymptotics we have that, in particular, one of the sides is "controlled" by the other. We will formalize this idea, since this will be used later.

Proposition 11.5.2. *Let $f, g \colon \mathbb{N} \to \mathbb{R}$ be strictly positive functions. If $f \sim g$, then there exist positive constants $c_0, c_1 \in \mathbb{R}$ such that*

$$c_0 g(n) \leq f(n) \leq c_1 g(n)$$

for every $n \in \mathbb{N}$.

Proof. We write $f \sim g$ when the ratio $f(n)/g(n)$ converges to 1 as $n \to +\infty$. Consequently, there exists $n_0 \in \mathbb{N}$ such that

$$\frac{1}{2} \leq \frac{f(n)}{g(n)} \leq \frac{3}{2}$$

for all $n > n_0$. Let $M = \max\left\{\frac{f(n)}{g(n)} : 1 \leq n \leq n_0\right\}$ and $m = \min\left\{\frac{f(n)}{g(n)} : 1 \leq n \leq n_0\right\}$. Then

$$\min\left\{\frac{1}{2}, m\right\} \leq \frac{f(n)}{g(n)} \leq \max\left\{\frac{3}{2}, M\right\}$$

for every $n \in \mathbb{N}$. Setting c_0 and c_1 as the constants in the left and right-hand sides above, we get the desired.

□

Proposition 11.5.3. *There exists a universal constant $c > 0$ such that for each $n \in \mathbb{N}$ and every convex body $K \subseteq \mathbb{R}^n$ centered at the origin we have that*

$$\frac{b(K)}{M(K)} \leq c\sqrt{n}.$$

Proof. Write $b := b(K)$ and $M := M(K)$, and let $x_0 \in \partial K \cap \frac{1}{b}\mathbb{S}^{n-1}$. Since both b and M are invariant under orthogonal transformations, we may assume that $bx_0 = e_1$. From Lemma 11.5.1, it follows that $h_K(e_1) = 1/b$, that is,

$$\max\{\langle x, e_1\rangle : x \in K\} = 1/b,$$

implying $\langle x, e_1\rangle \leq 1/b$ for every $x \in K$. Geometrically, this means that K is contained in the slab of width $2/b$ bounded by the hyperplanes orthogonal to e_1 and passing though e_1/b and $-e_1/b$. Now, if $x = (x_1, \ldots, x_n) \in \mathbb{S}^{n-1}$, then $x/||x||_K \in K$, and this gives that

$$\left\langle \frac{x}{||x||_K}, e_1 \right\rangle \leq 1/b,$$

which implies $||x||_K \geq bx_1$. Since the same holds for $-x$, we get that $||x||_K \geq b|x_1|$ for every $x \in \mathbb{S}^{n-1}$. Therefore,

$$M(K) = \int_{\mathbb{S}^{n-1}} ||x||_K \, d\sigma \geq b \int_{\mathbb{S}^{n-1}} |x_1| \, d\sigma \sim \frac{b}{\sqrt{n}} \frac{\sqrt{2}}{\sqrt{\pi}},$$

where the last step comes from (11.5.3). Now Proposition 11.5.2 gives that there exists a constant $c_0 > 0$ such that

$$b \int_{\mathbb{S}^{n-1}} |x_1| \, d\sigma \geq \frac{c_0\sqrt{2}}{\sqrt{\pi}} \frac{b}{\sqrt{n}},$$

and putting $c = \sqrt{\pi}/c_0\sqrt{2}$ we get

$$M(K) \geq \frac{b}{c\sqrt{n}},$$

from which the desired follows. Of course, c does not depend on n nor K.

□

Next we state, without a proof, the so-called *Pisier MM^*-estimate*. The proof of this result is beyond the scope of this book, and we refer the interested reader to Artstein-Avidan et al. [133].

Theorem 11.5.1. *There exists a universal constant $c > 0$ such that for any $n \in \mathbb{N}$ and every convex body $K \subseteq \mathbb{R}^n$ centered at the origin there exists $T \in \mathrm{GL}(n)$ for which the inequality*

$$M(TK)M((TK)^\circ) \leq c \log[d_{BM}(TK, B^n) + 1]$$

holds true. As a consequence, we have from Theorem 9.2.3 that $M(TK)M((TK)^\circ) \leq c \log n$. Moreover, in view of Remark 11.5.2, we may assume that $T \in \mathrm{SL}(n)$.

We also have a universal lower bound for $M(K)M(K^\circ)$, which can be proved by using the theory on dual mixed volumes (see Section 11.1) or directly via the classical Hölder inequality (Theorem 11.1.1). We prove it next.

Theorem 11.5.2. *For any convex body $K \subseteq \mathbb{R}^n$ having the origin as an interior point we have*

$$M(K)M(K^\circ) \geq 1.$$

Equality holds if and only if K is a ball centered at the origin.

Proof. This follows straightforwardly from inequality (11.1.13), once $M(K)$ is half the mean width of K° (and, of course, $M(K^\circ)$ is half the mean width of K).

We also can derive this inequality from Hölder's classical inequality. We will need the inequality $|x|^2 = \langle x, x\rangle \leq ||x||_K ||x||_{K^\circ}$, shown in item **(iii)** of Proposition 2.3.6. Using Hölder's inequality for the gauge functions of K and K°, with exponents p and q equal to $1/2$, we get

$$\begin{aligned}\left(M(K)M(K^\circ)\right)^{1/2} &= \left(\int_{\mathbb{S}^{n-1}} \left(||x||_K^{1/2}\right)^2 d\sigma(x)\right)^{1/2} \left(\int_{\mathbb{S}^{n-1}} \left(||x||_{K^\circ}^{1/2}\right)^2 d\sigma(x)\right)^{1/2} \geq \\ &\geq \int_{\mathbb{S}^{n-1}} \left(||x||_K ||x||_{K^\circ}\right)^{1/2} d\sigma(x) \geq \int_{\mathbb{S}^{n-1}} |x| \, d\sigma(x) = 1,\end{aligned}$$

and hence we have indeed that $M(K)M(K^\circ) \geq 1$.

The equality case can also be derived independently from the theory of dual mixed volumes. If K is a ball centered at the origin, then a straightforward calculation gives that $M(K)M(K^\circ) = 1$. Conversely, if we have that $M(K)M(K^\circ) = 1$, then from the equality case of Hölder's inequality it follows that $||\cdot||_K$ is a positive multiple of $||\cdot||_{K^\circ}$, and hence h_K is a positive multiple of h_{K°. Writing $h_{K^\circ} = \alpha h_K$ for some $\alpha > 0$, we get that $K^\circ = \alpha K$. Setting $L = \sqrt{\alpha}K$, we have

$$L^\circ = \frac{1}{\sqrt{\alpha}}K^\circ = \sqrt{\alpha}K = L.$$

Let $x \in L$. Then, $x \in L^\circ$, and it follows that $\langle x, x\rangle \leq 1$. Consequently $|x| \leq 1$ for every $x \in L$, and hence $L \subseteq B^n$. Since polarity reverses inclusion, it follows that $L = L^\circ \supseteq B^n$. Thus, we have that $L = B^n$, and hence K is a ball centered at the origin (compare with Exercise 2.43).

□

Lemma 11.5.4. *There exist universal constants $c, c_0, c_1 > 0$ such that for every $n \in \mathbb{N}$ and any convex body $K \subseteq \mathbb{R}^n$ there exists a map $T \in \mathrm{GL}(n)$ for which the position $\tilde{K} = TK$ of K has the following properties:*

(i) $M(\tilde{K}) = 1$,

(ii) $1 \leq b(\tilde{K}) \leq c_1\sqrt{n}$ *and*

(iii) $M(\tilde{K}) \cdot M(\tilde{K}^\circ) \leq c \log\Big(d_{BM}(\tilde{K}, B^n) + 1\Big)$ $(\leq c_0 \log n)$,

simultaneously.

Proof. From Theorem 11.5.1, let $T_0 \in \mathrm{GL}(n)$ be such that

$$M(T_0K) \cdot M((T_0K)^\circ) \leq c \log\Big(d_{BM}(T_0K, B^n) + 1\Big) \leq c_0 \log n.$$

For any $\lambda > 0$, we have

$$M(\lambda T_0K) = \int_{\mathbb{S}^{n-1}} ||x||_{\lambda T_0K}\, d\sigma(x) = \frac{1}{\lambda}\int_{\mathbb{S}^{n-1}} ||x||_{T_0K}\, d\sigma(x) = \frac{M(T_0K)}{\lambda},$$

and setting $\lambda = M(T_0K)^{-1}$, $T = \lambda T_0$ and $\tilde{K} = TK$, we get that $M(\tilde{K}) = 1$. From Lemma 11.5.3 we get that

$$M(\tilde{K}) \cdot M(\tilde{K}^\circ) \leq c \log\Big(d_{BM}(\tilde{K}, B^n) + 1\Big) \leq c_0 \log n$$

for some universal constant $c_0 > 0$, where we are also using the invariance of the Banach-Mazur distance under linear maps (see Corollary 9.2.1). Now, Proposition 11.5.3 and the fact that $M(\tilde{K}) = 1$ give that $b(\tilde{K}) \leq c_1\sqrt{n}$ for some universal constant $c_1 > 0$. The fact that $b(\tilde{K}) \geq 1$ comes from the trivial bound $b \geq M$.

Notice that the three involved constants can be taken to be the same, if we replace them by $c = \max\{c, c_0, c_1\}$ (abusing notation).

□

Remark 11.5.4. The importance of this lemma is that whenever we are dealing with linear invariant concepts (such as $s(K)$ and $\bar{s}(K)$, for example), we may assume that **(i)**, **(ii)** and **(iii)** hold.

Proposition 11.5.4. *Let $K \subseteq \mathbb{R}^n$ be a convex body centered at the origin such that $M(K) = 1$ and $b(K) \leq c_0\sqrt{n}$, and let $\alpha > 1$. Define $\lambda := M(K^\circ)\alpha$ and*

$$K_1 = \operatorname{conv}\left((K \cap \lambda B^n) \cup \frac{1}{\alpha}B^n\right). \tag{11.5.4}$$

Then we have the inequality

$$\operatorname{vol}_n(K_1) \geq \operatorname{vol}_n(K)\exp(-cn/\alpha^2),$$

where $c > 0$ is some universal constant which does not depend on n, α and K. Moreover, if $\alpha < n^{1/3}$, we also have the inequality

$$\operatorname{vol}_n(K_1) \leq \operatorname{vol}_n(K)\exp(cn/\alpha^2).$$

Proof. Applying Lemma 11.4.3 for $P = \{0\}$, we get

$$\operatorname{vol}_n(K) \leq N(K, sB^n)\operatorname{vol}_n(K \cap sB^n)$$

for every $s > 0$. Setting $s = \lambda$, then from the definition of K_1 it follows that

$$\operatorname{vol}_n(K_1) \geq \operatorname{vol}_n(K \cap \lambda B^n) \geq \frac{\operatorname{vol}_n(K)}{N(K, \lambda B)}.$$

Now we use Sudakov's inequality (Theorem 11.4.2) to obtain

$$\frac{1}{N(K, \lambda B^n)} \geq \exp\left(-cn\left(\frac{M(K^\circ)}{\lambda}\right)^2\right) = \exp(-cn/\alpha^2),$$

where $c > 0$ is a universal constant (given by Sudakov's inequality) and the last equality comes from the definition of λ. Plugging this in the previous inequality, we get

$$\operatorname{vol}_n(K_1) \geq \operatorname{vol}_n(K)\exp(-cn/\alpha^2),$$

which is the first inequality we wanted to prove. For the other inequality, first notice that

$$K_1 \subseteq \operatorname{conv}\left(K \cup \frac{1}{\alpha}B^n\right),$$

because $(K \cap \lambda B^n) \subseteq K$. Observe that $B^n \subseteq b(K)K$. Indeed, if $x \in B^n$, then $||x||_K \leq \max\{||x||_K : x \in B^n\} = b(K)$. It follows that $L = \frac{1}{\alpha}B^n \subseteq \frac{b(K)}{\alpha}K \subseteq b(K)K$ because $\alpha > 1$, and hence $b(K)/\alpha < b(K)$. Finally, since $b(K) \geq 1$ we can apply Corollary 11.4.1 for $L = \frac{1}{\alpha}B^n$, K and $b = b(K)$ to obtain

$$\operatorname{vol}_n(K_1) \leq \operatorname{vol}_n\left(\operatorname{conv}\left(K \cup \frac{1}{\alpha}B^n\right)\right) \leq 2eb(K)nN\left(\frac{1}{\alpha}B^n, K\right)\operatorname{vol}_n(K) =$$
$$= 2eb(K)nN(B^n, \alpha K)\operatorname{vol}_n(K).$$

Now we use the dual Sudakov's inequality (Theorem 11.4.1) to bound the term $N(B^n, \alpha K)$, where we also recall that we always have $N \leq \bar{N}$. We get

$$\mathrm{vol}_n(K_1) \leq 2eb(K)n\mathrm{vol}_n(K)\exp\left(cn\left(\frac{M(K)}{\alpha}\right)^2\right) \leq$$

$$\leq 2ec_0n^{3/2}\mathrm{vol}_n(K)\exp\left(cn\left(\frac{M(K)}{\alpha}\right)^2\right) = 2ec_0n^{3/2}\mathrm{vol}_n(K)\exp(cn/\alpha^2),$$

where in the second inequality we used the fact that $b(K) \leq c_0\sqrt{n}$, and the last equality comes from $M(K) = 1$. Now we appeal to some standard analysis and consider a universal constant $c_1 > 0$ such that $n^{3/2} \leq c_1\exp(n^{1/3})$ for every $n \in \mathbb{N}$. Thus,

$$\mathrm{vol}_n(K_1) \leq 2ec_0c_1\mathrm{vol}_n(K)\exp(n^{1/3})\exp(cn/\alpha^2).$$

Now, observe that if $c' > 0$, then

$$\exp(c'n^{1/3}) = \exp((c'-1)n^{1/3} + n^{1/3}) = \exp(n^{1/3})^{c'-1}\exp(n^{1/3}) \geq e^{c'-1}\exp(n^{1/3}).$$

Hence, choosing $c' > 0$ such that $e^{c'-1} \geq 2ec_0c_1$, we get that

$$\exp(c'n^{1/3}) \geq 2ec_0c_1\exp(n^{1/3}),$$

implying

$$\mathrm{vol}_n(K_1) \leq \mathrm{vol}_n(K)\exp(c'n^{1/3})\exp(cn/\alpha^2).$$

Since $\alpha < n^{1/3}$ we have that $c'n^{1/3} < c'n/\alpha^2$. Finally, setting $c'' = c + c'$ we have

$$\mathrm{vol}_n(K_1) \leq \mathrm{vol}_n(K)\exp(c'n/\alpha^2)\exp(cn/\alpha^2) = \mathrm{vol}_n(K)\exp(c''n/\alpha^2),$$

and this concludes the proof.

□

Remark 11.5.5. Notice that the involved constants do not depend on n, K and α. Indeed, the constant $c > 0$ of the first inequality comes from Sudakov's inequality. The constant c'' depends on c_0 and c_1, which are also universal.

Proposition 11.5.5. *Under the same assumptions as in Proposition 11.5.4, we have that*

$$\exp(-cn/\alpha^2)\mathrm{vol}_n(K_1^\circ) \leq \mathrm{vol}_n(K^\circ) \leq \exp(cn/\alpha^2)\mathrm{vol}_n(K_1^\circ)$$

for some universal constant $c > 0$.

Proof. The proof is very similar to the proof of Proposition 11.5.4, and we will skip it.

□

Corollary 11.5.1. *There exists a universal constant $c > 0$ such that*

$$\exp(-cn/\alpha^2) \leq \frac{s(K)}{s(K_1)} \leq \exp(cn/\alpha^2). \tag{11.5.5}$$

Proof. This is an immediate consequence of the inequalities in Propositions 11.5.4 and 11.5.5.

□

Remark 11.5.6. Recall very carefully that we are assuming $\alpha > 1$ and that one of the inequalities above holds when $\alpha < n^{1/3}$. In what follows, we will need the fact that $\log n < n^{1/3}$ for every natural number n.

Theorem 11.5.3 (Bourgain-Milman). *There exists a universal constant $c > 0$ such that*

$$\bar{s}(K) \geq c$$

for any $n \in \mathbb{N}$ and every convex body $K \subseteq \mathbb{R}^n$ centered at the origin.

Proof. We start with a convex body $K \subseteq \mathbb{R}^n$ centered at the origin, and up to a linear transformation which does not modify the value of $\bar{s}(K)$, we assume that conditions **(i)**, **(ii)** and **(iii)** in Lemma 11.5.4 hold. Let K_1 be as in (11.5.4). We first show how to "control" the Banach-Mazur distance $d_{BM}(K_1, B^n)$. If $\alpha > 1$ and $\lambda = M(K^\circ)\alpha$, we have from Theorem 11.5.2 that

$$1 \leq M(K)M(K^\circ)\alpha^2 = \alpha\lambda,$$

and from this inequality we get immediately that

$$\frac{1}{\alpha}B^n \subseteq K_1 \subseteq \lambda B^n.$$

Thus,

$$d_{BM}(K_1, B^n) \leq \lambda\alpha = M(K)M(K^\circ)\alpha^2 \leq \alpha^2 c_0 \log n,$$

for some universal constant $c_0 > 0$ given by property **(iii)** in Lemma 11.5.4. Now we set $\alpha = \alpha_1 = \log n$ (we assume $n \geq 3$, and for $n = 2$ we replace n by 3) to obtain $d_{BM}(K_1, B^n) \leq c_0(\log n)^3$. We may also assume that

$$\exp(-cn/(\log n)^2) \leq \frac{s(K)}{s(K_1)} \leq \exp(cn/(\log n)^2),$$

for a universal constant $c > 0$ given in Corollary 11.5.1. Up to a linear isomorphism (which does not modify the Banach-Mazur distance and the Mahler constant), we may assume that K_1 is such that conditions **(i)**, **(ii)** and **(iii)** in Lemma 11.5.4 hold.

Now we put $\alpha_2 := \log(\log n) := \log^{(2)} n$, $\alpha_3 := \log^{(3)} n$, and continue until the smallest positive integer t such that $\alpha_t := \log^{(t)} n < e$. Observe that this is a decreasing sequence, since we always have that $\log n < n$. Also, we guarantee that $\alpha_j > 1$ if $j \leq t$, because if $\alpha_j < 1$, then $\alpha_{j-1} < e$. Setting $\lambda_2 := M(K_1^\circ)\alpha_2$ we define

$$K_2 := \operatorname{conv}\left((K_1 \cap \lambda_2 B^n) \cup \frac{1}{\alpha_2}B^n\right).$$

As before, we have the estimate

$$d_{BM}(K_2, B^n) \leq M(K_1)M(K_1^\circ)\alpha_2^2 \leq c\alpha_2^2 \log\left(d_{BM}(K_1, B^n) + 1\right) \leq$$
$$\leq c\alpha_2^2 \log(c_0(\log n)^3 + 1) = c(\log^{(2)} n)^2 \log(c_0(\log n)^3 + 1) \leq c_1(\log^{(2)} n)^3,$$

for some universal constant $c_1 > 0$. Just to clarify the last inequality, we write

$$c_0(\log n)^3 + 1 \leq (c_0 + 1)(\log n)^3 \leq (\log n)^{c'}(\log n)^3 = (\log n)^{3+c'},$$

for some universal constant $c' > 0$. Indeed, since c_0 is a universal constant, we may choose $c' > 0$ such that $(\log n)^{c'} \geq c_0 + 1$. From the inequality above, it follows that we only need to take $c_1 = c(3 + c')$. As in the previous step, we have from (11.5.5) that

$$\exp\Big(- cn/(\log^{(2)} n)^2\Big) \leq \frac{s(K_1)}{s(K_2)} \leq \exp\Big(cn/(\log^{(2)} n)^2\Big),$$

where we are, again, considering the constant $c > 0$ given in Corollary 11.5.1.

We repeat this process t times until we reach a convex body K_t centered at the origin with the property that

$$d_{BM}(K_t, B^n) \leq c_{t-1}(\log^{(t)} n)^3 < e^3 c_{t-1} := c_{**},$$

where $c_{**} > 0$ is a constant which does not depend on K or n. Therefore,

$$\alpha_t \lambda_t = M(K_t)M(K_t^\circ)\alpha_t^2 \leq c\alpha_t^2 \log\Big(d_{BM}(K_t, B^n) + 1\Big) \leq ce^2 \log(c_{**} + 1) := c_*.$$

The construction of K_t gives that

$$\frac{1}{\alpha_t} B^n \subseteq K_t \subseteq \lambda_t B^n,$$

and this yields $B^n \subseteq \alpha_t K_t \subseteq \alpha_t \lambda_t B^n$. Since $c_* \geq \alpha_t \lambda_t$, setting $1/\alpha_t := r$ gives

$$rB^n \subseteq K_t \subseteq \lambda_t B^n \subseteq c_* r B^n.$$

By duality, we also have immediately

$$\frac{1}{c_* r} B^n \subseteq K_t^\circ \subseteq \frac{1}{r} B^n.$$

These inclusions give the following bounds for $s(K_t) = \mathrm{vol}_n(K_t) \cdot \mathrm{vol}_n(K_t^\circ)$:

$$(c_*)^{-n} s(B^n) \leq s(K_t) \leq (c_*)^n s(B^n). \tag{11.5.6}$$

Also, we have inequalities

$$\exp\left(-\frac{cn}{\alpha_{j+1}^2}\right) \leq \frac{s(K_j)}{s(K_{j+1})} \leq \exp\left(\frac{cn}{\alpha_{j+1}^2}\right)$$

whenever $j = 0, \ldots, t-1$, and where we are considering $K_0 := K$. We use these inequalities to bound each Mahler constant $s(K_j)$ in terms of the Mahler constant $s(K_{j-1})$ of the previous body, until we reach the final bounding $s(K)$ in terms of $s(K_1)$. This will lead to

$$\exp\left(-cn \sum_{j=1}^t \alpha_j^{-2}\right) \leq \frac{s(K)}{s(K_t)} \leq \exp\left(cn \sum_{j=1}^t \alpha_j^{-2}\right),$$

and together with (11.5.6) we obtain immediately that

$$\frac{1}{(c_*)^n} \exp\left(-cn \sum_{j=1}^t \alpha_j^{-2}\right) s(B^n) \leq s(K) \leq (c_*)^n \exp\left(cn \sum_{j=1}^t \alpha_j^{-2}\right) s(B^n). \tag{11.5.7}$$

Now, observe that since each α_j is a number which is greater than 1, we get

$$\sum_{j=1}^{t} \alpha_j^{-2} \leq \frac{1}{\alpha_1} + \frac{1}{\alpha_2} + \ldots + \frac{1}{\alpha_t},$$

and we have to bound this last sum. We notice that

$$\alpha_1 = \exp(\alpha_2) = \exp(\exp(\alpha_3)) = \ldots = \exp(\exp(\ldots(\alpha_t))),$$

and hence we may write

$$\frac{1}{\alpha_1} + \frac{1}{\alpha_2} + \ldots + \frac{1}{\alpha_t} = \frac{1}{\alpha_t} + \frac{1}{\exp(\alpha_t)} + \frac{1}{\exp(\exp(\alpha_t))} + \ldots + \frac{1}{\exp(\exp(\ldots(\alpha_t)))}.$$

Since $\alpha_t > 1$, we have the following inequalities:

$$\begin{aligned} \alpha_{t-1} &= \exp(\alpha_t) \geq e > 2, \\ \alpha_{t-2} &= \exp(\alpha_{t-1}) \geq e^2 > 4, \\ \alpha_{t-3} &= \exp(\alpha_{t-2}) \geq e^4 > 2^3 = 8 \\ &\vdots \\ \alpha_1 &= \exp(\alpha_2) > 2^{t-1}, \end{aligned}$$

from which we get that

$$\frac{1}{\alpha_1} + \ldots + \frac{1}{\alpha_t} \leq 1 + \frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^{t-1}} \leq 2.$$

Plugging this estimate into (11.5.7) yields immediately

$$\frac{1}{(c_*)^n} \exp(-2cn) s(B^n) \leq s(K) \leq (c_*)^n \exp(2cn) s(B^n),$$

and writing $C = c_* e^{2c}$ we get that

$$\frac{1}{C^n} s(B^n) \leq s(K) \leq C^n s(B^n).$$

The inequality on the left gives

$$\bar{s}(K) = \left(\frac{s(K)}{s(B^n)} \right)^{1/n} \geq \frac{1}{C},$$

which we wanted to prove.

□

Remark 11.5.7. From the usual multiplicative formula for the Gamma function, and using the fact that the Gamma function is non-decreasing for $z > 1$, we get the bound

$$\Gamma(z)^2 \leq \Gamma(z)\Gamma\left(z + \frac{1}{2}\right) = 2^{1-2z}\sqrt{\pi}\Gamma(2z).$$

Setting $z = \frac{n}{2} + 1$, we arrive at

$$\Gamma\left(\frac{n}{2}+1\right)^2 \leq 2^{1-n-2}\sqrt{\pi}\Gamma(n+2) = 2^{-n-1}\sqrt{\pi}(n+1)!$$

Therefore,

$$s(K) \geq c^n \frac{\pi^n}{\Gamma\left(\frac{n}{2}+1\right)^2} \geq \left(\frac{c\pi}{2}\right)^n \frac{1}{2\sqrt{\pi}(n+1)!} \geq \frac{c_0^n}{n!}$$

for some universal constant $c_0 > 0$. As it was mentioned in Section 5.4, the Mahler conjecture asks whether this holds for $c_0 = 4$. The best known value at this moment is $c_0 = \pi$, and it was obtained by Kuperberg in [1476] (see also [297]).

11.6 Exercises

The objective of the first exercise is to show that the Petty projection inequality (10.4.10) is, in some sense, stronger than the isoperimetric inequality for convex bodies (which is given in Theorem 5.2.1). Our main source for this exercise is Lutwak [1627]. For $p \neq 0$ and $K \in \mathcal{K}^n$, we define the *p-mean of the projection function of K* to be the number

$$S_p(K) := \left(\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)^p \, d\sigma(u)\right)^{\frac{1}{p}} = \left(\frac{1}{n\kappa_n}\int_{\mathbb{S}^{n-1}} \mathrm{vol}_{n-1}\big(\mathrm{proj}_{u^\perp}(K)\big)^p \, d\mathcal{H}^{n-1}(u)\right)^{\frac{1}{p}},$$

where we recall that σ is the standard probability measure on $\mathbb{S}^{n-1}$, which is just the normalized Hausdorff measure on $\mathbb{S}^{n-1}$, and that κ_n is the volume of B^n. Of course, the term $n\kappa_n$ appears as the normalization factor because from (4.3.3) we have $\mathcal{H}^{n-1}(\mathbb{S}^{n-1}) = n\kappa_n$.

Exercise 11.1. *Let $K \subseteq \mathbb{R}^n$ be an arbitrarily given convex body, and recall that we denote its usual surface area measure by $S(K)$.*

(a) *Verify that*

$$S(K) = \frac{n\kappa_n}{\kappa_{n-1}} S_1(K).$$

(b) *Show that*

$$S_{-n}(K) = \kappa_n^{\frac{1}{n}} \mathrm{vol}_n(\Pi^\circ K)^{-\frac{1}{n}}.$$

(c) *Prove that $S_1(K) \geq S_{-n}(K)$.*

(d) *Using the Petty projection inequality, check that*

$$S_{-n}(K) \geq \kappa_{n-1}\kappa_n^{(1-n)/n}\mathrm{vol}_n(K)^{(n-1)/n}.$$

(e) *Derive the isoperimetric inequality for convex bodies from the previous items.*

Exercise 11.2. *Give an example of a valuation $\varphi : \mathcal{S} \to \mathbb{R}$ which is not a fully additive functional.*

Hint: Let $\mathcal{S} = \Big\{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b, c\}\Big\}$.

A map $\varphi\colon CC^n \to CC^n$ is called a *Minkowski valuation* if whenever $K, L \in CC^n$ are such that $K \cup L \in CC^n$, it holds that

$$\varphi(K \cup L) + \varphi(K \cap L) = \varphi(K) + \varphi(L),$$

where we are considering the Minkowski sum in CC^n, of course. In the exercises below we briefly investigate some Minkowski valuations.

Exercise 11.3. *Let the* difference body map *be the map* $D\colon CC^n \to CC^n$ *defined as*

$$D(K) = K - K$$

for each $K \in CC^n$. *Show that* D *is a Minkowski valuation.*

Exercise 11.4. **(a)** *Let* $0 \leq j \leq n-1$ *and* $K, L, C_{j+1}, \ldots, C_{n-1} \in CC^n$, *and assume that* $K \cup L \in CC^n$. *Show that*

$$\begin{aligned} S(K \cup L[j], C_{j+1}, \ldots, C_{n-1}, \cdot) + S(K \cap L[j], C_{j+1}, \ldots, C_{n-1}.\cdot) = \\ = S(K[j], C_{j+1}, \ldots, C_{n-1}, \cdot) + S(L[j], C_{j+1}, \ldots, C_{n-1}, \cdot). \end{aligned}$$

That is, the mixed surface area measures have the valuation property.

(b) *For each* $1 \leq i \leq n-1$, *let* $\Pi_i\colon CC^n \to CC^n$ *be the map which associates each* $K \in CC^n$ *to the set* $\Pi_i K$ *as defined in Exercise 7.14. Prove that* Π_i *is a Minkowski valuation.*

Exercise 11.5. **(a)** *Let* $A, B \subseteq \mathbb{R}^n$. *Show that*

$$1_{A\cup B}(x) + 1_{A\cap B}(x) = 1_A(x) + 1_B(x)$$

for any $x \in \mathbb{R}^n$.

(b) *Let* $M\colon CC^n \to CC^n$ *be the map which associates each* $K \in CC^n$ *to its moment body* MK, *defined in (10.3.8). Prove that* M *is a Minkowski valuation.*

Recall that CC_0^n denotes the family of sets $C \in CC^n$ such that $0_{\mathbb{R}^n} \in C$. Let $p > 1$. A map $\varphi\colon CC_0^n \to CC_0^n$ is an L_p *Minkowski valuation* if it has the valuation property for the p-sum. That is, if

$$\varphi(K \cup L) +_p \varphi(K \cap L) = \varphi(K) +_p \varphi(L)$$

for every $K, L \in CC_0^n$ such that $K \cup L \in CC_0^n$.

Exercise 11.6. *Let* $p > 1$.

(a) *Show that*

$$(K \cup L) +_p (K \cap L) = K +_p L$$

whenever $K, L, K \cup L \in CC_0^n$.

(b) *Let* $D_p\colon CC_0^n \to CC_0^n$ *be the* L_p *difference body map, defined as*

$$D_p(K) := K +_p (-K)$$

for each $K \in CC_0^n$. Prove that D_p is an L_p Minkowski valuation.

Hint: See the proof of Lemma 6.2.2.

Exercise 11.7. **(a)** *For $p \geq 1$ and $K \in CC_0^n$, let*

$$h(M_pK, x) := \left(\int_K |\langle x, z \rangle|^p \, dz \right)^{1/p}$$

for each $x \in \mathbb{R}^n$. Show that the function $h(M_pK, \cdot)$ is sublinear, and conclude that this is the support function of a set $M_pK \in CC^n$ (we call it the L_p moment body of K).

(b) *Fix $p > 1$. Let $M_p \colon CC_0^n \to CC_0^n$ be the L_p* moment map*, which associates each $K \in CC_0^n$ to its L_p moment body M_pK. Prove that M_p is a L_p Minkowski valuation.*

Let $K \in CC^n$. A *rotation mean* of K is a set $K' \in CC^n$ having the form

$$K' = \frac{1}{m}(T_1K + \ldots + T_mK),$$

where $m \in \mathbb{N}$ and $T_1, \ldots, T_m \in \mathrm{SO}(n)$. The objective of the next exercise is to prove a theorem (due to Hadwiger) which states that for any $K \in CC^n$ there is a sequence of rotation means of K converging in the Hausdorff metric to the ball centered at the origin and whose radius is half the mean width of K.

Exercise 11.8. *Let $K \in CC^n$, and assume that $\dim K > 0$. Denote by $\mathcal{R}(K)$ the family of rotation means of K.*

(a) *Show that $\mathcal{R}(K)$ is a bounded family in $\mathbb{R}^n$.*

(b) *Conclude that $\mathrm{cl}(\mathcal{R}(K))$ is compact in the Hausdorff metric.*

(c) *Check that the map $r_0 \colon CC^n \to \mathbb{R}$ defined as*

$$r_0(L) := \min\{r \geq 0 : L \subseteq rB^n\}$$

is well-defined (meaning that the minimum indeed exists), and that it is continuous in the Hausdorff metric.

(d) *Prove that $w(K') = w(K)$ for any $K \in \mathcal{R}(K)$, where we remind the reader that $w(C)$ stands for the mean width of $C \in CC^n$.*

(e) *In view of* **(b)** *and* **(c)***, we may let*

$$m_0 := \min\{r_0(L) : L \in \mathrm{cl}(\mathcal{R}(K))\}.$$

Use item **(d)** *to verify that $m_0 > 0$. Assume that this minimum is attained for $L_0 \in \mathrm{cl}(\mathcal{R}(K))$. Prove that $L_0 = m_0B^n$.*

(f) *Conclude that there is a sequence of rotation means of K converging to $\frac{w(K)}{2}B^n$.*

Recall that a functional $\varphi \colon CC^n \to \mathbb{R}$ is *Minkowski additive* if

$$\varphi(K + L) = \varphi(K) + \varphi(L)$$

for any $K, L \in CC^n$. We also recall that φ is $\mathrm{SO}(n)$-invariant if

$$\varphi(TK) = \varphi(K)$$

for any $T \in \mathrm{SO}(n)$.

Exercise 11.9. *Assume that $n \geq 2$, and let $\varphi\colon CC^n \to \mathbb{R}$ be a functional which is Minkowski additive, $\mathrm{SO}(n)$-invariant and continuous at B^n. Prove that φ is a multiple of the mean width functional.*

We introduce some new concepts for the next exercise. A *dissection* of a polytope $P \in \mathcal{P}^n$ is a (finite) family of polytopes $\{P_1, \ldots, P_m\}$ such that $P = P_1 \cup \ldots \cup P_m$ and $\dim(P_i \cap P_j) < n$ whenever $i \neq j$. We say that polytopes P and Q are *T-equidissectable* if there exist dissections $\{P_1, \ldots, P_m\}$ of P and $\{Q_1, \ldots, Q_m\}$ of Q with the property that each Q_i is a translate of P_i.

Exercise 11.10. *Let $\varphi\colon \mathcal{P}^n \cup \{\emptyset\} \to \mathbb{R}$ be a simple and translation-invariant valuation. Prove that if $P, Q \in \mathcal{P}^n$ are T-equidissectable, then $\varphi(P) = \varphi(Q)$.*

Recall that $\mathbf{Val}_n$ is the vector space of valuations $\varphi\colon CC^n \cup \{\emptyset\} \to \mathbb{R}$ which are continuous (when restricted to CC^n, of course) and invariant under proper rigid motions.

Exercise 11.11. *Let $\varphi \in \mathbf{Val}_n$ be simple. Prove that $\varphi = cV_n$ for some constant $c \in \mathbb{R}$. In other words, φ is homogeneous of degree n.*

For the next exercise, we recall that a set $S \subseteq \mathbb{R}^n$ is called a *star body* if $\mathrm{seg}[0_{\mathbb{R}^n}, x] \subseteq S$ for every $x \in S$ and if the radial function $\rho_S\colon \mathbb{R}^n \setminus \{0_{\mathbb{R}^n}\} \to \mathbb{R}$ of S is positive and continuous. In what follows, we denote the family of star bodies of $\mathbb{R}^n$ by $\mathcal{S}^n$. (A comprehensive survey on star bodies and related notions is Hansen et al. [1157].) Notice that the definition of dual mixed volumes given in (11.1.1) can be naturally extended to star bodies.

We define the *radial sum* of vectors $x, y \in \mathbb{R}^n$ as

$$x \tilde{+} y := \begin{cases} x + y, & \text{if } x \text{ and } y \text{ lie in a same ray starting at } 0_{\mathbb{R}^n} \\ 0, & \text{otherwise} \end{cases}.$$

The *radial sum* of star bodies $K, L \in \mathcal{S}^n$ is defined as

$$K \tilde{+} L := \left\{ x \tilde{+} y : x \in K \text{ and } y \in L \right\}.$$

Exercise 11.12. **(a)** *Let $K, L \in \mathcal{S}^n$. Prove that*

$$\rho_{K \tilde{+} L} = \rho_K + \rho_L,$$

and show that $K \tilde{+} L \in \mathcal{S}^n$.

(b) *Prove that the dual mixed volume is additive for $\tilde{+}$ in each argument; that is, show that*

$$\tilde{V}(K \tilde{+} L, K_2, \ldots, K_n) = \tilde{V}(K, K_2, \ldots, K_n) + \tilde{V}(L, K_2, \ldots, K_n)$$

for any $K, L, K_2, \ldots, K_n \in \mathcal{S}^n$.

(c) *Check that equality (4.3.7) also holds for star bodies.*

Exercise 11.13. **(a)** *Using Hölder's inequality, prove the* Minkowski inequality for integrals*: let X be a topological space, and let μ be a finite Borel measure on X. For any positive continuous functions $f, g : X \to \mathbb{R}$ and every $p \in (1, +\infty)$ it holds that*

$$\left(\int_X (f+g)^p \, d\mu\right)^{1/p} \leq \left(\int_X f^p \, d\mu\right)^{1/p} + \left(\int_X g^p \, d\mu\right)^{1/p},$$

with equality if and only if $f = \lambda g$ for some $\lambda > 0$.

(b) *Prove the* dual Brunn–Minkowski inequality*, which states that*

$$\mathrm{vol}_n(K \tilde{+} L)^{1/n} \leq \mathrm{vol}_n(K)^{1/n} + \mathrm{vol}_n(L)^{1/n},$$

for any $K, L \in \mathcal{S}^n$. Show also that equality holds if and only if K and L are dilates.

In what follows, the reader must keep in mind the notations, hypotheses and definitions of Exercise 9.13, where the *Orlicz projection bodies* were defined.

Exercise 11.14. **(a)** *Let $\phi_1 : \mathbb{R} \to [0, +\infty)$ be given as $\phi_1(t) = |t|$. Show that for any $K \in \mathcal{K}_0^n$ we have*

$$\Pi_{\phi_1} K = \frac{2}{n\mathrm{vol}_n(K)} \Pi K.$$

(b) *Verify that if $p > 1$, and letting $\phi_p(t) = |t|^p$, we have*

$$h(\Pi_{\phi_p} K, u)^p = \frac{1}{n\mathrm{vol}_n(K)} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, dS_{p,0}(K, v). \tag{11.6.1}$$

Exercise 11.15. *Use item* **(a)** *of Exercise 11.14, the result stated in Remark 9.5.3, and Exercise 10.11 to give an alternative proof of the Petty projection inequality (Theorem 10.4.2).*

For $n \in \mathbb{N}$, $p > 1$, and $u \in \mathbb{S}^{n-1}$, define

$$c_{n,p} := \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, d\mathcal{H}^{n-1}(v).$$

Since $\mathcal{H}^{n-1}|_{\mathbb{S}^{n-1}}$ is invariant under orthogonal transformations, we have that this number is a constant which does not depend on $u \in \mathbb{S}^{n-1}$. Having this in mind, and for $K \in \mathcal{K}_0^n$, we then define the L_p *projection body* of K as the convex body $\Pi_p K$ whose support function over $\mathbb{S}^{n-1}$ is given by

$$h(\Pi_p K, u)^p = \frac{1}{c_{n,p}} \int_{\mathbb{S}^{n-1}} |\langle u, v\rangle|^p \, dS_{p,0}(K, v) \tag{11.6.2}$$

for each $u \in \mathbb{S}^{n-1}$. Of course, we extend it to $\mathbb{R}^n$ by positive homogeneity. Our main reference for the L_p projection bodies is [1638] by Lutwak, Yang, and Zhang.

Exercise 11.16. **(a)** *For $p > 1$, check that*

$$h(\Pi_p K, \cdot) = \left(\frac{n\mathrm{vol}_n(K)}{c_{n,p}}\right)^{1/p} h(\Pi_{\phi_p} K, \cdot)$$

for any $K \in \mathcal{K}_0^n$, where $\Pi_{\phi_p} K$ is the Orlicz projection body whose support function is given in (11.6.1). Conclude that the function defined in (11.6.2) is indeed the support function of a convex body having the origin as an interior point.

(b) *Verify that* $\Pi_p B^n = B^n$.

The objective of the next exercise is to give a proof (based on the result stated in Remark 9.5.3) of the so-called L_p *Petty projection inequality*, which can be seen as an extension of the Petty projection inequality for polar L_p projection bodies.

Exercise 11.17 (L_p Petty projection inequality)**.** *Prove that for any* $p > 1$ *and every* $K \in \mathcal{K}_0^n$ *it holds that*

$$\mathrm{vol}_n(K)^{\frac{n}{p}-1}\mathrm{vol}_n(\Pi_p^\circ K) \leq \kappa_n^{\frac{n}{p}},$$

where $\Pi_p^\circ K := (\Pi_p K)^\circ$. *Verify that the equality case holds if and only if* K *is an ellipsoid symmetric with respect to the origin.*

Exercise 11.18. *Prove that the definitions of separation number given in Definition 11.4.2 are indeed equivalent.*

Recall that $\mathcal{W}\colon CC^n \to \mathbb{R}$ stands for the *Wills functional* defined as

$$\mathcal{W}(K) := \sum_{j=0}^{n} V_j(K),$$

for each $K \in CC^n$. The next exercise is an integral representation for the Wills functional which was taken from Hadwiger [1135].

Exercise 11.19. *Prove that the Wills functional has the integral representation*

$$\mathcal{W}(K) = \int_{\mathbb{R}^n} e^{-\pi d(K,x)^2}\, dx,$$

where $d(K, x)$ *represents the distance from a point* $x \in \mathbb{R}^n$ *to* K, *and the integration is taken with respect to the Lebesgue measure of* $\mathbb{R}^n$.

Hint: Show that

$$\varphi(K) := \int_{\mathbb{R}^n} e^{-\pi d(K,x)^2}\, dx$$

defines a real-valued, continuous valuation on $CC^n \cup \{\emptyset\}$ which is invariant under proper rigid motions (extended as $\varphi(\emptyset) = 0$, of course). Then, use Hadwiger's theorem to show that φ is a combination of intrinsic volumes. To find the coefficients, compute $\varphi(\varepsilon B^n)$ for $\varepsilon > 0$. For that computation you can use the equality (5.6.2).

Exercise 11.20. *Extend the previous exercise and prove that for any* $a > 0$ *the functional*

$$\varphi_a(K) = \int_{\mathbb{R}^n} e^{-ad(K,x)^2}\, dx$$

defined for $K \in CC^n$ *(and extended as* 0 *to* $K = \emptyset$*) is a real-valued, continuous valuation invariant under proper rigid motions. Show that*

$$\varphi_a(K) = \sum_{j=0}^{n} \left(\frac{\pi}{a}\right)^{\frac{n-j}{2}} V_j(K)$$

for any $K \in CC^n$.

Exercise 11.21. *Using Exercise 6.33 in the definition (6.6.2), write*

$$w(1_K) := \frac{4}{n(2\pi)^{n/2}} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \left[\int_{\mathbb{R}^n} e^{-\frac{1}{2} d(\varepsilon K, x)^2} - \int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2}} \right].$$

From Exercise 11.20, conclude that

$$w(1_K) = \frac{2\kappa_n}{(2\pi)^{\frac{1}{2}} \kappa_{n-1}} w(K)$$

for every $K \in CC^n$.

Exercise 11.22. *Let $\tilde{w}(\cdot)$ be as defined in (5.6.1). Show that*

$$\tilde{w}(1_K) = w(1_K)$$

for any $K \in CC^n$.

Hint: Use Exercises 11.21, 5.30, and item **(a)** of Exercise 4.7.

Exercise 11.23. *Let $H \colon \mathbb{R}^n \to \mathbb{R}$ be defined as $H(x) := e^{-\pi |x|^2}$ for each $x \in \mathbb{R}^n$. Similarly to Exercise 6.33, show that*

$$[H \star (\varepsilon \cdot K)](x) = e^{-\pi d(\varepsilon K, x)^2}$$

for any $K \in CC^n$, $\varepsilon > 0$ and $x \in \mathbb{R}^n$. Using this equality, show that

$$\mathcal{W}(K) \geq \frac{(1-\varepsilon)^{-\frac{n}{2}(1-\varepsilon)}}{\varepsilon^{n\varepsilon}} \mathrm{vol}_n(K)^{\varepsilon} \tag{11.6.3}$$

for every $0 < \varepsilon < 1$ and any $K \in CC^n$.

Hint: Using Exercise 11.19, write the Wills functional of εK as the integral of $H \star (\varepsilon \cdot K)$. Then, use the version of the Prékopa-Leindler inequality given in Exercise 1.28.

Remark 11.6.1. The inequality (11.6.3) was originally proved by Alonso–Gutiérrez et al. [83].

Exercise 11.24. *Let $A, B \subseteq \mathbb{R}^n$ be convex bodies, and assume that B is centered at the origin. Let $x_1, \ldots, x_N \in A$ be such that $x_i \notin x_j + B$ whenever $i \neq j$. Prove that*

$$N \leq \frac{\mathrm{vol}_n(2A + B)}{\mathrm{vol}_n(B)}.$$

Conclude that the separation number $M(A, B)$ is well-defined as a natural number.

11.7 Notes

Among the themes of the five sections in this chapter here, we have chosen the following three ones for being discussed also in the notes below: dual mixed volumes, valuations of convex bodies, and the Bourgain–Milman inequality. The other two topics are partially discussed in other chapters of our book.

11.7.1 Dual mixed volumes

The concept of *dual mixed volumes* goes back to Lutwak's papers [1625] and [1630], and it is fundamental for Lutwak's *dual Brunn–Minkowski theory*. Therefore notions like e.g. *mixed intersection bodies*, *Orlicz dual mixed volumes* and many further ones are based on this concept.

THE NOTION IN BASIC ARTICLES AND BOOKS

In [1625] duals of the classical mixed-volume inequalities (of Minkowski, Fenchel, and A. D. Alexandrov) are proved, and related ball characterizations are given; the second paper [1630] is discussed in our subsection referring to *intersection bodies*, like also Lutwak's article [1631] in our part on *centroid bodies*. A first brief summary on the topic is given already in Section 24.3 of the monograph [531] of Burago and Zalgaller. In Lutwak's handbook survey [1633] dual mixed volumes occur for instance in connection with *Busemann's intersection inequality*. Different ways to and generalizations of dual mixed volumes are presented in Appendix A.7 of Gardner's book [930] and in Section 9.3 (and its notes) of Schneider's monograph [2167].

More precisely, Gardner gives a summary in the Appendix A.7 of [930], where the concept of dual mixed volumes is formulated *in terms of general star bodies*, and several statements are nicely compared with their analogues on mixed volumes given in appendix A.3. Some corresponding theorems are proven there, such as the *dual Kubota integral recursion* (see also [1626] and [942]). In the appendix A.7 of [930] it is also underlined that Klain takes in [1396] and [1397] a somewhat *different approach to dual mixed volumes*, which also allows to get a natural analogue of Hadwiger's famous characterization theorem in the dual theory.

RELATED FURTHER NOTIONS

The related notion of *mixed intersection bodies* is discussed in Section 10.10 of Schneider's book [2167]. Ghandehari [953] derives inequalities involving dual mixed volumes of a convex body and its dual. The important paper [2532], extending results of Lutwak and paving the way to the famous *Busemann–Petty problem*, is due to Zhang. Analogously to characterizations of projection bodies in terms of mixed volumes, Zhang derives *characterizations of intersection bodies via dual mixed volumes*. Applying the principle also to polar projection bodies and curvature images of projection bodies, *centroid bodies* and the *affine surface area* come into the game, yielding existence results on centered convex bodies that are *not* intersection bodies (which later proved to be very important for the Busemann–Petty prolem).

Using an analogue of the classical Steiner formula for the volume of a generalized outer parallel set, Sangwine-Yager [2099] obtains a geometric interpretation of Lutwak's concept of *dual quermass-integrals* from Lutwak's paper [1625]. Gardner and Vasallo [941] also treat statements dual to the equivalence of the Brunn–Minkowski inequality and Minkowski's first inequality for convex bodies, referring to star bodies in the sense of Lutwak's dual Brunn–Minkowski theory.

Investigating inequalities for mixed intersection bodies, Zhao and Leng [2548] establish *Brunn–Minkowski and Alexandrov–Fenchel inequalities for dual quermassintegrals* of such bodies, thus dualizing inequalities for projection bodies. Dualizing earlier results of Lutwak, Zhao and Leng, the same authors derive in [2549] inequalities for *dual quermassintegrals of polars of mixed projection bodies*. They combine geometric inequalities for mixed volumes (of Minkowski, Alexandrov-Fenchel, and Brunn-Minkowski), applied to the integrand, with integral inequalities (of Hölder and Minkowski).

APPLICATIONS AND EXTENSIONS

Using dual mixed volumes, Milman [1783] gives a valuable contribution to the famous slicing problem by explaining, unifying and extending known partial results due to Ball and Junge treating convex sets that are close to unit balls of subspaces or quotients of L_p spaces.

An important role play computations involving *radial functions of star bodies* and dual mixed volumes. Bastero et al. [250] optimize dual mixed volumes of convex bodies suitably under linear and affine maps. They prove an interesting extension of the *dual Kubota formula* which, for a star body, relates the dual mixed volumes with parallel sections. Convex bodies in the so-called Gauss-John position are considered, too.

Dulio et al. [774] characterize the dual mixed volume V^* defined on the class of n-tuples of star sets in n-space in the following sense: Let F be an additive functional on this class which is rotation invariant and vanishes if the intersection of two of its arguments is $\{0\}$. Then $F = cV^*$, where c is some positive constant. (Here, "additive" means additive in each argument with respect to radial addition, and "rotation invariant" means that F remains fixed if the same rotation is applied to each of its arguments.)

Jiménez and Villanueva [1315] extend these investigations with the help of the theory of multilinear operators on spaces of continuous functions, on this level *again characterizing dual mixed volumes*. Zhao [2541] investigates the *L_p-dual Minkowski inequality*, giving a lower bound of the L_p-dual mixed volume of two convex bodies in terms of suitable powers of their volumes. He proves that for $1 \leq p < \infty$ there is a (multiplicative) constant, strongly depending on the inradius and circumradius of both convex bodies, such that the L_p-dual Minkowski inequality can be reversed. And in the more recent paper [1642], Lutwak, Yang, and Zhang use L_p dual mixed volumes to show that the concepts of Minkowski's classical first mixed volume, L_p mixed volumes, L_p entropy, and dual mixed volumes are unified. Establishing geometric inequalities and variational integral formulas for the unified mixed volumes and for the new L_p dual curvature measures, a surprising *connection between the L_p Brunn–Minkowski theory and the dual Brunn–Minkowski theory* is shown. This yields the deep L_p dual Minkowski problem unifying the L_p Minkowski problem, the dual Minkowski problem, and the L_p Alexandrov problem.

Zhao presents with [2540] a thorough study of *Orlicz dual mixed volumes*, using a so-called Orlicz harmonic sum of star bodies. Here also Gardner et al. [938] and Zhu et al. [2555] have to be cited, both based on the new concept of *Orlicz radial addition*. With the help of this notion, the *dual Orlicz–Brunn–Minkowski inequality* and a *dual Orlicz–Minkowski inequality* are established, yielding also notions like the dual log-Brunn–Minkowski inequality. As it happens in the usual Brunn–Minkowski theory with M-addition and Orlicz addition, the concept of radial M-addition corresponding with the radial Orlicz addition is introduced. Ye [2499] continues with this setting introducing dual Orlicz L_ϕ affine and geominimal surface areas of star bodies; these are shown to be affine invariant, and *Orlicz affine isoperimetric, Santaló and Alexandrov–Fenchel type inequalities* are obtained. Also the paper [1563] of Li and Mou is related: the authors study the *dual Orlicz geominimal surface area* by the dual Orlicz mixed volume and thus the existence and uniqueness of *dual Orlicz–Petty bodies*.

Continuing investigations from [2540], Lutwak's notion of dual affine quermassintegral is extended by Zhao [2545] to the concept of *Orlicz dual affine quermassintegrals*, with the subcase of Orlicz dual mixed volume. It is shown to be invariant on SL(n) transformations, and the corresponding *Orlicz dual*

Brunn–Minkowski inequality is derived. We cite here also Xia's work [2480], in which Minkowski type inequalities for dual Orlicz mixed affine quermassintegrals are proved.

Huang et al. [1245] construct, for each n-dimensional convex body with the origin in its interior, a series of *dual curvature measures* on the unit sphere which is associated with the dual quermassintegrals. Important subcases are the cone-volume measure and Alexandrov's integral curvature of the polar body. A new approach to this part of the dual Brunn–Minkowski theory is used to extend Alexandrov's variational formula for volume to dual quermassintegrals, and it turns out that the *dual curvature measures are differentials of the dual quermassintegrals*. As analogue of the *Minkowski–Christoffel problem* from the classical Brunn–Minkowski theory, in the dual Brunn–Minkowski theory the following *dual Minkowski problem* is studied: For an integer k with $1 \leq k \leq n$ and a finite Borel measure on the unit sphere, find necessary and sufficient conditions on that measure such that it is the k-th dual curvature measure of a convex body. The authors solve the essential symmetric case, i.e., when the considered measure is even and the solution is within the class of origin-symmetric bodies.

11.7.2 Valuations of convex bodies

In convexity, a *valuation* is a finitely additive function from a family of subsets of a given set to an abelian semigroup. E.g., Lebesgue measure is a valuation on finite unions of n-dimensional convex bodies, and other examples of valuations on finite unions of convex bodies are surface area and mean width. A starting point was the dissection theory of polytopes (in particular, Hilbert's third problem), and in the 1930s, Blaschke obtained the first classification of real-valued, $\mathrm{SL}(n)$ invariant valuations. Hadwiger continued with his classification of continuous and rigid motion invariant valuations. Today the theory of valuations is a classical and, at the same time, also very modern subfield of convex geometry.

BOOK PARTS AND SURVEYS

Important cornerstones of its earlier development were given by Hadwiger, McMullen and Schneider (see Hadwiger's book [1134] and the excellent surveys [1753] of McMullen and Schneider as well as [1745] of McMullen). Here we also need to highlight the excellent survey [2170] of Schneider, presenting basic notions, methods, and geometric approaches which were important for later developments of this subfield. Modern developments are determined by deep contributions of Alesker and many others, until ca. 2013 summarized in Chapter 6 of Schneider's monograph [2167]. Also the even more recent book-like publications and proceedings [1300] of Jensen and Kiderlen and [49] of Alesker summarize related notions, methods, and important results, and they show also relations to neighboring disciplines (like, e.g., integral and stochastic geometry, see also [51] of Alesker and Fu). We also mention the survey [1748] of McMullen on *valuations and the polytope algebra*, and the notes [1318] of Jochemko on *valuations of lattice polytopes*. In her very recent and important survey [1617], Ludwig surveys and updates almost all aspects of valuation theory in convexity, and also of more general settings (like valuations on function spaces). Due to the stormy development of valuation theory, it is impossible to give here a complete overview. Thus, the following representation is only a selection of interesting publications in this direction, exemplarily showing parts of the current theory. We also refer to our notes in Chapter 10 where several papers treating valuations (and related to notions like projection bodies, intersection bodies, centroid bodies etc.) are mentioned; we do not cite them here again.

We start with discussing the historical development of the subfield represented by very basic, comprehensive publications.

HISTORICAL DEVELOPMENTS

In the classical book [1134], Hadwiger presents his famous characterization theorem for the quermassintegrals of n-dimensional convex bodies, which yields also many applications. It says that all continuous rigid motion invariant valuations on convex bodies are a linear combination of the quermassintegrals (a shorter, elegant proof is due to Klain [1395]). Hadwiger forced research on further functionals with related properties, and on the *relations between valuations and dissection theory*. The first comprehensive survey on *valuations in convexity* which appeared after Hadwiger's monograph [1134] is [1753] of McMullen and Schneider. There the *volume of convex bodies* is described as origin of the occurrence of these additive functions in two different ways. One is based on the functions coming from the *Brunn–Minkowski theory* (such as mixed volumes, surface area functions, and Hadwiger's characterization of quermassintegrals), and the other one came up with Hilbert's third problem and the related *dissection theory of polytopes*. Besides classical examples like the Euler characteristic on unions of convex bodies and lattice point enumerators, also the *polytope algebra and dissection theory* are discussed, continued by polynomial expansions for general valuations and deep characterization theorems (here also *Minkowski-additive functions and quermassintegrals* are essential). Regarding Euler-type relations, also the expositions [2093] of Sallee and [2224] of Shashkin (not cited by McMullen and Schneider in [1753]) might be mentioned. The next detailed overview is McMullen's brilliant handbook article [1745], which also highlights the *polytope algebra* as the universal group for translation-invariant valuations on the family of polytopes over an ordered field. McMullen gives in [1745] also an update of the state of the art in the whole subfield. The survey [1613] of Ludwig collects interesting results on *affine invariant, equi-affine invariant and* SL(n)*-covariant valuations on convex bodies*. First affine invariant and equi-affine invariant real-valued valuations are presented, followed by characterization theorems for the affine surface area, the centro-affine surface area and further valuations. Then SL(n)-covariant and SL(n)-contravariant valuations with values in the space of tensors, convex bodies and star bodies are studied, followed by characterization theorems for the projection body, the difference body, the moment body, and the intersection body. And in his booklet [49], Alesker summarizes developments by presenting (not only) his most important contributions to the field. He starts with classical results, like McMullen's polynomiality and decomposition theorems, and Hadwiger's characterization of n-homogeneous valuations. Then the foundations for further topics are given (e.g., generalized functions on manifolds), to continue with his basic *irreducibility theorem*, saying that the natural representation of GL(V) in the space of even, translation invariant, continuous and k-homogeneous valuations on a finite-dimensional vector space V is irreducible; also odd valuations are treated. The author finally surveys deep results on valuations which can be seen as applications of the irreducibility theorem; for example, the product and the convolution of valuations as well as the Fourier type transform on valuations are presented there.

One might say that also with these two publications (namely [1613] and [49]) a large part of modern valuation theory is presented. With the (already mentioned) very recent survey [1617] Ludwig excellently updates them.

We continue with presenting papers treating *special aspects of valuations*, and we start with tensor valuations and related topics.

SPECIAL ASPECTS: TENSOR VALUATIONS AND RELATED TOPICS

Hug and Schneider [1259] study *local Minkowski tensors* which are valuations on the space of convex bodies with values in a space of tensor measures. This way they generalize the intrinsic volumes, the curvature measures, and the isometry covariant Minkowski tensors (introduced by McMullen [1746] and characterized by Alesker [43]). The authors classify, analogously to Hadwiger's famous characterization theorem (see again also the results in [43] of Alesker), all locally defined tensor

measures on convex bodies which share the properties of isometry covariance and weak continuity with the local Minkowski tensors.

Bernig and Hug [300] prove *new kinematic formulas* for continuous and translation-invariant tensor valuations on convex bodies. The authors extend some of the algebraic notions of valuation theory (such as convolution, product, and Alesker-Fourier transform) from the scalar-valued to the tensor-valued case. These results are finally translated into kinematic formulas involving the addition or intersections of convex sets. Here we also mention the related exposition [299] by Bernig and Hug, containing an interesting overview of new algebraic structures on the space of *translation invariant, smooth tensor valuations and related kinematic formulas*.

The Lecture Notes [1300] by Jensen and Kiderlen give an introduction to tensor valuations and their applications, especially in *stochastic geometry and local stereology*. Also extensions to non-Euclidean concepts and non-convex sets are discussed.

SPECIAL ASPECTS: (LATTICE) POLYTOPES

As already said, McMullen surveys in [1748] the foundations of the theory of valuations on convex polytopes, including an outline of the theory of polytope and weight algebras. With the help of this theory also McMullen's nice approach to the necessity part of the well-known *g -theorem*, characterizing face-vectors of simple polytopes, is presented. Based on valuation theory for convex polytopes, also an algebraic proof of the *classical Brunn–Minkowski inequality for polytopes* is given.

We also mention the survey [1937] by Panina and Streinu, where *virtual polytopes* are treated (i.e., the elements of the Grothendieck group of the commutative monoid of convex polytopes under the operation "Minkowski sum"), mainly discussing their geometric representations and relations between different representations. The connections to McMullen's polytope algebra are discussed, and the two-dimensional case is investigated in a particularly detailed manner. The authors study also *applications of virtual polytopes*, e.g. referring to Alexandrov's problem on convex bodies with conditions on their curvature, or studying *valuations and mixed volumes of virtual polytopes*.

Ludwig and Reitzner [1620] give a *classification of* $\mathrm{SL}(n)$ *invariant valuations on the space of convex polytopes*, and of polytopes which contain the origin.

Ludwig and Silverstein [1622] present a comprehensive study of *tensor valuations on lattice polytopes*, proving analogues of classical theorems on lattice points and valuations on polytopes using so-called discrete moment tensors. Such results are Ehrhart's theorem, the Ehrhart reciprocity theorem, and the Betke-Kneser theorem. Also the notes [1318] of Jochemko and the survey [405] of Böröczky et al. give an overview to the theory of *valuations on lattice polytopes*. It is shown that the Betke-Kneser theorem establishes a fascinating discrete analogue of Hadwiger's theorem for lattice-invariant valuations on lattice polytopes. Here the number of lattice points plays a fundamental role as analogue of volume. Analogies and differences between the valuation theory of convex bodies and that of lattice polytopes are discussed, and relations to the *Ehrhart theory* are presented, too.

Classification results for convex-body valued valuations defined on lattice polytopes are obtained by Böröczky and Ludwig [406]. The authors present a complete classification for *Minkowski valuations on lattice polytopes* that are related to the special linear group over the integers and are translation invariant. Notions like projection bodies and difference bodies occur there.

We continue with relations to *integral geometry*.

SPECIAL ASPECTS: INTEGRAL GEOMETRY AND FURTHER TOPICS

First we mention here the Lectures [51] of Alesker and Fu, nicely describing the relations between *valuations and integral geometry*.

Of great importance for convex and integral geometry, Klain [1395] and Schneider [2157] classify the *continuous, simple, and translation-invariant valuations on the space of convex bodies*, Klain in

the even and Schneider in the odd case. An easier approach is presented in [1482] by Kusejko and Parapatits, where the authors prove also a necessary and sufficient condition for two polytopes to be *translative-equidecomposable*.

Ludwig and Reitzner [1619] generalize a classical result of Blaschke by characterizing linear combinations of affine surface area, volume, and the Euler characteristic as *upper semicontinuous and equi-affine invariant valuations*. This is continued by Schütt and Werner in [2199], where the so-called extended p-affine surface area is characterized.

Klain [1399] extends Hadwiger's characterization theorem to the hyperbolic plane, also investigating cases from the higher dimensional *hyperbolic geometry*. He derives partial results on the characterization of hyperbolic volume, and the spherical volume in even-dimensional spheres is completely characterized.

In a series of papers, Alesker develops a theory of *valuations on manifolds*. Corresponding results of him and other mathematicians are summarized in his survey [47].

Ludwig [1614] characterizes the *Holmes–Thompson area from Minkowski geometry* with the help of the useful notion of *bivaluation*.

Haberl [1120] characterizes valuations which are continuous with respect to the Hausdorff metric and based on *Blaschke addition* (adding surface area measures). And in [1562] Li, Yuan, and Leng give a complete classification of *homogeneous,* $\mathrm{SL}(n)$ *contravariant* L_p *Blaschke valuations*, for $1 < p, p \neq n$ and p not being an even integer.

Based on results of Schuster, who (for $n \geq 3$) characterized smooth translation invariant and $\mathrm{SO}(n)$ equivariant even Minkowski valuations (based on usual Minkowski addition) with the help of convolutions with spherical Crofton measures, Schuster and Wannerer [2207] investigate *even Minkowski valuations* (which are equal for K and $-K$) establishing equivalent integral representations in terms of associated convex bodies (so-called Klain bodies) and generating functions. They describe relations between spherical Crofton measures, Klain bodies and generating functions. Based on all this, the authors additionally demonstrate a nice variety of methods which are useful for investigating valuations.

In [2208], Schuster and Wannerer succeed with a complete characterization of the *Minkowski valuations being continuous with respect to the Hausdorff metric, translation invariant and* $\mathrm{SO}(n)$ *-equivariant*, without needing additional assumptions like, e.g., evenness. Using his irreducibility theorem and properties of the real Monge-Ampère operators due to A. D. Alexandrov and Błocki, Alesker [50] studies an explicit relation between continuous valuations on convex functions invariant under adding arbitrary linear functionals, and translation-invariant continuous valuations on convex bodies. This means that he constructs a linear map from the function-space to the space of valuations and proves that this map has dense image and infinite-dimensional kernel. He shows ways to construct examples of *continuous valuations on convex functions* in such ways (using also complex, quaternionic, and octonionic Monge-Ampère operators and their Hessian).

Leinster [1540] develops a theory of convex sets and integral geometry for the space l_1^n where convexity is defined by geodesic arcs rather than line segments; isometries consist here of permutations of the coordinates followed by translations, and the analogue of the intrinsic k-volume is the sum of the k-volumes of projections on coordinate k-subspaces. For example, the fact that a continuous isometry invariant valuation is a linear combination of intrinsic volumes is analogous to Hadwiger's famous characterization of quermassintegrals.

For n-dimensional convex bodies, Colesanti, Hug, and Gómez [639] obtain a characterization of *translation invariant, continuous valuations which are homogeneous of degree* $n-1$ *and satisfy a Brunn–Minkowski type inequality*.

Colesanti gives in [637] an up-to-date representation of the geometric theory of log-concave functions, discussing the main points of this theory and its applications. Topics discussed are, e.g., functional versions of area measure and mixed volumes, and valuations on such functions.

Averkov and Bianchi [160] study *real-valued, even, translation-invariant valuations in relation to covariograms*.

Finally we mention here a recent series of papers of Colesanti, Ludwig, and Mussnig, treating valuations and related topics on convex functions under the common title "The Hadwiger theorem on convex functions". All of them are referred to in the last, very recent Part IV, namely [640]. E.g., in [640] itself new proofs of Hadwiger's theorem for smooth and for continuous valuations on convex functions as well as the theorem of Klain–Schneider on convex functions are derived.

11.7.3 The Bourgain–Milman inequality

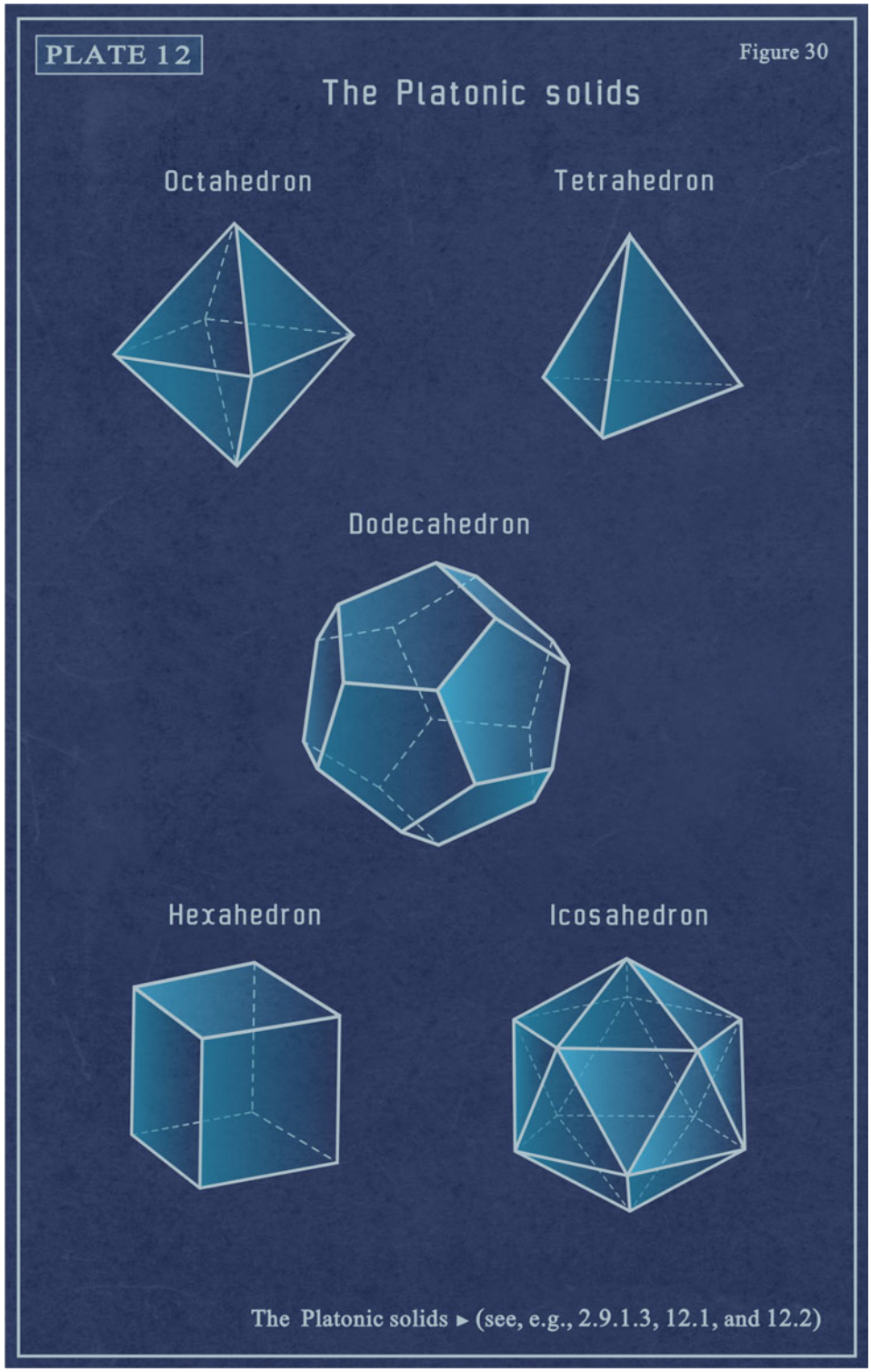
The *Bourgain–Milman inequality*, also known as an isomorphic version of the reverse Santaló inequality (see Bourgain and Milman [477] for the announcement, and then [478]) has, combined with results of Minkowski, also interesting applications in *geometric number theory*. The approach in [478] makes essential use of methods from the *(local) geometry of Banach spaces and convexity*, such as the isoperimetric inequality on the sphere, random subspaces, the Banach-Mazur distance, and the Rademacher projection.

Pisier [1973] found a new approach, using a striking theorem concerning the existence of certain isomorphisms from l_2^n onto an arbitrary n-dimensional (real or complex) Banach space; we refer also to the representation in the first part of his important monograph [1974] and Section 4.3 of the survey [1579] by Lindenstrauss and Milman.

Pisier's related approach to *Milman's inverse Brunn–Minkowski inequalities* and related topics is nicely extended by Carl and Edmunds in [576], using numbers which quantify the cardinality of coverings of convex sets or sections of convex sets by balls or cylinders. This tool, combined with the geometric viewpoint of convexity and that of linear operators, yields also a very nice approach to *Santaló's inequality and its reverse*. Klartag and Milman [1411] obtain functional versions of the reverse Santaló inequality.

Nazarov [1881] and G. Kuperberg [1476] improve the constant in the Bourgain–Milman inequality, and in the final part of the notes [2082] Nazarov's proof is discussed in detail. Also an exhaustive presentation of the state of the art of the whole subject is given there (see also the final part of Błocki's survey [374] and his paper [375], showing somehow how far Nazarov's method might reach). A simplification of Nazarov's approach is obtained by Błocki in [373]. In [961] Giannopoulos et al. give an alternative proof of the reverse Santaló inequality. It is based on some properties of the *isotropic position* of a convex body, particularly using the fact that every convex body with bounded isotropic constant satisfies the reverse Santaló inequality, see also Chapter 7 of Brazitikos et al. [496].

In Chapter 8 of the monograph [133] the M-ellipsoid and the M-position of a convex body are nicely discussed and used to approximate a convex body in many computations, thus getting also a key tool for proving the *Bourgain–Milman inequality*, as an isomorphic version of the reverse Santaló inequality.


PLATE 12
Figure 30
The Platonic solids
Octahedron
Tetrahedron
Dodecahedron
Hexahedron
Icosahedron
The Platonic solids ▸ (see, e.g., 2.9.1.3, 12.1, and 12.2)

Chapter 12
Historical steps of development of convexity as a field

"Not everything that counts can be counted. Not everything that can be counted counts."

— Albert Einstein

In this chapter we will show *historical steps of the development of convexity as a field* and, in addition, developments of the *relations between convexity and neighbouring disciplines* (like *differential geometry*) or fields that are partially subdisciplines (like the theory of *convex polytopes*).

We emphasize that the first part of this chapter is not meant as a "history of convexity"! Namely, to write such a history would demand many further and deeper studies, with more comparative considerations, and also with more presentations of simultaneous developments in other areas. History (of mathematics) as a subject has its own rules and methods, sometimes quite different from those of mathematics. Having this in mind, we limit ourselves to the (mainly chronological) representation of important historical articles, surveys, and books in which fundamental works and milestones of the field are derived, reflected, and summarized. Roughly the same style is continued in Section 12.4 of this chapter here, where the "growing together" with parts of the most important neighbouring disciplines is discussed. And here we note that it is also impossible to present the whole structure of such neighbouring disciplines; therefore the list given here is also not complete (one example of a missing discipline is *computational geometry*—we discuss it only very shortly in subsection 12.4.1).

It is clear that, going this way, many really important research papers cannot be taken into consideration since it is impossible to describe all important results of convexity and their origins. Especially in the last decades the field has grown extensively, and also the authors of this book were surprised to see the magnitude of these developments! Citing instead mainly historical articles, expository papers, research monographs, textbooks, and proceedings gives a chance to cover at least great parts of the field.

By these reasons we decided to present the material, which of course is mainly from the twentieth and the beginning of the twenty-first century, in preferably chronological form of *historical steps*, in addition ordered by sub-headlines like *research monographs, textbooks, proceedings*, and others. And in the second chapter part each of the neighbouring disciplines (like *discrete geometry*, or the *geometry of finite-dimensional real Banach spaces*) gets its own subsection.

Plate 12 shows the five convex regular 3-polytopes that are usually also called *Platonic solids*. Many facts about the history of these polyhedra are described in this chapter and in Chapter 2.

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12.1 Greek antiquity

Concrete citations of the ancient contributions discussed now can be found in the historical articles cited after this first paragraph here. In his *Timaeus* [1975], Plato (428/427–348/347 BC) studies properties of *regular convex polyhedra* (see Plate 11 above), and Aristotle (384–322 BC) uses *convexity and concavity* as metaphorical entities in several of his works; see, e.g., Chapter 9 of his *Physics* [125] and the corresponding discussion in Section 4.5 of Papadopoulos [1945]. Euclid (around 325–265 BC) presents in his *Elements* [131] basic knowledge on *equidissectable and equicomplementable polygons and polyhedra* as well as *regular polyhedra*. Although Euclid does not give an exact definition of the notion of convexity, he uses the words *convex and concave* appropriately.

In his treatise *On the Sphere and the Cylinder* (see [123]), Archimedes (287–212 BC) gives the *first precise definitions of convexity of curves* (this is also comprehensively discussed in [1477] and in [783]) and surfaces, using also concavity. He also proves that the *length of a convex curve* equals the limit of polygonal paths approximating that curve, extending this analogously to surfaces and surface areas. (This exhaustion method goes back even to Eudoxos of Cnidus (c.408–c.355 BC).) Archimedes continues in his treatise *On the Equilibrium of Planes* (see again [123]) with positions of *centers of gravity*. Due to Pappus (Book V of his *Collection*; see Papadopoulos [1949]), Archimedes was also the first who described all 13 *Archimedean solids*.

Apollonius of Perga (c.240–c.190 BC) uses the convexity notion to study *intersections of conics*; see his Book IV [113]. Ptolemy (c.100–c.170 AD) offers in his *Almagest* [2376] a famous necessary and sufficient condition for a convex quadrilateral to be cyclic. This is known as *Ptolemy's relation* between the side-lengths and the diagonal-lengths, and it is very useful due to many applications.

Zenodorus (between 200 BC and 90 AD) is the author of a lost treatise on *isoperimetric figures* in the plane (having equal perimeter but differing areas) and in space (with equal surface area, but differing volumes), showing also that among all *convex m-gons* of given perimeter the regular *m*-gon has maximal area (if such a maximizing *m*-gon exists). We know the work of Zenodorus only thanks to the *Collection* of Pappus [1949]. Summarizing, one can say that main insights on *convex sets, measurements of convex bodies, decomposition problems, the isoperimetric question, various properties of polyhedra, convex curves, and surfaces* were gained by Greek mathematicians already in ancient times.

Finally we mention here the interesting phenomena called *entasis* and *curvature* in the ancient Greek architecture, discussed in the essay [1963] of F. C. Penrose and in Section 4.6 of [1945]. Namely, there are famous buildings whose horizontal lines (lines of steps or of entablature) are, to suggest a better impression of straightness, replaced by slightly convex curves. This holds also for the "generators" in Doric columns.

These and more contributions to convexity from Greek antiquity are discussed and summarized in historical publications of Gruber (see [1064] and [1058]) as well as Papadopoulos (cf. [1945]); for the *history of the isoperimetric problem* we add Porter's article [1988]. These papers refer also to later periods, and Papadopoulos [1945] additionally treats the ancient (and later) history of the following further topics: *astronomy, mirrors and optics* (also in relation to conics), *billiards in convex domains*, and *convex and concave surfaces* in the writings of Aristotle.

We also mention the historical and mathematical exposition [1477] of Kupitz and Martini, in which they give a comprehensive discussion of the *history of rectification, using also ancient and later definitions of convexity*. The authors present in great detail the contributions of Hippocrates of Chios, Eudoxos, Euclid, Archimedes, and Pappus, which were paving the way for basic ideas from convexity (this is only the ancient part; the later development of rectification is also broadly discussed in [1477]). And also the two nice articles [952] of Gericke and [12] of A'Campo-Neuen and Papadopoulos should be considered here. They particularly refer to the *historical development of the notion of curvature in view of convexity*. Both papers start with corresponding works of Euclid, Apollonius, and Archimedes, and they discuss also further contributors in later times. We mention

that, of course, also historical surveys like Fenchel [856], Gruber [1064], and Dwilewicz [783] refer to efforts of ancient mathematicians. But the main contents of these works refer to later periods.

12.2 From the seventeenth to the nineteenth century

It seems that after the ancient Greeks for a long time no really fundamental, explicit contributions to convex geometry were made. But mainly starting with the seventeenth century many famous mathematicians started to contribute directly or indirectly to the theory of convex bodies and polytopes. Different aspects of related developments are emphasized in the mainly historical articles [856], [783], [1058], [1064], [1391], [1392], [1393], [1394], [887], [1477], and [12], from which we take a selection of main ideas for our representation. Thus, we will restrict ourselves to main lines of development, only at some places going into details (like now in case of Kepler).

Besides his scientific achievements in astronomy (like his three laws for planetary motion) and optics (seeing this field as an own topic of investigations), one of the most outstanding scientists of his time, namely Kepler (1571–1630), published also many results in pure mathematics. His comprehensive collected works [1367] (a project of 22 volumes, up to 1998 published by the Bavarian Academy of Sciences) are almost exclusively in Latin; the part which is translated into English or German is, step by step, increasing. A literature review showing these developments until 2002 is [973].

It is well-known that Kepler systematized and enriched all the existing knowledge on polyhedra and related topics by an approach which was mathematical in the strict sense: he defined *classes of polyhedra* and tried to verify their completeness (on this way finding, e.g., the class of *semiregular antiprisms*). He also understood that the small stellated dodecahedron and the great stellated dodecahedron (known already to Uccello and Jamnitzer) are *regular star polyhedra with non-convex faces*. Considered together with two other star polyhedra, today they are known as *the class of regular or Kepler–Poinsot star polyhedra*. Here we mention that the name of the "stella octangula", not belonging to this class (but being related), is also due to him.

In addition, Kepler successfully studied types of *zonohedra*, namely the *rhombic dodecahedron* and the *rhombic triacontahedron*, which are also known as *duals of Archimedean solids*. (The complete list of these 13 duals, sometimes also called *isohedra*, was given by Catalan in the nineteenth century, yielding also the name *Catalan solids*.) Continuing Archimedes and various Renaissance artists, Kepler was the first who systematically explored the class of *convex uniform polyhedra*, using a clear definition. Based on this, in his "Harmonice Mundi" (famous also by Hindemith's opera with the same title; see [1365]) he derived also the two *infinite families of semiregular prisms and antiprisms*, showing the completeness of this class.

Kepler's statement that there are 14 Archimedean solids (stated in page 11 of [1364] without explanation) is widely discussed in Grünbaum's clarifying article [1086]. Despite these logical investigations, Kepler was not free of the mysticism of his time. In [1365] the regular polyhedra are suitably presented with their Platonic associations fire, air, earth, water, and the cosmos. Also in [1365] Kepler nicely presents the interplanetary distances of all planets known at that time. For this he used a suitably ordered, nested system of in- and circumspheres of the five regular polyhedra, such that one Platonic solid fits between a pair of spheres (its in- and its circumsphere), these spheres containing the planetary orbits. (Of course, by not only one reason this system is false.)

Further scientific achievements of Kepler, which are related to convex geometry, refer to the *barrel volume* (historical development of integral calculus), the recently confirmed *Kepler conjecture* on the densest ball packings in Euclidean 3-space (a famous problem from discrete geometry; see the impressive collection [1146] of Hales and Ferguson), and various new ideas related to *crystal growth* (geometric crystallography; see, e.g., the survey [812] of Engel). Lagrange continued Kepler regarding densest lattice packings of circular discs in the plane (see Gruber [1064]). We mention a few additional

references which are close to the mathematical part of Kepler's work. Namely, in the papers [671] of Coxeter and [689] of Cromwell, Kepler's investigations of polyhedra are nicely discussed; for example, there occur the above-mentioned interplanetary distances, *Bilinski's second rhombic dodecahedron*, the *Archimedean solids and Kepler's two regular star polyhedra* (which were unknown to Poinsot, who 200 years later presented the two others; the completeness of this list of four was shown by Cauchy in 1813).

We continue with mentioning selected cornerstones of the development of convexity in the eighteenth and nineteenth century, based on the historical articles of Fenchel, Gruber, and Dwilewicz: [856], [1064], [1058], and [783].

It was Euler who gave, around 1753, important relations between the numbers of vertices, edges, and faces of convex 3-polytopes, among them being the famous *Euler formula* (rigorously proved by Legendre in 1794). However, it should be noted that already Descartes presented results on angles of polytopes which were direct forerunners of this relation (this is only known due to an annotation of Leibniz found in 1860; for details see Federico's book [835]). This development fruitfully continued in the direction of the combinatorial and the incidence structure of faces of convex polyhedra. For example, it was yielding statements like this one here, due to Euler: the number of faces with an odd vertex number is even. Regarding higher dimensions and (regular) convex polytopes, Schläfli's contributions from 1852 (not published until 1901) should be mentioned, and later related contributions were given by Steinitz (1920), and others.

Fourier (1789 and 1826) dealt with conditions for the solvability of linear inequalities, followed by Gordan (1873), Minkowski (1896), Farkas (1902), Motzkin (1936) and others; this led to the development of Linear Programming as an important field of the twentieth century (by Kantorovic and Dantzig). The historical way of this development is nicely traced by Fenchel [856].

Gauss (1831) found the densest lattice packing of balls in 3-space; this can be seen (especially regarding higher dimensions) as one starting point of a well-known and important subfield of discrete geometry.

The next famous mathematician also very important for convexity is Cauchy. He contributed, in the first half of the nineteenth century, with the following problems and concepts to convexity: First he investigated the *unique determination of convex polyhedral surfaces* up to proper or improper rigid motions if there is a homeomorphism between these surfaces whose restriction to any facet is a rigid motion; his proof (1813) had a minor gap, found by Hadamard (1907) and closed by Lebesgue (1909) and Steinitz (1934, posthumous). An excellent survey on Cauchy's theorem and the related rigidity theory is Connelly's handbook article [647], where also many generalizations and consequences (e.g., the related Alexandrov–Pogorelov theory, the infinitesimal version, higher dimensions) are presented. Cauchy also showed the completeness of the *list of Kepler–Poinsot polyhedra*. This result and his rigidity consideration discussed above are also presented in the monograph [690] of Cromwell. And another famous result of Cauchy, his *projection formula* or *surface area formula* (1850), is important in integral geometry and turned out to be a special case of *Kubota's integral recursion calculating quermassintegrals*; see, e.g., Section 3 of the handbook article [2182] of Schneider and Wieacker. Cauchy's contributions to convexity and their consequences (like Weyl's problem on the existence of closed convex surfaces with a given intrinsic metric from 1916) are additionally discussed in the historical article [856] of Fenchel, showing also the way to Alexandrov's related contributions from the forties.

The study of *projective convexity and polarity* goes back to von Staudt (1847) and Pasch (1882) (see, e.g., [1950]). The results of von Staudt were rediscovered by Minkowski (posthumous) and Steinitz (in the early twentieth century).

Also in the middle of the nineteenth century, Steiner contributed in different ways. We refer also to our remarks below under the headline "Selected and collected works" (see [2294]), and we add here his contributions to the following notions and concepts which directly or indirectly enriched the field of convexity: *projective duality*, *Steiner points* (as curvature centroids), and *Steiner symmetrization*

and the four hinge method (used for solving the *isoperimetric problem*, for which he gave a series of proofs with missing existence part, as Dirichlet pointed out; see also our Section 2.7 and our subsection 2.9.3). Further on, the so-called *Steiner space partitions by hyperplanes* (today called *arrangements of hyperplanes* (see Chapter 18 of Grünbaum's famous book [1085] and the monographs [1919] of Orlik and Terao as well as [23] by Aguiar and Mahajan) and his formula for the *volume of parallel bodies of convex bodies* (cf. Section 4.2 of [2167]) should be mentioned here. Steiner also studied various problems related to maxima and minima. For example, due to his contribution the famous *Fermat–Torricelli problem* from location science is sometimes called *Steiner–Weber problem*; see Chapter II of Boltyanski et al. [432]. And also the concept of *Steiner minimal trees* should be mentioned here; see, e.g., Cieslik [626]. The latter two problems are interesting here since they are broadly studied in normed spaces, using the convexity of the unit balls. The geometric properties of these convex unit balls are essential for the ways of solving these problems, and both monographs [432] and [626] contain also historical surveys. Regarding the *isoperimetric problem*, famous names followed that of Steiner, among them being Schwarz, Weierstrass, Minkowski, Blaschke, Bonnesen, E. Schmidt, and Dinghas. These developments are discussed in Chapters 5 and 6 of Hadwiger's book [1134], different chapters of the monograph [531] of Burago and Zalgaller, and Osserman's comprehensive survey [1923]. The *volume of parallel bodies* is combined with intrinsic volumes and led to the notion of "functional theorems" due to Blaschke (1937).

12.3 The twentieth century and the twenty-first century

12.3.1 Important contributors and authors of research monographs

We start here with famous authors of trend-setting monographs or, in exceptional cases, articles by which they strongly contributed to the development of *classical convex geometry* itself. It is clear that there are many further authors of such influential books who contributed to convexity, but their fundamental monographs are already or will be cited and described in other chapters of our book, under different headlines. For example, we mean here Coxeter's "Regular Polytopes" and the "Abstract Regular Polytopes" by McMullen and Schulte in Chapter 2, L. Fejes Tóth's "Lagerungen" and "Regular Figures" in Section 12.4.1, Grünbaum's "Convex Polytopes" and Ziegler's "Lectures on Polytopes" in Section 12.4.2, the "Introduction to Tropical Geometry" of Maclagan and Sturmfels in Section 12.4.3, Santaló's "Integral Geometry and Geometric Probability" like also the "Stochastic and Integral Geometry" of Schneider and Weil in Section 12.4.5, Pisier's book "The Volume of Convex Bodies and Banach Space Geometry", Thompson's "Minkowski Geometry", the monograph "Geometry of Isotropic Convex Bodies" of Brazitikos et al., as well as the two volumes of "Asymptotic Geometric Analysis" due to Artstein-Avidan, Giannopoulos, and V. D. Milman in Section 12.4.7. Of course, there are various further such important books. And besides these authors of trend-setting monographs, there is of course a long list of famous further contributors to the field—we would, however, not be able to reflect this in a complete manner, also due to the stormy development of the field within the last decades.

H. BRUNN (1882–1939) and H. MINKOWSKI (1864–1909)

We come now, as a starting point of the development in the twentieth century, to the pioneering work of Brunn and Minkowski. It is well known that their related contributions form somehow the birth of convex geometry as an own discipline, which then developed faster in the twentieth century.

Based on a strict definition of the notion of convexity, one starting point of convex geometry was the dissertation (see [510] and also [511]) of *H. Brunn* from the end of the nineteenth century, containing

also an *early version of the Brunn–Minkowski theorem*. Although Brunn listed results without proofs, his contribution was quite deep due to its refreshing originality.

Soon after, the importance of this theorem in more general settings was observed by *H. Minkowski*, who laid also the conceptual foundations of the new coming field of mathematics by introducing natural and efficient notions (like *support function*). In the historical article [1392] of Kjeldsen tribute is paid to the achievement of Minkowski not depending on Brunn's work; see also [1393]. Additionally, it is underlined in [1392] that ideas basic for convex geometry can be found already in Pasch's geometry book [1950] from 1882, which is also acknowledged in the monographs [2296] of Steinitz and Rademacher and [654] of Coppel. Minkowski's *theory of mixed volumes* and the *Brunn–Minkowski theorem* were investigated in the period from 1887 to 1904. Mainly in his works "Volumen und Oberfläche" (see [1807] and [1803]) and the unfinished treatise "Theorie der konvexen Körper, insbesondere Begründung ihres Oberflächenbegriffs" (for the first time published in [1807], pp. 131–229), Minkowski provided important related tools and showed ways to applications, such as *geometric inequalities referring to convex sets* (see also [1391]) or the *determination of polytopes or convex surfaces via curvature concepts/surface area functions* (we refer also to the "Literaturbericht" cited with [1807] below and the comment [2468] of Wilson).

In 1910 Hilbert connected Minkowski's studies with certain elliptic PDE's on the unit sphere and proved Minkowski's related inequalities using his theory of integral equations (see Fenchel [856]). In [1807] and [1803] one can find further works of Minkowski on convexity, e.g., theorems on *convex polyhedra* and *sets of constant width*. And also the appendices of [1803] are interesting: the first refers to the *geometry of numbers*, and the second to *mixed volumes*, existence and uniqueness theorems, and sets of constant width.

Minkowski's book [1804] has to be mentioned here, since the geometry of numbers is closely related to convexity and discrete geometry. Namely, based on his studies of quadratic forms associated with polyhedral regions he succeeded to separate the *geometry of numbers* as an own field from number theory (we discuss this more concretely in our subsection on discrete geometry). Minkowski realized that convexity and central symmetry of ellipsoids are important properties of ellipsoids, and this inspired him to prove the famous theorem that *every n-dimensional convex body centered at the origin and with volume larger than* 2^n *contains at least one further point with integral coordinates*.

We also emphasize one further aspect from [1804], not only referring to discrete geometry. As already mentioned by Dunford and Schwarz (see p. 372 in their book [775]), the theory of finite-dimensional real Banach spaces (or the geometry of finite-dimensional normed linear spaces) "is primarily due to Minkowski". Therefore Busemann and others use the name *Minkowski geometry* for it (in particular also Thompson in his book [2361]), which naturally leads also to the term *Minkowski spaces*. (One should note that "this Minkowski geometry" is clearly different from the Minkowskian Spacetime Geometry.)

More precisely, in [1805] and [1804] Minkowski introduces a class of functions, called *einhellige Strahldistanzen*, for modelling direction-dependent oriented distances in spaces whose unit balls are convex bodies having the origin as an interior point (gauge bodies). One motivation came from crystal growth: crystal shapes are mostly polyhedral, and one can model such situations by the assumption that units for measuring lengths are different in different directions, so meant in the Euclidean background metric. It is clear that with this notion he even introduces a concept more general than norms, since the symmetry assumption is dropped. That goes hand in hand with Gromov's statement (written in the introduction of his influential book [1048]): "Besides, one insists that the distance function be symmetric, that is, $d(x, x') = d(x', x)$. (This unpleasantly limits many applications: the effort of climbing up to the top of a mountain, in real life as well as in mathematics, is not at all the same as descending back to the starting point.)". This quote refers to the fact that in many applied situations the more general concept of *quasi-metric spaces* (introduced in Wilson's work [2468]) is more useful than the subcase of metric spaces. Thus, neglecting the symmetry of distance functions yields an

analogous transition from Minkowski spaces to so-called generalized Minkowski spaces. For example, these spaces are geometrically investigated by Zaustinsky [2526], in Chapter 15 of Rockafellar's book [2046], Cobzas' book [631], Jahn's dissertation [1292], and, to mention a discipline far away from analysis, in various papers from computational geometry.

Nowadays Minkowski's designation *einhellige Strahldistanzen* is usually replaced by *gauges*, *asymmetric distance functions*, *Minkowski functionals*, or *convex distance functions* (the latter mainly used in computational geometry). Hence Minkowski laid also the axiomatic foundations to this concept which is more general than that of finite-dimensional real Banach spaces; but of course he also considered the symmetric subcase (thus generally satisfying $d(x, x') = d(x', x)$; see above), and he called the respective distance concepts *wechselseitige Strahldistanzen*, which later were renamed *norms* and became popular in many mathematical disciplines. This development continued with Helly (1921) and Banach (1922) yielding the concept of Banach spaces and the Hahn–Banach theorem (1927).

Here we mention the historically interesting fact that Riemann, about 30 years before Minkowski, in his famous Habilitationsschrift [2033] already suggested one such geometry (namely that of the finite-dimensional ℓ_4 norm). We refer to Busemann's survey [547], where the motivation of Riemann is discussed in detail.

Summarizing all this in view of convexity, one sees that due to the axioms, which were proposed by Minkowski, both the concepts of Minkowski geometry (with unit balls which are convex bodies centered at the origin) and generalized Minkowski geometry (with unit balls which are convex bodies having the origin as interior point) go back to him, and that this is the starting point of developments creating the fields of *Finsler geometry* (via relative differential geometry; see Busemann's survey [547]), *Banach space theory* (later also adding the asymptotic theory and infinite-dimensional spaces; cf. the comprehensive historical representation [1972] of Pietsch, introductory parts of Thompson's book [2361], and great parts of *discrete geometry* (mainly with topics related to Minkowski's geometry of numbers; here the fourth part of Gruber's monograph [1070] is recommendable).

But also other fields are closely related to this, e.g., *distance geometry* (in those days also called metric topology) in the spirit of Menger and Blumenthal (see the article [1769] and the book [371], in the latter case not mentioning Busemann's work on metric methods in Finsler spaces), *combinatorial geometry of convex bodies* (Chapters II and V of Boltyanski et al. [431]), *optimization and location science* (see, e.g., Jahn et al. [1293]), and *computational convexity* (see the handbook articles [1025] and [1028] of Gritzmann and Klee).

Further mathematical achievements of Minkowski in convexity are nicely described in [856], namely his already mentioned theorem that a centered convex body in Euclidean n-space having volume at least 2^n must contain an opposite pair of integer lattice points, clearly different to the origin (this is also important for Diophantine application), and many additional efforts regarding convex sets themselves, such as linear combinations of convex sets or his generalized notion of surface area from 1901. Here the developments around his inequalities regarding mixed volumes have to be mentioned (Chapter 6 in this book here). Minkowski's contributions were also basic for the development of the concept of locally convex topological vector spaces (von Neumann, 1935), the duality concept and the theory of convex functions, unique determination of general convex bodies by surface area functions (Alexandrov's extension in the thirties), the weakened Brunn–Minkowski theorem for non-convex sets (Lyusternik, 1935), and non-Euclidean spaces (E. Schmidt, late forties). The importance of Minkowski's convexity concept for the development of mathematical programming (and vice versa) is nicely discussed in [1394]. And various basic notions and tools go back to him, e.g., the notion "support function".

Biographical notes and a short overview regarding Minkowski's mathematical work related (not only) to convexity can be found in the Note 4.13 of the fourth chapter of Gardner's book [930]. Special recognition to Minkowski's (and Peano's) contributions to the development of modern convexity, and its influence also to convex analysis, is given in the interesting historical works [887], [1392], and [783].

E. HELLY (1884–1943) and J. RADON (1887–1956)

In many books and surveys, directly or indirectly related to convexity, one can find *Helly's famous theorem on intersections of convex sets* and/or *Radon's partition theorem*, using the notion of *convex hull*. The two excellent surveys [708] of Danzer, Grünbaum, and Klee and [790] of Eckhoff, treating both theorems, might suffice as basic related references. But there are more convexity-related activities of Helly and Radon, and we discuss them now from the historical point of view.

Tikhomirov [2367] nicely appreciates the difficult life and comprehensive work of the Viennese mathematician Eduard Helly, whose first full-time academic appointment came to be only in 1943, in the United States. Helly was one of the early trailblazers of functional analysis using basic properties of convex sets. He carried over fundamental ideas of Minkowski to introduce a concept of norm for general sequence spaces (1921), suitably *generalizing the notion of Minkowski functional of a convex body* and investigating related *duality properties*. These and other results were found by him before Banach defined *complete normed spaces* (1922). In a more special setting he proved the Hahn–Banach theorem, before Hahn (1927) and independently Banach (1929) derived it in full generality! But in his early contributions to this field, Banach did not underline the close connection to convexity. This was done by collaborators like Mazur, explaining the connection between the Hahn–Banach theorem and notions like supporting and separating hyperplanes. All this is nicely described in three historically interesting articles, namely [556] and [557] by Butzer et al., and [1884] by Netuka and Veselý; the authors underline Helly's important role regarding the development of fundamental principles of functional analysis which, e.g., are related to the *extension theorem of Hahn–Banach* (see also Hochstadt [1205]) and to the *uniform boundedness concept*. In [556] also a list of Helly's publications and a list of the courses he gave in Vienna are appended, and [1884] contains the original proof of *Helly's famous theorem concerning intersections of convex sets*.

The name of Johann Radon occurs several times in convex geometry. In the collected works [2010] of Radon, his contributions to the field of convexity are completely presented and also reviewed by Gruber. The main contributions are *Radon's partition theorem* (saying that any $(n + 2)$-point set in n-space can be partitioned into two sets whose convex hulls intersect), results on *isoperimetric inequalities and generalized convexity*, the famous *Radon curves* (important also for the geometry of normed planes), and the *integral transforms* called after him. These *spherical Radon transforms* from integral geometry have many fruitful applications in convexity, and they go back to Funk (1911) and even to Minkowski (1904); Radon's original paper is [2012]. They present an important tool in modern tomography (see the interesting report [2388] of Ullrich describing the long way to this important application); within convexity, methods from *geometric tomography* follow that line. Thus, how spherical Radon transforms and related tools occur in convexity is excellently presented in Section 3.4 of Groemer's book [1042], in Groemer's survey [1039], in Appendix C of Gardner's monograph [930], and in Chapter 3 as well as the Appendix in Schneider's book [2167]. *Radon's partition theorem* is fundamental for *combinatorial convex geometry* (see again the surveys [708] and [790]). It should be noted that in many references, the approach to Helly's intersection theorem is taken via Radon's partition theorem. Further on, we emphasize that the notion of *Radon curves* is important for *Minkowski and Finsler geometry* by characterizing special normed planes via symmetry of *Birkhoff's orthogonality concept*; cf. the original paper [2011] and, for many further properties of Radon curves, the recent expositions [1722] (here especially Section 6) and [1721] of Martini, Swanepoel, and Weiss. It turns out that these normed planes are precisely the *Radon planes*, namely with Radon curves as circles. For interesting biographic notes on Radon, mainly from the viewpoint of convexity, we cite the contributions [912] of Funk, [1056] of Gruber, and the Notes 2.8 in Gardner's book [930].

W. BLASCHKE (1885–1962)

After Brunn and Minkowski, one of the main contributors to the theory of convex bodies was W. Blaschke. Designations like "Blaschke–Santaló inequality", "Blaschke addition", "Blasche-Lebesgue theorem", "Blaschke's selection theorem" for existence proofs, the "Blaschke diagram", "Blaschke's ellipsoid characterization via planar shadow boundaries", and the "Theorem of Blaschke and Deicke" send a clear signal regarding the traces he left behind. Listing 40 articles from the period 1912 to 1956, Volume 3 of Blaschke's Collected Works (i.e., the collection [360]) refers to his direct contributions to convex geometry, e.g., to his proofs of the *Blaschke–Santaló inequality* (volume product) and of the *affine isoperimetric inequality*. He investigated general convex bodies in Euclidean and affine geometry, and also smooth convex surfaces in Euclidean and equiaffine differential geometry. Many of his convexity results are also cited in the monograph [446] of Bonnesen and Fenchel. Together with these relations to differential geometry and also to the calculus of variations, he defined the field of *integral geometry* (to which, e.g., Cauchy's projection formula belongs), to reach a certain separation from the field of geometric probability. Since nevertheless there is overlap with convexity, we cite here also parts of his Collected Works more referring to these fields, namely [356] and [359]. For example, many topics discussed in Blaschke's book [356] on *integral geometry* refer directly to convex bodies.

For example, Blaschke's early and famous book "Kreis und Kugel" [358] shows substantial directions, fresh ideas, and new methods for that time, thus becoming a precursor of many later publications. Blaschke studies the *isoperimetric problem in two and three dimensions* and the *Brunn–Minkowski theorem*, based on clearly defined notions of the field and tools like his selection theorem and Steiner symmetrization. In the final chapter various extremum problems are presented. These are based on *curvature notions of convex surfaces* in the spirit of differential geometry. The appendix contains further nice problems and results from convexity, referring, e.g., to *bodies of constant width and of constant brightness* (here the first example of a body of constant brightness is depicted), as well as the famous *characterization of ellipsoids via planar shadow boundaries*. *Blaschke addition* and *Blaschke symmetrals* are discussed in Gardner's book [930], and for the *Blaschke diagram* see Hadwiger's book [1133] and the article [486] of Brandenberg and González Merino. The *Blaschke–Lebesgue problem* (asking for the bodies minimizing the volume among all bodies of the same constant width) is, for example, discussed in Section 14.2 of the monograph [1713].

Natural extensions of results in the above-mentioned collections [360] and [359] can still stimulate recent research, and having these collections today is particularly useful since several original papers of Blaschke were hard to get. For bibliographical facts we refer to Note 6.4 of Chapter 6 of Gardner's monograph [930]. Blaschke also wrote the autobiographical book [357] which reflects his time in an interesting manner, since he travelled a lot.

Also in Gruber's historical survey [1064] Blaschke's contributions to convexity are comprehensively discussed, namely *characterizations of balls and ellipsoids* (e.g., via planar sharp shadow boundaries), his *selection theorem*, the *Blaschke symmetral* and *Blaschke addition*, the relations between convexity and (global) *differential geometry* (e.g., curvature types, geodesics,...), *affine geometry of convex bodies* (for instance with notions like *affine surface area*, the *Blaschke–Santaló inequality*, the *affine isoperimetric problem*), and the development of *integral geometry*.

W. SÜSS (1885–1952)

Also the founder of the Mathematical Institute at Oberwolfach (1944), namely W. Süss, strongly contributed to convexity. His results using and enriching convexity mainly refer to *classical convexity* and the *differential geometry applicable to Minkowski geometry*. The latter direction was important for the development of *Finsler geometry*. A bibliography of his mathematical works has been summarized by Gericke [950]. It is important to mention that during a longer visit in Japan he came into fruitful contacts with the geometers S. Nakajima (Matsumura), M. Fujiwara, and T. Kubota. A large collection of results on convex sets which are due to Süss and to these excellent Japanese geometers is cited by Bonnesen and Fenchel in [446]. (This is the main reason to mention Süss exceptionally here, i.e.,

without a trend-setting monograph.) And interesting biographical notes on Süss are given by Gardner in the Note 5.8 of Chapter 5 of Gardner [930].

T. BONNESEN (1873–1935) AND W. FENCHEL (1905–1988)

We come now to the monograph [446] of T. Bonnesen and W. Fenchel, which gave the first comprehensive overview to the field till 1934. But first we mention Bonnesen's booklet [445] from 1929, dealing with *sharpenings of the isoperimetric inequality* in two and three dimensions (the latter case is also called *isepiphan problem*). The author collects and proves several results in this direction, and he prefers geometric methods like Steiner and Schwarz symmetrization. Continuing in this setting, he presents also results of Brunn and Minkowski, such as Minkowski's (sharpened) inequalities for the volumes and mixed volumes of two convex bodies; related results on convex polyhedra are also obtained. Then some basics on *convex functions* (in the sense of Jensen) are developed, and also questions on sharpenings of the isoperimetric problem in terms of calculus of variations are dealt with.

Thanks to the excellent monograph [446] of Bonnesen and Fenchel and its comprehensive list of references, we have a complete overview regarding the developments and literature in the field of convexity between Brunn (his dissertation "Über Ovale und Eiflächen" was finished in 1887) and 1934. In addition, for more than sixty years the book [446] was the main reference in the field of convexity that became widespread, also with many applications in various other branches of mathematics. And still today experts and non-experts like to look at it, supported also by the late translation (1987) into English. In the central part of it, a self-contained representation of the Brunn–Minkowski theory, with proofs and showing further developments, is given. Many further topics are discussed, see below; their level of detail depends on their proximity to the Brunn–Minkowski theorem. Some natural themes of the field are only discussed to a limited extent (e.g., problems closer to differential geometry, or the relations to affine geometry), but in view of the list below nevertheless an impressive completeness is presented. Thus, due to its refreshing style and this completeness, it is a milestone for researchers, but also suitable as a textbook for beginners in the field. The main guidelines discussed are basic properties of convex sets, convex hulls, boundary structure and supporting hyperplanes, polytopes, cap bodies and tangential bodies, representations via supporting and radial function, polarity, linear combinations of bodies, parallel bodies, homothets, concave families of convex bodies, convergence and Blaschke's selection theorem, approximations via polytopes or analytically bounded bodies, integral formulas for volume and mixed volumes, surface area, cross-section measures, projection bodies, circum- and inball, relative differential geometry, symmetrization procedures (of Steiner, Schwarz,...), extremal problems, Minkowski's inequalities and sharpenings, difference bodies, isoperimetric problems, unique determination via curvature functions, centrally symmetric convex bodies and lattice points in their interior, bodies of constant width, characterizations of circles, ellipses, balls and ellipsoids, and curvature properties of curves and surfaces. In 1971 appeared a reissue of the 1948 Chelsea reprint of the classical version from 1934, which was reviewed in the Zentralblatt. The English translation from 1987 was also reviewed, and the Springer reprint from 1974 contains a list of minor errata.

Bonnesen is particularly known for *Bonnesen's inequality*, a strengthening of the classical isoperimetric inequality with many variations and further extensions, and Fenchel published results which became fundamental for convex analysis and also nonlinear programming. Here we have to mention also Fenchel's historical article [856] that we cited already.

A. D. ALEXANDROV (1912–1999)

A. D. Alexandrov was one of the greatest geometers of the twentieth century and also a leading convex geometer, and his deep results in mathematics are considered to be among the best contributions of Russian science. Honors one can find in [1485] by Kutateladze, and in the preface of his Selected Works [38]. The latter book contains also a list of his scientific publications and biographical notes.

This collection, interesting for graduate and postgraduate students, researchers, and also historians in mathematics, gives also a nice overview on Alexandrov's scientific heritage (see, additionally, Note 3.13 of Gardner's monograph [930]).

Regarding convex geometry, Alexandrov's main work refers to the theory of (polyhedral) convex surfaces in 3-space; his two classical books [40] and [39] on the *intrinsic geometry of convex polytopes and convex surfaces* are most important, and papers by him, collected in Part I of his Selected Works [38], add further topics. These refer to *Minkowski's theorem on the determination of convex polytopes by the exterior normals and the areas of their facets* (an elementary proof is given) and its extensions, e.g., to convex bodies having analytic support functions and referring also to the *differential geometry of not necessarily differential convex and non-convex surfaces*. Further studied subjects are *mixed volumes, surface area measures* as well as various *modern notions of the Brunn–Minkowski theory*. The *intrinsic geometry of arbitrary convex surfaces* (introducing the concept of a metric of positive curvature which also helps to solve Weyl's related problem), *bending and rigidity problems of convex surfaces* (introducing the notion of infinitesimal bending, used already in the thirties), the already mentioned *differential geometry of not necessarily differentiable convex and non-convex surfaces* (here also Pogorelov and Zalgaller have to be mentioned), and the famous *Alexandrov–Fenchel inequality*.

Extending a result of Venkov on translative space-fillers to spaces of constant curvature, Alexandrov contributed also to the *theory of tilings* and to crystallography. Also publications showing relations to PDE and mathematical physics are presented in [38]. The survey [456] of Borisov et al. is not the latest, but a very good exposition reflecting Alexandrov's scientific biography and work. Coming back to Alexandrov's two monographs [40] and [39], we add the following summaries.

It is known that, due to various articles of Alexandrov, great parts of his book [40] were known already before its appearance in 1948, but that in contrast to these papers the work [40] contains complete proofs and deep additional material; it is a classical book on convex surfaces in 3-space, studied via their well-defined intrinsic metric and with the help of polyhedral approximation. The *axiomatic foundation of this intrinsic geometry of convex surfaces in 3-space* is an important issue of [40], and it is asked for necessary and sufficient demands that a metric space is isometric to a convex surface (i.e., *the realization of a metric space by a closed convex surface* is in the focus). Besides this central concept, also the following topics on convex surfaces in three-dimensional space are studied in [40]: *shortest curves in convex surfaces*, the related *concept of angle and curvature notions*, the subcase of convex polyhedra, the notion of *intrinsic area* (via triangulations), and extensions of these results to surfaces in spherical and hyperbolic spaces.

Alexandrov's monograph [39] is encyclopedic regarding questions of the following type: What geometric properties should be prescribed for a convex polyhedron P such that P is determined up to a certain transformation (translation, motion, similitude)? The author studies the *bounded and the unbounded situation*, in both cases with finite vertex, edge, and face numbers, but in the second one with possibly unbounded faces. Doubly covered planar regions which are convex and polygonal are considered as polyhedra to simplify the formulation of the theorems. Alexandrov considers plane schemes of polyhedra and develops from this their elementary topology (Euler's theorem and Cauchy's lemma on the distribution of plus and minus signs). It is proved that for a given plane scheme with positive curvature and satisfying Euler's theorem exactly one bounded convex polyhedron belonging to this scheme exists, up to motions, and that this also holds for the unbounded case (with additional hypotheses).

Minkowski's theorem (on polyhedra with the same systems of outer normal vectors and analogous face areas) is presented with a new proof (based on Brouwer's theorem of the invariance of domains). Polyhedra whose vertices lie on given rays with the same starting point are studied similarly, and also *infinitesimal rigidity* is contemporarily discussed. One can see [39] in the line of books reconsidering classical synthetic geometry in that time (a further famous example is the classical monograph [1197] of Hilbert and Cohn-Vossen). It should be noted that the edition of [39] cited here has been prepared by A. D. Alexandrov and Zalgaller. Since many open problems from the 1950 edition were solved in the

meantime, numerous related comments (also on the used methods) are added, like also historical notes and possibilities of further generalizations. As valuable supplements, also the contents of two articles (by Volkov and Shor) are added. With its 377 references, Alexandrov's book [39] reflects the state of research in those days. (Regarding completing contributions of Zalgaller to the *intrinsic geometry of not necessarily convex surfaces* and related *polyhedral approximation*, we mention the monograph [41] which also reflects many related results and tools.)

A. V. POGORELOV (1919–2002)

Many results of A. V. Pogorelov are strongly related to Alexandrov's work (see Sections 7.1 and 7.2 of [2167] and Section 10 of [1070]). More precisely, Pogorelov's monograph [1979] should be seen in close connection with Alexandrov's book [40], since many results from there are used in [1979]. Also most of the author's results treated shortly below can be found already in Pogorelov's earlier booklets and monographs, namely in [1976], [1977], [1978], and others. The terminology is as follows: A closed [open] convex hypersurface is the boundary of a bounded [unbounded] convex set with non-empty interior, and a convex hypersurface is a connected open subset of a closed [open] convex hypersurface. Two convex hypersurfaces are *isometric* if they are intrinsically isometric, and a convex hypersurface is *rigid* if any convex hypersurface isometric to it is congruent to it. The book refers to related results in n-dimensional space, but partially also presented for the three-dimensional case. It starts with a review on (mainly Alexandrov's) *intrinsic geometry of convex surfaces*, and it continues with regular convex hypersurfaces having a regular metric, *uniqueness of convex hypersurfaces* (regarding intrinsic isometry), the *rigidity of convex hypersurfaces without flat pieces*, *convex hypersurfaces in spaces of constant curvature*, spherical maps of convex hypersurfaces (*Minkowski's and Christoffel's problems, including corresponding stability results*), the geometric theory of Monge–Ampère equations, and hypersurfaces of bounded extrinsic curvature (including also their approximation by smooth hypersurfaces, and their relation to the hypersurfaces of bounded intrinsic curvature investigated by Alexandrov).

In the later booklet [1980], Pogorelov gives a complete and correct presentation of *Minkowski's multidimensional problem*, namely to determine a smooth convex hypersurface H if a sufficiently smooth and positive function f on the unit sphere with its center of mass at the origin is given which presents the Gaussian curvature of H. Related problems on Monge–Ampère equations are discussed, too. The author starts with a *survey on the differential geometry of convex surfaces* going back also to the monograph [446] of Bonnesen and Fenchel, and to papers and books of Alexandrov, the author himself (with corrections), and Busemann (see below). Pogorelov gives also a proof of Minkowski's theorem without considering smoothness properties and, in the final chapter, a related characterization of quadrics. Here one should cite also the deep paper [1899] by Nirenberg and the interesting, more recent exposition [1091] by Guan.

H. BUSEMANN (1905–1994)

H. Busemann is one of the main founders of *metric geometry, convexity in metric spaces*, and *convexity theory combined with (synthetic) differential geometry*; the corresponding references are given below. So he became also the leader of *synthetic Finsler geometry* in the twentieth century, considering Finsler spaces as certain geodesic metric spaces satisfying conditions that geodesics have to satisfy (e.g., local extensibility). He wrote important papers to create, besides other developments, the fundaments of *local Finsler geometry*, i.e., of the *geometry of finite-dimensional normed spaces* also called *Minkowski geometry* (see Busemann's nice survey [547] and the book [2361] of Thompson). His approach to Finsler spaces is discussed in the recent survey [109] of Andreev, where it is also shown how nicely this allows the study of singular Finsler spaces. (Here we also recommend to read the short article [1944] of Papadopoulos about Busemann's life; see also Part 8.14 in the Notes to Chapter 8 of Gardner's monograph [930].)

Combining metric spaces and convexity, Menger's classical theory on metric properties of spaces, continued by Blumenthal (see [1769] and Blumenthal's later book [371]), can also be seen as some starting point for Busemann's studies. Based on [1769] and [371], Busemann's book [544] refers to *metric geometry of manifolds*, more precisely having a locally Euclidean and mainly (but not only) two-dimensional setting in mind. An interesting mixture of methods from differential geometry, topology, and group theory is used, and the representation concentrates on *Finsler spaces* (thus also generalizing the concept of Riemannian spaces) which may be *locally defined as Minkowskian* (i.e., following the axioms of Minkowski geometry; see again Thompson's book [2361]). This means that sufficiently small neighborhoods of a point x are isometric to a Euclidean neighborhood of it, which is metrized by a *convex body B symmetric at x*. So also various own results on Minkowski spaces (i.e., on finite-dimensional real Banach spaces) are treated. In this framework, the geometry of B is the main link to convexity. For example, a number of *characterizations of (special) Minkowski spaces* (for instance, via the notion of perpendicularity) is given in [544] and papers cited there. Related to this, the author poses also interesting problems (for higher dimensions), some of them close to convexity. He continues such investigations also in later publications, thus stimulating research activities by deep questions.

Rewriting and extending these parts of [544] in his later book [548], he gives a *synthetic approach to global differential geometry* and its connections to *foundations of geometry*, and also to *calculus of variations*. This book is more general than [544] and is mainly devoted to the general concept of so-called *geodesic spaces with the metric convexity property* of Menger. Two more axioms are needed before the existence of geodesics with usual properties is assured: an axiom of local prolongability of segments, and a condition guaranteeing its uniqueness. Then G-spaces are finitely compact metric spaces which satisfy these axioms and have the convexity property. Many different geometries are presented by special G-spaces, and so Busemann's work is, as already said, also very important from the viewpoint of foundations of geometry. It turns out that the basic property of straight G-spaces (for studying perpendiculars there) is the convexity property of spheres. Based on the notions of perpendiculars and bisectors, one can find in Chapter III of [548] an approach to Minkowski geometry which is more elegant than that in [544], and again interesting characterizations of (special) Minkowski spaces and the Euclidean subcase via purely geometric notions and problems are given. We also mention that in Chapter II Hilbert geometries in convex sets of affine spaces are constructed. There exist different approaches to Finsler geometry, with that of Busemann from [548] being the one close to convexity. We recommend to read the nice exposition [1944] of Papadopoulos.

Rund [2077] published the first well-arranged, comprehensive book on such approaches and the (in those days roughly 30 years old) field which should be called *geometry of Finsler spaces*. We cite [2077] here mainly since it contains a large section on the metric properties of the local situation, i.e., on the *geometry of tangent spaces which are Minkowskian*. The fundamental metric concepts for this book, such as length, angle, trigonometric ratios, and areas, are introduced (not in terms of the Euclidean background metric, but) in terms of local Minkowski geometry. This holds also for the theory of subspaces, and therefore also *two-dimensional Finsler spaces* are particularly studied, with analogous tools. The book shows an almost complete list of publications in the described field up to 1957, referring also to valuable survey articles.

We mention here also the booklet [2526] of Zaustinsky; the author wants to verify to what extent results of Busemann from [548] depend upon the symmetry axiom of the metric (see also the long paper [1101] of Guggenheimer). In a detailed way Zaustinsky proves that nearly every theorem of the symmetric theory which can be formulated without the symmetry assumption holds true also without this assumption. This line of Busemann's research created many further publications, and in [550] Busemann brings all these efforts up to the 1970 state. This book contains many further generalizations; e.g., the problem of *characterizing Finsler spaces among G-spaces* is solved elegantly. (G-spaces are finitely compact metric spaces satisfying certain axioms and having the convexity property.) In Section 8 one can find *characterizations of Hilbert and Minkowski geometries, even with not necessarily symmetric distance*. Calling this geometry (due to possible non-symmetry of the unit

ball) also Minkowskian is permissible. Namely, Minkowski's first intention (when he started to create the axiomatic foundations of Minkowski spaces) was to use convex bodies as unit balls which are not necessarily centered at the origin; see above and also the Mathematical Review MR0296877 of Zaustinsky. In [2526] he shows that "nearly every theorem (so far attempted) of the symmetric theory which can be formulated at all without the assumption of symmetry holds without this assumption". (A book in this direction is [631] by Cobzas, and a recent related paper is Jahn et al. [1294].)

Like late papers of Busemann, also the commentary [1947] by Papadopoulos and Troyanov on three early papers of Busemann shows that characterizations and properties of Minkowski spaces (in the symmetric setting) accompanied him all his scientific life. And this is perhaps a right place to mention the two recent collections [551] and [552] of his works; see our discussion on selected works below. Here we also cite the comprehensive book [1943] by Papadopoulos, containing first a survey on geometric concepts referring to (metric) spaces of non-positive curvature. Various properties of such spaces considered in [1943] are built on the notion of *convexity and that of geodesic space*. For example, the distance functions there are convex (finally leading to Busemann spaces), and further convexity-related notions occurring in [1943] are *Hausdorff metric on subspaces, convexity of functions and point sets, geodesic spaces and geodesic convexity, Menger convexity*, and *locally convex spaces*. Interesting historical notes (also on Hadamard's related contributions) and a detailed presentation of some basics from convexity theory are also given. The book contains various interesting problems and many examples.

Also the later book [549] of Busemann contains a lot of material closely related to convexity. It can be roughly dissected into two parts. Namely, in the first two chapters (called "Extrinsic Geometry" and "The Brunn-Minkowski Theory and its Applications") the author nicely complements the influential book [446] of Bonnesen and Fenchel, mainly discussing topics conjectured or suggested there. The last two chapters (titled "Intrinsic Geometry" and "Realization of Intrinsic Metrics") also contain material coming from other books, like mainly Alexandrov's book [40]. In the first part, basic properties of *convex hypersurfaces, results related to curvature, mixed volumes, the general Brunn–Minkowski theorem, the Alexandrov–Fenchel inequality*, and further related topics are presented. Chapter III deals with concepts and notions from the *intrinsic geometry of convex hypersurfaces*, such as metrics, differentiability of geodesics, angles, intrinsic curvature, and the Gauss–Bonnet theorem. From Section 13 on (which refers to angles), the author restricts to two-dimensional convex surfaces in three-dimensional space. The fourth chapter contains deep results, for example referring to the following themes: *rigidity of convex polyhedra and the realization of polyhedral metrics, Weyl's problem* on the realization of a regular metric of positive curvature given on a sphere in 3-space, local realizations with non-negative curvature, and existence of open surfaces and monotypy of closed surfaces (including a famous related result of Pogorelov). Busemann uses in [549] an impressively broad spectrum of tools and settings, such as classical differential geometry, functions of reals and set functions, elliptic partial differential equations, integral equations, and quadratic forms.

Finally we mention also the famous paper [553] of Busemann and Petty. It contains ten problems formulated in terms of classical convexity whose solutions would yield deep results in Minkowski geometry. Nine of them are still unsettled, and the only solved one is the famous *Busemann–Petty problem* (but see Alfonseca et al. [57]).

H. HADWIGER (1908–1981)

The very nice booklet [1133] of H. Hadwiger refers to developments of the theory of *volume* V, *surface area* F, and the *mean curvature integral* M of a convex body K in 3-space. In the first chapter the concepts of *Minkowski addition, inner and outer parallel bodies* are introduced, and by means of Blaschke's selection theorem the *local compactness of the space of convex bodies* is proved. Further notions and tools discussed by the author are *approximation of general convex bodies by polyhedra, Steiner symmetrization, properties of the functionals* V, F, *and* M, *linear combinations*

of them, and *Steiner's formula* (first verified for polyhedra, and then extended to all convex bodies). As application, *Cauchy's integral formula* is obtained. Inequalities for V, F, and M yield the *Brunn–Minkowski theorem, Minkowskian inequalities*, and the *isoperimetric inequality in terms of V and F*. Alternative proofs (by integral-geometric methods), equality discussions, the extension of the Brunn–Minkowski theorem to *concave families of convex bodies*, and the *Blaschke–Santaló diagram* (visualizing inequalities between three quantities depending on the shapes of convex sets; see the recent paper [486]) are presented. In the final chapter an *introduction to integral geometry* from a general point of view is given, where famous *formulas of Crofton, Cauchy, Blaschke, and Santaló* occur as specializations. This booklet is a compact and readable introduction into the field of convexity; many basic topics from the heart of the field are presented in a very stimulating way.

In the important monograph [1134], Hadwiger summarizes measure-theoretic concepts referring to the *lattice of convex sets (Hadwiger's Konvexring)* in n-dimensional space in a way that is as elementary as possible. Particularly essential is his *characterization theorem for the quermassintegrals* of convex bodies in Euclidean n-space: *All continuous rigid motion invariant valuations on convex bodies are a linear combination of the quermassintegrals*.

The first part of the book [1134] is dedicated to the *axiomatic foundation of the concepts of volume and area* for bodies which are point sets without special regularity conditions for their boundaries. The second part treats several *isoperimetric inequalities* and *general integral geometry*. For any bounded and closed sets, the isoperimetric inequalities (with equality discussions) are given, and the notions of volume, area, and k-dimensional cross-section are introduced and used without any integral calculus. The author starts with elementary geometry of *(special types of) polytopes*, including subjects like *Minkowski addition* and notions like *parallelotopes (affine cubes), cylinders*, and so on. From here on, the concept of *invariance of classes of sets (and of several related functionals) under different groups of motions and more general transformations* is a central topic of the book. Then the existence and uniqueness of a translation-invariant, additive, non-negative content on the class of polyhedra (with the value 1 for the unit cube) is presented, where notions like *scissors congruence (Hilbert's third problem), Minkowski decompositions, content* and *surface area* are discussed. For Hilbert's third problem we mention here also the nice monograph [423] of Boltyanski; see the description below under his name. The Jordan content for bounded sets is obtained from the content for polyhedra, and the *Lebesgue measure* from the Jordan content (differing slightly from the usual way). Further *measure-theoretic aspects* (like also the *Banach–Tarski paradox*) are discussed without proofs.

In the chapter on the geometry of sets and set operations, *Jung's and Gale's inequality, Minkowskian set operations, parallel sets, Brunn's theorem on the kth root of the content of sets from concave (or linear) set families, outer and inner cross-section measures*, and *different types of symmetrizations* are treated.

The author continues with a chapter named like the whole book, namely "Content, Surface Area, and Isoperimetry". Here the *Brunn–Minkowski inequality* and the *isoperimetric inequality for relative surface areas and measures* of two arbitrary compact subsets of n-space are studied, and the *isoperimetric property of the sphere* is established. The very long final chapter refers to general integral geometry; many results on *functionals of convex bodies* like *mean width* and the whole series of *(mixed) quermassintegrals* are derived, and *formulas of Cauchy-, Crofton-, Blaschke-, and Santaló-type* are given.

Here we have to emphasize once more Hadwiger's characterization of the series of quermassintegrals in Chapter 6 (§1) by simple and natural properties, thus showing their fundamental importance (see also Theorems 4.2.6 and 4.2.7 in Schneider's book [2167], where it is also said that Hadwiger's respective proof is the only existing one). This book is a valuable source for experts and becoming experts in the field; it is a pity that until now the text was not translated into English! A translation into Russian (from 1966) exists.

The booklet [1137] of Hadwiger and Debrunner is in this version a translation (by V. Klee) of the original work from 1960 and contains (besides many problems from very different mathematical

fields) also results and problems on the *combinatorial geometry of convex sets*, such as *Helly-type problems, partition problems*, and *covering problems*. And the nice survey [1136] of the same authors can be seen as a successful precursor of [1137]. Natural continuations of parts from [1137] related to convexity can also be found in the excellent surveys [708] of Danzer et al. and [790] of Eckhoff.

Since much of Hadwiger's work is related to Minkowski addition, we may cite here also the booklet [755] of Dinghas. He treats the notion of *superadditive set functional based on Minkowski sums of sets*. In terms of this notion, the *inequalities of Minkowski, Hölder, and Jensen* as well as the *theorem of Brunn–Minkowski* are proved, and using a known theorem of Lyusternik also a generalization of the Brunn–Minkowski theorem (replacing the vector sum there by a Riemann–Minkowski integral over a parametrized set family) is given. Many further related topics are studied. For example, discussions of the *isoperimetric problem in spherical and hyperbolic space* are given.

V. L. KLEE (1925–2007)

Klee was a very versatile mathematician having broad interests and covering a wide range of mathematical areas. He gave essential contributions to *functional analysis, convexity, optimization, combinatorics, geometry, graph theory*, and further fields. For example, he successfully applied his investigations on polytopes to linear programming and showed (constructing together with Minty the Klee–Minty cube) that the solution of certain problems with the simplex algorithm needs (instead of polynomial growth) exponential computational effort. The progress that came from this in linear programming showed clearly how geometric methods can help in fields which are originally non-geometric. In convexity, Klee left deep marks mainly with the following activities:

He edited the very important proceedings [1415], containing trend-setting and fundamental articles like [708] on Helly's theorem and its relatives, [1080] on measures of symmetry, and [1079] on Borsuk's partition problem. These proceedings created and stimulated various important directions of research in the field.

Being one of the authors of the famous survey [708] (together with Danzer and Grünbaum), and continuing somehow the exposition [1136], Klee translated the known booklet of Hadwiger and Debrunner (treating combinatorial geometry of convex sets and related topics in the plane) into English; see again [1137]. But not only that. He added a chapter and further material. Also regarding plane geometry, he published with Wagon the nice problem book [1417]. Its first chapter contains a lot of material related to convexity and discrete geometry.

When Grünbaum wrote his famous monograph [1085] on convex polytopes, Klee's comprehensive research on polytopes led him to collaborate substantially with Grünbaum by writing Chapters 16 and 17 of that book. Therefore he is also one of the editors of the second and expanded edition of that book.

Klee was also interested in algorithmical problems. On the one hand, he was (collaborating with Gritzmann) one of the creators and promotors of the new subfield *computational convexity* (see, e.g., the handbook articles [1025] and [1028]), and, on the other hand he strongly influenced with his *art gallery problems* and related questions also the area of *computational geometry*.

The important work of Klee in classical and computational convexity, on polytopes and their relatives, in combinatorics, graph theory, etc. was deservedly honored by the "Victor Klee Festschrift", i.e., by the comprehensive proceedings [1030]. And it is very nice that, completing all his deep research work on convexity topics, Klee wrote also nice expository and inspiring articles suitable for a broad audience and showing clearly the beauty of the field; a very nice example is [1419].

V. G. BOLTYANSKI (1925–2019)

Boltyanski's first contribution to convexity in book form is the elementary introduction [2487] to planar convexity, written together with I. M. Yaglom. Below we describe this classical book in more detail (together with its "follower" [1646] by Lyusternik).

The booklet [425] by Boltyanski and Gohberg, nicely written in the Russian tradition of popular small books, presents mainly two famous, still unsettled problems from the combinatorial geometry of (compact sets or) convex bodies: *Borsuk's partition problem* and *Hadwiger's covering problem* (asking for the minimum number of smaller homothetical copies needed to cover a given convex body). Also the famous *illumination problem*, equivalent to the latter, is presented in an elementary way. The discussions are mainly referring to the plane and to three-dimensional space, but outlooks to higher dimensions are given, too. The Borsuk problem is also extended to *normed planes*, and various open problems are posed. Similarly, parts of the later collection [433] treat also the Borsuk and the Hadwiger problem as well as generalizations of Helly's theorem for convex bodies.

Boltyanski's monograph [423] refers to *scissors congruence* and *Hilbert's third problem* (to find polyhedra with equal volumes which are not *equi-decomposable*). By finding necessary conditions for equi-decomposability not satisfied by certain polyhedra of equal volumes, Dehn solved this problem already in 1900. This led to deep investigations of the connection between equality of volume and equi-decomposability (with respect to some group of motions); in [423] this topic is intensively studied, and in Sah's monograph [2087] more recent developments and related problems not covered by [423] are presented. The author emphasizes the algebraic approach, and also the spherical and hyperbolic scissors congruences are discussed. In the spherical case the respective classes form a ring, and a certain quotient ring of this ring is provided with a structure of a Hopf algebra. Further related results are obtained, and also in the exposition [1356] Kellerhals discusses connections between scissors congruence and the volume measurement in spherical, Euclidean, and hyperbolic spaces. In the broad survey [2518] Zakharevich studies *classical scissors congruence, multidimensional scissors congruence along McMullen's polytope algebra* (see [1743]), and *scissors congruence of varieties*. Here we should also refer to the book [2019], and regarding historical aspects of Hilbert's third problem we refer also to the book [690] of Cromwell.

Like the booklet [425], also the book [434] of Boltyanski and P. Soltan refers to the *combinatorial geometry of convex bodies*; it is in several parts an important forerunner of the monograph [431], treating topics from there. The types of problems summarized in [434] belong to the families of *Helly-type problems, illumination problems*, or *Borsuk-like partition problems*. But the authors go much further than it is done in [425]. In this framework, they consider also *generalized convexity notions* (namely the so-called d-convexity, suitable also for metric problems in normed spaces, and the concept of H-convexity) to get also *generalized Helly-type results*. Since it is closely related, we mention here also the monograph [1160] of Harazishvili (more details are discussed below).

In the monograph [431] of Boltyanski et al., the combinatorial geometry of convex bodies is extended by the following topics: *Helly dimensions of classes of convex bodies, generalized convexity notions in normed spaces, fixing and hindering systems of convex bodies, inner illumination problems*, and the *combinatorial geometry of belt bodies* (this latter notion generalizes that of *zonoids*, and various results obtained for zonoids are extended to belt bodies). Here we mention also the book [424] by Boltyanski and Baladze referring to Helly dimensions. The last chapter of [431] contains an extensive list of (also today still unsettled) problems from the combinatorial geometry of convex bodies. Here is also the right place to mention the expositions [1716] of Martini and V. Soltan, [316] of K. Bezdek, and [1714] of Martini et al., which interestingly extend the usually known illumination concepts. One of the most recent surveys on the problem of covering by smaller homothetical copies is [321]; all significant papers on this unsettled problem (till 2018) can be obtained from there, or from papers cited there.

It should be mentioned that Boltyanski's main research field was *optimal control*, mainly studying applications of differential equations to optimal control. He was awarded the Lenin Prize as one of the

discoverers of the *maximum principle* in the research group around Pontryagin, and he wrote really many books. Among them one finds the geometric monographs and booklets mentioned above, but also (partially popular) mathematical books on combinatorial topology, topological Boolean algebras, envelopes, geometry of masses, and several other themes.

K. LEICHTWEISS (1927–2013)

In his monograph [1531], Leichtweiss studies mainly *metric properties of convex sets in Euclidean spaces*. Based on the Euclidean metric-induced topology, he first discusses fundamental notions (duality, separation, polyhedral sets, and so on) and basic theorems from convexity, like *theorems of Helly, Krein–Milman, and Stasziewicz*). Accompanied by analytical representations and functionals of convex sets, one finds *Blaschke's selection theorem, mixed volumes and quermassintegrals*, famous inequalities with the related *theorems of Brunn–Minkowski, Busemann, and A. D. Alexandrov*, and tools like the usual *symmetrizations of Steiner and Schwarz*, and also *central symmetrization*. Coming from the notion of *distance function*, the author also studies *Minkowski geometry* and derives in this more general setting the respective *extension of Jung's theorem* and the well-known *inequalities for self-circumferences of unit circles* of normed planes. Although most of the notions and results discussed in this book are geometric in nature, Leichtweiss mainly prefers an analytic and algebraic approach. The style of the book is clear, and the contents are well organized. Each of the 24 sections is finished by inspiring exercises.

In the preface of his second book [1537], Leichtweiss says that "the affine geometry of convex bodies is the theory of the invariant entities and equivariant associated bodies with respect to the group of the volume-preserving affine transformations of the n-dimensional affine space, i.e., the *equiaffine geometry of convex bodies*." In this setting he studies also *differential geometry of convex bodies and surfaces*, laying special emphasis on *affine surface area* (comparison of different definitions, investigations regarding their equivalence, etc.) and *affine inequalities* (referring to *difference bodies, centroid bodies, projection bodies, intersection bodies*, and further concepts). Regarding the latter, equality cases lead to special types of convex bodies, such as to *simplices and ellipsoids* (see also our subsubsections 2.9.1.1 and 2.9.1.6). In his studies of the equiaffine differential geometry of the boundaries of smooth convex bodies, he presents analogues to the Euclidean case. Even practical applications in areas like shipbuilding are described, and interesting problems and conjectures are posed.

R. SCHNEIDER

The first version (for hyperplanar sections of one convex body K and for dimensions $n \leq 3$) of the famous *Brunn–Minkowski theorem* was obtained by Brunn (1887). Various generalizations by several mathematicians followed (Minkowski being the first to be mentioned), and so the theorem was extended to vector sums or to the more general polynomial expansion for the volume of linear combinations of k convex bodies, yielding also the notion of mixed volumes and deep related inequalities. In 1934, these and further related developments were summarized in the classical and inspiring monograph [446] of Bonnesen and Fenchel. They succeeded in giving an excellent and almost complete overview of the grown field, with many motivating open problems. In the following decades, no book replaced [446] in an analogous thematic breadth, although subsequently many new results and notions came to be (the *general Alexandrov–Fenchel inequalities, the axiomatic characterization of quermassintegrals due to Hadwiger*, as well as *dual mixed volumes and intersection bodies* are good examples).

Only Schneider's monograph [2167] from 1993 was able to live up to these claims, thus filling a gap adequately and becoming the main reference for the theory of convex bodies. And even theoretically higher expectations would have been satisfied by its second expanded version from 2014 (containing approximately 250 additional pages; see below)! It shows analogously and more recently the growth of the whole field of convexity, how it became more widespread and permanently more interesting for other disciplines (like *functional analysis, differential geometry, convex analysis, optimization*,

and further ones). On the other hand, the unbroken topicality of Schneider's monograph shows also that the *Brunn–Minkowski theory* has stably maintained its central place in the whole field. Recent developments mainly refer to topics like *local formulae of integral geometry, valuations, new deep inequalities for various functionals, applications of zonotopes and zonoids*, and *stability results and equality cases regarding classical and affine inequalities* (most noteworthy being the *Alexandrov–Fenchel inequality*).

Our description of contents at first refers to the first edition, and below we add contents and important aspects of the second one. The book is divided into chapters and sections, where theorems are elegantly proved, sometimes even in different ways. At the end of each section one can find survey-like notes bringing the reader up to the state of the art; the history and developments of the respective subfields, tools, or important notions are presented, and interesting open problems are posed.

After fundamental properties of *convex functions and sets*, clearly presented in the first chapter (e.g., characterizations of convex functions, duality, support functions, and Hausdorff metric), the *boundary structure of convex bodies* is broadly discussed in Chapter 2. Besides the Ewald–Larman–Rogers theorem concerning the measure of the set of directions of segments in the boundary of a convex body, also polytopes (often used as a stepping stone to theorems on general convex bodies) are treated. The *simultaneous approximation of k convex bodies by simple, strongly isomorphic polytopes* is used for the later *proof of the Alexandrov–Fenchel inequality*, and also *Baire category results* are confirmed. The third chapter treats *Minkowski (or vector) addition*, i.e., notions like *indecomposability, additive mappings and valuations, zonoids, and generalizations thereof* (as examples for classes that are close under addition). The fourth chapter refers to *curvature measures, area measures*, and the global analogues of them, called *quermassintegrals*. *Integral-geometric formulae* are given in their local version, *for curvature resp. area measures*. In Chapter 5 *mixed volumes and mixed area measures* are studied, using approximation with strongly isomorphic polytopes and a formula expressing mixed volumes as alternating sums of ordinary volumes (of sum sets). In Chapter 6 the author proves some of the most important theorems from convex geometry: *the Brunn–Minkowski theorem, Minkowski's inequalities between mixed volumes, and Alexandrov–Fenchel inequalities*. Also a difficult discussion on the equality cases in the Alexandrov–Fenchel inequalities and *related stability theorems* are given. In the last chapter, one can find *applications of the Brunn–Minkowski theorem and of the inequalities for mixed volumes*, including *Minkowski's existence theorem* for convex bodies with given area function; and the author also presents *uniqueness theorems for bodies with preassigned area measures*. The bibliography includes about 1400 items, among them being the major monographs and surveys from the field.

We now come to the 2014 edition in which the author added three new chapters on *valuations on convex bodies, extensions and analogues of the Brunn–Minkowski theory*, and *affine constructions and inequalities*. Moreover, the material on applications from the first edition is now extended and to be found within two chapters. One can study the L_p *Brunn–Minkowski theory, the dual Brunn–Minkowski theory of star-shaped sets*, and the *theory of log-concave functions*. To give at least some more precise notions treated in the (also new) final part, we mention here *ellipsoids associated with a convex body (like the John ellipsoid), projection and intersection bodies, the difference body, the covariogram*, and *shadow systems*. Also the *Blaschke–Santaló inequality, Mahler's conjecture, the problems of Busemann–Petty and of Shephard* are studied. Additionally, one can see many supplements to further chapters from the first edition. The notes at the end of the sections are substantially expanded, and the new list of references exceeds the number of 2000! Also this new edition of Schneider's book meets all expectations: after more than 20 years, it is again a perfect and the most comprehensive reference for readers being interested in convex geometry, in the broadest and in the deepest sense.

Schneider is also very active in stochastic and integral geometry, with many problems and results close to convexity. Together with Weil he published in 1992 the book [2177] on *integral geometry*, and in 2000 the book [2178] on *stochastic geometry*. Later they extended the corresponding material to publish in 2008 the comprehensive, very interesting monograph [2179] on both fields together. This

is one of the most important references for both directions. In particular, the contents of [2179] are described in our subsection 12.4.5 here.

The recent book [2174] of Schneider is devoted to *geometric applications of convex cones*, where basic geometric functionals of convex cones are studied, e.g., *conic intrinsic volumes* and Grassmann angles. The relation between the geometry of subsets of the sphere and the geometry of cones is investigated, which finally turns out to be equivalent. Analogues of important formulas (like Steiner's formula or the kinematic formula) are developed, and arrangements of hyperplanar subspaces and the respective tessellations of the space into polyhedral convex cones are taken into consideration, also in view of conic intrinsic volumes. Random cones based on random hyperplane arrangements are topic of investigation, and finally *coconvex sets* are discussed. The author develops parts of a Brunn–Minkowski theory for them.

H. GROEMER

Also for the geometry of convex bodies, *spherical harmonics and Fourier series* have proven to be important tools. Groemer carefully provides particularly developed methods as well as an impressive, so far complete collection of representative results till then in his self-contained monograph [1042] (see also the earlier handbook article [1039], completing the whole picture). It is the first book combining these two tools with *convex geometry* (and also with the concept of *star-shaped sets*). Also the historical development of this subfield (going back to the beginning of the twentieth century, namely to results of Hurwitz and Minkowski related to isoperimetric inequalities and convex bodies) is nicely presented, discussing also very basic contributions (by Alexandrov, Berg, Schneider, etc.). The collection of references has its own value, since before the appearance of this book the corresponding literature was really scattered.

After clarifying the fundaments from analysis (like *orthogonal function systems, the Laplace operator on spheres, and spherical integration*) and convex geometry (*support functions, metrics for sets of convex bodies, mixed volumes, difference bodies, projection bodies*, etc.), the author excellently explains how *spherical harmonics* can be understood as a higher dimensional analogue of Fourier series. *Legendre polynomials* and *associated Legendre functions* are presented, and to prepare applications coming later, the *Funk–Hecke theorem* is proved. Then, in the main chapters of the book, the author lists a large variety of geometric applications. In this spirit, Chapter 4 collects planar applications, where Fourier series are sufficient. Presented topics refer to the *isoperimetric and related inequalities, Wirtinger's inequality, rotors, tangential polygons, mixed areas*, and *polygonal approximations*. The author underlines that one reason for the usefulness of Fourier series for convex geometry is that various typical concepts (like *area, perimeter, Steiner points, mixed area*,...) can be expressed in terms of *coefficients of Fourier series of support functions*. This is also essential in Chapter 5, in view of harmonic expansions of support functions. Here an impressive variety of *higher dimensional problems* is presented, for example with the following subjects: *inequalities for mean projections and mixed volumes, the isoperimetric and Wirtinger's inequality, projections and sections* (with hyperplanes or half-spaces), *Cauchy's surface area formula*, an *ellipsoid characterization* via tangential boxes, *Steiner points*, and *Schneider's famous theorem on rotors in polytopal sets* (which the authors characterizes as one of the most sophisticated applications of spherical harmonics in geometry).

The book gives an excellent impression of the *interplay between harmonic analysis and geometry*. Since it is self-contained and nicely written (e.g., with clear proofs), it can be successfully used as a textbook, and due to the complete representation of related literature and particularly developed methods, it is also very useful for specialists and research beginners in this field.

R. J. GARDNER

In his monograph [930], Gardner defines *geometric tomography* as the area of mathematics studying the retrieval of information about a geometric object from data about its sections, or projections, or both. Thus, as a geometric counterpart to computerized tomography (using finitely many X-rays of human patients), it has overlaps with *convexity, stereology*, and *geometric probing* (in robotics); in addition it has also aspects of *inverse problems*.

The studied objects in [930] range from convex bodies (mainly for projection functions) via star-shaped sets (mainly for section functions) to compact or measurable point sets in Euclidean n-space. For our purpose here, the large overlap with convexity is most interesting. Looking into the classical book [446] of Bonnesen and Fenchel, one finds important notions and tools from [930] already there (we mention also a few additional, closely related aspects): *outer and inner cross-section measures of convex bodies* which present measures of projections and sections, and the correspondingly *associated bodies* (like *projection bodies, intersection bodies, cross-section bodies, difference bodies, Blaschke bodies*, etc.); this topic is also broadly discussed in Martini's survey [1710]. Furthermore, *Steiner symmetrizations* and other types of symmetrization procedures play an essential role, like also the *shapes* of projections or sections. Basic tools from the *Brunn–Minkowski theory of convex bodies* and from the more recent *dual Brunn–Minkowski theory for star bodies* are used, various *integral transforms* are applied (and basically discussed in a corresponding appendix), and powerful *geometric inequalities* are combined and presented, appropriate to the aims of the book. Besides this, notions like *X-rays and chord functions* are developed, and corresponding problems are investigated. The book gives the first complete overview to the field as it is described above. It contains many fascinating research problems which were, with difficult methods, meanwhile solved (like the famous *Busemann–Petty problem* or the *characterization of bodies of constant brightness and constant width*), and still unsolved problems (like famous inequalities, e.g., also discussed in Lutwak's handbook survey [1633], in great parts still current). The recent state of subfields is described in notes after each chapter, and a brief biography of the main contributor to each topic or chapter can also be found there. Written extremely motivatingly and stimulatingly, as well as nicely illustrated, this deep monograph is, as we believe, one of the best and influential books in convex geometry of the last decades.

We present now some selected (types of) results from the chapters. A *parallel X-ray in direction u* describes the length of each chord of the investigated set S parallel to u (one can take a Steiner symmetral at a hyperplane orthogonal to u). What sets of directions suffice such that all X-rays in these directions uniquely determine S? For example, it is shown that there are three directions which uniquely distinguish a plane convex body S, and such questions are also extended to n-space. Another problem: What can be said about a convex body S in n-space from its k-dimensional orthogonal projections? For $k = 1$ this is related to *difference bodies* and *bodies of constant width*, for $k = n - 1$ to *projection bodies, zonoids*, and *bodies of constant brightness*. Replacing projections by sections, we get analogous questions for *intersection bodies* and *cross-section bodies*, and these questions can be combined with each other. Combined with section functions, the framework of the studied objects can be naturally extended to *star-shaped sets*. Also *point X-rays* are introduced, and the study of *chordal functions and equichordal problems* follows. Reconstructions regarding projection functions and section functions are also discussed, and the book closes with three important appendices (discussing *mixed and dual mixed volumes*, important *inequalities*, and, as already mentioned, certain *integral transforms*). As the author says, the second edition from 2006 "incorporates....about 300 additional references. This expansion indicates the amazingly rapid development of geometric tomography over the decade since the first edition appeared."

P. M. GRUBER (1941–2017)

A basic contribution to many fundamental developments in convexity is the monograph [1070] of Gruber. Treating many topics from the *geometry of convex sets and polytopes*, the author presents at the same time many deep applications in other subfields of (pure and applied) mathematics. In

particular, this refers to *measure theory, number theory*, the *geometry of numbers* as well as *lattice polytopes* and *lattice point problems, calculus of variations, complex function theory, potential theory, (algorithmical) optimization, theory of polynomials, aspects of approximation theory, mathematical physics (e.g., isoperimetry)*, and *discrete mathematics/discrete geometry* (also crystallography and tilings). In each section, the main results are presented with proofs (and perhaps together with deep problems), and then the author discusses variants and applications in other fields, giving also basic references related to further developments.

The book has four chapters: 1. *Convex functions*; 2. *Convex bodies*; 3. *Convex polytopes*; 4. *Geometry of numbers and aspects of discrete geometry*. In the first (and shortest) chapter one finds for example *Jensen's inequality and Alexandrov's theorem on second-order differentiability*, in the second (besides basic properties of convex sets) *mixed volumes and quermassintegrals, valuations, the Brunn–Minkowski theorem* (e.g., with applications in crystallography, also via Wulff's theorem), *symmetrization procedures* (also used for fixing famous inequalities), the *intrinsic metric and Weyl's problem* (rigidity of convex surfaces), *approximation aspects*, and results on the *space of convex bodies* (e.g., Baire category results). The third chapter is devoted to *combinatorial properties of polytopes, Hilbert's third problem, rigidity, uniqueness and existence theorems (of Alexandrov and Minkowski), lattice polytopes*, and a section on *linear programming*. The final chapter, more related to discrete geometry, deals for instance with *lattices, Minkowski's first fundamental theorem* (on lattice points necessarily contained in origin-symmetric convex bodies), *successive minima, the Minkowski–Hlawka theorem, Mahler's selection theorem, Mordell's inverse problem, (lattice) packings and coverings, tilings with convex polytopes (Dirichlet–Voronoi and Delone tilings, translative space-fillers), optimum quantization (and sums of moments)*, and *Koebe's representation theorem for planar graphs*. This chapter contains also interesting applications to *Diophantine approximation, polynomials, error correcting codes, numerical integration*, and *graphs*.

Written in a very readable style, this book can be recommended to anyone interested in *convexity and discrete geometry*. It is an excellent graduate book for teaching courses under the mentioned headlines, but it is also important for experts in the field or people working in the given applied directions; they will find many nice problems and useful literature. Finally we mention that, increasing nicely the readability of the book, the author gives amusing historical notes and various quotations of famous mathematicians.

Two further books with Gruber as one author should be additionally mentioned: the monograph [1075] of Gruber and Lekkerkerker on the *geometry of numbers*, and the book [814] treating lattice point problems. Both these books are cited in our subsection on discrete geometry.

A. BARVINOK

Here, like also in our section on convex polytopes, we cite Barvinok's book [242] reflecting many facets of convexity. It is characteristic for this book that the author has also applications in mind (see, for instance, his discussions on *linear programming* and on the *ellipsoid method*). Together with this optimization background, *polarity and duality* play an important role throughout the book. The author's approach is mostly dimension-free, and (besides the Euclidean setting) also based on the framework of topological vector spaces.

For general convex sets, the *theorems of Carathéodory, Radon, and Helly* are derived, and the geometric construction of the *Euler characteristic* is given. This yields *positive multivariate polynomials*, insights regarding *uniform (Chebyshev) approximation*, and certain *valuations on convex sets* (e.g., the intrinsic volumes). Then (among other results) the *Krein–Milman Theorem* for locally convex topological vector spaces is proved. Furthermore, the *extreme points* of some convex sets, considered as infinite-dimensional extensions of familiar Euclidean objects, are explored. The *duality concepts* for linear programming in topological vector spaces ordered by cones are developed. In addition, it is shown that polarity in Euclidean space can be extended to a *valuation on the algebra of closed convex*

sets. This completes the proof of the *Weyl–Minkowski Theorem* and proves an exact condition for a point to belong to the moment cone. Besides many nice contributions to the theory of (also) *lattice polytopes*, mentioned in our polytope section, Barvinok presents in his seventh chapter "Lattices and convex bodies" various *discrete aspects of convexity*, including the corresponding Minkowski theorem and also the Minkowski–Hlawka theorem (see also Chapter 24 of Gruber's monograph [1070]), with relations to dense sphere packings, flatness results, and the Lenstra–Lenstra–Lovász basis reduction algorithm. Also Pick's (generalized) formula and *Doignon's lattice version of Helly's theorem* can be found in this chapter.

And here we have to mention also Barvinok's monograph [243] and the proceedings [246] with him as one of the editors. These publications refer to integer points in polytopes (like also the book [263] of Beck and Robins).

A. KOLDOBSKY

The monograph [1444] of Koldobsky treats a new *Fourier analytic approach to problems from convex geometry*. The starting points have to do with spherical harmonics and their applications in convex geometry. Based on the volumes of projections and sections of convex bodies and star-shaped sets, which are expressed in terms of the Fourier transform of powers of their Minkowski functionals, it is also important for problems from geometric tomography. This setting generalizes the well-known classical formulas of Laplace and Pólya, and with respect to sections of l_p-balls also results of Hensley, Ball, Meyer, and Pajor as well as Oleszkiewicz and Pelczynski. In the late 90s, the author recognized that existing approaches to such problems are special cases of a more general Fourier transform formula, thus yielding a unified approach.

For example, based on these elegant methods in the book a *relatively simple proof of Minkowski's uniqueness theorem* is given: every star-shaped body centred at the origin is uniquely determined by the volumes of its central hyperplane sections. Further on, a Fourier analytic *characterization of intersection bodies* is given, and some examples of intersection and non-intersection bodies are presented. This leads also to the famous *Busemann–Petty problem* from 1956, posed in [553]. The author presents his unified solution of this problem, explaining also clearly why there is a change from dimension 4 to dimension 5. A connection between intersection bodies and the theory of L_p-spaces is discussed, and (as already mentioned) also extremal sections of l_q-balls are studied. Analogously, the *Fourier analytic approach to projections of convex bodies* is presented. It turns out that this approach is surprisingly similar to that for sections; this shows the deep and fascinating *duality between sections and projections in convex geometry*.

In every chapter interesting historical remarks, open problems, and further related results are given. For more results and background material, valuable hints to cited monographs and articles are given. The booklet [1452] of Koldobsky and Yaskin is somehow a continuation of [1444], e.g., containing results which came to be after the appearance of [1444] (such as results on the Busemann–Petty problem in non-Euclidean spaces, or on the non-equivalence of certain generalizations of intersection bodies). Additionally, interesting relations between intersection bodies and projection bodies yield more insights about the mysterious duality mentioned already above. Further on, the main ideas to the general approach from [1444] are given in [1452] in a shorter form; so interested researchers can faster learn the subject.

S. ALESKER

The booklet [49] of Alesker gives a brief introduction into recent developments of the field of *valuations*, which is a central part of *analytic convex geometry*. This subject had its first culmination in the work of Hadwiger (see, e.g., [1134]), including his famous *characterization of quermassintegrals*

or (after renormalization) intrinsic volumes, as continuous valuations invariant under rigid motions. The author is one of the leading experts in the modern theory of valuations.

For example, he first presents basic extension theorems and results like *McMullen's polynomiality and decomposition theorems, Hadwiger's characterization of n-homogeneous valuations*, and the *Klain–Schneider characterization of simple valuations*. After this, generalized functions on manifolds, the *Goodey–Weil embedding*, and the famous *irreducibility theorem of Alesker* are presented. Applying results of this type, the author proves *McMullen's conjecture on the approximation of translation invariant, continuous valuations by linear combinations of suitable mixed volumes*. A survey of results on valuations obtained after his proof of the irreducibility theorem is given. These results refer to *products on valuations, the convolution of valuations, the Fourier type transform on valuations, valuations invariant under a subgroup of the rotation group transitive on the sphere*, and *valuations related to Monge–Ampère type operators*.

Although being small, this booklet gives a very nice overview on the state of the art. It is based on lectures and describes successfully parts of the important research by the author. The main results are prepared in detail, and the author presents a comprehensive bibliography.

We should mention here also a second book-like publication of Alesker, namely [51] referring to Integral Geometry and valuations. And it should be noted that a very recent update of related investigations is given by Ludwig in the comprehensive survey [1617].

12.3.2 Further books on convexity after 1950

In the fifties and sixties, the enriched material about convex sets reached a level such that many universities installed courses on them, also due to the increasing importance for neighbouring disciplines (like several optimization fields, discrete and combinatorial geometry, functional analysis, etc.; thus, some need for textbooks reflecting also teaching aspects, fruitful tools, and single aspects in a stronger measure came to be. In the following, we list such books (or parts of books) which take, in different ways, this viewpoint into account.

The book [2487] of Yaglom and Boltyanski is an excellent introduction to the *geometry of convex figures in the plane*, where a large part of the material can be extended to higher dimensions. The authors nicely use 310 figures to make proofs more understandable. In the first part a long series of problems is given, and solutions are collected in the second part. The book addresses to any high-school pupil, student (at university and college), and even mathematician, since anyone can benefit from the elegant geometric style of the authors. After presenting general properties of convex figures, *Helly-type theorems* are studied, and problems which might be treated with continuity arguments. *Minkowski addition of convex figures, isoperimetric-like problems, sets of constant width*, and *rotors* are further topics discussed in this classical book. A welcome completion to [2487] is the nice and elementary monograph [1646] of Lyusternik; it is discussed below in more detail.

In [802] Eggleston had collected basic problems more or less closely related to convexity and wrote up in book form. Among them being the reader will find *Borsuk's partition problem, Dowker-type theorems, multiple intersection points of diameters of convex bodies*, and various problems and results on *bodies of constant width*. Practically in contrast to [802], Eggleston continued one year later with the little monograph [803], more devoted to the *convexity property in general*. In the introductory part he says that simplicity and generality are characteristic for convexity, and he succeeds to verify this (with both books). Under this roof, he nicely presents *Helly-type theorem* (including also *Horn's theorem*), *support and distance functions, metric and convergence regarding convex bodies* together with *Blaschke's selection theorem, continuous functionals* (particularly, the volume as Lebesgue measure), *mixed volumes*, and the *Brunn–Minkowski theorem*. Again also *bodies*

of constant width are discussed, and, although the studied topics are in great parts elementary, the book is obviously committed to a higher position.

This development one could particularly see also in Soviet mathematics. For example, after the beautiful book [2487] of Yaglom and Boltyanski (see above), also the nice and elementary monograph [1646] of Lyusternik helped to increase the popularity of convex sets. It contains *characterizations of constant width sets and ellipsoids*, theorems from *lattice geometry*, and various nice results on *convex polytopes*. For example, *Cauchy's theorem on the congruence or symmetry of two convex polyhedra* having correspondingly congruent or similarly situated faces is presented, and also the *Brunn–Minkowski inequality* is treated, together with various consequences of it. A. D. Alexandrov wrote one chapter of this book, with an elementary proof of the well-known theorem that a convex polyhedron is determined by the areas and normal directions of its faces. Summarizing, one can say that [1646], although in great parts written on an "elementary level", geometrically stimulates any interested reader, also young students.

We see this as the right place to mention also the book [696] of Cundy and Rollett. It gives detailed instructions for producing mathematical models, e.g., by using paper, plastics, cardboard, or sheet metal, and having concepts like nets of polyhedra, paper folding, or constructing curves graphically and mechanically in mind. It proves extremely stimulating for teachers, and we mention here only model types close to convexity (there is much more in this book). Presented polyhedral models include *Platonic and Archimedean solids, stellated analogues* (like the Kepler–Poinsot star polyhedra), *Catalan solids, polyhedral compounds*, and *deltahedra* (see also our Section 2.9.1 here). Further on, there are quadrics and plane models of *dissections, tessellations*, and *curves of constant width*. Not only students, but also teachers in advanced mathematics could benefit from a thorough study of [696].

And we cite here also the outstanding book [2448] of Wells. As a dictionary of geometric figures, theorems, and phenomena, it collects perfect drawings and short descriptions of more than 200 respective entries. As topics close to convex geometry, the following notions are shortly presented: *billiards, dissections and tessellations, regular and semiregular polyhedra, compound polyhedra, Zalgaller polyhedra, star polyhedra, monostatic polyhedra, iterations of (suitably nested or otherwise related) polygons, curves of constant width and rotors, (Penrose) tilings, translative space-fillers*, and further types of *zonohedra*.

Dinghas published the booklet [755] that we mentioned already above when discussing Hadwiger's work. We cite it here once more as a contribution to classical convexity, where topics like *Minkowski sums of sets, the Brunn–Minkowski theorem, the isoperimetric inequality*, and also *symmetrization procedures* are treated. The booklet contains also interesting historical notes.

Valentine's introduction [2391] into the theory of convex sets is based on the initial assumption to be, from the very beginning, within topological (Hausdorff) vector spaces. Also, some of the topics which are typical for convexity books are not present. Like in Benson's book [273], also the *Minkowski metric* is considered, and interesting characterizations of *convexity and local convexity* are given. Notions like *separation, duality, support function, maximal convex subsets* and *upper semi-continuous decompositions* are treated, *types of boundary points* as well as *Helly-type theorems* (including *Krasnosel'skii's theorem*) are studied, and also the final part with interesting exercises and problems of different levels of difficulty should be mentioned here.

In the preface of Benson's textbook [273] the author interestingly says the following: "This text is intended to be used for an upper-division course in geometry suitable for any mathematics major whether he intends to teach, to work in industry, or to do graduate work in mathematics. In order to be of use to all three of these groups such a course must help train high school teachers, prepare students for further study in mathematics, and develop the student's geometric intuition." Benson also says that particularly a geometric course around convexity topics is admirably suited to meet these objectives. Contents presented in the book are *elementary basics from measure theory, equidecomposability, Helly-type theorems, sets of constant width, transformations* (also *duality, Minkowski addition of sets*, and *Steiner symmetrization*), *extremum problems* (like the isoperimetric one), and *Blaschke's selection*

theorem. The final chapter gives a nicely written introduction to *Minkowski geometry* (i.e., the geometry of finite-dimensional real Banach spaces), in those days, besides Valentine [2391], a rarity.

Guggenheimer wrote his book [1102] under the personal headline "what every applied analyst should know of geometry". To prepare the *theory of convex sets*, he starts with analytic geometry followed by affine geometry. Then topics like *Minkowski addition, properties of the Minkowski sum, Hausdorff metric*, and *Blaschke's selection theorem* are discussed. *Support functions* are presented, including their Gâteaux derivative, and together with *distance functions* also *polarity* goes hand in hand with this. Functionals defined by the *Steiner formula* and a section on *integral geometry (Kubota's formula)* follow, and the part on *Helly-type theorems* includes here also *Klee's theorem*. Finally, *Steiner symmetrization* is applied to the *Brunn–Minkowski theorem*, and even a section on *discrete geometry* (packing and covering problems) is added.

The carefully written booklet [2377] of Toranzos and Nanclares is devoted to *foundations of convexity*, containing material about *topological properties, boundary structure, separation and support theorems, polarity, combinatorial properties (Kirchberger's and Krasnosels'skii's theorem)*, and *applications in (linear) programming*, yielding also properties of *convex functions* (e.g., *Fenchel's theorem*). Selected further topics refer to *Blaschke's selection theorem, bodies of constant width*, and *Borsuk's partition problem*; in addition, various topics related to *star-shaped sets* are taken care of.

In the monograph [1160] of Harazishvili, several problems from the *combinatorial geometry of convex sets* (e.g., illumination problems) are presented. The book also refers to *affine diameters of convex bodies* and the study of properties of convex sets obtained from information about their intersections with related sets (such as the *Choquet simplex property* and how it characterizes usual n-dimensional simplices). Also *Hilbert's third problem* is discussed. In the second and related book [1161] of Harazishvili, again combinatorial properties of set families and the classical *Brunn–Minkowski inequality* are presented, with applications. The book contains also several results on *convex polytopes*. The author lays emphasis on showing relations between the considered problems and related topics in other mathematical disciplines (such as *combinatorial topology, analysis*, and *classical number theory*).

The textbook [1359] of Kelly and Weiss, which is mainly aimed at late undergraduates, gives a fundamental treatment of convex geometry, and especially *geometric methods* are nicely used. The notions, theorems, and proofs are clearly presented and completed by inspiring examples, the motivations carry themselves, and no special prior knowledge is necessary. Important contents presented there are *topological fundaments, basic structure and geometry of convex bodies and surfaces, types of hulls, linear combinations of sets*, and *properties of the Hausdorff metric*; various classical theorems (like *Helly-type theorems*) are added.

Also Lay's book [1521] is on the undergraduate level; only some basic knowledge on linear algebra and elementary topology is assumed. After introducing parts on *basic properties of convex sets* as well as *separation and support*, also *Kirchberger-type theorems* are presented, and studies of *sets of constant width, universal covers*, and the *isoperimetric problem* follow. As important further subject *Blaschke's selection theorem* is derived and used, and also three concepts of *duality* are presented, namely *polarity, duality of cones*, and a third type needed in linear programming. Further topics are *convex sets in finite matrix games*, and some introduction to *convex functions*. The author adds carefully selected exercises of three categories: easy, difficult, and open-ended (i.e., at least in those days unsolved).

The monograph [831] of Faro Rivas gives an interesting survey on the problem of *approximating n-dimensional centrally symmetric convex bodies by so-called polynomial bodies*, which can be presented via homogeneous polynomials of even degree. After introducing necessary notions (*support functions, Blaschke's selection theorem, approximation by convex polytopes*, etc.), basic properties of polynomial bodies and symmetric star-shaped bodies as well as different kinds of convergence for them are given. A special metric yields then the theorem that *any symmetric convex body can be approximated by polynomial bodies*. Different criteria for approximation from inside and outside and some existence

and uniqueness results for optimal approximations follow. Examples are given (such as *approximation by polytopes, ellipsoids, algebraic bodies*, etc.), and a collection of references on the approximation of convex bodies is added.

Although its title "Geometric Inequalities" describes a very broad field, only conditionally suitable for a course in convexity, we mention here the classical book [531] of Burago and Zalgaller, since it makes, on the other hand, extraordinary reference to convexity. Different types of *isoperimetric inequalities* (also obtained via *symmetrization procedures*), the *Brunn–Minkowski theorem*, and *mixed volumes* as well as their specializations to *quermassintegrals* are studied there; also inequalities involving *measures of sections, projections, and polars* of given convex compacta and *vector analogues of mixed volumes* (that is, Schneider's mixed target vectors of convex bodies) are taken into consideration. On the other hand, certain related topics are excluded; e.g., there are no inequalities related to packings and coverings, Minkowski's geometry of numbers, measures of symmetry, and the combinatorial geometry of convex bodies. Also isoperimetric inequalities important in mathematical physics are not discussed.

Another related publication on geometric inequalities interesting here (since the greatest part of it is related to convexity) is the nicely written booklet [1354] of Kazarinoff. It contains approaches to isoperimetric theorems, Dido's problem, a comprehensive discussion of *Steiner symmetrization*, and a section on "the reflection principle", yielding elegant methods which can also be used in convexity. The final chapter collects solutions or hints for each discussed problem of the book.

In the introduction of his book [2399], van de Vel says: "I decided to write a book that could serve as a general reference. This has resulted in the present volume, in which graphs appear side by side with Banach spaces, classical geometry with matroids, and ordered sets with metric spaces." Collecting notions and results from various mathematical fields and putting all of them somehow together, in a common perspective, van de Vel nicely demonstrates the *broad applicability of the convexity notion*. At the beginning he gives an overview to *convexity aspects of median algebras, convexity spaces, ordered sets, normed vector spaces, superextensions, semilattices, matroids, metric spaces, trees*, and more. The part "Abstract Convex Structures" is dedicated to notions and concepts like the *hull operator, separation, interval spaces, modular spaces*, and *Bryant–Webster spaces*. In the second chapter, the *classical theorems of Carathéodory, Helly, and Radon* follow. Also *Tverberg's generalizations* and respective types of independence are considered. The author continues with the study of *convex structures in topological spaces*, referring for instance to the question of continuity of the hull operator. In the final chapter things like the *affine representation of convexity spaces, dimension theory in combination with invariants*, and *fixed point theorems* are treated. As already said, the book is very broad. Among the themes not handled in it are geodesically convex sets and d-convex sets (see, e.g., Chapter II in [431]), both concepts going back to Menger and intensively studied by many authors in the sixties and seventies.

Webster's book [2437] can be taken as a well-written basis of undergraduate or graduate courses. After some fundamental notions and theorems from *classical convexity* (among them being the *facial structure of convex bodies* and *Blaschke's selection theorem*), polytopes are studied, e.g., the following subjects: *cyclic polytopes*, the *Dehn–Sommerville equations*, and also *Gale transforms*, leading also to the application field of *linear programming*. The author continues with *convex functions, analytic inequalities, support and gauge functions, convex programming* as application field, and *mixed volumes*, the latter being introduced first for polytopes and then extended to general convex bodies (using suitable approximation arguments). The *Brunn–Minkowski theorem* and the *isoperimetric* as well as *isodiametric inequalities* are derived, and the *symmetrization procedures of Schwarz and Steiner* are presented. At the end, a selection of further related topics is nicely treated, such as *Helly-type theorems, axiomatic convexity, convexity-preserving mappings*, and *sets of constant width*.

Hörmander's book [1209] presents a deep treatment of analysis aspects of convexity (we cite it suitably again). It covers also very different aspects of convexity. Examples are *projective convexity, C-convexity* (the natural complex analogue to usual convexity), further *generalized convexity notions, isoperimetric inequalities*, and *symplectic geometry*.

The field of C-convexity is also treated in the important monograph [108] of Andersson et al., where it is shown to what extent the analogy between usual convexity and complex convexity holds. This monograph is the first systematic and unified treatment referring to C-convexity. It contains (in Chapter 1) a detailed justification of why the projective space is the proper setting to understand and build the theory coming from that convexity notion.

We now discuss some textbooks which are close to *axiomatic convexity* and to *generalized convexity notions*. Prenowitz and Jantosciak [1993] develop an *axiomatic theory of convex sets* and linear spaces based on the notion of "join". This notion refers to pairs of points in an underlying set J yielding for each pair a subset of J. The join of two sets is then the join of two points from them. A set closed under "join" is called convex, and the family of the convex subsets of J endowed with the binary operation "join" forms a join-semilattice. Closedness under extension is called "linear", and based on this the authors present *representation theorems for convex and linear hulls of sets*. They study then notions like *half-spaces, cones, polytopes*, and the *facial structure of convex sets*. It turns out that further properties and proofs given in this framework are very algebraic in nature, and the axioms of this "join-geometry" do not deliver a dimension theory for linear spaces and a *combinatorial theory of convex sets*; the authors finally add the axioms necessary for this. The book is on the undergraduate level, and many examples like also detailed proofs help the reader to understand the independence of axioms and the abstract reasons for differences to usual geometric principles.

V. Soltan's booklet [2270] is addressed to researchers, to postgraduates, and to students interested in *convexity, convex analysis, and the geometry of normed spaces*. Various aspects of convex analysis that are related to an *axiomatic approach to convexity* are taken into consideration. Basic notions (like *convex spaces, convexity bases, separability*, etc.) from axiomatic convexity are introduced, and *convex and quasi-convex functions in convex spaces* are investigated. This yields generalizations of the *theorems of Kreĭn–Milman, Kuhn–Tucker, and Fenchel–Moreau*, and also *Helly-type theorems* are discussed. In a particularly detailed way, the generalized convexity notion of *d-convexity* (based on Menger's d-segments and suitable for solving metrical problems in normed spaces) is presented; see also Chapter II of the monograph [431].

Coppel's monograph [654] presents an *axiomatic approach to convex geometry* which is interesting for itself and well suited for graduate students of mathematics. The very basic notions of "point" and "line segment" are considered within a "convex geometry" only by two axioms. Then more (combinatorial) axioms define a "linear geometry" in which the *theorems of Helly, Radon, and Carathéodory* hold. Still far from Euclidean geometry (a line may contain only two points), *polytopes* are introduced, and on this step already many known results appear. Then further axioms require segments to have interior points and to be subsets of longer segments, yielding a geometry which in dimensions > 2 has to be *Desarguesian*, and isomorphic to a *subset of a projective space over an ordered division ring*. Then a completeness axiom (as an analogue of the Dedekind property for the real line) enforces that this ring embodies the real numbers, finally yielding *spaces of convex sets in a real vector space*. The book contains a lot of historical material. Perhaps it is interesting to mention here once more the booklet [2270].

The book [685] of Cristescu and Lupsa treats *generalized convexity notions* and their applications in a wide sense. The authors propose new convexity settings by neglecting connectedness and present also two classifications within a variety of about 100 such notions. For example, in the first classification *non-connected convexities* of such type are discussed in view of axiomatic convexity, convexity in nonlinear structures, via intersectional and separational approaches, and with respect to sets or given functions. The authors perfectly show that many branches of pure and applied sciences are close to this extended amount of convexity notions. In the second part they discuss *applications in pattern recognition, integer convex sets, generalizations of convex functions, optimization*, and *pharmaco-economics*. A large list of references is given, and also a brief historical part on convexity is contained.

In the later book [877] of Fink and Wood, two *generalized convexity notions* are comprehensively studied. One is defined by the connectedness of intersections of the respective point sets with lines

(not necessarily in all directions), and the other by getting the sets as intersections of suitably reduced families of half-spaces. Unfortunately, related concepts in the literature (like H-convexity; see Chapter III in [431]) are ignored. Nevertheless, the book [877] is nicely written, and it closes with interesting conjectures, possibly stimulating further investigations. Further important references with respect to generalized convexity notions are the famous survey [708] of Danzer, Grünbaum and Klee (see Section 9 there) and the Handbook of Generalized Convexity and Generalized Monotonicity [1131]. For generalized convexity notions in normed spaces, see also Section 3 of the survey [1720] of Martini and Swanepoel.

Moszyńska's textbook [1857] is very suitable for courses or seminars for students interested in fields of mathematics which are close to *convex geometry*. After introducing fundamental properties of *metric spaces, convex sets, convex polytopes*, and *transformations in the space of convex bodies*, the author takes care of *curvature notions, surface area measures, extensions of the class of convex bodies, selectors for convex bodies*, and *polarity*. In the third part, *star bodies* (also in view of geometric tomography) are comprehensively discussed, and related notions like *radial and star metrics, dual intrinsic volumes, star duality, the Shephard problem, the Busemann–Petty problem*, and *selectors of star bodies* are examined. The book is illustrated by many figures and completed also by carefully selected exercises.

The book [1518] of Lauritzen can be excellently used for reading an *undergraduate course on convexity* for students of mathematics, economics, and computer science, especially emphasizing aspects close to *linear programming*. Milestones presented nicely are the *Fourier–Motzkin elimination* for solving systems of linear inequalities, a deeper theory of polyhedra with the *Lemma of Farkas and Minkowski's theorem*, structural results as *Carathéodory's theorem, separation theorems* and the *Minkowski–Krein–Milman theorem*. The second large part of the book refers to *convex functions*, containing also discussions of real quadratic forms, the Hessian, and the cone of positive semi-definite matrices. Coming then to *convex optimization*, the author treats also the *Karush–Kuhn–Tucker conditions* and the *interior point methods*. The book contains also a lot of historical material.

With the textbook [2381] for graduates, Toth concentrates on two important subjects from convexity. First, he studies *measures of symmetry* (see also Grünbaum's well-known survey [1080]), which are suitably defined continuous real functions on the space of convex bodies, measuring the deviation from the centrally symmetric shape which is reached at one extremum. And second, *stability results of geometric inequalities for convex bodies* are studied (cf. also the survey [1040]); they give an estimate how far away a convex body is from an extremal body when the inequality is satisfied with approximate equality. After a nice introduction into typical tools and results from convexity (e.g., *Hausdorff distance, Blaschke's selection theorem, the Minkowski–Krein–Milman theorem*), a brief introduction to *mixed volumes*, and *Helly-type theorems*), first results on measures of symmetry are presented: *Minkowski's measure of symmetry* and its *critical set, Klee's estimate on the dimension of the critical set*, and investigations of *bodies of constant width*. The author continues with a deeper study of further measures of symmetry: *the centroid Minkowski measure, Winternitz measures*, and the famous *Rogers–Shephard volume ratio*. In connection with stability results, for example, *Groemer's stability of the Brunn–Minkowski inequality* and *Schneider's stability estimate for the Minkowski–Radon inequality* are discussed. Finally, stability results (due to Schneider) referring to the *inequalities of Jung and Steinhagen* and their *extensions to normed spaces* (due to Bohnenblust and Leichtweiss) are given. And in the appendix certain symmetry measures are applied to convex bodies arising as moduli spaces of certain maps from homogeneous Riemannian spaces to spheres.

Also in view of *needs in convex analysis*, V. Soltan [2277] presents a comprehensive, gap-filling book on *basic algebraic and topological properties of convex sets*. Very detailedly he provides notions and topics like *convex hulls, recession cones, normal and barrier cones, separation by hyperplanes and slabs, extreme and exposed faces*, as well as *polyhedral sets and polytopes*. One characteristic side of the book is that it bundles (with elegant, but sometimes also difficult and longer proofs) conveniently many useful theorems and lemmas of the "folklore type". This is meant as follows: if one needs them

as lemmas, the original references are hard to find or unknown, and hence they have to be proved again. The book contains also interesting notes after each chapter, and there is a valuable collection of exercises.

Suitable for an upper undergraduate course on convex sets is the nicely written book [1545] of Leonard and Lewis. It treats *classical convex geometry* and demands prior knowledge only in linear algebra and elementary geometry. It contains an interesting collection of inspiring exercises, and the following topics are (among others) explicitly considered: *basic properties of convex sets and hulls* (with Birkhoff's theorem on doubly stochastic matrices as interesting application), *Helly-type theorems, the art-gallery theorem*, and *sets of constant width*. A large variety of carefully selected examples and counterexamples illustrate the needed mathematical notions and concepts, and also useful problems are stated. This monograph [1545] got a very nice completion by the booklet [1546] of the same authors. This booklet contains extensive solutions of all odd-numbered problems posed in [1545]. The selection (of this numbering) is carefully done to find also a good approach to the problems indexed by even numbers. Thus, the works [1545] and [1546] together offer students an easier approach to the whole field of convexity, even via self-education.

A very popular class of convex bodies, having especially many nice properties and also applications, is that of *bodies of constant width*. This topic is mentioned or studied in many books; see, e.g., [2487]. The comprehensive survey [596] was presented by Chakerian and Groemer, and later Montejano wrote a booklet on this topic; see [1832]. Also the update of [596] given in the survey [1177] and the survey in [1720] on constant width sets in normed spaces showed finally that the large amount of existing results on sets of constant width and related topics is still not satisfactorily reproduced. Inspired by this, Martini, Montejano, and Oliveros decided to collect all the (old and new) material and to write up the book [1713]. Surprisingly, it came up with 1225 references, most of them directly related to sets of constant width. Thus, the notion of constant width amazingly shows a really big contrast between the relative narrowness of the definition and the surprisingly large variety and number of results. After a relatively long introduction, also to *convex geometry in general*, the authors of [1713] discuss a broad spectrum of very different aspects of constant width sets: *basic and also characteristic properties* of them; the two-dimensional case; *systems of lines* (e.g., envelopes); *spindle convexity and ball polytopes*; *completeness and reducedness* (as strongly related notions); *constructions*; *sections of bodies of constant width*; *constant width sets in normed and other non-Euclidean spaces*; bodies of constant width in *differential geometry, analysis, discrete geometry, and topology*; inequalities; related concepts (*rotors, billiards*); and applications (in art, designs, and engineering).

The most recent, graduate textbook on convexity is the excellently written monograph [1263] by Hug and Weil. After introductory parts on *convex sets and convex functions*, the *Brunn–Minkowski theory* (including extra sections on *mixed volumes, the Alexandrov–Fenchel inequality*, and *Steiner symmetrization*) and the *theory of area measures* (including a deep discussion of *projection bodies and zonoids*, then going further to a section on *valuations*) are developed. A chapter on *integral-geometric formulas* follows, divided into sections on *invariant measures, projection and section formulas, and kinematic formulas*. At the end of each section, useful and motivating exercises and supplements are given. The final chapter of the book presents then solutions to selected exercises. One can emphasis some very nice points in this book: e.g., the new *verification of the symmetry of the mixed volumes*, the *introduction of mixed surface area measures*, and the *proof of the Alexandrov–Fenchel inequality* are excellently done, and not in usual ways. Also the representations of the *local surface area measures* and the section on *valuations* benefit from interesting arguments.

12.3.3 Problem books and problem-oriented surveys

Unsolved mathematical questions put together in problem collections certainly occur in all parts of mathematics. But often the needed machinery of technical tools complicates already the understanding of the posed problems and makes them even inaccessible, e.g., for beginning graduate students. However, one of the oldest branches of mathematics, namely geometry, knows many stated and easily understood problems of great appeal which, however, are hard to solve. This is especially typical for *convex geometry*, and it holds together a large community of mathematicians which differ strongly by their used tools (e.g., methods from functional analysis, combinatorics, topology, etc.), but are interested in the same subjects: *convex bodies or polytopes*. Many problems there can be easily explained and create immediate interest, but are hard to solve. Examples are the Busemann–Petty problem (meanwhile solved, but in deep and different steps), Borsuk's partition problem (still unsettled, but showing very interesting partial results), Hadwiger's covering problem (still open, with even less partial results), or the question of whether bodies of constant width and constant brightness are balls (affirmatively solved). Good examples to show this principle are the proceedings of the Hungarian conference series "Intuitive Geometry" (for precise citations, see our part on discrete geometry), where the designation "Intuitive Geometry" is precisely meaning collections of problems of such a type.

Historically we have to start here with the famous monograph [446] of Bonnesen and Fenchel, in which the field of classical convexity was for the first time clearly described and, more or less, defined. Besides this important role that [446] played, it was also *the first problem book in the field of convexity*: many inspiring questions posed in it defined directions of research and were unsettled for a long time!

We also mention Eggleston's problem book [802] (we referred to it already above) which contains basic and stimulating problems from, or close to, convexity, e.g., related to *Borsuk's partition problem, intersection points of (affine) diameters of convex bodies*, and problems on *bodies of constant width*.

Also the book [2487] of Yaglom and Boltyanski should once more be mentioned here, giving many problem-oriented impulses to study convexity in the plane, e.g., regarding *rotors and sets of constant width*. Regarding the latter, it should also be mentioned that various exercises from [1713] can be taken as research problems.

Niven [1902] aims to collect problems on maxima and minima which are solvable in an elementary manner, without usual calculus and optimization methods. It turns out that his book [1902] contains a lot of geometric methods and problems, many of them referring also to convexity. For example, *Jensen's inequality* is derived and applied to *extremal problems for polygons* inscribed in or circumscribed about circles and ellipses. Further on, *isoperimetric problems* are comprehensively studied and, for instance, solved via *inner parallel figures*. Also, the *isoperimetric problem for tetrahedra* is nicely treated. It is proved that no convex m-gon with $m > 6$ can yield a *tiling* the plane, and Fagnano's problem on minimizing the perimeter of a triangle inscribed in a given triangle is proved. The book is addressed to college undergraduates, but also to a much wider audience; also mathematicians desiring geometric entertainment will find inspiring problems. The author offers almost 200 interesting and educational problems for the reader (with answers and, in most cases, complete solutions). One can find also much historical background and good suggestions and references for further reading.

In addition, we mention the problem-oriented book [2187] of Schoenberg discussing also interesting topics from convexity and discrete geometry; it is mentioned more explicitly in our section on discrete geometry.

The problem book [288] of Berger, Pansu, Berry, and Saint-Raymond completes Berger's two volumes [282] and [283]. We cite it here since it contains also problems from convexity or close to it, e.g., referring to polytopes, compact convex sets, conics and quadrics, and the sphere for its own sake.

The monograph [687] of Croft, Falconer, and Guy is a very typical source book for "intuitive problems" (which are "easy to state", but "not easy to solve") from convex, discrete, and combinatorial geometry. Topics discussed refer to the following fields (in each case, only one example is given): *classical convexity (isoperimetric problems), polytopes and polyhedral sets (shadows of polyhedra), tilings*

and dissections (equi-decomposability), packing and covering (packing balls in space), combinatorial geometry (sets that can be moved to cover sets of lattice points), finite point sets (minimum number of distinct distances), and *general geometric problems (Euclidean Ramsey problems)*. Impressively, there are 148 sections. Any reader should express great thanks to the authors for this service to the mathematical community.

Similarly, in Chapter 1 of the book [1417] Klee and Wagon collect problems in the spirit of convexity and close to those posed in [687]. For example, there one can find *illumination problems, questions on equichordal points, universal covers*, and also problems on *forming convex polygons*.

The problem book [492] of Brass, Moser, and Pach collects problems in discrete geometry, showing the explosive development of this field in the computer age. It is partially also interesting from the viewpoint of convexity. Namely, three chapters treat *packings, coverings, and tilings* (referring also to general convex bodies), problems on *convex polygons*, applications of *geometric graphs* and *geometric inequalities* in view of *isoperimetric problems, circum- and inscribed convex sets, universal covers* and, last but not the least, *approximation problems*.

Here we should also mention once more the problem booklet [1546] as addendum of the convexity book [1545] of Leonard and Lewis (see above fore more details).

There are also monographs which contain full chapters presenting problems. For example, the whole Chapter VIII of Boltyanski et al. [431] is filled with research problems, most of them referring to convex bodies. So the authors pose problems on *generalized convexity notions* (like d-convexity, based on Menger's metric segments) and different *generalizations of Helly-type results, Borsuk's problem, (primitive) fixing systems, inner and outer illumination, shadow boundaries*, and a variety of *antipodality types*.

There are also problem collections in the form of articles which were published "on their own" (i.e., not as final sections or problem corners within books with the same authors) and containing many questions which are still unsettled. An early example is [1414] by Klee, and one of the most interesting such collections is [1078] by Gruber and Schneider. There are several further ones; e.g., there are in fact "ten Busemann–Petty problems" (see [553]), and as it is well-known, only one of them is completely solved until now. Another problem collection, treating polytopes, consists of the two articles [2231] and [2232], and here we mention also Melzak's collection (see [1765] and [1766]) of altogether 60 interesting convexity problems (the latter articles are completed and perhaps also inspired by his monograph [1767]). Not all problems from these collections are solved; thus, to look at them might be still interesting.

We emphasize that, besides these collections, *proceedings of convexity meetings* very often contain *problem corners*; see the corresponding discussion below.

12.3.4 Related proceedings

As is common also in other fields of mathematics, in classical convexity proceedings of important conferences are used for publishing research as well as survey-like articles, which can show the state of the art regarding important subfields of the field and famous problems posed there. The survey-like and also a problem-oriented style of respective articles there strongly help to cultivate and structure the whole field. By citing such proceedings we cannot be complete, but will mention at least very essential examples. We start with examples from the sixties, when convexity started to structure itself to become a strong, lifely, and self-contained branch of mathematics. Some of these proceedings became really milestones in this development.

The important proceedings [1415], edited by V. Klee, brought together many famous mathematicians and contain also very basic articles like that on *Helly's theorem and its relatives* [708], on *measures of*

symmetry [1080], and on *Borsuk's partition problem* [1079], to mention only three examples (see also Klee's contributions above). Also a section with unsettled problems is there (see above).

The next essential step in this setting was given by the Copenhagen proceedings [515], containing again contributions of very famous mathematicians and an interesting problem section. Here fields like *convex analysis, functional analysis, and discrete and convex geometry* come together, and notions like *Blaschke sums* and *projection bodies* announce important lines of future research.

In the early seventies, the proceedings [528] (edited by Burago) appeared, containing articles on *convex surfaces in hyperbolic space, centrally symmetric convex bodies, deformation of closed convex surfaces depending on changes of their curvatures, convex polyhedra whose faces are equiangular polygons or suitable combinations of them, and geometric means of convex bodies*. In particular, the article [529] of Burago and Zalgaller should be emphasized, due to its broad and introductory character and discussing also extensions to Riemann spaces and functional analysis; see also [530].

Further proceedings in this sequence are [1358], containing various contributions on *Helly-type and Tverberg-type theorems, polygons* (in a deeper sense), *star-shaped sets*, and again a collection of research problems.

Toelke and Wills edited [2373], where 11 of the 25 articles refer to convex geometry. Among them are important and deep surveys on the *boundary structure and curvature of convex bodies, kinematic integral formulae for convex bodies, combinatorial aspects, polytope graphs, (Gale) diagrams, representations of polytopes, polyhedral sets*, and, in particular, *zonotopes* (already with links to *oriented matroids*), *geometry of numbers, continuous/discrete functionals of convex bodies*, as well as a selection of 105 deep problems, edited by Gruber and Schneider (cf. [1078]) and mentioned already above.

Next, we mention [1353] (also based on [1352]) edited by Kay and Breen. Regarding participating specialists and contributions, the fields *polytope theory, abstract convexity, tilings and packings, mixed volumes*, and some further concepts are presented.

Two convexity conferences (in Vienna 1981, and in Siegen 1982) motivated Gruber and Wills to edit the book [1076], in great parts specifying research directions due to the important articles in it. As examples we mention keywords from articles close to the heart of the field: *bodies of constant width, approximations of convex bodies*, convex geometry and *differential geometry, valuations, ellipsoids, zonotopes/zonoids* and related topics, *stereology* (also related to the forthcoming subfield of *geometric tomography*; see [930]), and *polyhedral manifolds with high symmetries*. Articles from neighbouring fields refer to *Banach space theory* (e.g., themes like closest points), *discrete geometry* (packing and covering), and *optimization*. The interesting contribution of Fenchel nicely illuminates parts of the history of convexity.

The article [1046] of Groemer on the history of Fourier series and spherical harmonics in convexity can be found in the proceedings [488] on Fourier analysis and convexity, where also Fourier analytic methods in view of *lattice point problems* and *projections/sections of convex bodies* are investigated.

We come now to proceedings whose contents mainly treat *overlaps between discrete geometry and convexity*. In the eighties, Hungarian geometers and specialists from combinatorics started a series of proceedings and respective conferences under the name "Intuitive Geometry". By this notion, which is due to L. Fejes Tóth, collections/solutions of usually hard problems are meant which, in Hilbert's words, on the other hand can be explained to the "man on the street", even to a non-mathematician. These books contain many papers treating discrete geometry and polytope theory, but many of these articles are also dedicated to (or have overlaps with) convexity. One can roughly say that "discrete geometry is somehow sharing this series with convexity". In the following short description of contents, we restrict to these two aspects, and also we announce that also here we cannot be complete. The first book in this series is [386], and it contains articles on *Brunn–Minkowski and Bonnesen-style inequalities, zonotopes and zonoids, packings and coverings* (referring to convex bodies), *universal covers, tilings, Hausdorff approximation, and convex polytopes* (e.g., equi-decomposable, noninscribable, non-Euclidean, and metrically extremal ones), *isoperimetric problems*, and *Minkowski geometry*.

The next book in this series is [387], edited by K. Böröczky and G. Fejes Tóth, and containing important contributions to the *Alexandrov–Fenchel inequality, geometric tomography, cross-section measures, illumination of convex bodies, shadow boundaries, affine diameters, tilings, Hamilton surfaces in polytopes*, and finally *Helly-type results*. In [219], essential papers on *projection functions of convex bodies, Carathéodory-type results, the polytope algebra, illumination problems, on-line packings and coverings of convex bodies*, and also *star-shaped sets* can be found.

Here we mention also the report [350]. And the proceedings [220] continue with papers on *dense packings of convex bodies, decompositions of multiple coverings, (colorful) transversal theory, extremal properties of mosaics, Voronoi diagrams*, and *uniform polyhedrals* (an extension of the notion of uniform polyhedra). As the last for the time being book in this series one has [103], in which contributions to the following themes are delivered: the *topological transversal Tverberg theorem, extremal packing densities of congruent convex bodies, Erdős–Szekeres theorems for families of convex sets, the twelve-spheres problem, spaces of convex m-partitions, four-dimensional regular compounds*, and *combinatorial distance geometry in Minkowski spaces*.

Although not belonging to the discussed series (being closer to the geometry of numbers), we mention here also the proceedings [188], in which one can find papers on *Minkowski's successive minima, covering problems of convex bodies, finite sphere packings*, and *isoperimetric problems*.

There are still three volumes of proceedings to be mentioned here. The first one, namely [320], edited by K. Bezdek, A. Deza, and Ye, contains articles on *polytope theory, generalizations of the colourful Carathéodory theorem, the dodecahedral conjecture* (on Voronoi cells of minimal volume in a packing of unit spheres in 3-space), *Minkowski geometry, minimal tilings* with convex cells each containing a unit ball, and on further related topics. An interesting problem section concludes the book. The second volume is [646]; related to convexity, there are papers on *polytope theory, homothetic coverings and illumination problems, zonotopes, symmetries of projections and sections of convex bodies*, and *volumes of convex hulls of different convex bodies*. And the third collection to be mentioned here is [310], collected by A. Bezdek. Articles in it which are close to convexity refer to *lattice point problems, special types of polytopes* (e.g., equifacetted, uniform, and edge-antipodal ones), *fixing systems of convex sets, packings and coverings, Helly-type theorems, common transversals, sets of constant width*, and various further themes.

Similar to the series "Intuitive Geometry", there are several proceedings on "Discrete and Computational Geometry". Here the term "convex geometry" no longer occurs, but the contents of the books (of this type) cited here have roughly the same percentage of articles on convexity as the books on "Intuitive Geometry". (We note that, regarding contents, also the "Festschrift" [129] discussed below would perfectly fit here.)

We start with the proceedings [1003], edited by Goodman, Pollack, and Steiger. They contain results on *Ehrhart polynomials of convex polytopes, h-vectors of simplicial complexes, spherical polytopes, infinite universal polytopes, lattice packings of convex polygons*, and *transversal theory*; also interesting connections to algebraic geometry are several times dealt with.

In [603], collected by Chazelle, Goodman, and Pollack, one can find interesting expositions on the *universality theorems for oriented matroids and polytopes, shadows of polytopes, the d-step conjecture, flag complexes and star shelling, transversal theory, packing and covering, polygonal covers, periodic and aperiodic tilings in n-dimensional space, special types of polyhedral sets*, and *important interactions between real algebraic geometry, and discrete and computational geometry*.

The collection [1001], due to Goodman, Pach, and Pollack, presents important surveys, e.g., on *extremal problems for lattice polytopes*, the *computational complexity of convex bodies*, again *transversal theory, pseudo-triangulations*, as well as *kissing, blocking, and covering numbers of convex bodies*. The proceedings [1002], collected by Goodman, Pach, and Welzl, contain papers on *polytope theory* (combinatorial structure like h-vectors, special polytopes like zonotopes, and edge-antipodal ones), *approximation theory, coverings and (cylindrical) partitions of convex bodies, Delaunay tessellations*, and *Hadwiger numbers*.

Further on, we cite here the proceedings [843] which were completely published in the journal "Discrete and Computational Geometry". They contain articles about *finite packings and coverings by congruent convex sets, the total curvature of shortest paths in the boundary of convex 3-polytopes* (that cannot be arbitrarily large), *lower bounds on diameters of convex polytopes with long edges and many vertices, sections of simplices*, and *Radon numbers* (of lattices).

Not really being proceedings (since there was no respective conference), but thematically fitting precisely here, also the book [1929], edited by Pach, should be mentioned. It consists of 12 survey-like chapters written by different authors, and some of these chapters are directly related to convexity: they refer to *geometric transversal theory, homothetic coverings, packings and coverings of convex bodies, applications of Borsuk's theorem* (e.g., for centrally symmetric polytopes), and, more algorithmically, *Delaunay triangulations and convex hulls* (in the plane) and the complexity of *polytope volume computation*.

Here we shortly mention also the proceedings [2060] and [19], which refer to interrelations between *convexity and graph theory*. They contain articles about *polytope graphs, lattice polytopes, decomposability of polytopes, neighbourliness of polytopes, Hanner polytopes*, and further related and interesting topics; interesting research problems are posed, too.

Also related to graph theory is the collection [1930] of 30 invited surveys on the theory of *geometric graphs*. Here one can find articles on *tangency numbers of convex regions, large distances in convex polygons, diameter graphs* (e.g., important for Borsuk's problem), *equilateral sets in certain normed spaces, and intersection patterns of convex sets*.

In [999], collected by Goodman et al., articles on *arrangements of hyperplanes* (and related *zonotopes*), important inequalities like those of *Blaschke–Santaló and Alexandrov–Fenchel*, results on *local convexity, of Baire category type, on star-shaped sets, tilings, coverings, and packings, f-vectors of polytopes, and polyhedral manifolds* are compiled.

The nice article collection [2221] due to Senechal and Fleck gives beautiful insights to *polyhedral geometry* from very different viewpoints. It addresses to mathematicians, artists, architects, designers, engineers, chemists, bioscientists, crystallographers, and philosophers, and at the same time to students, scholars, and teachers. Several prominent geometers submitted articles on polyhedra and their relations to abstract and applied mathematics, and also to various other disciplines (like molecular geometry and crystal structures). The proceedings contain many nice figures and photographs, encouraging mathematicians and non-mathematicians to think about the geometry of polyhedra and to try creating new objects and theories.

Before coming to the handbooks, we mention the important proceedings [351] referring to *polytopes in the abstract, convex, and computational sense*. Each of the three chapters (in the same sequence with one of these three adjectives as headlines) contains between four and eight survey articles of different authors, and we shortly describe some of them. So in the part "abstract polytopes" the groups which form the basis for the *classification of regular polytopes* are discussed by Cohen, and Coxeter himself describes the evolution of the *Coxeter–Dynkin diagrams*. Modern developments in these directions are discussed by McMullen, and Grünbaum considers very general polyhedral sets which can have hollow faces. In the part on "convex polytopes", Stanley surveys *Eulerian posets*, and also combinatorial properties of the *face structure* are discussed. However, also very different (metrical) topics are treated, such as *polytopes and the Brunn–Minkowski theory* (by Schneider), or *approximations of general convex bodies by polytopes* (by Gruber). Here also *volumes of non-Euclidean polytopes* are surveyed by Kellerhals. In the "computational part" also typical convexity results and notions are discussed, e.g., *volume and mixed volumes* in the article of Gritzmann and Klee. And at the end of the book, one can find an interesting problem collection.

Three important collections of articles (in style similar to proceedings, but in fact being "Festschrifts" containing also various important contributions to convexity) are [714], [1030], and [129]. In [714] one can find important contributions treating *polytopes and honeycombs* (tilings, spherical designs and polytopes, star polytopes, etc.), *extremal problems from convexity* (translative intersections,

Minkowski's lattice point theorem, etc.), and *combinatorics* (of convex polyhedra); some articles directly refer to the inspiring work of Coxeter.

The collection [1030] impressively presents the broad interests of Klee (reaching from many aspects of discrete mathematics to functional analysis, from computational geometry to mathematical programming, from theoretical computer science to combinatorics and graph theory, etc.). To a large part, his papers show that he understood *convex geometry* as a method to be used in all mathematical disciplines where it can be useful (a famous example in this direction is the *Klee–Minty cube*). And the papers in [1030] show how convexity is alive in the form of discrete, combinatorial, computational, and stochastic results and methods. To take polytopes as an example, we find there results on *random polytopes, approximation by polytopes, self-dualities, polyhedral realizations of regular maps, f-vector theory, triangulations, antipodality types, isoperimetric problems for polytopes, and chiral polytopes*.

Dedicated to Goodman and Pollack (the founders of the journal "Discrete and Computational Geometry"), the largest part of the Festschrift [129] treats typical results from *discrete and computational geometry*. But many of the articles are directly or closely related to *convexity*, e.g., referring to *lattice points in convex bodies, applications of geometric graphs in convexity, arrangements of hyperplanes and half-spaces, the Hadwiger–Debrunner (p, q) problem, concepts in polytope theory, and isoperimetric problems*.

Finally we mention here the proceedings [717], edited by Dawson et al., in which *topics between convexity and fractal geometry* are treated. Besides articles on fractal geometry (e.g., on fractal star bodies), they contain papers on the boundary structure, sections and projections of convex bodies, inner parallel bodies, and convex sets in normed spaces (two of the latter papers are also surveys).

12.3.5 Handbooks

We now come to the *Handbook of Convex Geometry* [1077], thankfully edited by Gruber and Wills. Regarding the amount of results, concepts, problems, and applications discussed in it, this is clearly the most comprehensive publication in the field of convex geometry. It contains 40 carefully written surveys on all important aspects of convexity, ordered (except for Gruber's historical article at the beginning) in five parts. The first two parts form Volume A, the other three Volume B. In the following list, we mention a selection of these surveys always meaning those parts of their contents which are in strong relation to convex geometry.

Part 1 Classical Convexity: *characterization theorems, mixed volumes, (stability of) isoperimetric and further geometric inequalities, rigidity, curvature and surface area measures, the space of convex bodies, approximation results, and special types of convex bodies* (special types of polytopes, simplices, ellipsoids, centrally symmetric bodies with measures of symmetry, and bodies of constant width).

Part 2 Combinatorial Aspects of Convexity: *Helly-type theorems and generalizations, problems in discrete and combinatorial geometry, combinatorial theory of convex polytopes, polyhedral manifolds, oriented matroids, relations to algebraic geometry, mathematical programming and discrete optimization, and computational geometry*.

Part 3 Discrete Aspects of Convexity: *Geometry of numbers and lattice point problems, packing and covering (also in the finite sense), tilings, valuations and dissections, and geometric crystallography*.

Part 4 Analytic Aspects of Convexity: *Convex functions and calculus of variations, convexity and differential geometry, relations to mathematical physics, local theory of normed spaces, nonexpansive maps and fixed points, critical exponents, Fourier series and spherical harmonics, zonoids and related topics, and Baire categories*.

Part 5 Stochastic Aspects of Convexity: *integral geometry and stochastic geometry*.

Every survey has its own bibliography, author index, and subject index.

Now we mention here some further handbooks from neighbouring disciplines which have important overlaps with the field of convex geometry.

There are many interrelations between convexity and Banach space theory, which are also subjects of several chapters in the two parts of the *Handbook of the Geometry of Banach Spaces* edited by Johnson and Lindenstrauss; see [1323] and [1324]. The two editors of this two-volume Handbook present in [1322] an introductory article for the whole project, namely presenting results and concepts of Banach space theory that are basic for later surveys in the Handbook, making them accessible to any interested reader from a graduate course in analysis and functional analysis. Moreover, this article is a very useful document for everyone interested in Banach spaces. The following topics in it are close to convex geometry: *geometry of the norm* (e.g., uniform convexity and the Bishop–Phelps theorem), *p-convexity, Banach–Mazur distances, Dvoretzky's theorem*, and further themes.

In the nice survey [200] of Ball (including full proofs) three topics are discussed which, as the author says, can be located in the intersection of functional analysis, harmonic analysis, probability theory, and convex geometry. First, *convolution inequalities* from convex geometry are treated, such as the *Brascamp–Lieb inequality and its reverse*, or the *reverse isoperimetric inequality*. Second, *volumes of sections of convex bodies* are studied, among them being Vaaler's related theorem and extensions of it, and *upper bounds for the volumes of sections of cubes*. And third, the author offers *plank problems* and related topics.

Although classical convexity "does happen" in finite-dimensional spaces, we cite also the important survey [888] of Fonf, Lindenstrauss, and Phelps on infinite-dimensional convexity, i.e., on important results concerning *convex sets in infinite-dimensional spaces*. Among other results in this direction, the reader finds a discussion of *exposed and strongly exposed points* of convex subsets of a Banach space in connection with the *properties of Radon–Nikodym and Krein–Milman*, and also boundaries and support are studied (Bishop–Phelps theorem, etc.). Properties of *convex bounded polytopes and polyhedral Banach spaces* are presented, like also *stable convex sets and convex tilings*. Interesting research problems are offered, and an extensive list of references concludes the survey.

Also the exposition [1447] of Koldobsky and König contains results from convex geometry. For example, *extensions of isometries in L_p-spaces, the Busemann–Petty problem, approximation of zonoids by zonotopes, and projection constants of finite-dimensional Banach spaces* are topics discussed here.

From the second volume of the "*Handbook of the Geometry of Banach Spaces*" [1324], we want to mention only two articles. Mankiewicz and Tomczak-Jaegerman [1692] study *properties of "random" subspaces/quotients of finite-dimensional real Banach spaces and applications thereof*. One starting point is *Gluskin's theorem* about the diameter of the Minkowski compactum. Famous examples of interesting spaces are offered, e.g., Szarek's example of a space without the well-bounded Dvoretzky–Rogers factorization. This paper shows at several places that *properties of general convex bodies* are very important. Clever ideas are used to present well-known results of Gluskin, Szarek, and Mankiewicz, where random quotients of l_1^m give examples of *spaces with extremal non-Euclidean behavior* with respect to the basis constant, the symmetry constant, the Banach–Mazur distance to l_1^n, the constant in the Dvoretzky–Rogers factorization, and so on.

The second article [1992] from Volume II of the Handbook, due to Preiss, refers to geometric measure theory in Banach spaces. Important notions, that here play a role, are *differentiability of convex functions, Asplund space, and the Radon–Nikodym property*.

We switch now to the *Handbook of Metric Fixed Point Theory* [1385], from which we will cite two surveys here.

In Fuster's paper [914], treating moduli and constants in metric fixed point theory, many *moduli of convexity* are discussed, where the *Clarkson modulus of convexity* is naturally taken as a starting point. Further moduli follow, among them being the *Lovaglia local modulus of convexity, Gurarii's modulus of convexity, Milman's k-dimensional modulus of convexity and smoothness*, and some others. Also properties in terms of the *Jung constant* are discussed. A metric space is said to be *hyperconvex* if it

is convex and its closed balls have the so-called binary Helly property involving pairwise intersection properties of closed balls of that space.

Espínola and Khamsi [819] survey results on *hyperconvex spaces*, including also a historical discussion. The (historically) first result on hyperconvex spaces (namely, the metric counterpart to the Hahn–Banach theorem) is proved in a conveniently simplified form. A notion of *hyperconvex hull* is introduced, and the following properties of it are derived: uniqueness (up to an isometry) and preservation of the *Hausdorff and Kuratowski measures of non-compactness*. Also the classical fixed point theorem on *nonexpansive mappings of bounded hyperconvex spaces* is proved, and many related results follow, including also approximation results of Ky Fan type.

The *Handbook of Generalized Convexity and Generalized Monotonicity* [1131], closer to our chapter referring to *convex analysis* (but we bring it here, under the Handbook headline), contains 14 surveys, the first eight of which treat *generalized convexity concepts* with the background of *convex and quasi-convex analysis*, and of optimization. The articles illuminate many different aspects of this branch; *generalizations of convex functions, differentiability properties, scalar and vector optimization for different types of generalized convex functions* (pseudo- and quasi-convex, pseudolinear and invex ones, etc.), extensions of the common convex conjugation to generalized convexity (through the *abstract Fenchel–Moreau theory of conjugation*), the field of *abstract convexity* (on functions that can be written as pointwise suprema of functions belonging to a given set of functions), and various further themes.

From the *Handbook of Geometric Analysis*, we cite the survey [2491] of Yang (and its continuation [2490]) dealing with the construction of *affine integral invariants of convex bodies*; see also our section on integral geometry.

In some articles of the *Handbook of Hilbert Geometry* [1946], at least *certain aspects of convexity* play a role. And also Volume V of the *Handbook of Applicable Mathematics* (see Ledermann and Vajda [1523]) contains, as chapter 4, an *introduction to convexity*.

A *Finsler metric* on a manifold M is a family of norms in tangent spaces of M, which vary smoothly with the base point; thus the local situation is described by a *Minkowski geometry*, giving a link to convexity. Therefore, we cite here also the survey [92] from the *Handbook of Differential Geometry*, Vol. II, treating the state of the art of and posing open problems to this field; see also [93].

The *Handbook on Semidefinite, Conic and Polynomial Optimization* [110] contains a few expository papers very close to convexity. As topics broadly discussed there, we enumerate *convexifications used for convex relaxations, semi-definite representation of convex sets, and convex hulls*, like also *convex hulls of algebraic sets*.

We continue with [1000], the *Handbook of Discrete and Computational Geometry* (first edition: 1997; third one: 2018). From the viewpoint of convexity, its first part *Combinatorial and Discrete Geometry*, its second *Polytopes and Polyhedra*, and its third *Algorithms and Complexity of Fundamental Geometric Objects* are most interesting. The first part contains essential surveys on *packings and coverings, tilings, Helly-type theorems, geometric transversals, oriented matroids, discrete aspects of stochastic geometry, lattice point problems and lattice polytopes*, and the second part starts with a survey on *basic properties of polytopes*, focusing on combinatorial (faces and their relations) and geometric (such as volume and surface area) properties, and it continues with articles on *subdivisions and triangulations of polytopes, face numbers of polytopes and complexes, symmetry of polytopes and polyhedra, polytope skeletons and paths, and polyhedral maps*. And in the third (more applied) part articles on *convex hull computations, Voronoi diagrams and Delaunay triangulations, arrangements, polygons, visibility, reconstruction problems, and computational convexity* can be found. From further parts of the book, applying convexity partially, we still mention expository work on (linear) programming, algorithmic motion planning, and crystals and quasi-crystals.

We mention that also the more applied *Handbook of Computational Geometry* [2084] contains important surveys on *Voronoi diagrams, art gallery problems, and visibility*. In other surveys being

more technical in the sense of algorithms, at least the background is sometimes still close to convexity, e.g., by treating for example *convex hulls*.

And from the *Handbook of Discrete and Combinatorial Mathematics* we mention the survey [2312] of Streinu et al. on discrete and computational geometry, almost 100 pages long.

Also the *Handbook of Combinatorics* [1015], edited by Graham, Grötschel and Lovász, presents articles close to convexity. For example, topics discussed there refer to *polytope theory* (e.g., polytopal graphs), *polyhedral combinatorics* (integer hulls of polyhedra, etc.), *lattice point problems, packings and coverings of convex bodies, extremal problems in combinatorial geometry, n-dimensional tilings*, and various applications in mathematical fields, and other sciences. For example, the surveys [1487] of Lagarias and [1421] of Klee and Kleinschmidt are worth highlighting! The first one treats problems related to point lattices; besides topics mentioned already above, the author discusses *lattice point problems on convex bodies, the Ehrhart polynomial, n-dimensional crystallography, and tilings*, as well as applications like algorithmical geometry of numbers and integer programming. The second survey comprehensively treats *polytopal realizability of complexes* (but see here the later reference [2030]), *enumeration of certain combinatorial classes, face vectors, and the d-step conjecture*.

12.3.6 Selected and collected works

Honouring several important mathematicians whose contributions to convex geometry were groundbreaking, within the last decades corresponding collected treatises or selected works appeared. Such works are very important for completing and cultivating mathematical fields, and one should thank the editors and collectors for their work. This way it is also made easier to get difficult-to-access references. Since most of these selected/collected works are already cited in this book, we shall keep their mention here short.

We start with the collected works [2294] of the great synthetic geometer Steiner. Due to Steiner's ingenious geometric talent, these contain an impressingly large variety of geometric ideas and results. His posthumously published "Lectures in Synthetic Geometry" influenced many mathematicians with geometric ideas. Somehow he is even known as an opponent of analytical methods. He was active in many directions, meaning that he knew a large part of the geometric ideas and disciplines known at that time. For example, he was one of the first using the *inversion in a circle*, and he developed the *projective theory of conic sections*. Close to convexity, he presented *five proofs of the isoperimetric inequality* (introducing also his famous *symmetrization procedure* to be used there), and within a *general concept of centroids* he introduced the idea of *curvature centroids* which now are called *Steiner points*. Steiner also gave a famous formula for the *volume of the outer parallel body* of a convex body; this can be seen as a forerunner of the notion of *mixed volumes*.

Minkowski's selected works [1807] contain also *Minkowski's contributions to the foundations and first applications of convexity* as a new coming field. In contrast to the selection [1803], it contains also the important unfinished work "Zur Theorie der konvexen Körper". Both [1807] and [1803] do not contain Minkowski's convexity-related geometry of numbers [1804] which appeared in book form; see the subsection on discrete geometry. The selection [1803] appeared on the occasion of the 125th anniversary of Hermann Minkowski's birth. It contains reprints of eight mathematical papers by Minkowski, the famous commemorative speech of Hilbert, two appendices by Krätzel and Weissbach, and two handwritten letters to Hilbert and Hurwitz. The eight reprints include a short introductory paper on Minkowski's monograph on the *geometry of numbers*, and his famous paper on the *reduction theory of integral quadratic forms*. The other reprints deal with *convex bodies*, e.g., with the notion of *mixed volume*. There are two appendices, the first of which covers the geometry of numbers and its arithmetical applications, for instance the famous *lattice point theorem*; the second appendix deals directly with convex bodies, e.g., with *mixed volumes and bodies of constant width*. A quite detailed

introduction to the mathematical contents of the papers and a survey of the following work up to the early 1980s are there, too.

The contributions of Radon to the field of convexity are gathered in his collected works [2010] and already discussed above.

The edition [360] collects 40 papers of Blaschke which are closely related to convex geometry and range from 1912 to 1956. There is an "in memoriam" of Reichhardt, and Blaschke's work in convexity is presented by Leichtweiss, who also gives hints and references to later developments. And then four sectors (with respective papers) to the themes *Euclidean convex geometry, equiaffine convex geometry, Euclidean convex differential geometry, and equiaffine convex differential geometry* follow. Very important and famous results can be found in this collection, such as the *affine isoperimetric inequality, the Blaschke–Santaló inequality, various characterizations of ellipsoids* (also important for the geometry of normed spaces), *Blaschke's selection theorem* (essential for existence proofs), and the successful *applications of symmetrization procedures*. This collection shows impressively the many directions for which Blaschke motivated later convex geometers, based on interesting problems and conjectures he posed and methods he developed. The edition [360] is particularly useful by the following reason: several of Blaschke's papers reprinted here are, in their original form, not easy to reach. The books of Blaschke were already presented above, under his name.

It is well known that Santaló was a member of Blaschke's Hamburg School, and therefore we mention here his selected works [2105]. These contain (among many other geometric papers) 26 papers in *integral geometry*, 9 papers in *classical convexity*, and also contributions to the fields of *stereology and affine geometry*; e.g., also the famous *Blaschke–Santaló inequality* can be found there.

We continue with mentioning Alexandrov's selected works in [38], presenting his articles mainly referring to convex sets. Besides his books [40] and [39], no English translations of these works were available so far. Thus one effect of [38] is that the reader can easily see that Alexandrov's known, basic results on bending and rigidity problems on convex surfaces come indeed from his school. The contents of [38] are comprehensively discussed already above, under his name.

The selected works of Busemann (see [551] and [552]) contain Busemann's most important contributions to convexity; see the broad discussion above. Shortly speaking, Volume I treats mainly *foundations of geodesic spaces* and *metric geometry of Finsler spaces*, where the aspect of *being locally Minkowskian* is always emphasized. Volume II contains his basic papers on *classical convexity, integral geometry*, and further topics (also being important, but not interesting for this book here).

We also mention the selections [615] and [614], collecting papers and lectures of Chern. In [615], Chapter 3, various *convexity theorems derived with methods from differential geometry* are discussed. On the other hand, the second part of [614] contains eight never-before-published lectures of Chern on *integral geometry*.

The collection [676] contains 26 fundamental papers of Coxeter, mainly referring to *regular polyhedra, regular polytopes, and regular star-polytopes*. Here his life-long fascination with the *semiregular polyhedra and uniform polytopes* has to be underlined. Also his papers on the groups named after him are presented; his complete description of the *finite reflection groups* was very important for the geometry in the twentieth century. Further on, his papers on groups of unitary reflections, basic for the *classification of the regular complex polytopes*, are in the collection. And also Coxeter's interest in *sphere packings and lattices* is sufficiently reflected. In addition, the selection [680] should be cited here; in this book twelve papers of Coxeter, mainly treating polytope theory and topics from discrete geometry, are collected.

Also Motzkin worked successfully in convexity, and his selected works [1859] contain five papers in this field.

12.3.7 Examples of classical geometry books containing sections about convexity

There are also various classical mathematical or geometric books that contain sections or chapters treating the geometry of convex bodies. We will give examples here.

A good starting point is the famous book [1197] of Hilbert and Cohn-Vossen, containing nice material about *higher dimensional regular polytopes*, properties of *balls* (partially formulated in the spirit of convexity, e.g., as special *sets of constant width, solution of the isoperimetric problem*, or by using the notion of *mean curvature*).

Among various other topics, in the popular booklet [2009] Rademacher and Toeplitz discuss themes from convexity: *regular polyhedra, Jung's theorem, the isoperimetric problem*, and a relatively long section on *curves of constant width*.

The classical book [847] of L. Fejes Tóth on geometric figures having high degrees of certain regularities in a very wide sense presents many topics from convexity, such as the *Platonic and Archimedean polyhedra*, the *translative space-fillers* (which, in three dimensions, are completely presented by five types of *zonotopes*), the derivation of *all regular polytopes in higher dimensions*, several types of *isoperimetric problems, and inequalities*, yielding interesting *extremal figures*; known results from *packings and coverings* are also there. Parts of the contents of [847] can also be found in the surveys [141] by Ashkinuze as well as [2059] by Rosenfeld and Yaglom.

Also in the nice geometry book [1767], which came out of Melzak's notes for a one-year course in geometry, one can find material from convex geometry.

It is essential to mention the three books [282], [283], and [286] of Berger, treating classical geometry in the widest sense. Chapters 10 and 11 of [282] contain material from convexity, such as *inequalities yielding extremal properties of geometric figures, separation and support, Helly-type theorems*, and so on. In [283] mainly *polytope theory* is presented: *volume and surface area, regular polytopes, Euler's formula, Cauchy's rigidity theorem, approximation of convex sets by polytopes, the isoperimetric inequality, sets of constant width*, and further topics. And also in the general geometry book [286], to which the author was also inspired by [1197], many topics from convexity are presented: theorems on *sections and projections, geometry of polygons, polyhedra, and polytopes, Cauchy's rigidity theorem, isoperimetric inequalities, packings and tilings in the plane, lattices, and packings in higher dimensions*; due to these latter themes Minkowski's *geometry of numbers* is also presented. The problem book [288] directly completes the monographs [282] and [283].

Based on Euclid's *Elements* (and particularly on the Books I–IV) and also on Hilbert's (slightly modified) axioms, Hartshorne's monograph [1167] gives, with complete proofs, a critical, rigorous modern treatment of Euclid's geometry and also non-Euclidean geometries. Parts of that book are interesting also from the viewpoint of convexity. This refers to the theory of *area* (via dissection into triangles), *regular, semiregular, and regular-faced polyhedra* (Zalgaller is not mentioned), *symmetry groups of polyhedra, and Euler's and Cauchy's theorems*.

And also the book [1991] of Prasolov and Tikhomirov, addressed to a readership interested in geometry in a broad sense, contains material from convexity: several properties of *conics and quadrics, polyhedra and polygons* (formulas of Euler–Poincaré, Gram–Sommerville, Gauss–Bonnet, and Steiner–Minkowski), and an inspiring related look into infinite-dimensional space.

The elementary book [89] of Alsina and Nelsen, mainly discussing geometry, contains nice passages treating problems from convexity. These refer to *Dido's isoperimetric problem, regular polyhedra, Reuleaux polygons, tilings, triangulations of polygons*, and also *star polygons*.

In his monograph [2257], Singer studies topics suitable for beginning undergraduates. For example, they refer to *regular and semiregular tilings, regular and semiregular polyhedra, the Euler characteristic, Cauchy's rigidity theorem, curvature of polyhedra*, and the *polyhedral version of the Gauss–Bonnet theorem*.

We also mention the nice geometry book [115] of Apostol and Mnatsakanian. Besides many topics away from convexity, it contains nice discussions on *areas, perimeters and centroids of polygons* which are circumscribed to circles and other figures, *polygons rolling around other polygons*, and a problem more general than the isoperimetric one. With nice geometric methods standard calculus problems are elegantly solved (this goal is also pursued in the nice book [1902] of Niven, mentioned above as problem-oriented book), and mechanical interpretations in the spirit of Archimedes referring to *centroids and moments* are extended also to higher dimensions.

12.3.8 A collection of surveys

Since it is impossible to be complete with presenting here important surveys on convexity topics, the following collection has to be seen as a selection following the taste of the authors. In addition we underline that all handbook surveys (see above), in particular those from the Handbook of Convex Geometry [1077], are (although being essential) not explicitly cited here. Since these expositions are easy to reach, we concentrate (not only) on papers that are somehow scattered in the literature. The same holds for many surveys contained in Proceedings and Selected Works (see again above). The sequence of the following surveys is (almost) chronologically chosen, further expositions can be found in the other chapters.

As a combinatorial abstraction of classical convexity, the wide notion of alignments (i.e., so-called *abstract convex geometries*) is defined as follows: An *alignment* L (also called a domain finite convexity structure) on a non-empty set X is the family of subsets of X (which are named *convex sets*) closed under an algebraic closure operator L. Such a pair (X, L) is called a *convex geometry*; X can be finite, and this concept can, for example, refer to convexity in graphs, or to partially ordered sets. In the survey [792] of Edelman and Jamison the foundations to this topic are developed, and also an update is given. Various characterization theorems are presented, and new examples are offered. *Operations on convex geometries* are studied, and the paper concludes with a list of open questions concerning convex geometries defined in this way. Here we cite also the related survey [1295] of Jamison, in which many further interesting examples of alignments and their constructions are presented.

The handbook survey [1183] by Henk et al. on polytopes first treats *combinatorial properties (referring to the face lattice) of polytopes*, with special emphasis on *low-dimensional polytopes* and also on *polytopes with few vertices*. In the second part more geometric and metrical properties are discussed, such as *volume, surface area, and mixed volumes*. Special keywords and headlines are *zonotopes, cyclic and neighborly polytopes, the universality theorem for 4-polytopes, Gale diagrams, quermassintegrals*, and also *intrinsic volumes*.

In the expository paper [2159], Schneider surveys some ways where the notion of measures is used in classical convexity. For example, it is shown how *Hausdorff measures* are applied to present the size of certain singular sets corresponding to a convex body K (like the set of directions of all line segments in the boundary of K). Further examples are *curvature measures, surface area measures, principal curvatures, and principal radii of curvature* (for smooth convex bodies), in suitably unified form yielding the notion of *support measures* and allowing the study of the local structure of a convex body. Also *uniqueness and existence results treating surface area measures* are presented, and in the final chapter the author discusses *zonoids*, due to the fact that they are ranges of non-atomic vector-valued measures. Further on, an application to Poisson processes of hyperplanes is given. All these concepts and notions from the expositions [2147], [2152], and [2159] (see below here) are also comprehensively discussed in Schneider's monograph [2167], particularly in the chapters 2, 4, and 8.

In [1419], Klee nicely surveys some concepts, results, and application areas of convexity which have been active in those days. After clarifying basic notions of the field, he discusses problems and motivations from and relations to *discrete geometry (packings), measurements of convex bodies and*

related inequalities (e.g., that of Jung), *geometric probability* (Sylvester's problem), *functional analysis* (characterizations of the Euclidean case among general normed spaces), *combinatorial geometry* (Helly-type statements), *polyhedral graphs, bodies of constant width, the equichordal problem, convex and concave functions* (importance of extreme points and the Krein–Milman theorem), and *fixed point theory* (e.g., the Brouwer–Schauder–Tychonov theorem). Important tools (like *polarity and mixed volumes*) are also discussed.

The nice survey [769] of Dubois provides several topics from *finite- and infinite-dimensional convexity*. As examples, we mention *measures of non-convexity, properties of convex functions, separation properties, the Krein–Milman theorem, Helly-type theorems*, and *bodies of constant width*.

In [2147] Schneider gives a complete overview on results treating *different types of boundary points and faces* of convex bodies, defined in terms of supporting hyperplanes or convex combinations in the boundary. A unified approach to *area functions and curvature measures* (of Federer, restricted to convex bodies) follows. In this way, information on the *local shapes* of the bodies under consideration are obtained, and nice characterization theorems as well as open problems are given.

In his handbook article [2152] Schneider deepens this theory, paying particular attention to differentiability conditions and also the case of their absence. So it is discussed to what extent a convex body can fail to be *strictly convex* or *smooth*. For example, Reidemeister's theorem is considered, namely that the set of boundary points at which a convex body is not smooth is of $(n-1)$-dimensional Hausdorff measure zero. And with respect to strict convexity, a wonderful result of Ewald, Larman, and Rogers is presented, namely that the set of directions of line segments in the boundary of a given convex body is of σ-finite $(n-2)$-dimensional Hausdorff measure. Applications of and further developments to these results are presented. Then the author considers the notion of *pointwise curvature in the nonsmooth case*, and he discusses how "incomplete" this framework is to describe the local shape of a convex body. Using the notions of *curvature measures and surface area measures*, some generalized differential geometry for convex hypersurfaces is developed. *Mixed volumes* occur, and the special case of *convex polytopes* (with correspondingly modified notions) is also discussed. Existence and uniqueness results for these measures and respective characterization theorems are given, and at the end the known results about uniqueness and stability of the surface area and curvature measures are presented.

Gruber [1054] collects seven attractive results from convexity (which, in fact, are eight), for example subjects like *Löwner ellipsoids*, the *Busemann–Petty problem, flexibility of frameworks* (systems of line segments connected at common endpoints with hinges), and *geometric tomography* (X-rays based on Steiner symmetrizations).

Also in the survey [1055] Gruber clearly shows, by many properties of convex sets excellently presented, how wide the field is. For example, the following topics are treated: *historical aspects, boundary structure and curvatures, intrinsic metric and rigidity, sums and decompositions, topological and metric properties of the space of convex bodies, integral geometry and geometric probability, discrete geometry and geometry of numbers, combinatorial geometry, differential geometry, functional analysis, mathematical physics, tomographic questions*, and even *statistics*.

In [1067] the same author surveys the progress regarding some topics from convexity, such as the *problem of Steinitz* treating the boundary complexes of convex polytopes, *packings, coverings and tilings*, the relations between *mixed volume inequalities and algebraic geometry*, and *Baire category problems*.

Schneider's survey [2148] collects results on the geometrical characterization of the *equality case of the Alexandrov–Fenchel inequality*. The author states three conjectures concerning equality in this famous inequality, presents special parts of his conjectures which are confirmed, and derives further results supporting these conjectures. He continues presenting the state of the art with the later survey [2154]; see also Section 4 of Groemer's handbook survey [1040] and Chapter 7 in Schneider's book [2167].

Based on investigations regarding the Brunn–Minkowski theory (namely treating *(mixed) volumes, quermassintegrals, and surface area functions*), on Hadwiger's *characterization of quermassintegrals*,

Dehn's solution of *Hilbert's third problem*, and typical problems from *combinatorial geometry*, the brilliant survey [1753] of McMullen and Schneider presents a first collection of all these (and more) notions having the *valuation property*. Besides the above notions, for convex bodies valuations also occur as *lattice point enumerators, Steiner points, angle sums*, and in various further forms. This survey gives classical examples and general results, treats *dissections and simple valuations*, and continues with *general valuations*.

A natural continuation of [1753] is McMullen's excellent handbook article [1745], summarizing extensively the development of the subject after the appearance of [1753] and adding nicely the *polytope algebra* introduced by McMullen himself in 1989.

Based on the stage reflected by [1753] and [1745], Alesker [44] gives a survey on *rotation invariant, translation invariant, and unitarily invariant valuations*. More expositions on different types and aspects of valuations on convex bodies and with *relations to integral and stochastic geometry* are collected in the Lecture Notes [51] by Alesker and Fu and the Proceedings [2398]. Also in [45], the written version of Alesker's invited talk at the ICM in Beijing (2002), the author summarizes results on continuous valuations mainly due to him, concentrating on translation invariant valuations and showing important relations to integral geometry.

Further important surveys on deep developments around valuations refer to *valuations on manifolds* (see Alesker [48]) and *Minkowski tensors* (defined via a Steiner formula from the moment tensors of convex bodies and being rotation covariant, continuous, translatively polynomial, and symmetric tensor-valued valuations; see Hug and Schneider [1259]).

In addition we refer to the survey [1613] of Ludwig, treating *valuations which are invariant or covariant under the action of the special linear group SL(n)*; e.g., the known characterizations of these valuations are collected.

Classification results for function spaces are surveyed by Ludwig in [1615], where *real-valued, matrix-valued, and convex body-valued valuations on Lebesgue and Sobolev spaces* are studied.

Also Section 17 of the survey Hansen et al. [1157] on star-shaped sets contains an overview treating valuations.

The survey [2170] of Schneider gives an excellent introduction (also for newcomers) to the *modern theory and technical tools of valuations* in convexity.

We mention once more that in book form we have overviews on valuations in [49] by Alesker (see above), and also in Chapter 4 of the very recent monograph [1263] of Hug and Weil. We finish this sequence of valuation-related publications by mentioning the very recent and basic survey [1617] of Ludwig in which many investigations of valuations in convex geometry and more general concepts are successfully updated.

Restricting to dimensions two and three, Leichtweiss surveys in [1533] nicely the *interplay between convexity and differential geometry*, e.g., discussing notions from differential geometry which can be extended to the boundaries of arbitrary convex bodies. Central topics discussed in an elementary manner are *affine surface area* and *quermassintegrals*.

Groemer surveys in [1036] results on *covering and packing problems* with respect to convex sets in Euclidean n-space of the following kind: Let K be a convex subset and (K_i) a finite or infinite sequence of closed convex subsets. What conditions ensure that it is possible to apply rigid motions (or only translations) to the sets K_i such that their images cover K, or are contained in K with mutually disjoint interiors. In addition, what is known on densities of such coverings or packings? Since strong restrictions on the relations between the sets K_i are not assumed, such problems differ from traditional packing and covering problems.

Although mainly being a differential geometer, Berger wrote a very nice survey in convexity. Namely, in [284] selected (classical and more recent) results from convexity, but also various relations to other areas (like Minkowski geometry), are presented. Here is a list of selected topics discussed there: *polytopes, characterizations and applications of ellipsoids* (e.g., John's ellipsoid), *Banach–*

Mazur metric, duality and vector addition (Mahler's conjecture, Brunn–Minkowski theorem), and *topology on the set of all convex bodies* (Baire category results which are "against intuition").

In his second survey [285] close to convexity, Berger nicely discusses *circle packings, Kepler's conjecture*, and the *Tammes problem* (treating arrangements of spherical caps on S^2). He presents also the problem whether every *combinatorial type of convex polyhedra* can be represented by a convex polyhedron inscribed in the sphere. He switches then to the related question to decide when a polyhedron is equivalent to one all whose edges are tangent to a sphere. This leads to *packings of non-congruent circles* and the famous related *theorem of Koebe–Andreev–Thurston* (yielding a complete answer). Unfortunately, the list of references related to these nice questions is not complete (e.g., important results of Rivin and Schramm are not cited). Nevertheless, this is a nice paper showing at the end also nice relations of circle packings to *conformal geometry*.

The survey [1710] of Martini exposes results on *cross-section measures of convex bodies* as they are introduced by Bonnesen and Fenchel in §7 of [446]. If K is an n-dimensional convex body and F an i-dimensional subspace, then the i-dimensional outer cross-section measure of K at F is the i-dimensional volume of the orthogonal projection of K into F, while the i-dimensional inner cross-section measure of K at F is the maximal i-dimensional volume of the intersection of K with translates of F. In [1710] results and open problems treating these cross-section measures and their associated bodies (having these quantities as support or radial function) are collected. The latter are known as *projection bodies, intersection bodies, cross-section bodies, difference bodies, Blaschke bodies*, and so on. The survey mainly (but not only) deals with the cases $i = 1$ and $i = n - 1$. In particular, using the case of $(n - 1)$-*dimensional inner cross-section measure* regarding radial function, the notion of *cross-section body* is introduced. All these topics are excellently presented and extended in Chapters 3, 4, 8, and 9 of Gardner's monograph [930]; see also Chapter 10 of [2167].

In [1339] Kalai gives an excellent survey on the recent progress regarding generally unsolved and famous problems located "between" combinatorics and convexity. These concern *Borsuk's conjecture, packings, Helly-type theorems, the piercing conjecture* of Hadwiger and Debrunner, *polytopes* (realization spaces, polytope algebras, rigidity, h-vectors, diameters of polytopes, and the complexity of linear programming), *shifting, triangulations*, and various further topics. At the end of each topic, the main conjectures are presented, and an informative list of references is given.

There are several types of *shadow boundaries of convex bodies*, and these have various applications. For example, there are different *illumination problems, combinatorial questions (for polytopes), covering problems*, and *characterizations of special types of convex bodies* for which shadow boundaries play an essential role. The survey [1712] of Martini collects many such results, and there are *types of antipodality* (see also the survey [1717] of Martini and V. Soltan) which can be interpreted as subcases of shadow boundaries of convex bodies.

The two excellent surveys [994] and [995] of Goodey, Schneider, and Weil discuss generalizations of the fact that support functions of projection bodies and difference bodies express projection functions of given n-dimensional convex bodies creating them (these ideas go back to Minkowski; see §7 of Bonnesen and Fenchel [446]). The authors study questions like: what can be said about the original convex bodies from knowing their *i-th projection functions*? Such problems are clearly tomographic in nature. The latter are defined by the i-dimensional volume of the orthogonal projections of the given convex bodies onto i-dimensional subspaces from the respective compact Grassmannian manifold of order i. They compare the situation in the intermediate cases $1 < i < n - 1$ to the extreme cases $i = 1$ and $n - 1$. Up to translation, centrally symmetric bodies are uniquely determined by any of their projection functions, and not in the non-symmetric case for the extreme cases. On the other hand, there are non-symmetric convex bodies which are uniquely determined (up to a translation and/or reflection) by their ith projection function if $1 < i < d - 1$. Also the relations between projection functions and Radon transforms on Grassmannians are presented. In [995], related *Baire category results* are presented, and the given bibliography is impressive. These tomographic topics are clearly related to

the notion of *cross-section measures*; see Chapter 3 of Gardner's monograph [930], Chapter 10 of [2167], and the survey [1710].

A beautiful survey of analytic aspects of convex geometry is presented by Peri [1964]. Among others, the author nicely discusses the following topics and tools: *Steiner's formula for parallel bodies, mixed volumes, valuations, Hadwiger's characterization of volume and his theorem on valuations, Blaschke's principal kinematic formula*, and the *inequalities of Brunn–Minkowski and Alexandrov–Fenchel*.

An important notion of convexity in linear spaces is that of *Choquet simplex*. This is a convex subset K with the following property: the intersection of any two non-empty homothetical copies of K is again a (possibly degenerate) homothet of K. Compact Choquet simplices in finite-dimensional spaces are simplices in the usual sense. V. Soltan [2272] surveys many results on this concept by reducing the given assumptions, characterizing also other classes of convex sets in finite dimensions (e.g., by weakening the homothety assumption for the intersections).

In [2273], V. Soltan surveys basic properties and applications of *affine diameters* which are proper chords of convex bodies whose two endpoints lie in two parallel supporting hyperplanes (examples are usual diameters or double normals). Equivalently, they are longest chords of convex bodies in their own directions. Important applications and a number of interesting open problems touch the following notions and concepts: *characterizations of special types of convex bodies, critical ratios and critical sets, measures of asymmetry, orthogonality in normed spaces*, and *Radon curves*.

If the unit circle C of a norm and that of its antinorm are homothets, then C is a *Radon norm*. Radon curves are precisely those unit circles of normed planes which guarantee that Birkhoff orthogonality is symmetric (these planes are called *Radon planes*). In the exposition [1721] of Martini and Swanepoel a collection of results on Radon planes and, more general, arbitrary Minkowski planes is given. The latter results occur if, for instance, the norm is replaced by the antinorm. On these lines several characterizations of Radon curves are derived, and also the *isoperimetric problem* for convex m-gons is investigated. The paper exposes old and new results, but most of them with new, streamlined proofs.

In the already mentioned historical survey [783], Dwilewicz nicely presents how the notion of convexity was developed in the works of Archimedes, Cauchy, Minkowski, and others. He continues with the *Hahn–Banach theorem* and how convexity became important in *real and complex analysis*. Finally, also the relation to *toric varieties in algebraic geometry* is emphasized.

The *central limit problem* states the following: for any n-dimensional convex body K, there exists a direction u such that the projection of the uniform measure on K onto U is close to the Gaussian distribution, where the error term tends to zero when $n \to \infty$. In the survey [236] of Barthe, this problem is put into the right historical context, and also the solutions to it in full generality as well as its relations to other typical results in *asymptotic convex geometry* (in particular, to *Dvoretzky's theorem* in Milman's form) are discussed. Related questions on *volume distribution on convex bodies* (like the *variance problem*, the *Kannan–Lovasz–Simonovits conjecture*, and the *hyperplane conjecture*) are presented in the final part. For central limit problems we also refer to Section 9.9 of Gardner's monograph [930].

Reduced bodies (see also our subsection 2.9.1.10) were introduced by Heil in 1974: A convex body K in Euclidean n-space is said to be *reduced* if the *minimal width* of each convex body properly contained in K is smaller than the minimal width of K. Examples are all bodies of constant width in arbitrary dimension, and the regular odd-gons in the planar case (a comprehensive survey on reduced bodies in the Euclidean case, with interesting open problems, is [1505] by Lassak and Martini). Therefore, the family of reduced bodies forms a proper superset of that of *diametrically complete bodies* which, in Euclidean spaces, is identical with the family of *bodies of constant width*. (The knowledge about bodies of constant width is presented in the book [1713] whose forerunner was the excellent survey [596]; see also our subsection 2.9.1.8 here.) Since the notion of *width function of convex bodies* can be suitably extended to normed spaces, also these notions can be successfully carried over to *Minkowski geometry*, but there the situation is more difficult. Namely, it turns out that in normed spaces completeness and constant width are no longer equivalent notions, and that there are even *complete bodies not being*

reduced. A survey on reduced bodies in normed spaces, also containing deep research problems, is [1506] by Lassak and Martini. For example, the question occurs whether, in general normed spaces, a convex body which is reduced and (diametrically) complete is necessarily of constant width! In general this is open, but Brandenberg et al. [487] have shown that the implication holds for a large class of convex bodies, namely for those having a smooth extreme point.

All three authors of [2081], namely Ryabogin, Yaskin, and Zvavitch, used successfully methods from harmonic analysis to solve deep tomographic problems in convex geometry and also referring to star-shaped sets. This paper contains an interesting list of related open problems suitable for being studied with possibly the same type of techniques and methods; also a survey of related results is given and compared with the still unsettled problems.

The nicely written notes [2082] of Ryabogin and Zvavitch are focused on duality of convex bodies and inequalities around the so-called *volume product* (Blaschke–Santaló inequality and Mahler's conjecture). The authors try to give a representation of the state of the art. For example, regarding Mahler's conjecture the authors present *Nazarov's proof of the Bougain–Milman inequality*, give a proof of the local minimality of the cube, and show the *validity of the conjecture for zonotopes and zonoids*. Koldobsky's method *to present spherical Radon and cosine transforms in terms of the Fourier transform and distributions* is brought to the reader, and also *tomographic problems* (determination of convex bodies by volumes of their sections and projections) are described, too. The paper [409] of Böröczky et al. contains also an overview to the volume product, regarding the symmetric and the general case.

Collecting almost 200 references, Nagy et al. [1872] succeed in combining *non-parametric statistics and convex geometry* via a link between the concepts of half-space depth (an important tool for the efficient design for multivariate and functional data) and the *floating body problem*. It is shown that the convex floating body corresponds to the upper level sets of the half-space depth. Using the notion of *measure of symmetry*, the maximum depth can be seen as a generalized such measure for convex bodies, and also the notion of *affine surface area* is used in this comprehensive representation. Thus, by the presented connection between two very different mathematical fields, the authors succeed in attacking partially some open problems regarding theoretical properties of the depth.

The survey [12] of A'Campo and Papadopoulos presents the *history of curvature*, starting from Greek antiquity (Euclid, Apollonius, Archimedes, etc.), showing then the contributions of Huygens, Euler, and Monge, and discussing the twentieth century works of Bonnesen, Fenchel, Busemann, Feller, and A. D. Alexandrov. The authors succeed in demonstrating how notions like *curvatures of curves and surfaces* were studied from the synthetic viewpoint by the Greeks, from the analytic one by Fermat, Descartes, Newton, and Leibniz, and, in the twentieth century, somehow again via synthetic methods (particularly visible in the work of Busemann and A. D. Alexandrov).

The nicely written survey [1235] of Horváth collects many long-standing open problems treating *convex-hull-type volume functions* and related topics, reflecting in this sense roughly a 70-year period. For example, these problems refer to *covariograms, width functions, difference bodies, brightness functions*, and various further topics.

The expository paper [1157] of Hansen et al. is dedicated to the fundamental mathematical notion of *starshapedness*, which is one of the most natural generalizations of the convexity concept. While convexity has a long history, going back even to ancient times (e.g., Archimedes), the geometry of star-shaped sets is a younger field, and no related historical overview or monograph exists. Due to this situation, the paper [1157] is the first systematic exposition discussing *geometric, analytical, combinatorial, and topological properties of star-shaped sets* in the broadest sense. The strong relations to notions like *radial functions, section functions, visibility, (support) cones, kernel, duality*, and many others are emphasized. In addition, also useful applications of this notion in mathematical fields like *classical convexity, convex analysis, functional analysis, discrete, combinatorial and computational geometry, differential geometry, approximation theory, PDE, optimization*, and further branches are presented in [1157].

In the survey [2279] of V. Soltan, the knowledge on *support and separation properties of convex sets* in Euclidean n-space are discussed. The necessary terminology and all known fundamental results are presented in a clear and uniform way, and the classical Minkowskian theorems on support and separation of convex bodies are taken as starting point, leading to the modern concepts. In addition, the analogues of these results concerning bounding and asymptotic hyperplanes are taken into consideration, like also more general separation types (e.g., by slabs and complementary convex sets).

12.4 Development of the overlaps with neighbouring disciplines

The intensive growth of convex geometry as a useful field in the heart of mathematics is also based on interesting relations to *neighbouring mathematical disciplines*. They always deliver important methods, problems, and tools, and (on the other hand) they fruitfully use approaches, refreshing ideas, and natural notions from convexity. More precisely, convex geometry is the branch of geometry where geometric properties of convex sets are studied, mainly in Euclidean space, but also in non-Euclidean geometries (like in non-Euclidean normed spaces, or in hyperbolic spaces). Since convex sets naturally occur in many areas of mathematics, this field influences, and is influenced by, various neighbouring disciplines and research fields. Due to the fundamental character of geometric ideas from convexity and, on the other hand, the expanding character of these fields, meanwhile parts of them belong already to convex geometry. On the other hand, various branches of convexity have split off and "belong" more to such neighbouring disciplines.

In the spirit of these developments, there grew strong relations to convex analysis, functional analysis, other parts of analysis, stochastic and integral geometry, differential geometry, the geometry of numbers, combinatorics, theory of polytopes and polyhedral combinatorics, discrete and computational geometry, convex and linear optimization, game theory, (geometric) graph theory, etc. One can see a very broad spectrum of used techniques and methods, such that in the Handbook of Convex Geometry [1077] the recommendation is given to structure the field also via methods and aspects, more precisely by classical, combinatorial, discrete, analytical, and stochastic aspects (see simply the structure of the Handbook [1077] itself).

And more modern developments reinforce these viewpoints even more, with further methods and aspects. In the following, we will present a selection of such disciplines and show the main developments in those parts of them which are close to convexity. Again we do this by citing mainly books, proceedings, and surveys, ordered partially by themes and partially in chronological sequence. The neighbouring disciplines chosen here are *discrete geometry, polytopes and polyhedral sets, algebraic geometry, differential geometry, integral and stochastic geometry, convex analysis, and the theory of Banach spaces*.

12.4.1 Discrete (and combinatorial) geometry of convex bodies

Discrete geometry and convexity have large overlaps; representatives of both fields very often met at conferences or organized them even together. It is impossible to give a complete description of these overlaps. Therefore this subsection here is in great parts created selectively. And as completion we recommend our subsection 2.9.1.2 treating the (discrete) geometry of the cube.

12.4.1.1 Description of the field and some history

Discrete geometry and also *combinatorial geometry* (as we mean it in our framework here) are subfields of geometry treating combinatorial properties of suitable geometric objects and constructive methods for studying finite or discrete sets of them; these objects can be points, planes, circles, spheres, convex polytopes, general convex bodies, and so on. For example, basically interesting questions are like this: how can the objects of such families be arranged to intersect, to cover something, or to form a packing? And there are many other types of suitable problems.

We give an overview to discrete geometry starting with comprehensive representations of selected partial fields (contributions of L. Fejes Tóth, geometry of numbers, combinatorial geometry of convex bodies, scissors congruence, and tilings). Then we continue with further related monographs (not cited before in this section), related proceedings, and handbooks as well as selected surveys. The field is too large for following also single research lines and citing research articles!

DELIMITATION OF FIELDS

Discrete geometry has a large overlap with *convex geometry*, and also with *computational geometry*. It is sometimes hard to separate these fields clearly. And for many mathematicians great parts of the large field of *(convex) polytopes/polyhedral combinatorics* belong to discrete geometry. Here, however, we treat polytopes and polyhedral combinatorics in an extra subsection (namely, the next one). To separate our considerations meaningfully from topics which are too far away from convexity, we emphasize that subfields like *finite geometries, digital geometry*, or *discrete differential geometry* are not discussed here; in a natural way we mainly limit ourselves to results and problems *treating convex sets as studied or used objects*! We note that various references and results discussed in Section 12.3 above could also be treated here. On the other hand, at the end we give at least a small list of *monographs in computational geometry*—to provide at least a small reference to this modern and very important area.

Perhaps one can say that discrete geometry started to exist with Kepler's famous book [1366] (where the *Kepler conjecture* on optimal ball packings, but also interesting facts about the *geometry of polyhedra* and further remarkable topics come from), followed by Thue's contribution [2363] (*optimality of the hexagonal circle packing*; see also in [2364]).

Although results on polyhedra, (regular) polytopes, rigidity questions, and tilings were known already when Kepler, Cauchy, Schläfli, Steiner, and others were active, the first deeper developments of discrete geometry (understood in the sense above) started only at the end of the nineteenth century. Instances of related subfields grown this way are the investigations on the density of circle packings (Thue), on the geometry of numbers (Minkowski), and on lattice packings of balls (Blichfeldt). Names of later famous *selected* contributors are L. Fejes Tóth, Coxeter, Rogers, Erdős, Conway, Gruber, Wills Lagarias, and many further ones. A good example for showing how merged the fields *convex geometry* and *discrete geometry* can be is Gruber's monograph [1070].

A HISTORICALLY IMPORTANT PULSE IN THE LATE FORTIES

In the late 1940s, L. Fejes Tóth started with systematic research about optimal arrangements of geometric figures (*packings or coverings* of balls or convex bodies, and also further topics like *tilings* or *extremal polyhedra*). So besides many other publications his classical book [851] came to be. It should be noted that this important book was recently translated into English and at the same time extensively expanded by G. Fejes Tóth and W. Kuperberg! The creation of the original version went hand in hand with the creation of the *Hungarian school in discrete geometry*. Bárány nicely describes in [213] the influential development of discrete and convex geometry in Hungary (twentieth century). It should be noted that his interesting representation additionally covers *incidence geometry, finite geometries*, and also *stochastic geometry*.

The classical book [851] contains many beautiful results in the spirit of discrete geometry in two and three dimensions. For example, results for the planar case which are nicely presented in this monograph are regular m-gons that have extremal area and perimeter among all convex m-gons inscribed in (or circumscribed about) a given circle; metric characterizations of the equilateral triangle; area approximation of convex regions by inscribed and circumscribed polygons (yielding a characterization of ellipses); length approximation of the boundary of a convex region by in-(circum-) scribed m-gons; packings of unit circles (yielding extremal lattice arrangements); packings of circles having different radii; finite circle packings and coverings; packings of congruent convex regions into a larger convex region; lattice packings and coverings, and so on. Also the *regular and Archimedean polyhedra*, including a proof of Euler's formula, are shortly discussed, with nice figures of these bodies. Later in the book, theorems on *extremal properties of regular and semiregular polyhedra* are proved, e.g., using congruent spherical caps which cover, or are packed on, the sphere. This yields point systems which are the vertex sets of the regular tetrahedron, octahedron, or icosahedron. Here also point systems on the sphere whose minimum distance is maximal are interesting, known from papers of van der Waerden, Habricht, and Schütte. Also the state regarding *optimal (lattice) arrangements of equal spheres* is comprehensively treated. The book is written in a problem-oriented and stimulating style, and each chapter contains a historical outline. The bibliography is rich, and directions of further research are clearly shown.

12.4.1.2 Geometry of numbers

We continue with Minkowski and the *geometry of numbers*, see the original work [1804] from 1896 and, showing from the today's point of view the large variety of concrete applications in discrete geometry, the fourth chapter (starting with Section 21) of Gruber's monograph [1070].

This field is *the part of number theory which uses geometry for the study of algebraic numbers*. More precisely, a ring of algebraic integers is viewed as a lattice in n-space, and studying these lattices one gets information on algebraic numbers. There are deep connections to functional analysis, Diophantine approximation, and other fields. Clearly, here we will discuss the geometric side of the field.

FUNDAMENTAL BOOKS

Siegel's book [2244] has, in its present form, the following resources: it comes from lectures that the author gave at New York University during 1945–46, and the original notes by Friedman have never been published, but were rewritten by Chandrasekharan, yielding a version that was finished in 1988. In Chapter I, "Minkowski's two theorems" are exposed in a detailed manner, and so Minkowski's basic work on the geometry of numbers, treating *convex bodies, gauge functions*, and problems around *lattice points*, is presented. *Minkowski's first theorem* is the theorem about the existence of lattice points in convex bodies, and *Minkowski's second theorem* is a refinement of the first one, dealing with the product of the *successive minima of the gauge function*. Deep applications in *number theory* and further mathematical disciplines are presented, and also the concept of *sphere packings* is discussed, including *Blichfeldt's upper bound for the packing density*. Related books which appeared after the New York lectures of Siegel (like Cassels [584], Rogers [2053], Gruber and Lekkerkerker [1075], Conway and Sloane [651], etc.) cover parts of the theory in more detail (see also the handbook article [1063] of Gruber). We mention also the monograph [1155] of Hancock, in which all contributions of Minkowski to the geometry of numbers are collected, in the style of a translation with comments. There is, however, no real presentation of later results, and also newer simplifications of the theory are not reflected. Nevertheless, the book is useful as a translation and for readers having no easy approach to Minkowski's original works. A large number of nice figures is given there.

The first advanced textbook treating the geometry of numbers also after Minkowski's basic and original work [1804] is the monograph [584] of Cassels. It contains fundamental results referring to the

following themes (we mention only topics related to our aims): *lattice theory, theorems of Blichfeldt and Minkowski, distance functions* (star-shaped sets, polarity), related theorems of Mahler, the theorem of Swinnerton-Dyer for convex bodies, theorems of Davenport and of Minkowski–Hlawka, successive minima, lattice constants for convex cylinders and balls, lattice points in star-shaped sets, etc.; the representation is clear and yields a timeless and readable book which contains also a large number of original proofs. Since the 1971 edition contains no substantial changes or completions (and the 1997 version is only a reprint of that), clearly all recent developments are not contained. (For example, the reader will find nothing about lattice polytopes including Ehrhart's polynomial and his reciprocity law; but it reflects the main stream of the geometry of numbers including the British school, with names like Mordell, Davenport, Rankin, Rogers, and Swinnerton-Dyer.) Due to its completeness (in the 1950s), clear style, and readability, the book will certainly remain one of the classical books on the geometry of numbers.

In the very readable and self-contained book [2053], Rogers collects many results on *lattice and non-lattice packings and coverings*, mainly treating (for large dimension n) upper and lower bounds on densities of best such arrangements of convex bodies which, in particular, can be balls, cylinders, or simplices. In the introductory part also other problems, beyond these restrictions, are exposed, and so also unsolved questions can be taken from this book. But also information on *classical packing densities of balls* are given there, including famous results of Blichfeldt, Rogers himself, and Davenport. The author gives then (also alternative) definitions of *lattice, periodic, and general types of packings and coverings*, and he presents proofs of the obvious *upper bounds for packing densities, lower bounds for covering densities*, and equalities for the best general and best periodic densities. Further results (which are harder to get) follow; they refer to the existence of reasonably *dense packings of arbitrary convex bodies*, and to reasonably good coverings. Difference bodies, central symmetry, the ratio between the volumes of the considered convex body and its difference body, and the inequalities of Blichfeldt play an essential role. Other tools that the author uses are *approximation methods, Voronoi polytopes*, and *Delaunay dissections*. The comprehensive list of references covers roughly the last 20 years before the book appeared.

Also the important monograph [1075] treats basic contributions to the *classical geometry of numbers in n-dimensional Euclidean space*, and together with the contents of the related monographs [584] and [2053] it gives a complete picture of this field until the appearance of its second edition. Also relations to other branches of mathematics, in particular to *convex geometry, packing and covering, Diophantine approximation, analytic number theory, coding theory, and numerical analysis*, are shown. Main topics of the book related to convexity are basics regarding convex bodies, star-shaped sets and lattices, Minkowski's fundamental theorem (his second theorem), generalizations of Blichfeldt, Mordell, and others, polarity and Mahler's compound convex bodies, Mahler's compactness theorem and the Minkowski–Hlawka theorem with improvements by Rogers and Schmidt, and further ones. In the second and third editions, much new material was added, e.g., results on *lattice polytopes and lattice point inequalities*, further applications (e.g., in coding theory), many nice open problems, and a comprehensively updated bibliography. The book is very useful as an excellent research reference, but can also be used as an advanced introduction to the geometry of numbers.

A very recent book on the geometry of numbers that we additionally mention is [1915] of Olds et al., in which mainly plane geometry is studied. Minkowski's theorem on convex bodies, with applications to Diophantine approximation and quadratic forms, and lattice packings (also in n-space, but without the Leech lattice) are presented, like also short biographies of Minkowski and Blichfeldt. The connections between geometric results and topics from number theory are nicely illustrated.

We finish here the list of monographs treating the geometry of numbers and its applications in discrete and convex geometry by citing the final and important chapter of Gruber's book [1070]. By writing, roughly speaking, a fourth part of this essential convexity book on the geometry of numbers, Gruber demonstrates how important he believes this area is in correspondence to classical convex

geometry. For example, one can find there the following topics: *Minkowski's first fundamental theorem, the Minkowski–Hlawka theorem, successive minima, lattice points in large convex bodies, packings and coverings of convex bodies, and tilings with convex polytopes* (e.g., Dirichlet–Voronoi tilings and translative space-fillers).

SURVEYS ON THE GEOMETRY OF NUMBERS

A survey nicely showing the connections between *sphere packings/coverings* and the *classical geometry of numbers* is [2460] by Wills. The author discusses topics like *finite and infinite packings/coverings and their connections to lattice point problems*, especially *finite sphere packings/coverings* together with *sausage problems*, connections to the growth of crystals, *lattice polytopes and lattice point enumerators* using quermassintegrals, and linear functionals like *diameter and minimal width*. This is continued in the survey [2461], again of Wills, which refers to *quermassintegrals, successive minima, and the lattice point enumerator* in the geometry of numbers.

Another important survey on the geometry of numbers and its convexity side is Gruber's handbook article [1063]. After a detailed description of fundamental theorems in this field (*theorems of Minkowski, Blichfeldt, Siegel, and Minkowski–Hlawka*), the author treats selected topics like, for instance, *successive minima*. Besides theoretical results he also discusses *computational aspects* of some topics, and an interesting collection of problems (e.g., *Voronoĭ's conjecture about parallelohedra*) is contained, too. Also the historical development of the field is nicely presented.

A further survey to be mentioned here is the handbook article [1034] of Gritzmann and Wills, dedicated to the part of the geometry of numbers which relates properties of a convex set K and a general lattice L. Good examples are *lattice point enumerators* with respect to the intersection of K and L, or the facial structure of the largest polytope contained in K with vertices from L, its *Euler characteristic, and quermassintegrals*. Also the following is discussed: for solving integer programming problems in polynomial time, one has to represent the convex hull of the integer lattice points in the feasible region in terms of suitable inequalities in polynomial time. Thus, in the paper also *computational aspects* are discussed, and an interesting collection of examples is presented.

12.4.1.3 Combinatorial geometry of convex bodies

The next subfield of discrete geometry discussed here could be called *combinatorial geometry of convex bodies*. It brings together principles from combinatorics and (convex) geometry, studying combinations and arrangements of geometric objects and discrete properties of them. We assume these objects as being (special types of) convex bodies. The term "combinatorial geometry" (in this sense) was apparently first used by Hadwiger and Debrunner in [1137]. This booklet is a translation of the original version in German and contains (besides many problems from very different mathematical fields) also results from the combinatorial geometry of convex sets, such as *Helly-type problems, partition problems*, and *covering problems* referring to (special types of) convex bodies. Natural continuations of such parts of the original version of [1137] can be found in the famous survey [708] of Danzer, Grünbaum, and Klee.

HADWIGER'S COVERING PROBLEM AND HELLY-TYPE RESULTS

We now come to publications of Boltyanski and several coauthors (see also subsection 12.3.1 above). The booklet [425], nicely written in the Russian tradition of popular small books, discusses mainly two famous, still unsettled problems from the combinatorial geometry of (compact sets or) convex bodies: *Borsuk's partition problem* (can every bounded subset C of the n-dimensional Euclidean space be partitioned into $n + 1$ sets each having diameter smaller than that of C?) and *Hadwiger's covering*

problem (asking for the minimum number of smaller homothetical copies of a given convex body K needed to cover K). Also the famous *illumination problem*, equivalent to the latter, is presented in an elementary way. It asks for the upper bound of the minimal number of oriented light directions (parallel light) necessary to illuminate the whole boundary of a convex body (a boundary point is *illuminated by a direction* if the light ray reaching it meets, after that, the interior of the body). The discussions are mainly referring to the planar and the three-dimensional case, but outlooks to higher dimensions are given, too. The Borsuk problem is also considered in normed planes, and various open problems are posed.

Also the later book [433] of Boltyanski and Soifer treats the Borsuk and the Hadwiger problem, and extensions of Helly's theorem for convex bodies are studied, too. Great parts of [433] are repeated in the expanded version [2266]. And also the book [434] refers to the combinatorial geometry of convex bodies; it is an important forerunner of the wider monograph [431], treating several topics from there. More extensively summarized in [434] of Boltyanski and P. Soltan and in [431] of Boltyanski et al. are *families of Helly-type problems*, or questions related to Borsuk's partition and Hadwiger's covering problem. Also in view of a complete combinatorial study of *belt polytopes and belt bodies* (which are, as natural generalizations of zonotopes and zonoids, dense in the family of all convex bodies), the authors of [431] go much further than it is done in [425]. They consider *generalized convexity notions* (namely the so-called d-convexity, suitable for metric problems in normed spaces, and also the concept of H-convexity) to get generalized Helly-type results, namely studying the *Szökefalvi-Nagy problem regarding the Helly dimension of convex bodies*. (A forerunner regarding the metric notion of d-convexity, useful in normed spaces, is the booklet [2268] of P. Soltan; see also again [434].) And regarding coverings by smaller homothetical copies and the equivalent classical illumination problem, the authors extend their studies also to so-called *inner illumination problems*. Problems on *fixing and hindering systems* of convex bodies (i.e., minimal subsets in their boundaries sufficient to fix the bodies against translations) are close to illumination problems (see §44 of the book [431] and [429]).

Here we also mention the books [1160] and [1161] which reflect many results from the combinatorial geometry of convex bodies (like illumination problems, or results on polytopes). And the monograph [424] collects problems in similar directions, e.g., about *Helly-variants for translative classes of convex bodies* or *special covering problems*.

SURVEYS ON BORSUK'S AND HADWIGER'S COVERING PROBLEM

We cite now mainly later surveys on the three most important problems from the combinatorial geometry of convex bodies, discussed or mentioned above.

We start with *Borsuk's partition problem*: Important surveys on this topic are given by Grünbaum [1079] (containing several unsolved generalizations), Raigorodski [2015] (with sketches of proofs showing, e.g., more detailed successive solutions in 3-space via shrinking the diameters of the four parts, and analyzing the 946-dimensional counterexample), once more Raigorodski [2016] (giving a historical survey and high-dimensional counterexamples, general partial results also in the low-dimensional case), and Kalai [1341] (presenting many old questions and also new ones still interesting after the counterexamples have been found). A very recent exposition on the problem is [2575] due to Zong, giving a good overview also on related subjects. Using a new reformulation, the author presents a computer proof program for this famous partition problem. With respect to problem books or problem chapters of books, the Borsuk problem and variants of it are presented in Section D14 of Croft et al. [687] and in Chapter VIII of Boltyanski et al. [431].

Second, we treat once more the *Levi–Hadwiger covering problem* which is equivalent to the *illumination problem of Hadwiger and Boltyanski*: The expositions [312], [1716], [316], and [1714] give many modifications of the illumination concept that is usually known, and corresponding results are systematized and enriched by various open problems. This is also done in the more recent monograph [317] of K. Bezdek, also otherwise containing a broad spectrum of subjects in discrete geometry. The

most recent survey on the covering problem and its relatives is [321] by K. Bezdek and Khan; all significant papers on this unsettled problem (till 2018) can be obtained from there or from papers cited there. A longer survey is also contained in §34 of the monograph Boltyanski et al. [431]. Regarding problem books we mention Section D17 of [687], Chapter VIII of [431], and Section 3.3 of [492].

GENERALIZATIONS OF HELLY-TYPE PROBLEMS

Third, by citing now important surveys related to *Helly-type problems*, we refer also to the *Helly numbers* (or Helly dimension) mentioned already above. The first comprehensive survey in this direction is the famous and encyclopedic paper [708] by Danzer, Grünbaum, and Klee. It collects an impressively large variety of beautiful generalizations, modifications, and applications of Helly's famous theorem, posing also many related unsolved problems. The next important survey, concentrated on the strongly related *Radon theorem* (yielding a convenient proof of Helly's theorem and so-called *Radon partitions*), is Eckhoff's paper [789], taking into consideration also partitions into more subsets (Tverberg's theorem), or presenting research with respect to other spaces. In Reay's survey [2024], various problems inspired by Randon's theorem and their history are presented, for example: convex polytopes and primitive Radon partitions; relations to abstract convexity spaces. Interesting open problems are collected, too. Similarly, Sierksma [2245] collects results and problems around Helly's theorem for aligned spaces (i.e., convexity spaces with domain finiteness) and presents relations between Helly numbers, Radon numbers, Carathéodory numbers, and their generalizations. It is again Eckhoff [790] who presents a brilliant handbook survey of the material contained in several hundred papers on generalizations, modifications, and consequences of Helly-type theorems (which appeared after the excellent survey [708]). Generalizations can refer to different types of considered sets (e.g., star-shaped ones or spheres), or the property of "having in common" may change (for instance, refer to common transversals). The author evaluated more than 300 references (research papers, books, and surveys), and his excellent writing style captivates the reader! Further information about Helly-type theorems can be found in subsection 12.3.1 above, under the names of Helly and Radon.

The survey [224] of Bárány and Soberón is related to *Tverberg's theorem* generalizing Radon's partition theorem. It states that any $(n+1)(r-1)+1$ points in n-space can be divided into r disjoint subsets such that the convex hulls of these subsets all have a point in common. The authors survey the last 20 years before the appearance of [224] regarding advances of topological, linear algebraic and combinatorial aspects of Tverberg's theorem; they also discuss the history of this famous theorem and pose various open problems. Main points are topological and colorful versions, the structure of (universal) Tverberg partitions, and (far-reaching) applications.

The topological Tverberg conjecture is in the focus of the related exposition [218] of Bárány, Blagojević, and Ziegler. It reflects exciting recent developments in this direction, which nicely combines algebraic topology with combinatorial convexity. The authors describe how various consequences and extensions of Tverberg's topological theorem can be obtained when r is a prime power. They also give a surprising construction of counterexamples to the topological Tverberg conjecture for all $r > 5$ which are not prime powers.

Another direction, e.g., replacing the passage "having a point in common" in Helly's theorem by "met by a line" (then called a *line transversal*), yields the *geometric transversal theory* having Helly's theorem as a subcase with "point transversals". We start with Wenger's related survey [2449], giving an overview of this subfield. The author discusses *Helly-type theorems and generalizations to nonconvex sets, Hadwiger's transversal theorem*, and *Gallai type problems*. Also the space of transversals is studied, and interesting problems are posed.

Continuing Wenger's survey [2449], Holmes [1225] summarizes the grown knowledge about Helly-type results and transversal theory, considering also geometric permutations, new planar results (and possibilities to obtain higher dimensional results), and a nice list of open problems. Another interesting survey is [984] due to Goaoc, treating the space of line transversals, and convexity properties of the

family of line transversals of families of convex sets in a fixed order. Further on, geometric permutations are defined as opposite pairs of orderings of the members of a family of disjoint balls induced by an intersecting line. Results on bounds for *pinning, Hadwiger, and Helly numbers* of disjoint balls are surveyed. (Note that the pinning number is the minimum number k such that if a transversal to a set of disjoint balls is isolated in the space of transversals, then it is isolated by a subset of at most k balls; and the Hadwiger number is the minimum number k such that a family must have a transversal if every subsequence of at most k balls has an order-respecting transversal.) Also computational aspects are discussed.

Of course, also older basic surveys like Danzer et al. [708] and Eckhoff [789] contain material about common transversals.

The very recent exposition [18] by Adiprasito et al. encourages the study of Helly-type theorems without dimension, also taking care for colourful and fractional consequences. We notice that also many convexity books contain representations of Helly-type results. Good examples are the problem book Croft et al. [687] (see Sections E1–E4 there) and Matoušek's monograph [1732], whose Chapter 8 is completely dedicated to modern generalizations of Helly-type theorems (see also Section 1.3 for the classical part). There the *fractional Helly theorem, the colorful Carathéodory theorem*, and *Tverberg's theorem* are nicely presented.

In the very recent survey of Bergold et al. [290] famous theorems from convexity are combined with topological drawings (using graph theory) whose triangles are interpreted as convex sets, giving the used link to convexity. So the authors obtain *generalizations of Kirchberger's and Carathéodory's Theorem*, and they also derive results on arbitrarily large Helly numbers.

A further related research direction deals with the combination of Helly-type theorems and *generalized convexity notions* which influence the Helly numbers of families of convex sets; see, e.g., Section 9 of [708] by Danzer et al. and Chapters II and III of Boltyanski et al. [431]. Strongly related to Helly's theorem and holding for *star-shaped sets* is *Krasnosel'skii's theorem*. It says that a compact subset S of n-dimensional space is star-shaped if and only if every $n+1$ points of S can "see" a common point via S. Breen [499] surveys results on this theorem, where conditions implying the dimension of the convex kernel of S are essential. More results on Krasnosel'skii-type theorems and starshapedness are collected in Section 6 of the fundamental survey Hansen et al. [1157]. And we also refer to §15 of the book [431], where an extension to normed spaces is discussed.

Recently, Bárány's booklet [217] on combinatorial properties of convex sets, families of convex sets, and finite point sets related to convexity appeared. It refers to *classical Helly-type theorems*, but also to very modern related research directions, namely around *Tverberg's theorem, the colourful versions of Helly's and Carathéodory's theorem, and the* (p,q)*-theorem* of Alon and Kleitman. The treated problems are frequently combinatorial in nature, but the proofs use also ideas from geometry, suitably combined with the *theory of graphs and hypergraphs*. Thus, the booklet should be located between convex geometry, combinatorics, and discrete geometry. It can be used as a textbook and is suitable for undergraduates and graduates. But also researchers in the field will certainly benefit from it. As mathematical pre-requisites, one needs only basic linear algebra and some knowledge about (hyper-)graph theory.

12.4.1.4 Scissor's congruence (Hilbert's third problem)

It is well known that the third of Hilbert's famous 23 problems asks the following for two arbitrarily given polyhedra of equal volume: is it always possible to dissect the first into a finite number of polyhedral pieces which can be reassembled to get the second polyhedron? Hilbert conjectured that this is not always possible, and Dehn confirmed this within a year; as a counterexample showed, in general the answer is "no".

This problem is basic for the question whether the volume of tetrahedra/polyhedra can be derived purely elementary (for polygons in the plane, equal area and equi-decomposability are equivalent properties). Dehn answered this problem via necessary conditions for equi-decomposability which are not satisfied by special polyhedra of equal volume. This created a little research field on polyhedra of equal volumes which are equi-decomposable regarding some group of motions (see, e.g., the final remarks of Chapter 2 in Hadwiger's monograph [1134]). With his monograph [423], Boltyanski successfully completes and updates an earlier booklet from 1956 on this theme; see the Mathematical Reviews (MR0230198). In particular, he takes results of Sydler (on the *sufficiency of Dehn's conditions for equi-decomposability* in 3-space) and Jessen (affecting Sydler's results, being important also for higher dimensions) into consideration, and in a table he collects all information about the current state of Hilbert's problem and relatives. The book also contains a precise and exciting description of the history of the problem. Further on, the elementary methods of measuring areas and volumes of polygons and polyhedra are presented, together with the theorem of Bolyai–Gerwien with also non-Euclidean results. With results of Gerling and Bricard also dissections of symmetric polyhedra are considered, and many further ideas are nicely discussed. It should also be mentioned that [423] was filling a gap in [505]; in these proceedings, edited by Browder, a discussion of the third Hilbert problem is missing.

Also the section C20 of the problem book [687] of Croft et al. contains a nice related discussion on equi-decomposability. Further (more recent) material around Hilbert's third problem is discussed in our Subsection 12.3.1 above, under the name of Boltyanski there. Repeatedly we mention the more recent monograph [2087] of Sah, where also non-Euclidean concepts are presented. And we recommend the article [1356] by Kellerhals.

In the nice expository article [583] of Cartier one can find historical material on Hilbert's third problem, Sydler's proof, some homological comments, relations to hyperbolic and spherical analogues, and the results of Jessen and Thorup that two polytopes are equivalent by decomposition and translation iff they have the same invariants of Hadwiger. And Zakharevich [2518] studies scissors congruence along McMullen's polytope algebra. Also the books [2019] of Rajwade and [690] of Cromwell should be cited, for instance in view of historical aspects of Hilbert's third problem. And the book [734] of Devadoss and O'Rourke clearly demonstrates that scissors congruence is also an important topic for computational geometers.

12.4.1.5 Tilings

It is well known that a *tiling* (or *tessellation*) of a surface S, using a well-defined family of shapes (yielding so the notion of *tiles*), is a covering of S by tiles with no gaps and no overlaps, thus combining suitably the notions of packings and coverings. The same words are used for higher dimensional analogues, but also *honeycombs* and *space filling* are suitable designations. And there are also various non-Euclidean variants of tessellations. One usually distinguishes *periodic tilings* (with repeating patterns, say) and *aperiodic tilings* (lacking a repeating pattern). Citing monographs and important surveys, we try to give now some overview to the field.

Among monographs on tilings, the book [25] of Ahuja and Schachter is an early example. The authors try to give a general approach to the subject, discussing also *regular* and *Archimedean tilings, Voronoi tessellations*, and *Delaunay triangulations*. The most comprehensive book treating tilings is the comprehensive monograph [1088] Grünbaum and Shephard.

This monograph [1088] is the first systematic mathematical treatment of the very old and comprehensive topic of *plane tilings*! The authors summarize all the knowledge about them (and about the neighbouring field *patterns*, which is not discussed here). They succeed in *unifying the necessary structure of notions and the notation*, very useful also for related future research. In addition, they win every reader by keeping the artistic flavour and aesthetic enjoyment of the subject. This is also

secured by a *facsinatingly large variety of wonderful figures and helpful diagrams*. From the large set of topics covered in the book, we select the following ones for mentioning them here: *tilings with regular polygons or star polygons*, theoretical aspects of a general theory of plane tilings, classification problems (using transitivity properties and symmetry groups), "well-behaved" (or "normal") tilings, colored tilings and groups of colored symmetries, classification for prototiles of monohedral tilings (all tiles are congruent), aperiodic tilings (based on the idea of m prototiles that cannot possibly be arranged in a periodic way; the authors deliver the historically first detailed representation of this very interesting field), Wang tiles (with interesting applications, e.g., regarding the Turing machine), tilings with unusual types of tiles, tilings without the assumption that their tiles are topological disks, and tilings that cover the plane not only once. Many nice exercises of different levels of difficulty (including also *research problems*) complete the fascinating picture conveyed by this book which sometimes is called the "bible" of this partial field of geometry.

Montesinos [1839] studies the relations between *classical tessellations* (Euclidean, spherical, and hyperbolic ones) and *3-manifolds*, where the space of positions of these classical tessellations is studied. One of the main aims is to give the reader geometric insights into low-dimensional topology. This leads the reader to basic ideas on orientable closed 3-manifolds, in particular the Poincaré homology sphere. In this book, known material is presented in an unusual, very interesting way. The impressively rich illustrations (29 colored illustrations and about 200 line drawings) heighten the positive impression left by this book; it is clearly dedicated to geometry and intention.

It is well known that tilings play also an essential role in Escher's work. There is an extensive literature on this subject. From the mathematical point of view, the proceedings [681] edited by Coxeter et al. present one of the most important compilations on this theme. Prominent authors discuss the following topics there (we give only a small selection of themes, directly related to tilings): *colored symmetries and groups, combinatorial types and classifications of several special kinds of tilings, applications* (e.g., in crystallography), problems on tilings initiated by Escher, *isohedral tilings and their colourings, and periodic and also non-periodic tilings of Euclidean and hyperbolic planes*. We should also mention that in the third part of [681], various articles deal with *polyhedral geometry* (e.g., polyhedra with interesting algebraic and combinatorial properties which are closely related to the Platonic solids and the Kepler–Poinsot star polyhedra).

In his book [1595] from the series "Design Science Collection", Loeb tries to combine analysis and intuition with respect to plane and spatial figures, to study the processes "discovery through experiments" and "development of generalized concepts" by using the results of these experiments. Based on these concepts, mathematicians and designers might come closer. Topics covered by [1595] related to tilings are *tessellations and Diophantine equations, Dirichlet domains, Penrose tilings*, and finally also *crystallography*.

And we also cite Bonner's book [444] discussing the interplay between Islamic geometric patterns, design, and related techniques for creating such patterns and related tessellations. The main goal there is to offer a classification by *design methodology* (instead of using symmetry groups). This really nice book contains more than 600 color figures and photographs.

There are three important handbook surveys on tilings, the first two in the "Handbook of Convex Geometry", and the third one in the "Handbook of Discrete and Computational Geometry". In the second part of his survey [2137], Schmitt discusses different concepts of *finite tilings*, such as tilings by similar or congruent tiles, fault-free tilings of rectangles by (congruent) rectangular tiles, tilings of polygons by certain types of polygons, tilings of compact, convex sets by smaller tiles regarding diameter (with applications to the Borsuk problem), and regarding smaller homothets (leading to Hadwiger's related covering problem). Here "tilings" mean coverings as well. In his handbook survey [2201], Schulte discusses results on tilings mainly related to convex geometry. These refer to classifications of various (combinatorial) types of planar tilings, Euler's theorem for tilings, lattice tilings, tilings by regular polygons, monohedral tilings (by convex polygons in the plane, and by three-dimensional polyhedra as space-fillers), tilings of space by polyhedra of a single combinatorial type (monotypic

tilings), and an interesting discussion of *quasi-periodic tilings, quasi-crystals, and Penrose tiles*. The third survey [2128], prepared by Schattschneider and Senechal in the special format of the Handbook [1000] edited by Goodman et al., lists the main results and open problems on tilings in four sections. These refer to *general properties of tilings, monohedral tilings, and periodic and non-periodic tilings*. The list of references is selected in such a way that the reader has a good entry into the field.

The first part of the proceedings [1116], edited by Guy and Woodrow and dedicated to "Recreational Mathematics and its History", contains several contributions on the theory of tilings. The authors, among them being famous geometers like Coxeter and Grünbaum, study *connections between ornaments of architectural friezes and triangulations of polygons, polygon theory and tilings, tessellations as a tool in proving theorems of geometry, colourings of tessellations, and famous tilings of rectangles*.

SPECIAL TILINGS: VORONOI DIAGRAMS

Very popular and special types of tilings (in the plane, and with higher dimensional analogues) are the so-called *Voronoi diagrams* (or *Dirichlet tessellations*) and their duals, called *Delaunay triangulations*. These notions became particularly popular in *Computational Geometry*. We list now only some of the most important books and expositions in this direction.

Klein [1423] introduces the general concept of *abstract Voronoi diagrams*, based on the concept of *bisecting curves* instead of notions like sites and distances. In this context, *Voronoi regions* are introduced by intersecting halfplanes. This approach allows to unify the theory on Voronoi diagrams, and to study, e.g., as a subcase, Voronoi diagrams for general metrics. Efficient algorithms for constructing abstract Voronoi diagrams are given, and the booklet contains also an excellent brief historical part and an exposition of known approaches and algorithms.

In the book [1020] of Grima and Márques algorithms for computing *closest and farthest point Voronoĭ diagrams* and *triangulations on surfaces like cylinders, cones, spheres, and tori* are studied. Repeatedly, Voronoi regions are subject of comprehensive surveys, as it is the case with Edelsbrunner [794], Fortune [892], Sugihara [2325], and Aurenhammer and Klein [154]. These surveys reflect, in each case, the correspondingly recent state of knowledge regarding *Delaunay triangulations, Voronoi diagrams, and their generalizations*, in particular clarifying structural and geometric properties.

Two comprehensive monographs on Voronoi diagrams and related topics are Okabe et al. [1914] (second extended edition from 2000, and suitable for graduates and researchers) and [155] of Aurenhammer, Klein, and Lee from 2013, useful for students like also researchers. In the first book mainly generalizations of the Euclidean setting and applications are discussed. For example, Voronoi diagrams for several points at a time, called *higher order Voronoi diagrams*, are discussed, and *farthest-point Voronoi diagrams, kth nearest-point Voronoi diagrams, and Voronoi diagrams with obstacles* follow. Non-Euclidean distance concepts are used, and application fields (like *pattern analysis*) are studied. The other book quickly moves from the planar Euclidean setting to extensions regarding *types of sites, distance or influence functions, higher dimensions, and geodesic surface*. The considered distance functions can, e.g., be Minkowskian or even non-symmetric convex distance functions (gauges), and also abstract Voronoi diagrams are presented.

In view of *applications*, the post office problem (finding the point in a given set that is closest to a given point) is considered, and subgraphs of Delaunay triangulations are discussed, in view of their applicability (e.g., for minimum spanning trees). Contrarily, Delaunay supergraphs (like higher order Delaunay graphs, witness Delaunay graphs, geometric clustering, partitional clustering, and hierarchical clustering) are presented, too. And topics like *motion planning, retraction, and polygonal robots* are also considered. In Section 4 of their survey [1720], Martini and Swanepoel comprehensively discuss the geometry of *bisectors and Voronoi diagrams* in normed linear spaces.

Here we also mention the undergraduate book [734] of Devadoss and O'Rourke, standing in a typical way *between discrete geometry and computational geometry*. There also Voronoi regions and Delaunay triangulations are studied.

We finish this part by citing two books written as help for the preparation of school teachers for Mathematics. These are Sibley's book [2243] and an introductory book of Kerins et al., namely [1369]. The first book contains, distributed over two chapters, material about *quasi-crystals, tilings in general, and Voronoi diagrams*. And the second monograph gives an introduction to *non-periodic tilings in two dimensions and space-filling polyhedra*. Quasi-crystals are not studied, but the underlying mathematics useful for their study is presented. *Penrose tilings, the irrationality of the golden ratio, the connections between tessellations and packing problems, and Voronoi diagrams in 2 and 3 dimensions* (themes that are combined with precollege mathematics) are there.

And we mention that Gruber's monograph [1070] contains also (in its Section 32.1) a discussion of Voronoi diagrams and Delaunay triangulations, which is then (still under the natural headline "Tilings with Convex Polytopes") extended to space-fillers (Section 32.2) and concluded with a nice collection of famous problems and conjectures related to these topics.

12.4.1.6 Further books (and directly related expositions) treating discrete geometry

Guggenheimer's book [1102] from 1977 contains a section on discrete geometry, namely referring to *packings and coverings*.

A problem-oriented book, with sections independent of each other and containing interesting topics from convexity and discrete geometry, is Schoenberg's monograph [2187]. One can find there *lattice point problems*, an interesting section on *Helly-type theorems, applications of finite Fourier series* (e.g., isoperimetric problems for polygons), and *billiard problems* emphasizing special types of polygons and polyhedra. The book contains many interesting problems.

The booklet [814], consisting of 17 small and mostly independent chapters (mainly without or only with sketchy proofs) treats different aspects and applications of *lattice point problems* and refers only in parts to convexity. However, we mention here the following selected themes, all of them meant for lattice points and related to convexity and discrete geometry: *equidissectable and lattice polytopes, different types of packings and coverings, tilings and crystallography, Minkowski's fundamental theorem (from the geometry of numbers), successive minima, visibility*, and connections with *integral geometry*.

The monograph [651] of Conway and Sloane is one of the most important cornerstones regarding *ball packings in n-dimensional Euclidean space*. Related topics are the *kissing number problem* (asking how many balls can be arranged so that they all touch a central one), the problem of *covering the space by congruent balls of least density*, and the quantizing problem having applications in data compression. Since repeatedly lattice arrangements occur as optimal ones, the *classification of lattices and quadratic forms* is a further central subject of the book. This yields connections to *error-correcting codes, Steiner systems, t-designs*, and *finite groups*. Further on, applications in *number theory* and also relations to disciplines outside of mathematics are treated, e.g., to *channel coding, chemistry and crystallography, quasi-crystals*, and *physics* (dual theory and superstring theory). The book contains also additional material, such as tables otherwise hard-to-access; these refer to bounds for kissing numbers, the even and odd unimodular lattices in dimensions up to 24, and groups associated with the Leech lattice.

The problem book [687] of Croft et al., already mentioned several times, contains also many interesting and deep research problems from *discrete and combinatorial geometry of convex bodies*, mainly in the chapters B (Polygons and Polyhedra), C (Tilings and Dissections), D (Packing and Covering), and E (Combinatorial Geometry).

Only in its first part the book [1931] of Pach and Agarwal contains results related to convexity: *Minkowski's theorem from the geometry of numbers, Fáry's theorem about lattice packing and covering*

of convex discs, and properties of *polygonal approximations of convex sets* are studied. Further on, the density of packings (resp. coverings), with extremal properties, upper (resp. lower) bounds, and related constructions are treated, and finally a chapter shows the connection between *circle packings and planar graphs* (e.g., Koebe's theorem).

Zong's nice book [2567] collects basic special problems of convex and discrete geometry which, partially, have been solved, but also new developments are shown. The selection principle yields mainly "intuitive problems" (which are easy to state but hard to solve). And also, sometimes the solutions of settled problems have been unexpected and surprising. In this sense, *Borsuk's partition problem, Fejes Tóth's sausage conjecture (finite packings), translative space-fillers, kissing and blocking numbers (including the 13-spheres problem), several Baire category problems, the Busemann–Petty problem*, and *Dvoretzky's theorem* are deeply discussed, and (fitting to the selection principle) many proofs are done in an intuitive way.

In a concise and elegant form, the book [2568] of Zong covers many aspects of the field *sphere packings* (finite or not, lattice or not, multiple or not, related topics like kissing numbers, etc.), comprehensively introducing the reader into the whole area. More precisely, the author discusses the *Gregory–Newton problem* and *Kepler's conjecture, lattice sphere packings* and also *positive quadratic forms, lower bounds for the packing densities of spheres* and for the *blocking and kissing numbers of spheres*. Also analogous *lower bounds* are presented, *multiple and also finite sphere packings, holes in sphere packings*, and further topics. The author also presents short biographies of mathematicians important in the field. This book gives a nice view on the field, demonstrating the steps of development (also mainly documented by the following sequence of discussed monographs): L. Fejes Tóth [851], Rogers [2053], Conway and Sloane [651], Gruber and Lekkerkerker [1075], Erdős, Gruber and Hammer [814], and Martinet [1699]; for finite packings and coverings we add K. J. Böröczky [391]). It can be recommended for researchers, lecturers, and advanced students.

Here we cite also the nice and informative surveys [2569], [2572], and [2573], extending the subject of the book [2568] to *packings of centrally symmetric convex bodies* (defining a used norm, and combined with the useful concepts of deep holes and free planes; for the latter, see, in particular, [2569]), and for *covering numbers* as well as *kissing and blocking numbers* the paper [2572]. In this survey, the blocking number is considered as a relatively new concept, interesting for itself but also connecting the framework of kissing and covering numbers. The author presents the basic problems and the most important results about these three types of numbers and their relations to each other. In the expository paper [2573] classical and modern results on *packings, coverings, and tilings by convex figures* in the plane are comprehensively presented. Exact formulas for packing and covering densities for certain types of convex figures are established, and also results on (classifications of) tilings are studied. More complicated topics (like congruent packings and coverings, simultaneous packings and coverings, and multiple packings and coverings) follow, and also some problems and results on *finite packings and coverings* are described.

Meant as introduction into the field of discrete geometry for graduates, the stimulating monograph [1732] of Matoušek in the Czech way aims to study combinatorial properties of arrangements of (usually finitely many) geometric objects, like points, hyperplanes, or even convex sets. These properties are related to *incidence problems, arrangements of hyperplanes, Davenport–Schinzel sequences, extensions of the theorems of Helly–Radon–Carathéodory*, and so on. Also asymptotic results are presented, since they are important for estimating the complexity of geometric algorithms needed to study such properties of configurations of objects. With its richness, the book is also an excellent reference for researchers from neighbouring disciplines, such as *combinatorial optimization, graph theory, or computational geometry*. Within the large variety of themes presented by the author, the following ones are closely related to convexity: *foundations of convexity and polytopes* (e.g., Minkowski's lattice theorem on convex bodies, hyperplane arrangements, and Voronoi diagrams), *intersection patterns and transversals of convex sets* (e.g., variations and generalizations of Helly-type theorems, (p, q)-theorems), *polyhedral combinatorics* (face structure of polytopes, upper bound theorem), and problems of *embedding finite*

metric spaces into normed spaces (approximate embeddings, lower and upper bounds for embeddings). These and further theorems are confirmed by detailed proofs, and tools are carefully explained, particularly also the more advanced methods like *Gale diagrams*. At the end of each section, one finds notes describing (historical) origins and applications. Interesting open questions and nice exercises (indicated by their difficulty) are there. At the end of the book, short summaries of all chapters are given, and an extensive list of more than 500 references is contained.

In his third book [2571], Zong elegantly shows that already an object like the cube yields many deep and difficult problems which are able to combine different fields of mathematics (the reader should compare this passage here with our subsection 2.9.1.2). It is interesting for geometers, analysts, algebraists, and mathematicians from combinatorics and coding theory. First the author studies *extremal central k-dimensional sections* of n-dimensional cubes, in many cases of $k < n - 1$ open, and solved by Vaaler and Ball for $k = n - 1$. Then the same is done with orthogonal projections (also onto orthogonally complementary subspaces). The (generally unsettled) problem of the *maximum volume of a k-dimensional simplex contained in the n-dimensional unit cube* follows, and upper and lower estimates on the minimum number of elements of a *triangulation of the cube* are discussed. The author continues with *0/1-polytopes*, i.e., convex hulls of subsets of the vertex set of the unit cube. Besides relations to coding theory, a classification of 0/1-polytopes and upper estimates for numbers of k-dimensional faces are given. Then Minkowski's conjecture on *lattice twins* (in any lattice tiling of the unit cube, there are two cubes which meet facet-to-facet), its solution by Hajos, and a more general conjecture of Furtwängler referring to multiple lattice tilings are treated. Finally, the author presents *Keller's conjecture* saying that in any translative lattice tiling of the unit cube, there are two cubes meeting facet-to-facet. Keller and Perron confirmed this for $n < 7$, and the known counterexamples for $n > 7$ are discussed, too (see, for further results, again subsection 2.9.1.2). The author nicely demonstrates a wide variety of methods: e.g., analysis in the form of log concave measures and the Brascamp–Lieb inequality regarding cross-sections, hyperbolic geometry with respect to the triangulation problem, and graph theory in the case of Keller's conjecture. As a forerunner of this book, we also mention the interesting survey [2570], analogously showing the rich geometry of cubes and, for understanding it, the used machineries from and the applications in number and group theory, probability and matrix theory, coding and design theory, combinatorics, and non-Euclidean geometry.

The book [1699] of Martinet refers to subjects from the geometry of numbers and presents progress in this area. The author focuses on *extreme lattices*, i.e., lattices for which the density of the canonical sphere packing attains a local maximum. First basic *geometric properties of lattices in Euclidean spaces* are introduced. Differing from other books like [651] of Conway and Sloane, the author is *not restricted to integral lattices*. He treats the *geometric inequalities* and lattice constants of Hermite, Minkowski, Mordell, Rankin, and Bergé-Martinet, continuing with an elegant account of *Voronoi's theorem on extreme lattices* (yielding a local maximum of the sphere packing density). Also *Voronoi's algorithm for perfect lattices* (resp. perfect quadratic forms) is given. Root lattices, their relatives, and related classifications are treated, and in the final parts more recent results are presented. These contain *Hermitian structures on lattices*, the classification of *eutectic lattices* (up to dimension 4), extremality for families of lattices, and also extensions of Voronoi's algorithm; four appendices give the needed algebraic and arithmetic basics. The book benefits from many nice exercises and notes (at the end of each chapter) on the origins of notions and results. In the extended edition a new appendix discusses an interesting recent development (largely due to Venkov) on *relations between extreme lattices, spherical designs, and modular forms*. Further changes in the new version are also available at the author's web page.

A basic contribution to many fundamental developments in convexity is the monograph [1070] of Gruber. Its contents are already described in our subsection 12.3.1; this holds also for the great parts treating topics from *discrete geometry*.

Also the book [317] of K. Bezdek collects material about interesting problems from discrete geometry and having (stronger or weaker) connections to convexity: *densest sphere packings, covering*

convex bodies by the smallest possible number of smaller homothets, coverings by planks and cylinders (in particular, the affinely invariant version of Tarski's plank problem), *volumes of finite arrangements of balls and the Kneser–Poulsen conjecture* (stating that if the centres of one arrangement of m equal balls are no farther apart than the corresponding centers of another arrangement, then the total volume of the first arrangement is not larger than that of the second), and *ball-polyhedra* (which are, roughly speaking, the intersection of finitely many unit balls having also relations to bodies of constant width; see also [1478] and Chapter 6 of the monograph [1713]). The book contains selected proofs and more than 40 research problems; thus one can use it also as small textbook and source of problems, suitable for advanced undergraduates, graduates, and researchers.

Similar to the contents of [317], the book [318] of the same author contains chapters covering *packings of balls, contractions of finite ball arrangements* (also the Kneser–Poulsen conjecture), *spindle convexity and ball-polyhedra* (their facial structure, the analogue of the Euler–Poincaré formula, and illumination of them; see again also Chapter 6 of [1713]), and *coverings by cylinders* (e.g., plank problems). At some places, extensions to *spherical or hyperbolic spaces* are considered.

In the very recent book [323] K. Bezdek and Lángi collect several research problems from discrete geometry where the *volume* plays a significant role, thus proposing the name "volumetric discrete geometry" for this little subfield. First *isoperimetric type inequalities for special types of polytopes* are studied; e.g., finding the n-dimensional polytopes of minimum/maximum volume circumscribed about/inscribed in the unit ball. Then inequalities treating the volume of the convex hull of a pair of convex bodies follow, close to classical results of Rogers and Shephard from the 1950s. Variants of these problems are given. Again the *Kneser–Poulsen conjecture* and related questions are discussed, and (as a generalization of the well-known Hadwiger number of a convex body) the so-called *contact number of a finite translative packing* of this body is introduced and studied. As subfamily of translative packings, the so-called *totally separable translative packings* are investigated. In this case any two packing elements can be separated by a hyperplane disjoint from their interiors. Here relations to *Oler's inequality* from the geometry of numbers and to the geometry of *normed planes* are discovered. Various open questions and also research problems are posed, and in the second part of the book some of the theorems presented in the first are proved.

With the monograph [391] of K. J. Böröczky we now come to *finite packings and coverings*. It is the first book completely dedicated to this subject. The tools of this subfield of discrete geometry come from classical convexity; these are, e.g., *intrinsic volumes and mixed volumes*. These methods yield a unified theory of *finite and infinite packings*. But staying with finite packings, at least for them the author gives a satisfying representation of what in this case "density" means, if no container is given (as also in the sausage conjecture of L. Fejes Tóth). A suitable volume-based density measure (called *parametric density* and making also higher dimensional results possible) is developed, and the way of ideas leading to this notion is long and combined with excellent contributors (like Thue, L. Fejes Tóth, Rogers, Groemer, Oler, Wills and his school in Siegen, and the author himself). First *planar packings (and coverings)* are studied, going then to higher dimensions and to *large ball packings of lattice and periodic type*; their global shape is very essential. And here also the *Wulff shape* comes in, with relations to physics. The book shows also that the theory of *finite coverings* is not so good developed since an analogue of the notion of parametric density is missing. Further topics and tools discussed in the book are *finite planar packings of minimal area, perimeter or diameter, packings and coverings on the sphere and in hyperbolic space, Newton- and Hadwiger-numbers, the famous covering problem by smaller homothetical copies, and sausage problems*.

Finite packings and coverings were also deeply investigated from the school of Wills in Siegen; important contributions and surveys, fundamental also for the book [391], come from there. For example, we mention the comprehensive handbook article [1033] of Gritzmann and Wills (with special emphasis on participating convex bodies), where for *sausage packing and bin packing problems* convex bodies are packed into a given convex body of smallest possible volume, this body coming from a certain family of convex bodies. *Sausage covering problems* are also considered, and also

the algorithmic side is discussed. In [2464], Wills surveys results on finite packings, especially the correlation between parametric density and *ideal crystals*, and in the exposition [1185] of Henk and Wills the emphasis is on the role of parametric densities for the unification of the theory of finite (free) packings and infinite packings; see also the nice representation [2465] of Wills. Clearly, results on finite packings are also related to *containment problems*. The recent and more applied book [2131] of Scheithauer refers to this field from the viewpoint of optimization.

Schürmann's book [2194] treats the *geometry of positive definite quadratic forms*, i.e., it deals with *periodic lattice packing problems and covering problems of spheres*. It is an important reference on *Voronoi's reduction theories* applied to *computational approaches to packing and covering problems of spheres*. The book is close to the geometry of numbers and to computational geometry. More precisely, the author presents *generalizations and applications of Voronoi's first and second reduction theory*. His investigations yield theoretical and algorithmic tools for studying periodic lattice problems. For example, it is shown that strongly eutectic and perfect lattices cannot be locally improved to a denser periodic set. Voronoi's second reduction theory is based on *Delaunay polyhedra and Delaunay subdivisions*. These topics are generalized to the concept of *m-periodic point sets*, i.e., unions of finitely many translates of a lattice. The last chapter refers to the local analysis of coverings, i.e., to the *lattice sphere covering problem* and the *lattice packing–covering problem*. The author studies local optimality of lattices getting new optimal covering and packing–covering lattices in small dimensions.

12.4.1.7 Handbooks and proceedings related to discrete geometry

Handbooks and proceedings containing also important articles on topics from discrete geometry close to convexity (like the Hungarian series "Intuitive Geometry") are cited already above, under 12.3.4 and 12.3.5. We repeat here shortly only a few of them, and a few new ones are added. Besides all this, one should note that additionally there are many Conference Proceedings dedicated to computational geometry in a related sense, numbered in series like "International Symposium on Computational Geometry". It is impossible to cite all of them here. And also they usually do not contain many articles close to convexity. Besides these proceedings, we mention also the symposia on *Discrete Algorithms* and on *Discrete Geometry for Computer Imagery*, which regularly create proceedings containing also interesting material on the discrete geometry of convex bodies.

HANDBOOKS

We start with the "Handbook of Convexity" [1077]. Two of its five parts (namely Part 2 and Part 3, treating *combinatorial and discrete aspects of convexity*) are directly related to discrete geometry! We do not analyze them now in a deeper sense, since their single surveys are themselves cited in our book.

Second, we mention the (third edition of) the "Handbook of Discrete and Computational Geometry" [1000]. Topics close to convexity and presented there by surveys are *packings & coverings, tilings, Helly-type theorems, polygons, discrete aspects of stochastic geometry, different aspects of polytopes* (also lattice polytopes), *convex hull computations, Voronoi diagrams and Delaunay triangulations, visibility questions, computational convexity, crystals and quasi-crystals, rigidity*, and also different kinds of *triangulations*.

The next is the "Handbook of Computational Geometry" [2084] collected by Sack and Urrutia. It contains expositions on *arrangements of hyperplanes and their applications, Voronoi diagrams, polygon decomposition*, and problems on *visibility, art gallery, and illumination problems*.

PROCEEDINGS

The proceedings [603], edited by Chazelle et al., contain such surveys on *polyhedra and polytopes, packings & coverings, visibility and illumination problems, tilings* (also aperiodic ones), and *transversal theory*.

Further on, in the proceedings [1001], edited by Goodman et al., one can find topics like *lattice polytopes, arrangements of lines, polygon theory, computational complexity of convex bodies, transversal theory, unfolding polyhedra, triangulations* as well as *kissing, blocking, and covering numbers* of convex bodies.

The anthology [1002], edited by Goodman et al., contains articles on *transversals, shellings of polytopes, edge-antipodal polytopes, inequalities for zonotopes, visibility, and covering problems*. Also the *Schönhardt polyhedron* as well as *Hadwiger numbers of direct products of convex bodies* are presented.

Finally we still mention here the proceedings [320] edited by Bezdek et al.; they contain articles on *Minkowski geometry, tilings, polytopes, the strong dodecahedral conjecture, sphere packings, and kissing numbers*, completed by *colourful generalizations of a Carathéodory-type theorem, symmetries in polyhedral computations*, and an interesting problem collection.

12.4.1.8 Important further surveys on discrete geometry

We now cite expositions important for the wide subfield called *packings and coverings*; see also the subsections 12.4.1.6 and 2.9.1.2.

SURVEYS ON PACKINGS AND COVERINGS

The first comprehensive survey in this direction after the books [851] and [2053] is [837] due to G. Fejes Tóth. Important topics till 1982 covered in its first part are *densest packing and thinnest covering of spheres in Euclidean space, packings of balls in spherical space, packing and covering in hyperbolic geometry, bounds for lattice packing and lattice covering* (also transparent and separable packings), and *multiple packings and coverings*. In the second part, the author treats *packing and covering by convex sets* and also *permeability of layers of convex sets*. Then problems connected with the number of neighbours in a packing of convex bodies and *packings of congruent convex bodies with maximal number of neighbours* are discussed. The bibliography (roughly 300 items) reflects a fast development of the field.

The chronologically next survey on packing and covering is [841] by L. Fejes Tóth and Wl. Kuperberg, where particular attention is given to the two-dimensional case. A good selection of important theoretical results on general convex bodies and centrally symmetric ones is discussed. The authors present results on *packing or covering by congruent copies of convex sets*, by *translative copies*, and by *copies centered at the points of a lattice*. Results on polygons are exposed, and multiple packings and coverings are studied, too. Finally also *packings with restriction regarding the number of neighbours* are treated.

We continue with the excellent handbook article [842] by G. Fejes Tóth and Wl. Kuperberg, in general referring to *arrangements of convex bodies*. First, *density bounds* in Euclidean n-space (Minkowski–Hlawka theorem and its refinements, results of Rogers about lattice coverings, density bounds for different types of packings and coverings with equal balls) are discussed. *Blichfeldt's method, Dirichlet cells and Delaunay triangulations, and lattice and multiple arrangements* are further discussed topics, and *spherical geometry* is also taken care of. Regarding *hyperbolic space*, the remained difficulties in defining the notion of density relative to the whole space are discussed. After further related results on density bounds, additional concepts are shown. For example, these are density bounds for arrangements with restricted properties, e.g., *separable or compact packings, m-neighbour*

packings, and Newton numbers (maximum number of congruent copies of a convex body which can touch this body without mutual overlaps and related topics).

This important survey is continued and updated by the exposition [838] of G. Fejes Tóth, in which also topics like *finite packings and coverings, container problems*, and (breakthrough results on) *sausage conjectures* are discussed.

And the exposition [839] by the same author gives an overview of the theory of *coverings with convex bodies*. It is visible that, compared with packings, this particular field is developed less extensively. Packing problems are very popular since they are combined with the name of Kepler and with results due to Lagrange and Gauss, treating quadratic forms; covering problems came later and do not have so many connections to other fields as packings have. So the geometry of numbers and possible applications in crystallography made packings more attractive; and also it turned out that covering problems are generally more difficult than the analogous packing problems. But this disadvantage of covering problems almost disappeared in the 1950s, and the above surveys show also this phenomenon.

We continue with the survey [315] by K. Bezdek. The author surveys (often surprising) results on *packings of congruent balls in n-dimensional spaces of constant curvature*. Among the topics discussed there are *Hadwiger numbers, touching numbers and Newton numbers of convex bodies, contact graphs of finite packings and the combinatorial Kepler conjecture, isoperimetric problems for Voronoi cells*, and also *the strong dodecahedral conjecture*. Furthermore, *the strong Kepler conjecture* and *estimates for densities of ball packings in higher dimensions* are discussed.

Berger's beautiful exposition [285] collects many problems related to arrangements of circles or disks. The survey [1073] of Gruber involves an overview to important developments around *lattice packings and coverings of balls and convex bodies*; second, results on the *uniqueness of densest lattice packings and thinnest lattice coverings of a generic convex body* are studied. Besides this, *local minima on the lattice covering density of balls and bounds for kissing numbers of densest lattice packings of generic convex bodies* are discussed, and it should be noticed that the author lays emphasis on results in arbitrary dimensions.

ADDITIONAL EXPOSITIONS

In the nice survey [1184] of Henk and Wills, the authors expose the knowledge about *Minkowski's successive minima of centrally symmetric convex bodies*. Based also on Minkowski's related first and second theorems and generalisations thereof, results connected with *intrinsic volumes, the lattice point enumerator, Ehrhart polynomials, and simultaneous Diophantine approximations* are discussed.

In the exposition [1666] of Maehara and Martini many basic and also intuitive results on the *geometry of integral lattices* are presented, most of them with new and streamlined proofs. Some of the results are also new. The authors take special care of *lattice polygons and lattice simplices*.

We mention also Seidel's nice exposition [2216] on problems from *discrete geometry in non-Euclidean spaces*. The topics studied in [2216] have common background, although they live in Euclidean, spherical, elliptic, and hyperbolic geometries. Some of the considered problems are related to *semiregular polytopes, the integral lattices of Witt and Leech*, and *finite groups*. In the chapter about hyperbolic geometry also *volumes of simplices* are treated, and the author lays emphasis on

demonstrating also the wide variety of used methods and tools (e.g., these are coming from group theory, harmonic analysis, topology, number theory, and computation).

Yaglom [2486] presents an interesting evaluation of combinatorial and discrete geometry within the evolution of mathematics, emphasizing also computer-aided solutions of important related problems.

12.4.1.9 Selected monographs from computational geometry

Although we have no extra subsection referring to the neighbouring field of *computational geometry*, we would like (at the end of this subsection) at least to mention some books which are important for its history or in closer connection to discrete geometry.

The first two monographs creating a breakthrough of basic ideas of this new field were [1994] by Preparata and Shamos, and [793] by Edelsbrunner. Dealing with the analysis and design of efficient algorithmical approaches to geometric problems which have a computational nature, the authors first clarify the needed fundaments from geometry, the theory of algorithms, and data structures. Two sides of the studied geometric problems are somehow *close to convexity*: they are intuitively rich, and they have promising applications in fields like *computer graphics, computer-aided geometric design, robotics, linear programming*, etc.; among these geometric problems, the following ones touch our interests here: computing the *convex hull of a point set in n-dimensional space*, studying *arrangements of hyperplanes and zonotopes, intersection problems for polygons, half-spaces, and polytopes*, and constructing *largest convex subsets, Voronoi diagrams, and Delaunay triangulations*. Also visibility questions and geometric transformations are presented. An important forerunner of both monographs was the book [1762] of Mehlhorn, in which several typical geometric problems of such types occur for the first time in book form.

A well-known topic from computational geometry is that of art gallery problems, strongly related also to the geometry of *star-shaped sets and Krasnosel'skii's theorem* (see the survey [1157] of Hansen et al.). In its original version, the *art gallery problem* asks for sets of points (= guards, static or moving) such that their union oversees the whole interior and boundary of a given star-shaped or even more general polygon. The book [1921] of O'Rourke discusses this problem and various extensions of it, leading to a nice representation of visibility, decomposition, and covering problems related to polygons. Also triangulation algorithms are given, and analogous investigations regarding three-dimensional polytopes are done (showing that various results are not extendable this way).

In a second book (see [1922]), O'Rourke presents many typical techniques used in computational geometry, combined with popular geometric problems and configurations, among them being also *polygon triangulation, convex hulls, Voronoĭ diagrams, arrangements of hyperplanes, intersection problems, and motion planning*. The author performs also interesting *randomized algorithms*, e.g., with respect to triangulations and convex hull in three dimensions. With many nice exercises, the book is suited for a successful course in computational geometry.

Boissonnat and Yvinec [419] discuss deterministic tools, random sampling, and (dynamic) randomized algorithms with respect to the geometric concepts of convex hulls, triangulations (also three-dimensional), arrangements of hyperplanes, Voronoi diagrams, and further ones. We mention here also once more the book [734] of Devadoss and O'Rourke which (besides topics like convex hulls, triangulations, and Voronoi diagrams) also takes more recent subjects into consideration, e.g., *pseudotriangulations, curve reconstruction, and locked chains*.

And also the beautiful monograph [280] of de Berg et al. is excellently suited for courses about computational geometry. It is combined with the website http://www.cs.ruu.nl/geobook/, where a lot of further material can be found. Topics treated here and close to convexity are *polygon triangulations* and applications, *art-gallery problems, linear programming, Voronoĭ diagrams and Delaunay triangulations, convex hulls, motion planning, and visibility*. Each chapter contains respective historical notes, useful exercises and problems, and a comprehensive list of 348 references is given which shows the main developments in the partial fields discussed in that book.

Har-Perled [1158] is a basic work on *geometric approximation algorithms* in the spirit of discrete and computational geometry; it has some overlap with [280] but is more theoretic in nature (e.g., knowledge of properties of Voronoi diagrams is assumed). Such algorithms can be found in many practical applications and are often fast. Topics close to convexity that are treated explicitly in this book are randomized algorithm for *linear programming in low dimensions*, an approach to the *smallest enclosing ball problem, approximating a convex body by an ellipsoid and approximating the minimum volume bounding box of a point set, geometric set cover problems* (such as interpreting the art gallery as a problem of covering an arbitrary polygon with polygonal visibility regions), and some basic *polyhedral and duality theory*. The book is nicely illustrated by about 200 well-chosen and colored figures; interesting examples and exercises as well as biographical notes at the end of each chapter ensure very good readability and suitability as a textbook as well as a research monograph. Thus, this book can be used by students, researchers, and engineers.

Joswig and Theobald [1328] wrote an interesting introduction to computational geometry (in the first seven chapters dealing with *polytopes and polyhedra, linear programming, convex hulls, Voronoi diagrams, and Delaunay triangulations*) and to computational algebraic geometry (in the last three chapters, calling this part *nonlinear computational geometry*). Topics of the second part are *Gröbner bases and solutions of systems of polynomial equations*, with applications in *computer graphics and robotics*. The authors use methods from discrete and elementary algebraic geometry, optimization, and numerical mathematics to present algorithms in polyhedral and algebraic geometry. Thus, the book gives mainly an introduction into "linear and nonlinear" computational geometry for advanced undergraduates in mathematics and computer sciences.

12.4.2 Polytopes, polyhedral sets, and related geometric figures

The notion of *convex polytopes* defines or yields

- a proper subset of the family of all convex bodies (but creating an own conceptual structure),

- in parts a subfield of discrete geometry, also strongly related to combinatorics and computational geometry,

- an aid or tool for many related and/or applied directions (like *approximation theory, linear programming, oriented matroids*, ...).

All this we will take into consideration; in the sense of these viewpoints, we will mainly have convexity in mind. The emphasized proximity to convex geometry is also a reason for not following the viewpoint that polytope theory is a part of discrete geometry. We treat both fields in different subsections. Furthermore, in our book here polytopes occur also in Chapter 3 and Sections 2.9, 4.4, 4.7, 10.2, and 10.6. Thus, especially here we underline that publications already cited in other parts of our book are here very seldomly cited again. Thus, one should not wonder why, for example, such important books like Coxeter's famous monograph [670] are not described here: this book is already commented in our Section 2.9! And the two famous books of L. Fejes Tóth are described in our part on discrete geometry.

Hence there are many overlaps of polytope theory with subfields and notions of convexity already treated. Examples are *zonotopes* (meant as polyhedral *projection bodies* in geometric tomography, with their correspondence to *arrangements of hyperplanes and oriented matroids*), or as *polytopes with high degrees of symmetry*. Or the important role that classes of polytopes with regularity properties (e.g., semiregular and uniform polytopes) play regarding our sense of symmetry in science. The field,

with all its natural extensions (e.g., to *polyhedral sets, star-shaped polytopes, polyhedral manifolds*, etc.) and applications (in *linear programming, topology, approximation theory*, and so on) has long been subject of rapid development. Its resulting diversity gives an additional reason not to present it under 12.4.1 (although many mathematicians would perhaps see it like this). Further references related to polytopes can be found in other parts of 12.4 and the above-mentioned sections. As in the other parts of the book, the abundance of existing literature dictates that we are able to cite only monographs, proceedings, and surveys (also from handbooks), and we do this predominantly (but not only) in chronological order.

12.4.2.1 Monographs on polytopes

In the following we cite different types of monographs treating polytopes. Due to the ancient Greeks and names like Kepler and Schläfli, the theory of polytopes with *symmetry properties* is somehow older than the theory of their *general properties* (e.g., topological or combinatorial ones). Therefore we begin with books treating the first type of properties. Of course there are also monographs containing both aspects. Then mostly the dominant part decides the assignment.

BOOKS ON POLYTOPES WITH SYMMETRY PROPERTIES

For the following list of books we refer also to our subsection 2.9.1.4.

Basic results of Schläfli to the theory of polytopes are summarized in his book [2135], and about his interesting life one can read in [1357]. His most famous contribution to geometry is the *discovery of the higher dimensional regular polytopes*, almost 50 years before the appearance of [2135] (here we also refer to [670] and to §4 of the nice survey [2059], where also an elementary derivation of the n-dimensional regular polytopes is given). Further on, his representation of *Euler's relation for convex polytopes of higher dimensions* came to be in 1852 and appeared therefore 50 years later, too. And also his studies of (higher dimensional and projective) *arrangements of hyperplanes*, following Steiner's ideas, should be mentioned here; see also Chapter 18 of Grünbaum's book [1085]. Shortly after the appearance of [2135], Schoute published his book [2189], containing a comprehensive treatment of the *regular polytopes*. Schlegel diagrams of the regular 4-polytopes are given (see also §23 of the popular book [1197] of Hilbert and Cohn-Vossen), and the two *Schläfli–Hess polytopes* (i.e., two of the *four-dimensional regular star polytopes*) found by Schläfli himself are presented, too. Unfortunately, the other eight Schläfli–Hess polytopes (found by Hess) are not given there, although they were found already in 1885. It should be mentioned that later Schoute also published a series of suggestive papers about interesting sections and projections of the four-dimensional regular polytopes, and Boole Stott [449] continued with similar constructions of *four-dimensional semiregular polytopes*. We refer to the survey [2059] of Rosenfeld and Yaglom, where a nice, elementary derivation of the n-dimensional regular polytopes is given. And an excellent (historical) summary of ways to the *higher dimensional analogues of Platonic solids, Archimedean solids, Kepler's infinite families of prisms and antiprisms, the Kepler–Poinsot polyhedra* and others, connected with names like Schoute, Barrau, Mulder, Levi van Oss, Boole Stott, Elte, and Wythoff, is the paper [672] of Coxeter. In the two surveys [1709] and [1711] Martini discusses also many such polytope classes, obtained from regular polytopes by systematic weakenings of the corresponding definitions.

A comprehensive, early, very basic, and nicely illustrated reference on the *theory of polygons and polyhedra in 2- and 3-space* is Brückner's monograph [507], containing also a lot of *historical material* and beautiful figures. Besides the restriction to dimensions not higher than 3, the author does also not use group-theoretic methods. He starts with discussing general polygons, where the development of clear notions by Girard (1626) and Meister (1770) and the construction of a general theory due to Wolf (1847) and Wiener (1864) are presented. Then *special types of polygons, like regular, semiregular, and constructible ones*, are studied. Here the contributions of Poinsot (1810) and Hess (1874) as

well as the 65.537-gon of Hermes should be emphasized. (Namely, Hermes found a compass-and-straightedge construction of the regular m-gon with $m = 65.537$, the largest known Fermat prime number. Its description (more than 200 large pages) can be seen in the mathematical library of the Göttingen university—a mathematical curiosity.) Regarding three-dimensional polyhedra, the author discusses *Euler's theorem with analogues for non-convex generalizations* (like *one-sided polyhedra*), and similar other equalities, e.g., in terms of k-gonal faces, are also derived. Then *Archimedean, isogonal, and isohedral polyhedra* are investigated, including *Kepler's regular star polyhedra* and their "followers" due to Poinsot, Cauchy, and Hess. Of great importance are the beautiful 12 picture boards in this book, where examples of the discussed polyhedra are shown. (They come from the author's impressive collection of models!) Many of the shown polyhedra find here their historically first pictorial representation.

A next essential book on polytopes is Schoute's monograph [2189] discussed already above. It contains a nice part on *regular polytopes and regular star polytopes*, and with the help of Schlegel diagrams a wonderful visual approach to types of such polytopes is given.

In the Chapters VII till X of Sommerville's well-known book [2281], many properties of *n-dimensional convex polytopes* are presented. For example, intersections with hyperplanes are described, and *calculations of volumes, surface areas, and angle measures* are given. *Euler's extended theorem* is studied, as well as the concept of *regular polytopes of n-space*. This latter concept was then expanded by Coxeter in his famous book [670], also developing notation and important notions for polytopes in general.

We add here a remark on the survey [141] by Ashkinuze, cited already above: besides some general theory of polyhedra, it contains *all Platonic, Archimedean, and dual-Archimedean (i.e., Catalan) polyhedra* as well as *all Kepler–Poinsot star polyhedra*, with corresponding figures and explanations on how to derive them.

After a long period in the first half of the twentieth century, mainly the strong development of *linear programming* (combined with the rediscovery of *neighbourly polytopes* by Gale in 1955, and also with Motzkin's *upper bound conjecture* posed in 1957) started to revitalize the *combinatorial theory of convex polytopes*, and so Grünbaum was inspired to write his excellent monograph [1085]. He has done a very extensive, laborious job to collect painstakingly all the related literature, going back also into the nineteenth century and further. Besides new insights, like Perles' new *Gale diagrams*, also old results like the *theorems of Euler, Eberhard, and Steinitz* are nicely presented. In connection with analogues of Euler's theorem, also the *Dehn–Sommerville equations* (expressing the linear relations between the number of faces of different dimensions of simplicial polytopes) are discussed. Further basic topics treated in this book are *neighbourly polytopes, extremal problems regarding the number of faces* (here also the *upper bound conjecture of Motzkin), properties of boundary complexes* (e.g., n-connectedness of n-polytopes), *angle-sums relations and the Steiner point, addition and decomposition* (Minkowski addition and Blaschke addition), *paths on polytopes* (including diameters), and concluding remarks (e.g., on *regular polytopes and antipodality types*). Carefully selected problems at the end of each chapter complete this nicely written book; in many cases they help to extend the viewpoints, like also richness of historical and generalizing remarks. The book became very influential, and this effect can be easily measured by the two very extensive surveys [1087] and [1081] of Grünbaum and Shephard, which appeared shortly after [1085]. The book fixed also a lot of notation and many notions in the field. Chapter 18 of [1085] is dedicated to Steiner's space partitions by discrete or continuous families of hyperplanes, shortly called *arrangements of hyperplanes*. This subject got its natural continuation (after [1085]) by Grünbaum's nice booklet [1083]. In this work also *arrangements of pseudolines* are considered, i.e., finite families of not necessarily straight "lines" which still have the property that any two of them have one common point and cross each other there. This opens some new doors, e.g., showing ways to known *geometric configurations*. And the author studies also *spreads*, being continuous families of curves which behave like pseudolines. Coming back to [1085], we have to mention its second edition (from 2003). This extended version shows impressively how the whole

story went on. The additional new material (altogether 74 further pages) was prepared by V. Kaibel, V. Klee, and G. M. Ziegler. It contains, for each chapter, a supplied section reflecting the important developments since then, and is enriched by additional references. Thus, the reader can easily combine the original text with the additional material about important subsequent developments.

Analyzing Johnson's enumeration of *convex, regular-faced polyhedra in 3-space* given in [1320], Zalgaller proved in his booklet [2519] the completeness of this list. It turns out that besides the 5 Platonic solids, the 13 Archimedean solids, and Kepler's infinite families of uniform prisms and antiprisms, there are precisely 92 types of such polyhedra. A recommendable exposition, giving a complete set of photographs of corresponding models, drawings of nets, and convenient tables, is Berman [292]. Calling a regular-faced polyhedron P *simple* if one cannot divide it into two regular-faced polyhedra by a plane intersecting the surface of P along its edges, Gurin and Zalgaller expose in [1114] an interesting classification of *simple regular-faced polyhedra*. In the second part of [1114], *tessellations of 3-space into regular-faced polyhedra* are investigated. Classical and related work due to Delaunay, Voronoi, and others is presented, and some later contributions of varied types of polyhedra (e.g., *polyhedra with regular vertex figures* and their generalizations) are discussed, like also higher dimensional classes. Here we should once more mention the three books [2450], [2451], and [2452] of Wenninger. They describe constructions of highly symmetric polyhedra, like the *uniform and isohedral polyhedra* with regular vertex figures and faces that can be star polygons, related decompositions of the 2-sphere and geodesic domes and honeycombs.

Roman wrote the nice booklet [2057] on the geometry of special three-dimensional convex polyhedra, their constructions, nets, and applications in crystallography. Besides a comprehensive representation of properties of the *Platonic and Archimedean solids*, also the *Catalan solids* are constructed. In addition, the author discusses the Kepler–Poinsot star polyhedra and an interesting collection of tasks and problems (with interesting hints) closes the book.

The book [2447] of Wells consists of two parts which, in ordinary 3-space, refer to *three-dimensional nets* (i.e., discrete graphs, finite or infinite) and three-dimensional polyhedra (meaning a *polyhedral surface, again finite or infinite*). Mainly uniform nets of type (m, p) are studied, which means the shortest circuit containing any two edges through any vertex has the same number m of edges, and each vertex valence is p. Analogously, a polyhedral surface is uniform of type (m, p) if each polygon composing it has the same number m of edges, and again each vertex valence is p. The author proposes these *uniform three-dimensional nets and uniform three-dimensional polyhedra* to be generalizations of the Platonic polyhedra and regular tessellations of the plane. Within this framework (which should be compared with that of Grünbaum in [1084], going much further regarding notions and mathematical depth) the author studies some special kinds of uniform nets, and he investigates those nets whose symmetry group contains a group of translations, under which the vertices of the net fall into few orbits. This interesting book is possibly written more for *crystallographers* than for mathematicians; it is nice that the author gives a systematic description of the kinds of configurations that arise this way in crystal structures.

Adam and Wyss [15] present in an elementary, beautiful way various famous *polyhedral forms with high symmetry*, namely regular and semiregular polyhedra and their relatives and extensions, such as *compound polyhedra* (suitable unions of such a solid and its polar), and also *stellated polyhedra*. The authors try to show complicated structures more easily; e.g., the compound figures occur on loose transparencies, thus better showing the overlaps. Constructive approaches are emphasized, colored figures are used, and the whole book can be recommended for teachers and students.

Robertson [2039] investigates special types of n-dimensional convex polytopes which are defined as follows: Two combinatorially equivalent polytopes P_1, P_2 are called *symmetrically equivalent* if they have the same symmetry group and if this group operates equally on the facet set of P_1 and on the facet set of P_2. To consider *similar polytopes* as the same objects, the author constructs the space of all symmetry classes of convex polytopes using the Hausdorff metric and a suitable quotient topology. Then a polytope is said to be *perfect* if all polytopes, which are symmetrically equivalent to this

polytope, are similar to each other. For instance, in 3-space the *Platonic solids are perfect, but also the cuboctahedron, the icosidodecahedron, and their duals*. Among other results in higher dimensions, the author shows that all regular polytopes are perfect, and that in spaces of even dimension there are countably many types of perfect polytopes.

In Chapter 3 of the beautiful book [382], Böhm and Quaisser present (with many nice figures) knowledge on the *Platonic, Archimedean polyhedra, and Catalan solids*, and also the six *regular polytopes in 4-space* and the *four Kepler–Poinsot and ten Schläfli–Hess star polytopes* are discussed. It is shown that there are no regular star polytopes in dimensions higher than 4.

Cromwell [690] treats *polyhedra* in 3-space in many directions: the author discusses comprehensively the *history, existence and uniqueness questions, construction methods, proof methods, extensions of definitions, and applications* (also in arts). He starts, in a motivating manner, with polyhedra in art, nature, architecture, and other fields, to present then ancient developments connected with Egyptians, Babylonians, and Chinese geometry. Classical Greek mathematics follows, including the existence and construction of *Platonic solids*. The *Archimedean solids*, the related achievements of Alexandrian and Arabic mathematicians as well as the *development of perspective*, connected with Alberti, Uccello, della Francesca, Pacioli, Dürer, and Jamnitzer, are discussed. Also *Hilbert's third problem* is studied, with Dehn's negative answer (but without Sydler's complete solution). The *Kepler–Poinsot star-polyhedra and Euler's relation* with its partial anticipation by Descartes are discussed, like also *Cauchy's rigidity theorem* and his proof of completeness of the family of *regular star-polyhedra*. One chapter is nicely dedicated to the development of the *symmetry notion via group theoretical aspects*, and also *colourings of polyhedra* are studied. The topics of the final chapter are *compound polyhedra and polyhedral metamorphosis* (via transitivity properties and colour symmetries). Besides deep historical studies, the book contains also proofs and mathematical backgrounds of the discussed topics. Many of the results are understandably presented and thus good to read also by non-specialists. Many proofs are simply, but nevertheless completely done; many illustrations and a large bibliography complete this nice reference.

The book [2565] of R. Ziegler combines general *crystal morphology* with the *morphology of symmetric convex polyhedra*. Basic *laws of crystallography and the seven crystal classes* are nicely presented, completed by *symmetry groups of crystals and polyhedra*. An extensive presentation of the morphology of symmetric polyhedra with many transformation series is given, and the *theory of lattice structures* is also taken into consideration. There are many beautiful figures and illustrations, and so this very detailed book is an excellent source of information in the described directions of geometry. Among other topics, the book [2566] contains discussions on and beautiful illustrations of *Platonic, Archimedean, and Catalan solids* (with Kepler's famous planetary model), *vertex-transitive polyhedra, isohedra, deltahedra, compounds, crystal forms, three-dimensional images of the tesseract, and projective images of regular polyhedra*. It somehow continues [2565], namely partially extending the variety of presented geometric figures from there.

In the book [2019], Rajwade studies *three-dimensional convex polyhedra satisfying strengthenings and weakenings of usual geometric properties*. The selection of the respective conditions (e.g., being regular-faced) is similar to that in Martini's survey [1709]. The second book part treats *Hilbert's third problem*; see also the monograph [423].

The encyclopedic book [1755] by McMullen and Schulte on *abstract regular polytopes* gives a thorough exposition of the subject itself, its applications, and connections with other fields. It starts with a valuable historical account about "Platonic" solids, their generalizations, and higher dimensional analogues, and it provides the foundations for forthcoming chapters, in the form of notions and concepts like *face lattices, Schläfli symbols, Coxeter groups, regular maps*, and the following basic regularity notion: *A geometric structure (which can be a polytope, a tessellation, or an abstract poset) is said to be regular if its symmetry group acts transitively on the family of all its flags (maximal chains)*. The notion of *abstract polytopes* is introduced, using suitable conditions for the incidence of faces, and finally "abstract regular polytopes" are defined as those abstract polytopes which

have *flag transitive automorphism groups*. Thus, abstract regular polytopes form highly symmetric combinatorial structures having many interesting geometric, algebraic, and topological properties. With this background, large parts of the book refer to group-theoretical aspects; *Coxeter groups (of Euclidean, spherical, and hyperbolic type), unitary reflection groups and Hermitian forms, groups associated with space-forms, and abstract regular polytopes related to linear groups* are examples. Based on these abstract concepts, the authors nevertheless show the geometric beauty of the topic when discussing realizations and constructions. So the "amalgamation problem" (finding polytopes whose facets and vertex figures are of given types) and "realizations" (embeddings of polytopes in Euclidean spaces) are demonstrated, and it is also discussed what is known about *polytopes which have natural embeddings in manifolds*. These polytopes are characterized by the fact that the facets and vertex figures have to be spherical. The authors present also many results on *spherical, projective, star, (higher) toroidal, and chiral polytopes* with regularity properties in the described sense, and also *hyperbolic) honeycombs* are studied. Further on, the book part treating the *classification of locally toroidal polytopes* (i.e., either the facets or the vertex figures are torus-like) with regularity properties, following Grünbaum's intuitions, should be mentioned. Summarizing, especially also in view of corresponding lines of research this book is an important milestone in the sequence of monographs treating *polytopes and polyhedral structures with regularity properties*.

The monograph [650] of Conway, Burgiel, and Goodman-Strauss is dedicated to the *symmetry notion* in general, and it contains also nice and interesting parts related to polytopes; partially the presented results are new or get at least new proofs. For example, these refer to *Archimedean tilings in the hyperbolic plane, generalized Schläfli symbols, certain types of 3-tessellations, and new space group notations*. In this book, for the first time the complete list of *four-dimensional Archimedean polytopes* (found by Conway and Guy in 1965) is comprehensively described. Further on, the authors explain the "Keplerian" names for the *Archimedean and Catalan solids* ("Keplerian", since the original work of Archimedes on this topic got lost); they use or extend such polyhedra to tessellations of the plane and of Euclidean 3-space. The *Euler characteristic* for polyhedra is discussed and generalized for classifying also surfaces. And the book also performs the *17 plane crystallographic groups* and the *219 crystallographic space groups*.

The book [29] of Akiyama and Matsunaga, written in the style of Plato's dialogues and suitable for a very broad readership, requires only mathematical background from school (deeper results and approaches can be found in independent appendices). The book covers an impressively broad spectrum of themes (we mention only the themes close to our aims here): *tessellations, special types of polygons and polyhedra (e.g., Platonic solids), cross-sections of the cube, minimal perimeter lengths, honeycombs and packings, a universal building-block for parallelohedra*, and so on. Further on, the results introduced include *all 17 groups related to periodic planar tilings, Penrose tilings, Conway polytopes, equi-decomposability, Hilbert's third problem, the Dehn–Hadwiger theorem, and art gallery problems*. This elementary, but very broad book is also suitable as reference for lectures on everyday mathematics.

Regular polytopes and their symmetries (see, e.g., [670]) present one of the oldest, nicest, and richest subjects in geometry, starting with *classical regular polygons, Platonic and Archimedean solids*, but continuing in modern times with *abstract regular polytopes* (cf. again the monograph [1755]). However, besides these developments also recent studies of the geometric side of this field can be registered, associated with topics again changed within that framework (e.g., by underlining *realizations and rigidity* and finding different ways to understand the *geometric and combinatorial symmetry of polytopes*). A very recent, deep, and comprehensive book reflecting the modern *geometric theory of polytopes* is McMullen's monograph [1751], including a wide discussion of applications and new techniques. This new book, deliberately developed for this field after [1755], nicely completes [1755] by studying concrete realizations of abstract structures. The author classifies *regular polytopes and apeirotopes of full or nearly full rank*. For example, concrete representatives of polytopes studied geometrically from new viewpoints and in detail are the *24-cell, the Platonic icosahedron and dodecahedron, the 600-cell and 120-cell, star polytopes, honeycombs, and regular compounds of polytopes*.

BOOKS ON GENERAL (E.G., TOPOLOGICAL OR COMBINATORIAL) PROPERTIES OF POLYTOPES

Historically interesting, the book [788] of Eberhard seems to be one of the first deep studies treating shapes of convex polyhedra, where the author mainly restricts to *simple polyhedra*. For example, Eberhard discusses *continuous deformability* of them into each other when certain information about all corresponding pairs of their vertex figures are given. If f_m denotes the number of m-gonal facets of a polyhedron, then also realizability questions (regarding allowed combinations of these numbers) are investigated. The book contains various results in this direction, and *Eberhard's theorem* says that for every sequence (f_m) with $3 \leq f \neq 6$ satisfying a certain equation containing these numbers, there are values of f_6 such that the sequence (f_m) with $f \geq 3$ can be realized by some convex, simple 3-polytope (see also Section 13.3 in Grünbaum's book [1085]).

For the sake of completeness, we mention here also Minkowski's explicit contribution on *general theorems on convex polyhedra*, which are published in his "Gesammelte Abhandlungen" [1803].

Based on his unfinished and encyclopedic article "Polyeder und Raumeinteilungen" (see [2295]), Steinitz grasped known and new results as well as own lecture notes to an almost complete book manuscript which was excellently supplemented and finished by Rademacher; so the monograph [2296] came to be, as a joint work of them. The authors discuss *Euler's theorem, Cauchy's rigidity theorem, and characterizations of combinatorial types of convex polyhedra* (also based on older investigations of Cayley, Kirkman, Möbius, and Eberhard referring to the *morphology of only simple and simplicial polyhedra*). *Polyhedral complexes* are topologically introduced, and their *geometric realizability as convex polyhedra* is a central topic of this book. It is well known that this stimulated many further research activities. The authors give three very interesting and different proofs of the corresponding fundamental theorem. For one of them, the notion of *projective convexity* is successfully introduced, another one strictly uses Hilbert's axiom system.

In his monograph [39], A. D. Alexandrov problems of the following type are studied: How far must certain data (like face areas, edge-lengths, area angles, etc.) for a convex polyhedron P in 3-space be prescribed such that P is determined up to some transformation (e.g., translation, motion, or similarity)? Clearly, here *uniqueness and existence* have to be considered, and thus *Cauchy's uniqueness theorem* (P is determined by its facets and the way they are put together) and *Minkowski's existence and uniqueness theorem* (formulated via prescribed normal directions and areas of the facets of P) play an essential role. With respect to such questions, the book is encyclopedic in nature. It opens also various opportunities to extend theorems about polyhedral surfaces to general surfaces, as studied in the author's book [40] (see also our discussion of Alexandrov's general role in the history of convexity). However, these possibilities are only mentioned in the form of hints at the end of the respective chapters. Giving many new proofs and even some new results, the author considers (besides the bounded) also the unbounded case, where the number of vertices, edges, and faces are still finite, but edges can be half-lines. After developing a *topology of convex polyhedra*, the author proves *Cauchy's lemma* on the distribution of plus and minus signs in his theorem on the uniqueness of convex polyhedra (obtaining them, e.g., from a net; see also §3 in [141]). Then it is shown that for a given plane scheme with positive curvature and satisfying the Euler theorem, up to motions exactly one bounded convex polyhedron exists which corresponds to this scheme. With additional assumptions, this is extended to *infinite polyhedra* and to convex polyhedra with boundary. In the following chapters, similar investigations regarding Minkowski's theorem follow. With another proof than Minkowski's, this theorem is given in a slightly generalized form: If two bounded convex polyhedra P and Q have parallel (oriented) facets, and if one can translate any face of $P[Q]$ into a proper subset of the parallel face of $Q[P]$, then Q may be obtained from P via translation. Again with additional assumptions, this is also extended to *unbounded polyhedra*. Investigations on *infinitesimal rigidity* are presented,

with interesting historical background. Similar problems are discussed, like *Let a bounded polyhedron be deformed in a way that the intrinsic vertex distances remain stationary. Then this deformation is initially (or infinitesimally) a motion*.

In many places of this understandably written book additional reports (for example, on *extensions to convex surfaces*, or to *spherical and hyperbolic space*) are given, without proofs. In 1958, the Russian original was translated into German, and in view of the 2005 edition one has to say that most problems listed in the book are meanwhile settled. Also several passages are presented in refined form or with new alternative proofs; the footnotes are important, too. New comments by V. A. Zalgaller (on the author's advice) are contained, and the list of references is enlarged. Supplementing Chapters 3, 4, and 5, the translations of two articles by Volkov and one by Shor are added and placed at the end of the book.

The famous *upper bound conjecture (UBC)* states that the maximum number of j-dimensional faces of an n-dimensional (convex) polytope P with fixed vertex number is attained when P is a *cyclic polytope*. This was confirmed by McMullen and published in 1970, when the greatest part of the strongly related book [1757] by McMullen and Shephard was already written. Thus, McMullen's proof (based on the "shelling procedure" of Bruggesser and Mani) is reproduced in the final chapter of [1757]. It perfectly fits into these notes on the *combinatorial theory of convex polytopes* (which came up as a result of a research seminar at the University of East Anglia in Norwich). After discussing the *facial structure of polytopes* in general, and of special types of them (like *cyclic and neighbourly polytopes*), the *Dehn–Sommerville equations, Gale transforms*, and further tools are discussed, which partially can also be used for extensions to *spherical complexes*. The authors do not directly use the numbers f_j of j-dimensional faces of the considered polytopes, which form their so-called f*-vectors*. Instead they use so-called g*-numbers* (which are linear combinations of the numbers f_j), furthermore important for conjectures formulated at the end of the book and referring to a complete characterization of the f-vectors.

Having the algorithmical description and representation of "polyhedra" from the viewpoint of *computer graphics* in mind, Nef [1883] uses a very general notion of "polyhedron": it is a point set in n-dimensional Euclidean space consisting of finitely many open half-spaces which are combined by the operations "intersection", "union", and "creating the complementary set". Therefore, also half-spaces and linear manifolds are polyhedra. In the first part of the book [1883] the author gives a very detailed representation of the theoretical side of this concept, therefore interestingly enriching the literature on *polytopes and polyhedral sets* already on the conceptual basis. For example, the author describes the subcase of convex polyhedra, *the face structures*, and also *vector sums of polyhedra*. All statements are carefully proved, and also motivations for all further notions are comprehensively given. Thus, independent of the preparation of the second (algorithmical) part of the book, this first part is theoretically interesting in itself.

The book [182] of Bair and Fourneau gives a profound treatment of the theory of convex polytopes and polyhedra in linear spaces. It contains many results and elaborates in great detail those topics the authors are interested in. The style allows an easy reading, and the proofs are clear and complete. After fundamental statements on *separation and faces of convex sets*, also the notions of *convex polytopes and polyhedra* are discussed, and further on the authors study *representations of polyhedra and their applications* (like again separation), *Choquet simplices* (see here also Soltan's survey [2272]), and related systems of linear inequalities.

Yemelichev et al. [2500] cover combinatorial problems from the theory of (general as well as special) *systems of linear inequalities with real or integer variables*. In particular, the connections between *combinatorial and topological aspects of the theory of polytopes* and the methods of their analytical description in view of *linear and discrete programming* are emphasized. With this in mind, *integer points in polyhedra, permutation, and transportation polyhedra* are studied, and the book ends with a collection of interesting related problems and conjectures.

In the booklet [1329], Jucovič discusses *graph-theoretic aspects of polytope theory* (e.g., Hamilton circuits, or the Theorem of Steinitz), but also *metric properties of special families of polytopes* (like the Archimedean solids, or the translative space-fillers) are presented there.

Federico's historically very interesting book [835] on the geometry of polyhedra shows that already about 1630 Descartes tried to work with an *algebraically oriented stereometry*, discovering for example a borderline case of the *Gauss–Bonnet formula for exterior angles of polyhedra*. The comparison with Euler's later discoveries is interesting and shows the difference of methods of two very famous mathematicians. It should also be mentioned that the starting point of the story, the little scripts "Progymnasmata de solidorum elementis" of Descartes, survived only as a copy made by Leibniz.

The book [514] of Brøndsted is suitable for graduates. The author develops the *theory of convex polytopes* from its fundaments to the research level of that time. First *general properties of convex sets* (like their facial structure and polarity) are discussed. Many well-known theorems from convexity are carefully proved, and in the second chapter *general properties of convex polytopes* are considered, including *polarity, combinatorial duality, polyhedral graphs, neighborly, and cyclic polytopes*. The *Dehn–Sommerville relations, the upper bound theorem, and the lower bound theorem* follow, where the upper and lower bound theorems are studied for the case of simple polytopes (rather than the simplicial ones, as most authors do). Due to this, the proof of the upper bound theorem really differs from other published versions. This nicely written book is recommendable for everybody interested in convexity, polytopes, and combinatorial geometry.

Since *commutative algebra* played an increasingly important role in combinatorics, Stanley was motivated to write the very interesting booklet [2288], discussing two main areas: *linear equations in non-negative integers, and number of faces in simplicial complexes*. Especially in connection with the second area, several problems and results from polytope theory are generalized, e.g., treating *rational convex polytopes, combinatorial reciprocity theorems, the Stanley–Reisner ring of a simplicial complex*, and (coming from this) the *upper bound theorem for spheres*. Although being very short, the booklet also contains previously unpublished results.

Edelsbrunner's monograph [793] was groundbreaking for the stormy development of *computational geometry*. Besides computational parts, *combinatorial structures in geometric problems* and also *geometric applications* play an essential role in this book. Topics discussed and being close to polytopes are the investigations on *hyperplane arrangements, convex hulls, and Voronoi diagrams*, and also the complexity of polytopes is discussed (here the notions *cyclic polytopes, Dehn–Sommerville relations, and an asymptotic version of the upper bound theorem* occur).

The important book [1047] of Grötschel, Lovász, and Schrijver on *combinatorial optimization* shows (mainly in Chapters 6 and 7) how important *polyhedral geometry* and, in particular, knowledge about *rational polyhedra* in this field are. And in the introductory part of the survey [1603], Lovász underlines the role that *polyhedral combinatorics* plays in *semi-definite optimization*. Here we refer also to Schrijver's three-volume book [2192] (see below), and to the booklet [1333] of Jünger, in which the foundations of polyhedral combinatorics (such as *cutting plane methods, matroid intersection, separation methods, total dual integrality, structures of faces and facets*) are discussed; its importance for solving optimization problems is emphasized by the study of the NP-hard acyclic subdigraph problem, with many applications in operations research and economics.

Oriented matroids are an interesting tool for studying the combinatorial (or abstract) theory of convex polytopes. Chapter 9 of the corresponding monograph [354] of Björner et al. deals with *polytopes*, discussing, e.g., *constructions of polytopes, cyclic and neighborly matroid polytopes, the Steinitz problem, polyhedral subdivisions, and triangulations*. Material on *arrangements of hyperplanes and zonotopes* can be found, too. The authors interestingly reach the research front by showing relations to *(generalised) hypergeometric functions, toric varieties, and combinatorial differential manifolds*. A nice completion of the monograph [354] is Bokowski's handbook article [420], where mainly the *Steinitz problem* (i.e., the classification of all combinatorial types of convex polytopes of given dimension and vertex number) is discussed from the viewpoint of oriented matroids.

With his excellent monograph [2560], Ziegler set, after Grünbaum's book [1085], a trademark in modern polytope theory. Although the book [2560] is not encyclopedic (as [1085] is), the author brings together a large part of the field and almost 500 important references, which are discussed and commented in extensive and very helpful notes. Due to this, the book covers many beautiful problems and subjects in the theory of polytopes, old and new. The reader can feel the author's enthusiasm, especially for the *geometric side of polytope theory*, and his didactical ability to present the material in a very stimulating way. Except for the fundamental subjects in Chapters 0 and 3, the main topics (i.e., lectures) of the book are independent from each other; the only needed background consists of real affine geometry and basics from vector spaces. The explicitly studied topics of the book are *basic geometric and combinatorial properties of convex polytopes* (e.g., the Fourier–Motzkin elimination and the Farkas lemma, face lattices, and polarity); *graphs of polytopes* (in particular, Steinitz' theorem for 3-polytopes); *polyhedral complexes, Schlegel diagrams of 4-polytopes and d-diagrams; oriented matroids, duality, Gale diagrams, and applications; fans, arrangements, zonotopes, nonrealizable oriented matroids, and zonotopal tilings; shellability and the Euler–Poincaré equation, h-vectors and Dehn–Sommerville equations, the upper bound theorem, and some extremal set theory* (Kruskal–Katona theorem, Macaulay's theorem); and *fiber polytopes and beyond, permuto-associahedra*. Although the connections with *commutative algebra and algebraic geometry* are not explicitly given, they can be found in the references. The book is nicely illustrated, each chapter ends with demanding problems and exercises which are carefully selected. The interested reader finds a brilliant approach to polytope theory.

The book [826] of Ewald, somehow prepared by the handbook survey [825], gives a nice introduction to the theory of *convex polytopes and polyhedral sets in relation to algebraic geometry* (see also our description in the subsection on convex algebraic geometry). Here as a combining link *toric varieties* (or torus embeddings) are used. In this setting, notions from both fields, which correspond to each other, are juxtaposed. With this starting point, mutually influencing stimuli are used to advance both theories. A similar reference is Oda's book [1909] (again we refer to our section on convex algebraic geometry).

Richter-Gebert investigates in [2030] the *realization space* of an n-dimensional convex polytope P, i.e., the set of all polytopes with the same face lattice as P. Since the polytope is given by a matrix of coordinates of its m vertices, the realization space of P is a subset of an (mn)-dimensional space. In contrast to a well-known result of Steinitz (namely that *realization spaces of 3-polytopes are contractible*), those of 4-polytopes can be arbitrarily complicated. The author proves in [2030] the *universality theorem for 4-polytopes*: For every primary basic semialgebraic set V there is a 4-polytope whose realization space is stably equivalent to V. It follows that all algebraic numbers are needed to coordinatize the vertices of 4-polytopes, and that the realizability problem for 4-polytopes is NP-hard. Also complexity questions and *universality theorems for related concepts (like diagrams, fans, and families of semialgebraic sets)* are studied. For example, a non-rational 4-polytope with 33 vertices is presented. Interesting open problems are posed, and the whole book is excellently written. It represents a great step in the development of polytope theory.

The excellently written textbook [2317] of Sturmfels underlines the interdisciplinary nature of *Gröbner bases*, which are distinguished bases of ideals in polynomial rings and an important tool for several *algorithms in algebraic geometry and commutative algebra* (see also our subsection on convex algebraic geometry). Based on them, the author shows the *relations between computational algebra and the theory of convex polytopes*, and he presents specific applications of this theory in the fields of *integer programming, computational statistics*, and further mathematical topics like *pure combinatorics, convexity, algebraic topology, and probability*. The main method is to associate geometric objects (like convex polytopes or simplicial complexes) with toric ideals or more general ideals, and to use then Gröbner bases to study these geometric objects, and vice versa.

A *cut*, as considered by M. Deza and Laurent in [746], is a semantic on a set S of m points of a vector space generated by the edges (pairs of points) from S, where S is partitioned by $S = U \cup V$. This is done such that pairs entirely within U or within V are at distance 0 (using a semimetric, for

which different points can have distance 0), while pairs intersecting both U and V are at distance 1. The *cut cone* is the set of positive linear combinations of cuts, and the *cut polytope* as then defined as the convex hull of the set of cuts. The cut polytope is the central object of the monograph [746]. It is shown that cuts, semimetrics, and the cut polytope occur in many areas of mathematics, such as *polyhedral combinatorics (embeddability of finite metrics in) Banach spaces, geometry of numbers* (Delaunay polytopes, Voronoi results for point lattices), *optimization* (semi-definite programming), *graph theory and designs* (embeddings into a Boolean cube), *computational probability and statistics, computer science, physics, and economics*. The *Delaunay polytopes and the geometry of numbers* play a particularly essential role in that book. And the authors study also the geometry of the cut polytope itself, showing one of its most striking applications: it is a cut polytope that provides (for large n) a *counterexample to Borsuk's long-standing partition conjecture* (referring particularly to the known construction of Kahn and Kalai from 1993).

The monograph [2085] of Sadoc and Mosseri (we recommend also to read its review in the Mathematical Reviews (MR1716082)) is a research monograph on *theoretical physics*, also with many applications. The class of related physical problems includes some complex structures which are not periodic crystals, i.e., both *quasi-crystals and disordered materials*. It refers also to applications like quasi-crystals, metallic glasses, atomic clusters, complex liquid crystals, amorphous semiconductors, polymers, phospholipid membranes, amphiphiles, and many others. The geometric part shows that in various situations, the geometry of the underlying space and of the system matter do not coincide and demand some "separate consideration". For example, the discovery of *quasi-crystals* (1984) showed that *crystallography* was much less developed than what physicists and chemists in those days would have expected. The insights corresponding with this general observation slowly develop. Also investigations in *non-Euclidean spaces* should be mentioned here. The helpful role of geometry presented in this book and coinciding with its philosophy contains subfields like *tilings, (regular) polytopes and symmetry groups, Schläfli's notation, the generalized Euler–Poincaré relation, quasi-crystals, and topological stability using homotopy groups*. Also due to a nice derivation of the *regular polytopes*, one understands that such mathematical concepts and notions help to find a unifying picture of areas in physics, mathematics, chemistry, metallurgy, and engineering.

Barvinok's convexity book [242], also cited in our part on history of general convexity, contains especially much material on polytopes, and therefore we mention it also here. After discussing general convex bodies (and, as example, also ellipsoids), he presents detailed material on the *upper bound theorem* and a brief discussion of *centrally symmetric polytopes*. In the final chapter, results on *lattice polytopes* are collected.

Chapter 5 of the monograph [1732] on discrete geometry is also dedicated to polytopes. Themes like *H- and V-representation as well as duality of convex polytopes, cyclic polytopes, the upper bound theorem, Gale transforms, and Voronoi diagrams* are studied there, and in the next chapter *arrangements of hyperplanes* are presented. Other chapters contain explicit discussions of further topics around polytopes (e.g., *volumes of polytopes, and symmetric polytopes*).

Schrijver's three-volume book [2192] excellently summarizes the state of the art (2003) of the *polyhedral approach to combinatorial optimization*. In its original form, a problem of combinatorial optimization asks for the minimum (maximum) of a real-valued function which is defined on a given finite set. Using convex hulls of these given finite sets (clearly yielding *convex polytopes*) and linearizing this function, problems from combinatorial optimization become problems posed in terms of *linear programming*. In a fascinatingly comprehensive way the author shows, besides many other important things, how specific knowledge on various properties of polytopes can help to solve such problems efficiently. The monumental work [2192] comprises almost 1900 pages and more than 4500 references; many interesting problems and conjectures are posed, and also very interesting historical notes are given. Here one should cite also (although it is older) the related handbook article [2191] of Schrijver on combinatorial problems studied with the help of polyhedral geometry. A survey on a large variety of related methods and techniques is given. Some keywords reflecting the contents of [2192]

and [2191] are *linear programming*, a geometric look at the *simplex method, primal-dual method and ellipsoid method*, applications of *total unimodularity in bipartite and general graphs, Edmonds' matroid intersection theorem, Fulkerson's blocking and antiblocking polyhedra, and generating integer hulls of polyhedra*.

Concentrated on *polytopes suitably arising from point configurations*, the book [2360] of Thomas presents an introduction into the modern theory of convex polytopes. After fundamental geometric and algebraic aspects from the theory of polytopes, the author deals with *Schlegel and Gale diagrams of polytopes*. This yields a construction of an *eight-dimensional non-rational polytope*. Then *triangulations of point configurations and their liftings to higher dimensions* are investigated, as well as *secondary polytopes* (secondary polytopes of point configurations are polytopes whose face poset is isomorphic to the poset of all regular subdivisions). As a concrete example of secondary polytopes, the *permutahedron* (also called Stasheff polytope) is described. The second part of the book contains a course on *Gröbner bases, state polytopes of point configurations*, and an application of polytope theory to *toric varieties*, thus showing interesting relations to *combinatorics and convex algebraic geometry*. The text is self-contained and nicely written; one can see it as successful quick introduction to the described field. Numerous figures and exercises are added. The book is based on an advanced undergraduate lecture course on *geometric combinatorics* and can be recommended as a textbook on modern aspects of polytope theory for undergraduates and graduates.

The famous *Theorem of Pick* (see, e.g., the expository paper [1666] of Maehara and Martini), giving the area of a convex lattice polygon P in terms of the number of integer points in the interior and on the boundary of P, reflects somehow the planar situation of what is the wider goal of the beautiful book [263] of Beck and Robins: using the number of integer points in a convex n-polytope P to estimate the volume of P. Thus, the monograph [263] can be located *in the intersection of geometry, number theory, and combinatorics*, giving a connection between "the discrete and the continuous", as also the title tells it. It contains many historical notes, nice problems, examples, and various applications. For example, it refers also to particular problems from combinatorics and number theory, like the problem of *counting all non-negative square matrices with equality of all row and column sums*, or like finding how many ways exist to change an amount of dollars into coins of smaller fixed denomination. The authors present the large variety of tools that occur in this framework, such as *Bernoulli polynomials, Ehrhart polynomials for the enumeration of lattice points, and Ehrhart quasi-polynomials for polytopes with rational vertices*. Further on, *Dehn–Sommerville relations on face numbers, finite Fourier series, Dedekind sums, suitable types of decomposing polytopes, volume via solid angles*, and further topics are discussed. In particular, the comprehensive introduction to properties of *Ehrhart functions* should be emphasized, like also the connections between *lattice points and decomposition techniques*, or the relations to *Fourier analysis*. Here we have to mention also the closely related proceedings [261] of Beck et al., containing various articles on this topic.

In Gruber's monograph [1070], polytope theory plays an essential role in several chapters. Its third chapter is even named "convex polytopes" and unifies many aspects, among them being the *combinatorial theory of convex polytopes, their face lattice and volume, isoperimetric problems for polytopes, Hilbert's third problem, rigidity of polytopal surfaces and frameworks, Alexandrov and Minkowski theorems for polytopes, lattice polytopes, linear programming, and tilings with polytopal cells*.

The main question investigated in the monograph [243] of Barvinok is well known: *how to count integer lattice points in a polytope in n-space?* The author presents an excellent and self-contained approach to some classical and also recent results (not only) in this direction. His unifying point of view is based on *valuations*. His methods are based on an algebra on polytopes and various properties of polyhedra (e.g., geometric ones). The reader finds the *Euler–Poincaré theorem, Pick's formula, Minkowski's theorem for convex bodies, and the Lenstra–Lenstra–Lovász basis reduction algorithm*. The so-called *exponential valuation on the algebra of polyhedra* is introduced and investigated. It can be interpreted as an *extension of the volume to unbounded polyhedra* (and specializes to volume in the bounded case), and one can evaluate with it exponential integrals over a polyhedron. In particular, it

is quite powerful and useful for *computing the volume of polytopes*, as demonstrated by Lawrence's volume formula for simple polytopes. Then the setting of *lattices, their bases and their fundamental domains* is established, and relations to the *classical geometry of numbers* are shown. In further parts the machinery of *exponential sums and generating functions* is used. Also, the author introduces a *counting valuation on the space of rational functions on* C^n, using it to state *Brion's theorem* relating the lattice point enumerator of a rational polytope to the lattice point enumerators of supporting cones subtended at its vertices. This counting valuation is related to the exponential valuation with the aim of *expressing the number of integer lattice points in a rational polytope in terms of the volumes of its faces and a certain valuation on the tangent cones at those faces*. This yields an exposition of the related recent *Berline–Vergne "local" formula*. Further contents of the nicely illustrated book are *unimodular polytopes, Ehrhart's theorems on the polynomial behavior of integer points in rational polytopes, decompositions for rational cones*, and related results. The book contains numerous exercises of various degrees of difficulty; many useful references are given, and it is a good source for graduate students and researchers. Its scope is wide, reaching from classical to very modern results.

Also the book [645] of Concini and Procesi is related to integer points in polytopes. Several topics "between" numerical analysis (*box-splines*), algebraic geometry, and polytope theory (hyperplane arrangements) are brought together, twisting around the central and merging topic of *computing the integer points in suitable families of variable polytopes*. The starting point is a finite list of non-zero elements (i.e., a finite spanning subset) of a finite-dimensional real vector space, from which the authors construct geometric, algebraic, and combinatorial objects like *polytopes* or even *zonotopes*. The already mentioned problem of computing all integer points in classes of polytopes is studied with the help of so-called partition functions and multivariate splines associated with the given list. (The simplest form of the problem is: compute the number of ways that a given non-negative integer can be expressed as the sum of m fixed positive integers.) The first book part collects *basics on convex sets, polytopes, and zonotopes* (the latter are needed to understand the nature of the partition function, and also their volume plays a role). Further topics treated in the book are splines, *toric arrangements (replacing usual hyperplane arrangements)*, and several aspects of *approximation theory*. In a very detailed style, the book is nicely written; it is completely self-contained and presents good examples.

Monoid algebras form an important part of algebra or K-theory (if one studies for instance the projective modules over these algebras), and the main part of the book [512] of Bruns and Gubeladze treats what the authors call "affine monoids"; they are isomorphic to finitely generated submonoids of Z^n, i.e., suitably generated by non-negative integral linear combinations of points from Z^n. These structures are also related to *polytope theory*, and this aspect yields a geometric meaning of the *affine monoid algebras*. Due to this, there are interesting *relations between convex geometry and combinatorial geometry, ring theory, K-theory, and toric geometry in commutative algebra*. So the book [512] refers also to the concept of *lattice polytopes*, including relations to *commutative algebra and algebraic geometry*, and it is natural that also geometric notions like *toric varieties, vector bundles, and triangulations* occur there.

In a very wide sense (also meaning higher dimensions), the book [1598] of De Loera et al. treats the *theory of triangulations of (convex hulls of) point sets*. Many nice applications are given, coming from fields like *algebra, computer science, combinatorics, and optimization*. From the methodological point of view, besides tools from computational geometry especially the interplay between geometry and combinatorics is successfully used. Presenting the *Gelfand–Kapranov–Zelevinsky central theorem on secondary polytopes* (see the second part of the basic monograph [948] of Gel'fand et al.), the authors give the first comprehensive treatment of *polytopes whose face lattices realize the posets of regular subdivisions of point configurations*. Further polytope classes and triangulation- and configuration-types investigated here are *cyclic polytopes, products of simplices, 0/1-polytopes, cubes, lattice and fiber polytopes, polytopal complexes, and Gröbner bases*. A chapter on computation and algorithms is contained, and a central goal of the authors is the use of the rich structure of the universe of triangulations to solve also computational questions (for example, *finding optimal triangulations* in

different settings). The level of the book is suitable for graduates, but also researchers will certainly find many suggestions. The special beauty of it is also manifested by almost 500 illustrations, and many nice examples and exercises are given.

In the monograph [1328] Joswig and Theobald give an overview on algebraic methods in computational geometry, where also the *notion of polytope*, considered in different directions, plays an important role. The first part presents concepts and techniques of linear structures like *polytopes and polyhedra, convex hulls, Voronoi diagrams, and Delaunay triangulations*. Topics from computational algebraic geometry (like Gröbner bases and polynomial systems) follow, and more applied parts of the book use software like polymake, Maple, and Singular. This interesting textbook (also cited in other subsections here) offers a nice mixture of algorithms in polyhedral and algebraic geometry.

BOOKS ON FURTHER TOPICS AROUND POLYTOPES

It is a natural motivation to generalize or to modify such an important notion like that of polytopes in various directions. We give now three examples.

First, one can stay in classical Euclidean space, but change the geometry of the polytope itself. A *ball polytope* is a non-trivial intersection of finitely many balls of equal radii, with many applications (for notions like ball convexity, ball hulls, etc.). For example, the boundary structure of ball polytopes is different to that of usual polytopes (see [1478]), but analogues for the theorems of Kirchberger and Carathéodory can be proved (cf. [324]; see also Chapter 6 and the corresponding notes in the monograph [1713]).

Another way of extensions is the consideration of *non-Euclidean polytopes*. Böhm and Hertel [381] give a deep representation of the state of knowledge around the theory of *volumes of convex polytopes in spaces of constant curvature*, including polytopes with asymptotic vertices. At first polytopes are represented in the standard models of the treated non-Euclidean geometries. Since an elementary approach to polyhedral volumes in *elliptic and hyperbolic spaces* requires suitable dissections into congruent pieces, topics like *equi-decomposability* become interesting, and so a strong relation to Hadwiger's investigations in his book [1134] is automatically given. Thus, dissecting polytopes suitably into simplices and therefore Hilbert's third problem are natural starting points. Here Schläfli's *orthoschemes* (simplices, which are suitable generalizations of right triangles) are particularly interesting, investigated, and used. (We refer here also to the nice survey [1356] of Kellerhals, focusing on the hyperbolic case.) The book also contains a very detailed study of *non-Euclidean simplices* whose volumes can be computed by using Schläfli's volume differential. Also *regular simplices in elliptic spaces* are studied, and for approaching their volumes a relatively simple integral (depending on edge-lengths) is constructed. We refer also to our part on Hilbert's third problem (scissors congruence) discussed in the subsection on discrete geometry above, and to the survey [9] of Abrosimov and Mednykh discussed here; see below.

One can also go from real to complex Euclidean space (using a unitary metric). A polytope is then a finite and connected configuration of points, lines, 2-planes, etc., being *regular* if its symmetry group behaves analogously to that in the real case. Shephard [2228] discovered and studied such *regular complex polytopes* already in the Fifties, and Coxeter published even (in 1974) a very nice book about them (see Coxeter [675]).

Or one investigates structures that are geometrically different to usual polytopes, but use the same methods to study them. An example here is given by the topic of *hyperplane arrangements*, considered as essential part of polytope theory. We present here a few monographs dedicated to it. One of the starting points in more modern times is Chapter 18 of Grünbaum's book [1085]. A classical book on this topic is Orlik's monograph [1918] where, e.g., *combinatorial, algebraic, topological, differential-geometric, group-theoretic, and further approaches* are discussed. In these directions the author studies *the intersection lattice, the Möbius function, and the Poincaré polynomial of the intersection lattice, the lattice homology, the cohomology algebra, and the algebra of logarithmic differential forms* related to (the complements of) hyperplane arrangements. Finally in [1918], the notions of *free arrangements*

and reflection arrangements are investigated. This book is completed by the comprehensive and deep monograph [1919] of Orlik and Terao, where again many useful concepts and techniques are presented, together with their interactions. And the used approaches are similar to those used in [1918]; the research front is reached by new results for arrangements in general position, and a historical overview (mentioning famous names like Arnol'd and Brieskorn) is given. Also the book [354] of Björner et al., treating *oriented matroids*, presents results on *hyperplane arrangements* and, in relation to them, *zonotopes*. In [1919] and [354] extensions to different types of spaces (also complex ones) are considered. Chapters 5 and 6 of Felsner's book [852] treat *arrangements of lines and pseudolines* (see also the booklet [1083] of Grünbaum) and are nicely written up. The presented investigations are restricted to the Euclidean situation, and the book contains many interesting notes and exercises.

The book [753] of Dimca on hyperplane arrangements enables even advanced undergraduate students to work with it, since it is inspiringly written and enriched by interesting research problems. The author is willing to deliver an enjoyable introduction to the theory based also on knowledge about very recent developments. So he gives a thorough survey of topics like *Milnor fibers, characteristic varieties, mixed Hodge structures, and free arrangements*, spanning a large arc to topics from *algebraic geometry*. A more recent book on arrangements is the monograph [23] of Aguiar and Mahajan, developing a rich theory of hyperplane arrangements over the field of real numbers, mostly using algebraic language. The book is divided into two parts and contains also five appendices. These refer to foundations of *regular cell complexes, posets, incidence algebras of posets, algebras and modules*, and further topics. The authors successfully demonstrate the applicability of arrangements in fields like *topology, algebraic geometry, or combinatorics*. Besides many very interesting examples, the authors present also notes with historical details, explanations, and references to important articles. Containing classical as well as new results, the book [23] can be seen, together with the recent monograph [753] of Dimca, as a modern introduction into the field. There are two modern developments treating arrangements of hyperplanes into more abstract directions of applications. One direction introduces *hypergeometric functions* from the viewpoint of arrangements of hyperplanes (see [1920]), and the other one (namely [24]) helps to develop *Hopf theory* in a new setting, where *combinatorial species, ideas from Tits' buildings theory*, and Rota's results on *incidence algebras* play an essential role. Also the fields of *semigroup theory, Hopf algebras, algebraic Lie theory, operands, and category theory* benefit from this comprehensive work.

12.4.2.2 Proceedings referring to polytope theory

The proceedings [2221], edited by Senechal and Fleck and shortly mentioned already in our general historical part, contain more than 20 articles on (mainly three-dimensional) geometry of polytopes. Particular themes occurring there are *special types of polyhedra (regular, semiregular and uniform ones, zonohedra, star polyhedra), polyhedra in crystal and molecular structures, prototiles and tessellations, duality, rigidity, polyhedral 2-manifolds*, and problems regarding *face numbers*.

Emmer et al. collected in [681] forty papers showing nicely the interplay that arts and mathematics (in particular studying geometric structures) can have. In those articles which are closer to mathematics, the authors nicely present various (mainly visual) aspects of tilings, complex polytopes, regular polyhedra, the fascination of visualizing higher dimensional figures (like polytopes), crystallography, and ornaments. But also artists very close to mathematics (like representatives of the Russian avant-garde or Pierro della Francesca) are nicely recognized.

The proceedings [1342] collected by Kalai and Ziegler contain important articles on polytopes, including the following themes: *0/1-polytopes, reconstructing polytopes from their graphs, flag vectors and numbers, volume computation*, and further (mainly computational) related topics.

And also the proceedings [27] edited by Akiyama, Kano, and Urabe collect many papers on polytopes and related topics. These treat topics like *packings with polytopes, types of dissections of*

polygons and polyhedra, foldings und unfoldings, triangulations, Delaunay diagrams, skeleta of metric polytopes, and various further problem areas.

We now come to proceedings related to *arrangements of hyperplanes* and related topics (for respective books see above), starting with the collection of Falk and Terao [829]. In this compilation one finds articles on *Coxeter arrangements of hyperplanes, regular tilings, nonlinear arrangements, polytopes and harmonic functions, homology and homotopy of the complement of (real and complex) arrangements, and reflection groups*. The collection [633] of Cohen and Suciu contains expositions on *cyclic covers of complements of arrangements, bundles of arrangements, cohomology and homotopy of the complement of real and complex arrangements, characteristic varieties of complex hyperplane arrangements, and combinatorially equivalent hyperplane arrangements*. This is followed by the proceedings [2353] of Terao and Yuzvinsky from 2009, offering articles on *hypersphere arrangements and applications, duality and Chern classes of logarithmic forms along hyperplane arrangements, a computational approach regarding average diameters of arrangements, hypergeometric integrals, discrete topological methods for and resonance webs of hyperplane arrangements, applications of arrangements to unfolding models*, and computations and conjectures. Also the proceedings [810] should be mentioned. Among other contributions, they contain expositions on *homological aspects of hyperplane arrangements, geometry and combinatorics of resonant weights, relations to toric varieties, and multi-arrangements*.

The collections [246] and also [261] refer mainly to *polytopes and lattice point problems*. Both proceedings contain articles treating *Ehrhart polynomials, valuations in connection to lattice point problems, Gröbner bases, triangulations of products of simplices*, and the *optimization of linear functions over the integer points of rational polytopes*. The proceedings [246] treating integer points in polyhedra and related topics collect interesting articles on *Ehrhart polynomials* (also of lattice polyhedral functions), *integral points in rational convex polytopes* (e.g., maximizing functions over them), the *lattice point generating function* (as a central tool in polyhedral lattice point counting), the extension of the *polar decomposition theorem* to arbitrary (not necessarily simple) polytopes, and polyhedral subdivisions of the product of two simplices. At the end, an interesting problem collection compiled by Haase and Reznick is given.

Also the anthology [713], edited by Devid and Elles and dedicated to Coxeter, collects various nice papers on polytopes. These refer to *(hyperbolic) Coxeter groups, regular and chiral polytopes, honeycombs, equivelar polyhedra, combinatorics of sections of hyperbolic polytopes, special types of polyhedra* (like golden rhombohedra), and many other geometric themes, such as *Tits buildings and geometric configurations*. It ends with the list of publications of Coxeter.

In the proceedings [146] combinatorial, algebraic, geometric, and topological aspects are merged. From the viewpoint of convexity, several papers on polytopes are especially interesting. They refer to *lattice polytopes and lattice triangulations, valuations on convex polytopes, the well-known g-theorem characterizing face vectors of simplicial polytopes, and the combinatorial classification of certain integral polytopes whose polars are also integral polytopes*.

The collection [1782] of Miller, Reiner, and Sturmfels does not only treat polytopes, but its thematic overlap with convexity consists mainly of articles on polytope theory (therefore we cite it here). There are expository papers on *lattice points in polytopes* (also the algebra of polytopes, valuations on this algebra, and Barvinok's algorithm to compute these points), *generalized associahedra* (for root systems arising from the theory of cluster algebra), *shellability of simplicial complexes and polytopes*, and on *characterizations of f-vectors of polytopes in view of the theorem of Steinitz*.

In [261] papers on *lattice points in polytopes, Ehrhart polynomials and inequalities for their roots, Delaunay polytopes, conic decompositions of rational polytopes, and polyhedral subdivisions of products of simplices* are collected; at the end, an interesting problem collection referring to related topics is given.

Also the proceedings [648], edited by Conder et al., on *rigidity and symmetry* contain several articles on polytope theory. They refer to *volumes of polytopes in spaces of constant curvature, the*

mixing construction for abstract polytopes (which provides a way to build a minimal common cover of two regular or chiral polytopes), *rigidity of regular polytopes, hereditary abstract polytopes* (i.e., every automorphism of each of their facets induces an automorphism of the polytope itself), and to the *classifications of regular polygonal complexes, chiral polyhedra, and two-orbit polyhedra* (with interesting open problems, e.g., stimulating related research on edge-transitive polyhedra).

The comprehensive collection [1195] of Hibi and Tsuchiya (articles treating mainly *lattice polytopes*) covers very many topics, among them being *lattice polytopes in toric geometry and relations to valuations, lattice point enumerators, lattice zonotopes, cubical polytopes, Eberhard-type theorems, and simple polytopes*. An interesting problem collection concludes the series of papers.

The very recent proceedings [697] edited by Cunningham, Mixer, and Schulte contain, among various further contributions, papers on *Meissner polytopes, self-dual and in particular self-polar polytopes, Tverberg-type theorems, geometric and combinatorial properties of primitive zonotopes, barycenters of point sets in faces of polytopes, realizations of the 120-cell, the rhombic triacontahedron in crystallography* (inspired by quasi-crystals), *groups occurring as combinatorial automorphism groups or symmetry groups of polytopes, and special tilings*.

For the following four proceedings, we were not able to find the editors. Therefore we cite them here in the main text.

The first two anthologies are

Special issue on research objectives and open problems. Part 1 (MR 83m:51003a) *Structural Topology* **4** (1980). Université du Québec à Montréal, Montréal, Que., 1980.

Special issue on research objectives and open problems. Part 2. (MR 83m:51003b) *Structural Topology* **5** (1980). Université du Québec à Montréal, Montréal, Que., 1980.

These two issues contain articles on *zonohedra, zonoids, spherical polyhedra, rigidity of polyhedral structures, stability of buckled icosahedral structures, further structures based on regular or semiregular convex and non-convex polygons and polyhedra*. A mathematical autobiography of M. Goldberg and a paper of M. Wenninger with the headline "avenues for polyhedral research" are also there.

The third proceedings are

Geometry of Toric Varieties and Convex Polytopes (partially in Japanese, MR 1206527). *Proceedings of a symposium held at the Research Institute for Mathematical Sciences, Kyoto University, Kyoto, June 17–21, 1991*. Sūrikaisekikenkyūsho Kōkyūroku No. 776 (1992). Kyoto University, Research Institute for Mathematical Sciences, Kyoto, 1992. pp. i–vi and 1–103.

Here one finds contributions on *rational points and polytopes, smooth convex approximations of polytopes, Ehrhart polynomials of star-shaped triangulations of balls, and duality questions*.

Last but not the least, we have as fourth proceedings:

Modern Aspects of Combinatorial Structure on Convex Polytopes (partially in Japanese, MR 1315999). *Proceedings of a conference held at the Research Institute for Mathematical Sciences, Kyoto University, Kyoto, June 14–18, 1993*. Sūrikaisekikenkyūsho Kōkyūroku No. 857 (1994). Kyoto University, Research Institute for Mathematical Sciences, Kyoto, 1994. pp. i–viii and 1–122.

This collection contains articles (partially in Japanese) treating *volumes of the dual polytopes of integral convex polytopes, strongly convex polyhedral cones, simplicial complexes, the intersection cohomology and toric varieties, and intersection homology groups of convex polytopes*.

12.4.2.3 Selected surveys on polytopes

An early exposition on polytope theory naturally inspired by *linear programming* is Klee's comprehensive survey [1418] (complete for that time). *Polytopal graphs and paths* in polytopes are naturally discussed, and many problems and conjectures (also close to this application field) are posed. It is a good sign for the stormy research development that started in those days regarding convex polytopes as the subject.

We give now a selection of important surveys on polytopes.

SPECIAL POLYTOPES WITH SYMMETRIES

The list of references given here in the following is also a completion of our subsections 2.9.1.3 and 2.9.1.4; see also above our book collection here.

Schulte [2202] collects the known definitions of *regular convex polytopes and their symmetry groups*, and he extends this point of view to the following polytope classes (which are perfectly represented by the section headings): *tessellations of n-space, regular star polytopes, regular skew polyhedra, Grünbaum–Dress polyhedra, semiregular and uniform polytopes, reflection groups, complex regular polytopes, and abstract regular polytopes*. And monographs reflecting this amount of knowledge are cited. The two surveys [1709]) and [1711] contain similar classifications, like also the book [2019].

The paper [1756] surveys the knowledge on geometric realizations of *regular and chiral polytopes* in low dimensions. A *realisation* of a regular abstract polytope is an embedding of its vertex set in Euclidean (or spherical, or hyperbolic) space, where the symmetries of the polytope are preserved. *Faithful realisations* are obtained when different vertices of the polytope map to different points in the space. The authors give a classification of *finite regular and chiral polytopes* which have faithful realisations in spaces whose dimension is as low as possible. They particularly enumerate the *regular and chiral polytopes in Euclidean 3-space*, and some new results regarding four dimensions are given, too. Also *regular and chiral apeirotopes* are discussed, and interesting related open problems are offered.

It is an interesting task to look for *realizations of abstract polytopes in Euclidean space*, which can clearly be considered as generalizations of face lattices of convex polytopes. Usually an abstract polytope has various realization polytopes. McMullen [1749] uses a so-called *fine Schläfli symbol* of a realization polytope, which "transforms" combinatorial information on that polytope into geometric ones, and says that this symbol is *rigid* if only one geometric realization is combined with that symbol, i.e., identifies it. He then examines several essential polytope classes deciding which representatives of them are rigid, above all the twelve *regular apeirohedra*.

Given a finite reflection group Γ, a Γ-*permutahedron* is the convex hull of the orbit under Γ of a point in a Weyl chamber of Γ (a connected component of the complement of the union of the reflection hyperplanes). And n-dimensional *associahedra* (also called *Stasheff polytopes*) are simple polytopes whose vertices correspond to the triangulations of a convex $(n+3)$-gon, and whose facets correspond to the diagonals of the $(n+3)$-gon. Suitably using finite reflection groups, one can generalize these notions, and both types of polytopes occur in various contexts and have interesting relations to other classes of polytopes. For example, *vertex-transitive zonotopes* are exactly those Γ-permutahedra that are zonotopes; and any *homogeneous zonotope* (i.e., an equilateral zonotope all whose vertices lie in a common sphere) is a Γ-permutahedron (see [2470]). The survey [1222] of Hohlweg collects various properties of and connections between these two important types of polytopes (*permutahedra* and *associahedra*). First permutahedra are derived from finite reflection groups through root systems, and then their faces are characterized. Based on this, the author constructs so-called Coxeter generalized associahedra and shows how to get, using this way, generalized associahedra from permutahedra. The paper contains also interesting open problems. And the exposition [585] presents two different

realizations of the associahedron with facet normal vectors given by small integer coordinates. Thus, two different ways (due to Hohlweg and Santos) of constructing exponential families of realizations of n-dimensional associahedra with normal vectors of such type are investigated. In particular, the authors classify associahedra in these families modulo linear equivalence of their normal fans.

We come now to a very special type of polyhedra, namely the *geometric fullerenes* (see [743]). In geometry, a fullerene is a convex 3-polytope all whose faces are either pentagonal or hexagonal. This notion became important due to applications (e.g., fullerenes occur as famous molecules or in architecture, the latter reminding us of the eponymous architect Buckminster Fuller). But this type of polyhedra became also mathematically interesting. For example, theoretical generalizations of the notion were given, leading to compact, simply connected, two-dimensional manifolds with boundary. An extensive study of such structures is given in [743]. The older short survey [1686] of Malkevitch on fullerenes collects the graph-theoretic and geometric tools which are useful in studying fullerenes.

Inspired by different generalizations of the concept of classical regular polytopes (as given in the beautiful paper [1084] of Grünbaum and in the monograph [1755]), Schulte and Weiss present in their nicely written survey [2203] a wonderful trip through the world of *abstract extensions of the notion of regular polyhedron in 3-space*. These extensions are mainly defined without the 2-faces themselves, but leaving in a regular sense the *skeleta of edges*. Classifications are presented, such as the complete enumeration of polyhedra regular in this sense, or the fixation of *six families of infinite polyhedra*, some of which create infinitely many different combinatorial types. Connections to *crystal symmetry* are clarified, too.

APPROXIMATION RESULTS

We now come to *approximation results concerning convex bodies and polytopes*, where the latter play the essential role (random approximation and random polytopes are treated in the historical part on stochastic geometry). Gruber [1053] presents a comprehensive survey on such approximation problems. Mainly referring to the *Hausdorff metric or symmetric difference metric*, he collects results on approximation of convex bodies by certain classes of polytopes (e.g., with bounded number of vertices). Similarly, he studies *properties of best-approximating polytopes, related Baire category results, classifications of polytopes due to approximation properties, and approximation by specific polytopes* (like simplices or parallelotopes). Here we also mention the booklet citefaro-riva by Faro Rivas, where, among other things, exterior and interior approximation of centrally symmetric convex bodies by polytopes is studied.

Gruber continues the work [1053] in his handbook article [1059] with similar special aims (for instance, using polytopes with a restricted number of vertices or facets), but extending the used metrics by considering also the *Eggleston metric, p-metric, surface deviation, and Banach–Mazur distance*. Properties of best approximating polytopes, symmetrizations (following suitable requirements), asymptotic estimates, and approximation of or by special classes of convex bodies or polytopes (like zonotopes) are headlines of this exposition. The survey [1066] completes [1059], more dedicated to polytopes. The author studies mainly *approximation of convex bodies by convex polytopes* with at most m vertices (m facets), primarily meaning approximation by *inscribed and circumscribed polytopes*. Mainly upper bounds are studied, and also related *asymptotic estimates and algorithmic approximation* are presented. The used distance measures are *Hausdorff metric, surface deviation, symmetric difference metric, Banach–Mazur distance*, and *Schneider distance*. New approximation results are cited, too.

This survey has a natural continuation in [1068], where the author discusses results on the *approximation of convex bodies by in- and circumscribed polytopes whose vertex number or facet number tends to infinity*, and this regarding *measures based on quermassintegrals, the Hausdorff metric, the Banach–Mazur distance*, and the *Schneider distance*. Also *approximations by floating bodies*, properties of *best and worst approximable convex bodies*, and algorithms for good *constructions of well*

approximating polytopes are presented. In Gruber's book [1070], only relatively short sections are devoted to approximation results.

Another update of [1059] and [1066] is the survey [504] of Bronshteĭn, citing about 230 references and studying different aspects of polyhedral approximation of convex bodies and related problems.

Our next survey to be mentioned here, namely [214] by Bárány, refers mainly to *random polytopes*. But in the final section the author applies the so-called cap-covering technique, which is strongly related to the *floating-body problem*, to present results on the approximation of convex bodies by polytopes and lattice polytopes. Werner [2455] treats also *floating bodies and their variants* (like surface bodies and weighted floating bodies) with applications, among them also being the *approximation of convex bodies by polytopes regarding volume difference*. Finally, we shortly want to mention here the book [2371] of Todd on minimum-volume ellipsoids, since in its final chapter maximum-volume ellipsoids inscribed in polytopes and also approximating parallelotopes are discussed.

Barvinok and Veomett [248] discuss *approximations of convex bodies by ellipsoids, algebraic hypersurfaces, projections of polytopes with a controlled number of facets, and by sections of the cone of positive semi-definite quadratic forms*. The *membership question* (whether or not a considered point belongs to the approximating set) is important in this expository paper.

LATTICE POINTS

A good starting point here is the handbook article [1034] of Gritzmann and Wills about lattice pioint problems. Its content is described above, under the headline GEOMETRY OF NUMBERS.

Brion [503] discusses nice results on *lattice points in convex lattice polytopes*, on the sum of values of *polynomial functions on lattice points of lattice polytopes*, and additive measures on lattice polytopes. He continues in [502] to survey results on lattice points in lattice polytopes in a much broader way, using for example also the *K-theory of toric varieties* as background. Also Lagarias [1487] presents results on *critical lattices, lattice polytopes, and the Ehrhart polynomial; in addition, also results from n-dimensional crystallography* are presented in [1487].

In the expository paper [247] Barvinok and Pommersheim study problems mainly on *lattice points in rational polytopes*, including the *efficient enumeration of lattice points, generating functions for lattice points in rational polytopes* and further related questions. Despite their mainly algorithmical motivation, also theoretical results are given, like connections with the theories of *valuations and toric varieties*.

The nice survey [1596] of De Loera presents *applications of lattice point enumeration in convex rational polytopes*. These applications include the following: *knapsack problems, integral network flows, transportation polytopes and contingency tables, magic squares*, and so on. A short discussion of lattice point problems for regions different to convex polytopes is given, too.

Bárány [215] collects results of the following type: What is the minimal area that a convex lattice polygon with exactly m vertices can have? Thus, questions on *minimal volumes and minimal surface areas of convex lattice polytopes*, on the *number of lattice-affine equivalence classes*, on the *limit shape of lattice polygons* and on further (extremal) problems for convex lattice polytopes are presented.

In [245], Barvinok presents two computational problems concerning *integer points in convex polytopes* whose dimension is given in advance: First, to compute the number of integer points in a polytope which is the intersection of an affine subspace and the non-negative orthant, and, second, treating the intersection of an affine subspace and the unit cube $[0,1]^n$. The author presents heuristic formulas for computing approximately these quantities, based on a probabilistic approach. Also a corresponding formula for 0/1-polytopes is obtained, and the author surveys all occurring formulas. He presents also interesting applications; e.g., these refer also to *transportation polytopes*.

In the exposition [1666], Maehara and Martini collect many results on the geometry of integral lattices, particularly referring to lattice polygons and simplices, and they add also some new results.

COMBINATORIAL PROPERTIES AND BOUNDARY STRUCTURE

Here we mention once more Klee's exposition [1418], motivated by developments in *linear programming*, discussing polytopal graphs and paths in polytopes and posing many deep problems. Similarly, a survey of results on *paths in polytopal graphs and extremal properties of such graphs* is given by Barnette [228]. For example, he presents questions around "long circuits" like Hamiltonian ones. Relations to "non-facets" are discussed (i.e., to n-polytopes which, in the combinatorial sense, cannot present a facet of an $(n+1)$-polytope that has only facets of this combinatorial type). Also solutions of the *upper bound conjecture* and the *lower bound conjecture* (by McMullen and the author, respectively) are briefly presented.

The basic idea of *Gale diagrams* is to represent a polytope P with m vertices in Euclidean n-space by an ordered set of m points in $(m-n-1)$-space whose combinatorial structure reflects the combinatorial properties of P, yielding often new viewpoints and insights. Citing 70 related papers and continuing Grünbaum's exposition in the third part of [1081], McMullen [1739] comprehensively surveys related results, giving a coordinate-free approach to corresponding techniques and a unified theory of transforms and diagrams. Applications and consequences in the following directions are also discussed: *combinatorial classification of polytopes, neighborly polytopes, zonotopes and arrangements of hyperplanes, polyhedral sets, and positive bases of sets*.

In his expository paper [1525], Lee aims to present new methods and techniques from the *combinatorial theory of polytopes* to specialists in *linear programming*. The most important topics he discusses are *Lawrence's volume formula, f-vectors and h-vectors, shellability, bistellar operations and p.1.-spheres, rigidity, triangulations, winding numbers, the moment map, and canonical convex combinations*. In their handbook survey [260], Bayer and Lee give a nice overview to combinatorial properties of convex polytopes. They start with *shellings* (shellable simplicial complexes, upper and lower bound theorem, etc.), relations to *commutative algebra and algebraic geometry* (toric varieties), *Gale transforms and relations to oriented matroids* (polytopes with few vertices, Lawrence polytopes, and so on), *polytope graphs* (here, e.g., Hamilton circuits and Steinitz' theorem), and *combinatorial structure and types* (also barycentric subdivisions). In Bayer's survey [257] two main themes are discussed: first a review on results about f-vectors of polytopes is given, treating recent developments around *flag vectors* and analogues of *h-vectors and the g-vectors of non-simplicial polytopes*. Second, the author discusses subdivisions of polytopes and their effect on the h-vectors. Here particular topics are *shallow subdivisions, local h-vectors, secondary polytopes, fiber polytopes, zonotopes, and arrangements of hyperplanes*.

Kalai [1338] exposes several nice topics from the combinatorial geometry of (mainly simplicial) polytopes. Clearly, the *k-skeleton of an n-dimensional polytope P* is the complex of its faces of dimension at most k. Perles showed that if b is fixed and n is sufficiently large, then an n-dimensional polytope with $b+n$ vertices is of such a type that its k-skeleton is isomorphic to the k-skeleton of some *pyramid*. It follows that if b and k are fixed, we have a fixed bound on the number of *combinatorially distinct k-skeleta of n-polytopes* with $b+n$ vertices, independent of n. Continuing this, the author considers problems around deeper invariants of polytopes, namely their so-called *g-numbers*. These numbers encode the number of faces of any dimension, and they are essential for simplicial polytopes. For an n-polytope there are $[n/2]$ such g-numbers, and they are of great importance in polyhedral combinatorics. Kalai studies problems for polytopes with small g-numbers.

Ziegler [2562] takes Steinitz' explicit description of all f-vectors of all 3-polytopes as a starting point. It is known that analogous investigations on f-vectors and flag vectors of 4-polytopes are not finished, and answers will certainly be very complicated. The author summarizes all that is known and discusses methods and tools that might help to come further. *Fatness of polytopes* is used, together with their complexity. And combinatorial, topological, discrete geometric, and arithmetic aspects play an essential role, too. Partial and deep results refer also to *rational convex 4-polytopes*.

The handbook article [1183] of Henk, Richter-Gebert, and Ziegler concentrates on two basic topics from polytope theory, the second of which is discussed in our subsection 4.7.4. The first one refers to the *face lattice* of convex polytopes with special emphasis on *low-dimensional polytopes* and *polytopes with few vertices*. Selected subthemes of these headlines refer to *representations as convex hulls of finite point sets and as bounded solution sets of finite systems of linear inequalities, basic constructions, polytope graphs*, *4-polytopes* (Schlegel diagrams and Richter-Gebert's universality theorem), and *Gale diagrams*. As nice examples *cubes, cross-polytopes, zonotopes, cyclic polytopes, neighborly polytopes*, and *0/1-polytopes* are discussed.

The famous *Hirsch Conjecture* states that the graph of an n-dimensional polytope or polyhedron P with m facet-defining inequalities cannot have *diameter* (i.e., length of the largest shortest path between a pair of vertices in the graph of P) larger than $m - n$. Ziegler [2564] gives a very nice survey of the history around this famous problem, starting with Hirsch himself (who formulated it in 1957, in a letter to Dantzig), continuing with the important work of Klee and many others, and coming to the counterexample constructed by Santos in 2010 (cf. [2107]). Also Santos himself wrote a survey about this (see [2108]), discussing recent progress with the problem and its background. He discusses activities referring to polytopes but also to simplicial complexes, and he offers also ideas around the web-based project polymath 3, as a possible tool to prove an upper bound possibly also holding for more general geometric objects. A further important survey is due to Eisenbrand [807]. He discusses the *polynomial Hirsch conjecture*, namely stating that the diameter can be bounded from above by a polynomial in m and n, and he concentrates on two ways of further research: distilling a combinatorial property of the edge graphs which could provide the required bound, and confirming the polynomial Hirsch conjecture for particular classes of polytopes.

McMullen presents in [1748] the foundations of the theory of *valuations on convex polytopes*, treating also the *polytope algebra and the theory of weight algebras*. He gives a detailed presentation of his approach to the g-theorem on the face numbers of simple polytopes (much more elementary than Stanley's first proof using toric varieties). He also offers a proof of the *Brunn–Minkowski theorem* and continues with *generalized Alexandrov–Fenchel inequalities*.

Rote [2063] surveys methods which help to construct a convex 3-polytope of given combinatorial structure, i.e., whose edges form a given 3-connected planar graph, focusing on efforts to get small integer coordinates.

It is known that the whole face lattice of a simple polytope can be determined from its graph. More generally, Bayer [258] surveys problems around *reconstructing face lattices from partial information*, in particular (as the headline says) also convex polyhedra from their skeleta. This survey contains a large variety of results and problems in this direction, treating particularly *simplicial polytopes, nearly simple polytopes, cubical polytopes, zonotopes, crosspolytopes, and Eulerian posets* (the latter do not allow counterparts of certain results holding for the more special settings). Finally, interesting open problems are posed.

Novik [1905] surveys recent progress in the study of *face numbers*, but not treating the general case. Presenting also interesting open problems, she restricts herself to *centrally symmetric simplicial polytopes and spheres*. More precisely, she goes from *neighborliness of centrally symmetric polytopes* and the *upper bound theorem for centrally symmetric simplicial spheres* to the *generalized lower bound theorem for centrally symmetric simplicial polytopes* and the *lower bound conjecture for centrally symmetric simplicial spheres and manifolds*.

Bayer [259] presents a comprehensive survey on the *cd-index of Eulerian posets* which is the most efficient encoding of the number of chains, specified by ranks, in such a poset. (A *Eulerian poset* is a graded poset in which every non-trivial interval has the same number of elements of even rank as that of odd rank; also the lattices of regions of oriented matroids are included here.) We start with a look at the history of the notion of cd-index. It started with the *combinatorial investigations of convex polytopes* (Steinitz' characterization of face vectors of 3-polytopes, analogous results in higher dimensions, also with relations to linear programming) and continued with discovered connections

between *polytopes and toric varieties* and, following a conjecture of McMullen, the *characterization of the face vectors of simplicial polytopes* (by Billera, Lee, and Stanley). When studying *ranked partially ordered sets*, Stanley proceeded to count chains in the poset with elements from a specified rank set (instead of counting the elements of each rank), and this step led to investigations of so-called *flag vectors* (whose complete characterization is still open) and *generalized Dehn–Sommerville equations* of polytopes (due to Bayer and Billera), which even apply to Eulerian posets. Then Fine discovered how to encode efficiently the flag vectors, namely with the *cd-index* which is also defined for Eulerian posets. For more about Eulerian posets, we refer to the excellent surveys [2290] and [343] of Stanley and Billera, respectively. In [2290], as special cases occur *face lattices of simplicial polytopes and triangulations of spheres*, and also subdivisions and the *Ehrhart polynomial* are there. And in [343] the *enumeration theory for flags in Eulerian posets* is discussed, yielding particularly the face posets of convex polytopes and regular CW-spheres, and also Bruhat intervals in Coxeter groups. Here occurs also the complete *cd-index*. In [344], Billera collects work of Stanley on f-vectors of polytopes, simplicial complexes, his proof of the upper bound conjecture for spheres, and the role of the g-theorem (using toric varieties). Also consequences of Stanley's results in polytope theory are presented, like *McMullen's proof of the g-theorem, toric h-vectors, flag f-vectors, the (complete) cd-index*, and the *generalized lower bound theorem*. All these things are summarized in the very recent survey [259] of Bayer, in which inequalities on the *cd*-index, connections with other combinatorial parameters, computational aspects, and algebraic approaches are presented.

SOME RESULTS ON POLYHEDRAL MANIFOLDS

The paper [2204] of Schulte and Wills is a survey on *polyhedral realizations of regular maps on surfaces of genus at least 2* in Euclidean 3-space (see also the older surveys [2458] and [2459] of Wills on regular cell decompositions of real 2-manifolds and their polyhedral realizations, yielding (in this sense) *regular polyhedra with hidden symmetries*). A brief introduction to the theory of regular maps and the associated (combinatorial) automorphism groups is given, and those regular maps, for which polyhedral realizations are known, are described, like also the polyhedra themselves. Nice figures show how they can be constructed.

The handbook article [500] of Brehm and Wills completes this picture: new developments and methods in this partial field are surveyed, and some history is given. The most powerful technique studied here is the *realizability of oriented matroids*. Also minimality problems are discussed, e.g., What is the minimal number of vertices of a triangulation of a manifold of given genus? In the final part, containing results on combinatorially regular polyhedra, also variants of regularity are presented: equivelarity, vertex transitivity, and further realization conditions of regular maps.

Schulz [2205] gives a survey on polyhedral manifolds which are contained in the boundary of convex polytopes. The main topics discussed are 2-manifolds with extremal properties (for example, again having minimal vertex number), Hamiltonian 2-manifolds, and corresponding higher dimensional manifolds. A later contribution to polyhedral 2-manifolds with planar faces and local regularity properties is given by Wills in [2467].

In the lecture notes [1470], Kühnel studies tight polyhedral manifolds with minimal total absolute curvature in Euclidean space, where tightness of the corresponding submanifolds is meant to generalize convexity such that the embeddings are embedded as convex as possible. In particular, tight triangulations are considered, and many problems and conjectures are posed; also an extensive list of references is given.

ARRANGEMENTS OF HYPERPLANES

Now we mention some basic surveys and book parts on *arrangements of hyperplanes* (i.e., on space partitions by finitely many hyperplanes, studying the generated face structure, etc.). These arrangements go back to Steiner and others. As already said above, the complete 18th chapter of Grünbaum's

basic book [1085] is dedicated to this nice topic. Grünbaum continues this overview with the updating survey [1082] (more than 80 references). This survey reflects the knowledge about *simplicial arrangements* (for instance, asking for maximal and conjectured minimal numbers of k-faces), the relation to the boundary structure of corresponding *zonotopes* is discussed, and a catalogue of types of simplicial 2-arrangements is presented. Grünbaum continues with the booklet [1083], which studies also arrangements of pseudolines and is described above. Cartier's nice survey [582] on arrangements gives an overview from Steiner's starting points via further history to recent developments important also for *combinatorics, topology, graph theory*, and other parts of mathematics. Without mentioning open problems, the author gives clear definitions, examples, and important theorems. Elementary approaches are not given (for this we recommend [1083]), but essential and deep connections to *matroid theory, algebraic geometry*, and the *theory of reflection groups* are presented. Zaslavsky [2525] surveys the knowledge about *extremal arrangements of hyperplanes* (i.e., about arrangements which have the most or the fewest cells when the number of hyperplanes and the dimension of the partitioned Euclidean or projective space are given). The tool mainly used there is that of *abstract matroid theory*.

A very important contribution to the theory of arrangements is Björner's survey [353] on *general subspace arrangements* (where the participating affine subspaces are not necessarily hyperplanes). For example, the author summarizes results on *Betti numbers* of complements of (general) subspace arrangements. It turns out that the collection of all intersections of non-empty sets of hyperplanes forms a *semilattice* when ordered by reverse inclusion. And when all the participating subspaces contain the origin, this semilattice becomes a lattice. The combinatorial and topological properties of subspace arrangements are comprehensively surveyed, and their relations to further topics (including computational complexity) are discussed.

Stanley's lecture notes [2291] form a collection of six lectures suitable for beginning graduates; it introduces nicely to hyperplane arrangements, posets, and matroids, and it shows relations to *geometric lattices and finite fields*. Interesting exercises are presented at the end of each lecture. Schenck [2132] gives a comprehensive survey on the results and problems treating the *homological approach to hyperplane arrangements*, and he presents also various computational examples. For instance, topics like *free (multi-)arrangements, addition–deletion theorems, and compactifications* are discussed. Also arrangements of plane curves are studied.

Treating arrangements of linear subspaces in connection with *commutative algebra and algebraic geometry*, the book chapter [2133] by Schenck and Sidman is a selective survey, also collecting interesting new problems and old conjectures. For example, *ideals* of such arrangements are studied. We finish this paragraph on surveys treating arrangements with the exposition [2352] of Terao and Yoshinaga, collecting many modern results about arrangements of hyperplanes, and focusing on notions like *free arrangements, free multi-arrangements*, and, in particular, on *Coxeter arrangements*. Further topics presented there are *Zaslavsky's formula on the number of cells, the topology of complex complements, and connections to oriented matroids*.

RESULTS RELATED TO ALGEBRAIC GEOMETRY

Bukhshtaber and Erokhovets [517] expose important bridges between the theory of convex polytopes (particularly, *flag numbers*) and disciplines like *algebraic geometry, topology, and commutative and homological algebra*. Among others, they discuss topics like *polytopes* (basics and constructions), the *Rota-Hopf algebra and flag vectors, topological realization of Hopf algebras, flag polynomials, the g- and h-polynomials of convex polytopes, homomorphisms of the rings of convex polytopes, and the cd-index of polytopes*.

Ohsugi [1913] presents a survey on the relationships between *Gröbner bases of toric ideals* and *triangulations of convex polytopes*. Gröbner bases give such triangulations, but (even more) "good" Gröbner bases give "good" triangulations (so-called *unimodular triangulations*). On the other hand,

for using polytopes to investigate *Gröbner bases of ideals of polynomial rings*, one needs the theory of *Gröbner fans and state polytopes*. Based on notions like *convex polytopes, fans, weight vectors, monomial orders, and Gröbner bases*, the author presents these things in a detailed form. Here we should also mention the paper [1031] of Gritzmann and Sturmfels, treating *computational aspects of Minkowski addition of convex polytopes* and their consequences for *Gröbner bases of polynomial ideals*. For fixed dimension n, they find a polynomial time algorithm for adding k convex n-polytopes with up to m vertices. *Minkowski addition of Newton polytopes* is also successfully used in this paper.

The survey [1326] of Joswig is a valuable contribution to *geometric combinatorics* and to *tropical convexity*. It is mainly devoted to the combinatorial side of *tropical polytopes*, and also related to algorithms. In the final section, the author explains how the software system polymake (designed for the *algorithmic treatment of polytopes and polyhedra*; see [944]) can be applied to study *tropical polytopes*. It is suitable for supporting research in this direction.

COMPUTATIONAL CONVEXITY

We come now to *computational convexity*, a discipline clearly defined by Gritzmann and Klee (see the handbook article [1028], giving a broad overview). Its methods come mainly from *convex geometry and discrete mathematics*, and its problems from *operations research, computer science, linear programming*, and further applied fields. It studies *computational and algorithmical aspects of high-dimensional convex sets, especially polytopes*, in the setting of *normed vector spaces of finite, but generally not restricted dimension* (rather than of fixed dimension). In computational convexity, the underlying model of computation is mainly the Turing machine model (suitable for the study of computational complexity). For investigating algorithmic aspects of general convex bodies (i.e., convex "non-polytopes"), this binary model is often augmented by additional devices called "oracles." For convex polytopes P, discussed problems can depend on the so-called V-representation (with P as convex hull of finitely many points with rational coordinates) or H-representation (as the intersection of finitely many closed half-spaces defined by linear inequalities with rational coefficients). The following research and expository works of Gritzmann and Klee are well suited as surveys to the field, showing also how it started growing in the early nineties. In [1024], inner and outer j-radii of polytopes in finite-dimensional normed spaces are studied. The inner j-radius of a convex body K is the supremum of the radii of the j-dimensional balls contained in K, and the outer j-radius of K measures how well K can be approximated, in a minimax sense, by an $(n\check{}j)$–flat. In [1027], the computation of *volumes and mixed volumes of convex polytopes and more general convex bodies* is investigated, and in [1026] recent results on computing, approximating, or measuring convex bodies which are, in a suitably defined class, the smallest containing a given convex body K, or the largest being contained in K. For example, this can refer to homothety classes or to affinity classes. Many applications of *containment problems* are given, coming from global optimization, pseudo-Boolean programming, sensitivity analysis of linear programming, set separation, the existence problem for Hadamard matrices, and so on. And also the handbook article [1025], mainly treating the *strong influence of convex geometry and polytope theory in convex and linear programming*, and referring also to the *ellipsoid method and Karmarkar's projective algorithm*, is an essential source of information.

The nicely written expository paper [540] of Burger et al. surveys topics which are close to contents of Gardner's book [930], namely on *projection bodies and shadow areas*. In particular, this is studied for polytopes, and the authors present also results on and computational approaches to *extremal shadow areas*, where the projected body is mostly a polytope.

APPLICATIONS: OPTIMIZATION

The basic and early expositions [1418] by Klee and [228] by Barnette, already cited above, show how important applications in optimization have been for the whole field. Also with the back-ground of

linear programming (LP), Klee and Kleinschmidt [1420] give a complete picture of results on the so-called *d-step conjecture* (which we here formulate for n instead of d): If $\triangle(n, m)$ denotes the *maximum diameter of the edge graph* of an n-dimensional convex polytope with m facets, then $\triangle(n, 2n) = n!$ This and related conjectures are important in view of studying the *computational complexity of edge-following LP algorithms*. Equivalent statements, relations to *linear programming*, low-dimensional results (like $\triangle(n, 2n) \leq n$ for $n \leq 3$, but not true for sufficiently large n), and *extensions to more general complexes* are given; lower and upper bounds on $\triangle(n, m)$ and results for special cases are also discussed. For a summary of more recent results in this direction and on the related *Hirsch conjecture*, we refer to Eisenbrand [807]. The d-step conjecture is also discussed in the handbook article [1421] of Klee and Kleinschmidt. In this survey the authors lay emphasis on combinatorial aspects of polytope theory. The *presentation of polytopes* via their vertices or the affine hulls of their facets are compared, and *polytopal realizability of complexes, enumeration of certain combinatorial classes, face-vectors with upper and lower bound theorems, the description of all face-vectors of polytopes, and of polytopal graphs* (in view of the d-step conjecture) are discussed themes. The authors present also a comprehensive list of related references.

We now come to *properties (of special types) of polytopes which are important in optimization theory*. We start with the surveys [169] of Bachem and Grötschel as well as [2004] of Pulleyblank, giving summaries of polyhedral methods developed for optimization problems. For example, in [169] faces and minimal representations of polyhedra are studied, and lattices associated with polyhedra are presented, too. Pulleyblank [2004] understands the term *polyhedral combinatorics* as an application of the theory of linear systems and linear algebra to combinatorial problems in this sense.

In the handbook article [542], Burkard discusses problems from (mixed) integer programming and combinatorial optimization. He shows the *strong influence of convexity* here, and how useful corresponding notions and tools from convex and discrete geometry are. These are *integral polytopes, polyhedral combinatorics, matroid polytopes, stable set polytopes, Delaunay triangulations, and rectilinear convexity*. The same author discusses in [543] problems from integer programming showing the important role that (generalized) convexity notions can play here. Similar to [542], *integral polyhedra, polyhedral combinatorics, and the use of convexity concepts* in algorithmic design are studied.

Kalai [1340] surveys nicely far-reaching applications of the fundaments of *linear programming* to the combinatorial theory of *simple polytopes*. He also describes subexponential randomized pivot rules and upper bounds on the diameter of graphs of polytopes. In this interesting survey, geometric fundaments and important developments around the *simplex algorithm* are nicely shown. A *0/1-polytope* is an n-dimensional convex polytope whose vertices are among those of the unit cube (and which occur always again in optimization theory). In the exposition [2561], Ziegler deals with the *combinatorics and geometry of 0/1-polytopes* collecting some very interesting properties of them. Among such properties of 0/1 polytopes, one could mention that there are doubly exponentially many of them (in terms of n), that they can have large number of facets, and that their defining inequalities can have very large integer coefficients. The article [2563] of Ziegler successfully aims to understand linear programs through geometric reasoning. Seeing a linear program as a polytope in space combined with an optimization direction, he convinces the reader that such geometric insights are indeed important for understanding several sides of *linear programming* in a better way. It is also shown that such a view is helpful in general, and particularly important also for the *simplex algorithm under various pivot rules*.

Polytopes being extremal with respect to path lengths in their edge graph (i.e., so-called *worst-case polytopes*) are also intensively studied in [2563]. It is shown that they all have (exponentially) long monotone paths of vertices in some direction. Hojny et al. [1223] give a comprehensive survey of 0/1 knapsack polytopes (i.e., the convex hull of all 0/1 vectors satisfying a single linear inequality which is given with non-negative coefficients). Among various other things, they discuss polyhedral properties, geometric properties for small dimensions, and extensions to (mixed-)integer knapsack polytopes and variants are also given.

FURTHER RESULTS

Sabitov [2083] surveys interesting results from the *metric theory of convex 3-polytopes*. These refer to *isometric immersions and embeddings of given polyhedral metrics in 3-space, rigidity questions, flexions of polyhedra without the face-rigidity condition*, a generalization of the Heron–Tartaglia volume formula for polyhedra, and an affirmative solution of the *bellows conjecture* on the volume of flexible polyhedra. (Note that this conjecture asserts that all flexible polyhedra keep a constant volume as they are flexed.)

Virtual polytopes have a long history, at least going back to Alexandrov's proof of the quadratic inequalities for mixed volumes. The set of all convex n-polytopes is a commutative monoid under the operation "+" of Minkowski addition. Then the group of *virtual polytopes* is the group of equivalence classes of pairs (P_1, P_2) of convex polytopes (identified with the difference of support functions of P_1 and P_2) under the operation $(P1, P2) + (Q1, Q2) = (P_1 + Q_1, P_2 + Q_2)$. Two pairs (P_1, P_2) and (Q_1, Q_2) are equivalent if $P_1 + Q_2 = Q_1 + P_2$. Thus one can say that the *set of virtual polytopes* is the *Grothendieck group of the commutative monoid of convex polytopes under Minkowski addition*. The survey [1937] of Panina and Streinu gives a comprehensive review on the theory of virtual polytopes. First the authors introduce the described concept, and then they study faces and different representations of virtual polytopes, especially taking care also of the two-dimensional case. These representations refer to the *algebra of polyhedral functions and the algebra of polytopes*, where *support functions and a system of translated cones* are used. More geometric representations refer to the three-dimensional case. Regarding applications, it is shown how virtual polytopes can be used to treat *Alexandrov's problem on the uniqueness of convex surfaces, valuations* (including volume and counting integer points in lattice polytopes), *mixed volumes, and decomposition of polytopes with respect to Minkowski addition*; even a link to *projective toric varieties* is given.

12.4.3 Convex algebraic geometry

Convex algebraic geometry is a young field between convex geometry and algebraic geometry. There are related notions which play from the very beginning an essential role: toric varieties, polyhedral structures, spectrahedra, tropical geometry,...

12.4.3.1 Some history

We start this subsection by mentioning the "field-defining article" [364] and the proceedings [363] containing it. Namely, in [364] Blekherman, Parrilo, and Thomas clearly describe the modern field of *convex algebraic geometry*: it arises from a successful synthesis of ideas and techniques from *optimization, convex geometry, and real algebraic geometry*. The studied objects are convex sets with algebraic structure which occurred, before discussing them from a unified viewpoint, independently in convex geometry, algebraic geometry, optimization, and also analysis. Recently they are investigated in a unified manner, with methods using interactions between algebra and convexity. This relatively young field opens interesting connections between mathematical disciplines and applied fields. Especially in [363] fundaments and progress of convex algebraic geometry are presented. And due to the important role that semi-definite optimization plays here, we have to mention also the "Handbook on Semidefinite,

Conic and Polynomial Optimization" [110] (edited by Anjos and Lasserre; see also below), in addition to all other handbooks cited in our book here.

Algebraic approaches to optimization problems, which are formulated in terms of multivariate polynomials, imply the necessity to understand the structure of non-negative polynomials, the efficiency and complexity of distinct representations of algebraic sets, and to advance the algorithmical side of this development. In this interplay, convexity created new ideas and tools which made mutual analogues visible and usable (e.g., within the conceptual apparatuses). Thus, *a natural intersection of the fields convex geometry, (real) algebraic geometry, and optimization came to be*, now known as *convex algebraic geometry*. Since *toric varieties* (i.e., normal varieties containing an algebraic torus T as an open dense set such that the natural action of T on itself extends the whole variety) are algebraic varieties whose geometry is already determined by combinatorial information, their theory connects algebraic geometry in a natural way with convex geometry and polytope theory.

Related developments started at the beginnings of the 1970s. We mention here the historical survey [783] by Dwilewicz, where also these *relations of convexity to toric varieties* are discussed. And Teissier's paper [2351] is an example of an early summary of such relations between algebraic geometry and convexity; e.g., he deduces the Alexandrov–Fenchel inequalities from the Hodge index theorem, and presents further results of such type treating algebraic varieties and integer points in cones or polytopes. This leads also to the notion of *tropical geometry* which is nicely described in the preface of the book [1658] by Maclagan and Sturmfels: "*Tropical geometry* is an exciting new field at the interface between algebraic geometry and combinatorics with connections to many other areas. At its heart it is geometry over the tropical semiring, which is $\mathbb{R} \cup \{\infty\}$ with the usual operations of addition and multiplication replaced by minimum and addition, respectively. This turns polynomials into piecewise-linear functions and replaces an algebraic variety by an object from polyhedral geometry, which can be regarded as a 'combinatorial shadow' of the original variety." A good introduction to ideas and methods from tropical geometry (besides the book [1658]) is given by the nice exposition [2031] of Richter-Gebert, Sturmfels, and Theobald.

As examples of typical geometric objects studied in convex algebraic geometry, we mention *spectrahedra* (and, more specific, e.g., *multifocal ellipsoids* which generalize usual ellipsoids by having finitely many foci) as well as *Newton polytopes*. Semi-definite programming is the computational problem of minimizing a linear function over a *spectrahedron* which is the intersection of the cone of positive semi-definite matrices with a linear space. Excellent surveys on properties and many applications of spectrahedra are [2405], [2406], and [2358]; multifocal ellipsoids are investigated in [1895] and [1294]; see also our extra-subsection at the end of subsection 2.9.1.6. *Newton polytopes* are integral polytopes associated with multivariate polynomials; they are of central interest in tropical geometry and characterize the Gröbner bases for an ideal; see the very recent survey [1355] by Kazarnovskii, Khovanskii, and Esterov.

Coming from convexity, we collect in the following some *literature related to convex algebraic geometry and related tropical geometry*. Again we concentrate on monographs, surveys, and handbook articles.

12.4.3.2 Books and book chapters

Most of the following books are related to the interplay between algebraic, convex, toric, and tropical geometry. In various cases also applications are motivating or at least very important issues, namely in fields like optimization and convex analysis.

POLYTOPES AND TORIC VARIETIES

The promising character for tremendous progress of convex algebraic geometry, connecting interesting fields and having interesting applications around, can be seen already in the important monograph [1909] of Oda. And similarly, one should also refer to Ewald's book [826], described below and mentioned also in our polytope part. After defining *fans and toric varieties*, Oda constructs various related examples of complex-analytic spaces and holomorphic maps, and interesting basic questions on the *birational geometry of toric varieties* are reduced into those on subdivisions of fans. He then deals with *integral convex polytopes and projective toric varieties*, discussing also cohomology aspects of compact toric varieties. Then holomorphic differential forms on toric varieties are treated, not only related to deformations and degenerations of complex-analytic spaces, but also interesting for *commutative algebra*. In the final part some applications are shown: *cusp surface singularities* and their higher dimensional analogues, and certain *complex compact manifolds as quotients of open subsets of toric varieties* with respect to the free actions of discrete groups are constructed. In the appendix some basic and deep results on convex polytopes are presented. Here we mention also the related survey [1910] of Oda himself, in its second part updating results from [1909], e.g., referring to *combinatorial properties of convex polytopes and toric varieties*. See also the related survey [1277] of Ishida, studying toric varieties whose fans are dual to polyhedral sets. This might be also the right place to mention a few forerunners of Oda [1909] suitably introducing toric varieties: these are the book [1360] of Kempf et al. and the survey [705] of Danilov.

Fulton's book [910] is devoted to toric varieties (toric embeddings) showing the interesting relations between *algebraic geometry and the geometry of convex cones* (together with lattice points). A basic dictionary, translating the notions of one field into those of the other one, was established already at the beginning of the 1970s. Besides an increasing field of applications, also the basic theory and the mentioned dictionary continued to grow. In one direction, this dictionary uses convex geometry to construct many nice examples of varieties and phenomena in algebraic geometry. This yields testing grounds for general theories, but serves also as good material for an introduction to algebraic geometry as well. In the other direction, theorems from algebraic geometry, when applied to toric varieties, very often lead to new insights in convex geometry. As a highlight, the closing chapter contains an application to the *geometry of convex polytopes* in terms of Stanley's proof of McMullen's upper bound theorem. This book should be directly compared with the monographs [1909] and [826] of Oda and Ewald, respectively.

The book [826] of Ewald, already announced above, gives a nice introduction to the theory of *convex polytopes and polyhedral sets in relation to algebraic geometry*, using the link of *toric varieties*. It was prepared somehow by Ewald's handbook article [825] and by [824]. The first part of the book is devoted to *combinatorial convexity* and starts with the basic notions related to convex bodies, the *combinatorial theory of polytopes and polyhedral sets* (also Gale transforms), basics of the *topology-oriented part of polytope theory* (e.g., Euler and Dehn–Sommerville equations), *stellar operations and shellings, Brunn–Minkowski theory, lattice polytopes, and fans*. The second part presents the *theory of toric varieties*. First affine toric varieties are introduced by lattice cones and the algebra given by monoids of all lattice points in cones, yielding then toric varieties in general. Fundamental notions like torus action, blow ups, and resolution of singularities are presented, leading the reader to *sheaves and projective toric varieties*; the author discusses then the usage of homology and cohomology to study *topological properties of toric varieties*. It is obvious that the nicely presented "dictionary" between notions from the convexity part and notions of algebraic geometry help readers, which are non-experts in one of both fields, to get an easier approach to the other field.

Containing more details with clear examples, more figures and more exercises than [1909] and [826], the book [668] of Cox, Little, and Schenck can be recommended as useful supplementary literature to these two monographs. It contains also parts directly related to convexity and convex polytopes, namely in Chapters 7 and 12. More comprehensively we discuss this book once more below.

Gröbner bases are distinguished bases of *ideals in polynomial rings*, and they became more and more a useful tool for certain algorithms in algebraic geometry and commutative algebra. Within this framework, the monograph [2317] of Sturmfels is devoted to the *interplay of computational algebra and polytope theory*. From the side of commutative algebra, toric ideals (as *defining ideals of toric varieties*) come into the game, and they are related to *polyhedral geometry*. Already in one of the first parts, the *state polytope* of an ideal (as a *generalization of the Newton polytope of a principal ideal*) is studied. More generally, the author associates geometric objects like convex polytopes or simplicial complexes with toric ideals, using *Gröbner bases* to study these geometric objects, and conversely. Motivations for this book came also from *integer programming and computational statistics*, and it shows that there are also further substantial relations to fields like *algebraic topology, probability and general convexity*.

The well-written monograph [668] by Cox et al. on *toric varieties* covers classical material as well as many results which, before its appearance, could only be located in research literature. The authors excellently present the powerful methods of algebraic geometry, combined with tools from related fields like also polytope theory. For example, the *relationships between toric varieties and convex polytopes* occur explicitly in Chapter 7, where the *relations between polytopes and projective toric varieties* are extended to those between *polyhedra and projective toric morphisms*. The next chapter refers to reflexive polytopes in relation to Gorenstein-Fano toric varieties, and Chapter 9 offers applications of cohomology theorems coming to the *Dehn–Sommerville equations for simple polytopes* and *lattice point enumerators via the Ehrhart polynomial*. Chapter 12 deals with *topological invariants of toric varieties*, yielding applications for polytopes. A survey on the *history of toric geometry* (from the 1970s on) can be found in Appendix A, and Appendix B offers computational methods. The book is useful for beginners as well as researchers in algebraic geometry, convexity, combinatorics, and commutative algebra; it has encyclopedic character and is, containing nice introductory parts, at the same time well suited as an excellent textbook.

Based on *toric metrized line bundles*, Burgos Gil, Philippon, and Sombra derive in their monograph [541] a closed formula for the *height of a toric variety*. Connected with this notion, they also study *models of toric varieties over a discrete valuation ring*, corresponding to *metrized line bundles, their associated measures*, and, local, as well as global, *heights*. Using tools like *polyhedral complexes, lattice polytopes, real Monge–Ampère measures, concave functions, and the Legendre-Fenchel duality*, they confirm that such heights can be expressed as a sum of integrals over a polytope of a family of concave functions. With this type of results, they succeed in presenting *close relations between the arithmetic geometry of toric varieties and convex analysis as well as convex geometry*. Varying the used metrics, the authors also derive further related height formulas. Thus, also further interesting metrics arising from polytopes are taken into consideration. The book contains a detailed introduction of the subject and also various examples, with helpful figures.

Toric topology generalizes toric geometry, roughly speaking, in the following sense: it can be considered as *the topological analogue of toric geometry in algebraic and symplectic geometry*, yielding many interesting research problems, e.g., regarding several types of (topological) rigidity problems or computing equivariant cohomology of manifolds with torus actions by combinatorial methods. The book [519] of Buchstaber and Panovis is a perfect reference for exploring that area, and in this sense; it establishes the foundations of the described issue and will certainly stimulate further applications. In the first chapter, *polytope theory* is presented, e.g., the *g-theorem* treating the face vectors of simple convex polytopes (and applying toric geometry to combinatorics), *Gale duality and Gale diagrams, Dehn–Sommerville relations, differential algebra of combinatorial polytopes*, and further related topics. The second chapter treats combinatorial structures in the orbit spaces of torus actions, leading also to *simplicial polytopes and complexes* as well as to *triangulations of manifolds*. Chapter 3 shows, among various other contents, also *face rings of simplicial complexes and generalized Dehn–Sommerville relations*. In the fifth chapter *toric varieties and manifolds* are studied, where the background is algebraic and symplectic geometry. And in Chapter 8 polyhedral geometry plays again a role, by introducing *polyhedral product spaces*.

TROPICAL GEOMETRY AND CONVEXITY

And in the book [2319] Sturmfels treats the topic of *polynomial equation systems and how to solve them*. In its third chapter the author presents once more the contents of the nice exposition [2318], where *mixed volumes of Newton polytopes* play an essential role (see, e.g., also Chapter 6 of [2319]). In Chapter 9, *ways to visualize systems of polynomial equations* are discussed, and it is shown how one of them (using also polyhedral complexes called "tropical varieties") leads to what is called now *tropical geometry*.

The notion of *tropicalization* replaces an algebraic variety by an object from polyhedral geometry and, thus, forms a *bridge between algebraic and convex geometry*; more precisely, it can be used to transform problems from algebraic geometry into corresponding problems typical for convex geometry, which are usually easier. Since polyhedra and polytopes play an essential role here, Maclagan and Sturmfels, introducing the reader with the essential book [1658] into this field, recommend Ziegler's book [2560] as the best suitable starting point to be studied first when coming from the side of *convex sets and polyhedral combinatorics*. Special topics in the book close to convexity are *polyhedral geometry and polyhedral complexes, mixed volumes* (of Newton polytopes), *combinatorics* (also of surfaces in 3-space), and *tropical convexity*. Further discussed or used topics are *multilinear algebra, Gröbner bases suitable for algebraic varieties, matroid theory, and tropical and toric varieties*. This well-written book is suitable for graduates and, measured at its rich contents, accessible with a minimal background from algebraic geometry.

The recent monograph [1327] of Joswig clearly shows, on the graduate level, the connections between tropical geometry and optimization. Related applied research directions (phylogenetics, neural networks, game theory, economics, etc.) are used to show this purpose; the author demonstrates how the combinatorics from tropical geometry can influence optimization. The core of the book (Chapters 5 through 9) refers to *tropical convexity*, i.e., to topics and notions like *tropical convex hulls, tropically convex sets, Carathéodry's theorem, arrangements of tropical hyperplanes, intersections of tropical half-spaces, matroids and their polytopes, related polyhedral geometry*, and so on. The author has chosen a nice expository style, and the reader need not have prior knowledge in algebraic geometry.

COMPUTATIONAL ALGEBRAIC GEOMETRY

The following three books combine notions like oriented matroids, polytopes, and relations also to algebraic geometry; one can classify them as representatives of *computational algebraic geometry*.

The field *computational synthetic geometry*, described by the book [421] of Bokowski and Sturmfels, deals with the algorithmic side of important *realizability problems from discrete geometry and convexity*, e.g., realizations of combinatorial structures (like *abstract polytopes, oriented matroids, hyperplane arrangements*, etc.) over certain fields. It is shown that algorithms for such realizations exist iff arbitrary polynomial equations are decidable over the corresponding fields. This gives many new discoveries, using the constructed objects (like polytopes). Particularly interesting in this setting is the discussion of the famous *Steinitz problem on characterizations of polytope graphs*. The used techniques come from *computational algebraic geometry, convexity, and automated theorem proving*.

In the monograph [354] of Björner et al., treating *oriented matroids*, the authors discuss the relation between matroids and *toric varieties* following Oda [1909]. The *lattice polytopes* associated with the toric varieties are the basis polytopes (defined as convex hulls of all incidence vectors of bases) of the considered matroids; see the end of Section 2.4.

The second part (three chapters) of the monograph [1328] of Joswig and Theobald, called "Nonlinear Computational Geometry", deals with *resultants, Gröbner bases, and the solution of polynomial systems with elimination*, creating *algorithms in polyhedral and algebraic geometry*. Using a nice and

wide mixture of methods and tools, this part of the book can be seen as an interesting example presenting the field of *computational algebraic geometry*. The authors use their representations in the final part for applications also in computer graphics, regarding curve reconstructions and robotics.

FURTHER RELATED BOOKS

The monograph [948] of Gel'fand, Kapranov, and Zelevinsky can be seen as a revival (and extension) of the *classical theory of resultants and discriminants*. Various types of polytopes, which became important in algebraic geometry and convex algebraic geometry, are studied. It turns out that especially in the (more combinatorial) second part of this book, consisting of seven chapters, *polytope theory* plays an essential role. Taking A as a finite set of (Laurent-)monomials, the authors study the vector space C^A of polynomials spanned by them. First they study *toric varieties*, and in the next chapter they present *Newton polytopes*, i.e., integral polytopes associated with a multivariate polynomial. These are a central object of study in *tropical geometry*, characterizing the *Gröbner bases for an ideal*. The authors also introduce *Chow polytopes of hypersurfaces*; they are Newton polytopes of the defining polynomials (i.e., the Chow polytope of an algebraic cycle is the Newton polytope of its Chow form being an element of the homogeneous coordinate ring of a Grassmannian). Examples of Chow polytopes (like *matroid polytopes and permutahedra*) are given. Then *secondary polytopes of given lattice polytopes* are introduced; their vertices correspond to certain triangulations of the given polytope into simplices. The authors give equivalent approaches to this notion via *fiber polytopes and Gale diagrams*, and also here examples (such as permutahedra and skew cubes) are given. Also in later parts of the book, interesting results on Newton polytopes and related notions are obtained. In addition, the authors present Schläfli's method for computing hyperdeterminants, a further nice link to polytope theory. We could only hint at a small part (topics related to polytope theory) of the broad spectrum of contents of [948]; the whole field treated in this book is much wider and covers also recent developments in algebraic geometry, homological algebra, and combinatorics.

In [1597] the authors start with natural basics from *polytope theory* (yielding the Minkowski–Weyl theorem, the Farkas lemma, and the ellipsoid method for systems of linear inequalities) and typical results and methods from the *geometry of numbers*. They continue with methods from *discrete and integer optimization* and present various applications, e.g., regarding integer polynomial optimization and interesting integer programming problems. Also the *computation of the convex hull* of various combinatorially defined sets of integer points plays a role. Showing advances in discrete optimization, the book successfully *connects optimization with the algebraic geometry over the reals*. This is done via *methods from algebraic geometry, commutative algebra, convex and discrete geometry*, and further tools.

In [1885], Netzer and Plaumann study *spectrahedra*, i.e., the solution sets to linear matrix inequalities, emerging in *convex and polynomial optimization, combinatorial geometry, and convex algebraic geometry*. They present interesting examples and show how to determine the convex sets that can be represented and approximated as spectrahedra and their projections. The authors show the developments (about 15 years) around this notion, and also the rich variety of methods is interestingly described.

12.4.3.3 Handbooks and proceedings

In addition to various handbooks cited in our book at other places, we mention here only one handbook, namely the "Handbook on Semidefinite, Conic and Polynomial Optimization" [110]. Among other things, the following topics close to convexity are discussed in articles from there: *Semi-definite representation of convex sets, convex hulls of algebraic sets, and further occurrences of convexity*

in semi-definite optimization. As an example, we mention the exposition [1178] of Helton and Nie, dealing, e.g., with *convex hulls of bounded semialgebraic sets*.

We continue with proceedings and start with the following ones (we were not able to find editors of them):

Geometry of toric varieties and convex polytopes (Japanese). In: *Proceedings of a Symposium held at the Research Institute for Mathematical Sciences, Kyoto University, Kyoto, June 5–9, 1995*. RIMS Kokyuroku **934** (1996). Kyoto University, Research Institute for Mathematical Sciences, Kyoto, 1996.

In these proceedings, which also emphasize the *connections between toric varieties and polytope theory*, 13 articles are collected (9 of them in Japanese), treating, e.g., *hyperplane arrangements and toric varieties, special types of convex sets, Voronoi diagrams in spaces of constant negative curvature, and triangulations of convex polytopes*.

The collection [146] contains articles on *polyhedra and polytopes, lattice polytopes with given polynomials, geometric combinatorics and compact toric surfaces, reflexive polytopes, and cones of flag vectors of 4-polytopes*.

In the proceedings [148], edited by Athorne, Maclagan, and Strachan, a nice introduction to *tropical algebraic geometry* is given, and articles on concrete themes from tropical geometry are there.

The book [363] by Blekherman, Parrilo, and Thomas collects articles (as chapters) which give an overview on research aspects arising from the interplay between classical algebraic geometry, optimization theory and convex geometry, leading to articles treating results typical for *convex algebraic geometry*. The study of structural properties of the feasible region of a semi-definite program (i.e., of a *spectrahedron*) clearly shows this interplay and the important role that convexity plays in this setting. For example, there are chapters on duality notions (considering *duality in convex geometry, Lagrangian duality in optimization, and projective duality for algebraic varieties*), identifications of classes of convex sets which are *spectrahedra or projections of spectrahedra*, and *spectahedral (outer) approximations of convex hulls of algebraic sets*. Using tools from algebraic geometry and convex analysis, also geometric properties of these and related set families are studied. This book shows also how applied disciplines like *control theory, combinatorial optimization, approximation theory, and theoretical computer science* benefit from convex algebraic geometry. And even more: The three editors and eleven authors succeeded in bringing together two fields (convex algebraic geometry and semi-definite optimization) which had no proper overlap before. This success is reached due to a successfully unified representation. And an impressively large number of refreshing exercises supports the readability of the book.

The proceedings [1195], edited by Hibi and Tsuchiya, collect contributions treating (several types of) *lattice polytopes and valuations, lattice zonotopes, the polytope algebra, mixed subdivisions of polytopes, moment polytopes of subvarieties of products of Grassmannians, Dehn–Sommerville equations, Eberhard-type theorems, projective toric surfaces, toric Fano varieties*, and many other results on polytopes and types varieties. The final paper discusses interesting problems left open from a 2018 workshop on lattice polytopes within this framework.

12.4.3.4 Selected surveys and expository papers

We start with expositions closely related to *toric varieties*.

TORIC VARIETIES AND POLYTOPES

The survey [1908] of Oda is a forerunner of the monograph [1909] and should also be studied in connection with the exposition [1277] of Ishida. In the survey [1910], Oda studies the theory of toric varieties combining algebraic geometry and the geometry of convex cones and lattices. He presents developments referring to *combinatorial properties of convex polytopes, intersection cohomology of*

toric varieties, subvarieties of toric varieties, and secondary polytopes and their conic versions. An early and similar exposition of Ewald related to convex algebraic geometry is [824]. Among other things the author *characterizes mixed volumes of lattice polytopes as intersection indices of certain varieties*, and *he deduces the Alexandrov–Fenchel inequality from the Hodge index theorem*.

In the exposition [2316], Sturmfels demonstrates how sparse systems of polynomial equations can be understood as a topic in the intersection of the fields *convex combinatorics, toric varieties*, and, further on, *computational algebra* and *invariant theory*. The author uses recent advances in combinatorics, derived by Gel'fand, Kapranov, and Zelevinsky in the context of generalized hypergeometric functions. He extends his former results on *Gröbner bases of toric varieties*, and notions like *secondary polytopes* and so-called *A-resultants* are of central importance. (Special cases of the latter are Sylvester resultants and hyperdeterminants.) Besides giving a nice introduction to this little field by emphasizing the significance of sparseness, the author also shows the importance of computing A-resultants. Interesting examples and also some new results are given.

Cox [667] presents a complete and very useful survey of old and new results on *toric varieties*, including different presentations and recent applications of them. In particular, also Batyrev's famous work on *reflexive polytopes and mirror symmetry* is presented. (Reflexive polytopes are those polytopes which have, together with their polars, only lattice vertices, and the mentioned mirror symmetry suitably exists between Calabi–Yau varieties.)

TROPICAL GEOMETRY AND CONVEXITY

The paper [2031] of Richter-Gebert, Sturmfels, and Theobald can be seen as an *exposition of fundaments in tropical algebraic geometry*. The additive group $(\mathbb{R}, +)$ forms together with the binary operation "min" a semiring, called the tropical semiring. This semiring is fundamental for a theory (the so-called *tropical geometry*) having many analogues with classical complex algebraic geometry. The min-operation plays the role of classical addition, and the classical addition that of tropical multiplication. The geometric objects of tropical geometry, called *tropical varieties*, are polyhedral cell complexes contained in tropical affine or tropical projective spaces. Topics discussed are *tropical planar curves, tropical linear subspaces, Bézout's theorem in the plane, classification of tropical planar quadrics, counterexamples to a tropical version of Pappus' theorem*, and so on. A preprint version of this paper became already a standard reference. This field has interesting overlaps with *convex geometry*, yielding the partial field of *tropical convexity*. This is also exposed in the work [735] of Develin and Sturmfels, with many "tropical notions" defined there. Among other things, the authors show that the combinatorial types of tropical polytopes are in bijection with regular triangulations of products of two simplices. Proving this result, various properties of tropically convex sets and tropical polytopes are proved, similar to analogous statements in classical convexity. Applications are discussed, e.g., with respect to phylogenetic trees.

The survey [1780] of Mikhalkin gives an overview to *tropical algebraic geometry*, e.g., discussing the notions of *tropical varieties and tropical morphisms*, and some *intersection theory on tropical varieties*. The main objects studied in tropical algebraic geometry are piecewise-linear in nature: for example, *tropical varieties are finite polyhedral complexes*, and so on. Hence one can demonstrate that tropical objects are degenerations of respective classical objects from algebraic geometry. This allows to translate some classical algebraic geometry problems into the language of tropical geometry, where they are often easier to solve. For example, one given example, nicely showing this principle, is the presented tropical solution to the problem of computing Gromov–Witten and Welschinger invariants of toric varieties. Relations to *enumerative geometry and computational geometry* are discussed, too.

In the expository paper [1585] Litvinov demonstrates that *tropical algebraic geometry* can be treated as a result of the Maslov dequantization applied to traditional algebraic geometry. From this point of view, one can see idempotent analysis as the limit of classical analysis over the tropical algebra; it is analysis over idempotent semirings and semifields. In this paper, the author shows how classical

nonlinear problems may become linear ones over the tropical semirings. One example of a semiring which is not a semifield is emphasized: the *family of n-dimensional compact, convex sets endowed with the operations of convex hull of the union, and Minkowski addition*. Further studied notions are *Newton polytopes of polynomials* (as subdifferentials of the dequantized polynomials) and, more general, *Newton convex bodies*.

Maclagan [1657] gives an expository introduction to the field of *tropical algebraic geometry*, based on his lectures at the *Workshop on Tropical Geometry and Integrable Systems* in Glasgow, July 4–8, 2011, and at the *ELGA 2011 School on Algebraic Geometry and Applications* in Buenos Aires, August 1–5, 2011. The text contains no proofs, but many nice examples. They present the main objects and methods of tropical geometry in an easy way.

Gubler [1093] generalizes basic results from tropical geometry to arbitrary non-Archimedean valued fields. He uses techniques from non-Archimedean analysis.

FURTHER RELATED TOPICS

Teissier [2351] surveys two important developments in *the field between convex polytopes and algebraic geometry*. Let $P_1, P_2, ..., P_m$ be a finite family of convex polytopes in Euclidean n-space; one can attach to this family an n-dimensional algebraic variety X such that to each P_i an inversible bundle L_i corresponds, thus obtaining a dictionary between combinatorial properties of P_i and properties (in the sense of algebraic geometry) of L_i. Based on this, the paper deals with properties of L_i and X in view of *Hodge theory*. The two announced main topics are first the deduction of the *Alexandrov–Fenchel inequalities* between mixed volumes of convex bodies, and, second, using the Lefschetz theorem and the Poincaré duality on X, necessary conditions on the *numbers of k-dimensional faces of simplicial polytopes*. The representation is nice and detailed, especially the correspondence between integer points in cones and polytopes and algebraic varieties.

Investigating an important class of semialgebraic varieties arising in combinatorial geometry, Vershik [2401] excellently exposes results on the *topology of the realization spaces of oriented matroids and of convex polytopes*. Order types are the natural equivalence classes of m-point sets in a projective space over an ordered field, and one central question is whether they are connected topological spaces. The answer is no, and it turns out that they can be *homotopy equivalent to semialgebraic varieties* (*Mnëv's universality theorem*). Extensions, applications, and different aspects of results in this spirit are collected and nicely presented, and ten deep open problems enrich this excellent publication. Similarly deep and related topics are discussed by Vershik in [2402], again combined with interesting open questions.

In the handbook article [260] of Bayer and Lee, cited already in the polytopes part of this chapter here, combinatorial aspects of convex polytopes are widely studied, and in Section 3 the authors discuss how successful also methods from algebraic geometry influenced the field.

In the nice expository paper [2318] of Sturmfels (the author won the 1999 Lester Ford Prize for its beautiful exposition), the *relationship between the number of solutions of a polynomial equation system and the Newton polytopes associated with the considered polynomials* is treated. Sturmfels starts with Descartes' rule of signs (on the number of positive real roots of a univariate polynomial) and Bézout's theorem (on bounds for the number of zeros in terms of the degrees of the polynomials). This yields the consideration of the *Newton polytopes of the polynomials*, which form then an important subject of the paper, together with their *Minkowski sums and the notion of mixed volume*. Here also Bernstein's theorem (roughly saying that the mixed volume counts the number of complex solutions of generic polynomials) is exposed.

In [2061] and [2062] Rostalski and Sturmfels study *duality notions* in a broad sense, namely in convex geometry, as Lagrangian duality in optimization, and as projective duality for algebraic varieties. Classes of convex sets which are *spectrahedra or projections of spectrahedra* are considered via semi-definite representability. And observations, like the following one on semialgebraic sets, are

deepend: The *polar dual* of such a set is necessarily semialgebraic and convex, but the correspondence between its faces and the faces of the polar dual is more complicated than in the classical situation.

The survey [1010] of Gouveia and Thomas deals with *approximation of convex sets* via inequalities with polynomials of bounded degree. Namely, the authors study *(outer) spectrahedral approximations of convex hulls of algebraic sets* which are parameterized by the degrees of the polynomials in the given representation.

Theobald [2358] exposes the developments and methods around *spectrahedra*, in the spirit of the interplay between algebraic geometry and convex programming. The author presents the algorithmic problems of deciding *emptiness or boundedness of spectrahedra*, and whether they contain each other. The approaches combine methods from real algebra and optimization.

Let X be an n-dimensional projective variety with a suitable flag of irreducible subvarieties, $L > 0$ an effective Cartier divisor on X, and consider non-zero global sections of L yielding vectors of non-negative integers which then, finally via closed convex hulls, allow to define a convex body called the *Newton–Okounkov body of L with respect to the flag*. In the surveys [1604] of Lozovanu and [1471] of Küronya and Lozovanu the authors present the knowledge about Newton–Okounkov bodies and the principle "to use convex geometry, through the theory of these bodies, to study the geometrical/algebraic/arithmetic properties of divisors on smooth projective varieties". The article [818] by Escobar and Kaveh nicely collects results concerning the use of convex polytopes in algebraic geometry and with respect to Newton–Okounkov bodies. It is shown how geometrical properties of a family of algebraic objects (e.g., toric varieties or projective varieties) can be studied in a family of combinatorial objects, for instance of polytopes. Examples are given, suitably also leading to the notion of *permutahedron* and to that of *associahedron*.

Rubinstein [2072] introduces some *asymptotic notion of positivity in algebraic geometry* which is related to *high-dimensional convex sets*. For example, their dimension grows with the number of birational operations; a useful overview is given.

12.4.4 Differential Geometry

In this subsection we will discuss publications which can be located *between convexity and differential geometry*. To a large extent, important ingredients coming from differential geometry to develop the corresponding side of convex geometry are already described in subsection 12.3.1, namely the contributions (mainly meant as books or collections) of Blaschke (e.g., [360] and [359]), Bonnesen and Fenchel (§17 of [446]), Süss (see the impressive list of his related papers presented in [446]), A. D. Alexandrov (cf. [40] and also [41]), Pogorelov (see [1976], [1977], [1978], [1979], and [1980]), Busemann (with [544], [548], [549], [550], and [551]), Leichtweiss (partially [1537] and his surveys discussed below), and corresponding parts in Schneider's book [2167] (e.g., Section 10.5) showing generalizations. We add also Chern (cf. [615] and [614]) and Rund [2077].

Many of these contributions reflect the following principle: assuming convexity of curves or surfaces allows to weaken other criteria (like smoothness assumptions), and therefore many of the results presented in these references *generalize* "typical" statements formulated in terms of differential geometry. On the other hand, there are books and surveys in differential geometry containing a lot of typical convexity results, like Chapter 3 of the selection [615]. (These refer to the *isoperimetric inequality, theorems of Fenchel and Crofton, Hadamard's theorem on convex surfaces, the rigidity theorem of Cohn-Vossen, and Minkowski's theorem on a closed convex surface with positive Gaussian curvature*.) Close to convexity, very often tools, methods, and results from differential geometry occur, combined with notions and settings like *curvature concepts, affine concepts, degrees of smoothness, non-Euclidean analogues or generalizations* (like related results in Banach spaces, with the special case of Minkowski geometry), and others. The latter concept of Minkowski geometry was successfully

extended to Finsler geometry. This development started early, and it is worth to have a closer historical look at it (see below).

12.4.4.1 Introduction and historical steps

Thus, before starting to present an overview on more recent publications between differential and convex geometry, we want to look at the earlier history of *Minkowski geometry, relative geometry, Finsler geometry, and the geometry of real Banach spaces*, hoping that this view allows a better understanding of the whole structure of this part of mathematics and its developments to modern areas. Hence we will go back to the beginning of the twentieth century (for the beginning of the story, see also Minkowski's contribution described in our subsection 12.3.1).

Inspired by *physical applications* (such as the geometry of crystal growth and the study of whiskers), Minkowski [1804] had the idea to replace the Euclidean unit sphere by the boundary of an arbitrary convex body (centered at the origin, or demanding only to have the origin as interior point). Thus, he defined *direction-dependent convex distance functions*, later in the first case called *norms*, and in the second case called *Minkowski functionals* or *gauges*. Minkowski gave the axiomatic fundaments for the theory of these distance functions. In particular, he created (in the centered case) the concept of *finite-dimensional real Banach spaces*, therefore also called *Minkowski spaces*.

This was the starting point for strong developments in (functional) analysis, yielding the notions of *norm* and *asymmetric norm* (or *gauge*), respectively. Also *differential geometry* became influenced by these new notions and ideas. Müller [1861], Duschek [778], Süss [2324], and (later) Gericke [949] overtook such concepts to the field of *classical differential geometry* (correctly citing Minkowski). Defining the boundary H of a smooth convex body centred at the origin as *unit sphere* (also called *indicatrix*), they defined the Gauss map of an immersed hypersurface G as a map via parallel tangent hyperplanes at H and G, still presenting a Peterson correspondence with the so-called *relative normal image* of G. On this way the classical theory of "principal curvatures" was generalized to "relative curvatures", and notions like "relative fundamental forms", "relative metric", and many others started to play the essential roles in this new field, called *relative differential geometry*. Witt [2471] and Rund [2076] continued these investigations by generalizing the concept of "angle", such that also notions like "Dupin indicatrix" and classical theorems (e.g., that of Meusnier) found analogues in relative geometry and for normed spaces. This can also be seen as a special case of the concept of relative normalization from affine differential geometry (see, e.g., Simon et al. [2251]). Also related to this framework, Schneider [2141] studied global theorems from the relative differential geometry of convex hypersurfaces. And for a wider view on related developments in relative differential geometry, the reader should consult the survey [238] of Barthel and Kern. For example, the monographs [2237] of Shirokov and Shirokov and [2251] of Simon et al., dedicated to affine differential geometry, contain sections on relative differential geometry. And here we also refer to Blaschke's related contributions collected in [361]. As an example, Heil's later paper [1176] on the four-vertex theorem nicely shows the relations between the notions of relative and Minkowski geometry.

To study the history of overlaps of *differential geometry and convexity* further, we recommend §17 of the monograph [446] of Bonnesen and Fenchel. It treats mainly properties of convex curves in the plane and surfaces of positive Gaussian curvature (e.g., boundaries of convex bodies) in higher dimensions. It shows at least the beginning of extensions from this setting with smoothness assumptions to general convexity assumptions, then not necessarily combined with differentiability properties. Regarding two dimensions, a survey on contributions treating *osculating circles, circles of curvature, and rolling circles/curves* is given there; famous names occurring there are Hjelmslev, Jessen, and Blaschke. Results on cyclic orders of convex curves, like the corresponding *four-vertex theorem*, follow, and extensions to *relative geometry and spherical geometry* are shown to be created or meaningfully enriched by contributions of Blaschke, Süss, Matsumura, Kneser, Kubota, Herglotz, Radon, and many

others. Also theorems like this example (due to Blaschke) are mentioned: *Each convex curve contains at least three pairs of points such that in the points of each such pair the tangents are parallel and the curvatures are equal*. Further theorems refer to *osculating conics and inequalities combining lengths and areas enclosed by convex curves*. Regarding higher dimensions and the theory of *convex surfaces*, results on *spatial curves* contained in the boundary of their convex hulls are presented, and early results on *surfaces of positive Gaussian curvature, rolling surfaces, infinitesimal bending, and rigidity* (e.g., of non-closed surfaces whose normal image is the whole sphere except for finitely many points). Here names like Cauchy, Minkowski, Liebmann, Hadamard, Hilbert, Weyl, Dehn, and Blaschke play a leading role, and the importance of *support functions* as a tool in this framework is underlined, too. Showing a starting point for future research, it is said that in those days no deep results on convex surfaces without differentiability assumptions were known.

We discuss now the development of those parts of *Finsler geometry* which came out of the research line described here (there are, of course, also other lines of development). As we said, the *geometry of (normed or) Minkowski spaces*, as a field also called *Minkowski geometry*, was born at the beginning of the twentieth century. (Modern basic references are the book [2361] of Thompson and the surveys [1722], [1720], and [190], and today it is a lively research field between *convexity, Banach space theory, Finsler geometry, discrete geometry*, and even *computational geometry*.) Its influence on classical differential geometry described above created the fields of *relative geometry* and *Finsler geometry*. A natural example of a notion showing these developments is that of *Gauss map of surfaces in Minkowski spaces*, which can be defined, e.g., with the help of *Birkhoff orthogonality* (see the expositions [69] and [70] of Alonso, Martini and Wu, and the recent contribution [192] by Balestro et al.). Although with the l_4-norm Riemann, in his famous inaugural lecture (1854), gave a first example of a Finsler metric, and although Minkowski created the concept of norms and gauges, the emerging area was officially named after Minkowski's academic grandson (via Carathéodory), namely Finsler. In his dissertation [878], Finsler studied already in 1918 the more general concept of *Finsler manifolds*. The name is due to Cartan (see [581]). A *varying Minkowski functional* is provided on each tangent space of such a manifold; thus every Finsler manifold becomes an intrinsic quasi-metric space when the distance between two points is defined as the infimum length of the curves that join them. The corresponding local situation is Minkowskian, and so these manifolds are more general than Riemannian manifolds (the tangent norms need not be induced by inner products). Always staying not far from the theory of convex bodies (important for the geometry of unit balls), such concepts were further developed in works of Busemann (see [546] and [547], where also relations to relative geometry and fruitful progress in Minkowski geometry and Finsler geometry are discussed), his student Petty who studied curvature types of curves in normed planes (cf. [1966] and [191]), Biberstein (see [337], where possibly for the first time differential geometry of surfaces in normed spaces was presented), and Guggenheimer (cf. [1101] and [1103]), who somehow rediscovered the differential geometry of curves and surfaces in Minkowski spaces in those days. In the papers [1101] and [1104] even the general case of gauges (disregarding the symmetry axiom for norms) is treated, see also the booklet [2526] of Zaustinsky which is based on the more general concept of Busemann's *geodesic spaces*, and the monograph [631] of Cobzas.

Busemann, academic grandson (via Courant) of Hilbert, recognized Finsler's concept as a special case of the *geometry of metric spaces*. (Here we also refer to a related overview of Laugwitz, presented in §15 of the book [1515].) Summarizing fundamental work of Busemann, we repeat that he is one of the main founders of *convexity in metric spaces* and *convexity theory combined with synthetic differential geometry*, later becoming even one leader of *synthetic Finsler geometry* in the twentieth century. Related books of him and further references are collected in subsection 12.3.1. Of course, there are different points of view on how to see and introduce Finsler geometry, but Busemann's way is essential and also interestingly close to convexity. It is natural that, studying the local situation in tangent spaces, Busemann also enriched the fundaments of Minkowski geometry. The recent survey [109] demonstrates the timeliness of his approach to Finsler spaces, and also the recent paper [1173] of

He, Yin, and Shen shows that the topic still yields many interesting geometric research problems. Here we should also mention the fundamental related contributions of A. D. Aleksandrov and Pogorelov (which are already discussed in subsection 12.3.1.).

The handbook [112], edited by Antonelli, reflects the stormy development that, further on, the geometry of Finsler manifolds has taken. Two important monographs containing also material on Minkowski spaces are [210] of Bao, Chern, and Shen, as well as [613] by Chern and Shen. Here one can mention also the following two expository papers on Finsler geometry, still close enough to normed spaces and convexity: [93] and [92] of Álvarez Paiva, the first with Thompson. In this first one the *theory of volumes and areas in normed and Finsler spaces* is developed, where the *Holmes–Thompson notion of volume* (see Chapter 6 of [2361]) plays a central role. For example, they say: "We believe that the most surprising feature of the Holmes-Thompson definition is the way in which it organizes a large portion of convexity into a coherent theory.". The second paper [92] also treats volume and area in Finsler spaces, but refers also to properties of unit spheres in Minkowski spaces and to Hilbert's fourth problem. In addition, we mention here the paper [1233] of Horváth which is related to the differential geometry of convex surfaces in normed spaces, but with another approach than discussed here.

12.4.4.2 Books and book chapters

We will now present more recent contributions, and we start with books or book chapters.

BOOKS OR CHAPTERS CONTAINING EXPLICIT CONVEXITY MATERIAL

There are many classical books on differential geometry containing chapters or sections on convex hypersurfaces. We give now some examples. Thorpe's classical book [2362] is an example of a textbook containing also a short chapter (namely Chapter 13) on *convex surfaces* and various further information useful also for convex geometers. The same holds for the book [532] of Burago and Zalgaller (see Chapter 5 there, and also the survey [530]). Further on, Guggenheimer's monograph [1102] contains some convexity-related differential geometry, like also Hsiung's classical textbook [1241]; in the latter book *curves of constant width, the isoperimetric inequality, the four-vertex theorem, and rigidity problems* are discussed. Walter treats in Section 3.8 of his monograph [2416] some *characterizations of convex hypersurfaces and spheres*.

The book [1515] of Laugwitz, an extended translation of the German original from 1960, gives a nice introduction to classical differential geometry, tensor algebra and analysis, Riemannian geometry, and differential geometry in the large. Especially in §15 on *metric differential geometry* and §16 on *differential geometry in the large*, the author comes close to subjects considered here. Namely, there he discusses *Minkowski geometry, Finsler spaces, the role of Löwner ellipsoids, characterizations of Riemannian spaces as special metric spaces* (like that of Weyl, or as the only pointwise isotropic spaces), and also *geodesics and systems of paths in Finsler spaces*. In particular, Section 15.9 offers an interesting hierarchical survey of types of spaces (containing, e.g., metric spaces, Riemannian spaces, Minkowski spaces, spaces of constant curvature, etc.) and related characterizing properties at their right place in this schema (like pointwise isotropy, existence of a similarity mapping, and so on) is given. In §16 the reader finds material on *support functions, curves of constant width, the four-vertex theorem, antipodality, convex hypersurfaces and spheres, the Gauss–Bonnet theorem*, and several *integral formulas referring to convex hypersurfaces*. In a short appendix, Laugwitz discusses the history of differential geometry, also emphasizing the importance of its *connections to convexity* (combined with names like Blaschke, A. D. Alexandrov, and Busemann).

Su's collection of notes [2320] contains a lot of convexity material, such as *isoperimetric questions, Steiner points, integral geometry in the sense of Blaschke, rigidity of convex surfaces, and Minkowski's uniqueness theorem*. Completed by many theorems on notions from classical differential geometry, it

should be very useful for graduates starting with research in global differential geometry and related convexity notions.

The monograph [287] of Berger and Gostiaux has also many connections to convexity, especially in Chapters 9 and 11. Themes discussed there refer to *convex curves and surfaces*. More precisely, the *four-vertex theorem, envelopes, isoperimetric inequalities* (also on surfaces), *Voronoi regions and convex polyhedra, cut loci, rigidity questions, Minkowski's uniqueness problem*, and the notions of *constant width, constant girth, and constant brightness* can be found there. The book contains many interesting examples and a rich bibliopraphy.

Especially in Chapters 5 and 11 of the monograph [1713] on *bodies of constant width*, methods and results from differential geometry are applied to investigate bodies of constant width. For example, these are *pedal functions, systems of lines and their envelopes, evolutes, floating figures, Gauss map, Gaussian curvature, local geometry, and affine geometry of constant width bodies*.

EQUIAFFINE CONVEX GEOMETRY

In the second edition [1557] of the classical book on affine differential geometry from 1993 (see also the important monograph [2237]), Li, Simon, Zhao, and Hu show excellently the further development of the field, going from elementary aspects to the research front. Chapter 7 is the important point for citing the book here. This chapter gives an introduction to *equiaffine convex geometry*, including the *affine isoperimetric inequality, mixed volumes, Steiner symmetrization*, and some *integral inequalities for affine curvature integrals*. And Chapter 4 contains interesting *characterizations of ellipsoids*.

In the preface of his book [1537], Leichtweiss says that "the affine geometry of convex bodies is the theory of the invariant entities and equivariant associated bodies with respect to the group of the volume-preserving affinities of the n-dimensional affine space, i.e., the *equi*affine geometry of the convex bodies." In this sense he studies the equiaffine geometry and the differential geometry of convex bodies and surfaces, laying special emphasize on *affine surface area* (comparison of different definitions, investigating their equivalence, etc.) and *affine inequalities* (referring to *difference bodies, centroid bodies, projection bodies, intersection bodies*, and further concepts). Regarding the latter, equality cases lead to special types of convex bodies, like to *simplices and ellipsoids*. After that, the author discusses the equiaffine differential geometry of the boundaries of *smooth convex bodies*, with analogues to the Euclidean case. Even practical applications in areas like shipbuilding are described, and interesting problems and conjectures are posed. Here we mention also the (partially historical) expositions [1533] and [1534] of Leichtweiss.

CURVATURE MEASURES

The monograph [1853] of Morvan brings together material from different areas of geometry in a very interesting way. Namely, the author presents many concepts of *curvature measures* which are, for example, interesting from the viewpoints of *classical differential and Riemannian geometry, polyhedral geometry*, and *general convexity*. Among the discussed topics the following ones are interesting for the interplay between differential and convex geometry: *polyhedral geometry* (in particular, the Gauss–Bonnet theorem for polyhedra, convex sets, classical differential geometry of curves and surfaces, Riemannian geometry, relative curvatures), *Hausdorff distance, approximations of volume, length and area, the Steiner formula for convex bodies, approximation problems for normal cycles of sequences of triangulated polyhedra, Voronoi diagrams*, and *Delaunay triangulations*. Many nice illustrations are given, some of them colored. The book is useful for researchers in differential geometry and convexity. Here might be also the right place to cite the exposition [36] of Alarcos and Santander, where the basic concepts related to curvature measures for convex bodies in the context of integral geometry are systematically derived. Used tools are *Kubota formulae, differential relations between k-volumes, parallel bodies, and Steiner formulae*.

The book [2023] of Rataj and Zähle could be cited at several places in our book. We decided to cite it here since it is also close to Morvan [1853] (see above). Being close also to the exposition [2145] of Schneider, it refers to *curvature functionals and their measure-theoretic extensions* regarding geometric properties which are invariant under certain transformation groups. For the "singular sets" studied in this framework, measure-theoretic methodologies are developed; these types of sets occur in many branches of mathematics and differ from classical concepts in convexity or differential geometry (typical examples are piecewise smooth manifolds and fractals). It is shown that these curvature functionals and measures form an *additive, continuous, and complete system of Euclidean invariants*. Suitable introductory parts for measure theory, convexity, and differential geometry are given; in later chapters locally finite unions of sets of positive reach, needed integral-geometric formulae, approximations as well as characterizations of curvature measures are carefully presented. Written in a clear style, the book is, on the one hand, suitable for graduates already familiar with basics from measure theory, convex analysis, and differential geometry. On the other hand, from this comprehensive collection of results, broadly representing a widespread subject, researchers will benefit by getting an excellent overview.

FURTHER RESULTS

The book [579] presents carefully selected papers of do Carmo in chronological order, highlighting deep and groundbreaking results in differential geometry and related fields. There is also a complete list of his publications (until 2010) and an overview of his scientific career with own commentaries. In addition, one should emphasize his contained paper on the development of differential geometry as a research field in Brazil. Three of the selected papers reflect important results "between" convexity and differential geometry. They refer to *rigidity and convexity of hypersurfaces, spherical images of convex hypersurfaces, and an isoperimetric inequality for surfaces*.

The convexity-related themes in the comprehensive collection [615] of Chern's works in differential geometry are collected in Chapter 3 on curves and surfaces in Euclidean space. There one can find the *four-vertex theorem* and the *isoperimetric inequality* for closed plane curves, various results on *curvature notions, the Gauss–Bonnet theorem, Hadamard's theorem on convex surfaces, Cohn-Vossen's rigidity theorem*, and *Minkowski's theorem on closed convex surfaces with positive Gaussian curvature*. As a fundament for the representation [615] (and also [614]), Chapter 1 of [615], overwritten with "From triangles to manifolds", is important since basic notions are clarified there. As examples, we mention here vector fields and their connection to the Euler–Poincaré characteristic, or the notion of tangent bundles. The second collection [614] of works of Chern starts similarly with a very general, introducing chapter entitled "What is geometry?", where Chern's personal view on geometry is reflected. He discusses the coordinatization of space by Descartes, projective geometry, affine and conformal geometries, as well as the importance of Klein's "Erlanger Programm". Further on, he treats own results and viewpoints in differential and integral geometry which are due to him. In this second book [614], the view is less concentrated on convexity-related results. But again the Euler–Poincaré characteristic and isoperimetric problems play a role.

12.4.4.3 Convexity-related proceedings

The comprehensive proceedings [612] of the 1980 Beijing Symposium on Differential Geometry and Differential Equations, edited by Chern and Wu, contain some articles close to convexity, namely treating *generalized support functions and their applications, convexity in Riemannian geometry, and rigidity problems of surfaces* with respect to *mixed Gauss curvature*.

In the proceedings [865] of the 1991 Berlin conference on global differential geometry and global analysis, collected by Ferus et al., one finds papers related to convexity. They refer to *isoptics*,

(hyperbolic) affine spheres, convex affine surfaces with constant affine mean curvature, and Radon's theorem (from differential geometry).

The Oberwolfach proceedings [1903], edited by Nomizu, Pinkall, and Simon, contain articles on the *Minkowski problem on curvature functions in affine geometry*, and also properties of certain types of *affine spheres* and *characterizations of ellipsoids* are studied. A nice article on the *history of affine surface area of convex bodies* is presented, too.

Also in [2326], collected by Suh, Berndt, and Lee, several articles close to convexity appeared. These refer to *affine geometry using homogeneous contour integrals* (a concept used to construct affine integral invariants of a convex body, which is important in affine integral geometry), the *Steiner formula* (here obtained via a differential geometric approach), the *principal kinematic formula for convex bodies, Bonnesen-type inequalities, the isoperimetric deficit of convex bodies*, and further interesting topics.

12.4.4.4 Selected surveys between convexity and differential geometry

We start with convex sets embedded in certain manifolds.

CONVEX SETS IN MANIFOLDS

The survey [530] collects results on complete Riemannian manifolds of non-negative curvature. The authors study open Riemann manifolds with non-negative curvature and emphasize the role of *convex sets* within this framework. A classification of possible types of convex sets in such spaces regarding their topological structure is given.

A natural continuation of this survey, discussing several aspects with overlaps, is the exposition [2415] of Walter. He surveys developments around geodesically convex sets and functions in (complete) Riemannian manifolds, considering both local and global aspects. He also concentrates on convex sets without additional smoothness assumptions and gives interesting applications and open problems. Further on, convexity in Riemannian geometry (e.g., used convexity techniques to obtain results in global Riemannian geometry, and properties of manifolds admitting convex functions) is surveyed in [2236] by Shiohama.

Berndtsson [294] surveys developments around the geometry of the space of *Kähler metrics*. The goal of the underlying course is the description of the role that convexity plays, both as a motivating field and as a tool. The author studies results in the local convex setting having complex-analytic analogues. In this framework, he treats for example the *Brunn–Minkowski inequality, Prékopa's theorem, and the Brascamp–Lieb inequality*.

CHARACTERIZATIONS AND UNIQUENESS THEOREMS

In his handbook article [1536], Leichtweiss follows the philosophy that certain differentiability assumptions may easily be removed if a differentiable hypersurface is *convex*, thus getting this way convex geometric generalizations of notions and results typical for differential geometry. Vice versa, it is not hard to derive convex geometric theorems in a differential geometric way. Leichtweiss sketches such proofs and shows the relations of both fields with combining remarks. The concrete topics used for this are *characterization theorems by Hadamard and Sacksteder on the convexity of a hypersurface*, the differential geometric counterpart to the *Alexandrov–Fenchel–Jessen inequalities for mixed volumes*, and also *uniqueness theorems of Süss and Liebmann*. The material is presented in an understandable manner, thus useful already for beginning graduates; on the other hand, it is certainly informative for researchers in convexity and differential geometry.

In Section 2 of the basic handbook survey [1691] on *characterizations of convex sets*, Mani discusses several theorems in the setting of differential geometry. They refer to *Weyl's problem* about sectional curvatures of the boundary of a convex body, *Minkowski's problem* on exact properties of a function to present the Gaussian curvature of the boundary of a convex body, and *convexity on Grassmannian cones*.

In his survey-like paper [1532], Leichtweiss discusses three classical topics referring to convex hypersurfaces. These share the property that *combined methods from convexity and from differential geometry* are useful to attack them: The *Blaschke–Santaló inequality, quermassintegrals under continuous symmetrizations*, and the *uniqueness theorem of Alexandrov–Fenchel–Jessen*.

BAIRE CATEGORY RESULTS

Zamfirescu's survey [2523] and Gruber's handbook article [1062] present a wide picture of *Baire category results in convexity* until 1993 (it seems that no comparable exposition appeared after [1062]). Baire category results exist in many directions of convexity and also in the spirit of differential geometry (therefore we cite the surveys here, in this subsection). It is usual to say that *most convex bodies have a certain property* if the convex bodies which do not have this property form a first category subset of the space of all convex bodies. (A first category set in a topological space is a countable union of nowhere dense sets.) It turns out that, based on this concept, the typical behavior of convex bodies frequently runs against our intuition. For example, in [1062] the following theorem is presented: *On the boundary of most three-dimensional convex bodies there are no closed geodesics, even if self-intersections are allowed*. This is in proper contrast to known results from classical differential geometry. Another example (due to Zamfirescu): *For most convex bodies at the most points of their boundary the lower sectional curvature in each direction is zero, and the upper sectional curvature in each direction is infinite*. This contrasts Alexandrov's result that the boundary of each n-dimensional convex body is twice differentiable at all points except those comprising a set of $(n-1)$-dimensional Hausdorff measure zero. In the two surveys [2523] and [1062] one can find many more results also from differential geometry. For example, they refer to *curvature concepts, smoothness, shadow boundaries, points of contact with circumspheres, ellipsoids, and billiards*.

FURTHER RESULTS

In the exposition [1983], Pólya illustrates some methods as they were presented in his famous earlier books, but here in the context of *global differential geometry*; however, *Steiner symmetrization* plays an important role here.

Two important surveys written by a differential geometer with related viewpoints on important inequalities from convexity (namely, *isoperimetric and Bonnesen-style inequalities*) are [1923] and [1924] due to Osserman.

Based on constructions in the sense of *equiaffine geometry*, Leichtweiss studies in his exposition [1535] the validity of several well-known properties of (differential geometric) *inner parallel hypersurfaces* of given convex hypersurfaces. There is an affine analogue from the differential geometric point of view going back to Chakerian, and one using typical methods from convexity theory. After discussions of these constructions, reconstruction problems for given *equiaffine differential geometric inner parallel hypersurfaces* are considered.

The comprehensive survey [1157] of Hansen et al. on *star-shaped sets* contains also a subsection (namely 19.3) on *starshapedness in differential geometry*. Topics discussed there are for instance: *equireciprocal points, isoptics, geometric flows, curvature properties, rigidity*, and geometry of symplectic spaces.

The historical article [2252] of Simon, Tjaden, and Wefelscheid honors Chern as one of the most active geometers of the twentieth century and, in particular, his work in Hamburg (1934–1936), also

with Blaschke who strongly influenced orientations of his later mathematical work. And the latter was also done regarding the study of *convex surfaces*.

Here we mention also the important contributions of Busemann (influenced by A. D. Alexandrov) to convex surfaces, mainly in view of synthetic differential geometry, Finsler geometry, geodesics, isoperimetric problems, etc.; see our discussion in subsection 12.3.1.

12.4.5 Integral geometry and stochastic geometry

In this subsection we survey the development of *integral geometry and stochastic geometry*, two disciplines that are closely intertwined and deeply related to convexity. Similar to other analogous subsections we subdivide the collected material roughly into represented monographs, proceedings, and surveys; independently, we present after this the subfield of *random polytopes and mosaics* in an extra way.

12.4.5.1 Introduction and important historical steps

Over centuries and decades, also *probability theory*, for many mathematicians going back to the studies of gambling by Pascal, has been completed by geometric aspects and problems. For example, in the eighteenth century *Buffon's famous needle problem* came in, and in the nineteenth century *Bertrand's paradoxon* was added. It became obvious how careful one already has to be when posing problems within this framework (for example, already the phrase "random chord of a circle" can lead to paradoxes). Crofton's early work (see, e.g., [688]) was important in this development; his classical theorem expresses the length of a curve as an expectation of the number of intersections with a random line. Such early contributions are nicely presented in the book [726] of Deltheil. The notion of *invariant measure* started to play a role, i.e., one studied measures on a geometric space invariant under the symmetry group of that space. More recently a broader view yields invariant (or equivariant) transformations from the space of functions on one geometric space to the space of functions on another such space. Such transformations can be integral transforms, like the Radon transform.

Because of the increasing necessity to use integration techniques, the emerging field of *integral geometry* became helpful to refine probability problems and statements referring to geometric objects. A very nice historical view on integral geometry, underlining also the roles that Cauchy, Crofton, Minkowski, Steiner, Blaschke, and Santaló played, is the work [1494] of Langevin. Among other things it shows that integral geometry in this sense can be seen as the application of probability theory (as it was axiomatized by Kolmogorov) in the context of Klein's Erlangen program. The development in this direction saw names like Poincaré (1912) and Deltheil [726], and it helped to establish the field of geometric probability more and more.

On the other hand, inspired by lectures of Herglotz (1933) in Göttingen, Blaschke and his Hamburg School (1935–1939, including Santaló and Chern) started strongly to develop integral geometry itself, based also on older results (like Cauchy's projection formula and Crofton's formula). The most interesting measures used in this framework were determined by corresponding group invariance requirements. Fundamental results (like the famous *kinematic main formula* and further developments) due to Blaschke, also the studied geometric objects (e.g., *affine subspaces or convex bodies* in Euclidean n-space) started to play an essential role. For instance, also *isoperimetric problems* were investigated in this setting. In the collection [359] of Blaschke papers one finds, together with a nice overview of Blaschke's pioneering work in integral geometry given by Santaló, altogether twenty papers of Blaschke himself treating integral geometry, and here we also mention the related booklet [356] of him. Natural extensions followed, such as analogous investigations in non-Euclidean spaces (e.g., due

to Santaló and Chern) or treating finite unions of convex bodies (Hadwiger's *convexring*; see the last chapter in Hadwiger's monograph [1134]). And so the *classical part of integral geometry* was established (thus fixed, besides the references in [359], by the monographs [2101] of Santaló, [1134] of Hadwiger, and [2305] of Stoka). A comprehensive monograph on the state of integral geometry in the middle seventies is the classic book [2103] of Santaló, showing also nice applications of stochastic problems. Here we mention also the strongly related monograph [2027] of Ren, whose forerunner in Chinese was written since there was no Chinese translation of Santaló's book [2103], and for Chinese students [2103] was also not easy to get. Therefore the book [2027] follows [2103] closely. But due to personal selection and the later appearance of [2027], this book contains also other and later results.

Here we also have to cite the monographs [1729] by Matheron and [96] by Ambartzumian, with different approaches to integral geometry. For example, in the latter book theoretical approaches in integral geometry and geometric probability are weakened, especially regarding the invariance of the used measures. In this sense, the book [96] contains *generalizations of integral-geometric results, some of them also related to convexity and to stereology*. The latter field was originally the three-dimensional interpretation of two-dimensional cross-sections of materials or tissues, but it has fruitfully expanded. The review [99] of Ambartzumian summarizes results obtained in the direction of [96] after that monograph; *valuations and tomographic problems* also play a role. One should also underline that the two books [96] and [98] of Ambartzumian, as well as [1401] of Klain and Rota, cover somehow the *combinatorial aspects of integral geometry*. For later developments in integral geometry close to convexity, we refer to the excellent survey [2182] of Schneider and Wieacker and the monographs [2177] and [2179] of Schneider and Weil. In contrast to [1729] and [96], these three references are based on the measure-theoretic approach coming from Federer's monograph [834], i.e., Federer's *curvature measures and area measures related to mixed volumes*.

As already mentioned, the field of *geometric probability* (dealing with mathematical models for random geometric structures and spatial data) started to grow separately (see the monographs of Santaló [2103], Kendall and Moran [1361], Solomon [2267], and Mathai [1728] as essential milestones), both fields having a lot of overlaps. Especially when the geometric consequences of obtained results dominated, the investigations became part of integral geometry (in the sense of Blaschke), and for a certain period geometric probability was seen as a field of applications of integral-geometric formulas; see again Santaló's classical book [2103], for applications and historical aspects Miles and Serra [1781] and, also continuing [2103], the survey [2443] of Weil and Wieacker, like also the monographs [2178] and [2179] of Schneider and Weil.

Around 1970, as a third and (from the beginning) independent development, the field of *stochastic geometry* came to be, which then was established in the last third of the twentieth century. This field is wider than geometric probability since it deals with *models for random sets* (not fixed ones, regarding shape) and collections of such sets combined with notions like *point processes*. The first milestones here were contributions by Harding and Kendall [1162] as well as Matheron's book [1729], and in addition the important thesis of R. E. Miles (on Poisson processes); see also Solomon [2267] and, regarding coverage processes, Hall [1147]. More recent basic references are [270], [2310], [619], [2179], [1362], and [665]. We add the "modern version" of Matheron's basic book [1729], namely the comprehensive, interdisciplinarily written book [1826] of Molchanov, containing also a chapter where tools like *Minkowski addition and support functions* are used. For emphasizing deep relations between stochastic and integral geometry (as well as convexity), we particularly refer once more to Matheron [1729] and its follower Molchanov [1826], the comprehensive surveys [2443] and [2162] of Schneider, Weil and Wieacker, and the books [2178] and [2179] of Schneider and Weil. For example, as typical topics of investigation close to convexity one can find there *random polytopes, intersections of convex bodies with random flats, random convex bodies, and random mosaics*.

One has to add also the fields of *stereology* (see, e.g., Chapter 16 in Santaló's monograph [2103] and Weil's survey [2441]) and, regarding later developments, *image analysis and spatial statistics*. For image analysis see the important book [2223] of Serra; also this field started to come up roughly

about 1970 (cf. additionally the two historical papers [693] and [694]) by Cruz-Orive, and for spatial statistics we refer to the books [2034] and [1825] by Ripley and Molchanov, respectively. In the classical sense, stereology concerns the three-dimensional interpretation of two-dimensional cross-sections of materials, at first glance seemingly similar to *geometric tomography* (see Gardner [930]). But it is quite different, since only a few representative sections (or projections) are used, followed by statistical extrapolation, and since the studied objects are often far from being convex. Important later references on stereology and image analysis being fundamental and/or not too far from our aims here are the last two chapters of Ambartzumjan's book [96] (not emphasizing the usual invariance of measure, but using general measures yielding interesting geometric properties), the proceedings [100] edited by Ambartzumian and Weil (containing various articles on stereology and convex sets), the book [2397] (taking into account advances in technical microscopy and results in geometric probability), Weil's survey [2441] (a comprehensive handbook article, mainly written for specialists in convex geometry), and the already mentioned monograph [2223] of Serra on mathematical morphology as a fusion of integral geometry, geometric probability, computer image processing, and image analysis (developments that go back to J. von Neumann). Finally, we mention the book [2410] of Voss treating methods from integral geometry and stereology in *image processing* (see below for more on its contents).

Now we take a more careful look at some books, proceedings and surveys which were particularly important for the development of the convexity-related parts of the three disciplines discussed above. And at the end, we present an extra overview to important publications treating *random polytopes and mosaics*.

12.4.5.2 Further monographs and book chapters on integral geometry

The 1955 edition of Blaschke's booklet [356] on *integral geometry* can be considered as the union of two notebooks (on planar and spatial integral geometry) which appeared already in the thirties. In several directions it shows also progress made in the meantime. For example, this holds for Nöbeling's extension of the main kinematic formula and Santaló's corresponding extensions to non-Euclidean geometries. The planar part contains various geometric results, e.g., *Crofton's theorem on the lengths of curves and the isoperimetric theorem* (following Santaló and Bonnesen's sharpening for not necessarily convex Jordan curves). The spatial part contains, e.g., the *surface area formula of Cauchy, Steiner's formula for the volume of parallel bodies, and Minkowski's inequality* (following from Bonnesen's planar sharpening via Cauchy's formula). Also spherical geometry is discussed in this setting, and many other results on convex bodies can be found there.

The book [1134] of Hadwiger is a milestone in the set of monographs on convexity, covering many different basic aspects of the field. Its main contents are comprehensively presented in subsection 12.3.1 above. Here we only emphasize once more that the very long final chapter of [1134] contains many beautiful results which are fundamental for *integral geometry*. And also Hadwiger's earlier booklet [1133] is a nice reference for integral geometry; its final chapter (including the remarks at the end of it) is completely dedicated to this field.

Another early reference treating integral geometry is Stoka's already mentioned book [2305], which presents many results from two- and three-dimensional integral geometry, such as *Crofton-type results, kinematic density, Blaschke's principal formula, contributions of Hadwiger, and applications of isoperimetric kind*. The spectrum of the book also refers to projective and centro-affine geometry, yielding *integral-geometric results on surfaces of constant curvature* as well as *applications of integral formulas and inequalities related to convex figures*.

The book [2327] of Sulanke and Wintgen can be excellently used as a fundament of an *advanced course in differential geometry*. The reason to cite it here is given by its last two chapters. They contain

applications of methods developed in this book for treating the integral formula of Gauss–Bonnet–Chern, and *n-dimensional Euclidean integral geometry* in general.

The book [1729] of Matheron is a first systematic representation of geometric probability in which fruitful ideas and methods from *probability, integral geometry, and set theory* are successfully combined. Choquet's notion of "random set" is motivated by practical applications of morphological character, e.g., coming from geology and technology, and the studied geometric object is also seen as a "random object". After presenting typical methods and notions (topologies in systems of open, closed, or compact sets, Hausdorff metric, Minkowski addition, homothets of convex sets, etc.), the author continues with covariance functions, continuity in probability, and weak stationarity concerned with the stochastic process determined by the indicator function of a random set. The fourth chapter is dedicated to integral geometry, reflecting large parts of the final chapter of Hadwiger's book [1134] treating *integral geometry and convexity*. Notions like *almost sure convexity, random sections and projections*, and the *characterization of the almost sure convex random closed sets* allow the randomization of different aspects of convexity. Based on a *generalized Steiner formula*, an alternative extension of the Minkowski functionals to unions of convex sets is given (unfortunately, in an ill-defined form, but see Mathematical Reviews (MR0385969)). Further investigated topics are the *convexity number and densities of Minkowski functionals* for random sets. Also set-valued maps are studied. In Chapter 6 the *Poisson networks of hyperplanes* and their *polyhedral cells* are studied. It should be noticed that various results from the presented material were obtained by Miles. Important for applications (e.g., regarding porous materials), also the theory of granulometries is treated. The author gives a rigorous fundament upon which geometric probability can be further developed, especially in the applied sense.
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As a "follower" of [2101], the classical book [2103] of Santaló (presenting in its first two parts the extended and updated contents of [2101]) is based on the *theory of differentiable manifolds and homogeneous spaces*, and it gives a much more comprehensive treatment of *integral geometry* than it was ever done before. It contains applications for stochastic problems and many results which were found after the appearance of [2101]. The third part covers *integral geometry in Euclidean n-space* including also *stereology* and problems like the determination of *particle size distributions* from the *size distributions in random sections*. And in the fourth and last part, *integral geometry in spaces of constant and non-constant curvature*, integral geometry in the sense of Gel'fand, and the reconstruction of functions from their integrals over manifolds of given families (Radon problems) are studied. The author gives notes to each chapter which present related developments; these show in particular how the fields of *integral geometry and geometric probability* influence each other. Here might also be the right place to mention Part II (26 papers in integral geometry), Part III (9 papers in convex geometry), and Part V (referring also to stereology) of Santaló's collected works [2105]. For example, Part III contains also the famous *Blaschke–Santaló inequality*. New applications of the so-developed field came to be in the seventies, when the applied fields of *stereology and geometric image analysis* were developed. As we said already: stereological questions are not far from those of *geometric tomography* (cf. [930]): information on full-dimensional convex bodies or particles are searched if information on lower dimensional sections or projections are given. However, the methods of both fields differ.

Since it contains also a short section on integral geometry, we shortly mention Guggenheimer's book [1102] once more; comprehensively it is already discussed in subsection 12.3.1 above. We also repeat here that the second part [359] of Blaschke's collected works contains 20 papers of him dedicated to integral geometry, with a description given by Santaló. And we mention that Su's collection [2320] of lectures and notes, already cited in 12.4.4, contains with lecture 2 also a part treating planar integral geometry. In classical integral geometry, *invariant measures* over rings of sets of geometric objects are investigated. Here the invariance is meant with respect to the transformation group acting on these objects induced by the point transformation group of the underlying geometry. Thus, a measure for point sets in the Euclidean plane, invariant under motions, is the usual area, being unique up to a scale factor. Similarly, there is a measure for sets of lines in the plane, invariant under motions of such sets,

which is unique up to a scale factor. Ambartzumian [96] found several important results involving measures lacking such traditional invariance.

The book [2310] of Stoyan et al. treats a large collection of themes from stochastic geometry, spatial statistics, and their applications. Until the appearance of [619], of Chiu et al., it was a fundamental source of inspiration, showing also the literature on applied probabilistics referring to spatial random structures. Readers can have various backgrounds: *pure probability, integral and convex geometry, material science, microscopy, and image analysis*. For example, the following topics are presented: *marked point processes, random closed sets, random measures, processes of geometrical objects, shape theory, random tessellations, and stereology*. Proofs are contained, but in various cases also referred to by the reader; many practical examples are given, and applications are shown. The book can be used as introductory text for statisticians starting in that field, as a helpful source for researchers in stochastic geometry, and as a collection of interesting applications, useful for applied sciences. As already noted, the book [619] is the extended third edition of [2310], modernized and strongly extended (e.g., more than 700 new references are added and, on the other hand, about 300 outdated references are deleted). Extended topics close to convexity are *geometrical networks and graphs, random tessellations, and stereology*, understood as application fields of stochastic geometry and spatial statistics.

In the classical sense, integral geometry and the corresponding geometric probability consider measures of geometric objects (lines, planes, congruent figures, etc.) which are invariant with respect to the group of motions. Then, later, the theory was extended to more general homogeneous spaces with a group of automorphisms, maintaining the invariance of measures. In the book [96], Ambartzumjan proposes to set aside the invariance condition and to use then general measures also yielding interesting geometric properties. If the group contains translations of the basic space, then the corresponding measures are necessarily totally infinite and cannot be normalized to become probability measures. Considering countably infinite sets of geometric objects yields another situation: spaces of such sets admit probability measures which are invariant. Based on such settings, Ambartzumjan published the book [96] and also a second monograph [98]. Due to their topological and combinatorial bases (see the first book [96]), the mentioned extensions to arbitrary measures and more general spaces of geometrical objects turned out to be promising concepts. So Ambartzumjan, continuing [2103] and [96], studies in [98] the so-called *factorization method in integral geometry and point processes* which is based on the idea of *Cavalieri's principle* (see our Section 4.2 and the subsection 4.7.1). (It should be noticed that there is a 2008 paperback reprint of [98].) Replacing areas of domains by a certain general measure, it turns out that this measure satisfies Cavalieri's principle iff it admits a certain factorization of the Lebesgue measure and a certain further measure. Combining this with more general spaces of geometrical objects, the author studies *geometrical random processes*, i.e., related Poisson point processes. The book is interesting for a *broad readership related to stochastic and integral geometry, from pure integral and convex geometry to random point processes, random sets, and their statistics*. And it is worth mentioning that the overlap with other related books is small. Besides Cavalieri's principle there are several topics discussed further, again close to convexity: *metrics and convexity, the invariant measure of the sets of lines separating two convex domains, the invariant measure of the space of tetrahedral shapes, the so-called flag representation of bounded centrally symmetric convex bodies* (see again [96], containing already a complete presentation), *isoperimetric inequalities, random chord length distribution for convex polygons, Voronoi mosaics*, and so on. We note that Part 1 of the more comprehensive monograph [101] of Ambartzumian, Mecke, and Stoyan (consisting of two parts, where Part 2 presents a revised version of a book by Stoyan and Mecke) is based on [98]. Part 2 of [101] starts with the concept of *random closed sets and the point processes*, continuing with a detailed representation of the most important models of stochastic geometry. Also stereological problems are contained.

Ren's book [2027] treats classical integral geometry in the sense of Blaschke going back to ideas of geometric probability due to Crofton, Herglotz, and Blaschke himself. Both fields are taken care of, and regarding form and content [2027] follows (as already mentioned above) in great parts the encyclopedic

monograph [2103] of Santaló. The reasons are understandable: the Chinese students needed such a book in their language, and the Chinese forerunner edition of [2027] satisfied that need. After an introduction on properties of convex sets, measure theory, and fundamental integral formulae, the author applies integral geometry to typical problems from convexity: *isoperimetric inequality, measure of segments within a convex domain, and random convex sets in a lattice of parallelograms*. Then the author continues with integral geometry in homogeneous spaces, particularly in Euclidean space, and applications; this part contains themes like *kinematic density for a Lie group, quermassintegrals, the fundamental kinematic formula, stereological problems, Hadwiger's conditions for a domain to contain another domain, and characterizations of simplices*. An important *reverse Petty projection-type inequality* of Zhang from convexity is discussed, using the polar of the projection body and characterizing simplices. This is a nicely written textbook for undergraduates and graduates, and it is also suitable for researchers who look for applications of integral geometry.

Mecke's book [1758] is dedicated to *applications of integral geometry to statistical physics*, in particular to percolation, complex fluids, and the structure of the universe. In the first two chapters needed fundaments from integral and stochastic geometry are provided, among them being *quermass-integrals, fundamental kinematic and Crofton formulas, geometry of lattices, Voronoi diagrams, and applications to stereology*. Then, in Chapters 3 to 6, the above themes from statistical physics are presented in detail. For example, it is shown how a Boolean model and Minkowski functionals can be used to describe the distribution of galaxies and related clusters.

In the monograph [1401] of Klain and Rota the concept of *invariant valuations* is used as fundament; the authors introduce the notion of valuation and prove *Groemer's integral theorem for valuations*, which then is used and applied systematically. Coming from Buffon's needle problem, the authors extend the concept of geometric probability to the *space of straight lines in the Euclidean space*. The *valuations invariant under the action of permutation groups* are characterized, and this new idea is used throughout the book. Starting with finite unions of parallelotopes in Euclidean space, the authors study the *integral geometry of finite unions of arbitrary compact, convex sets*. Classical theorems from convexity follow, such as *Helly-type theorems* (including those of Klee and Carathéodory), the *Euler–Schläfli–Poincaré theorem, and Cauchy's projection formula*. Extensions to Grassmannians take place. For example, in the fourth chapter *all continuous, invariant valuations of lattice polytopes and intrinsic i-dimensional measures* are studied, and related investigations for compact, convex sets are presented later. Invariant valuations are extended to *polyconvex sets* (i.e., finite unions of convex sets), and a *general Crofton-type valuation for intrinsic valuation in all dimensions* is obtained. In the ninth chapter Hadwiger's famous characterization theorem is discussed and also used to obtain elegant approaches to typical results from integral geometry and geometric probability. Interesting extended views on *mixed volume functions and the Brunn–Minkowski theory* in this framework are given, and the authors also speculate on *potential inequalities between various intrinsic volumes of a convex set*.

Part II of the collection [614] of papers and lectures of Chern contains notes from Chern's eight lectures on integral geometry which were not published before. He discusses two main periods of the field, the first one usually called *classical integral geometry*. Chern starts with this first period and its forerunners, thus presenting *Buffon's needle problem, Bertrand's paradox, Crofton-type results, Poincaré's concept of the kinematic measure, the isoperimetric inequality* within this framework, and results from Blaschke's School of Hamburg, including also Santaló's work. Here he points out that *Blaschke's kinematic formula*, which is also derived, is the most fundamental formula in integral geometry. He continues with *integral geometry in n-dimensional Euclidean, elliptic, and hyperbolic spaces*, using Grassmannians and investigating differential and integral invariants of k-dimensional submanifolds in n-dimensional Euclidean space. The second discussed period started in 1961 with contributions of Gel'fand and Helgason, yielding *applications of integral geometry in representation theory on non-compact Lie groups*. Also this is discussed, and, at the end of this Part II, also the *Radon transform* is briefly presented.

The monograph [270] contains problems nicely selected from the intersection between stochastic geometry, geometric measure theory, convex geometry, and others. Topics from convexity and related parts of integral geometry are hereby *intrinsic volumes, kinematic formulae, integral-geometric projection measures* (for random fibre and surface systems), *stereology*, and the study of anisotropic fibre and surface systems in dimensions two and three, for which they present various approaches, also using *typical methods from convex geometry*. The book is rich of interesting examples and exercises.

We mentioned already that Molchanov's monograph [1826] is seen as a "follower" of Matheron's classical book [1729]. Indeed, this even refers to notation and basic notions. Comparing with [1729], the modern parts of [1826] contain the following subjects (we select only those which are somehow related to convexity): *asymptotic results for unions of random sets and for Minkowski sums, techniques for probability theory in Banach spaces, and limits for unions and convex hulls of random sets* in locally compact Hausdorff separable space. For example, the results on Minkowski sums include strong laws of large numbers, central limit theorems, and large deviation principles. Also in the appendices convexity plays a role. In the book many interesting exercises are given, and applications (e.g., in finance and economics) play an essential role. Compared with the first edition (where this was already excellently done), in the second edition the principle of *giving important notes at the end of each chapter* is strengthened. These detailed notes give a comprehensive literature review on the historical origins and also the variety of recent research papers and applications.

The book [2410] by Voss gives an introduction to *classical integral geometry and its applications in stereology and image reconstruction* in two and three dimensions. After basics from convex geometry, integral geometry, and the Radon transform, examples for and transitions to image processing applications in fields like biology, medicine, petrography, and metallurgy are given. There are interesting biographical notes on known integral geometers and stereologists, and the book contains many exercises (with solutions at the end of it).

The excellent and bridge-building monograph [2179] of Schneider and Weil comprehensively treats the fundaments, relations, and applications of *stochastic and integral geometry*, also with particular emphasis on *notions and tools from convex geometry* (for the sake of completeness, we refer here also to the two earlier books [2178] and [2177] in German, written by the same authors). In the first part, *foundations for models in stochastic geometry and the use of integral geometry* are described. General *random closed sets in topological spaces* are introduced. *Point processes, general random measures, Poisson processes, and particle processes* are typical subjects here. In the second part, "Integral Geometry", the authors present the most important mathematical part that (besides probability theory) is used in stochastic geometry, namely the *background from integral and convex geometry*. Typical topics and tools treated here are the *kinematic formula, integral geometry for cylinders and thick sections, translative integral geometry and spherical integral geometry*. Notions from convex geometry that directly occur here are, e.g., *Hadwiger's general integral-geometric theorem, intrinsic volumes, polyconvex sets* (which can be represented as finite unions of convex bodies), *translative, kinematic and Crofton formulas, Hausdorff measures and their applications* in this setting, *mixed volumes and more general series of mixed measures, section and projection functions, support functions of convex bodies, spherical convexity, the Blaschke–Petkantschin formula* (with generalizations), and many more. In the third part, "Selected Topics from Stochastic Geometry", convexity is important, too. Typical notions studied here are *convex hulls of random points, random polytope projections, randomly moving convex bodies, randomly touching convex bodies, and geometric inequalities for convex bodies*. Also tessellations induced by stationary Poisson processes are investigated, e.g., as *Voronoi or Delaunay tessellations* with respect to Poisson point processes, or as *hyperplane tessellations* generated by Poisson point processes in the space of hyperplanes. There is even a chapter titled "Facts from Convex Geometry", where the rigid motion invariant *intrinsic volumes* and their local counterparts, the *curvature measures*, are treated. Every section of this book ends with carefully written bibliographical and historical notes, implying the encyclopedic character that it also has. Besides this, it is written

in a very reader-friendly style. The book is an excellent source not only for experts in stochastic geometry or theoretical and applied probability theory (referring to geometrically interesting objects). Also researchers in other fields of pure mathematics (somehow related to convexity), or in applied disciplines (like material science or spatial statistics) will benefit from it.

Finally we mention that also the very recent monograph [1263] of Hug and Weil contains a chapter on *integral-geometric formulas*. The special themes considered there are *invariant measures, projection and section formulas, and kinematic formulas*.

12.4.5.3 Monographs and book chapters closer to geometric probability and stochastic geometry

Of course, we have to mention also here the classical books [2103], [1729], and [2223] of Santaló, Matheron, and Serra, respectively. Also other books mentioned under 12.4.5.2 contain parts on geometric probability and stochastic geometry.

The monograph [526] of Buldygin and Kharazishvili concerns *convexity and its applications in measure theory, probability, and statistics* (also topological aspects play a role). In the introductory part, basic facts from *measure theory, convexity, and probability* are given. The first edition of [526] (from 1985) is concentrated on *applications of the Brunn–Minkowski inequality*, such as Minkowski's theorem for polytopes, isoperimetric inequalities, Anderson's inequality related to symmetric unimodal functions and probability tools for convex sets, and matchings of Gaussian processes. The largely extended second edition (from 2000, having also a new title) contains an interesting discussion of polytopes in finite-dimensional vector spaces. Then, for infinite-dimensional spaces, fundaments from topology, the convergence of random elements, and the theory of (also quasi-invariant) probability measures are discussed, and arguments from *convex geometry* are used to study results on oscillation properties, comparison principles for Gaussian processes, integration, and optimal estimation of stochastic processes. Also in the appendices convex geometry plays a role: *properties of convex curves and relations between convex geometry and number theory* are discussed there. Due to an excellent collection of exercises, the book can be conveniently used for a postgraduate course on probability measures.

The book [2310] of Stoyan et al. treats a large collection of themes from stochastic geometry, spatial statistics, and their applications. Until the appearance of [619] of Chiu et al., it was a fundamental source of inspiration, showing also the literature on applied probabilistics referring to spatial random structures. Readers can have various backgrounds: *pure probability, integral and convex geometry, material science, microscopy, and image analysis*. For example, the following topics are presented: *marked point processes, random closed sets, random measures, processes of geometrical objects, shape theory, random tessellations, and stereology*. Proofs are contained, but in various cases also referred to by the reader; many practical examples are given, and applications are shown. The book can be used as introductory text for statisticians starting in that field, as a helpful source for researchers in stochastic geometry, and as a collection of interesting applications, useful for applied sciences. As already noted, the book [619] is the extended third edition of [2310]. Extended topics close to convexity are *geometrical networks and graphs, random tessellations, and stereology*, understood as application fields of stochastic geometry and spatial statistics).

In his research monograph [380], Bocşan presents measurable multifunctions, limit theorem (like laws of large numbers) for random sets that are convex and bounded, and ergodic theorems for random sets, with related fixed point theorems.

Hall's monograph [1147] is dedicated to *coverage processes*, a notion quickly presented by the following simple example: On some circumference, place a given number of arcs of the same length

at random. How large is the uncovered part of the given circumference? This simple coverage problem could be generalized in many directions. The book [1147] contains large varieties of such problems and interesting tools, many of them being close to convexity. For example, we mention here further on: number of lattice points covered by a random disk, interpenetrating spheres distributed randomly in space, random packing problems, mosaics, stereological problems, number of sets that intersect a given set (e.g., isotropic convex grains), questions around curvatures, etc.; in the appendices the author discusses the Radon–Nikodým theorem and the central limit theorem. Each chapter contains a block of exercises and detailed notes.

The comprehensive book [2179] of Schneider and Weil (like also its forerunner [2178]) we described already above.

Finally, we mention here Schneider's very recent book [2174] on the geometry and probability problems posed for *convex cones*. The author develops the methods which are necessary for geometric applications of convex cones, and several notions from classical convexity are extended to this field. Finitely many hyperplanes through the origin generate a tessellation of the space into polyhedral convex cones. The author investigates such tessellations also for the case that these hyperplanes are random, with a distribution satisfying some mild conditions. General random cones and such cones in half-spaces are studied. For example, non-trivial intersections for isotropic random cones are investigated. Also random cones in high dimensions are taken into consideration.

12.4.5.4 Related proceedings

In several articles of the (relatively early) proceedings [1299], edited by Jensen and Gundersen, *stereological problems* are discussed.

The Oberwolfach proceedings [100], collected by Ambartzumian and Weil, contain articles directly related to convexity, with subjects like *random touching of convex bodies, stereology, quermassintegrals, distribution of chord lengths of convex sets, random tessellations*, and further ones.

Ambartzumian collected in [97] articles related to integral and stochastic geometry. Also convexity topics are presented, such as Ambartzumian's "combinatorial integral geometry" in connection with *metrics and zonoids, expected convex hulls, and Banach space probabilities* related to *nested sequences of convex bodies and their support functions*. Problems from image analysis and stereology are treated, too.

Now we mention the proceedings to six conferences held in Italy, the first two without names of official editors. All of them have the same title "Conference on Integral Geometry, Geometric Probability Theory and Convex Bodies" (or a similar one), and they appeared in different Italian journals (most articles therein have been reviewed individually). These proceedings are

Mathematical Reviews (MR1400688): *Proceedings of the Second Italian Conference on Integral Geometry, Geometric Probability Theory and Convex Bodies*. Held in Lecce, June 5-8, 1991. *Rend. Sem. Mat. Messina Ser. II* **1(15)** (1991), suppl., Università degli Studi di Messina, Messina, 1991. pp. i-viii and 1-104,

Mathematical Reviews (MR1407778): *Proceedings of the Third Italian Conference on Integral Geometry, Geometric Probability Theory and Convex Bodies*. Held in Cagliari, May 5-9, 1992. *Rend. Sem. Mat. Messina Ser. II* **2(17)** (1993), suppl., Università degli Studi di Messina, Messina, 1993. pp. i-vi and 1-128,

and [2306], [2307], [2308], as well as [2309], all four edited by Stoka. In these issues one can find also many articles on topics from (classical) convexity or very close to it, with themes like *Blaschke sums of convex bodies, Baire category results, geometry of numbers, intrinsic volumes in*

Minkowski spaces, random polytopes and random convex hulls, oriented matroids, (finite) packings and coverings, isoperimetric and other inequalities, diametrically complete sets (in Banach spaces), geometry of convex curves, volumes of pairs of polytopes, mixed volumes, the polytope algebra, extremum problems for convex polyhedra, approximation of compact sets in Minkowski spaces, and Steiner points. And of course various contributions to *geometric probability and integral geometry* are there.

The proceedings [2007], with editors Quinto, Cheney, and Kuchment, contain some articles on integral geometry, Radon transforms, and (X-ray) computerized tomography. Also non-Euclidean concepts are considered.

In the collection [1021] of Grinberg et al. integral-geometric aspects of convex bodies are mainly studied, and in the collection of deep and important expository articles therein topics like *valuations, floating and illumination bodies, containment measures* (related also to isoperimetric inequalities), *curvature notions and Crofton measures in (projective) Finsler spaces, inequalities for the volume of a convex body and the volume of its projection body or its L^p centroid body, chord-power integrals of convex bodies, and tomographic questions* are used, discussed, or generalized.

Also in the proceedings [173] on *stochastic geometry*, edited by Baddeley et al., one finds papers related to convexity. The corresponding topics are *random polytopes* (volume and combinatorial structure) and *approximation of convex bodies by polytopes, Crofton-type and Steiner-type formulas and Hadwiger's characterization theorem, curvature measures as local versions of the intrinsic volumes, and random (convex) mosaics* (in particular, Voronoĭ and Delaunay mosaics).

The book [1362] of Kendall and Molchanov collects many expository papers describing modern trends in stochastic geometry like also its classical fundament. Some of these articles are very close to convexity, referring, e.g., to *random polytopes* (convex hulls of random points concerning estimates for higher moments, limit theorems, and large deviation inequalities, recently driven forward by relations to the floating body and the visibility problem), *random tessellations, stereology*, and stochastic geometry models usable in image analysis.

The proceedings [2286] edited by Spodarev contain a dozen of important articles, e.g., treating *foundations of stochastic geometry, a rigorous introduction to integral geometry, random tessellations, random polytopes, and discrete stochastic geometry*. Some of them are extra cited below.

In [2323], edited by Suh et al., a nice lecture on integral geometry of convex bodies by Zhang is contained, but also articles on related affine geometric results, Bonnesen-style inequalities, and isoperimetric problems are there.

The 15 excellent articles of the collection [2398] by Jensen and Kiderlen reflect the recent state of knowledge about *(tensor) valuations* related to convexity, integral, and stochastic geometry as well as imaging. Examples of concrete subjects discussed in these Lecture Notes are *Minkowski tensors* (defined via a Steiner formula from the moment tensors of convex bodies), *Crofton formulas for tensor-valued curvature measures, lattice polytopes, random tessellations, stereology*, and image analysis.

The proceedings [331], edited by Bianchi, Colesanti and Gronchi, refer to analytical aspects and again topics of convexity very close to integral geometry. They contain articles dealing with *dual curvature measures, powered mean curvatures, Crofton formulae for tensorial curvature measures, reverse volume difference inequalities, discrete centro-affine curvatures of convex polygons, the simplex mean width conjecture*, and a *volume characterization with the help of a Brunn–Minkowski type inequality*.

12.4.5.5 Selected surveys on integral geometry

In the form of nice examples, the survey [2101] on *integral geometry* collects different aspects, extended to *convex bodies in unimodular affine space*, spaces of constant curvature, Hermitian spaces, and Riemannian spaces. An early survey on the field of *integral geometry*, as it was established by Blaschke,

Chern, the author himself, and others, is Santaló's paper [2102]. It consists of eight parts; among them one finds representations of *integral geometry in Euclidean 3-space, applications to convex bodies, the fundamental kinematic formulae, integral geometry in complex space and in Riemannian spaces*, and an important collection of bibliographical notes. Fundamental results are comprehensively explained, and many nice examples and applications are given. As an illustrative example of discussed results, we mention Fenchel's theorem on the total curvature of a closed space curve.

Based on integral-geometric concepts of Federer, Hadwiger, and Firey treating variants of the classical *principal kinematic formula and contact measures for convex bodies*, Weil [2439] surveys the related knowledge and succeeds in offering a complete theory around this. Additionally, he creates a different new concept with new results.

Whereas the authors of other basic contributions to integral geometry constructed the used invariant measures as integrals of suitable invariant differential forms, Schneider and Wieacker [2182] take the direct measure-theoretic way due to Federer [834]; see also [2177] and [2179]. Results comprehensively discussed (mainly in their local versions) are related to *intersections of fixed and moving geometric objects, Minkowski sums of convex bodies, mixed volumes, projections and sections, distance integrals due to Hadwiger, contact measures (related to quermassintegrals), and additive extensions of geometric functionals onto the convexring*. Also global versions and classical results are presented, such as *Crofton's formula, Cauchy's surface area formula, and Kubota's integral recursion*. In addition, the authors treat *translative integral geometry* (see also [2442]), with related auxiliary *zonoids*. Summarizing, the survey [2182] gives an excellent overview to integral geometry modern in those days. Completing the bibliography in Santaló's classical book [2103], the references give a complete picture of the field, roughly in the period from 1975 to 1990.

An excellent survey on the field of stereology, popularizing it for convex geometers, is [2441] due to Weil. The author shows the importance of notions like *convexring, curvature measures and centroids, intrinsic volumes, (random) section, and projection results* for this purpose, like also the relationship to questions from *geometric tomography*, with similarities and differences. *Mean value formulas*, also important for stereological applications and concentrated on curvature measures, are discussed in [2440] by the same author, who also wrote the survey [2442] on integral geometry under the group of translations. The latter is strongly motivated by problems from stereology and image analysis (e.g., in view of non-isotropic random structures). Among several touching points with convexity, there occur certain mixed functionals which can be seen as *analogues of mixed volumes*. In these surveys, also interesting open problems are posed.

In the exposition [99], Ambartzumian reviews results in combinatorial integral geometry that followed his monograph [96]. Special topics important in this development and related to convexity are the Wicksell problem for randomized convex shapes, Pleijel-type identities for convex domains, problems concerning random line processes, and *valuations in integral geometry*. The common combinatorial roots of these problems are discussed, and relations to other fields (like tomography, cos-transforms, etc.) are shown.

In his long survey [1759], Mecke emphasizes (from the viewpoint of integral geometry) the importance of certain geometric functionals in statistical physics, in particular of *Minkowski functionals* like volume, surface area, and integral mean curvature. The importance of their additivity in physical applications is emphasized, and it is demonstrated that the dependence of many physical phenomena on the geometry of spatial structures assures that *integral geometry* is a promising tool for studying physical systems. This holds particularly in combination with the Boolean model, well known in stochastic geometry. The author illustrates that an integral-geometric approach to several stochastic structures in physics can be very useful, e.g., by applying related measures to topics like complex fluids, porous media, and pattern formation in dissipative systems. This survey is strongly related to Mecke's book [1758].

As a notion from integral geometry, the *containment measure* is the kinematic measure of a random convex body contained in a fixed convex body. In the exposition [2530], Zhang and Zhou survey

results on containment measures which are related to *integral formulas, isoperimetric inequalities, and geometric probability*.

In view of Hilbert's fourth problem, also the construction of geometries of the following type is interesting: In n-space, define the distance between two points as the measure of the hyperplanes intersecting their connecting segment for any positive measure on the space of hyperplanes vanishing on the hyperplane sets through a point. Busemann's intention of generalizing Hilbert's fourth problem in view of k-dimensional areas, for which k-flats minimize area, led here to the concept of *Crofton measures*, and with the related survey [2161] Schneider gives a complete picture on *generalizations of the classical Crofton formula of integral geometry from the Euclidean situation to Minkowski and projective Finsler spaces*. The general nonsmooth case is emphasized, and the relations to the theory of *zonoids* are shown.

In the survey [2162], Schneider collects techniques which are typical for integral geometry and important in stochastic geometry, all of them referring to convex bodies. Hitting probabilities with respect to two convex bodies (yielding the iterated kinematic formula), *curvature measures as local versions of intrinsic volumes, the translative formulas* (via intersections of a convex body with translates of other convex bodies), and *generalizations of Crofton-type formulas* describe the presented type of problems. The latter leads to an existence result for *Crofton measures of Holmes–Thompson areas* in *certain Minkowski spaces*, further on extended to *projective Finsler spaces*.

In the two surveys [2491] and [2490] the author studies affine geometric invariants of convex bodies obtained by integration and averaging. Constructions of *affine integral invariants of convex bodies* are given, yielding notions like the affine and L^p-affine surface area and providing a natural way of dealing with the useful notion of *homogeneous contour integral* in this framework; e.g., in this way a type of *Cramér–Rao inequality* is extended to star bodies.

The lecture [2529] of Zhang can be seen as a short, compact course in integral geometry. Basic ideas of Crofton, Poincaré, Blaschke, and Santaló can be found like also the relationship between Lebesgue and Grassmannian measure are used *to characterize quantities like volume and surface area of convex sets in the form of integral formulae*. The useful concepts of *support functions, vector addition, outer parallel sets, quermassintegrals, and k-th curvature integrals* are used or presented, yielding (among other things) a *generalization of Steiner's formula* and the *principal kinematic formula for convex bodies*.

Kiderlen gives in [1376] a self-contained introduction to *integral geometry and stereological applications*. First integral-geometric theorems (mostly with proofs) are presented, such as *Hadwiger's characterization of intrinsic volumes, Crofton's formula, the principal kinematic formula for polyconvex sets, and Blaschke–Petkantschin formulae*. The proofs are based on invariance arguments and an axiomatic approach to intrinsic volumes. In model-based stereology these tools are applied, and this yields unbiased estimators of specific intrinsic volumes of stationary random sets. Design-based stereology is also discussed, where Crofton's formula yields unbiased estimators of intrinsic volumes from isotropic uniform random flats. In its clearly written form, this article gives indeed a mathematically rigorous introduction

Here we add also the very recent and beautiful paper [563] of Calka, where classical questions from recreational mathematics are collected which can directly show the way to modern problems in geometric probability and related fields. Not requiring much pre-requisite knowledge on stochastic geometry and introductory in style, the author nicely shows *the ways from Buffon's needle problem to integral geometry, from Bertrand's paradox to random tessellations, from Sylvester's four-point problem to random polytopes, and from Jeffrey's bicycle wheel to random coverings*.

12.4.5.6 Selected surveys on stochastic geometry

In his nice survey [172] Baddeley discusses *basic ideas in stochastic geometry* which are closely related to *image analysis*. Particularly, he treats (among various further themes) the *synthesis of stochastic models of images, general concepts of geometrical information, the theory of random sets, and techniques of geometric integration theory*.

Molchanov [1820] surveys several aspects of the theory of random sets, particularly results for compacta and relations between stochastic processes and random sets.

The important handbook survey [2443] by Weil and Wieacker nicely shows the interplay of geometry and probability theory, where the probability distributions are introduced on geometric objects and figures of increasing complexity. Important directions of investigation refer to *convex hulls of random points in a convex body, random flats intersecting a convex body and geometric functionals of the polyhedral sets determined by these flats, random convex bodies and general random closed sets, Minkowski sums, locally finite unions of convex bodies*, random point processes in the space of closed sets (the latter forming the "points"), *random tessellations*, and some *stereological problems*.

Once more we mention the survey [2162] of Schneider, for techniques which are typical for convexity-related integral geometry and important in stochastic geometry.

Schneider and Weil present in the chapter [2180] basic tools and structures of stochastic geometry. So they lay important foundations for the whole collection [1362] of Kendall and Molchanov. A historical review nicely shows the main points of development from elementary geometric probability to the advanced models used in modern stochastic geometry. After presenting basic geometric and stochastic concepts, nicely chosen examples demonstrate their interplay.

Molchanov [1822] gives a nice *exposition on foundations of and developments within stochastic geometry*. He applies nicely introduced measure notions to random closed sets, explains the characterization of distributions of random sets by means of capacity functionals, and presents the concept of selection. The expectation is introduced, the convexifying effect in view of Lyapunov's theorem for ranges of vector measures is discussed, and Minkowski sums of random sets are studied.

12.4.5.7 Surveys on random polytopes and mosaics

Random polytopes are geometric objects very important in stochastic geometry and have a history of about 150 years. In general, they are considered as *convex hulls of a random set of finitely many points in Euclidean space*, but sometimes they are also introduced as *intersections of a random family of half-spaces*. Their study is reliant on methods and tools from both convex geometry and probability theory.

In 1864, J. J. Sylvester [2333] formulated the following (ill-posed) question: "Show that the chance of four points forming the apices of a reentrant quadrilateral is 1/4 if they be taken at random in an indefinite plane." Understanding that the "indefinite plane" without a properly defined probability measure on it is the problem, he modified the question within a year, and in the following form it became well known as Sylvester's four-point problem: Let K be a compact, convex set in the plane with non-empty interior, and choose four random, independent points uniformly from K. What is the probability $p(K)$ that the convex hull of the four points is a triangle, and for what K is $p(K)$ largest or smallest? This must clearly yield characterizations of special types of convex sets, and therefore questions of this type are interesting for convex geometers. Fifty years later Blaschke showed $p(disk) \leq p(K) \leq p(triangle)$.

It turns out that random polytopes developed to an own, well-established field that yields various interesting results in the spirit of *geometric probability*. We want to discuss the most important surveys covering that area of research (a proper monograph, completely dedicated to this subject, does not exist). It is convenient to take, as starting points, the respective overviews in the monograph [2103] of

Santaló and in the survey [171] of Baddeley collecting all developments in geometrical probability, also those treating random polytopes. The seemingly first survey after [2103] and [171], namely [520] of Buchta, discusses results on mean values of characteristics (or random variables, like volume, surface area, vertex number, facet number, etc.) of the *convex hull of m random points in a convex body K* situated in Euclidean *n*-space (or even in a metric compact manifold). Also special agreements for K are made, such as being a ball or a special polyhedron. On the other hand, also polytopes as intersections of random half-spaces are considered in [520]. In particular, the known results of Rényi and Sulanke on *convex hulls of random points and random polygons enclosed in a generalized annulus between two convex curves* are presented.

In view of *random approximation*, Schneider [2150] studies random polygons and polytopes as intersections of random supporting half-spaces. This paper contains a survey on *convex hulls of random points in n-dimensional convex bodies*, and the asymptotic behaviour of the expectations of some geometrically interesting functionals are especially taken care of. Affentranger's survey [21] continues that of Buchta [520] mentioned above. More general related questions and respective contributions by Renyi and Sulanke, Wieacker, Bárány, Schneider, and others are excellently surveyed by Weil and Wieacker in [2443], and Bárány exposes with [212] the later state of knowledge regarding the following generalizations: Let K be an n-dimensional convex body, and consider (instead of a quadruple) a random sample X_m of m uniform, independent points from K, with $p(m, K)$ denoting the probability that X_m is in convex position. For example, the asymptotic behavior of $p(K, m)$ for $n = 2$ is discussed, and also higher dimensional results are given. The second exposed problem analogously refers to the integer points instead of X_m in K, motivated by *integer programming and the geometry of numbers*. The paper [212] was largely extended by the same author, yielding the wonderful survey [215] which contains more old and new results around Sylvester's problem and further developments on random polytopes. These refer also to the *asymptotic behaviour of the expectations* and the famous theorem of Rényi and Sulanke on mean volume asymptotics in the plane. It is also demonstrated that, although proof methods are different, random polytopes and lattice polytopes behave similarly in this setting. Like partially also in [215], the lecture [214] of the same author gives a thorough introduction to the theory of random polytopes, emphasizing also tools and methods. In particular, the importance of the *floating body* of K in this framework is discussed, like also its useful combination with the *economic cap covering theorem*. Results concerning *central limit theorems and large deviation inequalities for functionals of random polytopes* are also treated.

Gruber [1068] surveys results suitable for comparing *best and random approximation of n-dimensional convex bodies by inscribed polytopes* (whose vertex number m is fixed or tending to infinity), and by *circumscribed polytopes* with the same assumptions on their facet number m. This is done by using, e.g., the following distance measures: *the symmetric difference metric* (regarding volume), *analogous measures based on other quermassintegrals, and the distances of Hausdorff, Banach–Mazur, and Schneider*. In each case, the situations that m is fixed as well as satisfies the corresponding asymptotic situation (m tends to infinity) are studied. The author presents *the best and the worst approximable convex bodies*, and he discusses algorithms for constructing well approximating polytopes and *relations to the floating body problem*.

Surveying three themes of the eight ones presented by Weil and Wieacker in the survey [2443], the important exposition [2163] of Schneider is devoted to *random polytopes as convex hulls of randomly chosen points* (in most cases uniformly chosen in a convex body), *random projections of high-dimensional (regular) polytopes*, and also *intersections of random half-spaces*. Clearly, the first case, concerning volume approximation of convex bodies by random polytopes if the number of generating points tends to infinity, continues the contributions [214] and [216] by Bárány. Distributional aspects of these random polytopes are discussed, in particular *large deviation inequalities regarding volume and face numbers*. In the second case, purely combinatorial properties of random polytopes (namely neighborliness) are studied, and high-dimensional rotated regular polytopes are projected into lower dimensional random subspaces, generating random polytopes there. And in the third case *asymptotic*

shapes of random polytopes in hyperplanar tessellations of n-space are dealt with. The author offers joint results obtained with Hug and Reitzner referring to generalizations of Kendall's problem (from the 1940s) on the *shape of large cells in random tessellations*. It turns out that certain *inequalities of isoperimetric type* are useful in this setting.

Several parts of Reitzner's interesting overview [2026] continue and update directions from [2163]: again random polytopes as convex hulls of m points uniformly distributed in a convex body K and random polytopes as intersections of finitely many random half-spaces occur there. Choices of *random points from convex surfaces or the boundaries of convex polytopes and, particularly, from their vertex sets* are studied, like also a Gaussian polytope (namely, the convex hull of random points chosen accordingly to the n-dimensional standard normal distribution). Further related results refer to certain inequalities, estimates for higher moments, and limit theorems. As recent geometric tools again the floating body and the visibility regions are clearly described.

The survey [1255] of Hug is split into two parts: the first one refers to *convex hulls of random points*, thus yielding random polytopes; and the second treats *intersecting random half-spaces, hyperplane tessellations, and cells of random mosaics*. In the first case, the author gives an updating overview emphasizing that many classical results for random points in smooth convex bodies were extended to convex bodies with weaker differentiability assumptions, and the enrichment of classical results on expectations by distributional results. The dual model of random polyhedral sets obtained from samples of random hyperplanes leads to further results in this setting. In the second part, mainly results on the shape of large cells in random tessellations are studied. Also *random hyperplane and Poisson–Voronoi tessellations* are studied. The author nicely shows the mixture of methods from geometry and probability, e.g., using *geometric inequalities, stability results*, and even *zonoids*.

A very recent broad survey on random polytopes is Calka [562], mainly concentrated on their asymptotic properties and, in particular, expectation and variance estimates. Jointly with colleagues, the author recently investigated the scaling limit of several models (e.g., the uniform model in a smooth convex body) and derived limiting variances for geometric properties (like the number of k-dimensional faces and the volume). This survey gives a respective overview, particularly studying random polytopes in the unit ball and Gaussian polytopes (see also the survey [2026] of Reitzner cited above).

In the exposition [245], Barvinok studies two computational problems concerning polytopes that are defined as certain intersections (e.g., of the non-negative orthant and an affine subspace). To get approximately the number of integer points in these polytopes, heuristic formulas are used which are based on probabilistic methods.

As one can see already in the two surveys [1255] of Hug and [2163] of Schneider, it makes also sense to treat here the related field of *random mosaics* (or *random tessellations*).

A hypersurface (from the space of locally countable, $(n-1)$-rectifiable closed subsets of n-space) is called a *mosaic* if its complement is a locally finite union of bounded, disjoint, open convex sets, the closures of which then are the *cells* of the mosaic. *Local finiteness* means that every compact subset of n-space meets only a finite number of cells. This definition implies that all cells of a mosaic are convex polytopes whose k-faces can be called the k-faces of the mosaic. An early expository article about random mosaics is [2104] of Santaló. It summarizes lectures of him to the Royal Academy of Sciences in Madrid (1988) and introduces the reader into the field of planar random mosaics. The author discusses *geometric and topological properties of mosaics* (e.g., the average area or number of vertices per cell) defined either as averages regarding increasing balls, or as an expectation regarding a stochastic process generating the mosaic. Explicit *stochastic models for random mosaics* (for example, Voronoĭ–Dirichlet tessellations based on the Poisson point processes) are presented, and results on the distributions and moments of cell properties and general results for stationary random mosaics are surveyed. Another good starting point here is the last but one section of the handbook article [2443] of Weil and Wieacker, where classical examples obtained from stationary Poisson point processes (also

Voronoĭ tessellations and Delaunay tilings) are discussed. These are due to Miles, Matheron, Stoyan, Mecke, Zähle, and Cowan; also extensions are discussed in [2443].

We mention here three books containing parts referring (or completely dedicated) to random mosaics. These are Chapter 10 of [2310], [1914], and [1869], and they show that random mosaics in two or three dimensions yield good models for various real structures. Particularly we emphasize the comprehensive monograph Okabe et al. [1914] on very many aspects of (random) *Voronoi diagrams and Delaunay triangulations*. These aspects are the history of both notions, their (not only geometric) properties, various generalizations of both concepts, *spatial interpolation* (methods for approximating surfaces and ordering multivariate data), and the algorithmical side (yielding constructions and their complexities in the spirit of computational geometry). Also *Poisson–Voronoi diagrams*, respective *Delaunay tessellations*, other types of "random" Voronoi diagrams and application fields are studied. The part close to stochastic geometry can be found mainly in Chapter 5, but also Chapter 7 is interesting, where models of spatial processes are investigated. The second edition of [1914] contains more new and revised results on the Poisson–Voronoĭ diagram. In the lectures [1869] the author gives a rigorous mathematical treatment of *random Voronoi mosaics*. After giving basic geometric properties and the necessary *fundaments from integral geometry*, he studies *stationary Voronoi tessellations*. The classical Poisson–Voronoi tessellations allow computations of many interesting theoretical characteristics of the "typical" cell. (But many quantities are known only from simulations.) It should be emphasized that in these Lecture Notes especially many probabilistic results on Voronoi tessellations and their proofs can be found.

We cite now Schütt's two surveys [2196] and [2197] on *best and random approximation of n-dimensional convex bodies by polytopes* (see also Gruber [1068]). In the first one, also an elegant new proof of an approximation result regarding the *symmetric difference metric* is given, and in the second the dependence on the dimension n in asymptotic formulae is compared with results on *random approximation of (smooth) convex bodies by polytopes with given vertex number*.

In the exposition [1254] of Hug basic properties of *stationary random mosaics* (i.e., of random translation-invariant, locally finite tessellations of the Euclidean space) are studied. Geometric notions like typical k-face, typical cell, zero cell, and large cell referring to such mosaics are introduced. Basic properties of these geometric objects (such as their distribution, mean intrinsic volumes, etc.) and their interrelations are investigated, in particular for *Poisson–Voronoi, Poisson–Delaunay and Poisson-hyperplane mosaics*. For example, *mean values of intrinsic volumes of typical faces* are explicitly calculated in the Poisson–Voronoĭ case, whereas the *exact distribution density of the typical cell* is given in the Poisson–Delaunay case. And for Poisson hyperplane mosaics some calculations of mean values and an inequality for the mean volume of the zero-cell are presented. Also for these types of tessellations, the author discusses interesting problems related to the famous conjecture of Kendall (saying that "the shape of a large zero cell of a planar isotropic Poisson line tessellation is asymptotically a circular").

Also Calka's [561] is a survey on some recent methods and results on *specific models for the theory of random tessellations*. Two typical types of results are presented: *exact distributional calculations and asymptotic estimates*. First two basic constructions of random tessellations are given: those obtained by throwing random hyperplanes, and those obtained by constructing Voronoi cells around random points. Then the notion of *typical cell of a stationary tessellation* is introduced, and exact distributional results on geometrical characteristics (like the typical k-face) are derived. Some *asymptotic properties of cells* follow, and recent results to the above-mentioned conjecture of Kendall are exposed. And finally, new models of *iterated tessellations* appearing in applications like telecommunications are discussed. Here we mention also once more Calka's nice paper [563], showing conveniently the way from classical problems (like Bertrand's paradox and Sylvester's four-point problem) to modern research areas (like random tessellations and random polytopes, respectively).

It is well known that certain figures and also tessellations, solving extremal problems in discrete or convex geometry (for example, problems of isoperimetric type), often have a high degree of symmetry; see, e.g., the classical book [847] of L. Fejes Tóth. In the survey [2166] of Schneider, various results

on *extremal properties of mainly special (i.e., hyperplane, Voronoi or Delaunay) types of random mosaics* are exposed. Among those of geometric flavor, some results are related to extremal problems for expected sizes of average cells (defined in two ways) satisfying some additional condition. This leads to *random mosaics with high symmetry or of a very simple type*, namely made up of parallelepipeds only. High symmetry here means that the distribution of the random mosaic, which is usually assumed to be translation invariant, is also rotation invariant. Extremal problems for the sizes of average cells (not taking expectations) seem ill-posed at first, since extrema cannot be attained. Nevertheless, average cells of large size approximate in many cases certain definite shapes, for example balls, segments, and regular simplices, with high probability.

Werner [2455] gives a survey on *best and random approximation of convex bodies by polytopes in view of convex floating bodies*, their variants (like surface bodies and weighted floating bodies), and applications. The geometric quantities of the considered convex bodies are given by the affine surface areas. Extensions to convex bodies without smoothness assumptions are emphasized.

12.4.6 Convex analysis

Convex analysis is a field of mathematics in which properties of convex functions and convex sets are studied, often with applications in convex optimization as a subfield of optimization theory. Having many further application areas, the theory of *convex functions* (see our Chapter 1) is not a subfield of convex analysis, but has a large overlap with it. In several cases the question in which chapter something is cited or discussed depends only on nuances of emphasis. We try to give here a representation of methods and results from the part of convex analysis that is close to convex geometry, but without any doubt some passages from the (notes of our) first chapter would fit in this part here in a useful sense. But we decided from case to case where the material is better stored.

Since also in convex analysis there are thousands of related research articles in journals, it is necessary to restrict the representation here to books, proceedings, and survey-like papers. In addition, we will mainly refer to topics discussed in our Sections 1.3 and 1.4, such as *subgradients, subdifferentials, conjugacy, duality*, and related ones. Only some additional notions (near to convexity) will be also presented, like generalized convexity concepts.

12.4.6.1 Historical remarks and a short description of the field

Since the appearance of Rockafellar's classical book [2046], *convex analysis* is a well-established mathematical field. Its application areas present a broad range of mathematical fields, with optimization as the most important one. Here one can cite the historical paper [1394] of Kjeldsen, where the introduction of the concept of a *general convex body* by Minkowski and the development of *mathematical programming* are discussed. Kjeldsen nicely describes how and why these two new fields of the twentieth century developed by certain connections to each other, and how they inspired one another. Since the mid and late sixties the name "convex analysis" appeared in the mathematical language, and anyone working close to this field (e.g., in convex optimization) knew how deeply it has influenced several parts of mathematics. An interesting representation of the *history of convex analysis* is given in the introduction of the book [1669] of Magaril-Il'yaev and Tikhomirov; see also the third chapter of the article [761] of Dorofeeva and Tikhomirov. And the broad survey [2368] of Tikhomirov (see also [2366] of the same author), discussing basic principles and results of the *theory of extremal problems*, treats also the history of basic tools and results of functional and convex analysis needed to develop this theory (see also the monograph [1275] of Ioffe and Tikhomirov, which is described in more detail below). Tikhomirov also discusses the importance of the Lagrange principle within this framework.

Convex analysis has many overlaps with other mathematical fields. As we will see below, it became especially important for *several parts of optimization and analysis*. For example, by two main reasons methods from convex analysis turned out to be particularly efficient in *optimal control and calculus of variations*: First, due to the abundance of meaningful convex problems, especially involving multiple integrals, and, second, since we have the inherence of a certain portion of convexity in any variational problem, even if it is non-convex. Besides very many concrete touching points, there are various books clearly dedicated to fields from analysis and optimization, but containing nicely written full chapters on topics and methods from convex analysis and notions from convexity, *since they are simply needed*. Although many of the books cited more below or also in other parts of our book satisfy this criterion, we will mention here only *seven examples*.

First we look at the monograph [1275] of Ioffe and Tikhomirov, giving a unified approach to the *theory of extremal problems* in a very broad sense and comprising applications in mathematical programming, control theory, and calculus of variations (e.g., Kuhn–Tucker conditions, the Euler–Lagrange equation, and Pontryagin's maximum principle are covered). (Local) convex analysis is present already in the introductory parts, namely in the form of *convex sets and functions, Fréchet-derivatives, subdifferentials and the Moreau–Rockafellar theorem, conjugate functions*, which are applied to locally convex problems, and further notions and tools. Thus, besides topics from functional analysis, differential equations and some other fields, convex analysis is one of the fundaments of this book covering a wide range of mathematical topics.

Second, the book [291] of Berkovitz on optimization contains sections about *separation theorems, geometric inequalities, alternative theorems for systems of convex inequalities, subgradients and subdifferentials, and optimization of nonsmooth convex functions*.

Third, the monograph [732] of Denkowski et al. on *nonlinear analysis* has in its fifth chapter a nice representation of *conjugate functions and subdifferentials of convex functions*; in particular, the celebrated Clarke subdifferential is clearly presented.

Fourth, the book [2385] of Tuy on *global optimization* may be noted. It is divided into two parts, dedicated to convex analysis and global optimization, respectively. The first part contains four chapters, beginning with the study of *convex functions and convex sets*, followed by the basics on fixed points and equilibrium problems as well as *differences of convex functions and sets*, respectively. In the second part, a large variety of typical problems and methods from global optimization is presented.

The *fifth* example is [1374] of Khan and Tammer, presenting the field of *set-valued optimization* which includes, as special cases, classical and vector optimization, and which was influenced by nonlinear and nonsmooth optimization, set-valued analysis, and variational analysis. Interestingly, typical recent application fields are economical and financial models. In this book the authors give the first unified view onto the state of the art in this field, and in the chapters 5 to 8 typical notions and tools from convex analysis and convexity (such as conjugates, subdifferentials, and duality) are introduced, of course on the set-valued level.

The *sixth* example is Bagirov et al. [180] regarding *nonsmooth optimization*, strongly based on relations to nonsmooth analysis and variational analysis. Several subdifferential notions (related to that of Clarke) are introduced, and continuous subdifferential approximations usable in computations are particularly taken into consideration. The authors present a rich list of applications in different sciences. Also in *stochastic optimization* (i.e., optimization where decisions must be taken in the presence of uncertainty) basic ideas from convex analysis can be successfully used.

This can be also seen in our *seventh* example, the monograph [443] of Bonnans, which is mainly based on the setting of Banach spaces. It shows that the theoretical fundaments of *stochastic optimization* are strongly based on notions like *convex function, duality, Fenchel conjugacy, subdifferential*, etc. (see particularly the first and the last chapter). For example, it is shown that sometimes convex functions over measure spaces, constructed as Fenchel conjugates of integral functions of continuous functions, are equal to some integral of a function of their density. On the other hand, there are also books about convex analysis containing really much material from neighbouring disciplines like *convex geometry*.

One example we mean here is [1669] of Magaril-Il'yaev and Tikhomirov, containing an impressively large amount of theorems from classical convex geometry. For example, the authors discuss rigidity of polyhedra, the theorems of Helly, Radon, and Carathéodory, Cauchy and Steiner–Minkowski formulae, and the Brunn–Minkowski inequality.

We continue now with a survey on the literature in convex analysis which is ordered by several themes (like *duality*) and related to our work here.

12.4.6.2 Duality and conjugacy

As we will see below, conjugate duality and extensions of it are strongly connected with the origins of convex analysis.

CONJUGATE DUALITY AND EXTENSIONS

Duality mappings and conjugate duality are basic notions in convex analysis, and already Rockafellar himself paid careful attention to this topic (see Part III of Rockafellar [2046], but also his publications [2044] and [2047]). In [2046] he shows the key role of *conjugate duality*, a basic notion from convex analysis, to present the most powerful existence, uniqueness, and characterization theorems for solutions in convex optimization. Thus, *conjugacy of convex functions* (already introduced by Fenchel; see the historical paper [1387] of Kiselman) started to play also a very fundamental role in non-convex optimization. It turns out that conjugacy correspondence for convex functions generalizes the classical *polarity introduced in convex geometry for convex cones*. Rockafellar [2046] considers also several dual (regarding the conjugacy correspondence) pairs of functional operations. Based on the concept of *convex bifunction* and the adjoint of a convex bifunction, the notion of duality for generalized convex programs is obtained.

Due to these and even earlier contributions, *conjugate duality* started to play an essential role in convex optimization and various other fields. For example, the booklet [2047] of Rockafellar gives a thorough introduction to conjugate duality in finite- and infinite-dimensional problems. In this publication the author strictly deleted topics from convexity which are unnecessary for the duality concept; necessary ones remained of course, and they are measured regarding their significance in the study of duality. It is demonstrated what remains true if certain convexity or lower semicontinuity assumptions are deleted. Another early reference is the expository paper [2071] of Rubinstein, which is based on [2070] and contains an abstract model of duality for optimization problems. A collection of *pairs of dual problems* is given, and properties of convex sets related to programming are studied. Here also the booklet [2413] of Walk should be mentioned, which nicely introduces into the *geometry of convex sets and functions*; it presents also fundamental results in duality theory of mathematical programming, as needed in optimization and the corresponding standard teaching programs of universities.

CONCEPTS OF EKELAND, PENOT, AND OTHERS

In some situations one succeeds by reversing inequalities between functions, in other cases the needed transforms are classical in nature. We continue with the survey [1961] of Penot, having Ekeland's important contribution to variational analysis (see [808]) as a starting point. Penot's framework of duality developed in [1961] *generalizes classical duality theories*, and he shows (also surveying existing related results) the links of Ekeland's duality concept to other well-known schemes. He also verifies that his general concept has other well-known dualities as special cases, in particular also the Fenchel–Rockafellar duality and the Clarke duality. The lecture notes [472] are an important contribution to conjugate duality in convex optimization, building a bridge to Rockafellar's monograph [2047] by exposing the developments between both works. The author presents an approach to *modern duality*

theory based on perturbations. Further presented topics are sufficient conditions for strong duality which are as weak as possible, strong and total conjugate duality, totally Fenchel stable functions (in the sense of Simons [2254]), and applications of duality to *monotone operators*. Similarly, the monograph [1014] of Grad gives a deep insight into *duality theory in vector optimization* and related topics. The author particularly studies the Wolfe- and the Mond–Weir-type dualization, also dealing with *generalized convexity concepts*. As in [472], the final chapter offers results involving monotone operators. In the interesting paper [754], Dinca surveys results on duality mappings on Banach spaces. For smooth reflexive Banach spaces the paper contains a large collection of examples of functions and some continuity properties, and duality mappings are also put in relation to monotonicity and norm-attaining functionals. For example, the famous theorems of James, Bishop, and Phelps are stated in terms of a duality mapping. Also, the author nicely presents the relations of duality mappings to important mathematical fields and concepts, such as PDE or reflexivity and subreflexivity in Banach spaces.

FURTHER DUALITY TYPES

The survey paper [1486] of Kutateladze and Rubinov treats the so-called *Minkowski duality* permitting a one-to-one relation between a sublinear function and its subdifferential. Historical remarks and an extensive list of references are given. The book [1935] of Panik, offering the foundations of convex analysis to students of several disciplines, covers basic tools and topics of convex analysis, particularly *duality and separation*. Duality plays a key role, and separation is understood as the root for dual schemes. The case of *quasi-convex duality theory* is surveyed in [1700]. Given a family H of extended real-valued functions, closed under pointwise supremum, the author studies *H-convex functions* and an H-conjugation theory applied to H-duality in optimization problems (for the concept of H-convexity, see [2072] and below in this subsection).

Related to these notions, Rolewicz [2056] summarizes results on duality also referring to extended real-valued functions in the absence of a linear structure. He discusses basic results for respective generalized convexity notions, including generalizations of the Krein–Milman theorem, the Lagrangian duality, and the Fenchel conjugacy. Also extensions of classical results of Mazur on the differentiability of convex functions and of Asplund on relations between local uniform convexity and local uniform smoothness are presented. A unified approach to *duality in multi-objective optimization* is given in the survey [1873] by Nakayama. One of the difficulties is that the efficient solution to multi-objective problems is not necessarily unique. The definition of infimum (or supremum) of a set with a partial order plays a key role in the development of the duality theory in multi-objective optimization, and in [1873] this framework is also considered from a geometric viewpoint. The paper [137] of Artstein-Avidan and Milman surveys results which characterise the notion of duality in several spaces of functions and of sets, such as convex functions, s-concave functions, convex bodies, and log-concave functions. The authors minimize the information or transforms needed to derive the concrete formulas for duality in each of these cases.

We now come to the *duality principle in convex algebraic geometry* (see [2061] and [2062] of Rostalski and Sturmfels, as well as subsection 12.4.3 above), a field between convexity, optimization, and algebraic geometry. The authors compare three notions of duality which are essential in different parts of mathematics: *duality of convex bodies, duality of projective varieties*, and the *Karush–Kuhn–Tucker conditions in optimization*, derived from the Lagrange duality. Recently, structural properties of feasible regions of semi-definite programs, which are so-called *spectrahedra*, are intensively investigated. The authors summarize the problem of identifying types of convex sets which are spectrahedra or projections of them under semi-definite representability. Many inspiring examples in small dimensions are presented, illustrated also by nice figures. Finally to dualities, we refer to the publications [2233], [752], and [884], nicely presenting how useful dualities can be in economics, and discussed in more detail at the end of this subsection, among the application fields.

12.4.6.3 Subdifferentials, subgradients, and related notions

We continue with textbooks and surveys containing basic material on important *properties of convex functions and sets* and the notions of *subdifferential, subgradient, Fenchel conjugacy*, and once more *duality* in the widest sense. First we mention here the textbooks [2001], [13], [1912], and [692]. These books were published in the 1980s (or, in one case, in 2003), and they treat *classical convex analysis as a whole*. Nicely ordered and presented, classical tools and notions from convex analysis are given there. Unfortunately, these monographs were not translated into English. Rich in content, Kusraev and Kutateladze [1483] present a survey on tools and results in classical convex analysis, based on a very extensive bibliography. They discuss the role of cones, operators, subdifferentials, and duality in this field. The formula of Moreau–Rockafellar, the duality of Fenchel, and the variational principle of Ekeland are presented like also Clarke's tangent and normal cones. Similarly, the survey [2366] by Tikhomirov contains basics from convex analysis, i.e., duality principles, subdifferentials, and generalized convexity notions, also in *spaces of vector-valued functions*. In [731] optimization problems involving nonsmooth (nondifferentiable) functions of a finite number of variables are investigated. The authors discuss basic notions from convex analysis which are important for optimization theory. For example, they pay much attention to ϵ-subdifferentials of convex functions and subgradients; also quasi-differentiable functions are introduced and studied. The corresponding optimality conditions are given in terms of set inclusion. Also the textbook [174] is dedicated to many basic themes from or related to convex analysis, such as monotone operators, subgradients of convex functions, and duality theory via conjugate functions or, alternatively, via Lagrangian functions. An entirely geometric and self-contained approach to properties of convex functions and sets in the n-dimensional setting is given in the self-contained textbook [1906], showing their importance for convex analysis. It starts with geometrical and topological properties of convex sets and cones, *some separation theory, spherical images, and support functions and support hyperplanes* of such sets. A thorough treatment of convex functions and boundaries of convex sets is given, and the *Fenchel conjugate* together with its geometric properties is studied. Further on, a general theory of convex optimization is given, including the Kuhn–Tucker theory, the role that Fenchel duality theory plays there, a generalization to weaker regularity conditions, applications to classes of linearly constrained optimization problems, and a general notion of dual problems. The book does not mainly address to the researchers in this topic, but to the beginners who will certainly appreciate the sometimes comprehensive formulations and proofs. The interesting book [1484] on fundamental concepts of subdifferentials is mainly dedicated to explicit rules for the *calculation of subdifferentials of convex operators* in topological vector spaces. Minkowski's duality theory is used to describe the geometry of subdifferentials and to study them intrinsically. Generalizations of the chain rule as well as algebraic laws for their calculus are presented. The authors also present a chapter with the topological theory of subdifferentials, and another chapter refers to extremal problems, e.g., in vector optimization. In the last chapter, the Clarke subdifferential is comprehensively discussed; see also the surveys [1960] of Penot and [972] of Giles.

APPROXIMATE SUBDIFFERENTIALS

It is obvious that some of the references already cited above and being cited in the forthcoming paragraphs, especially on nonsmooth analysis, also refer to *approximate subdifferentials and related topics*. But the publications discussed now here are possibly more explicit in this sense. We mentioned Ekeland's theorem on the existence of approximate minimizers of lower semi-continuous functions bounded from below; see his own survey [808] and our discussion above on calculus of variations. In the well-written survey [1200] of Hiriart-Urruty, the author summarizes results pertaining to the *approximate subdifferential* and the *approximate first- and second-order directional derivatives* of a convex function, and the later survey [1203] of Hiriart-Urruty et al. refers also to approximate subdifferentials of convex proper functions on locally convex real topological vector spaces. The aim is to present calculus rules for the subdifferentials of convex functions constructed from other convex functions via known operations from convex analysis, without any qualification conditions.

Among other interesting points, the authors apply the presented results by calculating approximate subdifferentials for non-convex marginal functions and convex hulls of non-convex functions, and by expressing approximate normals to sublevel sets in terms of the defining convex functions.

We continue with the two-volume series [1201] and [1202] of Hiriart-Urruty and Lemaréchal. The first volume is an excellent exposition of *variational aspects of finite-dimensional convex analysis* in connection with continuous (convex) optimization. Nicely written and illustrated, it is an excellent reference for students, engineers, and many further representatives of applied sciences, but on the other hand also for experts in optimization and convex analysis. General properties of convex sets and convex functions as well as duality concepts are considered. The duality between sublinear functions and convex sets is examined, and applications of the subdifferential calculus to convex minimization problems are given. The authors use the *Lagrange duality* and the connection between Lagrange multipliers and saddle points of the Lagrangian. The subdifferential calculus is presented for finite-valued convex functions only, since this is sufficient to treat convex programming problems with explicit equality and inequality constraints. Extended-valued functions are dealt with in the second volume [1202], which excellently presents advanced numerical algorithms in convex optimization based on interesting developments in convex analysis. There are two major themes in this second volume: *bundle methods* and *approximate subdifferentials*. The authors explain carefully the bundling implementable mechanism to generate subgradients of convex functions at given points. Implementation difficulties there motivate the introduction of the so-called *approximate subdifferential construction*. A thorough study of approximate subdifferentials yields an extended calculus and involves the Legendre–Fenchel transform. The authors apply such basics to Lagrange relaxation, general decomposition in mathematical programming, and dual algorithms from convex optimization, and they give the necessary duality theory in various situations. Also dual forms of the presented bundle methods are considered, and the authors obtain algorithms for practical nonsmooth convex optimization. Also in the book [1847] due to Mordukhovich and Nam interesting approximation results are presented.

In the book [730] tools from convex analysis in *nonsmooth analysis and homogeneous approximation of functions, sets, and mappings* are studied. In [730] of Demyanov and Rubinov and in the survey [729] of Demyanov and Roshchina, different types of subdifferentials are treated. Based on exhaustive families of upper convex and lower concave approximations, the dual tools of the upper exhauster and the lower exhauster are studied, and the subdifferentials of Fréchet, Gâteaux, Clarke, and Michel-Penot are dealt with in terms of exhausters. Also in the survey [728] of Demyanov approximation problems are discussed. Due to the variety of subdifferentials and quasi-differentials, positively homogeneous local approximations of nonsmooth functions are shown to play an essential role. For codifferentials, non-homogeneous approximations are discussed. In view of nonsmooth analysis (see below), we mention also the two papers [728] of Demyanov, where approximation results play a role, and [2359] of Thibault in view of approximate convexity.

Finally, referring to approximation topics but not being related to subdifferentials, we mention shortly the book [2335] of Takahashi. It contains many interesting results from *convex analysis and approximation of fixed points in Banach spaces*, and it is a useful reference for mathematicians directly interested in the interplay of these fields, but also for researchers in physics, engineering, operations research, and mathematical economics.

SUBDIFFERENTIALS OF CLARKE, FRÉCHET, MICHEL-PENOT, MORDUKHOVICH, AND OTHERS

The book [464] of Borwein and Lewis provides a comprehensive presentation of the main features of convex analysis in its correspondence with *nonlinear optimization*. It gives a concise treatment of the area, showing in particular the *relevance of developments in nonsmooth analysis for optimization theory*. The material is presented in the Euclidean setting, but the coordinate-free style motivates extensions to Banach spaces, which are also studied in the last chapter. Optimality conditions are

discussed with the assumptions of *differentiability, nondifferentiability*, and also *local Lipschitz continuity*. Further on, after developing the *differentiability theory for convex functions*, the authors give a useful exposition of the *duality between a convex function and its Fenchel conjugate*, which in turn is significant for duality theory of convex programs. Studies in convex process duality lead to the introduction of the new concept of *multivalued maps*. In view of nonsmooth optimization, different types of subdifferentials are discussed: the well-known *Clarke subdifferential* introduced in 1973 (which inaugurated nonsmooth analysis, and reduces to the classical gradient for smooth functions and to the subdifferential for convex functions; see also Clarke [629] and Clarke et al. [630]), the *Michel-Penot subdifferential*, and the *limiting subdifferential of Mordukhovich*. The authors develop a calculus for these subdifferentials, applications in general nondifferentiable situations are also given, and subcases as well as further consequences of these advanced concepts are studied, e.g., treating *polyhedral convex functions*, eigenvalue functions, as well as duality for linear semi-definite programming.

Coming back to Clarke, Penot [1960] stresses compactness conditions which enable him to obtain *calculus rules for subdifferentials of lower semi-continuous functions, not necessarily locally Lipschitz or convex*. These rules are linked with calculus rules for normal cones or even with the computation of coderivatives of multimappings. The *Clarke subdifferential* (of a locally Lipschitz real-valued function on a Banach space) provides one of the most natural generalizations of the notion of subdifferential of a continuous convex function. In the article [972] of Giles some basic properties of *Clarke subdifferentials in Banach spaces* are surveyed, mainly based on notions like "regularity", "pseudo-regularity", "strict differentiability", and "uniform (Fréchet) strict differentiability".

The excellent research monograph [1962] of Penot on generalized differentiation and subdifferential concepts continues a series of related books consisting of Rockafellar and Wets [2048], Borwein and Zhu [470], Mordukhovich [1845], Schirotzek [2134], and further ones, e.g., treating Fréchet and limiting Fréchet subdifferentials on Fréchet smooth or Asplund spaces. But in [1962] Penot goes further. Namely, with concepts like Dini–Hadamard subdifferentials on Gâteaux smooth spaces, Michel-Penot (moderate) subdifferentials, and approximate subdifferentials on arbitrary Banach spaces, Penot presents a really broad spectrum of subdifferential concepts in nonsmooth analysis, in addition to a fairly complete account of the Fréchet calculus and limiting Fréchet subdifferentials on Asplund spaces. Also, he lays emphasis on the connections between basic ideas of all branches of local nonlinear analysis, from classical differentiation through convex analysis to nonsmooth analysis. Thus, as it is done also by Rockafellar and Wets [2048] as well as Schirotzek [2134], he strongly exposes the role of convex analysis in this framework. One important subject of the book is the *calculus of the subdifferentials on smooth spaces* (like Gâteaux smooth spaces and Fréchet smooth ones), another one is the careful study of the generalized gradient of Clarke and Michel-Penot subdifferentials (regarding generalized gradients the author is on the lines of Chapter 2 from Clarke et al. [630], but gives more attention to the geometric concepts of normal and tangent cones).

SUBDIFFERENTIALS IN BANACH SPACES

Also other publications on subdifferentials cited here refer to Banach spaces. But other dominating aspects force them under a different heading.

The largely self-contained monograph [970] of Giles treats *convexity in Banach spaces*, where the differentiability of convex functions plays a key role. After basics about convex sets and functions, linear topological spaces, and Banach spaces, the author presents a systematic study of *Gâteaux and Fréchet differentiability of convex functions on Banach spaces*, closely connected with (weak) Asplund spaces. In this context many results are proved, such as the Mazur theorem concerning the Gâteaux differentiability of continuous convex functions, and versions of the Bishop–Phelps theorem. In the final chapter two open problems of convexity of Banach spaces are discussed: the *Mazur intersection property* concerning the question whether any bounded closed convex set in a Banach space is an intersection of closed balls, and the problem *whether any Chebyshev subset of a normed linear*

space is convex. For many further results and observations on the differentiability of functions in our framework here we refer also to the surveys [971] of Giles and [465] of Borwein et al., to the books [464] of Borwein and Lewis and [466] of Borwein and Vanderwerff, and also to the related discussion of differentiability and Banach-space theory in the related subsection.

In the extensive expository paper [467] of Borwein and Zhu, which is dedicated to the mathematician Clarke, a first-order subdifferential calculus in Banach spaces with Fréchet-smooth norms is presented, using a subgradient approach based on Fréchet derivatives. At the beginning, four results in this theory are shown to be equivalent, among them being the nonlocal and the local fuzzy sum rule. Later on, constrained minimization problems and subdifferential calculus are discussed, further on based on the Fréchet-type concept and the fuzzy approach. Then the authors study main ideas from various fields where nonsmooth methods have been successfully used (e.g., *approximate mean value theorems*, integration of subgradient set-valued mappings, sensitivity analysis in mathematical programming, *fundamental results in functional analysis*, and metric regularity; see below). The paper contains *alternative constructions of subdifferentials* and a rich bibliography. We also refer to the addendum [469] of Borwein and Zhu, containing a correction and further references.

Zalinescu [2527] has the aim "to present the conjugate and subdifferential calculus using the method of perturbation functions in order to obtain the most general results in this field" and "to provide important applications of this calculus and the properties of convex functions". In the second chapter the notions of *convex functions, directional derivative, conjugate function, and subdifferential* are suitably introduced, together with the general *convex optimization problem* and its associated value or perturbed function. Due to a presented *broad duality formula* and its effects on the subdifferential calculus, the *Fenchel duality*, etc., many applications can be successfully studied. In particular, a *characterization of convex functions via an abstract subdifferential* is obtained, having as particular cases the subdifferentials of Clarke and Fenchel. The author continues with *applications in Banach spaces*, particularly studying subdifferentials there and presenting theorems like that of Bishop–Phelps, Rockafellar's monotonicity theorem (with a simple proof), Zagrodny's approximate mean value theorem, etc.; *various notions from the geometry of Banach spaces are characterized via convex analysis*.

The recent book [1272] of Ioffe summarizes also results on convex analysis in Banach spaces that were obtained by him and can also be found in some expository papers which appeared before [1272]; see below.

PUBLICATIONS STRONGLY BASED ON CONVEX GEOMETRY

The following two references contain a particularly large number of results from classical convexity.

Also the nice textbook [1847] by Mordhukovich and Nam refers to convex analysis in finite-dimensional spaces. Based on algebraic and topological properties of convex functions and sets, basics of the subdifferential calculus and Fenchel conjugation are studied. The authors present many further results from *convexity* (such as the *theorems of Radon, Helly*, and *Carathéodory*, approximation results, etc.), and various applications from nonlinear optimization with many suggestive examples are given. The finite-dimensional setting continues with Krantz [1463]; his booklet differs from many other books in the field by its brevity and the analytic material, for example using *functions defining convex domains, or functions for which the sublevel sets exhaust the convex sets from inside*. Different orders of convexity are studied, too. Various classical theorems and notions from convexity are also given, like again the theorems of Radon, Helly, and Carathéodory, but also the *Krein–Milman theorem, the inequalities of Brunn–Minkowski, Jensen, and Hölder*, the concept of *polarity, convex sets with nonsmooth boundaries, and convex functions without derivatives*.

Bakelman [183] studies *nonlinear geometric PDE*, especially elliptic ones. It turns out that the theory of convex bodies and hypersurfaces is closely related to the theory of convex solutions of nonlinear elliptic equations, in particular to the *Monge–Ampère equations*. In [183] various significant results in this direction are summarized. Since such results are scattered in different places, the book

is a particularly valuable contribution. Basic elements needed from convexity are properties of *mixed volumes, the inequalities of Brunn–Minkowski, Alexandrov, and Fenchel*. In addition, the Minkowski problem for convex polytopes and further results due to Minkowski are needed. The weak and general solutions of boundary value problems for Monge–Ampère equations are generally convex and concave functions. Dealing with the generalized solution of the Dirichlet problem for the n-dimensional Monge–Ampère equation, a special geometric functional in the class of general convex functions and certain profound geometric constructions and inequalities in the functional dual space are investigated.

FURTHER RESULTS BASED ON SUBDIFFERENTIALS AND RELATED TOPICS

The monograph [302] of Bertsekas is an introductory textbook, well designed for teaching purposes and complementing the mainstream of literature related to convex analysis. It covers *convexity in finite-dimensional spaces and related applications in optimization*, both in terms of characterizations of solutions and algorithms. *Subgradients and incremental subgradient methods* play an essential role. The author develops a theory and techniques applicable to non-convex problems; he derives weak duality conditions, minimax and saddle-point theorems, and constrained optimization duality, in the sense of Lagrange and Fenchel. A large portion of the text is concerned with *Lagrange multipliers and Lagrangian duality*, and a chapter on *polyhedral convexity* contains some forms of the Farkas lemma, including a nonlinear form, some linear programming duality, and unimodularity in the context of integer programming. The book [303], also due to Bertsekas, on convex analysis and optimization, is compared with [302], even more a textbook, written also in a good didactic style with many interesting examples and exercises, and illustrated geometrically by many figures showing excellently the visual treatments of many problems. Thus, the reference [302] is very suitable for self-study of the topics. All basic tools and notions from (the more theoretical side of) convex analysis which are necessary for the goals in optimization are widely discussed. In an own chapter, also *polyhedral convexity* is nicely presented.

Zagrodny [2516] surveys results and poses open problems on *directional convergence of subgradients, subdifferentials*, and Hessians. He also presents relations to further open problems, such as *Alexandrov's theorem in infinite dimensions, or the convexity of Chebyshev sets*. The book [747] of Dhara and Dutta treats optimality conditions in finite-dimensional convex optimization and contains an interesting variety of approaches to the field. The authors provide the *material on convex sets and functions* needed for preparing optimality conditions, i.e., the calculus of subdifferentials and conjugate functions. *Optimality conditions of Karush–Kuhn–Tucker type* for inequality-constrained convex programs follow, and a comparison of Lagrangian duality and Fenchel duality is given. After this, optimality conditions with inequality constraints are widely discussed. For example, a characterization of convexity for a set defined by smooth inequality constraints is derived and applied to characterize optimality for a smooth inequality-constrained nonlinear program. Going beyond convex programming (but staying with subdifferentials), problems of *maximizing a convex function over a convex set and minimizing a difference of convex functions over a convex set* are discussed.

12.4.6.4 Nonlinear analysis and calculus of variations

We continue with results strongly related to nonlinear analysis and calculus of variations.

RESULTS IN HILBERT SPACES

Continuing with Banach-space theory, we will cite now some references restricted to the *Hilbert-space setting*. In the first six chapters of the book [149] by Aubin, which mainly treats *nonlinear analysis* and its applications in economic science and game theory, *topics from convex analysis which*

are important for convex optimization in metric and Hilbert spaces are discussed. These are, for instance, *projection mappings, conjugate convex functions, subdifferentials*, and, in particular, an extension of Clarke's subdifferential for locally Lipschitz functions and further generalizations. The book is recommendable for students and researchers interested in nonlinear analysis with applications in the above-mentioned two fields. Also treating Hilbert spaces, Takahashi [2336] deals with the fundaments of nonlinear and convex analysis via convex functions, where *nonlinear mappings and fixed point theorems* are the most important notions and concepts. To make the book self-contained, basic results from topology and linear functional analysis are also presented, and the text is partially also built on [2335]. Also the monograph [256] of Bauschke and Combettes, discussing a large variety of topics from convex analysis (e.g., also monotone operators), is restricted to Hilbert spaces. And the book [698] of de Cursi and Sampaiocan is mainly seen as an *introduction of convex analysis in the Hilbert space setting*, where the authors take particularly care of perturbed minimization of certain (convex) functionals, penalty methods, and a thorough presentation of monotone and maximal monotone operators in Hilbert space, with applications to variational inequalities. Also the book [256] by Bauschke and Combettes should be mentioned here, since monotone operators are studied there in the Hilbert-space setting.

CALCULUS OF VARIATIONS

Now we will discuss the relation to the field of *calculus of variations*, where *Ekeland's variational principle* plays (also historically) an essential role. The book [809] of Ekeland and Temamis is one of the first general works introducing convex analysis into that field. First the authors discuss fundamental notions from general convex analysis like *conjugacy, subdifferential calculus, and duality theory for convex optimization problems*, the latter also in the infinite-dimensional setting. This beginning part can be seen as a good and sufficiently thorough introduction to (in those days modern) convex analysis. In addition, *duality theory for convex variational problems* (involving multiple integrals) is investigated, and this yields nice presentations of famous problems like the Dirichlet problem, the Neumann problem, the minimal hypersurface problem and other basic questions from mechanics, mathematical physics, and so on. With the example of the minimal hypersurface problem, the reader will see the wonderful interplay between methods from convex analysis and specific methods reflecting the nature of the problem. Known as his *variational principle*, Ekeland established in 1972 a theorem on the existence of approximate minimizers of lower semi-continuous functions bounded from below. In his own survey [808] from 1979, he shows already a surprising variety of examples, where his principle yields simple proofs of existing or new deep results in Banach spaces or complete metric spaces. He discusses forerunners, like Clarke's first-order necessary conditions for nonsmooth optimization, and later results, like the related and prominent Caristi–Kirk fixed point theorem. Ekeland's variational principle is *one of the most applicable results from nonlinear analysis*. As a tool, it is used for studying problems from optimal control, fixed point theory, game theory, dynamical systems, and many further applied fields. Several famous results (such as the Petal theorem and Takahashi's minimization theorem) from nonlinear analysis turned out to be equivalent to Ekeland's variational principle. It is clear that this development is also presented in [809].

The monograph [470] of Borwein and Zhu excellently summarizes later results in the spirit of variational approaches in nonlinear analysis, including examples like the Borwein–Preiss (smooth) variational principle, or simple proofs of the Banach and the already mentioned Caristi–Kirk fixed point theorems. There are chapters showing the power of (set-valued) subdifferential calculus in different nonsmooth settings, and also further tools from convex analysis are discussed, such as *Fenchel duality*. They are shown to have somehow an analogous position in a convex setting as variational principles have in a nonsmooth situation. Related books (with overlaps, but also enriching due to different aims) are [2048] by Rockafellar and Wets, [630] by Clarke et al., and [1845] as well as [1846], both by

Mordukhovich. Clearly, with all these steps of generalizations the "distance" to the original convex setting is increasing.

12.4.6.5 Functionals and operators

We continue with discussing *functionals* and *operators* in our framework.

As various references in this subsection demonstrate, the use of convexity ideas in nonlinear optimization is strongly based on notions like *convex and conjugate functions, subdifferentials of convex functions, and generalizations thereof*. A second and basically different way of development is based on *techniques treating convex functionals associated with operator equations* in applied mathematics and theoretical physics. Convexity helps here when global extremum principles, dual extremum principles, and the calculation of bounds on *functionals* are established. The book [2264] of Smith gives an update (1985) regarding this line of research. Fundamental properties of operators illustrated by examples of ordinary and partial differential operators are given, and after a survey of the calculus of operators and functionals including Gâteaux and Fréchet derivatives, variational principles are introduced. The inequalities related to the requirement that functionals be convex are discussed. The minimum principles follow, and it is shown how this structure can be exploited to obtain dual extremum principles for the functionals.

Here we should also mention once more the earlier survey [143] of Asplund on *convex functions in relation to monotone mappings*. Levin discusses in [1550] the calculation of subdifferentials and *conjugate functions of convex integral functionals in spaces of measurable vector functions*. Second, he takes the theory of lifting into consideration. The studied connection between *convex analysis and lifting theory* permits him to apply convex analysis in these spaces of measurable vector functions. To succeed in this way, *subdifferentials of convex integral functionals in spaces of measurable vector functions* are investigated, yielding applications in measure theory. Also the duality of convex integral functionals and applications in mathematical economics are presented. The book contains many deep and difficult results, partially published for the first time. It is a very interesting reference for experts in *functional analysis, optimization, and mathematical economics*.

Like also its ten years older first edition, entitled "Minimax and Monotonicity", the book [2254] by Simon summarizes results about *multifunctions on Banach spaces* (especially also nonreflexive spaces) and their relationship to minimax maximal monotonicity, strongly connected also with (sub)gradients and subdifferentials of convex functions. The survey [2045] of Rockafellar deals with *convex functions defined on real Banach spaces*, and it contains a proof of his own result that if a convex function is lower semi-continuous, then its subdifferential is a *maximal monotone operator*. Various results are stated that concern sums, domains, and ranges of maximal monotone operators. The later paper [460] of Borwein excellently surveys contributions to the theory of maximal monotone operators using representative convex functions and suitable tools from convex analysis for studying them. The developed theory treating reflexive Banach spaces is particularly taken care of. In his second related survey [461], Borwein gives an excellent and wide-ranging overview on the 50-year *history of monotone operator theory, also within the framework of convex analysis*, and with far-reaching applications (e.g., in PDE, nonlinear semigroups, optimization, and even in infinite-dimensional holomorphy). The monograph [527] of Burachik and Iusem is dedicated to *set-valued mappings* and so-called *enlargements of monotone operators*, which are natural extensions of the ε-subdifferentials from convex analysis. Important topics in the book are fixed point theory for set-valued mappings, subdifferentials of convex functions leading to maximal monotone operators, and the mentioned enlargements of monotone operators. Due to many exercises, the book can be nicely used for a graduate course.

We continue with the book [256] by Bauschke and Combettes, in which the authors collect many basic results from three areas of nonlinear analysis whose interrelations attracted many researchers: convex analysis, monotone operator theory (see above), and the theory of nonexpansive mappings.

Instead of the general concept of Banach spaces, the authors have chosen the *Hilbert-space setting*, to cover many interesting applications while avoiding the technical difficulties related to the general case. Topics close to our interests here and considered in [256] are *convex sets, functions and cones, polarity and conjugate duality, in particular the Fenchel duality, differentiability of convex functions, convex optimization*, as well as monotone operators. A high level of presentation is paired with careful and detailed discussions of many applications and algorithms, and in the second edition the authors have added about 140 new pages and over 270 new results. In the final chapter of the book [472] on *conjugate duality in optimization theory*, a link between convexity and monotonicity via convex representations (e.g., Fitzpatrick functions) and monotone operators is presented. Also treating monotone operators, Grad [1014] studies *duality theory in scalar and vector optimization* within the framework of convex analysis. Following the relations described above, his book contains a whole chapter on monotone operators and, via a new approach to maximal monotonicity of bifunctions, extensions of results known for the reflexive case to the setting of nonreflexive Banach spaces.

12.4.6.6 Generalized convexity notions in convex analysis

Our next aim is the discussion of the wide framework of *generalized convexity concepts in convex analysis*, which is formally a part of abstract convex analysis.

ONE MOTIVATION

The power of methods from convex analysis for solving problems from convex optimization motivated researchers to create generalized convexity notions in convex analysis, with the hope that they could yield more general concepts applicable to non-convex optimization problems. The respective branch "abstract convex analysis" deals with convex-like structures of different types, hopefully useful for this aim and further fields of application. Numerous ideas and constructions from classical convex analysis are extended to correspond with more general types of sets or functions.

First we mention proceedings like [2127] by Schaible and Ziemba as well as (later) [1455] by Konnov, Luc, and Rubinov which contain many papers interesting from the viewpoint of generalized convexity notions and showing a stormy development around these concepts. In the book [165], Avriel et al. try to present a unified treatment. They study *quasi-concave, pseudo-concave, and related functions*, and applications in fields like microeconomics are discussed. *Duality theorems* are applied, and functional forms (like composite functions, products, and ratios) are also investigated.

AN EMERGING STRUCTURE OF NOTIONS, E.G., INVEXITY

Further on, three surveys should be mentioned here: [1827] by Mond, [1959] by Penot, and [1348] by Kanniappan and Pandian. In [1827] various generalized convexity types are reviewed, and their applications in optimization are also discussed (these include quasi-convexity, invexity, k-convexity, and pseudo-convexity of functions). Note that *invex functions* are a generalization of convex functions with important properties; e.g., all their stationary points are global minimizers, which is of course useful in optimization. In the later exposition [1959] of Penot, generalized directional derivatives and subdifferentials are used to characterize generalized convexity notions, followed by a survey of nondifferentiable generalized convex functions in view of nonsmooth analysis. And in [1348], strongly motivated by requirements coming from optimization, also generalized convexity notions (like quasi-convexity, invexity, pseudo-invexity, and quasi-invexity) and their relations to each other and to convex functions are surveyed. Two later books refer analogously to generalized convexity concepts. Namely, the book [1810] of Mishra and Giorgiis is concentrated on the generalized convexity notion of *invexity* and its importance for *optimization*, and it seems to be the first book presenting a comprehensive

overview on invex and closely related functions in various types of optimization problems (with single-valued and vector-valued objective functions, and with static as well as dynamic problems). The links between invexity and generalized monotonicity, the role of invexity for nonsmooth functions (in particular, the Lipschitzian case using Clarke's theory of generalized gradients), and the important role of invexity regarding duality theory are carefully studied, and also the relations to other generalized convexity notions (like pseudo-convexity) are considered. Similarly, the book [1811] of Mishra, Wang, and Lai is dedicated to *generalized convexity concepts and their applications in optimization and multi-objective programming*. In particular, invexity, quasi- and pseudo-invex functions, and more types and modifications thereof are discussed, in differentiable and nondifferentiable forms.

A LOOK INTO ASTRACT CONVEX ANALYSIS, AND GLOBAL OPTIMIZATION

Under such generalized convexity assumptions, second- and higher order *duality results for minimax problems* and nondifferentiable optimization problems as well as relationships between vector variational-like inequality problems and vector optimization problems (smooth and nonsmooth) are considered. Also, the authors give a nice discussion on general *motivations for extended convexity notions*. Here we mention also [2121] and [2123]. One of the first deep books going into the direction discussed here is Singer [2258]. Therein the field *abstract convex analysis* is studied which turned into an important part of functional analysis with numerous and rich applications; the book can also be considered as a first exhaustive survey, and like various references cited above it is also strongly related to the *geometry of Banach spaces*. In this more general setting the author gives *generalized analogues of notions like subdifferential, duality, and conjugation*. For example, among dualities the conjugations are those appearing at most in the definition of dual optimization problems. It is demonstrated that generalized convexity notions studied in this field are not created for the sake of (artificial) generalization itself; *the goal is to find hidden convexity traits in non-convex problems*, to understand them in a better way. As already said, there is a large overlap with Banach-space geometry, and the book [2258] is certainly useful for all specialists in convex analysis and related parts of functional analysis, optimization, and applied disciplines. Here we should also cite the monograph [2072] of Rubinstein on *abstract convexity and global optimization*. Its main goal is roughly described as follows: From the classical setting of functional analysis we know that if f is a convex, lower semi-continuous function on a set X, then there is a subset U of the class A of all continuous affine functions on X such that f equals the supremum of all $h \in U$. Replacing A by some special class H of functions, one obtains abstract convex functions with respect to H, called *H-convex functions*. In abstract convex analysis, H-convex sets play the analogous role that closed convex sets play in convex analysis. This analogy allows to define notions like conjugacy, subdifferentials, etc., also for the H-setting of abstract convexity. For example, the *corresponding generalization of the convex subdifferential* based on abstract convexity means the representation of a function of a rather complicated nature as the upper envelope of a set of fairly simple functions. It turns out that the right choice of the classes H can yield an efficient framework for solving problems, especially in *global optimization*. In this sense, [2072] is an important reference for global optimization.

Regarding generalized convexity notions, here also the *Handbook of Generalized Convexity and Generalized Monotonicity* [1131], edited by Hadjisavvas, Komlósi, and Schaible, should be referred to. And also proceedings like Konnov et al. [1455] (see also the subsection above) should be mentioned. As Rubinstein [2072], also the book [2385] of Tuy shows how useful tools and notions from (finite-dimensional) convex analysis can be for global optimization. The notion of quasi-convexity and the concept of differences of convex functions are used therein. And also the monograph [691] of Crouzeix treats generalized convexity and also generalized monotonicity (the latter motivated by applications to variational inequalities and equilibrium problems). The author focuses particularly on *generalized convex functions (like quasi- and pseudo-convex ones)* and analogous *generalized monotone operators (like quasi- and pseudomonotone ones)*.

An important survey related to generalized convexity is [643] by Colombo and Thibault. Namely, *prox-regular sets* are non-empty subsets of Hilbert spaces that can be supported at any boundary point by a ball of a locally fixed radius in all possible normal directions. Such sets, also known as *weakly convex sets or sets of positive reach*, are obviously more general than convex sets, and this property is strongly related to semiconvexity of functions (via their epigraph) and hypomonotonicity of multivalued operators (considering truncated normal cones over sets of subdifferentials of functions). The work [643] completely surveys properties of prox-regular sets, also in infinite dimensions, and also regarding applications. Historical notes and an extensive list of references are given, too.

The book [111] of Ansari, Lalitha, and Mehta refers to *variational analysis and optimization*, where again generalized convexity and generalized monotonicity play the central role. In the first part standard notions from convex analysis are studied, and the generalized convexity notions mainly studied in the book are presented here. These are *quasi-convex, pseudo-convex, and pseudo-monotonic functions*, partially differentiable. One important property of these functions is that necessary optimality conditions for problems involving them are also sufficient for optimality. Generalized directional derivatives of extended real-valued functions are also discussed, and the role of Clarke's generalized directional derivative of locally Lipschitzian functions and the respective subdifferential are studied (via duality relations). Then, coming from generalized convexity concepts, analogous monotonicity and generalized monotonicity notions are presented, in smooth and nonsmooth frameworks (including the concept of *set-valued mappings*). Based on this, in further parts of the book *nonsmooth variational inequalities* and solution sets of constrained optimization problems with smooth objective functions of generalized convexity types are studied.

A somewhat different generalized convexity notion is discussed in the book [877] of Fink and Wood. There convexity is generalized (first in the planar case) by replacing the set of all lines by a smaller set of lines. An orientation set O is a set of given lines passing through the origin, an O-line is a translate of a line in O, and a set is called *O-convex* if its intersection with every O-line is connected. In order to get a natural definition also in higher dimensions, hyperplanes have to be similarly used. There are many similarities with the usual convexity notion, but O-convex sets may be disconnected (giving rise to study also O-connected sets). The mentioned book [877], written in an elementary way, collects results on such convexity types. For example, common notions from classical convexity (like visibility, hull, and kernel) are extended to this generalized convexity concept. Unfortunately, its authors do not cite references on similar convexity notions.

RELATED ALGEBRAIC STRUCTURES

In some sense *related to generalized convexity*, the following two references give links to *algebraic structures*. Discussing the possibility of introducing convexity notions in fundamental algebraic structures, Borwein and Giladi give, in their expository paper [463], an excellent insight how far one can go in this direction. The most general definition of convexity is given for sets in a semimodule over a semiring (e.g., a subcase is the notion of convexity for subsets of a monoid), and it is shown that on this abstract stage also convex functions can be defined. The authors succeed in extending much material from classical convex analysis onto this level, including various theorems on convex functions and sets, a subdifferential calculus, and a duality theorem of Fenchel type. Although (compared with [463]) having different goals, the authors of [1148] also consider an abstract algebraic framework. Namely, they embed several recent results in set optimization suitably into an abstract algebraic setting and thus succeed to study some fundamental notions from convex and variational analysis in a correspondingly extended setting. These notions are *subdifferential, Fenchel conjugate, and a powerful duality notion for vector- and set-valued optimization*. For example, it turns out that in this framework the class of *positively homogeneous and additive set-valued functions* plays the same role as the class of linear functions in classical convex analysis. Particular cases are carefully investigated, such as linear

vector optimization from a theoretical point of view, and the case of set-valued risk measures from the viewpoint of financial applications.

12.4.6.7 Nonsmooth analysis

Also the field of *nonsmooth analysis* is closely related to convex analysis.

NONSMOOTH ANALYSIS AND NONSMOOTH OPTIMIZATION

We start with the monograph [730] of Demyanov and Rubinov on *constructive nonsmooth analysis*, where tools from convex analysis in nonsmooth analysis are studied and different approaches to the theory of "differentiation" for nondifferentiable functions, and therefore nondifferentiable optimization problems, are investigated (a first version of the book, in Russian, appeared already in 1990). After studying homogeneous approximation of functions, sets, and mappings, the authors consider different types of generalized derivatives, in particular the generalized gradients of Clarke and the theory of quasi-differentiable and codifferentiable functions. Nice examples are collected, coming mainly from the theory of convex functions and min-max functions. We also cite once more the book [464] of Borwein and Lewis, in which important relations between nonsmooth analysis, convex analysis, and optimization theory are discussed. And Schirotzek [2134] gives a broad introduction into basic notions (e.g., conjugacy, duality principles and, in particular, Clarke and Michel-Penot subdifferentials), and developments of convex analysis. The author underlines that variational principles created something really new in the development of nonsmooth analysis and became important tools to work also beyond the Lipschitz case. In the survey [728] of Demyanov on nonsmooth optimization, the first chapter gives a brief introduction into the framework of nonsmooth analysis corresponding with subdifferentials of convex functions, such as the Shor and the Clarke subdifferential. Further on, the so-called Dem'yanov–Rubinov quasi-differentials and codifferentials are particularly studied. The paper reflects also approximation results.

SUBSMOOTH FUNCTIONS AND SETS

The comprehensive paper [2359] of Thibault exposes the concepts of subsmooth functions and subsmooth sets in nonsmooth analysis. They are defined for normed spaces, and it turns out that equivalence to the concept of *approximate convexity* holds. Various properties and characterizations of subsmooth functions are established, in particular a characterization through the submonotonicity of its subdifferential (which is defined axiomatically). Comparisons with other notions in the literature are also presented, and the whole paper gives a detailed overview of the subject, also with interesting bibliographical comments and references. And for the following results, the motivation came directly from nonsmooth analysis.

METRIC REGULARITY

Metric regularity as a traditional theory in analysis aims to understand deeply those properties of mappings and sets (in their whole variety) which capture a suitably defined regularity behavior. We cite here three surveys and one book of Ioffe, a leading expert in this direction, namely [1269], [1270], [1271], and [1272]. Extensions of results in nonsmooth analysis to set-valued settings and the explanation of basic results on subdifferential calculus as metric regularity properties of special set-valued mappings lead Ioffe [1269] to unify results from metric regularity theory, and to present their *applications to subdifferential calculus in Banach spaces*. In [1270] many further developments are discussed, e.g., necessary and sufficient conditions for metric regularity, and related stability

problems. In [1271] fundamental results on metric estimates of the distance from the sublevel set of a given function are used as starting point. For convex functions in Banach spaces, these estimates are formulated in terms of convex analysis, using notions like subdifferentials and normal cones. In the general case of lower semi-continuous functions defined on a complete metric space, the author moves to the regularity theory of *set-valued mappings between Banach spaces*, and to further steps. As the most comprehensive work on the theory of metric regularity, the research monograph [1272] by Ioffe summarizes the aspects discussed in these expositions. Its main goals are criteria for (and quantitative estimates of) rates of regularity, the effect of perturbations of a mapping on its regularity properties, and the role that metric regularity plays in analysis and optimization. For example, the relations between the investigated metric theory and the needed properties of Banach spaces are shown to be based on *tangential set-valued approximations and the theory of subdifferentials*. Last but not the least we mention here the monograph by Bagirov et al. [180]. The structure of this book perfectly shows the *fundamental role that nonsmooth analysis plays for nonsmooth optimization*.

12.4.6.8 Applications

Now we come to *applications* of results from convex analysis.

MECHANICS

Due to interesting generalizations in convex analysis, namely referring also to variational inequalities, in the mid-1970s important advances have been made in optimization theory and variational methods. Here Moreau, Rockafellar, and Clarke should be mentioned, and their ideas became fruitful fundaments for *nonsmooth applications in mechanics*, including dynamics, frictional and frictionless contact, structural optimization, unilateral contact, and plasticity. The proceedings [1850], edited by Moreau and Panagiotopoulos, contain a collection of mathematical papers on related applications in *mechanics*. We refer the reader's attention also to the book [698] of de Cursi and Sampaio, offering convex analysis in the Hilbert space setting. It contains nice examples from *mechanics and physics* which certainly attract applied scientists needing quick access to applicable tools and results from convex analysis. Parts of the text remind the reader of the ideas of Moreau who saw mechanics as the main application field of convex analysis.

ECONOMICS

Really important *applications of duality theory in economics* can be found in the classical, famous book [2233] of R. W. Shephard and the long survey [752] of Diewert. Shephard's (also historically very important) book is based on fundamental ideas from convex analysis, in particular the *duality concept between cost and production functions*. This duality as well as that between the (direct) utility function and the indirect utility function are comprehensively discussed in [752], and the author continues to show what else can be done with duality principles (all of them based on the Fenchel conjugacy) in different areas of *economics*. Here we refer to the survey [1701] due to Martínez-Legaz, where *duality theory is extended to generalized convexity*, again based on the Fenchel conjugacy; in particular, *quasi-convex conjugacy, duality, microeconomics duality*, and the monotonicity of demand functions are further studied topics. Also the book [884] of Florenzano and Le Van contains a lot of material on convex functions and from convex analysis, yielding the basic structure for *applications in economics and other application fields*. Also in view of further applications in economics, we refer to the proceedings [2127] with Schaible and Ziemba as editors, mainly dedicated to dualities and the generalized convexity aspect, and also Levin's book [1550] where subdifferentials of convex integral functionals in spaces of measurable vector functions yield *applications in economics*. In addition

Aubin [149] is related to economics, but also applications in *game theory* are studied there. In the monograph [1608] of Lucchetti, which is also interesting from the viewpoint of *economics and physics*, the author investigates how (properties of) convex functions behave under perturbations of data, and he lays special emphasis on the properties of stability, (generic) well-posedness, and the connections between them. The behavior of optimization problems under perturbations or small changes of data are studied, and Fenchel conjugacy as well as the development of a general duality theory are important issues of that book.

FURTHER APPLICATIONS

Here might be the right place to mention also the book [580] of Carr and Zhu, showing how helpful convex duality can be in *financial mathematics*, more precisely when the maximization of concave utility functions and the minimization of convex risk measures are investigated. For example, the authors study the role that *convex duality plays in the Markowitz portfolio theory*, in the study of a multi-period economy with finite status, and for continuous financial markets. Regarding applications of convex analysis in *electronics*, Goeleven's book [986] can be recommended. It shows the way how methods from convex analysis and from set-valued analysis can help to model electrical circuits involving non-regular devices like diodes and transistors. By means of convex subdifferentials, maximal monotone operators, and the Legendre–Fenchel duality it is shown that any Ampere–Volt characteristic in the plane, described by a maximal monotone graph, is formulated as a convex subdifferential relation. This yields descriptions of characteristics of different types of diodes and transistors. Various further related tools and applications are presented. The monograph [617] of Chi, Li, and Lin is a textbook on convex optimization, emphasizing applications in *signal processing and communications*. Despite this applied character, the second chapter contains a nice representation of fundaments of convex sets (their geometry, cone theory, duality, and convexity-preserving transformations), while the third part refers to convex and quasi-convex functions and their relations to each other. Chapter 9 is concerned with duality theory, especially of Fenchel and Lagrange type, and Chapter 4 with strongly related basics of convex and quasi-convex optimization. The applicability of these fundaments in signal processing and communications is nicely presented, and so the book can serve as a helpful textbook for students and researchers interested in convex optimization. Finally we mention also here the book [180] of Bagirov et al. on nonsmooth optimization and nonsmooth analysis since it contains a large variety of explained applications from *chemistry, biology, data analysis, engineering, and image restoration*.

12.4.6.9 Discrete convex analysis

A fundamental model in discrete optimization is the model of *matroids* (and its extensions to polymatroids and submodular rank functions). This is due to the fact that matroids can be viewed as certain "discrete" convex structures; many of their properties can be studied in terms of convexity. A general theory of "discrete convexity" based on this notion shows that convexity methods in (non)linear function minimization yield powerful tools in *discrete optimization*. Generalizing certain aspects of matroids, various models have been suggested. For example, Lovász suggested an extension of set functions with ground set V to functions on $\mathbb{R}^V$ being convex if and only if the set function is submodular (which is equivalent to the optimality guarantee of the greedy algorithm).

In his fundamental survey [1602] Lovász proves that, to a certain extent, *submodular set functions* play a role in discrete optimization analogous to that of convex functions in continuous optimization. He exposes the most important aspects of submodularity, comparing them with properties of convex functions and developing an even more fundamental connection between both concepts. He emphasizes the following observation regarding *combinatorial studies involving submodularity*. Take, for instance, a classical result from graph theory and replace certain linear functions occurring there (either in the

objective function or in the constraints) by submodular functions. Often the generalizations of the original results, obtained in this way, remain valid. In the last chapter of the influential book [1047] of Grötschel, Lovász, and Schrijver *submodular functions and polymatroids* are surveyed, and associated algorithmic problems are handled with the ellipsoid method.

With his book [907], Fujishige puts the scattered literature on submodular functions together and presents his unifying view onto the theory of these functions by means of *base polyhedra and duality* for submodular and supermodular systems. Also the different solutions (1999) of the long-standing open problem of *submodular function minimization*, clearly not contained in the first edition, are discussed. Another related monograph is [1874] by Narayanan, discussing many properties of matroids, submodular functions, and the principal partition, in view of their *applications in the efficient analysis of electrical networks*. Electrical engineers with background in algebra and graph theory will find it very useful, like also experts in *combinatorics* interested in applications. All the necessary background from matroids, graph theory, and electrical engineering is carefully presented, and many new results are exposed.

The survey [1865] of Murota updates the theory of *discrete convex analysis* developed for *integer-valued functions* defined on integer lattice points. It covers *discrete analogues of fundamental concepts like conjugacy, subgradients, and the Fenchel min-max duality*, and it confirms that the development was further on based on matroid-theoretic concepts like submodular functions and exchange axioms. The same author wrote the related, comprehensive book [1866]. The approach of Lovász (see above) yields the so-called L-convex functions. A further extension of set functions is based on the fact that the base vectors of a polymatroid satisfy an analogous exchange property as the bases of matroids, leading to the concept of M-convex functions. Murota [1866] presents the state of the art in the *matroid-based part of discrete convex analysis*, in particular showing that the duality between L- and M-convexity presents a manifestation of the general *Legendre–Fenchel duality in classical convexity*. For both cases, minimization algorithms are developed, and also applications (like models in economics and systems analysis by mixed matrices) are discussed.

In the survey [2341] of Tamura a concavity notion is discussed which generalizes submodularity, and applications of it in several *economic models*, among them being two-sided matchings and combinatorial auctions, are summarized. A further related survey is [1289], where fundamental properties of submodular functions and the corresponding developments of the algorithmic side (in view of optimization and approximation) are studied. Methods for minimization and maximization of submodular functions and for minimization of symmetric submodular functions are described.

Basic concepts of discrete convex analysis (in view of *discrete games*) are surveyed by Bilbao and Martínez-Legaz in [340]. Applications to n-person games by convex functions of n variables are presented, and a Fenchel-type duality as well as a subgradient concept are applied to a convex profit game and the dual concave cost game. Finally we mention here the surveys [1461] and [764] of Kovalev and Dragomir, respectively. In the first paper discrete analogues of gradient methods and related error estimates are discussed. In the second article recent discrete inequalities for functions defined on convex subsets of general linear spaces are studied. For example, several refinements and also reverses of Jensen's discrete inequality are surveyed.

Although *combinatorial optimization* (see, e.g., the important monograph [1047]) has relations to discrete convex analysis, and although it has also many overlaps with (polyhedral) convexity, we decided to give no survey on it in our book, since this would simply lead too far.

12.4.7 Convexity and Banach spaces

In this part of Chapter 12 we want to present basic results and notions from the geometry of Banach spaces which are close to the subcase of Minkowski geometry and to classical convexity. This means

that we mainly discuss (geometric) properties of convex bodies in finite-dimensional real Banach spaces.

12.4.7.1 Foundations and introductory remarks

A *Banach space* is a complete normed vector space. More precisely, such a space is a vector space with a metric allowing the computation of vector length and distance between vectors; with completeness we mean that a Cauchy sequence of vectors always converges to a well-defined limit which is within the space. As a generalization of *Minkowski spaces*, which today would be called *finite-dimensional real Banach spaces* and go back to Minkowski [1803], Banach spaces were introduced by the Polish mathematician Banach who developed this concept systematically in 1920–1922, together with Hahn and Helly (see Banach's famous monograph [205]). (About Minkowski's preparatory work we refer, in our book here, to the subsections 12.3.1 (Minkowski) and also 12.4.4.) Fréchet was the first who called these spaces "Banach spaces". The term grew out of investigations of function spaces (e.g., by Hilbert, Fréchet, and Riesz), and it became a central notion of *functional analysis* and important also for other branches of analysis; we refer to the comprehensive historical presentations due to Monna [1828], Dieudonné [751], and Pietsch [1972], and also to the books of Duda [771] and Mauldin [1733] which show, in particular, the mathematical life during the corresponding period in Lvov, where essential early steps of a fundamental development took place. Today the field of Banach spaces is of fundamental importance in many pure and applied areas. One central object studied or used in it is the *unit ball of a Banach space* which is a *convex body centred at the origin*, creating the notions of *norm* and *normed space* (such as strictly convex or rotund norms are studied). And often, besides unit balls, further objects studied in these spaces are convex sets, too. All this implies the strong connection of this field to convex geometry.

Answering a question of Grothendieck, Dvoretzky [781] proved that every sufficiently high-dimensional normed vector space has low-dimensional subspaces that are approximately Euclidean. In other words, every high-dimensional bounded convex set centred at the origin has low-dimensional centred sections which are approximately ellipsoids. With a new proof, Milman [1787] presented a surprisingly nice approach to that theorem, which became a very important milestone in the development of *asymptotic geometric analysis* (also called *asymptotic functional analysis* or the *local theory of Banach spaces*, sometimes also *asymptotic convexity*). Looking at the geometry of finite-dimensional real Banach spaces (also called *Minkowski geometry*) on the one hand, and infinite-dimensional Banach spaces on the other hand, asymptotic geometric analysis and its typical methods are "in between". This means that in this field one is concerned with geometric properties of finite-dimensional normed spaces, but particularly interested in the asymptotic behaviour of various quantitative parameters as the dimension goes to infinity. Many tools, theorems, and notions from classical convexity still occur there, such as *position theorems* (ellipsoids), *centroid bodies* (being zonoids), the *Blaschke–Santaló inequality*, and so on. But different to the main parts of classical convexity (and as shown by Doretzky's theorem), many results there are counter-intuitive and very surprising in nature. For example, also "concentration phenomena" occur there: As the dimension increases to infinity, families of metric probability spaces usually have the property that Lipschitz functions defined on them are ε-close to their median value, except on sets of exponentially small probability. Such phenomena yield the basic reason for the surprising degree of regularity that convex bodies in high dimensions reveal. Thus, in many volume and entropy calculations an arbitrary convex body can efficiently be replaced by a certain ellipsoid suitably associated with it. For example, consequences of this include *inverse Santaló and Brunn–Minkowski inequalities*.

Of course, also other subfields of functional analysis are convexity related and partially discussed here. Thus, in this survey part we want to reflect literature related to convexity from asymptotic

geometric analysis and from classical Banach space theory. We start with monographs, continue with proceedings, and finish with selected surveys.

12.4.7.2 Convexity-related monographs in the geometry of Banach spaces

We collect now monographs first treating the convexity-related geometry of Banach spaces, and then referring to the asymptotic theory of Banach spaces, also related to convexity.

CLASSICAL BOOKS FROM BANACH-SPACE GEOMETRY

The comprehensive monograph [775] of Dunford and Schwarz is a classical book on the *theory of linear operators (mainly in Banach spaces)* and various applications thereof. It is characterized by a very clear and attractive presentation of the field, suitable as a textbook like also a reference book. At several places one can find explicit use of aspects and notions from *convexity*, such as in Chapter IV (where 26 concrete types of spaces are discussed) and in Chapter V (titled "Convex sets and weak topologies"), where also *extremal points of convex sets* (Krein–Milman theorem), *functionals tangent to convex sets*, and fixed point theorems are treated. Chapter VI contains a proof of the *convexity theorem of Riesz* (about interpolation of operators, bounding the norms of linear maps between L^p spaces), in the extended form of his student Thorin.

Day wrote the relatively short, but classical and fundamental monograph [718]. It is said there that "this book contains a compressed introduction to the study of normed linear spaces and to that part of the theory of linear topological spaces, without which the main discussion could not well proceed". The main topics of representation are the spaces themselves, their structures, and their geometry. Fixed point theorems are also discussed. In Chapter V *classical convexity* plays an important role, including the *Krein–Milman theorem*, and in Chapter VII the author constructs the concept of *metric geometry in normed spaces*. Here properties of *smooth and strictly convex norms* are essential, and various *characterizations of inner product spaces* are derived, beginning with the *Jordan and von Neumann characterization*. Thus, partially the book is also a forerunner of Amir's book [104].

The investigations from the basic paper [2125] of Schäffer are continued in the monograph [2126]: Related to the *inner metric* of the unit sphere S of a Minkowski space the author mainly studies quantities like *lengths of curves, the girth of S, its inner radius, and its inner diameter*. The connections of these quantities to other geometric properties of the corresponding Banach spaces (such as superreflexivity and the Radon–Nikodým property, expressing the relationship between two measures defined on the same measurable space) are presented. The book contains a comprehensive survey on known results in that direction, many new results, and interesting open problems.

Diestel's Lecture Notes [750] from 1975 treat the background material from the geometry of Banach spaces that is necessary to understand the *Radon–Nikodým property* (meaning also vector-valued measures). Notions and tools from convexity used in [750] are *support functionals from convex analysis* (generalizing that of support functions), the *duality between smooth and strictly convex norms*, and the characterization of *non-separable duals of separable spaces*. The latter things come along with the proof of the *equivalence of the Krein–Milman property and the Radon–Nikodým property in dual spaces*, and the proof that spaces with the Radon–Nikodým property are precisely those for which closed bounded convex sets are the closed convex hulls of strongly exposed points (see also the next book). Further parts of [750] treat *convexity-dependent properties and differentiability of norms*, uniformly convex, and uniformly smooth Banach spaces.

Regarding *integral representations of points of compact convex sets* and the respective theory of *infinite-dimensional compact convex sets and Choquet simplices*, the following two monographs show basic developments. Somehow continuing Alfsen's book [58] on compact convex sets and boundary integrals, Asimow and Lewis published their monograph [142] on the geometry of ordered Banach

spaces in 1980. Corresponding to each compact convex set K, there is a partially ordered, real Banach space isomorphic to the space of real-valued affine continuous functions on K, equipped with the supremum norm. Among other topics, the authors study the duality theory for pairs of ordered Banach spaces, introduce Choquet simplices, give a comprehensive account of their properties, and apply convexity theory to complex Banach algebras, in particular function algebras.

It is well known that the *Radon–Nikodým theorem from measure theory* relates two measures defined on the same measurable space; namely, under certain conditions, any measure can be expressed in this way with respect to another measure on the same space. A Banach space X is said to have the *Radon–Nikodým property* if for vector measures with values in X an analogue of this property holds. The book [479] of Bourgin collects many results on such Banach spaces, showing also that notions from convexity (like *extreme, (strongly) exposed and support points*) are essential in this theory. Characterization theorems are presented, for example: A closed convex set has the Radon–Nikodým property iff each of its closed bounded convex subsets is the closed convex cover of its strongly exposed points. Further notions from convexity, explicitly needed in this book, are *dual spaces, separability types, and differentiability of convex functions*. The book nicely summarizes many related developments, and in its well-structured form it is also suitable as a good textbook.

The monograph [1278] on strict convexity and complex strict convexity gives a detailed and very broad view on geometric properties of the unit spheres in Banach spaces. After an introduction treating carefully selected parts of the general theory of normed spaces, the second chapter is dedicated to *strict convexity of norms* as the main topic of the book. Definitions and important properties of *strictly convex and uniformly convex Banach spaces* are considered, and also properties connected with *differentiability of the norm* are studied. The book contains also generalizations of uniform convexity and many characterization theorems of such properties. Examples in the form of "nice" and "equivalent" norms are presented, and various applications of all these notions are given. These refer to *approximation theory, fixed point theory, and probabilistic metric spaces*, and also *moduli of convexity* play an important role. The third chapter refers to *complex strictly convex spaces* and their applications to the theory of *vector-valued analytic functions*. In the appendix, classes of linear operators connected with geometrical structures and properties of Banach spaces are considered, and a comprehensive collection of references is collected.

Describing normed spaces geometrically, Amir's book [104] gives an excellent, self-contained, and exhaustive literature survey on many isometric *characterizations of inner product spaces*. After the introductory part, the author discusses planar characterizations, emphasizing *James and Birkhoff orthogonality characterizations* as well as *norm derivatives and best approximation characterizations*. (For James and Birkhoff orthogonalities the more recent exposition [69] of Alonso et al. updates the literature, and the even more comprehensive survey [70] by the same authors additionally collects related material about various further types of orthogonality.) Three- and higher dimensional characterizations of inner product spaces follow in [104], such as characterizations via *metric projections and proximinal maximal subspaces*. The author continues with a comprehensive survey on further topics, starting with *Blaschke's characterization of ellipses and ellipsoids* from the book [358], *Banach's introduction of normed linear spaces*, the famous *characterization of Jordan and von Neumann* (parallelogram equality), the *contractive projection characterization* (Kakutani), the summary in Day's monograph [718] (also referring to the concepts of Birkhoff and James orthogonality), contributions from the viewpoint of *approximation theory* (Phelps), and *Chebyshev centers* (Garkavi and Klee). The proofs are given in a convenient and unifying style, and great care is given to links between different (types of) characterizations, in particular also between some older, more geometric, and some modern, more sophisticated, ones. This book is still a useful source for researchers in the *theory of normed spaces*, but also in various related areas of functional analysis, convex geometry, and approximation theory.

Like in [104], many *characterizations of inner product spaces* are also collected and presented in the comprehensive book [1279]. After introductory sections containing *basic results on Banach and Hilbert spaces*, many *metric and operator characterizations of inner product spaces* occupy a really

large part of the monograph. Further sections are dedicated to Banach algebras, linear operators, and kernel spaces. Then the author comes back to inner product spaces, studying deep properties and also applications of them. Various relations of inner product structures to a wide variety of topics from other areas of mathematics are described, and in an extra chapter 28 related research problems, with given sources, are posed. In this book, one can really find impressively many results on and characterizations of inner product spaces—the bibliography includes about 3000 references! Like in the case of inner-product spaces, also *semi-inner products* give rise to suitable *orthogonality concepts* with interesting characterizations. Thus, we mention here also Dragomir's book [763] on *semi-inner products and their applications*. In Chapters 8 and 9 of it, different *orthogonality types for normed spaces* and their relations to semi-inner products are discussed.

Extending the concept of Euclidean norm to that of general normed spaces, various familiar facts might get lost. For example, in general the *orthogonality* of lines is no longer symmetric, *shortest segments* from a point to a hyperplane are no longer unique, and the *isoperimetrix* (i.e., the solution of the isoperimetric problem) is no longer a homothet of the unit ball. Inspired by such interesting facts, Thompson treats in his monograph [2361] the *geometry of finite-dimensional real Banach spaces* (also named "Minkowski geometry") strongly based on notions, results, tools, and problems coming from classical geometry and classical convexity. These are for example: the *Hahn–Banach theorem, Löwner ellipsoids and characterizations of ellipsoids* (and, therefore, of the Euclidean norm), the *Hausdorff metric and Banach–Mazur distance, mixed volumes, polarity*, and various related *inequalities*. As nicely presented in [2361], further reasons for this close relationship between Minkowski geometry and convexity are the unit balls that are convex bodies centered at the origin, and therefore many problems investigated in the Minkowski geometry of convex bodies are natural extensions of problems from "Euclidean convexity" to the more general concept of normed spaces. One of the central topics of the book, namely the notions of *surface area in the sense of Choquet–Busemann or of Holmes–Thompson measure*, leads in a natural way to the concepts of *intersection bodies and projection bodies* (central notions from classical convexity). The latter relation occurs also in the solution of the *isoperimetric problem*, yielding the isoperimetrix via these concepts suitably combined with polarity. For example, in the case of Holmes–Thompson measure the isoperimetrix can be presented as the *projection body of the polar body of the unit ball*, and is therefore always a *zonoid*. Special care is taken of the two-dimensional case: *sets of constant width, equichordal sets, self-perimeter, isoperimetry, transversality, and Radon curves* are typical topics presented in great detail. In addition, Thompson studies the *trigonometry, angle measures, and numerical parameters in Minkowski spaces*, finishing with Schäffer's work on girth, perimeter, and further numerical values of the unit sphere. All this is used for Thompson's concept of the respective *intrinsic geometry*. The book contains many research problems, and notes at the end of each chapter reflect historical facts and the corresponding state of knowledge.

We should mention that also the Chapters II, V, and VIII of the monograph [431] of Boltyanski et al. are dedicated to the geometry of normed linear spaces. The themes treated there are *d-convexity* (a generalized convexity notion helpful for solving metrical problems in such spaces), *Borsuk's partition problem*, and a collection of *open research problems*, in most cases still unsettled.

A very important and fundamental book on the *history of Banach spaces and bounded linear operators* between such spaces is the monograph [1972] of Pietsch. It contains not only history, but also a comprehensive and deep representation of *basic concepts, typical methods and results, the history of how they came to be, and the relations of the field to other parts of functional analysis, and further branches of mathematics*. For example, these are *set theory, topology, probability theory, combinatorics, and harmonic analysis*; this list is far from being complete. Following that book, we briefly mention now only aspects and a few mathematicians close to classical convexity. Namely, the author starts with the contributions of Minkowski (1896), Hilbert (1906), Lamson (1920, Banach spaces of scalar-valued functions on arbitrary sets), Wiener (1920, axioms), and also approaches by Helly (1921, normed linear spaces of sequences) and Hahn (1921, spaces with duality). Later developments (operators, functionals and dual operators, Hahn–Banach theorem, Banach's famous

monograph (1931-1932), etc.) follow, and in the fifth chapter (referring to the "post-Banach period") notions and methods from *convexity*, also going back to Minkowski, come in, such as *extremal points and related topics, the Krein–Milman theorem, geometry of the unit ball, strict convexity and smoothness, and approximation properties*. In the book part on the so-called modern period, with results like *Dvoretzky's theorem* and notions like the *Banach–Mazur distance, volume ratios*, etc., also the local theory of Banach spaces is discussed. This is, maybe, the right place to cite also Musielak's historical paper [1867] on the history of functional analysis. Although the content is highly selective, interesting contributions are described, e.g., the Bourbaki approach, also to the geometry of normed spaces.

Although only one chapter (namely Chapter 10) is meant, we cite here the monograph [1713] of Martini, Montejano, and Oliveros where properties of *bodies of constant width* in Minkowski spaces are derived and represented. The same is done with two further related classes: the family of *diametrically complete sets* and the family of *reduced sets*. The notes of this chapter form a comprehensive survey of the knowledge about constant width sets (and the two other classes mentioned above) in normed spaces. Also various further important notions (like *perfect norms*) occur there. (And an analogous outlook on these sets in *further non-Euclidean geometries* is given).

BOOKS ON THE ASYMPTOTIC THEORY OF BANACH SPACES

The Lecture Notes [1797] of V. D. Milman and Schechtman give an early overview to diverse methods and types of results of the *local theory of Banach spaces* (thus dealing with the geometrical structure of finite-dimensional normed spaces as the dimension grows to infinity). This work consists of two parts and five appendices. In the first part the authors study the known *concentration of measure phenomenon* studied comprehensively by Milman; he also gave a *proof of Dvoretzky's theorem based on Levy's isoperimetric inequality* for the Euclidean sphere. All this is presented in [1797], like also further deep applications. Also for Riemann manifolds this phenomenon is investigated. An important appendix contributed by Gromov is devoted to *isoperimetric inequalities in Riemann manifolds* (including a concise introduction to Riemann geometry), referring in particular to Ricci curvature in this framework. The second part of the Lecture Notes refers to the *theory of type and cotype of normed spaces* and its connection with the dimensions of the spherical sections of Banach spaces. Here the notion of *K-convexity of Banach spaces* and related links to Dvoretzky's theorem are essential.

The monograph [1974] of Pisier gives a self-contained presentation of results related to the *volume of n-dimensional centrally symmetric convex bodies* (balls of finite-dimensional normed spaces) and the geometry of the corresponding normed spaces. The author presents proofs of some deep related results (see again also [1797]). These are *a proof of Dvoretzky's theorem* using the concentration of measure phenomenon for Gaussian measures, the *inverse Brunn–Minkowski inequality* due to Milman, the *quotient of subspace theorem* due to Milman, and the *inverse Santaló inequality* derived by Bourgain and Milman. So-called *volume numbers* of certain operators T and the related entropy numbers of T are compared, and a detailed exposition of classes of *Banach spaces of weak cotype 2 or weak type 2* is given. *Weak Hilbert spaces* as the intersection of these classes are introduced and studied. A characterization and several properties of these spaces are presented, such as their *reflexivity* and *uniform approximation property*. In the final part, the author mentions some other open problems. This book successfully collects and combines various interesting results on *convex sets* and the related *local theory of Banach spaces*.

The book [2375] of Tomczak-Jaegermann continues, giving an overview till the end of the 1980s, the series of monographs referring to the *theory of finite-dimensional Banach spaces*, and it treats or generalizes also several basic notions and results from *convexity theory*. It concentrates mainly on operator-theoretic aspects of the *local theory of Banach spaces*, systematically presenting the theory of operator ideals and their metric properties in this setting. Various *parameters* of finite-dimensional Banach spaces are studied. Besides the comprehensive investigation of the *Banach–Mazur distance*, results like *Lewis' theorem* generalizing John's theorem on the maximum volume ellipsoid contained in the n-dimensional unit ball, *Gluskin's theorem* on the *diameter of the Minkowski compactum*, sharp estimates for *distances between special classes of Banach spaces, spaces with unconditional bases, projection constants with sharp estimates*, and several *inequalities* are presented. Using results about Gaussian processes, estimates for norms of random orthogonal factorizations are obtained. These estimates are used to investigate *Banach–Mazur distances within various classes of spaces*. The book contains many notes, open problems, and references; it is a recommendable reference for the modern theory of finite-dimensional Banach spaces, suitable for teaching and research aspects in the field.

Ledoux's monograph [1524] is focused on the concentration of measure phenomenon which is inspired by isoperimetric inequalities. A known example demonstrating it is the way that uniform measure on the standard sphere of n-space gets concentrated around the equator when the dimension becomes large. It is discussed in [1797], where also the state of the art (1986) is given, and Ledoux shows how the development continued (till 2001). Various new examples are presented, useful in disciplines like convex geometry, functional analysis, infinite-dimensional integration, discrete mathematics, complexity theory, and probability theory. The author poses also interesting related problems, referring to the logarithmic Sobolev inequality or to transportation costs.

Asymptotic convex geometry is also the topic of the book [496] of Brazitikos et al., understood (informally) as "a study of convex bodies with an emphasis on the dependence of various parameters on dimensions". It has strong relations to the *local theory of Banach spaces, probability theory, PDE, Riemannian geometry, harmonic analysis, and even combinatorics*. For various problems from asymptotic convex geometry, e.g., related to *mass distributions in high-dimensional convex bodies* (analysis of uniform measure on properly normalized such bodies), *isotropic positions, convex bodies, and measures* play an essential role. As a beautiful and self-contained introduction for new researchers, the monograph [496] can also be used as a handbook-like reference book for specialists, containing significant recent and deep results of the field. The first and introductory chapter contains a lot of material from (asymptotic) convexity, such as the *Brunn–Minkowski inequality and its functional forms, mixed volumes and classical geometric inequalities, classical positions of convex bodies* (minimal mean width and minimal surface area positions), the *Brascamp–Lieb inequality, its reverse* and its applications to *reverse isoperimetric inequalities, the hyperplane conjecture, covering numbers, Dvoretzky-type theorems, the Pisier inequality, the reverse Santaló inequality, and M-ellipsoids*. Later on, relations to *Busemann–Petty problems* are clarified, *Vaaler's inequality* and an extension by Meyer and Pajor, *Koldobsky's formula on hyperplane sections*, and upper bounds for the volume of *sections of the cube* are discussed. Many problems of asymptotic convex geometry have the natural setup in terms of the *isotropic position and isotropic convex bodies* (note that the affine class of any convex body contains, up to orthogonal transformations, a unique isotropic convex body). This concept is fundamental for the book, and is also based on it and amongst other deep questions, later book parts clarify the dependence structure of famous conjectures from asymptotic convex geometry: the *isotropic constant conjecture*, the *thin shell conjecture*, and the *Kannan–Lovász–Simonovits conjecture* (KLS). The authors comprehensively present many partial results and background information on these conjectures, their histories, and motivations. For example, Klartag's positive answer to the thin shell conjecture for unconditional isotropic convex bodies is presented, and it is shown how the thin shell conjecture is related to the *hyperplane conjecture*. In the final chapter it is shown how the KLS conjecture implies the hyperplane conjecture, and in the "further reading section" the monotonicity of the entropy is studied. Further notions presented (or used as tools) in this book and coming close to our book here are *centroid bodies, zonoids, random polytopes, polytopes with few vertices or facets, and*

0-1 polytopes. This book is certainly very useful for any reader interested in fields between geometry and analysis.

The book [133] of Artstein-Avidan, Giannopoulos, and Milman gives a detailed and comprehensive overview of *modern asymptotic geometric analysis*, and it contains also excellently selected material from *classical convexity*. The authors nicely present geometric, analytic, and probabilistic methods which yield enlightening approaches to many deep theorems, and various of them are elegantly proved in even different ways. We give here a representative selection of the presented results. The book starts with basics from classical convex geometry, such as the *Brunn–Minkowski inequality* with various proofs and applications, among them being the *Rogers–Shephard inequality on the volume of difference bodies* and the *isoperimetric inequality for convex bodies*. Discussed relationships between *duality and volume* lead to proofs of the *inequalities of Blaschke–Santaló and Urysohn*. The next chapter treats *classical positions* of convex bodies (John ellipsoid, minimal mean width, minimal surface area), and Ball's *reverse isoprerimetric inequality* is also proved. Chapter 3 deals with the *concentration of measure phenomenon* (which became important also in probability theory, asymptotic combinatorics, and other fields) and *isomorphic versions of isoperimetric inequalities*. Among the different spaces which the authors have chosen for this, the sphere is the starting point, followed, e.g., by Gaussian measure, the discrete cube, and various probability spaces. Here sometimes the extremal sets are first derived, followed by the computational consequence. Different types and generalizations of the *Khinchin inequality* are also studied, and the next part is mainly devoted to *covering and entropy numbers*, treating also their duality. The central topic of Chapter 5 is *Dvoretzky's theorem*, with different proofs, generalizations, and applications, and in Chapter 6, for an n-dimensional convex body, the *product of its mean width and the mean width of its polar body* are estimated. The authors show that this has an upper bound obtainable with the help of *Pisier's inequality*. The next chapter contains the following theorem of Milman (with different proofs): the diameter of a random section of proportional dimension of a high-dimensional symmetric convex body is controlled by the mean width of this body. Chapter 8 comes back to *positions*, more precisely to the *M-ellipsoids and M-positions of convex bodies*; these notions, introduced by Milman, ensure that, for any convex body, there exists an ellipsoid of the same volume which can replace the body in many computations (up to universal constants). This clearly presents a useful approximation tool, and it is important for the proof of the *reverse Brunn–Minkowski inequality and the Bourgain–Milman inequality*. Chapter 9 mainly refers to a Gaussian approach to results described above (like Dvoretzky's theorem), and Chapter 10 is a collection of recent observations and significant open problems on the *distribution of volume in high-dimensional convex bodies*. In this final part also the *hyperplane (or slicing) conjecture* is discussed, *centroid bodies* play a role, and the connection between isotropic log-concave measures and convex bodies is studied. Each chapter has a final section of notes and remarks, with respective main references and many applications and problems related to its content; altogether more than 600 references are given. The book gives an excellent introduction into the field for graduates, but it is also a top source for specialists in the field. Very interesting conjectures and discussions of typical methods are given. In [134], which is the second part of [133], the same authors continue to show surprising properties of geometric objects when their dimension goes up to infinity, and how these phenomena can influence other parts of mathematics. One of the important peculiarities of this field is the increasing ability to create methods which can help to investigate parametric families which become uncontrollably large. In this second part, the authors discuss the following topics (we give them in the form of shortened section headlines and understandable keywords): (*measure concentration* (for instance, the Poincaré inequality), *isoperimetric constants of log-concave measures* (for example, thin-shell estimates and the central limit theorem), *inequalities for Gaussian measures* (of dilates of centrally symmetric convex bodies), *volume inequalities* (also the Brascamp–Lieb inequality and applications in convexity), *local theory of finite-dimensional normed spaces* (e.g., type and cotype), the *Banach–Mazur compactum* (diameter, spaces with maximal distance to Euclidean space, Alon–Milman theorem), *asymptotic convex geometry and classical symmetrizations* (Steiner symmetrizations, (random) Minkowski symmetrizations, almost isometric symmetrization),

and *functional versions of geometric notions, and inequalities*. The authors excellently meet the special points which are certainly important also for future research. With interesting notes to each chapter and roughly 750 references, also this second part gives a brilliant overview on the mentioned themes.

In the Lecture Notes [75], Alonso-Gutiérrez and Bastero discuss two famous problems from the asymptotic theory of Banach spaces, treating the uniform measure on high-dimensional convex bodies: the *Kannan–Lovász–Simonovits conjecture* (KLS) and the *variance conjecture*, both connected with a certain concentration property which the uniform measure on a high-dimensional convex body should satisfy, and stating that this measure (under a normalization assumption regarding its covariance structure) shows the same *isoperimetry and concentration* as the standard Gaussian measure. The authors expose the two conjectures and the needed concepts (among them being *isoperimetric inequalities*), and they present in detail the cases for which the conjectures are confirmed, together with related methods. The connections between the two conjectures are discussed, like also their relations to some other famous conjectures from asymptotic convexity. For example, theorems by Eldan and Klartag and by Ball and Nguyen, relating the variance or the KLS conjecture with the *slicing problem*, and Eldan's work on the relation between the KLS conjecture and the *thin-shell conjecture* are presented. Various applications in probability and computer science are given. For example, it is demonstrated that the KLS conjecture is directly related to complexity problems of algorithms referring to high-dimensional convex bodies. This implies that it is also an *important conjecture in theoretical computer science*. The main notions, ideas, and methods from the whole Lecture Notes are successfully brought to the reader, with unifying and enlightening explanations and a self-contained, streamlined approach.

We continue with the interesting monograph "Alice and Bob meet Banach" [151] by Aubrun and Szarek. The names "Alice and Bob" come from information theory, where they represent the two parties which communicate. Quantum particles have various applications and can also help to develop new methods of *computation, communication, and cryptography*. The underlying theoretical aspects are compiled to form the so-called *quantum information theory*. Many branches of mathematics take seriously part in the study of this field. Although most of the underlying physics can be described by using finite-dimensional Hilbert spaces, also asymptotic viewpoints play an important role, and so also *asymptotic geometric analysis* comes in. The concept of *finite-dimensional convex bodies* is introduced, like also operations which can be performed on them, for example related *symmetrization procedures*. Cross-connections like that between the *mean width of a convex body* and stochastic processes are discussed. Further on, *packing and covering concepts* are treated, and in view of convexity related developments around *concentration of measure techniques, Dvoretzky's theorem, classical Banach space inequalities, random matrices, norms, and zonoids* are studied, like also various aspects of the already mentioned *quantum information theory*. By cleverly embedded exercises and a nice explanation style the authors succeed in making the needed technical material accessible, and to present key ideas and results elegantly. Parts of the book could be used for a graduate course on quantum information theory from the perspective of *functional analysis, convex geometry, random matrix theory, or applications of asymptotic geometric analysis*. To the large collection of exercises the authors added a related appendix of hints. Besides teaching aspects, the book is also useful as reference source for researchers. All chapters are concluded by interesting notes and references.

12.4.7.3 Proceedings devoted to the geometry of Banach spaces

We mention now important proceedings in the field. Again the selection depends on the proximity to the field of convexity.

Besides many further articles (mainly on metric geometry in general linear spaces), the proceedings [1358] (edited by Kelly) on the *geometry of metric and linear spaces* contain various articles on *Minkowski geometry and convexity*. More precisely, these articles refer to the following themes (also meant for convex bodies in normed spaces): a *general theory of polygons, k-equilateral sets*,

intersection and covering problems, Helly-type and Tverberg-type theorems in convex product structures, *smoothness classifications of types of Banach spaces, metric embedding techniques applied to geometric inequalities*, and further problems. An interesting problem section, combined with many prominent names, is given at the end.

Next we mention the *Proceedings of the Seminar on Random Series, Convex Sets and Geometry of Banach Spaces*, held at the Mathematisk Institut of the Aarhus Universitet (Aarhus, October 14-October 20, 1974) and collected in memory of E. Asplund (1931–1974). They appeared in the *Various Publications Series*, No. 24, published by the Aarhus Universitet, Matematisk Institut, Aarhus, 1975. These proceedings offer articles treating *geometric inequalities in Banach spaces, moduli of smoothness and convexity, intersection properties, convex sets with the Radon–Nikodým property*, and various further themes.

Among other topics, the proceedings [240] of Bartle et al. contain papers treating *Radon–Nikodým and Kreĭn–Mil'man properties of convex sets, minimax theorems without convexity, applications of the geometry of infinite-dimensional spaces to vector measures, girth and isomorphic classification of Banach spaces, exposed points, and orthogonality concepts*.

We now survey the important series of proceedings coming from the Israel seminar on *geometric aspects of functional analysis (GAFA)*. Also classical convex geometers have been invited there, and each volume contains some articles very close to central problems of (finite-dimensional) convexity. Every collection of this series appeared (except for one case) as an own volume of the Springer Lecture Notes in Mathematics, and besides many contributions which are not so close to convexity, in all these collections one can find also deep, convexity-related papers. It should be mentioned that the first five volumes were edited by J. Lindenstrauss and V. D. Milman; starting this series, they have done a meritorious job for the whole field. Later additional editors are G. Schechtman, B. Klartag, E. Milman, and S. Mendelson. We cite now shortly the whole sequence of GAFA Lecture Notes, ignoring the chronological order regarding two further proceedings (not really from this series). These additional proceedings we cite below, after the GAFA series. In all cases, we mention almost only the subthemes close to convexity. Over the years, an increase in work on asymptotic theory of Banach spaces can clearly be seen; but we decided to bring all GAFA seminars together here.

The very first proceedings here have no named editors; thus we cite them completely here in the text: *Israel Seminar on Geometrical Aspects of Functional Analysis, Papers from the seminar held in 1983/84*. Tel Aviv University, Tel Aviv, 1984. These proceedings contain convexity-related articles treating *Dvoretzky's theorem, Brunn's theorem and concentration of volume phenomena for symmetric convex bodies, combinatorial and probabilistic inequalities and their application to Banach space theory, and concentration of measure phenomena in the discrete case*.

The collection [1575] is the starting point of the described GAFA series published in the Springer Lecture Notes (the respective seminar was held between October 1985 and June 1986). This collection treats *lattice packings of symmetric convex bodies, Urysohn's inequality and volume ratios, covering numbers of convex bodies and Sudakov-type inequalities, dimension-free maximal inequalities for symmetric convex bodies, and Lipschitz embeddings of finite metric spaces in low-dimensional normed spaces*.

Articles in the continuation [1576], from the seminar held during 1986–1987, refer to *Minkowski sums and symmetrizations, isomorphic symmetrizations and geometric inequalities, finite-dimensional homogeneous Banach spaces, projection bodies, a generalization of the Krein–Milman theorem, properties of simplices, and a strengthening of Santaló's inequality for unit balls of normed spaces*.

The third collection [1577] gathers papers which treat *isotropic positions of unit balls of normed spaces, inertia ellipsoids and zonoids, volumes of sections of cubes and related problems, Santaló's inequality, Steiner symmetrizations, Dvoretzky's theorem*, and related topics.

In the fourth volume [1578] there are papers dealing with *duality relations, strengthened or new isoperimetric inequalities, coverings of sets by balls of the same diameter, and the volume of intersections of two balls*.

Results on *slicing convex bodies, the Banach–Mazur distance, projection functions on higher rank Grassmannians, volume of unions and intersections of balls in Euclidean space, and isoperimetric inequalities* can be found in the GAFA volume [1580].

In [1798] papers on *concentration of measure phenomena, slicing problems, convex bodies with minimal mean width, projections and symmetrizations of convex bodies, average volume of sections of star bodies, random aspects of high-dimensional convex bodies, the isotropic constant of non-symmetric convex bodies, and Shannon's entropy power inequality and Minkowski sums* are collected.

The *central limit property* of convex bodies, convex bodies and *log-concave probability measures*, the *maximal perimeter* of a convex set in n-space with respect to a Gaussian measure, *linear functionals on zonoids*, and *polytopes with vertices chosen randomly from the boundary of a convex body* are selected themes from the volume [1799].

The GAFA proceedings [1800] present papers treating *Brascamp–Lieb inequalities, exponential volume growth, and infinite-dimensional isoperimetric inequality, symmetrization and isotropic constants of convex bodies*, the multivariable version of *Ball's theorem on cube slicing, John-type ellipsoids, isomorphic random subspaces and quotients of convex and quasi-convex bodies, uniqueness of an M-ellipsoid of a given convex body, and an isomorphic version of the Busemann–Petty problem* for Gaussian measure.

The series continues with [1801], containing articles on *the surface Brunn–Minkowski-type inequality, isoperimetric constants, the maximal number of facets of 0/1 polytopes, functional versions of Santaló's inequality and of the Alexandrov–Fenchel inequalities, Euclidean metric entropy of convex bodies, the low-dimensional Busemann–Petty problem, random convex bodies lacking symmetric projections, a random version of Dvoretzky's theorem, and the central limit theorem for generalized Orlicz balls*.

In [1405] one can find results on *valuations, geometric inequalities related to log-concavity such as Brunn–Minkowski, Santaló or Busemann inequalities, the mean width of isotropic convex bodies, the vertex index of convex bodies, a variant of Dvoretzky's theorem, a Loomis–Whitney type inequality, Brunn–Minkowski and isoperimetric inequalities for convex bodies, and contact points of convex bodies with minimal volume ellipsoids*.

Topics close to convexity and studied in [1407] are *mixed volumes, isotropic convex bodies and centroid bodies, central limit theorem for non-convex bodies, estimates for measures of sections of convex bodies, surface area of a convex set with respect to log concave rotation invariant measures, diameter of random sections of convex bodies, Minkowski symmetrization and stability results, and a variant of Dvoretzky's theorem*.

And the GAFA proceedings [1408] present results on *valuations, random cube projections, measures of slices of unit balls, a sharp Poincaré-type inequality for the restriction of the Gaussian measure on the boundary of a convex set*, and also *radial sums for star bodies*.

The GAFA seminars 2017–2019 are collected in even two volumes of Springer Lecture Notes, namely [1409] and [1410]. In the first part, e.g., the following convexity-related themes are studied: *an extension of Zhang's reverse affine isoperimetric inequality and geometric structures in large random graphs with applications*. In the second volume, convexity-related topics are *a generalized central limit conjecture for convex bodies, Koldobsky's slicing inequality for symmetric convex bodies, intrinsic volumes, random polytopes, new classes of convex and star bodies* (obtained by using some reciprocality in connection with support and radial functions), the *Alon–Milman theorem on projections of symmetric convex bodies* (here extended to the non-symmetric case), *Ehrhard's Brunn–Minkowski type inequality, high-dimensional convex sets arising in algebraic geometry, and Esterov's formula for the volume of polytopes*.

For the sake of completeness regarding the name GAFA, we shortly cite the two very important references [61] and [62] which (under the headline "GAFA 2000") contain excellent and much more general contributions from very different branches of Mathematics, and with famous authors.

Nevertheless, two articles in Part II of this are dedicated also to convexity. They refer to *asymptotic geometric analysis* and (at least partially) to the *combinatorial side* of the field.

We now come back to proceedings not from the GAFA series which partially appeared not after them. The first, namely [880] edited in 1988 by Fitzpatrick and Giles, present works on *differentiability of distance functions in normed linear spaces, and of convex functions, counter-intuitive results from asymptotic geometric analysis, and nearest points in convex sets in normed spaces*.

The collection [1862] from 1990 contains articles on *moduli of (complex) convexity, characteristics of convexity, related fixed point theorems, the entropy of convex bodies with "few" extreme points, related inequalities, and differentiability of convex functions in Banach spaces*.

Ball and Milman edited the interesting volume [201] where, among various further results, the following themes close to convexity are discussed: *integrals of smooth and analytic functions over Minkowski sums of convex sets, the Gromov–Milman theorem on the concentration phenomenon on uniformly convex spheres, random points in isotropic convex sets, the Busemann–Petty problem, polytope approximations in spaces with good duality and interpolation properties, the reverse Brunn–Minkowski inequality, an "isomorphic" version of Dvoretzky's theorem, extremal properties of the regular simplex, and volume approximation of convex bodies by polytopes* (where also notions like *floating bodies and illumination bodies* are helpful).

The proceedings [1238], edited by Houdré et al., show important interactions between *concentration phenomena, isoperimetry, and functional inequalities*, which implied progress in functional analysis and probability theory flanked by and combined with developments in *combinatorics, convex geometry, harmonic analysis, and mathematical physics*. Particular convexity related topics treated in [1238] are, e.g., *Milman's ellipsoids and M-position of convex bodies, isoperimetric sets of radially symmetric measures, the hyperplane conjecture, quantitative isoperimetric inequalities, and isoperimetric bounds on convex manifolds*. Measure concentration bounds are usually obtained from isoperimetric type inequalities, and recently E. Milman reversed this direction confirming lower bounds on the isoperimetric profile under the assumption that the concentration function is infinitesimal. In a paper of Ledoux (also contained in [1238]), elegant alternative proofs of these results based on semigroup arguments are presented.

Also the collection [1618] of Ludwig et al. on *asymptotic geometric analysis* presents several papers which are especially interesting for convex geometry. One refers to the *variance conjecture* (equivalent to the thin shell width conjecture, and being a strengthening of the hyperplane conjecture on the minimal volume section of a convex body), where also linear images of hyperplane projections of the unit cube and hyperplane projections of the cross-polytope play a role. Further contributions treat *stability properties of the generalized spherical Radon transform, flag measures* (which describe local properties of convex bodies) and their applications to several convexity problems (e.g., treating zonoids), *extremal non-central sections of the unit cube, Euclidean sections of convex bodies* (deeply discussing Milman's version of Dvoretsky's theorem), *duality on convex sets in generalized regions* (where the class of all closed convex sets bounded between two convex regions is considered), *inequalities for SL(n) invariants of smooth convex bodies*, and *connections between information theory and convex geometric analysis*. (It should be noticed that *SL(n) invariant valuations* occur there, satisfying isoperimetric type inequalities.)

We continue with the proceedings [578] collected by Carlen et al.; they refer to convexity and concentration, where *randomized isoperimetric inequalities, non-standard constructions in convex geometry* (to get from two given convex bodies new types, e.g., as harmonic and geometric means), *measures of sections of convex bodies, entropy power inequalities in convex geometry and their reverse forms for distributions in the setting of convexity constraints, stability results for geometric inequalities and their functional counterparts* (for instance, the Blaschke–Santaló inequality and affine isoperimetric inequalities), and *counting integer points in polytopes of high dimensions*.

12.4.7.4 Selected surveys on the geometry of real Banach spaces

We finish this part of our book with a selection of convexity related surveys treating geometric results in real Banach spaces.

ELEMENTARY GEOMETRY IN MINKOWSKI SPACES

Many older survey papers on *Minkowski geometry* are not easily accessible, and basic facts from the *elementary geometry* in this (in general non-Euclidean) field have still proofs which are messy, or are considered as "folklore". And often one even cannot find clear proofs! This situation led Schäffer, an expert in the field, to formulate in [2125]: "The amazing amount of underbrush that has to be cleared away,..., indicates, to this author at least, that the geometry of finite-dimensional convex sets is still quite imperfectly known." As examples of such "older papers", which are not easy to reach, we mention Gołąb and Härlen [1012], Gołąb [1013], Petty [1966], and Yaglom [2485].

In the much more recent surveys [1722], [1720], and [190], the authors try to fill here a gap. Namely, in [1722] Martini et al. present many streamlined proofs and a lot of material referring to basic notions and results (on the level of elementary geometry) in this field. Shortly described, these efforts refer to the *triangle inequality, strict convexity* (of the norm), the *monotonicity lemma, extremal norms, equilaterality and affine regular hexagons, normality and Radon normes, circumscribed and inscribed Minkowskian circles, and self-perimeters and areas of unit circles*. In [1720], Martini and Swanepoel concentrate on four themes which are well studied in Minkowski spaces: *bodies of constant width* and various related notions, *generalized convexity notions* (such as d-convexity, B-convexity, Mayer's hyperconvexity, and Boltyanski's H-convexity) which are well suited for combinations with the concept of normed spaces, *the geometry of bisectors* (here direct generalizations of perpendicular bisectors in Euclidean space are meant) and, closely related to the latter notion, *Voronoi diagrams*. These notions are comprehensively surveyed, in several cases also in the more general setting of *gauges* (also called asymmetric norms). And in the overview [190] Balestro and Martini gather many references on *elementary geometry of polygons and circles, metric problems, differential geometry of curves and surfaces*, as well as *billiards* in Minkowski spaces are discussed. A further related paper is [1718].

The expository paper [1721] of Martini and Swanepoel particularly treats *Radon curves* and shows their importance for the geometry of normed planes; the authors use a symplectic approach to planar norms and antinorms, in this way conveniently getting the solution of the isoperimetric problem. Another related expository paper is Balestro et al. [189]; it deals with *angles, angle functions, and angular measures in normed planes and spaces*. Since there is no other survey or book in this direction, it really fills a gap. *Orthogonality concepts* in Minkowski geometry, exposed in the two papers [69] and [70] by Alonso et al., were mentioned already above. Continuing with expositions on elementary geometry in normed planes we cite also the papers [67] and [68] by Alonso, Martini, and Spirova. In the first one, all possible locations of *circumcenters of three non-collinear points* in an arbitrary normed plane are given, and this is based on the complete classification of all possible shapes of the intersection of two norm circles, also obtained there. And the second paper gives a complete picture of the location of *minimal enclosing discs* for arbitrary triangles in normed planes. This is based on a new classification of triangles in such planes using their "angles".

In the papers [1548] and [1547] Leopold and Martini present analogous (and further) results in higher dimensions, i.e., on simplices in n-dimensional Minkowski spaces (results on polygons in normed planes are also given). Thus, notions like *circumcenters, Euler lines, Monge points, and Feuerbach spheres of simplices in normed spaces* are introduced, and results on them are derived (see also our subsection 2.9.1.1 on simplices). Assuming the coincidence of circumcenters and vertex centroids, a suitable type of *regularity of simplices* in normed spaces is introduced and studied.

INEQUALITIES

In our partial survey collection on asymptotic convexity, given more below, inequalities will also play an essential role, but in the papers directly here they are more dominating.

The partially expository paper [1788] mainly aims to show the importance of several geometric inequalities used to get results on concentration of measure phenomena. *Medians of functions on the sphere and Grassmann manifolds, volume ratios, and geometric inequalities involving mixed volumes* play an essential role. Further on, *Minkowski's theorem on the volume of convex combinations of convex bodies, Steiner's theorem, Brunn–Minkowski inequalities for volumes and mixed volumes*, and the *inequalities of Urysohn and Santaló* are studied within this framework. Also the author estimates the dimensions of well-complemented Euclidean sections of arbitrary n-dimensional spaces in terms of the mixed volumes of corresponding unit balls.

In [1794] V. D. Milman and Pajor excellently survey properties of convex bodies which are associated with a given centrally symmetric convex body K. These are the *Legendre ellipsoid* L of K, whose moments of inertia are those of K, the *Binet ellipsoid* (a suitable homothet of the dual of L, well studied also in mechanics), and the *zonoid of inertia* being Petty's *centroid body* (the integral arithmetic mean of the support functions of the segments joining x and $-x$ in K). Using also suitably the notion of isotropic position, several inequalities (like inverse Hölder inequalities and classical inequalities going back to Blaschke and Busemann) are treated. For example, the volume of K is bounded from above by that of L, with equality iff K is an ellipsoid. Interesting open problems can be found in the final part.

In the comprehensive survey [929] of Gardner a network of many geometric and functional inequalities is presented having the Brunn–Minkowski inequality in its center. The relationships between these inequalities are the main topic of this exposition, considering (besides Brunn–Minkowski) the *Prékopa–Leindler inequality, the Brascamp–Lieb inequality, and Barthe's reverse Brascamp–Lieb inequality*. The author gives sketches of proofs, and he shows applications, analogues, and variants of the Brunn–Minkowski inequality. The reason to cite this nice paper here is the possibility of extensions, to more general geometric and functional inequalities in functional analysis as it is done by Klartag [1404]. Namely, based on [929], he deals in [1404], in an analogous sense, with functional inequalities which have, in different ways, a geometric background. First, *integral inequalities of generalized Prékopa–Leindler and log-Sobolev type* are derived from applications of the Brunn–Minkowski inequalitiy, and second, consequences of a *functional Blaschke–Santaló inequality* are obtained. Finally, two different functional analogues of the *Alexandrov–Fenchel inequality for mixed volumes* are presented. The author succeeds impressively in extending the paper [929] of Gardner.

The high-dimensional phenomena presented in the survey [1784] by Milman particularly concentrate on the relations between *geometry and measure theory*. The author emphasizes that such investigations can be extended in a very broad sense to *log-concave measures*. Therefore such extensions of various *geometric inequalities* are discussed and, conversely, it is shown that some typical probabilistic results have been successfully proved in such a geometric setting. Various related examples, recent results, and open problems are gathered. The discussed results and problems refer to the *inequalities of Blaschke–Santaló and Prékopa–Leindler, polarity, the central limit theorem, isotropic position*, and the possible extension to a *convex measure*.

SURVEYS ON ASYMPTOTIC CONVEXITY AND RELATED INEQUALITIES

As the title says, the survey [1790] of V. D. Milman is dedicated to Dvoretzky's famous theorem and its consequences over roughly 30 years. Containing no proofs and only a few technical pre-requisites, the paper is easy to read. After presenting the theorem itself, the spectrum of *uniformly continuous functions on the sphere of a Banach space and on Grassmann manifolds* is discussed, and consequences for centrally symmetric convex bodies and their duals follow. The author demonstrates the importance of *ellipsoids* in the study of convex bodies from the global (and not a local) structural point of view,

with the fact that ellipsoids can substitute for arbitrary convex bodies in the essential part of any volume computation. It turns out that these results are related to the *reverse Blaschke–Santaló inequality*. In the final part, the (intuitively surprising) existence of *lattice tilings by isomorphic Euclidean balls* is discussed, which the author observed jointly with Alon.

The very important handbook article [1579] of Lindenstrauss and V. D. Milman deals comprehensively with convex bodies in the local theory of Banach spaces. One famous example of a treated topic is *Dvoretzky's theorem*, due to modern proofs strongly related to isoperimetric inequalities. Sharp estimates on the dimension of nearly spherical sections of unit balls of n-dimensional spaces are obtained. Further on, the ellipsoid of maximal volume contained in a symmetric convex body (*John's ellipsoid*), the ellipsoid of inertia (*Legendre ellipsoid*), and Milman's *M-ellipsoids* (important for proofs of the *direct and reverse Blaschke–Santaló and Brunn–Minkowski inequalities*) are further discussed topics. These latter ellipsoids play a fundamental role for volume estimates. Open problems, such as the existence of "large" (in volume) hyperplane sections, are presented. Concerning this latter problem, the solution of the *Busemann–Petty problem* is essential. Also Gluskin's probabilistic proof of the existence of *pairs of n-dimensional spaces whose Banach–Mazur distance is proportional to n* is discussed, and applications to classical convexity are studied. The following results are emphasized, too. For example, it is shown how the sharp estimates on the *approximation of n-dimensional spheres by affine images of cubes*, obtained by Bourgain, Lindenstrauss, and Milman, relate to the number of projections needed to approximate the surface area of a convex body, and the number of Steiner symmetrisations needed to transform convex bodies into nearly spherical ones. Various basic notions from convexity occur as topics of investigation or as tools: *zonotopes, zonoids, Minkowski sums, and symmetrization types*. Additional results are given, including Ball's positive solution to the *plank problem for symmetric bodies*.

In a brilliant style, Ball combines in the expositions [199] and [200] impressive results from classical and asymptotic convexity. In [199], excellent lectures (with proofs or explanations) are presented to show developments in convexity which were induced by Dvoretzky's theorem. For example, these refer to *almost spherical slices of cubes and generalized octahedra, John's ellipsoids*, presentation of the *Brunn–Minkowski inequality* as an integral inequality and thus as a special case of the *Prekopa–Leindler inequality for integrals*, the *reverse isoperimetric inequality* for the non-symmetric and for the symmetric case (extremal for simplices and cubes, respectively), and, finally, *Dvoretzky's theorem* (in the version due to Milman). The author gives detailed descriptions of the most recent state of the art regarding each result, and he offers also a nice view on the ideas needed to handle it (therefore inviting a broad audience to see the methods of modern convex geometry). Various nice figures support the reader's approach to the field and show clearly how geometric in nature the topic is. In the handbook article [200], Ball treats topics from the *intersection of functional analysis, harmonic analysis, probability theory, and convex geometry*. He starts with inequalities involving volumes of convex bodies, their sections, and coverings. More precisely, the *Brascamp–Lieb inequality, its reverse, and the reverse isoperimetric inequality* are discussed. The exposition continues with results on *volumes of sections and projections* (e.g., Vaaler's theorem) and *plank problems*, i.e., coverings of convex bodies by planks (which are strips lying between two parallel hyperplanes whose width is essential). Given a convex body K, the relative width of a plank S is the width of S divided by the width of K in the direction perpendicular to S. This gives the opportunity to consider *Bang's affine plank problem* and to study plank problems also for (symmetric) norms and non-symmetric gauges. The author gives a comprehensive discussion of these extensions and offers finally also related problems (like densities of sphere packings).

In [960] and [965], Giannopoulos and Milman discuss important developments of asymptotic convex geometry, showing structural properties of high-dimensional convex bodies which typically can be combined with the following general principles: for increasing dimension, basic geometric properties of convex bodies differ less and less, and most bodies look more and more similar. This phenomenon is demonstrated for the family of all convex bodies via results which are, for example,

related to the following topics and keywords (again we mention only selected themes closer to classical convexity): *the Brunn–Minkowski inequality and volume-preserving transformations* (the latter also yielding new proofs of the Brunn–Minkowski theorem), *classical and isomorphic symmetrizations* (Minkowski and Steiner symmetrizations), *centroid and floating bodies*, proofs of inequalities like the *reverse Blaschke–Santaló inequality, Banach–Mazur distance estimates* (starting with Gluskin's theorem), *classical position problems* (e.g., John's position theorem on the maximum volume ellipsoid, and the minimal mean width position), *isotropic position problems* (in connection with the slicing problem; see also the survey [1794] of Milman and Pajor), the *Brascamp–Lieb inequality and its reverse*, as well as various further (sharp) geometric inequalities.

As in other surveys, in [1792] and [1791], V. D. Milman nicely presents counter-intuitive asymptotic phenomena for high-dimensional objects. In the first paper [1792] these phenomena are supported by examples involving, besides convex bodies, also homogeneous structures (both continuous and discrete) for which other parameters play the role of dimension. The notion of so-called *isomorphic geometry*, as the study of common geometric properties of suitably defined families of convex bodies, and the unexpected asymptotic behavior of them (when the dimension increases to infinity) are discussed. Here the geometry of *floating bodies and ellipsoids* plays an important role. And the second paper [1791] treats quantitative aspects of high-dimensional convex geometry, where no proofs are given, but the author excellently explains the fascination of such results. Also the tools and methods are nicely brought to the reader. To achieve this objective convincingly, he presents really many results. Examples refer to *radii of largest balls contained in random projections, the Milman ellipsoid, Brascamp–Lieb inequalities, extensions of the Dvoretzky–Rogers lemma, and isotropic positions in convex geometry*. A central observation of the author is that (as he says in the last remark) in the world of very high degrees of freedom (here very high dimension n is meant), the diversity does not increase exponentially with n. Instead of this "expectation", an asymptotic pattern regulates things and avoids such diversity.

The survey [1793] of Milman is in fact that survey where the author suggests the new name "asymptotic geometric analysis" (instead of "local theory of Banach spaces"), to give a better description for what is going on there. Instead of being isometric, results of this field are "isomorphic", with isomorphism constants not depending on the dimension. Based on the *concentration phenomenon*, various examples of a related regularity degree of convex bodies in high dimensions are discussed, showing also that in many volume and entropy calculations with other bodies, an arbitrary convex body can effectively be replaced by a certain ellipsoid associated with it, leading to deep inequalities (like the *reverse Santaló and Brunn–Minkowski inequalities*). Further topics are discussed, which have to do with the *expected diameter* of a k-dimensional random projection of an n-dimensional convex body, and interesting directions of future research (dealing with the concentration phenomenon) and open problems (related to *entropy duality, large quotients of finite-dimensional Banach spaces, and variants of Dvoretzky's theorem*) are given.

In their handbook article [960] (see also above), Giannopoulos and Milman study Euclidean structures on high-dimensional normed linear spaces; the unit balls of these Euclidean structures are ellipsoids corresponding with the unit ball B of the considered space, occurring, e.g., as largest ellipsoids contained in or smallest ellipsoids containing B, and also various further Euclidean structures are discussed. The introductory part is dedicated to the background from convex geometry, and then Dvoretzky's theorem; these Euclidean structures and applications thereof in convexity follow. In the appendix, also the geometry of the *Banach–Mazur compactum* is discussed, and a short survey regarding the *hyperplane conjecture* is given, too. This paper shows in a particularly nice manner the connections between convex geometry and Banach space theory.

Mankiewicz and Tomczak-Jaegermann [1692] survey results on *random phenomena* (being non-Euclidean in nature) *in asymptotic convexity*. One motivating point is Gluskin's theorem (from 1981) on the diameter of the Banach–Mazur compactum (a complete proof of this theorem is given in the article). "Random" subspaces/quotients of normed spaces became an important research topic since then, and many convex bodies with extremal behavior have been constructed, e.g., regarding

the Banach–Mazur distance. In various cases it has been verified that quantitative estimates, trivially obtained by comparison with the Euclidean case, are unimprovable. The authors extend respective methods to general convex bodies and obtain quantitative estimates using corresponding volumetric parameters of such bodies. They prove that there exists a separable Banach space without Schauder basis and give a characterization of Hilbert spaces connected with the existence of Schauder bases. Also very recent developments are taken into consideration.

Barthe [235] surveys developments on the *isoperimetric problem for product probability measures*. First, the isoperimetric problem is formulated in the general setting of a metric space, supplied with a Borel (probability) measure. After recalling some fundamental facts around the isoperimetric problem in *Riemannian manifolds* and the *concentration of measure phenomenon in Banach spaces*, the survey concentrates on product probability measures. The presented results combine *differential geometry, Banach space theory, and probability theory*, and basic notions that play an essential role are *Steiner symmetrization, Gaussian measure, and Sobolev-type inequalities*. At the end of the article, an interesting problem on isoperimetric properties of half-spaces related to isotropic convex bodies is posed.

Klartag [1403] surveys results on *isomorphic and almost isometric problems in high-dimensional convexity*. The author presents various examples, including results on Steiner and Minkowski symmetrizations that are necessary to transform a given convex body into another one whose boundary is suitably placed "between" the boundaries of two homothets of balls. Also results on volume distribution and the proof of the *thin shell conjecture* are discussed.

Also in the spirit of asymptotic geometric analysis, Szarek [2334] exposes *different interpretations of "complexity" of convex sets*. He starts with studying *metric entropy* (i.e., the question of how many balls one needs to cover a given convex set). The recent progress regarding the *Pietsch duality conjecture*, which is basic there, is discussed. Then problems on "geometric complexity" follow, e.g., referring to the *Banach–Mazur compactum*, and to the non-trivial *projection problem*. Finally, the "algorithmic complexity" of explicit constructions of interesting objects (whose existence is usually made credible with the help of probabilistic arguments) are considered.

In [236], Barthe surveys the works of Klartag, Fleury, Guédon, and Paouris on the solutions of the *central limit conjecture for convex bodies*. This conjecture asked whether the projections of a random variable, uniformly distributed on an isotropic convex body, are almost Gaussian, with projections of a "large" dimension. This famous conjecture was first proved for certain families of convex bodies, and then, by Klartag, in full generality. It is actually proved via a concentration inequality for log-concave random variables. Barthe describes all contributions, places the problem in a historical context, and explains proof ideas and the analogy to other basic results in asymptotic convexity, in particular, to *Dvoretzky's theorem in Milman's form*.

Guédon [1095] nicely surveys problems from high-dimensional convexity with log-concave measures as starting points. He then goes to the *hyperplane conjecture and the thin shell conjecture*, continuing with algorithmic convex geometry and *concentration phenomena* for probabilities in Banach spaces.

The extended notes [1097] of Gudéon et al. (based on lectures given by Gudéon at the Polish Academy of Sciences in 2011) culminate in the presentation of *Paouris' theorem* which, roughly speaking, says that most of the volume of an isotropic convex body (see Brazitikos et al. [496]) in Euclidean n-space is contained in a multiple of the Euclidean ball of radius $\sqrt{n}$. A comprehensive framework of needed tools and related deep results is carefully provided. The authors derive Ball's *geometric version of the Brascamp–Lieb inequalities* with a mass transport proof due to Barthe; they show how this yields Ball's *inverse isoperimetric inequality* for symmetric bodies. The notion of *Minkowski sum* and the *Brunn–Minkowski inequality with applications* (Brunn's theorem on the volume of parallel sections) follow. The authors continue with *Borell and Prékopa–Leindler type inequality consequences* for log-concave functions. Also a subsection on the *concentration of measure phenomenon, Dvoretzky's theorem*, and *reverse Hölder inequalities and volumes of sections of convex*

bodies is contained and shows also deeper related results on log-concave functions and probability measures. This finally leads to Paouris' theorem in a probabilistic version. Each section is concluded by excellent notes and comments. The whole notes can be excellently used as a starting point for graduates to enter the field; they present central topics from the local theory of Banach spaces in a brilliant way.

Here we mention also Guédon's nice representation of problems and results from high-dimensional convex geometry, which can be found in Appendix B of the basic book [1560] on Banach space theory.

INFINITE-DIMENSIONAL BANACH SPACES

Coming from finite-dimensional polyhedral spaces (i.e., their unit balls are polyhedral), Durier and Papini [777] study analogous infinite-dimensional spaces. Dropping the assumption to be finite-dimensional, one may consider several different definitions, all of them extending the above finite-dimensional property. Going this way, the authors review and compare various definitions of this type and the resulting classes of *polyhedral spaces in an infinite-dimensional setting*, and open problems are also posed.

In their handbook article [888], Fonf, Lindenstrauss, and Phelps give an excellent overview to the most important results treating *convex sets in infinite-dimensional Banach spaces*. One guideline that the authors follow is: what can be brought to experience about convex sets via topological methods rather than computations or probabilistic techniques? The paper contains many nice examples, also some proofs, and it is written in fluent style with as few technicalities as possible. The first section starts with the *Krein–Milman theorem* and important consequences thereof. In the second part the classical *Choquet representation theory* is discussed, with many applications, and section 3 describes *simplices and their structure* (in particular, the Poulsen simplex). The fourth part is devoted to the *Radon–Nikodým property* from a geometrical point of view, and its complex version called the *analytic Radon–Nikodým property*. Then, in section 5, the authors investigate *support points and boundaries*, and also the *Bishop–Phelps theorem*. The sixth section refers to convex bounded polytopes and the *theory of polyhedral spaces* (see also the survey [777] of Durier and Papini), which is partly *isometric*. But the authors present also nice characterizations of spaces which are *isomorphic* to polyhedral spaces. The final section collects various topics, such as *stable convex sets, nearest points, convex tilings*, and *proximinal sets*. Also important open problems are presented.

Mankiewicz and Tomczak-Jaegermann deal in their survey [1692] also with structural properties of finite- and infinite-dimensional Banach spaces. For example, they show how to use finite-dimensional constructions in the infinite-dimensional case.

FIXED POINT THEOREMS

Papini summarizes in his handbook article [1948] notions and results from the *fixed point theory of nonexpansive mappings* and their relations to *convexity*. The author gives already at the beginning a nice collection of related examples and also a short historical representation. Regarding fields of further extensions, the author recommends to study, besides convex sets, within that framework also *star-shaped sets, general non-convex sets, and unbounded sets*.

In his handbook article [2000], Prus surveys topics from the *geometry of Banach spaces* which have strong relations to and also applications in *metric fixed point theory*. He starts with properties like *strict convexity and smoothness*, and he characterizes strict convexity in terms of the fixed point set of nonexpansive mappings. Then he continues with *uniform convexity, uniform smoothness*, and extensions, and he considers various *moduli* associated with such convexity notions. Then the *normal structure in Banach spaces* is studied, again in connection with *fixed point theorems for nonexpansive mappings*. It is remarkable that the article [2000] contains also various elegant and complete proofs; the successful style in this publication shows the proof techniques in a nice way. Summarizing, one can say that this survey is a valuable resource for researchers in both participating fields (metric fixed point

theory and Banach space theory), and that it also shows a nice *application field for convex geometry*. The same can be said about the survey [914] of Fuster, from the same handbook. This related survey can also be located in the *interplay of metric fixed point theory and Banach space theory*. The author concentrates on related *moduli* (e.g., of convexity) and the Banach space properties associated with them. Occurring notions close to convexity and covered by the survey are *uniform and near-uniform convexity, uniform and near-uniform smoothness, non-compact convexity, and Jung's constant*.

SURVEYS ON FURTHER RELATED TOPICS

In the exposition [1382] of Kim and Kirk the interplay between *abstract convexity and metric convexity* are studied, and their relations to *metric fixed point theory* is also investigated. Nonexpansive mappings, convexity structures, and ball intersections play the important roles in that paper.

Extending the underlying Minkowskian gauge functional of a not necessarily finite-dimensional real normed vector space X to the general non-symmetric case, Révész and Sarantopoulos systematically study in their survey [2028] geometric concepts and results related to the *Minkowski measure of symmetry*. They combine geometric and approximation theoretic results referring to this topic, and they collect a number of related results with respect to the extended Minkowski functional, including *approximation of the classical Minkowski functional of the central symmetral* and also a *Chebyshev-type extremal problem for real-valued polynomial functions* on X.

The survey [558] of Cabezas-Rivas and Miquel contains results on *mean curvature of submanifolds in normed spaces*. Different to the Euclidean subcase, in the general case (of such spaces) we have many choices for the definition of mean curvature of a submanifold. The authors concentrate on *smooth and strictly convex norms*. Using different notions of volume and mean curvature, and considering the mean curvature as the first variation of the area, one can focus on fresh related results which also show the advantages and disadvantages of the definitions of mean curvature arising from the *Hausdorff and Holmes–Thompson notions of volume*.

In the interesting survey [1925], Ostrover discusses interesting and recently developed relations between *symplectic geometry* on the one hand, and the fields of *asymptotic geometric analysis, classical convexity, and the geometry of normed spaces* on the other hand. Topics close to convexity in this discussion are relations between an open *symplectic isoperimetric-type conjecture* due to Viterbo and *Mahler's volume product conjecture, Milman's reverse Brunn–Minkowski inequality* (bounding the volume of the vector sum of two convex bodies from above in terms of their volumes), *billiards in smooth convex bodies, Hanner polytopes*, and some *geometry of ellipsoids*.

The survey [1506] of Lassak and Martini shows the state of the art (2014) regarding the notion of reduced bodies in normed spaces (the Euclidean subcase is surveyed by the same authors in [1505]). A convex body K in an n-dimensional Minkowski space is said to be *reduced* if the minimal width of each convex body properly contained in K is strictly smaller than that of K. This definition is somehow "dual" to that of *diametrically complete sets*; in Euclidean space, the latter notion is equivalent to that of *constant width*. Thus, it is natural to study in normed spaces these three notions together. The first contribution presenting the knowledge around these notions on an earlier stage is the brilliant survey [596] by Chakerian and Groemer. It mainly concentrates on bodies of constant width in Euclidean space (but also in normed spaces). This is completed by the handbook article [1177] and section 2 of the comprehensive survey [1720] (state: 2004) on bodies of constant width in normed spaces, presenting also results on complete and reduced bodies. Also Chapter 10 of the monograph [1713] is dedicated to these classes of convex bodies, and its notes reflect the current state of knowledge (2019) on relations between sets of constant width, reducedness, and completeness in normed spaces. (An outlook on geometric properties of these sets in *other non-Euclidean geometries* is also given there.)

It should be noted that the structure of the overviews [1177], [1720] and in Chapter 10 of [1713] (headlines of subsections, etc.) follows that of the excellent forerunner [596]. It turns out that reduced bodies in normed spaces are not so easy to handle, and therefore the survey [1506] and the notes

to Chapter 10 of [1713] contain interesting related research problems. The existing material on diametrically complete sets and completions in normed spaces is covered by the very recent survey [1172] due to He, Martini, and Wu. And we refer to our above book parts 2.9.1.8, 2.9.1.9, and 2.9.1.10 for more information.

In the recent exposition [1786] of Milman, Milman and Rote, the authors study new classes of convex and star bodies called reciprocal bodies and flowers. For example, for a given convex body K, its *flower* is the star body whose radial function is given by the support function of K, and *reciprocal bodies* appear as the image of a new "quasi-duality" operation on the class of convex bodies. A wide range of properties of these sets and dualities in this framework are described. For instance, they prove that K is a reciprocal body if and only if the flower of K is convex. The authors also derive an Alexandrov–Fenchel type inequality, and results related to the *dual Brunn–Minkowski theory* are given, too. Based on [1786], in [1796] the theory of flowers is further developed. There one can find results like this: every flower centred at the origin has a "large"-dimensional projection which is "almost" a Euclidean ball.

Appendix A
Measure theory for convex geometers

"Millions saw the apple fall, but Newton asked why."

— Bernard Baruch

We add this appendix for readers who do not have a deeper background in the theory of measure (which, for us, also encompasses Lebesgue integration). This theory yields the mathematical tool needed for a complete understanding of notions like "area" and "volume", which are central in convexity. As it was made clear in Section 4.1, it is possible to read the book without going through the details of measure theory, but nevertheless we included this appendix with the motivation that the interested reader may find (almost) everything in a single place. Our main references for this chapter are the books [239] of Bartle, [255] of Bauer, [834] by Federer, [2090] of Salamon, and [2249] of Simon. A few results of measure theory are not dealt with here; they are only stated in the main text. We also do not discuss *densities* of measures, which are briefly mentioned in Section 7.4. Moreover, in Chapter 10 some results from measure theory are stated and proved. They were not included here since their applications are too specific for us, and in this appendix we are concerned with the general theory.

In Section A.1, we introduce general *outer measures*, *measures*, and *Borel measures*, with the due formal proofs (Lebesgue and Hausdorff measures are specializations of these). For the reader who is not interested in going through all the details of this more general theory, we inserted a summary of all the needed notions in the beginning of Section A.2, where we study the basics of Lebesgue integration.

The "standard" tool for defining a notion of "volume" in $\mathbb{R}^n$ is the classical *Lebesgue measure*. In Section A.3 we introduce this measure and prove its main properties. As it was mentioned in Section 4.1, we also can use the Lebesgue measures in lower dimension to define (lower dimensional) "volumes" of sets contained in affine subspaces of $\mathbb{R}^n$. The strategy for that construction is simple, and we also explain it formally in Section A.3. However, we also have the intuition of "non-flat lower dimensional" sets, like, e.g., a curve or the boundary of a convex body. To define "volumes" of these objects, we use *Hausdorff measures*, which we introduce and investigate in Section A.4 (these measures also bring a definition of what the "dimension" of a non-flat set really is).

In Section A.5, we introduce *product measures* and prove the classical *Fubini theorem*. As we recall, a specialization of this theorem for Lebesgue measures (which we also provide here) leads to Cavalieri's principle studied in Section 4.2. Last, in Section A.6 we state the *area* and *co-area formulas*, which are important in some parts of convexity (for example, the co-area formula provides a formal justification for the usual integration by polar coordinates and allows us to extend it to convex bodies other than the ball). Unlike in the rest of this appendix, we do not give proofs of these formulas. For them, the interested reader is referred to Evans and Gariepy [822].

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A.1 Outer measures and measures

An *outer measure* on a set X is a function $\mu\colon 2^X \to [0,\infty]$ such that $\mu(\emptyset) = 0$ and

$$\mu(E) \leq \sum_{j=1}^{\infty} \mu(E_j) \tag{A.1.1}$$

whenever E and $(E_j)_{j\in\mathbb{N}}$ are subsets of X such that $E \subseteq \bigcup_{j=1}^{\infty} E_j$ (that is, μ is *monotone subadditive*). We say that a subset $E \subseteq X$ is *μ-measurable* if

$$\mu(A) = \mu(A \setminus E) + \mu(A \cap E) \tag{A.1.2}$$

for every subset $A \subseteq X$. Notice that since $A \subseteq (A \setminus E) \cup (A \cap E)$ we already have the inequality $\mu(A) \leq \mu(A \setminus E) + \mu(A \cap E)$. Hence a subset $E \subseteq X$ is μ-measurable if and only if $\mu(A) \geq \mu(A \setminus E) + \mu(A \cap E)$ for any $A \subseteq X$ such that $\mu(A) < +\infty$ (if $\mu(A) = +\infty$, then the inequality holds trivially).

Remark A.1.1. Throughout this appendix, *countable* actually means *at most countable*, that is, any countable collection can be finite. This is why we use the notation $(E_j)_{j\in\mathbb{N}}$: by doing so, we may "complete" any finite collection of sets with finitely many empty sets.

A *σ-algebra* of a set X is a collection $\Sigma \subseteq 2^X$ of subsets of X satisfying:

(i) $\emptyset \in \Sigma$,

(ii) if $E \in \Sigma$, then $X \setminus E \in \Sigma$,

(iii) If $(E_j)_{j=1}^{\infty}$ is a countable collection of sets of Σ, then $\bigcup_{j=1}^{\infty} E_j \in \Sigma$.

That is, a σ-algebra is a collection of subsets of X which contains the empty set and is closed under complementation and countable unions. It is immediate to observe that, in this case, $X \in \Sigma$ and Σ is also closed under countable intersections.

Proposition A.1.1. *Let μ be an outer measure on a set X. The collection Σ_μ of μ-measurable sets is a σ-algebra of X.*

Proof. We have to prove that Σ_μ satisfies the properties **(i)**, **(ii)**, and **(iii)** above. The first one is obvious, since $\mu(A \setminus \emptyset) + \mu(A \cap \emptyset) = \mu(A) + 0 = \mu(A)$. Now let $E \in \Sigma_\mu$, and observe that

$$\mu(A \setminus (X \setminus E)) + \mu(A \cap (X \setminus E)) = \mu(A \cap E) + \mu(A \setminus E) = \mu(A),$$

for any $A \subseteq X$. Hence $X \setminus E \in \Sigma_\mu$, and this settles **(ii)**.

Now we prove **(iii)**. We start by proving that if $E_1, E_2 \in \Sigma_\mu$, then $E_2 \setminus E_1 \in \Sigma_\mu$. For any subset $A \subseteq X$, since $E_1 \in \Sigma_\mu$, we get that

$$\mu(A \cap E_2) = \mu((A \cap E_2) \setminus E_1) + \mu((A \cap E_2) \cap E_1) = \mu(A \cap (E_2 \setminus E_1)) + \mu(A \cap E_2 \cap E_1).$$

On the other hand, we have that $\mu(A \cap E_2) = \mu(A) - \mu(A \setminus E_2)$, because $E_2 \in \Sigma_\mu$. From these equalities, we get

$$\begin{aligned}\mu(A \cap (E_2 \setminus E_1)) = \mu(A \cap E_2) - \mu(A \cap E_2 \cap E_1) = \mu(A) - \mu(A \setminus E_2) + \mu(A \cap E_2 \cap E_1) \leq \\ \leq \mu(A) - \mu((A \setminus E_2) \cup (A \cap E_2 \cap E_1)) = \mu(A) - \mu((A \setminus E_2) \cup (A \cap E_1)) = \\ = \mu(A) - \mu(A \setminus (E_2 \setminus E_1)),\end{aligned}$$

from where we get that $\mu(A) \geq \mu(A \setminus (E_2 \setminus E_1)) + \mu(A \cap (E_2 \setminus E_1))$ for every $A \subseteq X$.

As a consequence, we get that if $E_1, E_2 \in \Sigma_\mu$, then $E_1 \cup E_2 \in \Sigma_\mu$. Indeed, using what we have just proved twice we get that $X \setminus E_2 \in \Sigma_\mu$ and $(X \setminus E_2) \setminus E_1 \in \Sigma_\mu$. Hence,

$$E := (X \setminus E_1) \cap (X \setminus E_2) = (X \setminus E_2) \setminus E_1 \in \Sigma_\mu.$$

Then, $E_1 \cup E_2 = X \setminus E \in \Sigma_\mu$. Even more, if E_1 and E_2 are disjoint, then replacing A by $A \cap (E_1 \cup E_2)$ and E by E_1 in equality (A.1.2) we get

$$\begin{aligned}\mu(A \cap (E_1 \cup E_2)) = \mu((A \cap (E_1 \cup E_2) \setminus E_1) + \mu((A \cap (E_1 \cup E_2)) \cap E_1) = \\ = \mu(A \cap E_2) + \mu(A \cap E_1),\end{aligned}$$

for each $A \subseteq X$. By induction, it follows immediately that any finite union of sets of Σ_μ is a set of Σ_μ. Moreover, if $E_1, \ldots, E_k \in \Sigma_\mu$ are pairwise disjoint (that is, $E_i \cap E_j = \emptyset$ whenever $i \neq j$), then we can apply the equality above inductively to obtain

$$\mu\Big(A \cap \Big(\bigcup_{j=1}^{k} E_j\Big)\Big) = \sum_{j=1}^{k} \mu(A \cap E_j), \tag{A.1.3}$$

for any $A \subseteq X$. Our next task is to prove that the equality above holds also for an infinite countable collection of pairwise disjoint sets in Σ_μ. Let $(E_j)_{j\in\mathbb{N}}$ be such a collection. For every $A \subseteq X$ and any $k \in \mathbb{N}$, we have

$$\begin{aligned}\mu(A) = \mu\Big(A \cap \Big(\bigcup_{j=1}^{k} E_j\Big)\Big) + \mu\Big(A \setminus \Big(\bigcup_{j=1}^{k} E_j\Big)\Big) \geq \mu\Big(A \cap \Big(\bigcup_{j=1}^{k} E_j\Big)\Big) + \mu\Big(A \setminus \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big) = \\ = \sum_{j=1}^{k} \mu(A \cap E_j) + \mu\Big(A \setminus \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big).\end{aligned}$$

Since this holds for any $k \in \mathbb{N}$, we can make $k \to \infty$ to obtain

$$\mu(A) \geq \sum_{j=1}^{\infty} \mu(A \cap E_j) + \mu\Big(A \setminus \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big) \geq \mu\Big(A \cap \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big) + \mu\Big(A \setminus \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big), \tag{A.1.4}$$

where the last inequality comes immediately from the monotone subadditivity of μ. The reverse inequality

$$\mu(A) \leq \mu\Big(A \cap \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big) + \mu\Big(A \setminus \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big)$$

is also immediate from the monotone subadditivity of μ. It follows that $\bigcup_{j=1}^{\infty} E_j \in \Sigma_\mu$, and since the left- and right-hand sides of (A.1.4) are equal, we get that they are also equal to the term in the middle, from where we obtain

$$\mu\Big(A \cap \Big(\bigcup_{j=1}^{\infty} E_j\Big)\Big) = \sum_{j=1}^{\infty} \mu(A \cap E_j), \tag{A.1.5}$$

for every $A \subseteq X$.

It remains to prove that if $(E_j)_{j\in\mathbb{N}}$ is a countable collection of μ-measurable sets not necessarily pairwise disjoint, then $\bigcup_{j=1}^{\infty} E_j$. For this sake, define $F_1 = E_1$ and

$$F_j = E_j \setminus \Big(\bigcup_{i=1}^{j-1} E_i\Big)$$

for $j \geq 2$. Hence, the sets $(F_j)_{j\in\mathbb{N}}$ are pairwise disjoint, and we clearly have

$$\bigcup_{j=1}^{\infty} E_j = \bigcup_{j=1}^{\infty} F_j,$$

and this concludes the proof, since we already have that the countable union of pairwise disjoint μ-measurable sets is μ-measurable.

□

Remark A.1.2. Any subset $E \subseteq X$ such that $\mu(E) = 0$ is μ-measurable. Indeed, if $\mu(E) = 0$, then for any $A \subseteq 2^X$ we have that $\mu(A\cap E) = 0$. Hence we have $\mu(A) \leq \mu(A\setminus E)+\mu(A\cap E) = \mu(A\setminus E) \leq \mu(A)$, where the first and last inequalities come from the monotonicity of μ. This gives the equality (A.1.1).

Remark A.1.3. Letting $A = X$ in (A.1.5), we obtain that

$$\mu\Big(\bigcup_{j=1}^{\infty} E_j\Big) = \sum_{j=1}^{\infty} \mu(E_j) \tag{A.1.6}$$

for any collection $(E_j)_{j\in\mathbb{N}}$ of pairwise disjoint μ-measurable sets. Therefore, an outer measure μ is a *measure* (in the classical sense) when restricted to the σ-algebra of μ-measurable sets. We recall the reader that a *measure* on a set X is a function $\mu\colon \Sigma \to [0,\infty]$ of some σ-algebra Σ of X with the property that $\mu(\emptyset) = 0$, and such that (A.1.6) holds whenever $(E_j)_{j=1}^{\infty}$ is a countable collection of sets of Σ which are pairwise disjoint.

Lemma A.1.1. *Let $(E_j)_{j\in\mathbb{N}}$ be a collection of μ-measurable sets. Then, we have the following statements:*

(i) *If $E_j \subseteq E_{j+1}$ for each $j \in \mathbb{N}$, then*

$$\lim_{k\to\infty} \mu(E_k) = \mu\Big(\bigcup_{j=1}^{\infty} E_j\Big),$$

(ii) *If $E_j \supseteq E_{j+1}$ for every $j \in \mathbb{N}$ and $\mu(E_j) < \infty$ for some j, then*

$$\lim_{k\to\infty} \mu(E_k) = \mu\Big(\bigcap_{j=1}^{\infty} E_j\Big).$$

Proof. For **(i)**, first observe that from the definition of μ-measurability, and since $E_j \subseteq E_{j+1}$, we get that $\mu(E_{j+1} \setminus E_j) = \mu(E_{j+1}) - \mu(E_j)$. Let $E_0 := \emptyset$ and notice that the sets of the form $E_j \setminus E_{j-1}$ are pairwise disjoint. Then, from (A.1.6) we have

$$\mu\Big(\bigcup_{j=1}^{\infty} E_j\Big) = \mu\Big(\bigcup_{j=1}^{\infty} E_j \setminus E_{j-1}\Big) = \sum_{j=1}^{\infty} \mu(E_j \setminus E_{j-1}) = \lim_{k\to\infty} \sum_{j=1}^{k} \mu(E_j \setminus E_{j-1}) =$$

$$= \lim_{k\to\infty} \sum_{j=1}^{k} \Big(\mu(E_j) - \mu(E_{j-1})\Big) = \lim_{k\to\infty} \mu(E_k).$$

Now we prove **(ii)**. Let $\mu(E_k) < \infty$. Elementary set theory gives the equalities

$$E_k \setminus \bigcap_{j=1}^{\infty} E_j = \bigcup_{j=1}^{\infty} E_k \setminus E_j.$$

The fact that $E_j \supseteq E_{j+1}$ for each j gives $E_k \setminus E_j \subseteq E_k \setminus E_{j+1}$ for each $j \in \mathbb{N}$. Hence we may apply **(i)** in the right-hand side union above to obtain

$$\mu(E_k) - \mu\Big(\bigcap_{j=1}^{\infty} E_j\Big) = \mu\Big(E_k \setminus \bigcap_{j=1}^{\infty} E_j\Big) = \mu\Big(\bigcup_{j=1}^{\infty} E_k \setminus E_j\Big) =$$

$$= \lim_{j\to\infty} \mu(E_k \setminus E_j) = \lim_{j\to\infty} (\mu(E_k) - \mu(E_j)),$$

where the first and the last equality justify because $E_j \subseteq E_k$ whenever $j \geq k$. Since $\mu(E_k) < \infty$, $\mu(E_k \setminus E_j)$ is a non-decreasing sequence of real numbers having $\mu(E_k)$ as an upper bound. Hence the last limit exists and becomes $\mu(E_k) - \lim_{j\to\infty} \mu(E_j)$. Substituting this equality in the above gives the desired.

□

Remark A.1.4. We proved Lemma A.1.1 in the case where we have an outer measure and restrict ourselves to μ-measurable sets. It is immediate to verify that the result also holds if we start with a measure μ_0 on some σ-algebra Σ and assume that $(E_j)_{j\in\mathbb{N}}$ is a sequence of sets in Σ. Indeed, in the proof of the lemma we used precisely the fact that any outer measure μ is a measure when restricted to the σ-algebra of μ-measurable sets.

The *Borel σ-algebra* of a topological space (X, τ) is the intersection of all σ-algebras of X which contain τ (that is, the "smallest" σ-algebra of X which contains the topology τ). The sets of the Borel σ-algebra are called *Borel sets*. An outer measure μ on a topological space (X, τ) is said to be *Borel regular* if all Borel sets are μ-measurable and for each subset $A \subseteq X$ there exists a Borel set $B \supseteq A$ such that $\mu(B) = \mu(A)$. Next we investigate the relation between Borel regular outer measures and the space topology. But first we show how to induce an outer measure to a subspace, which will be important later.

Lemma A.1.2. *Let $\mu\colon 2^X \to [0,\infty]$ be an outer measure on a set X, and assume that $Y \subseteq X$. The function $\mu_Y\colon 2^X \to [0,\infty]$ defined by*

$$\mu_Y(A) := \mu(Y \cap A)$$

for each $A \subseteq X$ is an outer measure on X, and every μ-measurable subset of X is also μ_Y-measurable. Moreover, if μ is Borel regular, Y is μ-measurable, and $\mu(Y) < \infty$, then μ_Y is also Borel regular.

Remark A.1.5. We often refer to μ_Y as the *induced outer measure* on Y. Despite being defined on X, we can clearly restrict it to the subsets of Y, when it is convenient (this happens very often). In this case, μ_Y is clearly the usual restriction of μ to 2^Y, and we will call μ_Y the *restriction* of μ to Y. We will often use the alternative notation $\mu|_Y$.

Proof. First observe that $\mu_Y(\emptyset) = \mu(Y \cap \emptyset) = \mu(\emptyset) = 0$. Now let $E \subseteq X$ and $(E_j)_{j=1}^{\infty}$ be a collection of subset of X such that $E \subseteq \bigcup_{j=1}^{\infty} E_j$. Then, since $Y \cap E \subseteq \bigcup_{j=1}^{\infty} Y \cap E_j$, we obtain

$$\mu_Y(E) = \mu(Y \cap E) \leq \sum_{j=1}^{\infty} \mu(Y \cap E_j) = \sum_{j=1}^{\infty} \mu_Y(E_j),$$

and this shows that μ_Y is indeed an outer measure. Now let E be a μ-measurable subset of X. For any $A \subseteq X$ we have

$$\begin{aligned} \mu_Y(A) = \mu(A \cap Y) = \mu((A \cap Y) \setminus E) + \mu(A \cap Y \cap E) = \\ = \mu(Y \cap (A \setminus E)) + \mu(Y \cap (A \cap E)) = \mu_Y(A \setminus E) + \mu_Y(A \cap E), \end{aligned}$$

from where we get that E is μ_Y-measurable. Observe that, until this point, is was not necessary to assume that Y is μ-measurable.

Now assume that μ is Borel regular, and that Y is a μ-measurable set with finite measure. Since we already know that any μ-measurable set is also μ_Y-measurable, it follows that every Borel sets are μ_Y-measurable. To check the remaining feature, choose a Borel set $B_1 \supseteq Y$ such that $\mu(B_1) = \mu(Y)$, and observe that since $\mu(Y) < \infty$ this leads to $\mu(B_1 \setminus Y) = 0$. We also choose a Borel set $B_2 \supseteq (B_1 \setminus Y)$ such that $\mu(B_2) = \mu(B_1 \setminus Y) = 0$. Now let $A \subseteq X$ be a subset. It is readily seen that

$$A \subseteq (A \cap Y) \cup (X \setminus B_1) \cup B_2.$$

Pick a Borel set $B_3 \supseteq A \cap Y$ such that $\mu(B_3) = \mu(A \cap Y)$. Hence $(X \setminus B_1) \cup B_2 \cup B_3$ is a Borel set which contains A, and

$$\begin{aligned} \mu_Y(A) \leq \mu_Y((X \setminus B_1) \cup B_2 \cup B_3) \leq \mu_Y(X \setminus B_1) + \mu_Y(B_2) + \mu_Y(B_3) = \\ = \mu(Y \setminus B_1) + \mu(B_2 \cap Y) + \mu(B_3 \cap Y) = \mu(B_3 \cap Y) = \\ = \mu(B_3) - \mu(B_3 \setminus Y) = \mu(A \cap Y) = \mu_Y(A). \end{aligned}$$

Here, the equality $\mu(Y \setminus B_1) = 0$ comes from the fact that $Y \setminus B_1 = \emptyset$. Also, $\mu(B_2 \cap Y) = 0$ because $\mu(B_2) = 0$. Finally, to verify equality $\mu(B_3 \setminus Y) = 0$, first observe that since $A \cap Y \subseteq B_3 \cap Y$, we have $\mu(B_3 \cap Y) \leq \mu(B_3) = \mu(A \cap Y) \leq \mu(B_3 \cap Y)$. Therefore, we get $\mu(B_3) = \mu(B_3 \cap Y)$. On the other hand, since Y is μ-measurable we have $\mu(B_3) = \mu(B_3 \cap Y) + \mu(B_3 \setminus Y)$. Since all these sets have finite, the desired statement follows.

□

Theorem A.1.1 (Carathéodory Criterion)**.** *Let (X, d) be a metric space. If μ is an outer measure on X such that*

$$\mu(A \cup B) = \mu(A) + \mu(B)$$

for any sets $A, B \subseteq X$ such that $\operatorname{dist}(A, B) > 0$*, then all Borel sets of (X, d) are μ-measurable.*

Proof. It is enough to prove that all closed sets are μ-measurable. Indeed, in this case Σ_μ is a σ-algebra containing all the open sets of (X, d), and the Borel σ-algebra of (X, d) is the intersection of all σ-algebras with this property. Hence it suffices to check that if $C \subseteq X$ is a closed subset, then

$$\mu(A) \geq \mu(A \setminus C) + \mu(A \cap C)$$

for every subset $A \subseteq X$ of finite measure. To prove that, first define, for each $j \in \mathbb{N}$, the set $C_j := \{x \in X : \operatorname{dist}(x, C) \leq 1/j\}$. We claim that

$$\operatorname{dist}(A \cap C, A \setminus C_j) \geq 1/j > 0$$

for every $j \in \mathbb{N}$. Indeed, if $z \in A \setminus C_j$, then $\operatorname{dist}(z, C) > 1/j$. In particular, for any point $y \in A \cap C$ we have $d(y, z) > 1/j$. Passing to the infimum when $z \in A \setminus C_j$ and $y \in A \cap C$ we get the desired inequality. Since $(A \cap C) \cup (A \setminus C_j) \subseteq A$, and from the hypothesis of the statement, we get

$$\mu(A) \geq \mu((A \cap C) \cup (A \setminus C_J) = \mu(A \cap C) + \mu(A \setminus C_j),$$

and hence all we have to prove is that $\mu(A \setminus C_j) \to \mu(A \setminus C)$ as $j \to \infty$. To verify that, first observe that the closedness of C gives

$$A \setminus C = \{x \in A : \operatorname{dist}(x, C) > 0\} = (A \setminus C_j) \cup \Big(\bigcup_{k=j}^{\infty} V_k\Big),$$

where the last equality holds for any $j \in \mathbb{N}$, where $V_k := \left\{x \in A : \frac{1}{k+1} < \operatorname{dist}(x, C) \leq \frac{1}{k}\right\}$ for each $k \in \mathbb{N}$. This equality, together with the subadditivity of μ and the fact that $A \setminus C_j \subseteq A \setminus C$, gives

$$\mu(A \setminus C_j) \leq \mu(A \setminus C) \leq \mu(A \setminus C_j) + \sum_{k=j}^{\infty} \mu(V_k),$$

for every $j \in \mathbb{N}$. It follows that it is enough to prove that $\sum_{k=1}^{\infty} \mu(V_k) < \infty$, because in this case standard analysis arguments give that $\sum_{k=j}^{\infty} \mu(V_k) \to 0$ as $j \to \infty$, and hence we will obtain from the above that $\mu(A \setminus C_j) \to \mu(A \setminus C)$ when $j \to \infty$. Each V_k is a subset of A, and hence has finite measure. Also, we claim that $\operatorname{dist}(V_k, V_{k+2}) > 0$ for every $k \in \mathbb{N}$. Indeed, if $x \in V_k$ and $y \in V_{k+2}$, then

$$d(x, y) \geq \operatorname{dist}(x, C) - \operatorname{dist}(y, C) > \frac{1}{k+1} - \frac{1}{k+2} = \frac{1}{(k+1)(k+2)} > 0,$$

where the first inequality comes from adapting the first inequality of Proposition 2.2.1 to arbitrary metric spaces dropping the convexity hypothesis (this is immediately done). Now we separate the odd and even values of k and use the hypothesis of the statement to obtain

$$\sum_{m=1}^{\infty} \mu(V_{2m}) = \mu\Big(\bigcup_{m=1}^{\infty} V_m\Big) \leq \mu(A) < \infty \text{ and}$$
$$\sum_{i=1}^{\infty} \mu(V_{2i-1}) = \mu\Big(\bigcup_{i=1}^{\infty} V_{2i-1}\Big) \leq \mu(A) < \infty,$$

from where we get that $\sum_{k=1}^{\infty} \mu(V_k) < \infty$, as we wanted to prove.

□

Metric topologies have a useful feature: if $C \subseteq (X, d)$ is a closed set, then C can be written as a countable intersection of open sets. Indeed, for each $j \in \mathbb{N}$ we define $C_j := \{x \in X : \operatorname{dist}(x, C) < 1/j\}$. It is easy to see that each of these sets is open, and that

$$C = \bigcap_{j=1}^{\infty} C_j.$$

In what follows, we say that a Borel regular outer measure μ on a topological space (X, τ) is *σ-finite* if there exists a countable collection of open sets $\{U_j : j \in \mathbb{N}\} \subseteq \tau$ such that

$$X = \bigcup_{j=1}^{\infty} U_j,$$

and $\mu(U_j) < \infty$ for each $j \in \mathbb{N}$. In this case, of course the sets U_j can be taken to be pairwise disjoint. Simply replace U_m by $U_m \setminus \left(\bigcup_{j=1}^{m-1} U_j \right)$ for each $m \geq 2$.

Theorem A.1.2. *Let (X, d) be a metric space. If $\mu \colon 2^X \to [0, \infty]$ is a σ-finite Borel regular outer measure on X, then we have*

$$\mu(A) = \inf\{\mu(U) : U \text{ is open and } A \subseteq U\} \tag{A.1.7}$$

for every subset $A \subseteq X$, and

$$\mu(E) = \sup\{\mu(C) : C \text{ is closed and } C \subseteq E\}, \tag{A.1.8}$$

for every μ-measurable set E.

Remark A.1.6. Of course, saying that equality (A.1.7) holds is the same as saying that for every $\varepsilon > 0$ there exists an open set $U \supseteq A$ such that $\mu(U) < \mu(A) + \varepsilon$. Analogously, equality (A.1.8) is true if and only if for any $\varepsilon > 0$ there exists a closed set $C \subseteq E$ such that $\mu(C) > \mu(E) - \varepsilon$. These viewpoints will be used without further comments in the proof below.

Proof. To prove (A.1.7) we notice that, since μ is Borel regular, it is enough to assume that A is a Borel set. We first consider the case where $\mu(X) < \infty$. Let

$$\mathcal{A} := \{A \subseteq X : A \text{ is a Borel set for which (A.1.7) holds}\}.$$

It is immediate to notice that this collection contains all open sets of X. Our task is to show that $\mathcal{A}$ is precisely the collection $\mathcal{B}_X$ of Borel sets of X. Of course, we have $\mathcal{A} \subseteq \mathcal{B}_X$, and we prove now that $\mathcal{A}$ is closed under countable unions. Let $(A_j)_{j \in \mathbb{N}}$ be a collection of sets of $\mathcal{A}$, and let $A = \bigcup_{j=1}^{\infty} A_j$. For an arbitrarily given $\varepsilon > 0$ and for each $j \in \mathbb{N}$, let U_j be an open set such that $U_j \supseteq A_j$ and

$$\mu(U_j) < \mu(A_j) + \frac{\varepsilon}{2^j}.$$

Once μ is Borel regular and each A_j is a Borel set, we get that every A_j is μ-measurable. Hence $\mu(U_j) = \mu(U_j \cap A_j) + \mu(U_j \setminus A_j) = \mu(A_j) + \mu(U_j \setminus A_j)$ for each j. Thus,

$$\mu(U_j \setminus A_j) < \frac{\varepsilon}{2^j}$$

for every $j \in \mathbb{N}$. Let $U = \bigcup_{j=1}^{\infty} U_j$. Then, U is clearly an open set which contains A, and it is also easy to see that

$$U \setminus A \subseteq \bigcup_{j=1}^{\infty} (U_j \setminus A_j).$$

Therefore, from the monotone subadditivity of μ we get that

$$\mu(U \setminus A) \leq \sum_{j=1}^{\infty} \mu(U_j \setminus A_j) < \sum_{j=1}^{\infty} \frac{\varepsilon}{2^j} = \varepsilon.$$

Now notice that A is a Borel set, because it is a countable union of Borel sets. Since μ is Borel regular, we get that A is μ-measurable, from where we have the equality $\mu(U) = \mu(U \cap A) + \mu(U \setminus A) = \mu(A) + \mu(U \setminus A)$. It follows that $\mu(U) < \mu(A) + \varepsilon$, and once U is open and $\varepsilon > 0$ is arbitrary, we get $A \in \mathcal{A}$.

We also claim that $\mathcal{A}$ is closed under countable intersections. Indeed, if $\varepsilon > 0$, A_j and U_j are as previously, then we also have the inclusion

$$\left(\bigcap_{j=1}^{\infty} U_j\right) \setminus \left(\bigcap_{j=1}^{\infty} A_j\right) \subseteq \bigcup_{j=1}^{\infty} (U_j \setminus A_j).$$

Denoting $\tilde{U} = \bigcap_{j=1}^{\infty} U_j$ and $\tilde{A} = \bigcap_{j=1}^{\infty} A_j$, we have from the above that

$$\mu\big(\tilde{U} \setminus \tilde{A}\big) < \varepsilon,$$

by the same argument used previously. Since $\tilde{A}$ is a Borel set (because it is a countable intersection of Borel sets), we get that $\tilde{A}$ is μ-regular (again, recall that we are assuming that μ is Borel regular) and hence the inequality above gives $\mu\big(\tilde{U}\big) < \mu\big(\tilde{A}\big) + \varepsilon$ (it is clear that $\tilde{A} \subseteq \tilde{U}$). Observe that this do not guarantee that $\tilde{A} \in \mathcal{A}$ yet, since $\tilde{U}$ is not necessarily open. What we do to overcome this problem is the following: start by observing that

$$\tilde{A} = \bigcap_{j=1}^{\infty} A_j \subseteq A_k \subseteq U_k$$

for each $k \in \mathbb{N}$. Therefore, we have that

$$\tilde{A} \subseteq \bigcap_{j=1}^{m} U_j$$

for each $m \in \mathbb{N}$, and the intersection in the right-hand side is an open set. Since

$$\left(\bigcap_{j=1}^{m} U_j\right) \setminus \tilde{A} \supseteq \left(\bigcap_{j=1}^{m+1} U_j\right) \setminus \tilde{A}$$

for every $m \in \mathbb{N}$, and once we are assuming that $\mu(X) < \infty$, we get from item **(ii)** of Lemma A.1.1 that

$$\lim_{m\to\infty} \mu\left(\left(\bigcap_{j=1}^{m} U_j\right) \setminus \tilde{A}\right) = \mu\left(\left(\bigcap_{j=1}^{\infty} U_j\right) \setminus \tilde{A}\right) = \mu\left(\tilde{U} \setminus \tilde{A}\right) < \varepsilon.$$

Hence we can choose a number $m_0 \in \mathbb{N}$ such that

$$\mu\left(\left(\bigcap_{j=1}^{m_0} U_j\right) \setminus \tilde{A}\right) < 2\varepsilon,$$

and this shows that $\tilde{A} \in \mathcal{A}$.

As a consequence of the fact that $\mathcal{A}$ is closed under intersections, we have that $\mathcal{A}$ contains all the closed sets of X. To check that, recall that any closed set of a metric space can be written as a countable intersection of open sets (and also remember that $\mathcal{A}$ contains all the open sets of X). Now we define

$$\mathcal{A}_c := \{A \in \mathcal{A} : X \setminus A \in \mathcal{A}\} \subseteq \mathcal{A},$$

which, in other words, is the subcollection of $\mathcal{A}$ consisting of sets in $\mathcal{A}$ having the property that their respective complements are also in $\mathcal{A}$. About this collection, we have the following:

(i) $\emptyset, X \in \mathcal{A}_c$. This is because $\emptyset, X \in \mathcal{A}$.

(ii) $\mathcal{A}_c$ is closed under complementation. Indeed, if $A \in \mathcal{A}_c$, then $X \setminus (X \setminus A) = A \in \mathcal{A}$, from where we get that $X \setminus A \in \mathcal{A}_c$.

(iii) $\mathcal{A}_c$ is closed under countable unions. In fact, let $(A_j)_{j\in\mathbb{N}}$ be a collection of sets in $\mathcal{A}_c$. Then $X \setminus A_j \in \mathcal{A}$ for each $j \in \mathbb{N}$, and hence

$$X \setminus \left(\bigcup_{j=1}^{\infty} A_j\right) = \bigcap_{j=1}^{\infty}(X \setminus A_j) \in \mathcal{A},$$

because $\mathcal{A}$ is closed under countable intersections. Thus, $\bigcup_{j=1}^{\infty} A_j \in \mathcal{A}$.

(iv) $\mathcal{A}_c$ contains all closed sets of X. To prove that, let $C \subseteq X$ be a closed set. We already know that $C \in \mathcal{A}$, because $\mathcal{A}$ contains all closed sets of X. Writing $C = \bigcap_{j=1}^{\infty} C_j$, where each C_j is an open set, we get

$$X \setminus C = X \setminus \left(\bigcap_{j=1}^{\infty} C_j\right) = \bigcup_{j=1}^{\infty}(X \setminus C_j) \in \mathcal{A},$$

where the last justifies because $\mathcal{A}$ is closed under countable intersections, and since each $X \setminus C_j$ is a closed set (thus, an element of $\mathcal{A}$).

As a consequence of **(i)**, **(ii)**, and **(iii)** we have that $\mathcal{A}_c$ is a σ-algebra. From **(ii)** and **(iv)**, we get that $\mathcal{A}_c$ contains all open sets of X, from where it follows that $\mathcal{B}_X \subseteq \mathcal{A}_c$. Summarizing, we have

$$\mathcal{B}_X \subseteq \mathcal{A}_c \subseteq \mathcal{A} \subseteq \mathcal{B}_X,$$

from where we obtain that $\mathcal{B}_X = \mathcal{A}$, as we wished. This settles the proof of (A.1.7) when $\mu(X) < \infty$.

It remains to consider the case where $\mu(X) = \infty$. We are assuming that μ is σ-finite, and hence we may write $X = \bigcup_{j=1}^{\infty} X_j$ where each X_j is an open set with finite measure (we also assume that the X_js are pairwise disjoint). Let $A \subseteq X$ be a Borel set (it is clearly still sufficient to prove the desired equality for Borel sets, because of the Borel regularity of μ). For each $j \in \mathbb{N}$, we define $\mu_j \colon 2^X \to [0, \infty]$ by

$$\mu_j(E) := \mu(E \cap X_j),$$

for every $E \subseteq X$. From Lemma A.1.2, each of these is a Borel regular outer measure on X, which also has the feature of being finite. Therefore, using the finite case that we have just proved, for any $\varepsilon > 0$ and for each $j \in \mathbb{N}$ we can select an open set U_j such that $U_j \supseteq A$ and

$$\mu_j(U_j) < \mu_j(A) + \varepsilon/2^j.$$

Summing on j we get

$$\sum_{j=1}^{\infty} \mu(U_j \cap X_j) \leq \varepsilon + \sum_{j=1}^{\infty} \mu(A \cap X_j).$$

Let $U = \bigcup_{j=1}^{\infty}(U_j \cap X_j)$. From (A.1.5) we get that the sum on the left-hand side of the inequality above equals $\mu(U)$, and that the sum in the right-hand side is $\mu(A)$, once all the involved sets are Borel sets, and hence μ-measurable. Thus, we get that $\mu(U) \leq \mu(A) + \varepsilon$. Since U is clearly open, and $U \supseteq A$ (because $A \subseteq U_j$ for every $j \in \mathbb{N}$), we obtain equality (A.1.7) also for the case where $\mu(X) = \infty$.

Now we prove (A.1.8) as a consequence of (A.1.7). Again, we start by the case where $\mu(X) < \infty$. For a μ-measurable set $E \subseteq X$ and for any $\varepsilon > 0$ we can select an open set $U \subseteq X$ such that $U \supseteq (X \setminus E)$ and

$$\mu(U) - \mu(X \setminus E) < \varepsilon. \tag{A.1.9}$$

Consequently, $X \setminus U$ is a closed set contained in E, and using the equalities $\mu(X) = \mu(U) + \mu(X \setminus U) = \mu(E) + \mu(X \setminus E)$, which are true because E and U are μ-measurable, we get

$$\mu(E) - \mu(X \setminus U) = \mu(X) - \mu(X \setminus E) - (\mu(X) - \mu(U)) = \mu(U) - \mu(X \setminus E) < \varepsilon,$$

and this settles (A.1.8) for the case where μ is finite.

Last, we prove that (A.1.8) holds also when $\mu(X) = \infty$. Again, we use the σ-finiteness of μ and write $X = \bigcup_{j=1}^{\infty} X_j$ as a disjoint union of open sets of finite measure. Also, we will work again with the (finite) outer measures μ_j defined in (A.1.9). Let $\varepsilon > 0$ be arbitrarily given. Since (A.1.8) holds for every measure μ_j, for each $j \in \mathbb{N}$ we choose a closed set $C_j \subseteq E$ such that $\mu_j(C_j) > \mu_j(E) - \varepsilon/2^j$. Summing in j, using (A.1.5) again, and considering the obvious fact that $C_j \supseteq C_j \cap X_j$ for each j, we get

$$\mu\left(\bigcup_{j=1}^{\infty} C_j\right) \geq \mu\left(\bigcup_{j=1}^{\infty}(C_j \cap X_j)\right) > \mu(E) - \varepsilon.$$

Applying item **(ii)** of Lemma A.1.1 to the sets $\bigcup_{j=1}^{m} C_j$, $m \in \mathbb{N}$, we obtain

$$\lim_{m\to\infty} \mu\left(\bigcup_{j=1}^{m} C_j\right) = \mu\left(\bigcup_{j=1}^{\infty} C_j\right) > \mu(E) - \varepsilon.$$

It is clear that each set $\bigcup_{j=1}^{m} C_j$ is a closed set which is contained in E. If $\mu(E) = \infty$, then the inequality above guarantees that these sets are a sequence of closed sets contained in E whose measures go to ∞, and hence (A.1.8) holds. If $\mu(E) < \infty$, then the inequality above yields that we may choose a number $m_0 \in \mathbb{N}$ such that

$$\mu\left(\bigcup_{j=1}^{m_0} C_j\right) > \mu(E) - 2\varepsilon,$$

and this concludes the proof, once $\varepsilon > 0$ is arbitrary.

□

Corollary A.1.1. *If, in addition to the hypothesis of Theorem A.1.2, we assume that X can be written as a union of countably many compact subsets, then we also have*

$$\mu(E) = \sup\{\mu(K) : K \text{ is compact and } K \subseteq E\} \tag{A.1.10}$$

for every μ-measurable set E such that $\mu(E) < \infty$.

Proof. Assume that $X = \bigcup_{j=1}^{\infty} K_j$, where each K_j is a compact subset of X. For each closed set $C \subseteq E$ and every $m \in \mathbb{N}$, let

$$F_m = \left(\bigcup_{j=1}^{m} K_j \right) \cap C.$$

Each of these sets is compact (because it is the intersection between a compact set and a closed set). Moreover, we have $F_m \subseteq F_{m+1}$ for each $m \in \mathbb{N}$, and

$$C = \bigcup_{m=1}^{\infty} F_m.$$

From Lemma A.1.1, we get that $\lim_{m\to\infty} \mu(F_m) = \mu(C)$. Hence for any $\varepsilon > 0$ we can choose a closed subset $C \subseteq E$ such that $\mu(E) - \mu(C) < \varepsilon/2$, and a number $m_0 \in \mathbb{N}$ with the property that $\mu(C) - \mu(F_{m_0}) < \varepsilon/2$. It follows that $\mu(E) - \mu(F_{m_0}) < \varepsilon$, from where we get that

$$\sup\{\mu(K) : K \text{ is compact and } K \subseteq E\} \geq \mu(E).$$

Since the reverse inequality is obvious, the desired follows.

□

A *Borel measure* on a topological space (X, τ) is a measure (in the classical sense, not an outer measure) defined on the Borel σ-algebra of (X, τ). If μ_0 is a Borel measure on (X, τ), then it is not difficult to see that the map $\mu\colon 2^X \to [0, +\infty)$ defined as

$$\mu(A) = \inf\{\mu_0(B) : B \text{ Borel}, \ B \supseteq A\}$$

is a Borel regular outer measure on X (which, of course, agrees with μ_0 on the Borel sets). Conversely, if μ is a Borel regular outer measure on X, then it is also easy to check that the restriction of μ to the Borel σ-algebra of X is a Borel measure.

A Borel measure μ_0 on a topological space (X, τ) is said to be *regular* (not to be confused with the concept of Borel regular **outer** measure introduced before) if for every Borel set $B \subseteq X$ and any $\varepsilon > 0$ there exist an open set $A \supseteq B$ and a closed set $F \subseteq B$ such that $\mu_0(A \setminus F) < \varepsilon$. Notice very carefully that it is immediate from this definition that (A.1.7) and (A.1.8) hold for any Borel set if we replace μ by a regular Borel measure.

Theorem A.1.3. *Every σ-finite Borel measure on a metric space (X, d) is regular.*

Proof. Let μ_0 be such a measure, and let μ be its extension to a Borel regular outer measure as described above. For any $\varepsilon > 0$, Theorem A.1.2 gives that there exist an open set $A \supseteq B$ such that $\mu(A) - \mu(B) < \varepsilon/2$ and a closed set $F \subseteq B$ such that $\mu(B) - \mu(A) < \varepsilon/2$. Since A, B, and F are Borel sets, these inequalities still hold if we replace μ by μ_0. Observing that $A \setminus F$ is the disjoint union of $A \setminus B$ and $B \setminus F$ we get

$$\mu_0(A \setminus F) = \mu_0(A \setminus B) + \mu_0(B \setminus F) < \varepsilon,$$

and this concludes the proof.

□

Remark A.1.7. In particular, if μ_0 is a finite Borel measure on a metric space, then μ_0 is regular. Sometimes it is useful to have this in mind. For example, in Section 7.1 we deal with finite Borel measures on $\mathbb{S}^{n-1}$, and they are, consequently, regular.

For us, it is also important to understand what is the Borel σ-algebra of a topological subspace. We clarify this next.

Proposition A.1.2. *Let (X, τ) be a topological space, and denote by $\mathcal{B}_X$ its Borel σ-algebra. Assume that $Y \subseteq X$ is endowed with the usual induced topology, and denote by $\mathcal{B}_Y$ the associated Borel σ-algebra. Then*

$$\mathcal{B}_Y = \{B \cap Y : B \in \mathcal{B}_X\}.$$

In particular, if $Y \in \mathcal{B}_X$, then $\mathcal{B}_Y \subseteq \mathcal{B}_X$. In this case, if μ_0 is a Borel measure on X, then its restriction to $\mathcal{B}_Y$ is a Borel measure on Y.

Proof. It is easy to verify that $\{B \cap Y : B \in \mathcal{B}_X\}$ is a σ-algebra of Y which clearly contains all open sets of Y. Then we have the inclusion

$$\mathcal{B}_Y \subseteq \{B \cap Y : B \in \mathcal{B}_X\}.$$

On the other hand, let $\mathcal{B} := \{B \in \mathcal{B}_X : B \cap Y \in \mathcal{B}_Y\}$. It is also easily seen that $\mathcal{B}$ is a σ-algebra on X which contains all open sets of X. Hence $\mathcal{B} = \mathcal{B}_X$. This gives the inclusion $\{B \cap Y : B \in \mathcal{B}_X\} \subseteq \mathcal{B}_Y$.

Assume that $Y \in \mathcal{B}_X$. If $B \in \mathcal{B}_Y$, then there exists $B_0 \in \mathcal{B}_X$ such that $B = B_0 \cap Y$. Therefore, B is the intersection of two Borel sets of X, from where we get that $B \in \mathcal{B}_X$. In this case, we can restrict a given measure $\mu_0 \colon \mathcal{B}_X \to [0, +\infty)$ to $\mathcal{B}_Y$, and of course this restriction is a measure on Y.

□

Remark A.1.8. We denote the restriction of the Borel measure $\mu \colon \mathcal{B}_X \to [0, +\infty)$ to $\mathcal{B}_Y$ as $\mu|_Y$, the same notation as we introduced for restrictions of outer measures (see Remark A.1.5).

A.2 Measurable functions and integration

A set X endowed with a σ-algebra Σ_X is often called a *measurable space*. A map $f \colon (X, \Sigma_X) \to (Y, \Sigma_Y)$ between measurable spaces is called *measurable* if $f^{-1}(E) \in \Sigma_X$ for all $E \in \Sigma_Y$. A *measure space* is a measurable space (X, Σ_X) endowed with a measure $\mu \colon \Sigma_X \to [0, +\infty]$. As an example of measurable maps, we have the continuous maps (with respect to the Borel σ-algebras). This is explained next.

Lemma A.2.1. *Let $f \colon X \to Y$ be a map between topological spaces X and Y. If f is continuous, then f is* Borel measurable*, meaning that it is measurable with respect to the Borel σ-algebras of X and Y, respectively.*

Proof. We recall that $\mathcal{B}_X$ and $\mathcal{B}_Y$ denote the Borel σ-algebras of X and Y, respectively. Let

$$f_*\mathcal{B}_X := \{B \subseteq Y : f^{-1}(B) \in \mathcal{B}_X\}.$$

This is clearly a σ-algebra of Y, known as the *push-forward σ-algebra* of $\mathcal{B}_X$ under f, and since f is continuous we have that $f_*\mathcal{B}_X$ contains all open sets of Y. Therefore, once $\mathcal{B}_Y$ is the smallest σ-algebra of Y containing all the open sets, we get the inclusion

$$\mathcal{B}_Y \subseteq f_*\mathcal{B}_X.$$

It follows that if $B \in \mathcal{B}_Y$, then we have that $B \in f_*\mathcal{B}_X$, and this gives $f^{-1}(B) \in \mathcal{B}_X$. This shows that f is measurable.

□

When it comes to functions, it is useful for us working with functions $f\colon X \to [-\infty, +\infty]$, since this is more general than working with functions $f\colon X \to \mathbb{R}$. The topology considered in $[-\infty, +\infty]$ is the following

$$\tau_{[-\infty,+\infty]} := \left\{ \begin{array}{l} \text{arbitrary unions of open intervals of } \mathbb{R} \text{ and} \\ \text{sets having the form } (-a, +\infty] \text{ and } [-\infty, a) \end{array} \right\}.$$

Using standard terminology of topology, the sets of $\tau_{[-\infty,+\infty]}$ are the *open sets* of $[-\infty, +\infty]$. The *intervals* of $[-\infty, +\infty]$ are the intervals of the $\mathbb{R}$ and the sets of the form $(a, +\infty]$, $[a, +\infty]$, $[-\infty, a)$ and $[-\infty, a]$ for $a \in \mathbb{R}$. The *open intervals* are the intervals which are open sets.

It is clear that the usual topology of $\mathbb{R}$ is the topology of $[-\infty, +\infty]$ described above induced on $\mathbb{R} \subseteq [-\infty, +\infty]$. Also, the usual relation of order of $\mathbb{R}$ is naturally extended to $[-\infty, +\infty]$ by saying that $a < +\infty$ and $-\infty > a$ for every $a \in \mathbb{R}$. We also extend the sum operation of $\mathbb{R}$ by putting

$$a + \infty = \infty + a = \infty \text{ and } a - (+\infty) = -\infty + a = -\infty,$$

and

$$\infty + \infty = \infty \text{ and } -\infty + (-\infty) = -\infty,$$

where the reader may notice that we sometimes denote $+\infty$ simply by ∞. We do not define the sum of $+\infty$ and $-\infty$. With respect to the product operation, we get the convention that $0 \cdot \infty = 0$ and $a \cdot \pm\infty = \pm\infty$, where the sign in the right-hand side depends naturally on the signs of a and ∞ in the left-hand side.

The Borel σ-algebra of $[-\infty, +\infty]$ is the Borel σ-algebra of the topology $\tau_{[-\infty,+\infty]}$ defined above, and we denote it by $\mathcal{B}_{[-\infty,+\infty]}$. Denoting by $\mathcal{B}_{\mathbb{R}}$ the Borel σ-algebra of $\mathbb{R}$ we clearly have

$$\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{B}_{[-\infty,+\infty]},$$

since any open set of $\mathbb{R}$ is also an open set of $[-\infty, +\infty]$. A function $f\colon (X, \Sigma_X) \to [-\infty, +\infty]$ is *measurable* if it is measurable with respect to $\mathcal{B}_{[-\infty,+\infty]}$. A function $f\colon (X, \Sigma_X) \to \mathbb{R}$ is *measurable* if it is measurable with respect to $\mathcal{B}_{\mathbb{R}}$ or, equivalently, if it is measurable when seen as a function onto $[-\infty, +\infty]$. We can give an easier characterization.

Lemma A.2.2. *Let (X, Σ_X) be a measurable space and $f\colon X \to [-\infty, +\infty]$. Then the following statements are equivalent:*

(a) *f is measurable,*

(b) *$f^{-1}((a, +\infty]) \in \Sigma_X$ for every $a \in \mathbb{R}$.*

(c) *$f^{-1}([-\infty, a)) \in \Sigma_X$ for each $a \in \mathbb{R}$.*

Proof. The implications **(a)⇒(b)** and **(a)⇒(c)** are immediate from the definitions. The implication **(b)⇒(c)** follows from writing

$$f^{-1}([-\infty, a)) = X \setminus f^{-1}([a, +\infty]) = X \setminus \bigcap_{j=1}^{\infty} f^{-1}\big((a - 1/n, +\infty]\big) = \bigcup_{j=1}^{\infty} \Big(X \setminus f^{-1}\big((a - 1/n, +\infty]\big)\Big),$$

and recalling that Σ_X is closed under countable unions and complementation. The converse **(c)⇒(b)** follows analogously.

To conclude the proof, it suffices to prove that **(b)** implies **(a)**. Assume that **(b)** holds and let

$$\mathcal{A} := \left\{E \in \mathcal{B}_{[-\infty,+\infty]} : f^{-1}(E) \in \Sigma_X\right\}.$$

By the hypothesis, we have that $(a, +\infty] \in \mathcal{A}$ for every $a \in \mathbb{R}$, and repeating the argument used to prove the implication **(b)⇒(c)** we also have that $[-\infty, a) \in \mathcal{A}$ for each $a \in \mathbb{R}$. Moreover, if (a, b) is any open interval of $\mathbb{R}$, then

$$f^{-1}((a,b)) = f^{-1}\big([-\infty,b) \cap (a,+\infty]\big) = f^{-1}([-\infty,b)) \cap f^{-1}((a,+\infty]) \in \Sigma_X,$$

once Σ_X is closed under countable intersections. Hence $\mathcal{A}$ contains all open intervals of $\mathbb{R}$. Therefore, $\mathcal{A}$ contains all open intervals of $[-\infty, +\infty]$. It is clear of $[-\infty, +\infty]$ that the intervals having rational endpoints form a countable basis of $\tau_{[-\infty,+\infty]}$, and hence any open set $A \subseteq [-\infty, +\infty]$ can be written as a countable union of open intervals:

$$A = \bigcup_{j=1}^{\infty} A_j.$$

We already know that $f^{-1}(A_j) \in \Sigma_X$ for each $j \in \mathbb{N}$. Thus,

$$f^{-1}(A) = \bigcup_{j=1}^{\infty} f^{-1}(A_j) \in \Sigma_X,$$

which shows that $\mathcal{A}$ contains all open sets of $[-\infty, +\infty]$. Since $\mathcal{A}$ is clearly a σ-algebra of $[-\infty, +\infty]$, we get that $\mathcal{A}$ contains $\mathcal{B}_{[-\infty,+\infty]}$. This concludes the proof.

□

Remark A.2.1. Notice that **(b)** holds if and only if $f^{-1}([-\infty, a]) \in \Sigma_X$ for every $a \in \mathbb{R}$ (and similarly for **(c)**). Hence the intervals in **(b)** and **(c)** can be replaced by closed ones. In particular, from item **(b)** of Proposition 1.1.3, we get that any lower semi-continuous function $f : \mathbb{R}^n \to [-\infty, +\infty]$ is Borel measurable.

Corollary A.2.1. *Let $f, g : (X, \Sigma_X) \to [-\infty, +\infty]$ be measurable functions, and let $c \in \mathbb{R}$. Then the functions cf, $f + g$, and fg are measurable.*

Proof. If $c = 0$, then cf is the zero function, which is clearly measurable. If $c > 0$, then we simply notice that $(cf)^{-1}((a, +\infty]) = f^{-1}((a/c, +\infty])$. For $c < 0$, we observe that $(cf)^{-1}((a, +\infty]) = f^{-1}([-\infty, a/c))$. Hence cf is measurable.

To prove that $f + g$ is measurable, notice that if $a \in \mathbb{R}$ and $q \in \mathbb{Q}$, then the set

$$E_q := f^{-1}((q, +\infty]) \cap g^{-1}((a - q, +\infty])$$

is measurable. We claim that

$$(f + g)^{-1}((a, +\infty]) = \bigcup_{q \in \mathbb{Q}} E_q.$$

Indeed, if $(f + g)(x) > a$, then $(f + g)(x) = a + \varepsilon$ for some $\varepsilon > 0$. Choosing $q \in \mathbb{Q}$ such that $q < f(x) < q + \varepsilon$ we get immediately that $x \in E_q$. Since the right-hand side of the above is a countable union of measurable sets we obtain that $(f + g)$ is measurable.

It remains to prove that fg is measurable. The easier path to do that is to prove that the square of a measurable function is measurable. If $a < 0$, then $(f^2)^{-1}((a, +\infty]) = X$, and if $a \geq 0$, then

$$(f^2)^{-1}((a, +\infty]) = f^{-1}([-\infty, -\sqrt{a})) \cup f^{-1}((\sqrt{a}, +\infty]).$$

Thus, f^2 is measurable whenever f is measurable. Now we write

$$fg = \frac{(f+g)^2 - (f-g)^2}{4},$$

and this shows that fg is measurable.

□

Proposition A.2.1. *Let* $\left(f_j \colon (X, \Sigma_X) \to [-\infty, +\infty]\right)_{j\in\mathbb{N}}$ *be a sequence of measurable functions. The functions* $\inf_{j\in\mathbb{N}} f_j$, $\sup_{j\in\mathbb{N}} f_j \colon (X, \Sigma_X) \to [-\infty, +\infty]$ *defined respectively as*

$$\inf_{j\in\mathbb{N}} f_j(x) := \inf\{f_j(x) : j \in \mathbb{N}\} \text{ and}$$
$$\sup_{j\in\mathbb{N}} f_j(x) = \sup\{f_j(x) : j \in \mathbb{N}\},$$

for each $x \in X$*, are measurable.*

Proof. Let $a \in \mathbb{R}$. For any given $x \in X$ we have that $\sup_{j\in\mathbb{N}} f_j(x) > a$ if and only if there exists $k \in \mathbb{N}$ such that $f_k(x) > a$. Consequently, we have

$$\left(\sup_{j\in\mathbb{N}} f_j\right)^{-1}((a, +\infty]) = \bigcup_{j=1}^{\infty} f_j^{-1}((a, +\infty]).$$

Every set in the union of the right-hand side is in Σ_X, once each function f_j is measurable. Hence the union itself is a set of Σ_X, and this shows that $\sup_{j\in\mathbb{N}} f_j$ is measurable.

To prove that the function $\inf_{j\in\mathbb{N}} f_j$ is also measurable, we simply write

$$\inf_{j\in\mathbb{N}} f_j = -\sup_{j\in\mathbb{N}}(-f_j).$$

Since it is clear that $-f$ is measurable whenever f is measurable the proof is complete.

□

Corollary A.2.2. *Under the same hypothesis of the last proposition, the functions*

$$\liminf_{j\to\infty} f_j,\ \limsup_{j\to\infty} f_j \colon X \to [-\infty, +\infty]$$

defined as

$$\operatorname{liminf}_{j\to\infty} f_j(x) = \operatorname{liminf}_{j\to\infty}(f_j(x)) \text{ and}$$
$$\operatorname{limsup}_{j\to\infty} f_j(x) = \operatorname{limsup}_{j\to\infty}(f_j(x)),$$

for each $x \in X$*, are measurable.*

Proof. Just recall that

$$\liminf_{j\to\infty} f_j = \sup_{k\in\mathbb{N}}\left(\inf_{j\geq k} f_j\right) \text{ and}$$
$$\limsup_{j\to\infty} f_j = \inf_{k\in\mathbb{N}}\left(\sup_{j\geq k} f_j\right).$$

Now the result follows immediately from the last proposition.

□

A function $s\colon X\to\mathbb{R}$ is said to be *simple* if its range $\mathrm{im}(s)$ is a finite set. Notice carefully that here we are considering real-valued functions. If s is such a function and $\mathrm{im}(s)=\{a_1,\ldots,a_m\}$ then writing $s^{-1}(a_j)=E_j$ for each $1\leq j\leq m$, we have the *canonical representation*

$$s=\sum_{j=1}^{m} a_j 1_{E_j},$$

where we recall the reader that $1_{E_j}\colon X\to\{0,1\}$ denotes the *characteristic function* of E_j. Notice carefully that the sets E_j are mutually disjoint. If a simple function is given in this representation, then we assume that $\mathrm{im}(s)=\{a_1,\ldots,a_m\}$ and $s^{-1}(a_j)=E_j$ for each $1\leq j\leq m$.

It is not difficult to see that a simple function $s\colon (X,\Sigma_X)\to\mathbb{R}$ is measurable if and only if $s^{-1}(a)\in\Sigma_X$ for every $a\in\mathrm{im}(s)$ or, equivalently, for any $a\in\mathbb{R}$ (we have that $s^{-1}(a)=\emptyset$ whenever a is not in the range of f). Indeed, if s is measurable, then since $\{a\}=\bigcap_{k=1}^{\infty}(a-1/k,a+1/k)$ is a Borel set for any $a\in\mathbb{R}$ we get that $s^{-1}(a)\in\Sigma_X$. On the other hand, if $\mathrm{im}(s)=\{a_1,\ldots,a_m\}$ and $s^{-1}(a_j)\in\Sigma_X$ for each $1\leq j\leq m$, then noticing that the pre-image of *any* subset of $\mathbb{R}$ under s is a finite union of some of the $s^{-1}(a_j)$s we get that s is measurable.

Next we prove that measurable (non-negative) functions can be approximated by measurable simple functions.

Theorem A.2.1. *Let (X,Σ_X) be a measurable function and $f\colon X\to[0,+\infty]$ a function. Then f is measurable if and only if there exists a sequence $\left(s_m\colon X\to[0,+\infty)\right)_{m\in\mathbb{N}}$ of simple functions with the following properties:*

i. [pointwise monotonicity] *$s_m(x)\leq s_{m+1}(x)$ for any $m\in\mathbb{N}$ and every $x\in X$.*

ii. [dominance] *$s_m(x)\leq f(x)$ for every $m\in\mathbb{N}$ and $x\in X$.*

iii. [pointwise convergence] *$\lim_{m\to\infty} s_m(x)=f(x)$ for each $x\in X$.*

Remark A.2.2. The limit in **iii** is taken with respect to the topology of $[-\infty,+\infty]$. In particular, $\lim_{m\to\infty} s_m(x)=+\infty$ if for any $\beta>0$ there exists $m_0\in\mathbb{N}$ such that $s_m(x)>\beta$ whenever $m>m_0$. Also, notice carefully that **ii** is a consequence of **i** and **iii**, and then our statement is somewhat redundant, but we opted by writing the property in **ii** to emphasize that this is a *dominated* convergence.

Proof. First, if such a sequence of simple functions exists for a given function $f\colon X\to[-\infty,+\infty]$, then it is clear from **i** and **iii** that

$$f(x)=\sup_{m\in\mathbb{N}} s_m(x),$$

for each $x\in X$, and hence f is measurable (see Proposition A.2.1).

Now we prove the converse. Let $f\colon X\to[-\infty,+\infty]$ be measurable. For each $m\in\mathbb{N}$, define $\phi_m\colon[0,\infty]\to[0,+\infty)$ by

$$\phi_m(t)=\begin{cases} 2^{-m}k, & \text{if } k=0,1,\ldots 2^m m-1 \text{ and } 2^{-m}k\leq t<2^{-m}(k+1),\\ m, & \text{if } t\geq m\end{cases}.$$

In Figure A.2.1 we illustrate the function ϕ_1. For a better comprehension of this construction, notice that for each $m \in \mathbb{N}$ we are "partitioning" $[0, +\infty]$ into the intervals of the form $[2^{-m}k, 2^{-m}(k+1))$ for $k = 0, \ldots, 2^{-m}m - 1$, together with $[m, +\infty]$. The function ϕ_m is constant in each of these intervals, and since they are finitely many, we get that ϕ_m is a simple function.

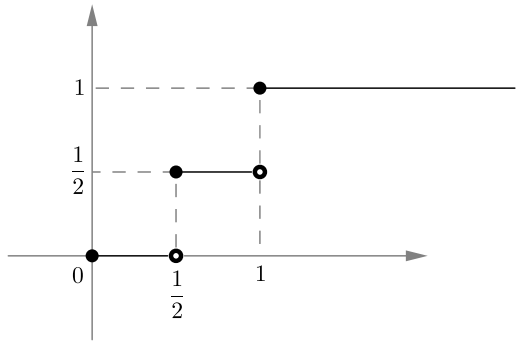


Fig. A.2.1 The graph of ϕ_1

Once the pre-image of any point of the range of ϕ_m is an interval, we get that ϕ_m is measurable. It is also easy to see that

$$\phi_m(t) \leq \phi_{m+1}(t) \leq t \tag{A.2.1}$$

for each $t \in [0, +\infty]$ and $m \in \mathbb{N}$. Now fix an arbitrary $t \in [0, +\infty)$, and for each natural number $m \geq t$ choose $k(m,t) \in \{0, 1, \ldots, 2^{-m}m - 1\}$ such that

$$k(m,t)2^{-m} \leq t < (k(m,t) + 1)2^{-m}.$$

To verify that $k(m,t)$ indeed exists, simply let $k(m,t) = \sup\{k \in \mathbb{Z} : k2^{-m} \leq t\}$. Observe that the inequalities above lead to

$$|t - k(m,t)2^{-m}| = t - k(m,t)2^{-m} \leq 2^{-m},$$

from where we get that $k(m,t)2^{-m} \to t$ as $m \to \infty$. Morover, from the definition of ϕ_m we have that $\phi_m(t) = k(m,t)2^{-m}$ for each $m \in \mathbb{N}$ and $t \leq m$. Therefore,

$$\lim_{m\to\infty} \phi_m(t) = t \tag{A.2.2}$$

for each $t \in [0, +\infty]$. This also holds for $t = \infty$, since from the definition we have that $\phi_m(\infty) = m$ for each $m \in \mathbb{N}$. Finally, for each $m \in \mathbb{N}$, define $s_m \colon X \to [0, +\infty)$ by

$$s_m = \phi_m \circ f.$$

Each function s_m is clearly simple (since its range is contained in the range of ϕ_m) and measurable (because the composition of measurable functions is measurable). From (A.2.1) and (A.2.2) we get that the sequence $(s_m)_{m\in\mathbb{N}}$ has the properties **i** and **iii** (and hence also **ii**). This concludes the proof. □

Remark A.2.3. We may derive an analogous result for a given non-positive function $f \colon (X, \Sigma_X) \to [-\infty, 0]$ by applying the theorem for $-f$.

At this point, we emphasize that simple functions are **always** finite, meaning that they are onto $[0, +\infty)$. The previous theorem states that even functions onto $[0, +\infty]$ can be approximated by these.

Let (X, Σ_X, μ) be a measure space and let $s\colon (X, \Sigma_X, \mu) \to [0, +\infty)$ be a simple non-negative measurable function whose canonical representation is $s = \sum_{j=1}^{m} a_j 1_{E_j}$. For any measurable set $E \subseteq X$, the *integral of s over E* is the number

$$\int_E s \, d\mu := \sum_{j=1}^{m} a_j \mu(E \cap E_j).$$

Notice that this can be infinite (in the case where there exists a number $j \in \{1, \ldots, m\}$ such that $\mu(E \cap E_j) = \infty$ and $a_j \neq 0$). By restricting ourselves to non-negative functions we avoid the sum $\infty + (-\infty)$, and hence our definition is coherent.

We extend this concept for an arbitrary measurable non-negative function f defined on (X, Σ_X, μ) as follows. Consider the collection

$$\mathcal{S}_f := \left\{ s\colon X \to [0, +\infty) : s \text{ is simple, measurable and } s \leq f \right\},$$

where $s \leq f$ means that $s(x) \leq f(x)$ for every $x \in X$, of course. This collection is non-empty since the zero function is clearly an element. For any set $E \in \Sigma_X$, the *integral of f over E* is defined as

$$\int_E f \, d\mu := \sup \left\{ \int_E s \, d\mu : s \in \mathcal{S}_f \right\}.$$

It is clear that this may be infinite. We move forward to state and prove some important properties of the integral of non-negative measurable functions.

Proposition A.2.2. *Let (X, Σ_X, μ) be a measure space.*

(i) *For any non-negative measurable function $f\colon X \to [0, +\infty]$ and every $E \in \Sigma_X$ we have*

$$\int_E f \, d\mu = \int_X f 1_E \, d\mu.$$

(ii) *If $f, g\colon X \to [0, +\infty]$ are non-negative measurable functions such that $f \leq g$ in a measurable set $E \subseteq X$, then*

$$\int_E f \, d\mu \leq \int_E g \, d\mu.$$

(iii) *For every $E, F \in \Sigma_X$ such that $E \subseteq F$ and each non-negative measurable function $f\colon X \to [0, +\infty]$ it holds that*

$$\int_E f \, d\mu \leq \int_F f \, d\mu.$$

(iv) *If $f\colon X \to [0, +\infty]$ is non-negative and measurable, then*

$$\int_E cf \, d\mu = c \int_E f \, d\mu$$

for any measurable set $E \subseteq X$ and every number $c \geq 0$.

Proof. For **(i)**, we first observe that $(f1_E)^{-1}(a) = f^{-1}(a) \cap E$ for any function $f\colon X \to [0,+\infty]$ and $a \in [0,+\infty]$. Hence if $s\colon X \to [0,+\infty)$ is a simple measurable function whose canonical representation is $s = \sum_{j=1}^m a_j 1_{E_j}$, then

$$\int_X s1_E \, d\mu = \sum_{j=1}^m a_j \mu((s1_E)^{-1}(a_j)) = \sum_{j=1}^m a_j \mu(E_j \cap E) = \int_E s \, d\mu \tag{A.2.3}$$

for every $E \in \Sigma_X$. If $s \in \mathcal{S}_f$, then we clearly have that $s1_E \in \mathcal{S}_{f1_E}$. Thus,

$$\int_E f \, d\mu = \sup\left\{\int_X s1_E \, d\mu : s \in \mathcal{S}_f\right\} \leq \sup\left\{\int_X p \, d\mu : p \in \mathcal{S}_{f1_E}\right\} = \int_X f1_E \, d\mu.$$

On the other hand, if $p \in \mathcal{S}_{f1_E}$, then because $1_E \cdot 1_E = 1_E$ we get that $p1_E \in \mathcal{S}_f$ and

$$\int_E f \, d\mu \geq \sup\left\{\int_E p1_E \, d\mu : p \in \mathcal{S}_{f1_E}\right\} = \sup\left\{\int_X p1_E \, d\mu : p \in \mathcal{S}_{f1_E}\right\},$$

where in the equality we used (A.2.3) again. Any simple non-negative function p such that $p \leq f1_E$ is such that $p(x) = 0$ whenever $x \notin E$, from where $p = p1_E$. That is, for any $p \in \mathcal{S}_{f1_E}$ we have $p = p1_E$. Then

$$\sup\left\{\int_X p1_E \, d\mu : p \in \mathcal{S}_{f1_E}\right\} = \sup\left\{\int_X p \, d\mu : p \in \mathcal{S}_{f1_E}\right\} = \int_X f1_E \, d\mu,$$

and this concludes the proof of **(i)**.

For **(ii)**, simply notice that if $f \leq g$ in E and $s \in \mathcal{S}_f$, then $s1_E \in \mathcal{S}_g$. Therefore, since $s = s1_E$ in E we get

$$\int_E s \, d\mu = \int_E s1_E \, d\mu \leq \int_E g \, d\mu.$$

Passing to the supremum as $s \in \mathcal{S}_f$ in the left-hand side we get the desired.

We obtain **(iii)** as an immediate consequence of **(i)** and **(ii)**. If $E \subseteq F$, then $f1_E \leq f1_F$ and hence

$$\int_E f \, d\mu = \int_X f1_E \, d\mu \leq \int_X f1_F \, d\mu = \int_F f \, d\mu.$$

Last, observe that **(iv)** is straightforward if f and g are simple functions, and from that the general case follows.

□

Going further, we now state and prove some important results concerning the convergence of integrals for pointwise converging functions. Here an observation is due. The advantage of working with $[-\infty,+\infty]$ (instead of with $\mathbb{R}$) is that *any* monotone sequence in this set converges. Indeed, it is well-known that any bounded monotone sequence of real numbers converges, and in $[-\infty,+\infty]$ any unbounded monotone convergent sequence converges to $+\infty$ (if it is non-decreasing) of $-\infty$ (if it is non-increasing).

Theorem A.2.2 (Monotone convergence)**.** *Let (X, Σ_X, μ) be a measure space and let $\left(f_m\colon X \to [0,+\infty]\right)_{m\in\mathbb{N}}$ be a sequence of measurable functions with the property that*

$$f_m(x) \leq f_{m+1}(x)$$

for each $x \in X$. Let $f\colon X \to [0, +\infty]$ be defined as

$$f(x) = \lim_{m\to\infty} f_m(x)$$

for each $x \in X$. Then f is measurable and

$$\int_E f\, d\mu = \lim_{m\to\infty} \int_E f_m\, d\mu$$

for any measurable set $E \subseteq X$.

Proof. We start considering the case where $E = X$, since the general case will follow easily from that. The pointwise monotonicity of (f_m) guarantees that

$$f(x) = \lim_{m\to\infty} f_m(x) = \sup_{m\in\mathbb{N}} f_m(x)$$

for each $x \in X$, and hence it follows from Proposition A.2.1 that f is measurable. It is also clear from the above that $f_m(x) \leq f(x)$ for each $x \in X$. Thus, from item **(ii)** of Proposition A.2.2 we get

$$\int_X f_m\, d\mu \leq \int_X f_{m+1}\, d\mu \leq \int_X f\, d\mu$$

for each $m \in \mathbb{N}$. The first inequality gives that

$$L := \lim_{m\to\infty} \int_X f_m\, d\mu$$

exists in $[0, +\infty]$, while the second inequality guarantees that

$$L \leq \int_X f\, d\mu.$$

Then all we have to prove is the reverse inequality. Fix a number $0 < \alpha < 1$ and let $s\colon X \to [0, +\infty)$ be a simple measurable function such that $s \leq f$. For each $m \in \mathbb{N}$ let

$$E_m := \{x \in X : f_m(x) \geq \alpha s(x)\}.$$

We have that $E_m \subseteq E_{m+1}$ for each $m \in \mathbb{N}$, and it is also immediate that $X = \bigcup_{m\in\mathbb{N}} E_m$. Moreover, observe that each set E_m is measurable, since we may write $E_m = (f_m - \alpha s)^{-1}([0, +\infty))$, and the function $f_m - \alpha s$ is clearly measurable. Using item **(ii)** of Proposition A.2.2 again we get

$$\int_{E_m} f_m\, d\mu \geq \int_{E_m} \alpha s\, d\mu = \alpha \int_{E_m} s\, d\mu$$

for each $m \in \mathbb{N}$, where we also used item **(iv)** of the mentioned proposition, of course. Consequently,

$$L \geq \int_X f_m\, d\mu \geq \int_{E_m} f_m\, d\mu \geq \alpha \int_{E_m} s\, d\mu,$$

which can be re-written as

$$\int_{E_m} s\, d\mu \leq \frac{L}{\alpha}.$$

Now let $s = \sum_{j=1}^{k} a_j 1_{A_j}$ be the canonical representation of s. Using item **(i)** of Lemma A.1.1 (and Remark A.1.4) we get

$$\mu(A_j) = \mu(A_j \cap X) = \mu\left(A_j \cap \left(\bigcup_{m=1}^{\infty} E_m\right)\right) = \mu\left(\bigcup_{m=1}^{\infty} A_j \cap E_m\right) = \lim_{m\to\infty} \mu(A_j \cap E_m)$$

for each $1 \leq j \leq k$. Thus,

$$\int_X s\, d\mu = \sum_{j=1}^{k} a_j \mu(A_j) = \sum_{j=1}^{k} a_j \lim_{m\to\infty} \mu(A_j \cap E_m) = \lim_{m\to\infty} \sum_{j=1}^{k} a_j \mu(A_j \cap E_m) =$$
$$= \lim_{m\to\infty} \int_{E_m} s\, d\mu \leq \frac{L}{\alpha}.$$

Once s is an arbitrary function of $\mathcal{S}_f$, we arrive at

$$\int_X f\, d\mu = \sup\left\{\int_X s\, d\mu : s \in \mathcal{S}_f\right\} \leq \frac{L}{\alpha},$$

and because $0 < \alpha < 1$ was arbitrarily fixed we get the desired remaining inequality.

Now we prove the case where $E \subseteq X$ is an arbitrary measurable set. In this case, observe that $f_m 1_E \leq f_{m+1} 1_E$ for each $m \in \mathbb{N}$, and

$$\lim_{m\to\infty} f_m(x) 1_E(x) = f(x) 1_E(x).$$

Hence we apply the case $E = X$ for the functions $f_m 1_E$, $f 1_E$ to get

$$\int_E f\, d\mu = \int_X f 1_E\, d\mu = \lim_{m\to\infty} \int_X f_m 1_E\, d\mu = \lim_{m\to\infty} \int_E f_m\, d\mu,$$

and the proof is complete.

□

Corollary A.2.3. *Let $f, g\colon (X, \Sigma_X) \to [0, +\infty]$ be measurable functions. For any measurable set $E \subseteq X$ we have the equality*

$$\int_E f + g\, d\mu = \int_E f\, d\mu + \int_E g\, d\mu.$$

Proof. Our strategy is to consider first simple functions and then extend for the general case by approximations. Let $s, p\colon X \to [0, +\infty)$ be simple measurable functions whose canonical representations are

$$s = \sum_{j=1}^{m} a_j 1_{E_j} \quad \text{and} \quad p = \sum_{i=1}^{k} b_i 1_{A_i},$$

respectively. It is clear that we have the equality

$$s + p = \sum_{j=1}^{m} \sum_{i=1}^{k} (a_j + b_i) 1_{E_j \cap A_i},$$

and consequently the integral of $s + p$ over a measurable set $E \subseteq X$ is given as

$$\int_E s + p \, d\mu = \sum_{j=1}^{m} \sum_{i=1}^{k} (a_j + b_i)\mu(E \cap E_j \cap A_i).$$

Once the sets E_j are mutually disjoint and their union is X, we may write

$$\sum_{j=1}^{m} \mu(E_j \cap A_i \cap E) = \mu(A_i \cap E)$$

for each $1 \leq i \leq k$. Of course the same holds if we interchange the roles of the E_js with that of the A_is. As a consequence, re-organizing the terms in the expansion of the integral of $s + p$ above we get

$$\int_E s + p \, d\mu = \sum_{j=1}^{m} \sum_{i=1}^{k} (a_j + b_i)\mu(E \cap E_j \cap A_i) =$$

$$= \sum_{j=1}^{m} a_j \sum_{i=1}^{i} \mu(E_j \cap A_i \cap E) + \sum_{i=1}^{k} \sum_{j=1}^{m} \mu(E_j \cap A_i \cap E) =$$

$$= \sum_{j=1}^{m} a_j \mu(E_j \cap E) + \sum_{i=1}^{k} b_i \mu(A_i \cap E) = \int_E s \, d\mu + \int_E p \, d\mu,$$

and this settles the case where the involved functions are simple. Now let $f, g \colon X \to [0, +\infty]$ be measurable functions. We let $(s_m)_{m\in\mathbb{N}}$ and $(p_m)_{m\in\mathbb{N}}$ be non-negative simple measurable functions converging to f and g, respectively, as in Theorem A.2.1. It is immediate that the sequence $(s_m + p_m)$ converges to $f + g$ in the same sense. Thus, from Theorem A.2.2, we get

$$\int_E f + g \, d\mu = \lim_{m\to\infty} \int_E s_m + p_m \, d\mu = \lim_{m\to\infty} \int_E s_m \, d\mu + \lim_{m\to\infty} \int_E p_m \, d\mu = \int_E f \, d\mu + \int_E g \, d\mu,$$

and this concludes the proof.

□

Theorem A.2.3 (Fatou's lemma). *Let (X, Σ_X, μ) be a measure space and $(f_n)_{n\in\mathbb{N}}$ be a sequence of non-negative measurable functions. Then*

$$\int_X \liminf_{n\to\infty} f_n \, d\mu \leq \liminf_{n\to\infty} \int_X f_n \, d\mu,$$

where $\liminf_{n\to\infty} f_n$ denotes the function which associates each $x \in X$ to the inferior limit of the sequence $(f_n(x))_{n\in\mathbb{N}}$, as in Corollary A.2.2.

Proof. For simplicity, denote

$$f := \liminf_{m\to\infty} f_m.$$

For each $k \in \mathbb{N}$ let $g_k := \inf_{j\geq k} f_j$. Then g_k is pointwise monotone and

$$\lim_{k\to\infty} g_k(x) = \sup_{k\in\mathbb{N}} g_k(x) = \liminf_{m\to\infty} f_m(x) = f(x)$$

for each $x \in X$. From Theorem A.2.2 it follows that

$$\int_X f(x)\,d\mu = \lim_{k\to\infty} \int_X g_k\,d\mu.$$

Now observe that for any fixed $k \in \mathbb{N}$ we have the inequality

$$g_k(x) = \inf_{j\geq k} f_j(x) \leq f_i(x)$$

for each $i \geq k$ and every $x \in X$. Consequently, we have the inequality

$$\int_X g_k\,d\mu \leq \int_X f_i\,d\mu$$

whenever $i \geq k$. Hence

$$\int_X g_k\,d\mu \leq \inf_{i\geq k} \int_X f_i\,d\mu.$$

Finally, this inequality leads to

$$\int_X f\,d\mu = \lim_{k\to\infty} \int_X g_k\,d\mu \leq \lim_{m\to\infty} \left(\inf_{i\geq k} \int_X f_i\,d\mu\right) = \sup_{k\in\mathbb{N}} \left(\inf_{i\geq k} \int_X f_i\,d\mu\right) = \\ = \liminf_{m\to\infty} \int_X f_m\,d\mu,$$

and the proof is complete.

□

Remark A.2.4. It is not difficult to see that Fatou's lemma holds if we integrate over any measurable set $E \subseteq X$. This comes from the fact that

$$\liminf_{m\to\infty}(f_m 1_E) = \left(\liminf_{m\to\infty} f_m\right) 1_E,$$

which is obtained trivially by investigating the behavior of the functions on both sides for points of E and points of $X \setminus E$. Hence

$$\int_E \liminf_{m\to\infty} f_m\,d\mu = \int_X \left(\liminf_{m\to\infty} f_m\right) 1_E\,d\mu = \int_X \liminf_{m\to\infty}(f_m 1_E)\,d\mu \leq \\ \leq \liminf_{m\to\infty} \int_X f_m 1_E\,d\mu = \liminf_{m\to\infty} \int_E f_m\,d\mu.$$

Thus far, we only have defined the integral of non-negative functions. Now we move forward to the general case. For the sake of convenience, we return to work with functions which takes values on the real line. Any such function $f\colon X \to \mathbb{R}$ can be written as

$$f(x) = f^+(x) - f^-(x),$$

where

$$f^+(x) = \max\{f(x), 0\} \text{ and} \\ f^-(x) = \max\{-f(x), 0\}.$$

The functions $f^+, f^-\colon X \to [0, +\infty]$ are called the *positive part* and *negative part* of f, respectively. It is easy to see that if f is measurable, then f^+ and f^- are also measurable. We also have immediately

the equality

$$|f| = f^+ + f^-,$$

and since these functions are non-negative, in the case where f is measurable (and hence also f^+, f^-, and $|f|$ are measurable) their respective integrals are already defined. Moreover, from Corollary A.2.3 we get

$$\int_X |f|\, d\mu = \int_X f^+\, d\mu + \int_X f^-\, d\mu,$$

and of course the same holds if we replace X by any other measurable set $E \subseteq X$. A measurable function $f\colon (X, \Sigma_X, \mu) \to \mathbb{R}$ is called *integrable* if

$$\int_X |f|\, d\mu < +\infty.$$

In this case, it is common to say that f is an L^1 *function*, and it is also common to write $f \in L^1(\mu)$. If f is integrable, then we obviously have

$$\int_E f^+\, d\mu < +\infty \quad \text{and} \quad \int_E f^-\, d\mu < +\infty$$

for any measurable set $E \subseteq X$. Thus, we define the *integral of f over E* to be the number

$$\int_E f\, d\mu := \int_E f^+\, d\mu - \int_E f^-\, d\mu.$$

Observe that if f is not integrable, then this number may not be well-defined, since we may have a sum $\infty - \infty$ in the right-hand side of the above. This justifies our definition of what is an "integrable" function. Notice also that this clearly coincides with the previous definition in the case where f is non-negative. Next we investigate which properties the integral still has when one replaces measurable non-negative functions by integrable functions.

Proposition A.2.3. *Let $f, g\colon (X, \Sigma_X, \mu) \to \mathbb{R}$ be an integrable function. We have the following.*

(a) *For any measurable set $E \subseteq X$,*

$$\int_E f\, d\mu = \int_X f 1_E\, d\mu.$$

(b) *If $c \in \mathbb{R}$, then $f + cg$ is integrable and*

$$\int_E f + cg\, d\mu = \int_E f\, d\mu + c \int_E g\, d\mu,$$

for every $E \in \Sigma_X$.

(c) *If $f \leq g$ in a measurable subset $E \subseteq X$, then*

$$\int_E f\, d\mu \leq \int_E g\, d\mu.$$

(d) *For any set $E \in \Sigma_X$ it holds that*

$$\left|\int_E f\,d\mu\right| \leq \int_E |f|\,d\mu.$$

Proof. Item **(a)** comes straightforwardly from the equality

$$f1_E = f^+1_E - f^-1_E.$$

For **(b)**, we observe that $|f + cg| \leq |f| + |c||g|$. Since these functions are non-negative, we use the properties of the integral of non-negative functions to obtain

$$\int_E |f + cg|\,d\mu \leq \int_E |f|\,d\mu + |c| \int_E |g|\,d\mu < +\infty,$$

from where we get that $f + cg$ is integrable. To prove the equality, notice that the case for $c = 0$ is immediate. If $c > 0$, then $(cg)^+ = cg^+$ and $(cg)^- = cg^-$, and

$$f^+ - f^- + cg^+ - cg^- = f^+ - f^- + (cg)^+ - (cg)^- = f + cg = (f + cg)^+ - (f + cg)^-.$$

Consequently, we have $f^+ + cg^+ + (f + cg)^- = f^- + cg^- + (f + cg)^+$, and each of these functions is non-negative. Hence using the properties of the integral of non-negative measurable functions we get

$$\int_E f^+\,d\mu + c\int_E g\,d\mu + \int_E (f + cg)^-\,d\mu = \int_E f^-\,d\mu + c\int_E g^-\,d\mu + \int_E (f + cg)^+\,d\mu,$$

and re-organizing the terms yields the desired. The case for $c < 0$ is similar: we only need to notice that $(cf)^+ = (-c)f^-$ and $(cf)^- = (-c)f^+$. The details are left to the reader.

Now we prove **(c)**. We have $f^+ - f^- = f \leq g = g^+ - g^-$, and this yields the inequality $f^+ + g^- \leq g^+ + f^-$. The advantage of this last inequality is that all functions involved are non-negative. Thus, using again the properties of the integral of these functions we get

$$\int_E f^+\,d\mu + \int_E g^-\,d\mu = \int_E f^+ + g^-\,d\mu \leq \int_E g^+ + f^-\,d\mu = \int_E g^+\,d\mu + \int_E f^-\,d\mu,$$

which can be re-written as

$$\int_E f^+\,d\mu - \int_E f^-\,d\mu \leq \int_E g^+\,d\mu - \int_E g^-\,d\mu,$$

and this is the desired inequality.

Finally, to prove **(d)** we simply write

$$\begin{aligned}\left|\int_E f\,d\mu\right| = \left|\int_E f^+\,d\mu - \int_E f^-\,d\mu\right| \leq \left|\int_E f^+\,d\mu\right| + \left|\int_E f^-\,d\mu\right| = \\ = \int_E f^+\,d\mu + \int_E f^-\,d\mu = \int_E |f|\,d\mu,\end{aligned}$$

and the proof is complete.

□

Theorem A.2.4 (Dominated convergence)**.** *Let (X, Σ_X, μ) be a measure space, and let $g\colon X \to [0, +\infty)$ be an integrable function. Let $\left(f_m\colon X \to \mathbb{R}\right)_{m\in\mathbb{N}}$ be a sequence of integrable functions satisfying:*

(i) $|f_m(x)| \leq g(x)$ *for any* $m \in \mathbb{N}$ *and every* $x \in X$ *(that is, the sequence* (f_m) *is* dominated *by* g*), and*

(ii) *for each* $x \in X$ *the sequence* $(f_m(x))_{m\in\mathbb{N}}$ *converges in* $\mathbb{R}$.

Then the function $f\colon X \to \mathbb{R}$ *defined as*

$$f(x) := \lim_{m\to\infty} f_m(x)$$

for each $x \in X$ *is integrable and we have the equality*

$$\int_X f\, d\mu = \lim_{m\to\infty} \int_X f_m\, d\mu.$$

Proof. First observe that for each $x \in X$ we have

$$f(x) = \lim_{m\to\infty} f_m(x) = \liminf_{m\to\infty} f_m(x),$$

because $(f_m(x))_{m\in\mathbb{N}}$ is a sequence that converges, and hence its limit coincides with its inferior limit. Thus, it follows from Corollary A.2.2 that f is measurable. By Fatou's lemma (Theorem A.2.3), item **(c)** of Proposition A.2.3, and from the fact that $|f_m| \leq g$ for every m, we get

$$\int_X |f|\, d\mu = \int_X \liminf_{m\to\infty} |f_m|\, d\mu \leq \liminf_{m\to\infty} \int_X |f_m|\, d\mu \leq \int_X g\, d\mu < +\infty,$$

where the last inequality follows from the integrability of g. Hence f is integrable. Moving forward, since $f_m(x) \to g(x)$ for each $x \in X$ we get the inequality $|f| \leq g$. Therefore, we obtain $|f_m - f| \leq |f_m| + |f| \leq 2g$, and this leads to

$$2g - |f_m - f| \geq 0$$

for each $m \in \mathbb{N}$. On the other hand, once $f_m \to f$ pointwise, we have that

$$\liminf_{m\to\infty} \big(2g - |f_m - f|\big) = \lim_{m\to\infty} \big(2g - |f_m - f|\big) = 2g.$$

Thus, applying Fatou's lemma to the sequence $2g - |f_m - f|$ we get

$$\begin{aligned}\int_X 2g\, d\mu &= \int_X \liminf_{m\to\infty} \big(2g - |f_m - f|\big)\, d\mu \leq \liminf_{m\to\infty} \int_X 2g - |f_m - f|\, d\mu = \\ &= \int_X 2g\, d\mu + \liminf_{m\to\infty} \int_X -|f_m - f|\, d\mu = \int_X 2g\, d\mu - \limsup_{m\to\infty} \int_X |f_m - f|\, d\mu.\end{aligned}$$

Consequently,

$$\limsup_{m\to\infty} \int_X |f_m - f|\, d\mu \leq 0,$$

and we notice that we actually have equality, since the left-hand side is a superior limit of non-negative numbers. Moreover, using again the fact that the integral of $|f_m - f|$ is always non-negative we obtain

$$0 \leq \liminf_{m\to\infty} \int_X |f_m - f|\, d\mu \leq \limsup_{m\to\infty} \int_X |f_m - f|\, d\mu = 0,$$

and this yields

$$0 = \lim_{m\to\infty} \int_X |f_m - f|\, d\mu.$$

Finally,

$$\lim_{m\to\infty} \left| \int_X f_m\, d\mu - \int_X f\, d\mu \right| \leq \lim_{m\to\infty} \int_X |f_m - f|\, d\mu = 0,$$

and this concludes the proof.

□

At this point, we present a useful equivalence to the coincidence of finite Borel measures on metric spaces.

Proposition A.2.4. *Let (X, d) be a metric space, and let μ and ν be finite Borel measures on X. Then there are equivalent:*

(i)

$$\int_X f\, d\mu = \int_X f\, d\nu,$$

for every bounded continuous function $f\colon X \to \mathbb{R}$.

(ii)

$$\int_X f\, d\mu = \int_X f\, d\nu$$

for every bounded positive continuous function $f\colon X \to \mathbb{R}$.

(iii) $\mu = \nu$.

Proof. The implications **(iii)⇒(i)**, **(iii)⇒(ii)**, and **(i)⇒(ii)** are trivial. We prove that **(i)⇒(iii)**. Let $F \subseteq X$ be any closed (non-empty) set, and let $d(\cdot, F)\colon X \to \mathbb{R}$ be the distance function to F, defined as

$$d(x, F) := \inf \big\{ d(y, x) : y \in F \big\}.$$

Consider the function $h\colon \mathbb{R} \to [0, 1]$ defined as $h(t) = 0$ if $t \geq 1$, $h(t) = 1$ if $t \leq 0$, and $h(t) = 1 - t$ if $0 \leq t \leq 1$. Notice that h is clearly continuous, and for each $\delta > 0$ define $g_\delta\colon X \to [0, 1]$ as

$$g_\delta(x) := h\left(\frac{1}{\delta} d(x, F)\right).$$

We have that g_δ is continuous, since it is a composition of continuous functions. If **(i)** holds, then

$$\int_X g_\delta\, d\mu = \int_X g_\delta\, d\nu$$

for every $\delta > 0$. We write

$$X = \{d(x, F) = 0\} \cup \{0 < d(x, F) < \delta\} \cup \{d(x, F) \geq \delta\},$$

and since F is closed, we have that $\{d(x, F) = 0\} = F$. Thus,

$$\int_X g_\delta \, d\mu = \int_F g_\delta \, d\mu + \int_{\{0<d(x,F)<\delta\}} g_\delta \, d\mu + \int_{\{d(x,F)\geq\delta\}} g_\delta \, d\mu =$$
$$= \int_F 1 \, d\mu + \int_{\{0<d(x,F)<\delta\}} 1 - \frac{1}{\delta} d(x,F) \, d\mu + \int_{\{d(x,F)\geq\delta\}} 0 \, d\mu =$$
$$= \mu(F) + \int_X \left(1 - \frac{1}{\delta} d(x,F)\right) 1_{\{0<d(x,F)<\delta\}} \, d\mu$$

for every $\delta > 0$. The functions

$$d_\delta(x) := \left(1 - \frac{1}{\delta} d(x,F)\right) 1_{\{0<d(x,F)<\delta\}}$$

clearly converge pointwise to 0 as $\delta \to 0$. In fact, $d_\delta(x) = 0$ for $x \in F$ and $\delta > 0$, and if $x \notin F$, then $d(x,F) > \delta$ for δ small enough. Hence for each $x \notin F$ we have that $d_\delta(x) = 0$ for δ small enough. Once all these functions are bounded by 1, from the dominated convergence theorem (Theorem A.2.4) we get that

$$\lim_{\delta \to 0^+} \int_X g_\delta \, d\mu = \mu(F).$$

Repeating this argument for the measure ν, we obtain that $\mu(F) = \nu(F)$ for every closed set $F \subseteq X$. Since μ and ν are finite, we have from Remark A.1.7 that they are regular. Hence (A.1.8) holds both measures, and since they coincide for closed sets we get that they are the same measure.

It remains to prove that **(ii)⇒(i)**. We first consider the case where $f\colon X \to \mathbb{R}$ is a bounded continuous function such that $f \geq 0$. Recall that the set $F := \{f = 0\}$ is closed. For each $\delta > 0$, define $h_\delta\colon \mathbb{R} \to [0,\delta]$ letting $h_\delta(t) = 0$ if $t \geq 1$, $h_\delta(t) = \delta$ if $t \leq 0$, and $h_\delta(t) = \delta(1-t)$ if $0 \leq t \leq 1$. Define $f_\delta\colon X \to [0,\delta]$ as

$$f_\delta(x) = h_\delta\left(\frac{1}{\delta} d(x,F)\right).$$

It is clear that f_δ is continuous. Once $f_\delta \geq 0$ in X and $f_\delta(x) = \delta > 0$ for any $x \in F$, it follows that $f + f_\delta$ is positive. From the hypothesis **(ii)** we get that

$$\int_X f + f_\delta \, d\mu = \int_X f + f_\delta \, d\nu$$

for any $\delta > 0$. Now notice that, because $0 \leq f_\delta \leq \delta$, we have the estimates

$$0 \leq \int_X f_\delta \, d\mu \leq \delta\mu(X).$$

Therefore,

$$\int_X f_\delta \, d\mu \to 0$$

as $\delta \to 0$. Of course, the same holds if we replace μ by ν, and it follows that

$$\int_X f \, d\mu = \int_X d\nu$$

for any non-negative bounded continuous function.

Finally, let $f\colon X \to \mathbb{R}$ be any bounded continuous function. Writing $f = f^+ - f^-$, we have that f^+ and f^- are non-negative, bounded, and continuous. Therefore,

$$\int_X f\,d\mu = \int_X f^+\,d\mu - \int_X f^-\,d\mu = \int_X f^+\,d\nu - \int_X f^-\,d\nu = \int_X f\,d\nu,$$

and this concludes the proof.

□

Let (X, Σ_X) and (Y, Σ_Y) be measure spaces, and let $\mu\colon \Sigma_X \to [0,\infty]$ be a measure on X. If $f\colon X \to Y$ is a measurable map, then the *push-forward measure* of μ is the measure on Y given by

$$f_*\mu(E) := \mu(f^{-1}(E))$$

for each $E \in \Sigma_Y$. This defines a map $f_*\mu\colon \Sigma_Y \to [0,\infty]$ which is clearly a measure. Indeed, for any countable disjoint collection $(E_j)_{j\in\mathbb{N}}$ of sets in Σ_Y we have that $(f^{-1}(E_j))_{j\in\mathbb{N}}$ is a countable disjoint collection of sets in Σ_X, and hence

$$f_*\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \mu\left(f^{-1}\left(\bigcup_{j=1}^{\infty} E_j\right)\right) = \mu\left(\bigcup_{j=1}^{\infty} f^{-1}(E_j)\right) = \sum_{j=1}^{\infty} \mu(f^{-1}(E_j)) = \sum_{j=1}^{\infty} f_*\mu(E_j).$$

This concept is very important for us. Next we characterize the integration with respect to a push-forward measure. This leads to an abstract version of the change of variables formula.

Proposition A.2.5. *If $g\colon (Y, \Sigma_Y) \to \mathbb{R}$ is integrable with respect to $f_*\mu$, then*

$$\int_Y g\,d(f_*\mu) = \int_X g \circ f\,d\mu. \tag{A.2.4}$$

Proof. We start by considering the case where $s\colon Y \to [0, +\infty)$ is a simple measurable function. Let $s = \sum_{j=1}^m a_j 1_{E_j}$ be the canonical representation of s. On the one hand, we have

$$\int_Y s\,d(f_*\mu) = \sum_{j=1}^m a_j f_*\mu(E_j) = \sum_{j=1}^m a_j \mu(f^{-1}(E_j)).$$

On the other hand, it is easy to see that $s \circ f\colon X \to [0, +\infty)$ is a simple function which can be written as $s = \sum_{j=1}^m a_j 1_{f^{-1}(E_j)}$, because $s \circ f(x) = a_j$ if and only if $x \in f^{-1}(E_j)$. Thus,

$$\int_X s \circ f\,d\mu = \sum_{j=1}^m a_j \mu(f^{-1}(E_j)),$$

and hence we get that

$$\int_Y s\,d(f_*\mu) = \int_X s \circ f\,d\mu.$$

Now let $g\colon Y \to \mathbb{R}$ be an integrable function. Let $(s_m)_{m\in\mathbb{N}}$ be a sequence of simple measurable functions converging to g^+ as in Theorem A.2.2. Then

$$\int_Y s_m\,d(f_*\mu) = \int_X s_m \circ f\,d\mu$$

for each $m \in \mathbb{N}$, and since $s_m \circ f$ clearly converges to $g^+ \circ f$ in the sense of the mentioned theorem, letting $m \to \infty$ on both sides of the equality above we get

$$\int_Y g^+ \, d(f_*\mu) = \int_X g^+ \circ f \, d\mu.$$

It is clear that the same argument can be applied for g^-. Thus,

$$\int_Y g \, d(f_*\mu) = \int_Y g^+ \, d(f_*\mu) - \int_Y g^- \, d(f_*\mu) = \int_X g^+ \circ f \, d\mu - \int_X g^- \circ f \, d\mu = \\ = \int_X (g \circ f)^+ \, d\mu + \int_X (g \circ f)^- \, d\mu = \int_X g \circ f \, d\mu,$$

where we used the straightforward equalities $g^+ \circ f = (g \circ f)^+$ and $g^- \circ f = (g \circ f)^-$.

□

Remark A.2.5. In the same conditions of the last proposition, we have

$$\int_E g \, d(f_*\mu) = \int_{f^{-1}(E)} g \circ f \, d\mu$$

for any measurable set $E \subseteq Y$. In fact, it is easy to see that $g1_E \circ f = (g \circ f)1_{f^{-1}(E)}$. Therefore,

$$\int_E g \, d(f_*\mu) = \int_Y g1_E \, d(f_*\mu) = \int_X g1_E \circ f \, d\mu = \int_X (g \circ f)1_{f^{-1}(E)} \, d\mu = \int_{f^{-1}(E)} g \circ f \, d\mu.$$

In a measure space (X, Σ_X, μ), we say that a property holds *μ-almost everywhere* (or simply *almost everywhere*, when there is no possibility of confusion regarding which measure is being considered) if it holds for a measurable set $E \subseteq X$ such that $\mu(X \setminus E) = 0$. Notice that, in particular, two integrable functions which coincide almost everywhere have the same integral.

Sometimes we deal with functions which are not defined in the entire regarded space, but only up to a set of zero measure (for example, the Gauss map of a nonsmooth convex body K is defined only $\mathcal{H}^{n-1}$-almost everywhere in ∂K; see Corollary 2.4.2). For this reason, we discuss this idea a little further. A measurable function $f\colon (X, \Sigma_X, \mu) \to [-\infty, \infty]$ is said to be *defined almost everywhere* if it is only defined over a measurable set $E \subseteq X$ such that $\mu(X \setminus E) = 0$. In this case, we will **always** replace f by the extension obtained setting $f(x) = 0$ whenever $x \in X \setminus E$. By doing that, we obviously have

$$\int_X f \, d\mu = \int_E f \, d\mu$$

and all properties of integration naturally extend to this context, as well as the definition of integrability, of course.

This viewpoint can also be adopted for measurable maps. If (X, Σ_X, μ) is a measure space and (Y, Σ_Y) is a measurable space, then a map $f\colon Z \to Y$ is said to be *defined almost everywhere in X* if $Z \in \Sigma_X$ and $\mu(X \setminus Z) = 0$. In this case, we can still define the push-forward measure $f_*\mu$ in the same way as before: for each $E \in \Sigma_Y$, we let $f_*\mu(E) := \mu(f^{-1}(E))$. To verify that this still yields a measure in Y, we just have to observe that the collection $\Sigma_Z := \{E \in \Sigma_X : E \subseteq Z\}$ is a σ-algebra of Z and that the restriction of μ to Σ_Z is a measure (hence $f_*\mu$ is σ-additive). If $g\colon Y \to \mathbb{R}$ is an integrable function, then $g \circ f\colon Z \to \mathbb{R}$ is a function which is defined almost everywhere in X. In view of what was established above for the integration of functions defined almost everywhere, we get that the formula (A.2.4) still holds in the case where f is defined almost everywhere in X.

We finish this section with an observation. The integral of a function $f\colon X \to \mathbb{R}$ clearly depends on the measure considered on X, but it also depends on the considered σ-algebra of X. In particular, even if we consider the *same* measure but replacing the original σ-algebra by a *smaller* σ-algebra the integral may change. This happens because the collection $\mathcal{S}_f$ depends on the considered σ-algebra. It is important for us to notice that this difference does not happen in the particular case of σ-finite Borel regular outer measures. We do it next.

Proposition A.2.6. *Let X be a topological space endowed with a σ-finite Borel regular outer measure $\mu\colon 2^X \to [0,+\infty]$, and let Σ be the σ-algebra of μ-measurable sets. Let $f\colon X \to \mathbb{R}$ be a function which is measurable with respect to $\mathcal{B}_X$ (and hence also with respect to Σ). Then f is integrable with respect to (X,Σ,μ) if and only if it is integrable with respect to $(X,\mathcal{B}_X,\mu)$. In this case, we have the equality*

$$\int_{X,\mathcal{B}_X} f\,d\mu = \int_{X,\Sigma} f\,d\mu,$$

where $\mathcal{B}_X$ and Σ stands for the respective σ-algebras with respect to which the integral is computed.

Proof. Let $g\colon X \to [0,+\infty)$ be a non-negative function which is $\mathcal{B}_X$-measurable (of course, this terminology means that g is measurable with respect to $\mathcal{B}_X$). We introduce the following collections:

$$\mathcal{S}_{g,\mathcal{B}_X} := \left\{ s\colon X \to [0,+\infty) : s \text{ is simple, } \mathcal{B}_X\text{−measurable and such that } s \le g \right\}$$

and

$$\mathcal{S}_{g,\Sigma} := \left\{ p\colon X \to [0,+\infty) : p \text{ is simple, } \Sigma\text{−measurable and such that } p \le g \right\}.$$

Then, we have

$$\int_{X,\mathcal{B}_X} g\,d\mu = \sup\left\{ \int_{X,\mathcal{B}_X} s\,d\mu : s \in \mathcal{S}_{g,\mathcal{B}_X} \right\} \quad \text{and}$$
$$\int_{X,\Sigma} g\,d\mu = \sup\left\{ \int_{X,\Sigma} p\,d\mu : p \in \mathcal{S}_{g,\Sigma} \right\}.$$

Once it is clear that every function which is $\mathcal{B}_X$-measurable is also Σ-measurable, we get that $\mathcal{S}_{g,\mathcal{B}_X} \subseteq \mathcal{S}_{g,\Sigma}$. This leads immediately to

$$\int_{X,\mathcal{B}_X} g\,d\mu \le \int_{X,\Sigma} g\,d\mu.$$

To prove the reverse inequality, let $p \in \mathcal{S}_{g,\Sigma}$ have the canonical representation

$$p = \sum_{j=1}^{m} a_j 1_{E_j}.$$

Put $a := \max\{a_j : 1 \le j \le m\}$, and let $\varepsilon > 0$ be arbitrarily given. Since μ is a σ-finite Borel regular outer measure, (A.1.8) gives that for any $1 \le j \le m$ there exists a closed set $C_j \subseteq E_j$ such that

$$\mu(C_j) > \mu(E_j) - \frac{\varepsilon}{a2^j}.$$

Notice that the sets $(C_j)_{j=1}^m$ are mutually disjoint because the sets $(E_j)_{j=1}^m$ are mutually disjoint. Hence we may define a function $s\colon X \to [0, +\infty)$ by

$$s := \sum_{j=1}^{m} a_j 1_{C_j}.$$

Then s is clearly $\mathcal{B}_X$-measurable (because every closed set is a Borel set), and we have

$$\int_{X,\mathcal{B}_X} s\,d\mu = \sum_{j=1}^{m} a_j \mu(C_j) > \sum_{j=1}^{m} a_j \mu(E_j) - \sum_{j=1}^{m} a_j \frac{\varepsilon}{a 2^j} \geq \int_{X,\Sigma} p\,d\mu - \sum_{j=1}^{m} \frac{\varepsilon}{2^j} > \\ > \int_{X,\Sigma} p\,d\mu - \varepsilon.$$

That is, for any function $p \in \mathcal{S}_{g,\Sigma}$ and any $\varepsilon > 0$ there exists a function $s \in \mathcal{S}_{g,\mathcal{B}_X}$ with the property that

$$\int_{X,\mathcal{B}_X} s\,d\mu > \int_{X,\Sigma} p\,d\mu - \varepsilon.$$

It follows that

$$\int_{X,\mathcal{B}_X} g\,d\mu = \sup\left\{\int_{X,\mathcal{B}_X} s\,d\mu : s \in \mathcal{S}_{g,\mathcal{B}_X}\right\} \geq \sup\left\{\int_{X,\Sigma} p\,d\mu : p \in \mathcal{S}_{g,\Sigma}\right\} = \int_{X,\Sigma} g\,d\mu,$$

and we have the desired equality for the case of non-negative functions. This equality also gives immediately that integrability with respect to $\mathcal{B}_X$ is equivalent integrability with respect to Σ.

Now the general case reduces to applying this particular case to the positive and negative parts. If $f\colon X \to \mathbb{R}$ is a $\mathcal{B}_X$-measurable function, then f^+ and f^- are non-negative $\mathcal{B}_X$-measurable functions, and hence

$$\int_{X,\mathcal{B}_X} f\,d\mu = \int_{X,\mathcal{B}_X} f^+\,d\mu - \int_{X,\mathcal{B}_X} f^-\,d\mu = \int_{X,\Sigma} f^+\,d\mu - \int_{X,\Sigma} f^-\,d\mu = \int_{X,\Sigma} f\,d\mu.$$

This equality makes sense whenever f is $\mathcal{B}_X$-integrable of Σ-integrable. Hence the proof is complete. □

A.3 Lebesgue measures

In this section we study the classical *Lebesgue measures*. Together with the *Hausdorff measures*, which will be introduced later, these are the main tools that we use to measure volumes and areas in convex geometry.

It is convenient to summarize some definitions and results which we introduced in Section A.1. Recall that an *outer measure* on a set X is a function $\mu\colon 2^X \to [0, +\infty]$ such that $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} \mu(E_j)$$

for any sequence $(E_j)_{j\in\mathbb{N}}$ of subsets of X. A *σ-algebra* of a set X is a collection $\Sigma \subseteq 2^X$ such that

(i) $\emptyset, X \in \Sigma$,

(ii) if $E \in \Sigma$, then $X \setminus E \in \Sigma$, and

(iii) for any sequence $(E_j)_{j\in\mathbb{N}}$ we have that $\bigcup_{j=1}^{\infty} E_j \in \Sigma$.

A function $\mu_0 \colon \Sigma \to [0,\infty]$ defined on a σ-algebra Σ of a set X is called a *measure* on X if $\mu(\emptyset) = 0$ and

$$\mu_0\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu_0(E_j)$$

for any *pairwise disjoint* sequence $(E_j)_{j\in\mathbb{N}}$ of sets of Σ (that is, $E_j \cap E_i = \emptyset$ whenever $i \neq j$; sometimes we call it simply *disjoint*). That is, an outer measure is defined for all subsets of X, whereas a measure is defined only on a σ-algebra of X.

Let $\mu \colon 2^X \to [0,\infty]$ be an outer measure. A subset $E \subseteq X$ is called *μ-measurable* if

$$\mu(A) = \mu(A \cap E) + \mu(A \setminus E)$$

for any $A \subseteq 2^X$. The collection of μ-measurable sets is a σ-algebra (Σ_μ, say), and the restriction of μ to Σ_μ is a measure on X.

If X is endowed with a topology τ, then the *Borel σ-algebra* of X is defined to be the intersection of all σ-algebras of X which contain τ (once the intersection of σ-algebras is clearly a σ-algebra, it is the smallest σ-algebra containing the topology). We often denote the Borel σ-algebra of a topological space X by $\mathcal{B}_X$, and the sets of $\mathcal{B}_X$ are called the *Borel sets* of X.

A *Borel regular outer measure* is an outer measure $\mu \colon 2^X \to [0,\infty]$ on a topological space X for which the following properties hold:

(i) $\mathcal{B}_X \subseteq \Sigma_\mu$, that is, all Borel sets are μ-measurable; and

(ii) for each $A \subseteq X$ there exists a set $B \in \mathcal{B}_X$ such that $A \subseteq B$ and $\mu(A) = \mu(B)$.

A *Borel measure* on a topological space X is a measure defined on the Borel σ-algebra of X. Notice that if X is a topological space, then the restriction of a Borel regular outer measure $\mu \colon 2^X \to [0,\infty]$ to $\mathcal{B}_X$ is a Borel measure on X. On the other hand, a Borel measure $\mu_0 \colon \mathcal{B}_X \to [0,\infty]$ is said to be *regular* if for any $B \in \mathcal{B}_X$ and every $\varepsilon > 0$ there exist an open set $A \supseteq B$ and a closed set $F \subseteq B$ such that $\mu(A \setminus F) < \varepsilon$. Observe very carefully that we have two distinct notions appearing here: "Borel regular **outer** measures" and "regular Borel measures". These definitions should not be confused.

The *Lebesgue outer measure* on $\mathbb{R}^n$ is an outer measure defined as follows. For $1 \leq k < n$, an *open k-cell* is a set of the form $C = (a_1, b_1) \times (a_2, b_2) \times \ldots \times (a_k, b_k)$ where $a_j, b_j \in \mathbb{R}$ and $b_j > a_j$, for each $j = 1, \ldots, k$. Analogously, we define a *closed k-cell* by considering closed intervals. If neither all intervals are open or all intervals are closed, then we simply have a *k-cell*. It is immediate to see that the interior (closure) of any k-cell is obtained simply by replacing the intervals by their respective interiors (closures). The *k-dimensional volume* of a k-cell (open, closed, or none) as above is defined as

$$\mathrm{vol}_k(C) := \prod_{j=1}^{k} (b_k - a_k).$$

Notice that, in particular, we have $\mathrm{vol}_k(C) = \mathrm{vol}_k(\mathrm{int}(C)) = \mathrm{vol}_k(\mathrm{cl}(C))$ for any k-cell C. We denote the collection of all open k-cells of $\mathbb{R}^n$ by $\mathcal{C}_k$. The (*outer*) *Lebesgue measure* $\mathcal{L}^n$ in $\mathbb{R}^n$ is defined as

$$\mathcal{L}^n(A) = \inf\left\{ \sum_{j=1}^{\infty} \mathrm{vol}_n(C_j) : \begin{array}{c} (C_j)_{j\in\mathbb{N}} \text{ is a collection of open} \\ n\text{−cells whose union contains } A \end{array} \right\}$$

for every subset $A \subseteq \mathbb{R}^n$. Observe that if A is compact, then the infimum above can be taken only on finite collections of open n-cells which cover A (because every open covering of a compact set admits a finite subcovering). In particular, if A is compact, then $\mathcal{L}^n(A) < \infty$. Notice also that we have $\mathcal{L}^n(\emptyset) = 0$ straightforwardly from the definition. To show that $\mathcal{L}^n$ is indeed an outer measure, we have to prove that

$$\mathcal{L}^n\left(\bigcup_{j=1}^{\infty} E_j \right) \leq \sum_{j=1}^{\infty} \mathcal{L}^n(E_j) \tag{A.3.1}$$

for any sequence $(E_j)_{j\in\mathbb{N}}$ of subsets of $\mathbb{R}^n$. The inequality is trivial if the right-hand side of the above is ∞. Hence we may assume that $\mathcal{L}^n(E_j) < \infty$ for each $j \in \mathbb{N}$, and that $\sum_{j=1}^{\infty} \mathcal{L}^n(E_j) < \infty$. For each $j \in \mathbb{N}$, choose a covering $(C_k^j)_{k\in\mathbb{N}}$ of E_j by open n-cells such that

$$\sum_{k=1}^{\infty} \mathrm{vol}_n(C_k^j) < \mathcal{L}^n(E_j) + \frac{\varepsilon}{2^j}.$$

Then $\bigcup_{j,k=1}^{\infty} C_k^j$ is a (countable) covering of $\bigcup_{j=1}^{\infty} E_j$ by open n-cells, and hence

$$\mathcal{L}^n\left(\bigcup_{j=1}^{\infty} E_j \right) \leq \sum_{j,k=1}^{\infty} \mathrm{vol}_n(C_k^j) = \sum_{j=1}^{\infty} \left(\sum_{k=1}^{\infty} \mathrm{vol}_n(C_j^k) \right) \leq \sum_{j=1}^{\infty} \left(\mathcal{L}^n(E_j) + \frac{\varepsilon}{2^j} \right) = \varepsilon + \sum_{j=1}^{\infty} \mathcal{L}^n(E_j).$$

Once $\varepsilon > 0$ is arbitrary, the desired inequality follows. Therefore, we have indeed that $\mathcal{L}^n$ is an outer measure on $\mathbb{R}^n$.

At this point, it is useful to introduce equivalent definitions of $\mathcal{L}^n$ by means of closed n-cells and general n-cells.

Lemma A.3.1. *For $A \subseteq \mathbb{R}^n$ let*

$$\bar{\mathcal{L}}^n(A) = \inf\left\{ \sum_{j=1}^{\infty} \mathrm{vol}_n(C_j) : \begin{array}{c} (C_j)_{j\in\mathbb{N}} \text{ is a collection of closed} \\ n\text{−cells whose union contains } A \end{array} \right\},$$

and

$$\tilde{\mathcal{L}}^n(A) = \inf\left\{ \sum_{j=1}^{\infty} \mathrm{vol}_n(C_j) : \begin{array}{c} (C_j)_{j\in\mathbb{N}} \text{ is a collection of } n\text{−cells} \\ \text{whose union contains } A \end{array} \right\}.$$

Then $\mathcal{L}^n(A) = \bar{\mathcal{L}}^n(A) = \tilde{\mathcal{L}}^n(A)$ for every $A \subseteq \mathbb{R}^n$.

Proof. We first prove that $\bar{\mathcal{L}}^n(A) = \mathcal{L}^n(A)$. For a given $\varepsilon > 0$, let $(C_j)_{j\in\mathbb{N}}$ be a covering of A by open n-cells such that

$$\sum_{j=1}^{\infty} \mathrm{vol}_n(C_j) < \mathcal{L}^n(A) + \varepsilon.$$

Then $(\mathrm{cl}(C_j))_{j\in\mathbb{N}}$ is a covering of A by closed n-cells, and since $\mathrm{vol}_n(\mathrm{cl}(C_j)) = \mathrm{vol}_n(C_j)$ we get

$$\sum_{j=1}^{n} \mathrm{vol}_n(\mathrm{cl}(C_j)) < \mathcal{L}^n(A) + \varepsilon,$$

from where it follows the inequality $\bar{\mathcal{L}}^n(A) \leq \mathcal{L}^n(A) + \varepsilon$. Once $\varepsilon > 0$ is arbitrary, we obtain that $\bar{\mathcal{L}}^n(A) \leq \mathcal{L}^n(A)$.

Now let $\varepsilon > 0$ be arbitrarily given, and let $(C_j)_{j\in\mathbb{N}}$ be a covering of A by closed n-cells, with $C_j = [a_1^j, b_1^j] \times \ldots \times [a_n^j, b_n^j]$, and such that

$$\sum_{j=1}^{\infty} \mathrm{vol}_n(C_j) < \bar{\mathcal{L}}^n(A) + \varepsilon.$$

It is readily seen that for each $j \in \mathbb{N}$ we may let $\delta(j) > 0$ be such that the open n-cell

$$C_j^{\delta(j)} := (a_1^j - \delta(j), b_1^j + \delta(j)) \times \ldots \times (a_n^j - \delta(j), b_n^j + \delta(j))$$

has the property that

$$\mathrm{vol}_n\big(C_j^{\delta(j)}\big) < \mathrm{vol}_n(C_j) + \frac{\varepsilon}{2^j}.$$

Then $(C_j^{\delta(j)})_{j\in\mathbb{N}}$ is a covering of A by open n-cells, and we have

$$\sum_{j=1}^{\infty} \mathrm{vol}_n\big(C_j^{\delta(j)}\big) \leq \sum_{j=1}^{\infty} \mathrm{vol}_n(C_j) + \varepsilon < \bar{\mathcal{L}}^n(A) + 2\varepsilon,$$

from where we get that $\mathcal{L}^n(A) \leq \bar{\mathcal{L}}^n(A) + 2\varepsilon$. Again, since $\varepsilon > 0$ is arbitrary we have the inequality $\mathcal{L}^n(A) \leq \bar{\mathcal{L}}^n(A)$. Hence we have equality $\mathcal{L}^n(A) = \bar{\mathcal{L}}^n(A)$. The proof of the other inequality is completely analogous.

□

An important property of the Lebesgue outer measure is that it is invariant under translations. Although this is quite immediate from the definition, we state and prove it now.

Lemma A.3.2 (Translation-invariance). *The Lebesgue outer measure is translation-invariant, that is, for any subset $A \subseteq \mathbb{R}^n$ and every $x \in \mathbb{R}^n$ we have $\mathcal{L}^n(x + A) = \mathcal{L}^n(A)$.*

Proof. If $C = (a_1, b_1) \times \ldots \times (a_n, b_n)$ is an open n-cell and $x = (x_1, \ldots, x_n)$ is a point of $\mathbb{R}^n$, then

$$x + C = (a_1 + x_1, b_1 + x_1) \times \ldots \times (a_n + x_n, b_n + x_n),$$

from where it follows that $\mathrm{vol}_n(x + C) = \mathrm{vol}_n(C)$. Moreover, if $(C_j)_{j\in\mathbb{N}}$ is a covering of A by open n-cells, we obviously have that $(x + C_j)_{j\in\mathbb{N}}$ is a covering of $x + A$ by open n-cells, and

$$\sum_{j=1}^{\infty} \mathrm{vol}_n(x + C_j) = \sum_{j=1}^{\infty} \mathrm{vol}_n(C_j).$$

This gives $\mathcal{L}^n(x + A) \leq \mathcal{L}^n(A)$. Similariy, if $(D_j)_{j\in\mathbb{N}}$ is a covering of $x + A$ by open n-cells, then $(D_j - x)_{j\in\mathbb{N}}$ is a covering of A by open n-cells, and

$$\sum_{j=1}^{\infty} \mathrm{vol}_n(D_j - x) = \sum_{j=1}^{\infty} \mathrm{vol}_n(D_j).$$

Consequently, we have the inequality $\mathcal{L}^n(A) \leq \mathcal{L}^n(x + A)$, and the proof is complete.

□

Next we prove that $\mathrm{vol}_n(\cdot)$ and $\mathcal{L}^n(\cdot)$ coincide for n-cells (this justifies why we adopt the notation $\mathrm{vol}_n(\cdot)$ for the Lebesgue measure throughout the entire book).

Theorem A.3.1. *If C is an n-cell of $\mathbb{R}^n$, then* $\mathrm{vol}_n(C) = \mathcal{L}^n(C)$.

Proof. We start with the case where C is closed. First, once C is a covering of itself by n-cells, we get the inequality $\mathcal{L}^n(C) \leq \mathrm{vol}_n(C)$.

For the reverse inequality we follow an argument which is attributed to von Neumann in [2090]. Of course, a closed 1-cell is a compact interval, and the one-dimensional volume of a compact interval $I = [a, b]$ is the usual length $\mathrm{vol}_1(I) = b - a$. We have

$$\mathrm{vol}_1(I) - 1 \leq \mathrm{card}(I \cap \mathbb{Z}) \leq \mathrm{vol}_1(I) + 1. \tag{A.3.2}$$

In fact, if $m_I = \min(I\cap\mathbb{Z})$ and $M_I = \max(I\cap\mathbb{Z})$, then it is clear that $a \leq m_I \leq a+1$ and $b-1 \leq M_I \leq b$. Since $\mathrm{card}(I \cap \mathbb{Z}) = M_I - m_I + 1$ these inequalities yield (A.3.2).

Let N be any natural number. Replacing I by $NI = [Na, Nb]$ in (A.3.2) we get

$$N\mathrm{vol}_1(I) - 1 \leq \mathrm{card}(NI \cap \mathbb{Z}) \leq N\mathrm{vol}_1(I) + 1,$$

while dividing this by N and observing that $\mathrm{card}(NI \cap \mathbb{Z}) = \mathrm{card}\left(I \cap \frac{1}{N}\mathbb{Z}\right)$, because $x \in NI \cap \mathbb{Z}$ if and only if $x/N \in I \cap \frac{1}{N}\mathbb{Z}$, lead to

$$\mathrm{vol}_1(I) - \frac{1}{N} \leq \frac{1}{N}\mathrm{card}\left(I \cap \frac{1}{N}\mathbb{Z}\right) \leq \mathrm{vol}_1(I) + \frac{1}{N}.$$

Letting $N \to \infty$ we obtain

$$\mathrm{vol}_1(I) = \lim_{N\to\infty} \frac{1}{N}\mathrm{card}\left(I \cap \frac{1}{N}\mathbb{Z}\right).$$

Thus, if we write $C = I_1 \times I_2 \times \ldots \times I_n$ for closed intervals $I_j = [a_j, b_j]$ we get

$$\mathrm{vol}_n(C) = \prod_{j=1}^{n} \mathrm{vol}_1(I_j) = \prod_{j=1}^{n} \lim_{N\to\infty} \frac{1}{N}\mathrm{card}\left(I_j \cap \frac{1}{N}\mathbb{Z}\right) =$$

$$= \lim_{N\to\infty} \prod_{j=1}^{n} \frac{1}{N}\mathrm{card}\left(I_j \cap \frac{1}{N}\mathbb{Z}\right) = \lim_{N\to\infty} \frac{1}{N^n}\mathrm{card}\left(C \cap \frac{1}{N}\mathbb{Z}^n\right).$$

The last equality comes from the observation that each point of $C \cap \frac{1}{N}\mathbb{Z}^n$ corresponds to a n-tuple of points $(c_1, \ldots, c_n)$ such that $c_j \in I_j \cap \frac{1}{N}\mathbb{Z}$ for each $1 \leq j \leq n$, and vice-versa.

Now let $(C_k)_{k\in\mathbb{N}}$ be a sequence of open n-cells such that

$$C \subseteq \bigcup_{k=1}^{\infty} C_k.$$

Since C is compact, there exist finitely many C_ks covering C, and renumbering the sets if necessary we may assume that

$$C \subseteq \bigcup_{k=1}^{m} C_k.$$

It follows that any point of $C \cap \frac{1}{N}\mathbb{Z}^n$ is contained in some $C_k \cap \frac{1}{N}\mathbb{Z}^n$, $1 \leq k \leq m$. Therefore,

$$\operatorname{card}\left(C \cap \frac{1}{N}\mathbb{Z}^n\right) \leq \sum_{k=1}^{m} \operatorname{card}\left(C_k \cap \frac{1}{N}\mathbb{Z}^n\right)$$

Consequently,

$$\operatorname{vol}_n(C) = \lim_{N\to\infty} \frac{1}{N^n} \operatorname{card}\left(C \cap \frac{1}{N}\mathbb{Z}^n\right) \leq$$

$$\leq \lim_{N\to\infty} \frac{1}{N^n} \sum_{k=1}^{m} \operatorname{card}\left(C_k \cap \frac{1}{N}\mathbb{Z}^n\right) = \sum_{k=1}^{m} \lim_{N\to\infty} \frac{1}{N^n} \operatorname{card}\left(C_k \cap \frac{1}{N}\mathbb{Z}^n\right) = \sum_{k=1}^{m} \operatorname{vol}_n(C_k) \leq \sum_{k=1}^{\infty} \operatorname{vol}_n(C_k).$$

Finally, since this inequality holds for any covering of C by a sequence of open n-cells $(C_k)_{k\in\mathbb{N}}$ we get that $\operatorname{vol}_n(C) \leq \mathcal{L}^n(C)$.

Now assume that C is an n-cell which is not necessarily closed. For any arbitrarily given $\varepsilon > 0$, let $D \subseteq \operatorname{int} C$ be a closed n-cell such that

$$\operatorname{vol}_n(D) > \operatorname{vol}_n(C) - \varepsilon.$$

Notice that the definition of $\mathcal{L}^n$ gives $\mathcal{L}^n(D) \leq \mathcal{L}^n(\operatorname{int} C)$, because any covering of $\operatorname{int} C$ is also a covering of D. Hence,

$$\operatorname{vol}_n(C) - \varepsilon < \operatorname{vol}_n(D) = \mathcal{L}^n(D) \leq \mathcal{L}^n(\operatorname{int} C),$$

and since $\varepsilon > 0$ is arbitrary, this yields $\operatorname{vol}_n(C) \leq \mathcal{L}^n(\operatorname{int} C)$. It follows that $\operatorname{vol}_n(C) \leq \mathcal{L}^n(C)$, because $\operatorname{int} C \subseteq C$ (and again we argue that every covering of C is also a covering of $\operatorname{int} C$). Since C is a covering of itself by n-cells we get that $\mathcal{L}^n(C) \leq \operatorname{vol}_n(C)$. This concludes the proof.

□

Corollary A.3.1. *We have*

$$\mathcal{L}^n(C) = \mathcal{L}^n(\operatorname{int} C) = \mathcal{L}^n(\operatorname{cl}(C))$$

for every closed n-cell $C \subseteq \mathbb{R}^n$.

Proof. Notice that $\mathcal{L}^n(C) = \operatorname{vol}_n(C) = \operatorname{vol}_n(\operatorname{int} C) = \mathcal{L}^n(\operatorname{int} C)$. The first and last equalities come from Theorem A.3.1, and the other equality follows from the original definition of the n-dimensional volume of an n-cell. The proof of the equality $\mathcal{L}^n(C) = \mathcal{L}^n(\operatorname{cl}(C))$ is analogous, because $\operatorname{vol}_n(\operatorname{cl}(C)) = \operatorname{vol}_n(C)$.

□

From now on, we will denote the σ-algebra of the $\mathcal{L}^n$-measurable subsets of $\mathbb{R}^n$ by $\Sigma_{\mathcal{L}^n}$. These sets are also called *Lebesgue measurable sets*. The restriction of the Lebesgue outer measure to $\Sigma_{\mathcal{L}^n}$ is the *Lebesgue measure* of $\mathbb{R}^n$. Our next objective is to show that the Borel sets of $\mathbb{R}^n$ are Lebesgue measurable, that is, to prove the inclusion $\mathcal{B}_{\mathbb{R}^n} \subseteq \Sigma_{\mathcal{L}^n}$. The tool that we use to prove that is Carathéodory Criterion (Theorem A.1.1).

Proposition A.3.1. *Let $A, B \subseteq \mathbb{R}^n$. If $\operatorname{dist}(A, B) > 0$, then*

$$\mathcal{L}^n(A \cup B) = \mathcal{L}^n(A) + \mathcal{L}^n(B).$$

As a consequence, we have that all Borel sets of $\mathbb{R}^n$ are Lebesgue measurable.

Proof. First notice that from (A.3.1) we already have the inequality

$$\mathcal{L}^n(A \cup B) \leq \mathcal{L}^n(A) + \mathcal{L}^n(B),$$

and hence all we have to prove is that reverse inequality holds in the case where $\operatorname{dist}(A, B) > 0$. To do so, assume that $\mathcal{L}^n(A \cup B) < \infty$ (if $\mathcal{L}^n(A \cup B) = \infty$ there is nothing to prove) and let $\varepsilon > 0$. Choose a covering $(C_k)_{k\in\mathbb{N}}$ of $A \cup B$ by closed n-cells such that

$$\sum_{k=1}^{\infty} \operatorname{vol}_n(C_k) < \mathcal{L}^n(A \cup B) + \varepsilon.$$

For each $k \in \mathbb{N}$ we decompose C_k as a union of finitely many closed n-cells each of which having diameter smaller than $\operatorname{dist}(A, B)/2$. That is, we write

$$C_k = C_k^1 \cup C_k^2 \cup \ldots \cup C_k^{m(k)}$$

for some $m(k) \in \mathbb{N}$, where each C_k^j is a closed n-cell and

$$\operatorname{diam}(C_k^j) < \frac{\operatorname{dist}(A, B)}{2}.$$

It is actually easy to show that such a decomposition exists: one just has to divide each interval of C_k in closed sub-intervals with length smaller than $\left(\frac{\operatorname{dist}(A,B)}{2}\right)^{1/n}$ and consider the cartesian products of n-tuples of these sub-intervals, each of which taken in one of the original intervals of C_k. By doing so, we also have

$$\operatorname{vol}_n(C_k) = \operatorname{vol}_n(C_k^1) + \ldots + \operatorname{vol}_n(C_k^{m(k)}).$$

The collection

$$\{C_k^j : k \in \mathbb{N} \text{ and } 1 \leq j \leq m(k)\}$$

is clearly countable (because it is a countable union of finite collections), and hence we may let $(D_i)_{i\in\mathbb{N}}$ be an enumeration of this collection. It is also clear that

$$\sum_{i=1}^{\infty} \mathrm{vol}_n(D_i) = \sum_{k=1}^{\infty} \left(\mathrm{vol}_n(C_k^1) + \ldots + \mathrm{vol}_n(C_k^{m(k)})\right) = \sum_{k=1}^{\infty} \mathrm{vol}_n(C_k).$$

Now we let $I_A := \{i \in \mathbb{N} : D_i \cap A \neq \emptyset\}$, and $I_B := \{i \in \mathbb{N} : D_i \cap B \neq \emptyset\}$. We have that $I_A \cap I_B = \emptyset$, because if there exist points $a \in D_i \cap A$ and $b \in D_i \cap B$, then $|a - b| \leq \mathrm{diam}(D_i) < \mathrm{dist}(A, B)$, a contradiction. Also, since the union of all sets D_i is a covering of $A \cup B$, we have

$$A \subseteq \bigcup_{i \in I_A} D_i \quad \text{and} \quad B \subseteq \bigcup_{i \in I_B} D_i.$$

In other words, $\{D_i : i \in I_A\}$ and $\{D_i : i \in I_B\}$ are coverings of A and B, respectively, by closed n-cells. Therefore,

$$\mathcal{L}^n(A) + \mathcal{L}^n(B) \leq \sum_{i \in I_A} \mathrm{vol}_n(D_i) + \sum_{i \in I_B} \mathrm{vol}_n(D_i) = \sum_{i=1}^{\infty} \mathrm{vol}_n(D_i) =$$
$$= \sum_{k=1}^{\infty} \mathrm{vol}_n(C_k) < \mathcal{L}^n(A \cup B) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we have the desired inequality. Now, the fact that all Borel sets of $\mathbb{R}^n$ are Lebesgue measurable comes immediately from Theorem A.1.1.

□

It follows that the restriction of $\mathcal{L}^n$ to $\mathcal{B}_{\mathbb{R}^n}$ is a Borel measure on $\mathbb{R}^n$ (because it is a measure when restricted to $\Sigma_{\mathcal{L}^n}$). As a consequence of Theorem A.1.3 we have that this Borel measure is regular. In fact, it is clear that $\mathcal{L}^n$ is σ-finite because

$$\mathbb{R}^n = \bigcup_{k=1}^{\infty} [-k, k] \times \ldots \times [-k, k], \tag{A.3.3}$$

and $\mathrm{vol}_n\left([-k, k] \times \ldots \times [-k, k]\right) = (2k)^n < \infty$. Notice carefully that we did not discuss the regularity of the Lebesgue **outer** measure (this is indeed true, as it will be explained in Corollary A.4.2 ahead), but only of the Borel measure obtained by restricting it to the Borel σ-algebra of $\mathbb{R}^n$. Thus, what we have is that for every Borel set $B \subseteq \mathbb{R}^n$ and any $\varepsilon > 0$ there exist an open set $A \supseteq B$ and a closed set $F \subseteq B$ such that $\mathcal{L}^n(A \setminus F) < \varepsilon$. As immediate consequences of this regularity property, we have the equalities

$$\mathcal{L}^n(B) = \inf\{\mathcal{L}^n(A) : A \text{ is open and } A \supseteq B\}, \tag{A.3.4}$$

and

$$\mathcal{L}^n(B) = \sup\{\mathcal{L}^n(F) : F \text{ is closed and } F \subseteq B\}. \tag{A.3.5}$$

Moreover, we can actually improve the second equality by considering compact sets instead of closed sets. For Borel sets with finite measure, we can obtain that as a consequence of Corollary A.1.1, but in the case of the Lebesgue measure this holds for all sets, as we show next.

Lemma A.3.3. *For any Borel set $E \subseteq \mathbb{R}^n$ we have*

$$\mathcal{L}^n(E) = \sup\{\mathcal{L}^n(K) : K \text{ is compact and } K \subseteq E\}. \tag{A.3.6}$$

Proof. Notice that the inequality

$$\sup\{\mathcal{L}^n(K) : K \text{ is compact and } K \subseteq E\} \leq \mathcal{L}^n(E)$$

holds trivially because we have $\mathcal{L}^n(A) \subseteq \mathcal{L}^n(E)$ whenever $A \subseteq E$. Hence all we have to prove is the reverse inequality, and it is clearly enough to prove that for any $\varepsilon > 0$ there exists a compact set $K \subseteq E$ such that $\mathcal{L}^n(K) \geq \mathcal{L}^n(E) - \varepsilon$.

First consider the case where E is bounded. If this is the case, then we may fix a number $\rho > 0$ such that $E \subseteq B(0, \rho)$. To simplify the notation, recall that $\mathrm{cl}(B(0, \rho)) = \rho B^n$, and observe that $\rho B^n \setminus E$ is obviously a Borel set of $\mathbb{R}^n$. Let $\varepsilon > 0$ be arbitrarily given. From (A.3.4) we may choose an open set $U \subseteq \mathbb{R}^n$ such that $\rho B^n \setminus E \subseteq U$ and

$$\mathcal{L}^n(U) \leq \mathcal{L}^n(\rho B^n \setminus E) + \varepsilon.$$

It is easy to see that $K := \rho B^n \setminus U$ is a compact set contained in E. Writing ρB^n as the disjoint union $\rho B^n = (\rho B^n \setminus U) \cup (\rho B^n \cap U) = K \cup (\rho B^n \cap U)$, and noticing that $\mathcal{L}^n(\rho B^n \cap U) \leq \mathcal{L}^n(U)$ because $\rho B^n \cap U \subseteq U$, we get

$$\begin{aligned}\mathcal{L}^n(K) = \mathcal{L}^n(\rho B^n) - \mathcal{L}^n(\rho B^n \cap U) &\geq \mathcal{L}^n(\rho B^n) - \mathcal{L}^n(U) \geq \\ &\geq \mathcal{L}^n(\rho B^n) - \mathcal{L}^n(\rho B^n \setminus E) - \varepsilon = \mathcal{L}^n(E) - \varepsilon,\end{aligned}$$

where the last equality comes from $\mathcal{L}^n(\rho B^n) = \mathcal{L}^n(\rho B^n \setminus E) + \mathcal{L}^n(E)$, which is justified because $E \subseteq \rho B^n$. Thus, for every $\varepsilon > 0$ there exists a compact set $K \subseteq E$ such that $\mathcal{L}^n(K) \leq \mathcal{L}^n(E) - \varepsilon$, and this gives (A.3.6) for the case where E is bounded.

Now assume that $E \subseteq \mathbb{R}^n$ is a Borel set which is not bounded. We may write E as the following union of bounded sets

$$E = \bigcup_{j=1}^{\infty} E \cap jB^n.$$

Once we have $E \cap jB^n \subseteq E \cap (j+1)B^n$ for each $j \in \mathbb{N}$, we may use item **(i)** Lemma A.1.1 to write

$$\mathcal{L}^n(E) = \lim_{j\to\infty} \mathcal{L}^n(E \cap jB^n) = \sup_{j\in\mathbb{N}} \mathcal{L}^n(E \cap jB^n),$$

where the last equality follows from the fact that the sequence $\big(\mathcal{L}^n(E \cap jB^n)\big)_{j\in\mathbb{N}}$ is non-increasing. Since each set $E \cap jB^n$ is a bounded Borel set, we have

$$\mathcal{L}^n(E) = \sup_{j\in\mathbb{N}} \Big(\sup\big\{\mathcal{L}^n(K) : K \text{ is compact and } K \subseteq E \cap jB^n\big\}\Big).$$

Since every subset of some $E \cap jB^n$ is also a subset of E, we obtain

$$\sup_{j\in\mathbb{N}} \Big(\sup\big\{\mathcal{L}^n(K) : K \text{ is compact and } K \subseteq E \cap jB^n\big\}\Big) \leq \sup\big\{\mathcal{L}^n(K) : K \text{ is compact and } K \subseteq E\big\}.$$

On the other hand, if $K \subseteq E$ is compact, then there exists a number j_0 such that $K \subseteq j_0 B^n$. Hence $K \subseteq E \cap j_0 B^n$, and the reverse of the inequality above also holds. This concludes the proof.

□

Corollary A.3.2. *Let* $\mathbf{g}$ *be a proper affine subspace of* $\mathbb{R}^n$. *For any Borel subset* $A \subseteq \mathbf{g}$ *we have* $\mathcal{L}^n(A) = 0$.

Proof. Let u be any unit normal vector of $\mathbf{g}$ (it is clear that $\mathbf{g}$ is contained in some hyperplane $\mathbf{h} = \mathbf{h}(u, \alpha)$), and let $K \subseteq \mathbf{g}$ be compact. Assume that $\mathcal{L}^n(K) > 0$. Consider the set

$$K(u) := \{x + tu : x \in K \text{ and } t \in [0,1]\}.$$

Then $K(u)$ is clearly compact, from where $\mathcal{L}^n(K(u)) < \infty$ (recall that any covering of a compact set by open n-cells admits a finite subcovering). Also, we have

$$\bigcup_{j=1}^{\infty} \left(K + \frac{1}{j}u\right) \subseteq K(u),$$

and this union is clearly disjoint. Geometrically, these sets lie in parallel distinct affine subspaces (namely, $\mathbf{g} + \frac{1}{j}u$). A formal argument is also easy to derive: assume that $x + \frac{1}{j}u = y + \frac{1}{m}u$ for $x, y \in K$ and $j \neq m$. Then

$$0 = \langle x - y, u\rangle = \left\langle \left(\frac{1}{m} - \frac{1}{j}\right)u, u \right\rangle = \frac{1}{m} - \frac{1}{j},$$

a contradiction. The first equality comes from the fact that $x, y \in \mathbf{g} \subseteq \mathbf{h}(u, \alpha)$. Finally, since $K + \frac{1}{j}u$ is a translation of K for each $j \in \mathbb{N}$, using the fact that $\mathcal{L}^n$ is translation-invariant we get

$$\mathcal{L}^n(K(u)) \geq \mathcal{L}^n\left(\bigcup_{j=1}^{\infty}\left(K + \frac{1}{j}u\right)\right) = \sum_{j=1}^{\infty} \mathcal{L}^n\left(K + \frac{1}{j}u\right) = \sum_{j=1}^{\infty} \mathcal{L}^n(K) = \infty,$$

and this is a contradiction. Hence $\mathcal{L}^n(K) = 0$. The general case follows immediately: if $A \subseteq \mathbf{g}$ is a Borel set, then any compact $K \subseteq A$ is contained in $\mathbf{g}$, and hence $\mathcal{L}^n(K) = 0$. Thus, from (A.3.6) we have the desired.

□

Remark A.3.1. Regarding again $\mathcal{L}^n$ as an outer measure, it is indeed true that $\mathcal{L}^n(A) = 0$ for any set $A \subseteq \mathbf{g}$. In fact, any of these sets is contained in a countable union of Borel sets contained in $\mathbf{g}$ (cover A by intersections of open balls with $\mathbf{g}$, for example).

From now on, we will use the expression "Lebesgue measure" to refer to the restriction of $\mathcal{L}^n$ to both $\Sigma_{\mathcal{L}^n}$ and $\mathcal{B}_{\mathbb{R}^n}$. But we warn the reader that the second variant must be assumed unless otherwise stated.

We also have that the Lebesgue measure is the unique translation-invariant Borel measure of $\mathbb{R}^n$ such that the measure of $[0,1]^n$ equals 1. This can be proved using the theory on *Haar measures*, but once this topic is not covered in this book we follow the constructive proof given in [2090].

Theorem A.3.2. *If $\mu\colon \mathcal{B}_{\mathbb{R}^n} \to [0,\infty]$ is a translation-invariant Borel measure on $\mathbb{R}^n$ such that $\mu([0,1]^n) = 1$, then $\mu = \mathcal{L}^n$.*

Remark A.3.2. Another way of stating this result is that any translation-invariant Borel measure on $\mathbb{R}^n$ which attributes a positive value to $[0,1]^n$ is a positive scalar multiple of $\mathcal{L}^n$.

Proof. Roughly speaking, our strategy is to decompose open sets of $\mathbb{R}^n$ as countable disjoint unions of sets having the form

$$\begin{aligned} x + [0, 2^{-k})^n &= \\ = x + [0, 2^{-k}) \times \ldots \times [0, 2^{-k}) &= [x_1, x_1 + 2^{-k}) \times \ldots \times [x_n, x_n + 2^{-k}) = \\ &= \left\{(y_1, \ldots, y_n) \in \mathbb{R}^n : x_j \leq y_j < x_j + 2^{-k} \text{ for all } 1 \leq j \leq n\right\}, \end{aligned}$$

for $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $k \in \mathbb{N} \cup \{0\}$. We start by collecting some facts about these sets. For the sake of simplicity, for each $k \in \mathbb{N} \cup \{0\}$ we denote

$$C(k) := [0, 2^{-k})^n = [0, 2^{-k}) \times \ldots \times [0, 2^{-k}).$$

(i) Let $z, w \in \mathbb{Z}^n$, with $z = (z_1, \ldots, z_n)$ and $w = (w_1, \ldots, w_n)$, and let $k \in \mathbb{N} \cup \{0\}$. If $z \neq w$, then

$$\big(2^{-k}z + C(k)\big) \cap \big(2^{-k}w + C(k)\big) = \emptyset. \tag{A.3.7}$$

In fact, if $z \neq w$, then at least one of the coordinates of z is distinct from the respective coordinate of w. Without loss of generality, we may assume that this difference occurs in the first coordinate, and that $z_1 > w_1$. Since both are integers, we have that $z_1 - w_1 \geq 1$. For arbitrarily given points $x = (x_1, \ldots, x_n) \in 2^{-k}z + C(k)$ and $y = (y_1, \ldots, y_n) \in 2^{-k}w + C(k)$ we have

$$y_1 < 2^{-k}w_1 + 2^{-k} = 2^{-k}(w_1 + 1) \leq 2^{-k}z_1 \leq x_1,$$

from where we get that $y_1 < x_1$. It follows that the first coordinate of any point of $2^{-k}w + C(k)$ is strictly smaller than the first coordinate of any point of $2^{-k}z + C(k)$, and hence (A.3.7) holds.

(ii) For each $k \in \mathbb{N} \cup \{0\}$, the cell $[0, 1)^n$ can be written as the disjoint union

$$[0, 1)^n = \bigcup_{z \in \mathbb{Z}^n,\, 0 \leq z_j \leq 2^k - 1} \big(2^{-k}z + C(k)\big).$$

That is, the union is taken over the points of $\mathbb{Z}^n$ having all coordinates between 0 and $2^k - 1$. To prove that, observe that the fact that the union is disjoint comes from **(i)**, and the inclusion

$$\bigcup_{z \in \mathbb{Z}^n,\, 0 \leq z_j \leq 2^k - 1} \big(2^{-k}z + C(k)\big) \subseteq [0, 1)^n$$

is immediate. Hence all we have to prove is the reverse inclusion. Let $x = (x_1, \ldots, x_n) \in [0, 1)^n$. It is clear that for each $j \in \mathbb{N}$, we have $\ell(j) \in \mathbb{N}$ with $0 \leq \ell(j) \leq 2^k - 1$ such that

$$\frac{\ell(j)}{2^k} \leq x_j < \frac{\ell(j) + 1}{2^k}.$$

Thus, we may write

$$x_j = \frac{\ell(j)}{2^k} + \delta(j)$$

for some $0 \leq \delta(j) < 2^{-k}$. It follows the equality

$$x = \left(\frac{\ell(1)}{2^k}, \ldots, \frac{\ell(n)}{2^k}\right) + (\delta(1), \ldots, \delta(n)) = 2^{-k}(\ell(1), \ldots, \ell(n)) + (\delta(1), \ldots, \delta(n)),$$

and since $z := (\ell(1), \ldots, \ell(n))$ is a point of $\mathbb{Z}^n$ having all coordinates between 0 and $2^k - 1$, and $(\delta(1), \ldots, \delta(n)) \in C(k)$ we get that $x \in 2^{-k}z + C(k)$. This proves the desired inclusion.

(iii) Immediately from **(ii)**, we have that for every $x \in \mathbb{R}^n$ the set $x + [0, 1)^n$ can be written as the disjoint union

$$x + [0, 1)^n = \bigcup_{z \in \mathbb{Z}^n,\, 0 \leq z_j \leq 2^k - 1} \big(x + 2^{-k}z + C(k)\big).$$

Now let μ be a translation-invariant Borel measure on $\mathbb{R}^n$ such that $\mu\big([0,1]^n\big) = 1$. For each $k \in \mathbb{N} \cup \{0\}$, let $\mu\big([0,2^{-k})\big) = \mu(C(k)) = \lambda_k$ and observe that $0 \leq \lambda_k \leq 1$. Once μ is translation-invariant, it follows from **(i)** that

$$\lambda_0 = \mu\big([0,1)^n\big) = \sum_{z\in\mathbb{Z}^n,\, 0\leq z_j\leq 2^k-1} \mu\big(2^{-k}z + C(k)\big) = \sum_{z\in\mathbb{Z}^n,\, 0\leq z_j\leq 2^k-1} \mu\big(C(k)\big) = \\ = 2^{nk}\mu(C(k)) = 2^{nk}\lambda_k,$$

where we also observe that there are 2^{nk} points in $\mathbb{Z}^n$ having coordinates between 0 and $2^k - 1$. Once $(0,2^{-k})^n \subseteq C(k) \subseteq [0,2^{-k}]^n$, we have

$$2^{-nk} = \mathrm{vol}_n\big((0,2^{-k})^n\big) = \mathcal{L}^n\big((0,2^{-k})^n\big) \leq \mathcal{L}^n(C(k)) \leq \mathcal{L}^n\big([0,2^{-k}]^n\big) = \\ = \mathrm{vol}_n\big([0,2^{-k}]^n\big) = 2^{-nk}.$$

This gives that $\mathcal{L}^n(C(k)) = 2^{-nk}$. Consequently, we have that

$$\mu(C(k)) = \lambda_k = 2^{-nk}\lambda_0 = \lambda_0\mathcal{L}^n(C(k)).$$

Now let $U \subseteq \mathbb{R}^n$ be a non-empty open set. We define collections of subsets of U by

$$C_0(U) := \big\{z + C(0) : z \in \mathbb{Z}^n \text{ and } z + C(0) \subseteq U\big\},$$
$$C_1(U) := \big\{2^{-1}z + C(1) : z \in \mathbb{Z}^n,\ z + C(1) \subseteq U \text{ and } 2^{-1}z + C(1) \nsubseteq E\ \forall\, E \in C_0\big\}$$

and, inductively, for $k \geq 2$,

$$C_k(U) := \big\{2^{-k}z + C(k) : z \in \mathbb{Z}^n,\ 2^{-k}z + C(k) \subseteq U \text{ and} \\ 2^{-k}z + C(k) \nsubseteq E \text{ for all } E \in C_0 \cup \ldots \cup C_k\big\}.$$

From **(i)**, we have that each collection $C_k(U)$ is disjoint. We claim that, even more, the collection

$$C(U) = \bigcup_{k=0}^{\infty} C_k(U)$$

is also disjoint. To prove that, let $m, k \in \mathbb{N}\cup\{0\}$ with $m > k$. Let $z, w \in \mathbb{Z}^n$ be such that $(2^{-k}z + C(k)) \cap (2^{-m}w + C(m)) \neq \emptyset$, and let $y = (y_1, \ldots, y_n)$ be a point of this intersection. Writing $z = (z_1, \ldots, z_n)$ and $w = (w_1, \ldots, w_n)$ we have that for each $1 \leq j \leq n$ there exist numbers $0 \leq \delta_k(j) < 2^{-k}$ and $0 \leq \delta_m(j) < 2^{-m}$ such that

$$y_j = 2^{-k}z_j + \delta_k(j) = 2^{-m}w_j + \delta_m(j).$$

Multiplying both sides of the second equality by 2^m, rearranging the terms, and taking absolute values yield

$$|w_j - 2^{m-k}z_j| = 2^m|\delta_k(j) - \delta_m(j)| < 2^m 2^{-k} = 2^{m-k},$$

where the inequality is given immediately from the inequalities which hold for $\delta_k(j)$ and $\delta_m(j)$. Since the left-hand side of the above is an integer (recall that $m > k$), we obtain

$$|w_j - 2^{m-k}z_j| \leq 2^{m-k} - 1,$$

which can be written as

$$1 + 2^{m-k}(z_j - 1) \leq w_j \leq -1 + 2^{m-k}(z_j + 1).$$

Multiplying the second inequality by 2^{-m} we get

$$2^{-m}w_j \leq 2^{-k}z_j + 2^{-k} - 2^{-m}. \tag{A.3.8}$$

Now we claim that

$$w_j > -1 + 2^{m-k}z_j. \tag{A.3.9}$$

Indeed, if this does not hold, then

$$y_j < 2^{-m}w_j + 2^{-m} \leq 2^{-m}(-1 + 2^{m-k}z_j) + 2^m = 2^{-k}z_j,$$

and this is a contradiction with the fact that $y \in 2^{-k}z + C(k)$. Once both sides of (A.3.9) are integer we get

$$w_j \geq 2^{m-k}z_j.$$

Multiplying this inequality by 2^{-m} and repeating (A.3.8) we obtain

$$2^{-k} \leq 2^{-m}w_j \leq 2^{-k}z_j + 2^{-k} - 2^{-m}.$$

Let $x = (x_1, \ldots, x_n) \in 2^{-m}w + C(m)$. Then for each $1 \leq j \leq n$ there exists a number $0 \leq \varepsilon_m(j) < 2^{-m}$ such that $x_j = 2^{-m}w_j + \varepsilon_m(j)$. Adding $\varepsilon_m(j)$ in the inequality above we get

$$2^{-k}z_j \leq x_j \leq 2^{-k}z_j + 2^{-k}\varepsilon_m(j) - 2^{-m} < 2^{-k}z_j + 2^{-k} + 2^{-m} - 2^{-m} = 2^{-k}z_j + 2^{-k}.$$

Once this holds for every $1 \leq j \leq n$, we get that $x \in 2^{-k}z + C(k)$. Because x was taken arbitrarily in $2^{-m}w + C(m)$ we get the inclusion $2^{-m}w + C(m) \subseteq 2^{-k}z + C(k)$.

As a consequence, it follows that if a set of $E \in \mathcal{C}_m(U)$ has non-empty intersection with a set $F \in \mathcal{C}_k(U)$, then $E \subseteq F$. But this is a contradiction with the construction of $\mathcal{C}_m(U)$. Hence $\mathcal{C}(U)$ is indeed a disjoint collection.

Notice also that $\mathcal{C}(U)$ is a countable collection since

$$\mathcal{C}_k(U) \subseteq \left\{2^{-k}z + C(k) : z \in \mathbb{Z}^n\right\}$$

for any $k \in \mathbb{N} \cup \{0\}$, and the collection in the right-hand side is countable because it is indexed by $\mathbb{Z}^n$. Then each collection $\mathcal{C}_k(U)$ is countable, and hence their union is also countable.

We also have that

$$U = \bigcup_{E \in \mathcal{C}(U)} E.$$

In fact, the inclusion "$\supseteq$" comes immediately from the construction of $\mathcal{C}(U)$. To prove the reverse inclusion, let $x \in U$. Since U is open, it follows that there exist a number $k \in \mathbb{N} \cup \{0\}$ and a point $z \in \mathbb{Z}^n$ such that

$$x \in 2^{-k}z + C(k) \subseteq U.$$

Hence if $k_0 \in \mathbb{N} \cup \{0\}$ is the minimum among all ks with this property, we have that $x \in C_{k_0}(U)$. Once μ is translation-invariant, we have that it assumes the same value for all sets of $C_k(U)$, and this value is $\mu(C(k))$. Recalling that $\mu(C(k)) = \lambda_0 \mathcal{L}^n(C(k))$ we get

$$\mu(U) = \sum_{E \in C(U)} \mu(E) = \sum_{k=0}^{\infty} \left(\sum_{E \in C_k(U)} \mu(E) \right) = \sum_{k=0}^{\infty} \left(\sum_{E \in C_k(U)} \lambda_0 \mathcal{L}^n(C(k)) \right) =$$
$$= \lambda_0 \sum_{k=0}^{\infty} \mathcal{L}^n \left(\bigcup_{E \in C_k(U)} E \right) = \lambda_0 \mathcal{L}^n \left(\bigcup_{E \in C(U)} E \right) = \lambda_0 \mathcal{L}^n(U).$$

Summarizing, we have proved that $\mu(U) = \lambda_0 \mathcal{L}^n(U)$ for every open set $U \subseteq \mathbb{R}^n$. Now we will show that this equality also holds for compact sets. This is actually very simple: if $K \subseteq \mathbb{R}^n$ is compact, then we may choose a number $\rho > 0$ such that $K \subseteq (-\rho, \rho)^n =: U$, and once both U and $U \setminus K$ are open sets we get

$$\mu(K) = \mu(U) - \mu(U \setminus K) = \lambda_0 \mathcal{L}^n(U) - \lambda_0 \mathcal{L}^n(U \setminus K) = \lambda_0 \mathcal{L}^n(K).$$

Once equality $\mu(\cdot) = \lambda_0 \mathcal{L}^n(\cdot)$ holds for open sets and for compact sets, we claim that it holds for any Borel set. This is obtained as a consequence of (A.3.4) and (A.3.6). If $B \subseteq \mathbb{R}^n$ is a Borel set, then

$$\mu(B) \leq \inf\{\mu(U) : U \text{ is open and } B \subseteq U\} = \inf\{\lambda_0 \mathcal{L}^n(U) : U \text{ is open and } B \subseteq U\} =$$
$$= \lambda_0 \mathcal{L}^n(B) = \sup\{\lambda_0 \mathcal{L}^n(K) : K \text{ is compact and } K \subseteq B\} =$$
$$= \sup\{\mu(K) : K \text{ is compact and } K \subseteq B\} \leq \mu(B).$$

Notice carefully that in the first and last inequalities we are not assuming any regularity property for μ (but only the inclusions $B \subseteq U$ for the first and $K \subseteq B$ in the last). From the inequalities above we get that $\mu(B) = \lambda_0 \mathcal{L}^n(B)$, and hence μ and $\lambda_0 \mathcal{L}^n$ agree on the Borel sets of $\mathbb{R}^n$.

Now it is enough to prove that $\lambda_0 = 1$ (because in this case we have $\mu = \mathcal{L}^n$ in $\mathcal{B}_{\mathbb{R}^n}$, and this is what we want to prove). To do so, recall that $\mathcal{L}^n\big([0,1]^n\big) = \mu\big([0,1]^n\big) = 1$. Since $\mu = \lambda_0 \mathcal{L}^n$ for compact sets we get

$$\lambda_0 = \lambda_0 \mathcal{L}^n\big([0,1]^n\big) = \mu\big([0,1]^n\big) = 1,$$

and this concludes the proof.

□

It is also very important for us to understand the behavior of the Lebesgue measure under a linear map. Before we give a sufficient topological condition under which the image of a Borel set by a continuous function is also a Borel set.

Lemma A.3.4. *If X is a* σ-compact topological space*, that is, if X can be written as a countable union of compact subsets, then the image of any Borel set $A \subseteq X$ by a continuous map $f : X \to Y$ is a Borel subset of Y.*

Proof. The strategy to prove that is similar to that of Lemma A.2.1: we let $\mathcal{A} = \{A \subseteq X : f(A) \in \mathcal{B}_Y\}$. It is easily seen that this family is a σ-algebra of X, and hence it suffices to show that it contains all open sets. To do so, let $(D_j)_{j \in \mathbb{N}}$ be a countable collection of compact subsets of X such that

$$X = \bigcup_{j=1}^{\infty} D_j.$$

Hence if $A \subseteq X$ is open, then we may write A as a countable union of compact sets as

$$A = \bigcup_{j=1}^{\infty} A \cap D_j.$$

Then the image of A under f is written as

$$f(A) = \bigcup_{j=1}^{\infty} f(A \cap D_j),$$

and the latter is a countable union of compact (hence Borel) subsets of Y. Thus, every open set of X lies in $\mathcal{A}$.

□

Theorem A.3.3. *If $A \subseteq \mathbb{R}^n$ is a Borel set and $T\colon \mathbb{R}^n \to \mathbb{R}^n$ is a linear map, then $T(A)$ is a Borel set and*

$$\mathcal{L}^n(T(A)) = |\det(A)|\mathcal{L}^n(A). \tag{A.3.10}$$

Proof. We start by proving the first claim. First, if T is an isomorphism, then T^{-1} is also an isomorphism (hence continuous), and from Lemma A.2.1 it follows that if $A \subseteq \mathbb{R}^n$ is a Borel set, then $\left(T^{-1}\right)^{-1}(A) = T(A)$ is also a Borel set.

In the case where T is not an isomorphism, we start by verifying that any vector subspace of $\mathbb{R}^n$ is a Borel set. To verify that, let $\mathbf{g}$ be such a subspace and write

$$\mathbf{g} = \bigcup_{j=1}^{\infty} \mathbf{g} \cap jB^n.$$

The right-hand side is a countable union of compact sets (hence Borel), from where we get that $\mathbf{g}$ is a Borel set (it is worth noticing that same argument can be used for the case where $\mathbf{g}$ is an affine subspace). Now we define the collection

$$\Sigma := \{A \subseteq \mathbb{R}^n : T(A) \text{ is a Borel set}\},$$

and we claim that this is a σ-algebra of $\mathbb{R}^n$. In fact, it is clear that $\emptyset, \mathbb{R}^n \in \Sigma$ since $T(\emptyset) = \emptyset$ and $T(\mathbb{R}^n) = \operatorname{im}(T)$, which is a vector subspace of $\mathbb{R}^n$. If $A \in \Sigma$, then $T(\mathbb{R}^n \setminus A) = \operatorname{im}(T) \setminus T(A)$, which is a difference between Borel sets, and hence a Borel set. Therefore, $\mathbb{R}^n \setminus A \in \Sigma$. To prove that Σ is closed under countable unions, simply observe that

$$T\left(\bigcup_{m=1}^{\infty} A_m\right) = \bigcup_{m=1}^{\infty} T(A_m).$$

Let $A \subseteq \mathbb{R}^n$ be an open set. It is easy to see that $T(A)$ is relatively open in $\operatorname{im}(T)$, and hence there exists an open set $U \subseteq \mathbb{R}^n$ such that $T(A) = \operatorname{im}(T) \cap U$. The latter is an intersection of Borel sets, from where we get that $T(A)$ is a Borel set. This gives that Σ contains all open sets of $\mathbb{R}^n$, and then we have that $\mathcal{B}_{\mathbb{R}^n} \subseteq \Sigma$. This means that the image of any Borel set by T is also a Borel set.

Next we prove the equality. If T is not an isomorphism, then $\operatorname{im}(T)$ is a proper subspace of $\mathbb{R}^n$. Consequently, in view of Corollary A.3.2 we have that the left-hand side of (A.3.10) vanishes for any Borel set $A \subseteq \mathbb{R}^n$. Since in this case we also have that $\det(T) = 0$, the right-hand side also vanishes for each Borel set $A \subseteq \mathbb{R}^n$ and we have the desired.

Now we consider the case where T is an isomorphism. Let $f\colon \mathrm{GL}(n) \to (0,\infty)$ be given by

$$f(T) = \mathcal{L}^n\big(T\big([0,1]^n\big)\big),$$

and notice carefully that we have indeed $0 < f(T) < \infty$, because $T\big([0,1]^n\big)$ is compact and has non-empty interior (because T is an isomorphism). Consider the map $\mu_T\colon \mathcal{B}_{\mathbb{R}^n} \to [0,\infty]$ defined as

$$\mu_T(B) = \frac{\mathcal{L}^n(T(B))}{f(T)}.$$

It is readily seen that μ_T is a translation-invariant Borel measure such that $\mu_T\big([0,1]^n\big) = 1$. Therefore, we have that $\mu_T = \mathcal{L}^n$, and this gives

$$\mathcal{L}^n\big(T(B)\big) = f(T)\mathcal{L}^n(B) \tag{A.3.11}$$

for each Borel set $B \subseteq \mathbb{R}^n$ and any $T \in \mathrm{GL}(n)$. Even more, it is immediate that f is the unique map from $\mathrm{GL}(n)$ onto $(0,\infty)$ with this property. Indeed, letting $B = [0,1]^n$ we see that the equality above completely determines f.

We also have that f is a group homomorphism between $\mathrm{GL}(n)$ (endowed with standard composition, of course) and the multiplicative group $(0,+\infty)$. If $S, T \in \mathrm{GL}(n)$, then using (A.3.11) twice we have

$$f(S\circ T) = \mathcal{L}^n\big(S\big(T\big([0,1]^n\big)\big)\big) = f(S)\mathcal{L}^n\big(T\big([0,1]^n\big)\big) = f(S)f(T)\mathcal{L}^n\big([0,1]^n\big) = f(S)f(T).$$

What we have to prove is that f is the absolute value of the usual determinant. To do so, we start by considering the case where the matricial representation of T in the canonical basis of $\mathbb{R}^n$ is a real diagonal matrix. We write $T = \mathrm{diag}(\lambda_1,\ldots,\lambda_n)$ and notice that $\lambda_j \neq 0$ for each $1 \leq j \leq n$, because T is an isomorphism. We claim that

$$f(T) = |\det(T)|.$$

To verify this, simply notice that

$$T\big([0,1]^n\big) = \big[\min\{0,\lambda_1\},\max\{0,\lambda_1\}\big] \times \ldots \times \big[\min\{0,\lambda_n\},\max\{0,\lambda_n\}\big],$$

and this yields

$$f(T) = \mathcal{L}^n\big(T\big([0,1]^n\big)\big) =$$
$$= \mathrm{vol}_n\Big(\big[\min\{0,\lambda_1\},\max\{0,\lambda_1\}\big] \times \ldots \times \big[\min\{0,\lambda_n\},\max\{0,\lambda_n\}\big]\Big) = |\lambda_1|\ldots|\lambda_n| = |\det(T)|.$$

Once f is a group homomorphism, we also have that $f(T) = |\det(T)|$ whenever T is a diagonalizable isomorphism. In fact, if $T \in \mathrm{GL}(n)$ is diagonalizable, then there exist $S, D \in \mathrm{GL}(n)$ such that $T = S^{-1}DS$ and the matricial representation of D in the canonical basis is a diagonal matrix. Hence

$$f(T) = f\big(S^{-1}DS\big) = f(S)^{-1}|\det(D)|f(S) = |\det(D)| = |\det(T)|.$$

Now we go to the case where $T \in \mathrm{GL}(n)$ has only real eigenvalues (but is not necessarily diagonalizable). We will use without a proof the fact that any of these isomorphisms can be approximated by diagonalizable isomorphisms (roughly speaking, this happens because one may find arbitrarily small perturbations which make all the eigenvalues distinct). Therefore, it is enough to show that f is continuous, since we obviously have that the usual determinant is continuous as a function of $\mathrm{GL}(n)$. To prove that, we can use any norm in $\mathrm{GL}(n)$, because all norms in a finite-dimensional vector space are equivalent (and hence provide the same topology). We endow $\mathrm{GL}(n)$ with the norm

$$||T|| := \max_{v \neq 0_{\mathbb{R}^n}} \frac{||Tv||_\infty}{||v||_\infty},$$

where $||\cdot||_\infty$ is the norm in $\mathbb{R}^n$ given by

$$||(v_1, \ldots, v_n)||_\infty := \max\{|v_1|, \ldots, |v_n|\}.$$

It is readily seen that the (closed) unit ball of $||\cdot||_\infty$ is $[-1, 1]^n$. We start by proving that f is continuous at $\mathrm{id}_{\mathbb{R}^n}$. It is well-known from the theory on linear maps that if $T\colon \mathbb{R}^n \to \mathbb{R}^n$ is a linear map such with $||T - \mathrm{id}_{\mathbb{R}^n}|| < 1$, then T is an isomorphism, and

$$T^{-1} := \sum_{j=1}^{\infty} (\mathrm{id}_{\mathbb{R}^n} - T)^j.$$

Even more, if we fix a number $0 < \delta < 1$ and let $||T - \mathrm{id}_{\mathbb{R}^n}|| < \delta$, then once it is clear that $||T \circ S|| = ||T|| \cdot ||S||$, it follows that

$$1 = ||\mathrm{id}_{\mathbb{R}^n}|| = ||T^{-1}|| \cdot ||T|| \geq ||T^{-1}|| \cdot \big(||\mathrm{id}_{\mathbb{R}^n}|| - ||T - \mathrm{id}_{\mathbb{R}^n}||\big) > ||T^{-1}|| \cdot (1 - \delta),$$

from where we get the inequality

$$||T^{-1}|| < \frac{1}{1 - \delta}.$$

The inequality $||T - \mathrm{id}_{\mathbb{R}^n}|| < \delta$ gives that $||Tx||_\infty \leq 1 + \delta$ for every $x \in [-1, 1]^n$. This means that each coordinate of Tx lies in the interval $[-(1 + \delta), 1 + \delta]^n$, and this gives

$$T\big([-1, 1]^n\big) \subseteq (1 + \delta)[-1, 1]^n.$$

On the other hand, inequality $||T^{-1}|| < 1/(1 - \delta)$ can be written as $||(1 - \delta)T^{-1}|| < 1$, and this yields that $||(1 - \delta)T^{-1}x||_\infty < 1$ for every $x \in [1, 1]^n$. This gives directly the inclusion

$$(1 - \delta)T^{-1}\big([-1, 1]^n\big) \subseteq [-1, 1]^n.$$

Applying T on both sides and repeating the inclusion derived before (only for the sake of summarizing) we get

$$(1 - \delta)[-1, 1]^n \subseteq T\big([-1, 1]^n\big) \subseteq (1 + \delta)[-1, 1]^n.$$

The left- and right-hand sides are n-cells, and hence we have

$$\mathcal{L}^n\big((1 - \delta)[-1, 1]^n\big) = 2^n(1 - \delta)^n \text{ and}$$
$$\mathcal{L}^n\big((1 + \delta)[-1, 1]^n\big) = 2^n(1 + \delta)^n.$$

Thus, since $\mathcal{L}^n\big([-1, 1]^n\big) = 2^n$, we get from the inclusions and equalities above that

$$(1 - \delta)^n \leq \frac{\mathcal{L}^n\big(T\big([-1, 1]^n\big)\big)}{\mathcal{L}^n\big([-1, 1]^n\big)} \leq (1 + \delta)^n,$$

and letting $B = [-1, 1]^n$ in (A.3.11) we get that the term in the middle is precisely $f(T)$. It follows that, for a given $\varepsilon > 0$, choosing $0 < \delta < 1$ such that $1 - \varepsilon < (1 - \delta)^n < (1 + \delta)^n < 1 + \varepsilon$ we get that

$$|f(T) - f(\mathrm{id}_{\mathbb{R}^n})| < \varepsilon$$

whenever $||T - \mathrm{id}_{\mathbb{R}^n}|| < \delta$. This proves that f is continuous at $\mathrm{id}_{\mathbb{R}^n}$.

To check that f is continuous at any $S \in \mathrm{GL}(n)$, simply notice that if $S_n \to S$, then $S^{-1}S_n \to \mathrm{id}_{\mathbb{R}^n}$, and hence $f(S^{-1}S_n) \to f(\mathrm{id}_{\mathbb{R}^n})$. Since $f(S^{-1}S_n) = f(S)^{-1}f(S_n)$ we obtain that $f(S_n) \to f(S)$.

As it was mentioned, the continuity of f and of the usual determinant together with the facts that $f(T) = |\det(T)|$ for every diagonalizable isomorphism gives that this equality also holds for every isomorphism having only real eigenvalues. This happens because these isomorphisms can be approximated by those. Finally, standard linear algebra theory guarantees that any isomorphism $T\colon \mathbb{R}^n \to \mathbb{R}^n$ can be written as a finite composition $T = T_1 \dots T_m$ of isomorphisms whose respective matrix representations are elementary matrices. Once these matrices only have real eigenvalues, we get

$$f(T) = f(T_1 \dots T_m) = f(T_1) \dots f(T_m) = |\det(T_1)| \dots |\det(T_m)| = |\det(T)|,$$

and this concludes the proof.

□

The Lebesgue measure in $\mathbb{R}^n$ is very useful when it comes to "full-dimensional sets", but it is zero over proper affine subsets (see Corollary A.3.2). In other words, the Lebesgue measure of $\mathbb{R}^n$ gives us a good notion of *volume*, but not of *area*. To overcome this problem, we finish this section with a very important construction which provides a way of endowing any affine subspace of $\mathbb{R}^n$ with a "reasonable" Lebesgue measure.

Construction A.3.1. We can endow any k-dimensional affine subspace of $\mathbb{R}^n$ with a "k-dimensional Lebesgue measure", and now we explain how to do this. First, consider the case where V is a k-dimensional vector subspace of $\mathbb{R}^n$, and let $T\colon \mathbb{R}^k \to V$ be any isomorphism which is also an isometry (we are assuming that $\mathbb{R}^k$ is endowed with the usual Euclidean metric, and that V inherits the metric of $\mathbb{R}^n$). For example, let T take the canonical basis of $\mathbb{R}^k$ to any orthonormal basis spanning V. Then we define

$$\mathcal{L}^k_V(B) := \mathcal{L}^k\big(T^{-1}(B)\big)$$

for any Borel set $B \subseteq V$. Since T is continuous, it follows that T is measurable with respect to the Borel σ-algebras of V and $\mathbb{R}^k$, and hence $\mathcal{L}^k_V$ is simply the push-forward measure of the Lebesgue measure on $\mathbb{R}^k$ by T.

It is important to verify that $\mathcal{L}^k_V$ is independent of the choice of the isometric isomorphism T. In fact, if $S\colon \mathbb{R}^k \to V$ is another isometric isomorphism, then $S^{-1}T\colon \mathbb{R}^k \to \mathbb{R}^k$ is an isometric isomorphism, and hence an orthogonal transformation. Thus, $|\det(S^{-1} \circ T)| = 1$, and consequently

$$\mathcal{L}^k\big(S^{-1}(B)\big) = \mathcal{L}^k\big(S^{-1}T\big(T^{-1}(B)\big)\big) = |\det(S^{-1}T)|\mathcal{L}^k\big(T^{-1}(B)\big) = \mathcal{L}^k\big(T^{-1}(B)\big).$$

It is not difficult to see that $\mathcal{L}^k_V$ is translation-invariant (within V, of course). Indeed, if $x \in V$ and B is a Borel set of V, then

$$\mathcal{L}^k_V(B + x) = \mathcal{L}^k\big(T^{-1}(B + x)\big) = \mathcal{L}^k\big(T^{-1}(B) + T^{-1}x\big) = \mathcal{L}^k\big(T^{-1}(B)\big) = \mathcal{L}^k_V(B),$$

where we used the translation-invariance of $\mathcal{L}^k$.

Now let $\mathbf{g}$ be any k-dimensional affine subspace of $\mathbb{R}^n$. Then there exist a k-dimensional vector subspace V and a point x_0 such that $\mathbf{g} = x_0 + V$. We define

$$\mathcal{L}^k_{\mathbf{g}}(B) := \mathcal{L}^k_V(B - x_0)$$

for each Borel set $B \subseteq \mathbf{g}$. The fact that this is independent of x_0 comes immediately from the translation-invariance of $\mathcal{L}^k_V$. Moreover, it is straightforward that B is a Borel set of $\mathbf{g}$ if and only if $B - x_0$ is a Borel set of V. Indeed being a Borel set of $\mathbb{R}^n$ is a translation-invariant property, and the Borel sets of V (resp. $\mathbf{g}$) are the intersections of the sets of $\mathcal{B}_{\mathbb{R}^n}$ with V (resp. $\mathbf{g}$); see Proposition A.1.2.

Remark A.3.3. Equivalently, we can also make a direct construction by considering an affine isometry $f\colon \mathbb{R}^k \to \mathbf{g}$ and letting $\mathcal{L}^k_{\mathbf{g}}(B) := \mathcal{L}^k\big(f^{-1}(B)\big)$ for each Borel set $B \subseteq \mathbf{g}$.

A.4 Hausdorff measures

We start this section by defining some constants that will be important later. In what follows, we recall that we use the notation

$$\kappa_m := \frac{\pi^{m/2}}{\Gamma\left(\frac{m}{2}+1\right)},$$

where Γ denotes the usual Gamma function:

$$\Gamma(x) := \int_0^\infty t^{x-1}e^{-t}\,dt, \quad x > 0$$

to denote the Lebesgue volume of the m-dimensional Euclidean unit ball.

Let (X, d) be a metric space. For $\delta > 0$, a (*countable*) δ-*covering* of a subset $A \subseteq X$ is a countable collection $(C_j)_{j\in\mathbb{N}} \subseteq 2^X$ of subsets of X such that $A \subseteq \bigcup_{j=1}^\infty C_j$ and $\operatorname{diam} C_j < \delta$ for each $j \in \mathbb{N}$. For any $m \geq 0$ we define the *m-dimensional Hausdorff measure* to be the function $\mathcal{H}^m\colon 2^X \to [0,\infty]$

$$\mathcal{H}^m(A) = \lim_{\delta\to 0^+} \mathcal{H}^m_\delta(A) \tag{A.4.1}$$

for each subset $A \subseteq X$, where we define

$$\mathcal{H}^m_\delta(A) := \kappa_m \inf\left\{\sum_{j=1}^\infty \left(\frac{\operatorname{diam} C_j}{2}\right)^m : (C_j)_{j\in\mathbb{N}} \subseteq 2^X \text{ is a countable } \delta-\text{covering of } A\right\},$$

for $A \neq \emptyset$, and $\mathcal{H}^m_\delta(\emptyset) = 0$. If a non-empty set $A \subseteq X$ does not admit a countable δ-covering, then the infimum above is defined to be ∞, and consequently in this case we have $\mathcal{H}^m_\delta(A) = \infty$. Moreover, it is readily seen that $\mathcal{H}^0$ is the *counting measure*, that is, $\mathcal{H}^0(A)$ is the number of elements of A in the case where A is finite, and $\mathcal{H}^0(A) = \infty$ if A has infinite elements.

Notice that if $\delta_1 \leq \delta_2$, then any δ_1-covering of A is also a δ_2-covering of A, from where we get that $\mathcal{H}^m_{\delta_2}(A) \leq \mathcal{H}^m_{\delta_1}(A)$. Hence, the limit in (A.4.1) can be written as

$$\lim_{\delta\to 0^+} \mathcal{H}^m_\delta(A) = \sup\{\mathcal{H}^m_\delta(A) : \delta > 0\},$$

from where we get that $\mathcal{H}^m(A)$ is well-defined, although if may be infinite. Even more, of course $\mathcal{H}^m(A) < \infty$ if and only if there exists $c \in \mathbb{R}$ such that $\mathcal{H}^m_\delta(A) < c$ for every $\delta > 0$.

From standard metric spaces theory arguments, we have that the diameter of a set is the same as the diameter of its closure. Hence we may consider δ-coverings by closed sets in the definition of $\mathcal{H}^m_\delta$ (and, consequently, in the definition of the m-dimensional Hausdorff measure). We also may assume that the δ-coverings are taken by open sets, since for any $\varepsilon > 0$ and every subset $A \subseteq X$ there exists an open set $U \supseteq A$ such that $\operatorname{diam}(U) \leq \operatorname{diam}(A)+\varepsilon$. Indeed, simply let $U = \{x \in X : \operatorname{dist}(x, A) < \varepsilon/2\}$. Moreover,

we may restrict ourselves to δ-coverings where each set intersects A (in fact, we can "discard" the sets which do not intersect A in any given covering).

Lemma A.4.1. *For each $m \geq 0$ and $\delta > 0$, $\mathcal{H}^m_\delta \colon 2^X \to [0,\infty]$ is an outer measure on (X,d).*

Proof. We have that $\mathcal{H}^m_\delta(\emptyset) = 0$ by definition, and hence we only have to prove that inequality (A.1.1) holds. Let $A \subseteq X$ and $(A_j)_{j\in\mathbb{N}}$ be a collection of subsets of X such that $A \subseteq \bigcup_{j\in\mathbb{N}} A_j$. Of course we may assume that $\mathcal{H}^m_\delta(A_j) < \infty$ for every $j \in \mathbb{N}$, since otherwise the desired inequality is trivial. For every $\varepsilon > 0$ and for each $j \in \mathbb{N}$, we may pick a countable δ-covering $(C^k_j)_{k\in\mathbb{N}}$ of A_j such that

$$\kappa_m \sum_{k=1}^{\infty} \left(\frac{\operatorname{diam} C^k_j}{2}\right)^m \leq \mathcal{H}^m_\delta(A_j) + \frac{\varepsilon}{2^j}.$$

Now observe that

$$A \subseteq \bigcup_{j=1}^{\infty} A_j \subseteq \bigcup_{j,k=1}^{\infty} C^k_j,$$

from where we get that $(C^k_j)_{j,k\in\mathbb{N}}$ is a countable δ-covering of A. Thus,

$$\begin{aligned}\mathcal{H}^m_\delta(A) \leq \kappa_m \sum_{j,k=1}^{\infty} \left(\frac{\operatorname{diam} C^k_j}{2}\right)^m &= \sum_{j=1}^{\infty} \left[\kappa_m \sum_{k=1}^{\infty} \left(\frac{\operatorname{diam} C^k_j}{2}\right)^m\right] \leq \\ &\leq \sum_{j=1}^{\infty} \left[\mathcal{H}^m_\delta(A_j) + \frac{\varepsilon}{2^j}\right] = \sum_{j=1}^{\infty} \mathcal{H}^m_\delta(A_j) + \varepsilon,\end{aligned}$$

and since $\varepsilon > 0$ is arbitrary we get (A.1.1).

□

Corollary A.4.1. *For every $m \geq 0$, the m-dimensional Hausdorff measure $\mathcal{H}^m \colon 2^X \to [0,\infty]$ is an outer measure on (X,d).*

Proof. We have $\mathcal{H}^m(\emptyset) = 0$ because $\mathcal{H}^m_\delta(\emptyset) = 0$ for every $\delta > 0$. To prove that inequality (A.1.1) holds for $\mathcal{H}^m$, notice that if $A \subseteq \bigcup_{j=1}^{\infty}$, then

$$\begin{aligned}\mathcal{H}^m(A) = \sup\{\mathcal{H}^m_\delta(A) : \delta > 0\} &\leq \sup\left\{\sum_{j=1}^{\infty} \mathcal{H}^m_\delta(A_j) : \delta > 0\right\} \leq \\ &\leq \sum_{j=1}^{\infty} \sup\{\mathcal{H}^m_\delta(A_j) : \delta > 0\} = \sum_{j=1}^{\infty} \mathcal{H}^m(A_j),\end{aligned}$$

where we are obviously using the fact that each $\mathcal{H}^m_\delta$ is an outer measure.

□

Let (X,d) be any metric space, and let $A \subseteq X$. The *Hausdorff dimension* of A is defined to be the number

$$\dim_{\mathcal{H}}(A) := \inf\{m \geq 0 : \mathcal{H}^m(A) = 0\},$$

provided there exists some number $m \geq 0$ such that $\mathcal{H}^m(A) = 0$. If such a number does not exist, then we set $\dim_{\mathcal{H}}(A) = \infty$.

Proposition A.4.1. *Let $A \subseteq (X, d)$ and $p \geq 0$. Then we have*

(i) *if $\mathcal{H}^p(A) < \infty$, then $\mathcal{H}^m(A) = 0$ for every $m > p$,*

(ii) *if $\mathcal{H}^p(A) > 0$, then $\mathcal{H}^m(A) = \infty$ for every $m < p$.*

As a consequence, if $0 < \mathcal{H}^p(A) < \infty$, then $\dim_{\mathcal{H}}(A) = p$.

Proof. For **(i)**, assume that $\mathcal{H}^p(A) < \infty$. Then there exists $c > 0$ such that $\mathcal{H}^p_\delta(A) < c$ for every $\delta > 0$. For any $m > p$, $\delta > 0$, and $(C_j)_{j\in\mathbb{N}}$ δ-covering of A, we have

$$\sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^m = \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^{m-p} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^p \leq \left(\frac{\delta}{2}\right)^{m-p} \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^p,$$

and hence

$$\begin{aligned}\mathcal{H}^m_\delta(A) = \kappa_m \inf\left\{\sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^m : (C_j)_{j\in\mathbb{N}} \text{ is a } \delta - \text{covering of } A\right\} \leq \\ \leq \kappa_m \left(\frac{\delta}{2}\right)^{m-p} \inf\left\{\sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^p : (C_j)_{j\in\mathbb{N}} \text{ is a } \delta - \text{covering of } A\right\} = \\ = \frac{\kappa_m}{\kappa_p} \left(\frac{\delta}{2}\right)^{m-p} \mathcal{H}^p_\delta(A) \leq c \frac{\kappa_m}{\kappa_p} \left(\frac{\delta}{2}\right)^{m-p}.\end{aligned}$$

Thus, we simply notice that

$$\mathcal{H}^m(A) = \lim_{\delta\to 0^+} \mathcal{H}^m_\delta(A) \leq \lim_{\delta\to 0^+} c \frac{\kappa_m}{\kappa_p} \left(\frac{\delta}{2}\right)^{m-p} = 0.$$

Now assume that $\mathcal{H}^p(A) > 0$. Then there exist numbers $c, \varepsilon > 0$ such that $\mathcal{H}^p_\delta(A) > c$ for $0 < \delta < \varepsilon$. Of course, this is because $\mathcal{H}^p(A) = \lim_{\delta\to 0^+} \mathcal{H}^p_\delta(A)$. Hence, for each $m < p$, any $\delta > 0$, and every δ-covering $(C_j)_{j\in\mathbb{N}}$ of A, we have

$$\sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^m = \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^p \left(\frac{\operatorname{diam}(C_j)}{2}\right)^{m-p} \geq \left(\frac{2}{\delta}\right)^{p-m} \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^p,$$

and similarly to what we did before, this leads to

$$\mathcal{H}^m_\delta(A) \geq \frac{\kappa_m}{\kappa_p} \left(\frac{2}{\delta}\right)^{p-m} \mathcal{H}^p_\delta(A).$$

If we restrict ourselves to the case where $0 < \delta < \varepsilon$, we get

$$\mathcal{H}^m_\delta(A) > c \frac{\kappa_m}{\kappa_p} \left(\frac{2}{\delta}\right)^{p-m}.$$

Finally, we obtain

$$\mathcal{H}^m(A) = \lim_{\delta\to 0^+} \mathcal{H}^m_\delta(A) \geq \lim_{\delta\to 0^+} c \frac{\kappa_m}{\kappa_p} \left(\frac{2}{\delta}\right)^{p-m} = \infty,$$

and the proof of **(ii)** is complete. The last claim comes from the fact that if $0 < \mathcal{H}^p(A) < \infty$, then the conditions in both **(i)** and **(ii)** are satisfied.

□

The next important proposition states that the m-dimensional Hausdorff measures are Borel regular outer measures.

Proposition A.4.2. *For any $m \geq 0$ we have*

$$\mathcal{H}^m_\delta(A \cup B) = \mathcal{H}^m_\delta(A) + \mathcal{H}^m_\delta(B)$$

whenever $2\delta < \operatorname{dist}(A, B)$. Consequently,

$$\mathcal{H}^m(A \cup B) = \mathcal{H}^m(A) + \mathcal{H}^m(B)$$

if $\operatorname{dist}(A, B) > 0$. In particular, $\mathcal{H}^m$ is a Borel regular outer measure.

Proof. If $(C_j)_{j\in\mathbb{N}}$ is a δ-covering of A and $(D_k)_{k\in\mathbb{N}}$ is a δ-covering of B, then the collection consisting of all C_js and D_ks is clearly a δ-covering of $A \cup B$, and

$$\mathcal{H}^m_\delta(A \cup B) \leq \kappa_m \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam} C_j}{2}\right)^m + \kappa_m \sum_{k=1}^{\infty} \left(\frac{\operatorname{diam} D_k}{2}\right)^m.$$

Since this holds for any δ-coverings of A and B, we can pass to the infimum on the right-hand side and this gives the inequality

$$\mathcal{H}^m_\delta(A \cup B) \leq \mathcal{H}^m_\delta(A) + \mathcal{H}^m_\delta(B).$$

Now assume that $(C_j)_{j\in\mathbb{N}}$ is a δ-covering of $A \cup B$ such that $C_j \cap (A \cup B) \neq \emptyset$ for every j. Define the sets of natural numbers

$$I_A := \{j \in \mathbb{N} : C_j \cap A \neq \emptyset\} \text{ and}$$
$$I_B := \{j \in \mathbb{N} : C_j \cap B \neq \emptyset\}.$$

Since $\operatorname{dist}(A, B) > 2\delta$, we get that $I_A \cap I_B = \emptyset$. The collection $(C_j)_{j\in I_A}$ is a δ-covering of A, and $(C_j)_{j\in I_B}$ is a δ-covering of B. Therefore,

$$\mathcal{H}^m_\delta(A) + \mathcal{H}^m_\delta(B) \leq \kappa_m \sum_{j\in I_A} \left(\frac{\operatorname{diam} C_j}{2}\right)^m + \kappa_m \sum_{j\in I_B} \left(\frac{\operatorname{diam} C_j}{2}\right)^m = \kappa_m \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam} C_j}{2}\right)^m,$$

because $I_A \cup I_B = \mathbb{N}$. The inequality above holds for any δ-covering $(C_j)_{j\in\mathbb{N}}$ of $A \cup B$ whose all sets intersect A or B. Hence we may pass to the infimum in the right-hand side above and obtain the reverse inequality

$$\mathcal{H}^m_\delta(A) + \mathcal{H}^m_\delta(B) \leq \mathcal{H}^m_\delta(A \cup B).$$

The second claim comes straightforwardly. To prove that $\mathcal{H}^m$ is Borel regular, first observe that from the Caratheodóry Criterion (Theorem 2.1.2), we already have that all Borel sets of (X, d) are $\mathcal{H}^m$-measurable. It remains to prove that for every subset $A \subseteq X$ there exists a Borel set $B \supseteq A$ with $\mathcal{H}^m(B) = \mathcal{H}^m(A)$. We clearly may assume that $\mathcal{H}^m(A) < \infty$, since otherwise we simply put $B = X$. For each $n \in \mathbb{N}$ there exists an open $1/n$-covering $\mathcal{U}_n = \{U_{n,j}\}_{j\in\mathbb{N}}$ of A such that

$$\kappa_m \sum_{j=1}^{\infty} \left[\frac{\operatorname{diam} U_{n,j}}{2}\right]^m < \mathcal{H}^m_{1/n}(A) + \frac{1}{n}.$$

Now we define

$$B = \bigcap_{n=1}^{\infty} \bigcup_{j=1}^{\infty} U_{n,j},$$

which is clearly a Borel set, because it is the countable intersection of open sets. It is also immediate that $B \supseteq A$. In particular, this gives the inequality $\mathcal{H}^m(A) \leq \mathcal{H}^m(B)$. We claim that equality $\mathcal{H}^m(B) = \mathcal{H}^m(A)$ holds. First, observe that for every set $E \subseteq 2^X$ we have

$$\mathcal{H}^m(E) = \lim_{n\to\infty} \mathcal{H}^m_{1/n}(E).$$

Now notice that $\mathcal{U}_n$ is a $1/n$-covering of B. Hence

$$\mathcal{H}^m_{1/n}(B) \leq \kappa_m \sum_{j=1}^{\infty} \left[\frac{\operatorname{diam} U_{n,j}}{2}\right]^m < \mathcal{H}^m_{1/n}(A) + \frac{1}{n}.$$

Making $n \to \infty$ on both sides we get $\mathcal{H}^m(B) \leq \mathcal{H}^m(A)$. This concludes the proof.

□

Remark A.4.1. It follows that any Borel set of (X, d) is $\mathcal{H}^m$-measurable and that the restriction of a Hausdorff measure to the Borel σ-algebra of (X, d) is a Borel measure. This is important for us because in most part of the applications we regard the Hausdorff measures as Borel measures, thus forgetting the fact that it is actually a measure on a possibly bigger collection of sets.

Next we prove the very important property of isometry invariance of Hausdorff measures. From now on, when there is possibility of confusion we denote the m-dimensional Hausdorff measure of a metric space (X, d) by $\mathcal{H}^m_X(\cdot)$.

Lemma A.4.2. *If $\varphi\colon (X, d_X) \to (Y, d_Y)$ is a bijective isometry between metric spaces, then*

$$\mathcal{H}^m_Y = \varphi_* \mathcal{H}^m_X$$

for every $m \geq 0$.

Proof. This follows immediately from the fact that $(C_j)_{j\in\mathbb{N}}$ is a δ-covering of a set $A \subseteq X$ if and only if $(\varphi(C_j))_{j\in\mathbb{N}}$ is a δ-covering of $\varphi(A) \subseteq Y$.

□

At this point, it is important to understand how a Hausdorff measure behaves when restricted to subsets. If (X, d) is a metric space, and $Y \subseteq X$, then the *restriction* of the Hausdorff measure $\mathcal{H}^m_X$ to Y is the usual restriction $\mathcal{H}^m_X\big|_Y\colon 2^Y \to [0, +\infty]$ defined as

$$\mathcal{H}^m_X\big|_Y(A) = \mathcal{H}^m_X(A),$$

for any $A \subseteq Y$ (this was introduced for general outer measures in Remark A.1.5). However, being a metric space with the induced metric, Y itself has an (intrinsic) m-dimensional Hausdorff measure $\mathcal{H}^m_Y(\cdot)\colon 2^Y \to [0, \infty]$.

Proposition A.4.3. *We have $\mathcal{H}^m_X(A) = \mathcal{H}^m_Y(A)$ for every $A \subseteq Y$. Consequently, $\mathcal{H}^m_X\big|_Y(\cdot) = \mathcal{H}^m_Y(\cdot)$.*

Proof. At the first moment, the difference between the measures $\mathcal{H}^m_X\big|_Y(\cdot)$ and $\mathcal{H}^m_Y(\cdot)$ is the following: to obtain the first one we consider δ-coverings by subsets of X, while the second one is defined via δ-coverings consisting of subsets of Y. Since any δ-covering by subsets of Y is, in particular, a δ-covering of subsets of X, we have the inequality

$$\mathcal{H}^m_X(A) \leq \mathcal{H}^m_Y(A)$$

for any $A \subseteq Y$. On the other hand, if $(C_j)_{j\in\mathbb{N}}$ is a δ-covering of $A \subseteq Y$ by subsets of X, then $(C_j \cap Y)_{j\in\mathbb{N}}$ is a δ-covering of A by subsets of Y. In other words, for each δ-covering by subsets of X it corresponds to a δ-covering by subsets of Y. Thus,

$$\mathcal{H}^m_Y(A) \leq \mathcal{H}^m_X(A),$$

for each $A \subseteq Y$. This concludes the proof.

□

Next we prove that the n-dimensional Hausdorff measure coincides with the Lebesgue measure in $\mathbb{R}^n$. We start with a result which bounds the Lebesgue measure of a set in terms of its diameter.

Theorem A.4.1 (Isodiametric inequality)**.** *Let $A \subseteq \mathbb{R}^n$. Then,*

$$\mathcal{L}^n(A) \leq \omega_n\Big(\frac{\mathrm{diam}A}{2}\Big)^n.$$

Proof. The inequality is trivial if $\mathrm{diam}A = \infty$, and hence we may assume that A is bounded. Even more, it is actually enough to prove that A is also closed (hence compact), because $\mathrm{diam}A = \mathrm{diam}(\mathrm{cl}(A))$, and $\mathcal{L}^n(A) \leq \mathcal{L}^n(\mathrm{cl}(A))$.

We will use Steiner symmetrizations with respect to the coordinates of $\mathbb{R}^n$, that is, with respect to the vectors of the canonical basis $\{e_1, \ldots, e_n\}$ of $\mathbb{R}^n$ (see the Section 2.7 for the definition). For $1 \leq j \leq n$, we denote the Steiner symmetral of A with respect to e_j as $S_j(A)$. It is readily seen that $S_j(A)$ is also compact, and from Fubini's theorem, we get that $\mathcal{L}^n(S_j(A)) = \mathcal{L}^n(A)$ (see also Proposition 2.7.1). Moreover, by item **(iv)** of Proposition 2.7.2, whose proof clearly does not rely on convexity, we have that $\mathrm{diam}(S_j(A)) \leq \mathrm{diam}A$. Therefore, applying the Steiner symmetrizations iteratively for all coordinate directions, we obtain the compact set $S(A) = S_1(S_2(\ldots(S_n(A))\ldots)$, for which we have $\mathcal{L}^n(S(A)) = \mathcal{L}^n(A)$ and $\mathrm{diam}(S(A)) \leq \mathrm{diam}(A)$. Since $S(A)$ is clearly symmetric with respect to all coordinates (because $\{e_1, \ldots, e_n\}$ is an orthogonal basis, and then the symmetrization with respect to a given coordinate does not flaw any already existing symmetry with respect to another coordinate), we get that $S(A)$ is centered at the origin. It follows that $S(A)$ is contained in the closed ball of radius $\frac{1}{2}\mathrm{diam}(S(A))$ centered at $0_{\mathbb{R}^n}$. Once κ_n is the volume of the unit ball, and using the derived relations between the measures and diameters of A and $S(A)$, we get that

$$\mathcal{L}^n(A) = \mathcal{L}^n(S(A)) \leq \kappa_n\Big(\frac{\mathrm{diam}S(A)}{2}\Big)^n \leq \kappa_n\Big(\frac{\mathrm{diam}A}{2}\Big)^n,$$

as we wanted to prove.

□

Lemma A.4.3. *If $A \subseteq \mathbb{R}^n$ is such that $\mathcal{L}^n(A) = 0$, then $\mathcal{H}^n_\delta(A) = 0$ for every $\delta > 0$.*

Remark A.4.2. It is clear that, in this case, we also have $\mathcal{H}^n(A) = 0$.

Proof. If $\mathcal{L}^n(A) = 0$, then for every $\varepsilon > 0$ there exists a collection $(C_j)_{j\in\mathbb{N}}$ of A by open n-cells such that $A \subseteq \bigcup_{j=1}^\infty C_j$ and

$$\sum_{j=1}^{\infty} \mathrm{vol}_n(C_j) < \varepsilon.$$

Let $\delta > 0$. Of course, each C_j can be written as the union of a finite collection of cubes with diameter smaller than δ, and with pairwise disjoint interiors. We denote this collection as $(C_j^k)_{k=1,\ldots,k(j)}$. Since each C_j^k is a cube, we have

$$\mathrm{vol}_n(C_j^k) = \Big(\frac{\mathrm{diam} C_j^k}{\sqrt{n}}\Big)^n.$$

The collection (C_j^k) is a δ-covering of A, and hence

$$\mathcal{H}^n_\delta(A) \leq \kappa_n \sum_{j=1}^{\infty}\sum_{k=1}^{k(j)} \Big(\frac{\mathrm{diam} C_j^k}{2}\Big)^n = \kappa_n \Big(\frac{\sqrt{n}}{2}\Big)^n \sum_{j=1}^{\infty}\sum_{k=1}^{k(j)} \mathrm{vol}_n C_j^k < \kappa_n \Big(\frac{\sqrt{n}}{2}\Big)^n \varepsilon.$$

Since $\omega_n(\sqrt{n}/2)^n$ is a constant and $\varepsilon > 0$ is arbitrary, we get that $\mathcal{H}^n_\delta(A) = 0$.

□

Theorem A.4.2. *We have*

$$\mathcal{L}^n(A) = \mathcal{H}^n(A) = \mathcal{H}^n_\delta(A)$$

for any $A \subseteq \mathbb{R}^n$ and every $\delta > 0$.

Proof. First, let $\delta > 0$ and pick any δ-covering $(C_j)_{j\in\mathbb{N}}$ of A. From the isodiametric inequality, and using standard properties of the Lebesgue measure, we get

$$\mathcal{L}^n(A) \leq \mathcal{L}^n\Big(\bigcup_{j=1}^{\infty} C_j\Big) \leq \sum_{j=1}^{\infty} \mathcal{L}^n(C_j) \leq \sum_{j=1}^{\infty} \kappa_n \Big(\frac{\mathrm{diam} C_j}{2}\Big)^n.$$

Passing to the infimum of the sums in the right-hand side above taken over all δ-coverings of A, we get that $\mathcal{L}^n(A) \leq \mathcal{H}^n_\delta(A)$.

Now we prove the reverse inequality. We say that a *k-dimensional cube* is a k-cell whose all sides have equal length. We will use a known fact that for any $\delta > 0$, any bounded open set $U \subseteq \mathbb{R}^n$ can be written as a countable union of cubes $(C_j)_{j\in\mathbb{N}}$ whose interiors are pairwise disjoint and such that $\mathrm{diam} C_j < \delta$. In this case, since the Lebesgue outer measure is a measure when restricted to the Borel σ-algebra of $\mathbb{R}^n$, and because the Lebesgue measure of a closed cube equals the Lebesgue measure of its interior, we also have that $\mathcal{L}^n(U) = \sum_{j=1}^{\infty} \mathcal{L}^n(C_j)$. Having this in mind, for each $j \in \mathbb{N}$ we may choose an open ball B_j contained in the interior of C_j such that $\mathrm{diam} B_j$ is greater than half the edge-length of C_j. Denoting by $\ell(C_j)$ the edge-length of C_j, we have that $\mathcal{L}^n(C_j) = \ell(C_j)^n$, and hence

$$\mathcal{L}^n(B_j) = \kappa_n \Big(\frac{\mathrm{diam} B_j}{2}\Big)^n > \kappa_n \Big(\frac{\ell(C_j)}{4}\Big)^n = \frac{\kappa_n}{4^n} \mathcal{L}^n(C_j),$$

from where we get that

$$\mathcal{L}^n(C_j \setminus B_j) = \mathcal{L}^n(C_j) - \mathcal{L}^n(B_j) < \Big(1 - \frac{\kappa_n}{4^n}\Big)\mathcal{L}^n(C_j).$$

Consequently, we get the inequality

$$\mathcal{L}^n\Big(U \setminus \bigcup_{j=1}^{\infty} B_j\Big) = \mathcal{L}^n\Big(\bigcup_{j=1}^{\infty} C_j \setminus B_j\Big) = \sum_{j=1}^{\infty} \mathcal{L}^n(C_j \setminus B_j) < \Big(1 - \frac{\kappa_n}{4^n}\Big) \sum_{j=1}^{\infty} \mathcal{L}^n(C_j) = \Big(1 - \frac{\kappa_n}{4^n}\Big)\mathcal{L}^n(U).$$

It follows that if $N_0 \in \mathbb{N}$ is large enough, we have the inequality

$$\mathcal{L}^n\Big(U \setminus \bigcup_{j=1}^{N_0} B_j\Big) \leq \Big(1 - \frac{\kappa_n}{4^n}\Big)\mathcal{L}^n(U).$$

Replacing U by $U \setminus \bigcup_{j=1}^{N} B_j$ (which is open and bounded) and repeating the argument, we find a finite collection $\{B_{N_0+1}, B_{N_0+2}, \ldots, B_{N_1}\}$ of pairwise disjoint open balls of radius smaller than δ such that

$$\mathcal{L}^n\Big(U \setminus \bigcup_{j=1}^{N_1} B_j\Big) = \mathcal{L}^n\Big(\Big(U \setminus \bigcup_{j=1}^{N_0} B_j\Big) \setminus \bigcup_{j=N_0+1}^{N_1} B_j\Big) \leq$$

$$\leq \Big(1 - \frac{\kappa_n}{4^n}\Big)\mathcal{L}^n\Big(U \setminus \bigcup_{j=1}^{N_0} B_j\Big) \leq \Big(1 - \frac{\kappa_n}{4^n}\Big)^2 \mathcal{L}^n(U).$$

Since $1 - \omega_n/4^n < 1$, repeating the construction inductively we obtain a pairwise disjoint family $(B_j)_{j\in\mathbb{N}}$ of open balls such that $B := \bigcup_{j=1}^{\infty} B_j \subseteq U$ and $\mathcal{L}^n(U \setminus B) = 0$.

Now let $(C_k)_{k\in\mathbb{N}}$ be a collection of open n-cells covering A. By the previous argument, for each k there is a collection $(B_j)_{j\in\mathbb{N}}$ of pairwise disjoint balls with diameter smaller than δ such that $B := \bigcup_{j=1}^{\infty} \subseteq C_k$ and $\mathcal{L}^n(C_k \setminus B) = 0$. From the previous lemma, we get that $\mathcal{H}^n_\delta(C_k \setminus B) = 0$, and hence, denoting by r_j the radius of the ball B_j, we get

$$\mathcal{H}^n_\delta(C_k) = \mathcal{H}^n_\delta(B) = \mathcal{H}^n_\delta\Big(\bigcup_{j=1}^{\infty} B_j\Big) \leq \sum_{j=1}^{\infty} \kappa_n r_j^n = \sum_{j=1}^{\infty} \mathcal{L}^n(B_j) = \mathcal{L}^n(B) = \mathcal{L}^n(C_k),$$

where we have to notice that B_j is a δ-covering for itself, and for that reason, $\mathcal{H}^n_\delta(B_j) \leq \omega_n r_j^n$. Consequently,

$$\mathcal{H}^n_\delta(A) \leq \mathcal{H}^n_\delta\Big(\bigcup_{k=1}^{\infty} C_k\Big) \leq \sum_{k=1}^{\infty} \mathcal{H}^n_\delta(C_k) \leq \sum_{k=1}^{\infty} \mathcal{L}^n(C_k),$$

and passing to the infimum on the right-hand side above among all coverings of A by open n-cells we get that $\mathcal{H}^n_\delta(A) \leq \mathcal{L}^n(A)$. This concludes the proof.

□

The coincidence of $\mathcal{L}^n$ and $\mathcal{H}^n$ has several consequences which we investigate next.

Corollary A.4.2. *The Lebesgue outer measure $\mathcal{L}^n$ is a Borel regular outer measure, and the restriction of $\mathcal{H}^n$ to $\mathcal{B}_{\mathbb{R}^n}$ is a regular Borel measure.*

Proof. The first claim follows straightforwadly from the equality $\mathcal{H}^n = \mathcal{L}^n$ since the n-dimensional Hausdorff outer measure is a Borel regular outer measure. For the second claim we only need to remember that $\mathcal{L}^n$ is a regular Borel measure when restricted to $\mathcal{B}_{\mathbb{R}^n}$.

□

Remark A.4.3. Observe that Proposition A.2.6 holds when the considered outer measure is $\mathcal{L}^n$. That is, if we integrate a Borel measurable function $f\colon \mathbb{R}^n \to (-\infty, +\infty]$ in the Lebesgue measure with respect to $\mathcal{B}_{\mathbb{R}^n}$ or $\Sigma_{\mathcal{L}^n}$ the result is the same.

Corollary A.4.3. *Any convex set $C \subseteq \mathbb{R}^n$ is Lebesgue measurable.*

Proof. If C does not have maximal affine dimension, then C is contained in some proper affine subspace of $\mathbb{R}^n$, and hence $\mathcal{L}^n(C) = 0$ (see Corollary A.3.2 and Remark A.3.1). Then, C is Lebesgue measurable (every set of zero measure is measurable; see Remark A.1.2). Thus, we may restrict ourselves to the case where C has non-empty interior. We first assume that C is bounded. Once we have $C = \operatorname{int}(C) \cup (C \cap \partial C)$, it suffices to prove that $C \cap \partial C$ is Lebesgue measurable (because $\operatorname{int}(C)$ is open, and hence Lebesgue measurable). Let $K = \operatorname{cl}(C)$. Then K is a convex body, and Corollary 2.1.2 gives that $C \cap \partial C \subseteq \partial K$. Thus,

$$\mathcal{L}^n(C \cap \partial C) \leq \mathcal{L}^n(\partial K) = \mathcal{H}^n(\partial K) = 0,$$

where the last equality comes from the fact that $\mathcal{H}^{n-1}(\partial K)$ is finite (proved in Theorem 4.4.1), together with Proposition A.4.1. Therefore, $\mathcal{L}^n(C \cap \partial C) = 0$, and this gives that $C \cap \partial C$ is $\mathcal{L}^n$-measurable.

In the case where C is not bounded, we cover C by countably many closed balls, and the intersection of C with each of these closed balls is a bounded convex set, hence Lebesgue measurable. It follows that C can be written as a countable union of Lebesgue measurable sets, and then C is Lebesgue measurable.

□

Remark A.4.4. It is not true that any convex subset of $\mathbb{R}^n$ is a Borel set (unless $n = 1$, of course: in this case all convex sets are intervals). We will not give a formal counterexample here, but intuitively one can think of taking the union of the interior of a convex body with a non-Borel subset of its boundary.

Corollary A.4.4. *Let $\mathbf{g}$ be a k-dimensional affine subspace of $\mathbb{R}^n$. The restriction of $\mathcal{H}^k$ to $\mathbf{g}$ coincides with the Lebesgue measure of $\mathbb{R}^k$ induced by any isometric isomorphism between $\mathbf{g}$ and $\mathbb{R}^k$ (see Construction A.3.1).*

Proof. Fixing any $x_0 \in \mathbf{g}$, we have that $V := \mathbf{g} - x_0$ is a vector subspace of $\mathbb{R}^n$. Since any Hausdorff measure on $\mathbb{R}^n$ is clearly translation-invariant, we may assume that $V = \mathbf{g}$. Let $T\colon \mathbf{g} \to \mathbb{R}^k$ be an isometric isomorphism. In view of Construction A.3.1, the Lebesgue measure of $\mathbb{R}^k$ is induced on $\mathbf{g}$ by

$$\mathcal{L}^k_{\mathbf{g}}(E) := \mathcal{L}^k(T(E))$$

for each Borel subset $E \subseteq \mathbf{g}$ (in this case, of course $T(E)$ is a Borel subset of $\mathbb{R}^k$), where $\mathcal{L}^k$ is the Lebesgue measure on $\mathbb{R}^k$. By the previous theorem, we have

$$\mathcal{H}^k_{\mathbb{R}^k}(T(E)) = \mathcal{L}^k(T(E)),$$

where $\mathcal{H}^k_{\mathbb{R}^k}$ denotes the k-dimensional Hausdorff measures on $\mathbb{R}^k$ (the notation $\mathcal{H}^k$ is reserved for the k-dimensional Hausdorff measure on $\mathbb{R}^n$). From Proposition A.4.3, it follows that $\mathcal{H}^k(E)$ can be calculated by taking δ-coverings of E by subsets of $\mathbf{g}$. Once T is an isometry between $\mathbf{g}$ and $\mathbb{R}^k$, each δ-covering of E by subsets of $\mathbf{g}$ corresponds to a δ-covering of $T(E)$ by subsets of $\mathbb{R}^k$, and vice-versa. Therefore,

$$\mathcal{H}^k(E) = \mathcal{H}^k_{\mathbb{R}^k}(T(E)).$$

From these equalities we obtain $\mathcal{L}^k_{\mathbf{g}}(E) = \mathcal{H}^k(E)$, and this is what we wanted to prove.

□

It is quite well-known, and easy to see from the definition, that the Lebesgue measure vanishes when restricted to an affine subspace (whose dimension is not total, of course). Once we have showed

that the restriction of a Hausdorff measure to a k-dimensional affine subspace can be obtained from the Lebesgue measure of $\mathbb{R}^k$, we will have something similar for Hausdorff measures. This is explained and proved below.

Corollary A.4.5. *Let* $\mathbf{g}$ *be an affine space of* $\mathbb{R}^n$ *such that* $\dim(\mathbf{g}) = k$. *Then for any bounded measurable subset* $E \subseteq \mathbf{g}$ *and every integer* $m > k$ *we have* $\mathcal{H}^m(E) = 0$.

Proof. Again, it is clear that we may assume that $\mathbf{g}$ is a vector subspace of $\mathbb{R}^n$. Let $\mathbf{h}$ be any m-dimensional vector subspace of $\mathbb{R}^n$ containing $\mathbf{g}$, and identify $\mathbf{h}$ with $\mathbb{R}^m$ by an isometric isomorphism T. By doing so, $T(\mathbf{g})$ is a vector subspace of $\mathbb{R}^m$ which does not have maximal dimension (because $m > k$). We let $\{u_1, \ldots, u_m\}$ be an orthonormal basis of $\mathbb{R}^m$ such that $\{u_1, \ldots, u_k\}$ is a basis of $T(\mathbf{g})$. Up to an orthogonal transformation (which does not modify any measure), we may assume that $\{u_1, \ldots, u_m\}$ is the canonical basis of $\mathbb{R}^m$.

If E is a bounded subset of $\mathbf{g}$, then $T(E)$ is a bounded subset of $T(\mathbf{g})$. Once we are assuming that $T(\mathbf{g})$ is spanned by the k first vectors of the canonical basis of $\mathbb{R}^m$, there is a k-cell $[a_1, b_1] \times \ldots \times [a_k, b_k]$ containing $T(E)$. Hence for every $\varepsilon > 0$ the m-cell

$$C_\varepsilon := [a_1, b_1] \times \ldots \times [a_k, b_k] \times [-\varepsilon, \varepsilon] \times [-\varepsilon, \varepsilon]$$

contains $T(E)$. Once and the m-dimensional volume of C_ε goes to 0 as $\varepsilon \to 0$, we obtain that $\mathcal{L}^m(T(E)) = 0$. Now we recall that $\mathcal{H}^m(E) = \mathcal{H}^m|_{\mathbf{g}}(E) = \mathcal{L}^m(T(E)) = 0$ and the proof is complete. □

We also need some results about the behavior of Hausdorff measures under Lipschitz mappings. These are stated (and sometimes proved) below. Recall that a map $f\colon (X, d_X) \to (Y, d_Y)$ between metric spaces is said to be a *Lipschitz map* if there exists a constant $c \geq 0$ such that

$$d_Y(f(x), f(z)) \leq c d_X(x, z)$$

for every $x, z \in X$. The *Lipschitz constant* of a Lipschitz map is the number

$$\mathrm{Lip}(f) := \sup \left\{ \frac{d_Y(f(x), f(z))}{d_X(x, z)} : x, z \in X,\ x \neq z \right\}.$$

Notice that if we allow that $\mathrm{Lip}(f) \in [0, +\infty]$, then we can define the Lipschitz constant for any map between metric spaces, and the Lipschitz maps are precisely the ones for which $\mathrm{Lip}(f) < +\infty$.

Theorem A.4.3. *Let* X, Y *be compact metric spaces, and let* $f\colon X \to Y$ *be a Lipschitz map. For any* $m \geq 0$, *and every subset* $A \subseteq X$ *we have the inequality*

$$\mathcal{H}^m\big(f(A)\big) \leq (\mathrm{Lip}(f))^m \mathcal{H}^m(A).$$

Remark A.4.5. Here we denote $\mathcal{H}^m_X$ and $\mathcal{H}^m_Y$ simply by $\mathcal{H}^m$ because it is clear from the context where each Hausdorff measure is being taken.

Proof. For an arbitrarily given $\delta > 0$, let $(C_j)_{j \in \mathbb{N}}$ be a δ-covering of A. Then it is clear that $(f(C_j))_{j \in \mathbb{N}}$ is a covering of $f(A)$ and that

$$\mathrm{diam}(f(C_j)) \leq \mathrm{Lip}(f)\mathrm{diam}(C_j)$$

for each $j \in \mathrm{N}$. Thus,

$$\mathcal{H}^m_\delta(f(A)) \le \kappa_m \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(f(C_j))}{2}\right)^m \le \operatorname{Lip}(f)^m \kappa_m \sum_{j=1}^{\infty} \left(\frac{\operatorname{diam}(C_j)}{2}\right)^m.$$

Passing to the supremum on the right-hand side (over the δ-coverings of A) we get that

$$\mathcal{H}^m_\delta(f(A)) \le \operatorname{Lip}(f)^m \mathcal{H}^m_\delta(A),$$

and since this holds for any $\delta > 0$ we obtain the desired inequality.

□

Corollary A.4.6. *For any $t > 0$, $A \subseteq \mathbb{R}^n$, and $m \ge 0$ we have*

$$\mathcal{H}^m(tA) = t^m \mathcal{H}^m(A).$$

Proof. For any $t > 0$ the *dilation map* $\varphi_t(x) = tx$ is a Lipschitz bijection of $\mathbb{R}^n$ whose Lipschitz constant is t. Moreover, its inverse is also a Lipschitz map, having Lipschitz constant $1/t$. Therefore, for each $m \ge 0$ and $A \subseteq \mathbb{R}^n$ we get

$$\mathcal{H}^m(tA) = \mathcal{H}^m(\varphi_t(A)) \le t^m \mathcal{H}^m(A) = t^m \mathcal{H}^{n-1}(\varphi_t^{-1}(tA)) \le t^m \frac{1}{t^m} \mathcal{H}^m(tA) = \\ = \mathcal{H}^m(tA),$$

and this concludes the proof.

□

As another corollary, we have that the $(n-1)$-dimensional Hausdorff measure is "monotone" under inclusion of convex bodies (notice that these measures are finite; this is discussed in Section 4.5).

Corollary A.4.7. *Let $K, L \subseteq \mathbb{R}^n$ be convex bodies such that $L \subseteq K$. Then $\mathcal{H}^{n-1}(\partial L) \le \mathcal{H}^{n-1}(\partial K)$.*

Proof. The restriction of the metric projection $p_L \colon \mathbb{R}^n \to L$ to ∂K is a Lipschitz map onto ∂L whose Lipschitz constant is not greater than 1 (this comes from Proposition 2.2.1 and from the clear fact that the metric projection onto L is realized on ∂L for any point which is not in the interior of L). Moreover, we have that $\partial L \subseteq p_L(\partial K)$, meaning that any point of ∂L is the metric projection of some point of ∂K. In fact, for any given $x \in \partial L$ consider the intersection z of the half-line $x + \mathbb{R}_+ u_L(x)$ with ∂K, where $u_L(x)$ is any outer normal of L at x. Then $p_L(z) = x$. From Theorem A.4.3 we get

$$\mathcal{H}^{n-1}(\partial L) \le 1^{n-1} \mathcal{H}^{n-1}(p_L(\partial K)) \le \mathcal{H}^{n-1}(\partial K),$$

and thus we have the desired inequality.

□

A.5 Product measures

In this section, we will define a σ-algebra and a measure in the Cartesian product of measurable spaces, and investigate integration with respect to this measure (this is the classical Fubini's theorem, which is very important for convexity). For this section, we follow mainly Salamon [2090].

Let (X, Σ_X) and (Y, Σ_Y) be measurable spaces. A *measurable rectangle* of $X \times Y$ is a set of the form $A \times B$ where $A \in \Sigma_X$ and $B \in \Sigma_Y$. The *product σ-algebra* of $X \times Y$, denoted by $\Sigma_X \otimes \Sigma_Y$, is the smaller σ-algebra on $X \times Y$ which contain all measurable rectangles. Formally, $\Sigma_X \otimes \Sigma_Y$ is the intersection

of all σ-algebras of $X \times Y$ containing every measurable rectangle of $X \times Y$. We start investigating the product σ-algebra by showing that the "sections" of measurable sets and functions of $\Sigma_X \otimes \Sigma_Y$ are measurable.

Lemma A.5.1. *Let $E \in \Sigma_X \otimes \Sigma_Y$, and let $x \in X$. Then the set*

$$E_x := \big\{ y \in Y : (x, y) \in E \big\}$$

is a set of Σ_Y. Also, if $y \in Y$, then the set

$$E^y := \big\{ x \in X : (x, y) \in E \big\}$$

is a set of Σ_X.

Remark A.5.1. We will often refer to set E_x as the *x-section* of E, and to set E^y as the *y-section* of E.

Proof. We prove the first claim. Consider the family $\mathcal{A}$ of subsets of $X \times Y$ given as

$$\mathcal{A} := \big\{ E \subseteq X \times Y : E_x \in \Sigma_Y \text{ for all } x \in X \big\}.$$

We show that $\mathcal{A}$ is a σ-algebra. It is immediate that $X \times Y \in \mathcal{A}$, and that $\emptyset \in \mathcal{A}$. Moreover, if $E \in \mathcal{A}$, then $(X \times Y) \setminus E$ is also a set of $\mathcal{A}$. Indeed, for any $x \in X$, we have

$$[(X \times Y) \setminus E]_x = \big\{ y \in Y : (x, y) \in (X \times Y) \setminus E \big\} = \big\{ y \in Y : (x, y) \notin E \big\} = Y \setminus E_x$$

and since $E_x \in \Sigma_Y$ we get that $Y \setminus E_x \in \Sigma_Y$. It remains to prove that $\mathcal{A}$ is closed under countable unions. Let $(E_m)_{m \in \mathbb{N}}$ be a sequence of sets in $\mathcal{A}$, and let

$$E := \bigcup_{m=1}^{\infty} E_m.$$

Then, for each $x \in X$,

$$E_x = \big\{ y \in Y : (x, y) \in E \big\} = \big\{ y \in Y : (x, y) \in E_m \text{ for some } m \in \mathbb{N} \big\} = \bigcup_{m=1}^{\infty} (E_m)_x,$$

and the latter is a set of Σ_Y because each set $(E_m)_x$ is in Σ_Y. Hence we have that $\mathcal{A}$ is a σ-algebra of $X \times Y$.

Now we observe that $\mathcal{A}$ contains all measurable rectangles of $X \times Y$. Indeed, if $A \in \Sigma_X$ and $B \in \Sigma_Y$, then for any $x \in X$ we have that either $E_x = \emptyset$ (if $x \notin A$) or $E_x = B$ (in the case where $x \in A$). Hence E_x is always a set of Σ_Y.

Finally, once $\mathcal{A}$ is a σ-algebra of $X \times Y$ which contains all measurable rectangles, we have that $\Sigma_X \otimes \Sigma_Y \in \mathcal{A}$. As a consequence, for every $E \in \Sigma_X \otimes \Sigma_Y$ and any $x \in X$ it holds that $E_x \in \Sigma_Y$. The second claim is completely analogous.

□

Lemma A.5.2. *Let $f : X \times Y \to [-\infty, +\infty]$ be a function which is measurable with respect to $\Sigma_X \otimes \Sigma_Y$. For any fixed $x \in X$, the function $f_x : Y \to [-\infty, +\infty]$ defined as*

$$f_x(y) := f(x, y)$$

for each $y \in Y$ is measurable with respect to Σ_Y. Similarly, for any fixed $y \in Y$ the function $f^y : X \to [-\infty, +\infty]$ given as $f^y(x) = f(x, y)$ for $x \in X$ is measurable with respect to Σ_X.

Remark A.5.2. The functions f_x and f^y are called the *x-section* and the *y-section* of f, respectively.

Proof. Fix $x \in X$ and let $U := (a, +\infty]$ for some $a \in \mathbb{R}$. Since f is measurable, we have that $f^{-1}(U) \in \Sigma_X \otimes \Sigma_Y$, from where we have that

$$[f^{-1}(U)]_x := \left\{y \in Y : (x, y) \in f^{-1}(U)\right\} \in \Sigma_Y.$$

We claim that $[f^{-1}(U)]_x = f_x^{-1}(U)$. Indeed, if $y \in [f^{-1}(U)]_x$, then $f(x, y) \in U$, and hence $f_x(y) \in U$. Thus, $y \in f_x^{-1}(U)$. On the other hand, if $y \in f_x^{-1}(U)$, then $f(x, y) = f_x(y) \in U$, and hence $(x, y) \in f^{-1}(U)$. This leads to $y \in [f^{-1}(U)]_x$. Consequently, we get that $f_x^{-1}(U) \in \Sigma_Y$ for any set $U = (a, +\infty]$, and this gives that f_x is measurable with respect to Σ_Y. The case for the functions f^y is analogous.

□

We proceed to a characterization of the product σ-algebra which will be useful later. A *monotone class* of a set X is a family $\mathcal{A}$ of subsets of X having the following properties:

(i) For any sequence $(A_m)_{m\in\mathbb{N}}$ in $\mathcal{A}$ with the property that $A_m \subseteq A_{m+1}$ for every $m \in \mathbb{N}$ we have $\bigcup_{m=1}^{\infty} A_m \in \mathcal{A}$.

(ii) If $(B_m)_{m\in\mathbb{N}}$ is a sequence in $\mathcal{A}$ such that $B_m \supseteq B_{m+1}$ for each $m \in \mathbb{N}$, then $\bigcap_{m=1}^{\infty} B_m \in \mathcal{A}$.

We will often refer to the unions and intersections as above as *monotone unions* and *monotone intersections*. With this terminology, a monotone class is a collection which is closed under monotone unions and monotone intersections.

It is readily seen that any σ-algebra is a monotone class. Returning to the case of the product σ-algebra, we say that a set $E \subseteq X \times Y$ is *elementary* if it can be expressed as a finite union of disjoint measurable rectangles.

Lemma A.5.3. *Let $E, F \subseteq X \times Y$ be elementary sets. Then, $E \cap F$, $E \cup F$, and $E \setminus F$ are also elementary sets.*

Proof. First of all, it is immediate to check that the union of **disjoint** elementary sets is an elementary set. In fact, if E and F are disjoint elementary sets, then expressing E and F as respective finite unions of measurable rectangles, we have that all of these rectangles must be disjoint, and hence the union $E \cup F$ is a finite union of disjoint measurable rectangles. Of course this extends to any finite union of disjoint elementary sets.

Let $A \times B$ and $C \times D$ be a measurable rectangles of $X \times Y$. Then,

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D),$$

and hence $(A \times B) \cap (C \times D)$ is a measurable rectangle. Now write

$$E = \bigcup_{m=1}^{N} A_m \times B_m \text{ and } F = \bigcup_{j=1}^{N} C_j \times D_j,$$

where each union is disjoint, and we assume for the sake of simplicity that both E and F are expressed as unions of the same number of rectangles. We may do that by adding sets $\emptyset \times \emptyset$ in the union which has the smallest number of rectangles. We write

$$E \cap F = \left(\bigcup_{m=1}^{N} A_m \times B_m \right) \cap \left(\bigcup_{j=1}^{N} C_j \times D_j \right) = \bigcup_{m,j=1}^{N} (A_m \times B_m) \cap (C_j \cap D_j) = \\ = \bigcup_{m,j=1}^{N} (A_m \cap C_j) \times (B_m \cap D_j),$$

and the latter is clearly a union of disjoint measurable rectangles, hence an elementary set.

Now we move forward to the set difference. Letting again $A \times B$ and $C \times D$ be measurable rectangles, we may write

$$(A \times B) \setminus (C \times D) = \big((A \setminus C) \times B\big) \cup ((A \cap C) \times (B \setminus D)),$$

and this shows that the difference of measurable rectangles is an elementary set (because the right-hand side of the above is a union of disjoint measurable rectangles). Using the same expression for E and F as before, we write

$$E \setminus F = \left(\bigcup_{m=1}^{N} A_m \times B_m \right) \setminus \left(\bigcup_{j=1}^{N} C_j \times D_j \right) = \bigcup_{m=1}^{N} \left[(A_m \times B_m) \setminus \left(\bigcup_{j=1}^{N} C_j \times D_j \right) \right] = \\ = \bigcup_{m=1}^{N} \left[\bigcap_{j=1}^{N} \big((A_m \times B_m) \setminus (C_j \times D_j) \big) \right].$$

Each set $(A_m \times B_m) \setminus (C_j \times D_j)$ is an elementary set, and we already have that the intersection of elementary sets is an elementary set. Thus, for each $m = 1, \ldots, N$, the set

$$\bigcap_{j=1}^{N} \big((A_m \times B_m) \setminus (C_j \times D_j) \big)$$

is elementary. If $m_1, m_2 \in \{1, \ldots, N\}$ are distinct, then

$$\left[\bigcap_{j=1}^{N} \big((A_{m_1} \times B_{m_1}) \setminus (C_j \times D_j) \big) \right] \cap \left[\bigcap_{j=1}^{N} \big((A_{m_2} \times B_{m_2}) \setminus (C_j \times D_j) \big) \right] = \emptyset,$$

because $(A_{m_1} \times B_{m_1}) \cap (A_{m_2} \times B_{m_2}) = \emptyset$ (since E is expressed as a union of disjoint measurable rectangles). It follows that $E \setminus F$ is a finite union of disjoint elementary sets, and by the comment in the beginning of the proof it follows that $E \setminus F$ is elementary.

Last, to prove that $E \cup F$ is elementary, simply write $E \cup F = (E \setminus F) \cup F$. The sets $E \setminus F$ and F are elementary and disjoint, and hence their union is an elementary set (using again the comment in the beginning of the proof).

□

Proposition A.5.1. *Let (X, Σ_X) and (Y, Σ_Y) be measurable spaces. The product σ-algebra $\Sigma_X \otimes \Sigma_Y$ is the smallest monotone class of $X \times Y$ which contains all elementary sets.*

Proof. Let $\mathcal{R} \subseteq 2^{X \times Y}$ be the collection of all elementary subsets of $X \times Y$, and let $\mathcal{A}$ be the smallest monotone class which contains $\mathcal{R}$ (that is, $\mathcal{A}$ is the intersection of all monotone classes of $X \times Y$ which contain $\mathcal{R}$). Because $\Sigma_X \otimes \Sigma_Y$ is a σ-algebra, it is a monotone class. Moreover, $\Sigma_X \otimes \Sigma_Y$ contains all measurable rectangles of $X \times Y$, and hence it also contains all elementary sets (any σ-algebra is closed

under finite unions, in particular). Thus, $\Sigma_X \otimes \Sigma_Y$ is a monotone class containing $\mathcal{R}$, and this gives the inclusion

$$\mathcal{A} \subseteq \Sigma_X \otimes \Sigma_Y,$$

and all we have to do is to prove the reverse inclusion. To do so, it is sufficient to show that $\mathcal{A}$ is a σ-algebra (indeed, any σ-algebra which contains all measurable rectangles of $X \times Y$ also contains $\Sigma_X \times \Sigma_Y$). We start by defining, for each $E \subseteq X \times A$, the collection

$$\mathcal{A}(E) := \big\{F \subseteq X \times Y : E \setminus F,\ F \setminus F \text{ and } E \cup F \text{ are sets of } \mathcal{A}\big\}.$$

Notice that if E_1 and E_2 are subsets of $X \times Y$, then it is immediate from the definition above that $E_1 \in \mathcal{A}(E_2)$ if and only if $E_2 \in \mathcal{A}(E_1)$.

We claim that $\mathcal{A}(E)$ is a monotone class. To verify that, first let $(F_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{A}(E)$ such that $F_m \subseteq F_{m+1}$ for any $m \in \mathbb{N}$. Since each F_m is a set of $\mathcal{A}(E)$, we have that $E \setminus F_m$, $F_m \setminus E$, and $E \cup F_m$ are sets of $\mathcal{A}(E)$ for each $m \in \mathbb{N}$. If we let

$$F := \bigcup_{m=1}^{\infty} F_m,$$

then we may write

$$E \setminus F = E \setminus \bigcup_{m=1}^{\infty} F_m = \bigcap_{m=1}^{\infty} (E \setminus F_m),$$

and the inclusion $F_m \subseteq F_{m+1}$ leads to $E \setminus F_m \supseteq E \setminus F_{m+1}$. Therefore, the intersection in the right-hand side of the above equality is a set of $\mathcal{A}$ (recall that $\mathcal{A}$ is a monotone class). Now we write

$$F \setminus E = \left(\bigcup_{m=1}^{\infty} F_m\right) \setminus E = \bigcup_{m=1}^{\infty} (F_m \setminus E),$$

and $F_m \setminus E \subseteq F_{m+1} \setminus E$ for each $m \in \mathbb{N}$. Then the right-hand side of the above is also a set of $\mathcal{A}$. Finally, we have

$$E \cup F = \bigcup_{m=1}^{\infty} (E \cup F_m) \in \mathcal{A},$$

where the latter holds because $E \cup F_m \subseteq E \cup F_{m+1}$ for each $m \in \mathbb{N}$. Hence we have $F \in \mathcal{A}(E)$. Similarly, we prove that the intersection of a sequence $(G_m)_{m\in\mathbb{N}}$ in $\mathcal{A}(E)$ with the property that $G_m \supseteq G_{m+1}$ for each $m \in \mathbb{N}$ is also a set of $\mathcal{A}(E)$. The details are left for the reader. This gives that $\mathcal{A}(E)$ is in fact a monotone class.

Now assume that $E \subseteq X \times Y$ is an elementary set. Then we notice that $\mathcal{A}(E)$ is a monotone class which contains $\mathcal{R}$. Indeed, if $F \in \mathcal{R}$, then by Lemma A.5.3 the sets $E \setminus F$, $F \setminus E$, and $E \cap F$ are in $\mathcal{A}$, and hence $F \in \mathcal{A}(E)$. Since $\mathcal{A}$ is the smallest monotone class of $X \times Y$ containing $\mathcal{R}$, we get that $\mathcal{A} \subseteq \mathcal{A}(E)$. Even more, we claim that if $F \subseteq \mathcal{A}$, then $\mathcal{A} \subseteq \mathcal{A}(F)$. In fact, once $\mathcal{A} \subseteq \mathcal{A}(E)$ for every elementary set $E \subseteq X \times Y$, we get that $F \in \mathcal{A}(E)$ for any elementary set $E \subseteq X \times Y$ (recall that $F \in \mathcal{A}$). It follows that $E \subseteq \mathcal{A}(F)$, and this shows that $\mathcal{A}(F)$ contains all elementary sets of $X \times Y$. Using again the fact that $\mathcal{A}$ is the smallest monotone class containing all elementary sets of $X \times Y$, we get that $\mathcal{A} \subseteq \mathcal{A}(F)$.

Now let $F, G \in \mathcal{A}$. From what we have just proved, we get that $G \in \mathcal{A}(F)$, and from the definition of $\mathcal{A}(F)$, it follows that $G \setminus F \in \mathcal{A}$ and $G \cup F \in \mathcal{A}$. It follows that $\mathcal{A}$ is closed under unions and set differences. As a consequence, we obtain that $\mathcal{A}$ is a σ-algebra in $X \times Y$. In fact, it is clear that $\emptyset, X \times Y \in \mathcal{A}$. Because $\mathcal{A}$ is closed under set differences and $X \times Y \in \mathcal{A}$, we obtain that $\mathcal{A}$ is closed under complementation (with respect to $X \times Y$). Finally, if $(F_m)_{m\in\mathbb{N}}$ is a sequence in $\mathcal{A}$, then defining

$$G_j := \bigcup_{m=1}^{j} F_m$$

for each $j \in \mathbb{N}$ we have a sequence in $\mathcal{A}$ (we proved that $\mathcal{A}$ is closed under finite unions) with the property that $G_j \subseteq G_{j+1}$ for each $j \in \mathbb{N}$. It is clear that

$$\bigcup_{m=1}^{\infty} F_m = \bigcup_{j=1}^{\infty} G_j,$$

and since $\mathcal{A}$ is a monotone class, we get that

$$\bigcup_{j=1}^{\infty} G_j \in \mathcal{A}.$$

This proves that $\mathcal{A}$ is closed under countable unions.

Since $\mathcal{A}$ is a σ-algebra which contains all measurable rectangles (because it contains all elementary sets), and once $\Sigma_X \otimes \Sigma_Y$ is the smallest σ-algebra of $X \times Y$ having this property, we obtain that $\Sigma_X \otimes \Sigma_Y \subseteq \mathcal{A}$, and this concludes the proof.

□

Now that we have a natural way of endowing a Cartesian product of measurable spaces with a σ-algebra, we move forward to investigate how to endow the product of measure spaces with a natural measure. In what follows, recall that a measure space (X, Σ_X, μ_X) is said to be *σ-finite* if there exists a collection $(X_m)_{m\in\mathbb{N}}$ of sets of Σ_X such that $X = \bigcup_{m=1}^{\infty} X_m$ and $\mu_X(X_m) < \infty$ for every $m \in \mathbb{N}$.

Theorem A.5.1. *Let (X, Σ_X, μ_X) and (Y, Σ_Y, μ_Y) be σ-finite measure spaces. For any given set $E \in \Sigma_X \otimes \Sigma_Y$, consider the functions $\varphi_E : X \to [0, +\infty]$ and $\psi_E : Y \to [0, +\infty]$ given as*

$$\varphi_E(x) = \mu_Y(E_x) \text{ and}$$
$$\psi_E(y) = \mu_X(E^y),$$

respectively, where the sets E_x and E^y are defined as in Lemma A.5.1. Then the functions φ_E and ψ_E are measurable, and

$$\int_X \varphi_E \, d\mu_X = \int_Y \psi_E \, d\mu_Y. \tag{A.5.1}$$

This equality can also be written as

$$\int_X \mu_Y(E_x) \, d\mu_X = \int_Y \mu_X(E^y) \, d\mu_Y. \tag{A.5.2}$$

Proof. Consider the collection

$$\mathcal{A} := \Big\{ E \in \Sigma_X \otimes \Sigma_Y : \varphi_E \text{ and } \psi_E \text{ are measurable, and (A.5.1) holds} \Big\}.$$

We claim that $\mathcal{A}$ contains any measurable rectangle of $X \times Y$. In fact, if $A \in \Sigma_X$ and $B \in \Sigma_Y$, then it is immediate to see that $\varphi_E = \mu_Y(B)1_A$ and $\psi_E = \mu_X(A)1_B$. Hence φ_E and ψ_E are measurable. Also,

$$\int_X \varphi_E \, d\mu_X = \int_X \mu_Y(B)1_A \, d\mu_X = \mu_Y(B)\mu_X(A) = \mu_X(A) \int_Y 1_B \, d\mu_Y = \\ = \int_Y \mu_X(A)1_B \, d\mu_Y = \int_Y \psi_E \, d\mu_Y.$$

We also have that $\mathcal{A}$ is closed under finite disjoint unions. To verify that, let $E_1, E_2 \in \mathcal{A}$ be such that $E_1 \cap E_2 = \emptyset$. For the sake of simplicity, denote $E = E_1 \cup E_2$. Then for each $x \in X$, we have

$$\varphi_E(x) = \mu_Y(E_x) = \mu_Y((E_1 \cup E_2)_x) = \mu_Y((E_1)_x \cup (E_2)_x) = \\ = \mu_Y((E_1)_x) + \mu_Y((E_2)_x) = \varphi_{E_1}(x) + \varphi_{E_2}(x),$$

where we notice that $(E_1)_x \cap (E_2)_x = \emptyset$ since otherwise we would have some point $(x, y) \in E_1 \cap E_2$. It follows that φ_E is measurable, once the sum of measurable functions is measurable. Similarly, we have that $\psi_E = \psi_{E_1} + \psi_{E_2}$, and hence the function ψ_E is also measurable. Since E_1 and E_2 are sets of $\mathcal{A}$, we have that equality (A.5.1) holds for (the functions associated with) these sets, and then

$$\int_X \varphi_E \, d\mu_X = \int_X \varphi_{E_1} + \varphi_{E_2} \, d\mu_X = \int_Y \psi_{E_1} + \psi_{E_2} \, d\mu_Y = \int_Y \psi_E \, d\mu_Y,$$

and this proves that $E \in \mathcal{A}$. As a consequence, we get that any elementary set of $X \times Y$ is contained in $\mathcal{A}$.

Next we show that $\mathcal{A}$ is closed under monotone unions. For that sake, first let $(E_m)_{m\in\mathbb{N}}$ be a sequence in $\mathcal{A}$ having the property that $E_m \subseteq E_{m+1}$ for each $m \in \mathbb{N}$, and let $E := \bigcup_{m\in\mathbb{N}} E_m$. Then for each $x \in X$ it is clear that $(E_m)_x \subseteq (E_{m+1})_x$ for every $m \in \mathbb{N}$. Also, it is immediate that

$$E_x = \left(\bigcup_{m=1}^{\infty} E_m\right)_x = \bigcup_{m=1}^{\infty} (E_m)_x$$

and using Lemma A.1.1 (see also Remark A.1.4) we get

$$\varphi_E(x) = \mu_Y(E_x) = \mu_Y\left(\bigcup_{m=1}^{\infty} (E_m)_x\right) = \lim_{m\to\infty} \mu_Y((E_m)_x) = \lim_{m\to\infty} \varphi_{E_m}(x)$$

for any $x \in X$. Thus, φ_E is the limit of measurable functions, and hence it is also measurable. Analogously, we have that

$$\psi_E(y) = \lim_{m\to\infty} \psi_{E_m}(y)$$

for each $y \in Y$. Therefore, ψ_E is also measurable.

The pointwise convergences $\varphi_{E_m} \to \varphi_E$ and $\psi_{E_m} \to \psi_E$ are monotone (non-decreasing). Indeed, the inclusion $(E_m)_x \subseteq (E_{m+1})_x$ leads to $\varphi_{E_m}(x) = \mu_Y((E_m)_x) \leq \mu_Y((E_{m+1})_x) = \varphi_{E_{m+1}}(x)$ for every $x \in X$, and the argument for ψ_{E_m} is the same. From the monotone convergence theorem (Theorem A.2.2), it follows that

$$\int_X \varphi_E \, d\mu_X = \int_X \lim_{m\to\infty} \varphi_{E_m} \, d\mu_X = \lim_{m\to\infty} \int_X \varphi_{E_m} \, d\mu_X = \lim_{m\to\infty} \int_Y \psi_{E_m} \, d\mu_Y = \\ = \int_Y \lim_{m\to\infty} \psi_{E_m} \, d\mu_Y = \int_Y \psi_E \, d\mu_Y,$$

and this shows that $E \in \mathcal{A}$.

The case of monotone intersections is not so straightforward. Once X and Y are σ-finite measure spaces, we may let $(X_m)_{m\in\mathbb{N}}$ and $(Y_m)_{m\in\mathbb{N}}$ be sequences in X and Y respectively such that $\mu_X(X_m) < \infty$ and $\mu_Y(Y_m) < \infty$ for each $m \in \mathbb{N}$, and with the property that

$$X = \bigcup_{m=1}^{\infty} X_m \quad \text{and} \quad Y = \bigcup_{m=1}^{\infty} Y_m.$$

By replacing each X_m by $X_m \cup X_{m-1} \cup \ldots \cup X_1$, we may assume that $X_m \subseteq X_{m+1}$ for any $m \in \mathbb{N}$, and we do the same for the sequence (Y_m).

Let $(F_j)_{j\in\mathbb{N}}$ be a sequence in $\mathcal{A}$ with the property that $F_j \supseteq F_{j+1}$ for each $j \in \mathbb{N}$, and define

$$F := \bigcap_{j=1}^{\infty} F_j.$$

For each $x \in X$, we have $(F_j)_x \supseteq (F_{j+1})_x$, and

$$F_x = \left(\bigcap_{j=1}^{\infty} F_j\right)_x = \bigcap_{j=1}^{\infty} (F_j)_x.$$

Assume that $F_1 \subseteq X_{m_0} \times Y_{m_0}$ for some $m_0 \in \mathbb{N}$. Then for every $x \in X$, we have that $(F_1)_x \subseteq (X_{m_0})_x = Y_{m_0}$, and hence $\varphi_{F_1}(x) = \mu_Y((F_1)_x) \leq \mu_Y(Y_{m_0}) < \infty$. As a consequence of this inequality, we may use item **(ii)** of Lemma A.1.1 to obtain

$$\varphi_F(x) = \mu_Y(F_x) = \mu_Y\left(\bigcap_{j=1}^{\infty} (F_j)_x\right) = \lim_{j\to\infty} \mu_Y((F_j)_x) = \lim_{j\to\infty} \varphi_{F_j}(x),$$

and it follows that φ_F is measurable (the same argument shows that ψ_F is also measurable). Now notice that since $\varphi_{F_1}(x) = 0$ whenever $x \notin X_{m_0}$, and $\varphi_{F_1}(x) \leq \mu_Y(Y_{m_0})$ for every $x \in X$, we get

$$\varphi_{F_1} \leq \mu_Y(Y_{m_0}) 1_{X_{m_0}},$$

and because $\mu_X(X_{m_0}) < \infty$ we have that the function in the right-hand side is integrable. Moreover, once $F_j \supseteq F_{j+1}$ for each $j \in \mathbb{N}$ we get in particular that

$$\varphi_{F_j} \leq \varphi_{F_1} \leq \mu_Y(Y_{m_0}) 1_{X_{m_0}}$$

for every $j \in \mathbb{N}$. From this, we obtain that all the functions φ_{F_j} are real-valued, that φ_F is real-valued, and that the pointwise convergence $\varphi_{F_j} \to \varphi_F$ is dominated by an integrable function. Of course, repeating the argument we get the same for $\psi_{F_j} \to \psi_F$. Therefore, we may use the dominated convergence theorem (Theorem A.2.4) to arrive at

$$\int_X \varphi_F \, d\mu_X = \lim_{j\to\infty} \int_X \varphi_{F_j} \, d\mu_X = \lim_{j\to\infty} \int_Y \psi_{F_j} \, d\mu_Y = \int_Y \psi_F \, d\mu_Y,$$

from where we conclude that $F \in \mathcal{A}$. Summarizing, we have proved that a monotone intersection $F := \bigcap_{j\in\mathbb{N}} F_j$ of sets of $\mathcal{A}$ is contained in $\mathcal{A}$ provided $F_1 \subseteq X_{m_0} \times Y_{m_0}$ for some $m_0 \in \mathbb{N}$.

Now let

$$C := \Big\{ E \in \Sigma_X \otimes \Sigma_Y : E \cap (X_m \times Y_m) \in \mathcal{A} \text{ for every } m \in \mathbb{N} \Big\}.$$

If E is an elementary set, then since all sets $X_m \times Y_m$ are also elementary, Lemma A.5.3 gives that $E \cap (X_m \cap Y_m)$ is an elementary set for each $m \in \mathbb{N}$. Once $\mathcal{A}$ contains all elementary sets of $X \times Y$, it follows that $E \cap (X_m \times Y_m) \in \mathcal{A}$ for every $m \in \mathbb{N}$, from where we get that $E \in C$. Thus, C contains all elementary sets of $X \times Y$.

It turns out that C is a monotone class, and we prove it next. First, if $(E_k)_{k\in\mathbb{N}}$ is a sequence in C such that $E_k \subseteq E_{k+1}$ for every $k \in \mathbb{N}$, then for each fixed $m \in \mathbb{N}$ the sequence $(E_k \cap (X_m \times Y_m))_{k\in\mathbb{N}}$ is a sequence in $\mathcal{A}$ with the property that $E_k \cap (X_m \times Y_m) \subseteq E_{k+1} \cap (X_m \times Y_m)$. Hence the union of each of these sequences is a set of $\mathcal{A}$ (recall that $\mathcal{A}$ is closed under monotone unions). Letting $E = \bigcup_{k\in\mathbb{N}} E_k$, we have

$$E \cap (X_m \times Y_m) = \left(\bigcup_{k=1}^{\infty} E_k \right) \cap (X_m \times Y_m) = \bigcup_{k=1}^{\infty} \big(E_k \cap (X_m \times Y_m) \big) \in \mathcal{A}$$

for each $m \in \mathbb{N}$. Thus, $E \in C$.

On the other hand, let $(G_j)_{j\in\mathbb{N}}$ be a sequence in C having the property that $G_j \supseteq G_{j+1}$ for any $j \in \mathbb{N}$, and let $G := \bigcap_{j\in\mathbb{N}} G_j$. For each $m, j \in \mathbb{N}$, let

$$F_j^m := G_j \cap (X_m \times Y_m),$$

and observe that since $G_j \in C$ for any $j \in \mathbb{N}$, we have that $F_j^m \in \mathcal{A}$ for every $m, j \in \mathbb{N}$. Fix an arbitrary number $m \in \mathbb{N}$. The sequence $(F_j^m)_{j\in\mathbb{N}}$ is a sequence in $\mathcal{A}$ such that $F_j^m \supseteq F_{j+1}^m$ for any $j \in \mathbb{N}$, and $F_1^m \subseteq X_1 \times Y_1$. Therefore,

$$F^m := \bigcap_{j=1}^{\infty} F_j^m \in \mathcal{A}.$$

We may also write F^m as

$$F^m = \bigcap_{j=1}^{\infty} \big(G_j \cap (X_m \times Y_m) \big) = \left(\bigcap_{j=1}^{\infty} G_j \right) \cap (X_m \times Y_m) = G \cap (X_m \times Y_m).$$

Thus, $(F^m)_{m\in\mathbb{N}}$ is a sequence in $\mathcal{A}$, and it holds that $F^m \subseteq F^{m+1}$ for each $m \in \mathbb{N}$ (because we always have that $X_m \times Y_m \subseteq X_{m+1} \times Y_{m+1}$). Using again the fact that $\mathcal{A}$ is closed under monotone unions, we arrive at

$$\bigcup_{m=1}^{\infty} F^m \in \mathcal{A}.$$

Now notice that once $X_m \subseteq X_{m+1}$ and $Y_m \subseteq Y_{m+1}$ for every $m \in \mathbb{N}$, we may write

$$X \times Y = \bigcup_{m=1}^{\infty} X_m \times Y_m, \tag{A.5.3}$$

and consequently

$$G = \bigcup_{m=1}^{\infty} \big(G \cap (X_m \times Y_m)\big) = \bigcup_{m=1}^{\infty} F^m.$$

This proves that $G \in \mathcal{A}$, and hence C is closed under monotone intersections.

It follows that C is a monotone class which contains all elementary sets. From Proposition A.5.1, this leads to $\Sigma_X \otimes \Sigma_Y \subseteq C$. The reverse inclusion comes trivially from the definition of C, and hence we get the equality $C = \Sigma_X \otimes \Sigma_Y$.

As a consequence, for each fixed $E \in \Sigma_X \otimes \Sigma_Y$ we obtain that $E \cap (X_m \times Y_m) \in \mathcal{A}$ for any $m \in \mathbb{N}$. Once it is clear that $E \cap (X_m \times Y_m) \subseteq E \cap (X_{m+1} \times Y_{m+1})$ for each $m \in \mathbb{N}$, and remembering once again that $\mathcal{A}$ is closed under monotone unions, we get

$$E = \bigcup_{m=1}^{\infty} (E \cap (X_m \times Y_m)) \in \mathcal{A}.$$

Therefore, $\Sigma_X \otimes \Sigma_Y \subseteq \mathcal{A}$. From the definition of $\mathcal{A}$, we have that the reverse inclusion also holds, and hence $\mathcal{A} = \Sigma_X \otimes \Sigma_Y$. This concludes the proof.

□

Under the same hypotheses and notation as in the previous theorem, we define the function $\mu_X \otimes \mu_Y \colon \Sigma_X \otimes \Sigma_Y \to [0, +\infty]$ as

$$\mu_X \otimes \mu_Y(E) := \int_X \varphi_E \, d\mu_X = \int_X \mu_Y(E_x) \, d\mu_X(x), \tag{A.5.4}$$

for each $E \in \Sigma_X \otimes \Sigma_Y$. From what we just have proved, we also can express $\mu_X \otimes \mu_Y$ as

$$\mu_X \otimes \mu_Y(E) = \int_Y \psi_E \, d\mu_Y = \int_Y \mu_X(E^y) \, d\mu_Y(y) \tag{A.5.5}$$

for any $E \in \Sigma_X \otimes \Sigma_Y$. The function $\mu_X \otimes \mu_Y$ is called the *product measure* of the measure spaces (X, Σ_X, μ_X) and (Y, Σ_Y, μ_Y). Next we prove that this is indeed a measure in the product σ-algebra $\Sigma_X \otimes \Sigma_Y$.

Proposition A.5.2. *Let (X, Σ_X, μ_X) and (Y, Σ_Y, μ_Y) be σ-finite measure spaces. Then the function $\mu_X \otimes \mu_Y \colon \Sigma_X \otimes \Sigma_Y \to [0, \infty]$ defined above is a measure.*

Proof. The fact that $\mu_X \otimes \mu_Y(\emptyset) = 0$ is immediate. Thus, all we have to prove is the σ-additivity. Let $(E_m)_{m \in \mathbb{N}}$ be a disjoint sequence in $\Sigma_X \otimes \Sigma_Y$, and define $E := \bigcup_{m=1}^{\infty} E_m$. We already know that for each $x \in X$ the sets $(E_m)_x$ form a disjoint sequence in Σ_Y, and

$$E_x = \bigcup_{m=1}^{\infty} (E_m)_x.$$

Thus, once μ_Y is σ-additive, we obtain

$$\begin{aligned} \mu_X \otimes \mu_Y(E) = \int_X \mu_Y(E) \, d\mu_X = \int_X \sum_{m=1}^{\infty} \mu_Y((E_m)_x) \, d\mu_X = \\ = \sum_{m=1}^{\infty} \int_X \mu_Y((E_m)_x) \, d\mu_X = \sum_{m=1}^{\infty} \mu_X \otimes \mu_Y(E_m), \end{aligned}$$

where the reader may notice that the summation can be moved out of the integral in the third equality because of a straightforward variant of the monotone convergence theorem for summations (define the functions to be the partial sums of the summation; if each function is non-negative, then this leads to a monotone non-decreasing sequence). This proves that $\mu_X \otimes \mu_Y$ is σ-additive.

□

Remark A.5.3. The product measure is a somewhat "natural" measure to consider in $X \times Y$ because

$$\mu_X \otimes \mu_Y(A \times B) = \mu_X(A)\mu_Y(B)$$

for any measurable rectangle $A \times B \subseteq X \times Y$. In fact, this is the unique measure on $\Sigma_X \otimes \Sigma_Y$ having this property (see Bartle [239] for a proof).

We move forward to state and prove important theorems which expresses the integral of a function defined in a product $X \times Y$ of measure spaces in terms of iterated integrals over X and Y. We start with the so-called *Tonelli theorem*, which holds for non-negative measurable functions. In what follows, we still denote the x-sections and y-sections of a measurable set $E \subseteq X \times Y$ and of a function f over $X \times Y$ by E_x, E^y, f_x, and f^y, respectively.

Theorem A.5.2 (Tonelli's theorem)**.** *Let (X, Σ_X, μ_X) and (Y, Σ_Y, μ_Y) be σ-finite measure spaces, and assume that $f\colon X \times Y \to [0, +\infty]$ is a non-negative function which is measurable with respect to $\Sigma_X \otimes \Sigma_Y$. The the functions $\Phi_f\colon X \to [0, +\infty]$ and $\Psi_f\colon Y \to [0, +\infty]$ defined as*

$$\Phi_f(x) := \int_Y f_x(y)\, d\mu_Y(y) = \int_Y f(x, y)\, d\mu_Y(y) \quad \textit{and}$$
$$\Psi_f(y) := \int_X f^y(x)\, d\mu_X(x) = \int_X f(x, y)\, d\mu_X(x)$$

are measurable with respect to Σ_X and Σ_Y, respectively. Moreover, we have the equality

$$\int_X \Phi_f\, d\mu_X = \int_{X\times Y} f\, d(\mu_X \otimes \mu_Y) = \int_Y \Psi_f\, d\mu_Y,$$

which we also can write as

$$\int_X \left(\int_Y f(x, y)\, d\mu_Y(y) \right) d\mu_X(x) = \int_{X\times Y} f(x, y)\, d(\mu_X \otimes \mu_Y) = \int_Y \left(\int_X f(x, y)\, d\mu_X(x) \right) d\mu_Y(y).$$

Proof. First consider the case where $f = 1_E$ for some set $E \subseteq \Sigma_X \otimes \Sigma_Y$. In this case, it is clear that for each $x \in X$ we have $1_E(x, y) = 1$ if $y \in E_x$ and $1_E(x, y) = 0$ if $y \notin E_x$, where E_x is the x-section of E as defined in Lemma A.5.1. Thus, it follows that $(1_E)_x = 1_{E_x}$. In other words, for any $x \in X$, the x-section of 1_E is the characteristic function 1_{E_x} of the x-section E_x of E. As a consequence, we have

$$\Phi_{1_E}(x) = \int_Y 1_E(x, y)\, d\mu_Y = \int_Y 1_{E_x}\, d\mu_Y = \mu_Y(E_x).$$

Hence it follows straightforwardly from Theorem A.5.1 that Φ_{1_E} is measurable with respect to Σ_X, since Φ_{1_E} is the function φ_E defined there. Analogously, we get that Ψ_{1_E} is measurable with respect to Σ_Y. From this same theorem and from the definition of the product measure given in (A.5.4), we also get that

$$\int_Y \Psi_{1_E}\, d\mu_Y = \int_X \Phi_{1_E}\, d\mu_X = \int_X \mu_Y(E_x)\, d\mu_X(x) = \mu_X \otimes \mu_Y(E) = \int_{X\times Y} 1_E\, d(\mu_X \otimes \mu_Y),$$

and this concludes the proof for the case of characteristic functions of measurable sets of $X \times Y$. Notice that it follows immediately that if f is a non-negative measurable simple function of $X \times Y$, then Φ_f and Ψ_f are measurable. Indeed, if $s = \sum_{j=1}^m a_j 1_{E_j}$ is a canonical representation of such a function, then

$$\Phi_s(x) = \int_Y \sum_{j=1}^m a_j 1_{E_j}(x,y)\, d\mu_Y(y) = \sum_{j=1}^m a_j \int_Y 1_{E_j}(x,y)\, d\mu_Y(y) = \sum_{j=1}^m a_j \mu_Y((E_j)_x)$$

and

$$\Psi_s(y) = \int_X \sum_{j=1}^m a_j 1_{E_j}(x,y)\, d\mu_X(x) = \sum_{j=1}^m a_j \int_X 1_{E_j}(x,y)\, d\mu_X(x) = \sum_{j=1}^m a_j \mu_X((E_j)^y).$$

Once we already know that the functions $x \mapsto \mu_Y((E_j)_x)$ and $y \mapsto \mu_X((E_j)^y)$ are measurable for each j, and because any linear combination of measurable functions is measurable, we get that Φ_s and Ψ_s are measurable. The linearity of the integral, the equality (A.5.2), and the expressions for Φ_s and Ψ_s above yield

$$\int_X \Phi_s\, d\mu_X = \int_X \sum_{j=1}^m a_j \mu_Y((E_j)_x)\, d\mu_X(x) = \sum_{j=1}^m a_j \int_X \mu_Y((E_j)_x)\, d\mu_X(x) =$$
$$= \sum_{j=1}^m a_j \int_Y \mu_X((E_j)^y)\, d\mu_Y(y) = \int_Y \sum_{j=1}^m a_j \mu_X((E_j)^y)\, d\mu_Y(y) = \int_Y \Psi_s\, d\mu_Y.$$

Even more, the expression for the integral of Φ_s over X given above together with the definition of the product measure (A.5.4) lead to

$$\int_X \Phi_s\, d\mu_X = \sum_{j=1}^m a_j \int_X \mu_Y((E_j)_x)\, d\mu_X = \sum_{j=1}^m a_j \mu_X \otimes \mu_Y(E_j) =$$
$$= \sum_{j=1}^m a_j \int_{X\times Y} 1_{E_j}\, d(\mu_X \otimes \mu_Y) = \int_{X\times Y} \sum_{j=1}^m a_j 1_{E_j}\, d(\mu_X \otimes \mu_Y) = \int_{X\times Y} s\, d(\mu_X \otimes \mu_Y),$$

and the result is proved for simple functions.

Now we go for the general case. If $f\colon X \times Y \to [0,+\infty]$ is a non-negative measurable function, then we may choose a sequence $(s_m\colon X \times Y \to [0,+\infty))_{m\in\mathbb{N}}$ of non-negative simple measurable functions converging (pointwise and monotonically) to f as in Theorem A.2.1. For each $m \in \mathbb{N}$ and every $x \in X$, we have that

$$\Phi_{s_m}(x) = \int_Y s_m(x,y)\, d\mu_Y(y) \le \int_Y s_{m+1}(x,y)\, d\mu_Y(y) = \Phi_{s_{m+1}}(x),$$

from where we get that $(\Phi_{s_m})_{m\in\mathbb{N}}$ is a monotone sequence of (Σ_X-measurable) functions of X. Analogously, $(\Psi_{s_m})_{m\in\mathbb{N}}$ is a monotone sequence of (Σ_Y-measurable) functions of Y. Thus, defining

$$\Phi := \lim_{m\to\infty} \Phi_{s_m} \quad \text{and}$$
$$\Psi := \lim_{m\to\infty} \Psi_{s_m},$$

we get from the monotone convergence theorem (Theorem A.2.2) that these functions are measurable, and

$$\int_X \Phi \, d\mu_X = \lim_{m\to\infty} \int_X \Phi_{s_m} \, d\mu_X \quad \text{and}$$
$$\int_Y \Psi \, d\mu_Y = \lim_{m\to\infty} \int_Y \Psi_{s_m} \, d\mu_Y.$$

Even more, once we know that the integral of Φ_{s_m} over X coincides with the integral of Ψ_{s_m} over Y for each $m \in \mathbb{N}$ (since all functions s_m are simple), we obtain the equality

$$\int_X \Phi \, d\mu_X = \int_Y \Psi \, d\mu_Y. \tag{A.5.6}$$

Now we claim that $\Phi = \Phi_f$ and $\Psi = \Psi_f$. We prove the first of these equalities. For each fixed $x \in X$, the sequence $((s_m)_x)_{m\in\mathbb{N}}$ of the x-sections of the functions s_m is a sequence of measurable functions of Y which converges pointwise and monotonically to f_x (because $s_m \to f$ pointwise and monotonically). Then, using the monotone convergence theorem again, we get

$$\Phi_f(x) = \int_Y f_x(y) \, d\mu_Y(y) = \lim_{m\to\infty} \int_Y (s_m)_x(y) \, d\mu_Y(y) =$$
$$= \lim_{m\to\infty} \int_Y s_m(x, y) \, d\mu_Y(y) = \lim_{m\to\infty} \Phi_{s_m}(x) = \Phi(x)$$

for each $x \in X$. Thus, $\Phi_f = \Phi$, and the equality $\Psi_f = \Psi$ follows from the same argument. Hence Φ_f and Ψ_f are measurable (because we already have that Φ and Ψ are measurable). From (A.5.6) we also get the equality

$$\int_X \Phi_f \, d\mu_X = \int_Y \Psi_f \, d\mu_Y,$$

and to prove the remaining equality we recall that we already proved that the theorem holds true for simple functions and use the monotone convergence theorem for the original sequence (s_m) and to write

$$\int_X \Phi_f \, d\mu_X = \lim_{m\to\infty} \int_X \Phi_{s_m} \, d\mu_X = \lim_{m\to\infty} \int_{X\times Y} s_m \, d(\mu_X \otimes \mu_Y) = \int_{X\times Y} f \, d(\mu_X \otimes \mu_Y).$$

This concludes the proof.

□

Now we move forward to consider functions which are not necessarily non-negative. Recall that a measurable function $f : (Z, \Sigma, \mu) \to \mathbb{R}$ is said to be *integrable* (with respect to μ) if

$$\int_Z |f| \, d\mu < +\infty,$$

and in this case we write $f \in L^1(\mu)$. For definitions and results regarding integrable functions, we refer the reader to Section A.2.

Theorem A.5.3 (Fubini's theorem)**.** *Let (X, Σ_X, μ_X) and (Y, Σ_Y, μ_Y) be σ-finite measure spaces, and let $f : X \times Y \to \mathbb{R}$ be a function which is integrable with respect to the product measure $\mu_X \otimes \mu_Y$. Then*

(i) *$f_x \in L^1(\mu_Y)$ for μ_X-almost every $x \in X$, $f^y \in L^1(\mu_X)$ for μ_y-almost every $y \in Y$.*

(ii) *Define the function $\Phi_f : X \to \mathbb{R}$ by letting*

$$\Phi_f(x) := \int_Y f_x \, d\mu_Y$$

whenever $f_x \in L^1(\mu_Y)$*, and* $\Phi_f(x) := 0$ *if* $f_x \notin L^1(\mu_Y)$*. Similarly, let* $\Psi_f : Y \to \mathbb{R}$ *be defined as*

$$\Psi_f(y) = \int_X f^y \, d\mu_X$$

if $f^y \in L^1(\mu_X)$ *and* $\Psi_f(y) := 0$ *otherwise. Then* Φ_f *and* Ψ_f *are* μ_X*-integrable and* μ_Y*-integrable, respectively.*

(iii) *We have the equalities*

$$\int_X \Phi_f \, d\mu_X = \int_{X\times Y} f \, d(\mu_X \otimes \mu_Y) = \int_Y \Psi_f \, d\mu_Y,$$

which can be re-written as

$$\int_X \left(\int_Y f(x,y) \, d\mu_Y(y) \right) d\mu_X(x) = \int_{X\times Y} f \, d(\mu_X \otimes \mu_Y) = \int_Y \left(\int_X f(x,y) \, d\mu_X(x) \right) d\mu_Y(y).$$

Proof. We prove the claims only for the x-sections of f, since the proof for the y-sections is completely analogous. As usual, let f^+ and f^- be the positive and negative parts of f, respectively. Notice that for every fixed $x \in X$, the positive part of f_x is given as

$$f_x^+(y) := (f_x)^+(y) = \max\{f_x(y), 0\} = \max\{f(x,y), 0\} = f^+(x,y) = (f^+)_x(y)$$

for each $y \in Y$. Similarly, $f_x^-(y) := (f_x)^-(y) = f^-(x,y) = (f^-)_x(y)$ for each $y \in Y$. Once f is measurable with respect to $\Sigma_X \otimes \Sigma_Y$, we have that f^+ and f^- are also $\Sigma_X \otimes \Sigma_Y$-measurable. Hence applying Tonelli's theorem for these functions we have that the functions

$$\Phi_f^+(x) := \int_Y (f^+)_x \, d\mu_Y = \int_Y f_x^+ \, d\mu_Y = \int_Y f^+(x,y) \, d\mu_Y \quad \text{and}$$
$$\Phi_x^-(x) := \int_Y (f^-)_x \, d\mu_Y = \int_X f_x^- \, d\mu_Y = \int_X f^-(x,y) \, d\mu_Y$$

defined for every $x \in X$ are μ_X-measurable, and that

$$\int_X \Phi_f^+ \, d\mu_X = \int_{X\times Y} f^+ \, d(\mu_X \otimes \mu_Y) \quad \text{and}$$
$$\int_X \Phi_f^- \, d\mu_X = \int_{X\times Y} f^- \, d(\mu_X \otimes \mu_Y).$$

We warn the reader that Φ_f^+ and Φ_f^- are **not** the positive and negative parts of the function Φ_f defined in the statement (although this holds for the points $x \in X$ where $f_x \in L^1(\mu_Y)$, as it is clear). Since f is integrable, the right-hand sides of the equalities above are finite. This leads to

$$\int_X \left(\int_Y f_x^+ \, d\mu_Y \right) d\mu_X = \int_X \Phi_f^+ \, d\mu_X < +\infty \quad \text{and}$$
$$\int_X \left(\int_Y f_x^- \, d\mu_Y \right) d\mu_X = \int_X \Phi_f^- \, d\mu_X < +\infty.$$

Consequently, the sets

$$N_+ := \left\{x \in X : \int_Y f_x^+ \, d\mu_Y = +\infty\right\} = \left\{x \in X : \Phi_f^+(x) = +\infty\right\} \quad \text{and}$$
$$N_- := \left\{x \in X : \int_Y f_x^- \, d\mu_Y = +\infty\right\} = \left\{x \in X : \Phi_f^-(x) = +\infty\right\}$$

are such that $\mu_X(N_+) = \mu_X(N_-) = 0$ (indeed, if N_+ has positive measure, then the integral of Φ_f^+ would be infinite; the case of N_- is analogous). Hence $N_+ \cup N_-$ also have zero measure, and for any point $x \notin N_+ \cup N_-$ we have that

$$\int_Y |f_x| \, d\mu_Y = \int_Y f_x^+ \, d\mu_Y + \int_Y f_x^- \, d\mu_Y < +\infty,$$

and this shows that $f_x \in L^1(\mu_Y)$ for μ_X-almost every point of X. This proves **(i)**.

Now we prove **(ii)**. If $x \notin N_+ \cup N_-$, then $f_x \in L^1(\mu_Y)$, and hence we have by the definition of Φ_f that

$$\Phi_f(x) = \int_Y f_x \, d\mu_Y.$$

From the definition of integral, it follows that

$$\Phi_f(x) = \int_Y f_x^+ \, d\mu_Y - \int_Y f_x^- \, d\mu_Y = \Phi_f^+(x) - \Phi_f^-(x).$$

Thus, in $X \setminus (N_+ \cup N_-)$, the function Φ_f is expressed as the difference of two μ_X-measurable functions. On the other hand, if $x \in N_+ \cup N_-$, then

$$\int_Y |f_x| \, d\mu_Y = \int_Y f_x^+ \, d\mu_Y + \int_Y f_x^- \, d\mu_Y = +\infty,$$

and hence in this case $f_x \notin L^1(\mu_Y)$. As a consequence, we get that $\Phi_f(x) = 0$ whenever $x \in N_+ \cup N_-$. The fact that Φ_f is measurable follows immediately: if $a > 0$, then $(\Phi_f)^{-1}((a, +\infty])$ coincides with $(\Phi_f^+ - \Phi_f^-)^{-1}((a, +\infty])$, and this latter set is measurable because $\Phi_f^+ - \Phi_f^-$ is measurable. If $a \leq 0$, then we simply write

$$(\Phi_f)^{-1}((a, +\infty]) = (\Phi_f)^{-1}((a, 0)) \cup N_+ \cup N_- \cup (\Phi_f)^{-1}((0, +\infty]) =$$
$$= (\Phi_f^+ - \Phi_f^-)^{-1}((a, 0)) \cup (N_+ \cup N_-) \cup (\Phi_f^+ - \Phi_f^-)^{-1}((0, +\infty]),$$

and the latter is a union of measurable sets (hence measurable). To verify that Φ_f is integrable, notice that since Φ_f coincides with $\Phi_f^+ - \Phi_f^-$ up to a set of zero measure (namely, $N_+ \cup N_-$), we have

$$\int_X |\Phi_f| \, d\mu_X = \int_X |\Phi_f^+ - \Phi_f^-| \, d\mu_X \leq \int_X \Phi_f^+ \, d\mu_X + \int_X \Phi_f^- \, d\mu_X =$$
$$= \int_{X \times Y} f^+ d(\mu_X \otimes \mu_Y) + \int_{X \times Y} f^- \, d(\mu_X \otimes \mu_Y) = \int_{X \times Y} |f| \, d(\mu_X \otimes \mu_Y) < +\infty,$$

where the last inequality comes from the fact that $f \in L^1(\mu_X \otimes \mu_Y)$. Observe that we also used the fact that Φ_f^+ and Φ_f^- are non-negative. This gives that Φ_f is integrable.

To prove **(iii)**, denote by $(\Phi_f)^+$ and $(\Phi_f)^-$ the positive and negative parts of Φ_f, respectively. Then in $X \setminus (N_+ \cup N_-)$ we clearly have that $(\Phi_f)^+ = \Phi_f^+$ and $(\Phi_f)^- = \Phi_f^-$. Using again the fact that the

measure of $N_+ \cup N_-$ is zero, we finally arrive at

$$\int_X \Phi_f \, d\mu_X = \int_X (\Phi_f)^+ \, d\mu_X - \int_X (\Phi_f)^- \, d\mu_X = \int_X \Phi_f^+ \, d\mu_X - \int_X \Phi_f^- \, d\mu_X =$$
$$= \int_{X\times Y} f^+ \, d(\mu_X \otimes \mu_Y) - \int_{X\times Y} f^- \, d(\mu_X \otimes \mu_Y) = \int_{X\times Y} f \, d(\mu_X \otimes \mu_Y).$$

As it was mentioned, the case for the y-sections of f is completely analogous, and hence we are done. □

As it is intuitive, Fubini's theorem gives "volumes" in terms of "areas of sections". This has important applications in convexity, such as *Cavalieri's principles*, for example (see Section 4.2). It turns out that if $m = k + j$, then the Lebesgue measure $\mathcal{L}^m$ (of $\mathbb{R}^m$) can be seen as the product measure of the Lebesgue measures $\mathcal{L}^k$ (of $\mathbb{R}^k$) and $\mathcal{L}^j$ (of $\mathbb{R}^j$), at least in the case where these measures are defined over the respective Borel σ-algebras. We see it next.

Proposition A.5.3. *Let $m, k, j \in \mathbb{N}$ be such that $m = k + j$, and assume that the Cartesian product $\mathbb{R}^k \times \mathbb{R}^j$ is naturally identified with $\mathbb{R}^m$. Hence we denote the points of $\mathbb{R}^m$ as (x, y), where $x \in \mathbb{R}^k$ and $y \in \mathbb{R}^j$. The following holds.*

(i) $\mathcal{B}_{\mathbb{R}^k} \otimes \mathcal{B}_{\mathbb{R}^j} = \mathcal{B}_{\mathbb{R}^m}$ *and*

(ii) *Regarding $\mathcal{L}^k$, $\mathcal{L}^j$, and $\mathcal{L}^m$ as measures over $\mathcal{B}_{\mathbb{R}^k}$, $\mathcal{B}_{\mathbb{R}^j}$, and $\mathcal{B}_{\mathbb{R}^m}$, respectively, we have that $\mathcal{L}^k \otimes \mathcal{L}^j = \mathcal{L}^m$.*

Proof. To prove **(i)**, let $\pi_1 \colon \mathbb{R}^m \to \mathbb{R}^k$ and $\pi_2 \colon \mathbb{R}^m \to \mathbb{R}^j$ be the projections defined as $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$ for any $(x, y) \in \mathbb{R}^m$. These maps are clearly continuous, and hence Borel measurable. Let $A \in \mathcal{B}_{\mathbb{R}^k}$ and $B \in \mathcal{B}_{\mathbb{R}^j}$ be arbitrarily given. Once the projections are Borel measurable, we have that

$$A \times \mathbb{R}^j = (\pi_1)^{-1}(A) \in \mathcal{B}_{\mathbb{R}^m} \quad \text{and}$$
$$\mathbb{R}^k \times B = (\pi_2)^{-1}(B) \in \mathcal{B}_{\mathbb{R}^m}.$$

Therefore, $A \times B = (A \times \mathbb{R}^j) \cap (\mathbb{R}^k \times B) \in \mathcal{B}_{\mathbb{R}^m}$, and this shows that $\mathcal{B}_{\mathbb{R}^m}$ contains all measurable rectangles of $\mathbb{R}^k \times \mathbb{R}^j$. Since the product σ-algebra $\mathcal{B}_{\mathbb{R}^k} \otimes \mathcal{B}_{\mathbb{R}^j}$ is by definition the smallest σ-algebra of $\mathbb{R}^k \times \mathbb{R}^j$ containing these rectangles, it follows that

$$\mathcal{B}_{\mathbb{R}^k} \otimes \mathcal{B}_{\mathbb{R}^j} \subseteq \mathcal{B}_{\mathbb{R}^m}.$$

For the reverse inclusion, recall that in the proof of Theorem A.3.2 we have proved that any open set $U \subseteq \mathbb{R}^m$ can be written as a countable union of sets having the form $x + [0, 2^{-i})^{-m}$. Each of these sets is a measurable rectangle, because writing in coordinates $x = (x_1, \ldots, x_m)$ we have

$$x + [0, 2^{-i})^{-m} =$$
$$= \Big([x_1, x_1 + 2^{-k}) \times \ldots \times [x_k, x_k + 2^{-i})\Big) \times \Big([x_{k+1}, x_{k+1} + 2^{-i}) \times \ldots \times [x_m, x_m + 2^{-i})\Big),$$

and the right-hand side is the Cartesian product of a Borel measurable set of $\mathbb{R}^k$ with a Borel measurable set of $\mathbb{R}^j$. The σ-algebra $\mathcal{B}_{\mathbb{R}^k} \otimes \mathcal{B}_{\mathbb{R}^j}$ contains all measurable rectangles of $\mathbb{R}^k \times \mathbb{R}^j$, and hence it also contains all countable unions of these rectangles. Consequently, $\mathcal{B}_{\mathbb{R}^k} \otimes \mathcal{B}_{\mathbb{R}^j}$ contains all open sets of $\mathbb{R}^m$. Since $\mathcal{B}_{\mathbb{R}^m}$ is the smallest σ-algebra of $\mathbb{R}^m$ containing all the open sets, we get that

$$\mathcal{B}_{\mathbb{R}^m} \subseteq \mathcal{B}_{\mathbb{R}^k} \otimes \mathcal{B}_{\mathbb{R}^j},$$

and this concludes the proof of **(i)**.

Now we prove **(ii)**. We already know from the previous item that $\mathcal{L}^k \otimes \mathcal{L}^j$ is a measure defined on the Borel σ-algebra of $\mathbb{R}^m$ (recall that we are considering the natural identification between $\mathbb{R}^k \times \mathbb{R}^j$ and $\mathbb{R}^m$). Hence it suffices to prove that $\mathcal{L}^k \otimes \mathcal{L}^j$ is translation-invariant and attributes the value 1 for the cell $[0,1]^m$. Observe first that writing $[0,1]^m = [0,1]^k \times [0,1]^j$ (that is, as a measurable rectangle) we have that

$$\mathcal{L}^k \otimes \mathcal{L}^j\big([0,1]^m\big) = \mathcal{L}^k\big([0,1]^k\big)\mathcal{L}^j\big([0,1]^j\big) = 1.$$

To show that $\mathcal{L}^k \otimes \mathcal{L}^j$ is translation-invariant, let $E \in \mathcal{B}_{\mathbb{R}^m}$ and $(x_0, y_0) \in \mathbb{R}^m$ be given. Notice that for any $x \in \mathbb{R}^k$ and $y \in \mathbb{R}^j$ we have

$$y \in \big((x_0,y_0)+E\big)_x \iff (x,y) \in (x_0,y_0)+E \iff (x-x_0, y-y_0) \in E \iff$$
$$\iff y \in y_0 + E_{x-x_0},$$

and hence $\big((x_0,y_0)+E\big)_x = y_0 + E_{x-x_0}$. This, together with the definition of the product measure and the translation-invariance of $\mathcal{L}^k$ and $\mathcal{L}^j$, leads to

$$\mathcal{L}^k \otimes \mathcal{L}^j\big(((x_0,y_0)+E)_x\big) = \int_{\mathbb{R}^k} \mathcal{L}^j\big(((x_0,y_0)+E)_x\big)\, d\mathcal{L}^k(x) =$$
$$= \int_{\mathbb{R}^k} \mathcal{L}^j\big(y_0 + E_{x-x_0}\big)\, d\mathcal{L}^k(x) = \int_{\mathbb{R}^k} \mathcal{L}^j\big(E_{x-x_0}\big)\, d\mathcal{L}^k(x) = \int_{\mathbb{R}^k} \mathcal{L}^j\big(E_z\big)\, d\mathcal{L}^k(z) = \mathcal{L}^k \otimes \mathcal{L}^j(E),$$

where the change of variables $z = x - x_0$ in the fourth equality is justified since $\mathcal{L}^k$ is translation-invariant (then the push-forward measure of $\mathcal{L}^k$ under any translation map is $\mathcal{L}^k$ itself). This shows that $\mathcal{L}^k \otimes \mathcal{L}^j$ is indeed translation-invariant and the proof is complete.

□

Remark A.5.4. Notice carefully that we regard the Lebesgue measures as measures defined over the respective Borel σ-algebras. The case where we consider the Lebesgue measures as measures over the respective σ-algebras of the Lebesgue measurable sets is slightly more complicated. We will not study this case here because this is not needed for our purposes, and we refer again to Salamon [2090] for more in this direction.

Corollary A.5.1 (Fubini's theorem for Lebesgue measures). *Let $f\colon \mathbb{R}^m \to \mathbb{R}$ be a function which is integrable with respect to the restriction of the Lebesgue measure $\mathcal{L}^m$ of $\mathbb{R}^m$ to the Borel σ-algebra $\mathcal{B}_{\mathbb{R}^m}$ of $\mathbb{R}^m$. Then, whenever $k, j \in \mathbb{N}$ are such that $k + j = m$, we have*

$$\int_{\mathbb{R}^k} \left(\int_{\mathbb{R}^j} f\, d\mathcal{L}^j \right) d\mathcal{L}^k = \int_{\mathbb{R}^m} f\, d\mathcal{L}^m.$$

Proof. Since $\mathcal{L}^k \otimes \mathcal{L}^j$ equals $\mathcal{L}^m$ when these measures are considered over the respective Borel σ-algebras the result follows immediately from Fubini's theorem.

□

This result settles the measure-theoretic formalism to explain the Cavalieri principles discussed in Section 4.2. However, sometimes we need to consider multiple integrals over other domains than the $\mathbb{R}^k$s (for example, over unit spheres). It turns out that item **(i)** of Proposition A.5.3 holds in greater generality, and this is what we need to formalize the application of Fubini's theorem in these other contexts.

Proposition A.5.4. *If X and Y are topological spaces, then $\mathcal{B}_X \otimes \mathcal{B}_Y \subseteq \mathcal{B}_{X\times Y}$. Equality holds if X is second countable, Hausdorff, and locally compact.*

Proof. In this proof, we assume some familiarity with standard topology concepts (mainly the *product topology*). A reference for the reader who does not have this "background" is Munkres [1864].

To prove the inclusion $\mathcal{B}_X \otimes \mathcal{B}_Y \subseteq \mathcal{B}_{X\times Y}$, we recall that the natural projections $\pi_1 \colon X \times Y \to X$ and $\pi_2 \colon X \times Y \to Y$ are continuous in the product topology of $X \times Y$ (hence Borel measurable). Then, we may proceed similarly to the proof of item **(i)** of Proposition A.5.3. For $A \in \mathcal{B}_X$ and $B \in \mathcal{B}_Y$ arbitrarily given, we have

$$A \times Y = (\pi_1)^{-1}(A) \in \mathcal{B}_{X\times Y} \text{ and}$$
$$X \times B = (\pi_2)^{-1}(B) \in \mathcal{B}_{X\times Y}.$$

Thus, $A \times B = (A \times Y) \cap (B \times X) \in \mathcal{B}_{X\times Y}$, and this shows that $\mathcal{B}_{X\times Y}$ contains all measurable rectangles of $X \times Y$. Once $\mathcal{B}_X \otimes \mathcal{B}_Y$ is the smallest σ-algebra of $X \times Y$ containing all measurable rectangles, it follows that $\mathcal{B}_X \otimes \mathcal{B}_Y \subseteq \mathcal{B}_{X\times Y}$.

Now we prove that the reverse inclusion holds in the case where X is second countable, Hausdorff, and second countable. Denote the topologies of X and Y by τ_X and τ_Y, respectively. Also, denote the product topology of $X \times Y$ by $\tau_{X\times Y}$. The assumption of X being second countable, Hausdorff, and locally compact gives that there exists a countable basis $\{U_j : j \in \mathbb{N}\}$ of τ_X such that $\mathrm{cl}(U_j)$ is compact for each $j \in \mathbb{N}$ (the closure is taken in the topology of X, of course).

Let $A \in \tau_{X\times Y}$ be arbitrarily given. For each $j \in \mathbb{N}$, we define

$$V_j(A) := \{y \in Y : (x, y) \in A \text{ for all } x \in \mathrm{cl}(U_j)\}.$$

We claim that each $V_j(A)$ is open. To verify that, let $j \in \mathbb{N}$ and $y_0 \in V_j(A)$ be arbitrary. Then, $(x, y_0) \in A$ for every $x \in \mathrm{cl}(U_j)$. Once A is open, this gives that for each $x \in \mathrm{cl}(U_j)$ there exist open sets $U(x) \subseteq X$ and $V(x) \subseteq Y$ such that $(x, y_0) \in U(x) \times V(x) \subseteq A$ (this happens because the product topology of a finite Cartesian product coincides with the usual box topology). It follows that $\{U(x) : x \in \mathrm{cl}(U_j)\}$ is an open cover of $\mathrm{cl}(U_j)$, and from compactness we get that there exist finitely many points $x_1, \ldots, x_m \in \mathrm{cl}(U_j)$ such that

$$\mathrm{cl}(U_j) \subseteq U(x_1) \cup \ldots \cup U(x_m).$$

Now we define $V := V(x_1) \cap \ldots \cap V(x_m) \subseteq Y$. Then V is non-empty (because $y_0 \in V$) and open (because it is a finite intersection of open sets). Moreover, we have that $\mathrm{cl}(U_j) \times V \subseteq A$. In fact, if $(x, y) \in \mathrm{cl}(U_j) \times V$, then $x \in U(x_k)$ for some $k \in \{1, \ldots, m\}$, and $y \in V(x_k)$ by the definition of V. Thus, $(x, y) \in U(x_k) \times V(x_k) \subseteq A$. As a consequence of this inclusion, we have that $V \subseteq V_j(A)$. Since V is an open set containing y_0, this proves that y_0 is an interior point of $V_j(A)$. Because $y_0 \in V_j(A)$ was taken arbitrarily, we obtain that $V_j(A)$ is open.

Moving forward, we show that we have the equality

$$A = \bigcup_{j=1}^{\infty} U_j \times V_j(A). \tag{A.5.7}$$

The inclusion "$\supseteq$" follows immediately from the definition of $V_j(A)$. For the reverse inclusion, observe that if $(x_0, y_0) \in A$, then there exist open sets $U \subseteq X$ and $V \subseteq Y$ such that $(x_0, y_0) \in U \times V \subseteq A$ (here, we are using again the fact that the product topology of $X \times Y$ is the box topology). Once X is locally compact, there exists an open set $U_0 \subseteq X$ such that $x_0 \in U_0 \subseteq \mathrm{cl}(U_0) \subseteq U$. Because $\{U_j : j \in \mathbb{N}\}$ is a basis, there exists a number $j_0 \in \mathbb{N}$ such that $x_0 \in U_{j_0} \subseteq U_0$. Hence $x_0 \in \mathrm{cl}(U_{j_0}) \subseteq U$, and consequently $y_0 \in V_{j_0}(A)$. It follows that $(x_0, y_0) \in U_{j_0} \times V_{j_0}(A)$.

Equality (A.5.7) and the fact that each $V_j(A)$ is open (and also each U_j, of course) prove that any open set of $X \times Y$ can be written as a countable union of measurable rectangles. Thus, $\tau_{X\times Y} \subseteq \mathcal{B}_X \otimes \mathcal{B}_Y$. Since $\mathcal{B}_{X\times Y}$ is the smallest σ-algebra of $X \times Y$ containing $\tau_{X\times Y}$, we get that $\mathcal{B}_{X\times Y} \subseteq \mathcal{B}_X \otimes \mathcal{B}_Y$.

□

This proposition has the following immediate consequence: if X and Y are topological spaces such that X is second countable, Hausdorff, and locally compact, and if $f\colon X \times Y \to \mathbb{R}$ is continuous with respect to the product topology of $X \times Y$, then f is measurable with respect to $\mathcal{B}_X \otimes \mathcal{B}_Y$. In this case, if μ_X and μ_Y are Borel measures on X and Y, respectively, and if

$$\int_{X\times Y} |f| d(\mu_X \otimes \mu_Y) < +\infty,$$

then we may apply Fubini's theorem and obtain

$$\int_X \int_Y f \, d\mu_Y d\mu_X = \int_Y \int_X f \, d\mu_X d\mu_Y.$$

In other words, a continuous function defined on a product $X \times Y$ of topological spaces (such that one of them is second countable, Hausdorff, and locally compact) is always measurable with respect to $\mathcal{B}_X \otimes \mathcal{B}_Y$, and hence if it is integrable (with respect to $\mu_X \otimes \mu_Y$), then we may use Fubini's theorem to change the order of integration. Notice also that, in particular, the finiteness of the integral of $|f|$ holds trivially if the considered spaces are compact and the considered measures are finite.

We finish this section with another consequence of Fubini's theorem, known as the *layer cake representation* of an integral. We use it to prove the Prékopa–Leindler inequality.

Corollary A.5.2. *Let $f\colon \mathbb{R} \to \mathbb{R}$ be a measurable function. Then*

$$\int_{\mathbb{R}} |f(x)| \, dx = \int_0^\infty \mathcal{L}^1(\{x \in \mathbb{R}\colon |f(x)| \geq t\}) \, dt,$$

where the integrals are taken with respect to the Lebesgue measure $\mathcal{L}^1$ in $\mathbb{R}$.

Proof. First notice that for every $t \geq 0$ and for any $x \in \mathbb{R}$ we have that

$$1_{[0,|f(x)|]}(t) = 1_{\{|f(x)|\geq t\}}(x),$$

where, as usual, 1_A denotes the characteristic function of the set $A \subseteq \mathbb{R}$, and $\{f(x) \geq t\}$ is a shortcut for $\{x \in \mathbb{R}\colon |f(x)| \geq t\}$. This equality leads immediately to

$$\int_{\mathbb{R}} \int_{\mathbb{R}} 1_{[0,|f(x)|]}(t) \, dxdt = \int_{\mathbb{R}} \int_{\mathbb{R}} 1_{\{|f(x)|\geq t\}}(x) \, dxdt.$$

The right-hand side can be directly computed:

$$\int_{\mathbb{R}} \int_{\mathbb{R}} 1_{\{|f(x)|\geq t\}}(x) \, dxdt = \int_{\mathbb{R}} \mathcal{L}^1(\{|f(x)| \geq t\}) \, dt = \int_0^\infty \mathcal{L}^1(\{|f(x)| \geq t\}) \, dt,$$

where the last equality justifies since the set $\{|f(x)| \geq t\}$ is obviously empty if $t < 0$. For the left-hand side, we use Fubini's theorem and write

$$\int_{\mathbb{R}} \int_{\mathbb{R}} 1_{[0,|f(x)|]}(t) \, dxdt = \int_{\mathbb{R}} \int_{\mathbb{R}} 1_{[0,|f(x)|]}(t) \, dtdx = \int_{\mathbb{R}} \mathcal{L}^1([0, |f(x)|]) \, dx = \int_{\mathbb{R}} |f(x)| \, dx,$$

and the proof is complete.

□

A.6 Area and co-area formulas

In this section we present some versions of famous formulas from geometric measure theory, namely the *area* and *co-area formulas*. We say that we will present "versions" because we need these results in less generality than it is usually studied in mentioned theory. For the proofs we refer the reader to Evans and Gariepy [822], Federer [834], and Simon [2249].

We also explain that differently from what we have been doing thus far, in this section the proofs of the main results will be omitted. This choice has two mutually complementary reasons: first, we would have to develop too much theory to present the proofs of the area and co-area formulas; second, we actually have very little use of these formulas throughout the text. In fact, the area formula is only used in Section 7.4 to derive a relation between the usual surface area measure μ_K given on $\mathbb{S}^{n-1}$ by a convex body $K \subseteq \mathbb{R}^n$ and the Hausdorff $(n-1)$-dimensional measure on $\mathbb{S}^{n-1}$. We also emphasize that the most part of the applications which we derive from the co-area formula can be also obtained from Fubini's theorem.

Roughly speaking, the version we consider of the area formula explains the behavior of Hausdorff measures under diffeomorphisms of hypersurfaces of $\mathbb{R}^n$, and it is related to the fact that the "differential geometric way" to measure area coincides with the $(n-1)$-dimensional Hausdorff measure on an oriented hypersurface of $\mathbb{R}^n$ (see Evans and Gariepy [822]).

Theorem A.6.1 (Area formula)**.** *Let $F\colon M \to F(M)$ is a diffeomorphism between compact, orientable, and without boundary hypersurfaces of $\mathbb{R}^n$. Then*

$$\int_{F(M)} f \circ F^{-1}(y)\, d\mathcal{H}^{n-1}(y) = \int_M f(x)\sqrt{\det\big((dF_x)\cdot(dF_x)^*\big)}\, d\mathcal{H}^{n-1}(x) \tag{A.6.1}$$

for any measurable function $f\colon M \to \mathbb{R}$, where $(dF_x)^*$ is the usual adjoint of dF_x with respect to the inner product of $\mathbb{R}^n$.

For a proof (of a more general version), we refer to Simon [2249]. Of course, the integrations in the left-hand side and in the right-hand side are taken with respect to the restrictions of $\mathcal{H}^{n-1}$ to $F(M)$ and M, respectively.

Remark A.6.1. Some might prefer to say that (A.6.1) is simply a change of variables formula. This viewpoint arises once the relation between the restriction of the $(n-1)$-dimensional Hausdorff measure to a smooth hypersurface M and the usual differential geometric definition of area of M is clarified. Since this is not the case in this book, we do not adopt this terminology. For us, the real meaning of the area formula is that it describes the relation between the $(n-1)$-dimensional Hausdorff measure respectively restricted to two diffeomophic hypersurfaces.

The co-area formula gives the integral (with respect to the Lebesgue measure) of an integrable function over an open subset of $\mathbb{R}^n$ in terms of integrals of this function over the level sets of a given Lipschitz function.

Theorem A.6.2 (Co-area formula)**.** *Let $U \subseteq \mathbb{R}^n$ be an open set and $u\colon U \to \mathbb{R}$ be a Lipschitz function. For any integrable function $f\colon U \to \mathbb{R}$ we have*

$$\int_U f(x)|\nabla u(x)|\, dx = \int_{\mathbb{R}} \left(\int_{u^{-1}(t)} f(y)\, d\mathcal{H}^{n-1}(y) \right) dt, \tag{A.6.2}$$

where dx stands for the integration with respect to the common Lebesgue measure, as usual.

Remark A.6.2. By Rademacher's theorem (see, e.g., Simon [2249]), every Lipschitz function $u\colon U \to \mathbb{R}$ is differentiable almost everywhere (with respect to the Lebesgue measure). Hence the integral in the left-hand side of the above is well-defined. In the right-hand side, $\mathcal{H}^{n-1}$ stands for the restriction of the $(n-1)$-dimensional Hausdorff measure to $u^{-1}(t)$.

Proof. See [822].

□

For us, the most important consequence of the co-area formula is the formula for integration by polar coordinates. We derive it now.

Corollary A.6.1 (Integration by polar coordinates). *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be an integrable function. Then*

$$\int_{\mathbb{R}^n} f(x)\,dx = \int_0^{+\infty} \left(\int_{t\mathbb{S}^{n-1}} f(y)\,d\mathcal{H}^{n-1}(y) \right) dt = \int_0^{+\infty} \int_{\mathbb{S}^{n-1}} f(tz) t^{n-1}\,d\mathcal{H}^{n-1}(z)dt. \tag{A.6.3}$$

Proof. The first equality comes from the co-area formula for $u(x) = |x|$. Indeed, we have that $u^{-1}(t) = t\mathbb{S}^{n-1}$ for each $t \geq 0$, and it is immediate that $|\nabla u(x)| = 1$ for every $x \neq 0^n_{\mathbb{R}}$. For the second equality, we observe that Corollary A.4.6 gives

$$\mathcal{H}^{n-1}(tA) = t^{n-1}\mathcal{H}^{n-1}(A)$$

for every Borel subset $A \subseteq \mathbb{S}^{n-1}$ and $t > 0$. Hence we have that the $(n-1)$-dimensional Hausdorff measure on $t\mathbb{S}^{n-1}$ is the push-forward measure of the $(n-1)$-dimensional Hausdorff measure on $\mathbb{S}^{n-1}$ under the dilation map $\varphi_t(z) = tz$ multiplied by t^{n-1}, that is,

$$\mathcal{H}^{n-1}\big|_{t\mathbb{S}^{n-1}} = t^{n-1}(\varphi_t)_*\mathcal{H}^{n-1}\big|_{\mathbb{S}^{n-1}}.$$

Thus, by the change of variables formula (A.2.4) we obtain

$$\int_{t\mathbb{S}^{n-1}} f(y)d\mathcal{H}^{n-1}(y) = \int_{\mathbb{S}^{n-1}} f(tz)t^{n-1}\,d\mathcal{H}^{n-1}(z)$$

for each $t \geq 0$. This gives the second equality.

□

Remark A.6.3. By Fubini's theorem we have that the order of integration in the right-hand side of (A.6.3) can be inverted when all maps involved are integrable. We use this without further comments.

The co-area formula can also be used to give an alternative proof of the Cavalieri principle (4.2.1).

Theorem A.6.3. *Let $K \subseteq \mathbb{R}^n$ be a convex body. For any $\theta \in \mathbb{S}^{n-1}$ we have*

$$\mathrm{vol}_n(K) = \int_{-h_K(-\theta)}^{h_K(\theta)} \mathrm{vol}_{n-1}\big(K \cap \{\theta^\perp + t\theta\}\big)\,dt.$$

Proof. We apply the co-area formula for $u(\cdot) = \langle \cdot, \theta\rangle$ and $f = 1_K$. Once it is clear that $u^{-1}(t) = \theta^\perp + t\theta$ and $\nabla u = \theta$, and writing $K(t) := K \cap \{\theta^\perp + t\theta\}$, we get

$$\mathrm{vol}_n(K) = \int_{\mathbb{R}^n} 1_K(x)\,dx = \int_{-\infty}^{+\infty} \int_{\theta^\perp + t\theta} 1_K(y)\,d\mathcal{H}^{n-1}(y)dt =$$

$$= \int_{-\infty}^{+\infty} \mathcal{H}^{n-1}\big(K \cap \{\theta^\perp + t\theta\}\big)\,dt = \int_{-h_K(-\theta)}^{h_K(\theta)} \mathrm{vol}_{n-1}\big(K \cap \{\theta^\perp + t\theta\}\big)\,dt.$$

In the last equality we simply recall that $K \cap \{\theta^\perp + t\theta\} = \emptyset$ whenever $t < -h_K(-\theta)$ or $t > h_K(\theta)$.

□

Remark A.6.4. We can actually re-obtain also the "slicing-by-lines" Cavalieri principle (4.2.2), but we need a stronger form of the co-area formula. If $U \subseteq \mathbb{R}^n$ is an open set, and $f : U \to \mathbb{R}^m$ is a Lipschitz map where $m \leq n$, then

$$\int_U g(x) J_f(x)\, d\mathcal{L}^n(x) = \int_{\mathbb{R}^m} \left(\int_{f^{-1}(y)} g(z)\, d\mathcal{H}^{n-m}(z) \right) d\mathcal{L}^m(y),$$

where $J_j(x) = \sqrt{\det(df_x) \cdot (df_x)^*}$ is the usual *Jacobian* of f, which is defined almost everywhere (Radamacher's theorem). To get (4.2.2) we simply identify $u^\perp$ with $\mathbb{R}^{n-1}$ and let $f : \mathbb{R}^n \to \mathbb{R}^{n-1}$ the natural projection onto the $(n-1)$-first entries, and $g = 1_K$.

A consequence of the co-area formula which is often useful in convexity is that one can derive a integration by polar coordinates-type formula where the unit sphere $\mathbb{S}^{n-1}$ is replaced by a convex body having the origin as an interior point. To explore this idea, we first need a technical lemma regarding the gradient of a gauge function.

Recall that if $K \subseteq \mathbb{R}^n$ is a convex body having the origin as an interior point and whose gauge function is γ_K, then from Proposition 2.3.7 we have that $\nabla \gamma_K$ is defined almost everywhere (with respect to the Lebesgue measure of $\mathbb{R}^n$, of course).

Lemma A.6.1. *Let K be a convex body having the origin as an interior point, and denote by γ_K its gauge function. Then the map $|\nabla \gamma_K(\cdot)|^{-1}$ is defined almost everywhere in $\mathbb{R}^n$, and it is* essentially bounded*, meaning that there exists a number $c \in \mathbb{R}$ such that*

$$\mathcal{L}^n\big(\{z \in \mathbb{R}^n : |\nabla \gamma_K(z)|^{-1} > c\}\big) = 0.$$

Proof. From Proposition 2.3.7 we have that $\nabla \gamma_K(z) \neq 0$ whenever γ_K is differentiable at z. Then $|\nabla \gamma_K(\cdot)|^{-1}$ is defined for all points of $\mathbb{R}^n$ where γ_K is differentiable (thus almost everywhere). We know that the support function $h_K : \mathbb{S}^{n-1} \to \mathbb{R}$ is continuous, and hence

$$m := \sup\{h_K(u) : u \in \mathbb{S}^{n-1}\} < +\infty.$$

Let $z \in \mathbb{R}^n$ be any non-zero point where γ_K is differentiable. Then, writing $\hat{z} = z/||z||$, from (2.3.2) and (2.3.4) we have

$$|\nabla \gamma_K(z)|^{-1} = h_K(u_K(\hat{z})) \leq m.$$

Once this holds for every $z \in \mathbb{R}^n$ where γ_K is differentiable, and because γ_K is differentiable almost everywhere, we finally arrive at

$$\mathcal{L}^n\big(\{z \in \mathbb{R}^n : |\nabla \gamma_K(z)|^{-1} > m\}\big) = 0.$$

This concludes the proof.

□

Now we are almost ready to state and prove the extension of the integration by polar coordinates formula. From Remark 2.3.12 we have, for every $t > 0$,

$$\big\{\text{points of } t\partial K \text{ where } \gamma_K \text{ is differentiable}\big\} = \big\{\text{smooth points of } t\partial K\big\},$$

and by Corollary 2.4.2 the latter has total $\mathcal{H}^{n-1}$ measure on $t\partial K$. It follows that γ_K is differentiable $\mathcal{H}^{n-1}$-almost everywhere in $t\partial K$. Even more, if γ_K is differentiable at $y \in t\partial K$ and we write $y = tz$

for $z \in \partial K$, then Proposition 2.3.7 guarantees that γ_K is differentiable at z, and

$$|\nabla \gamma_K(y)| = |\nabla \gamma_K(z)| = h_K(u_K(z))^{-1},$$

where we recall that $h_K(u_K(z)) > 0$ (we are assuming that K has the origin as an interior point). Thus, $|\nabla \gamma_K(\cdot)|^{-1}$ is well-defined $\mathcal{H}^{n-1}$-almost everywhere in $t\partial K$. This should be kept in mind in what follows.

Proposition A.6.1. *Let $K \subseteq \mathbb{R}^n$ be a convex body which has the origin as an interior point. Then*

$$\int_{\mathbb{R}^n} f(x)\,dx = \int_0^{+\infty} \int_{\partial K} f(tz) t^{n-1} h_K(u_K(z))\,d\mathcal{H}^{n-1}(z)dt$$

for any integrable function $f\colon \mathbb{R}^n \to \mathbb{R}$.

Proof. From Lemma A.6.1, we may assume that $|\nabla \gamma_K(\cdot)|^{-1} \leq m$ almost everywhere for some constant $m \in \mathbb{R}$. Then for any integrable function $f\colon \mathbb{R}^n \to \mathbb{R}$ we have

$$\int_{\mathbb{R}^n} \left| f(x)|\nabla \gamma_K(x)|^{-1} \right| dx \leq m \int_{\mathbb{R}^n} |f(x)|\,dx,$$

from where we get that the function $f|\nabla \gamma_K|^{-1}$ is also integrable. Since $\gamma_K^{-1}(t) = t\partial K$ for every $t \geq 0$, we apply the co-area formula for $u = \gamma_K$ and $f|\nabla \gamma_K|^{-1}$ to obtain

$$\int_{\mathbb{R}^n} f(x)\,dx = \int_{\mathbb{R}^n} \left(f(x)|\nabla \gamma_K(x)|^{-1}\right)|\nabla \gamma_K(x)|\,dx = \int_0^{+\infty} \int_{t\partial K} f(y)|\nabla \gamma_K(y)|^{-1}\,d\mathcal{H}^{n-1}(y)dt,$$

where the right-hand makes sense because $|\nabla \gamma_K|^{-1}$ is well-defined $\mathcal{H}^{n-1}$-almost everywhere in $t\partial K$. Similarly to what we have done in the proof of the integration by polar coordinates formula, from Corollary A.4.6, we can relate the $(n-1)$-dimensional Hausdorff measures on ∂K and $t\partial K$ as

$$\mathcal{H}^{n-1}\big|_{t\partial K} = t^{n-1}(\varphi_t)_* \mathcal{H}^{n-1}\big|_{\partial K},$$

where $\varphi_t(z) = tz$ is the usual dilation map. Thus, using the change of variables formula (A.2.4) we get

$$\int_{t\partial K} f(y)|\nabla_K(y)|^{-1}\,d\mathcal{H}^{n-1}(y) = \int_{\partial K} f(tz)|\nabla \gamma_K(tz)|^{-1} t^{n-1}\,d\mathcal{H}^{n-1}(z)$$

for any $t > 0$. We emphasize once again that the integral in the left-hand side is being taken over the points of $t\partial K$ where γ_K is differentiable. These are precisely the points of the form tz where $z \in \partial K$ is a point where γ_K is differentiable, and the integral in the right-hand side is taken with respect to these points. Moreover, we know from Proposition 2.3.7 that, in this case,

$$|\nabla \gamma_K(tz)|^{-1} = |\nabla \gamma_K(z)|^{-1} = h_K(u_K(z)).$$

Putting all these equalities together yields the desired.

□

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